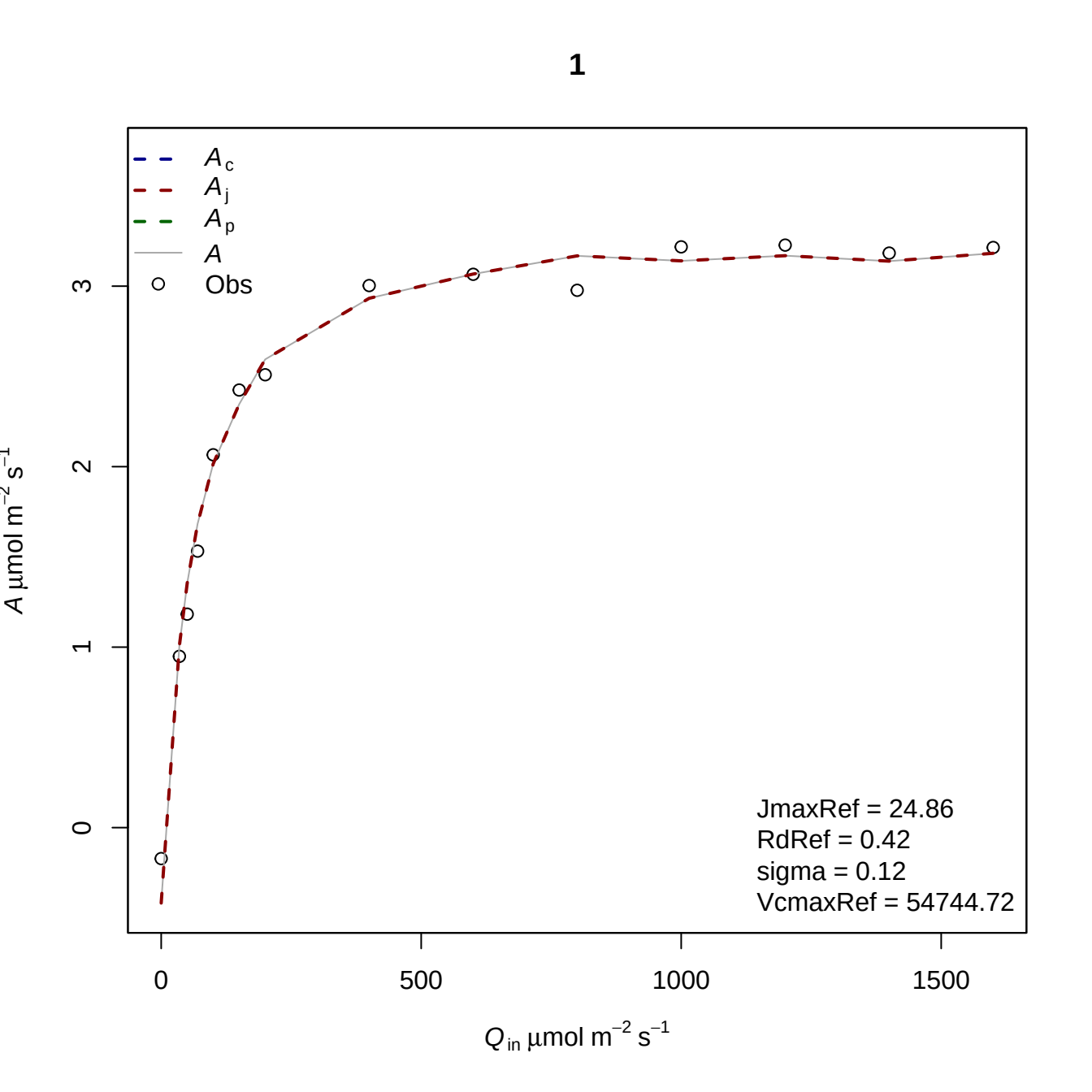
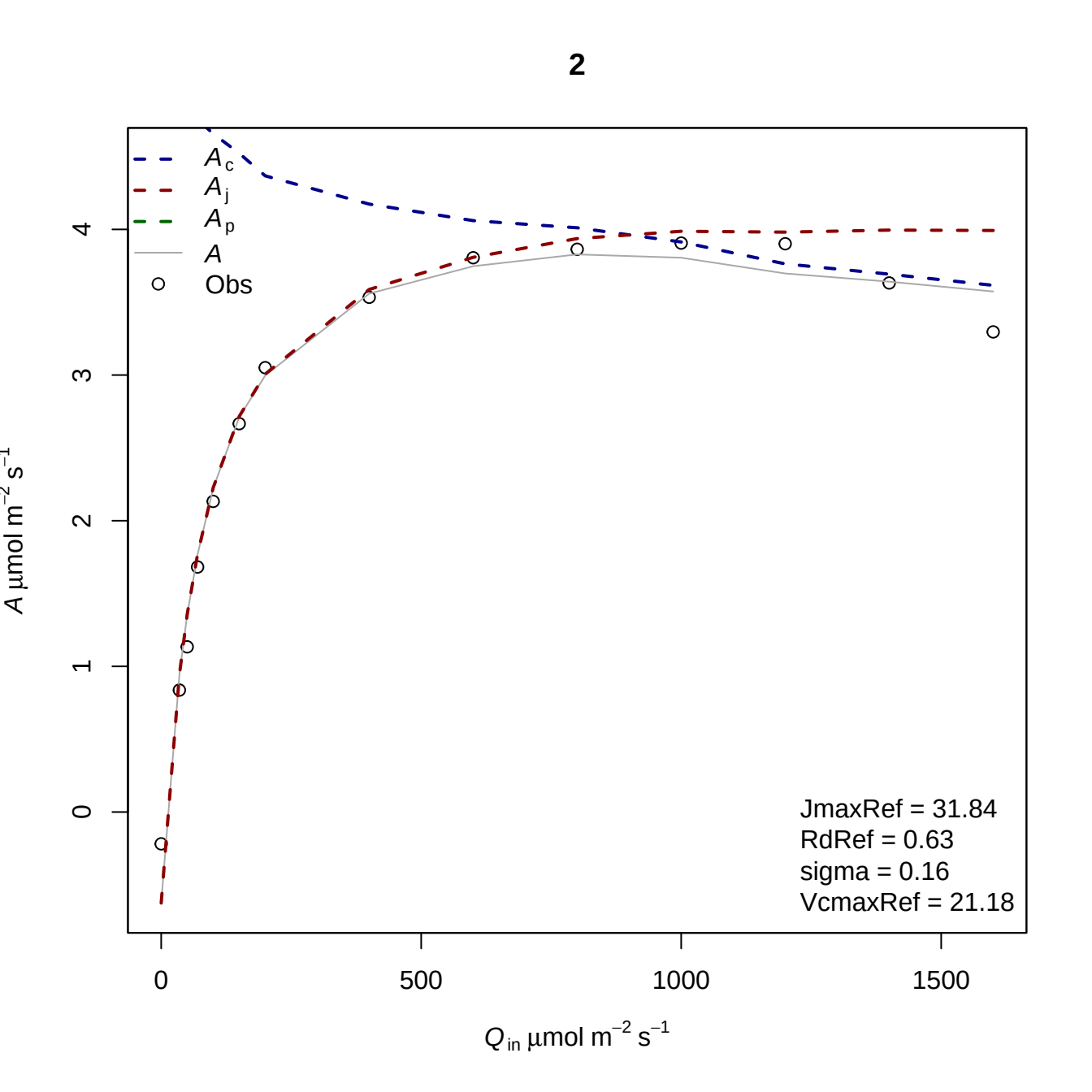
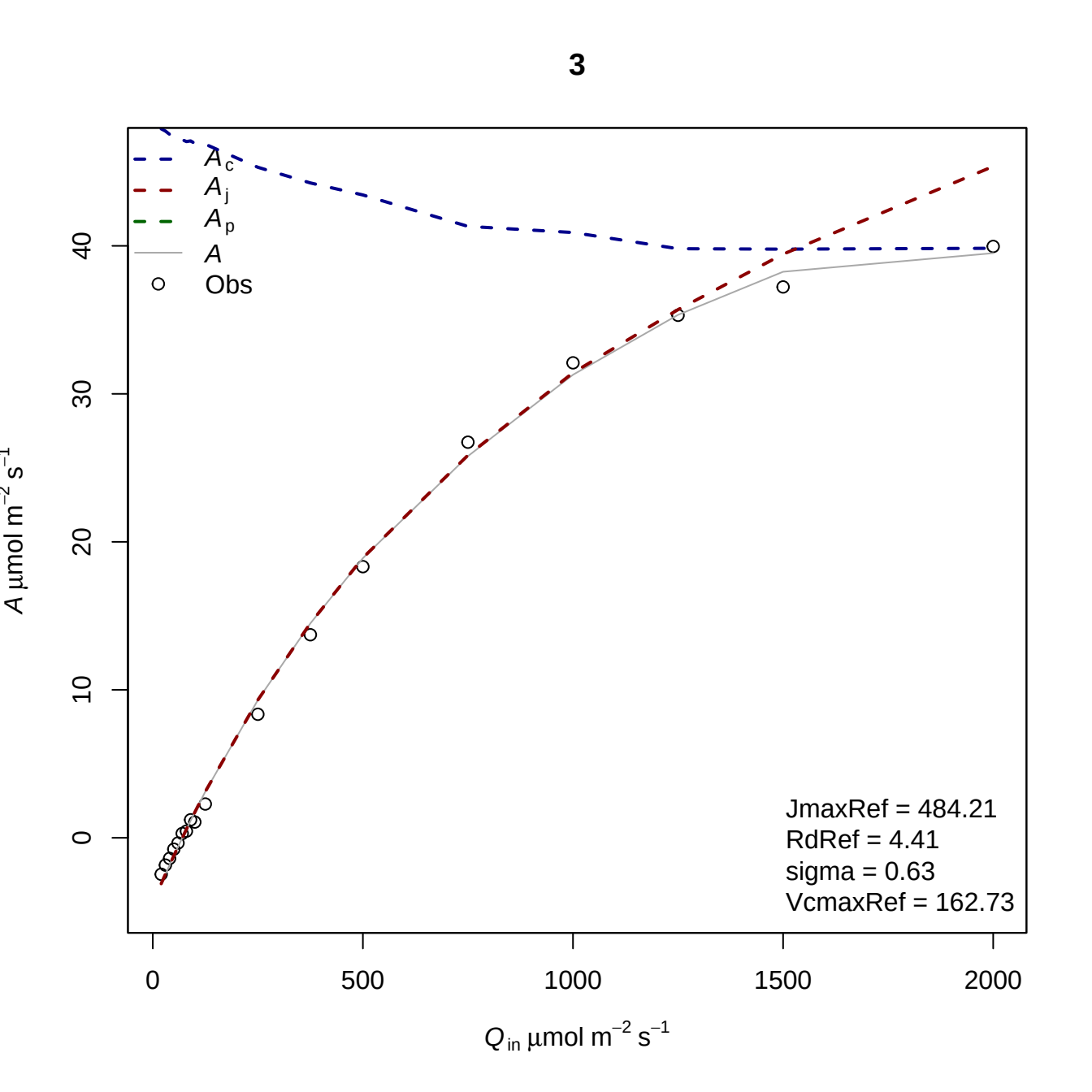
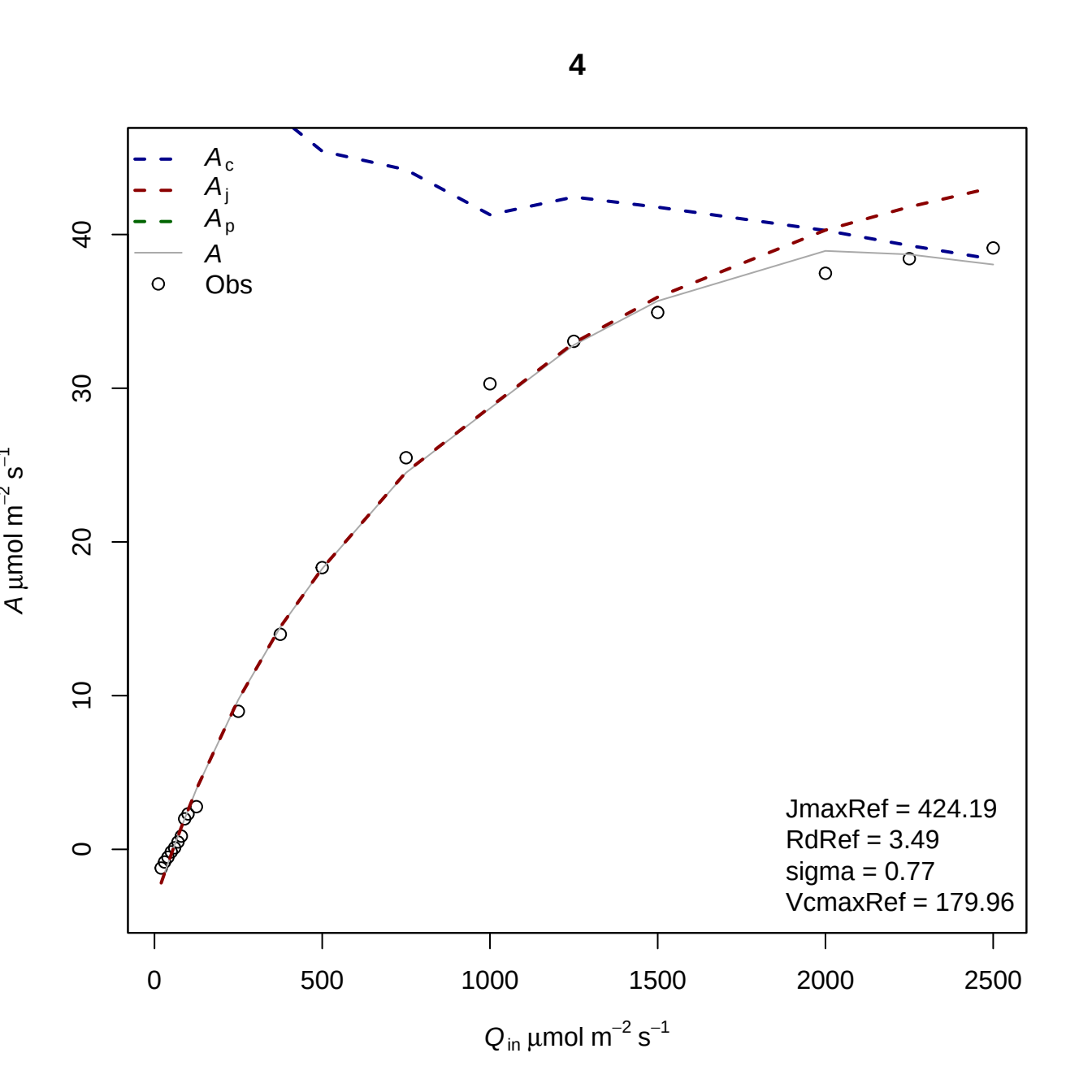
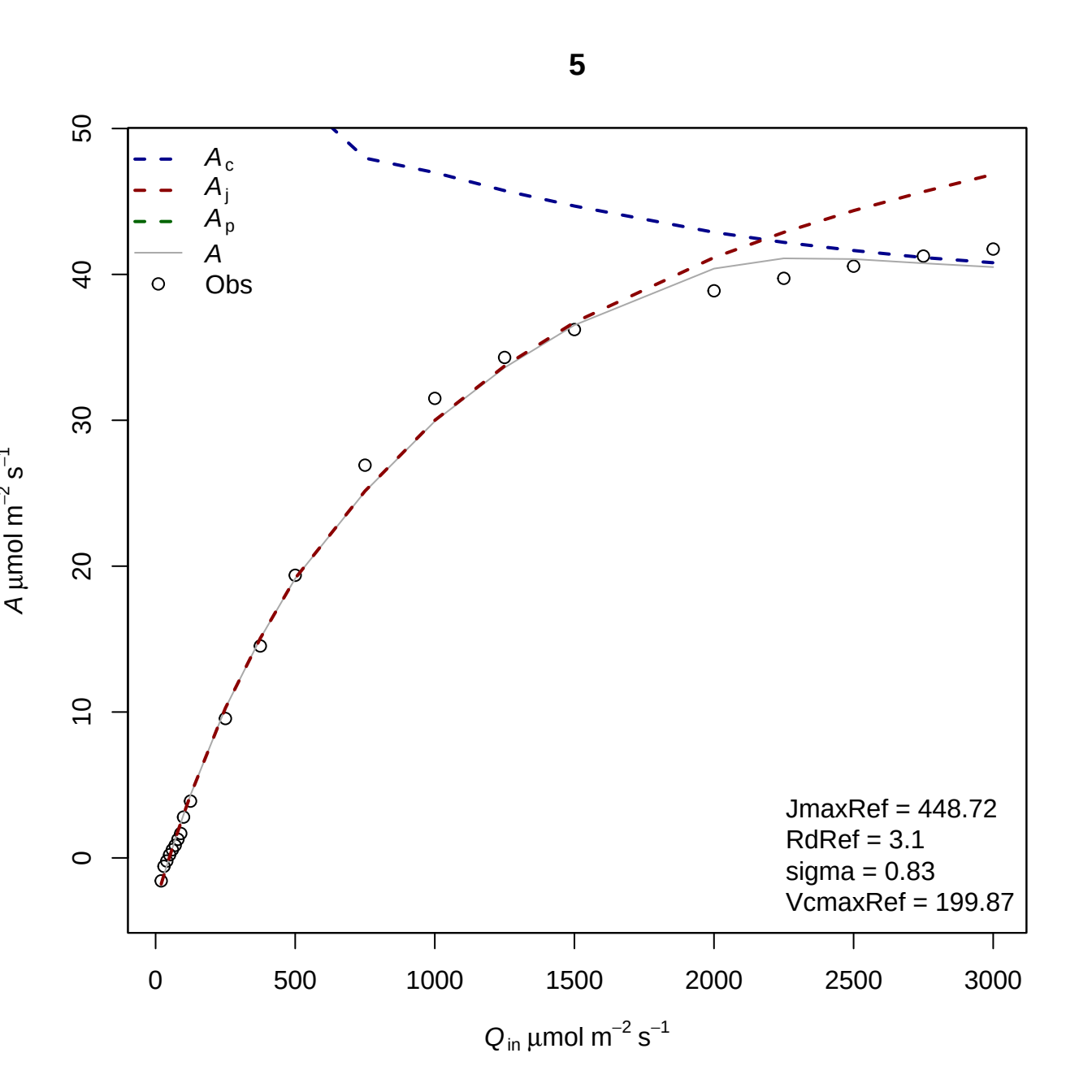
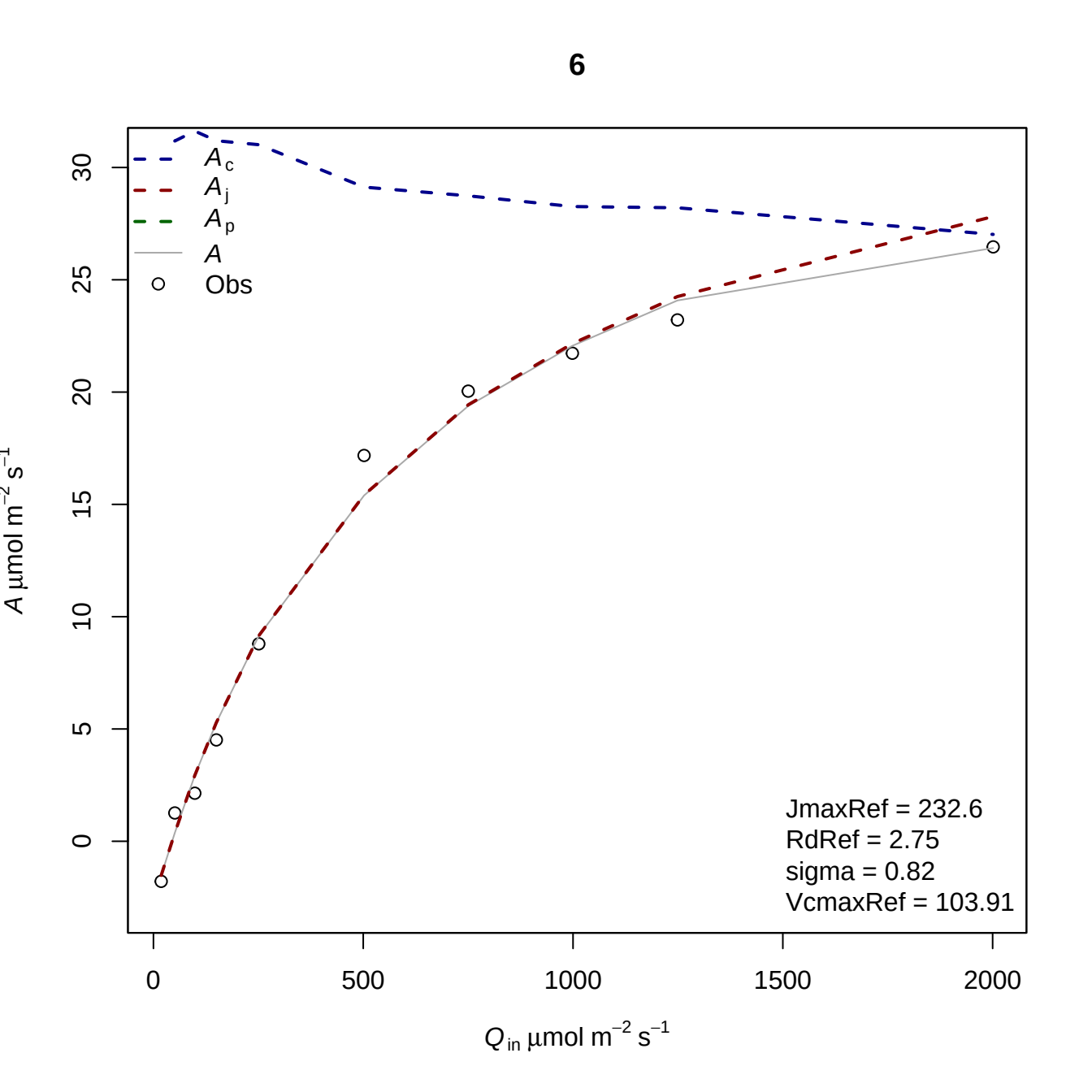

1
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 24.86
RdRef = 0.42
sigma = 0.12
VcmaxRef = 54744.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500

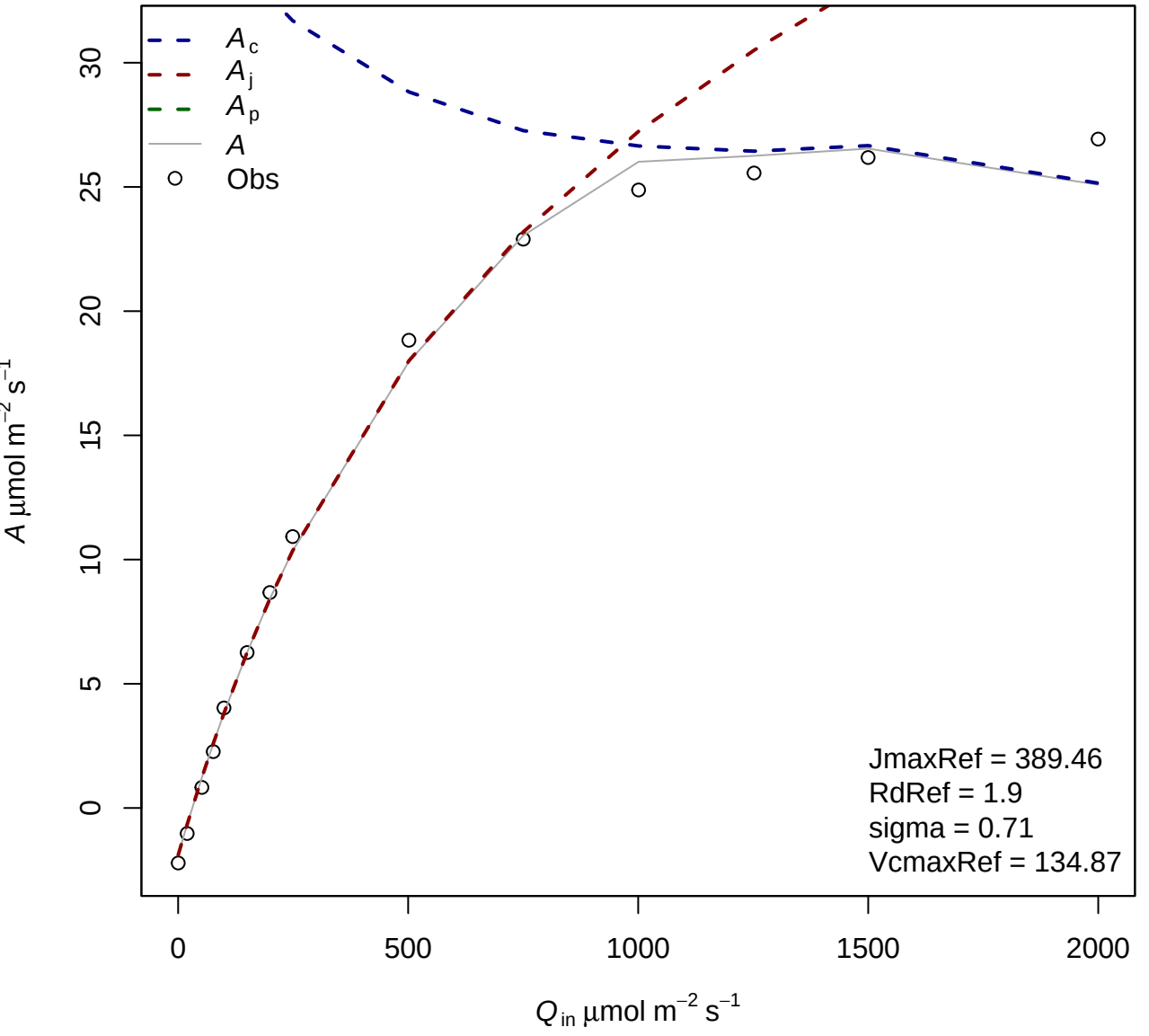

2
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.84
RdRef = 0.63
sigma = 0.16
VcmaxRef = 21.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

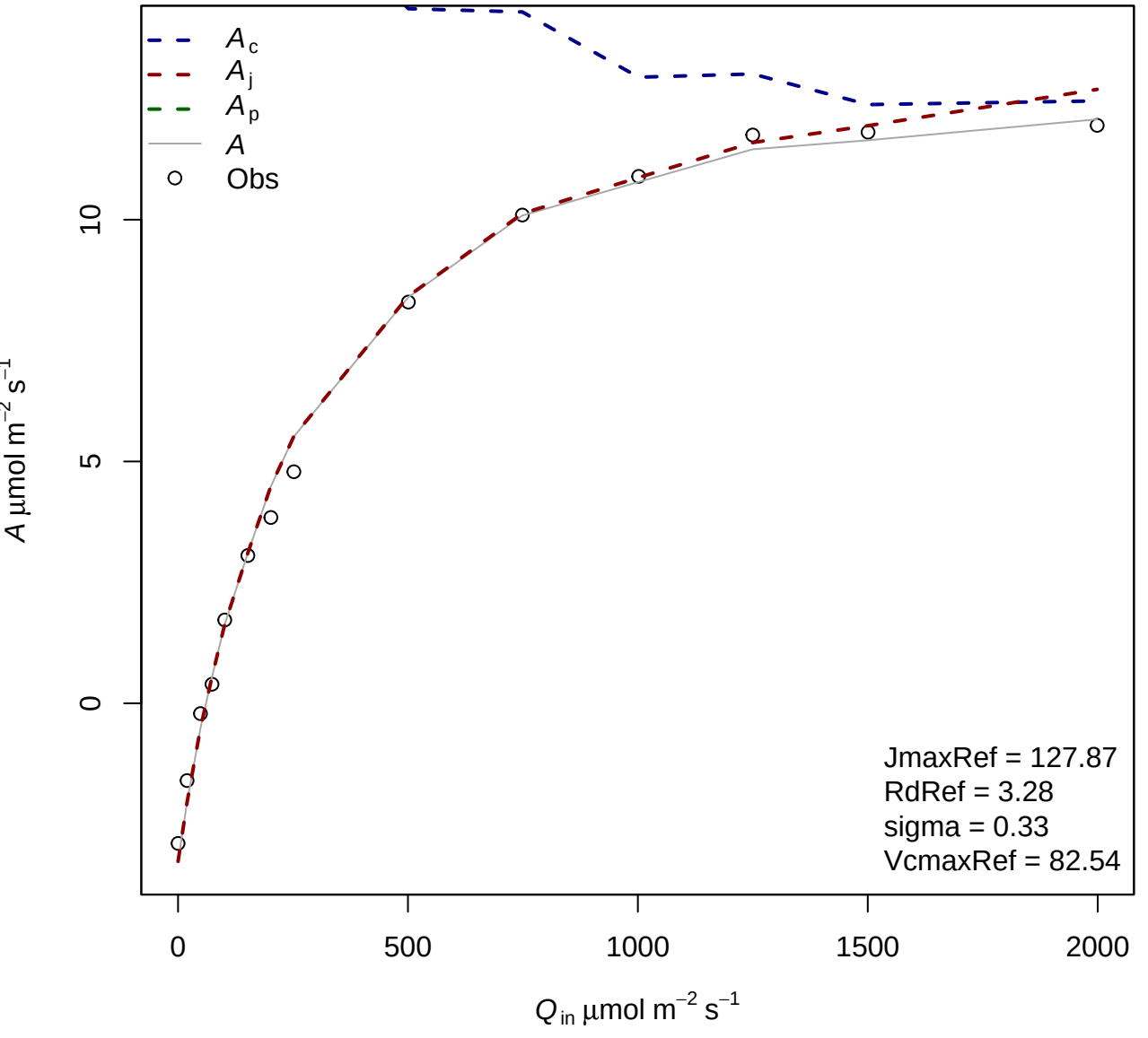

3
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 484.21
RdRef = 4.41
sigma = 0.63
VcmaxRef = 162.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000

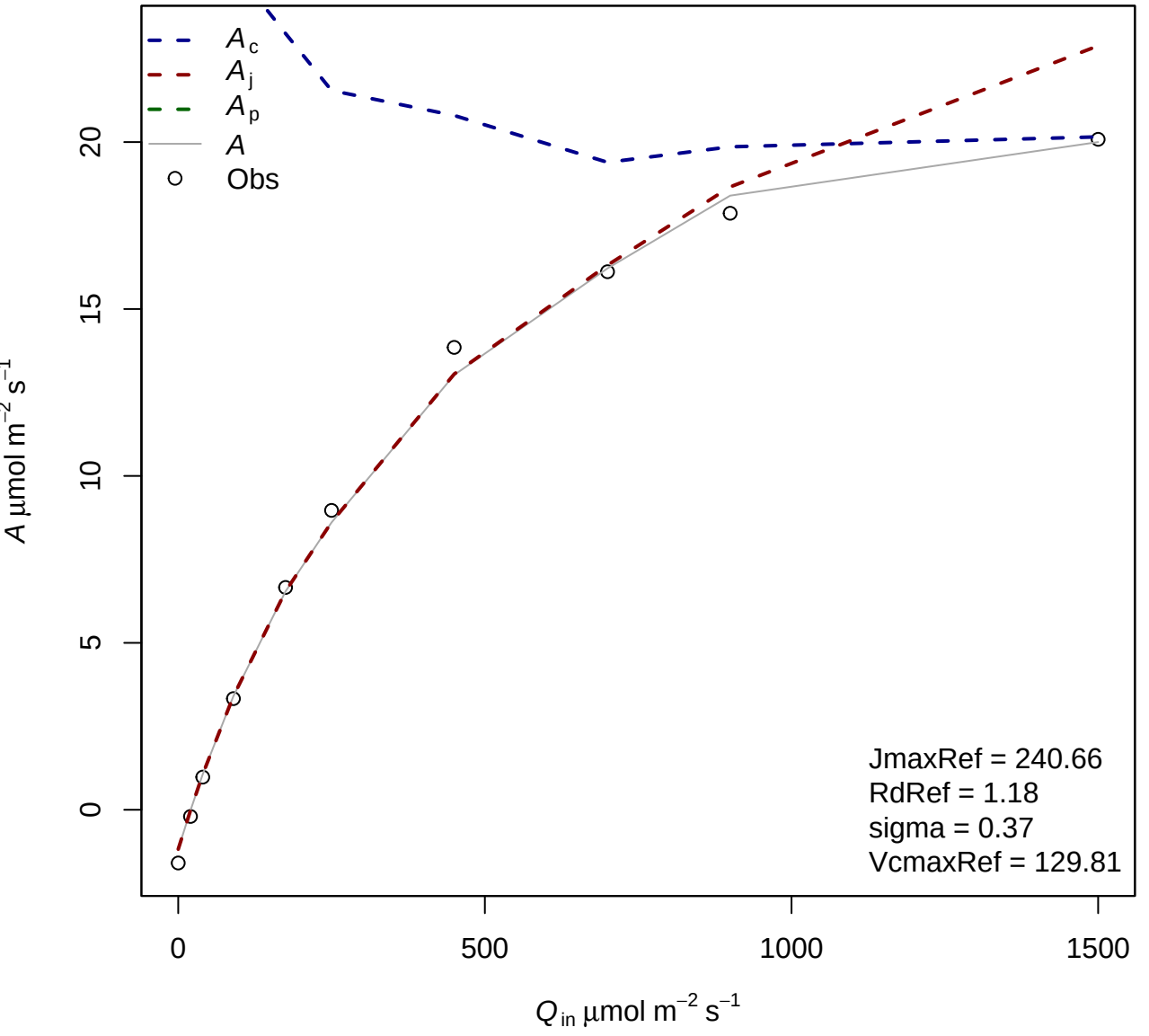

4
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 424.19
RdRef = 3.49
sigma = 0.77
VcmaxRef = 179.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000
2500

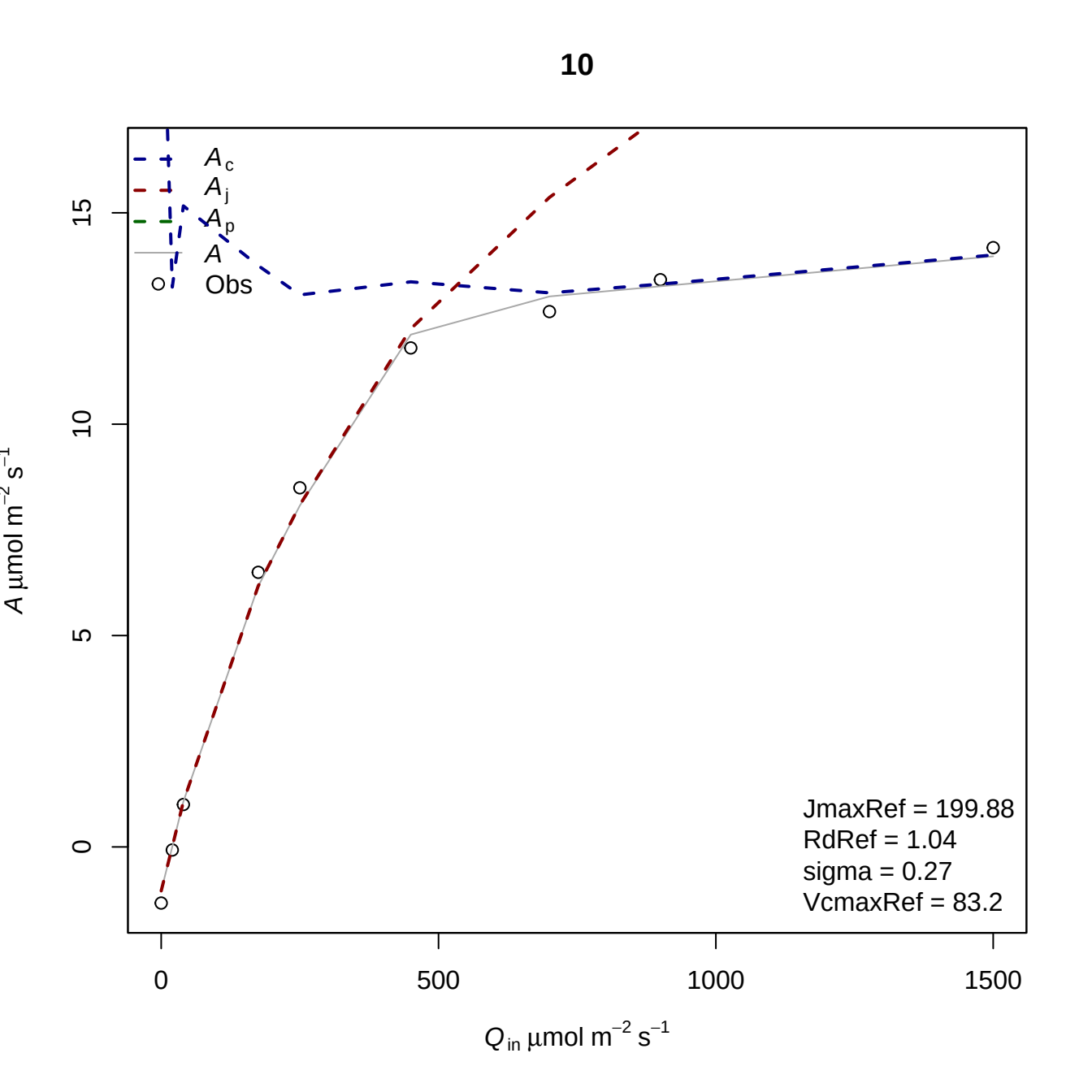

5
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 448.72
RdRef = 3.1
sigma = 0.83
VcmaxRef = 199.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

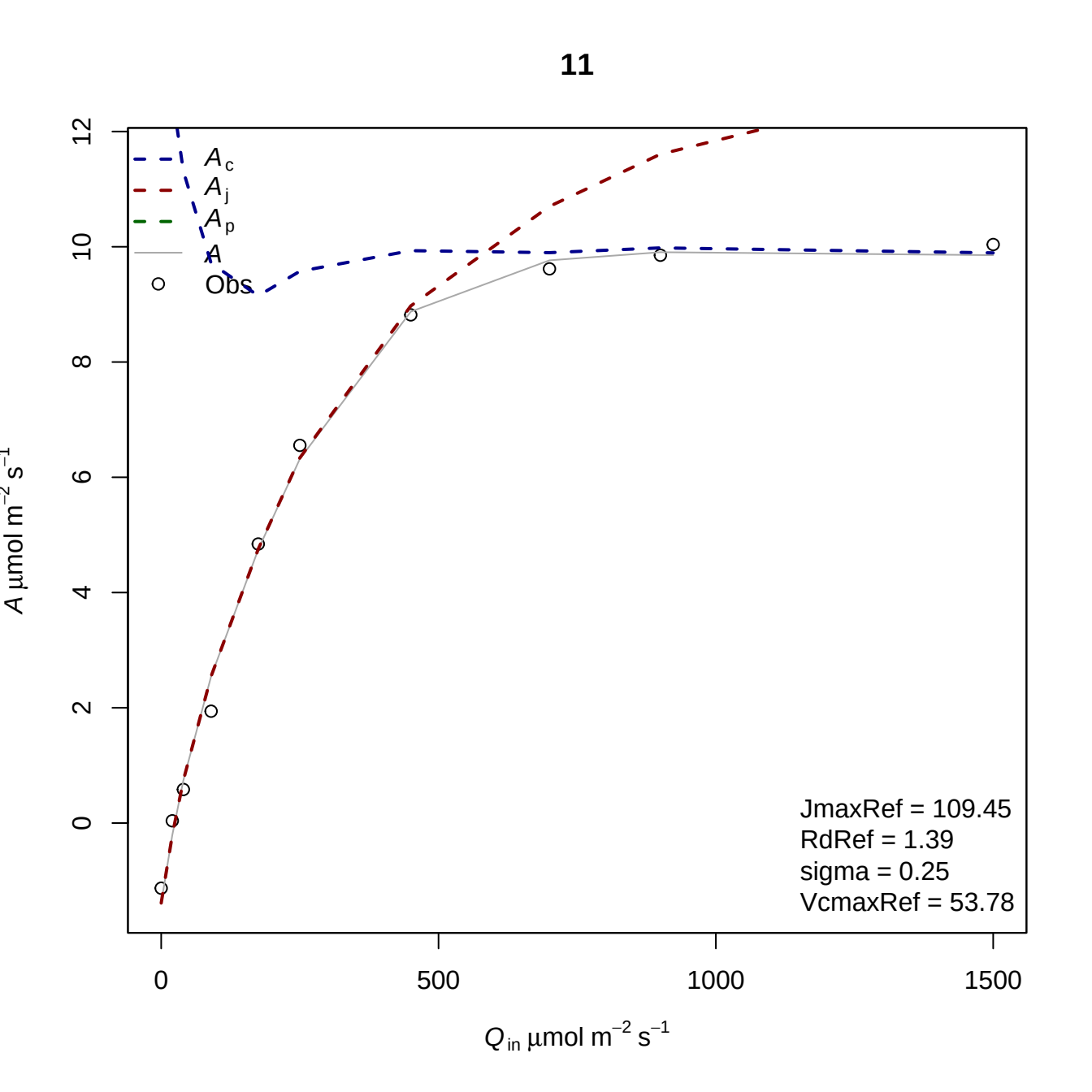

6
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 232.6
RdRef = 2.75
sigma = 0.82
VcmaxRef = 103.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

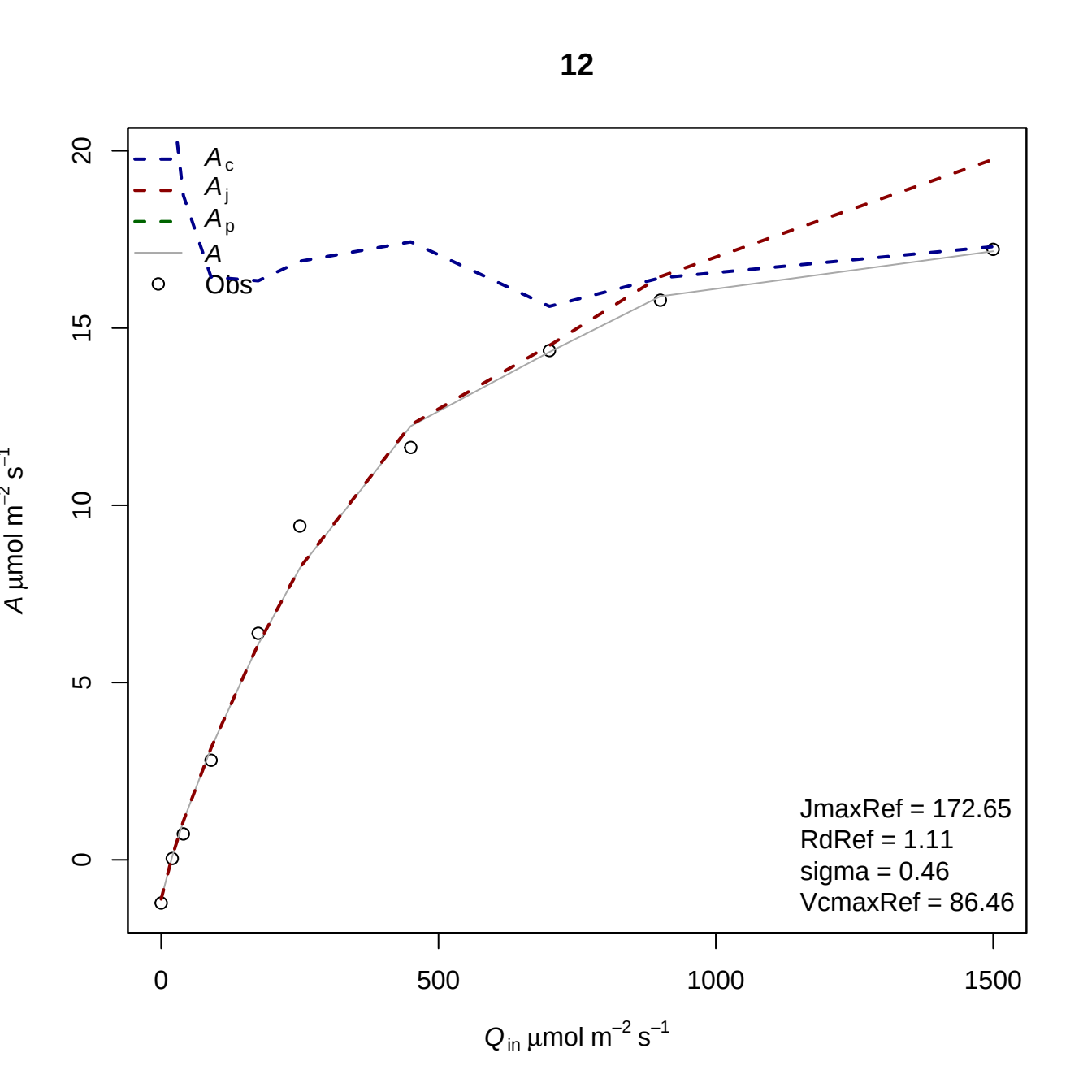

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 389.46
RdRef = 1.9
sigma = 0.71
VcmaxRef = 134.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

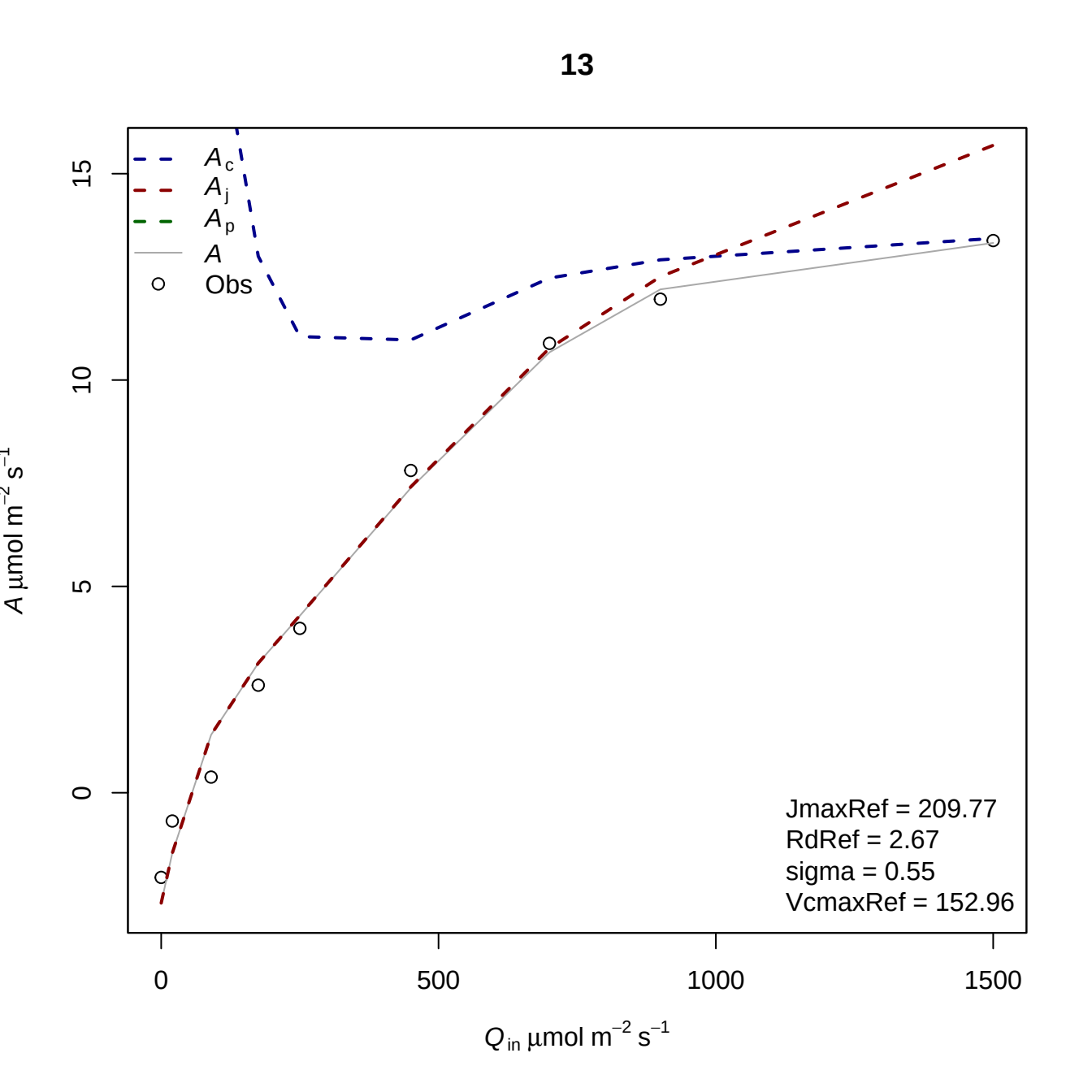

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.87
RdRef = 3.28
sigma = 0.33
VcmaxRef = 82.54
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

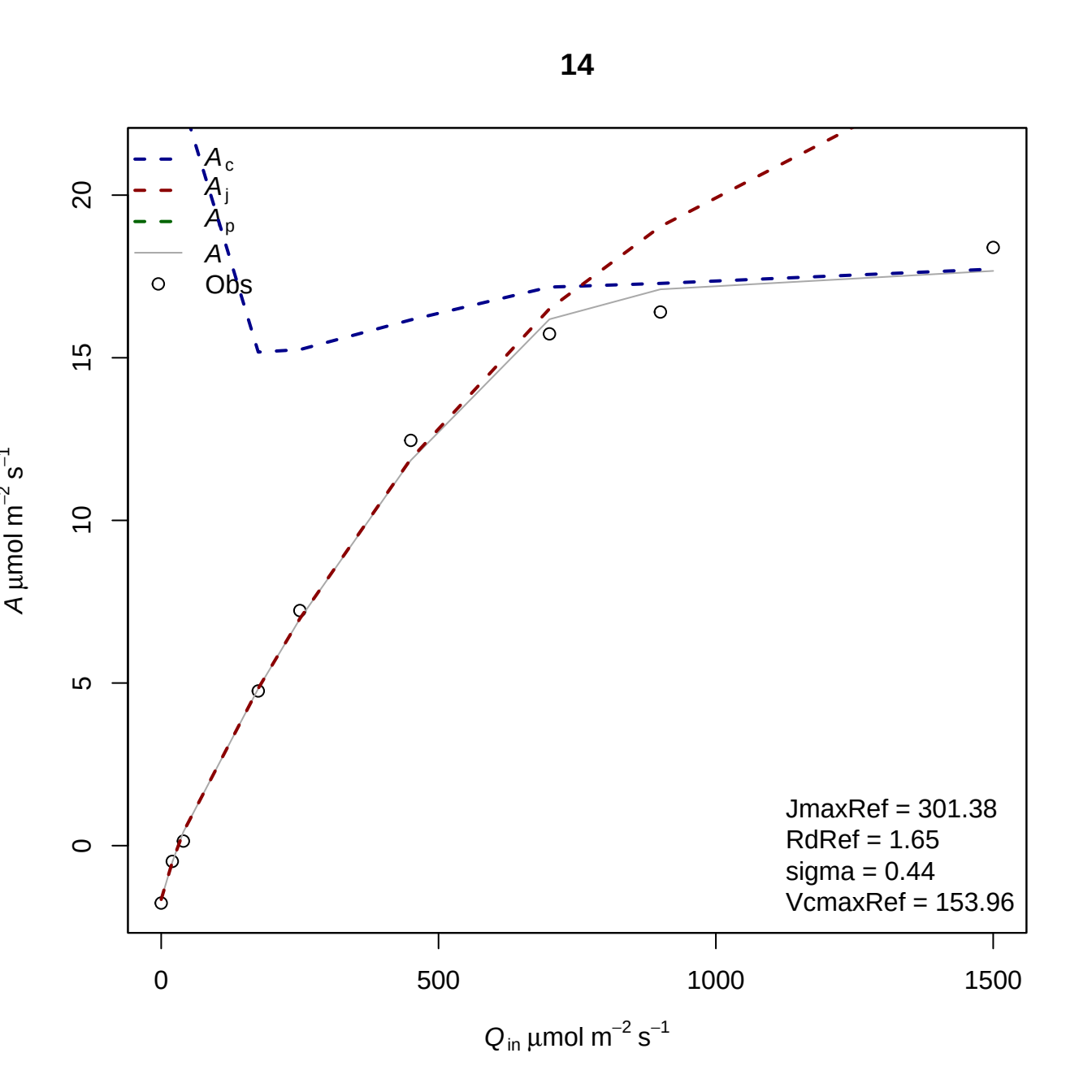

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 240.66
RdRef = 1.18
sigma = 0.37
VcmaxRef = 129.81
0
5
10
15
20
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

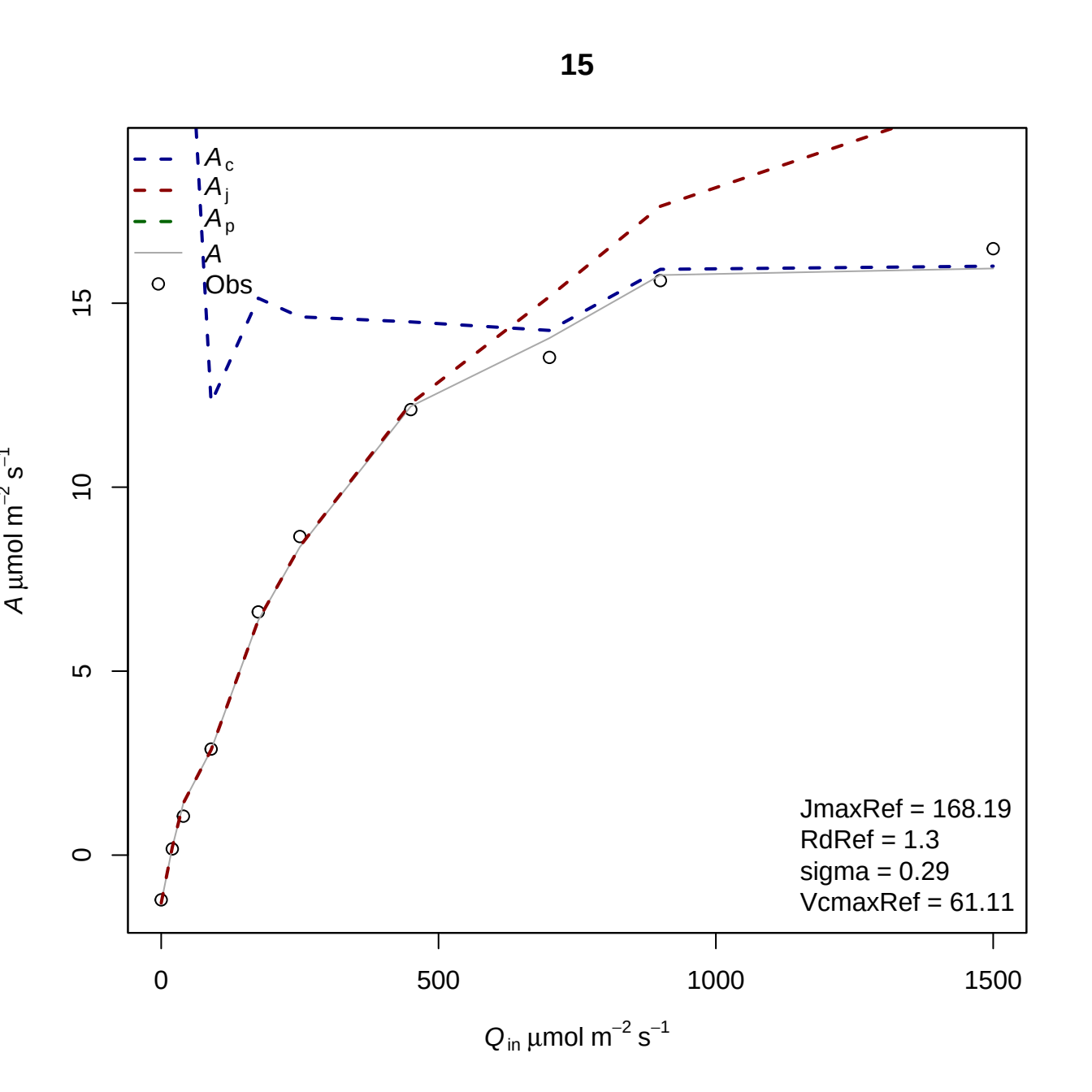

10
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 199.88
RdRef = 1.04
sigma = 0.27
VcmaxRef = 83.2
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

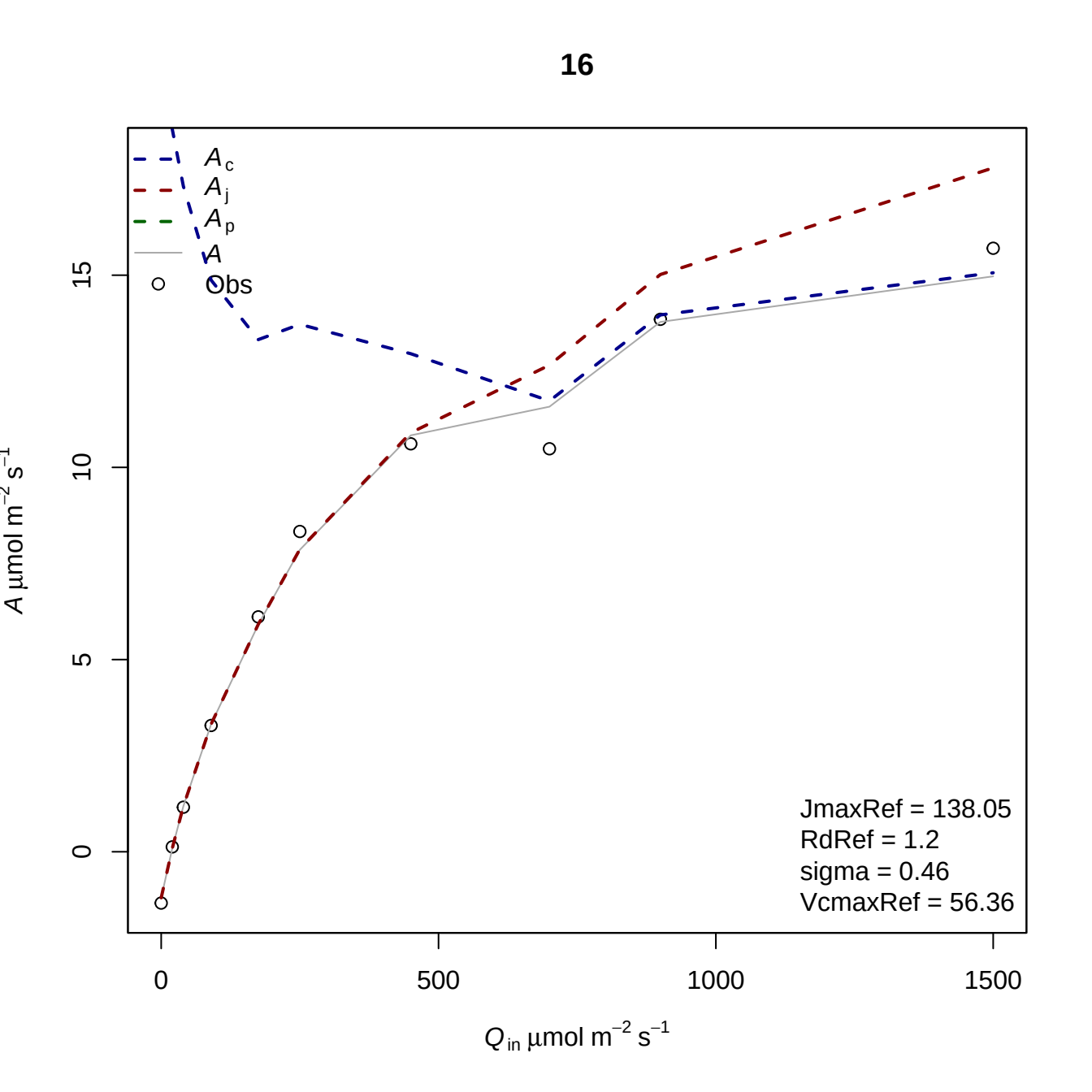

11
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.45
RdRef = 1.39
sigma = 0.25
VcmaxRef = 53.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

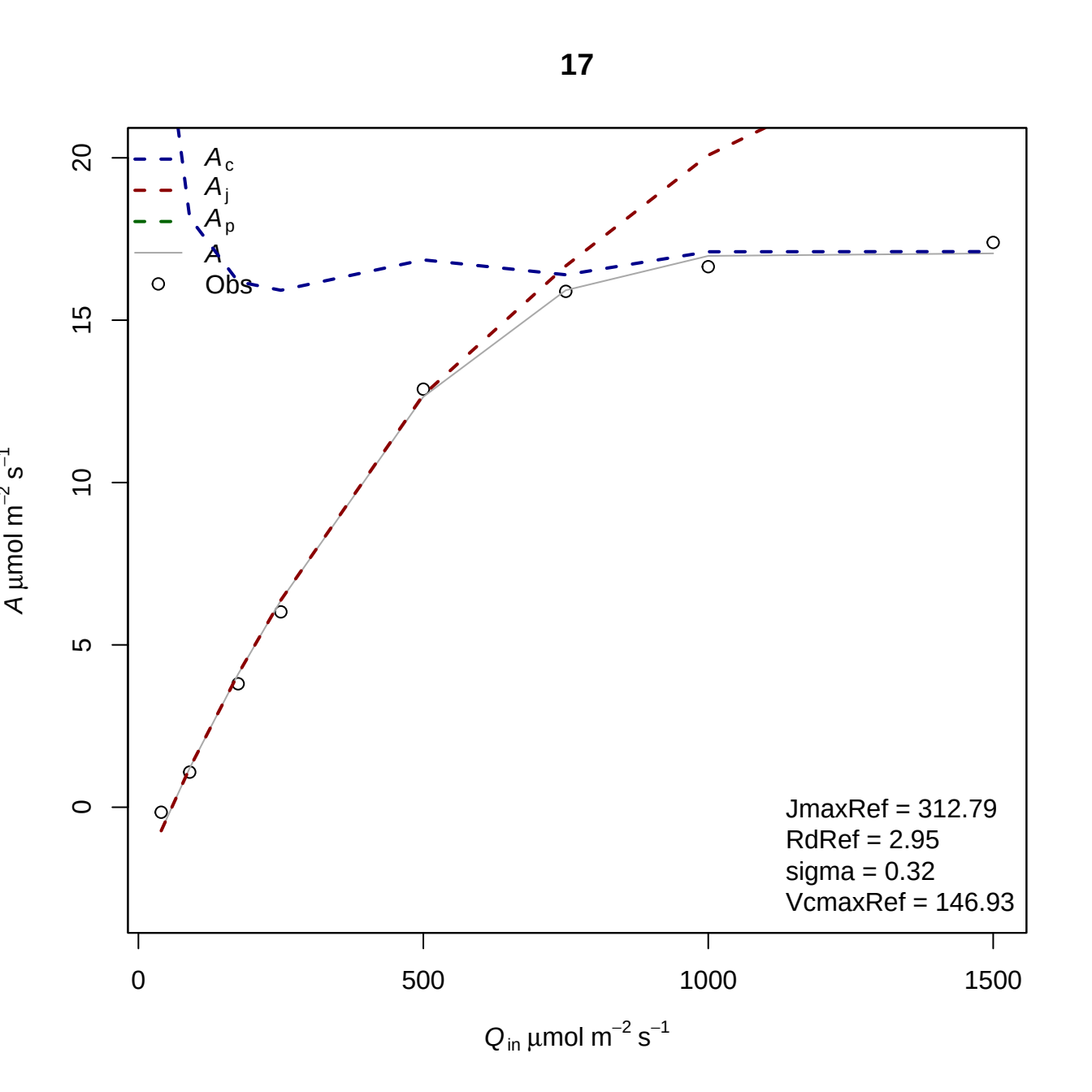

12
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 172.65
RdRef = 1.11
sigma = 0.46
VcmaxRef = 86.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

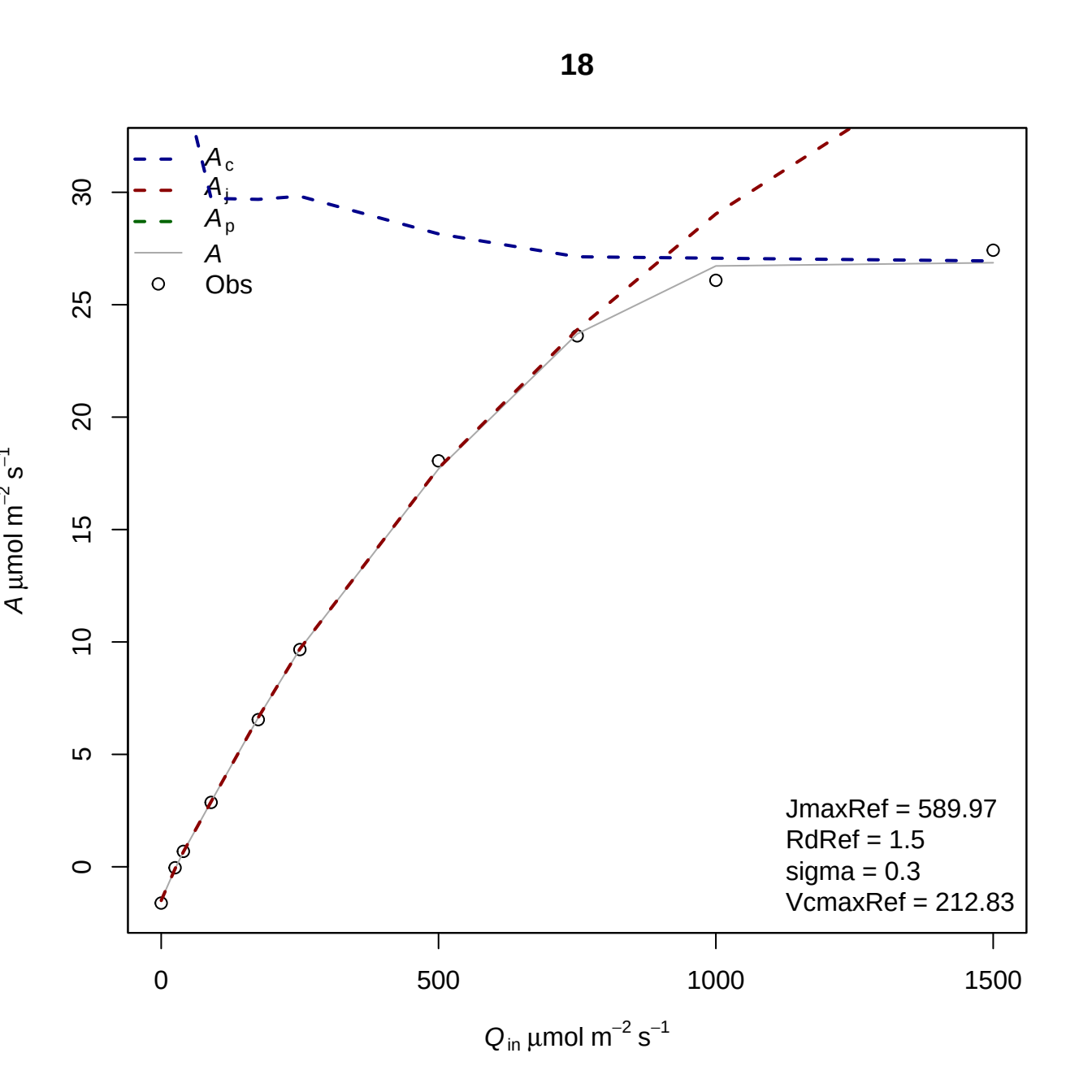

13
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 209.77
RdRef = 2.67
sigma = 0.55
VcmaxRef = 152.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

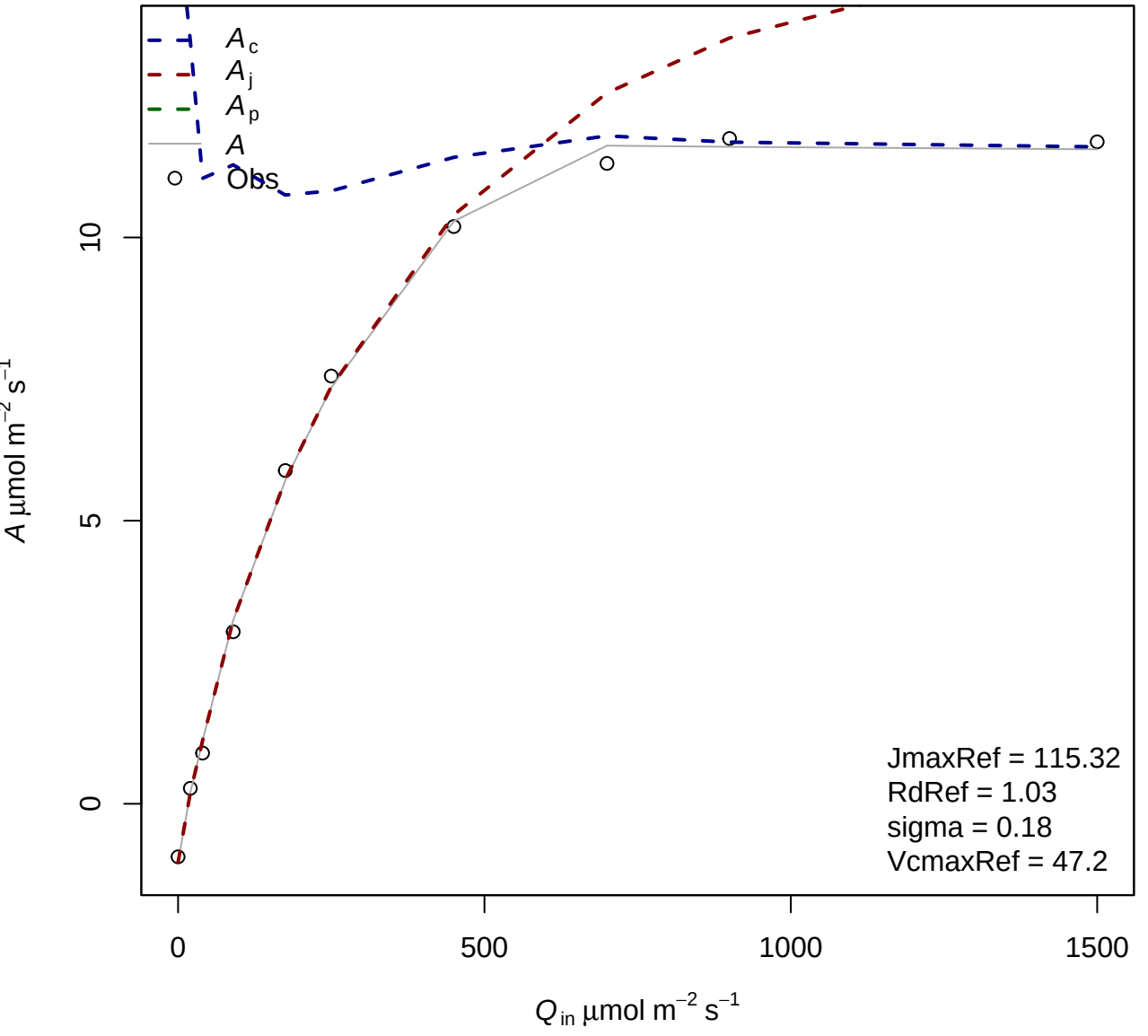

14
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 301.38
RdRef = 1.65
sigma = 0.44
VcmaxRef = 153.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

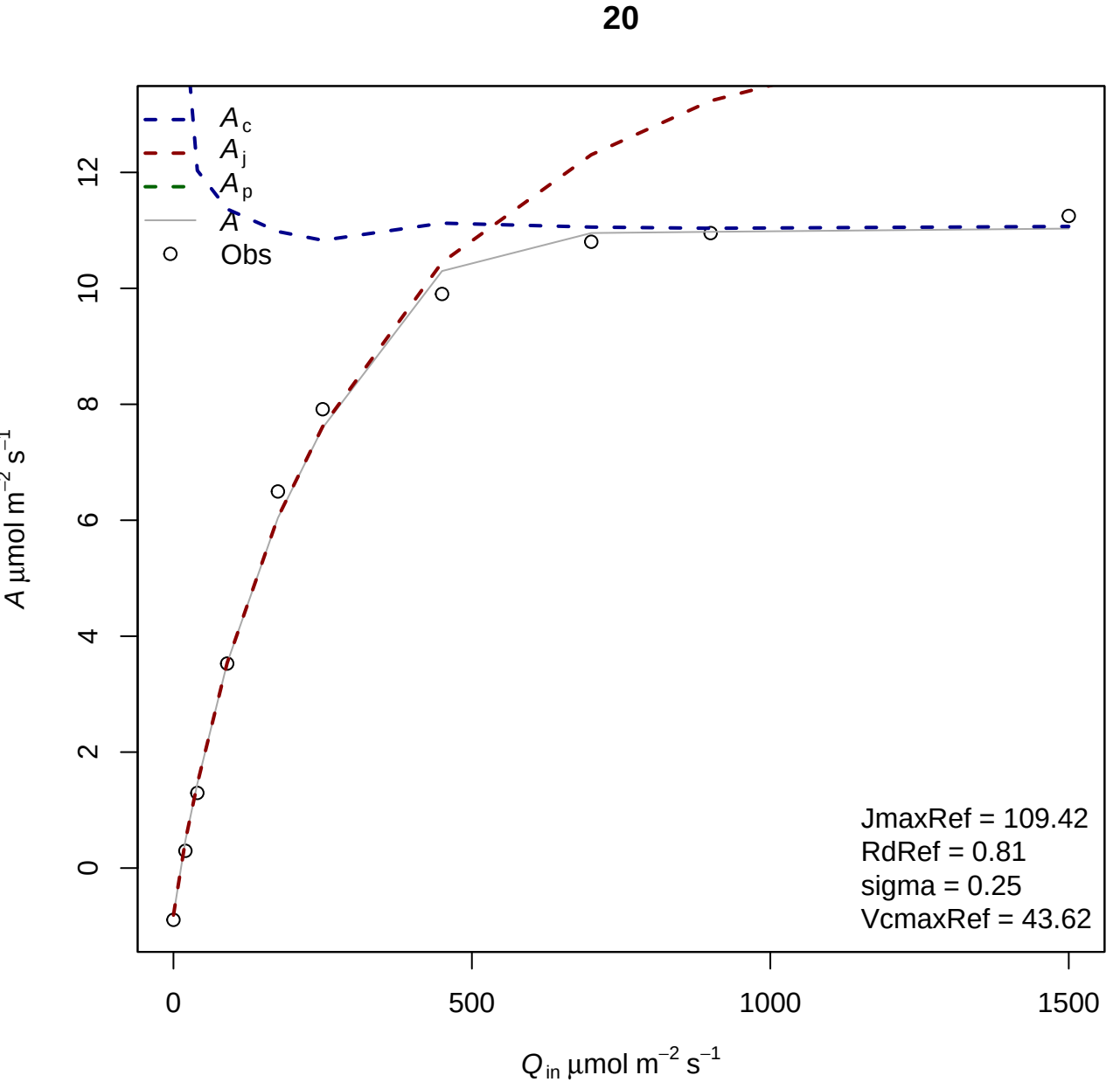

15
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 168.19
RdRef = 1.3
sigma = 0.29
VcmaxRef = 61.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

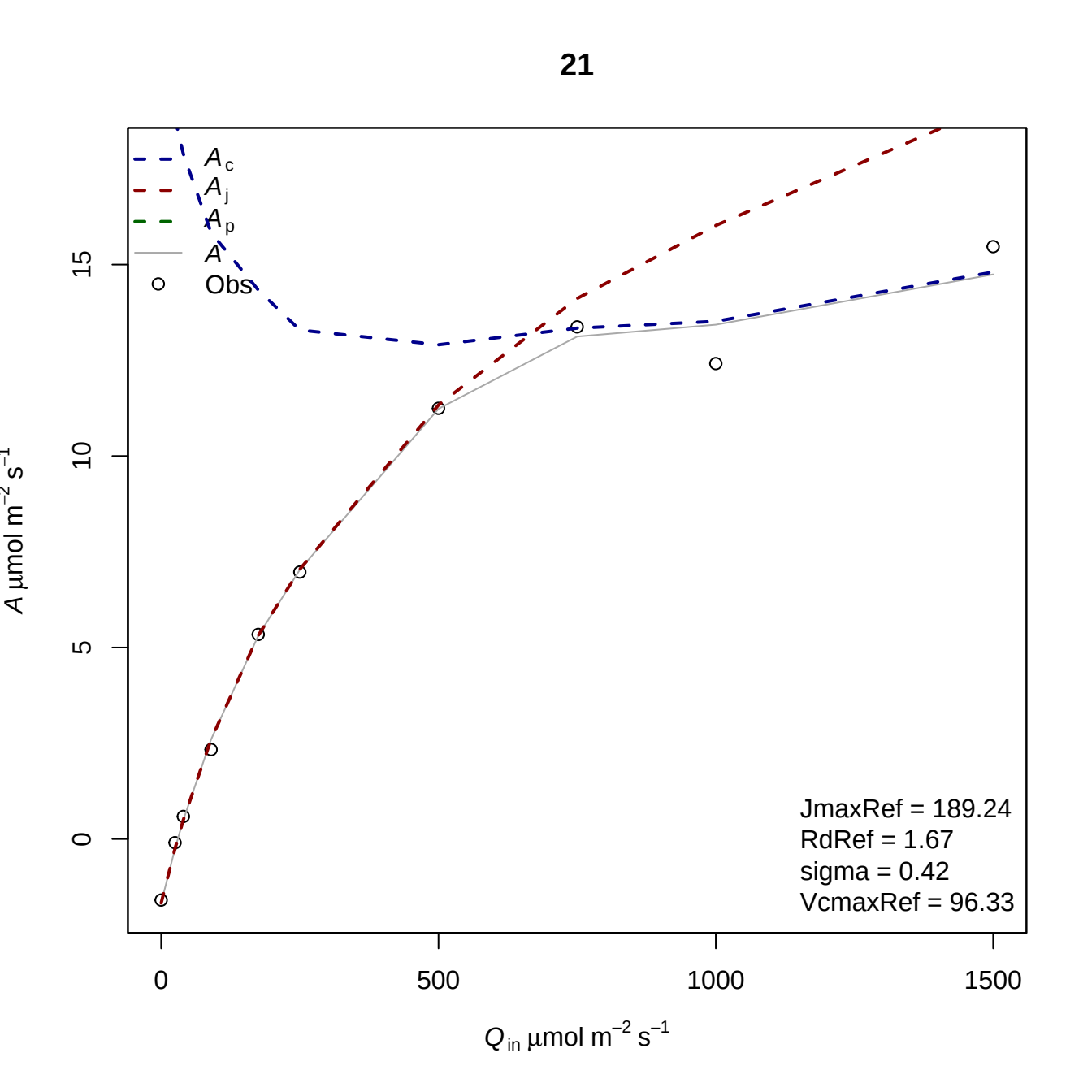

16
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.05
RdRef = 1.2
sigma = 0.46
VcmaxRef = 56.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

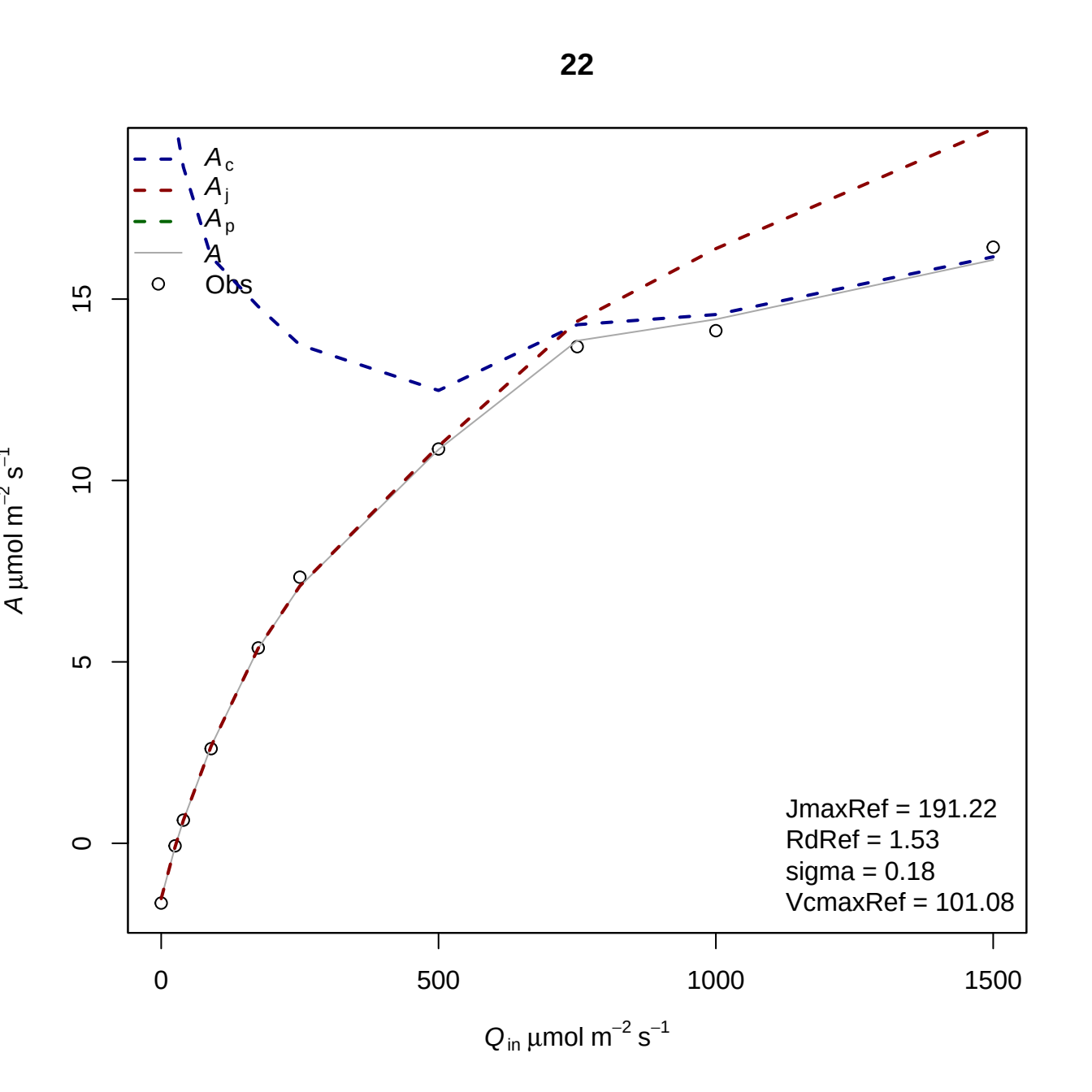

17
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 312.79
RdRef = 2.95
sigma = 0.32
VcmaxRef = 146.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

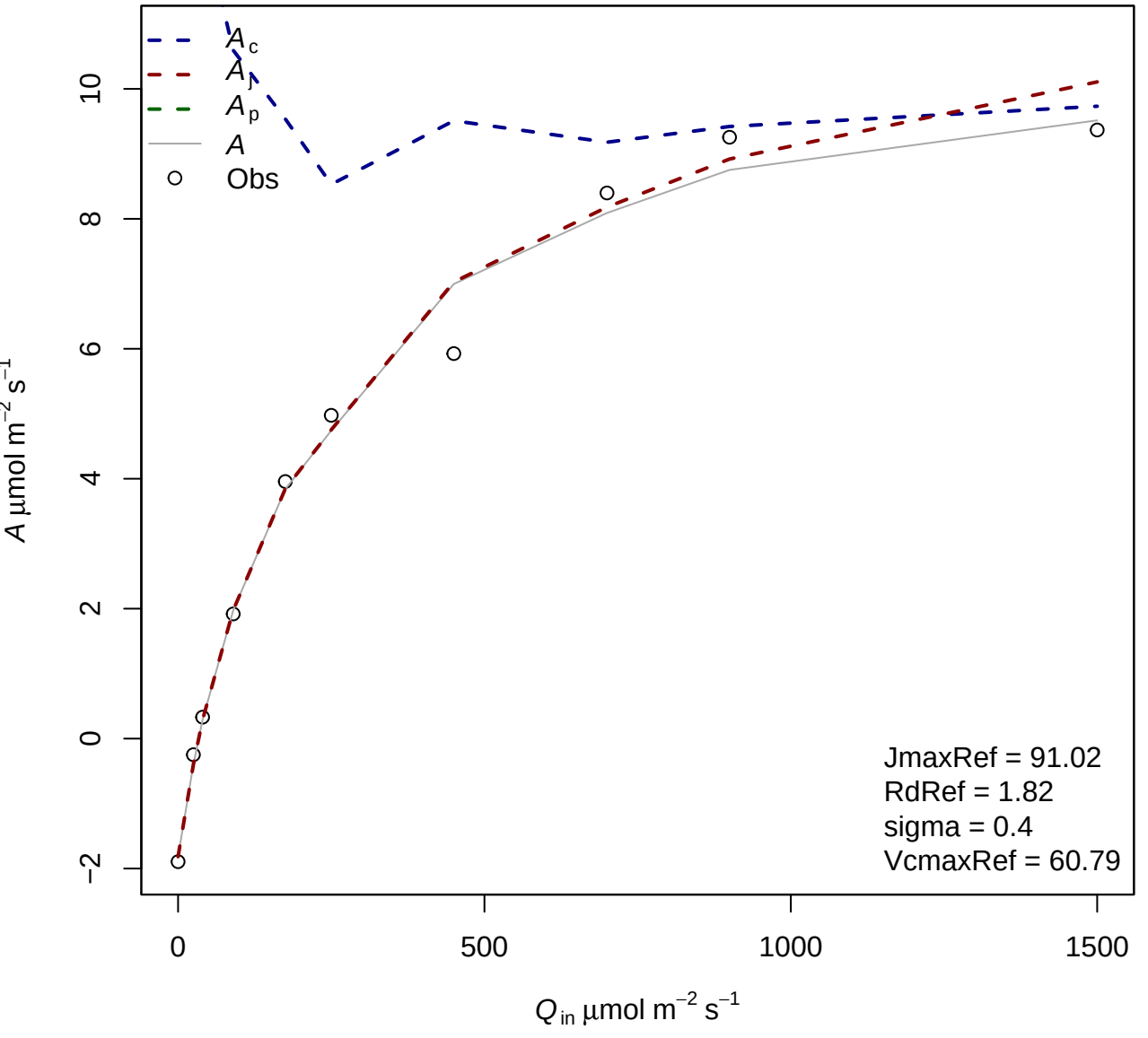

18
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 589.97
RdRef = 1.5
sigma = 0.3
VcmaxRef = 212.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500

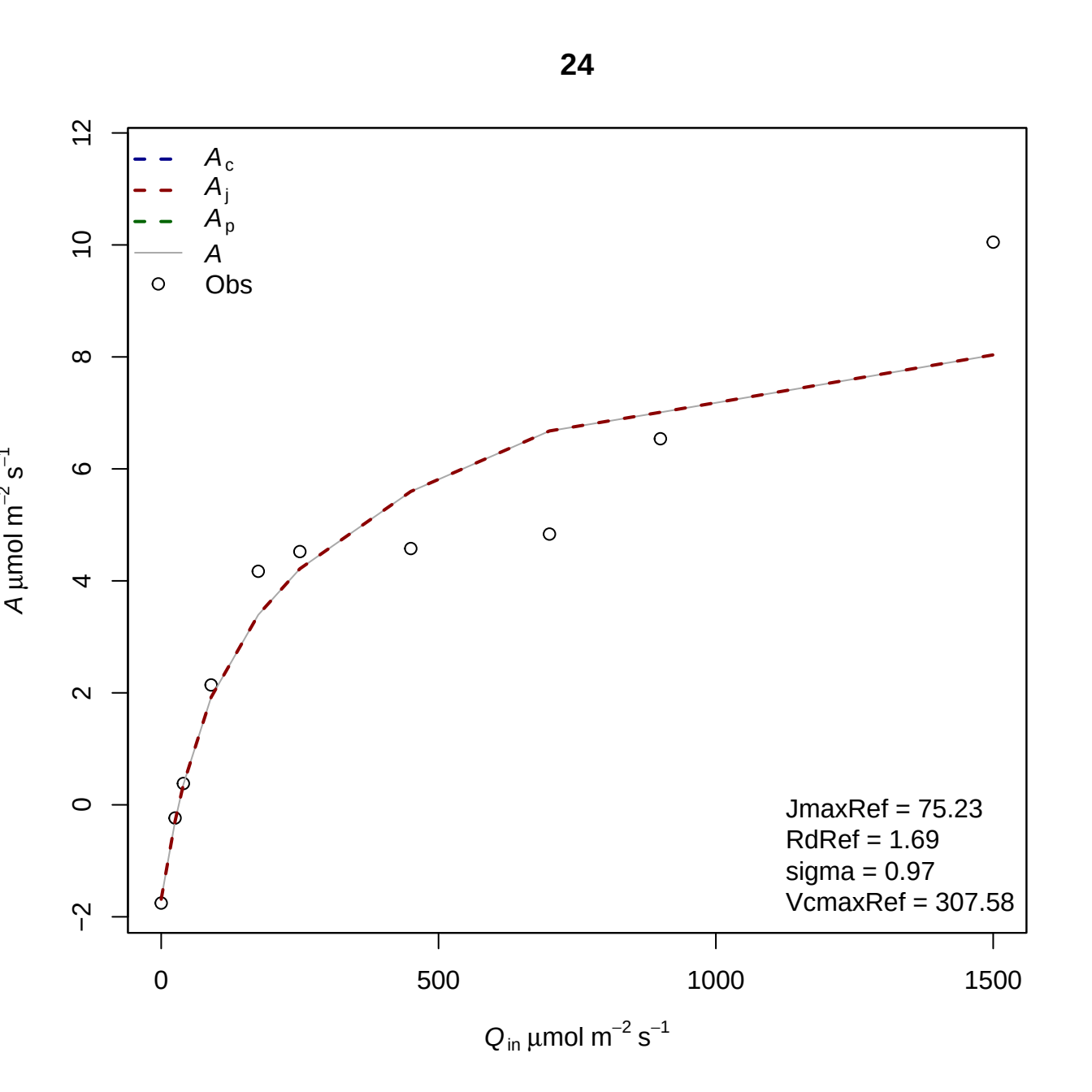

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.32
RdRef = 1.03
sigma = 0.18
VcmaxRef = 47.2
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

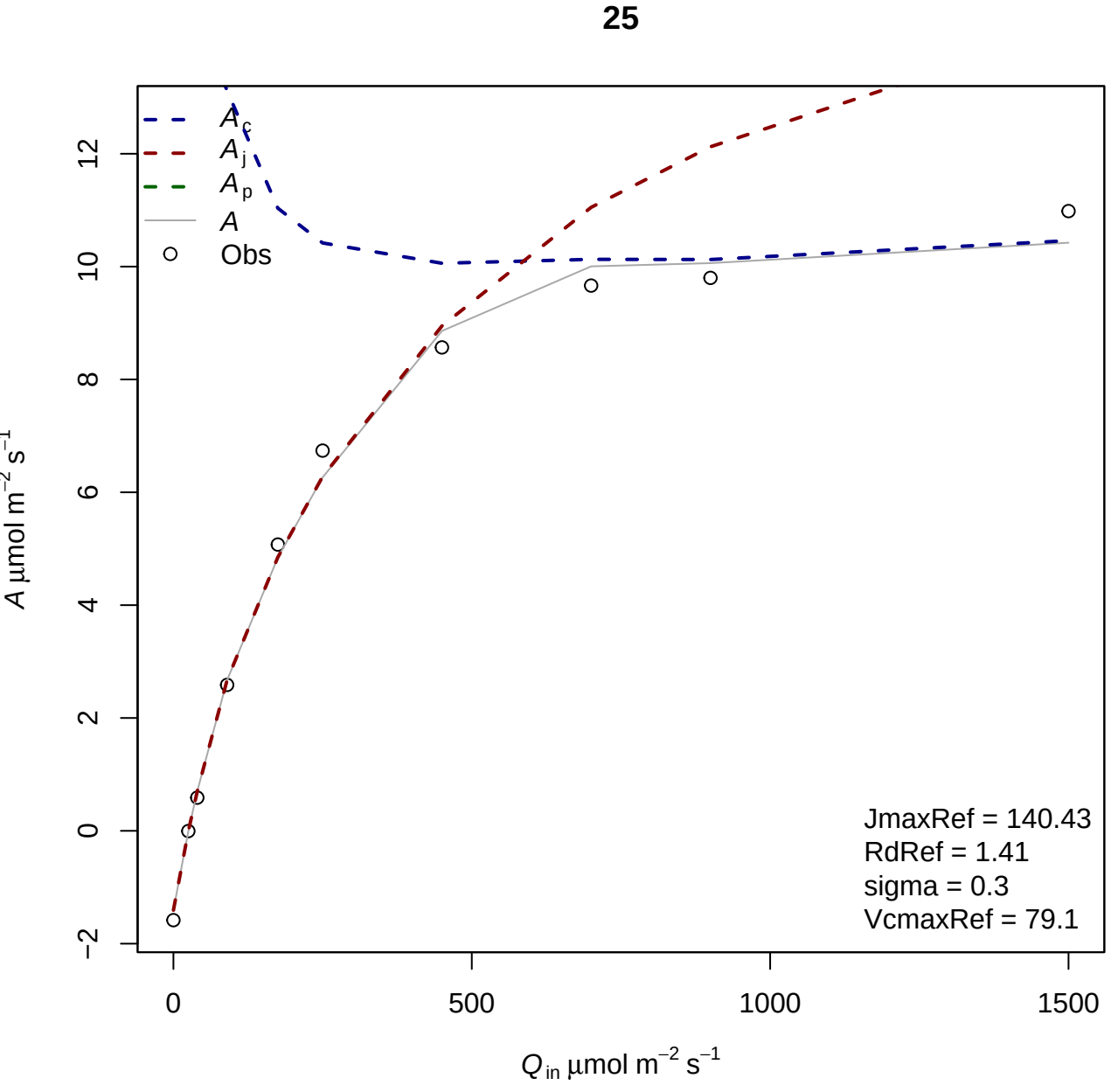

20
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.42
RdRef = 0.81
sigma = 0.25
VcmaxRef = 43.62
0
2
4
6
8
10
12
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

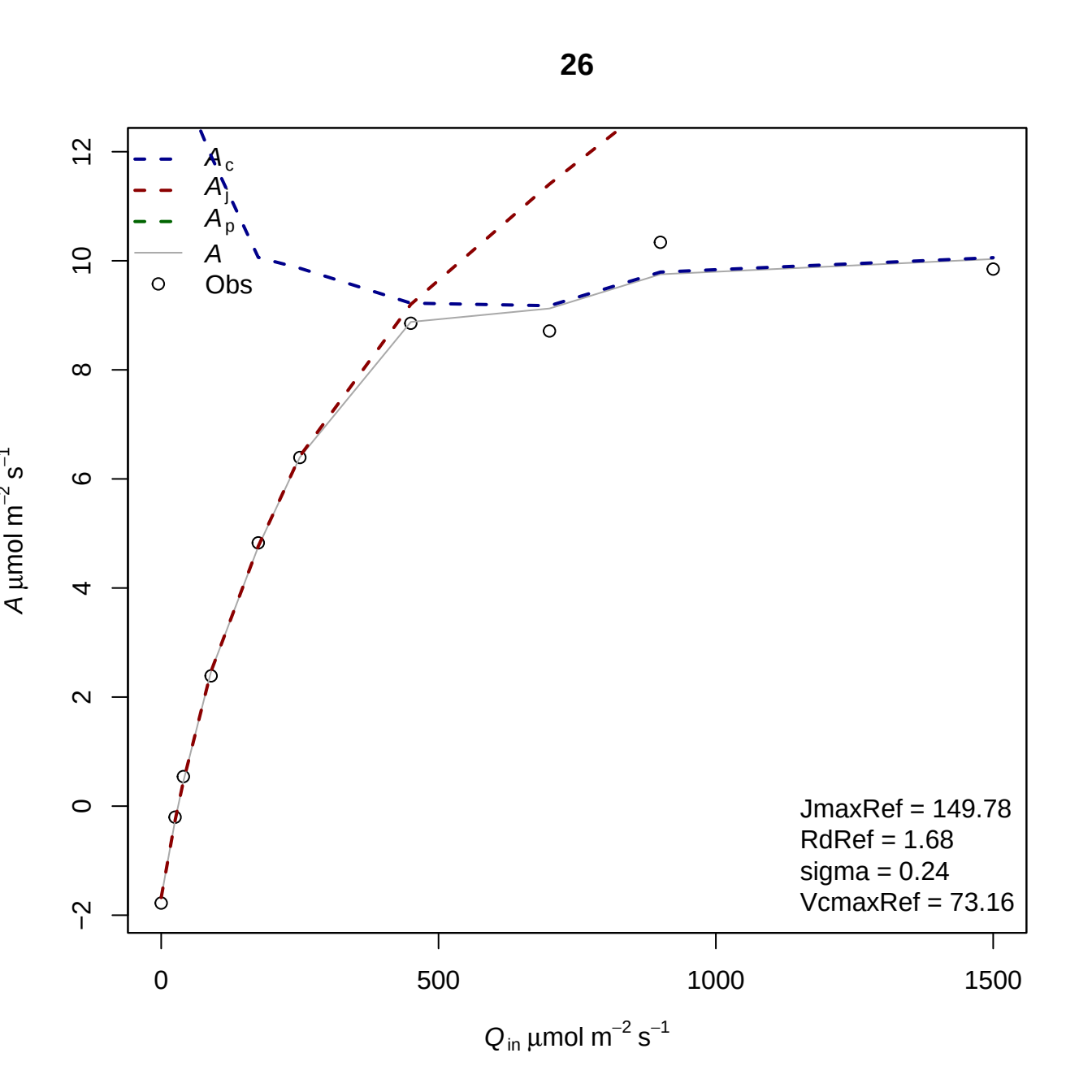

21
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 189.24
RdRef = 1.67
sigma = 0.42
VcmaxRef = 96.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

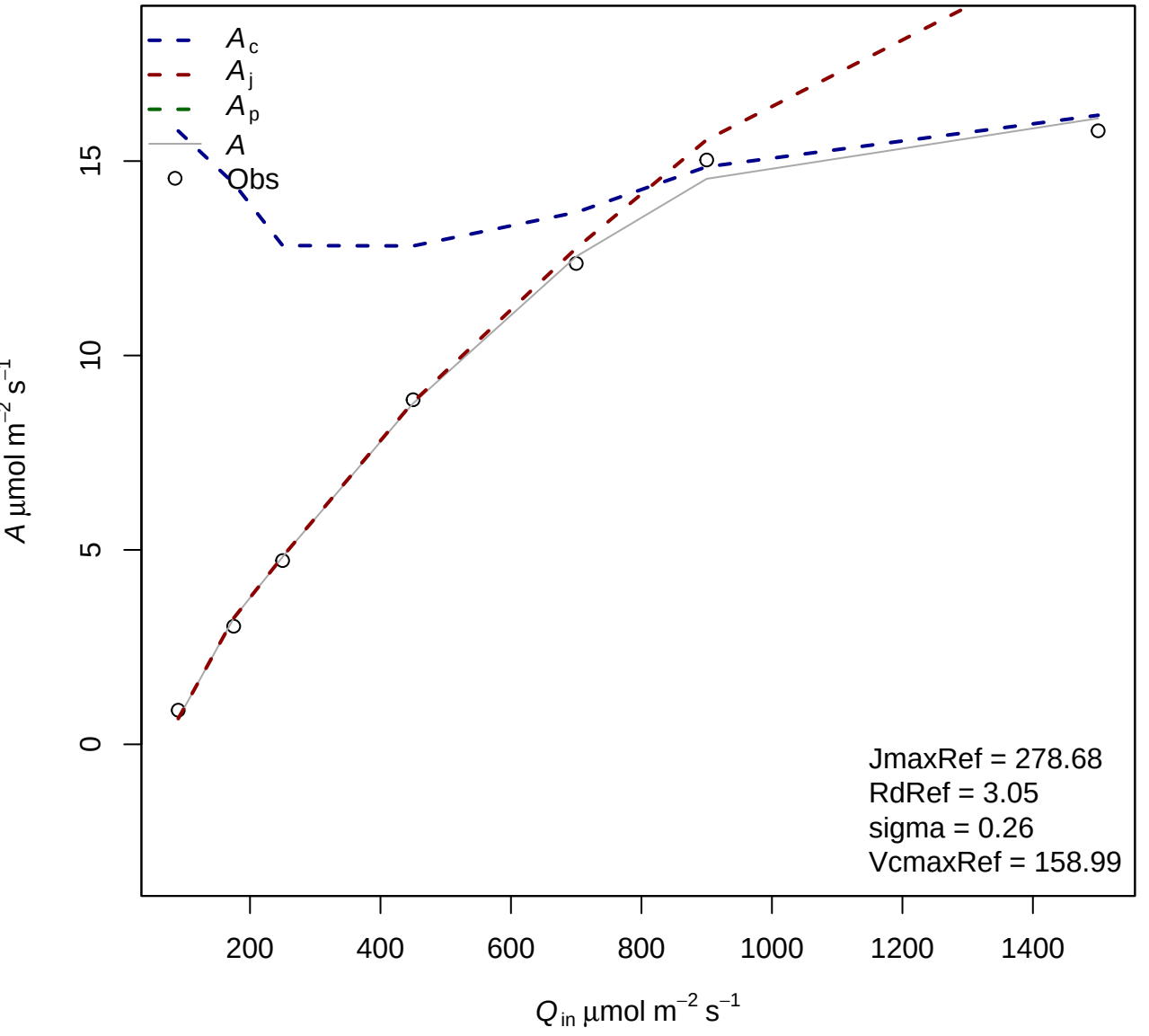

22
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 191.22
RdRef = 1.53
sigma = 0.18
VcmaxRef = 101.08
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

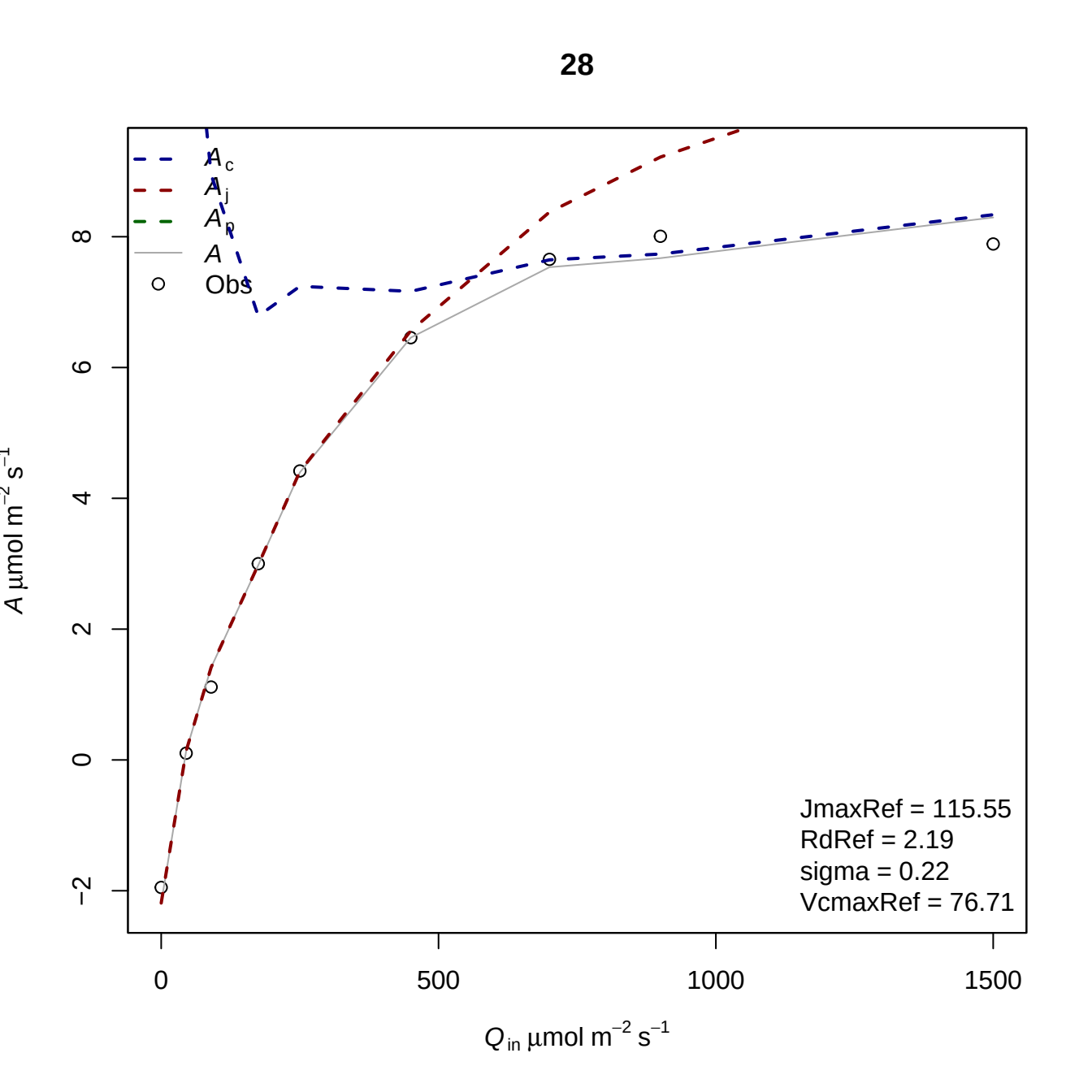

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.02
RdRef = 1.82
sigma = 0.4
VcmaxRef = 60.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

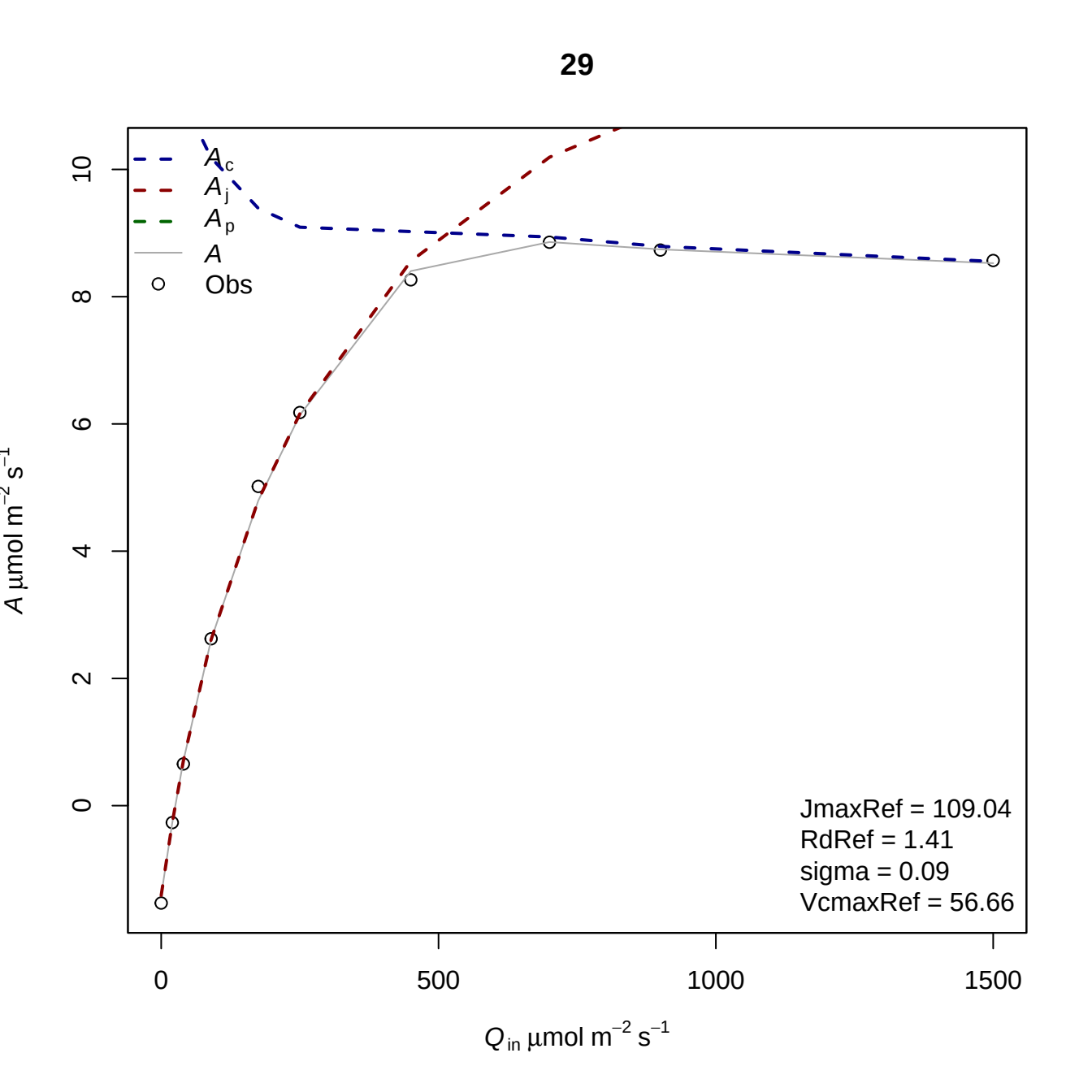

24
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.23
RdRef = 1.69
sigma = 0.97
VcmaxRef = 307.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

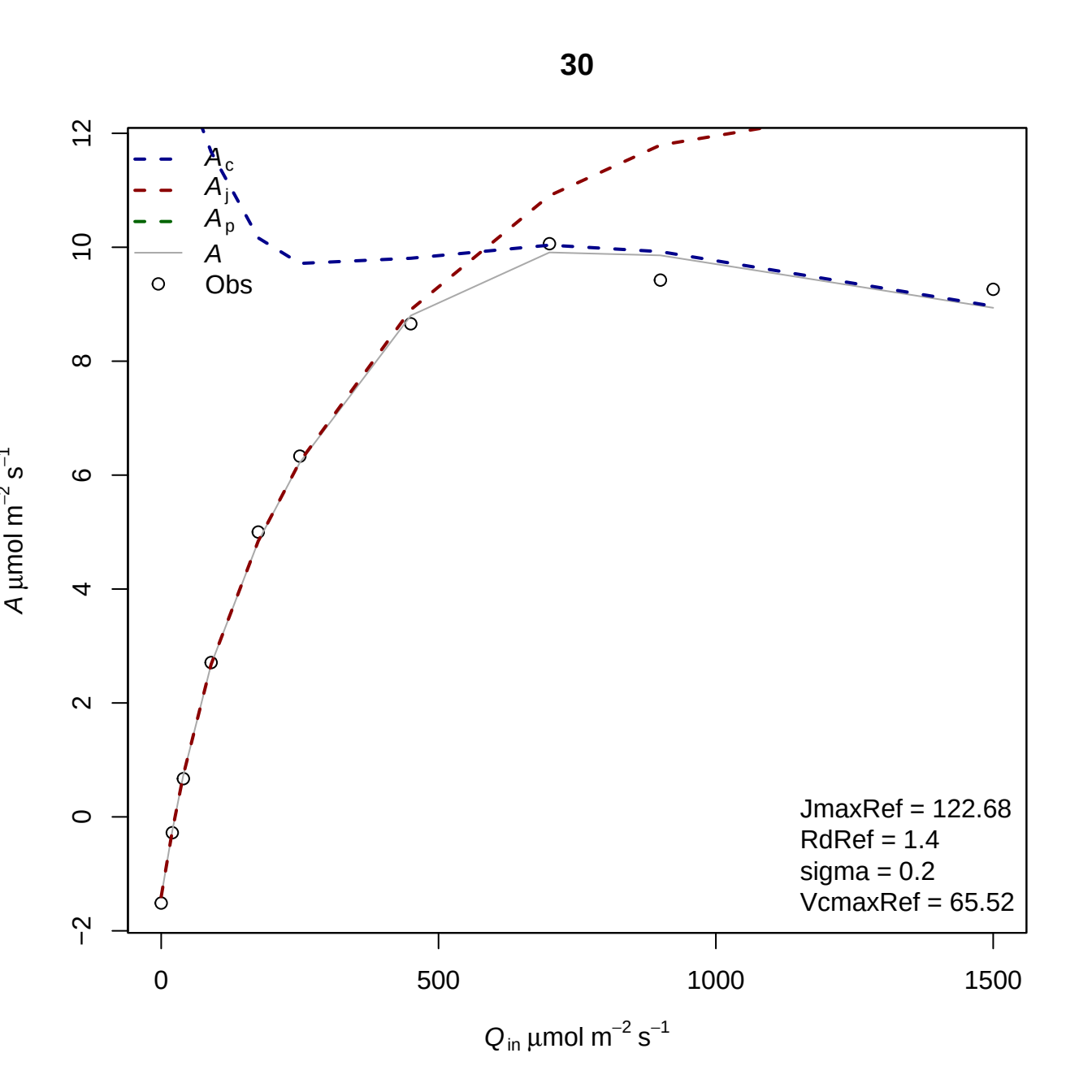

25
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.43
RdRef = 1.41
sigma = 0.3
VcmaxRef = 79.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500

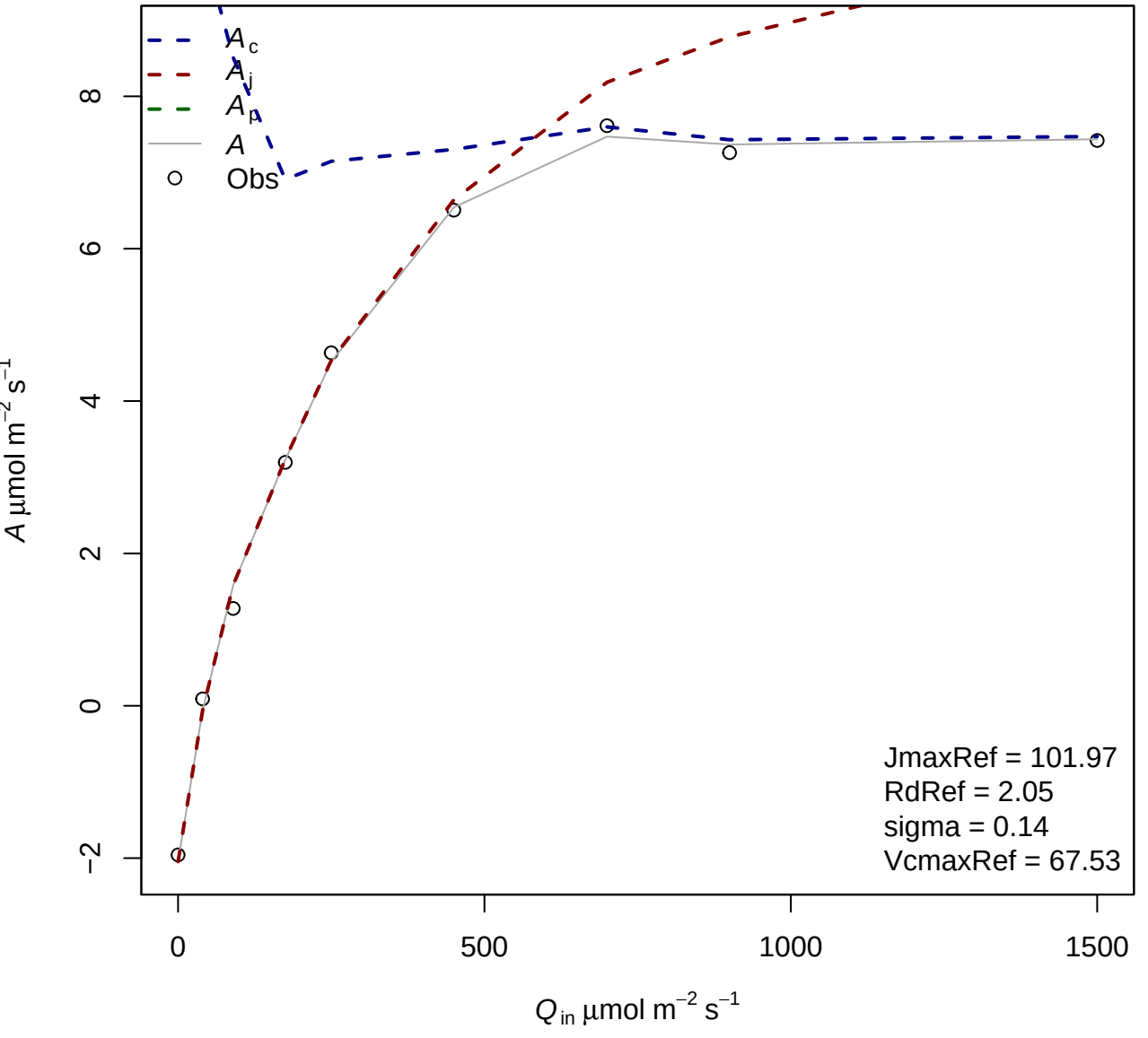

26
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 149.78
RdRef = 1.68
sigma = 0.24
VcmaxRef = 73.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

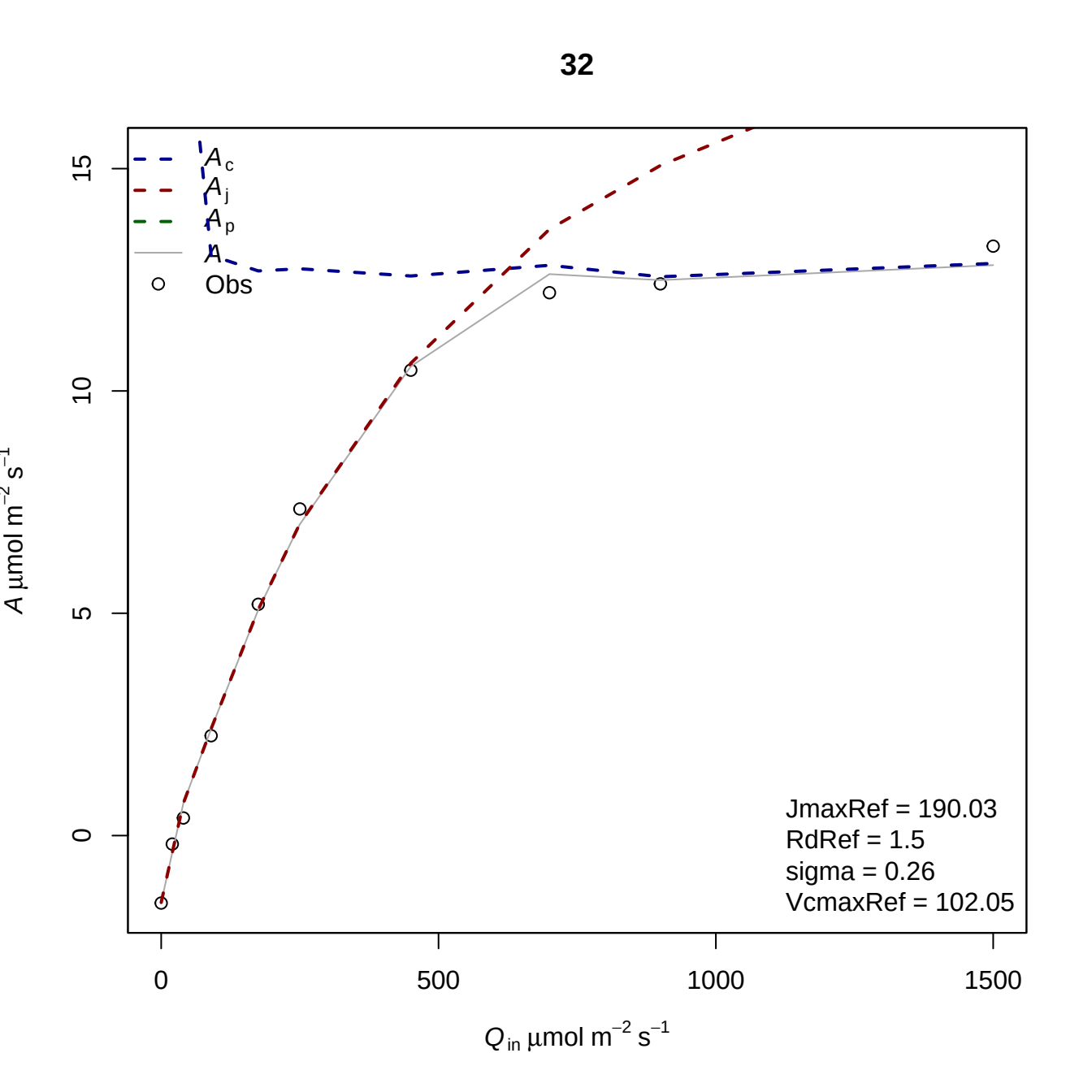

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 278.68
RdRef = 3.05
sigma = 0.26
VcmaxRef = 158.99
0
5
10
15
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

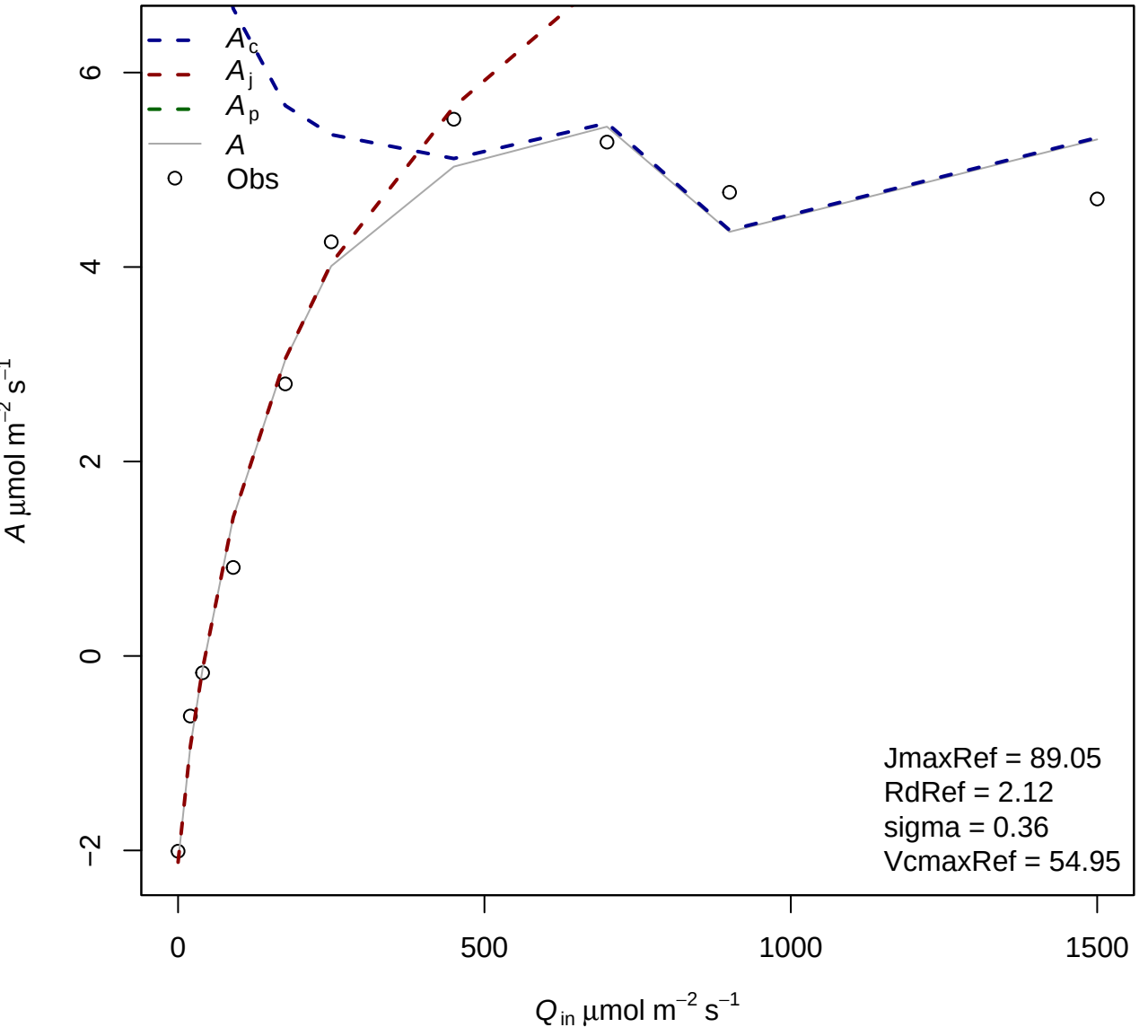

28
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.55
RdRef = 2.19
sigma = 0.22
VcmaxRef = 76.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

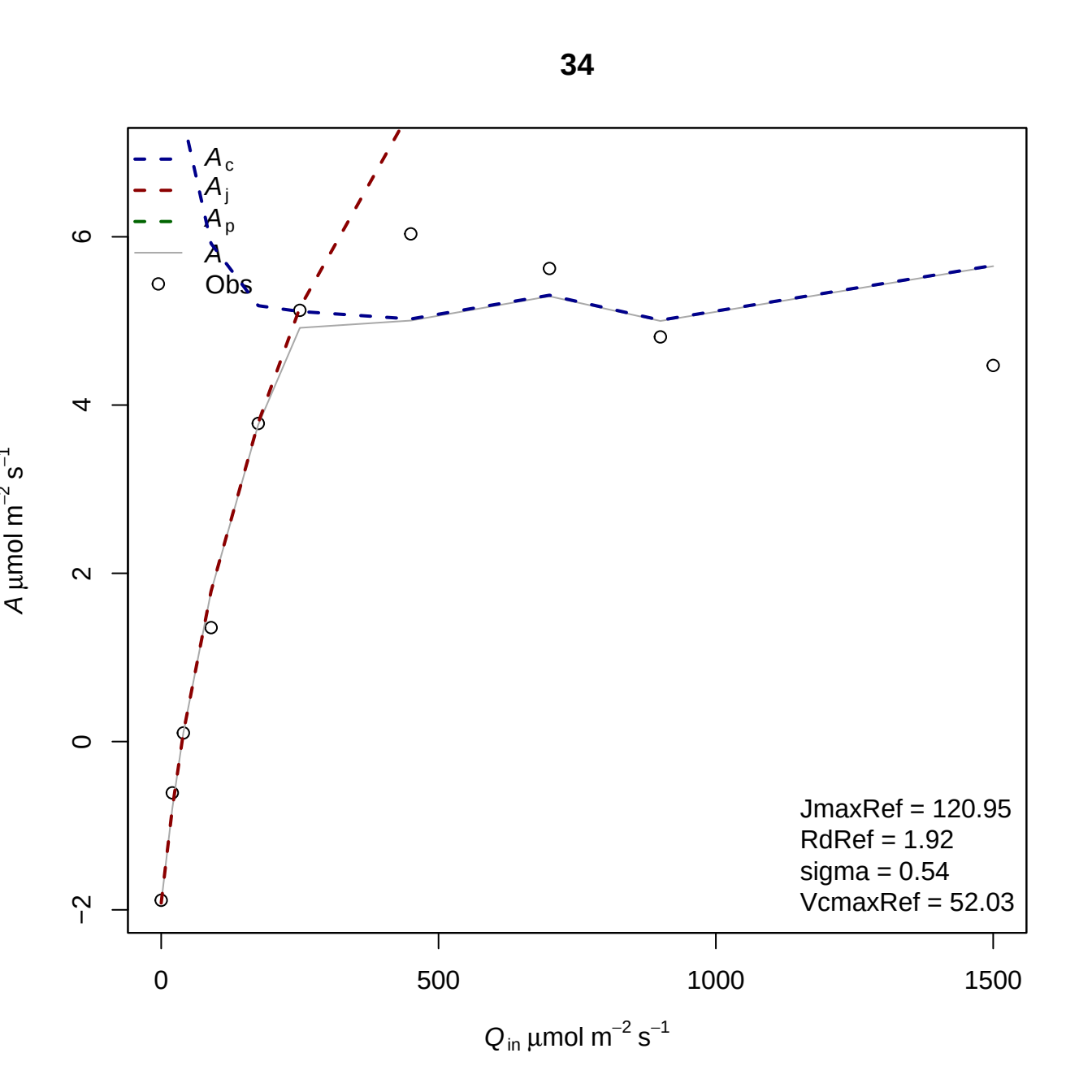

29
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.04
RdRef = 1.41
sigma = 0.09
VcmaxRef = 56.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

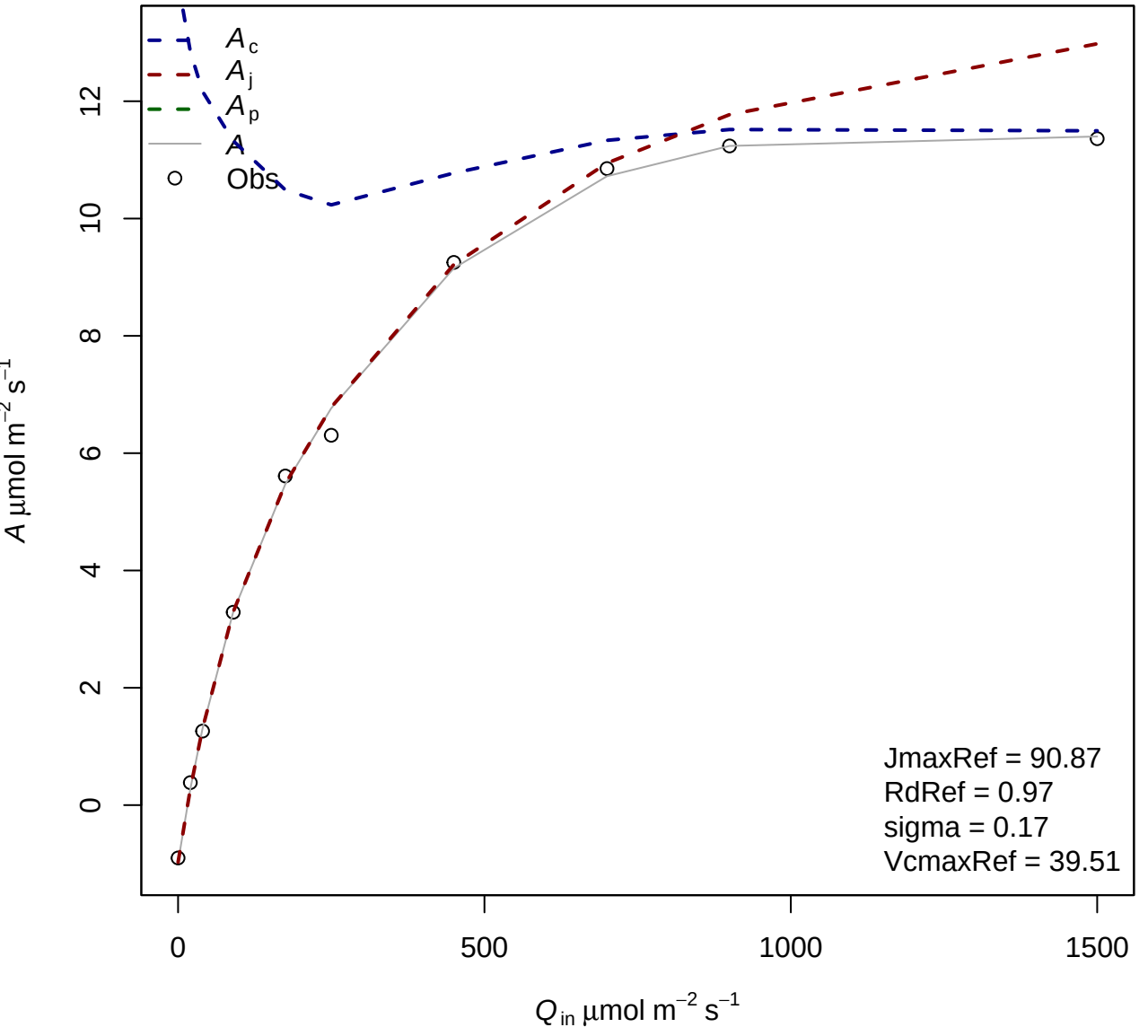

30
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.68
RdRef = 1.4
sigma = 0.2
VcmaxRef = 65.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

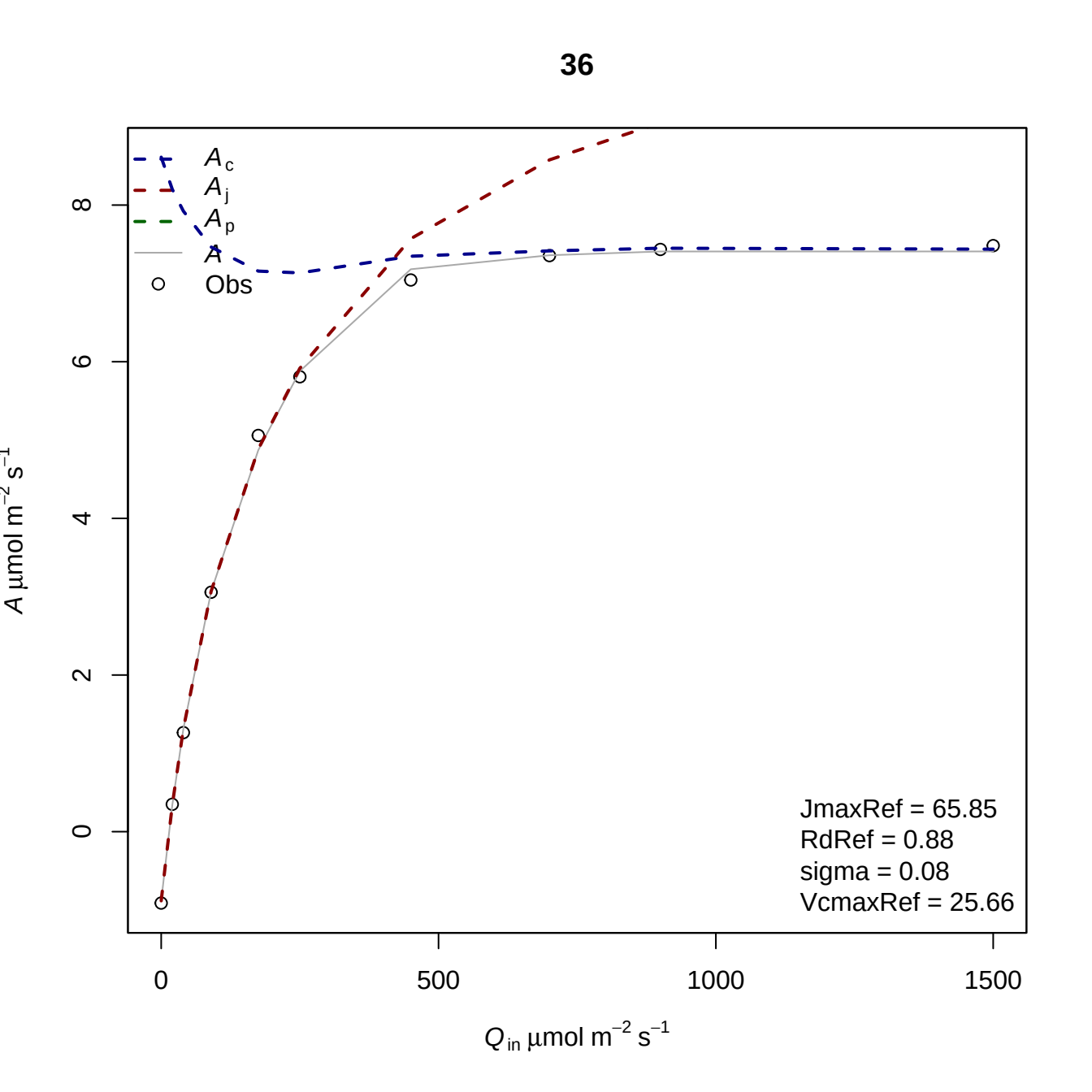

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.97
RdRef = 2.05
sigma = 0.14
VcmaxRef = 67.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

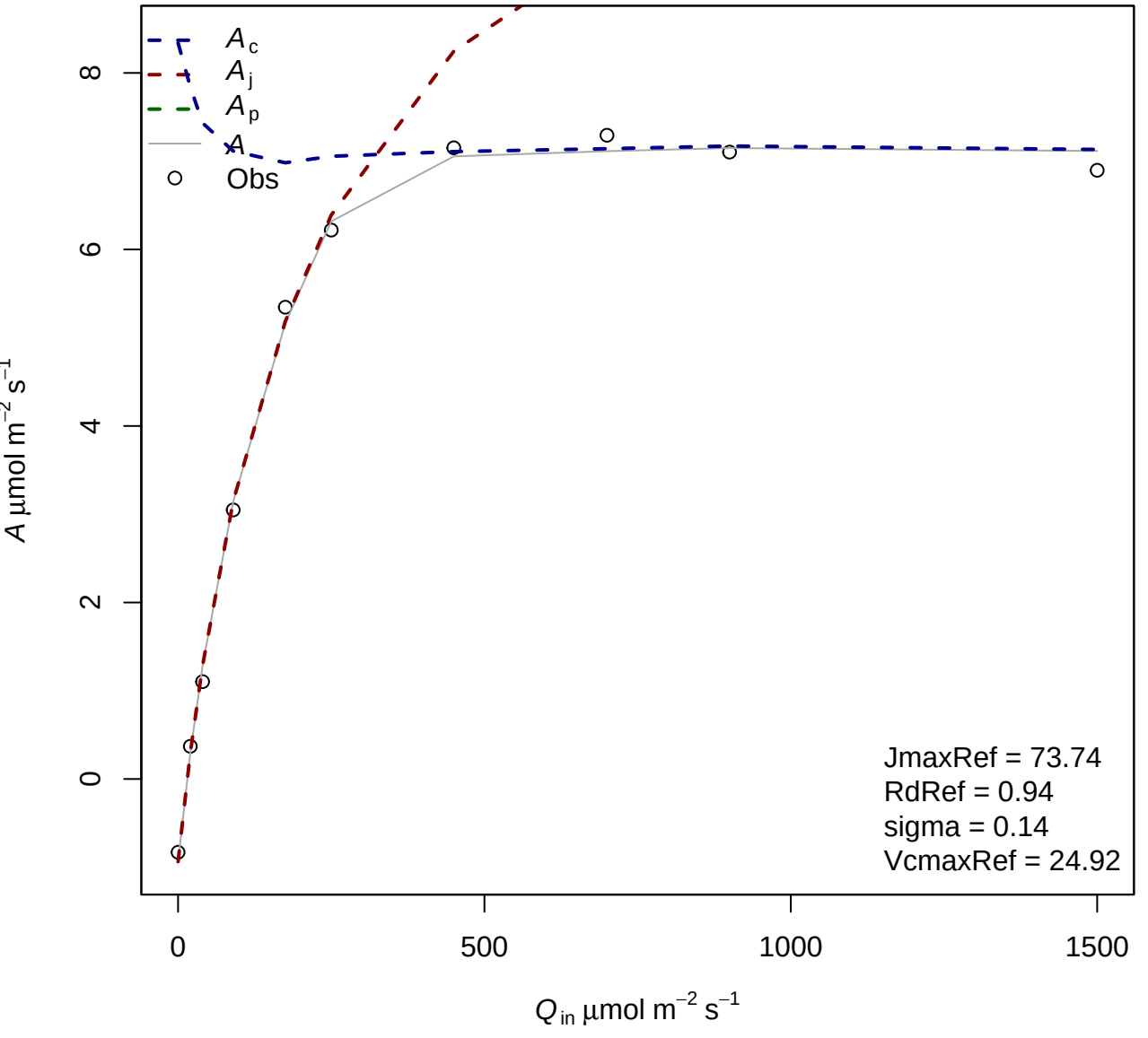

32
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 190.03
RdRef = 1.5
sigma = 0.26
VcmaxRef = 102.05
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

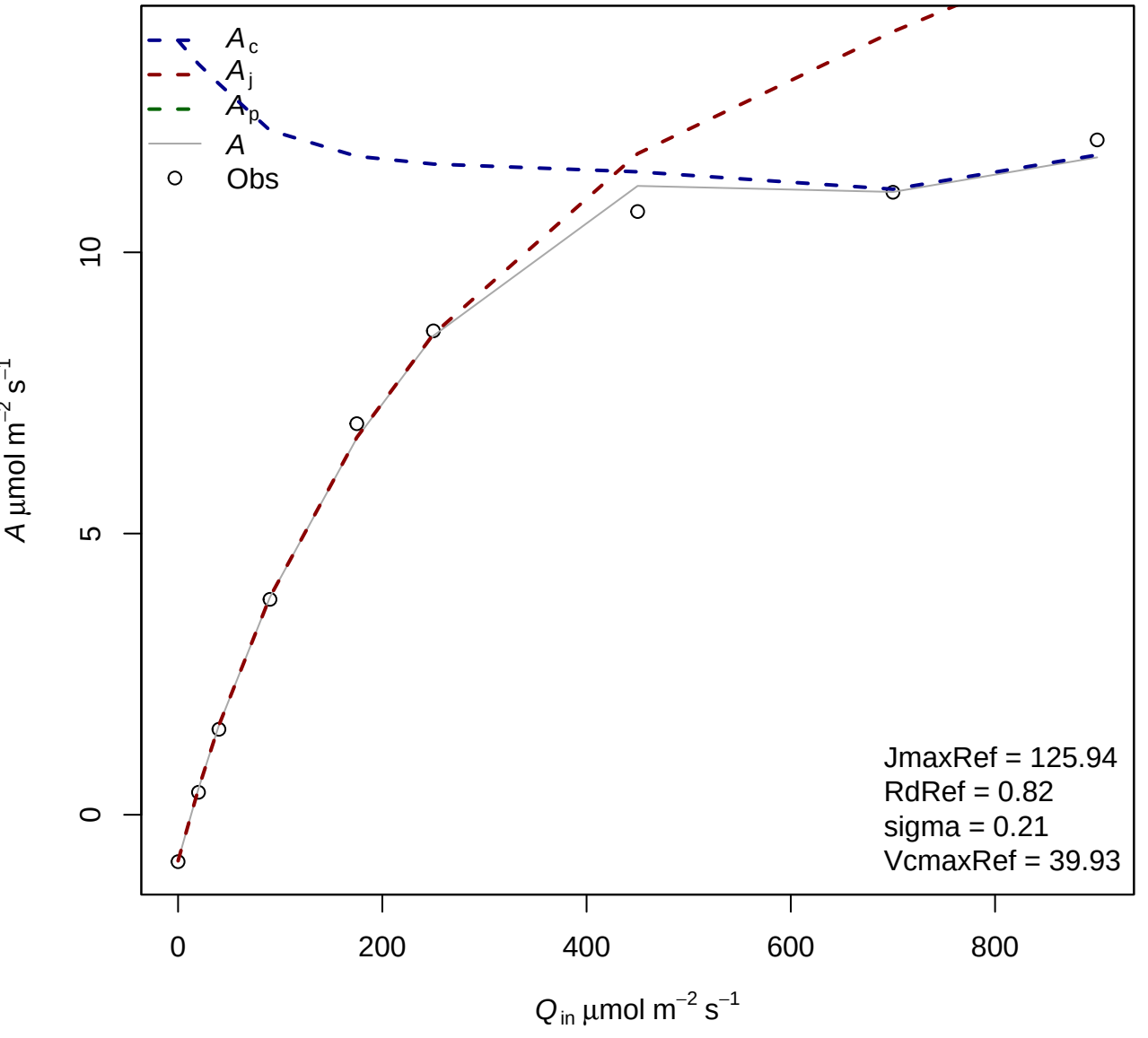

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.05
RdRef = 2.12
sigma = 0.36
VcmaxRef = 54.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

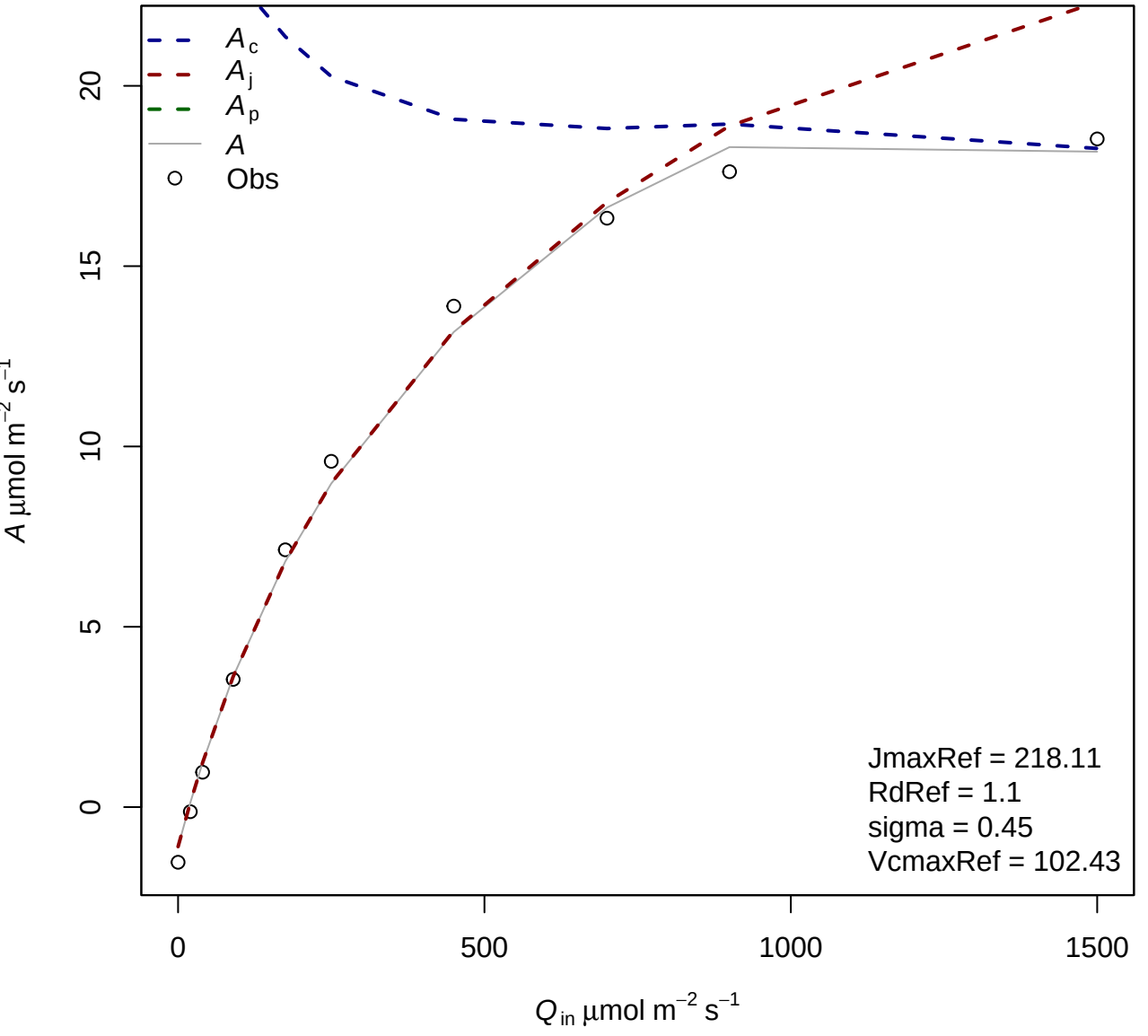

34
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.95
RdRef = 1.92
sigma = 0.54
VcmaxRef = 52.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

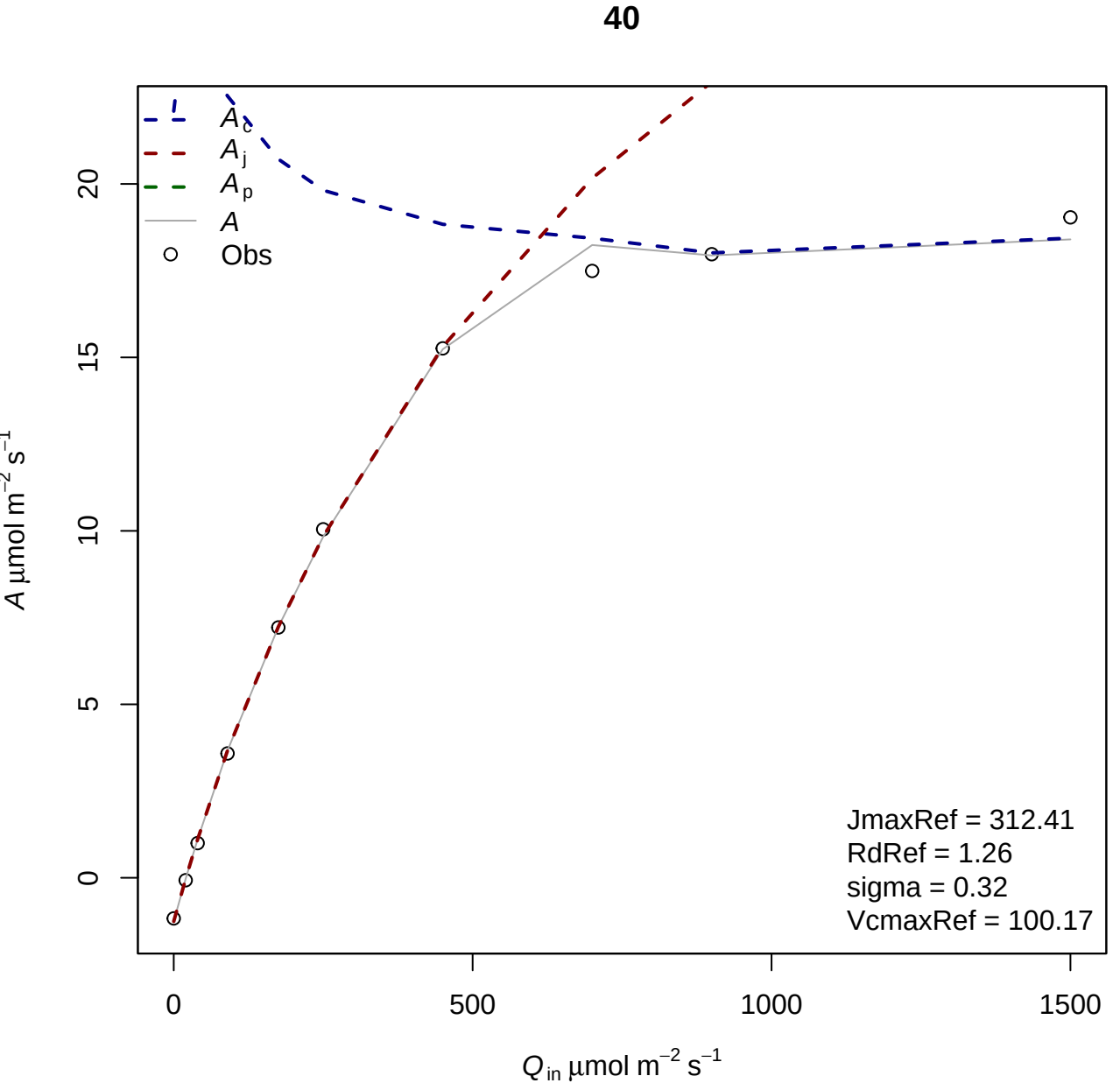

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.87
RdRef = 0.97
sigma = 0.17
VcmaxRef = 39.51
0
2
4
6
8
10
12
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

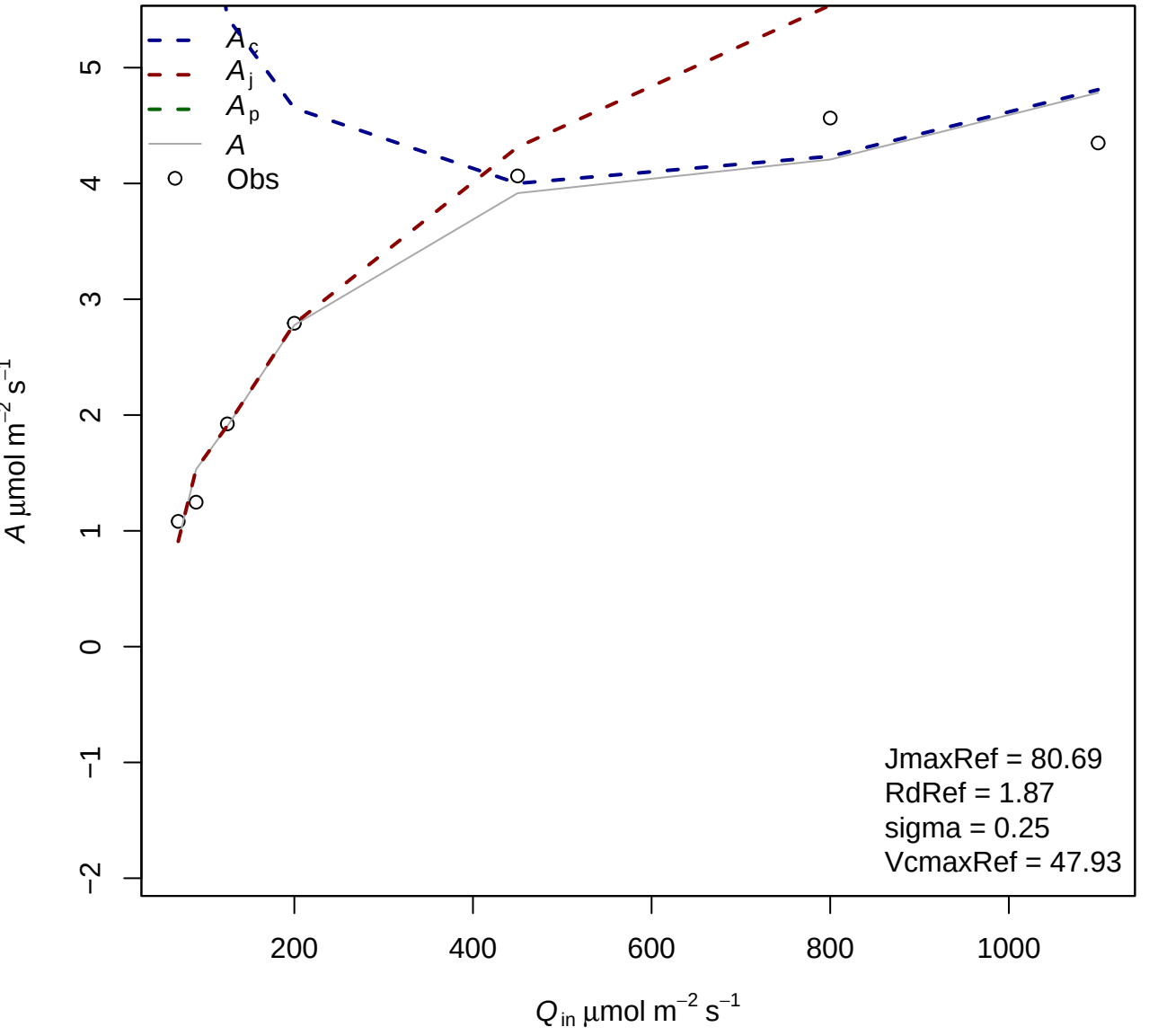

36
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.85
RdRef = 0.88
sigma = 0.08
VcmaxRef = 25.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

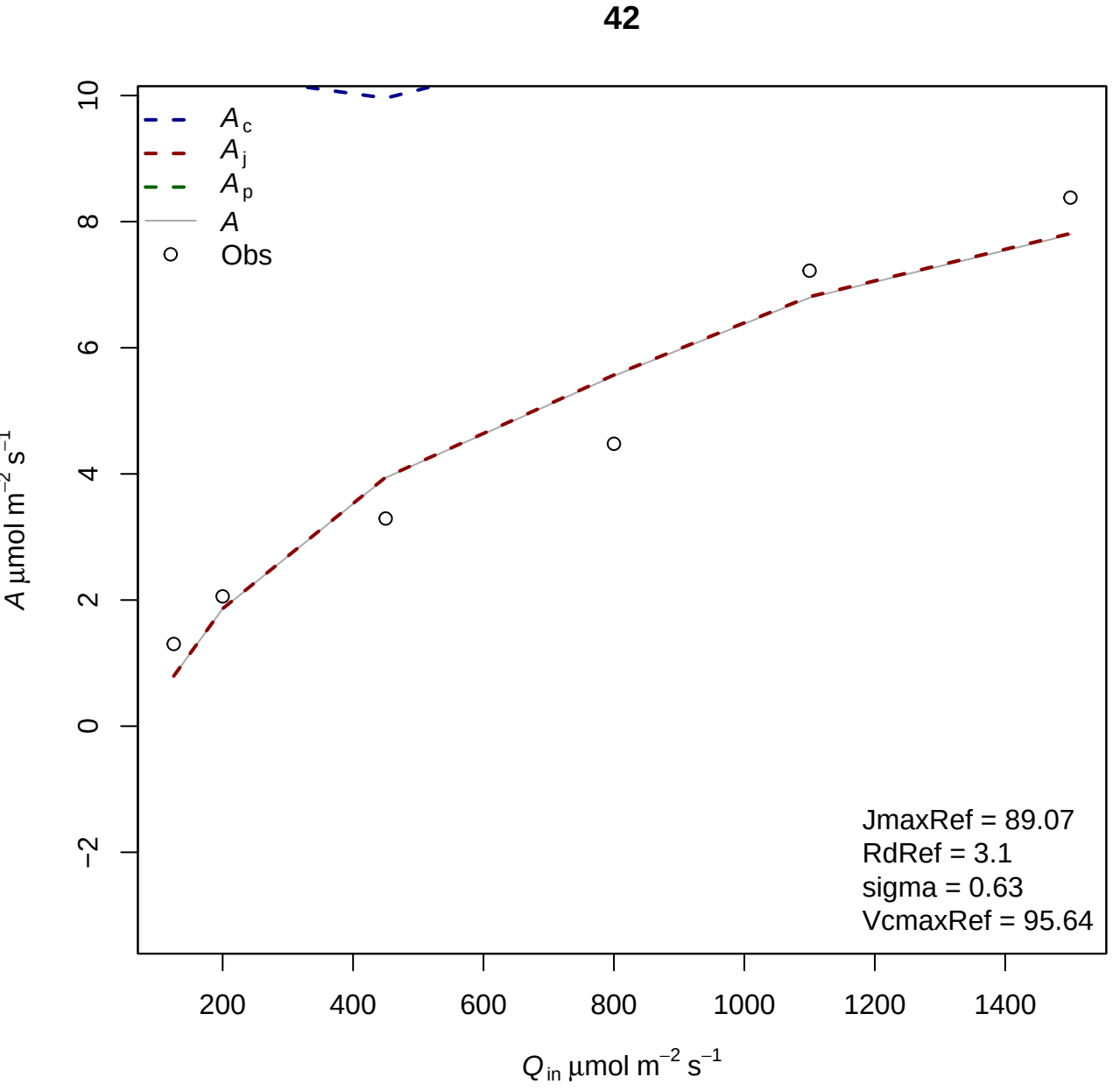

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.74
RdRef = 0.94
sigma = 0.14
VcmaxRef = 24.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

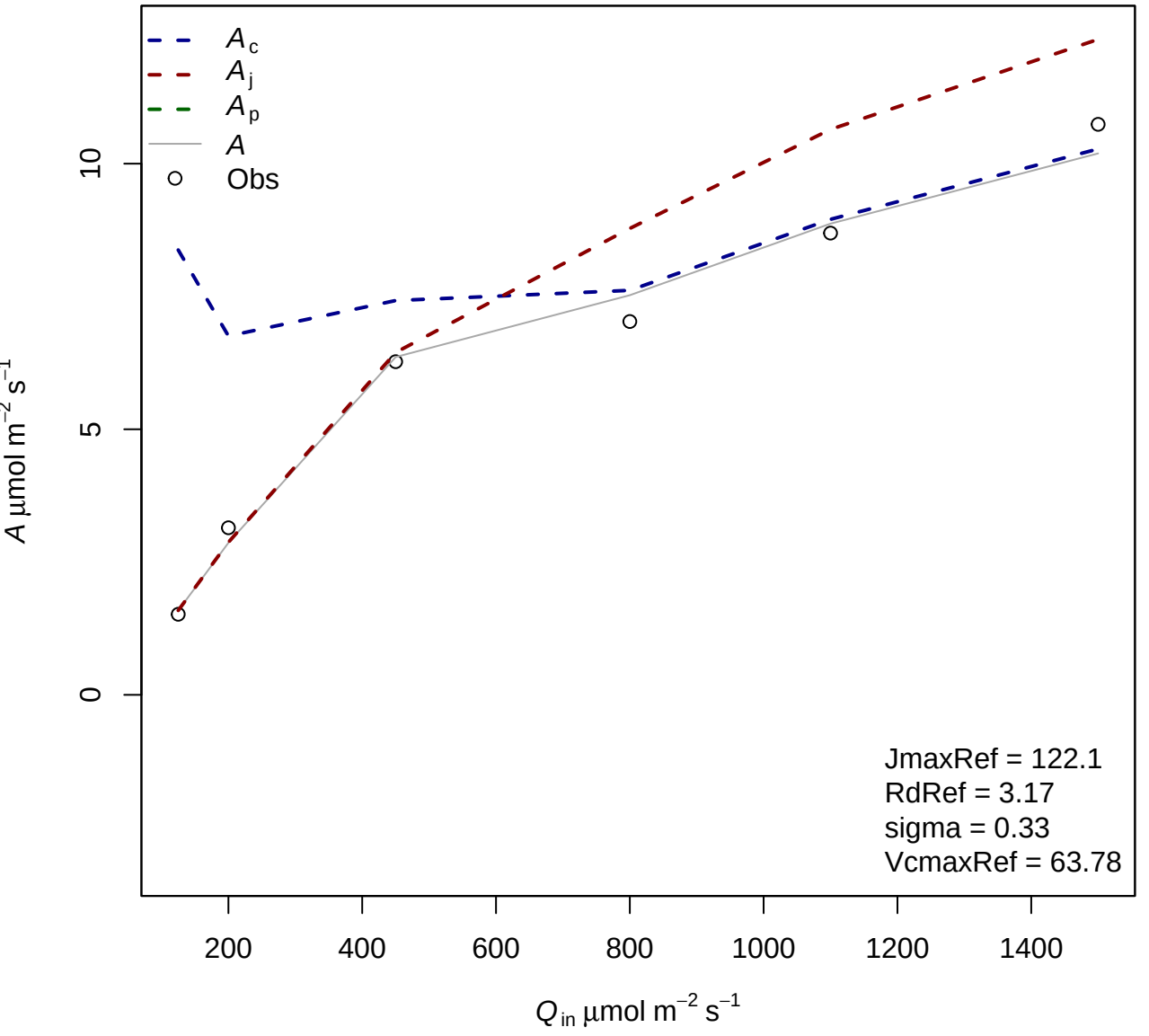

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.94
RdRef = 0.82
sigma = 0.21
VcmaxRef = 39.93
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
200
400
600
800
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

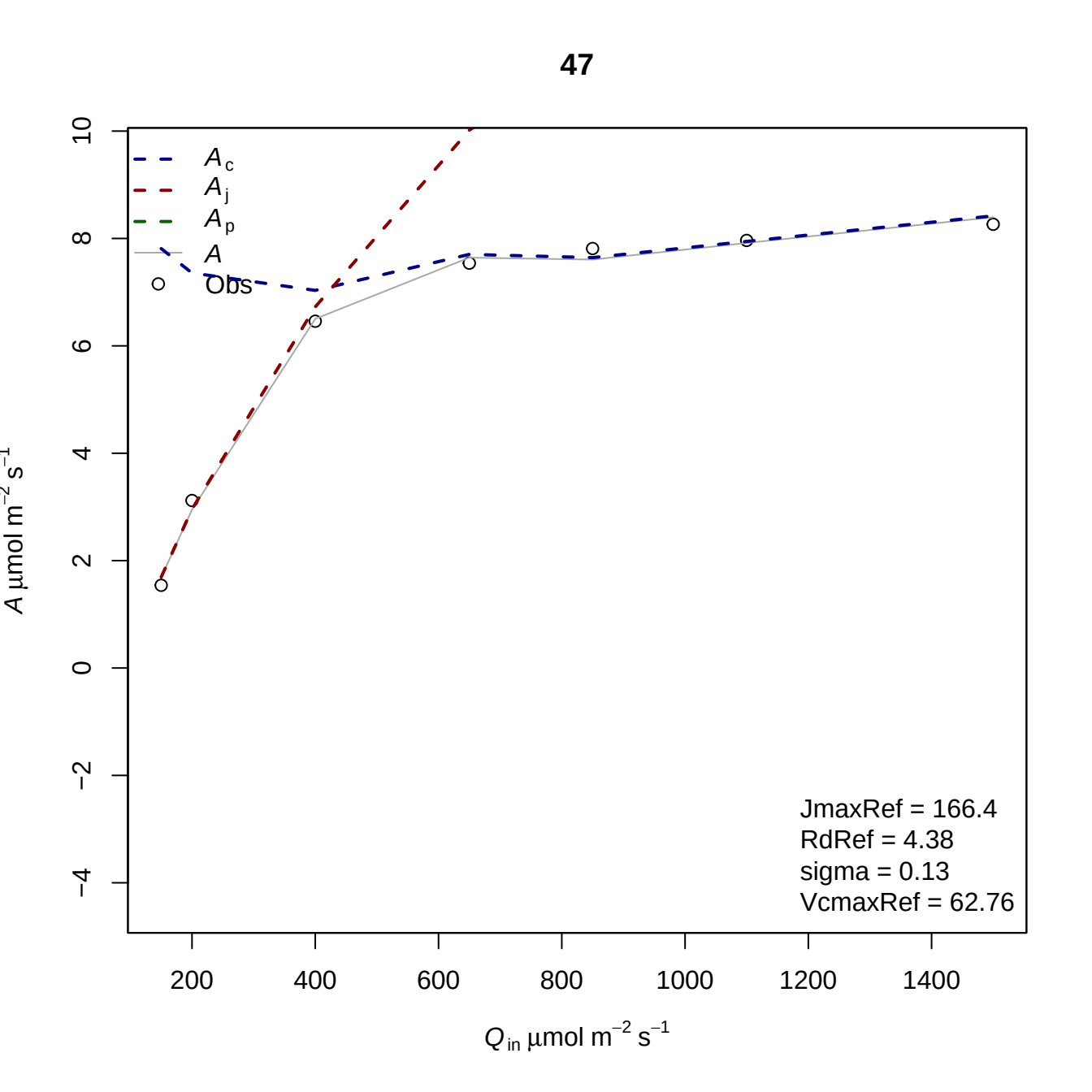

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 218.11
RdRef = 1.1
sigma = 0.45
VcmaxRef = 102.43
0
5
10
15
20
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

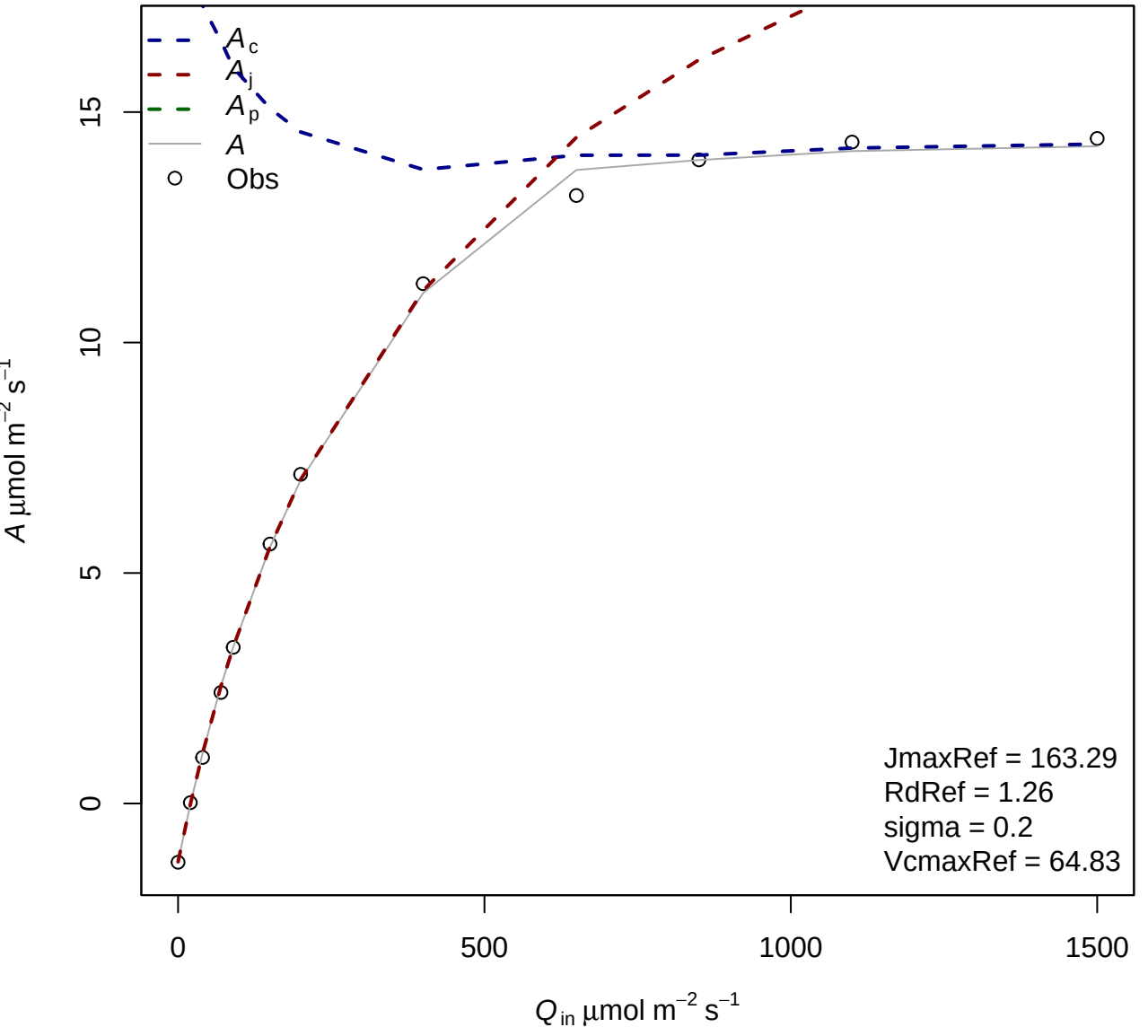

40
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 312.41
RdRef = 1.26
sigma = 0.32
VcmaxRef = 100.17
0
5
10
15
20
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

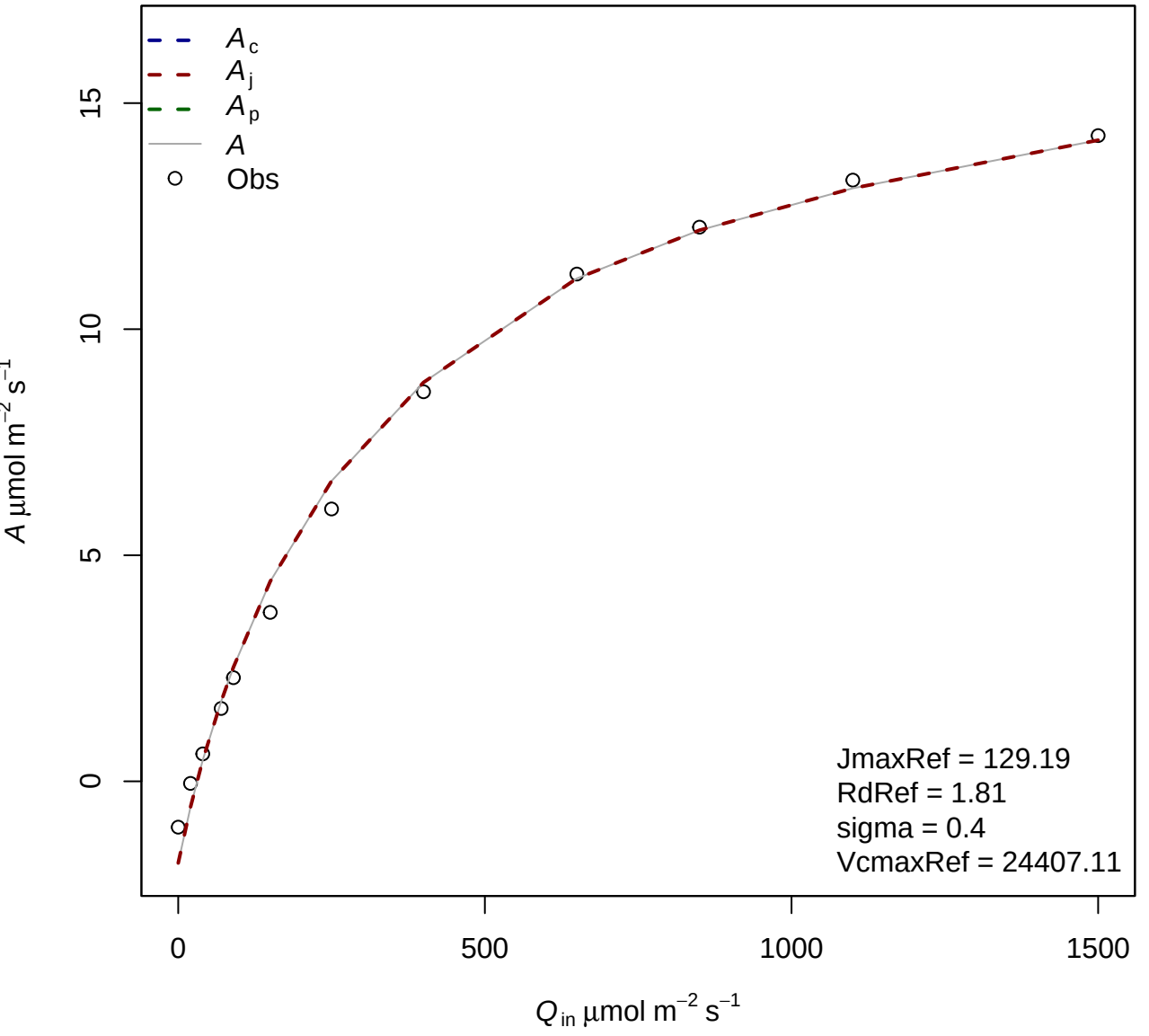

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.69
RdRef = 1.87
sigma = 0.25
VcmaxRef = 47.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
−1
0
1
2
3
4
5
200
400
600
800
1000

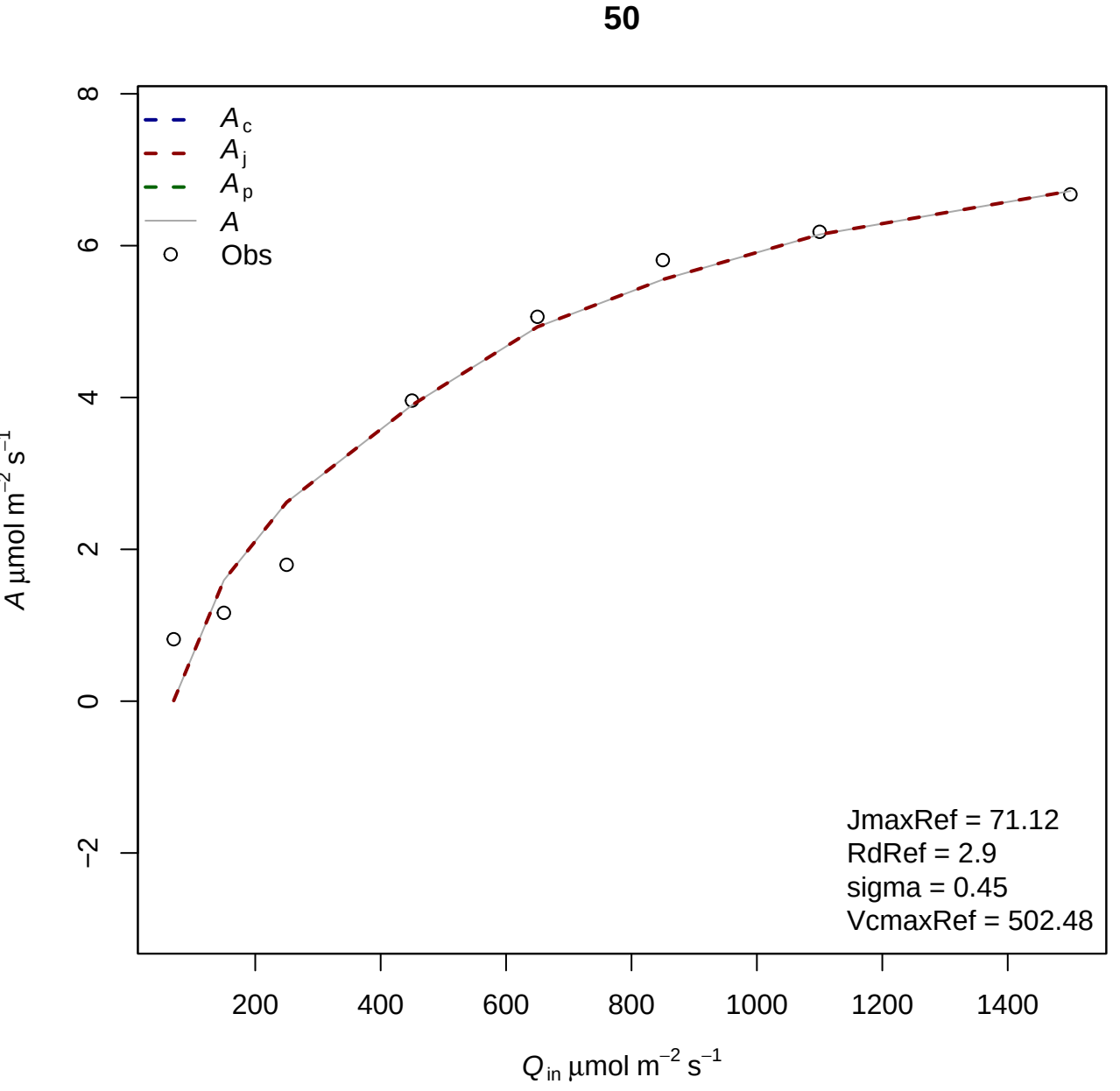

42
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.07
RdRef = 3.1
sigma = 0.63
VcmaxRef = 95.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

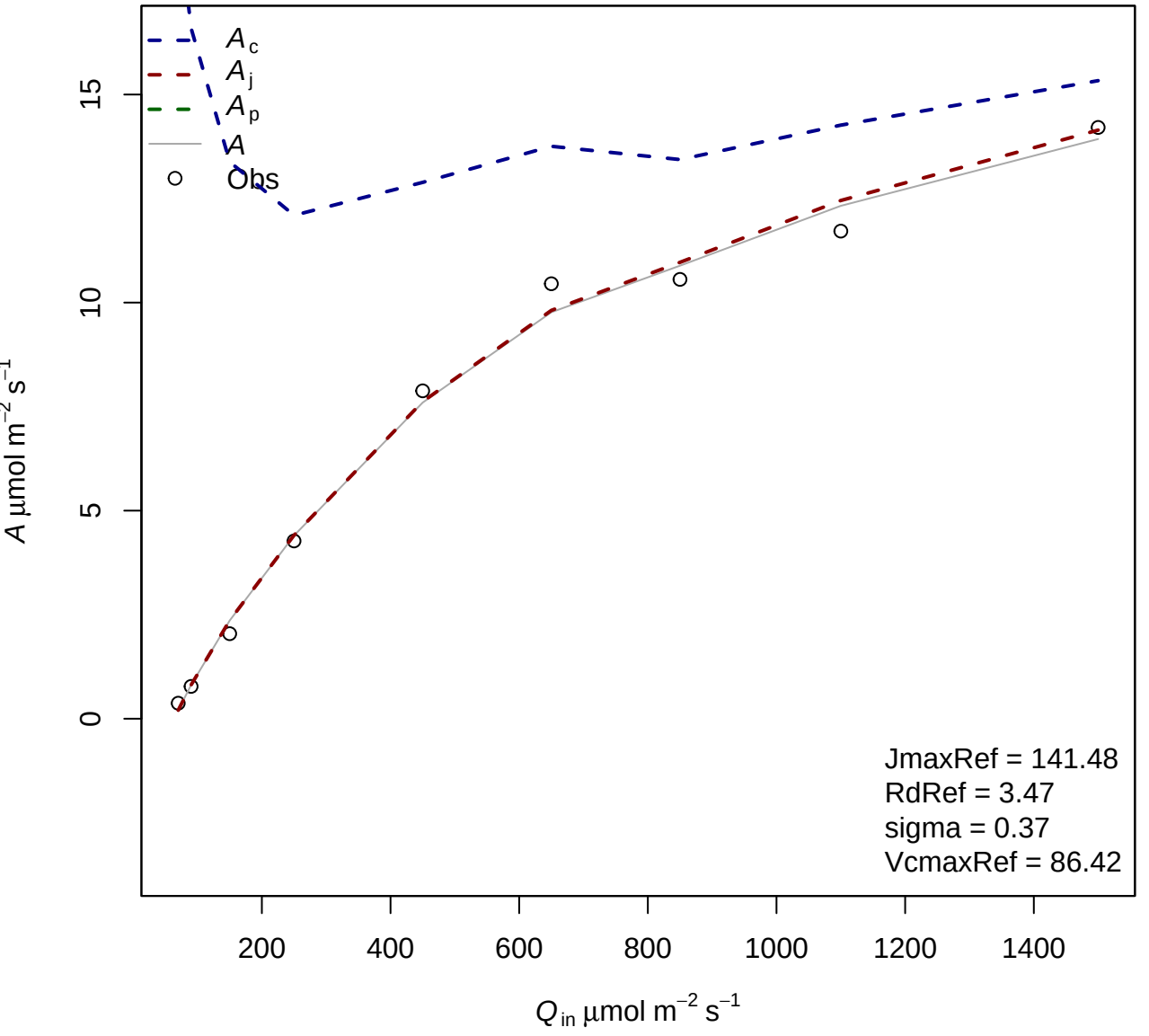

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.1
RdRef = 3.17
sigma = 0.33
VcmaxRef = 63.78
0
5
10
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

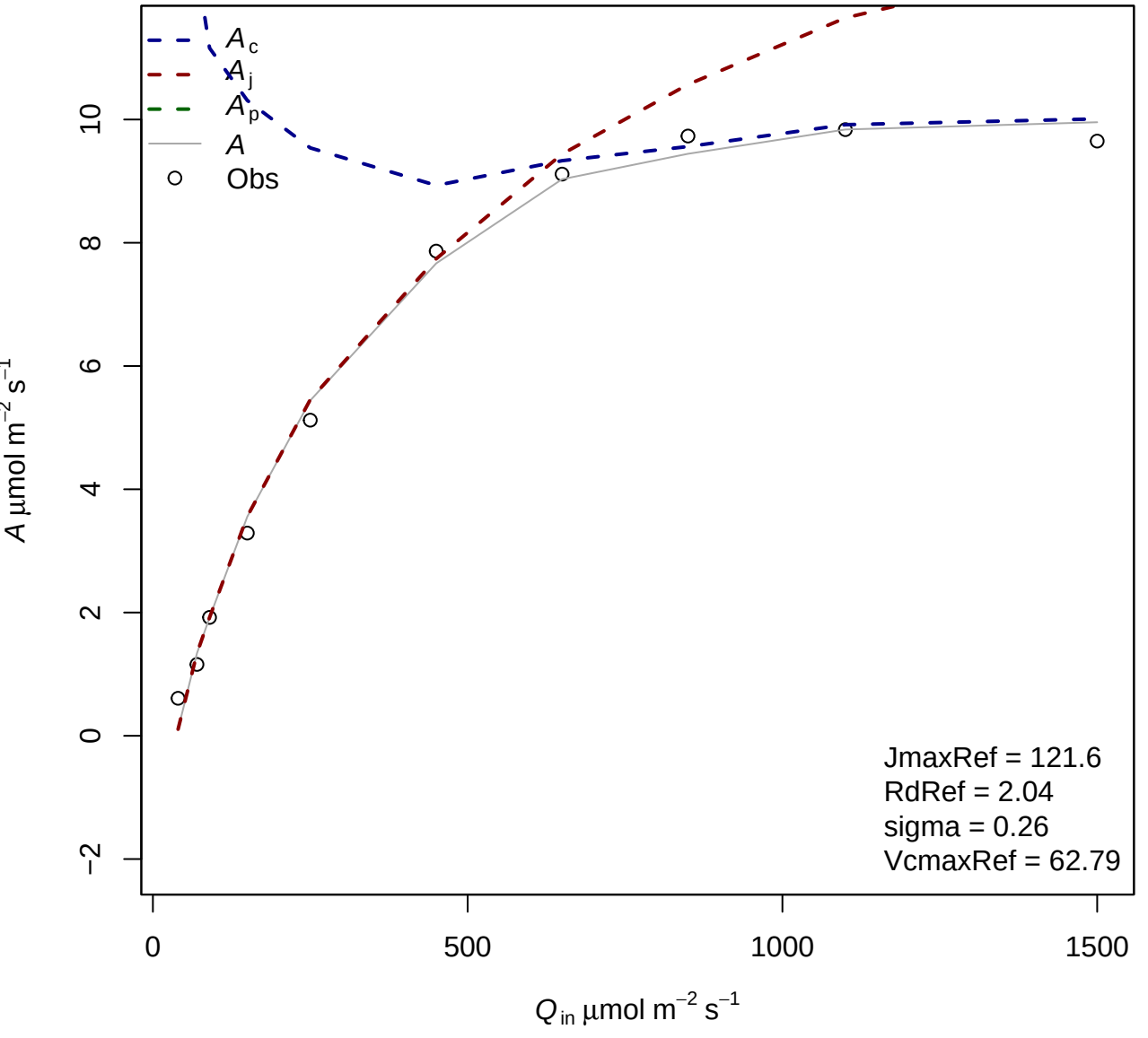

47
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 166.4
RdRef = 4.38
sigma = 0.13
VcmaxRef = 62.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

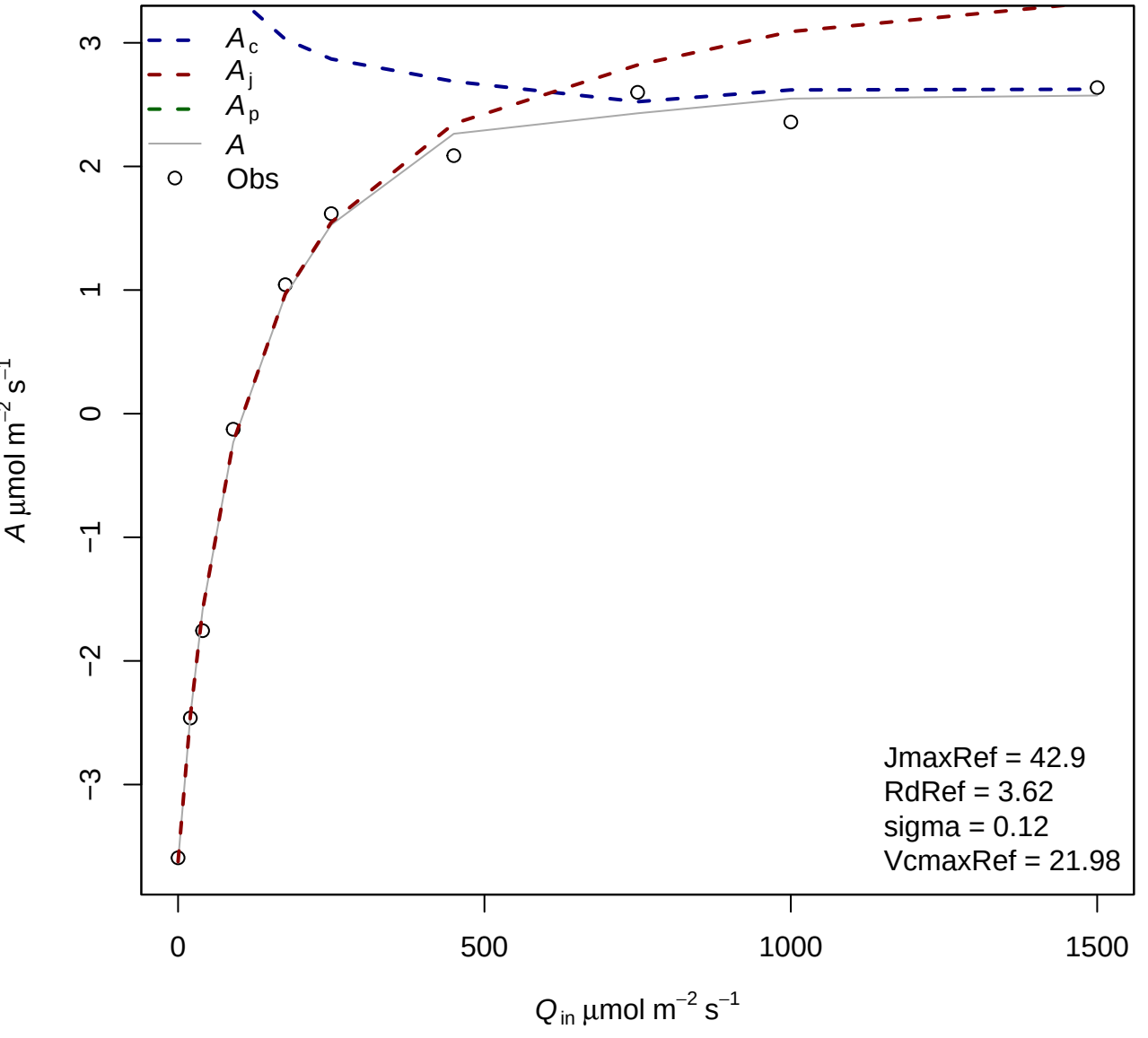

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163.29
RdRef = 1.26
sigma = 0.2
VcmaxRef = 64.83
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

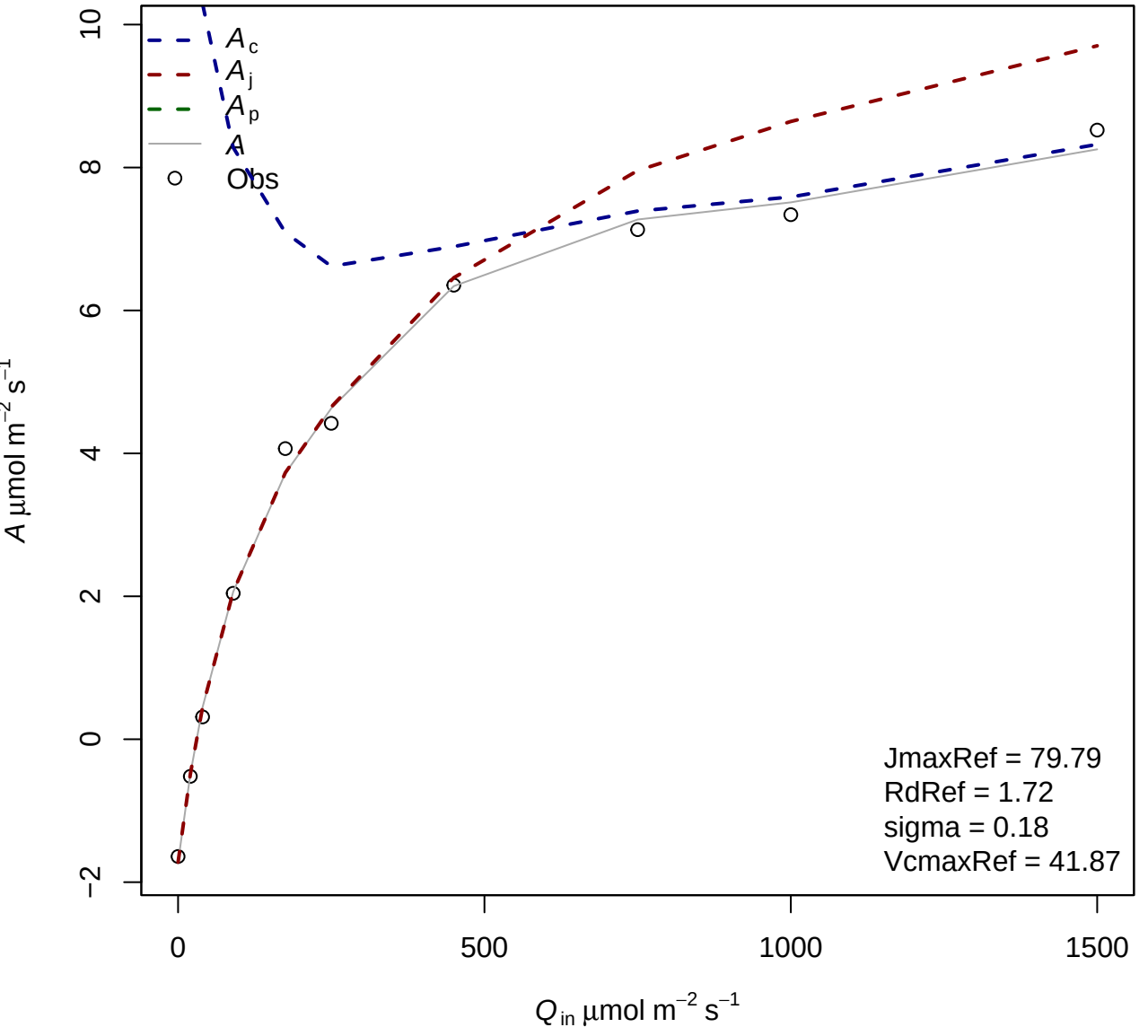

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.19
RdRef = 1.81
sigma = 0.4
VcmaxRef = 24407.11
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

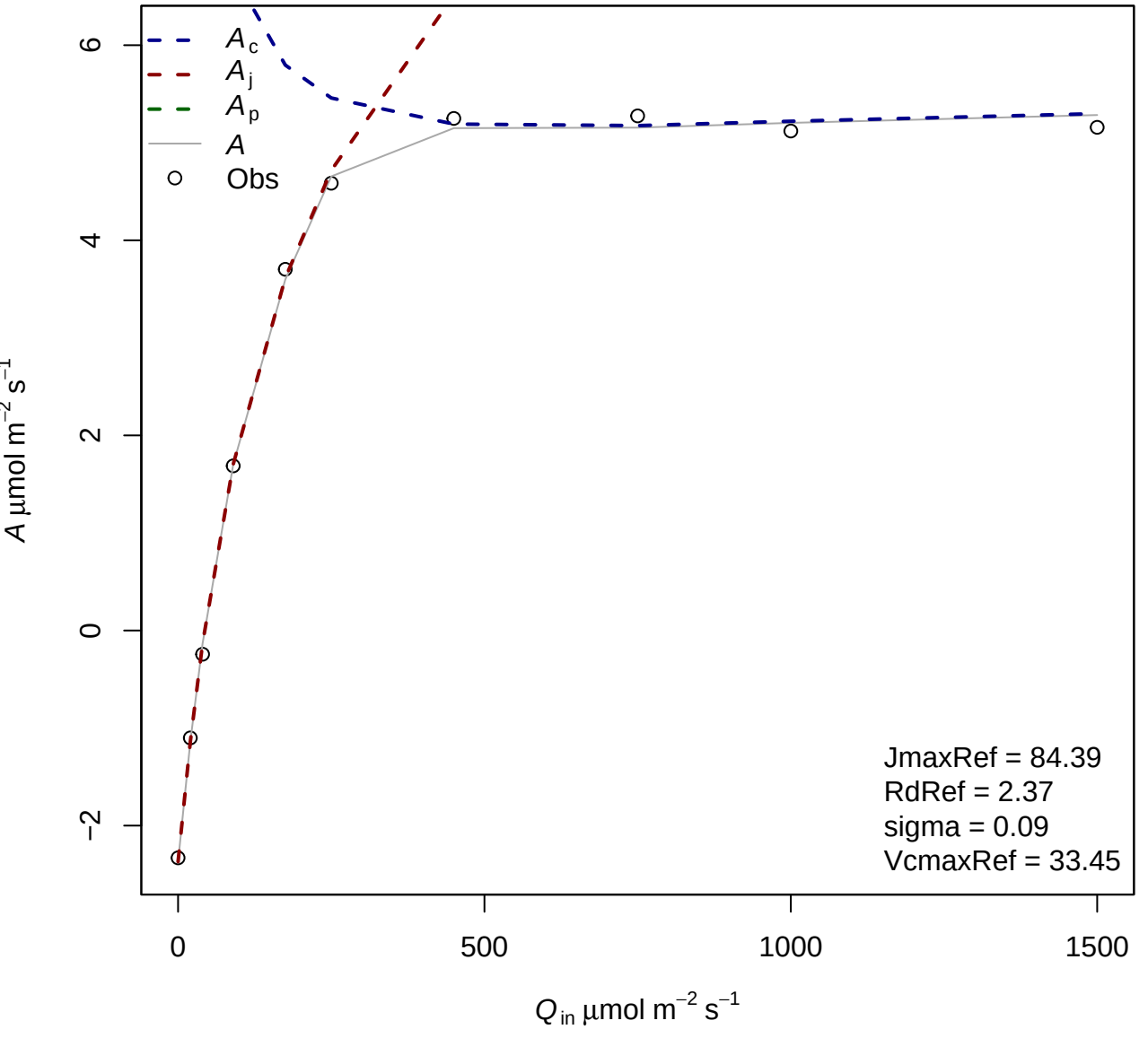

50
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.12
RdRef = 2.9
sigma = 0.45
VcmaxRef = 502.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

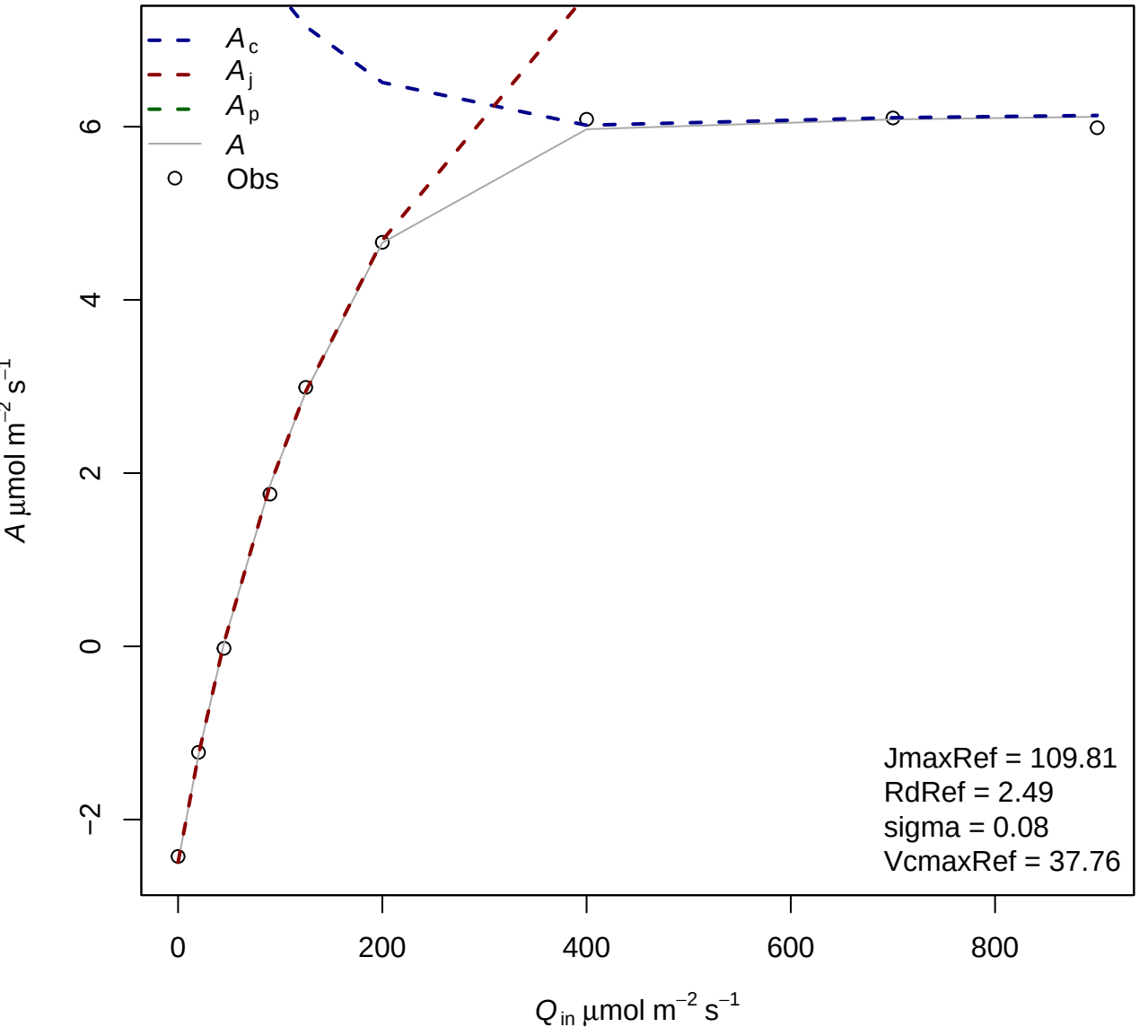

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 141.48
RdRef = 3.47
sigma = 0.37
VcmaxRef = 86.42
0
5
10
15
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

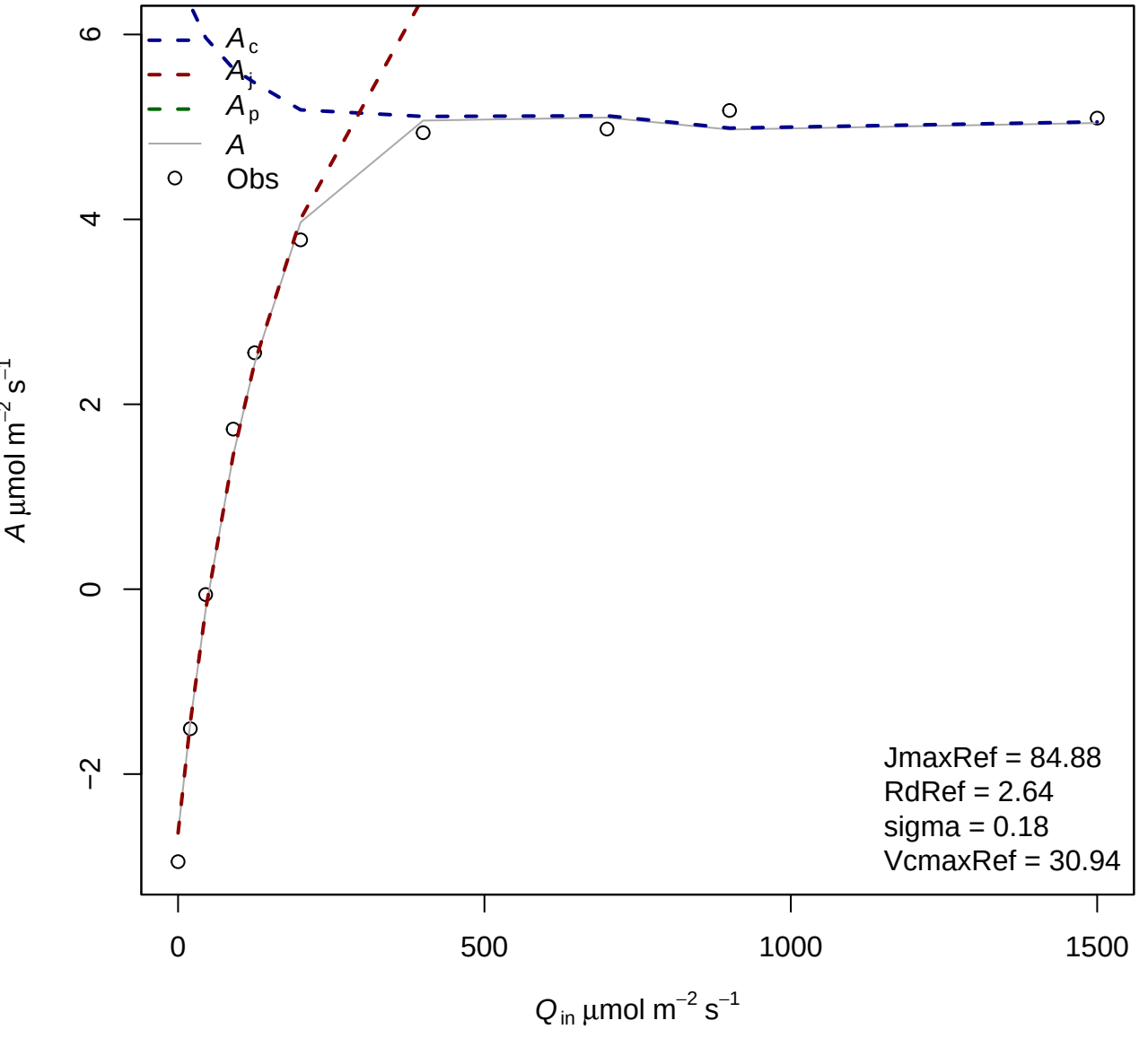

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.6
RdRef = 2.04
sigma = 0.26
VcmaxRef = 62.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

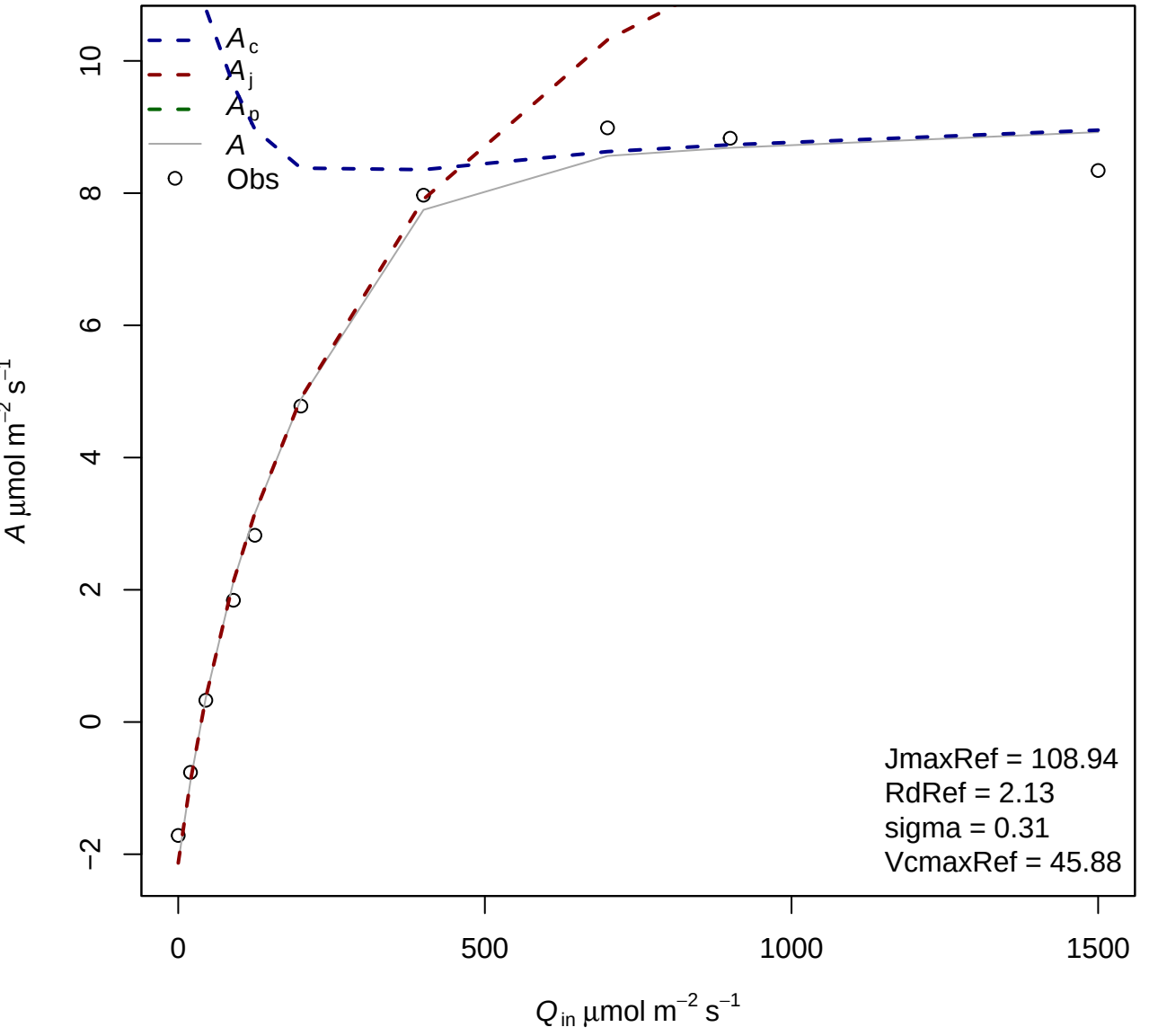

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.9
RdRef = 3.62
sigma = 0.12
VcmaxRef = 21.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
−2
−3
0
500
1000
1500

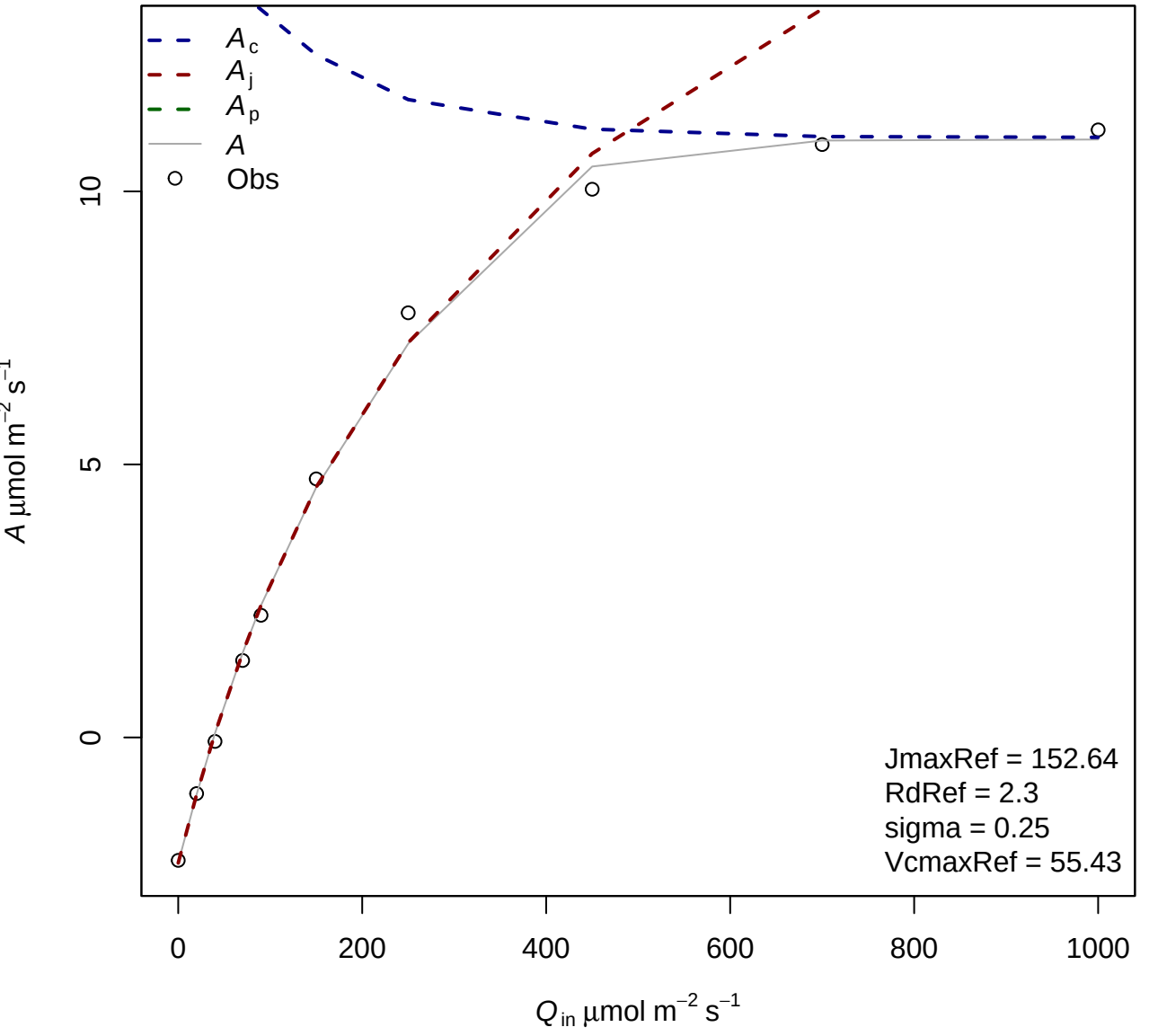

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.79
RdRef = 1.72
sigma = 0.18
VcmaxRef = 41.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

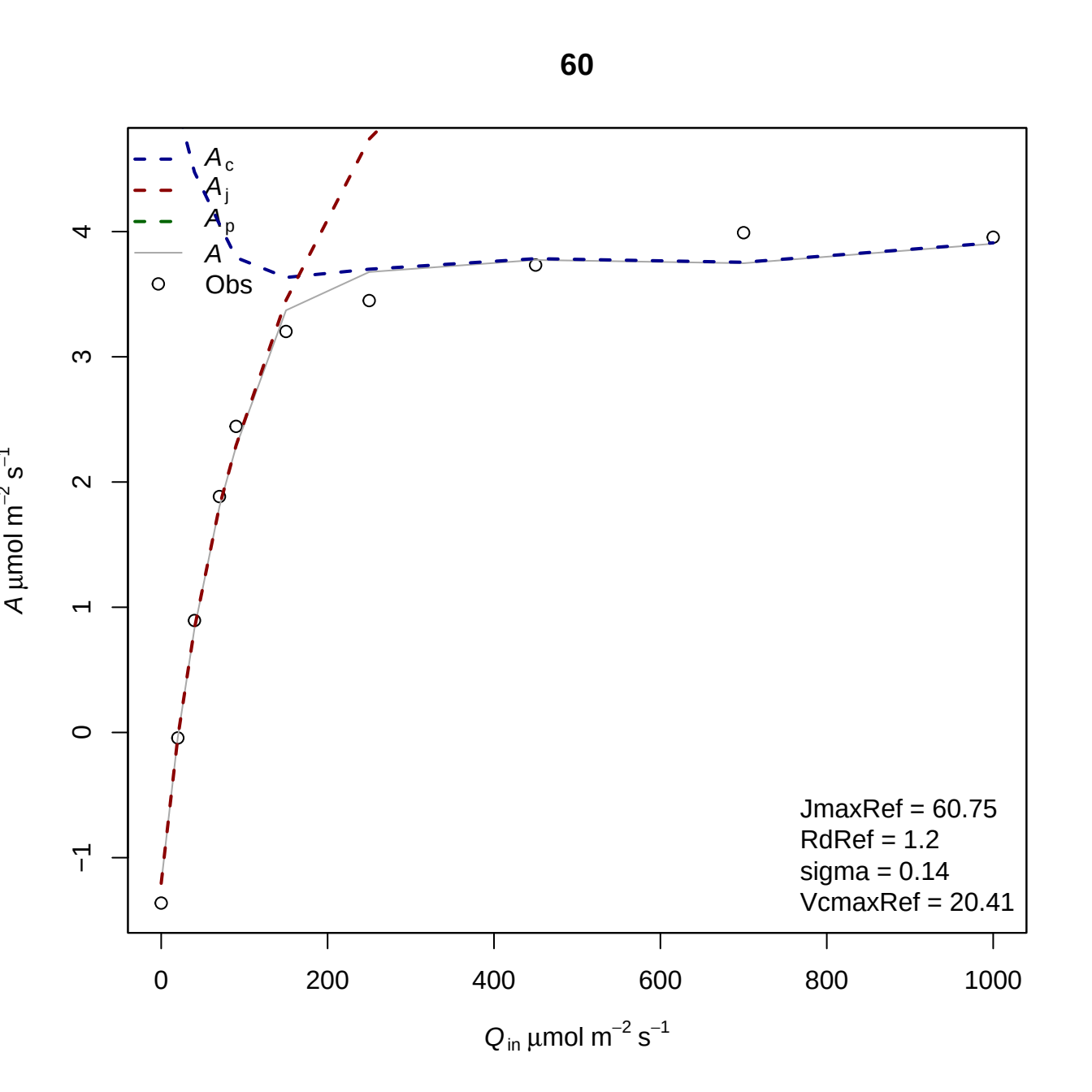

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.39
RdRef = 2.37
sigma = 0.09
VcmaxRef = 33.45
6
4
2
0
−2
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

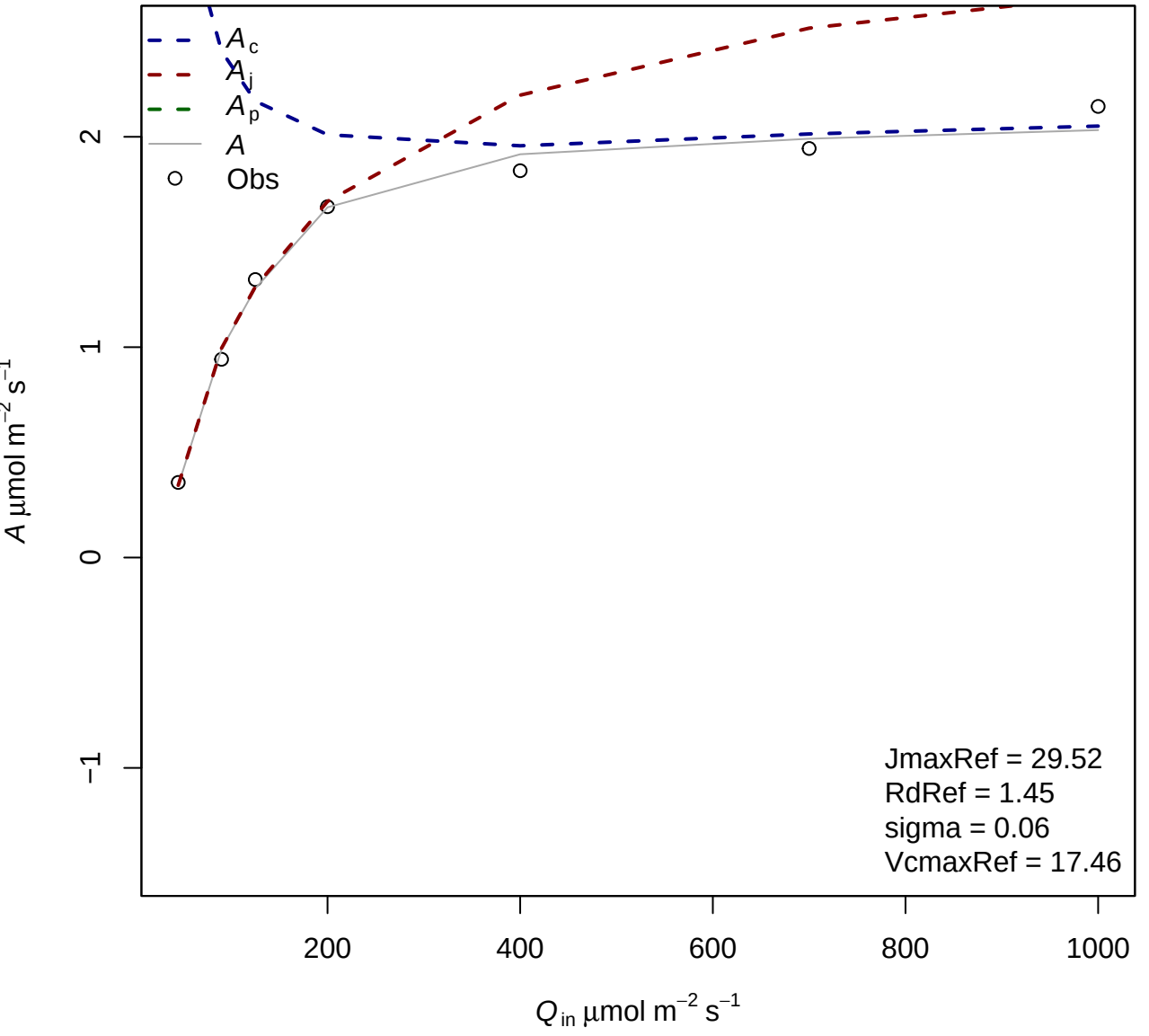

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.81
RdRef = 2.49
sigma = 0.08
VcmaxRef = 37.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800

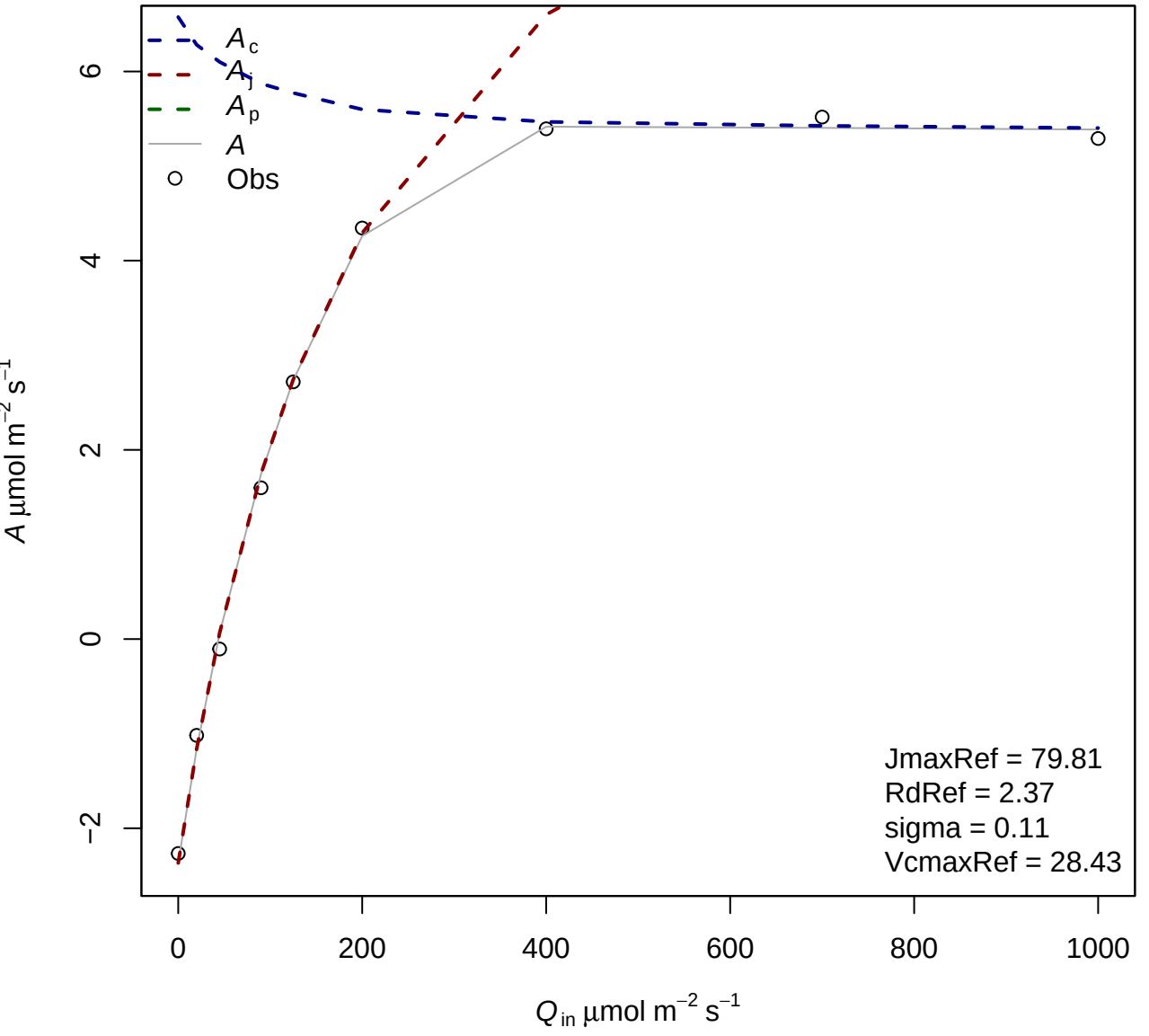

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.88
RdRef = 2.64
sigma = 0.18
VcmaxRef = 30.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

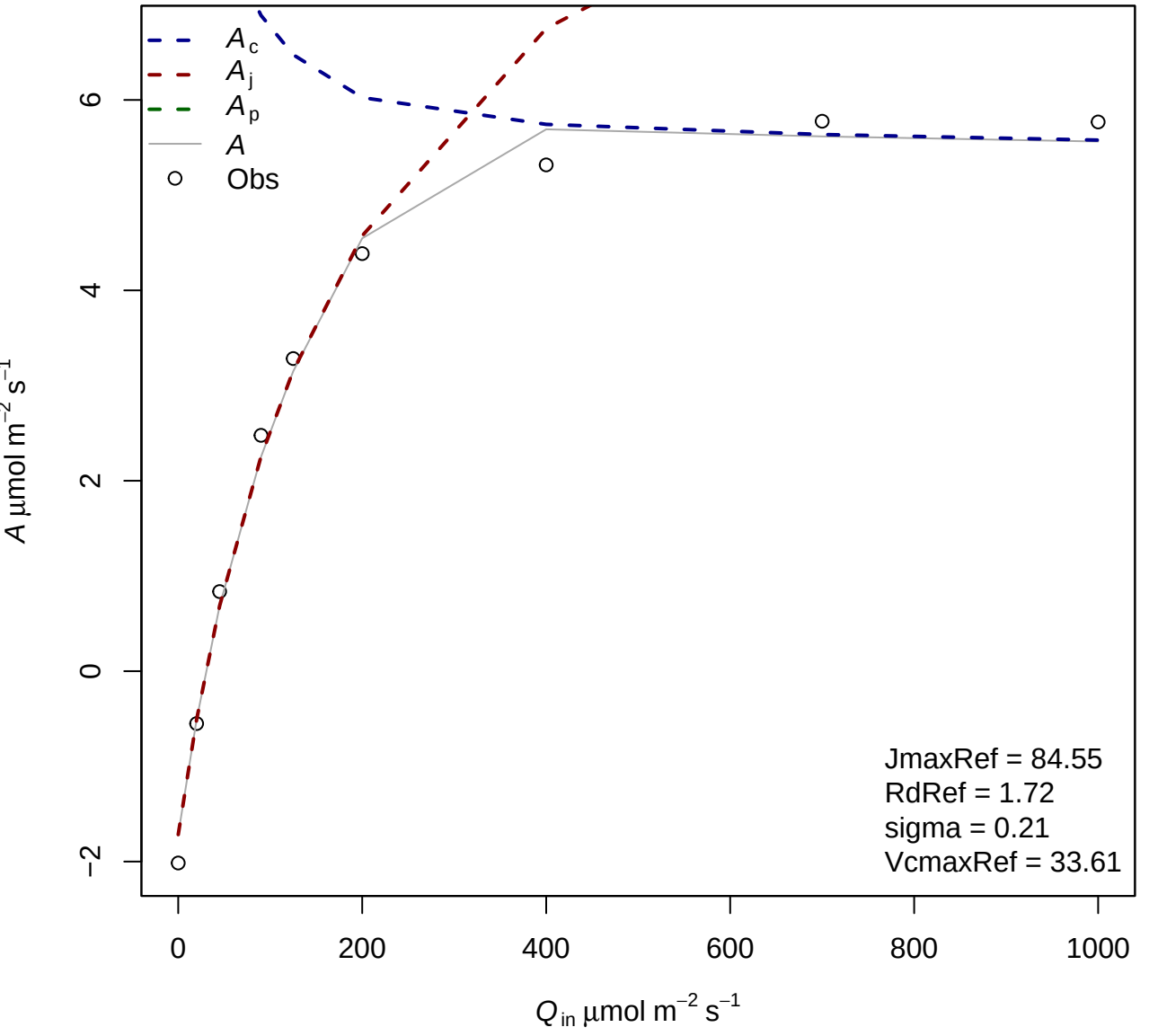

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.94
RdRef = 2.13
sigma = 0.31
VcmaxRef = 45.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

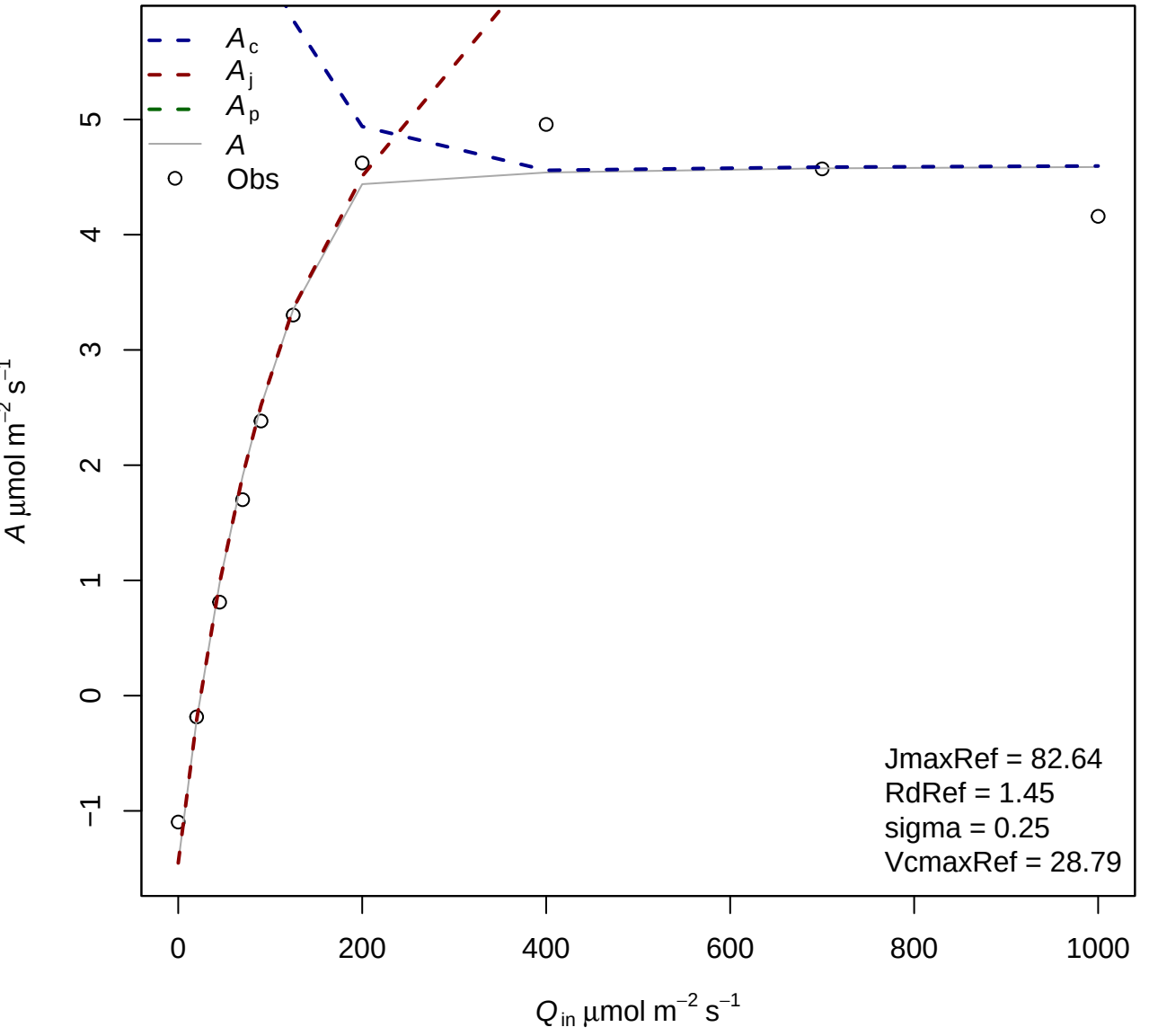

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.64
RdRef = 2.3
sigma = 0.25
VcmaxRef = 55.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000

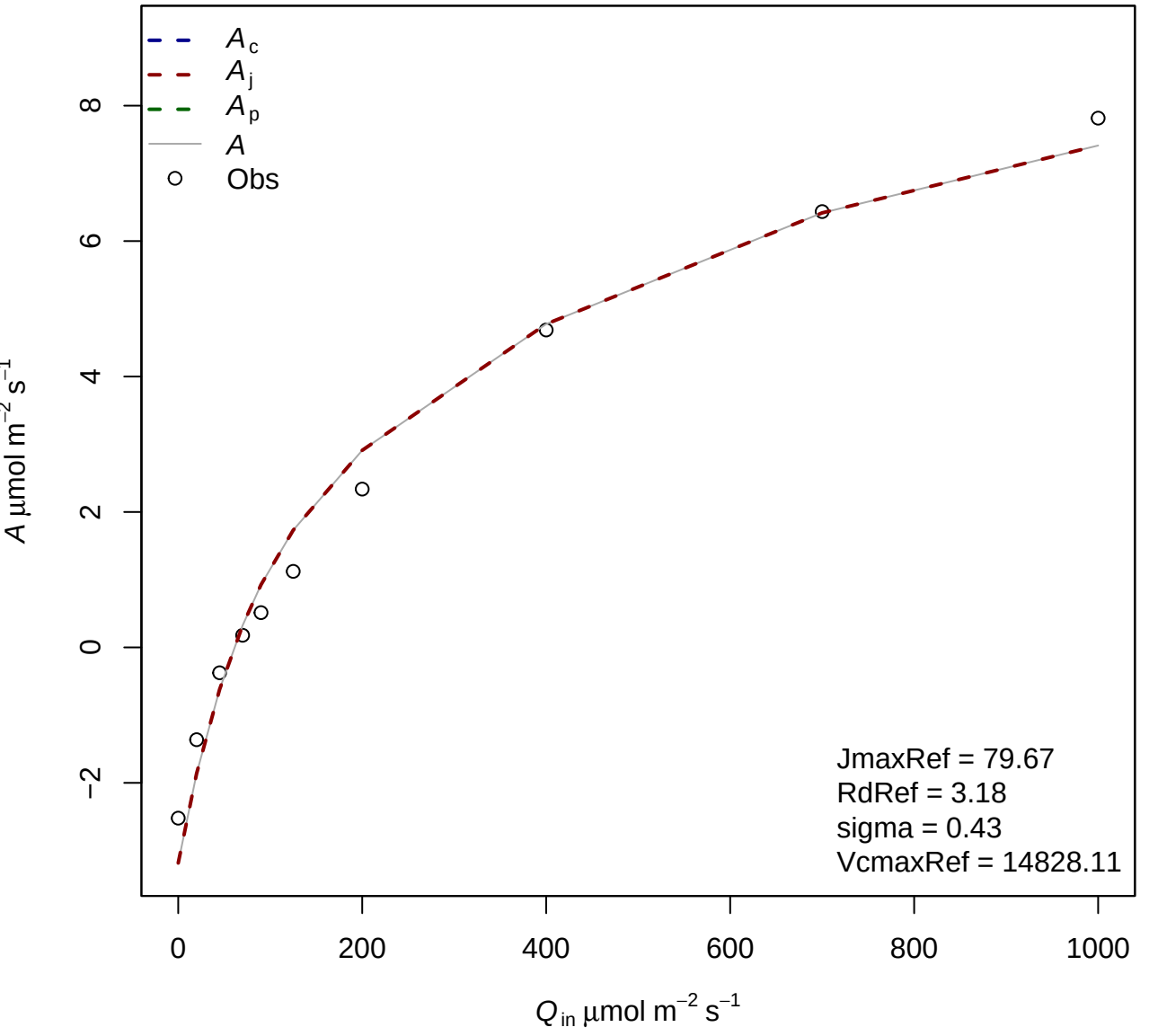

60
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.75
RdRef = 1.2
sigma = 0.14
VcmaxRef = 20.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000

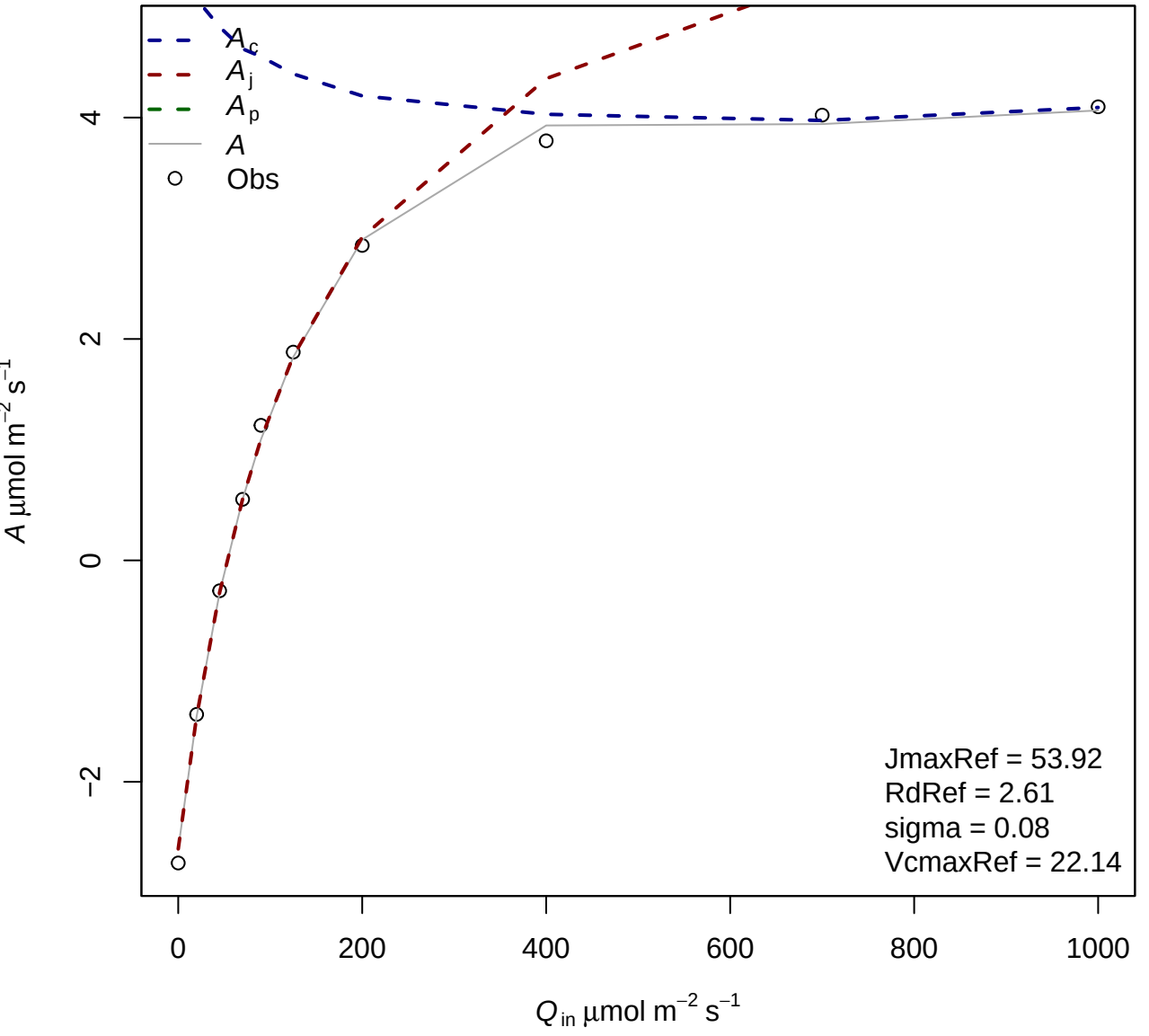

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.52
RdRef = 1.45
sigma = 0.06
VcmaxRef = 17.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
200
400
600
800
1000

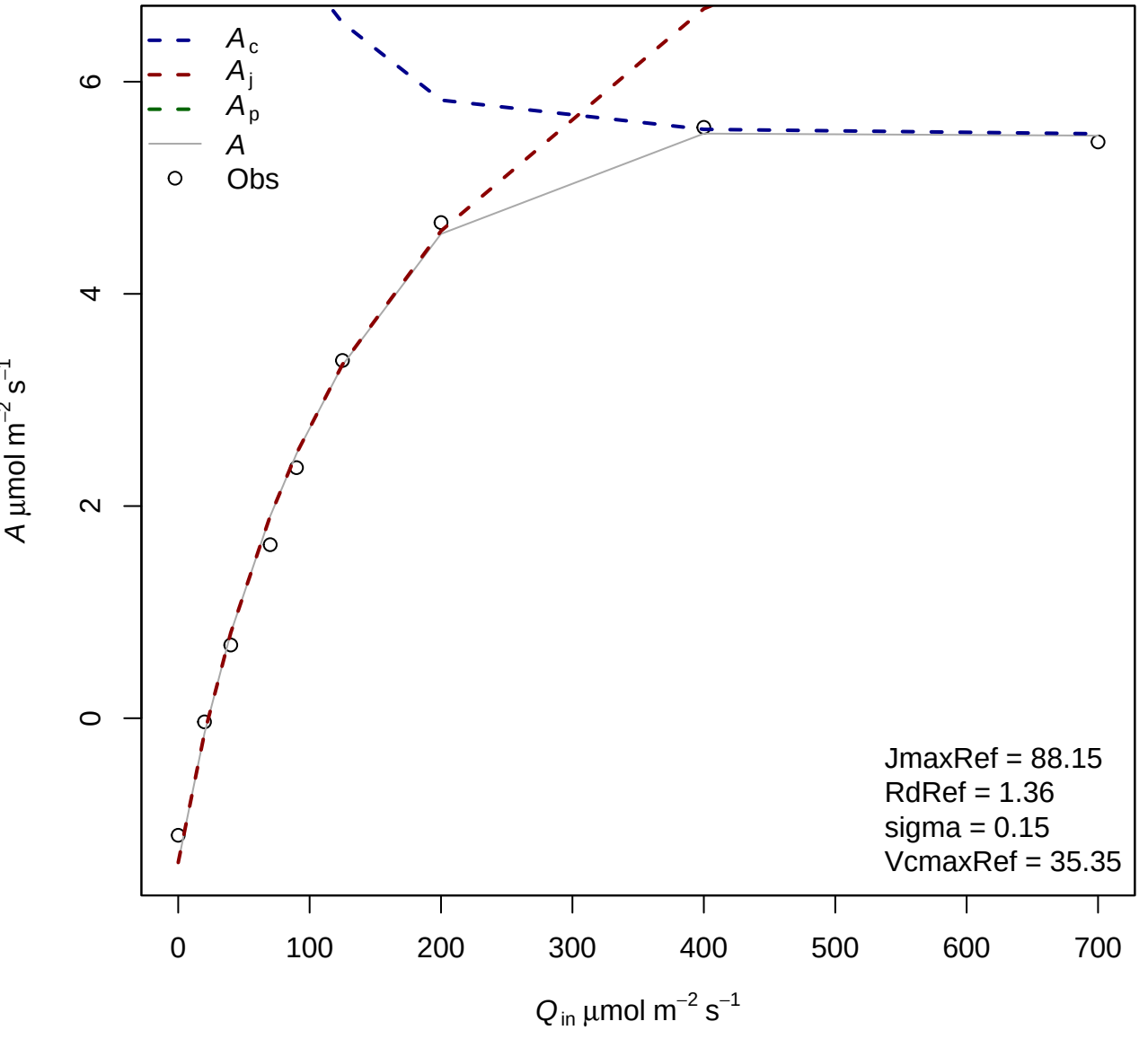

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.81
RdRef = 2.37
sigma = 0.11
VcmaxRef = 28.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000

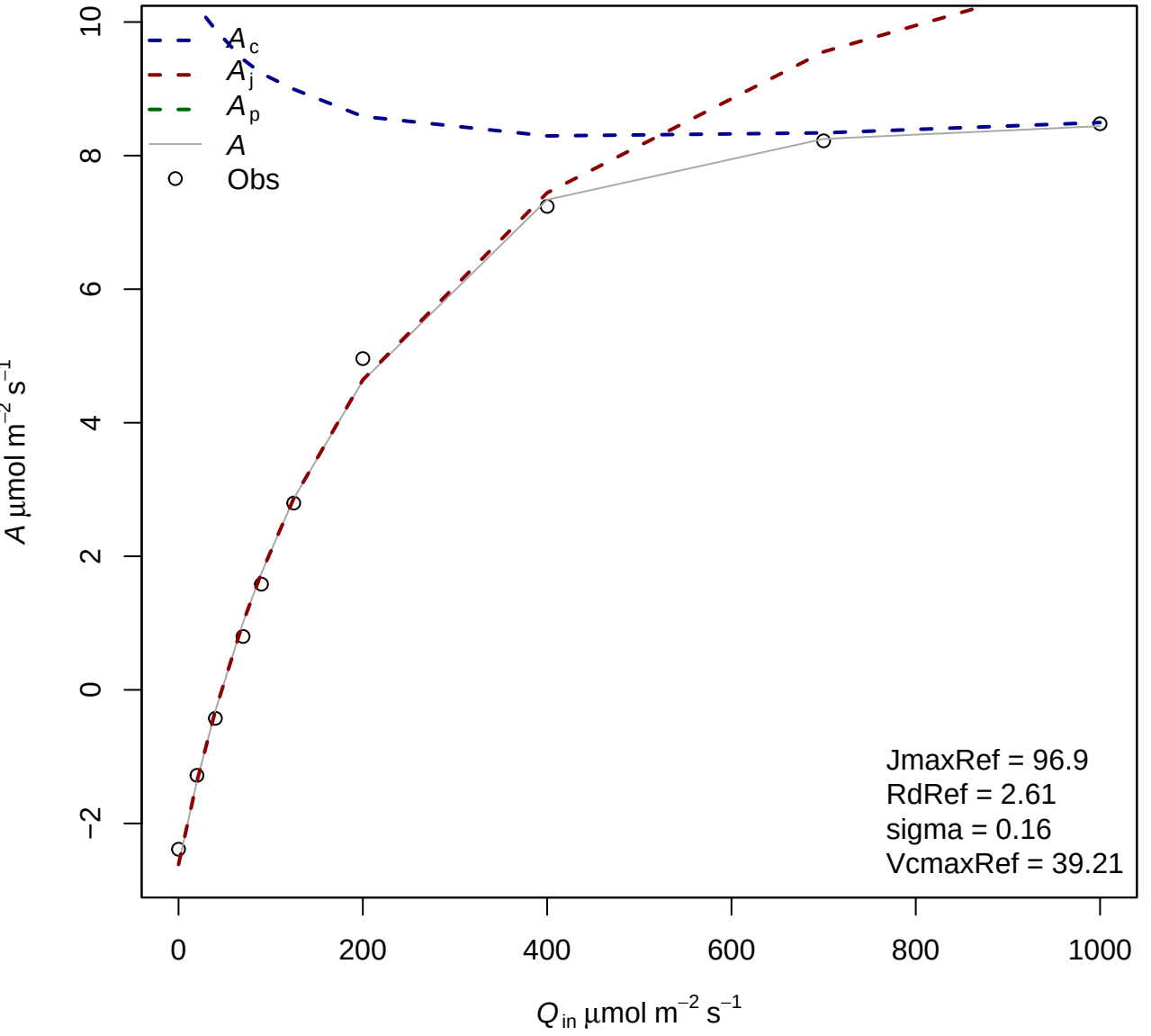

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.55
RdRef = 1.72
sigma = 0.21
VcmaxRef = 33.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000

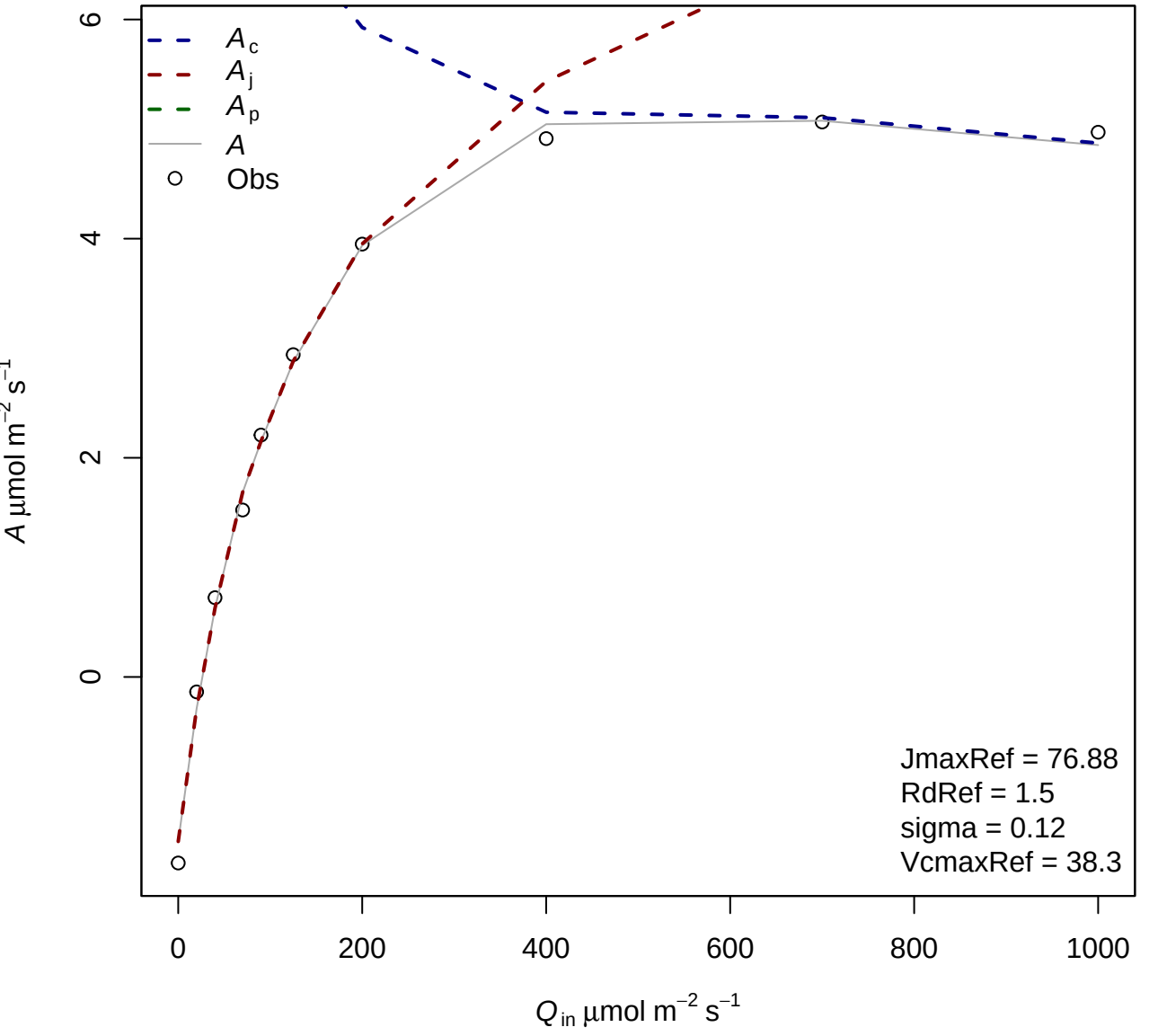

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.64
RdRef = 1.45
sigma = 0.25
VcmaxRef = 28.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
200
400
600
800
1000

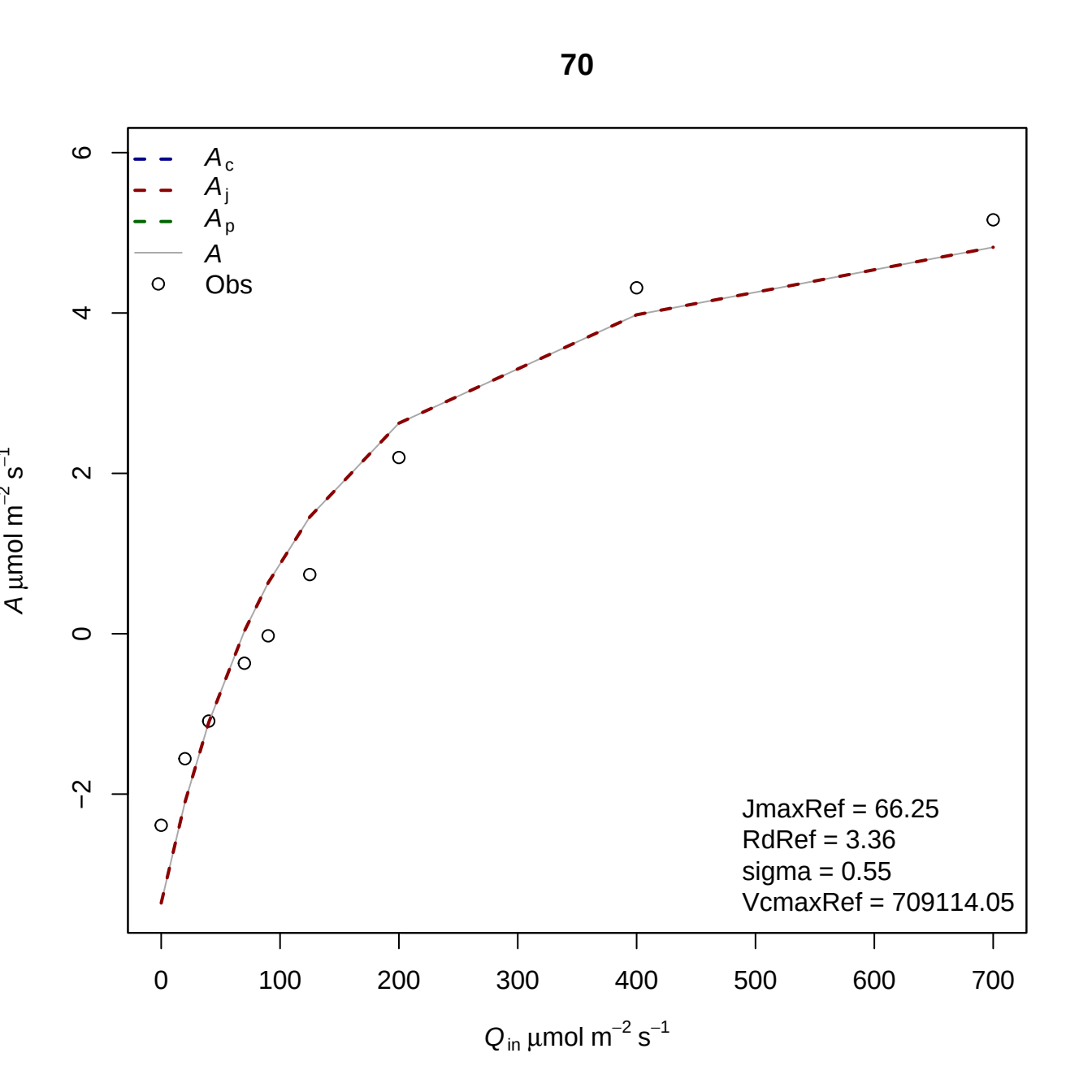

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.67
RdRef = 3.18
sigma = 0.43
VcmaxRef = 14828.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
200
400
600
800
1000

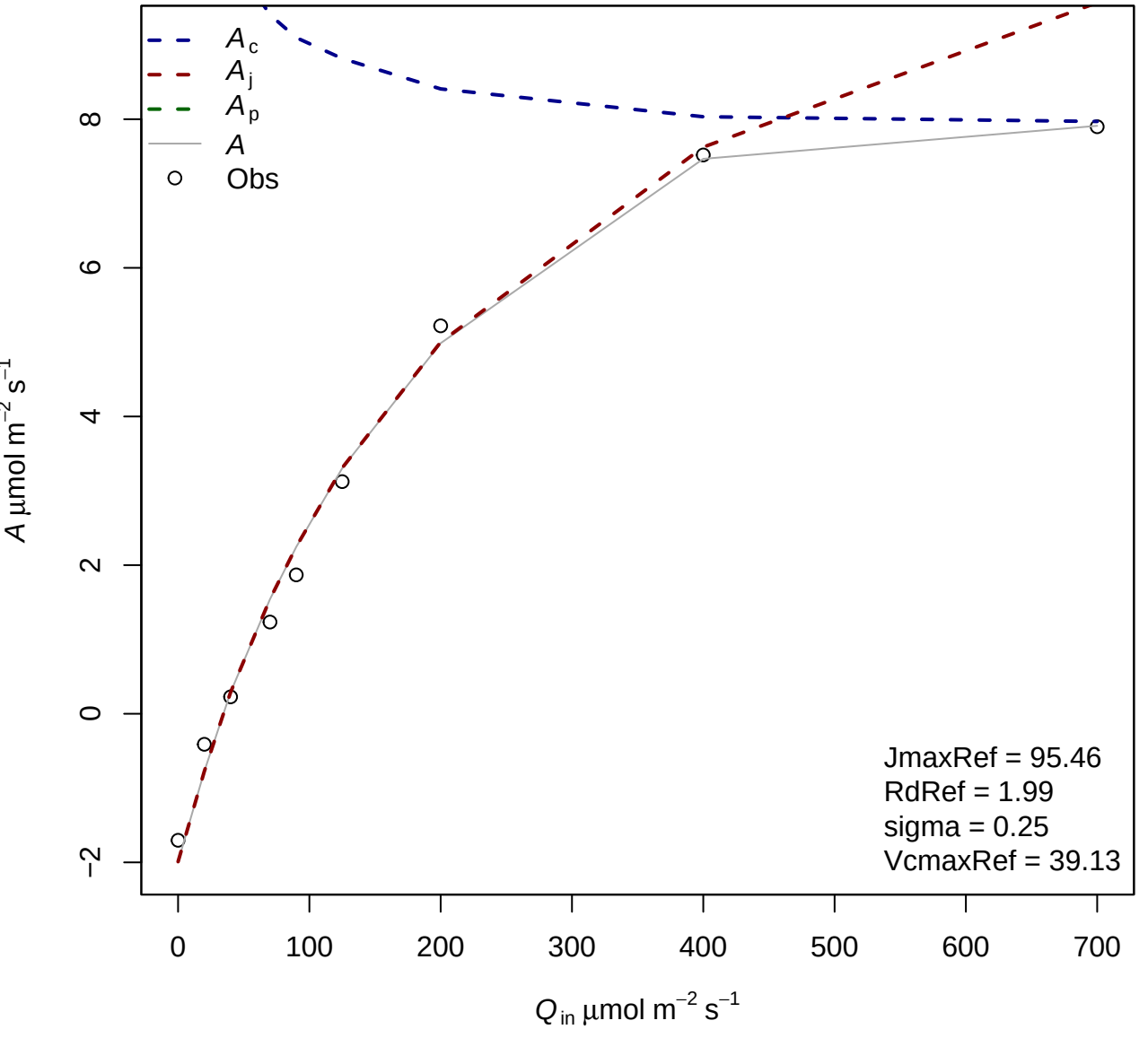

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 53.92
RdRef = 2.61
sigma = 0.08
VcmaxRef = 22.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000

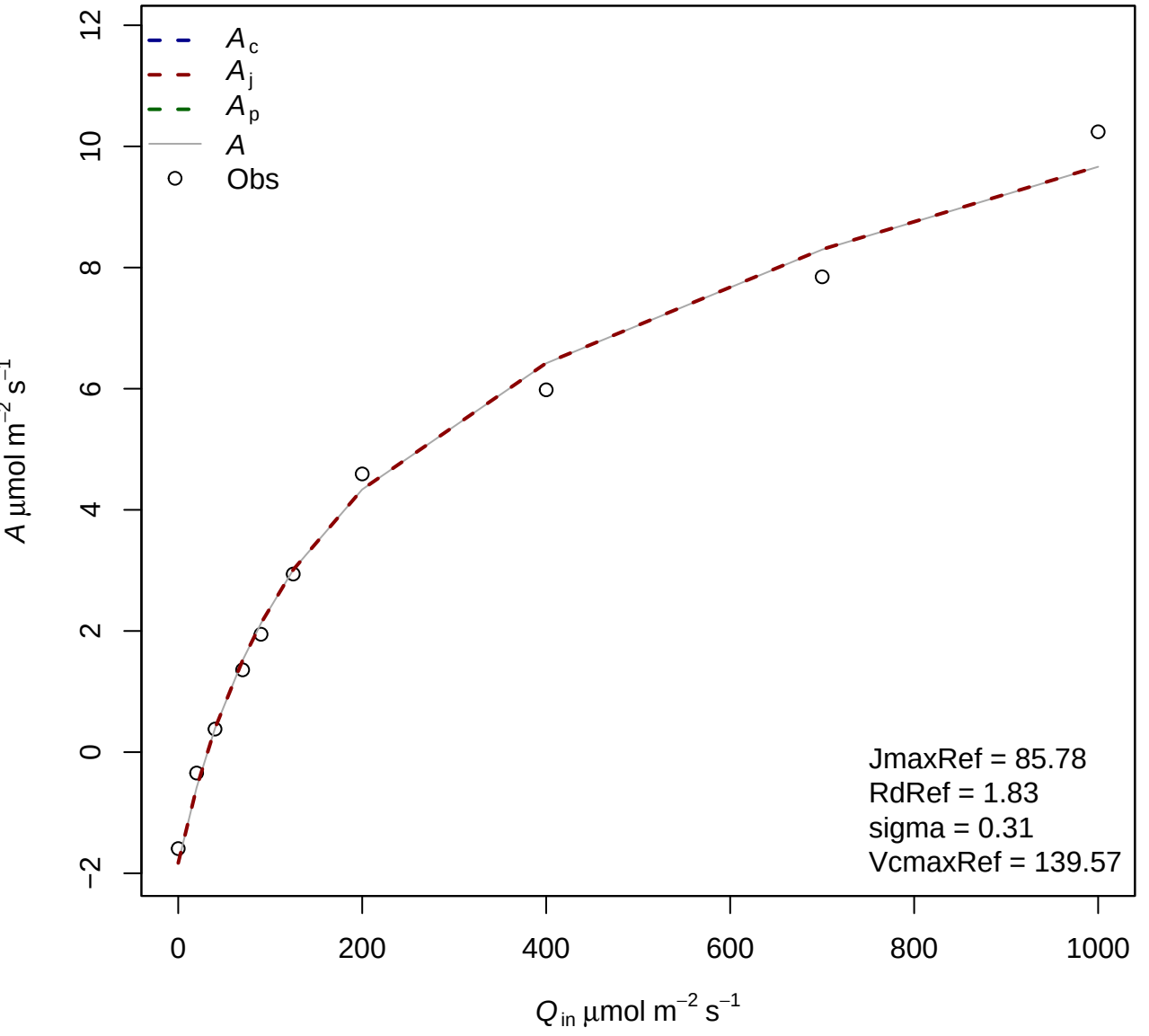

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.15
RdRef = 1.36
sigma = 0.15
VcmaxRef = 35.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
100
200
300
400
500
600
700

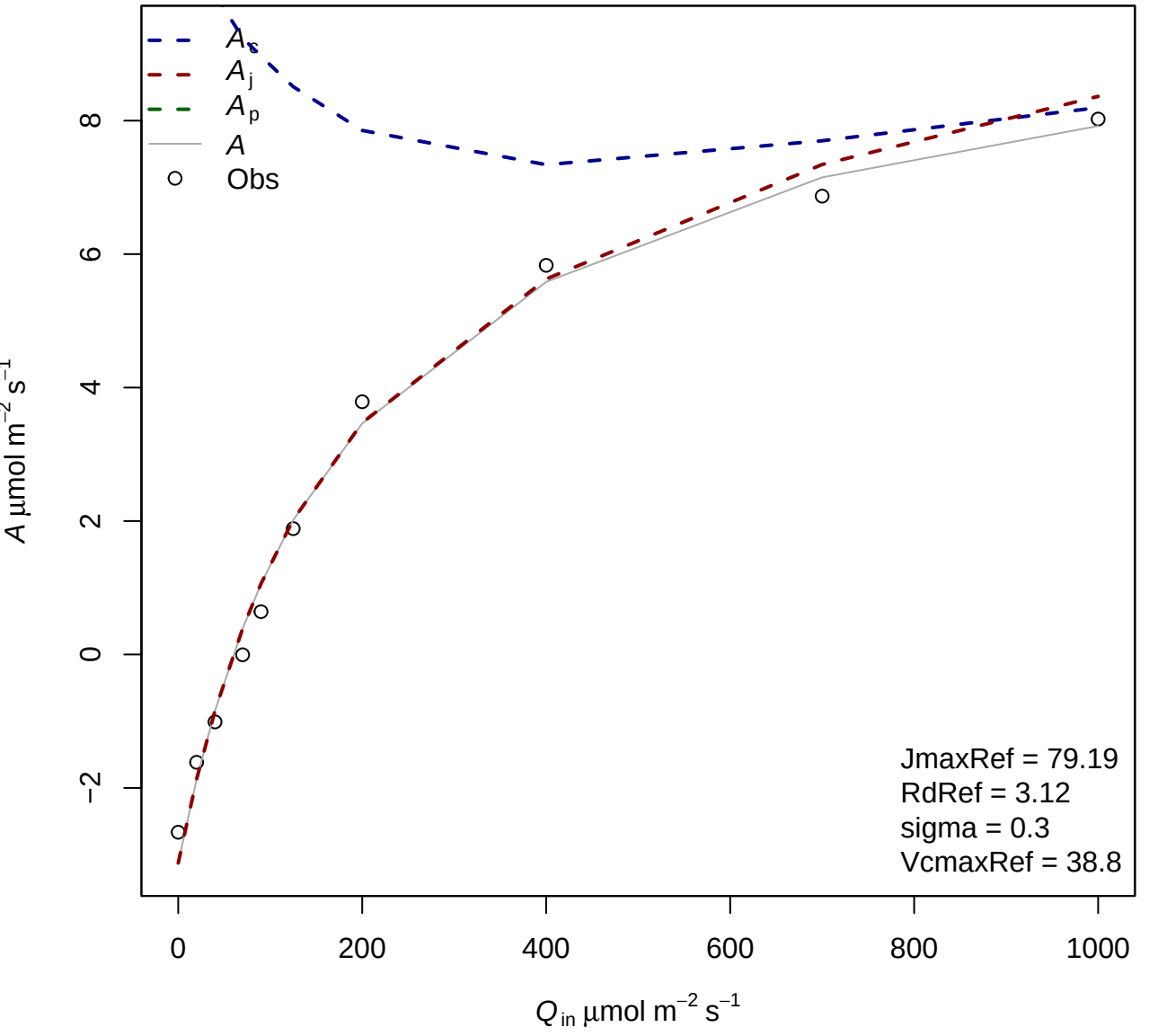

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.9
RdRef = 2.61
sigma = 0.16
VcmaxRef = 39.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
200
400
600
800
1000

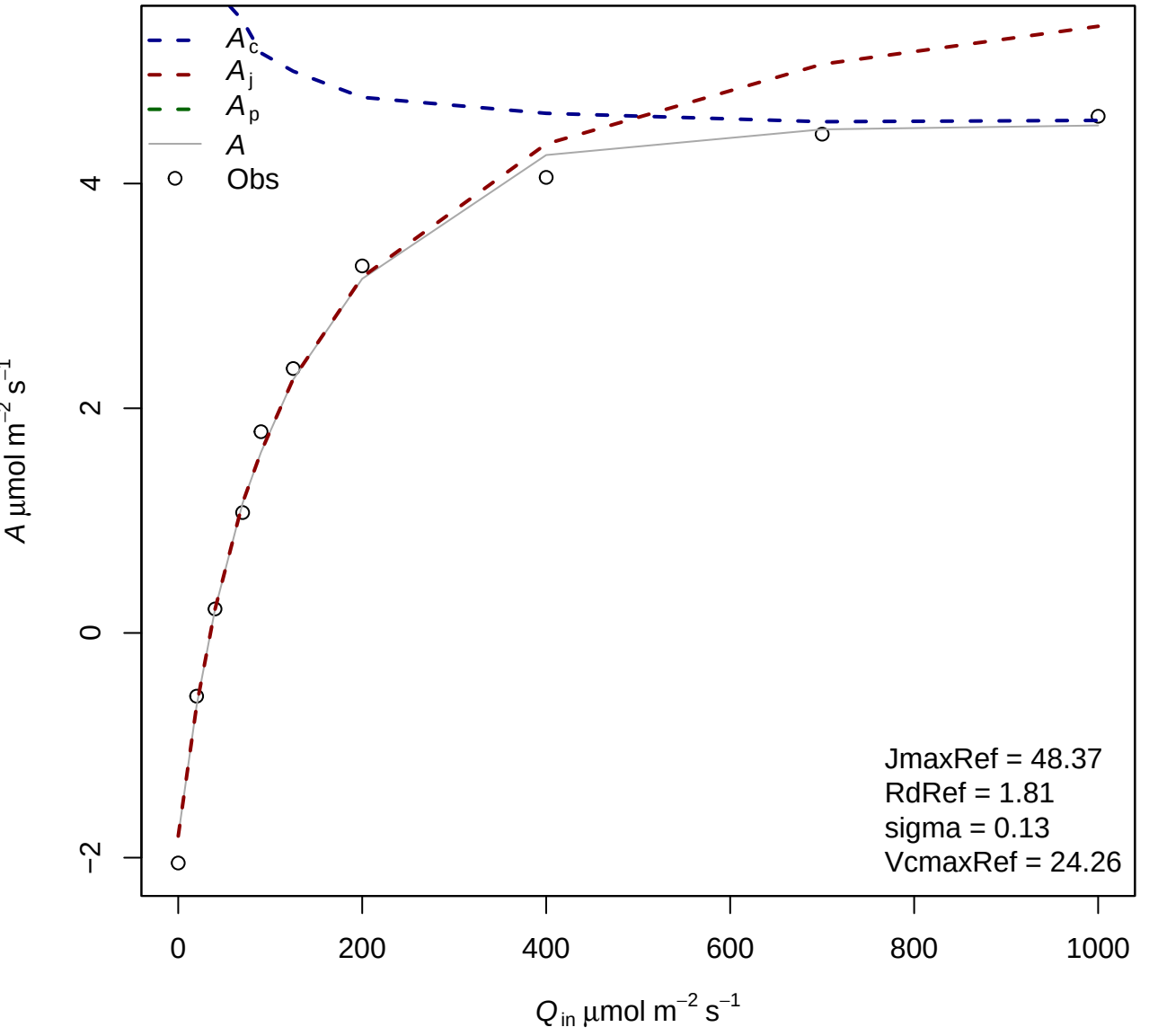

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.88
RdRef = 1.5
sigma = 0.12
VcmaxRef = 38.3
0
2
4
6
$A$ µmol m$^{-2}$ s$^{-1}$
0
200
400
600
800
1000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

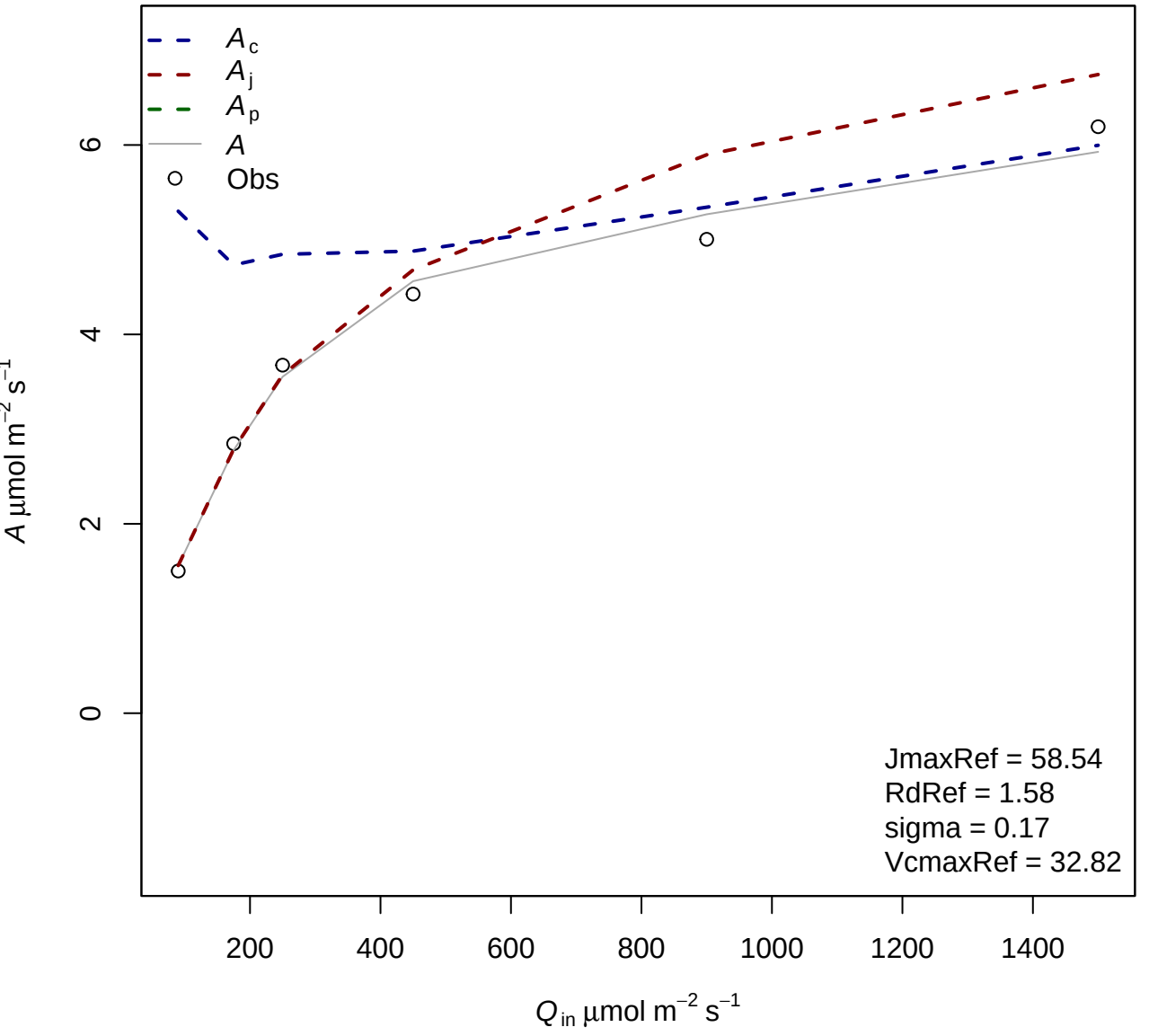

70
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.25
RdRef = 3.36
sigma = 0.55
VcmaxRef = 709114.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
100
200
300
400
500
600
700

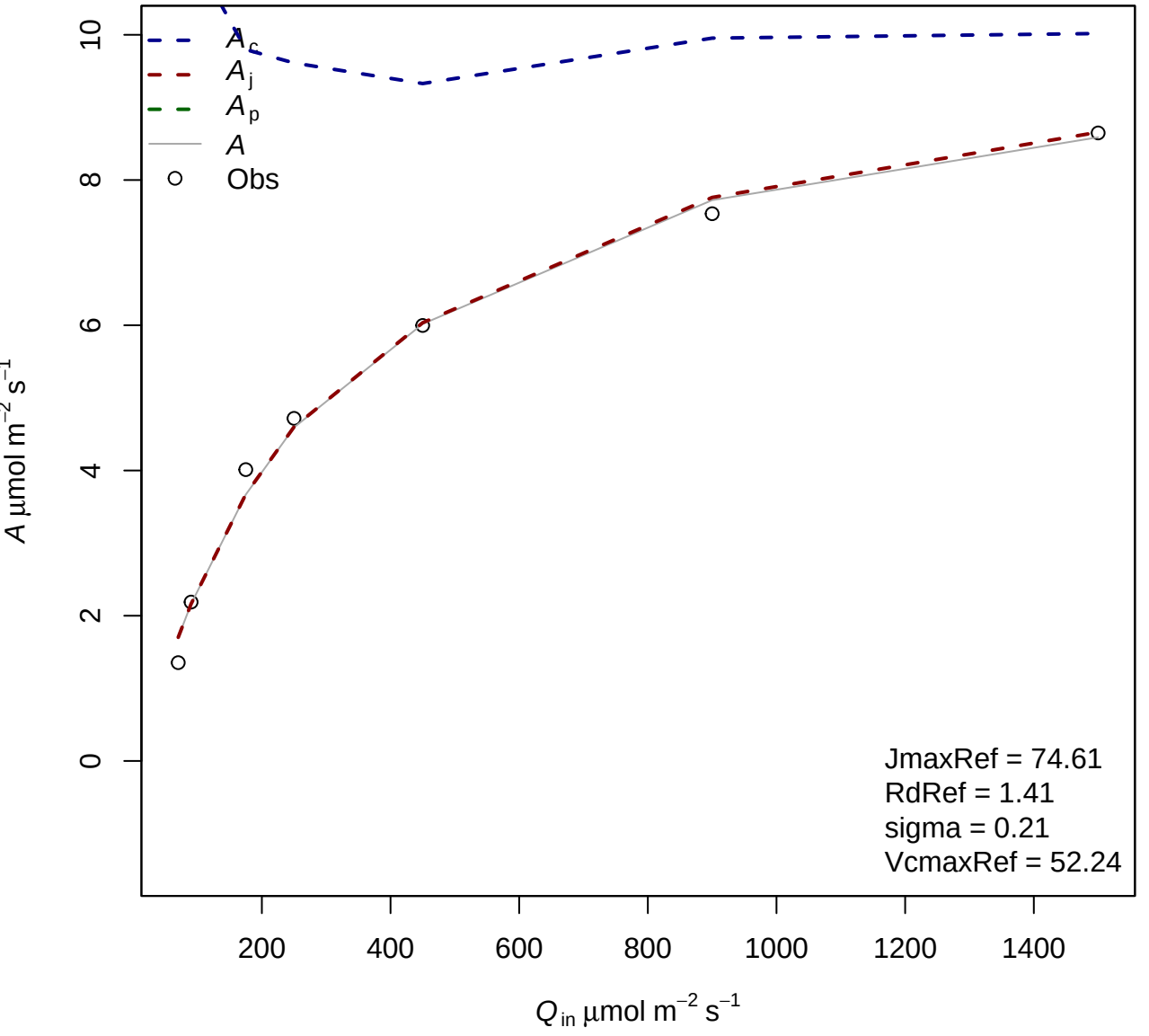

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.46
RdRef = 1.99
sigma = 0.25
VcmaxRef = 39.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

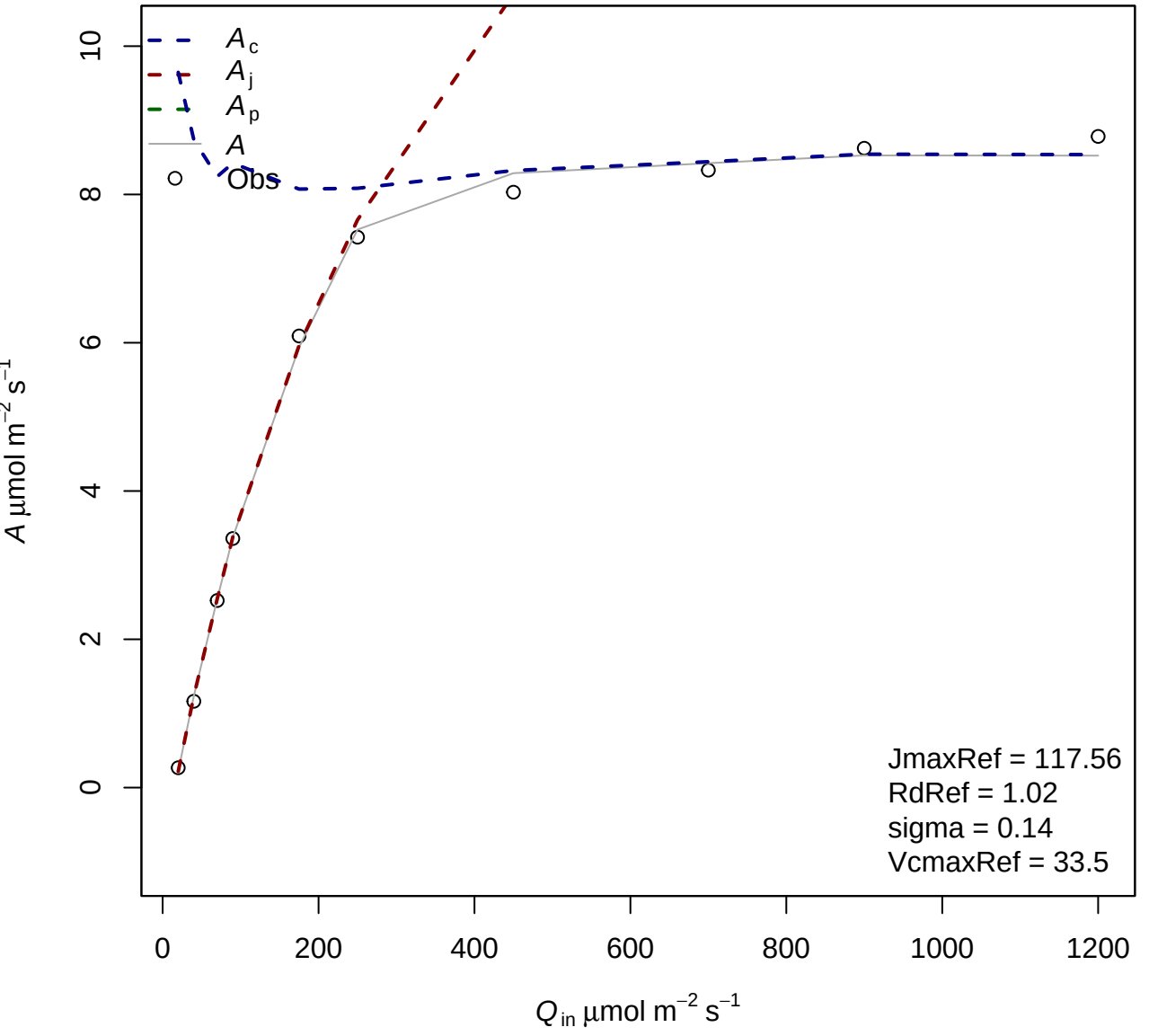

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.78
RdRef = 1.83
sigma = 0.31
VcmaxRef = 139.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000

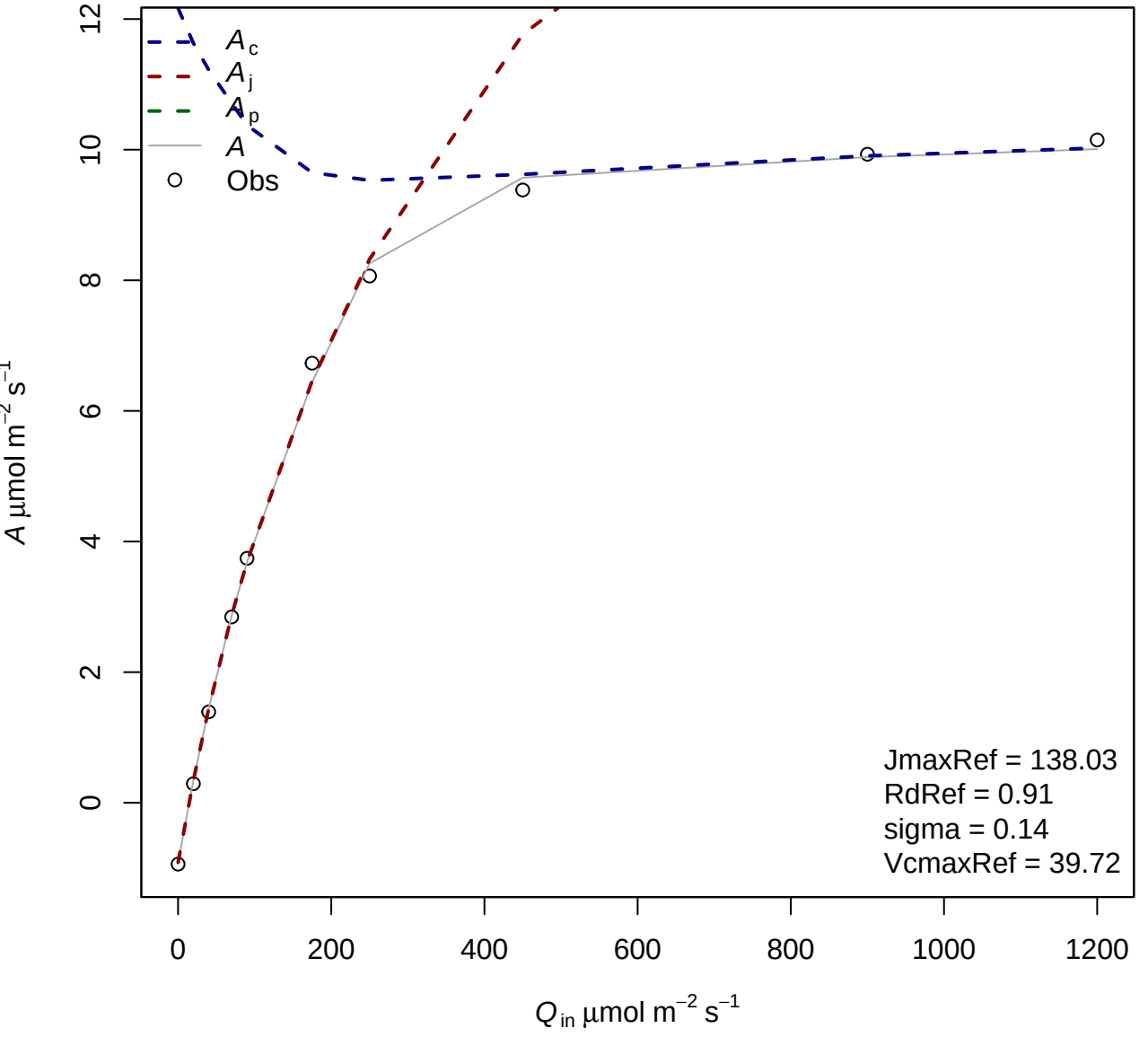

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.19
RdRef = 3.12
sigma = 0.3
VcmaxRef = 38.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
200
400
600
800
1000

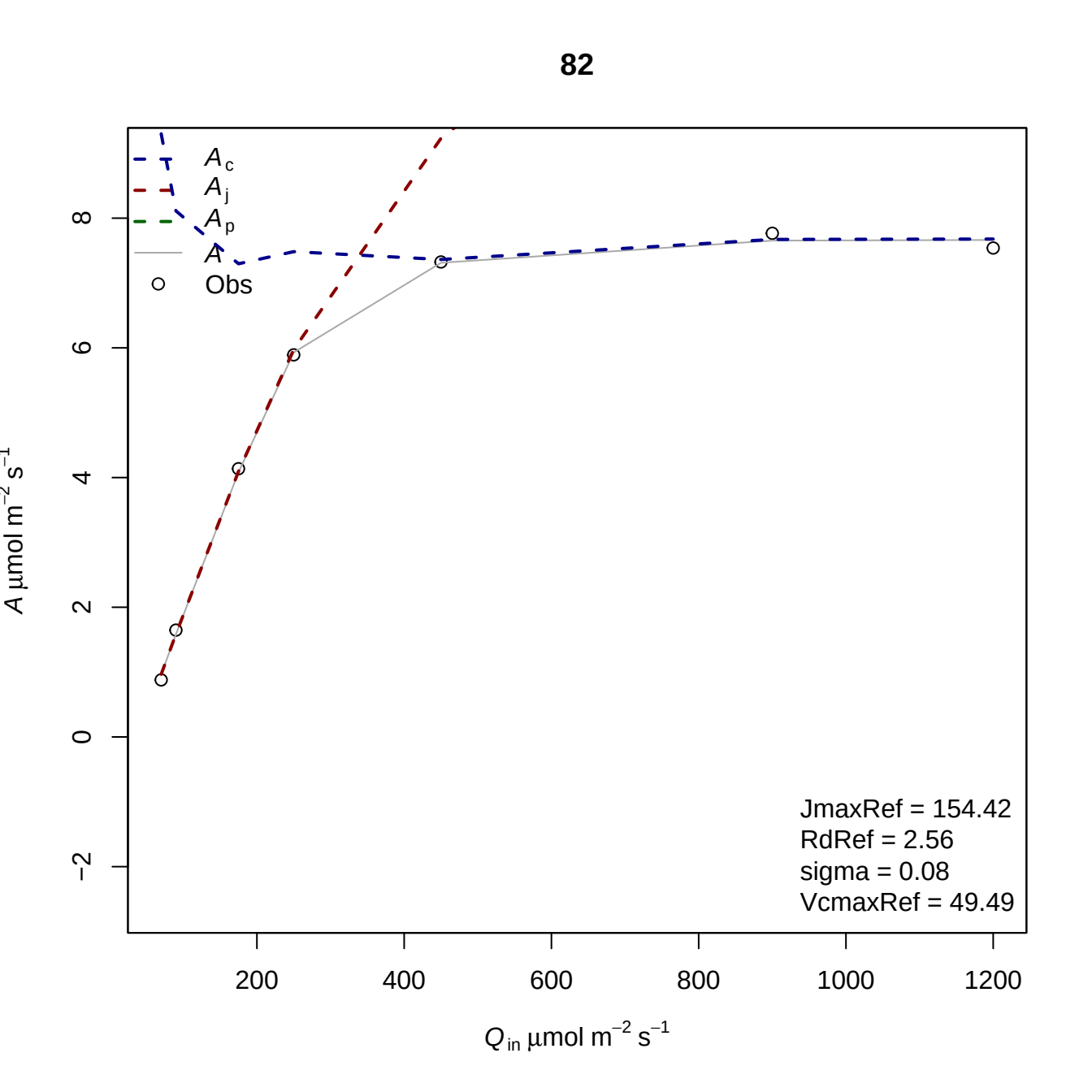

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.37
RdRef = 1.81
sigma = 0.13
VcmaxRef = 24.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000

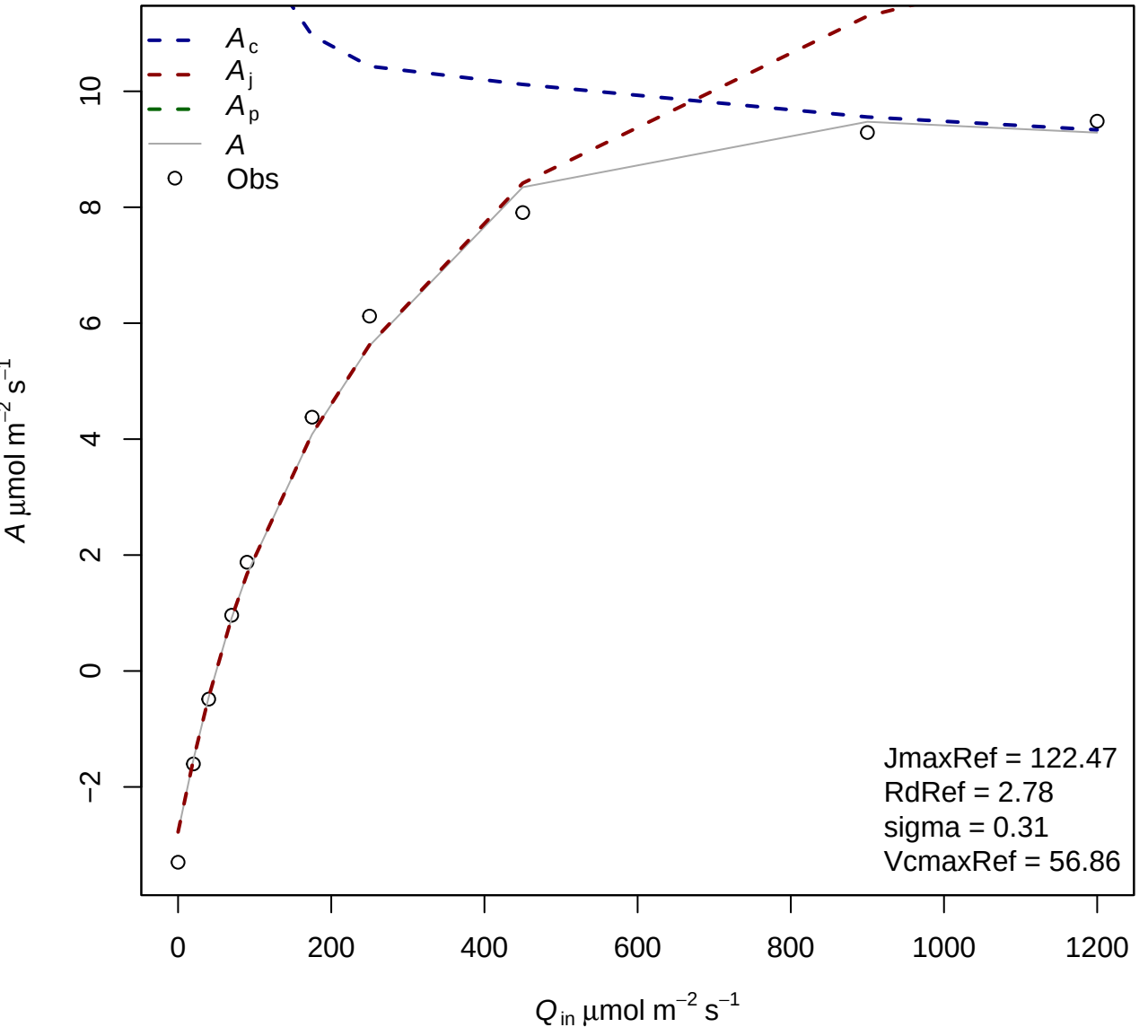

$A_c$
$A_j$
$A_p$
$A$
Obs
0
2
4
6
$A$ µmol m$^{-2}$ s$^{-1}$
200
400
600
800
1000
1200
1400
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 58.54
RdRef = 1.58
sigma = 0.17
VcmaxRef = 32.82

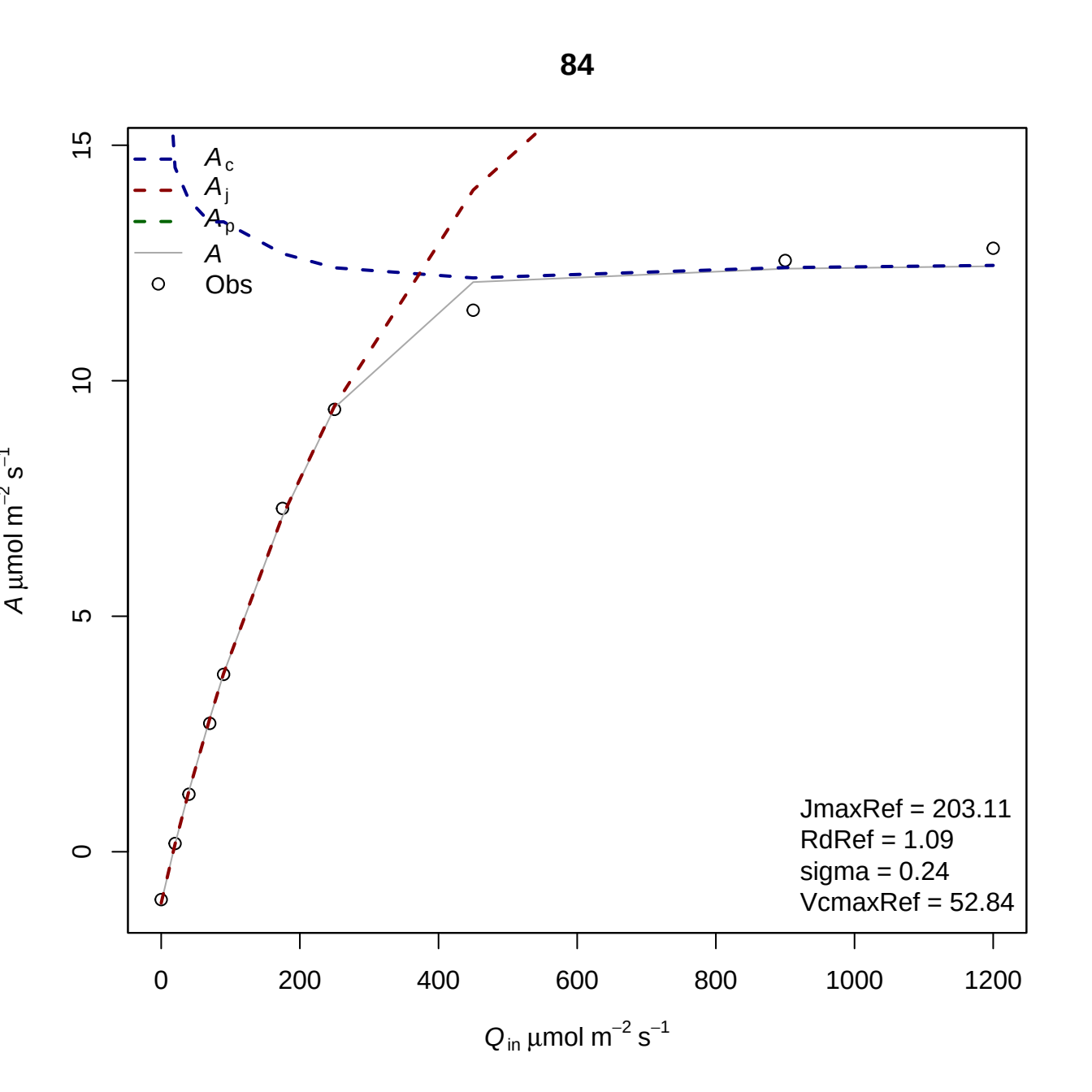

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.61
RdRef = 1.41
sigma = 0.21
VcmaxRef = 52.24
0
2
4
6
8
10
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

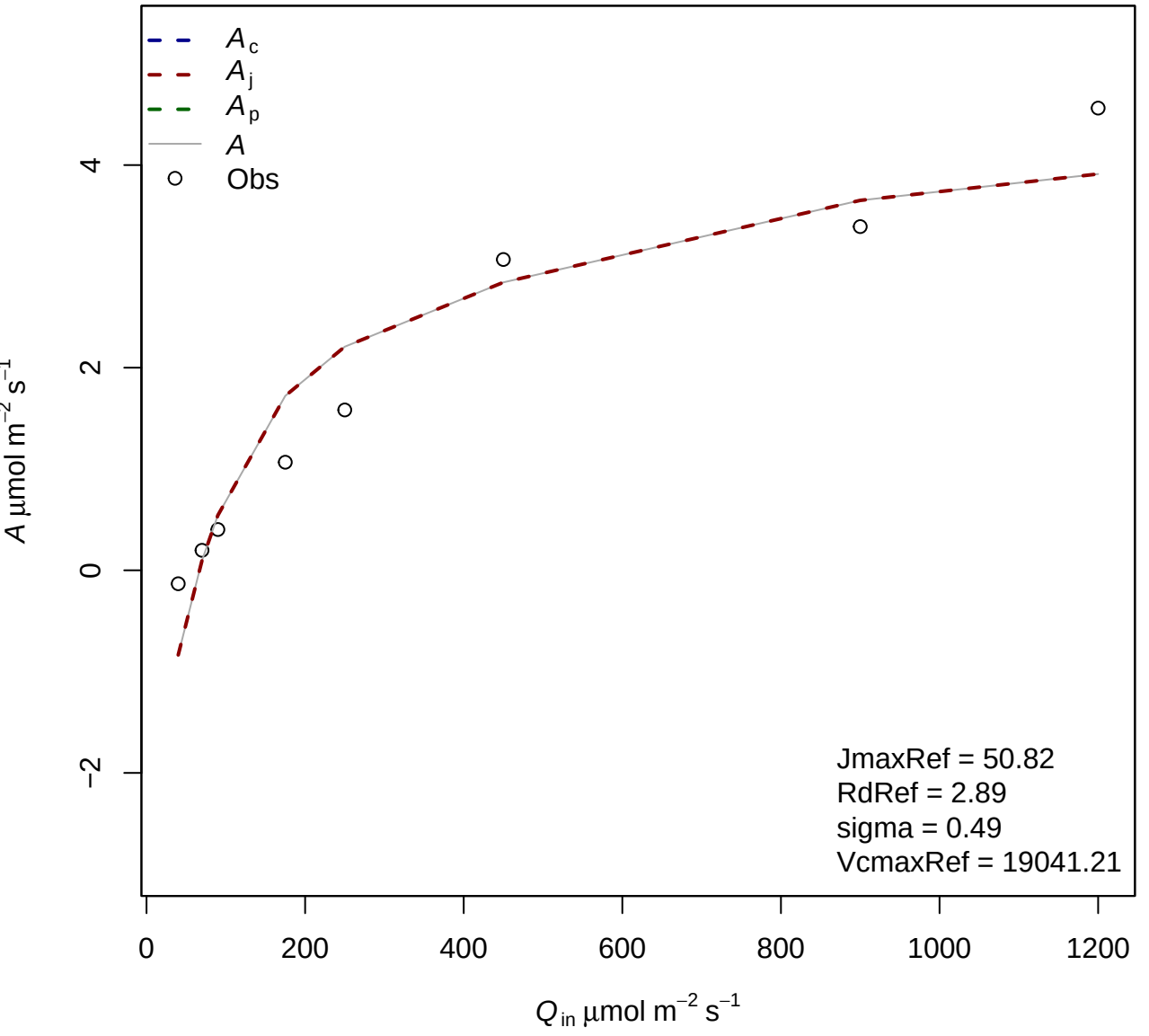

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.56
RdRef = 1.02
sigma = 0.14
VcmaxRef = 33.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
200
400
600
800
1000
1200

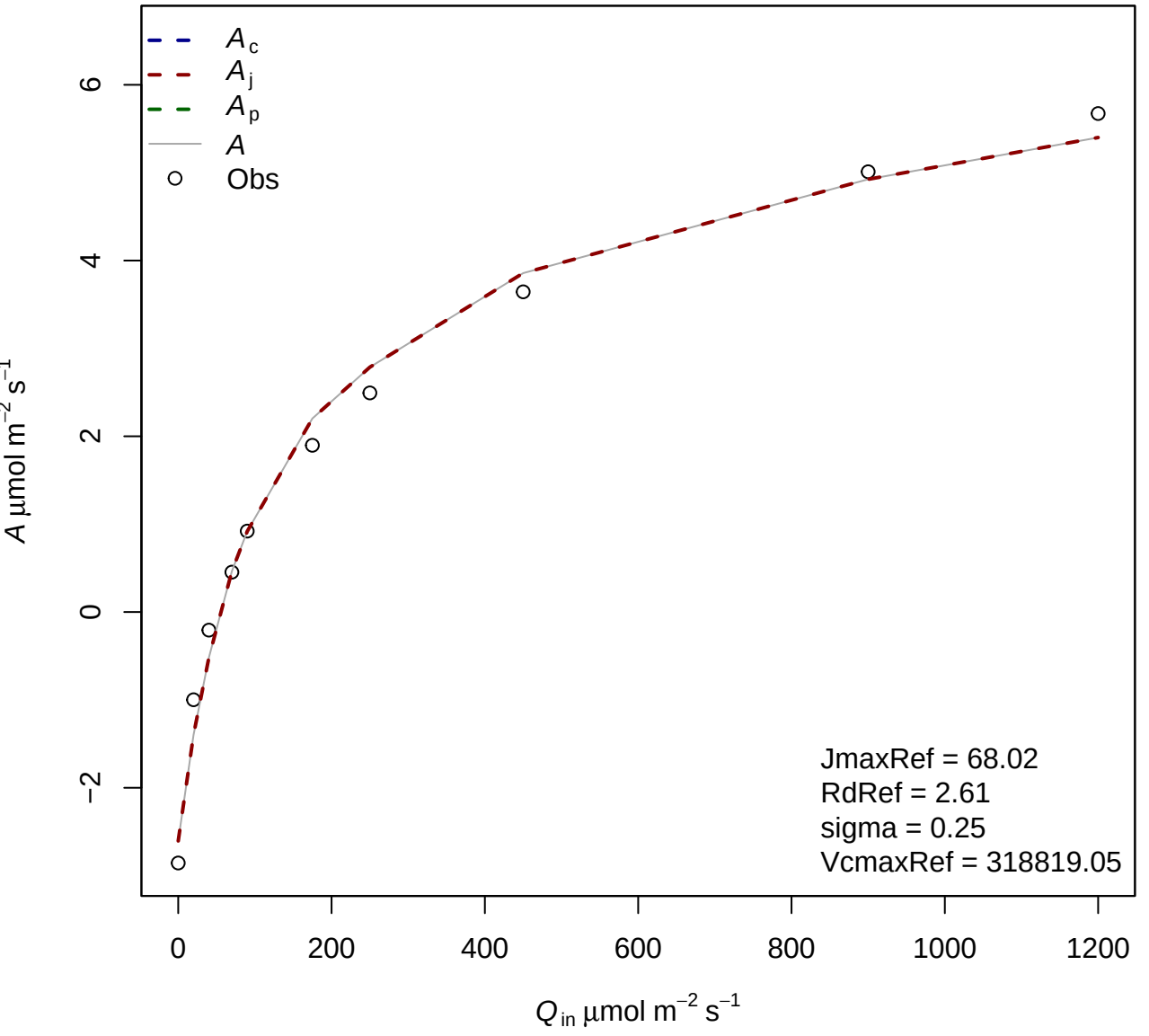

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.03
RdRef = 0.91
sigma = 0.14
VcmaxRef = 39.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

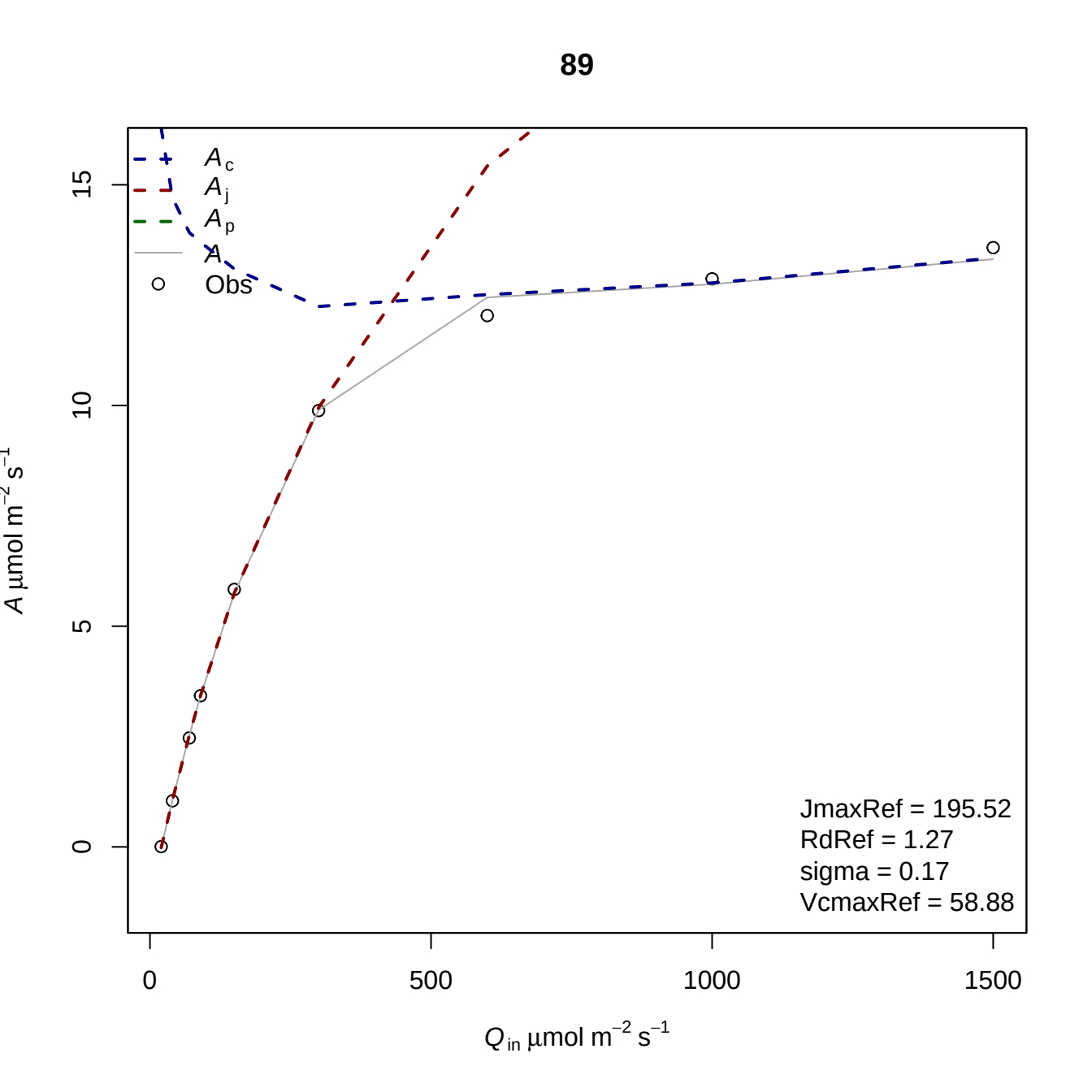

82
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 154.42
RdRef = 2.56
sigma = 0.08
VcmaxRef = 49.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
200
400
600
800
1000
1200

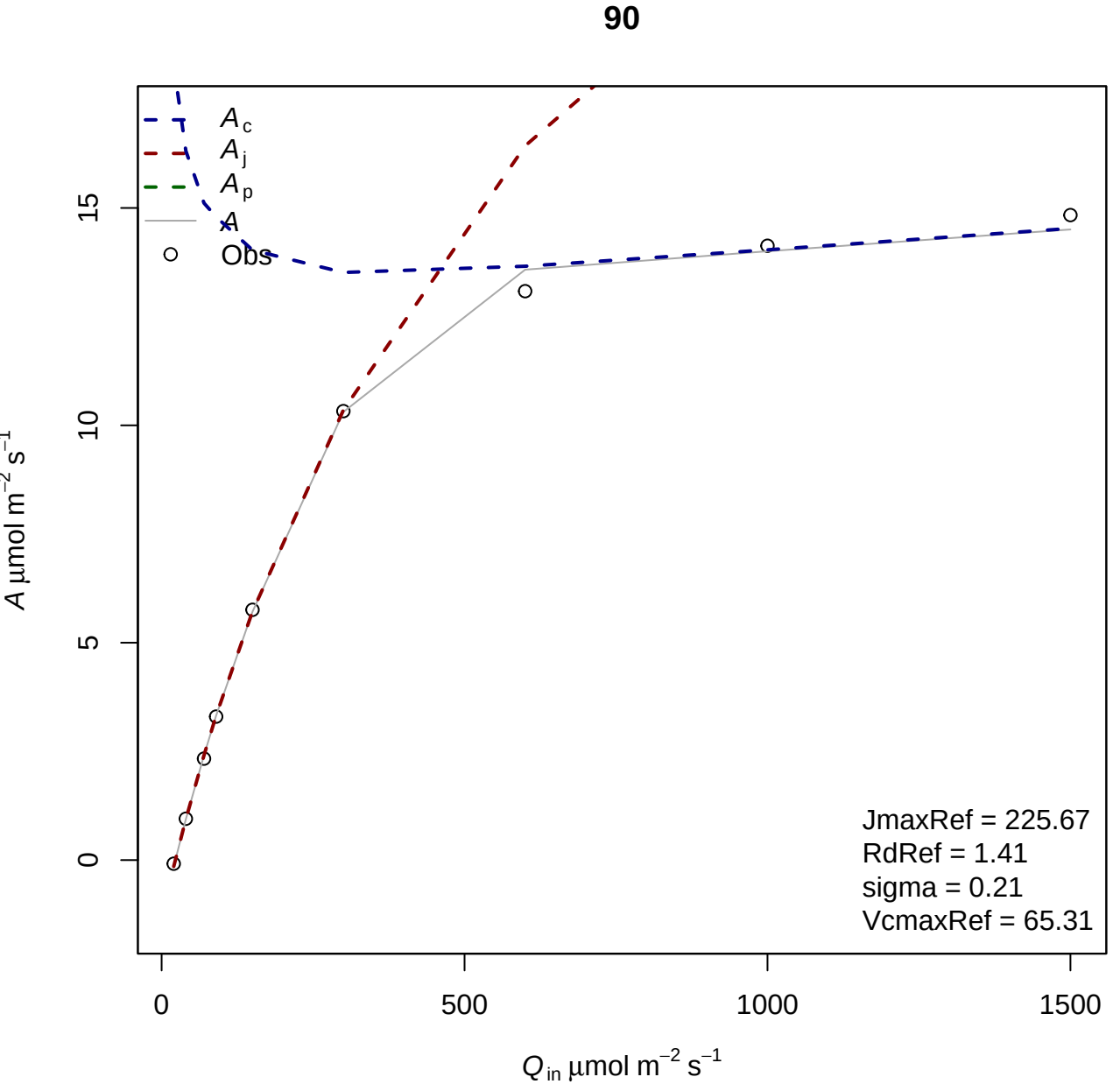

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.47
RdRef = 2.78
sigma = 0.31
VcmaxRef = 56.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

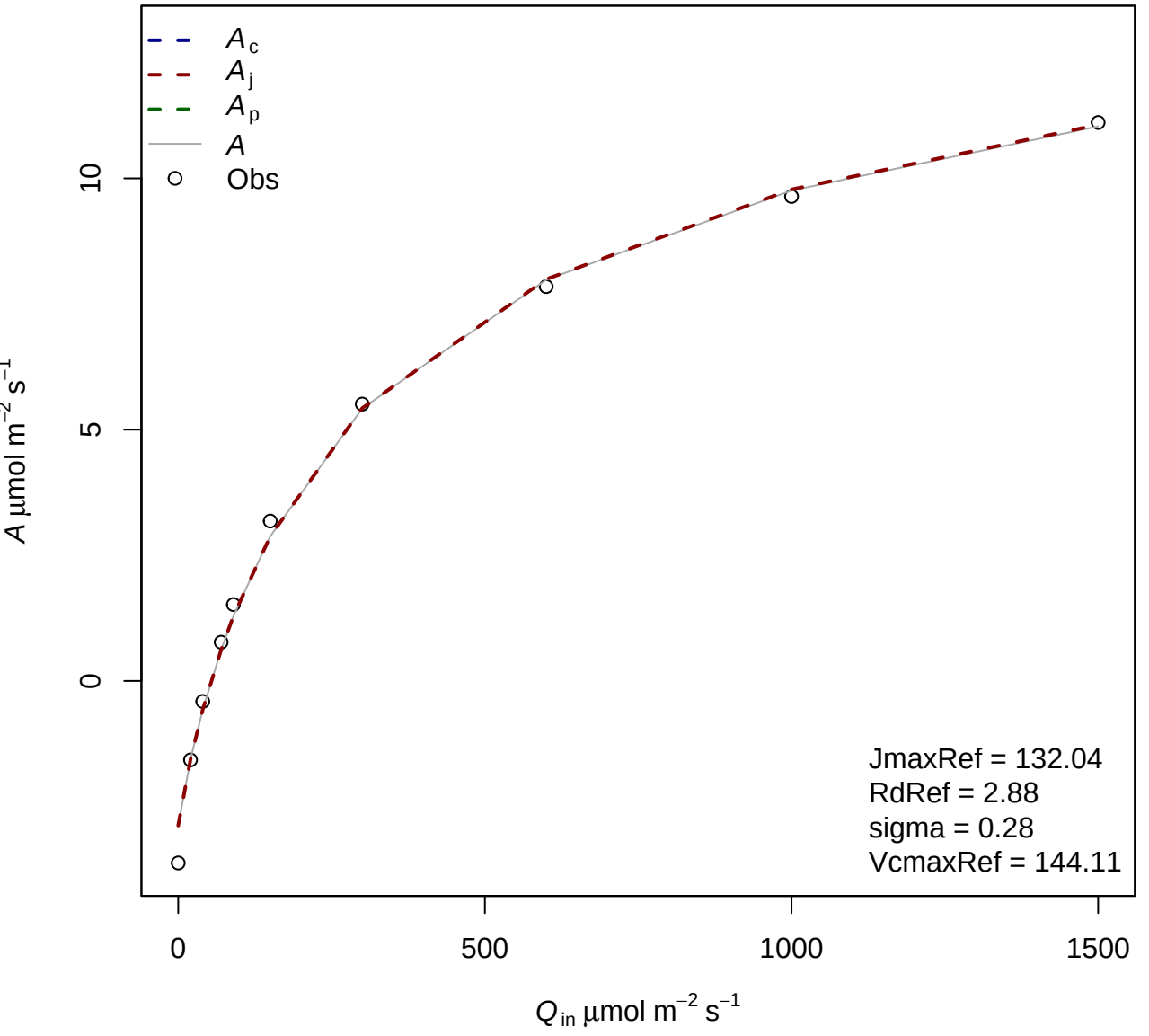

84
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 203.11
RdRef = 1.09
sigma = 0.24
VcmaxRef = 52.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000
1200

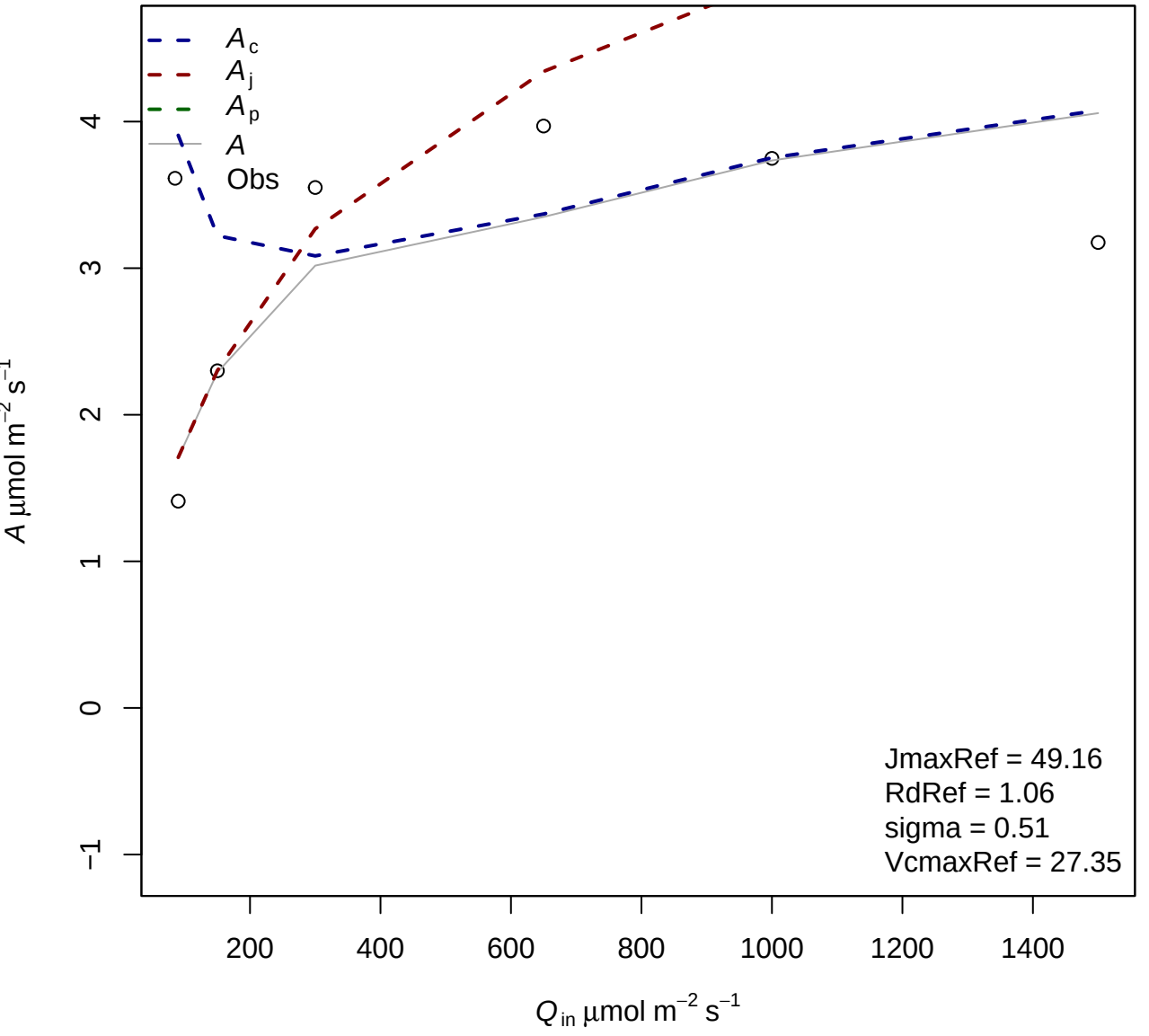

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.82
RdRef = 2.89
sigma = 0.49
VcmaxRef = 19041.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000
1200

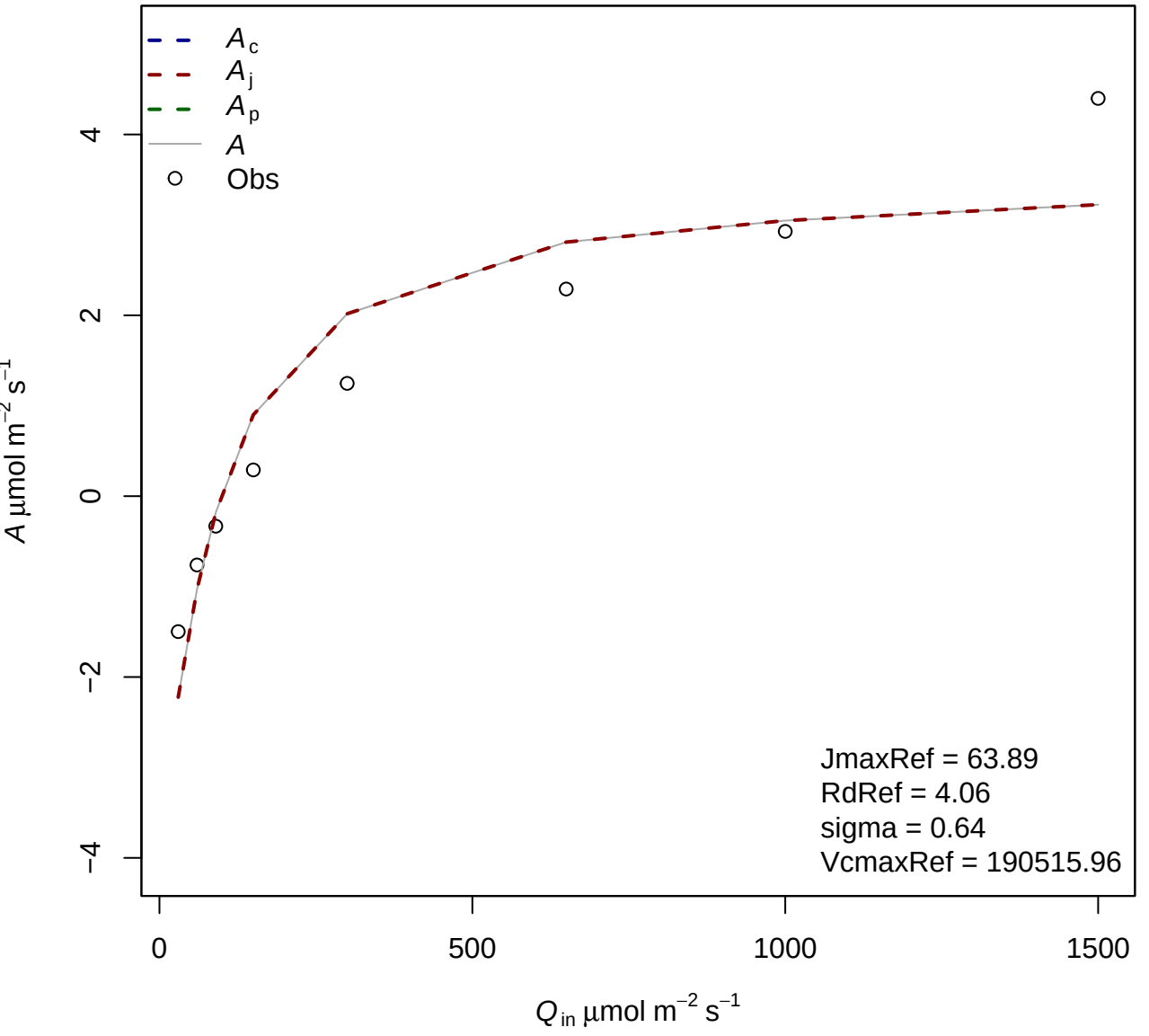

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.02
RdRef = 2.61
sigma = 0.25
VcmaxRef = 318819.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200

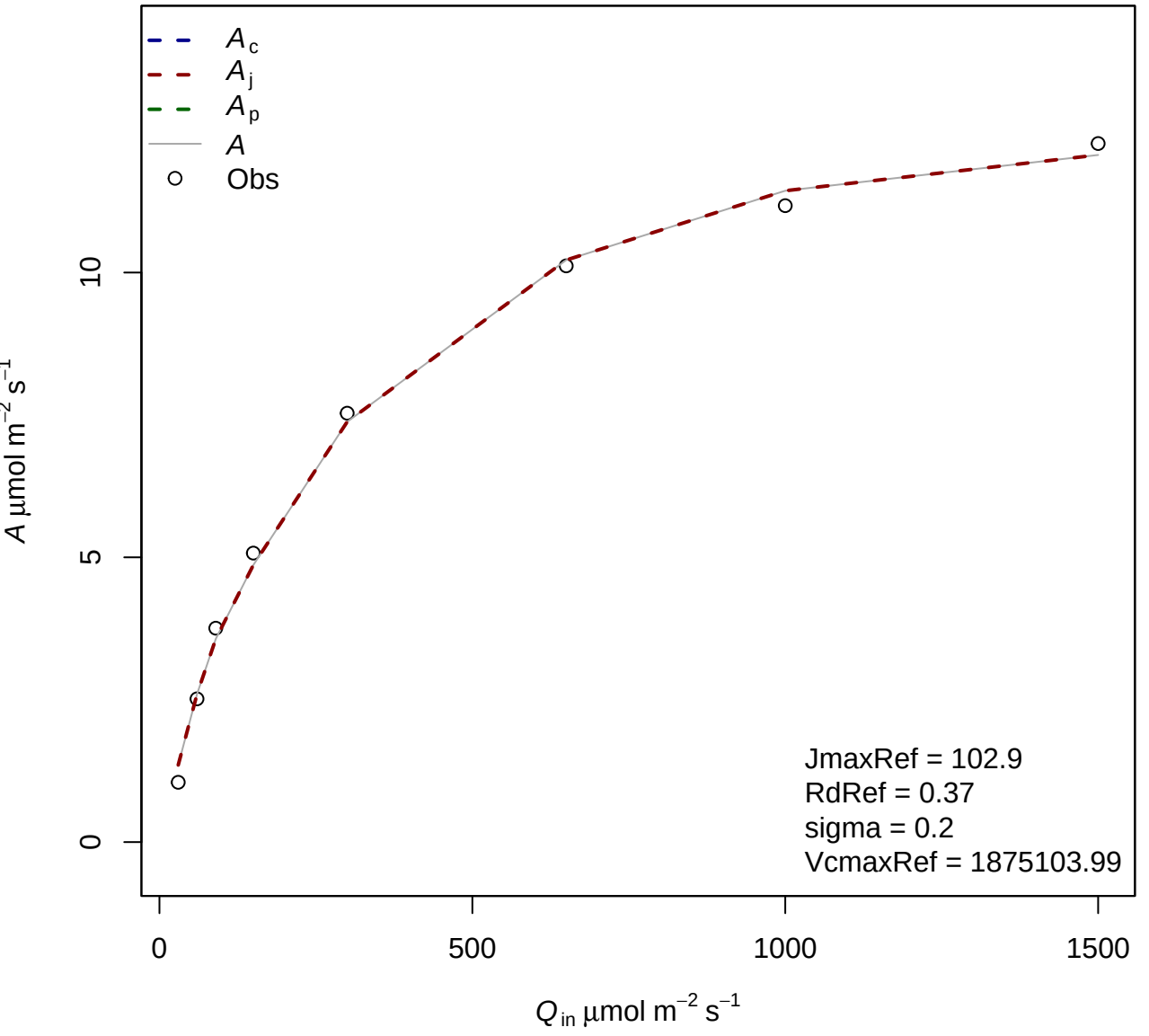

89
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 195.52
RdRef = 1.27
sigma = 0.17
VcmaxRef = 58.88
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

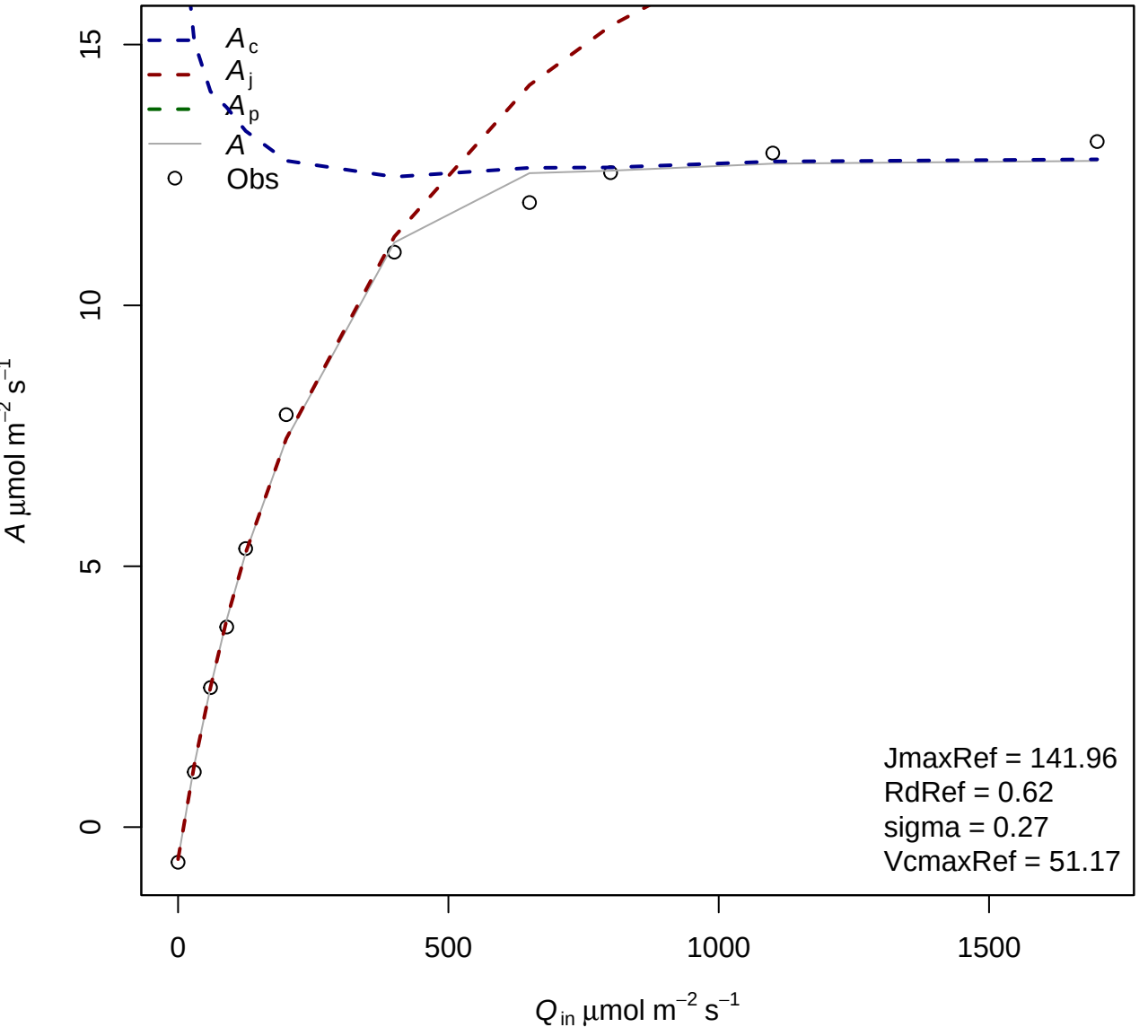

90
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 225.67
RdRef = 1.41
sigma = 0.21
VcmaxRef = 65.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

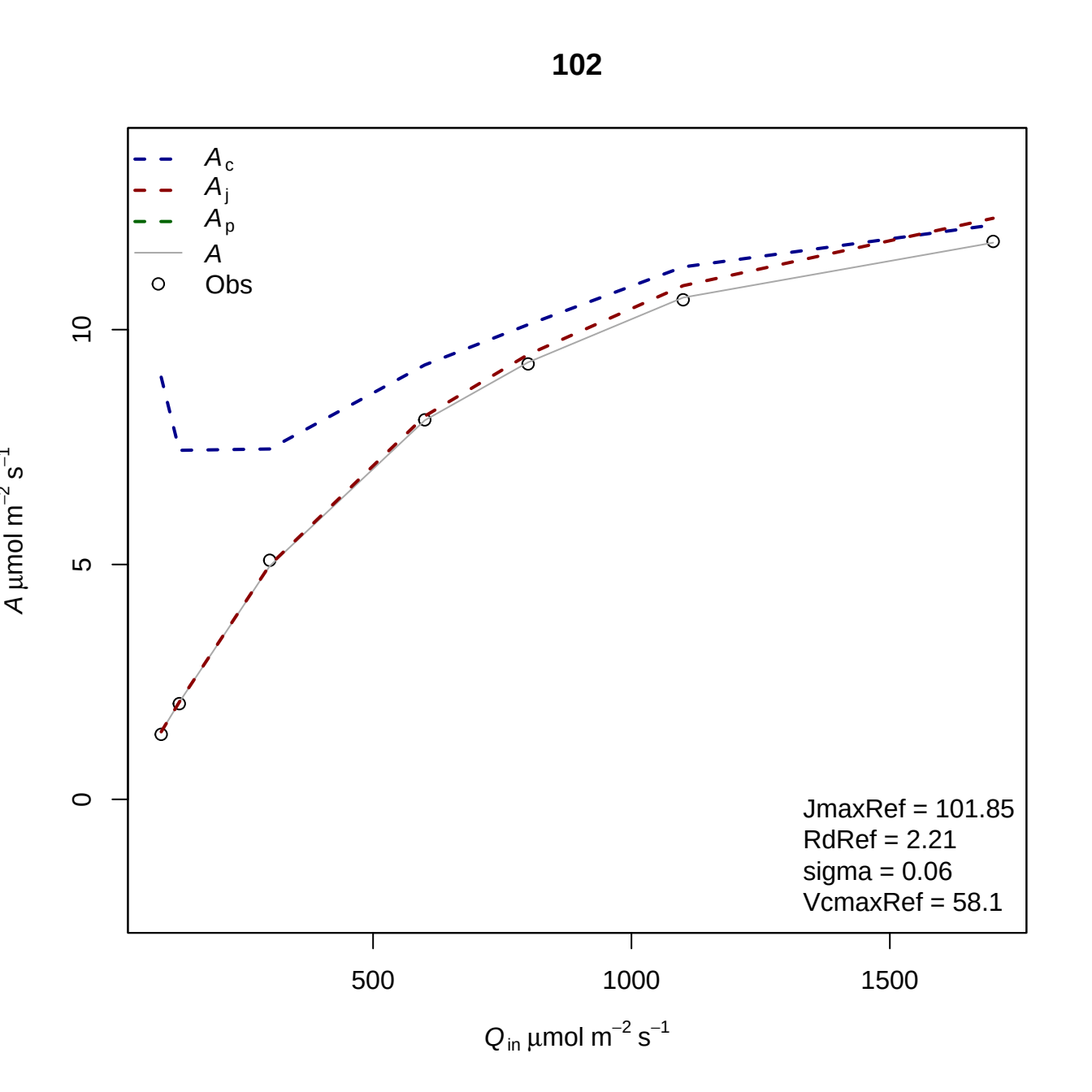

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.04
RdRef = 2.88
sigma = 0.28
VcmaxRef = 144.11
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

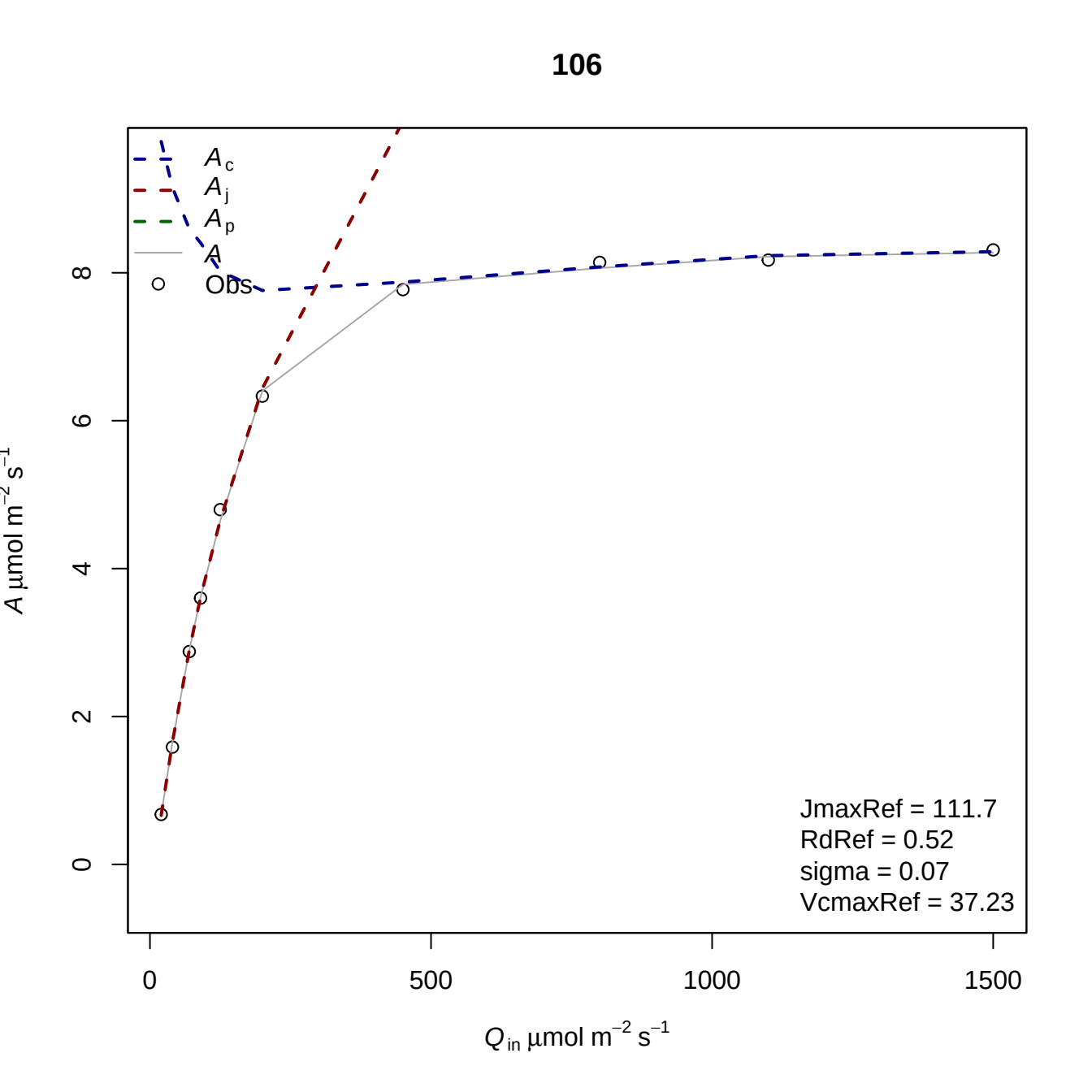

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.16
RdRef = 1.06
sigma = 0.51
VcmaxRef = 27.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

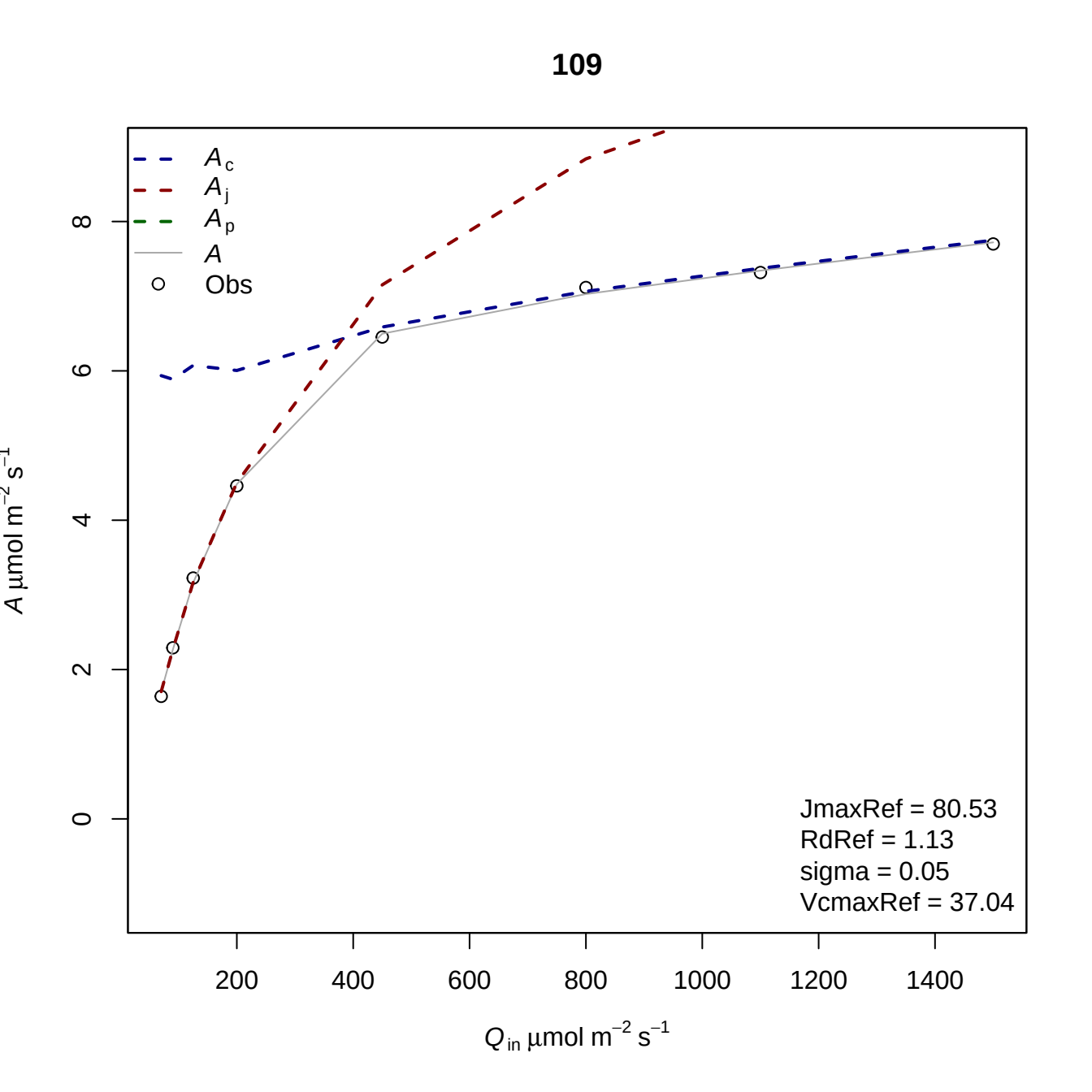

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.89
RdRef = 4.06
sigma = 0.64
VcmaxRef = 190515.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
−4
0
500
1000
1500

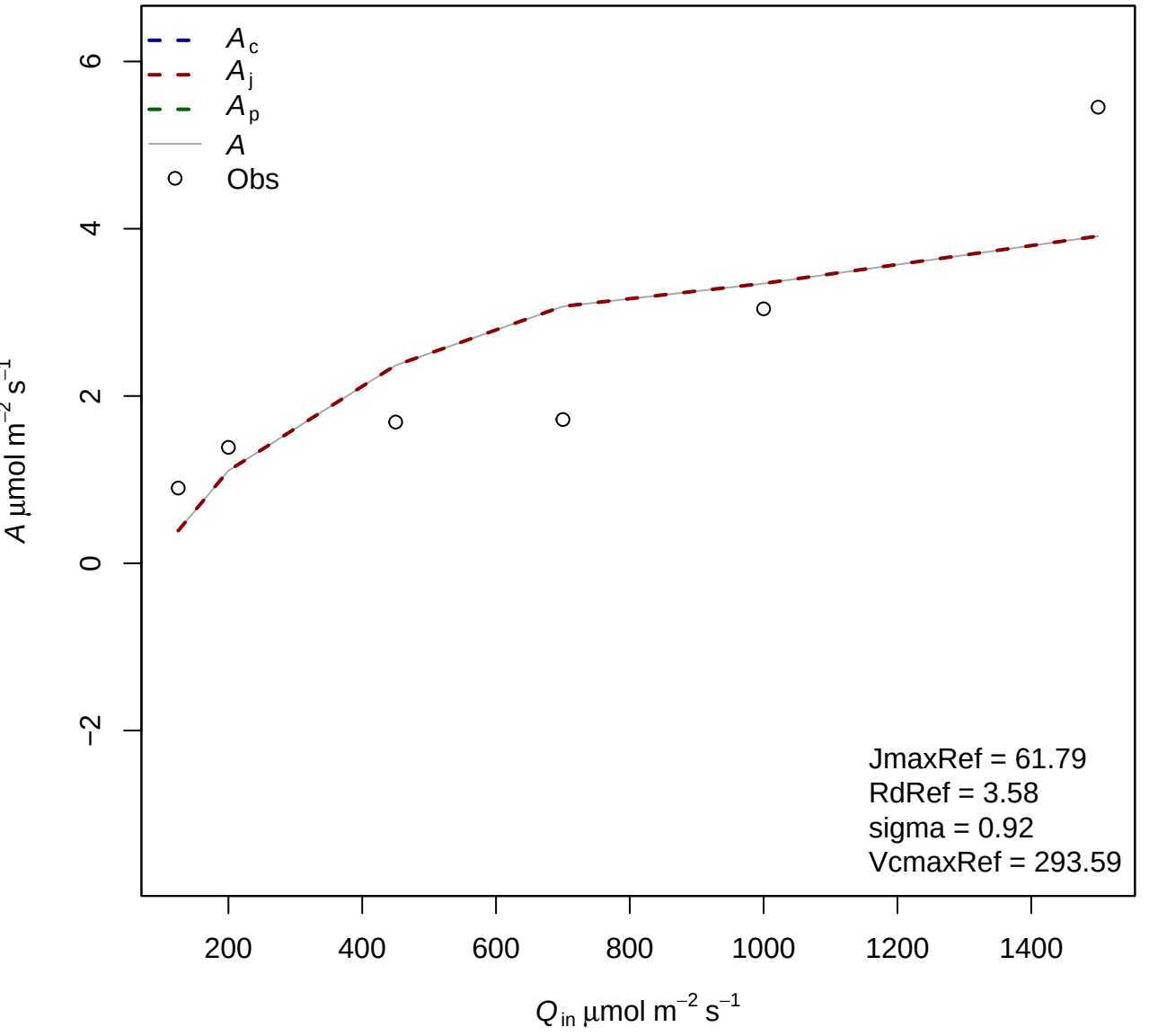

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.9
RdRef = 0.37
sigma = 0.2
VcmaxRef = 1875103.99
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

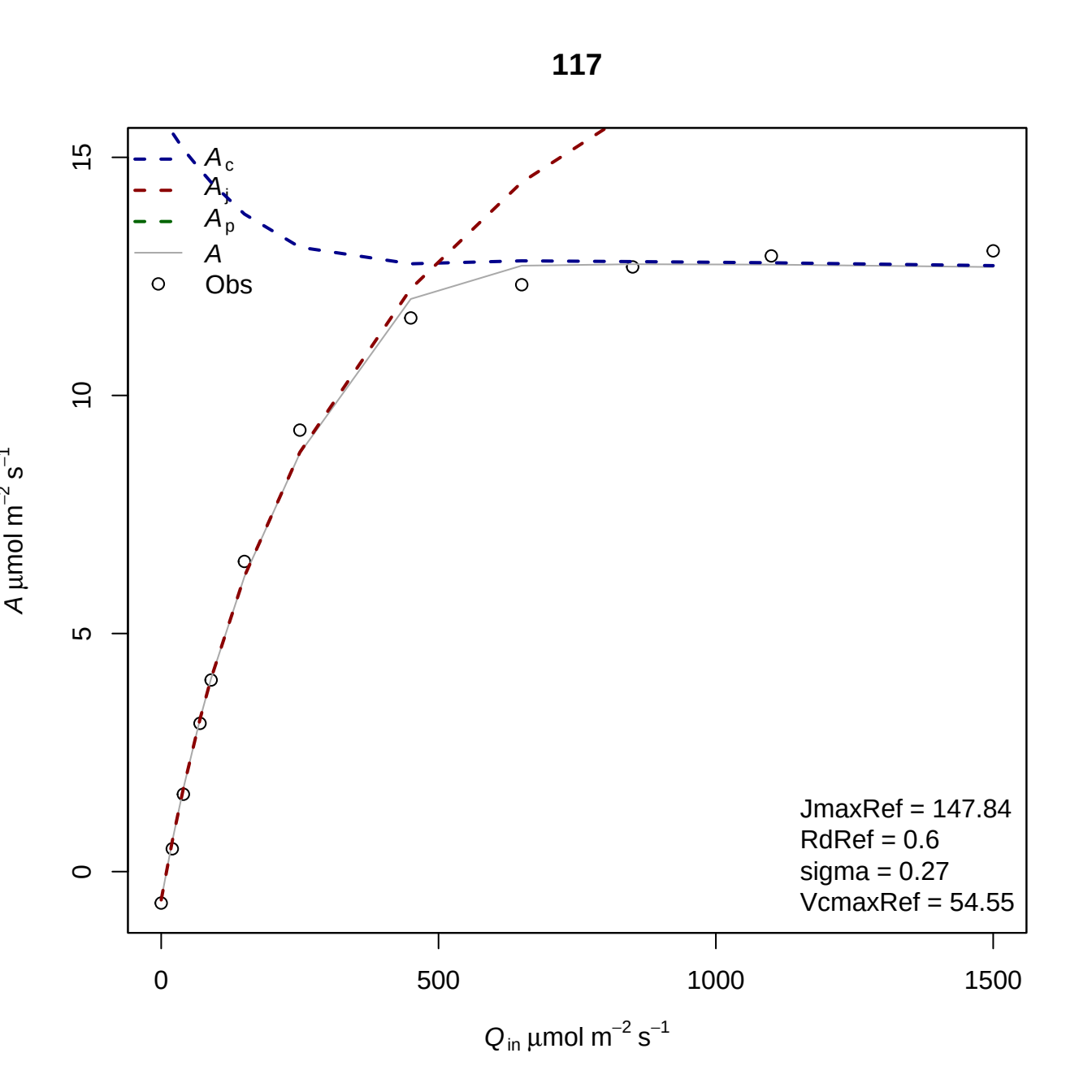

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 141.96
RdRef = 0.62
sigma = 0.27
VcmaxRef = 51.17
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

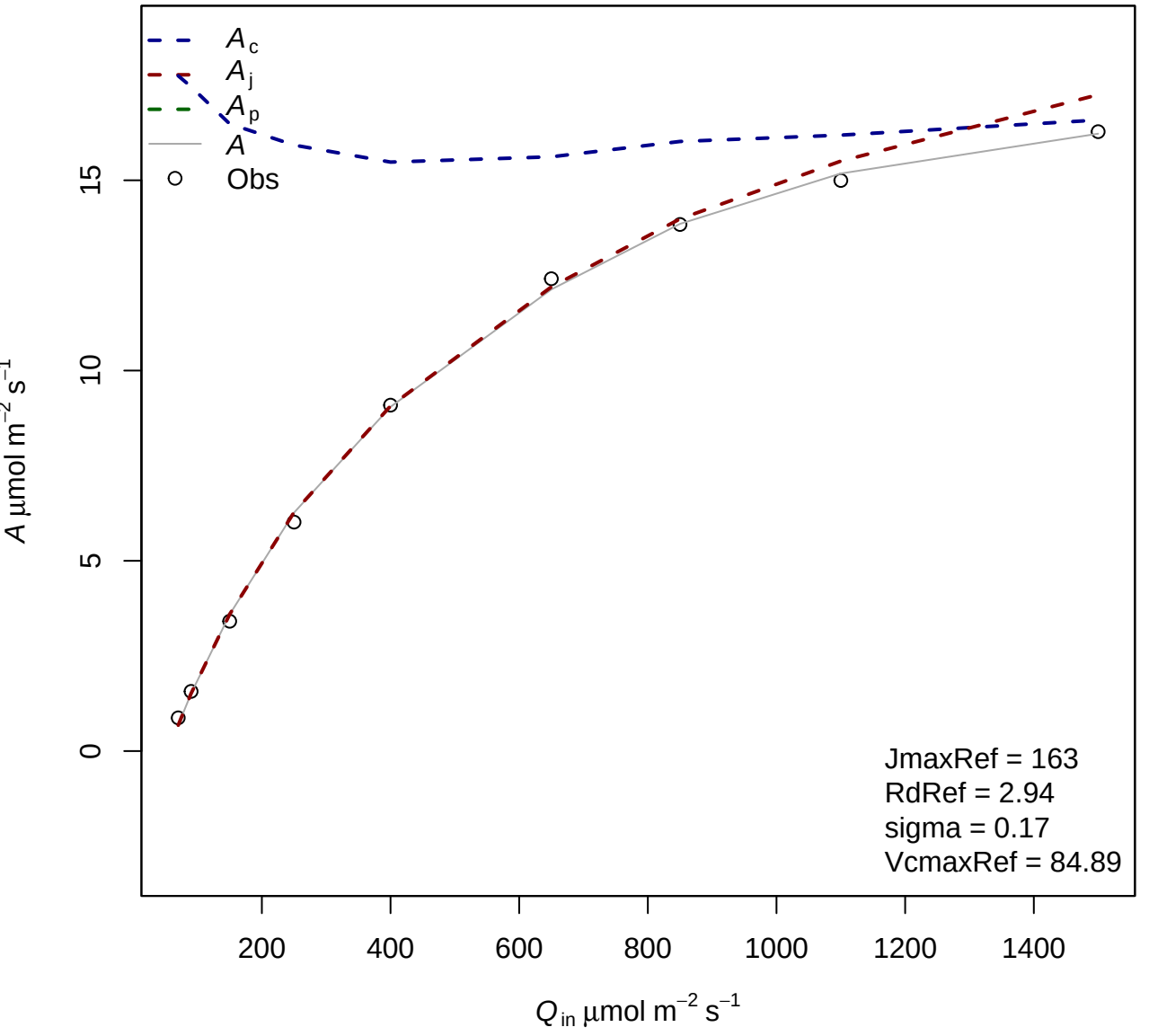

102
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.85
RdRef = 2.21
sigma = 0.06
VcmaxRef = 58.1
0
5
10
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

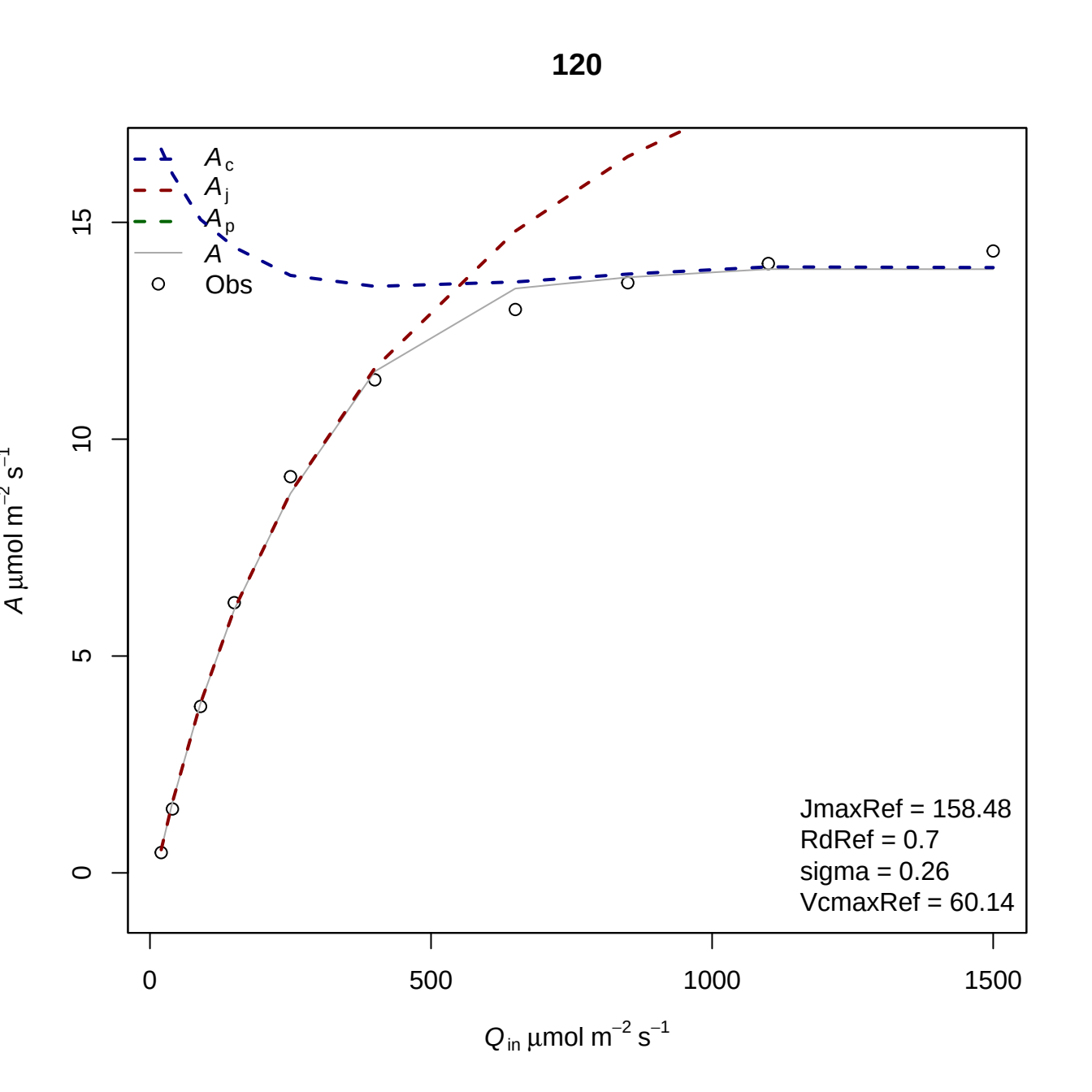

106
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.7
RdRef = 0.52
sigma = 0.07
VcmaxRef = 37.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

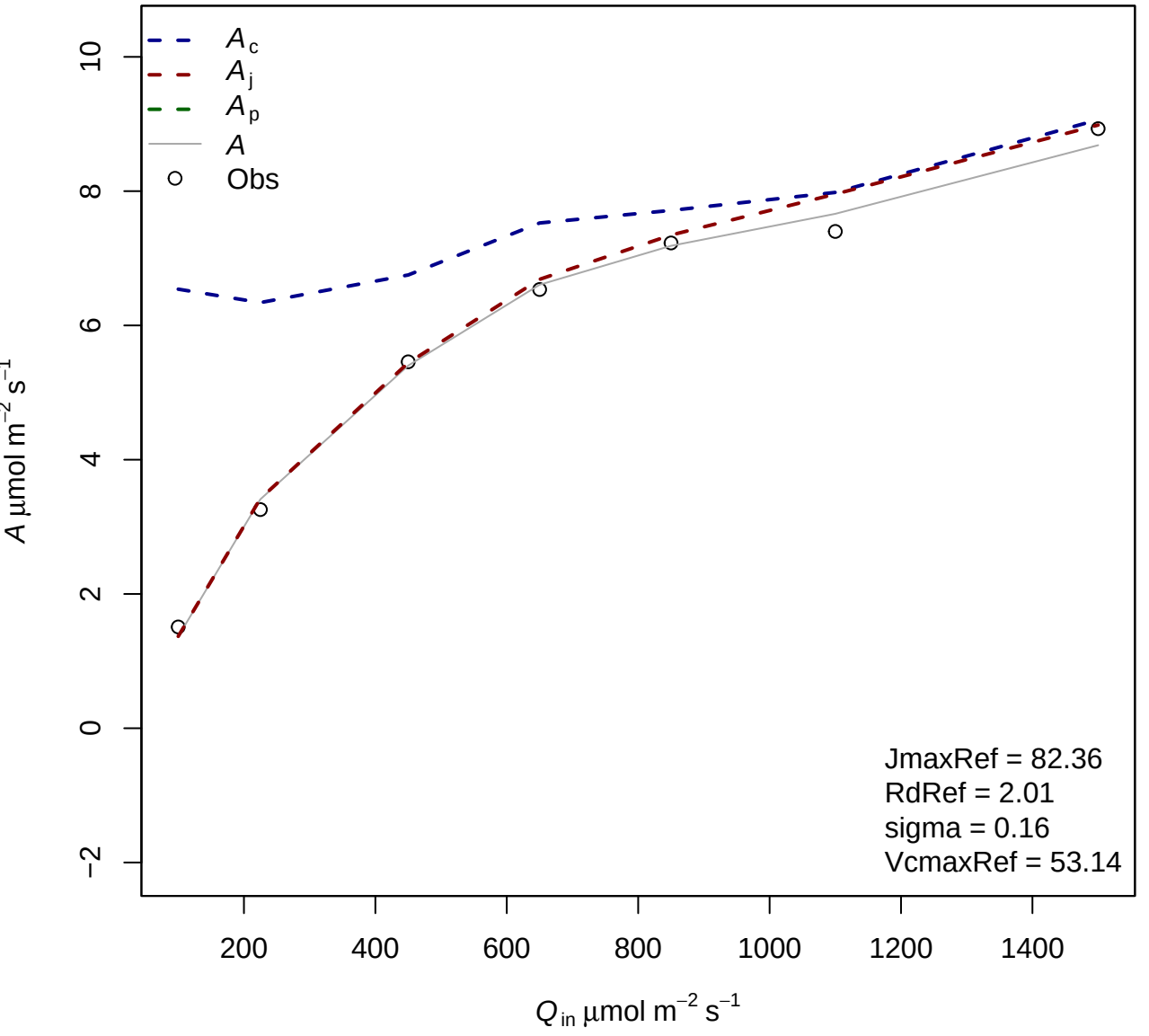

109
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.53
RdRef = 1.13
sigma = 0.05
VcmaxRef = 37.04
0
2
4
6
8
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

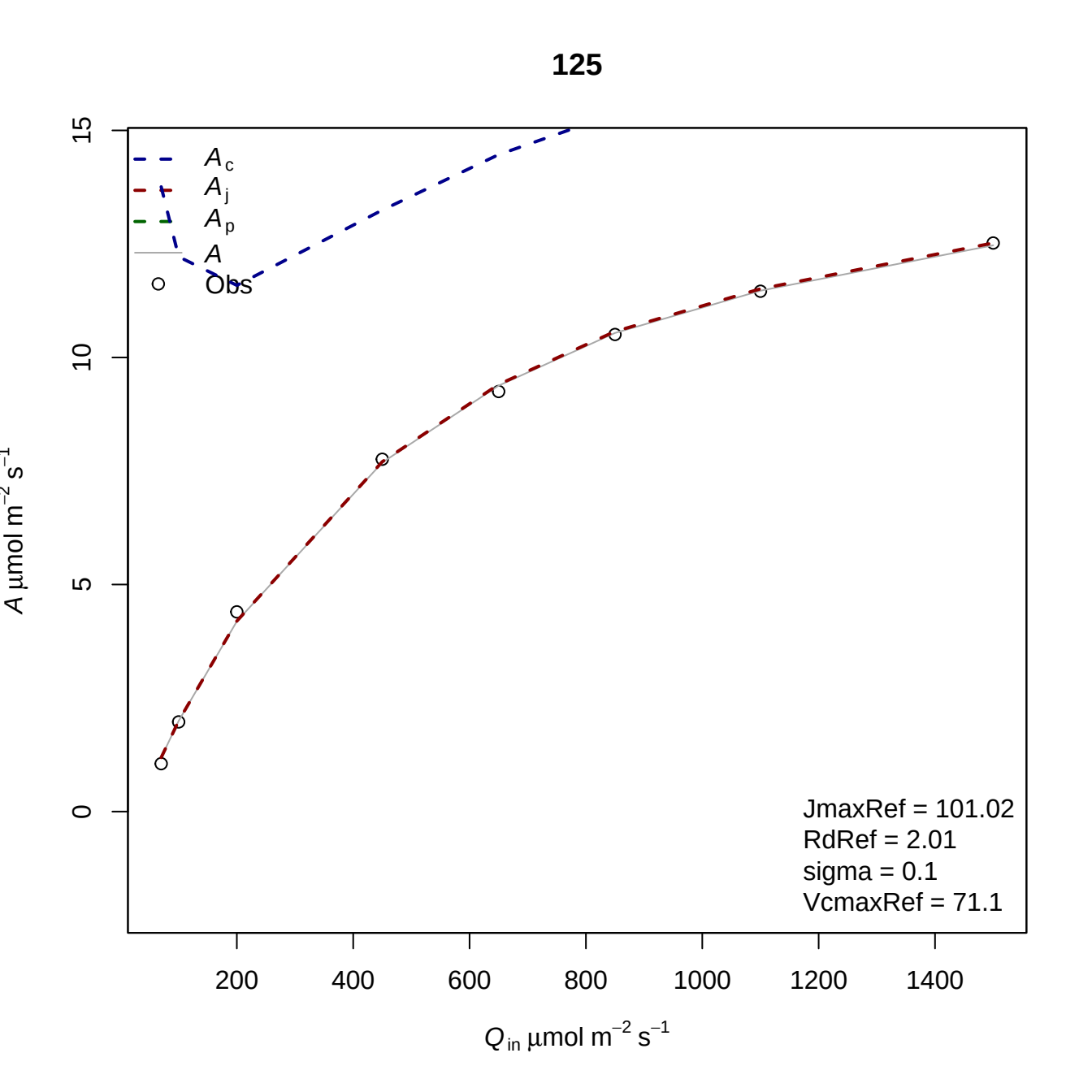

$A_c$
$A_j$
$A_p$
$A$
Obs
6
4
2
0
−2
$A$ µmol m$^{-2}$ s$^{-1}$
200
400
600
800
1000
1200
1400
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 61.79
RdRef = 3.58
sigma = 0.92
VcmaxRef = 293.59

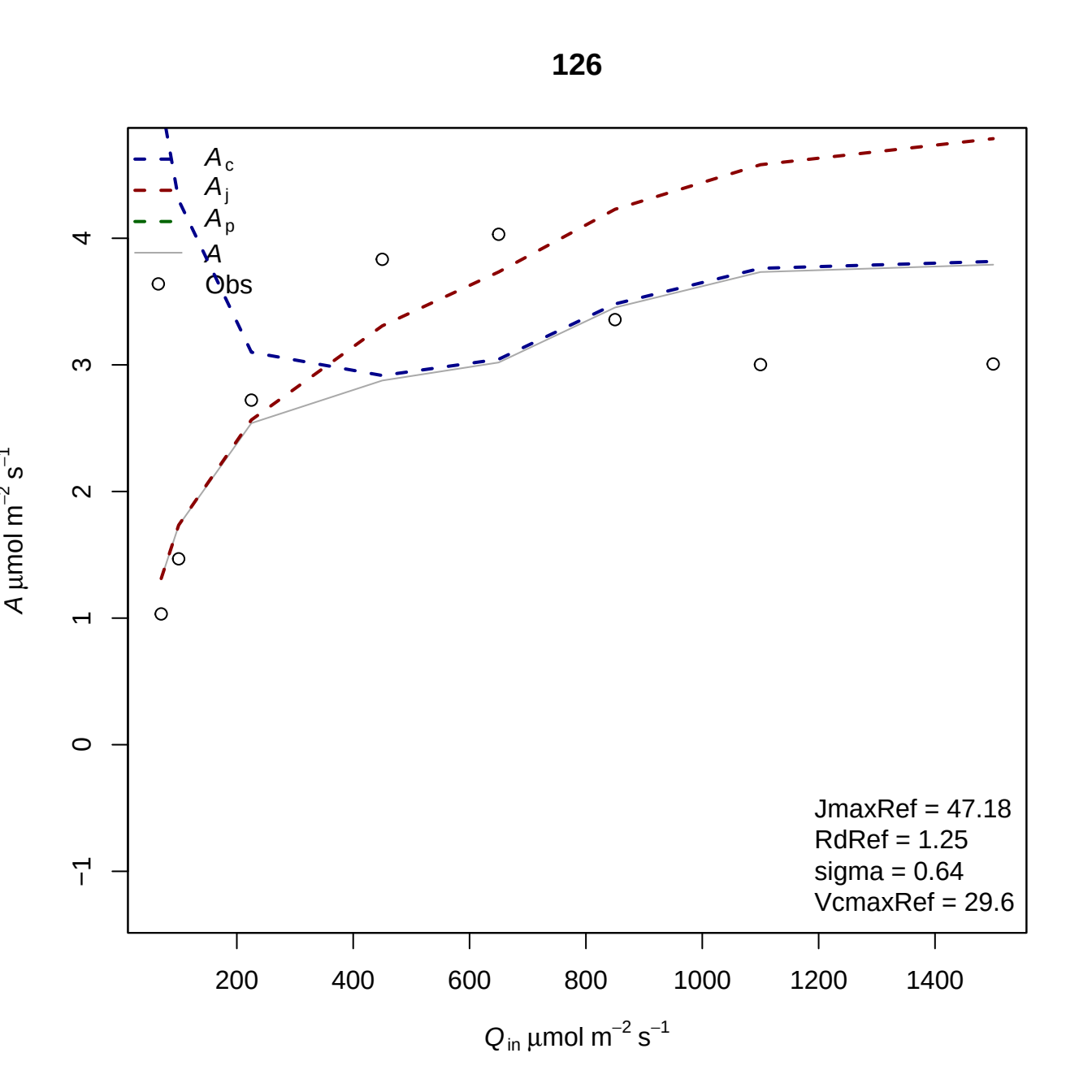

117
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.84
RdRef = 0.6
sigma = 0.27
VcmaxRef = 54.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

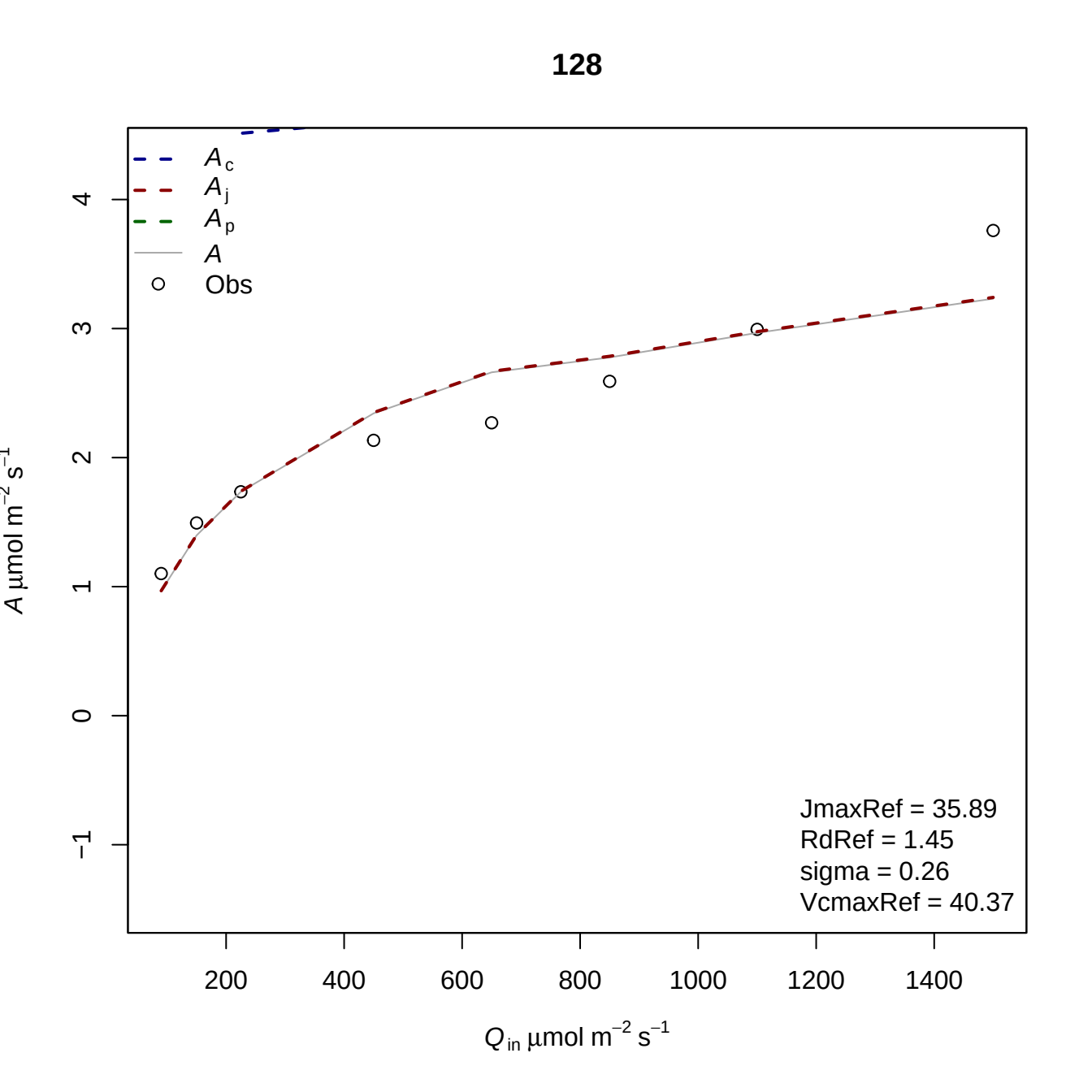

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163
RdRef = 2.94
sigma = 0.17
VcmaxRef = 84.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
200
400
600
800
1000
1200
1400

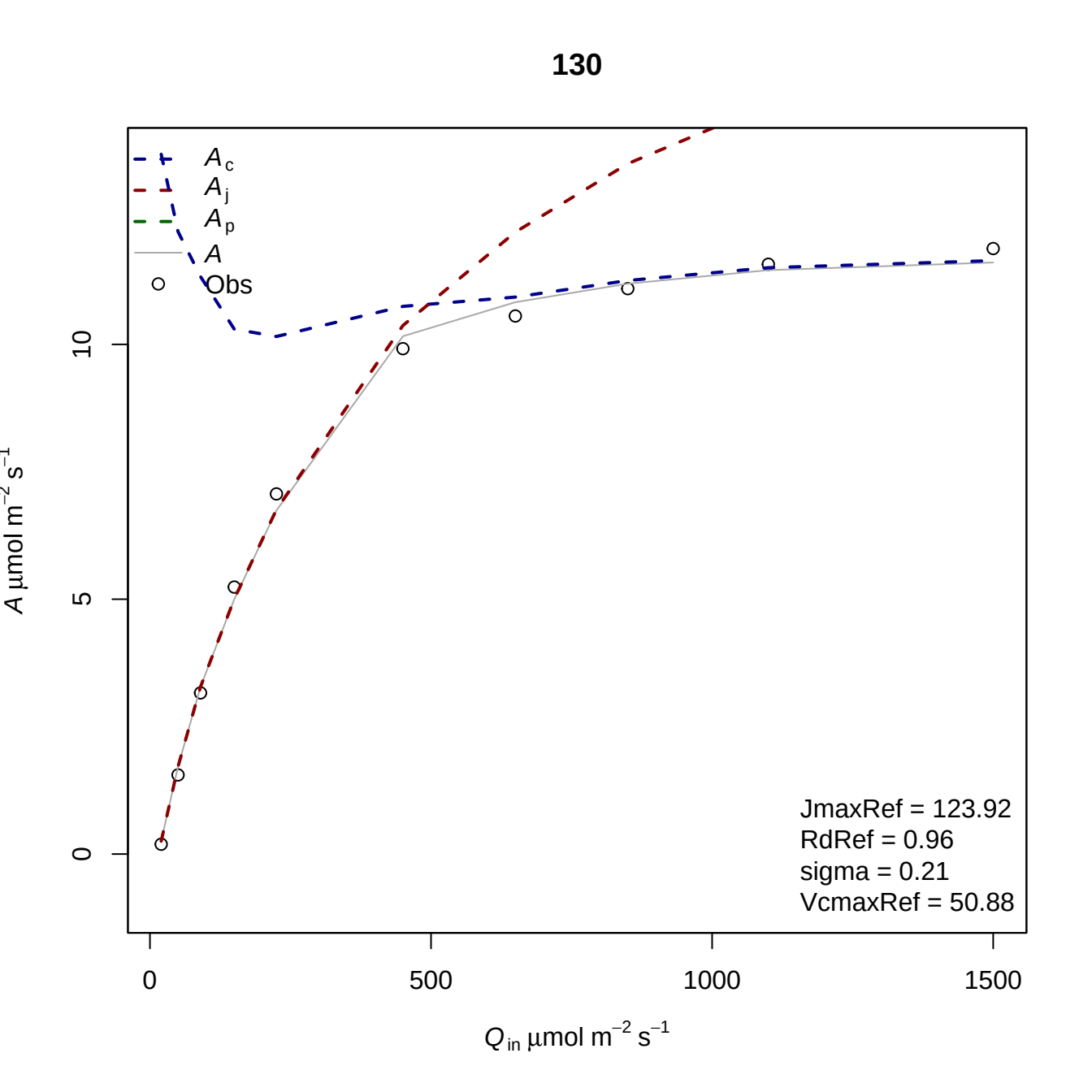

120
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.48
RdRef = 0.7
sigma = 0.26
VcmaxRef = 60.14
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

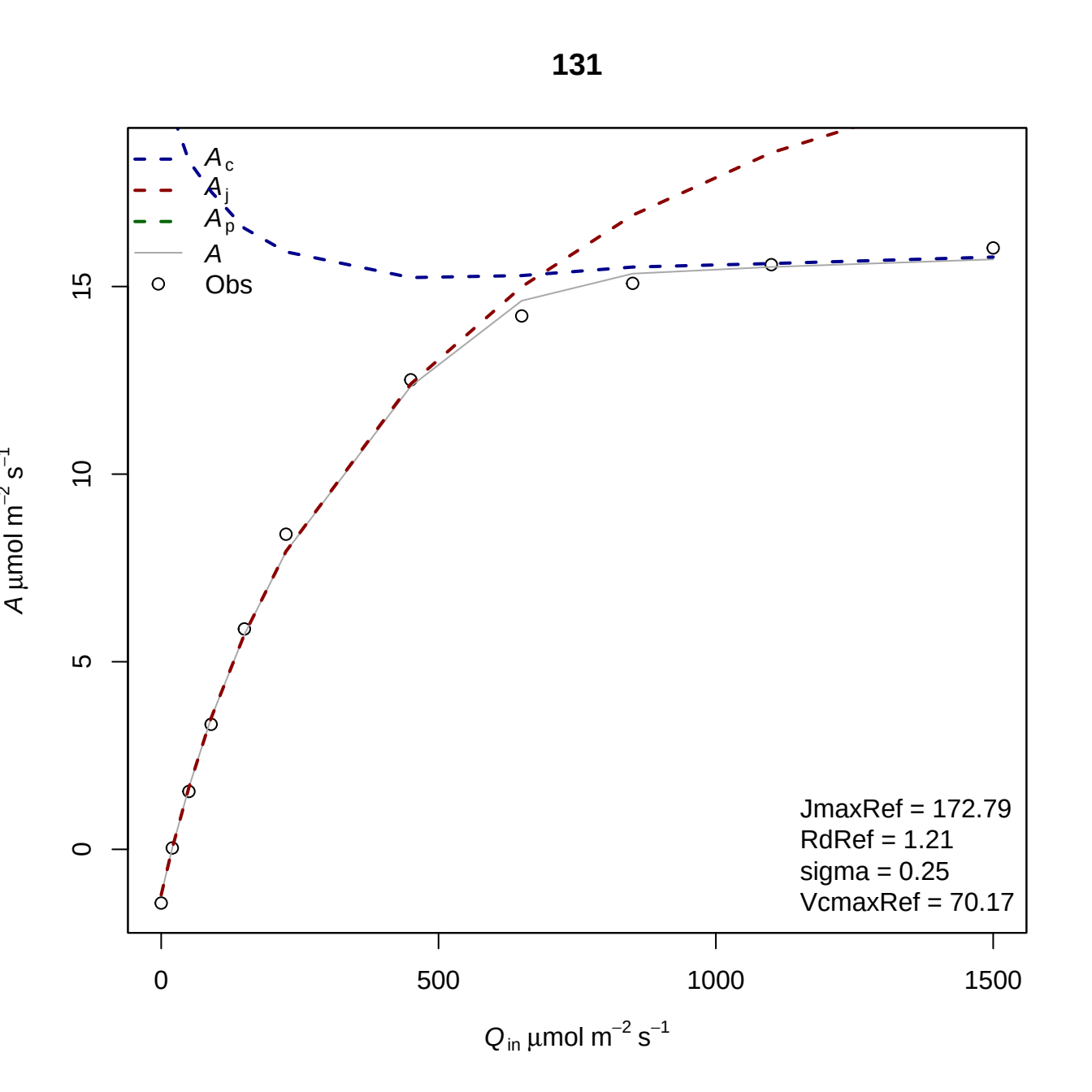

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.36
RdRef = 2.01
sigma = 0.16
VcmaxRef = 53.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

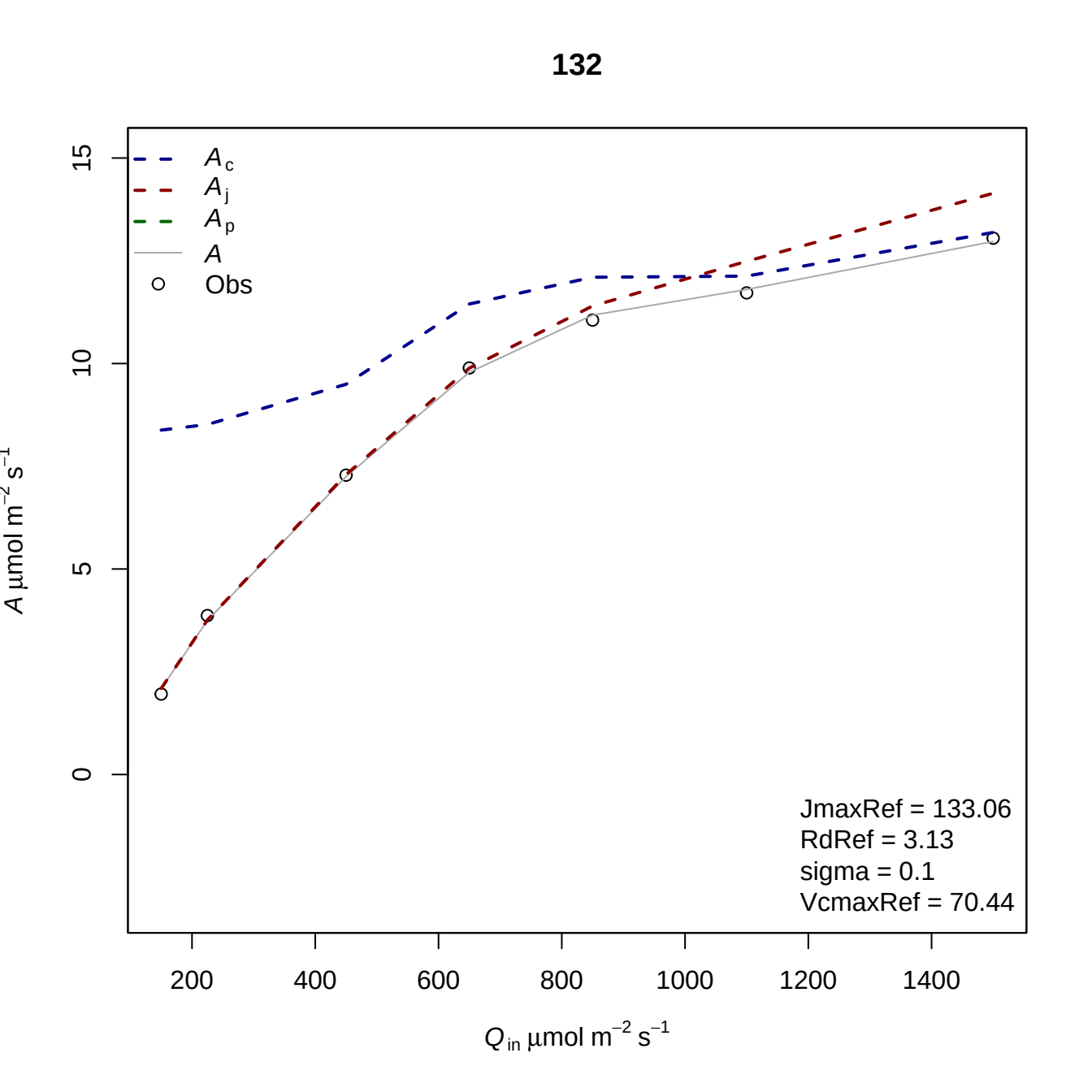

125
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.02
RdRef = 2.01
sigma = 0.1
VcmaxRef = 71.1
0
5
10
15
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

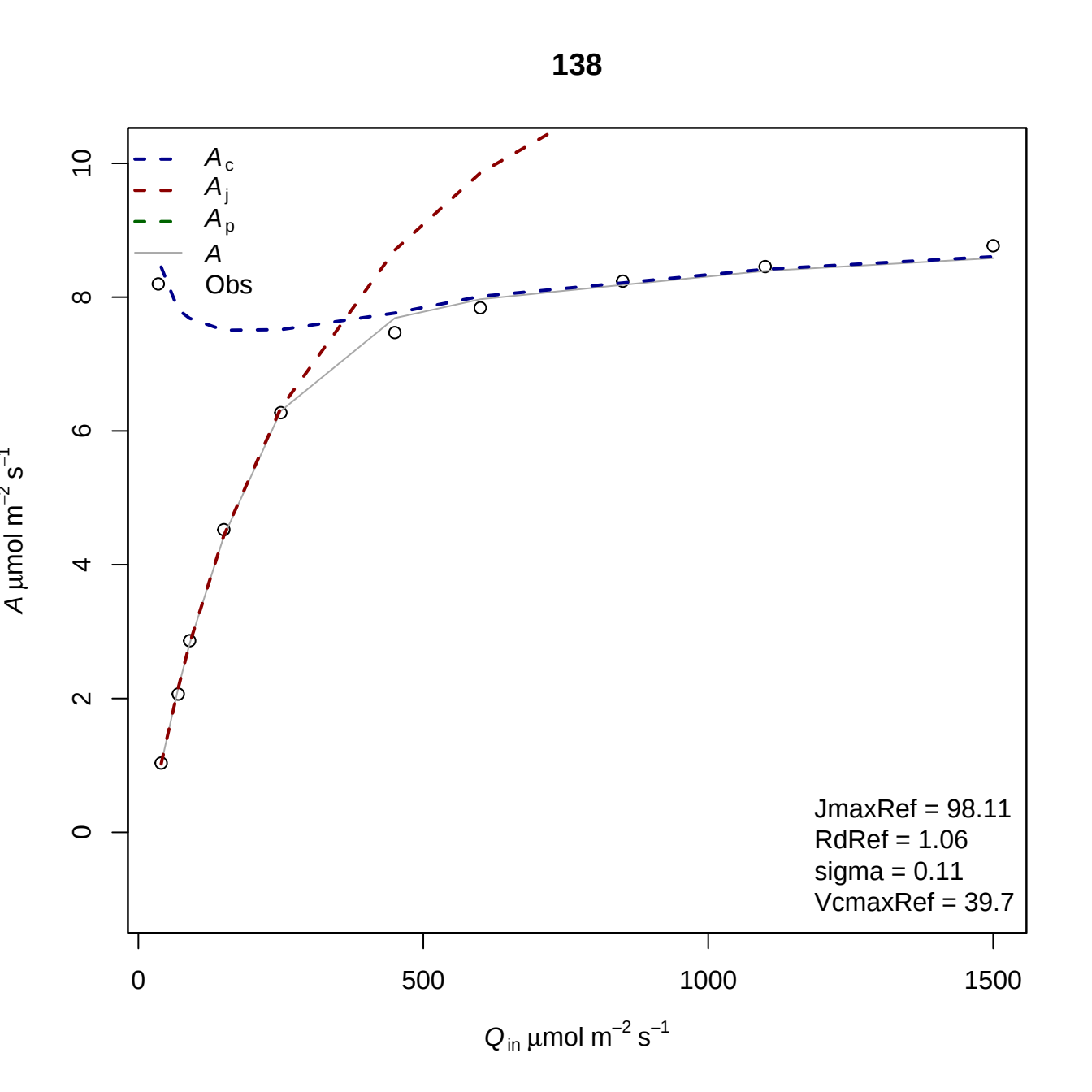

126
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.18
RdRef = 1.25
sigma = 0.64
VcmaxRef = 29.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

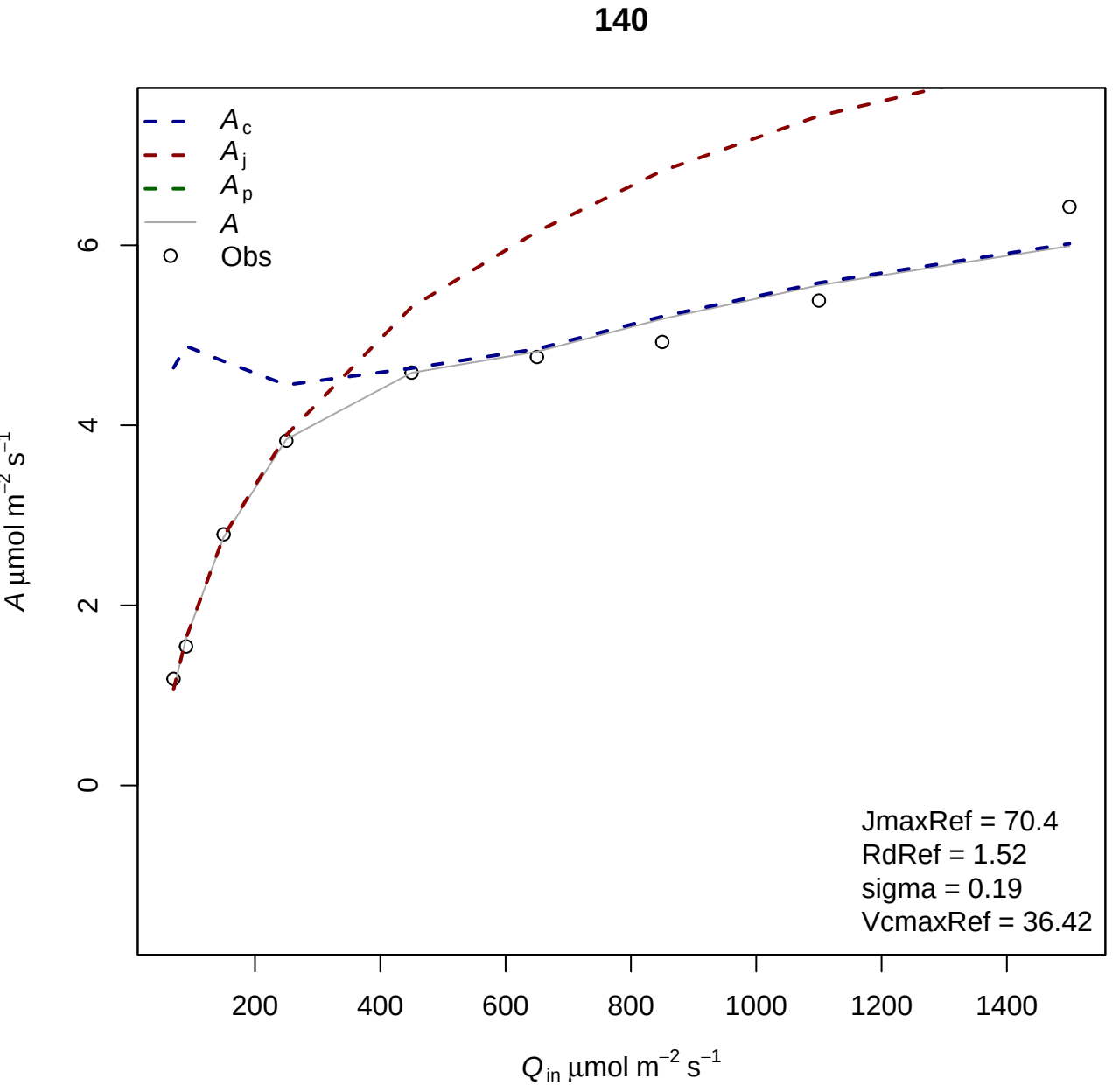

128
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 35.89
RdRef = 1.45
sigma = 0.26
VcmaxRef = 40.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

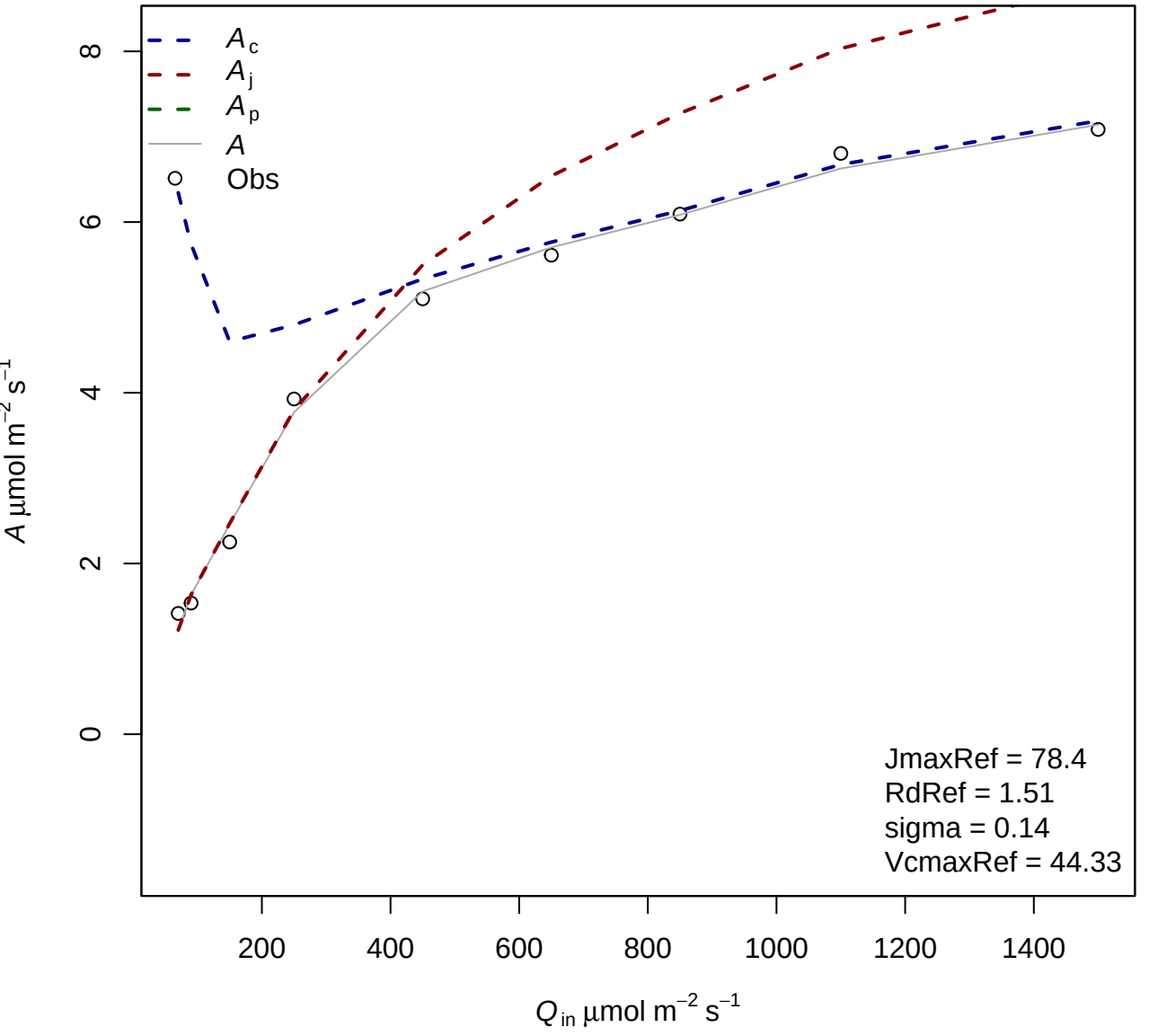

130
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.92
RdRef = 0.96
sigma = 0.21
VcmaxRef = 50.88
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

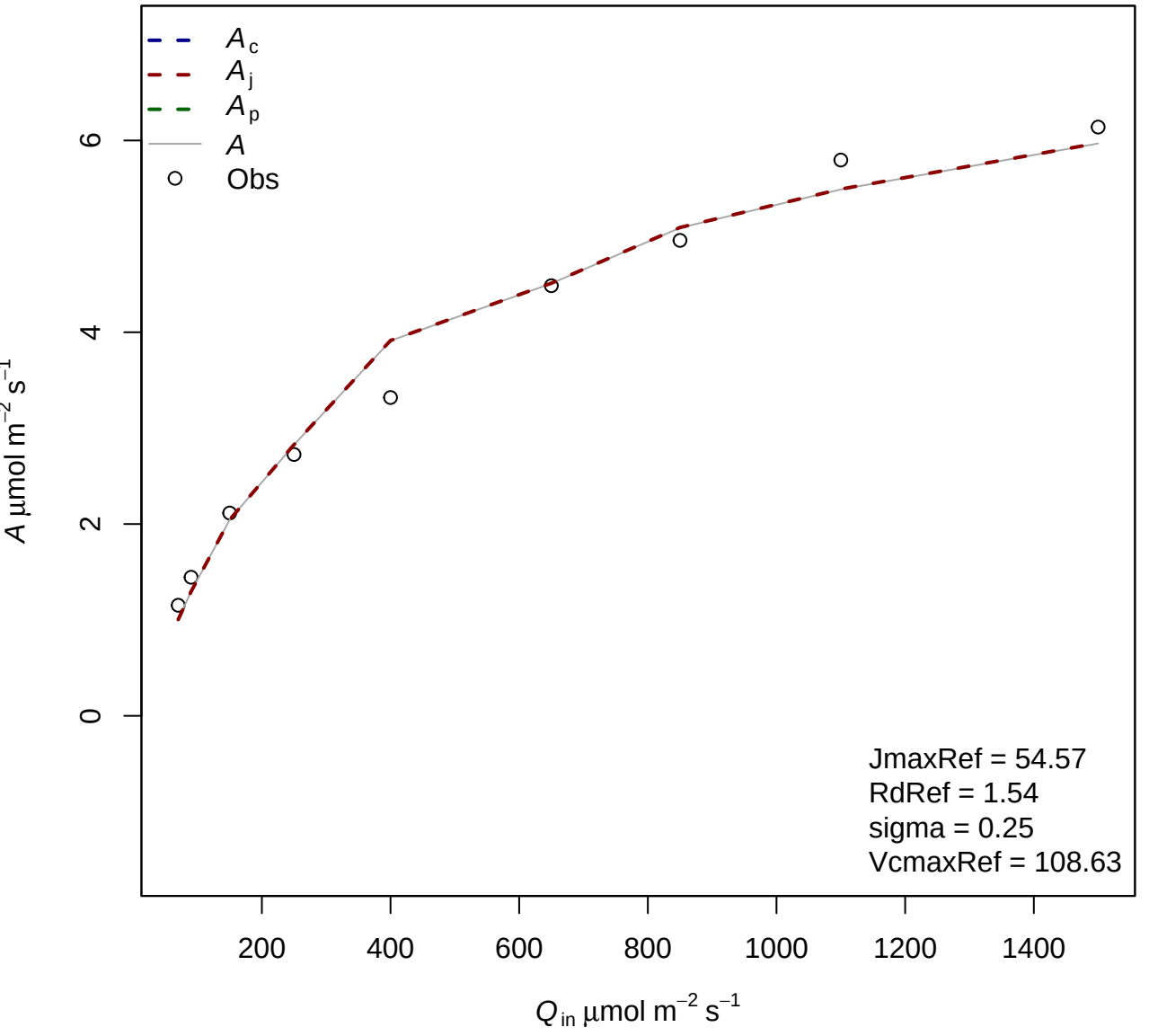

131
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 172.79
RdRef = 1.21
sigma = 0.25
VcmaxRef = 70.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

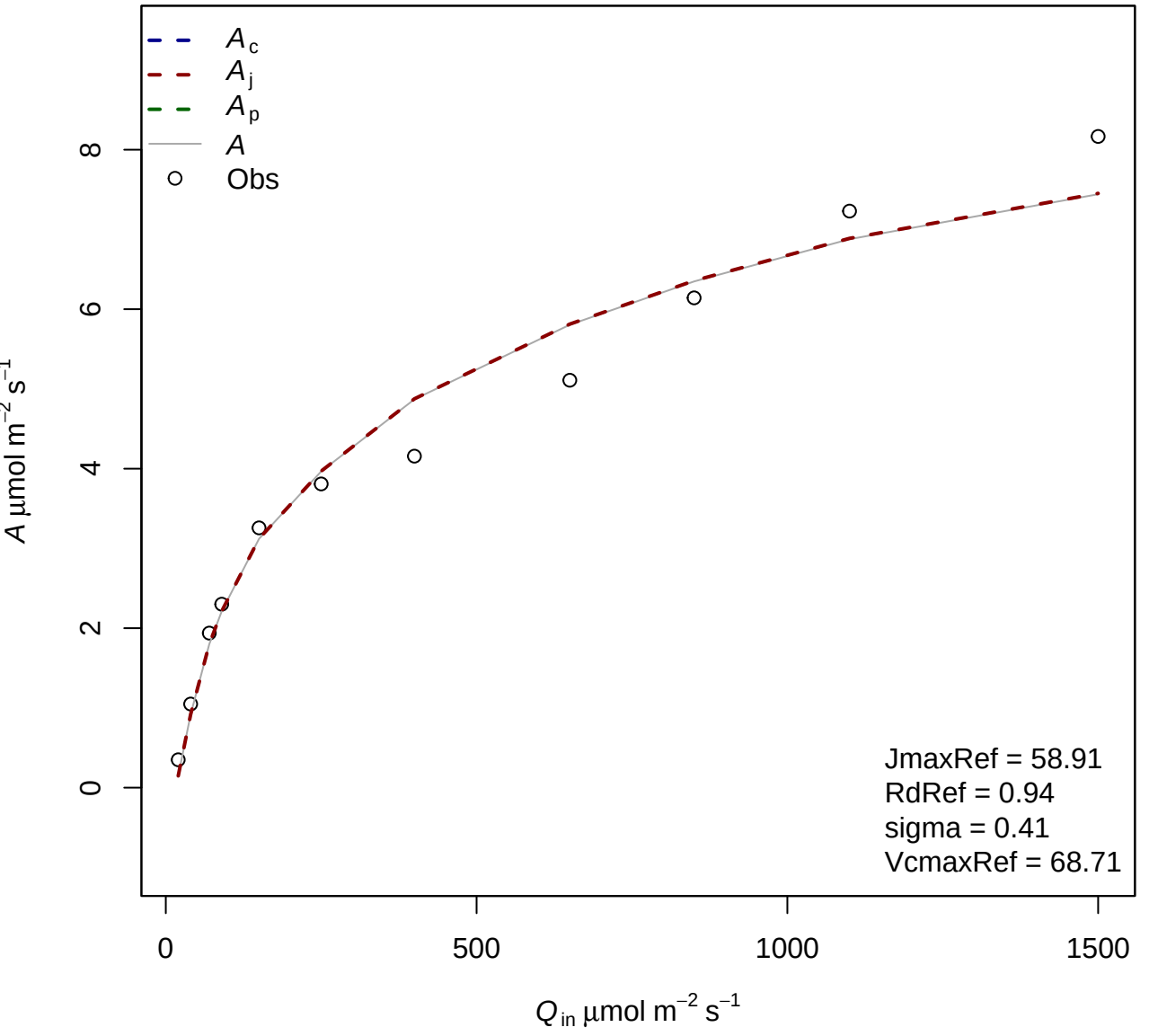

132
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.06
RdRef = 3.13
sigma = 0.1
VcmaxRef = 70.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
200
400
600
800
1000
1200
1400

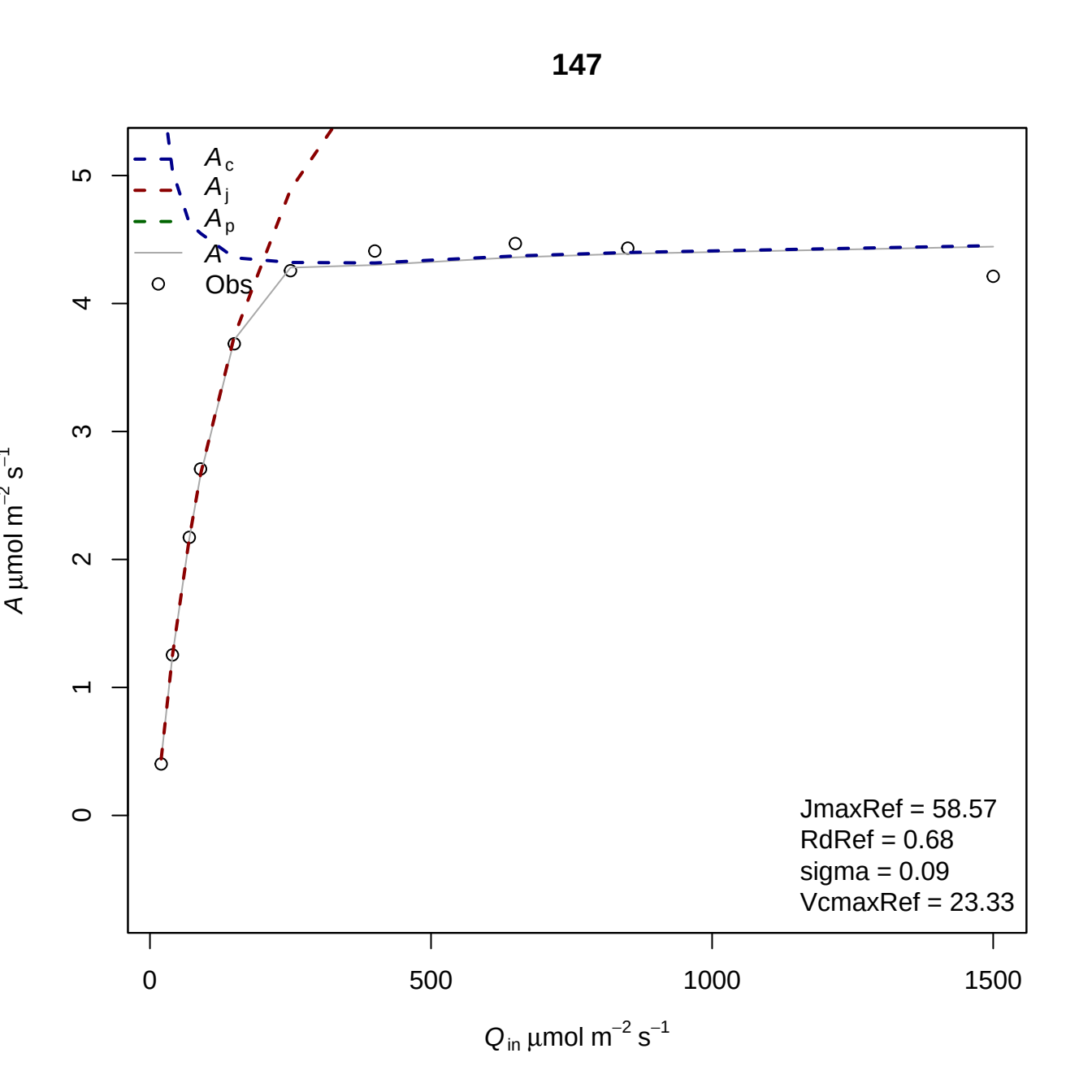

138
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.11
RdRef = 1.06
sigma = 0.11
VcmaxRef = 39.7
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

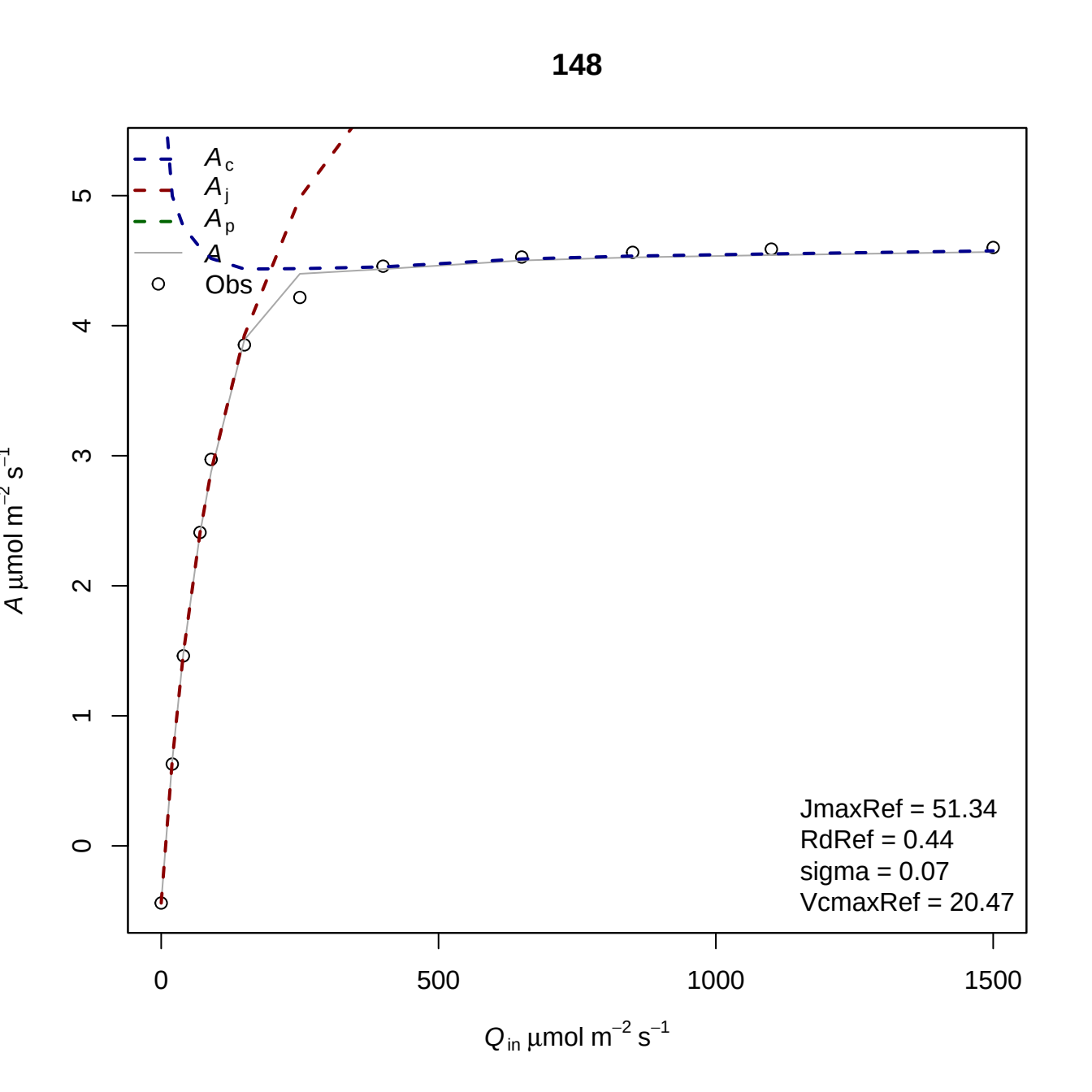

140
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.4
RdRef = 1.52
sigma = 0.19
VcmaxRef = 36.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
200
400
600
800
1000
1200
1400

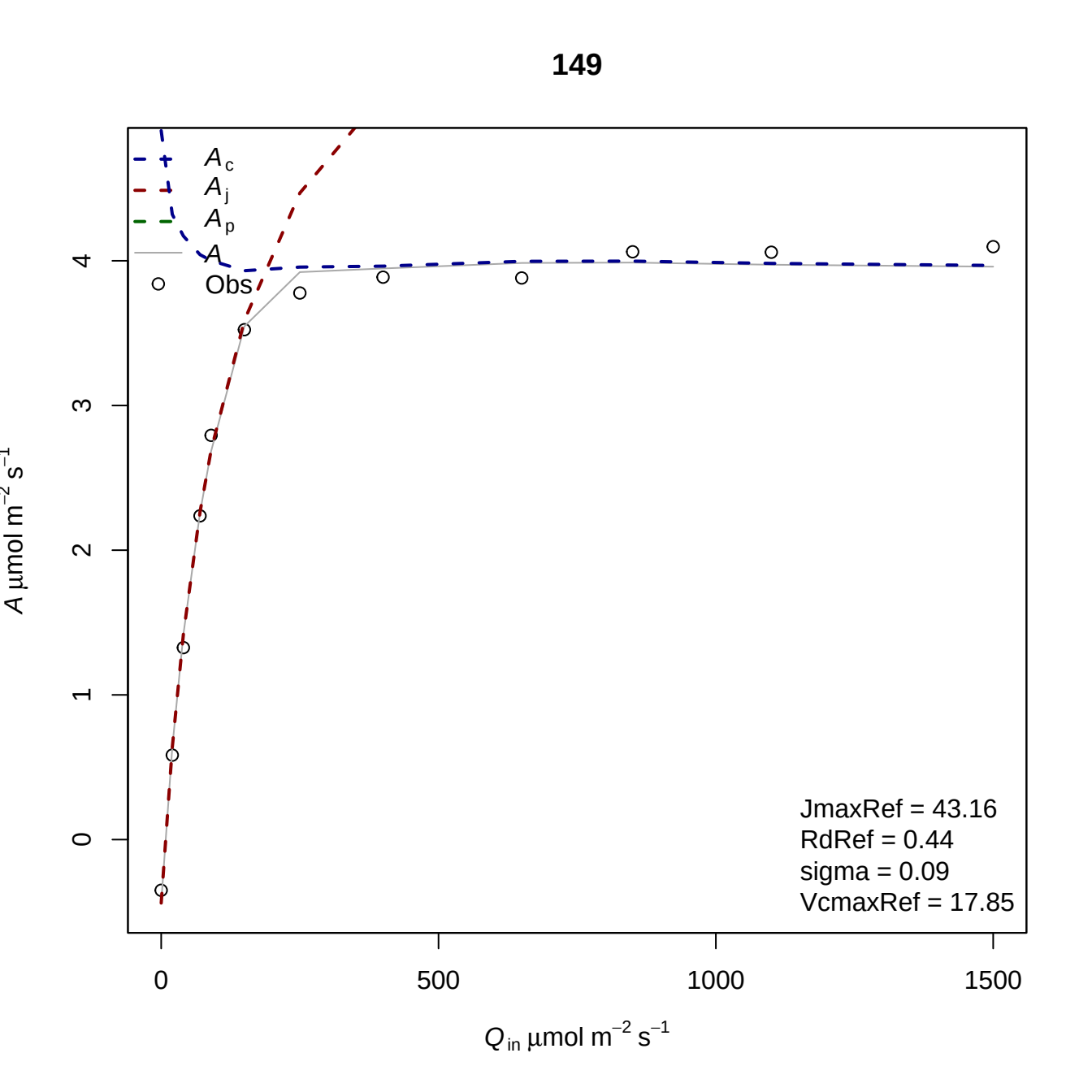

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.4
RdRef = 1.51
sigma = 0.14
VcmaxRef = 44.33
0
2
4
6
8
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

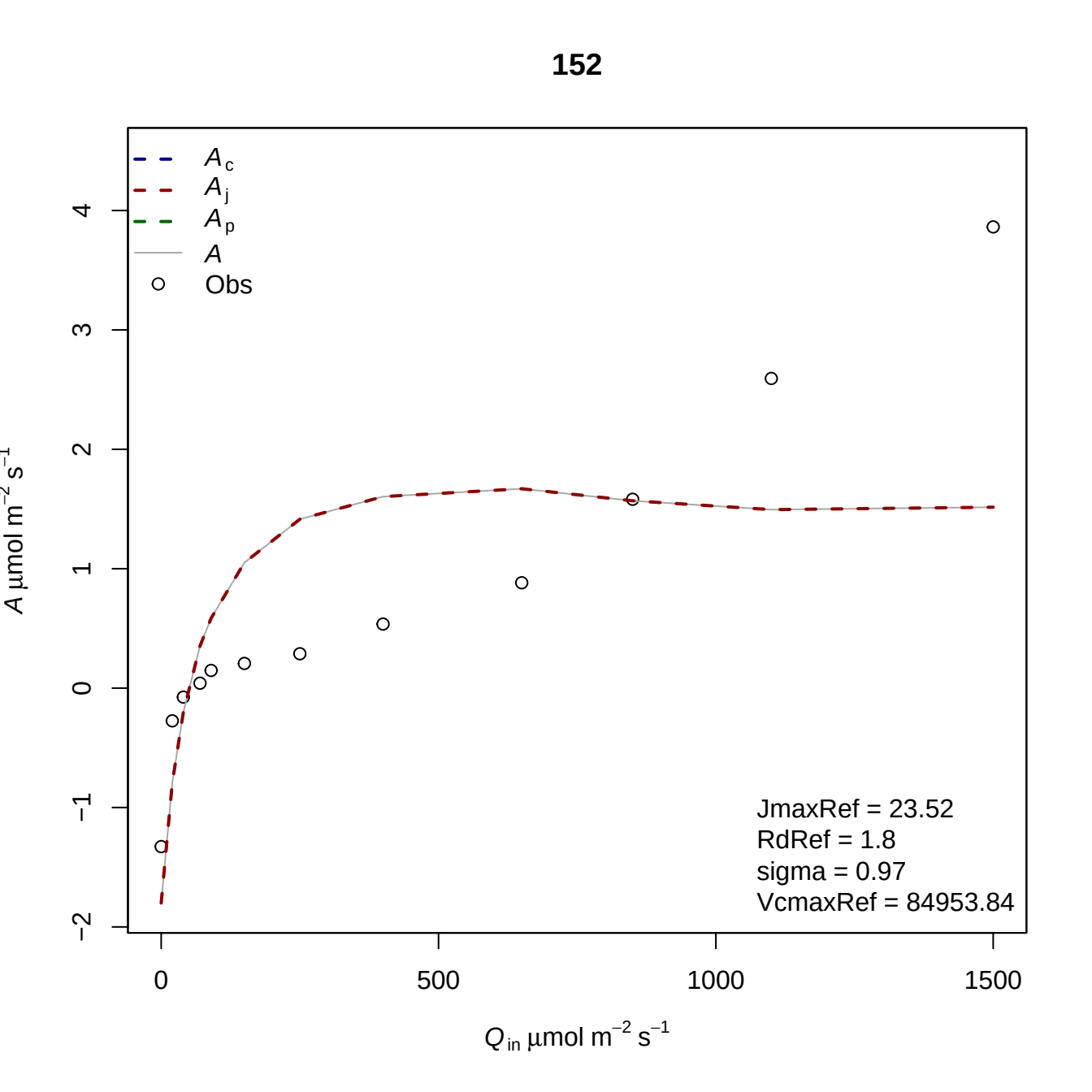

$A_c$
$A_j$
$A_p$
$A$
Obs
0
2
4
6
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 54.57
RdRef = 1.54
sigma = 0.25
VcmaxRef = 108.63

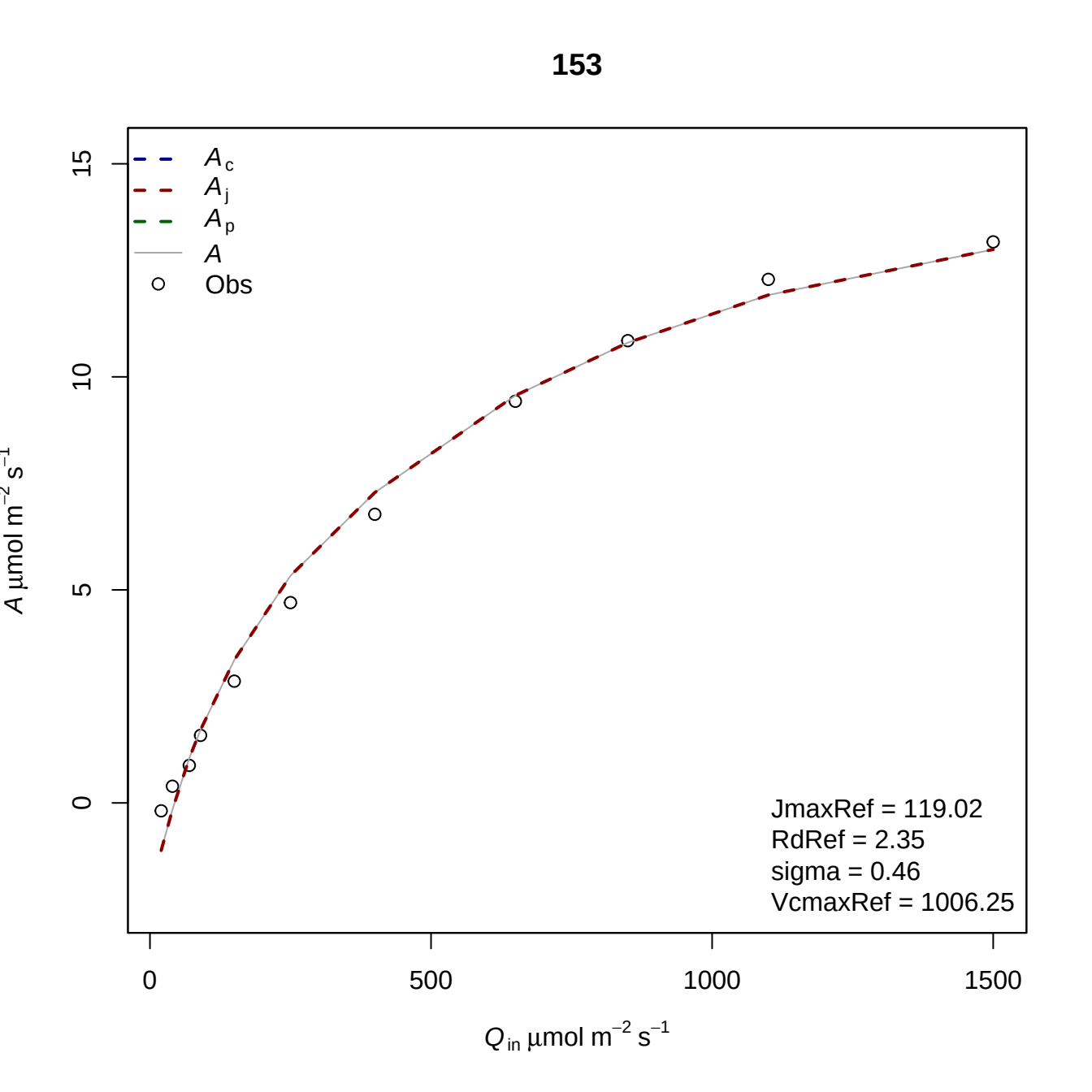

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.91
RdRef = 0.94
sigma = 0.41
VcmaxRef = 68.71
0
2
4
6
8
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

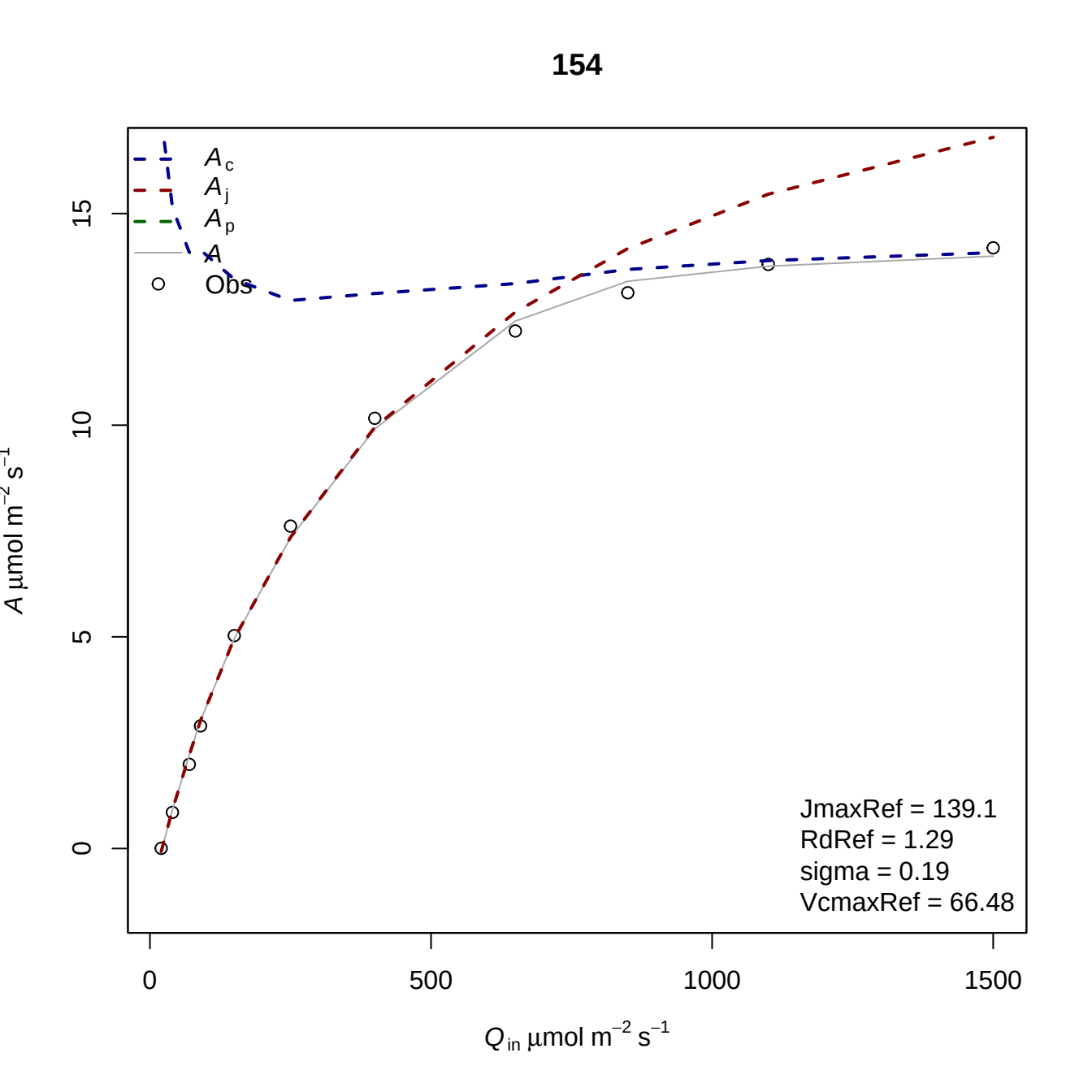

147
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.57
RdRef = 0.68
sigma = 0.09
VcmaxRef = 23.33
0
1
2
3
4
5
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

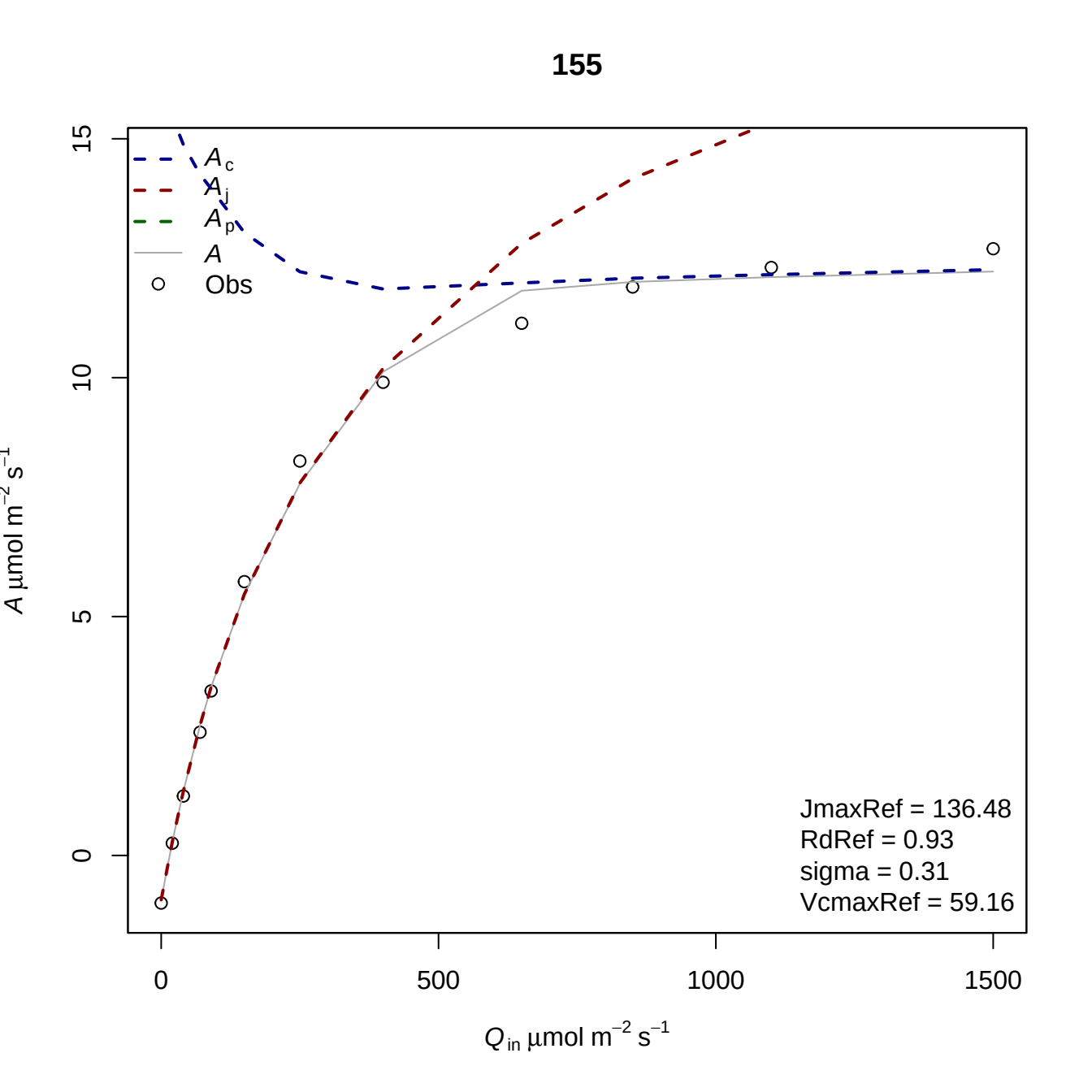

148
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.34
RdRef = 0.44
sigma = 0.07
VcmaxRef = 20.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

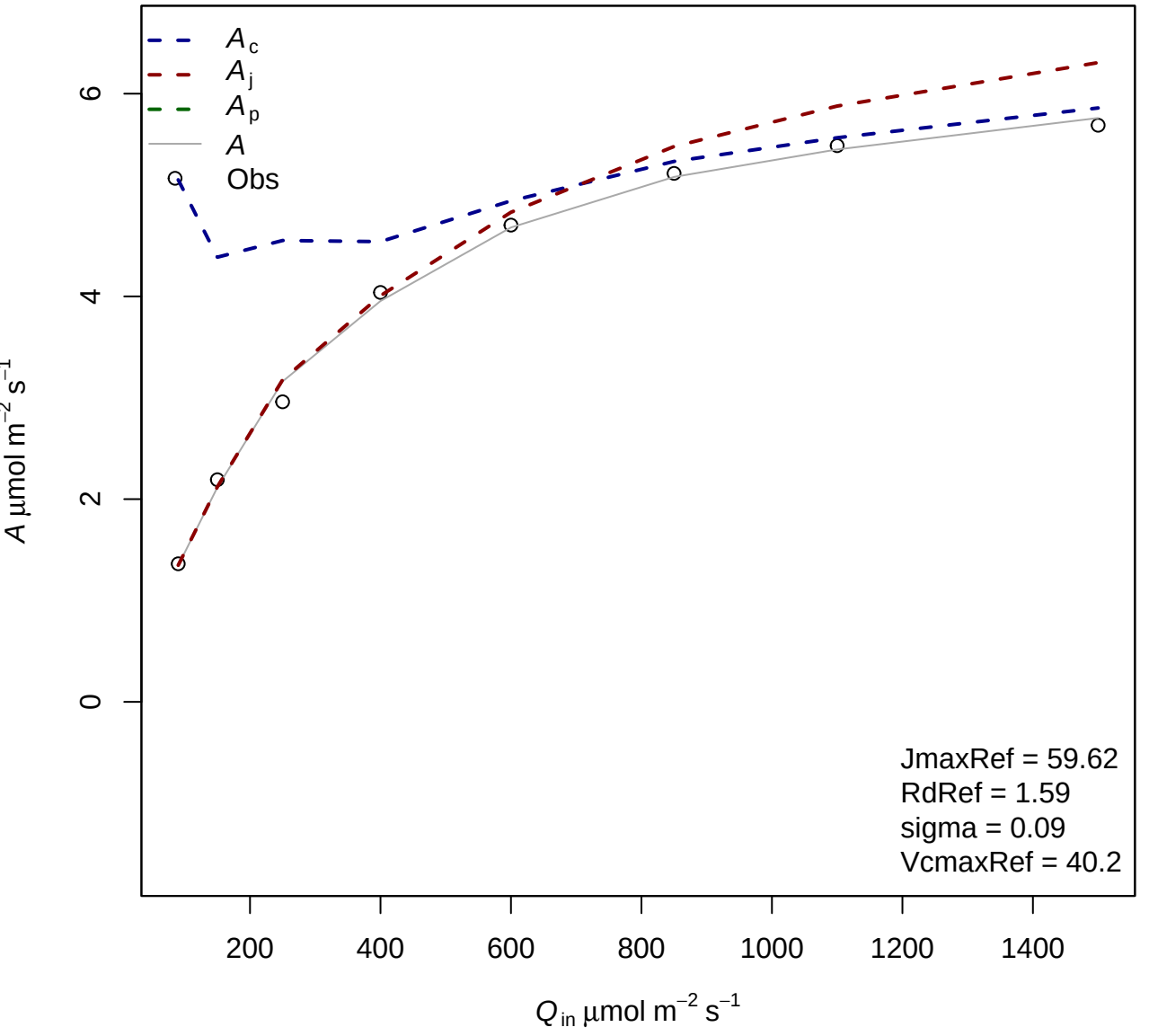

149
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.16
RdRef = 0.44
sigma = 0.09
VcmaxRef = 17.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

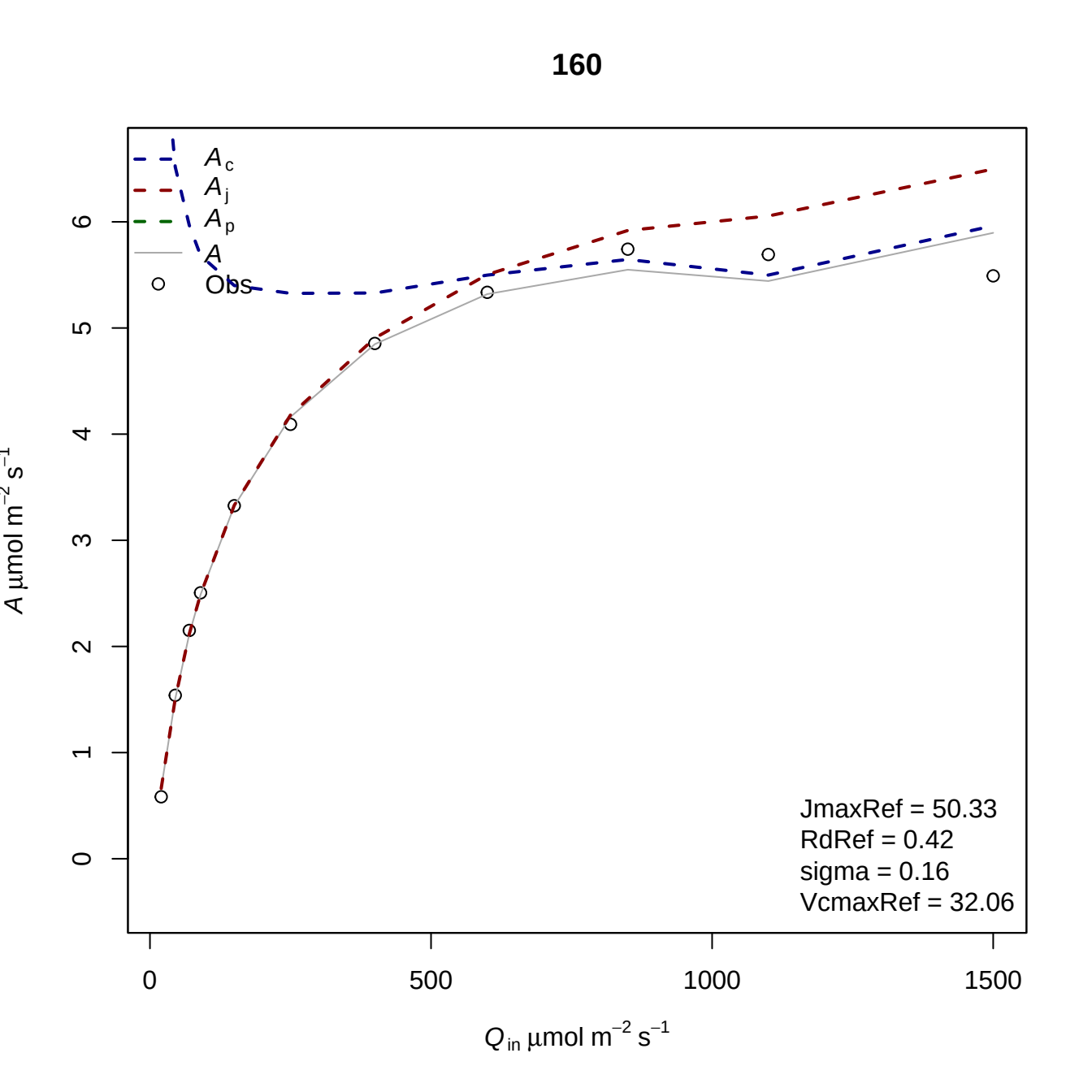

152
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 23.52
RdRef = 1.8
sigma = 0.97
VcmaxRef = 84953.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
−1
0
1
2
3
4
0
500
1000
1500

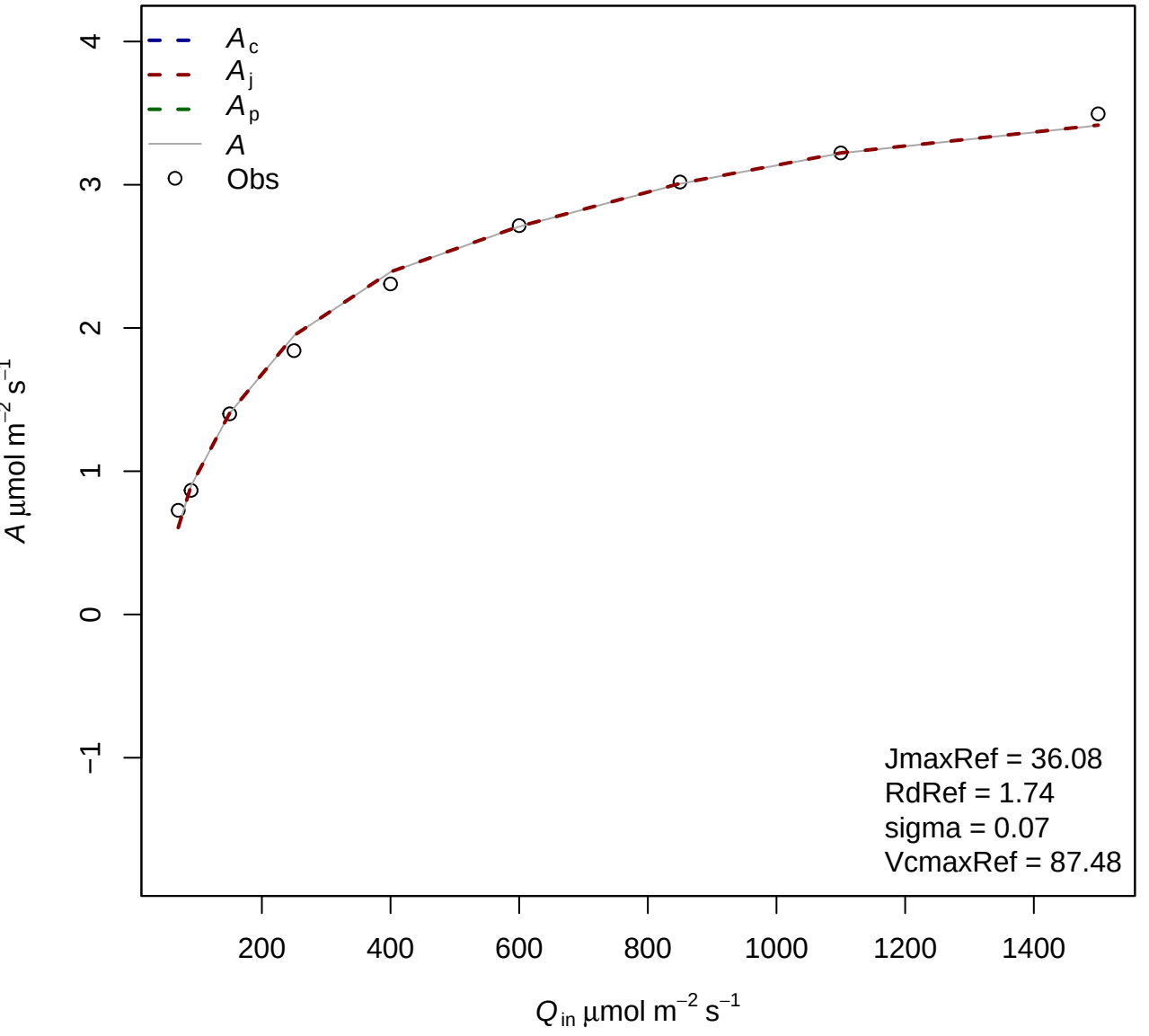

153
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.02
RdRef = 2.35
sigma = 0.46
VcmaxRef = 1006.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

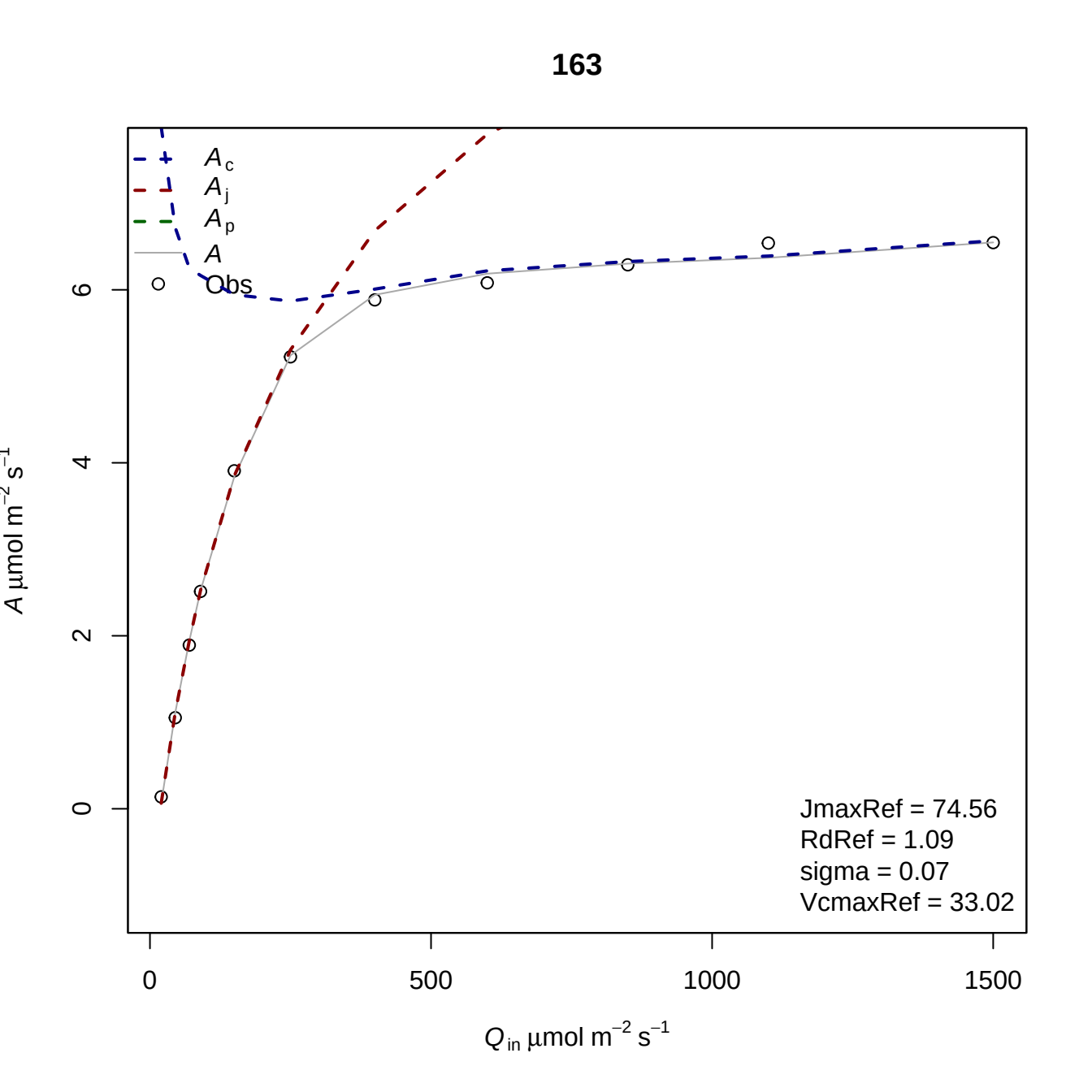

154
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.1
RdRef = 1.29
sigma = 0.19
VcmaxRef = 66.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

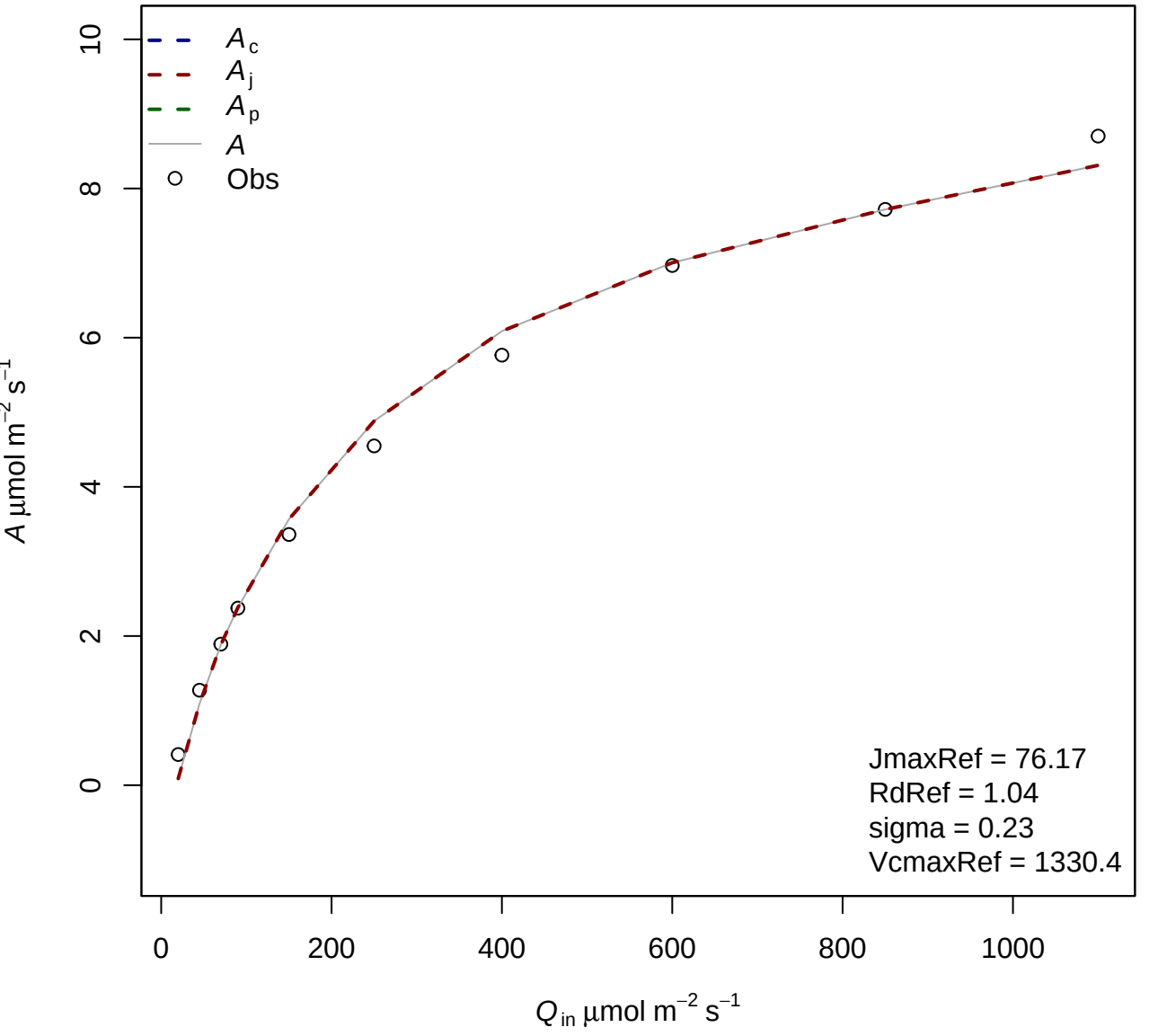

155
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.48
RdRef = 0.93
sigma = 0.31
VcmaxRef = 59.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

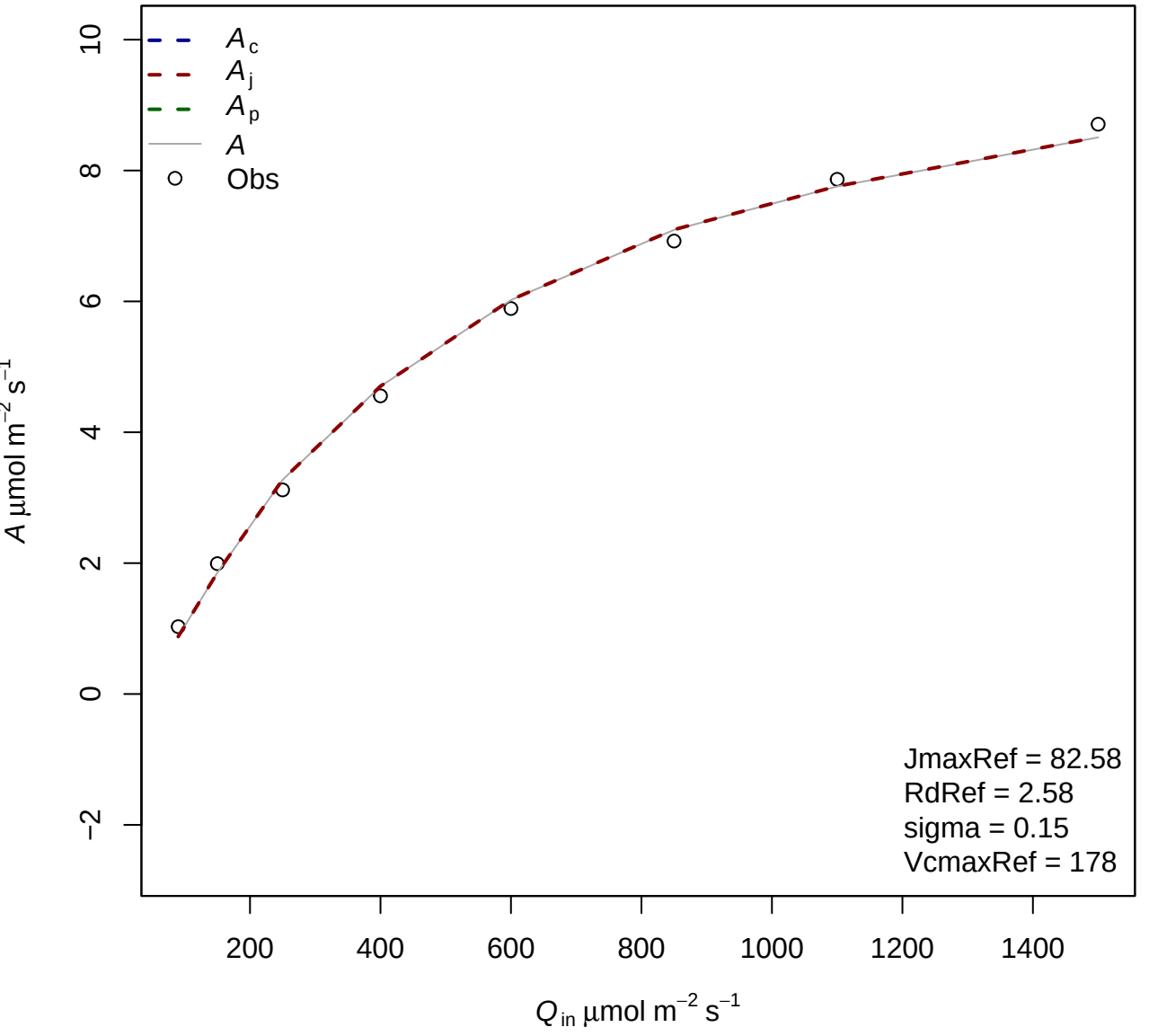

$A_c$
$A_j$
$A_p$
$A$
Obs
0
2
4
6
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 59.62
RdRef = 1.59
sigma = 0.09
VcmaxRef = 40.2

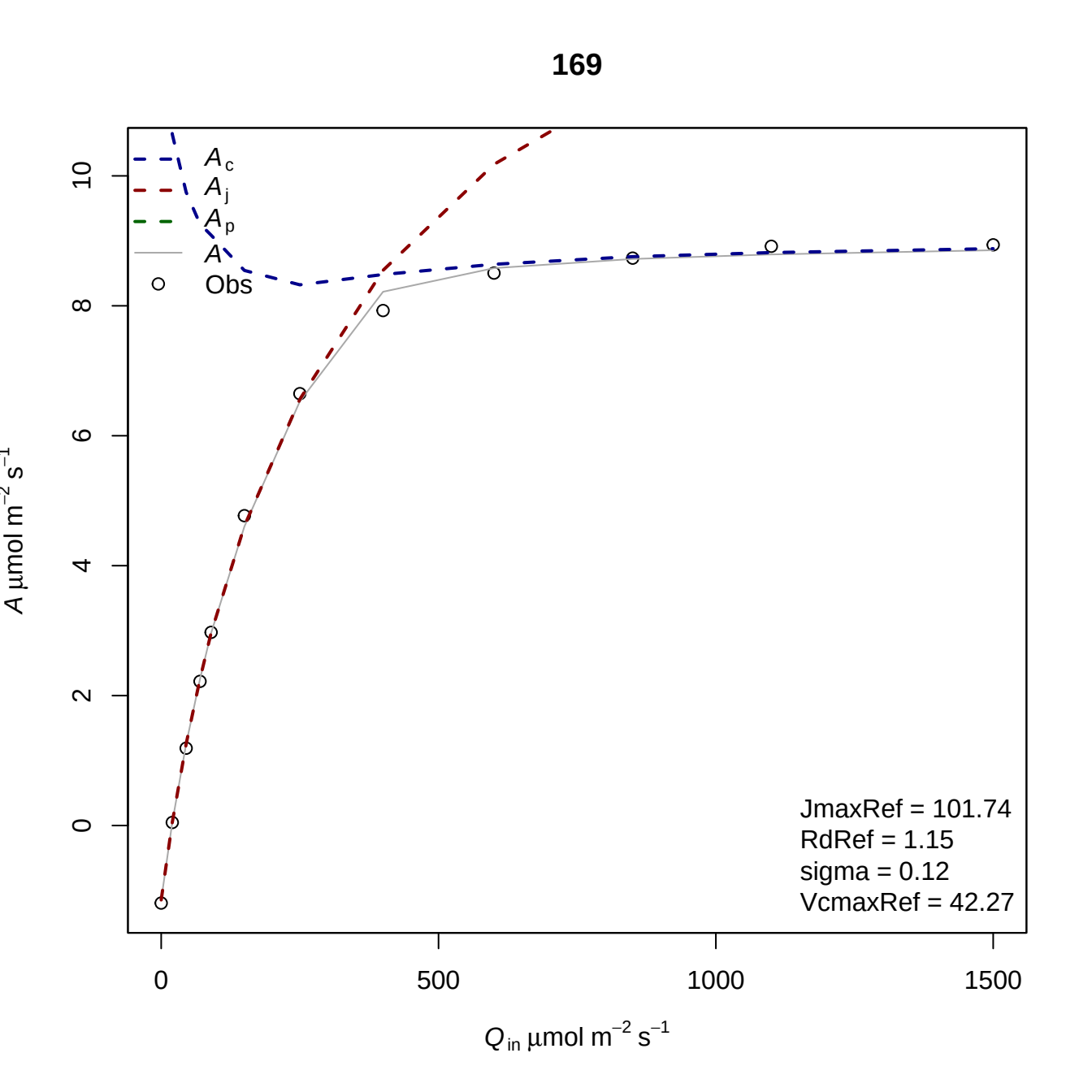

160
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.33
RdRef = 0.42
sigma = 0.16
VcmaxRef = 32.06
0
1
2
3
4
5
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

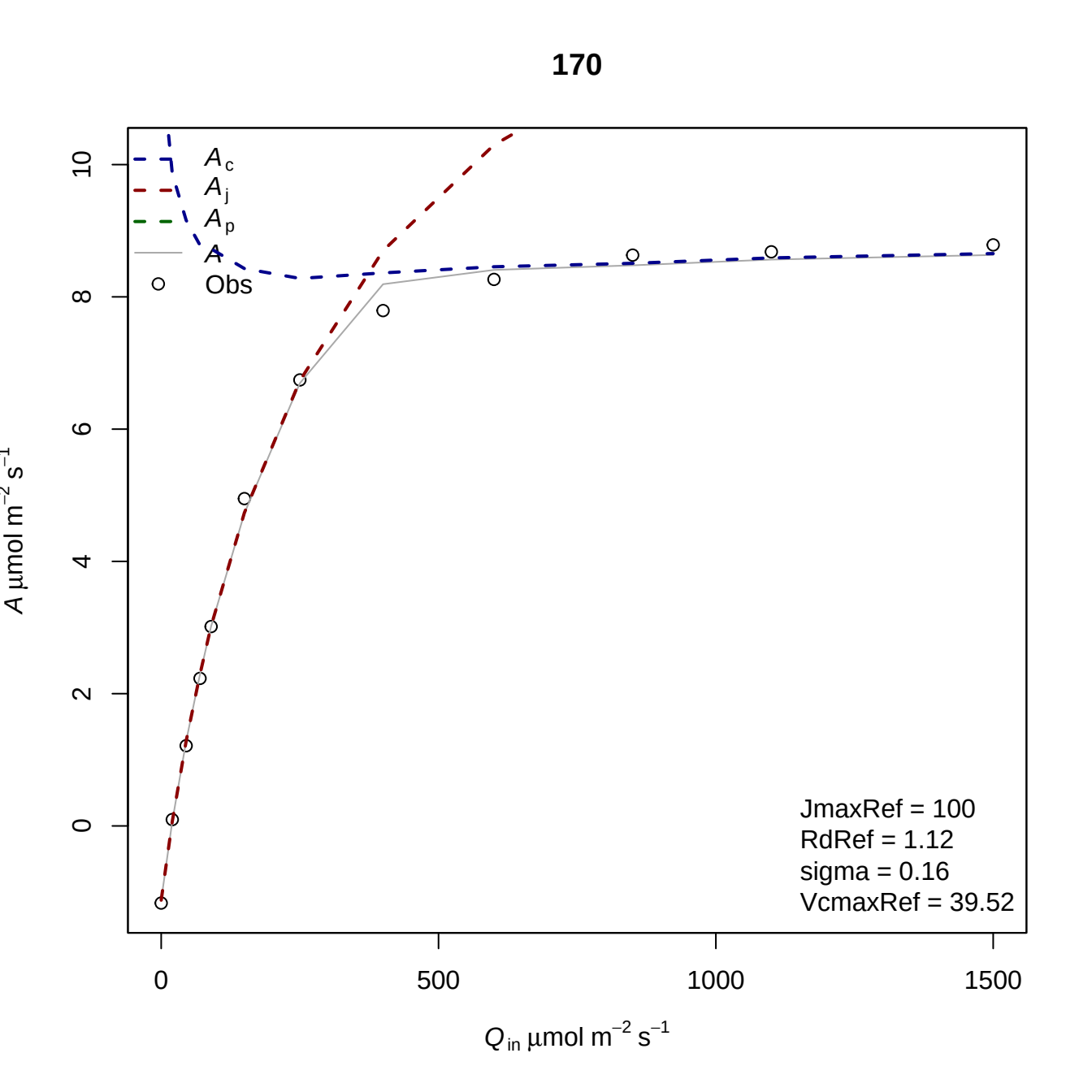

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.08
RdRef = 1.74
sigma = 0.07
VcmaxRef = 87.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

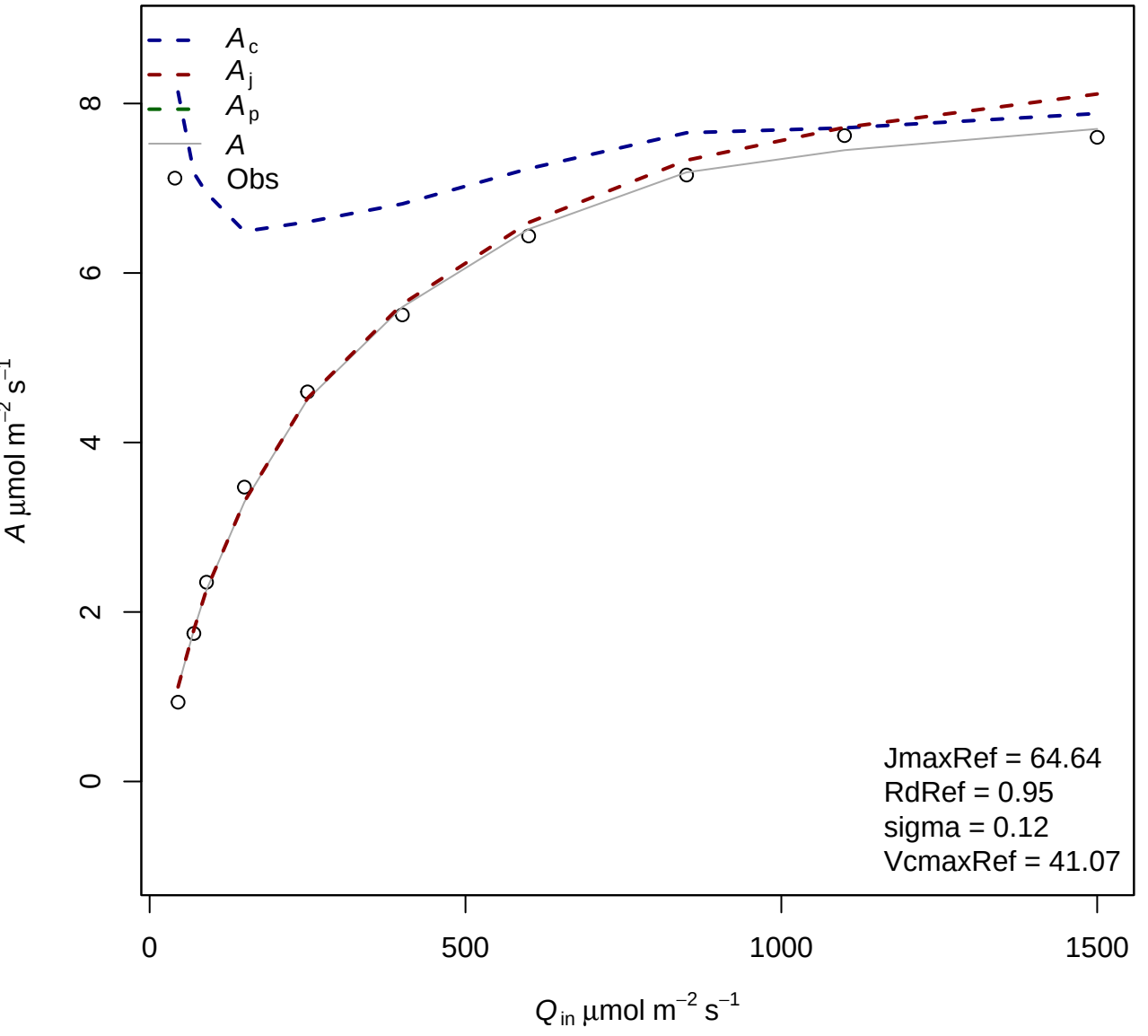

163
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.56
RdRef = 1.09
sigma = 0.07
VcmaxRef = 33.02
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

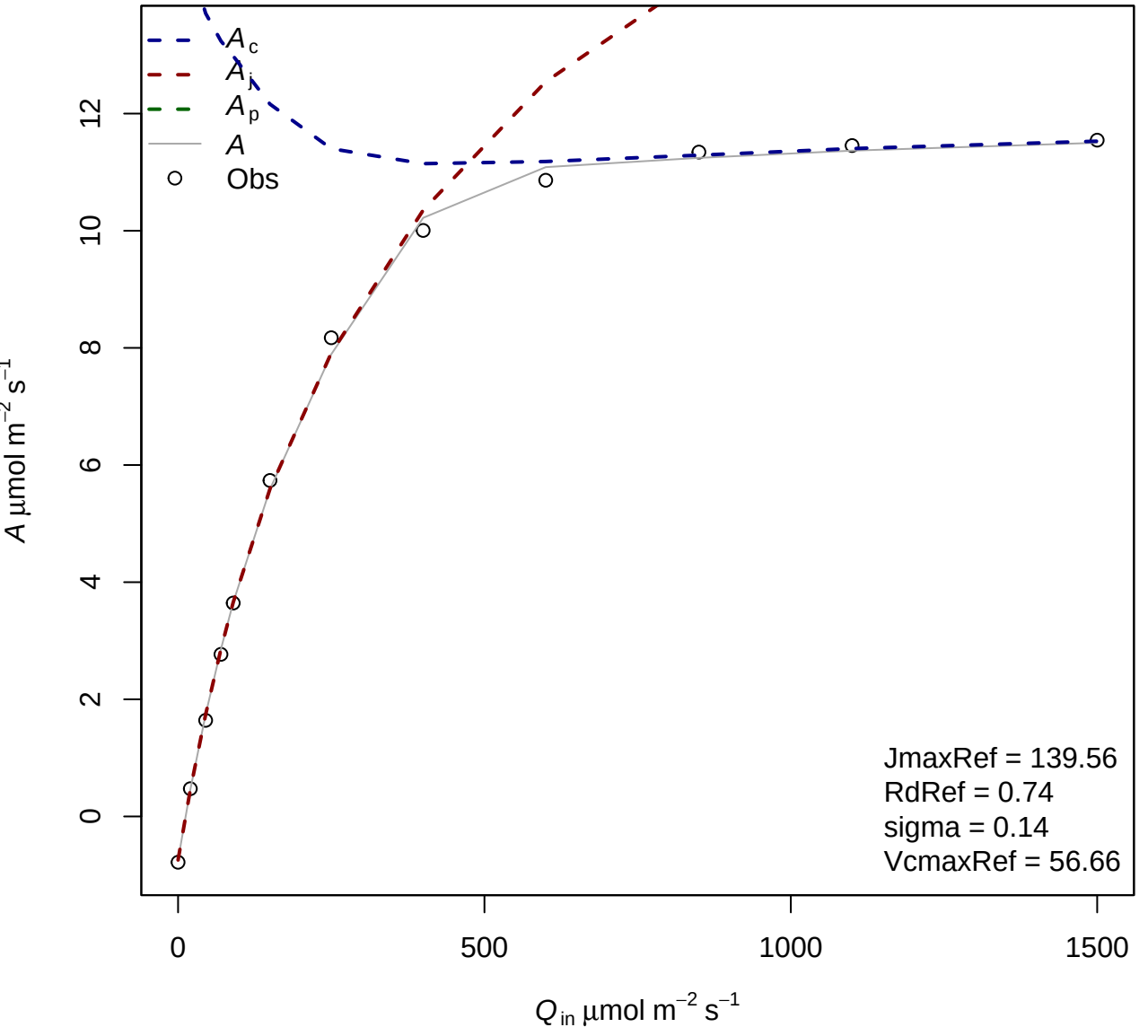

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.17
RdRef = 1.04
sigma = 0.23
VcmaxRef = 1330.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
200
400
600
800
1000

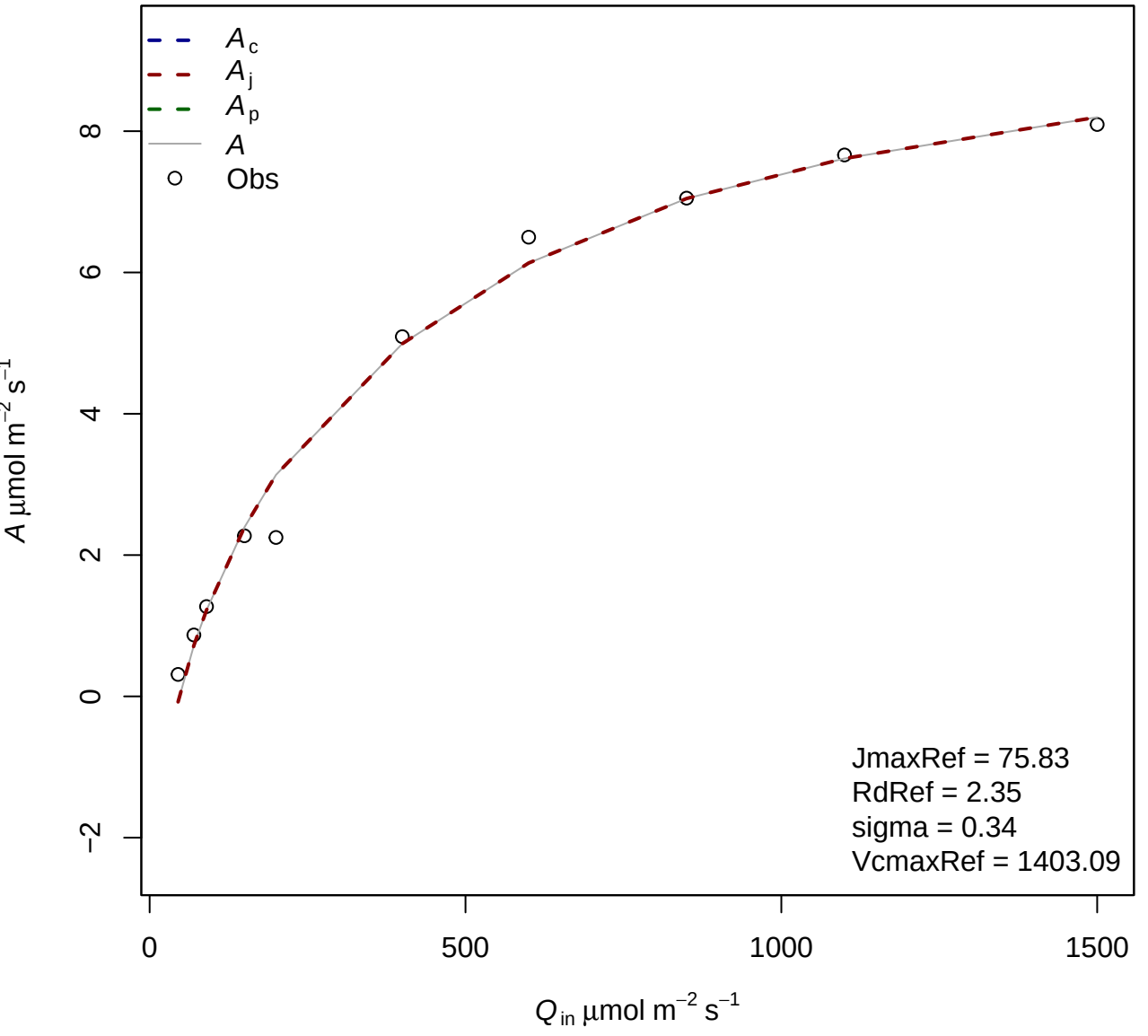

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.58
RdRef = 2.58
sigma = 0.15
VcmaxRef = 178
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

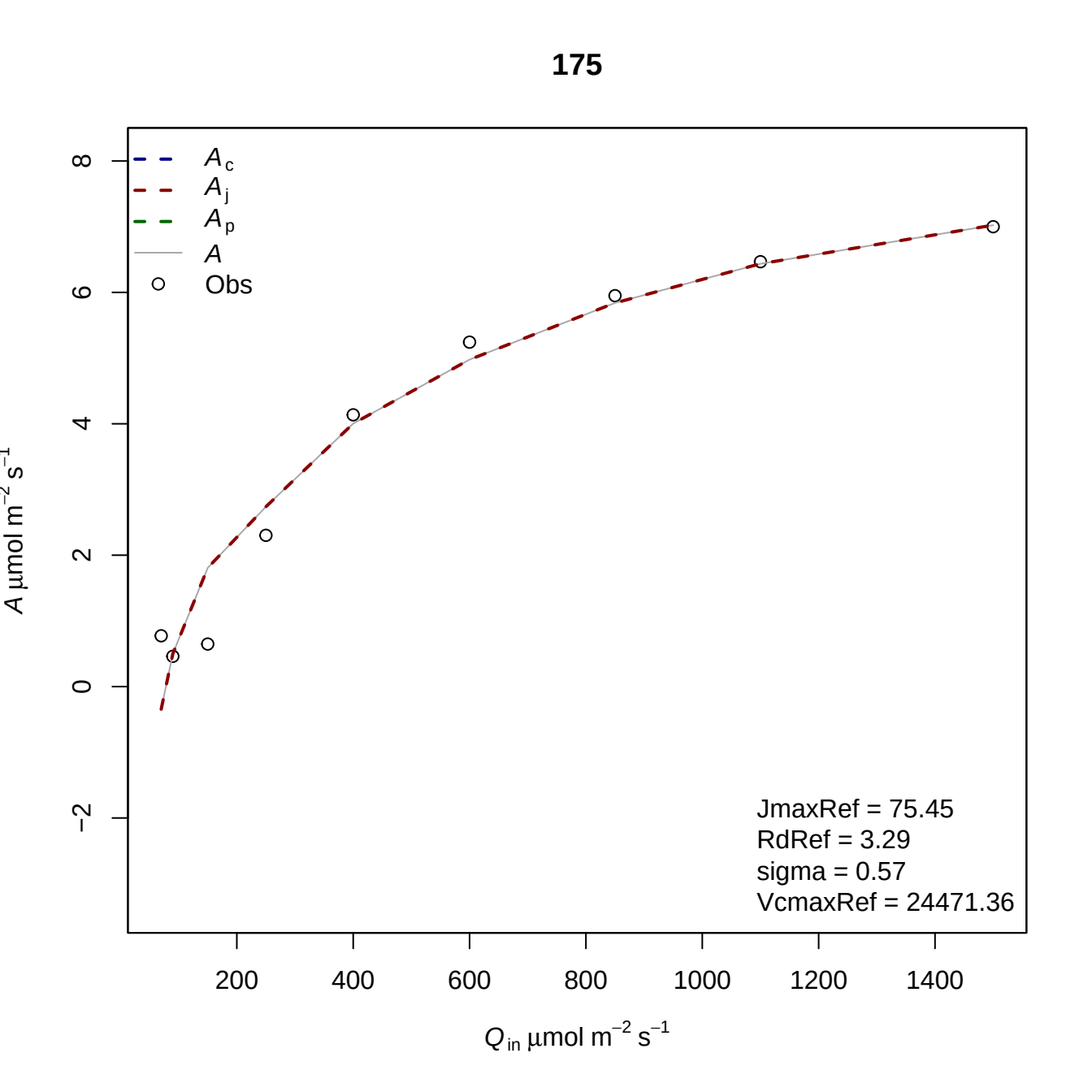

169
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.74
RdRef = 1.15
sigma = 0.12
VcmaxRef = 42.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

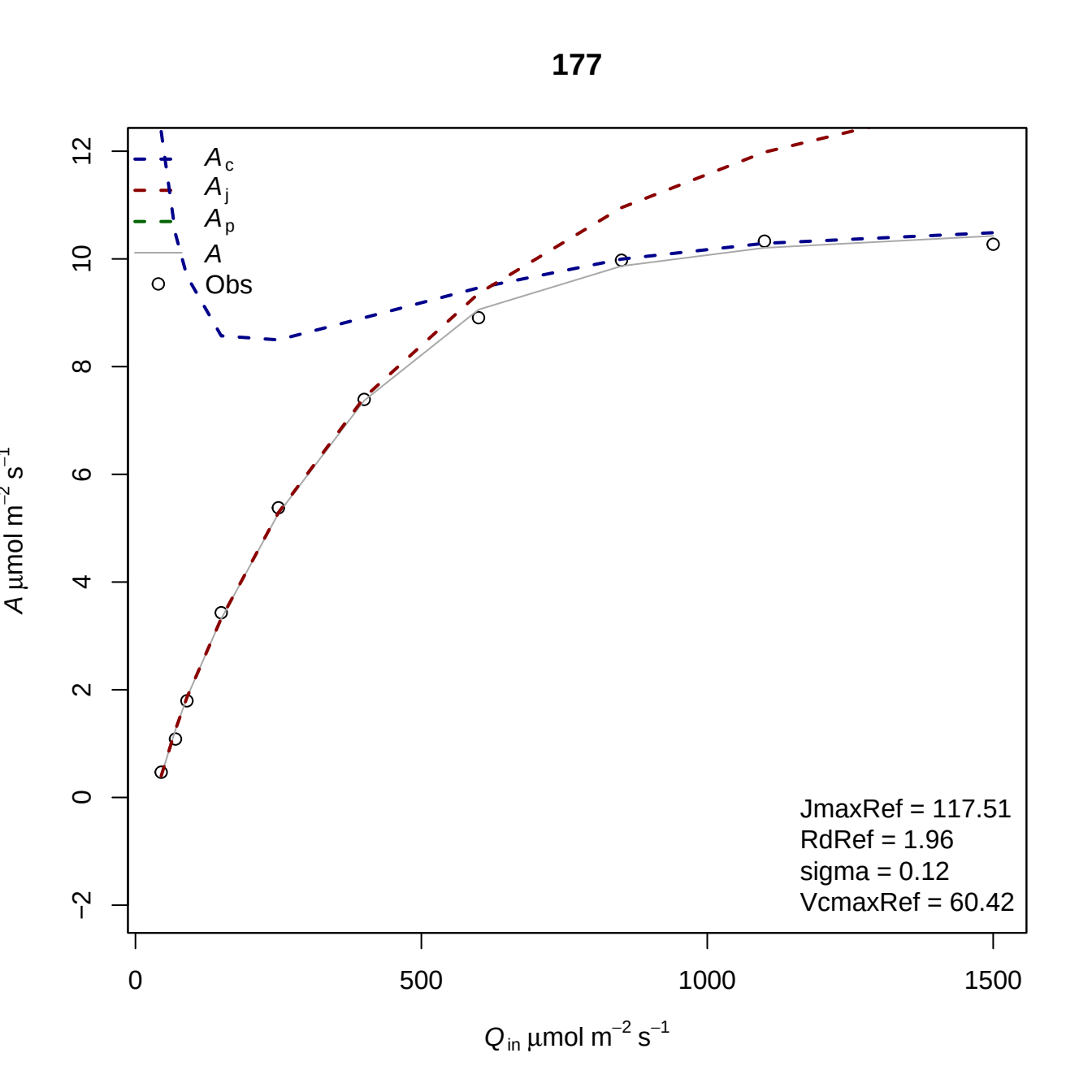

170
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100
RdRef = 1.12
sigma = 0.16
VcmaxRef = 39.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

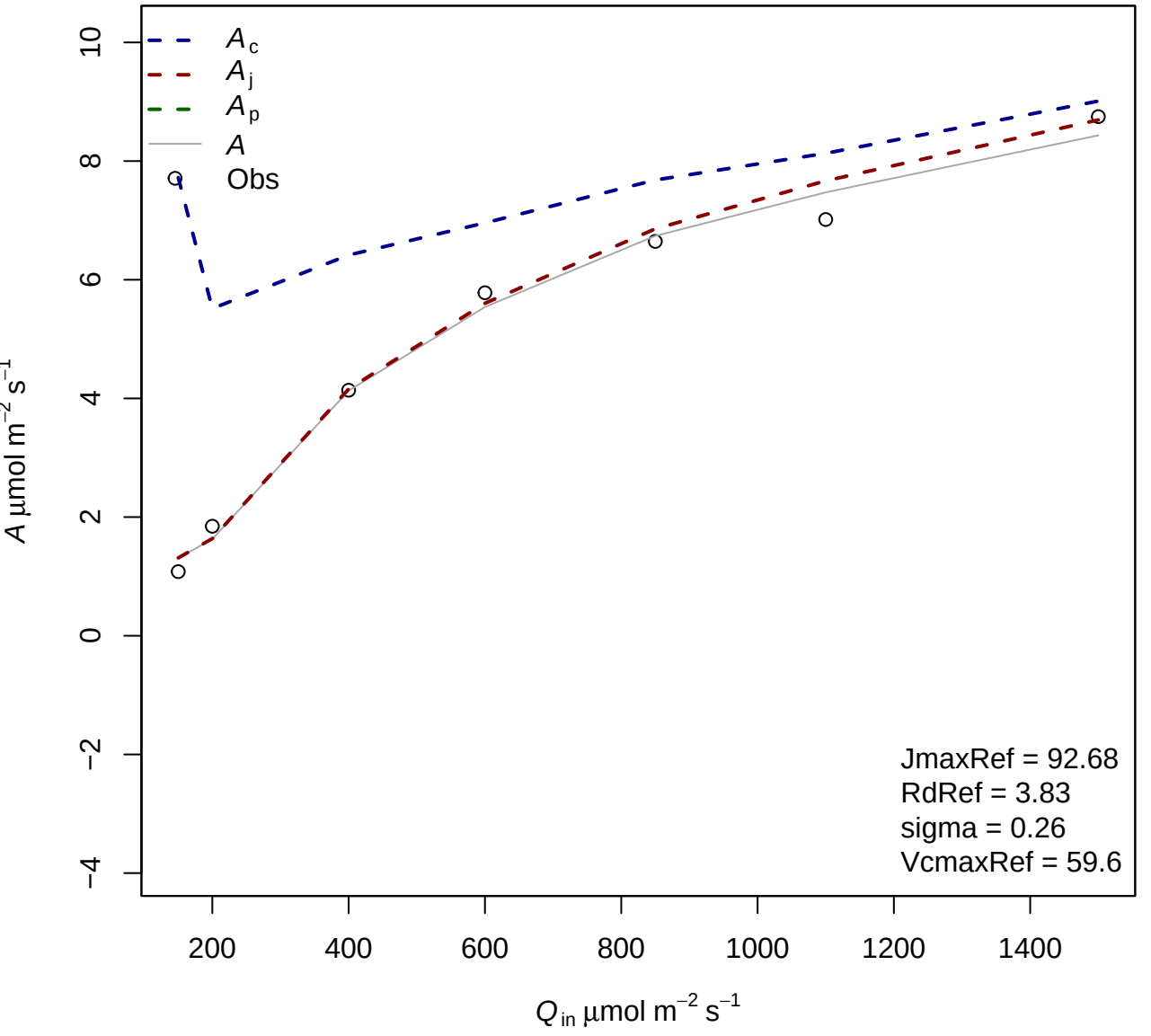

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.64
RdRef = 0.95
sigma = 0.12
VcmaxRef = 41.07
0
2
4
6
8
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

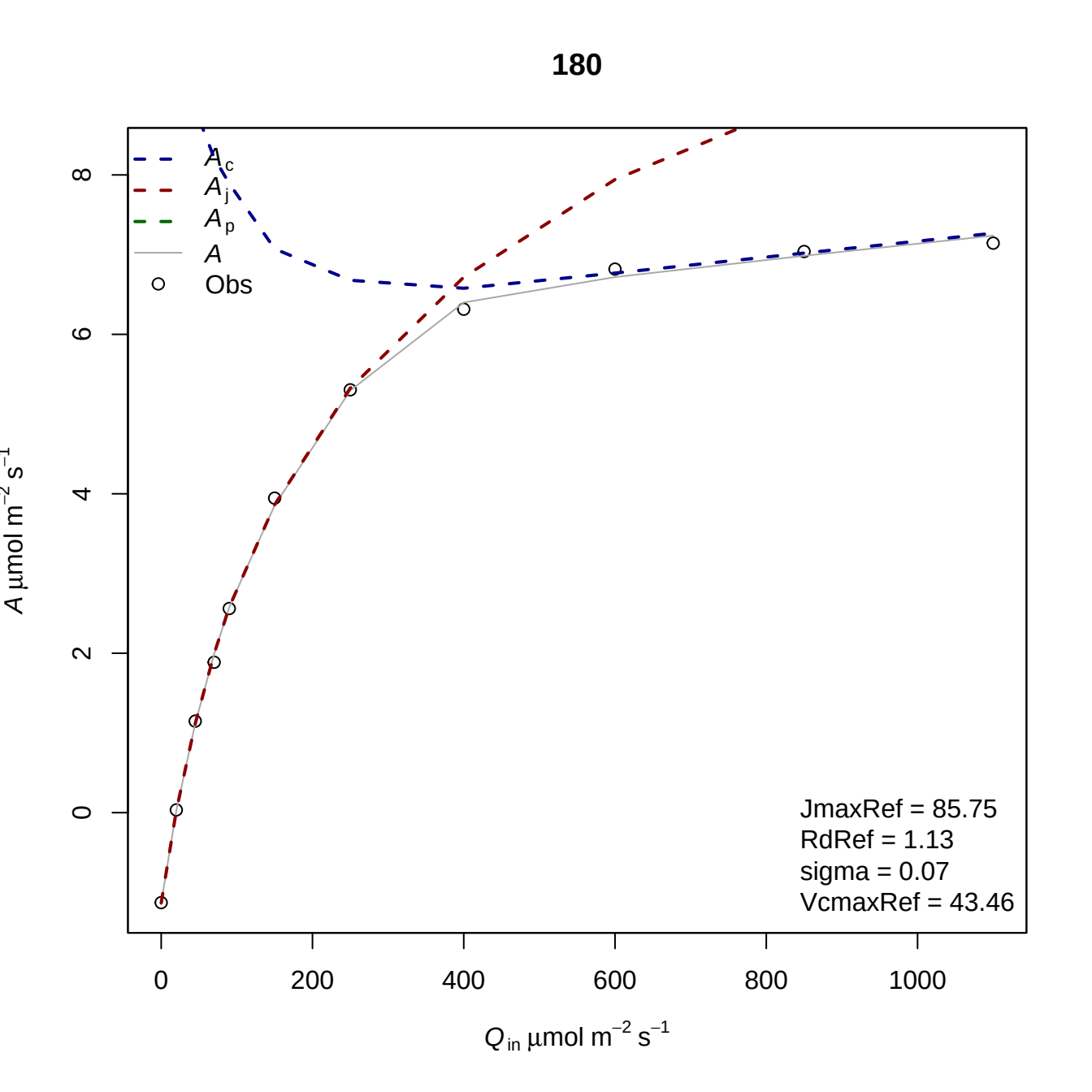

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.56
RdRef = 0.74
sigma = 0.14
VcmaxRef = 56.66
0
2
4
6
8
10
12
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

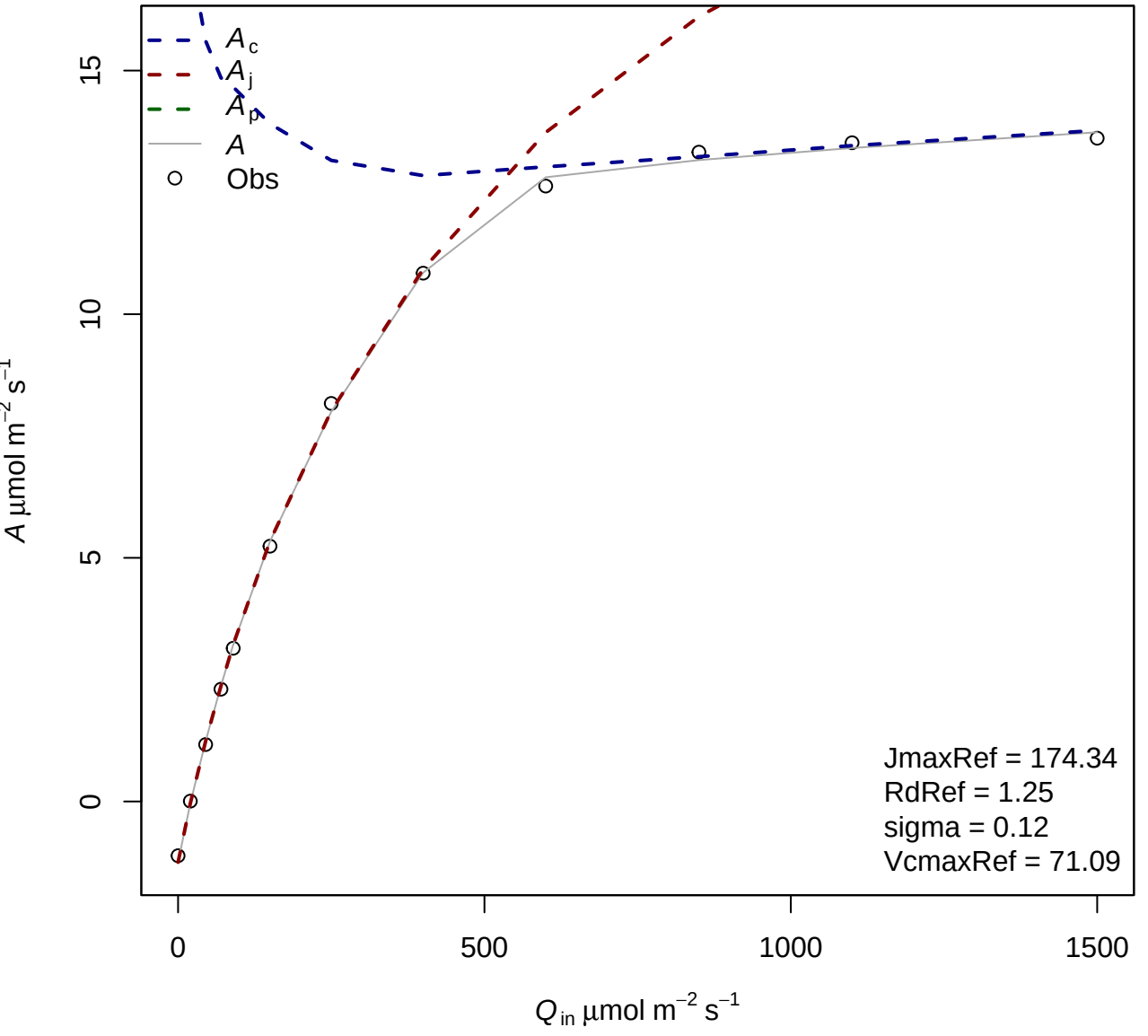

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.83
RdRef = 2.35
sigma = 0.34
VcmaxRef = 1403.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

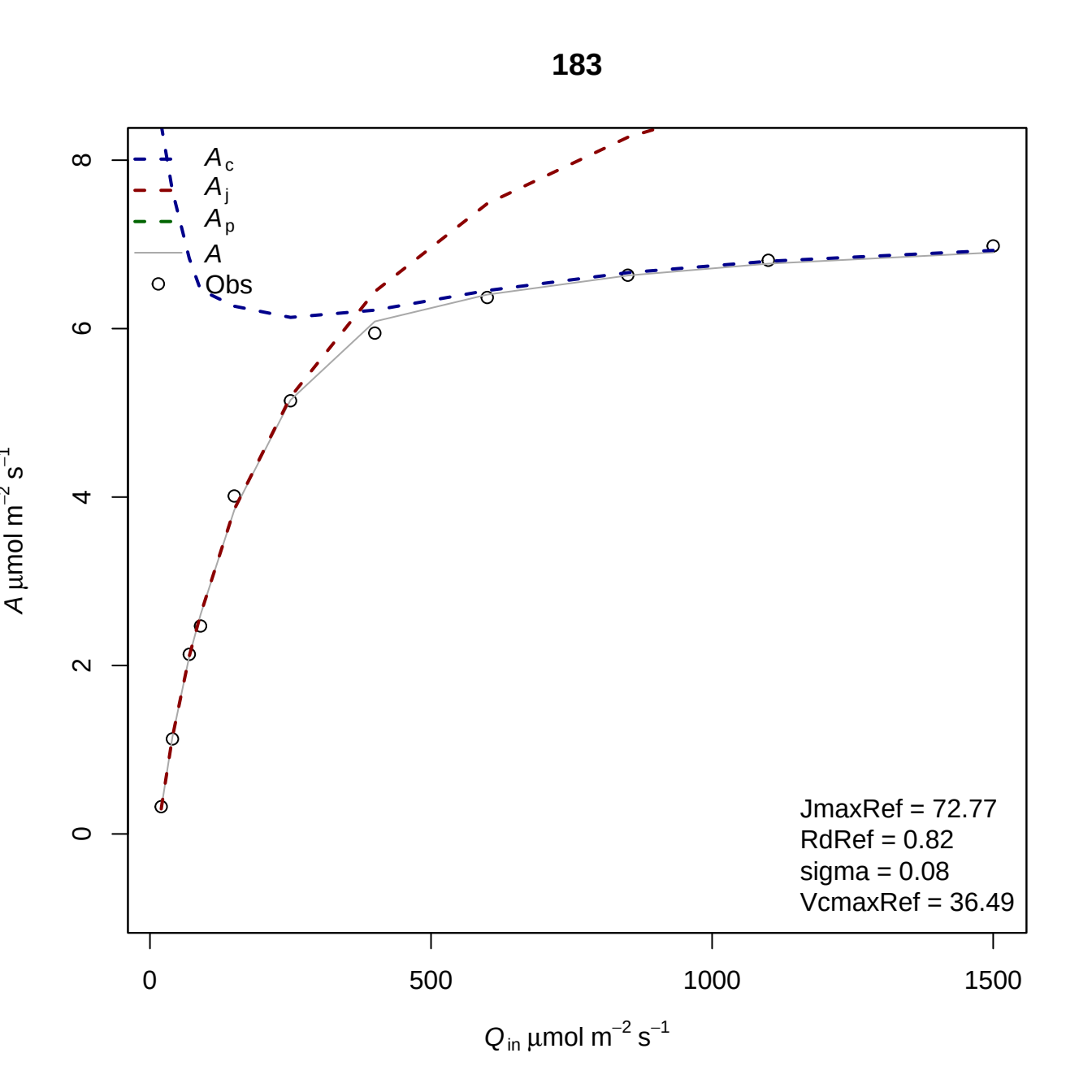

175
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.45
RdRef = 3.29
sigma = 0.57
VcmaxRef = 24471.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

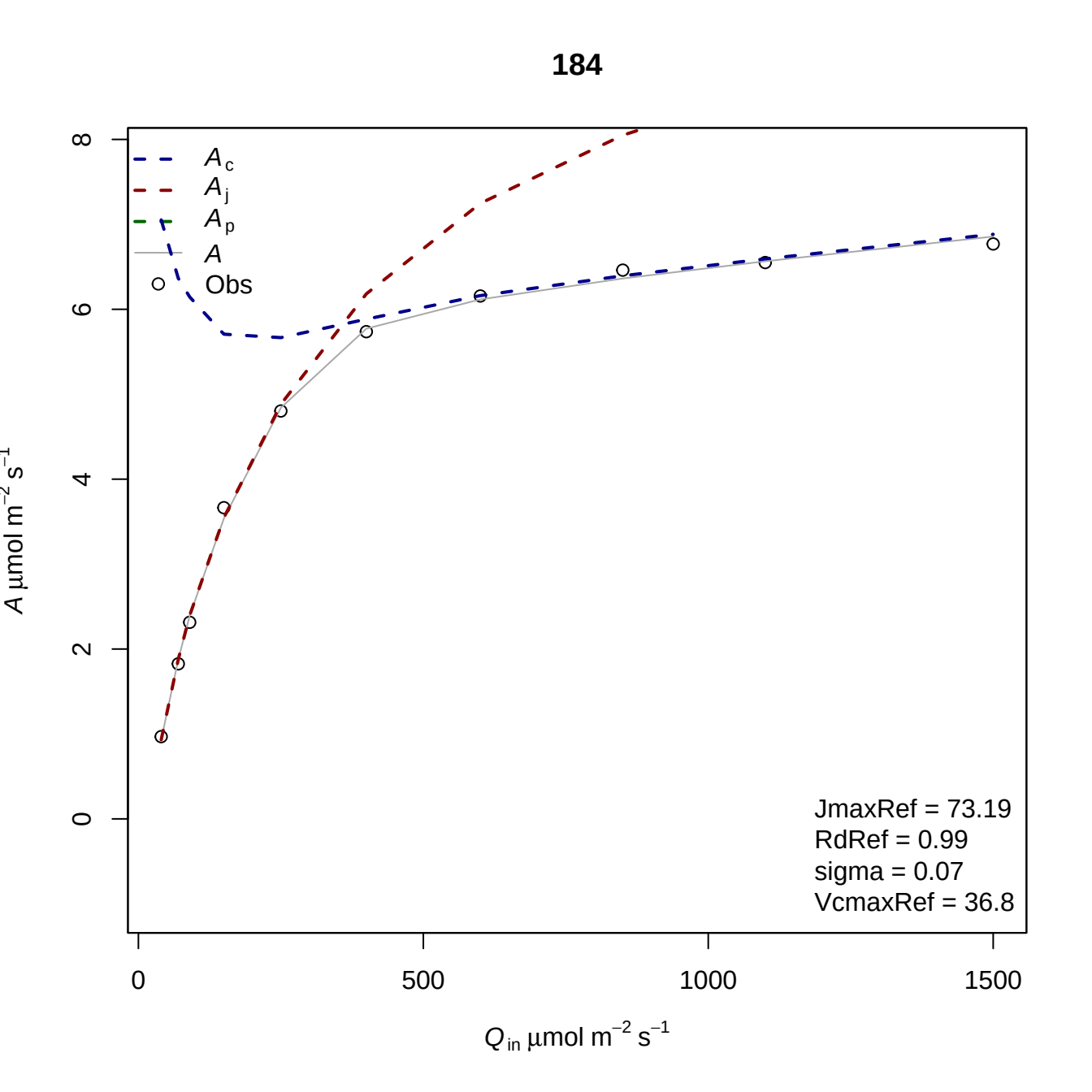

177
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.51
RdRef = 1.96
sigma = 0.12
VcmaxRef = 60.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

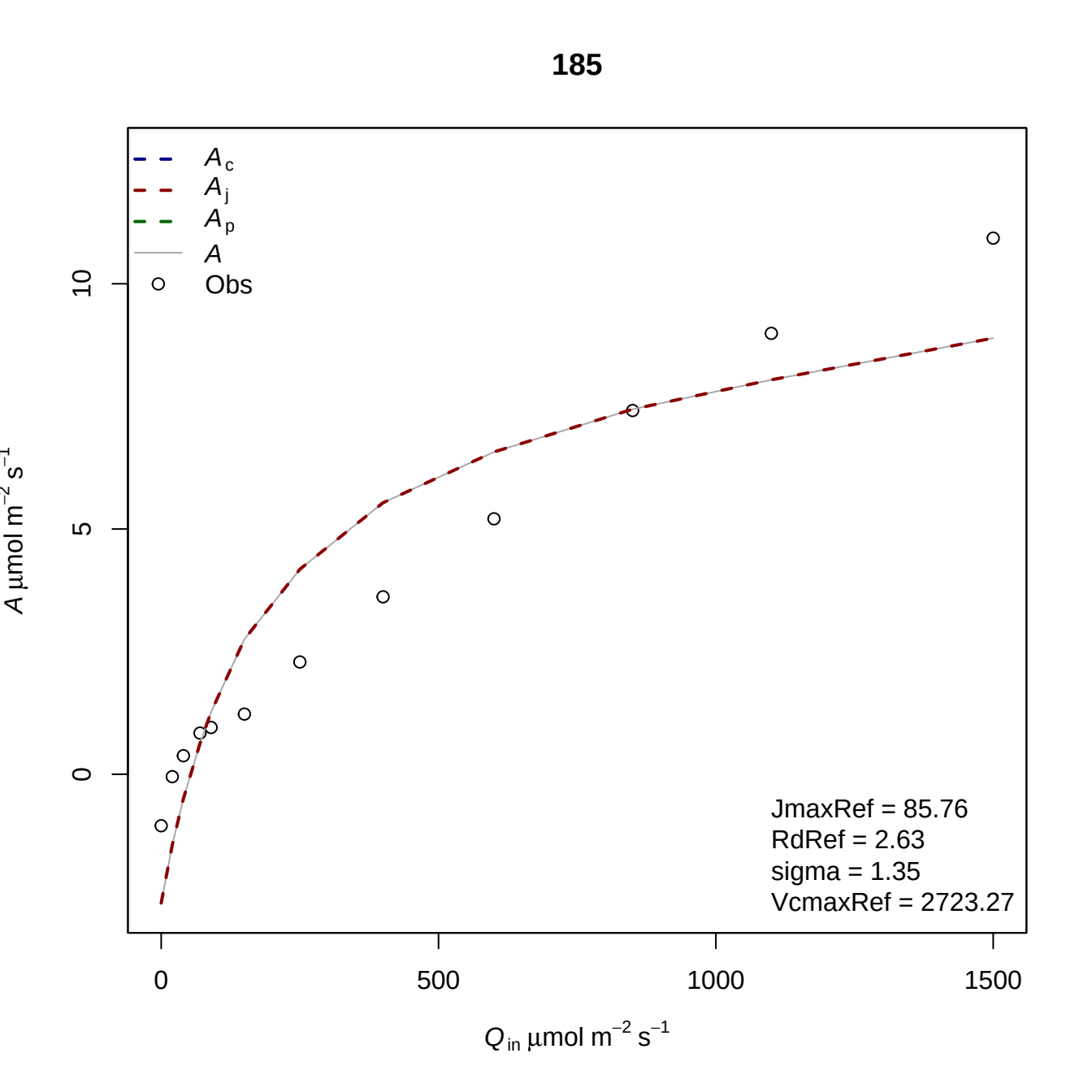

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.68
RdRef = 3.83
sigma = 0.26
VcmaxRef = 59.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

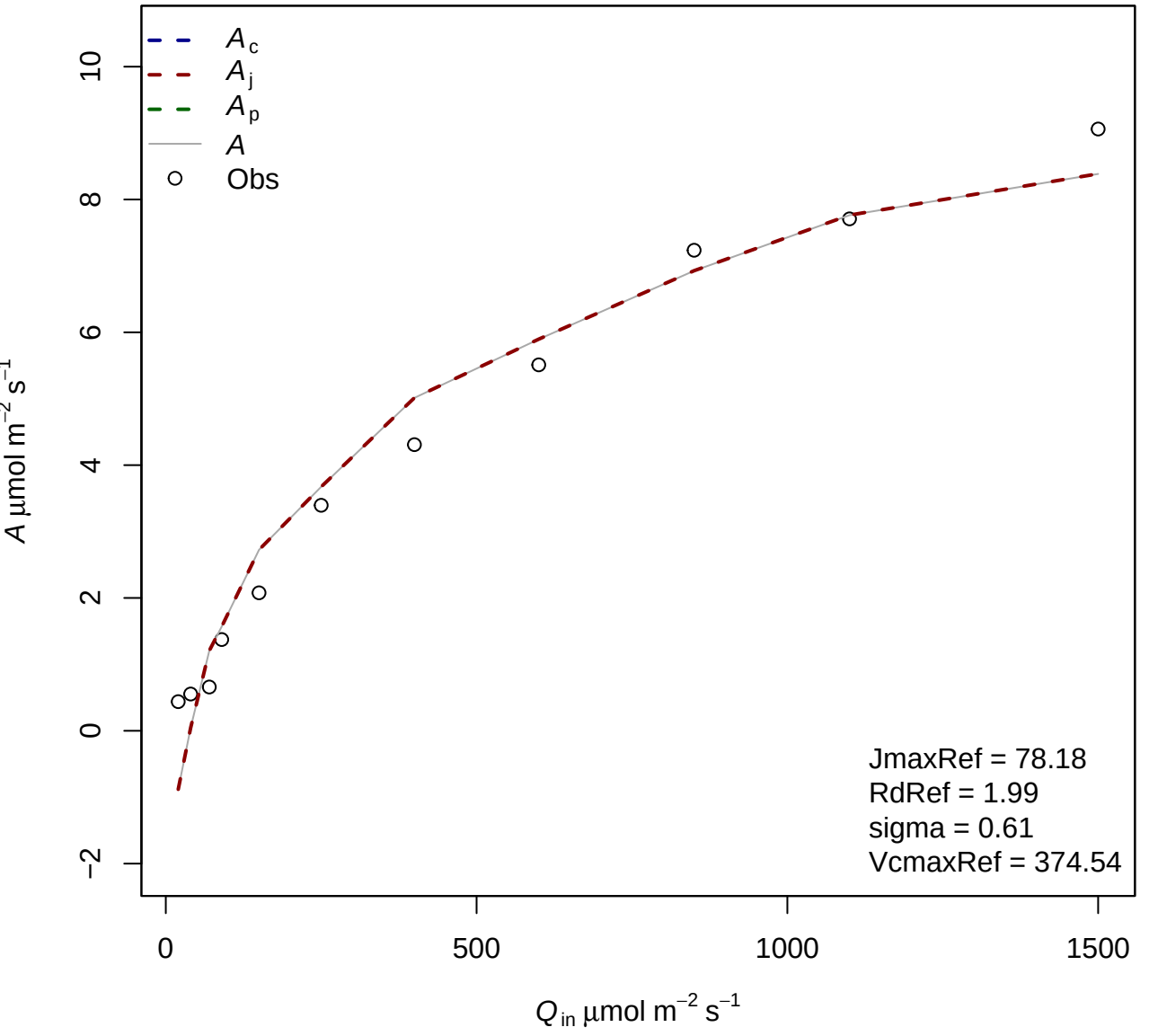

180
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.75
RdRef = 1.13
sigma = 0.07
VcmaxRef = 43.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
200
400
600
800
1000

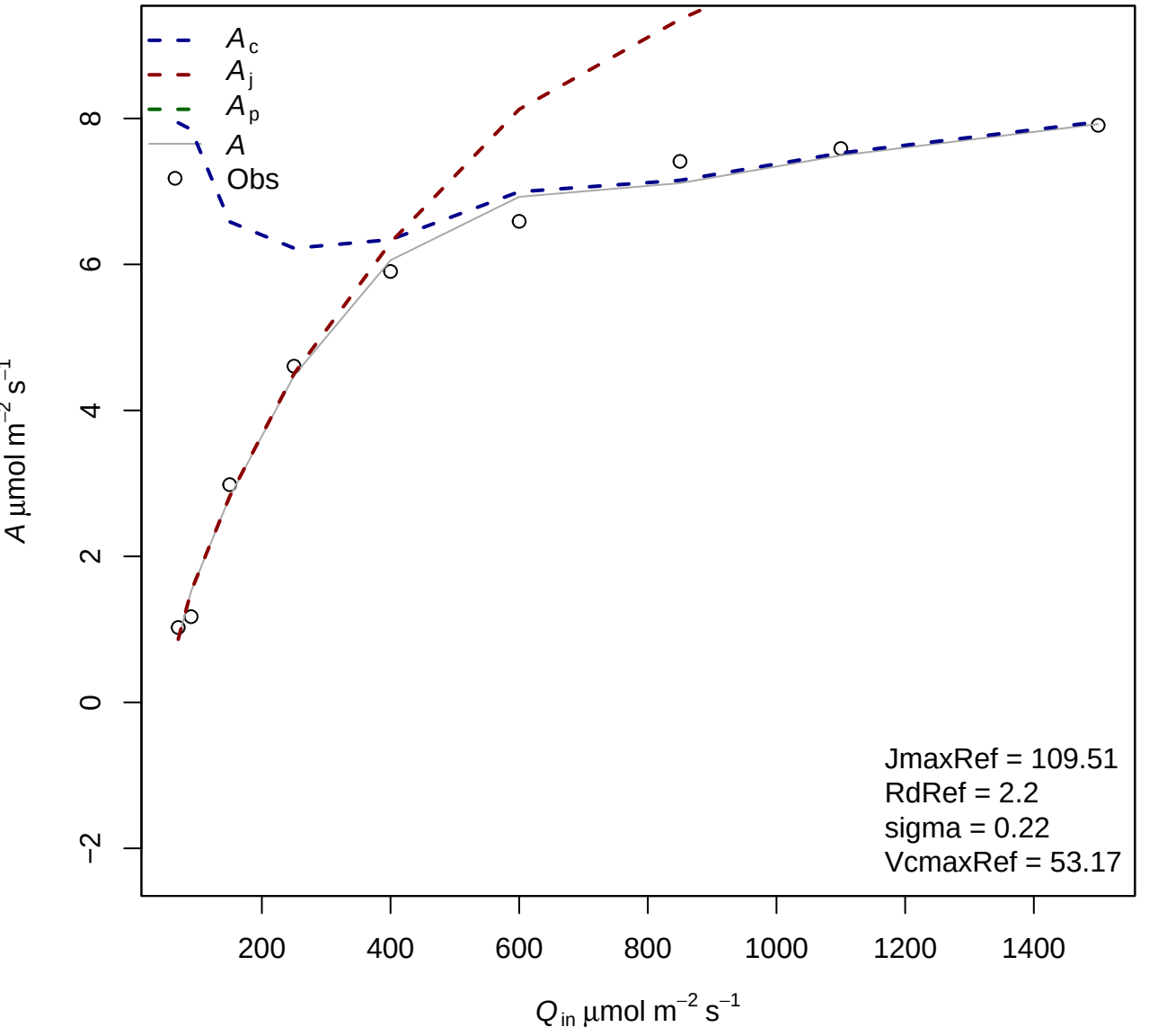

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 174.34
RdRef = 1.25
sigma = 0.12
VcmaxRef = 71.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

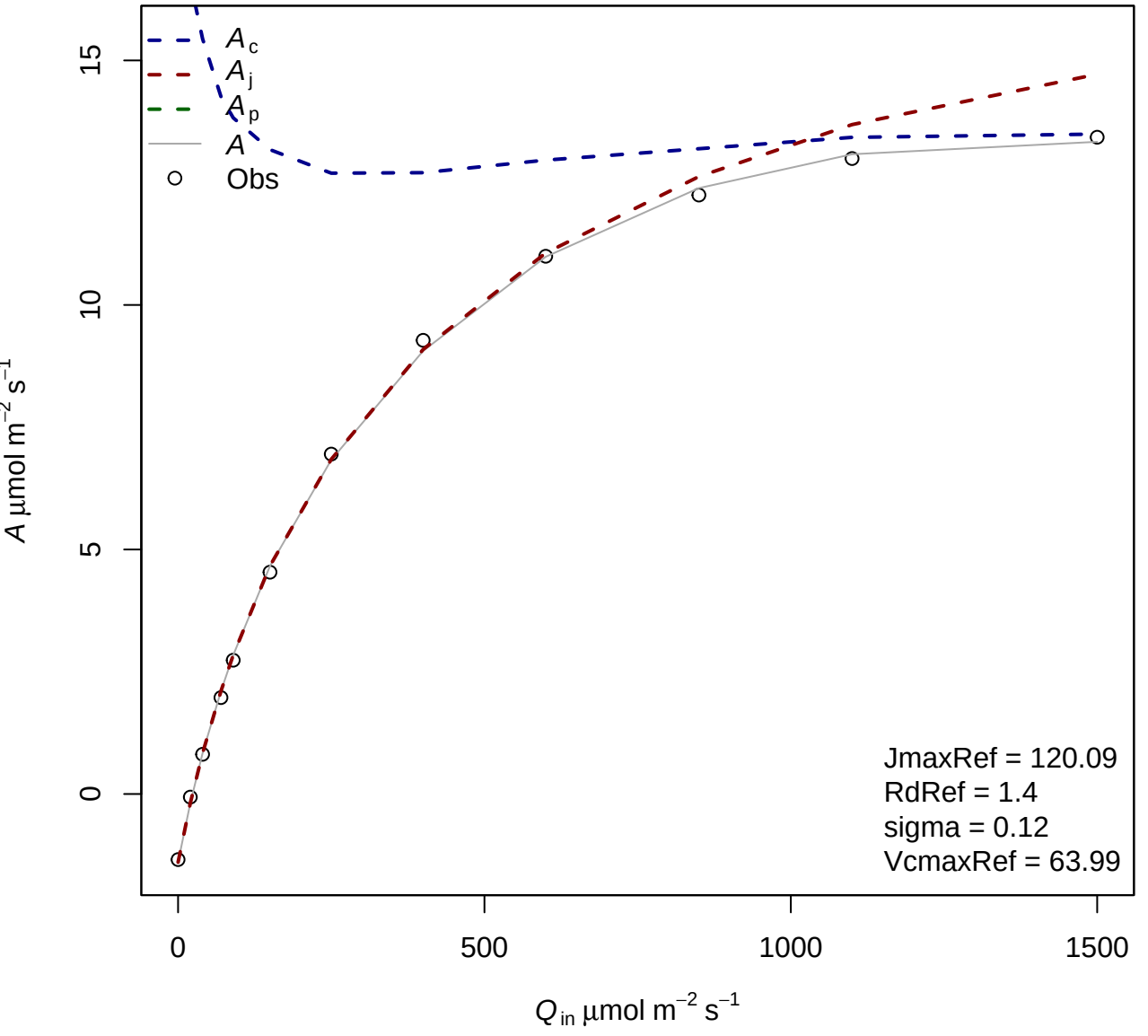

183
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.77
RdRef = 0.82
sigma = 0.08
VcmaxRef = 36.49
0
2
4
6
8
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

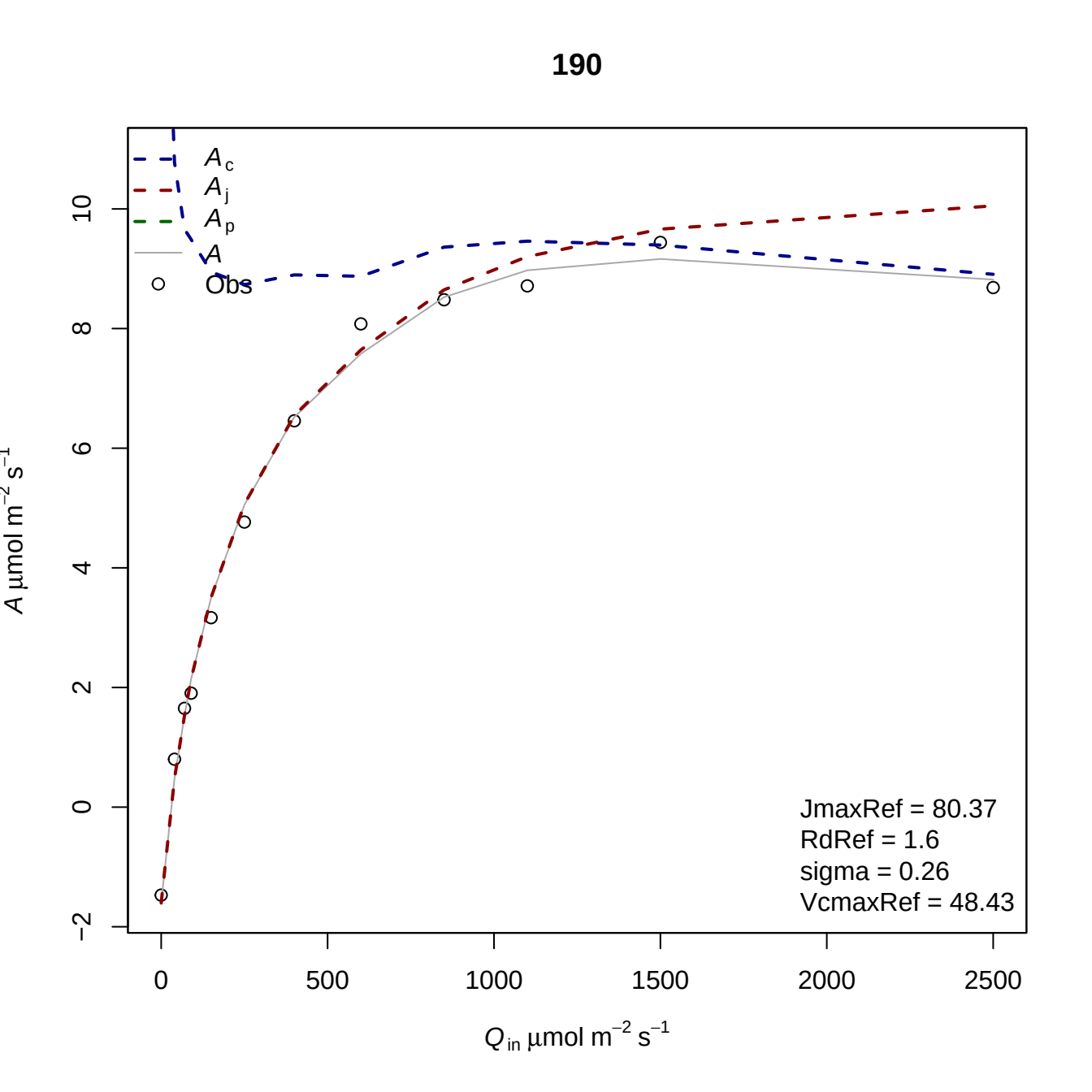

184
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.19
RdRef = 0.99
sigma = 0.07
VcmaxRef = 36.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

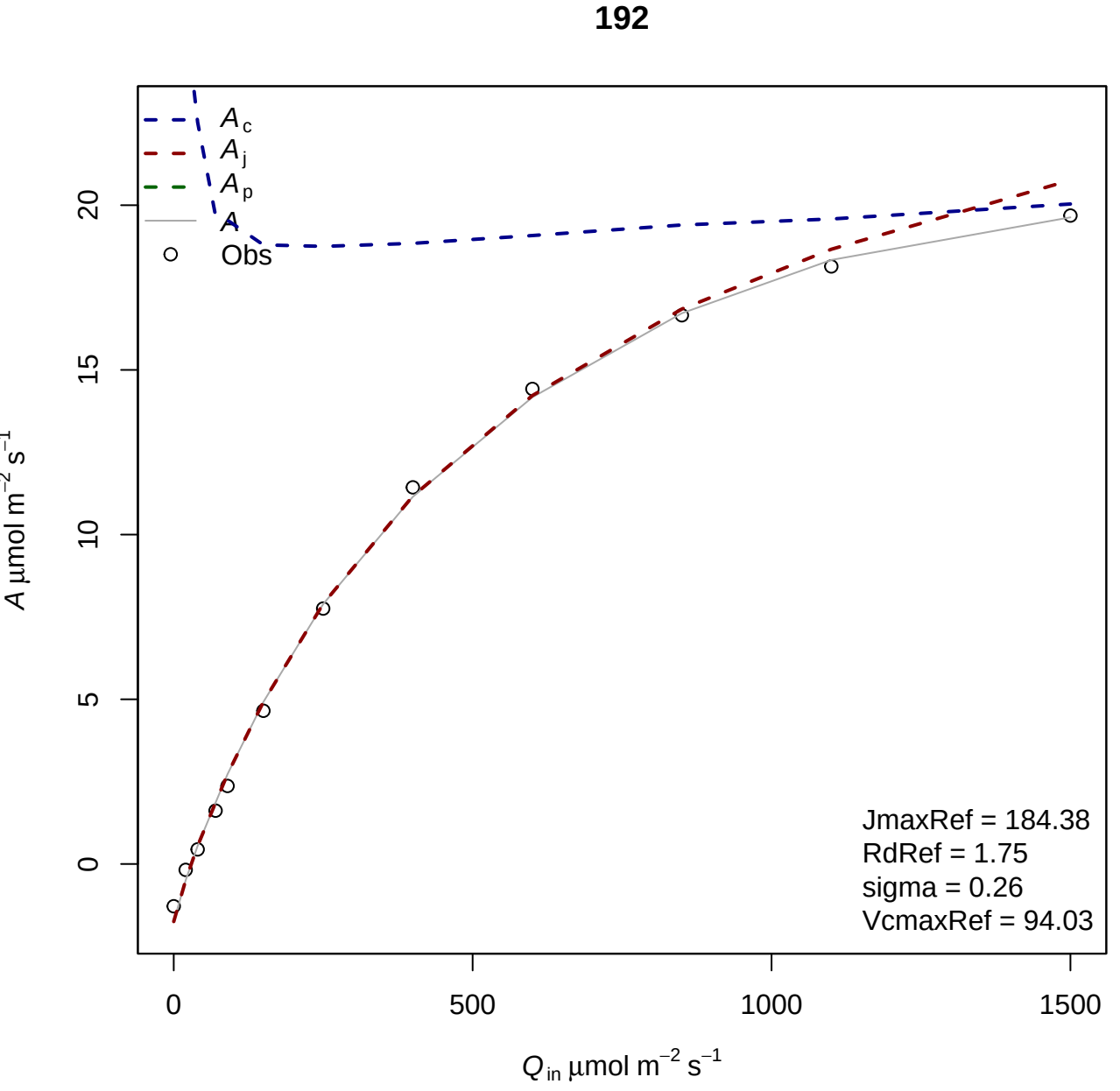

185
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.76
RdRef = 2.63
sigma = 1.35
VcmaxRef = 2723.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

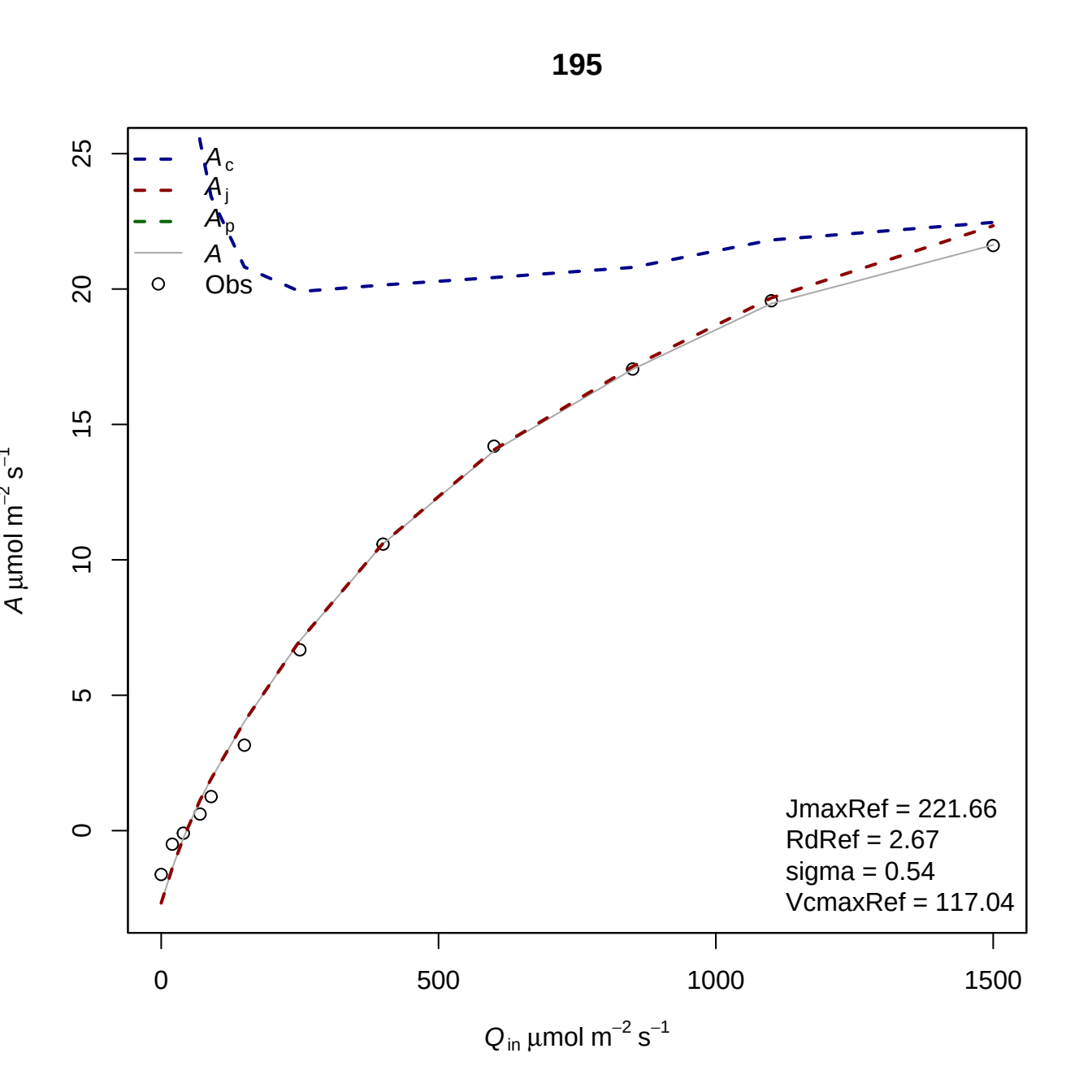

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.18
RdRef = 1.99
sigma = 0.61
VcmaxRef = 374.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

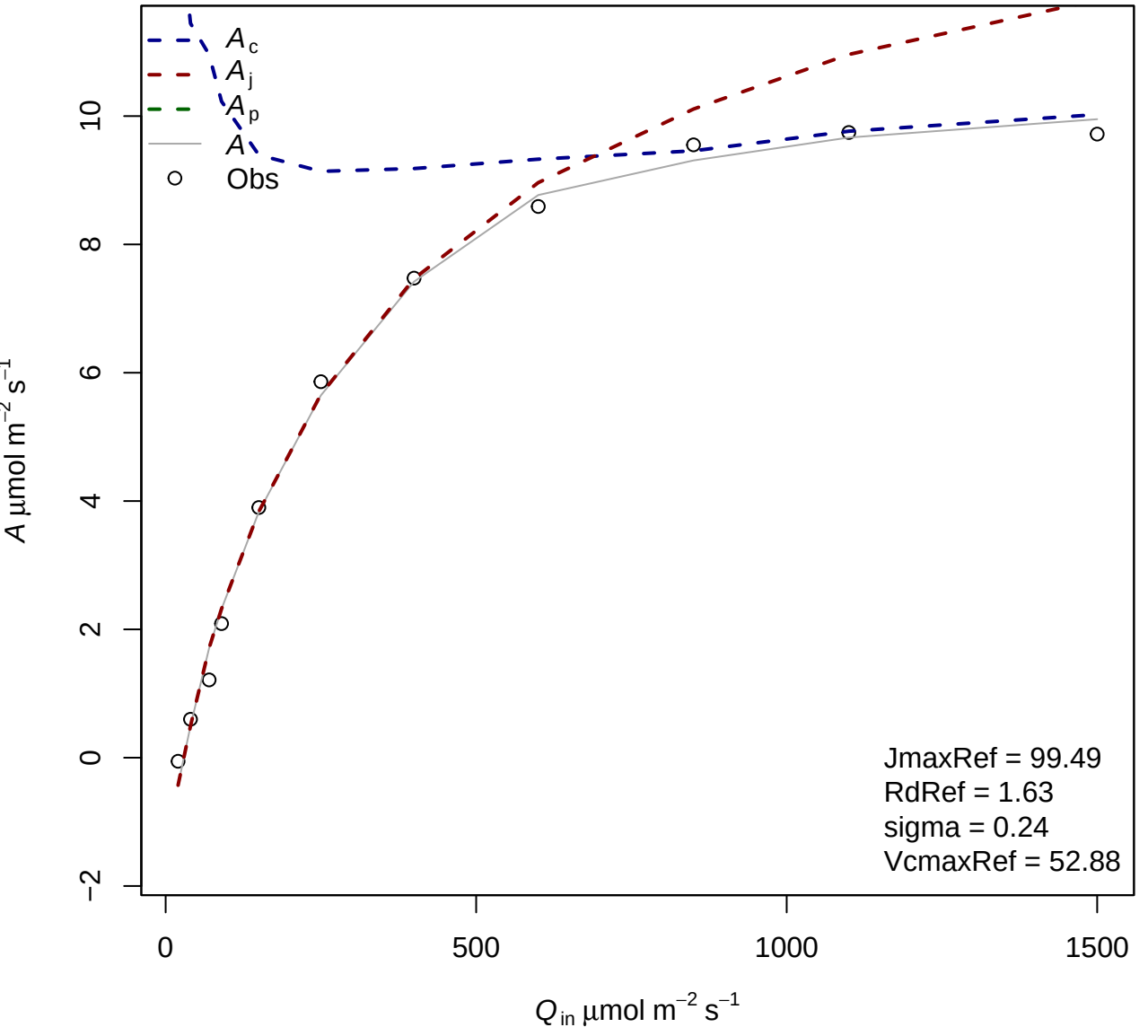

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.51
RdRef = 2.2
sigma = 0.22
VcmaxRef = 53.17
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

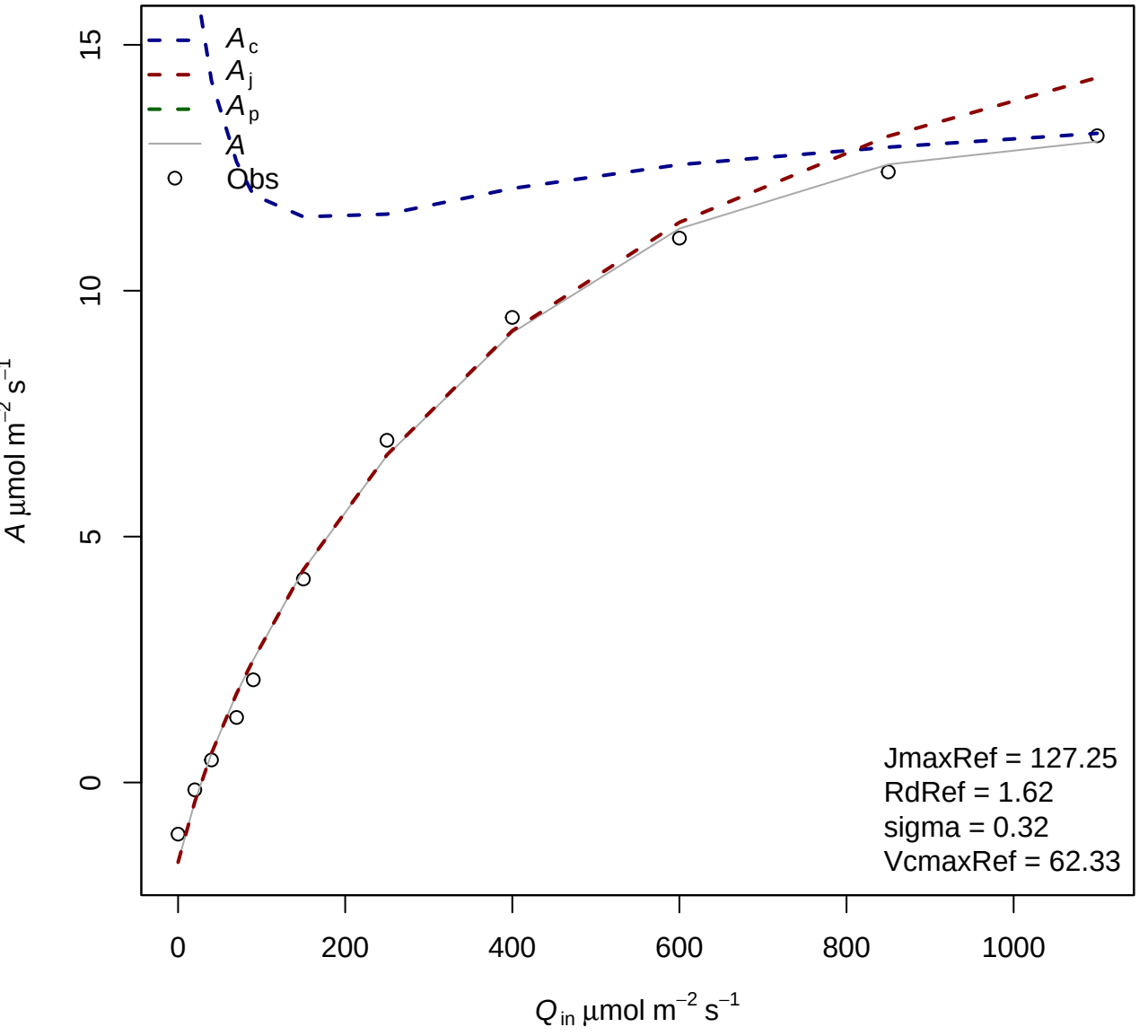

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.09
RdRef = 1.4
sigma = 0.12
VcmaxRef = 63.99
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

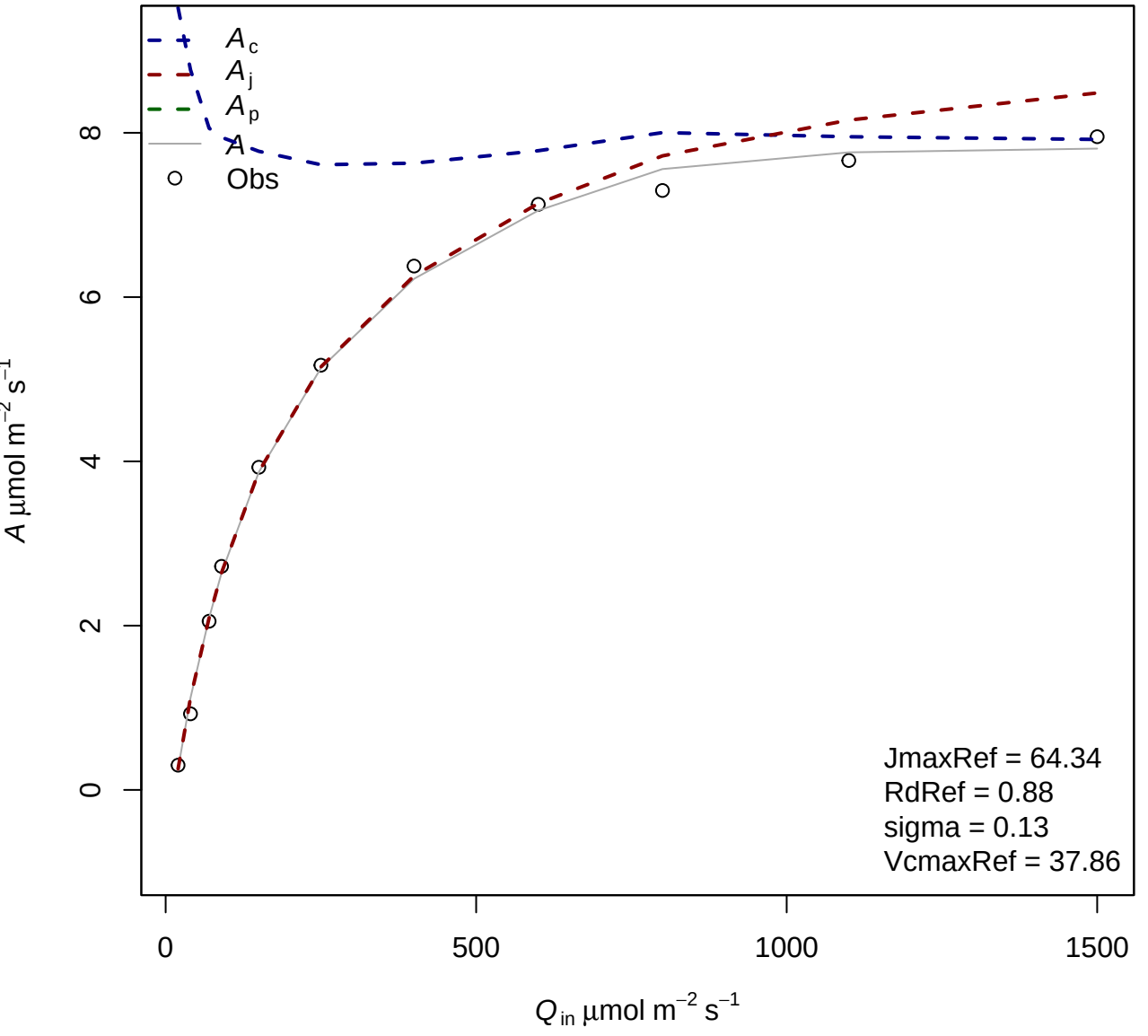

190
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.37
RdRef = 1.6
sigma = 0.26
VcmaxRef = 48.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

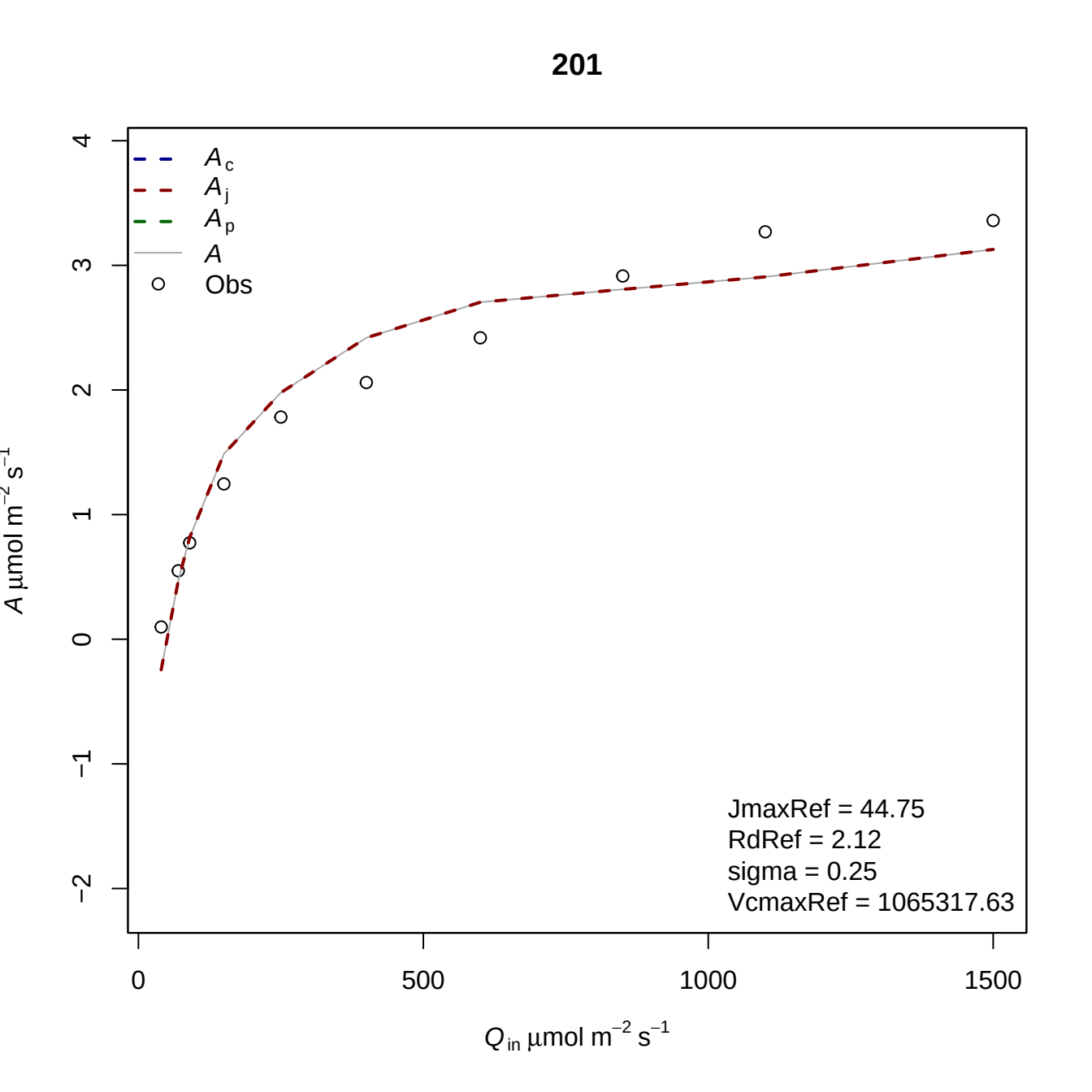

192
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 184.38
RdRef = 1.75
sigma = 0.26
VcmaxRef = 94.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

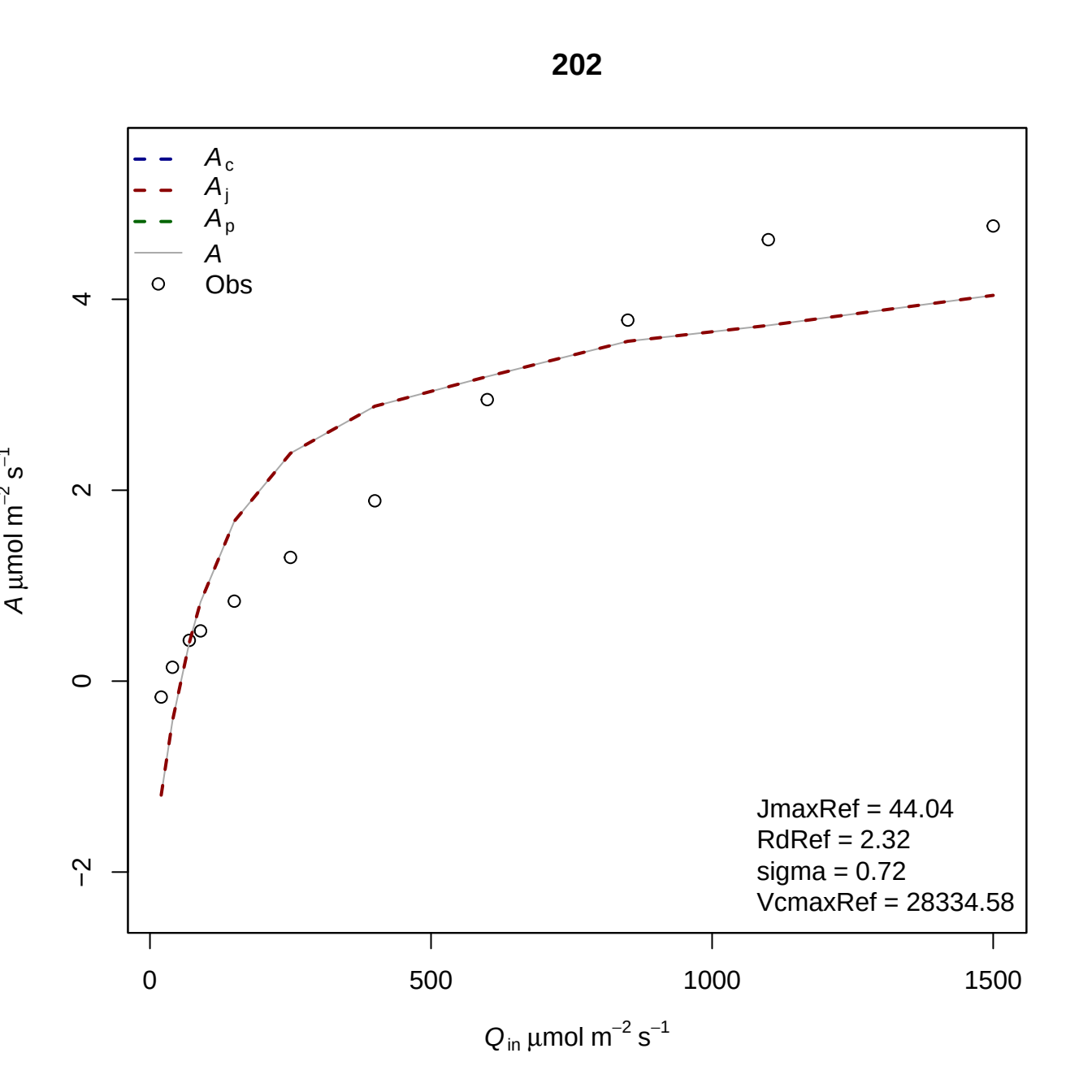

195
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 221.66
RdRef = 2.67
sigma = 0.54
VcmaxRef = 117.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

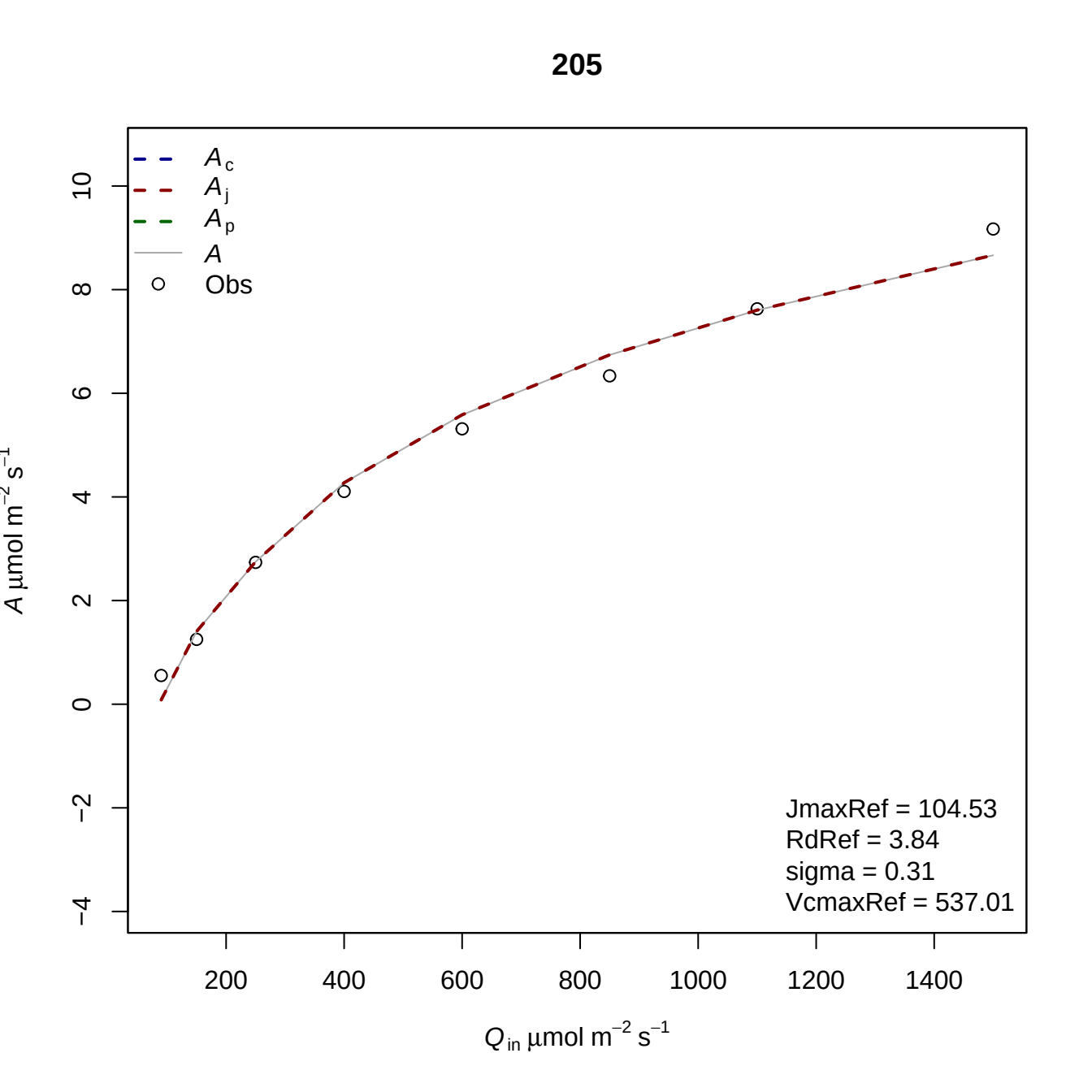

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.49
RdRef = 1.63
sigma = 0.24
VcmaxRef = 52.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

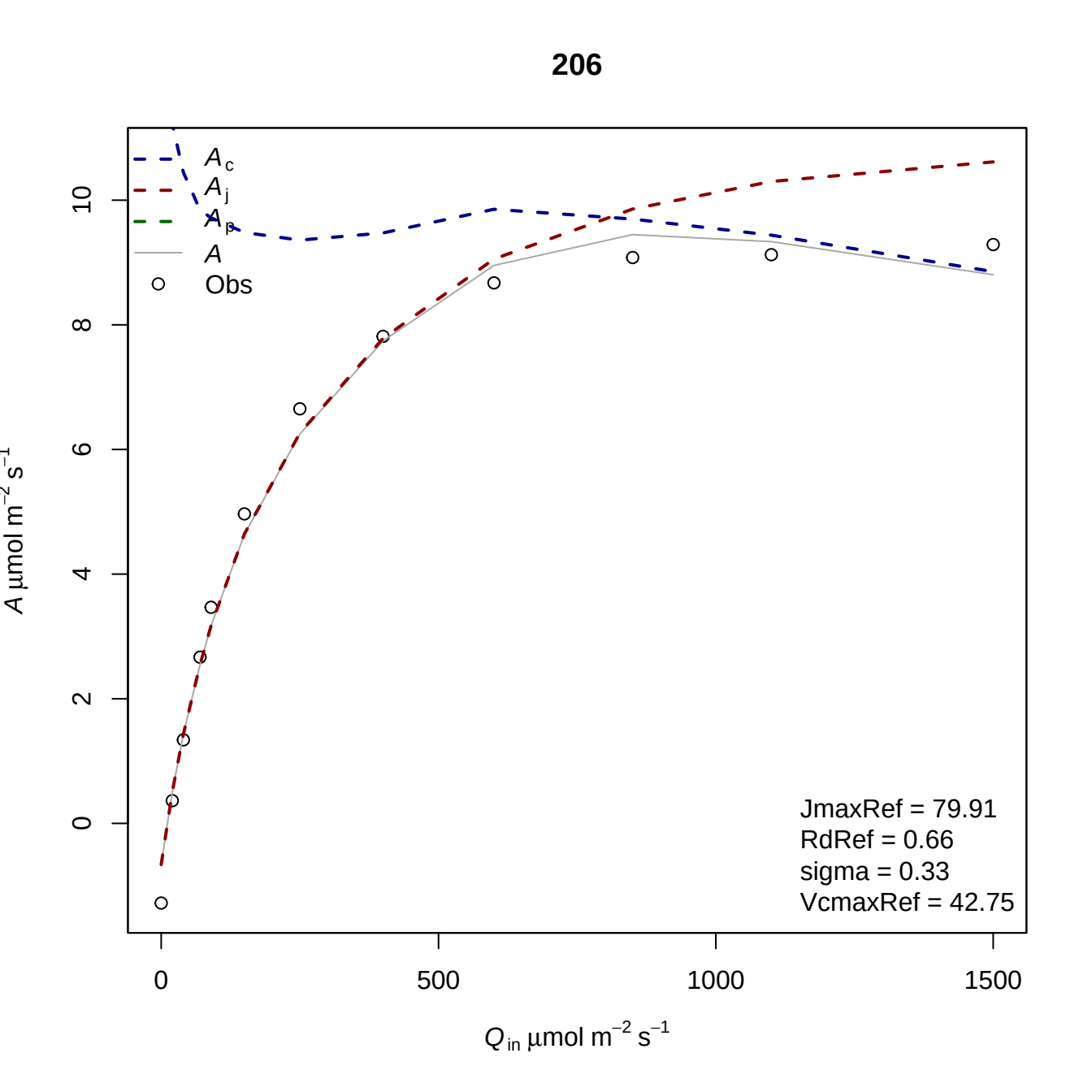

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.25
RdRef = 1.62
sigma = 0.32
VcmaxRef = 62.33
0
5
10
15
0
200
400
600
800
1000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

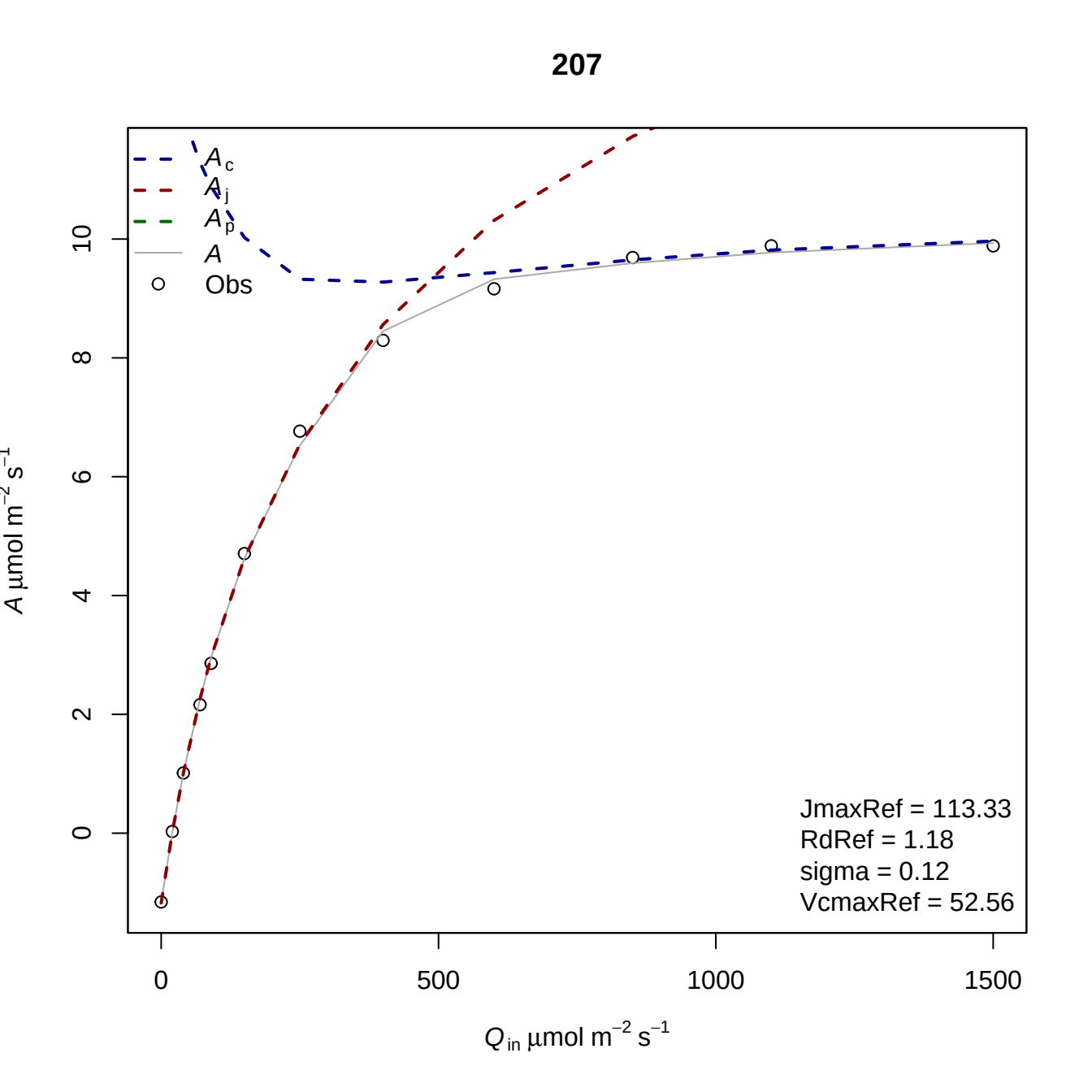

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.34
RdRef = 0.88
sigma = 0.13
VcmaxRef = 37.86
0
2
4
6
8
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

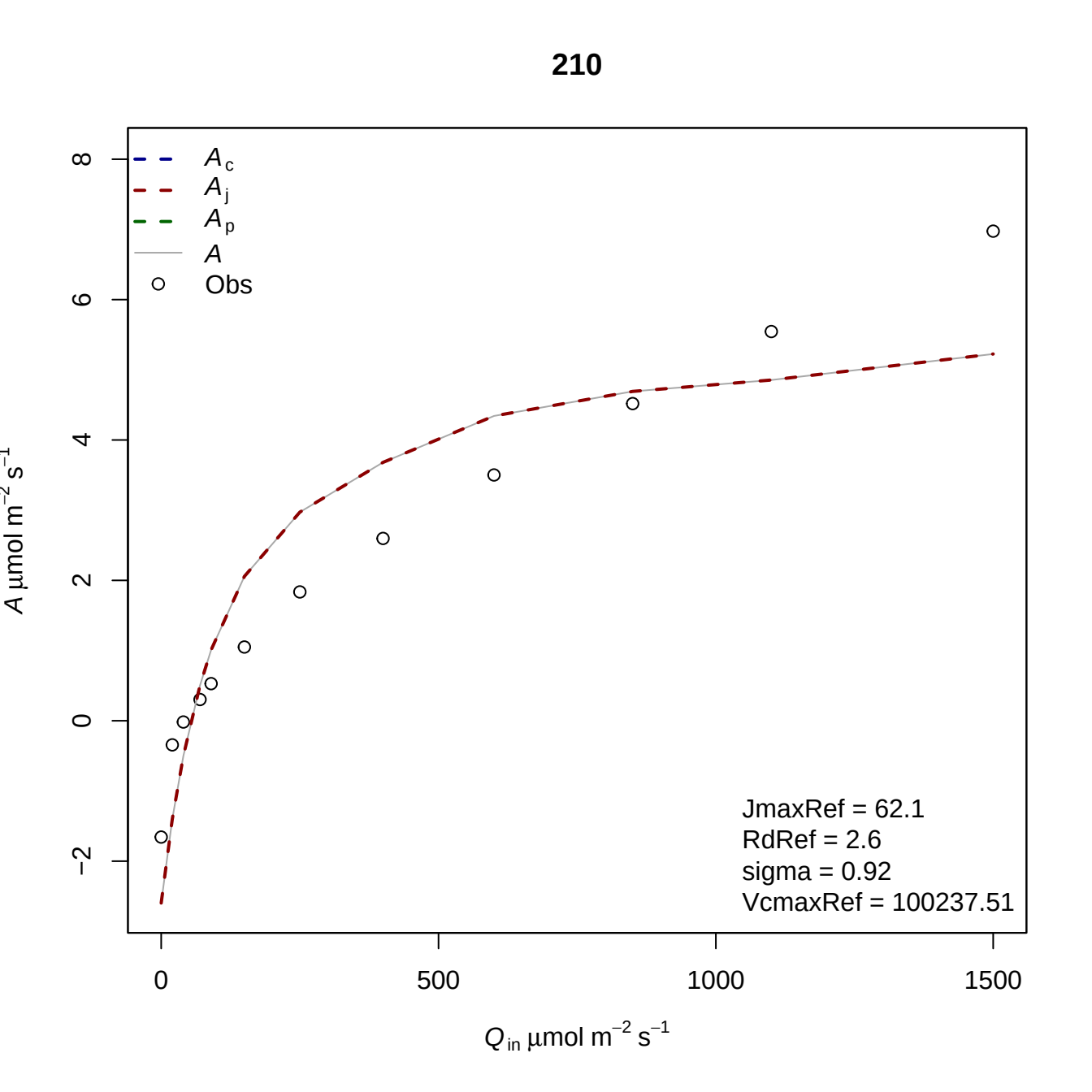

201
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.75
RdRef = 2.12
sigma = 0.25
VcmaxRef = 1065317.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
0
500
1000
1500

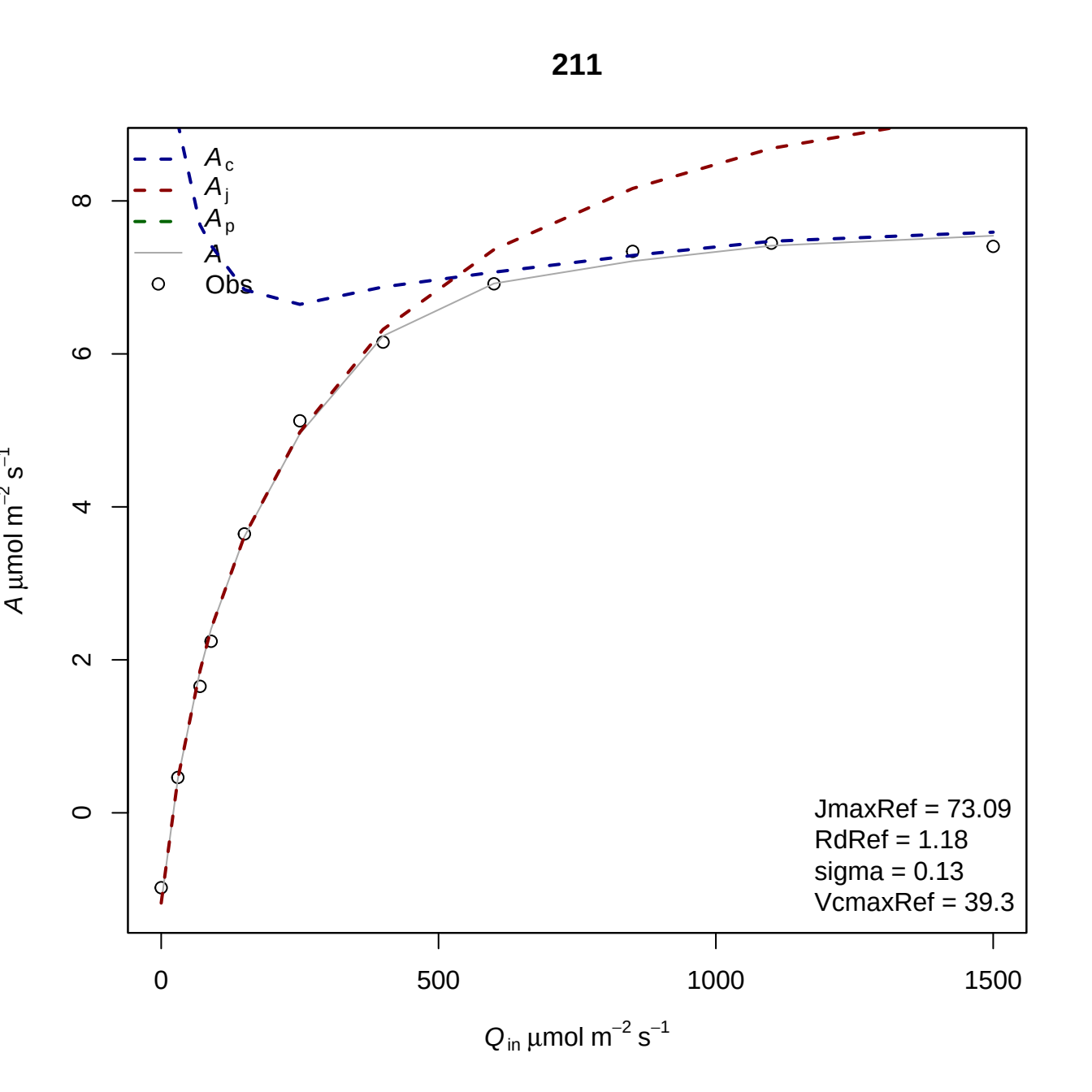

202
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.04
RdRef = 2.32
sigma = 0.72
VcmaxRef = 28334.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500

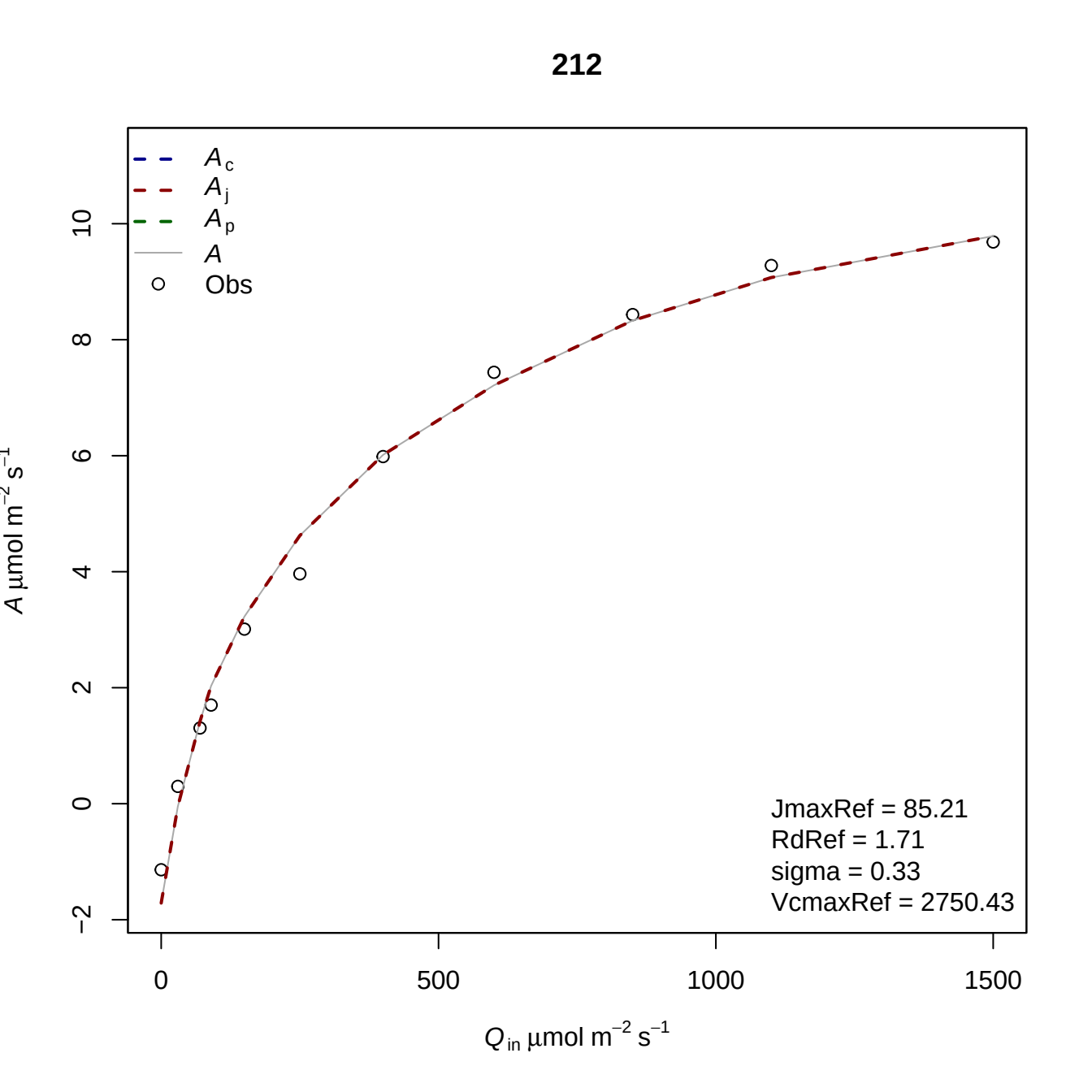

205
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.53
RdRef = 3.84
sigma = 0.31
VcmaxRef = 537.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

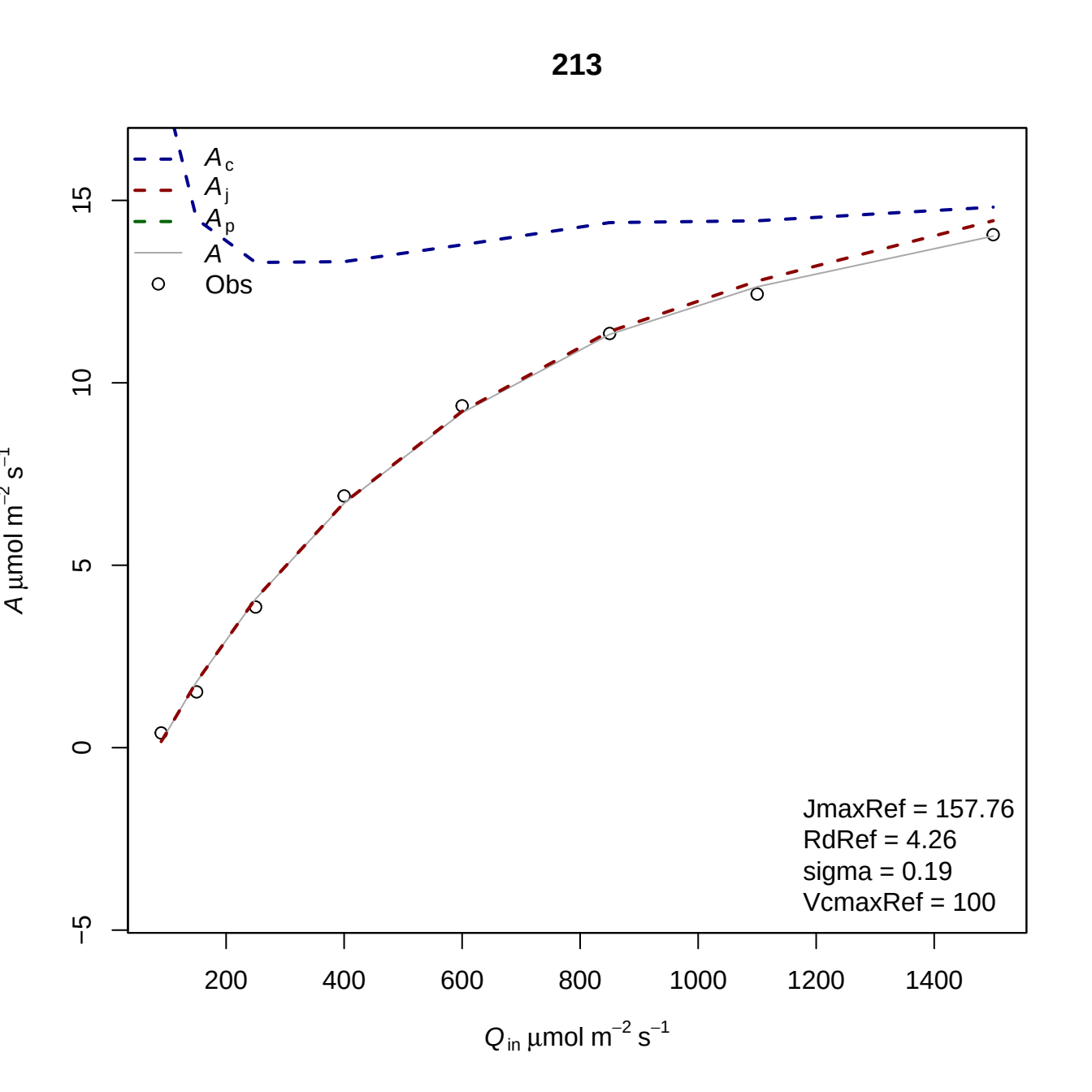

206
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.91
RdRef = 0.66
sigma = 0.33
VcmaxRef = 42.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

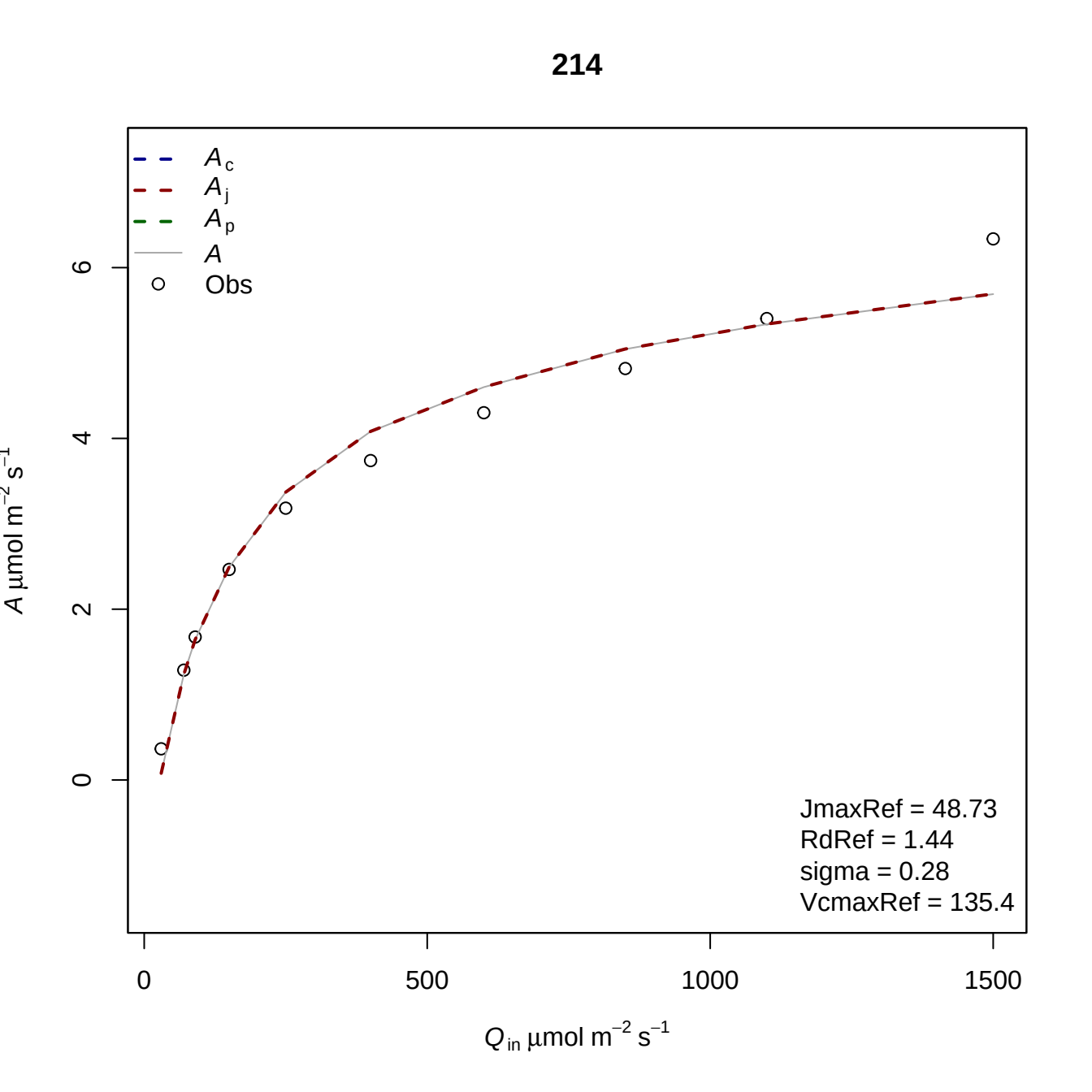

207
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 113.33
RdRef = 1.18
sigma = 0.12
VcmaxRef = 52.56
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

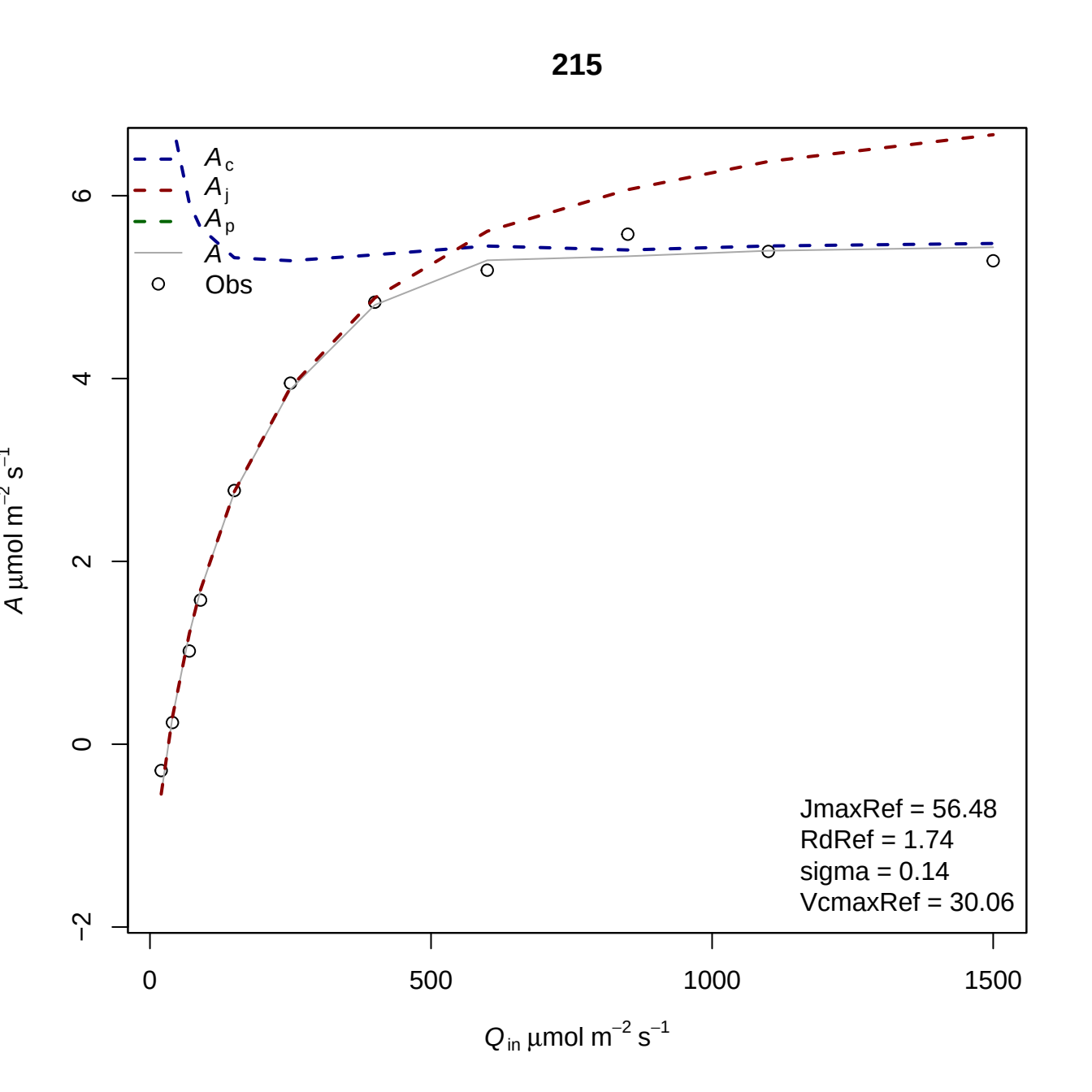

210
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.1
RdRef = 2.6
sigma = 0.92
VcmaxRef = 100237.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

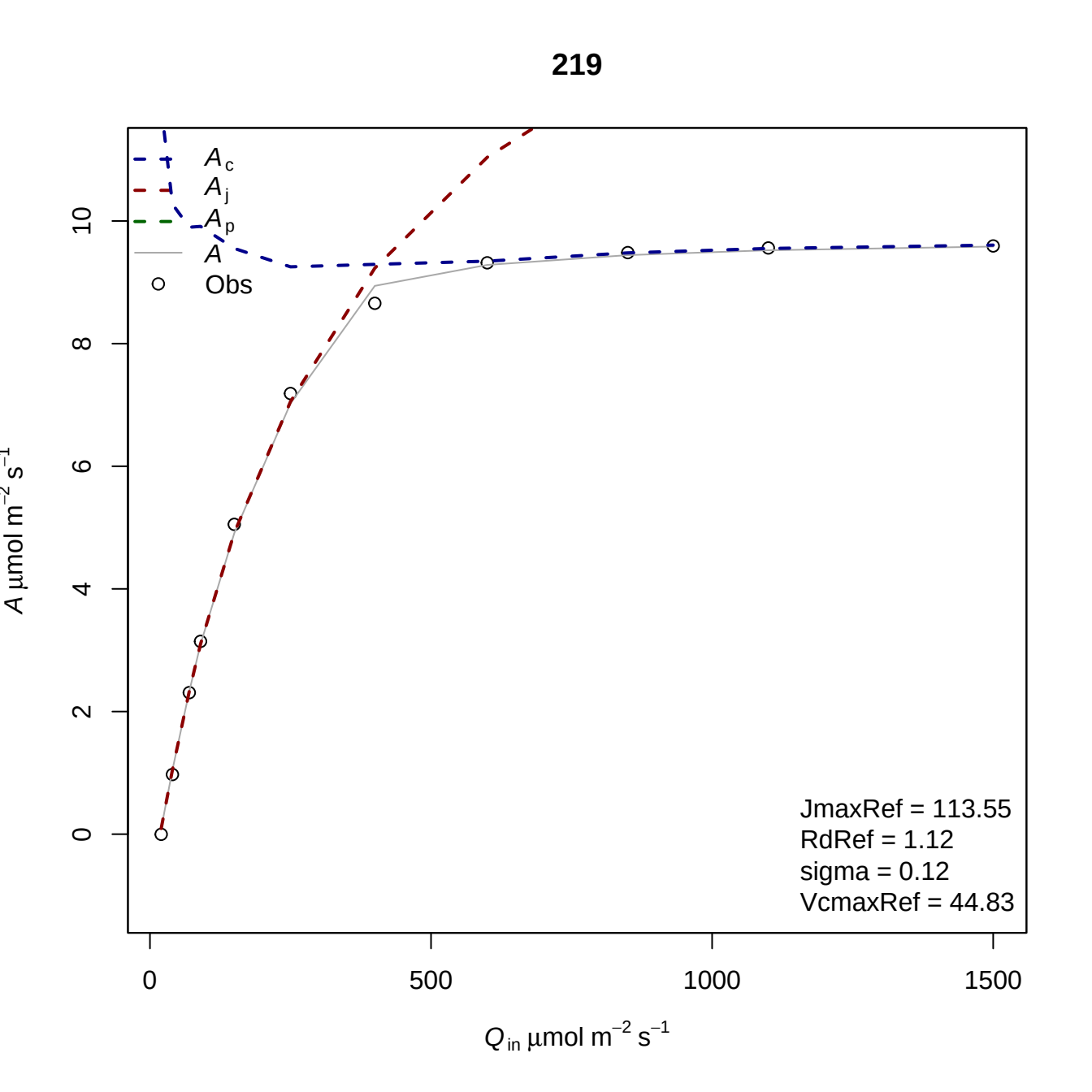

211
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.09
RdRef = 1.18
sigma = 0.13
VcmaxRef = 39.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

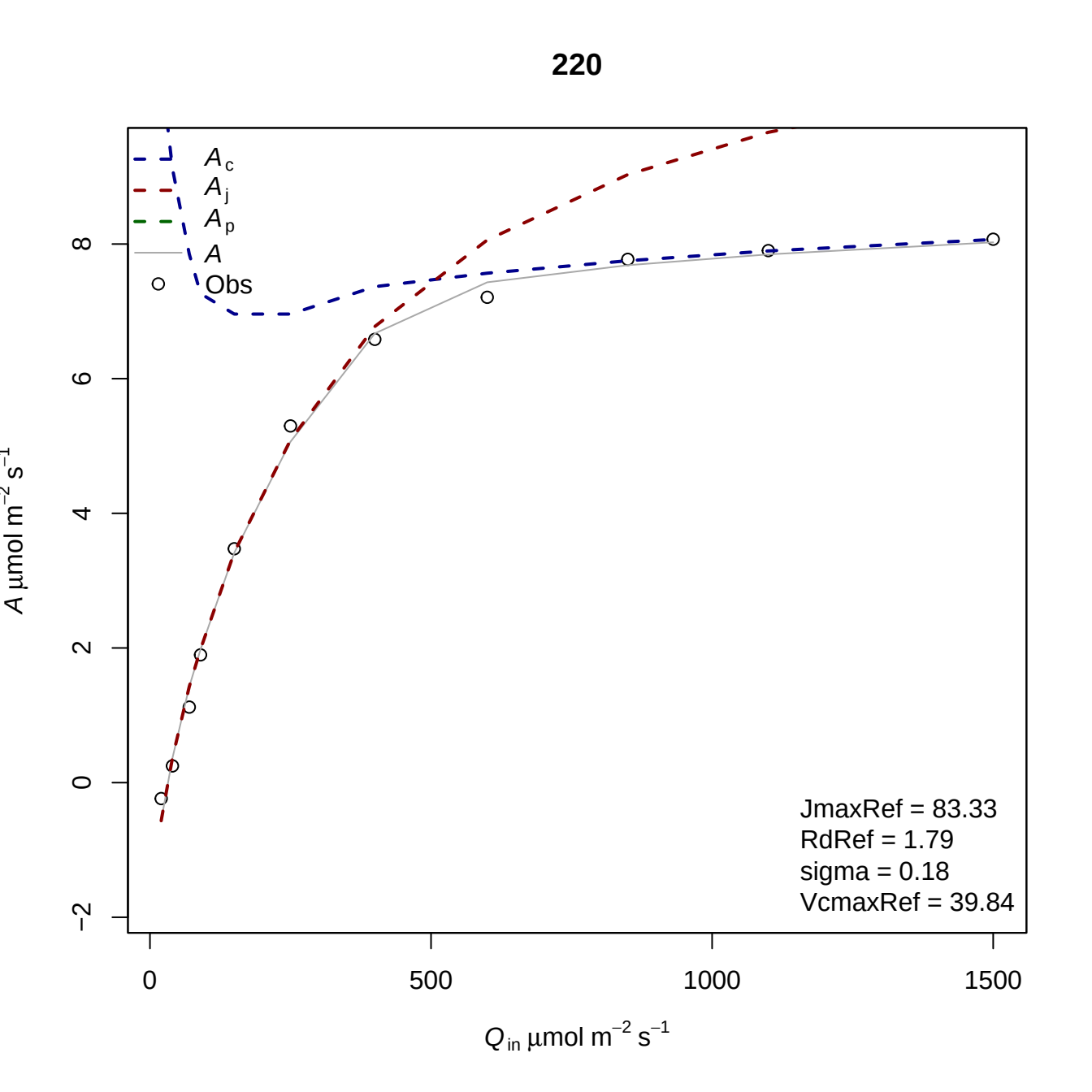

212
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.21
RdRef = 1.71
sigma = 0.33
VcmaxRef = 2750.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

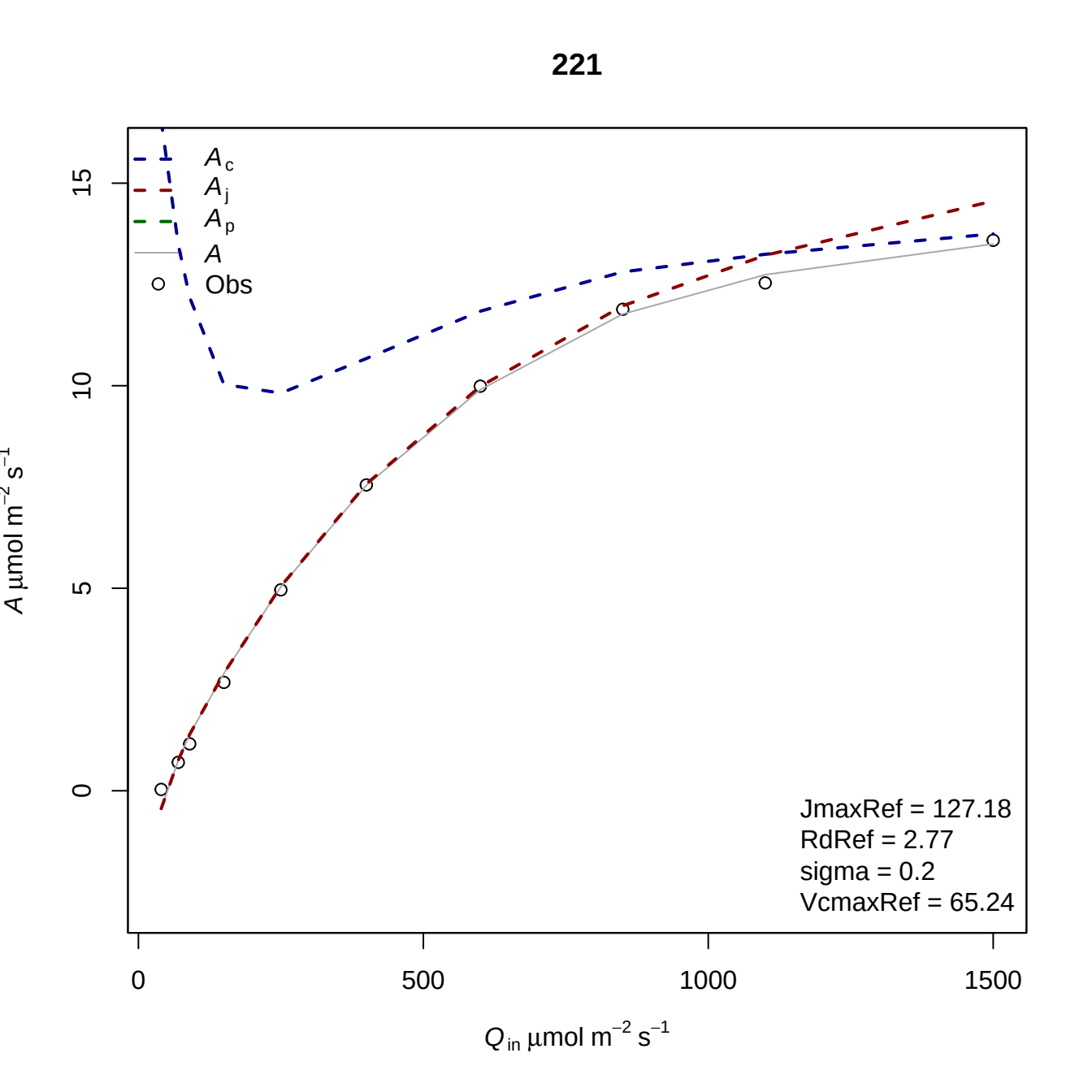

213
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 157.76
RdRef = 4.26
sigma = 0.19
VcmaxRef = 100
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
200
400
600
800
1000
1200
1400

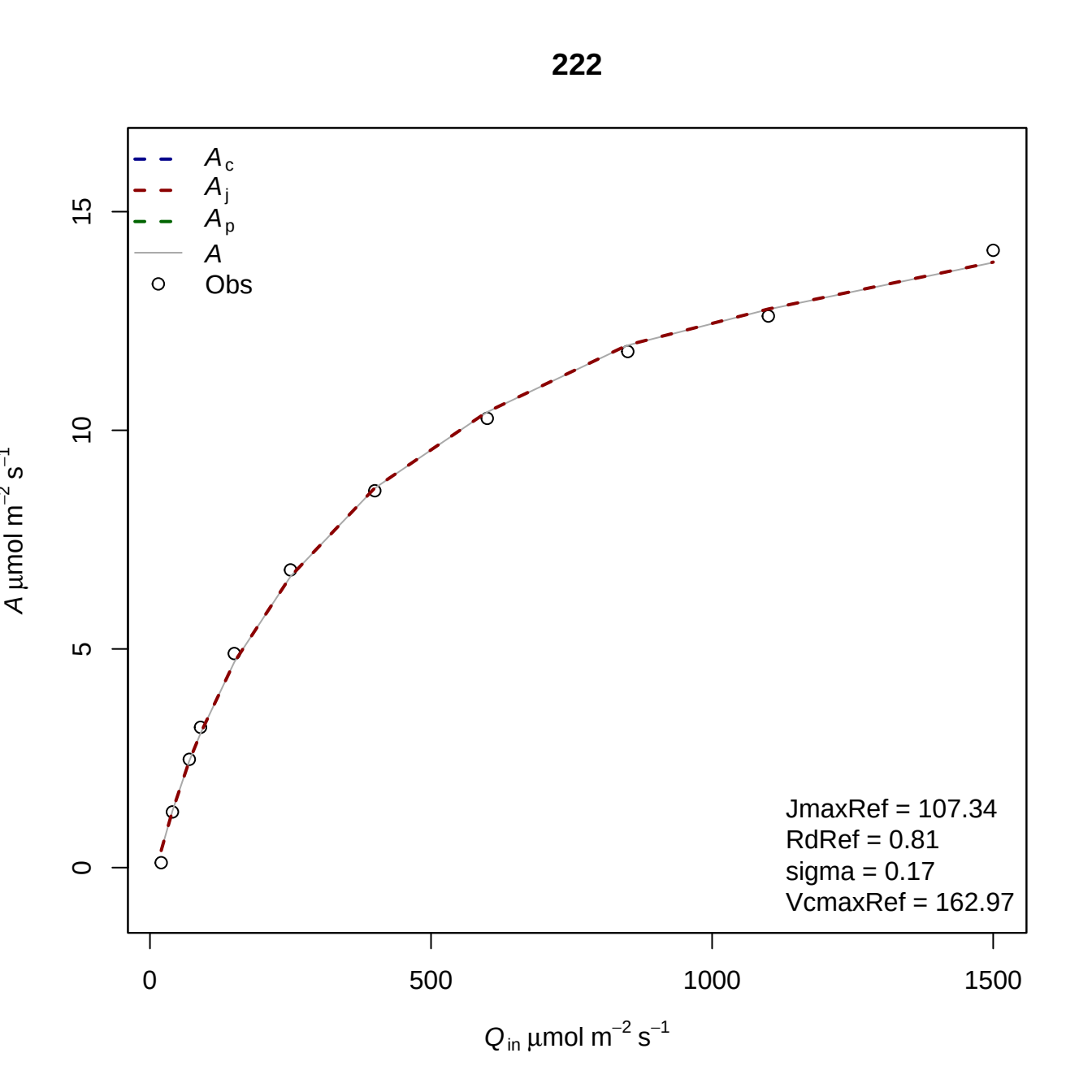

214
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.73
RdRef = 1.44
sigma = 0.28
VcmaxRef = 135.4
$A$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

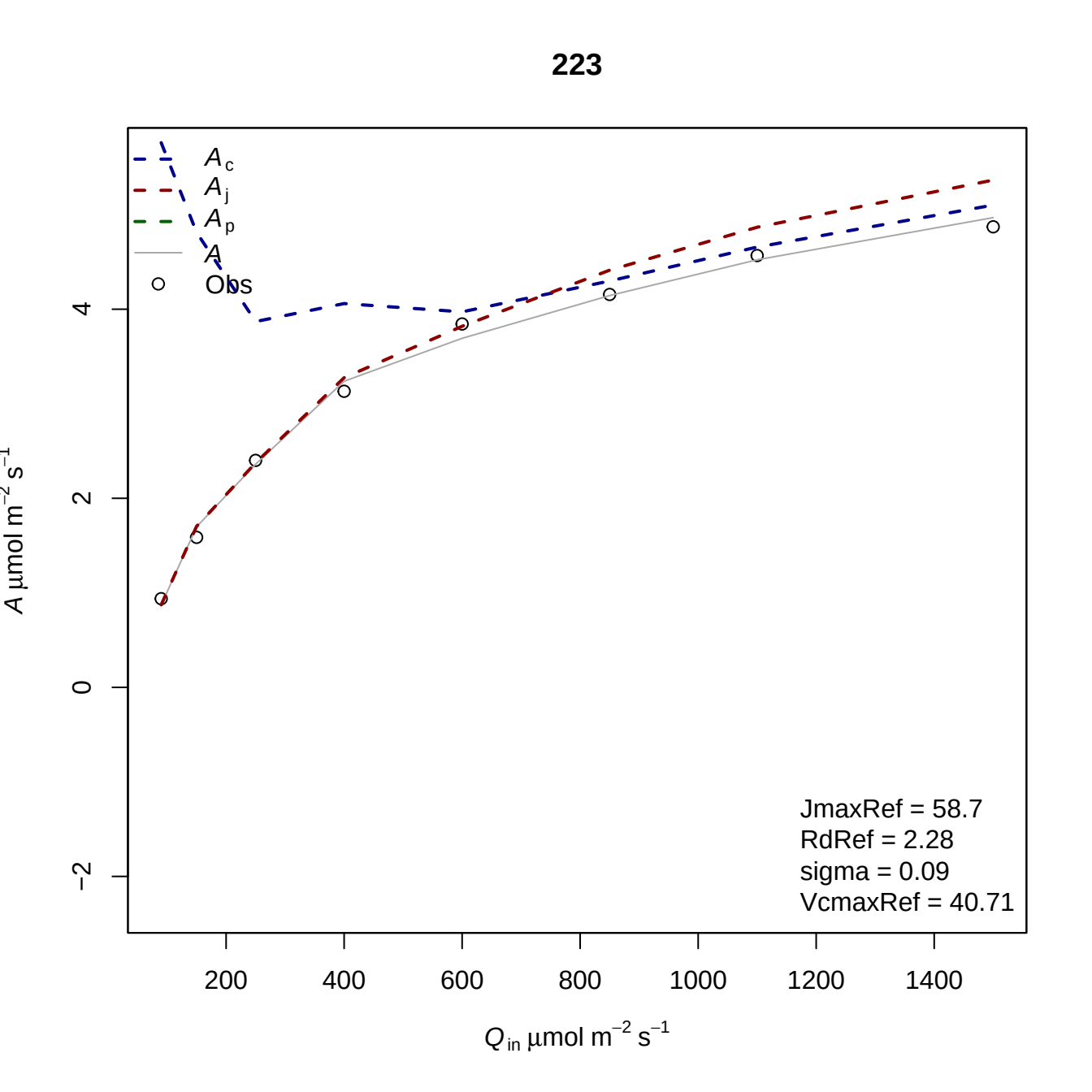

215
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.48
RdRef = 1.74
sigma = 0.14
VcmaxRef = 30.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

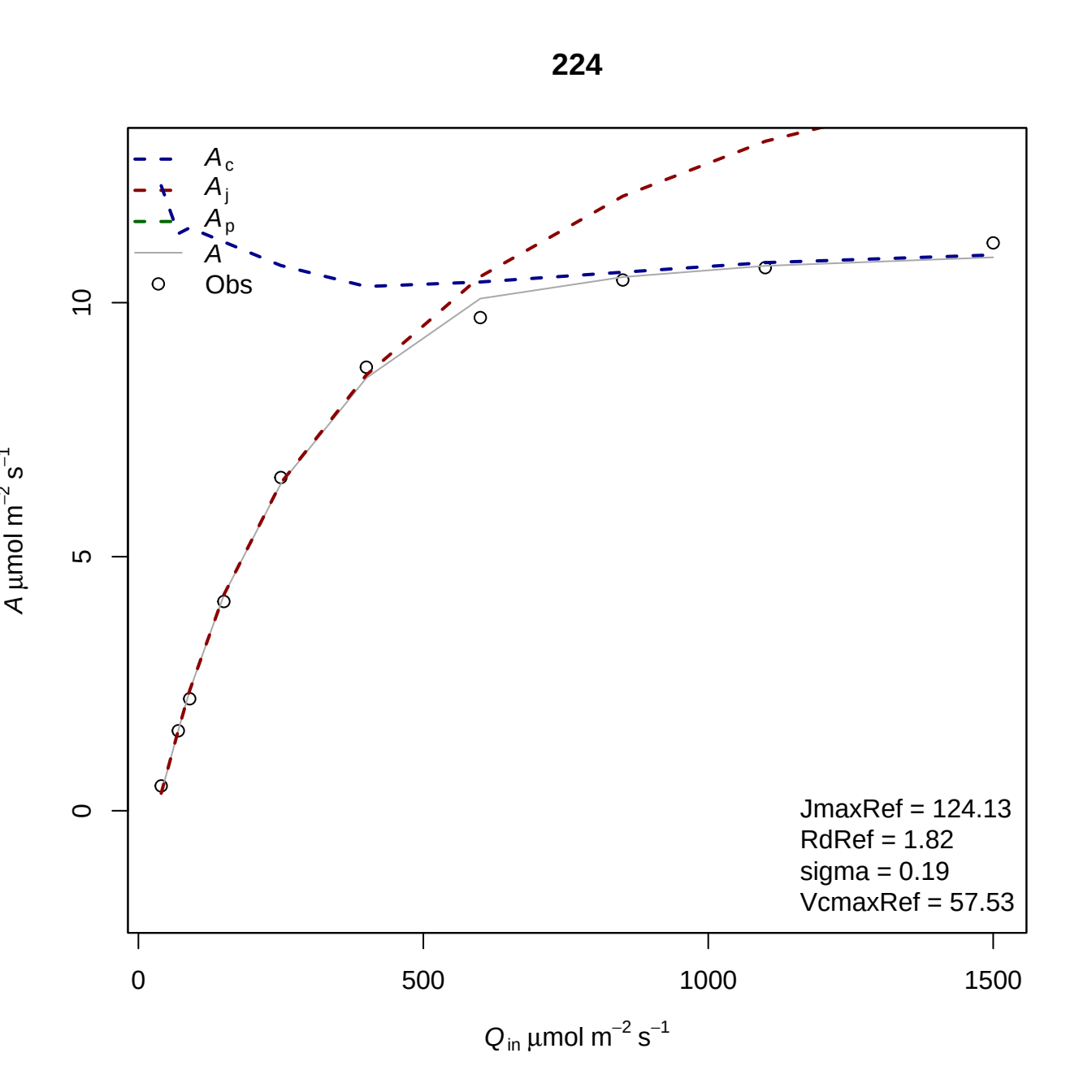

219
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 113.55
RdRef = 1.12
sigma = 0.12
VcmaxRef = 44.83
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

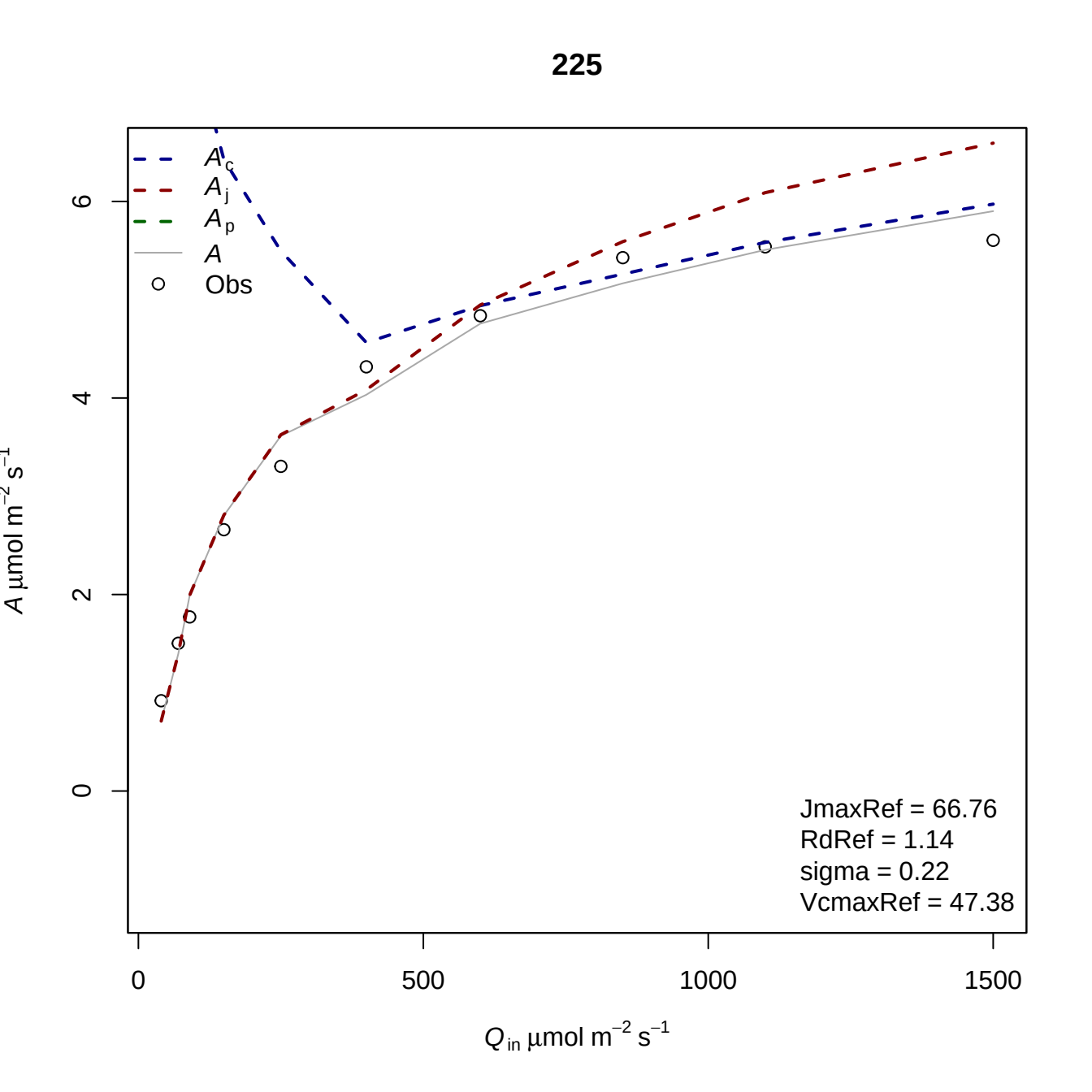

220
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.33
RdRef = 1.79
sigma = 0.18
VcmaxRef = 39.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

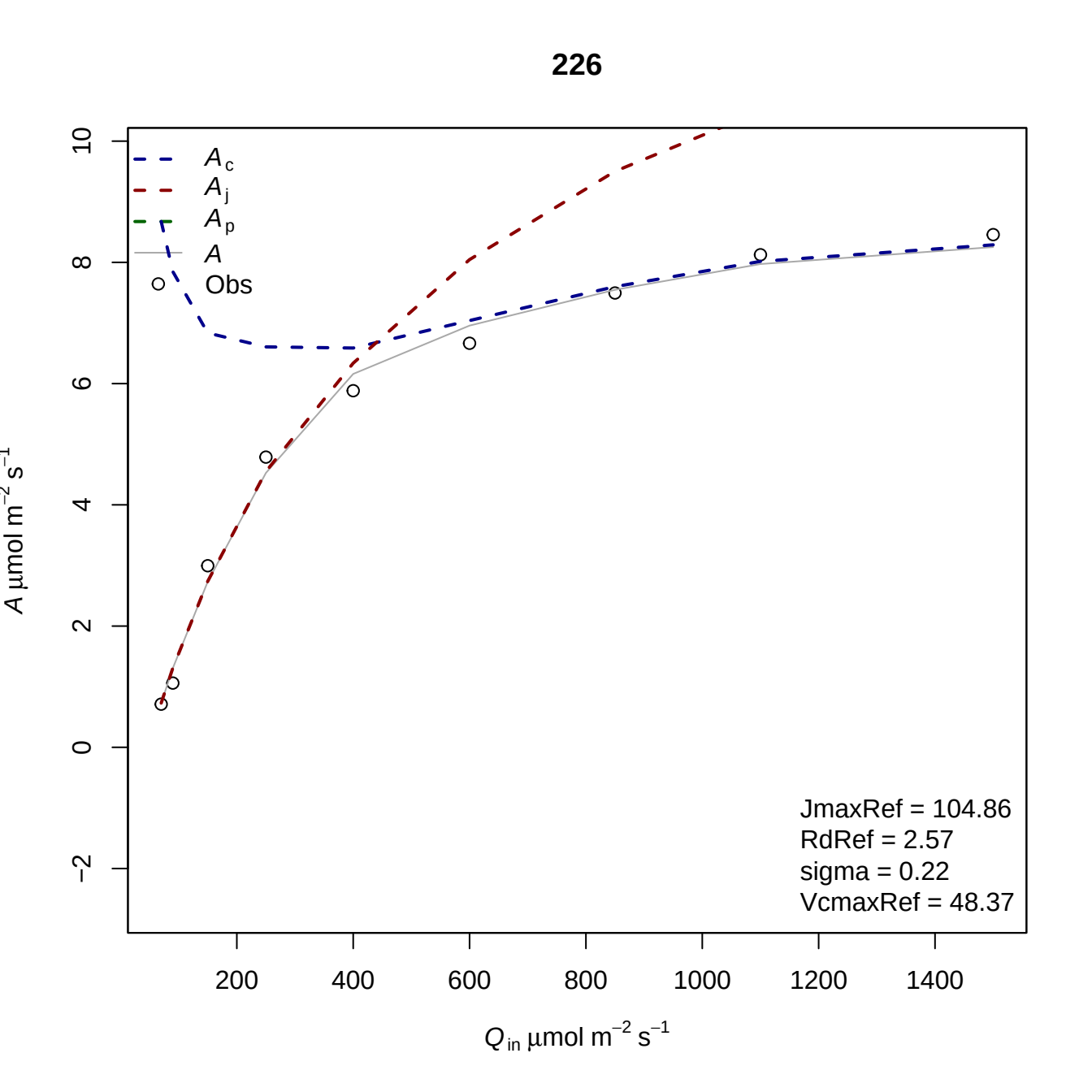

221
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.18
RdRef = 2.77
sigma = 0.2
VcmaxRef = 65.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

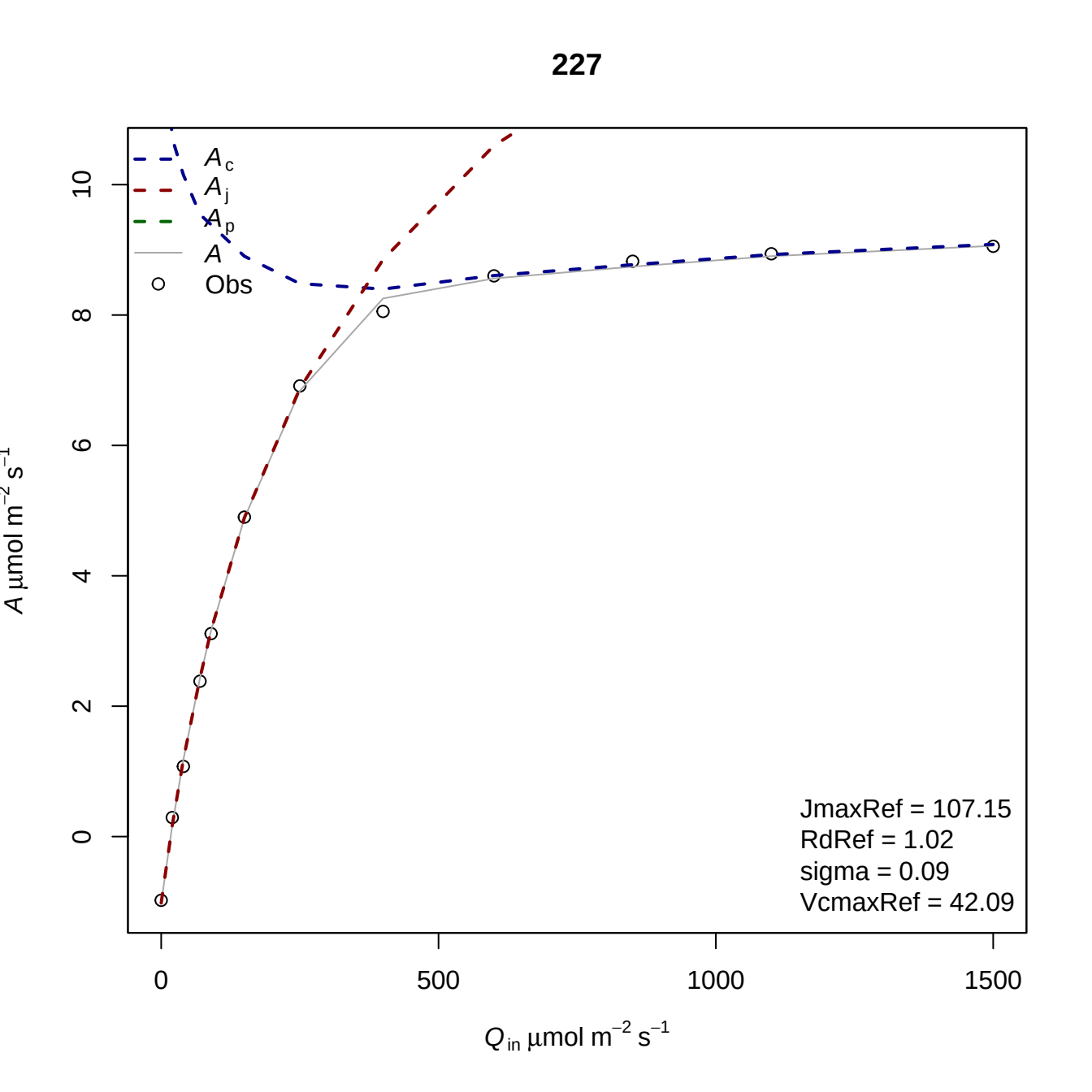

222
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.34
RdRef = 0.81
sigma = 0.17
VcmaxRef = 162.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

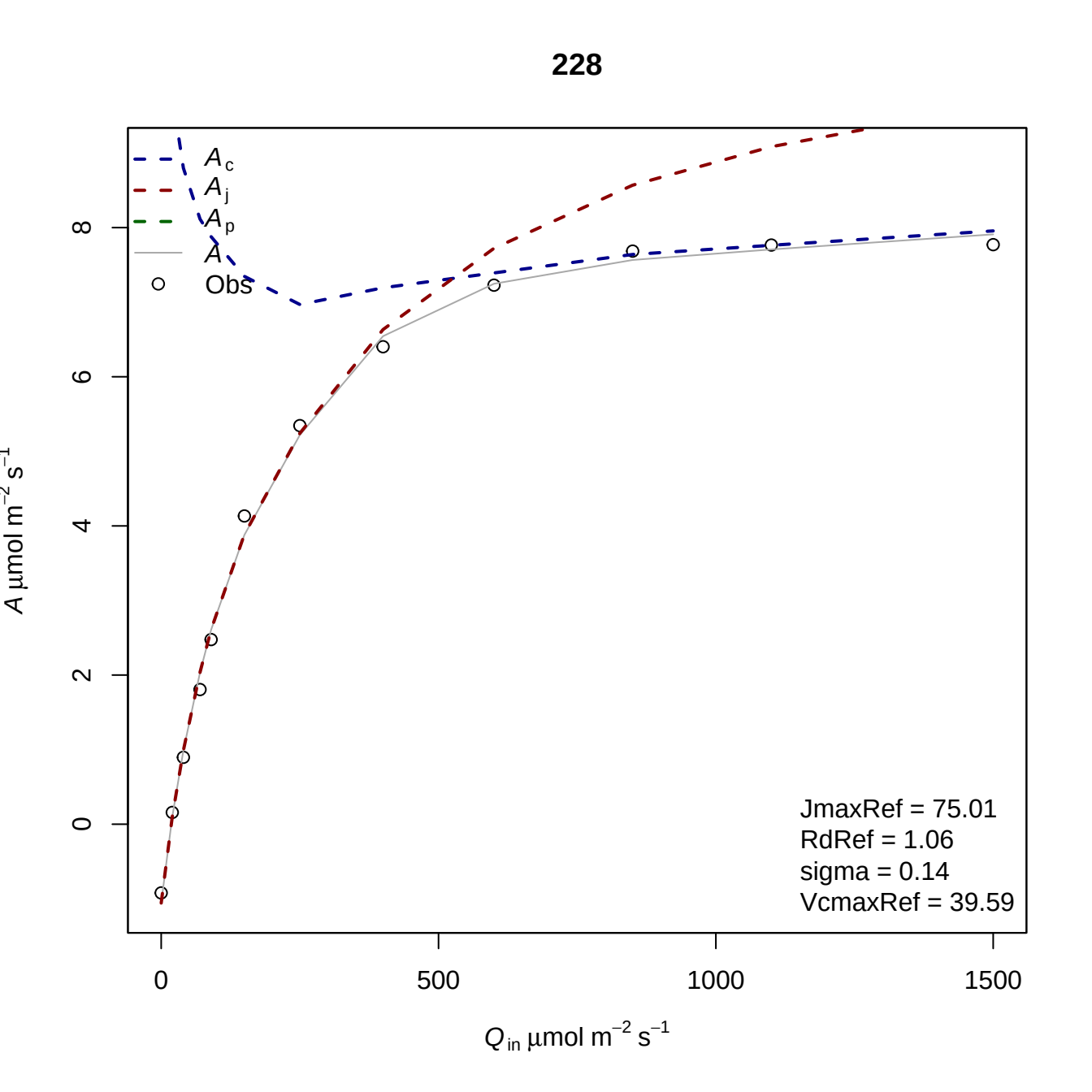

223
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.7
RdRef = 2.28
sigma = 0.09
VcmaxRef = 40.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
200
400
600
800
1000
1200
1400

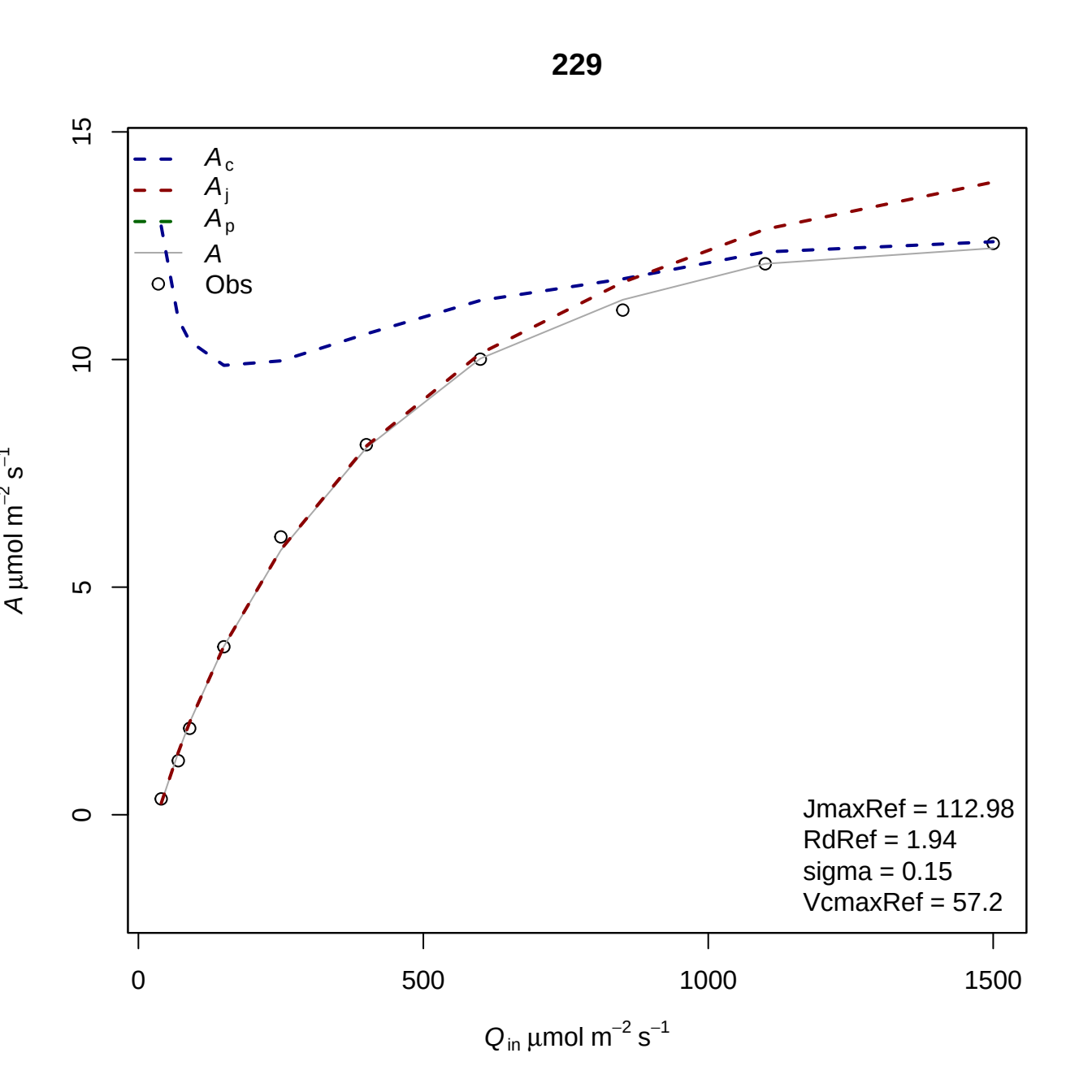

224
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.13
RdRef = 1.82
sigma = 0.19
VcmaxRef = 57.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

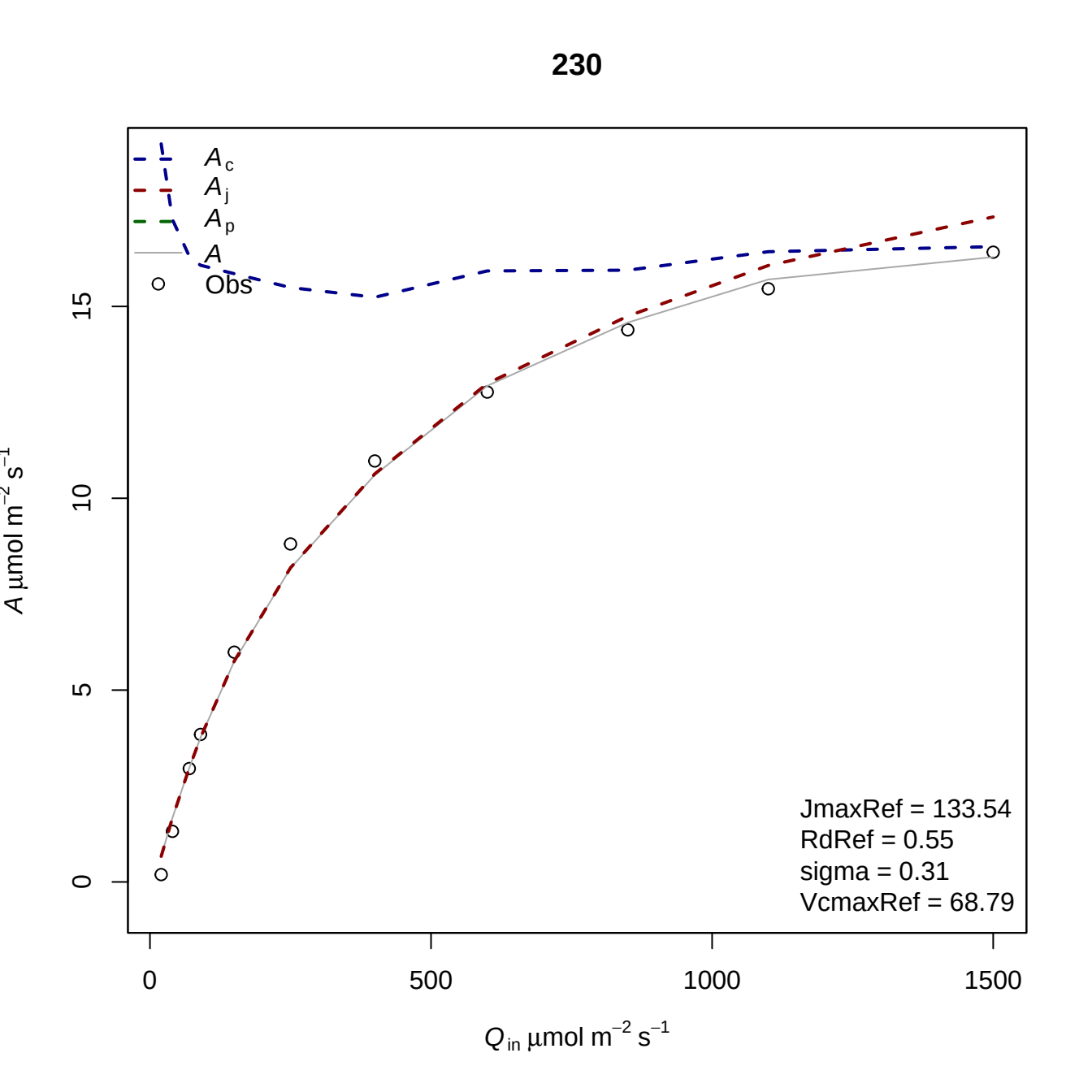

225
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.76
RdRef = 1.14
sigma = 0.22
VcmaxRef = 47.38
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

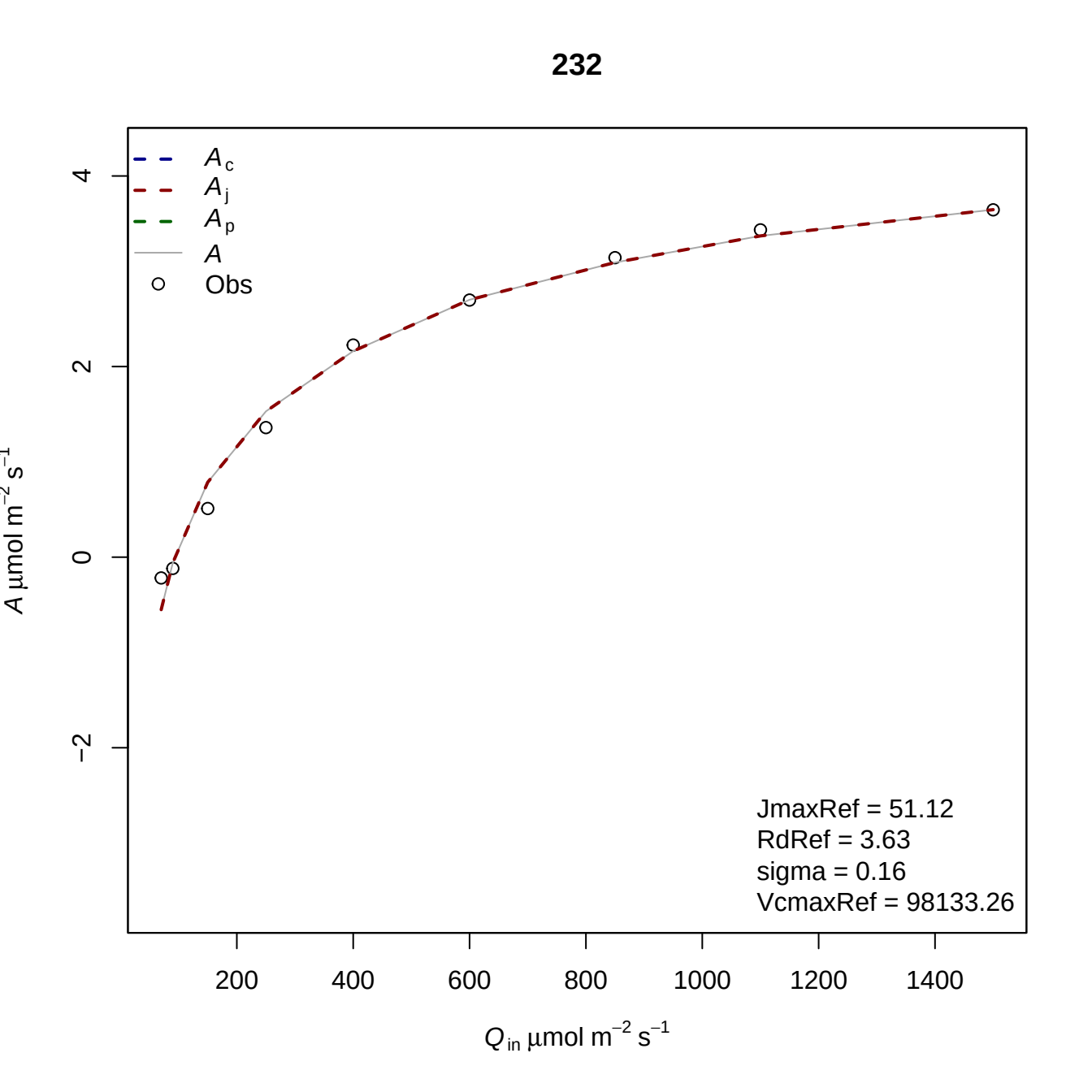

226
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.86
RdRef = 2.57
sigma = 0.22
VcmaxRef = 48.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

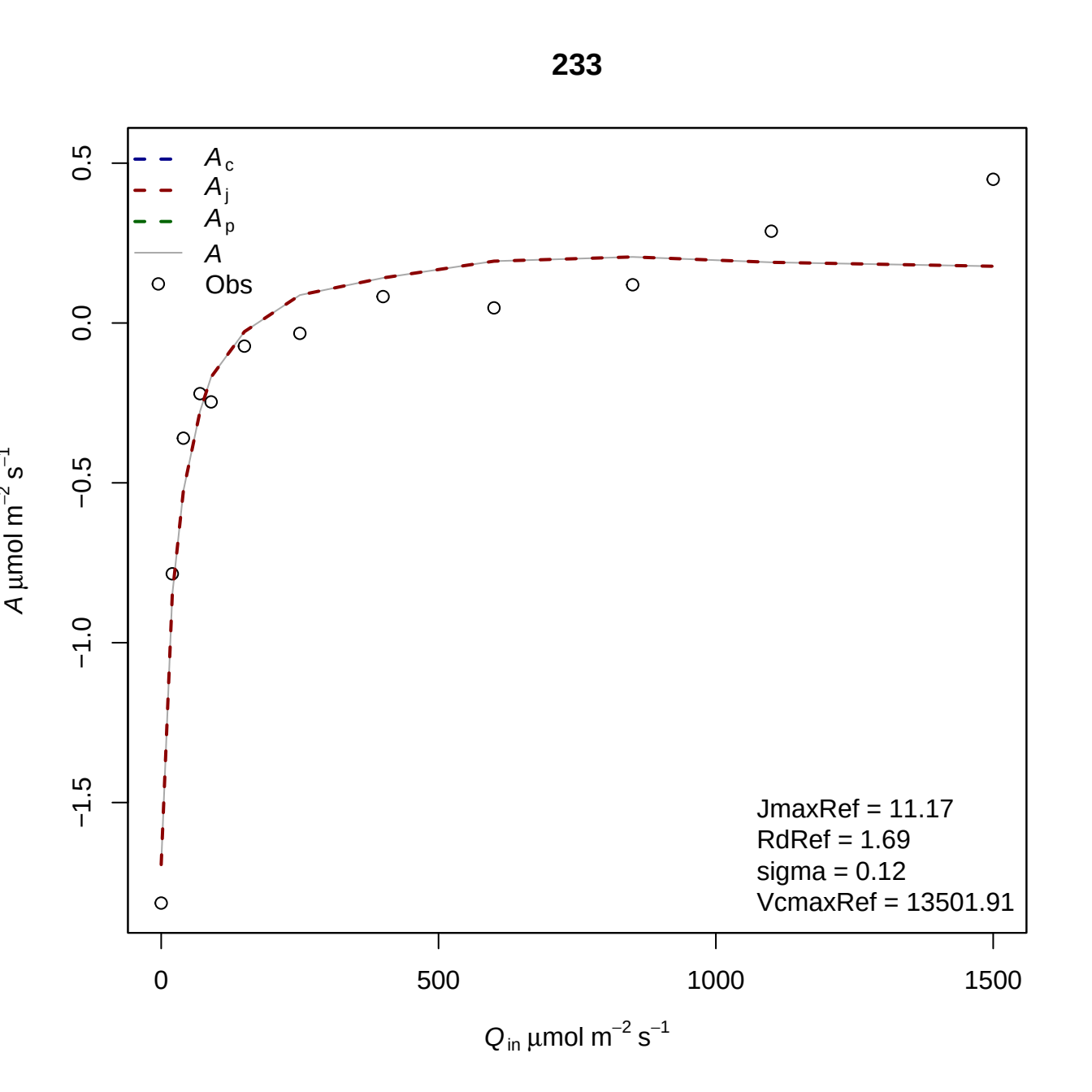

227
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.15
RdRef = 1.02
sigma = 0.09
VcmaxRef = 42.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

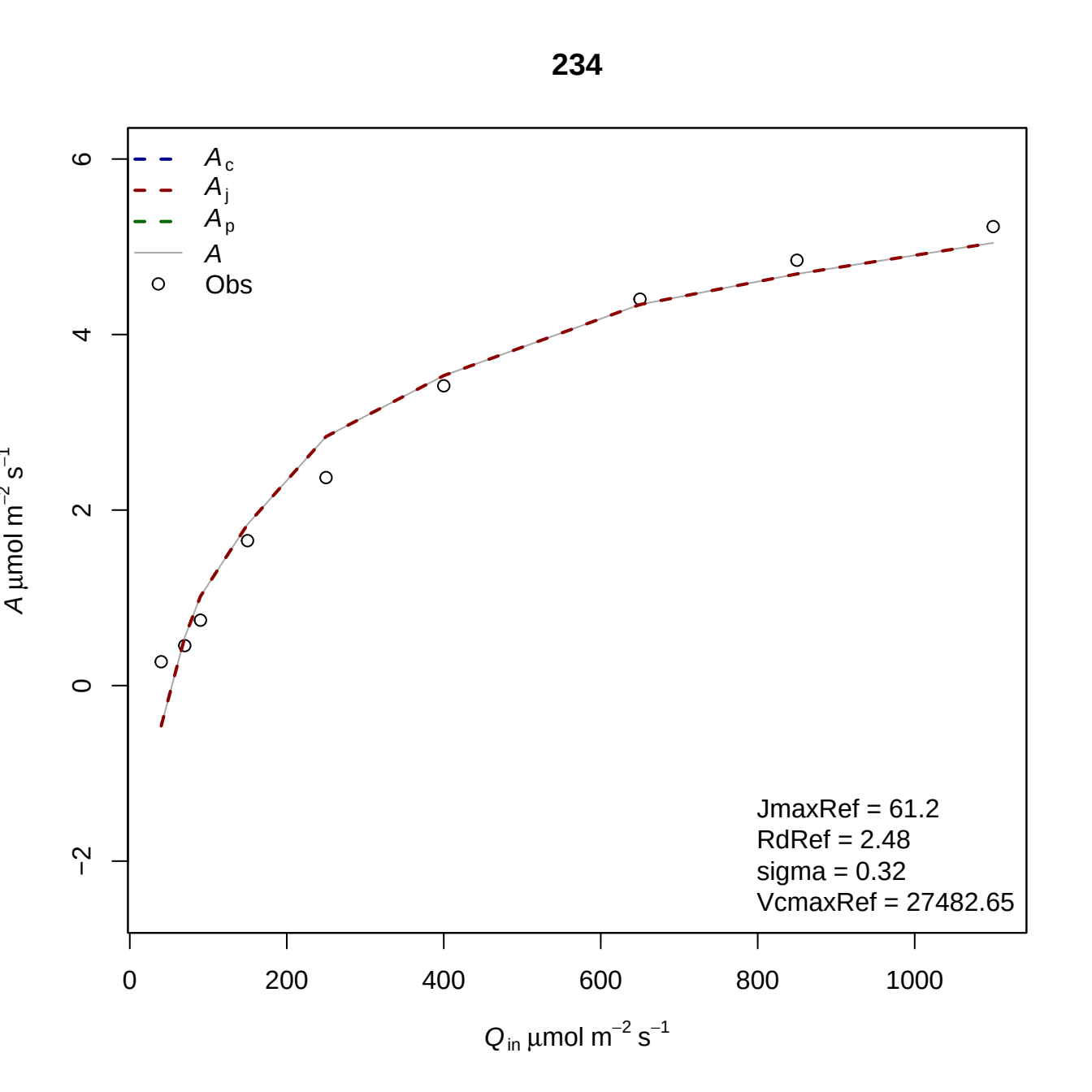

228
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.01
RdRef = 1.06
sigma = 0.14
VcmaxRef = 39.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

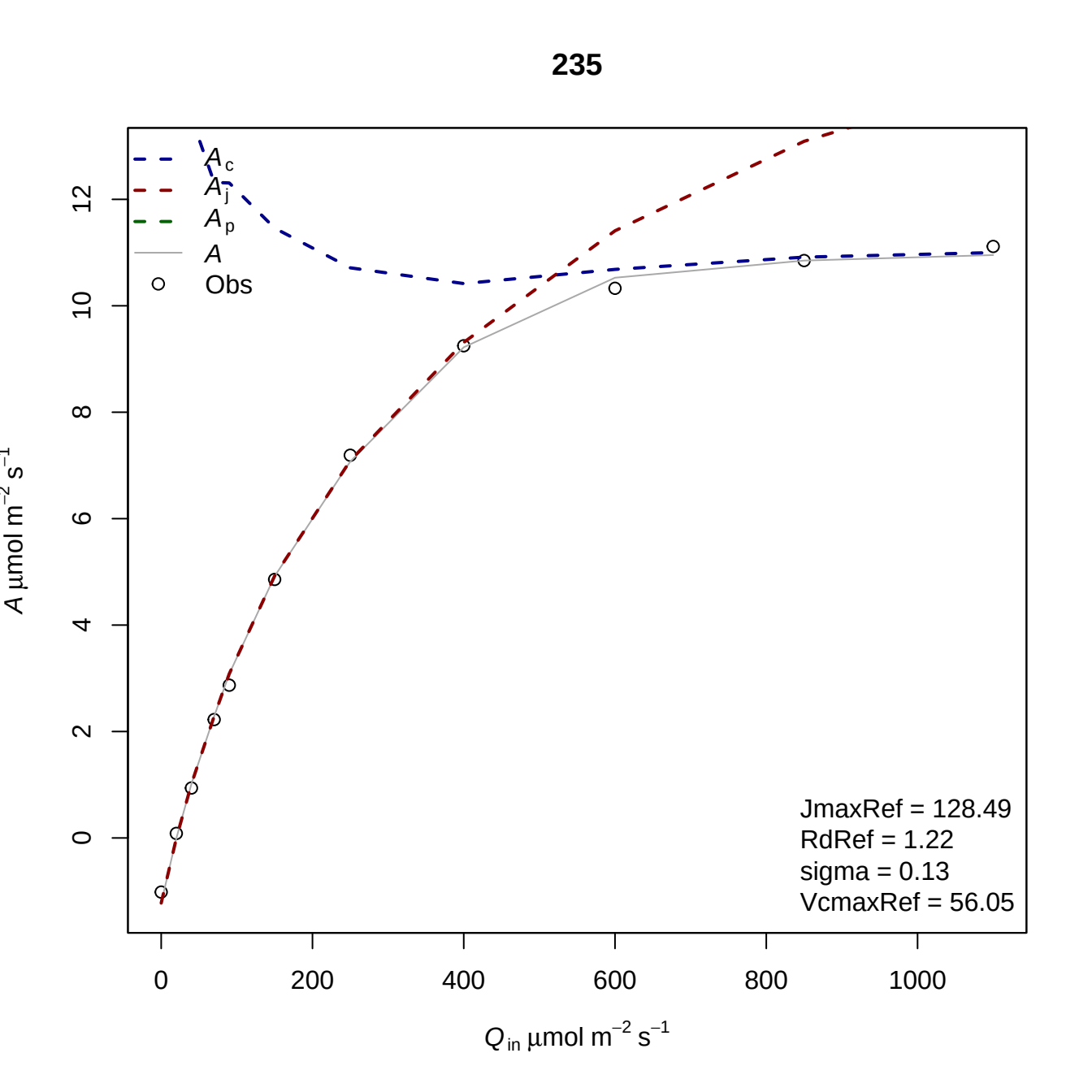

229
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.98
RdRef = 1.94
sigma = 0.15
VcmaxRef = 57.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

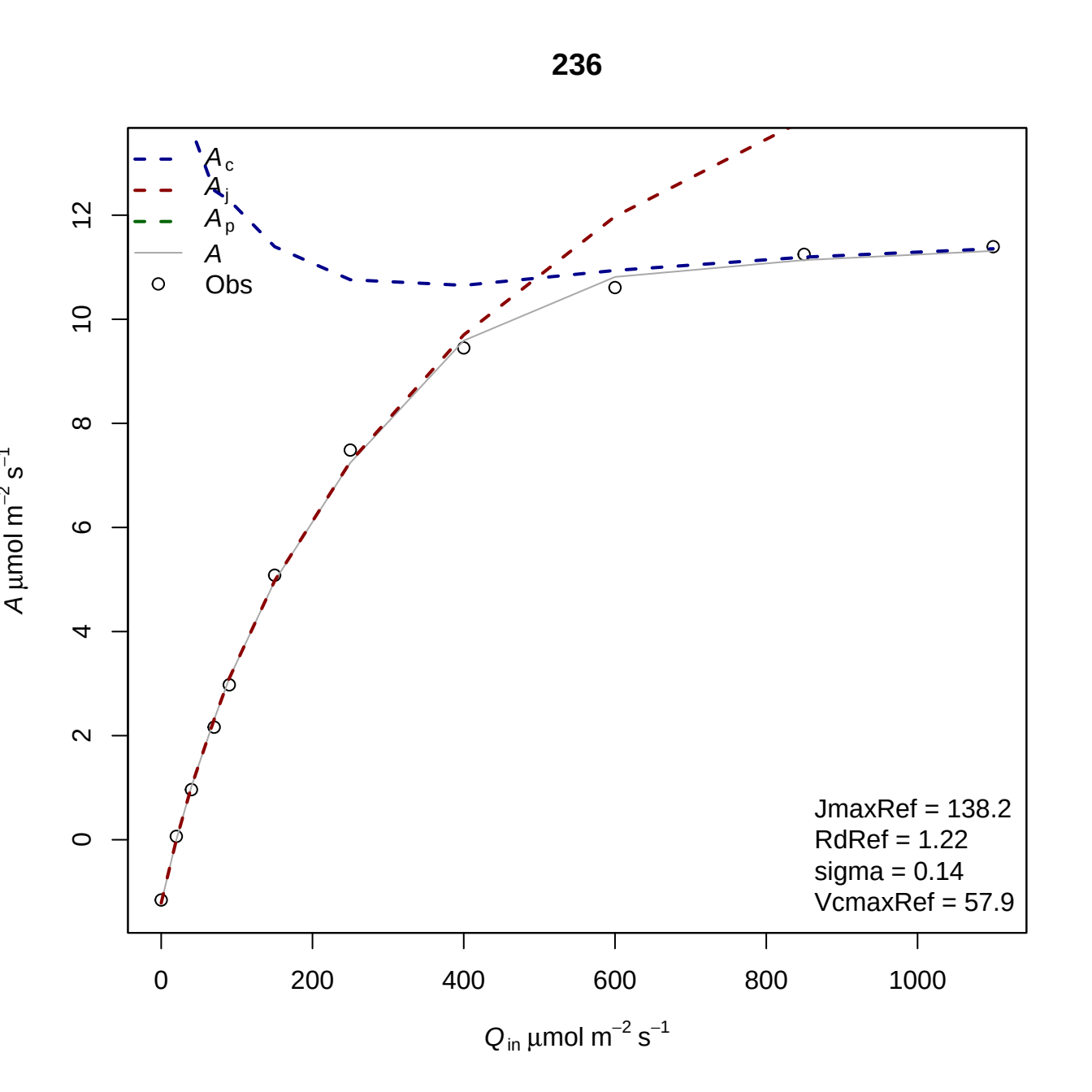

230
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.54
RdRef = 0.55
sigma = 0.31
VcmaxRef = 68.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

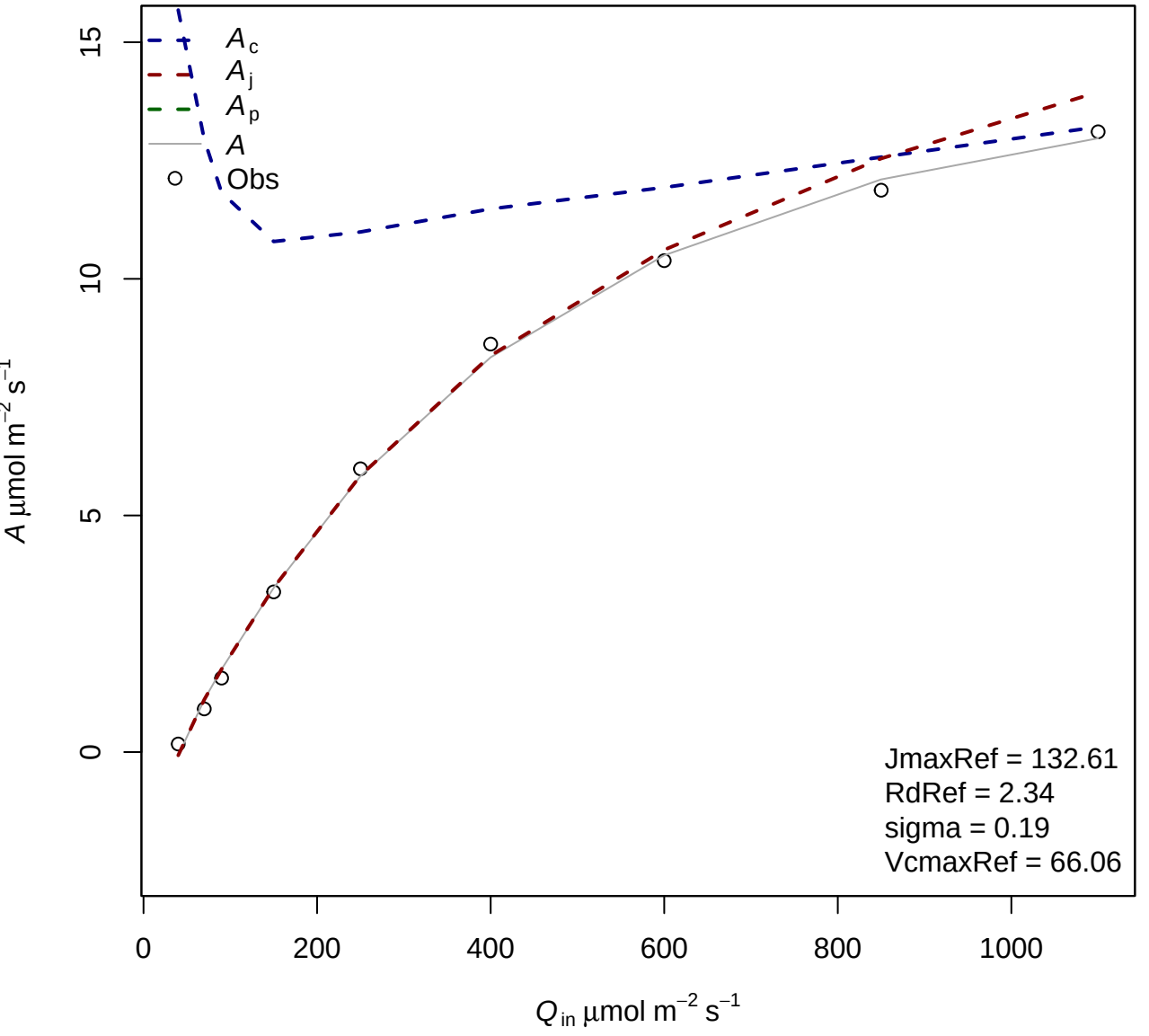

232
$A_c$
$A_j$
$A_p$
$A$
Obs
4
2
0
−2
$A$ µmol m$^{-2}$ s$^{-1}$
200
400
600
800
1000
1200
1400
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 51.12
RdRef = 3.63
sigma = 0.16
VcmaxRef = 98133.26

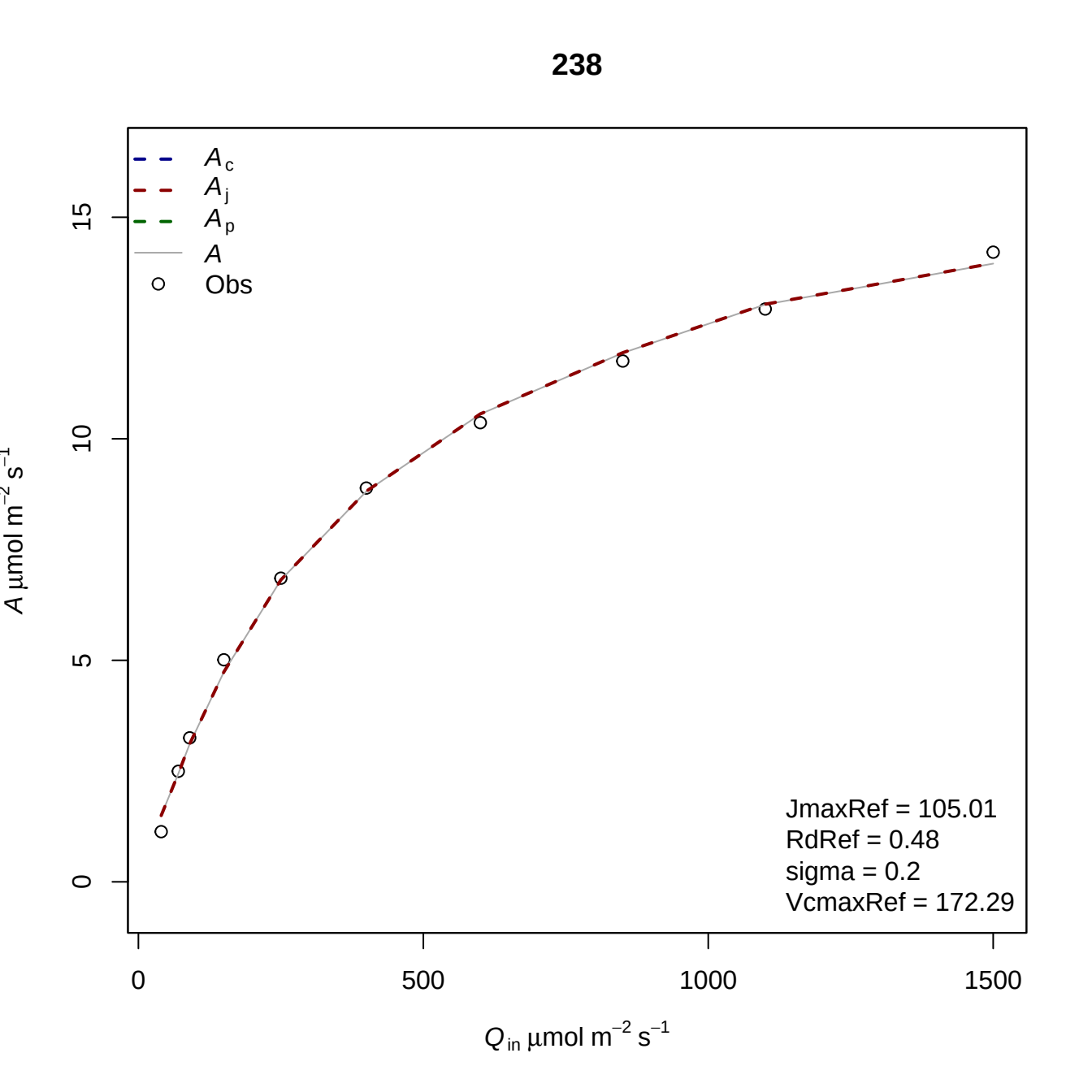

233
$A_c$
$A_j$
$A_p$
$A$
Obs
0.5
0.0
−0.5
−1.0
−1.5
$A$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 11.17
RdRef = 1.69
sigma = 0.12
VcmaxRef = 13501.91
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

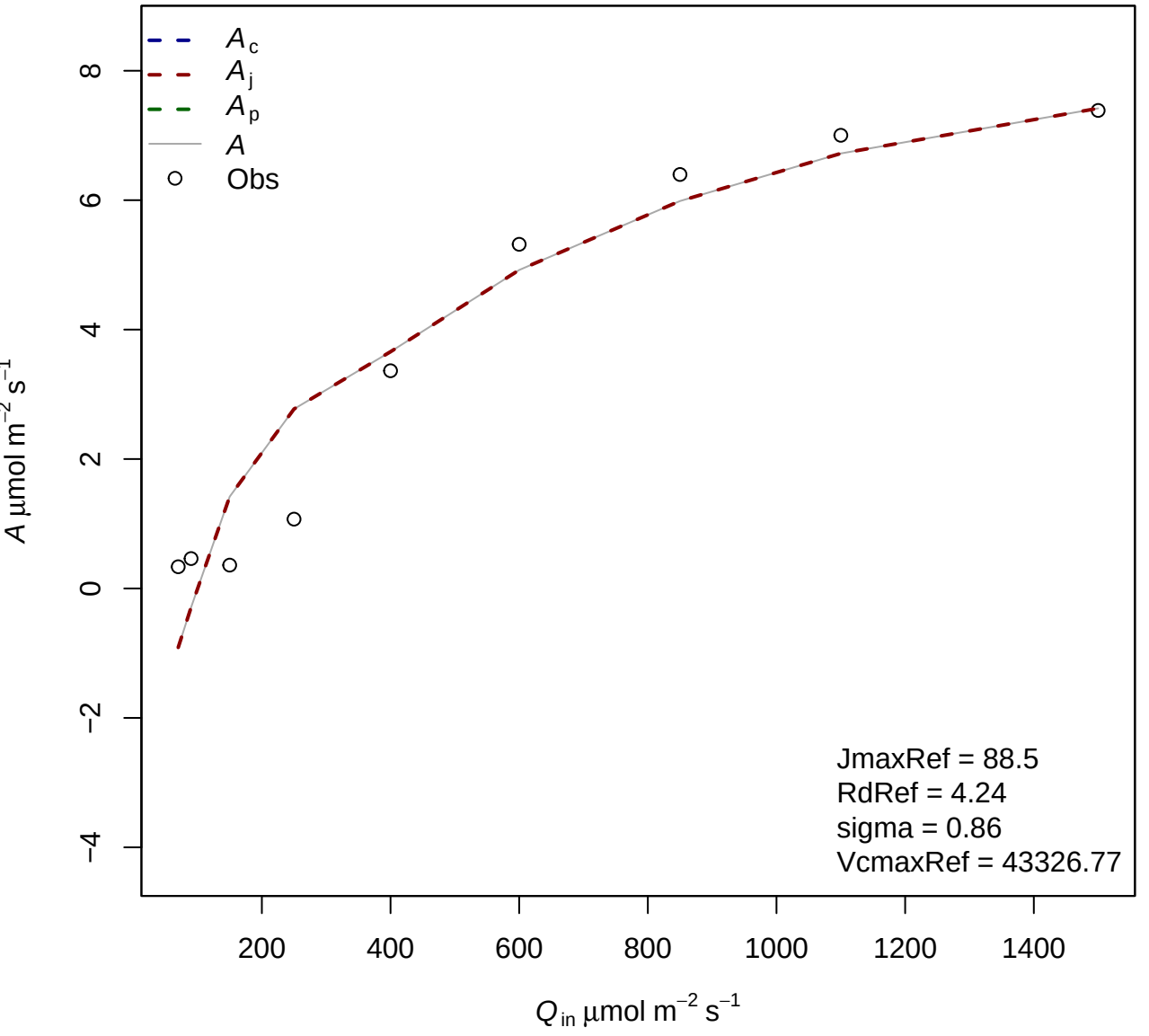

234
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.2
RdRef = 2.48
sigma = 0.32
VcmaxRef = 27482.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800
1000

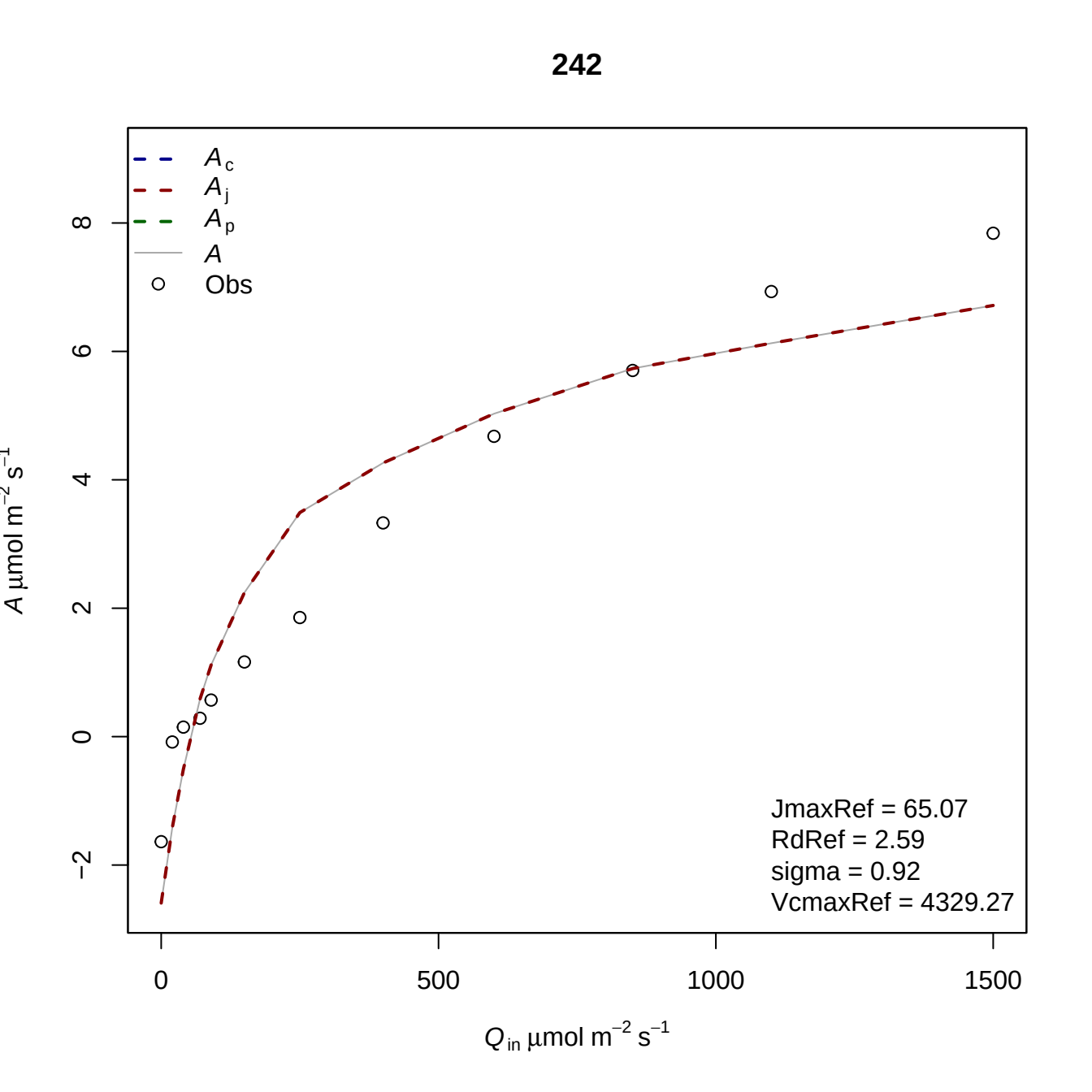

235
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.49
RdRef = 1.22
sigma = 0.13
VcmaxRef = 56.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000

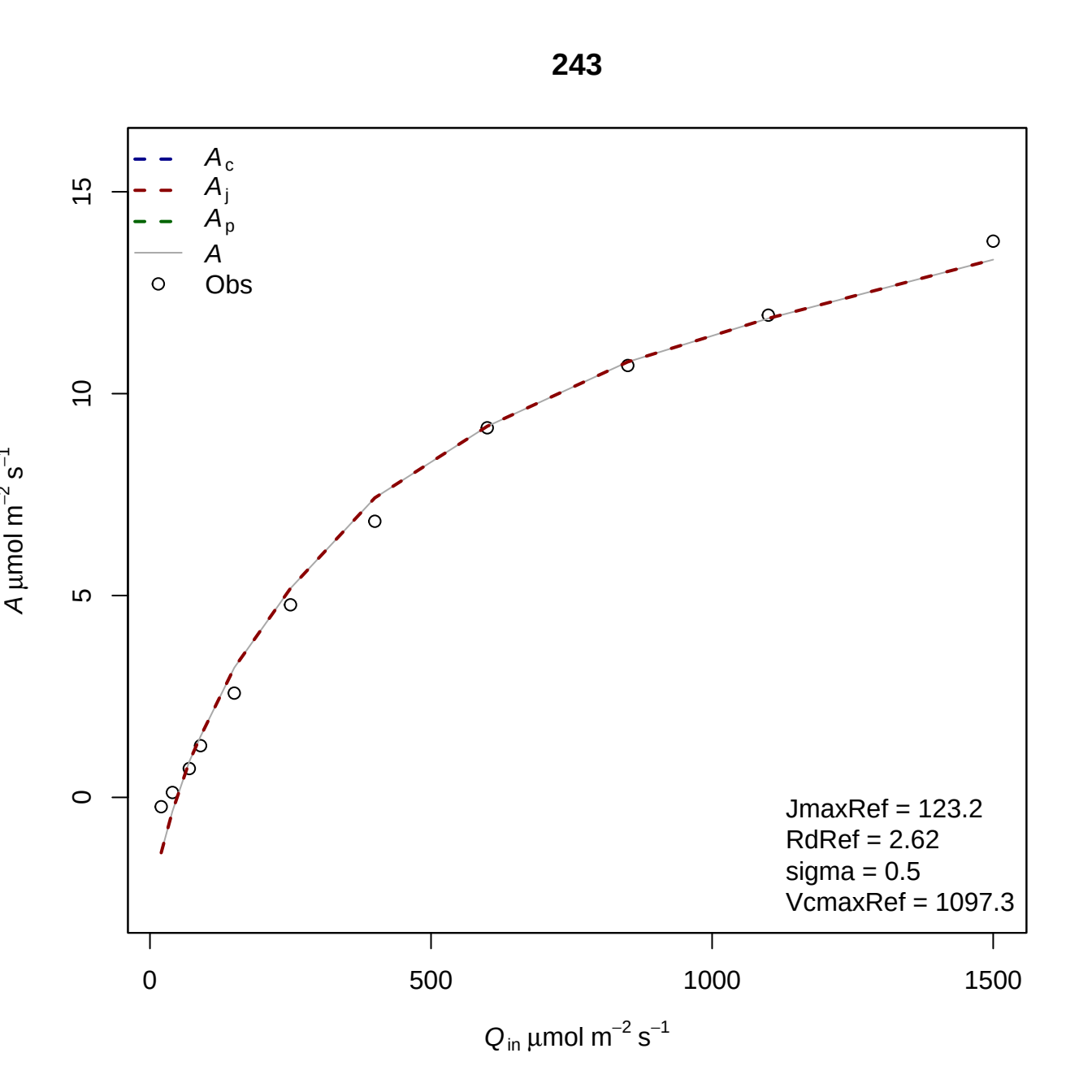

236
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.2
RdRef = 1.22
sigma = 0.14
VcmaxRef = 57.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000

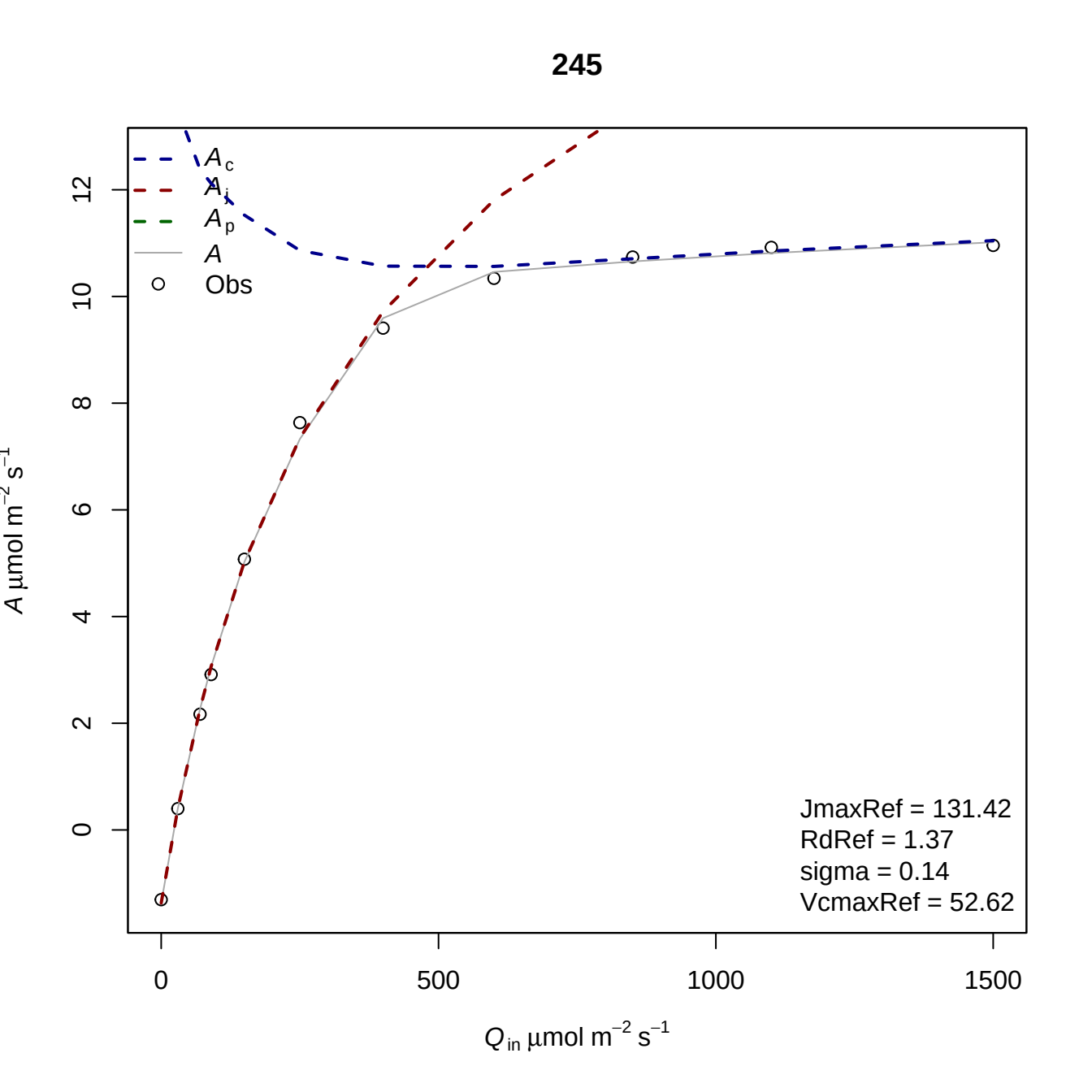

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.61
RdRef = 2.34
sigma = 0.19
VcmaxRef = 66.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000

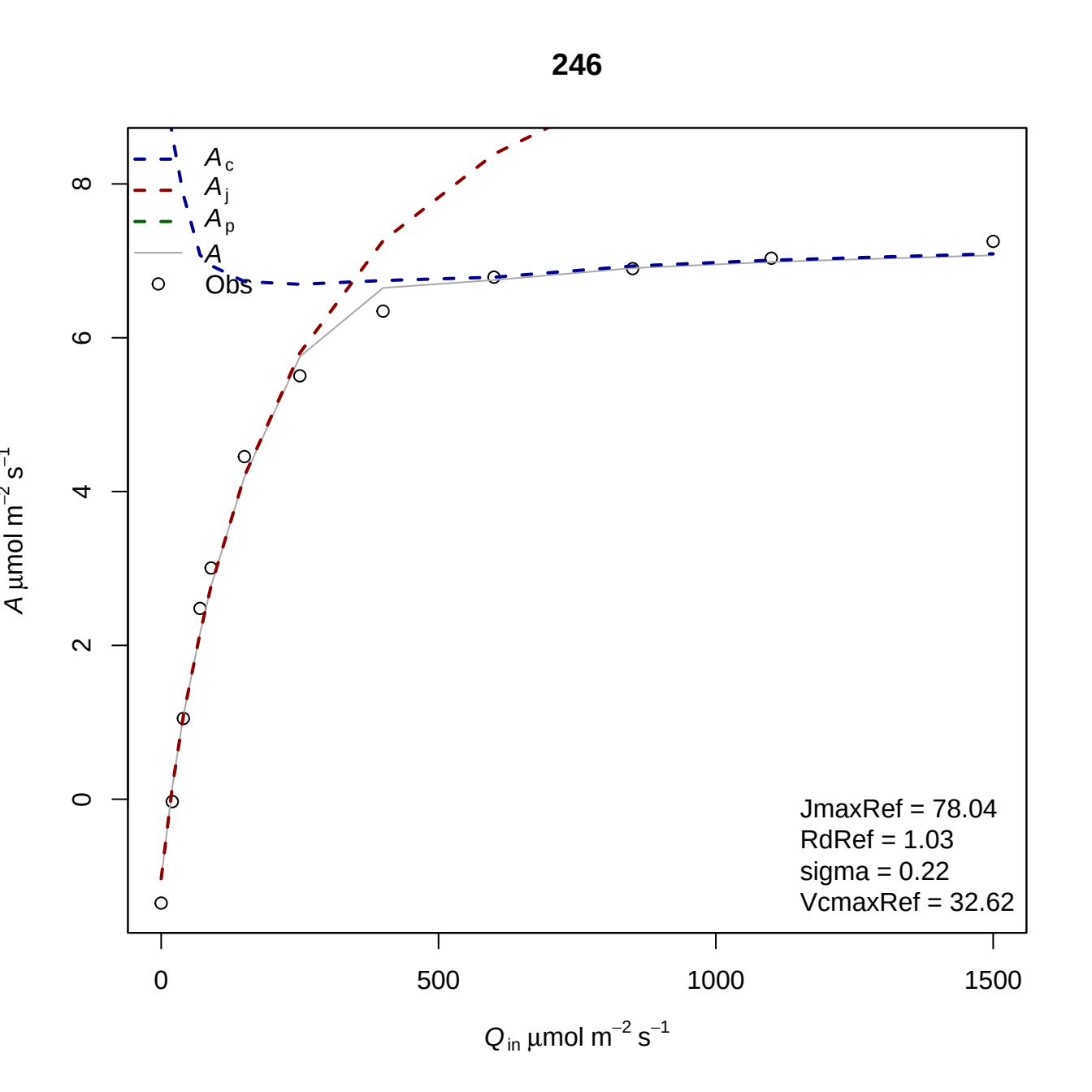

238
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.01
RdRef = 0.48
sigma = 0.2
VcmaxRef = 172.29
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

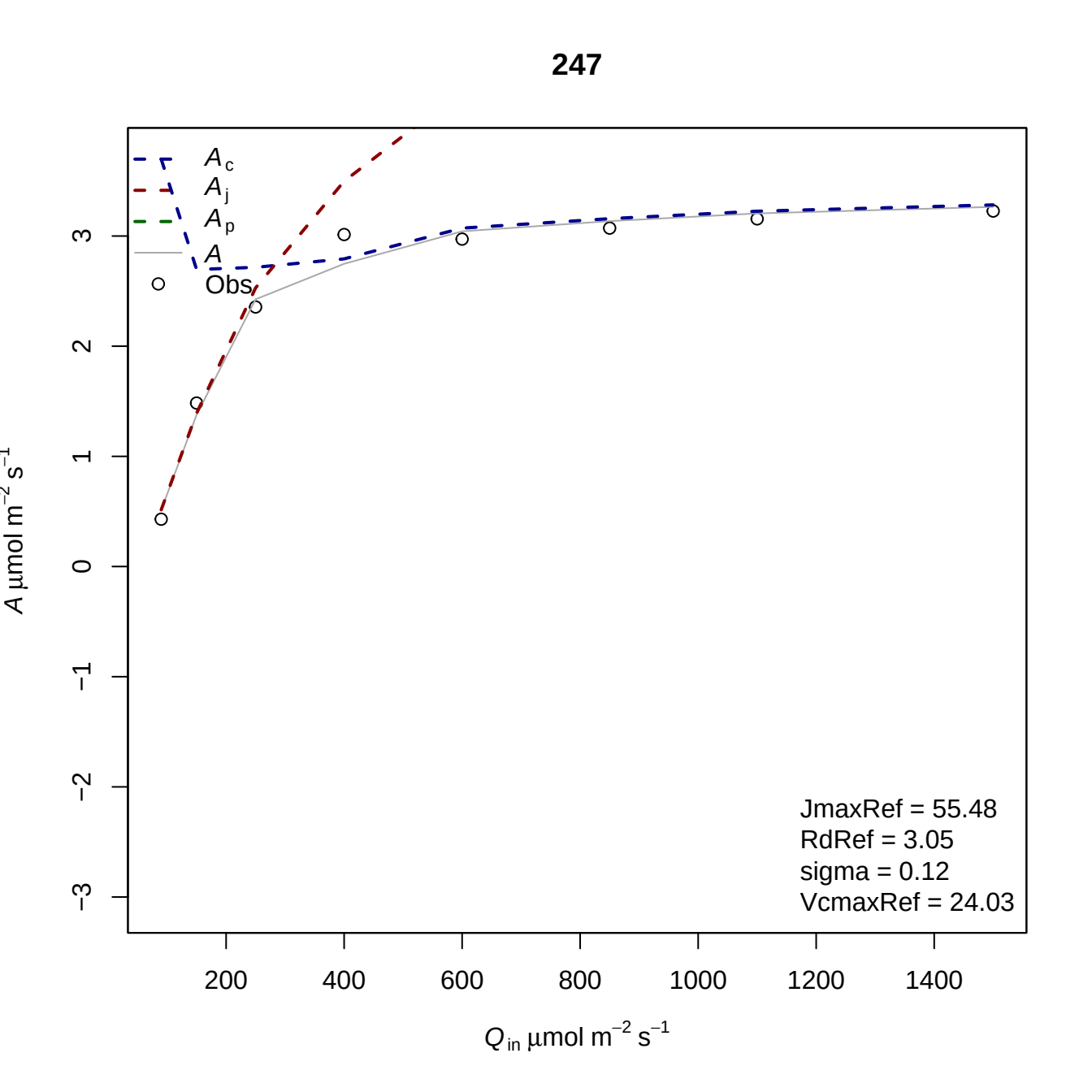

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.5
RdRef = 4.24
sigma = 0.86
VcmaxRef = 43326.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400

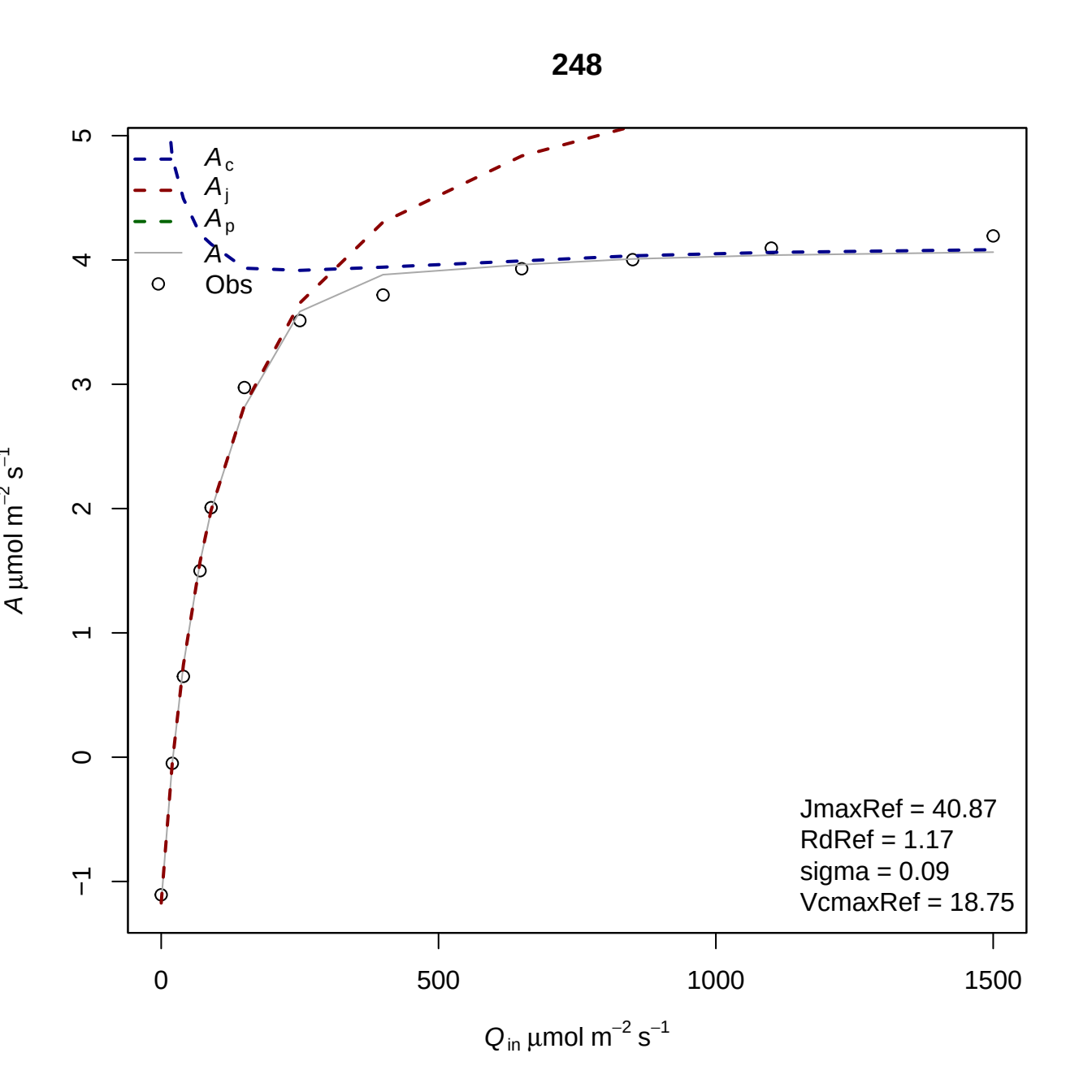

242
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.07
RdRef = 2.59
sigma = 0.92
VcmaxRef = 4329.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

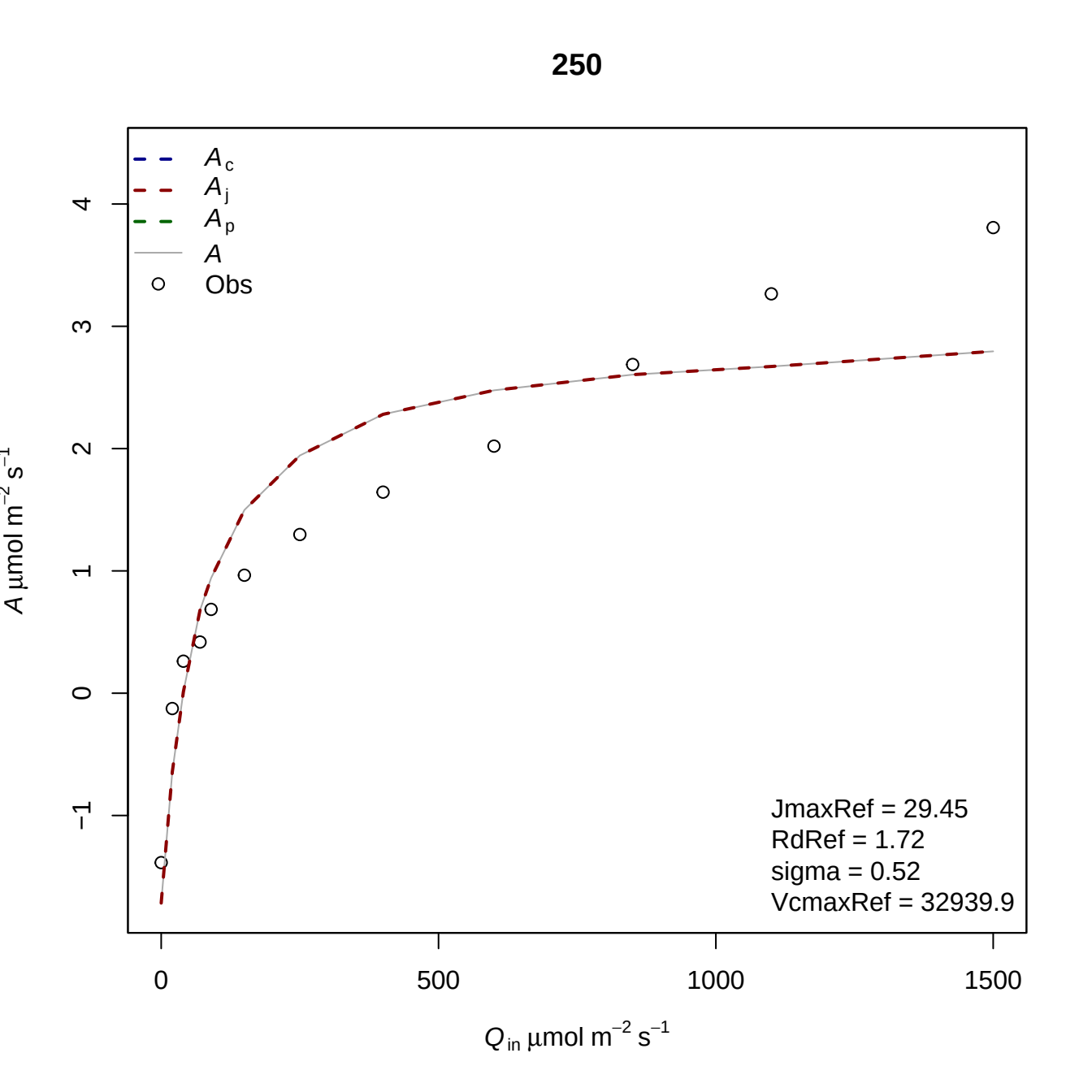

243
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.2
RdRef = 2.62
sigma = 0.5
VcmaxRef = 1097.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

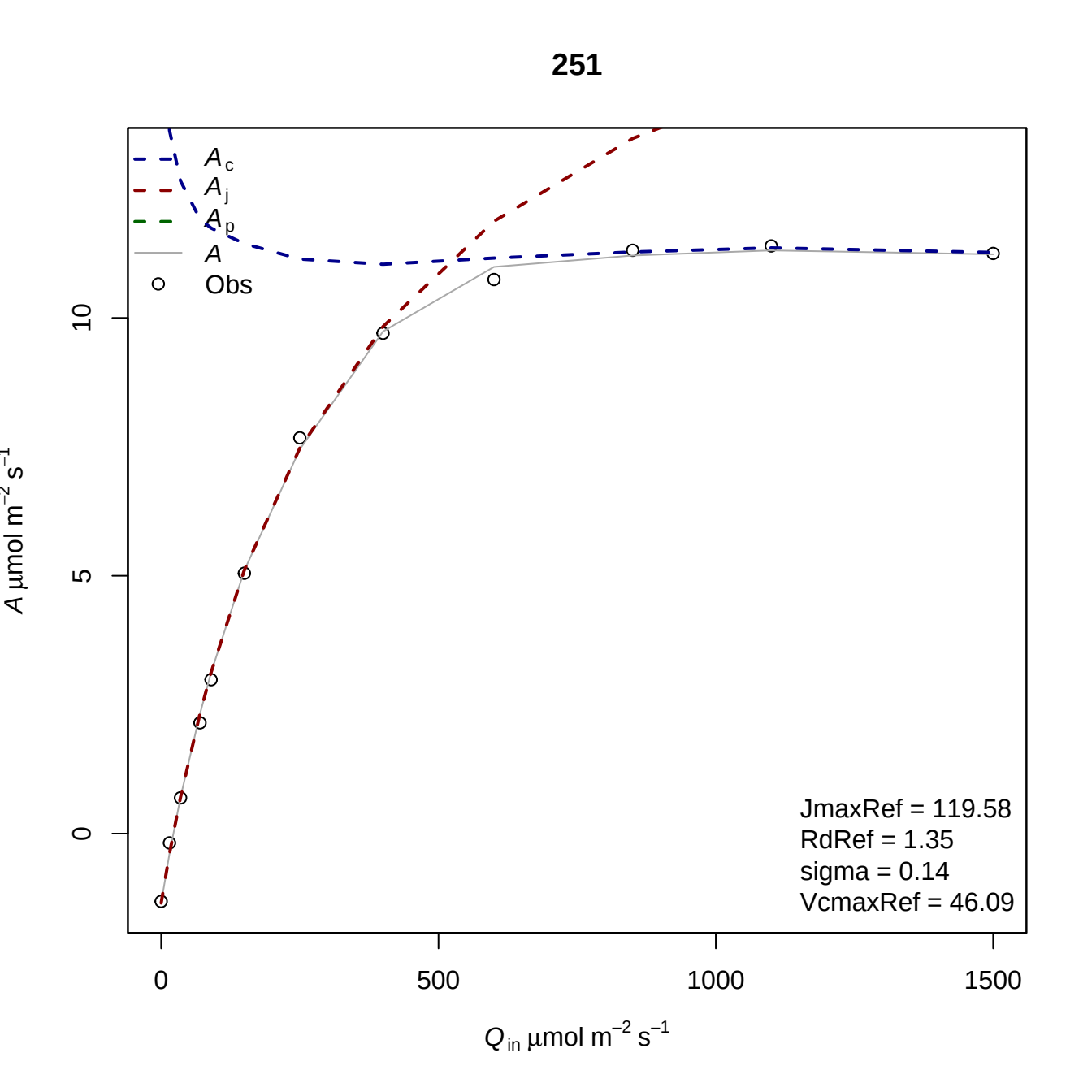

245
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.42
RdRef = 1.37
sigma = 0.14
VcmaxRef = 52.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

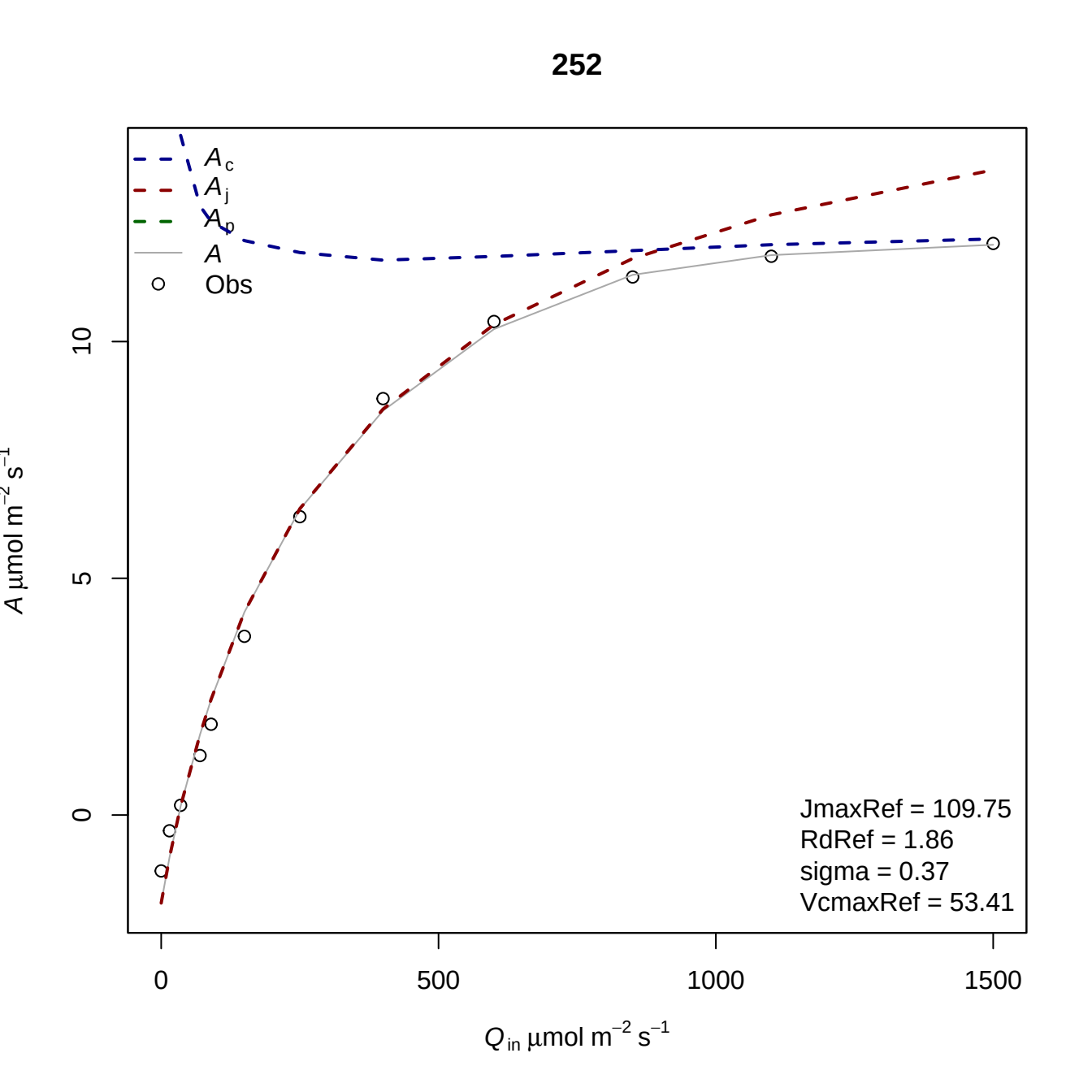

246
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.04
RdRef = 1.03
sigma = 0.22
VcmaxRef = 32.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

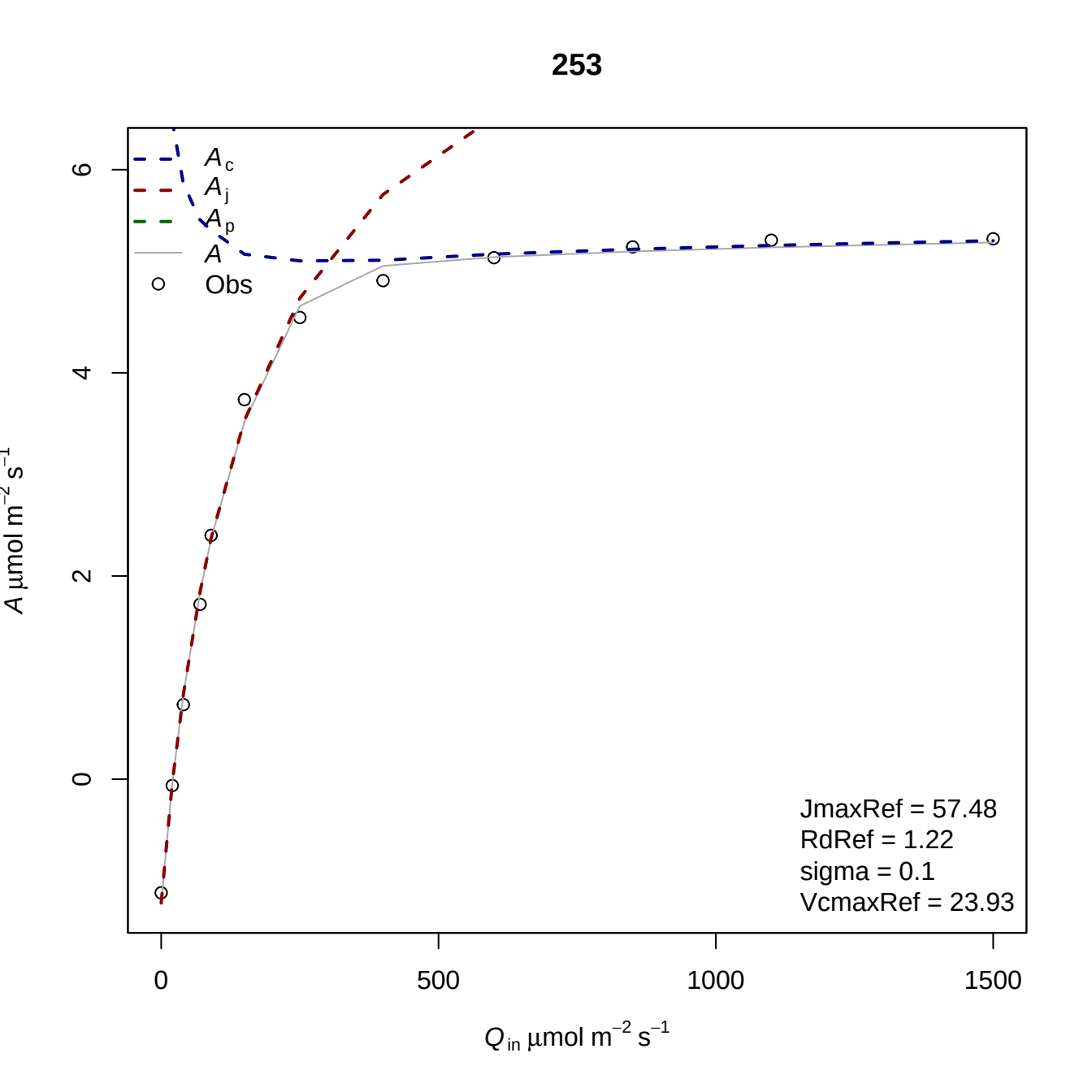

247
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.48
RdRef = 3.05
sigma = 0.12
VcmaxRef = 24.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
−2
−3
200
400
600
800
1000
1200
1400

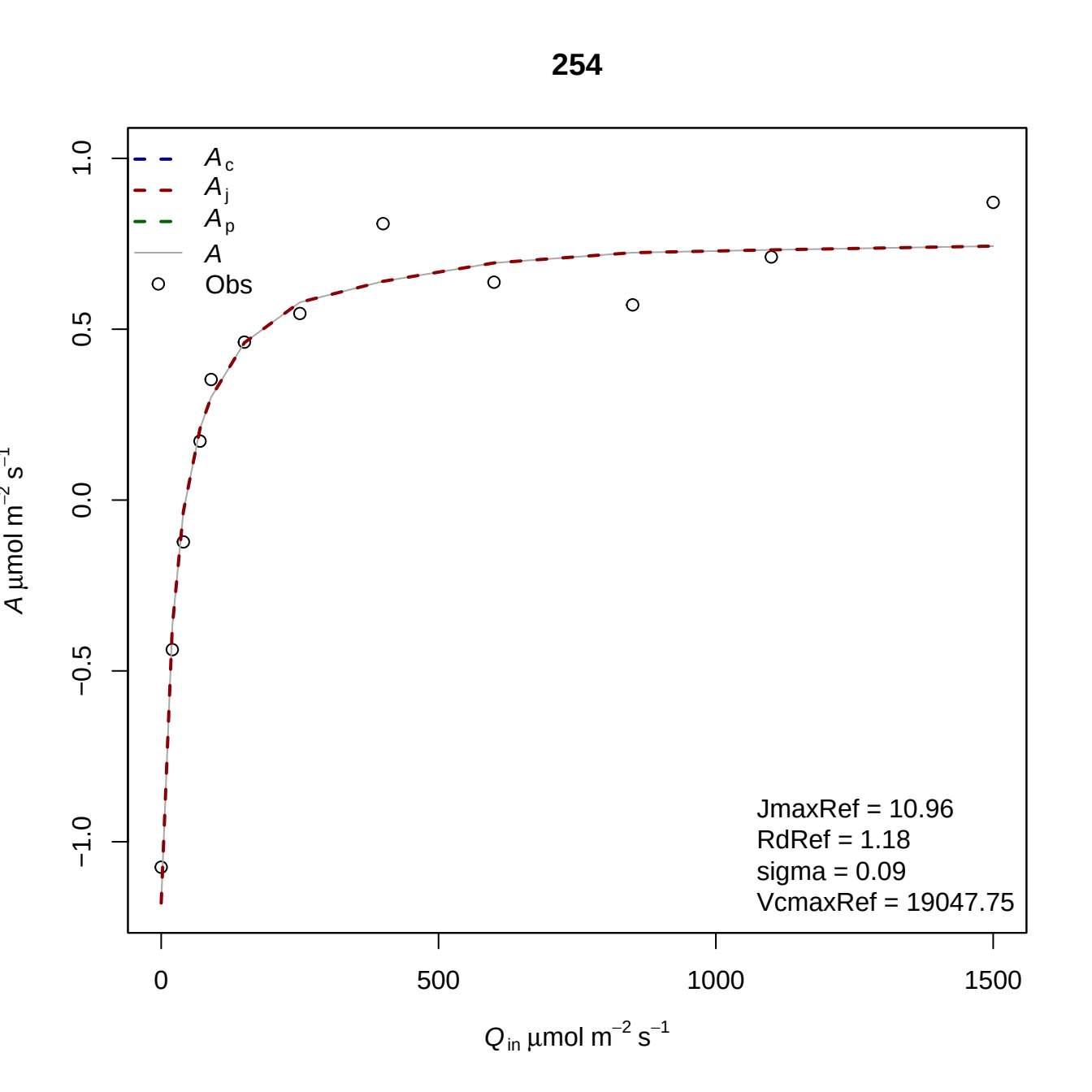

248
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.87
RdRef = 1.17
sigma = 0.09
VcmaxRef = 18.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

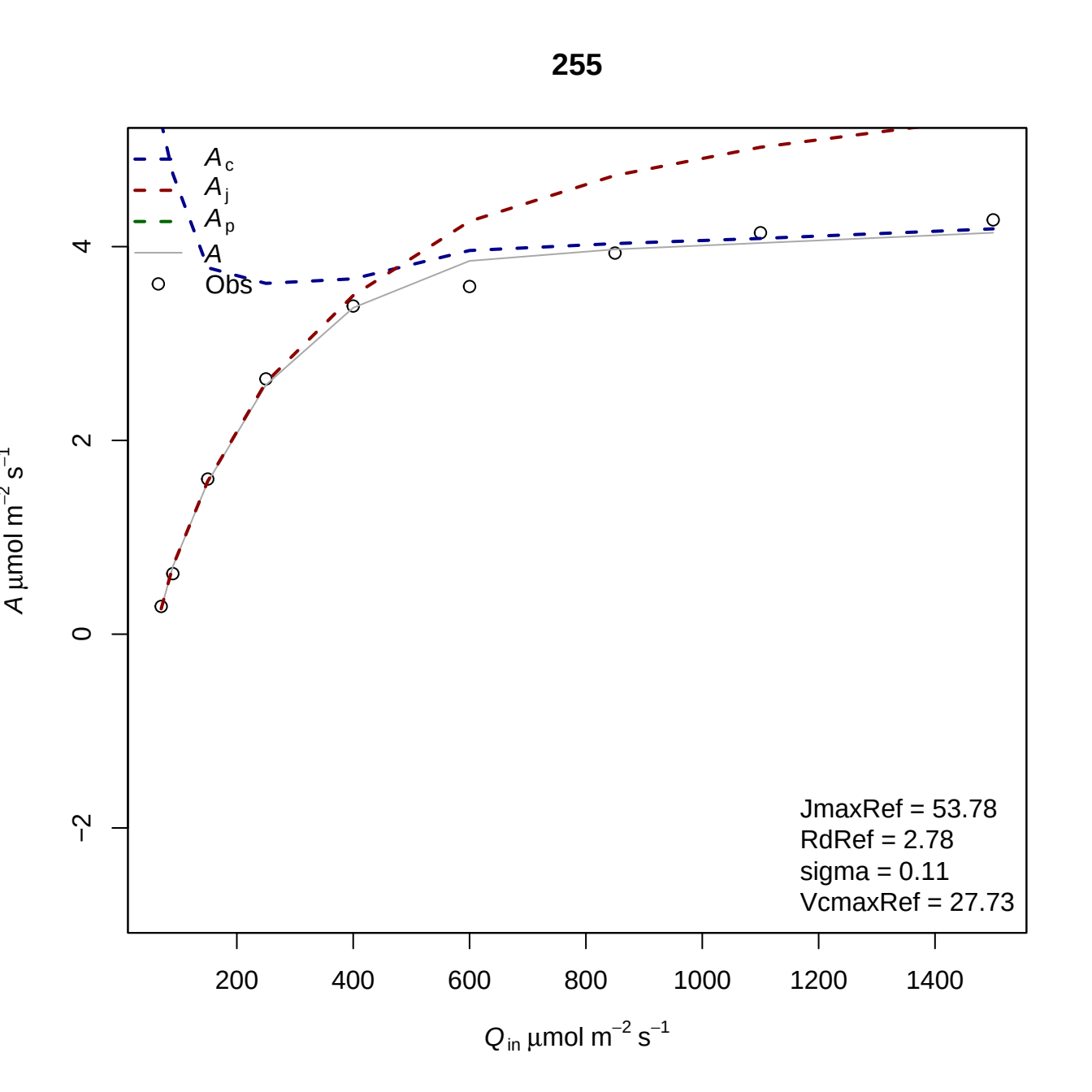

250
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.45
RdRef = 1.72
sigma = 0.52
VcmaxRef = 32939.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

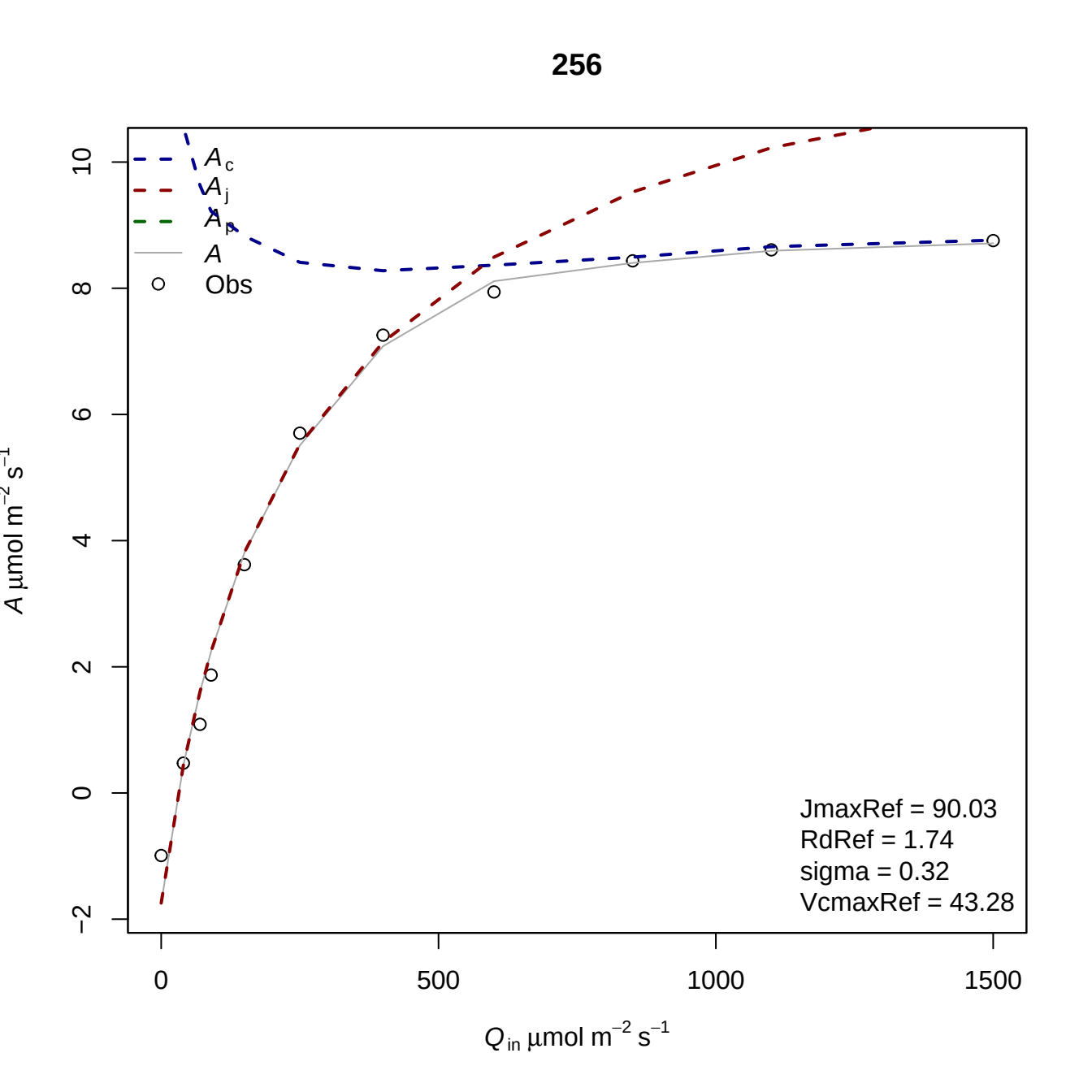

251
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.58
RdRef = 1.35
sigma = 0.14
VcmaxRef = 46.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

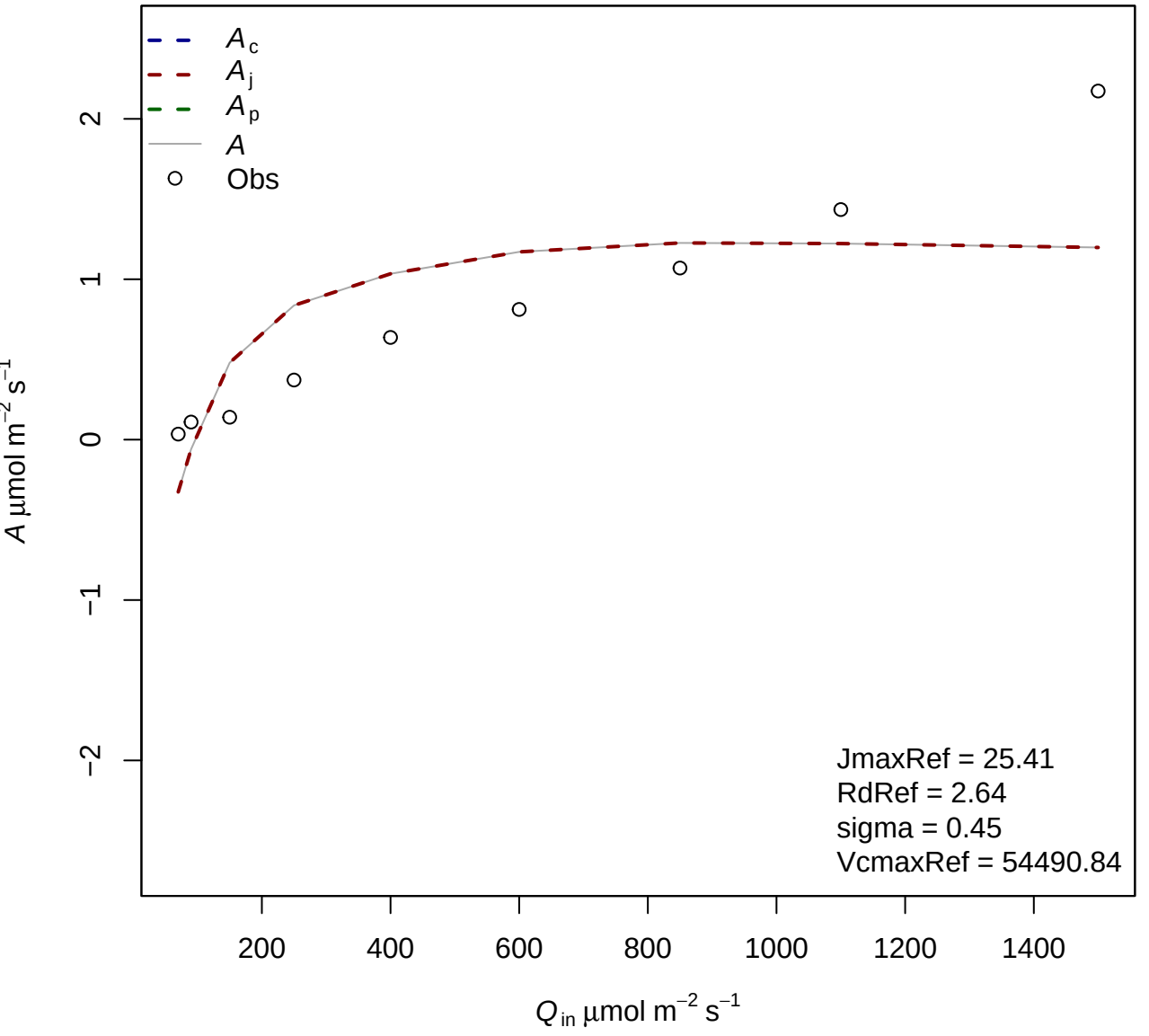

252
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.75
RdRef = 1.86
sigma = 0.37
VcmaxRef = 53.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

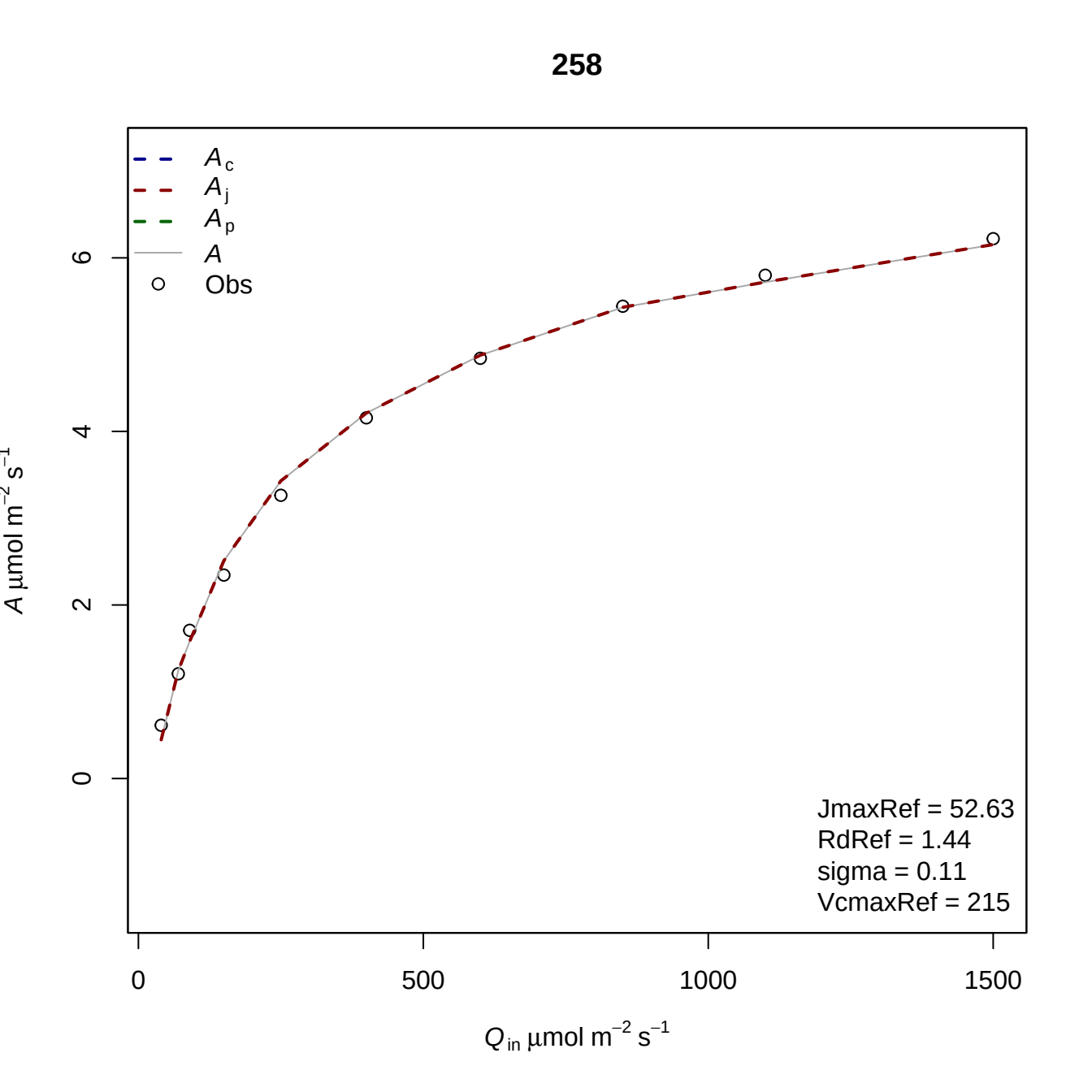

253
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.48
RdRef = 1.22
sigma = 0.1
VcmaxRef = 23.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

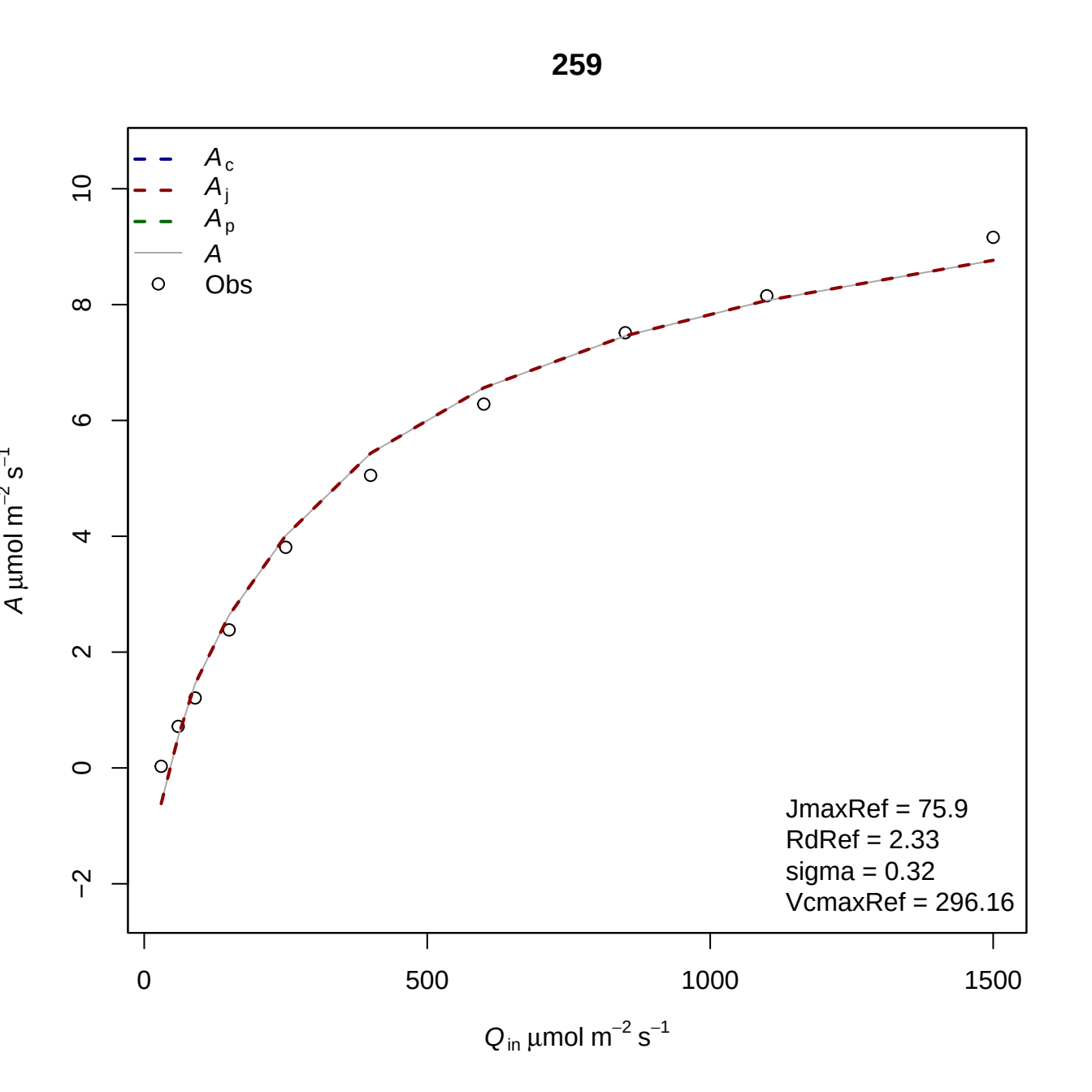

254
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 10.96
RdRef = 1.18
sigma = 0.09
VcmaxRef = 19047.75
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

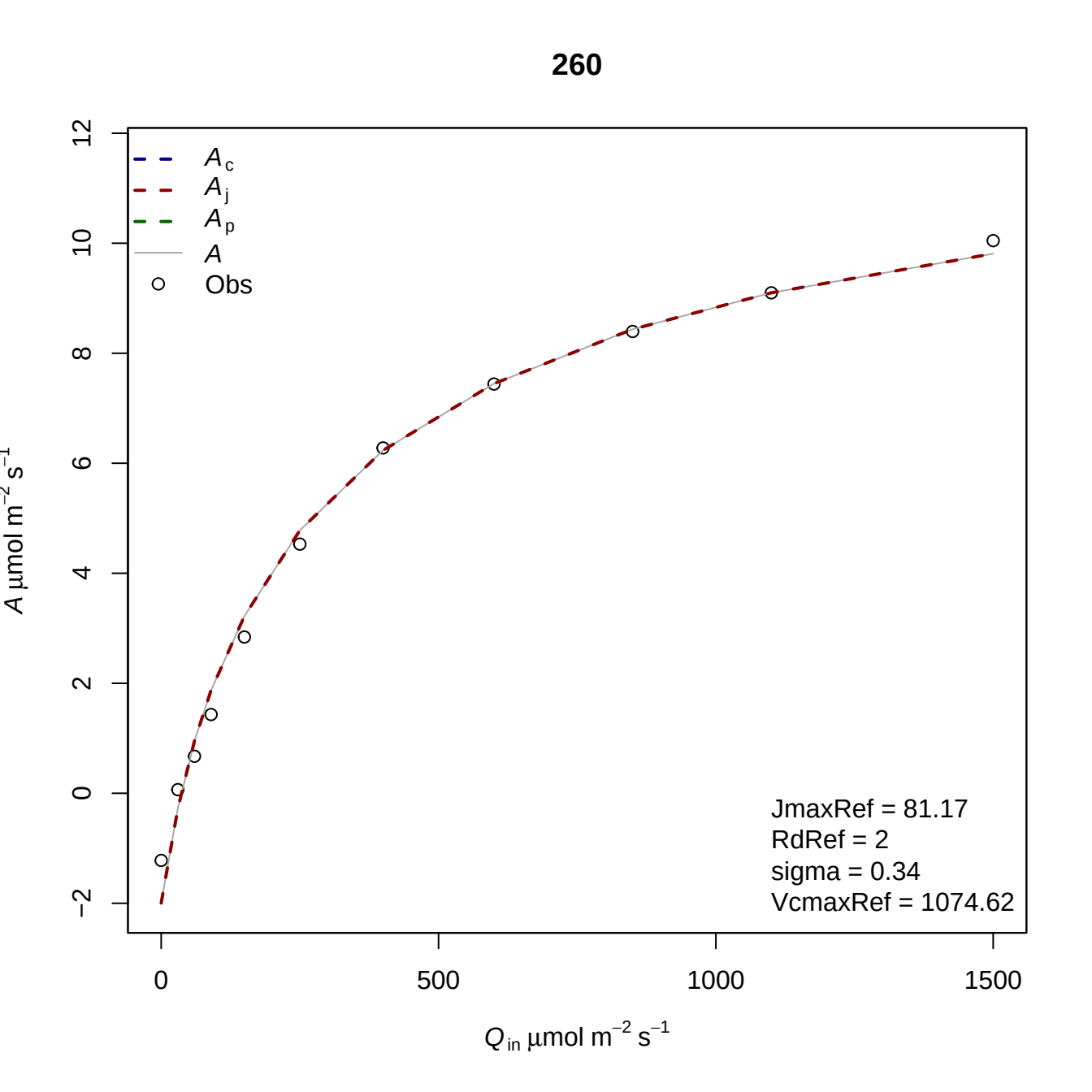

255
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 53.78
RdRef = 2.78
sigma = 0.11
VcmaxRef = 27.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
200
400
600
800
1000
1200
1400

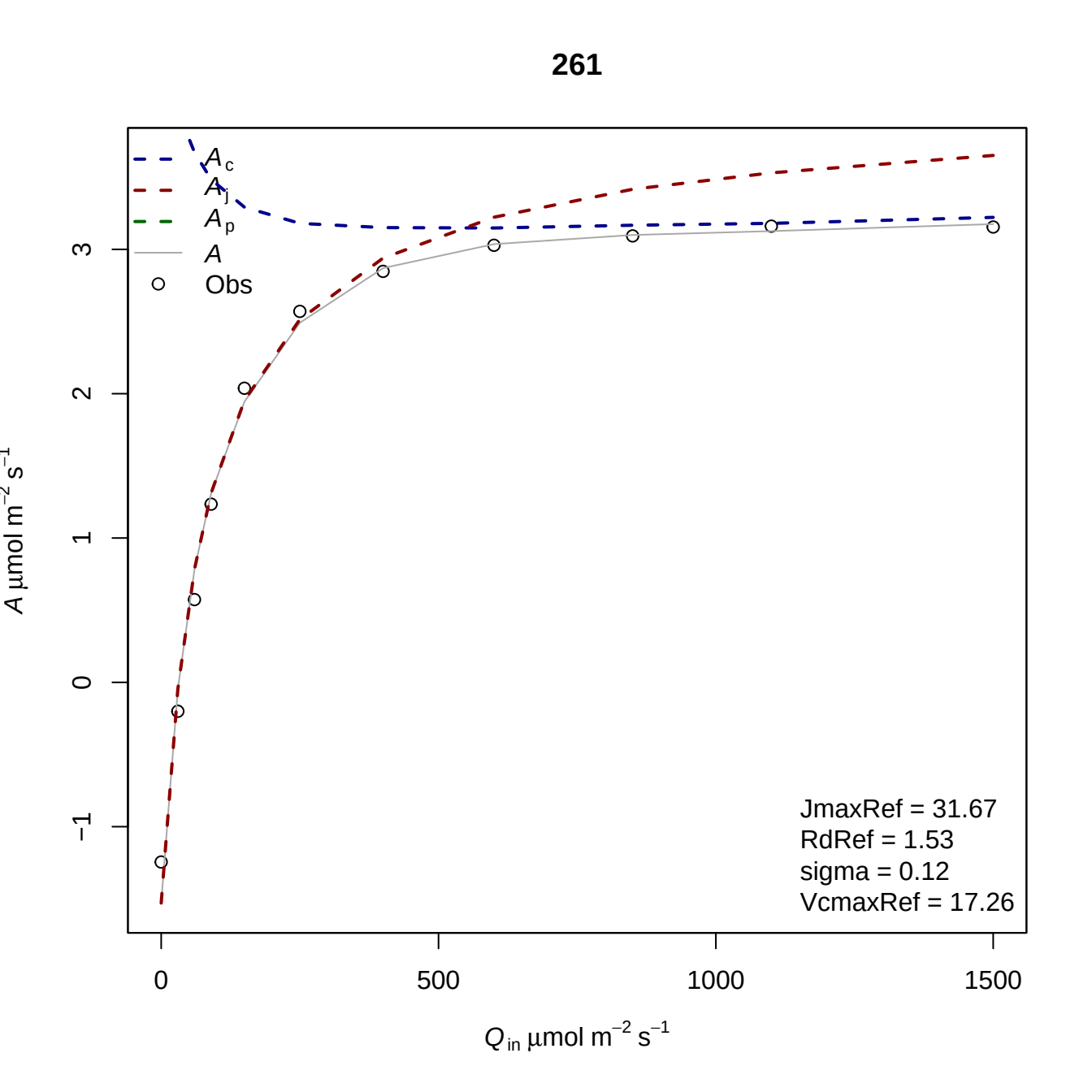

256
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.03
RdRef = 1.74
sigma = 0.32
VcmaxRef = 43.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

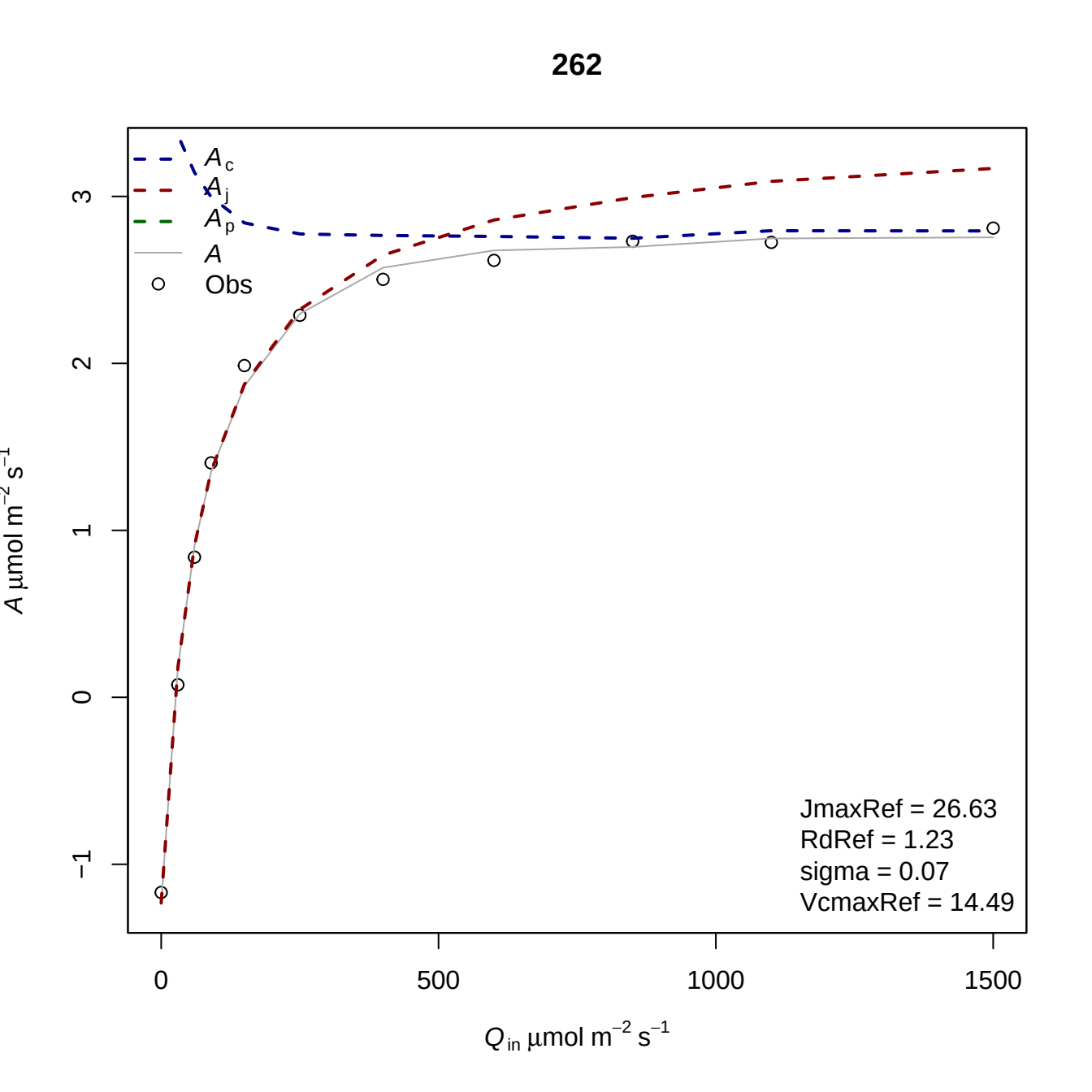

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.41
RdRef = 2.64
sigma = 0.45
VcmaxRef = 54490.84
2
1
0
−1
−2
200
400
600
800
1000
1200
1400
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

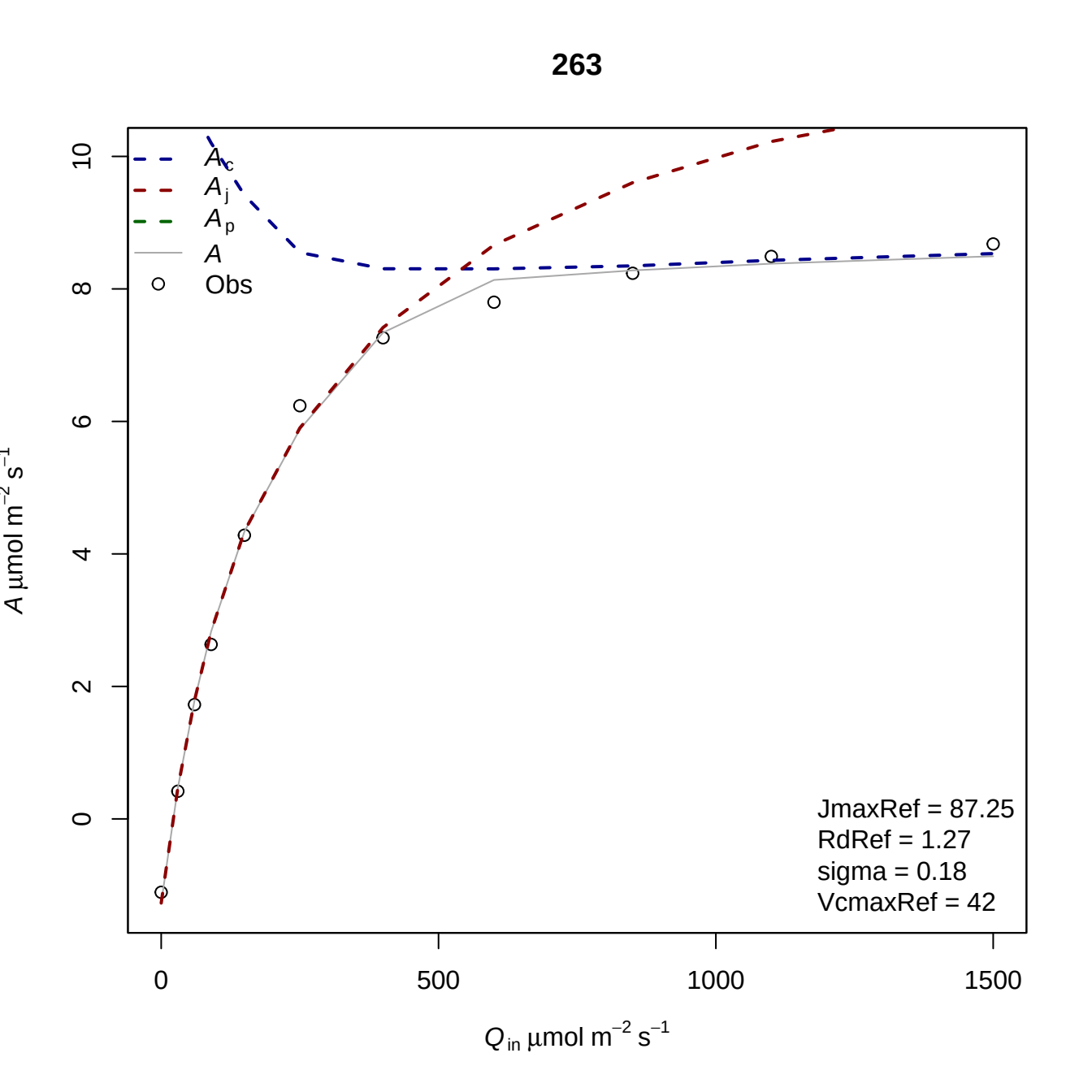

258
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 52.63
RdRef = 1.44
sigma = 0.11
VcmaxRef = 215
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

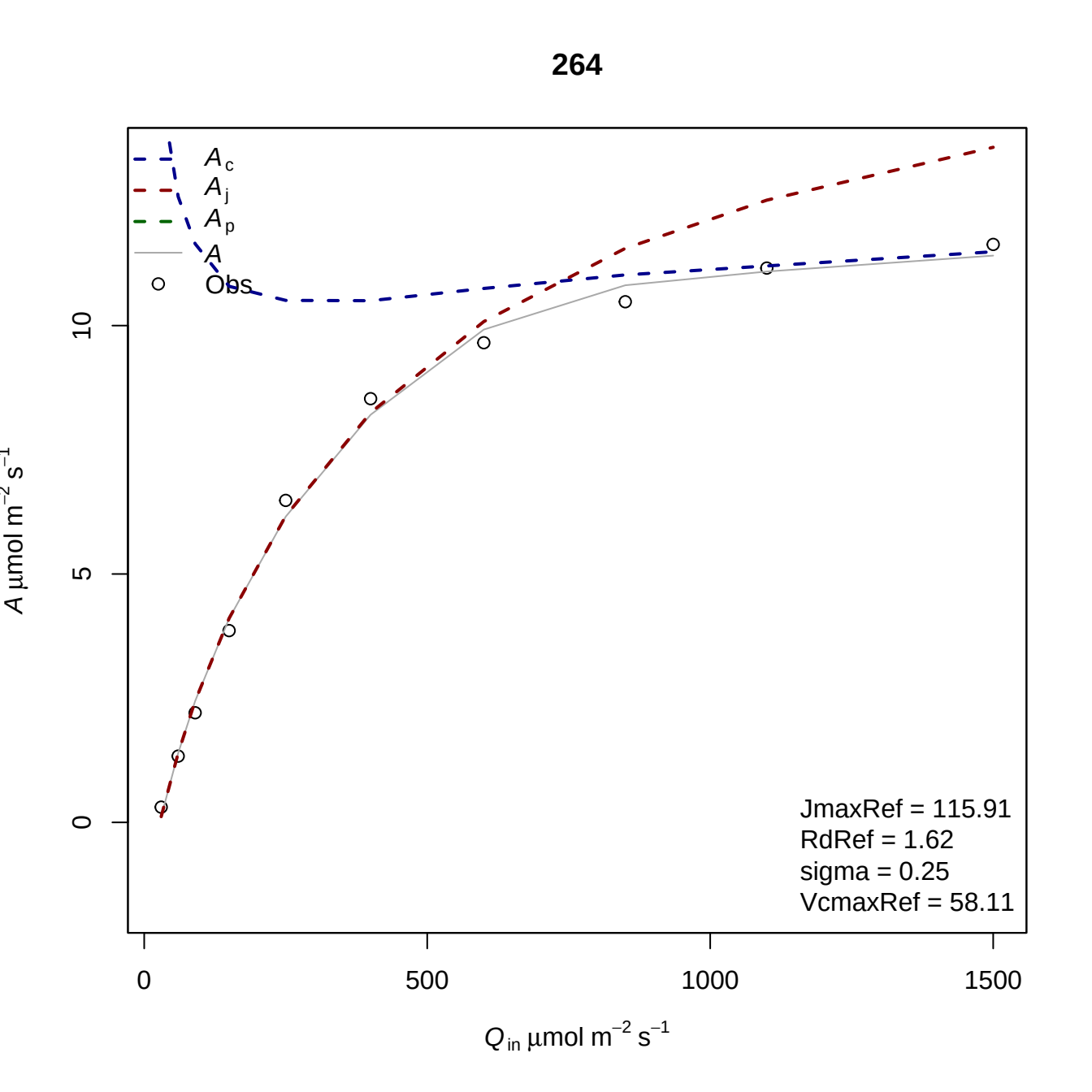

259
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.9
RdRef = 2.33
sigma = 0.32
VcmaxRef = 296.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

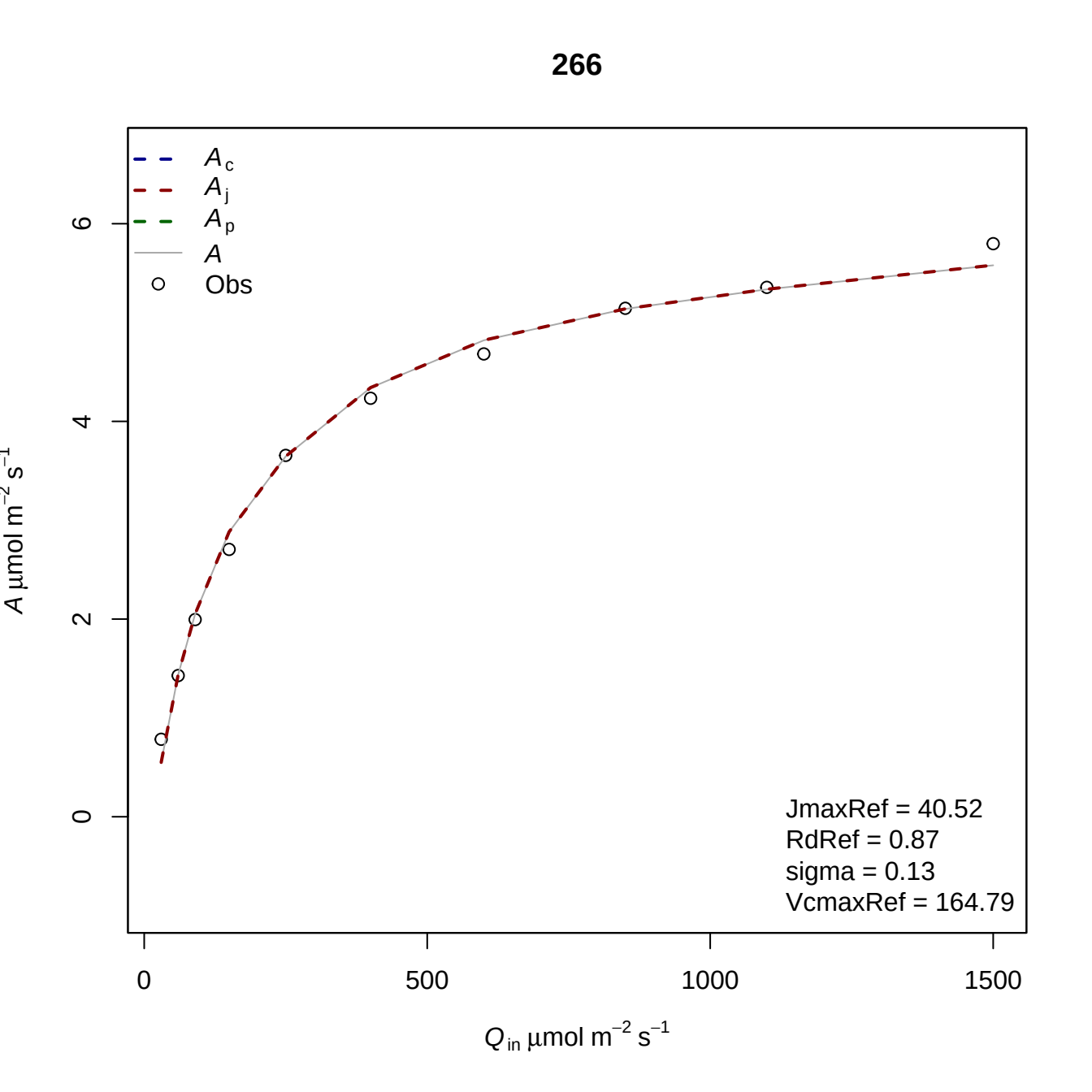

260
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.17
RdRef = 2
sigma = 0.34
VcmaxRef = 1074.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

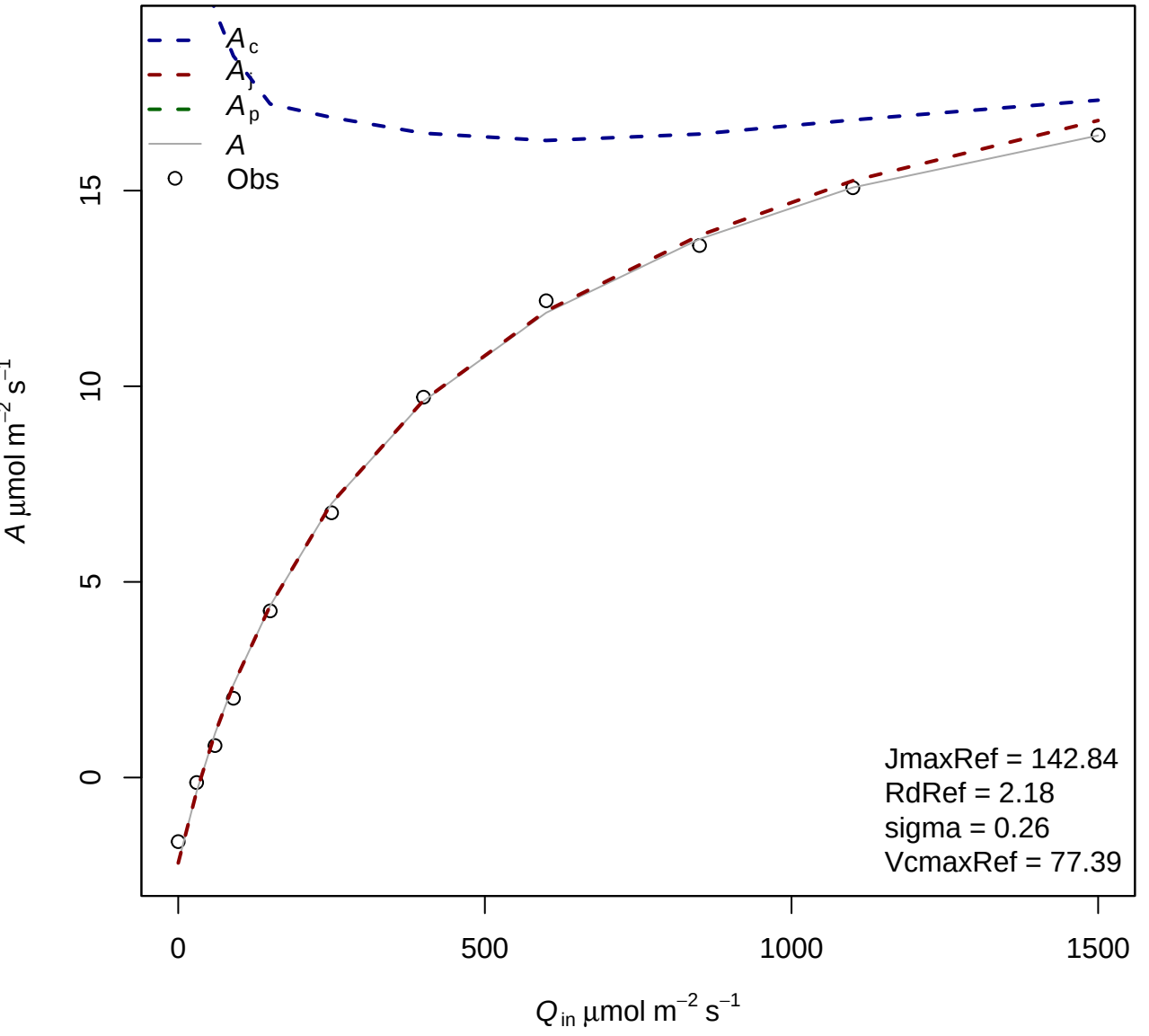

261
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.67
RdRef = 1.53
sigma = 0.12
VcmaxRef = 17.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

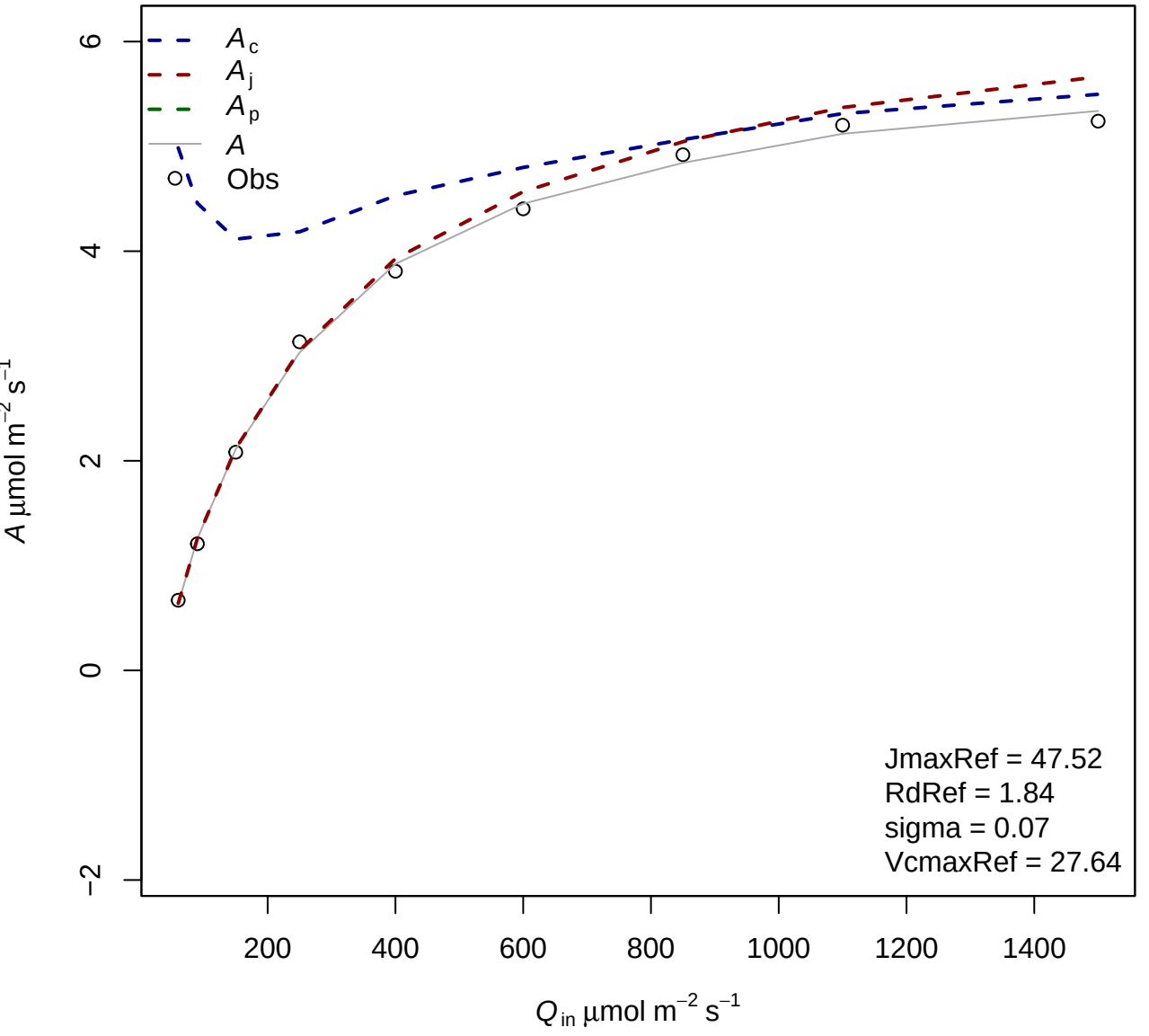

262
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.63
RdRef = 1.23
sigma = 0.07
VcmaxRef = 14.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

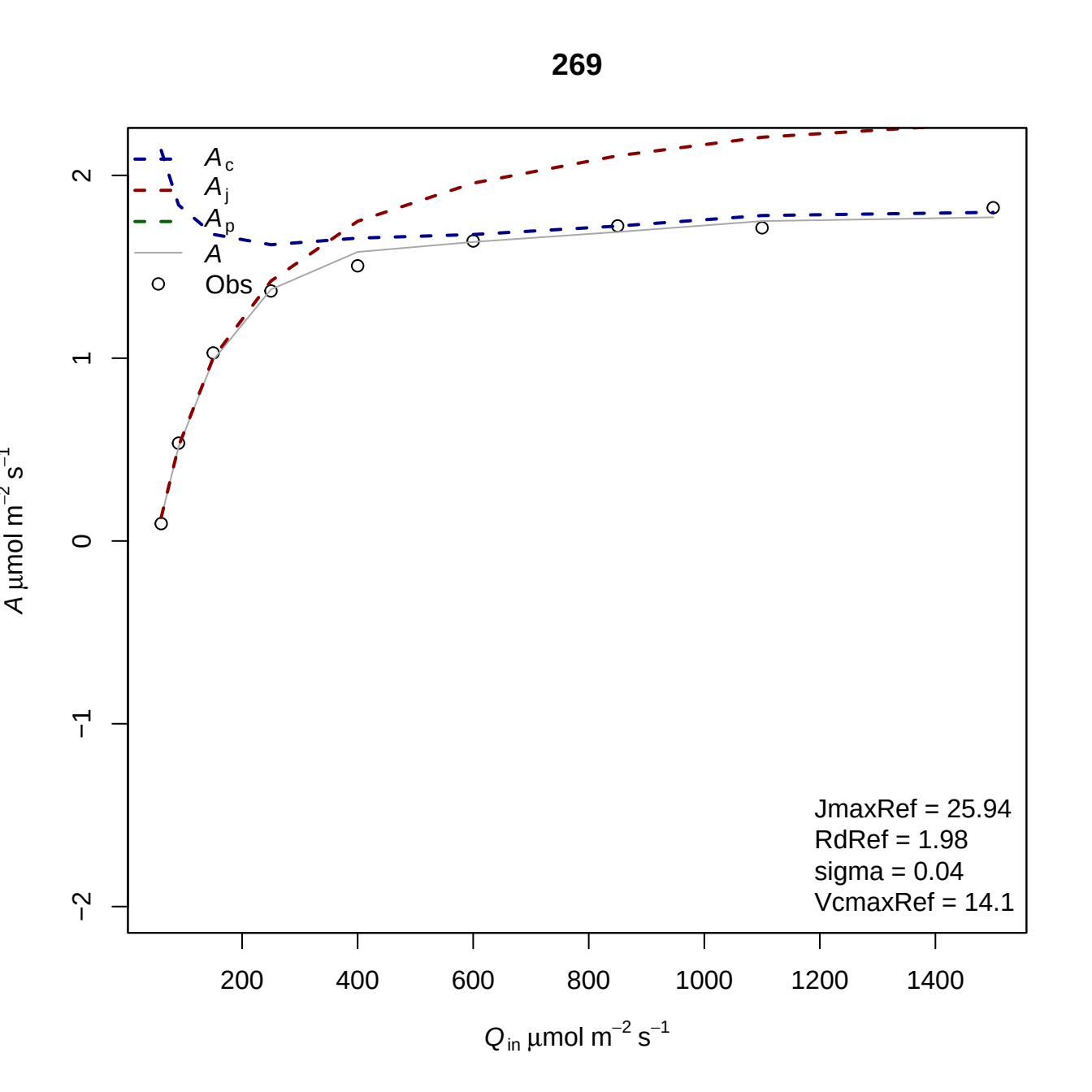

263
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.25
RdRef = 1.27
sigma = 0.18
VcmaxRef = 42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

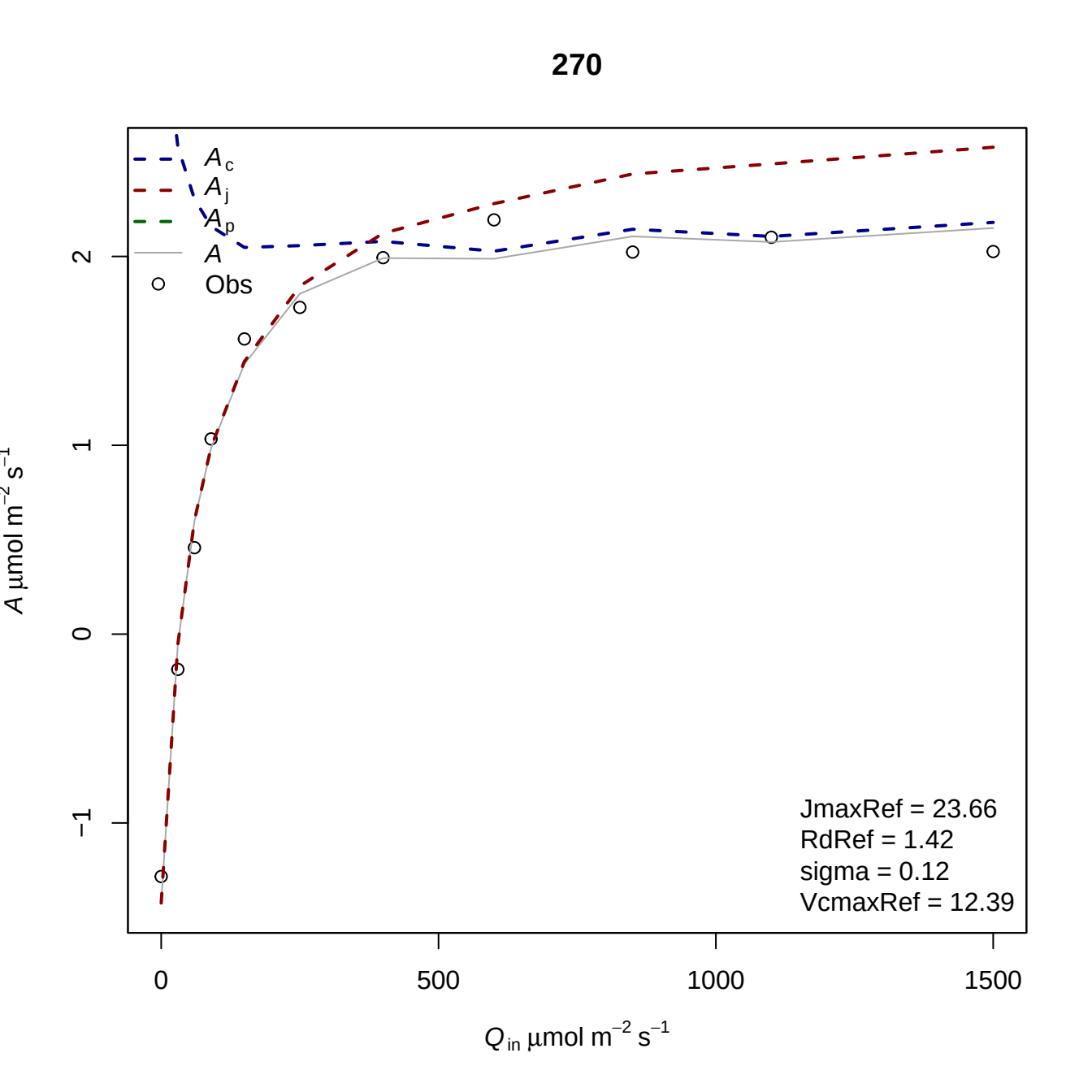

264
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.91
RdRef = 1.62
sigma = 0.25
VcmaxRef = 58.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

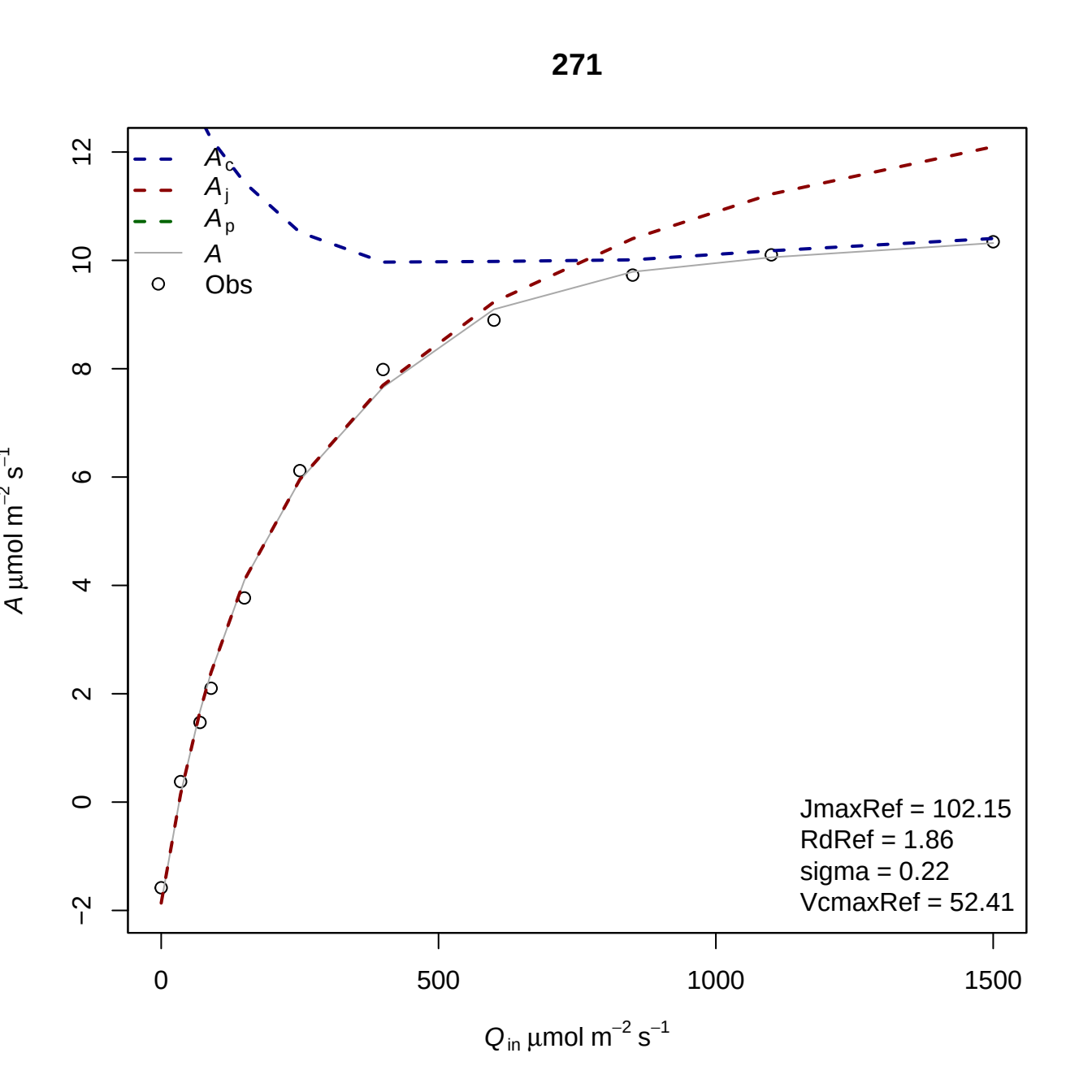

266
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.52
RdRef = 0.87
sigma = 0.13
VcmaxRef = 164.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

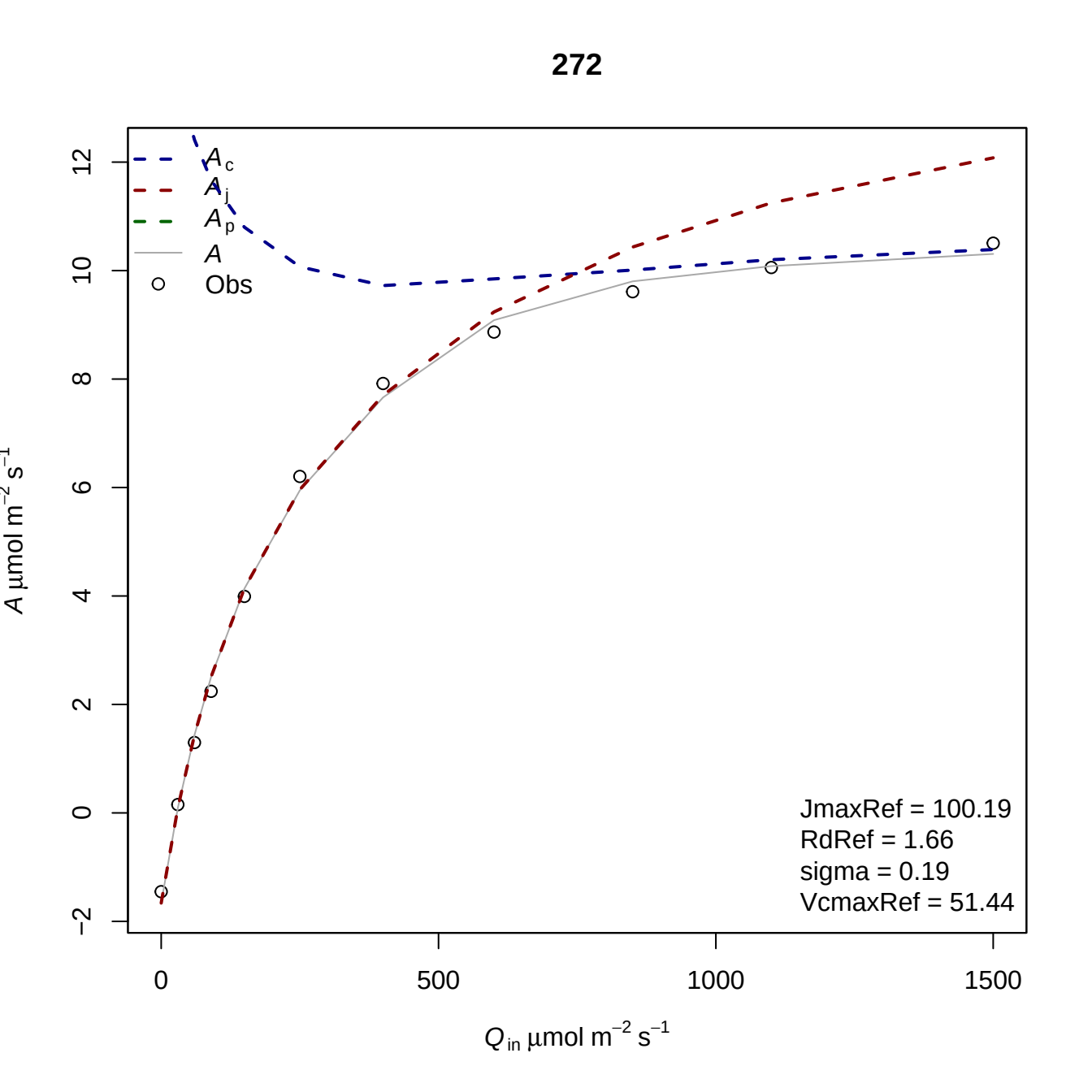

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.84
RdRef = 2.18
sigma = 0.26
VcmaxRef = 77.39
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

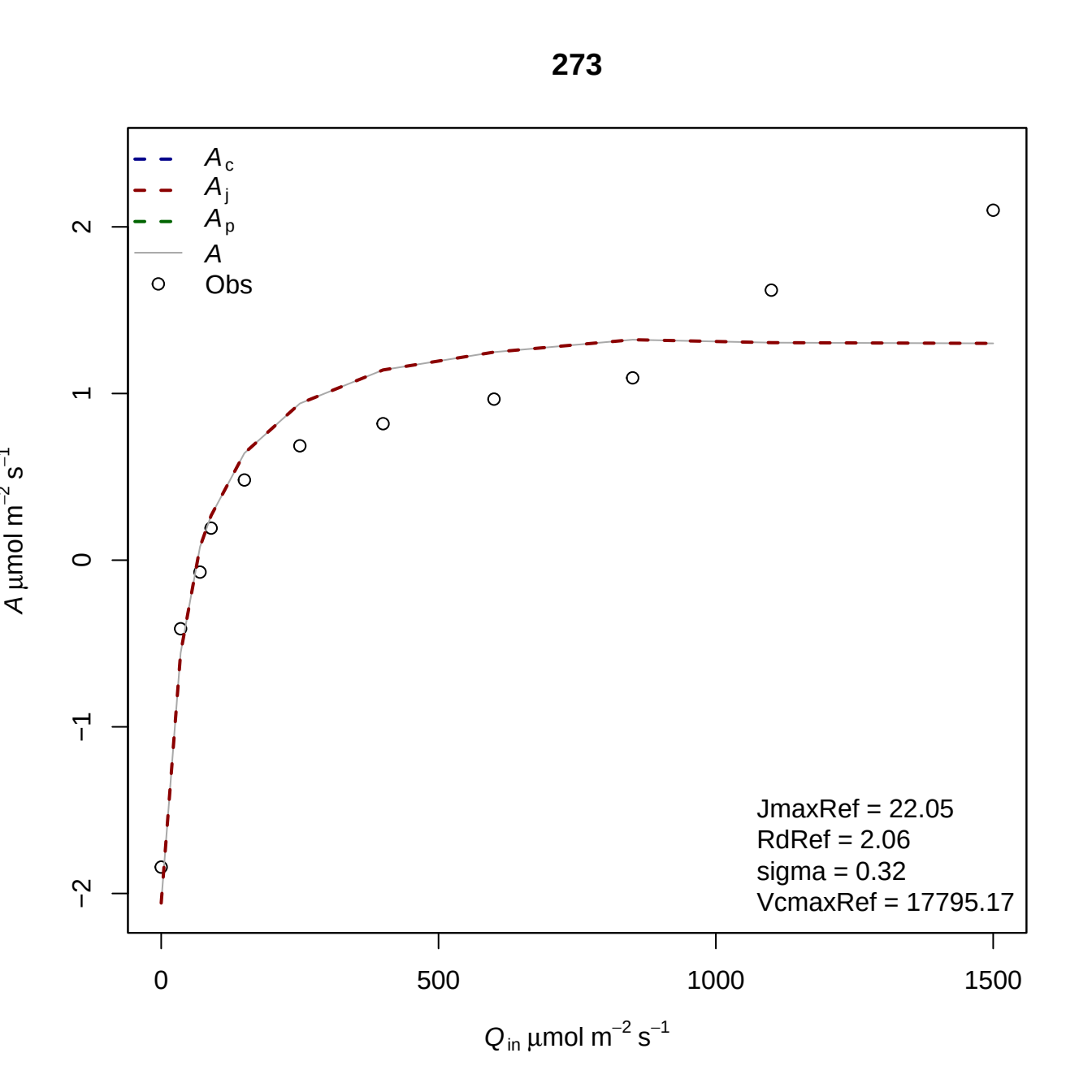

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.52
RdRef = 1.84
sigma = 0.07
VcmaxRef = 27.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
200
400
600
800
1000
1200
1400

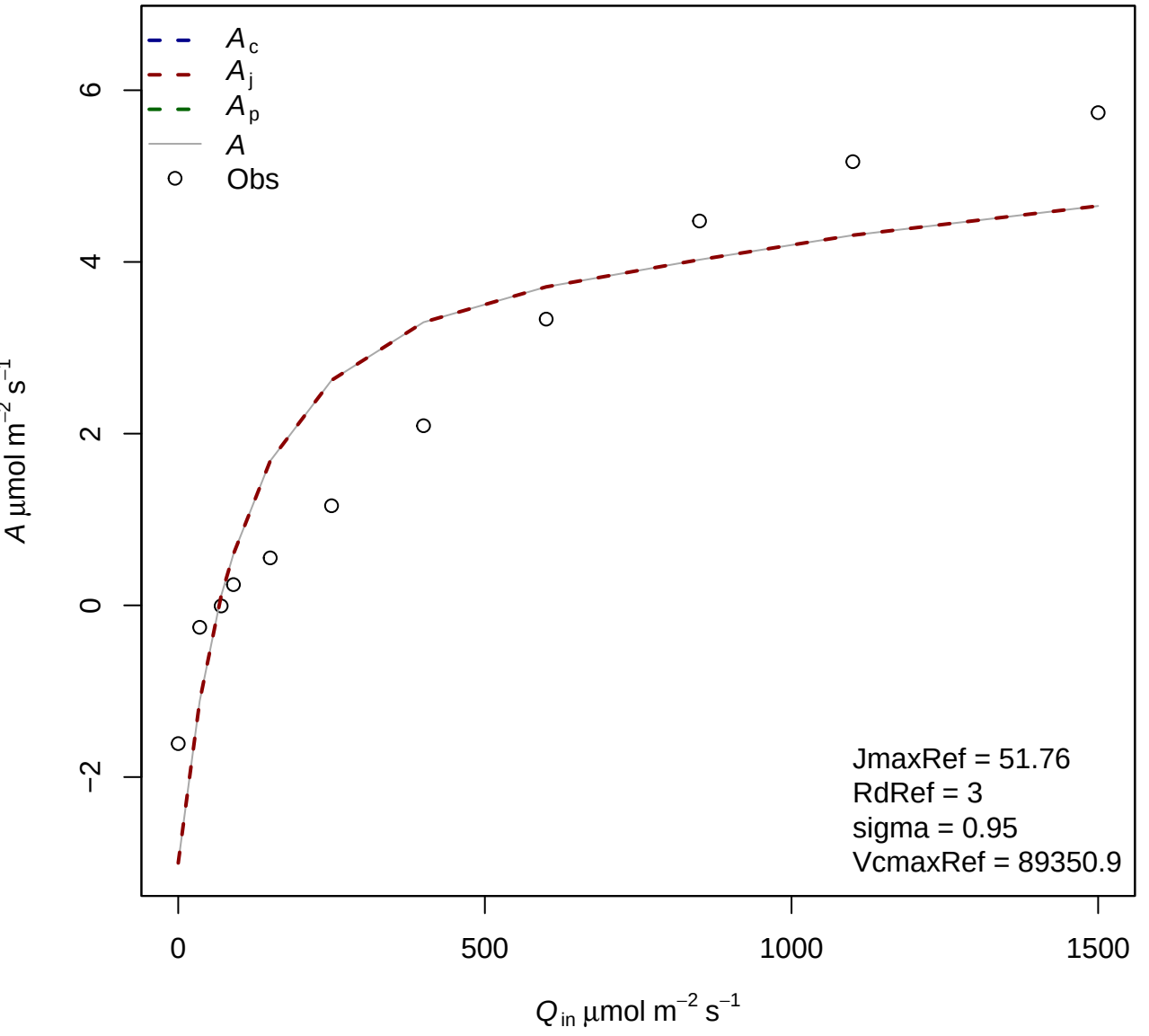

269
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.94
RdRef = 1.98
sigma = 0.04
VcmaxRef = 14.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2
1
0
−1
−2
200
400
600
800
1000
1200
1400

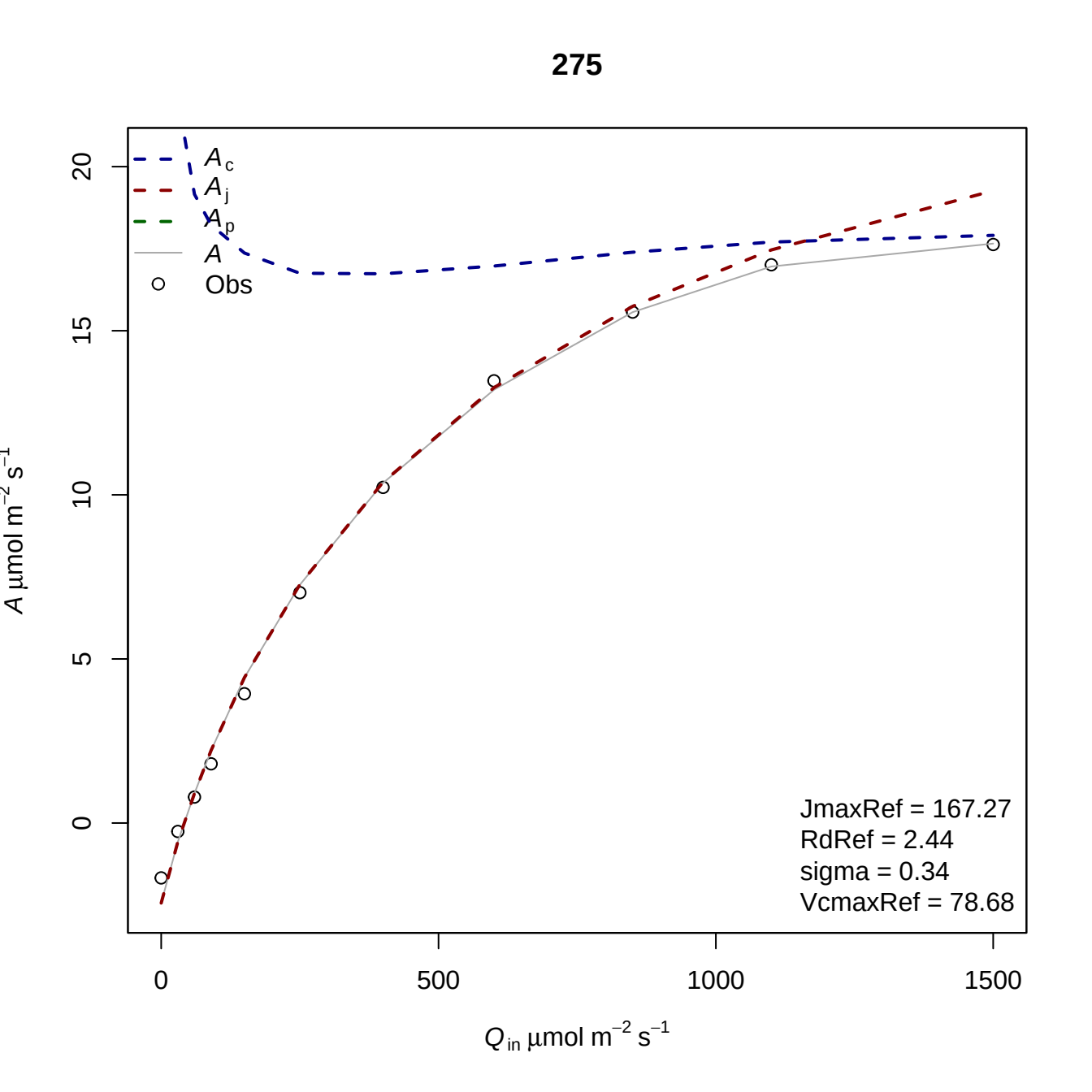

270
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 23.66
RdRef = 1.42
sigma = 0.12
VcmaxRef = 12.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
500
1000
1500

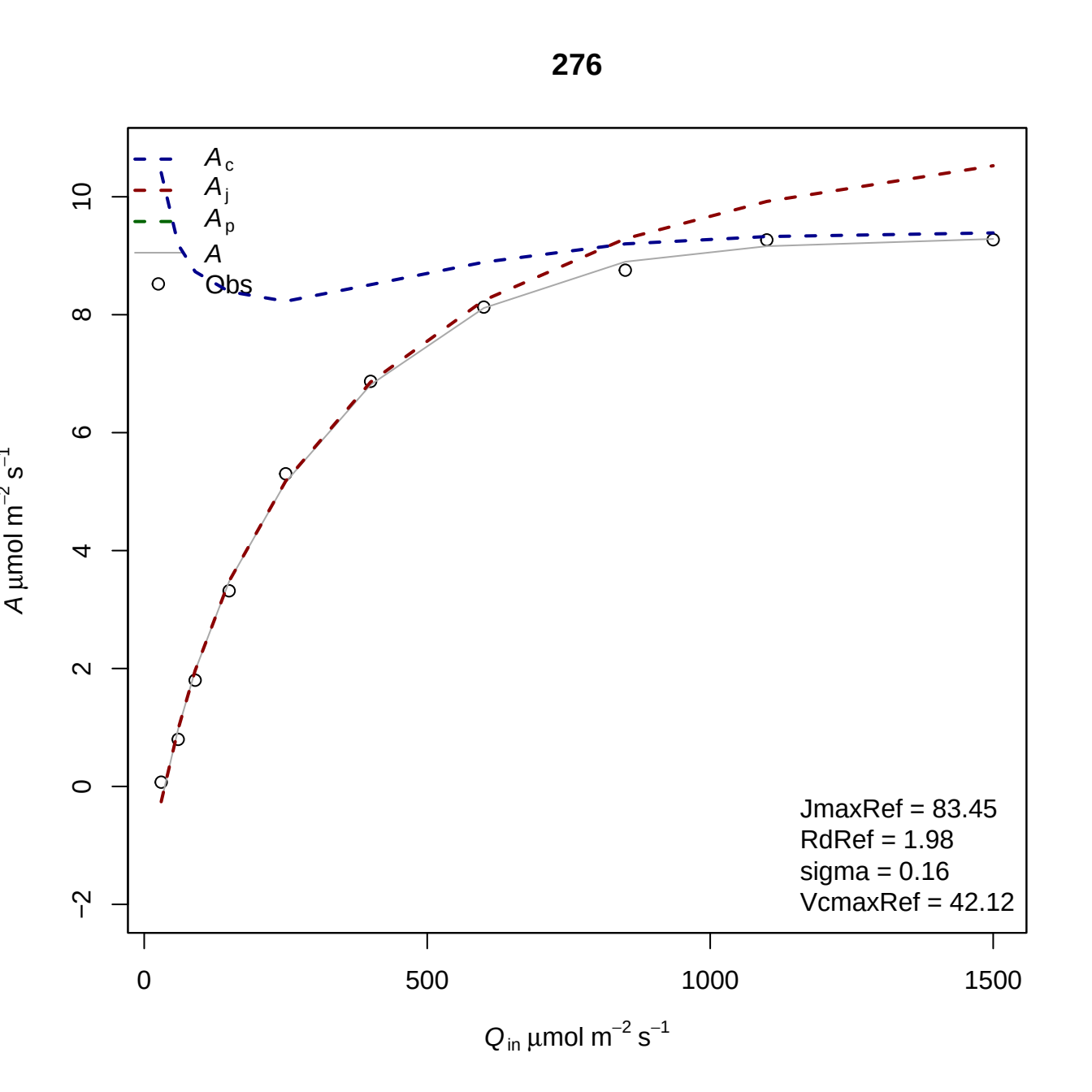

271
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.15
RdRef = 1.86
sigma = 0.22
VcmaxRef = 52.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

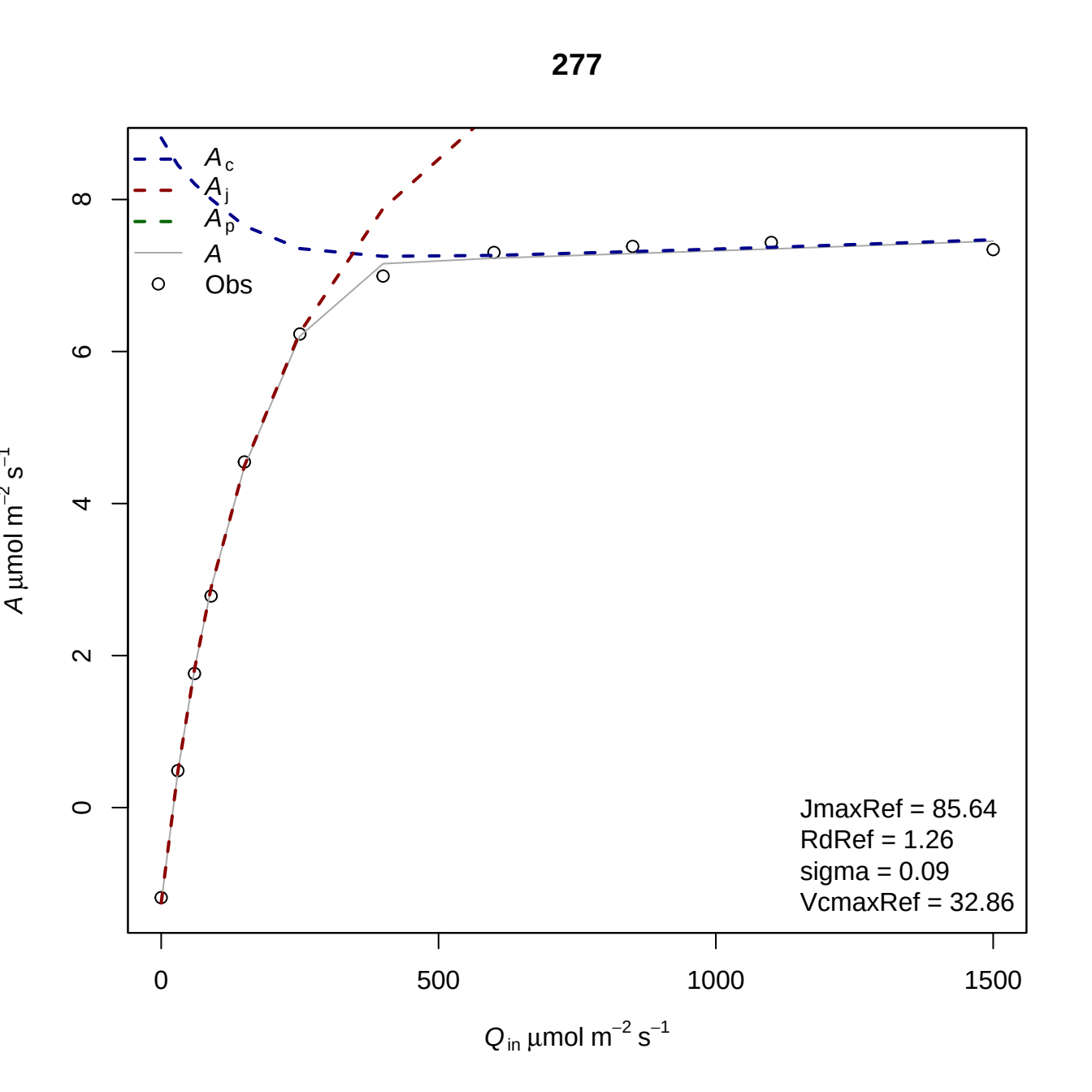

272
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.19
RdRef = 1.66
sigma = 0.19
VcmaxRef = 51.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500

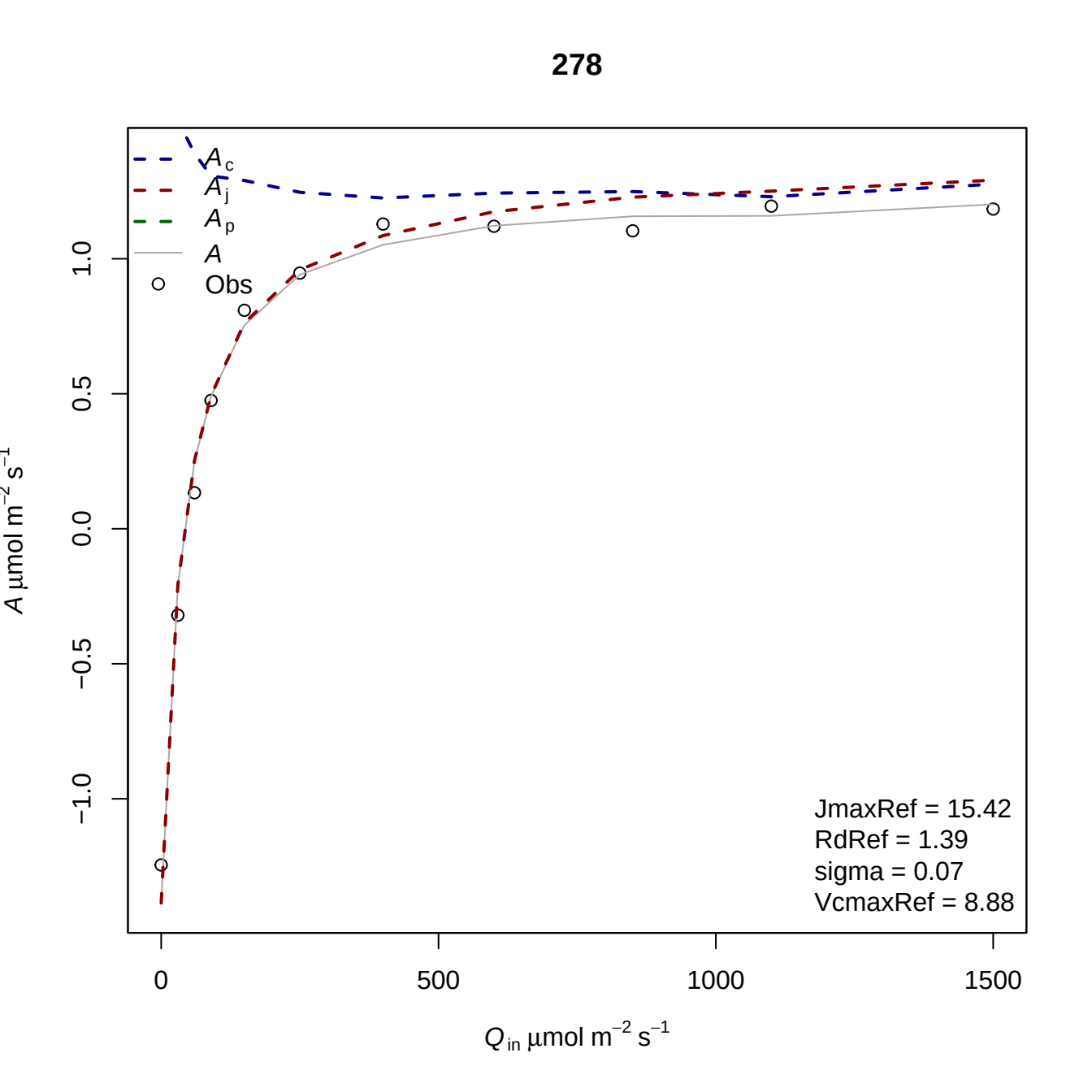

273
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.05
RdRef = 2.06
sigma = 0.32
VcmaxRef = 17795.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2
1
0
−1
−2
0
500
1000
1500

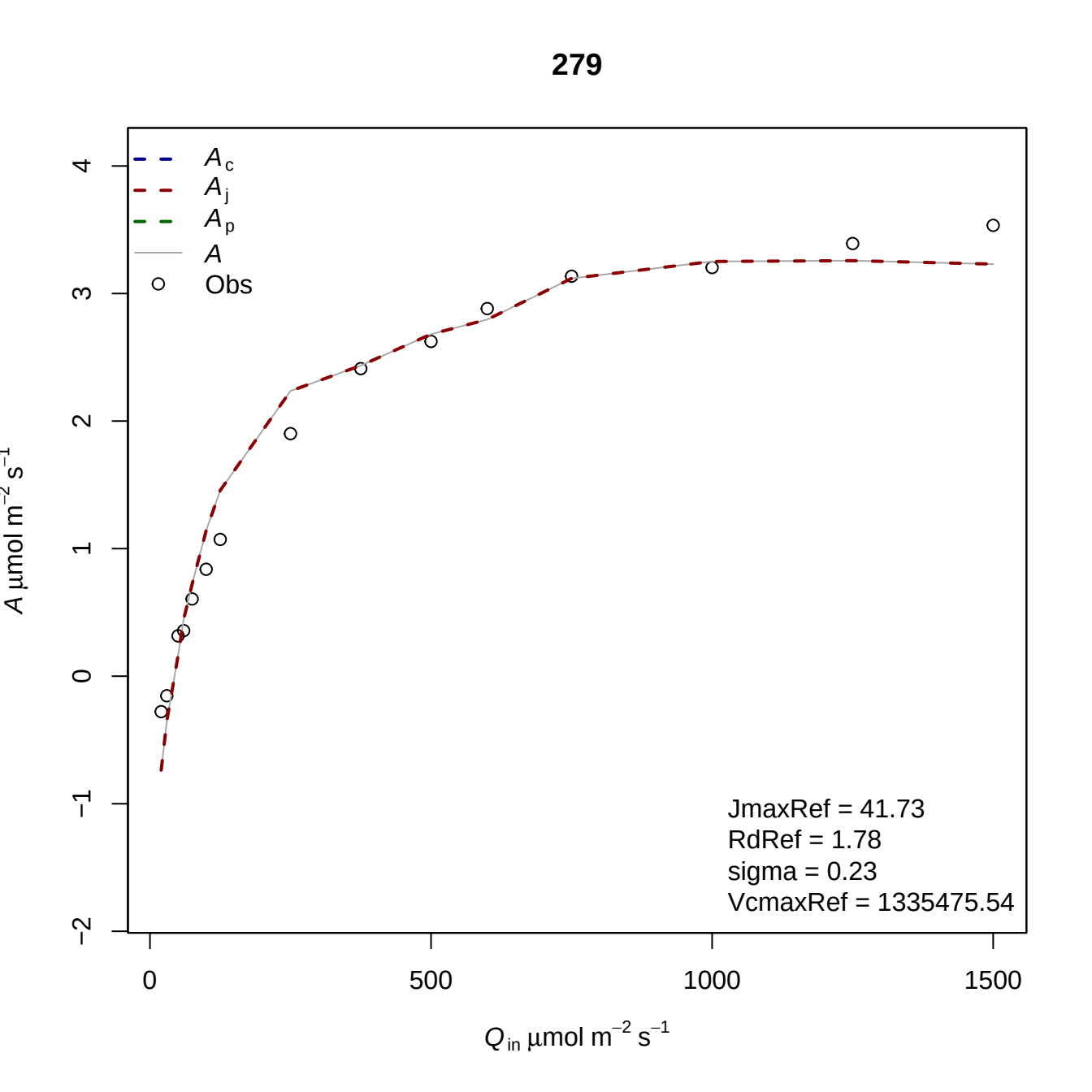

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.76
RdRef = 3
sigma = 0.95
VcmaxRef = 89350.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

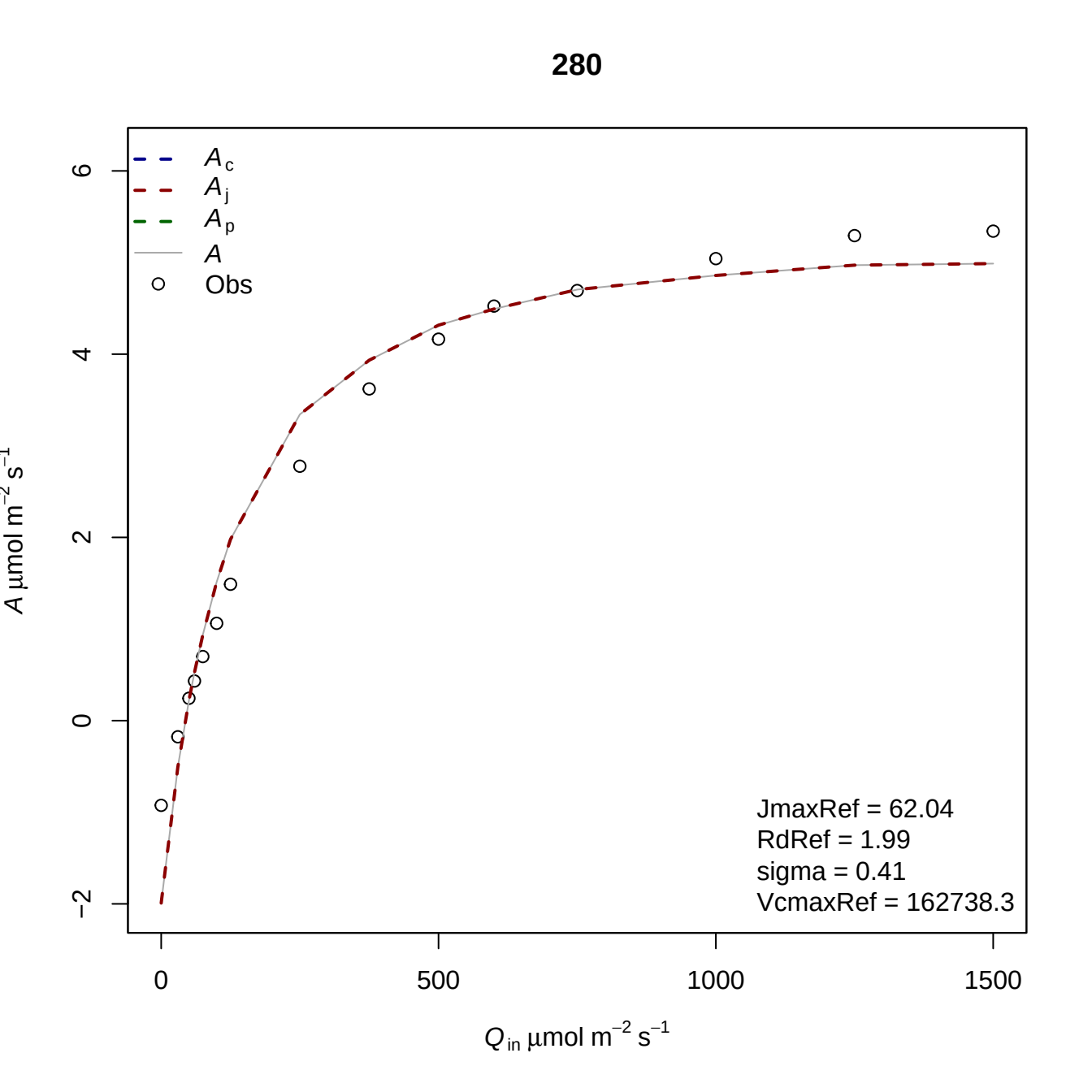

275
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 167.27
RdRef = 2.44
sigma = 0.34
VcmaxRef = 78.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

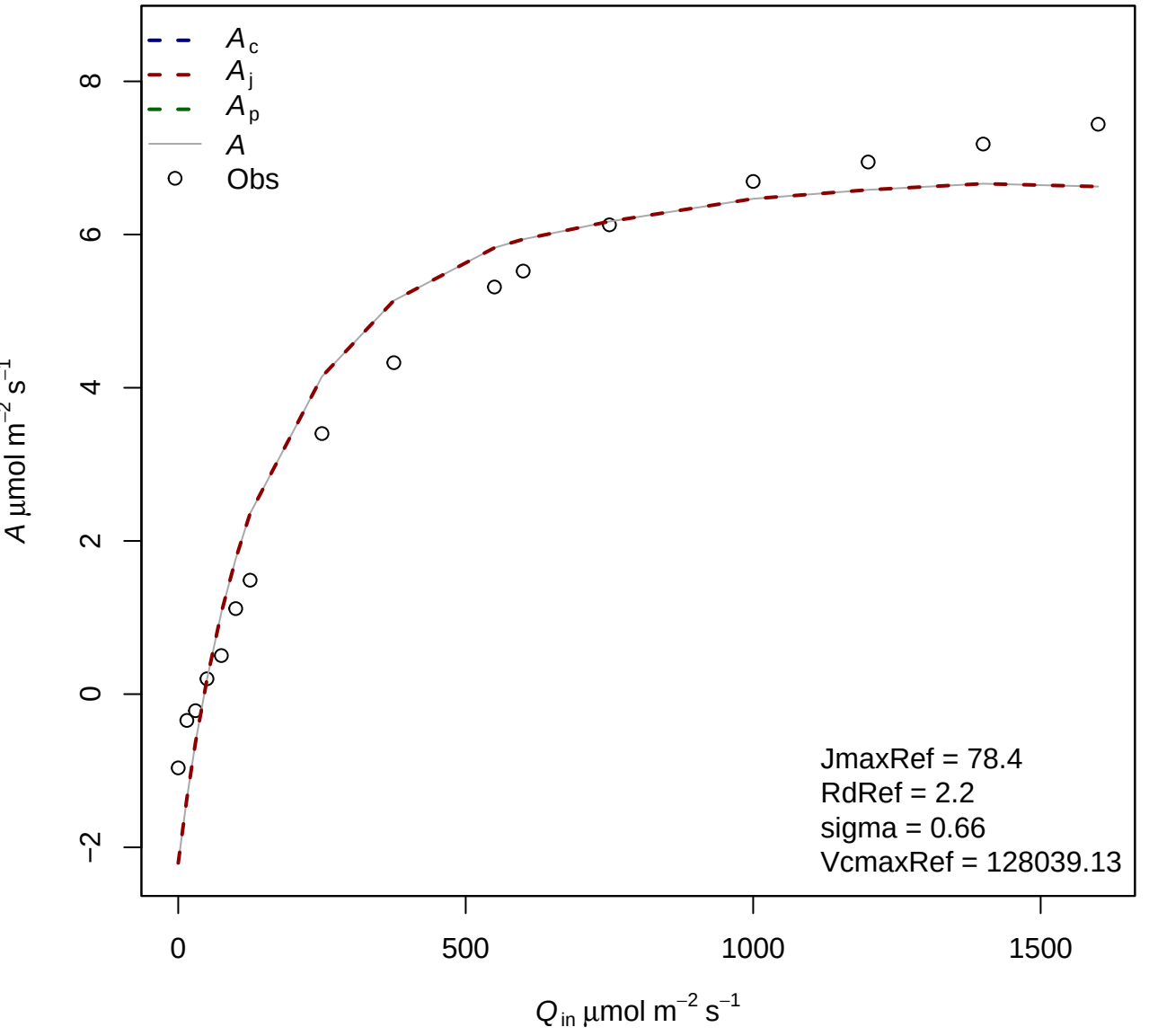

276
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.45
RdRef = 1.98
sigma = 0.16
VcmaxRef = 42.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

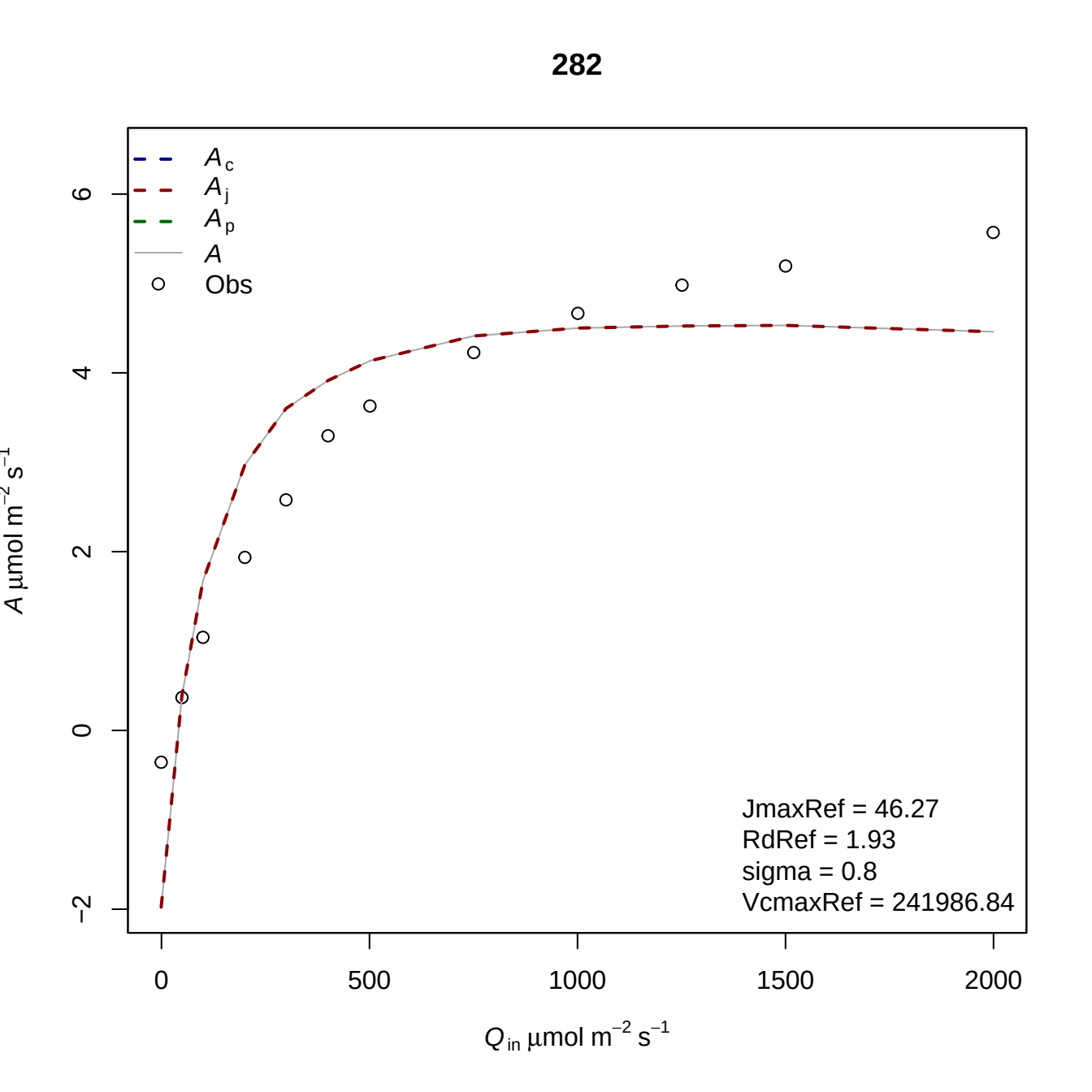

277
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.64
RdRef = 1.26
sigma = 0.09
VcmaxRef = 32.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

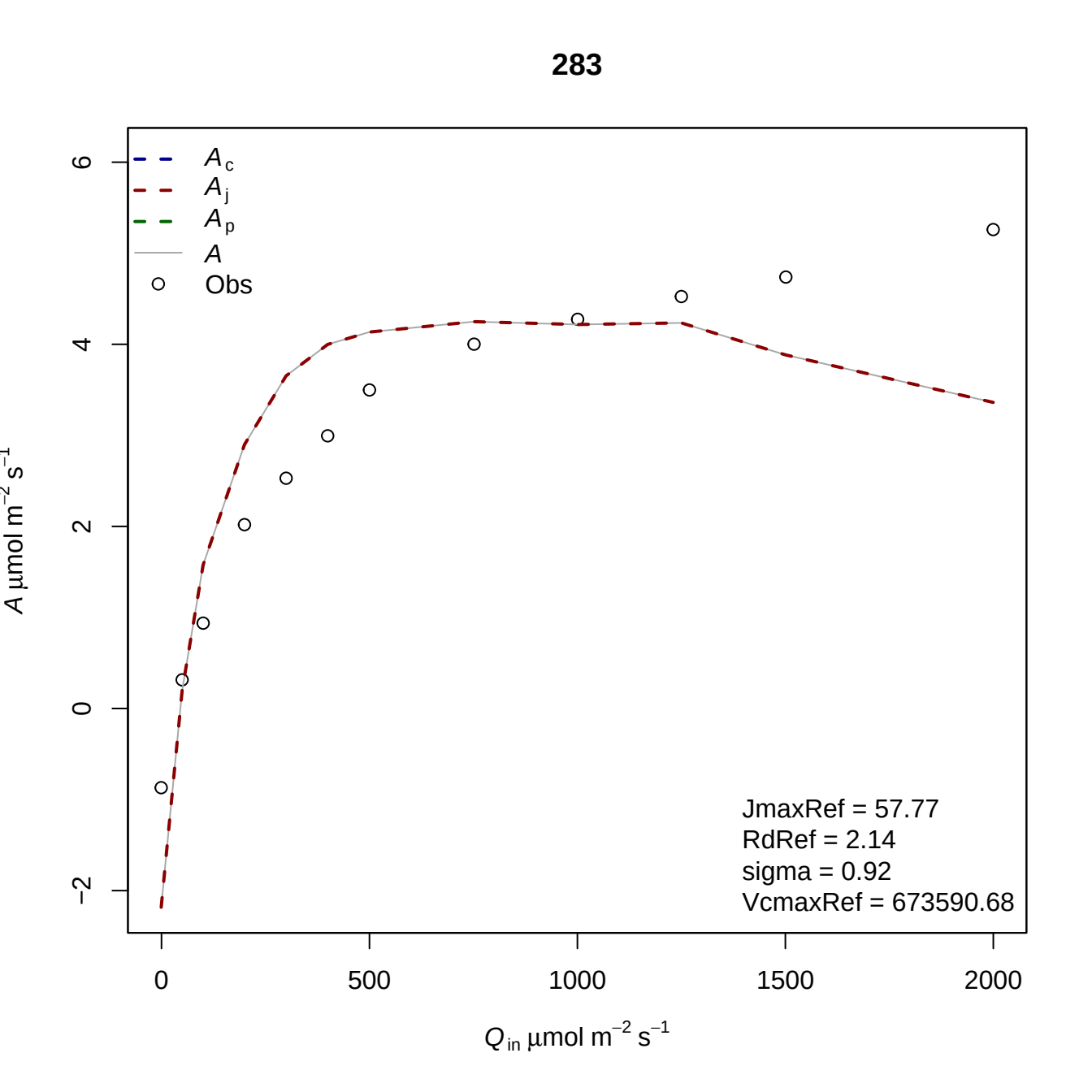

278
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 15.42
RdRef = 1.39
sigma = 0.07
VcmaxRef = 8.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

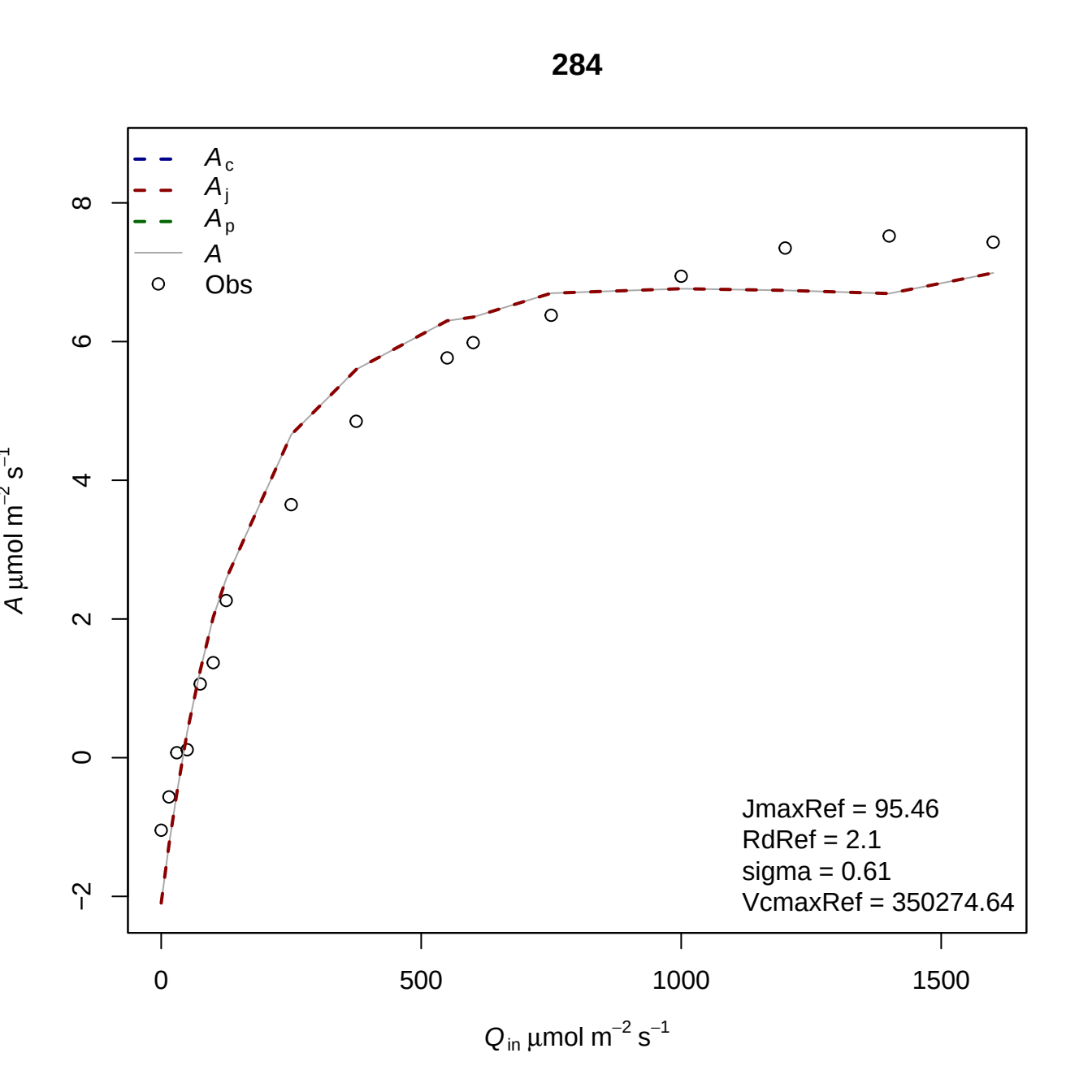

279
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.73
RdRef = 1.78
sigma = 0.23
VcmaxRef = 1335475.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
0
500
1000
1500

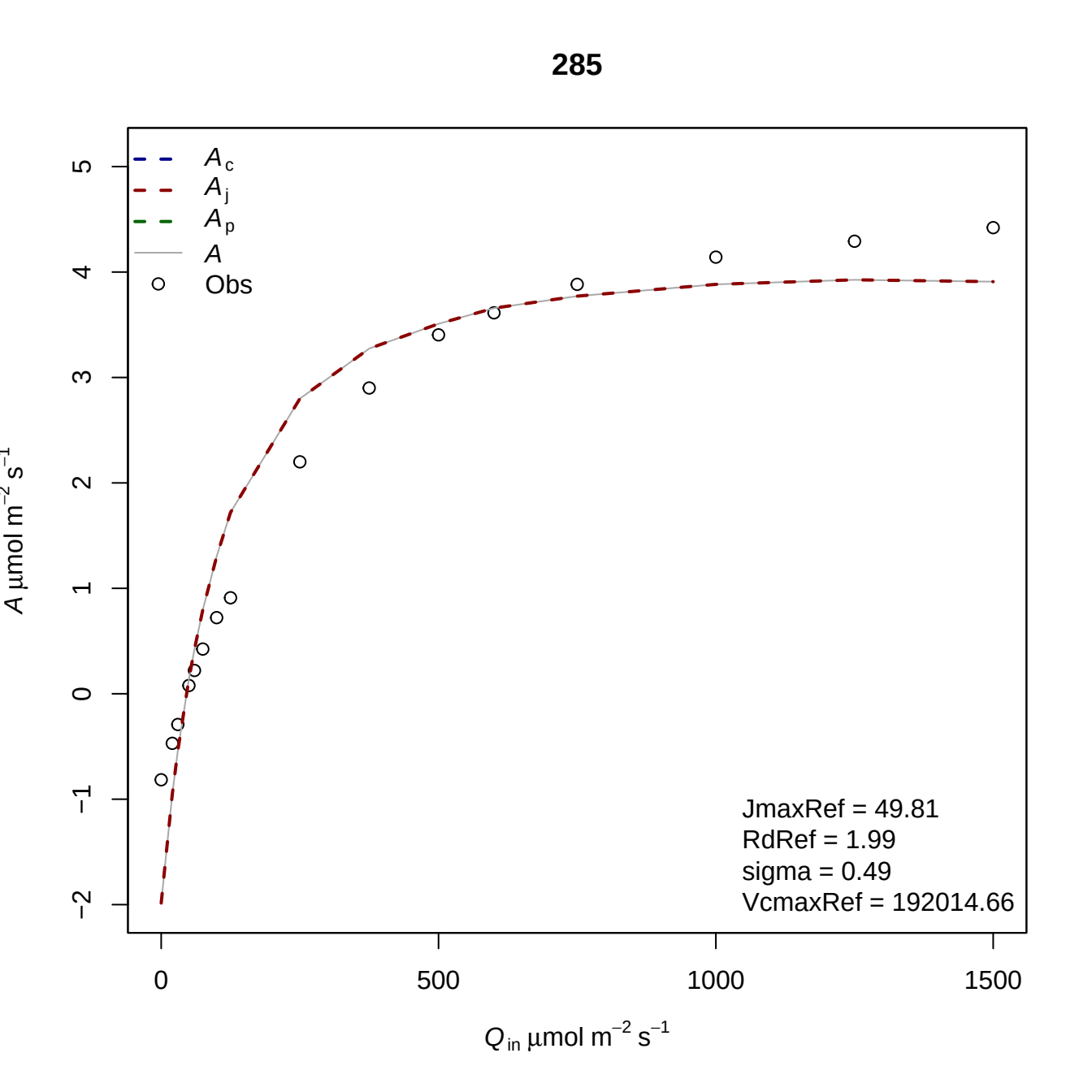

280
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.04
RdRef = 1.99
sigma = 0.41
VcmaxRef = 162738.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

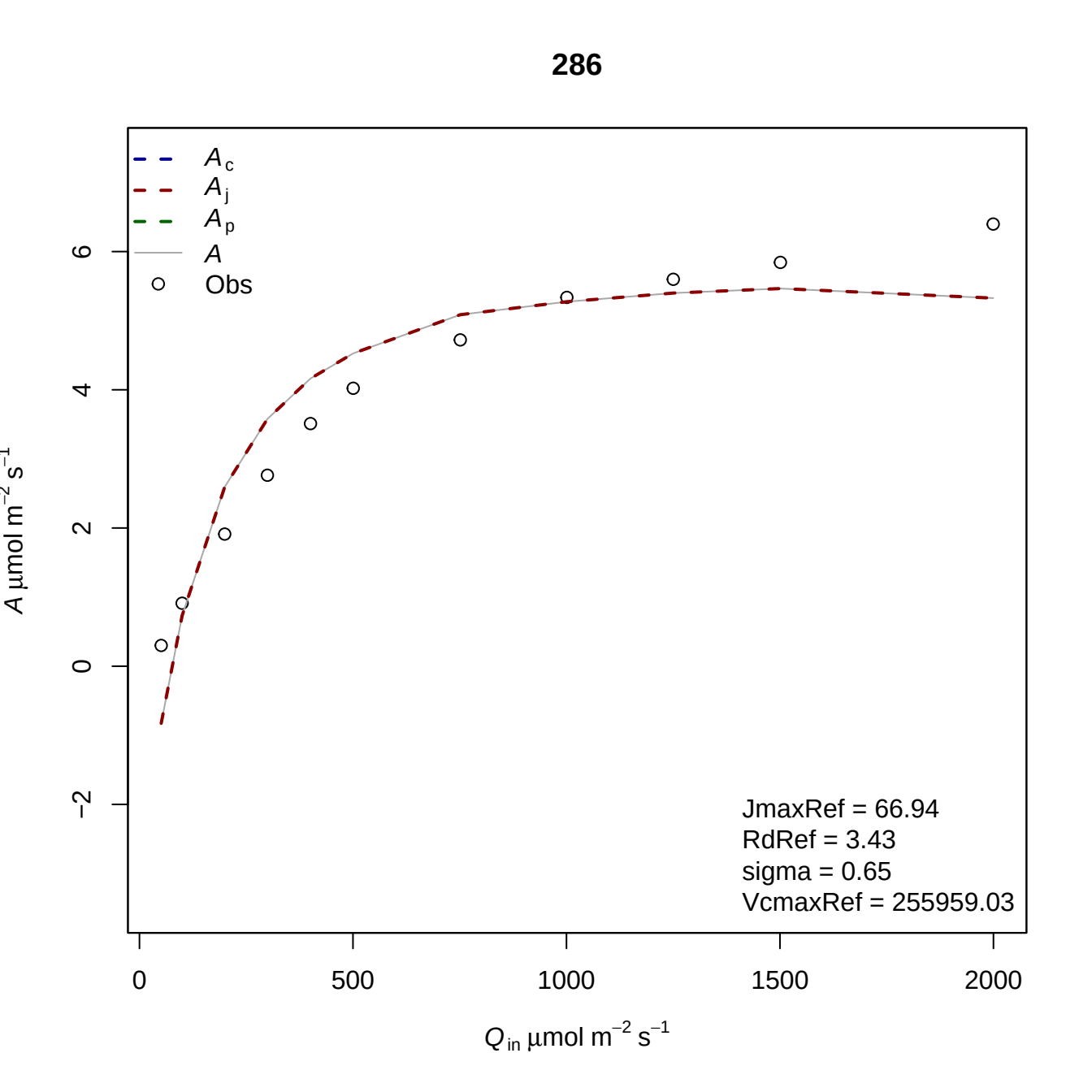

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.4
RdRef = 2.2
sigma = 0.66
VcmaxRef = 128039.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

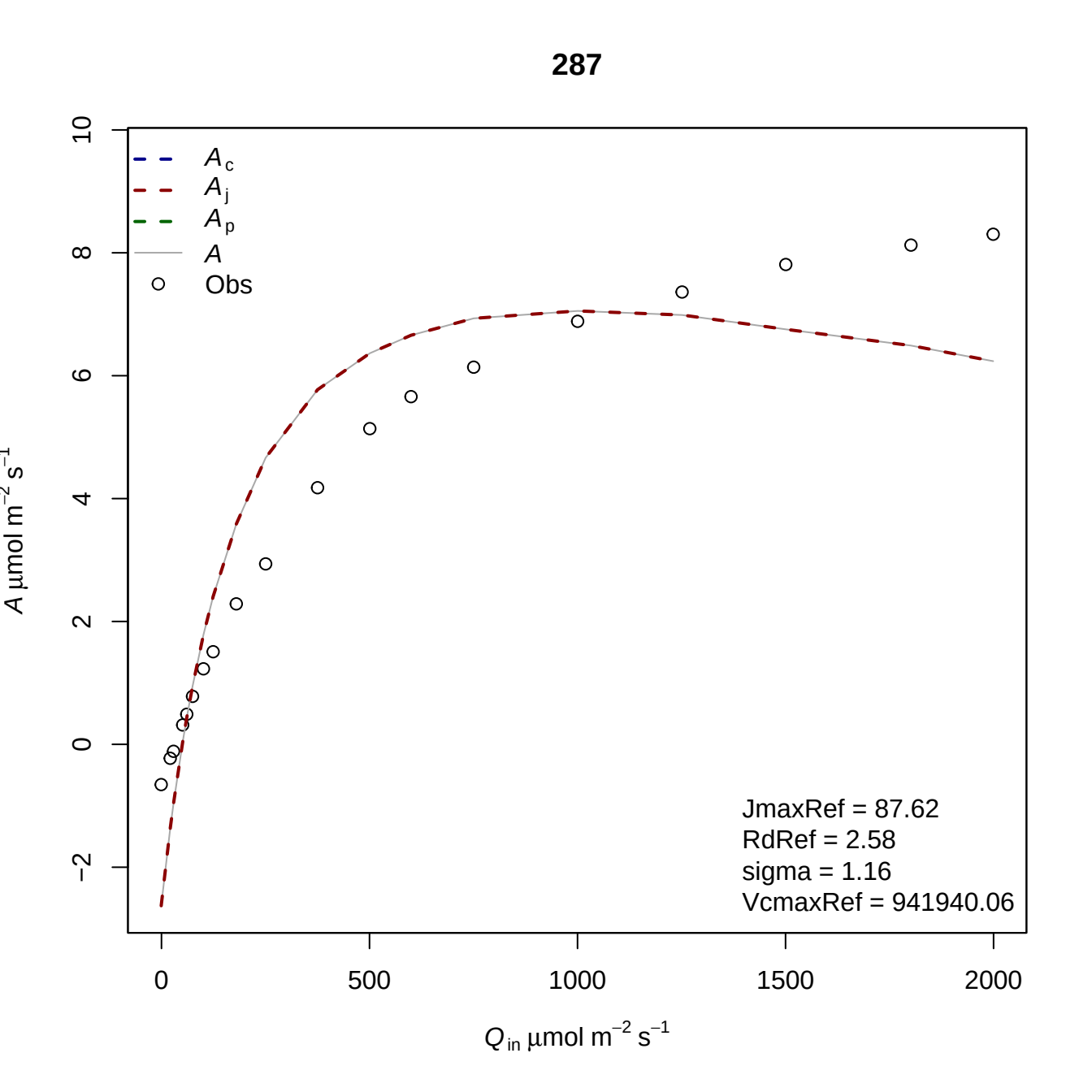

282
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.27
RdRef = 1.93
sigma = 0.8
VcmaxRef = 241986.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

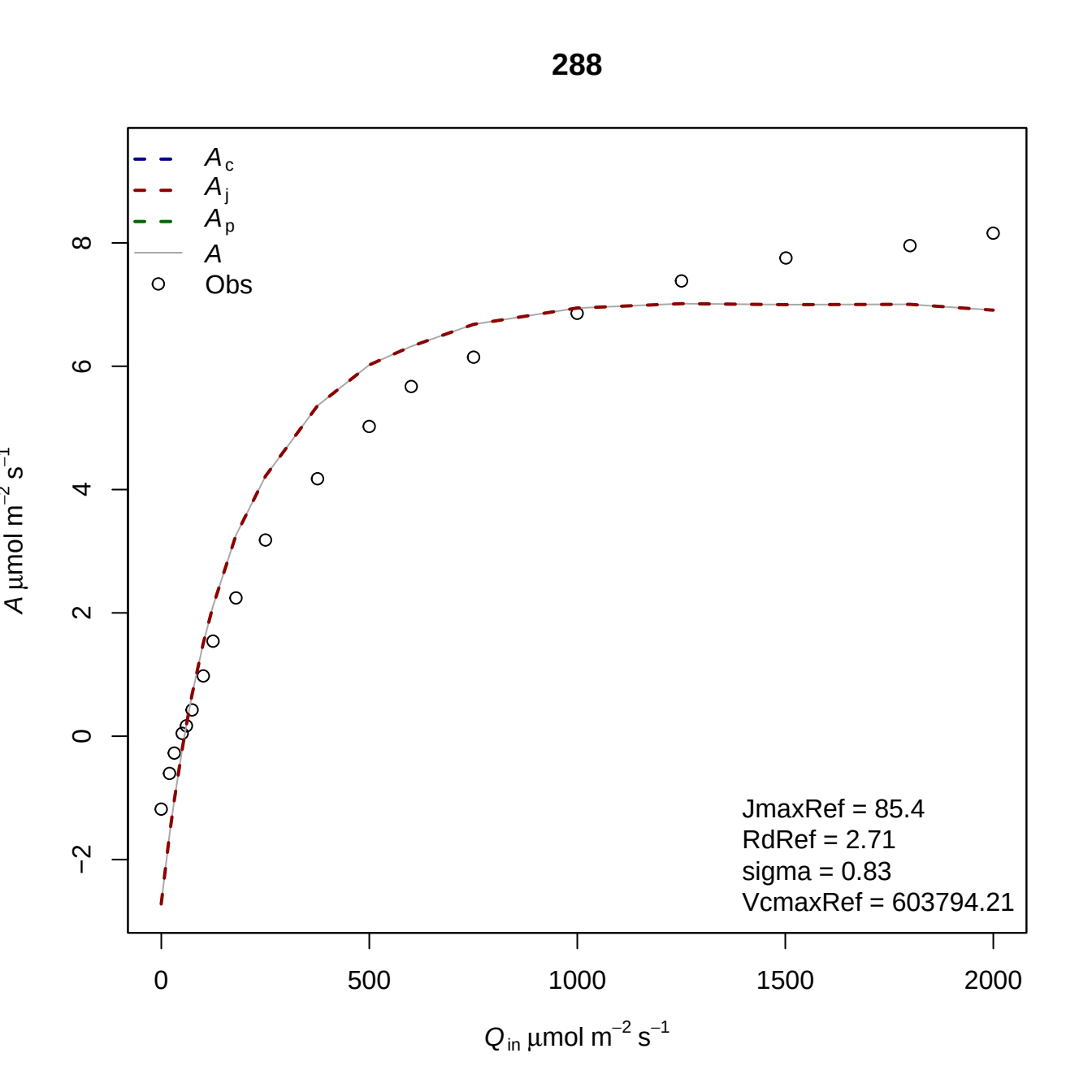

283
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.77
RdRef = 2.14
sigma = 0.92
VcmaxRef = 673590.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

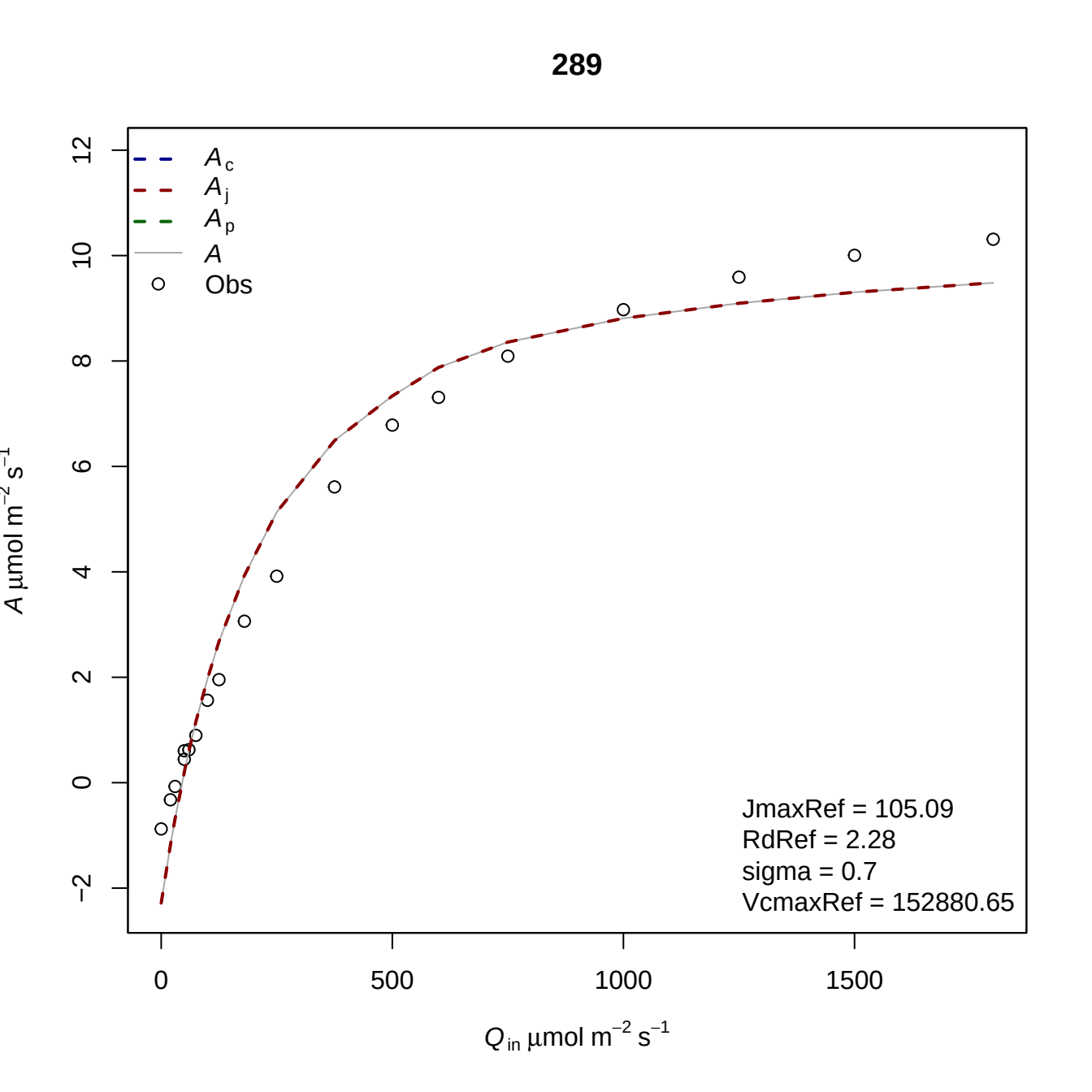

284
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.46
RdRef = 2.1
sigma = 0.61
VcmaxRef = 350274.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

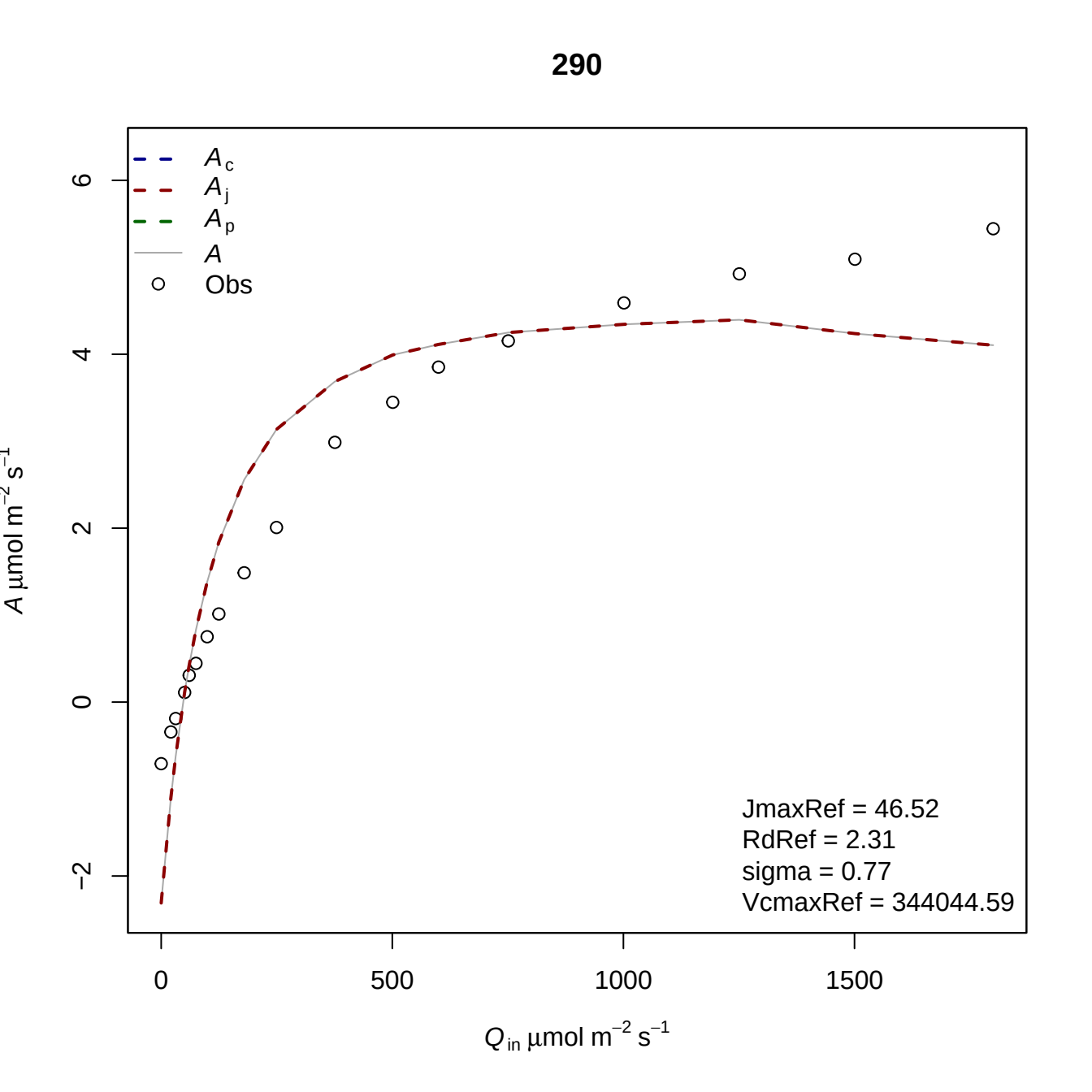

285
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.81
RdRef = 1.99
sigma = 0.49
VcmaxRef = 192014.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
5
4
3
2
1
0
−1
−2
0
500
1000
1500

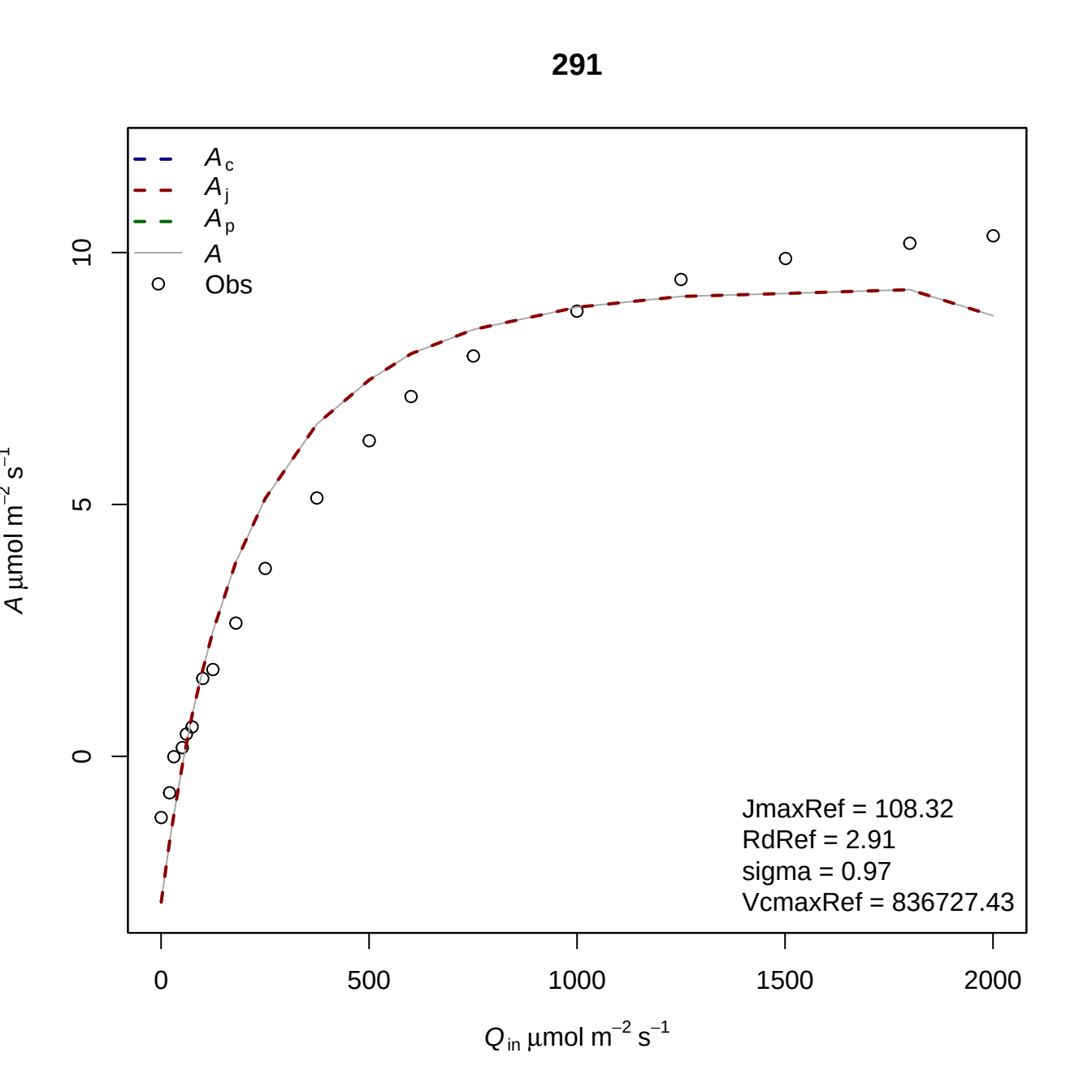

286
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.94
RdRef = 3.43
sigma = 0.65
VcmaxRef = 255959.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

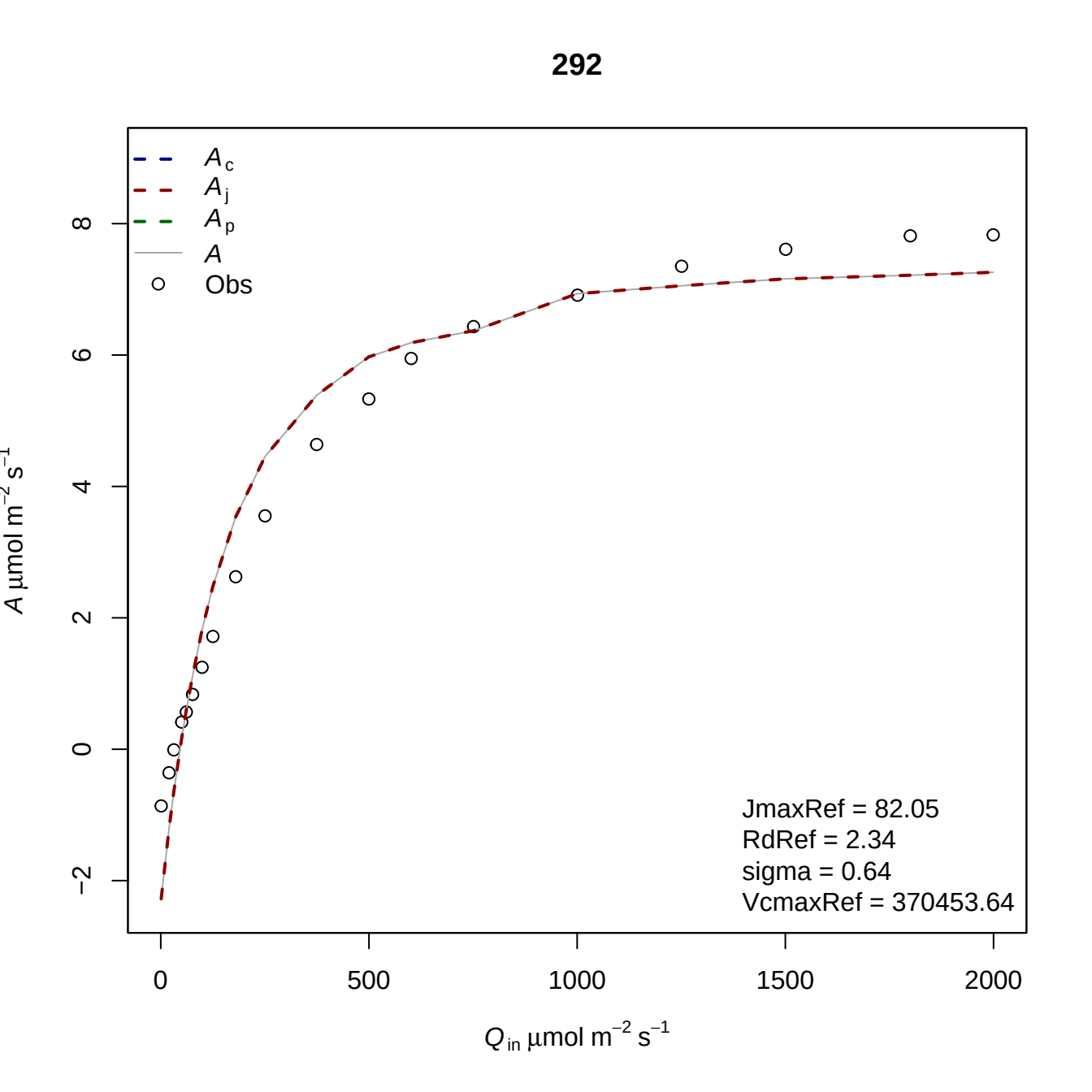

287
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.62
RdRef = 2.58
sigma = 1.16
VcmaxRef = 941940.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

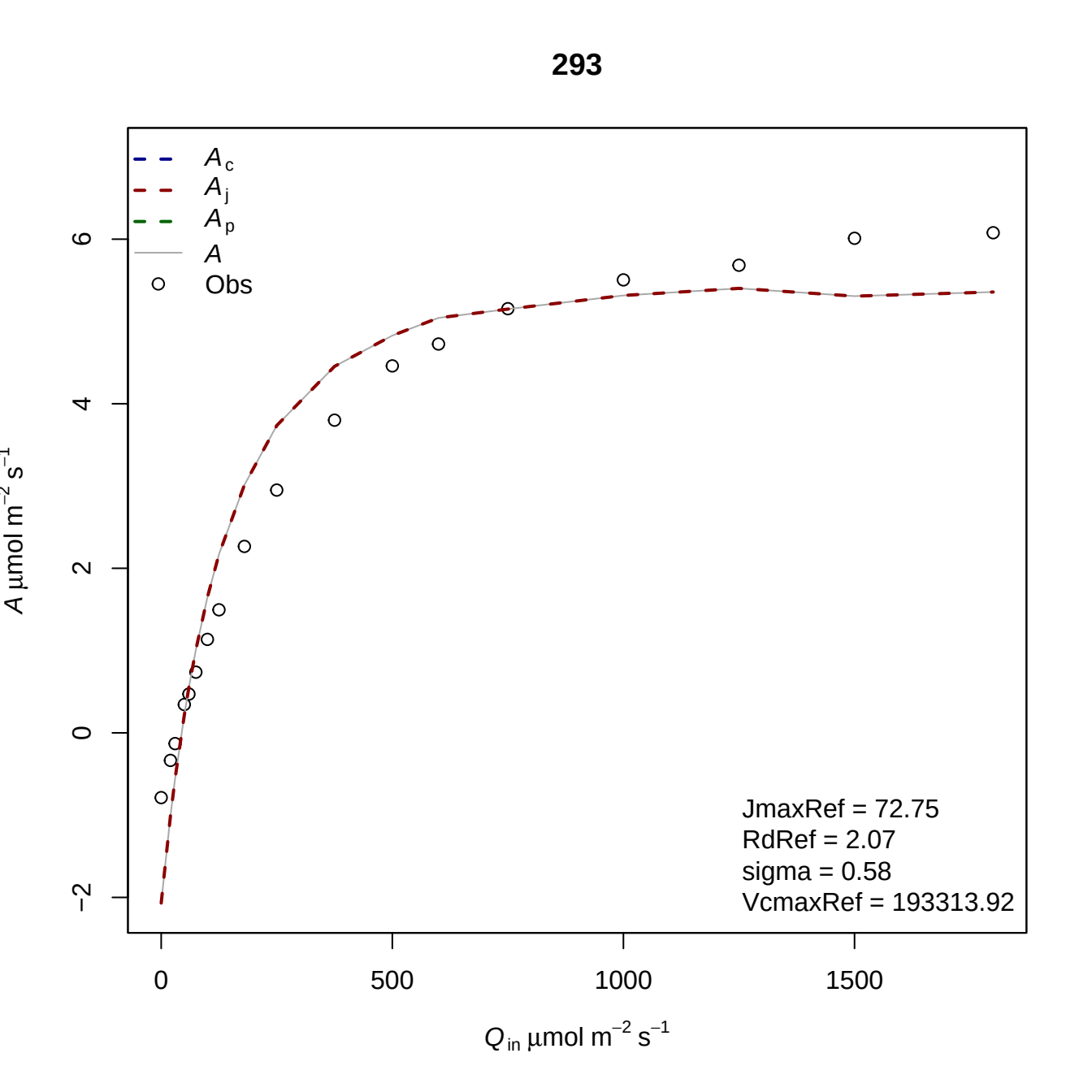

288
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.4
RdRef = 2.71
sigma = 0.83
VcmaxRef = 603794.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

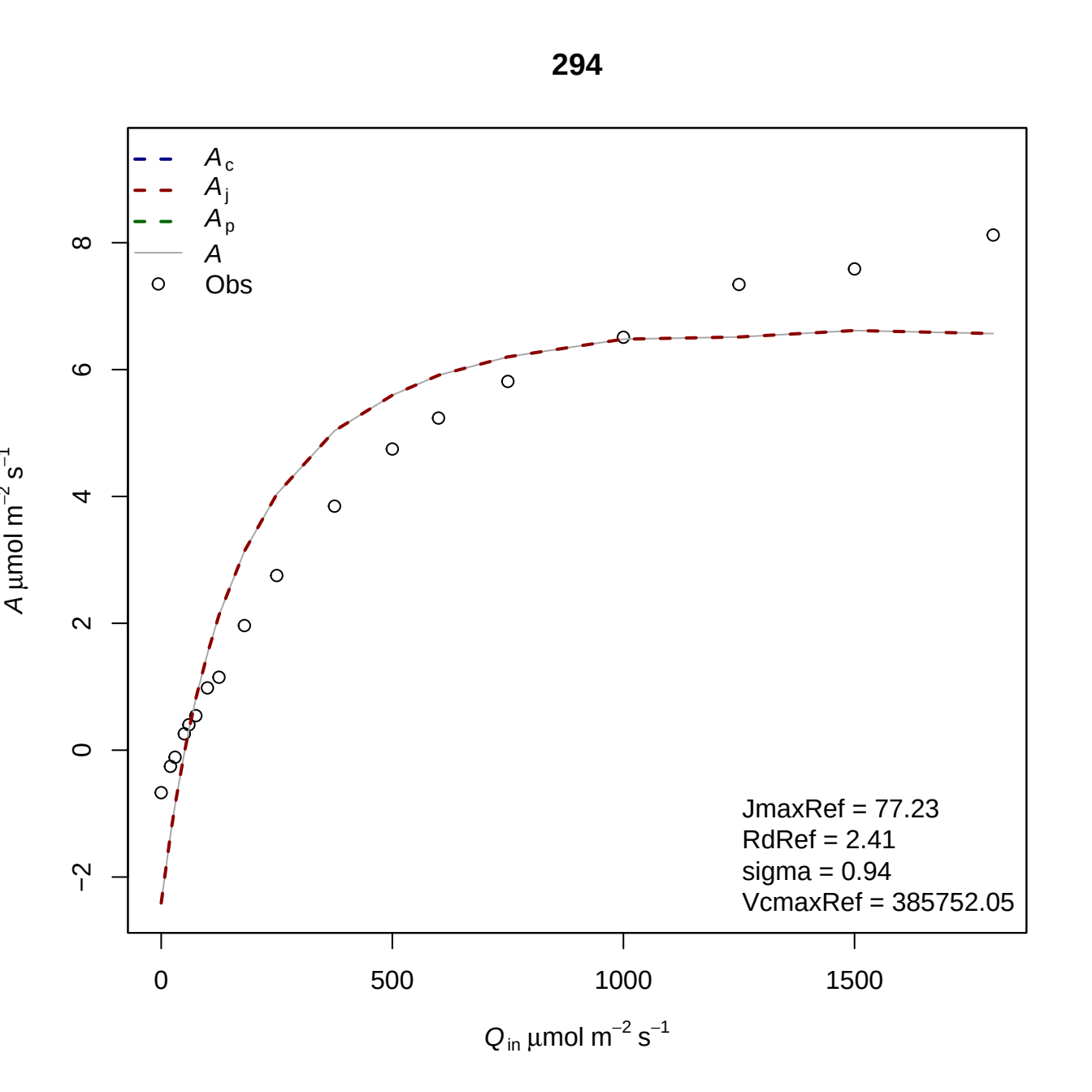

289
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.09
RdRef = 2.28
sigma = 0.7
VcmaxRef = 152880.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500

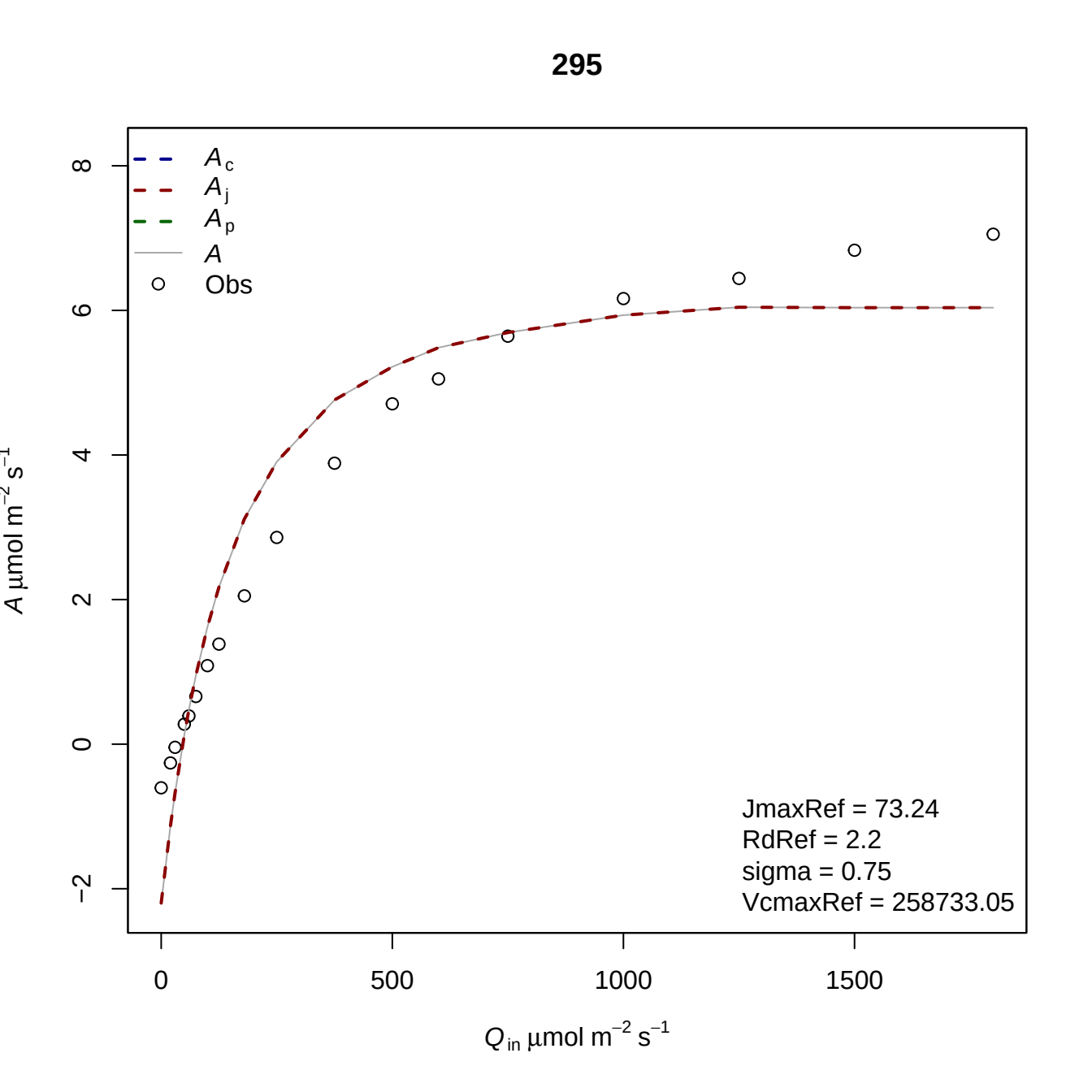

290
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.52
RdRef = 2.31
sigma = 0.77
VcmaxRef = 344044.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

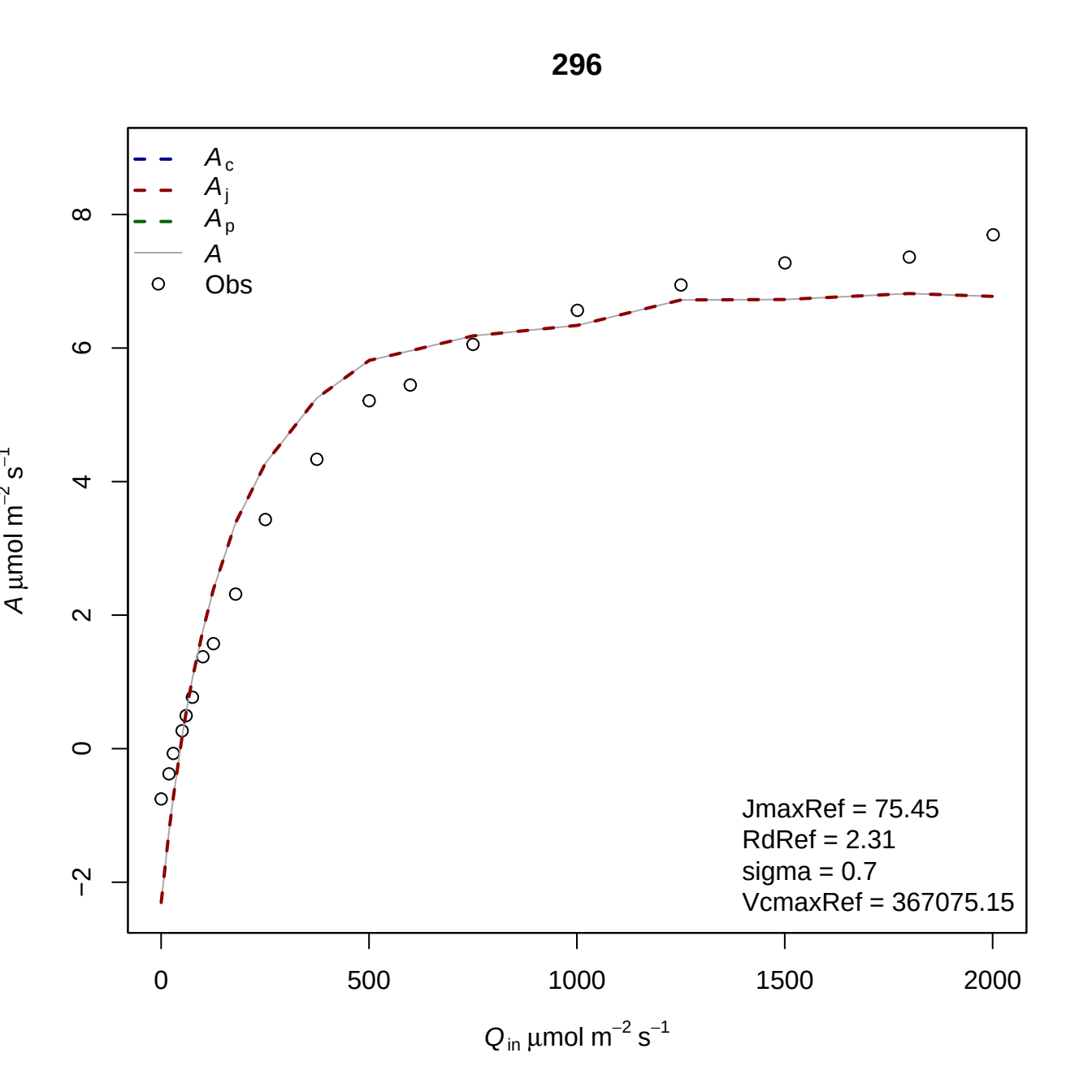

291
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.32
RdRef = 2.91
sigma = 0.97
VcmaxRef = 836727.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

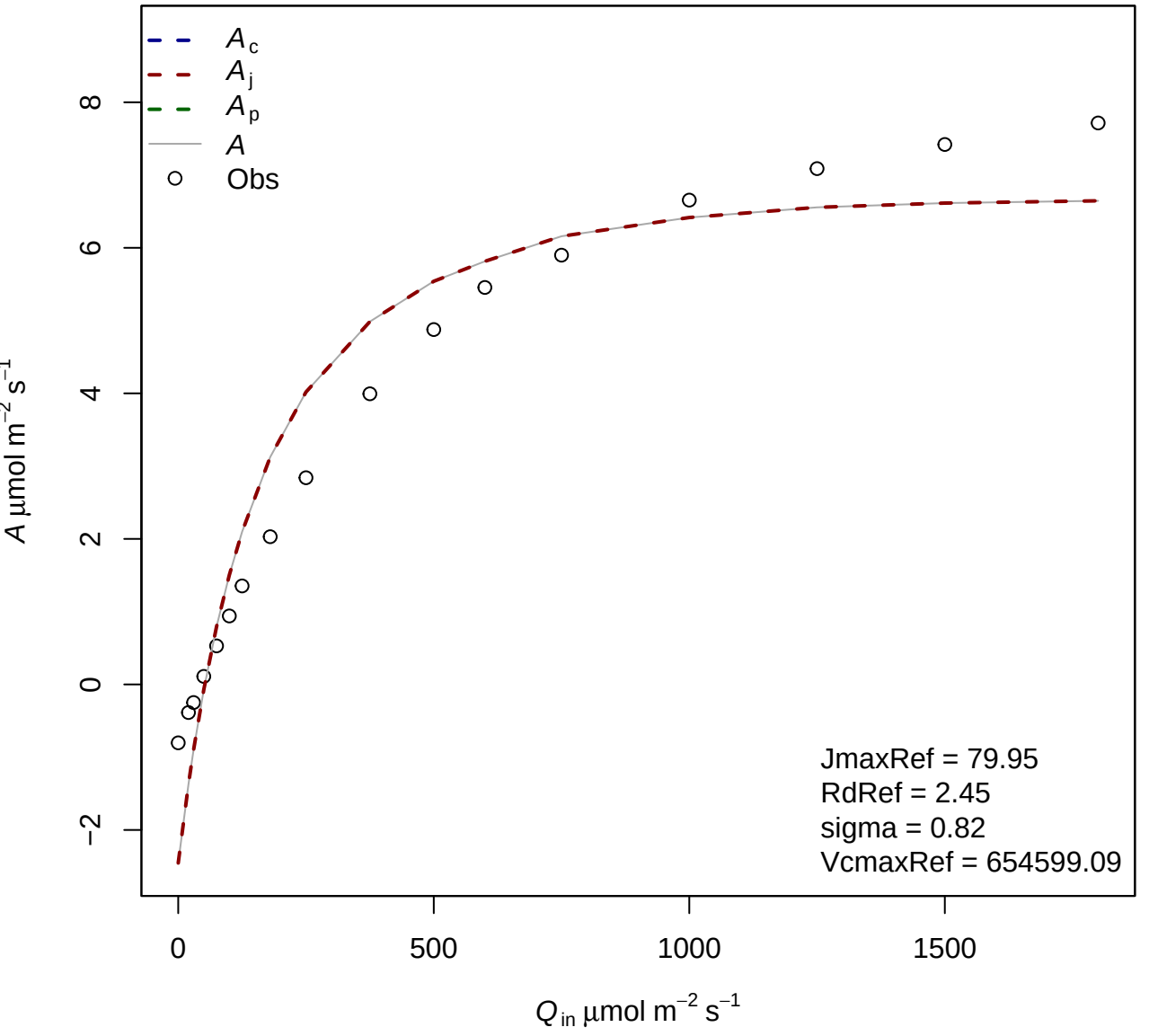

292
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.05
RdRef = 2.34
sigma = 0.64
VcmaxRef = 370453.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

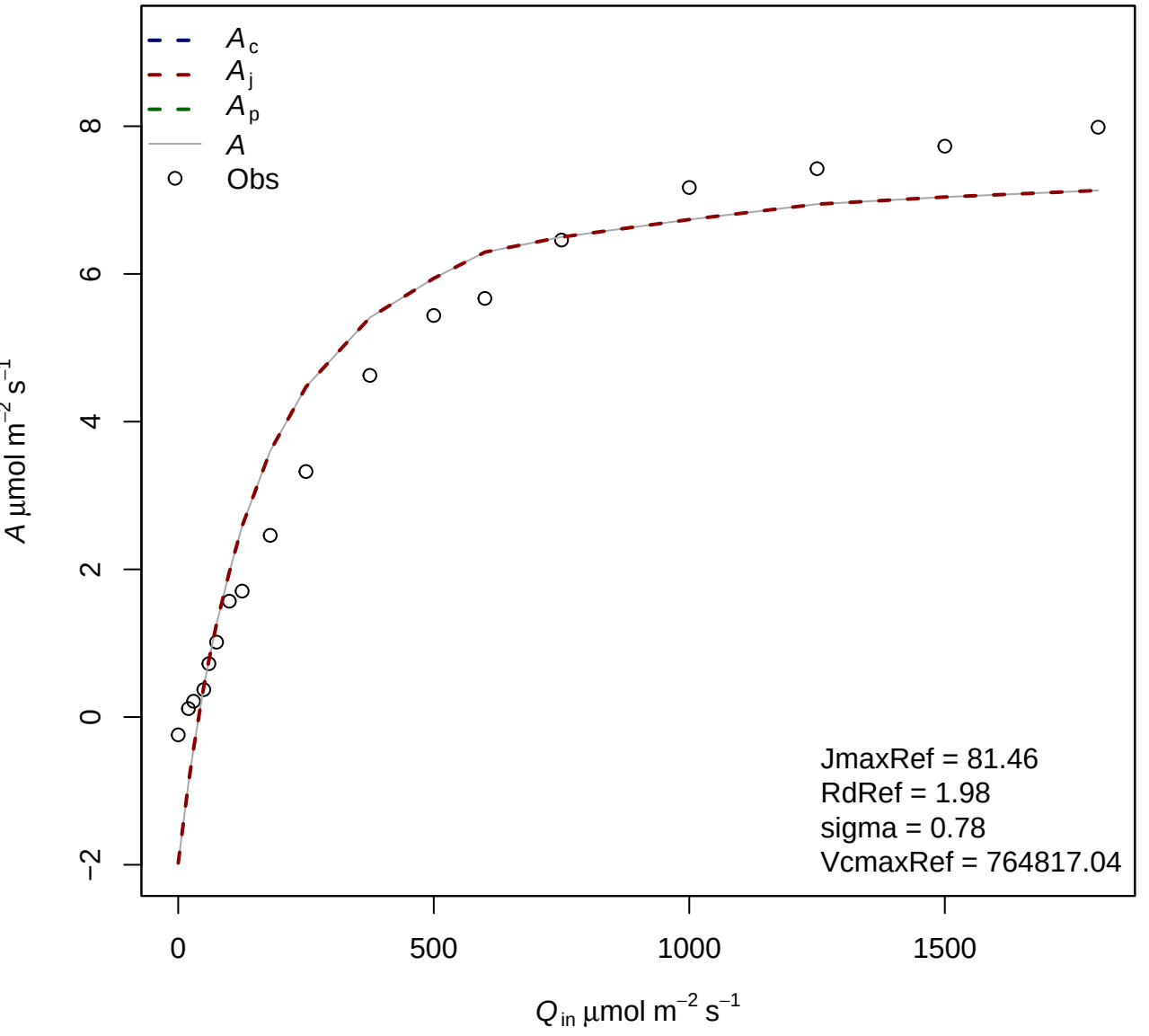

293
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.75
RdRef = 2.07
sigma = 0.58
VcmaxRef = 193313.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

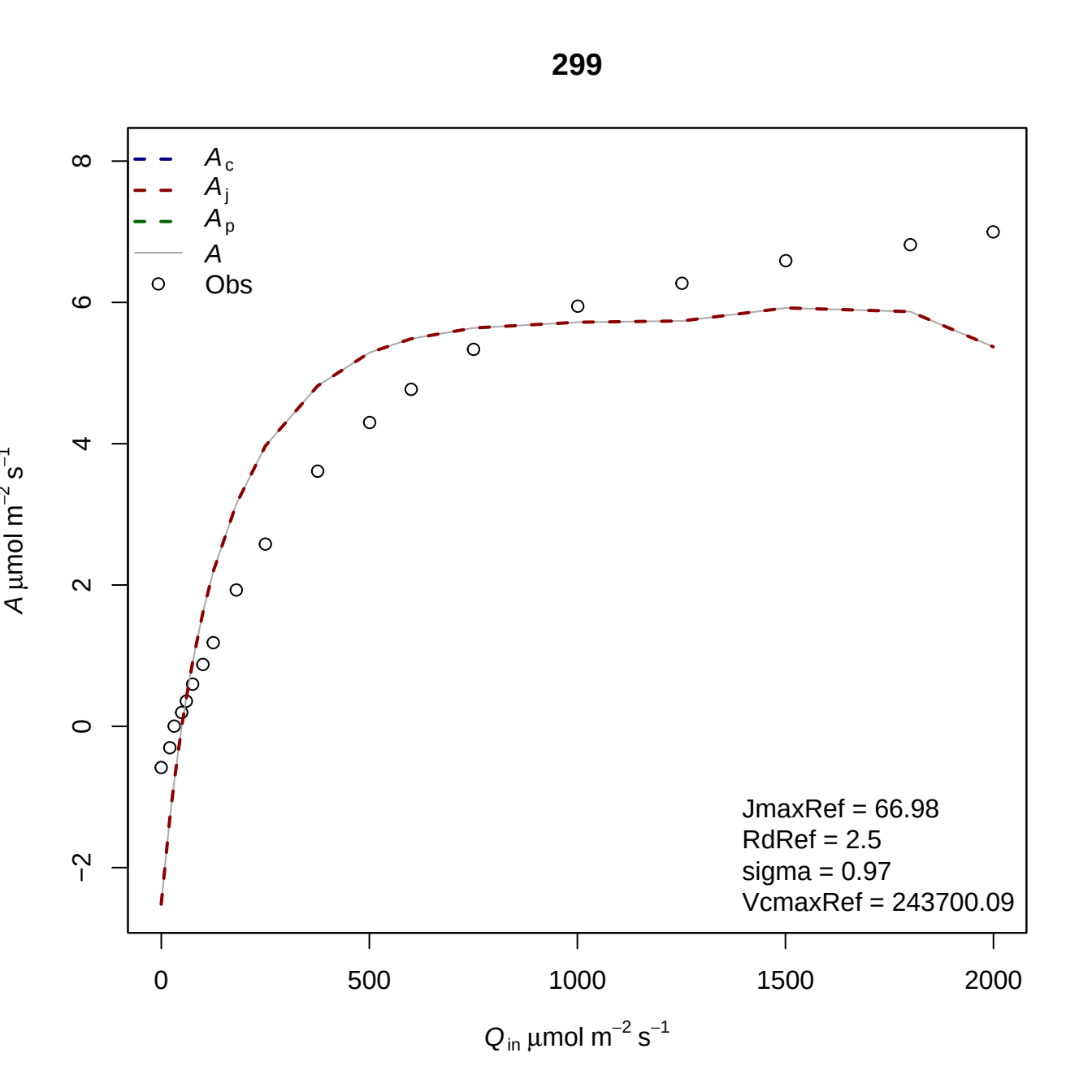

294
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.23
RdRef = 2.41
sigma = 0.94
VcmaxRef = 385752.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

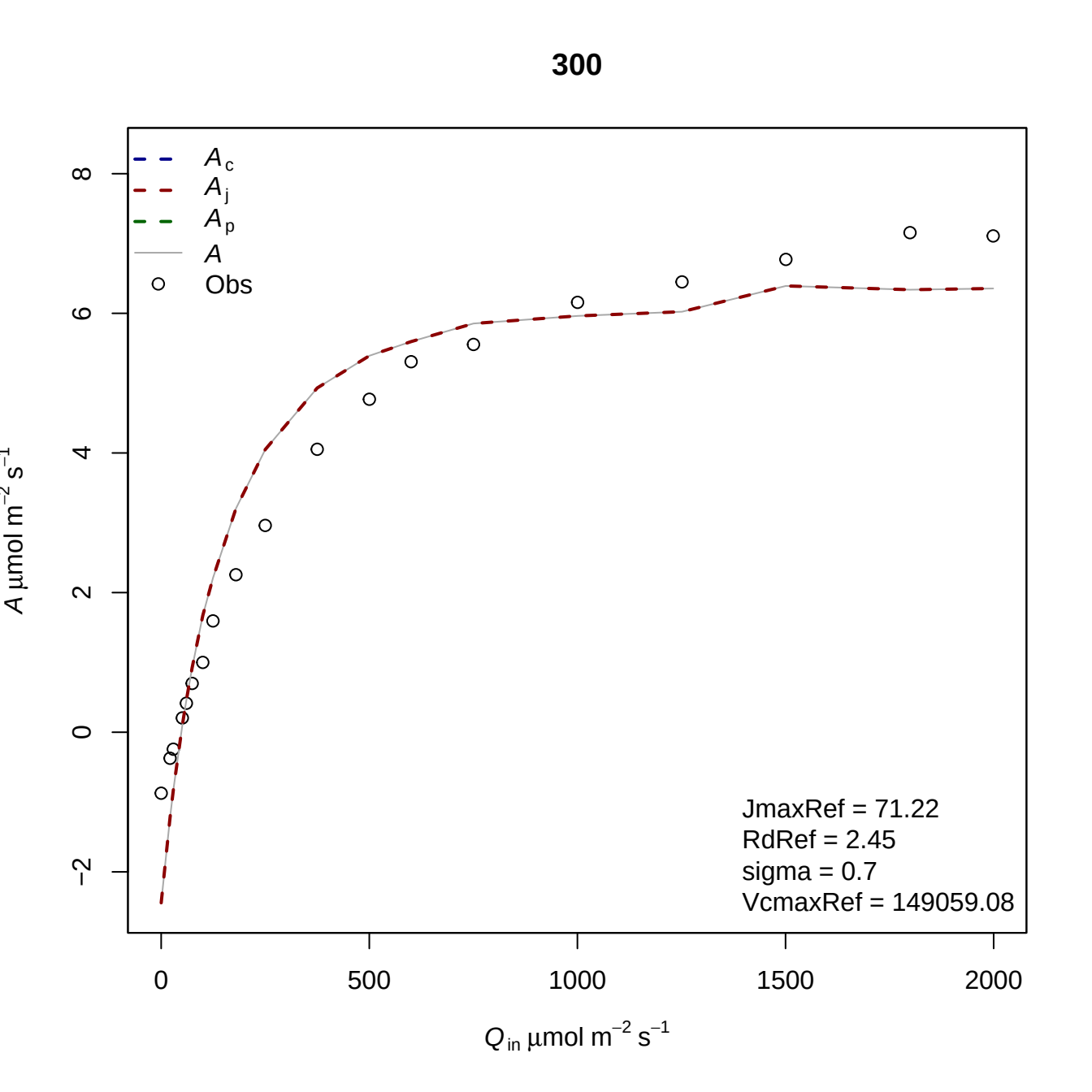

295
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.24
RdRef = 2.2
sigma = 0.75
VcmaxRef = 258733.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

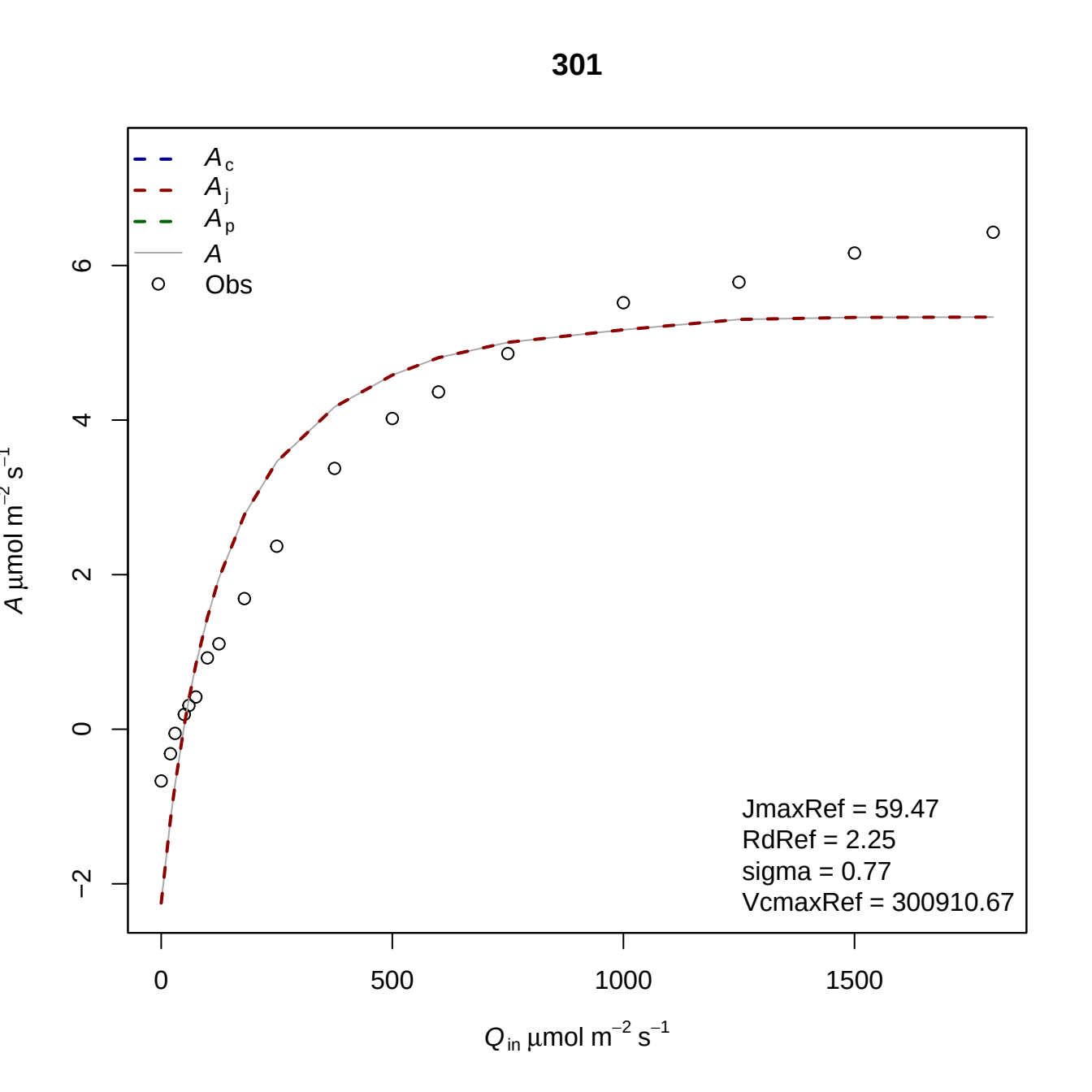

296
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.45
RdRef = 2.31
sigma = 0.7
VcmaxRef = 367075.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

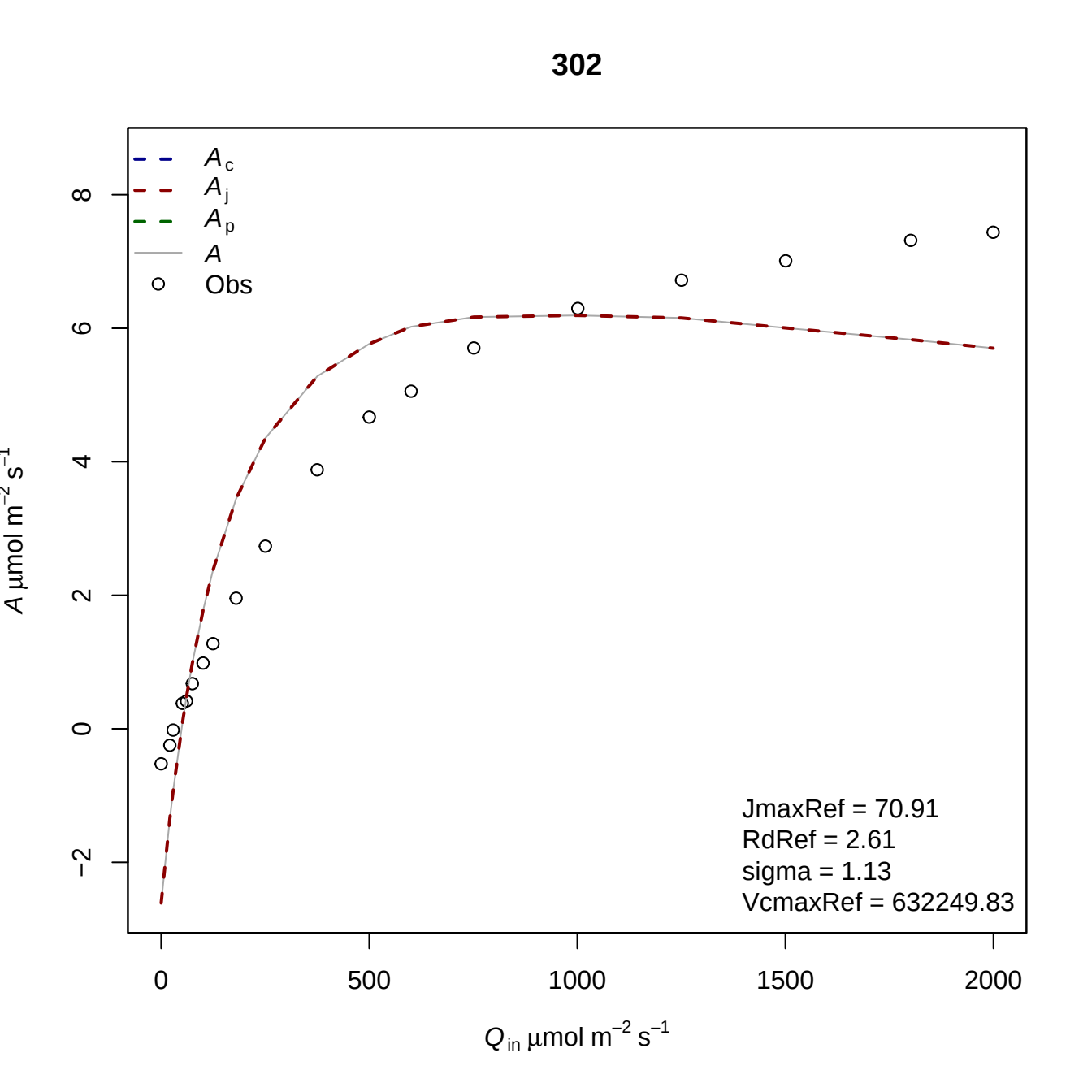

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.95
RdRef = 2.45
sigma = 0.82
VcmaxRef = 654599.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

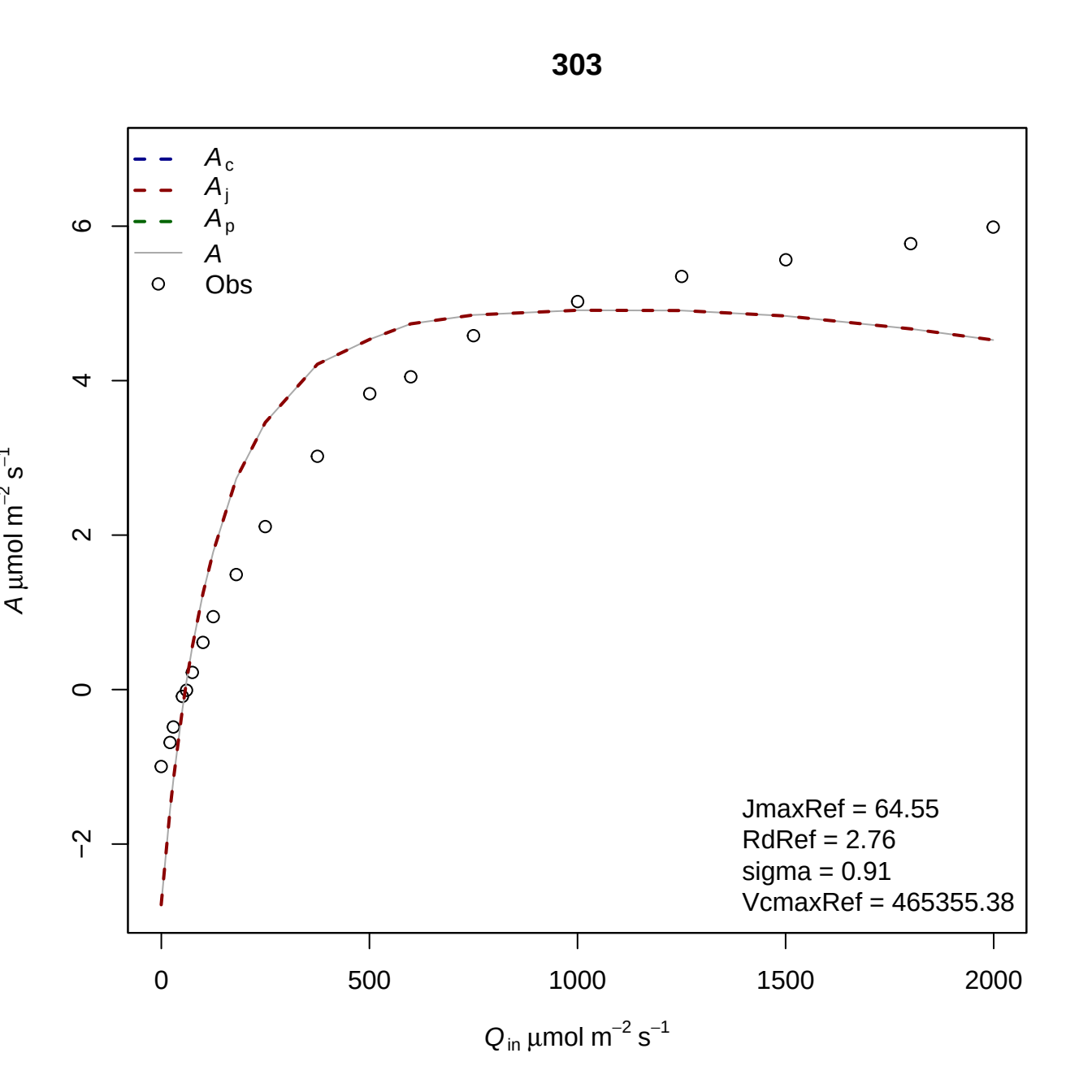

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.46
RdRef = 1.98
sigma = 0.78
VcmaxRef = 764817.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

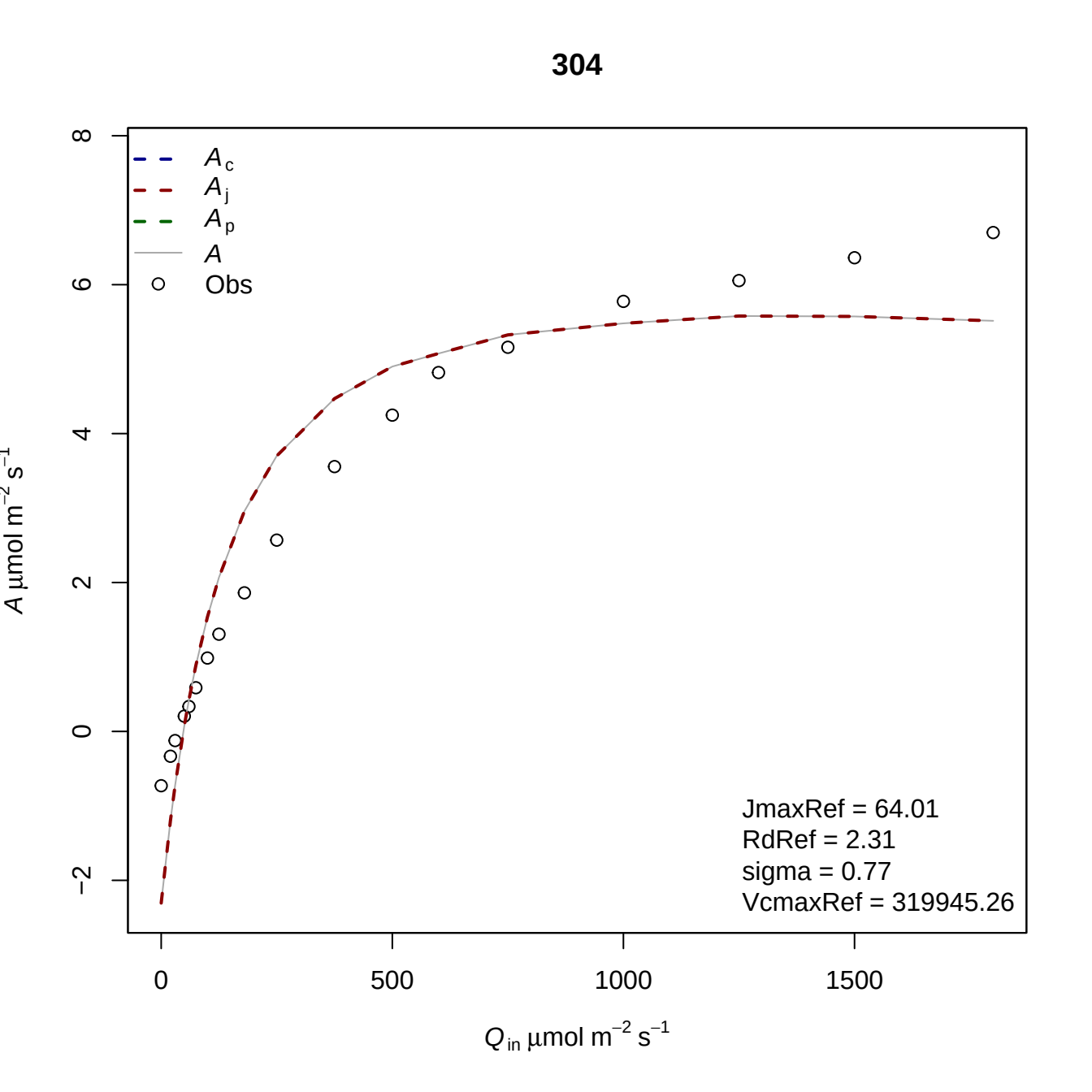

299
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.98
RdRef = 2.5
sigma = 0.97
VcmaxRef = 243700.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

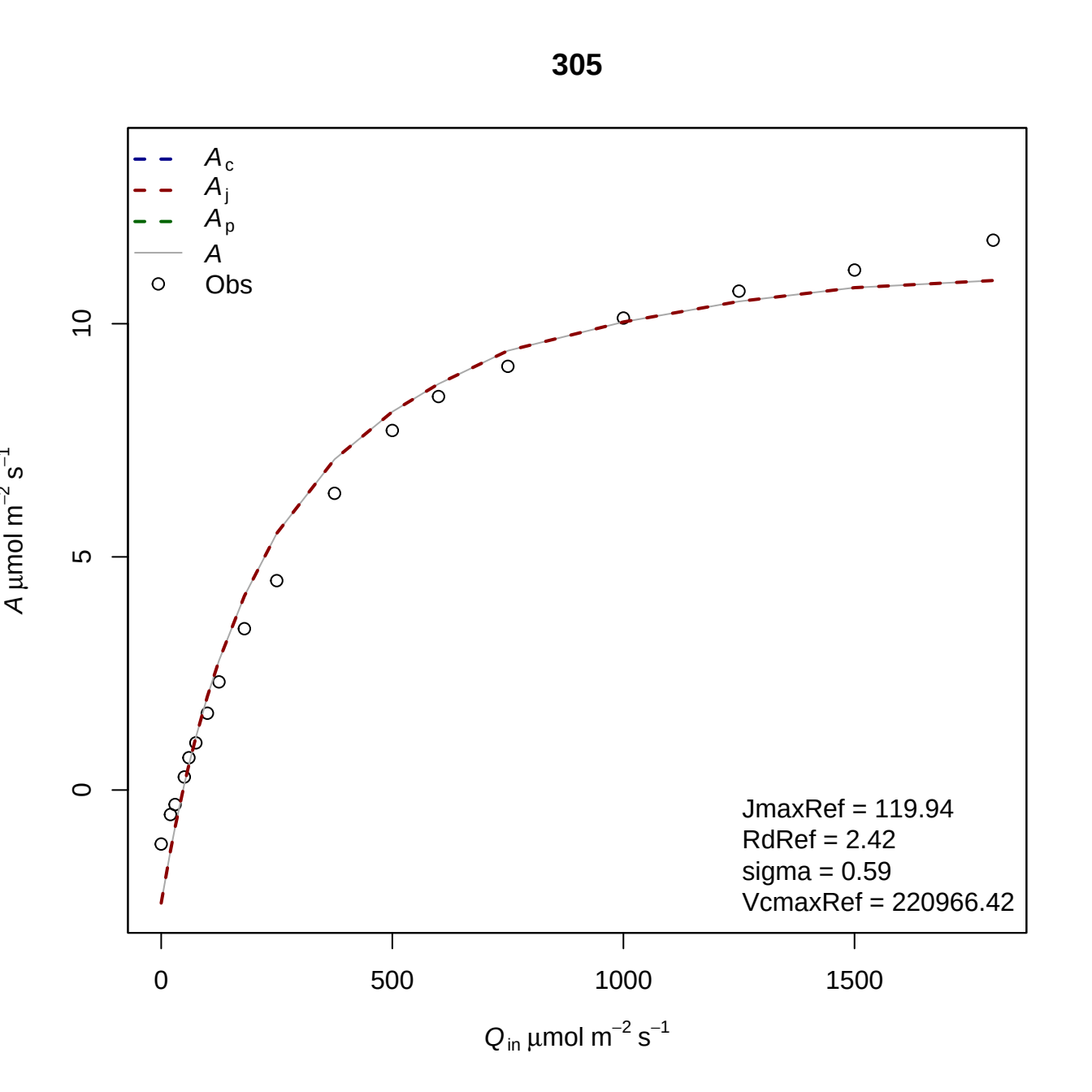

300
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.22
RdRef = 2.45
sigma = 0.7
VcmaxRef = 149059.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

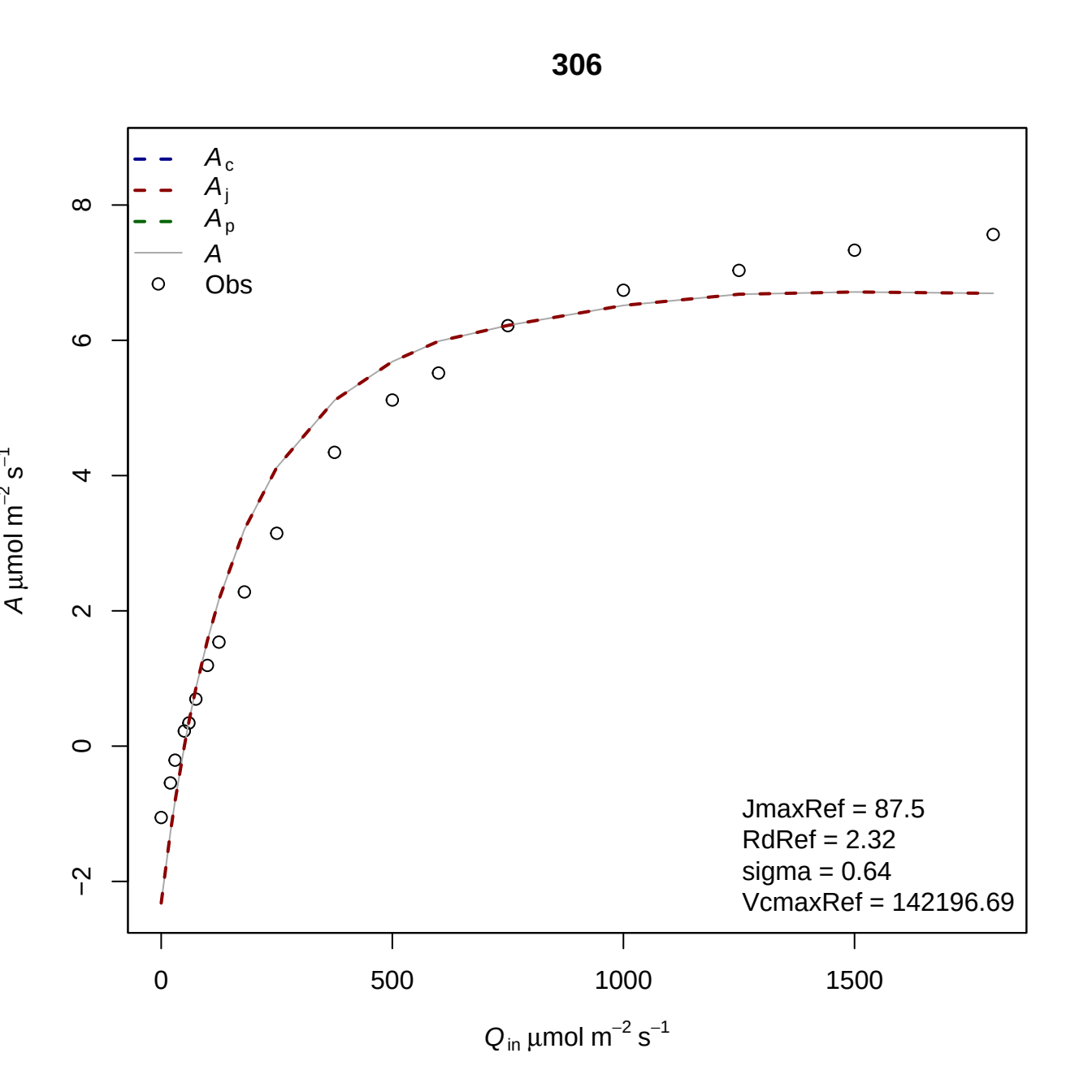

301
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.47
RdRef = 2.25
sigma = 0.77
VcmaxRef = 300910.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

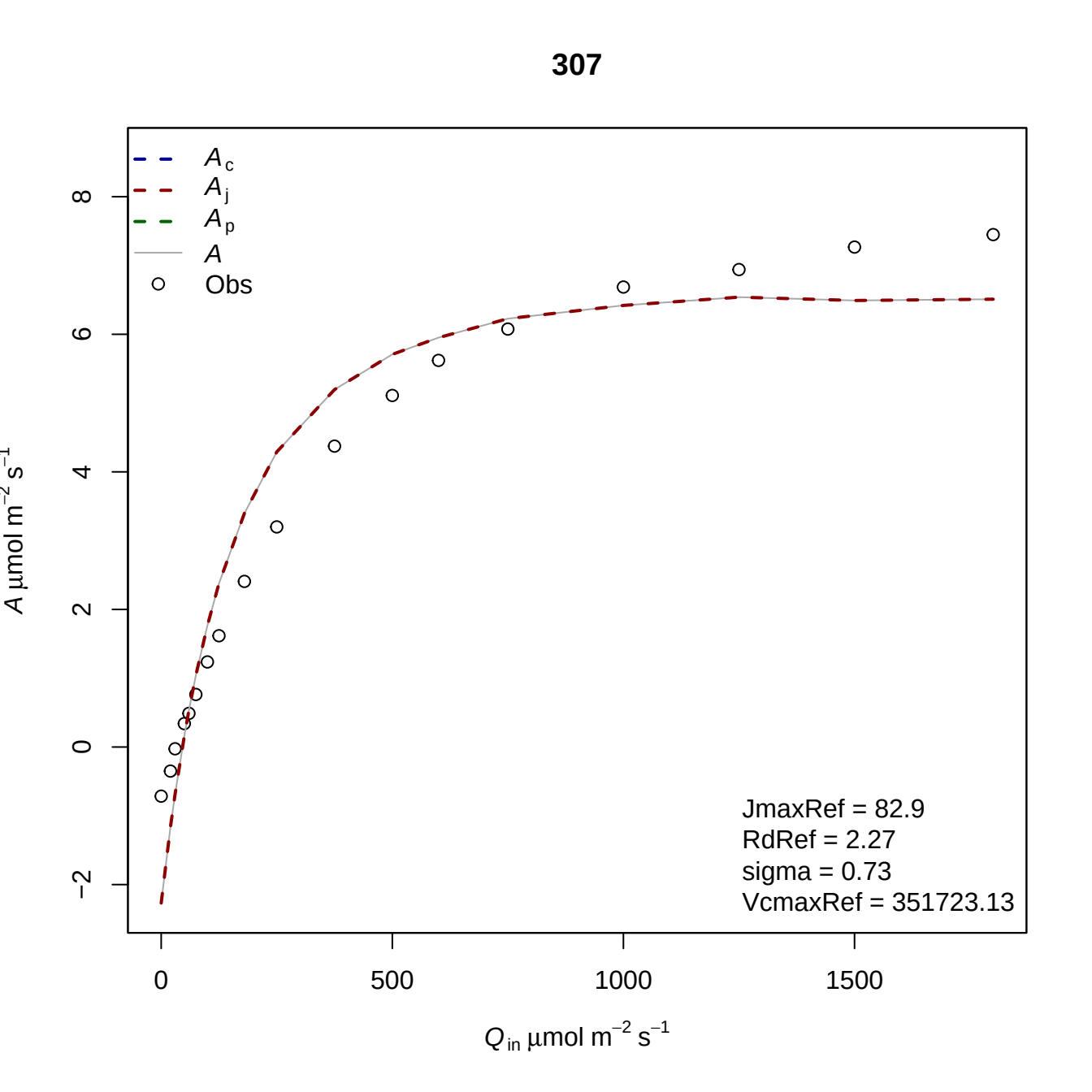

302
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.91
RdRef = 2.61
sigma = 1.13
VcmaxRef = 632249.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

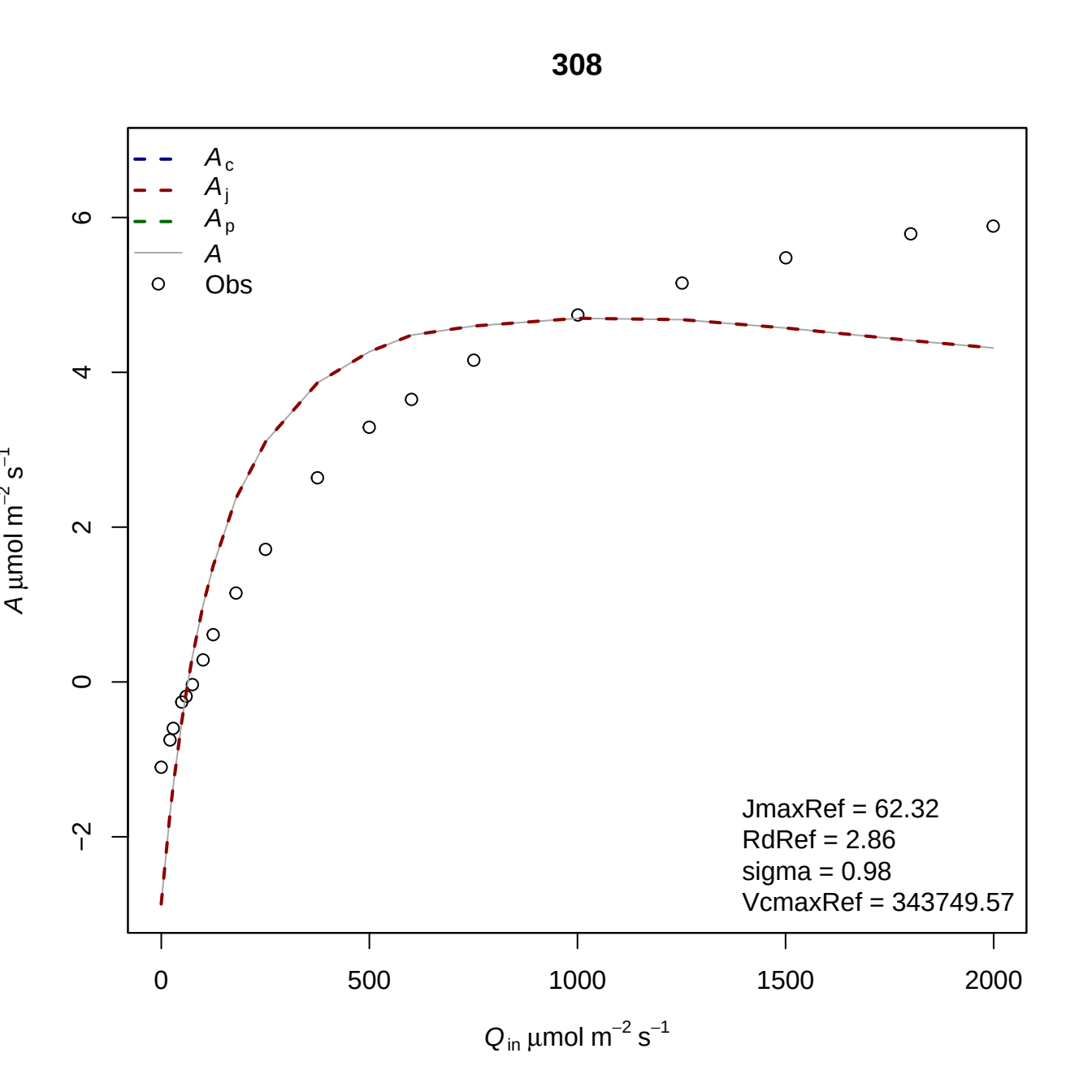

303
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.55
RdRef = 2.76
sigma = 0.91
VcmaxRef = 465355.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

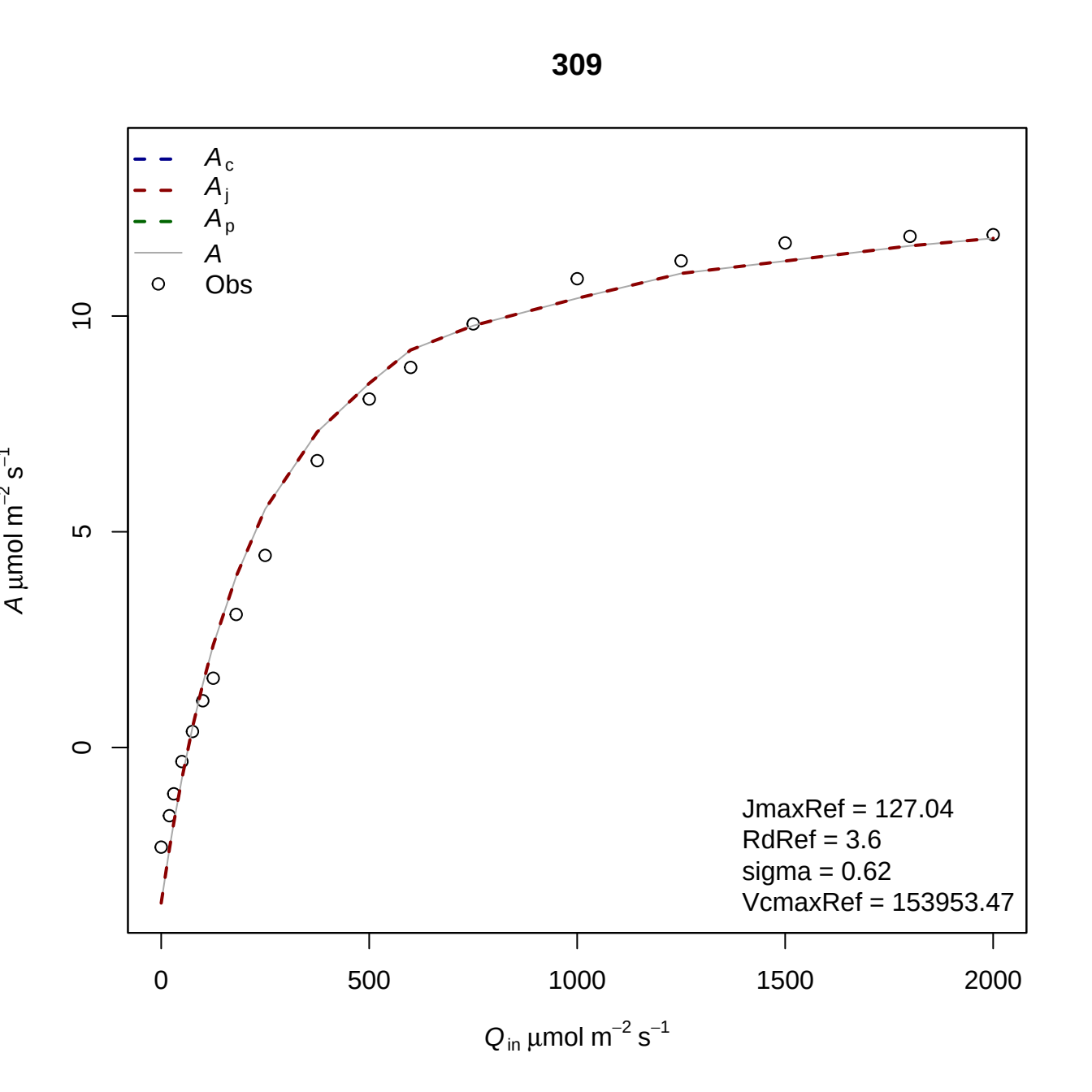

304
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.01
RdRef = 2.31
sigma = 0.77
VcmaxRef = 319945.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

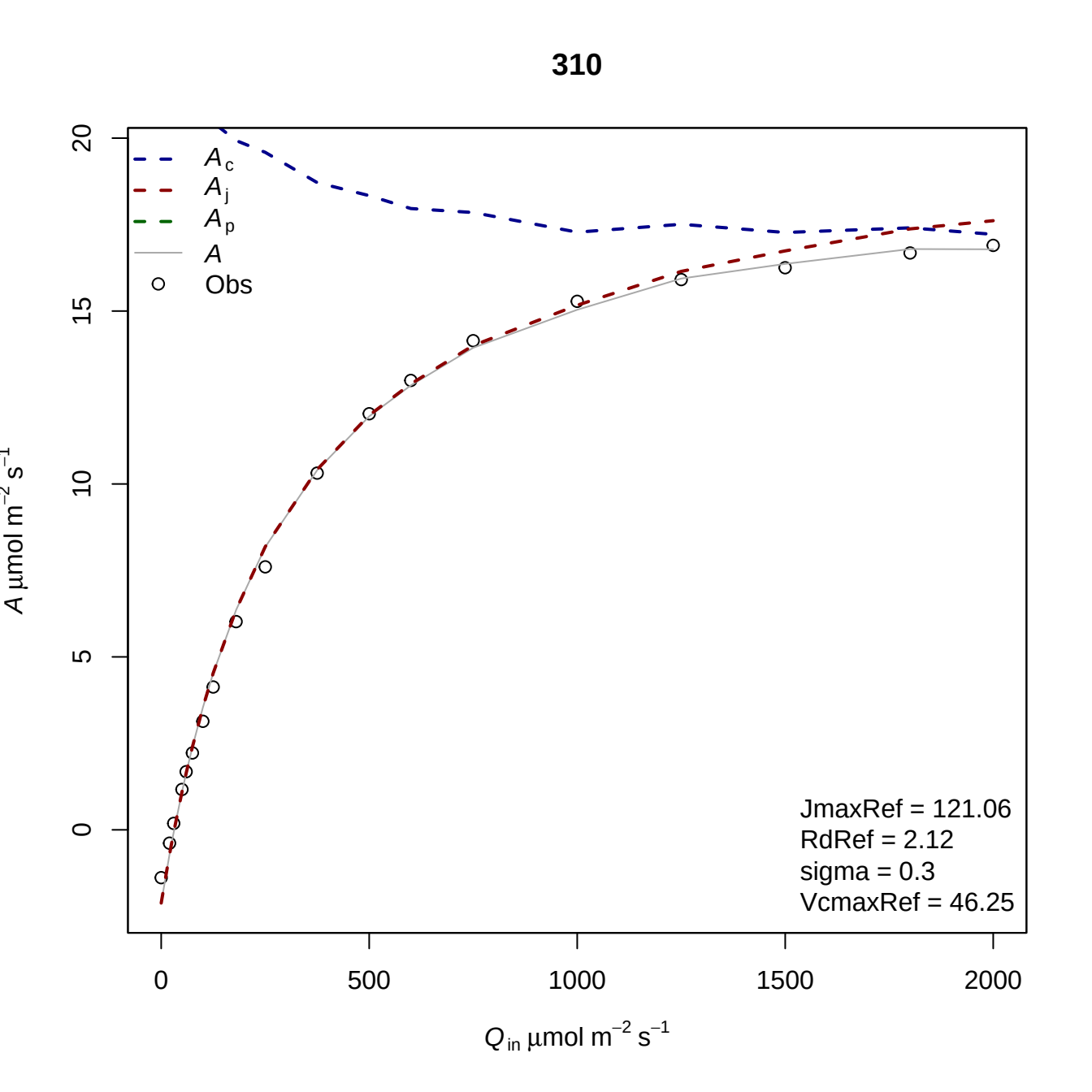

305
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.94
RdRef = 2.42
sigma = 0.59
VcmaxRef = 220966.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

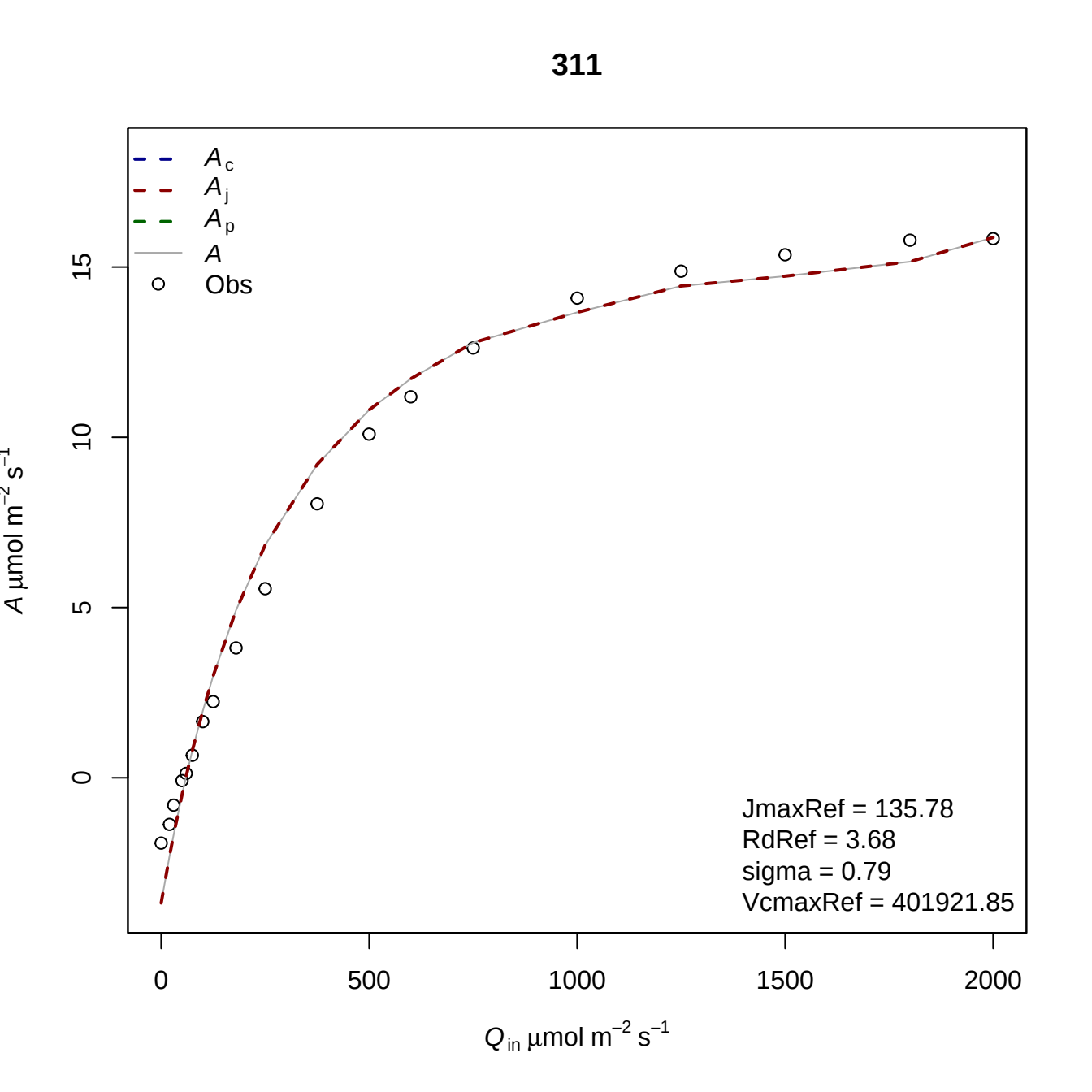

306
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.5
RdRef = 2.32
sigma = 0.64
VcmaxRef = 142196.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

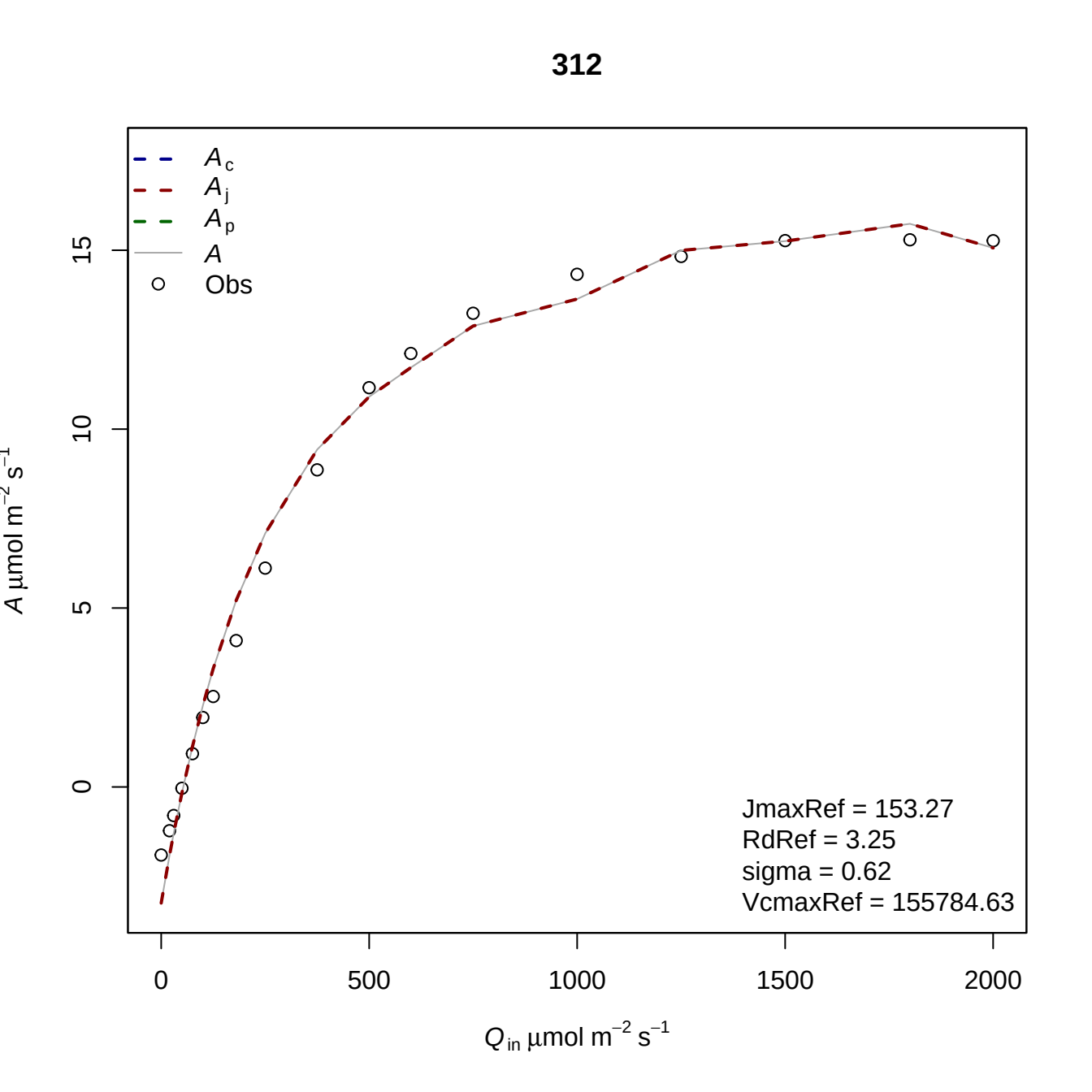

307
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.9
RdRef = 2.27
sigma = 0.73
VcmaxRef = 351723.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

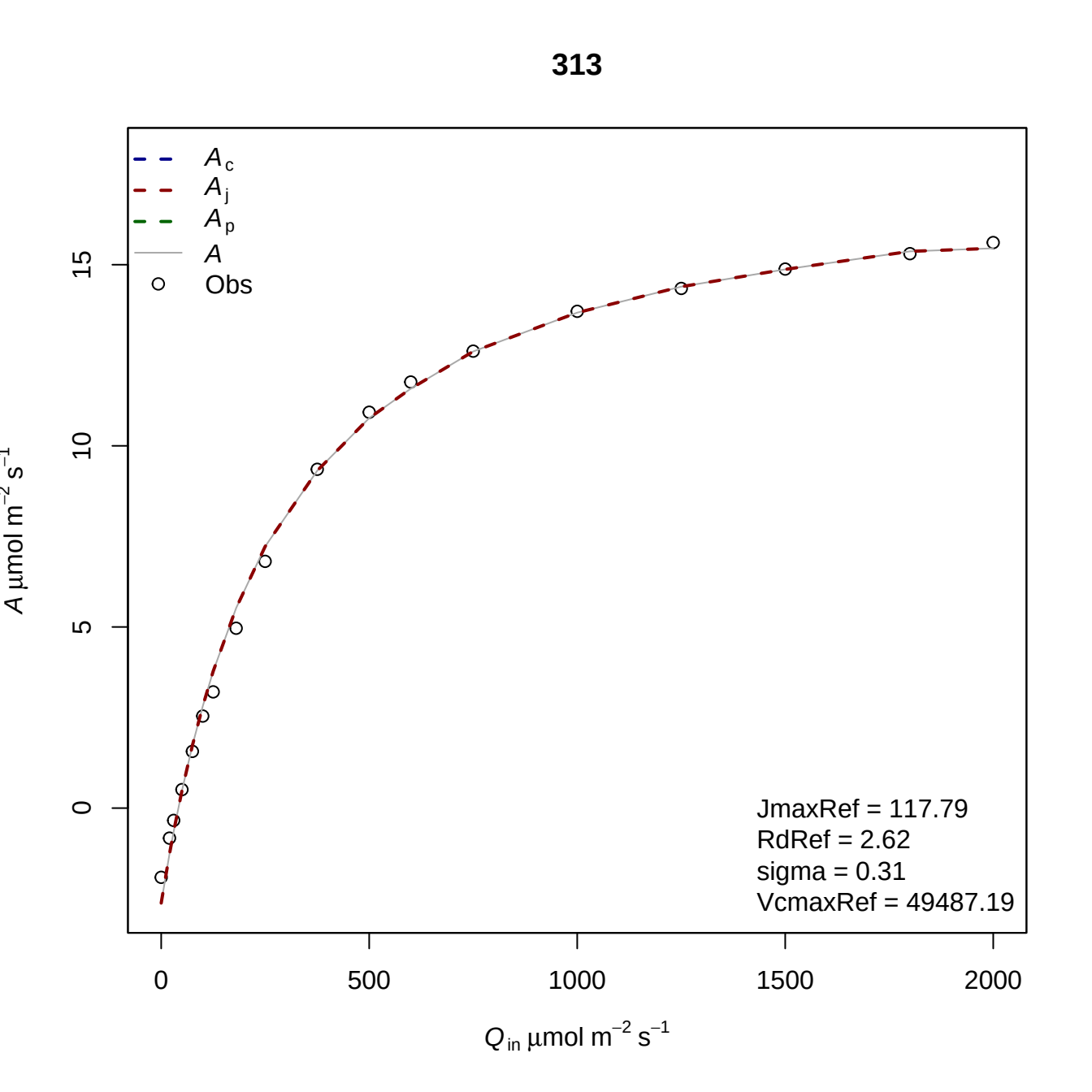

308
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.32
RdRef = 2.86
sigma = 0.98
VcmaxRef = 343749.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

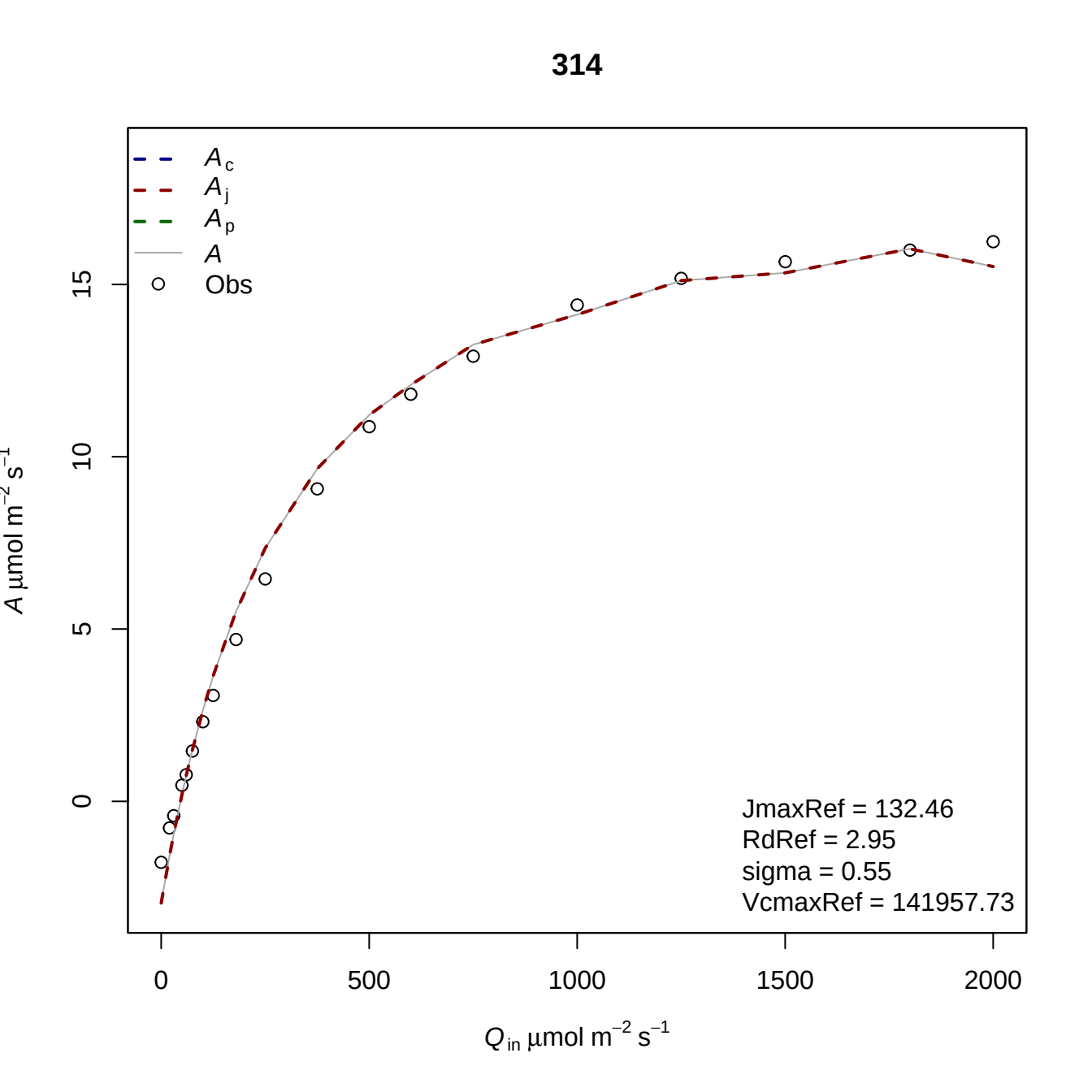

309
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.04
RdRef = 3.6
sigma = 0.62
VcmaxRef = 153953.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

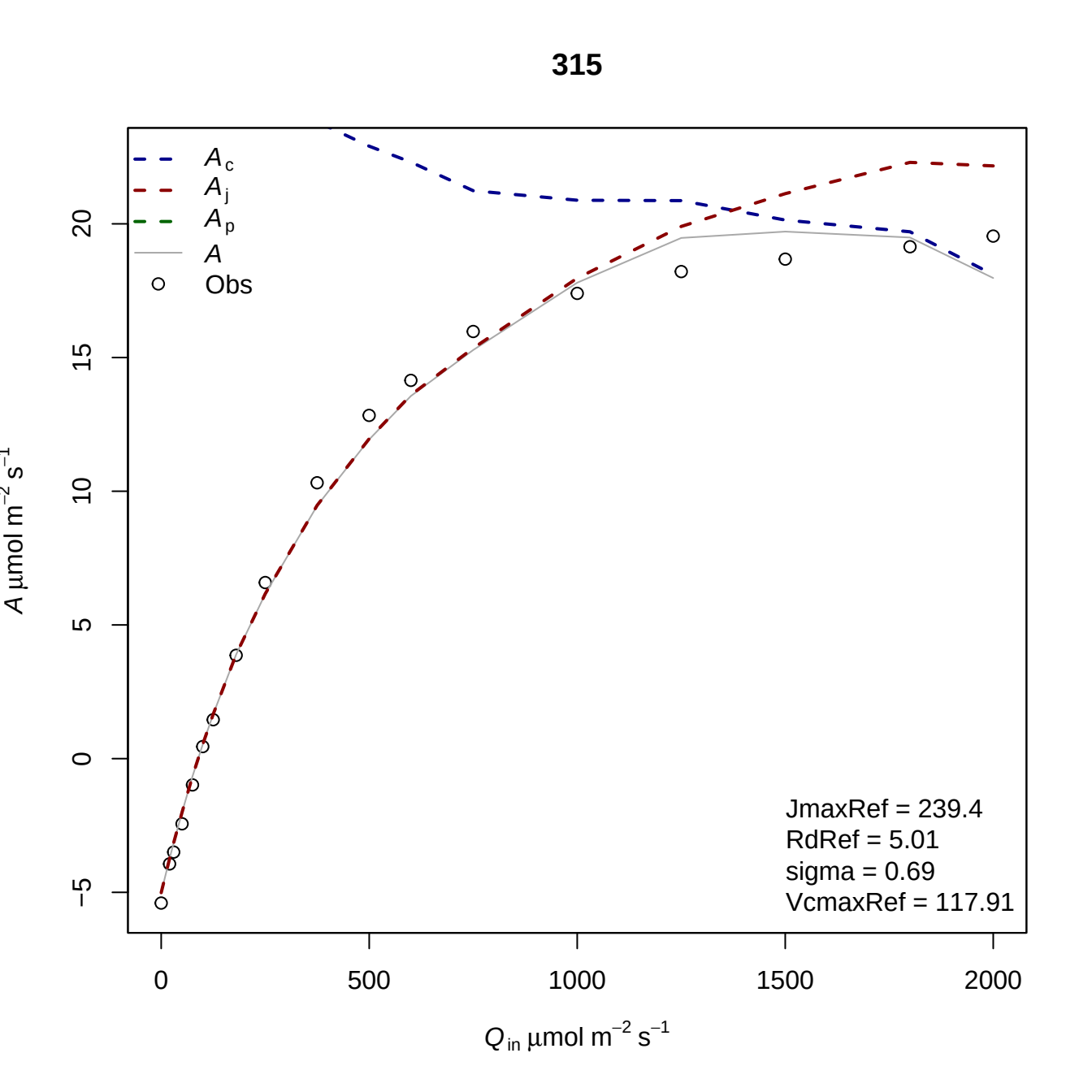

310
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.06
RdRef = 2.12
sigma = 0.3
VcmaxRef = 46.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

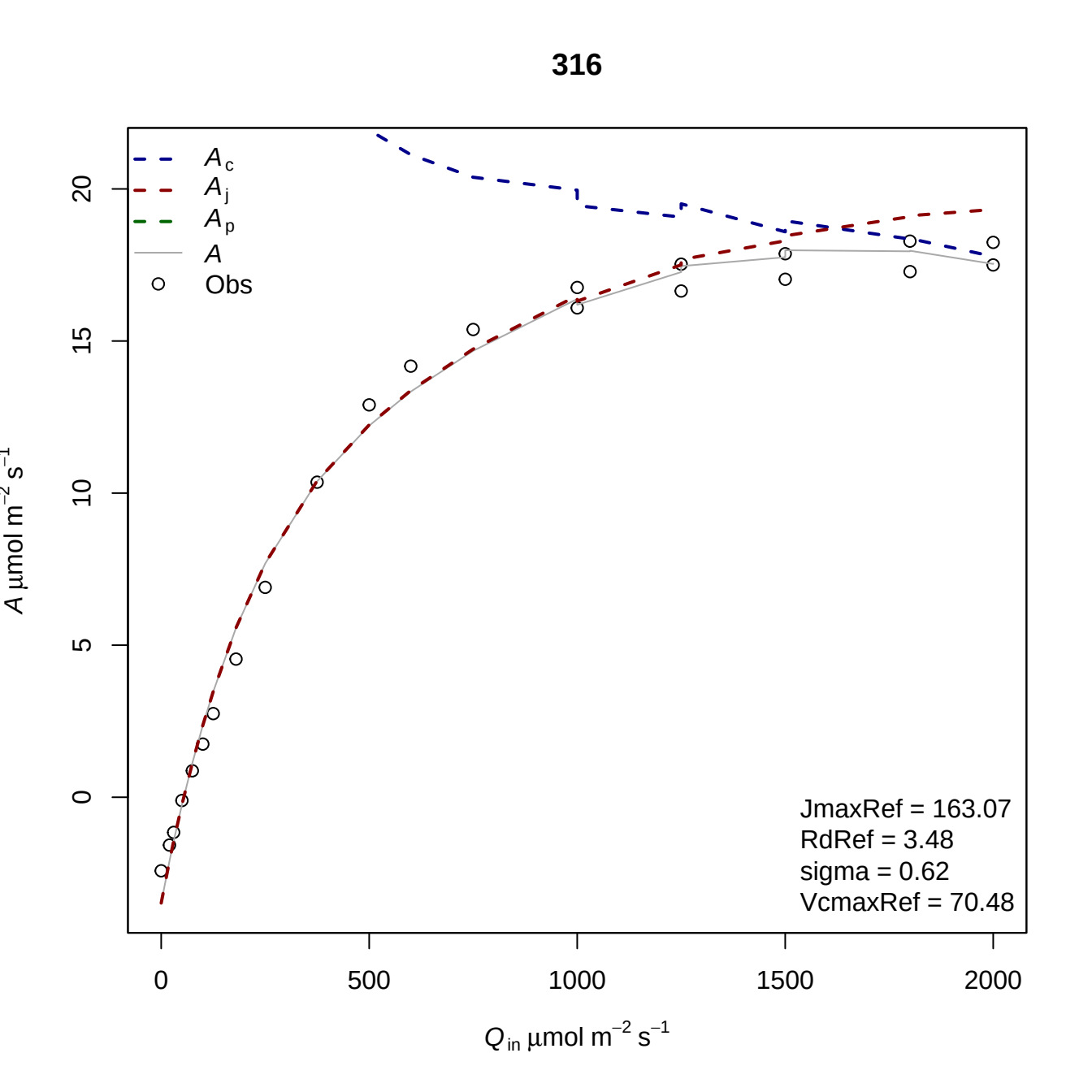

311
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.78
RdRef = 3.68
sigma = 0.79
VcmaxRef = 401921.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

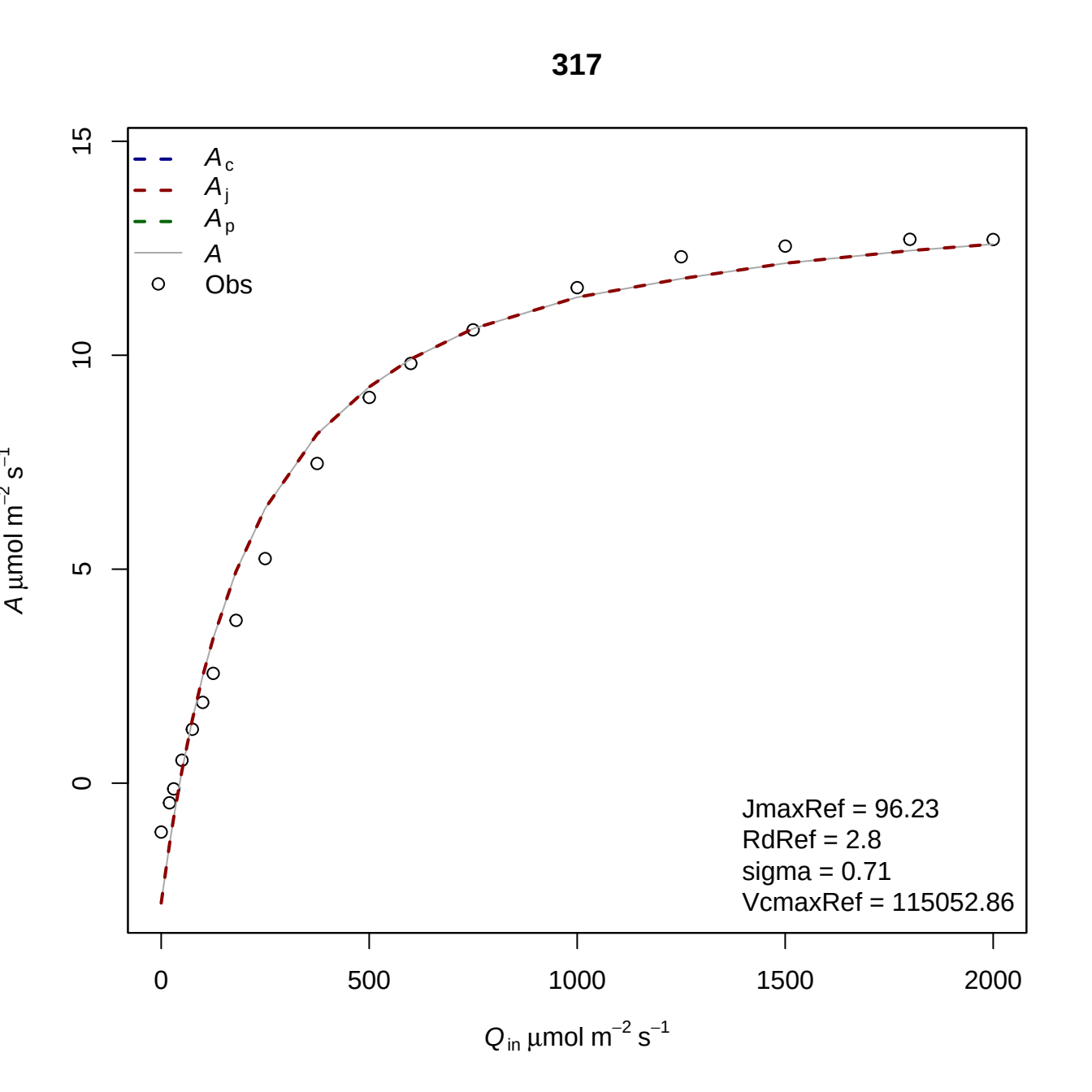

312
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.27
RdRef = 3.25
sigma = 0.62
VcmaxRef = 155784.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

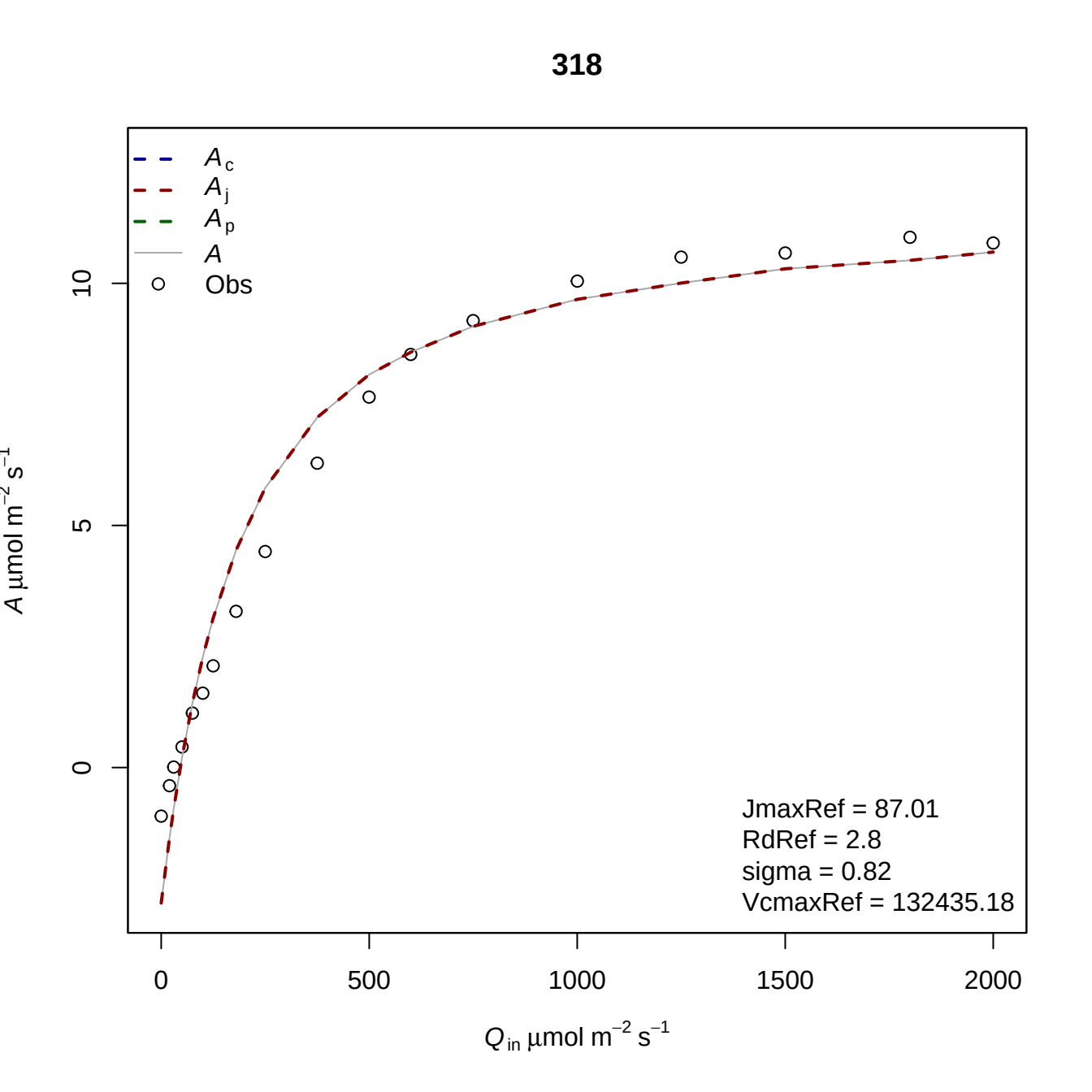

313
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.79
RdRef = 2.62
sigma = 0.31
VcmaxRef = 49487.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

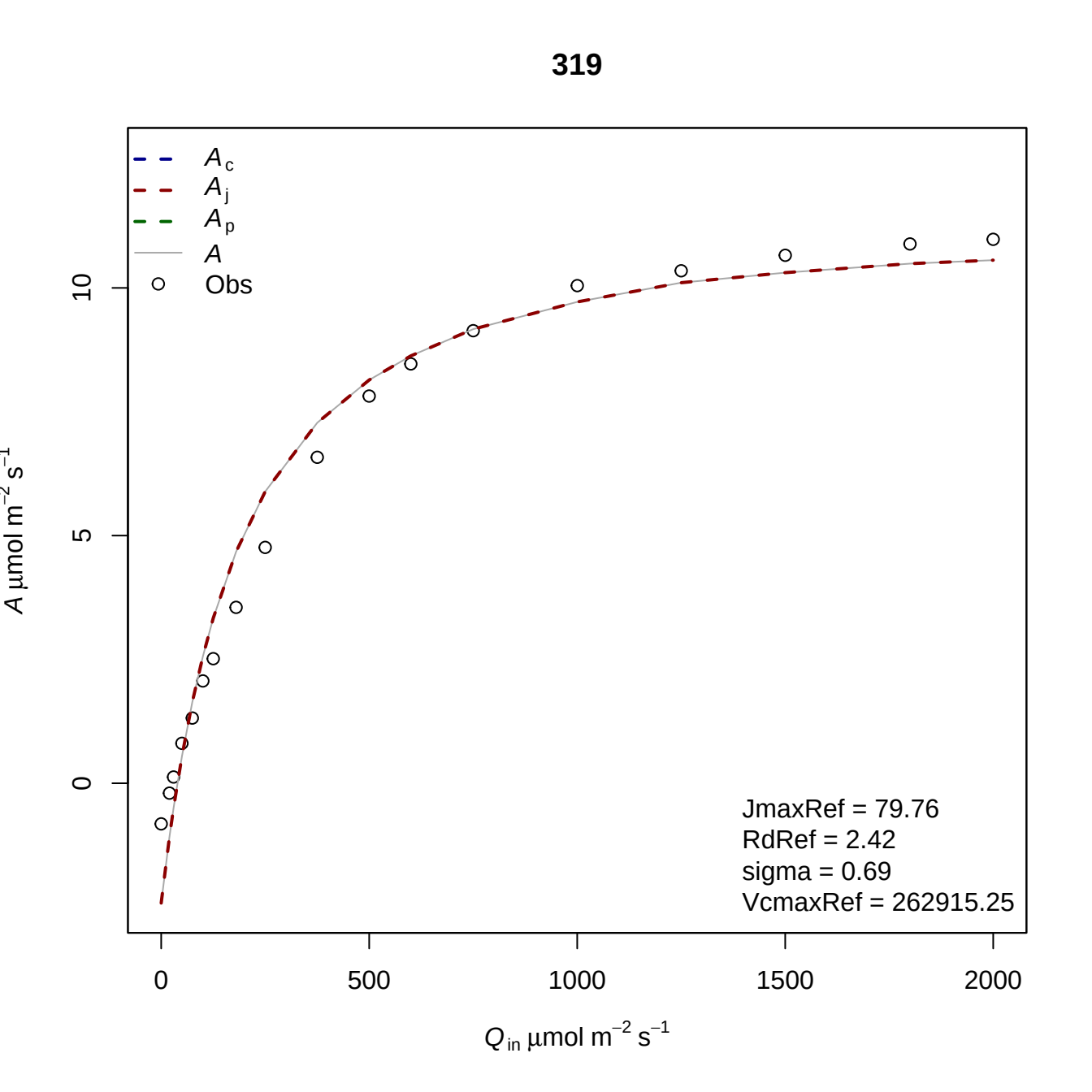

314
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.46
RdRef = 2.95
sigma = 0.55
VcmaxRef = 141957.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

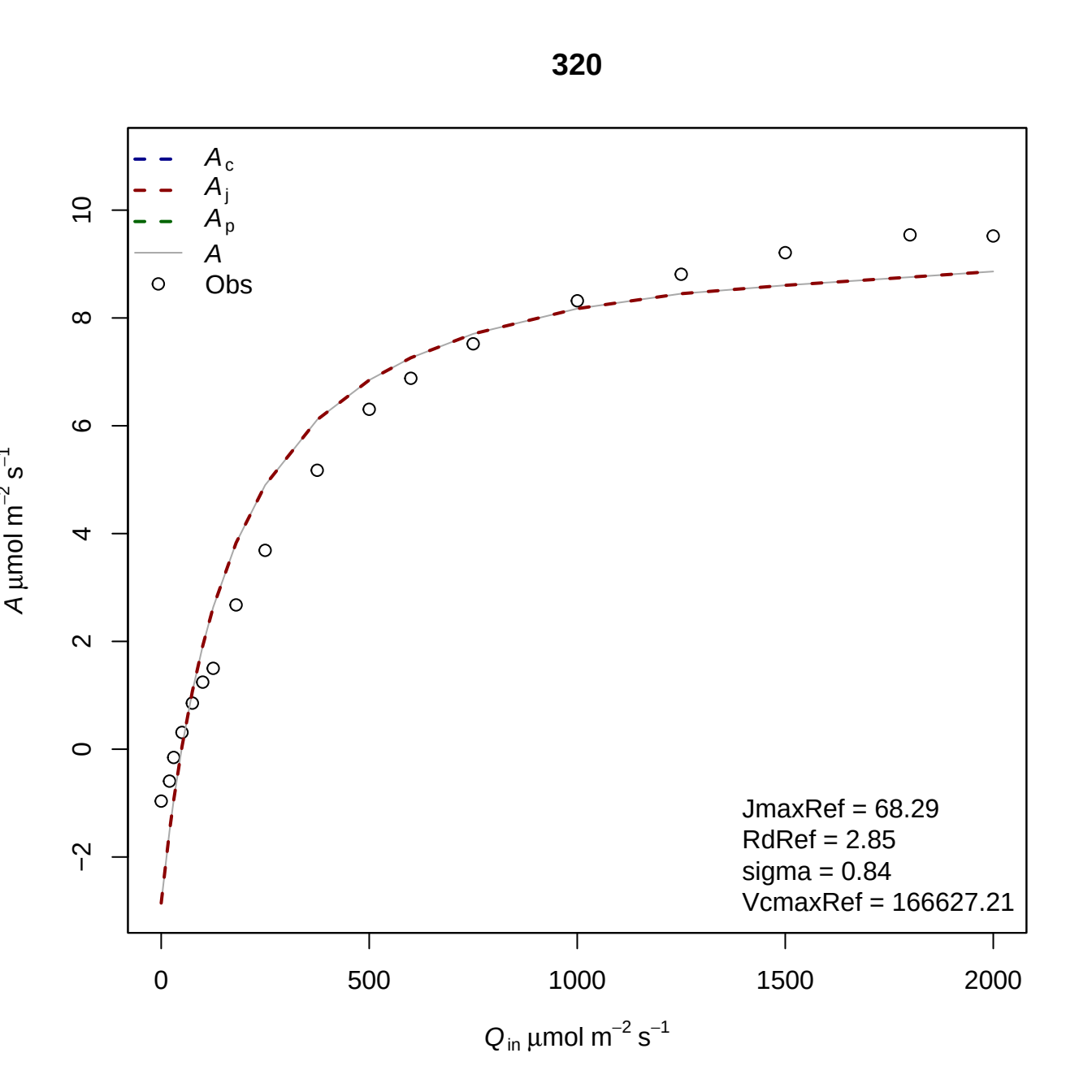

315
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 239.4
RdRef = 5.01
sigma = 0.69
VcmaxRef = 117.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

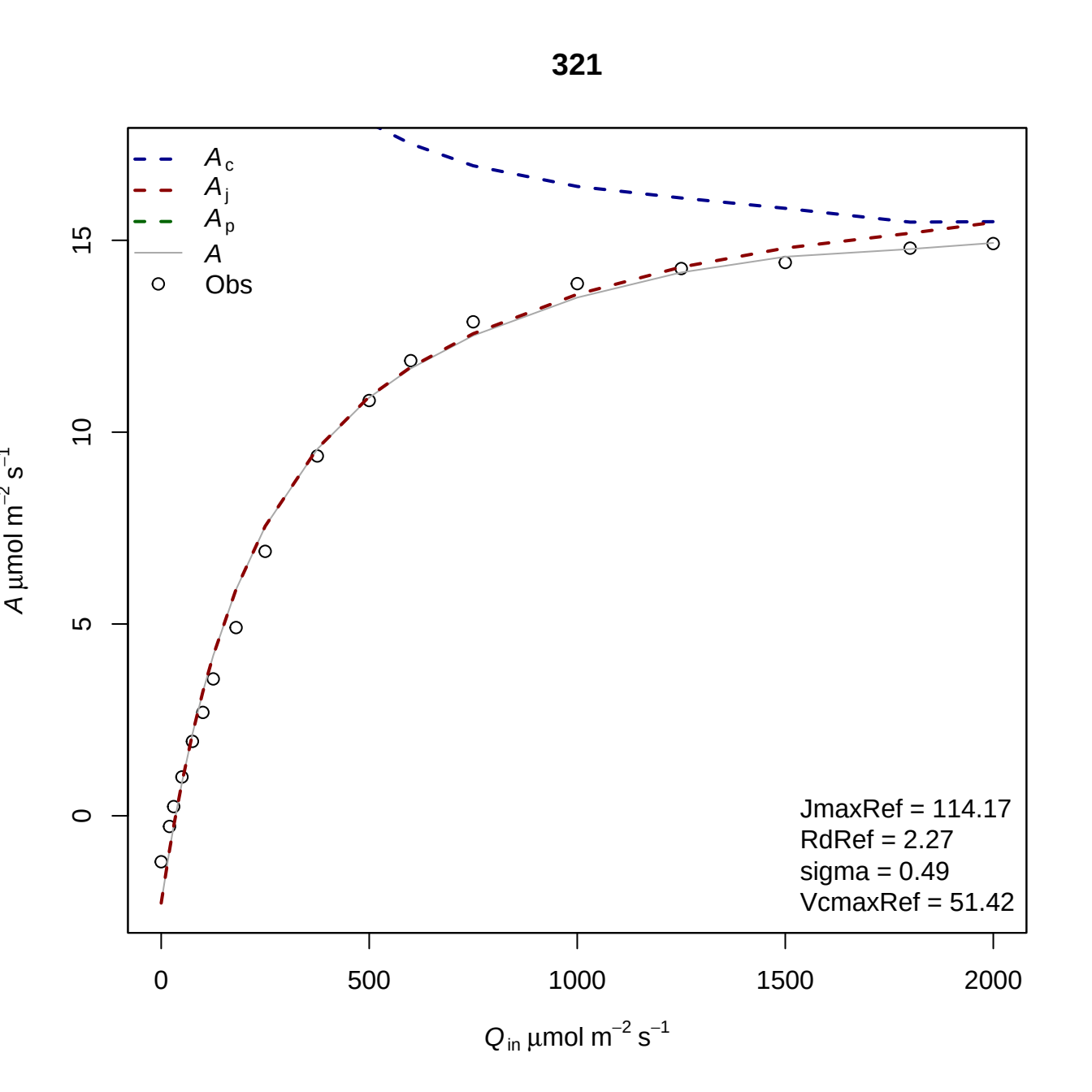

316
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163.07
RdRef = 3.48
sigma = 0.62
VcmaxRef = 70.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

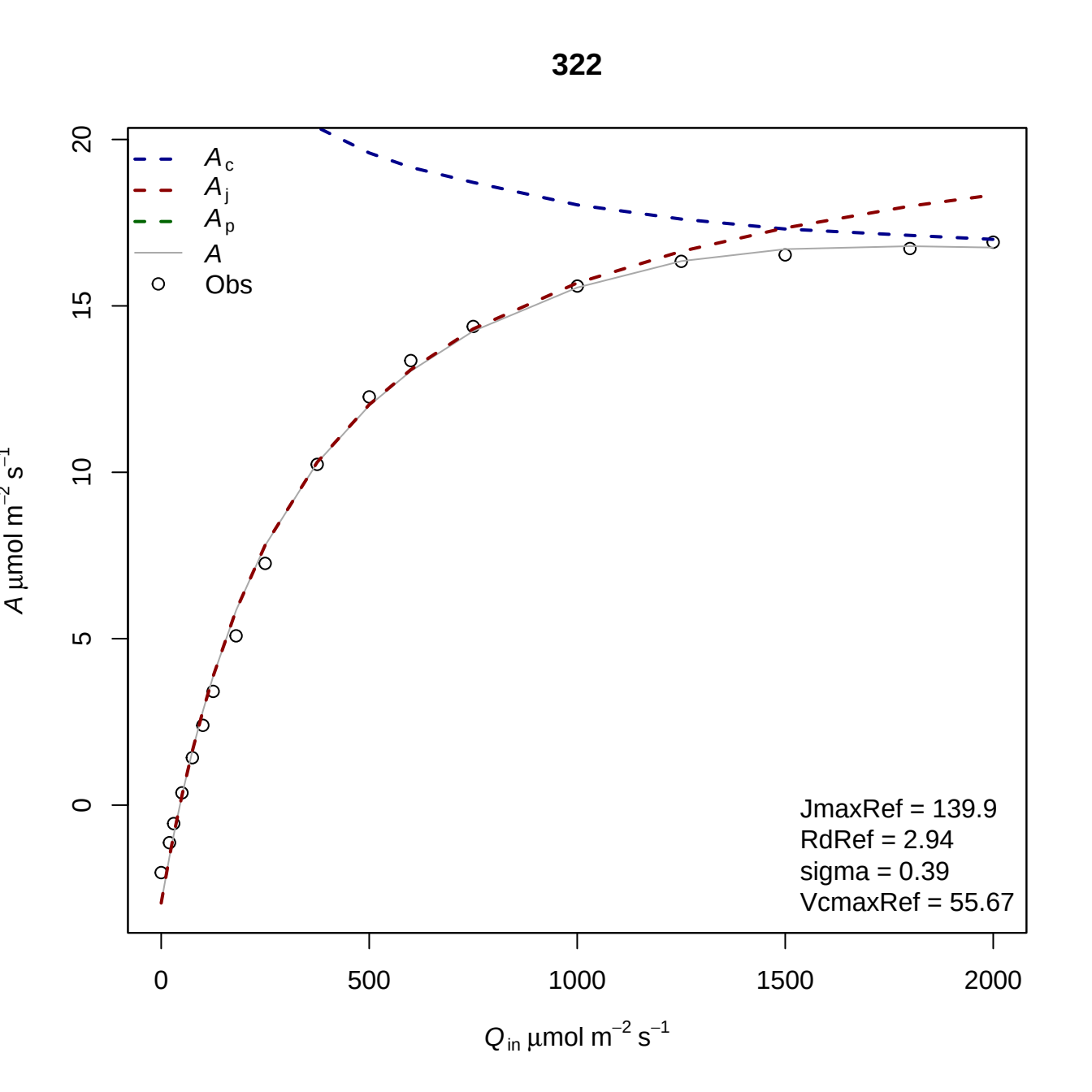

317
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.23
RdRef = 2.8
sigma = 0.71
VcmaxRef = 115052.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

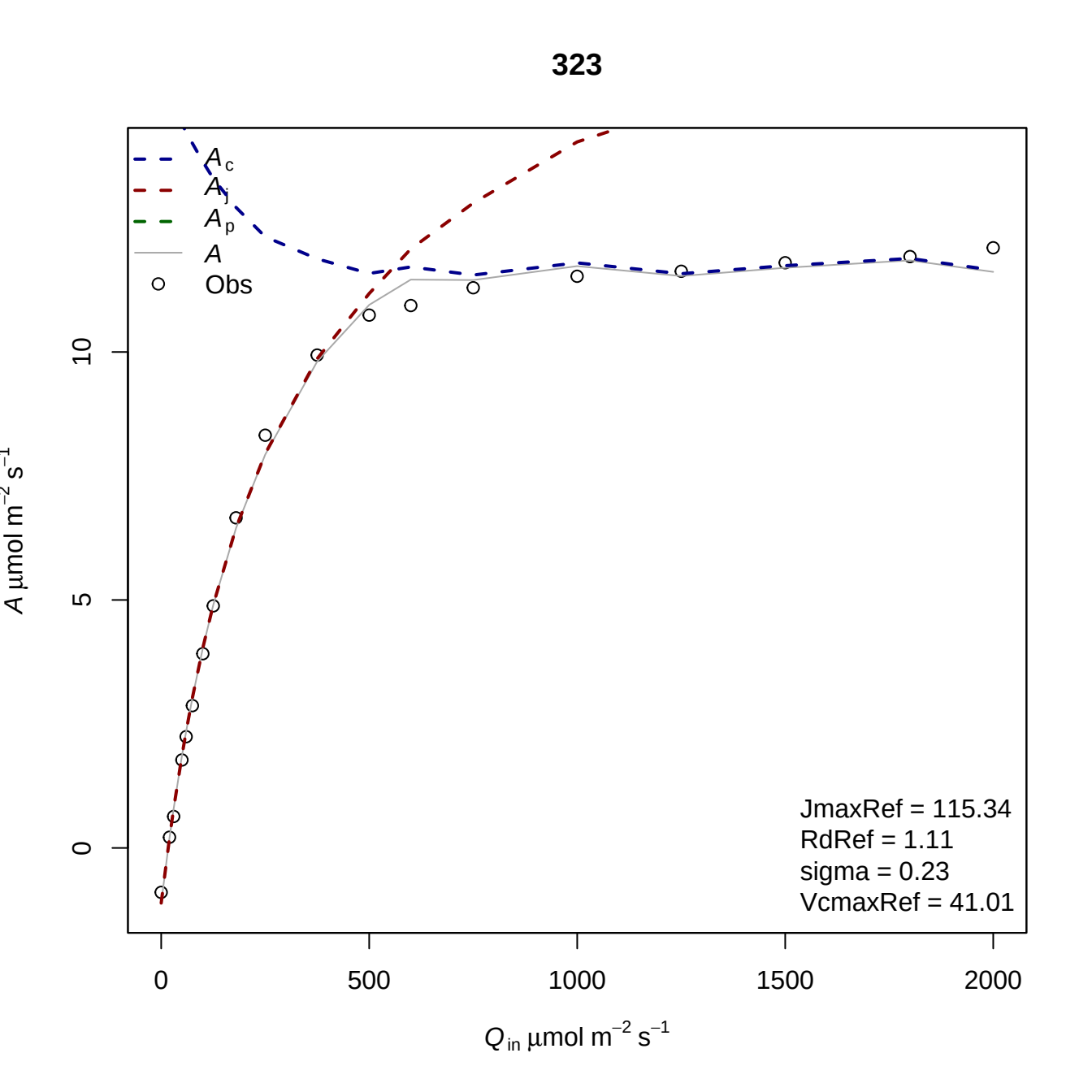

318
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.01
RdRef = 2.8
sigma = 0.82
VcmaxRef = 132435.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

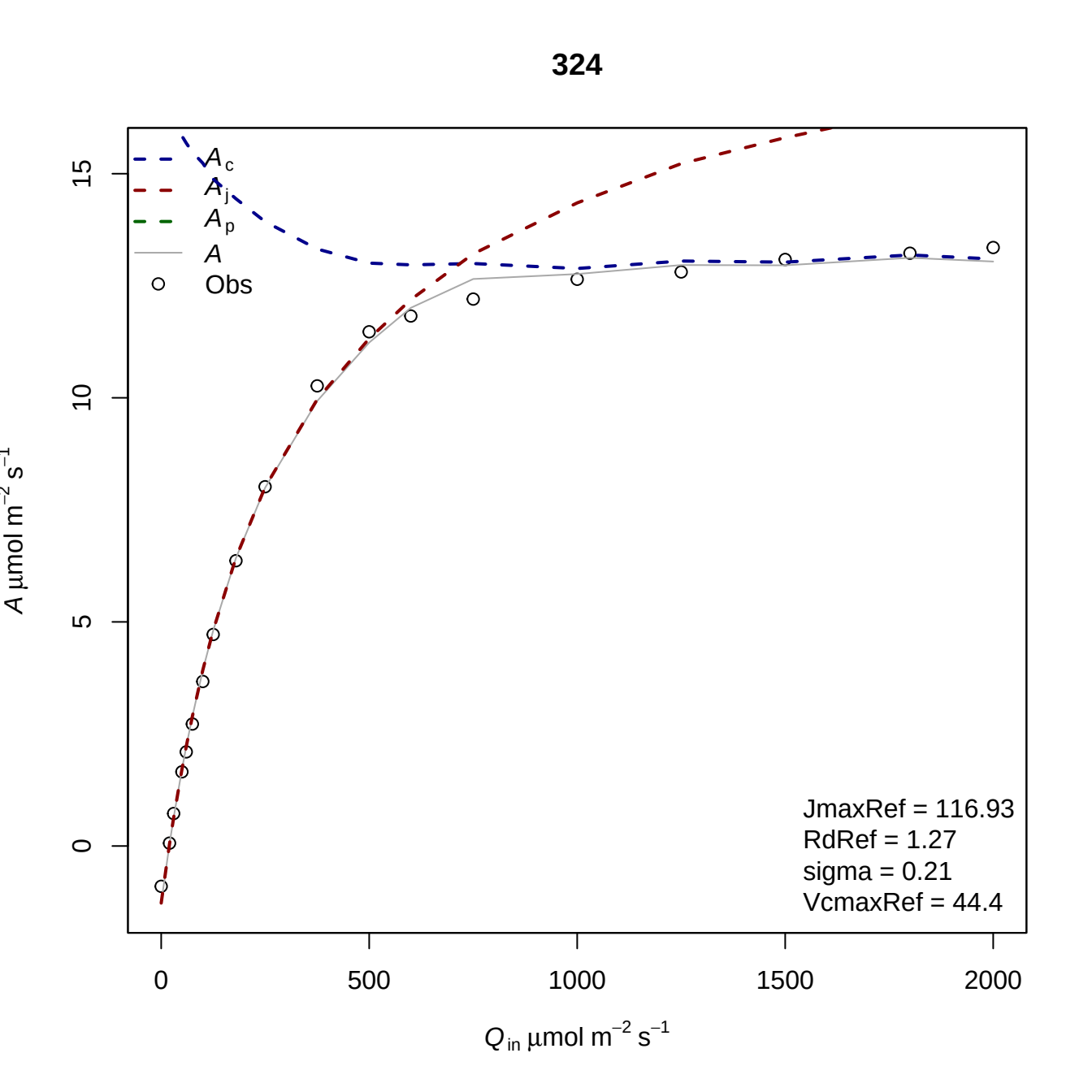

319
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.76
RdRef = 2.42
sigma = 0.69
VcmaxRef = 262915.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

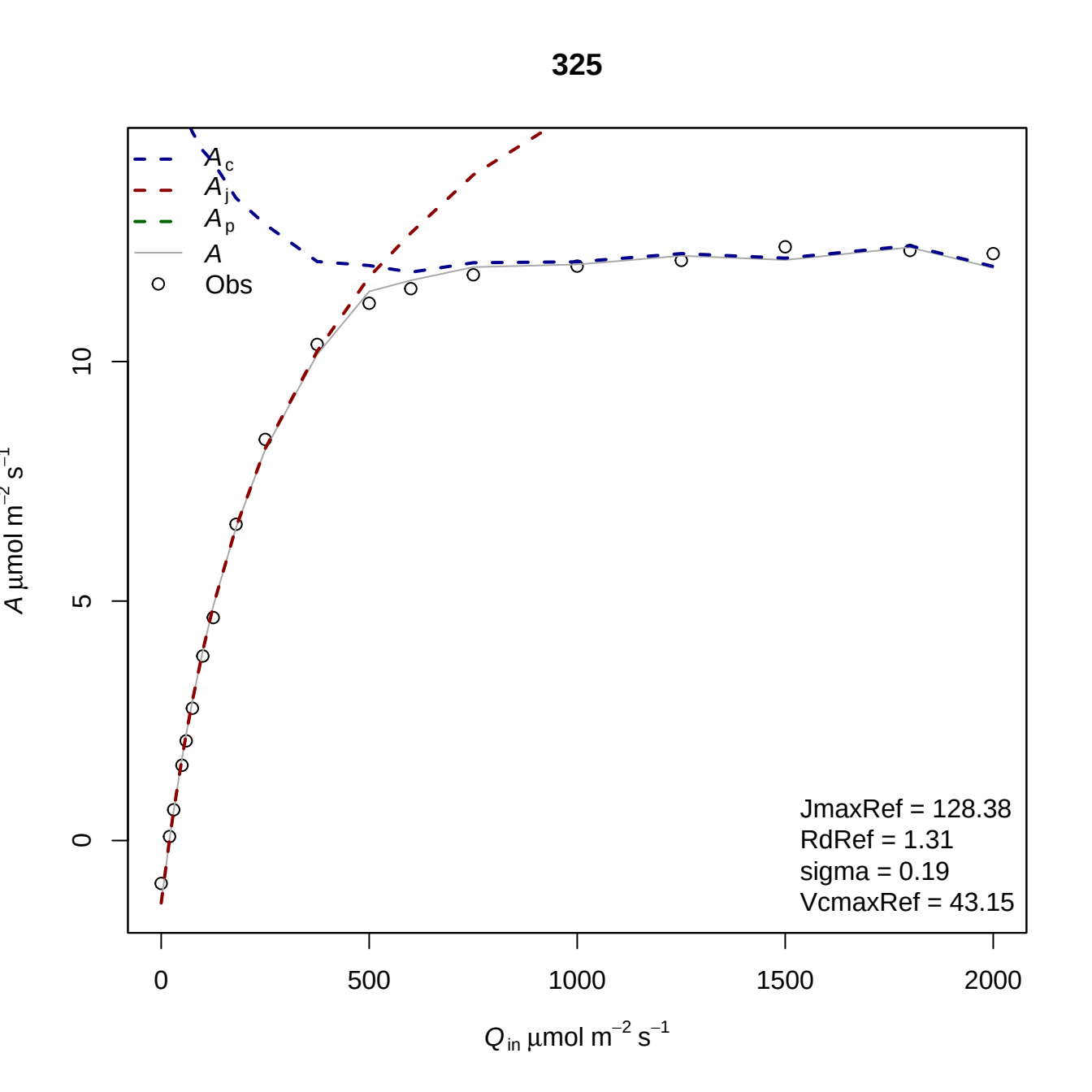

320
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.29
RdRef = 2.85
sigma = 0.84
VcmaxRef = 166627.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

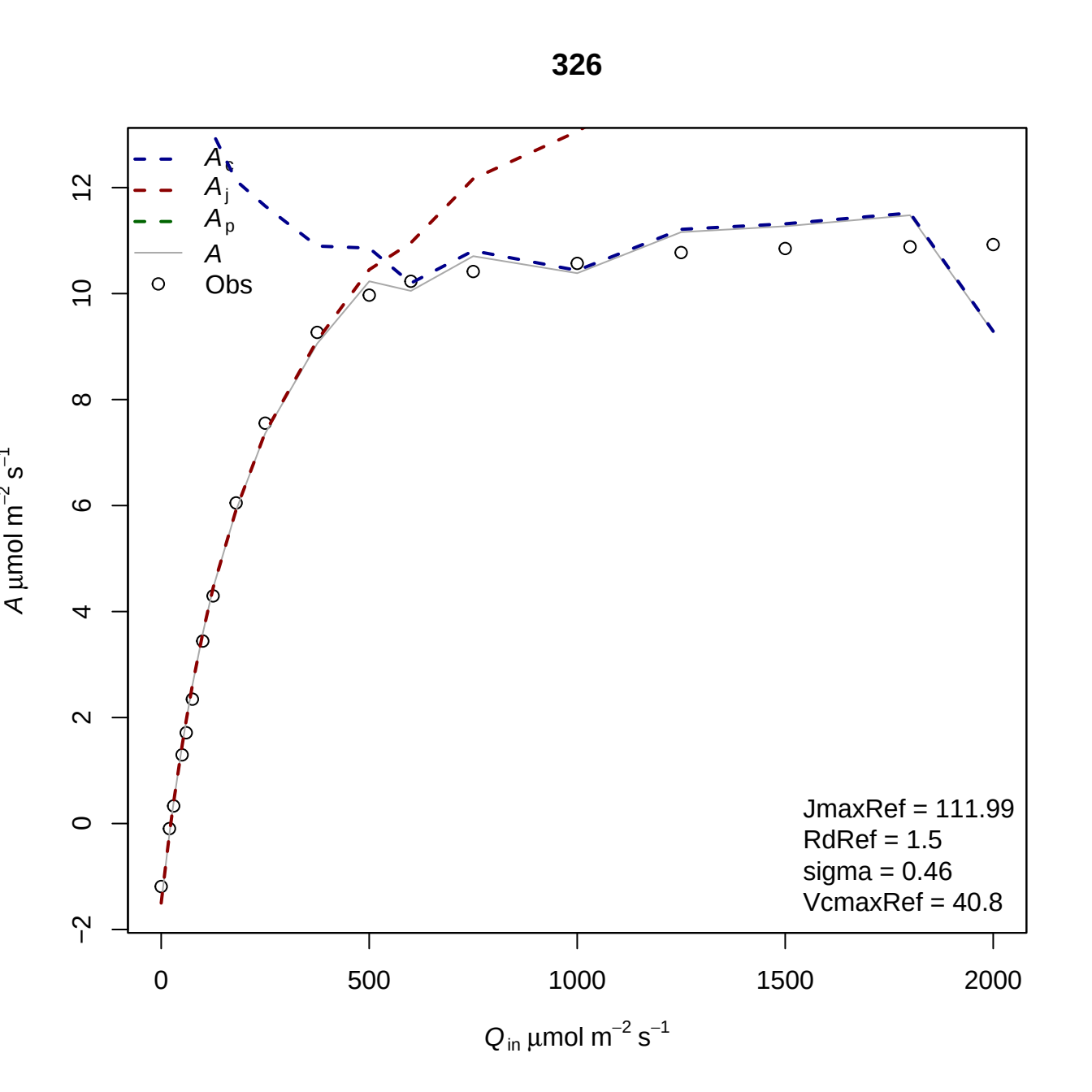

321
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.17
RdRef = 2.27
sigma = 0.49
VcmaxRef = 51.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

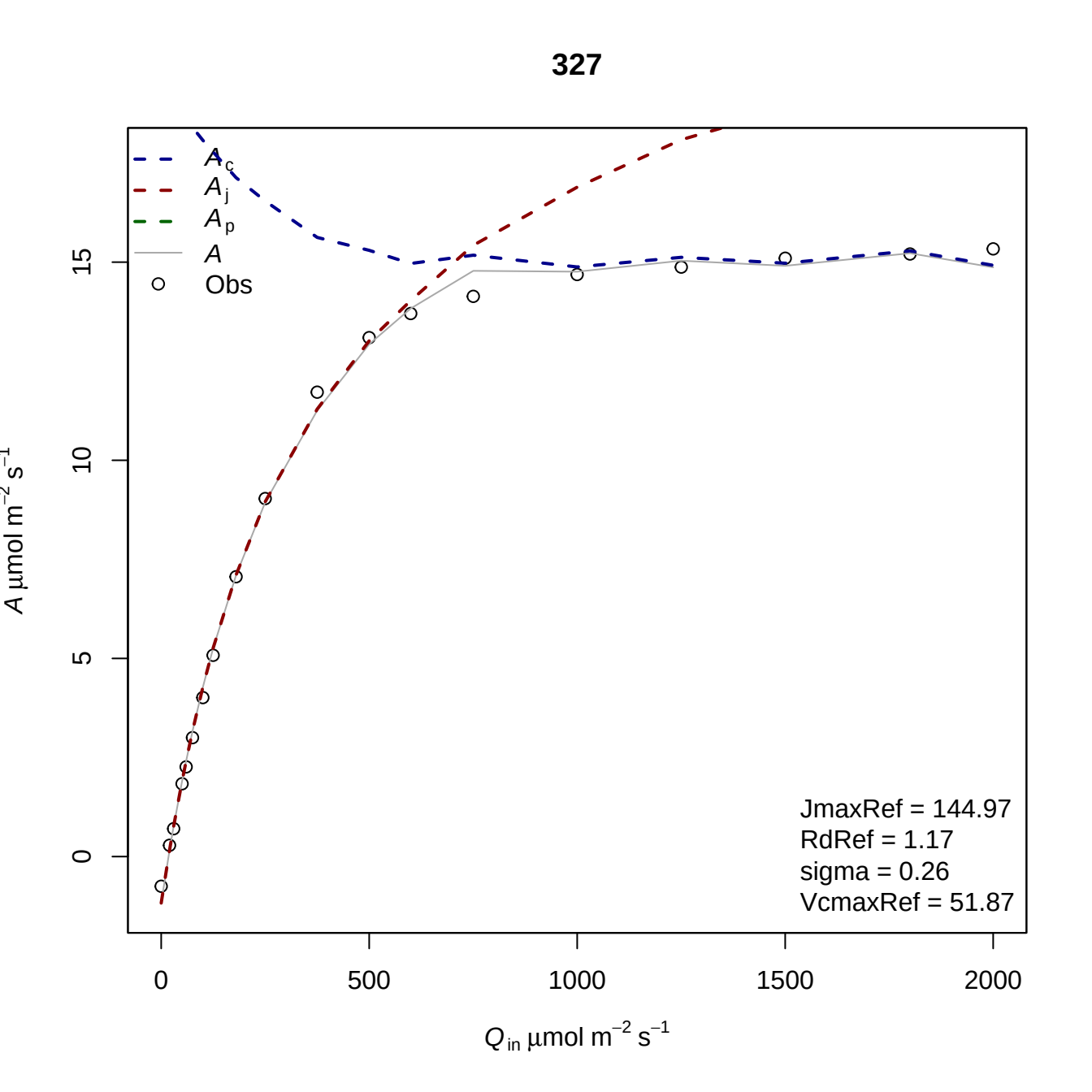

322
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.9
RdRef = 2.94
sigma = 0.39
VcmaxRef = 55.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

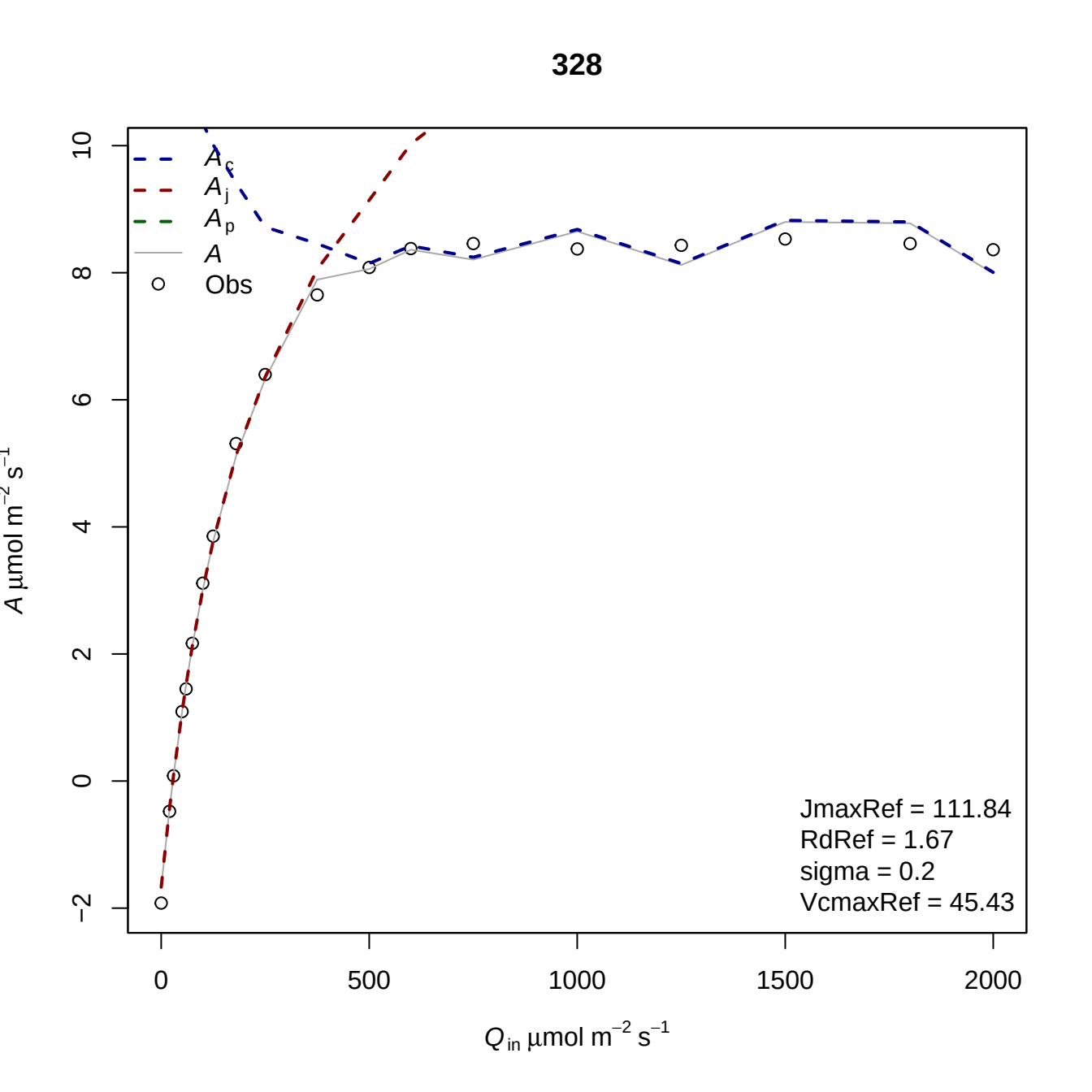

323
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.34
RdRef = 1.11
sigma = 0.23
VcmaxRef = 41.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

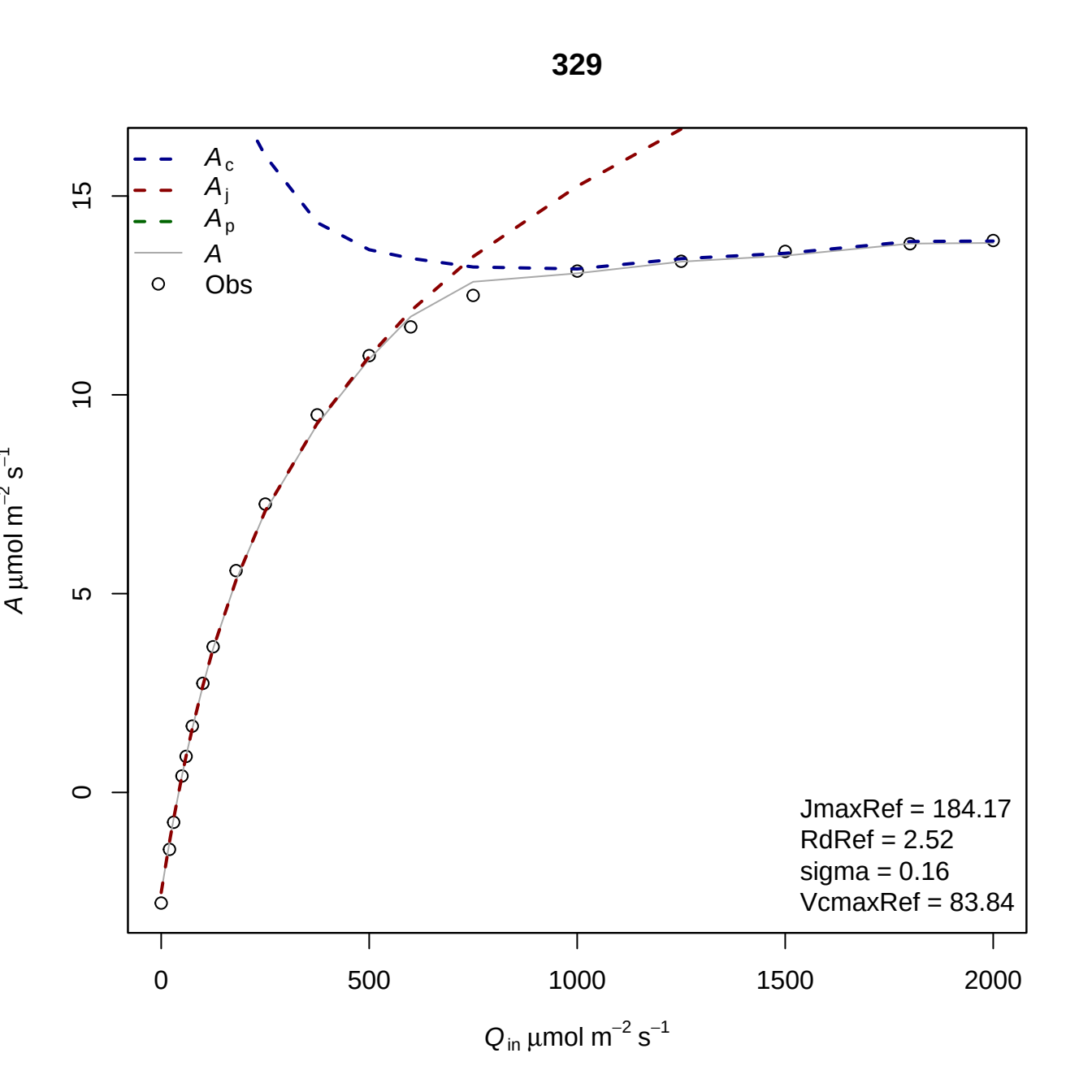

324
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.93
RdRef = 1.27
sigma = 0.21
VcmaxRef = 44.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

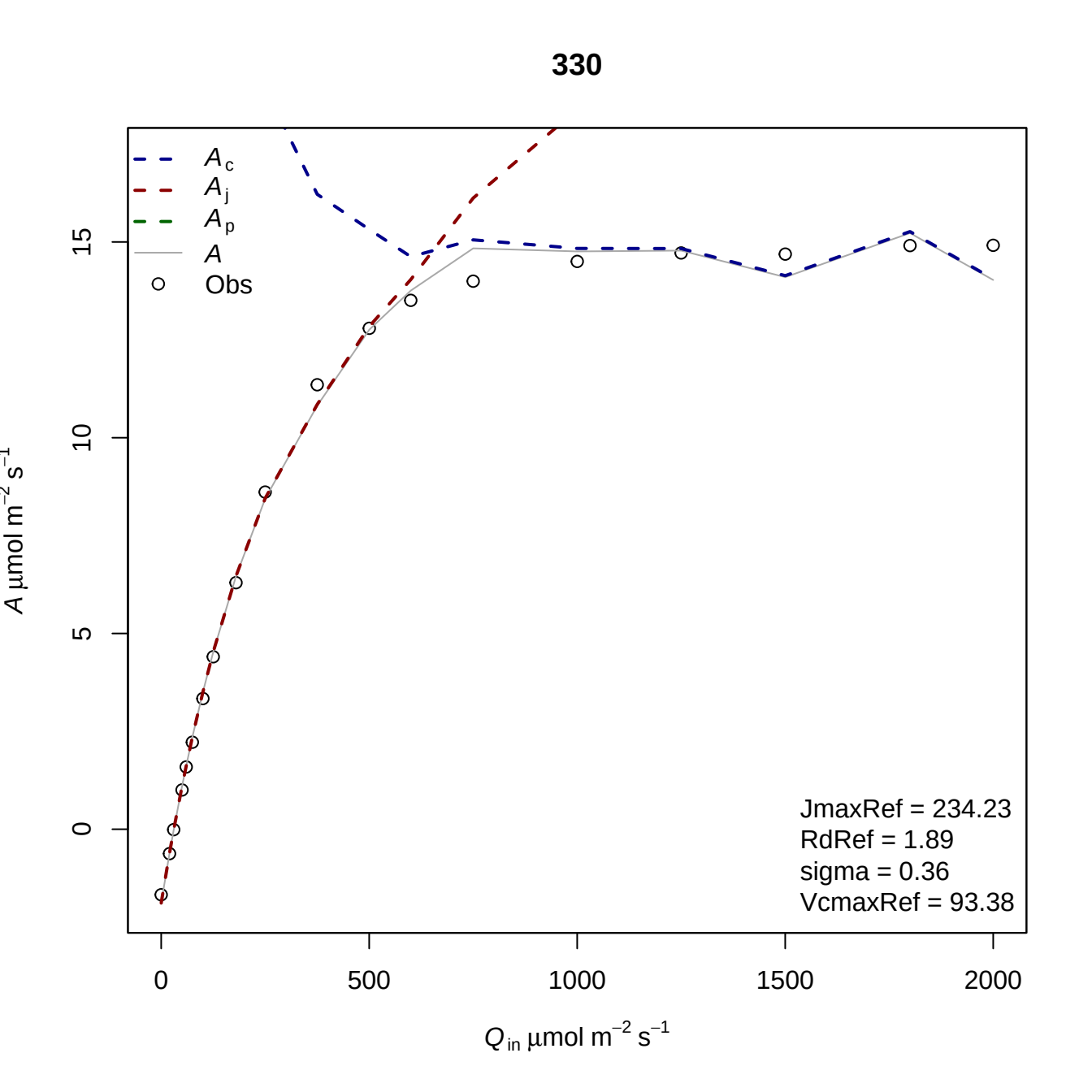

325
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.38
RdRef = 1.31
sigma = 0.19
VcmaxRef = 43.15
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

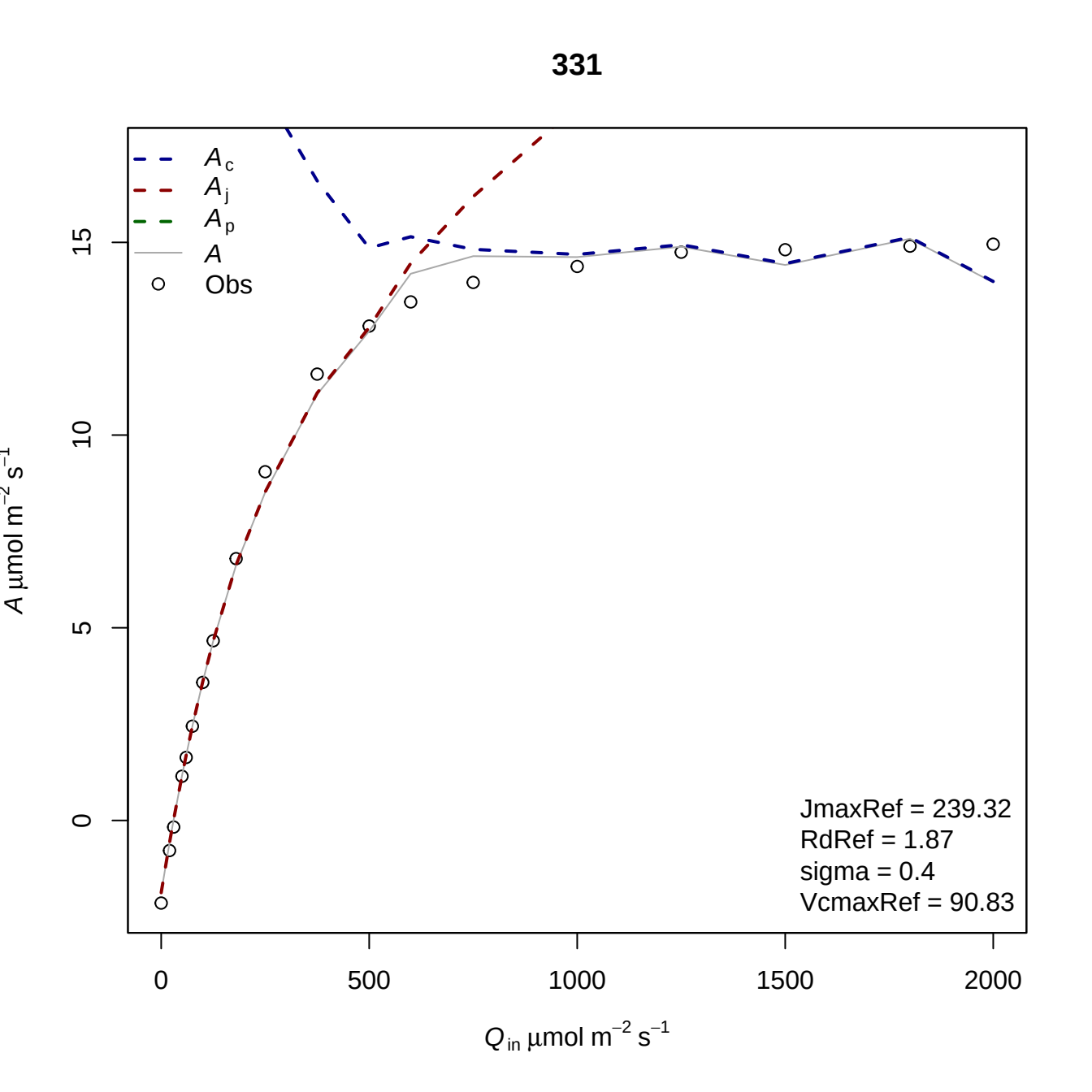

326
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.99
RdRef = 1.5
sigma = 0.46
VcmaxRef = 40.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

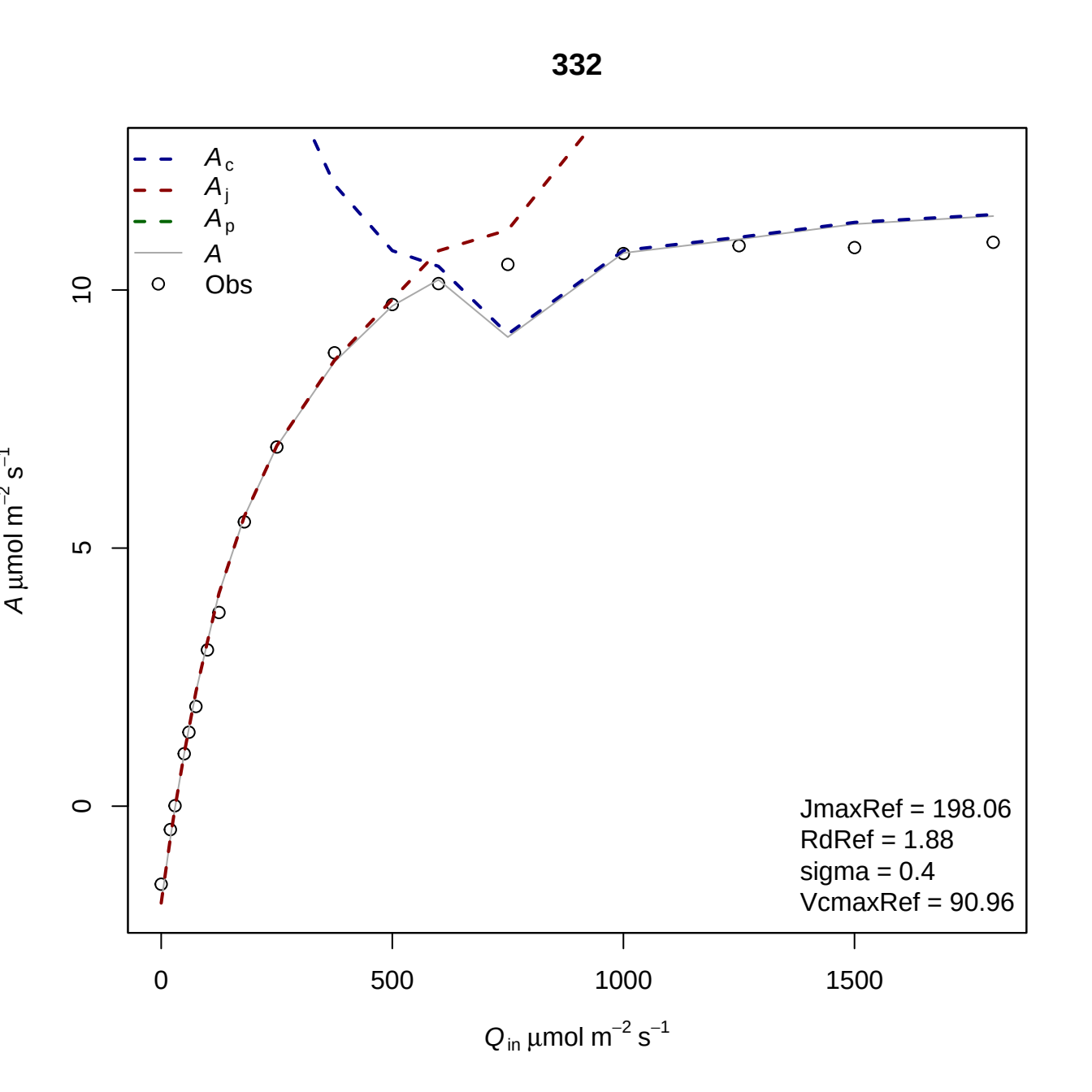

327
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 144.97
RdRef = 1.17
sigma = 0.26
VcmaxRef = 51.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

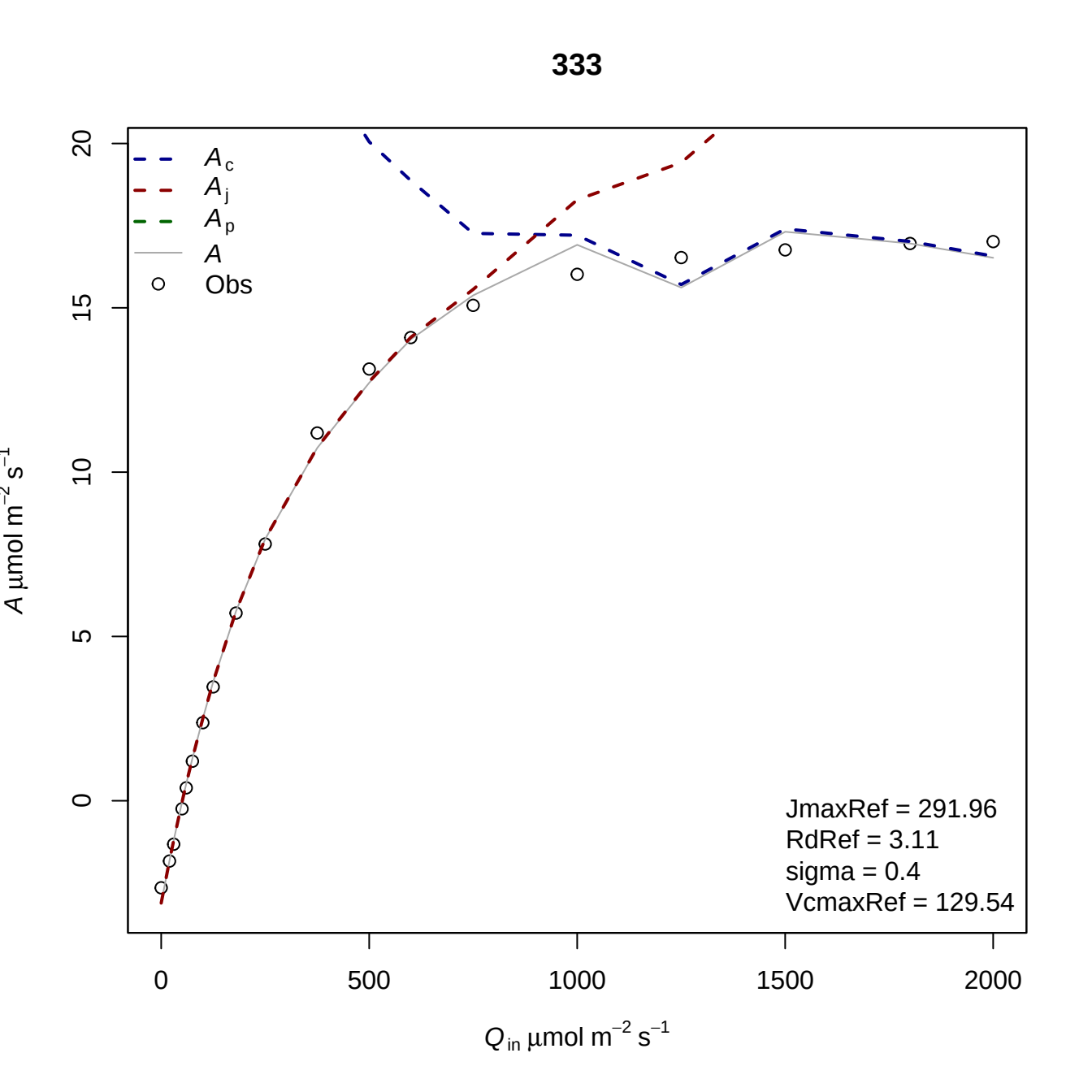

328
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.84
RdRef = 1.67
sigma = 0.2
VcmaxRef = 45.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

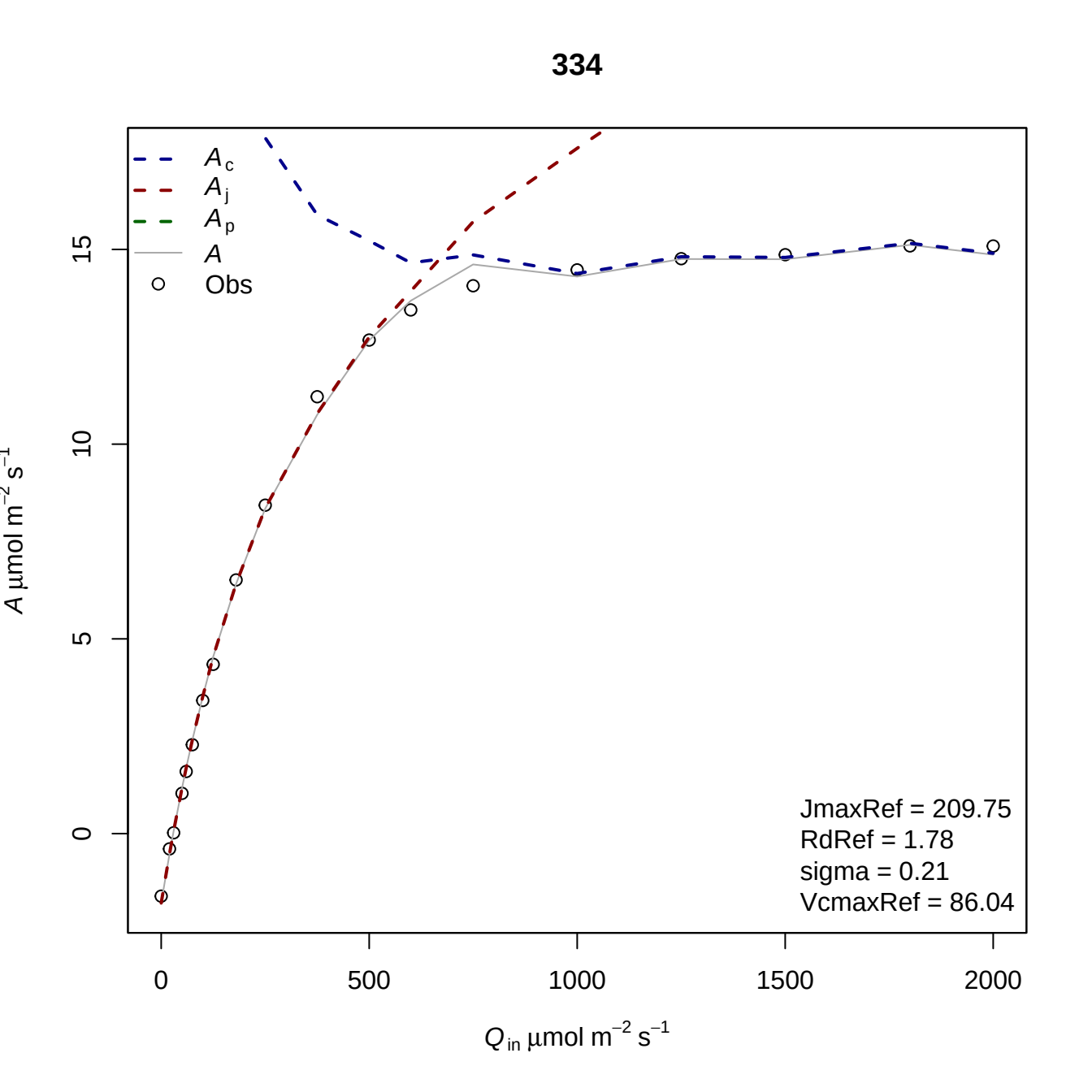

329
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 184.17
RdRef = 2.52
sigma = 0.16
VcmaxRef = 83.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

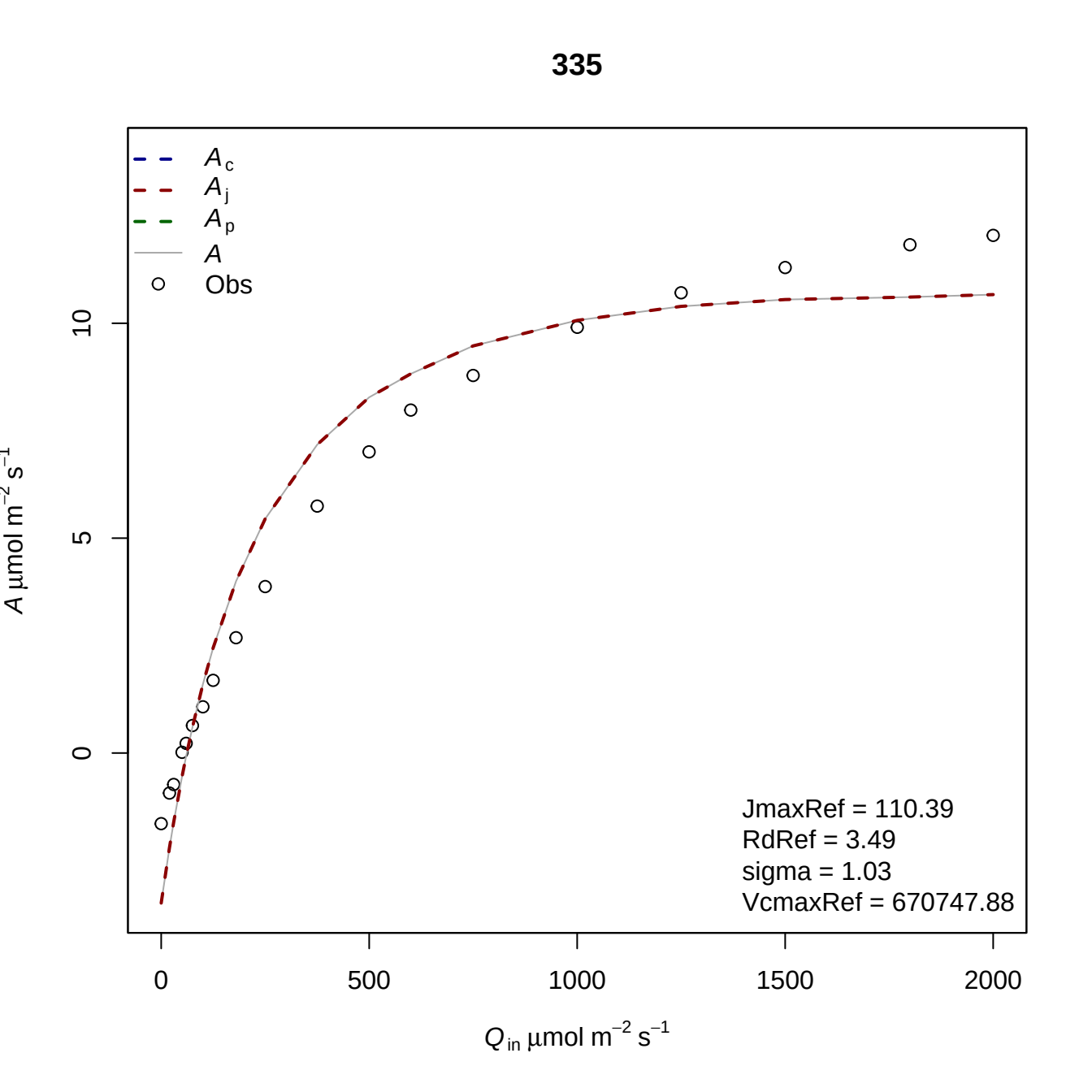

330
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 234.23
RdRef = 1.89
sigma = 0.36
VcmaxRef = 93.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

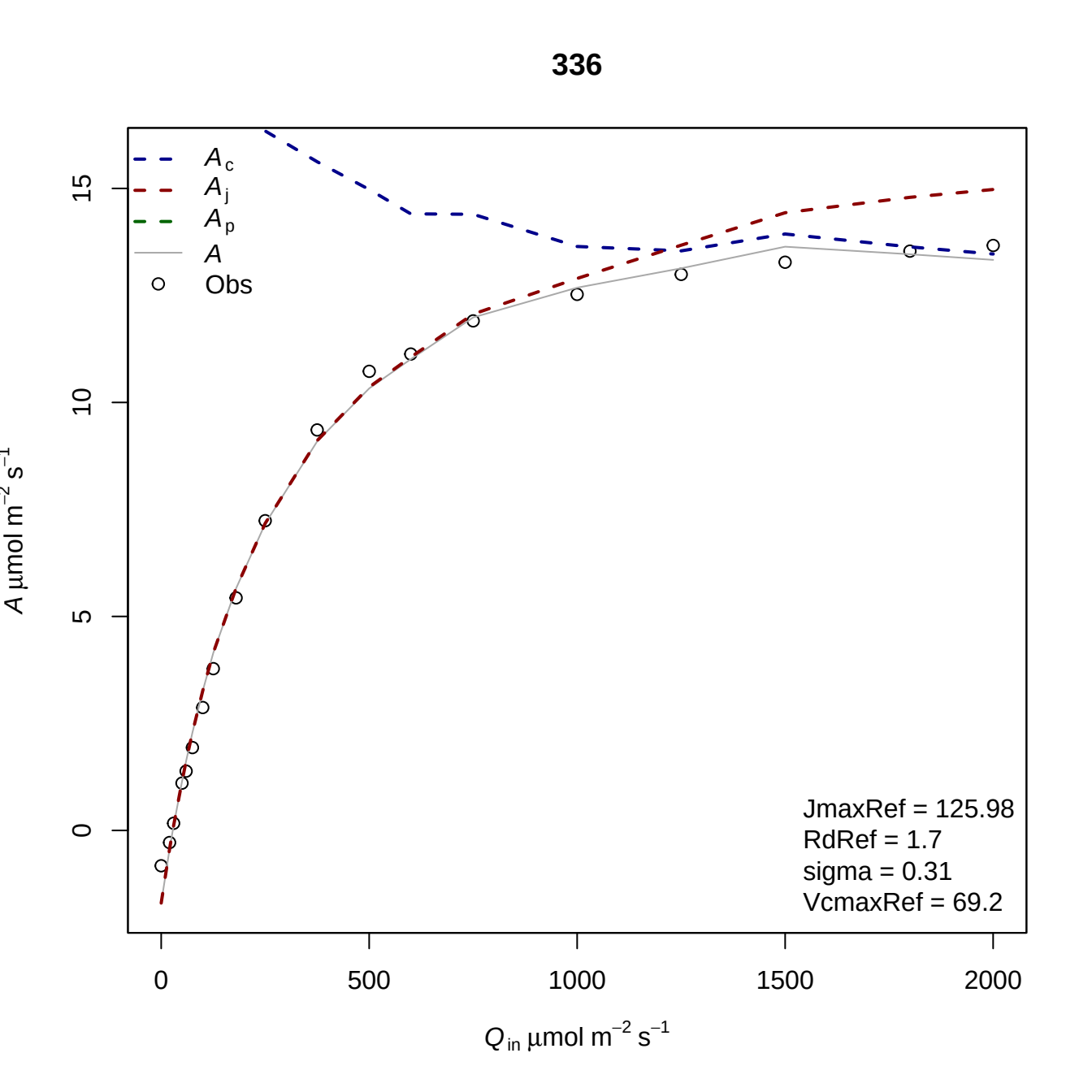

331
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 239.32
RdRef = 1.87
sigma = 0.4
VcmaxRef = 90.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

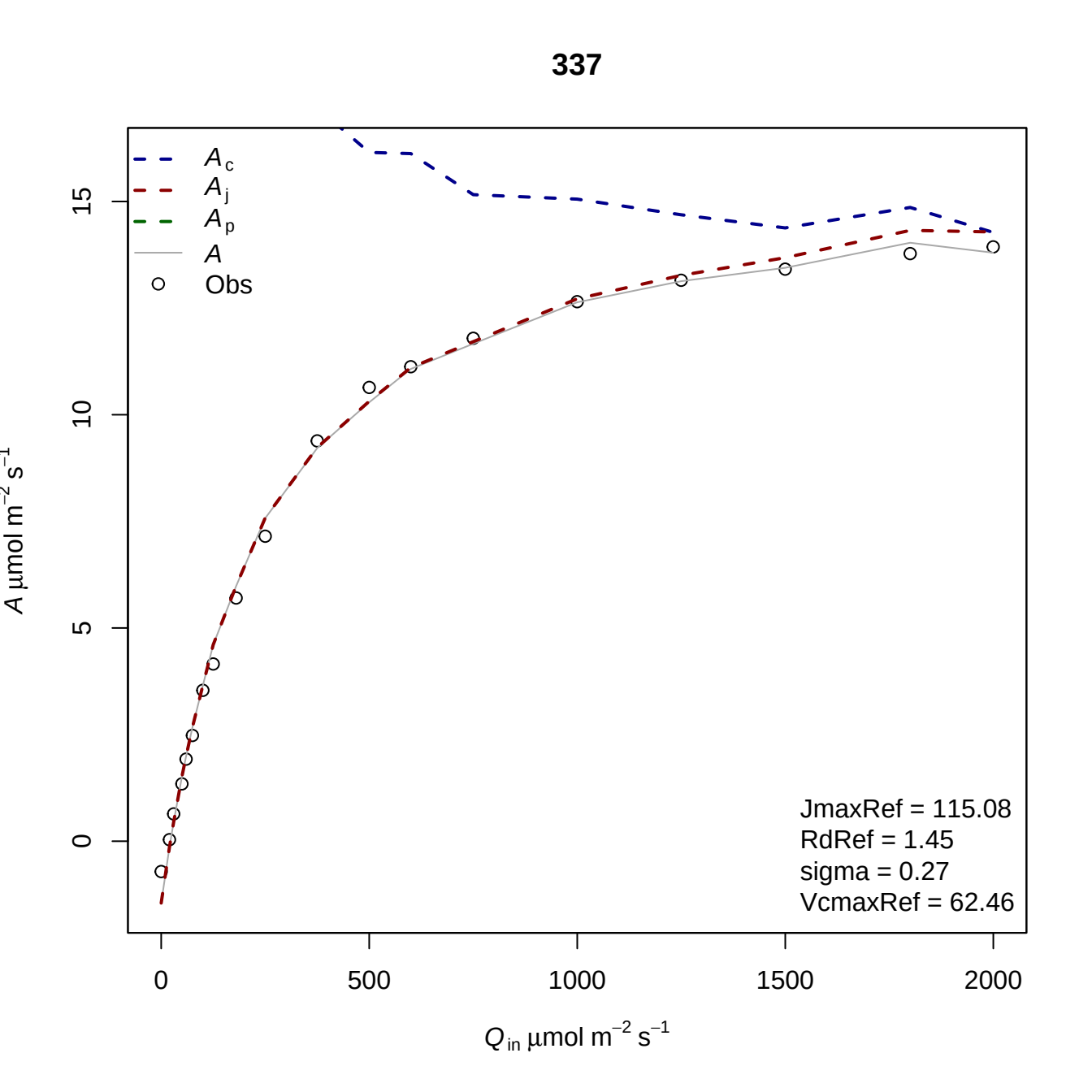

332
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 198.06
RdRef = 1.88
sigma = 0.4
VcmaxRef = 90.96
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

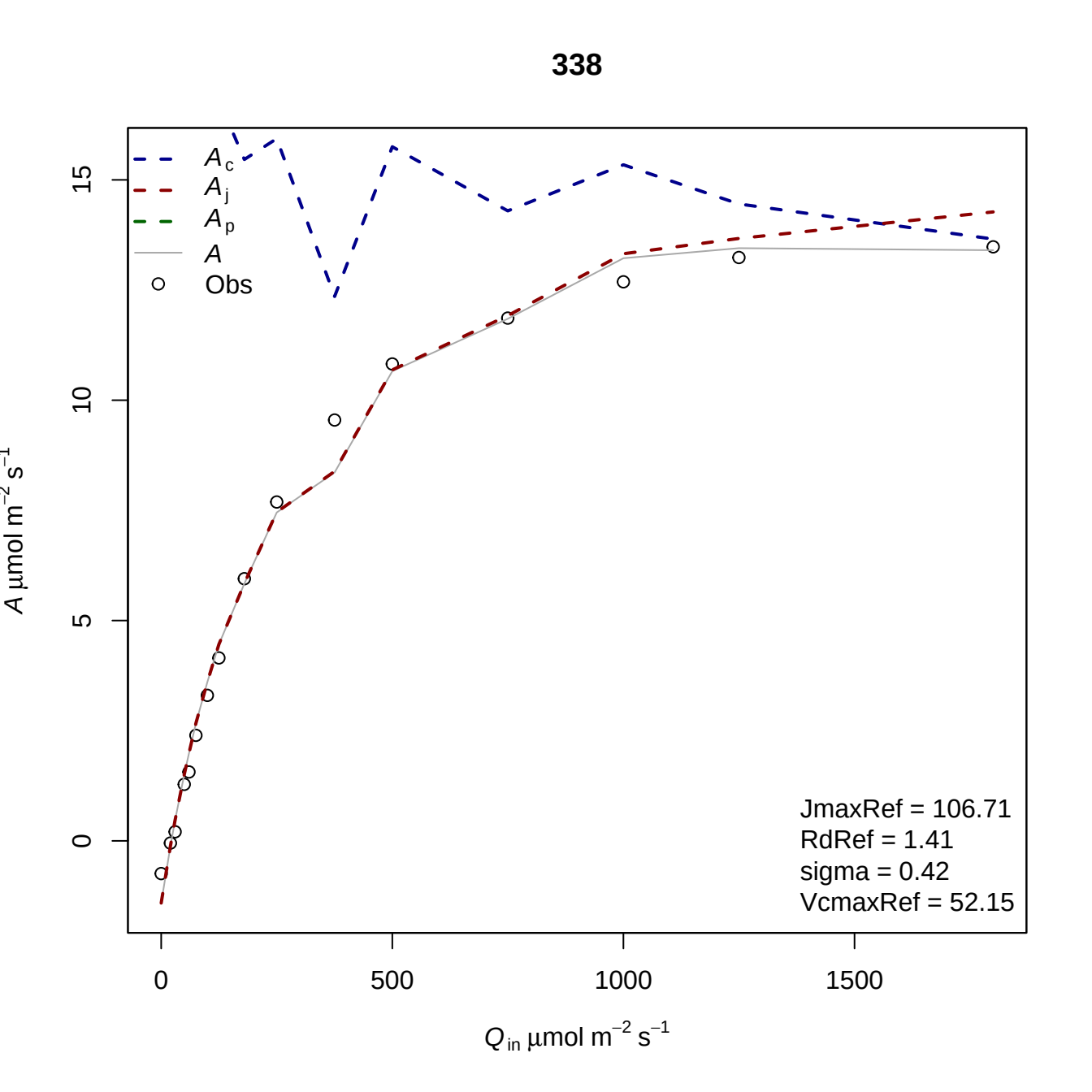

333
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 291.96
RdRef = 3.11
sigma = 0.4
VcmaxRef = 129.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

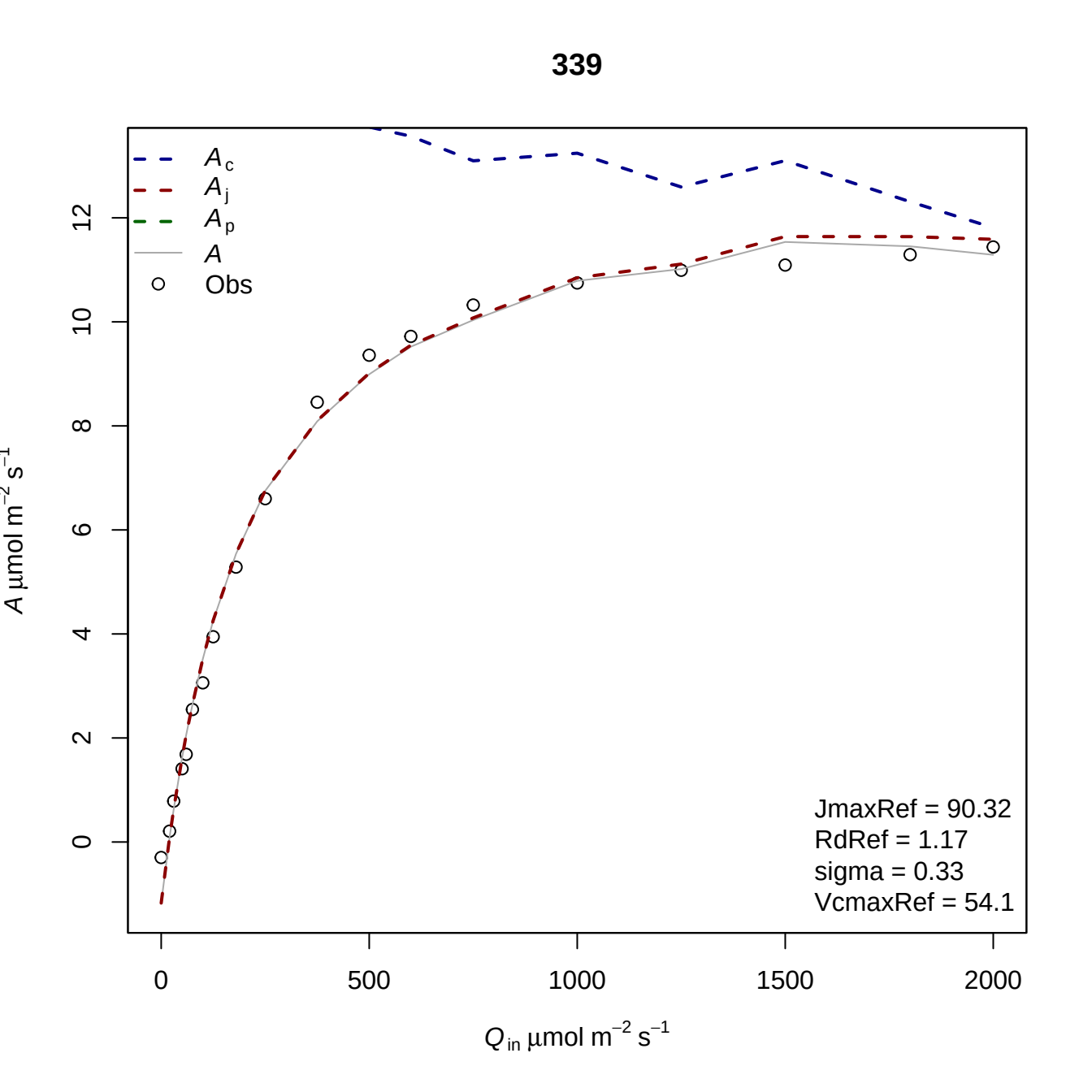

334
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 209.75
RdRef = 1.78
sigma = 0.21
VcmaxRef = 86.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

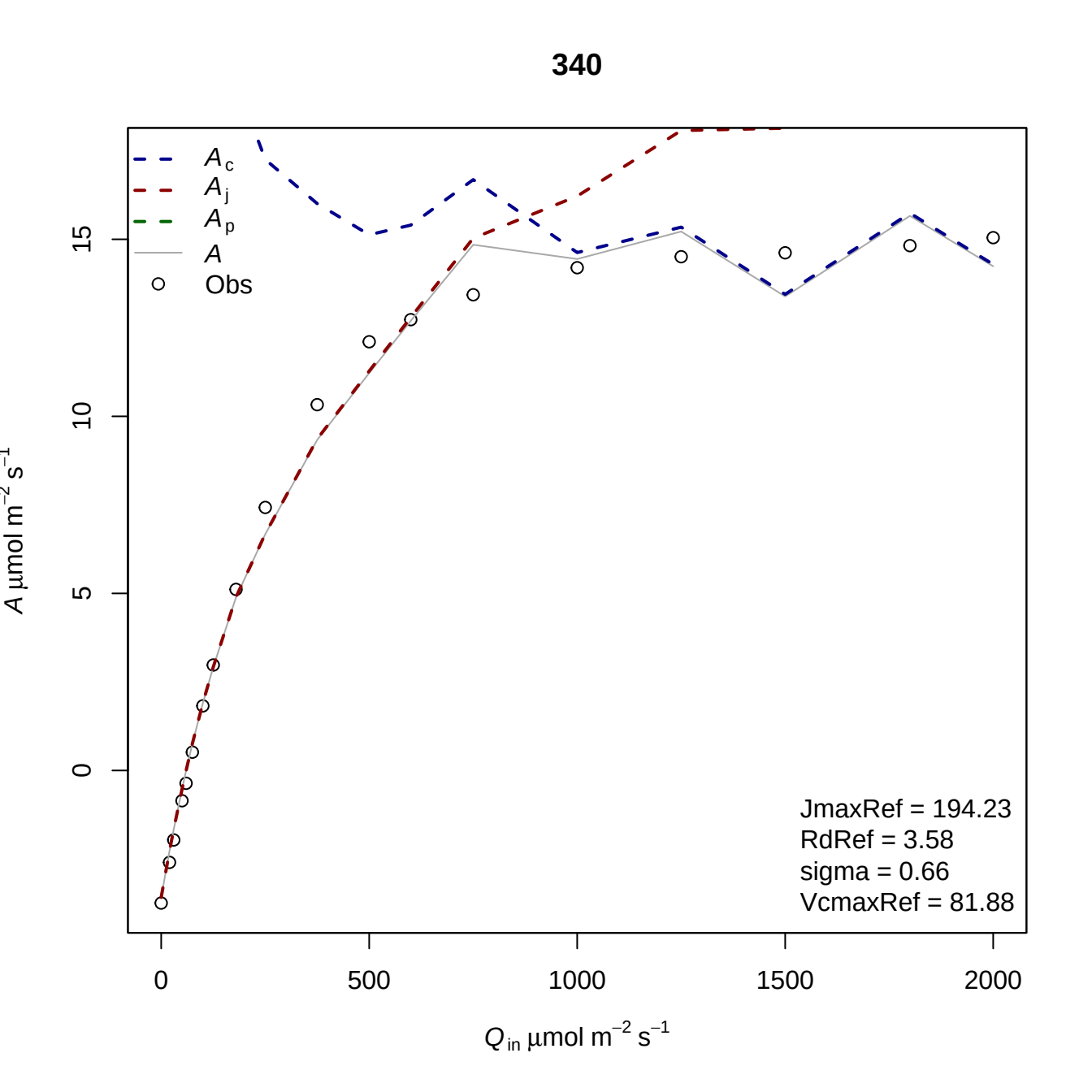

335
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.39
RdRef = 3.49
sigma = 1.03
VcmaxRef = 670747.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

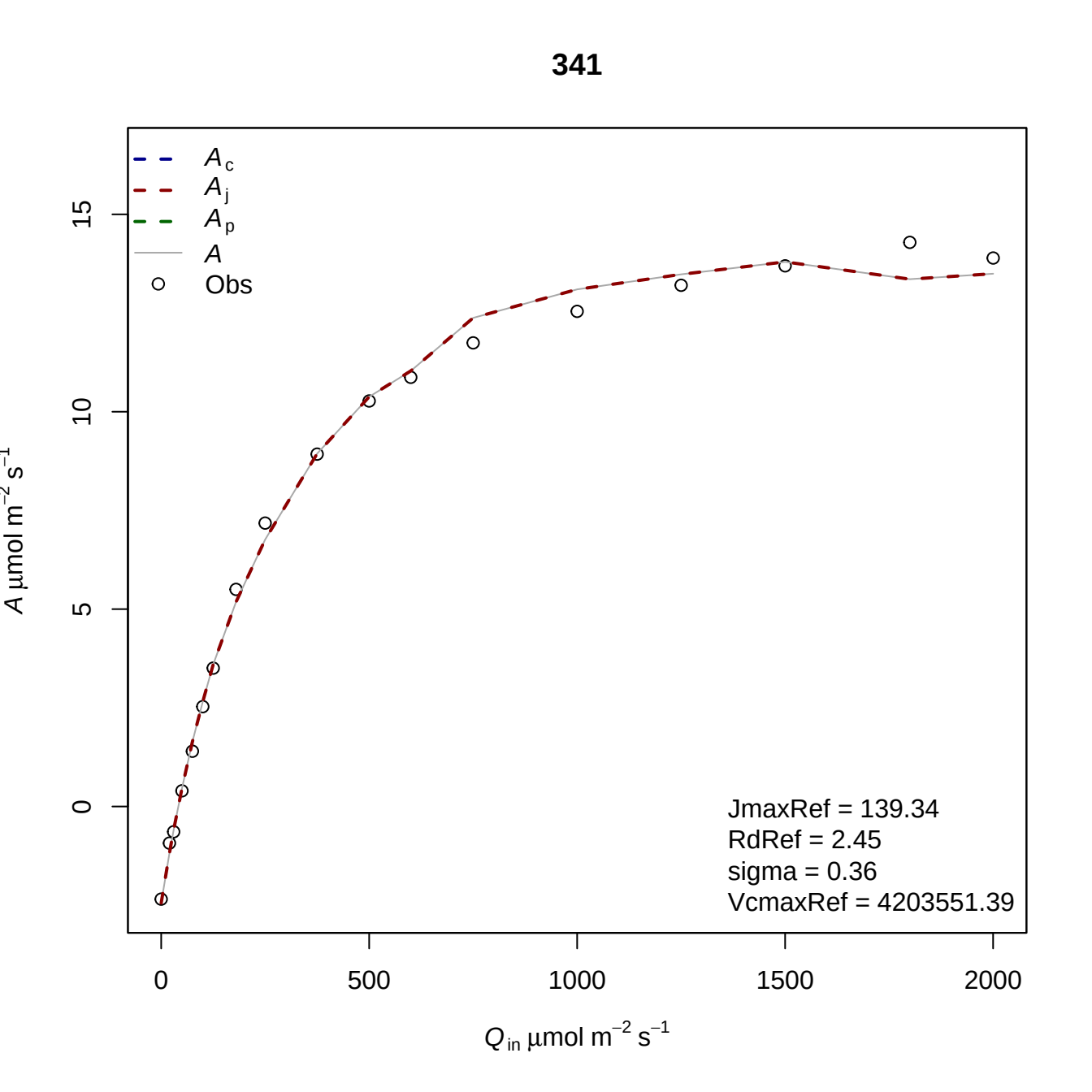

336
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.98
RdRef = 1.7
sigma = 0.31
VcmaxRef = 69.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

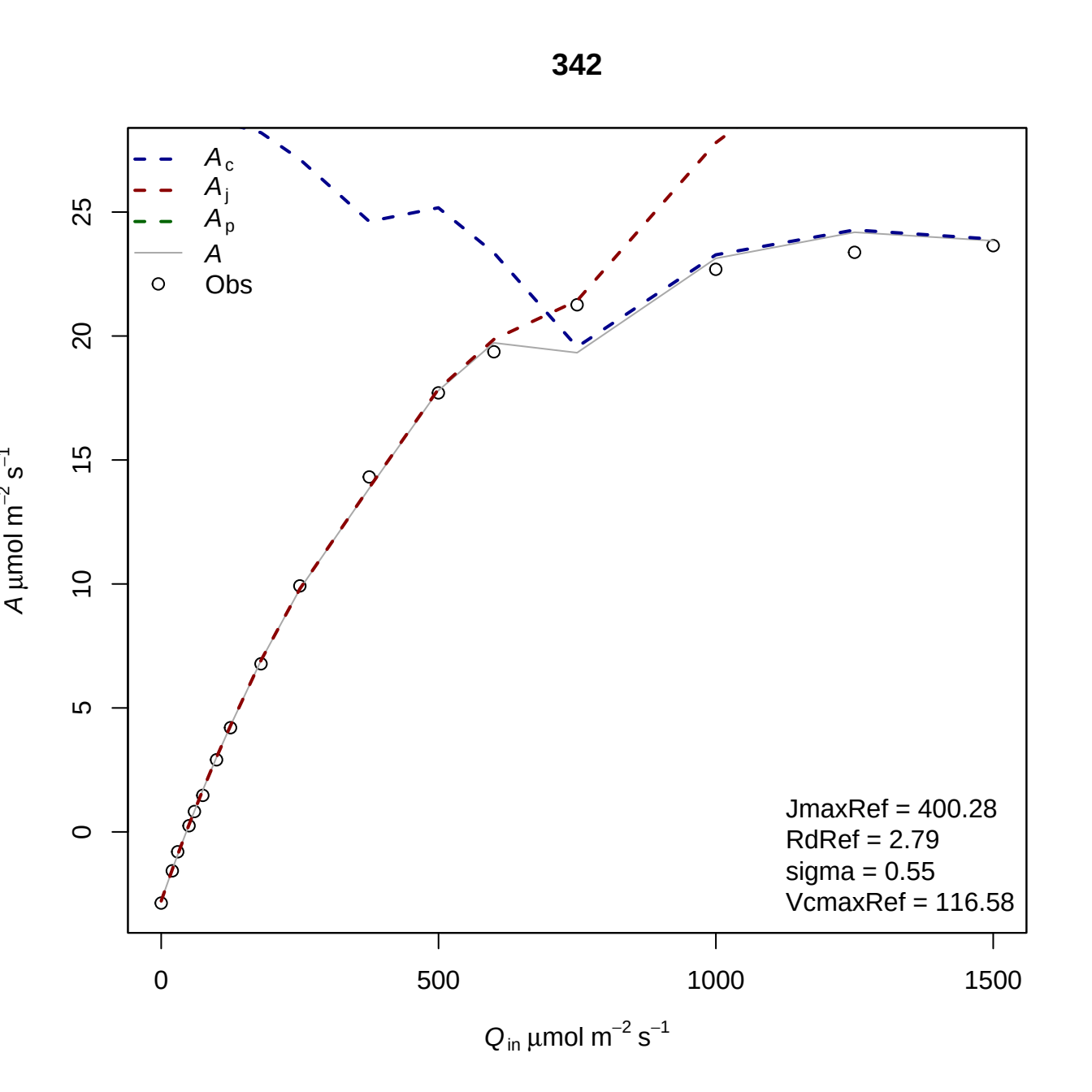

337
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.08
RdRef = 1.45
sigma = 0.27
VcmaxRef = 62.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

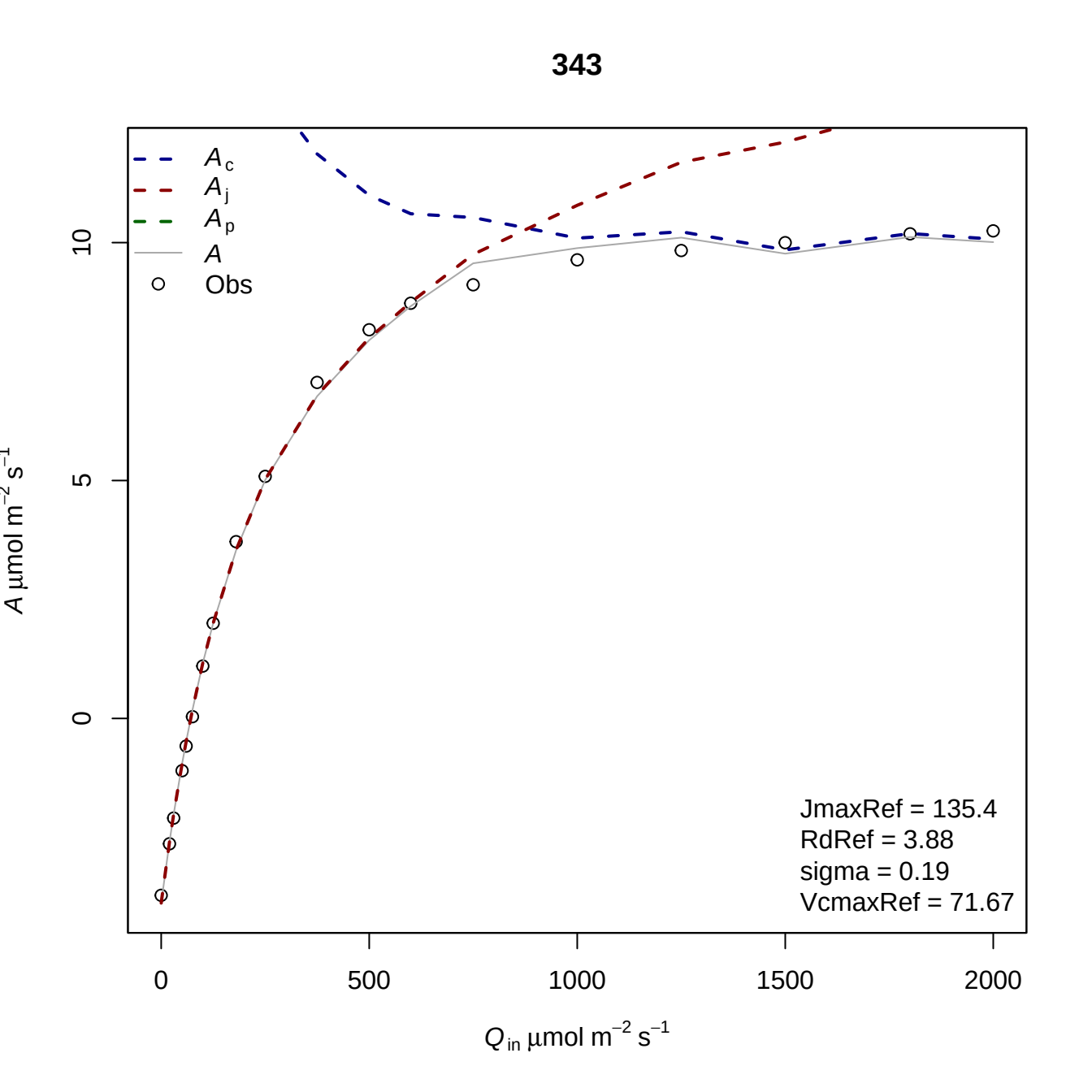

338
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 106.71
RdRef = 1.41
sigma = 0.42
VcmaxRef = 52.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

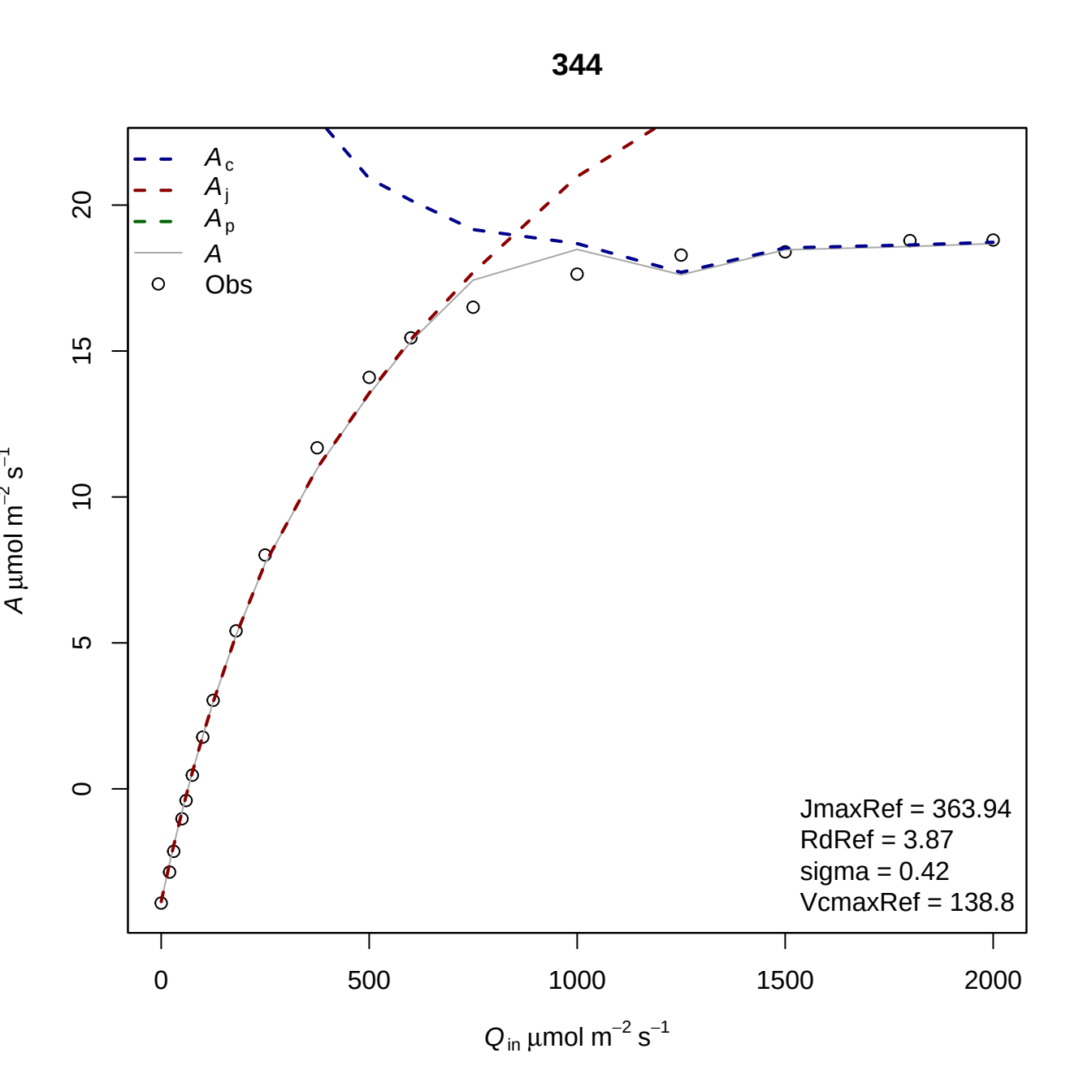

339
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.32
RdRef = 1.17
sigma = 0.33
VcmaxRef = 54.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

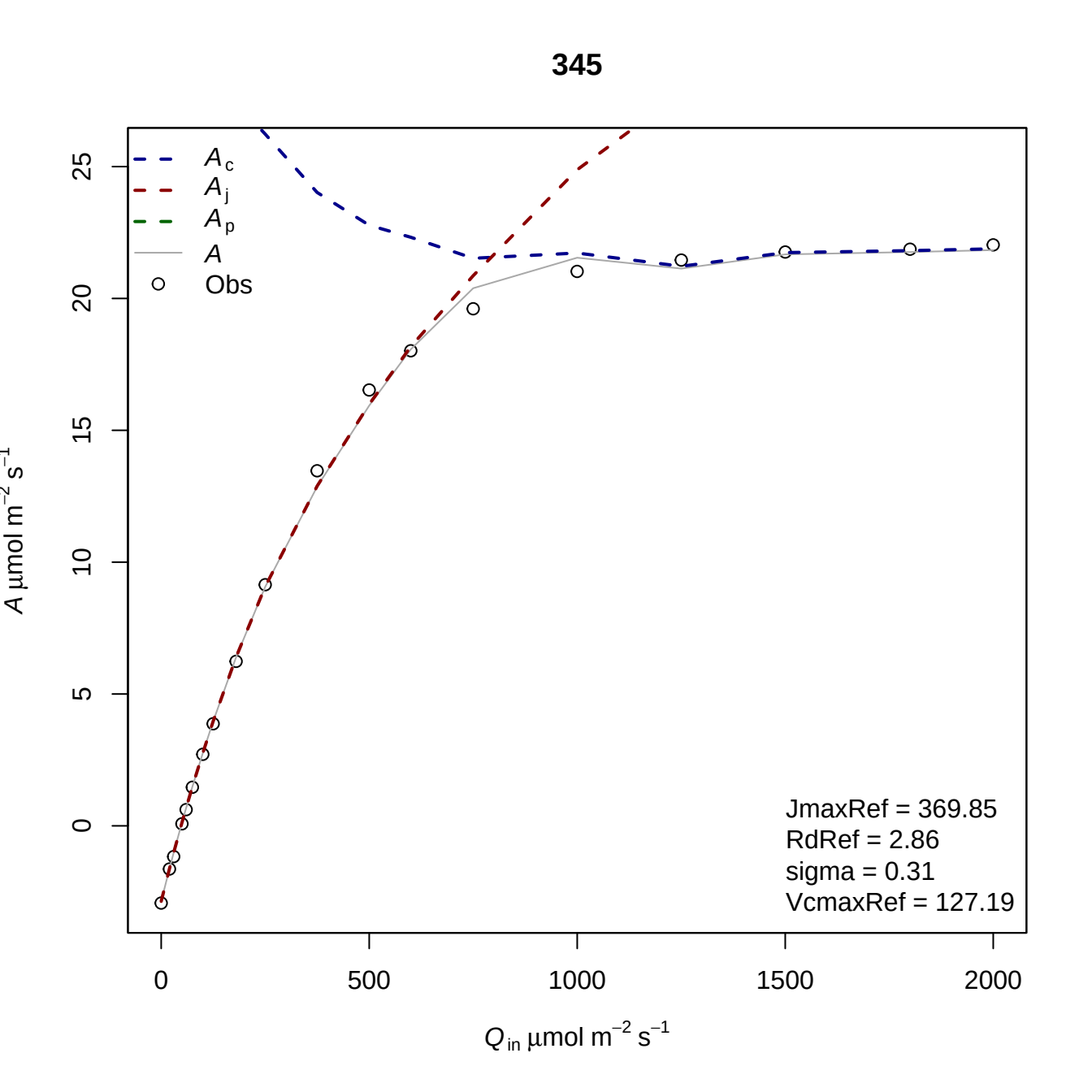

340
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 194.23
RdRef = 3.58
sigma = 0.66
VcmaxRef = 81.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

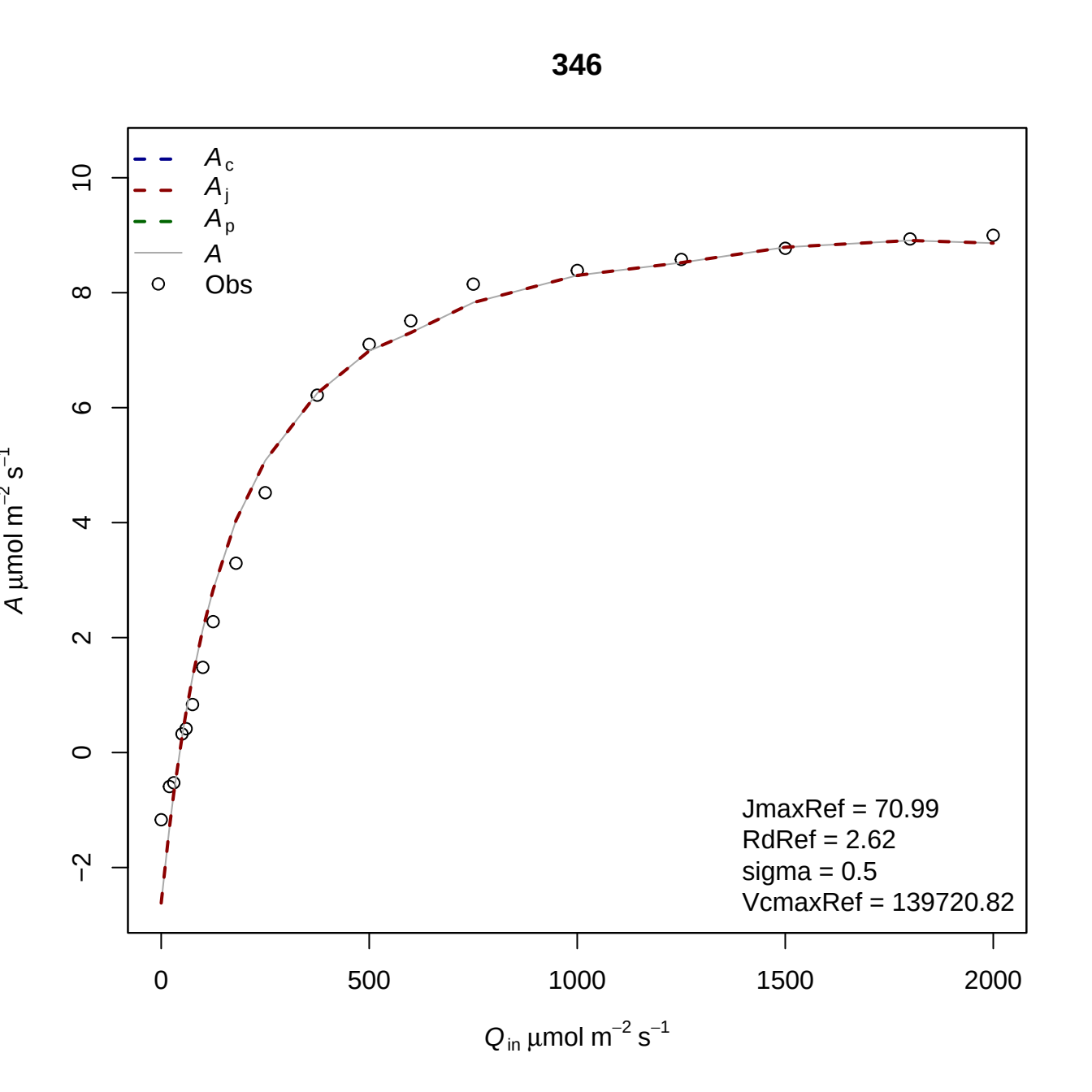

341
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.34
RdRef = 2.45
sigma = 0.36
VcmaxRef = 4203551.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

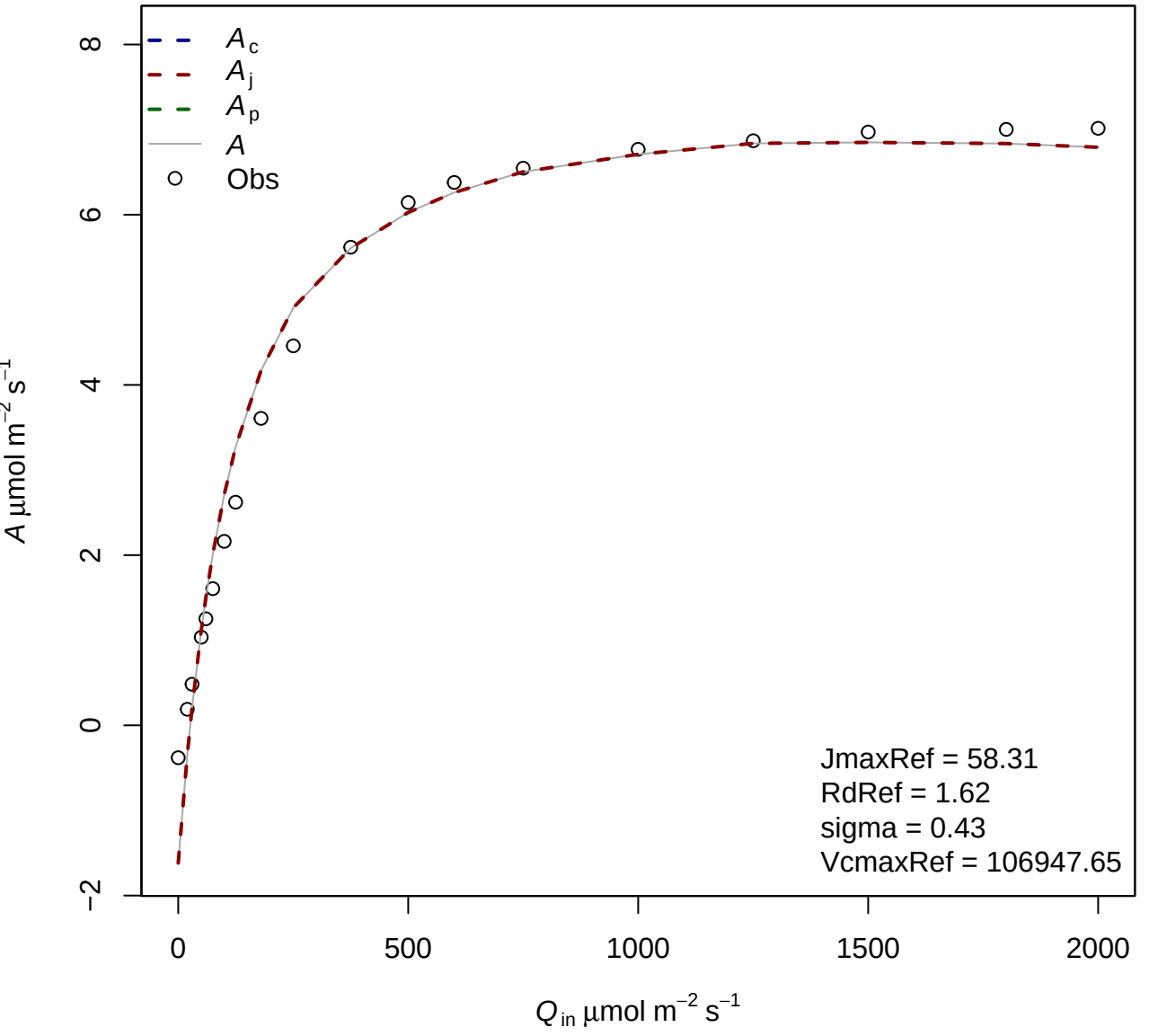

342
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 400.28
RdRef = 2.79
sigma = 0.55
VcmaxRef = 116.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

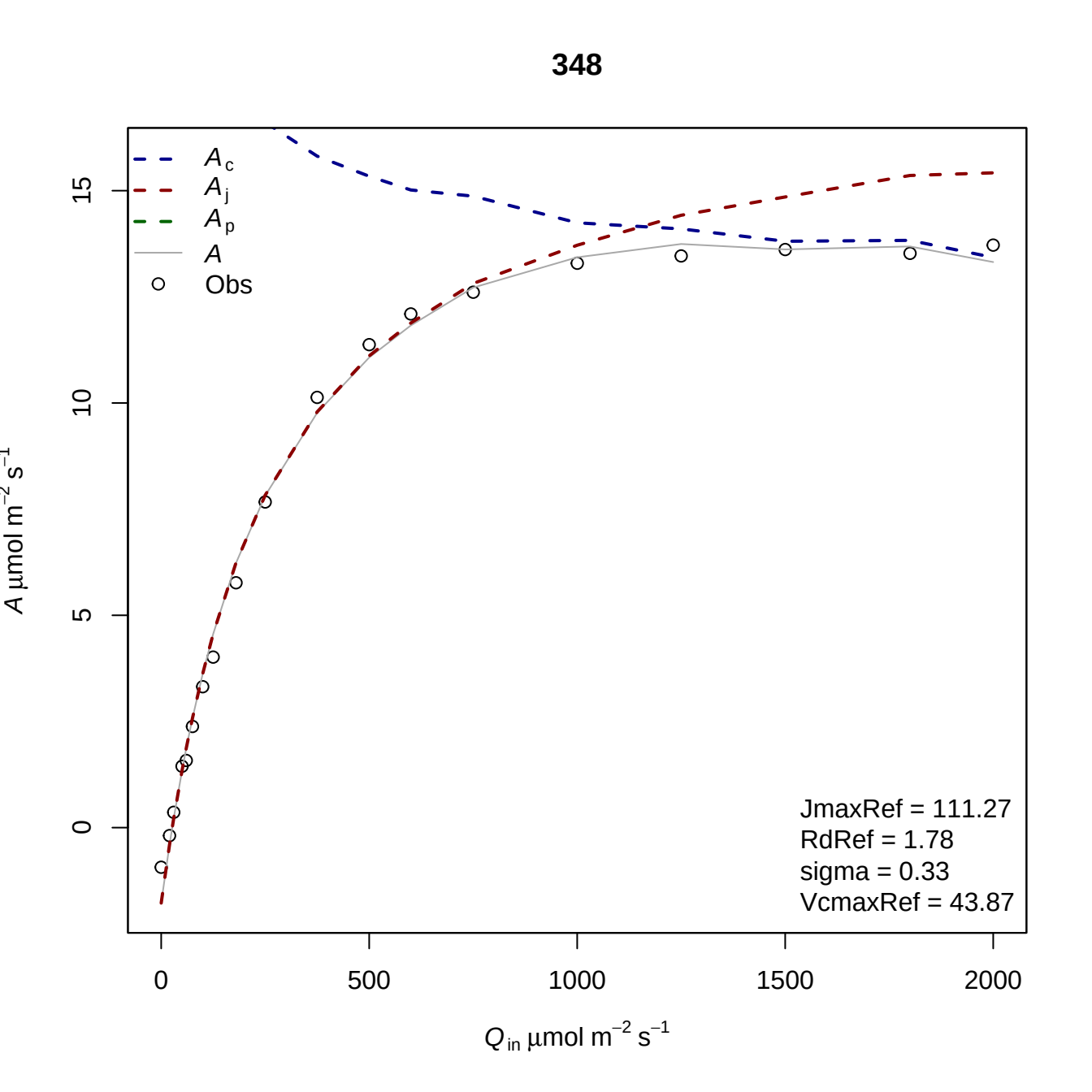

343
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.4
RdRef = 3.88
sigma = 0.19
VcmaxRef = 71.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

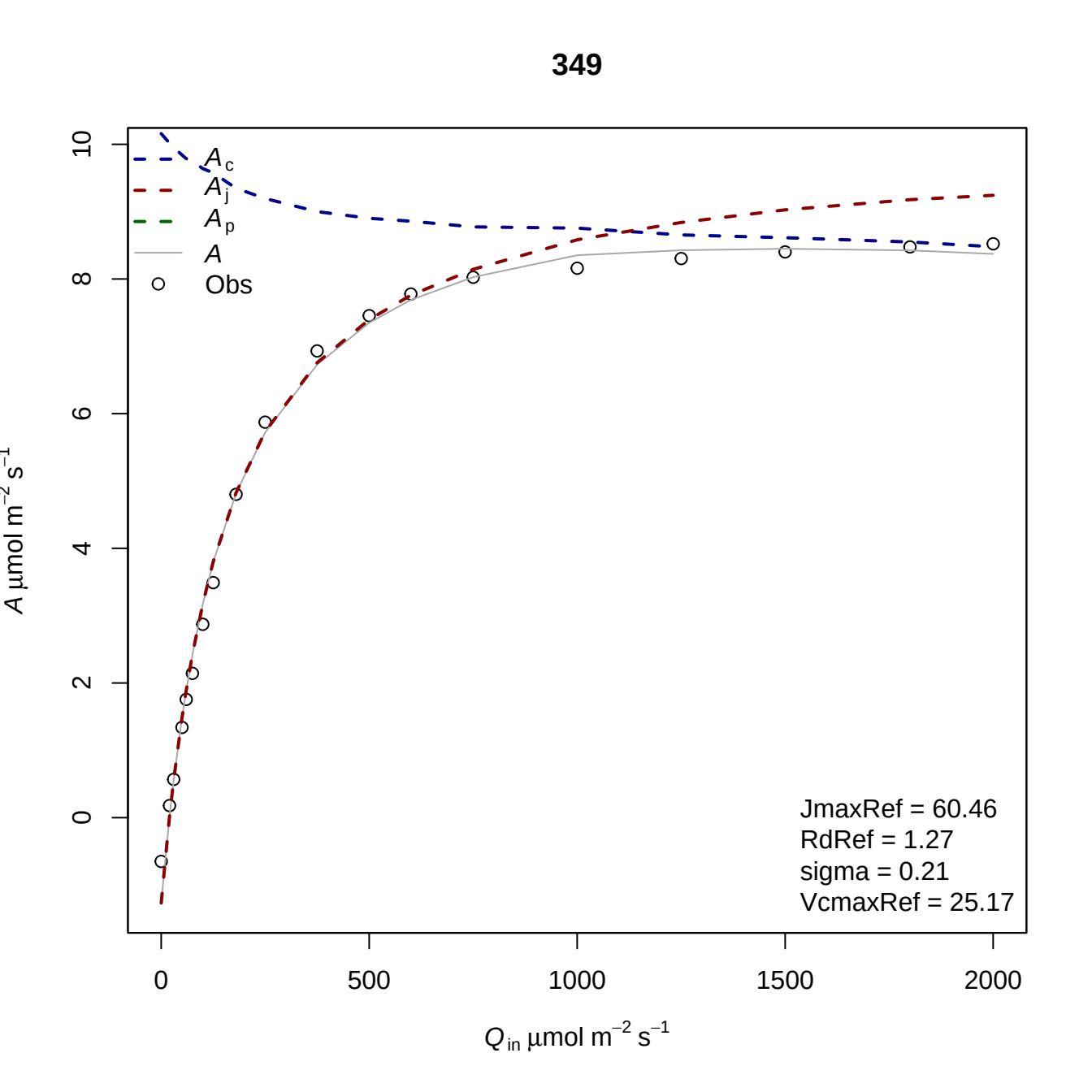

344
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 363.94
RdRef = 3.87
sigma = 0.42
VcmaxRef = 138.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

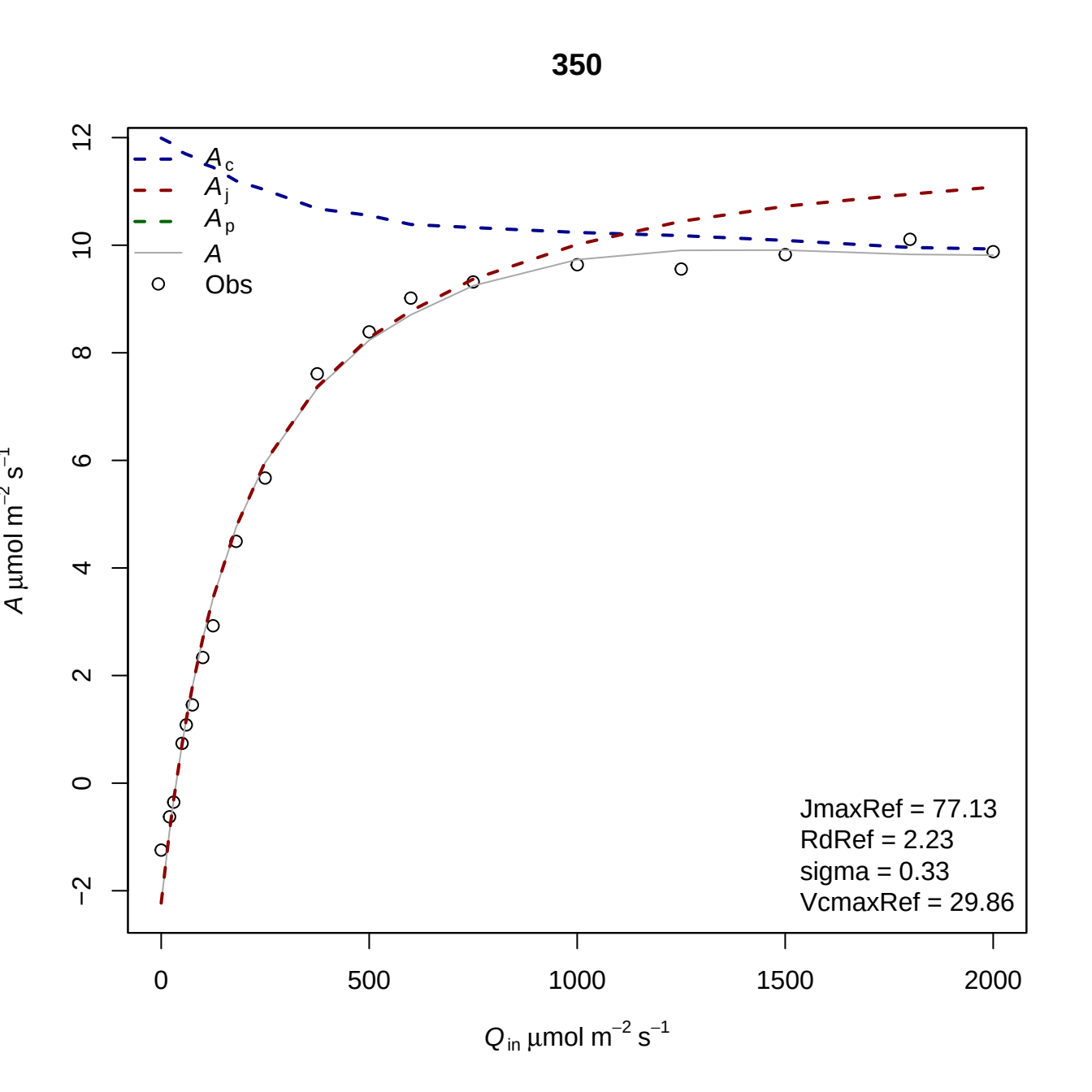

345
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 369.85
RdRef = 2.86
sigma = 0.31
VcmaxRef = 127.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

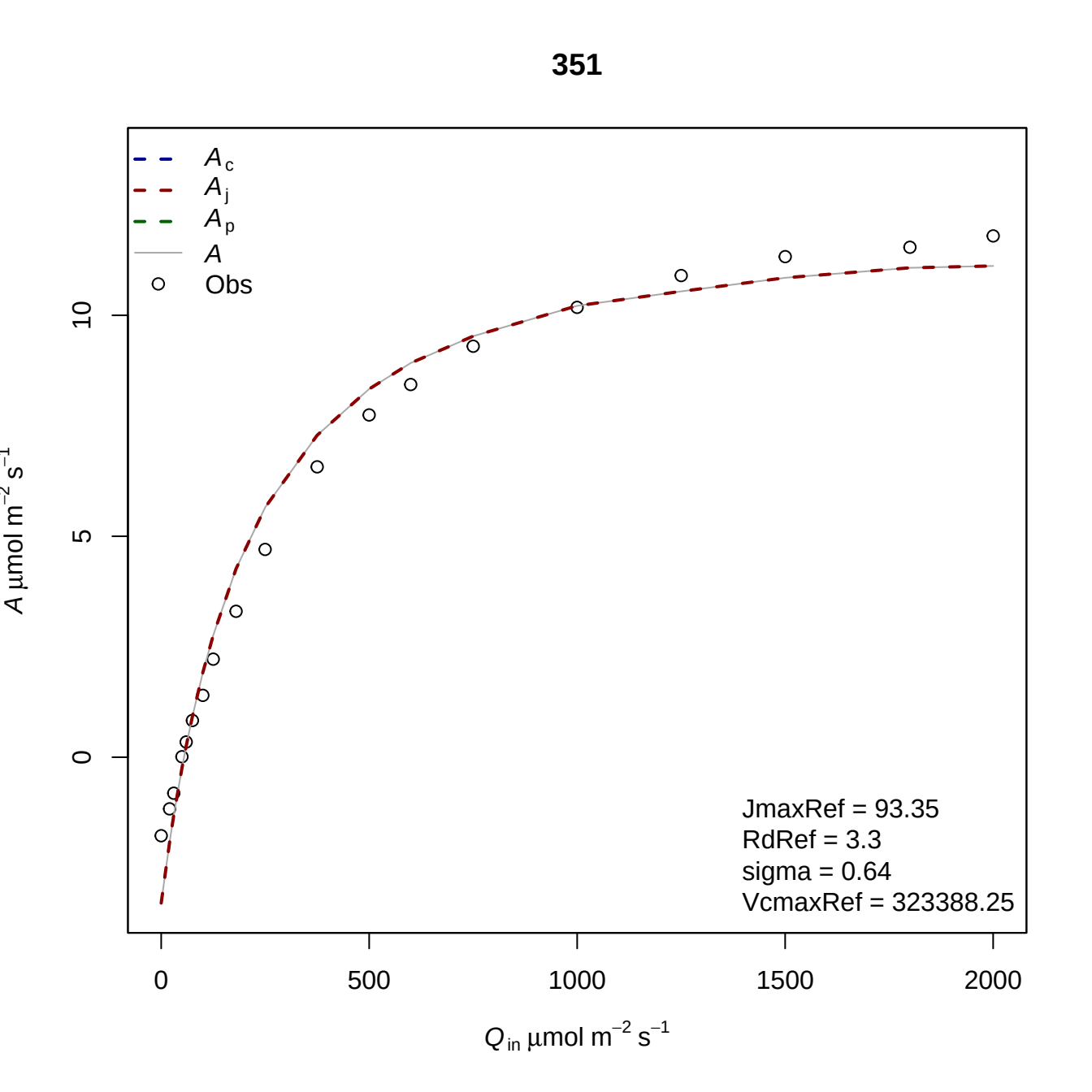

346
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.99
RdRef = 2.62
sigma = 0.5
VcmaxRef = 139720.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

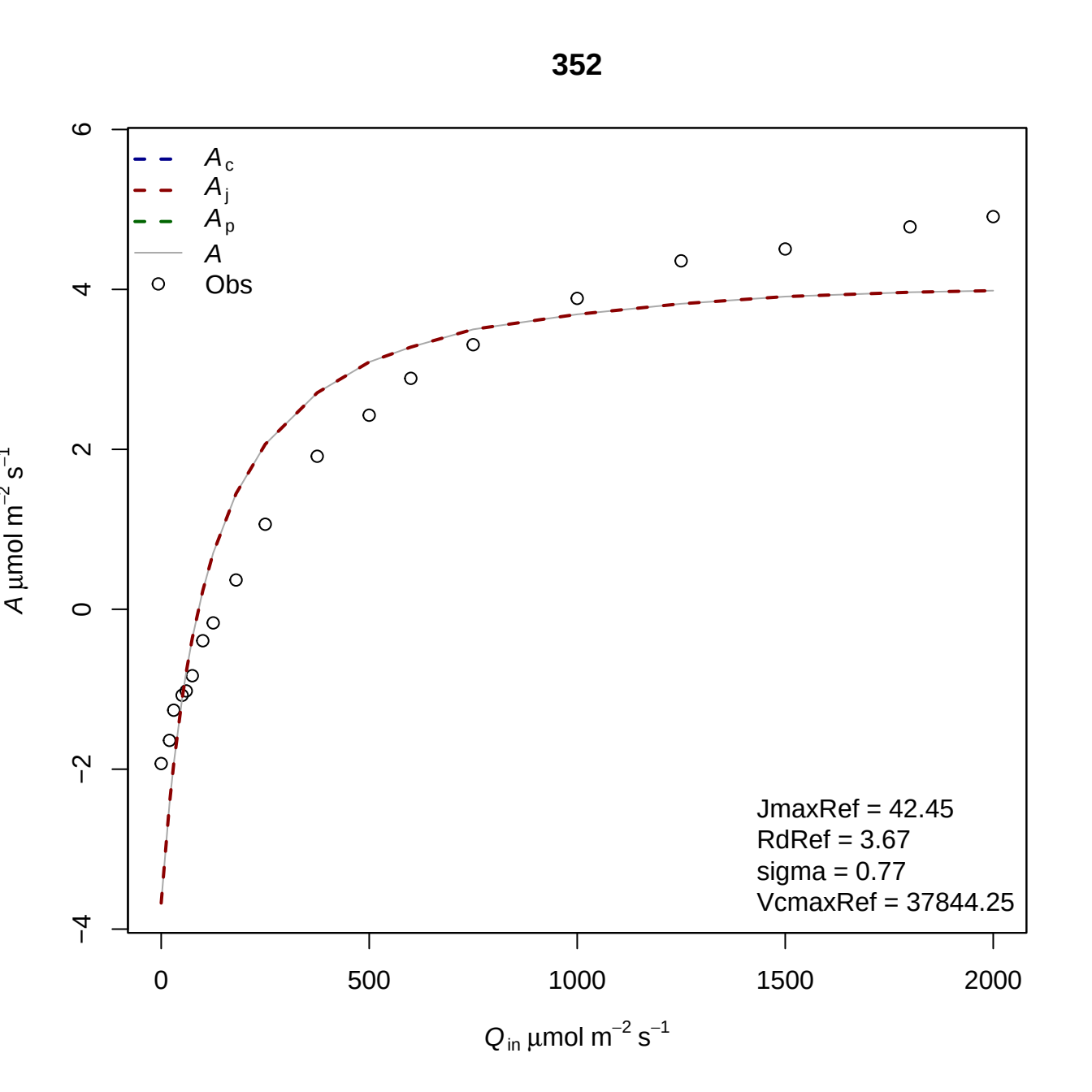

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.31
RdRef = 1.62
sigma = 0.43
VcmaxRef = 106947.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

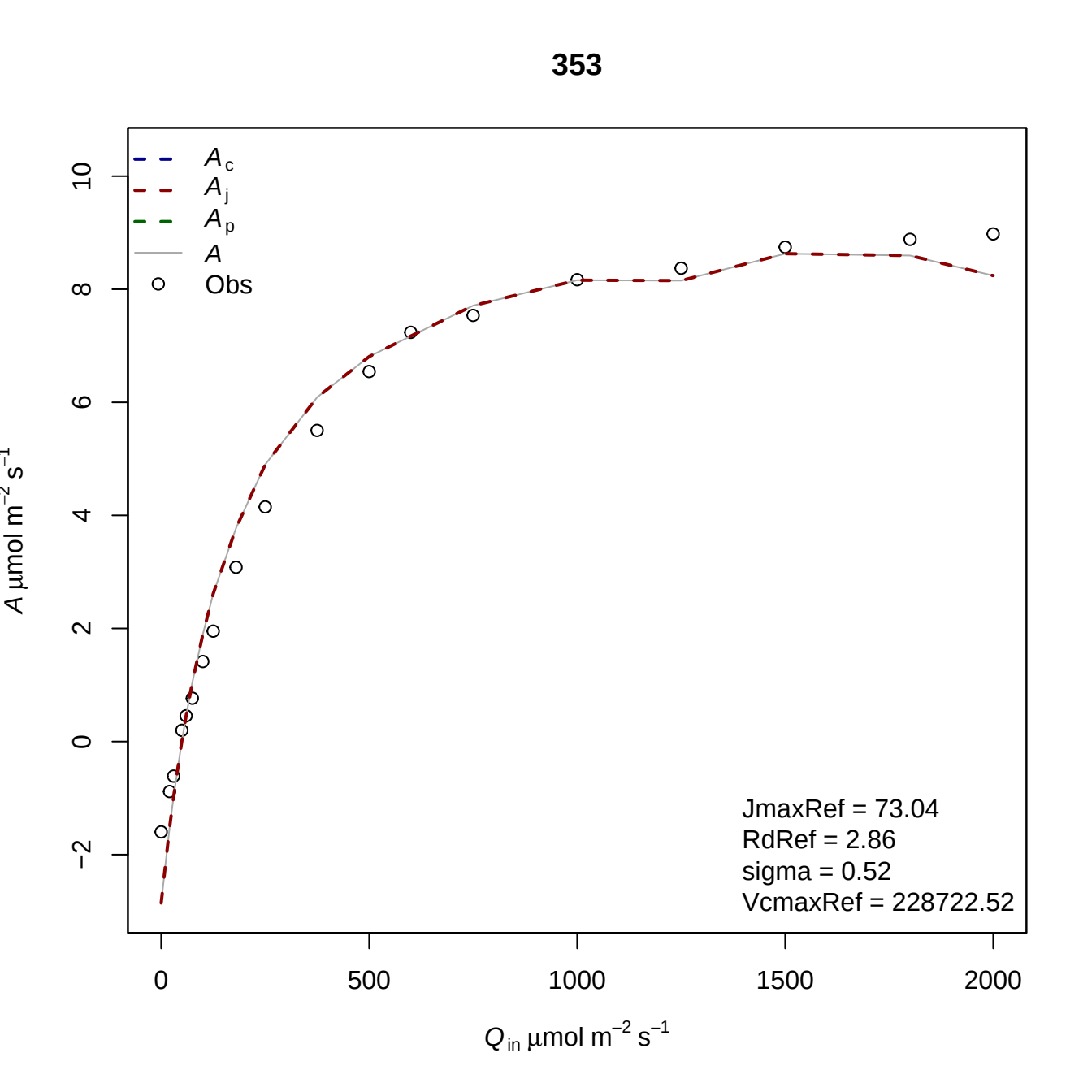

348
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.27
RdRef = 1.78
sigma = 0.33
VcmaxRef = 43.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

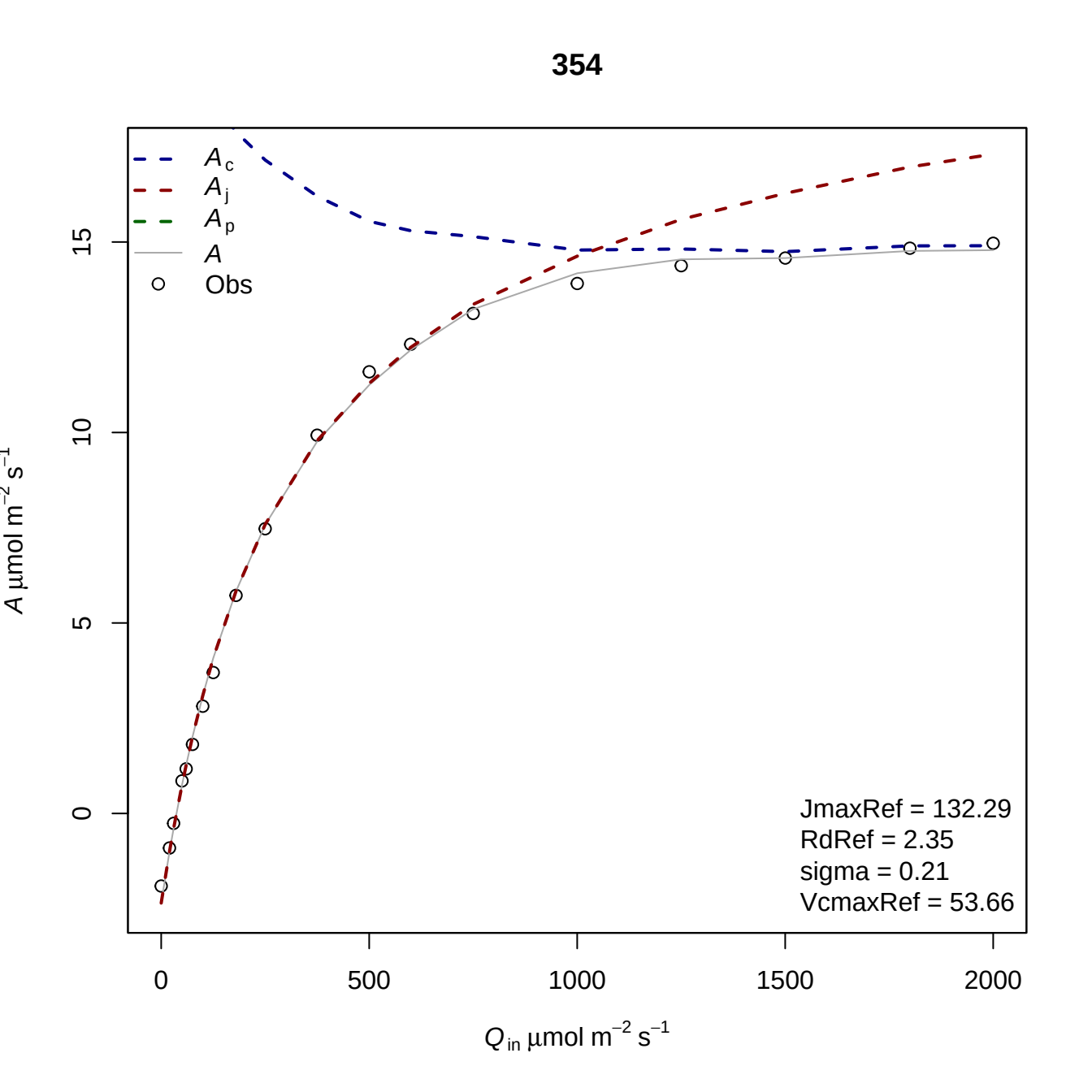

349
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.46
RdRef = 1.27
sigma = 0.21
VcmaxRef = 25.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

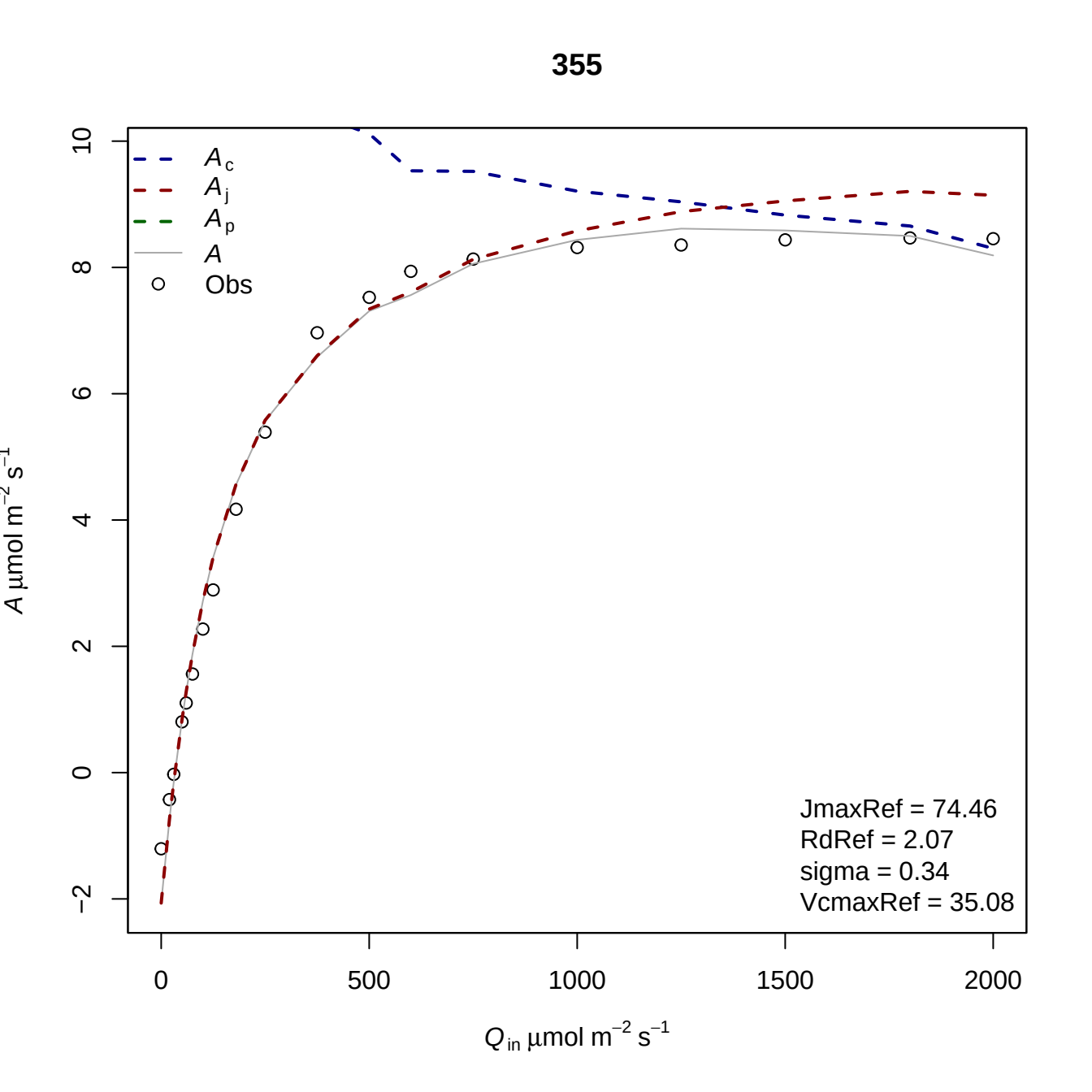

350
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.13
RdRef = 2.23
sigma = 0.33
VcmaxRef = 29.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

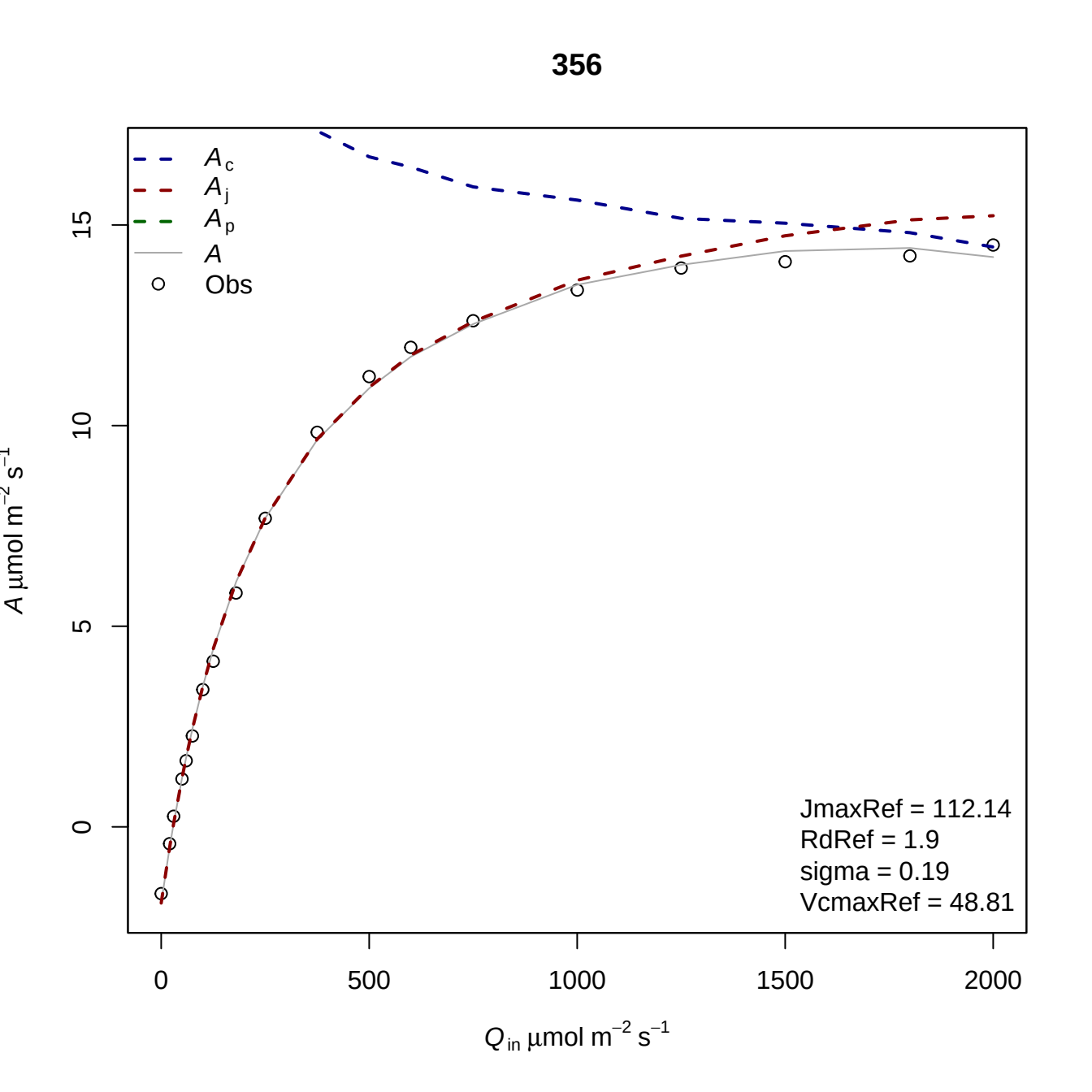

351
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.35
RdRef = 3.3
sigma = 0.64
VcmaxRef = 323388.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

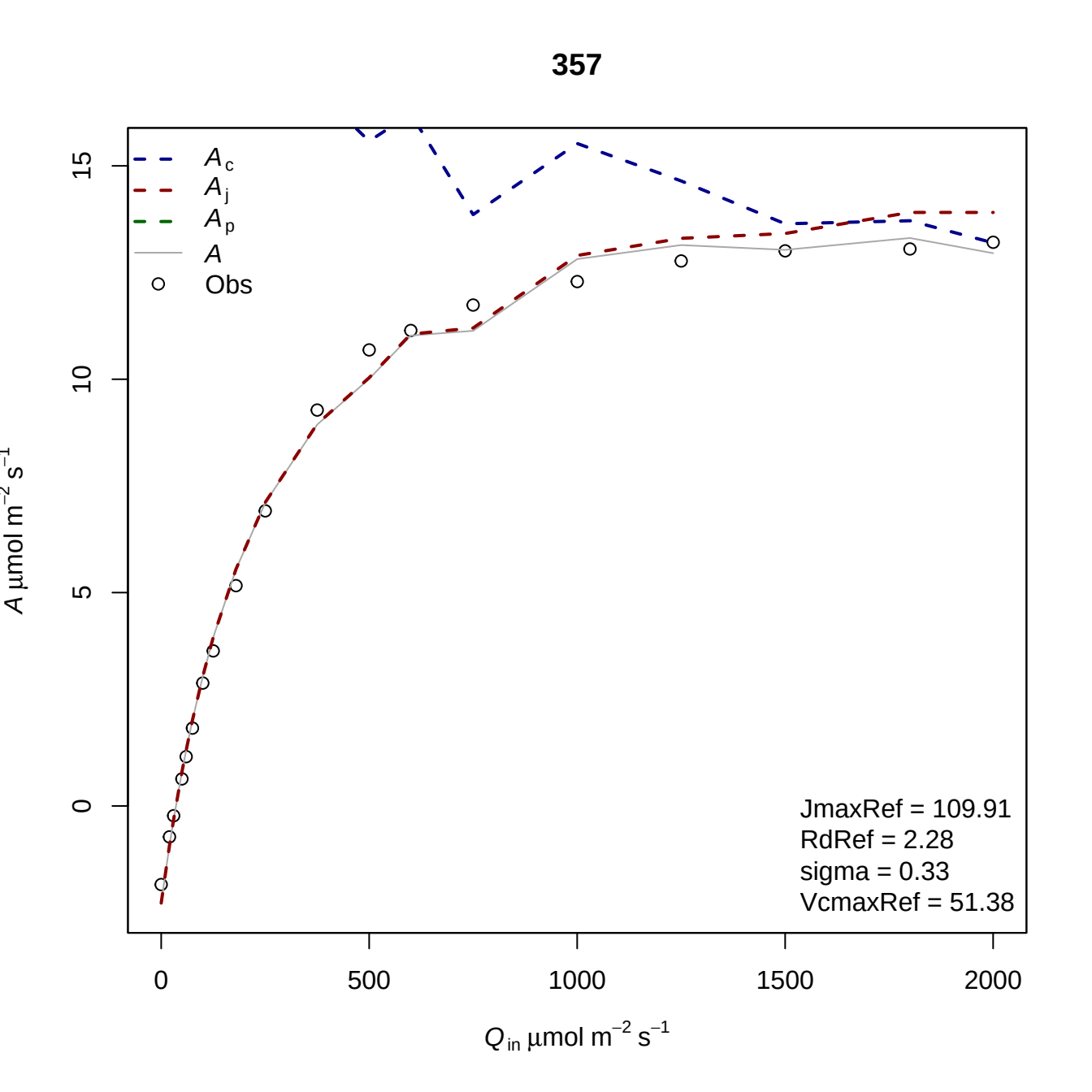

352
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.45
RdRef = 3.67
sigma = 0.77
VcmaxRef = 37844.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500
2000

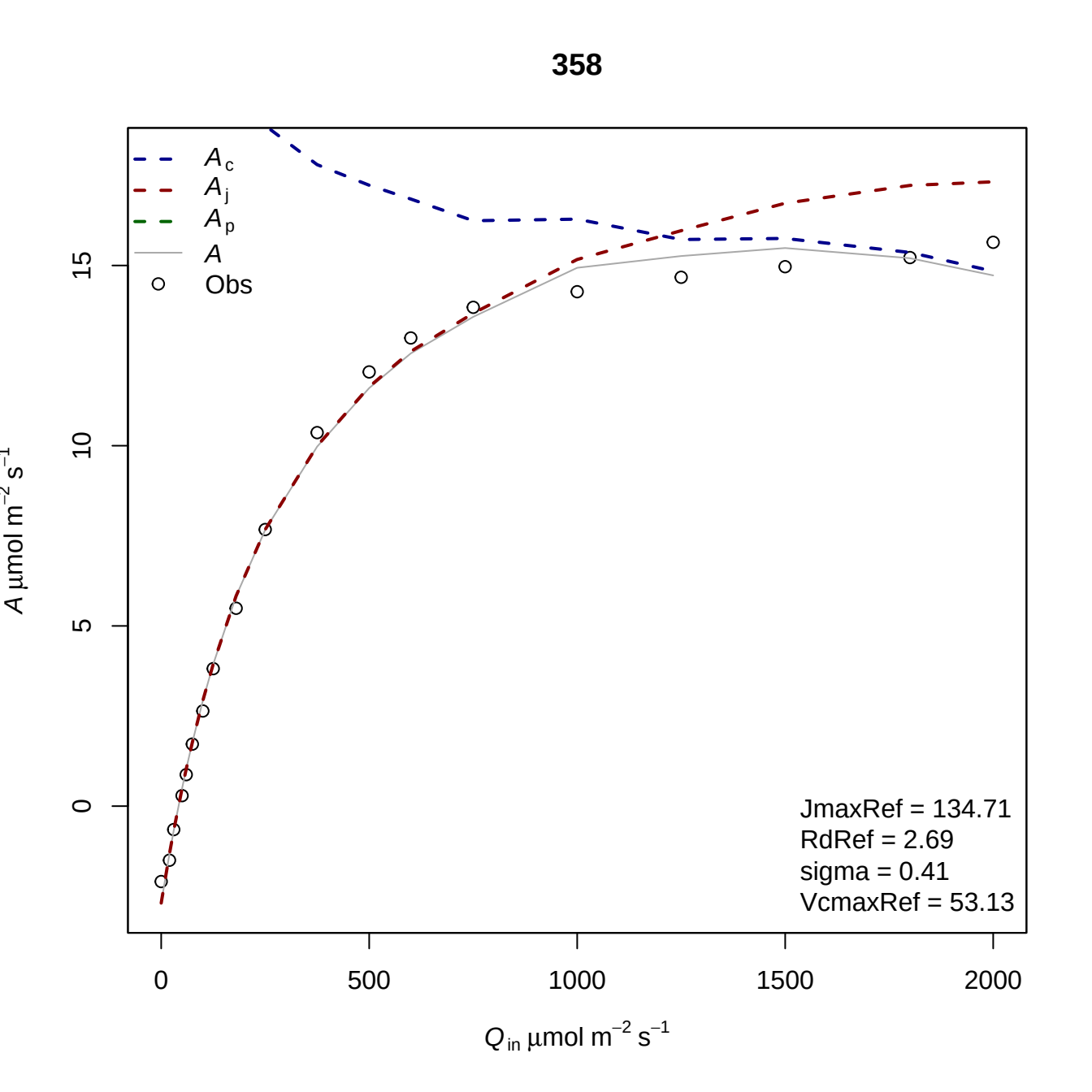

353
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.04
RdRef = 2.86
sigma = 0.52
VcmaxRef = 228722.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

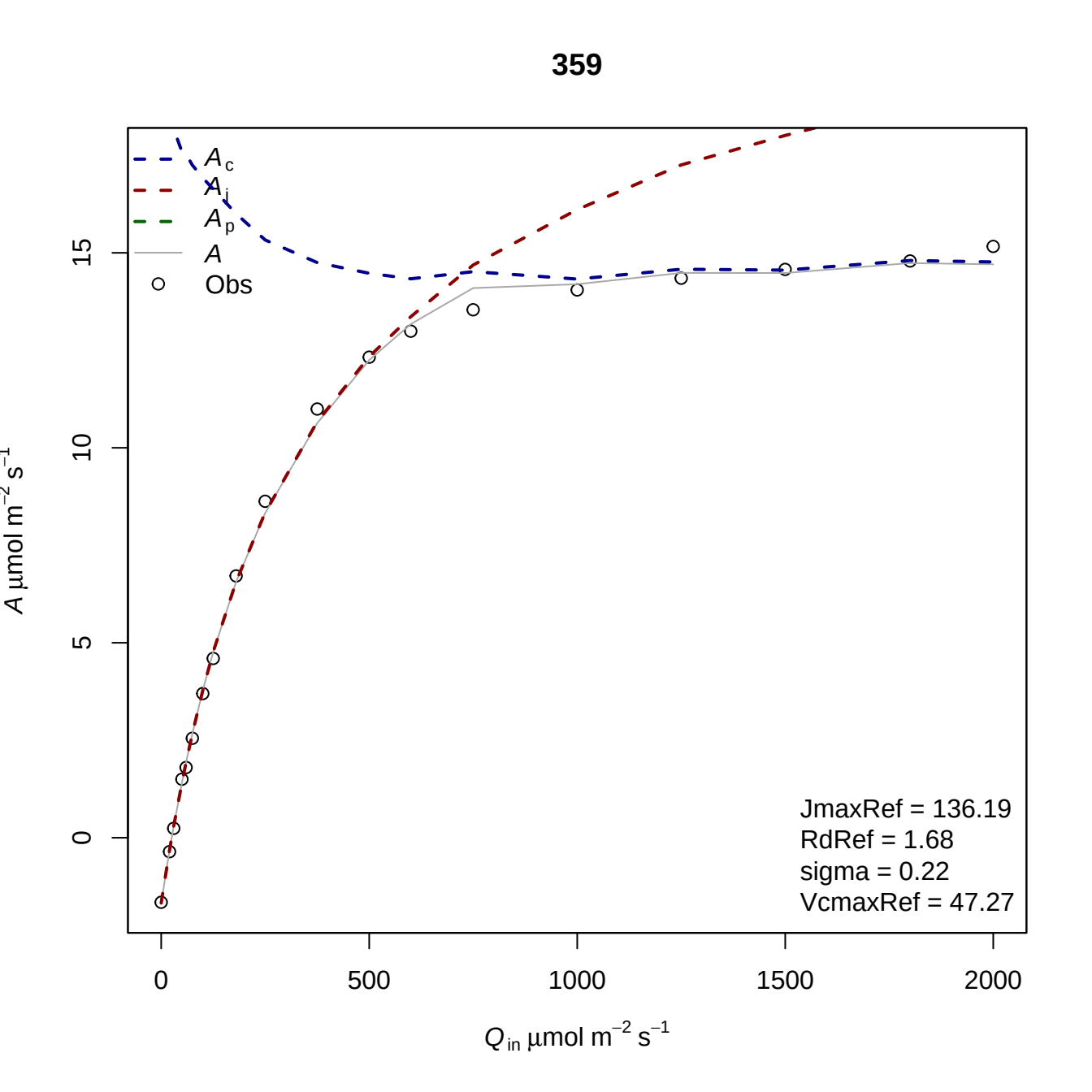

354
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.29
RdRef = 2.35
sigma = 0.21
VcmaxRef = 53.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

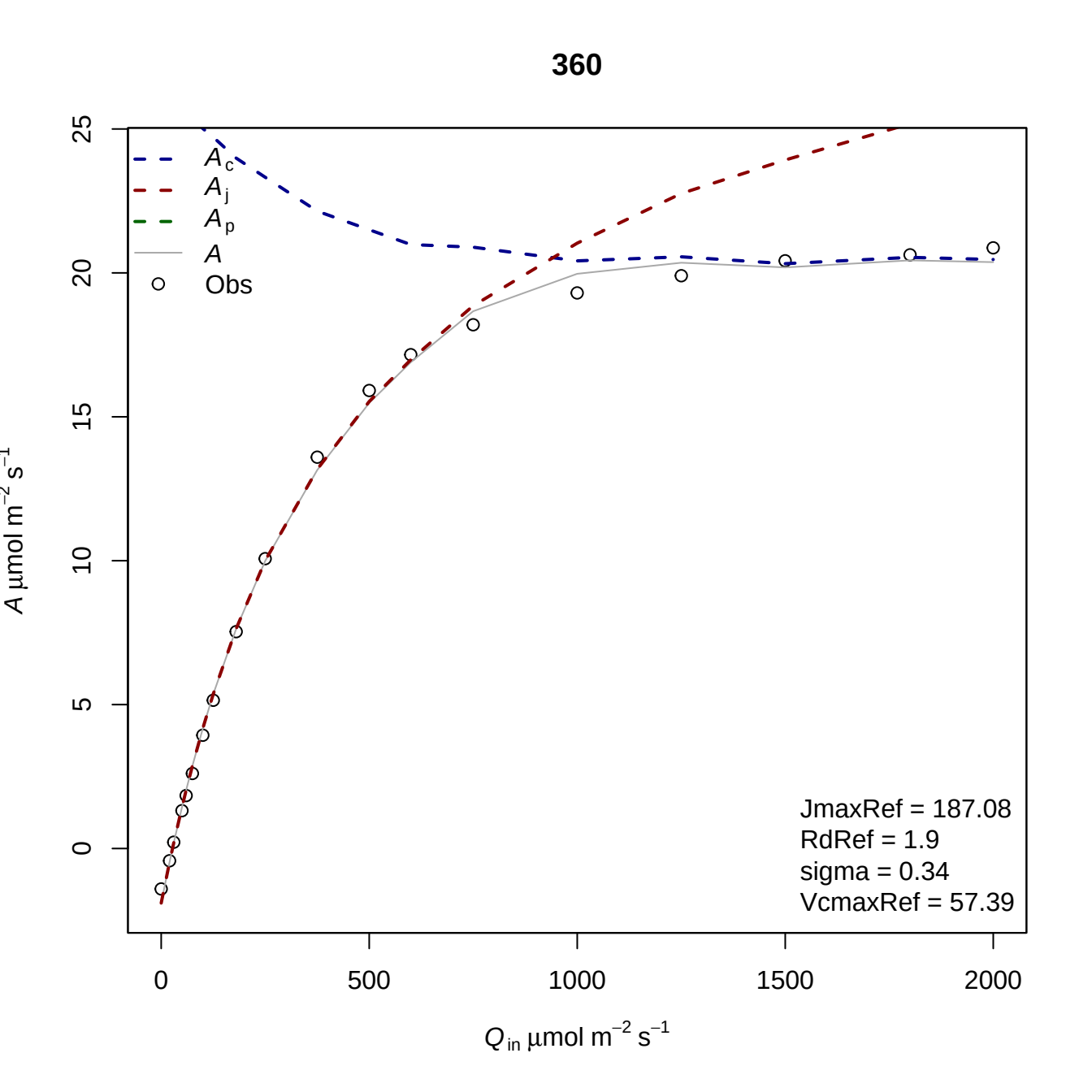

355
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.46
RdRef = 2.07
sigma = 0.34
VcmaxRef = 35.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

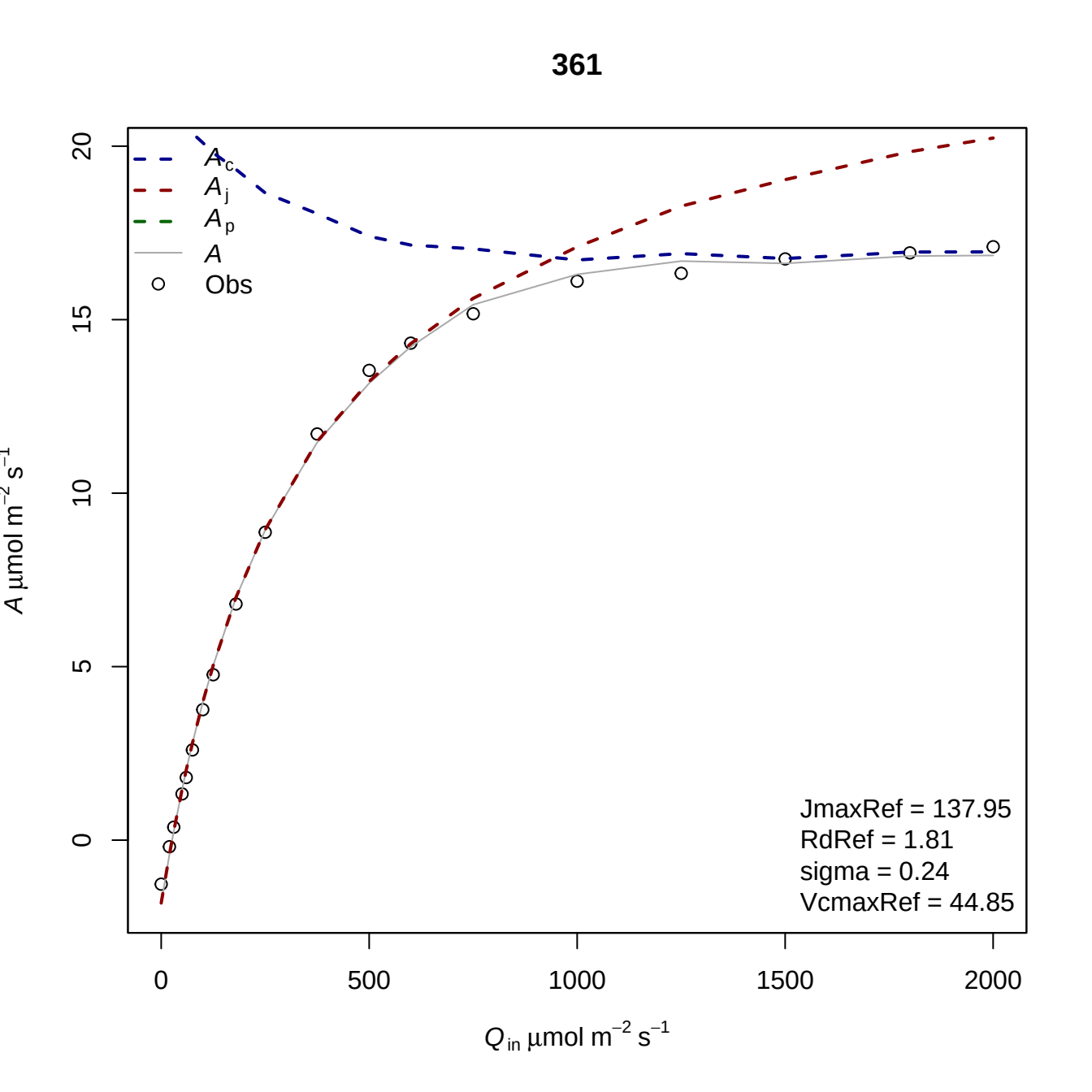

356
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.14
RdRef = 1.9
sigma = 0.19
VcmaxRef = 48.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

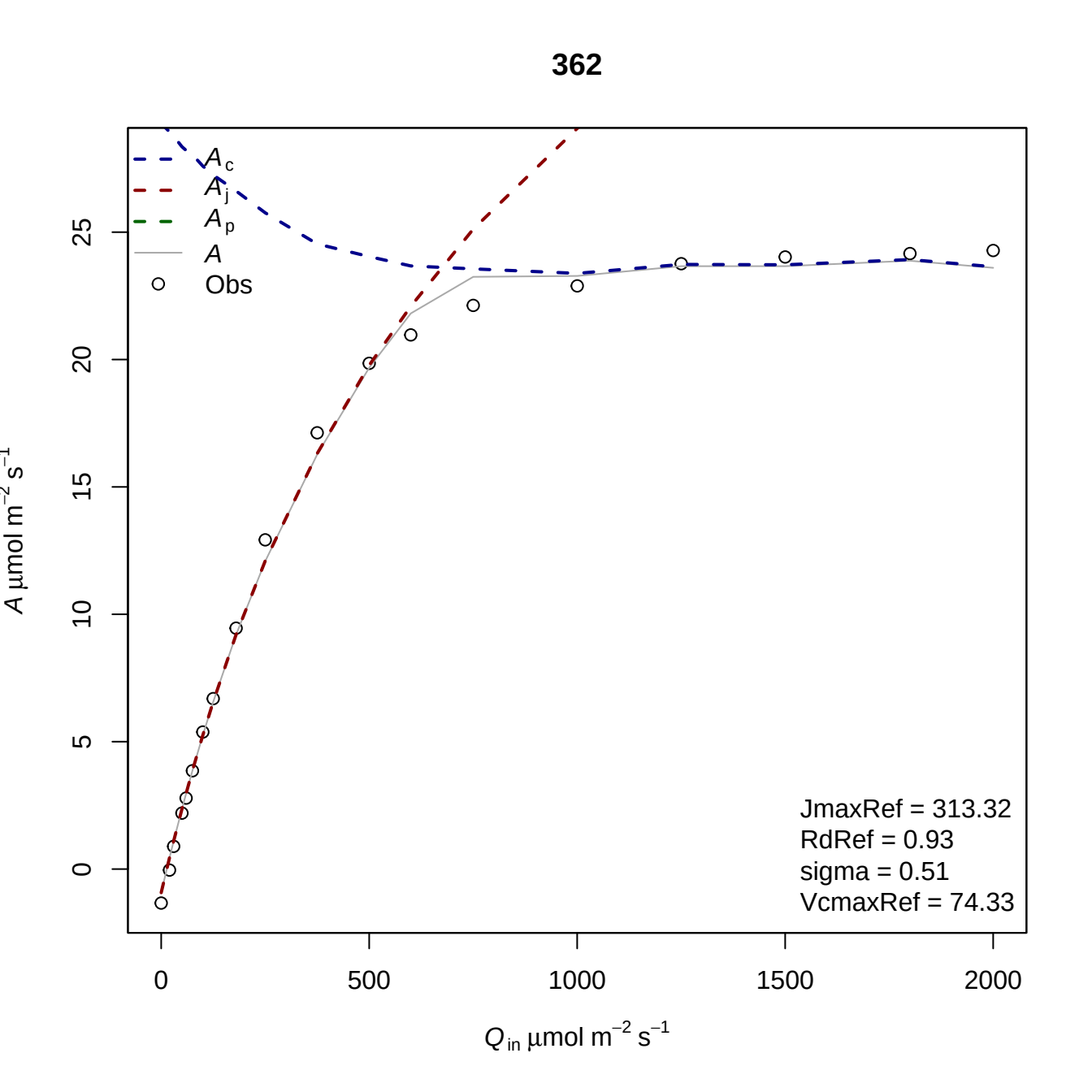

357
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.91
RdRef = 2.28
sigma = 0.33
VcmaxRef = 51.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

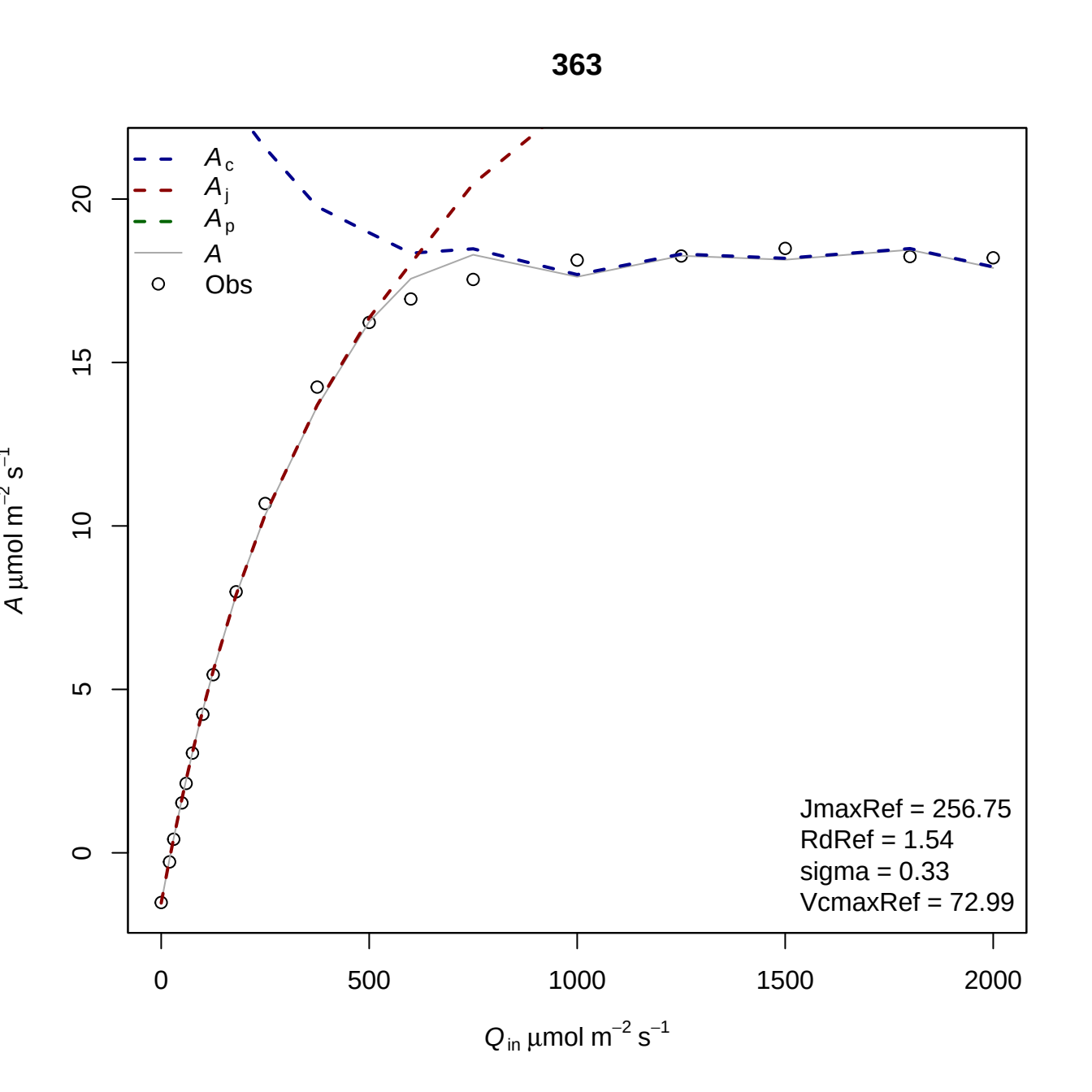

358
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.71
RdRef = 2.69
sigma = 0.41
VcmaxRef = 53.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

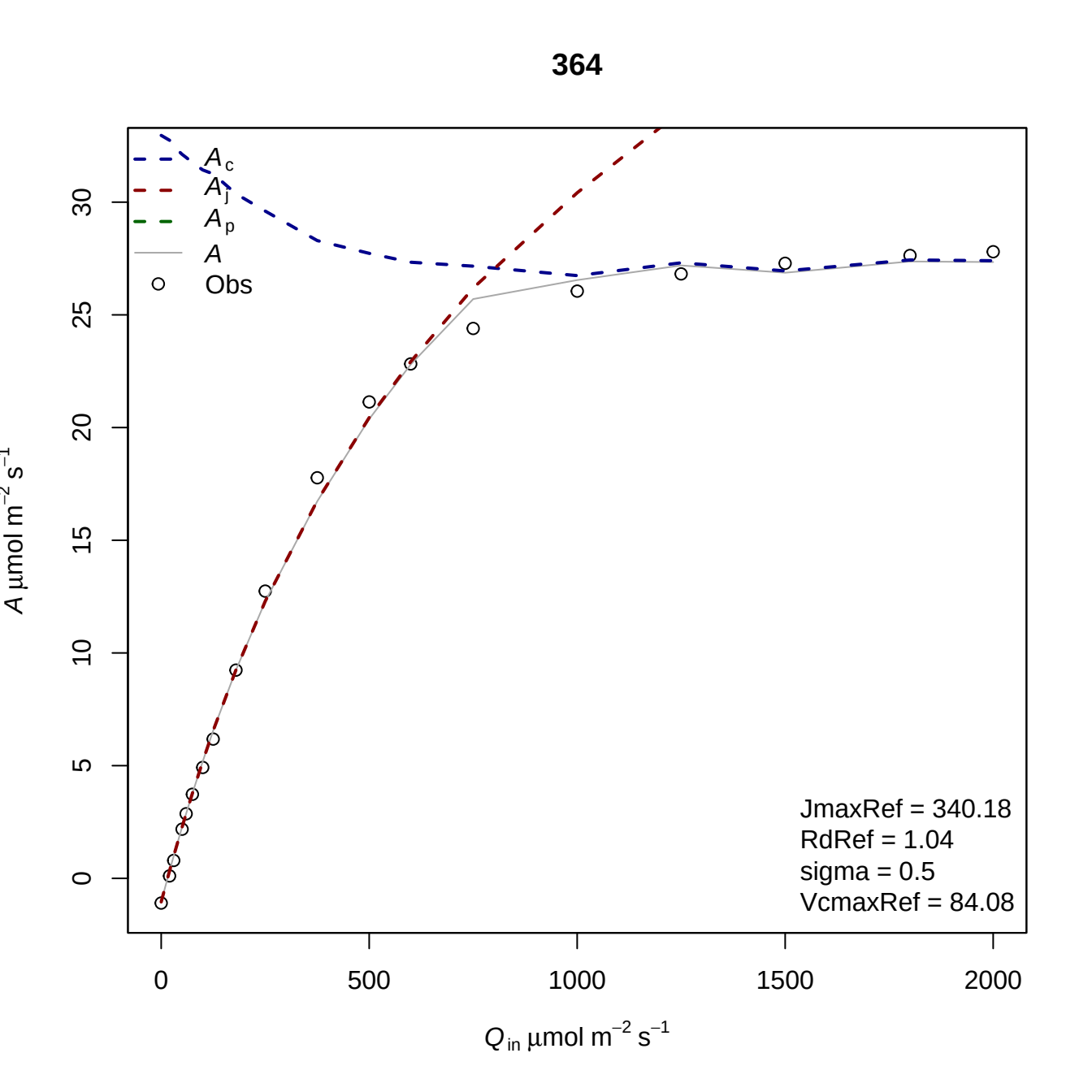

359
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.19
RdRef = 1.68
sigma = 0.22
VcmaxRef = 47.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

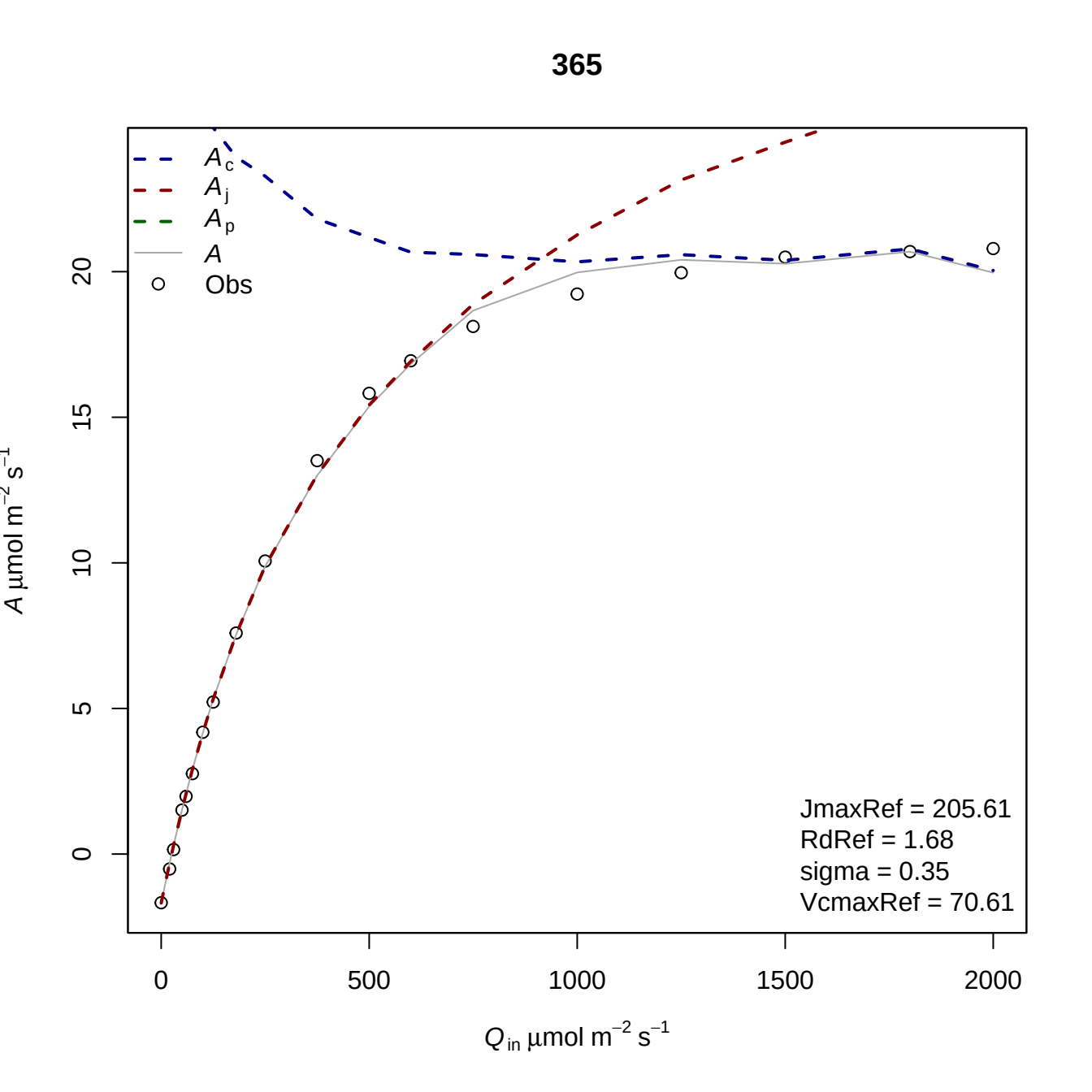

360
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 187.08
RdRef = 1.9
sigma = 0.34
VcmaxRef = 57.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

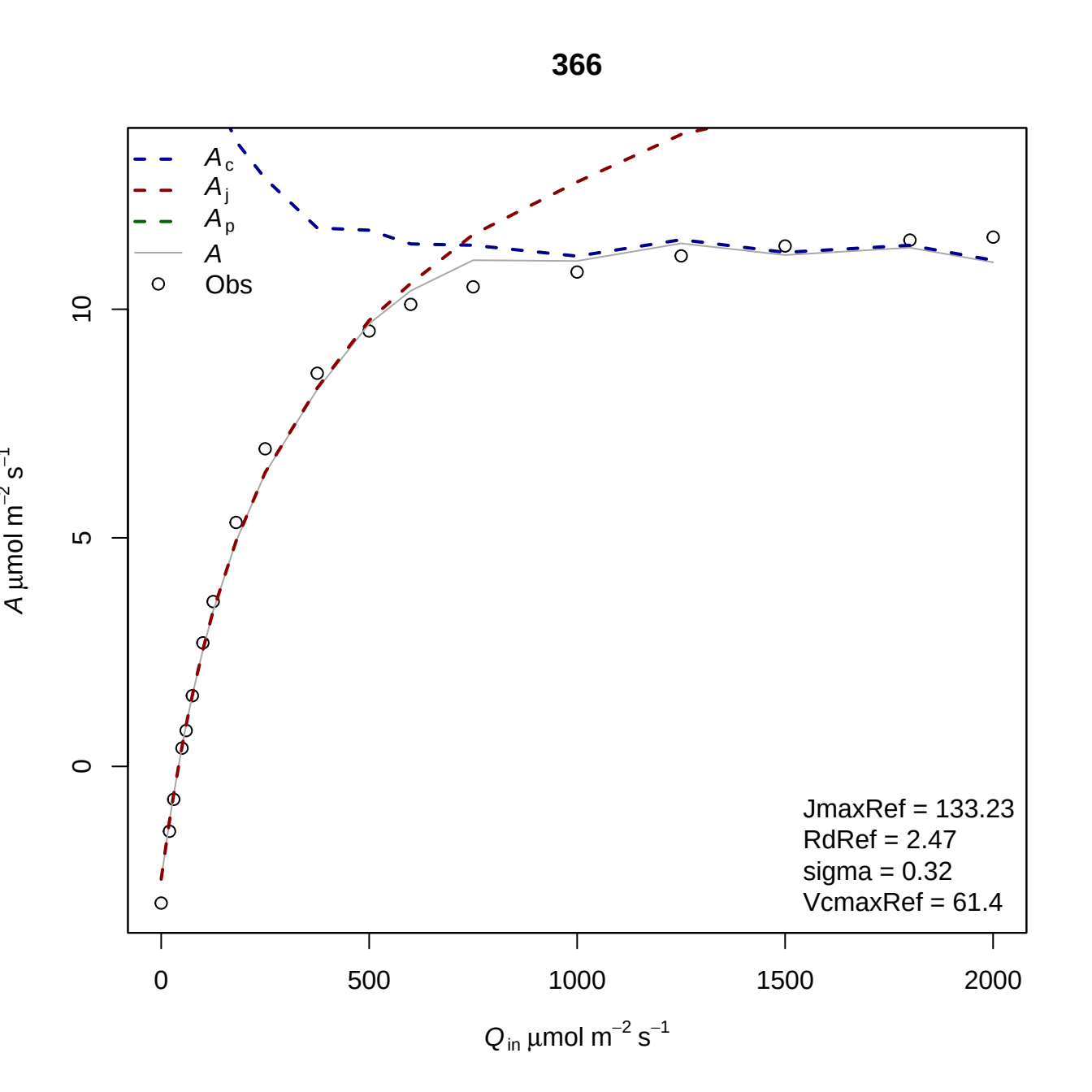

361
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.95
RdRef = 1.81
sigma = 0.24
VcmaxRef = 44.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

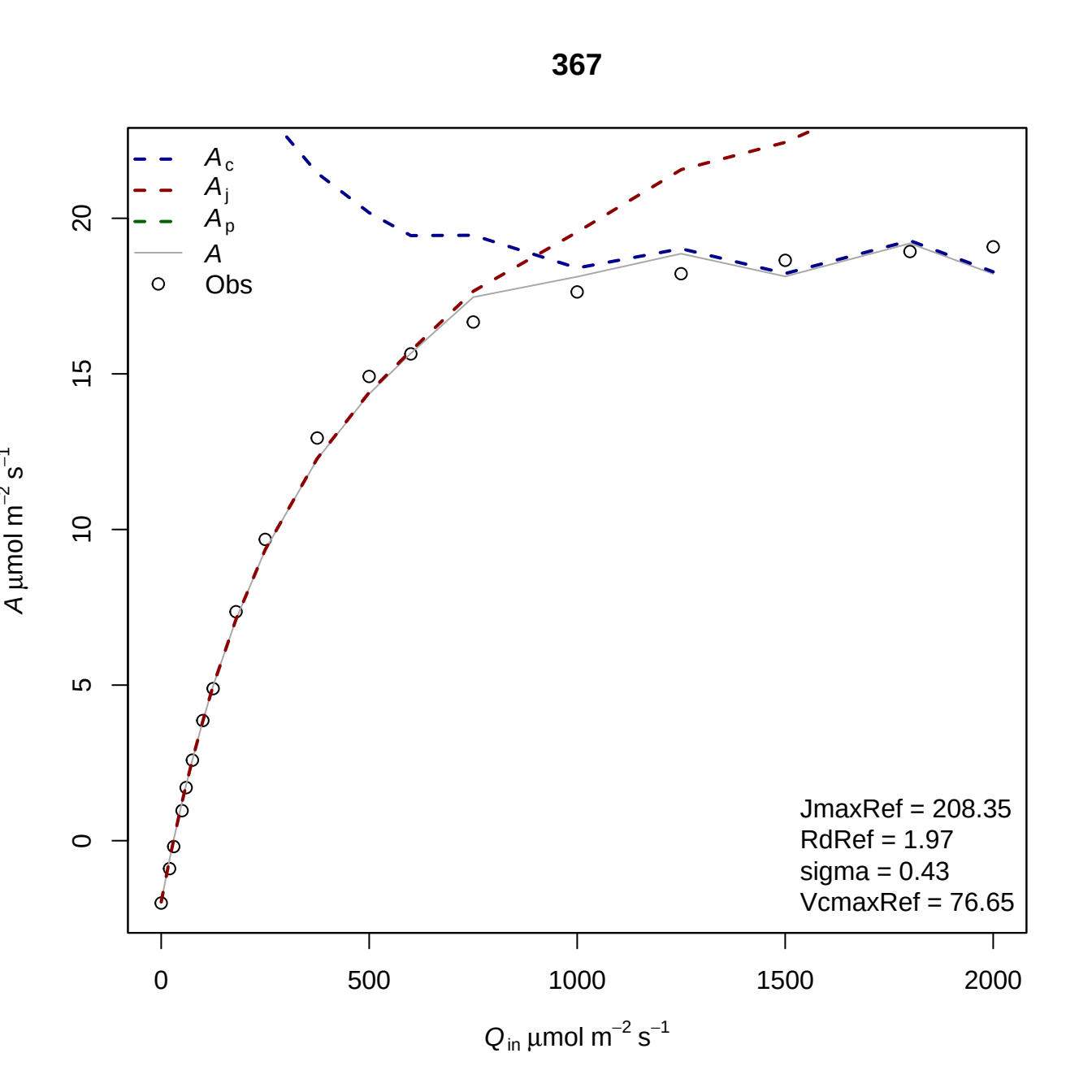

362
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 313.32
RdRef = 0.93
sigma = 0.51
VcmaxRef = 74.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

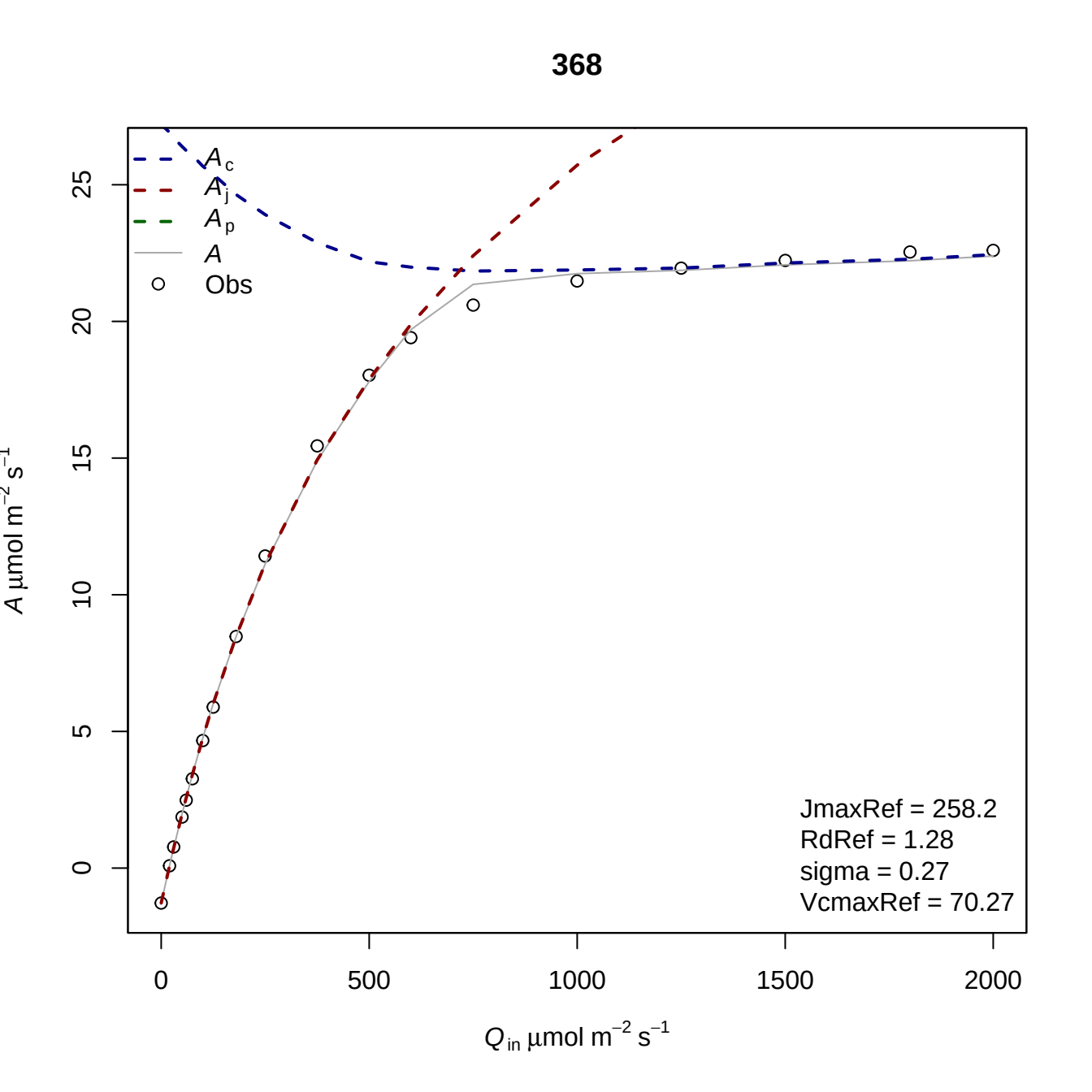

363
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 256.75
RdRef = 1.54
sigma = 0.33
VcmaxRef = 72.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

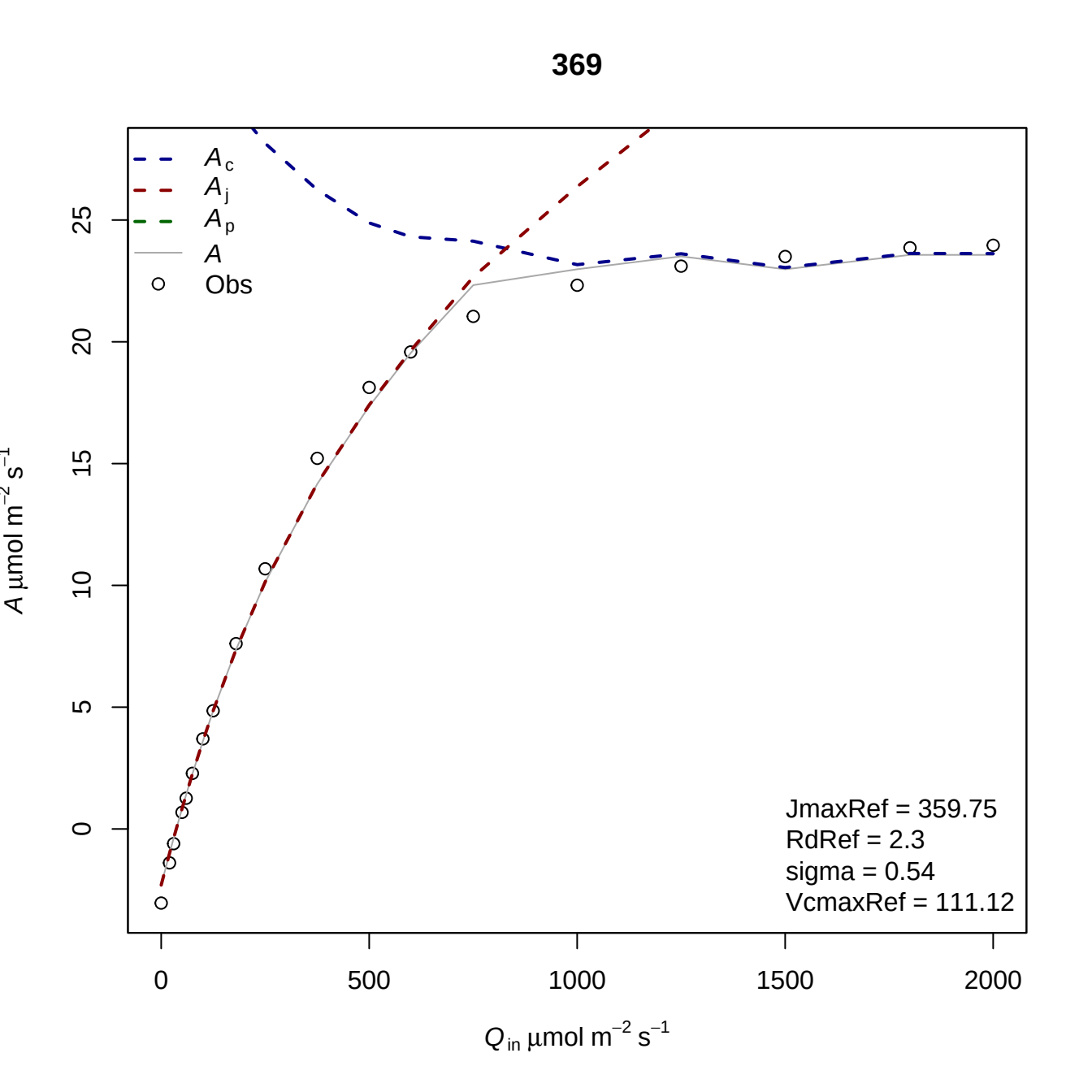

364
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 340.18
RdRef = 1.04
sigma = 0.5
VcmaxRef = 84.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

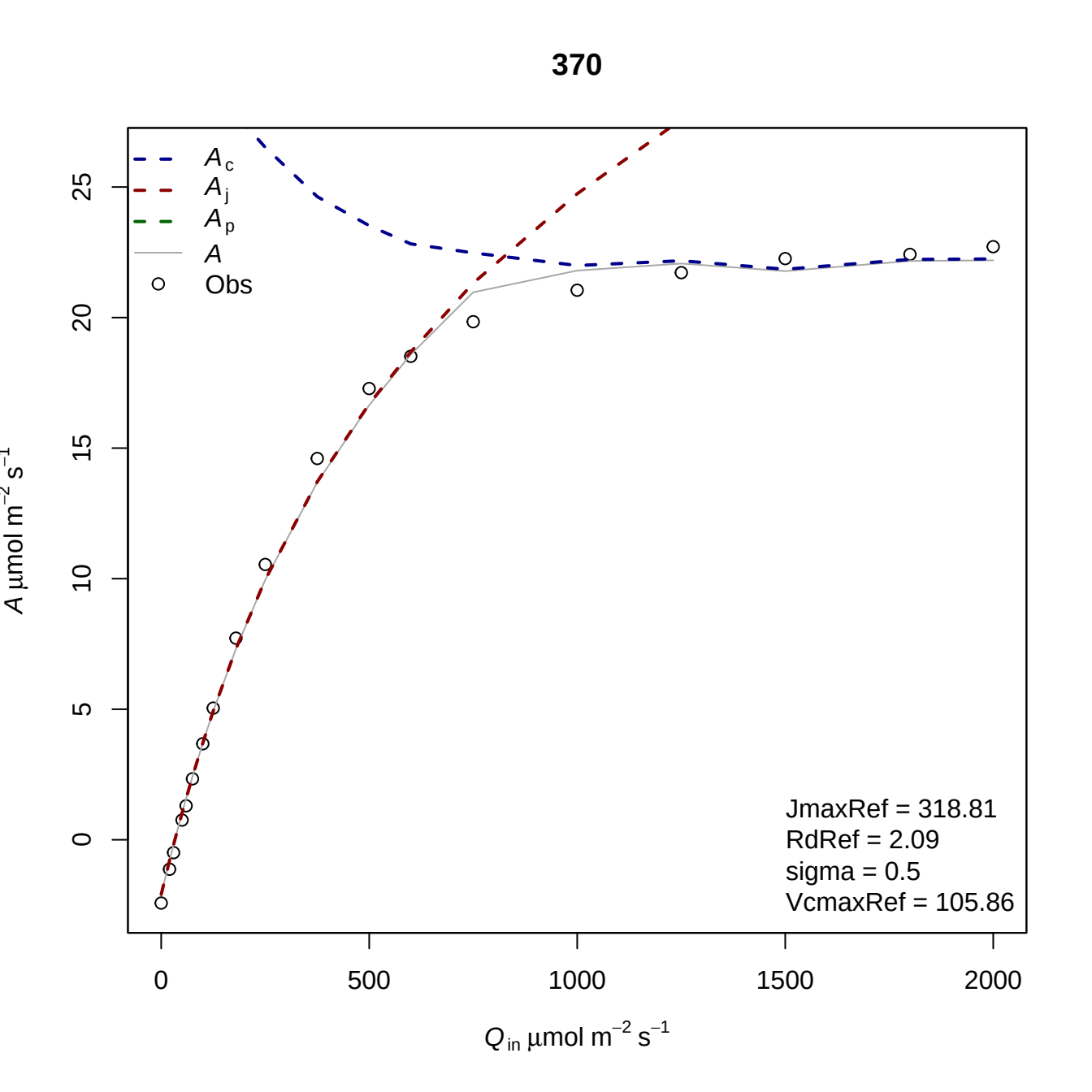

365
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 205.61
RdRef = 1.68
sigma = 0.35
VcmaxRef = 70.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

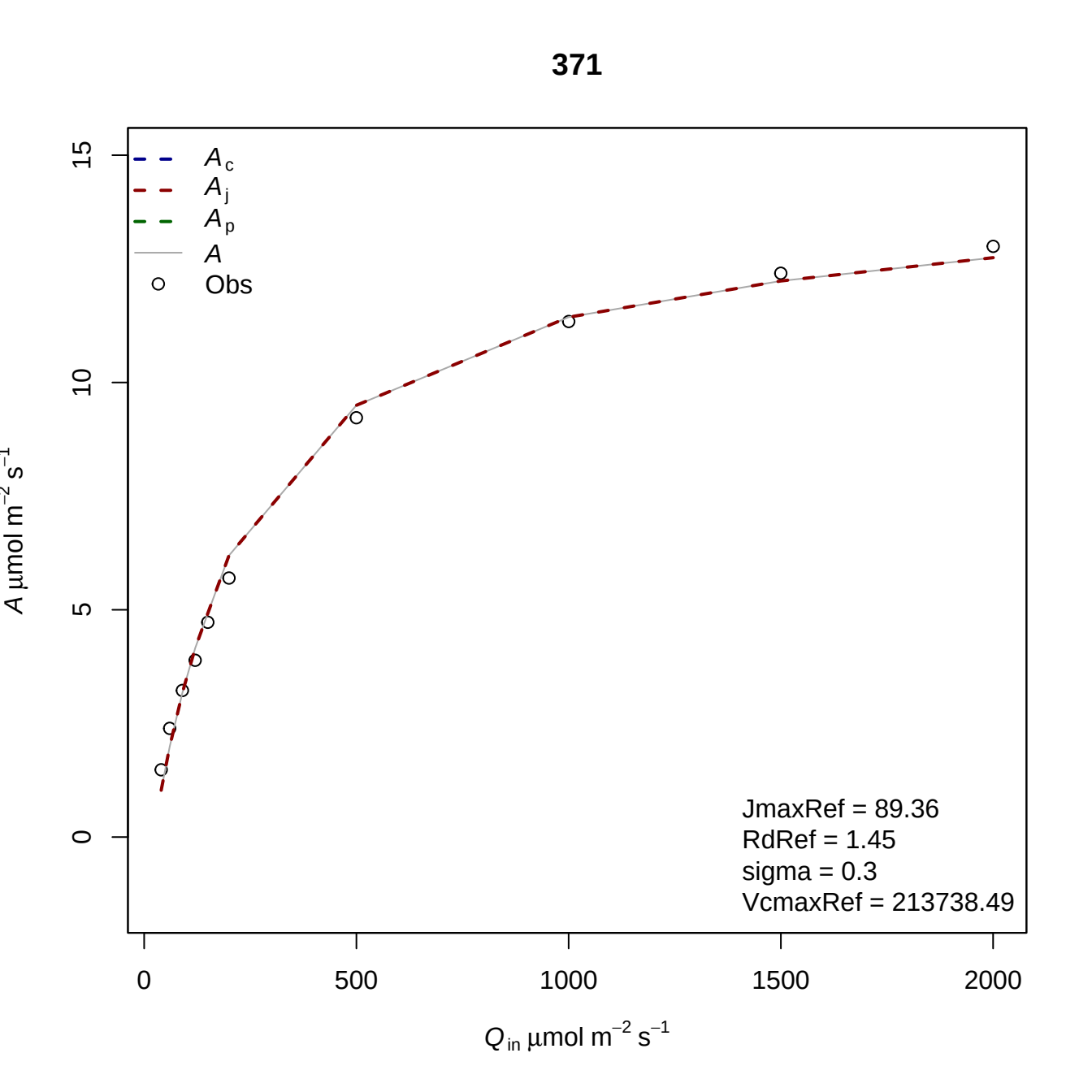

366
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.23
RdRef = 2.47
sigma = 0.32
VcmaxRef = 61.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

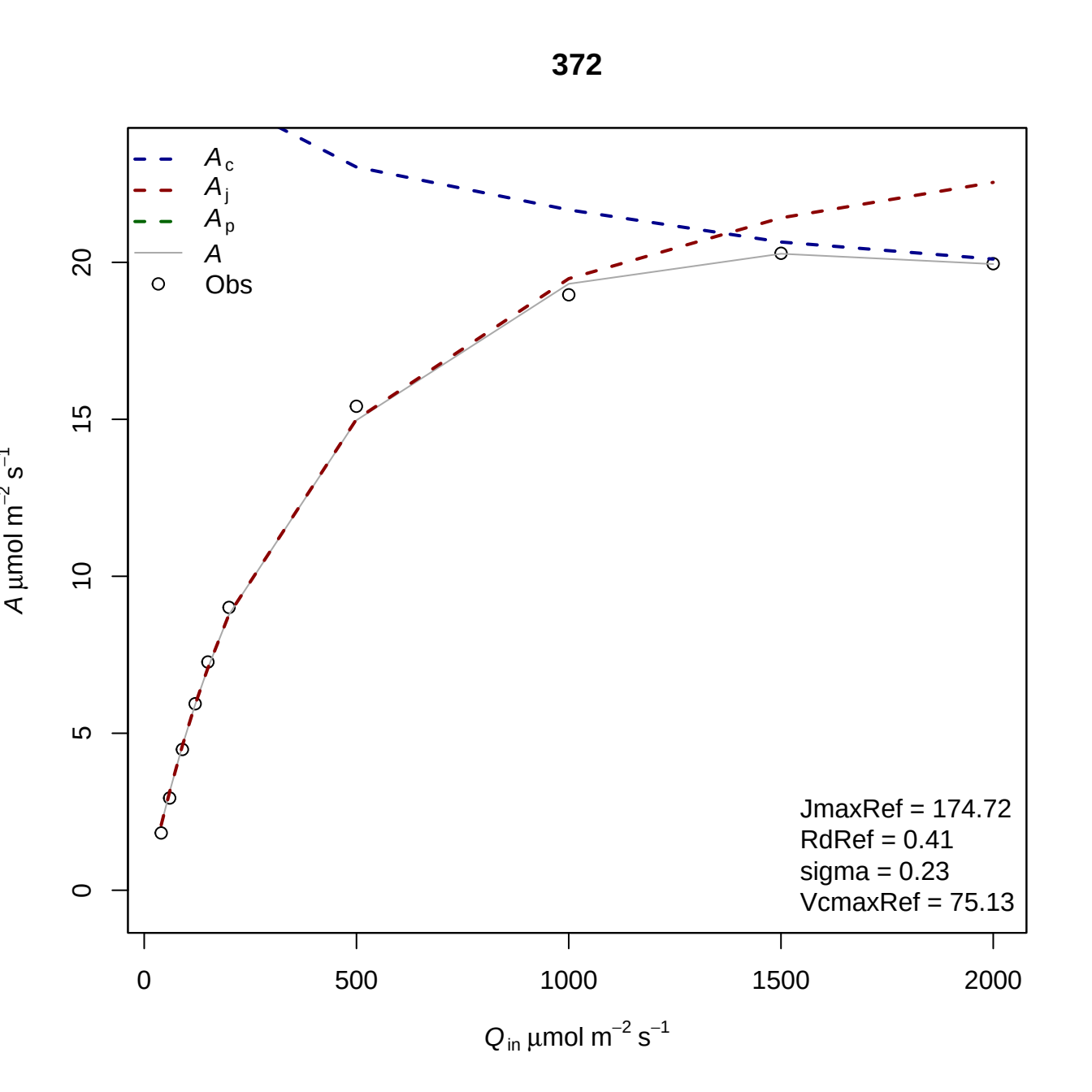

367
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 208.35
RdRef = 1.97
sigma = 0.43
VcmaxRef = 76.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

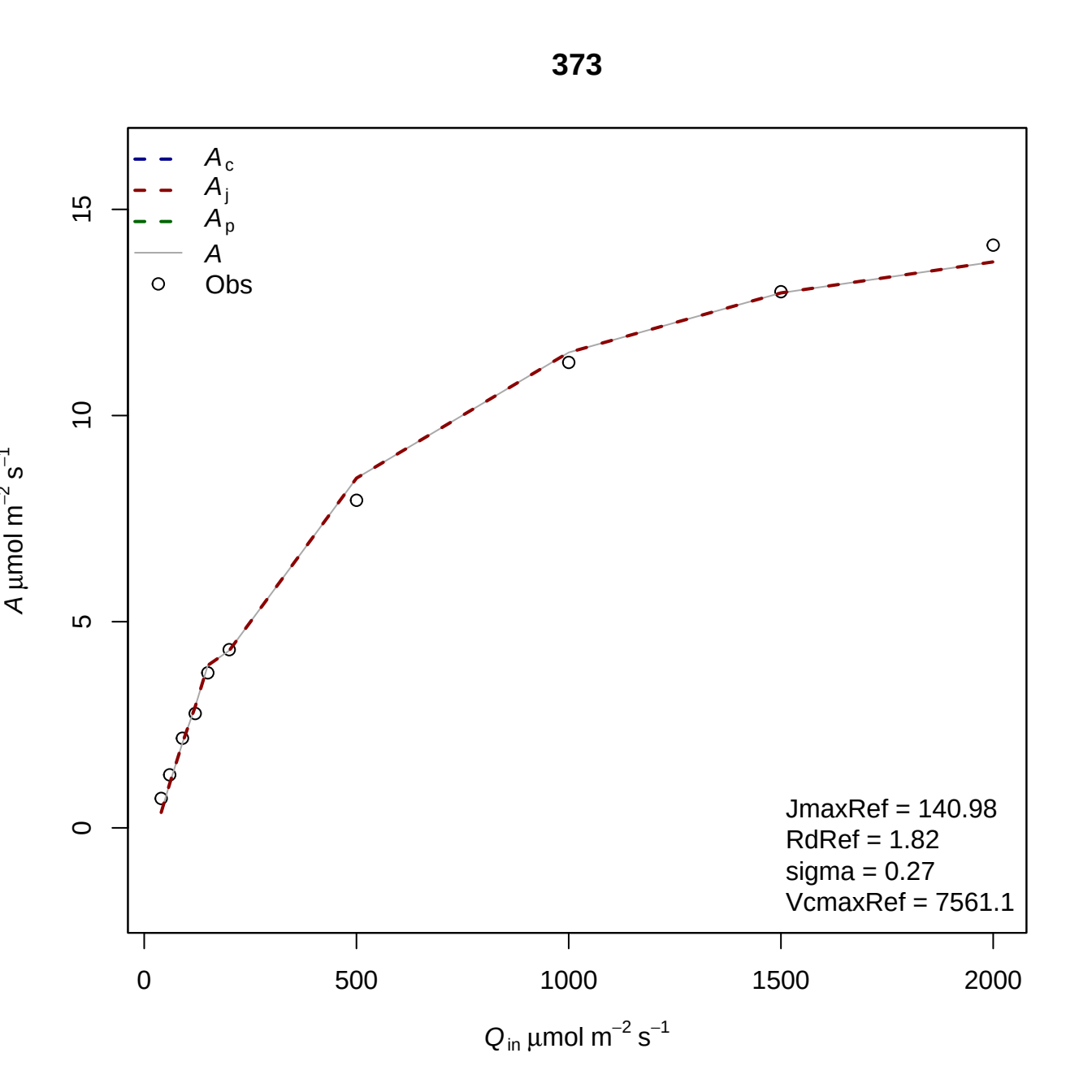

368
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 258.2
RdRef = 1.28
sigma = 0.27
VcmaxRef = 70.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

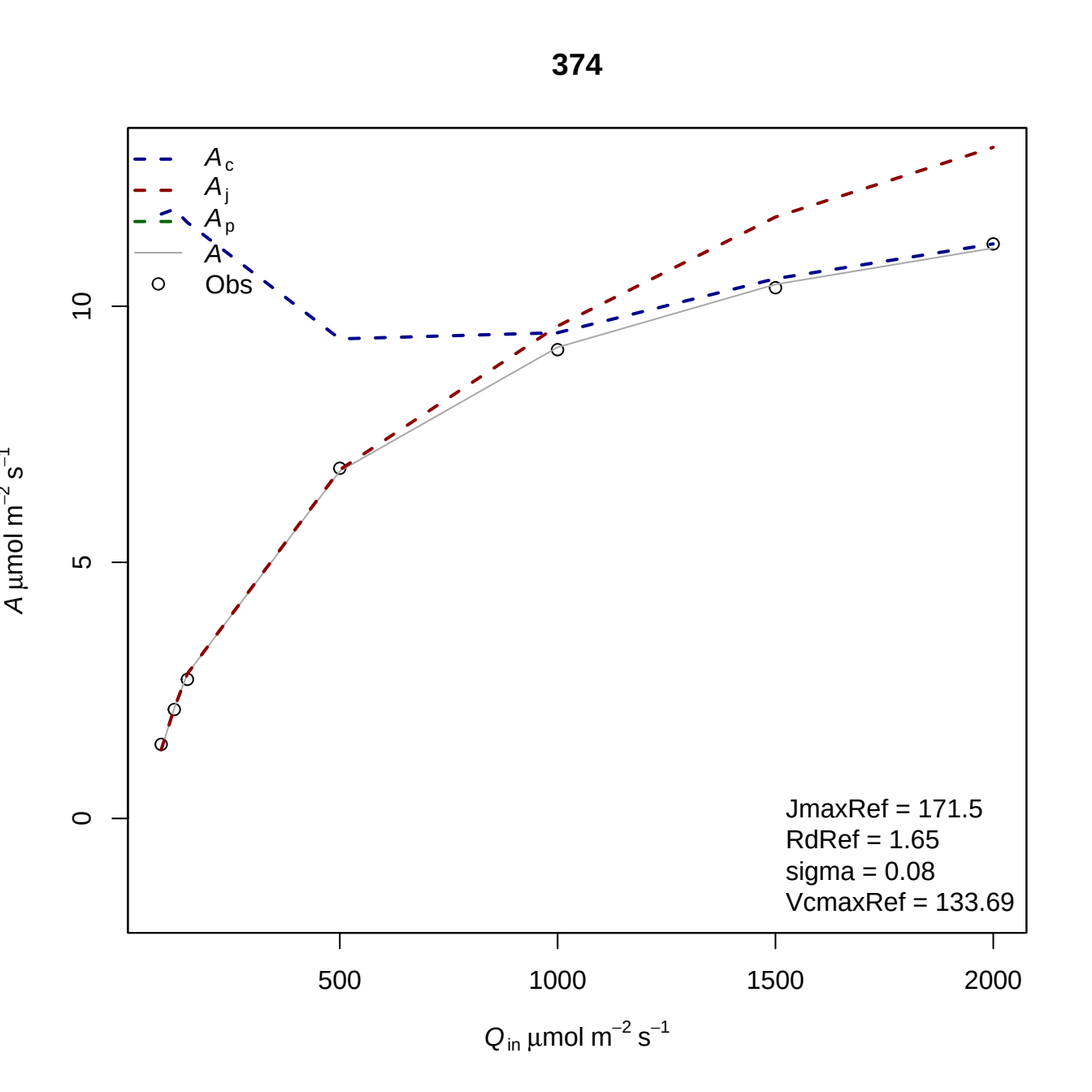

369
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 359.75
RdRef = 2.3
sigma = 0.54
VcmaxRef = 111.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

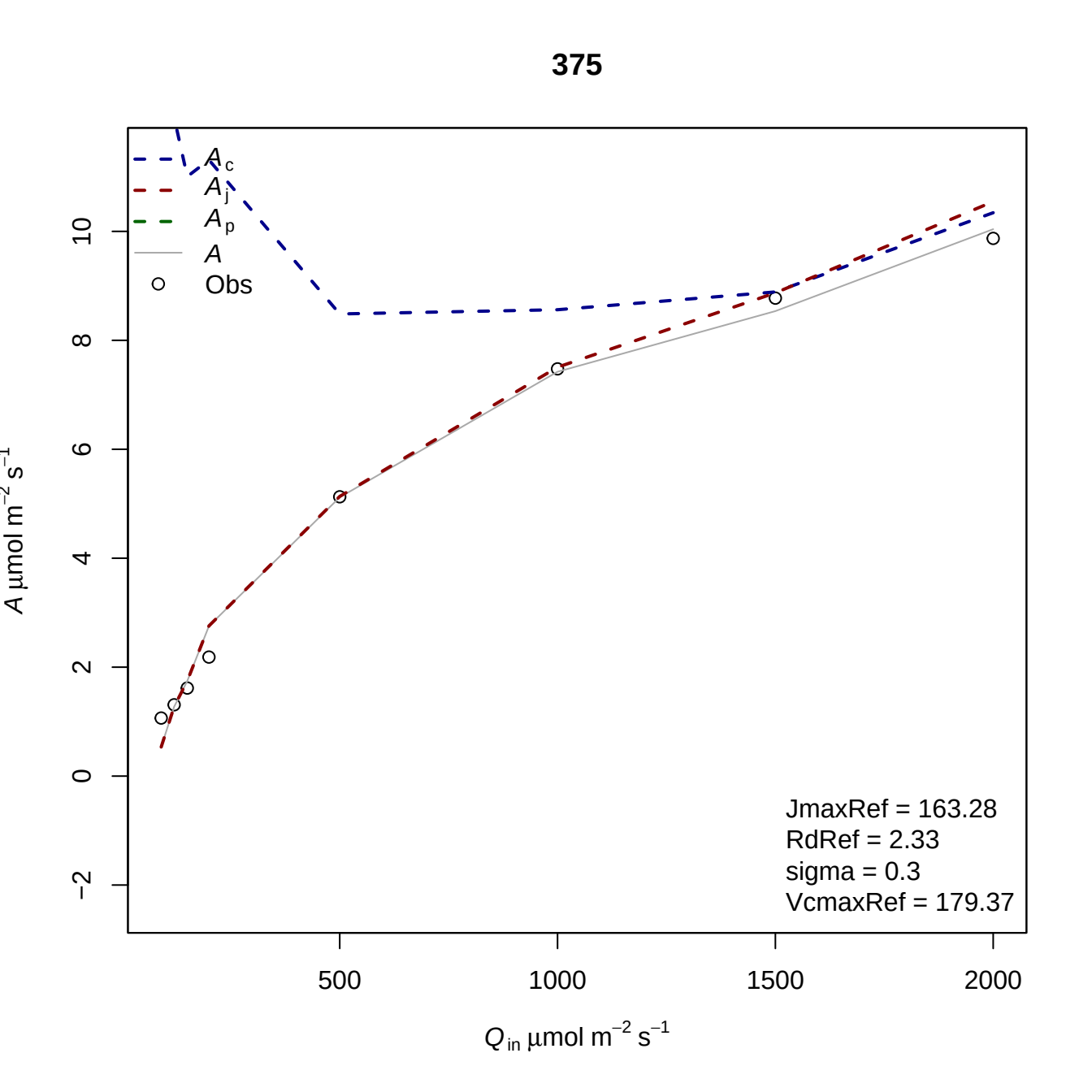

370
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 318.81
RdRef = 2.09
sigma = 0.5
VcmaxRef = 105.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

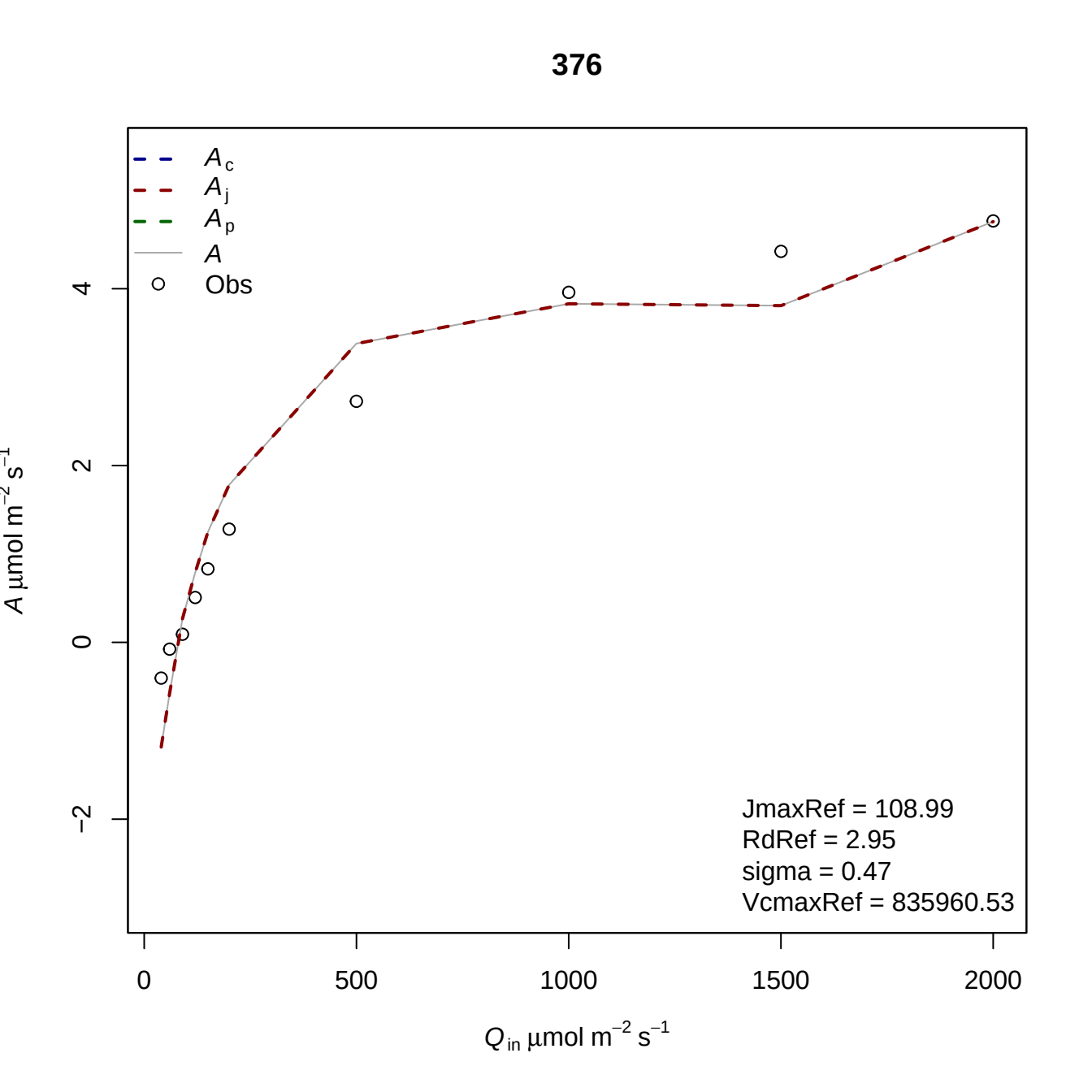

371
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.36
RdRef = 1.45
sigma = 0.3
VcmaxRef = 213738.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

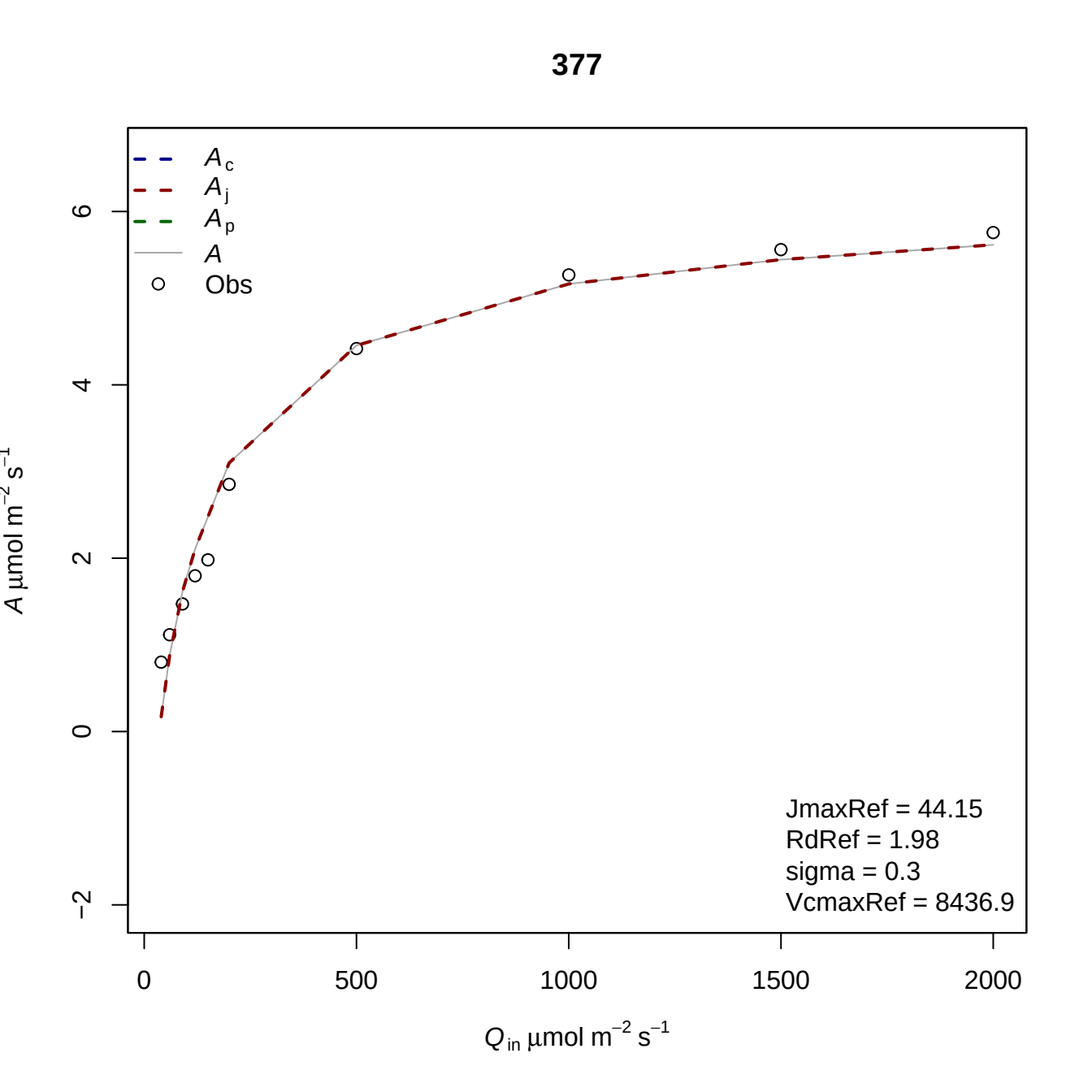

372
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 174.72
RdRef = 0.41
sigma = 0.23
VcmaxRef = 75.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

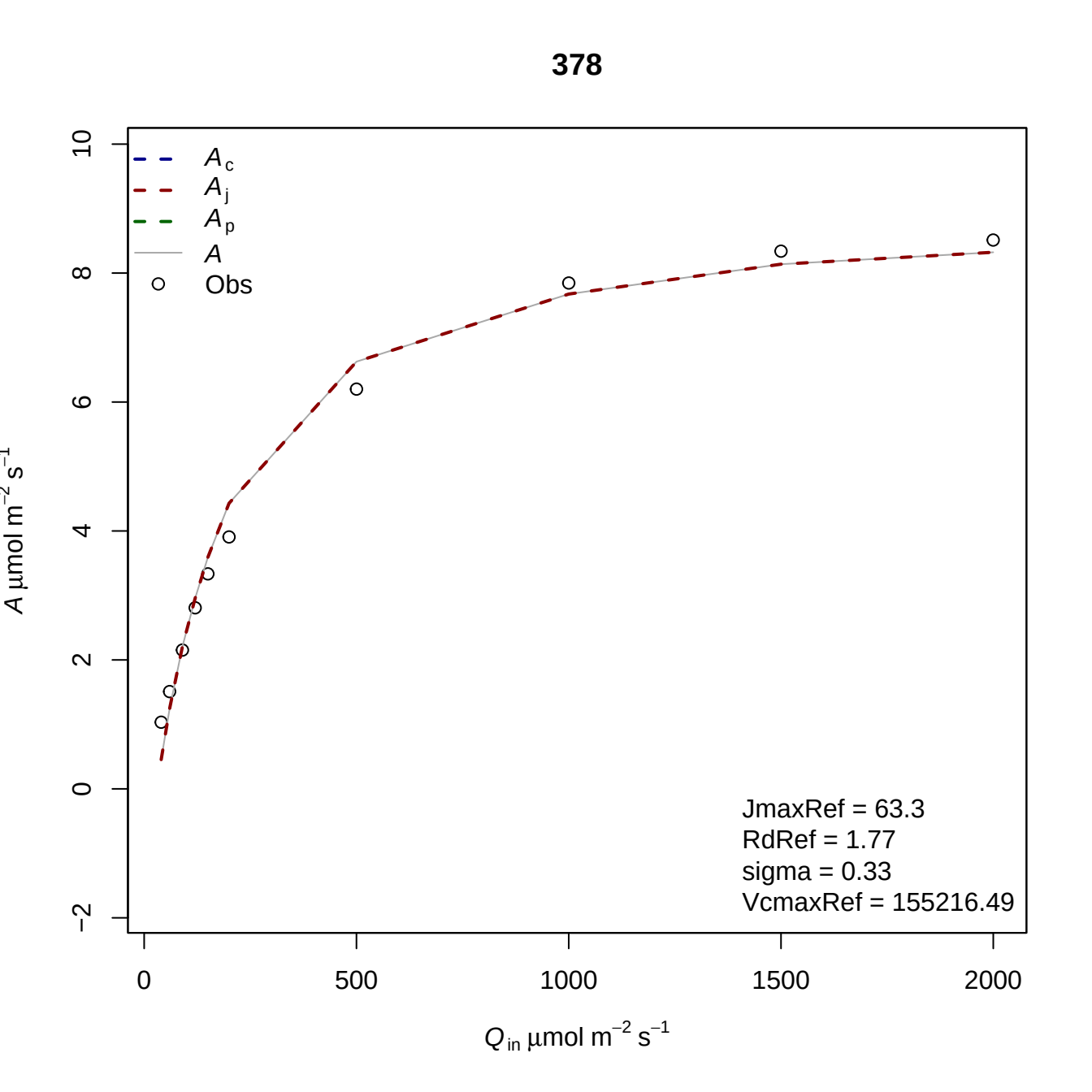

373
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.98
RdRef = 1.82
sigma = 0.27
VcmaxRef = 7561.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

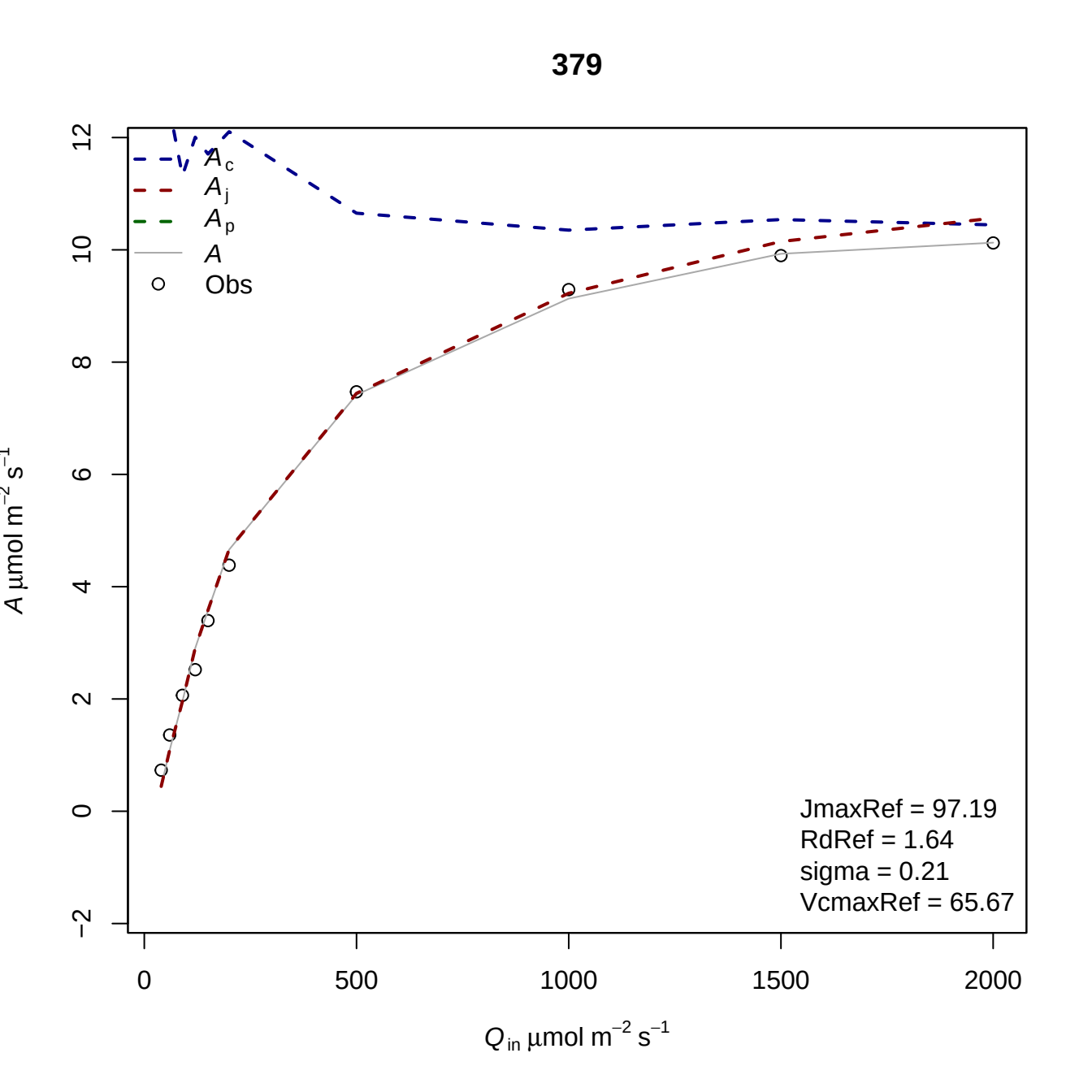

374
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 171.5
RdRef = 1.65
sigma = 0.08
VcmaxRef = 133.69
0
5
10
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

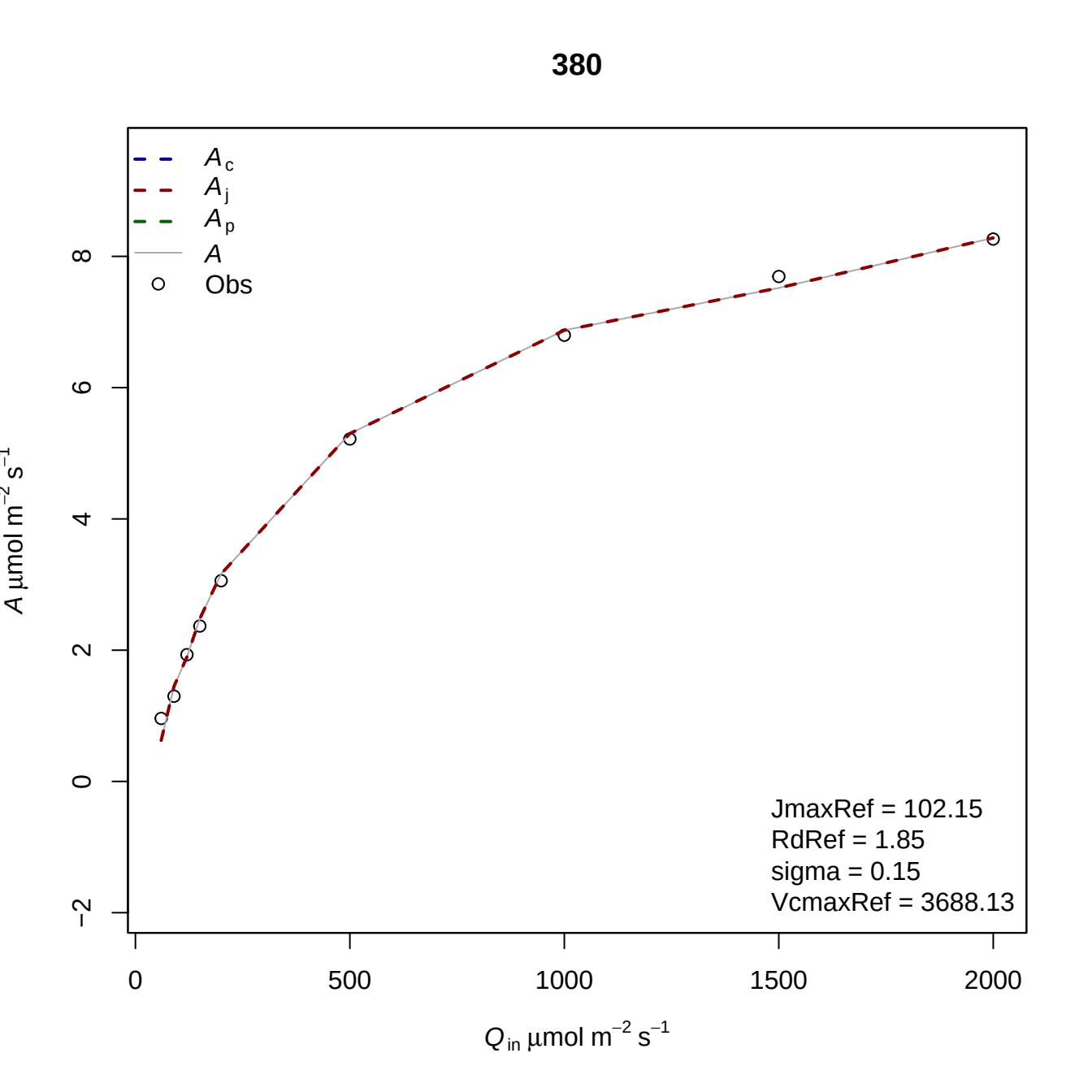

375
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163.28
RdRef = 2.33
sigma = 0.3
VcmaxRef = 179.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
500
1000
1500
2000

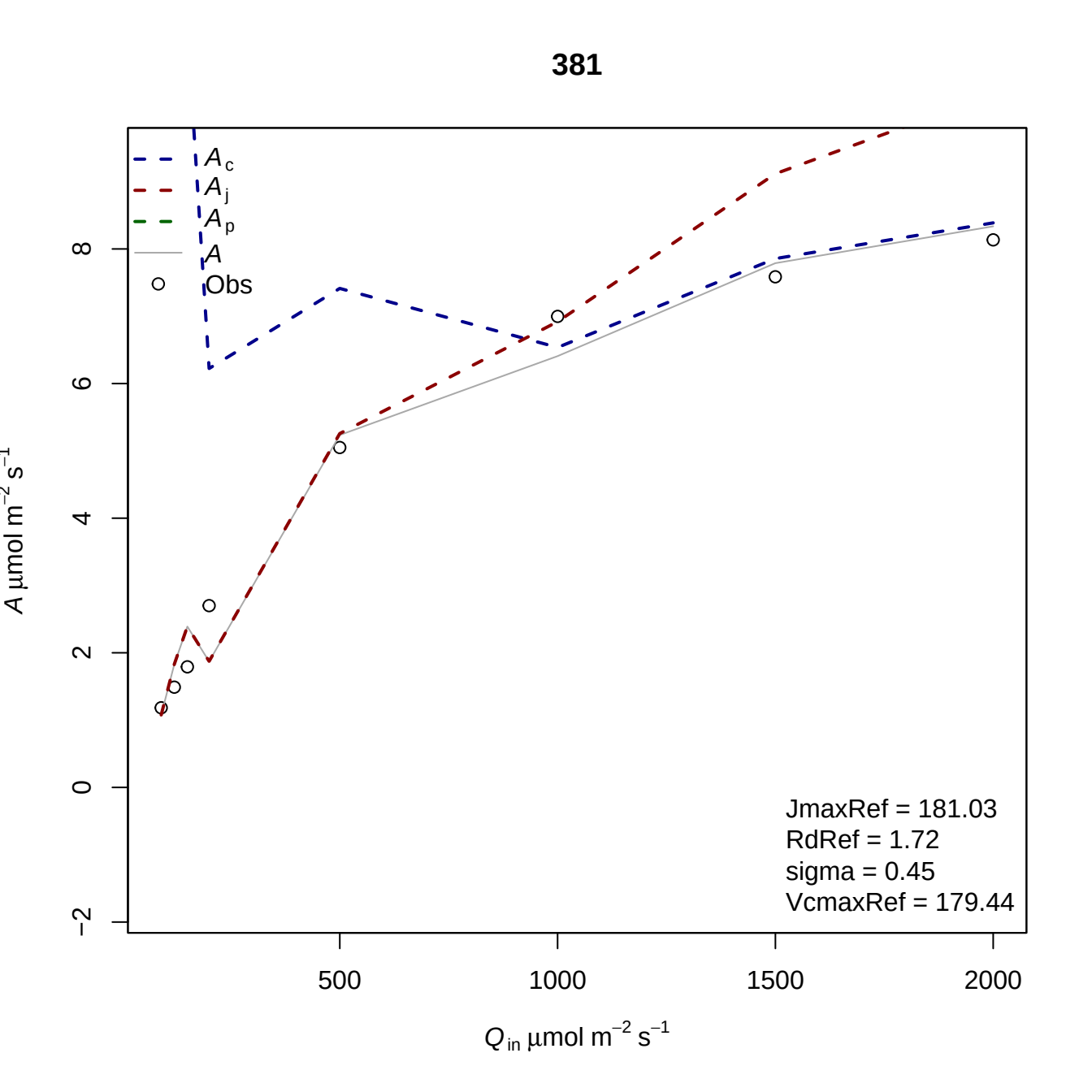

376
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.99
RdRef = 2.95
sigma = 0.47
VcmaxRef = 835960.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

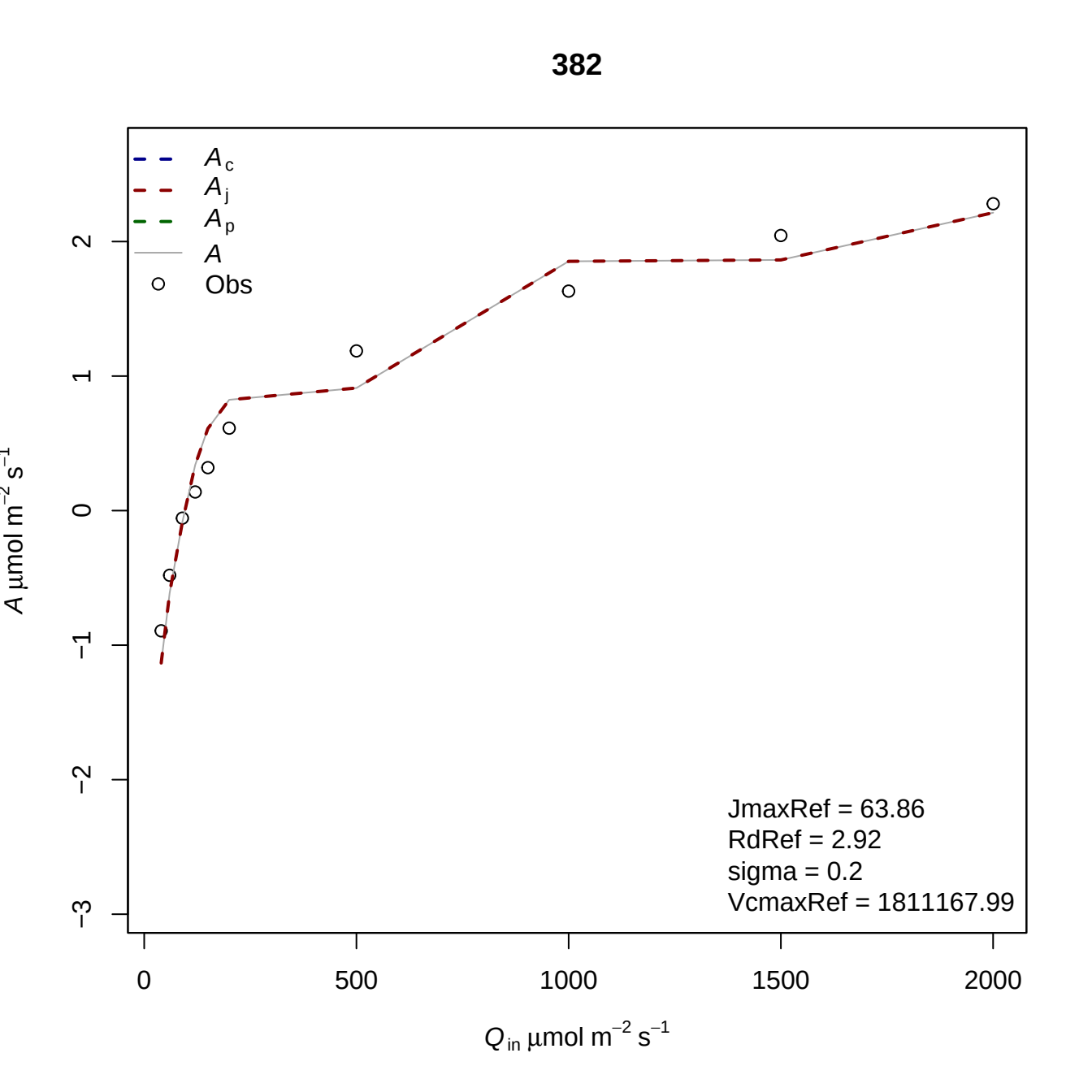

377
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.15
RdRef = 1.98
sigma = 0.3
VcmaxRef = 8436.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

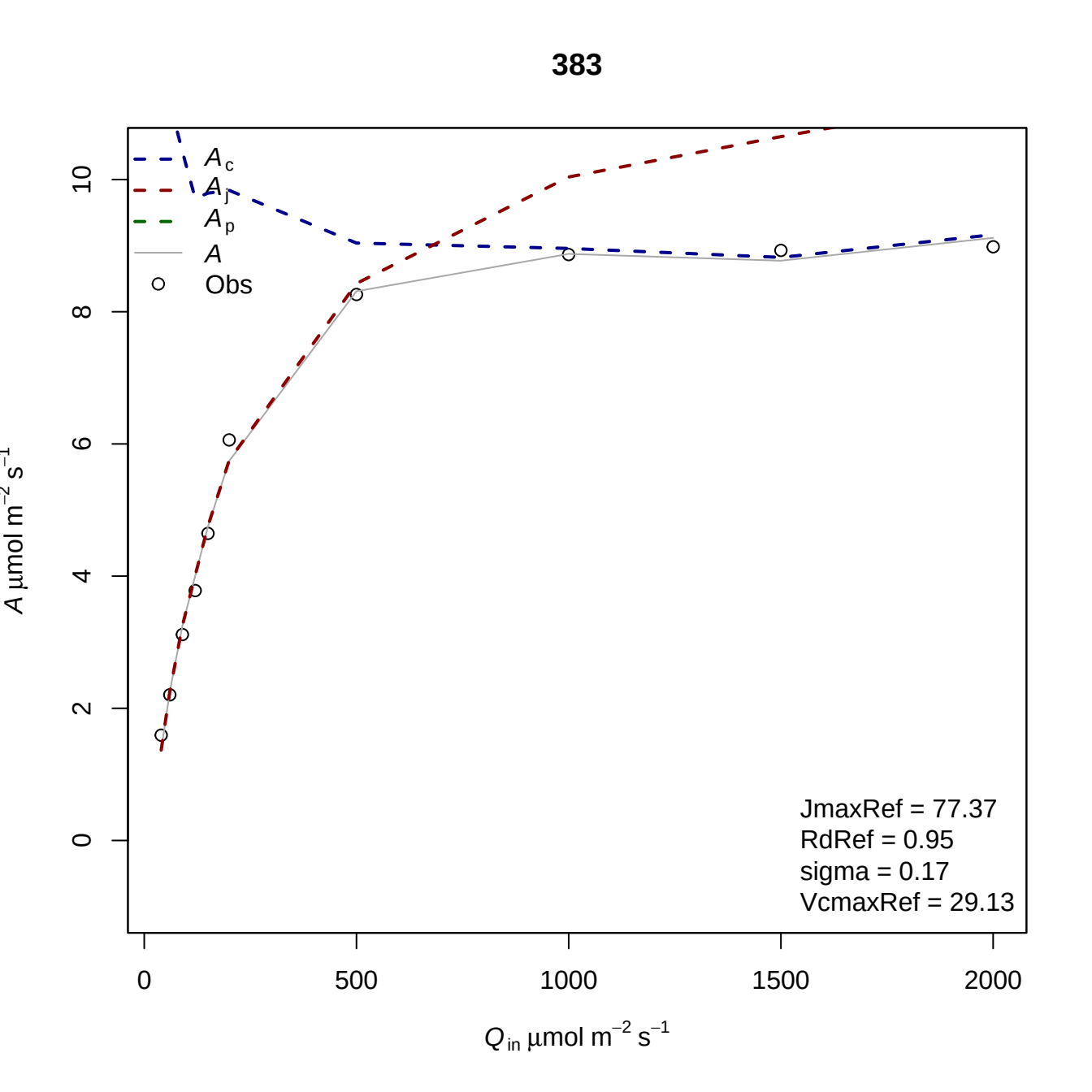

378
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.3
RdRef = 1.77
sigma = 0.33
VcmaxRef = 155216.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

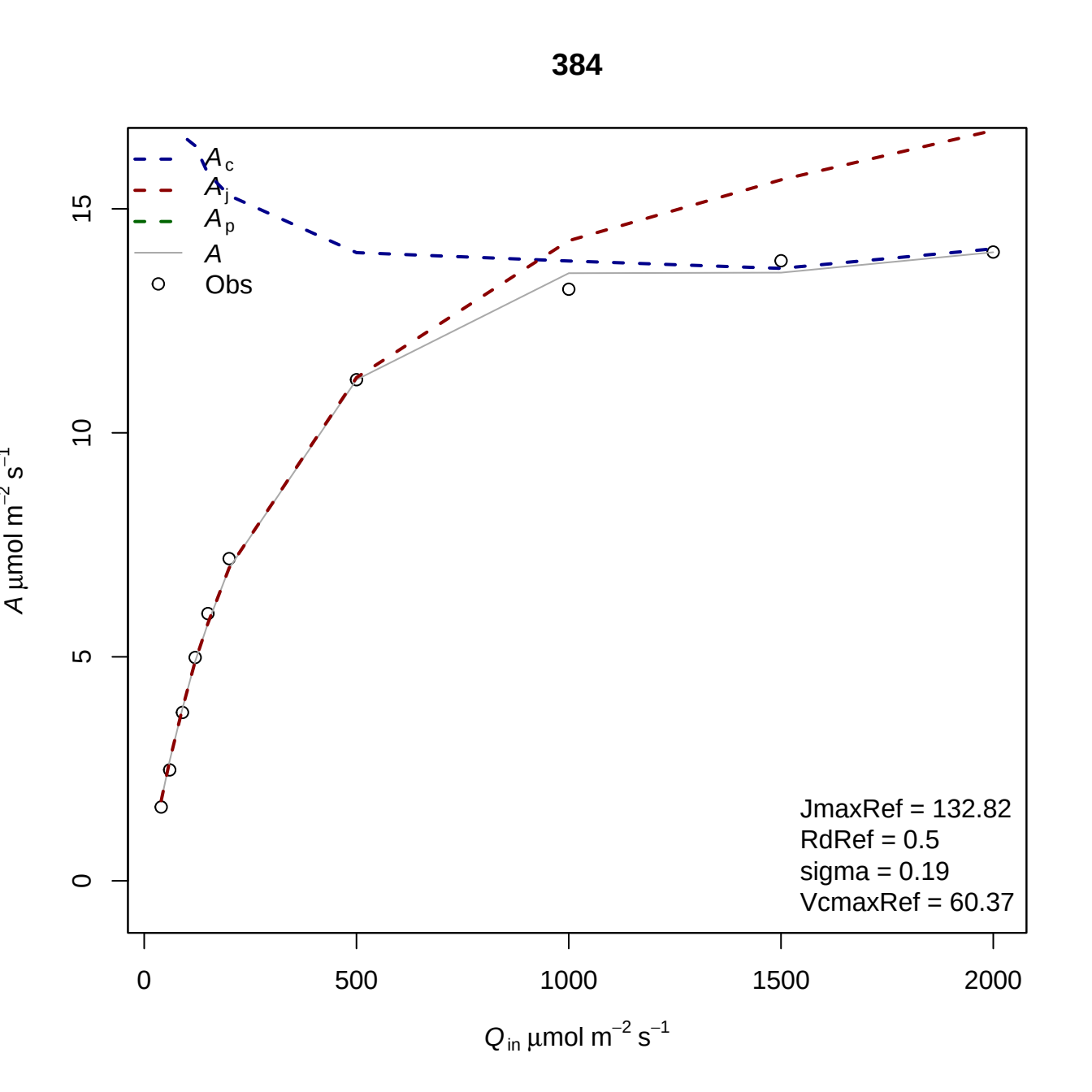

379
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.19
RdRef = 1.64
sigma = 0.21
VcmaxRef = 65.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

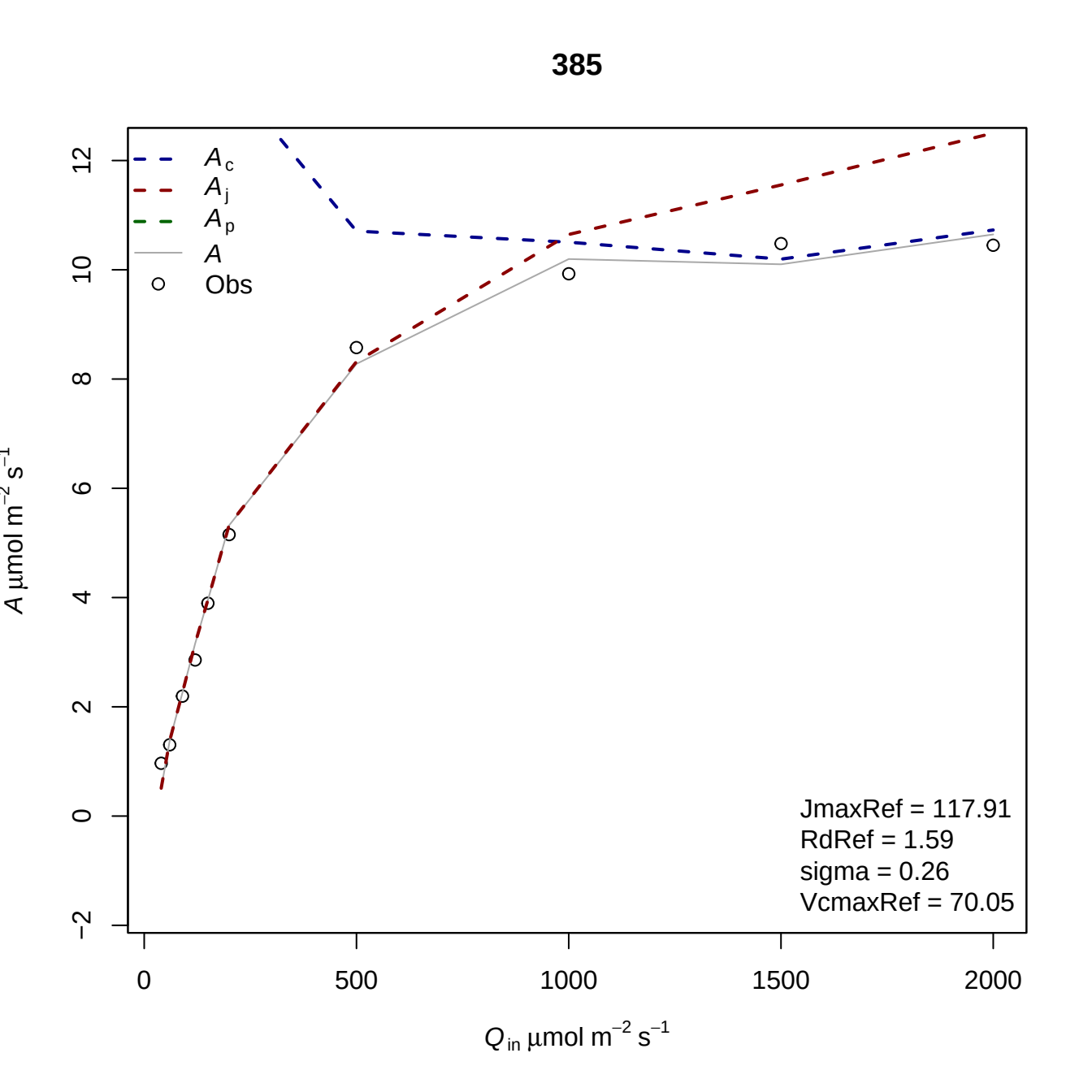

380
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.15
RdRef = 1.85
sigma = 0.15
VcmaxRef = 3688.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

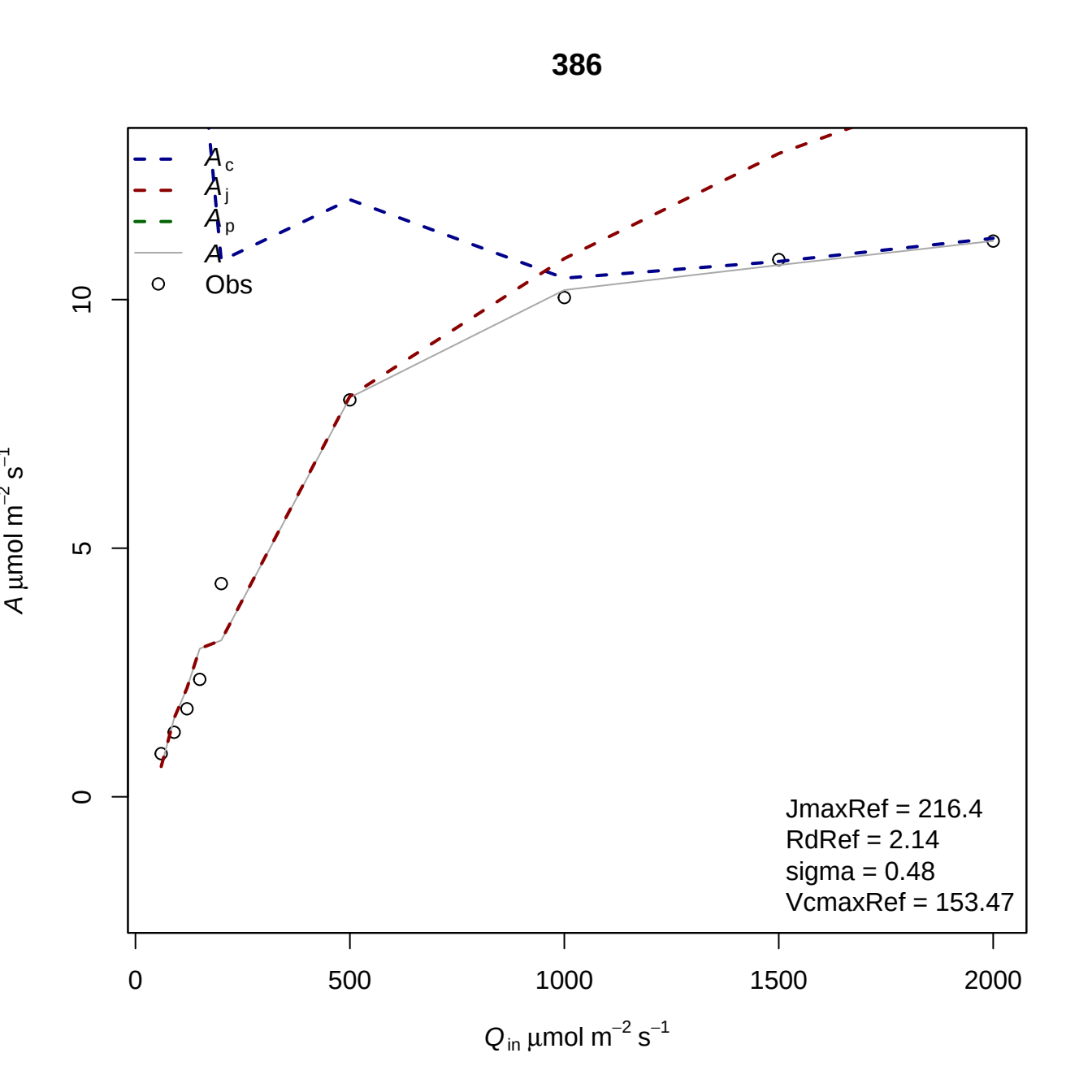

381
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 181.03
RdRef = 1.72
sigma = 0.45
VcmaxRef = 179.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
500
1000
1500
2000

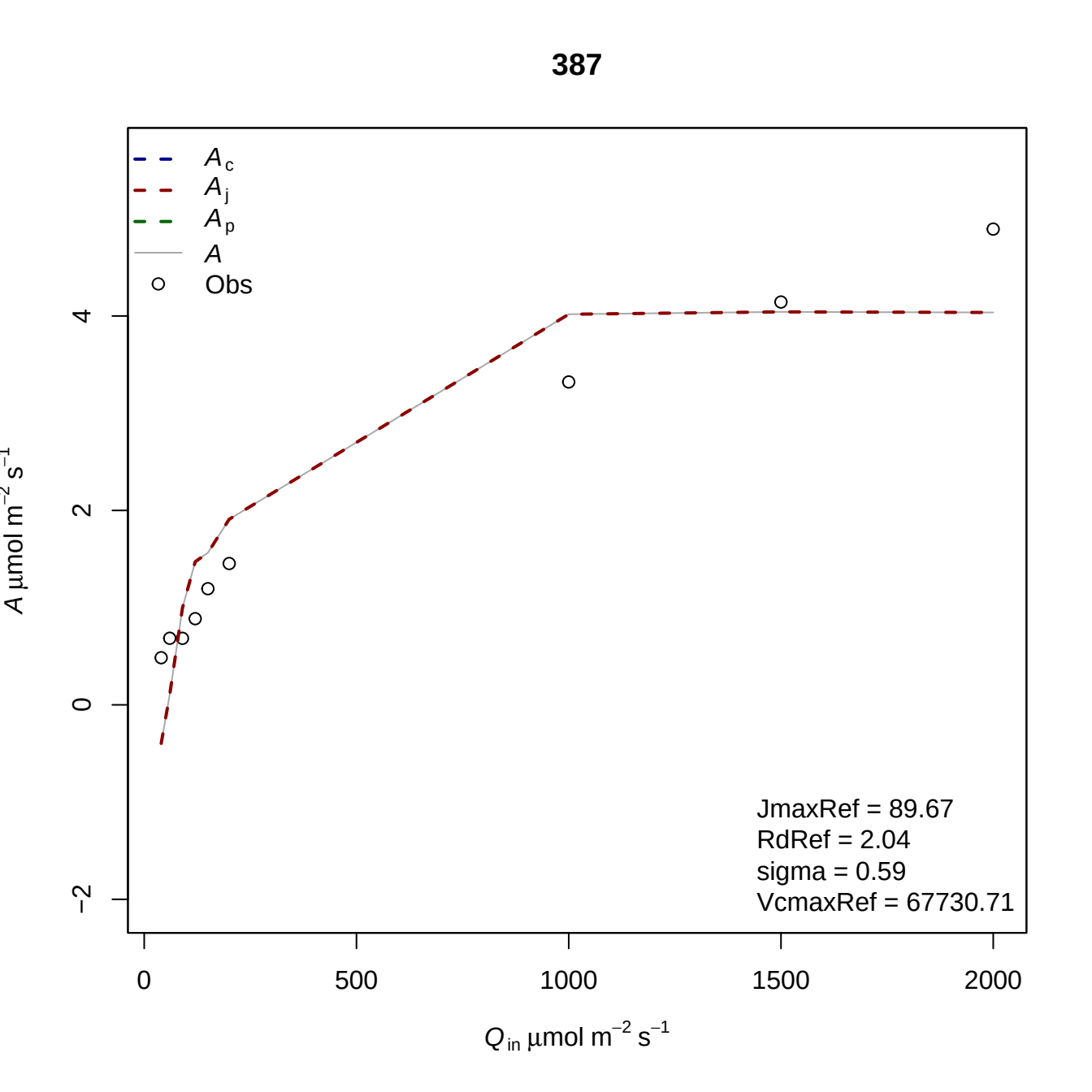

382
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.86
RdRef = 2.92
sigma = 0.2
VcmaxRef = 1811167.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−3
−2
−1
0
1
2
0
500
1000
1500
2000

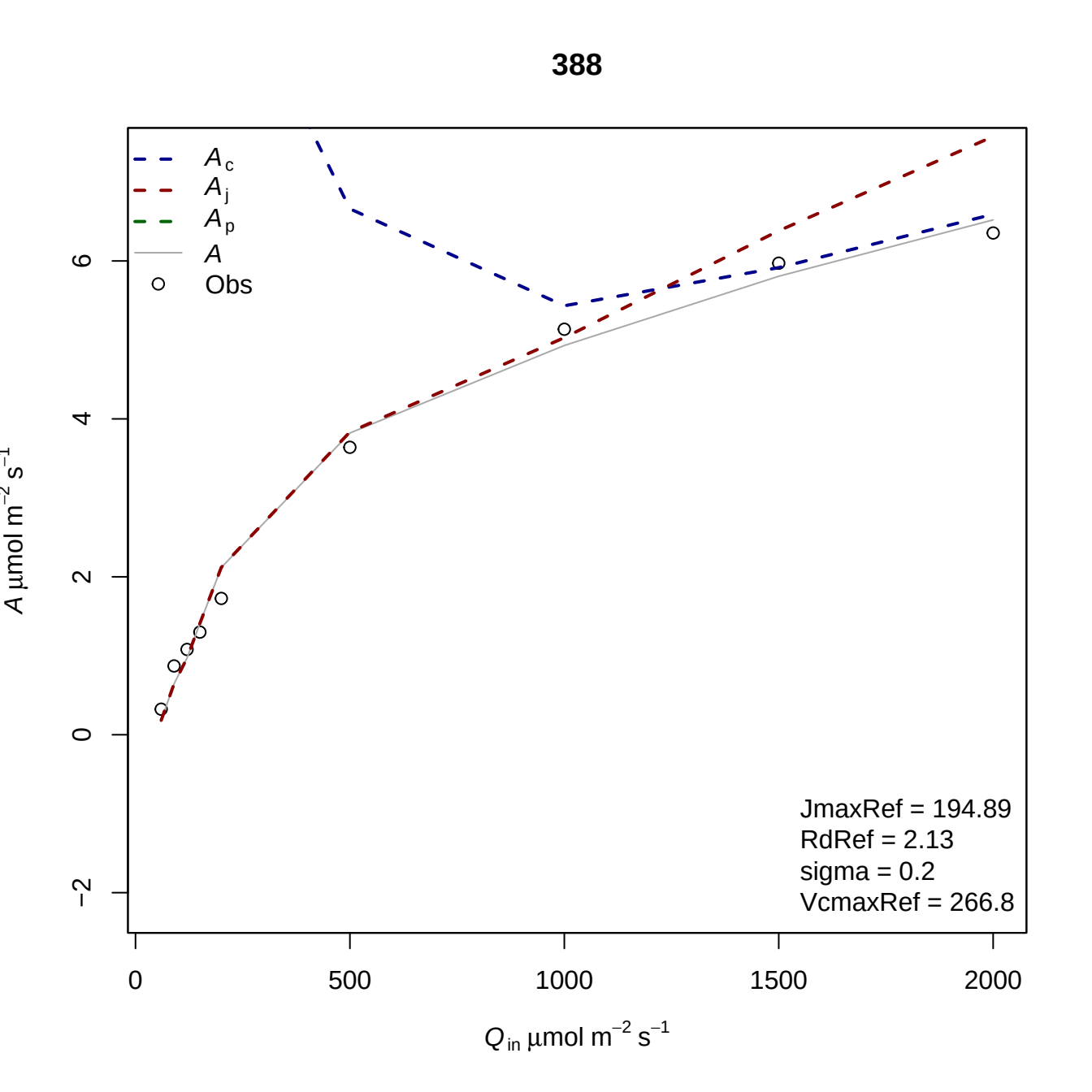

383
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.37
RdRef = 0.95
sigma = 0.17
VcmaxRef = 29.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

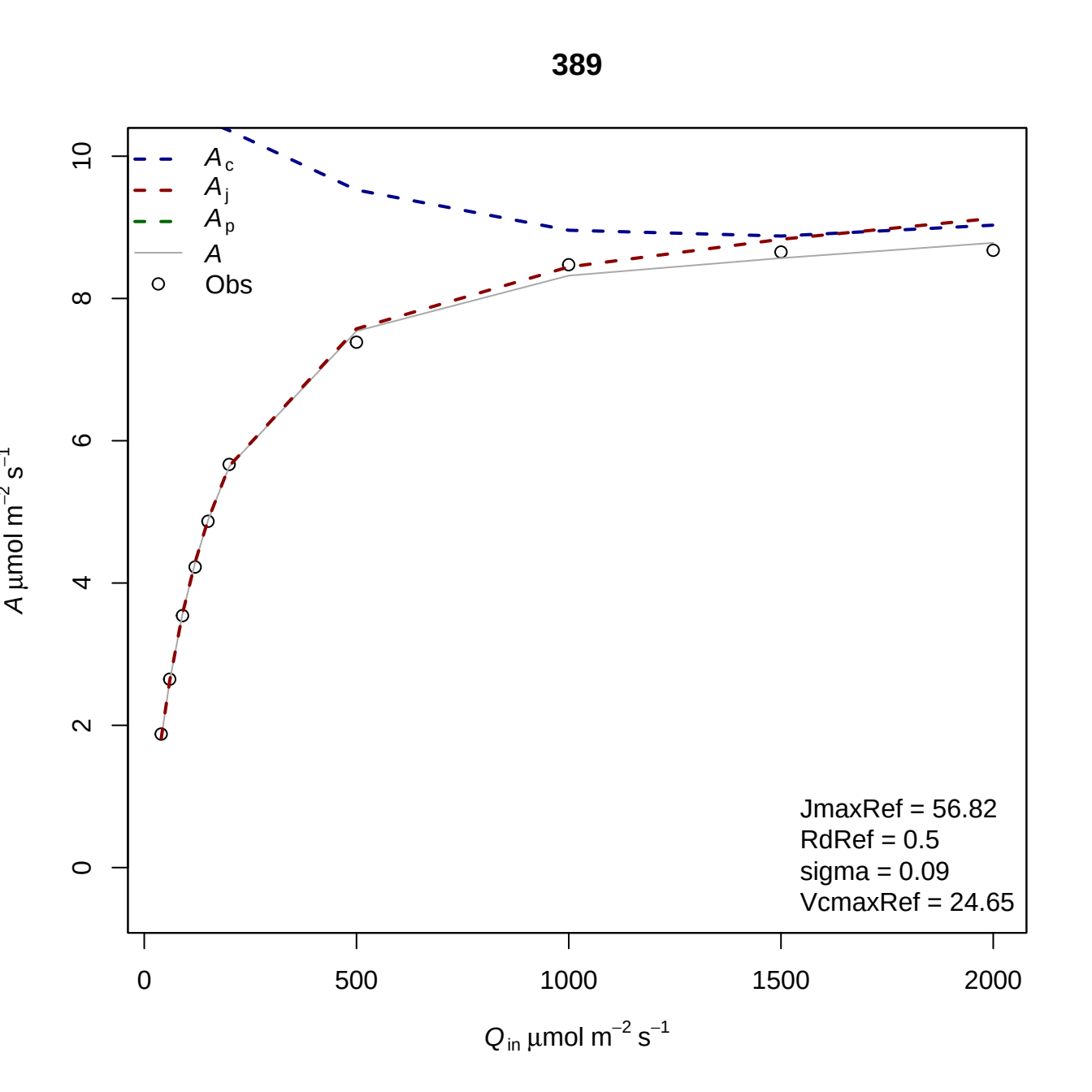

384
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.82
RdRef = 0.5
sigma = 0.19
VcmaxRef = 60.37
0
5
10
15
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

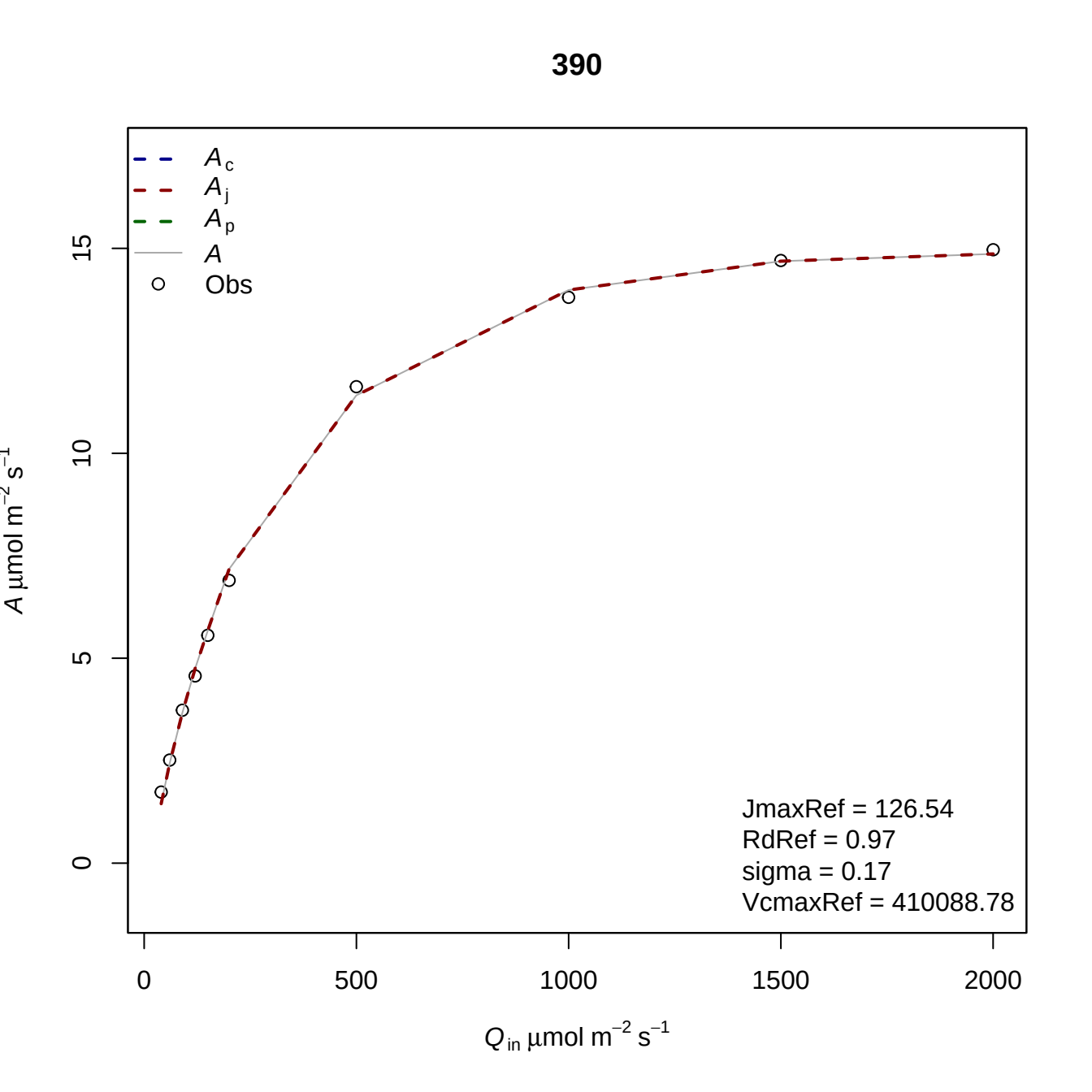

385
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.91
RdRef = 1.59
sigma = 0.26
VcmaxRef = 70.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

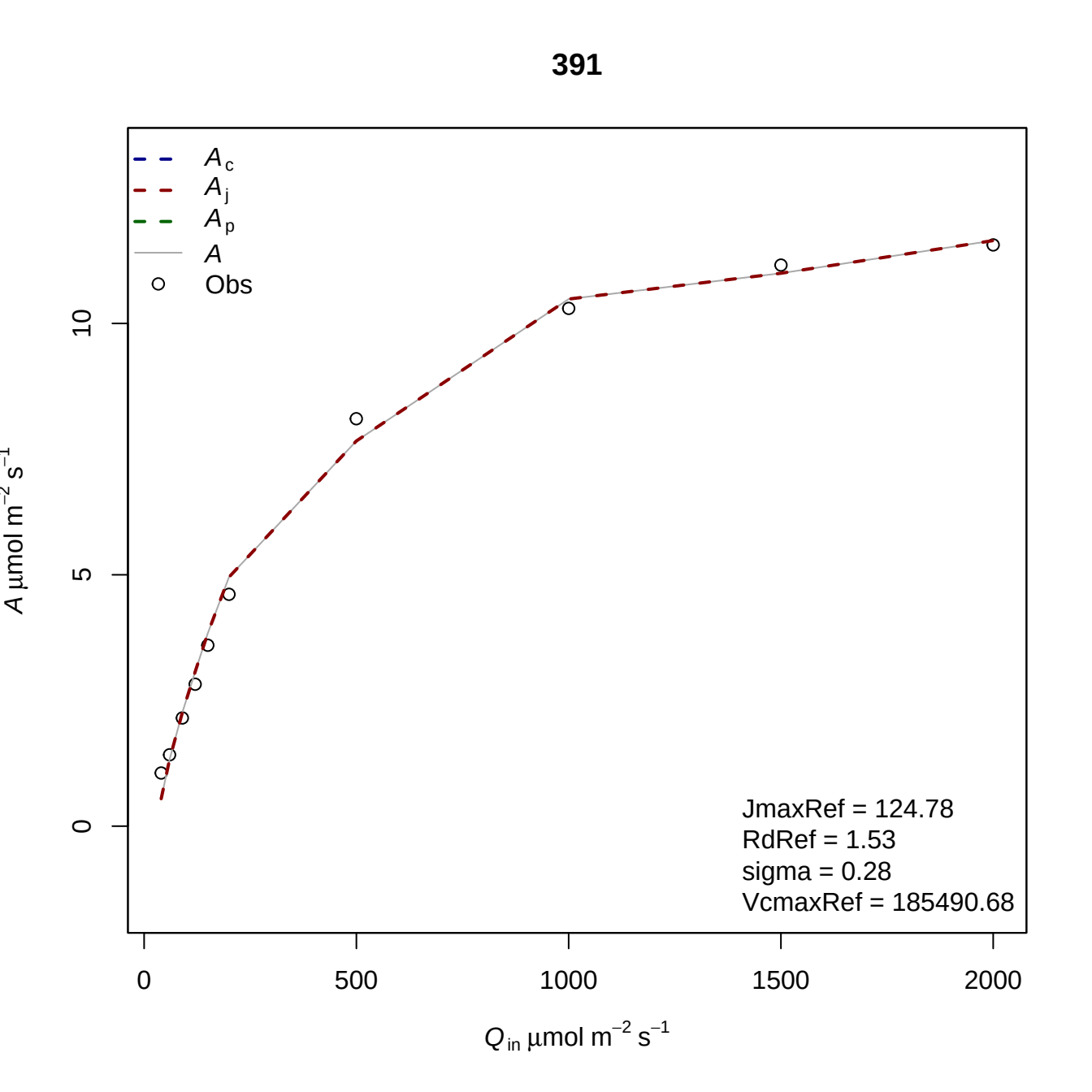

386
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 216.4
RdRef = 2.14
sigma = 0.48
VcmaxRef = 153.47
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

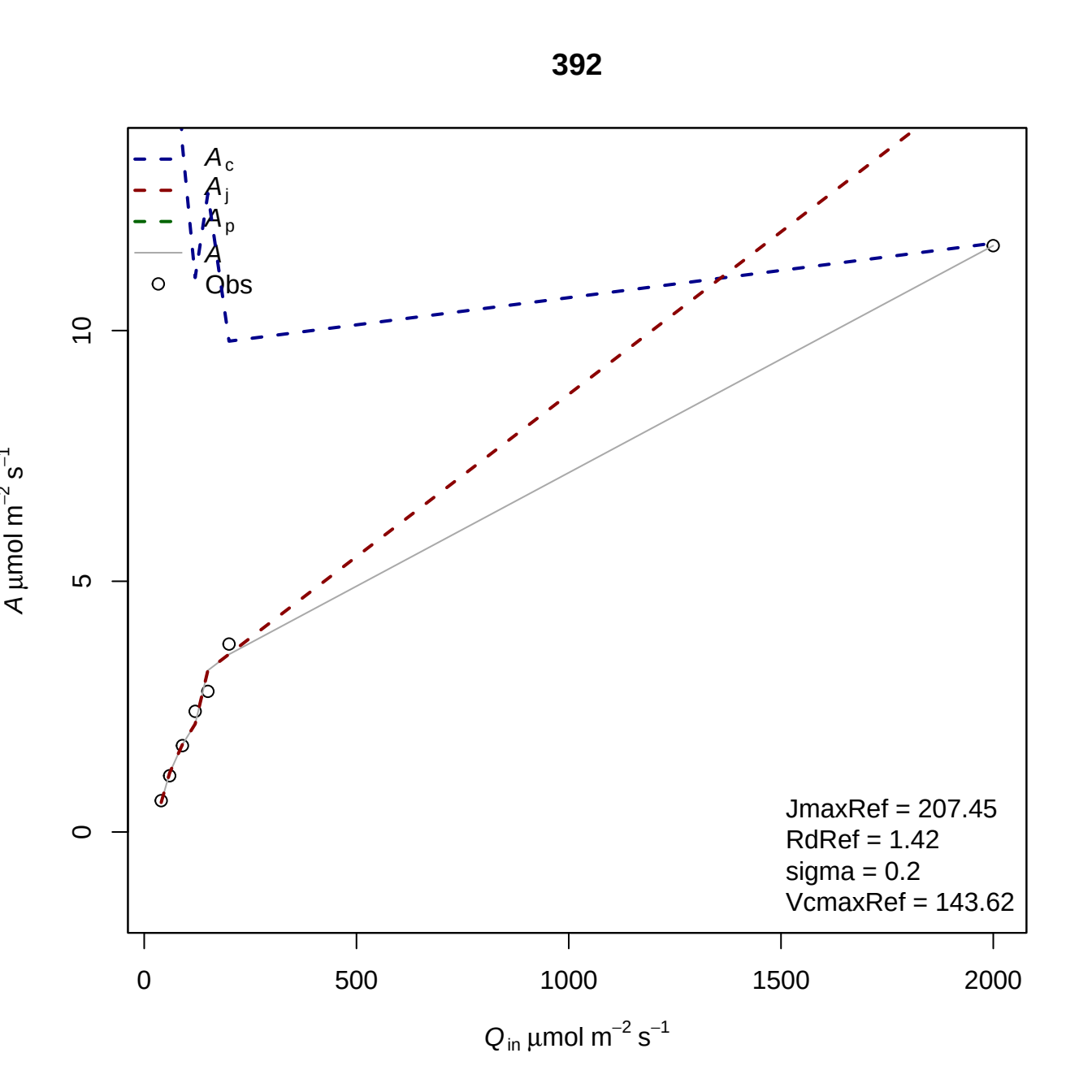

387
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.67
RdRef = 2.04
sigma = 0.59
VcmaxRef = 67730.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

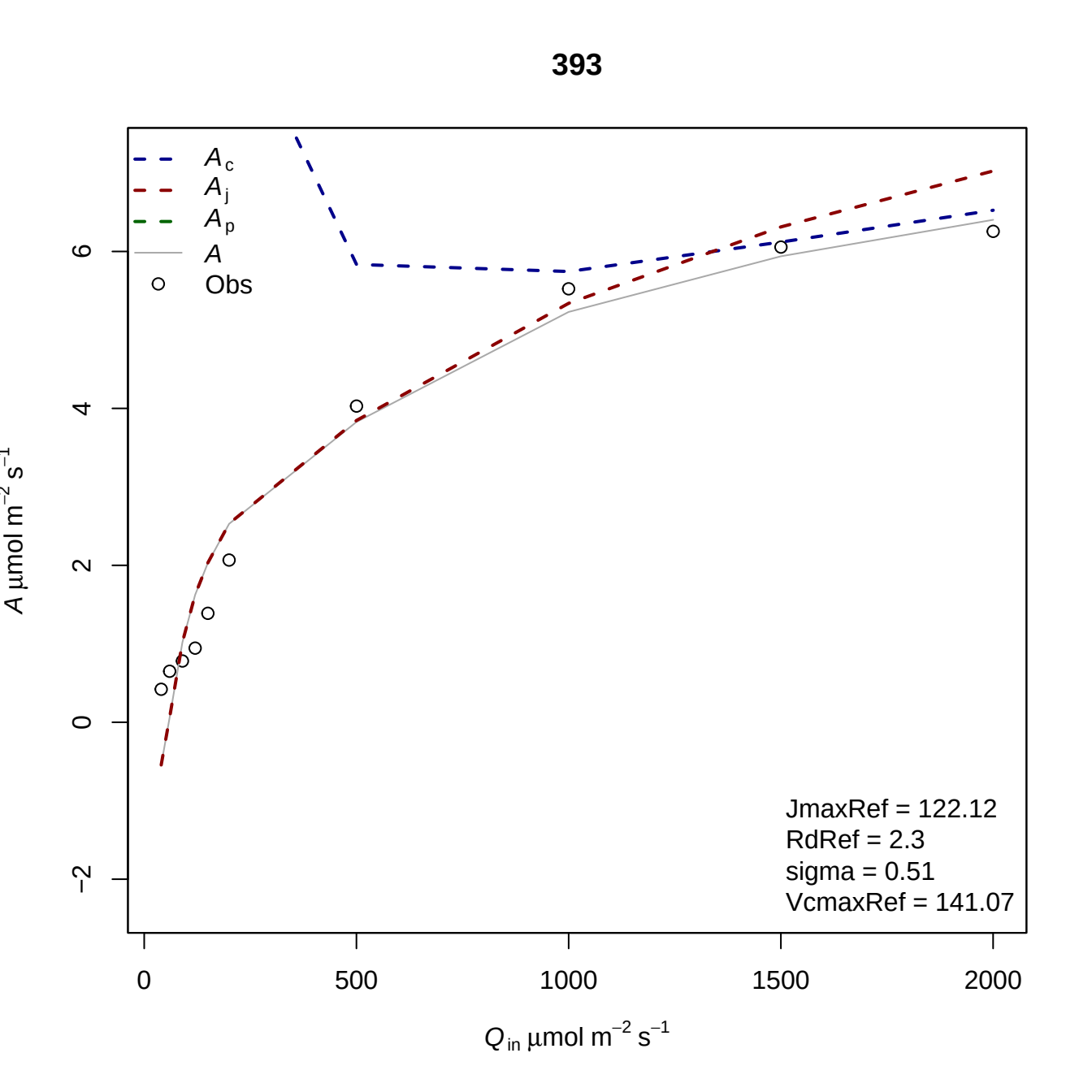

388
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 194.89
RdRef = 2.13
sigma = 0.2
VcmaxRef = 266.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

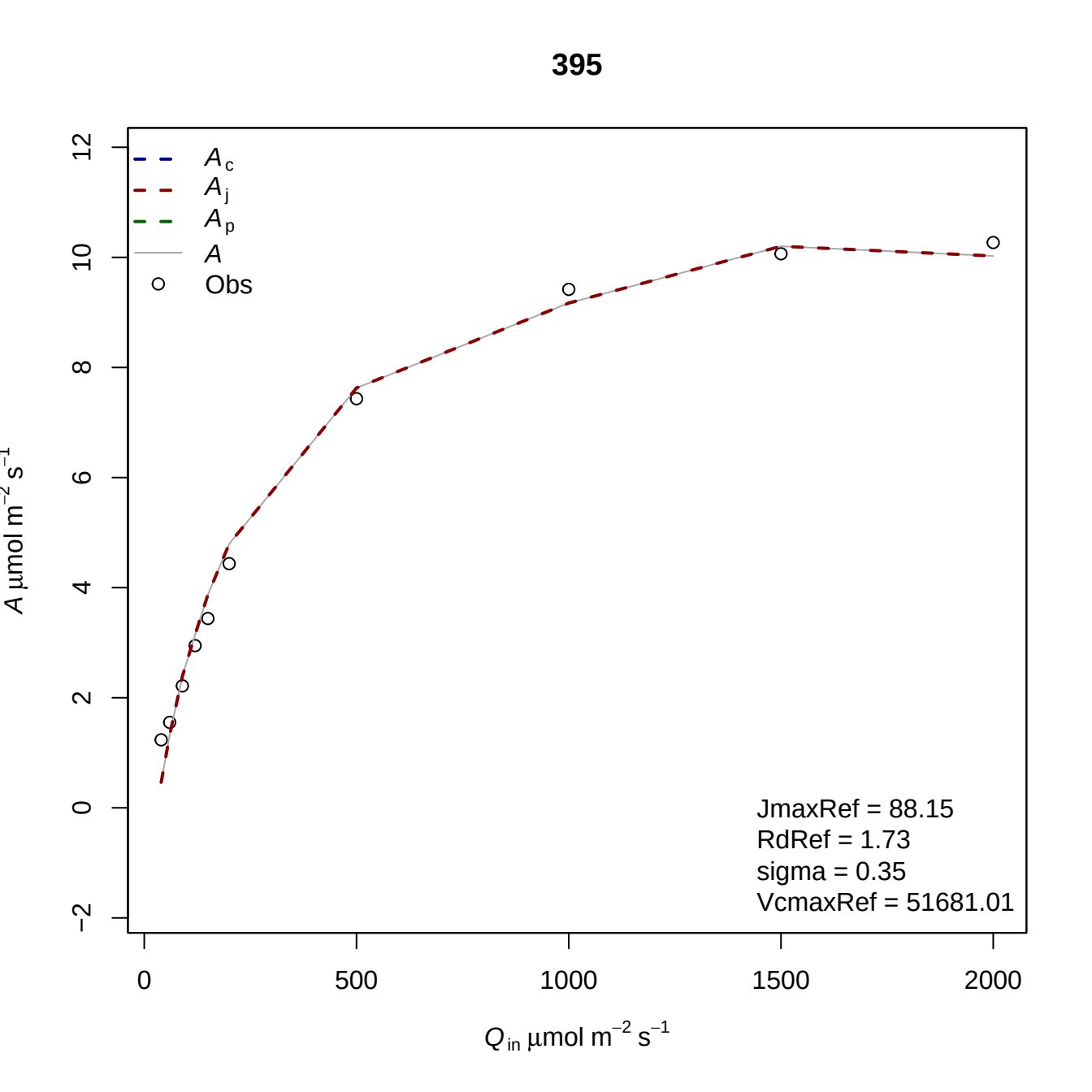

389
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.82
RdRef = 0.5
sigma = 0.09
VcmaxRef = 24.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

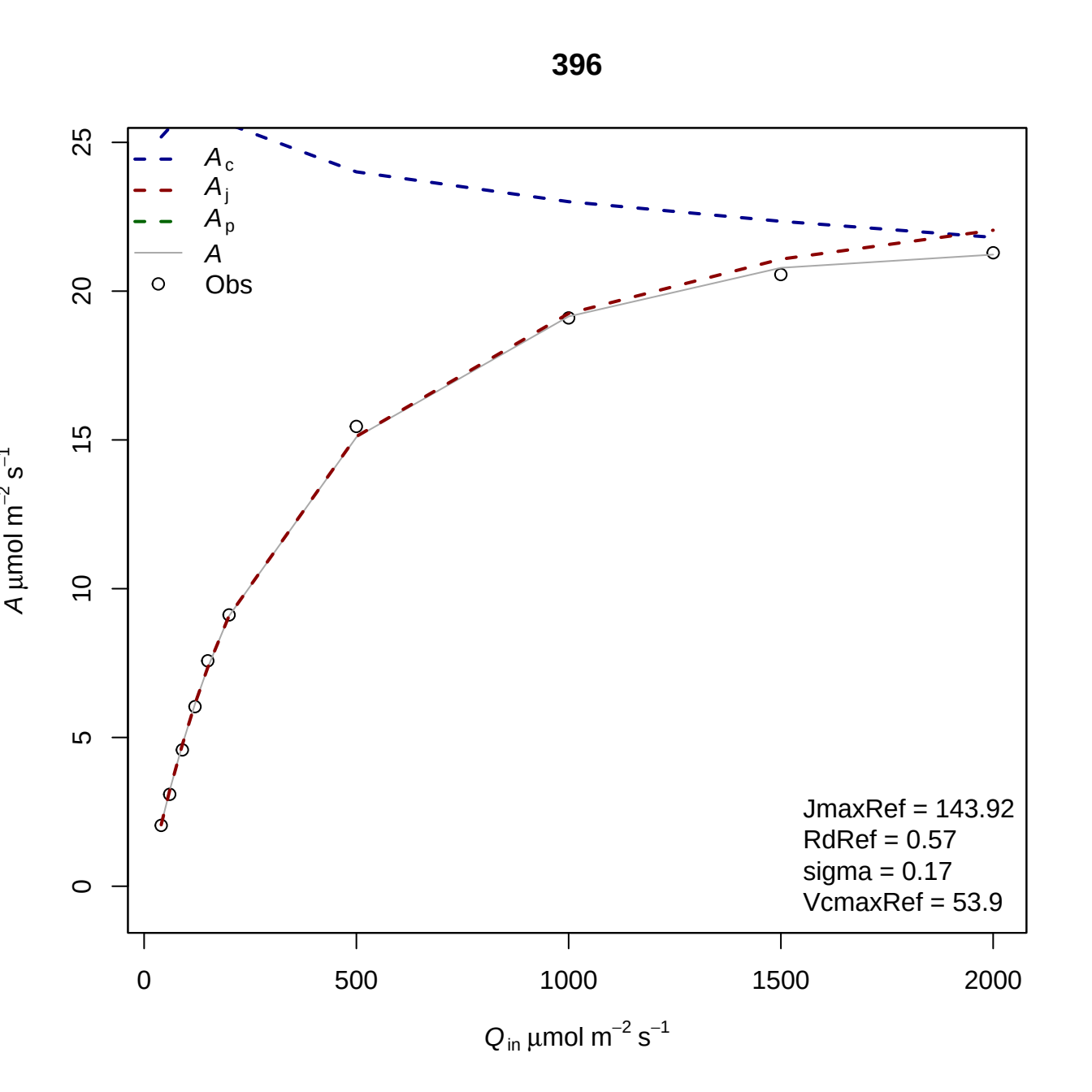

390
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 126.54
RdRef = 0.97
sigma = 0.17
VcmaxRef = 410088.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

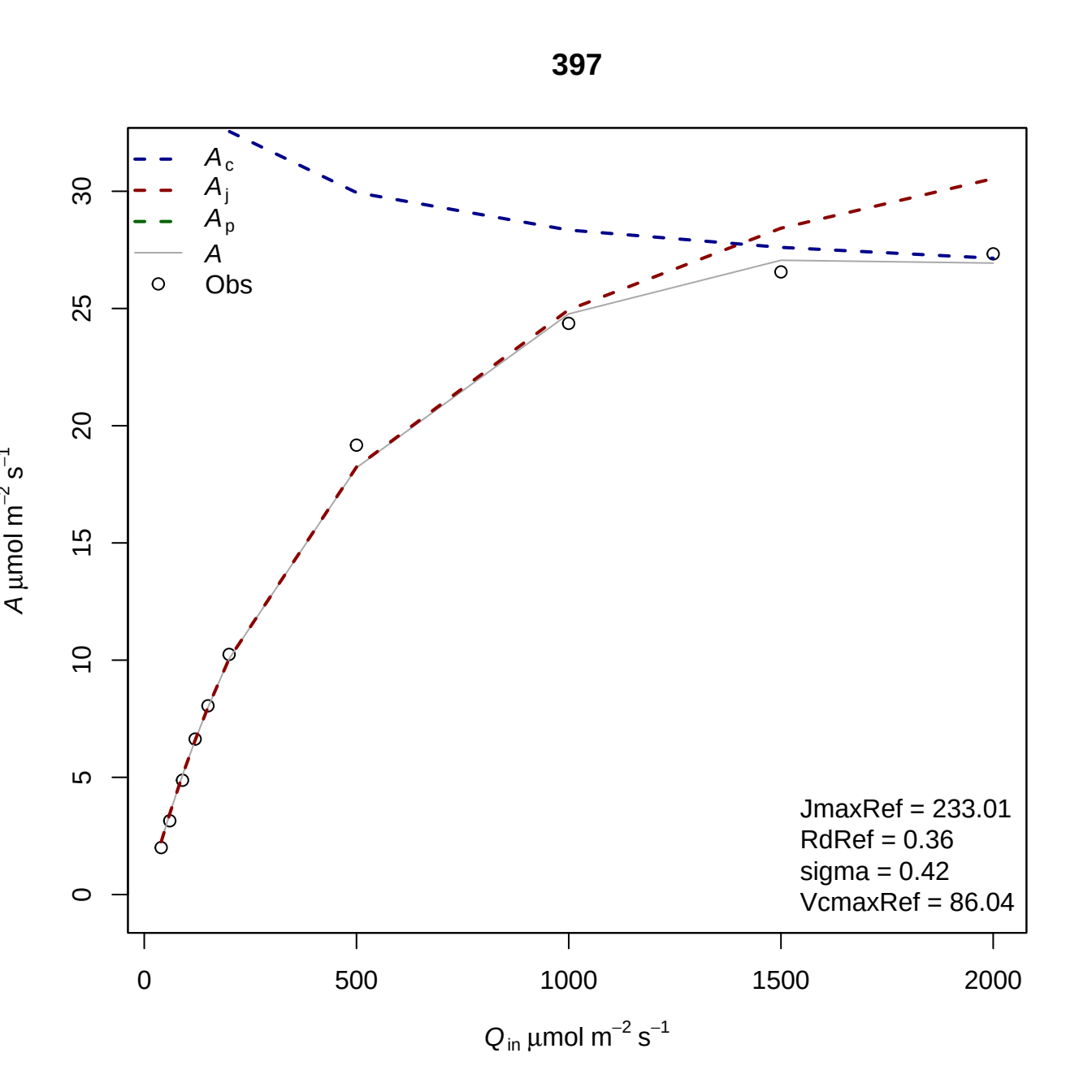

391
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.78
RdRef = 1.53
sigma = 0.28
VcmaxRef = 185490.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

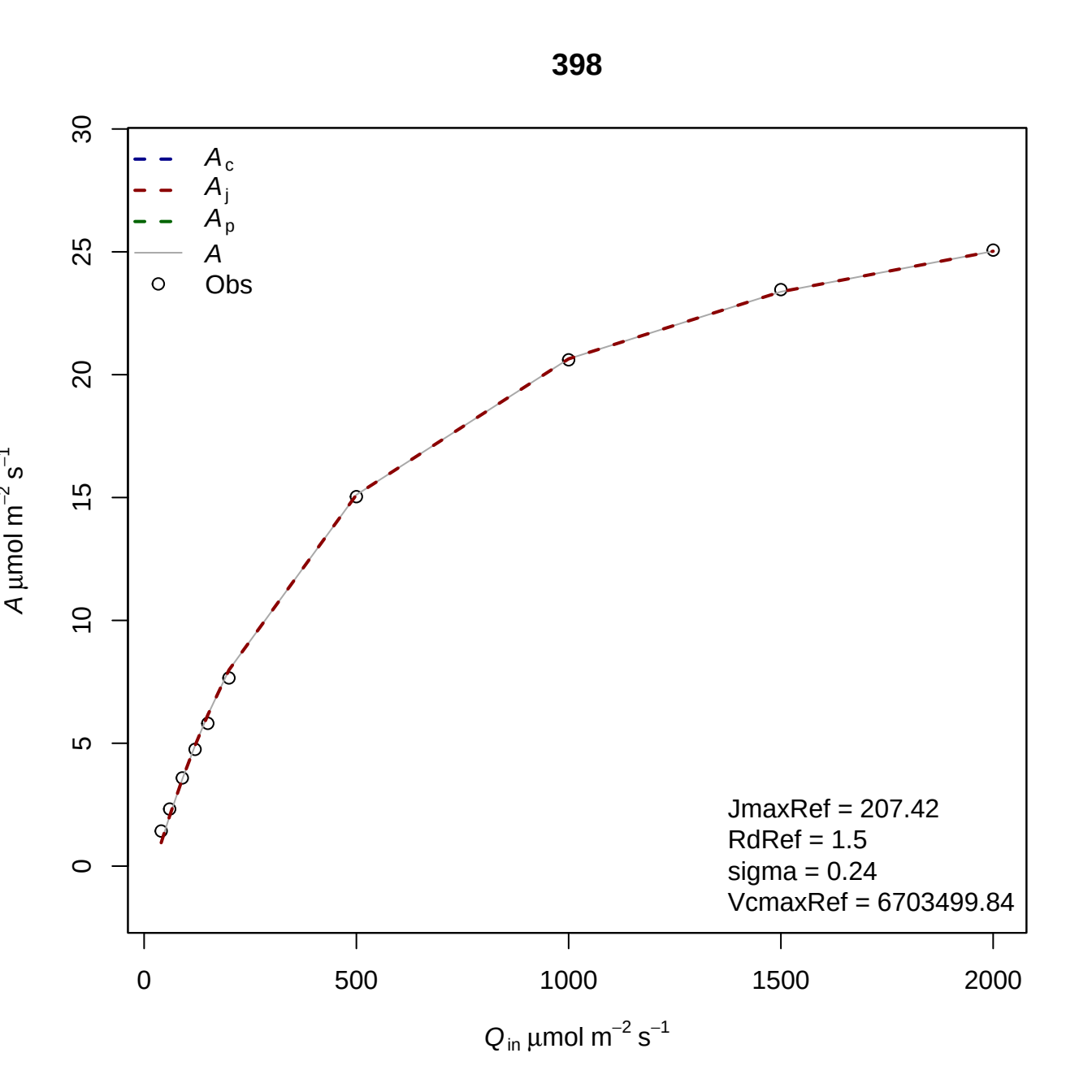

392
$A_c$
$A_j$
$A_p$
$A$
Qbs
JmaxRef = 207.45
RdRef = 1.42
sigma = 0.2
VcmaxRef = 143.62
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

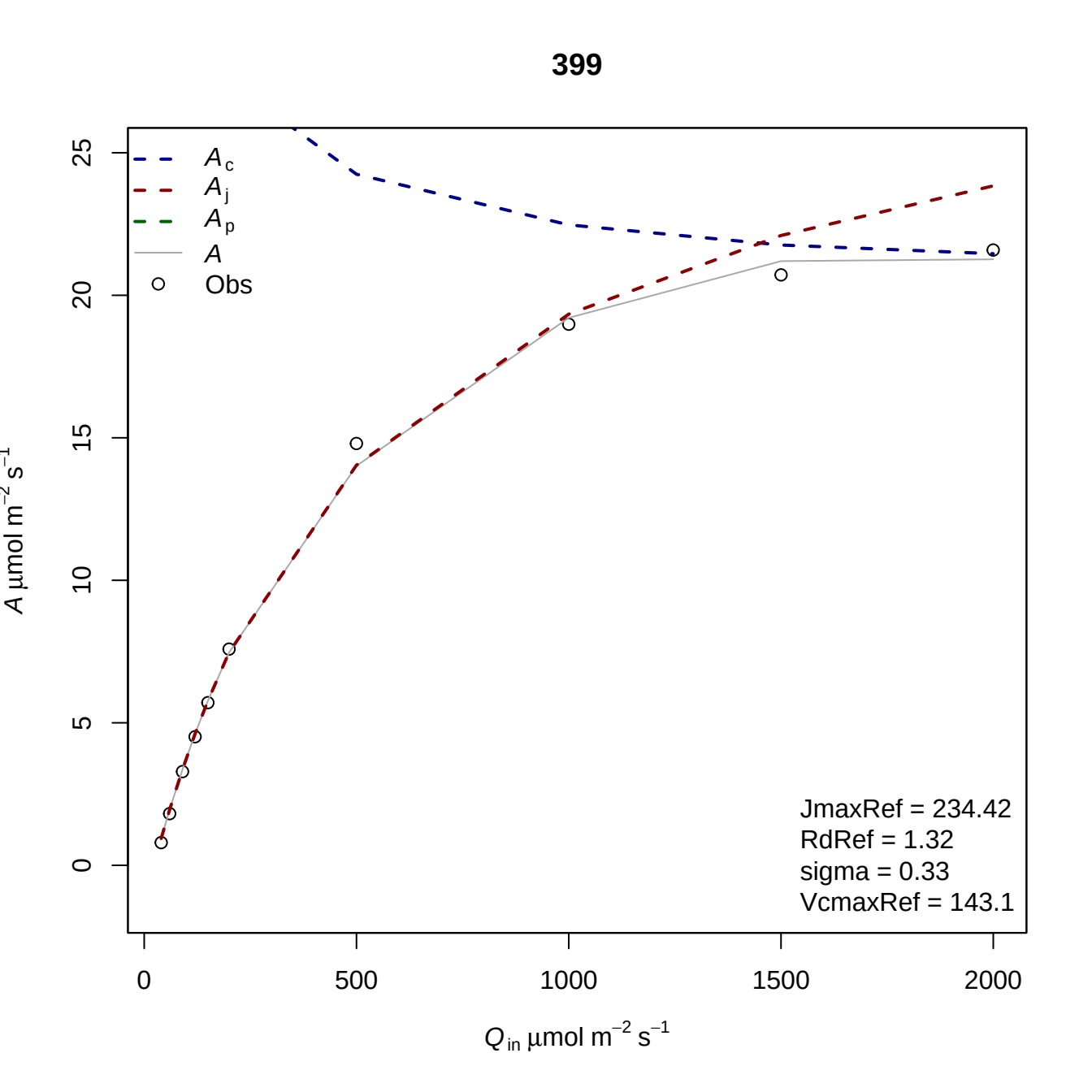

393
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.12
RdRef = 2.3
sigma = 0.51
VcmaxRef = 141.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

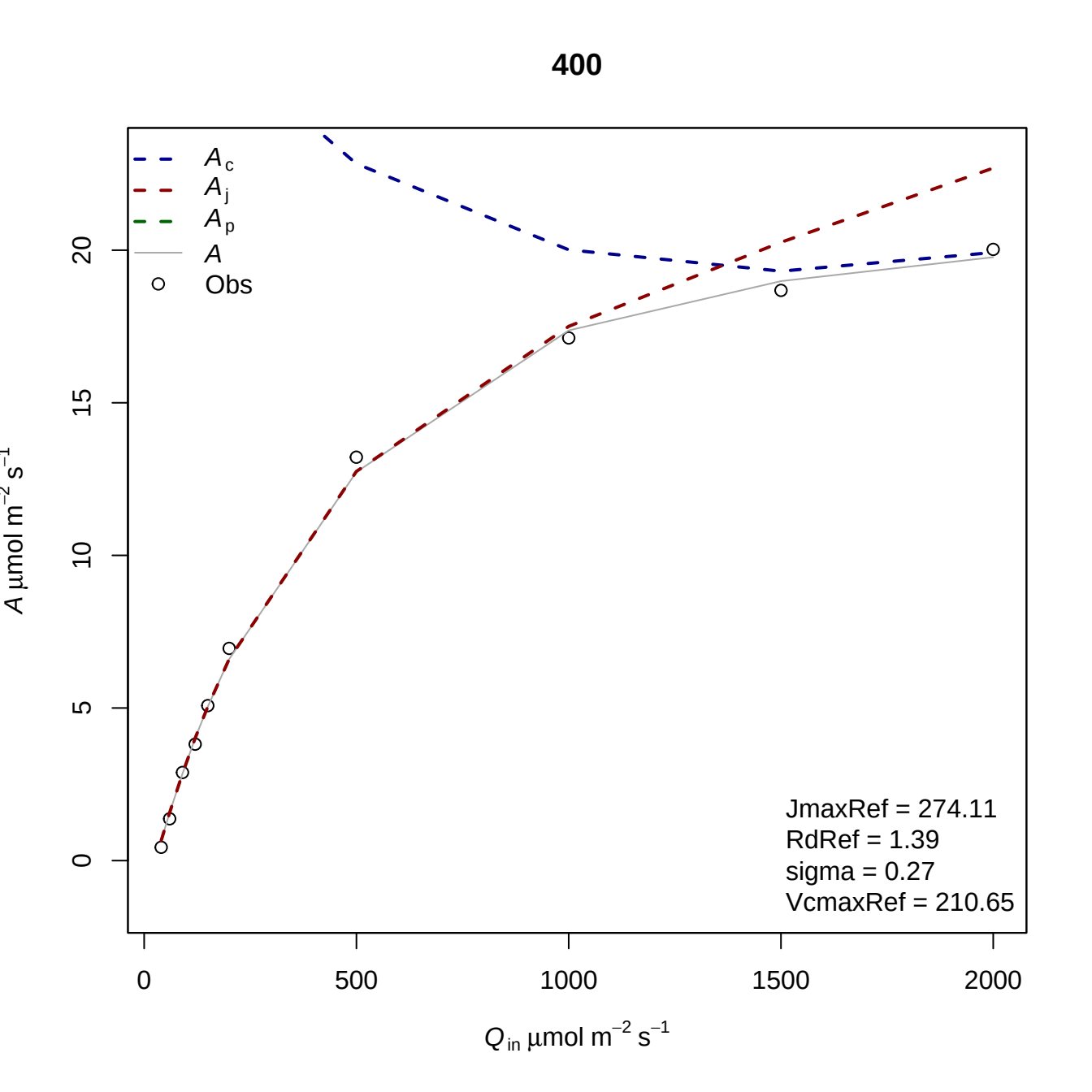

395
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.15
RdRef = 1.73
sigma = 0.35
VcmaxRef = 51681.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

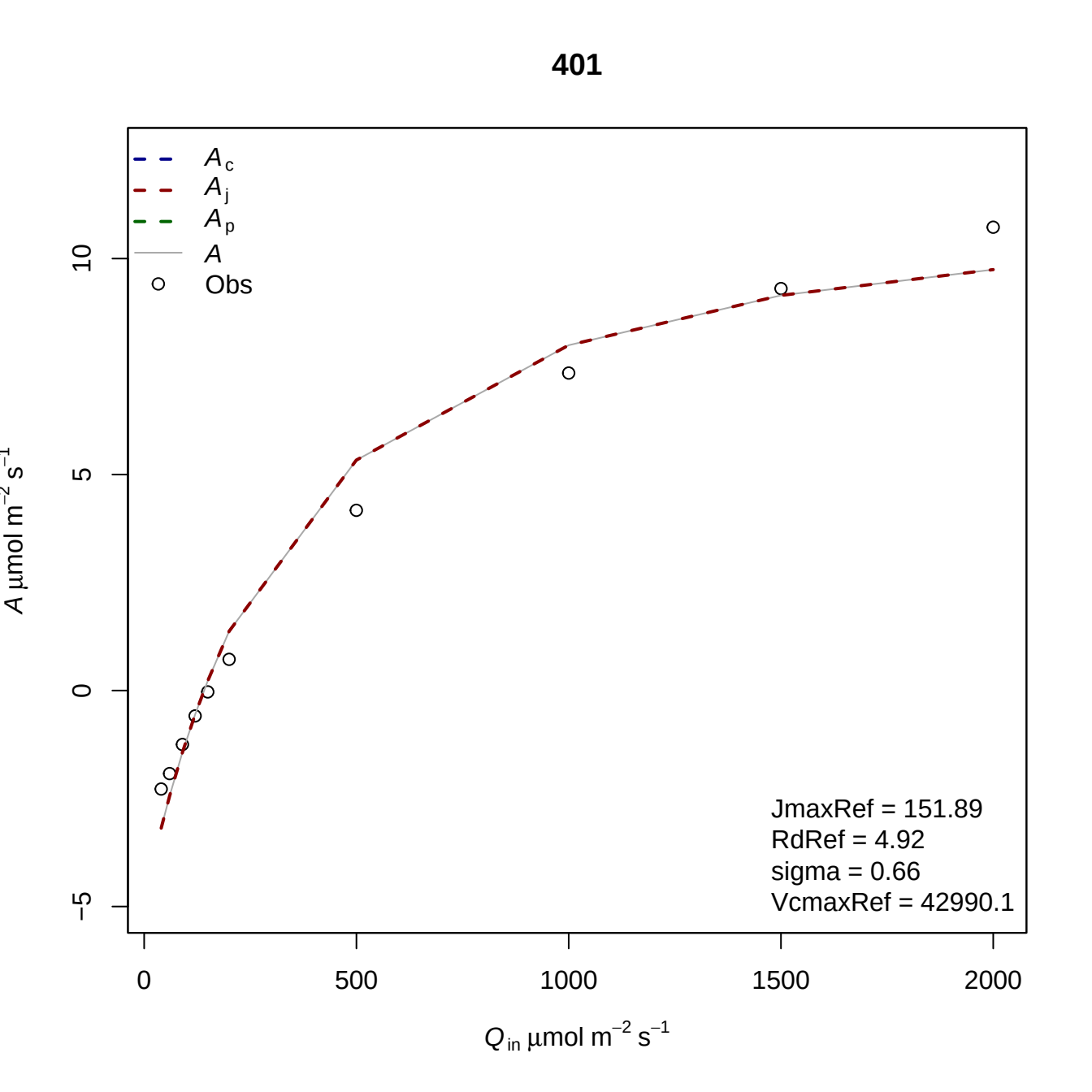

396
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.92
RdRef = 0.57
sigma = 0.17
VcmaxRef = 53.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

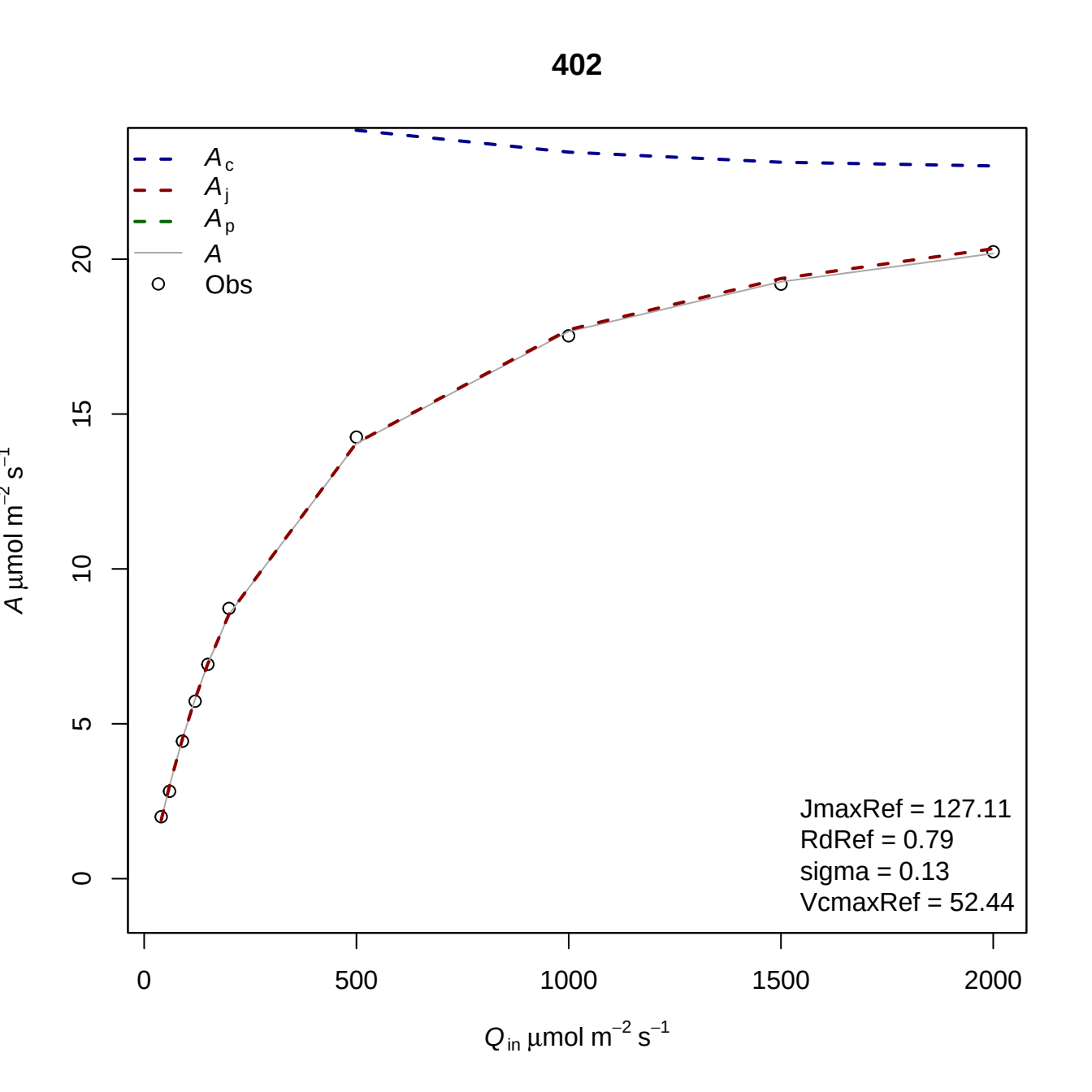

397
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 233.01
RdRef = 0.36
sigma = 0.42
VcmaxRef = 86.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

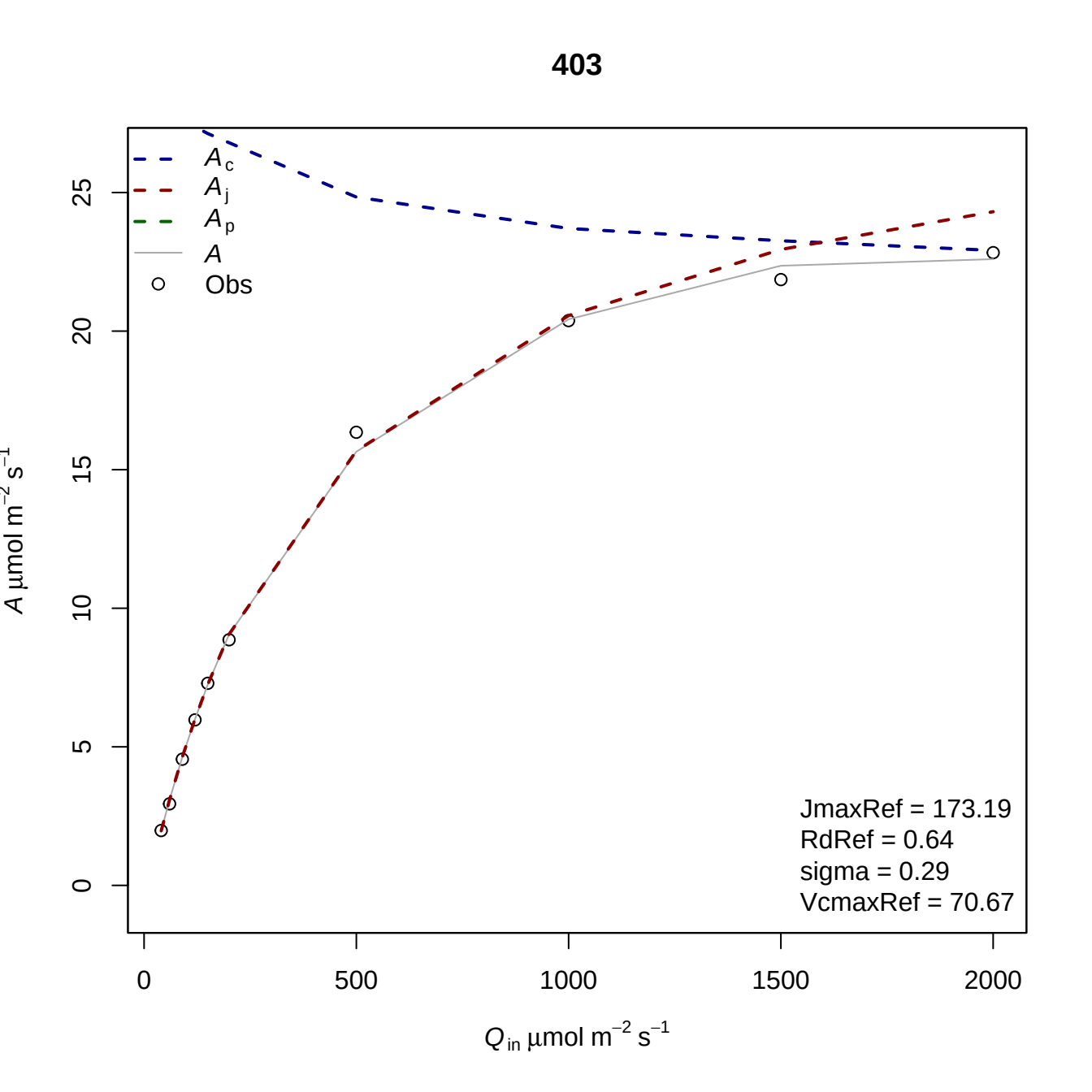

398
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 207.42
RdRef = 1.5
sigma = 0.24
VcmaxRef = 6703499.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

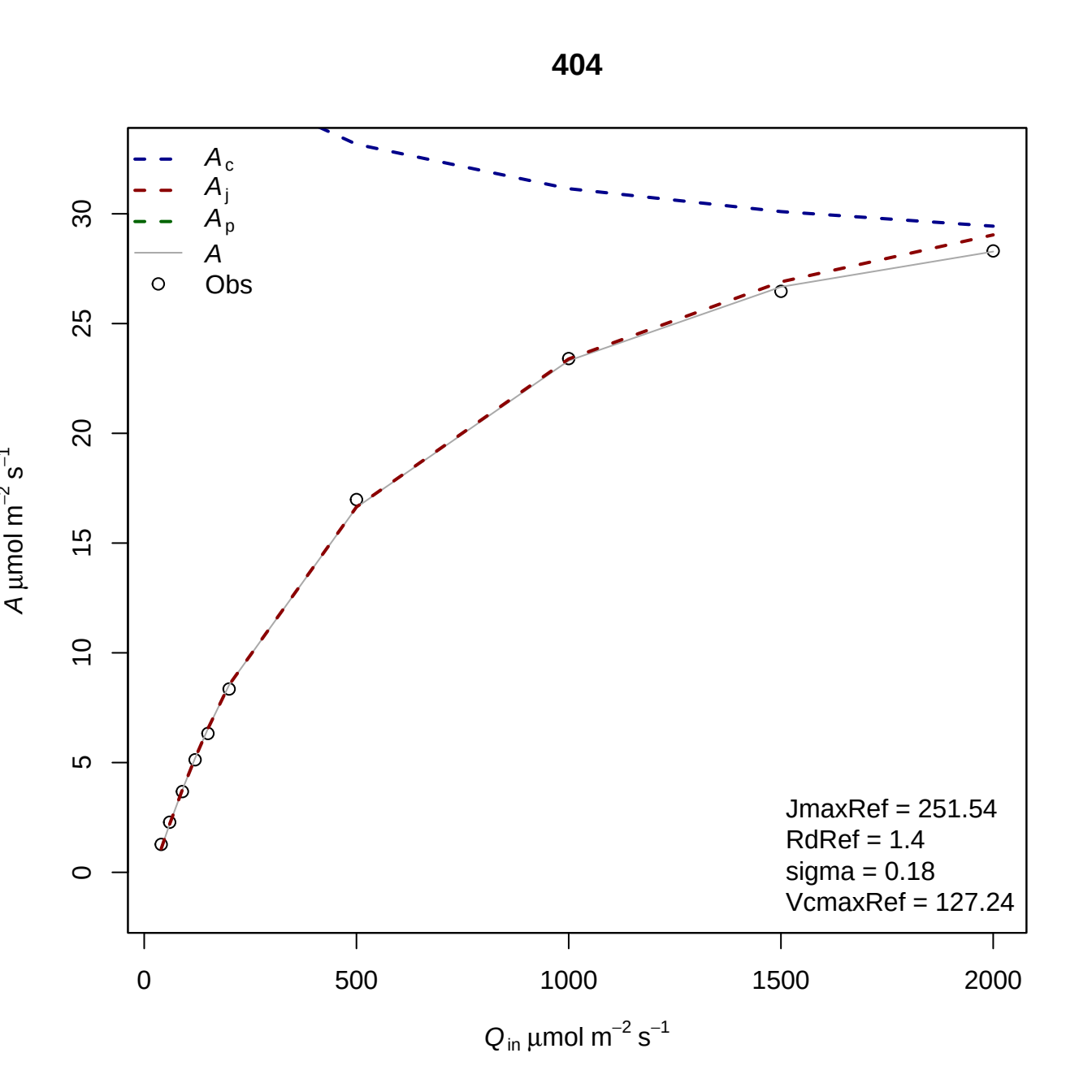

399
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 234.42
RdRef = 1.32
sigma = 0.33
VcmaxRef = 143.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

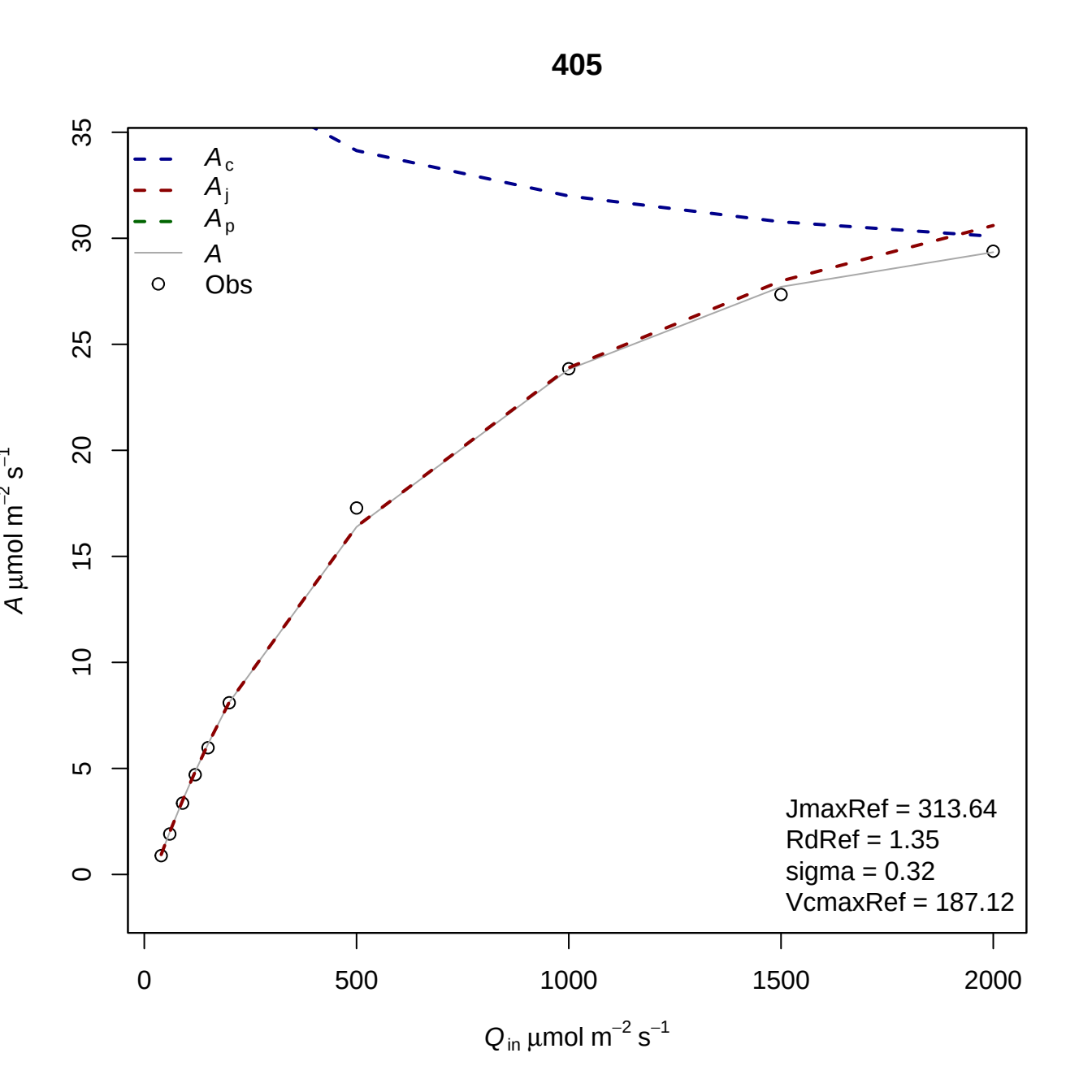

400
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 274.11
RdRef = 1.39
sigma = 0.27
VcmaxRef = 210.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

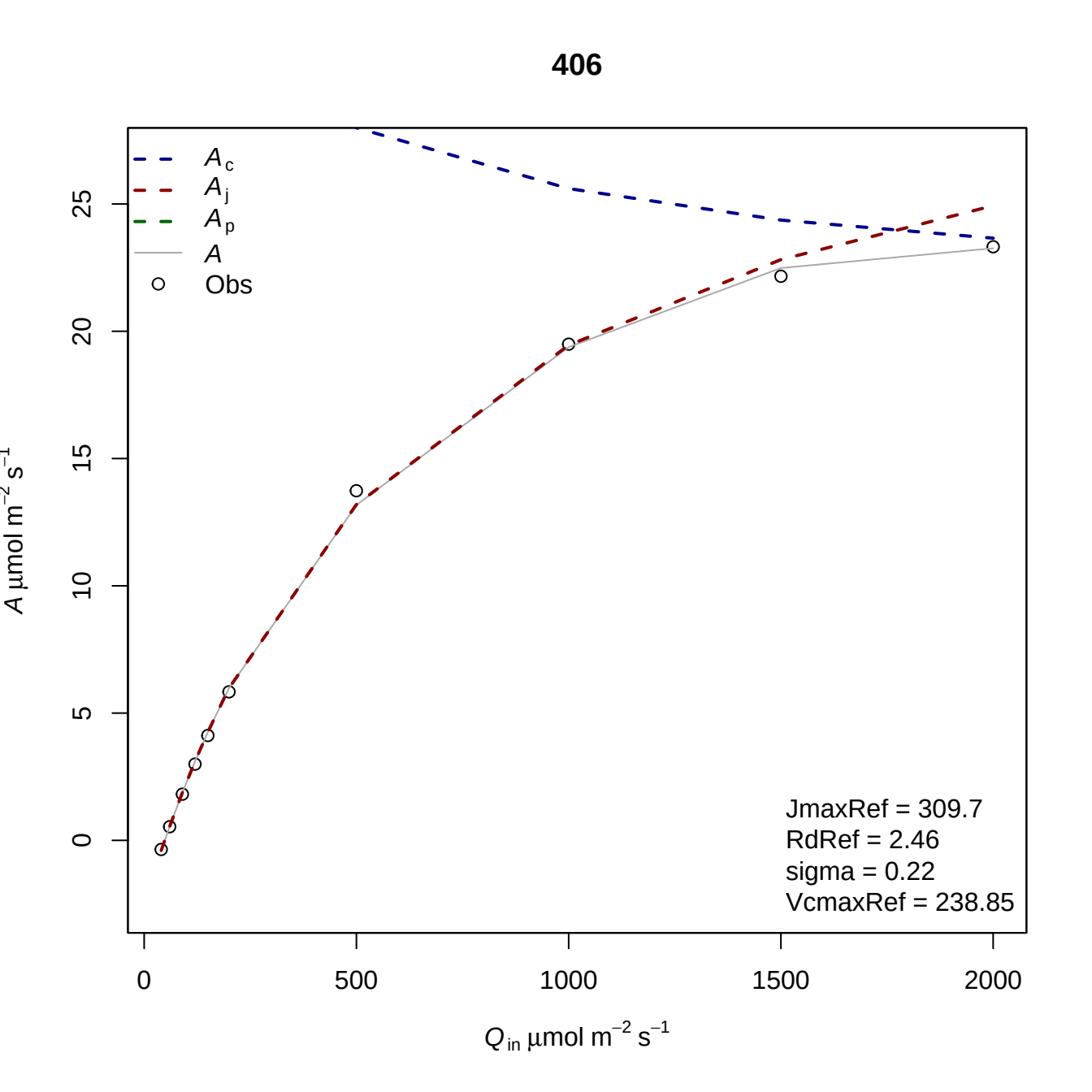

401
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 151.89
RdRef = 4.92
sigma = 0.66
VcmaxRef = 42990.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

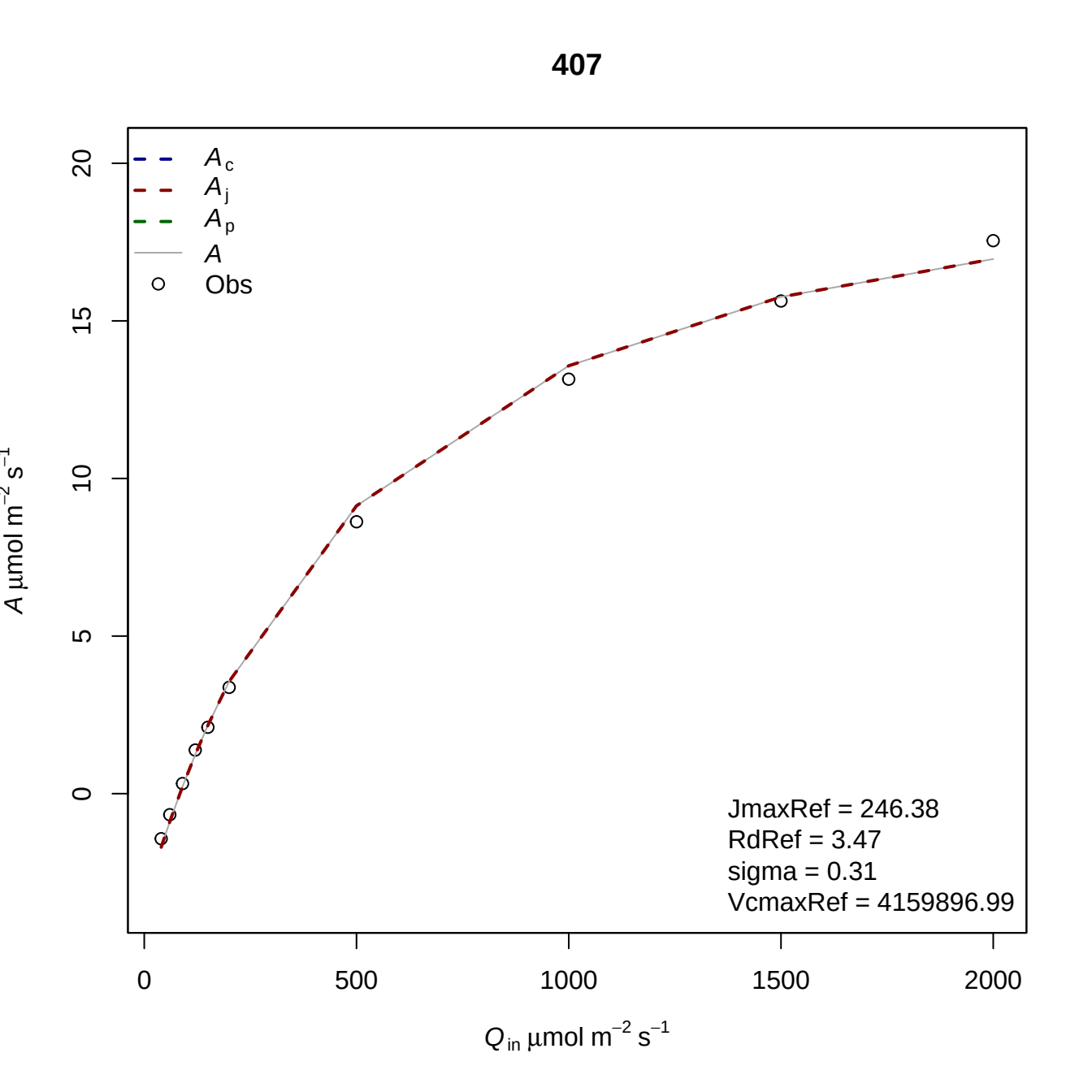

402
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.11
RdRef = 0.79
sigma = 0.13
VcmaxRef = 52.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

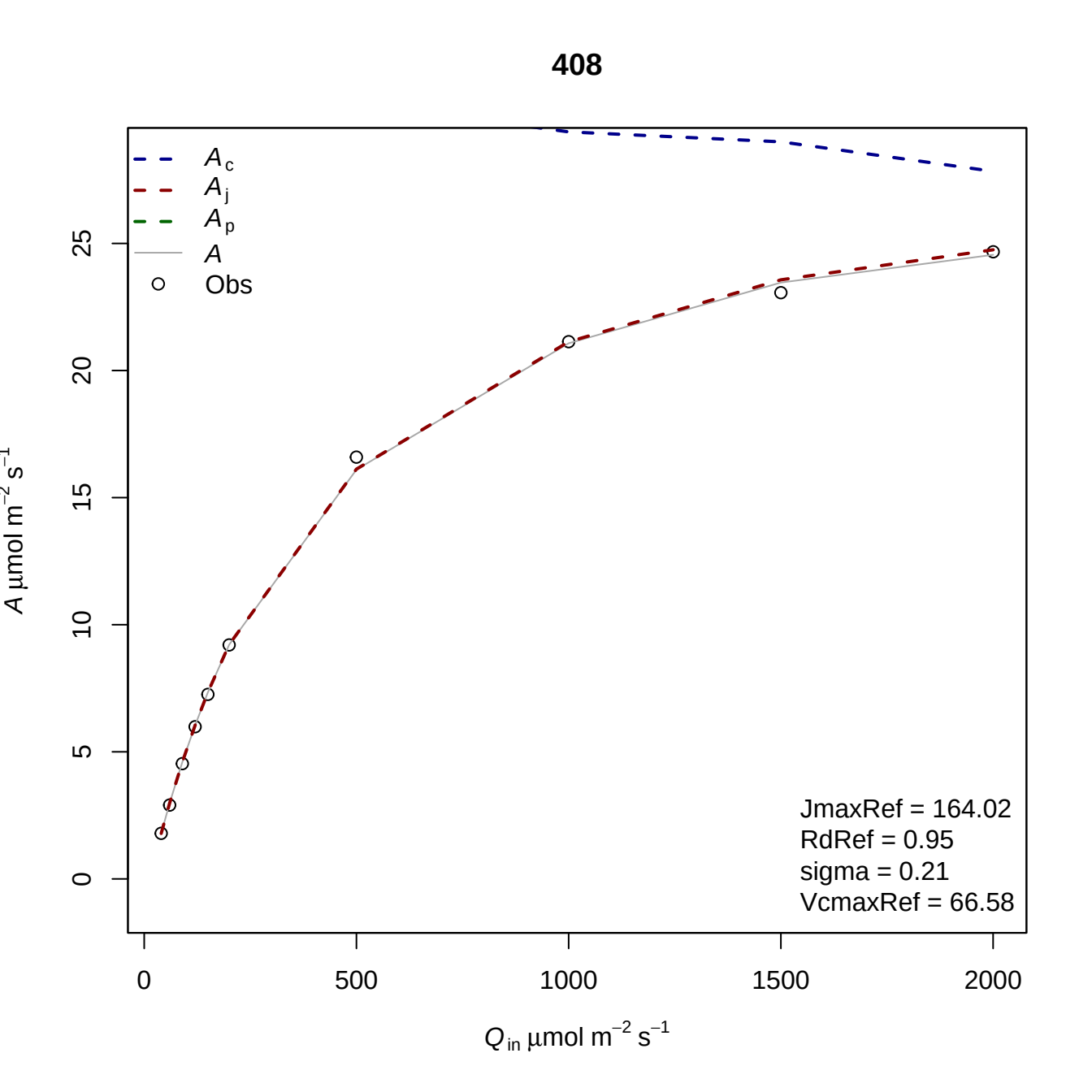

403
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 173.19
RdRef = 0.64
sigma = 0.29
VcmaxRef = 70.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

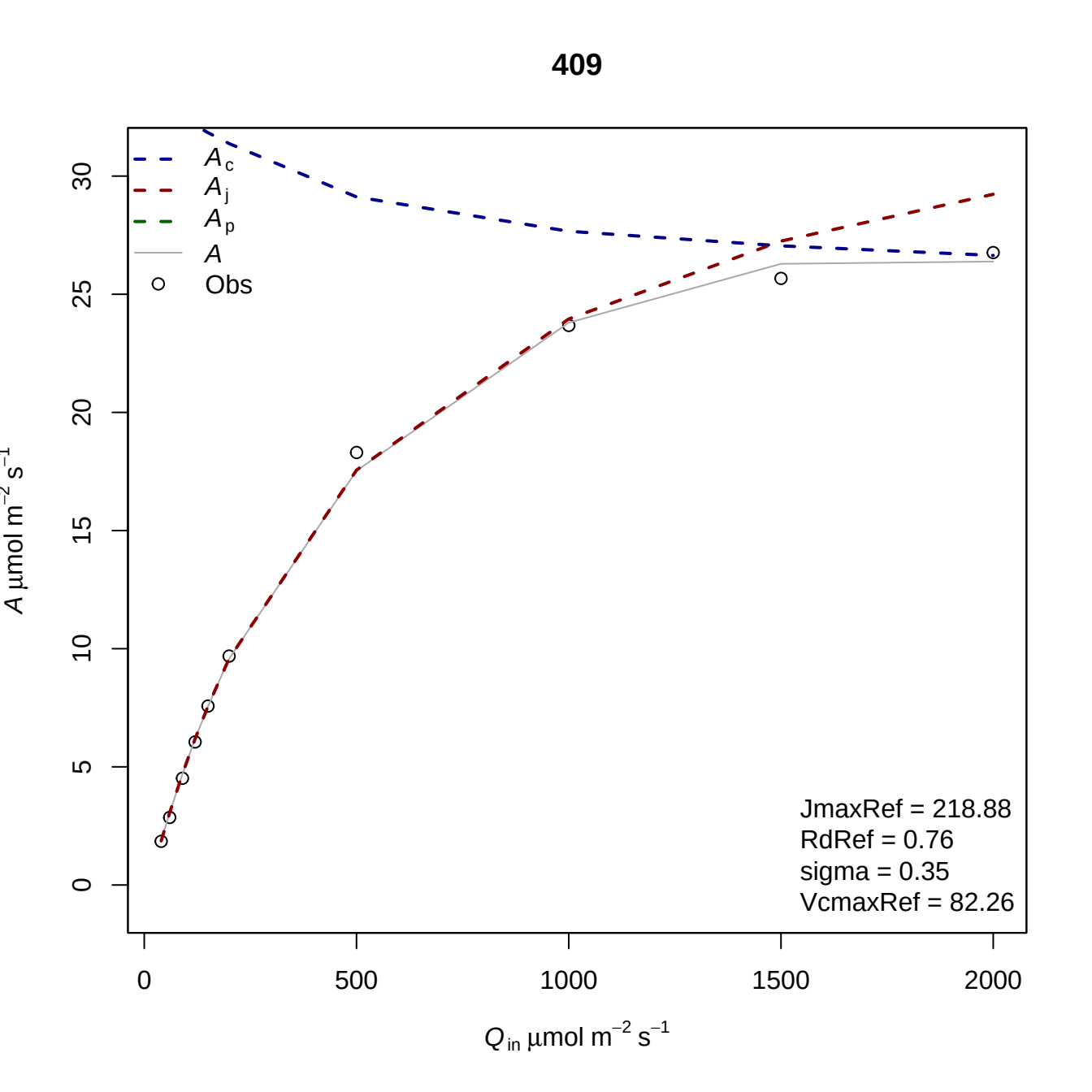

404
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 251.54
RdRef = 1.4
sigma = 0.18
VcmaxRef = 127.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

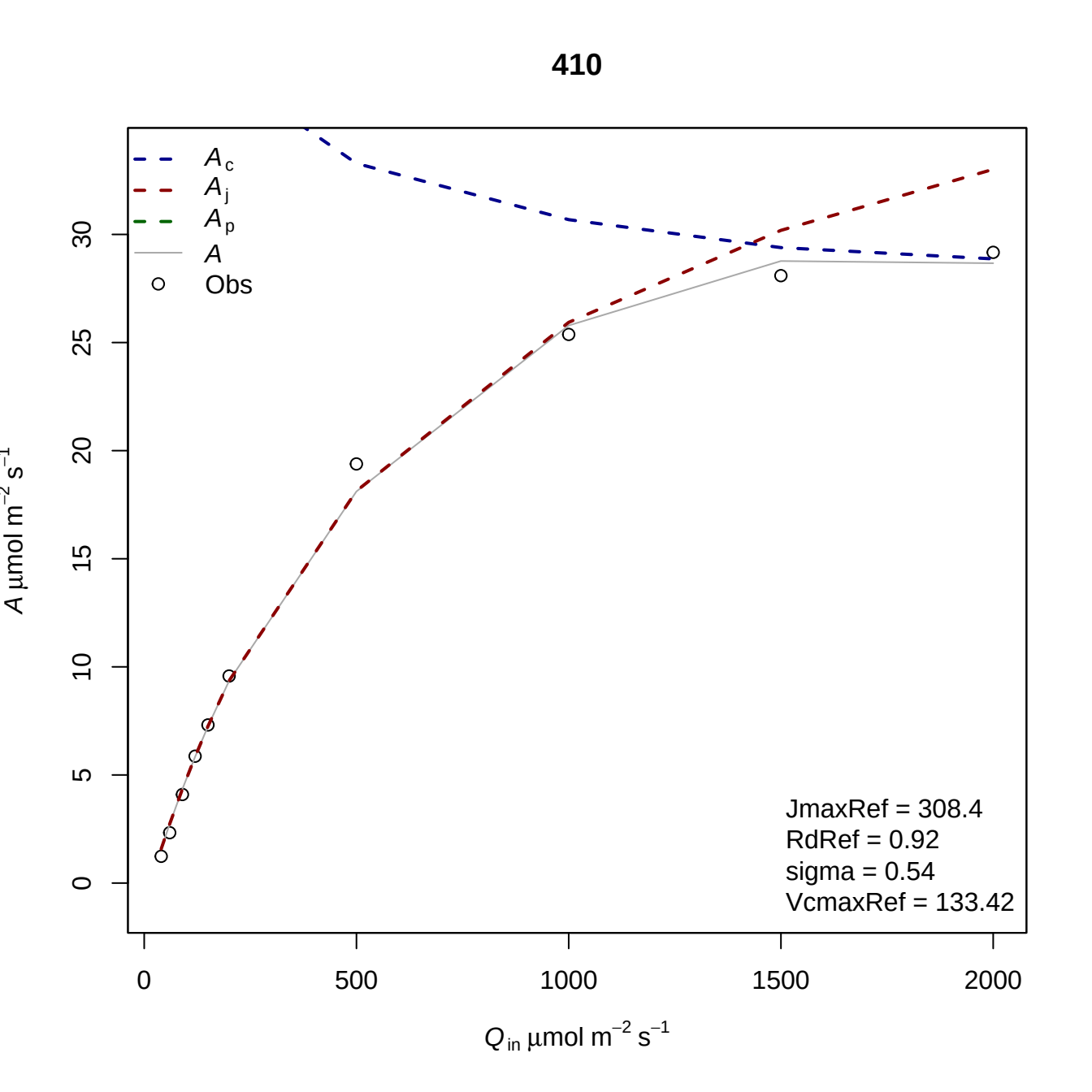

405
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 313.64
RdRef = 1.35
sigma = 0.32
VcmaxRef = 187.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

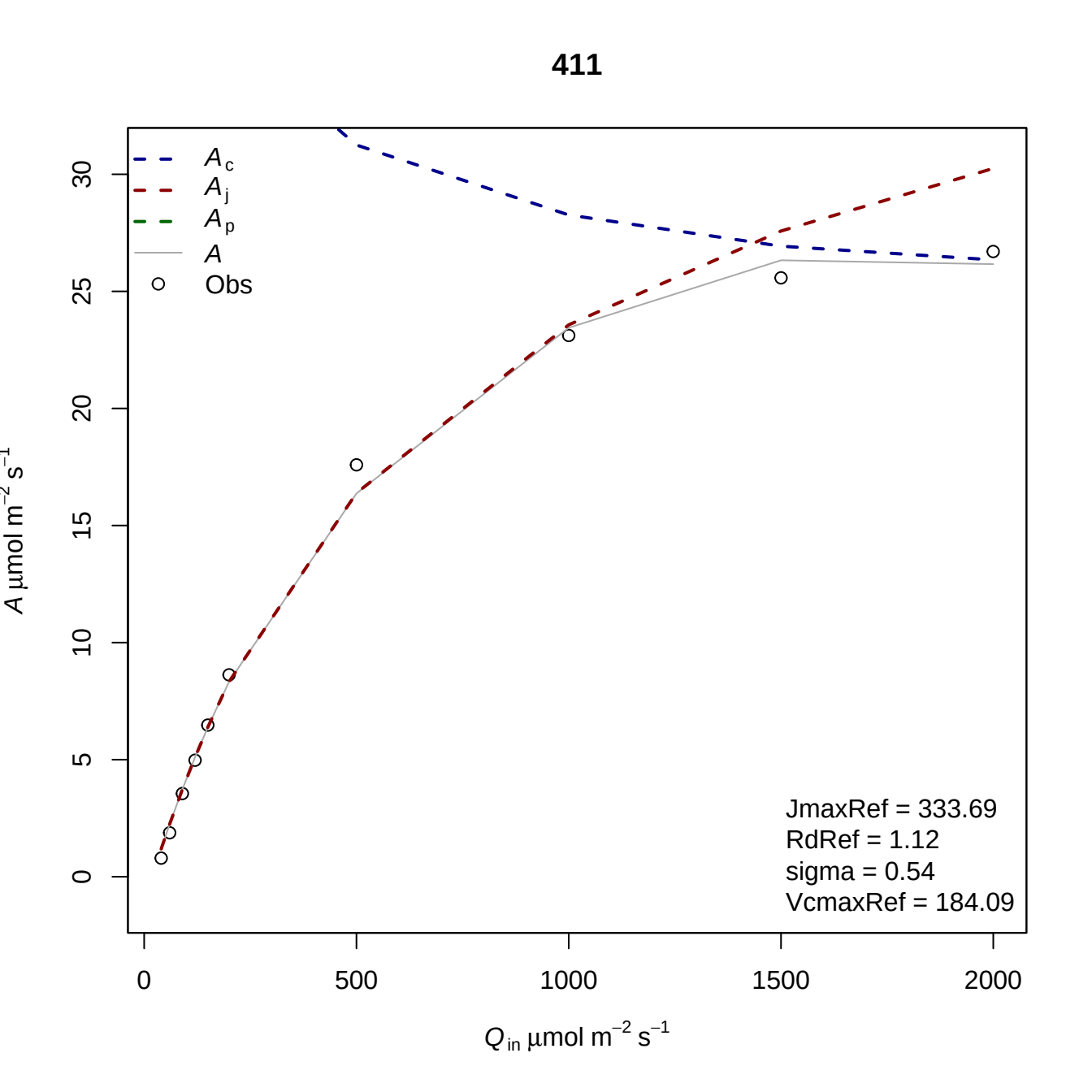

406
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 309.7
RdRef = 2.46
sigma = 0.22
VcmaxRef = 238.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

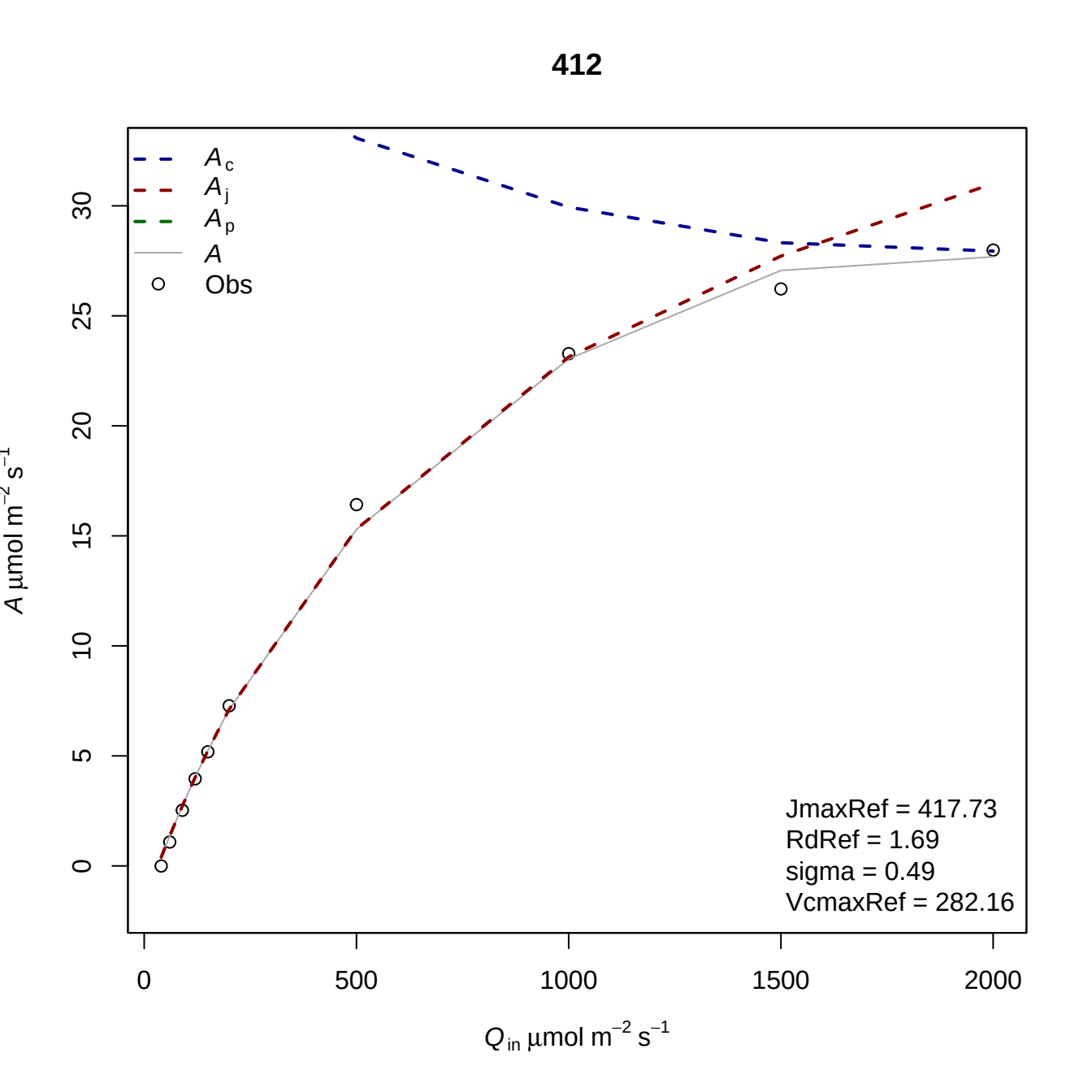

407
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 246.38
RdRef = 3.47
sigma = 0.31
VcmaxRef = 4159896.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

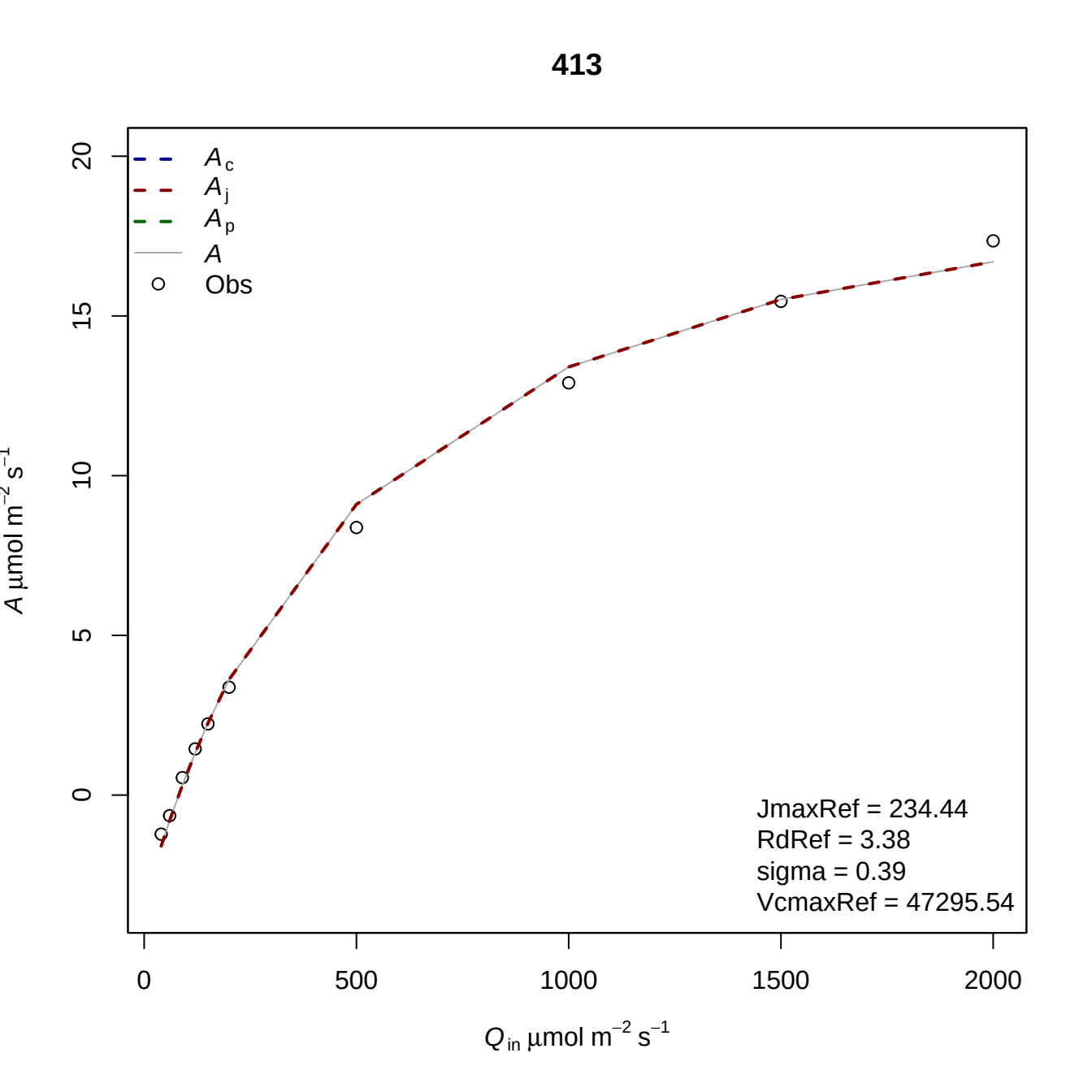

408
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.02
RdRef = 0.95
sigma = 0.21
VcmaxRef = 66.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

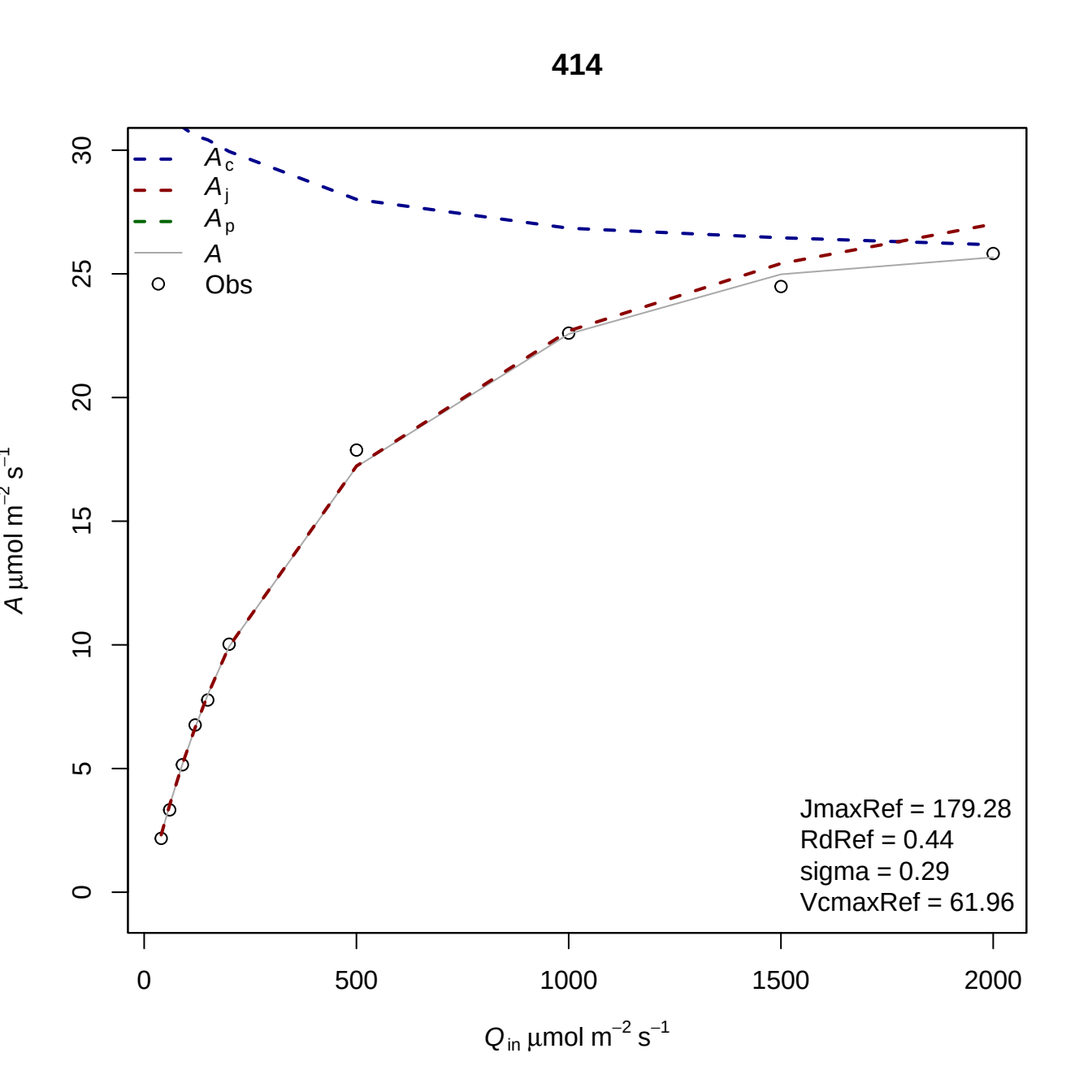

409
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 218.88
RdRef = 0.76
sigma = 0.35
VcmaxRef = 82.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

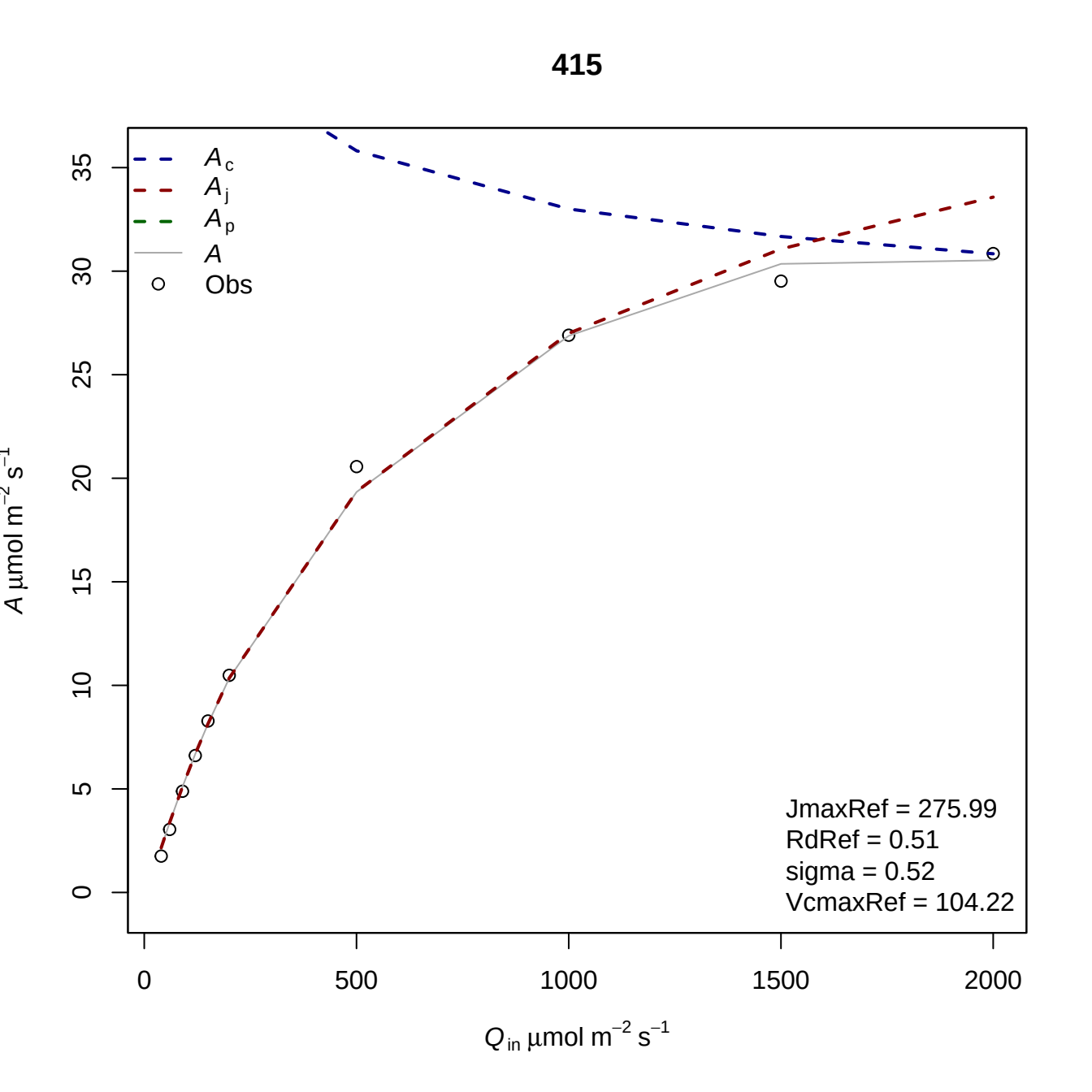

410
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 308.4
RdRef = 0.92
sigma = 0.54
VcmaxRef = 133.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

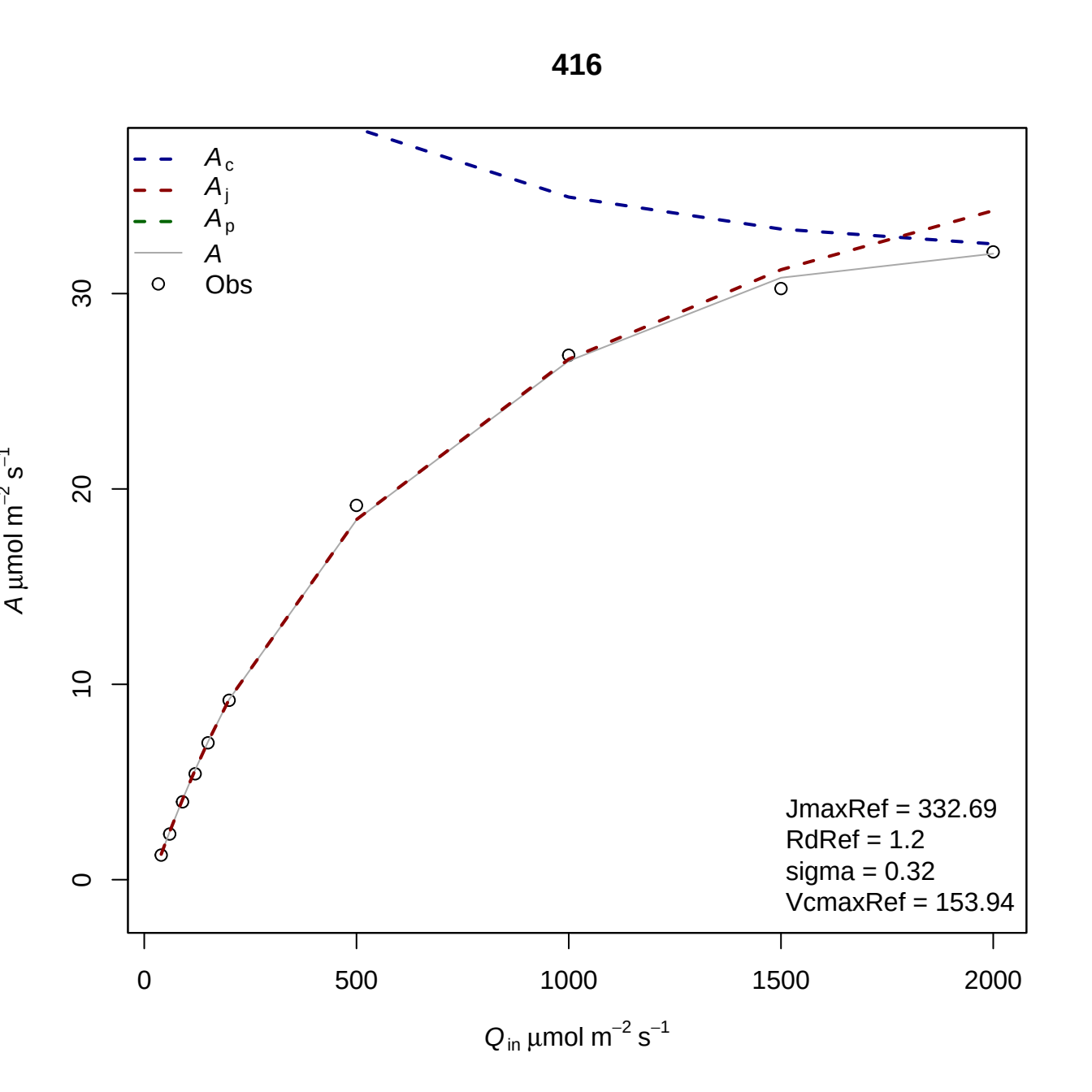

411
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 333.69
RdRef = 1.12
sigma = 0.54
VcmaxRef = 184.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

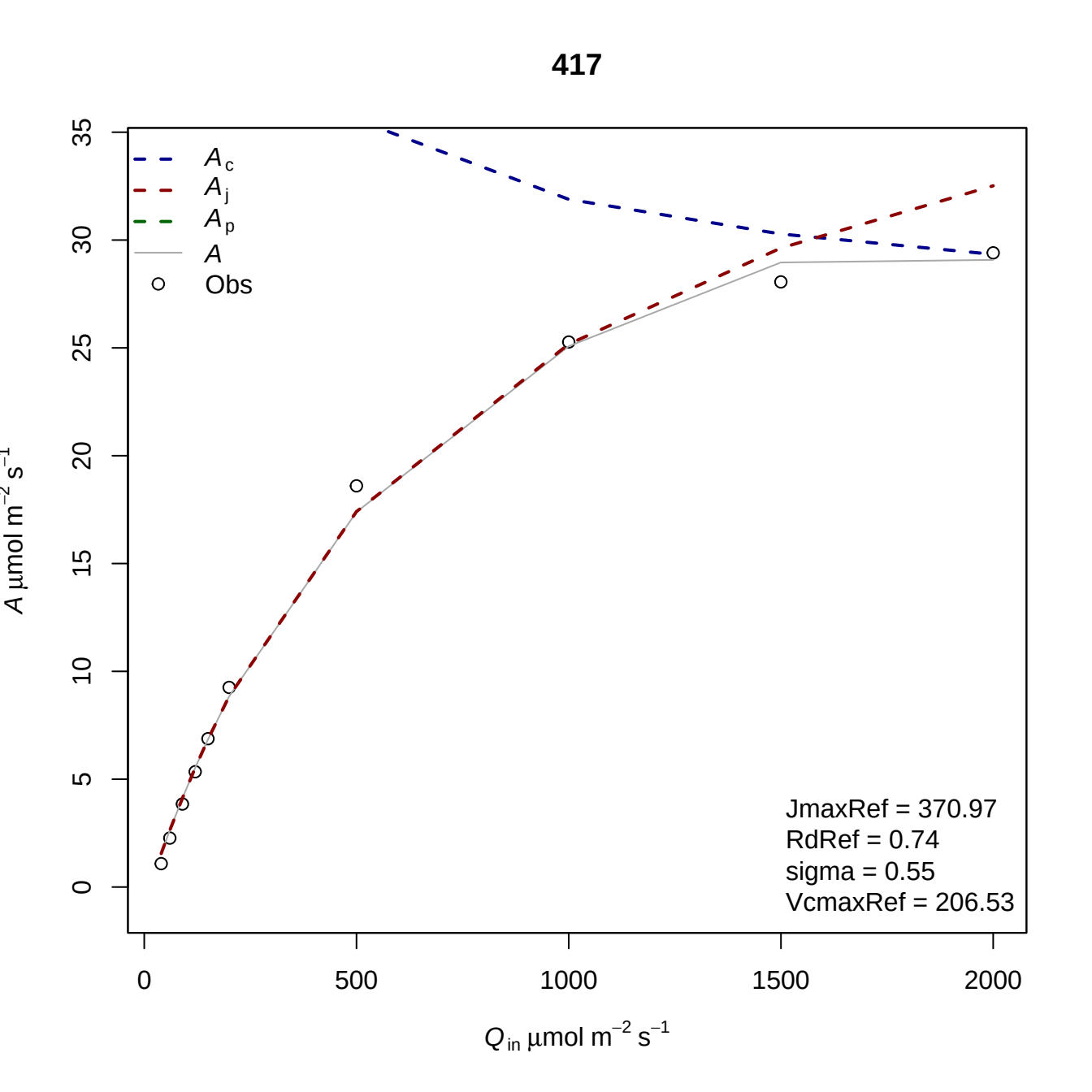

412
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 417.73
RdRef = 1.69
sigma = 0.49
VcmaxRef = 282.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

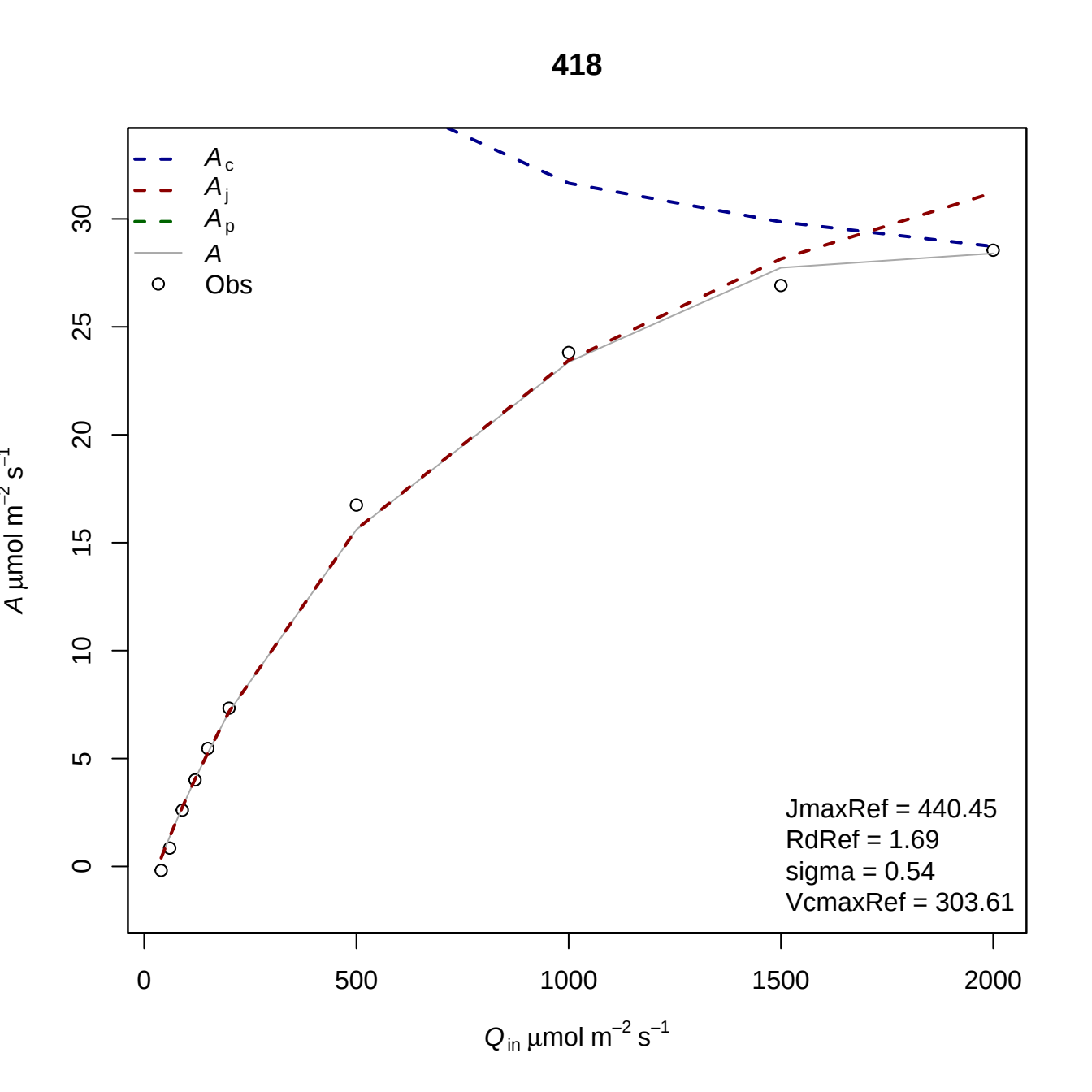

413
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 234.44
RdRef = 3.38
sigma = 0.39
VcmaxRef = 47295.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

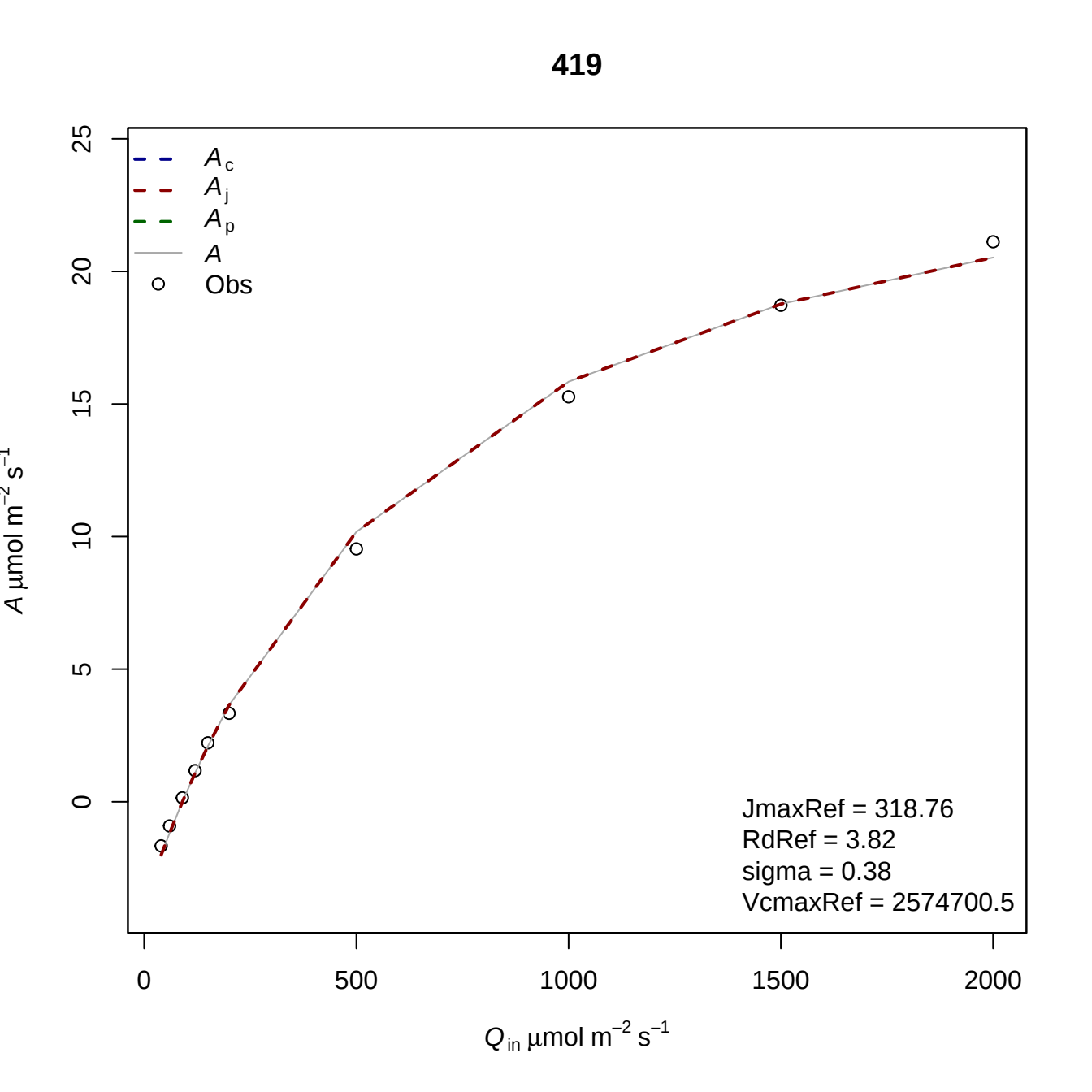

414
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 179.28
RdRef = 0.44
sigma = 0.29
VcmaxRef = 61.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

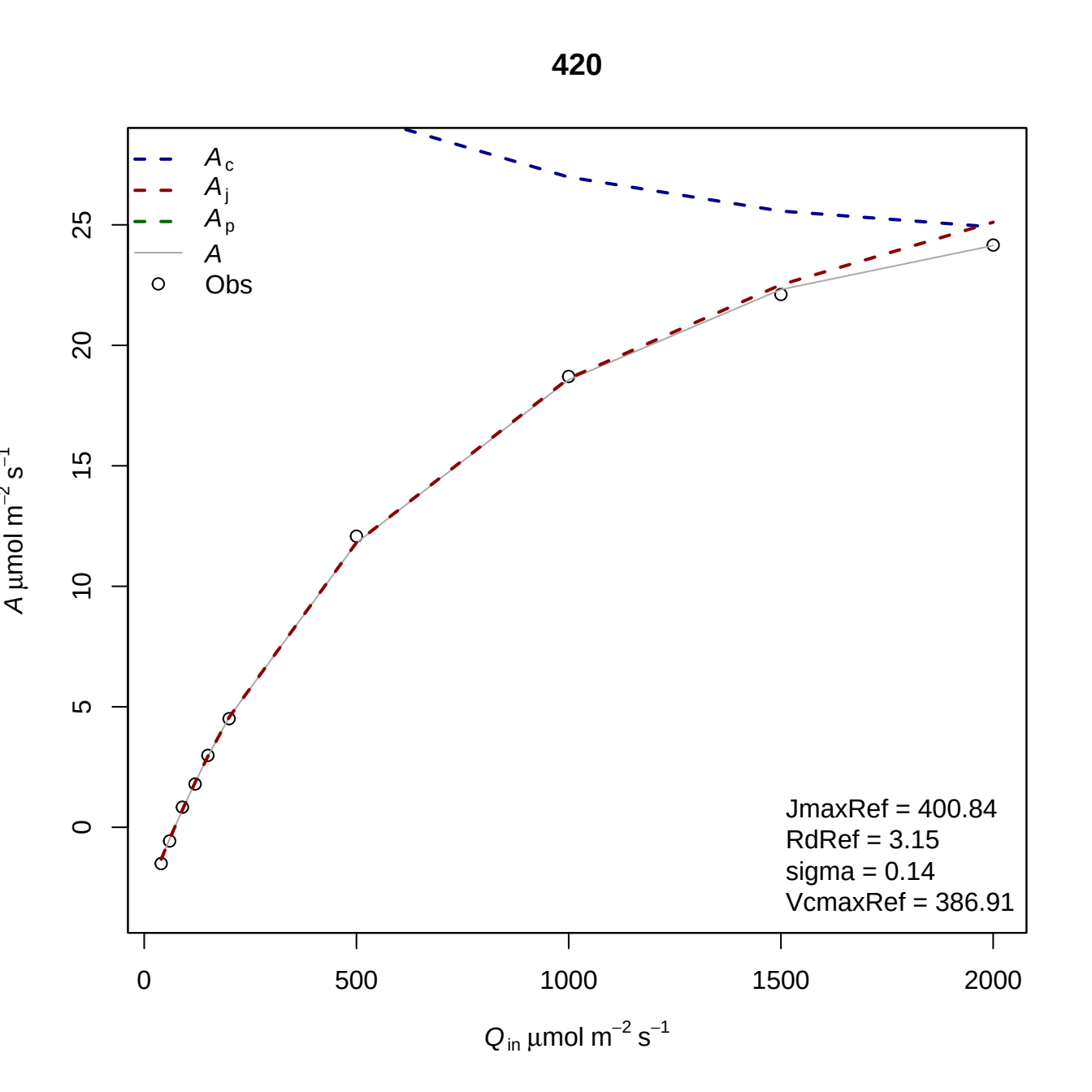

415
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 275.99
RdRef = 0.51
sigma = 0.52
VcmaxRef = 104.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

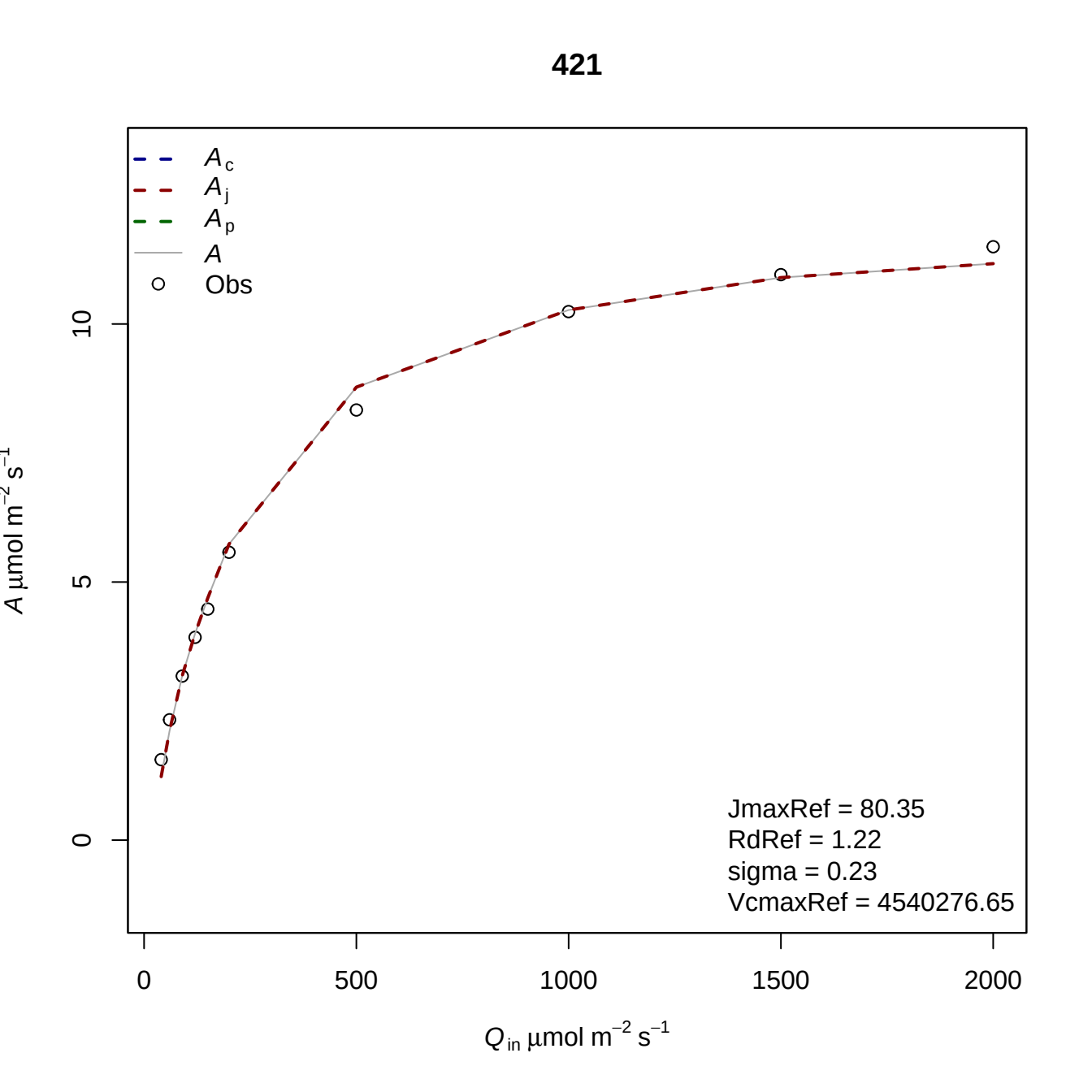

416
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 332.69
RdRef = 1.2
sigma = 0.32
VcmaxRef = 153.94
0
10
20
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

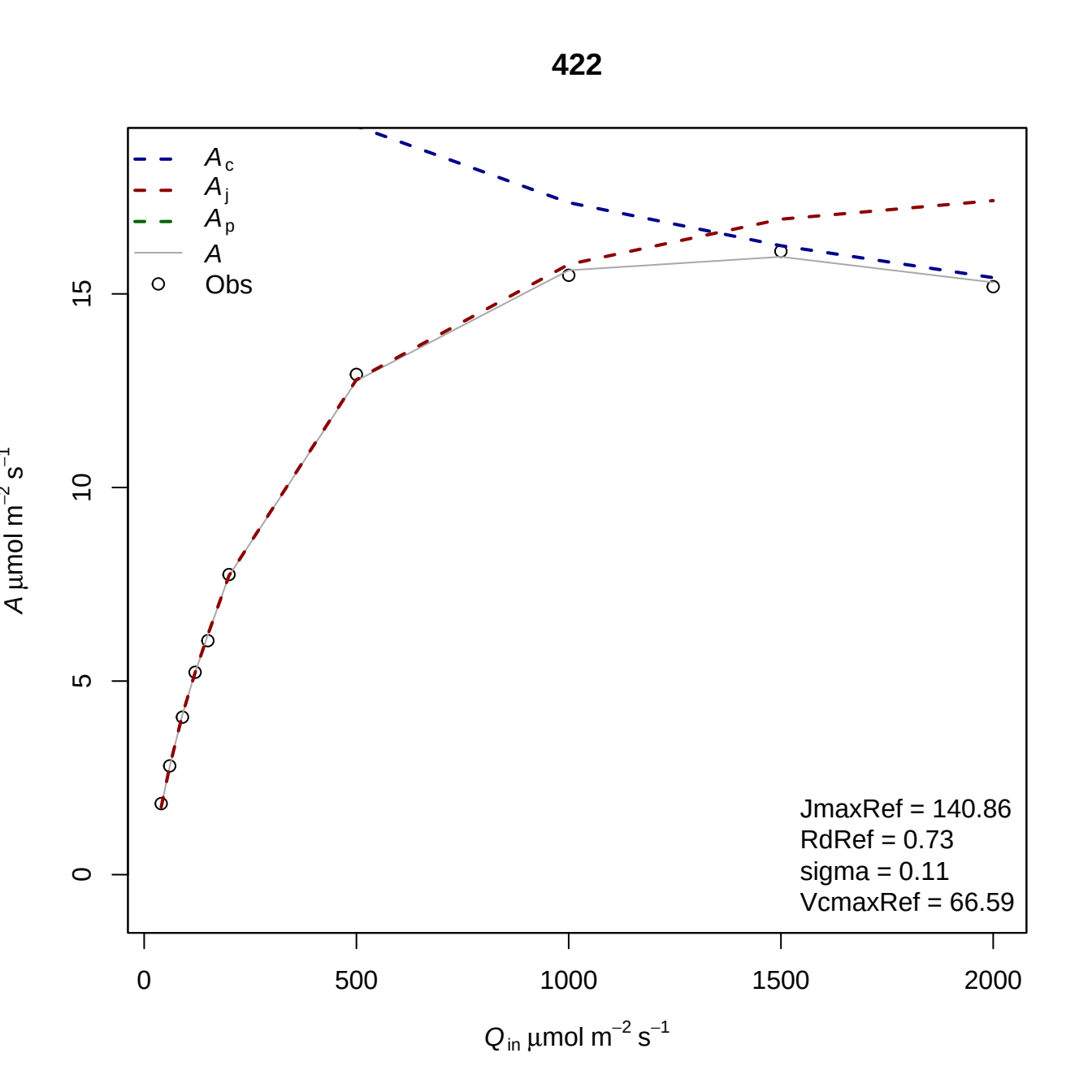

417
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 370.97
RdRef = 0.74
sigma = 0.55
VcmaxRef = 206.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

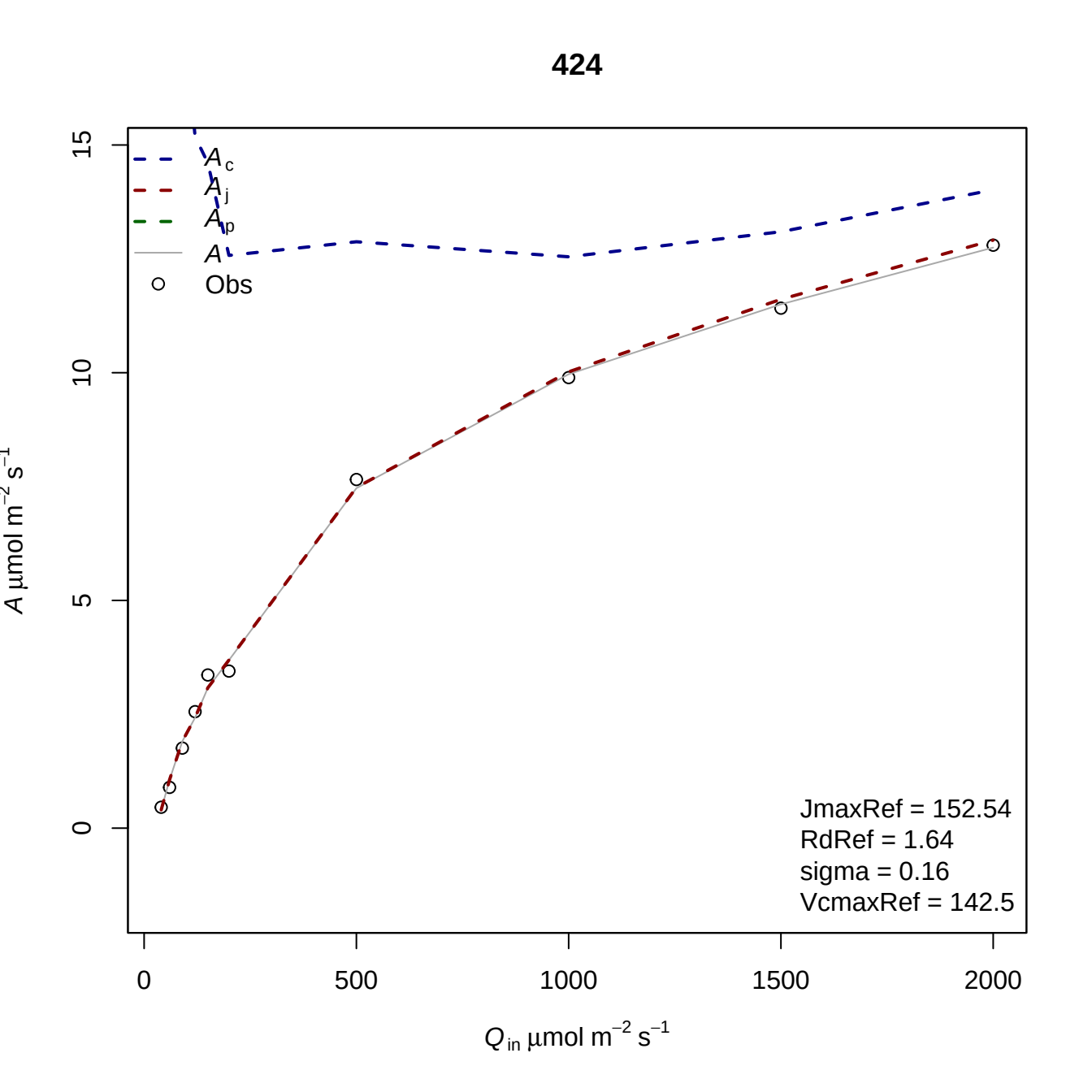

418
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 440.45
RdRef = 1.69
sigma = 0.54
VcmaxRef = 303.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

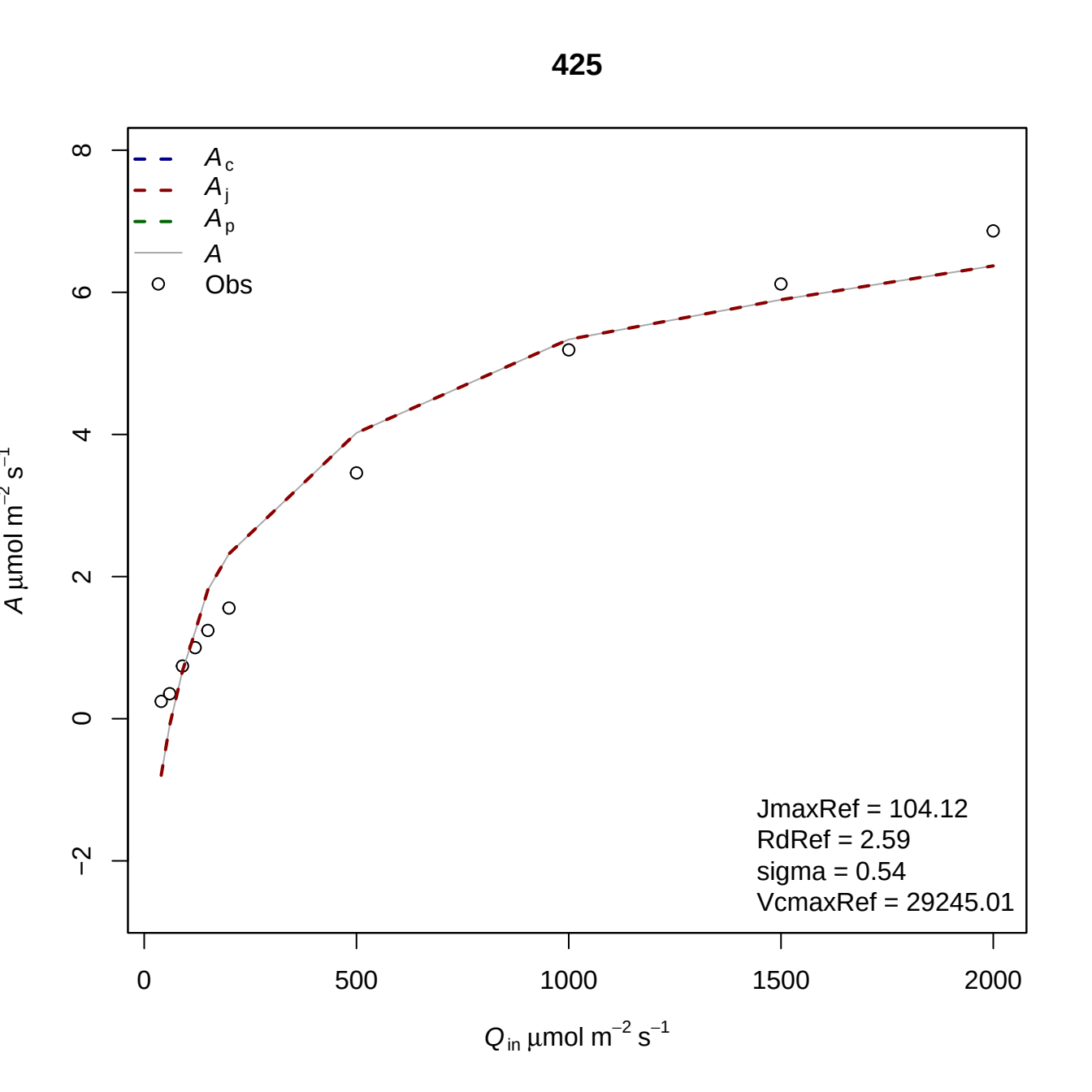

419
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 318.76
RdRef = 3.82
sigma = 0.38
VcmaxRef = 2574700.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

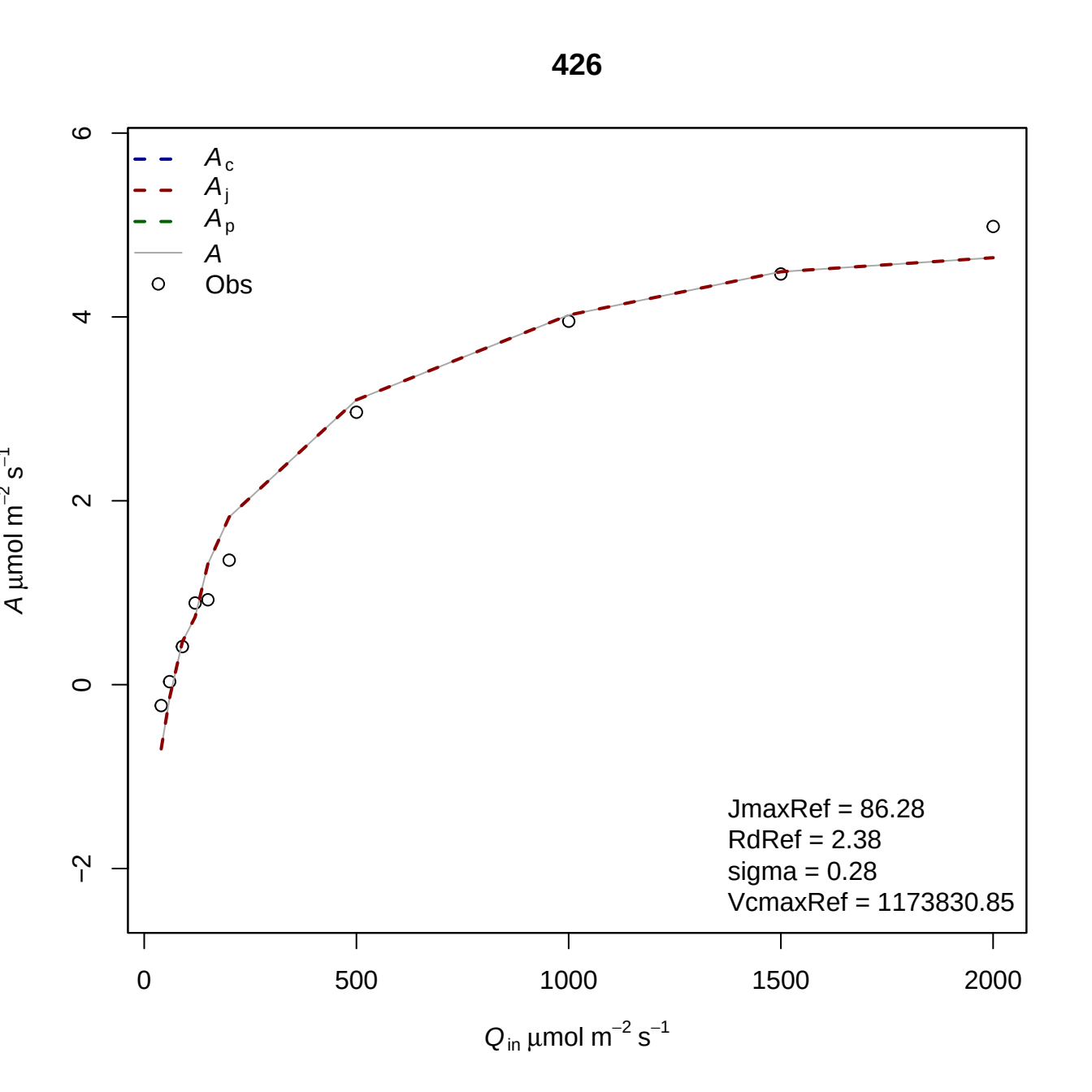

420
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 400.84
RdRef = 3.15
sigma = 0.14
VcmaxRef = 386.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

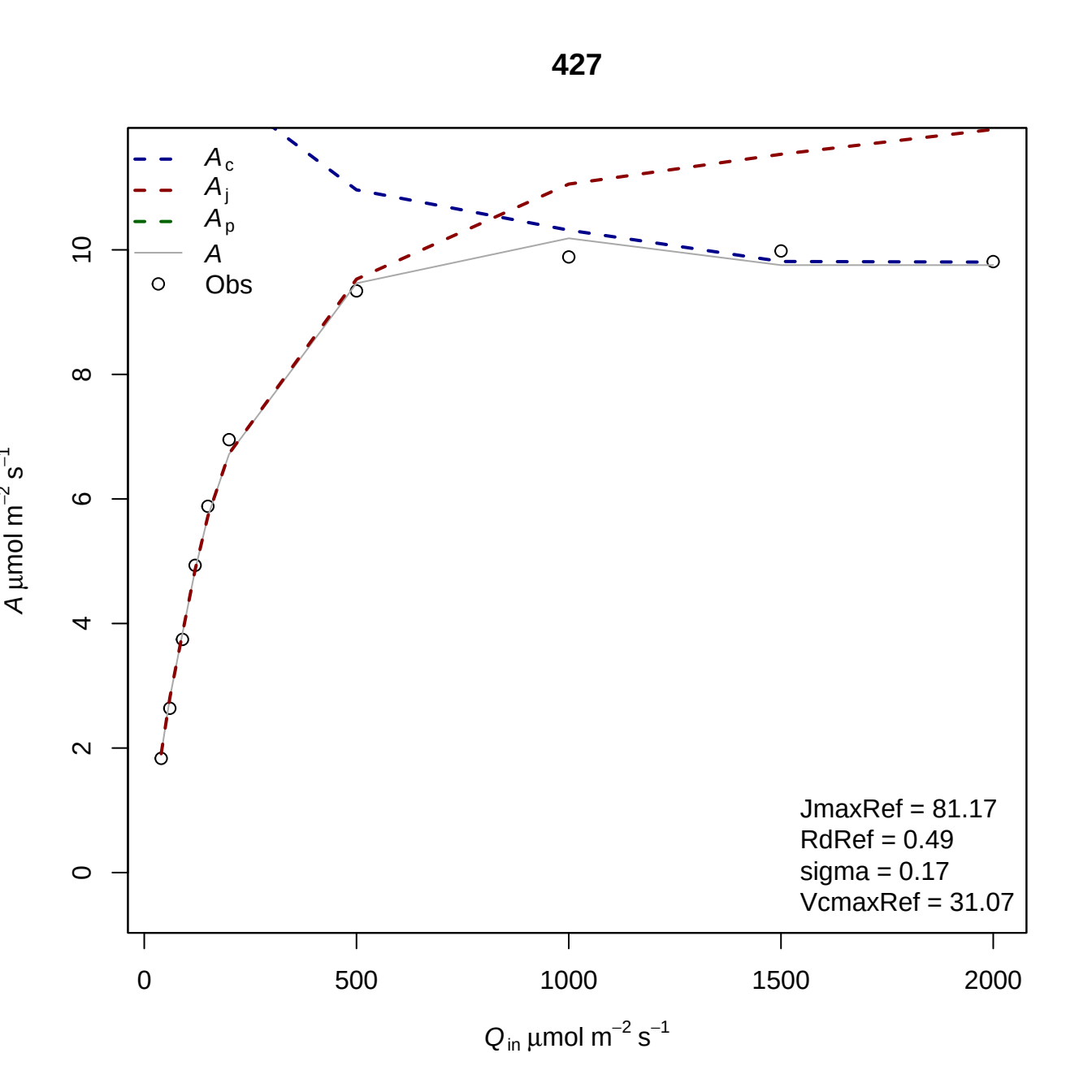

421
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.35
RdRef = 1.22
sigma = 0.23
VcmaxRef = 4540276.65
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

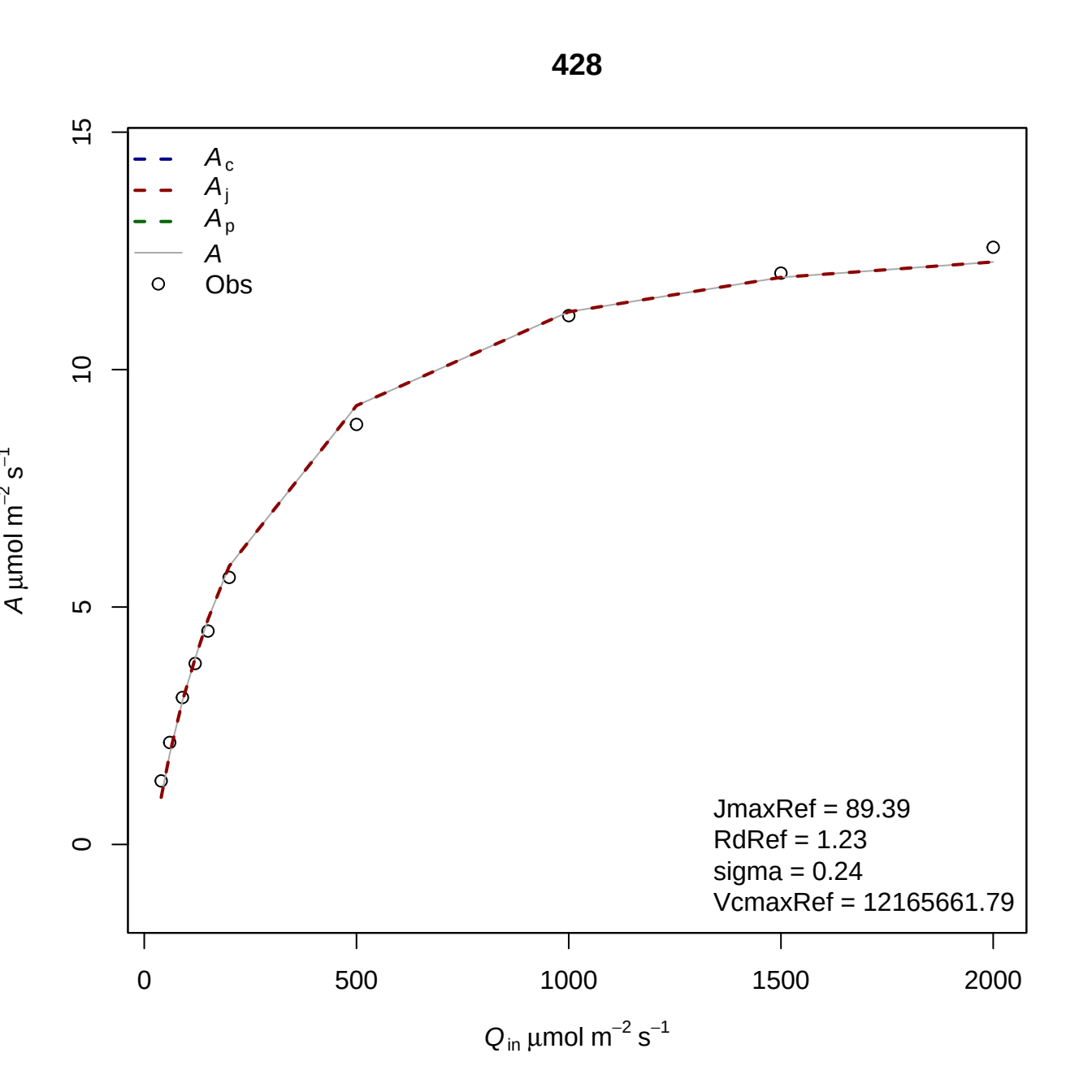

422
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.86
RdRef = 0.73
sigma = 0.11
VcmaxRef = 66.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

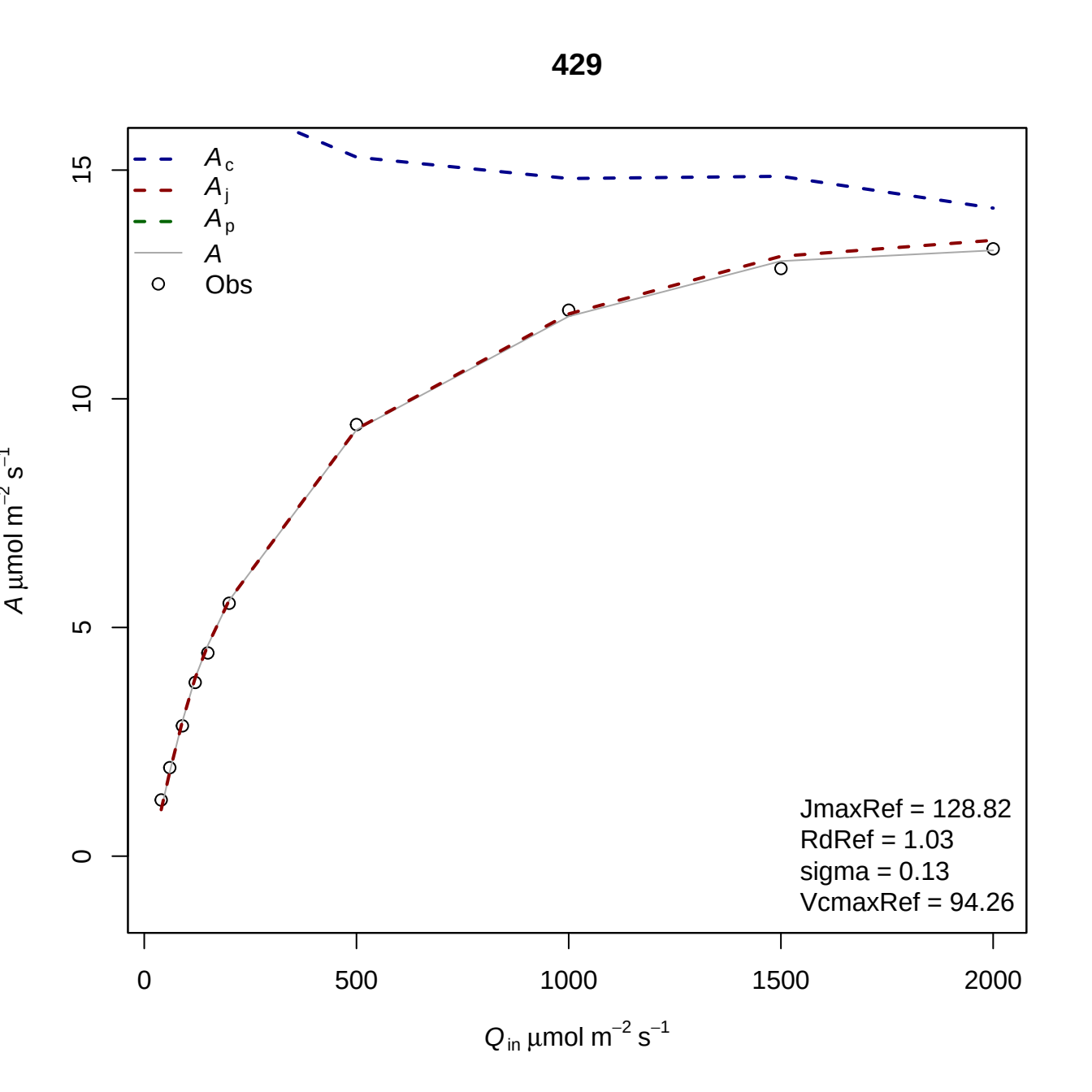

424
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.54
RdRef = 1.64
sigma = 0.16
VcmaxRef = 142.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

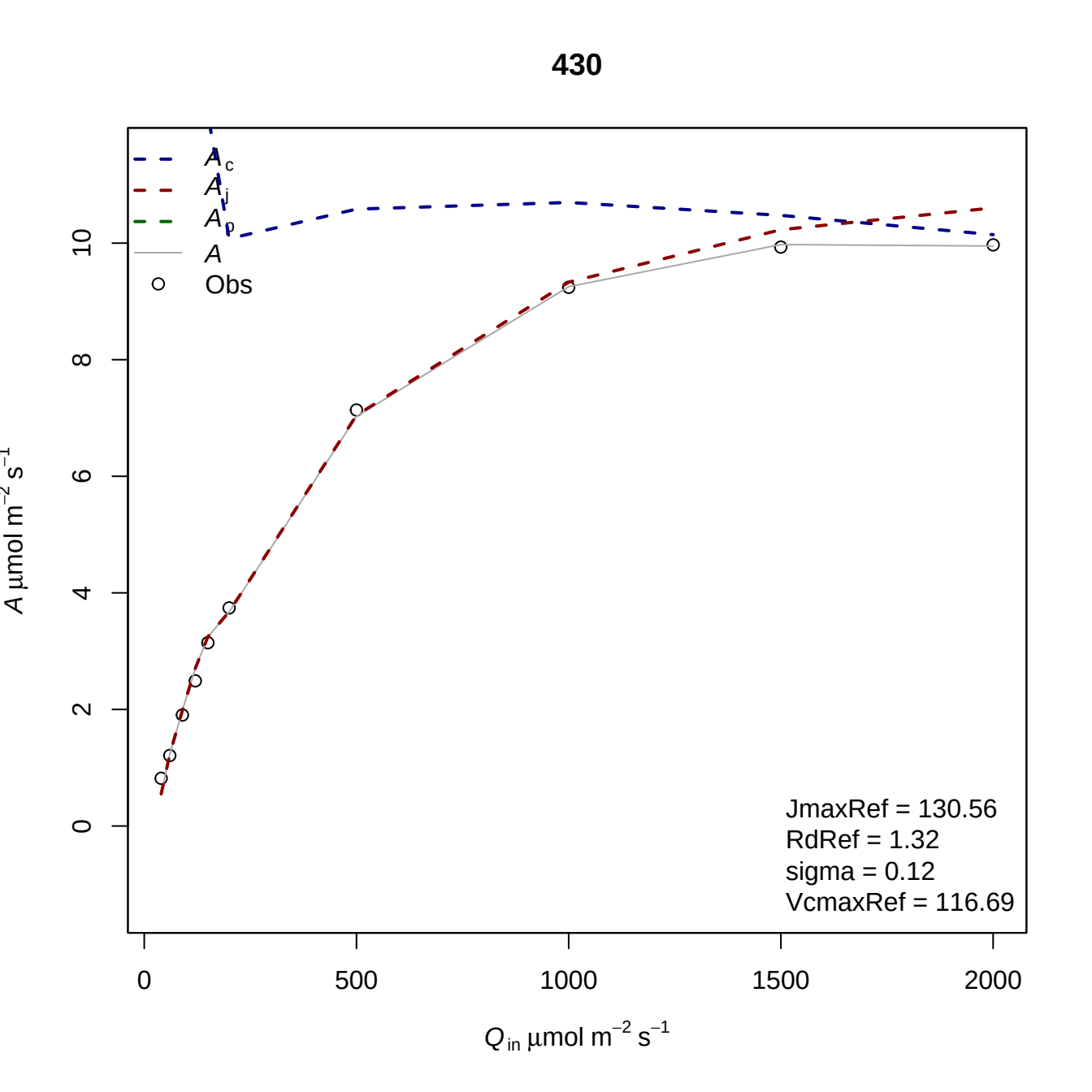

425
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.12
RdRef = 2.59
sigma = 0.54
VcmaxRef = 29245.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

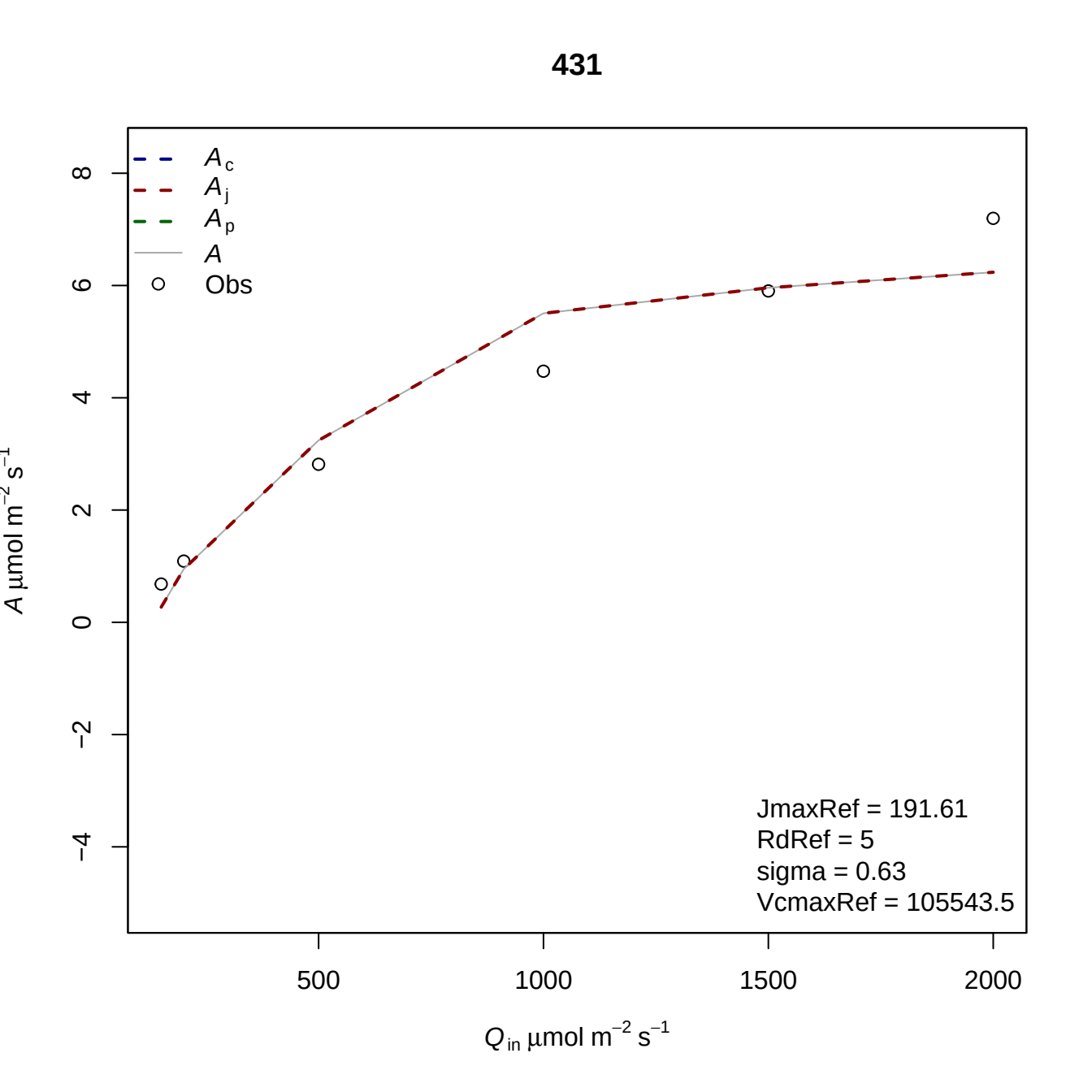

426
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.28
RdRef = 2.38
sigma = 0.28
VcmaxRef = 1173830.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500
2000

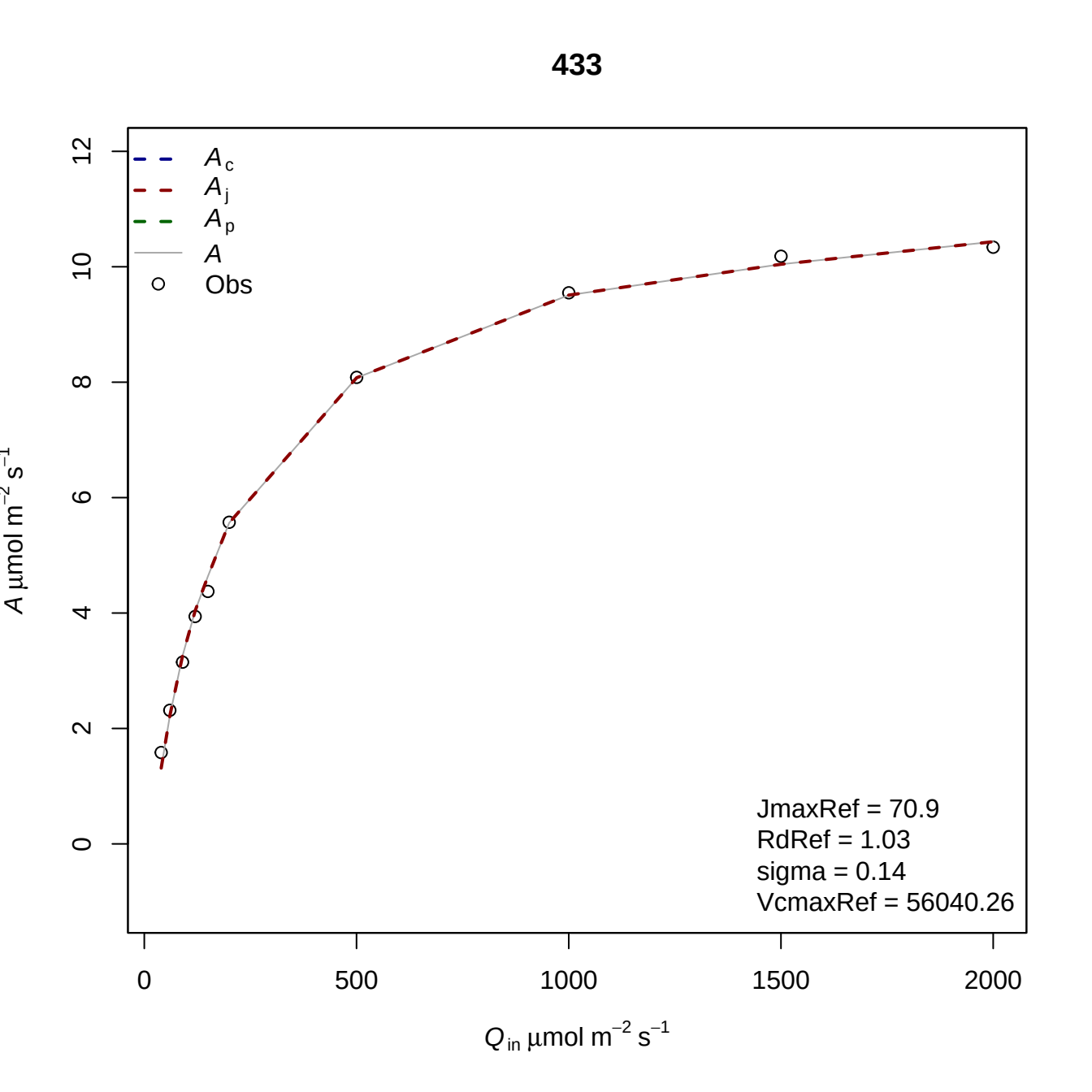

427
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.17
RdRef = 0.49
sigma = 0.17
VcmaxRef = 31.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

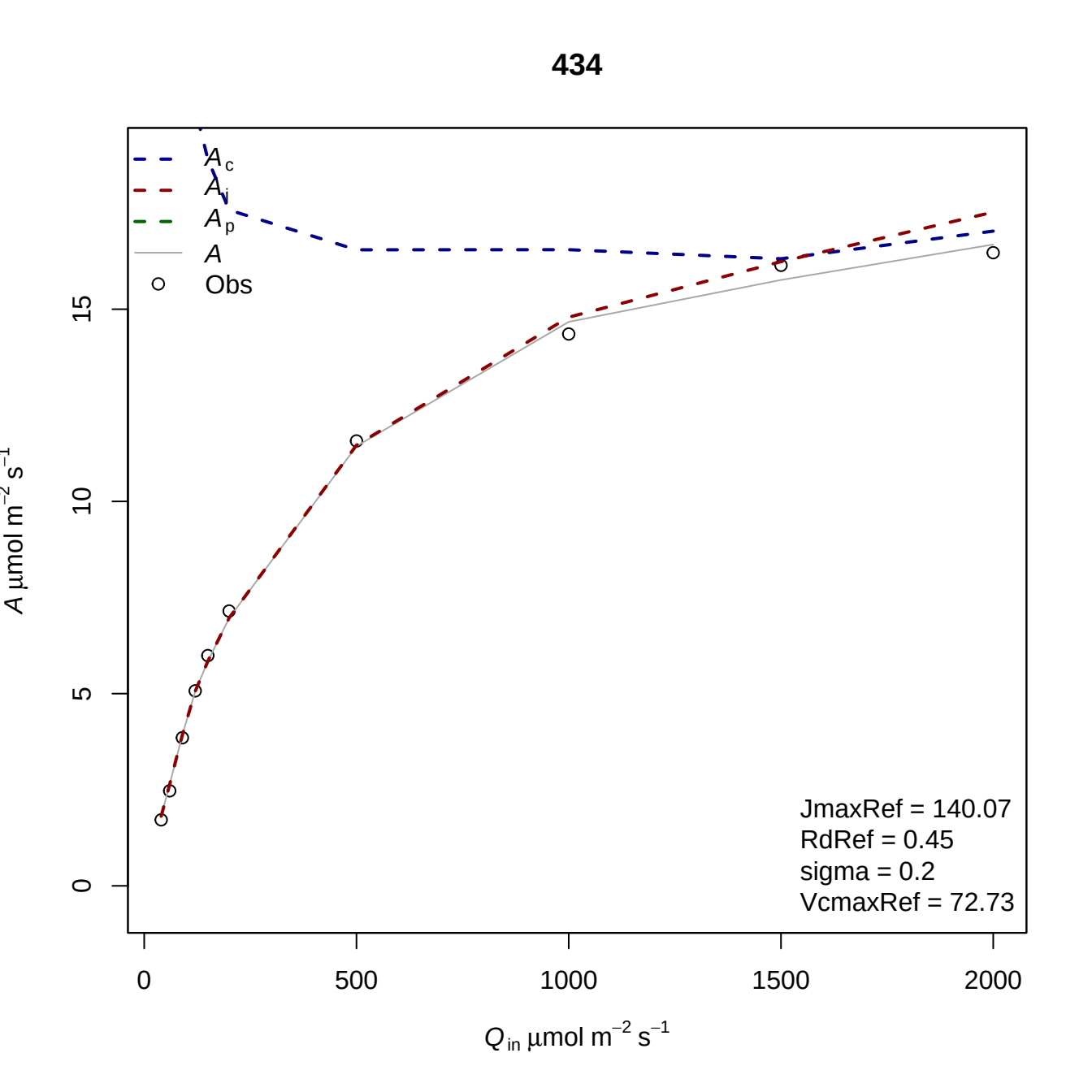

428
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.39
RdRef = 1.23
sigma = 0.24
VcmaxRef = 12165661.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

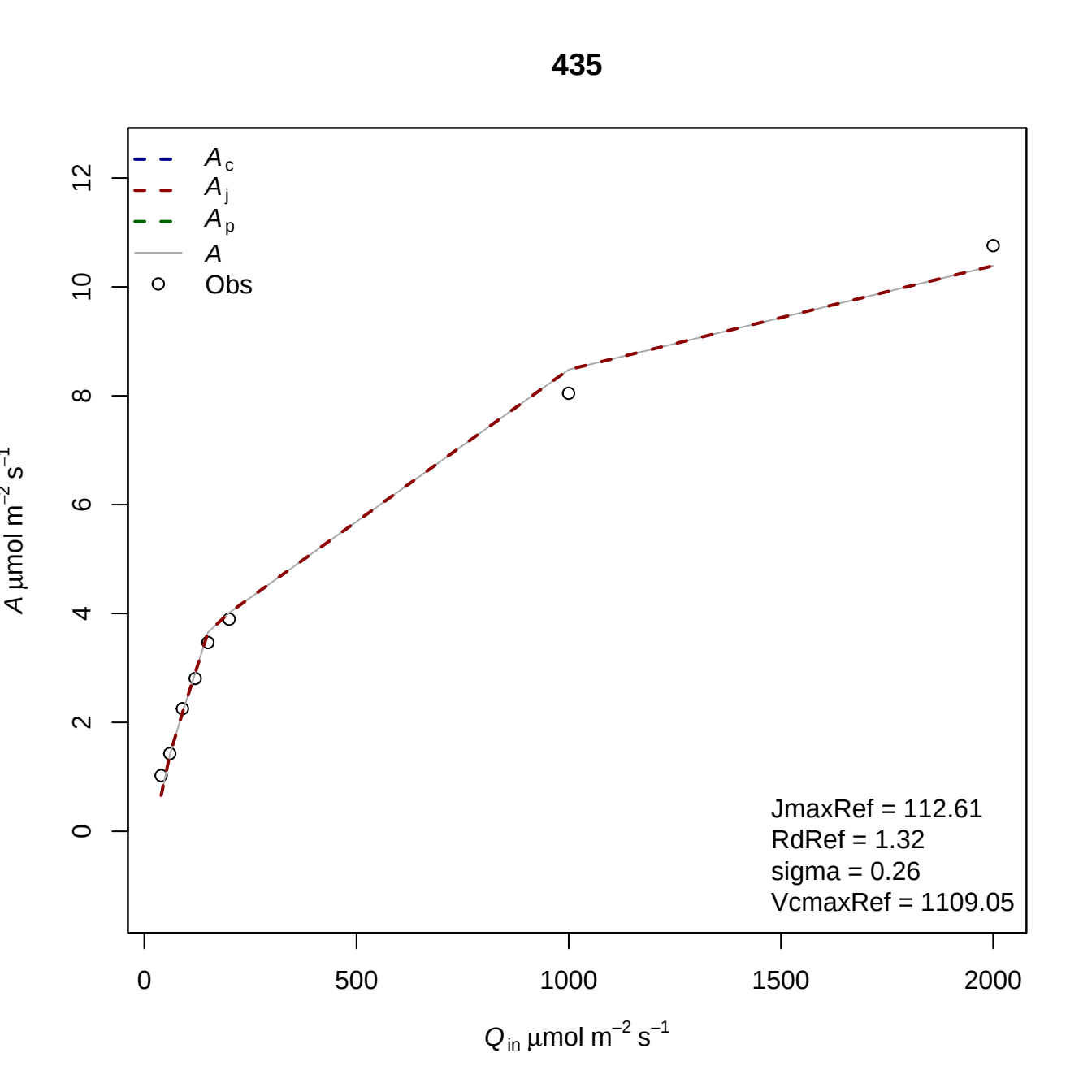

429
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.82
RdRef = 1.03
sigma = 0.13
VcmaxRef = 94.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

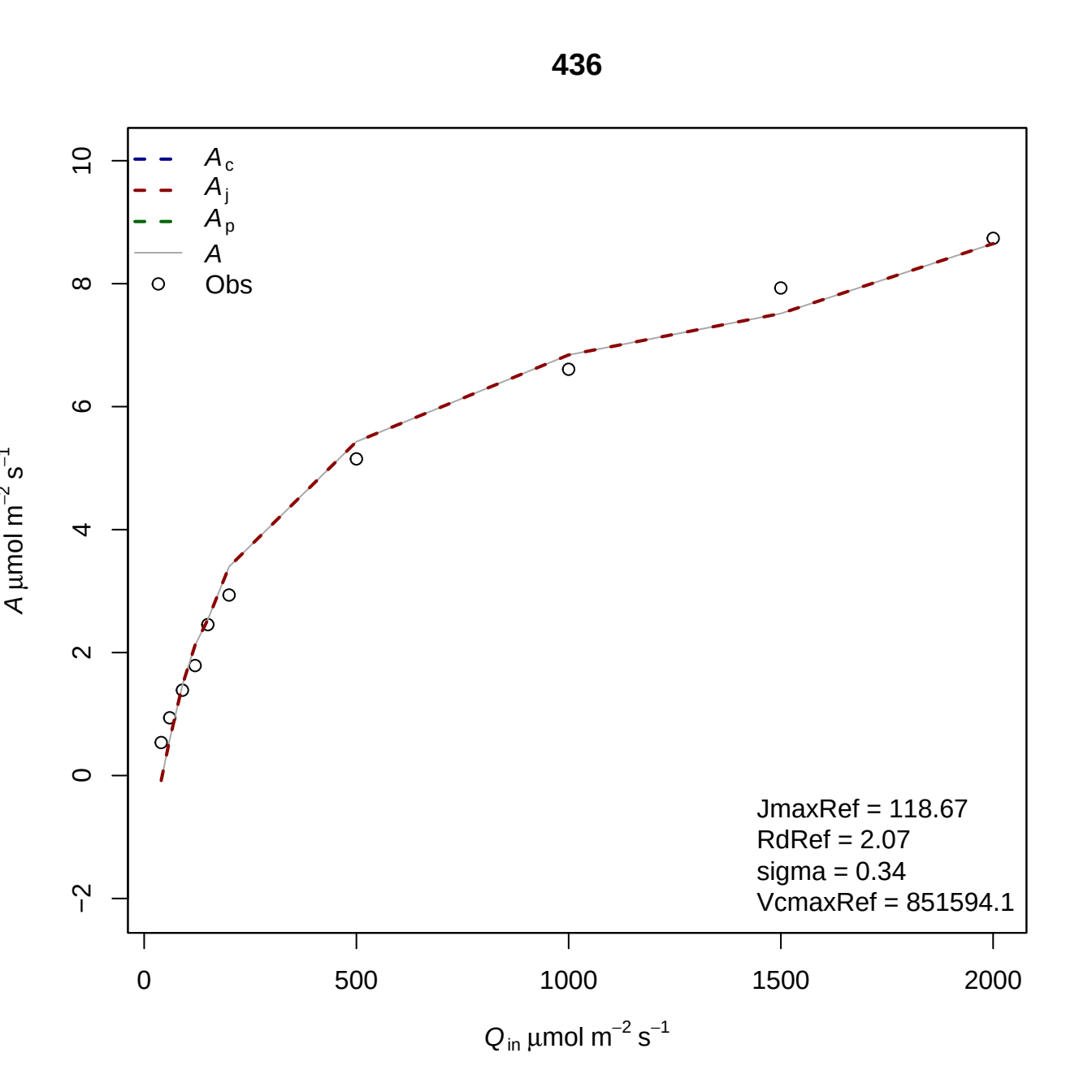

430
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.56
RdRef = 1.32
sigma = 0.12
VcmaxRef = 116.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

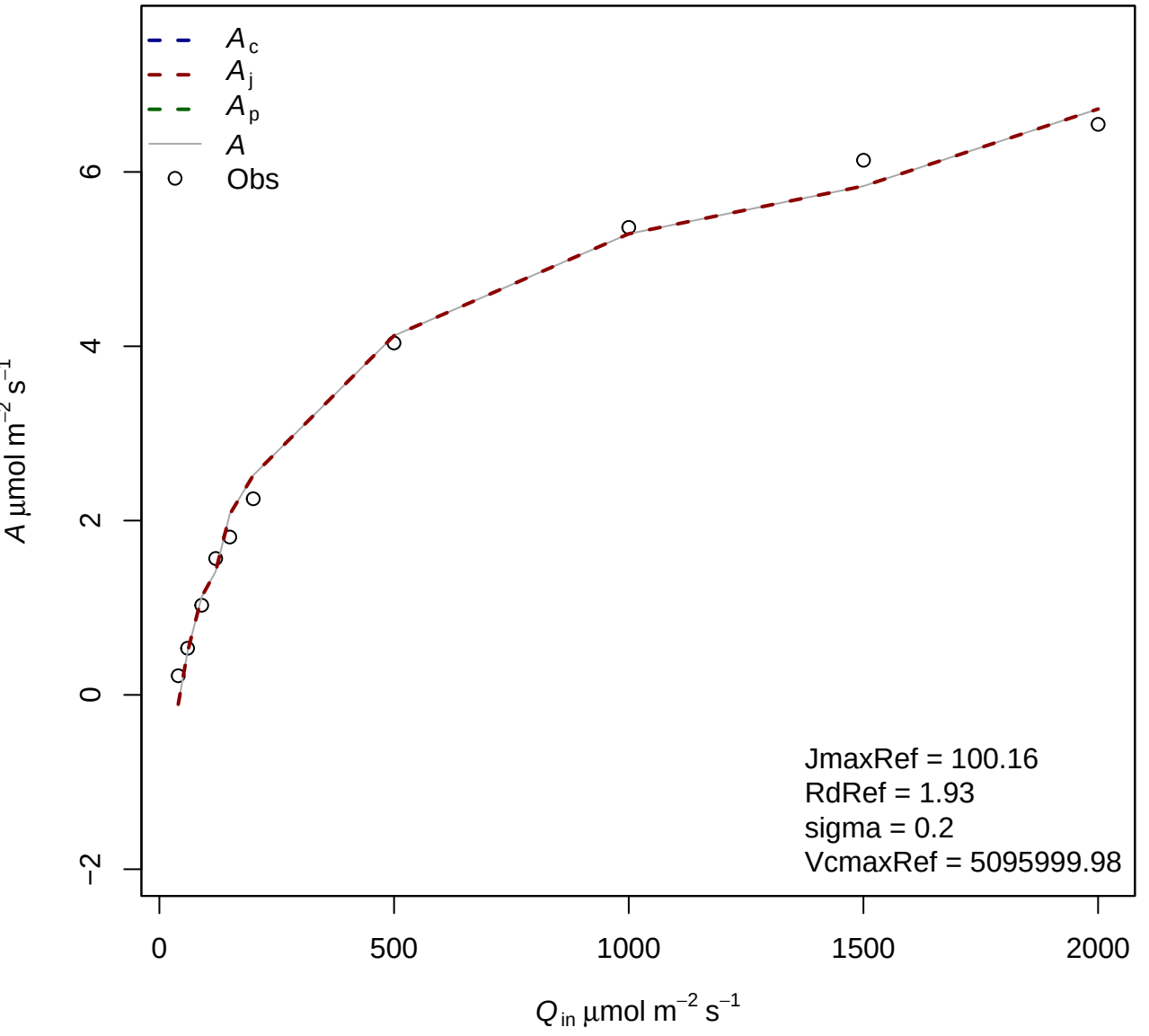

431
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 191.61
RdRef = 5
sigma = 0.63
VcmaxRef = 105543.5
8
6
4
2
0
−2
−4
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

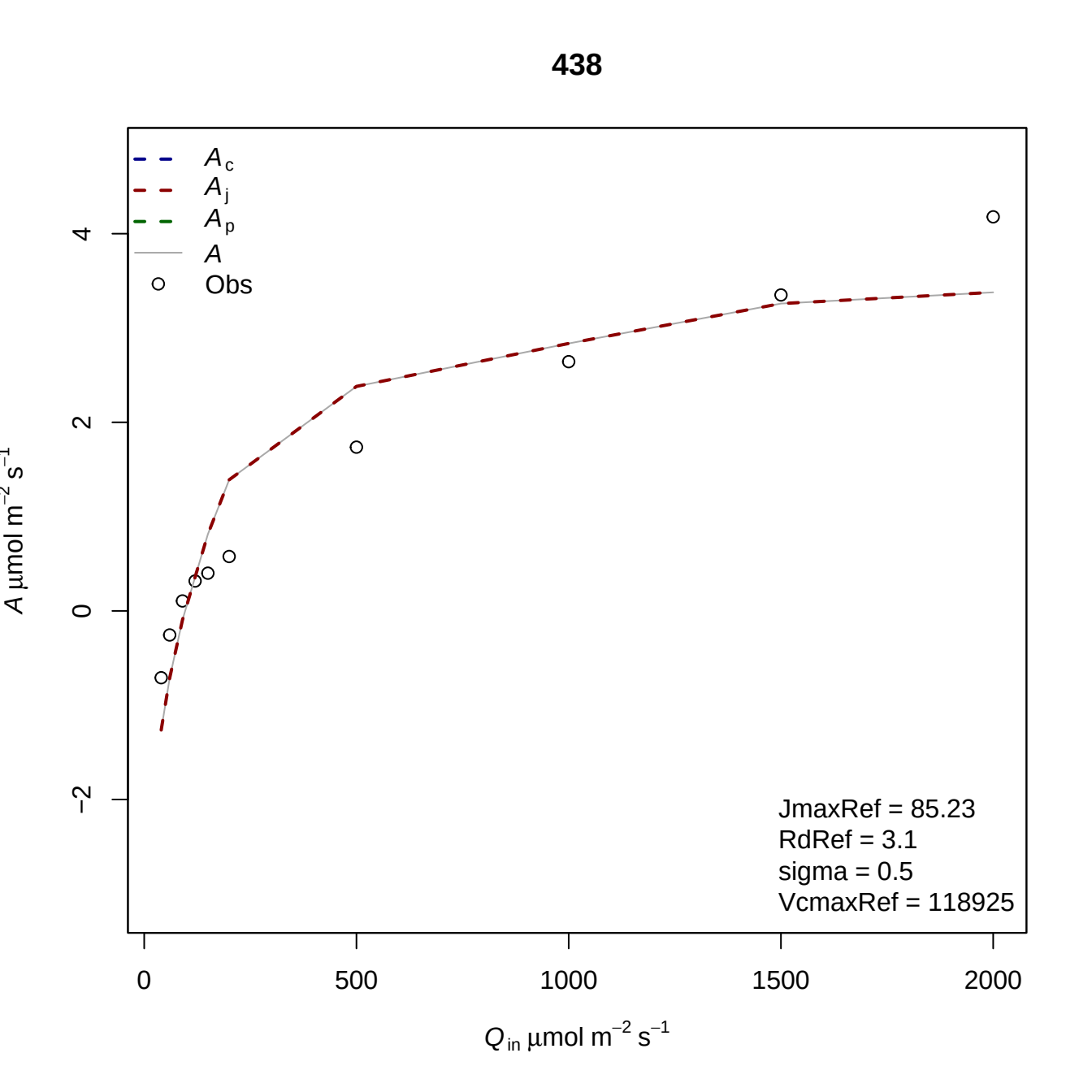

433
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.9
RdRef = 1.03
sigma = 0.14
VcmaxRef = 56040.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

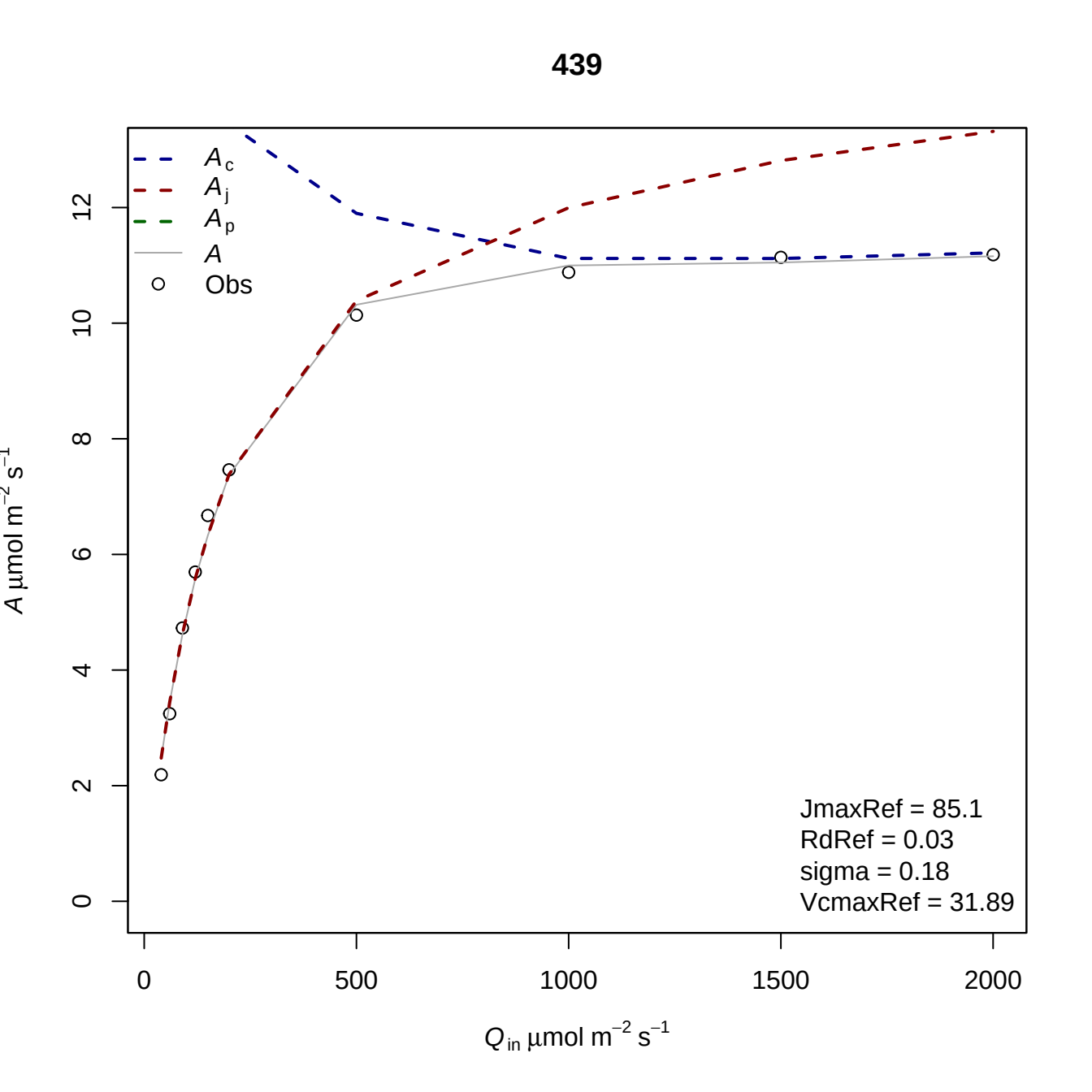

434
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.07
RdRef = 0.45
sigma = 0.2
VcmaxRef = 72.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

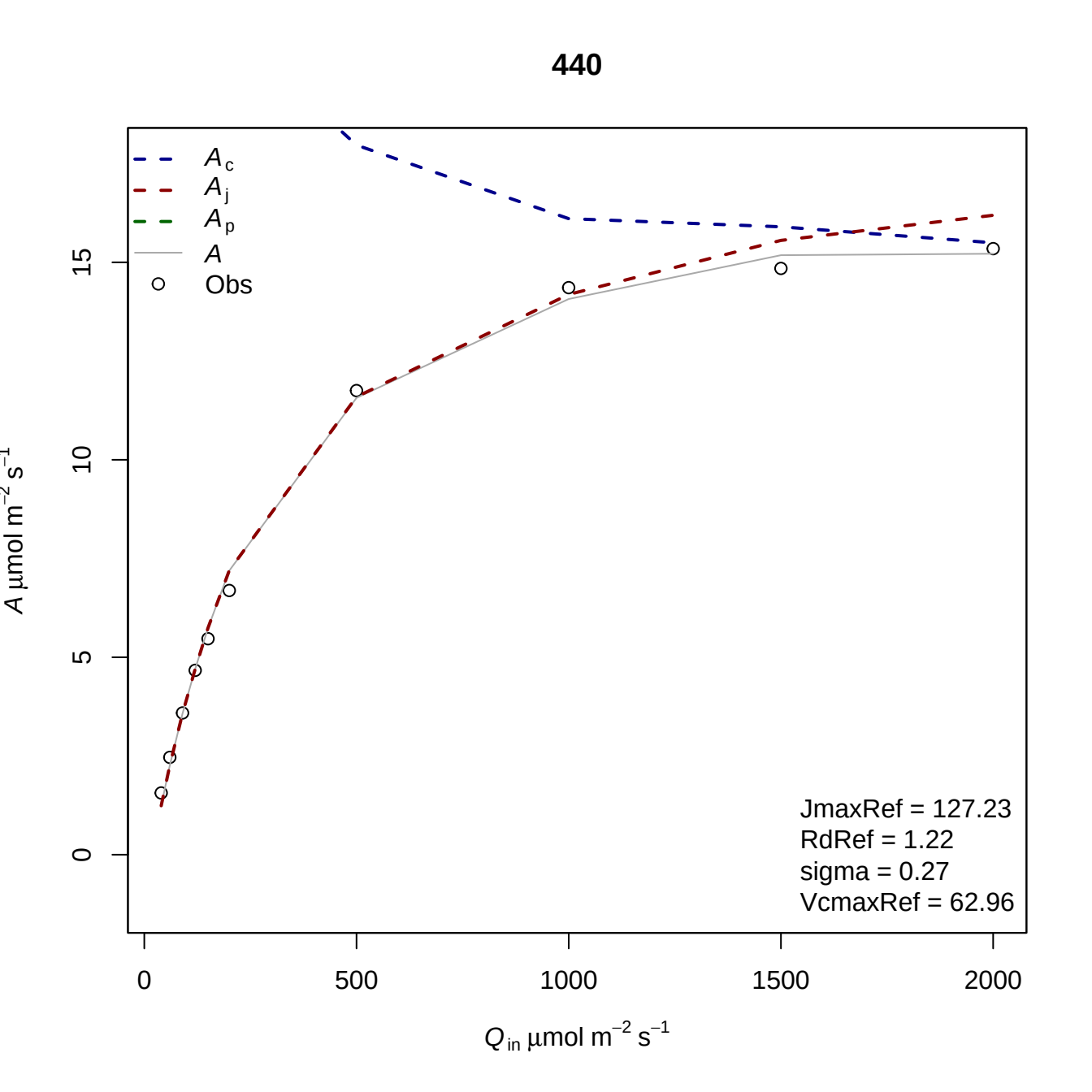

435
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.61
RdRef = 1.32
sigma = 0.26
VcmaxRef = 1109.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

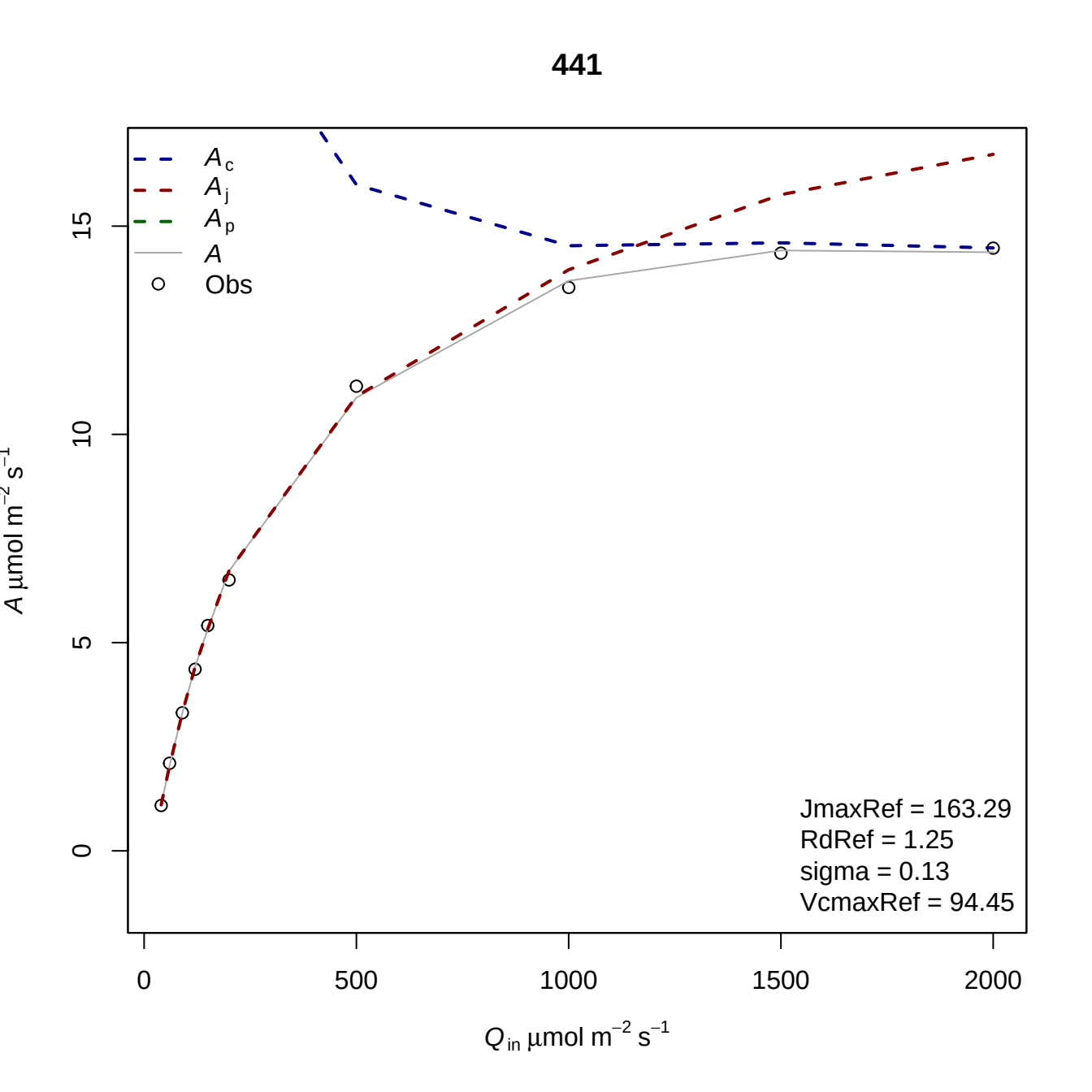

436
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.67
RdRef = 2.07
sigma = 0.34
VcmaxRef = 851594.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

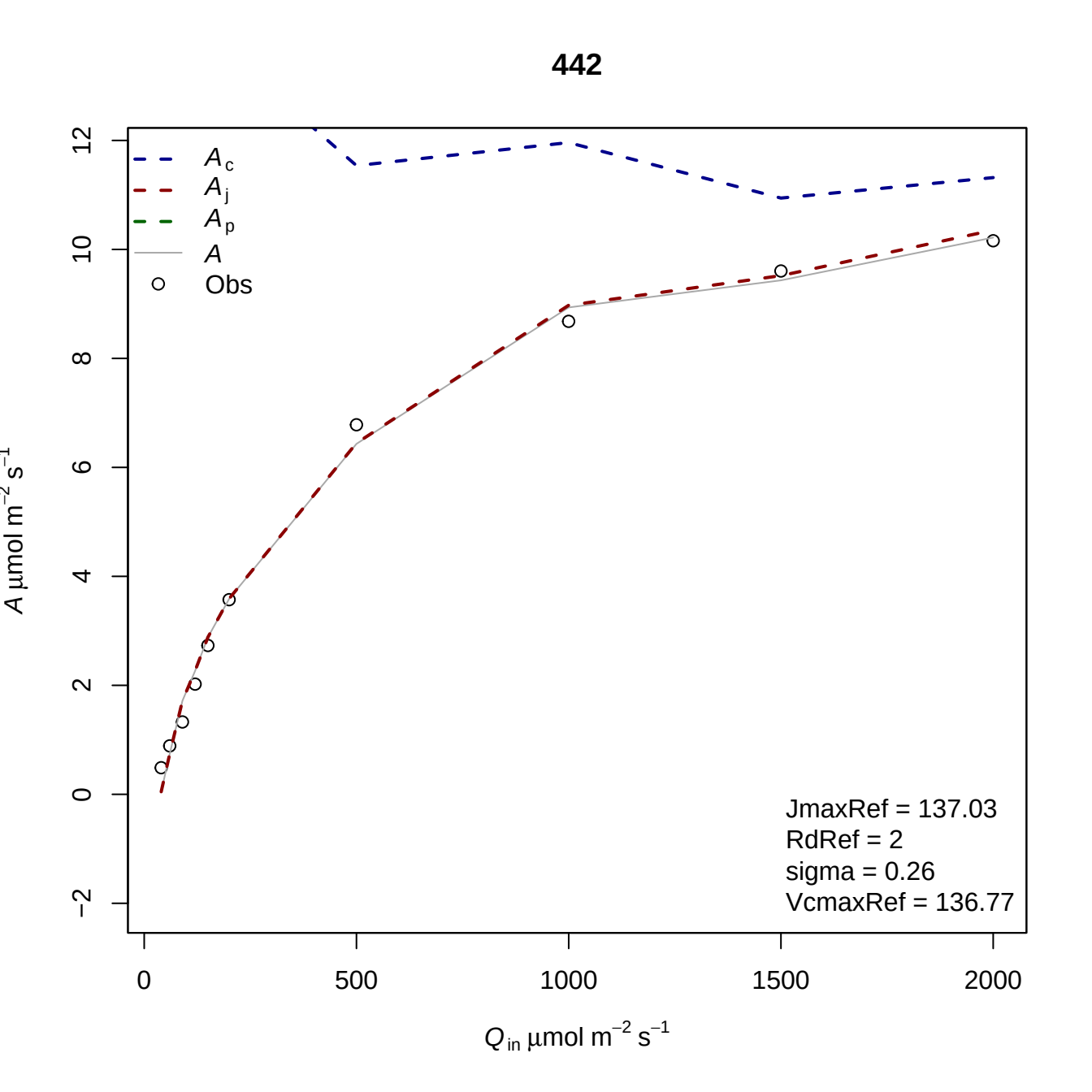

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.16
RdRef = 1.93
sigma = 0.2
VcmaxRef = 5095999.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

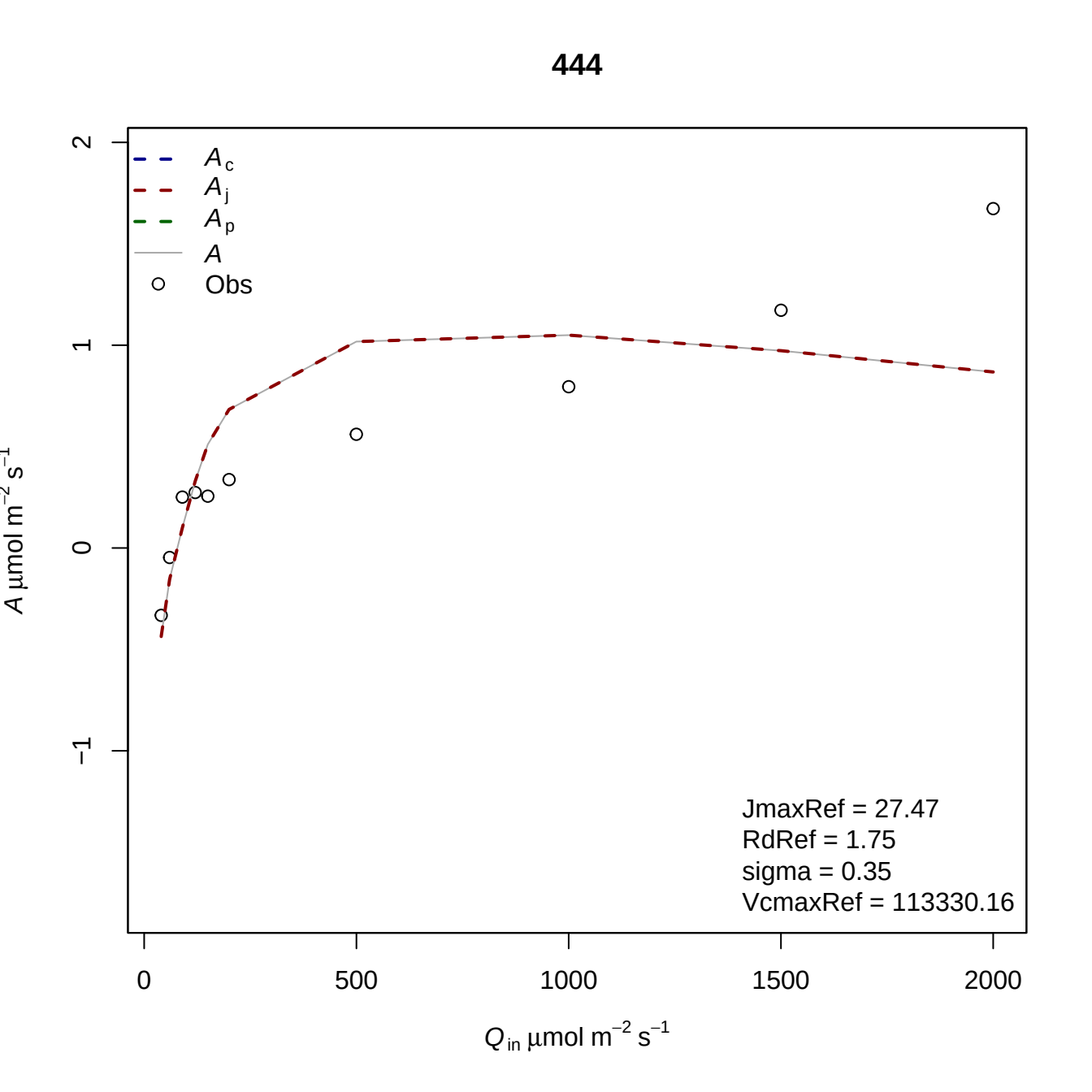

438
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.23
RdRef = 3.1
sigma = 0.5
VcmaxRef = 118925
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

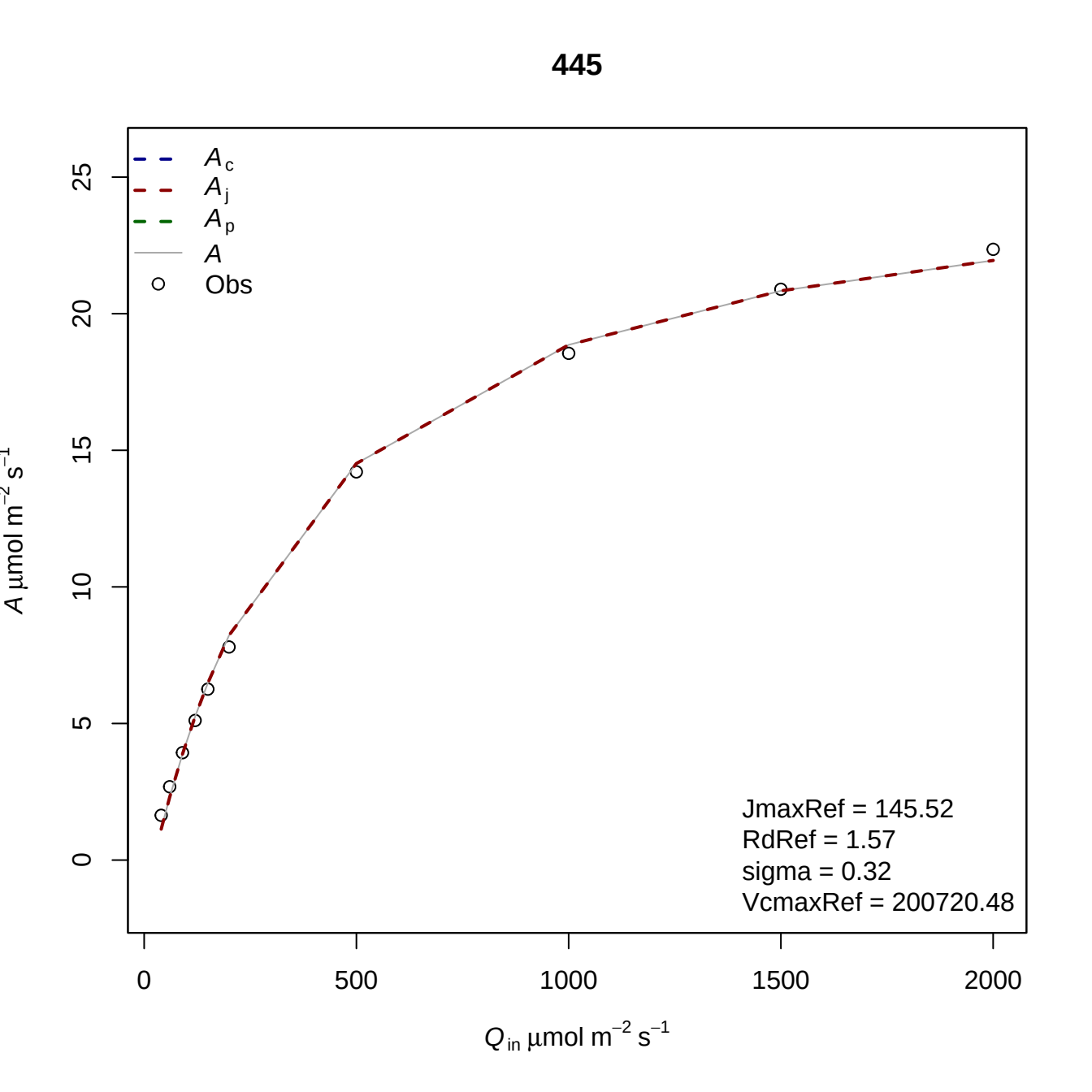

439
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.1
RdRef = 0.03
sigma = 0.18
VcmaxRef = 31.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

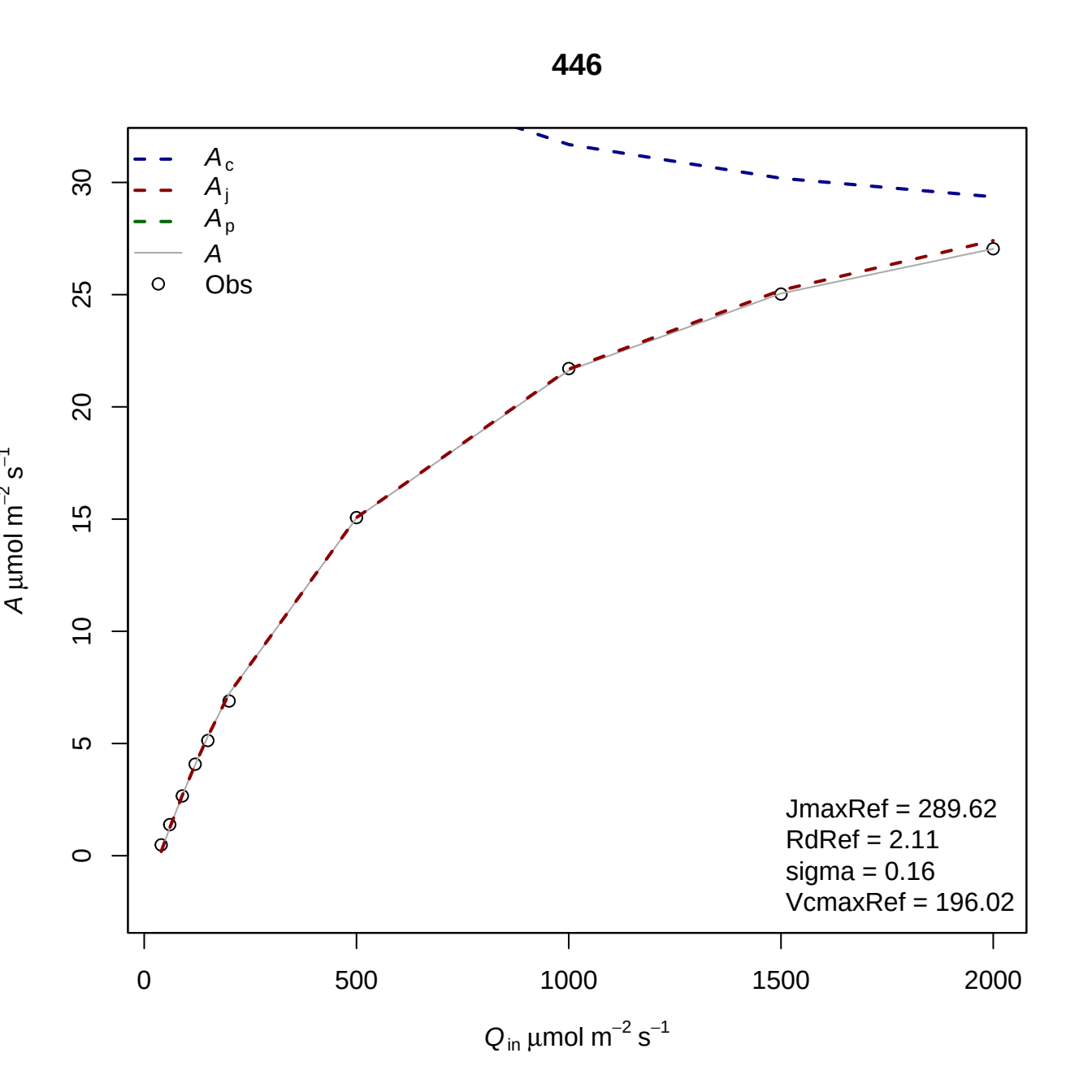

440
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.23
RdRef = 1.22
sigma = 0.27
VcmaxRef = 62.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

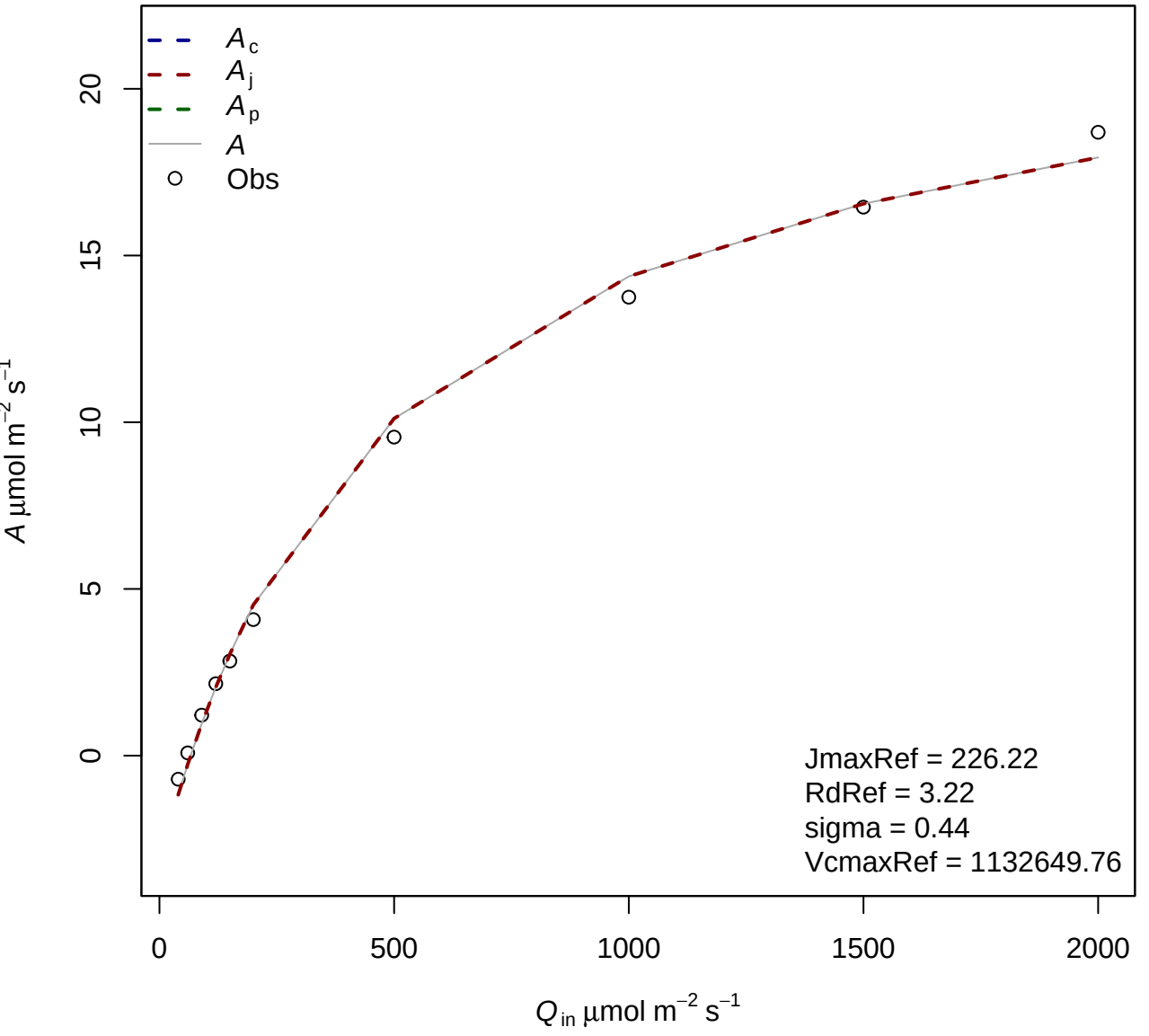

441
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163.29
RdRef = 1.25
sigma = 0.13
VcmaxRef = 94.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

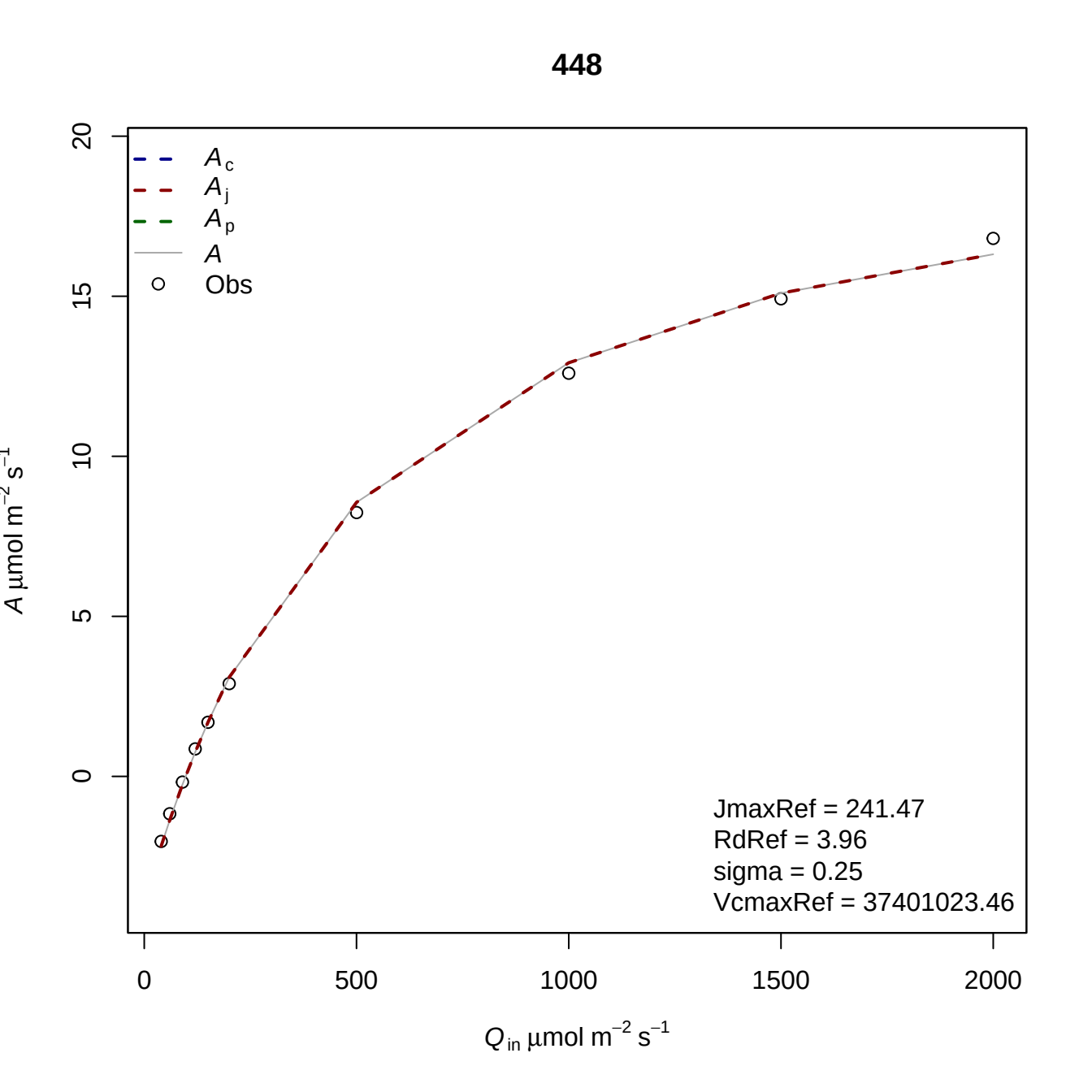

442
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.03
RdRef = 2
sigma = 0.26
VcmaxRef = 136.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

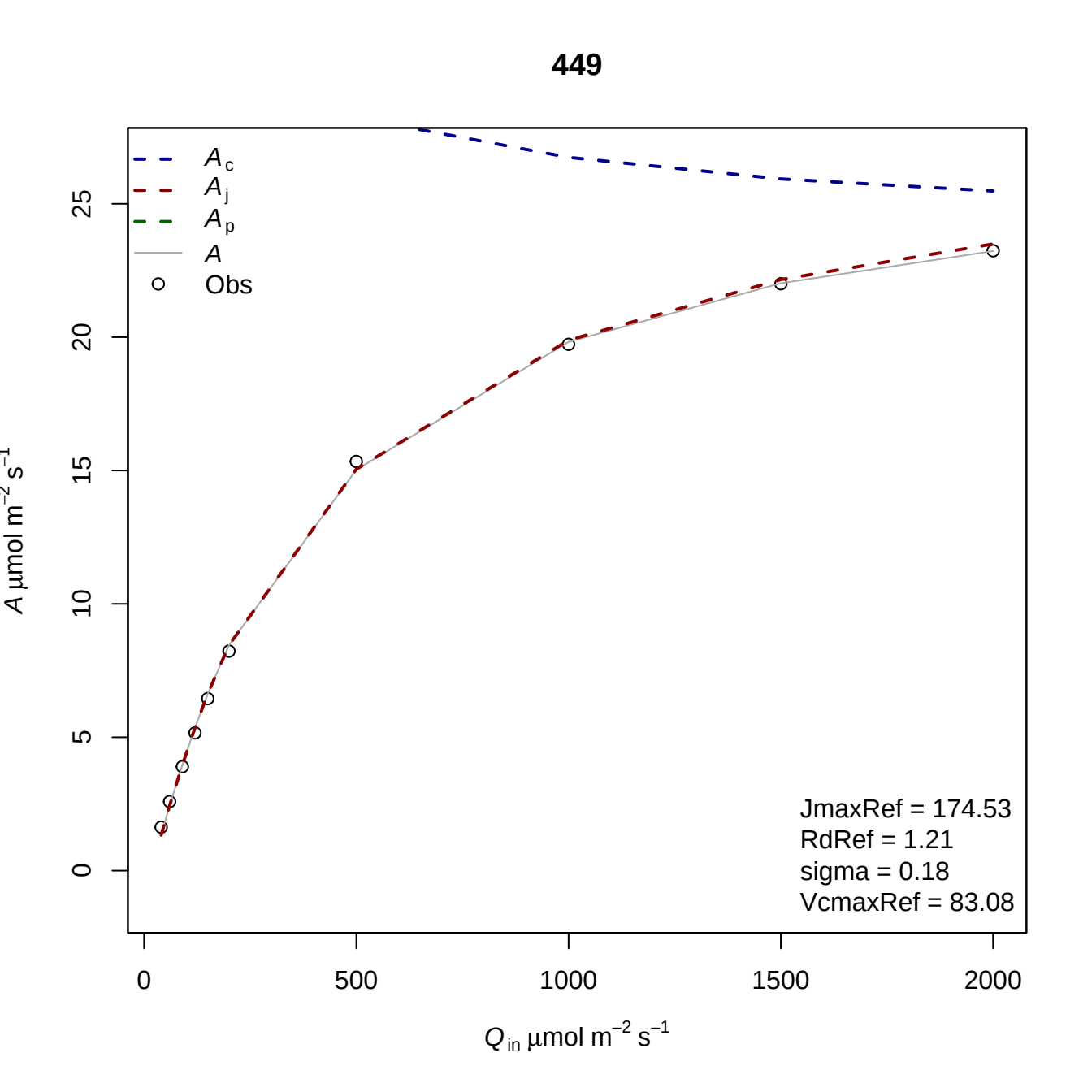

444
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.47
RdRef = 1.75
sigma = 0.35
VcmaxRef = 113330.16
2
1
0
−1
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

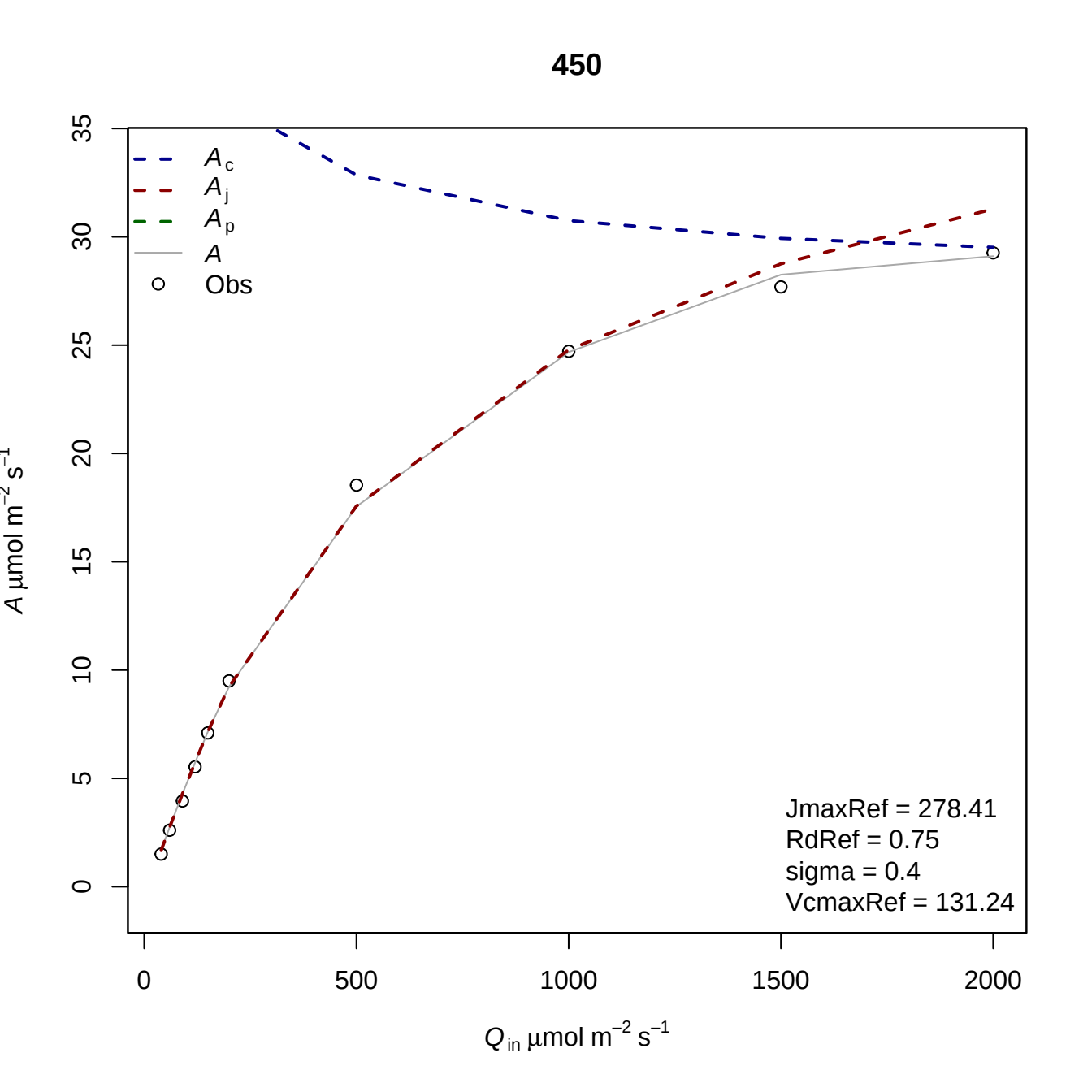

445
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.52
RdRef = 1.57
sigma = 0.32
VcmaxRef = 200720.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

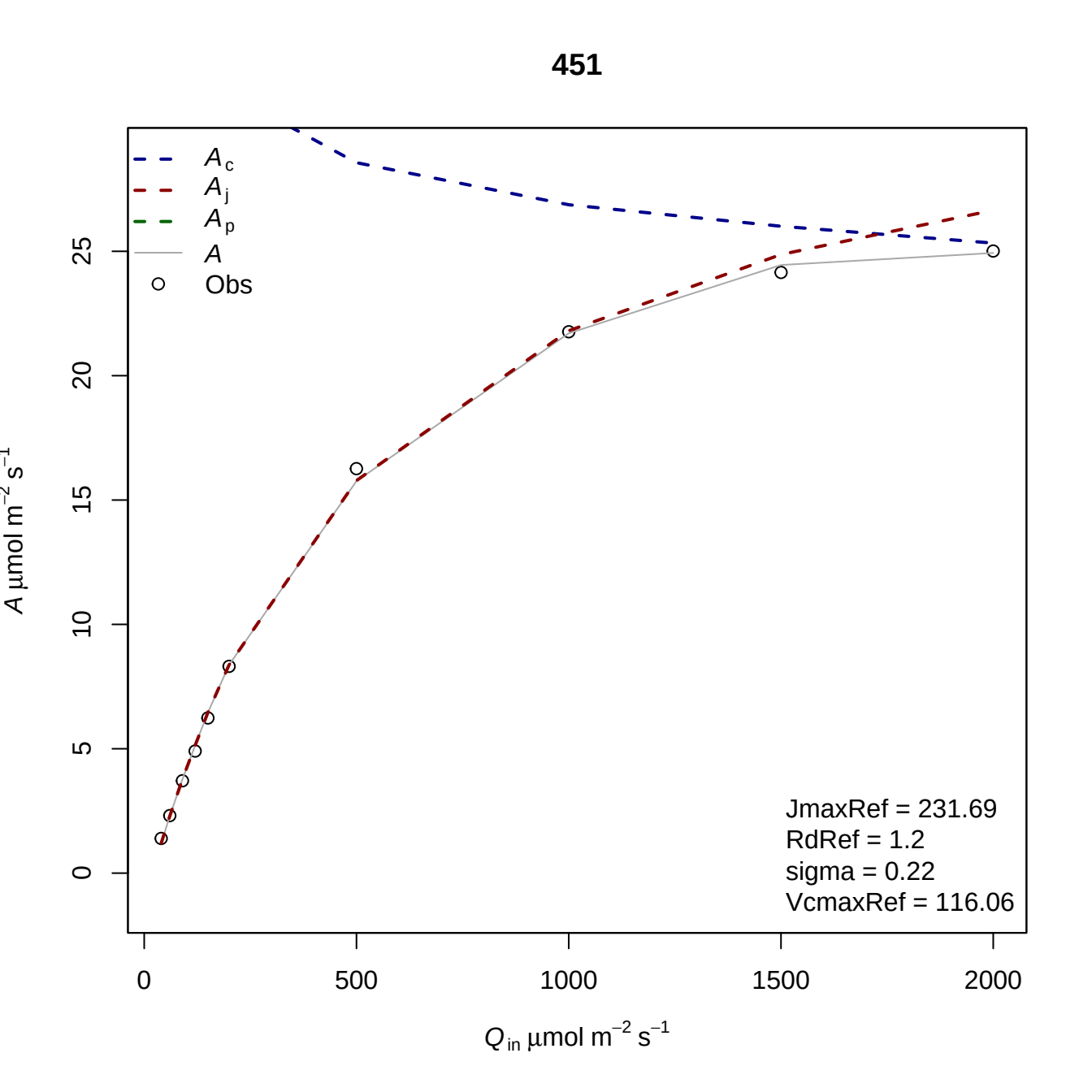

446
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 289.62
RdRef = 2.11
sigma = 0.16
VcmaxRef = 196.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

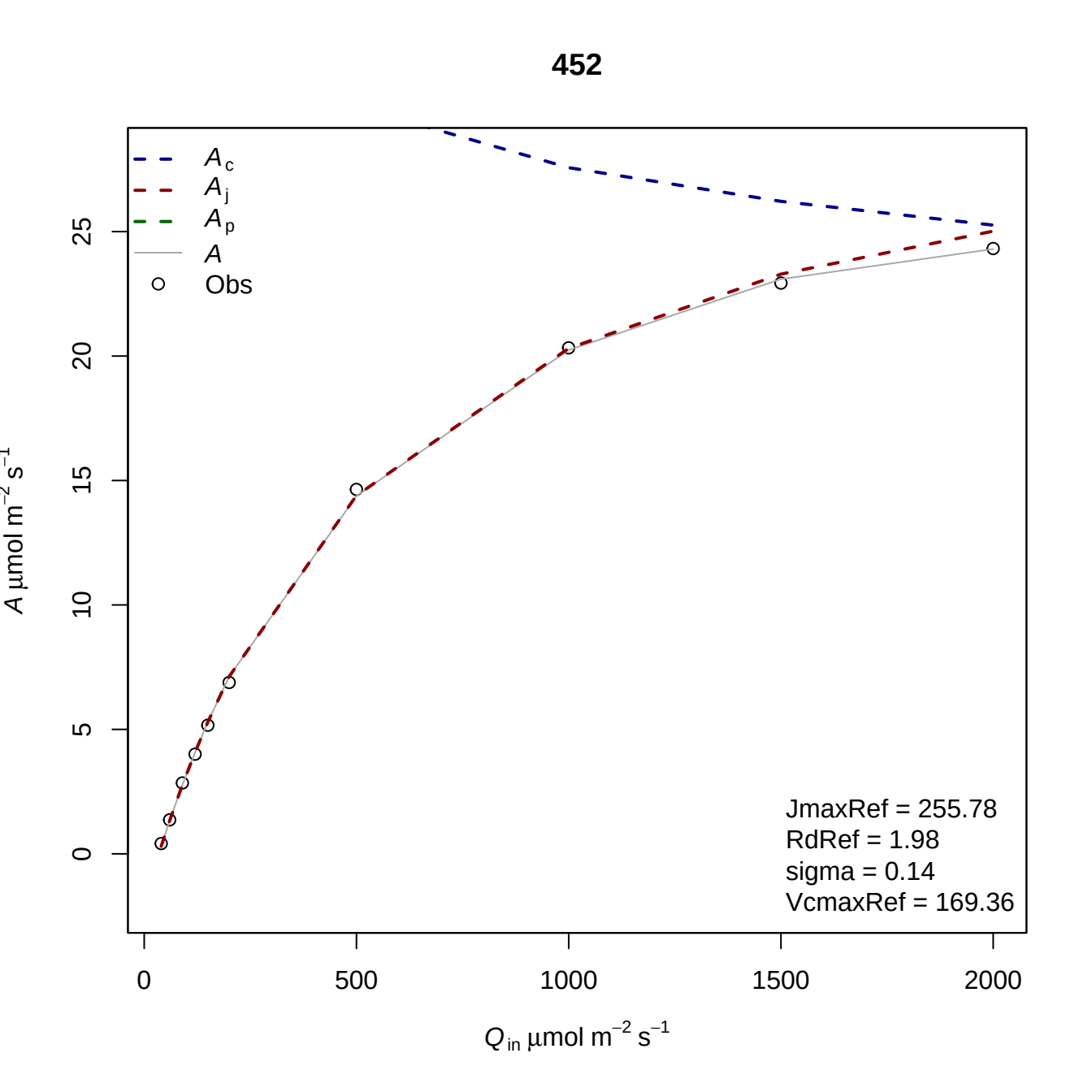

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 226.22
RdRef = 3.22
sigma = 0.44
VcmaxRef = 1132649.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

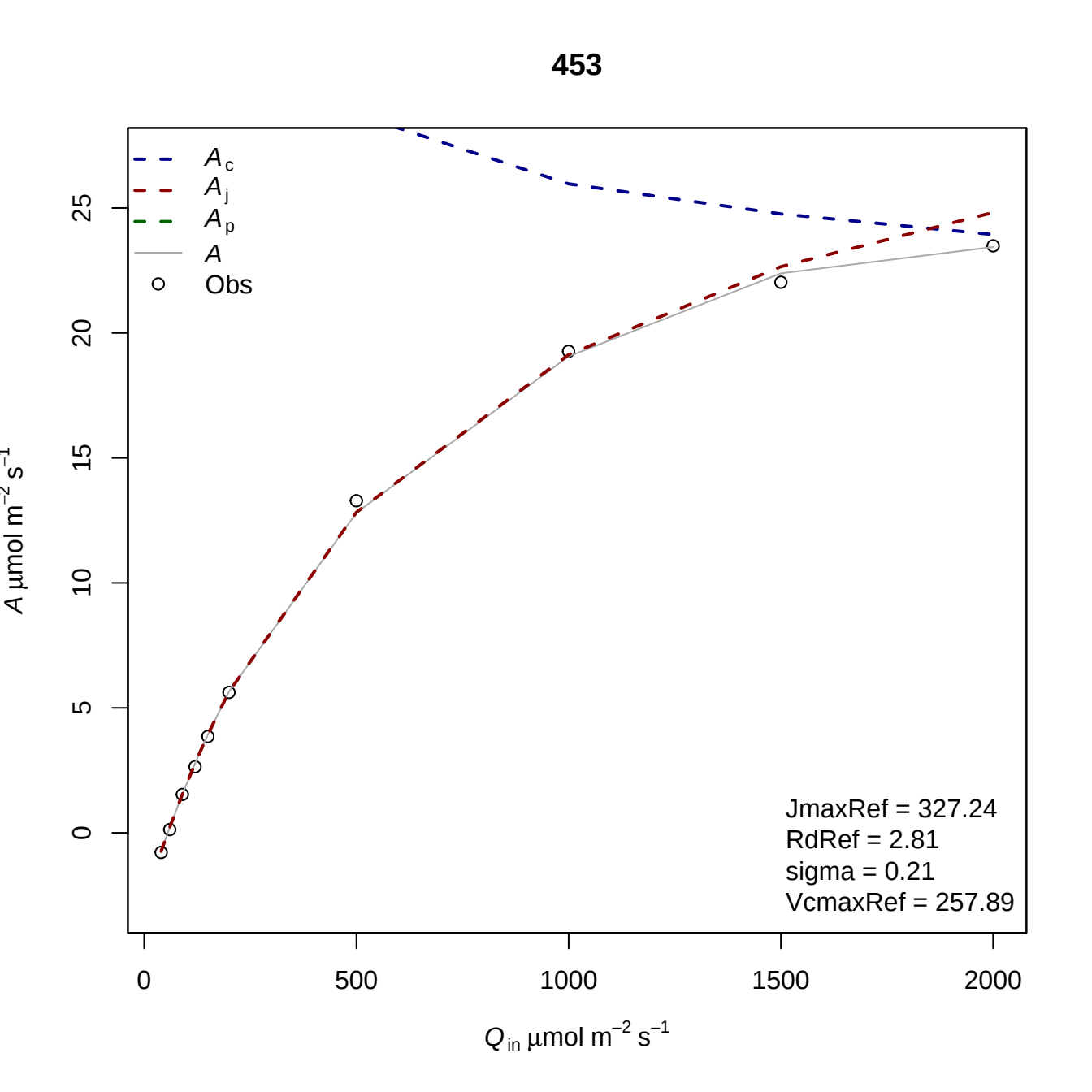

448
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 241.47
RdRef = 3.96
sigma = 0.25
VcmaxRef = 37401023.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

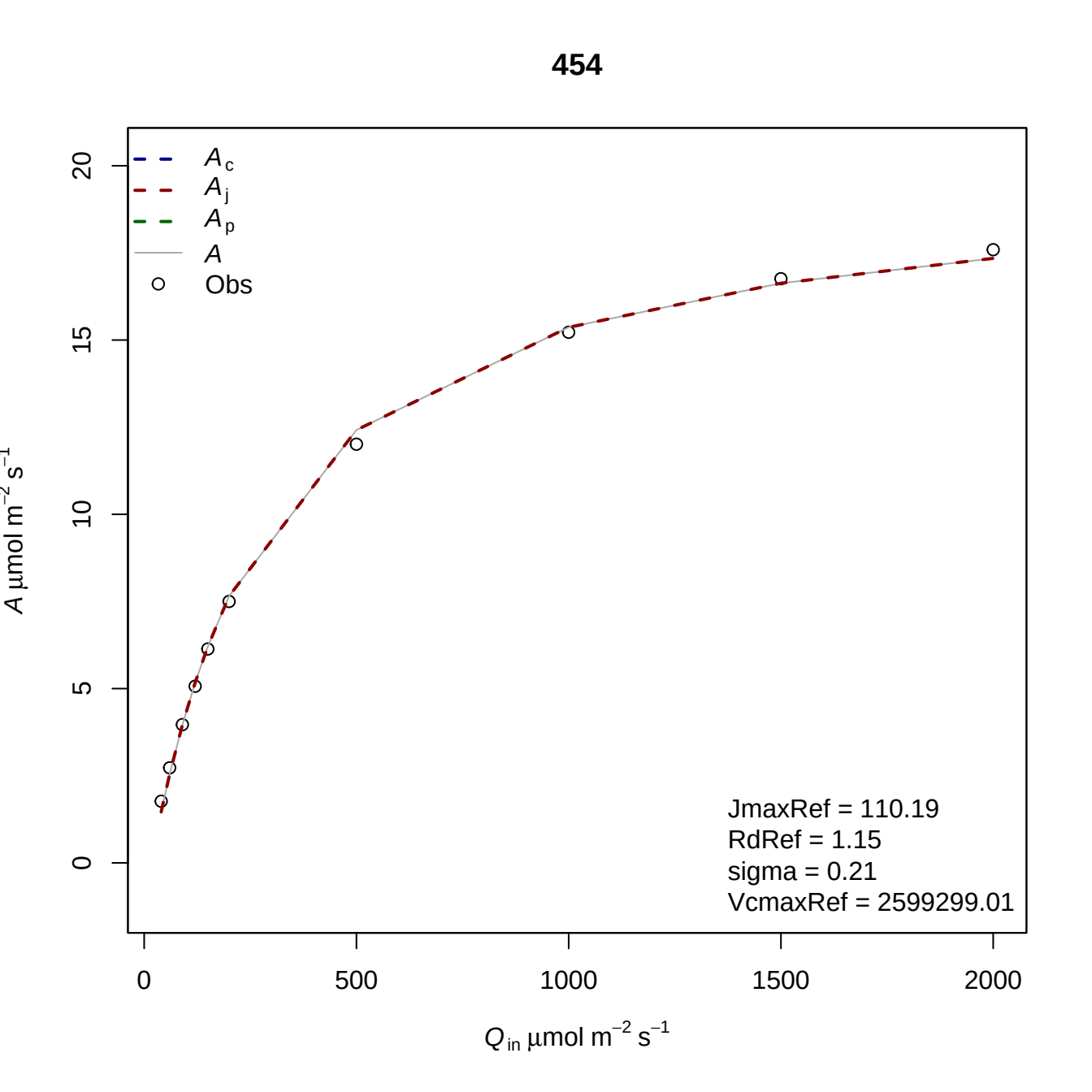

449
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 174.53
RdRef = 1.21
sigma = 0.18
VcmaxRef = 83.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

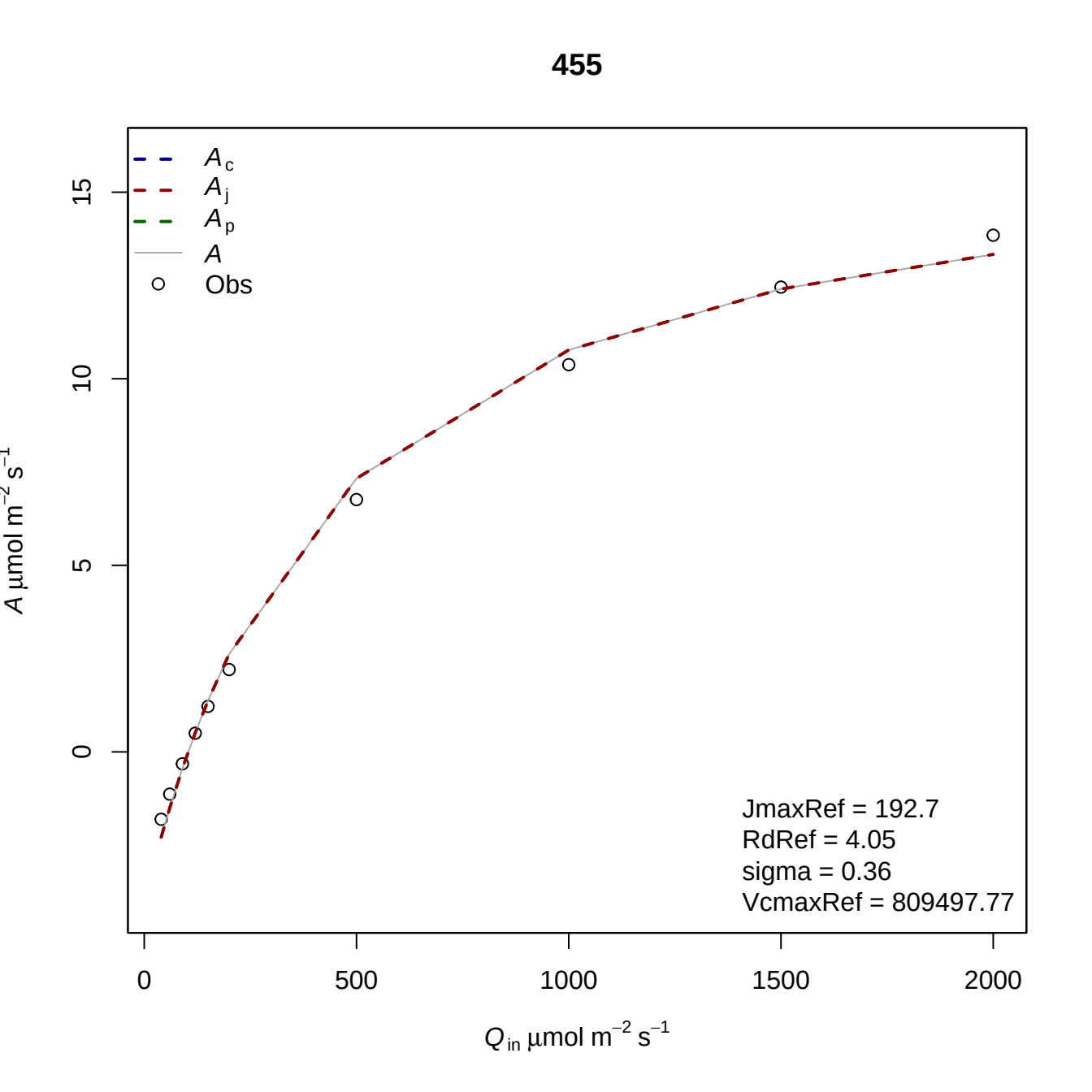

450
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 278.41
RdRef = 0.75
sigma = 0.4
VcmaxRef = 131.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

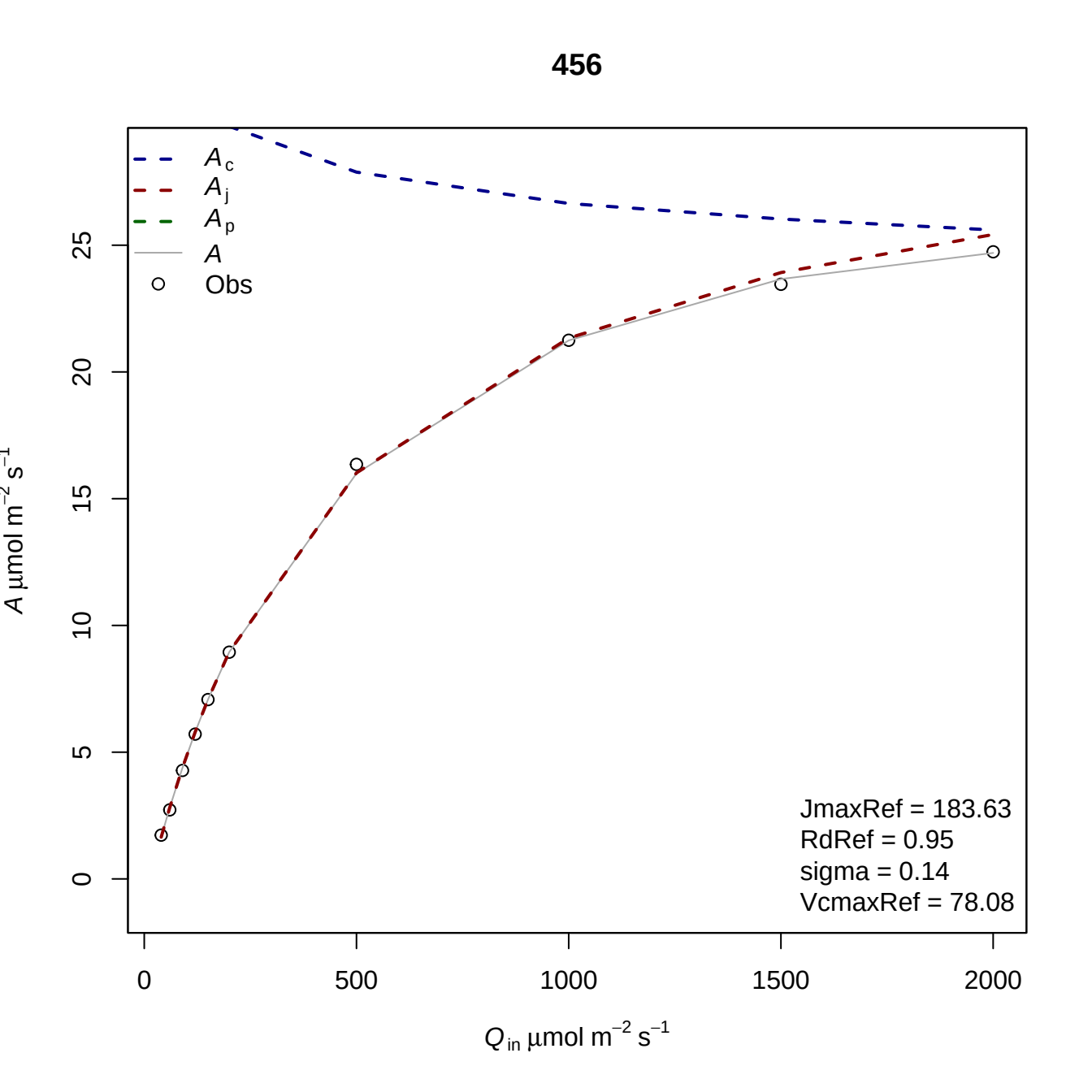

451
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 231.69
RdRef = 1.2
sigma = 0.22
VcmaxRef = 116.06
0
5
10
15
20
25
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

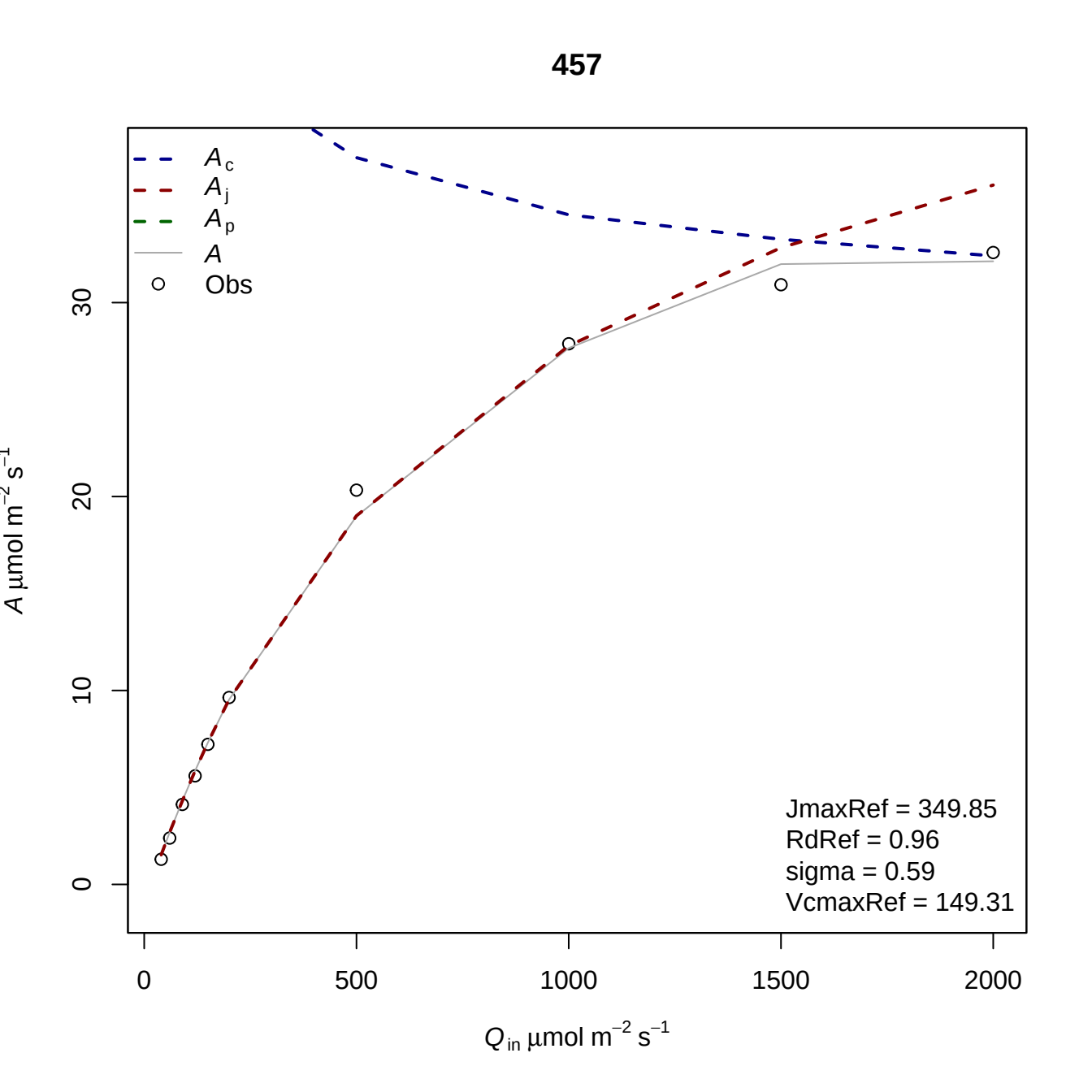

452
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 255.78
RdRef = 1.98
sigma = 0.14
VcmaxRef = 169.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

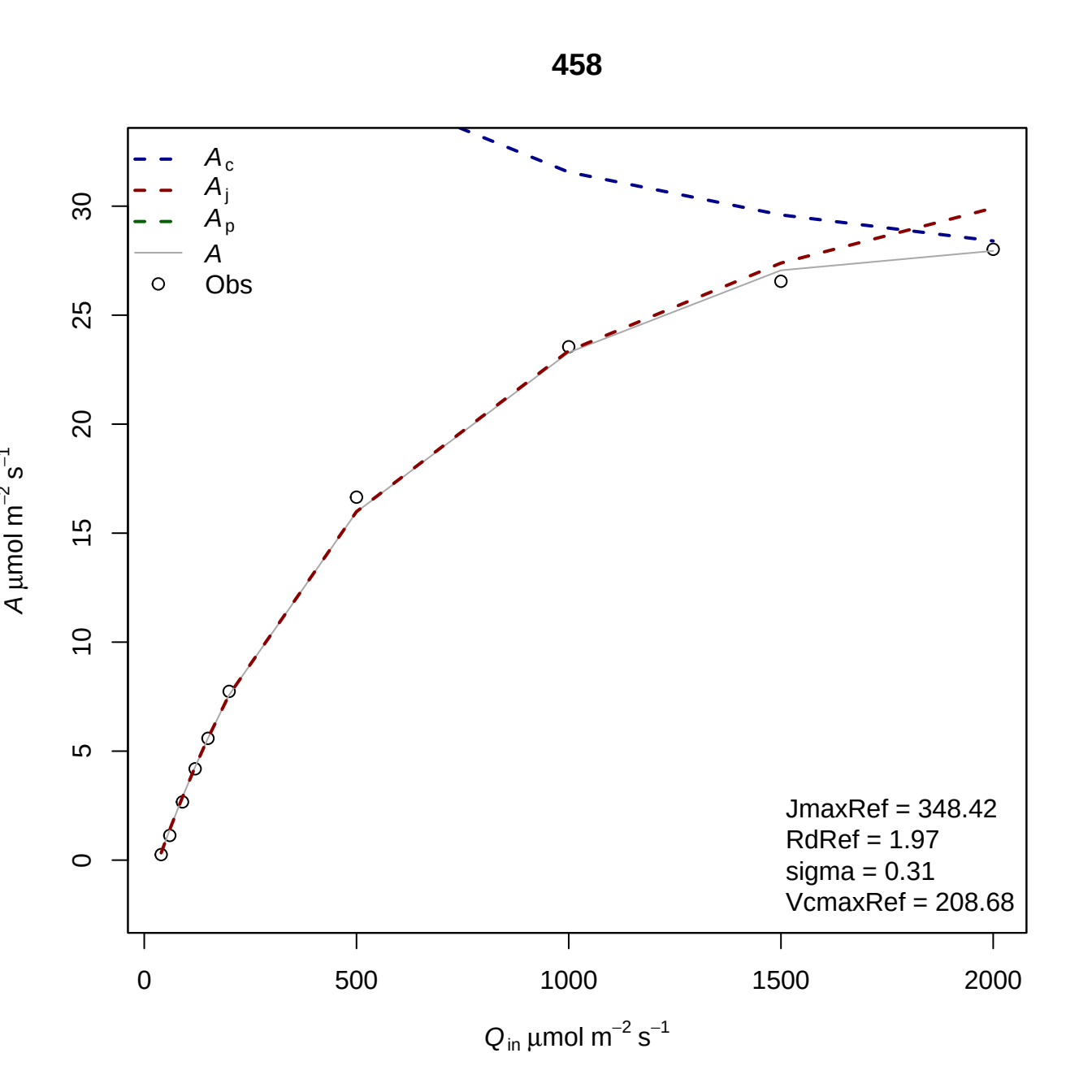

453
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 327.24
RdRef = 2.81
sigma = 0.21
VcmaxRef = 257.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

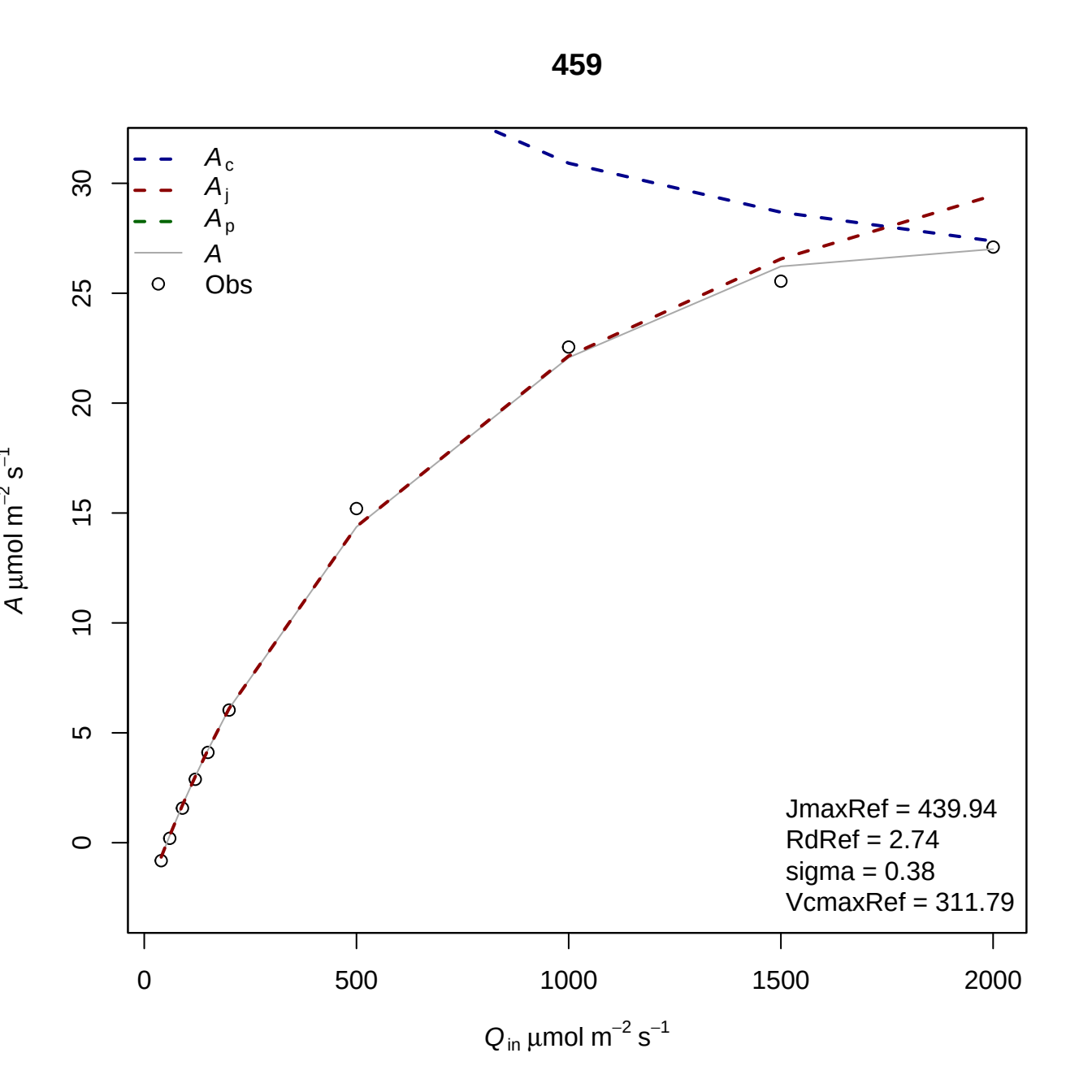

454
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.19
RdRef = 1.15
sigma = 0.21
VcmaxRef = 2599299.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

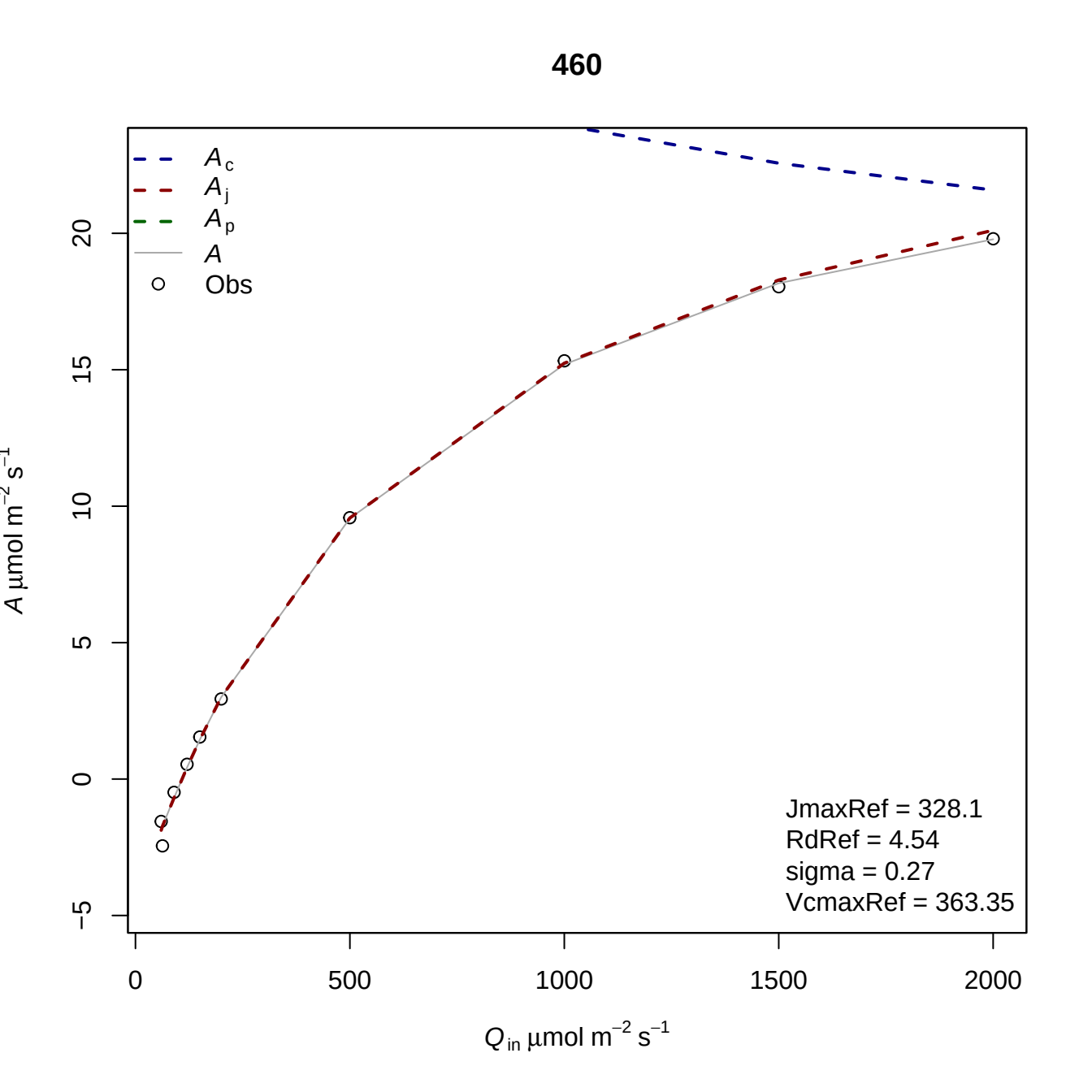

455
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 192.7
RdRef = 4.05
sigma = 0.36
VcmaxRef = 809497.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

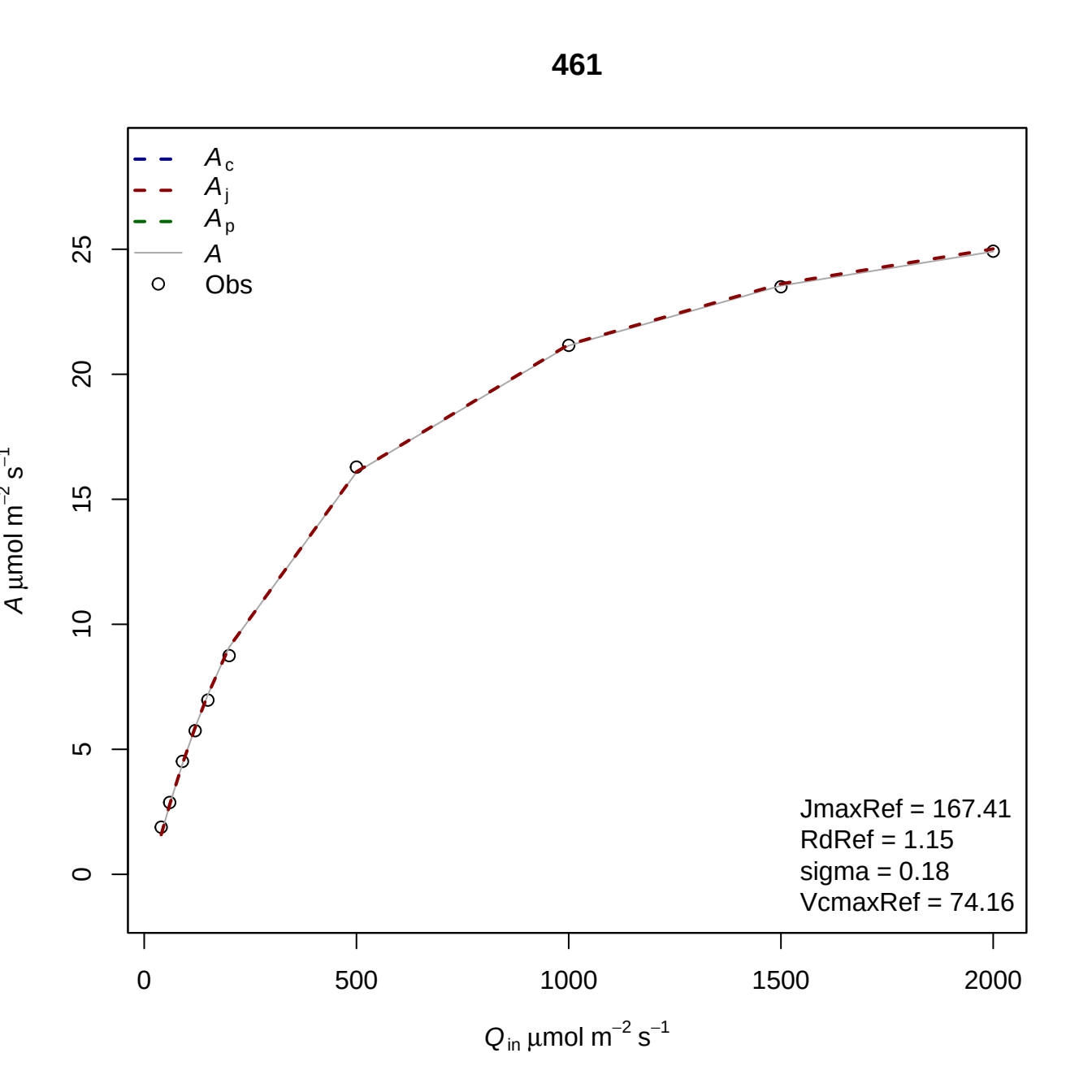

456
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 183.63
RdRef = 0.95
sigma = 0.14
VcmaxRef = 78.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

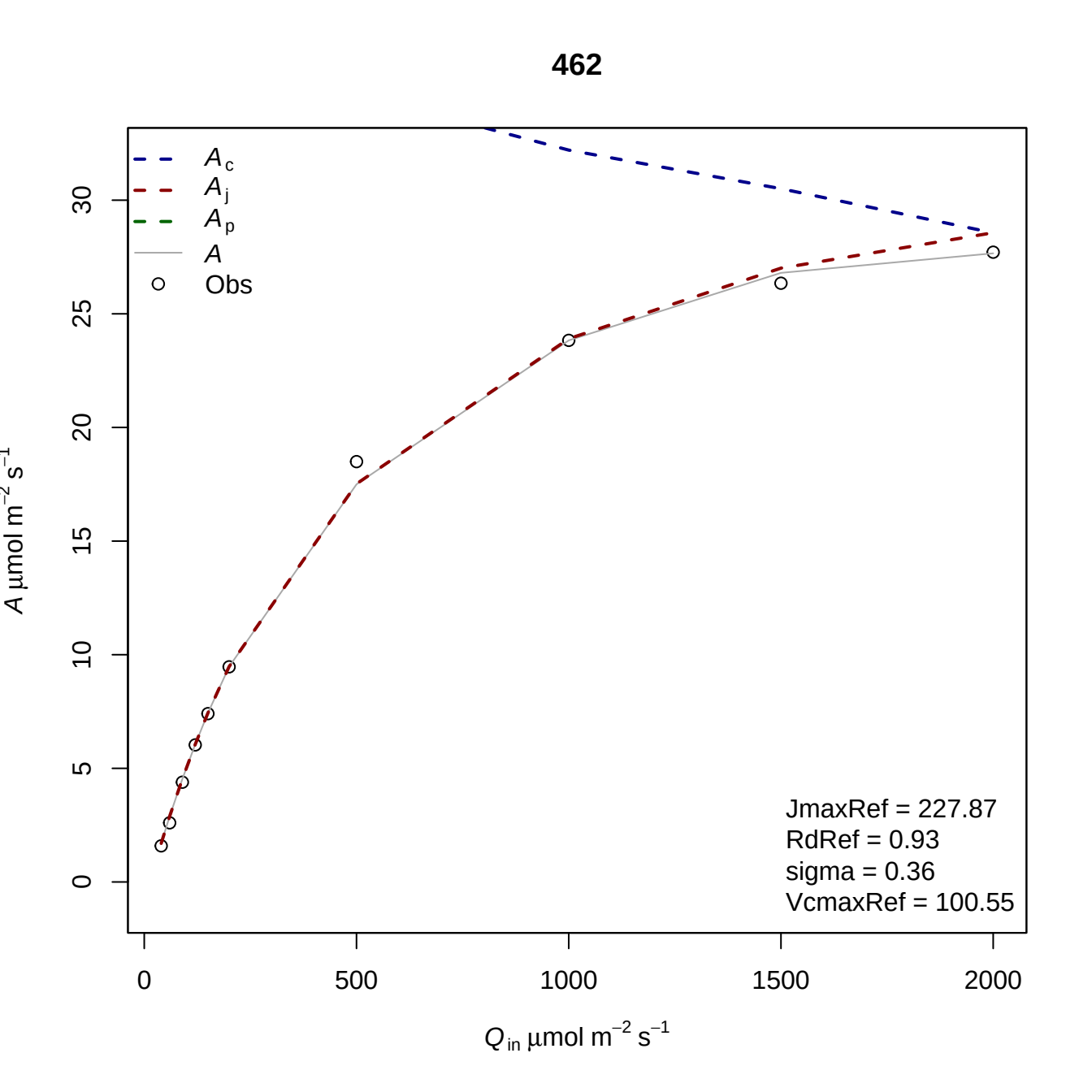

457
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 349.85
RdRef = 0.96
sigma = 0.59
VcmaxRef = 149.31
0
10
20
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

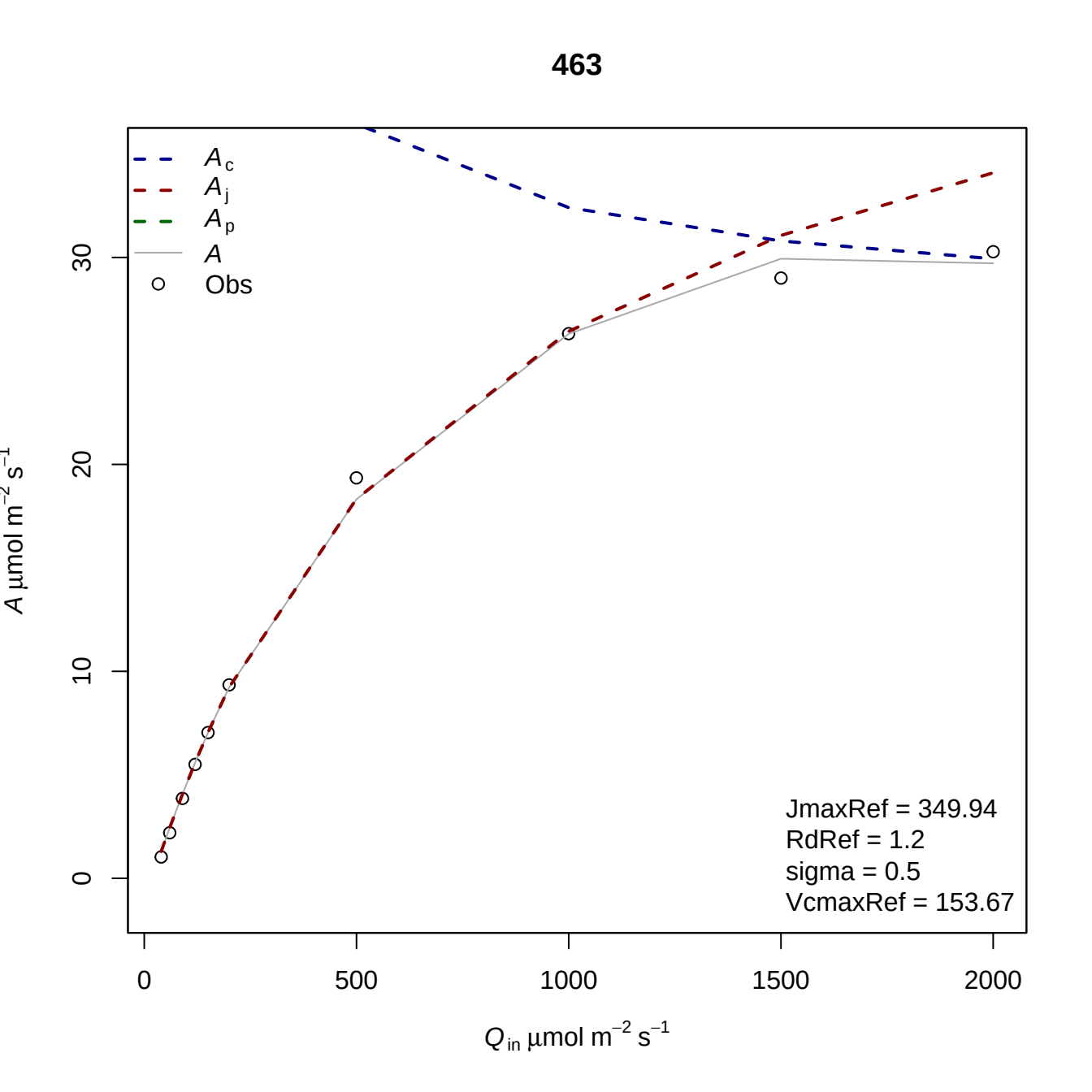

458
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 348.42
RdRef = 1.97
sigma = 0.31
VcmaxRef = 208.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

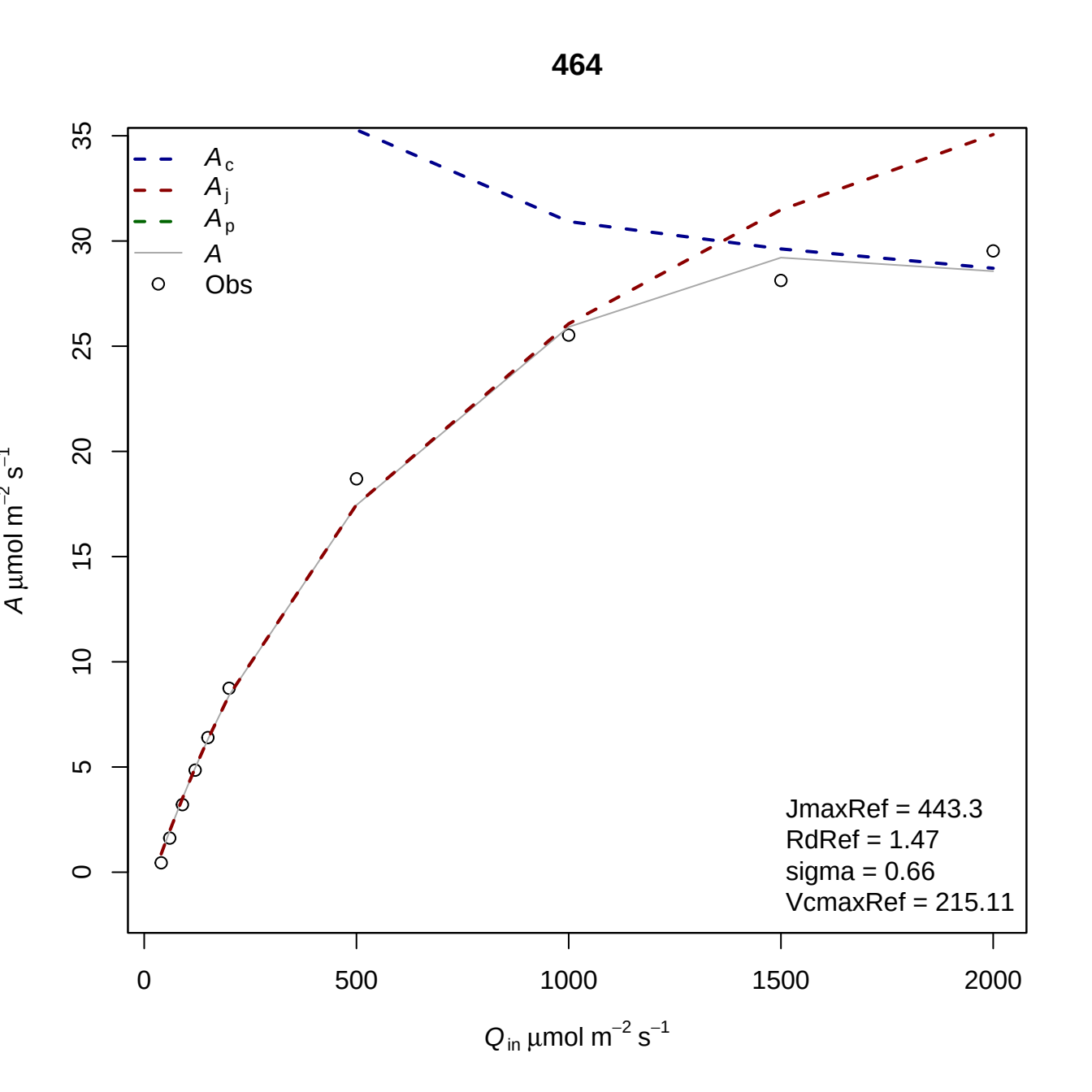

459
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 439.94
RdRef = 2.74
sigma = 0.38
VcmaxRef = 311.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

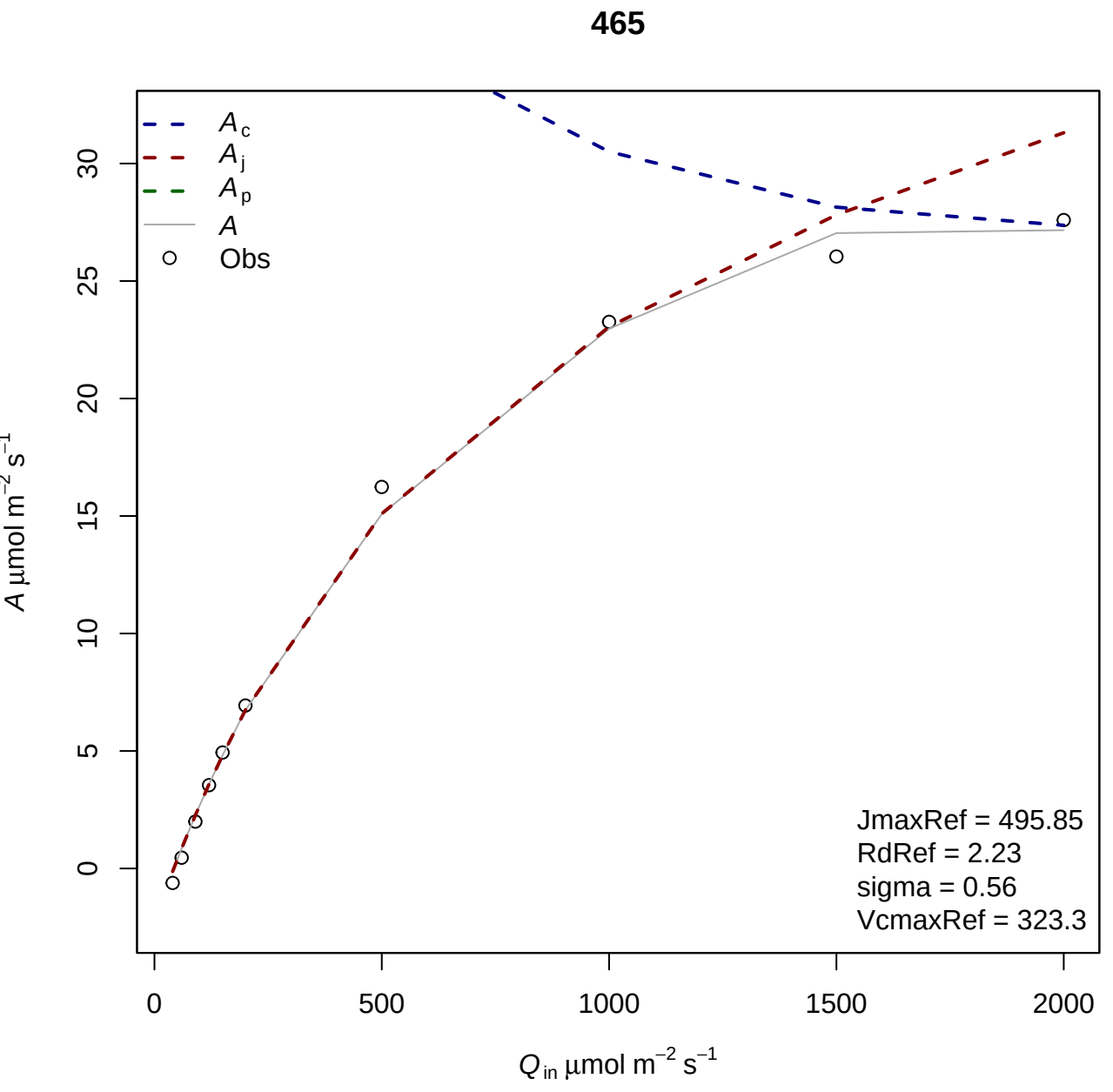

460
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 328.1
RdRef = 4.54
sigma = 0.27
VcmaxRef = 363.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

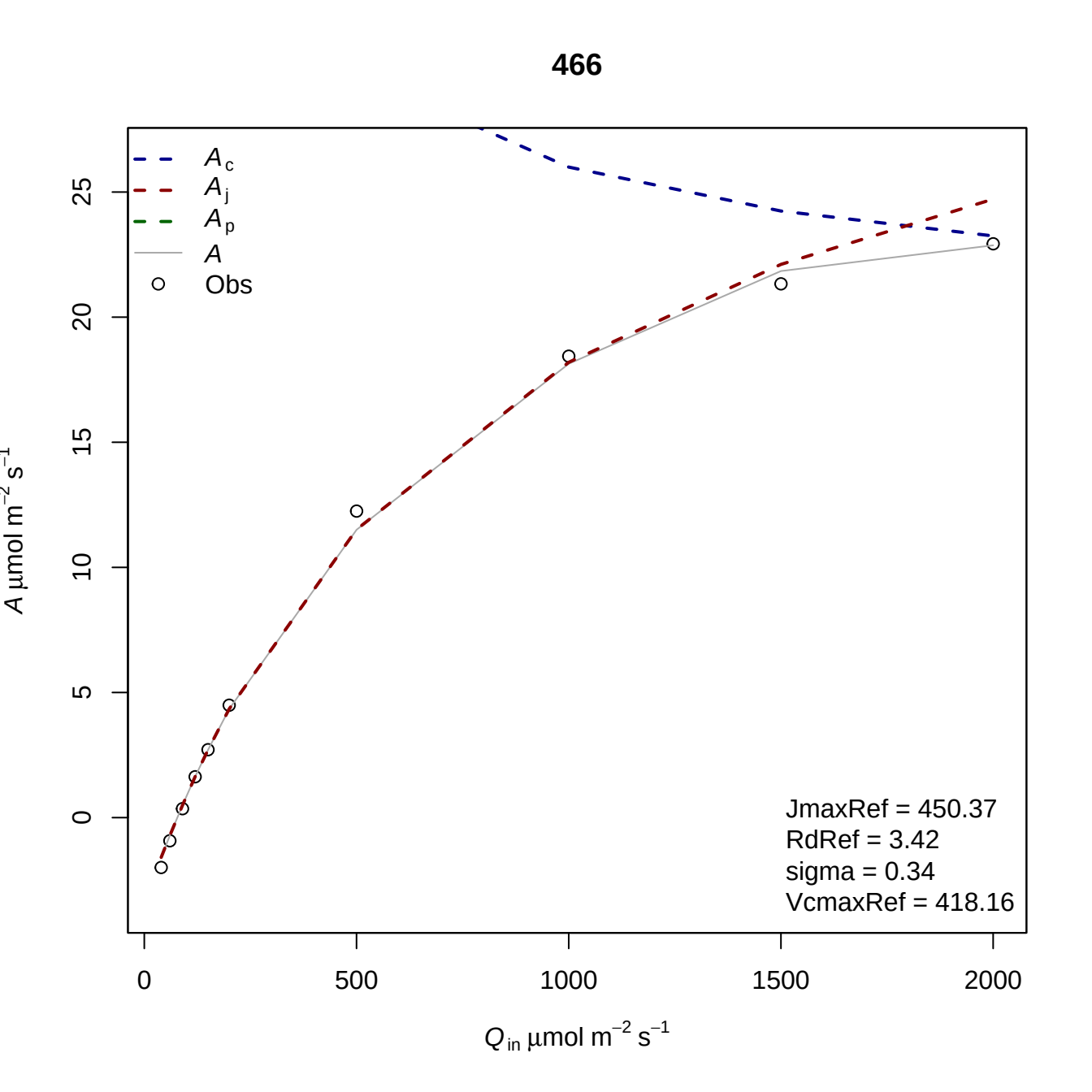

461
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 167.41
RdRef = 1.15
sigma = 0.18
VcmaxRef = 74.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

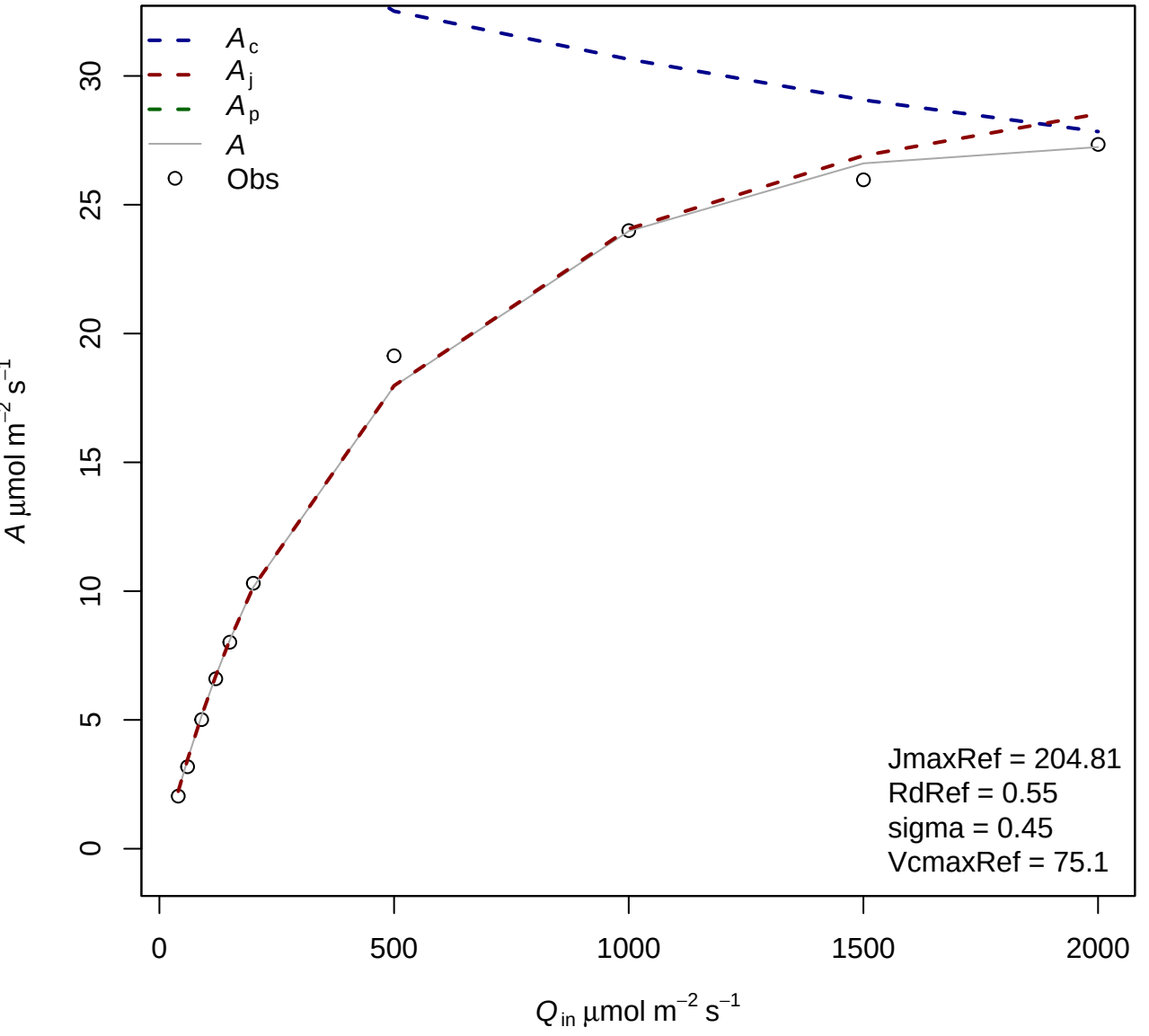

462
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 227.87
RdRef = 0.93
sigma = 0.36
VcmaxRef = 100.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

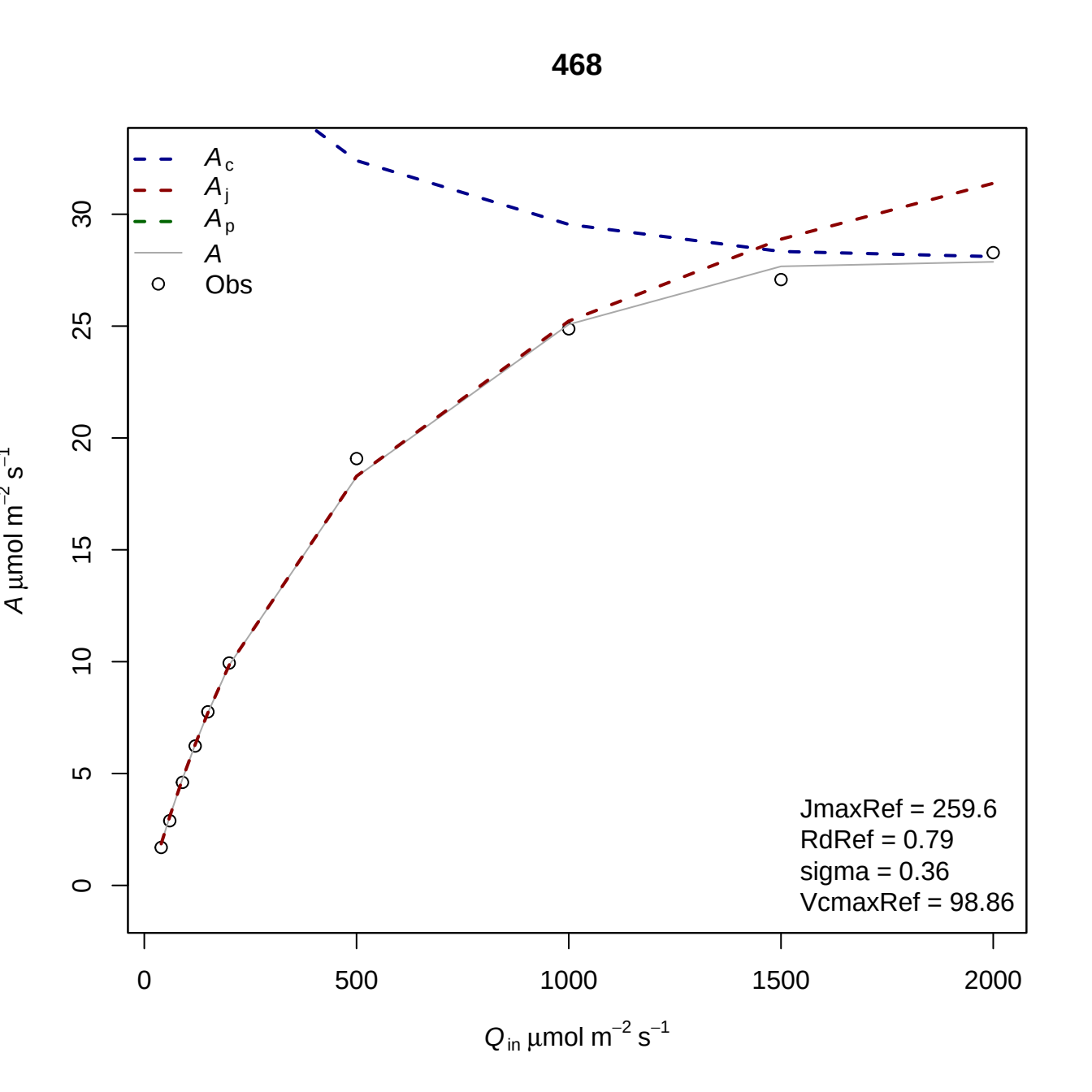

463
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 349.94
RdRef = 1.2
sigma = 0.5
VcmaxRef = 153.67
0
10
20
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

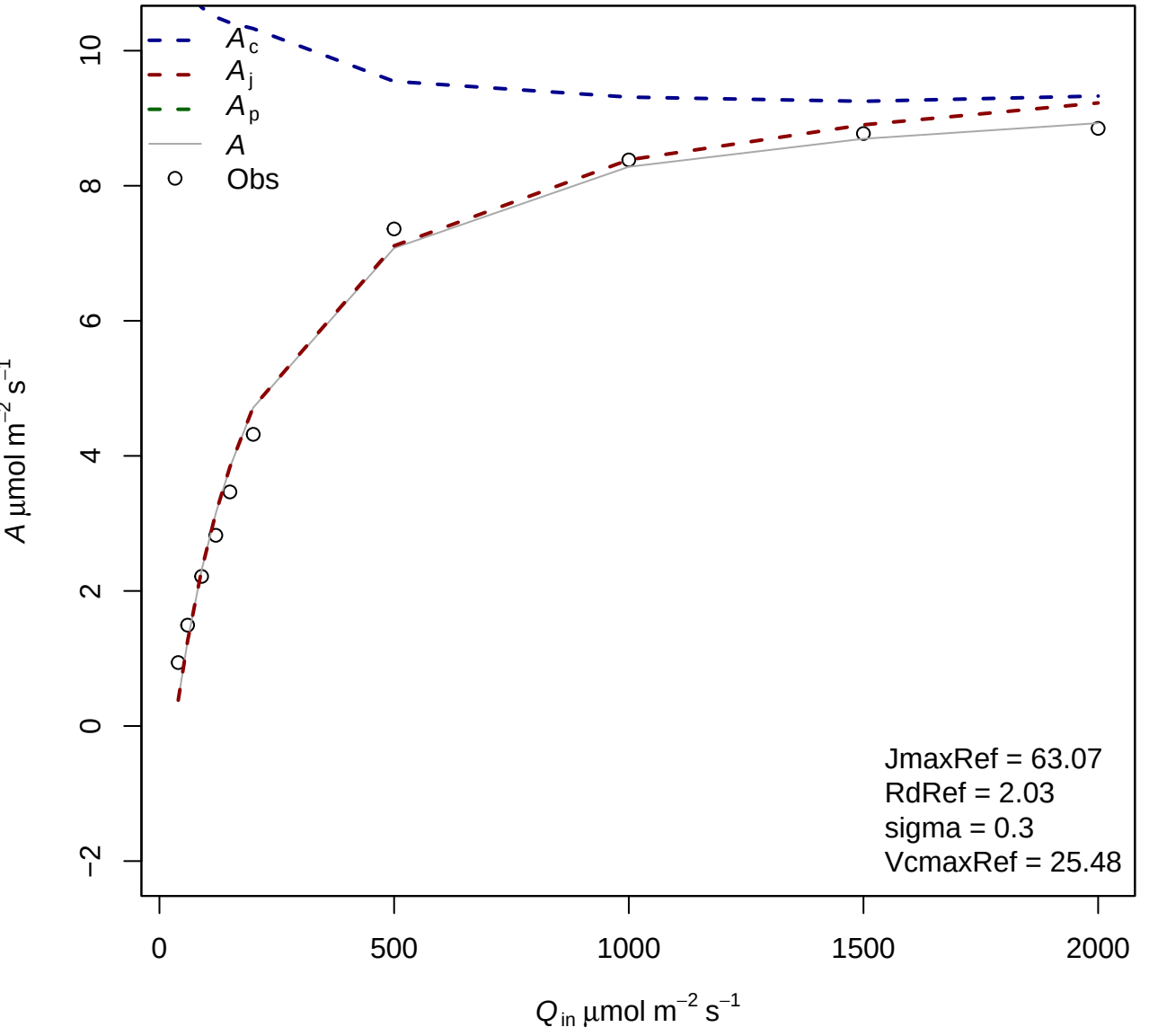

464
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 443.3
RdRef = 1.47
sigma = 0.66
VcmaxRef = 215.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

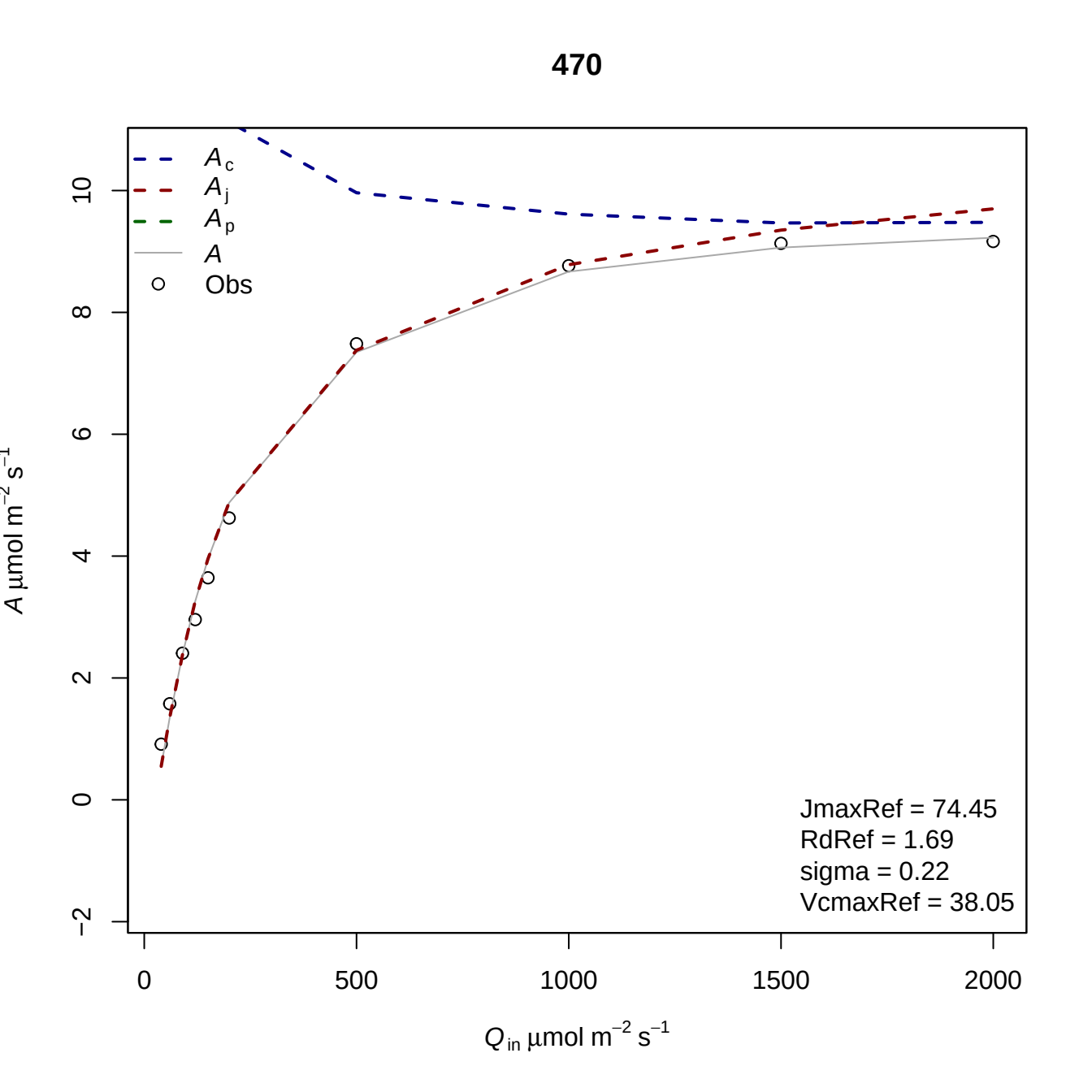

465
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 495.85
RdRef = 2.23
sigma = 0.56
VcmaxRef = 323.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

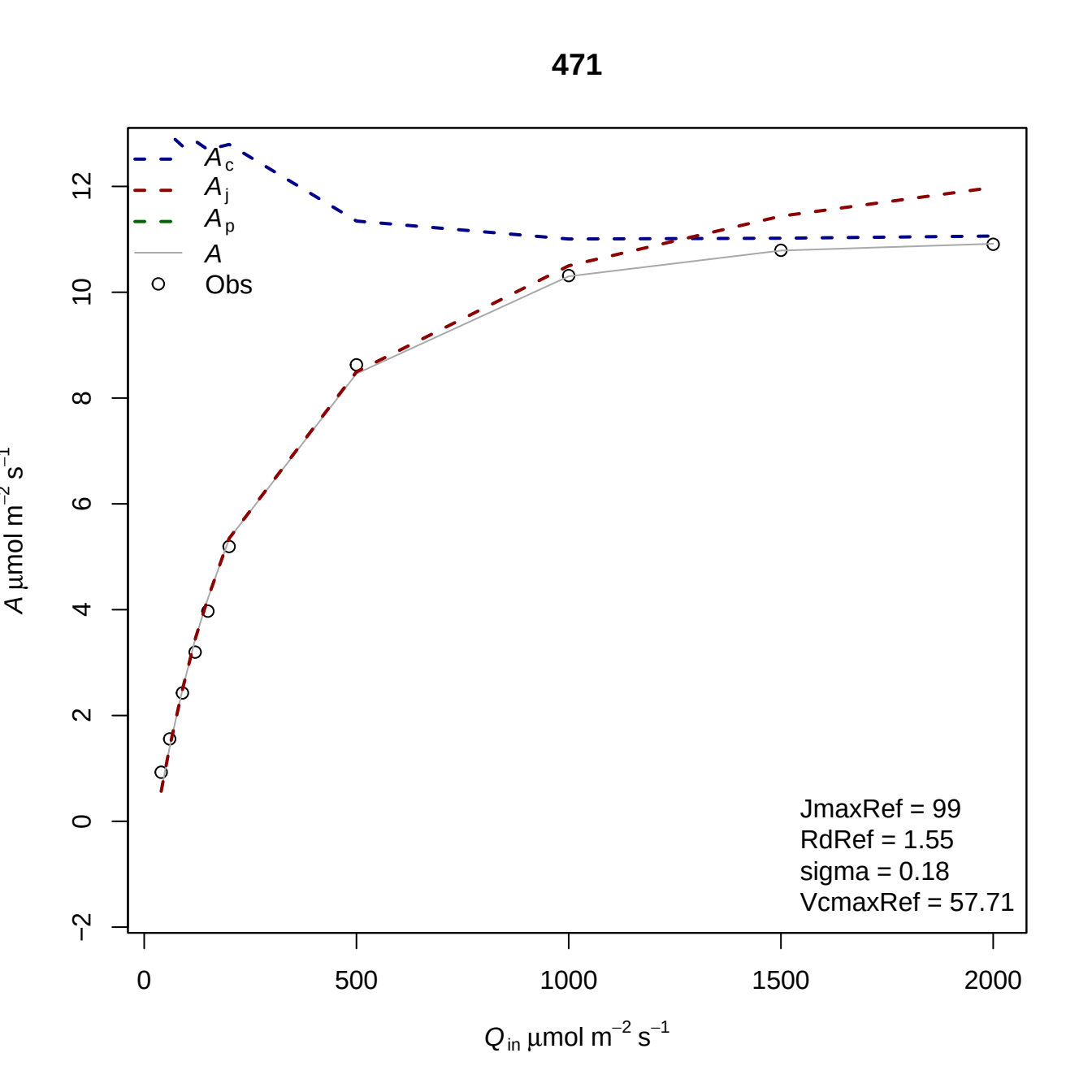

466
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 450.37
RdRef = 3.42
sigma = 0.34
VcmaxRef = 418.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

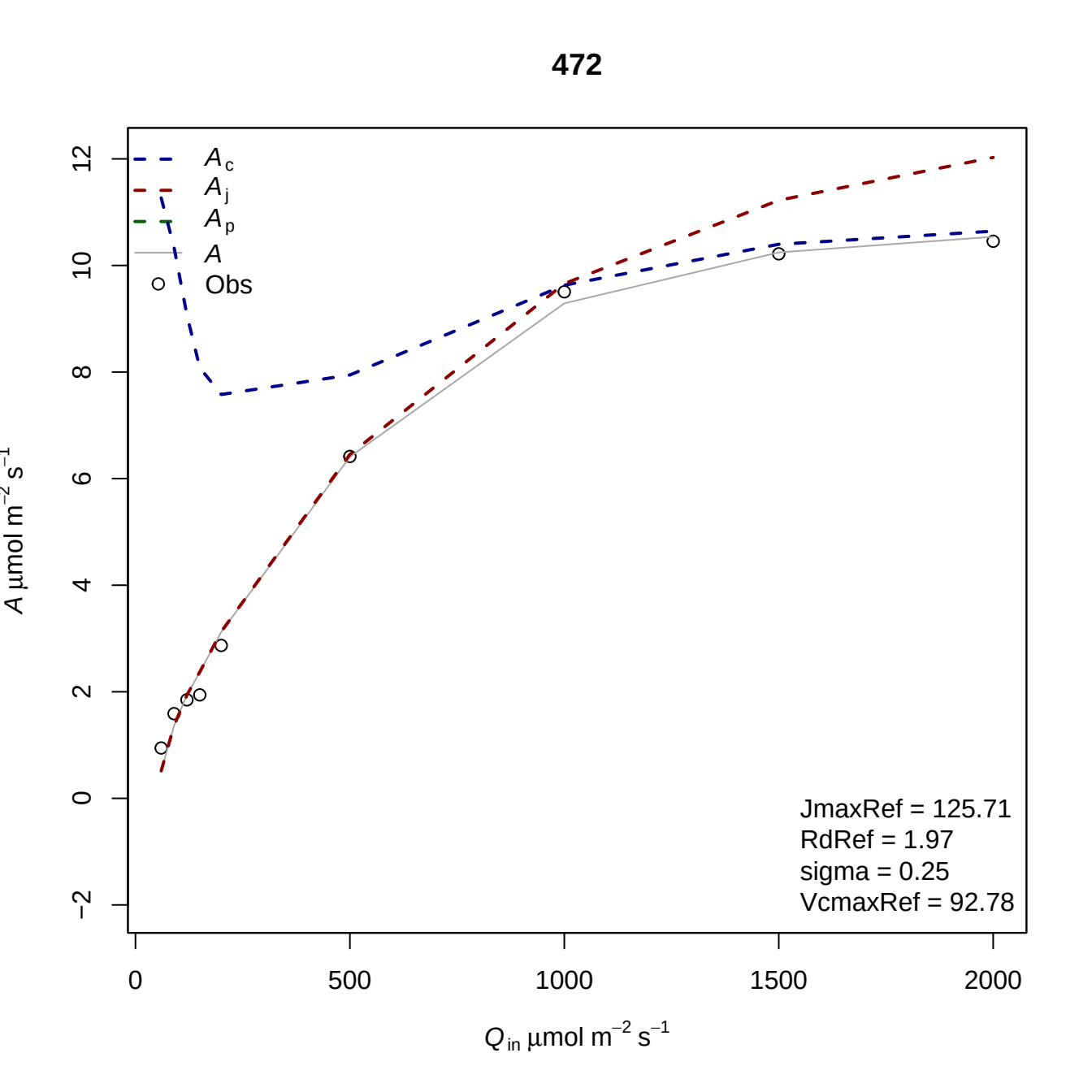

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 204.81
RdRef = 0.55
sigma = 0.45
VcmaxRef = 75.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

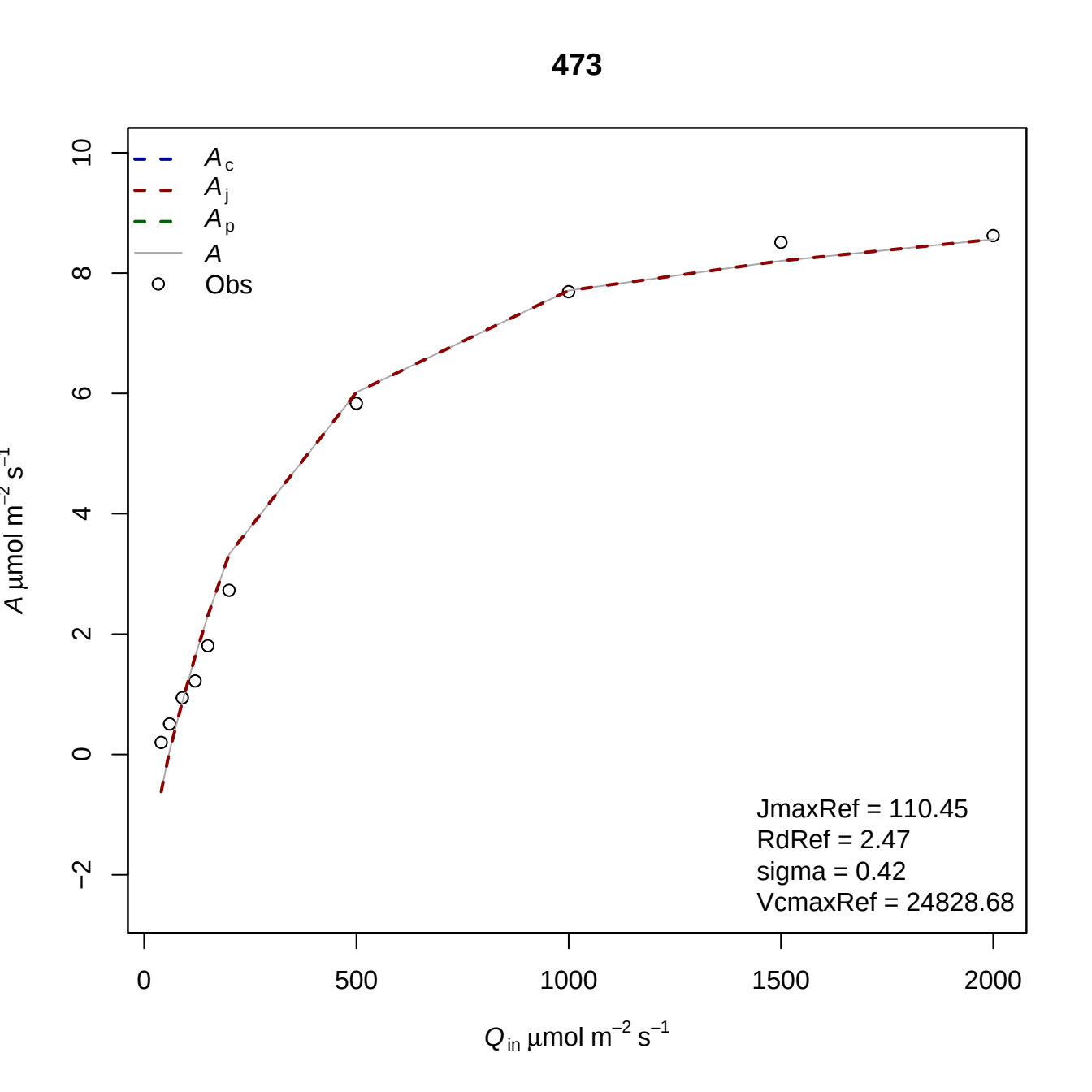

468
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 259.6
RdRef = 0.79
sigma = 0.36
VcmaxRef = 98.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

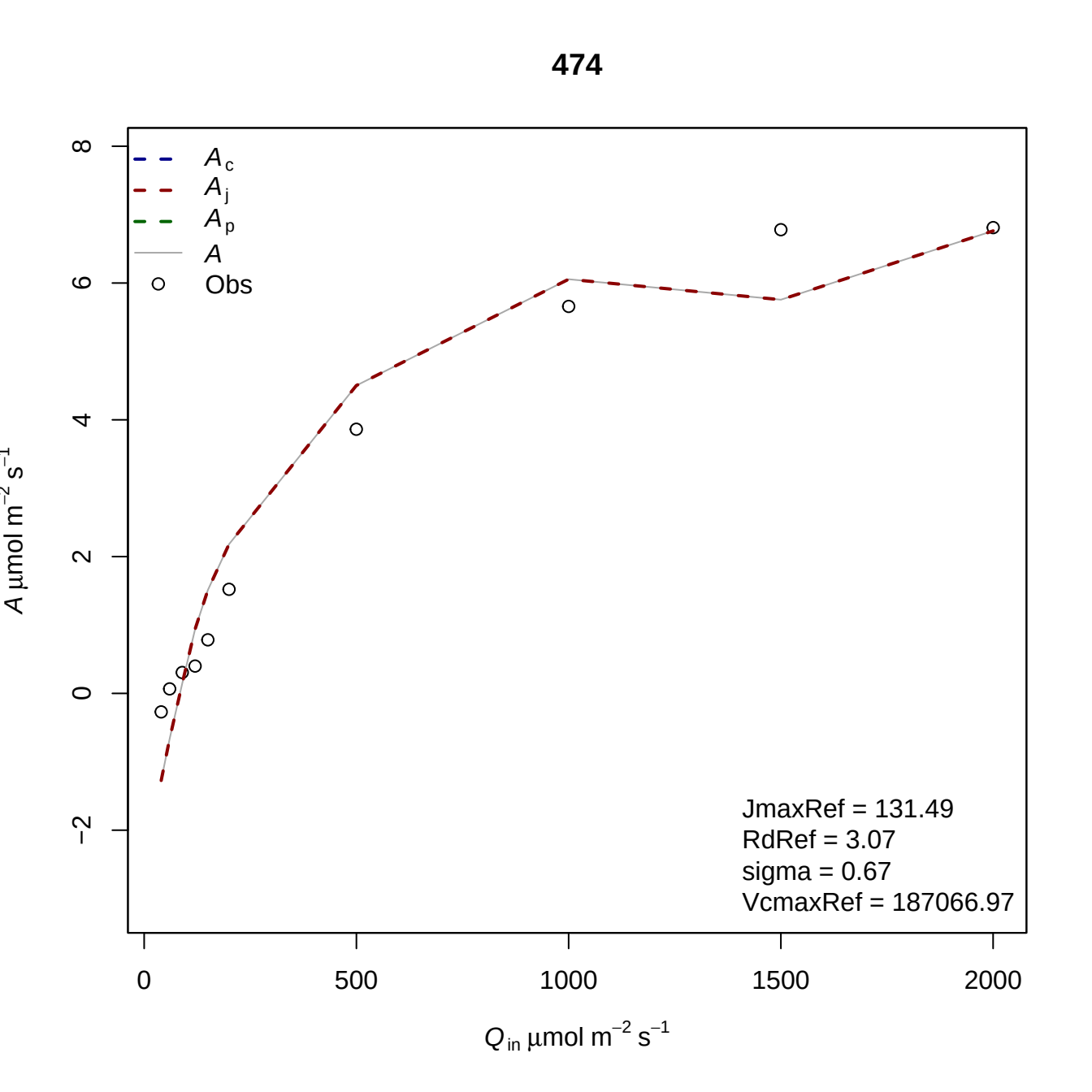

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.07
RdRef = 2.03
sigma = 0.3
VcmaxRef = 25.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

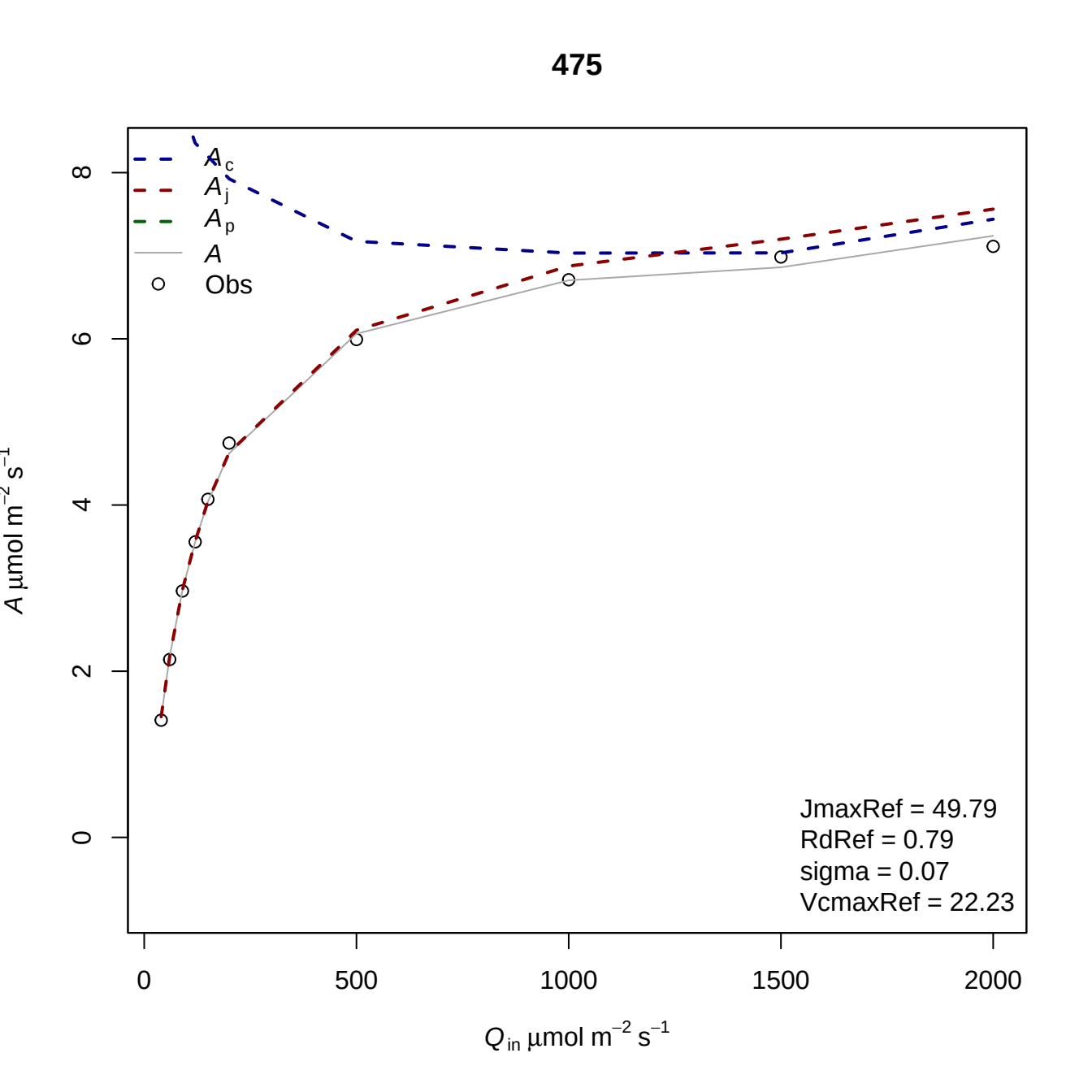

470
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.45
RdRef = 1.69
sigma = 0.22
VcmaxRef = 38.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

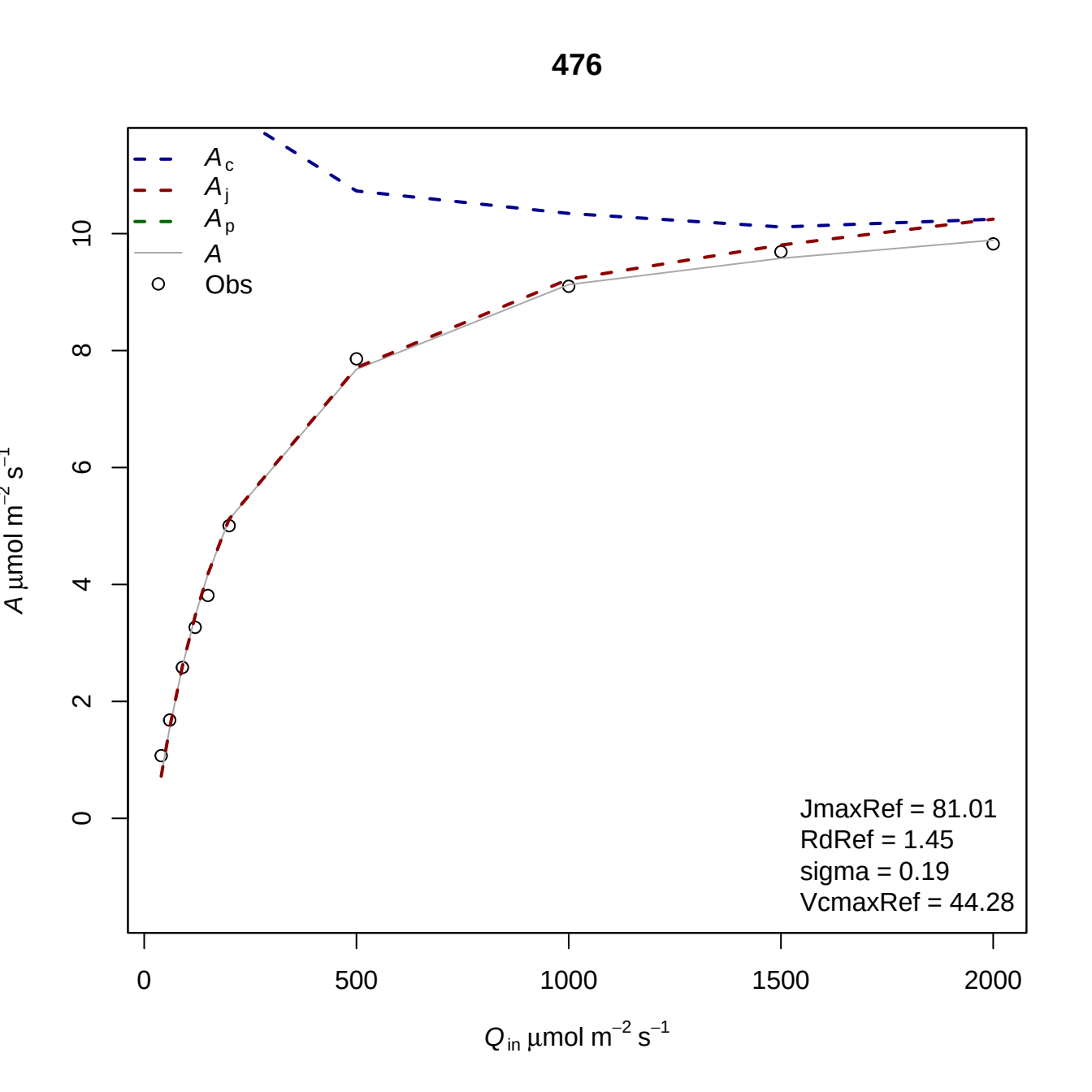

471
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99
RdRef = 1.55
sigma = 0.18
VcmaxRef = 57.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

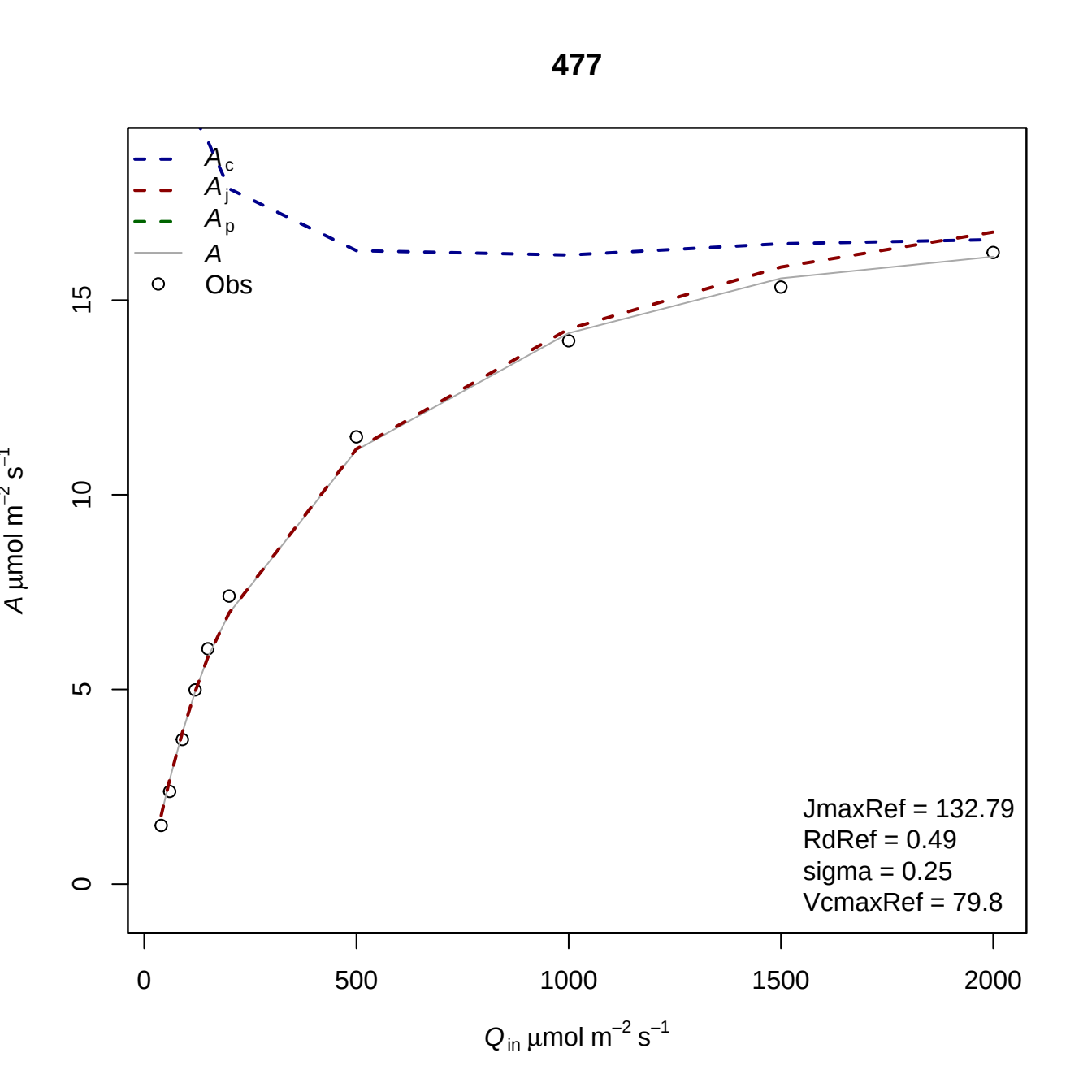

472
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.71
RdRef = 1.97
sigma = 0.25
VcmaxRef = 92.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

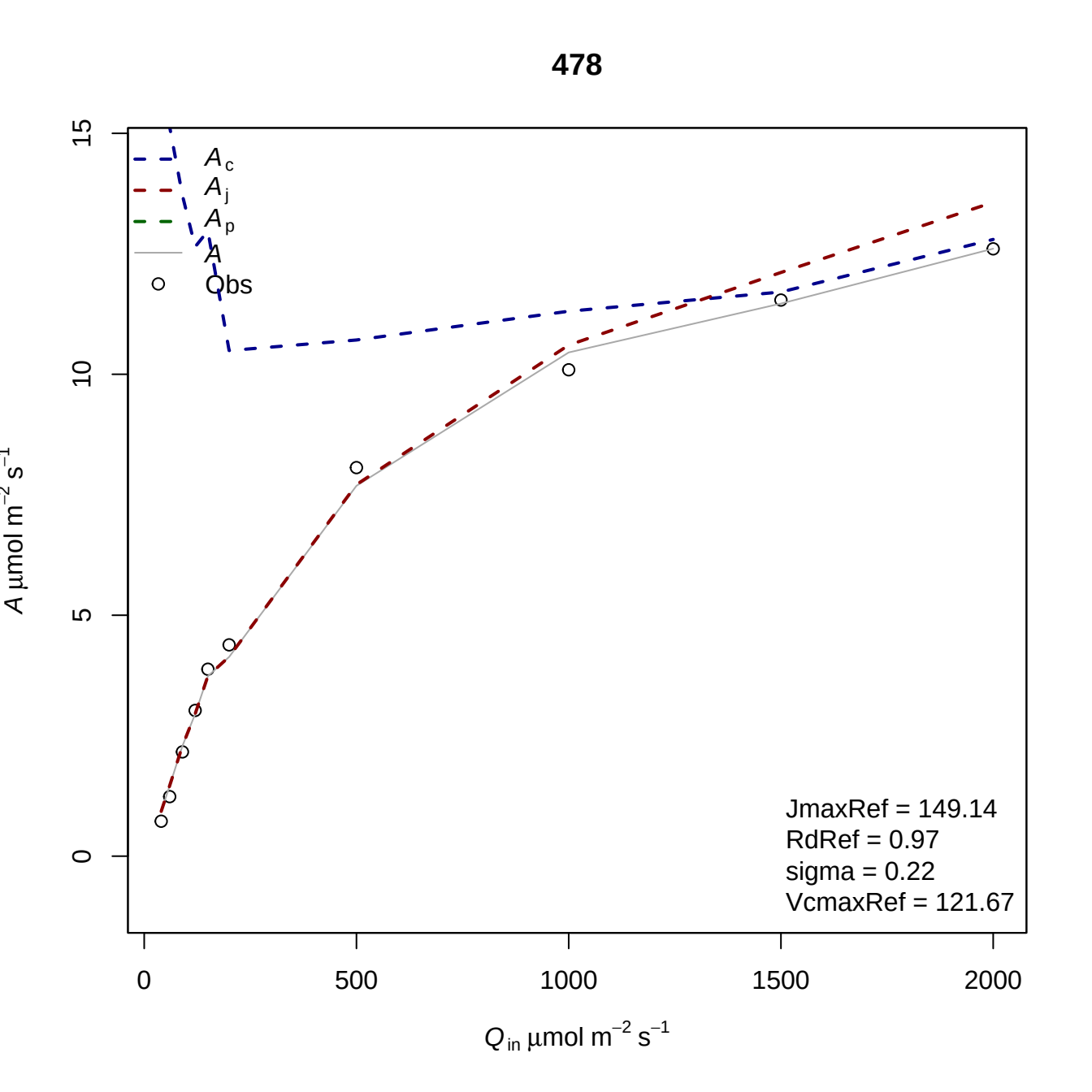

473
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.45
RdRef = 2.47
sigma = 0.42
VcmaxRef = 24828.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

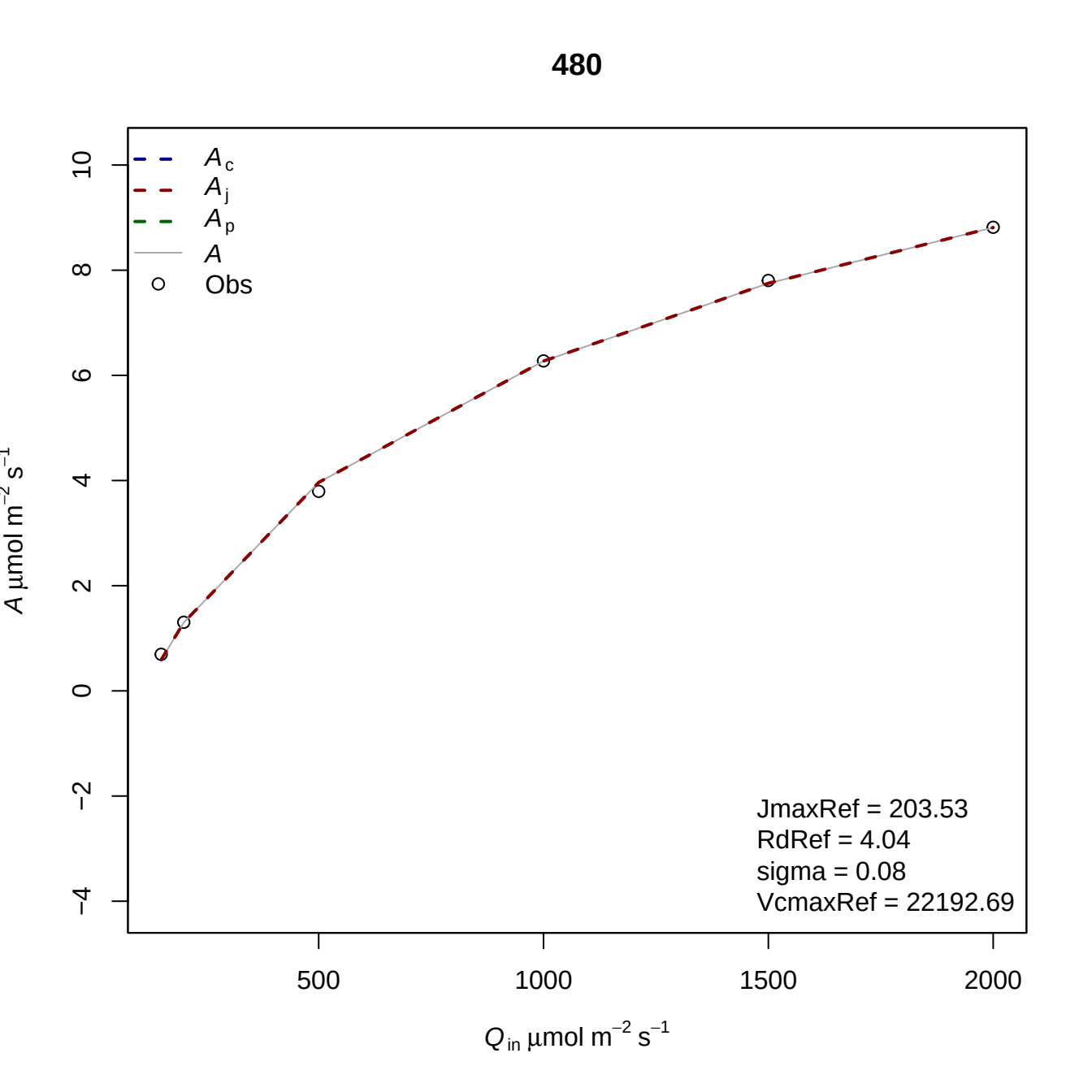

474
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.49
RdRef = 3.07
sigma = 0.67
VcmaxRef = 187066.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

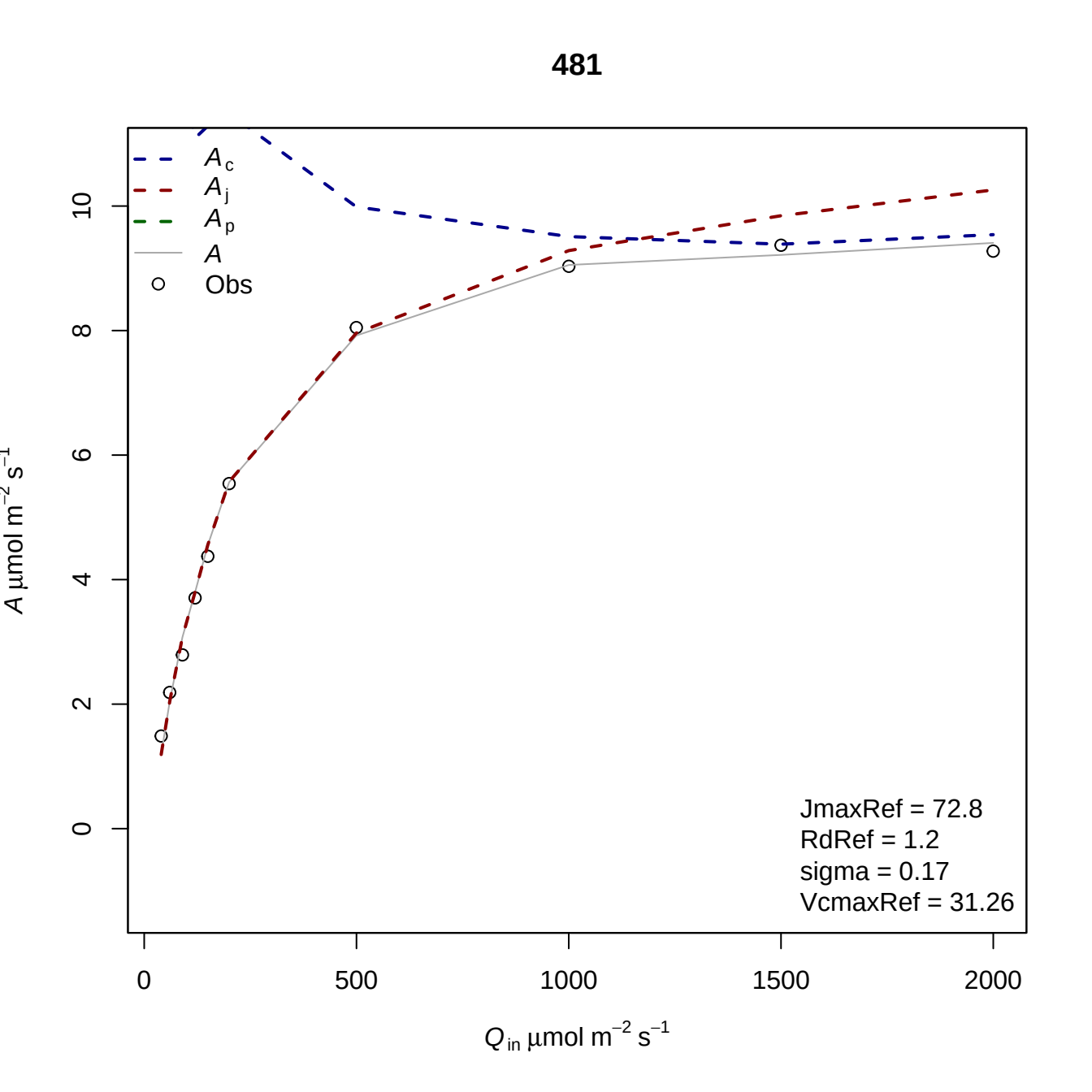

475
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.79
RdRef = 0.79
sigma = 0.07
VcmaxRef = 22.23
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

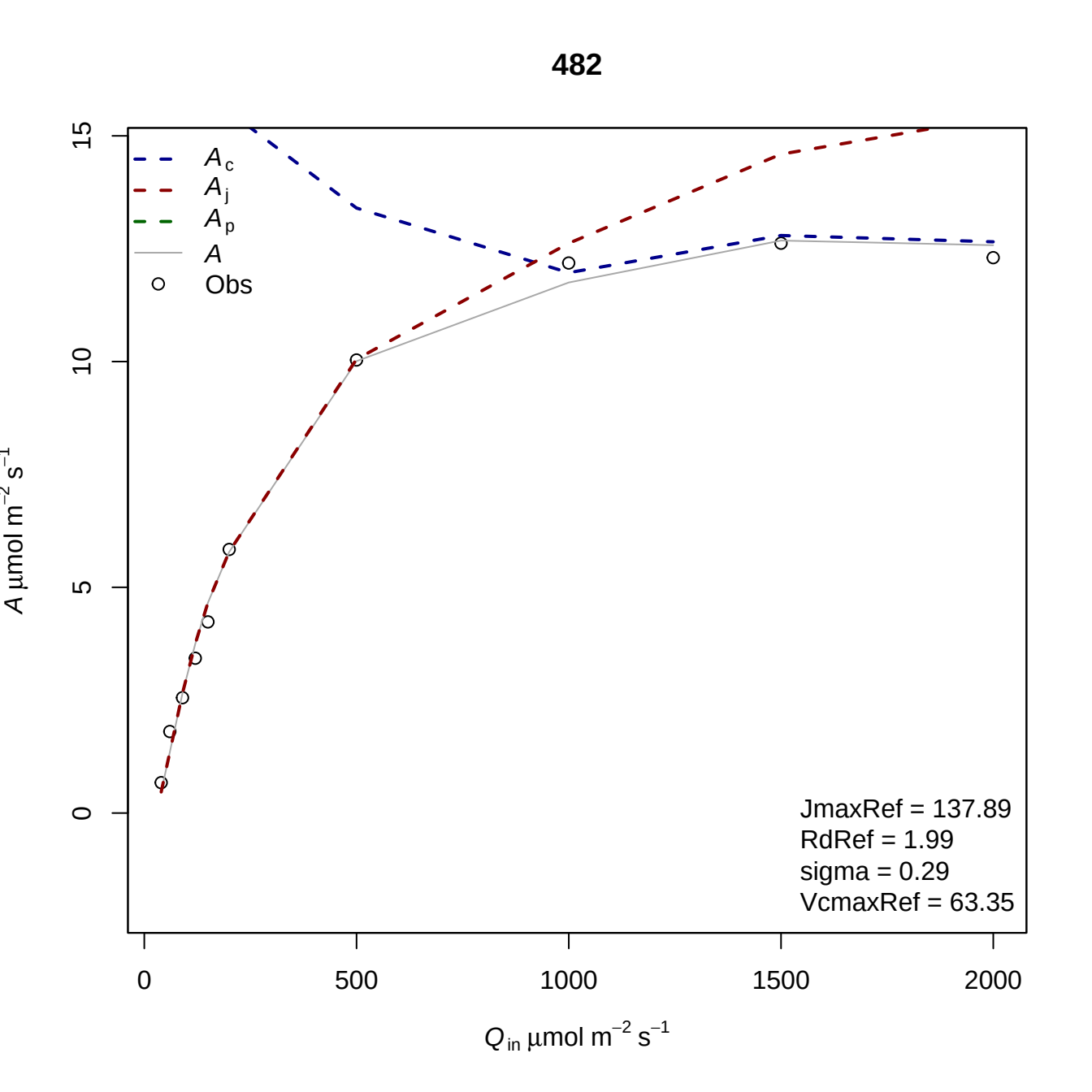

476
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.01
RdRef = 1.45
sigma = 0.19
VcmaxRef = 44.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

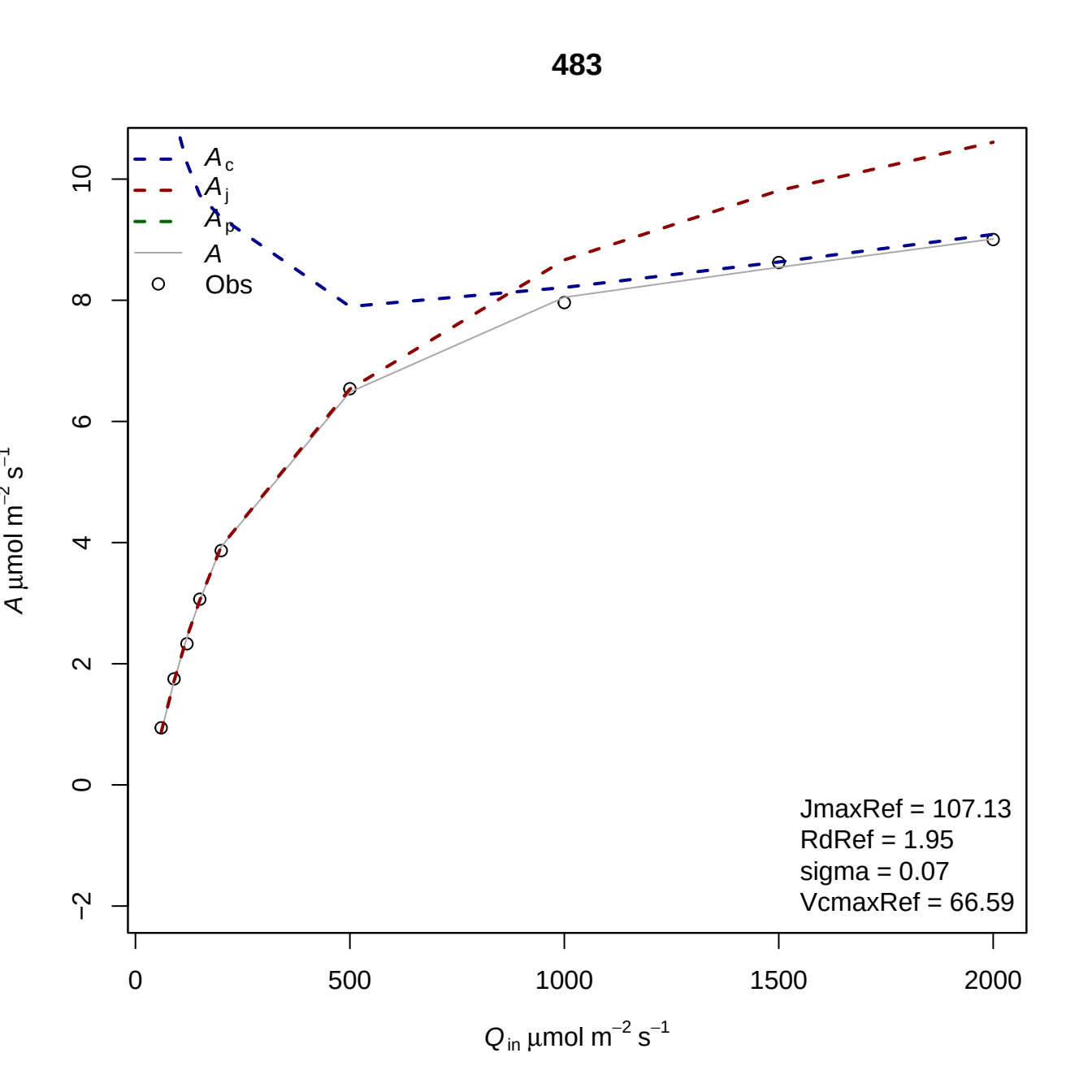

477
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.79
RdRef = 0.49
sigma = 0.25
VcmaxRef = 79.8
0
5
10
15
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

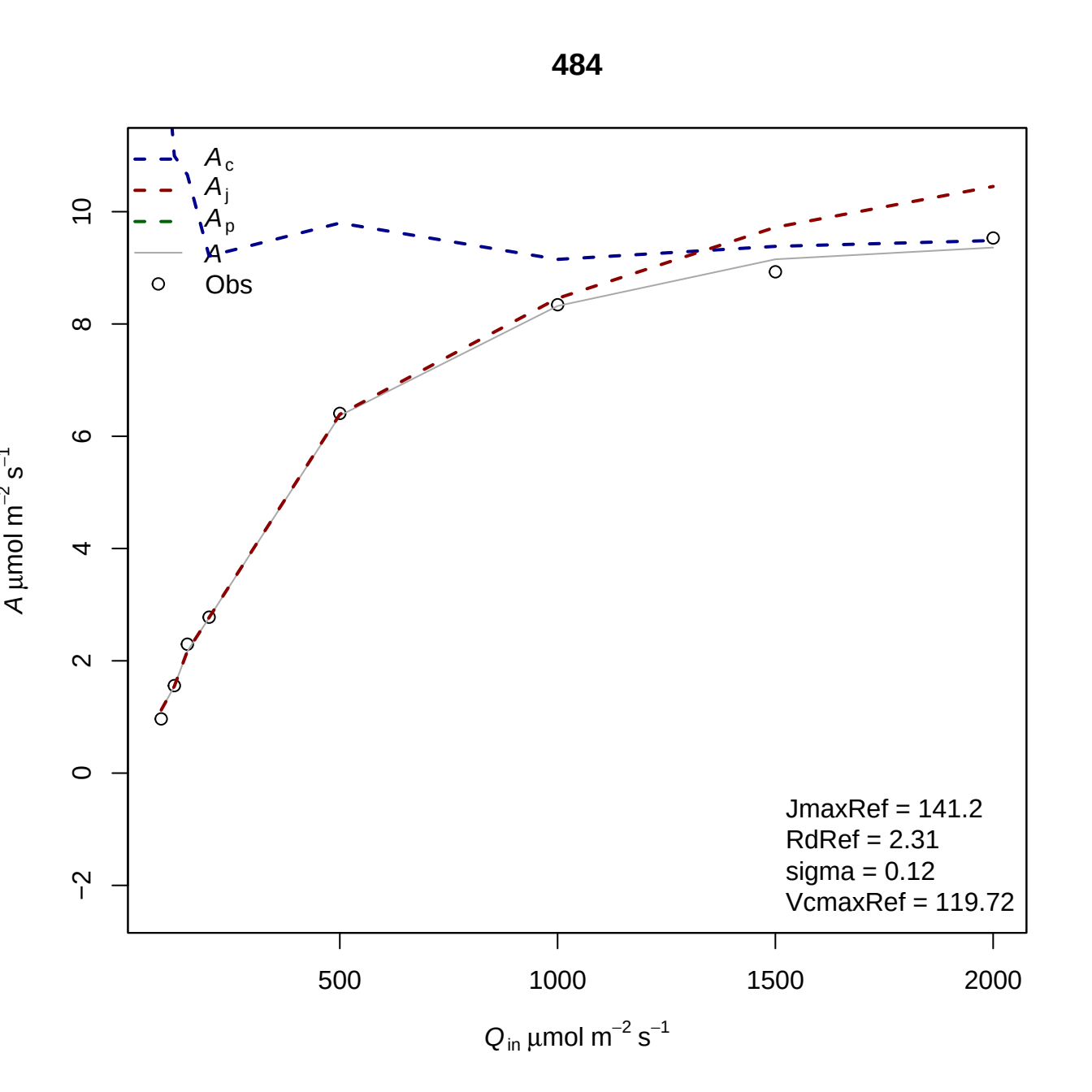

478
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 149.14
RdRef = 0.97
sigma = 0.22
VcmaxRef = 121.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

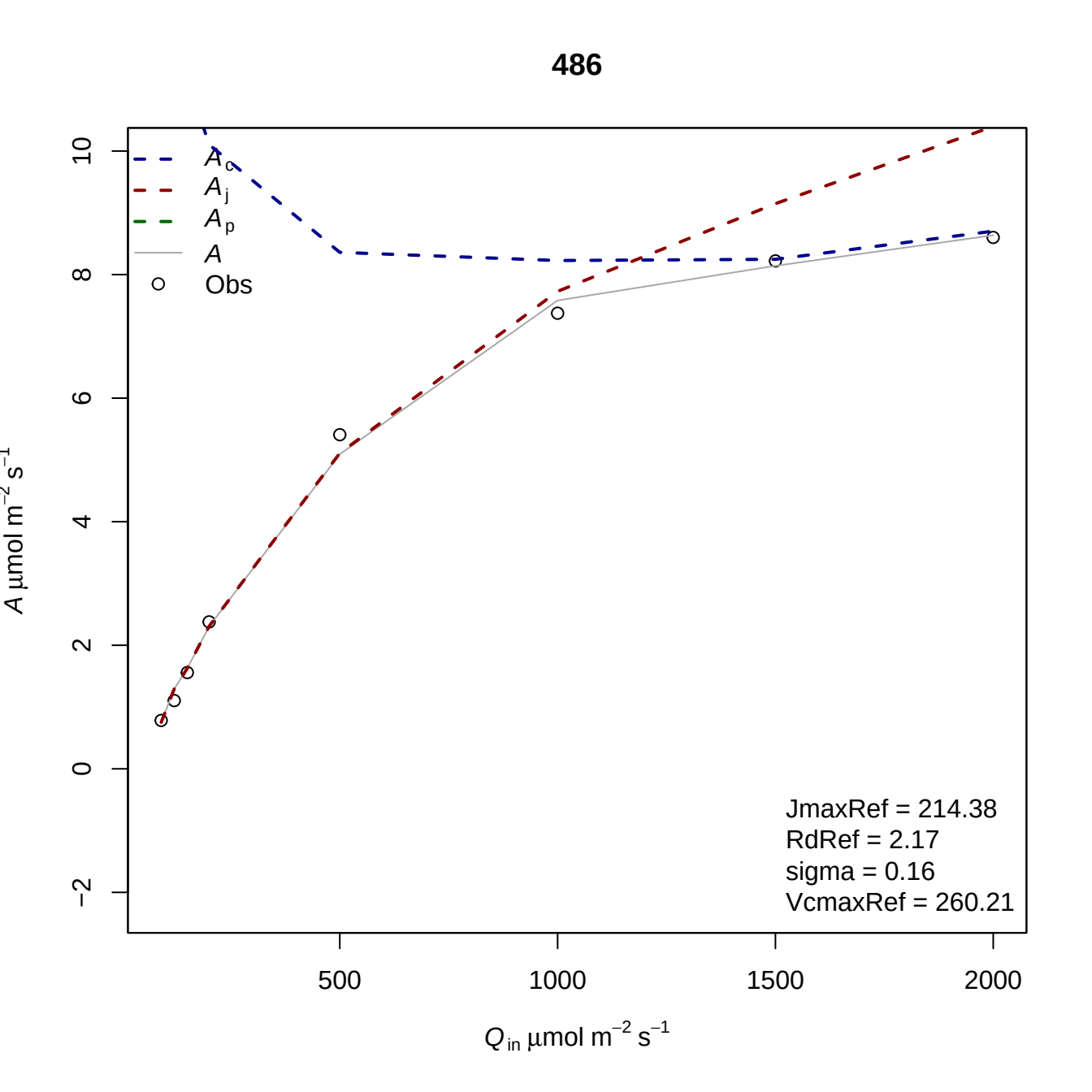

480
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 203.53
RdRef = 4.04
sigma = 0.08
VcmaxRef = 22192.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
500
1000
1500
2000

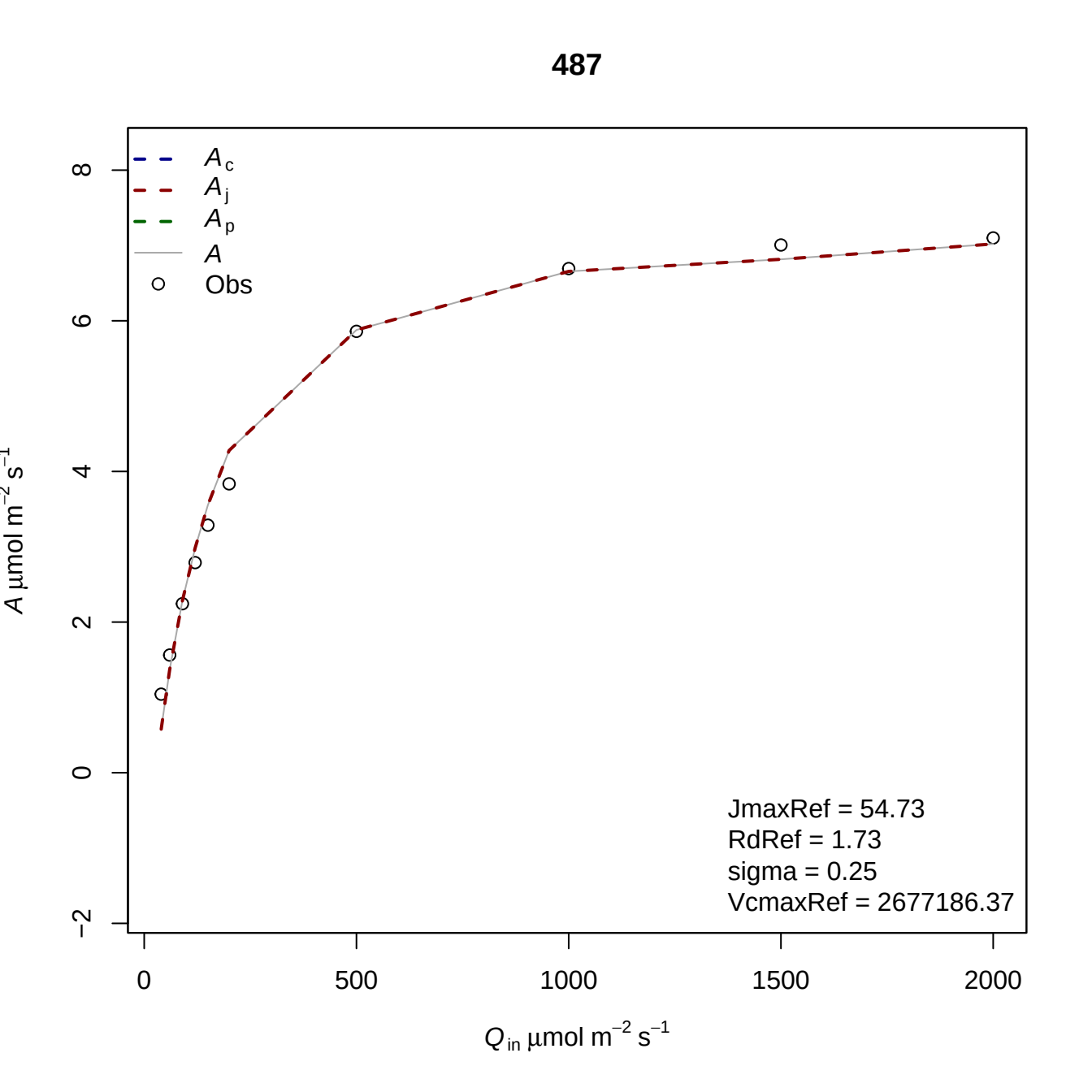

481
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.8
RdRef = 1.2
sigma = 0.17
VcmaxRef = 31.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

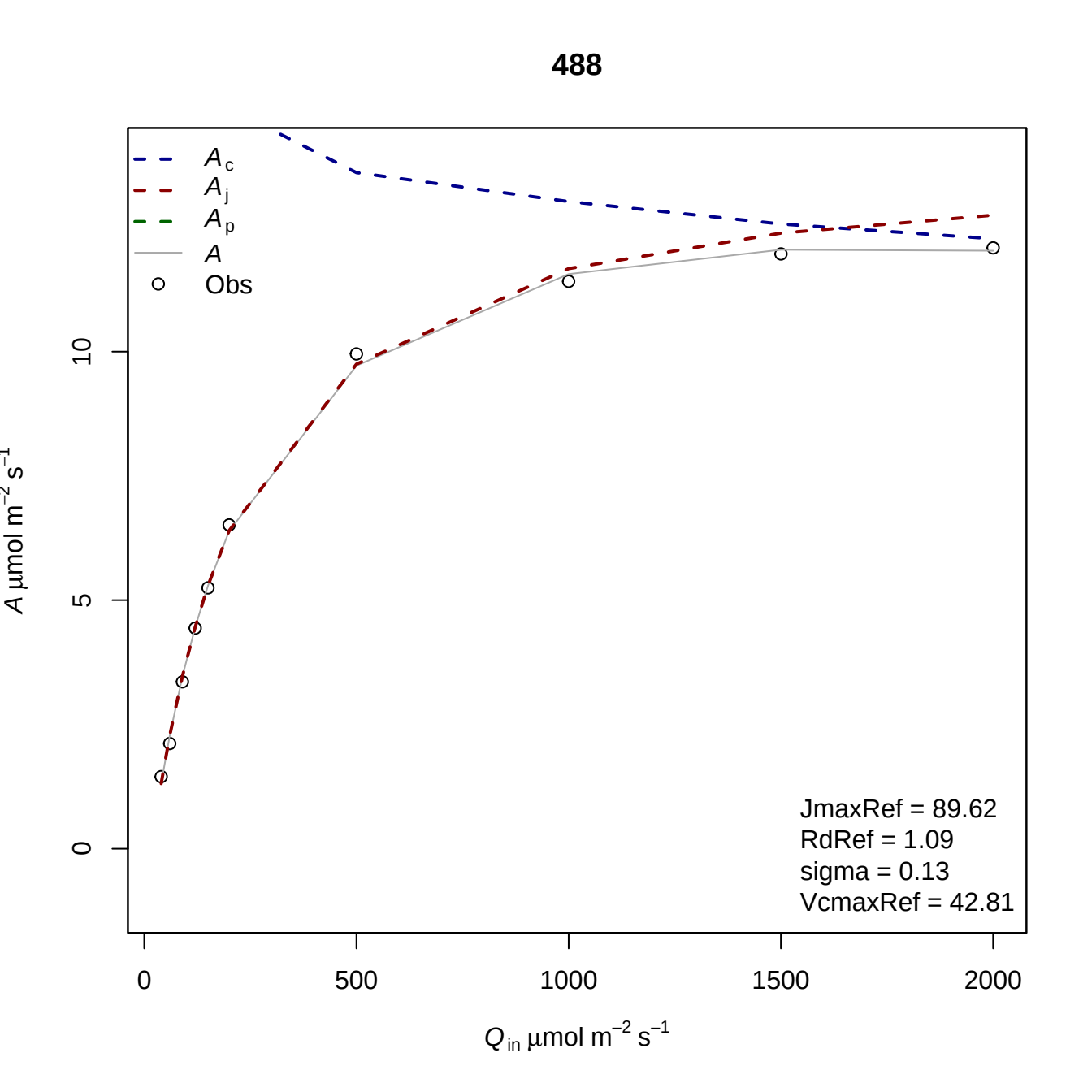

482
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.89
RdRef = 1.99
sigma = 0.29
VcmaxRef = 63.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

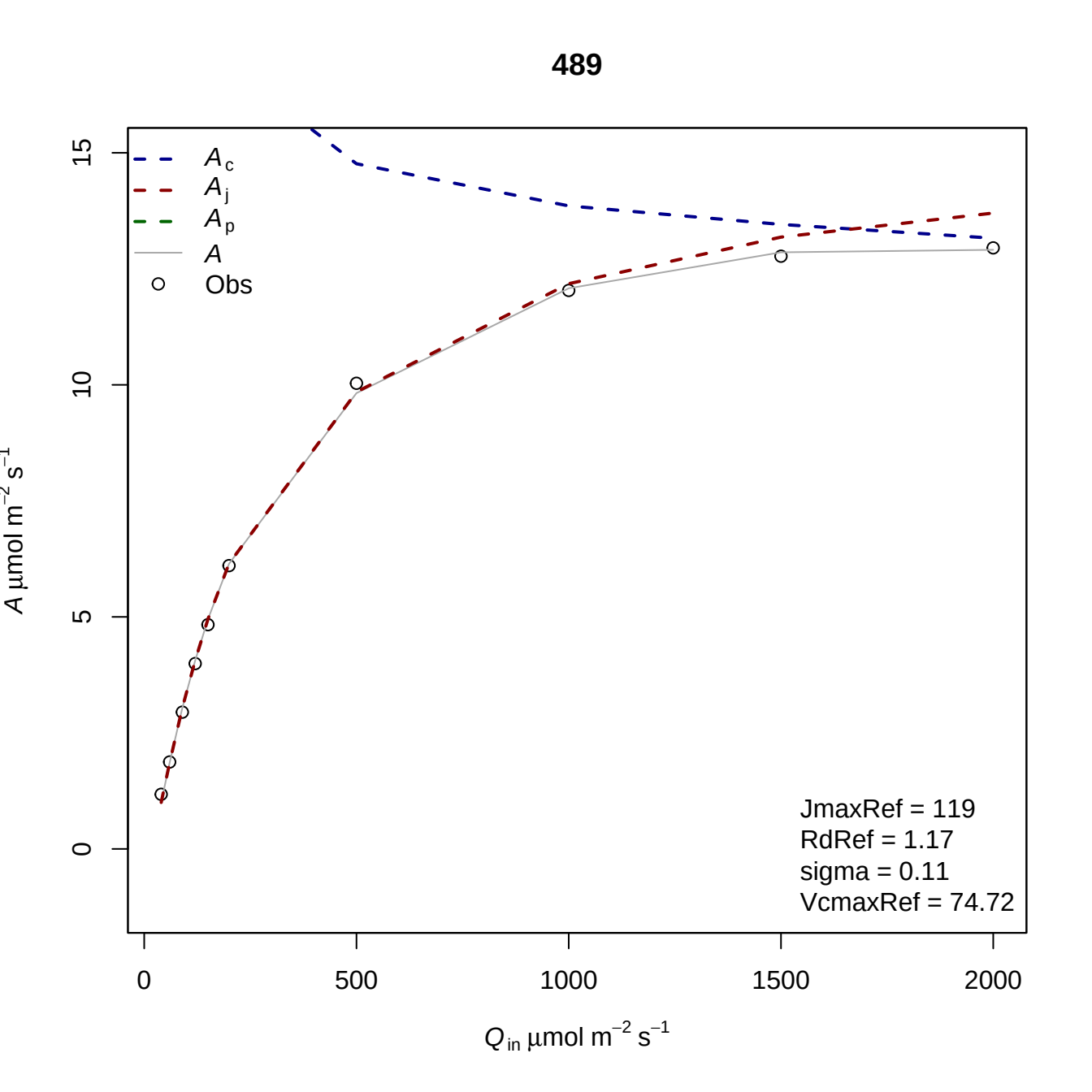

483
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.13
RdRef = 1.95
sigma = 0.07
VcmaxRef = 66.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

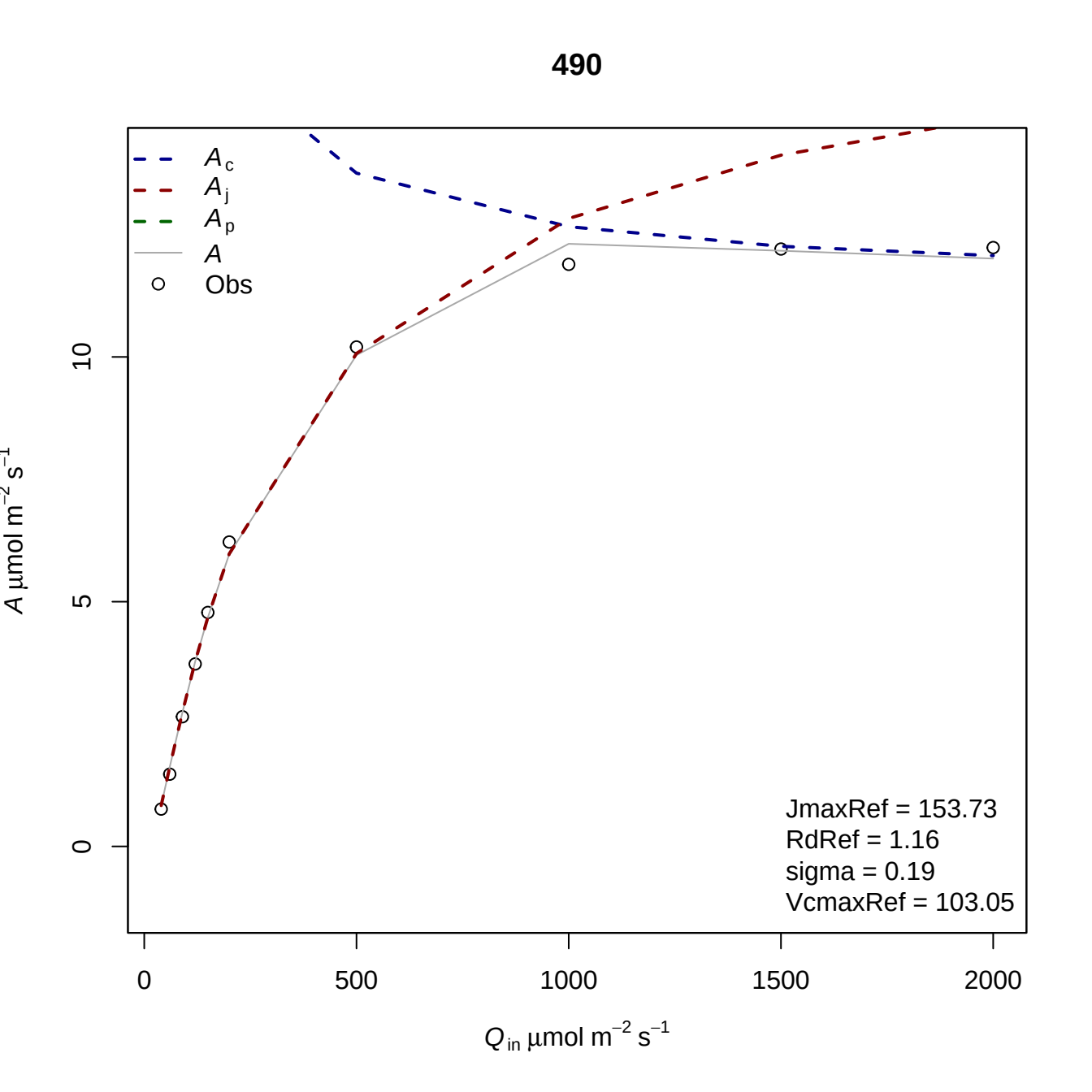

484
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 141.2
RdRef = 2.31
sigma = 0.12
VcmaxRef = 119.72
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
500
1000
1500
2000

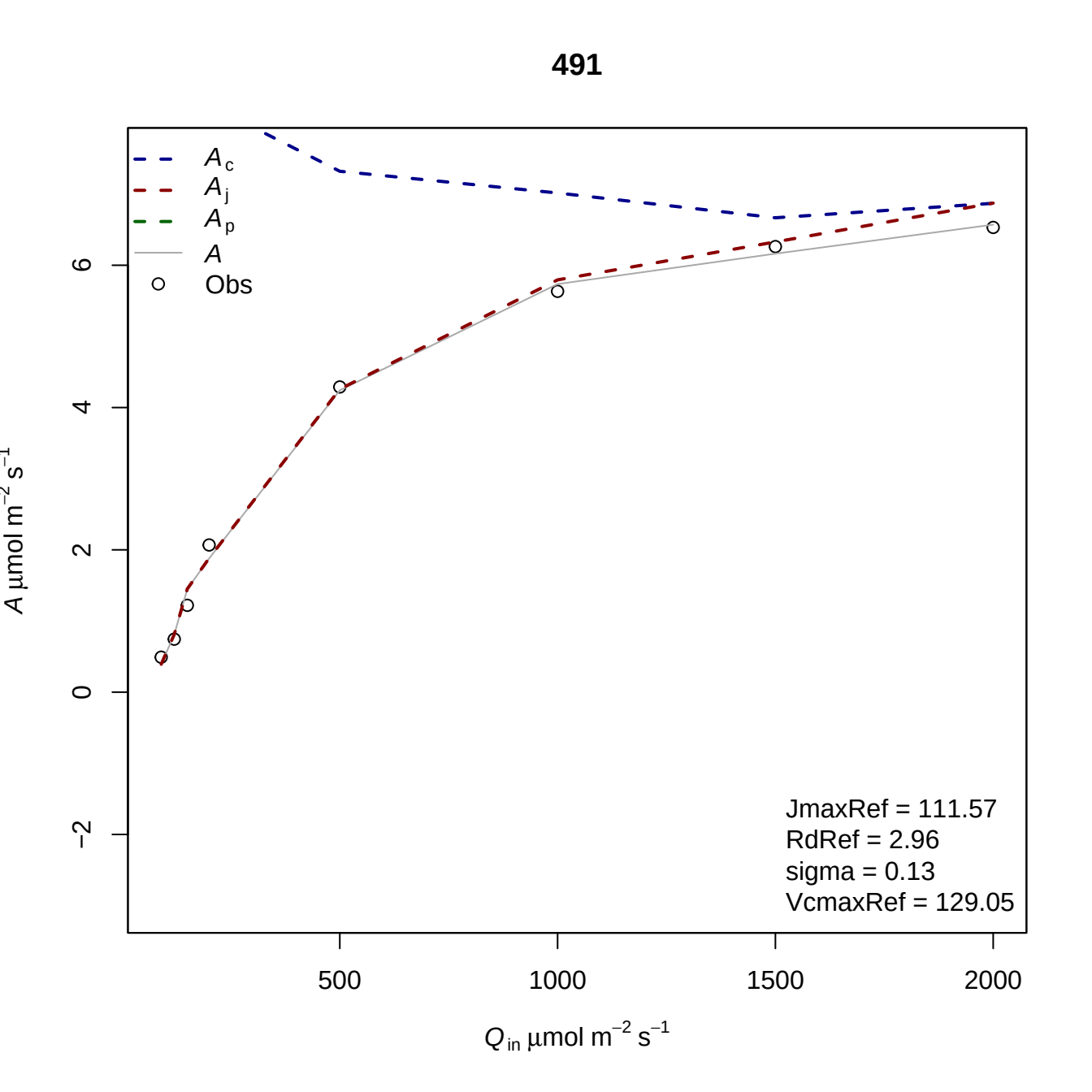

486
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 214.38
RdRef = 2.17
sigma = 0.16
VcmaxRef = 260.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
500
1000
1500
2000

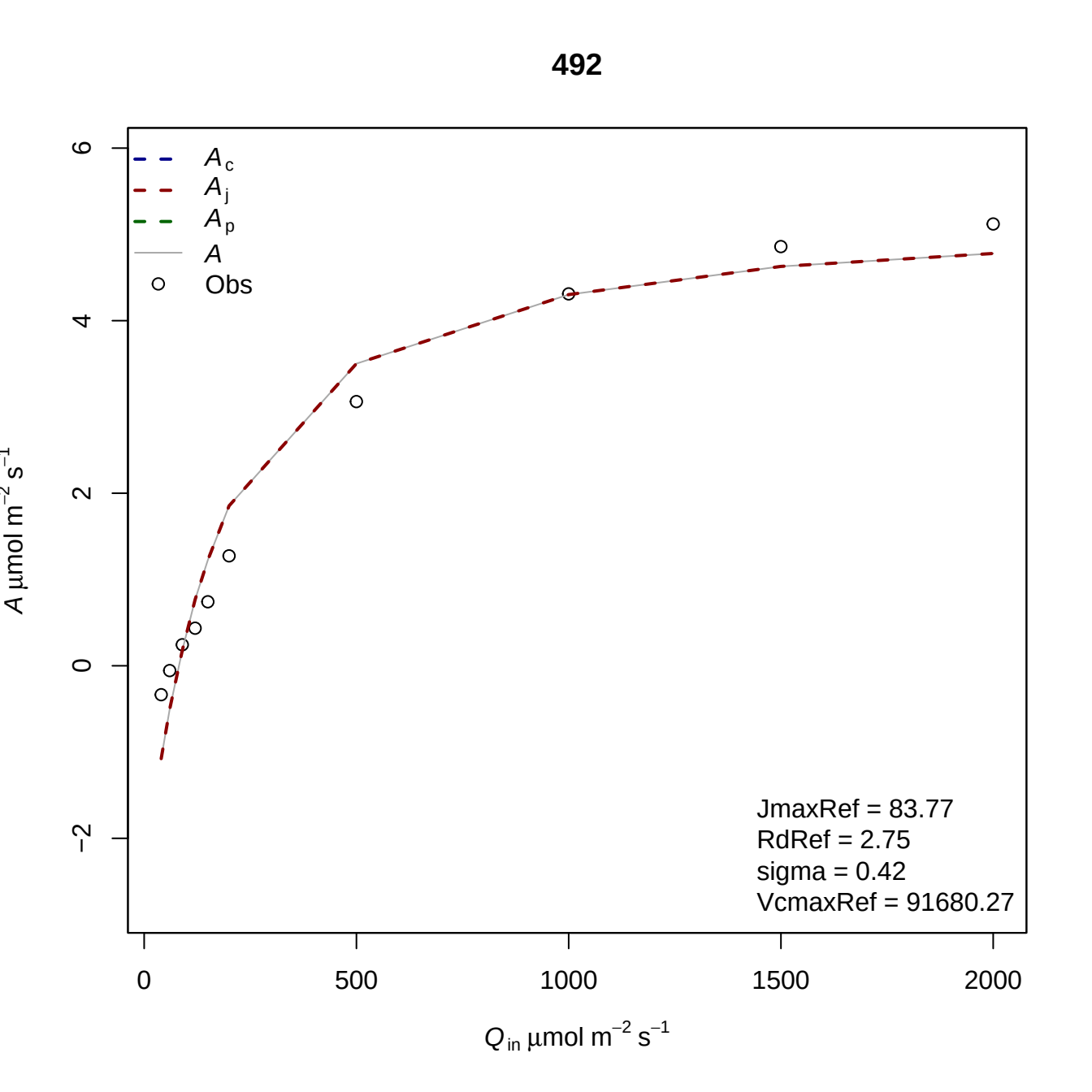

487
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.73
RdRef = 1.73
sigma = 0.25
VcmaxRef = 2677186.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

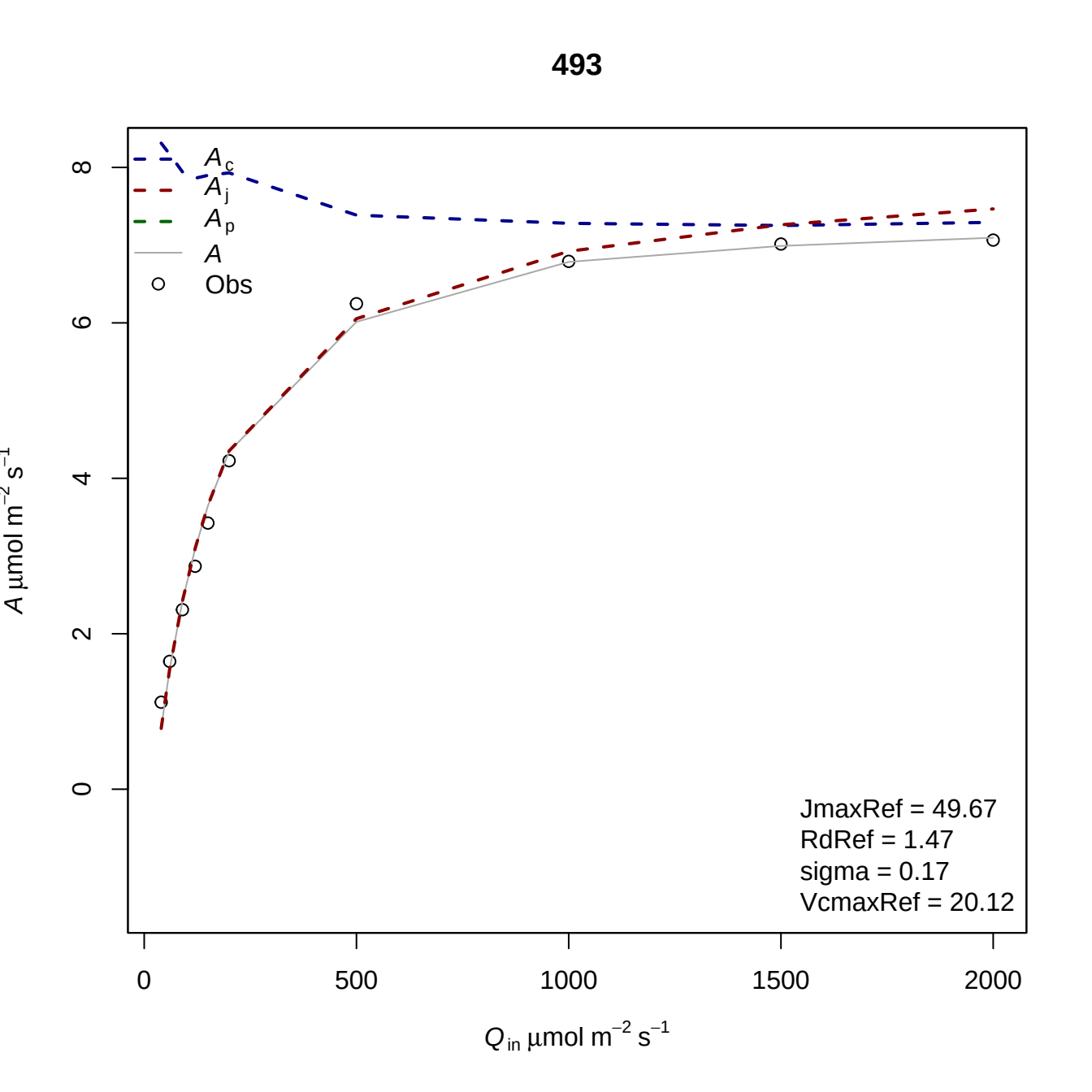

488
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.62
RdRef = 1.09
sigma = 0.13
VcmaxRef = 42.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

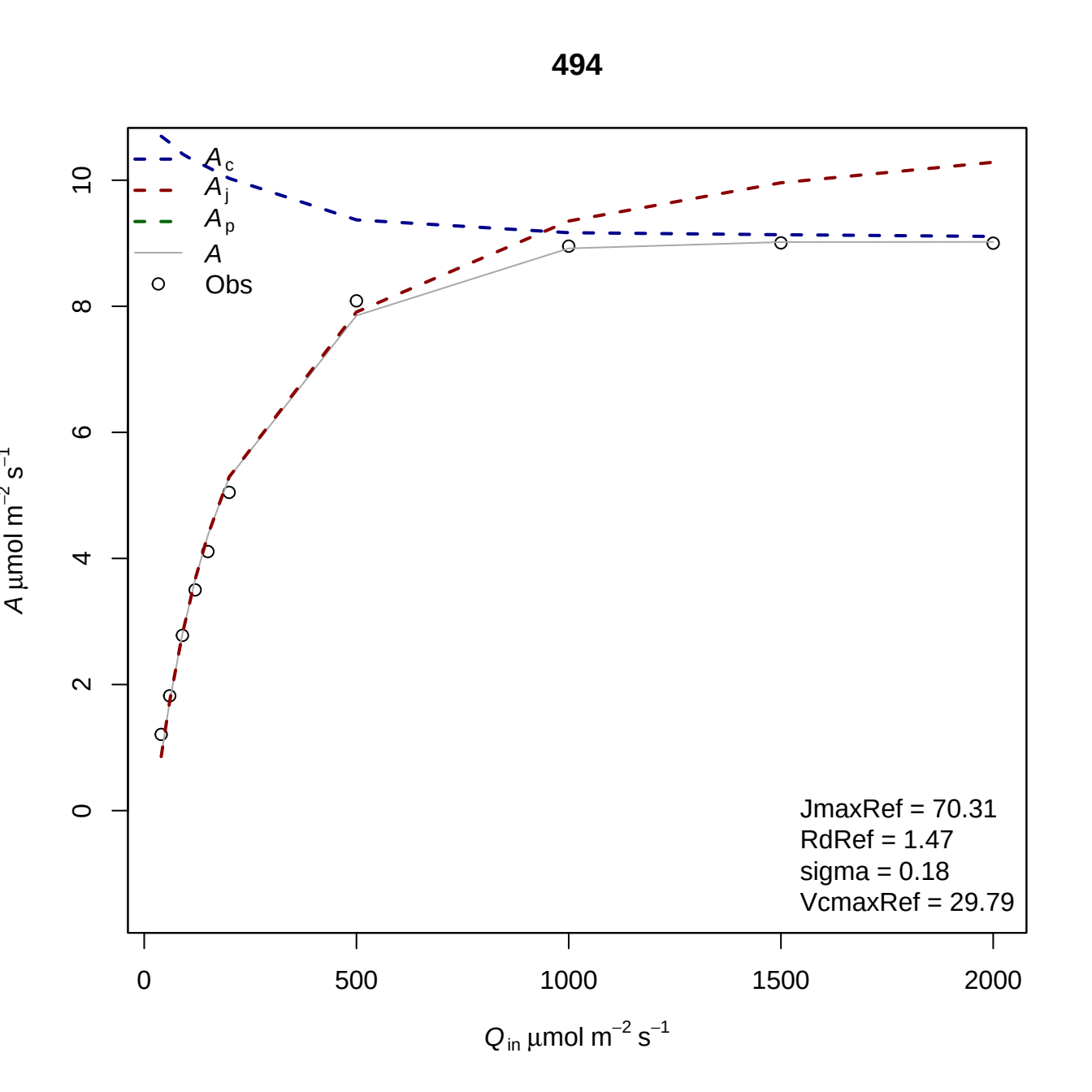

489
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119
RdRef = 1.17
sigma = 0.11
VcmaxRef = 74.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

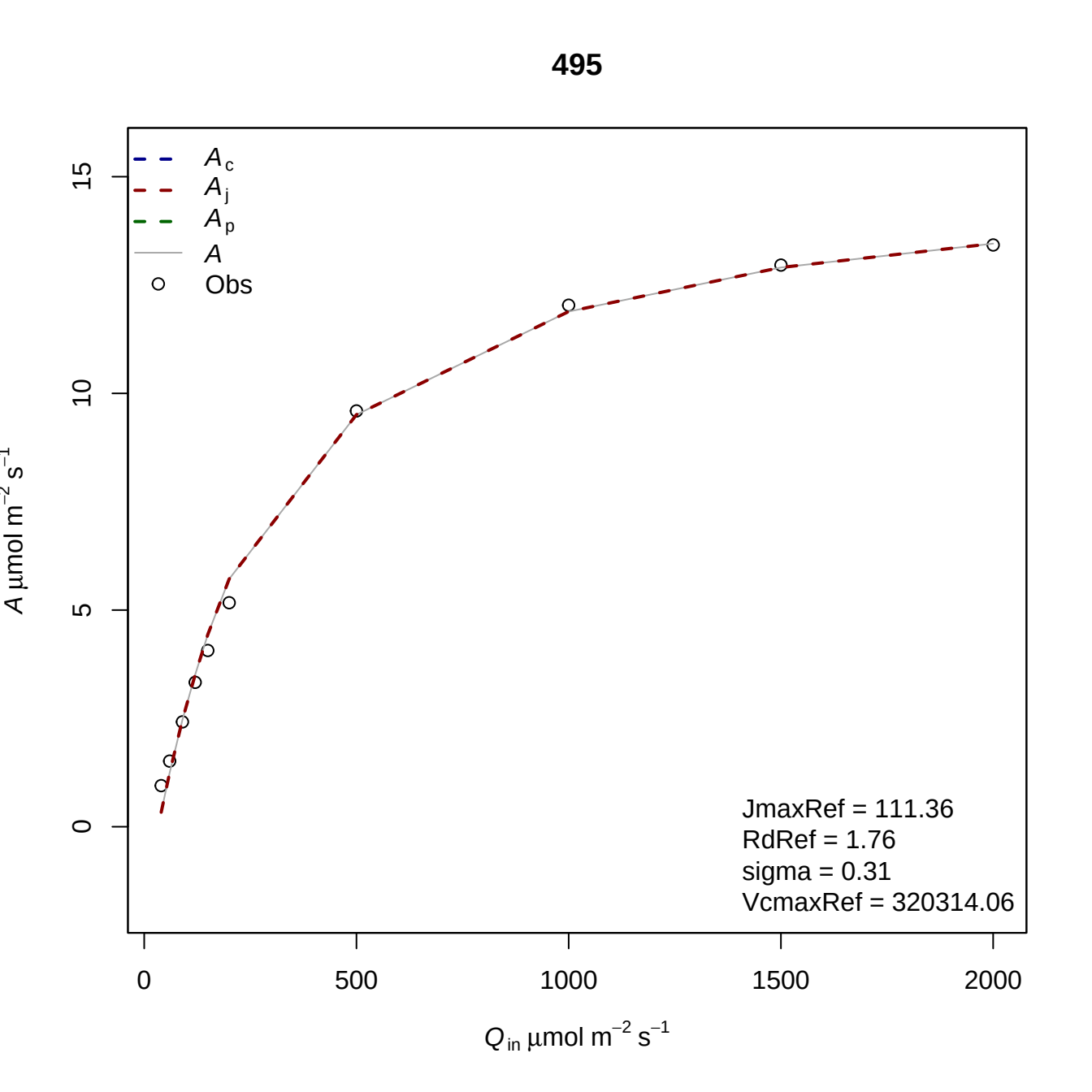

490
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.73
RdRef = 1.16
sigma = 0.19
VcmaxRef = 103.05
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

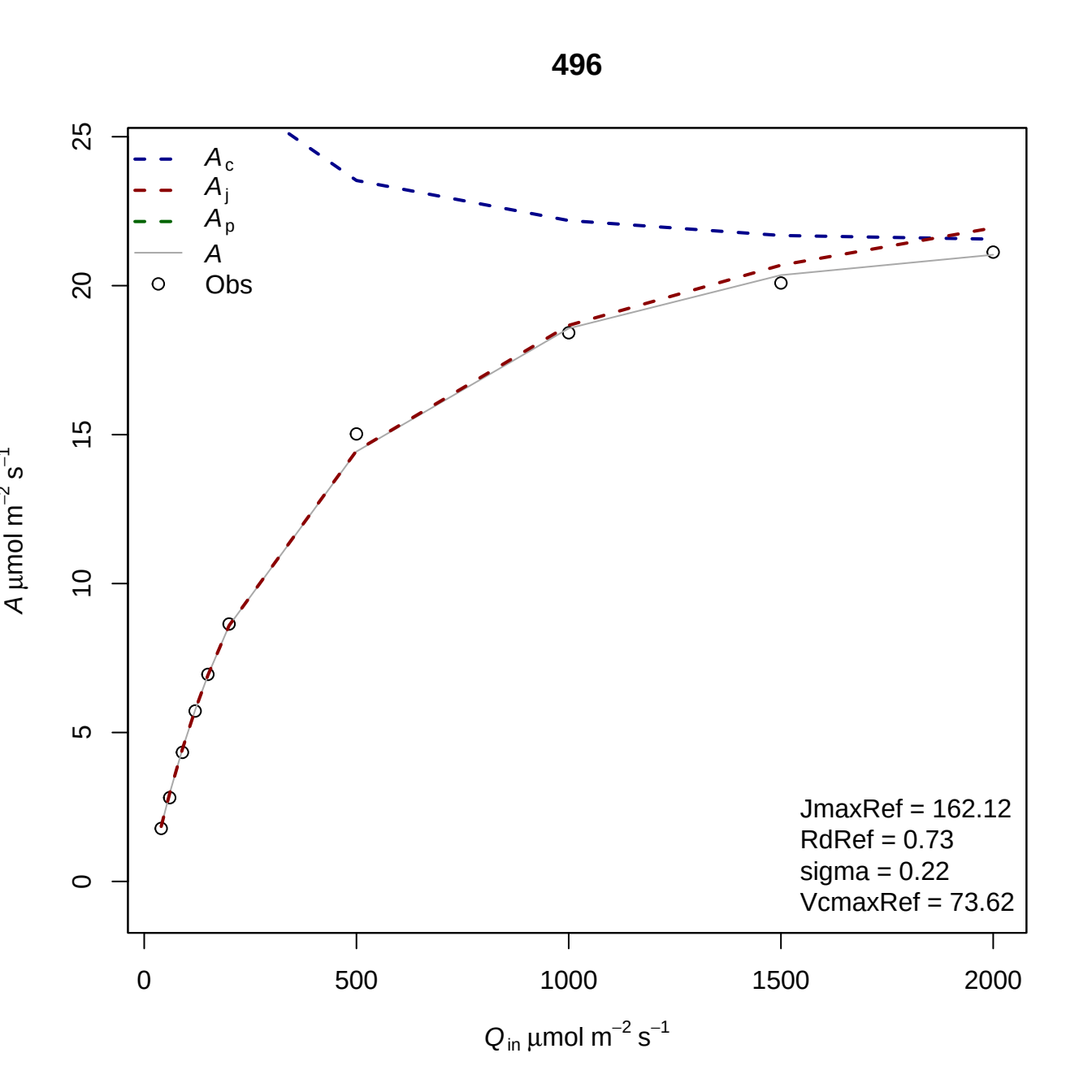

491
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.57
RdRef = 2.96
sigma = 0.13
VcmaxRef = 129.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
500
1000
1500
2000

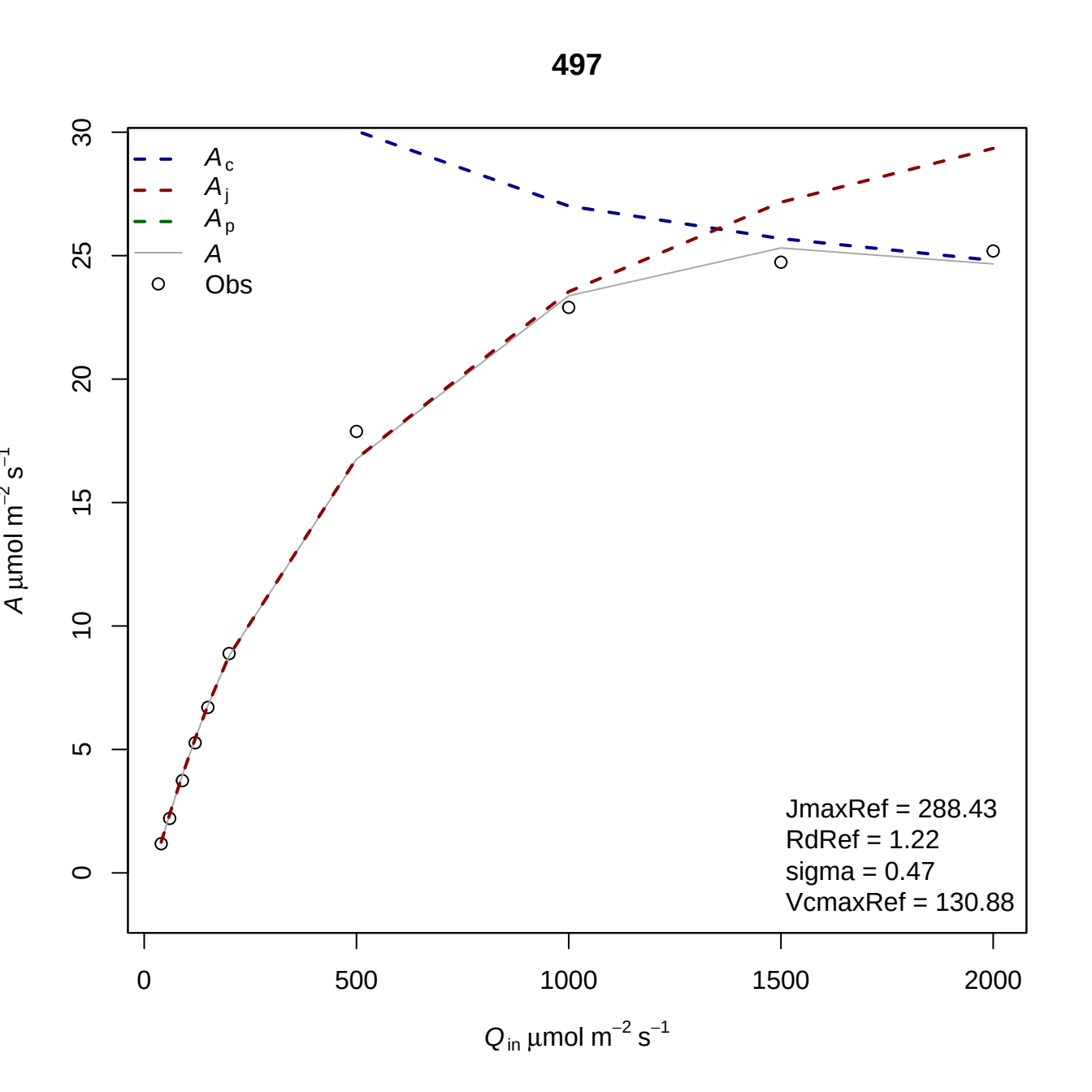

492
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.77
RdRef = 2.75
sigma = 0.42
VcmaxRef = 91680.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500
2000

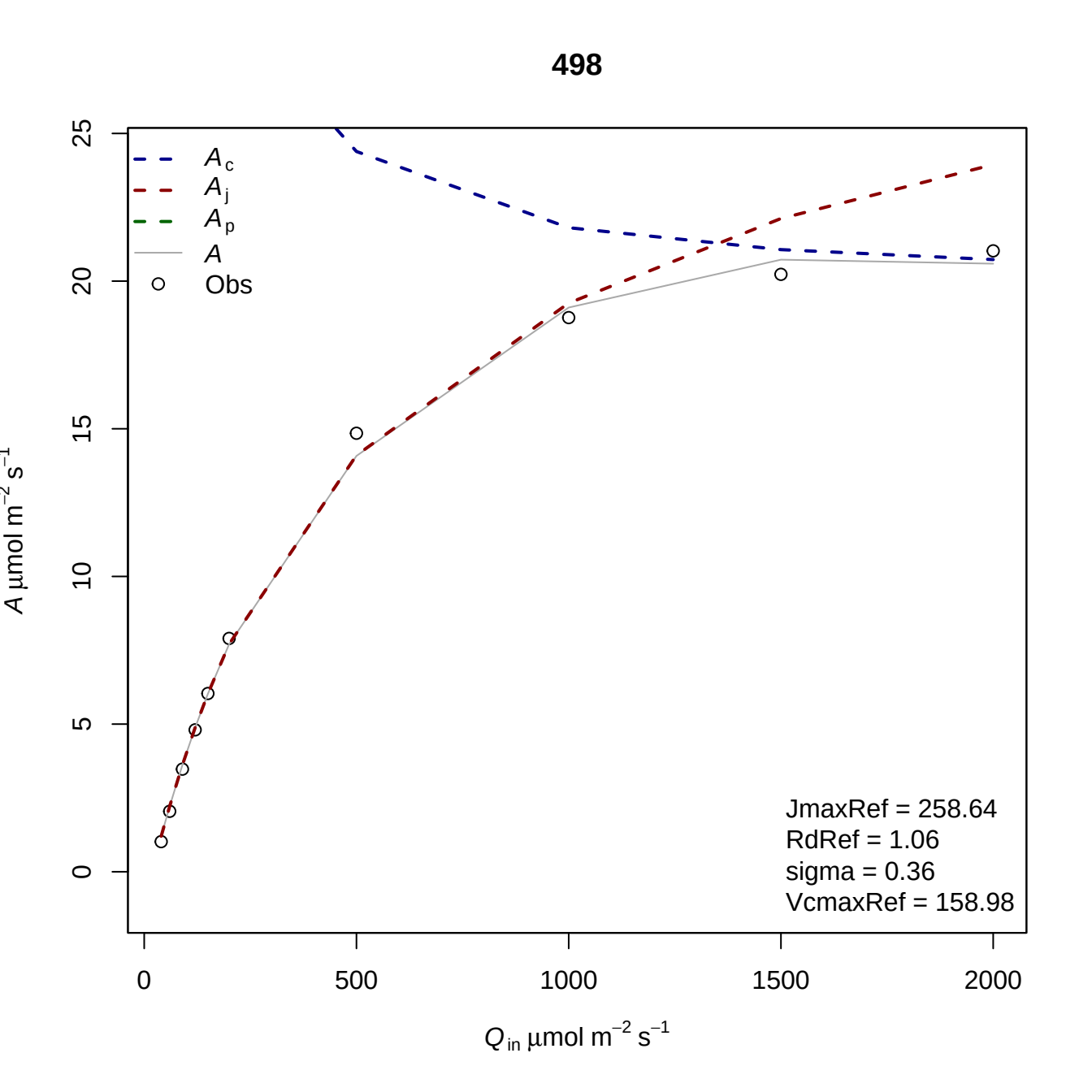

493
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.67
RdRef = 1.47
sigma = 0.17
VcmaxRef = 20.12
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

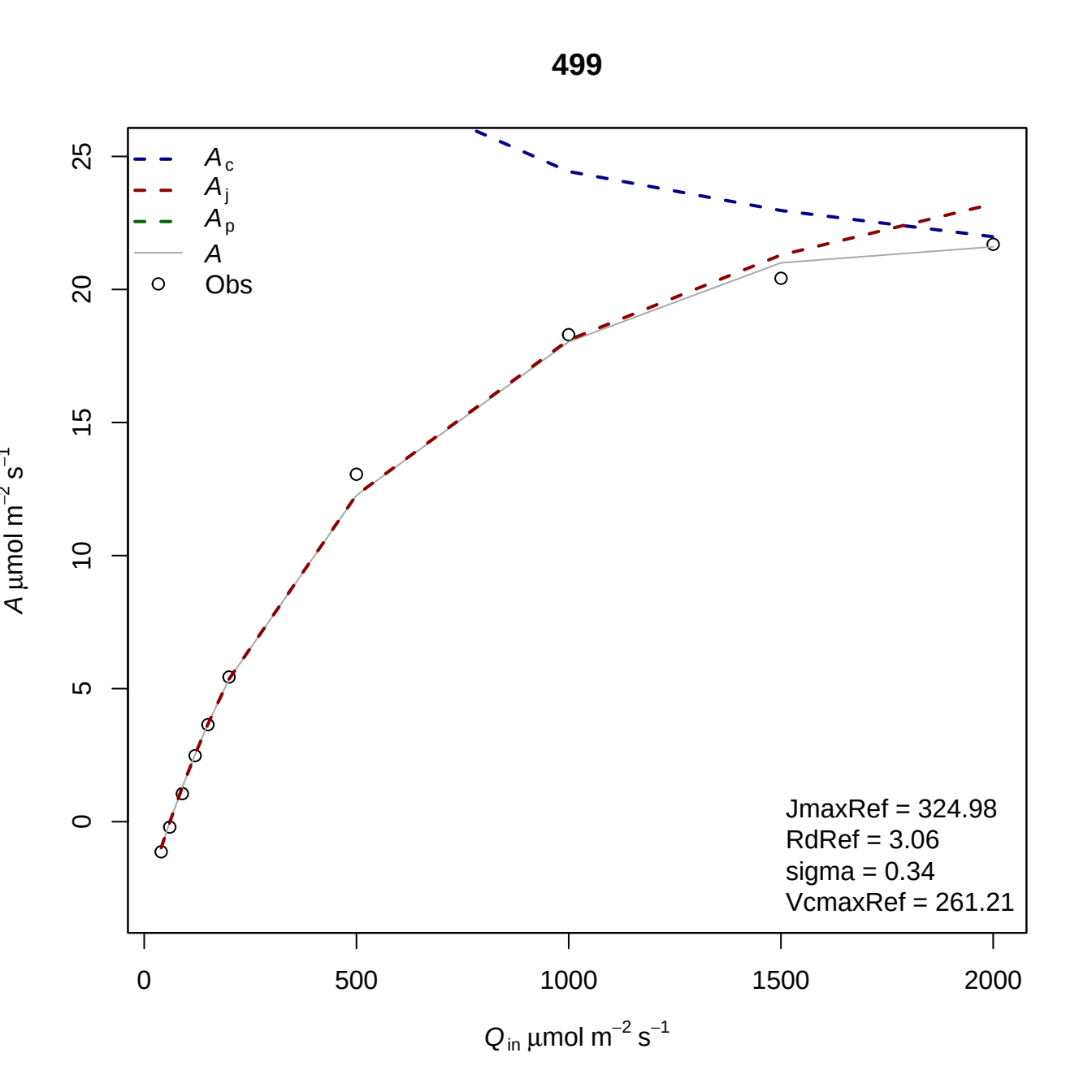

494
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.31
RdRef = 1.47
sigma = 0.18
VcmaxRef = 29.79
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

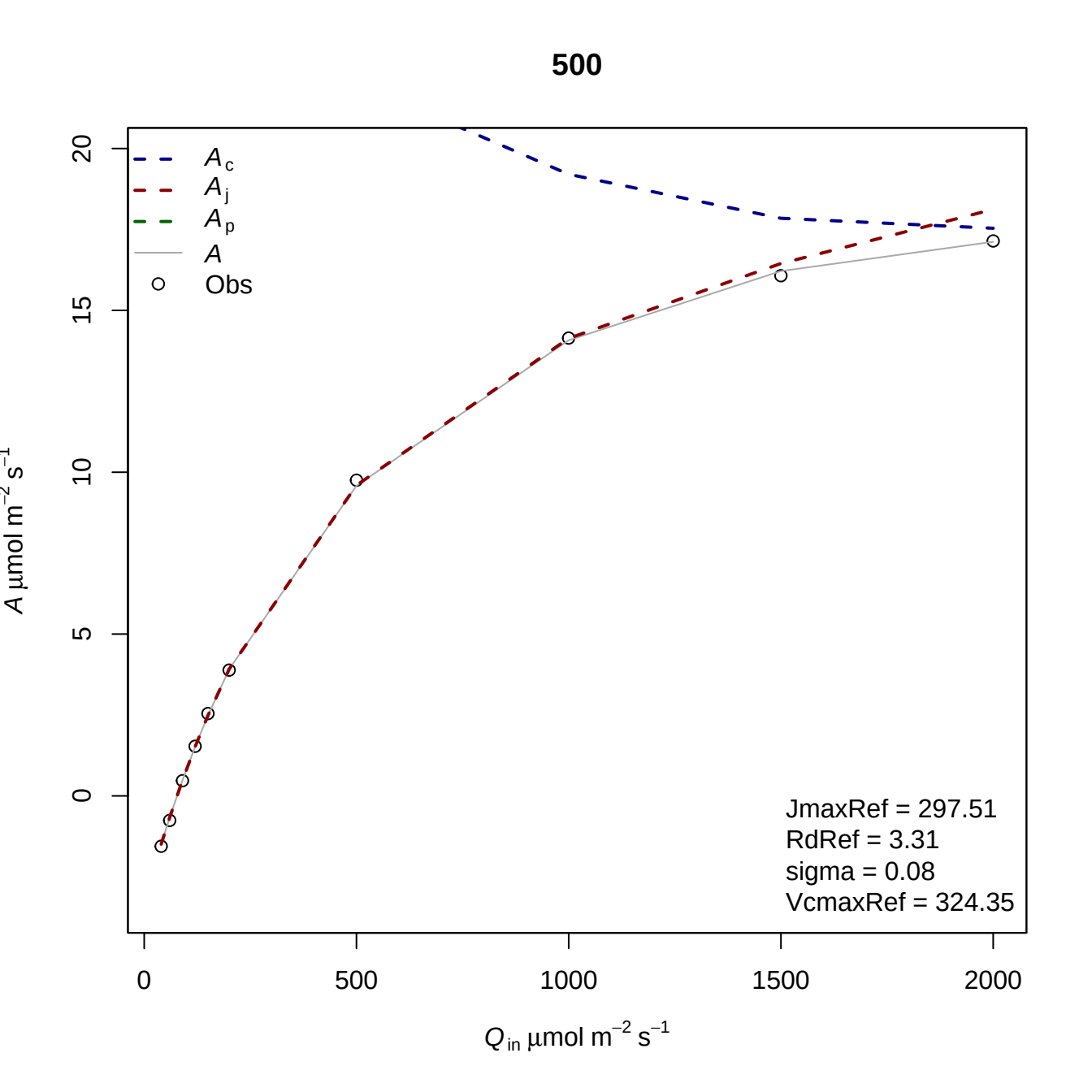

495
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.36
RdRef = 1.76
sigma = 0.31
VcmaxRef = 320314.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

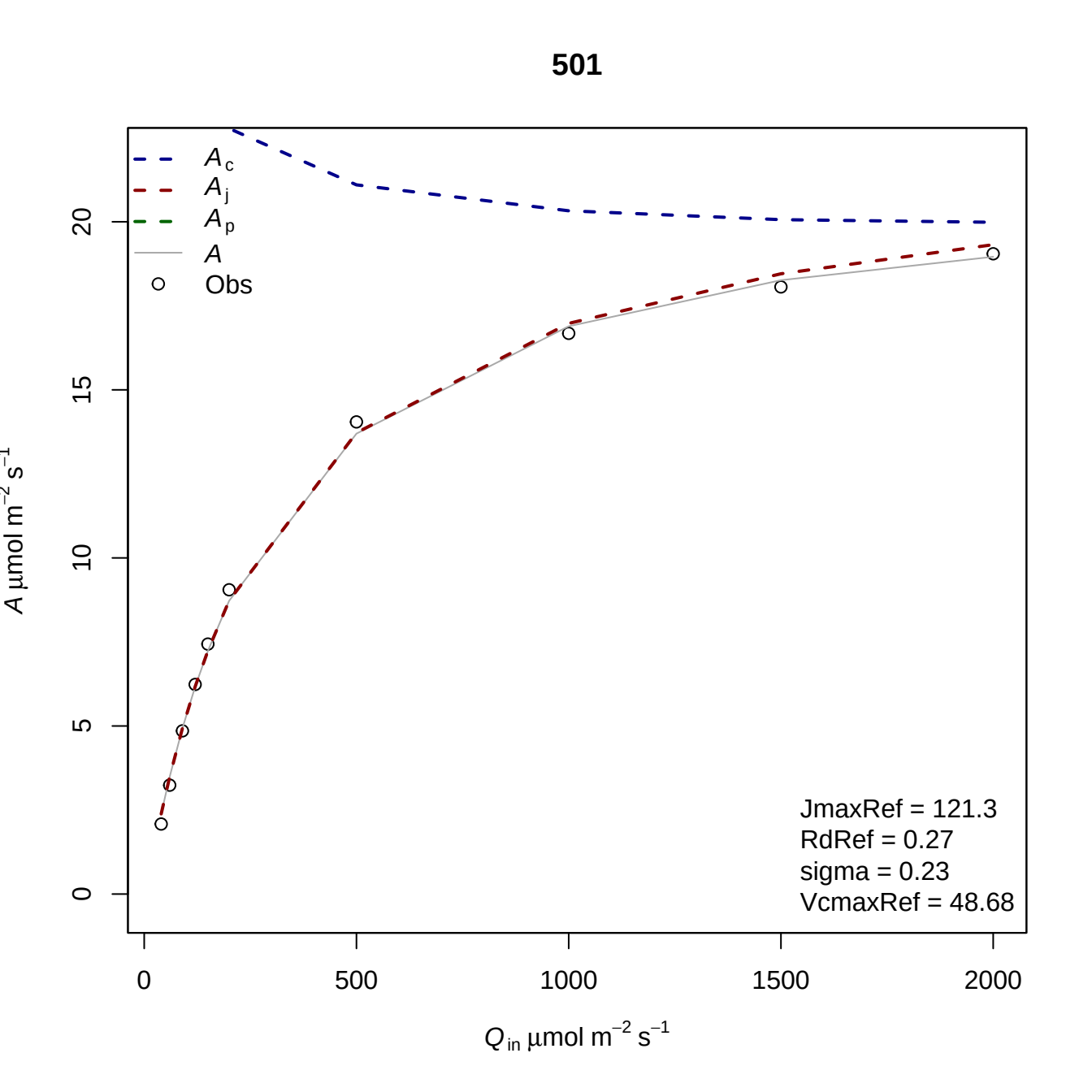

496
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 162.12
RdRef = 0.73
sigma = 0.22
VcmaxRef = 73.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

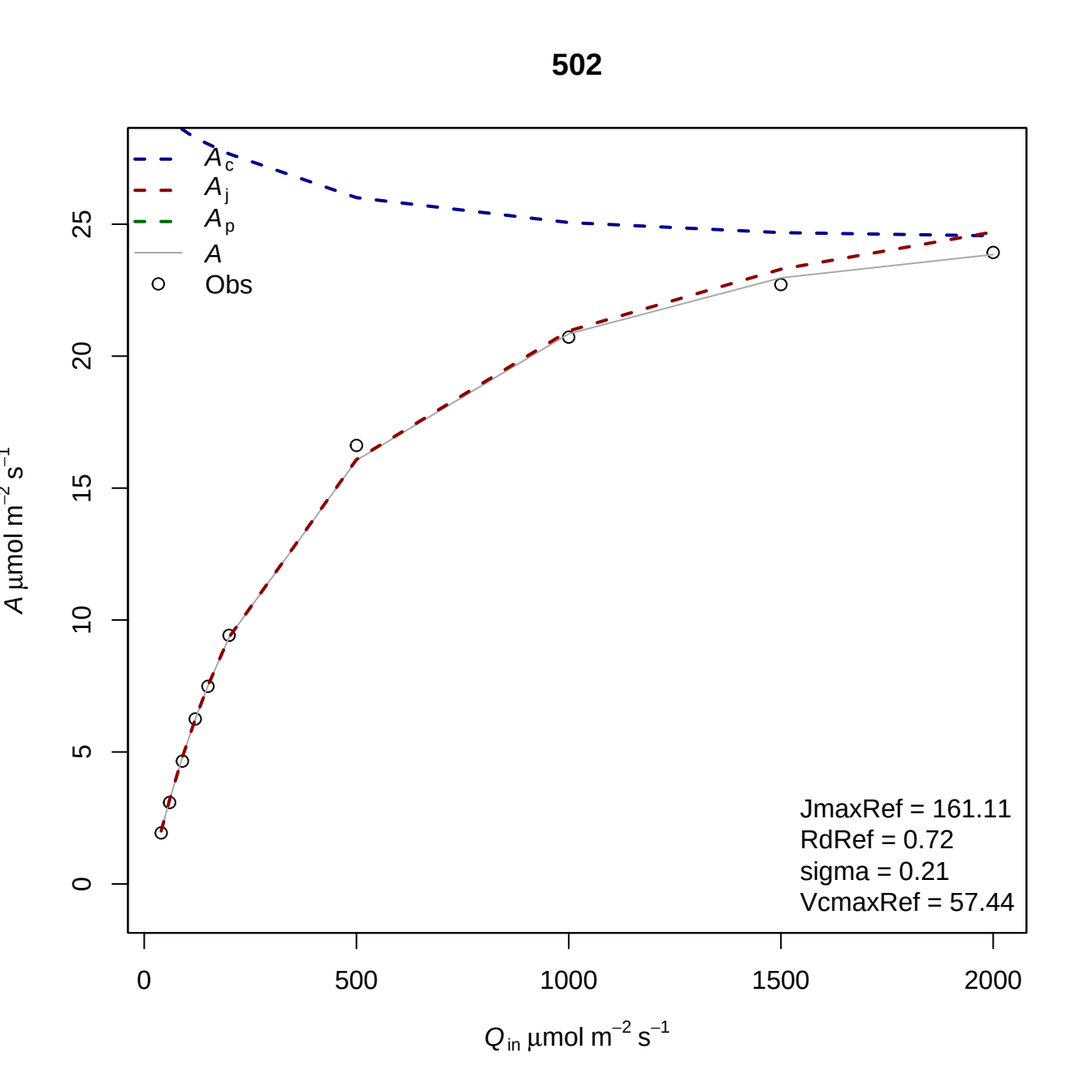

497
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 288.43
RdRef = 1.22
sigma = 0.47
VcmaxRef = 130.88
0
5
10
15
20
25
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

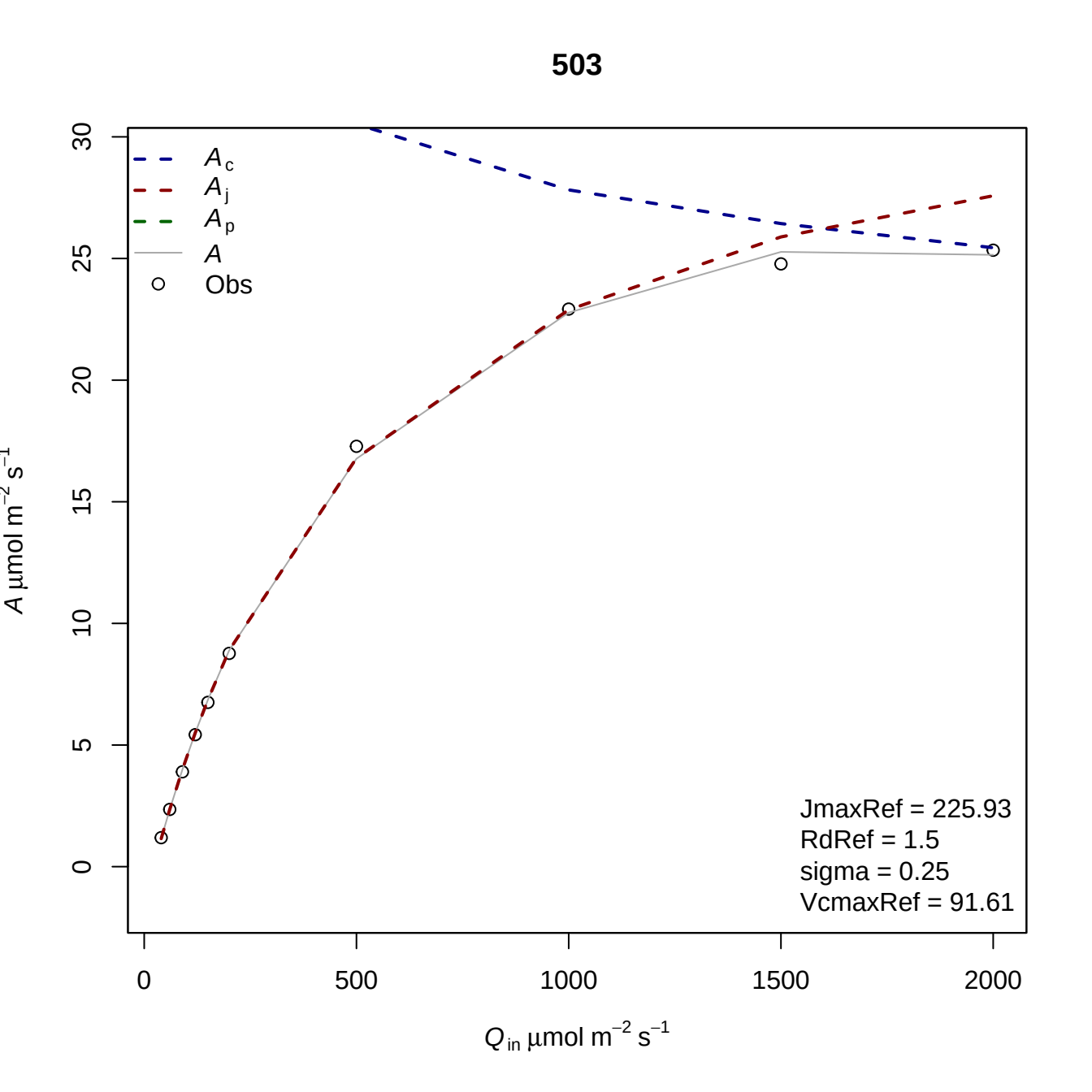

498
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 258.64
RdRef = 1.06
sigma = 0.36
VcmaxRef = 158.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

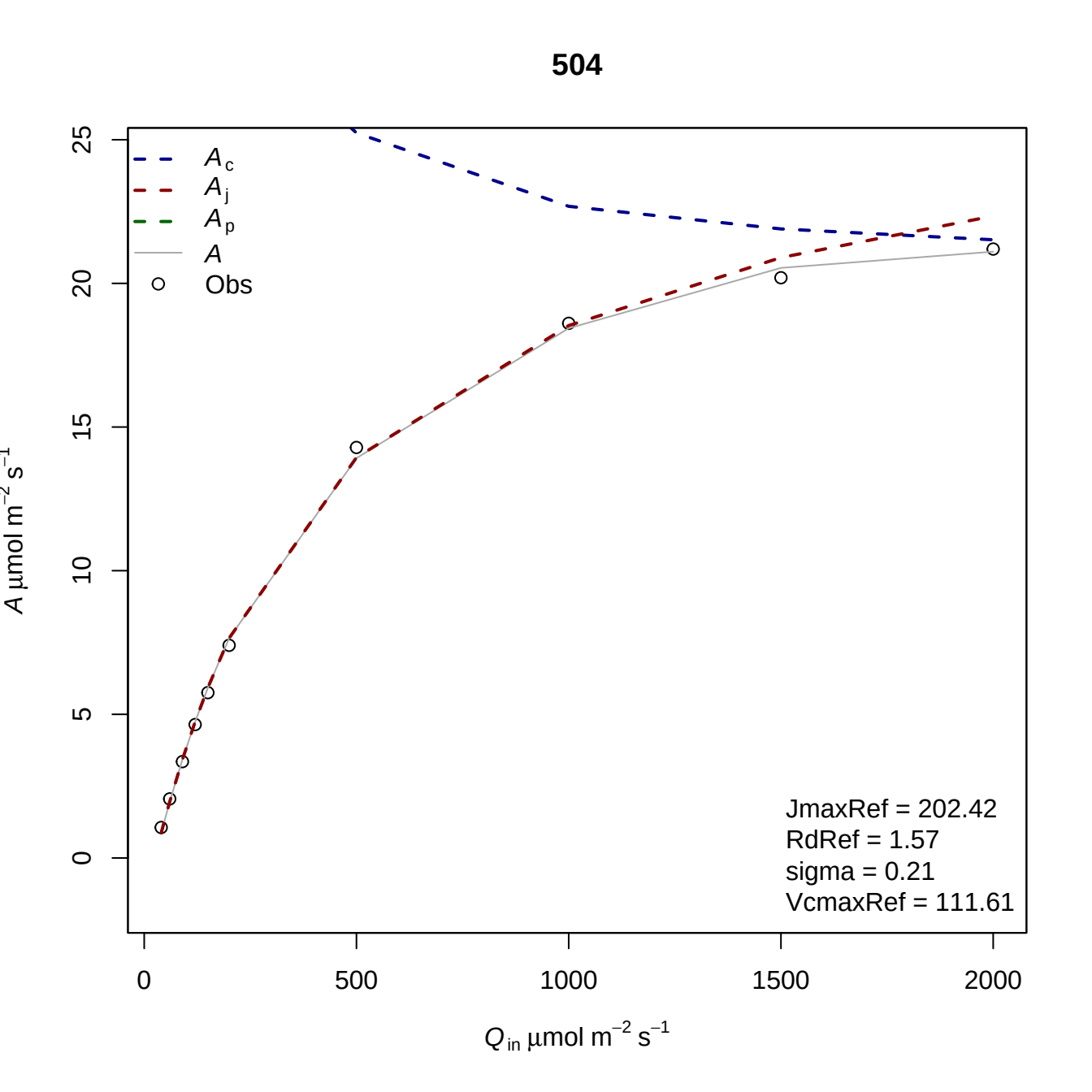

499
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 324.98
RdRef = 3.06
sigma = 0.34
VcmaxRef = 261.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

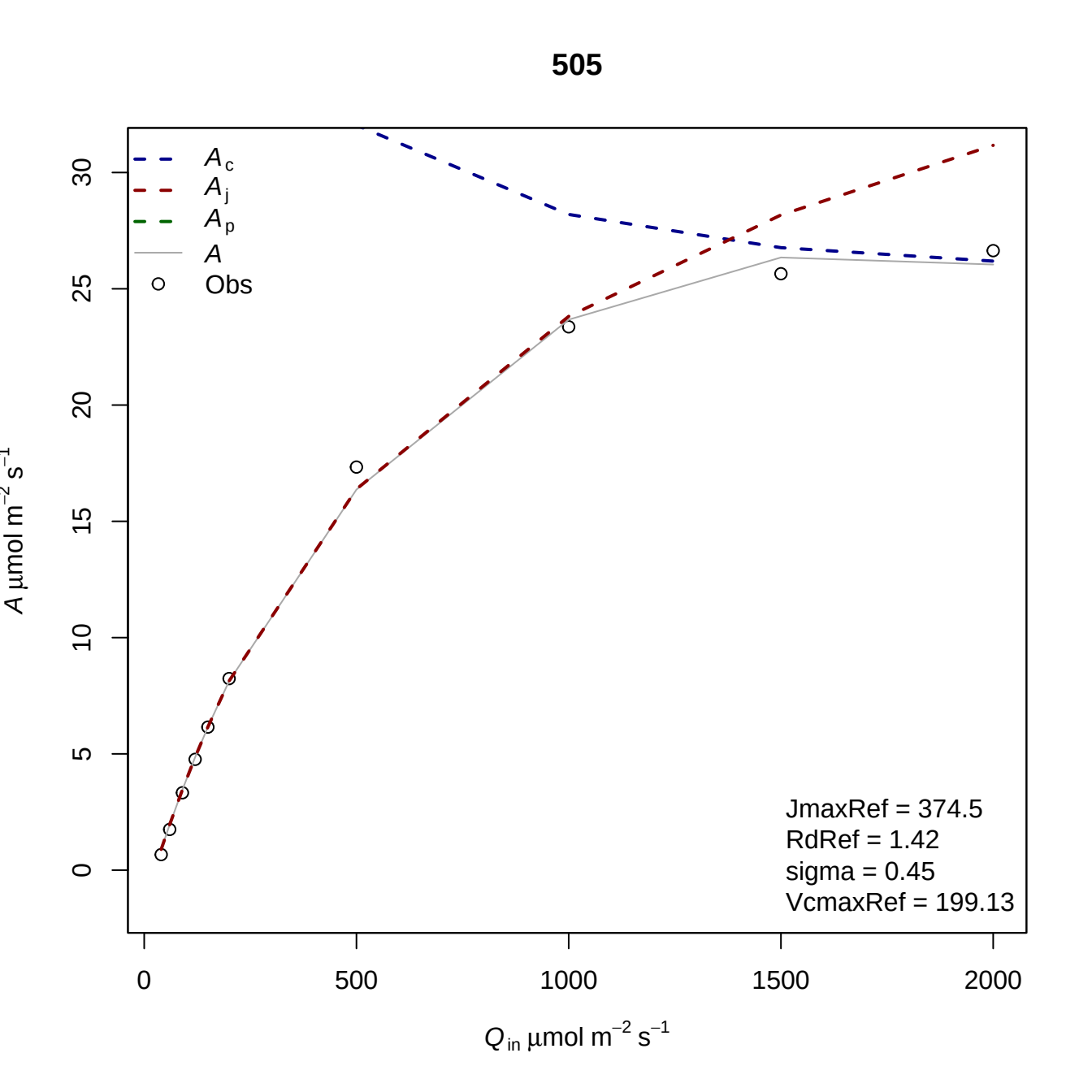

500
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 297.51
RdRef = 3.31
sigma = 0.08
VcmaxRef = 324.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

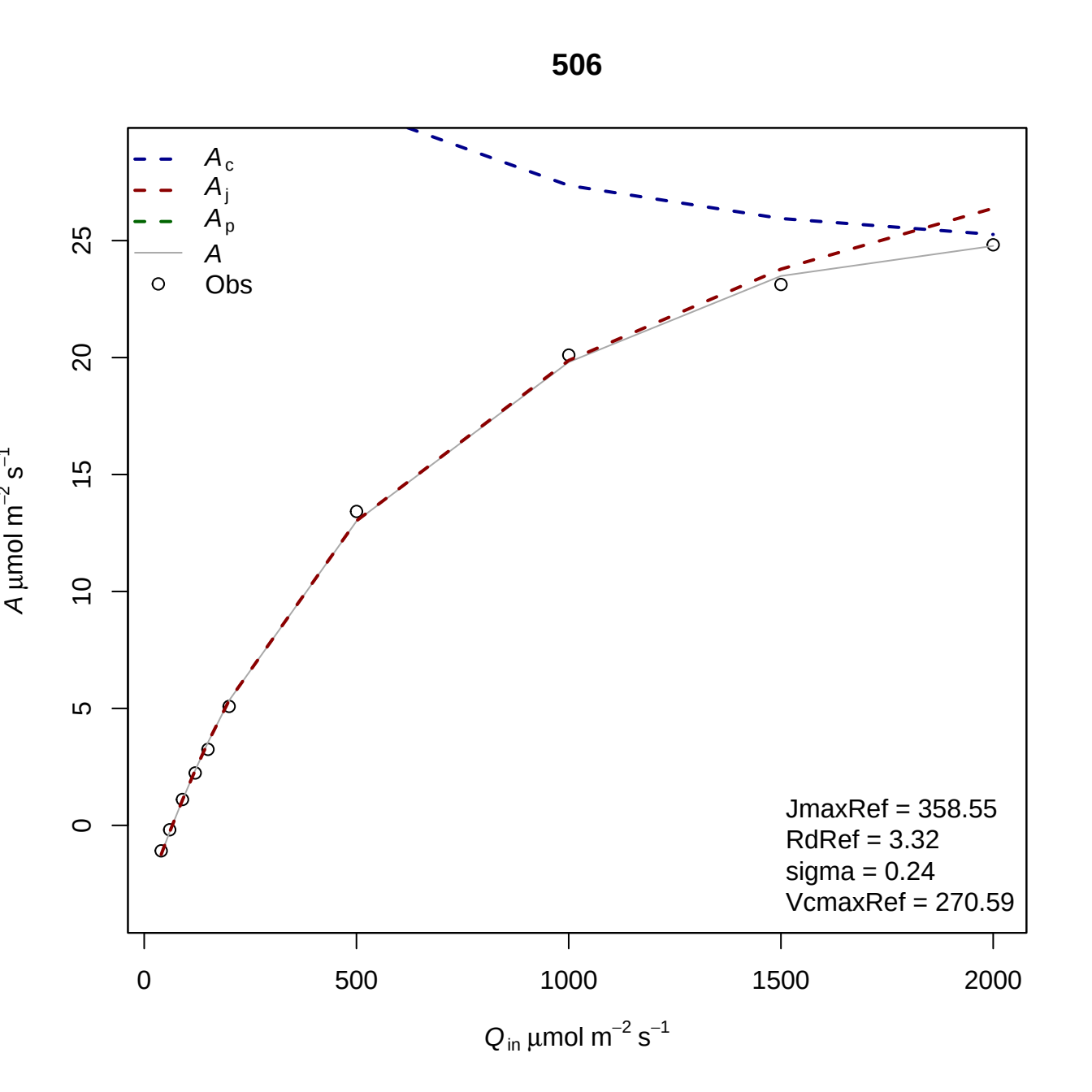

501
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.3
RdRef = 0.27
sigma = 0.23
VcmaxRef = 48.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

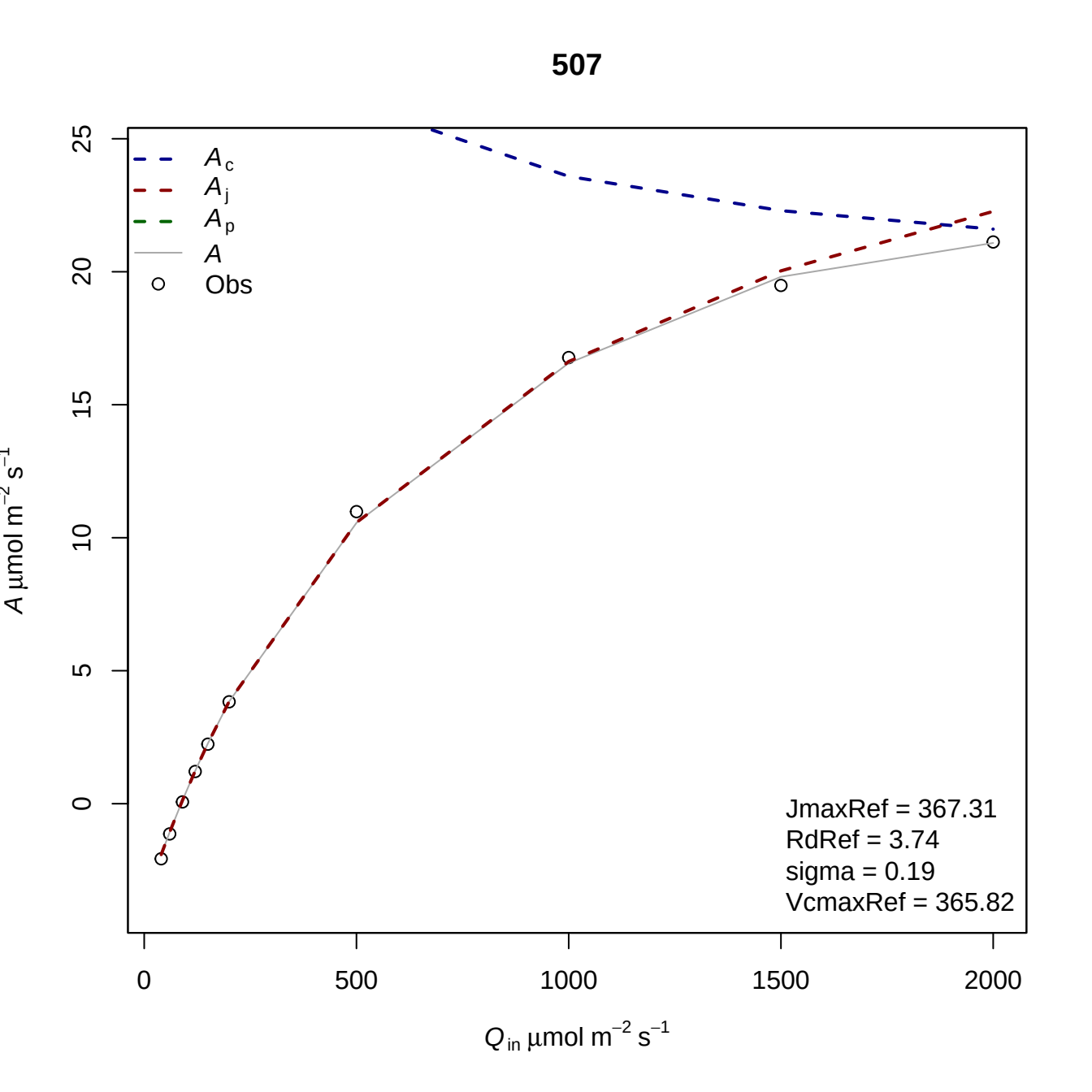

502
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.11
RdRef = 0.72
sigma = 0.21
VcmaxRef = 57.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

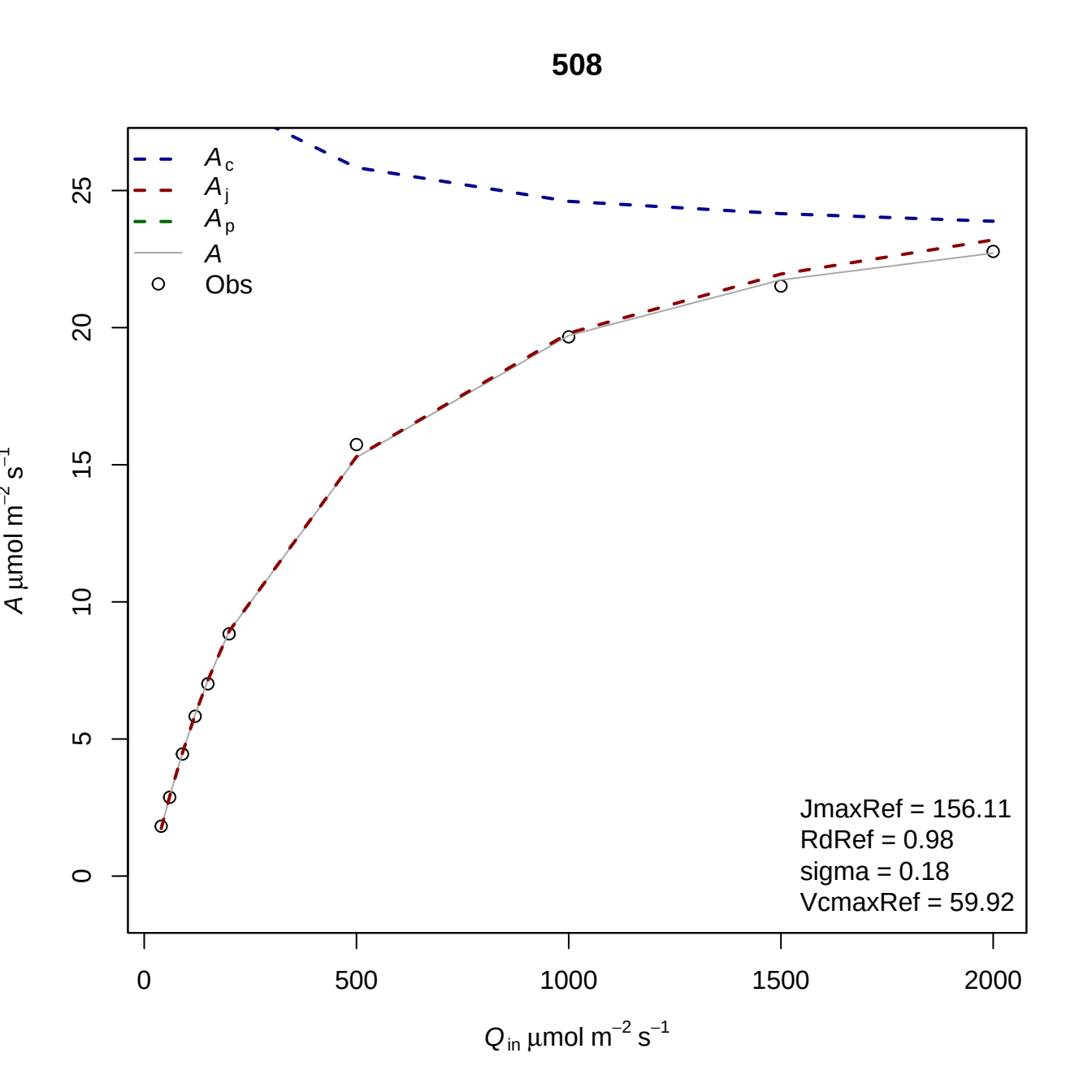

503
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 225.93
RdRef = 1.5
sigma = 0.25
VcmaxRef = 91.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

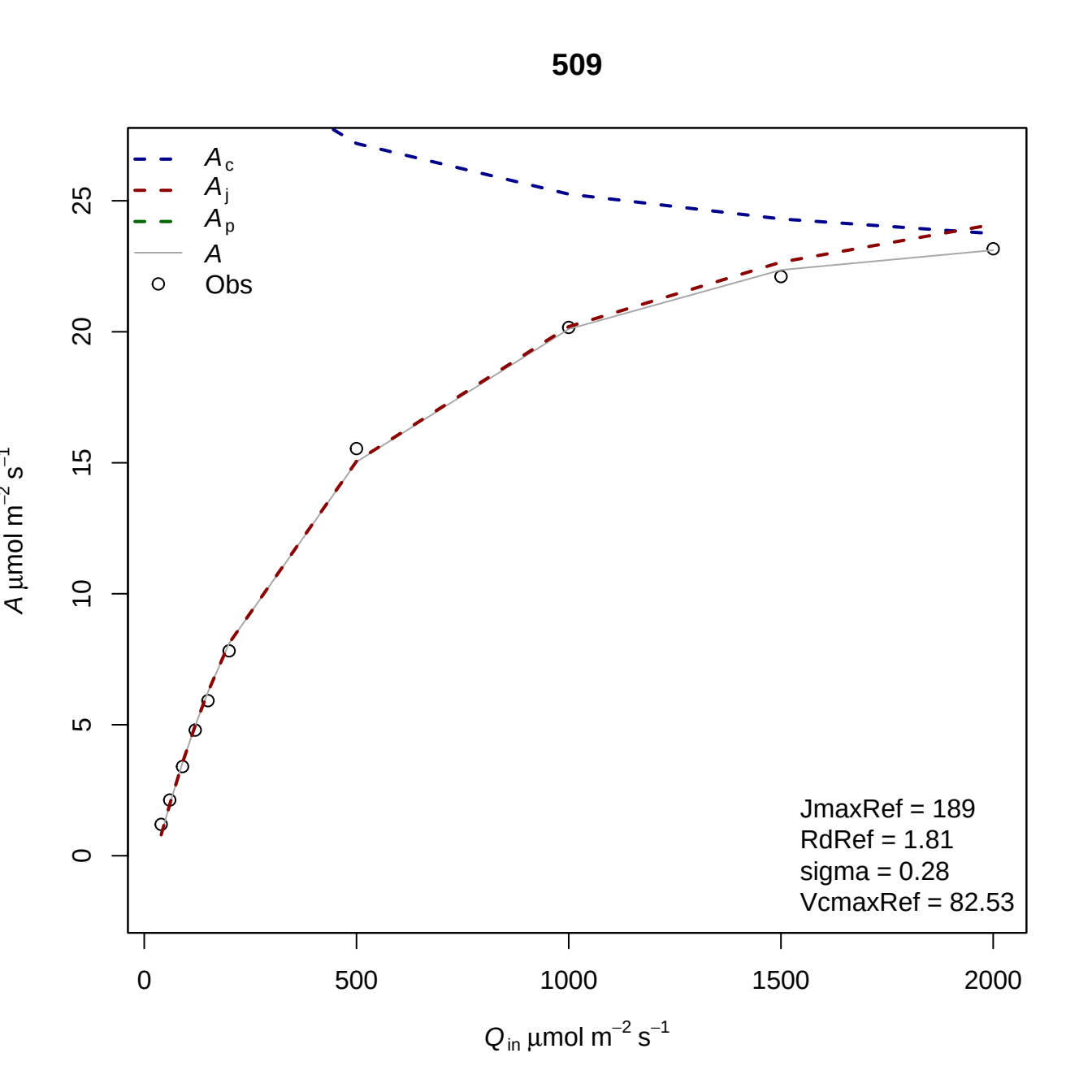

504
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 202.42
RdRef = 1.57
sigma = 0.21
VcmaxRef = 111.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

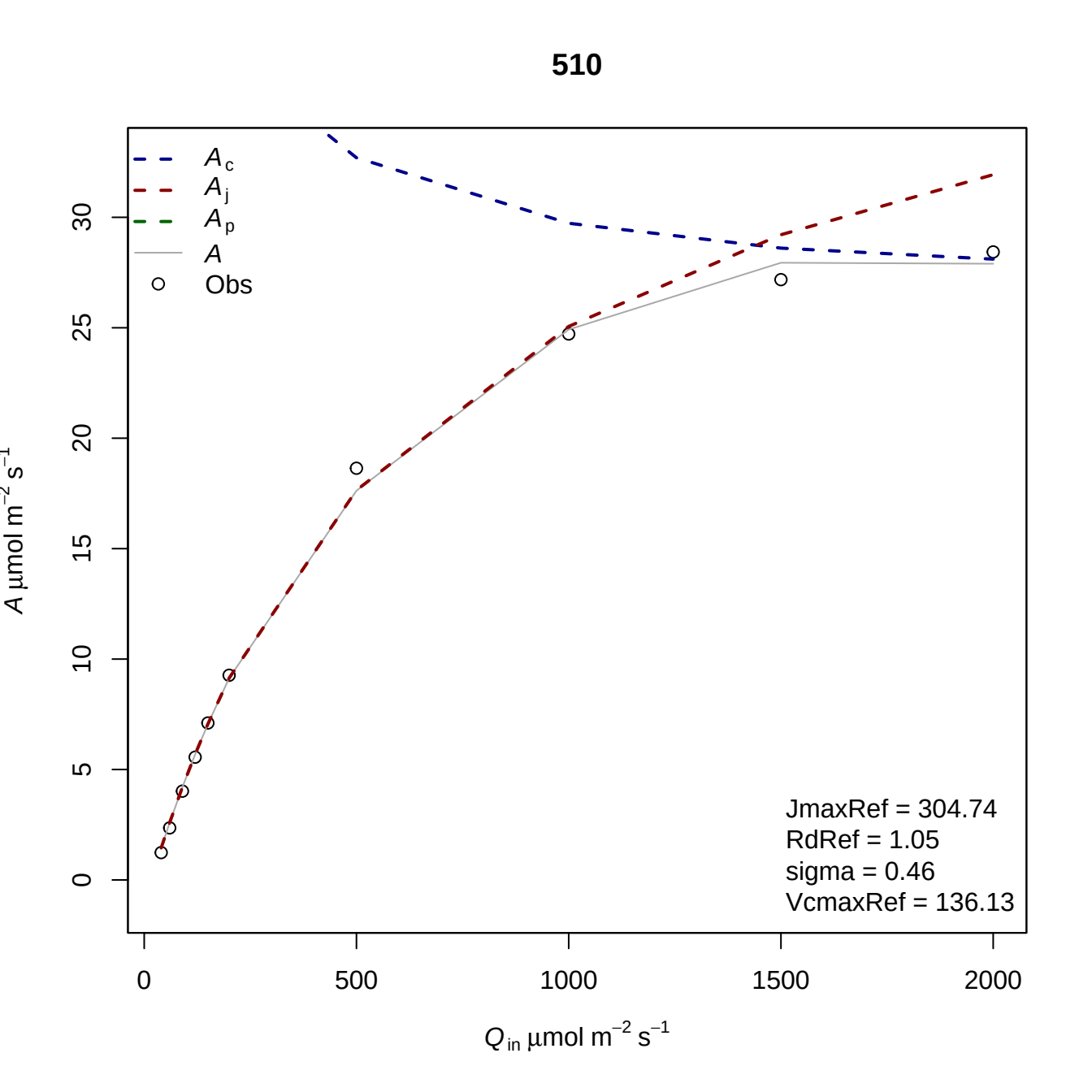

505
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 374.5
RdRef = 1.42
sigma = 0.45
VcmaxRef = 199.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

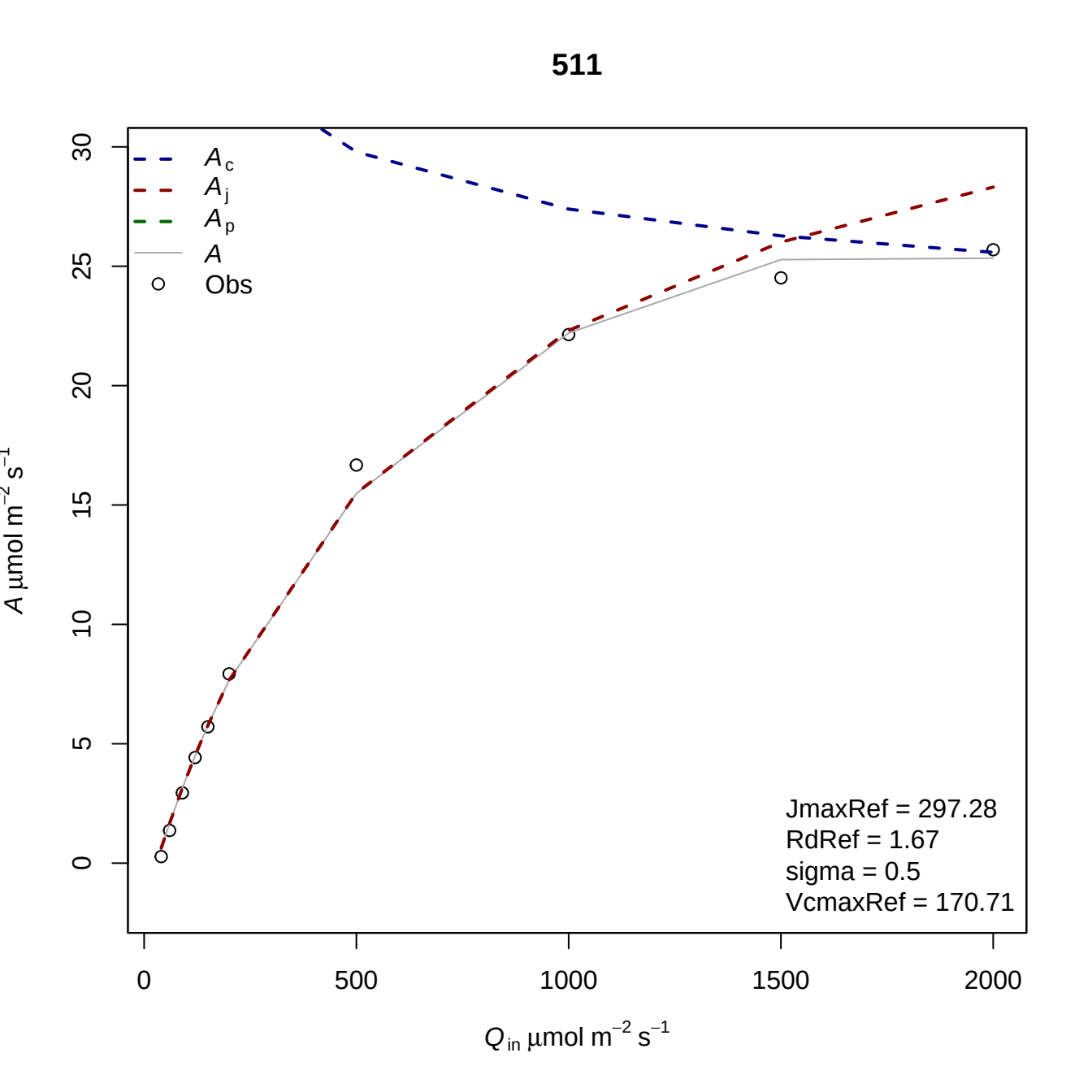

506
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 358.55
RdRef = 3.32
sigma = 0.24
VcmaxRef = 270.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

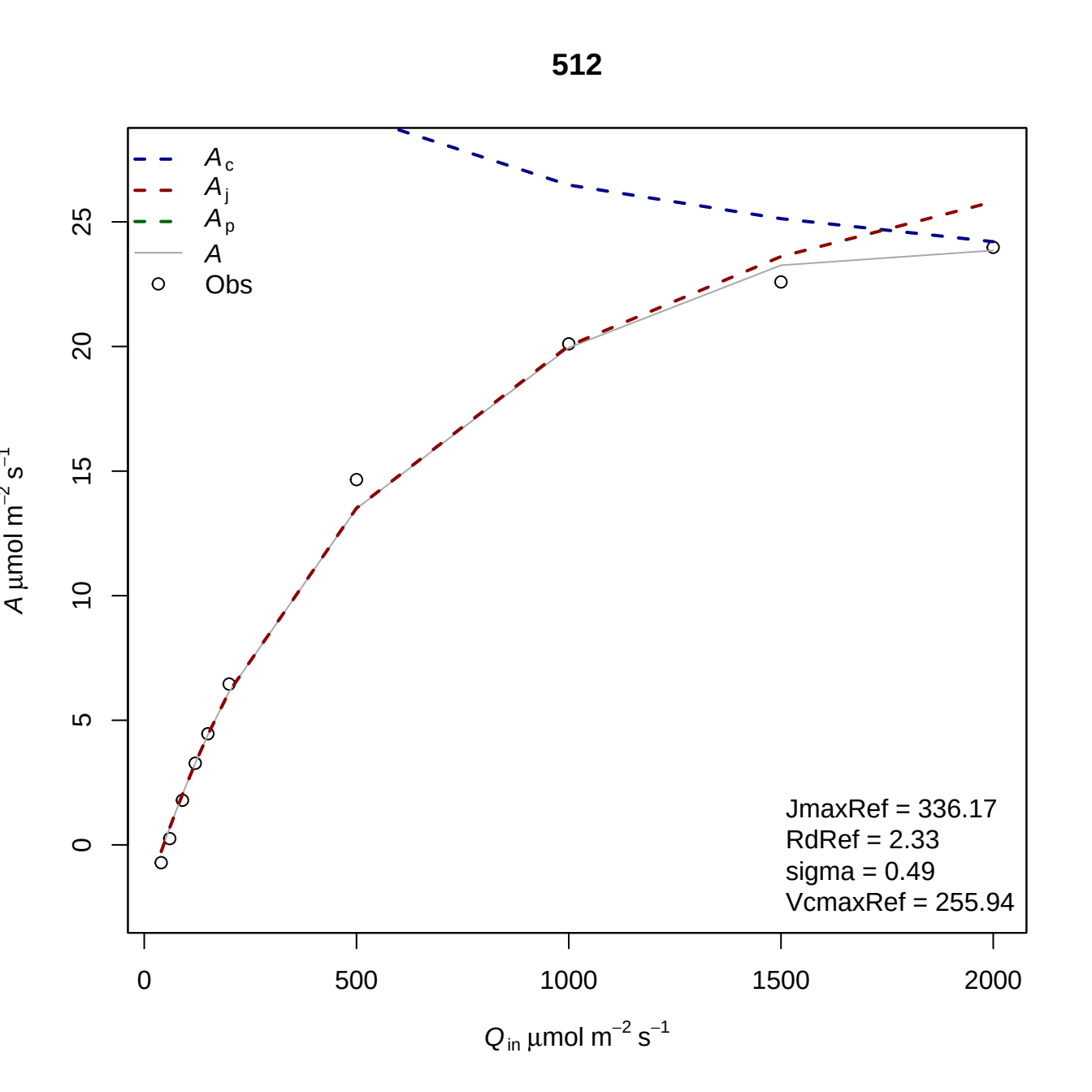

507
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 367.31
RdRef = 3.74
sigma = 0.19
VcmaxRef = 365.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

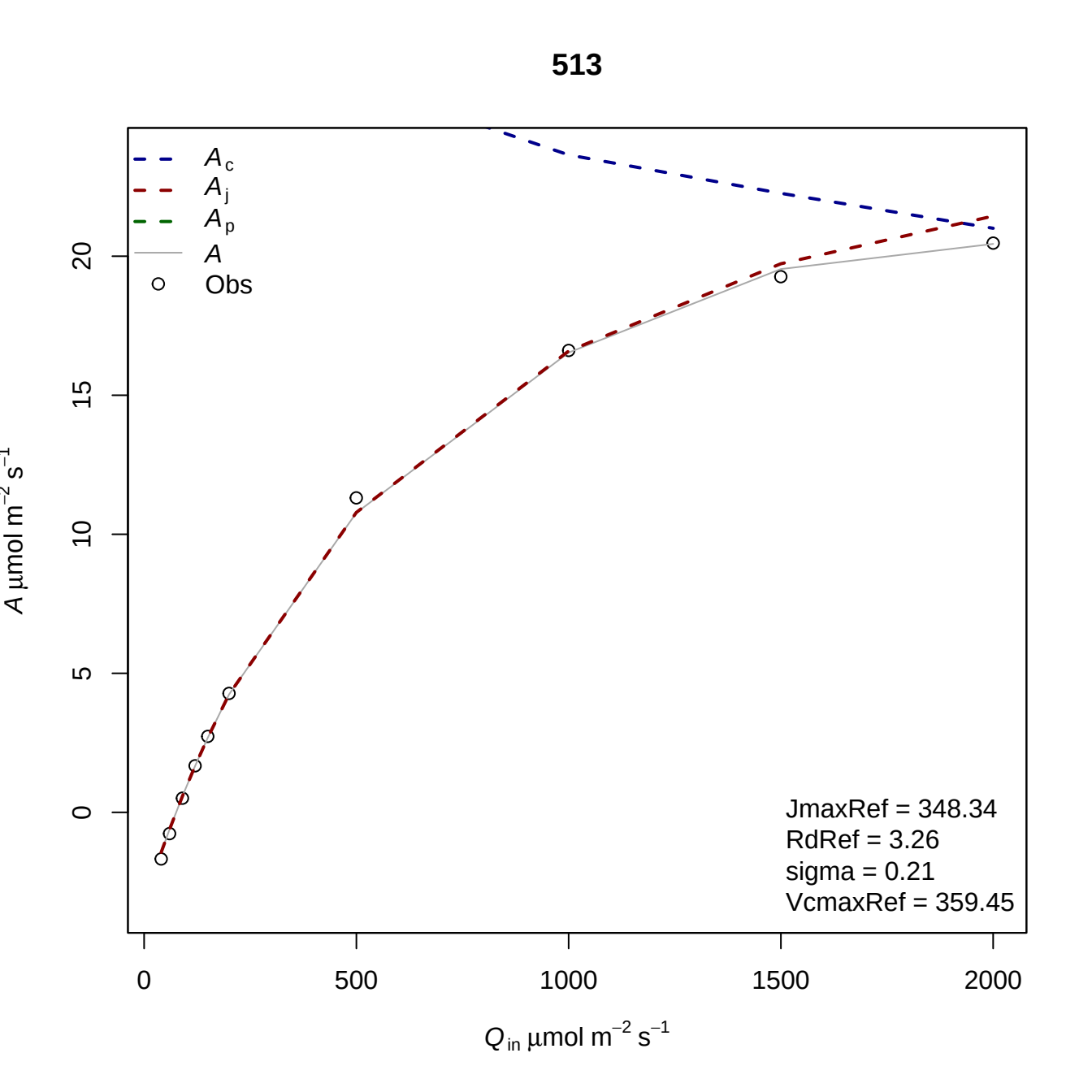

508
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 156.11
RdRef = 0.98
sigma = 0.18
VcmaxRef = 59.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

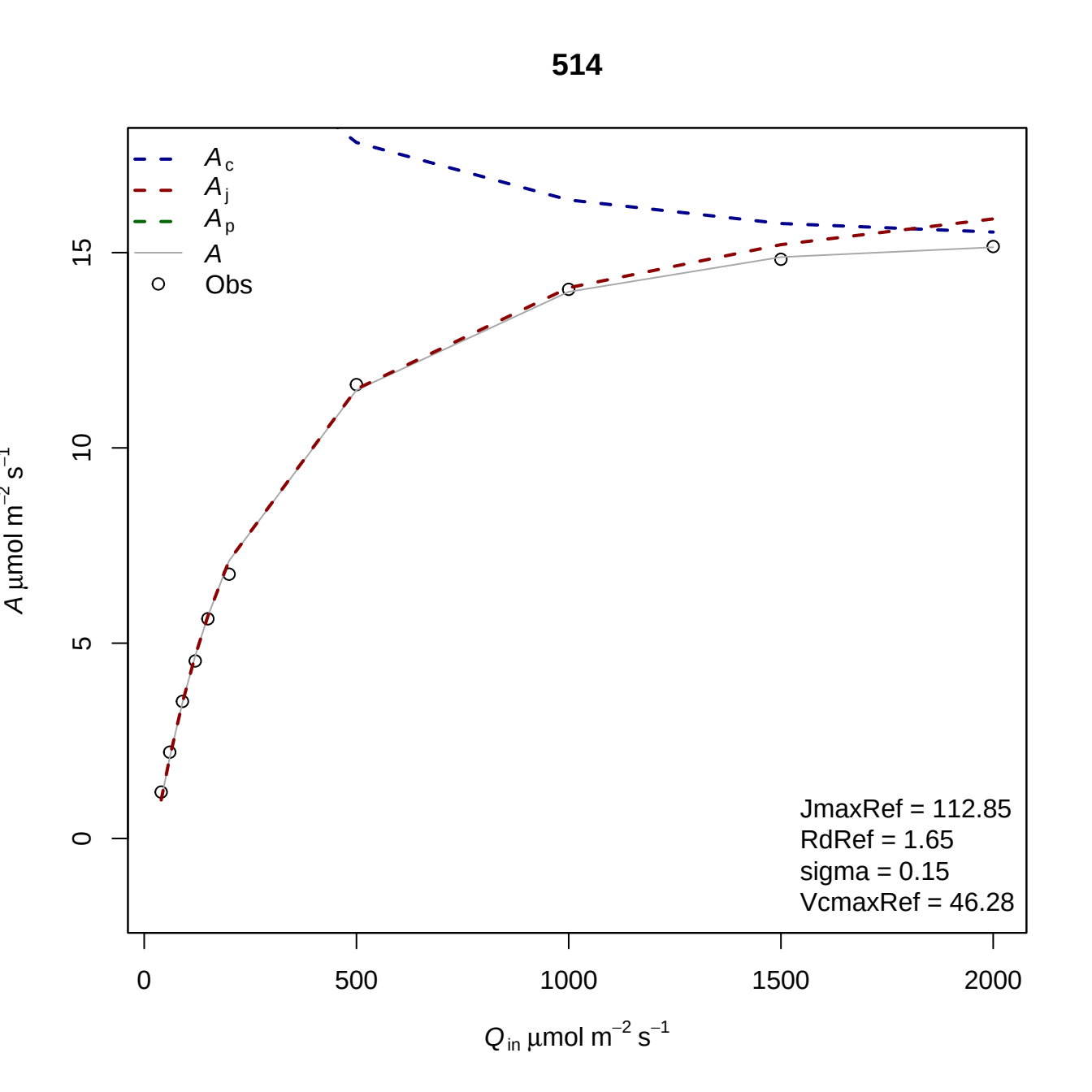

509
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 189
RdRef = 1.81
sigma = 0.28
VcmaxRef = 82.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

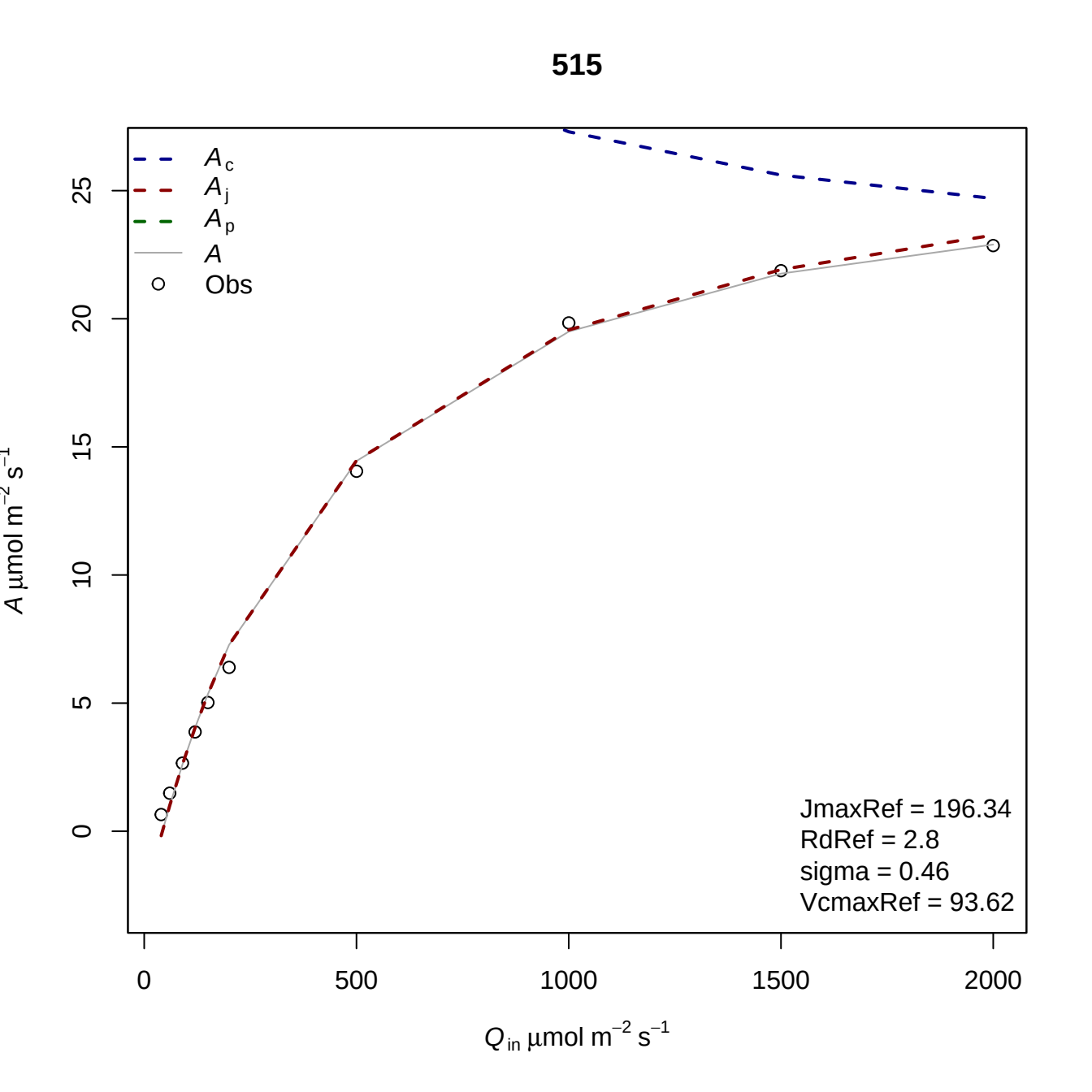

510
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 304.74
RdRef = 1.05
sigma = 0.46
VcmaxRef = 136.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

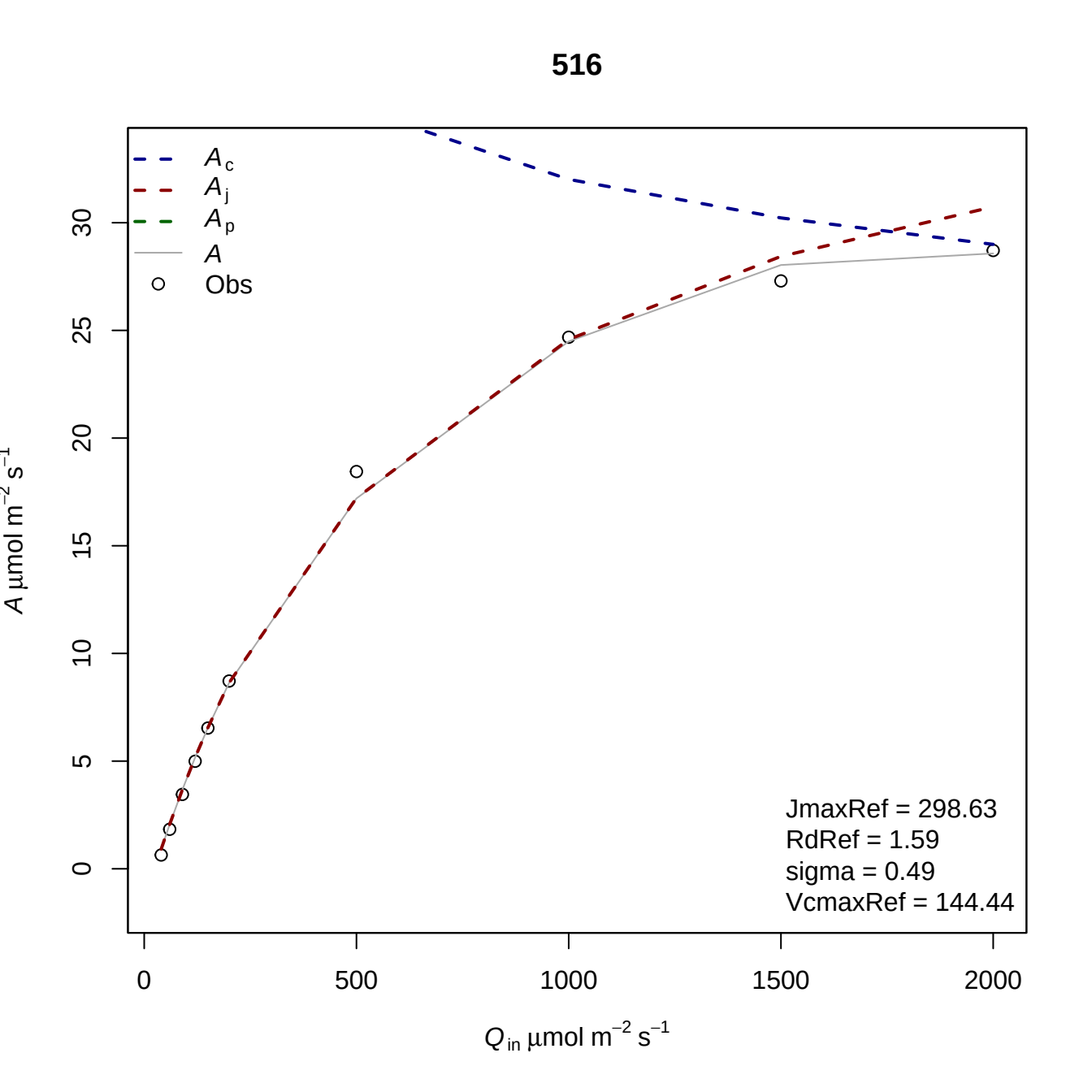

511
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 297.28
RdRef = 1.67
sigma = 0.5
VcmaxRef = 170.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

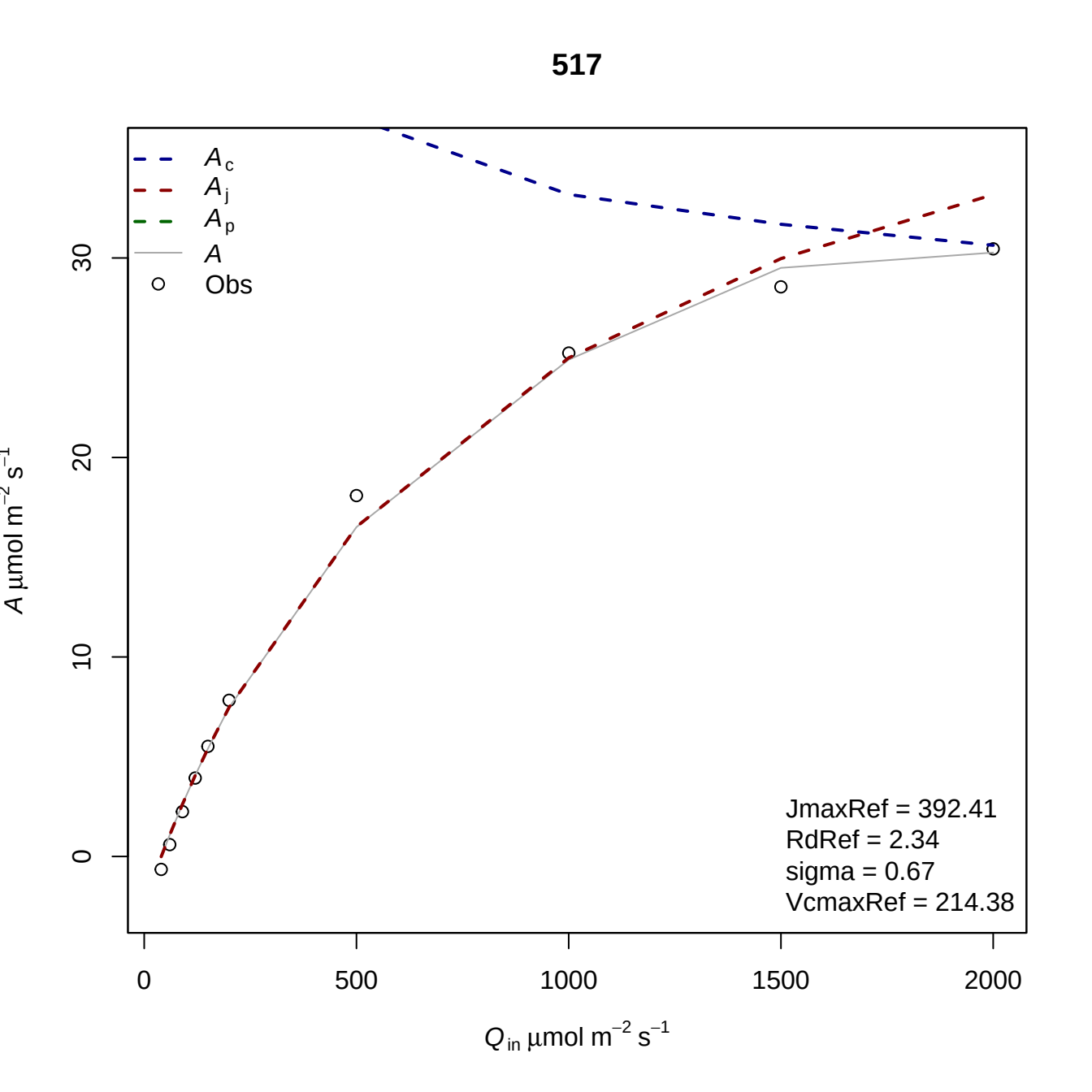

512
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 336.17
RdRef = 2.33
sigma = 0.49
VcmaxRef = 255.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

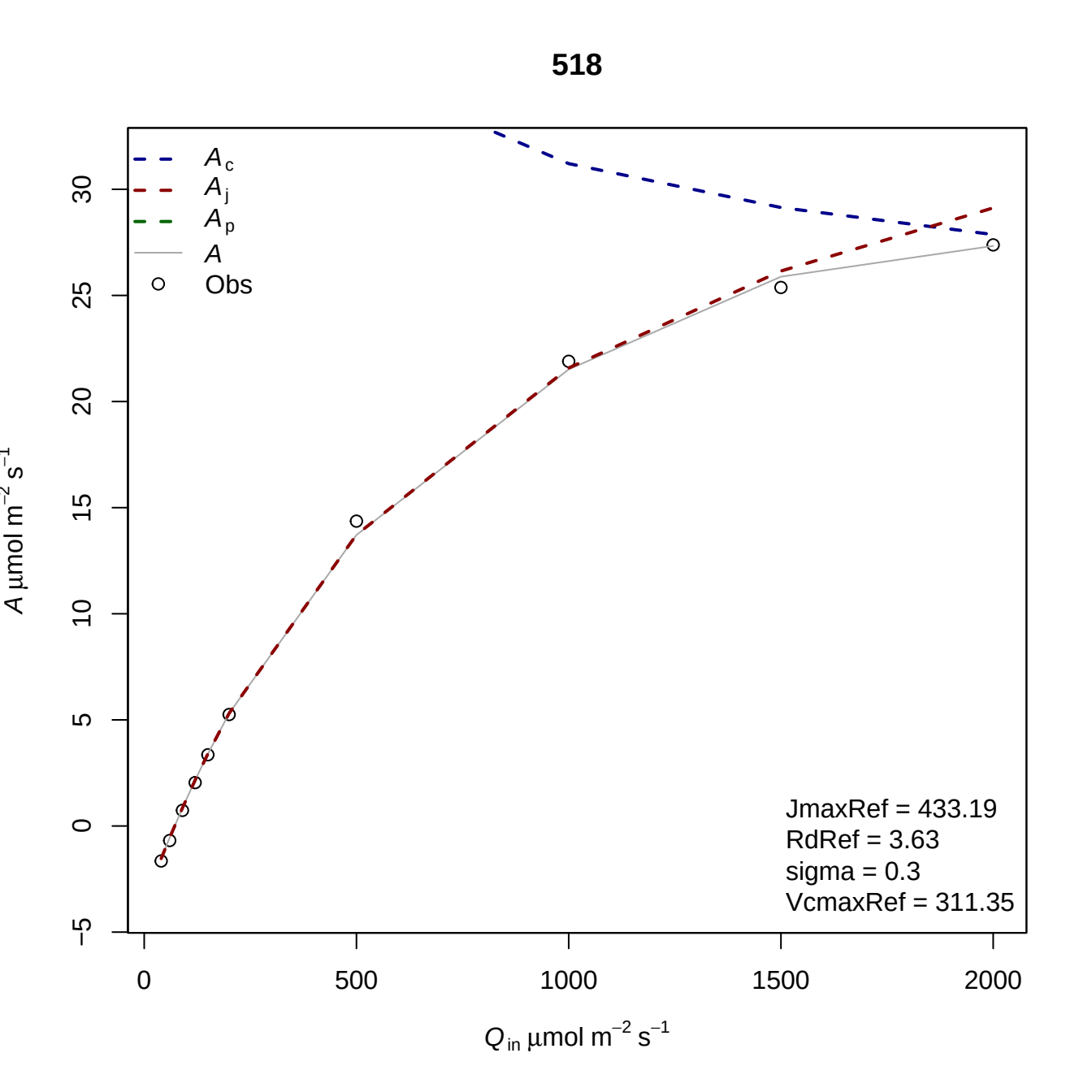

513
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 348.34
RdRef = 3.26
sigma = 0.21
VcmaxRef = 359.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

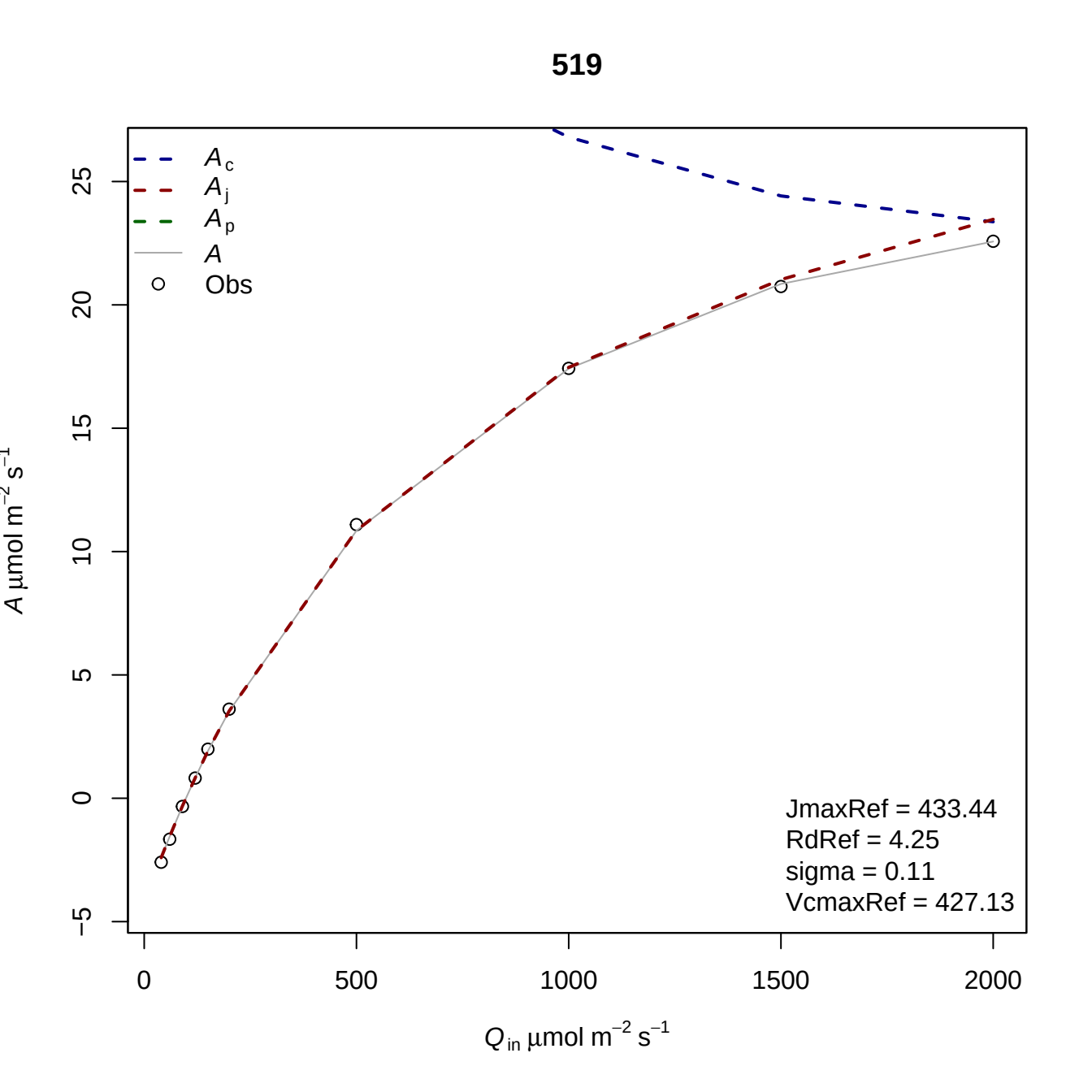

514
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.85
RdRef = 1.65
sigma = 0.15
VcmaxRef = 46.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

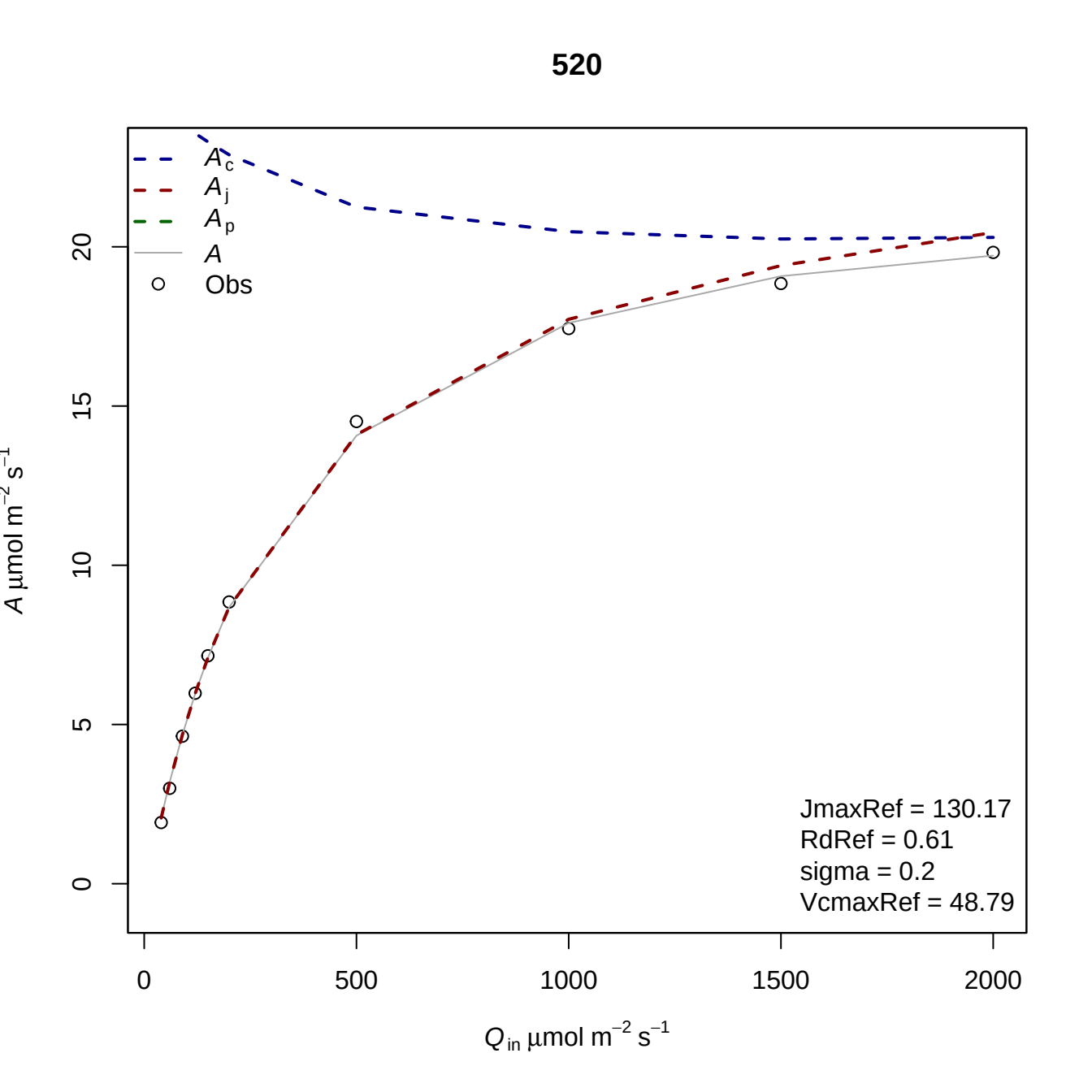

515
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 196.34
RdRef = 2.8
sigma = 0.46
VcmaxRef = 93.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

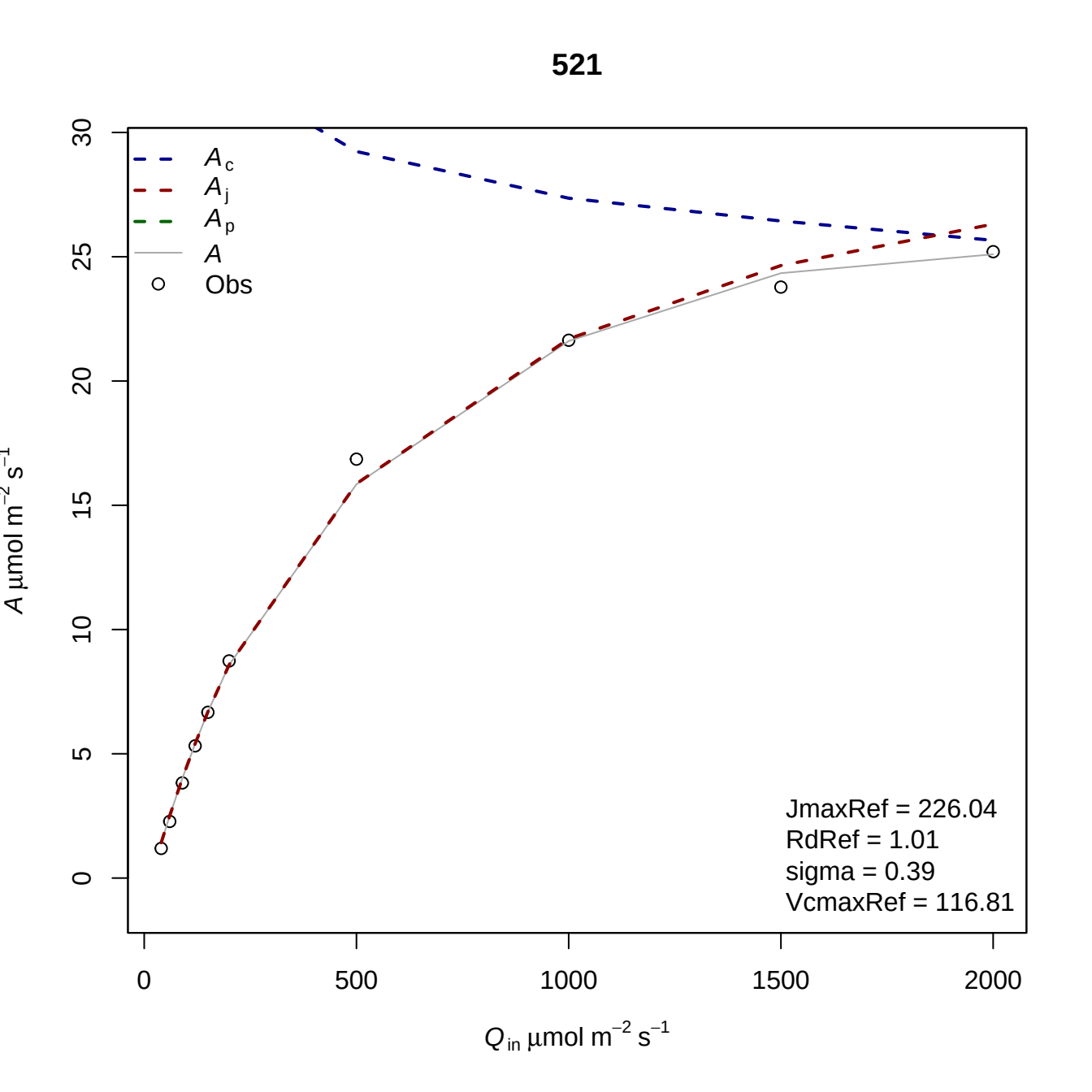

516
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 298.63
RdRef = 1.59
sigma = 0.49
VcmaxRef = 144.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

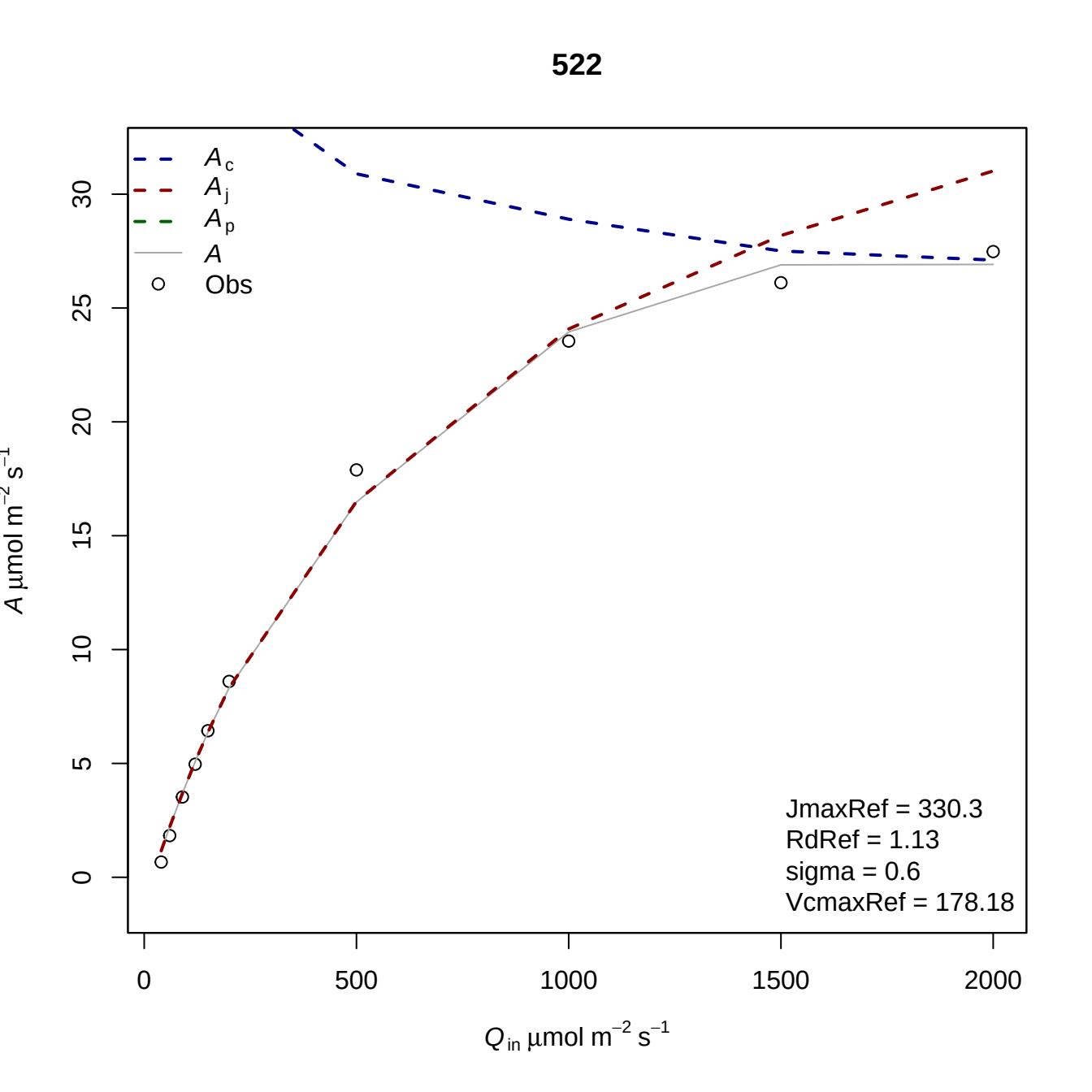

517
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 392.41
RdRef = 2.34
sigma = 0.67
VcmaxRef = 214.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

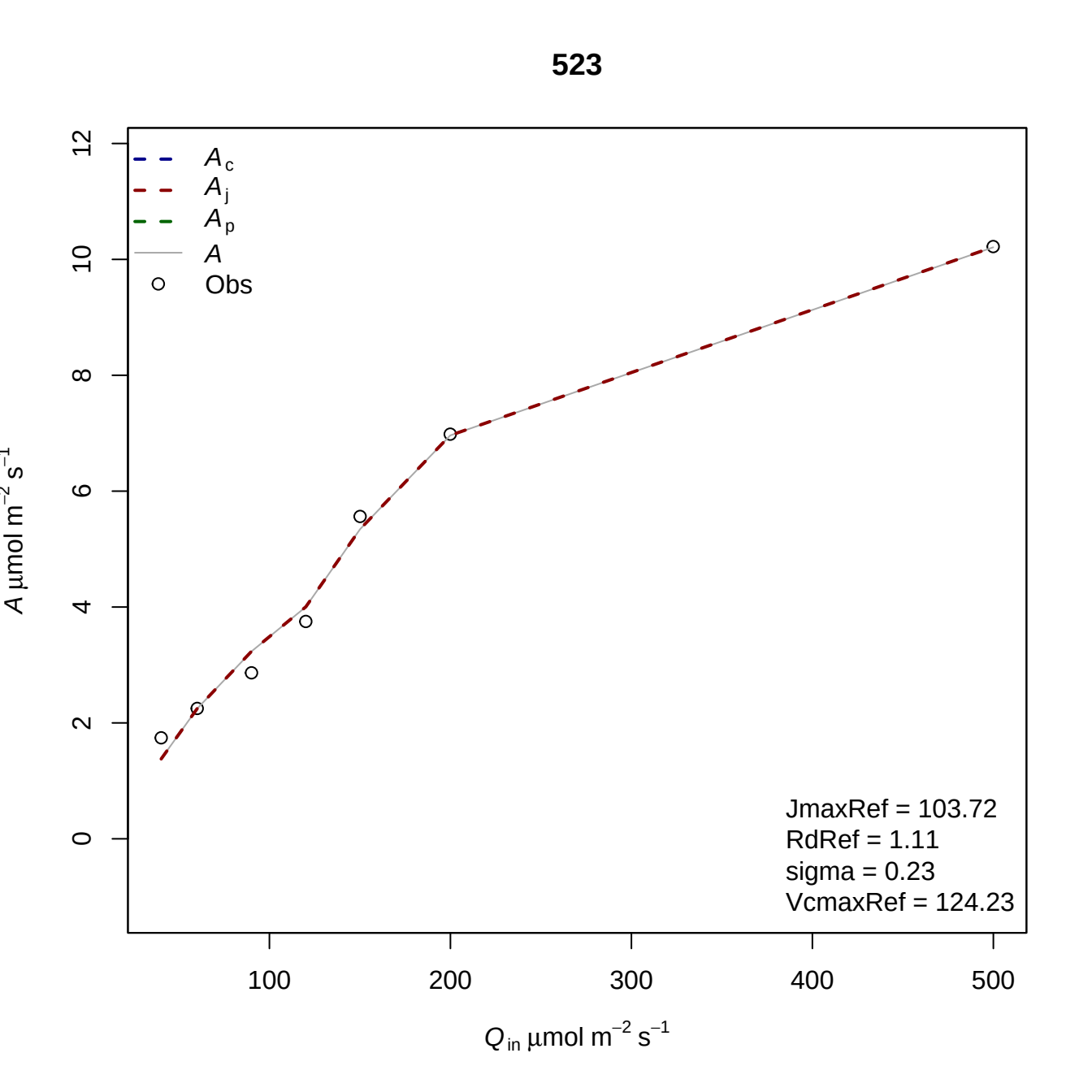

518
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 433.19
RdRef = 3.63
sigma = 0.3
VcmaxRef = 311.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

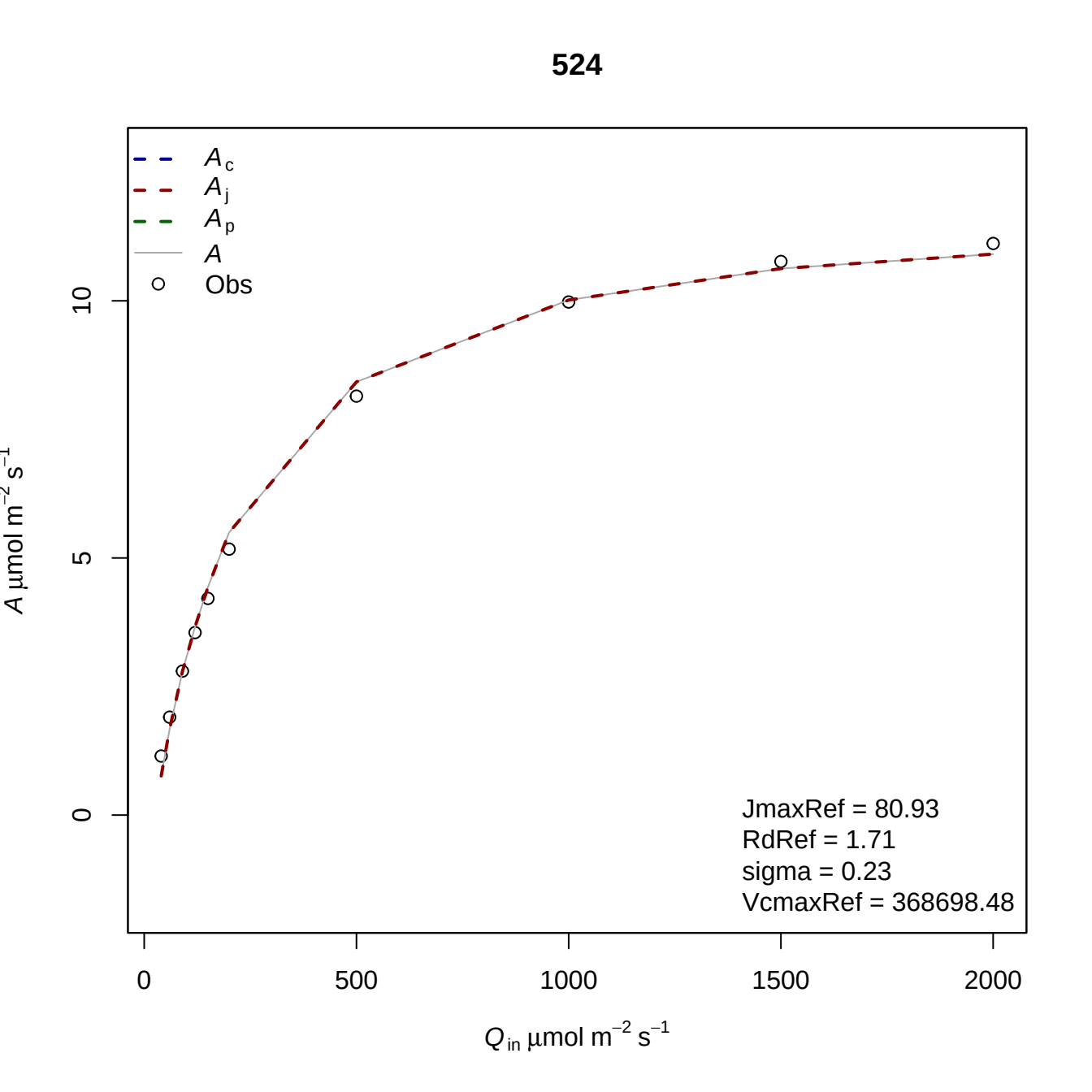

519
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 433.44
RdRef = 4.25
sigma = 0.11
VcmaxRef = 427.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
25
0
500
1000
1500
2000

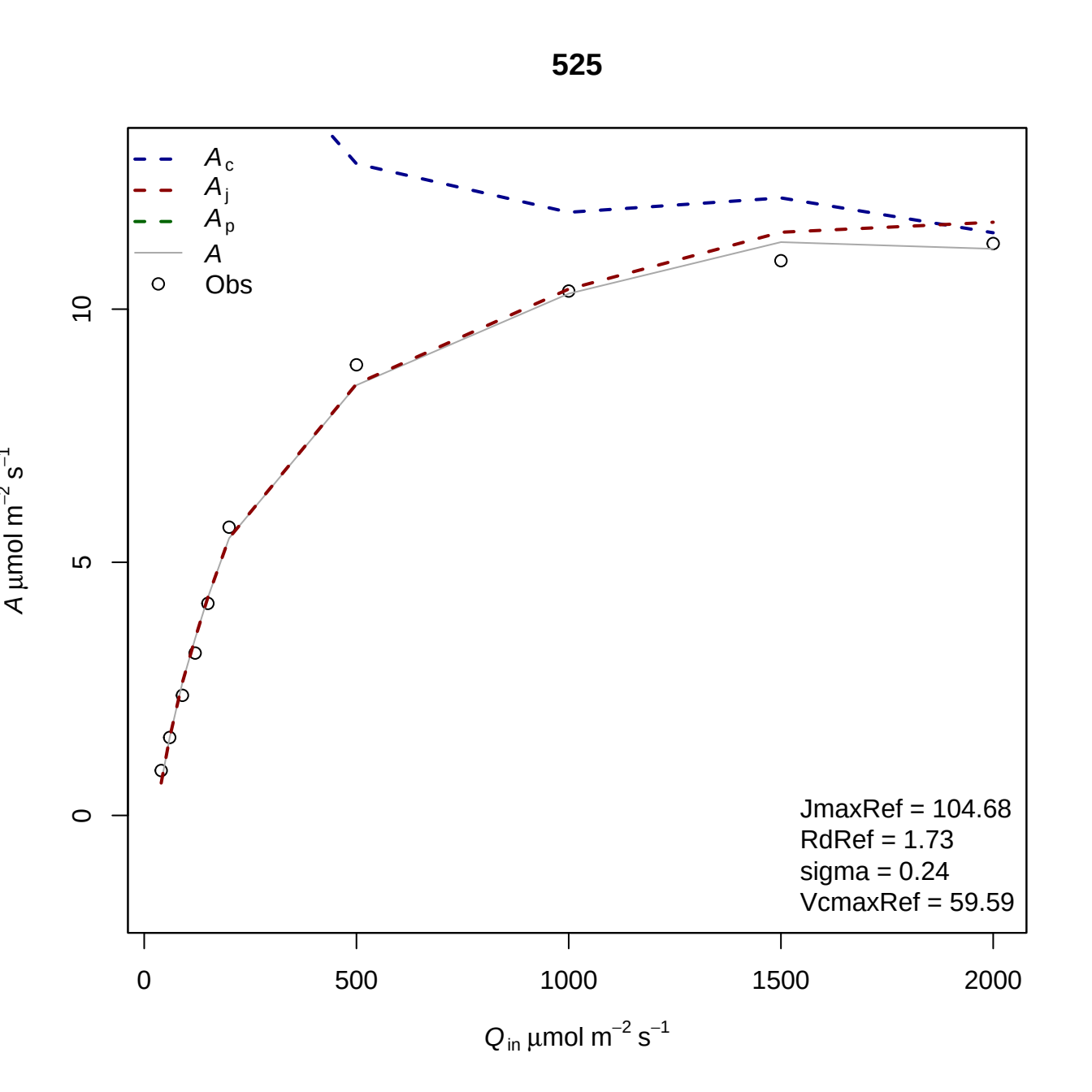

520
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.17
RdRef = 0.61
sigma = 0.2
VcmaxRef = 48.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

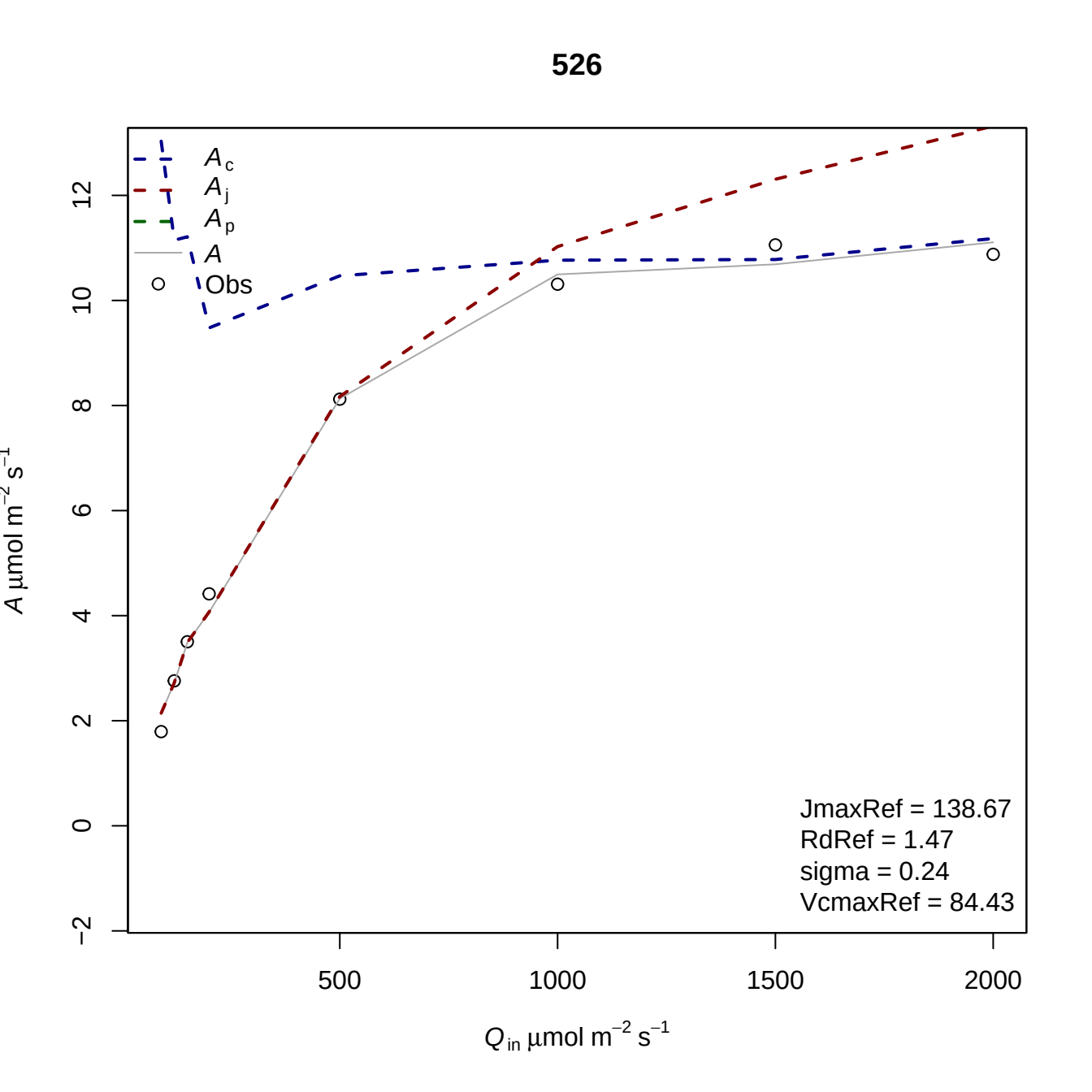

521
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 226.04
RdRef = 1.01
sigma = 0.39
VcmaxRef = 116.81
0
5
10
15
20
25
30
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

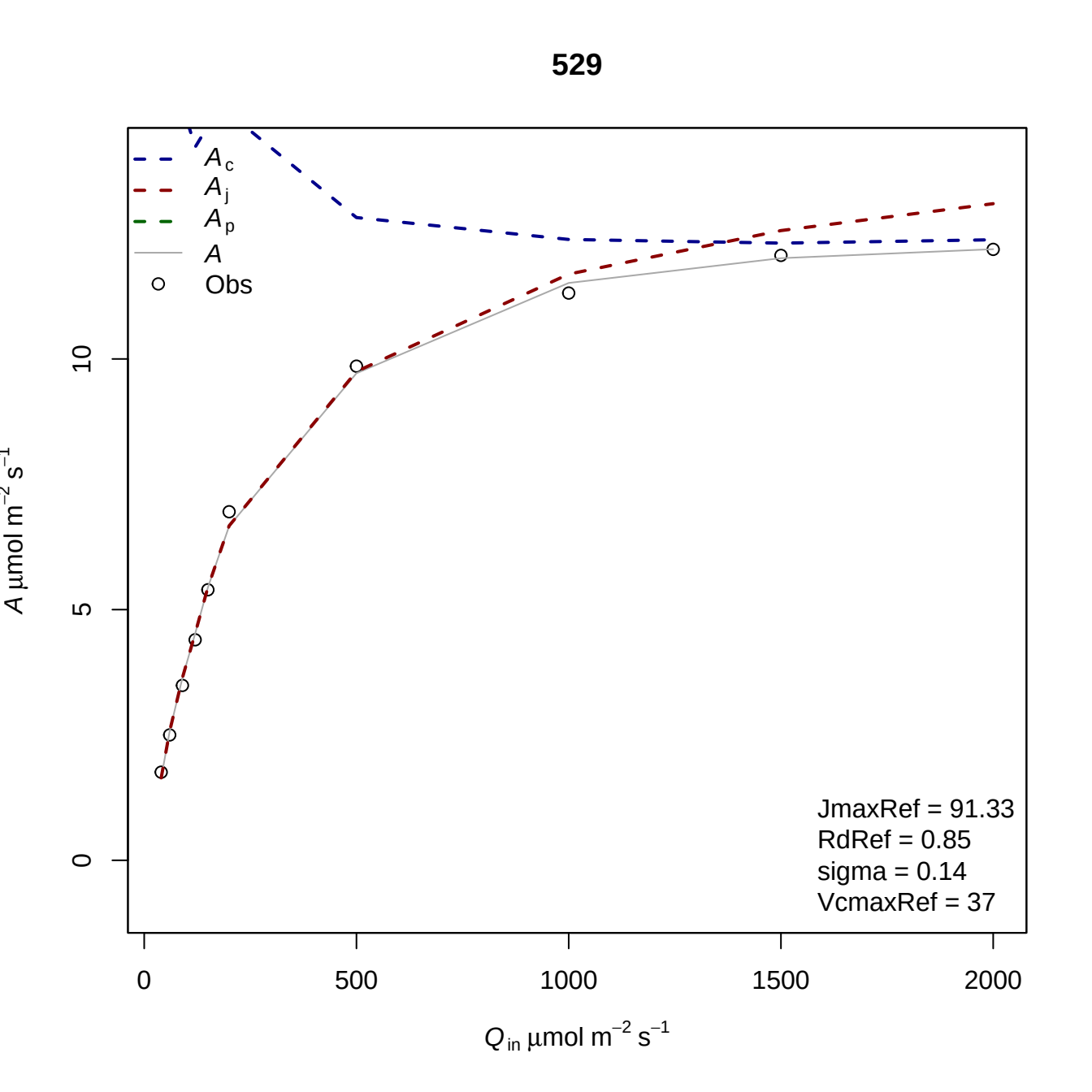

522
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 330.3
RdRef = 1.13
sigma = 0.6
VcmaxRef = 178.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

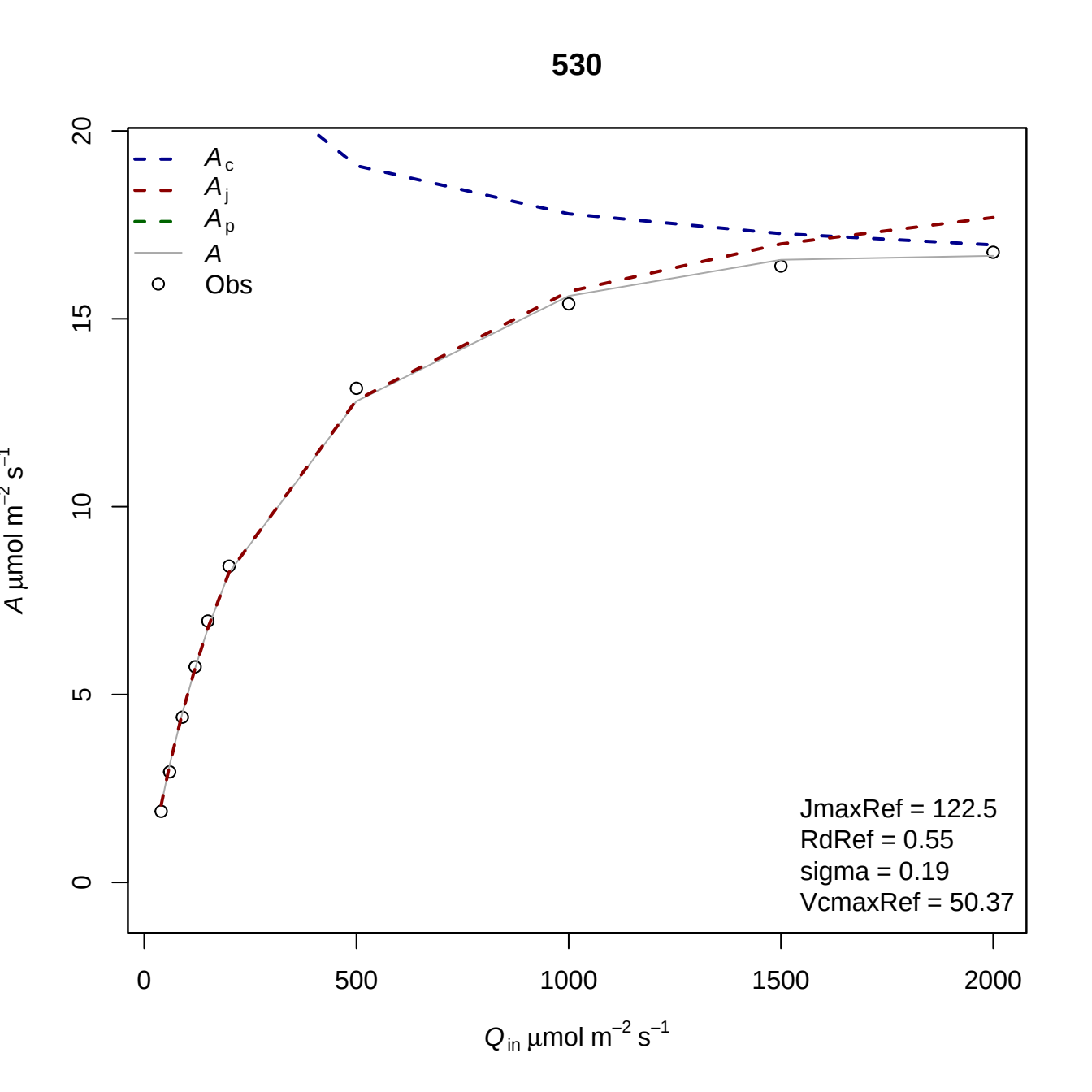

523
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.72
RdRef = 1.11
sigma = 0.23
VcmaxRef = 124.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
100
200
300
400
500

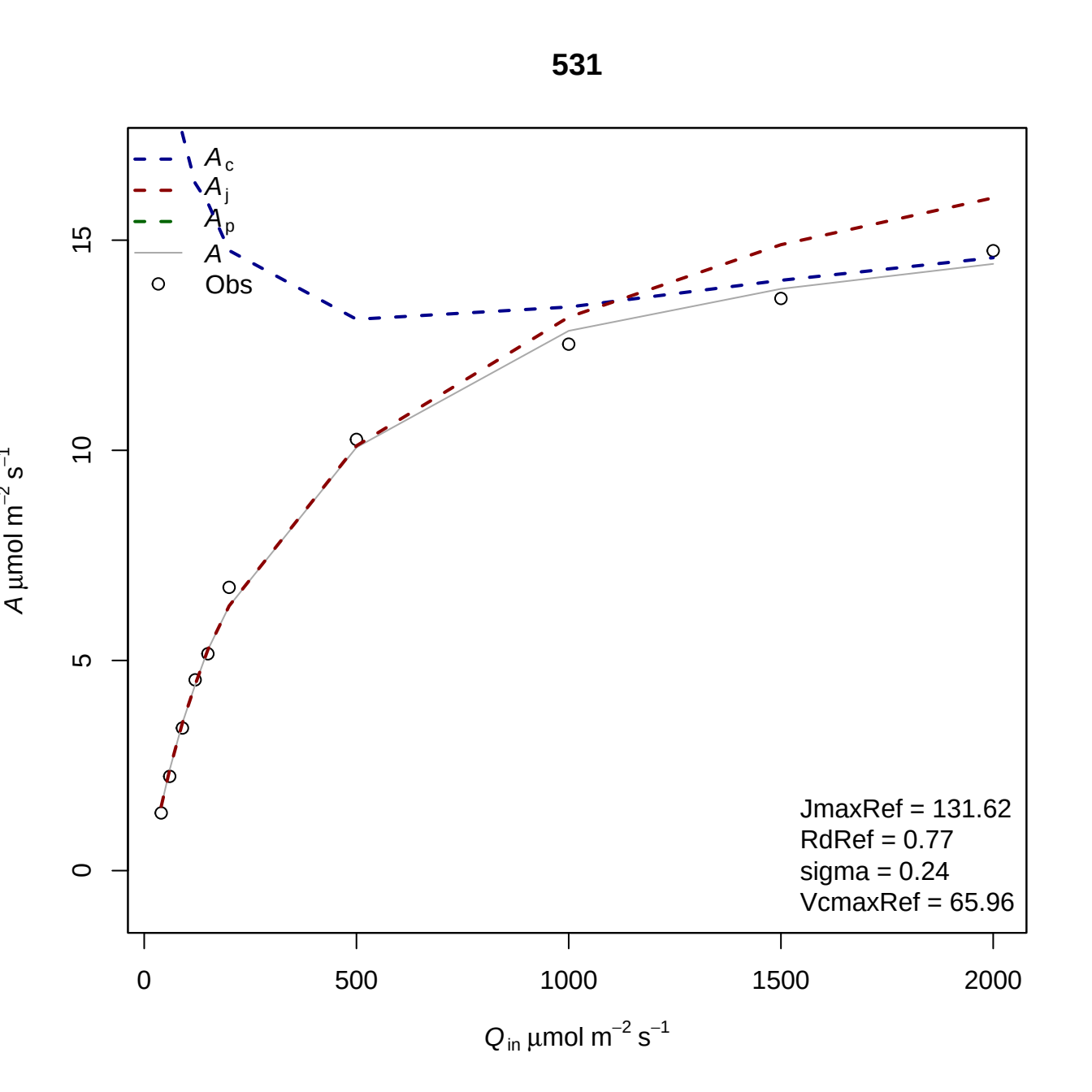

524
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.93
RdRef = 1.71
sigma = 0.23
VcmaxRef = 368698.48
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

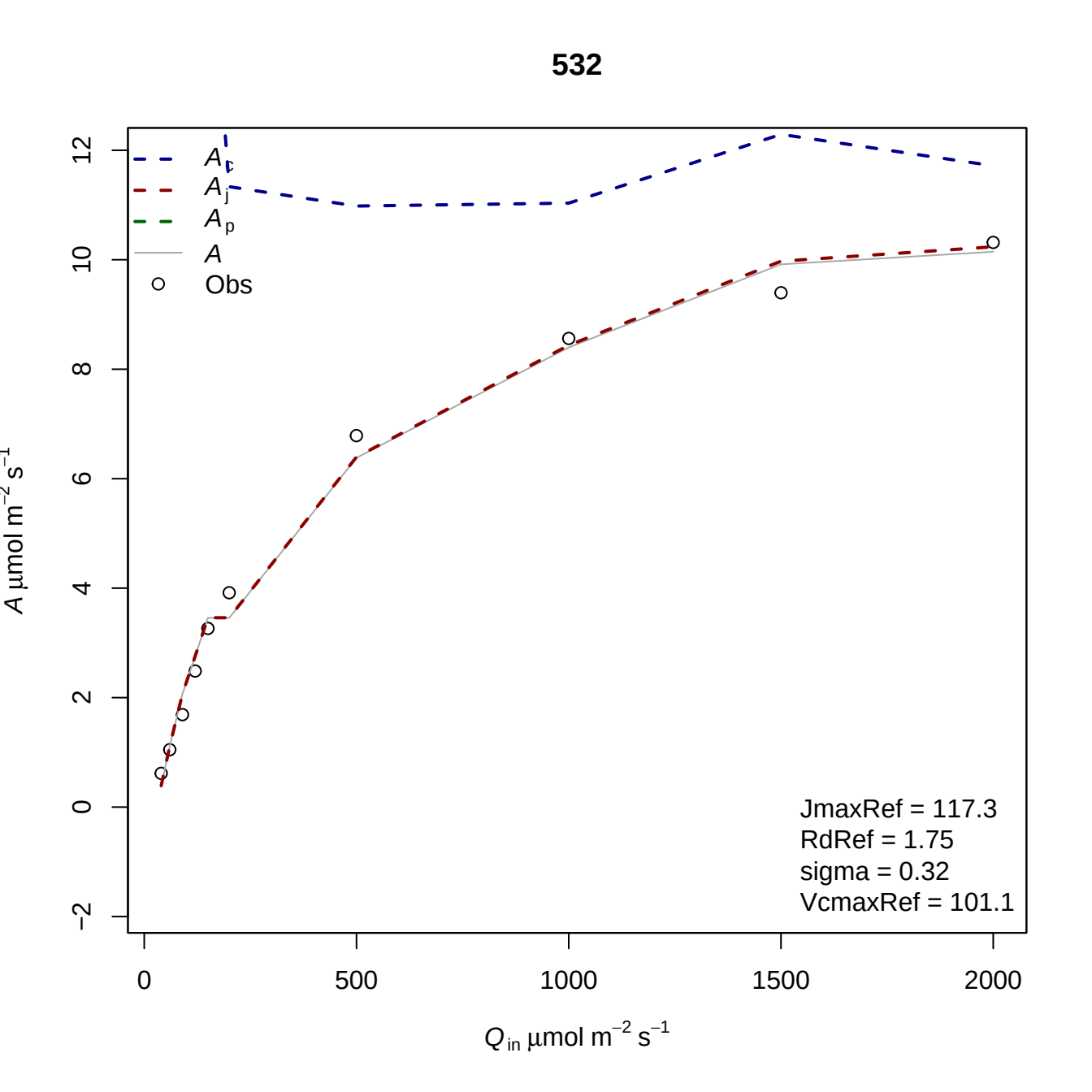

525
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.68
RdRef = 1.73
sigma = 0.24
VcmaxRef = 59.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

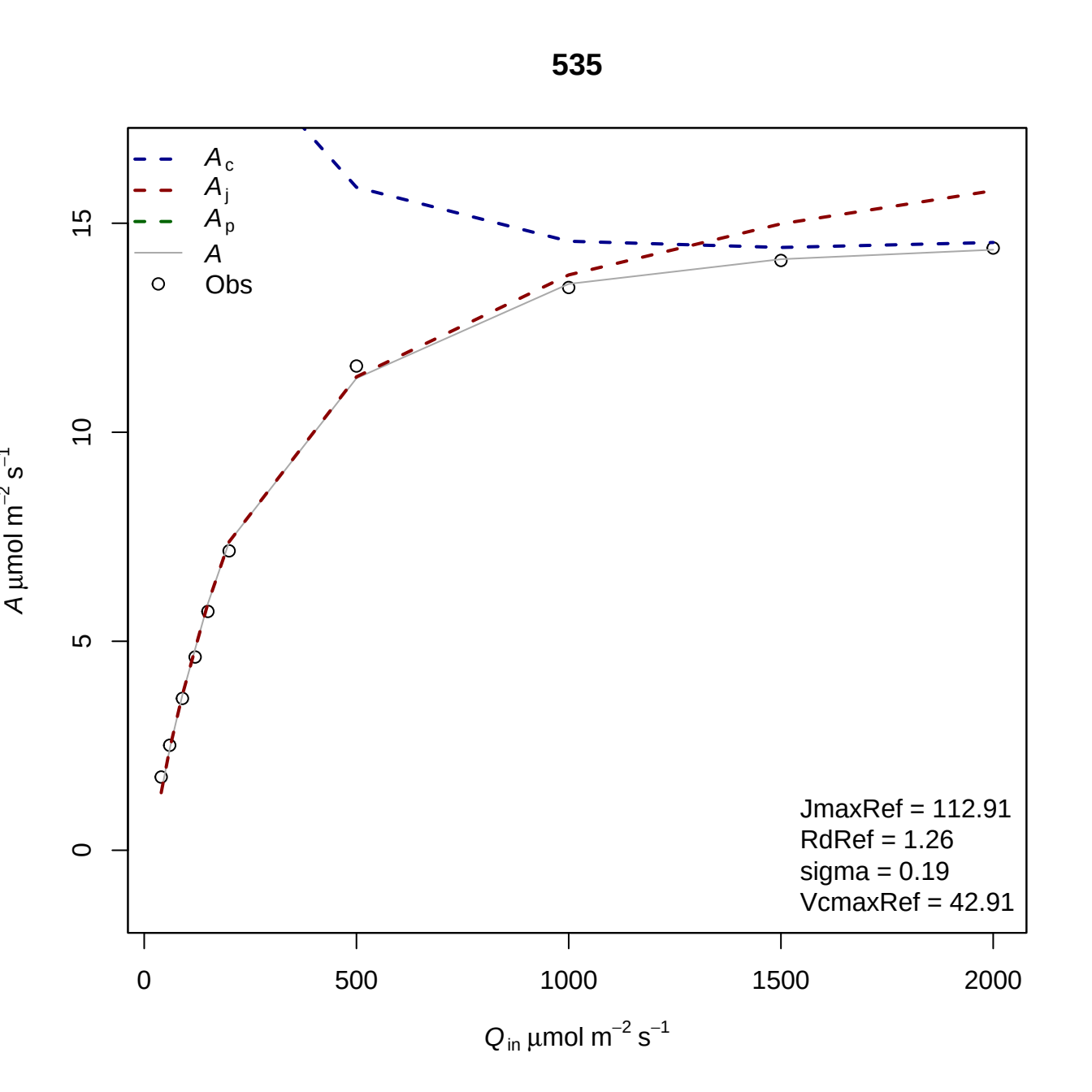

526
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.67
RdRef = 1.47
sigma = 0.24
VcmaxRef = 84.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
500
1000
1500
2000

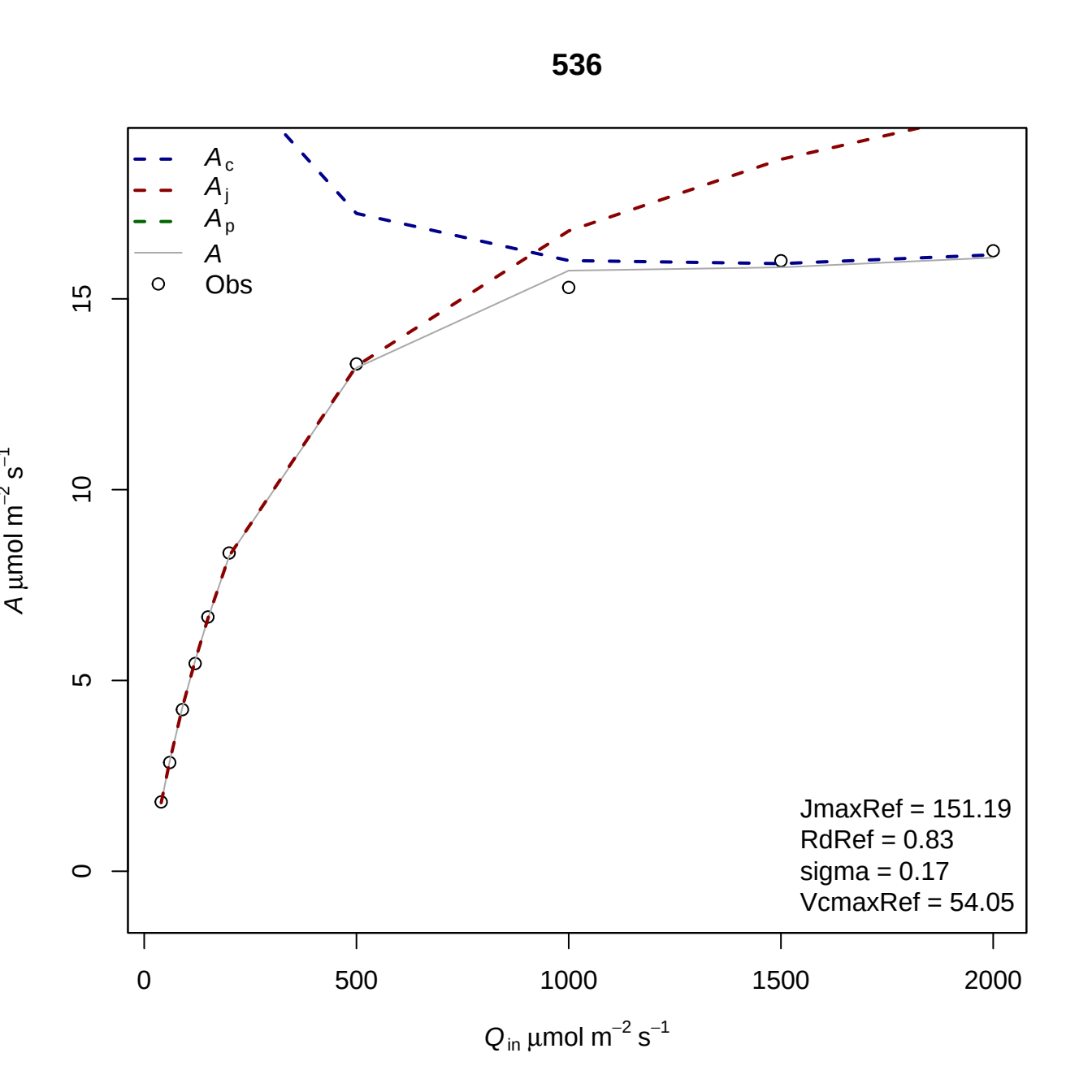

529
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.33
RdRef = 0.85
sigma = 0.14
VcmaxRef = 37
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

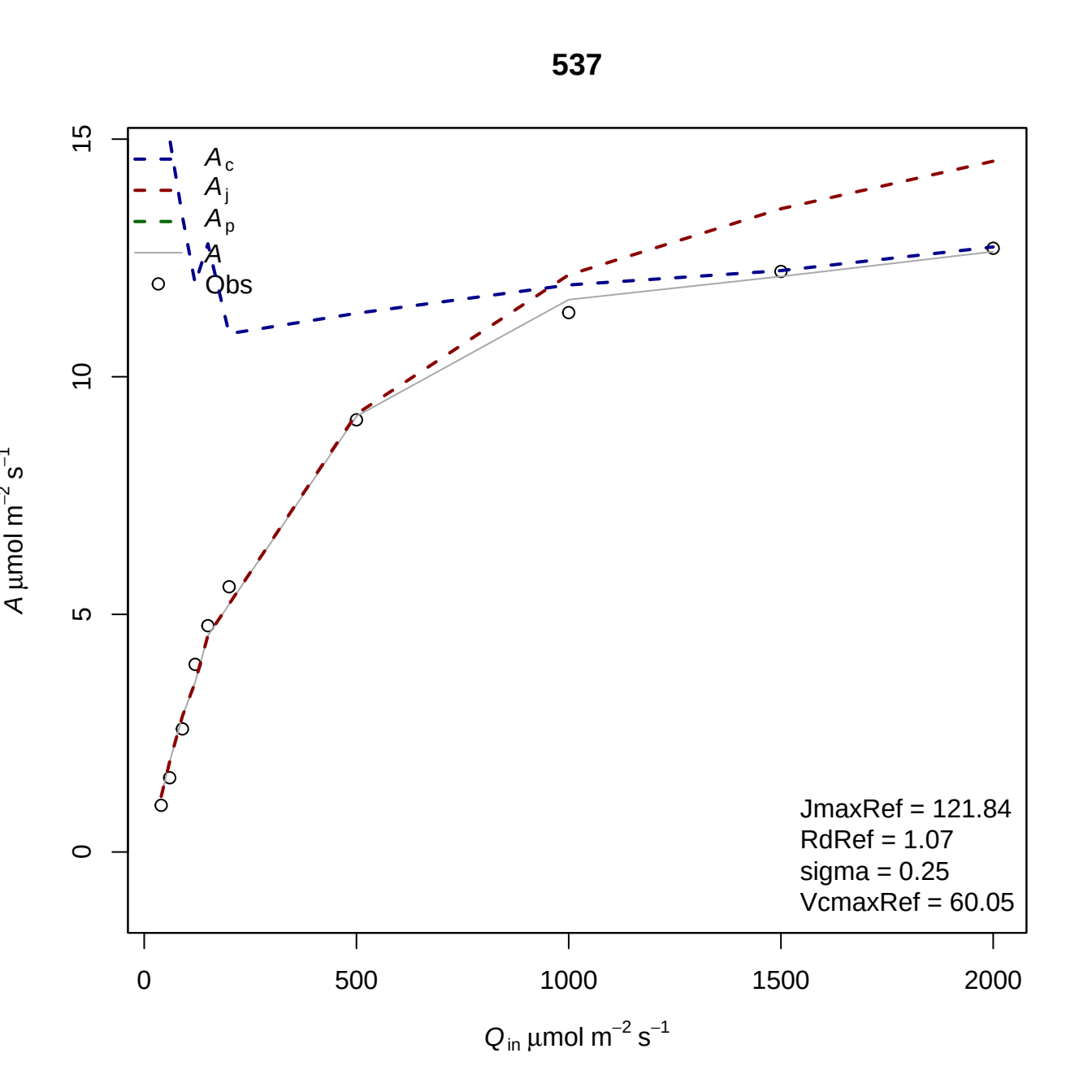

530
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.5
RdRef = 0.55
sigma = 0.19
VcmaxRef = 50.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

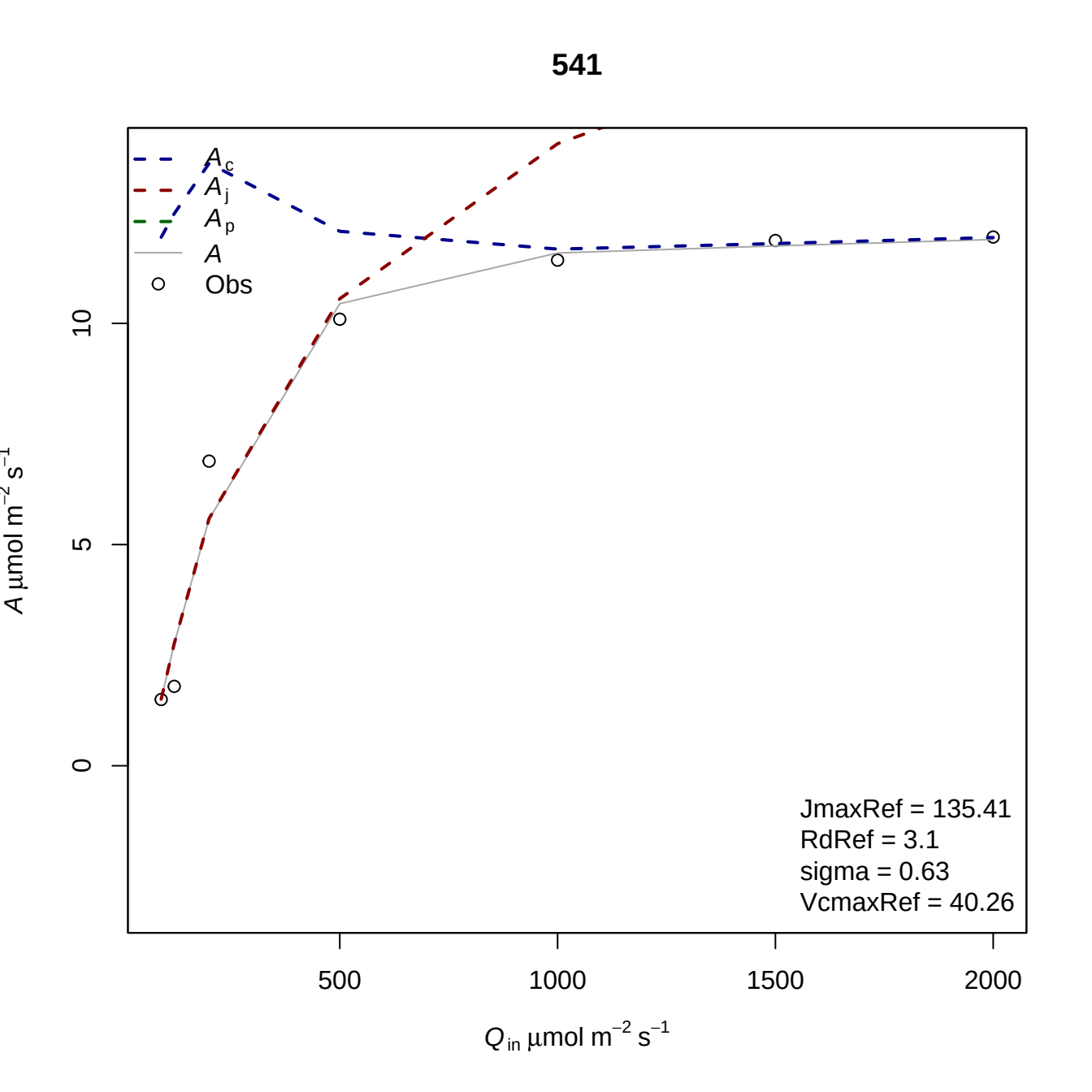

531
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.62
RdRef = 0.77
sigma = 0.24
VcmaxRef = 65.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

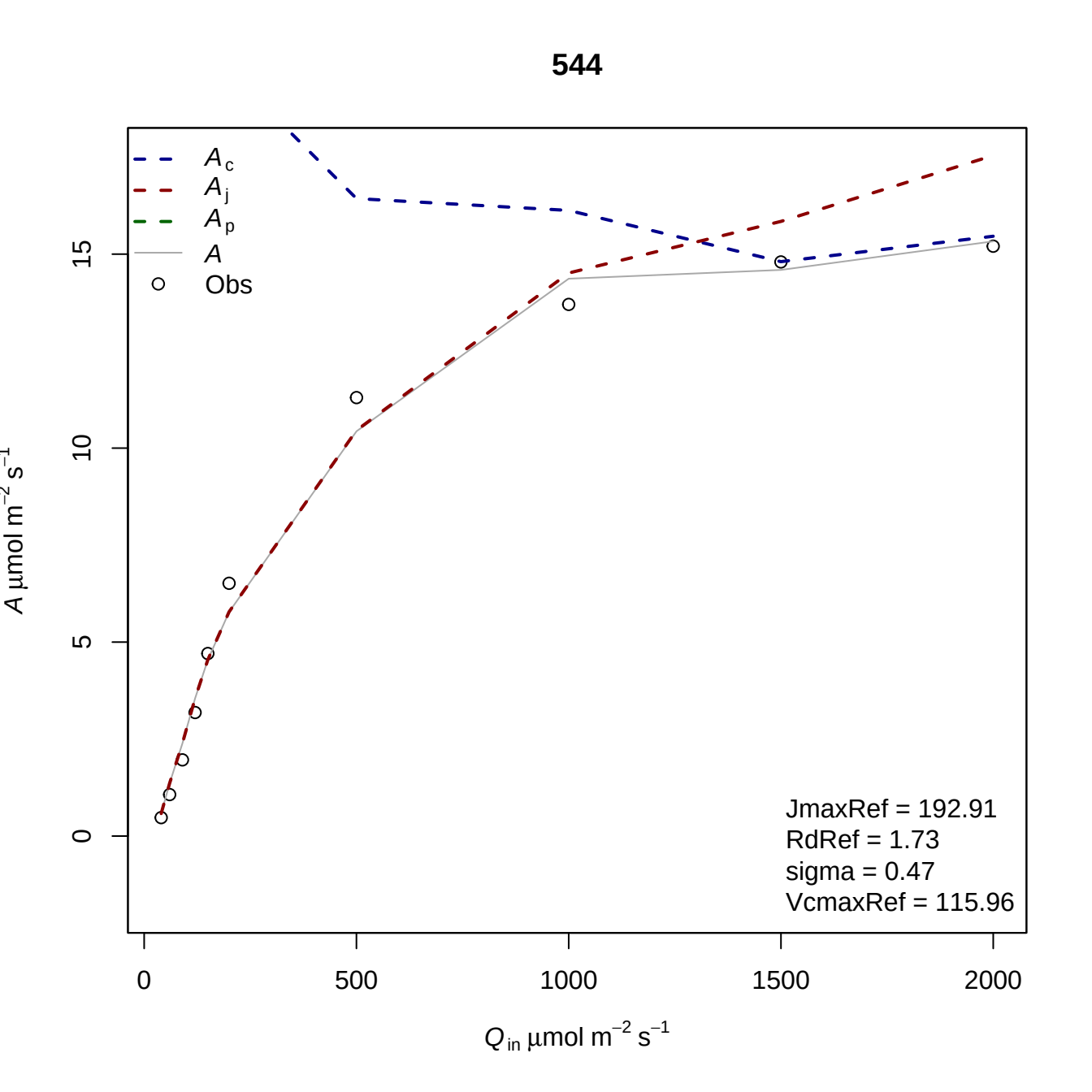

532
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.3
RdRef = 1.75
sigma = 0.32
VcmaxRef = 101.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

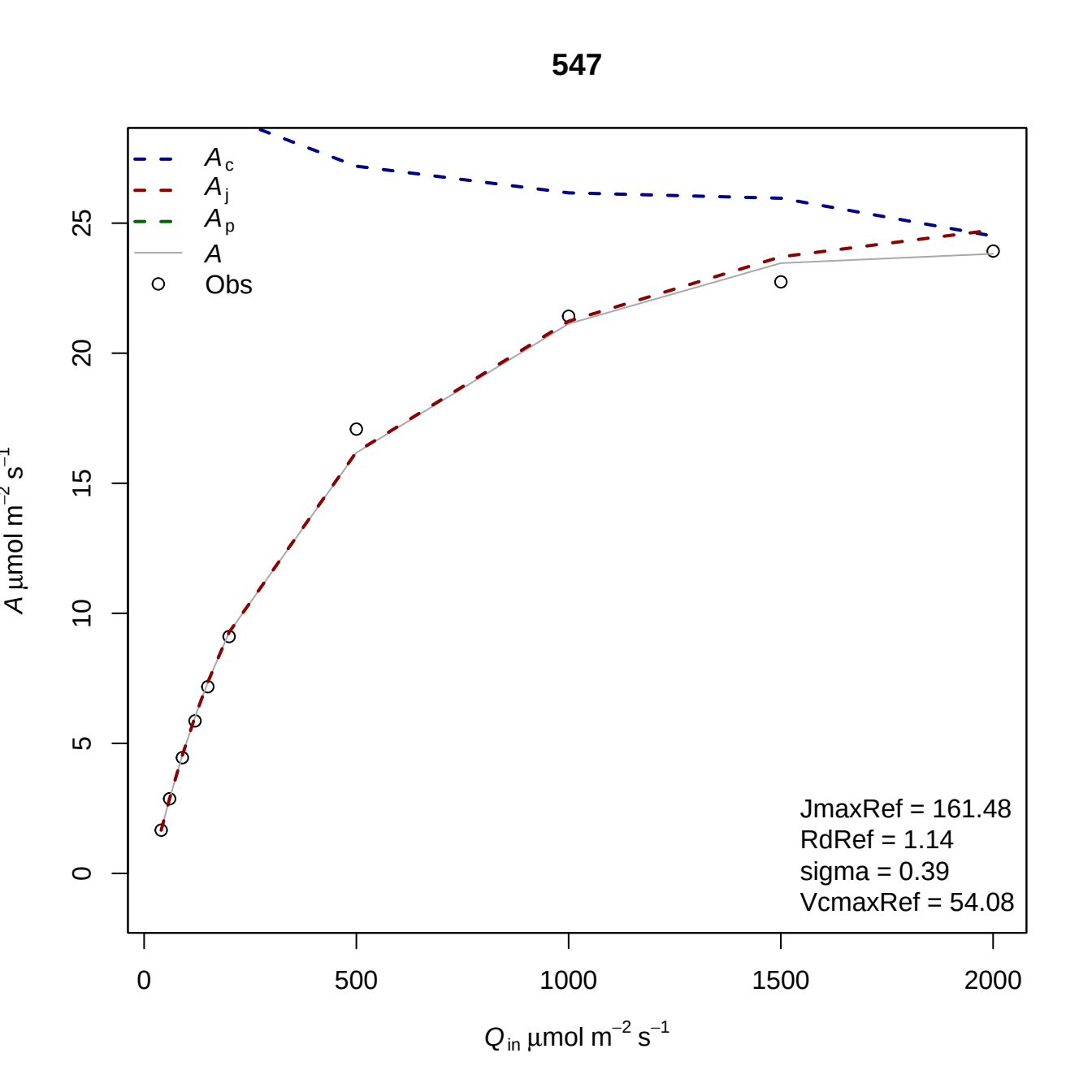

535
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.91
RdRef = 1.26
sigma = 0.19
VcmaxRef = 42.91
0
5
10
15
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

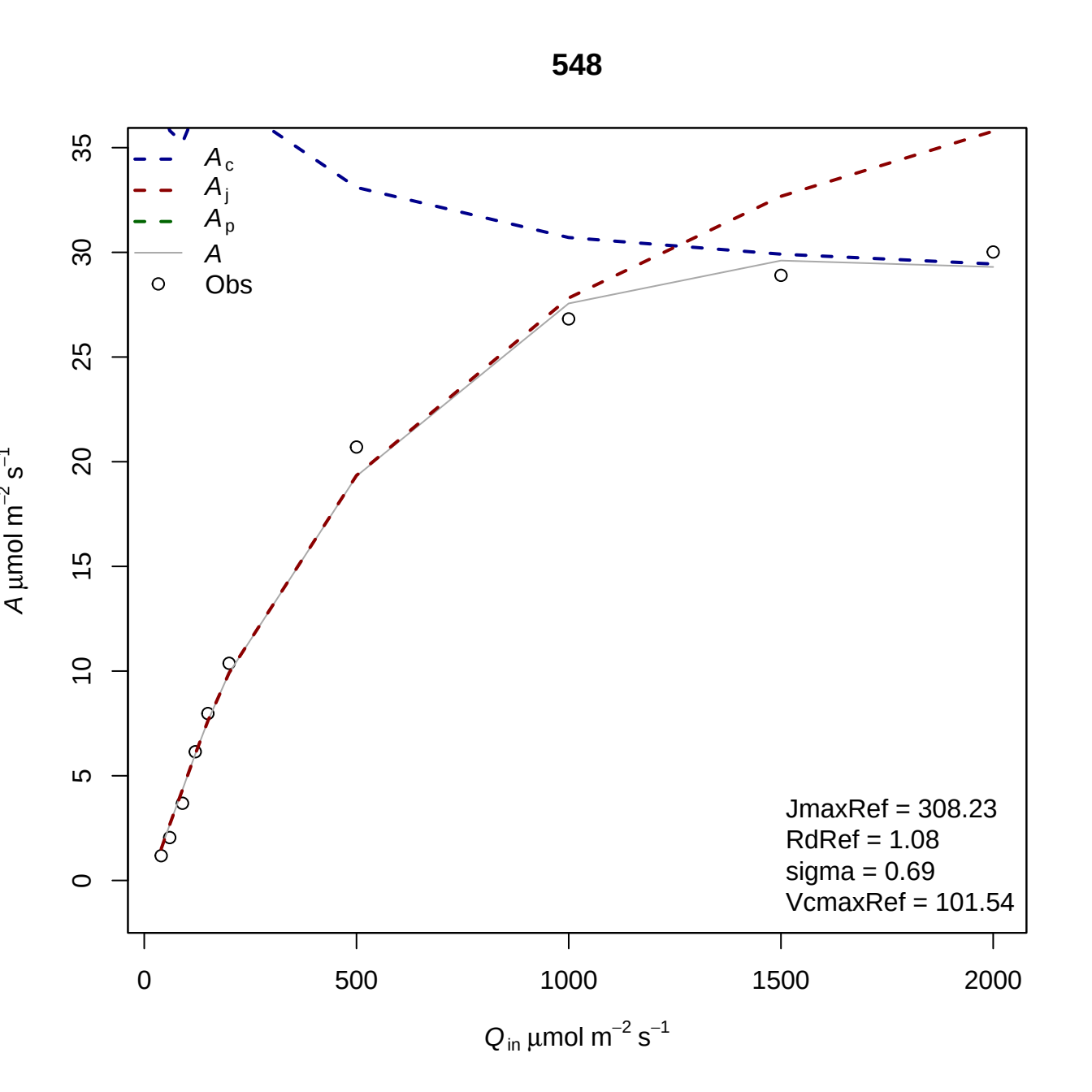

536
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 151.19
RdRef = 0.83
sigma = 0.17
VcmaxRef = 54.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

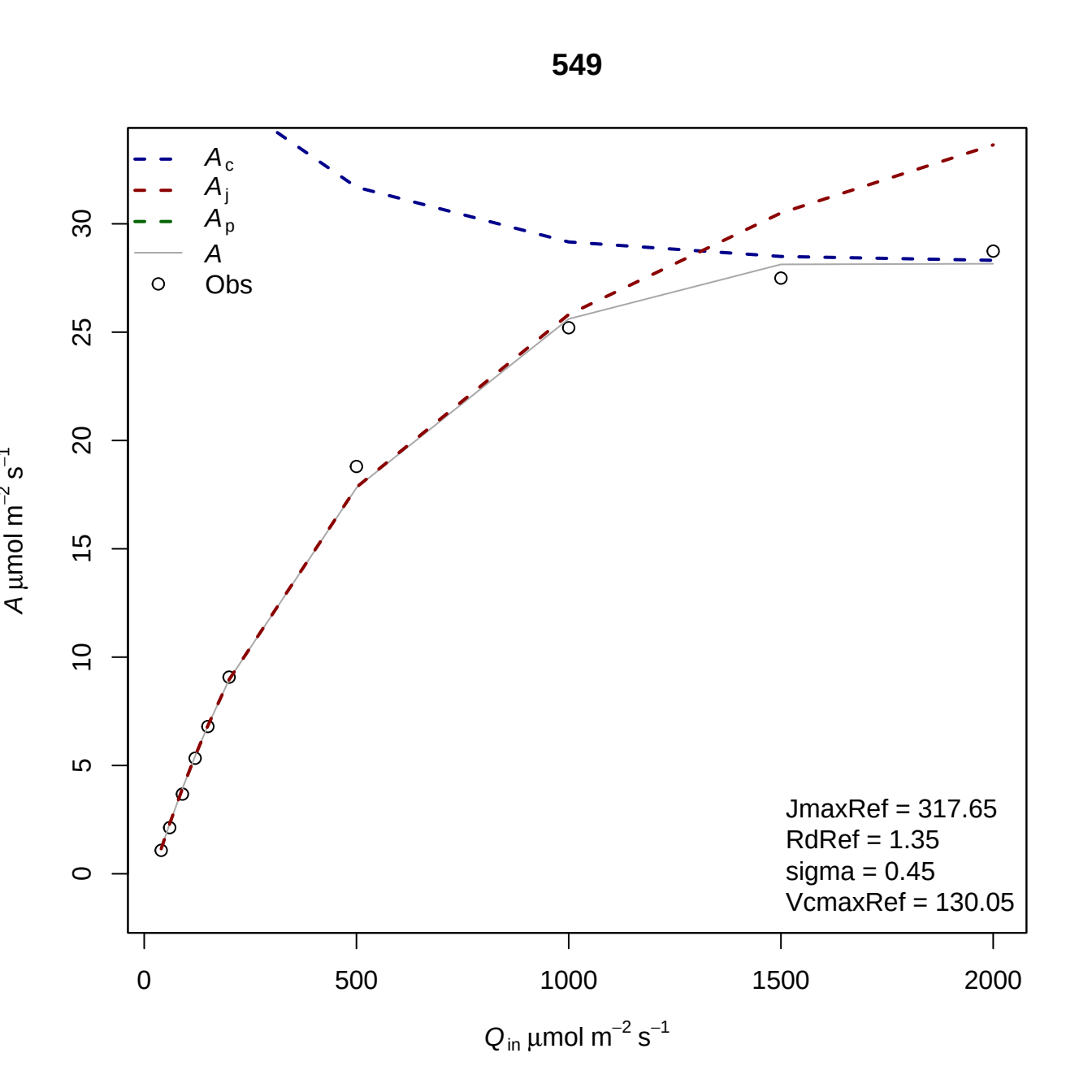

537
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.84
RdRef = 1.07
sigma = 0.25
VcmaxRef = 60.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

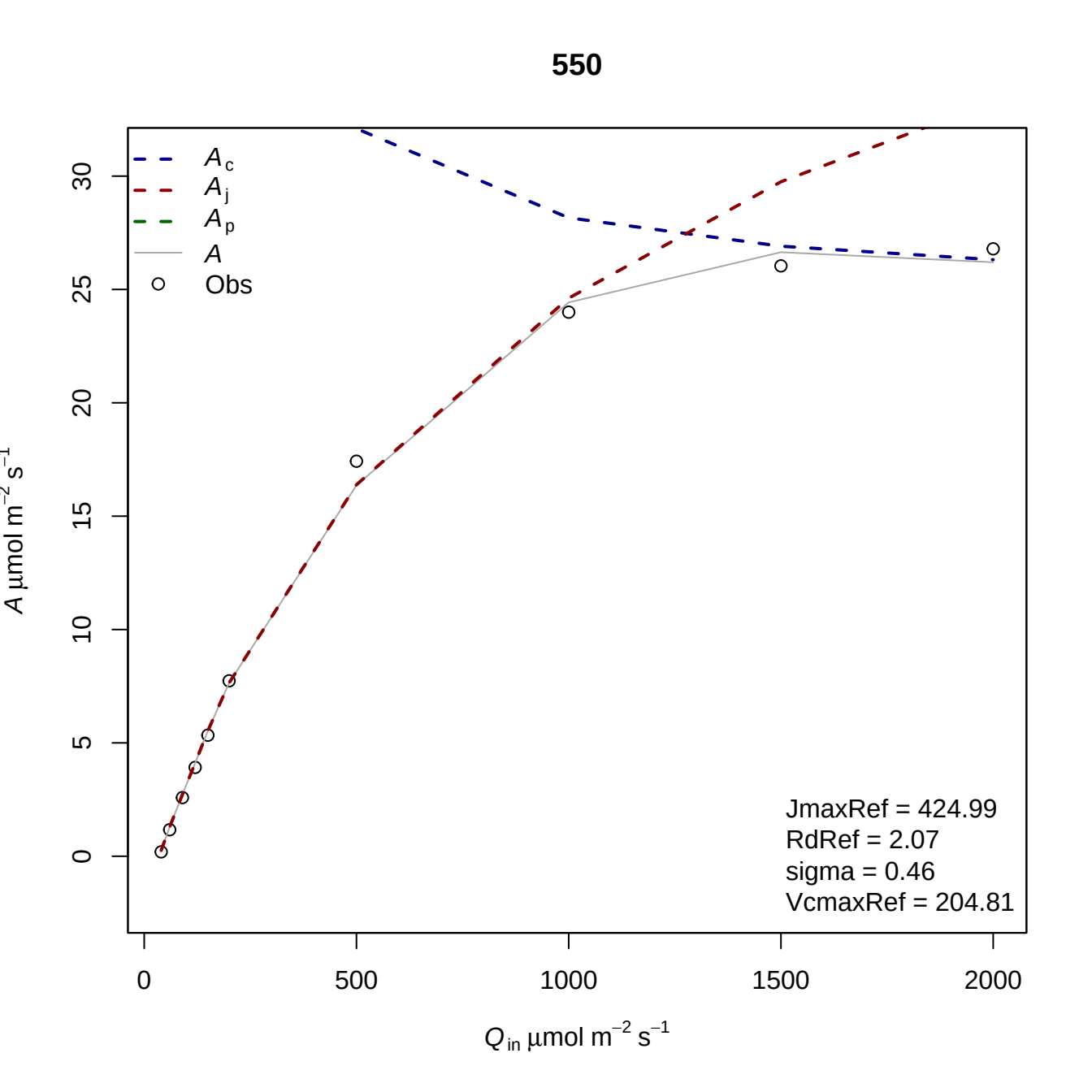

541
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.41
RdRef = 3.1
sigma = 0.63
VcmaxRef = 40.26
0
5
10
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

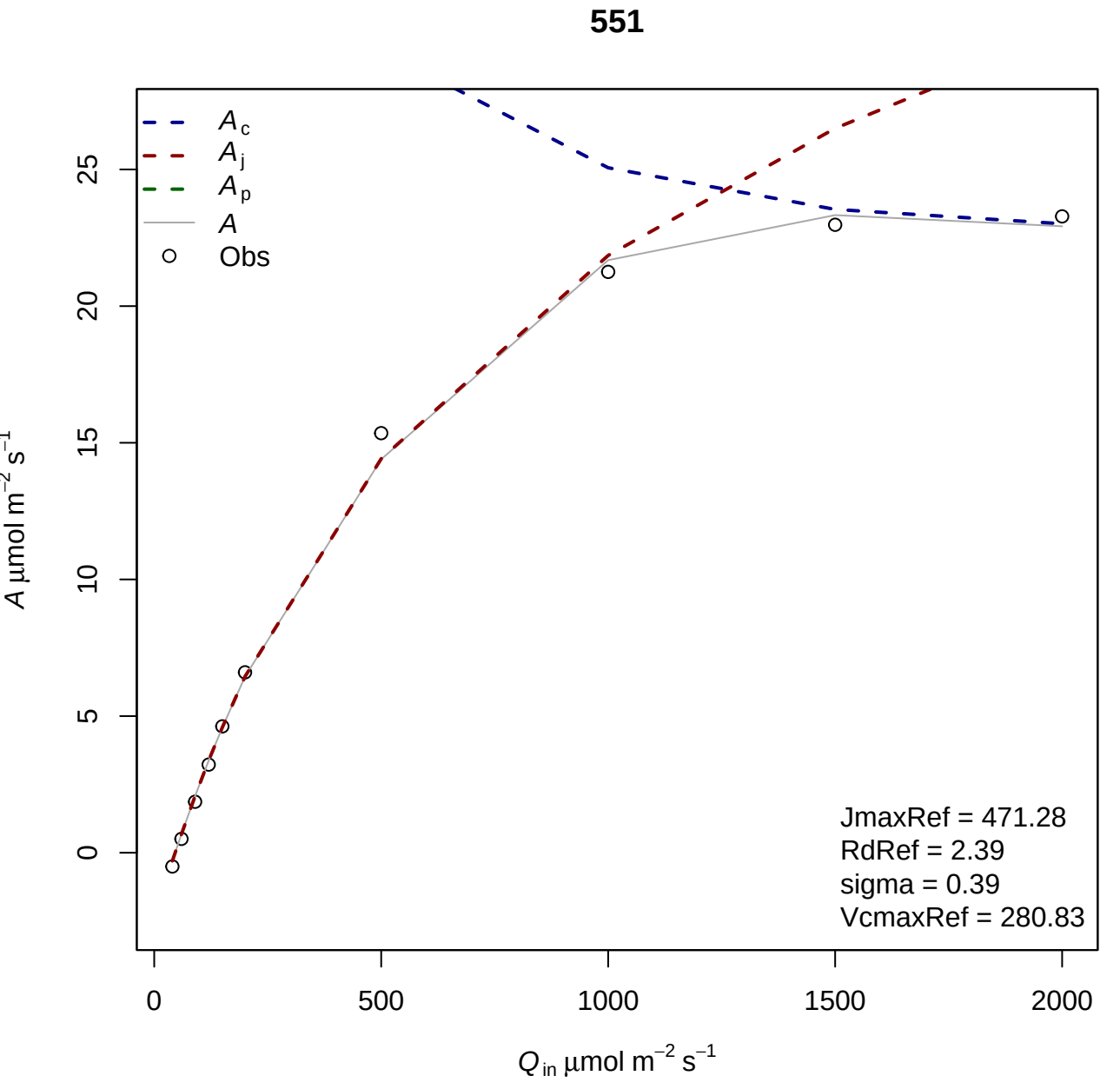

544
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 192.91
RdRef = 1.73
sigma = 0.47
VcmaxRef = 115.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

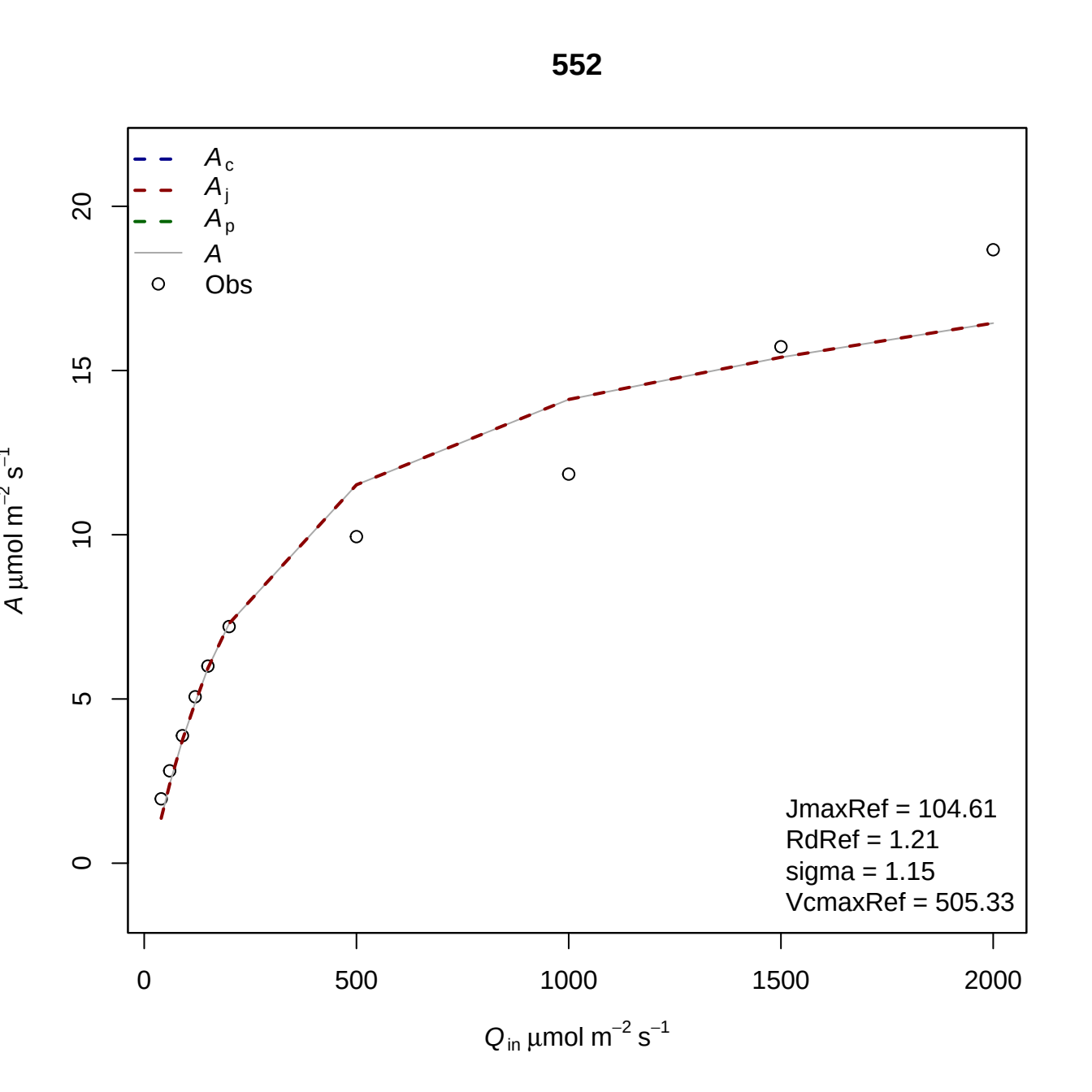

547
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.48
RdRef = 1.14
sigma = 0.39
VcmaxRef = 54.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

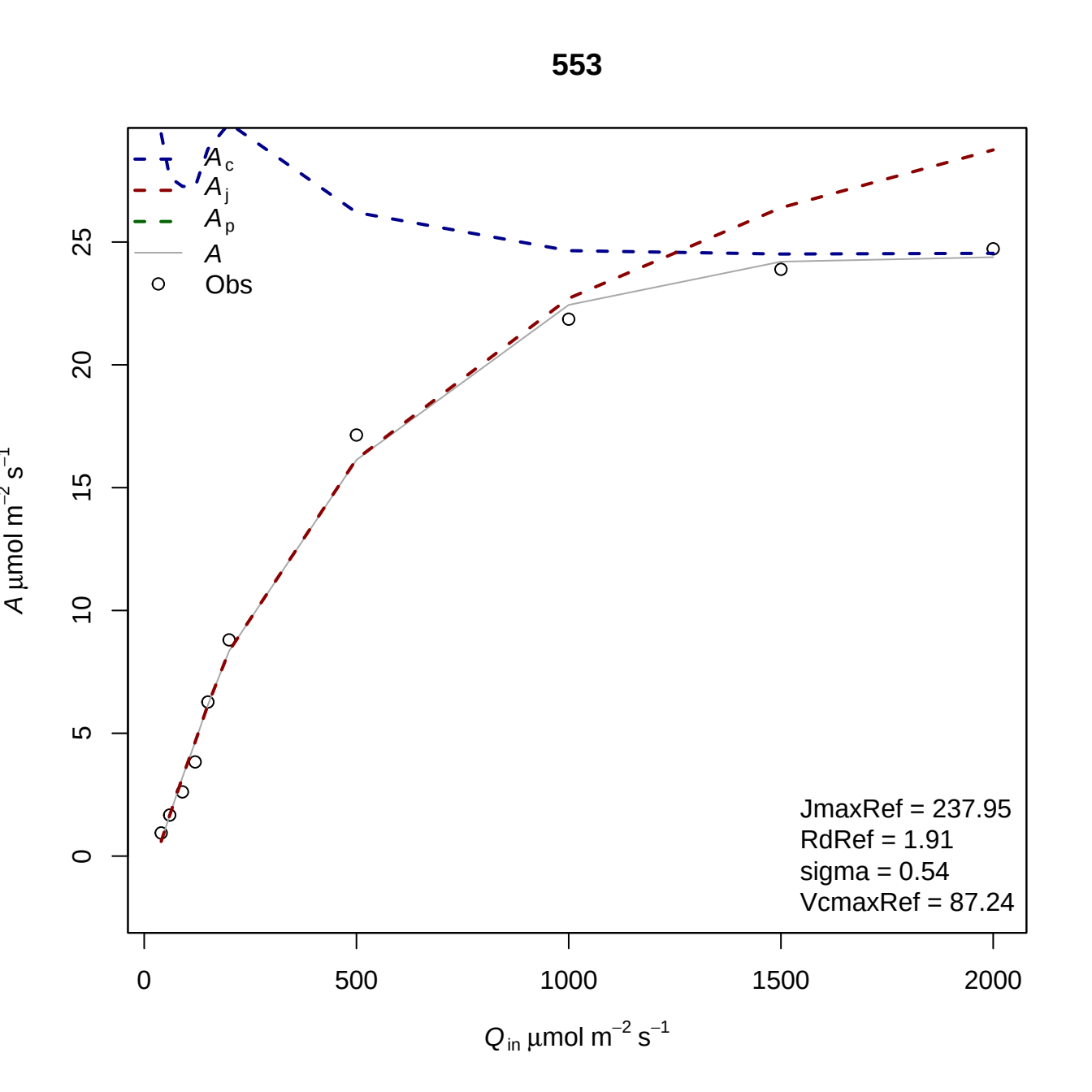

548
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 308.23
RdRef = 1.08
sigma = 0.69
VcmaxRef = 101.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

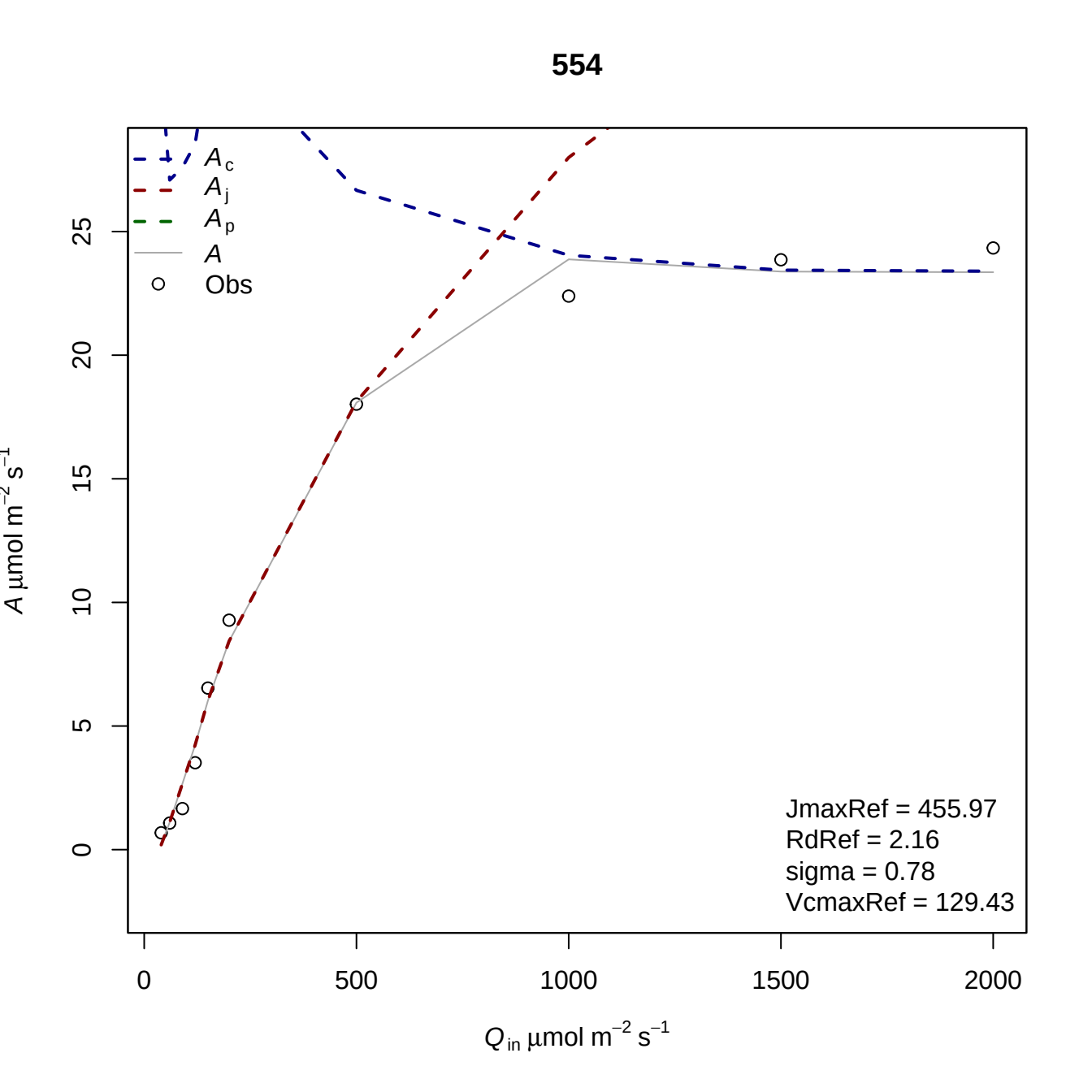

549
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 317.65
RdRef = 1.35
sigma = 0.45
VcmaxRef = 130.05
0
5
10
15
20
25
30
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

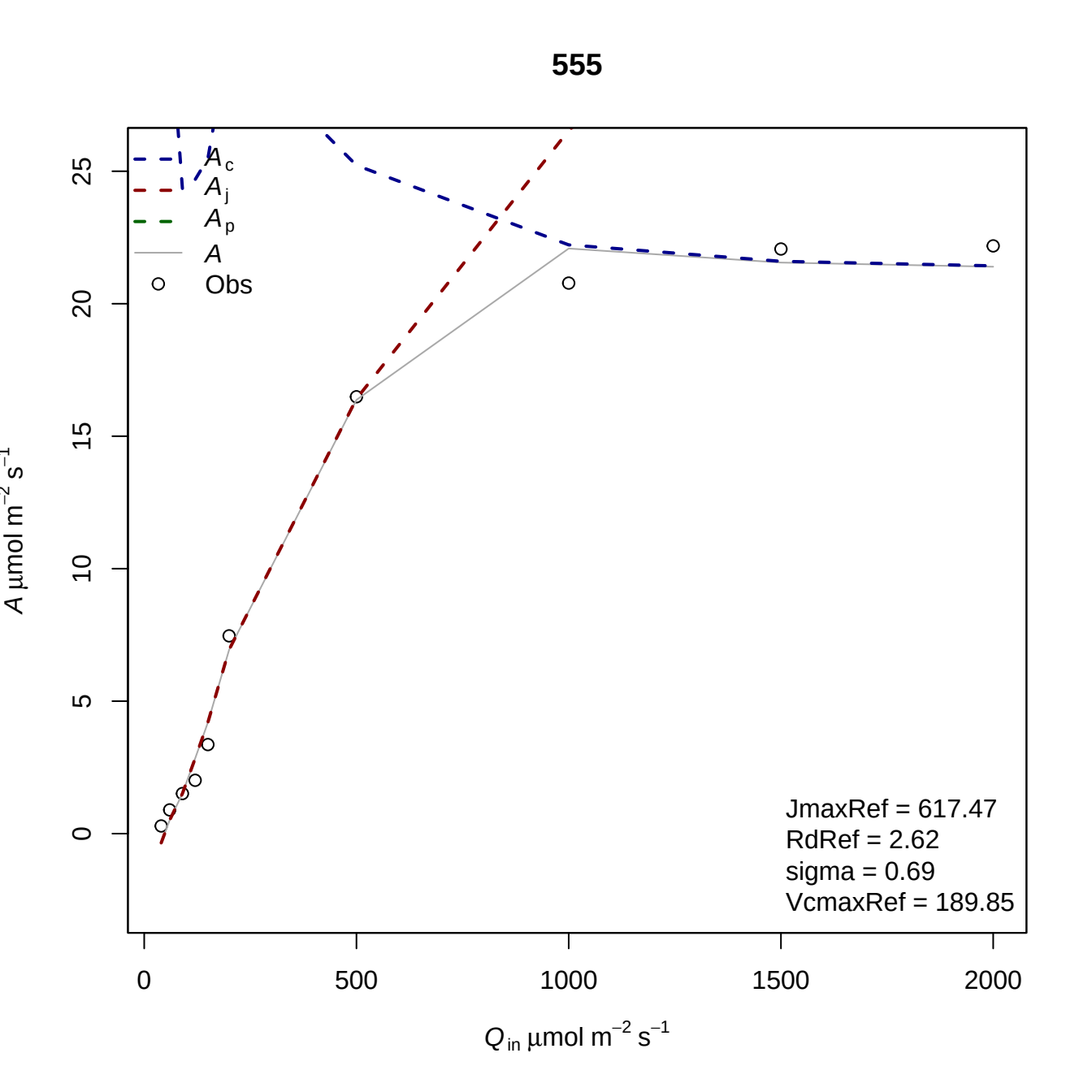

550
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 424.99
RdRef = 2.07
sigma = 0.46
VcmaxRef = 204.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

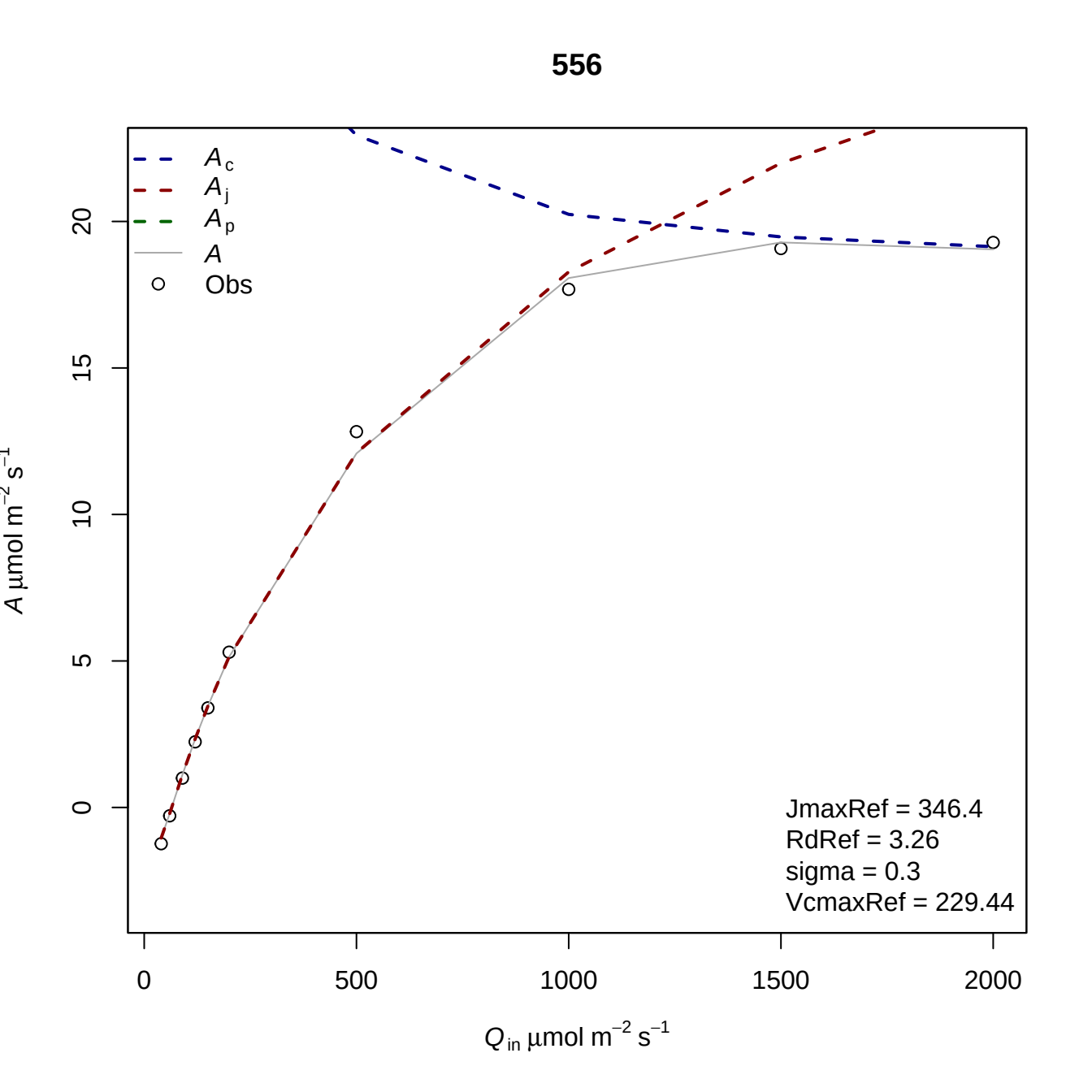

551
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 471.28
RdRef = 2.39
sigma = 0.39
VcmaxRef = 280.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

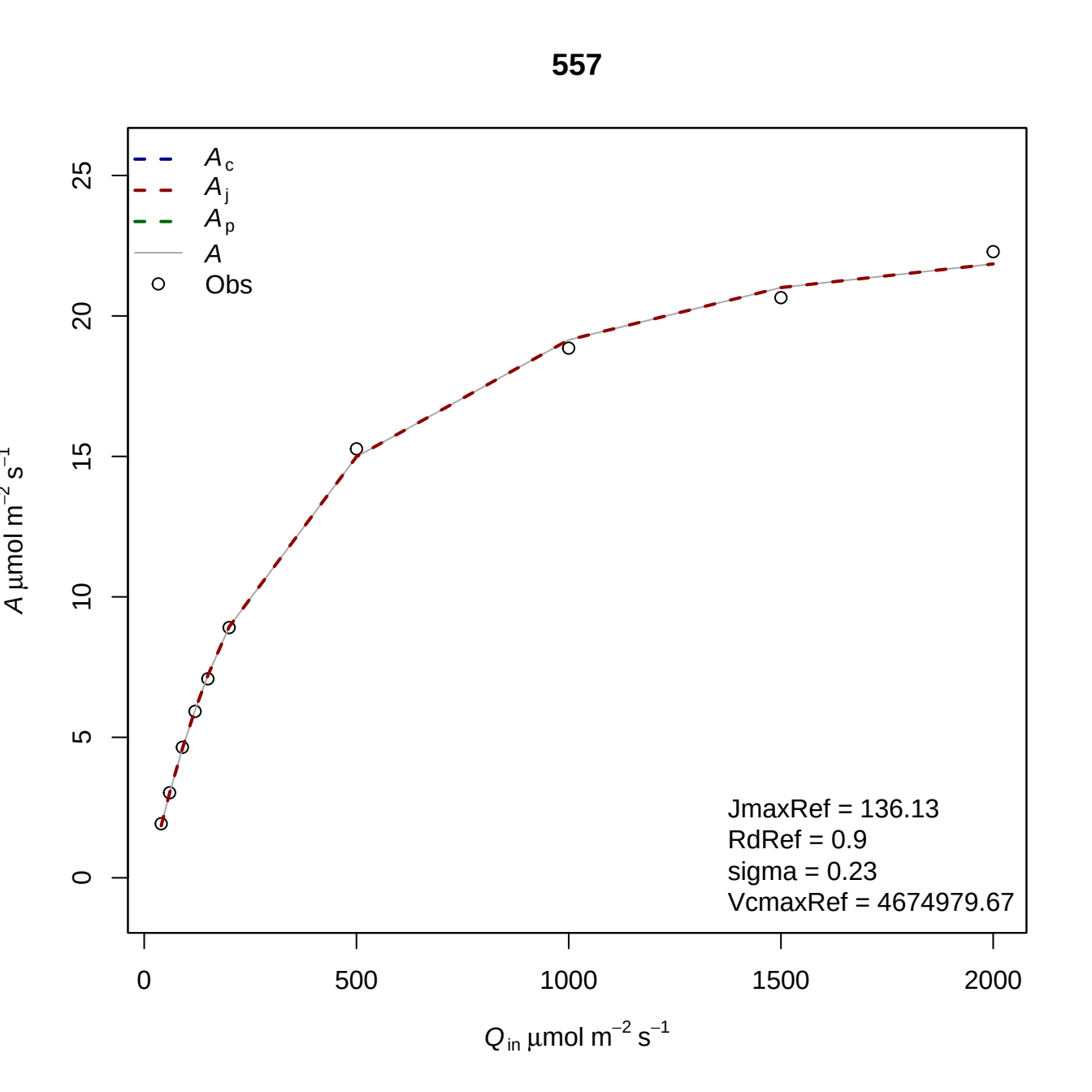

552
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.61
RdRef = 1.21
sigma = 1.15
VcmaxRef = 505.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

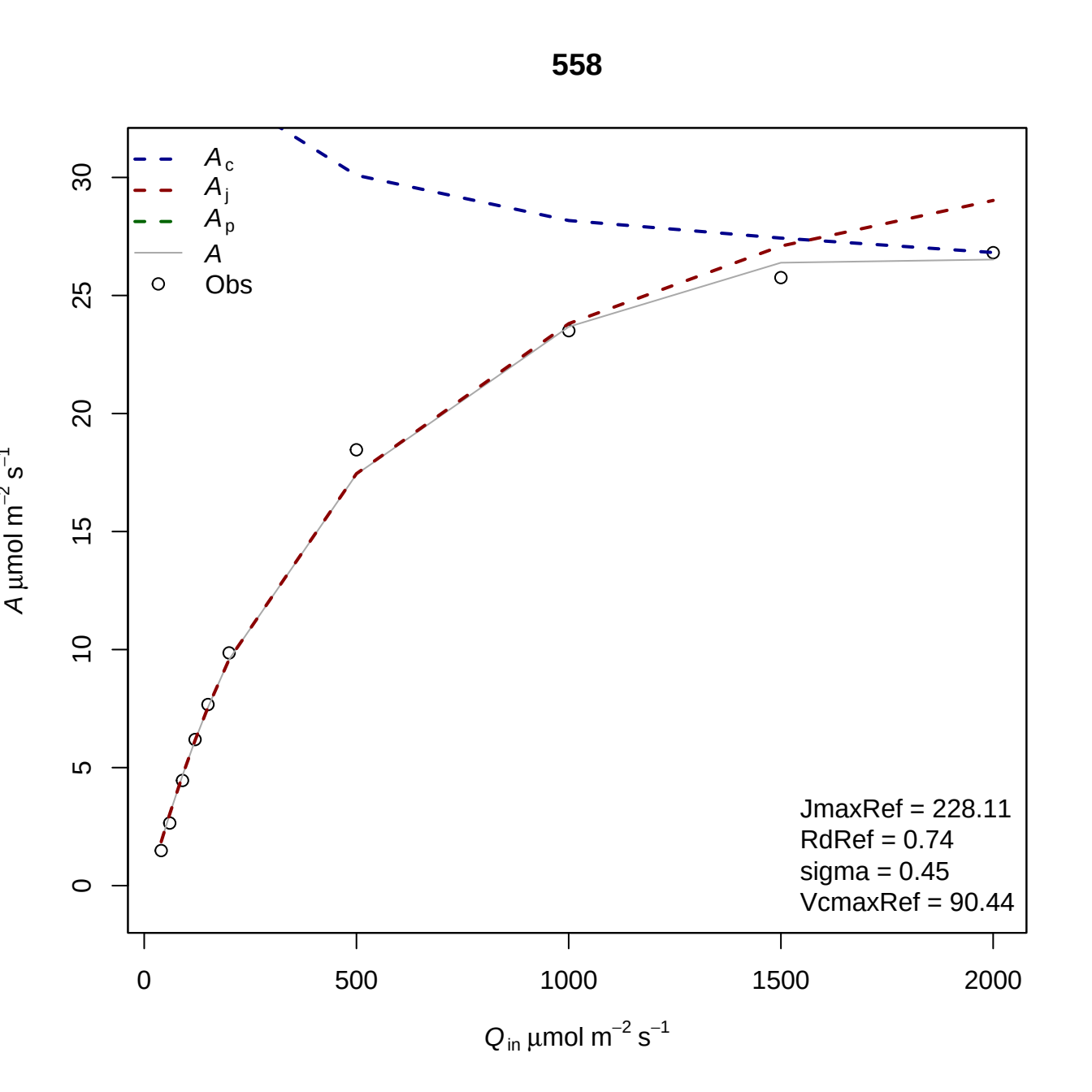

553
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 237.95
RdRef = 1.91
sigma = 0.54
VcmaxRef = 87.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

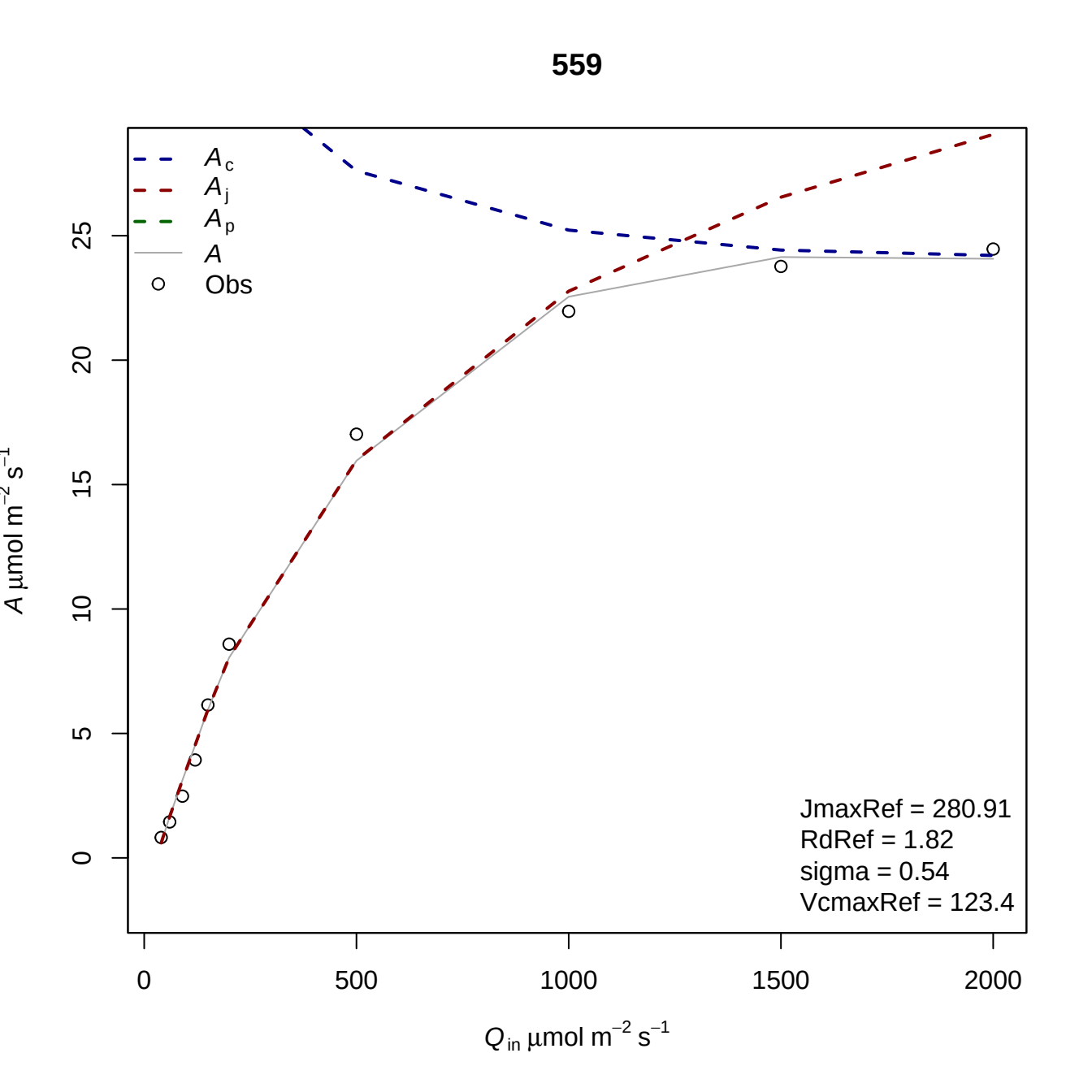

554
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 455.97
RdRef = 2.16
sigma = 0.78
VcmaxRef = 129.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

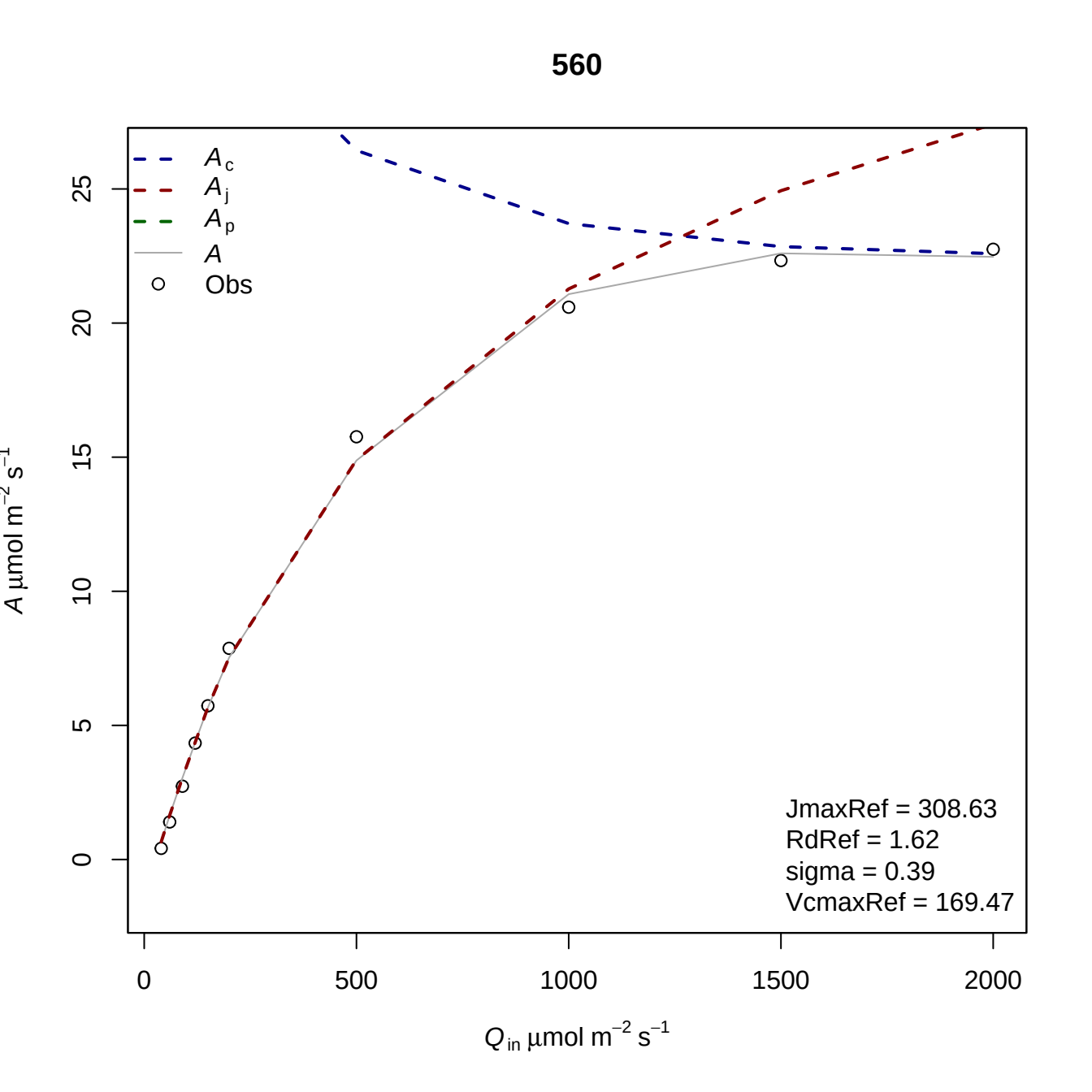

555
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 617.47
RdRef = 2.62
sigma = 0.69
VcmaxRef = 189.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

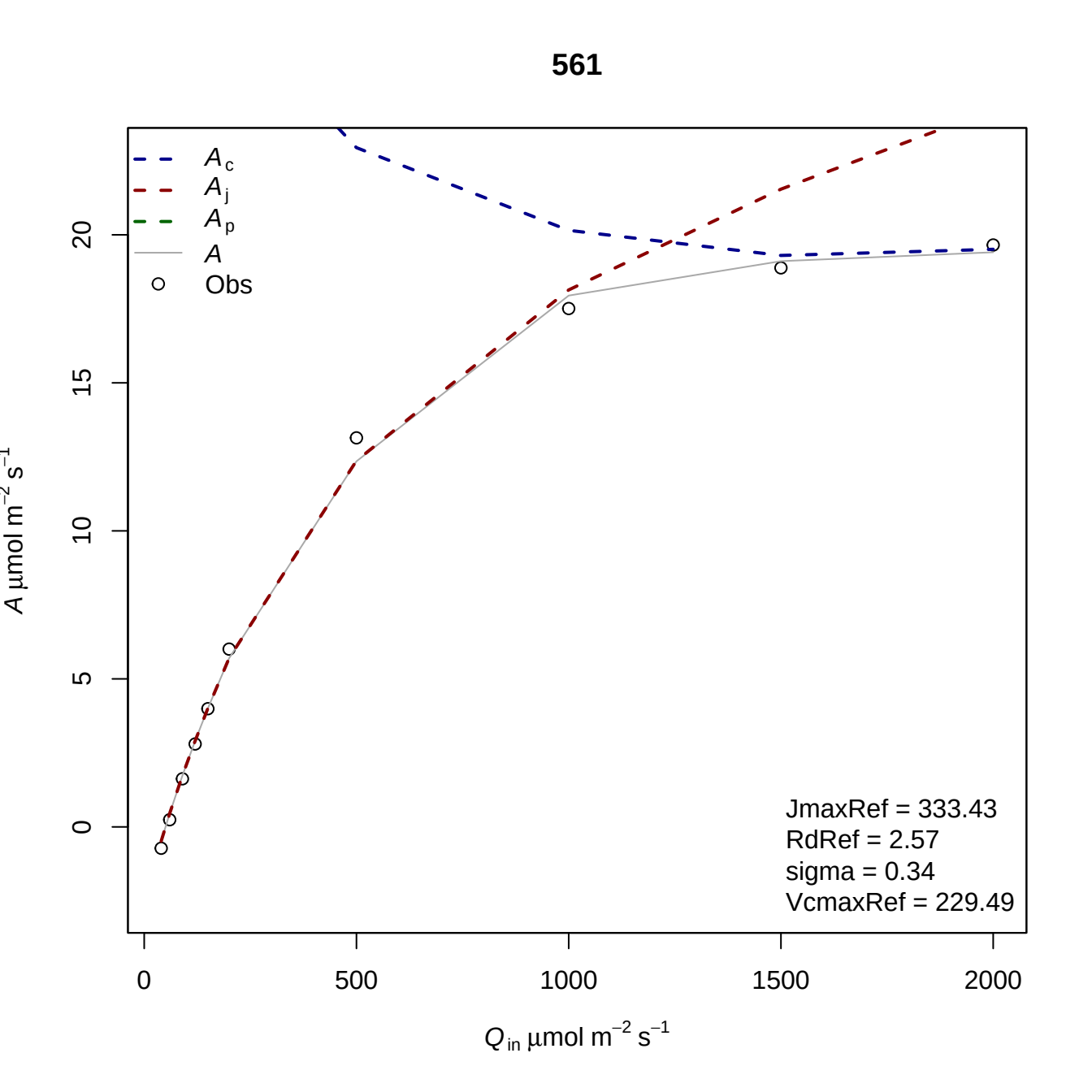

556
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 346.4
RdRef = 3.26
sigma = 0.3
VcmaxRef = 229.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

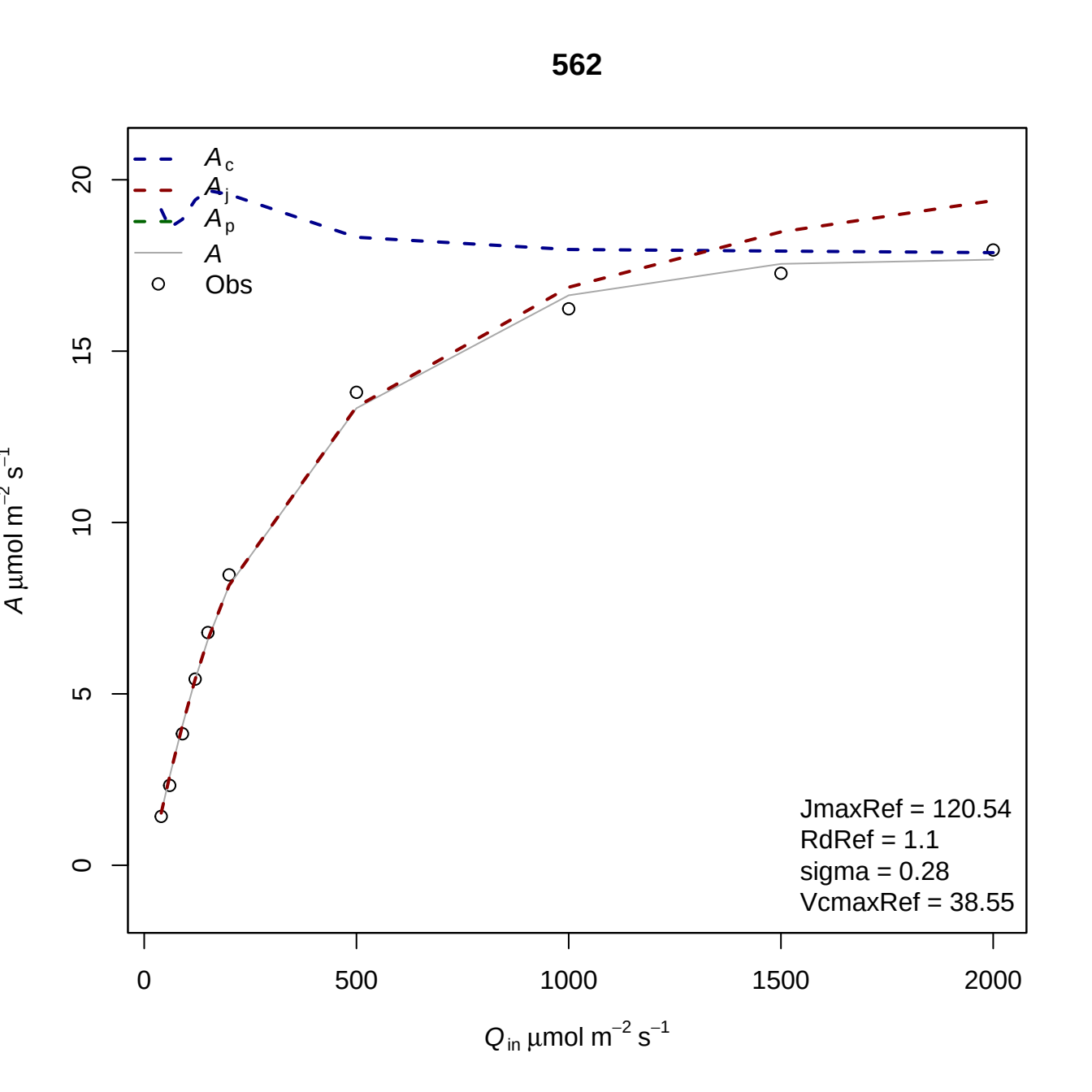

557
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.13
RdRef = 0.9
sigma = 0.23
VcmaxRef = 4674979.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

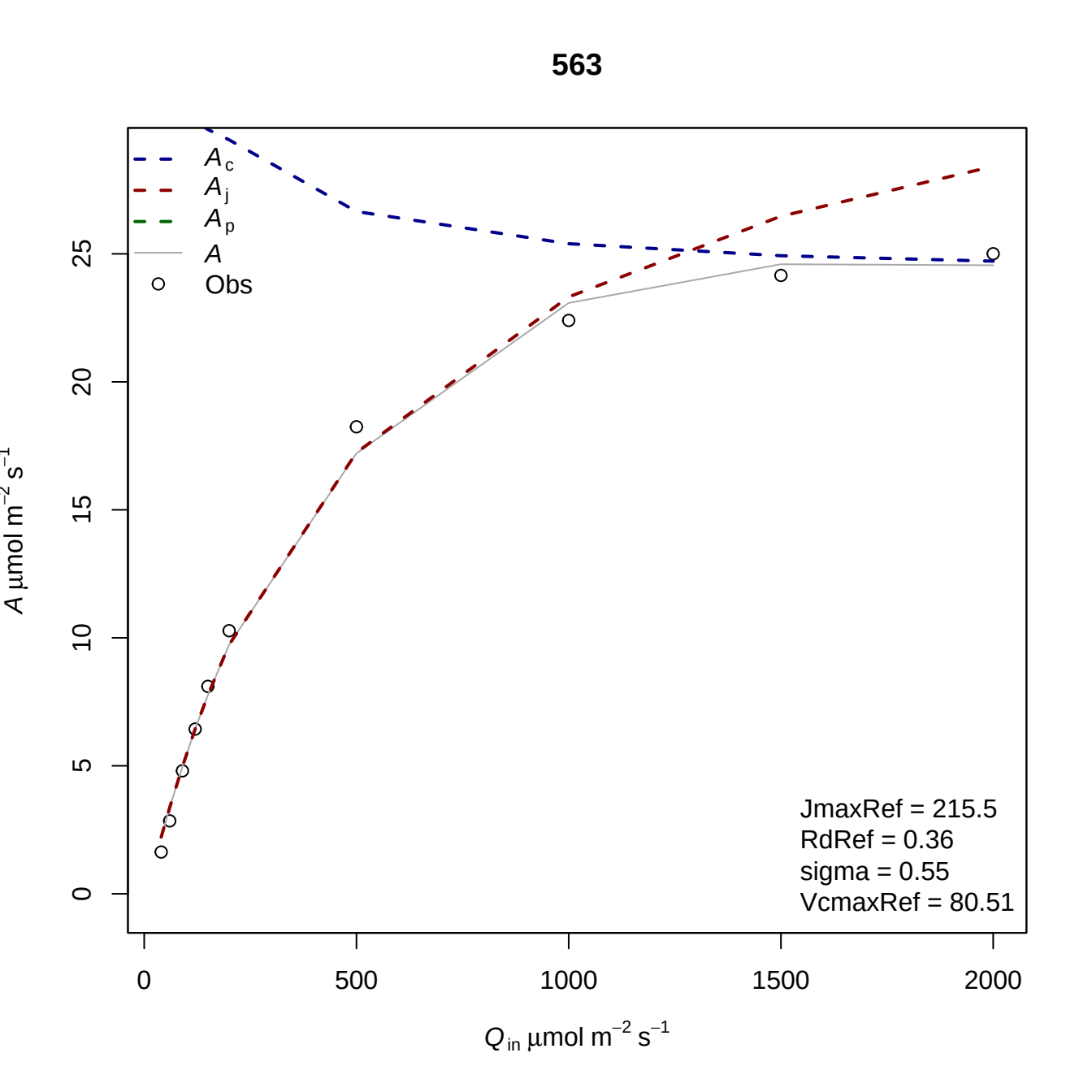

558
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 228.11
RdRef = 0.74
sigma = 0.45
VcmaxRef = 90.44
0
5
10
15
20
25
30
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

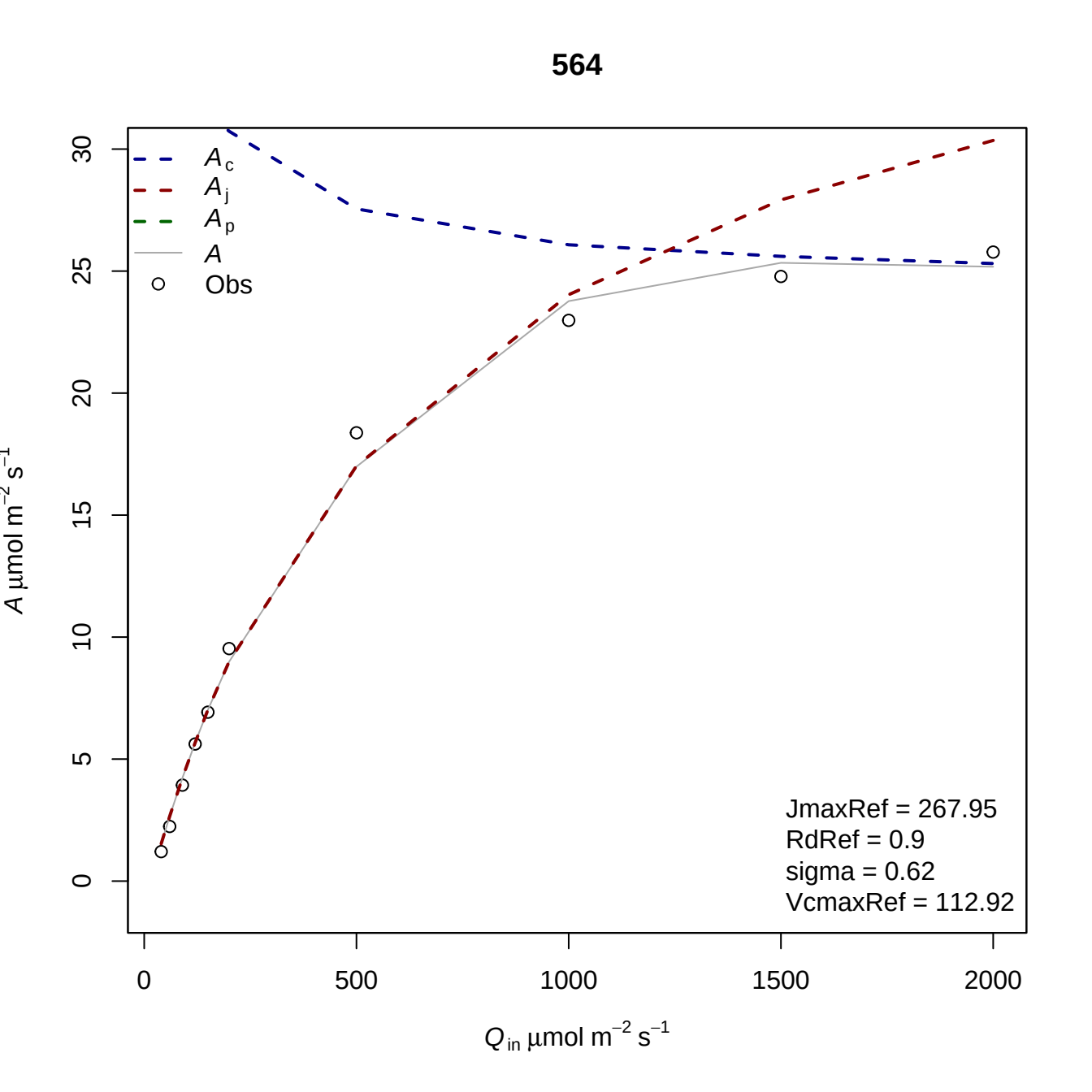

559
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 280.91
RdRef = 1.82
sigma = 0.54
VcmaxRef = 123.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

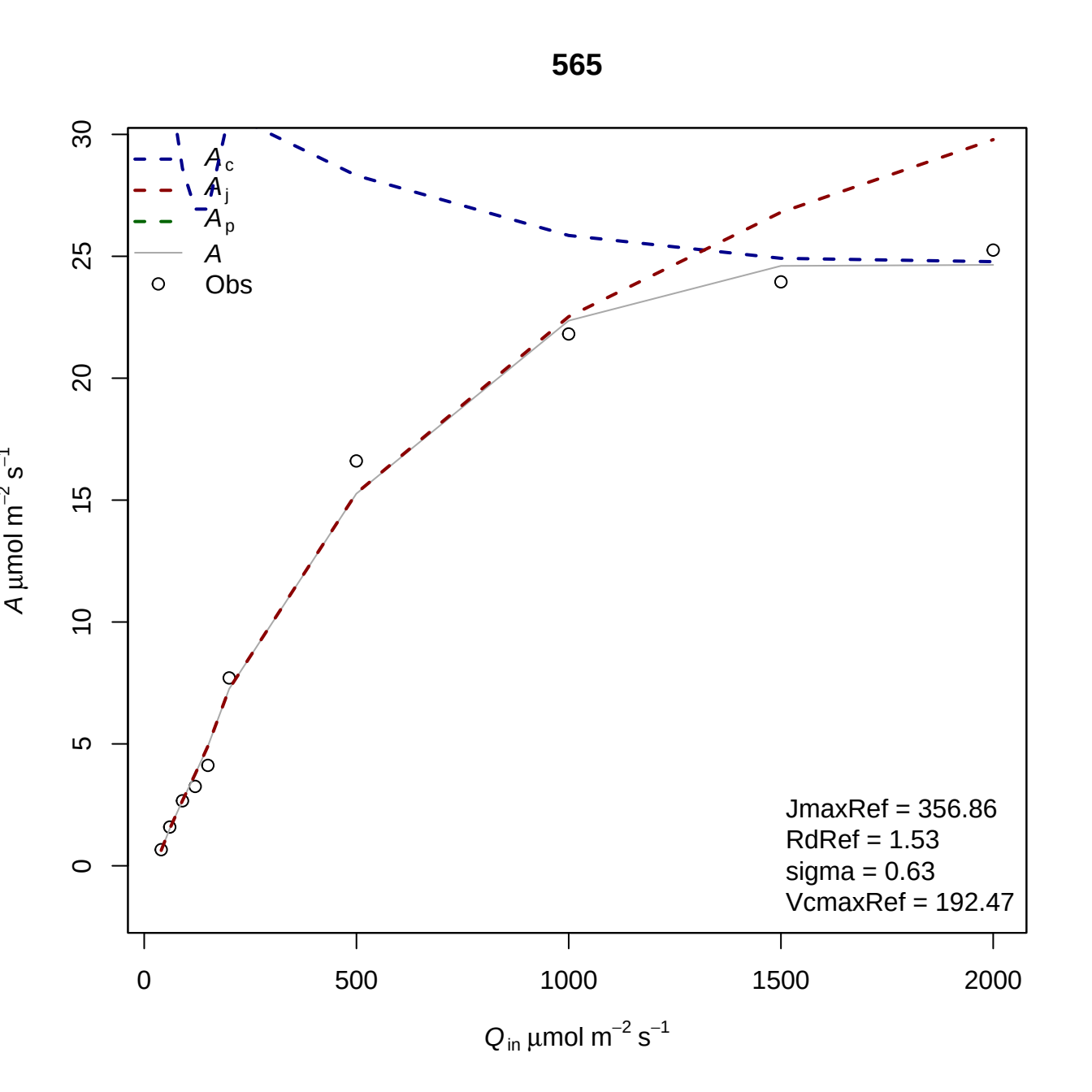

560
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 308.63
RdRef = 1.62
sigma = 0.39
VcmaxRef = 169.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

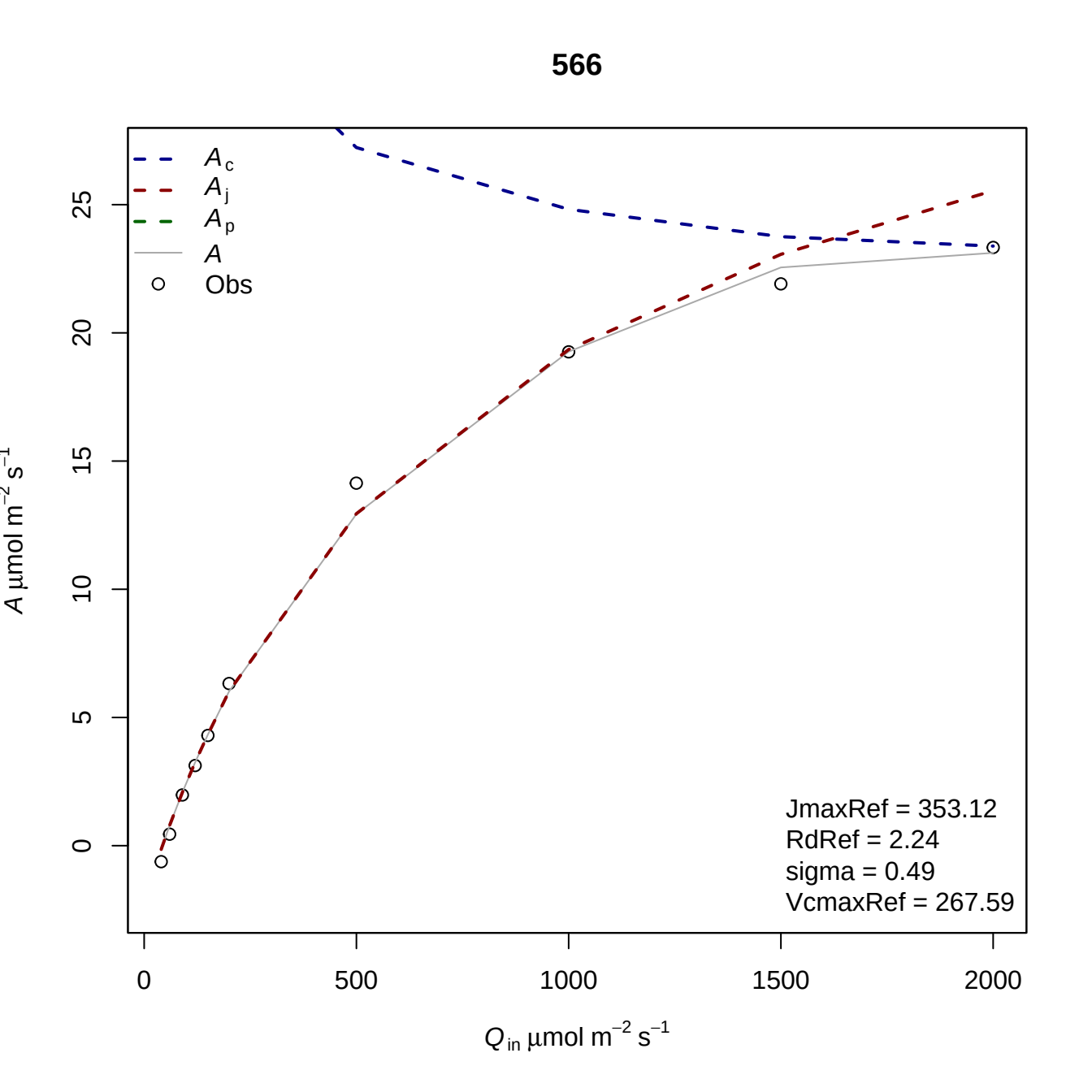

561
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 333.43
RdRef = 2.57
sigma = 0.34
VcmaxRef = 229.49
0
5
10
15
20
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

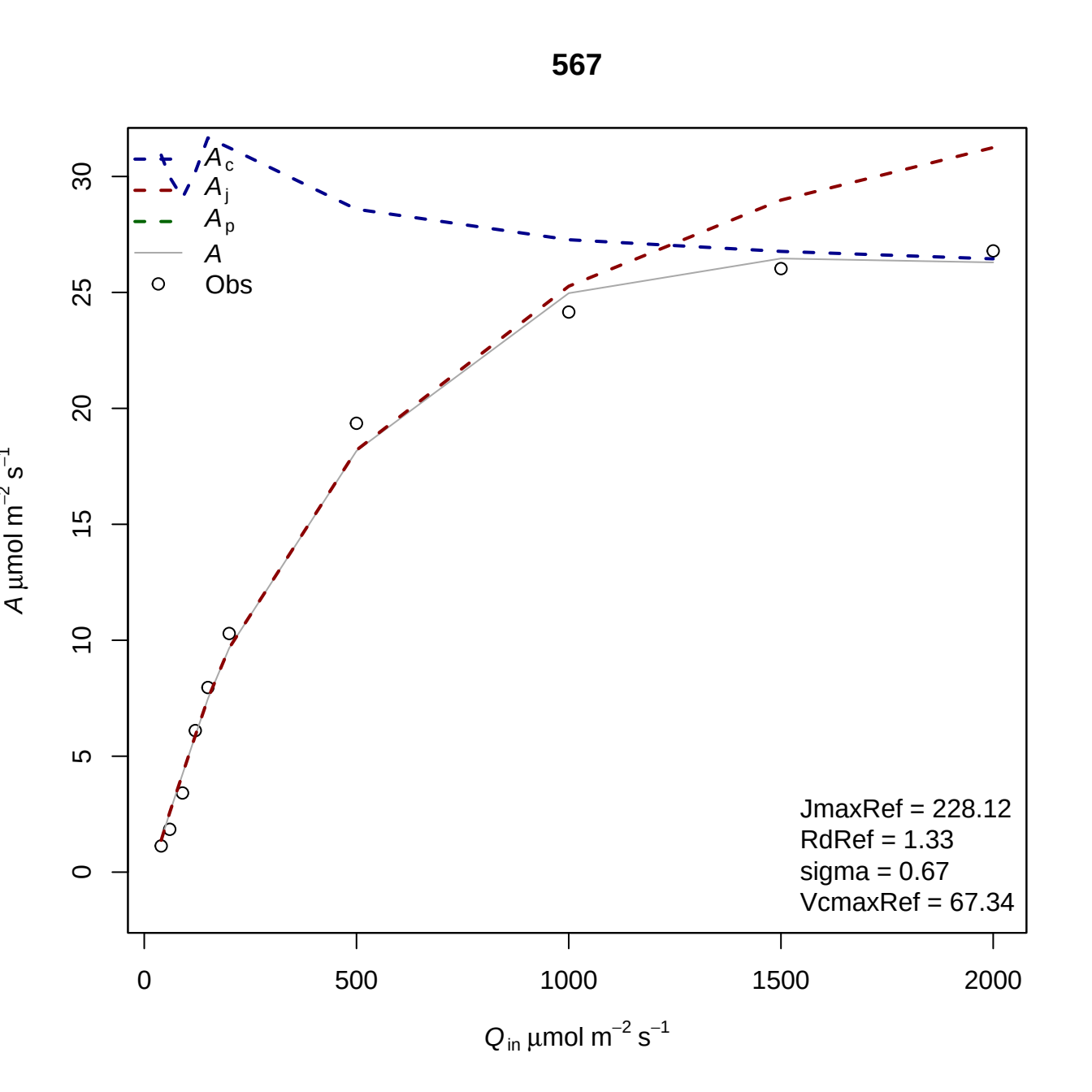

562
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.54
RdRef = 1.1
sigma = 0.28
VcmaxRef = 38.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

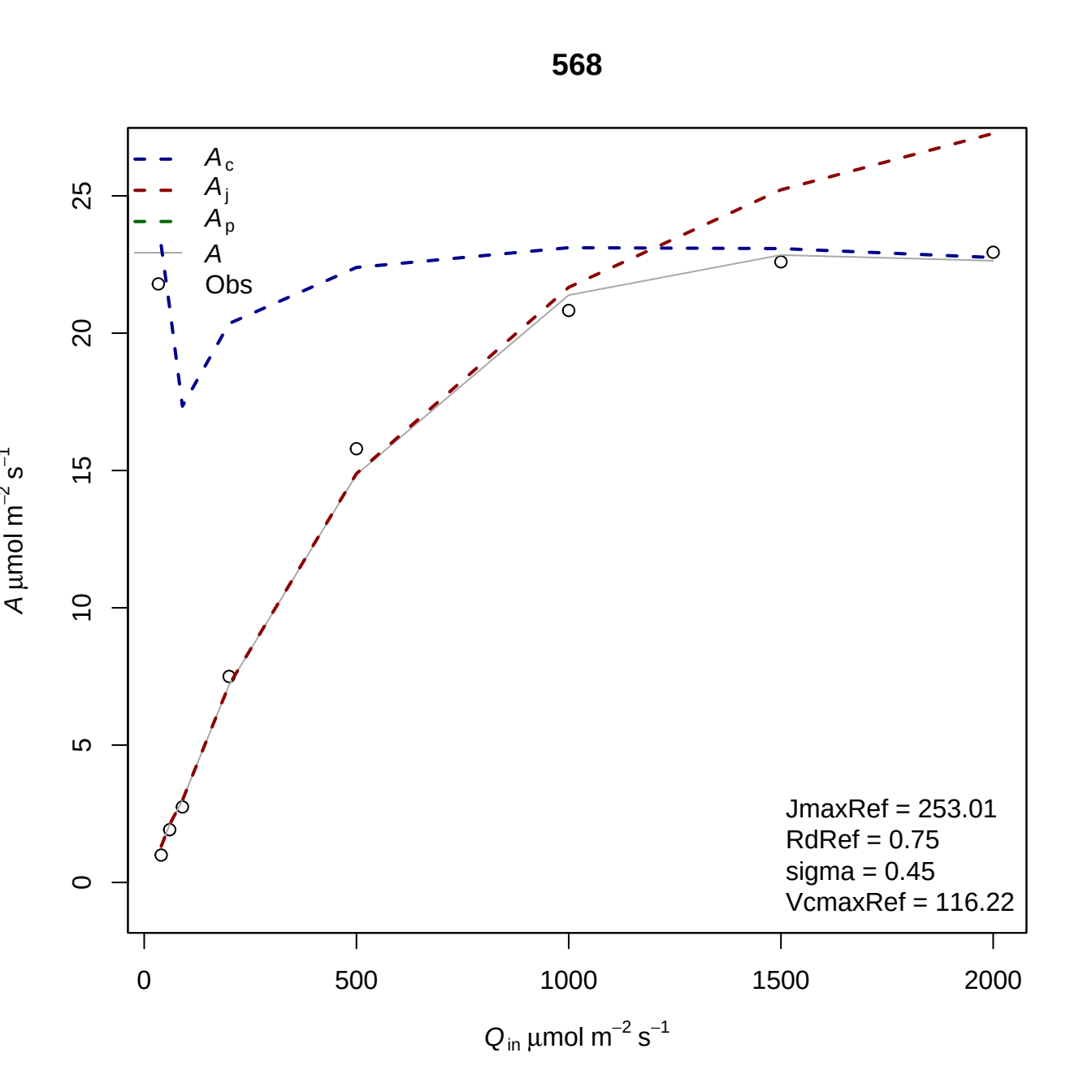

563
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 215.5
RdRef = 0.36
sigma = 0.55
VcmaxRef = 80.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

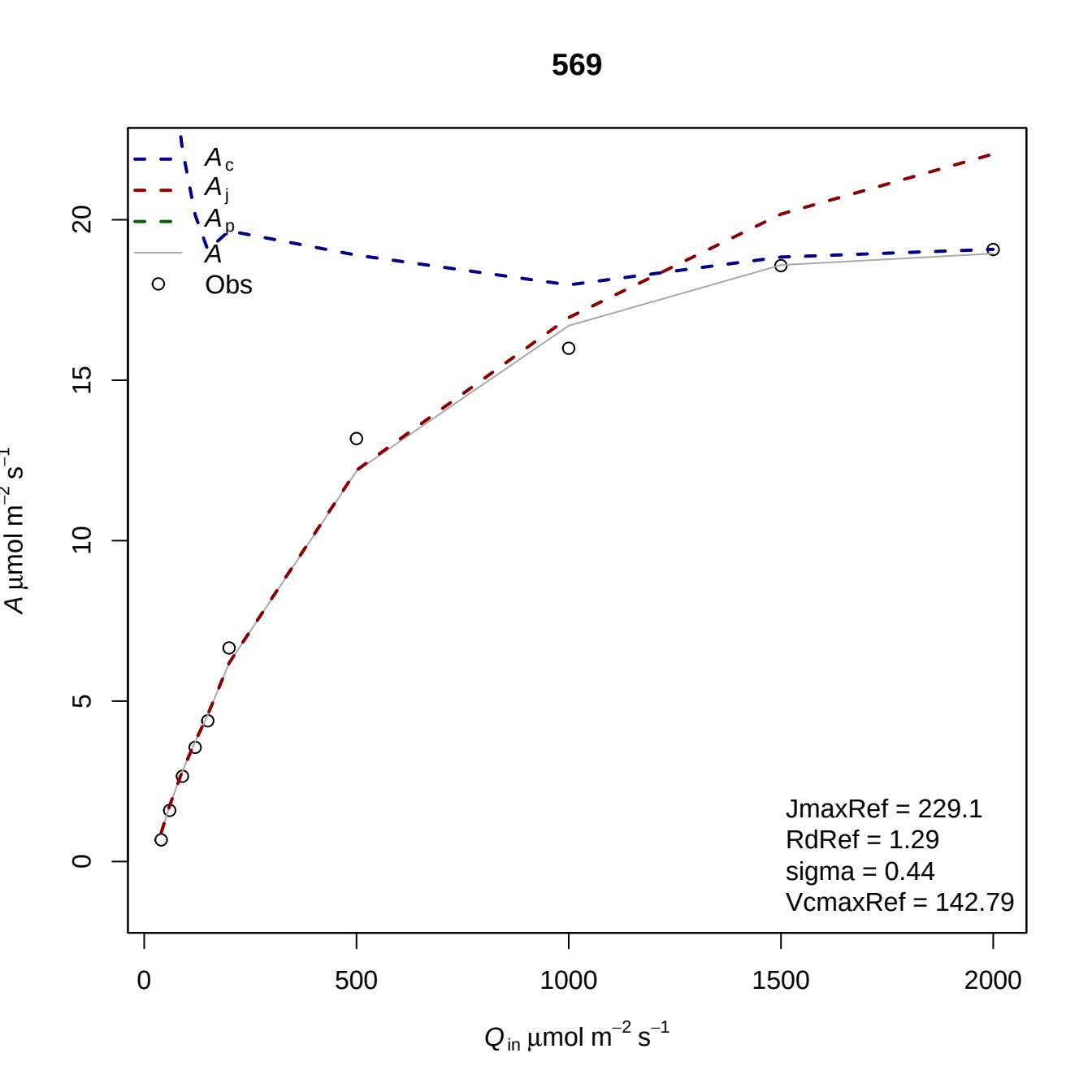

564
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 267.95
RdRef = 0.9
sigma = 0.62
VcmaxRef = 112.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

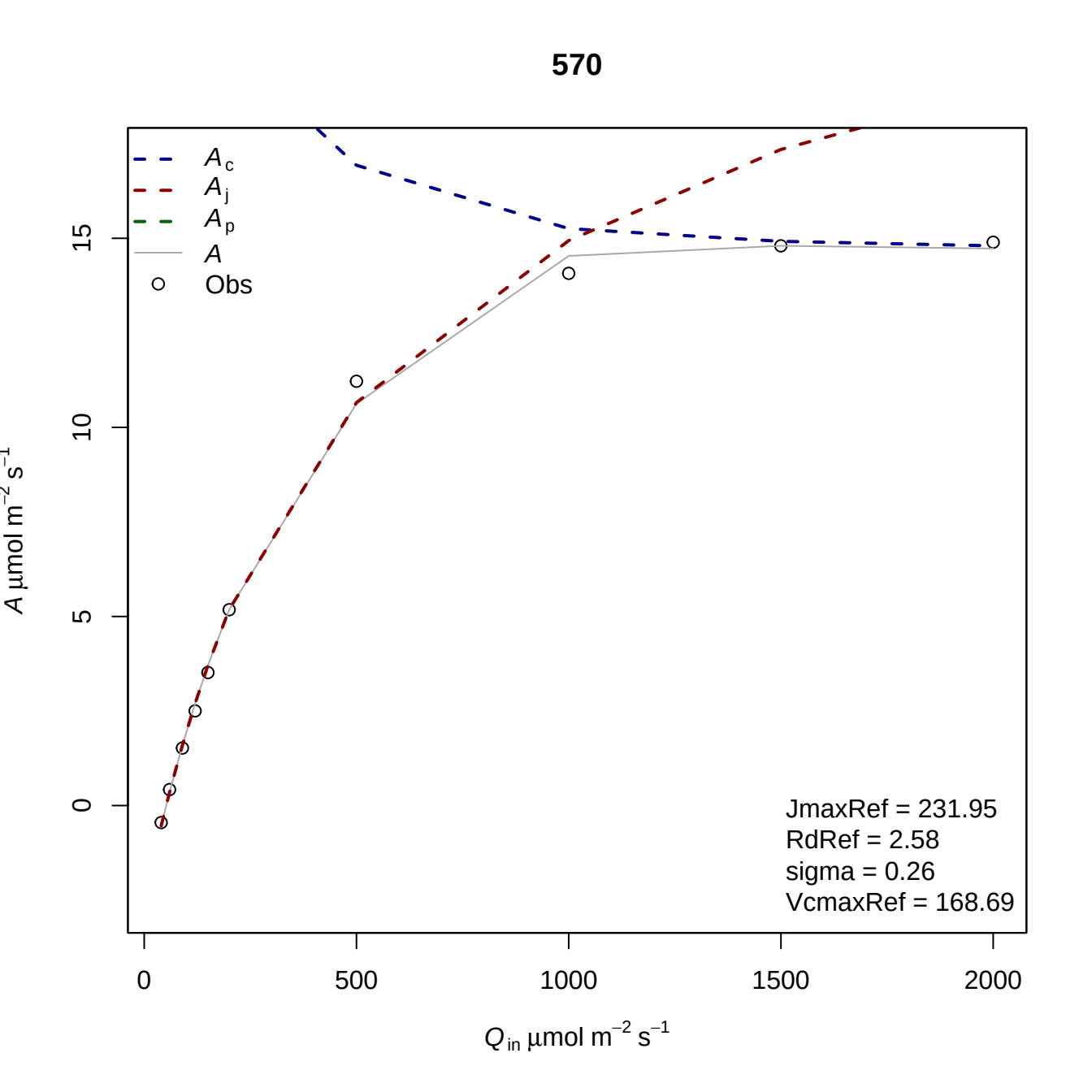

565
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 356.86
RdRef = 1.53
sigma = 0.63
VcmaxRef = 192.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

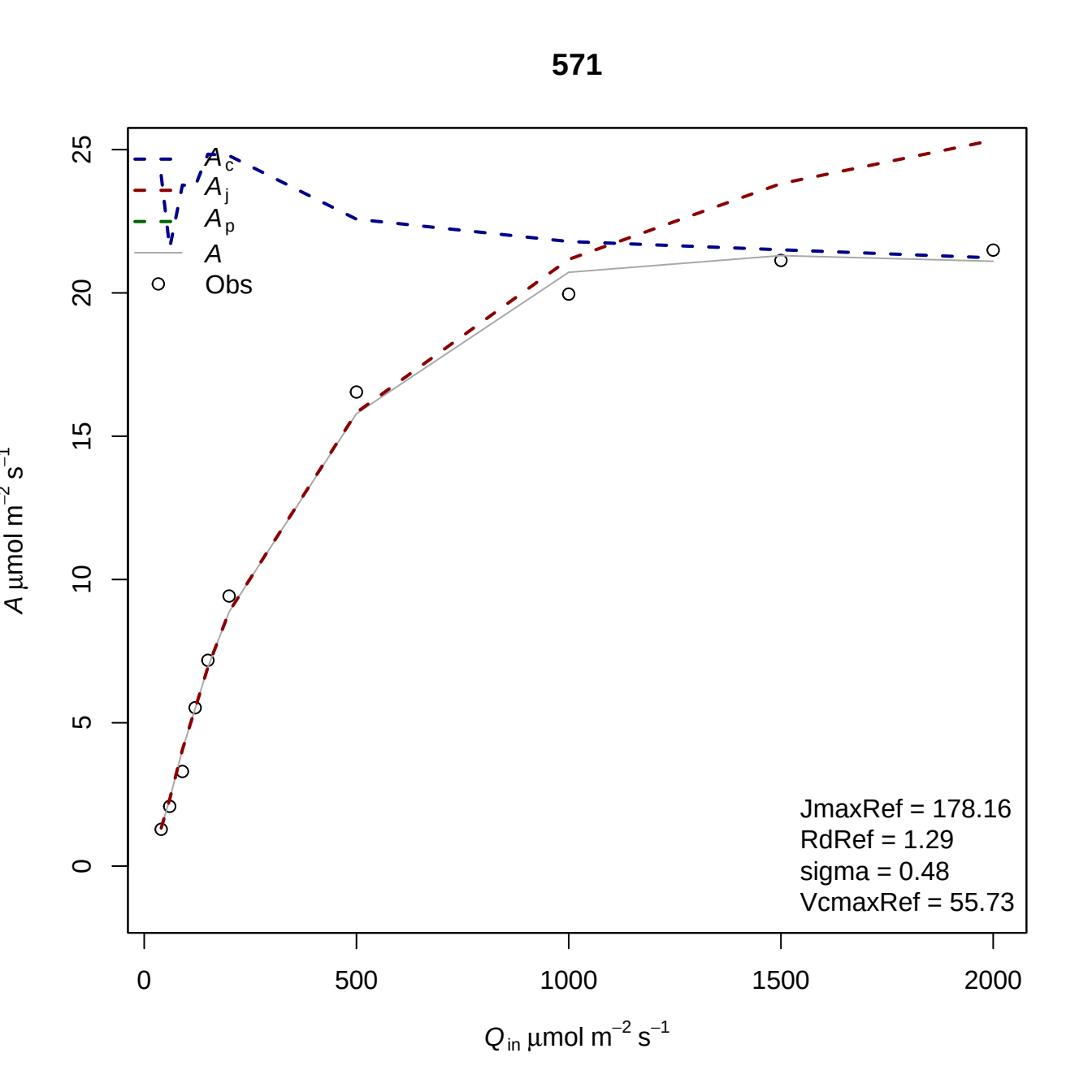

566
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 353.12
RdRef = 2.24
sigma = 0.49
VcmaxRef = 267.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

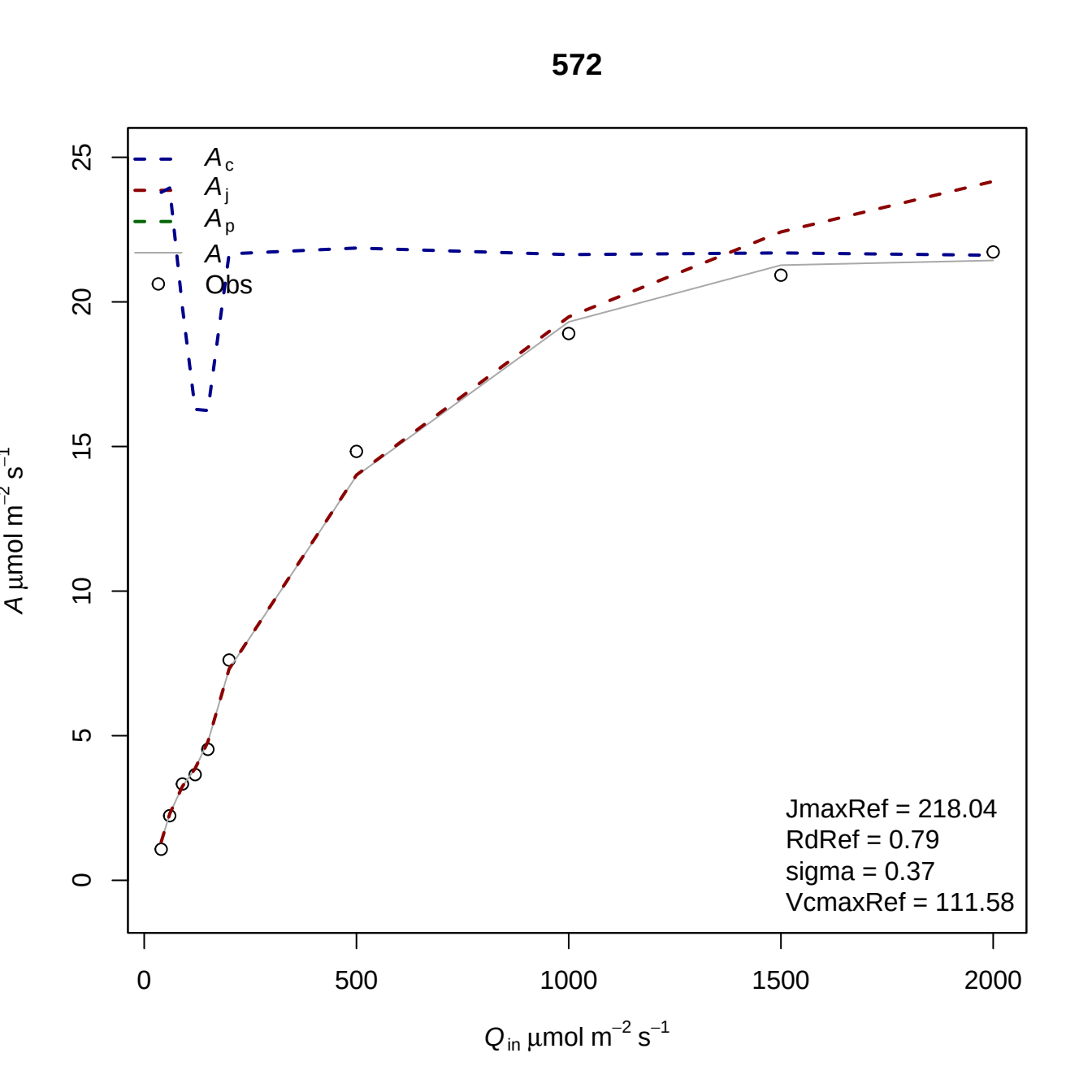

567
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 228.12
RdRef = 1.33
sigma = 0.67
VcmaxRef = 67.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

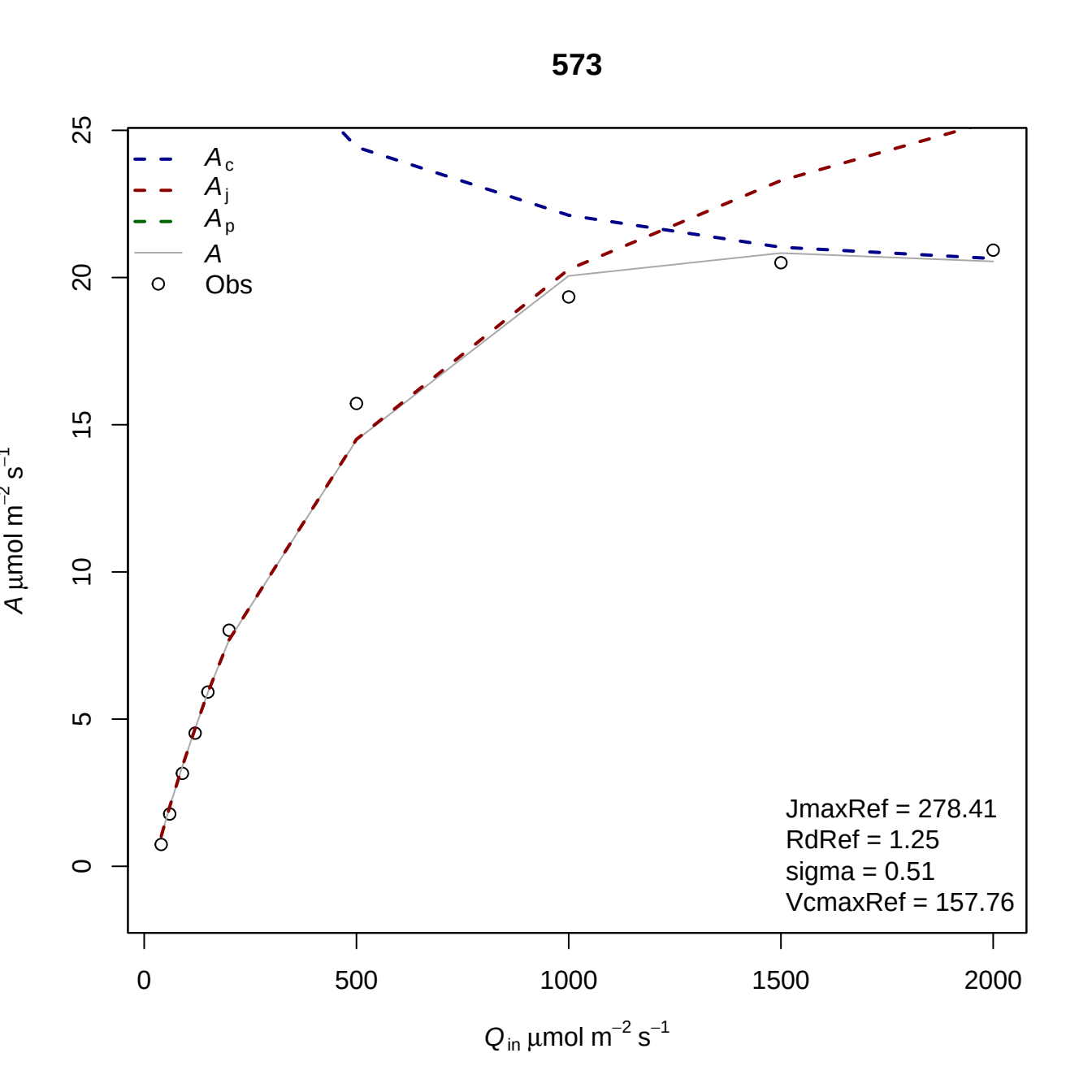

568
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 253.01
RdRef = 0.75
sigma = 0.45
VcmaxRef = 116.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

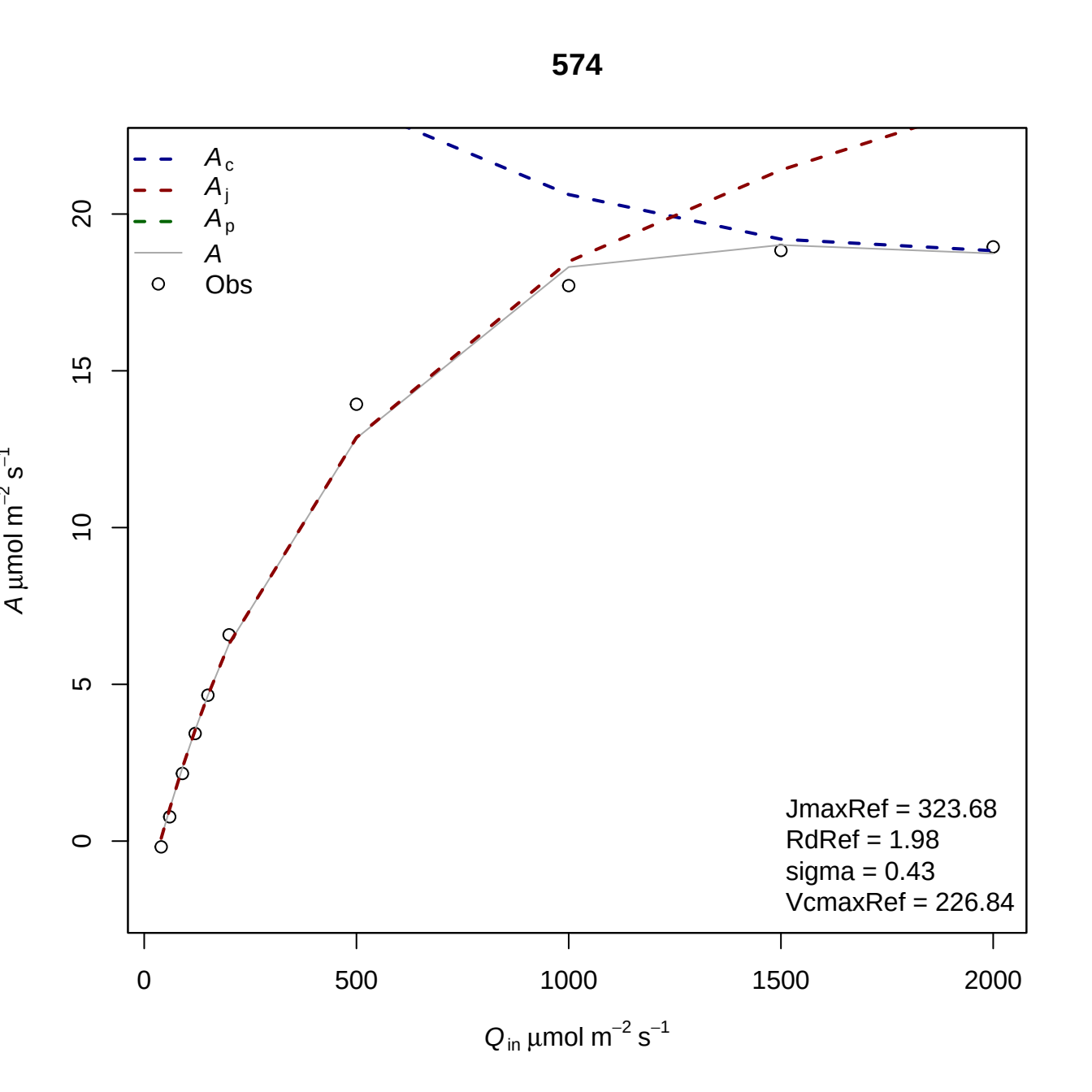

569
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 229.1
RdRef = 1.29
sigma = 0.44
VcmaxRef = 142.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

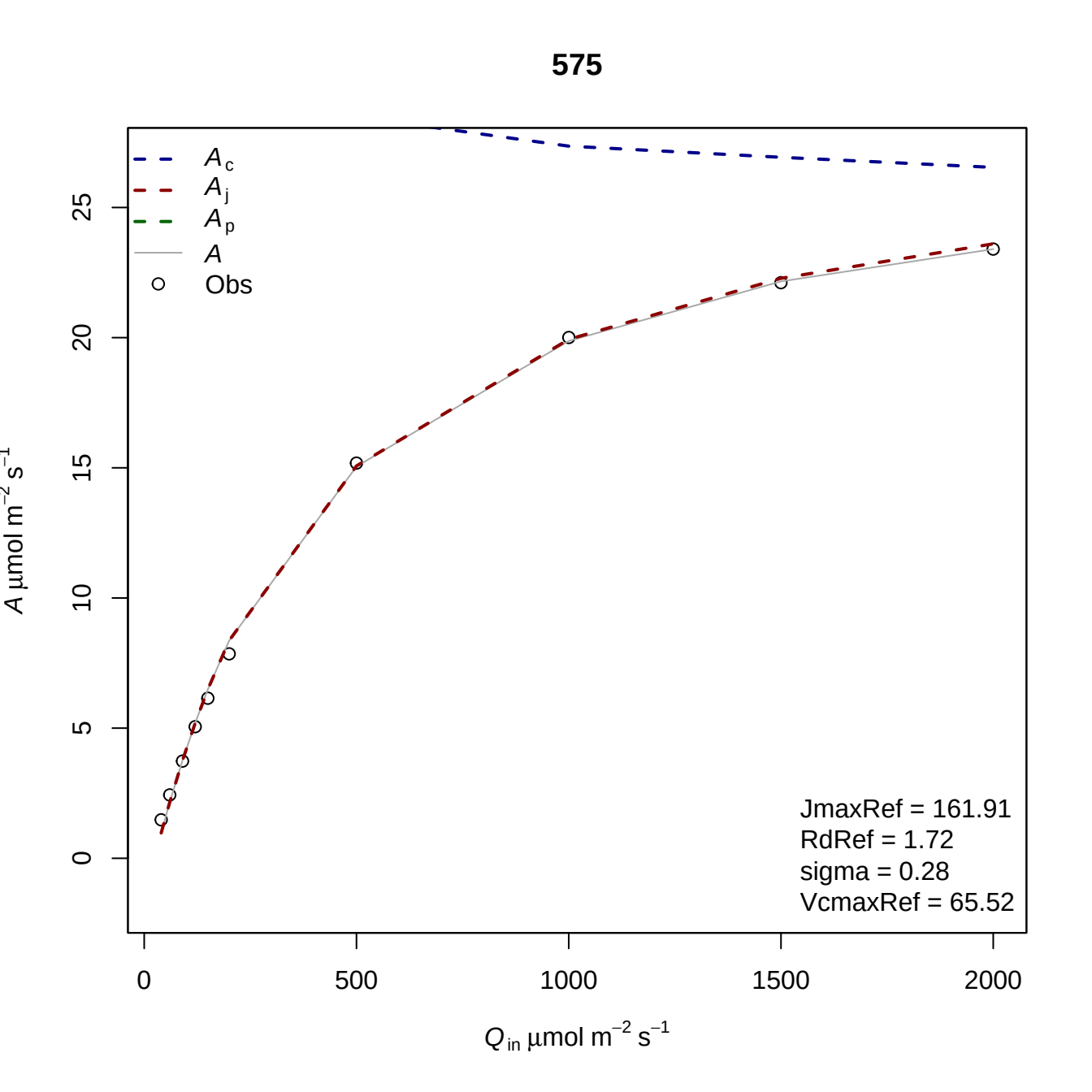

570
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 231.95
RdRef = 2.58
sigma = 0.26
VcmaxRef = 168.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

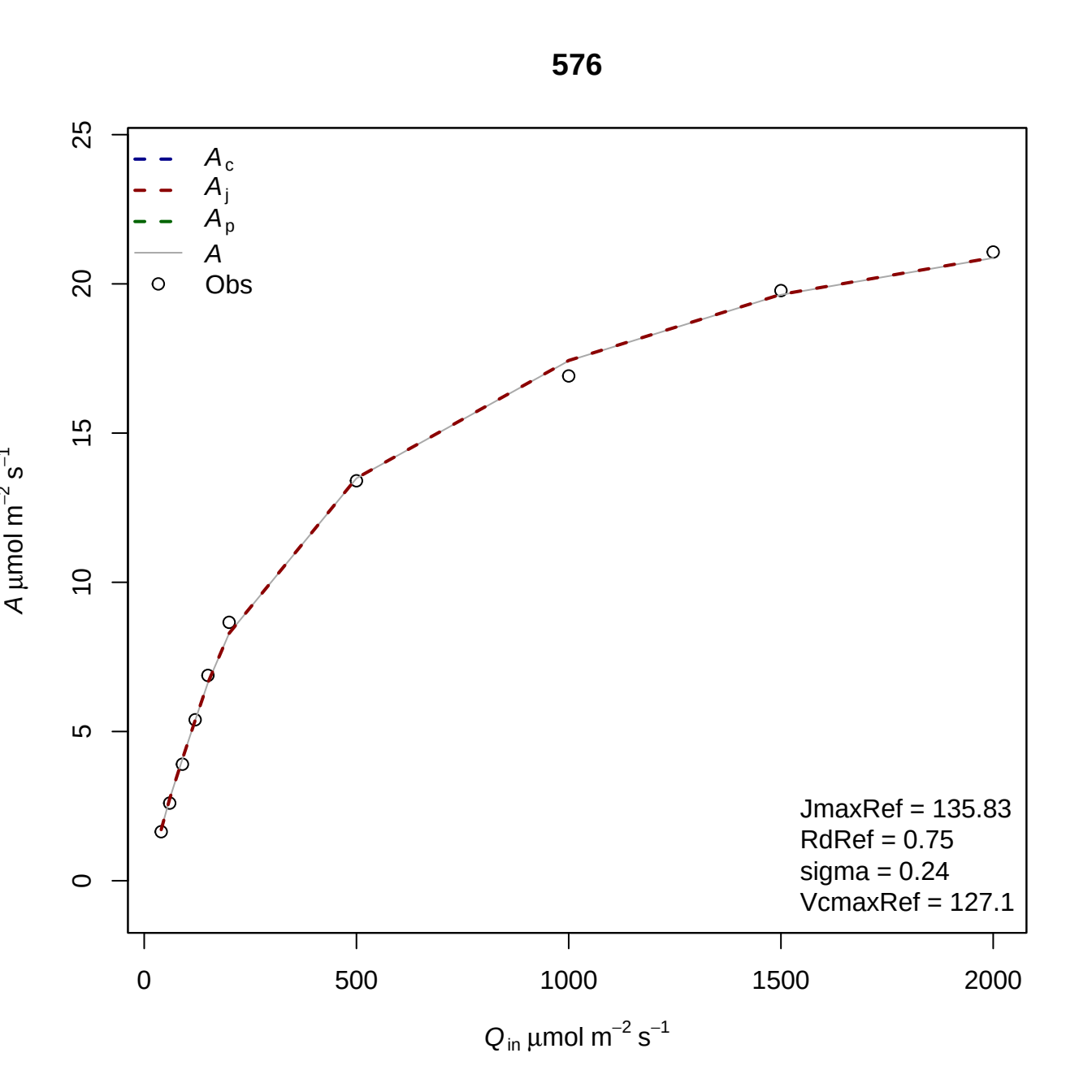

571
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 178.16
RdRef = 1.29
sigma = 0.48
VcmaxRef = 55.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

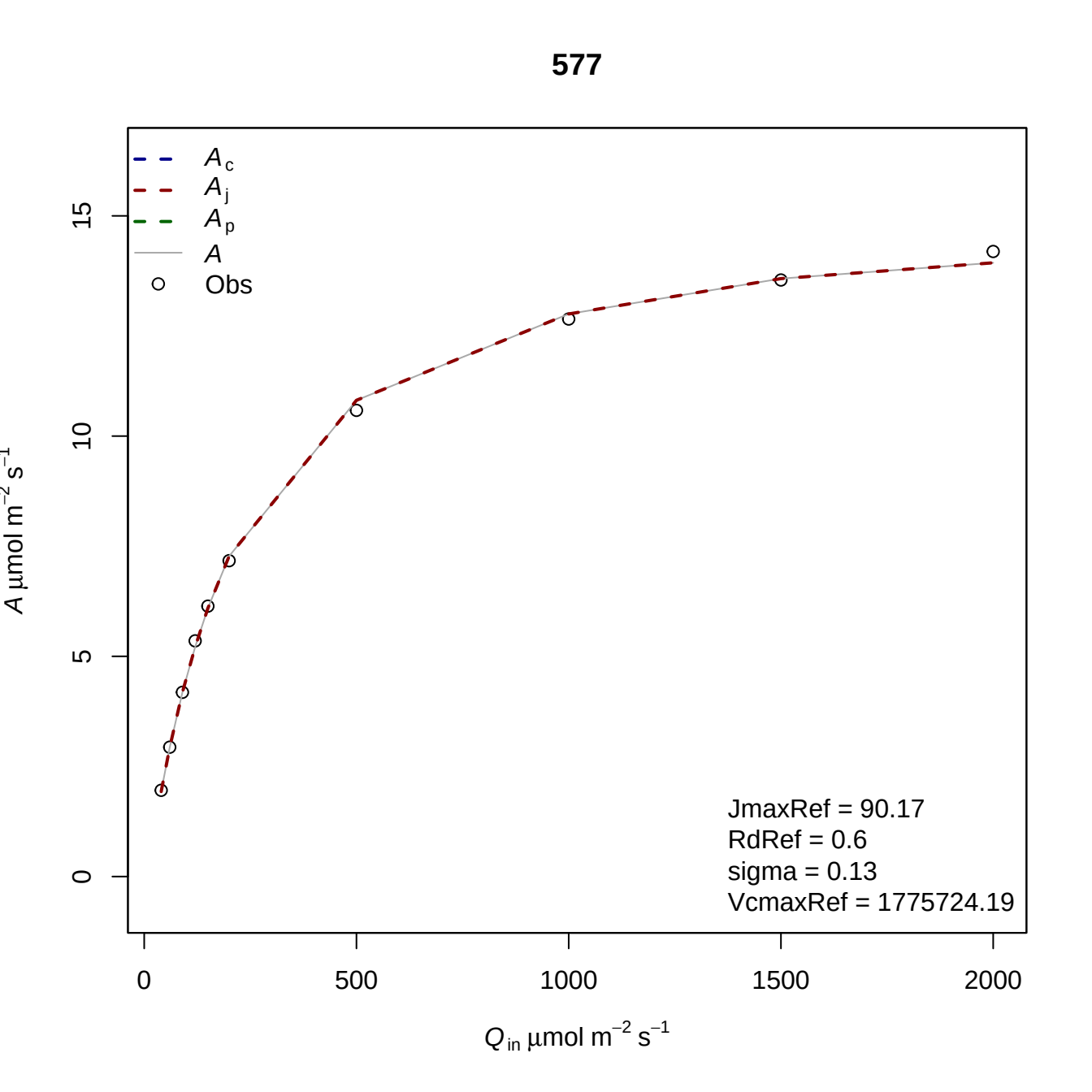

572
$A_c$
$A_j$
$A_p$
$A$
Obs
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
JmaxRef = 218.04
RdRef = 0.79
sigma = 0.37
VcmaxRef = 111.58

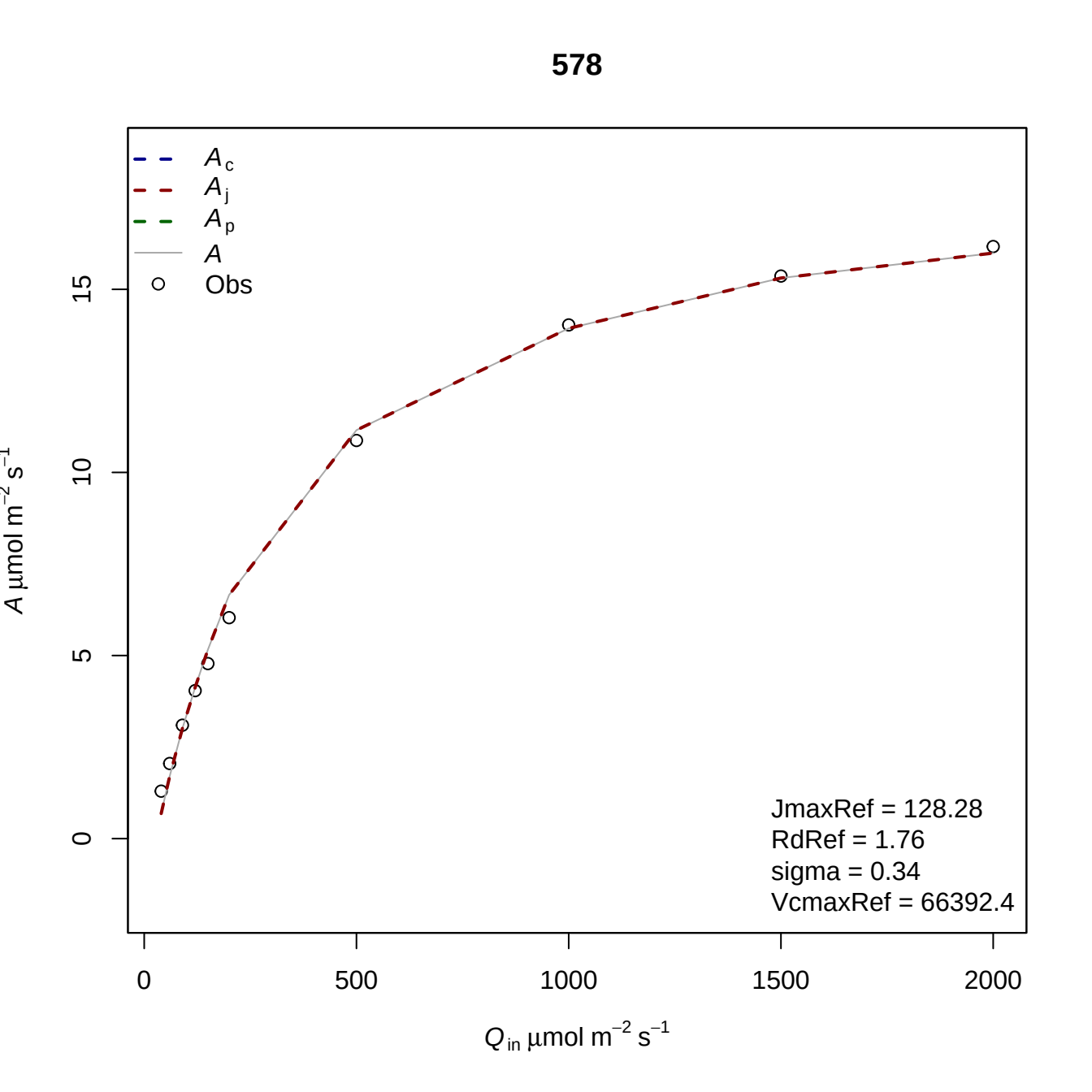

573
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 278.41
RdRef = 1.25
sigma = 0.51
VcmaxRef = 157.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

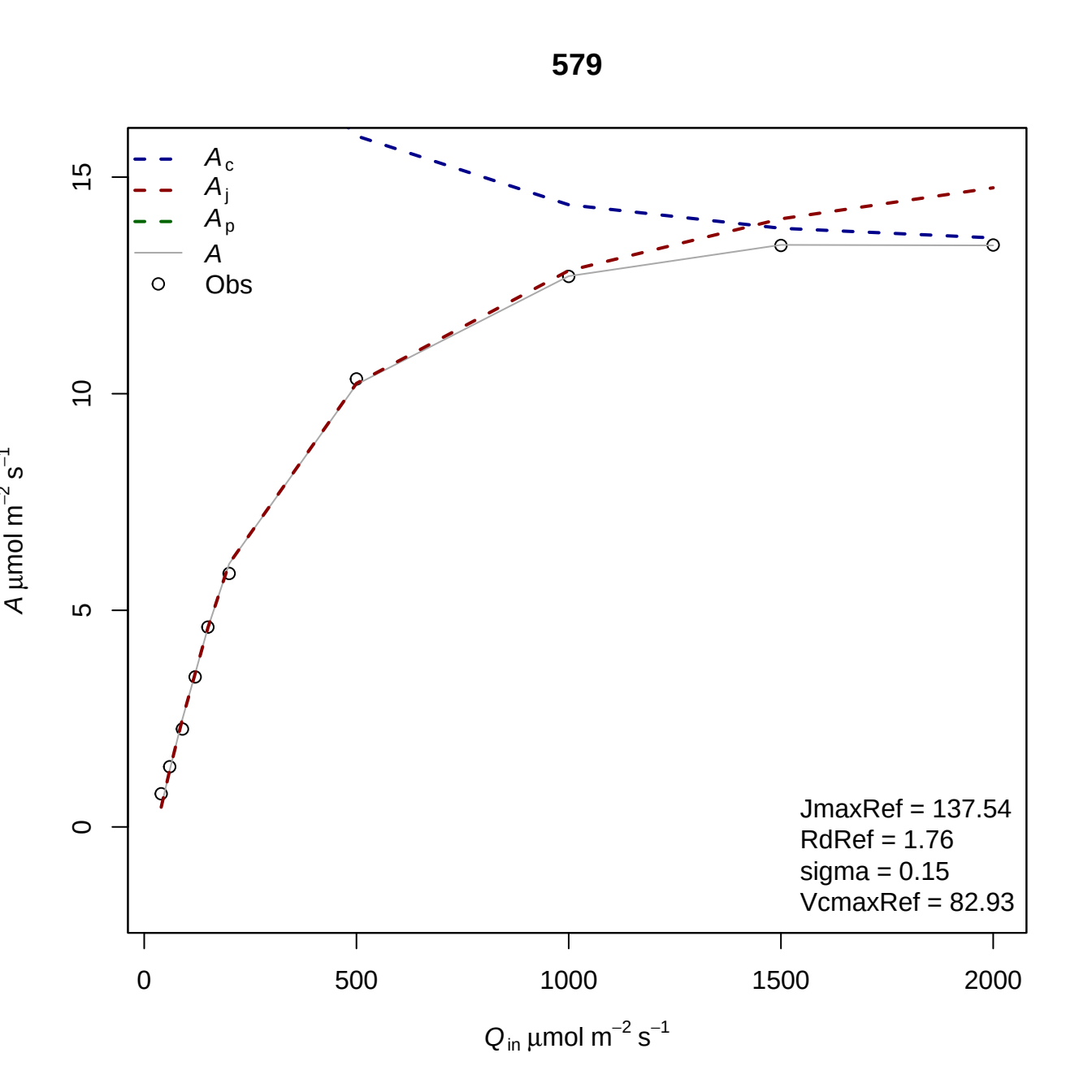

574
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 323.68
RdRef = 1.98
sigma = 0.43
VcmaxRef = 226.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

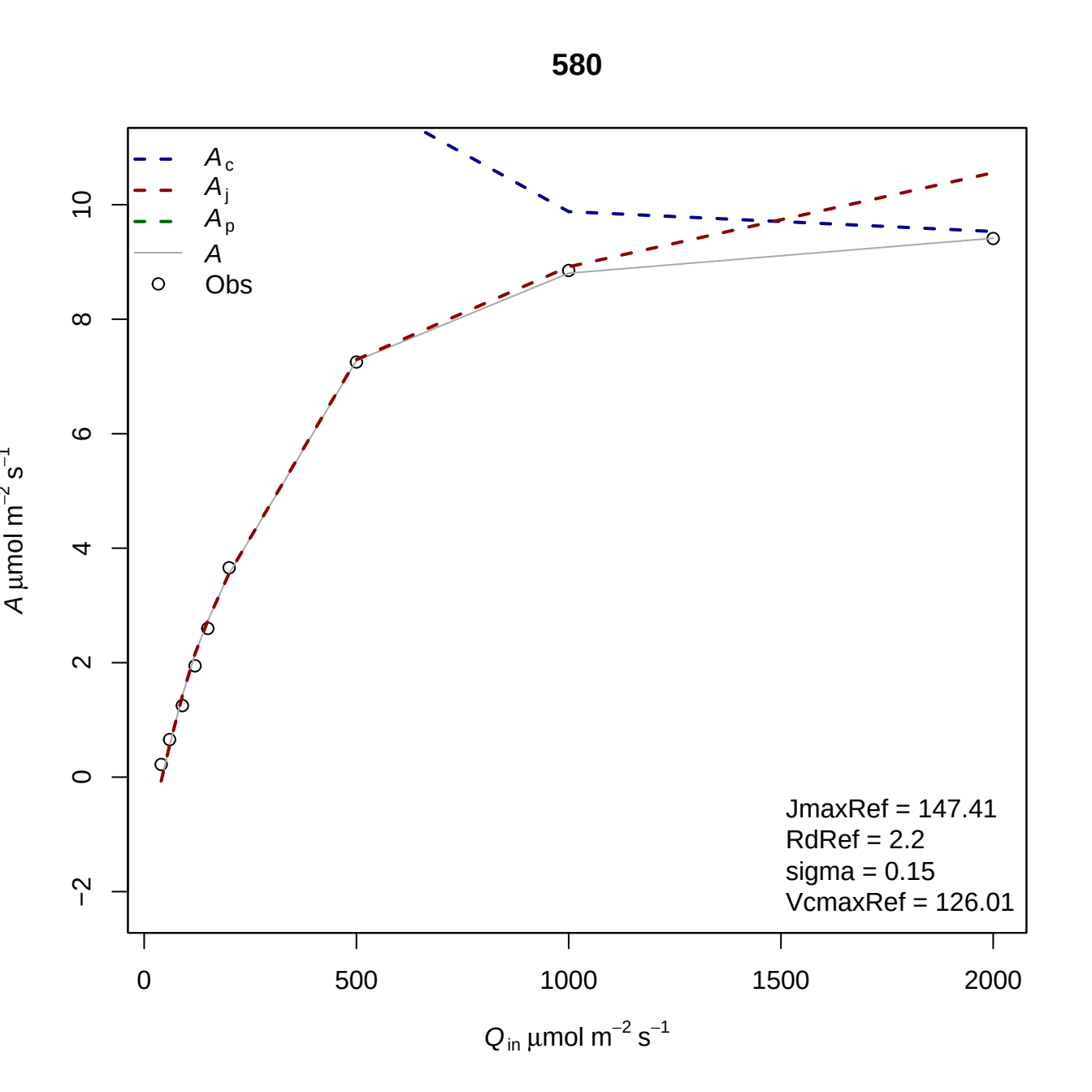

575
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.91
RdRef = 1.72
sigma = 0.28
VcmaxRef = 65.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

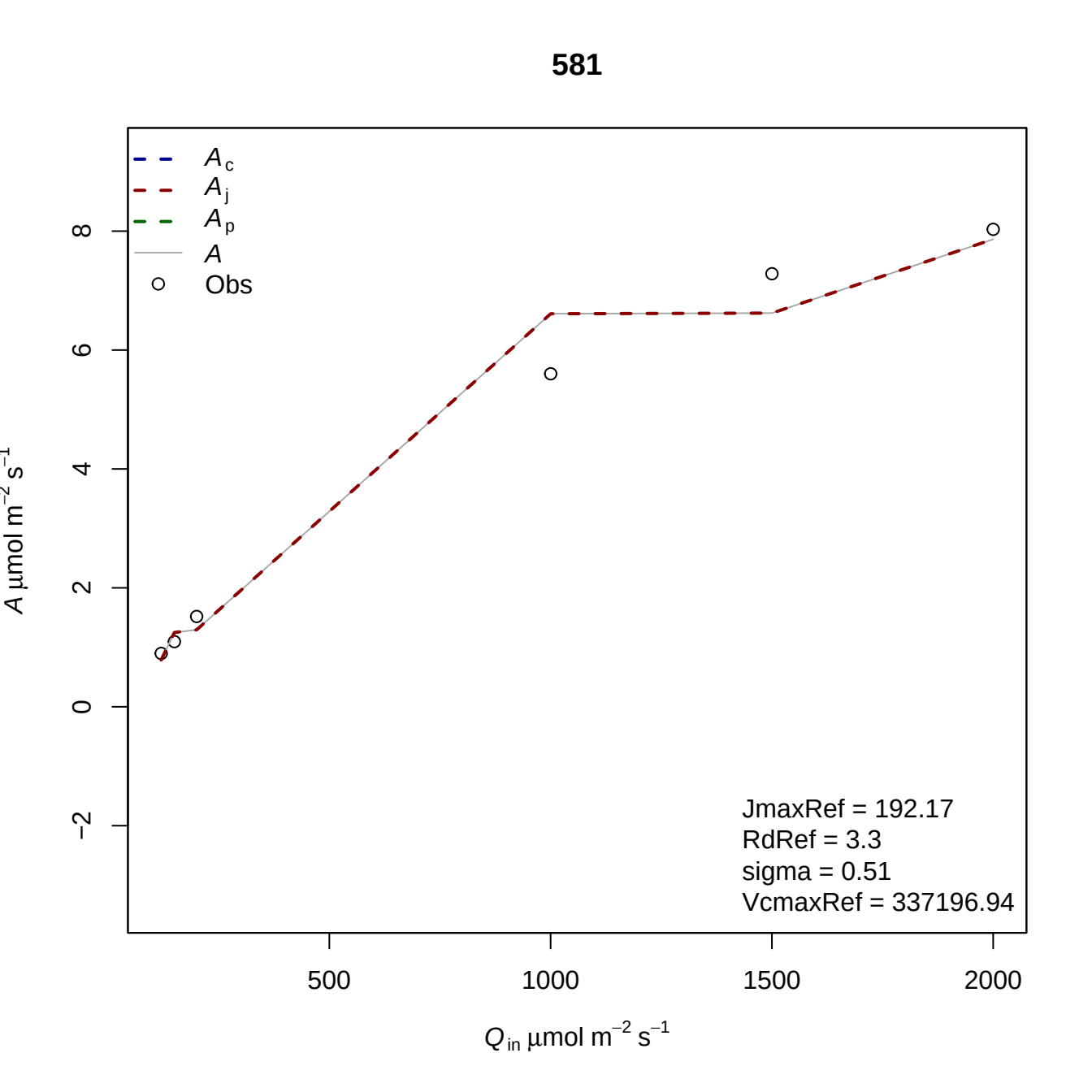

576
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.83
RdRef = 0.75
sigma = 0.24
VcmaxRef = 127.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

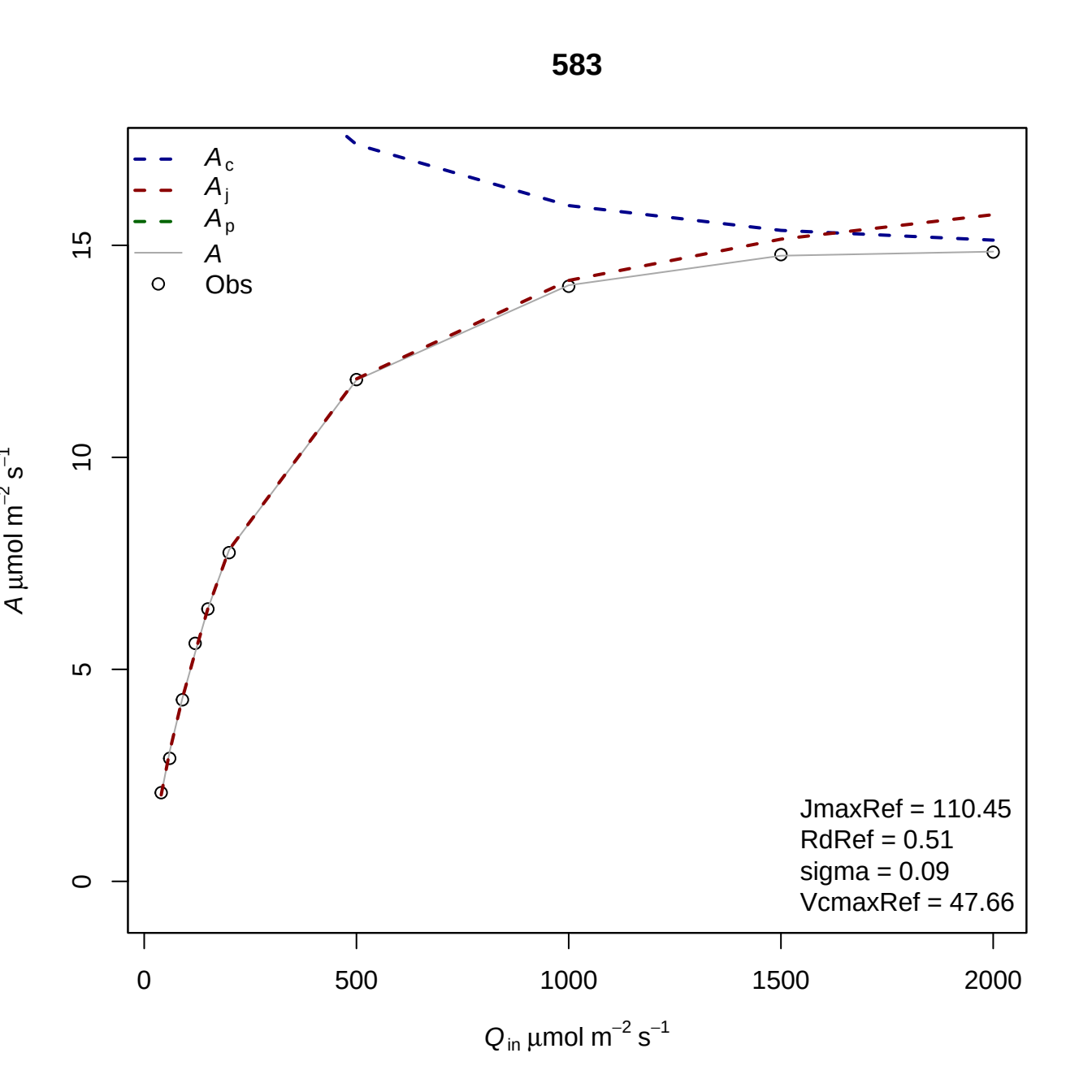

577
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.17
RdRef = 0.6
sigma = 0.13
VcmaxRef = 1775724.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

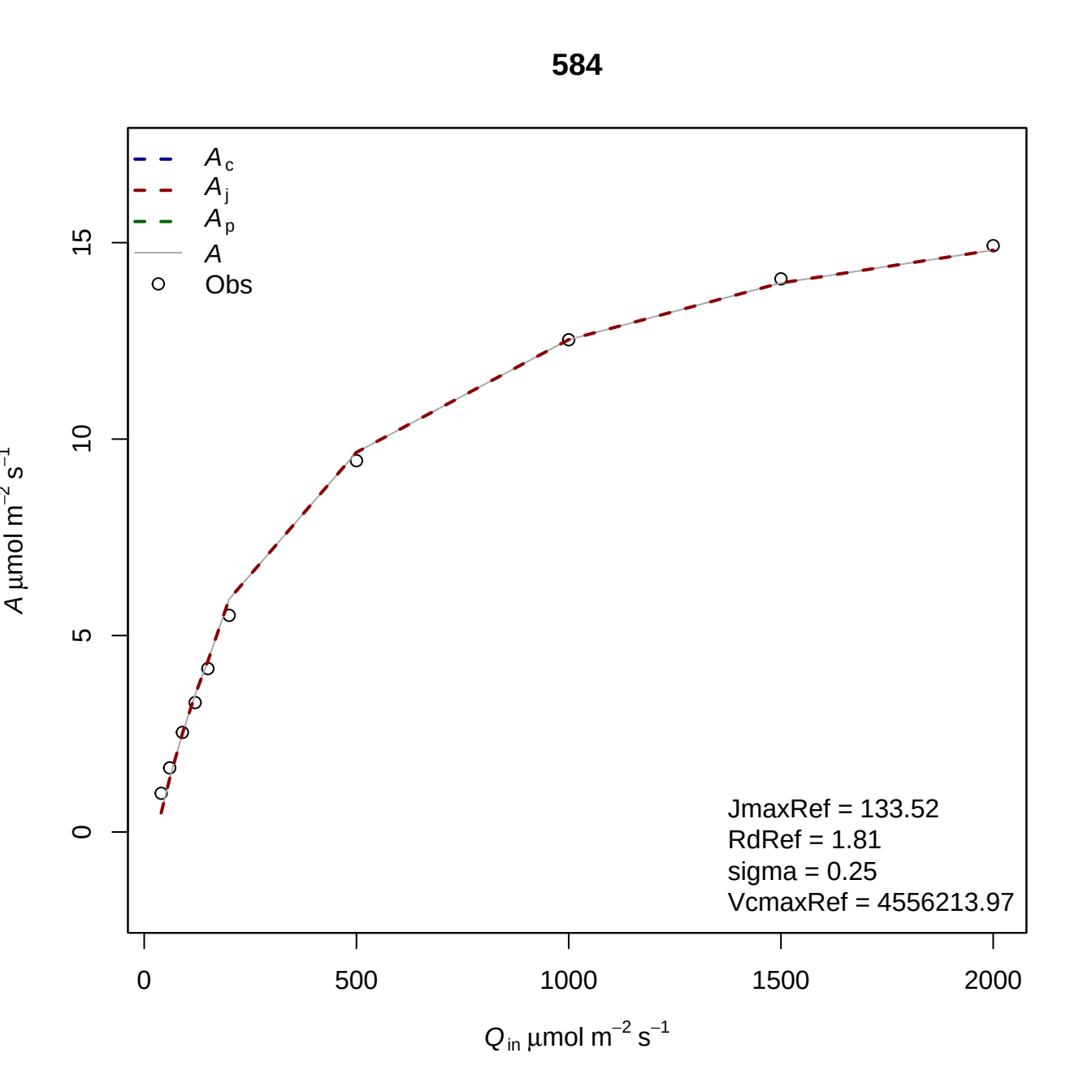

578
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.28
RdRef = 1.76
sigma = 0.34
VcmaxRef = 66392.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

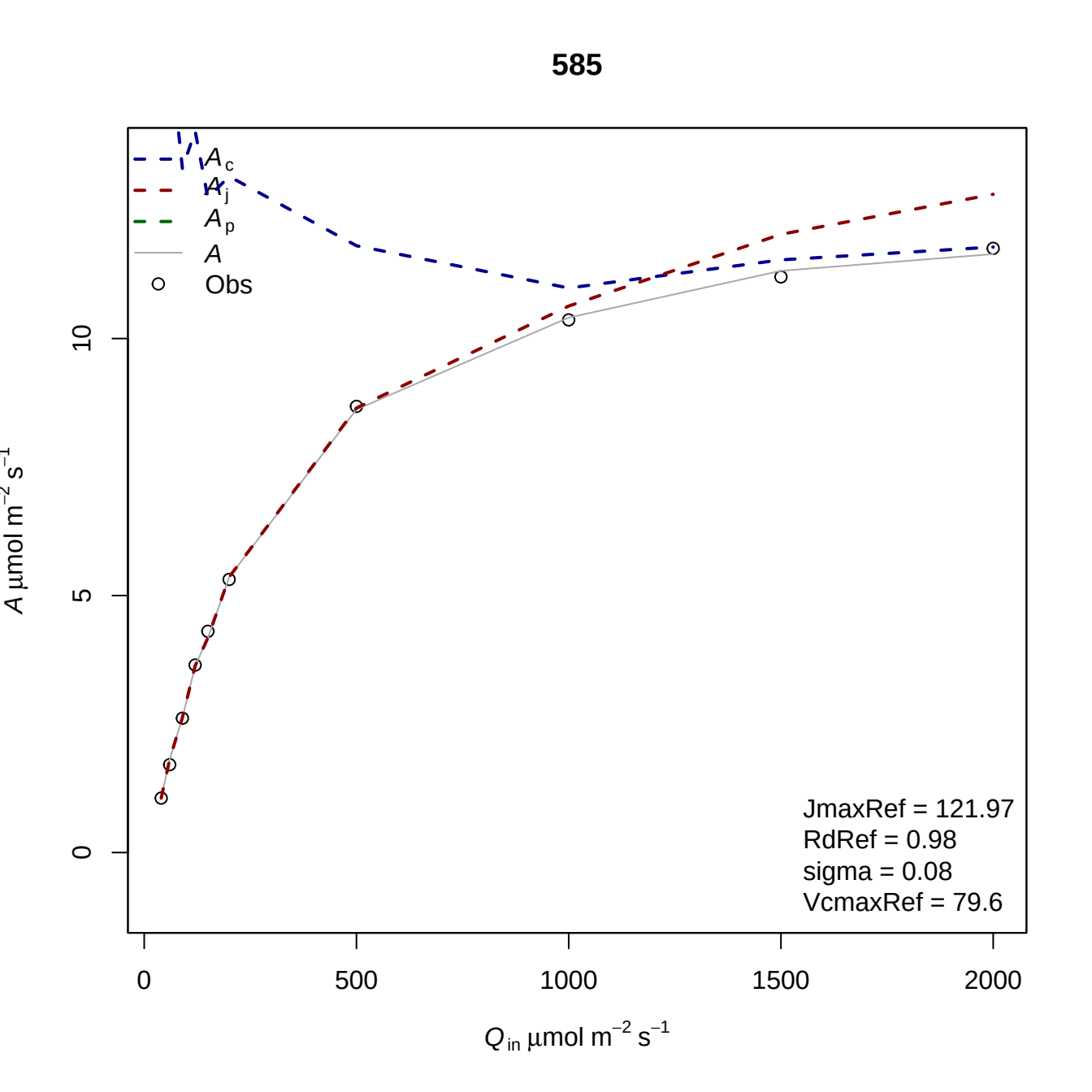

579
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.54
RdRef = 1.76
sigma = 0.15
VcmaxRef = 82.93
0
5
10
15
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

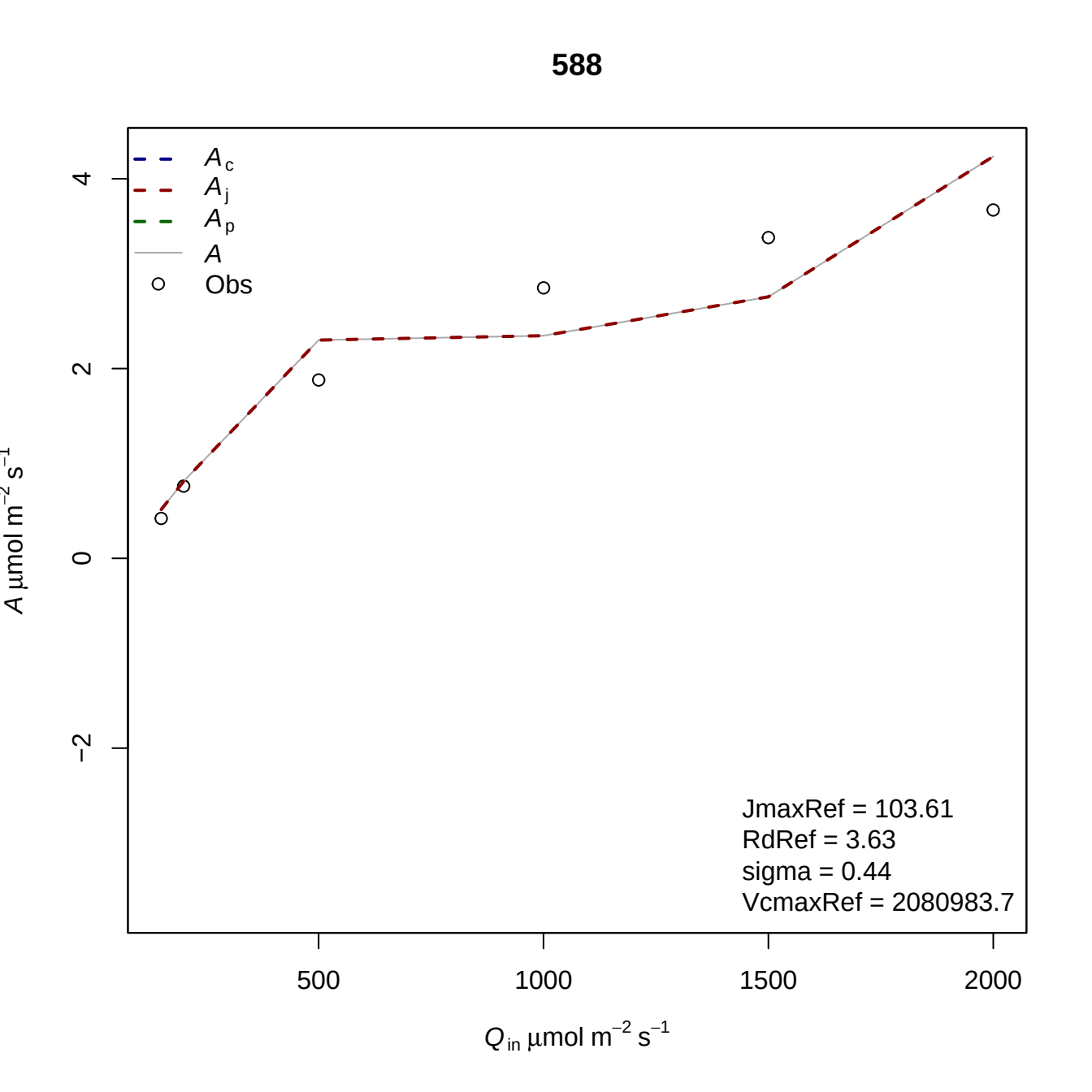

580
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.41
RdRef = 2.2
sigma = 0.15
VcmaxRef = 126.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

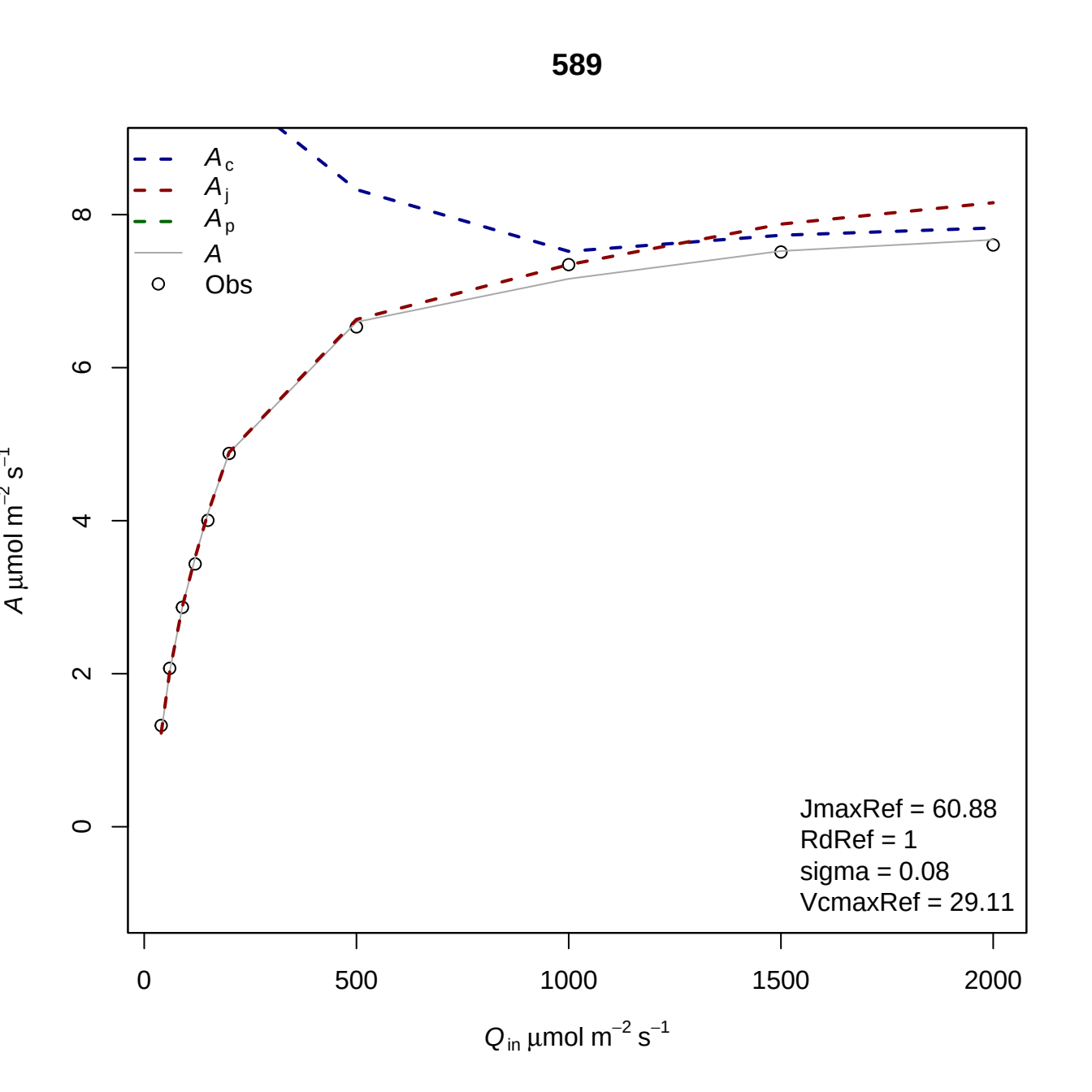

581
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 192.17
RdRef = 3.3
sigma = 0.51
VcmaxRef = 337196.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
500
1000
1500
2000

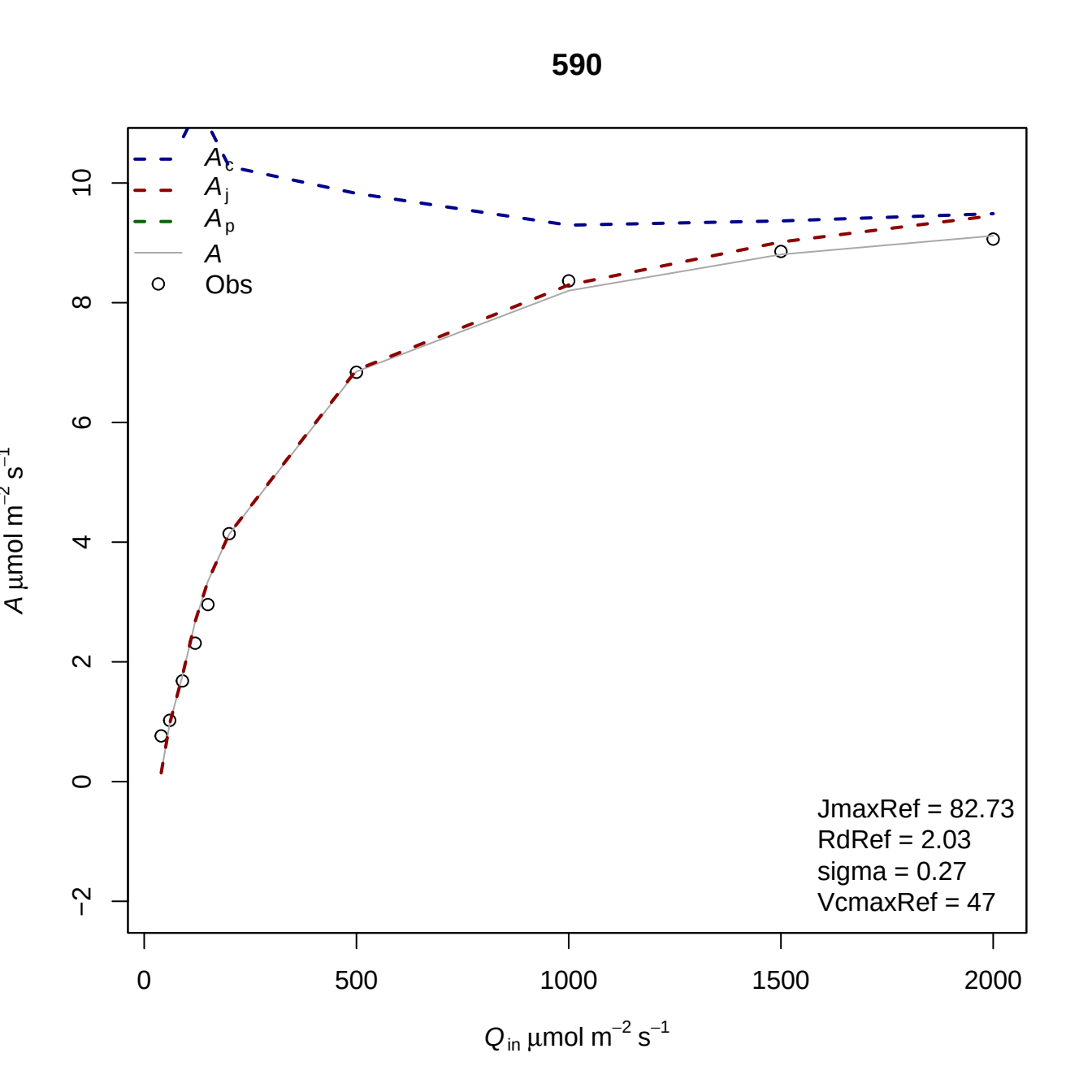

583
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.45
RdRef = 0.51
sigma = 0.09
VcmaxRef = 47.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

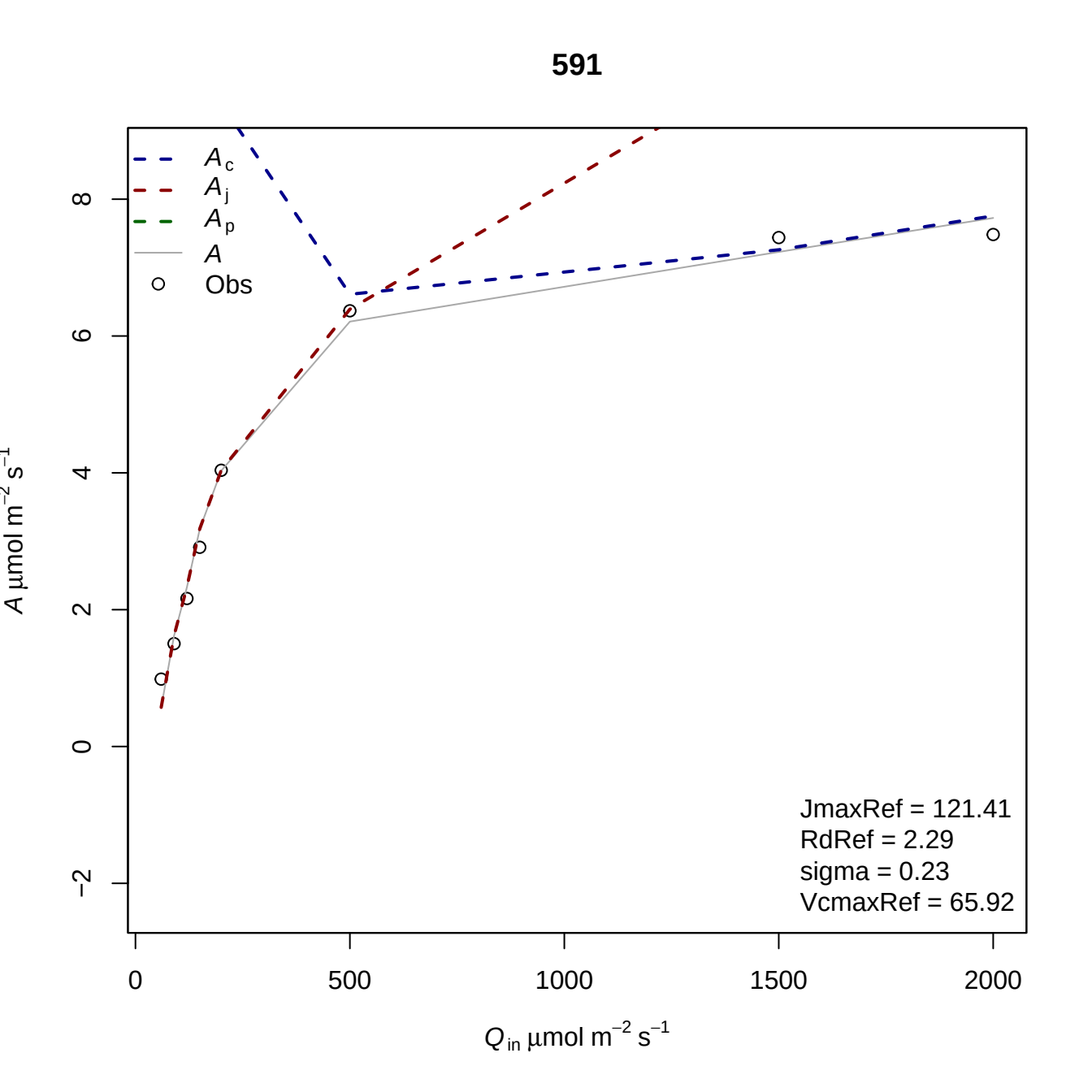

584
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.52
RdRef = 1.81
sigma = 0.25
VcmaxRef = 4556213.97
0
5
10
15
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

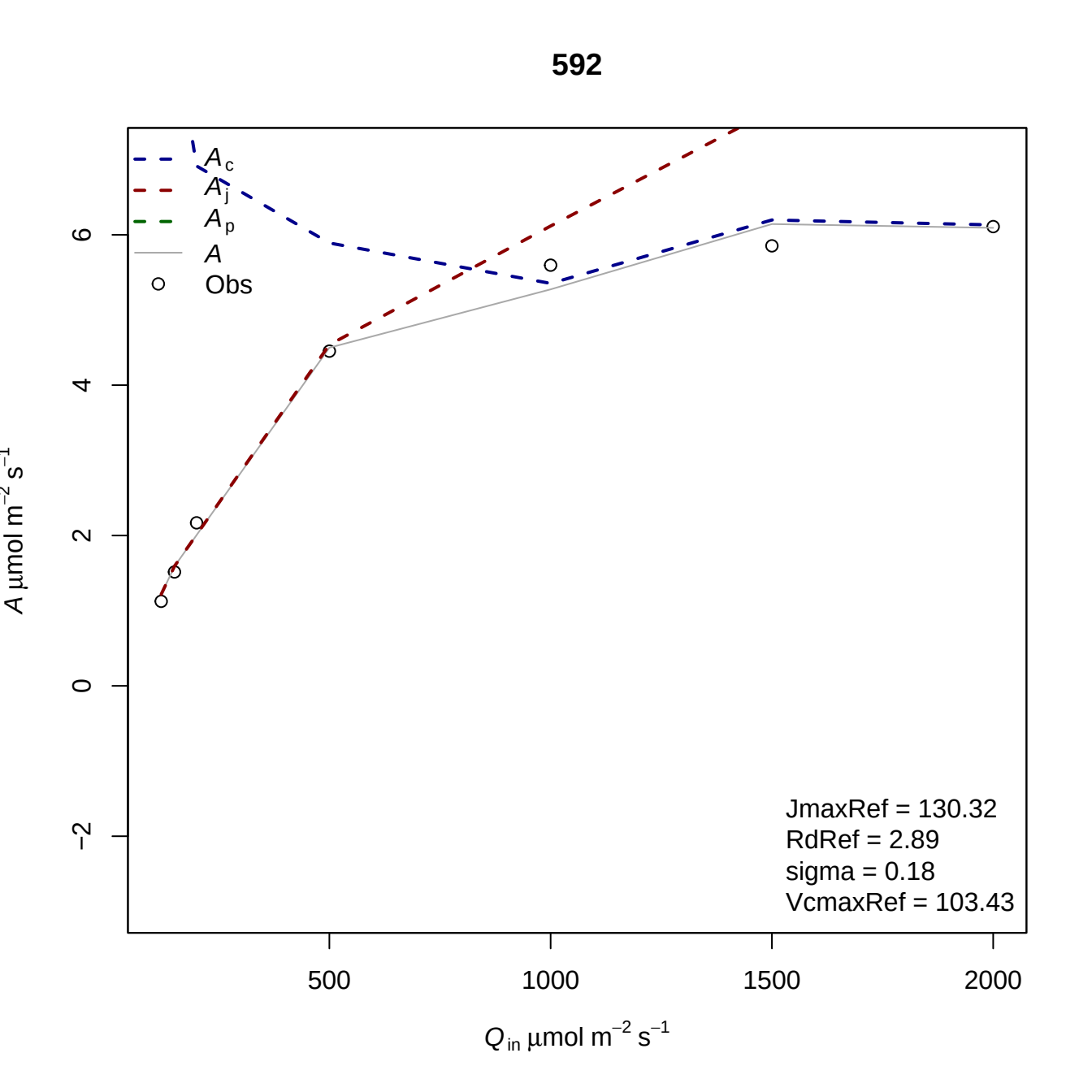

585
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.97
RdRef = 0.98
sigma = 0.08
VcmaxRef = 79.6
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

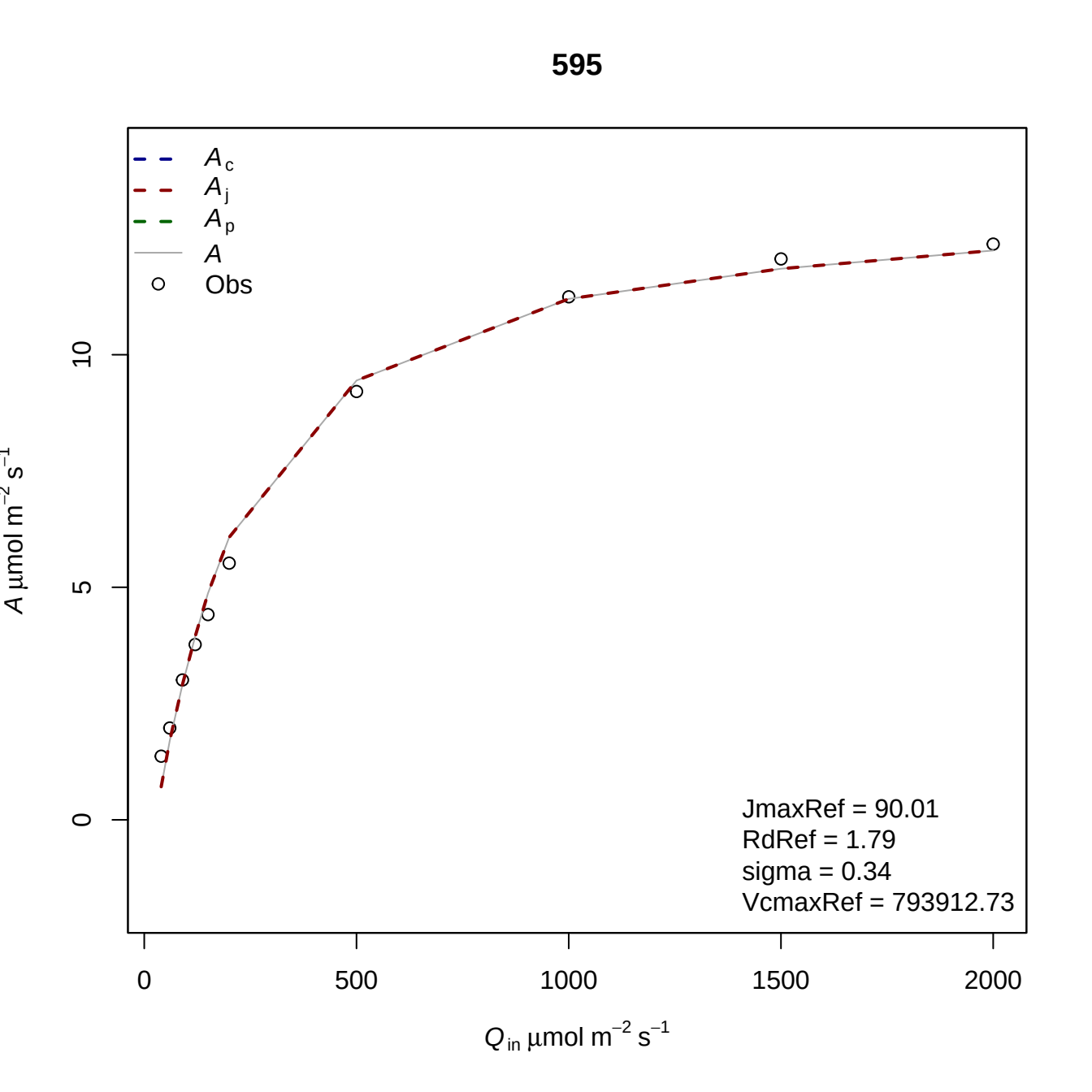

588
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.61
RdRef = 3.63
sigma = 0.44
VcmaxRef = 2080983.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
500
1000
1500
2000

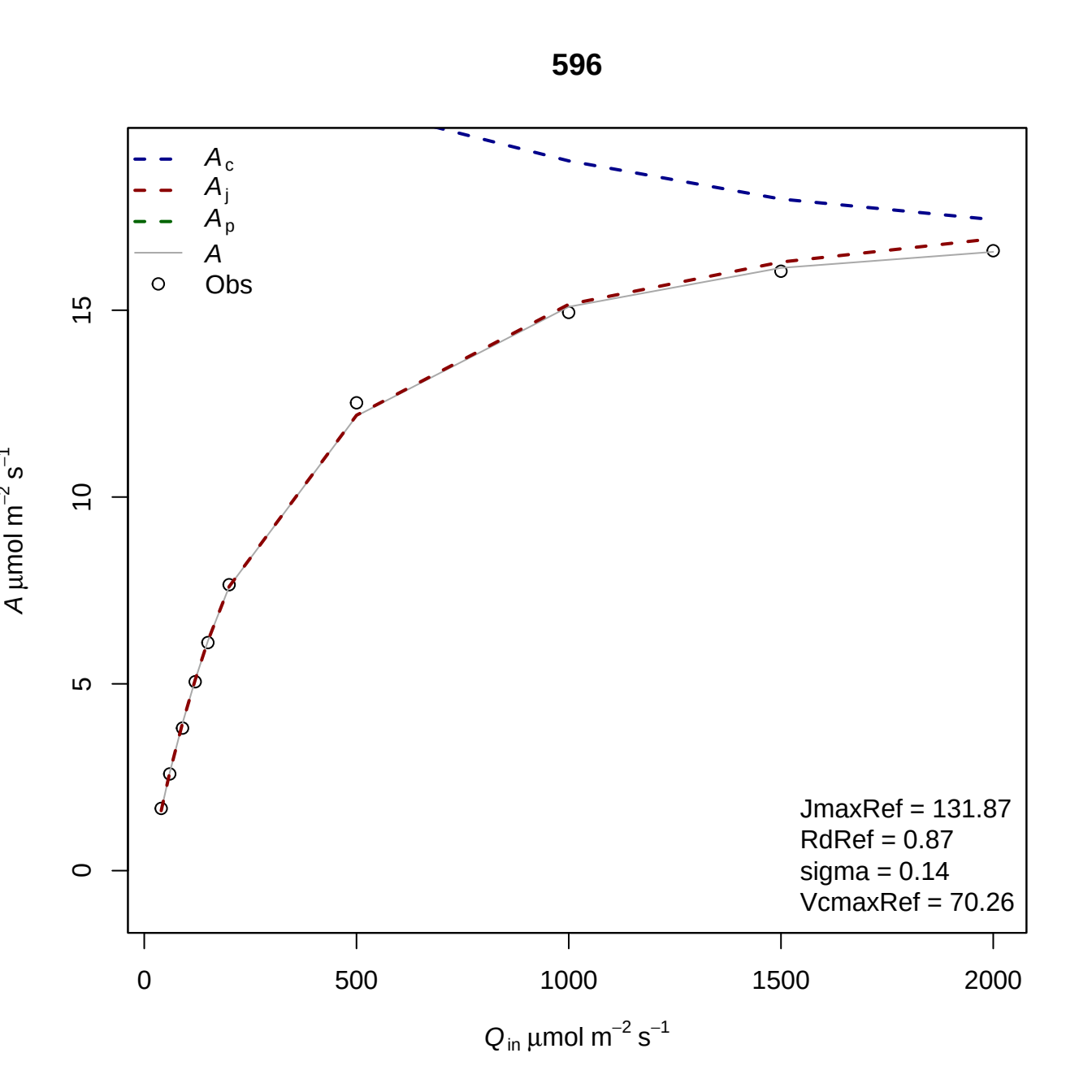

589
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.88
RdRef = 1
sigma = 0.08
VcmaxRef = 29.11
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

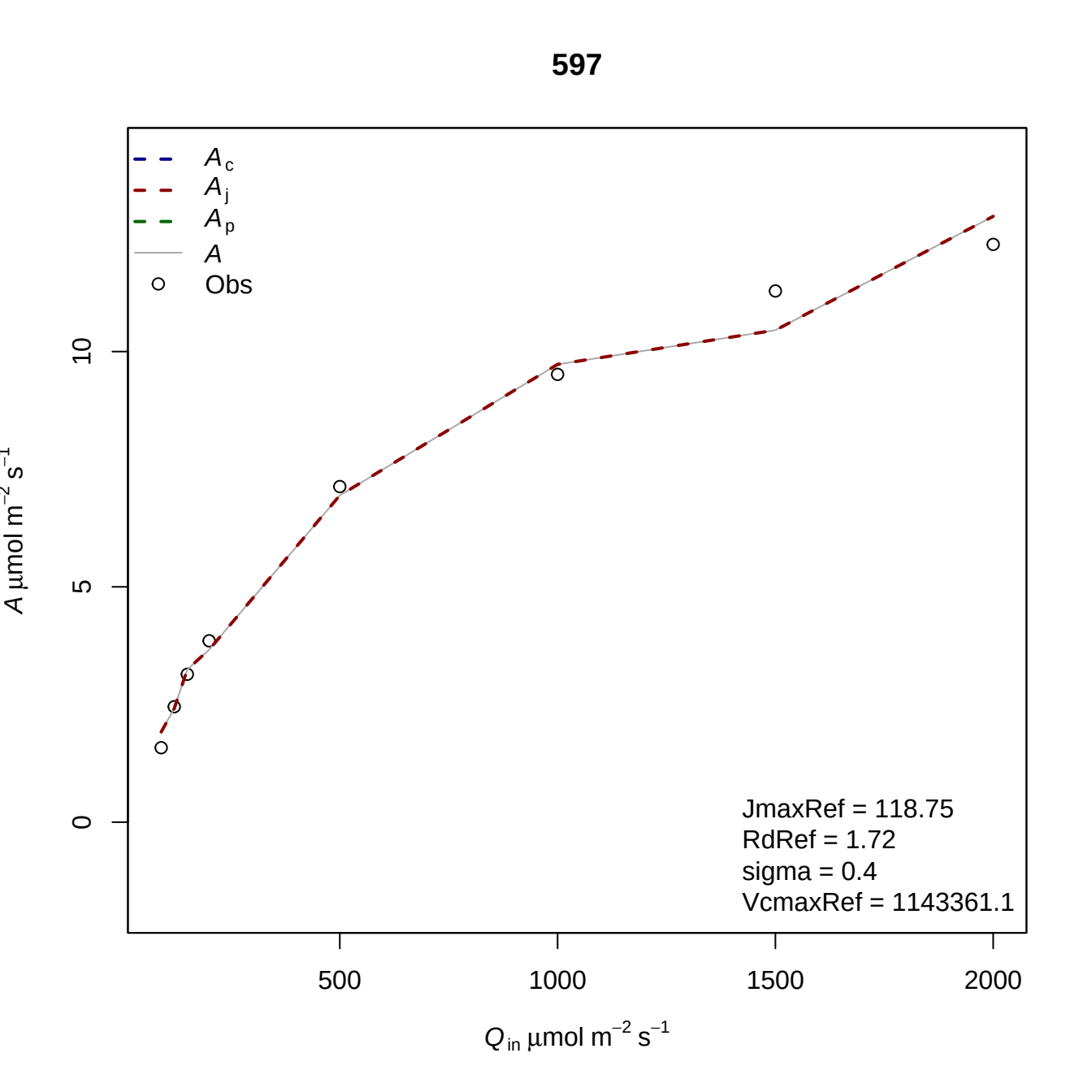

590
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.73
RdRef = 2.03
sigma = 0.27
VcmaxRef = 47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

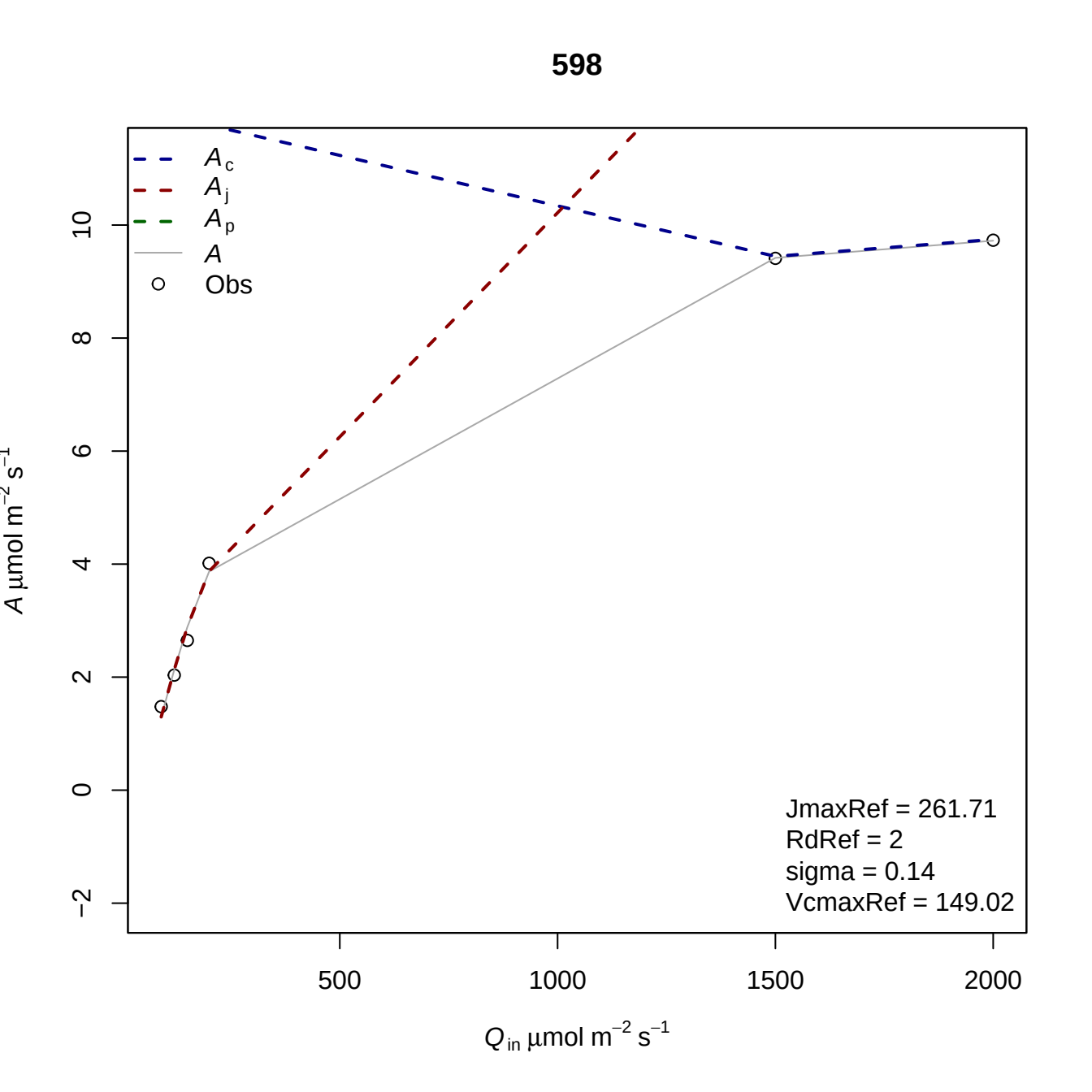

591
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.41
RdRef = 2.29
sigma = 0.23
VcmaxRef = 65.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

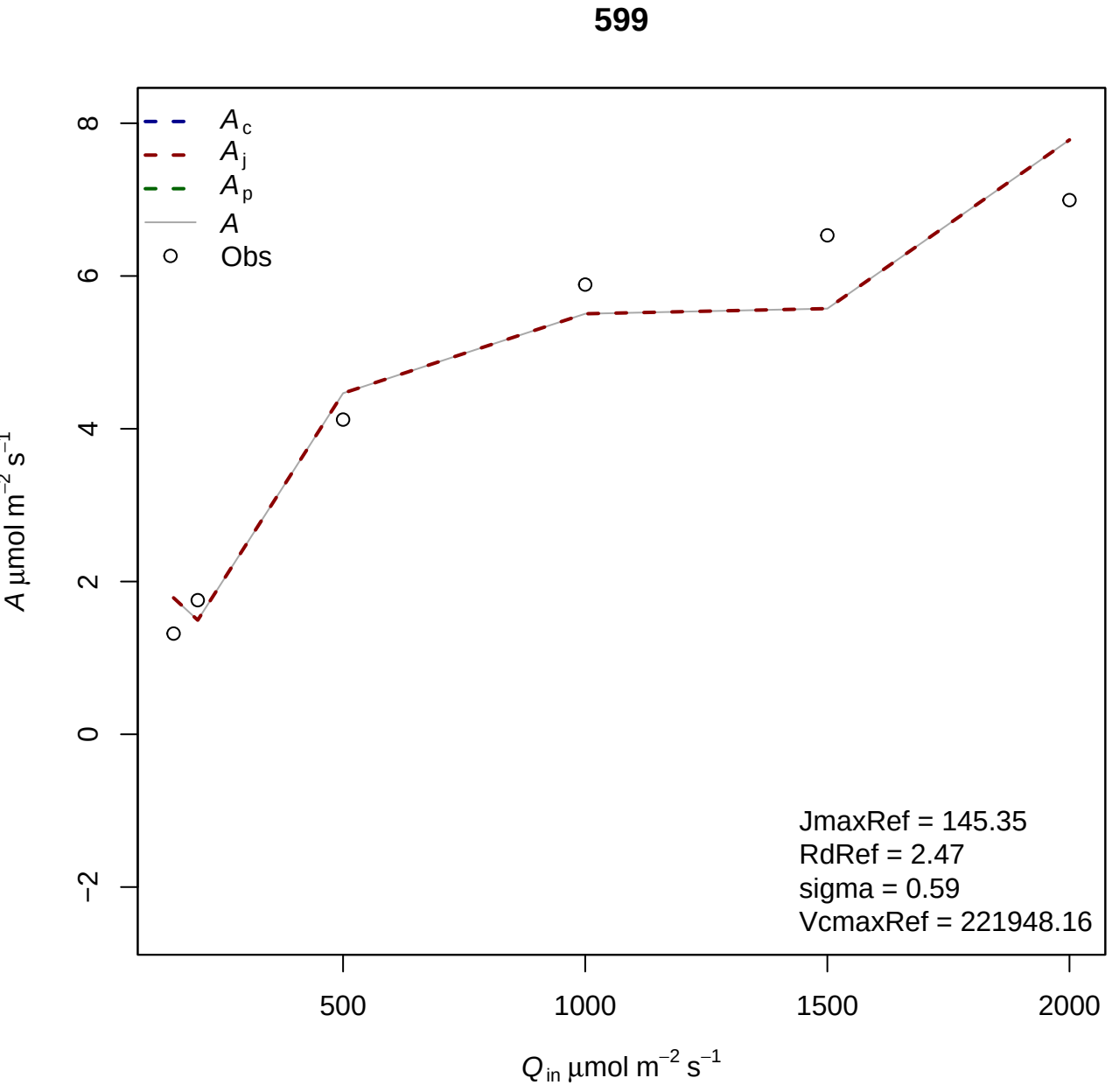

592
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.32
RdRef = 2.89
sigma = 0.18
VcmaxRef = 103.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
500
1000
1500
2000

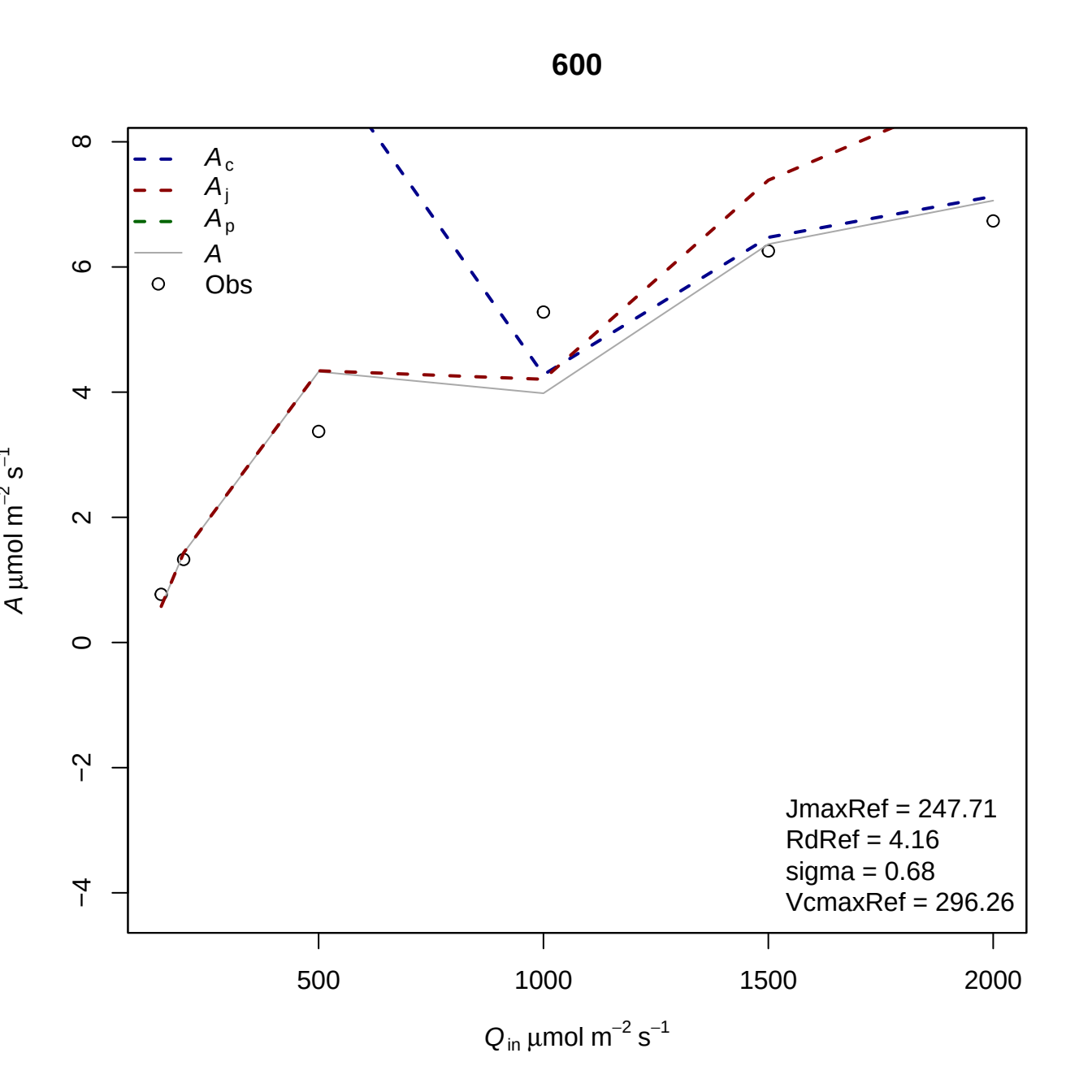

595
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.01
RdRef = 1.79
sigma = 0.34
VcmaxRef = 793912.73
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

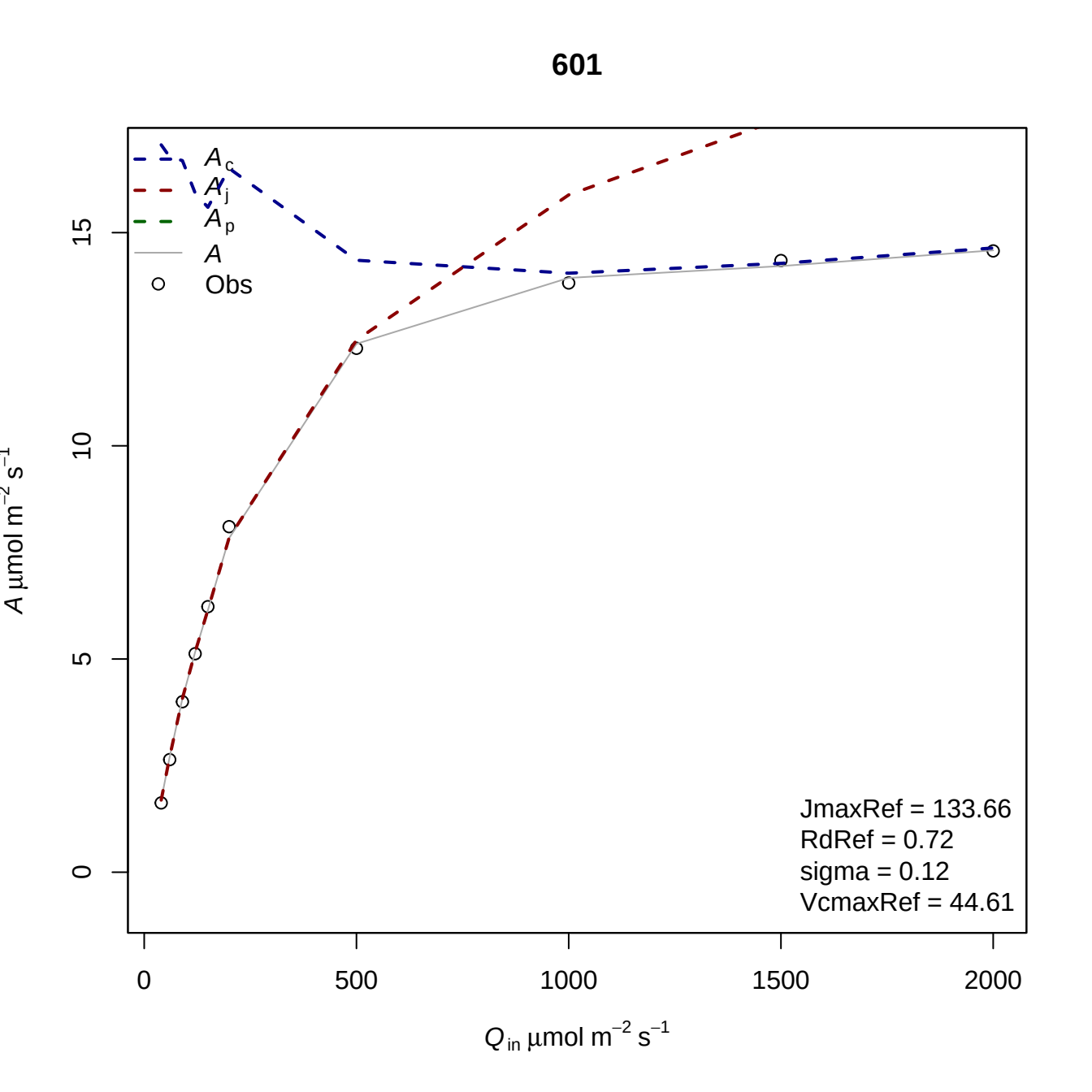

596
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.87
RdRef = 0.87
sigma = 0.14
VcmaxRef = 70.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

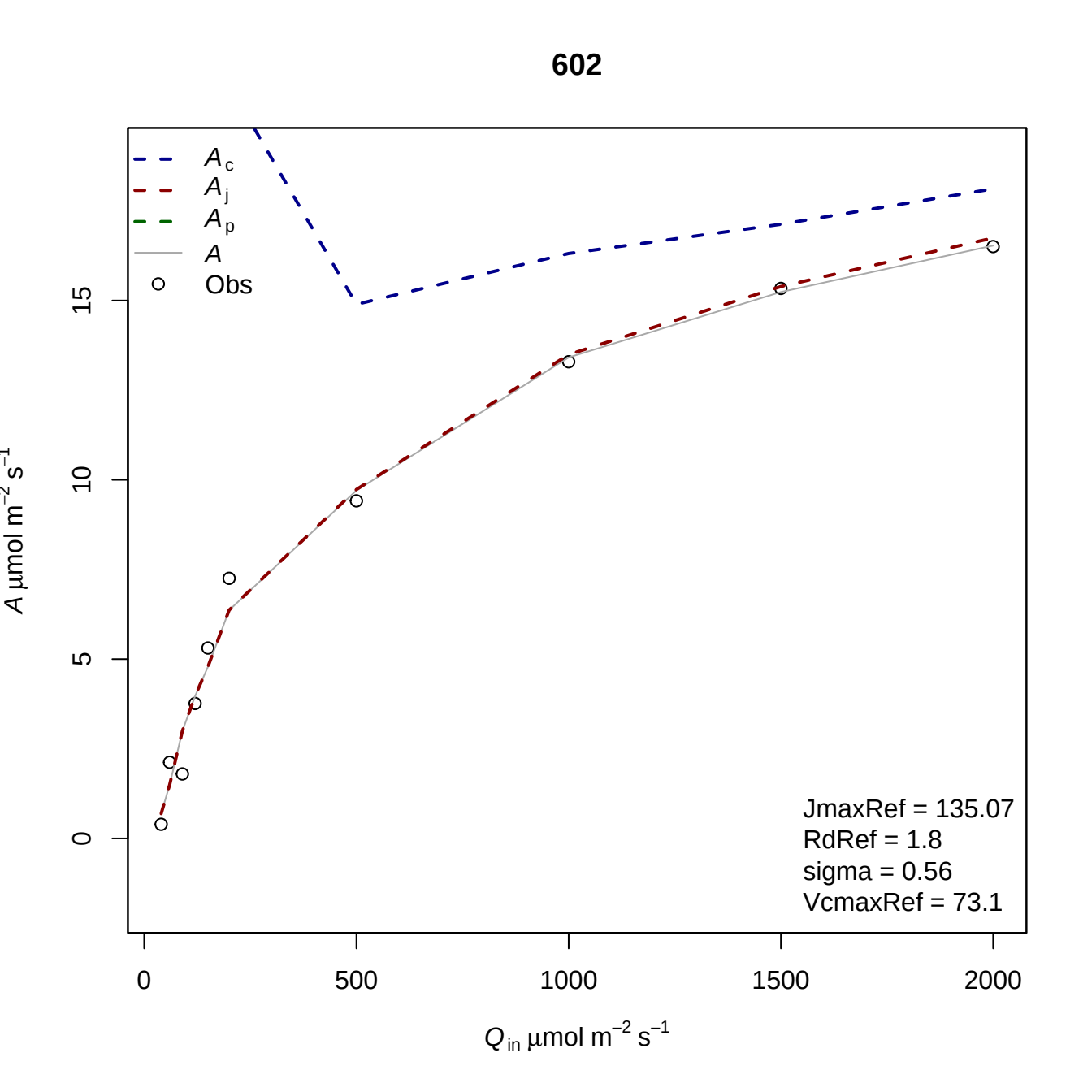

597
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.75
RdRef = 1.72
sigma = 0.4
VcmaxRef = 1143361.1
0
5
10
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

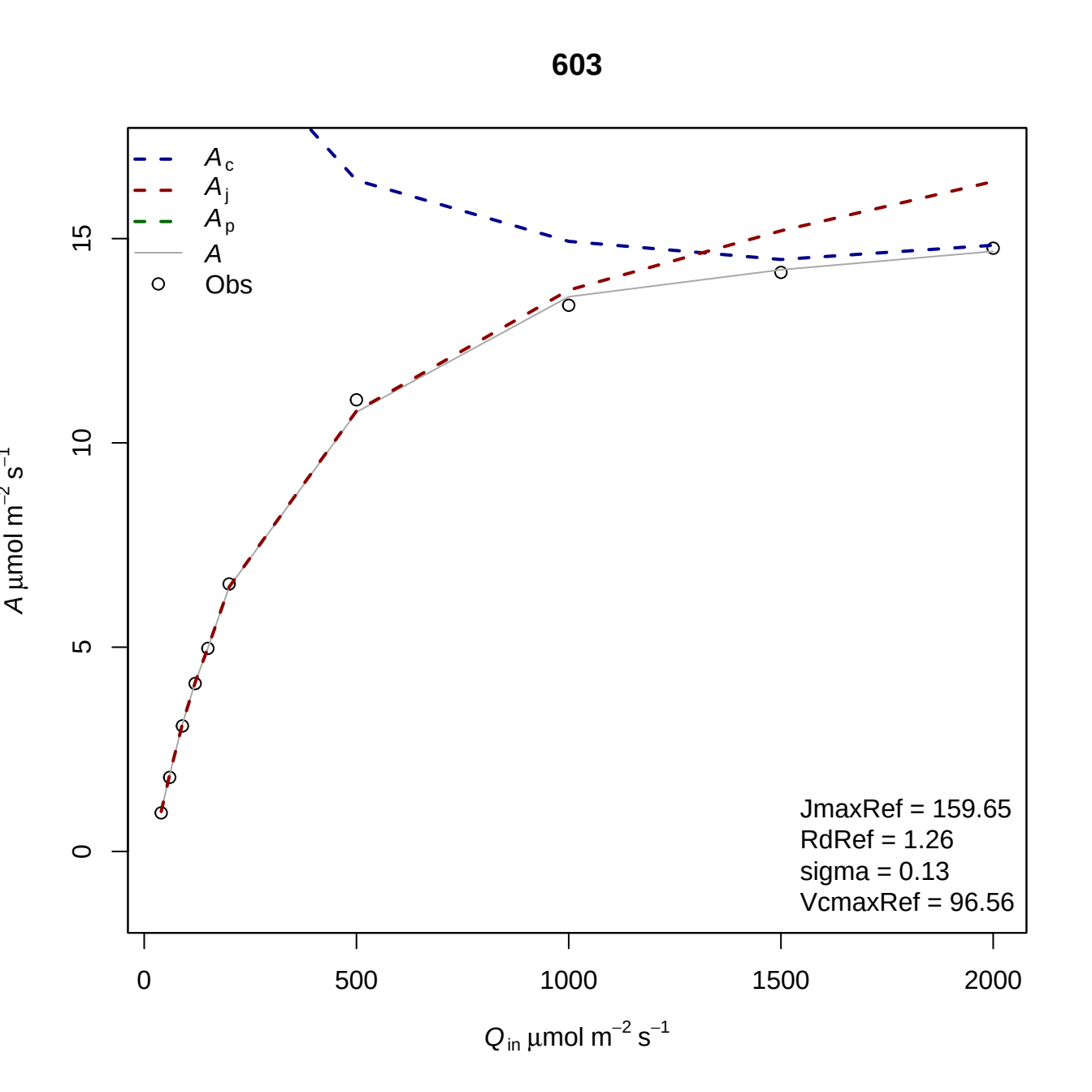

598
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 261.71
RdRef = 2
sigma = 0.14
VcmaxRef = 149.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
500
1000
1500
2000

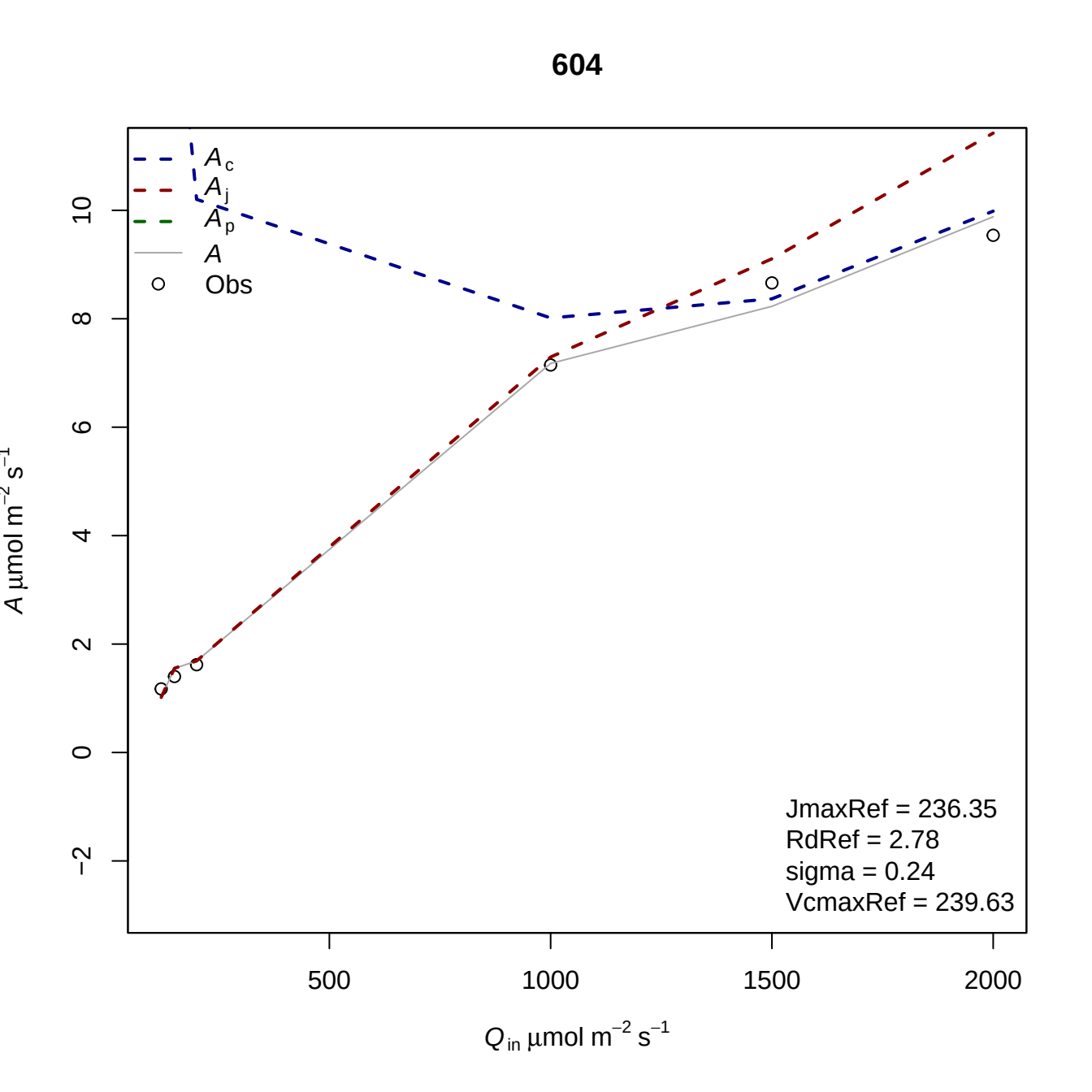

599
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.35
RdRef = 2.47
sigma = 0.59
VcmaxRef = 221948.16
8
6
4
2
0
−2
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

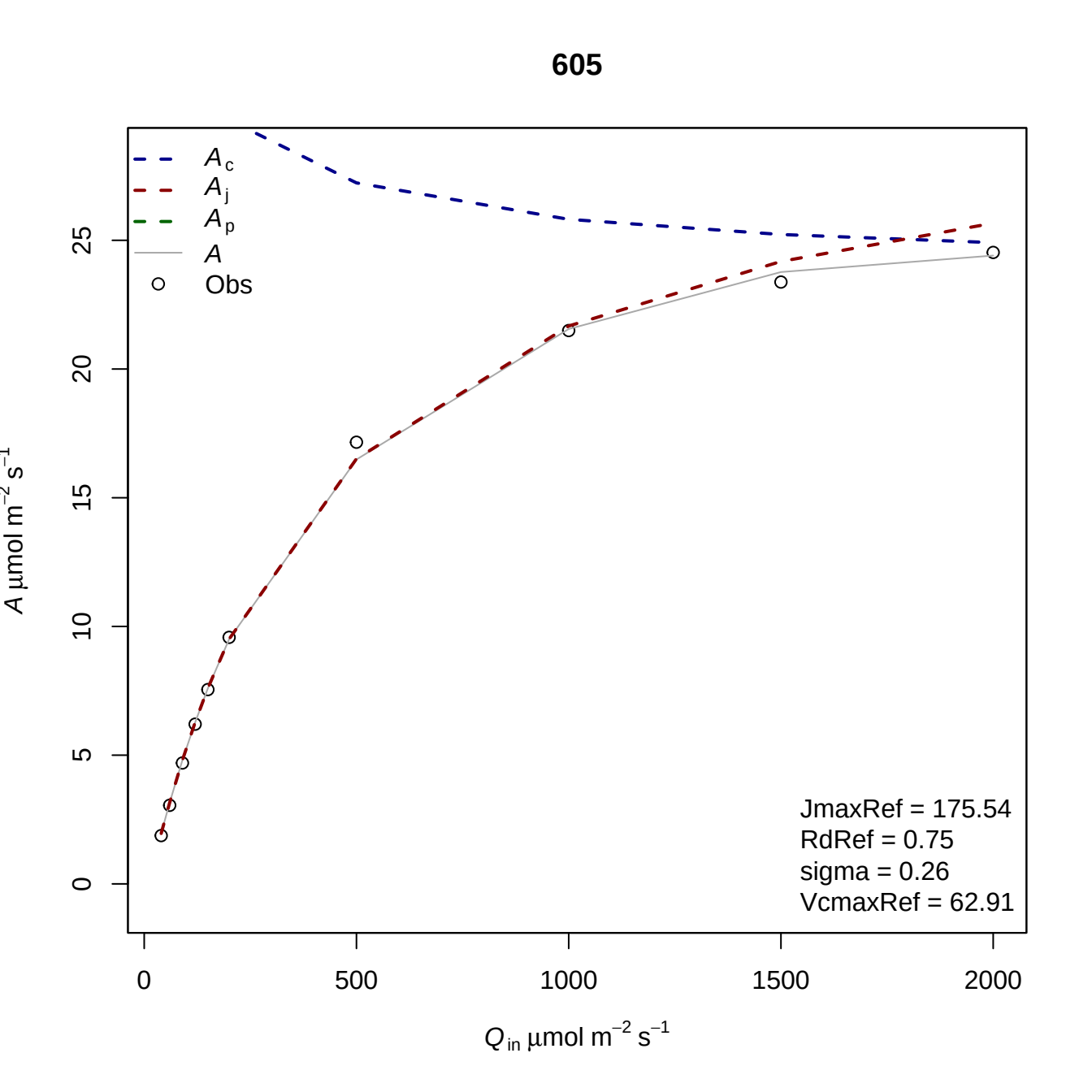

600
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 247.71
RdRef = 4.16
sigma = 0.68
VcmaxRef = 296.26
8
6
4
2
0
−2
−4
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

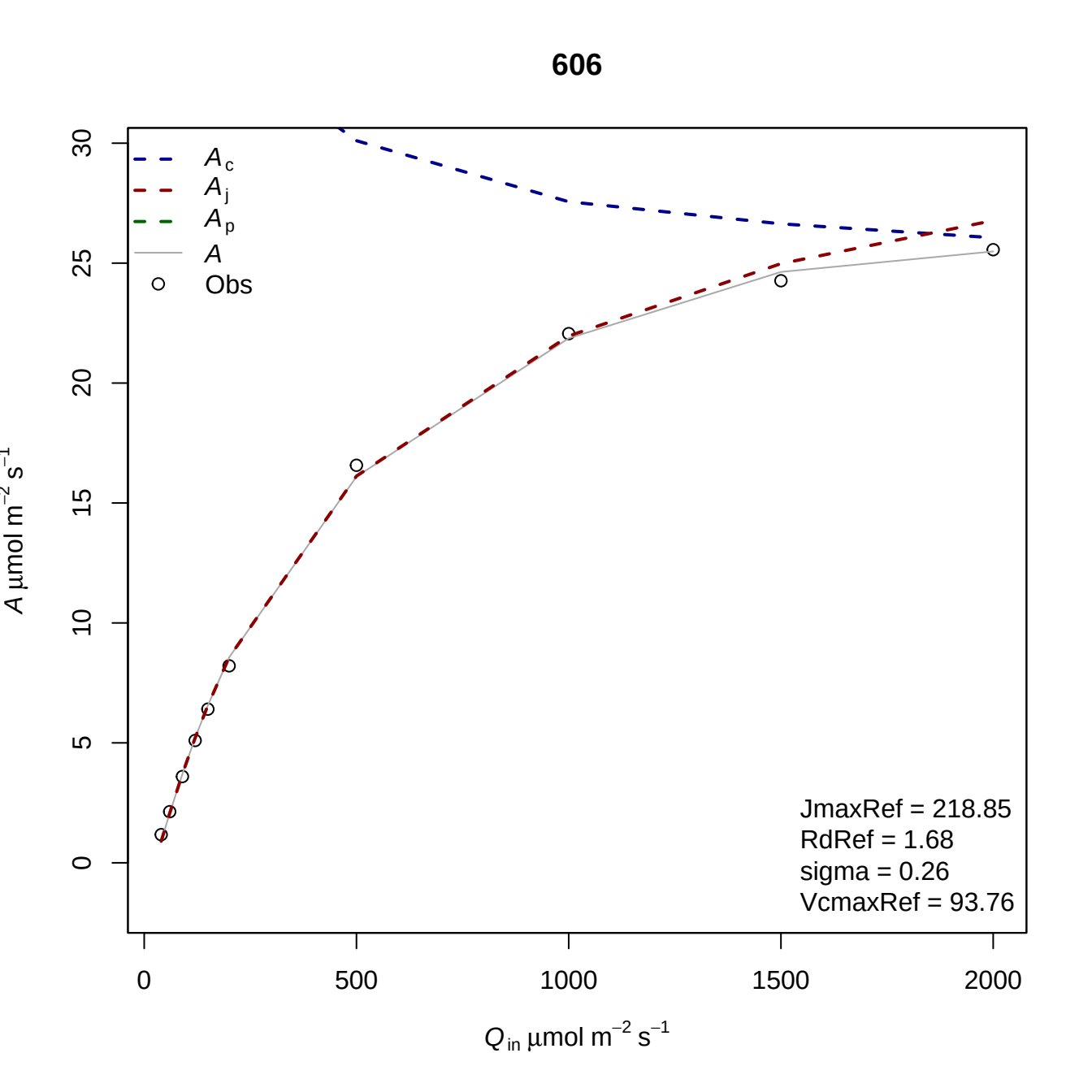

601
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.66
RdRef = 0.72
sigma = 0.12
VcmaxRef = 44.61
0
5
10
15
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

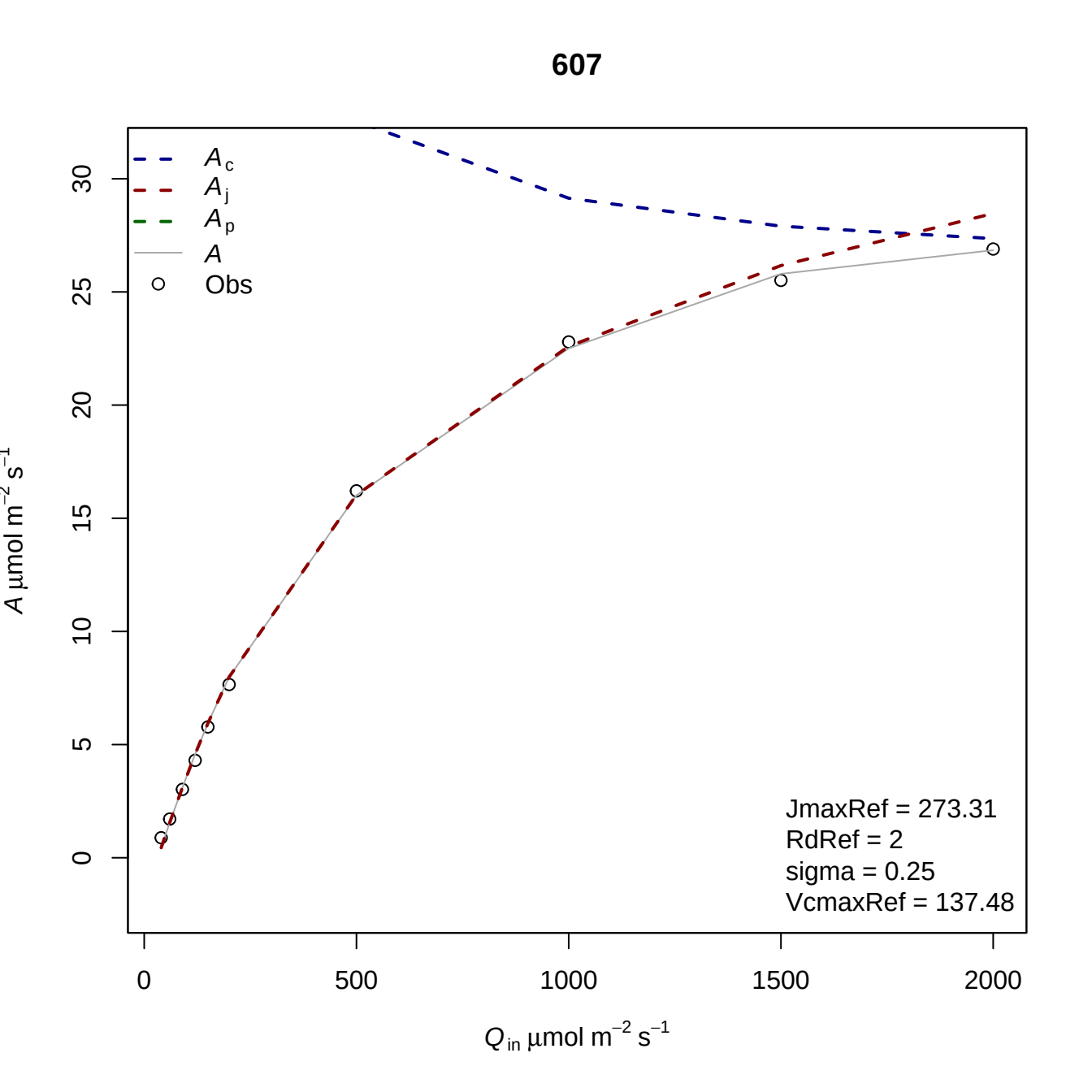

602
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.07
RdRef = 1.8
sigma = 0.56
VcmaxRef = 73.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

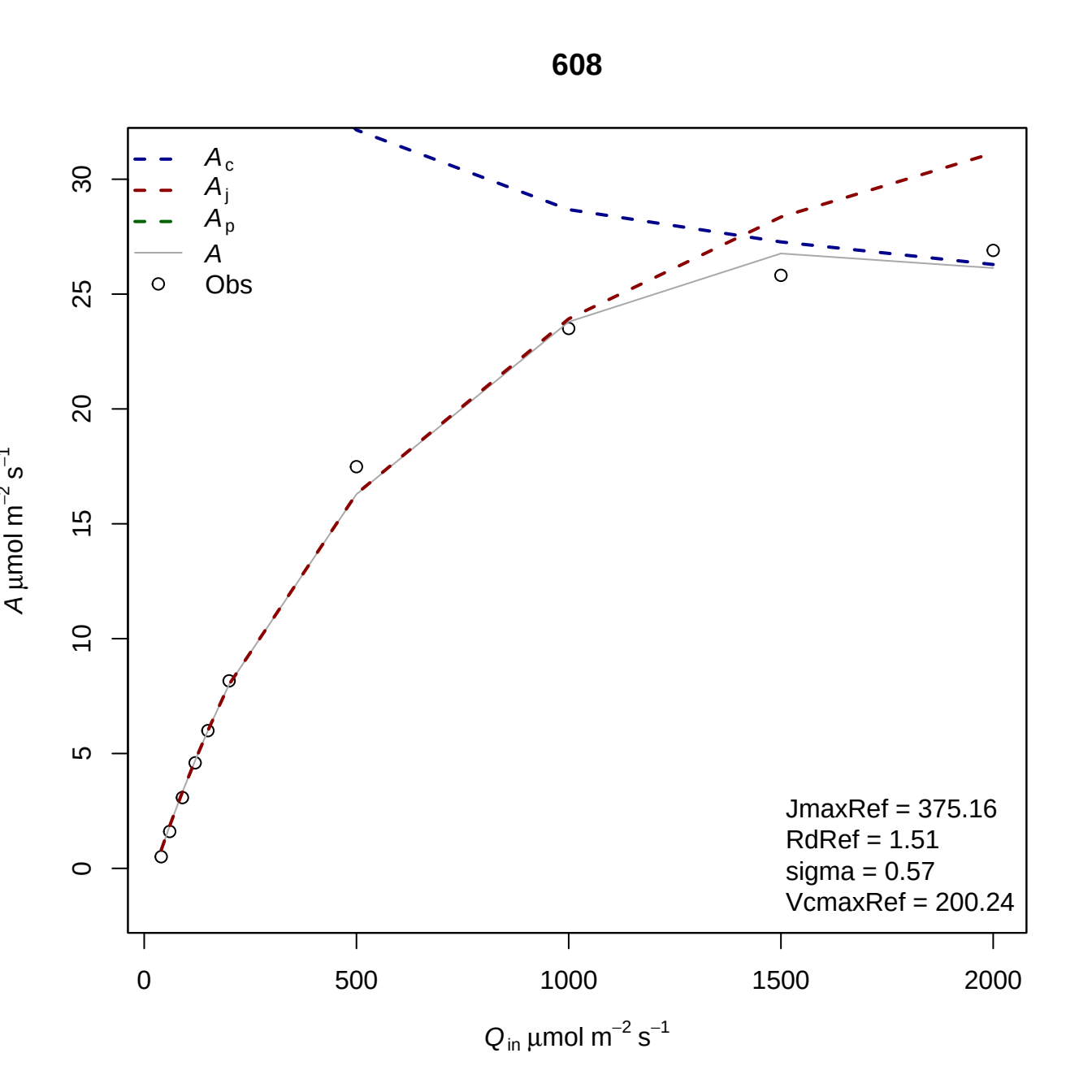

603
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 159.65
RdRef = 1.26
sigma = 0.13
VcmaxRef = 96.56
0
5
10
15
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

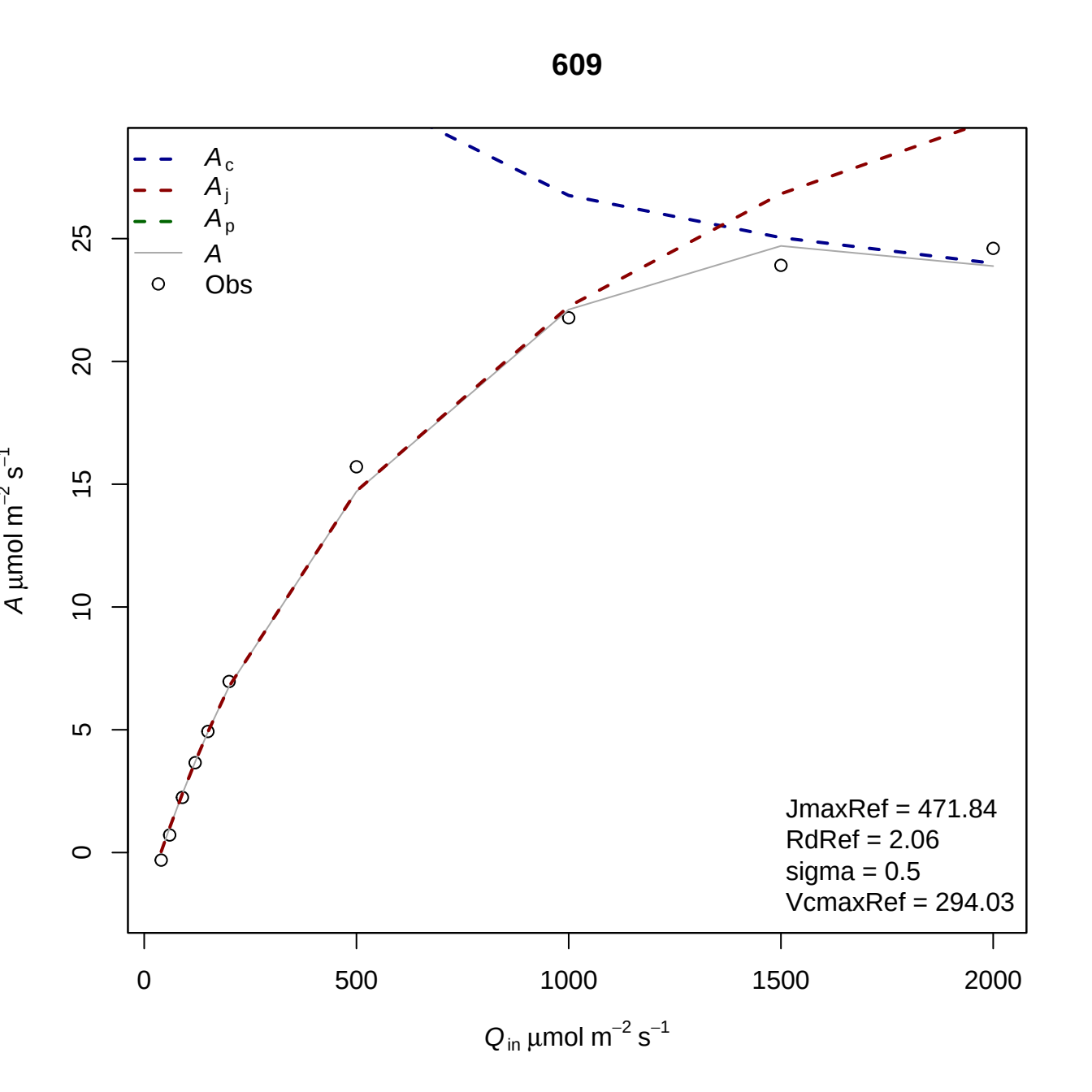

604
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 236.35
RdRef = 2.78
sigma = 0.24
VcmaxRef = 239.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
500
1000
1500
2000

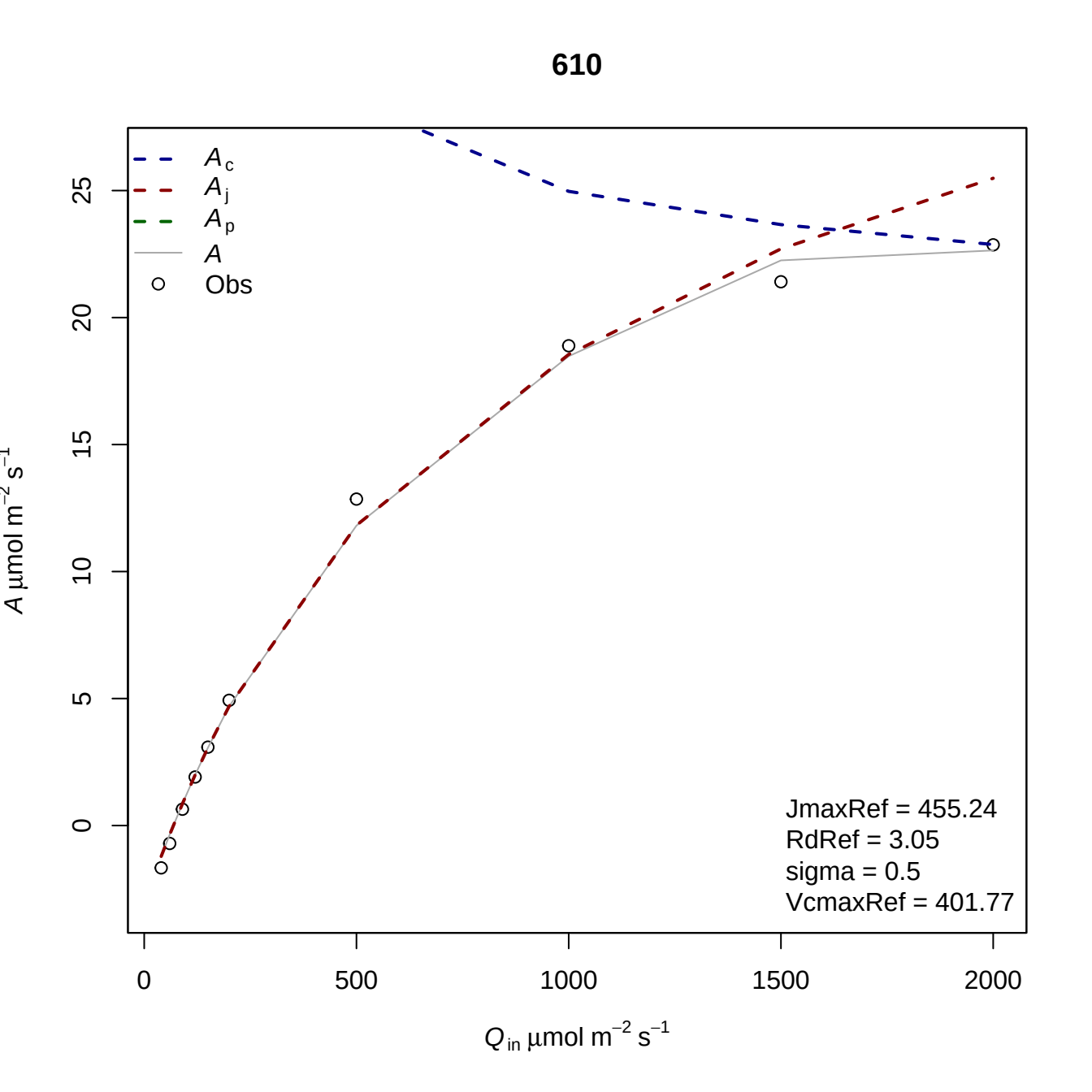

605
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 175.54
RdRef = 0.75
sigma = 0.26
VcmaxRef = 62.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

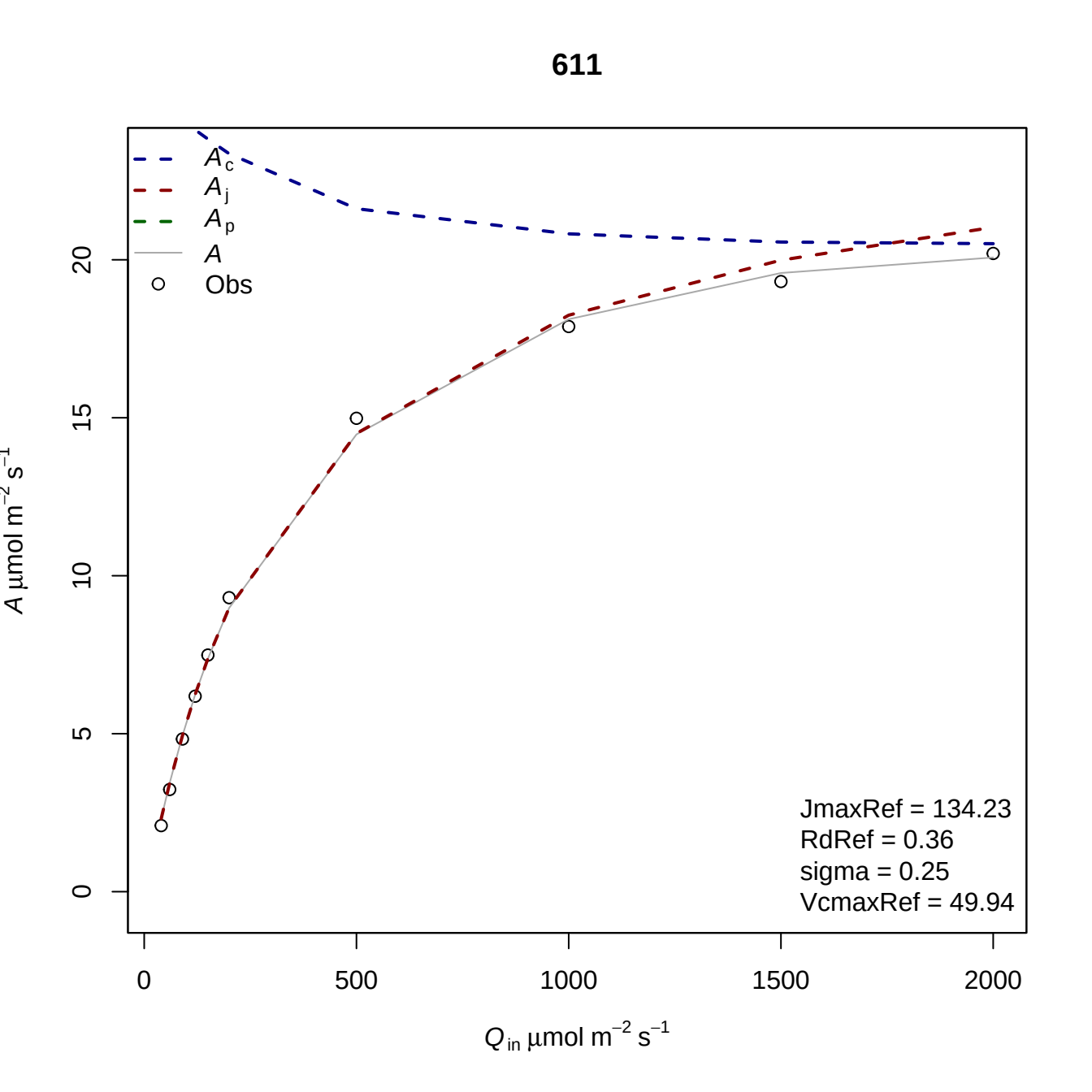

606
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 218.85
RdRef = 1.68
sigma = 0.26
VcmaxRef = 93.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

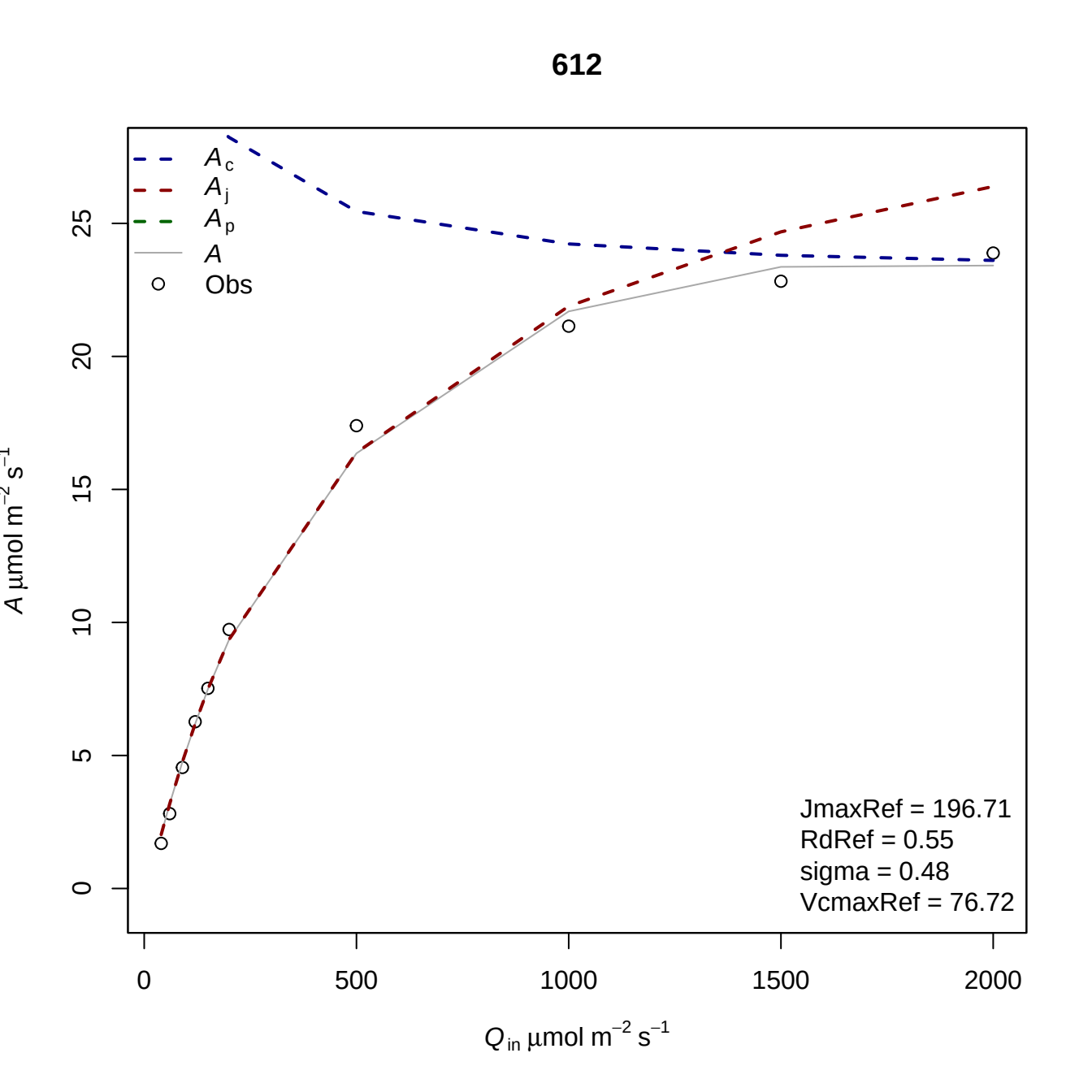

607
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 273.31
RdRef = 2
sigma = 0.25
VcmaxRef = 137.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

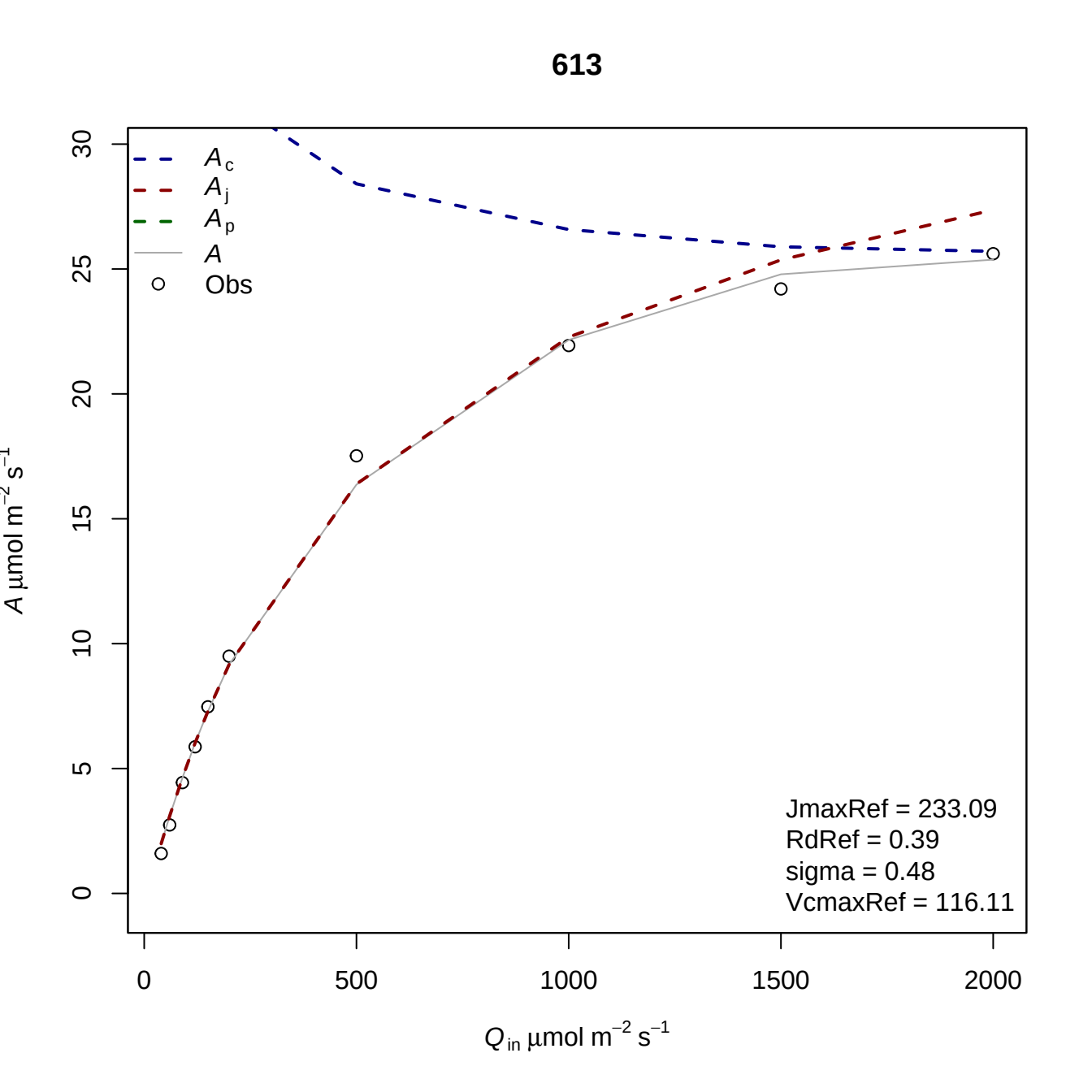

608
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 375.16
RdRef = 1.51
sigma = 0.57
VcmaxRef = 200.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

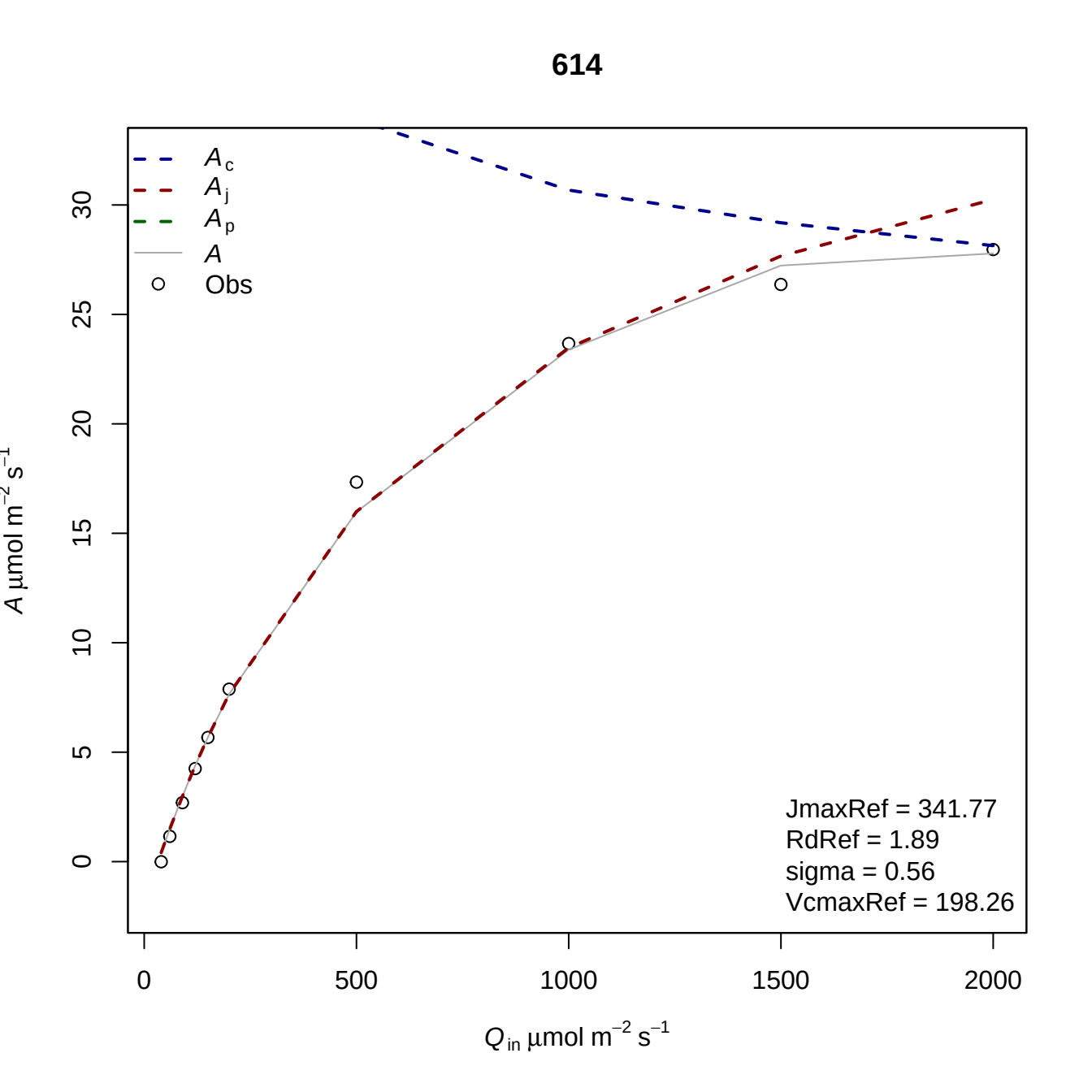

609
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 471.84
RdRef = 2.06
sigma = 0.5
VcmaxRef = 294.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

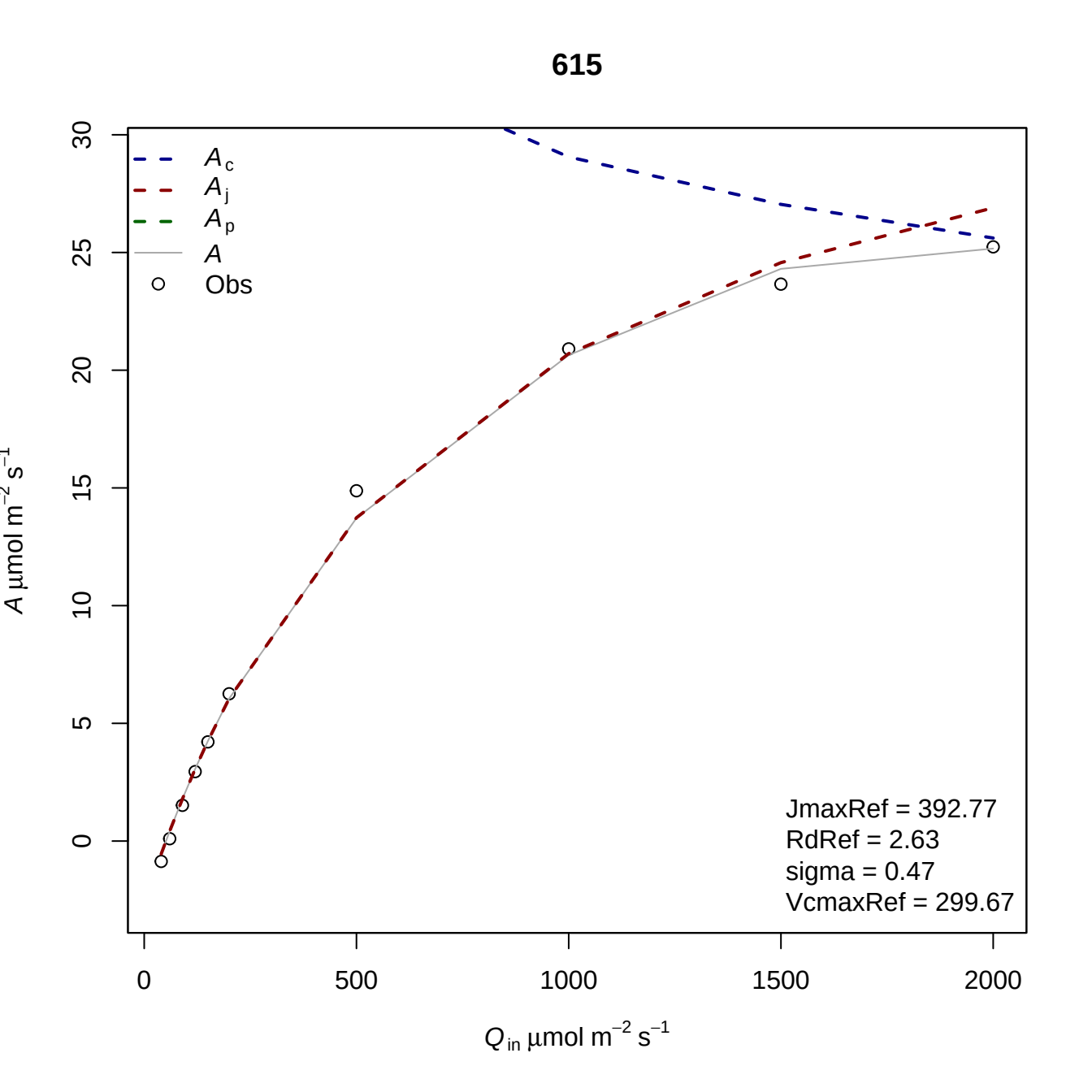

610
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 455.24
RdRef = 3.05
sigma = 0.5
VcmaxRef = 401.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

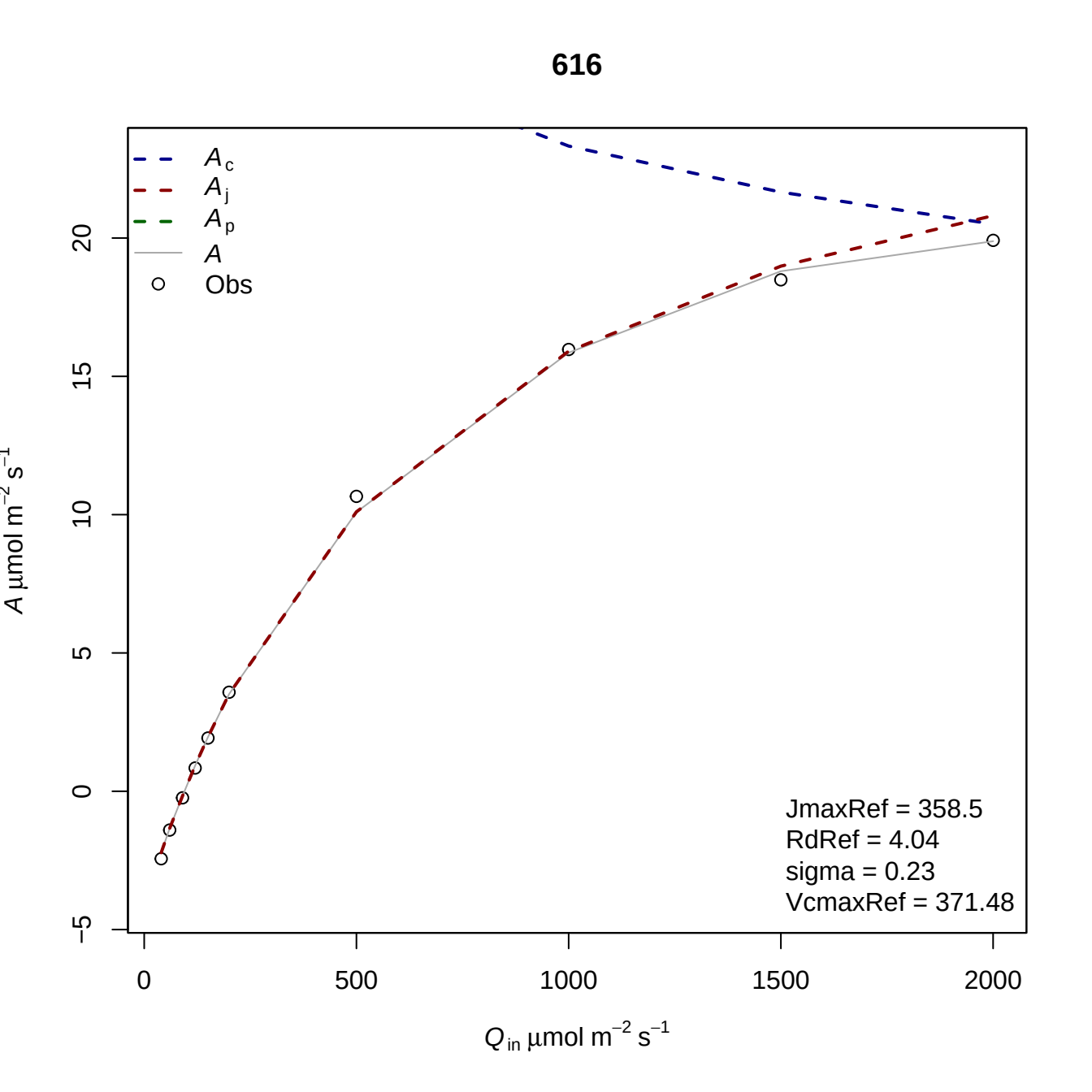

611
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.23
RdRef = 0.36
sigma = 0.25
VcmaxRef = 49.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

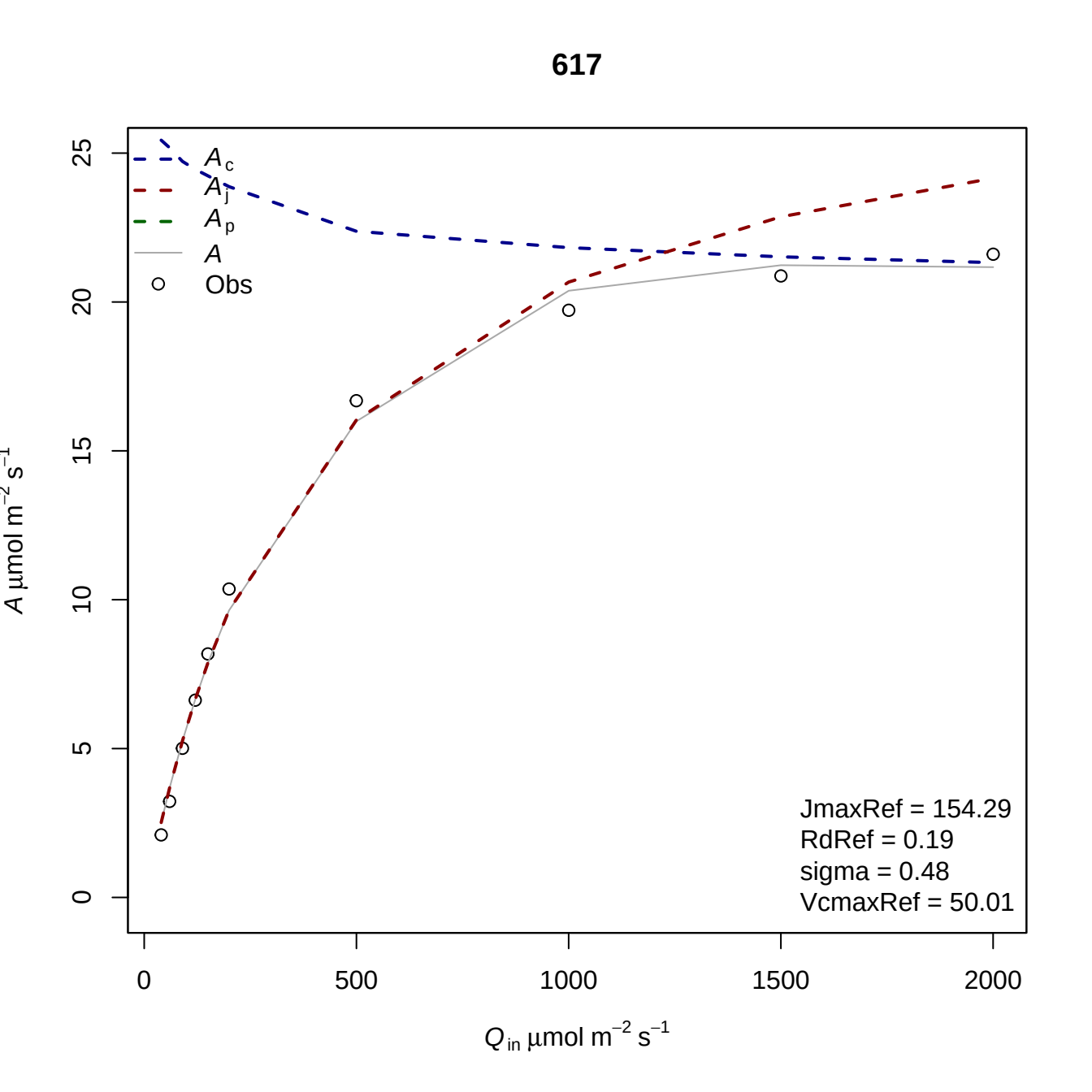

612
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 196.71
RdRef = 0.55
sigma = 0.48
VcmaxRef = 76.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

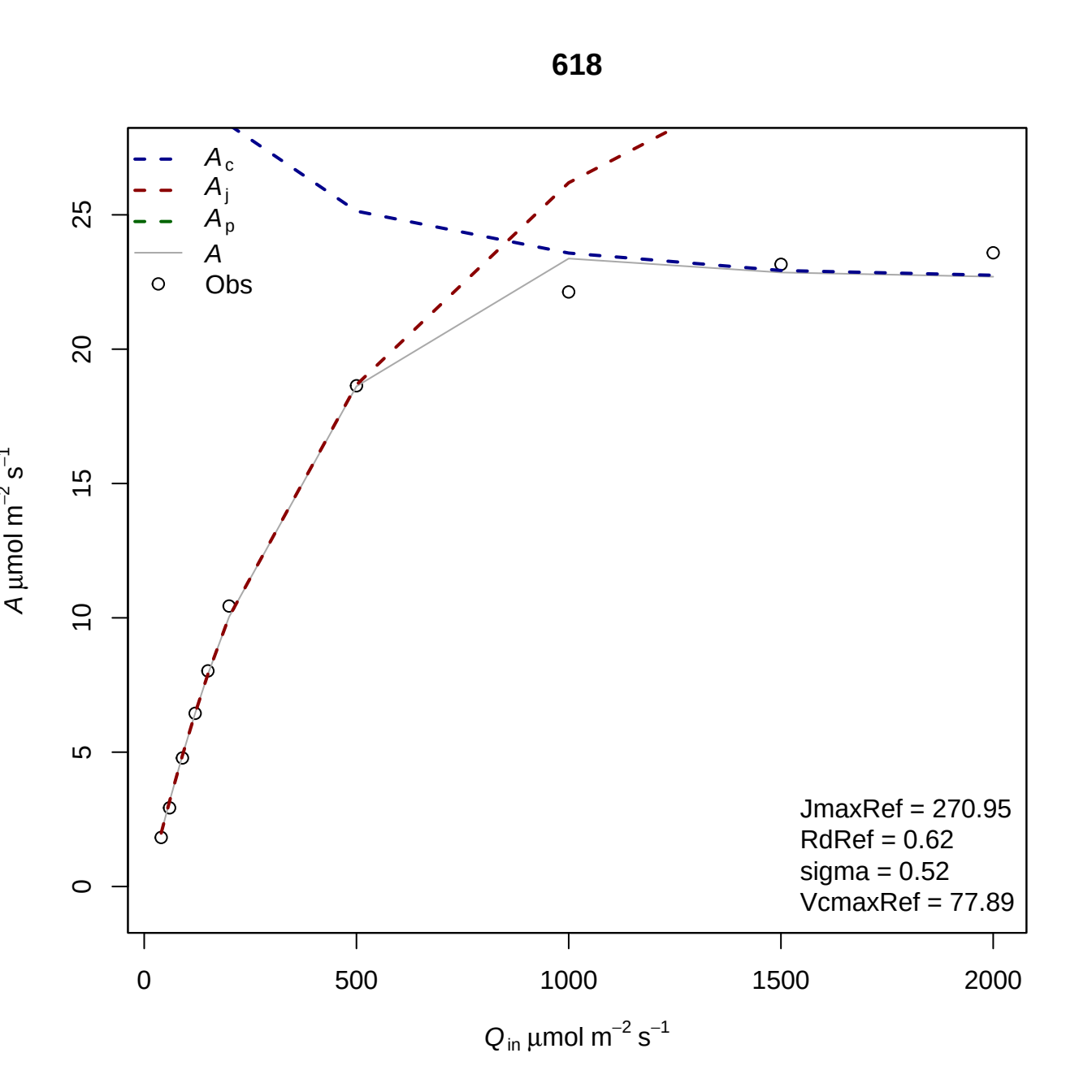

613
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 233.09
RdRef = 0.39
sigma = 0.48
VcmaxRef = 116.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

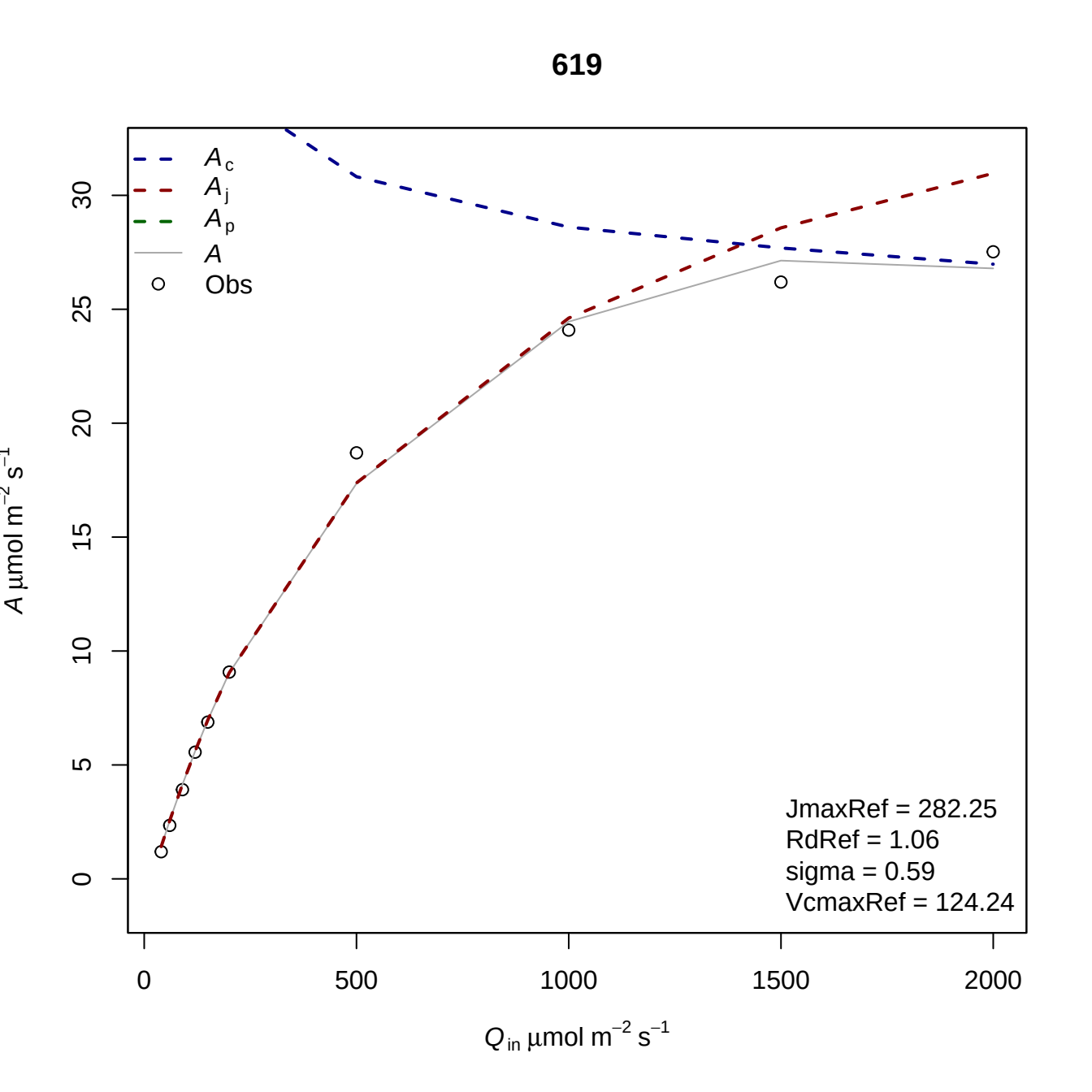

614
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 341.77
RdRef = 1.89
sigma = 0.56
VcmaxRef = 198.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

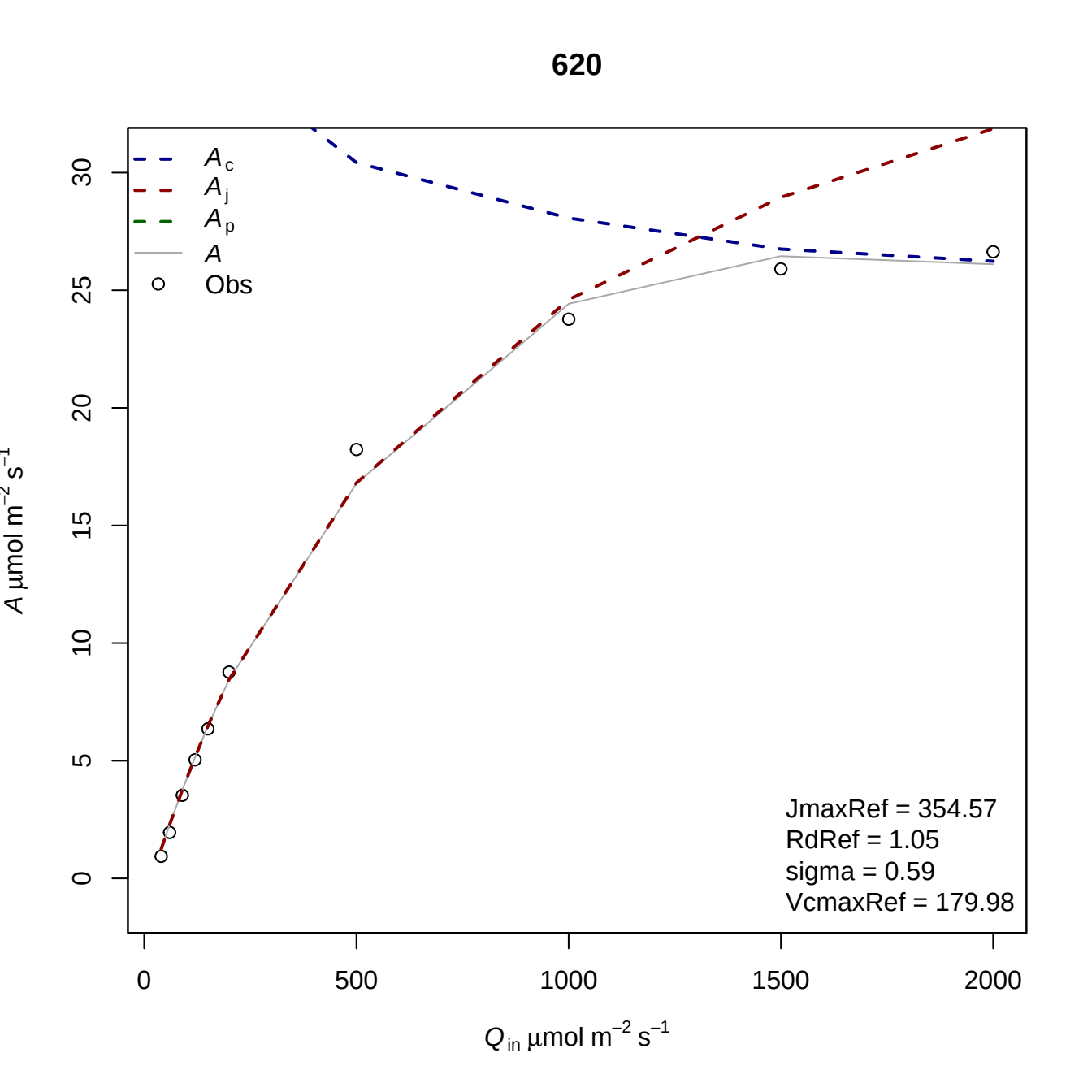

615
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 392.77
RdRef = 2.63
sigma = 0.47
VcmaxRef = 299.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

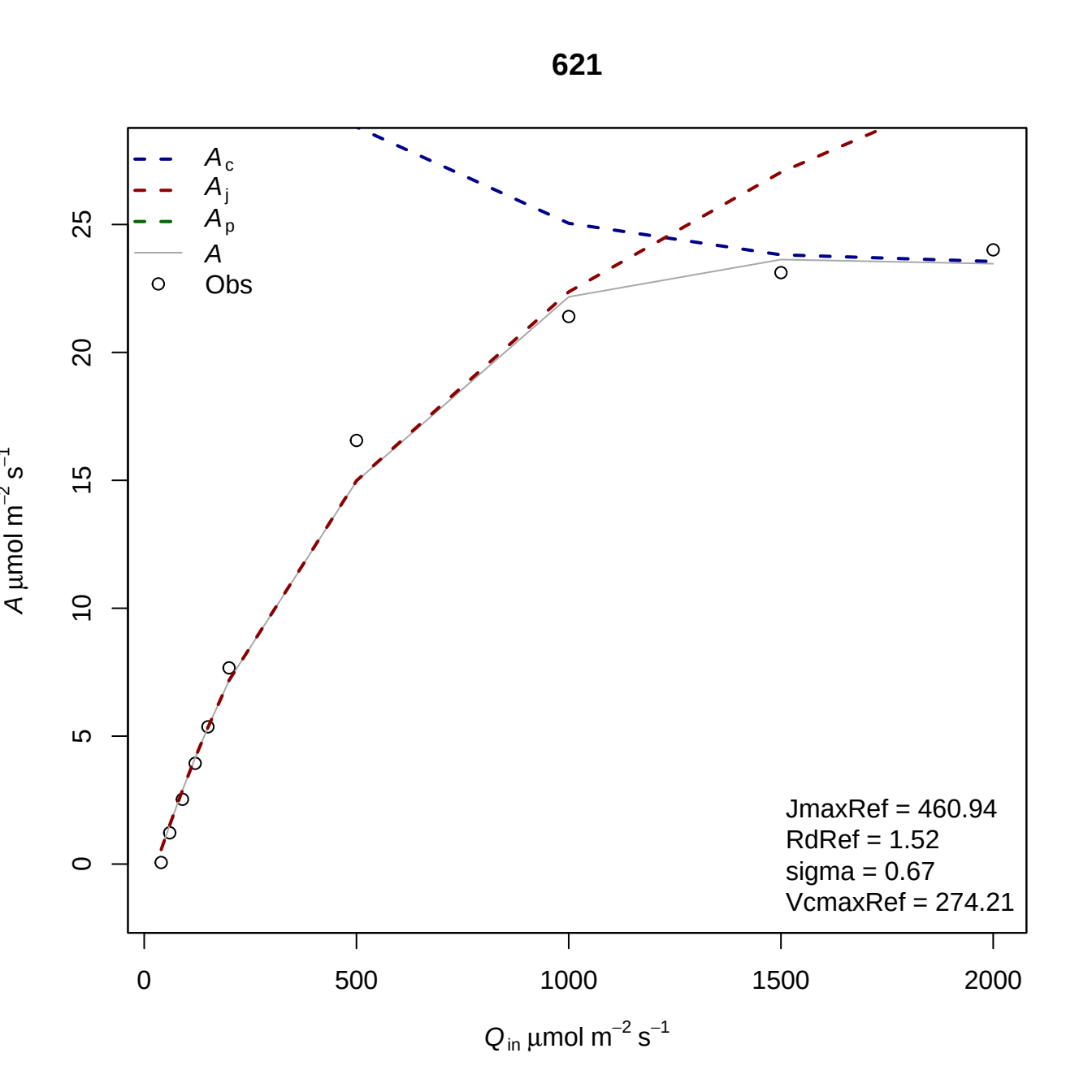

616
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 358.5
RdRef = 4.04
sigma = 0.23
VcmaxRef = 371.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

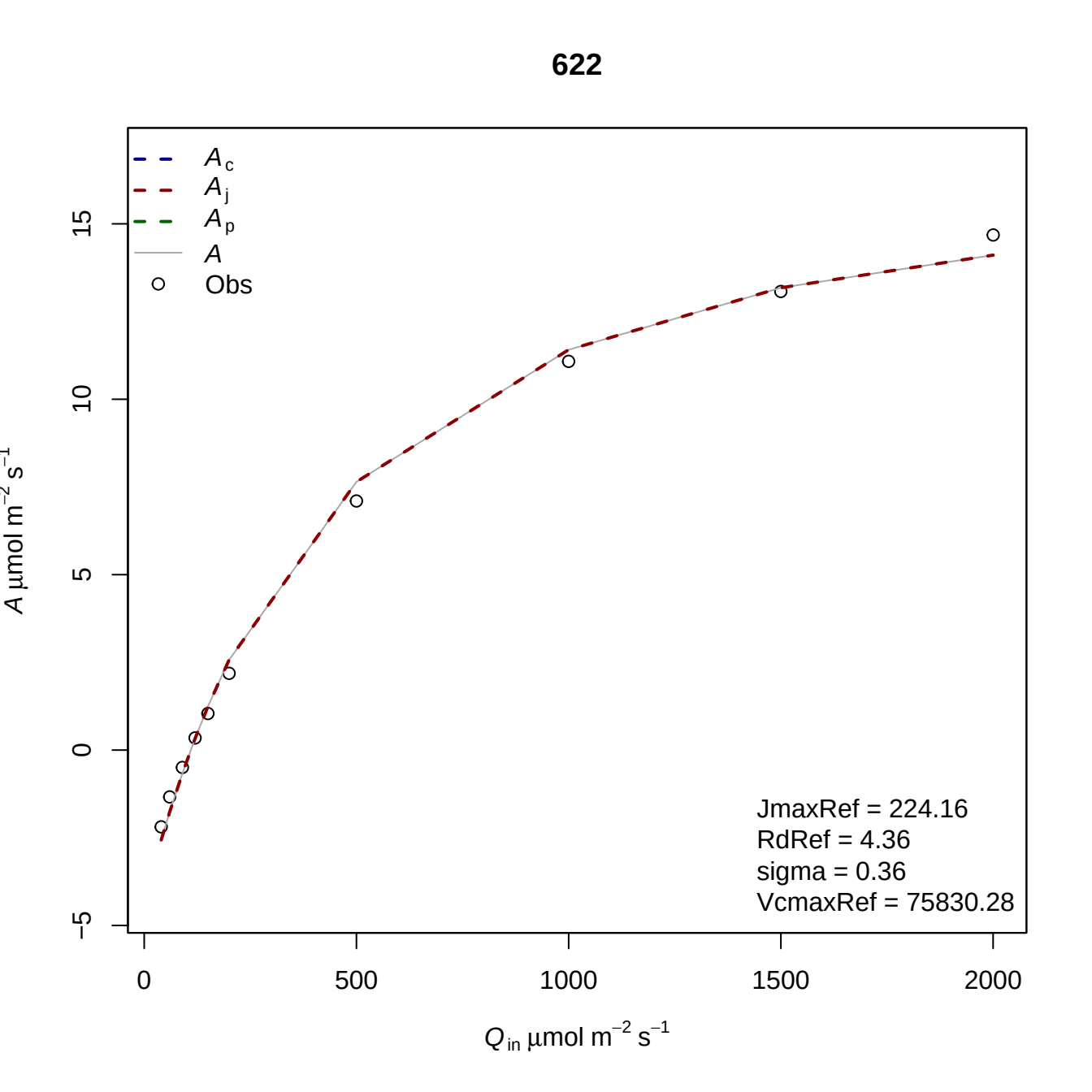

617
$A_c$
$A_J$
$A_p$
$A$
Obs
JmaxRef = 154.29
RdRef = 0.19
sigma = 0.48
VcmaxRef = 50.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

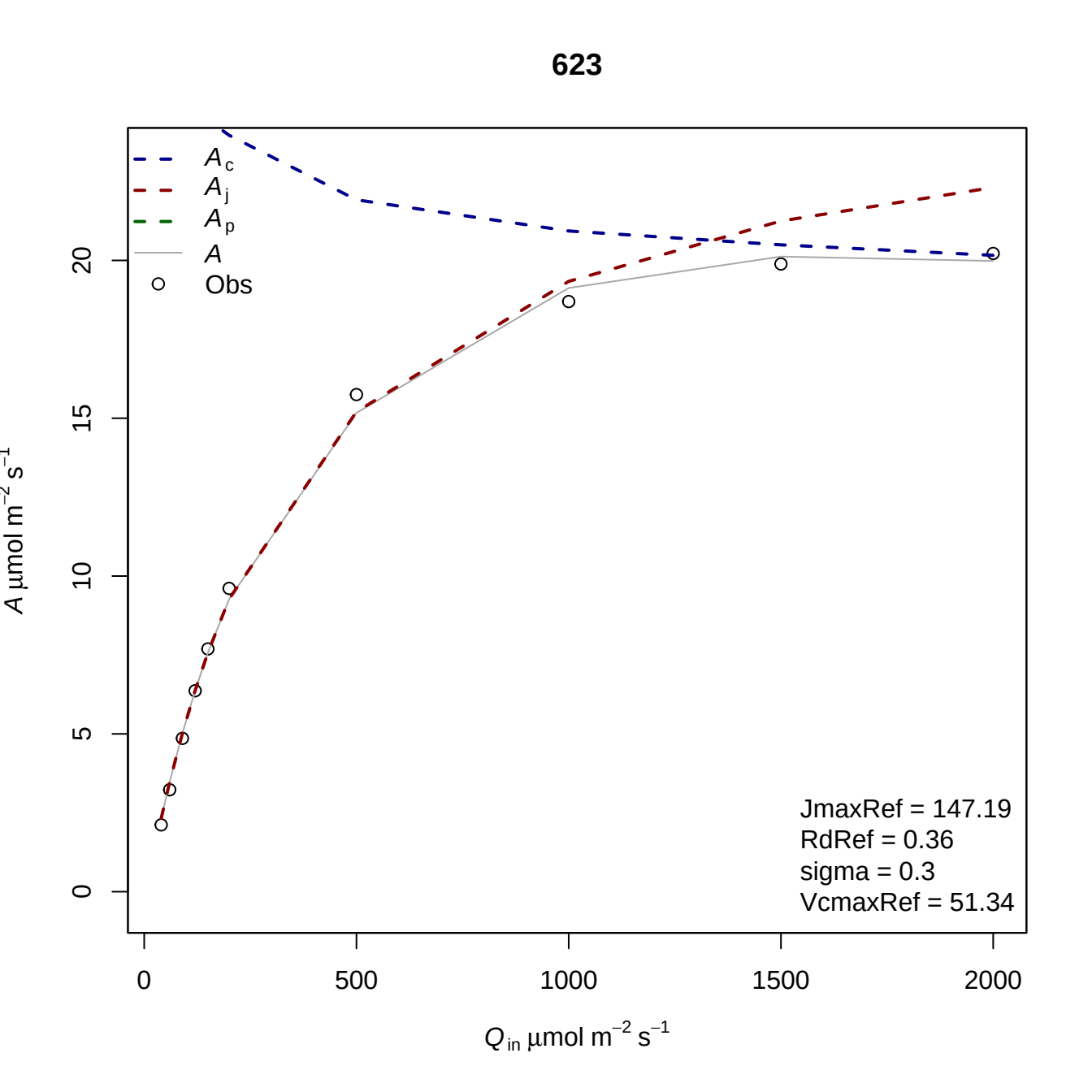

618
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 270.95
RdRef = 0.62
sigma = 0.52
VcmaxRef = 77.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

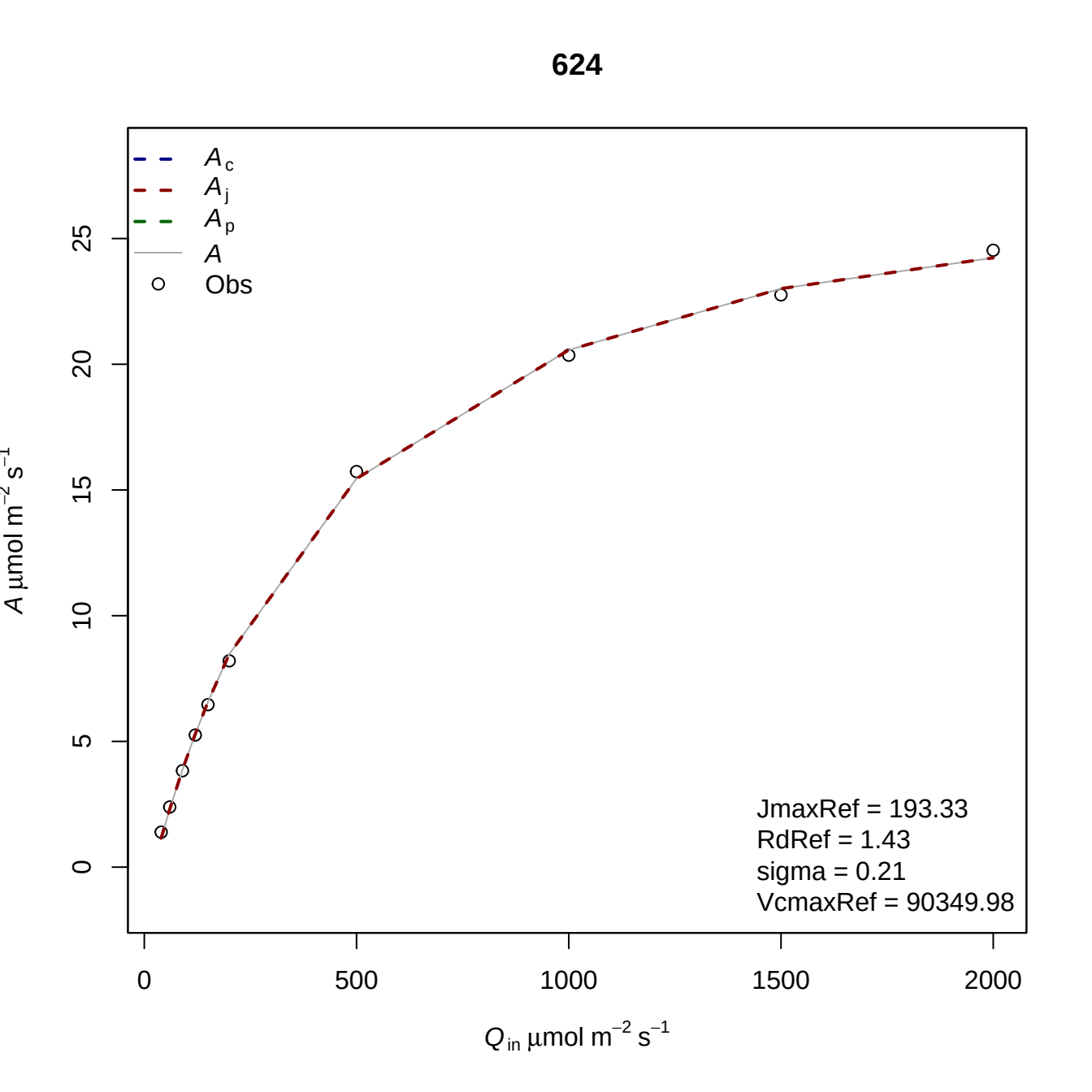

619
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 282.25
RdRef = 1.06
sigma = 0.59
VcmaxRef = 124.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

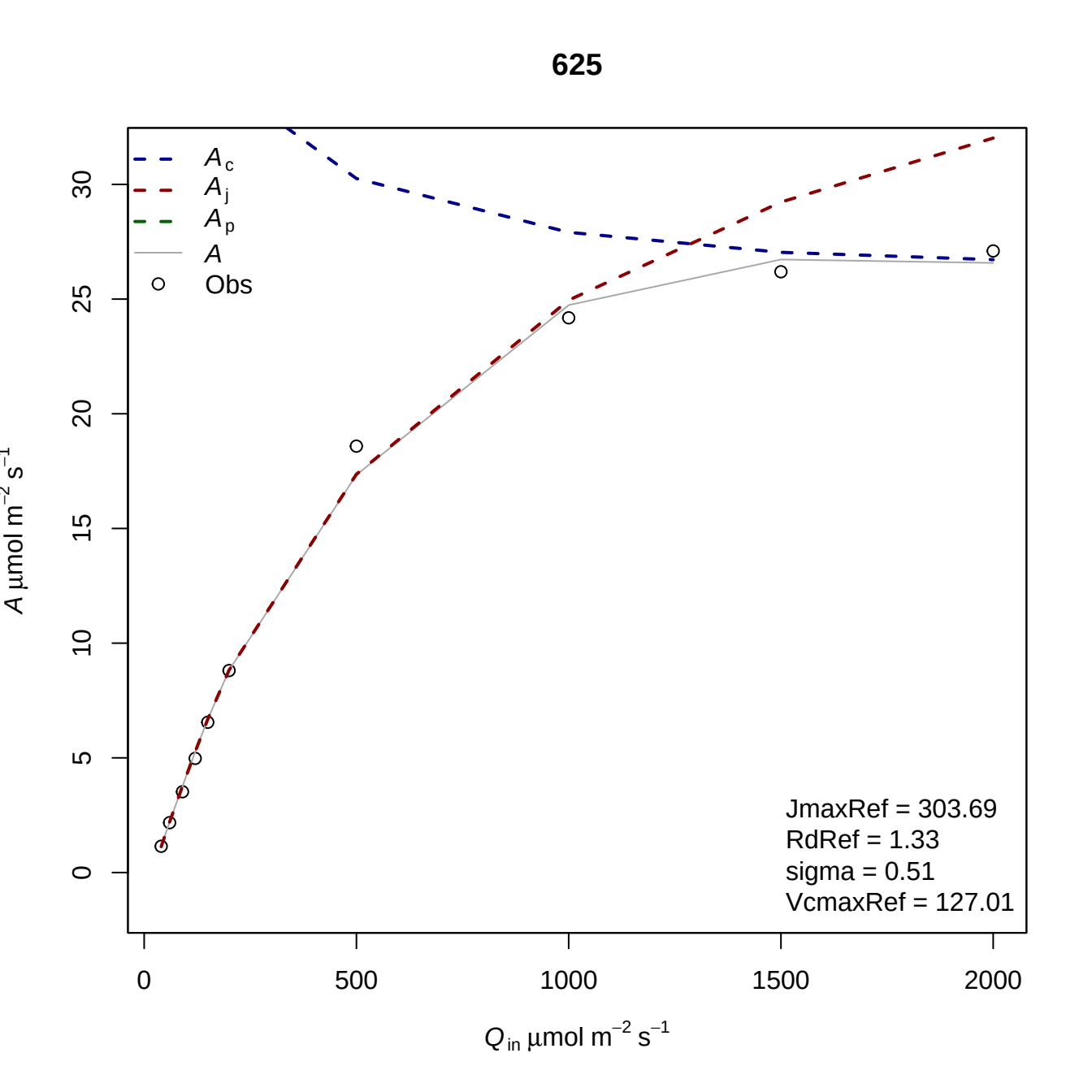

620
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 354.57
RdRef = 1.05
sigma = 0.59
VcmaxRef = 179.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

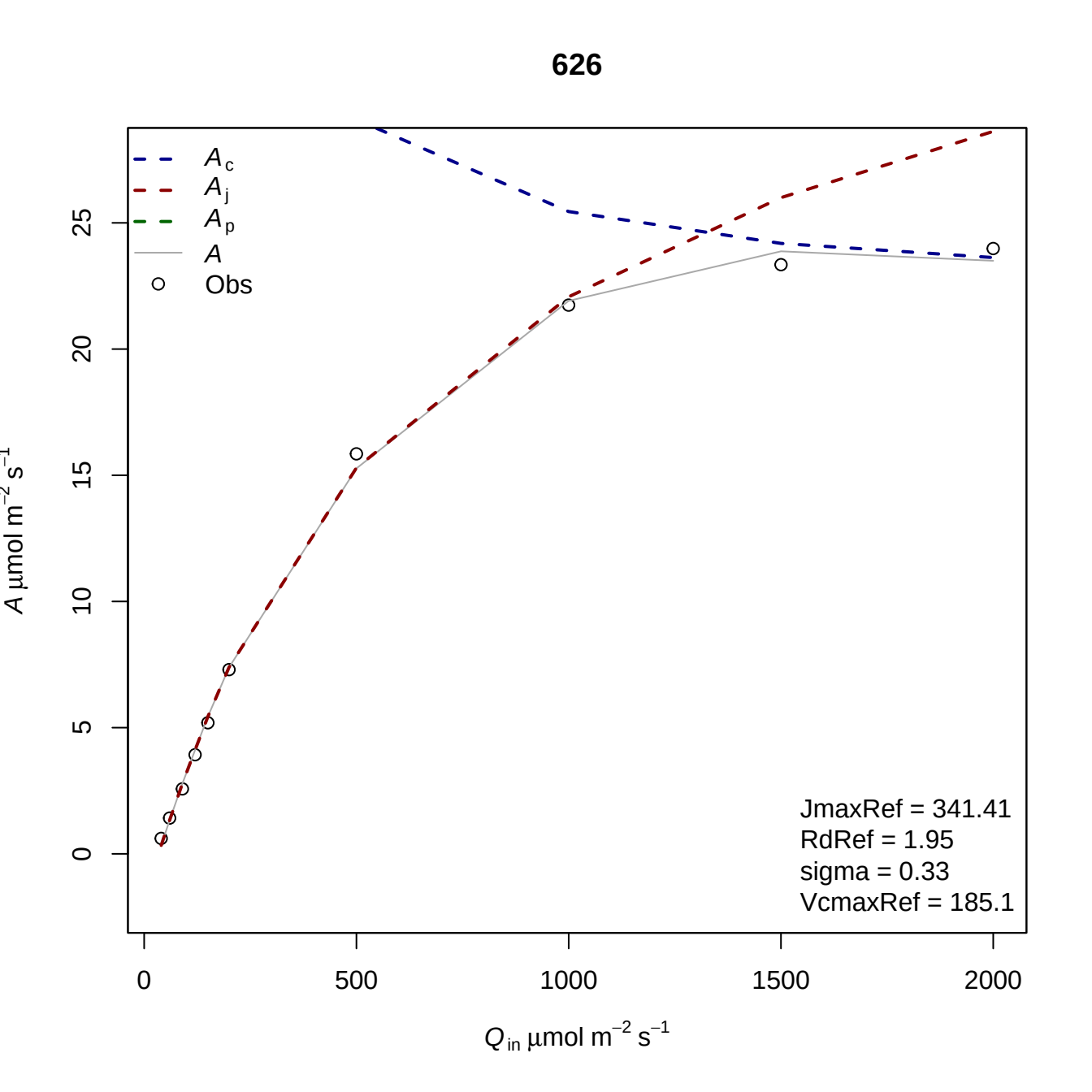

621
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 460.94
RdRef = 1.52
sigma = 0.67
VcmaxRef = 274.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

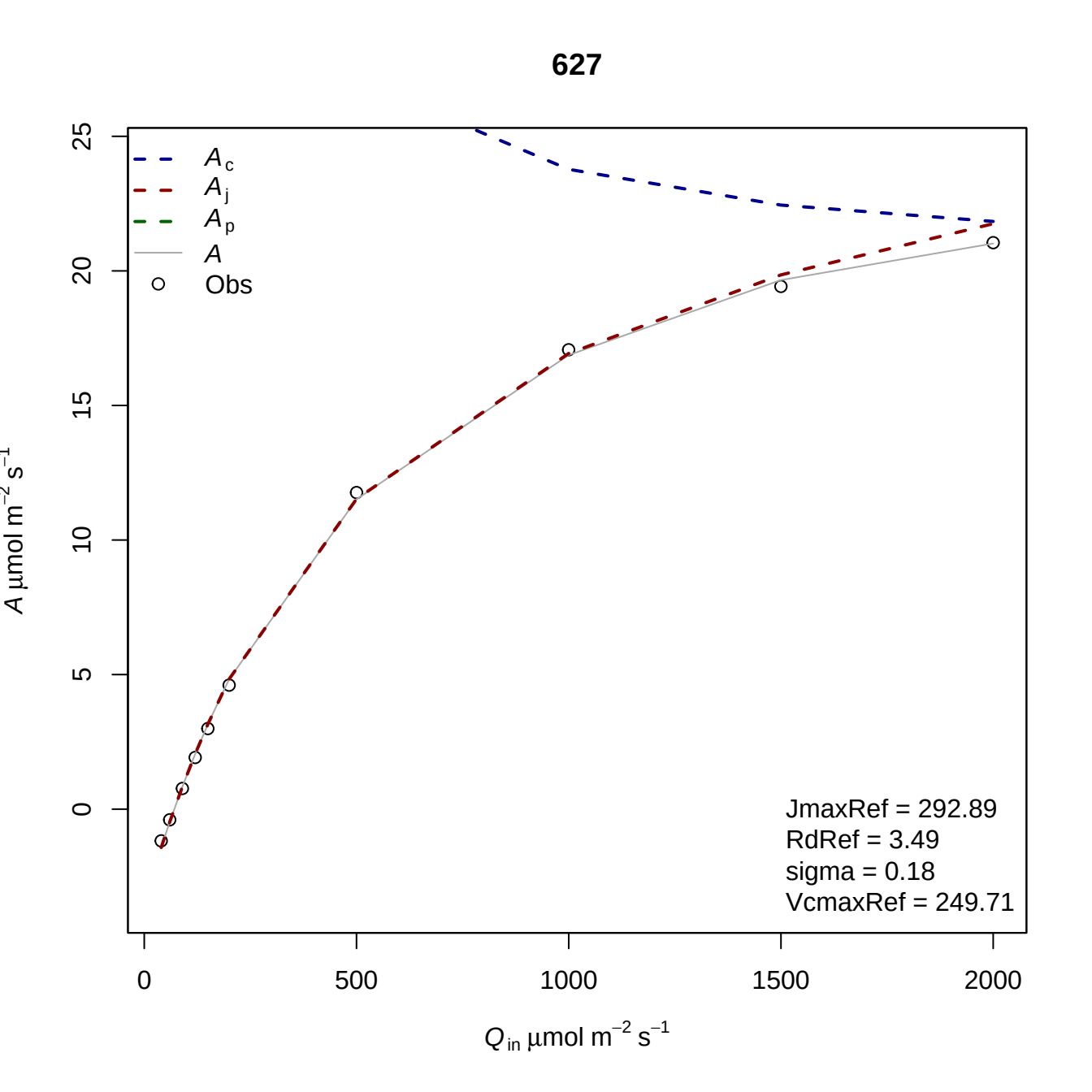

622
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 224.16
RdRef = 4.36
sigma = 0.36
VcmaxRef = 75830.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

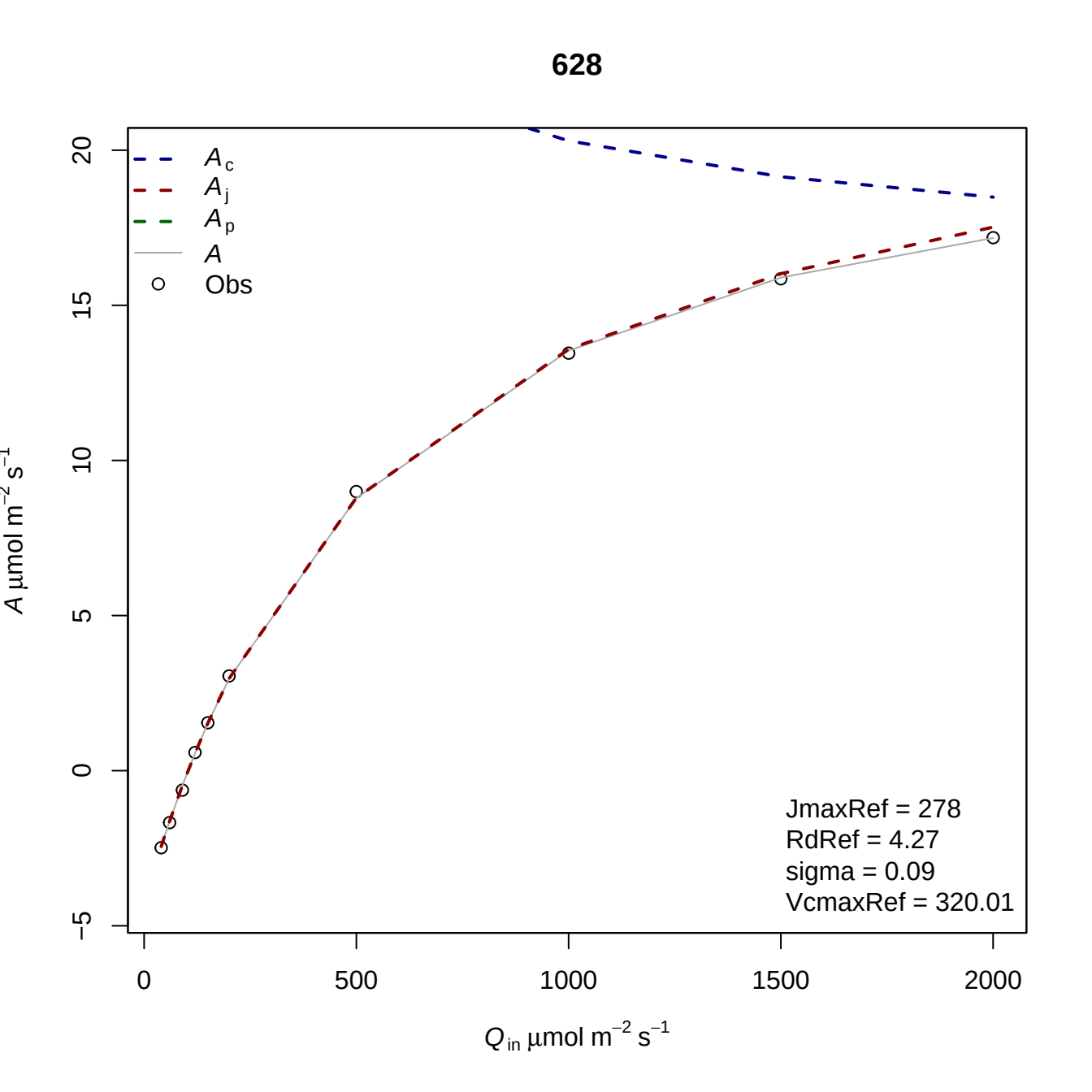

623
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.19
RdRef = 0.36
sigma = 0.3
VcmaxRef = 51.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

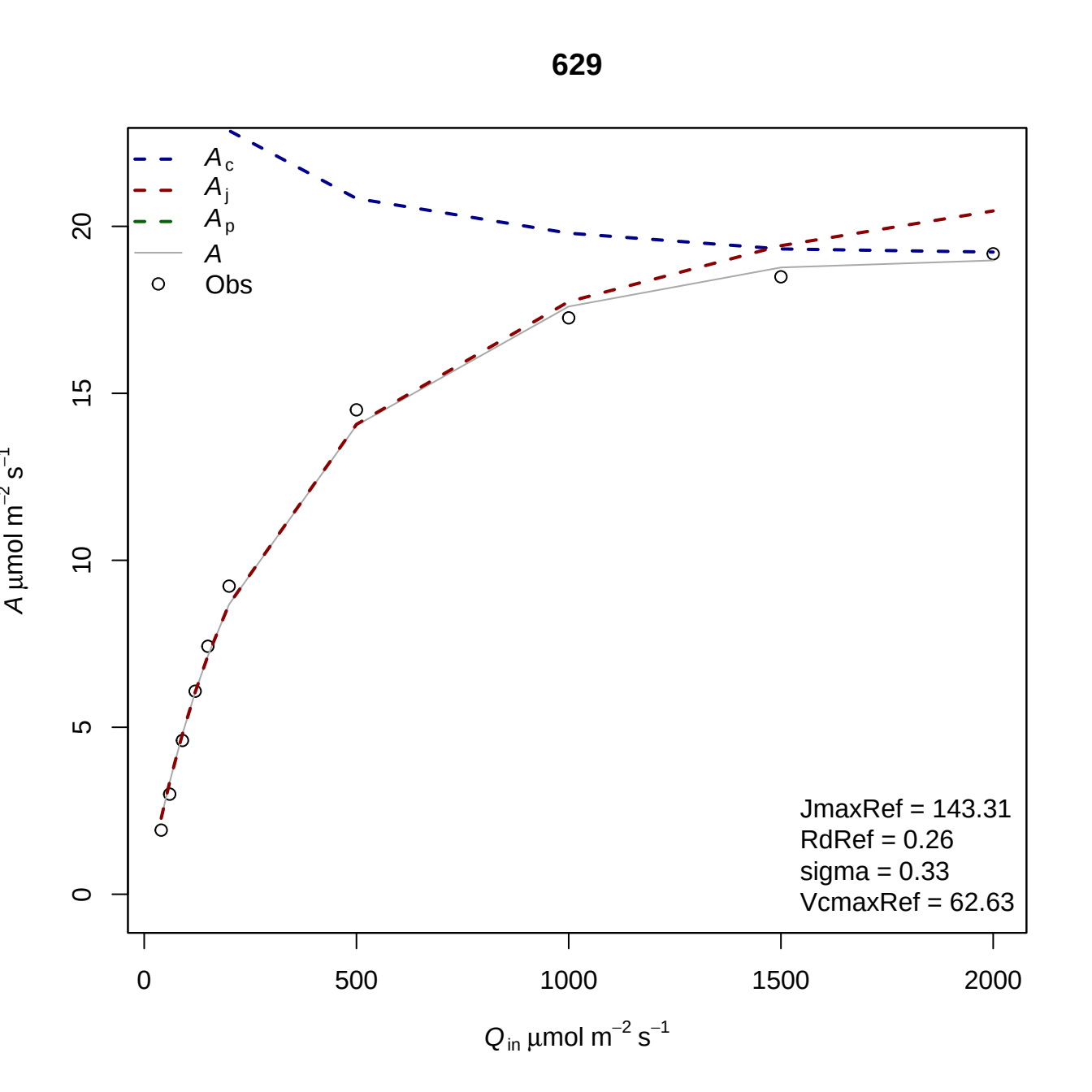

624
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 193.33
RdRef = 1.43
sigma = 0.21
VcmaxRef = 90349.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

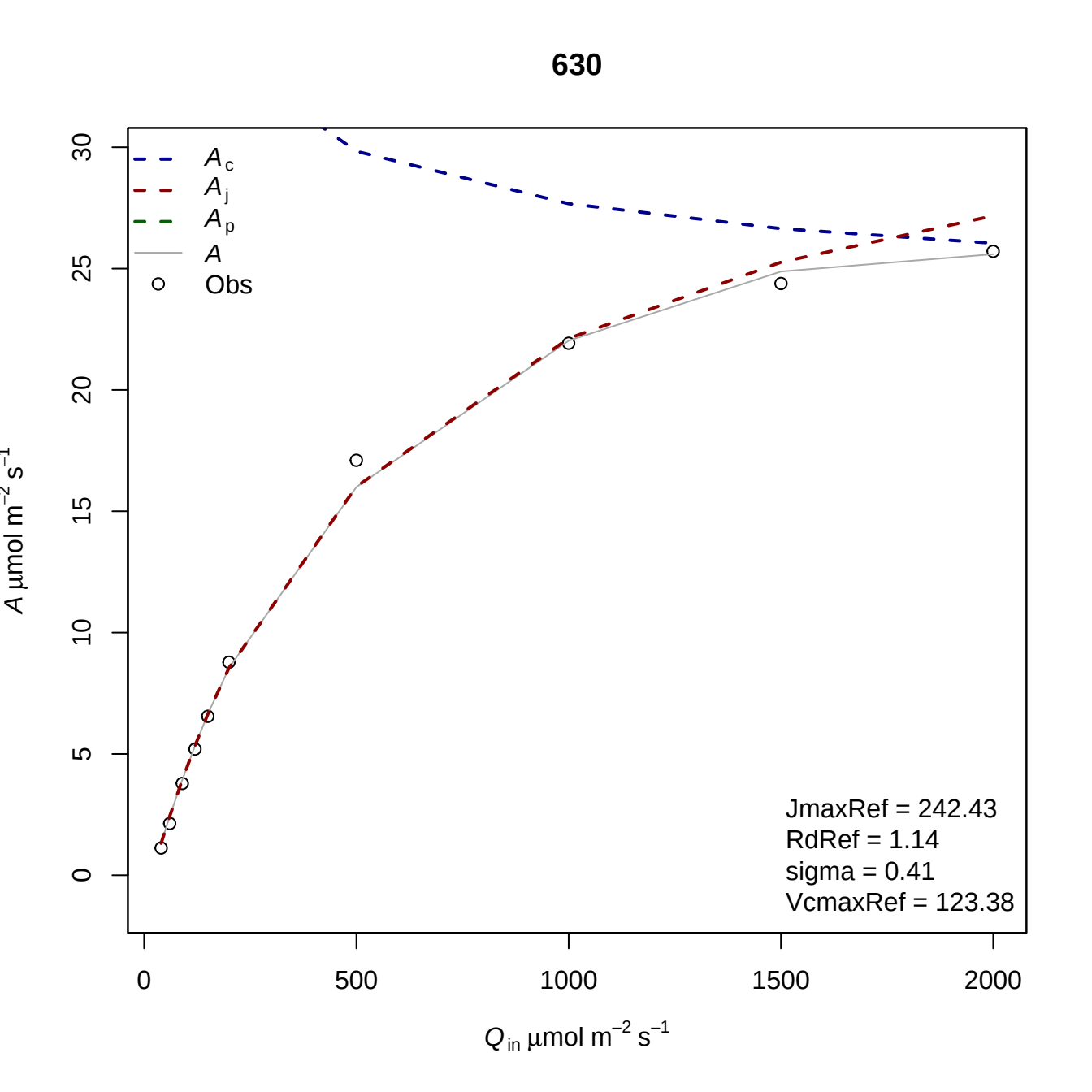

625
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 303.69
RdRef = 1.33
sigma = 0.51
VcmaxRef = 127.01
0
5
10
15
20
25
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

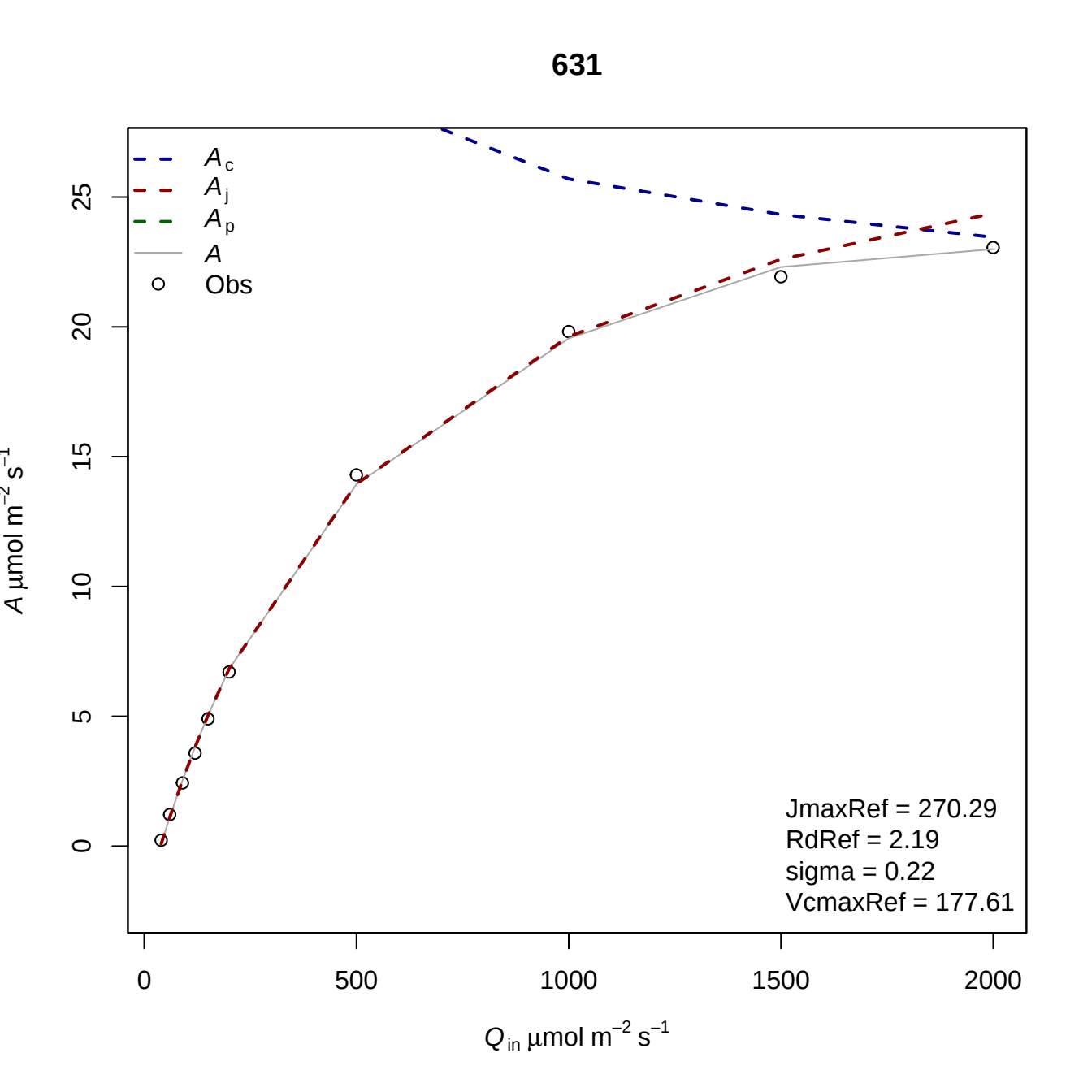

626
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 341.41
RdRef = 1.95
sigma = 0.33
VcmaxRef = 185.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

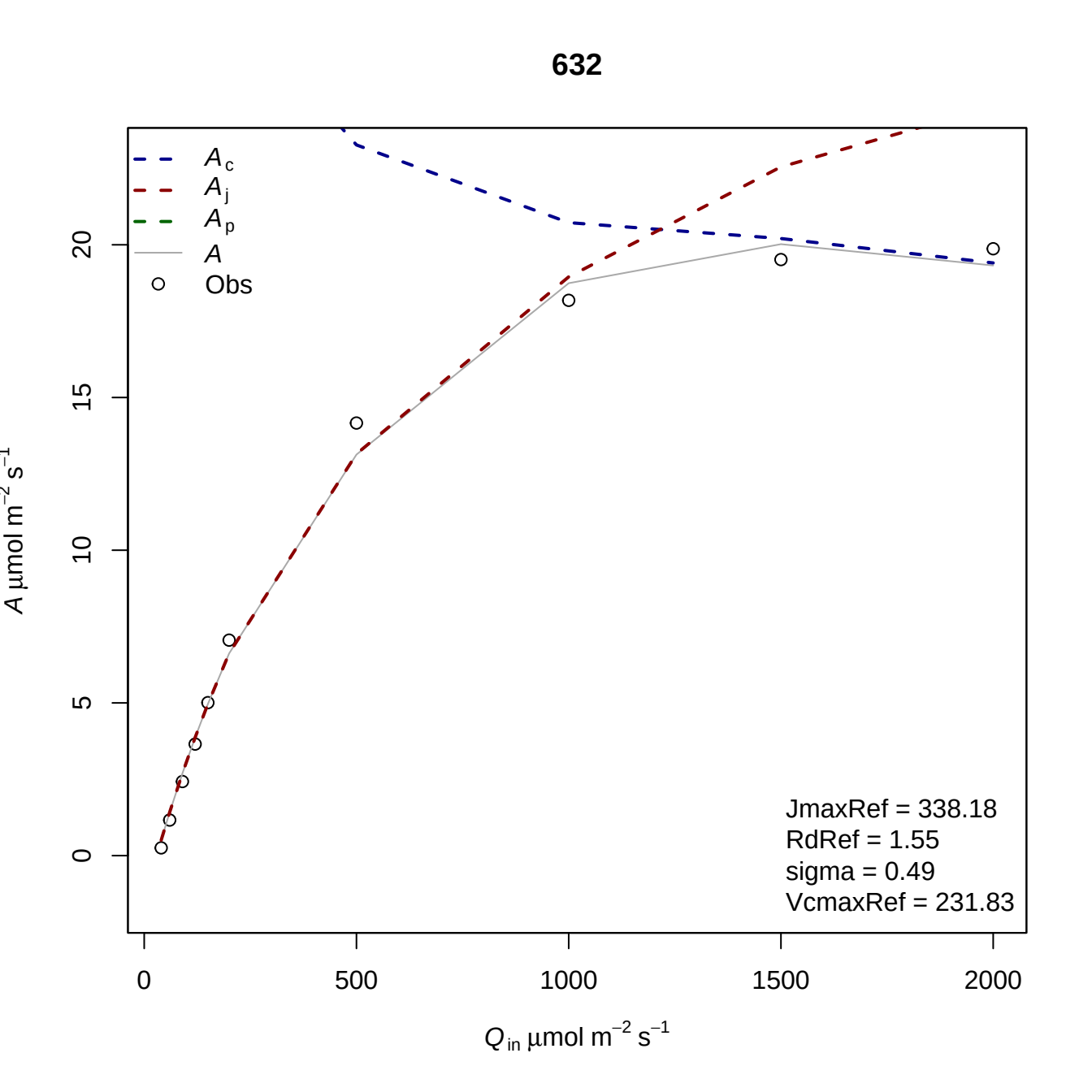

627
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 292.89
RdRef = 3.49
sigma = 0.18
VcmaxRef = 249.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

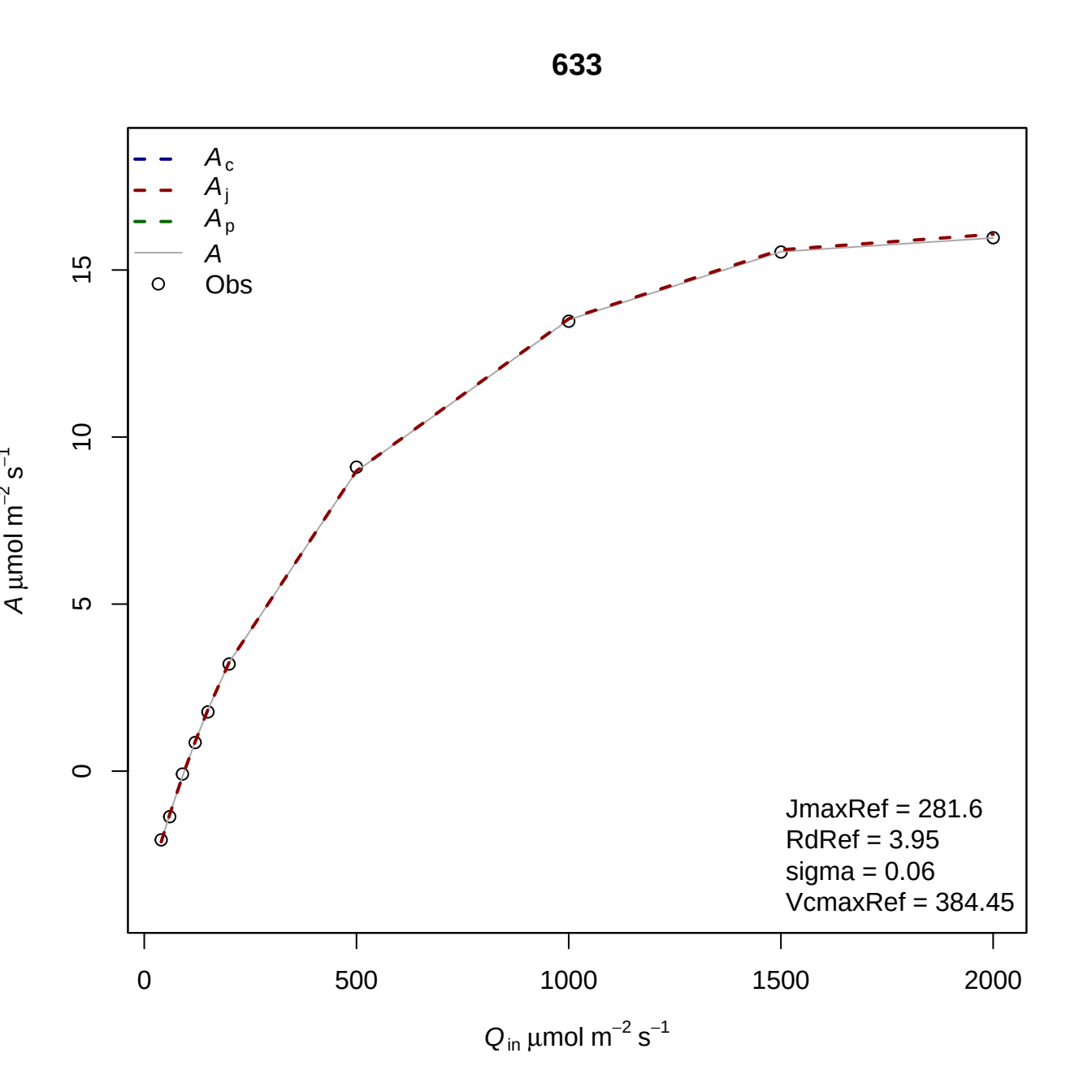

628
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 278
RdRef = 4.27
sigma = 0.09
VcmaxRef = 320.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
20
15
10
5
0
−5
0
500
1000
1500
2000

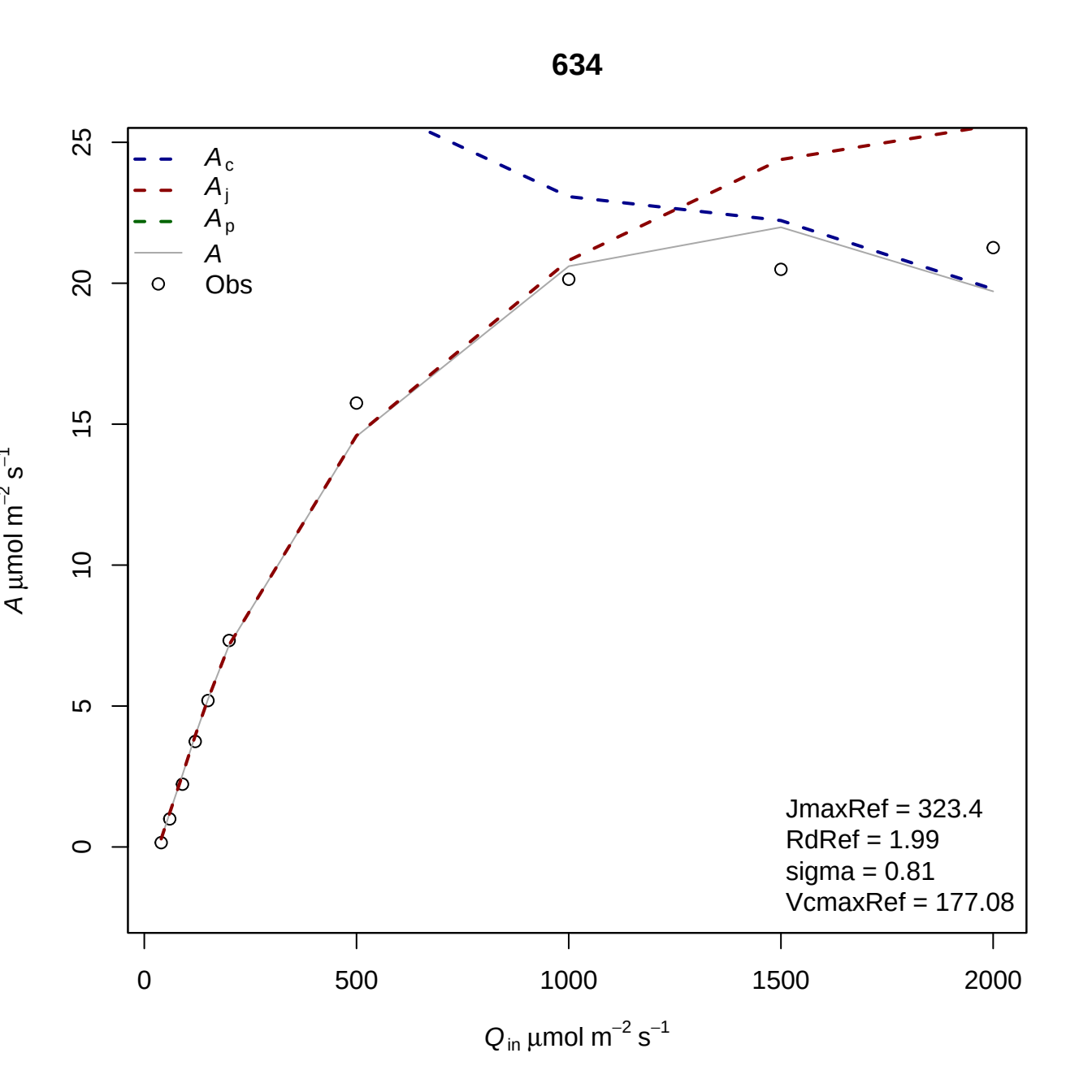

629
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.31
RdRef = 0.26
sigma = 0.33
VcmaxRef = 62.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

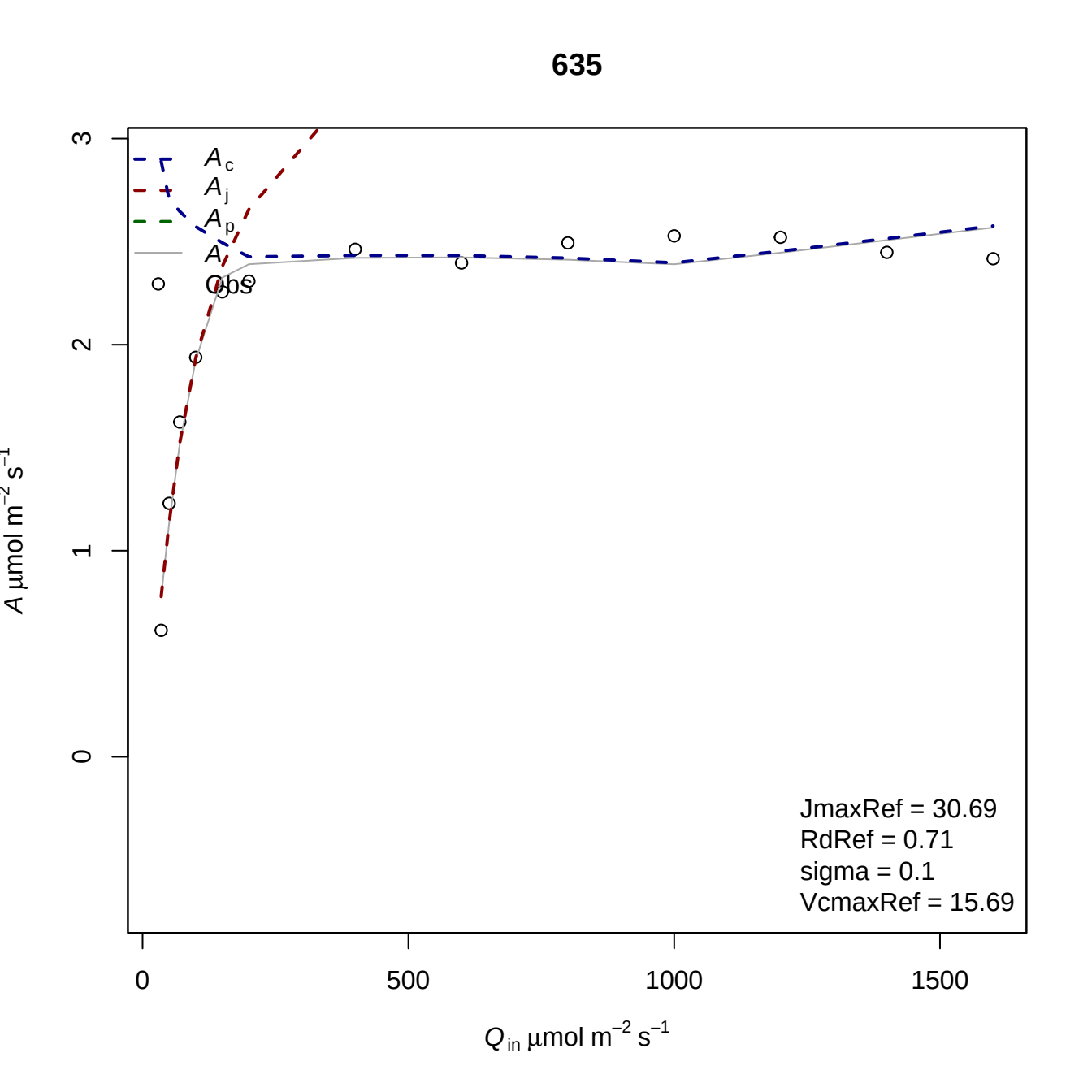

630
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 242.43
RdRef = 1.14
sigma = 0.41
VcmaxRef = 123.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

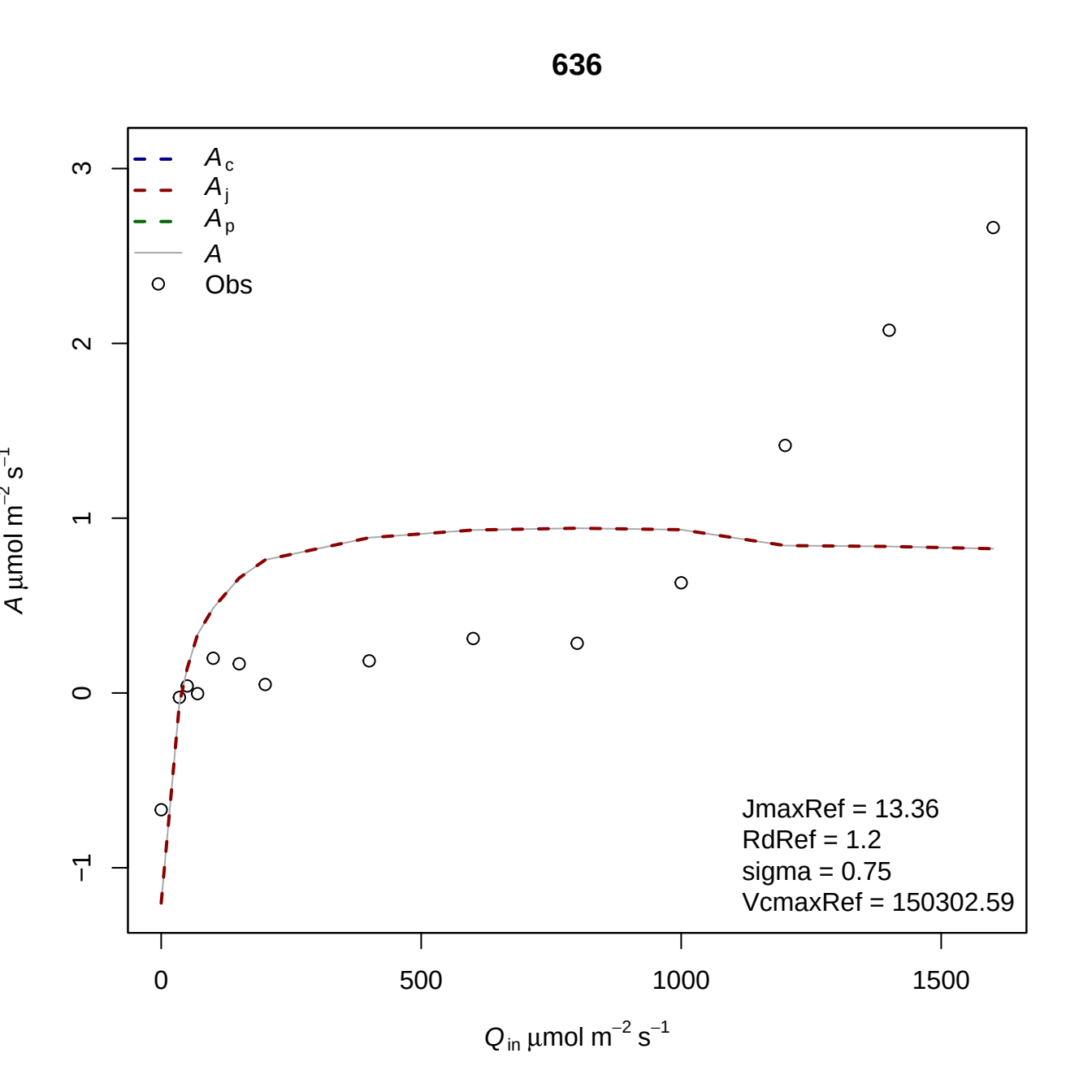

631
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 270.29
RdRef = 2.19
sigma = 0.22
VcmaxRef = 177.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

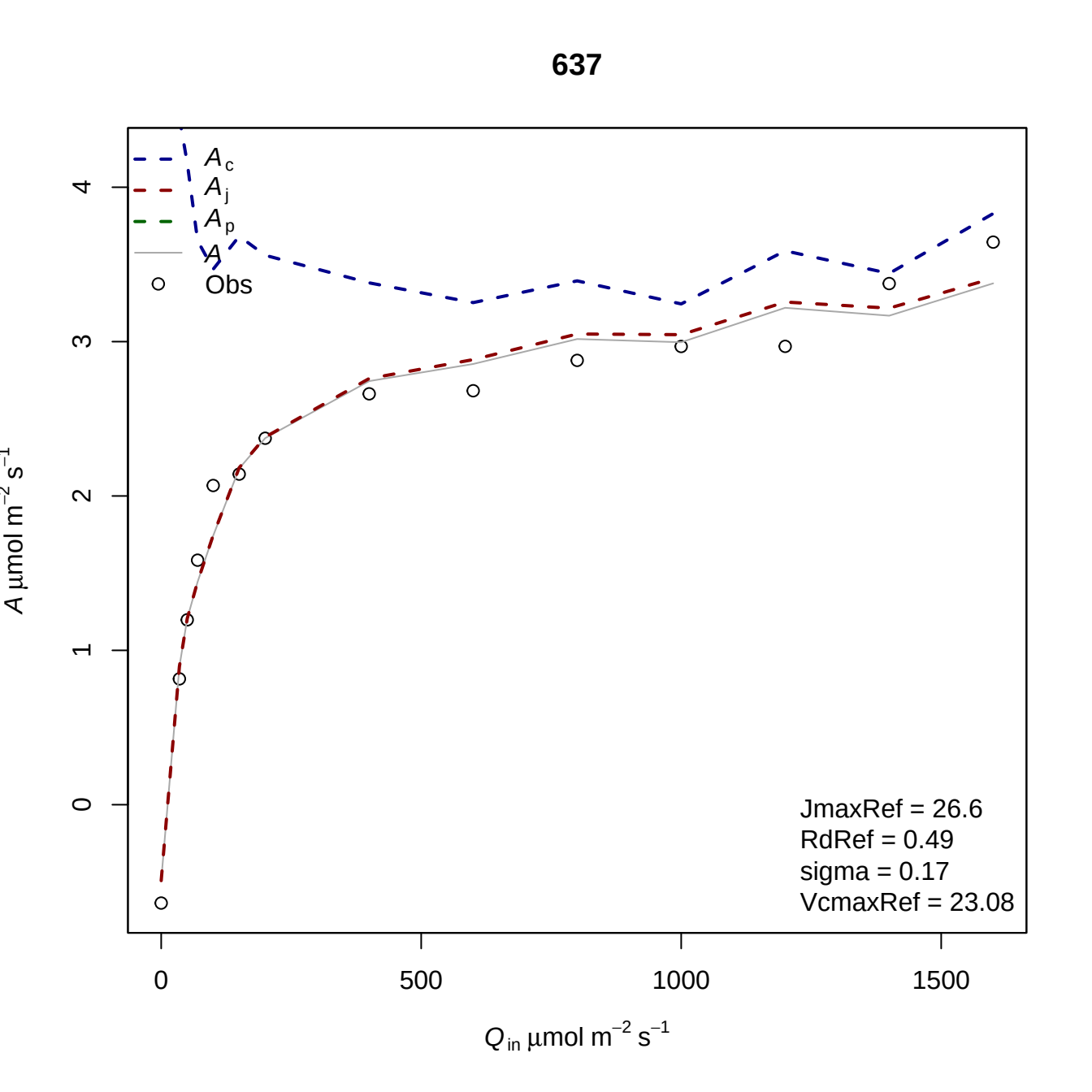

632
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 338.18
RdRef = 1.55
sigma = 0.49
VcmaxRef = 231.83
0
5
10
15
20
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

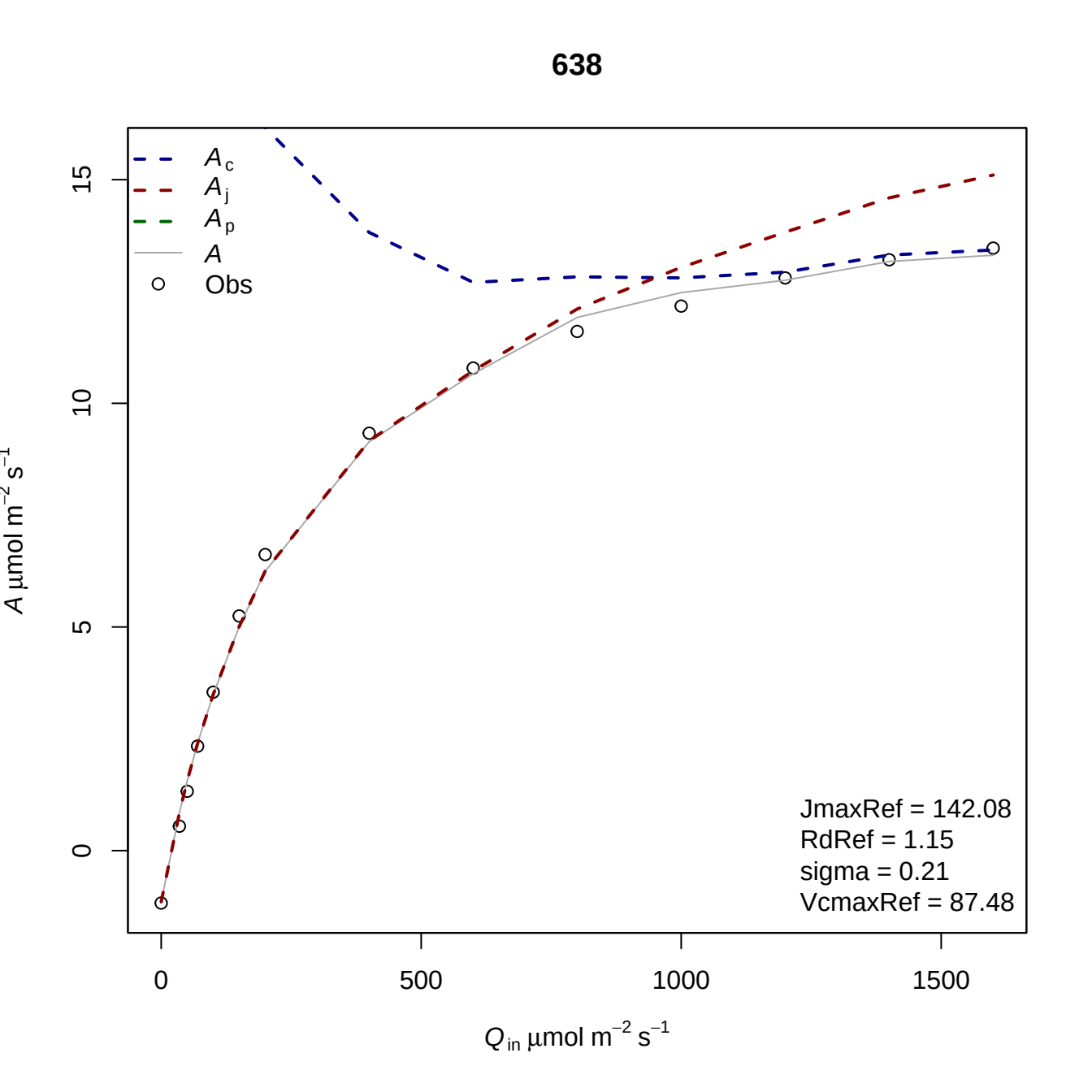

633
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 281.6
RdRef = 3.95
sigma = 0.06
VcmaxRef = 384.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

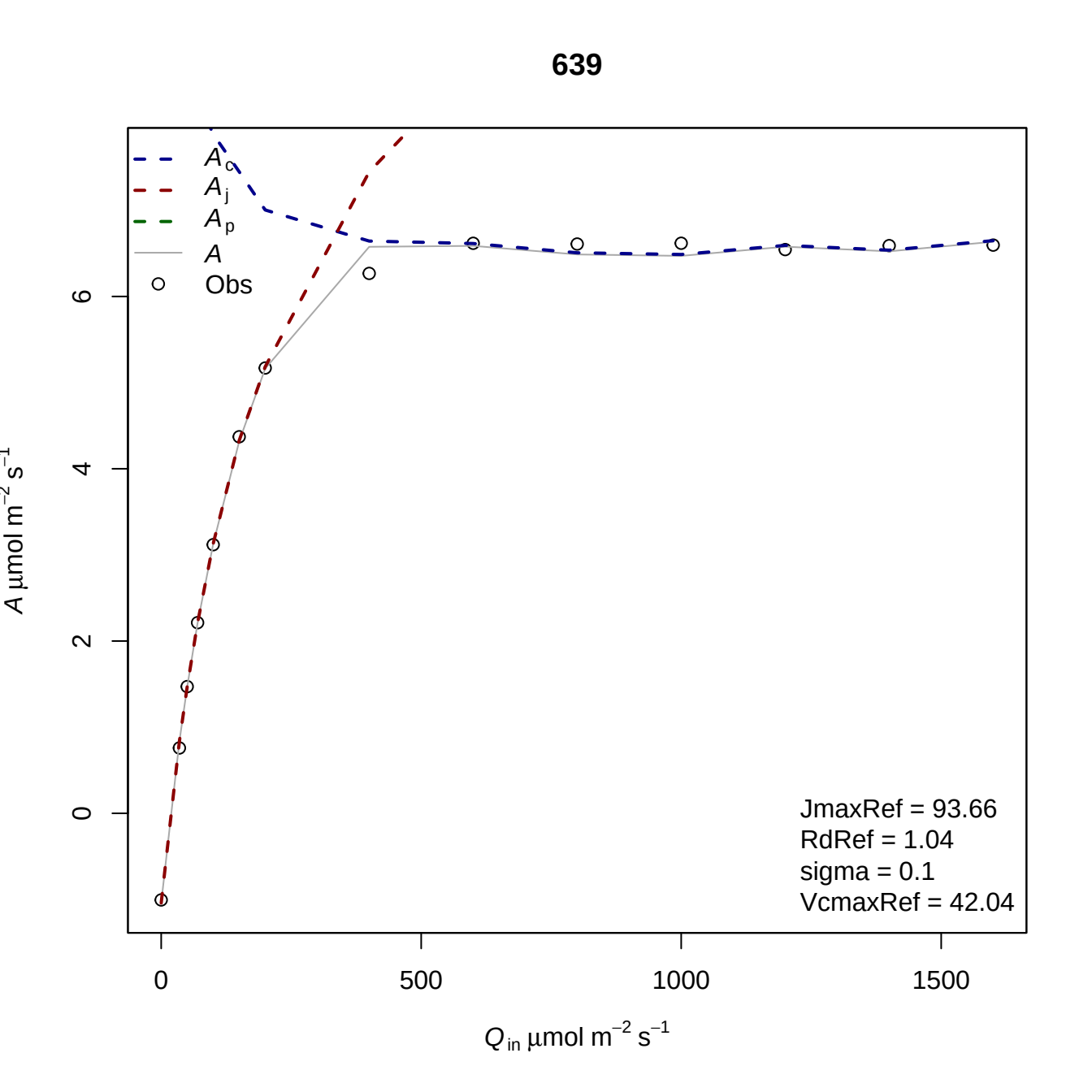

634
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 323.4
RdRef = 1.99
sigma = 0.81
VcmaxRef = 177.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

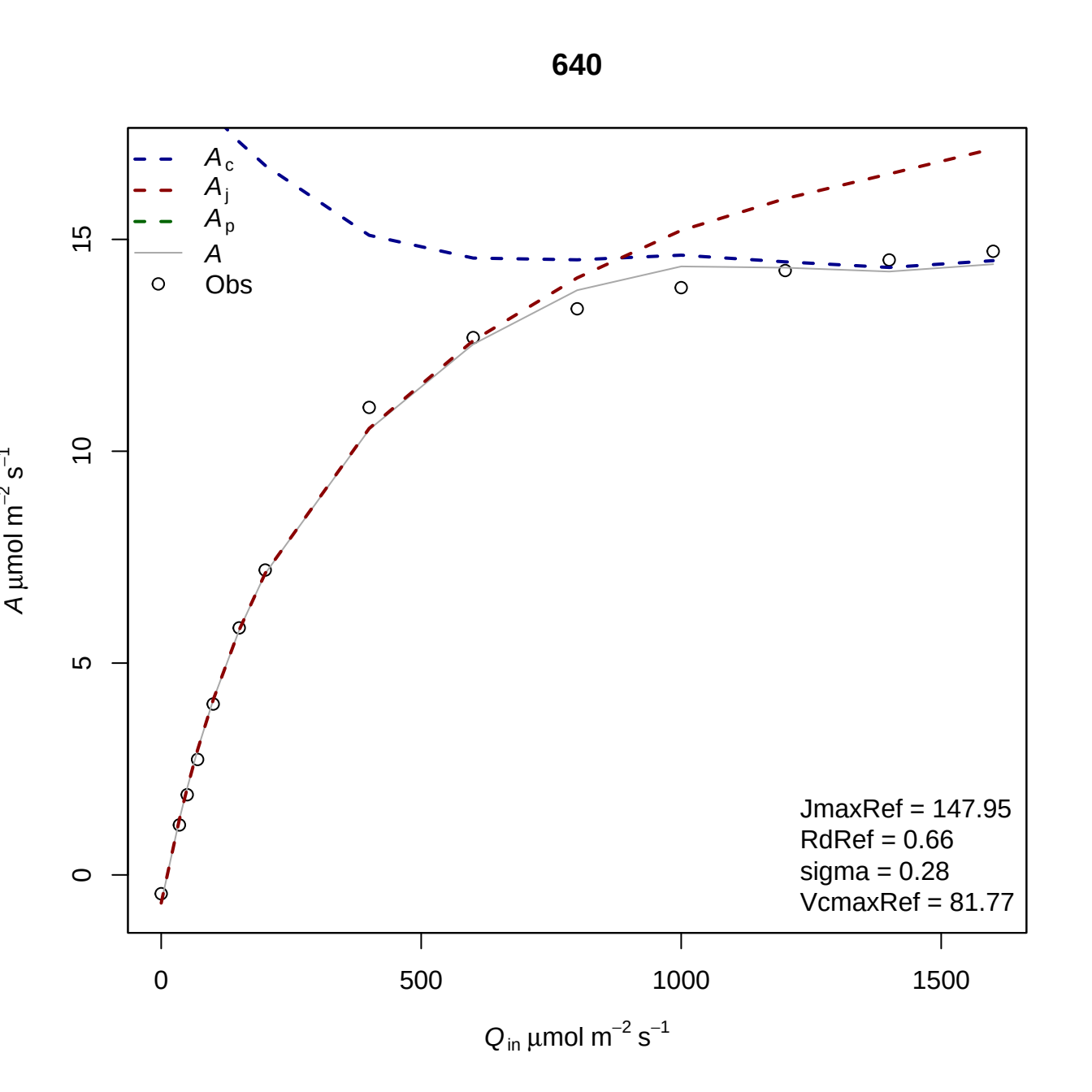

635
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.69
RdRef = 0.71
sigma = 0.1
VcmaxRef = 15.69
0
1
2
3
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

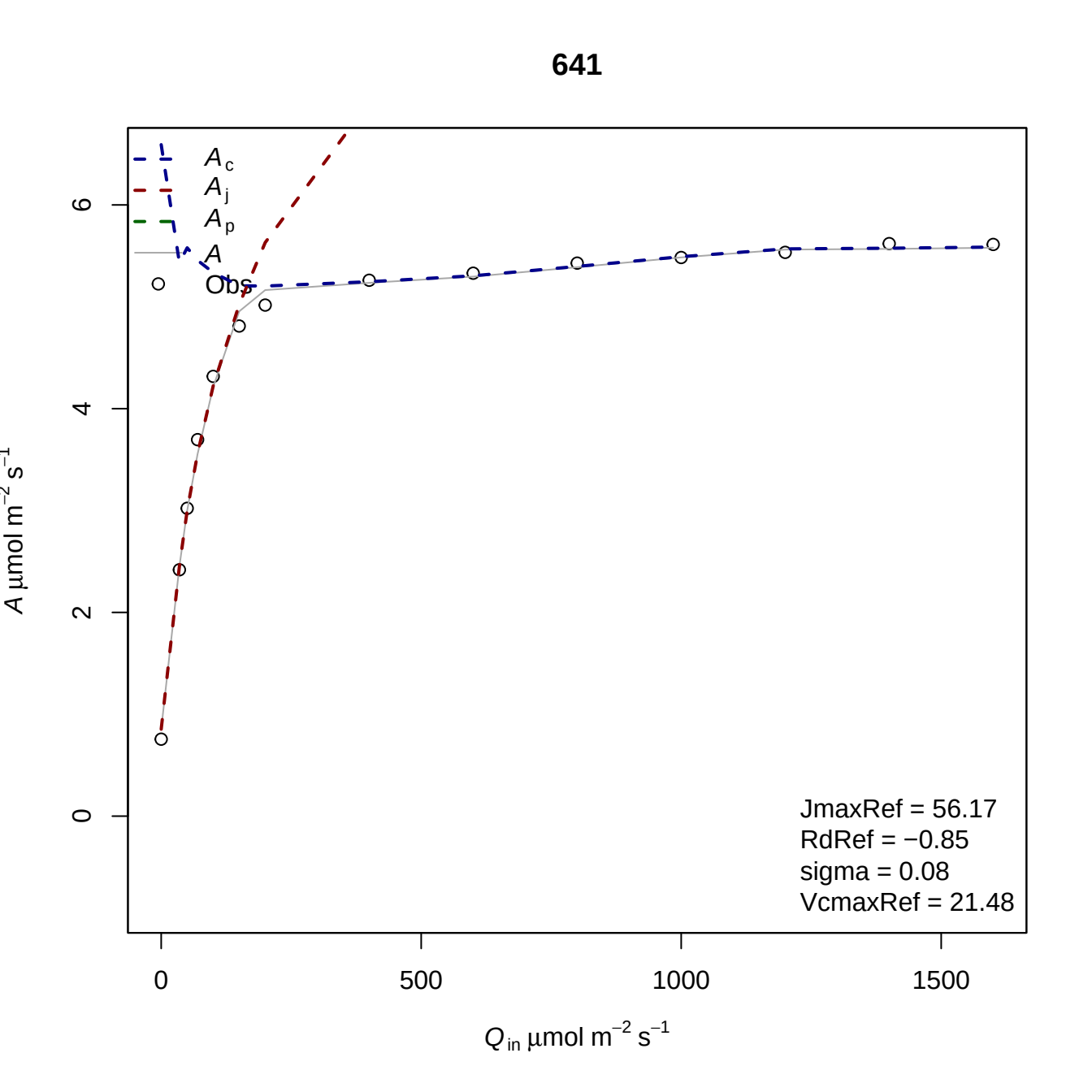

636
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 13.36
RdRef = 1.2
sigma = 0.75
VcmaxRef = 150302.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
0
500
1000
1500

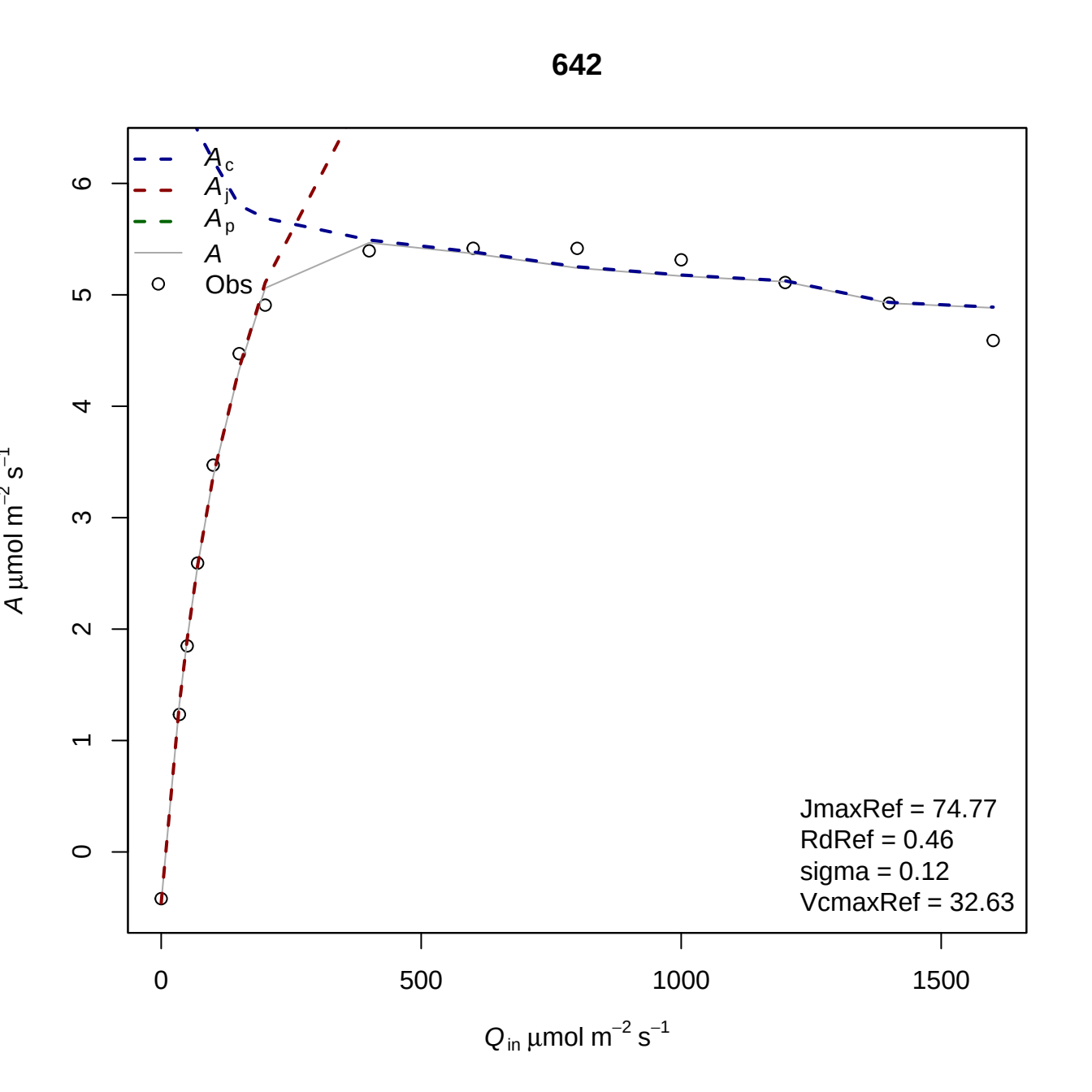

637
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 26.6
RdRef = 0.49
sigma = 0.17
VcmaxRef = 23.08
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

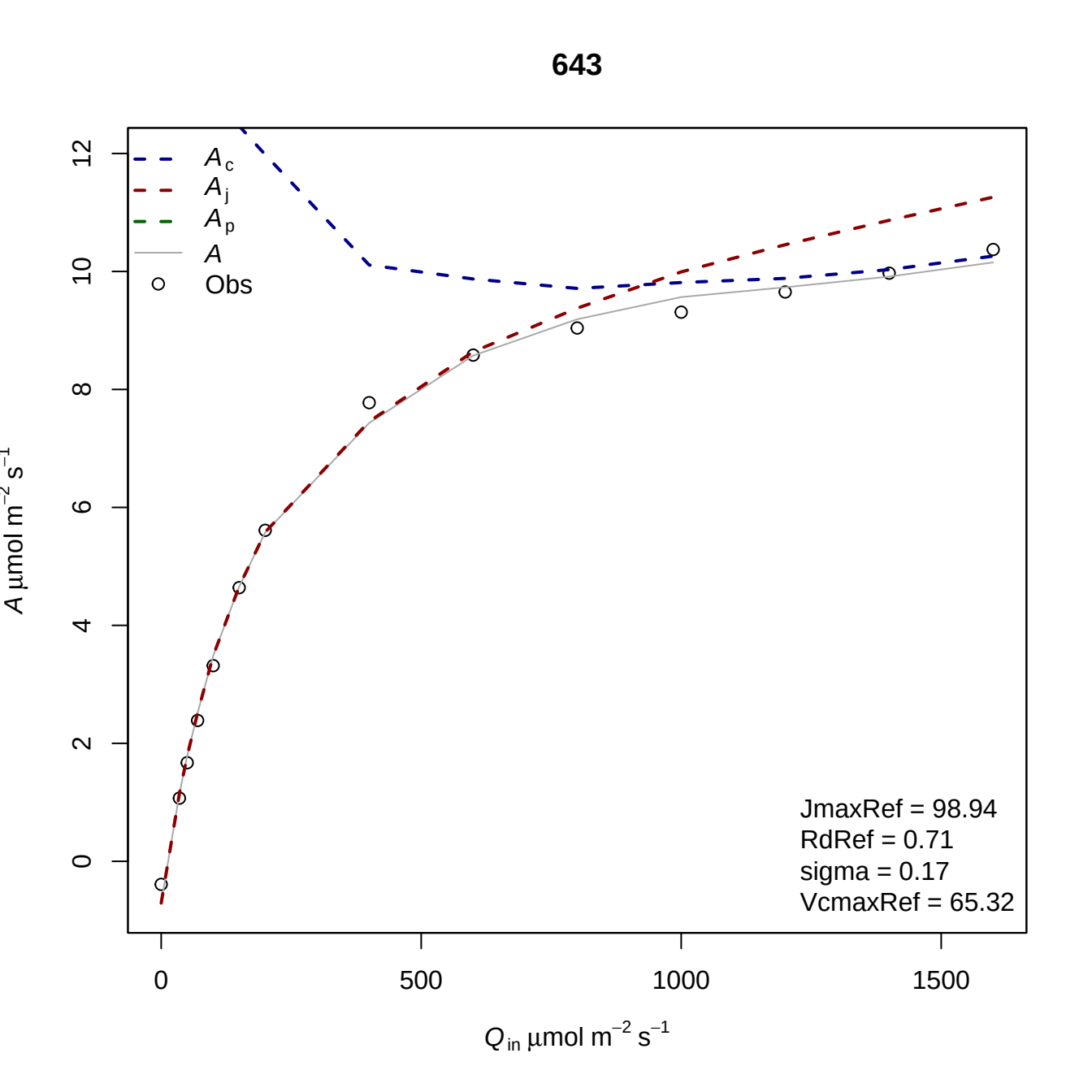

638
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.08
RdRef = 1.15
sigma = 0.21
VcmaxRef = 87.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

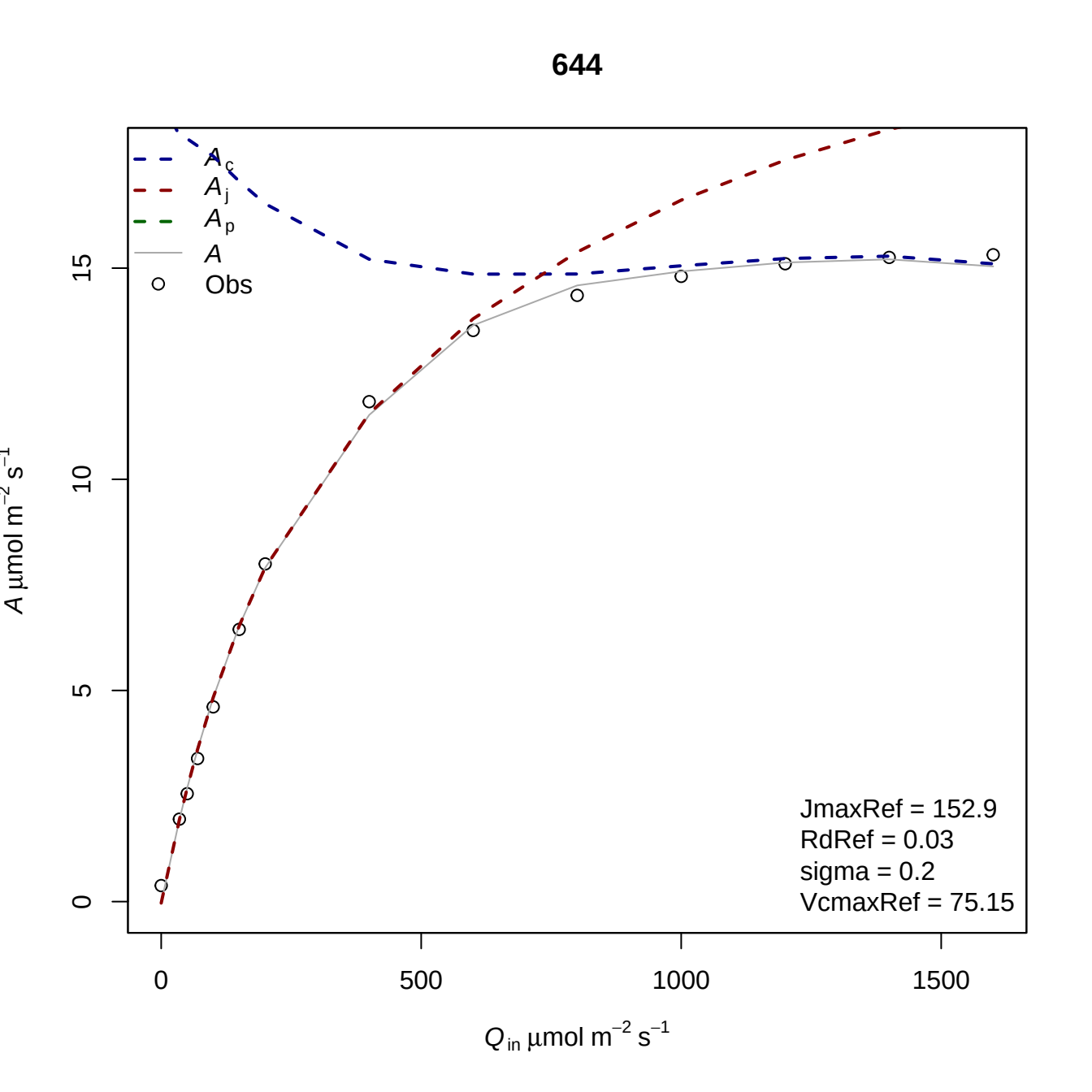

639
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.66
RdRef = 1.04
sigma = 0.1
VcmaxRef = 42.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

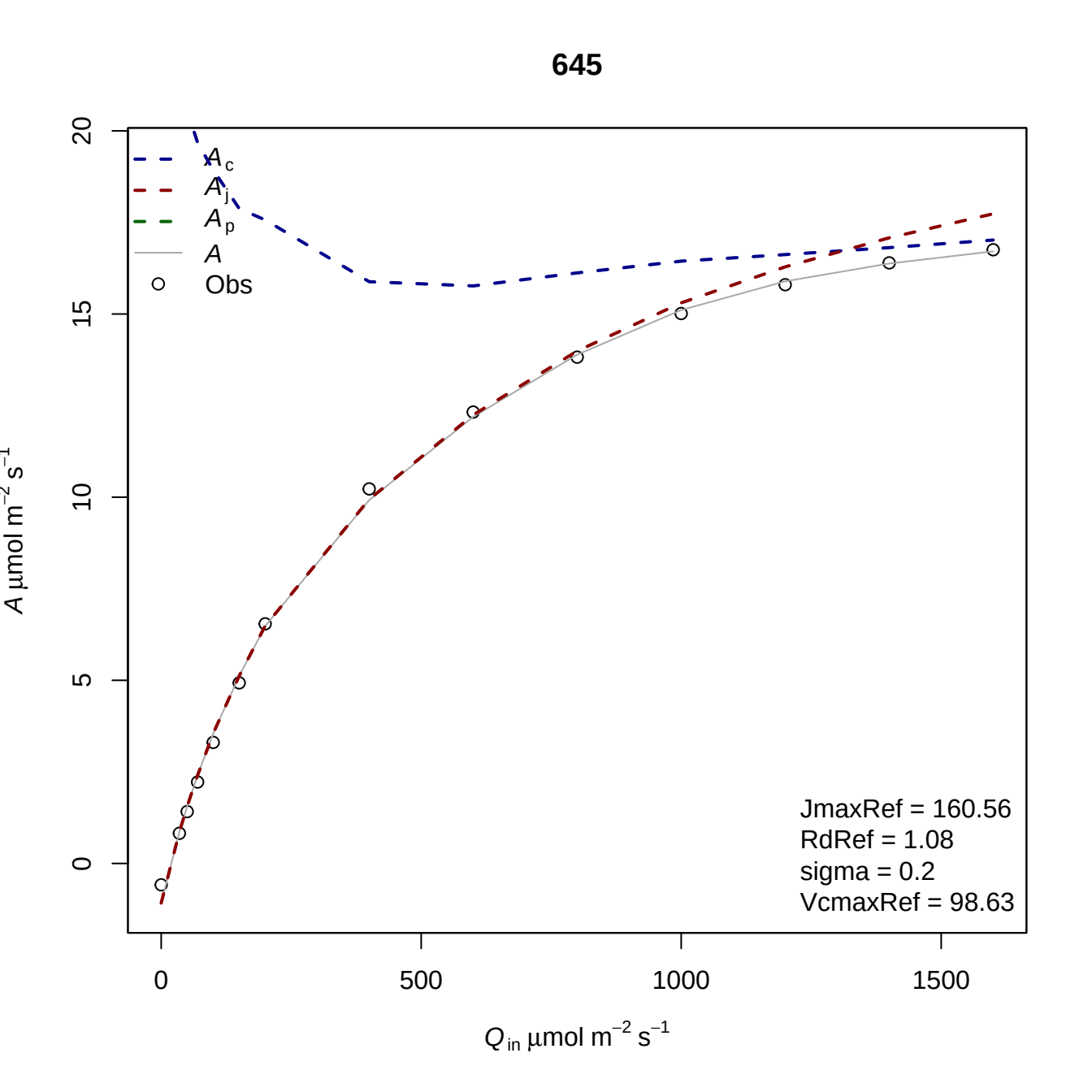

640
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.95
RdRef = 0.66
sigma = 0.28
VcmaxRef = 81.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

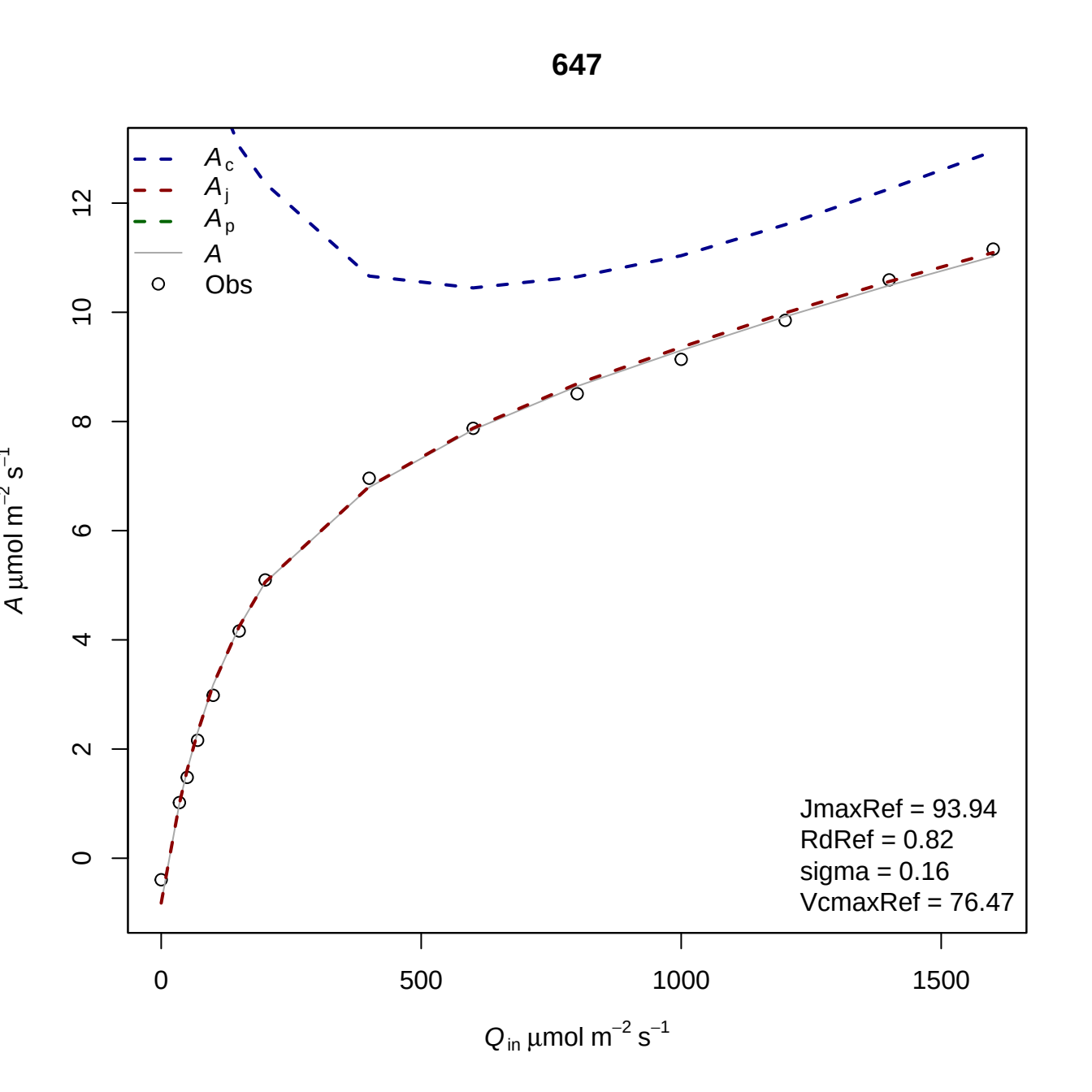

641
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.17
RdRef = −0.85
sigma = 0.08
VcmaxRef = 21.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

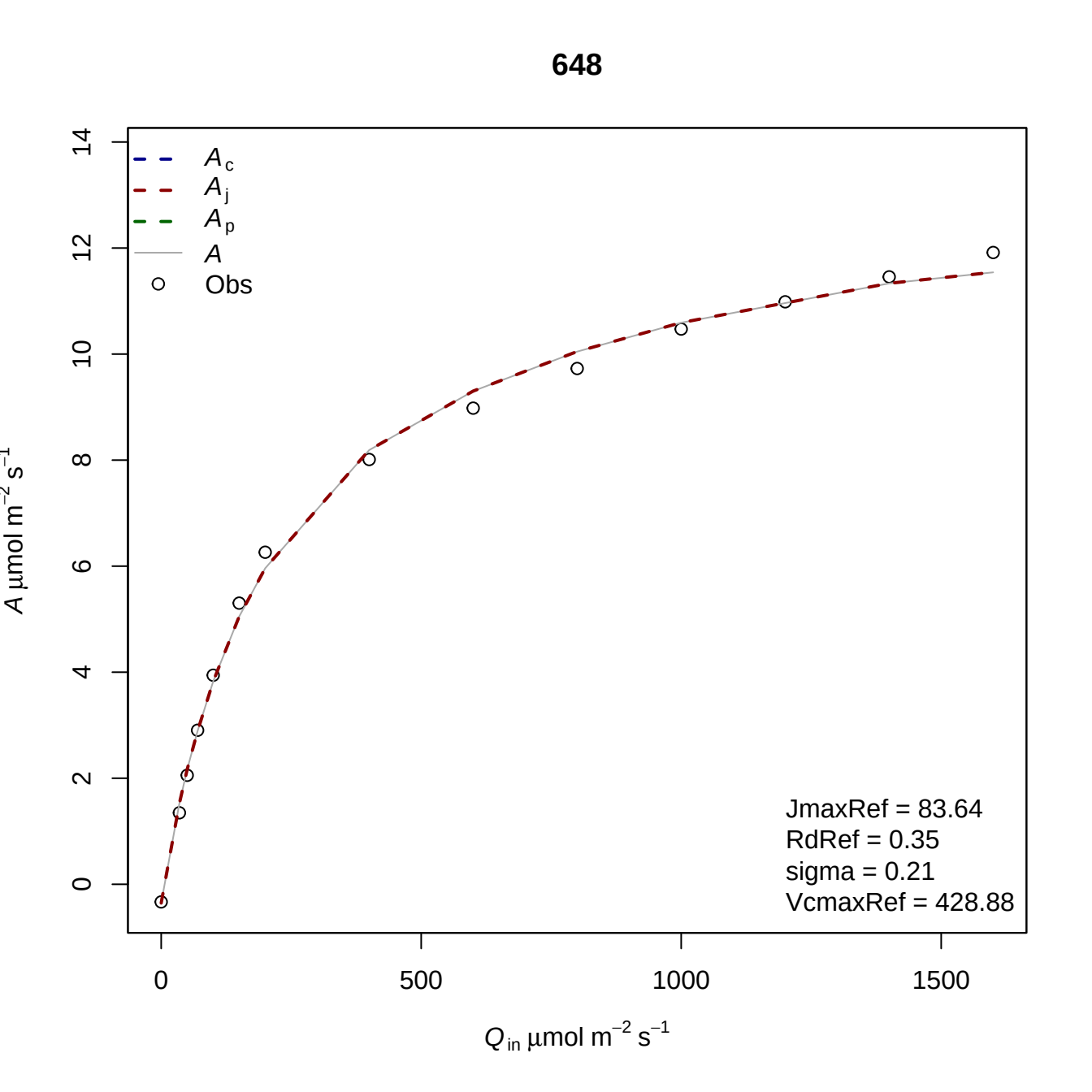

642
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.77
RdRef = 0.46
sigma = 0.12
VcmaxRef = 32.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

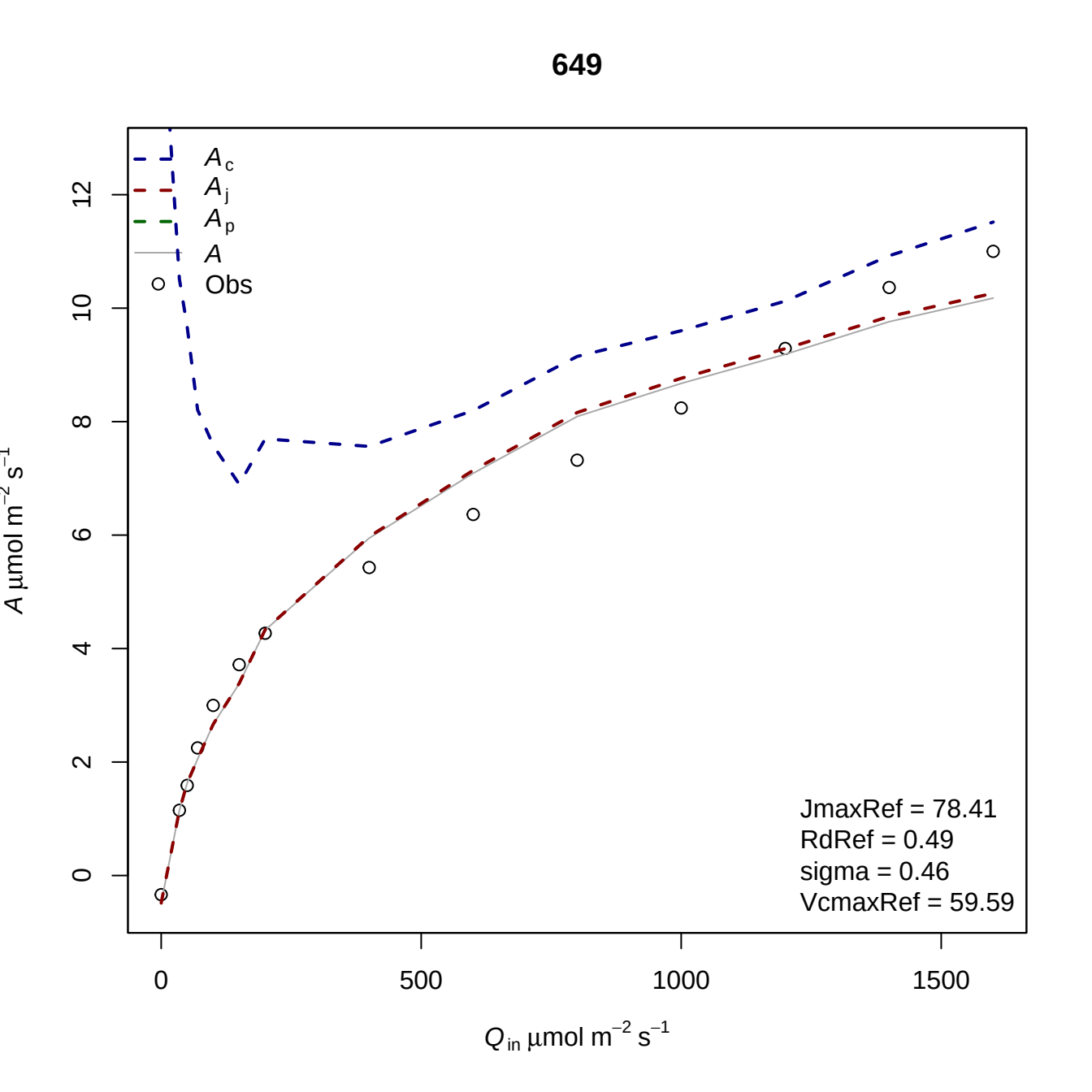

643
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.94
RdRef = 0.71
sigma = 0.17
VcmaxRef = 65.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

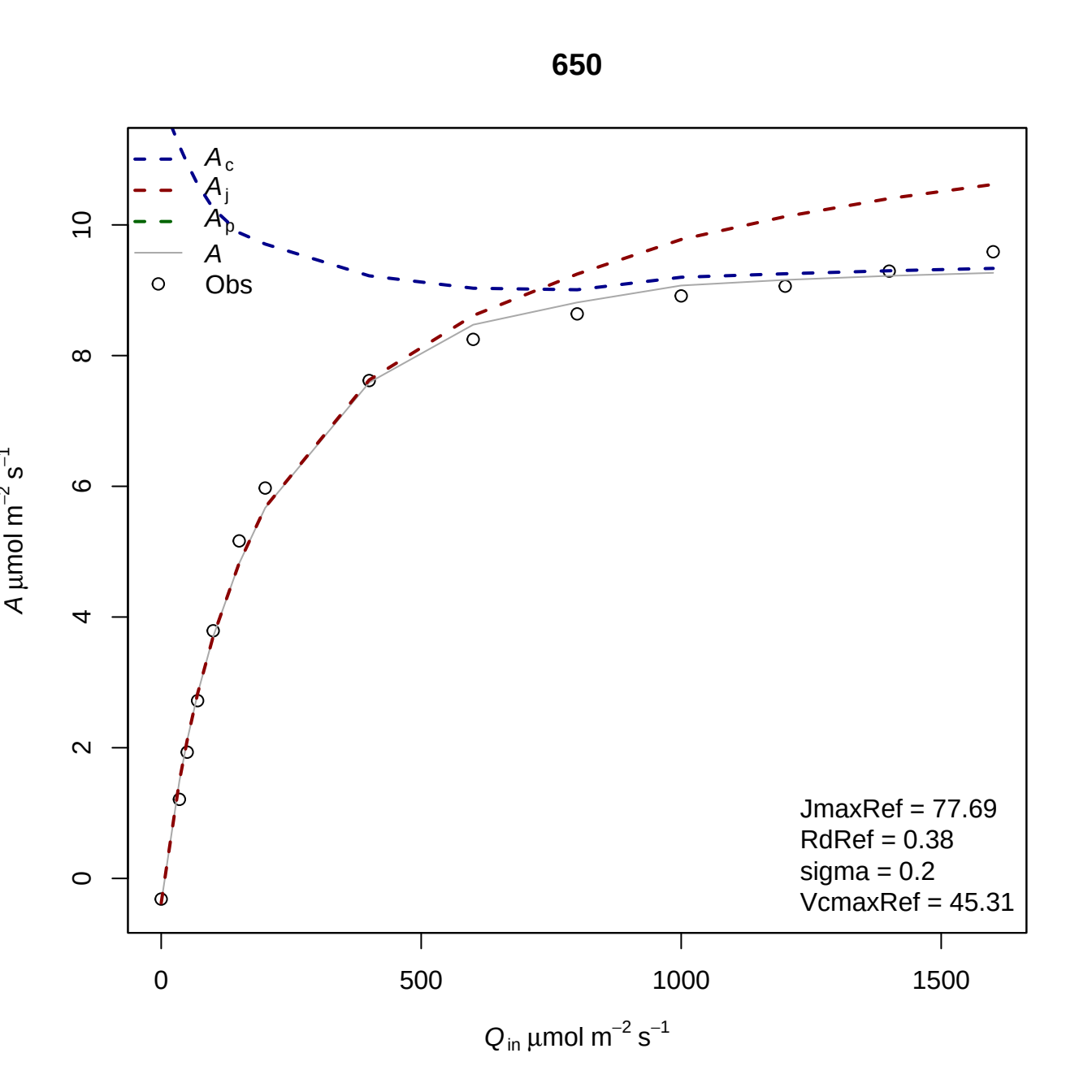

644
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.9
RdRef = 0.03
sigma = 0.2
VcmaxRef = 75.15
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

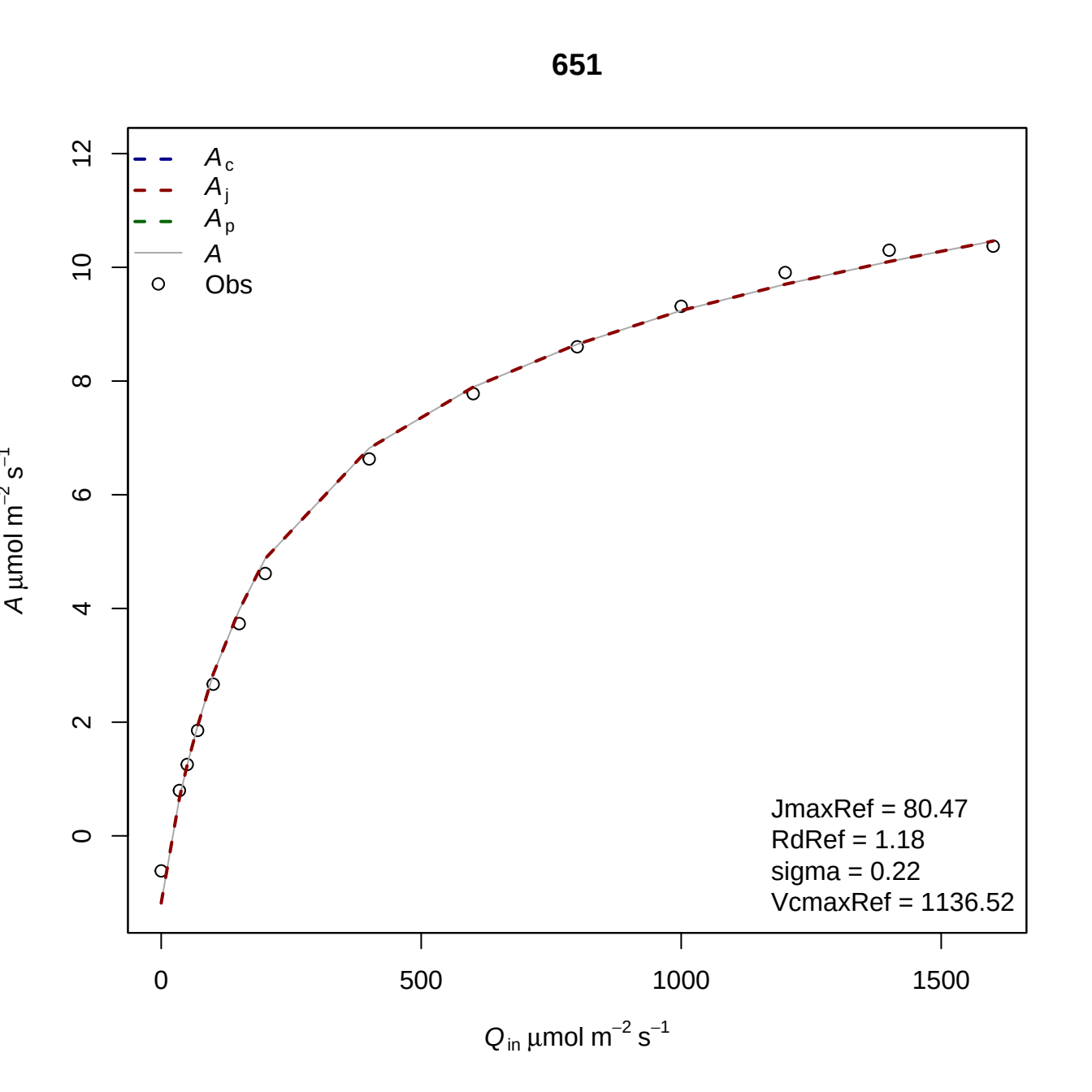

645
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.56
RdRef = 1.08
sigma = 0.2
VcmaxRef = 98.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

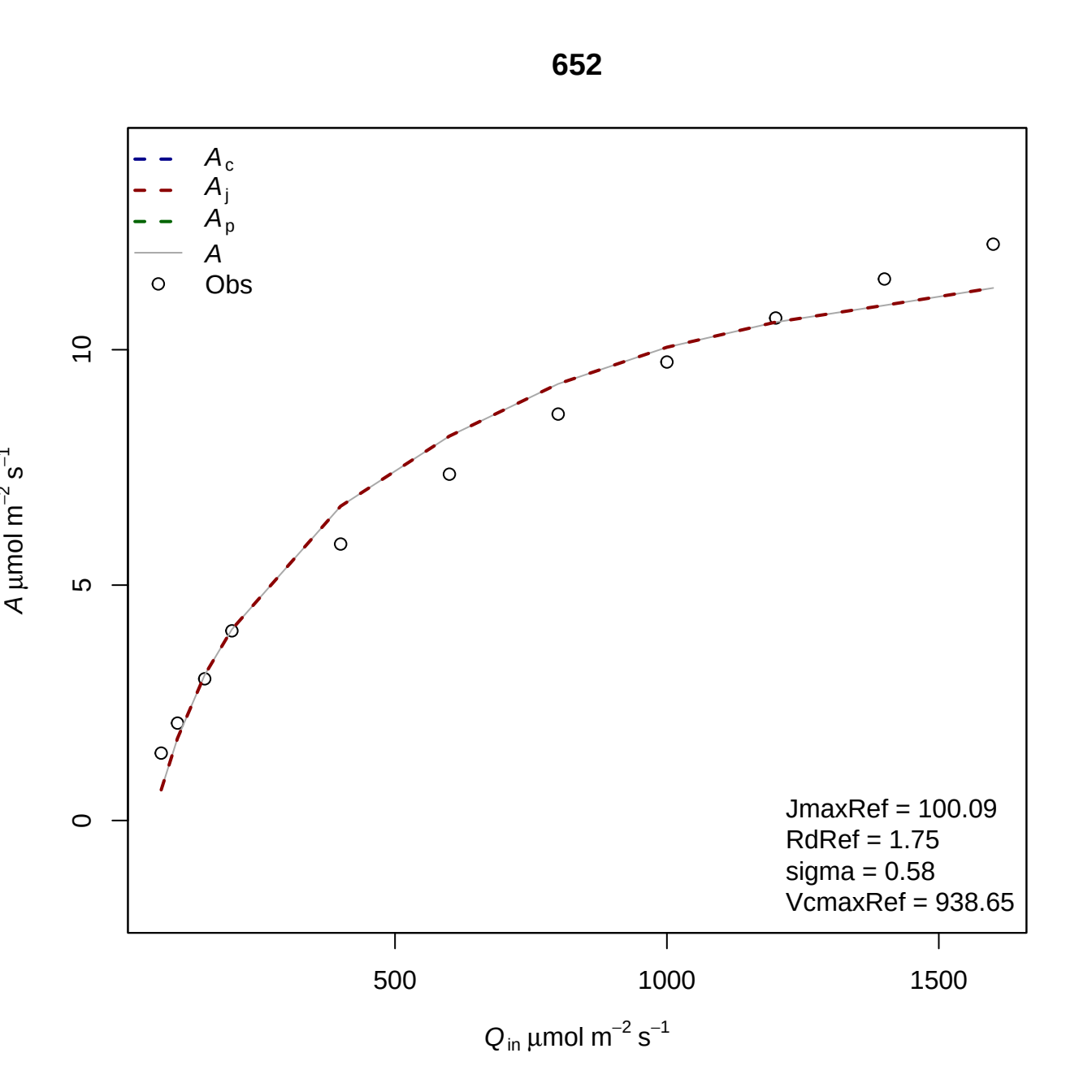

647
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.94
RdRef = 0.82
sigma = 0.16
VcmaxRef = 76.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

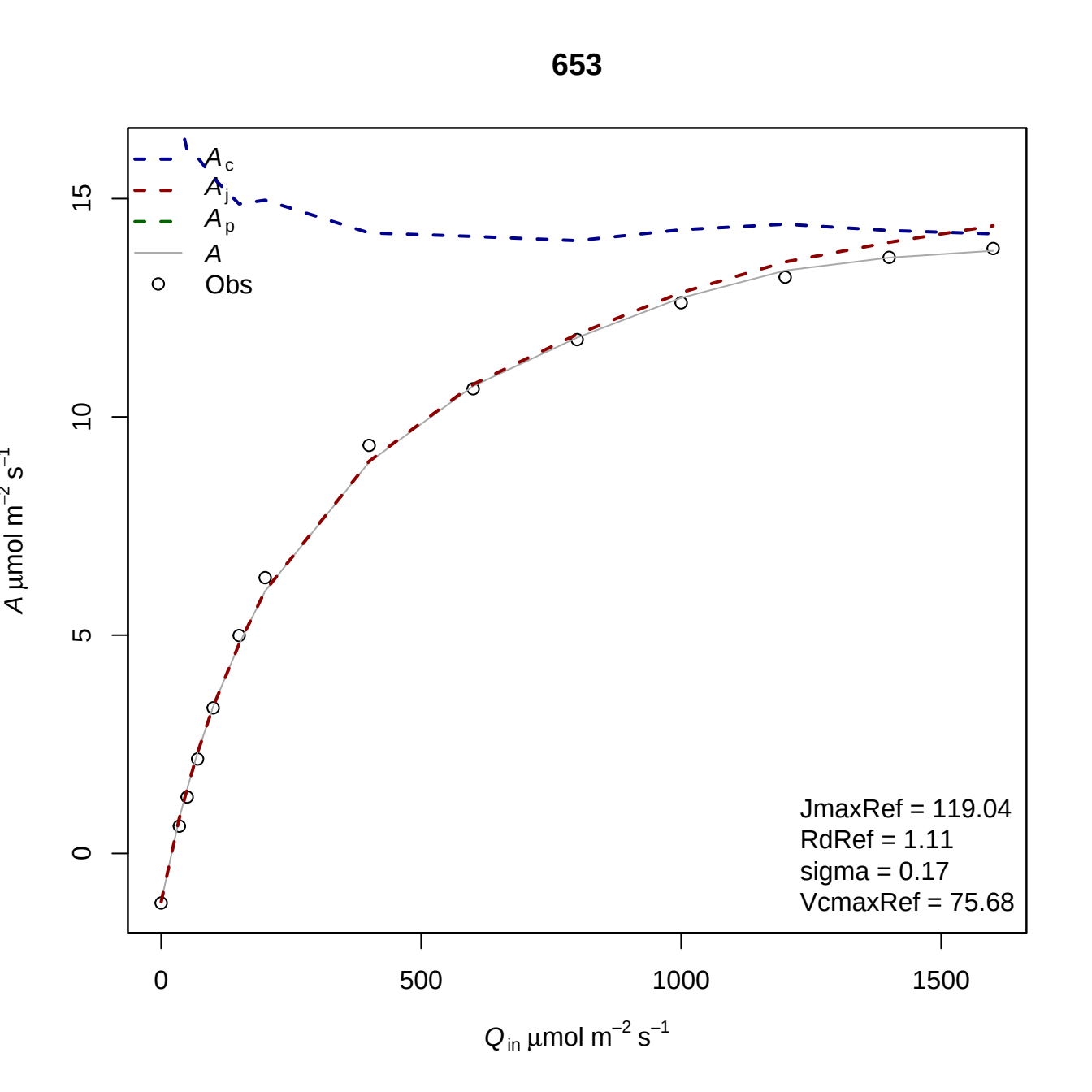

648
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.64
RdRef = 0.35
sigma = 0.21
VcmaxRef = 428.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

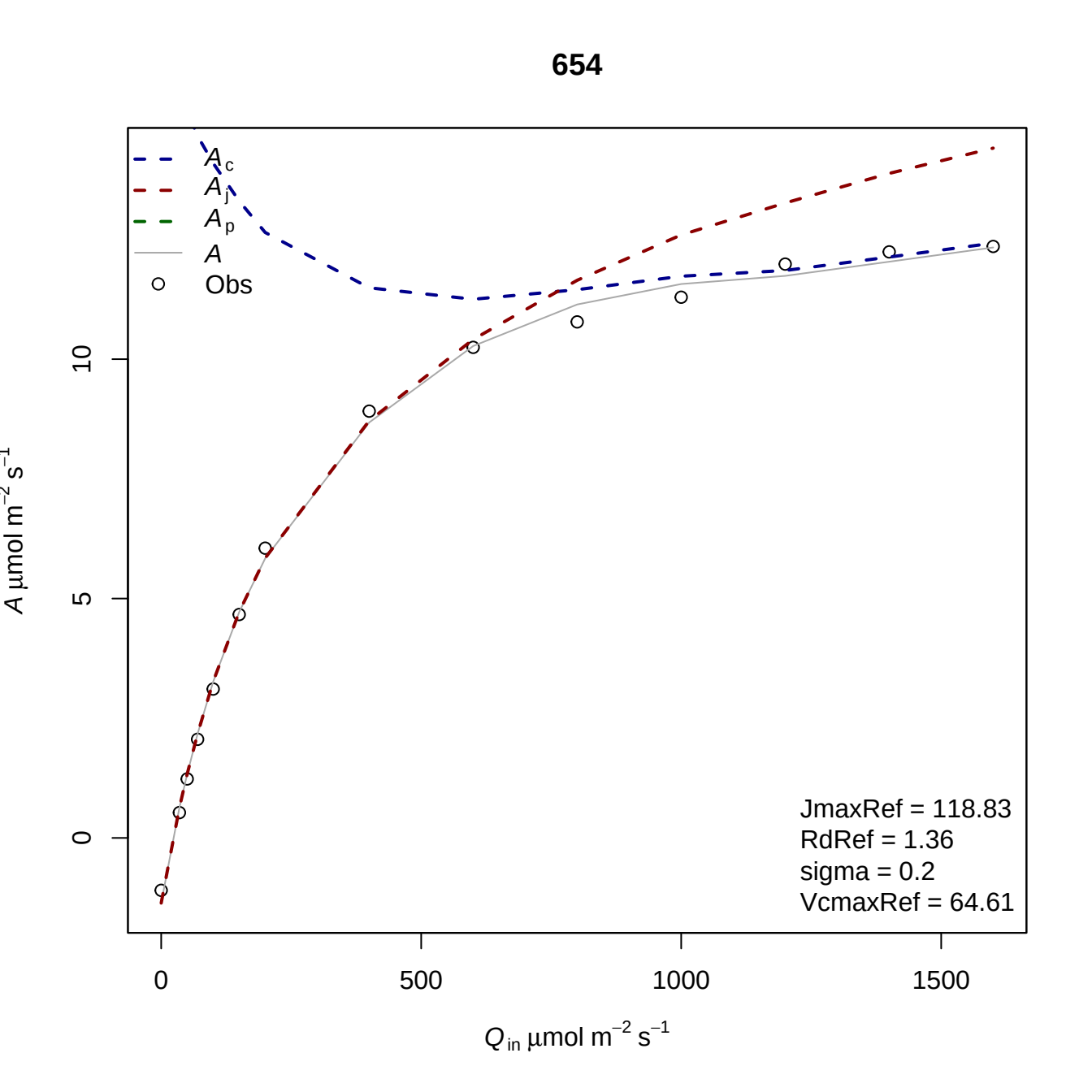

649
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.41
RdRef = 0.49
sigma = 0.46
VcmaxRef = 59.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

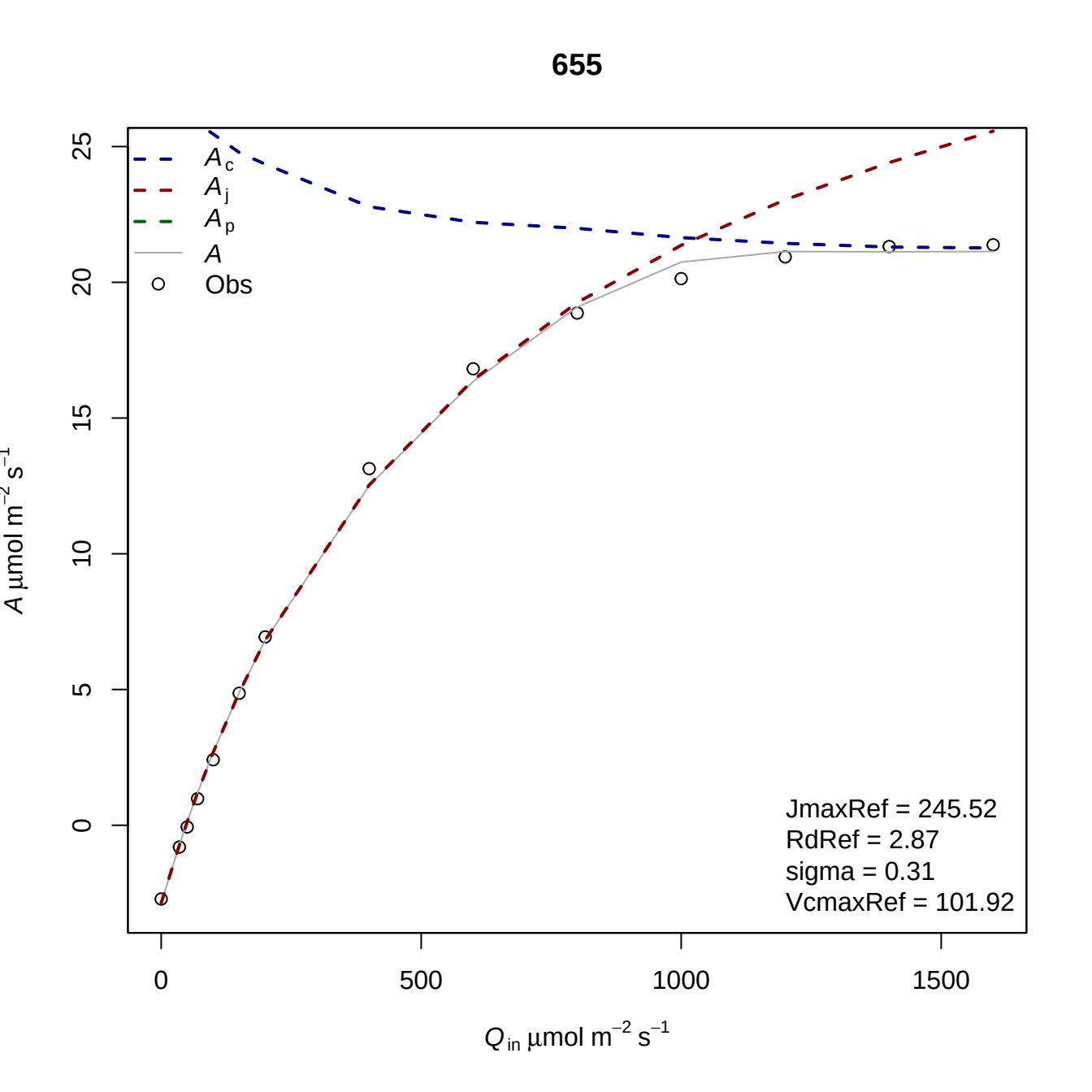

650
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.69
RdRef = 0.38
sigma = 0.2
VcmaxRef = 45.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

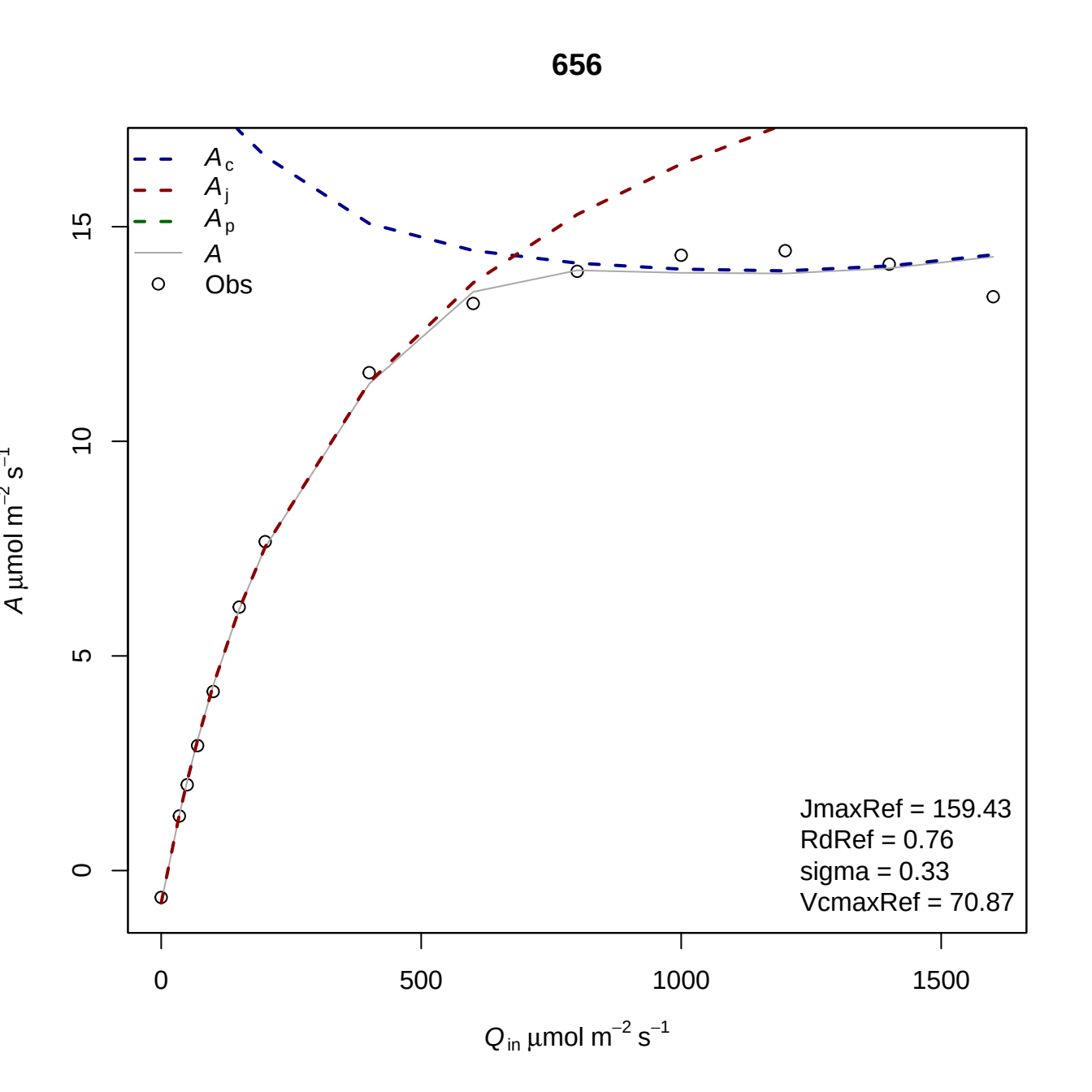

651
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.47
RdRef = 1.18
sigma = 0.22
VcmaxRef = 1136.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

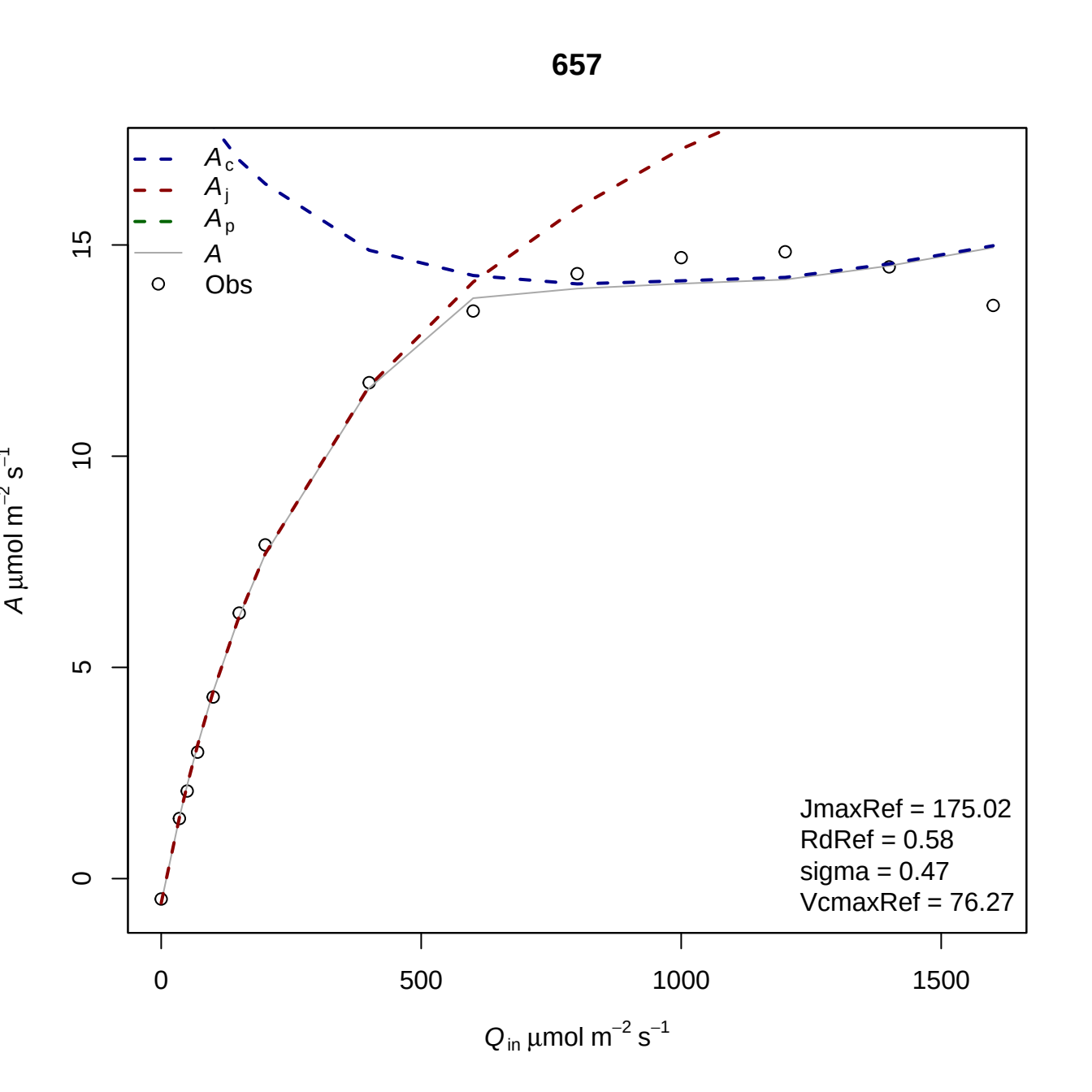

652
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.09
RdRef = 1.75
sigma = 0.58
VcmaxRef = 938.65
0
5
10
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

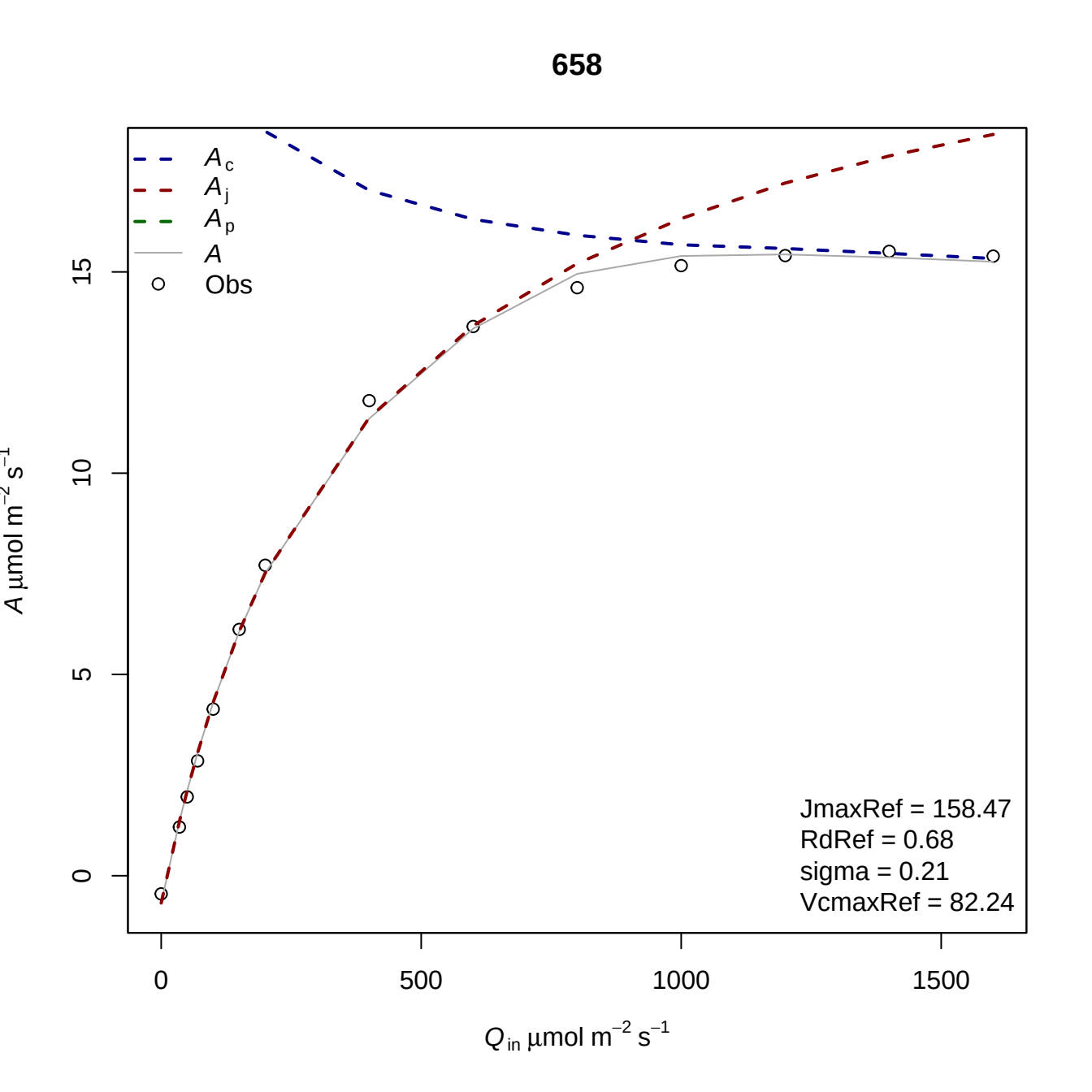

653
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.04
RdRef = 1.11
sigma = 0.17
VcmaxRef = 75.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

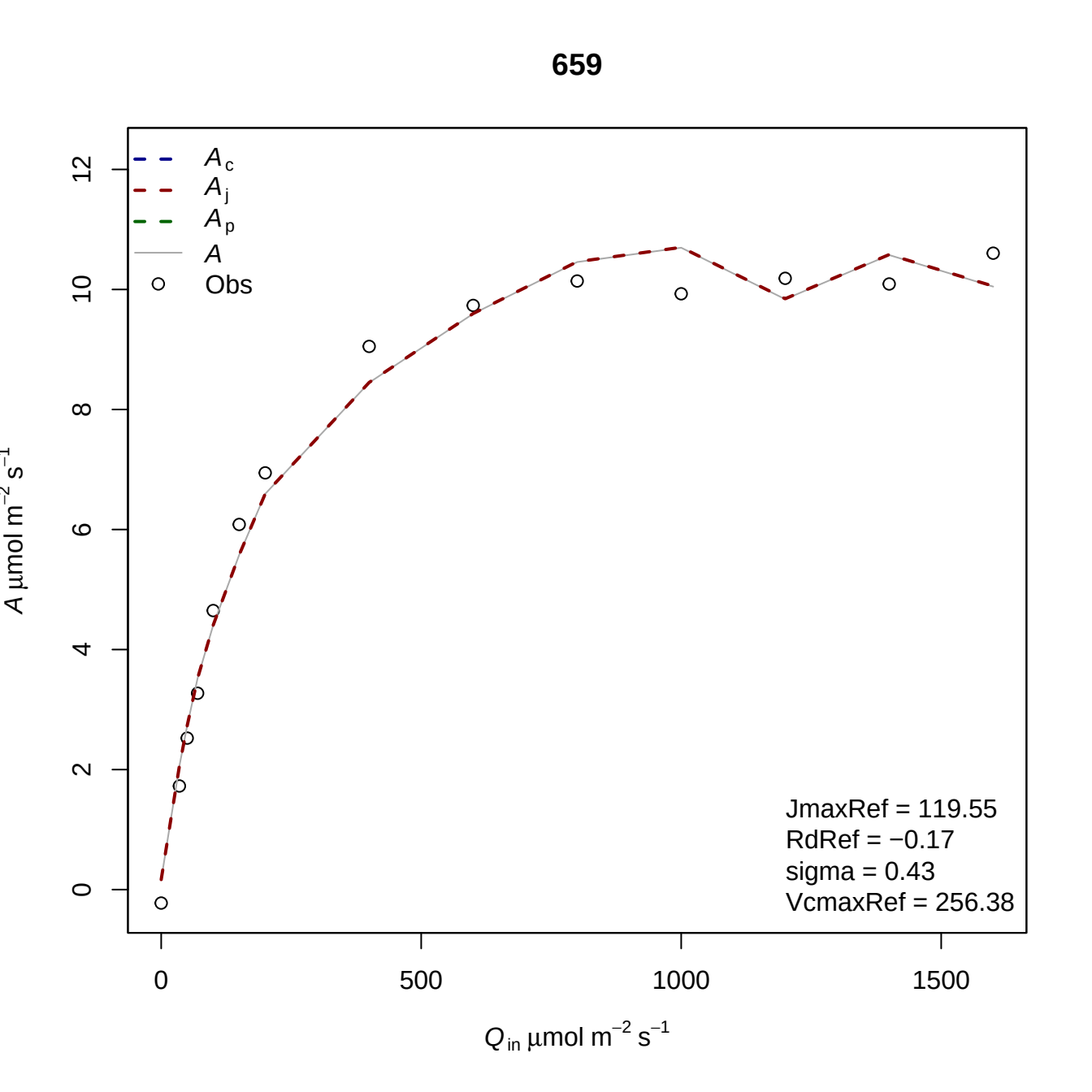

654
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.83
RdRef = 1.36
sigma = 0.2
VcmaxRef = 64.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

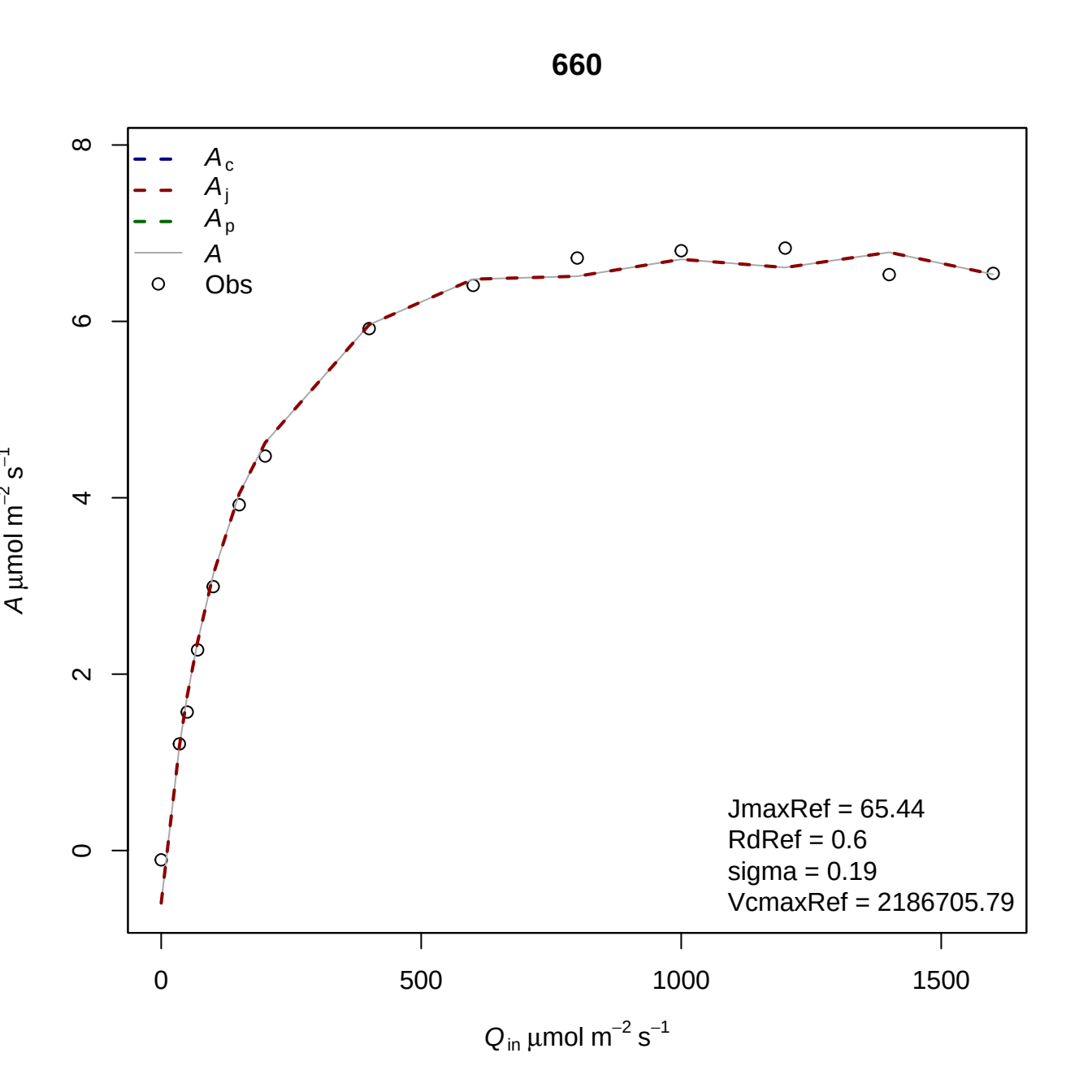

655
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 245.52
RdRef = 2.87
sigma = 0.31
VcmaxRef = 101.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

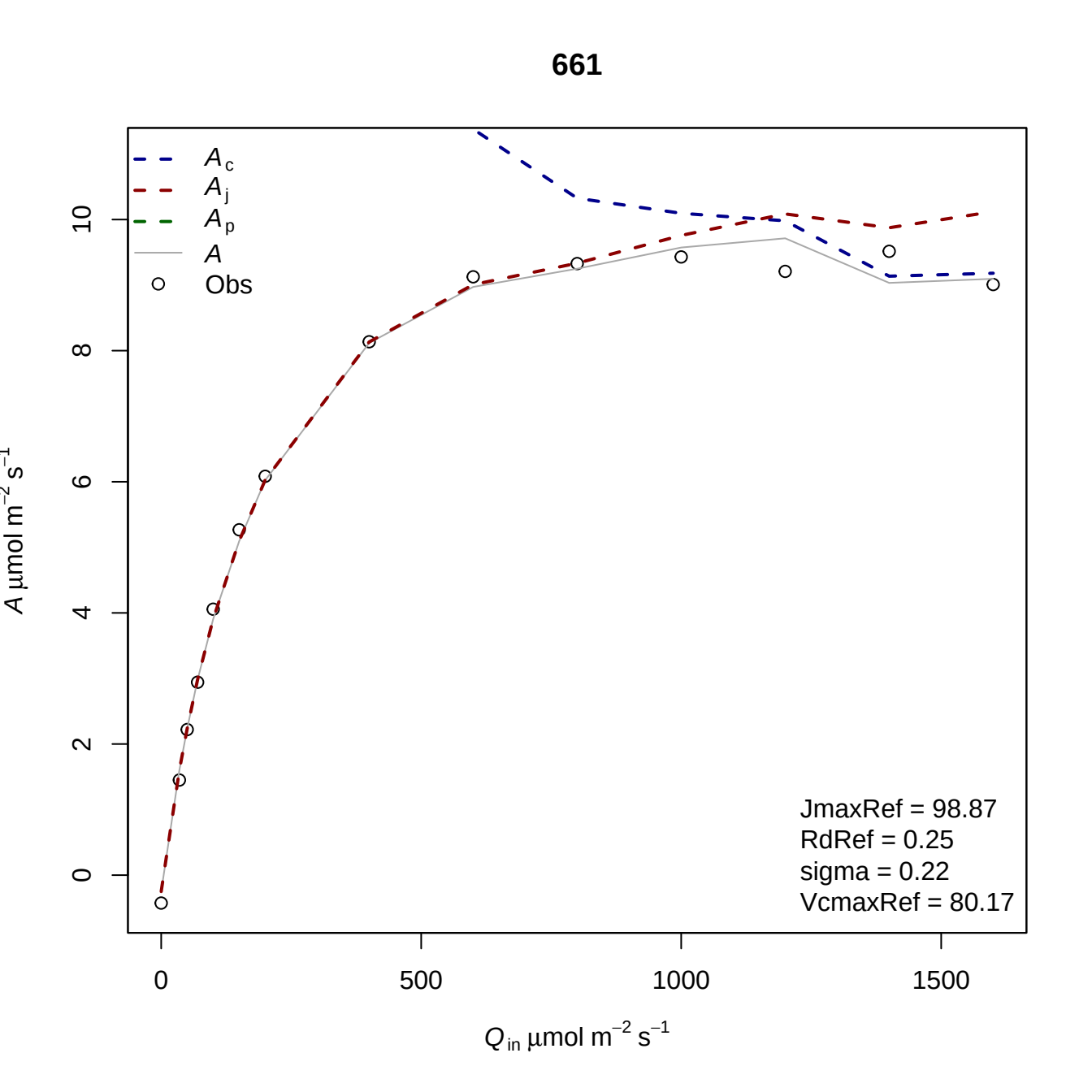

656
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 159.43
RdRef = 0.76
sigma = 0.33
VcmaxRef = 70.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

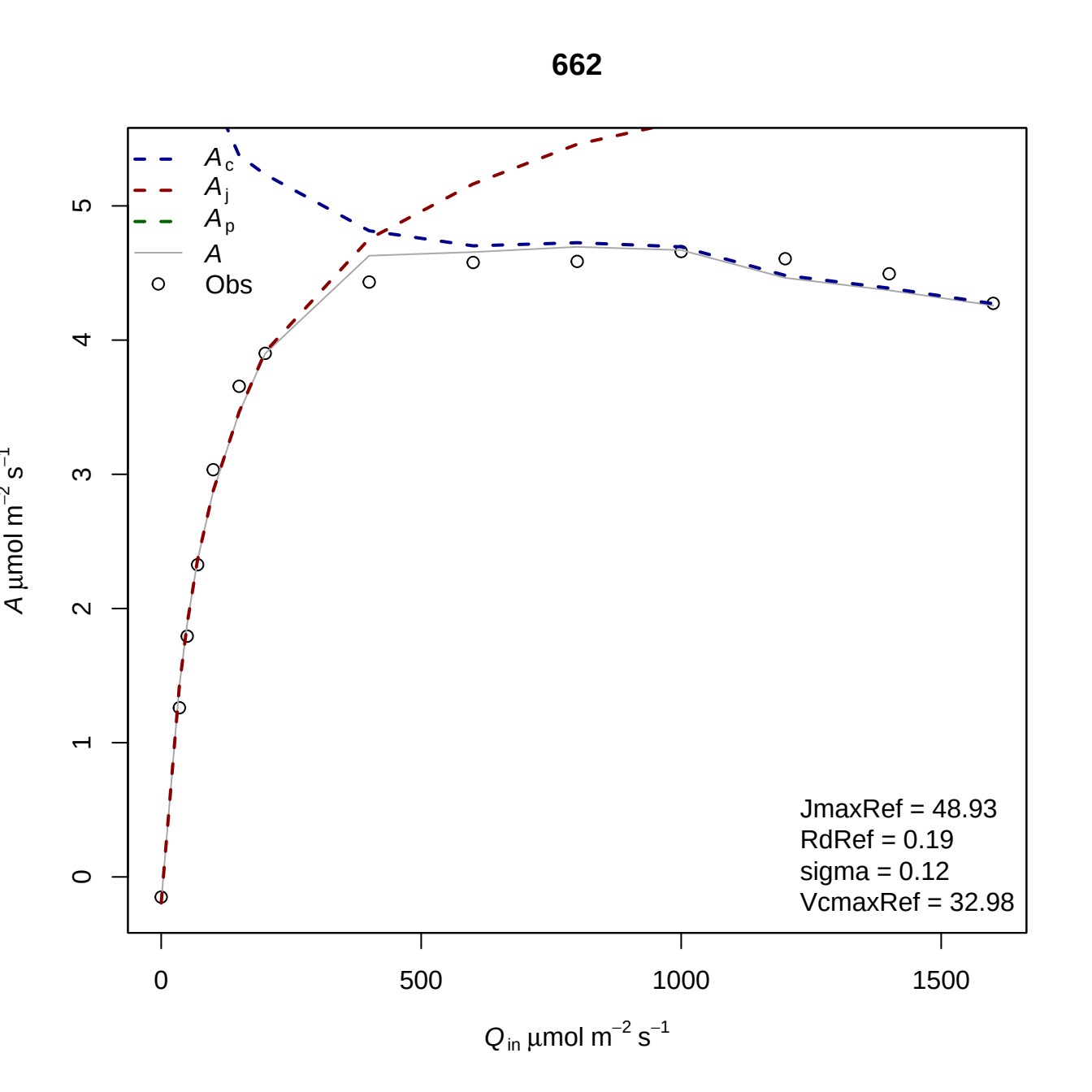

657
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 175.02
RdRef = 0.58
sigma = 0.47
VcmaxRef = 76.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

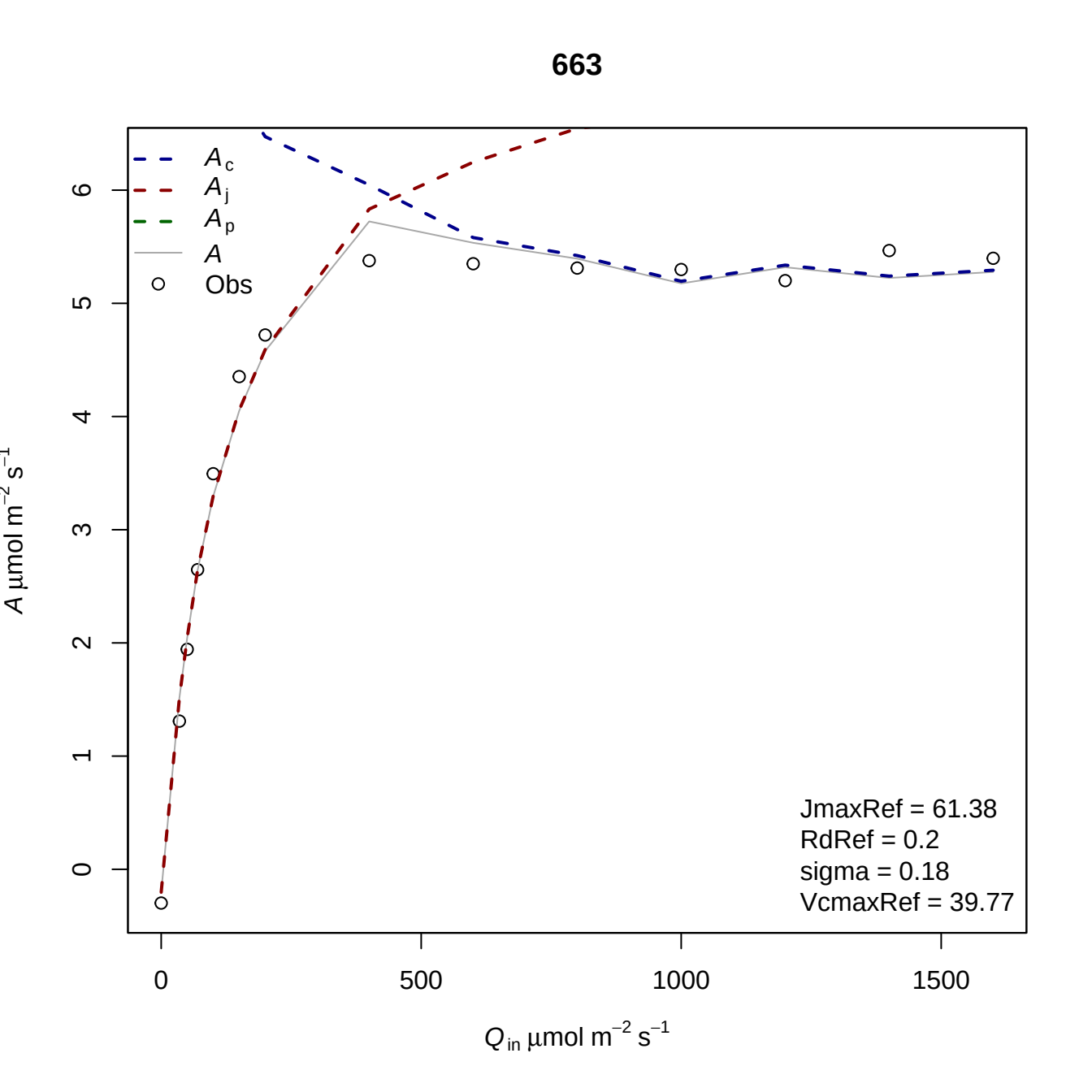

658
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.47
RdRef = 0.68
sigma = 0.21
VcmaxRef = 82.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

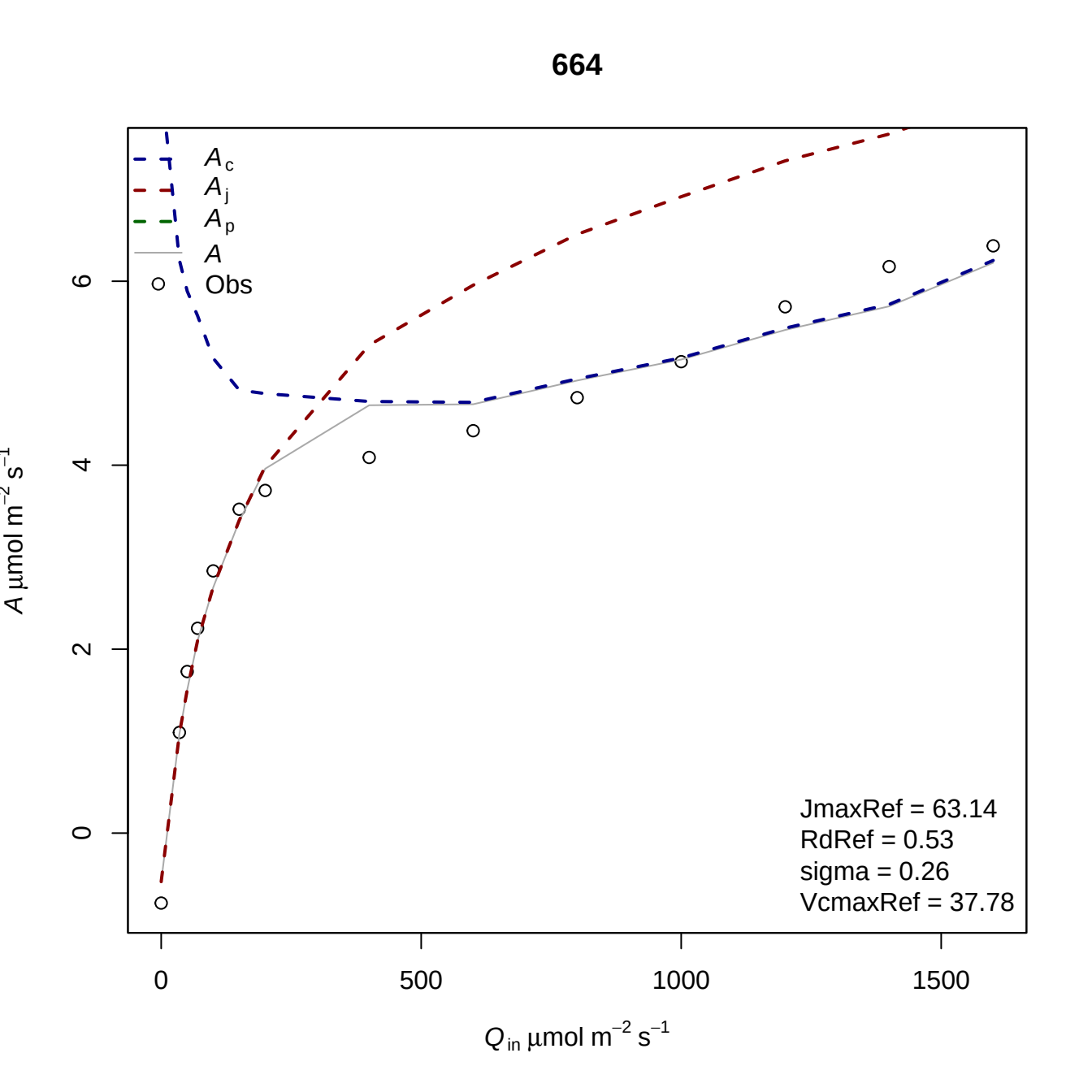

659
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.55
RdRef = −0.17
sigma = 0.43
VcmaxRef = 256.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

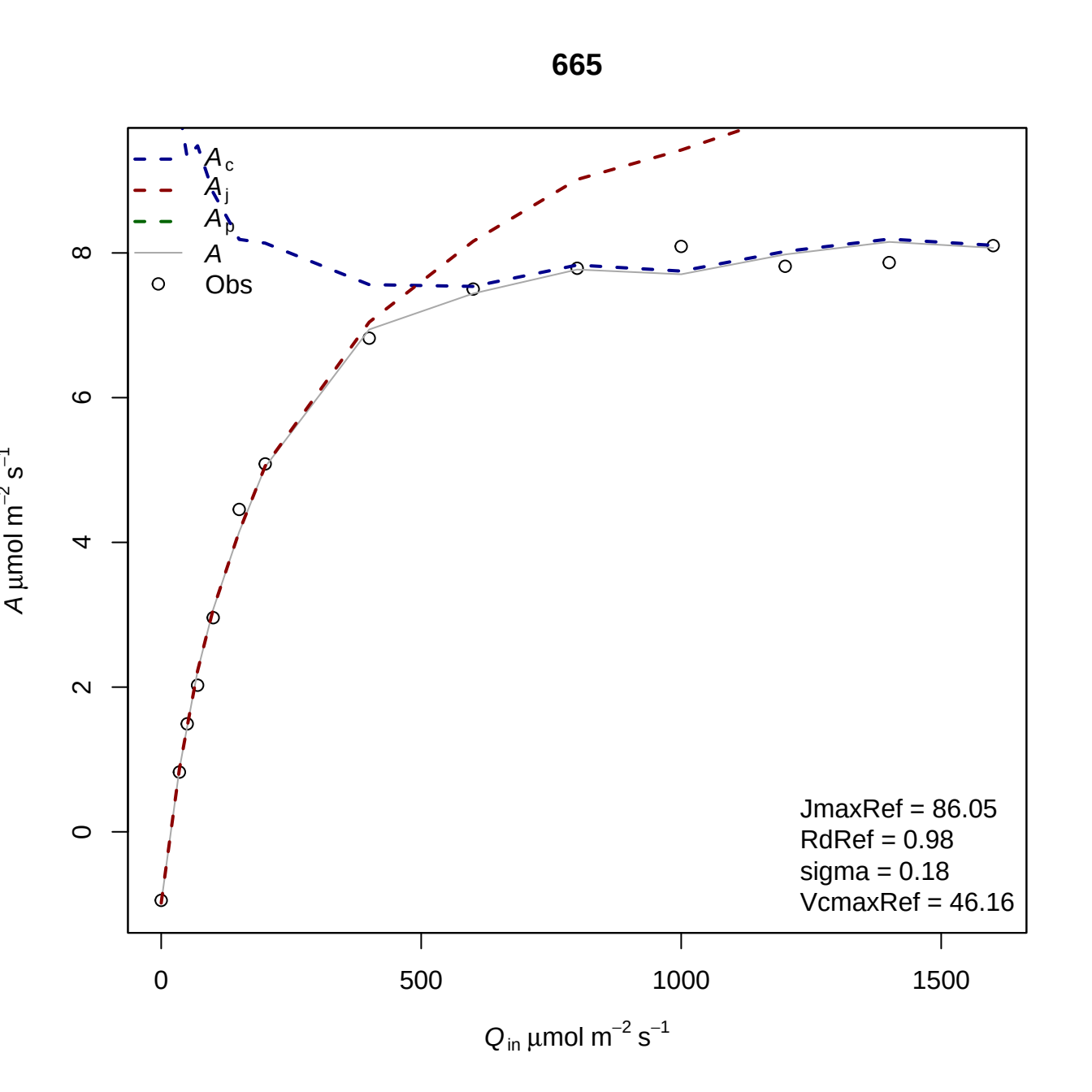

660
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.44
RdRef = 0.6
sigma = 0.19
VcmaxRef = 2186705.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

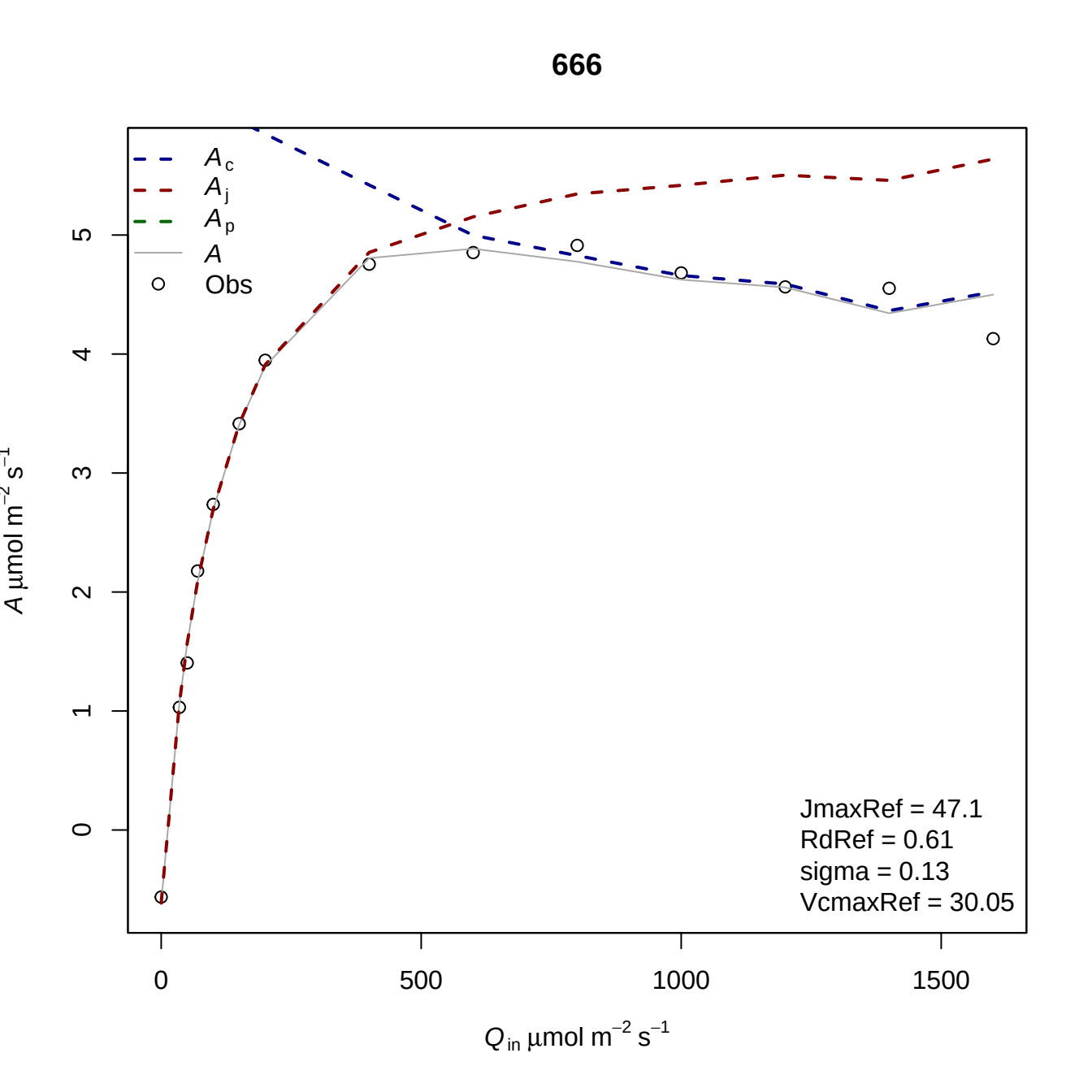

661
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.87
RdRef = 0.25
sigma = 0.22
VcmaxRef = 80.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

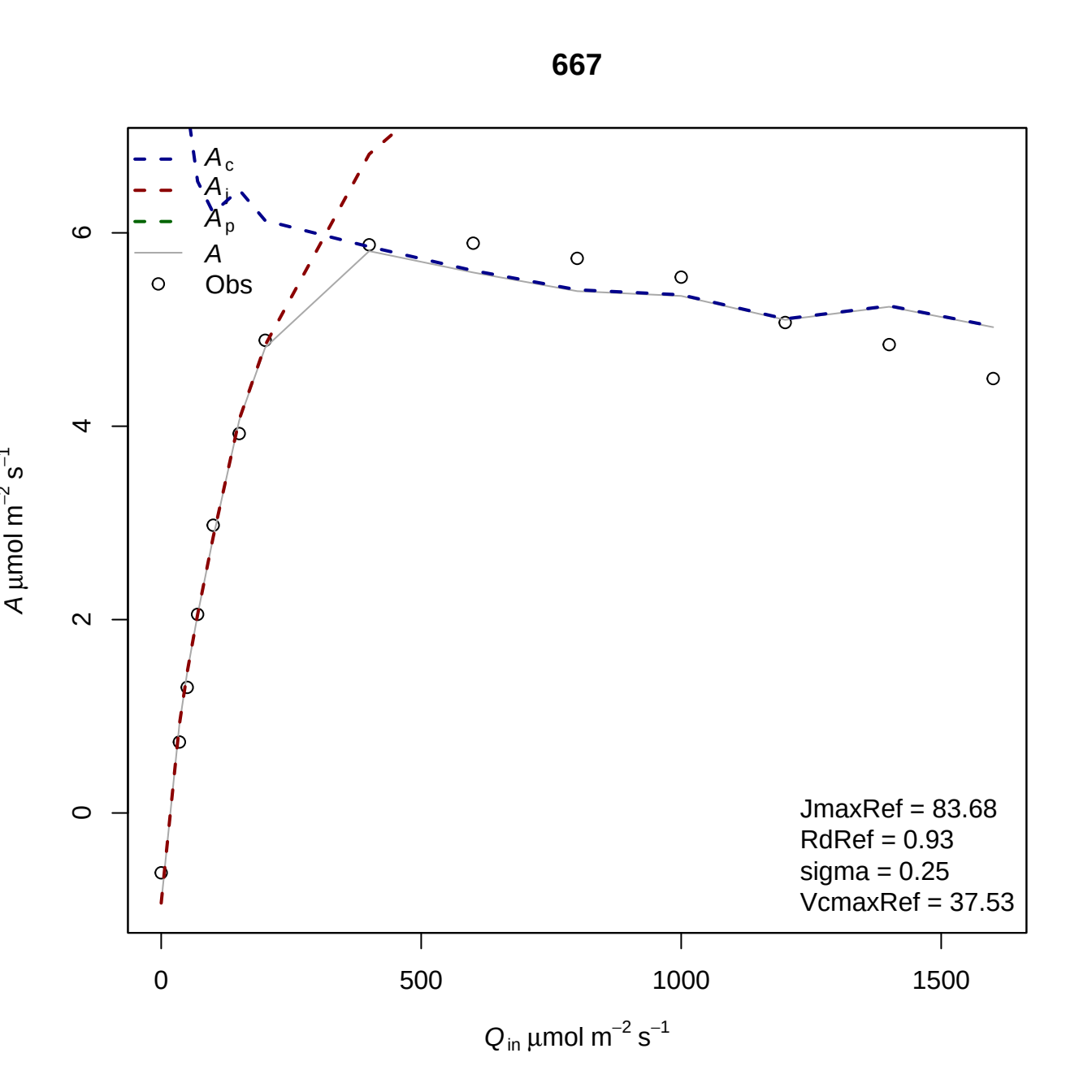

662
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.93
RdRef = 0.19
sigma = 0.12
VcmaxRef = 32.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

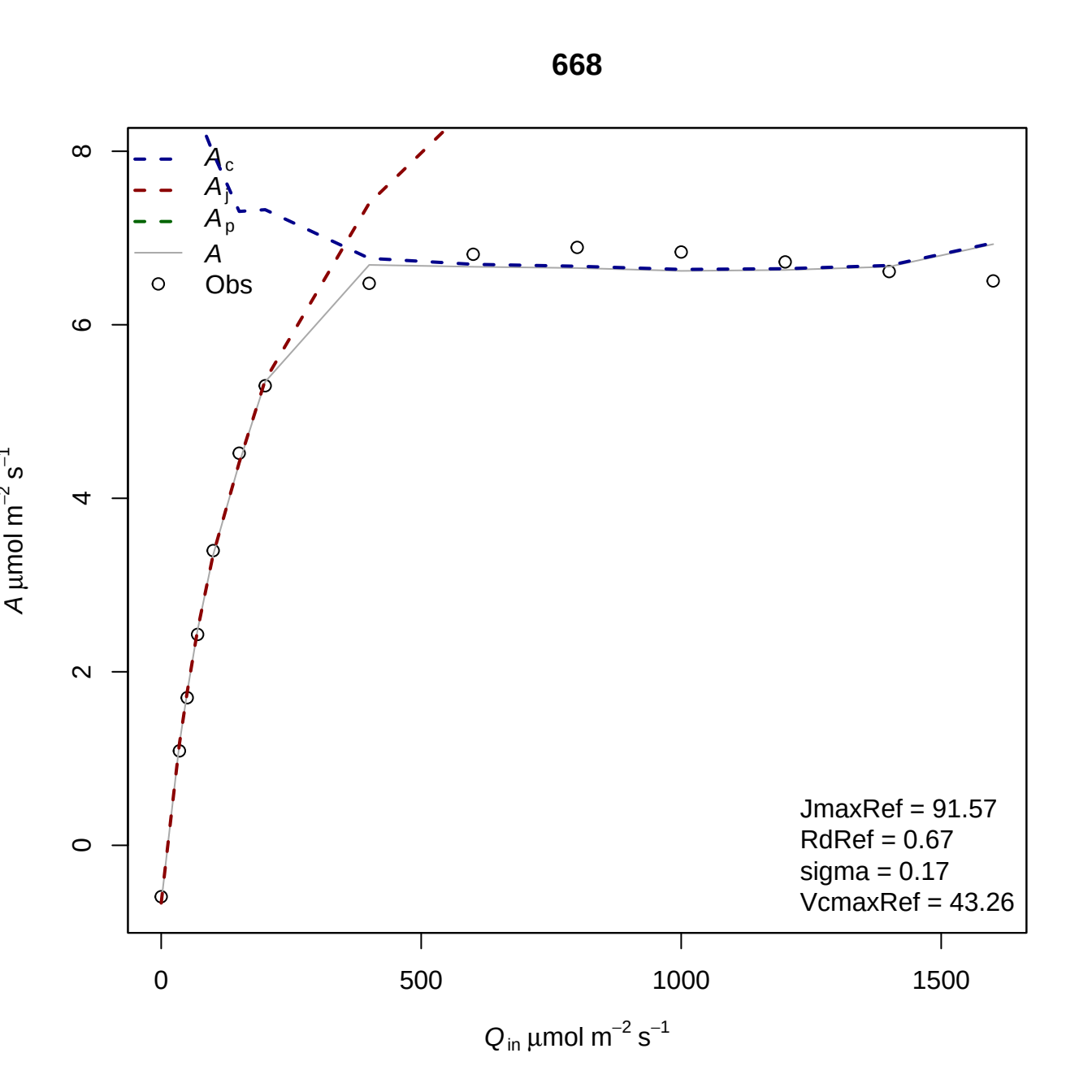

663
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.38
RdRef = 0.2
sigma = 0.18
VcmaxRef = 39.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

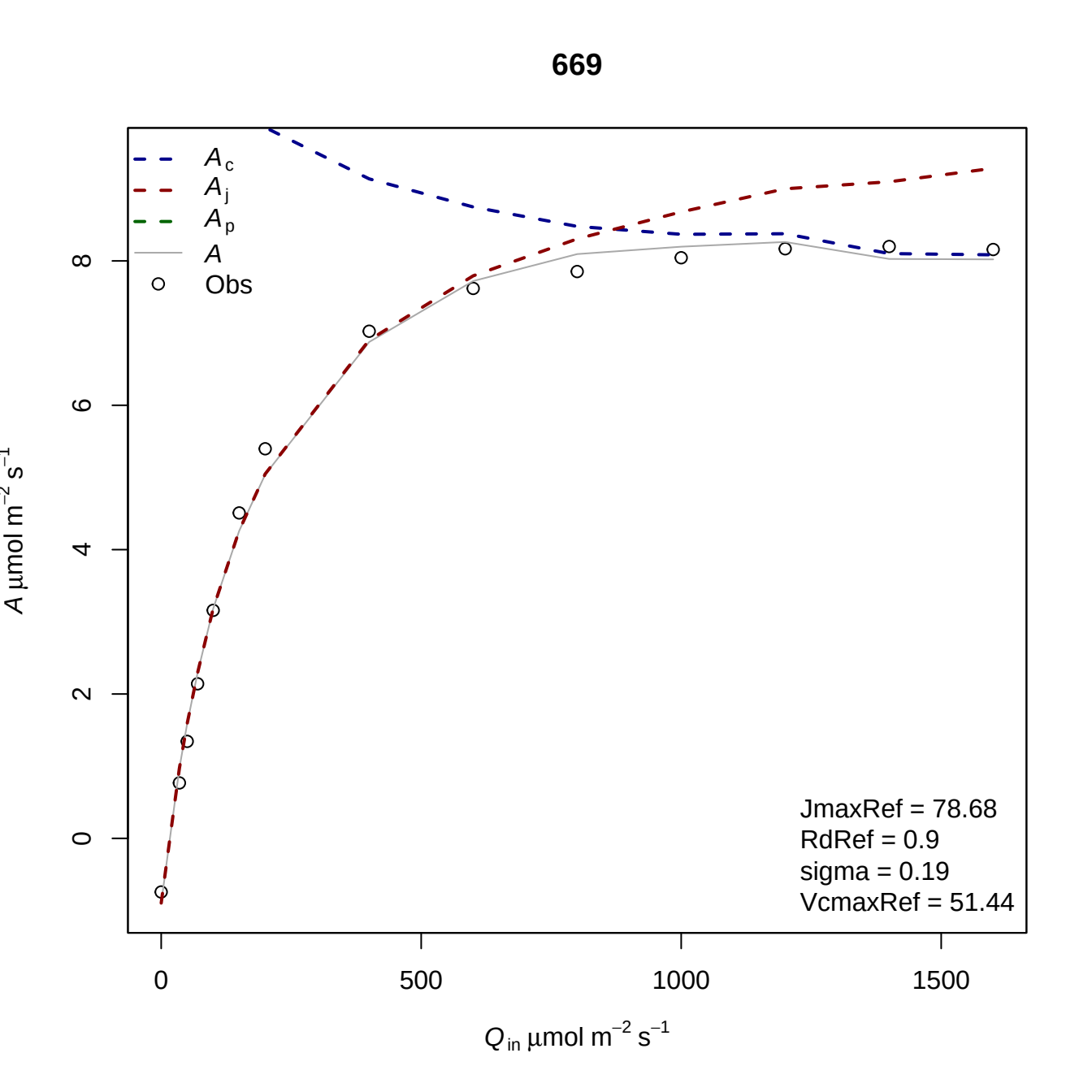

664
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.14
RdRef = 0.53
sigma = 0.26
VcmaxRef = 37.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

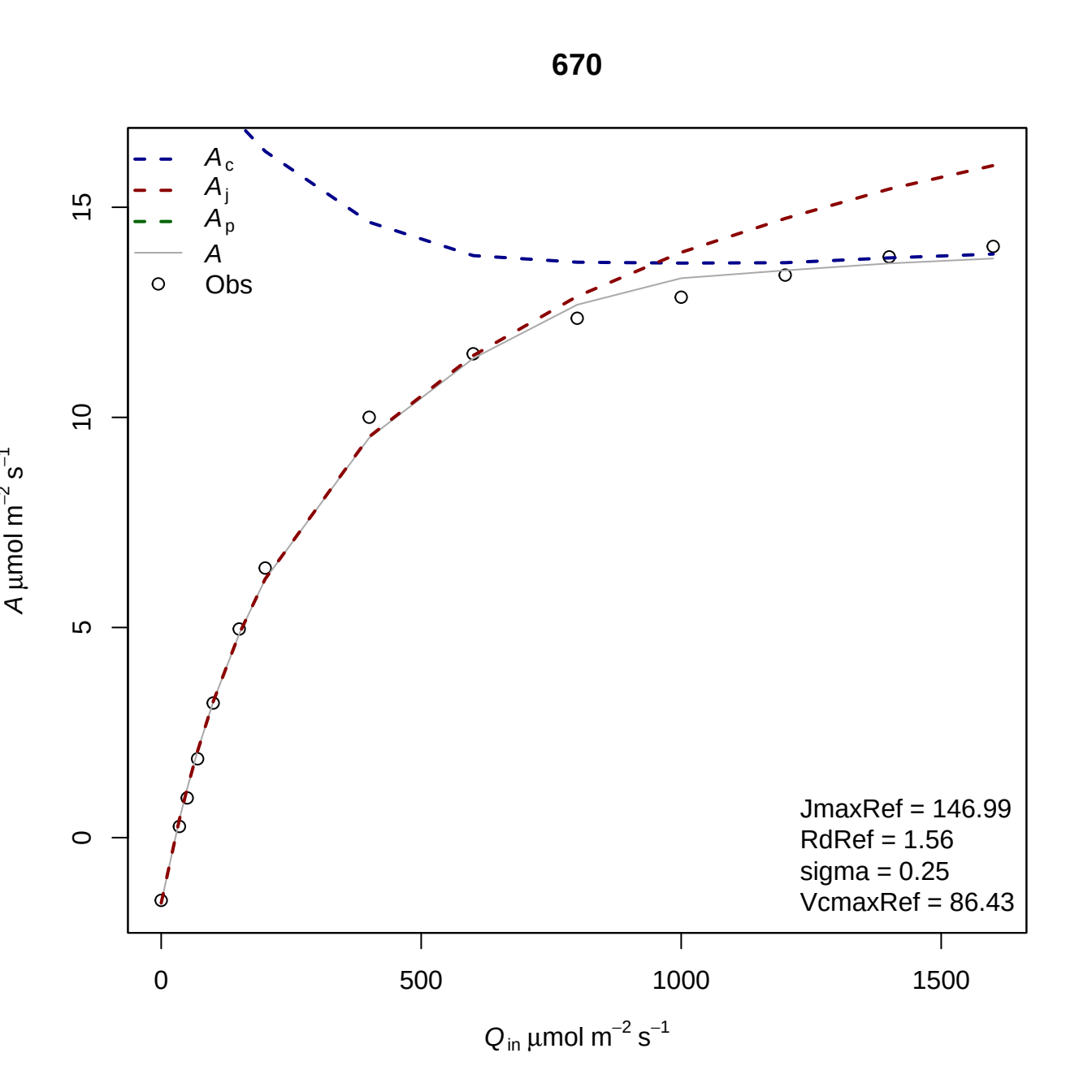

665
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.05
RdRef = 0.98
sigma = 0.18
VcmaxRef = 46.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

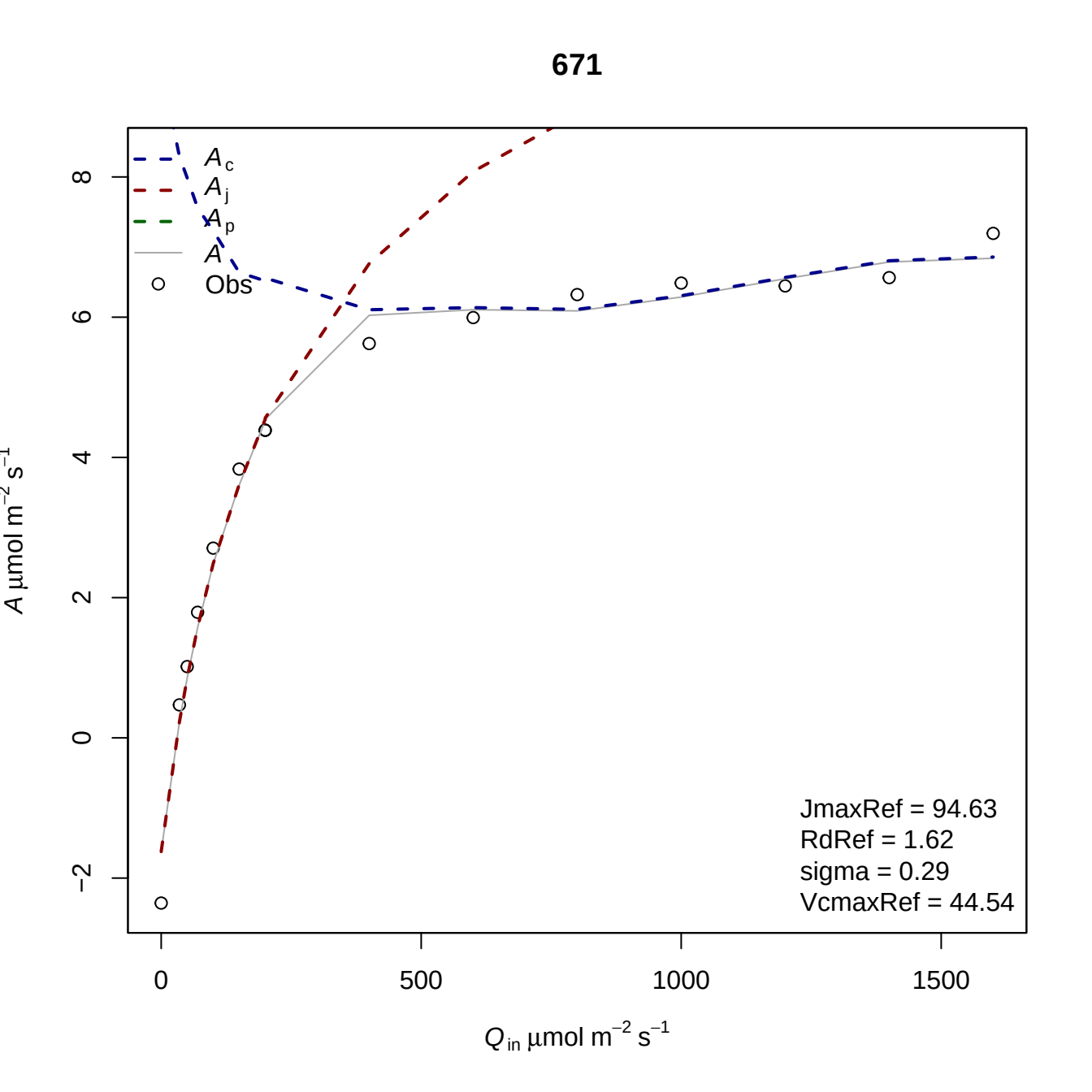

666
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.1
RdRef = 0.61
sigma = 0.13
VcmaxRef = 30.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

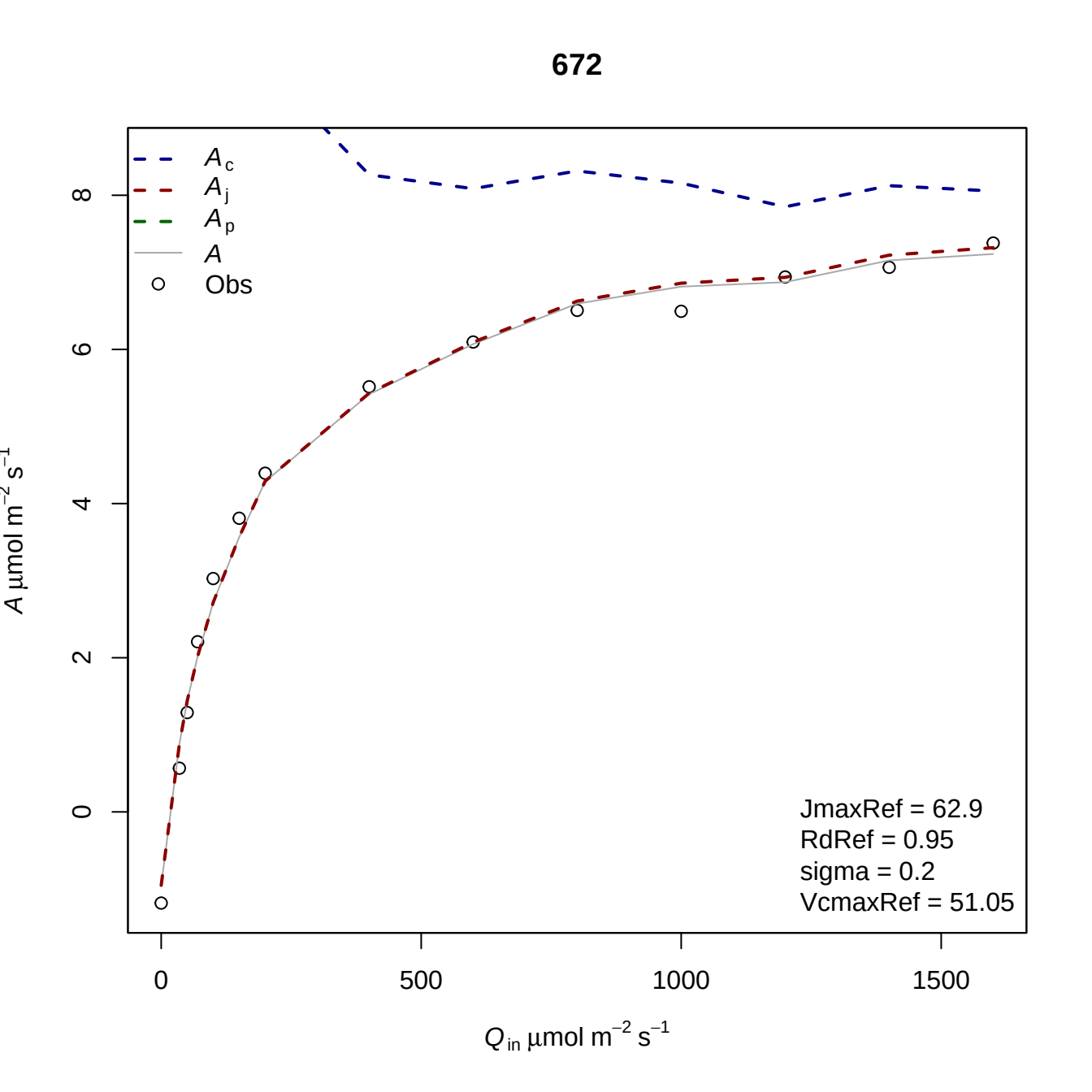

667
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.68
RdRef = 0.93
sigma = 0.25
VcmaxRef = 37.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

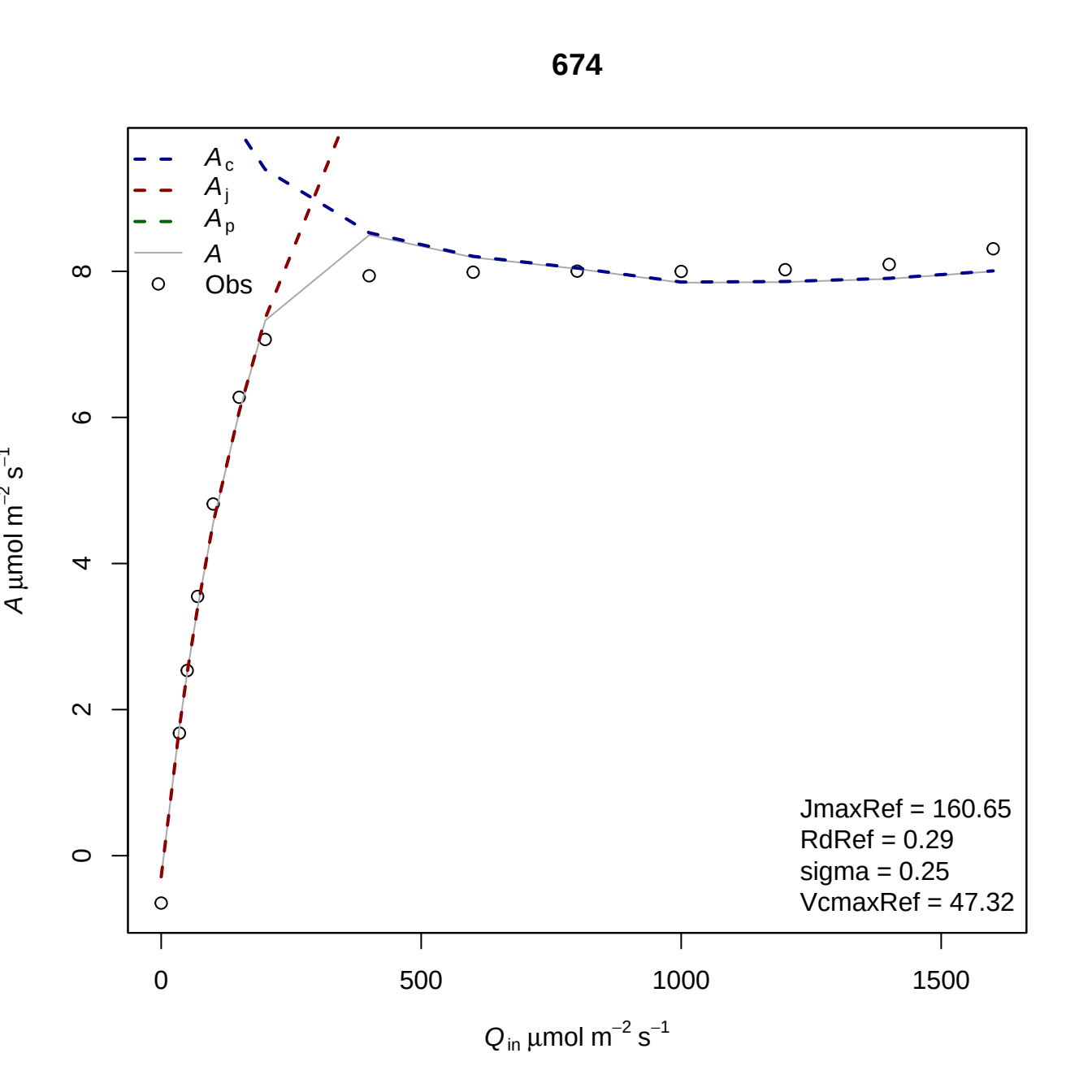

668
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.57
RdRef = 0.67
sigma = 0.17
VcmaxRef = 43.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

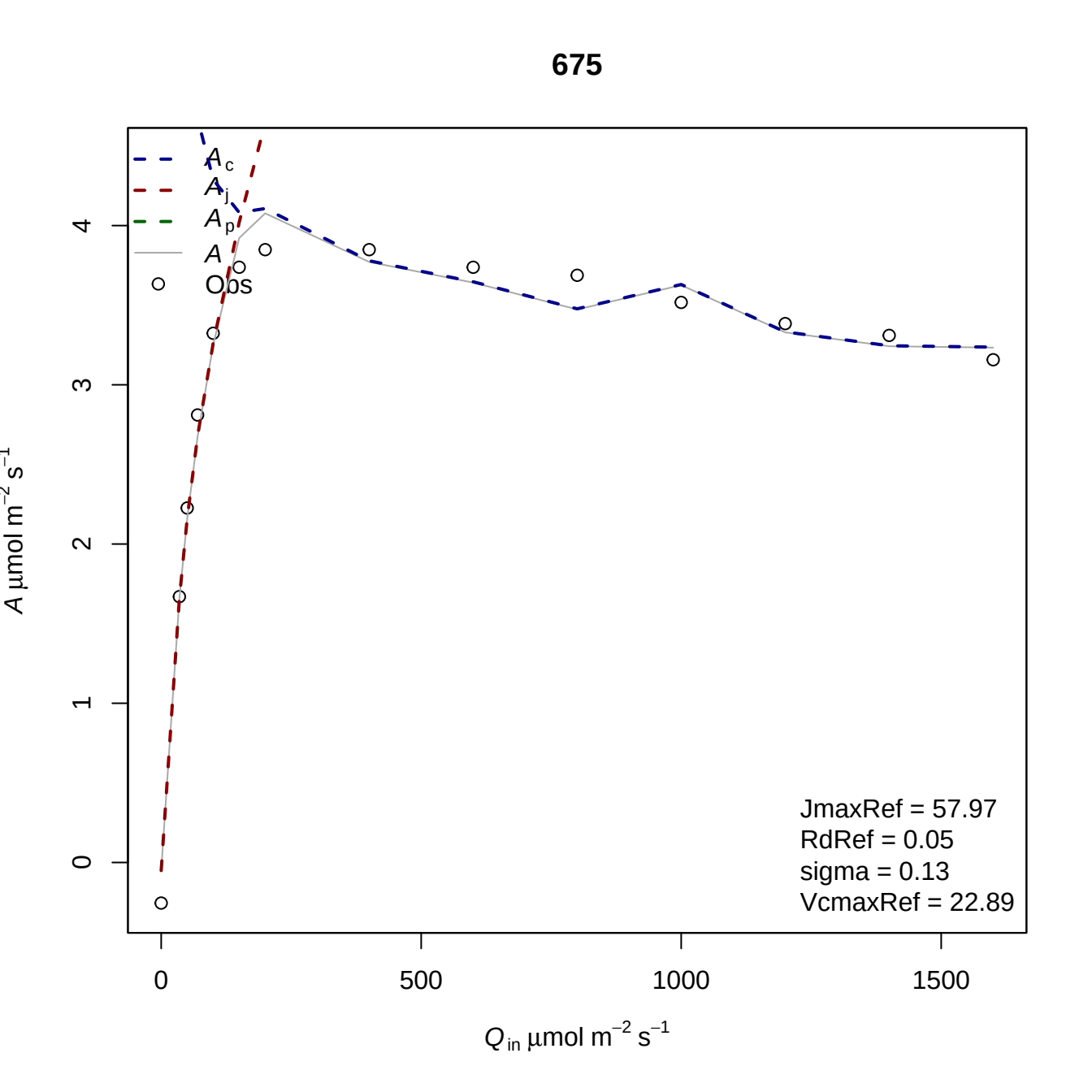

669
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.68
RdRef = 0.9
sigma = 0.19
VcmaxRef = 51.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

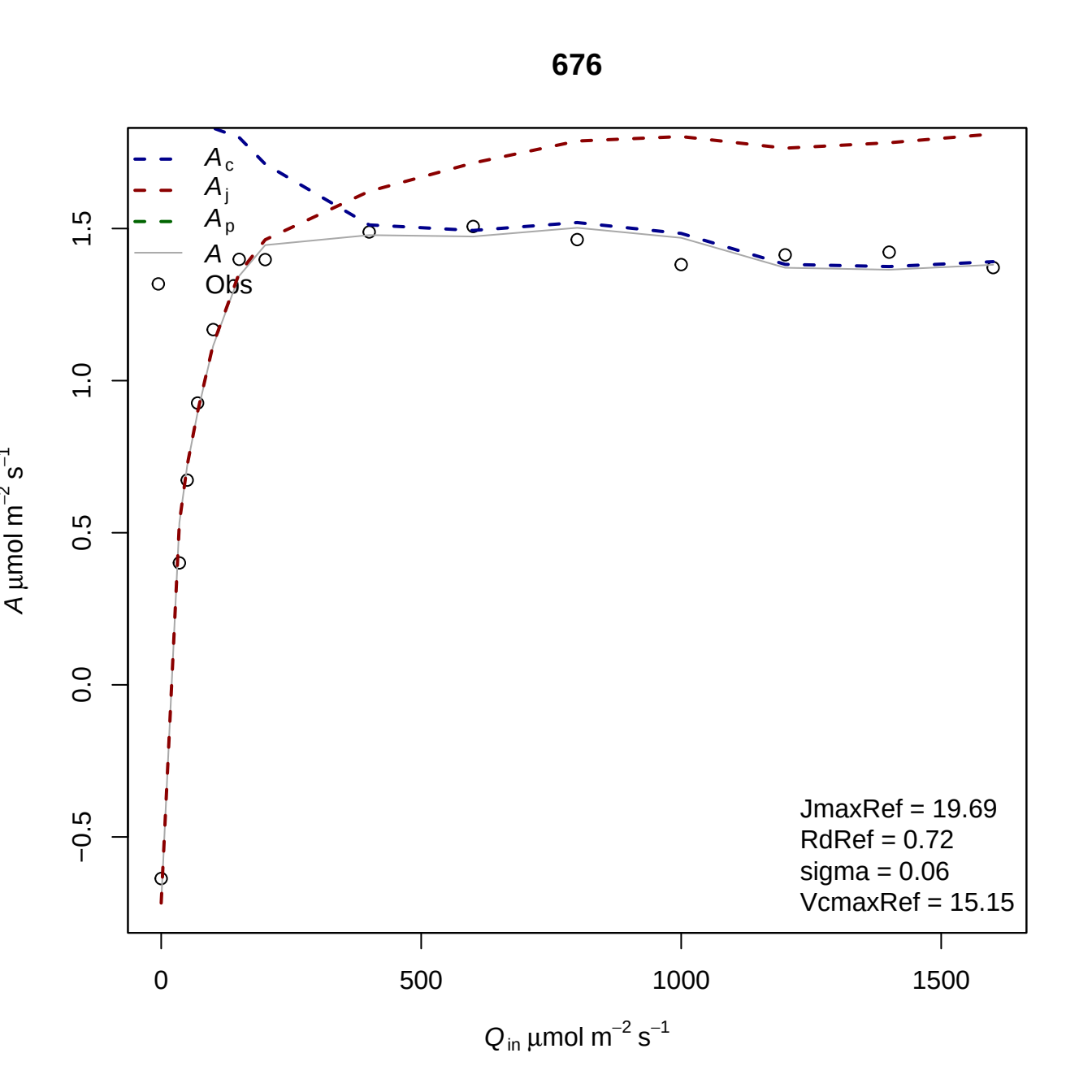

670
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 146.99
RdRef = 1.56
sigma = 0.25
VcmaxRef = 86.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

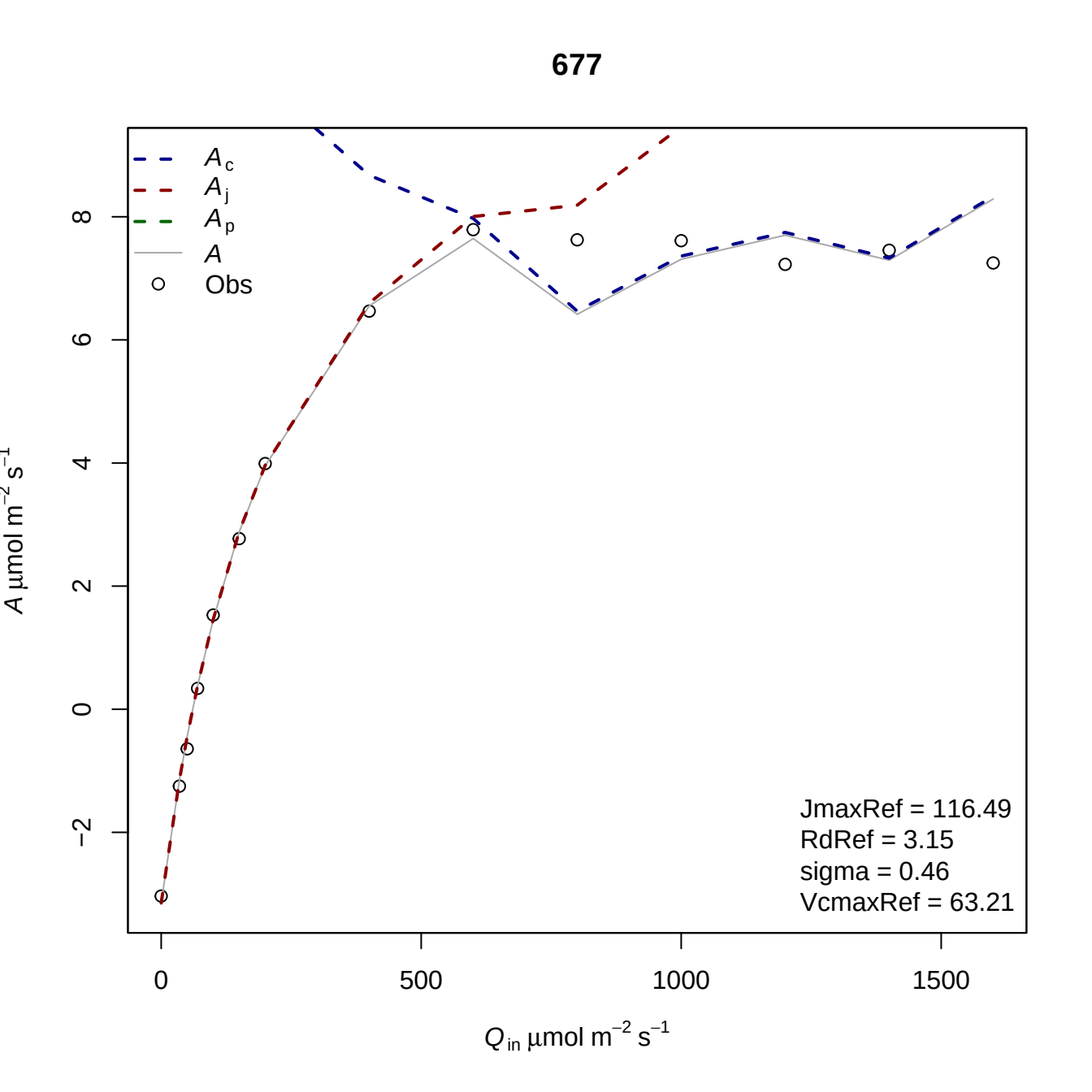

671
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.63
RdRef = 1.62
sigma = 0.29
VcmaxRef = 44.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

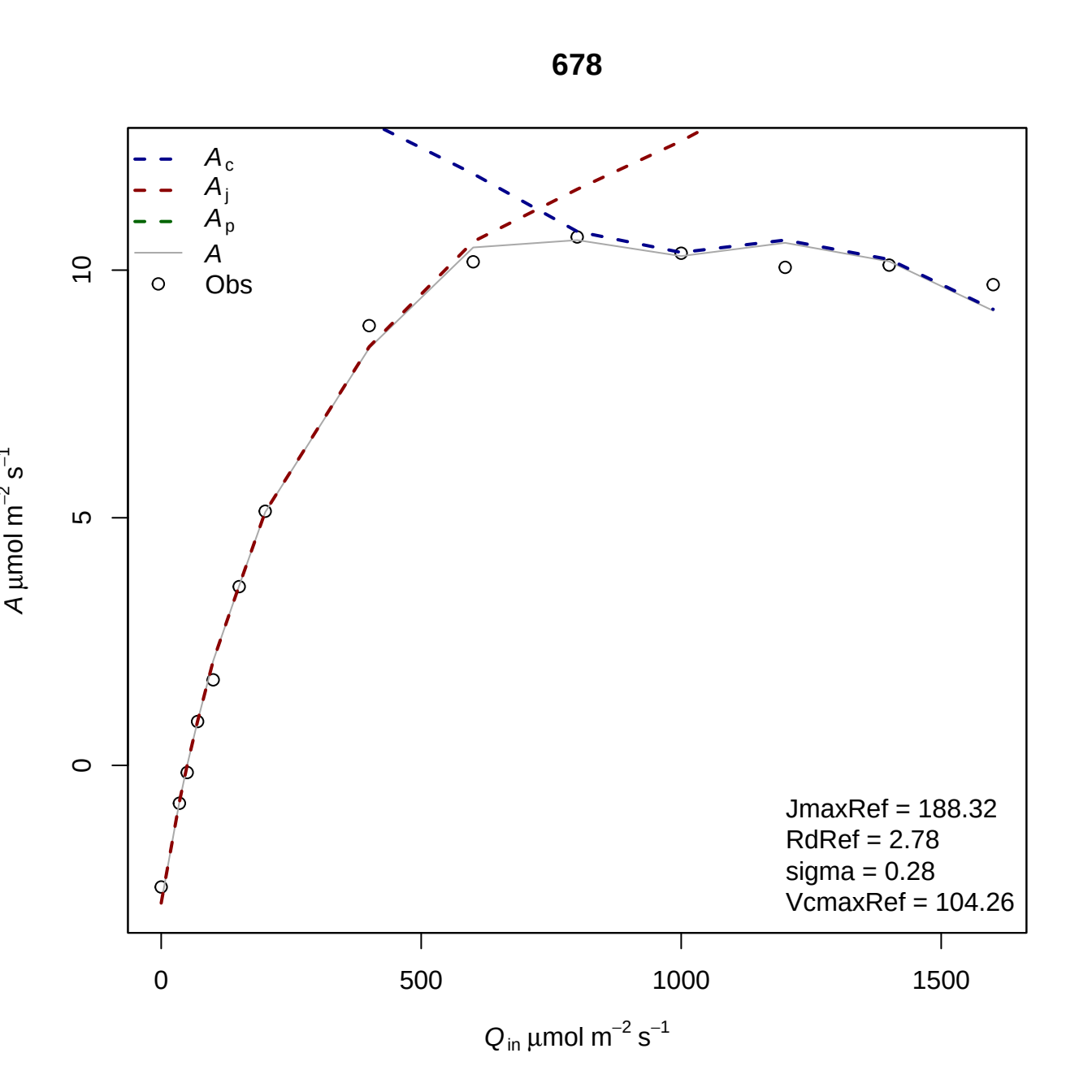

672
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.9
RdRef = 0.95
sigma = 0.2
VcmaxRef = 51.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

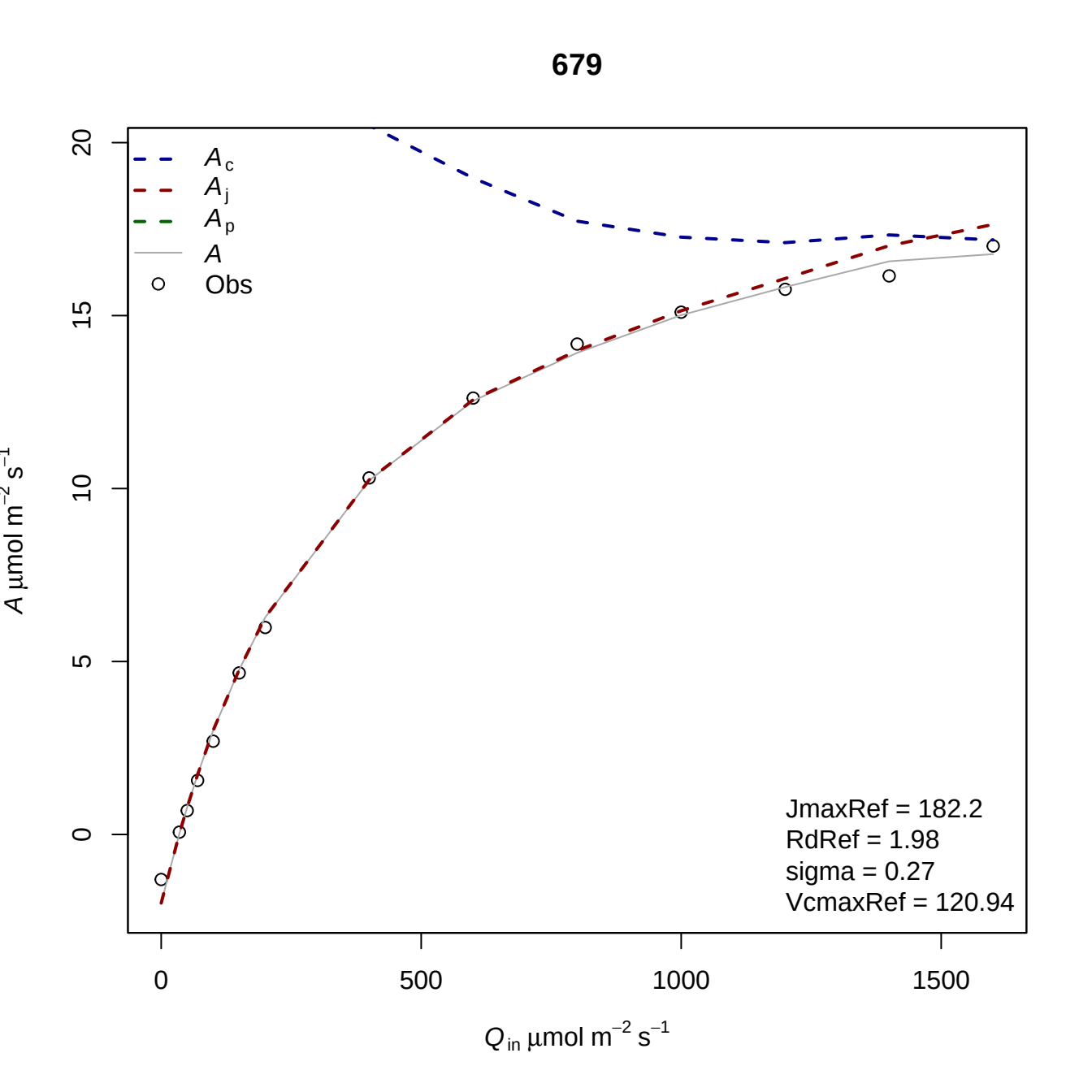

674
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.65
RdRef = 0.29
sigma = 0.25
VcmaxRef = 47.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

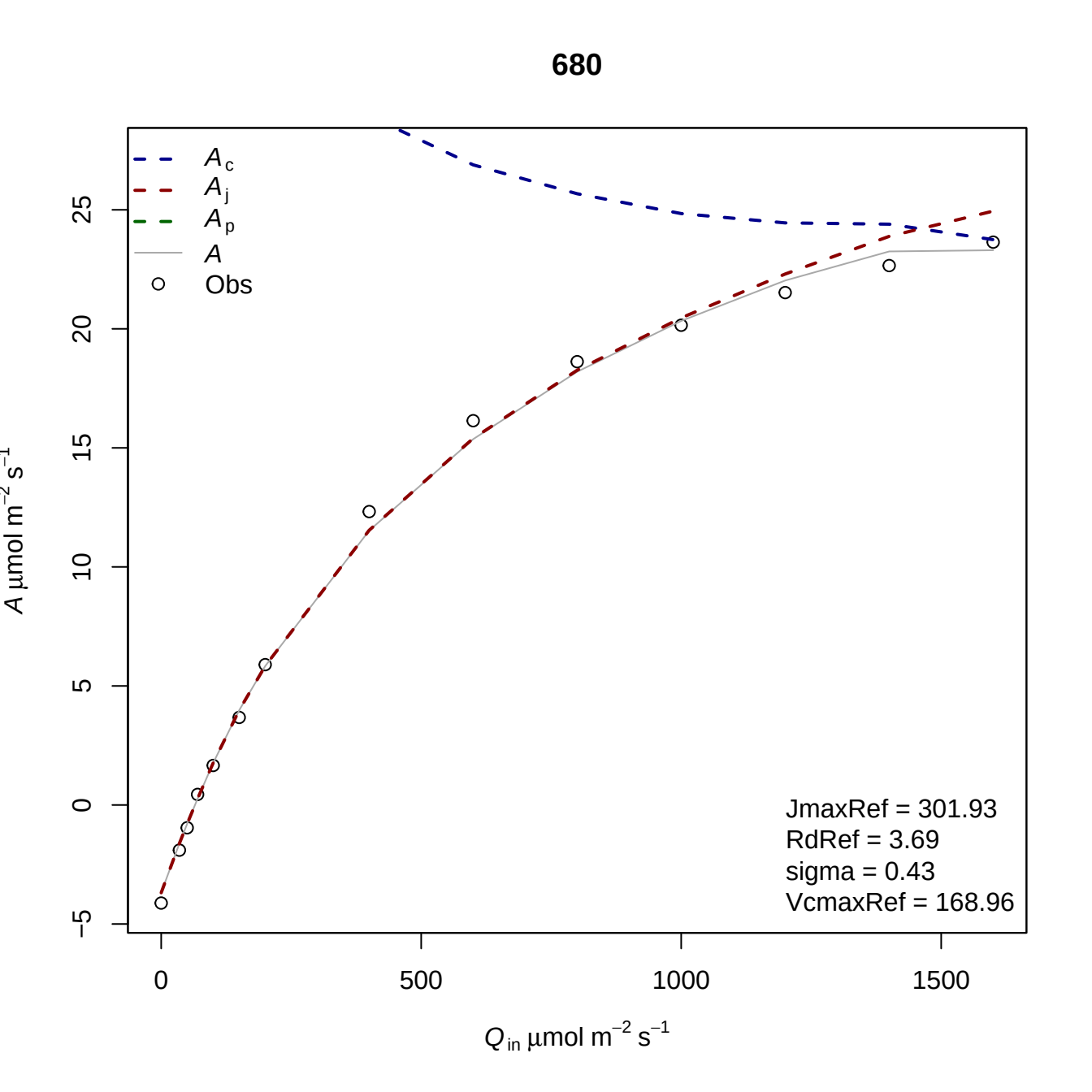

675
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.97
RdRef = 0.05
sigma = 0.13
VcmaxRef = 22.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

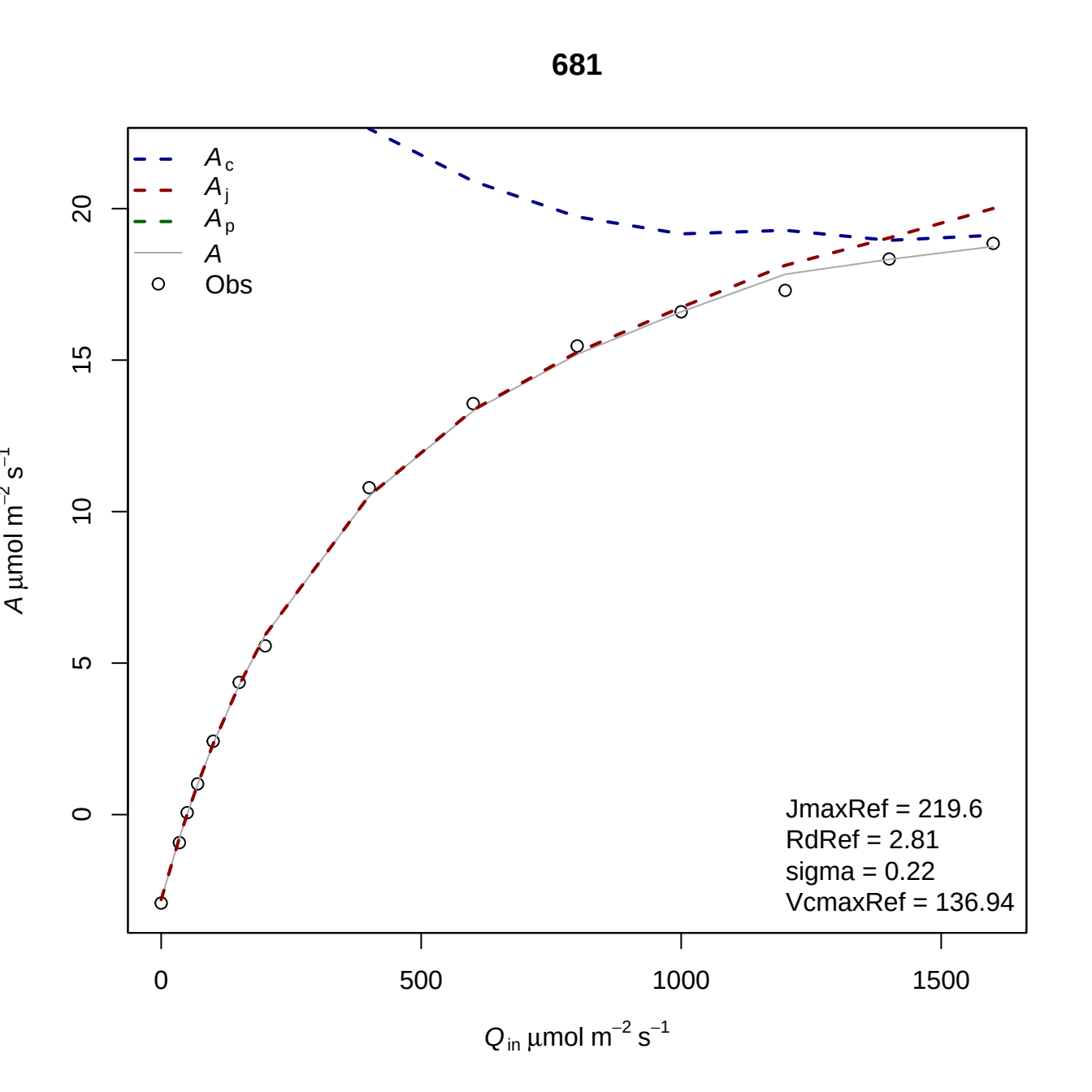

676
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 19.69
RdRef = 0.72
sigma = 0.06
VcmaxRef = 15.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

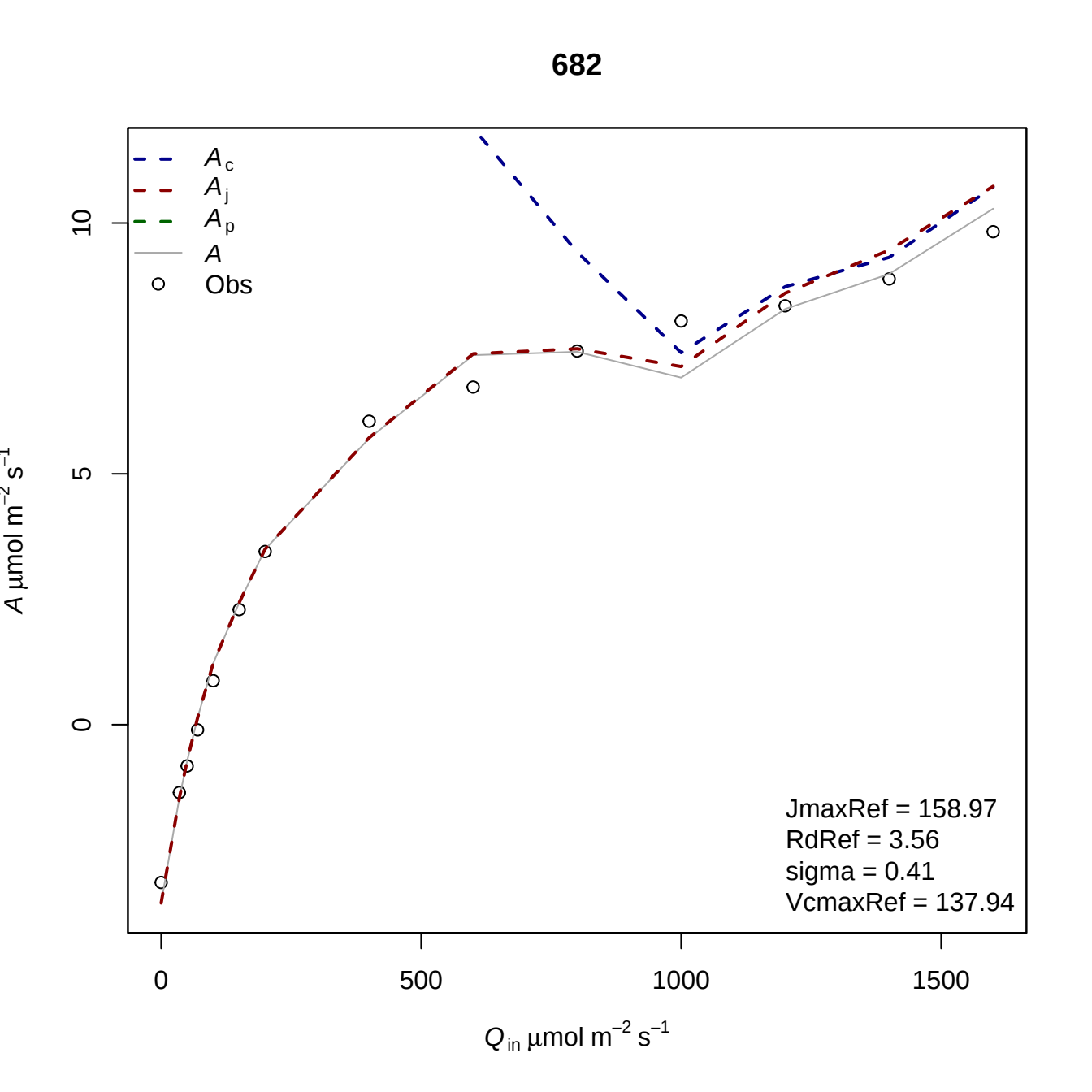

677
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.49
RdRef = 3.15
sigma = 0.46
VcmaxRef = 63.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

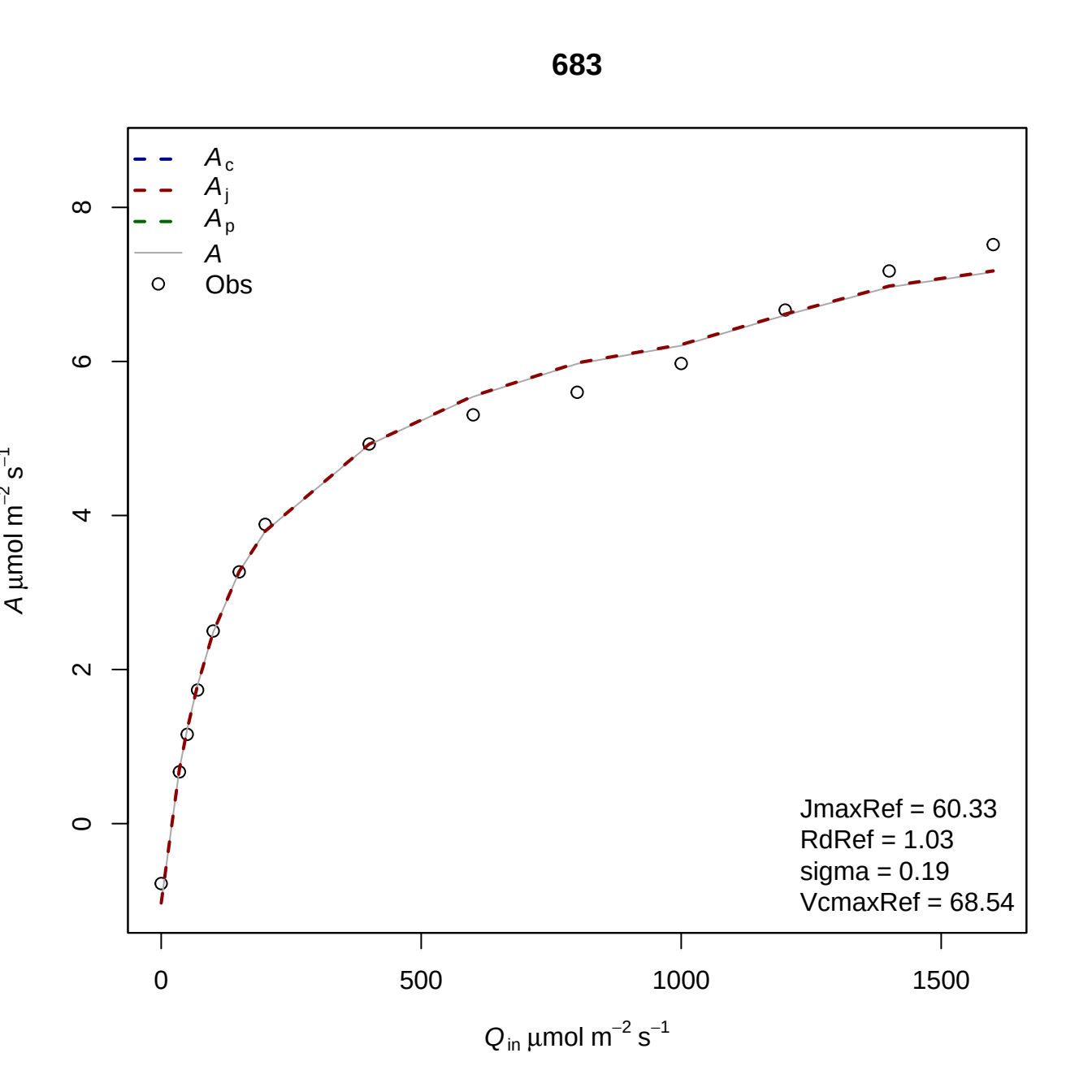

678
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 188.32
RdRef = 2.78
sigma = 0.28
VcmaxRef = 104.26
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

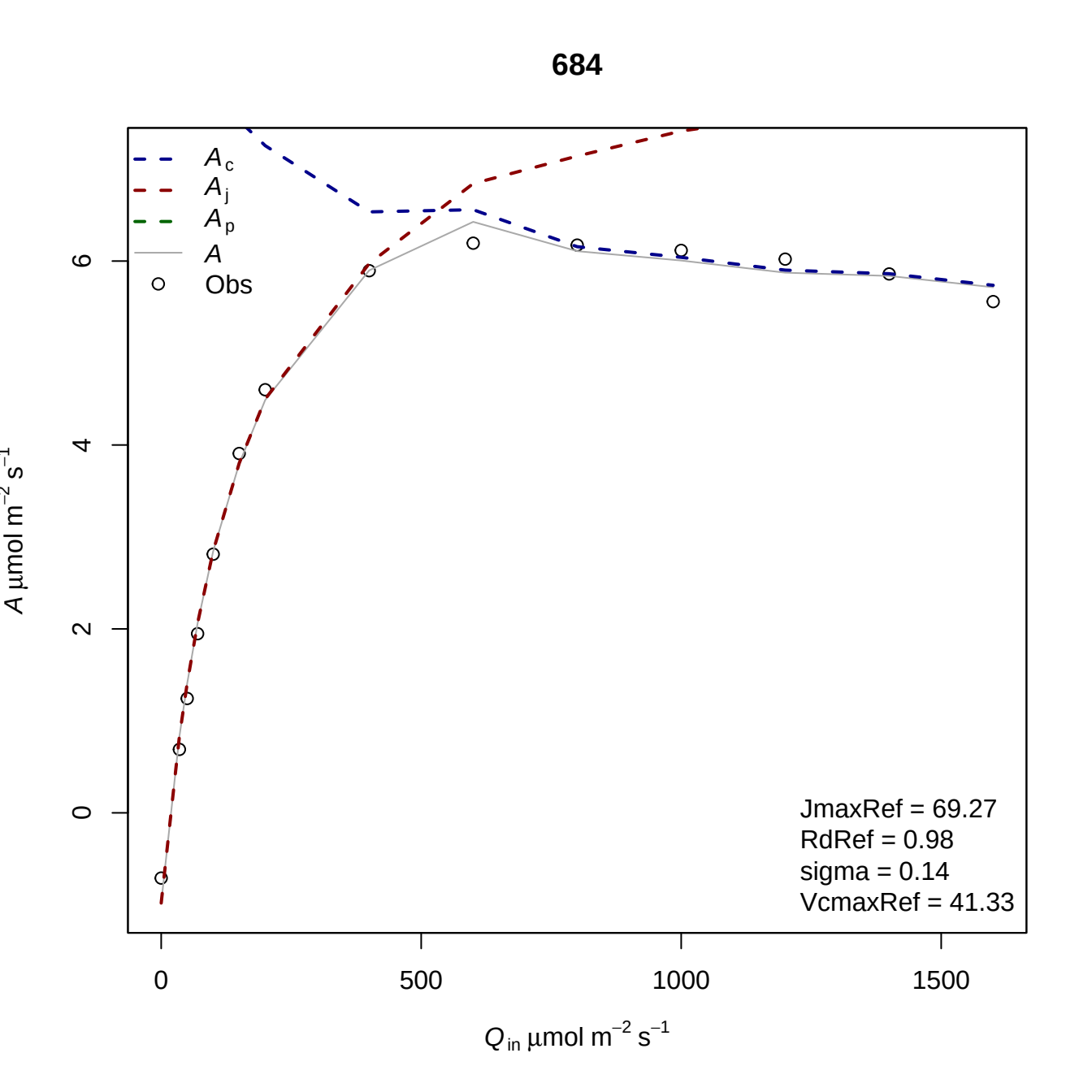

679
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 182.2
RdRef = 1.98
sigma = 0.27
VcmaxRef = 120.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

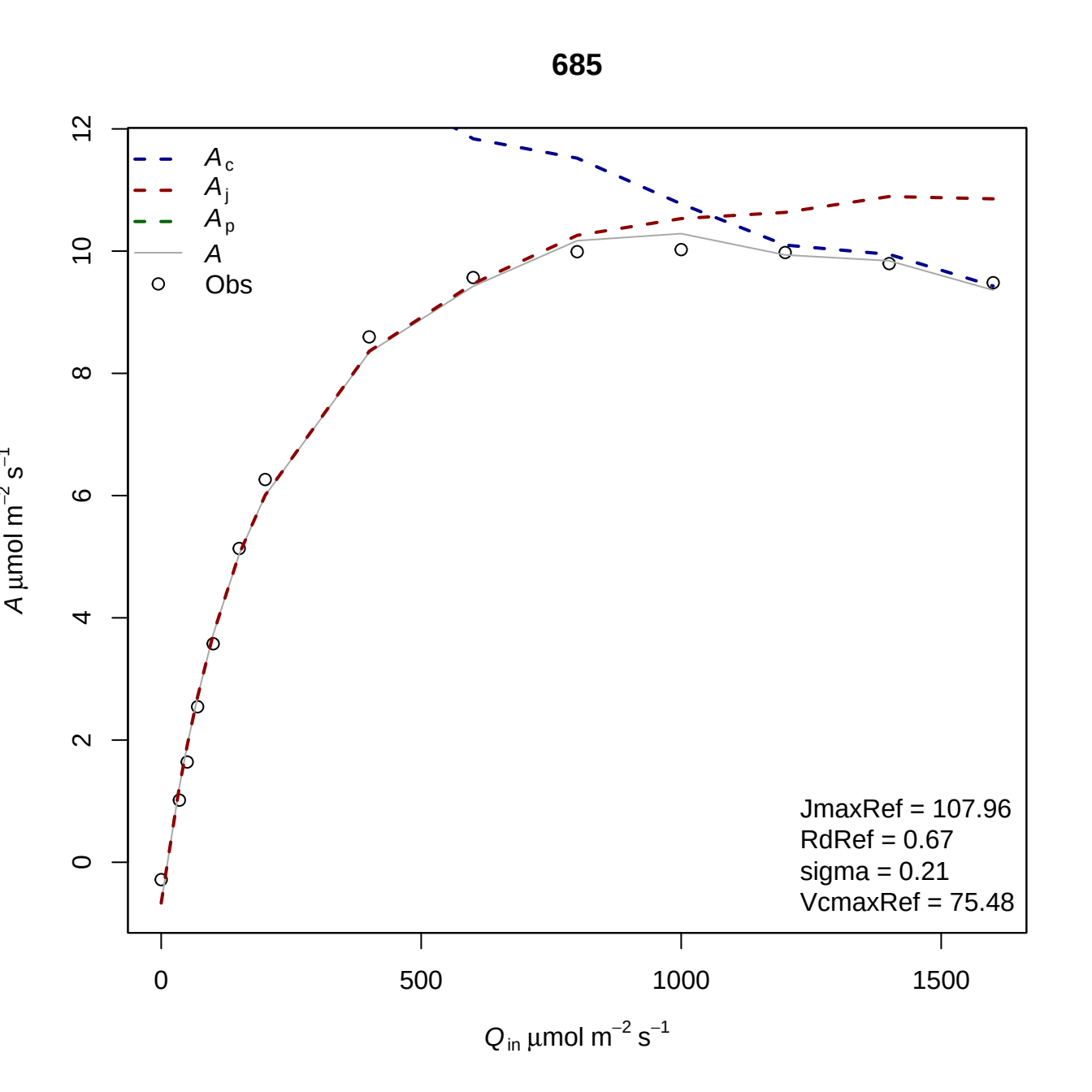

680
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 301.93
RdRef = 3.69
sigma = 0.43
VcmaxRef = 168.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
25
0
500
1000
1500

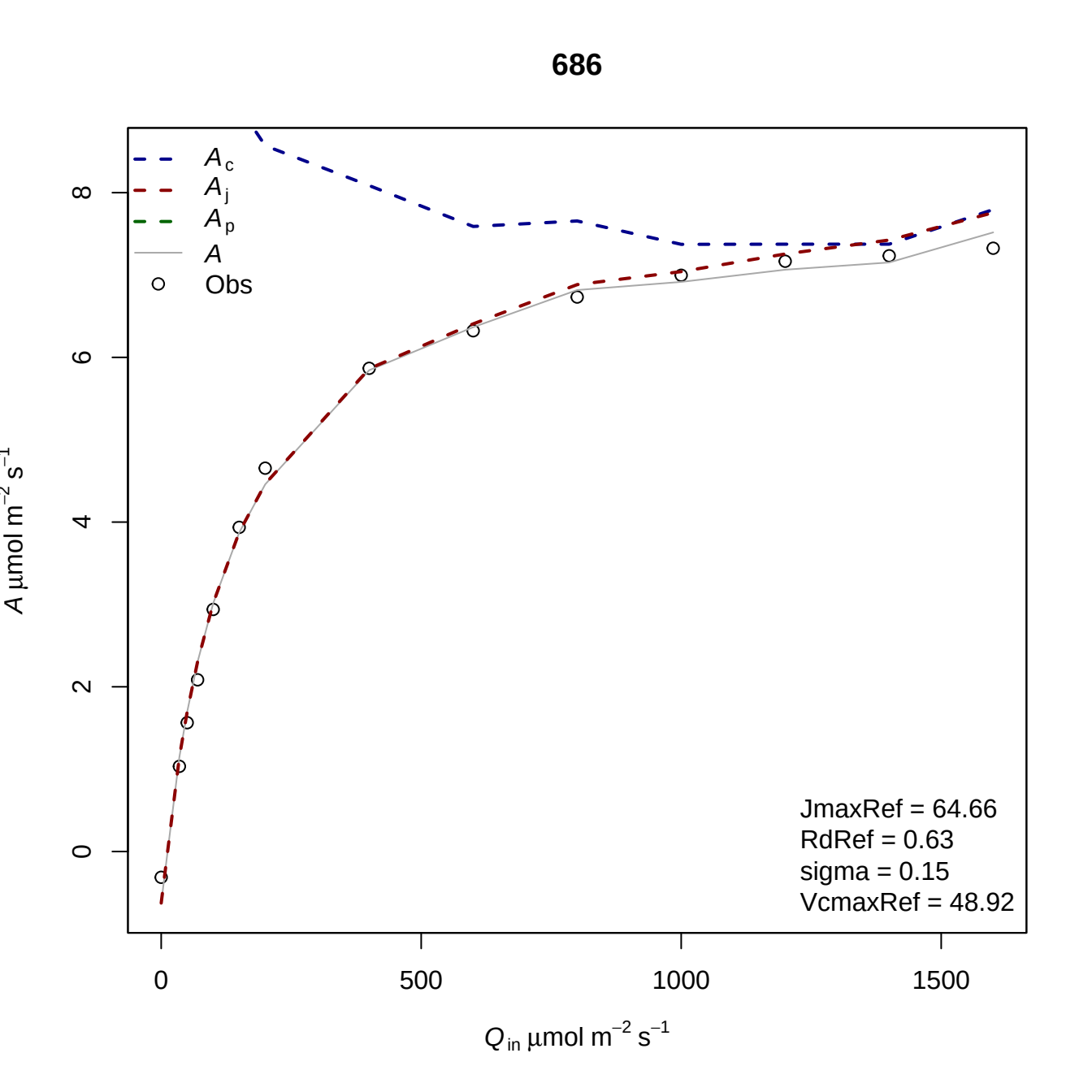

681
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 219.6
RdRef = 2.81
sigma = 0.22
VcmaxRef = 136.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

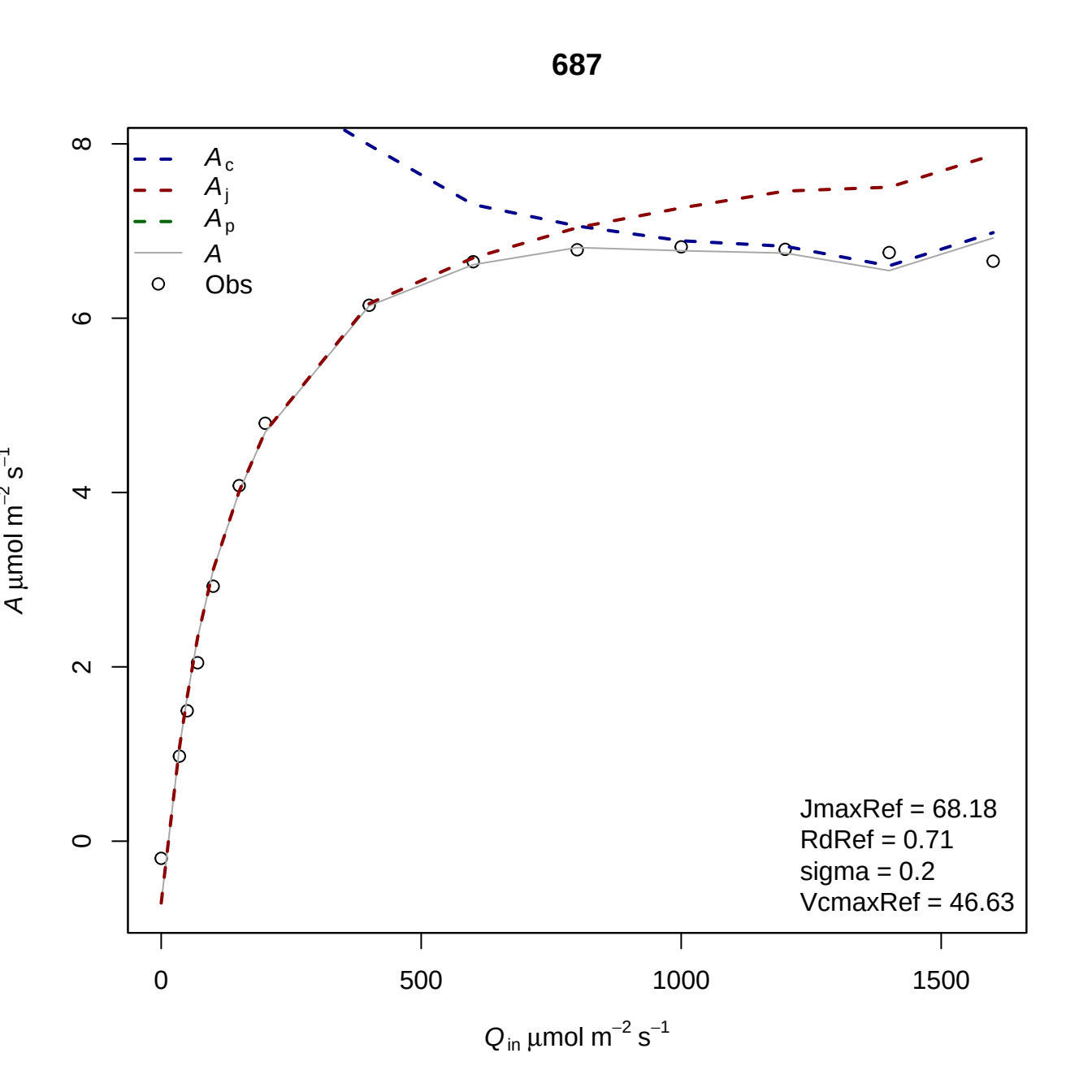

682
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.97
RdRef = 3.56
sigma = 0.41
VcmaxRef = 137.94
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

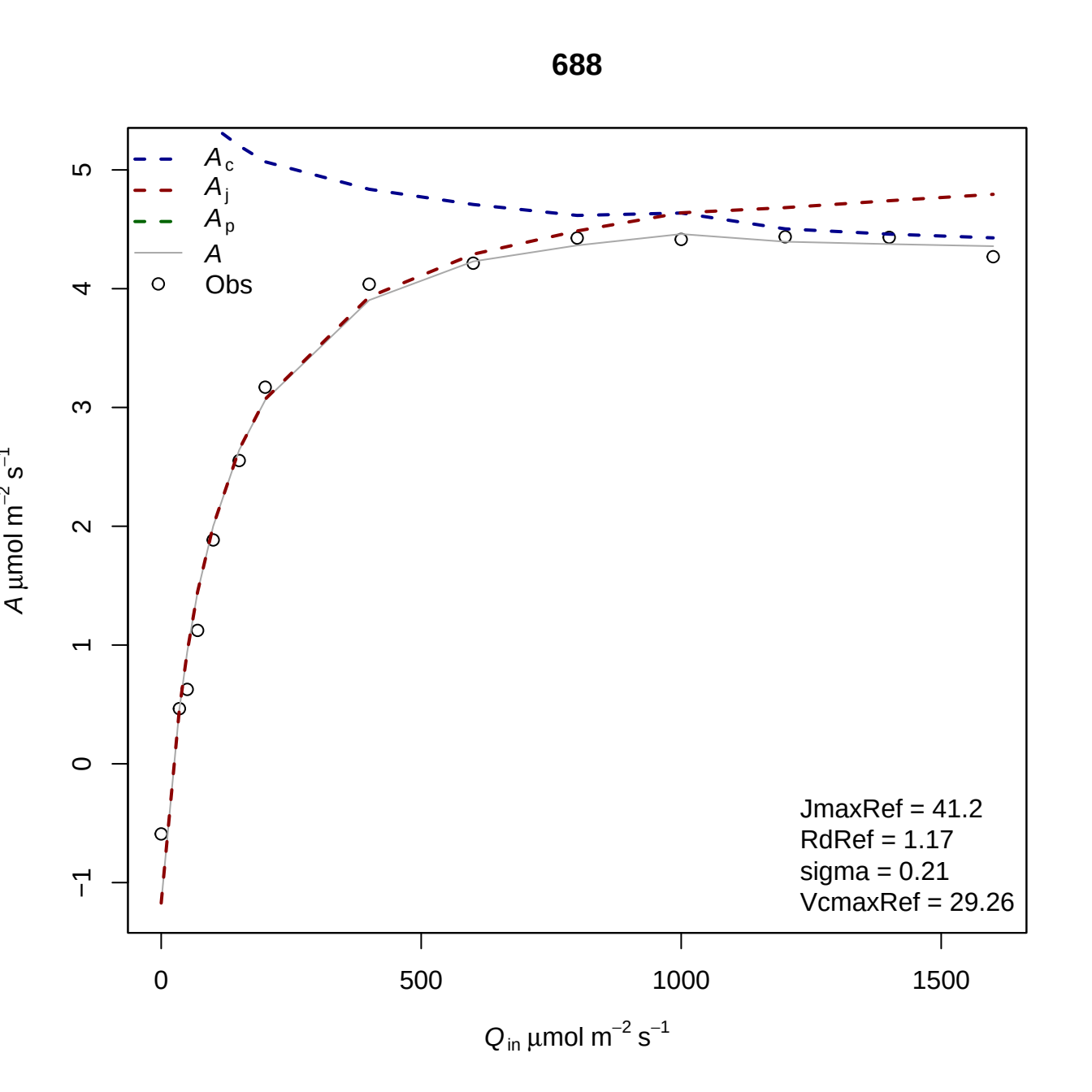

683
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.33
RdRef = 1.03
sigma = 0.19
VcmaxRef = 68.54
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

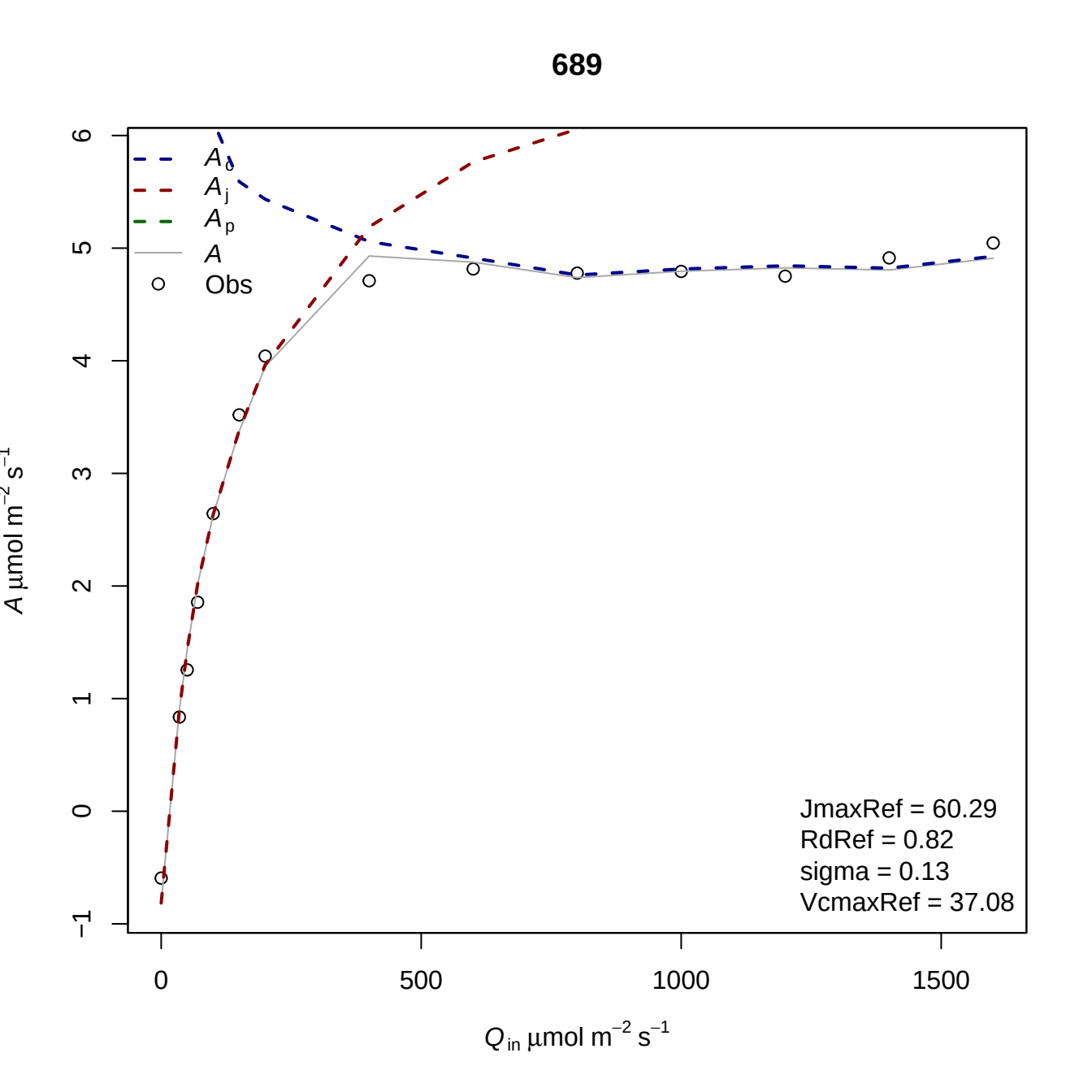

684
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.27
RdRef = 0.98
sigma = 0.14
VcmaxRef = 41.33
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

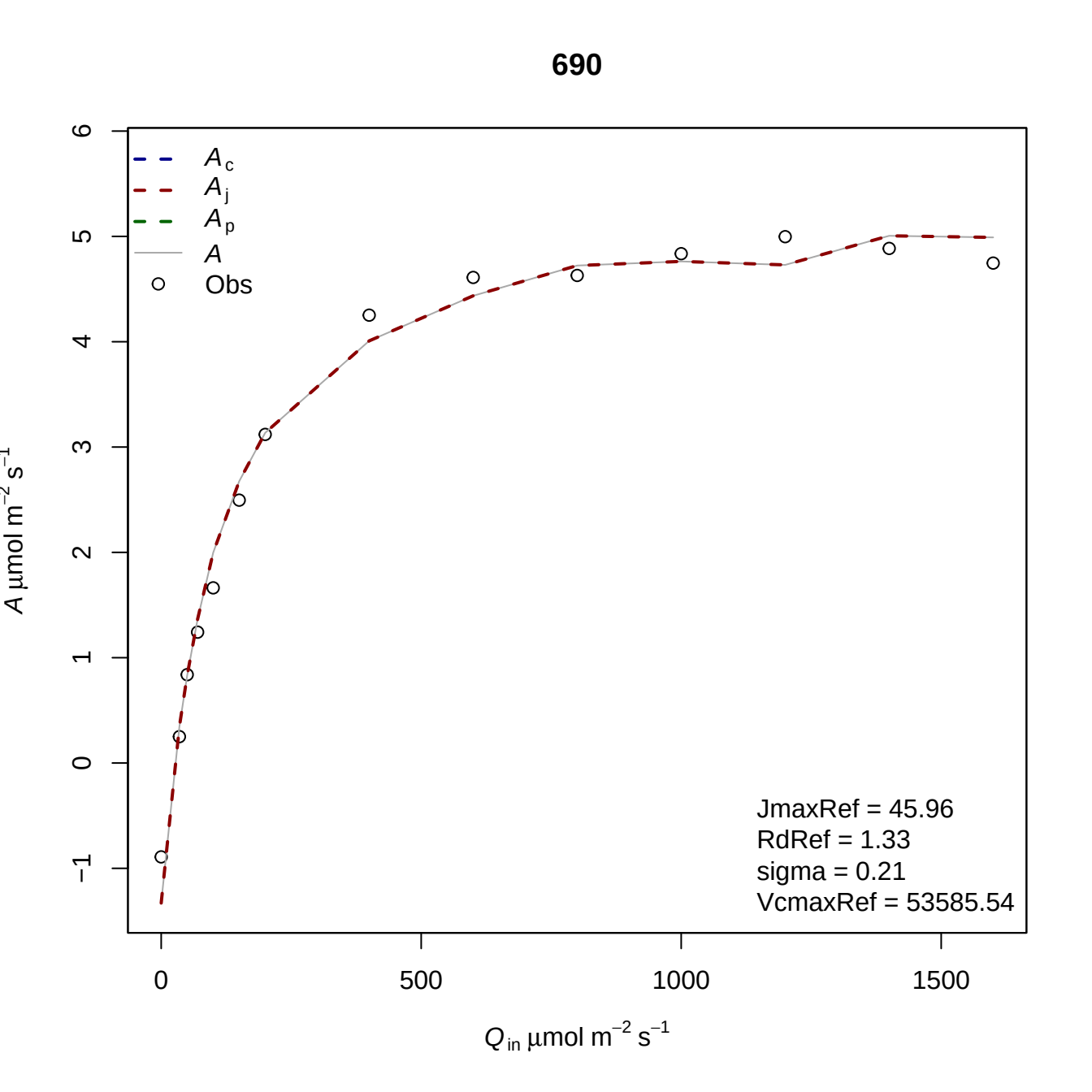

685
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.96
RdRef = 0.67
sigma = 0.21
VcmaxRef = 75.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

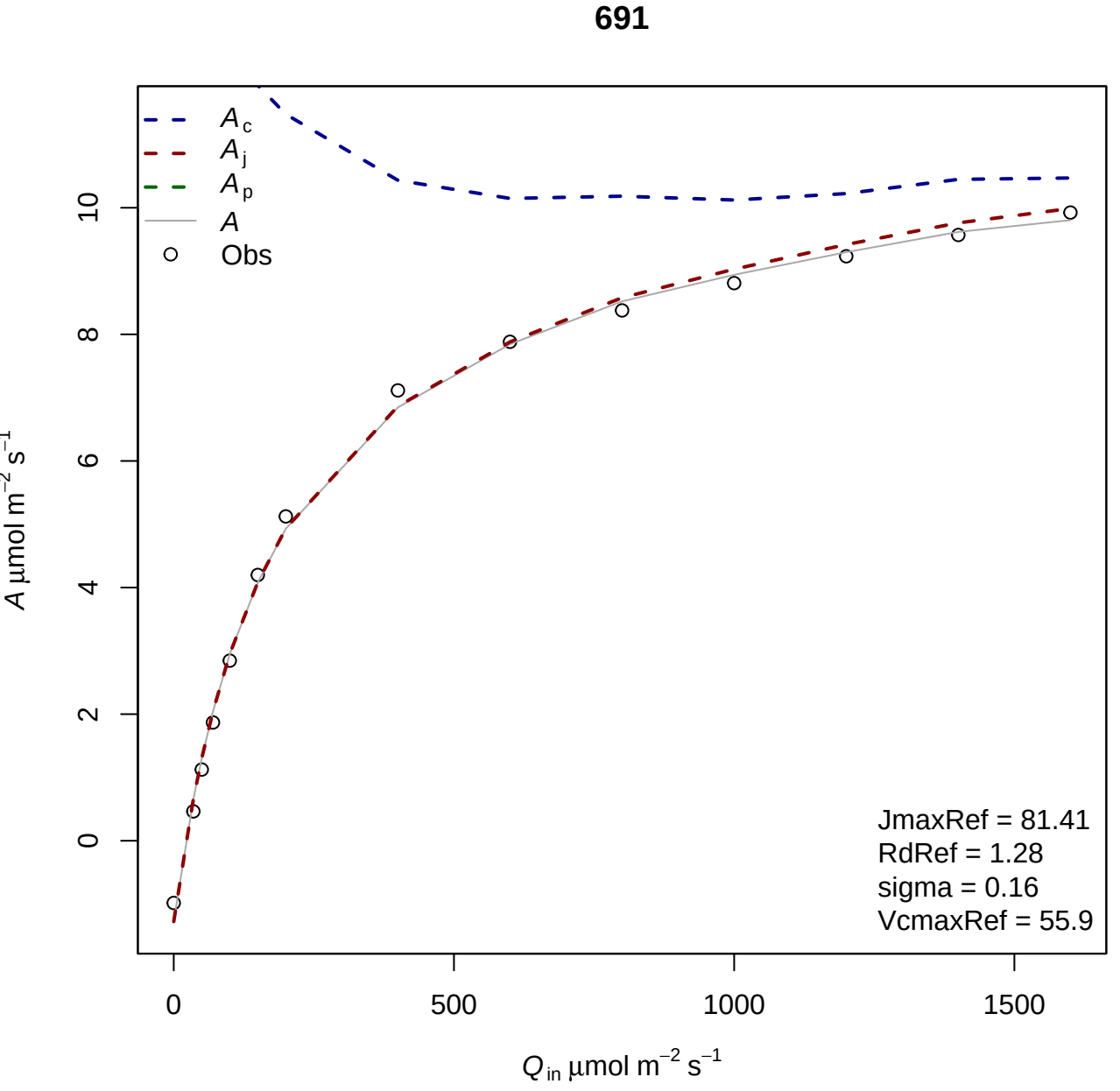

686
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.66
RdRef = 0.63
sigma = 0.15
VcmaxRef = 48.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

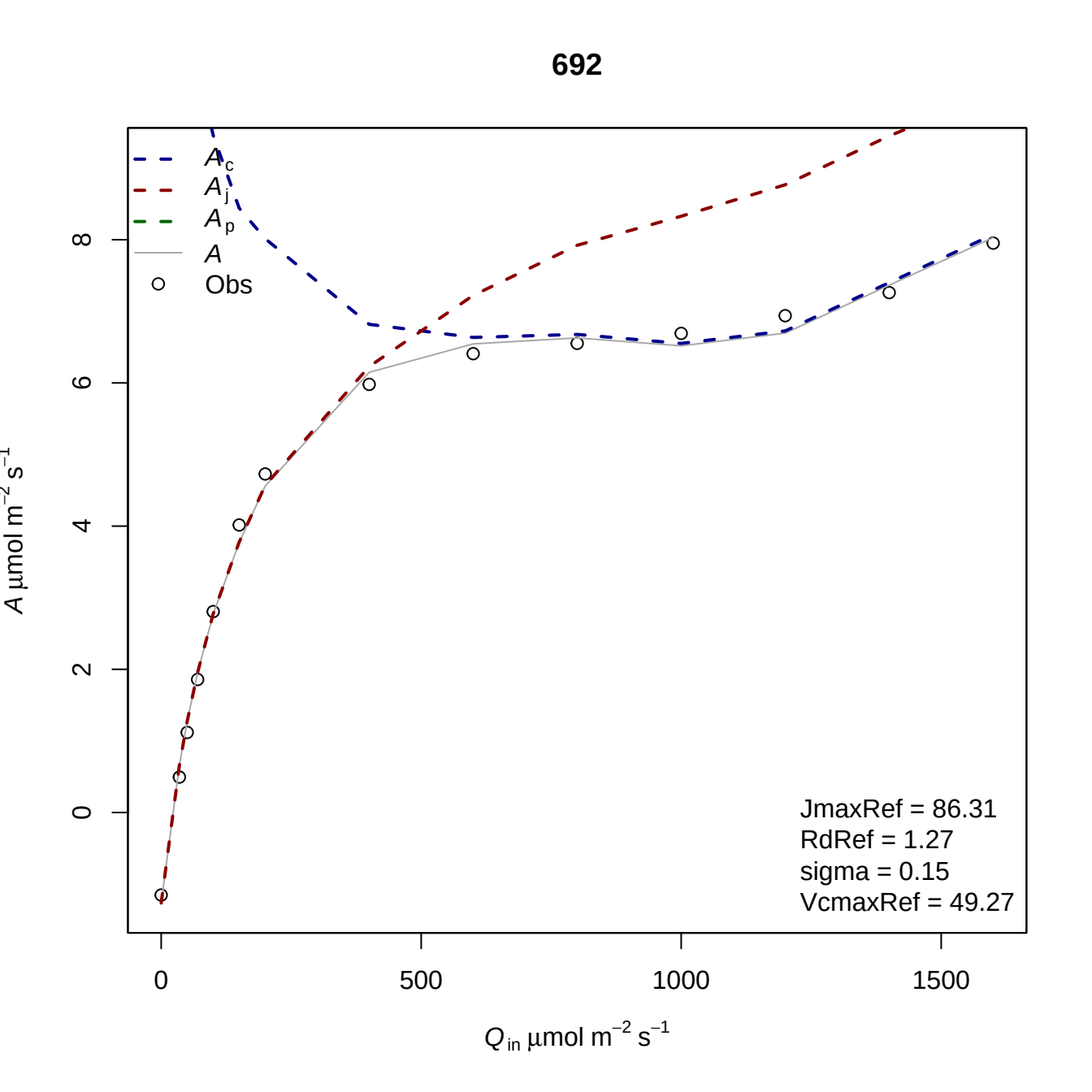

687
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.18
RdRef = 0.71
sigma = 0.2
VcmaxRef = 46.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

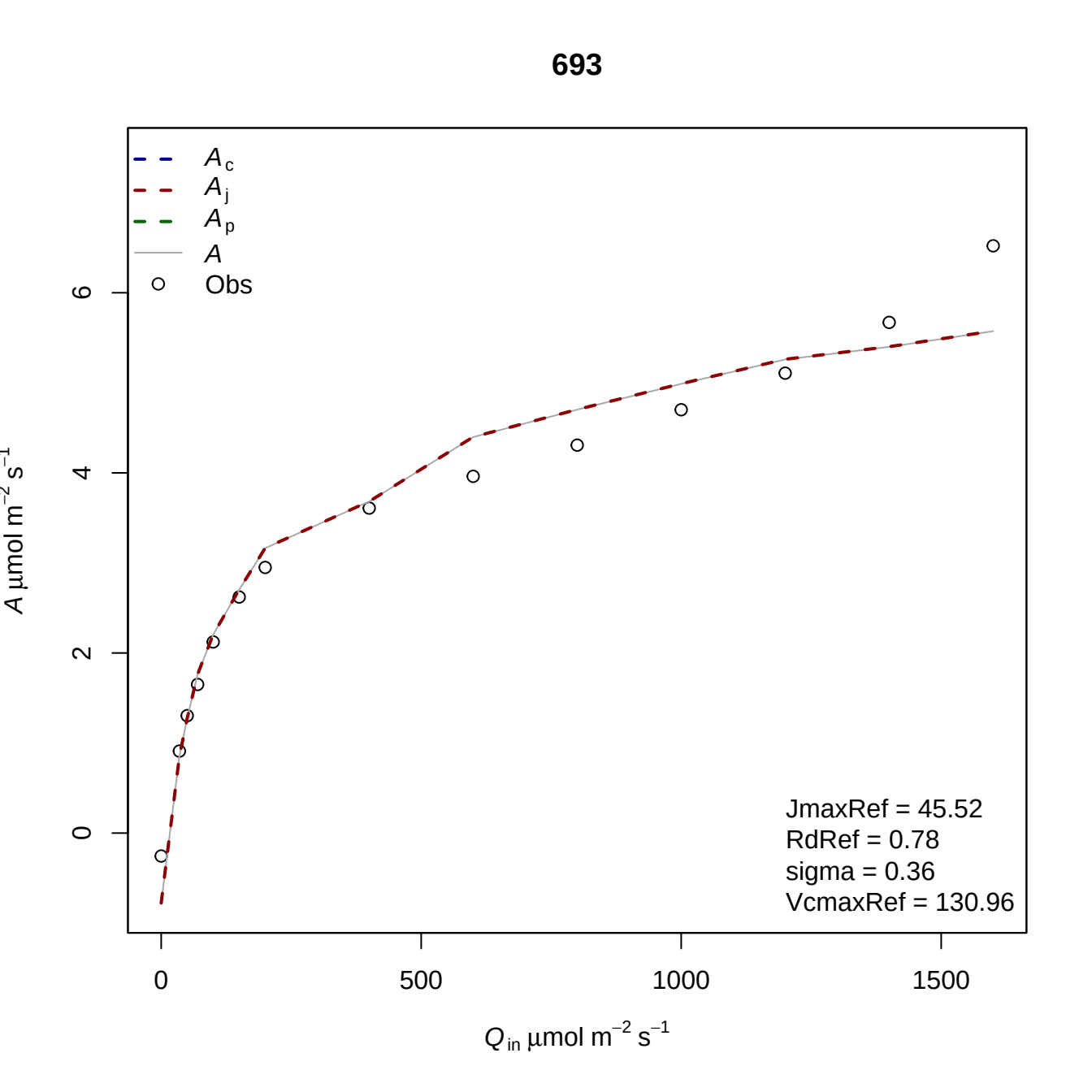

688
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.2
RdRef = 1.17
sigma = 0.21
VcmaxRef = 29.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

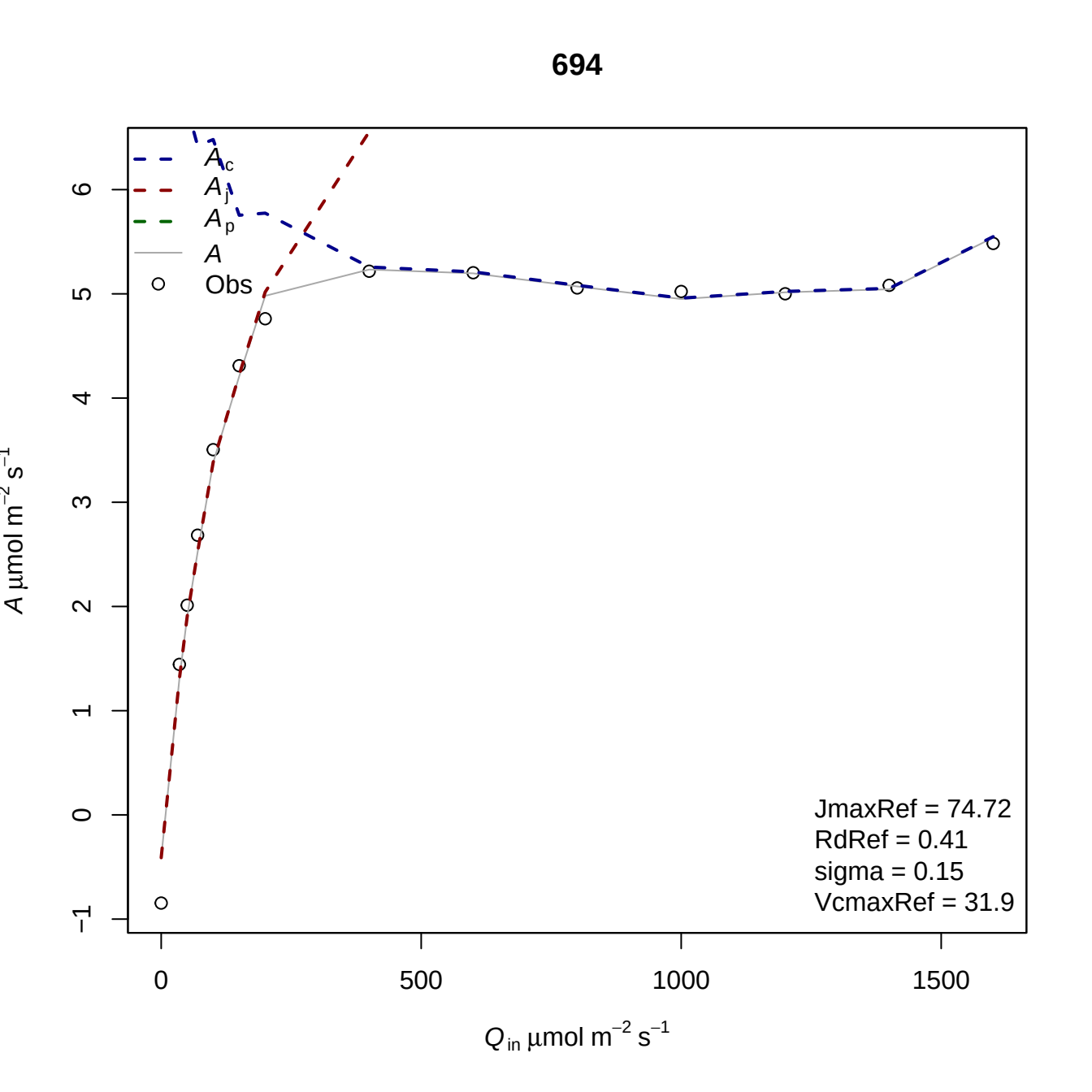

689
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.29
RdRef = 0.82
sigma = 0.13
VcmaxRef = 37.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

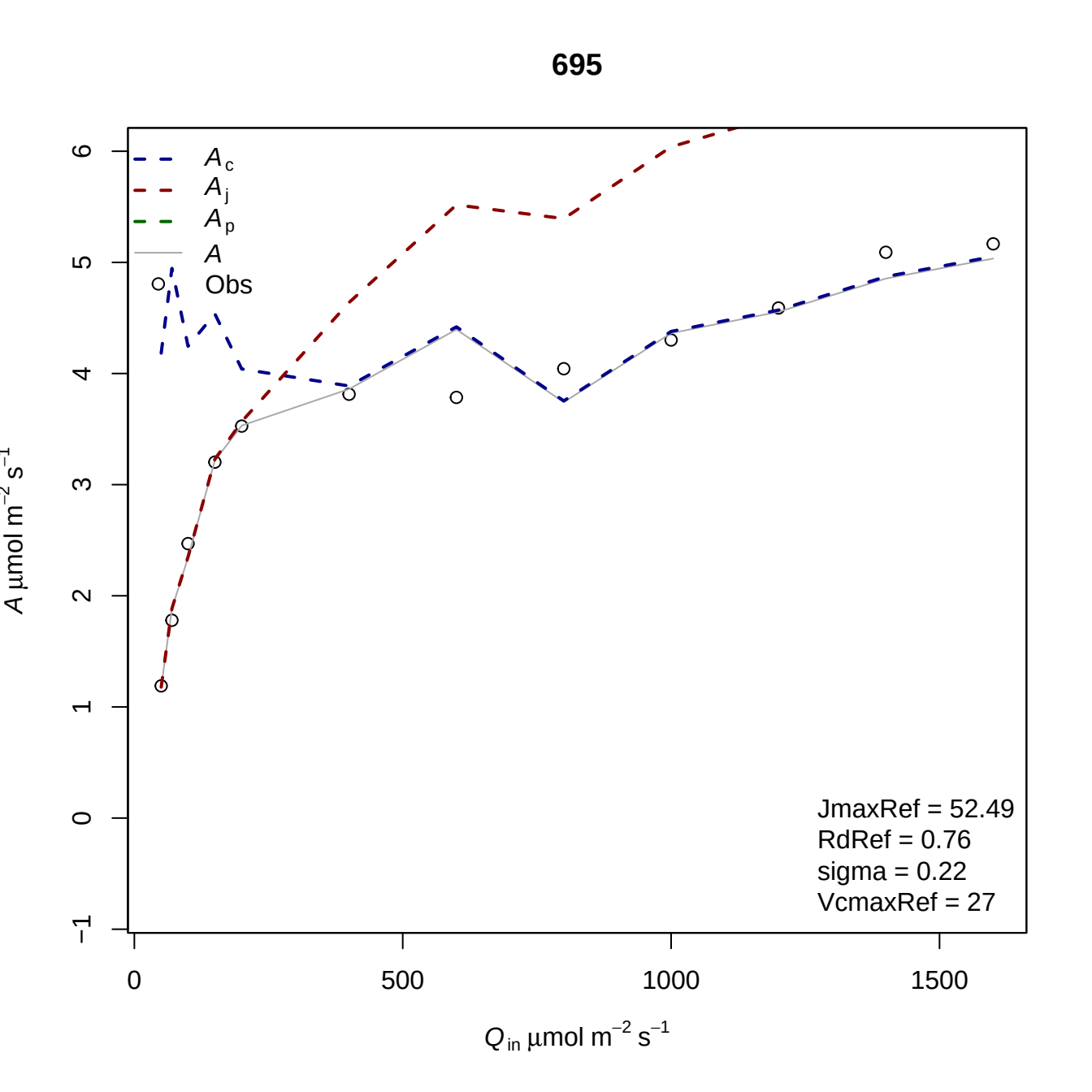

690
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.96
RdRef = 1.33
sigma = 0.21
VcmaxRef = 53585.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

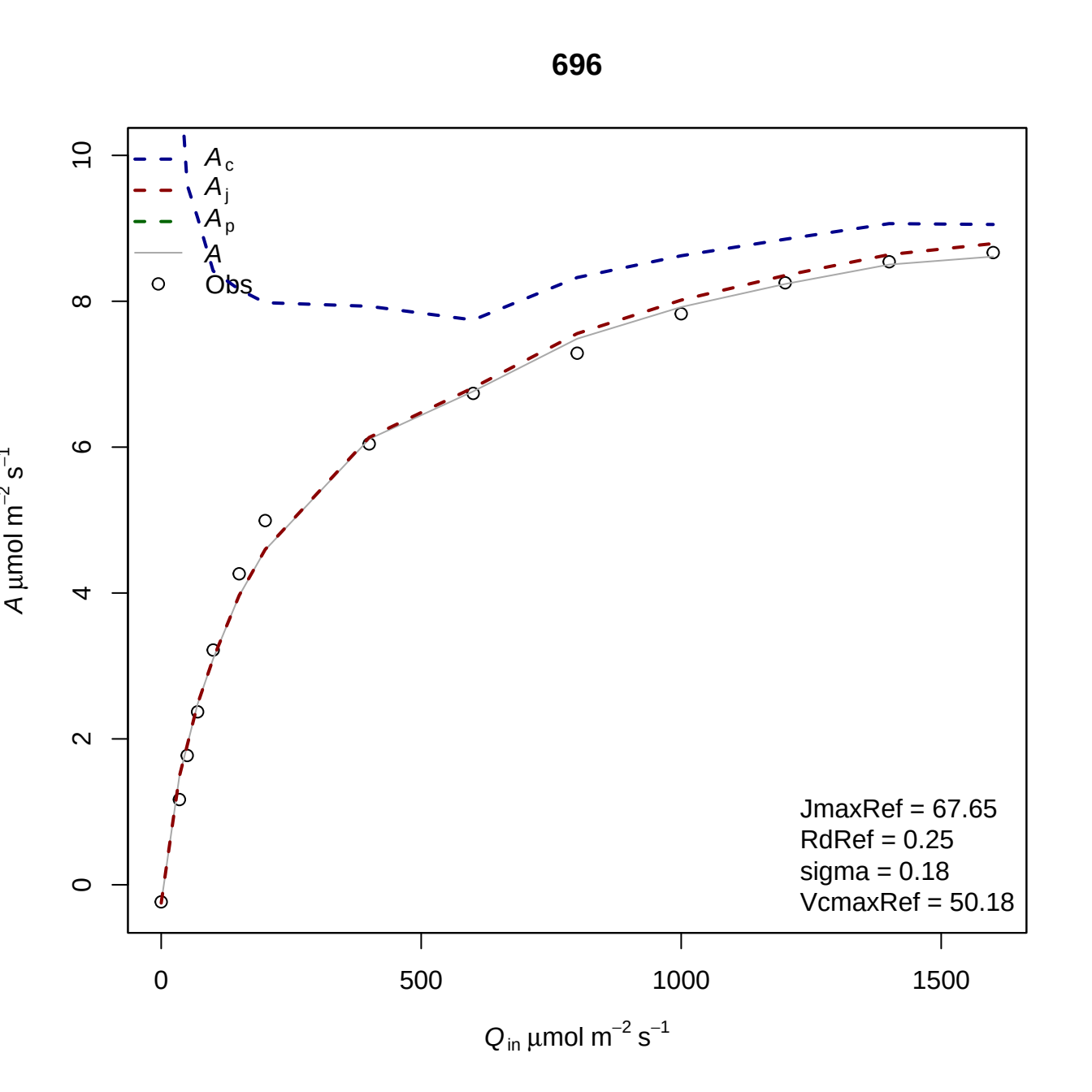

691
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.41
RdRef = 1.28
sigma = 0.16
VcmaxRef = 55.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

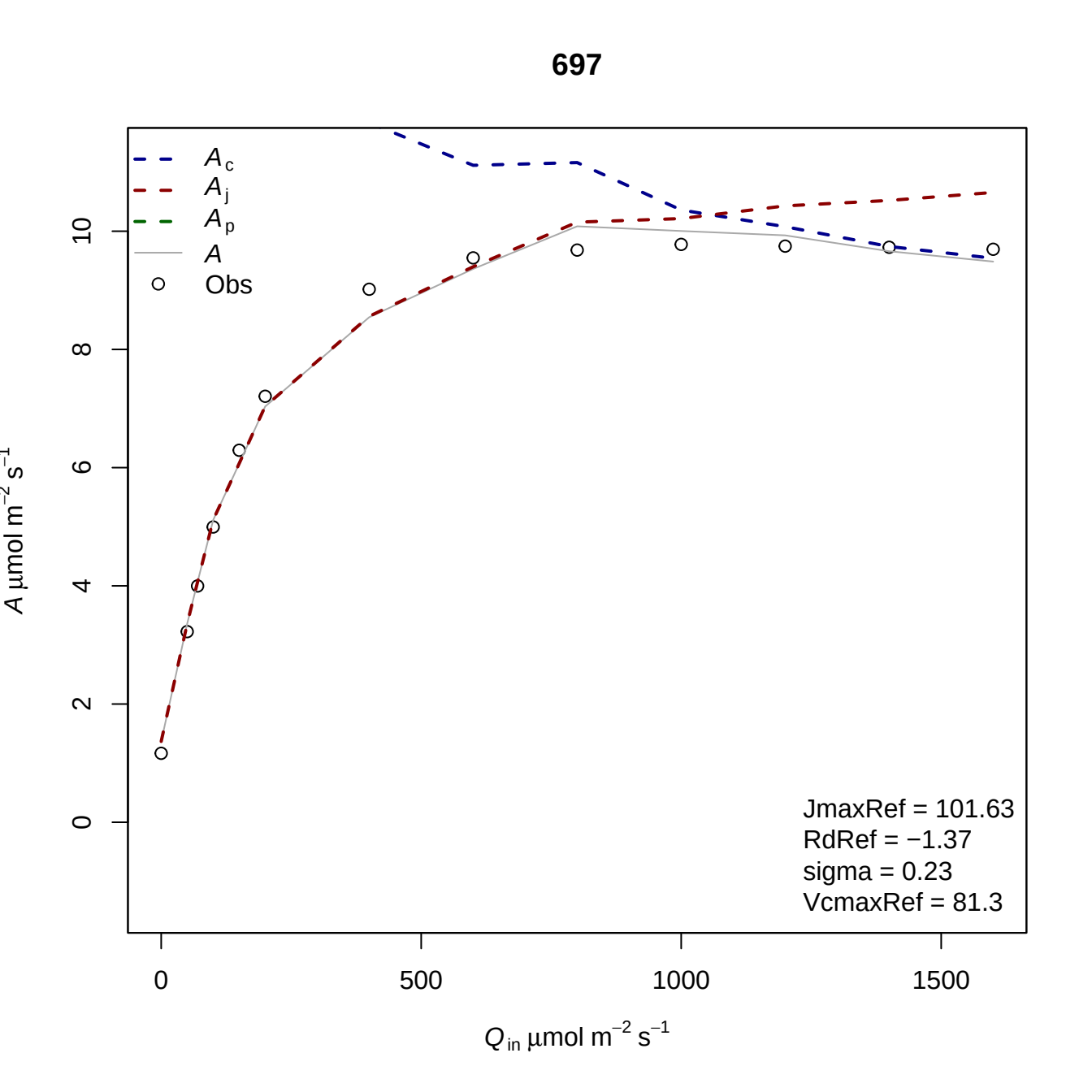

692
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.31
RdRef = 1.27
sigma = 0.15
VcmaxRef = 49.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

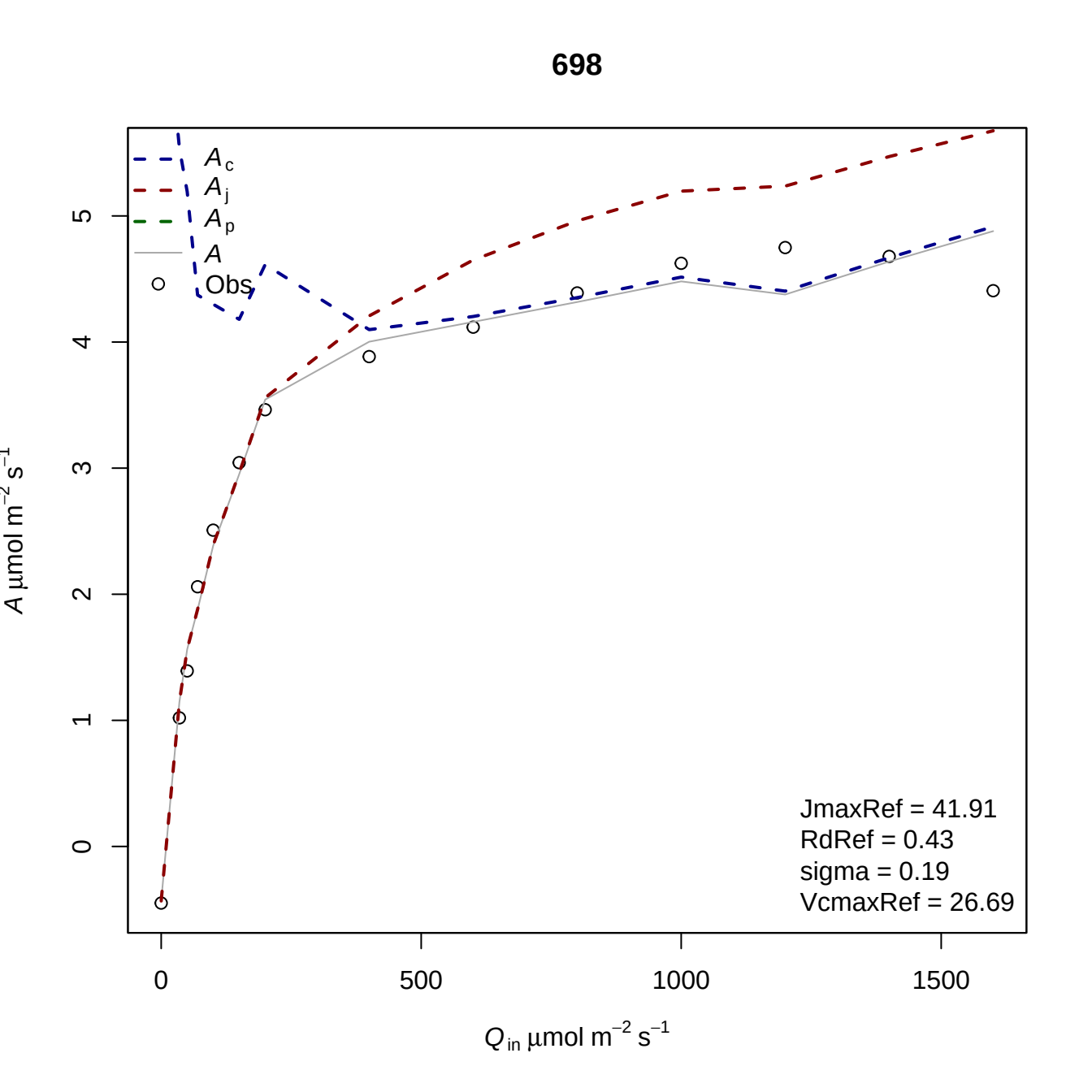

693
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.52
RdRef = 0.78
sigma = 0.36
VcmaxRef = 130.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

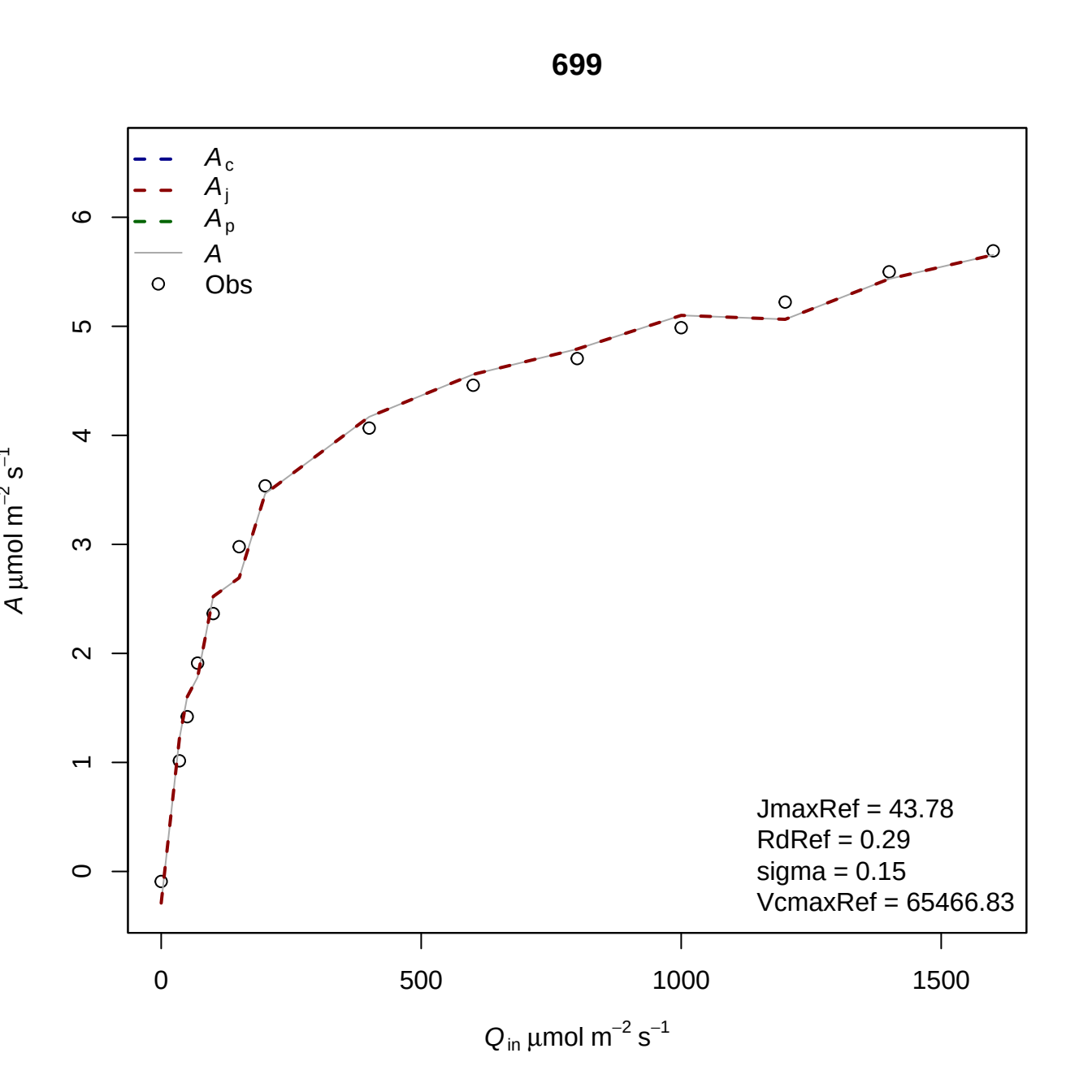

694
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.72
RdRef = 0.41
sigma = 0.15
VcmaxRef = 31.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

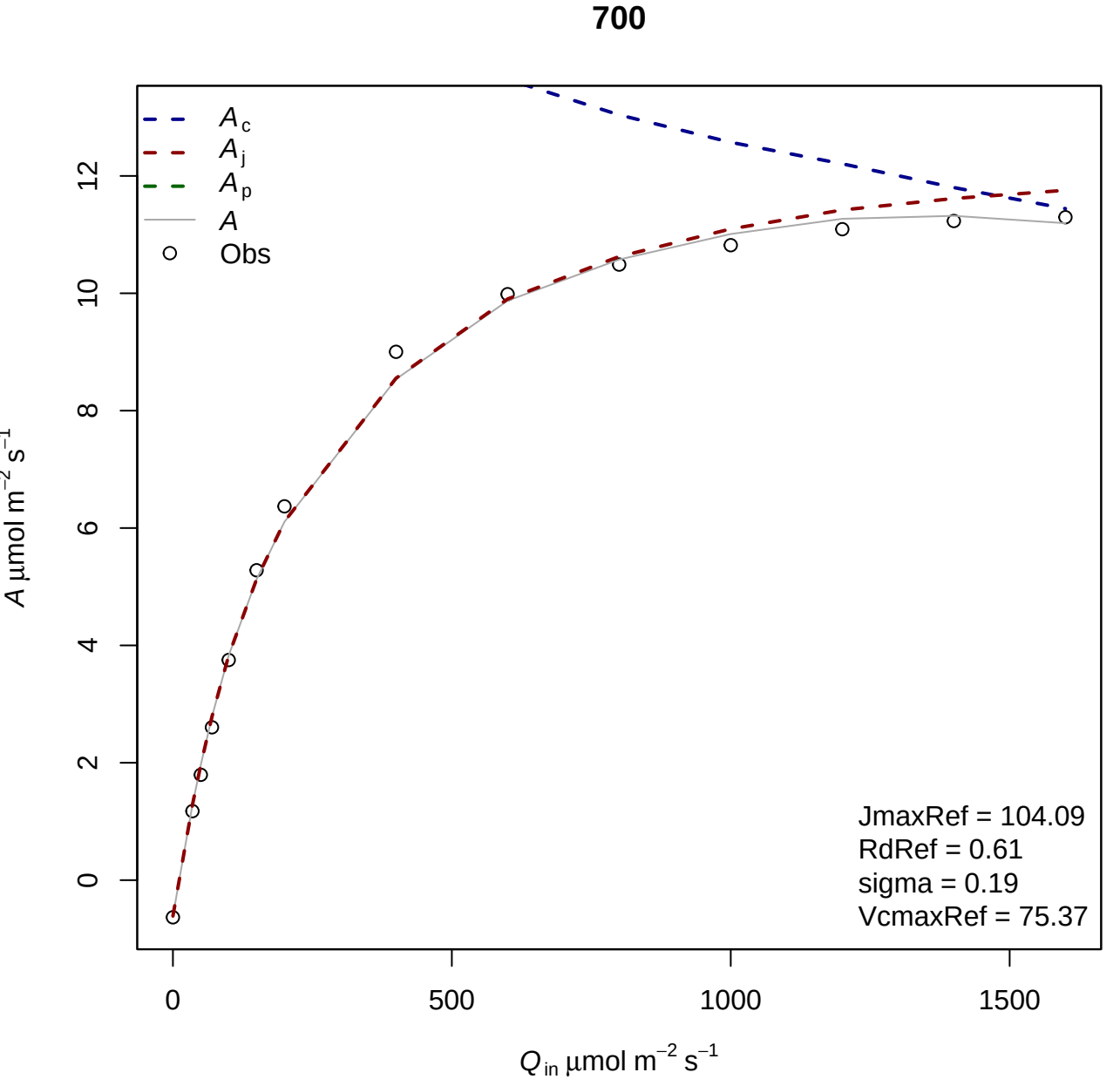

695
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 52.49
RdRef = 0.76
sigma = 0.22
VcmaxRef = 27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

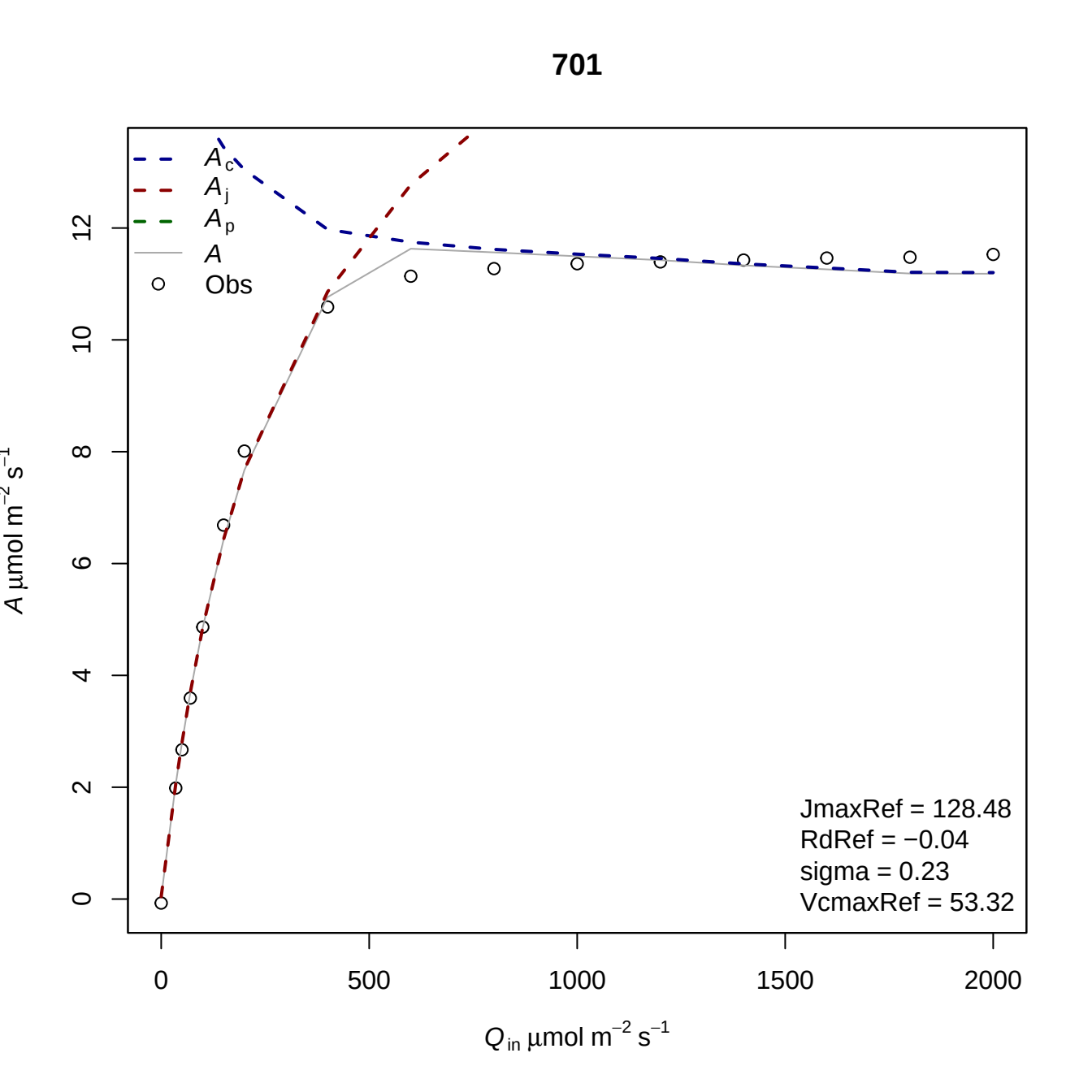

696
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.65
RdRef = 0.25
sigma = 0.18
VcmaxRef = 50.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

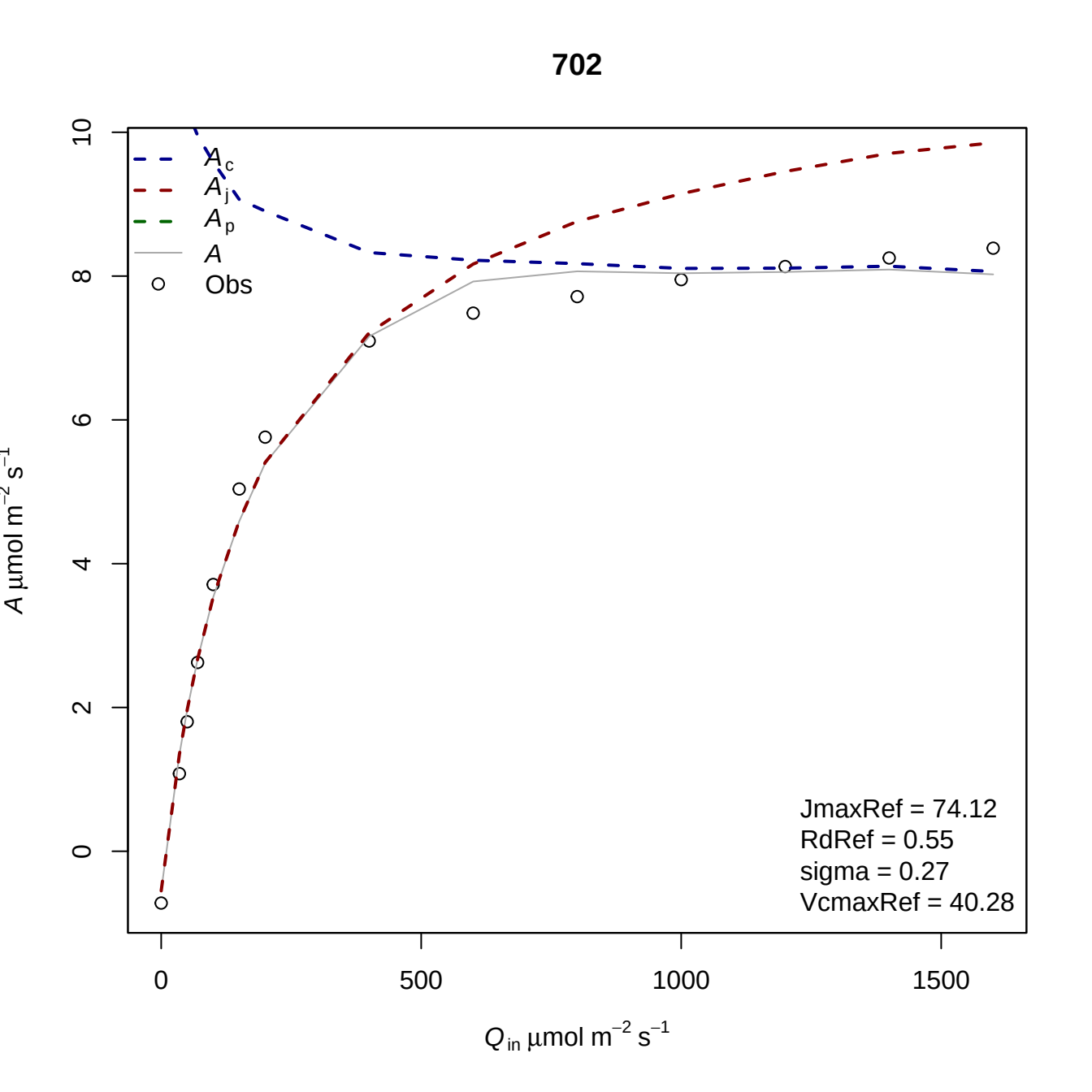

697
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.63
RdRef = −1.37
sigma = 0.23
VcmaxRef = 81.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

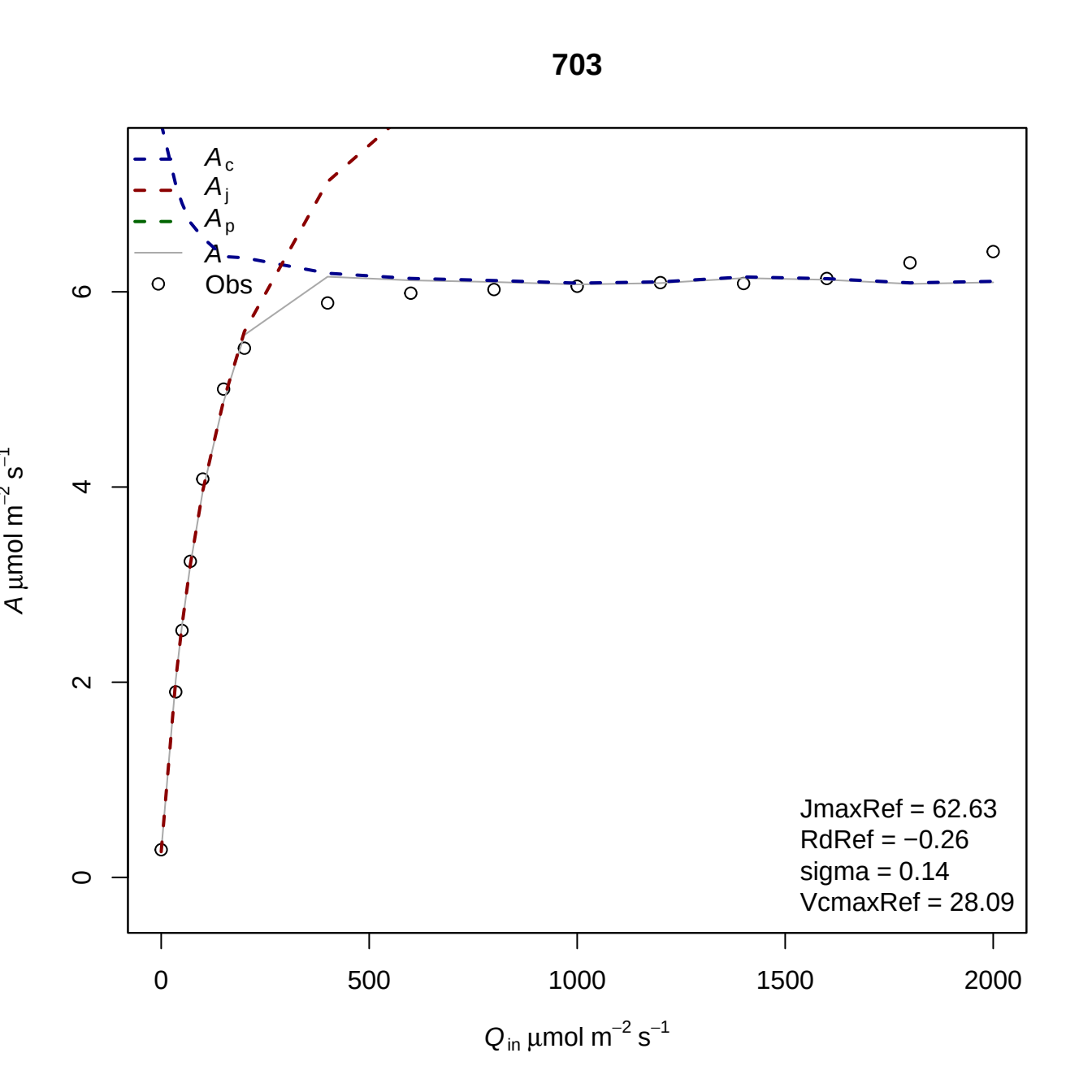

698
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.91
RdRef = 0.43
sigma = 0.19
VcmaxRef = 26.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

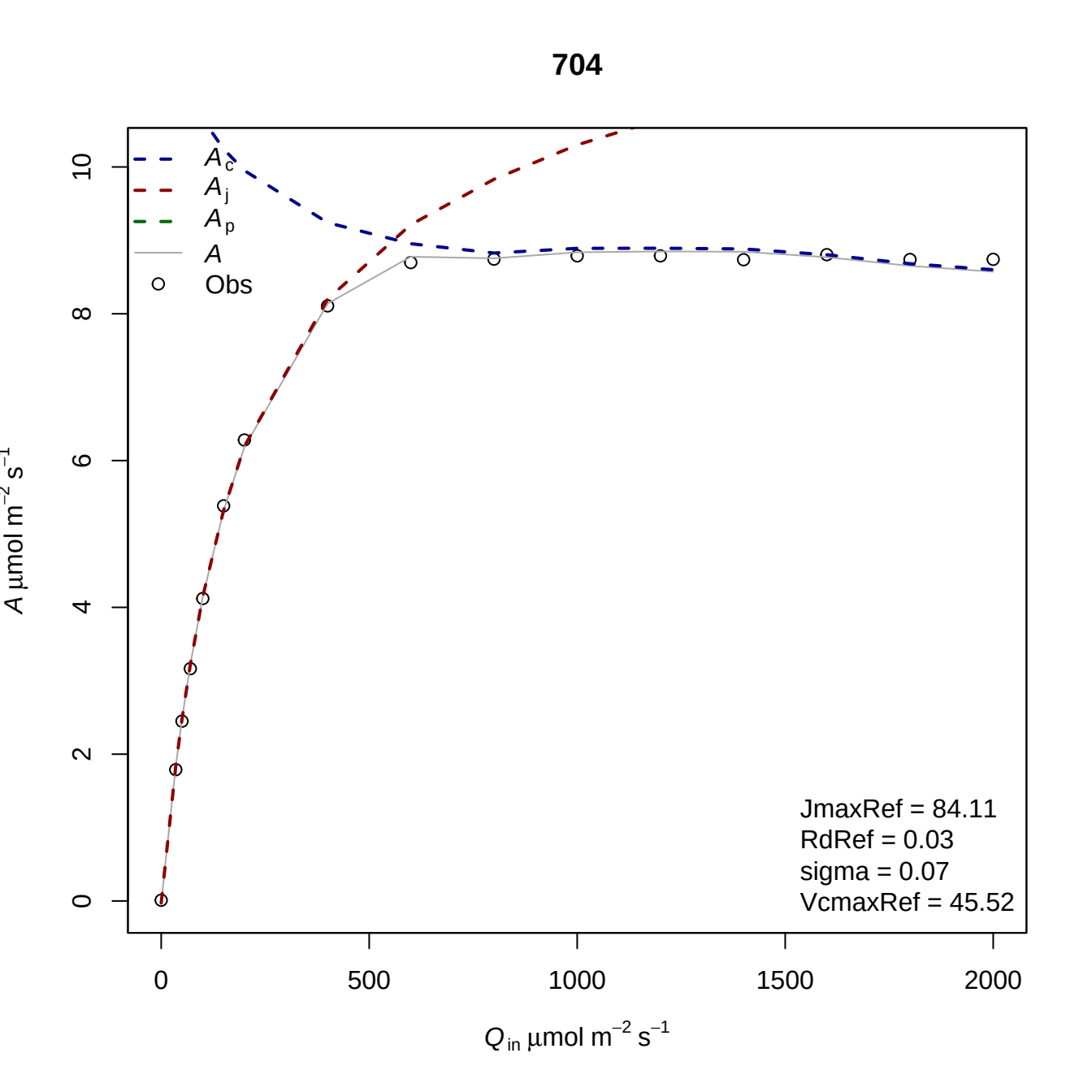

699
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.78
RdRef = 0.29
sigma = 0.15
VcmaxRef = 65466.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

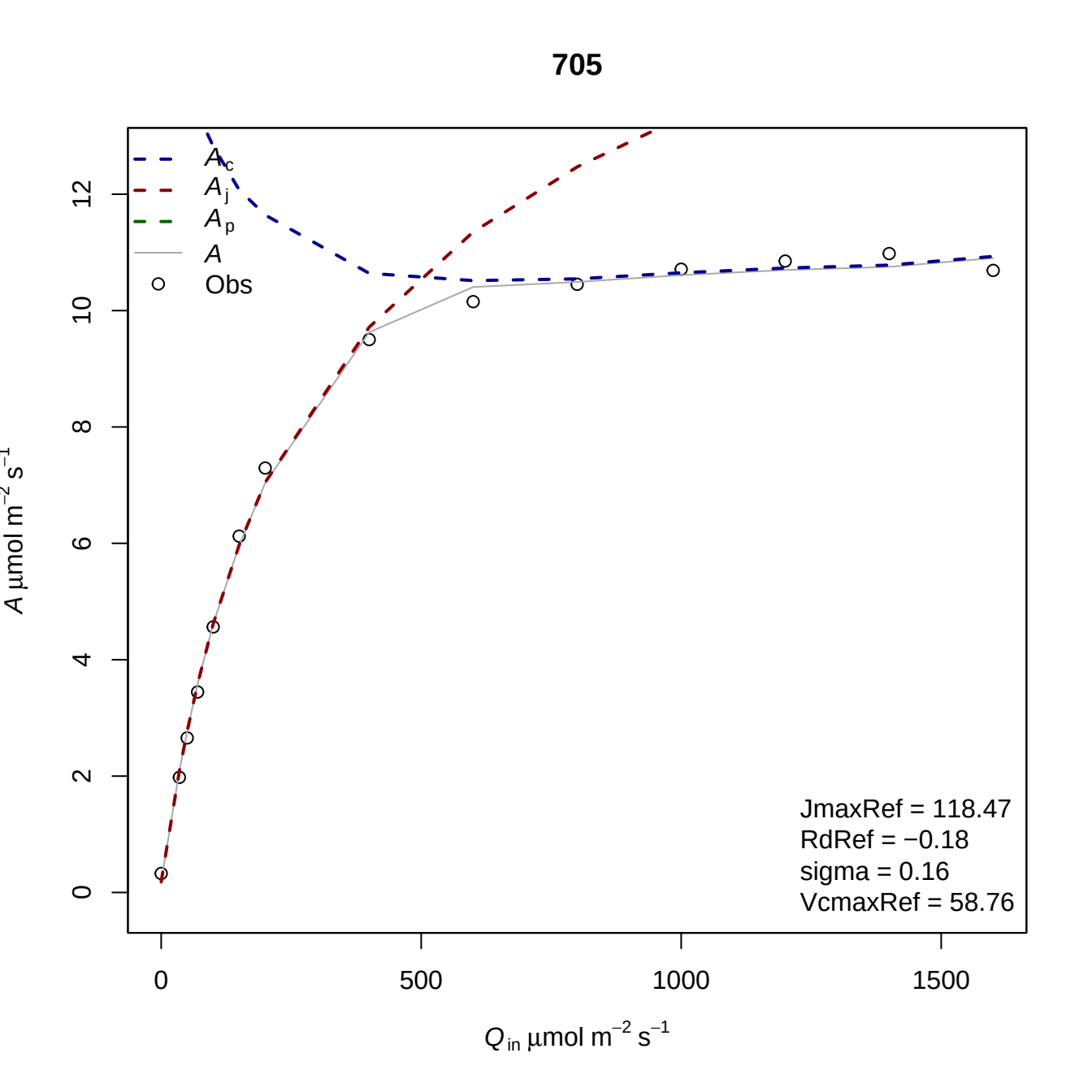

700
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.09
RdRef = 0.61
sigma = 0.19
VcmaxRef = 75.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

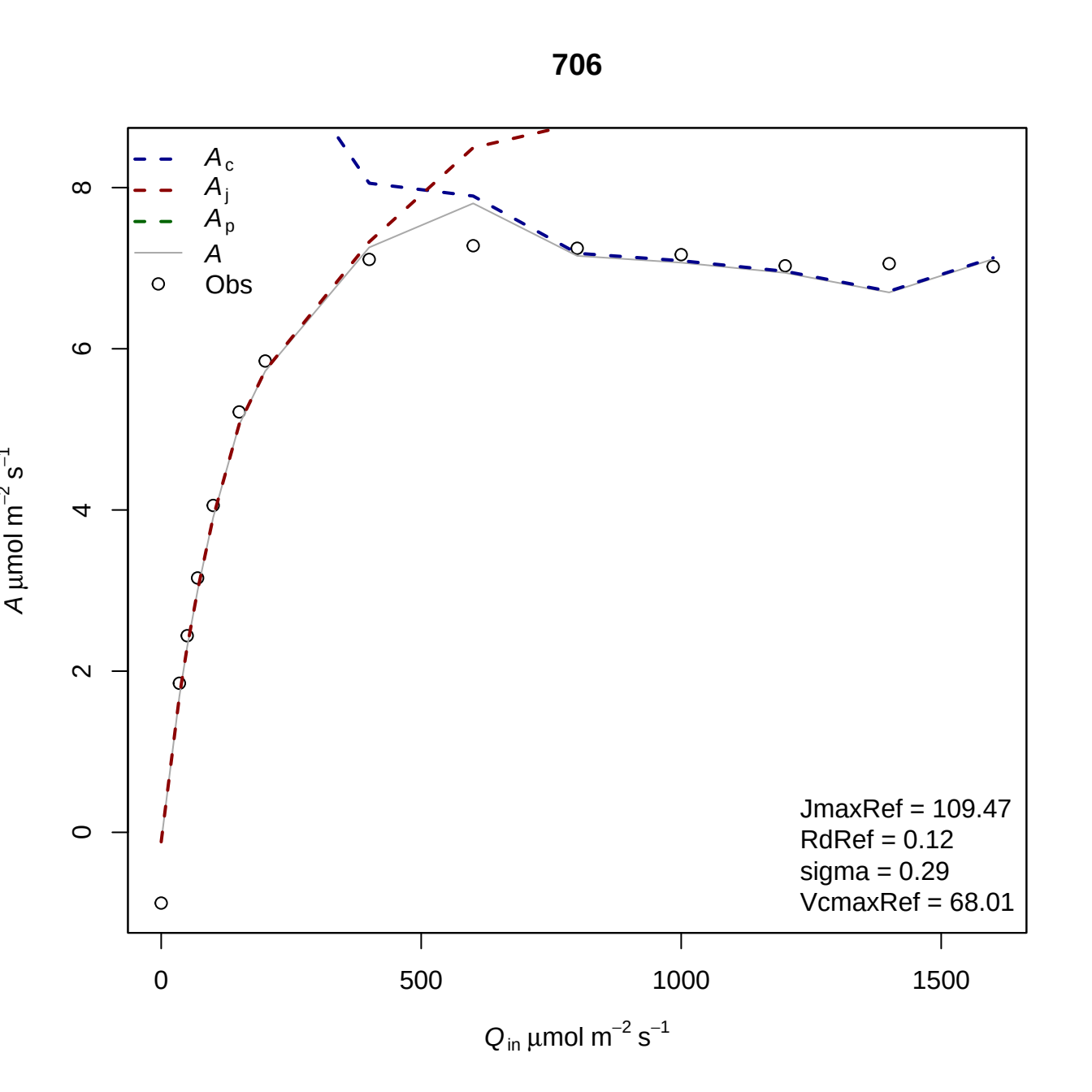

701
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.48
RdRef = −0.04
sigma = 0.23
VcmaxRef = 53.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

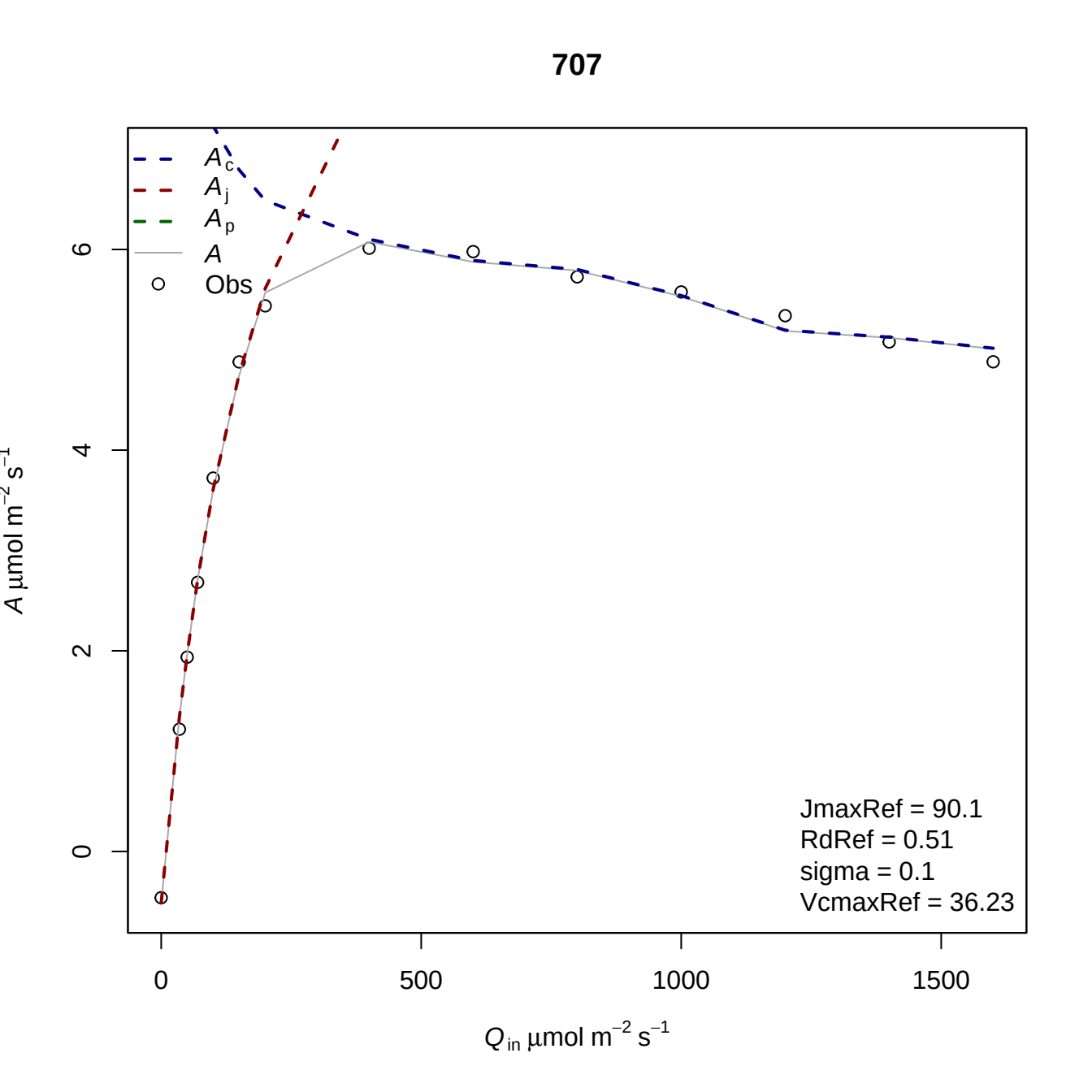

702
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.12
RdRef = 0.55
sigma = 0.27
VcmaxRef = 40.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

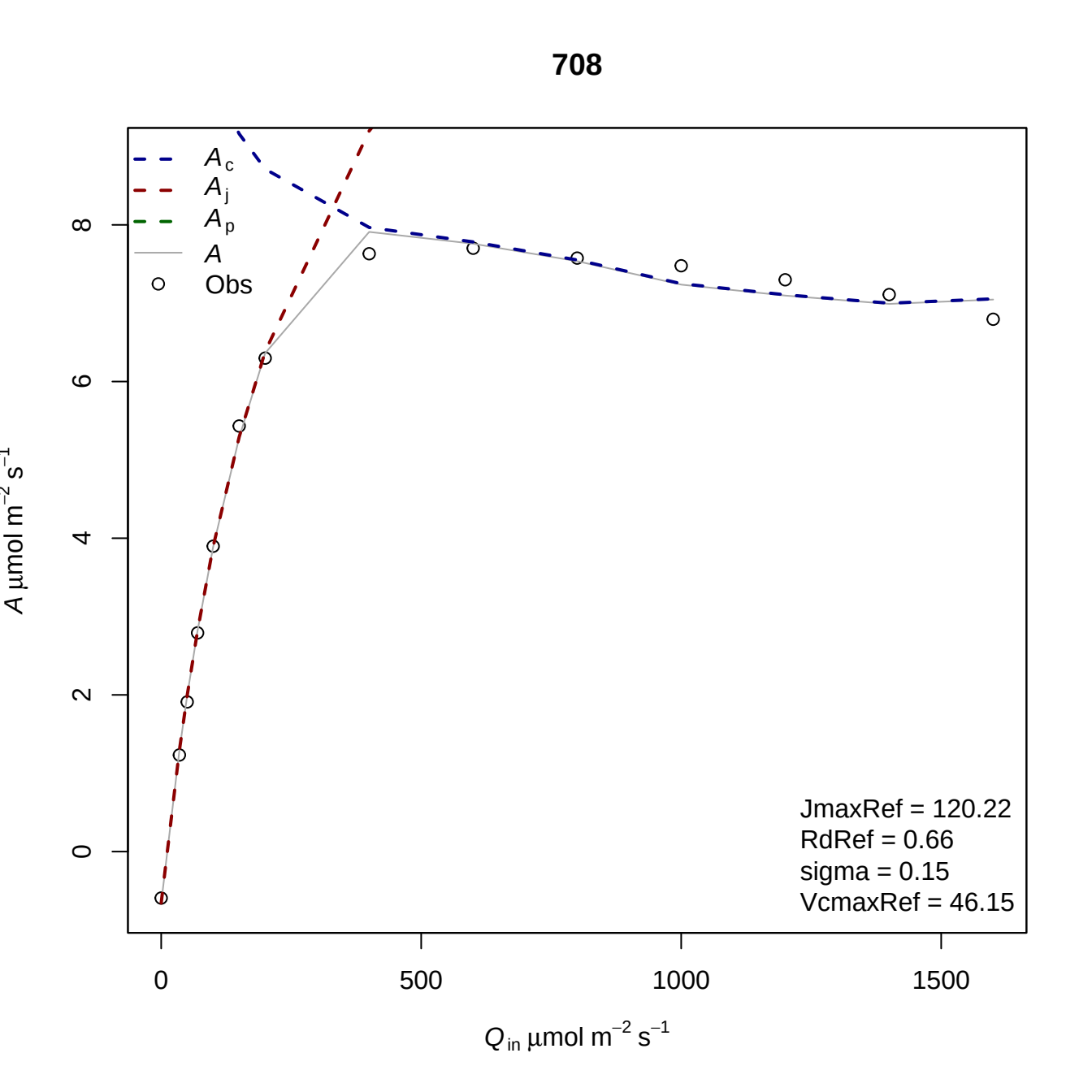

703
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.63
RdRef = −0.26
sigma = 0.14
VcmaxRef = 28.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

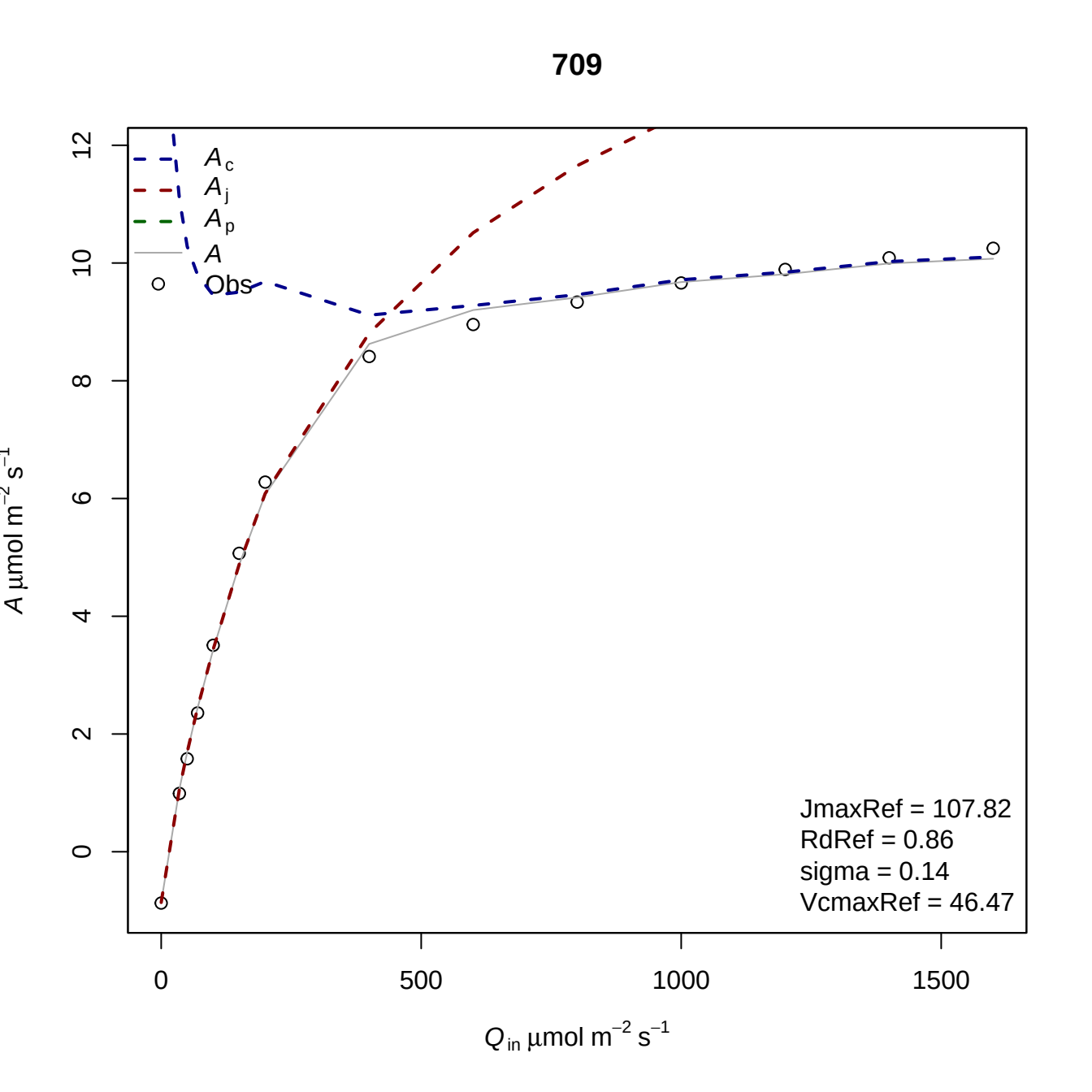

704
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.11
RdRef = 0.03
sigma = 0.07
VcmaxRef = 45.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

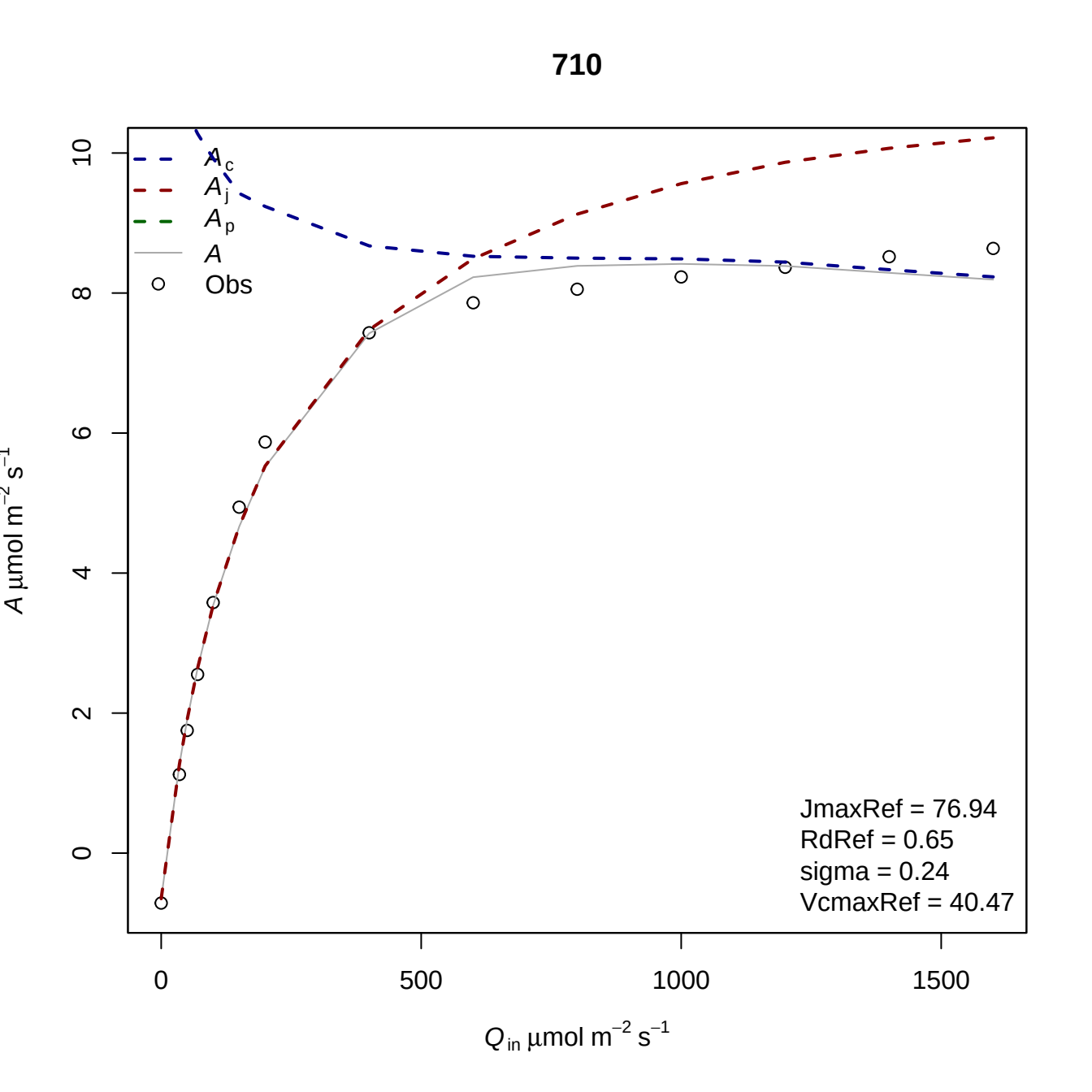

705
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.47
RdRef = −0.18
sigma = 0.16
VcmaxRef = 58.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

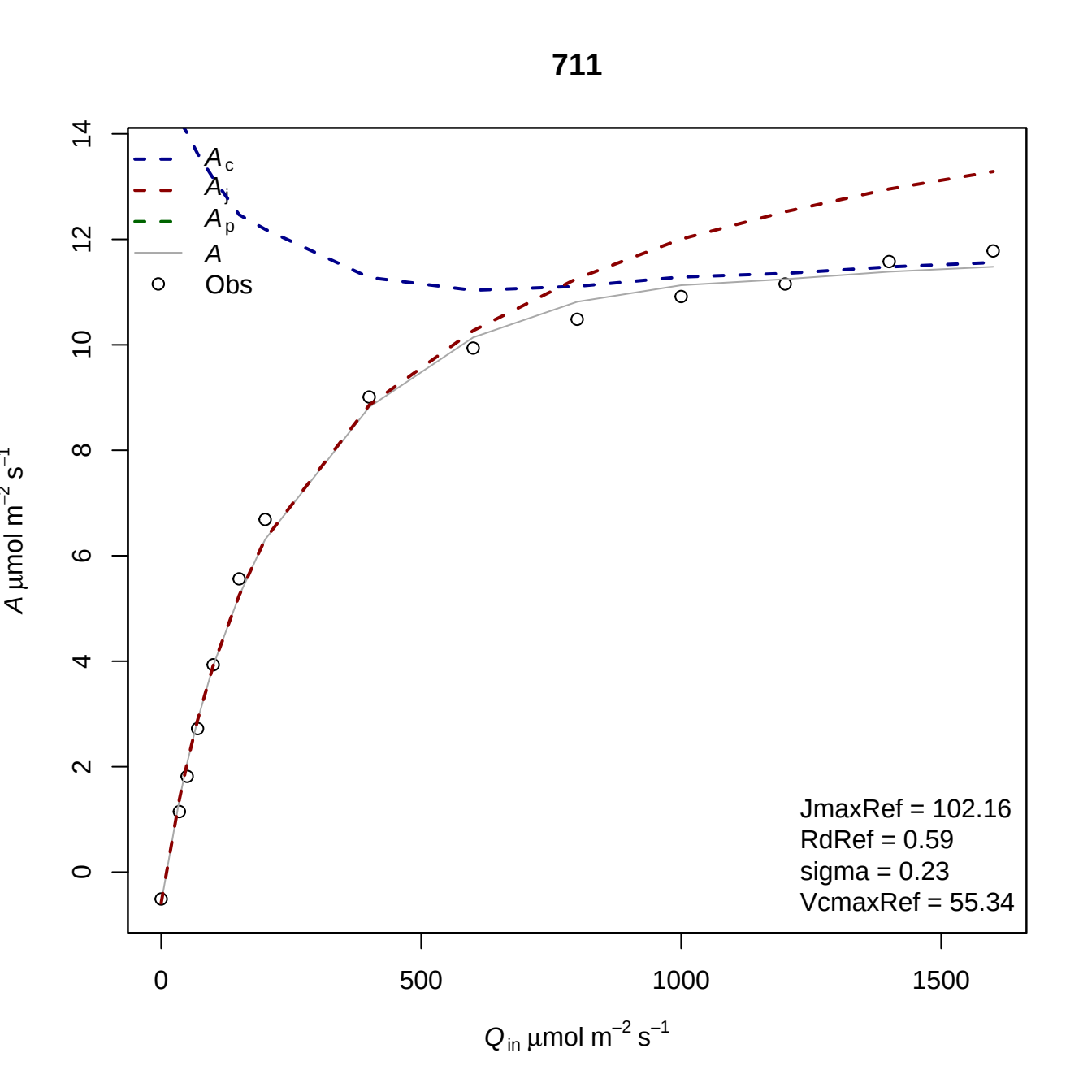

706
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.47
RdRef = 0.12
sigma = 0.29
VcmaxRef = 68.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

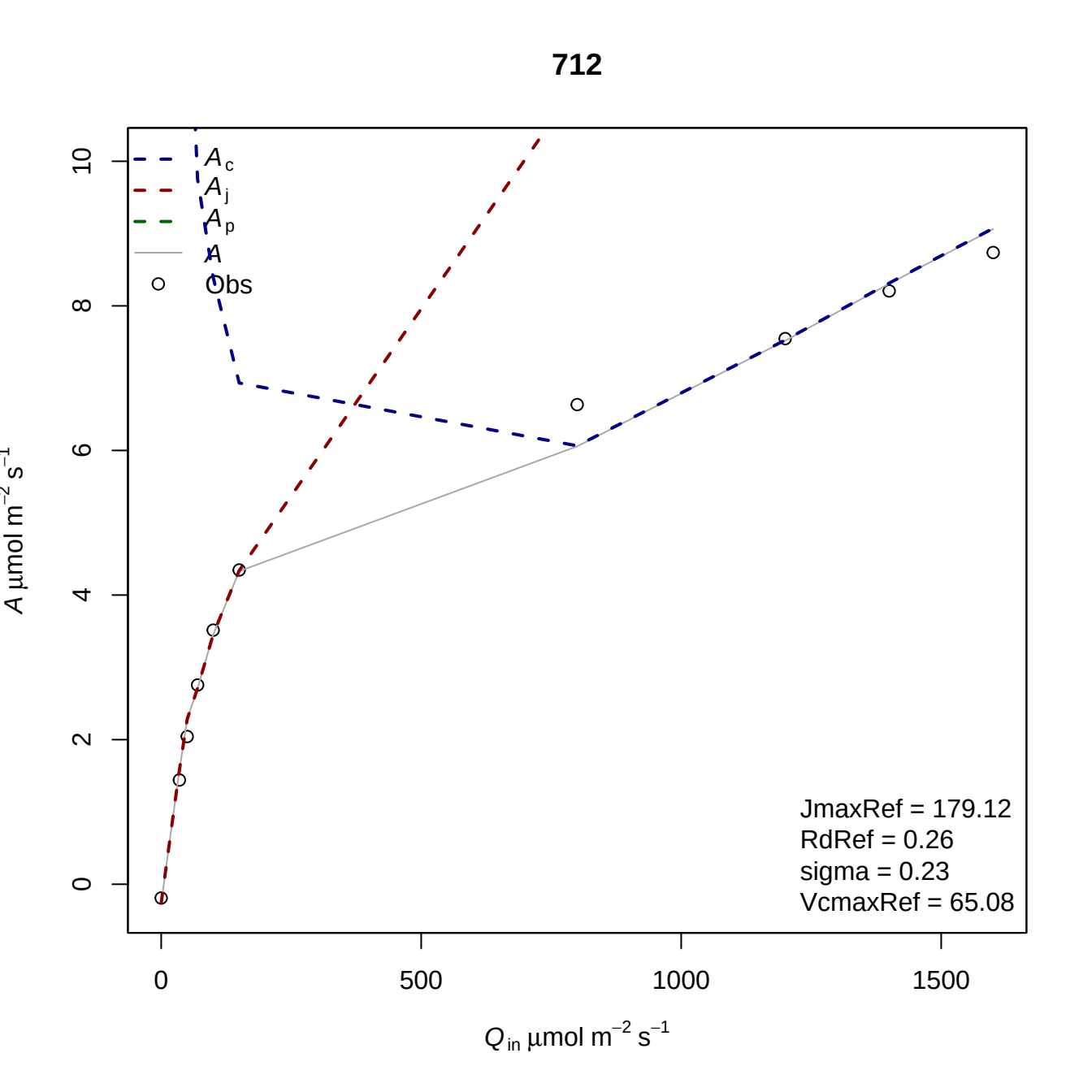

707
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.1
RdRef = 0.51
sigma = 0.1
VcmaxRef = 36.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

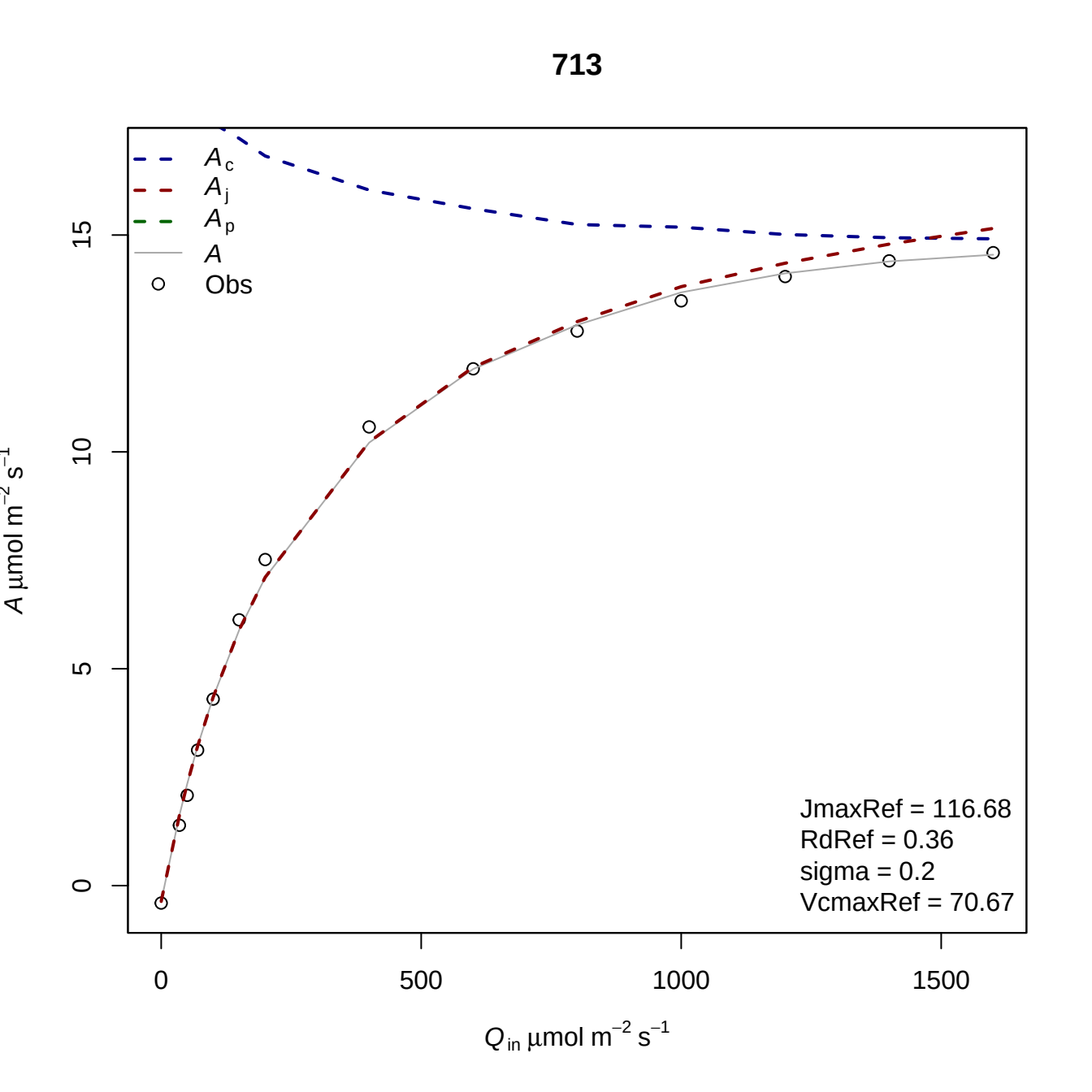

708
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.22
RdRef = 0.66
sigma = 0.15
VcmaxRef = 46.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

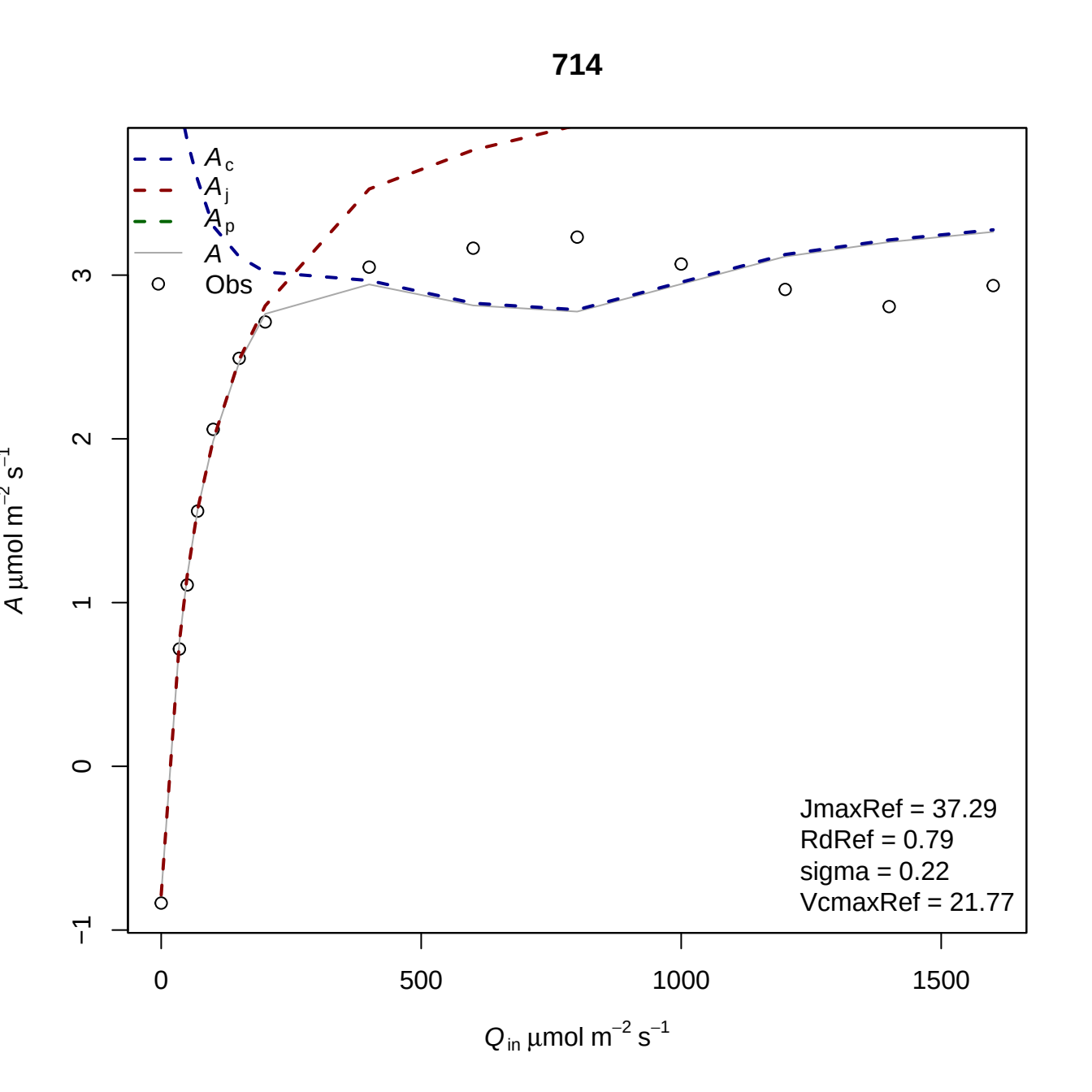

709
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.82
RdRef = 0.86
sigma = 0.14
VcmaxRef = 46.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

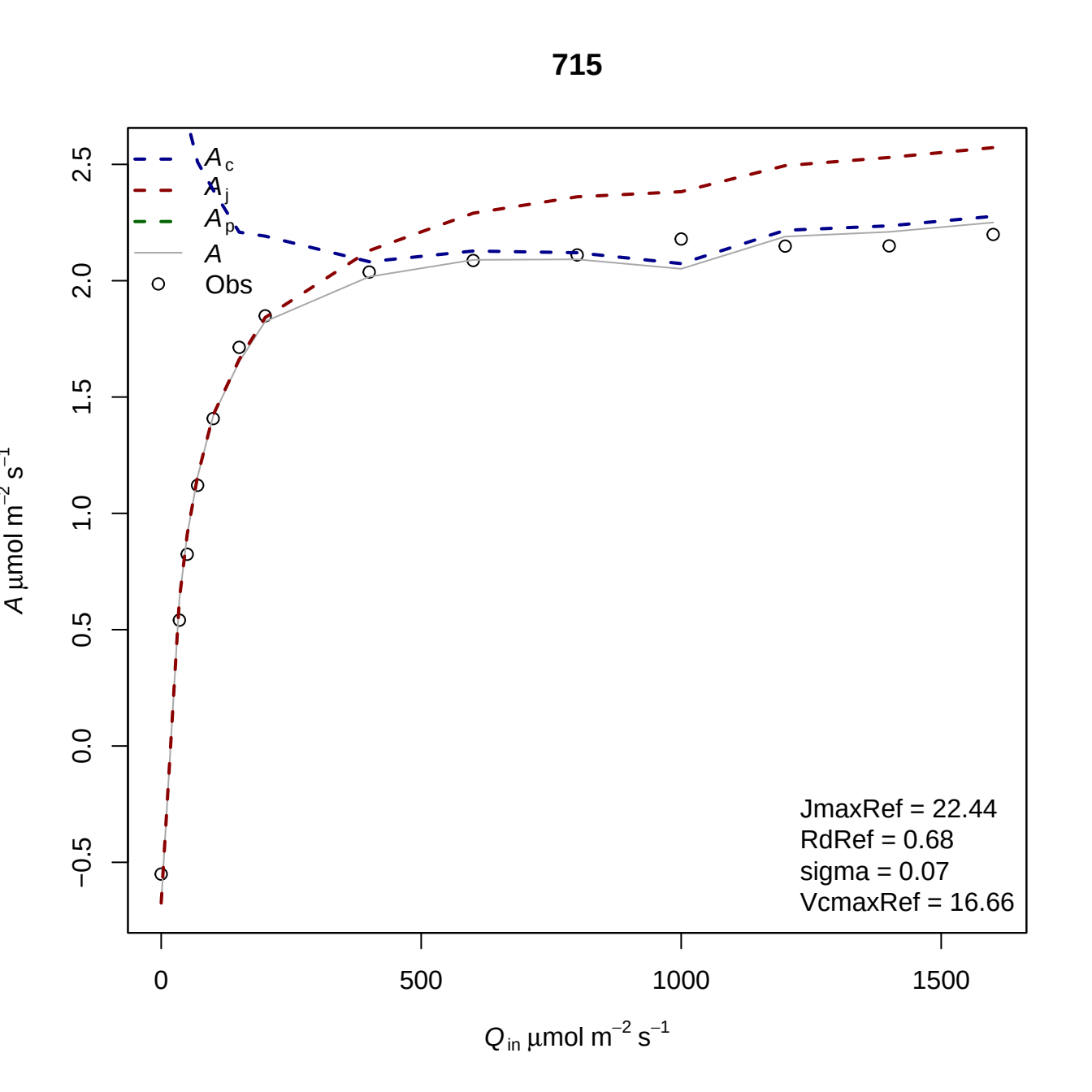

710
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.94
RdRef = 0.65
sigma = 0.24
VcmaxRef = 40.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

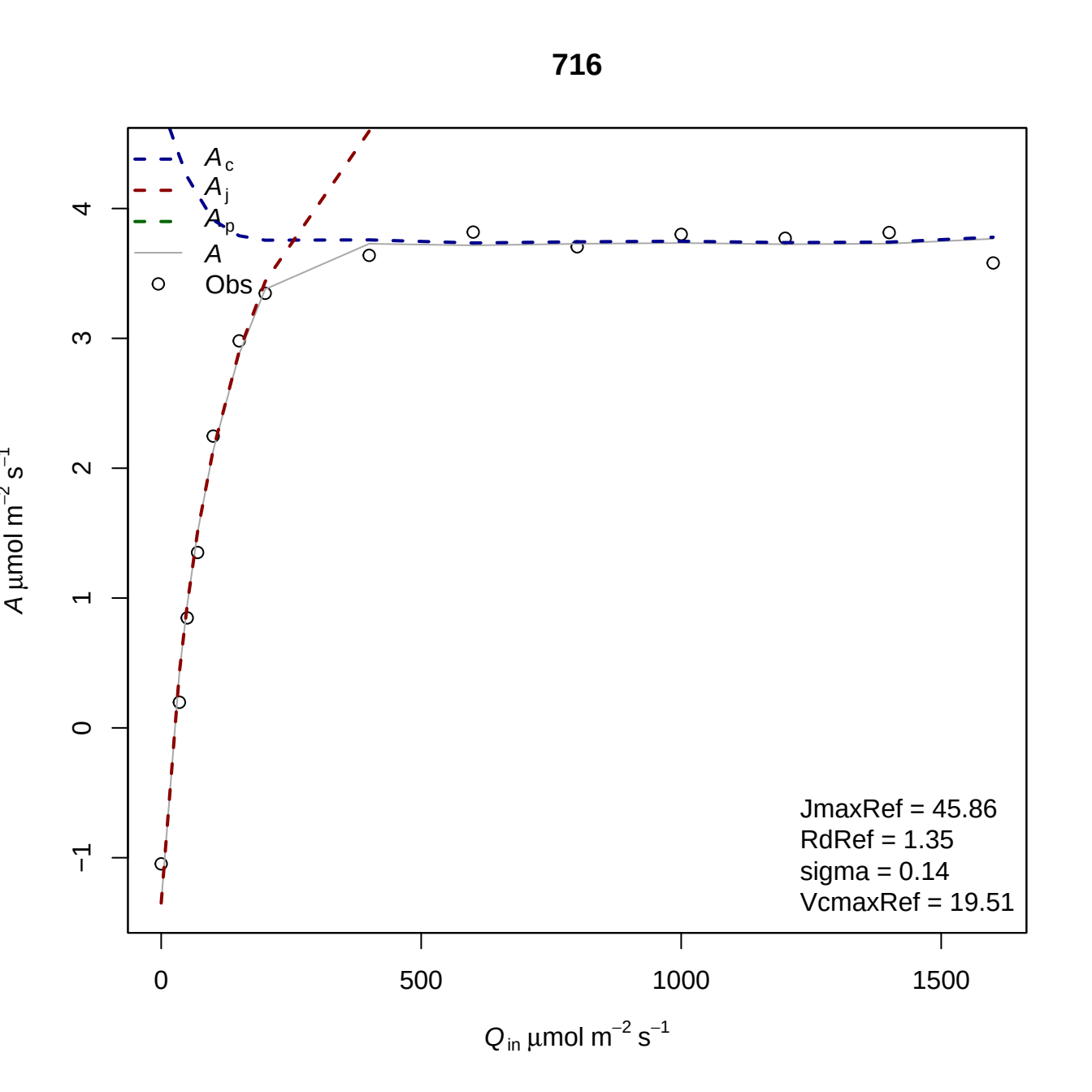

711
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.16
RdRef = 0.59
sigma = 0.23
VcmaxRef = 55.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

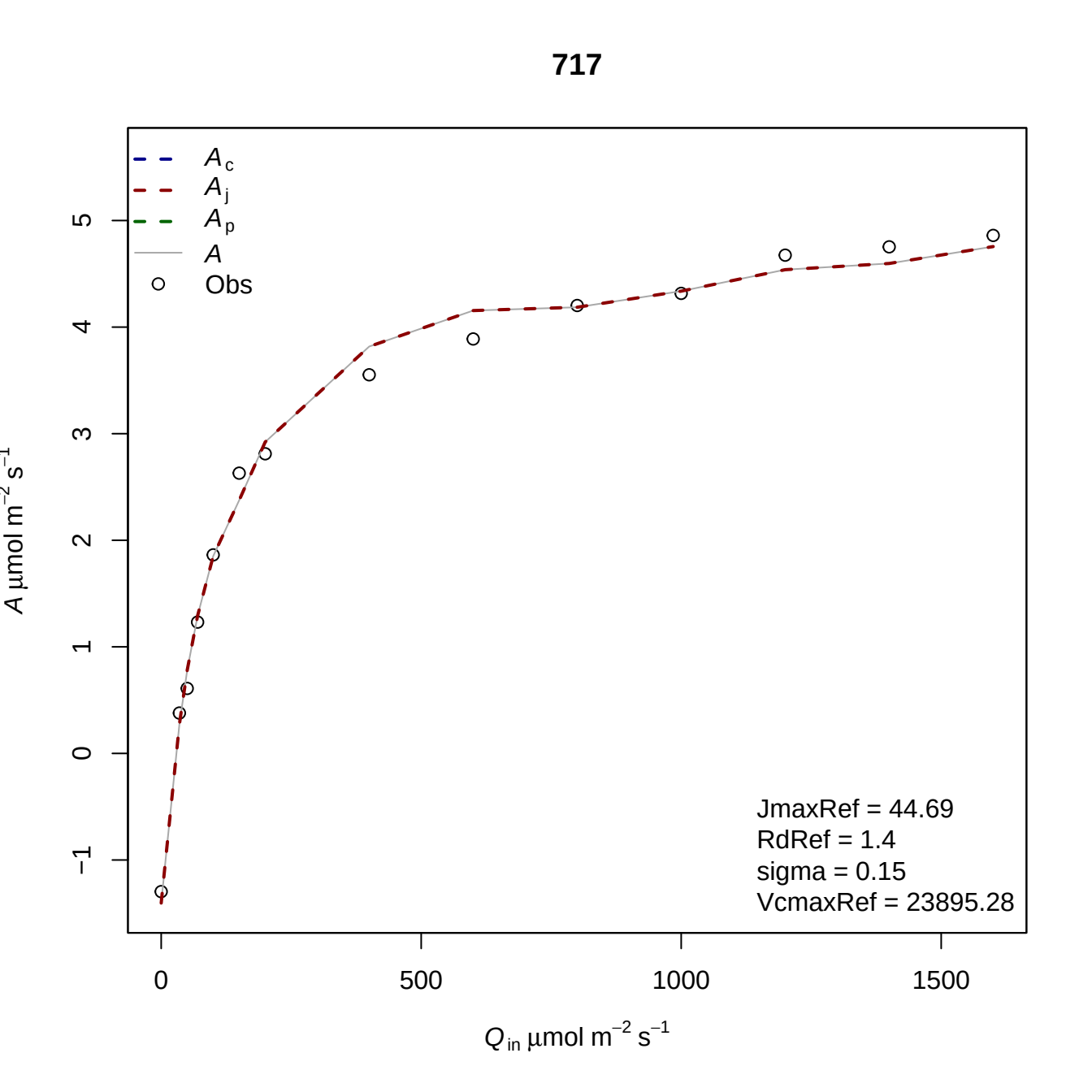

712
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 179.12
RdRef = 0.26
sigma = 0.23
VcmaxRef = 65.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

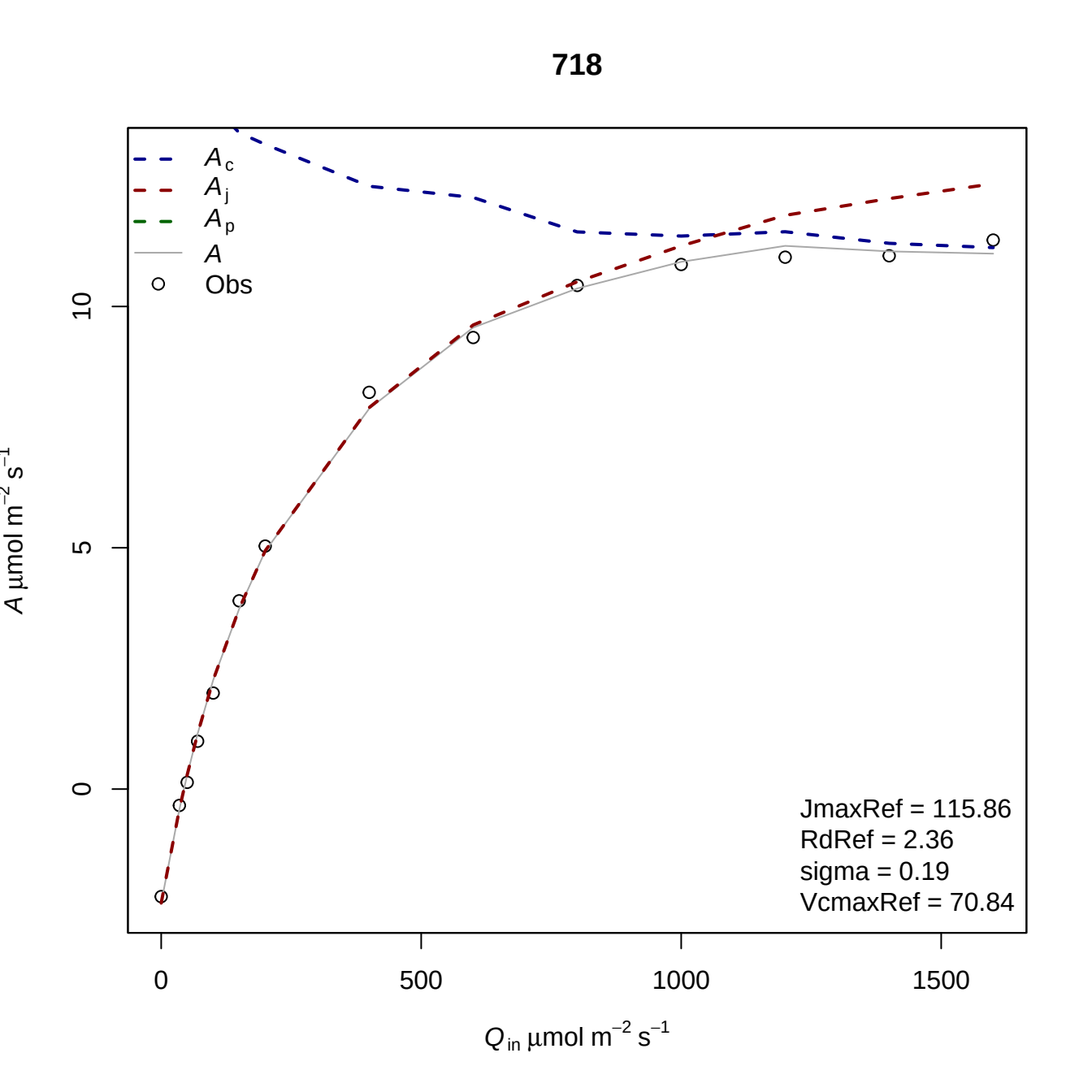

713
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.68
RdRef = 0.36
sigma = 0.2
VcmaxRef = 70.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

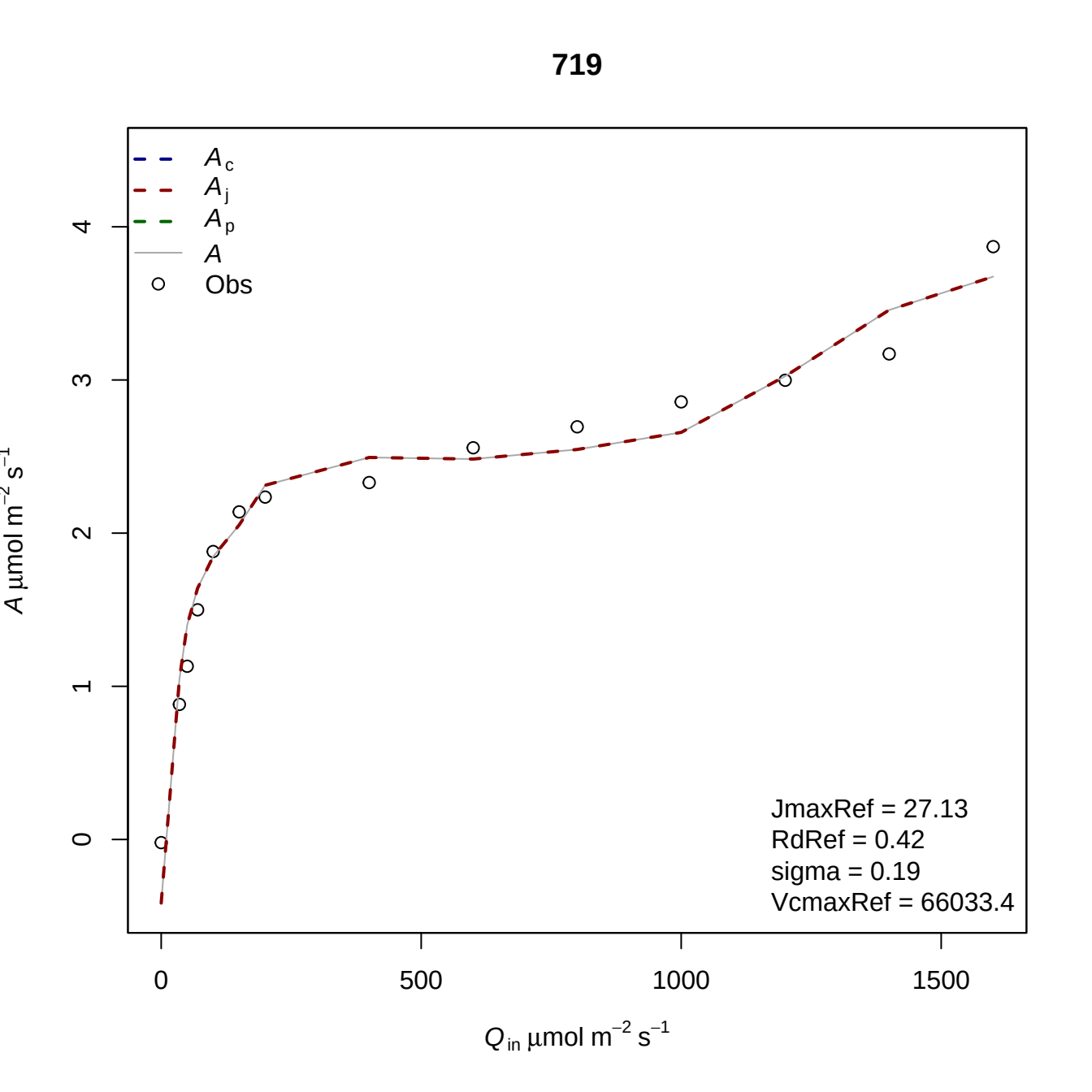

714
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.29
RdRef = 0.79
sigma = 0.22
VcmaxRef = 21.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

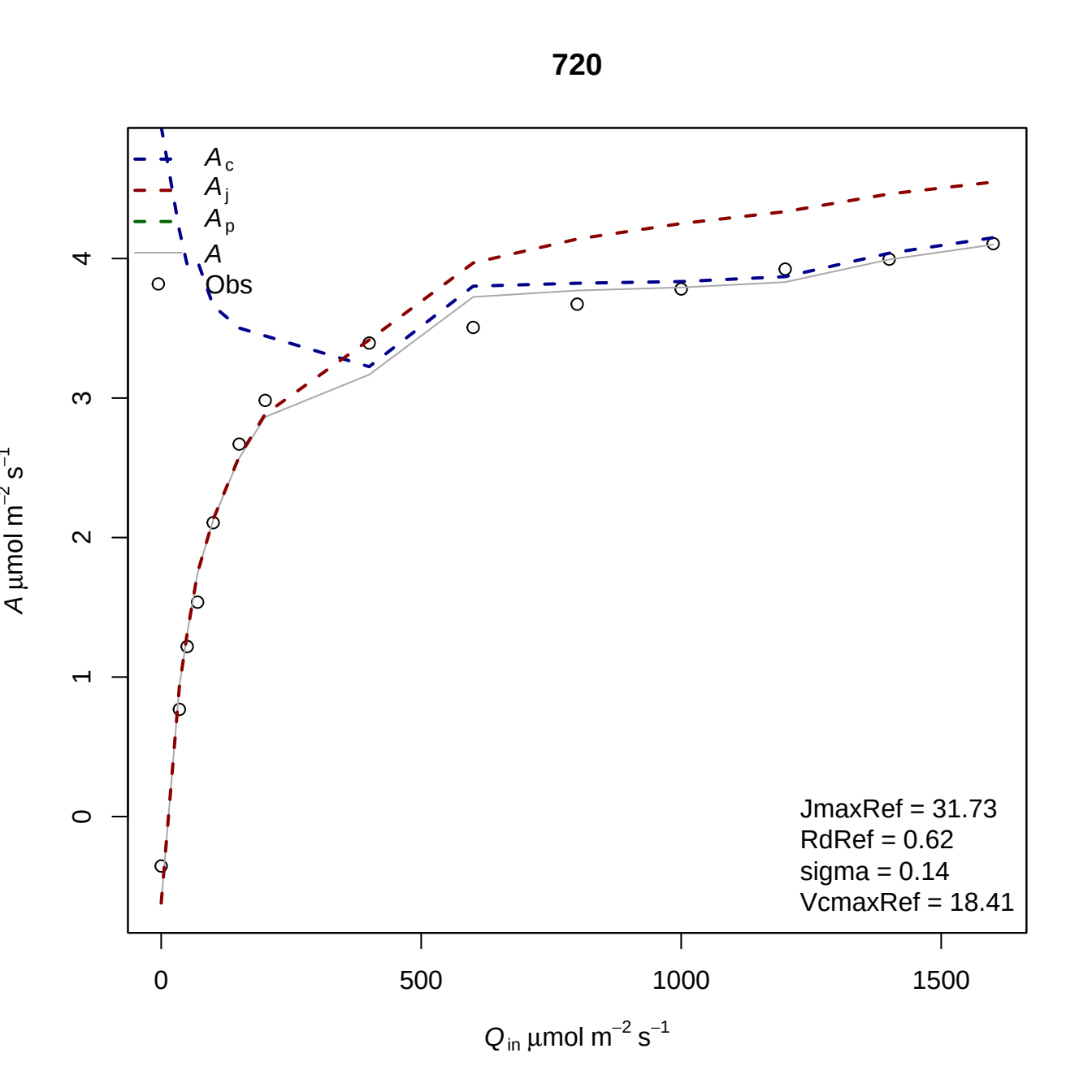

715
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.44
RdRef = 0.68
sigma = 0.07
VcmaxRef = 16.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

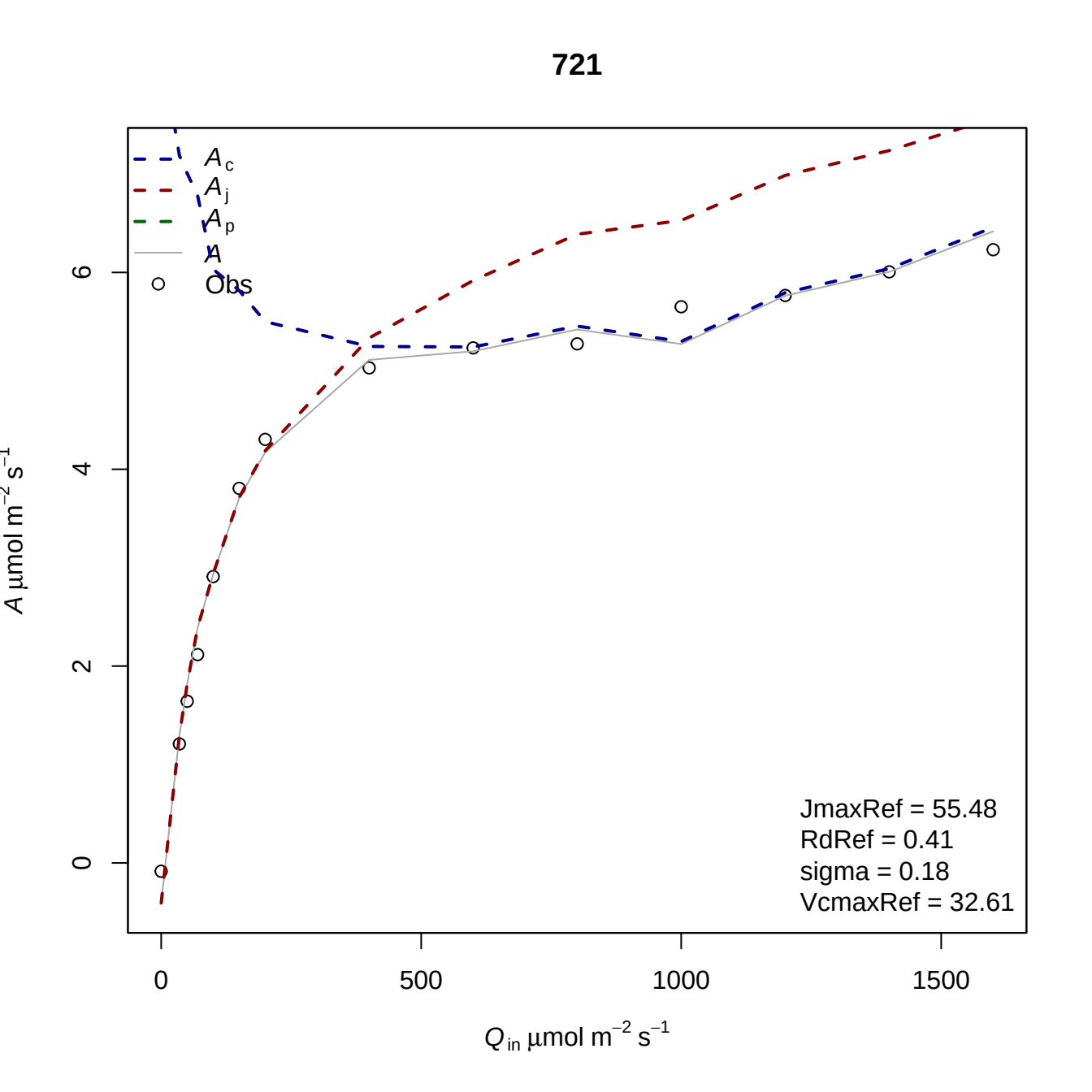

716
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.86
RdRef = 1.35
sigma = 0.14
VcmaxRef = 19.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

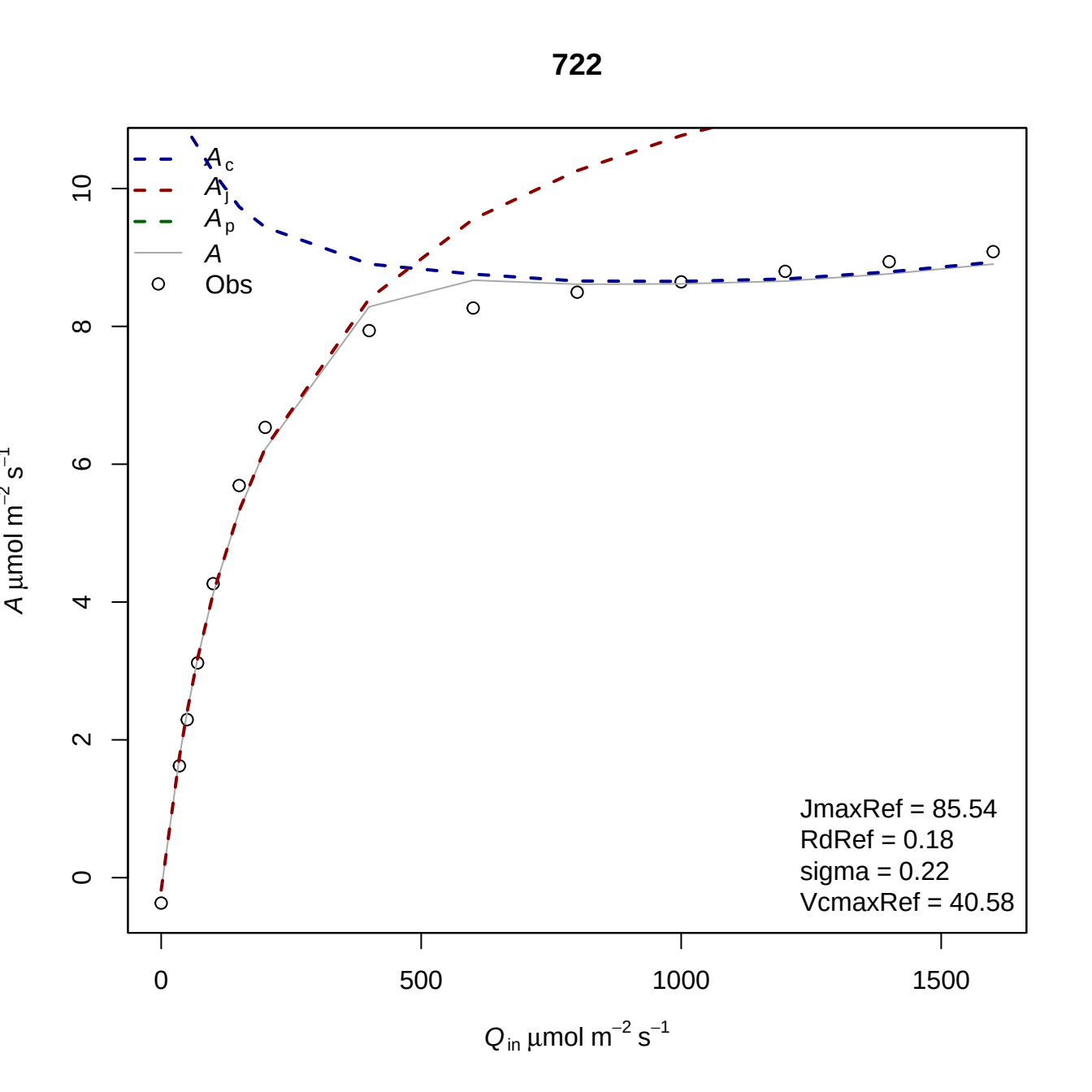

717
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.69
RdRef = 1.4
sigma = 0.15
VcmaxRef = 23895.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

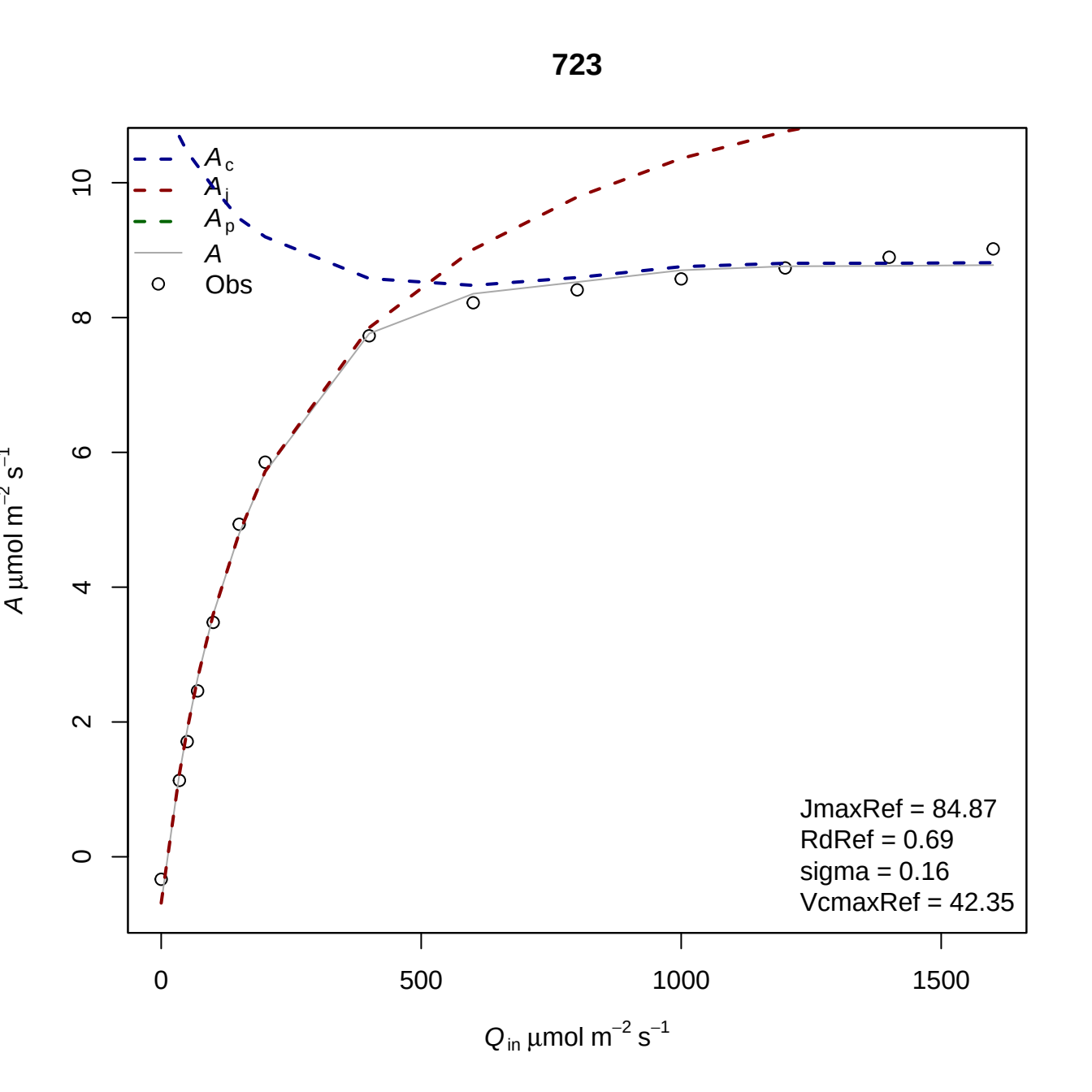

718
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.86
RdRef = 2.36
sigma = 0.19
VcmaxRef = 70.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

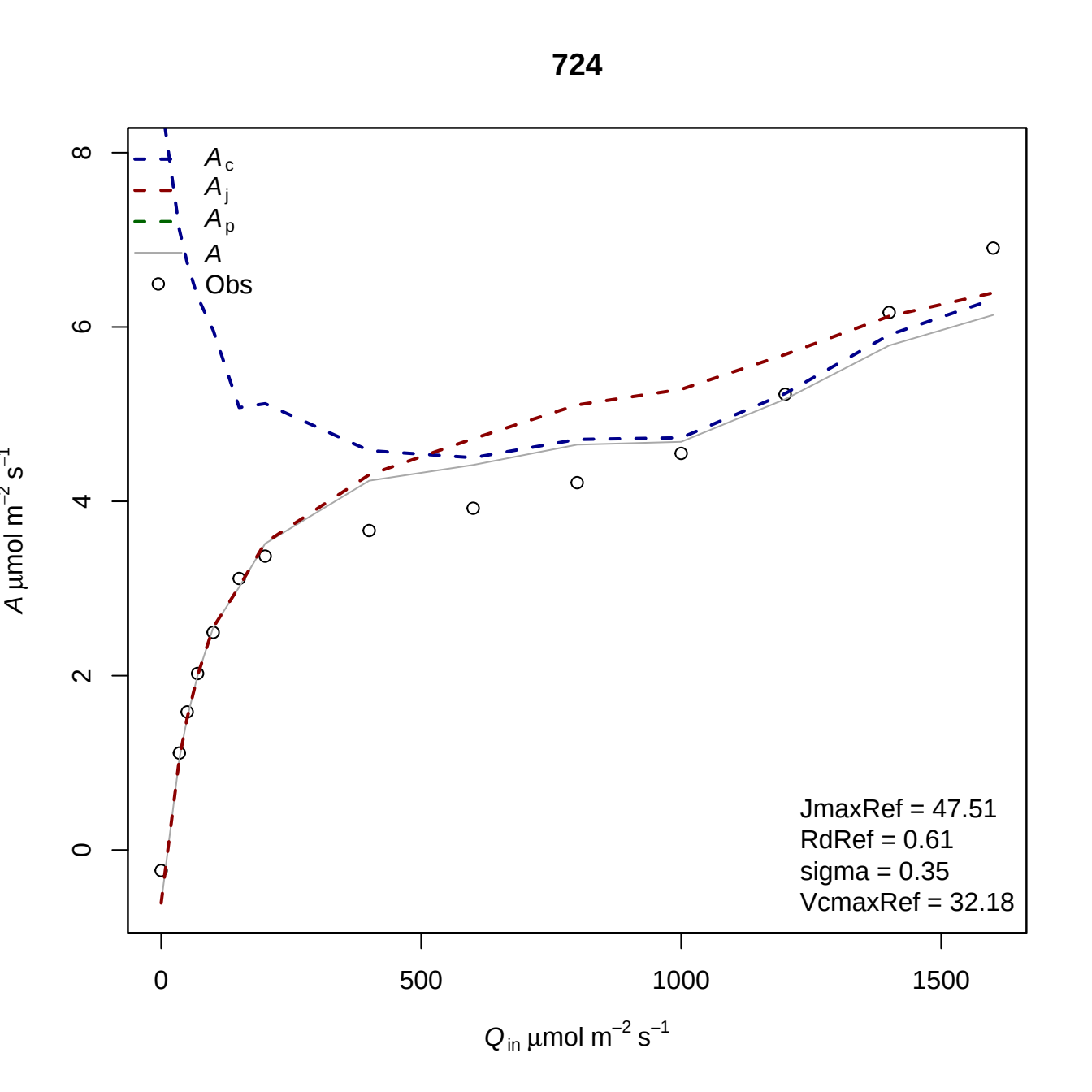

719
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.13
RdRef = 0.42
sigma = 0.19
VcmaxRef = 66033.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

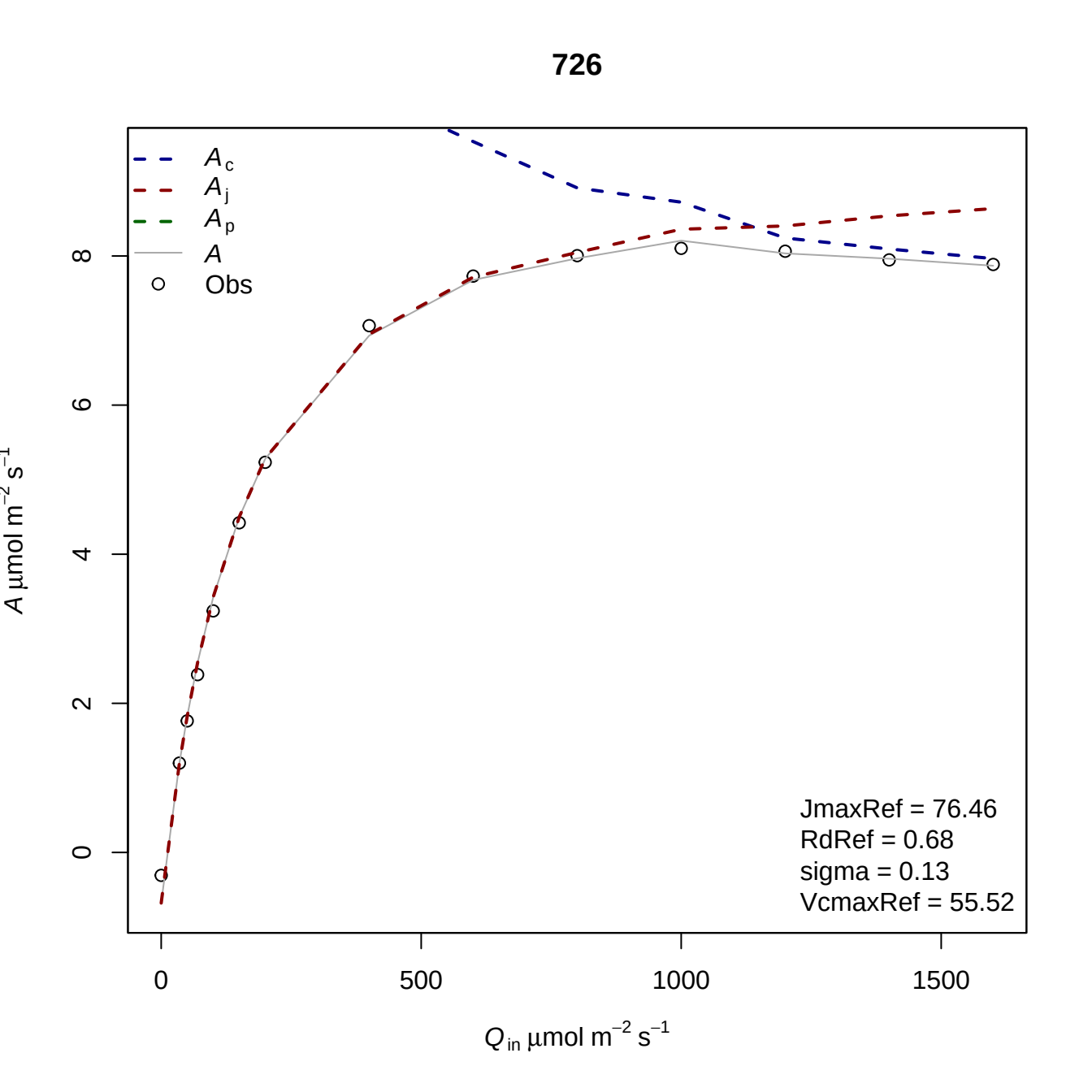

720
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.73
RdRef = 0.62
sigma = 0.14
VcmaxRef = 18.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

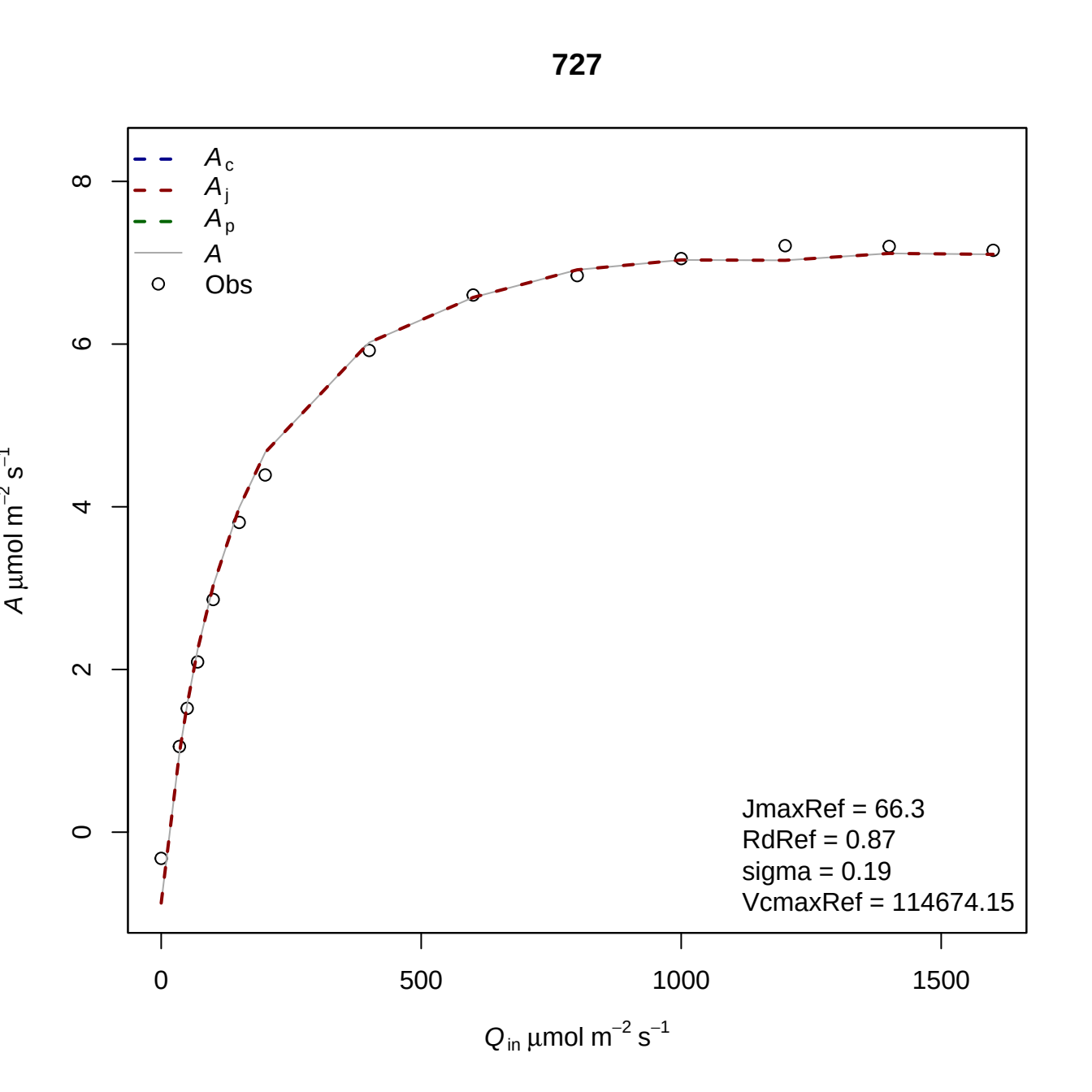

721
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.48
RdRef = 0.41
sigma = 0.18
VcmaxRef = 32.61
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

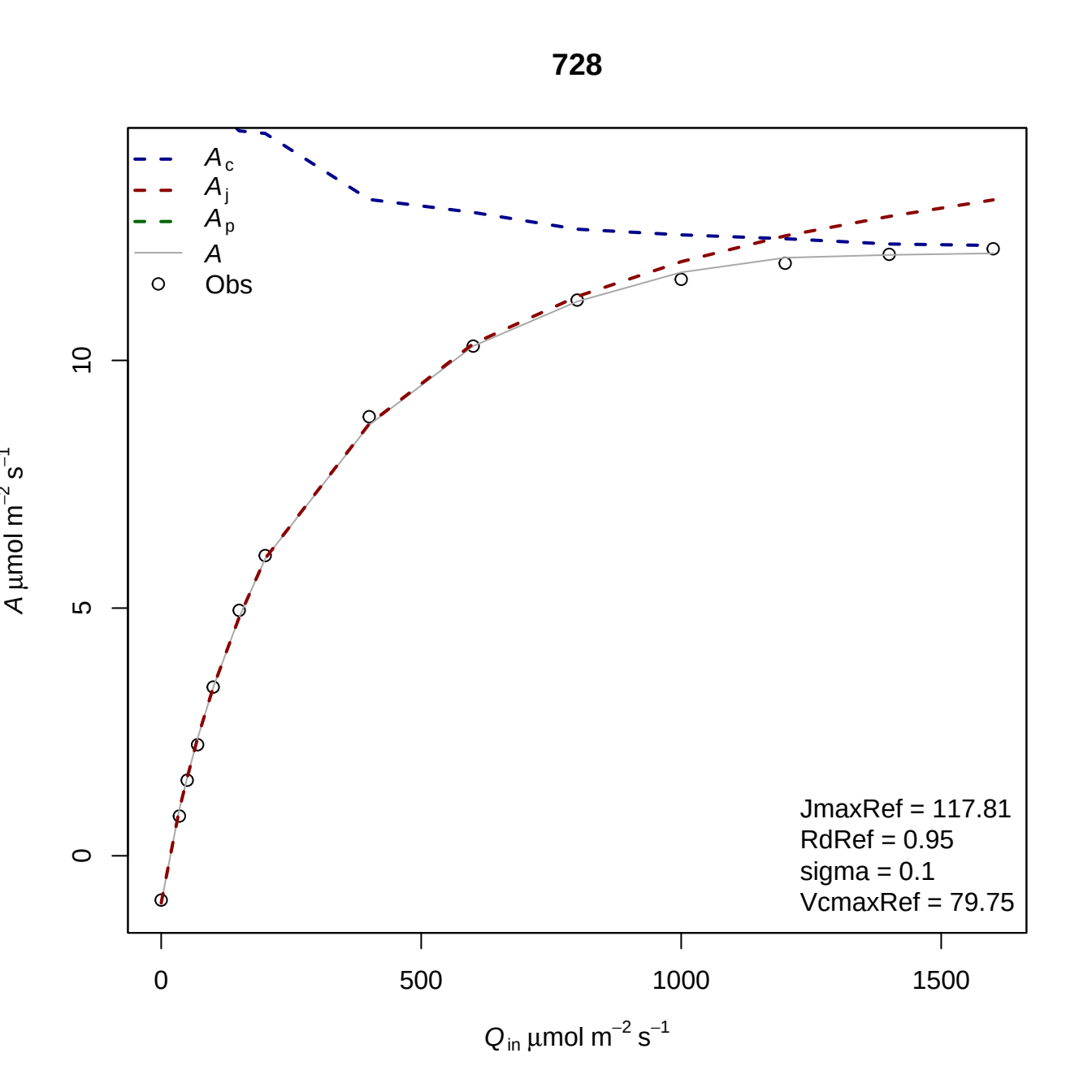

722
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.54
RdRef = 0.18
sigma = 0.22
VcmaxRef = 40.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

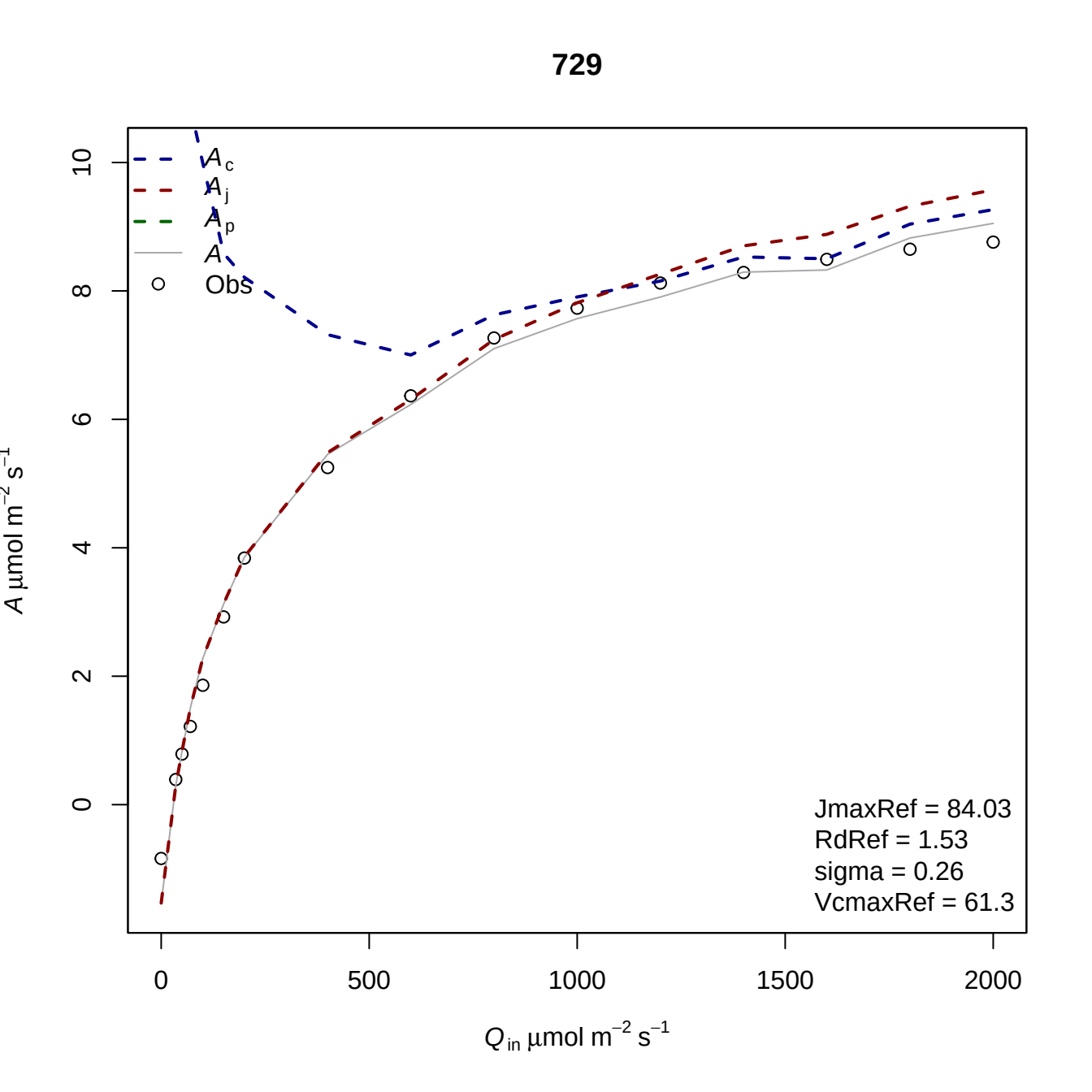

723
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.87
RdRef = 0.69
sigma = 0.16
VcmaxRef = 42.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

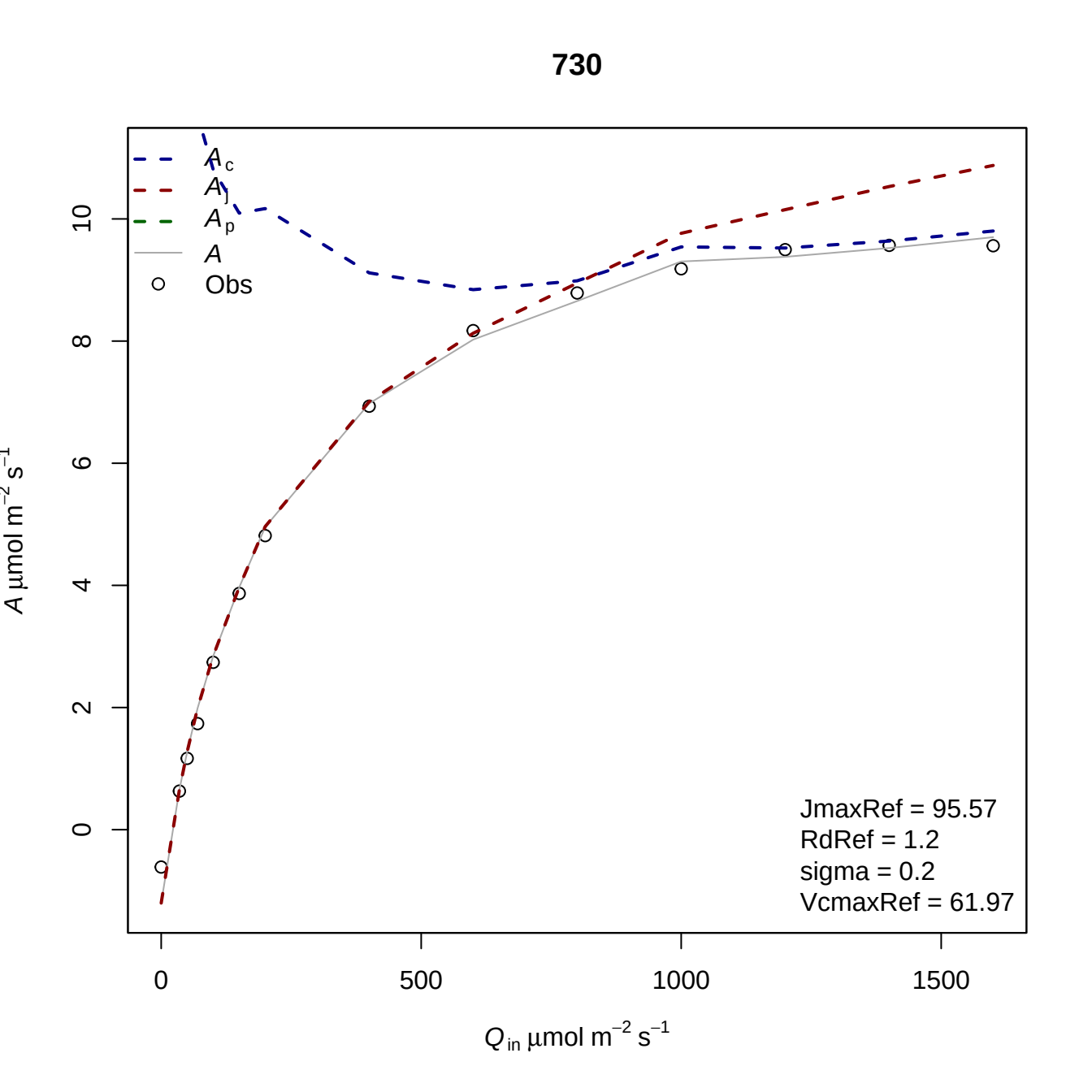

724
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.51
RdRef = 0.61
sigma = 0.35
VcmaxRef = 32.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

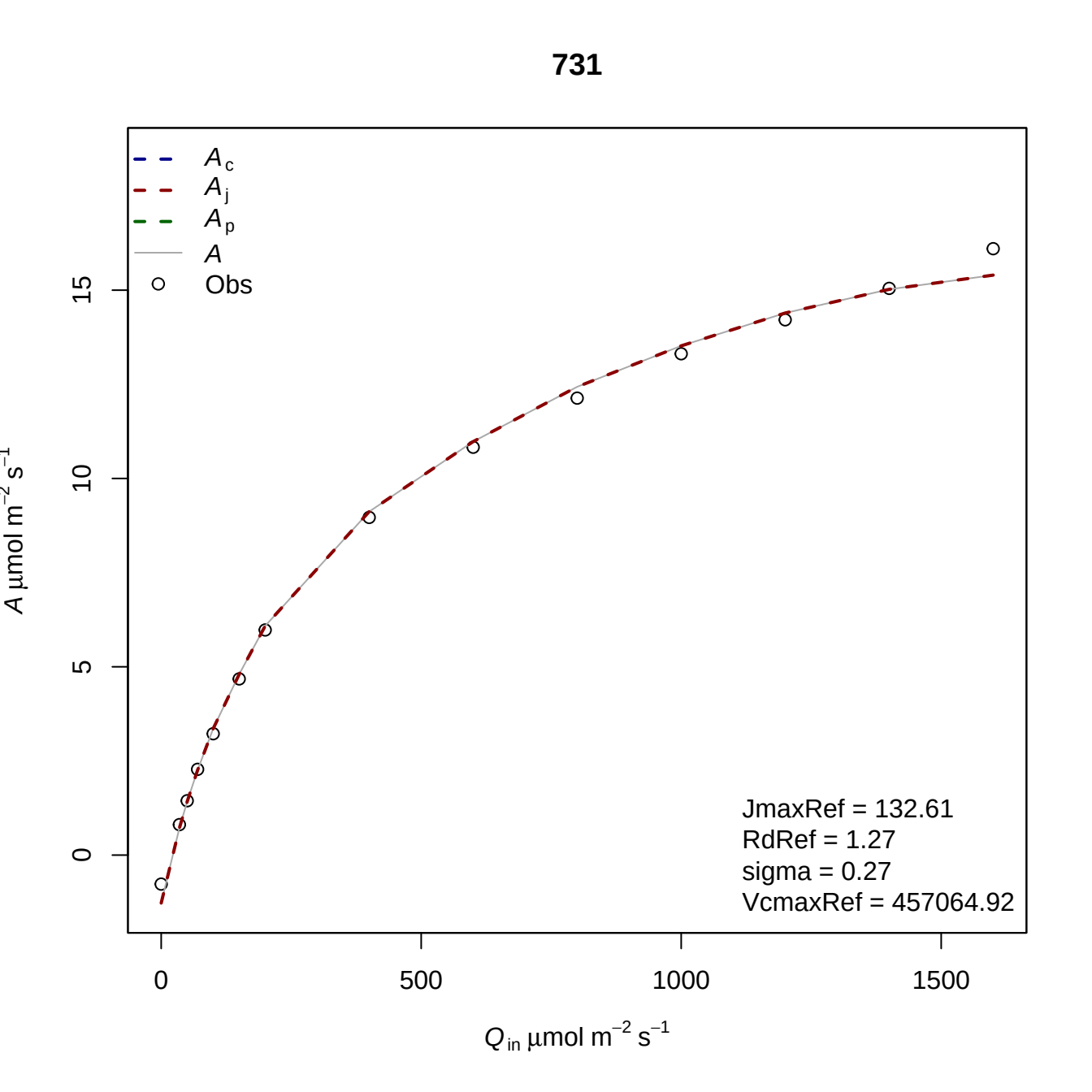

726
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.46
RdRef = 0.68
sigma = 0.13
VcmaxRef = 55.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

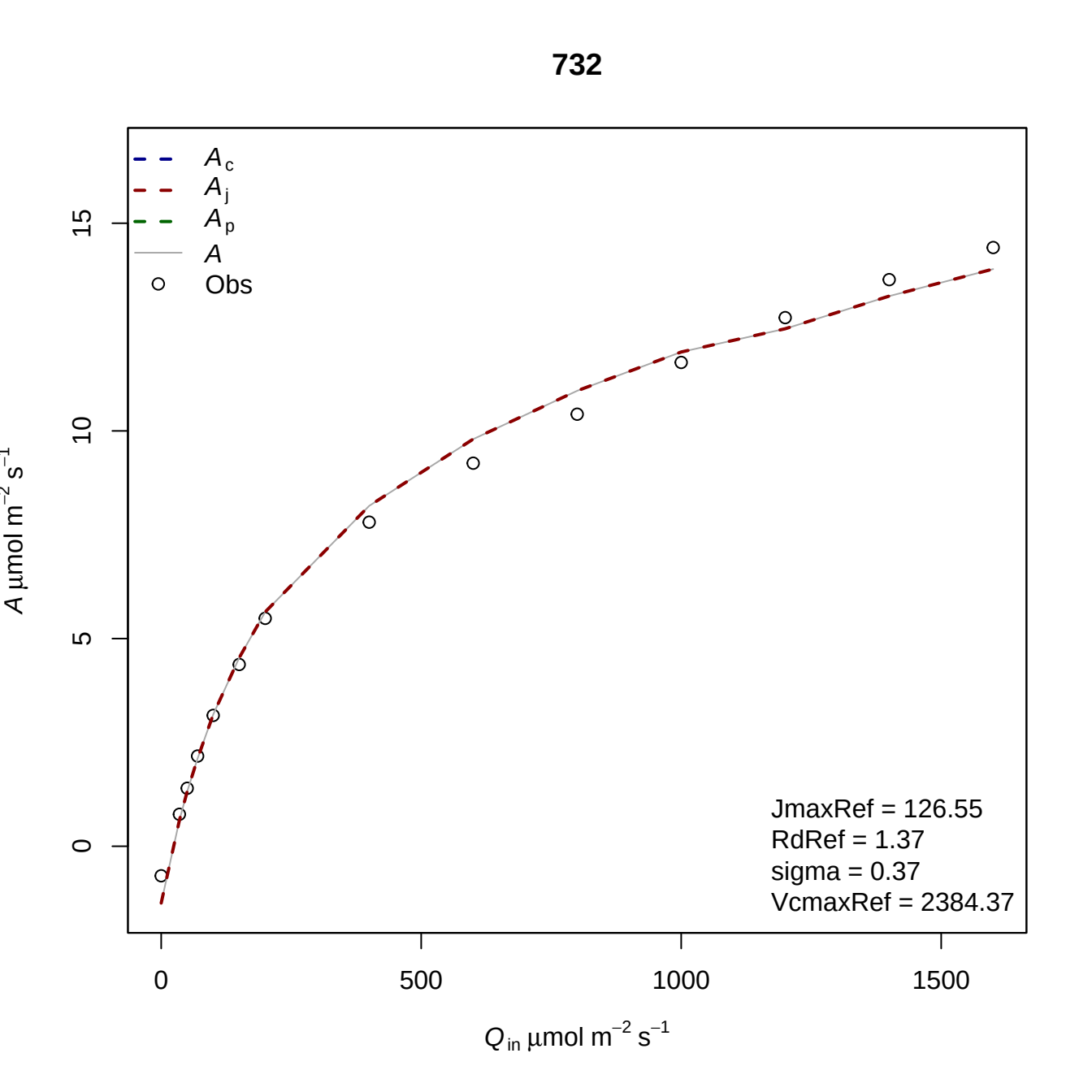

727
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.3
RdRef = 0.87
sigma = 0.19
VcmaxRef = 114674.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

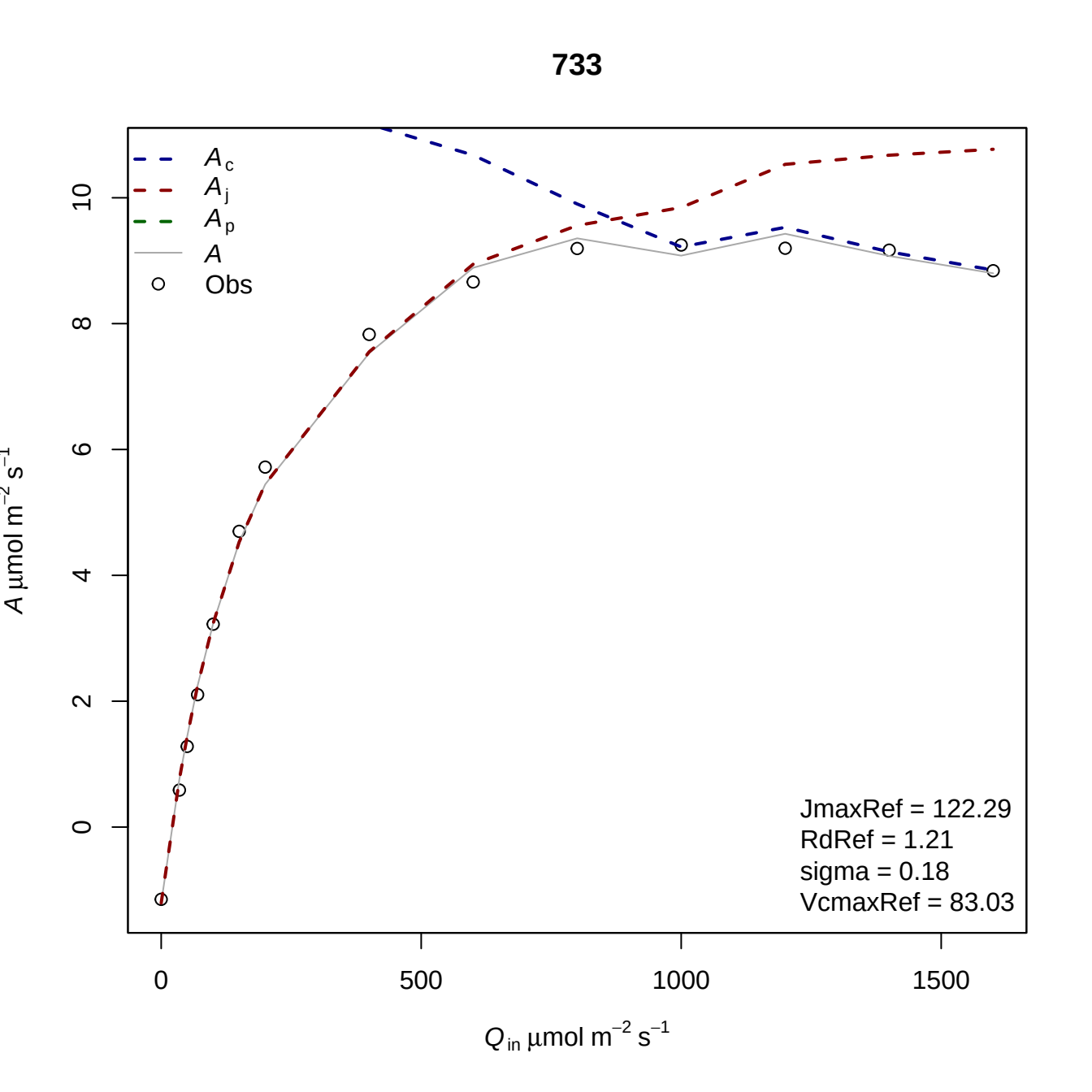

728
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.81
RdRef = 0.95
sigma = 0.1
VcmaxRef = 79.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

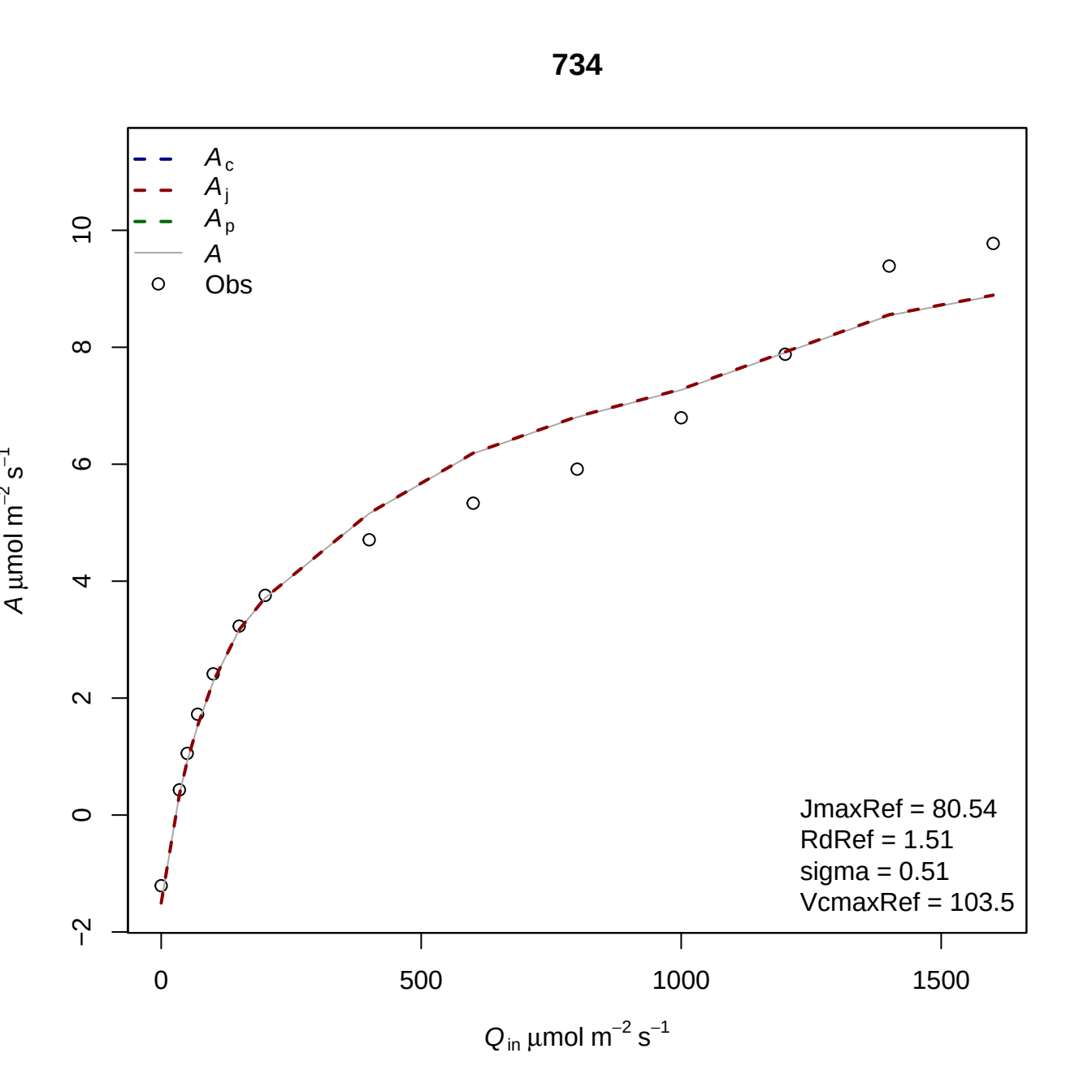

729
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.03
RdRef = 1.53
sigma = 0.26
VcmaxRef = 61.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

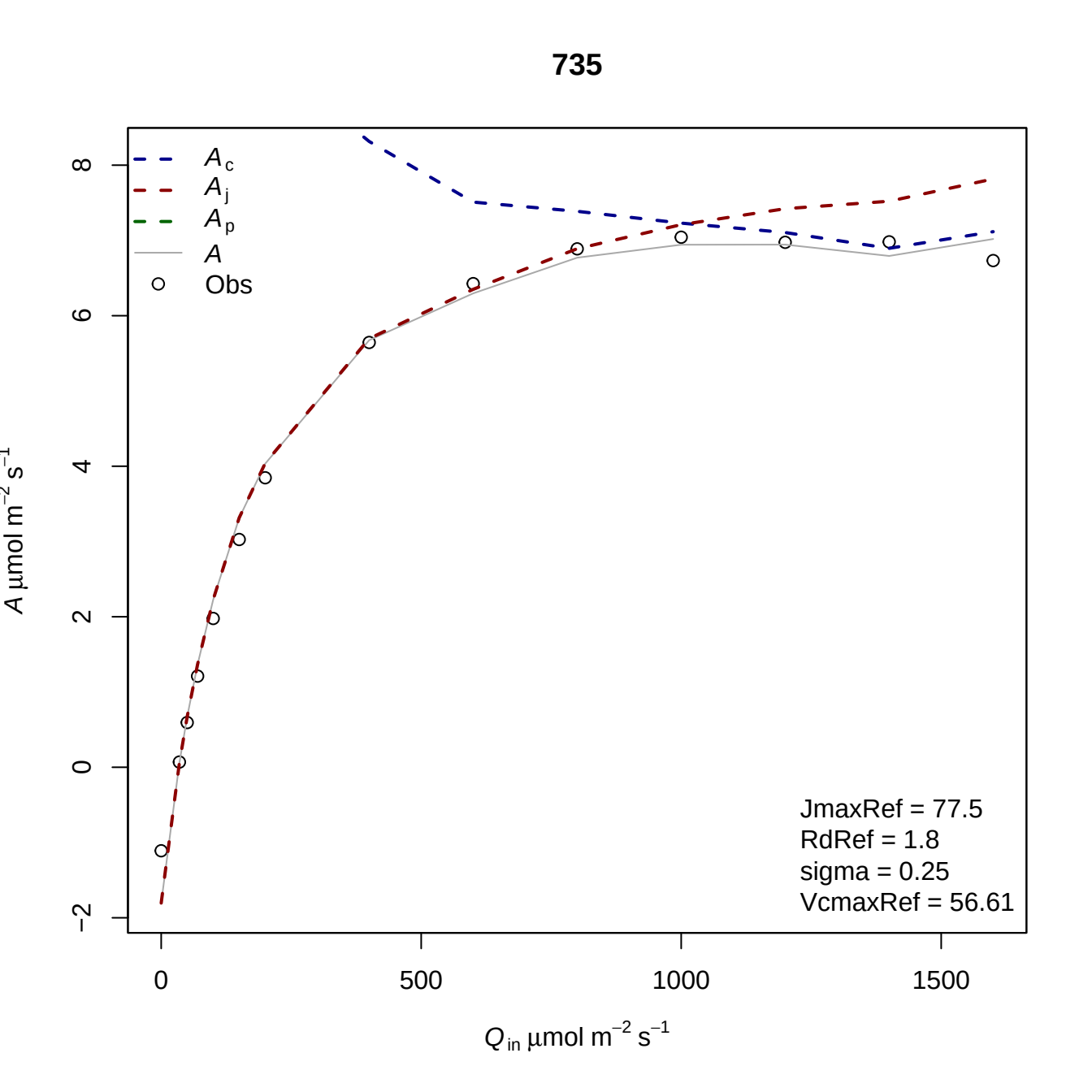

730
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.57
RdRef = 1.2
sigma = 0.2
VcmaxRef = 61.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

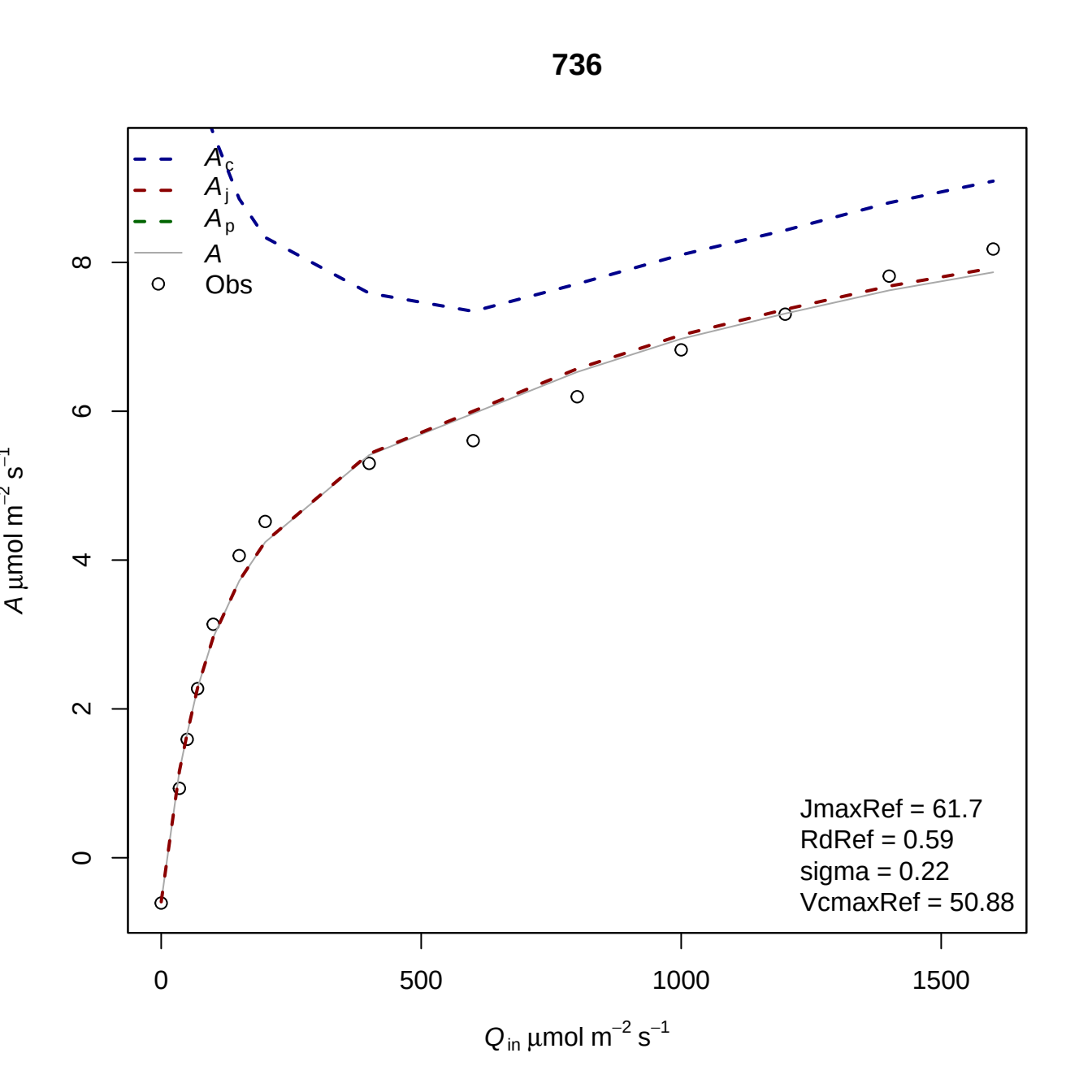

731
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.61
RdRef = 1.27
sigma = 0.27
VcmaxRef = 457064.92
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

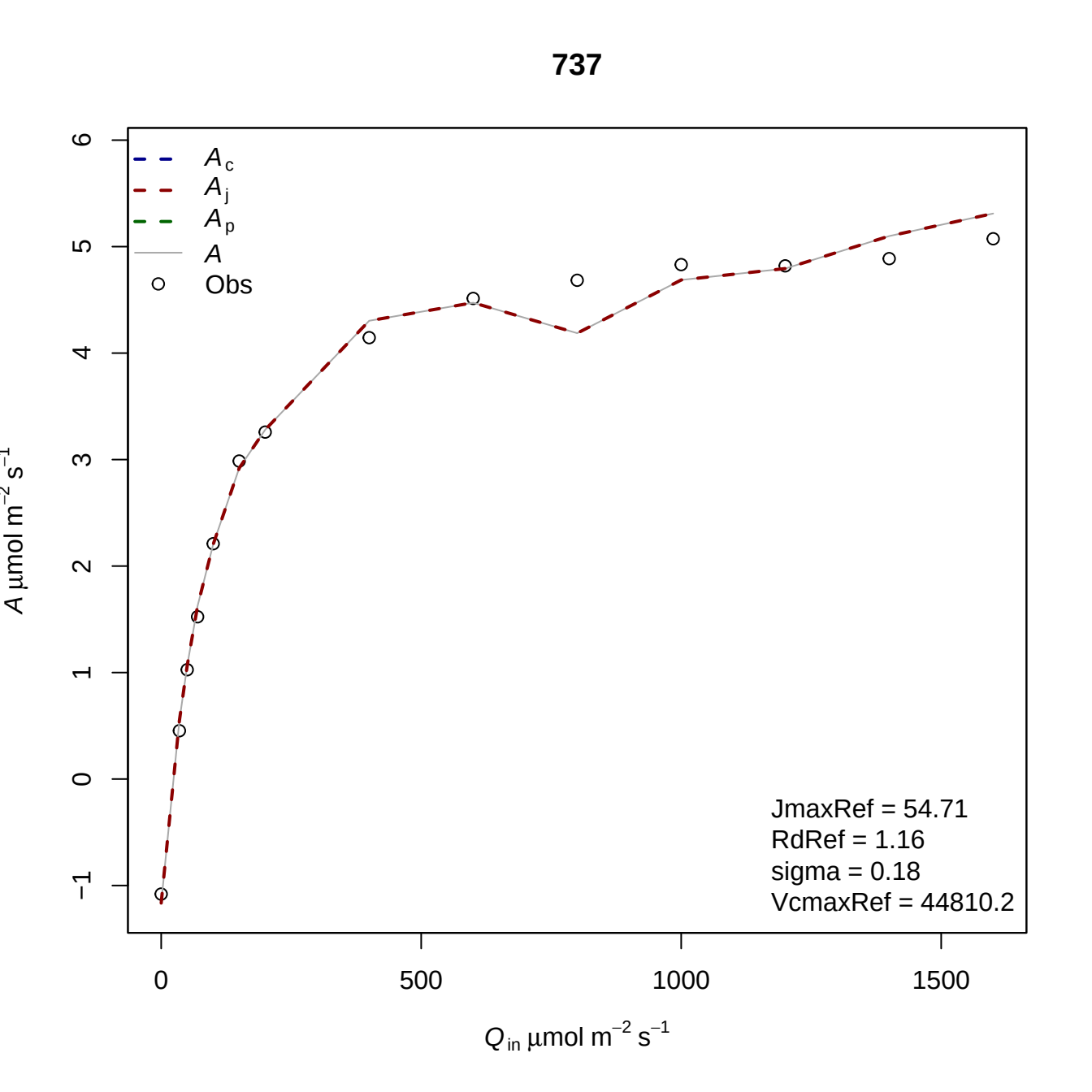

732
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 126.55
RdRef = 1.37
sigma = 0.37
VcmaxRef = 2384.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

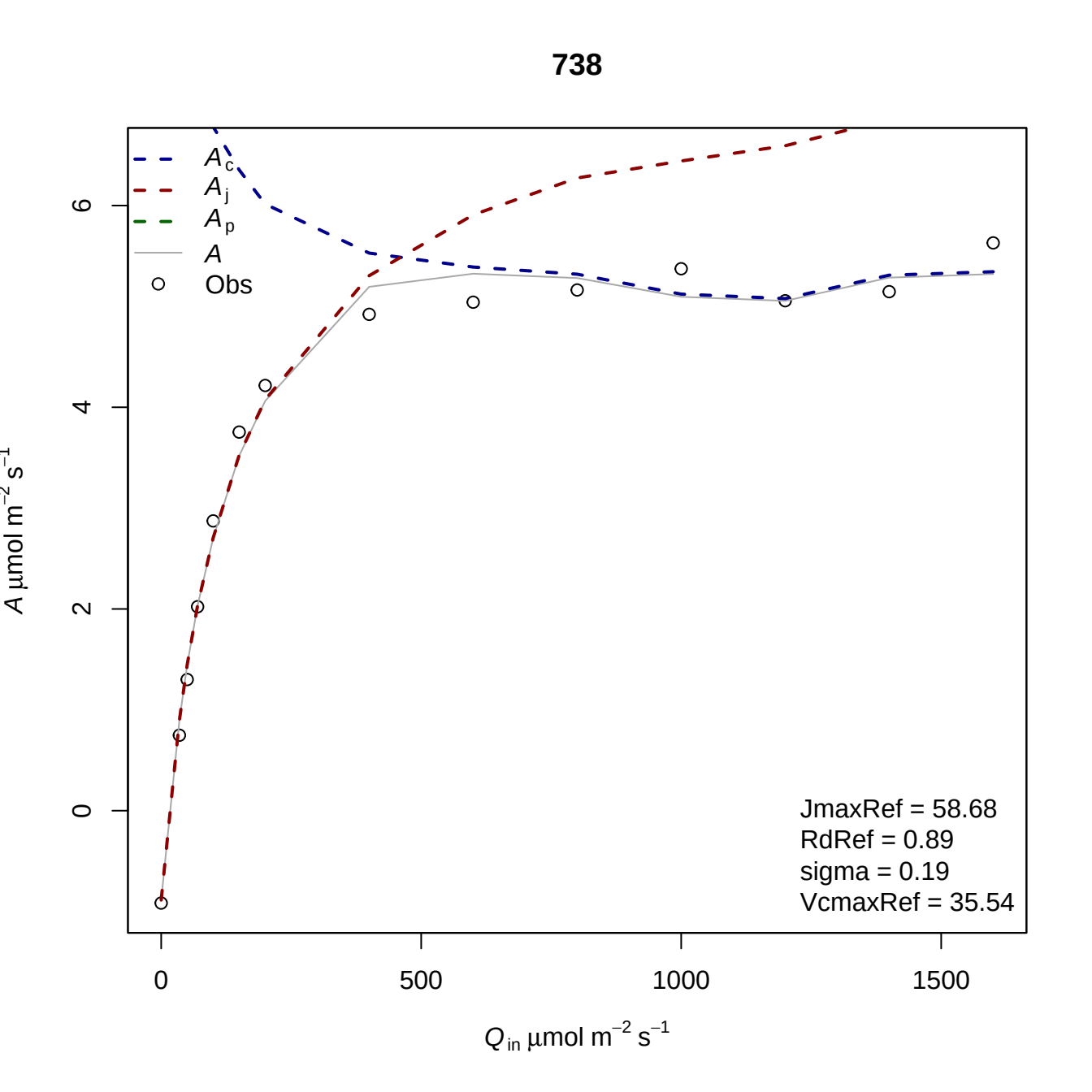

733
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.29
RdRef = 1.21
sigma = 0.18
VcmaxRef = 83.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

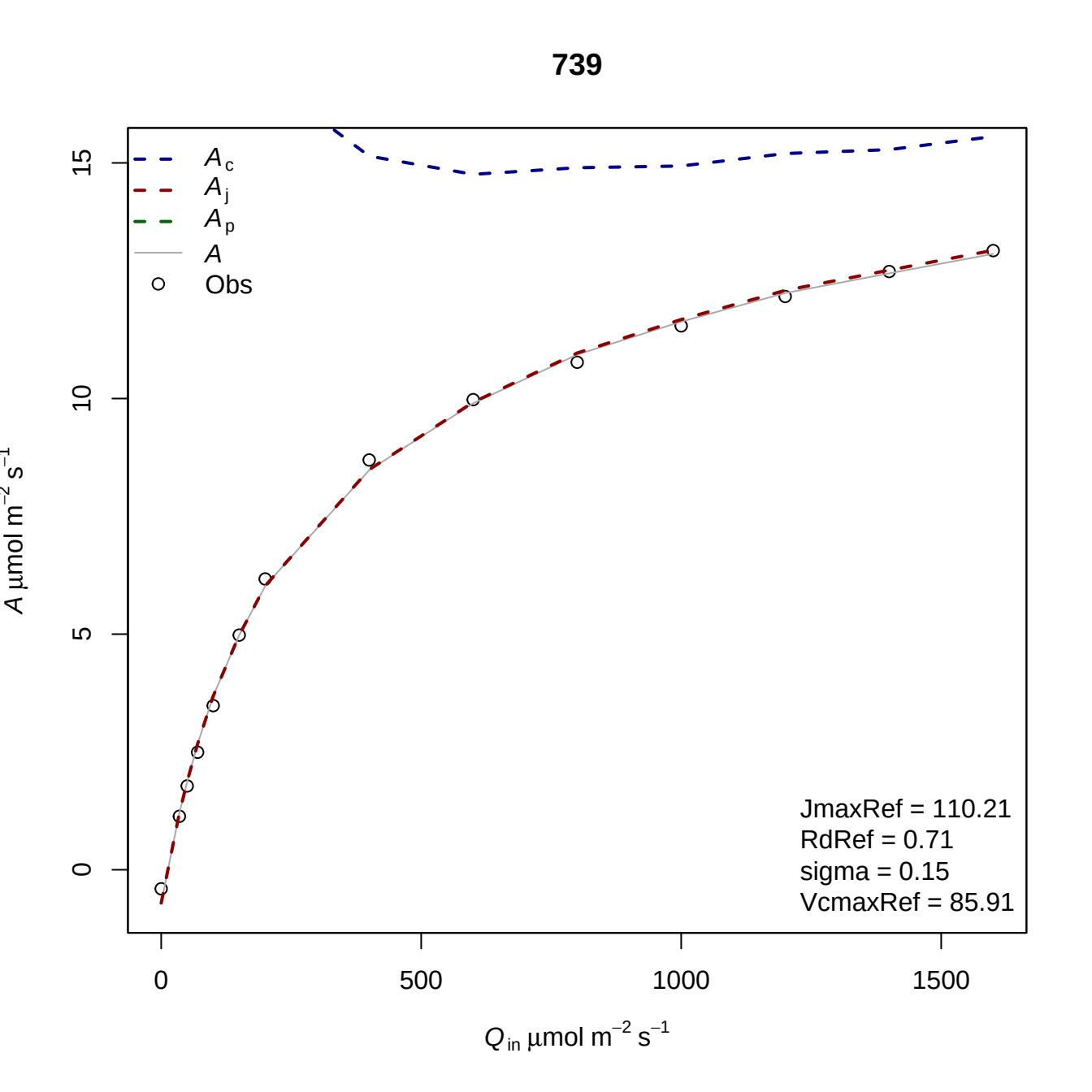

734
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.54
RdRef = 1.51
sigma = 0.51
VcmaxRef = 103.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

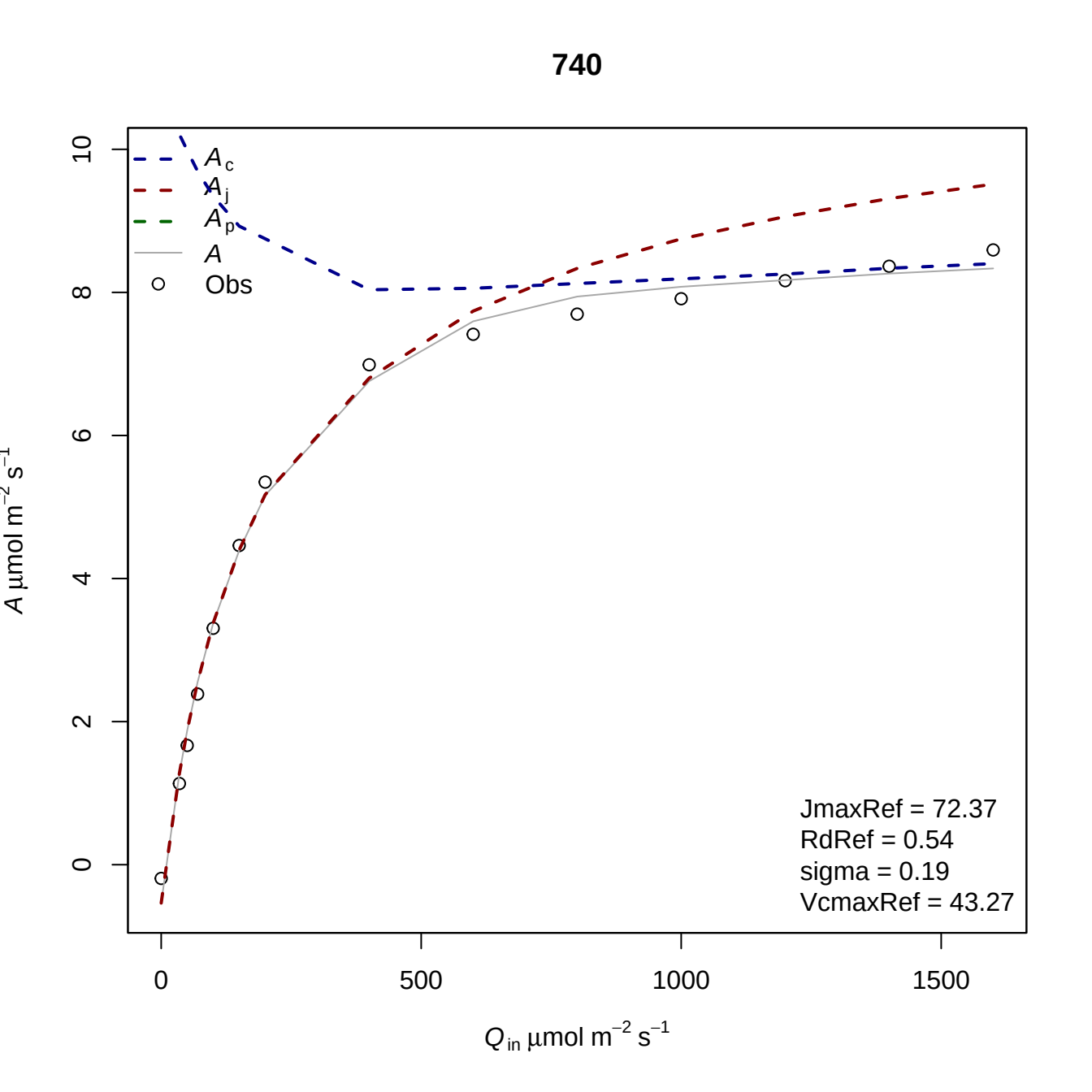

735
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.5
RdRef = 1.8
sigma = 0.25
VcmaxRef = 56.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

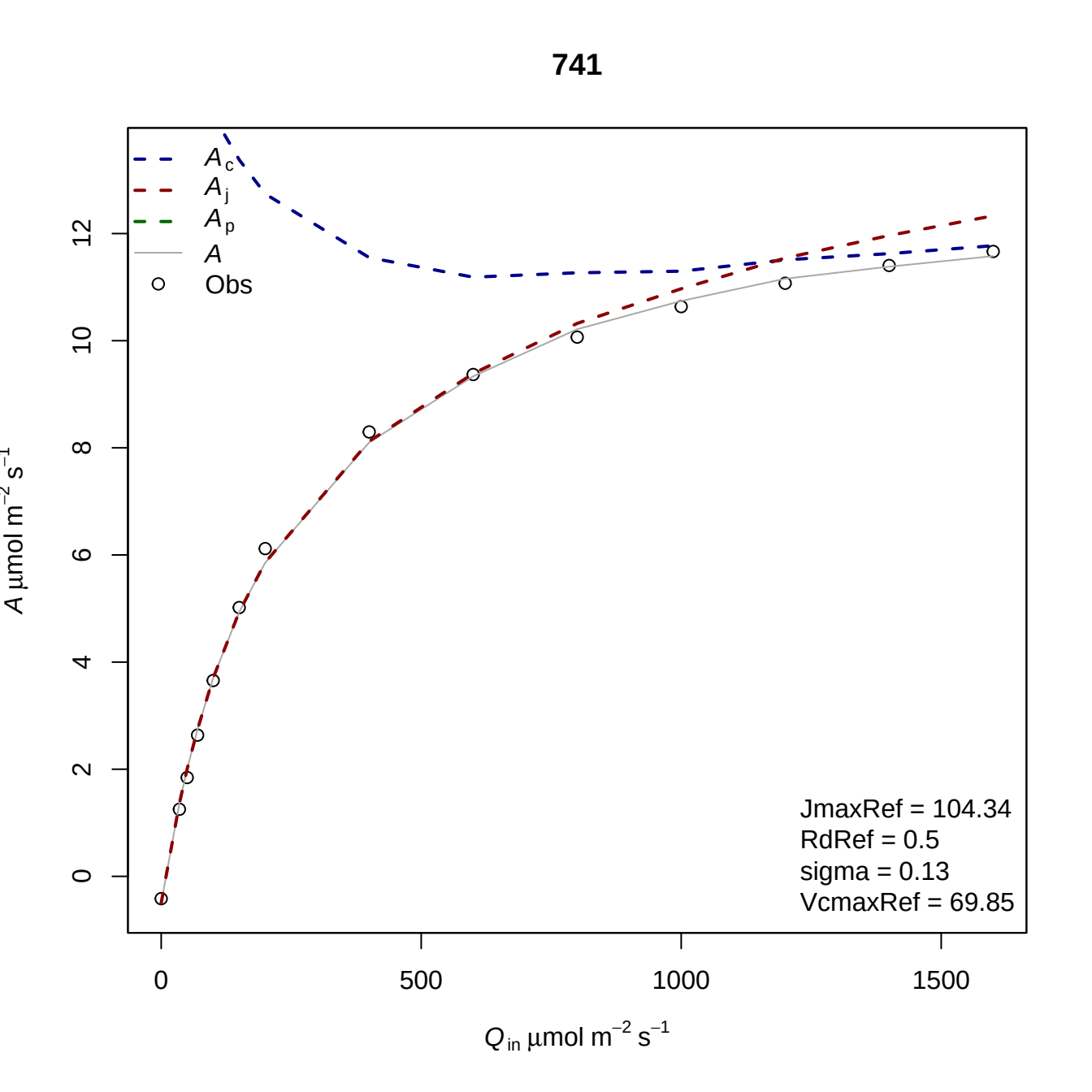

736
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.7
RdRef = 0.59
sigma = 0.22
VcmaxRef = 50.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

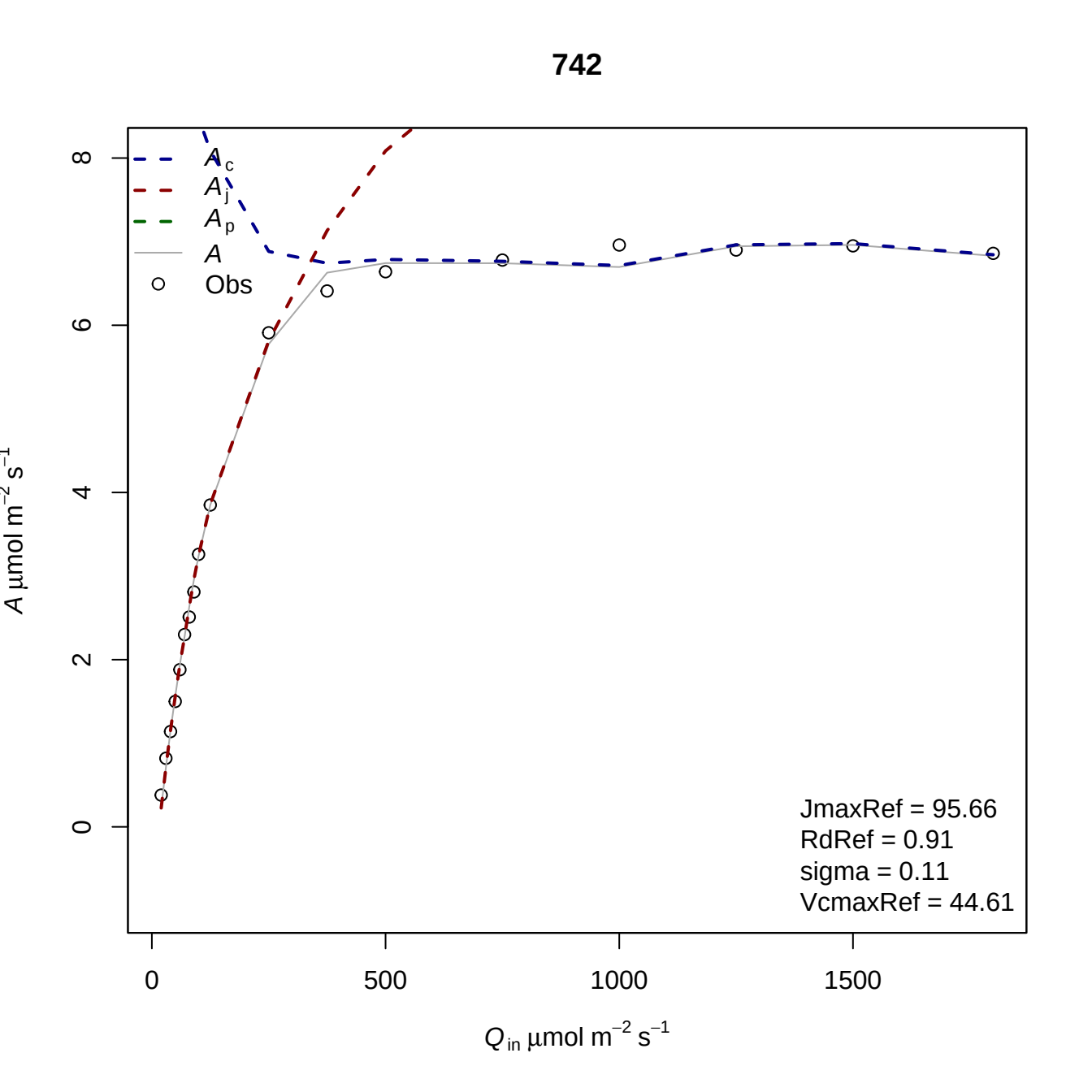

737
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.71
RdRef = 1.16
sigma = 0.18
VcmaxRef = 44810.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

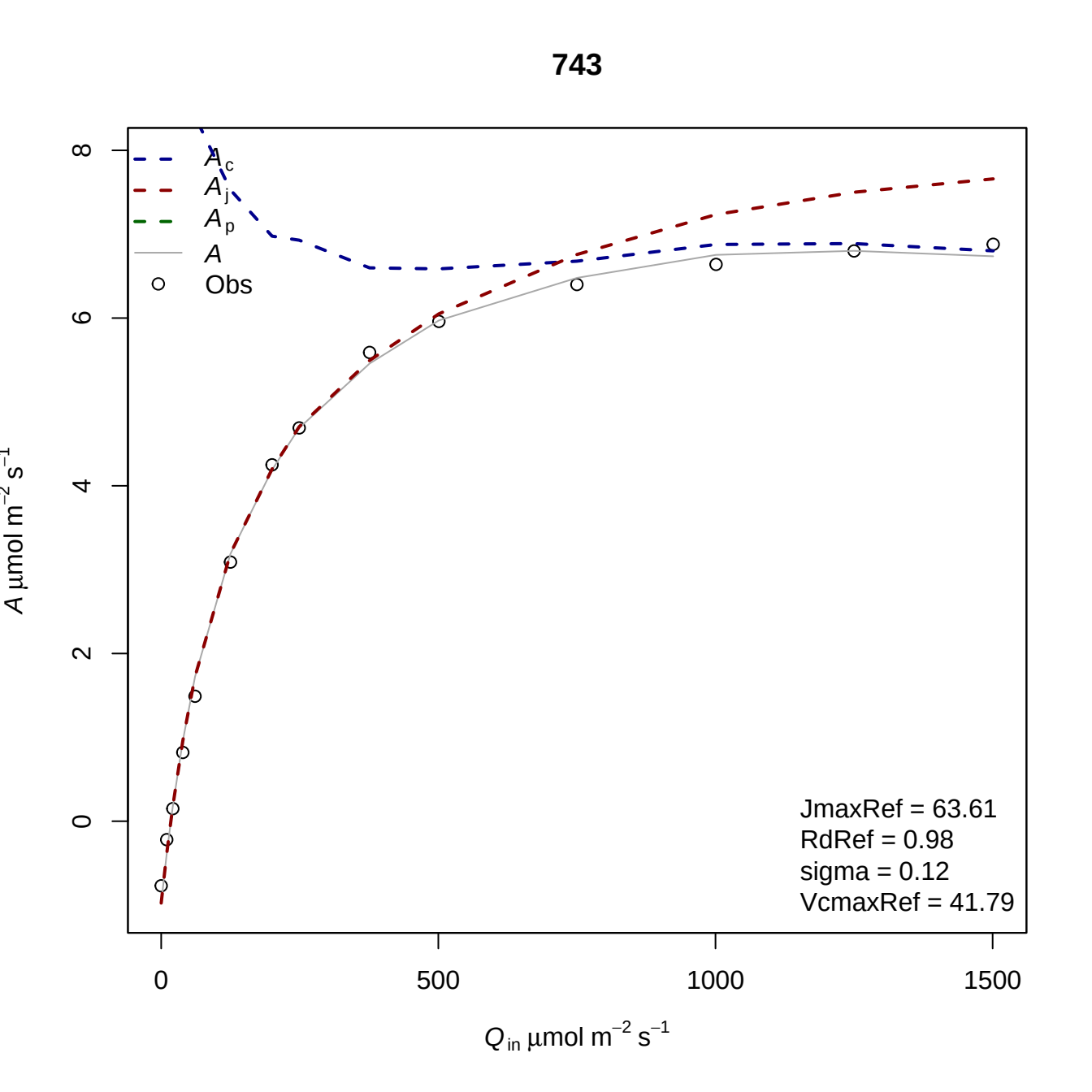

738
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.68
RdRef = 0.89
sigma = 0.19
VcmaxRef = 35.54
0
2
4
6
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

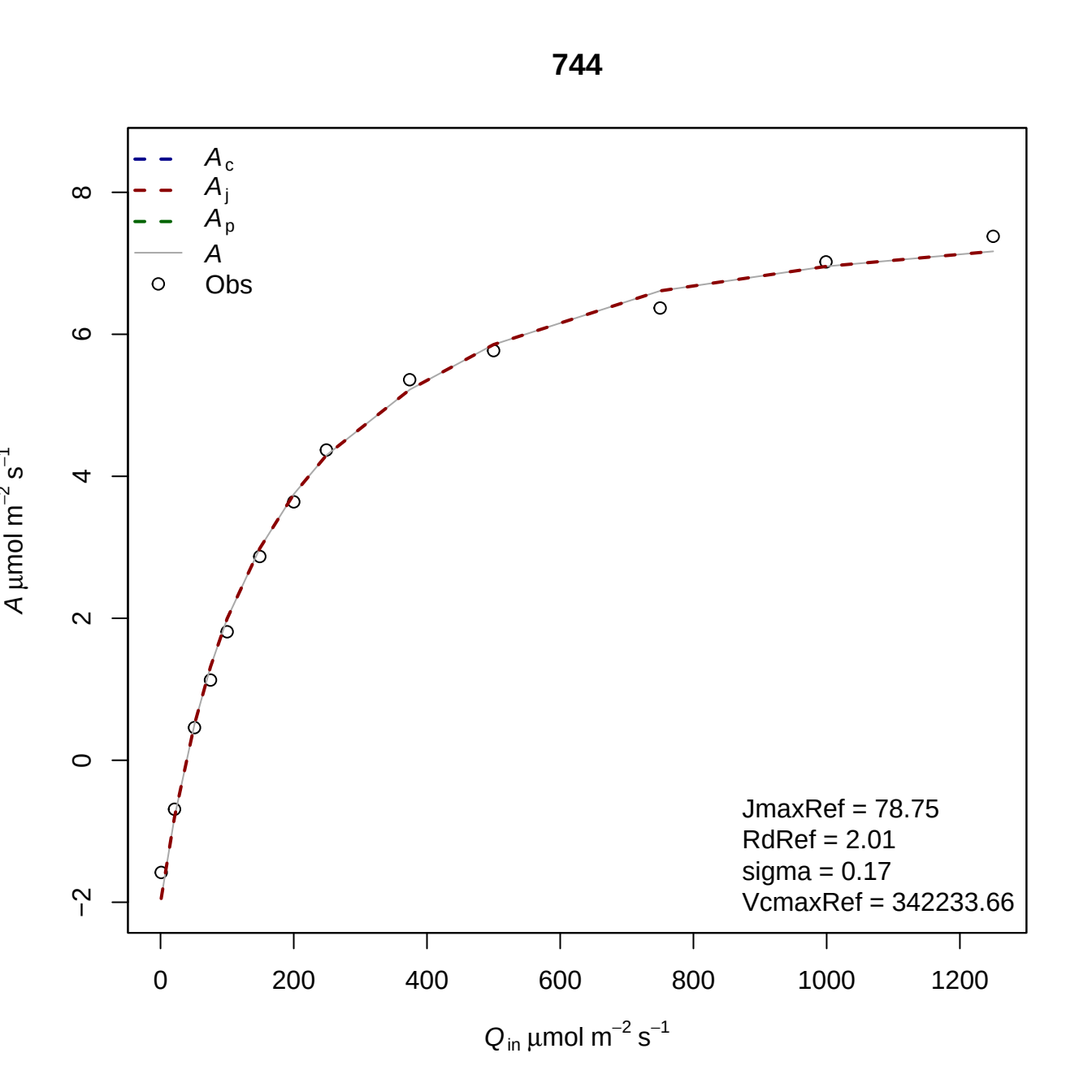

739
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.21
RdRef = 0.71
sigma = 0.15
VcmaxRef = 85.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

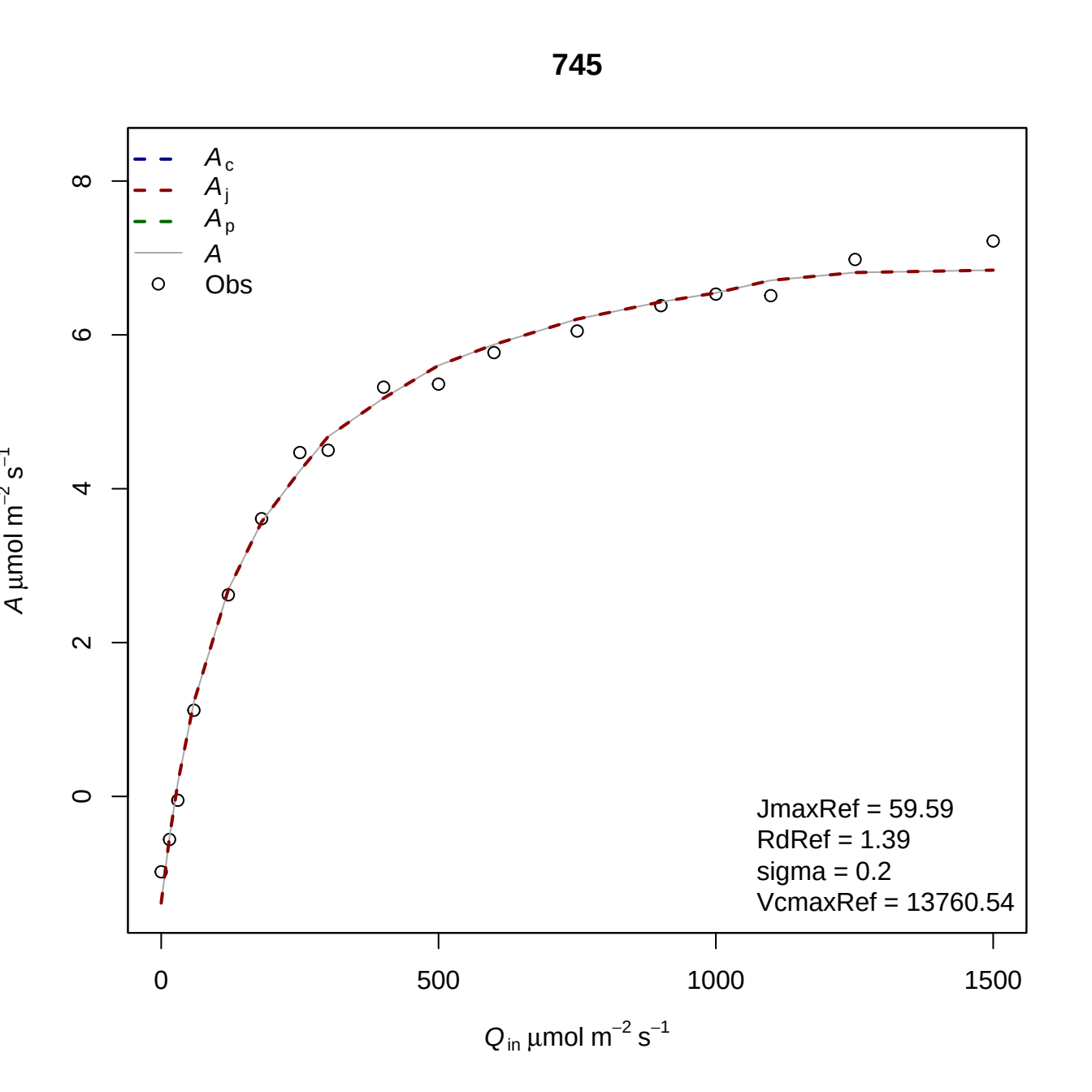

740
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.37
RdRef = 0.54
sigma = 0.19
VcmaxRef = 43.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

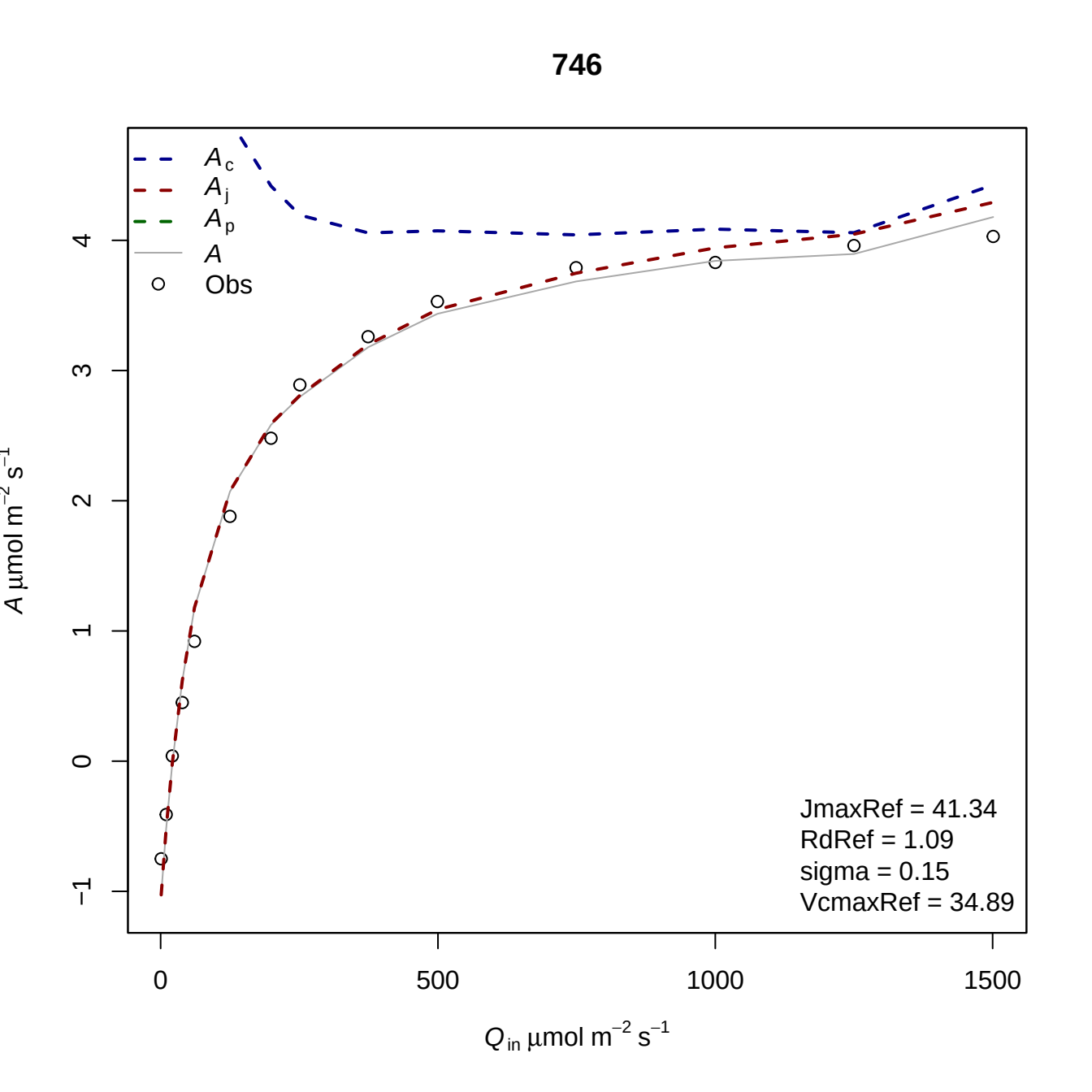

741
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.34
RdRef = 0.5
sigma = 0.13
VcmaxRef = 69.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

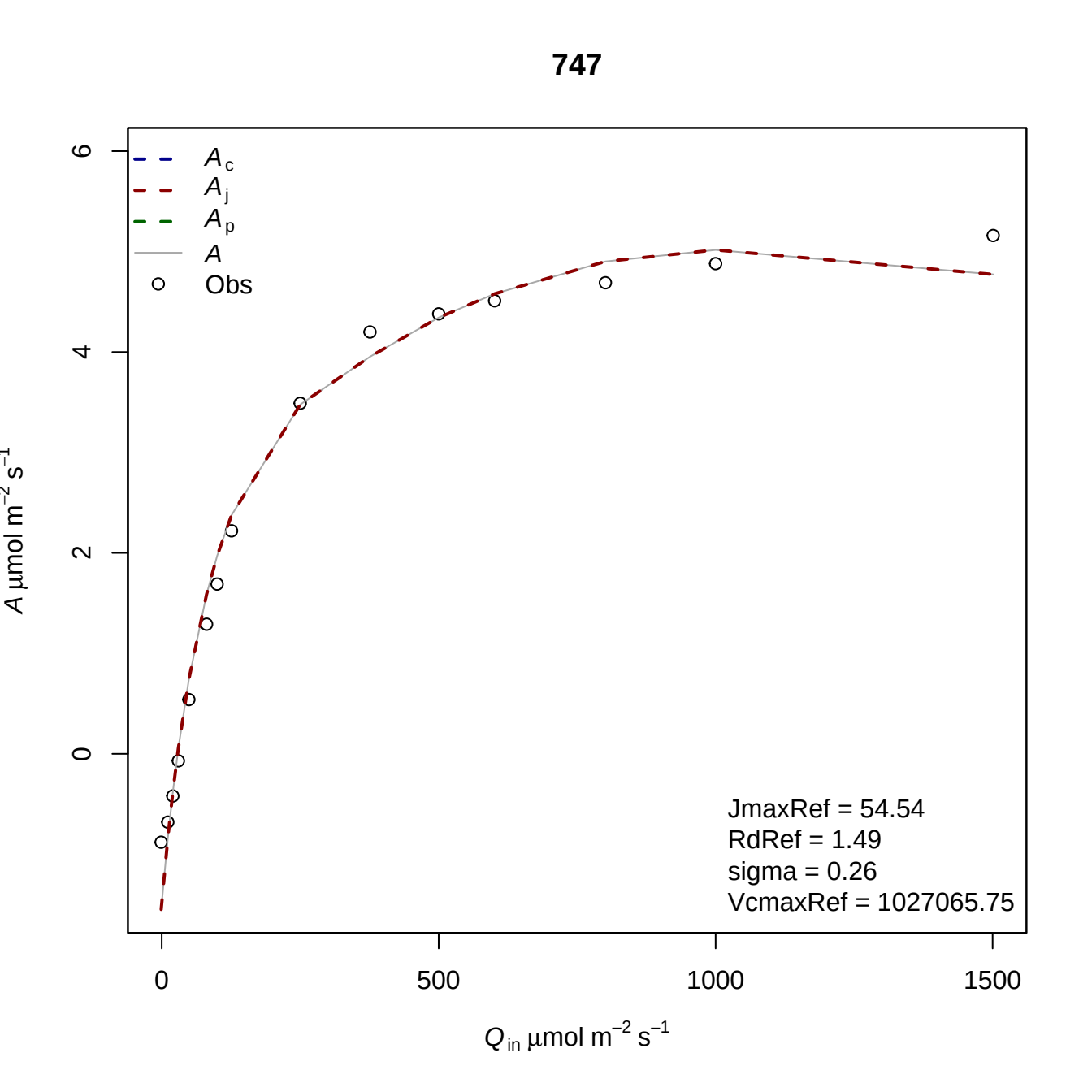

742
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.66
RdRef = 0.91
sigma = 0.11
VcmaxRef = 44.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

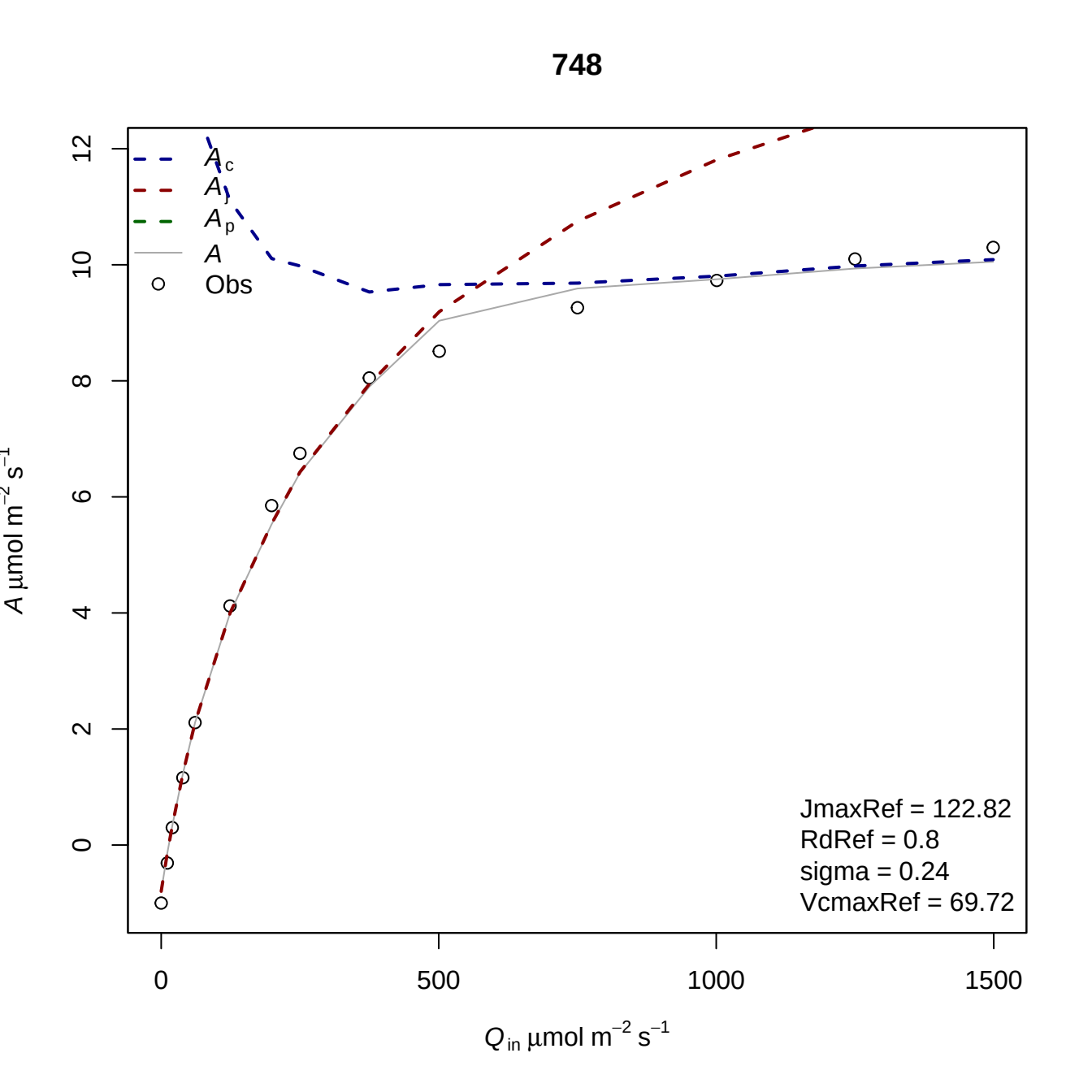

743
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.61
RdRef = 0.98
sigma = 0.12
VcmaxRef = 41.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

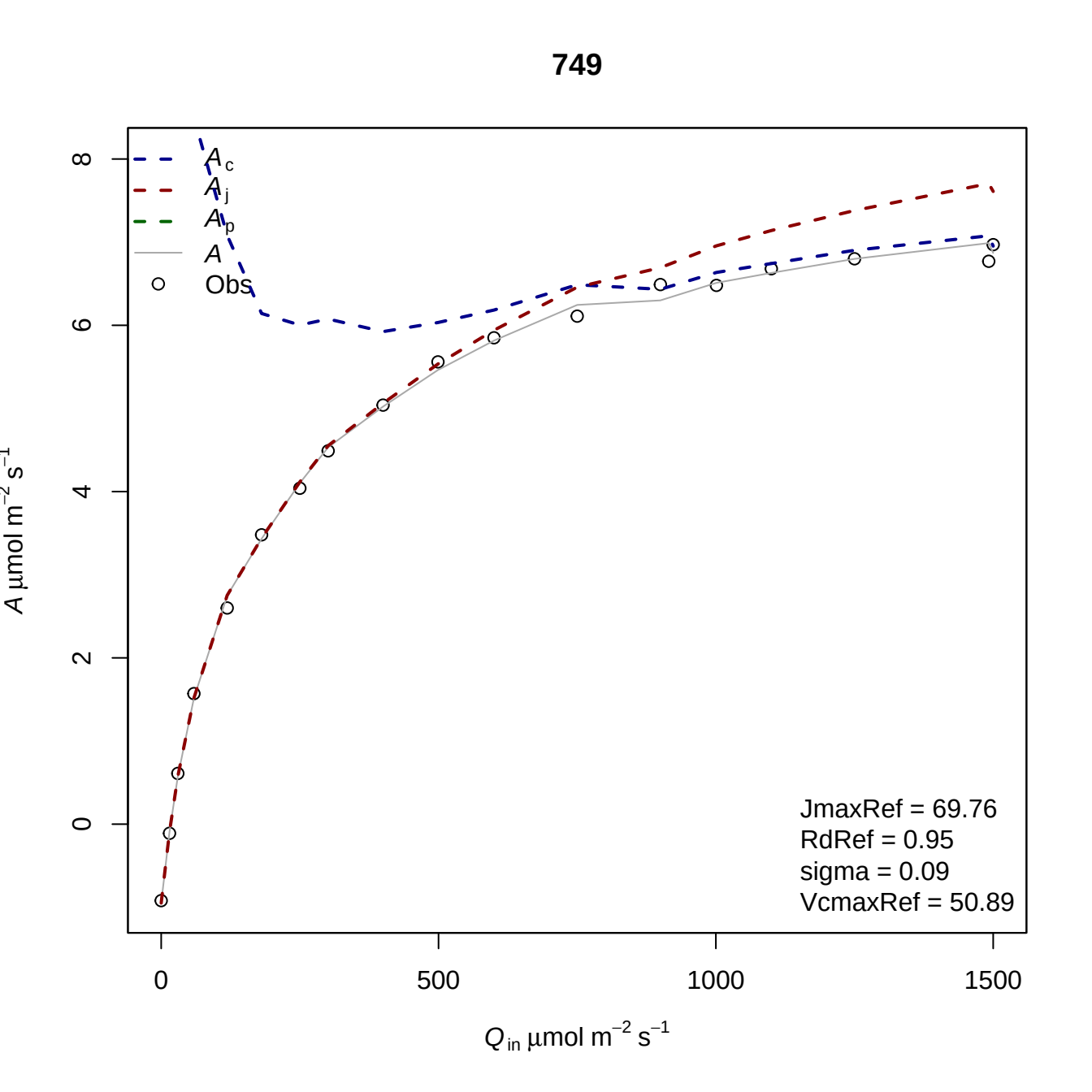

744
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.75
RdRef = 2.01
sigma = 0.17
VcmaxRef = 342233.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

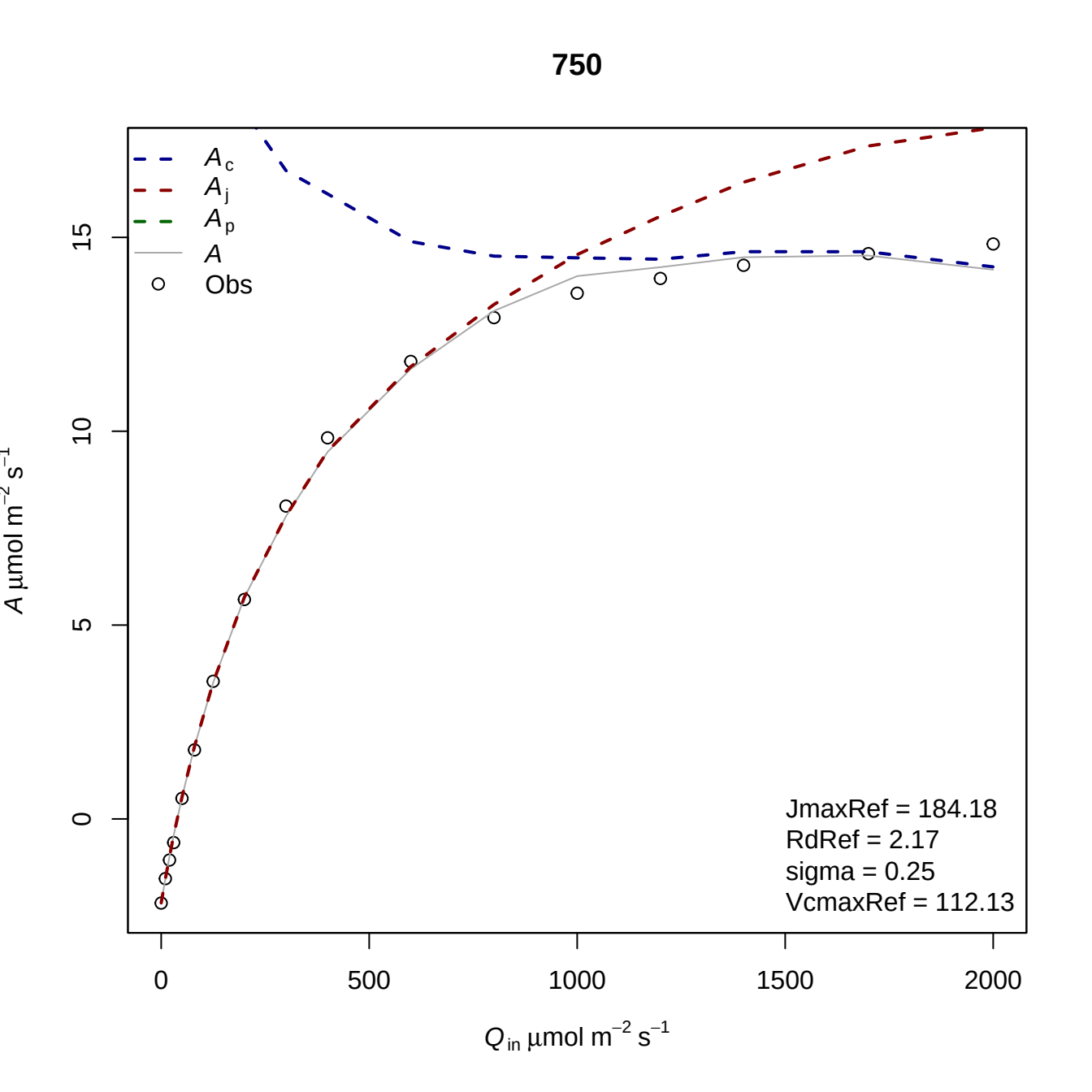

745
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.59
RdRef = 1.39
sigma = 0.2
VcmaxRef = 13760.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

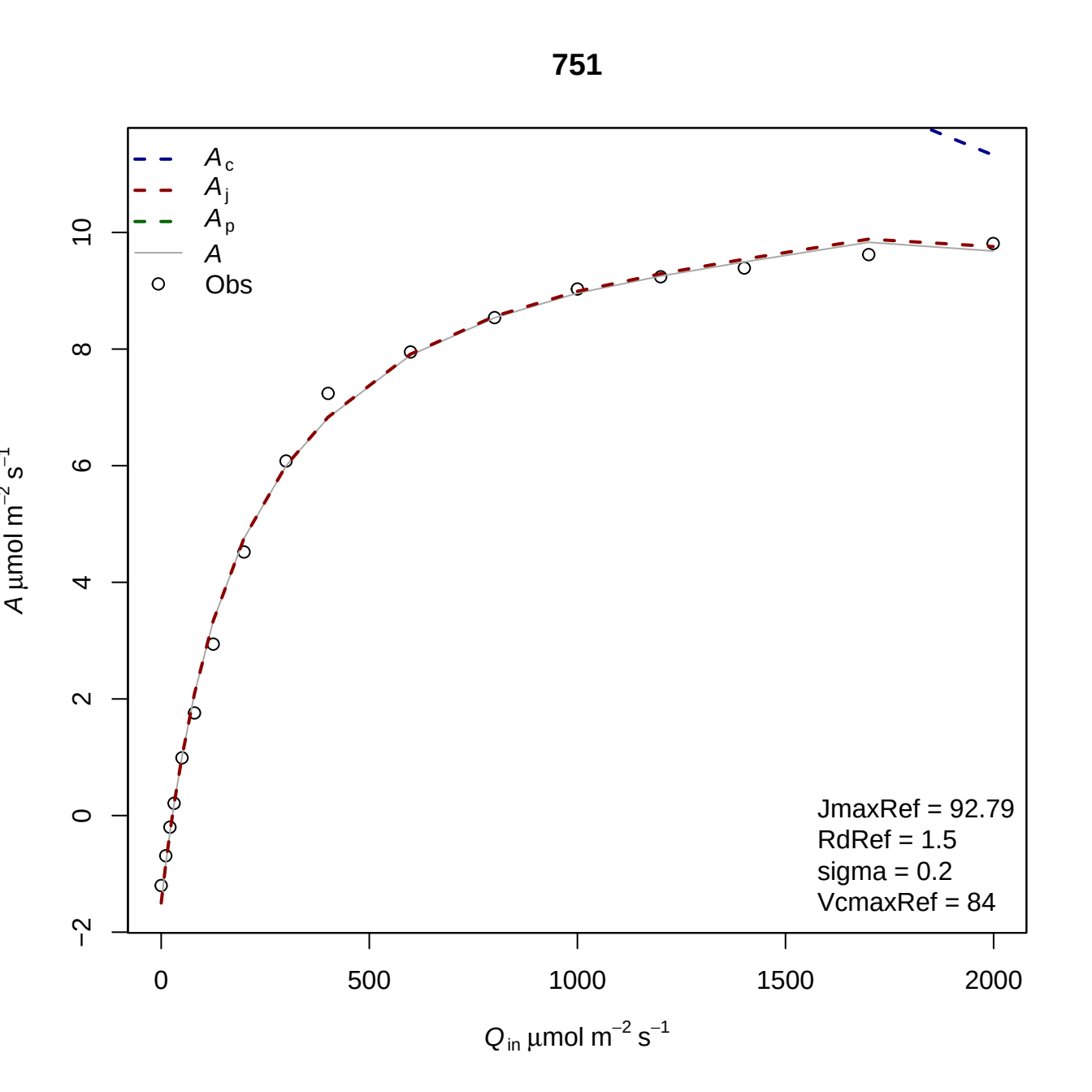

746
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.34
RdRef = 1.09
sigma = 0.15
VcmaxRef = 34.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

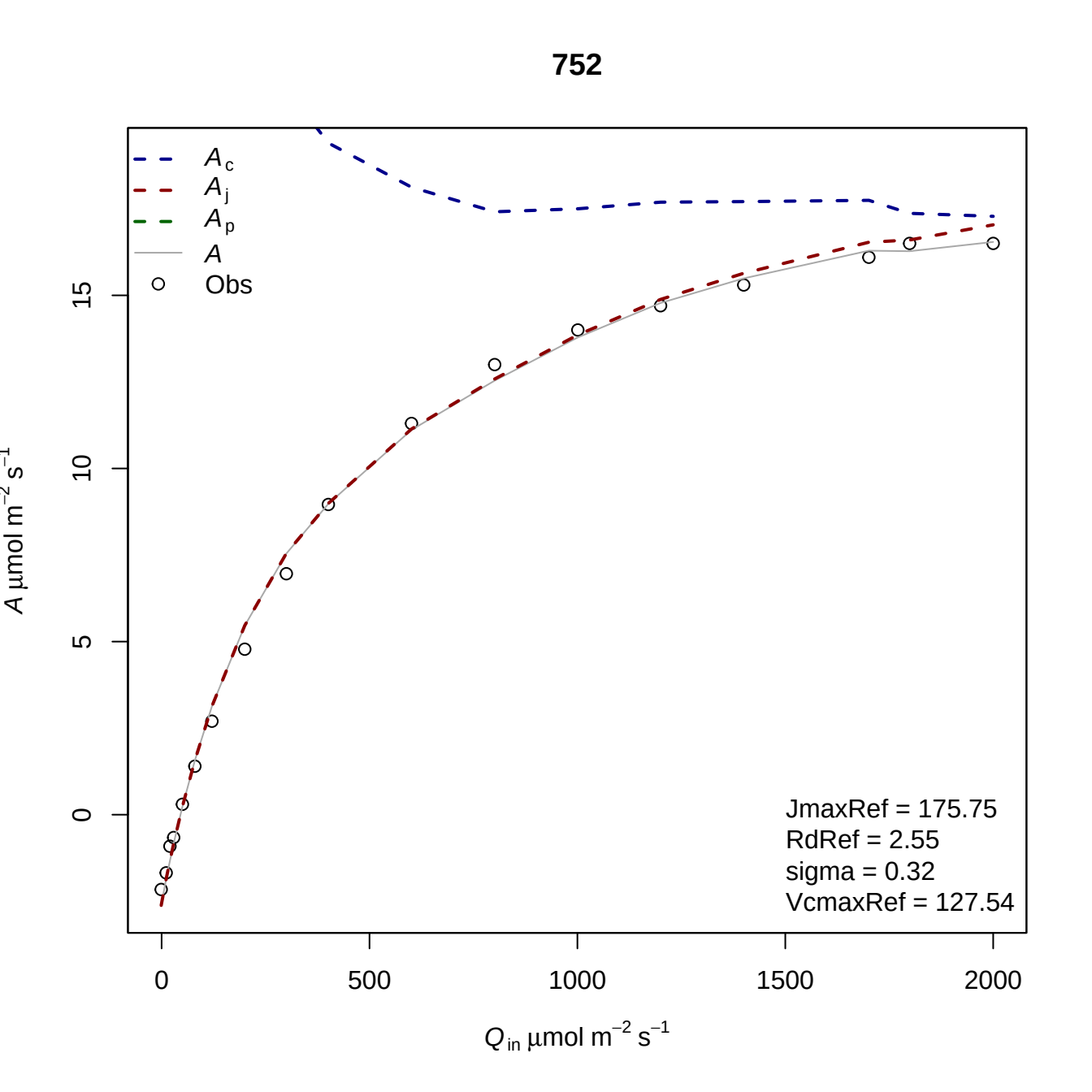

747
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.54
RdRef = 1.49
sigma = 0.26
VcmaxRef = 1027065.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

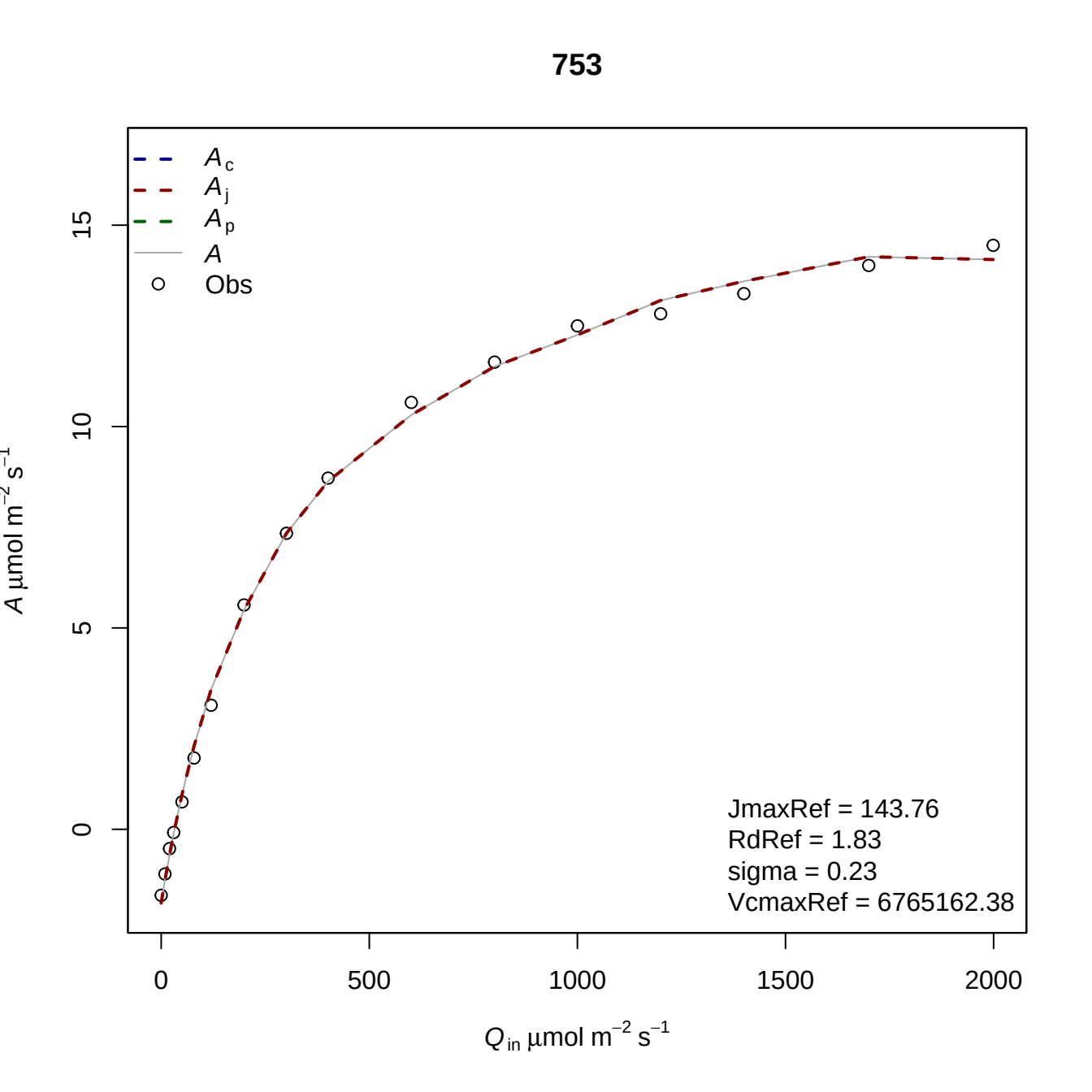

748
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.82
RdRef = 0.8
sigma = 0.24
VcmaxRef = 69.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

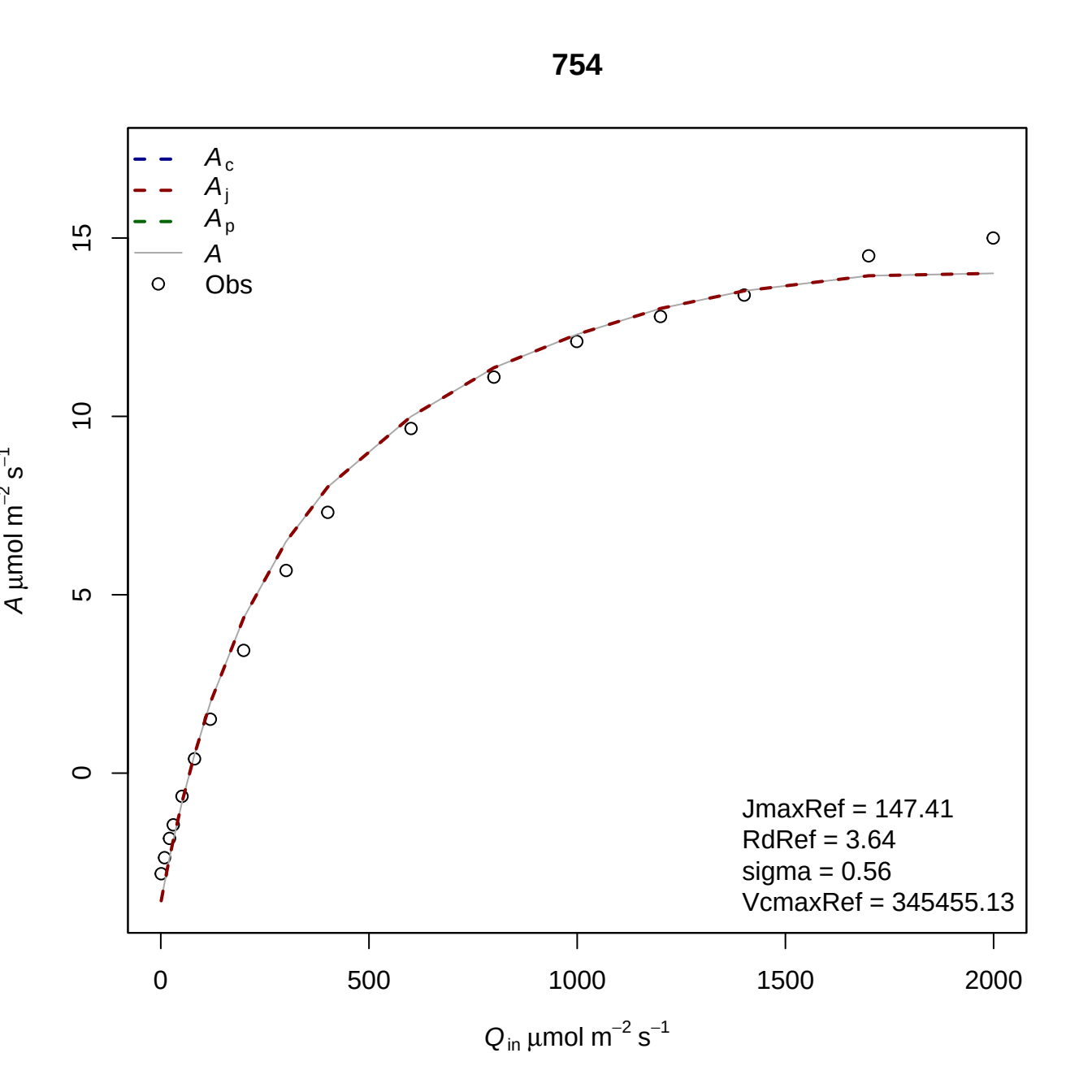

749
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.76
RdRef = 0.95
sigma = 0.09
VcmaxRef = 50.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

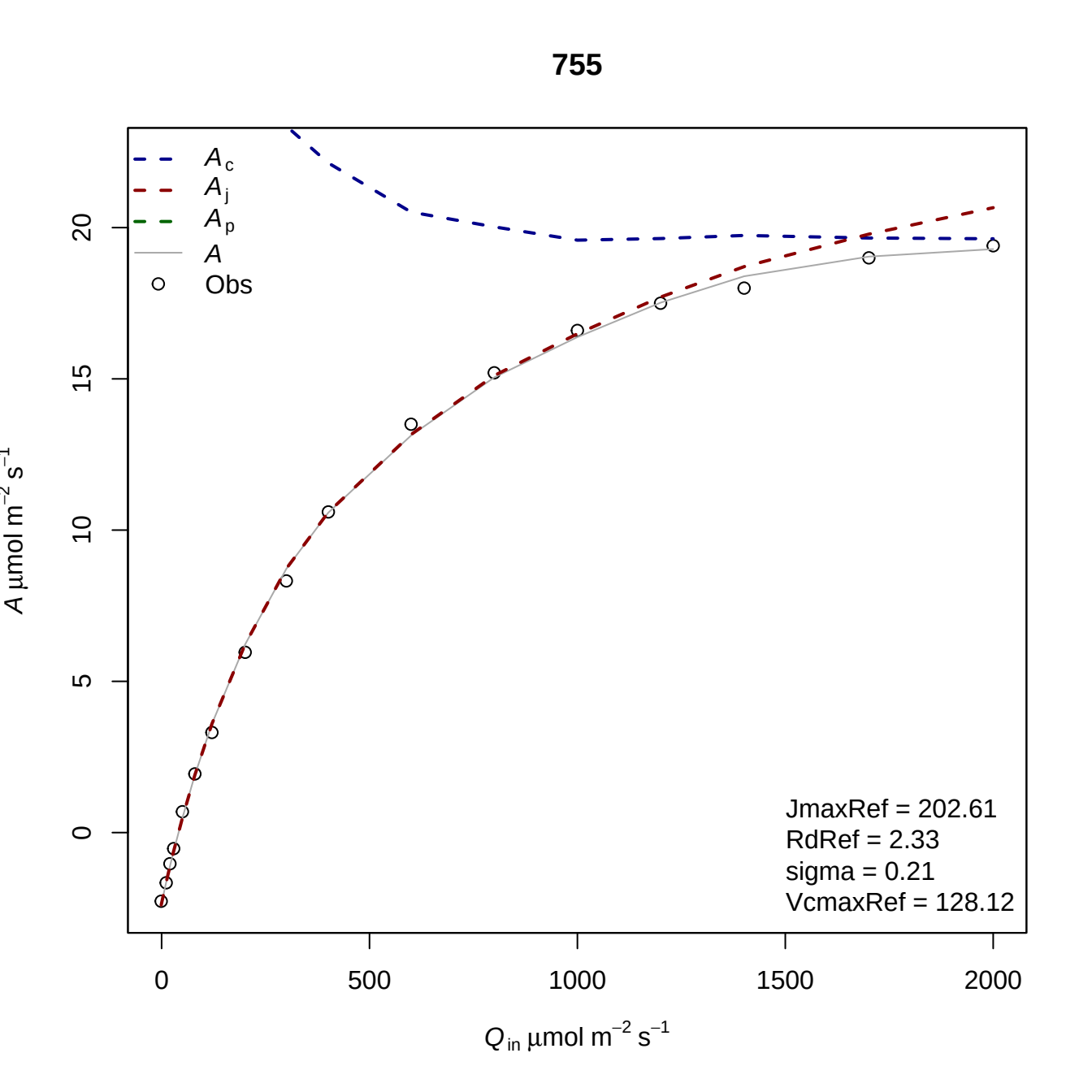

750
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 184.18
RdRef = 2.17
sigma = 0.25
VcmaxRef = 112.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

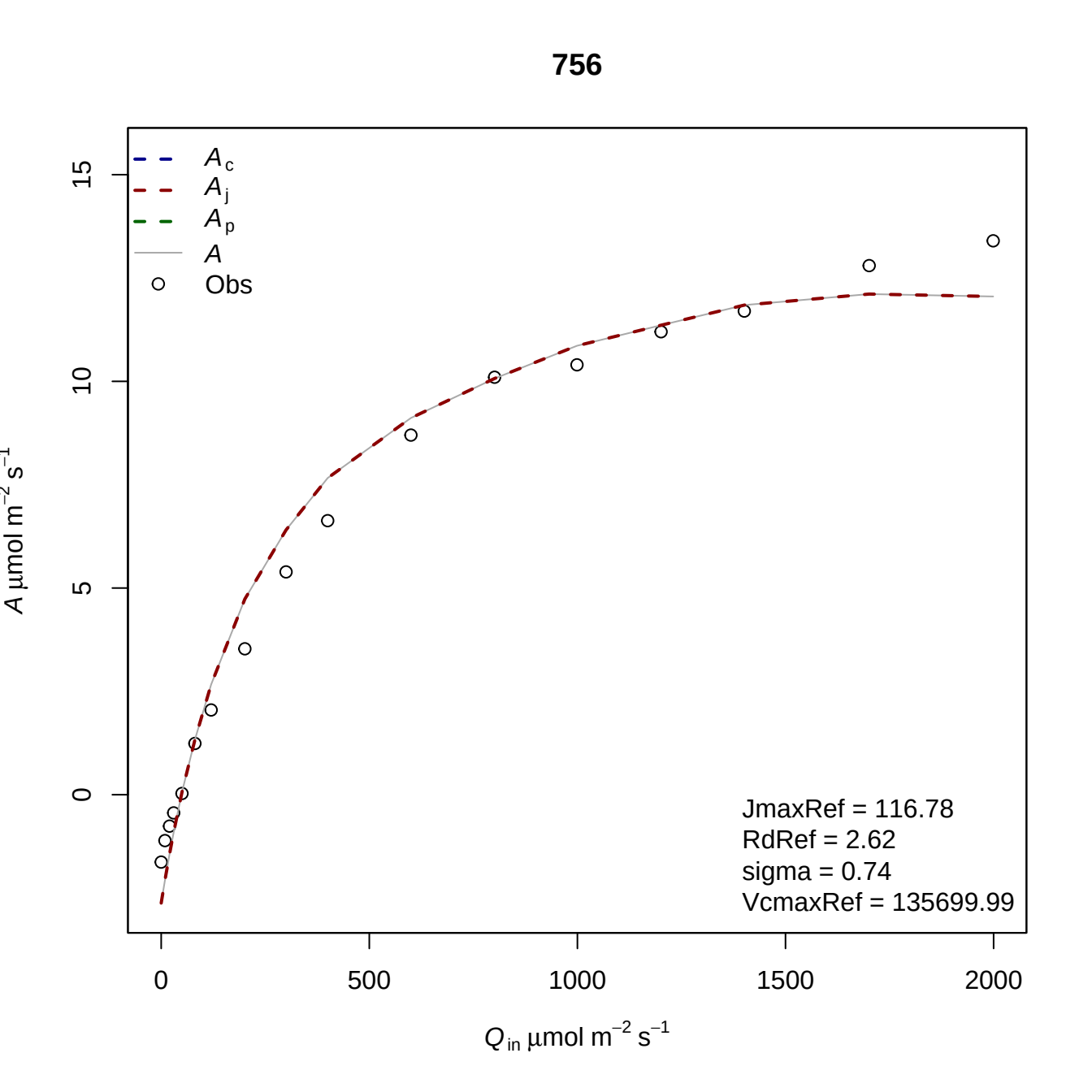

751
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.79
RdRef = 1.5
sigma = 0.2
VcmaxRef = 84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

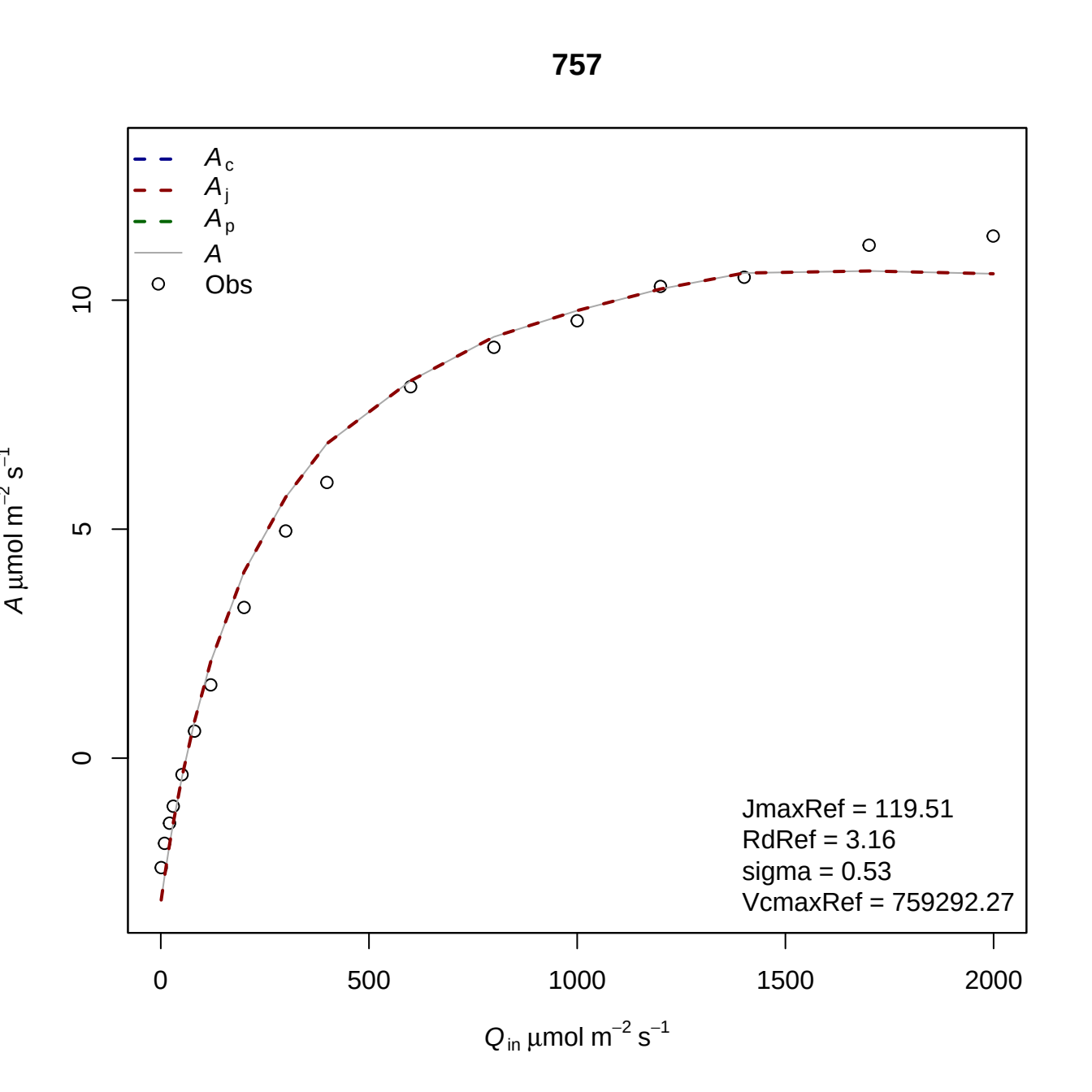

752
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 175.75
RdRef = 2.55
sigma = 0.32
VcmaxRef = 127.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

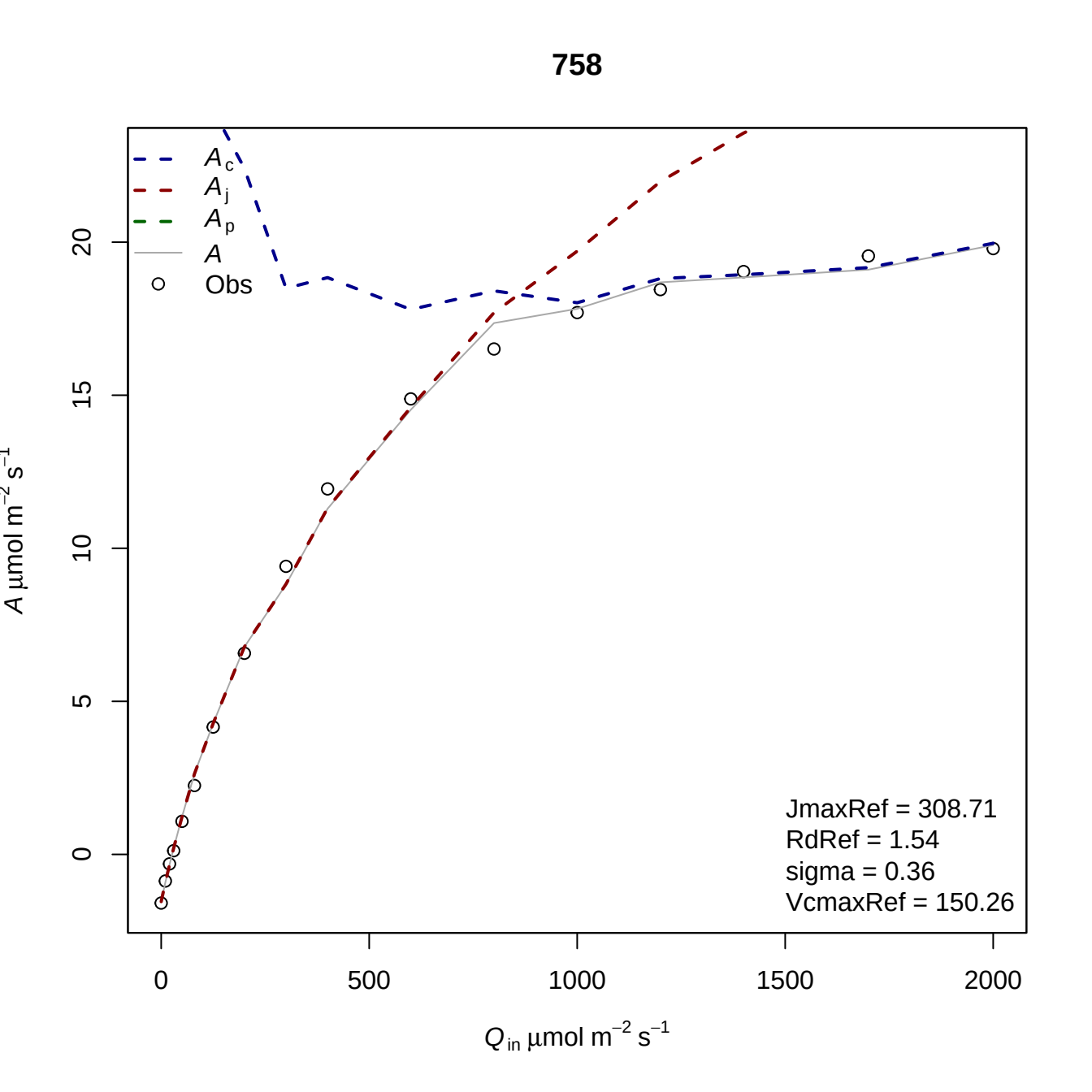

753
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.76
RdRef = 1.83
sigma = 0.23
VcmaxRef = 6765162.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

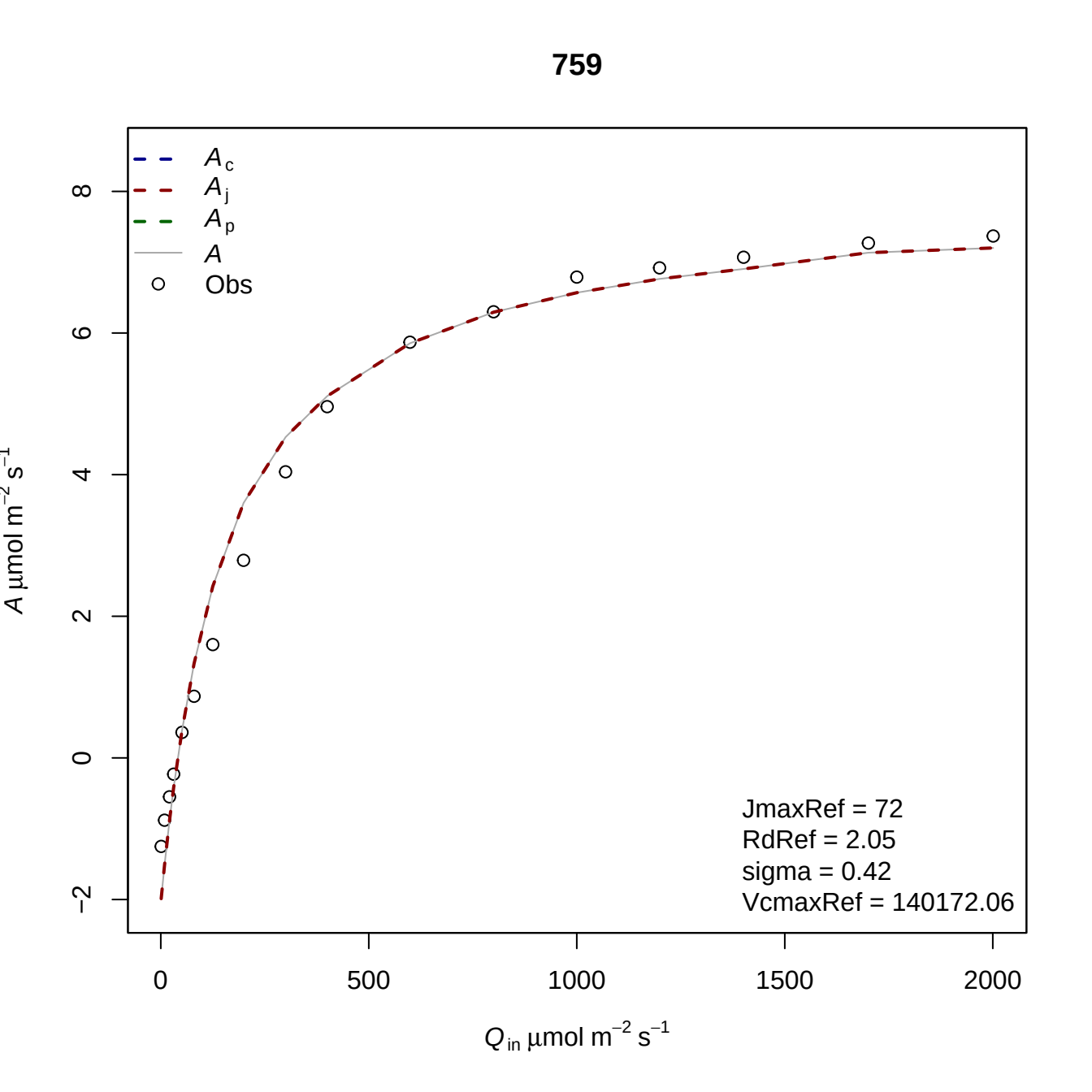

754
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.41
RdRef = 3.64
sigma = 0.56
VcmaxRef = 345455.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

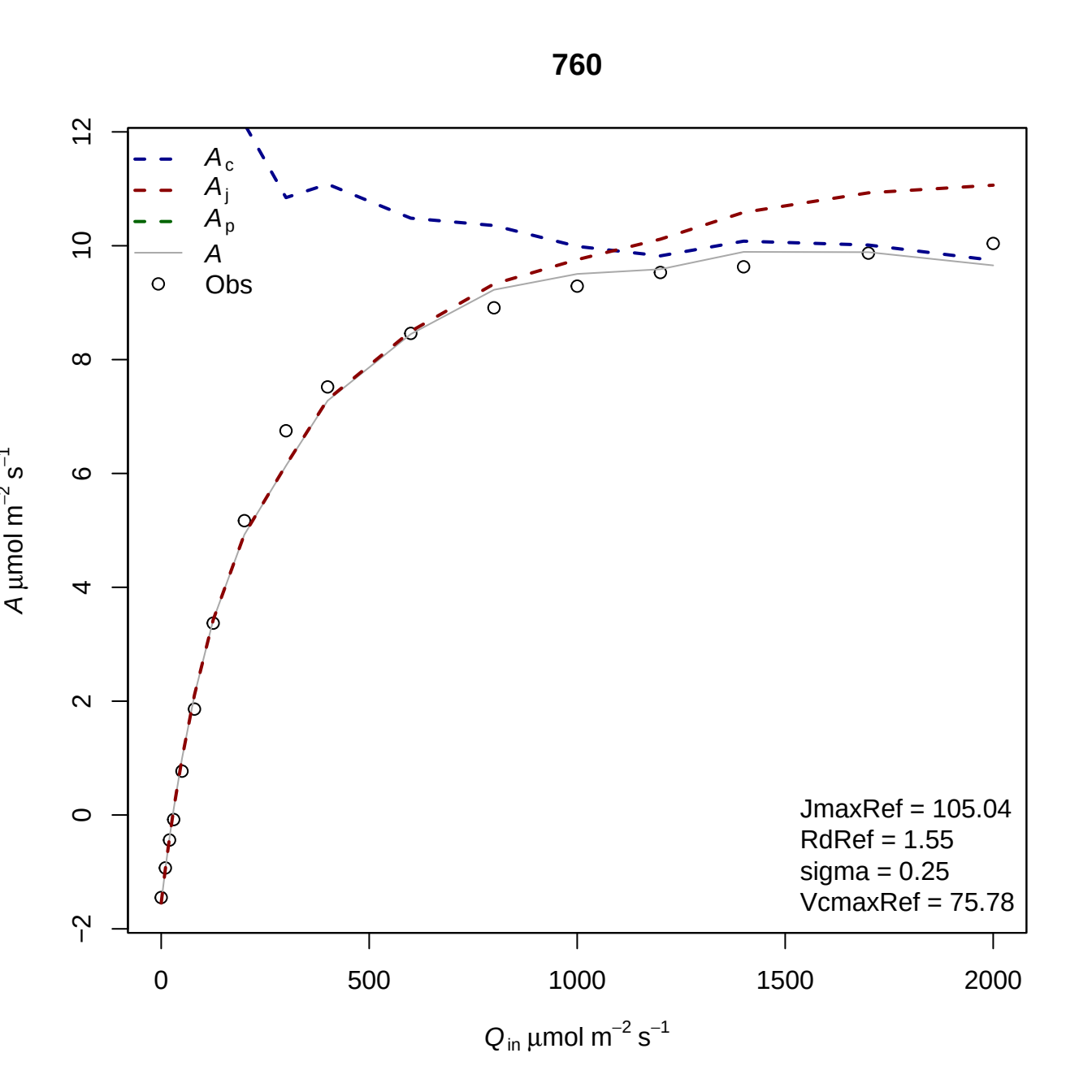

755
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 202.61
RdRef = 2.33
sigma = 0.21
VcmaxRef = 128.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

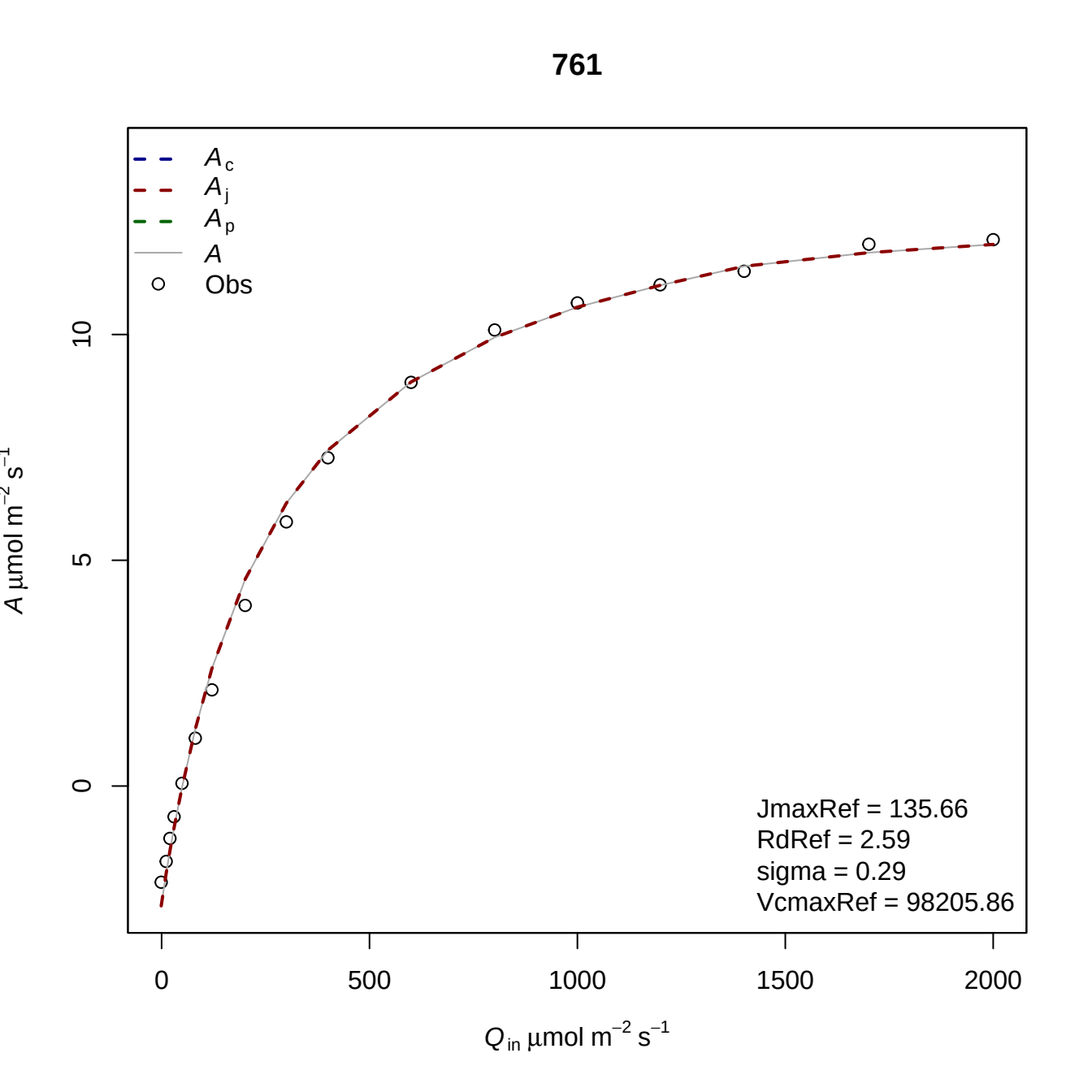

756
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.78
RdRef = 2.62
sigma = 0.74
VcmaxRef = 135699.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

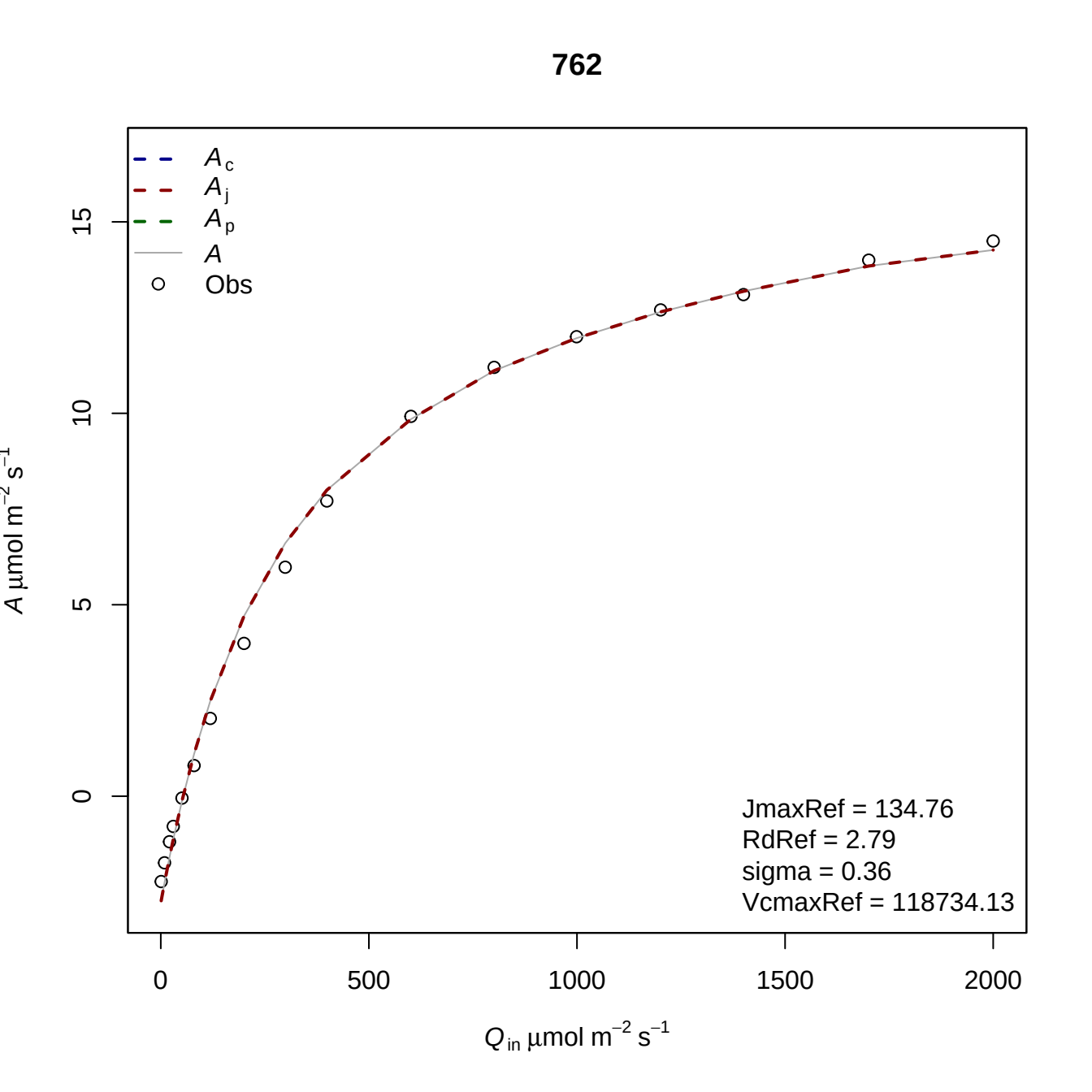

757
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.51
RdRef = 3.16
sigma = 0.53
VcmaxRef = 759292.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

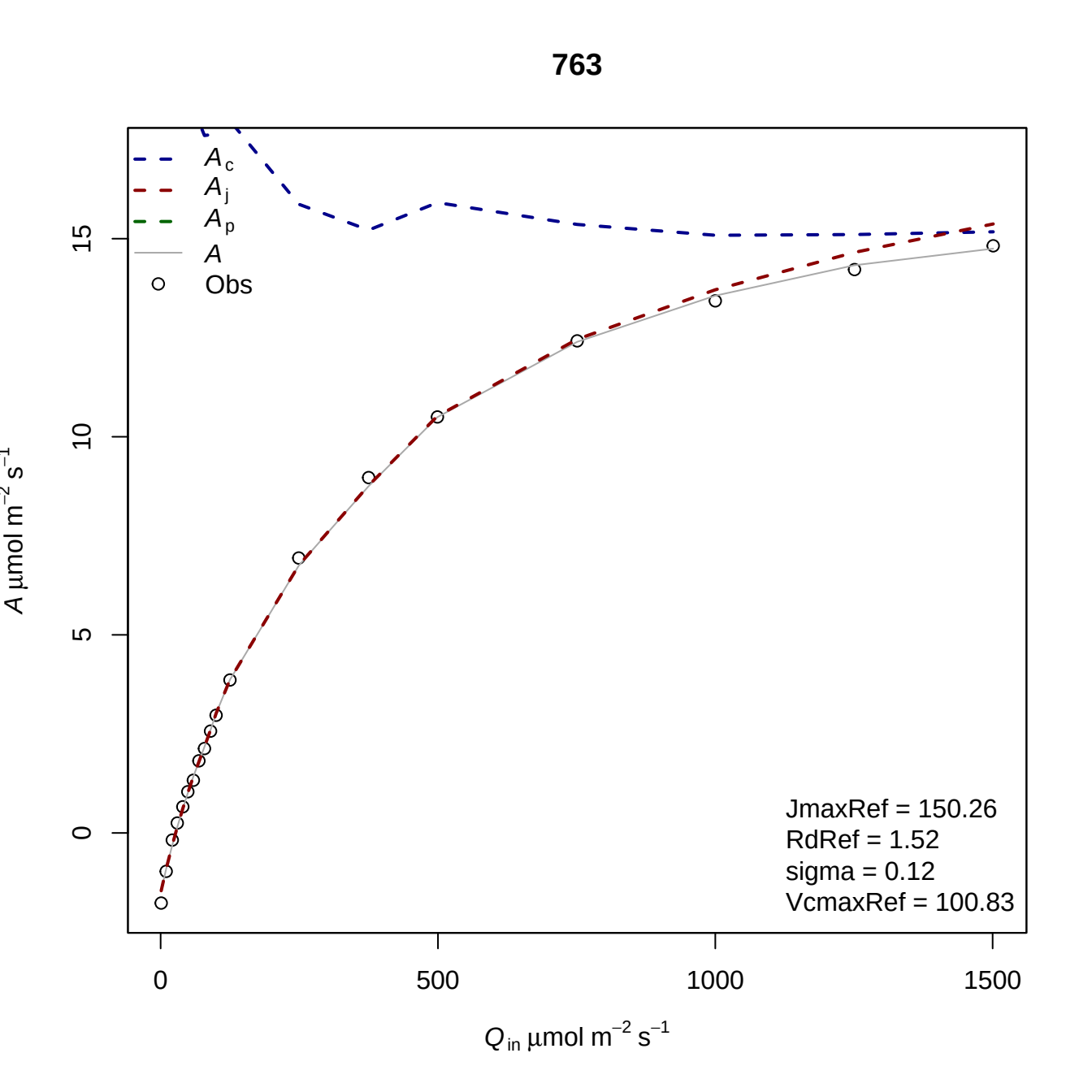

758
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 308.71
RdRef = 1.54
sigma = 0.36
VcmaxRef = 150.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

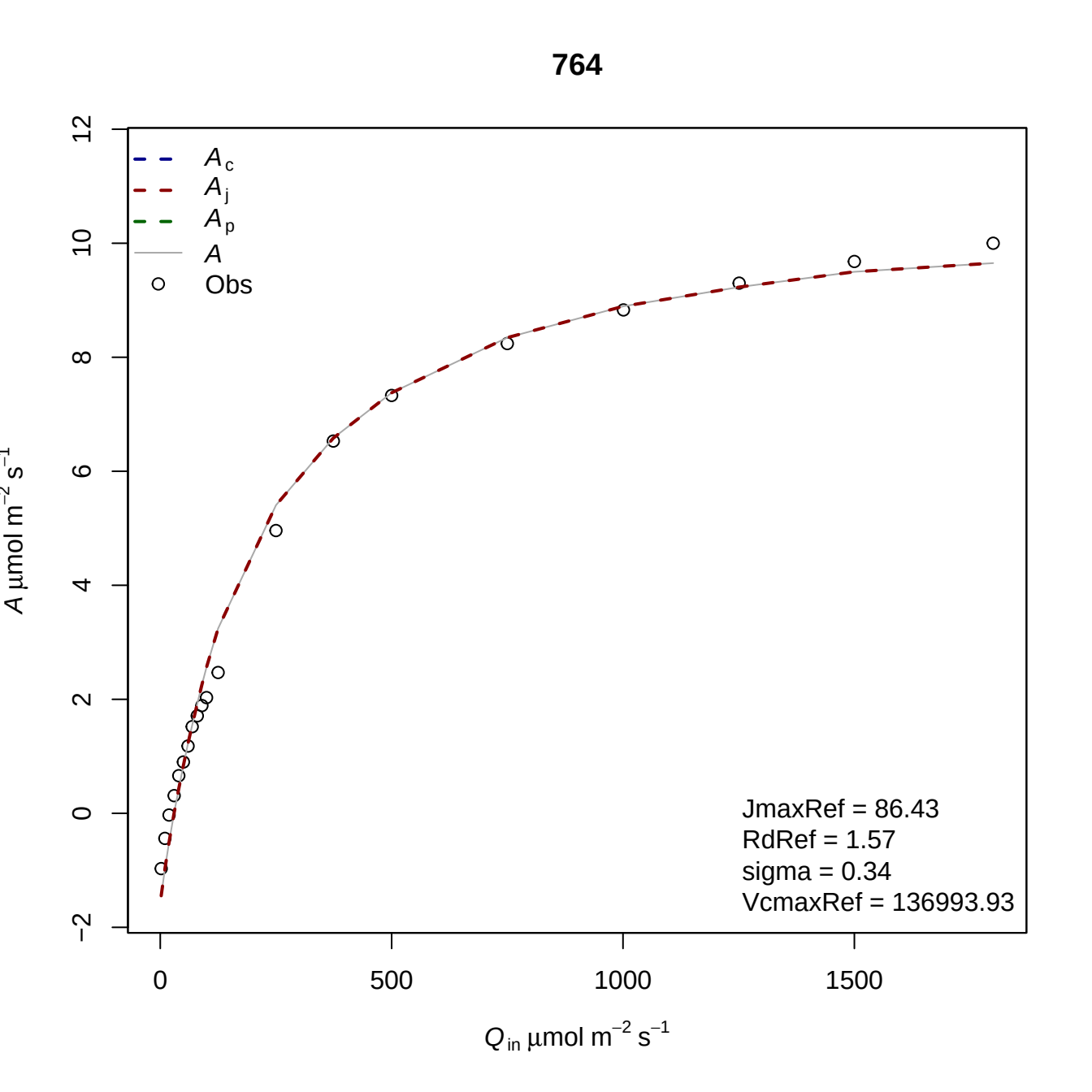

759
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72
RdRef = 2.05
sigma = 0.42
VcmaxRef = 140172.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

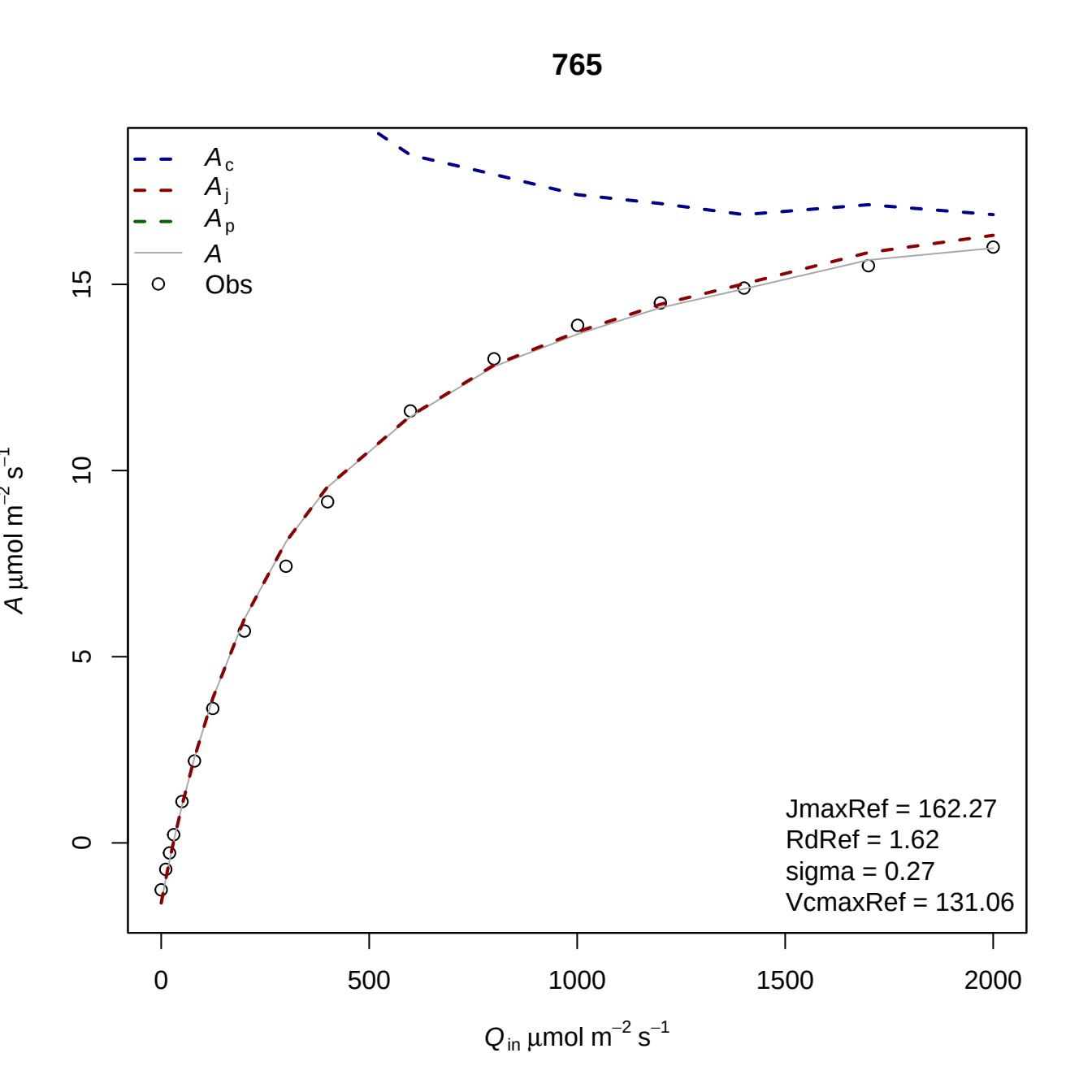

760
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.04
RdRef = 1.55
sigma = 0.25
VcmaxRef = 75.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

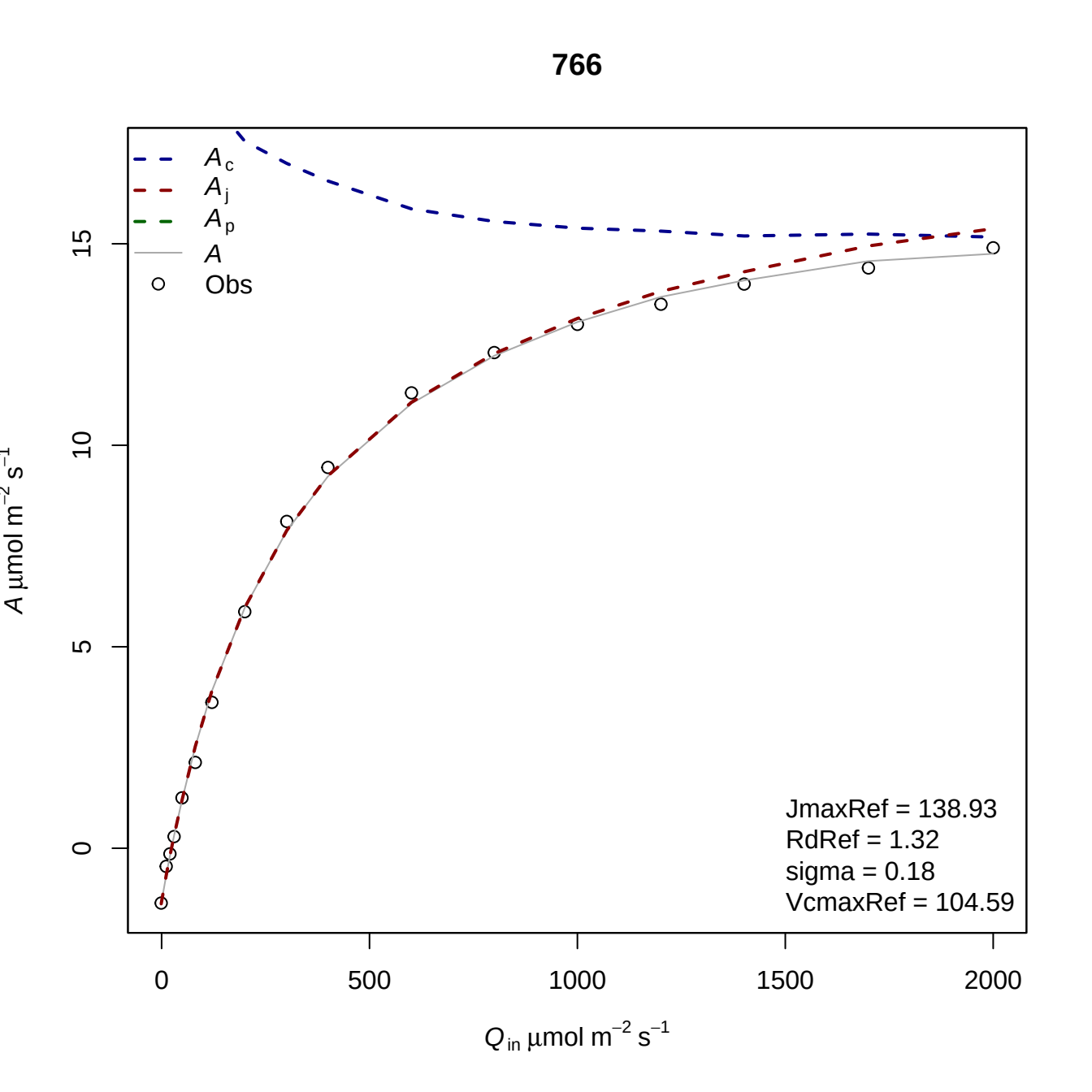

761
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.66
RdRef = 2.59
sigma = 0.29
VcmaxRef = 98205.86
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

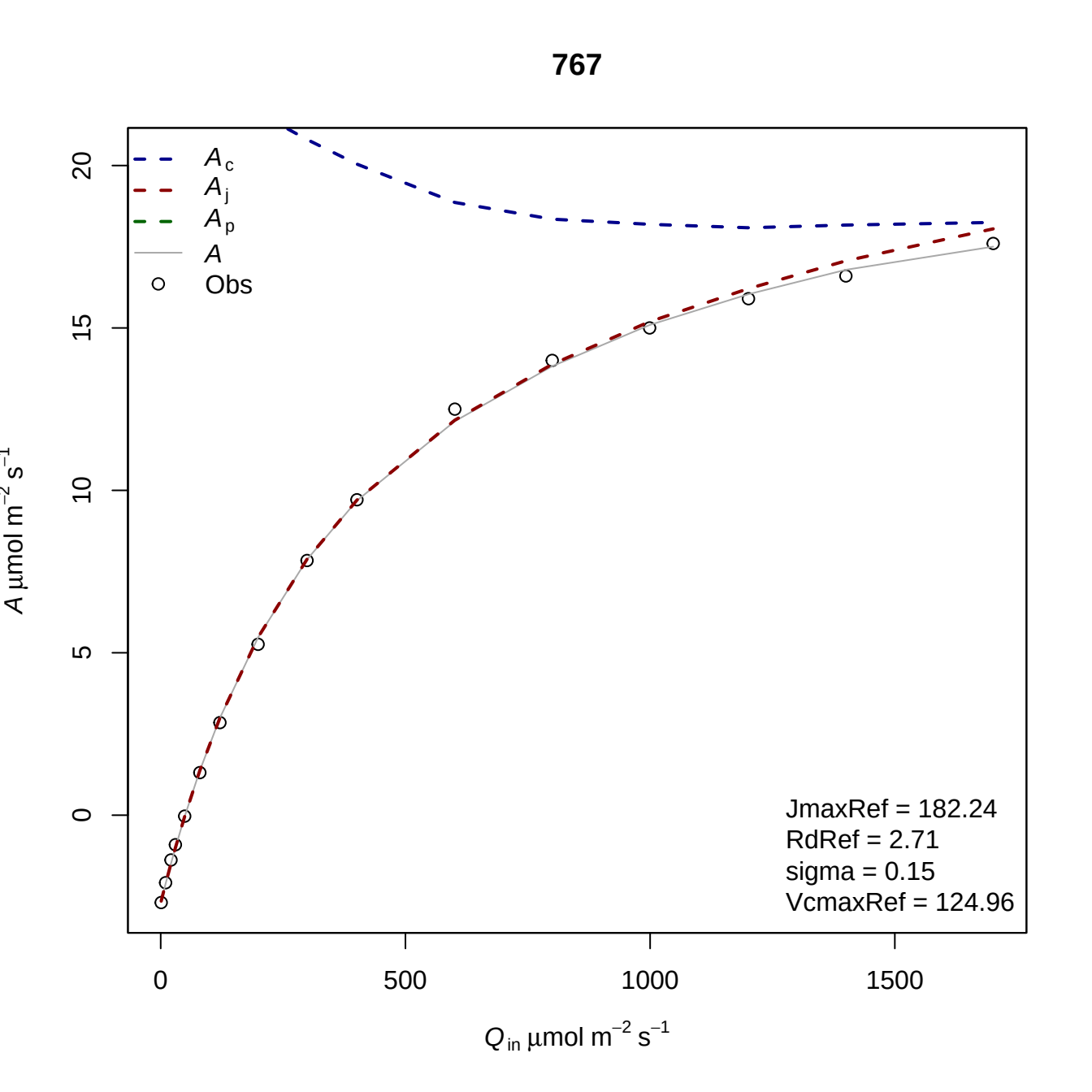

762
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.76
RdRef = 2.79
sigma = 0.36
VcmaxRef = 118734.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

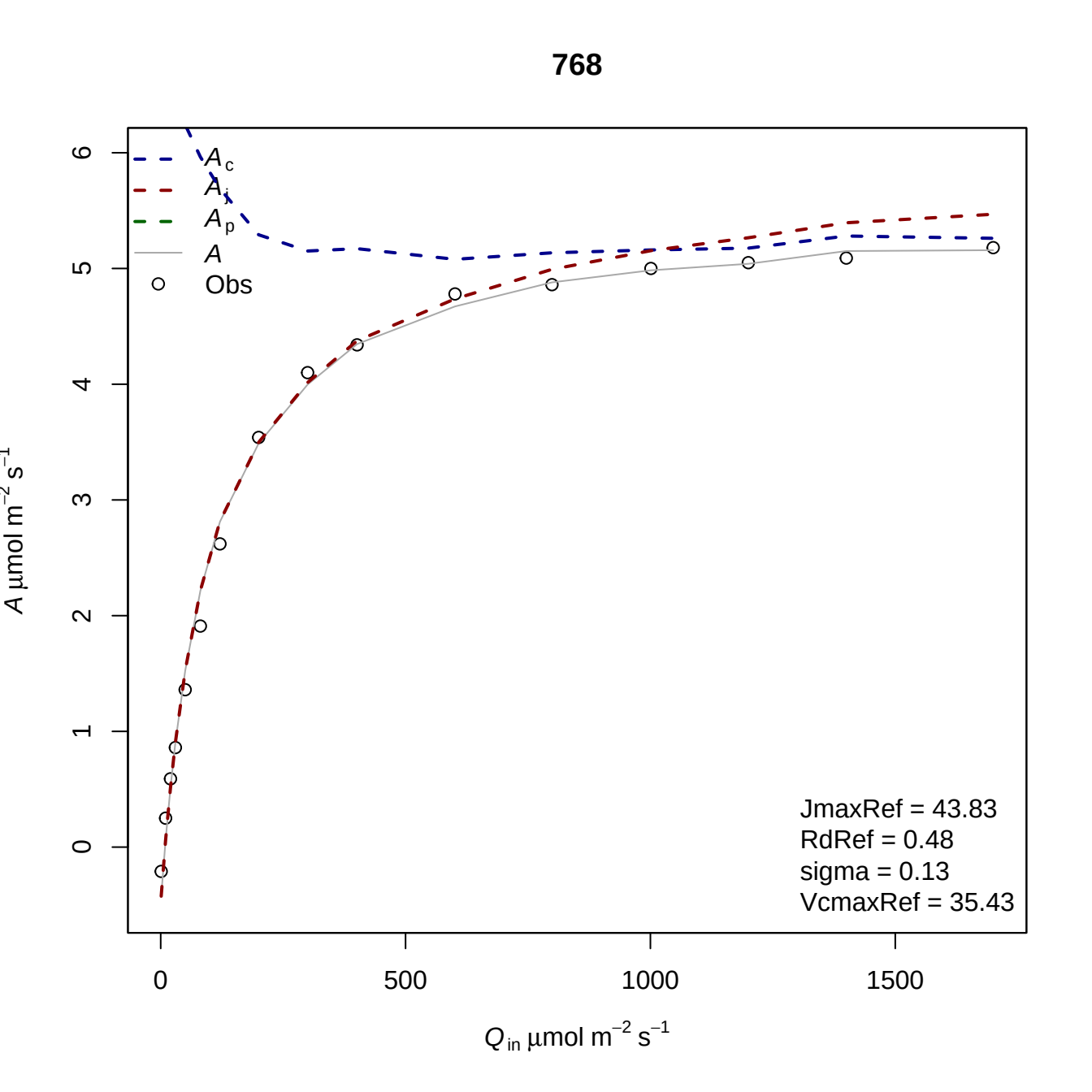

763
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 150.26
RdRef = 1.52
sigma = 0.12
VcmaxRef = 100.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

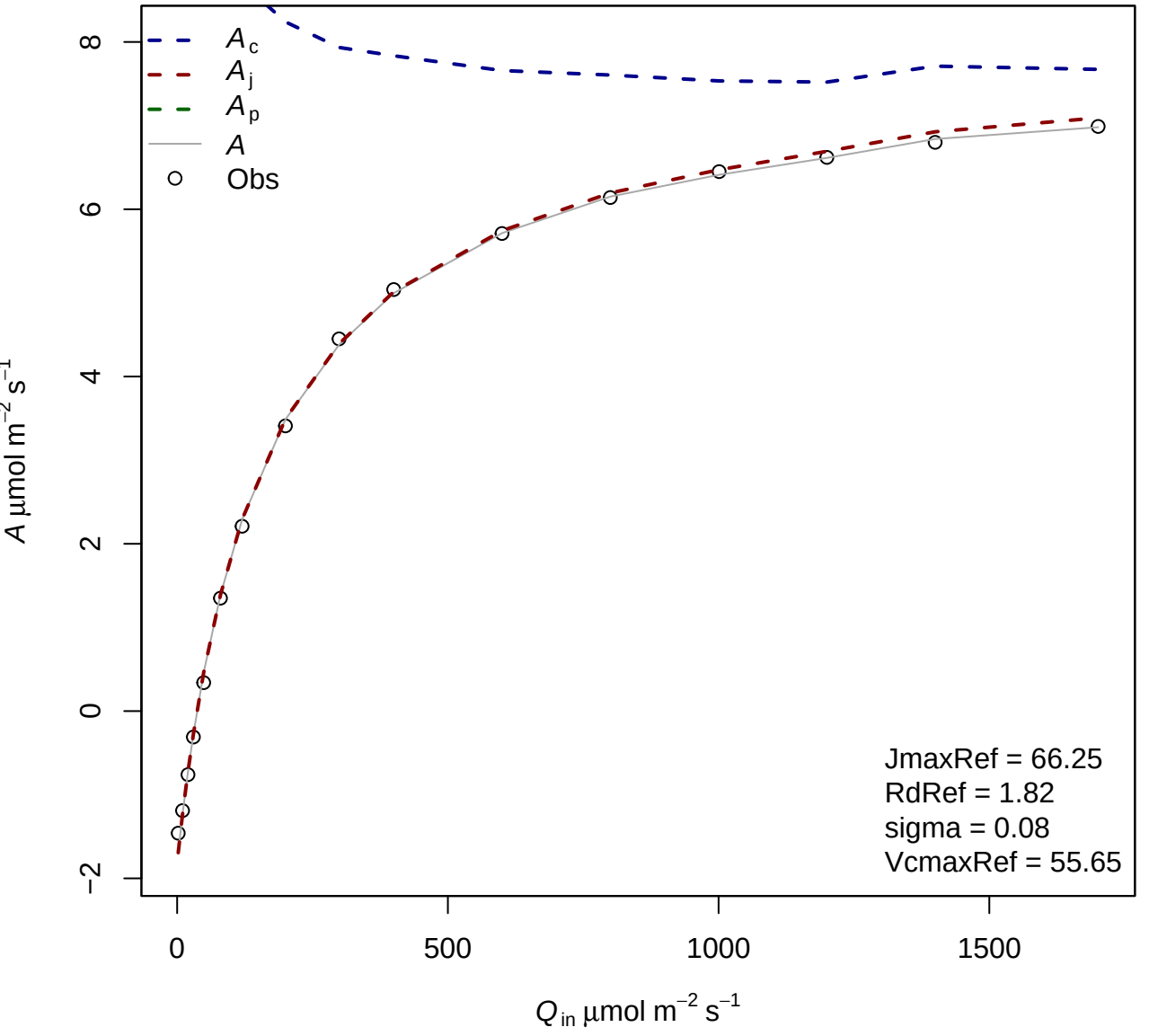

764
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.43
RdRef = 1.57
sigma = 0.34
VcmaxRef = 136993.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

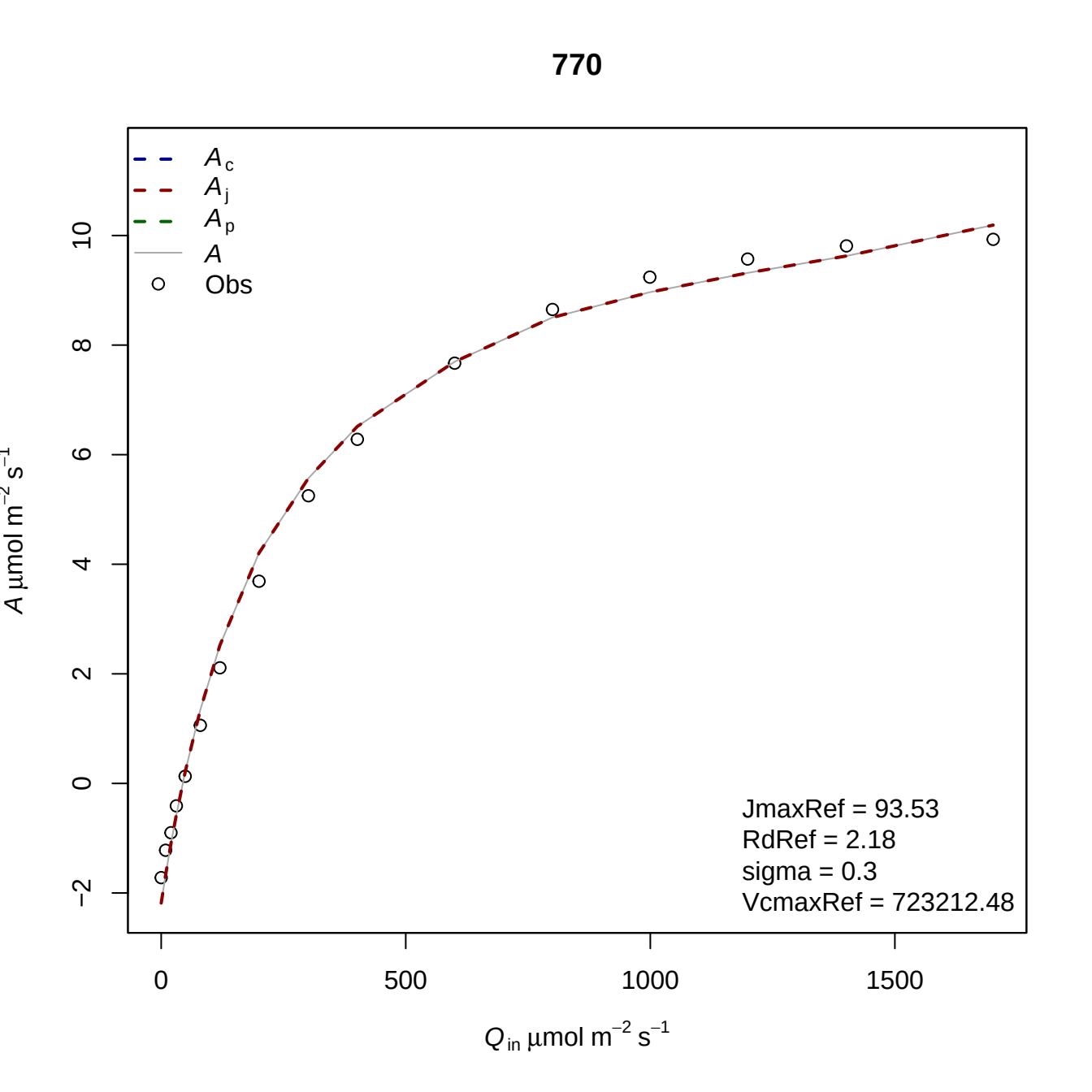

765
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 162.27
RdRef = 1.62
sigma = 0.27
VcmaxRef = 131.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

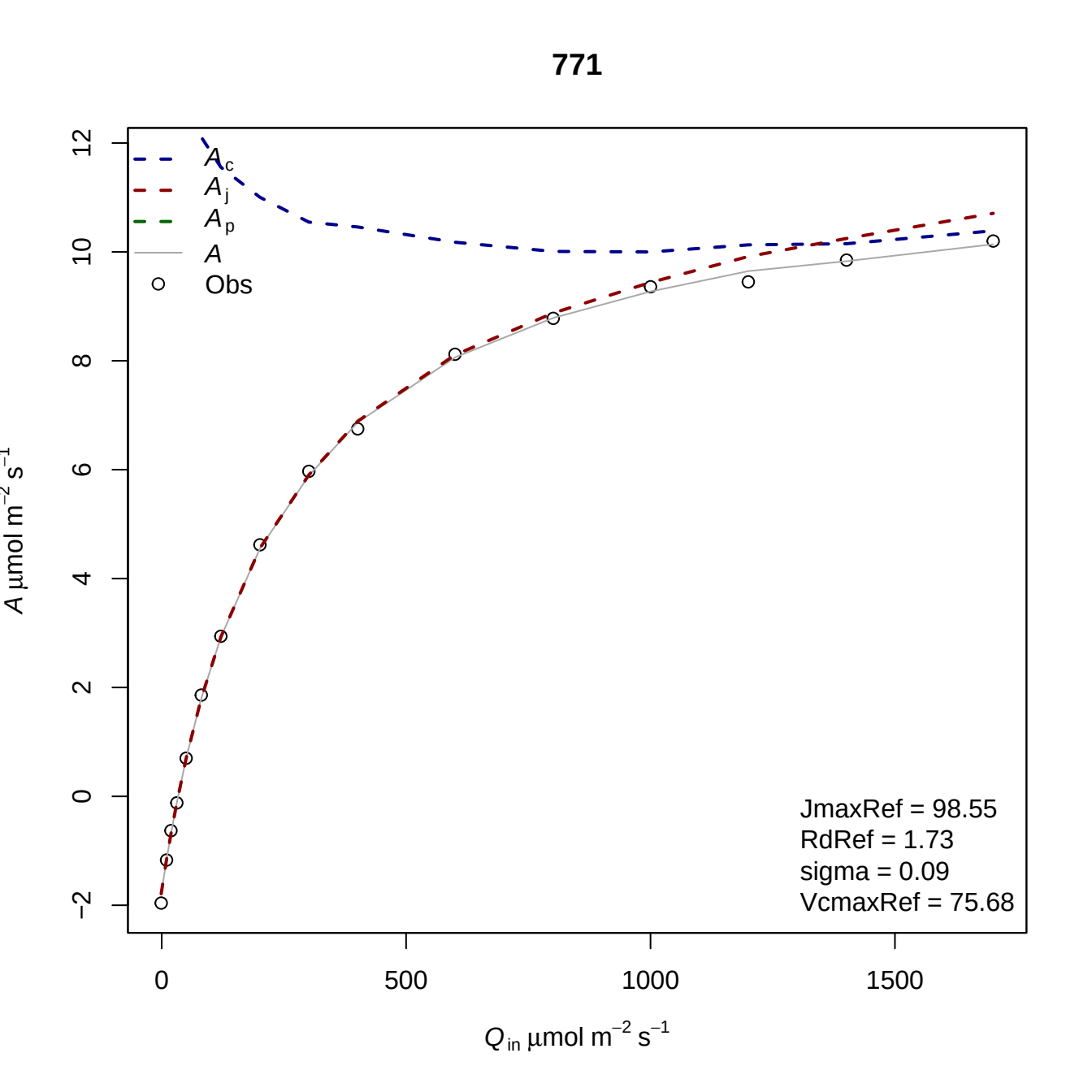

766
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.93
RdRef = 1.32
sigma = 0.18
VcmaxRef = 104.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

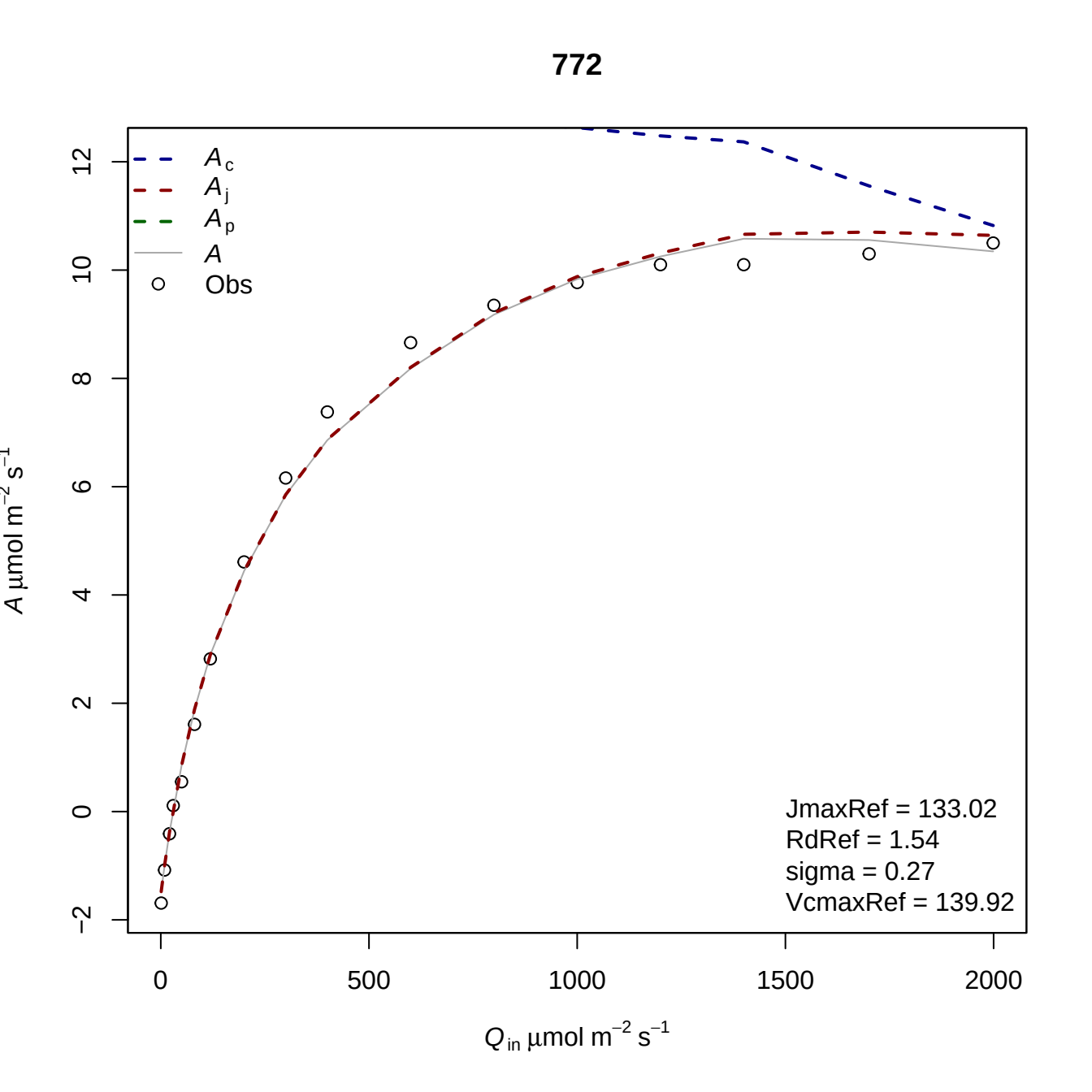

767
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 182.24
RdRef = 2.71
sigma = 0.15
VcmaxRef = 124.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

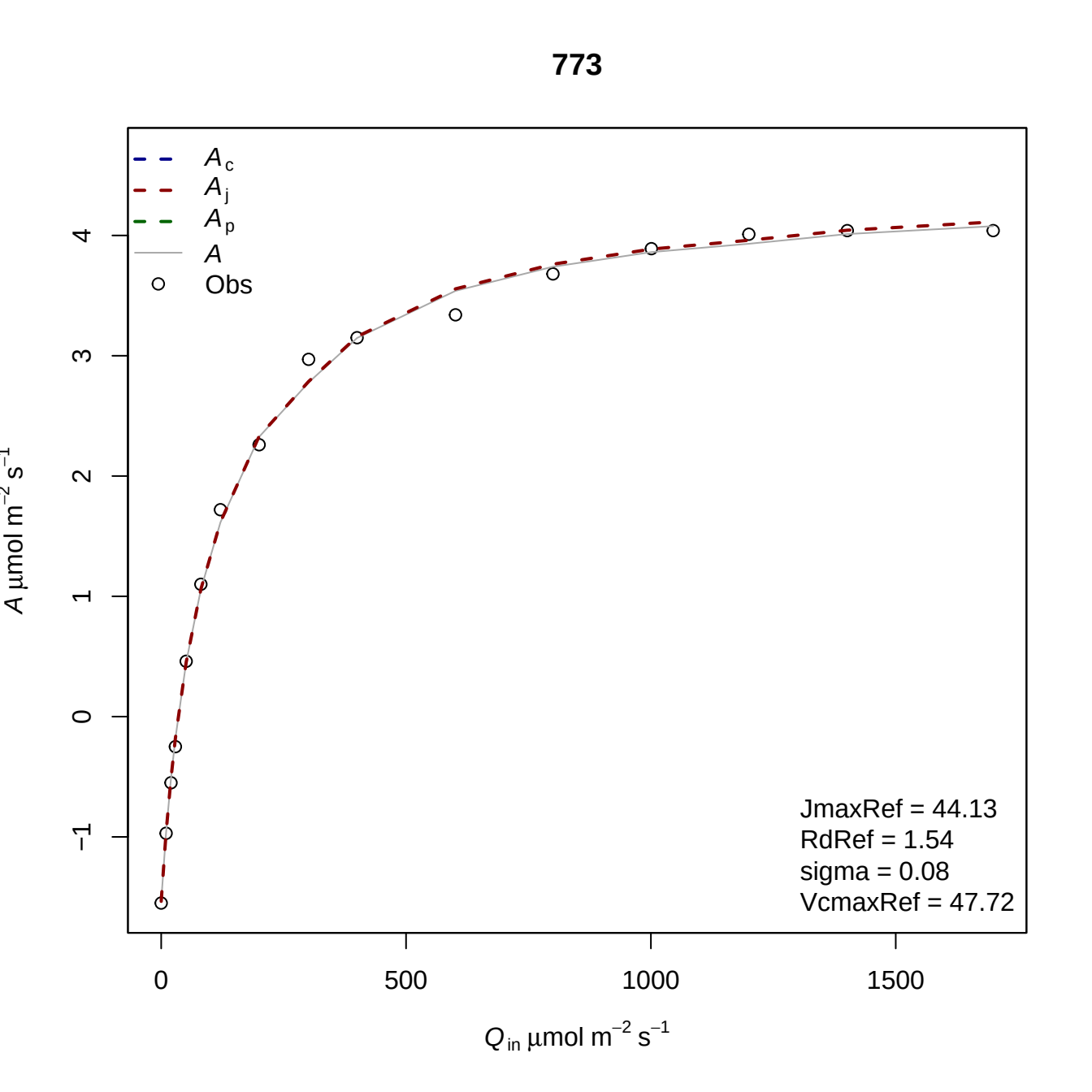

768
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.83
RdRef = 0.48
sigma = 0.13
VcmaxRef = 35.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

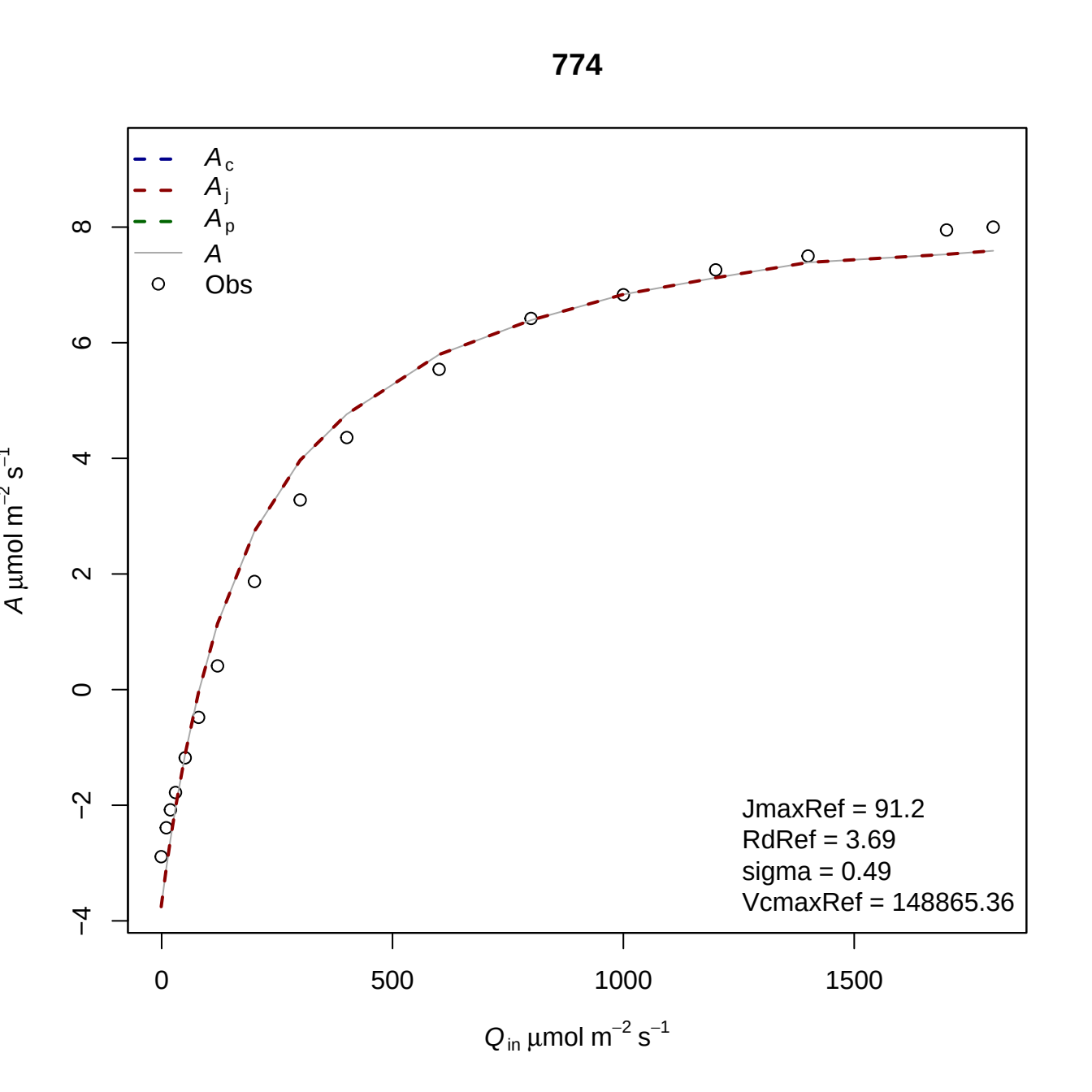

$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.25
RdRef = 1.82
sigma = 0.08
VcmaxRef = 55.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

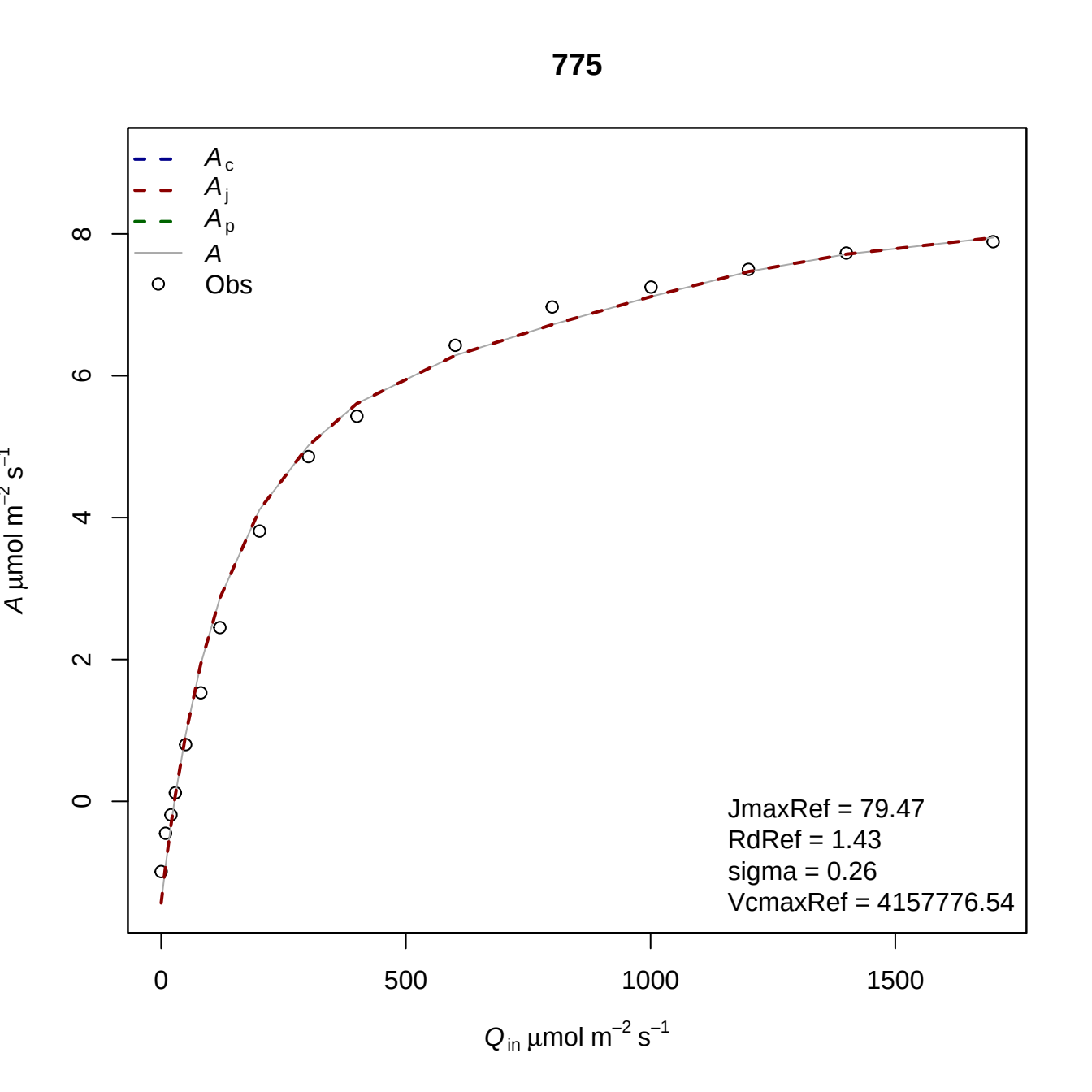

770
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.53
RdRef = 2.18
sigma = 0.3
VcmaxRef = 723212.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

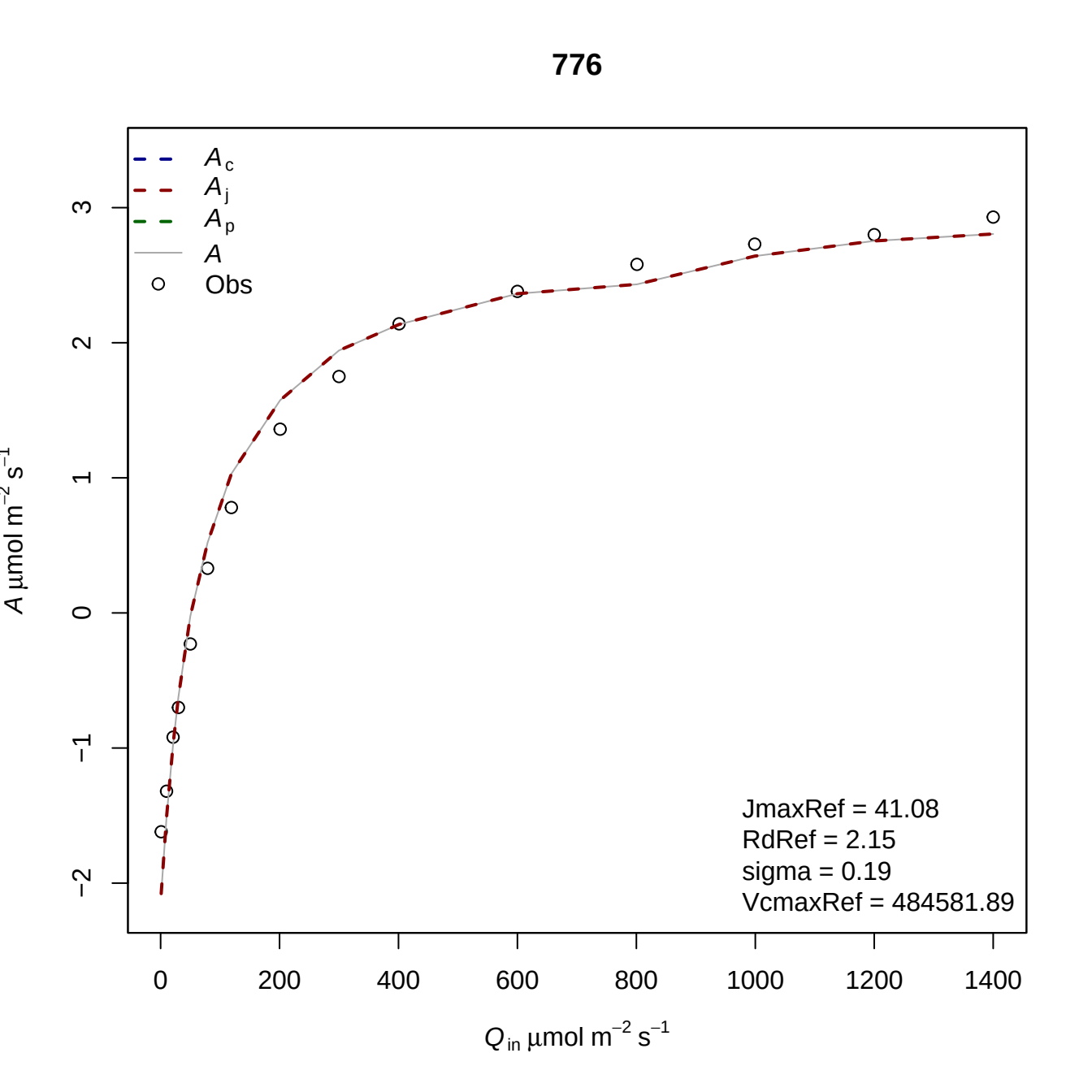

771
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.55
RdRef = 1.73
sigma = 0.09
VcmaxRef = 75.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

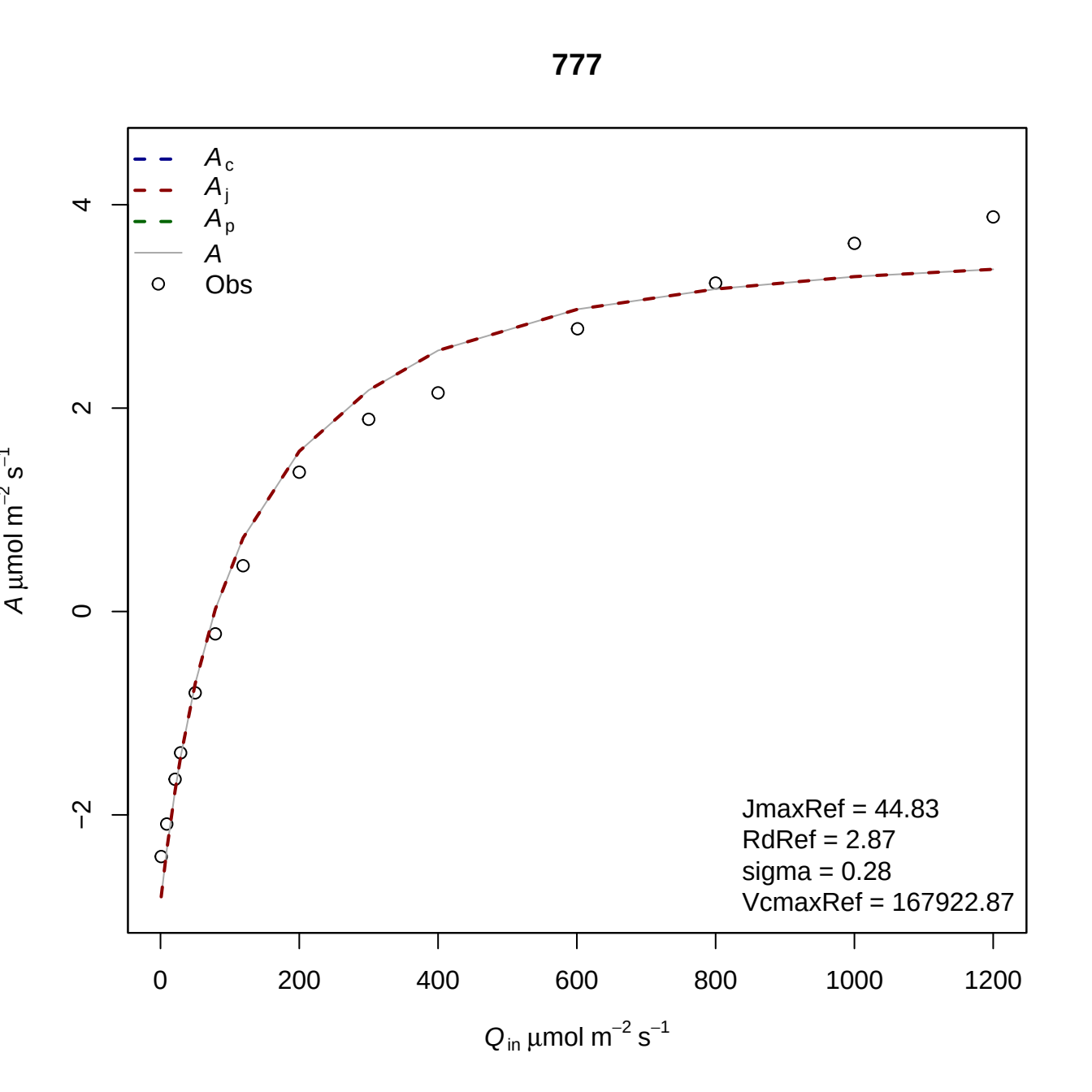

772
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.02
RdRef = 1.54
sigma = 0.27
VcmaxRef = 139.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

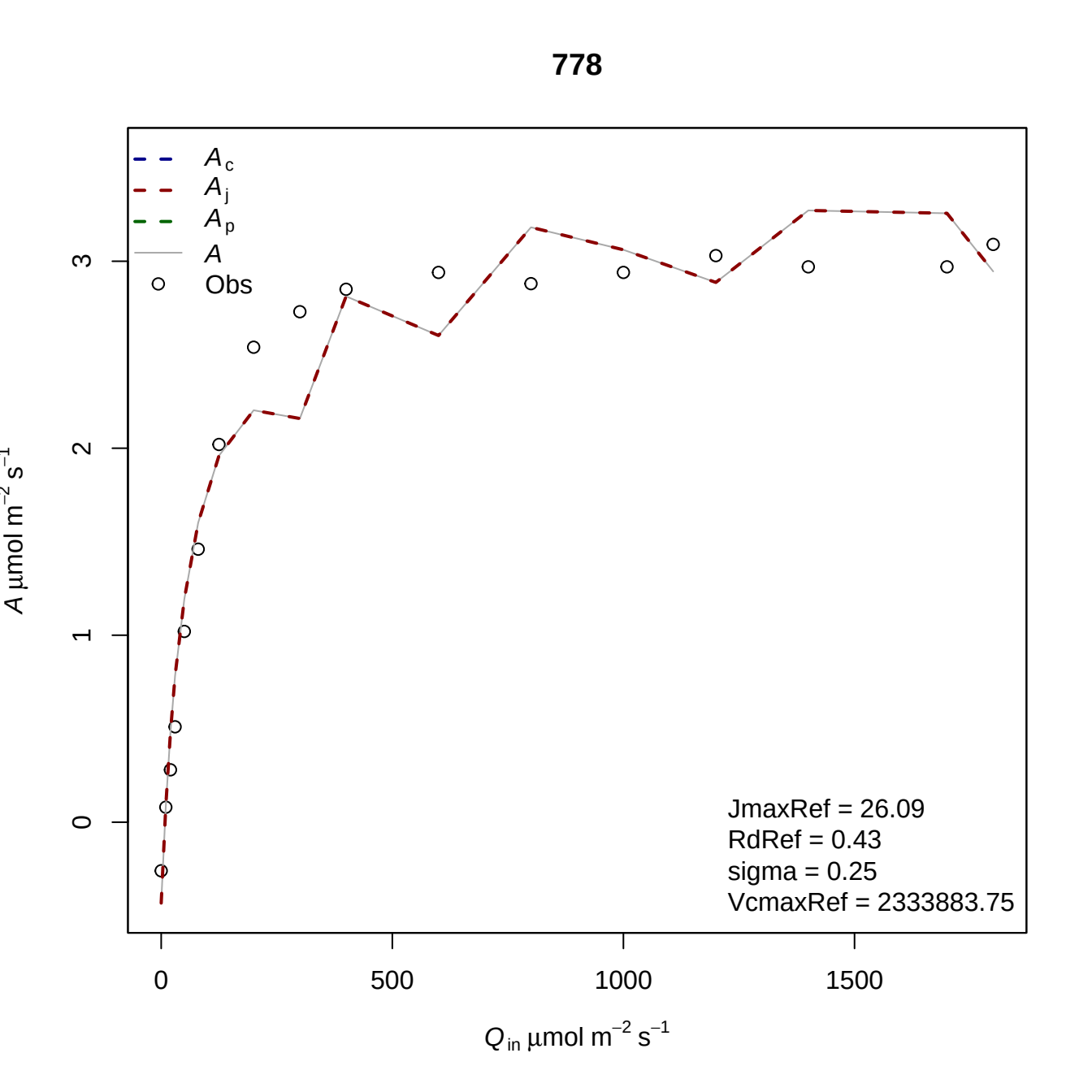

773
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.13
RdRef = 1.54
sigma = 0.08
VcmaxRef = 47.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

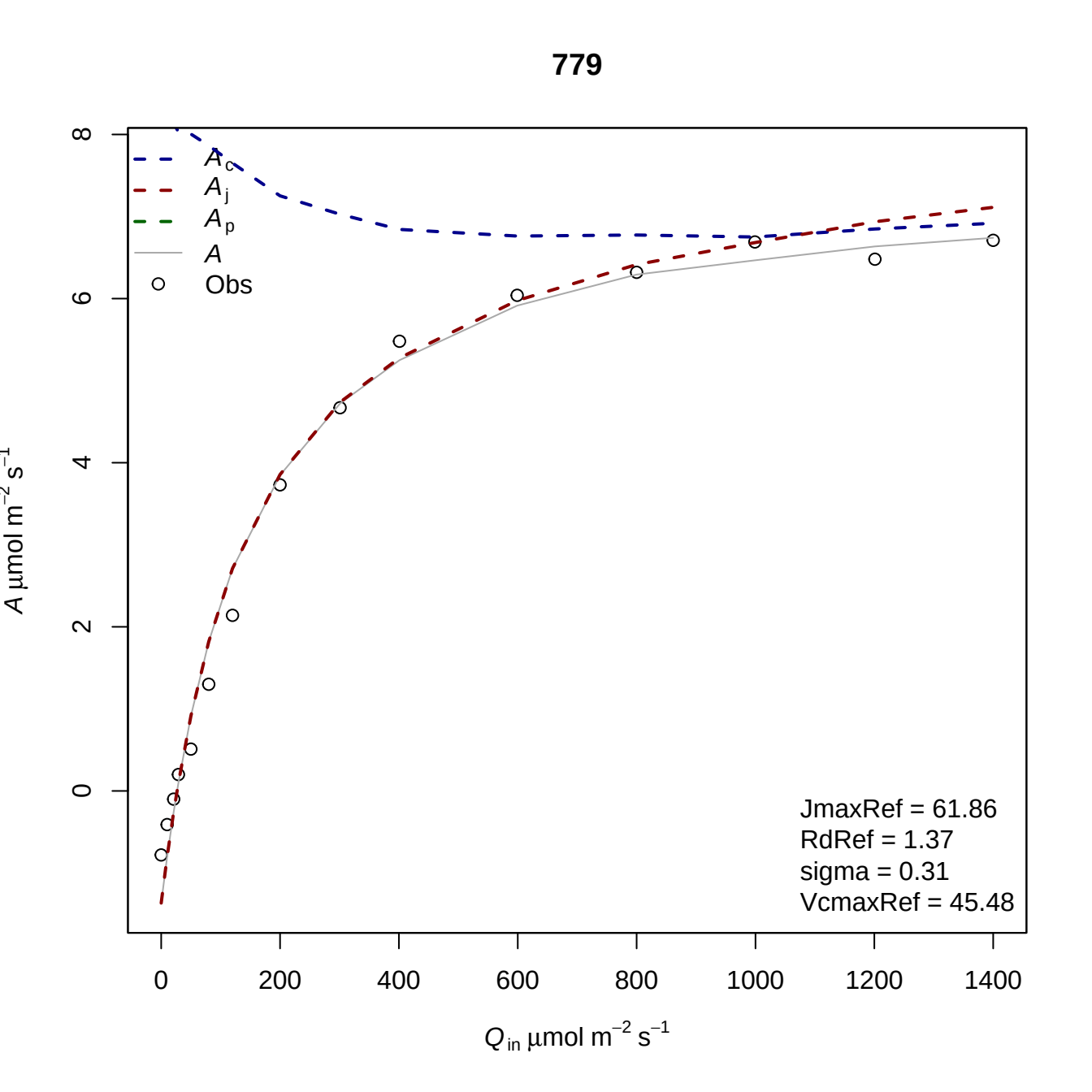

774
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.2
RdRef = 3.69
sigma = 0.49
VcmaxRef = 148865.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−4
−2
0
2
4
6
8
0
500
1000
1500

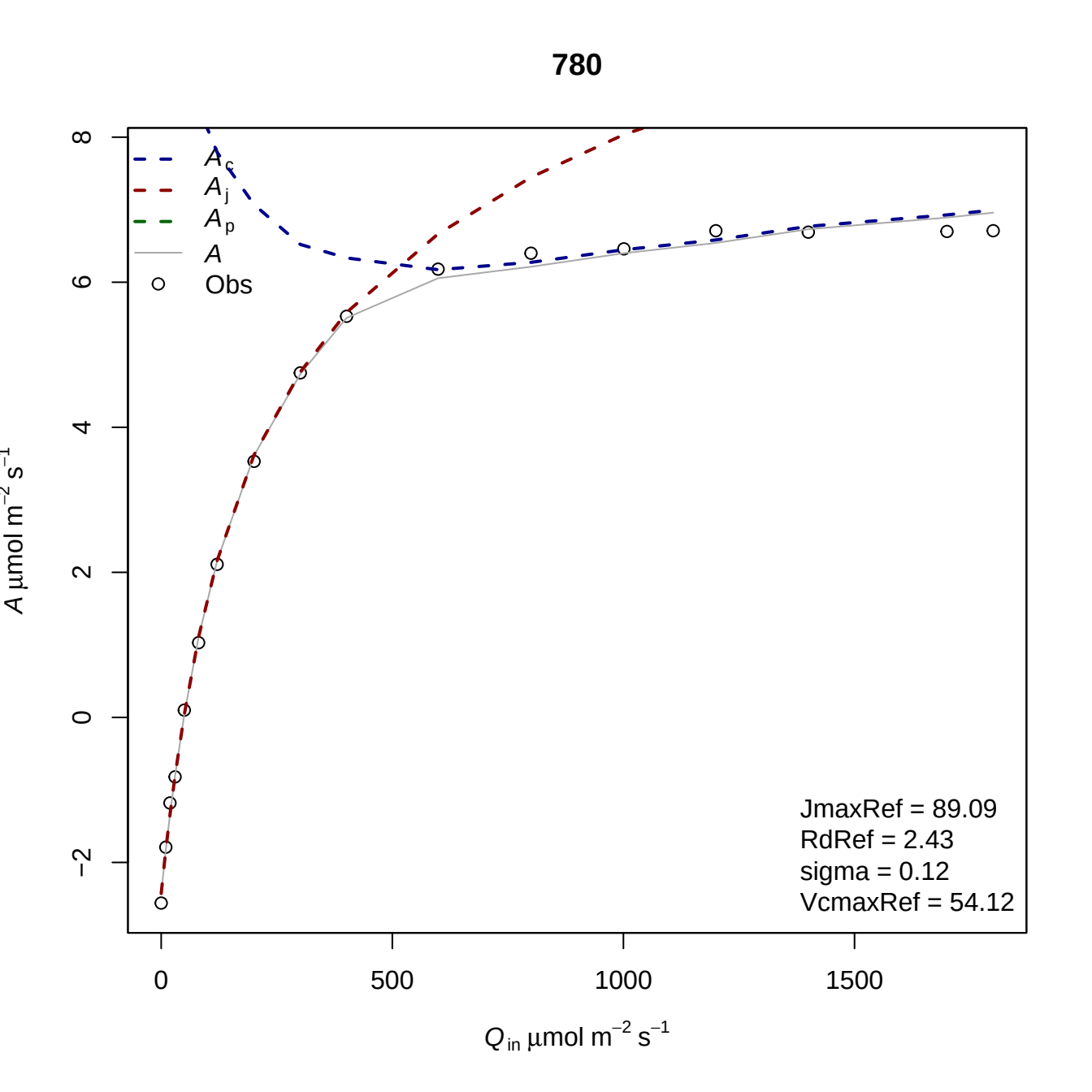

775
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.47
RdRef = 1.43
sigma = 0.26
VcmaxRef = 4157776.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

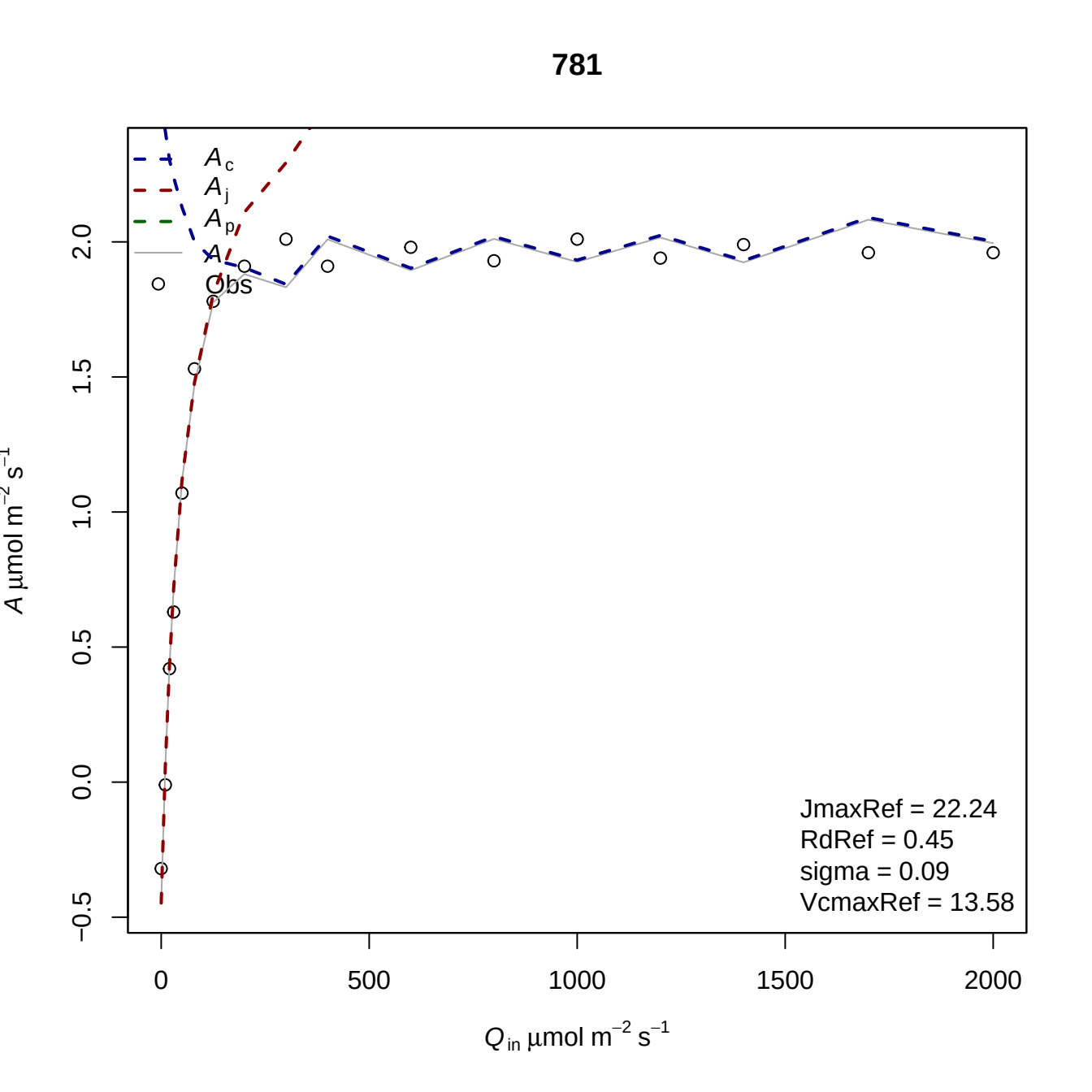

776
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.08
RdRef = 2.15
sigma = 0.19
VcmaxRef = 484581.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

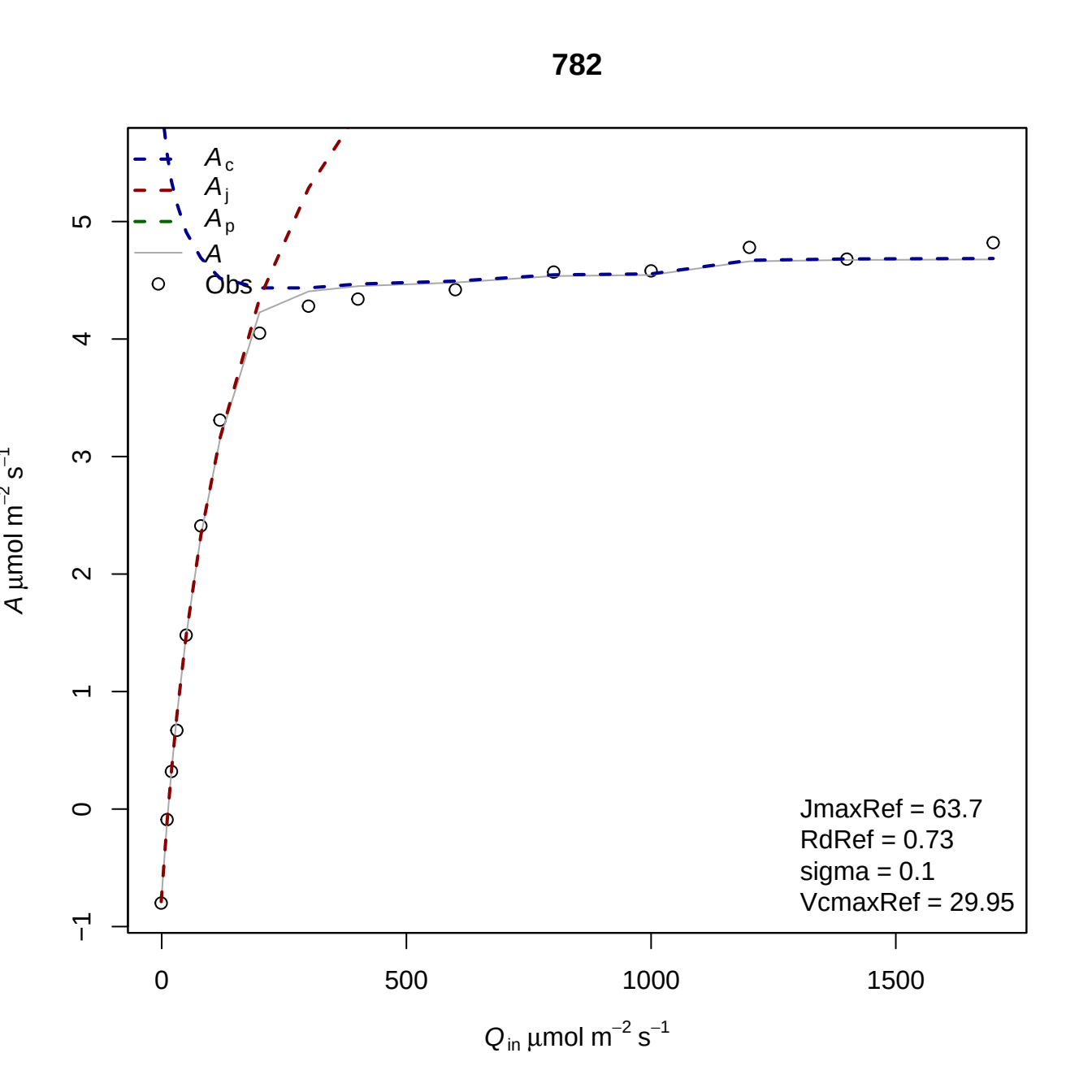

777
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.83
RdRef = 2.87
sigma = 0.28
VcmaxRef = 167922.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000
1200

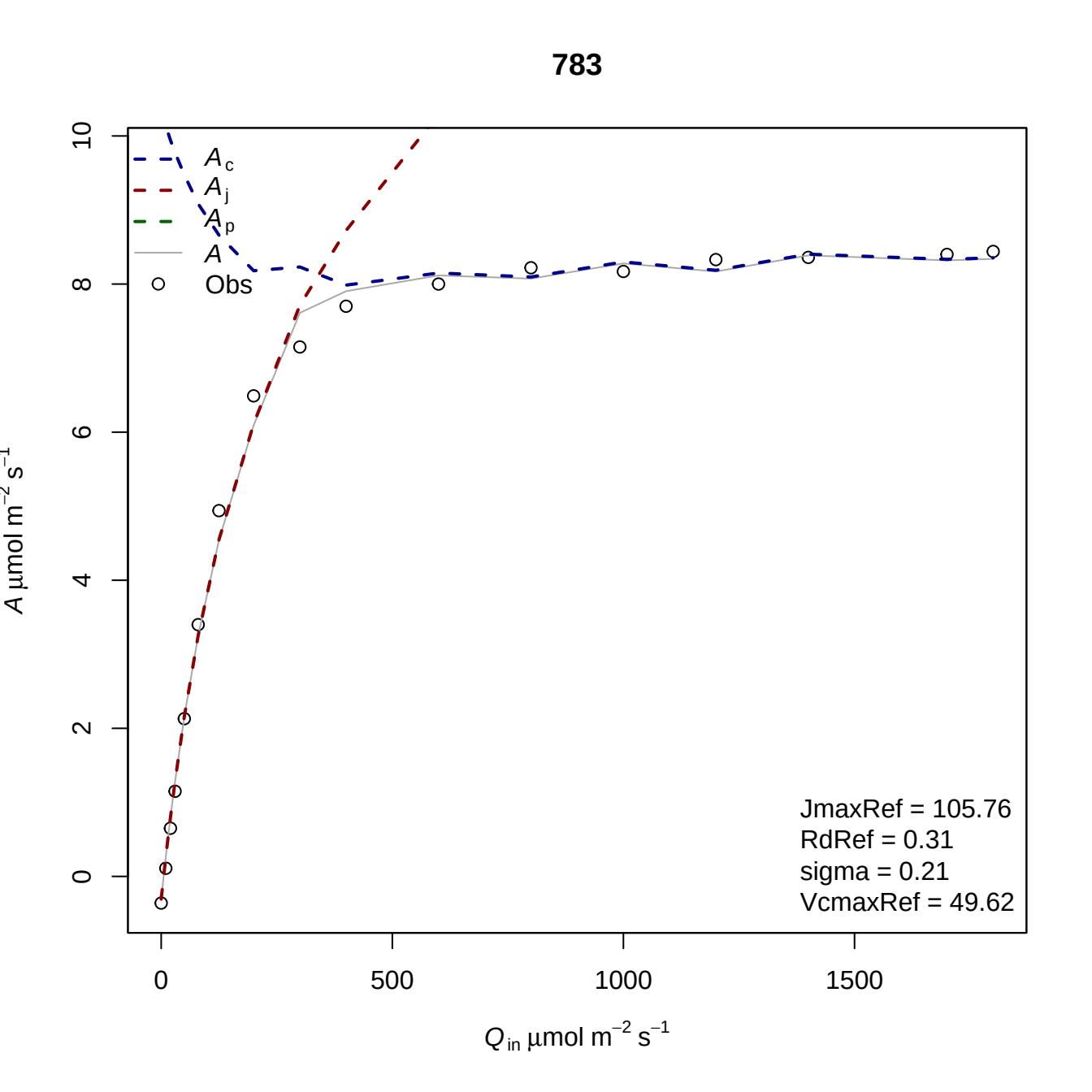

778
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.09
RdRef = 0.43
sigma = 0.25
VcmaxRef = 2333883.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500

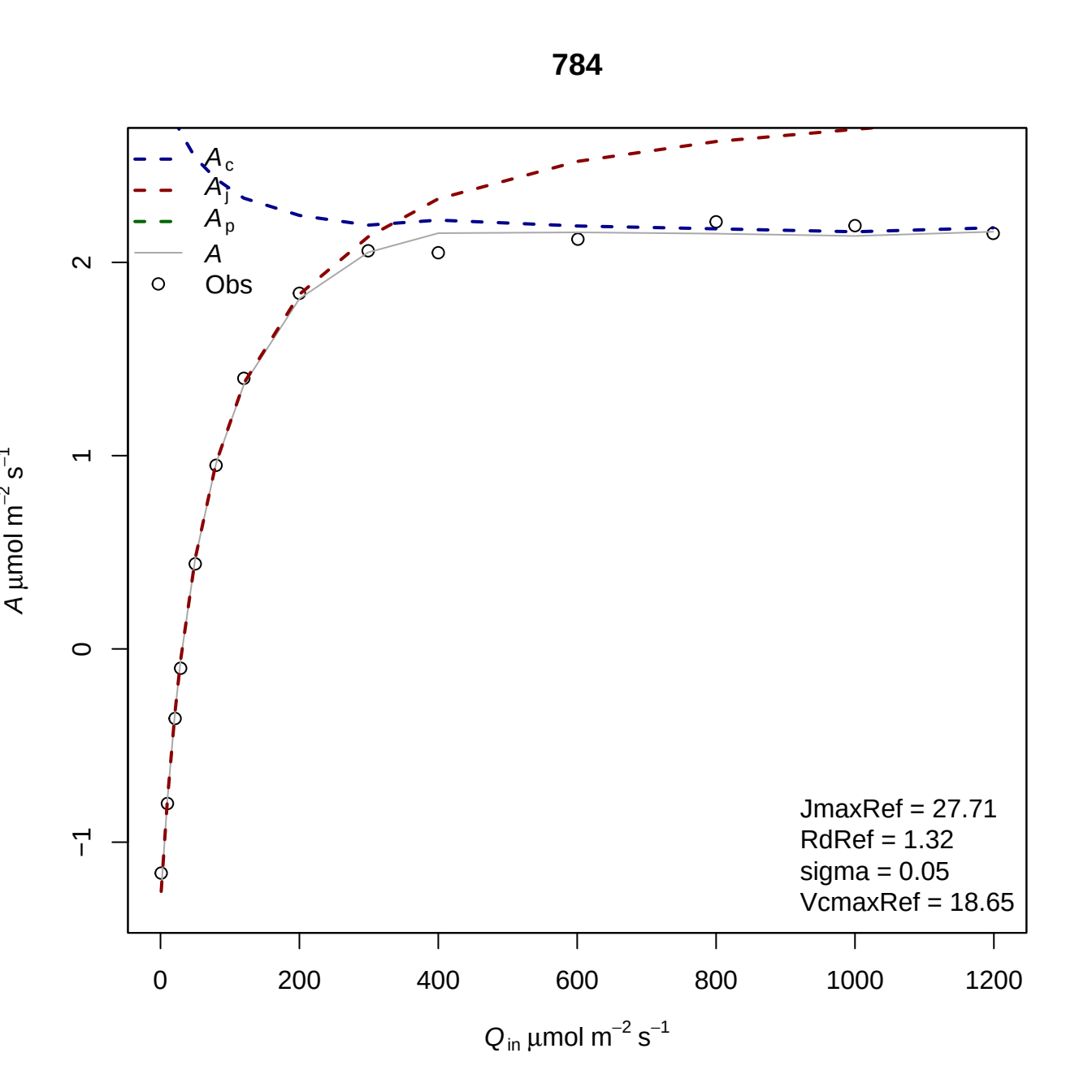

779
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.86
RdRef = 1.37
sigma = 0.31
VcmaxRef = 45.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

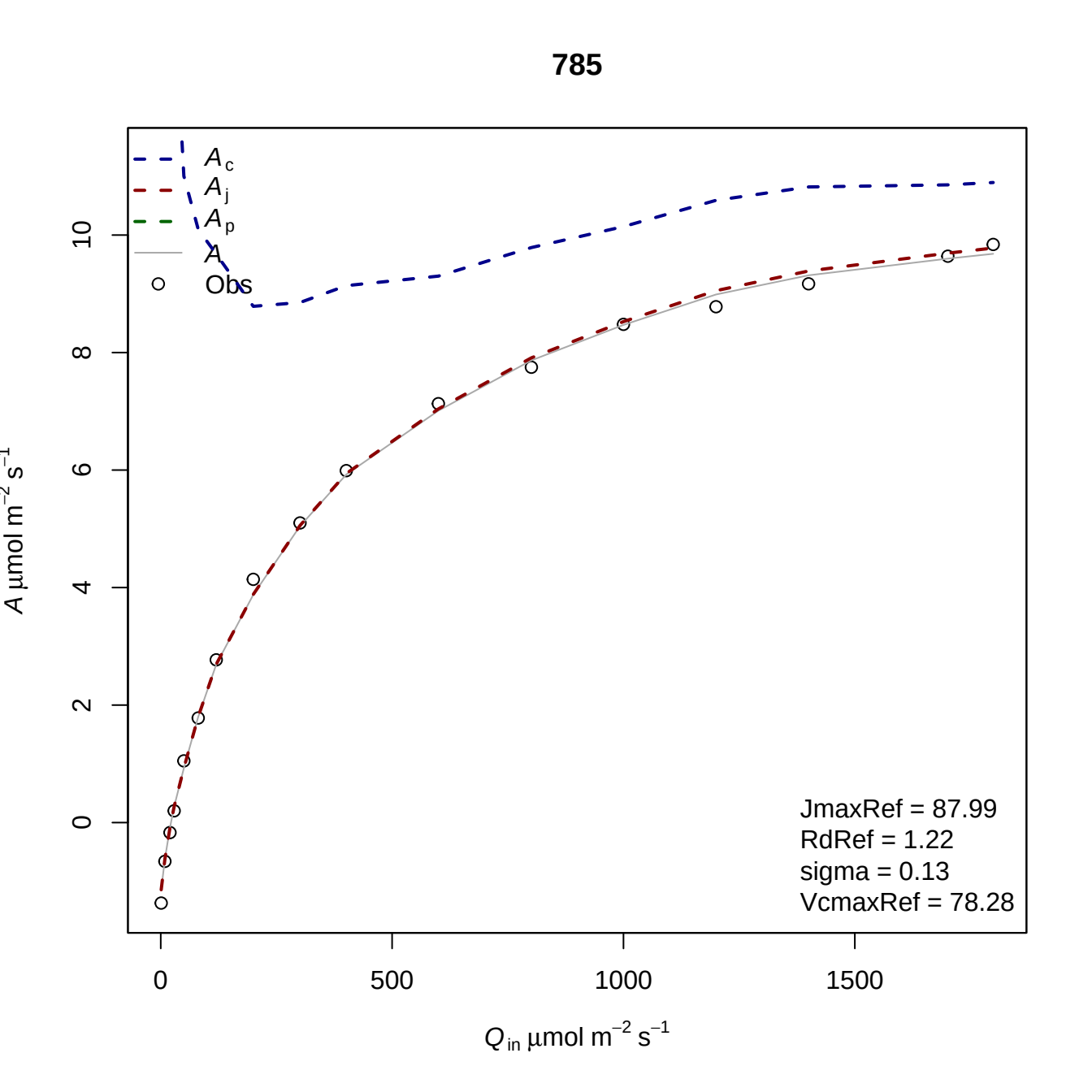

780
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.09
RdRef = 2.43
sigma = 0.12
VcmaxRef = 54.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

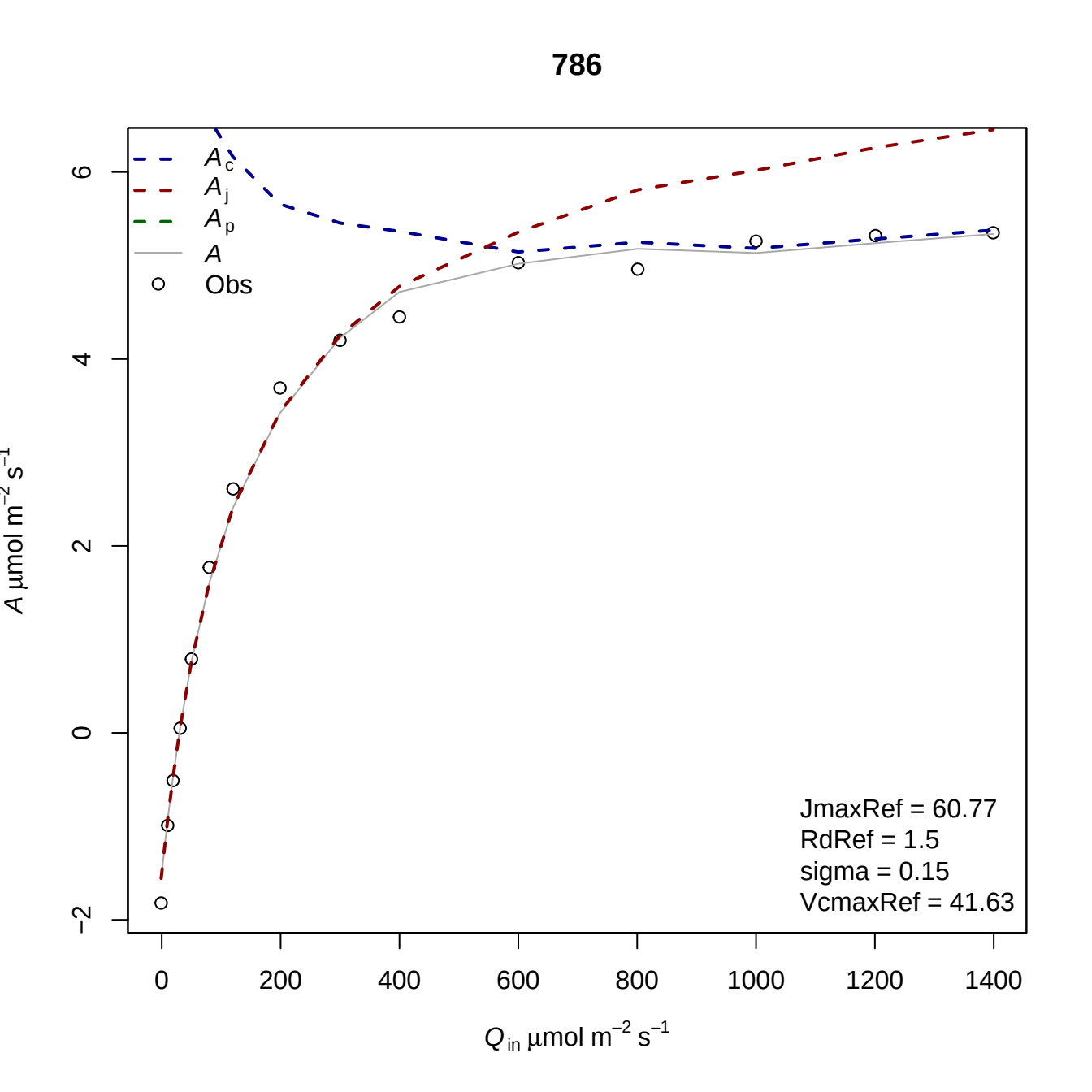

781
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.24
RdRef = 0.45
sigma = 0.09
VcmaxRef = 13.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

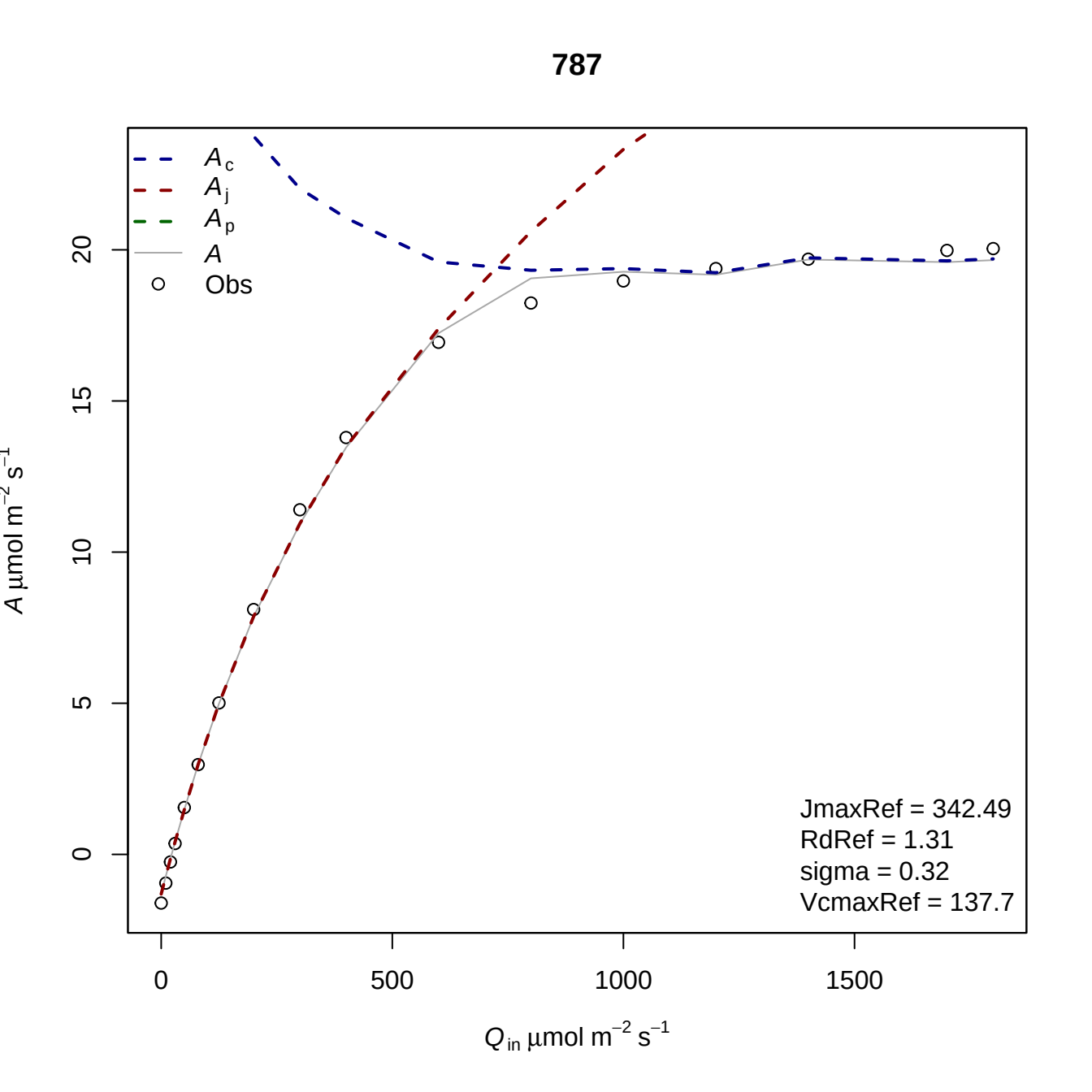

782
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.7
RdRef = 0.73
sigma = 0.1
VcmaxRef = 29.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

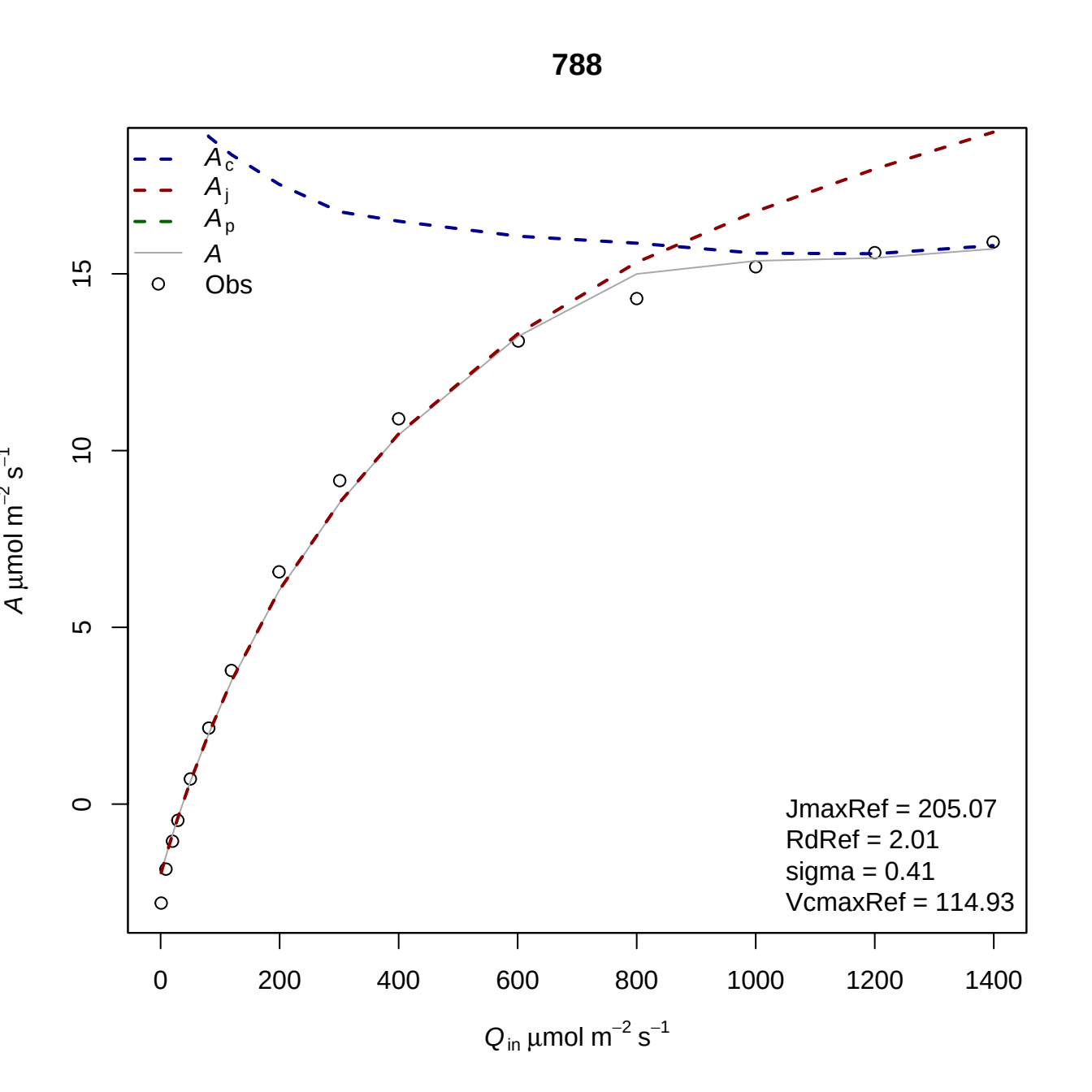

783
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.76
RdRef = 0.31
sigma = 0.21
VcmaxRef = 49.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

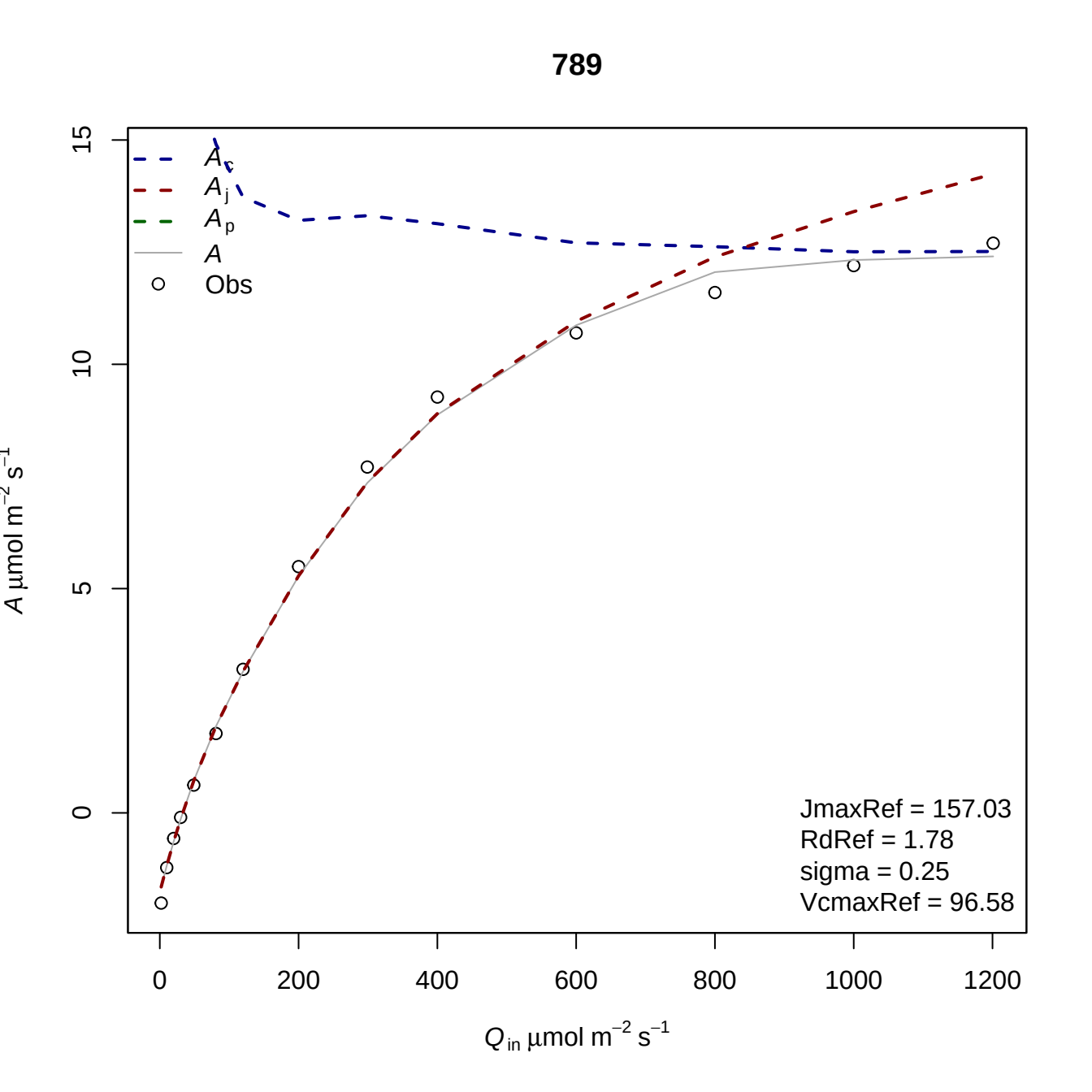

784
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.71
RdRef = 1.32
sigma = 0.05
VcmaxRef = 18.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
200
400
600
800
1000
1200

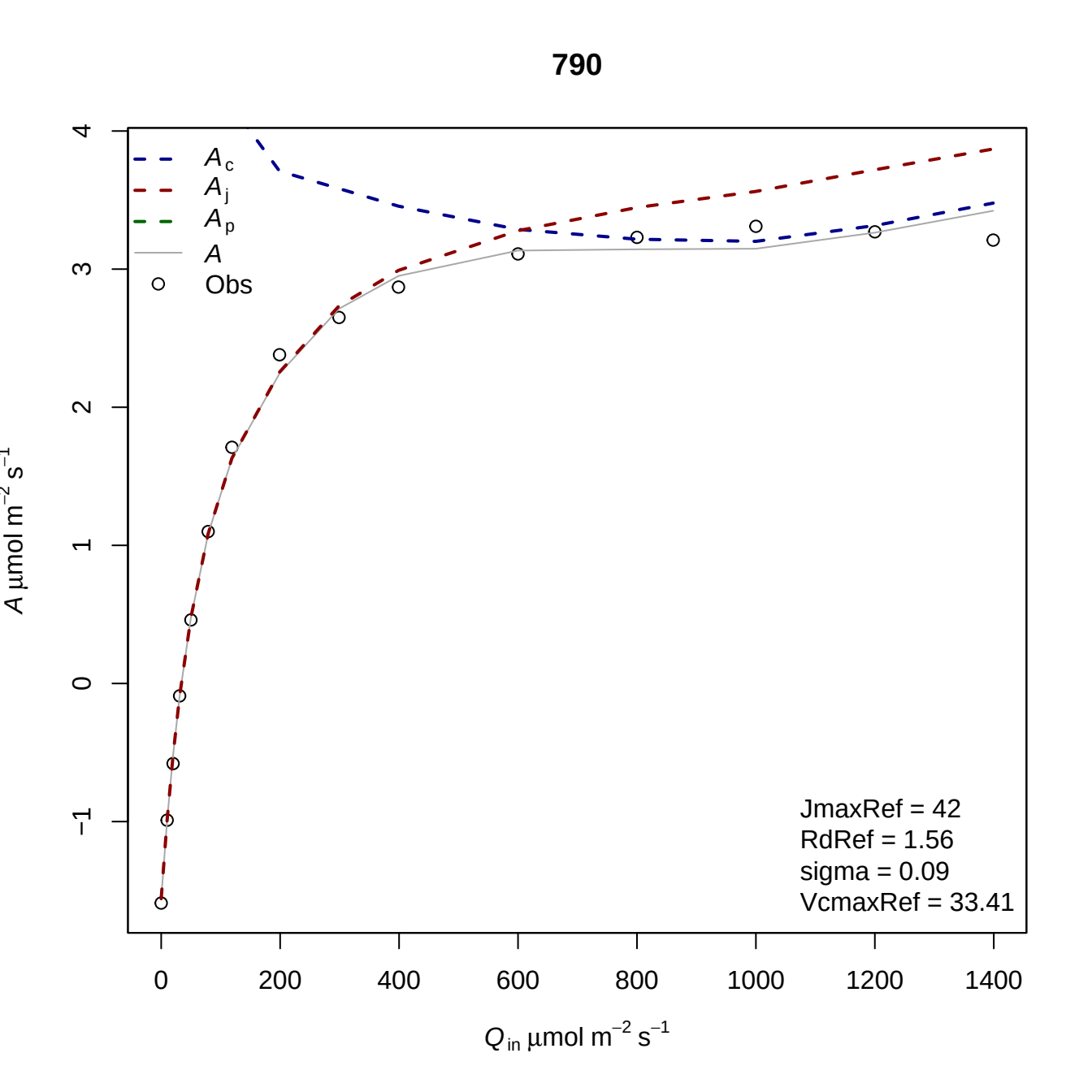

785
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 87.99
RdRef = 1.22
sigma = 0.13
VcmaxRef = 78.28
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

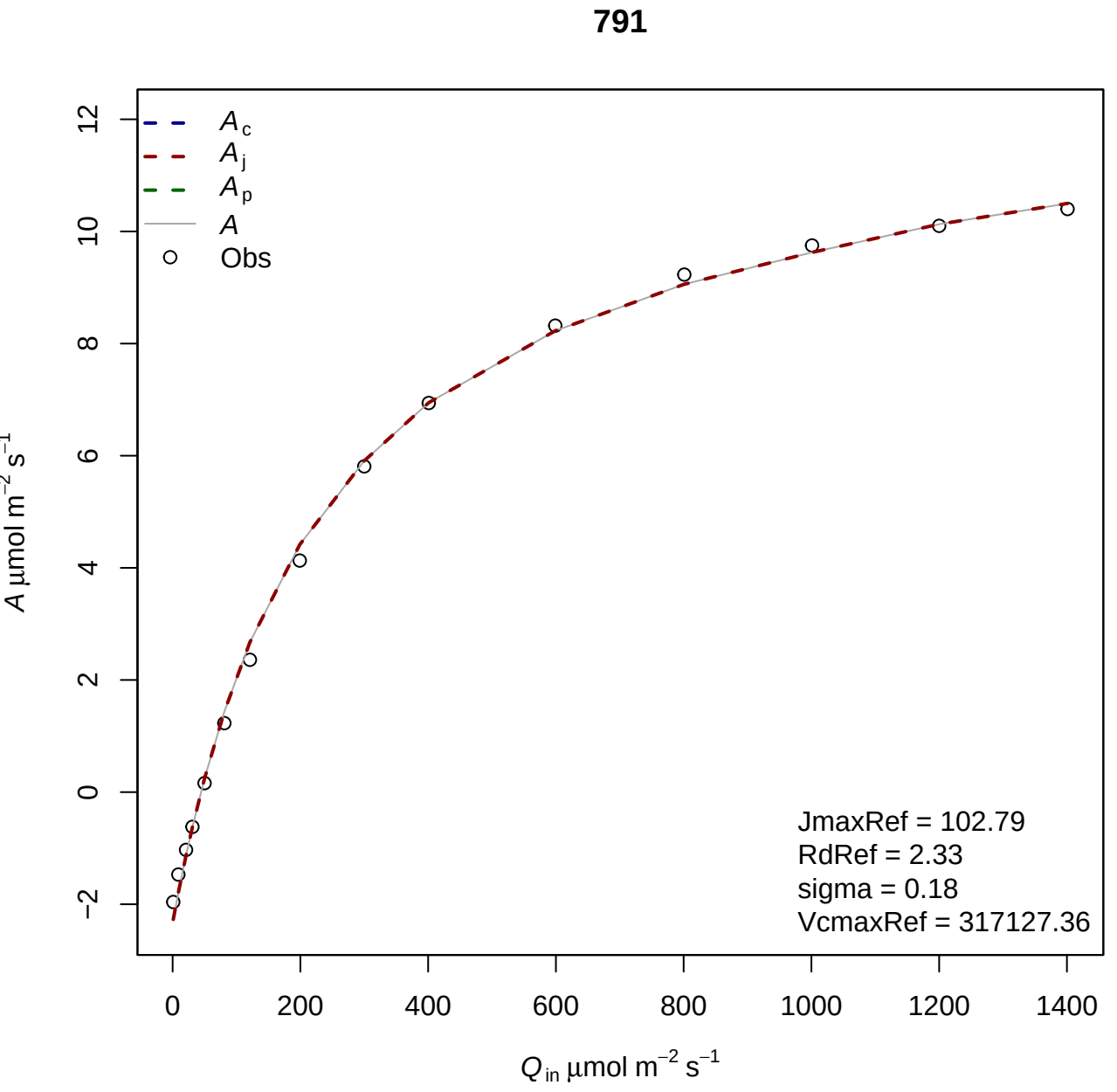

786
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.77
RdRef = 1.5
sigma = 0.15
VcmaxRef = 41.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

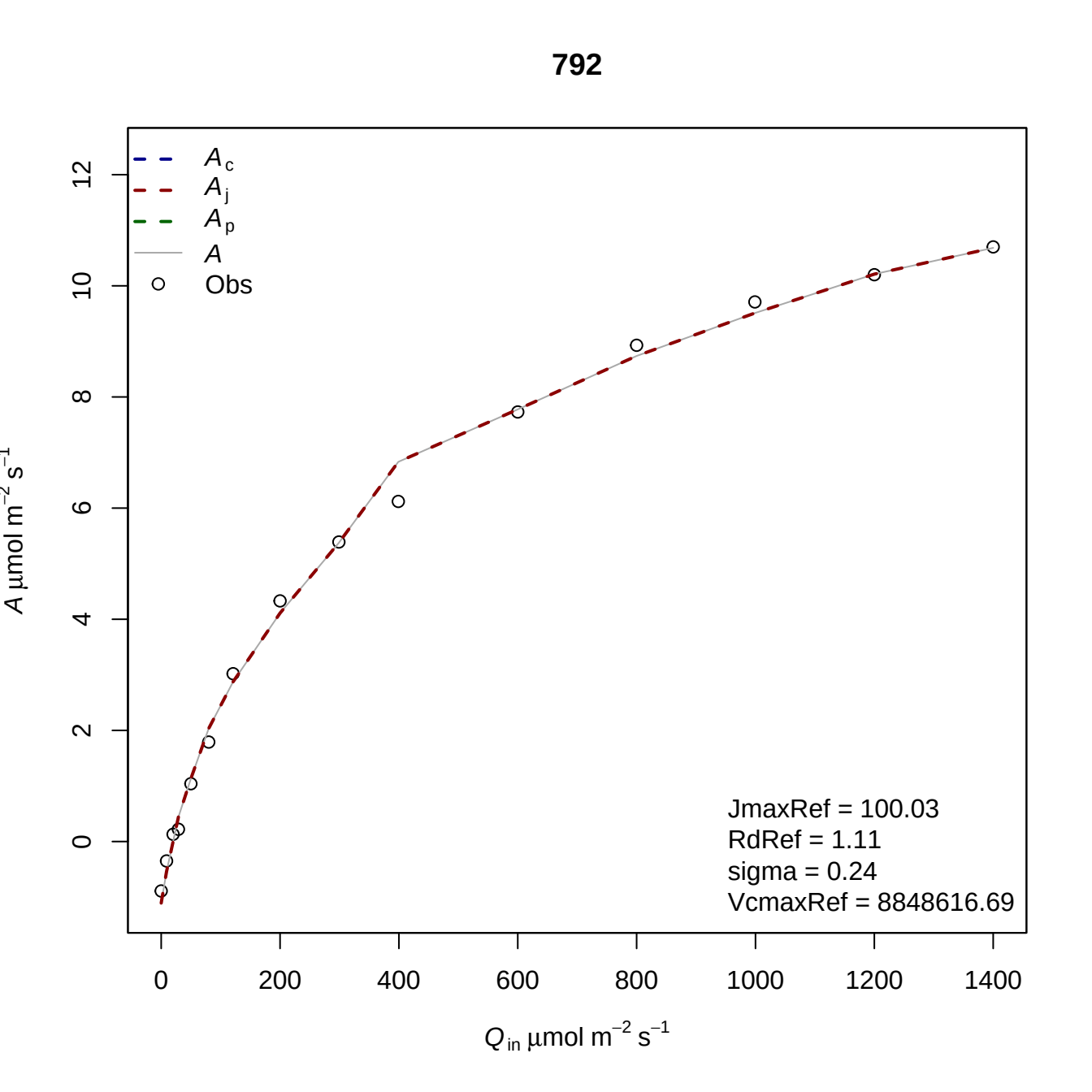

787
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 342.49
RdRef = 1.31
sigma = 0.32
VcmaxRef = 137.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

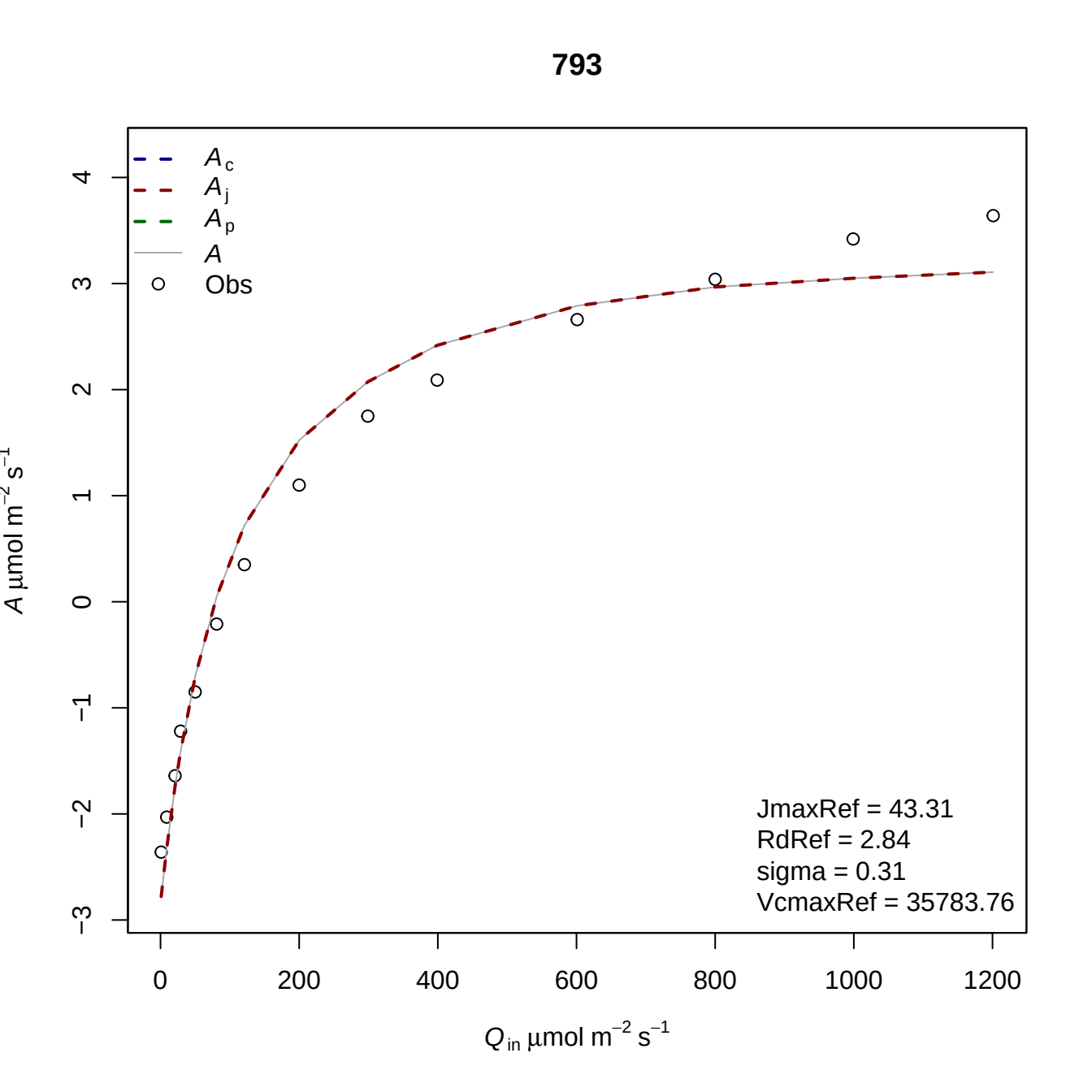

788
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 205.07
RdRef = 2.01
sigma = 0.41
VcmaxRef = 114.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000
1200
1400

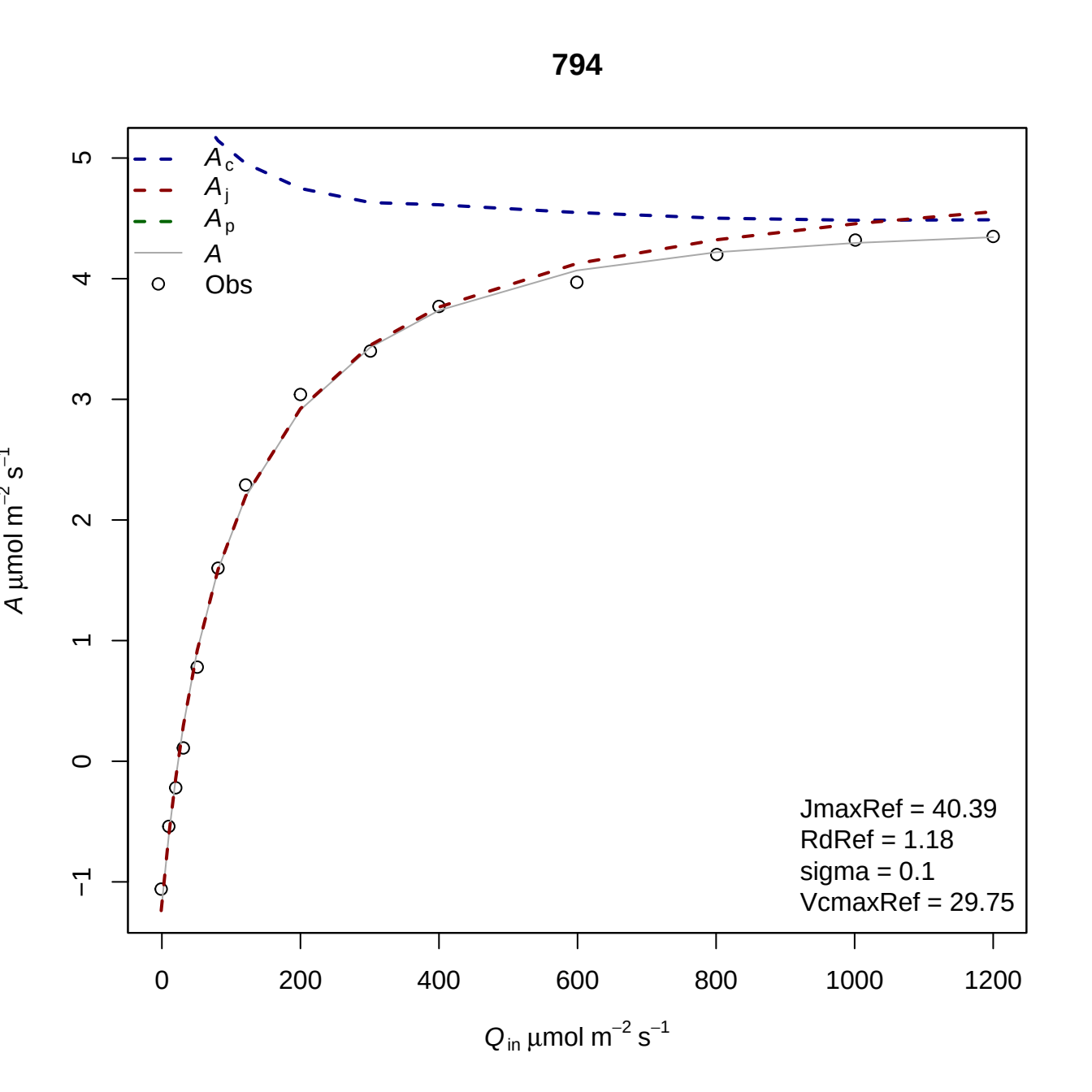

789
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 157.03
RdRef = 1.78
sigma = 0.25
VcmaxRef = 96.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000
1200

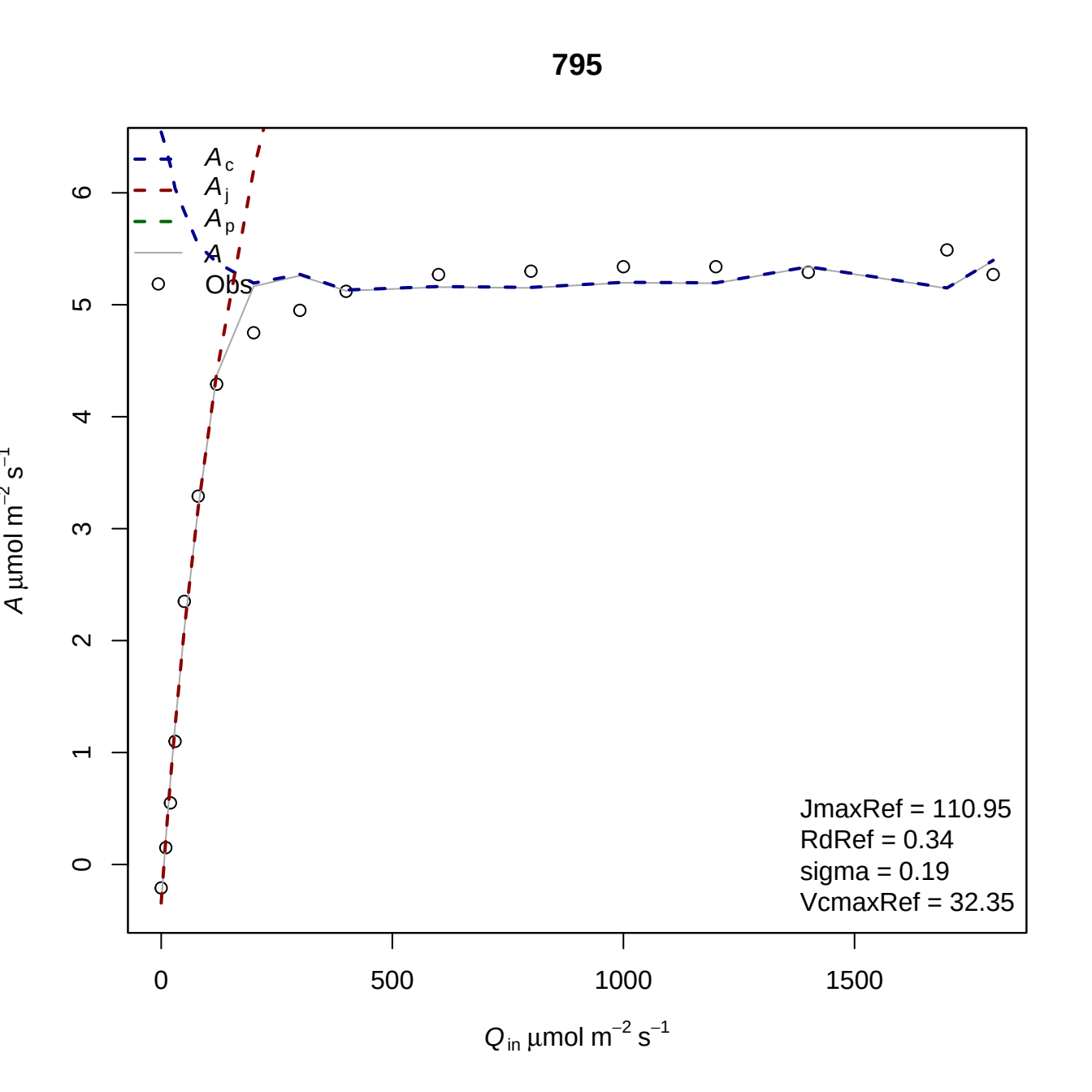

790
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42
RdRef = 1.56
sigma = 0.09
VcmaxRef = 33.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

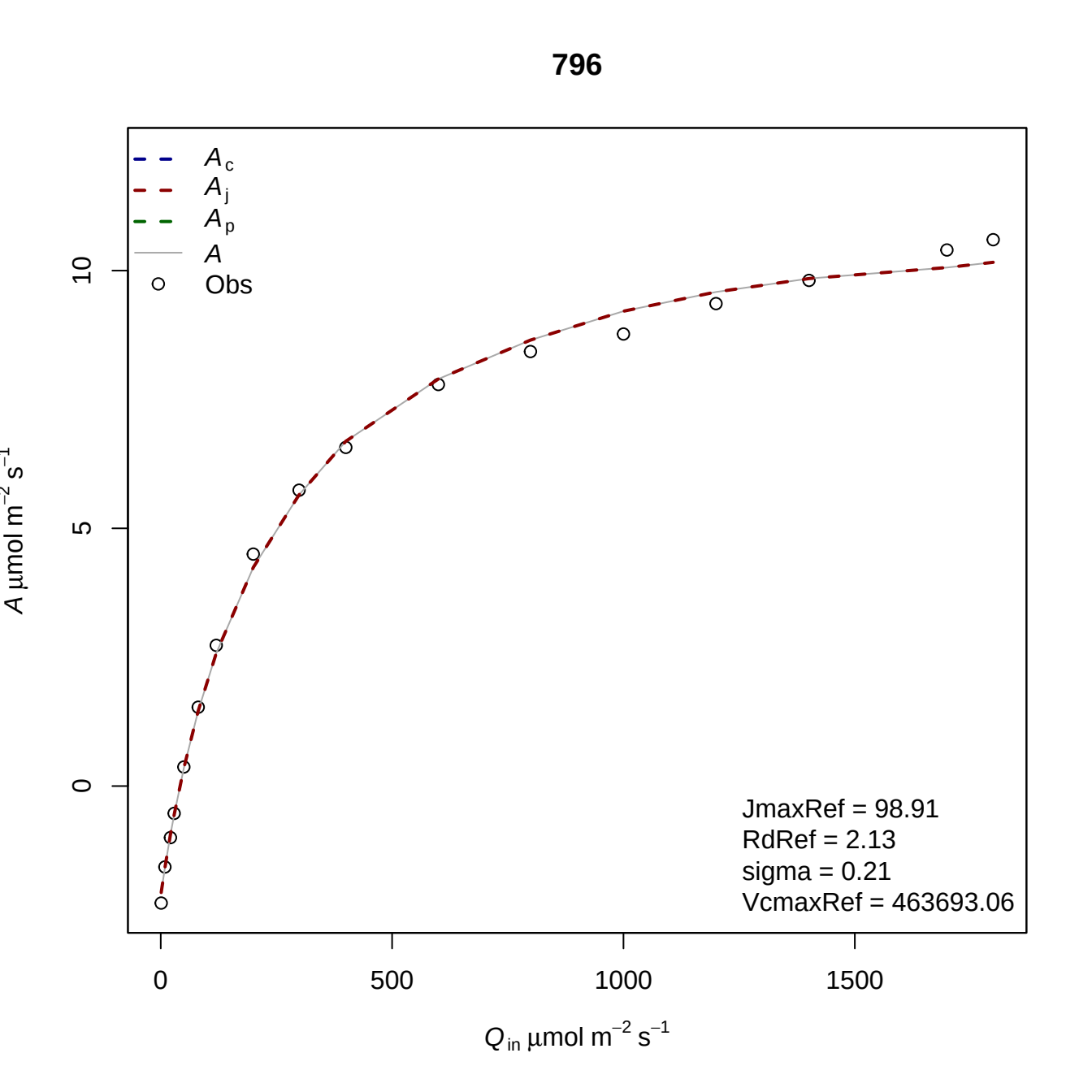

791
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.79
RdRef = 2.33
sigma = 0.18
VcmaxRef = 317127.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

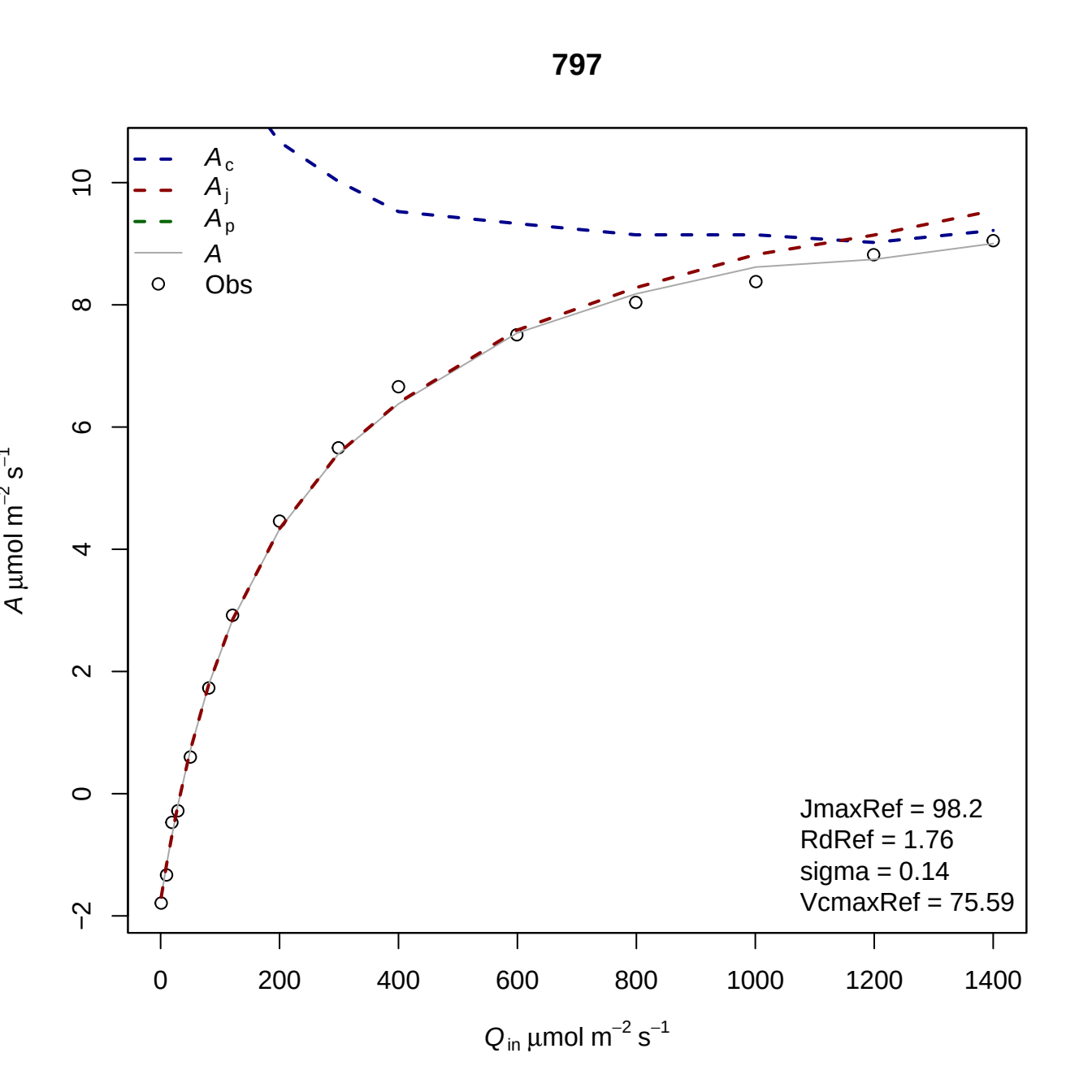

792
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.03
RdRef = 1.11
sigma = 0.24
VcmaxRef = 8848616.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

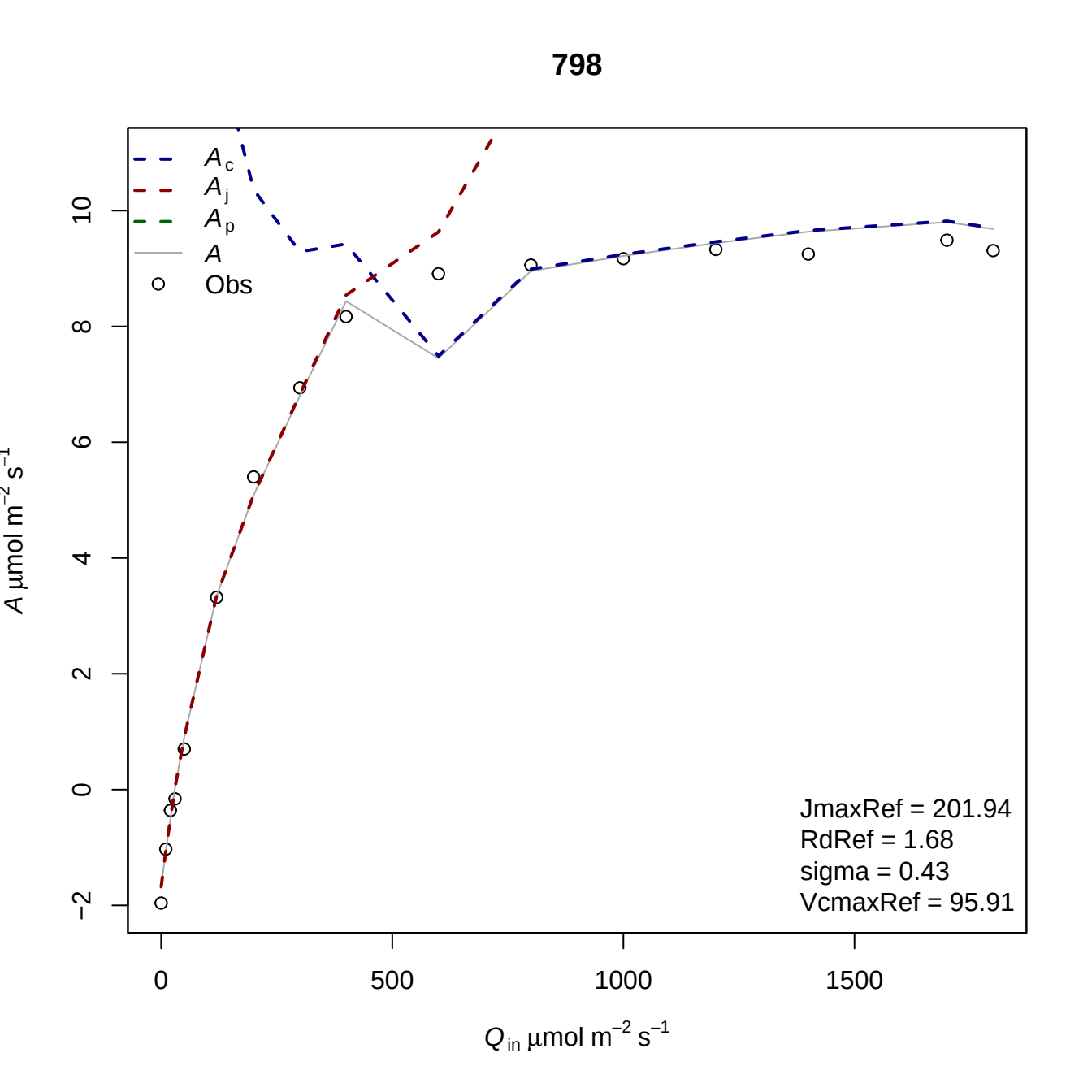

793
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.31
RdRef = 2.84
sigma = 0.31
VcmaxRef = 35783.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
−3
0
200
400
600
800
1000
1200

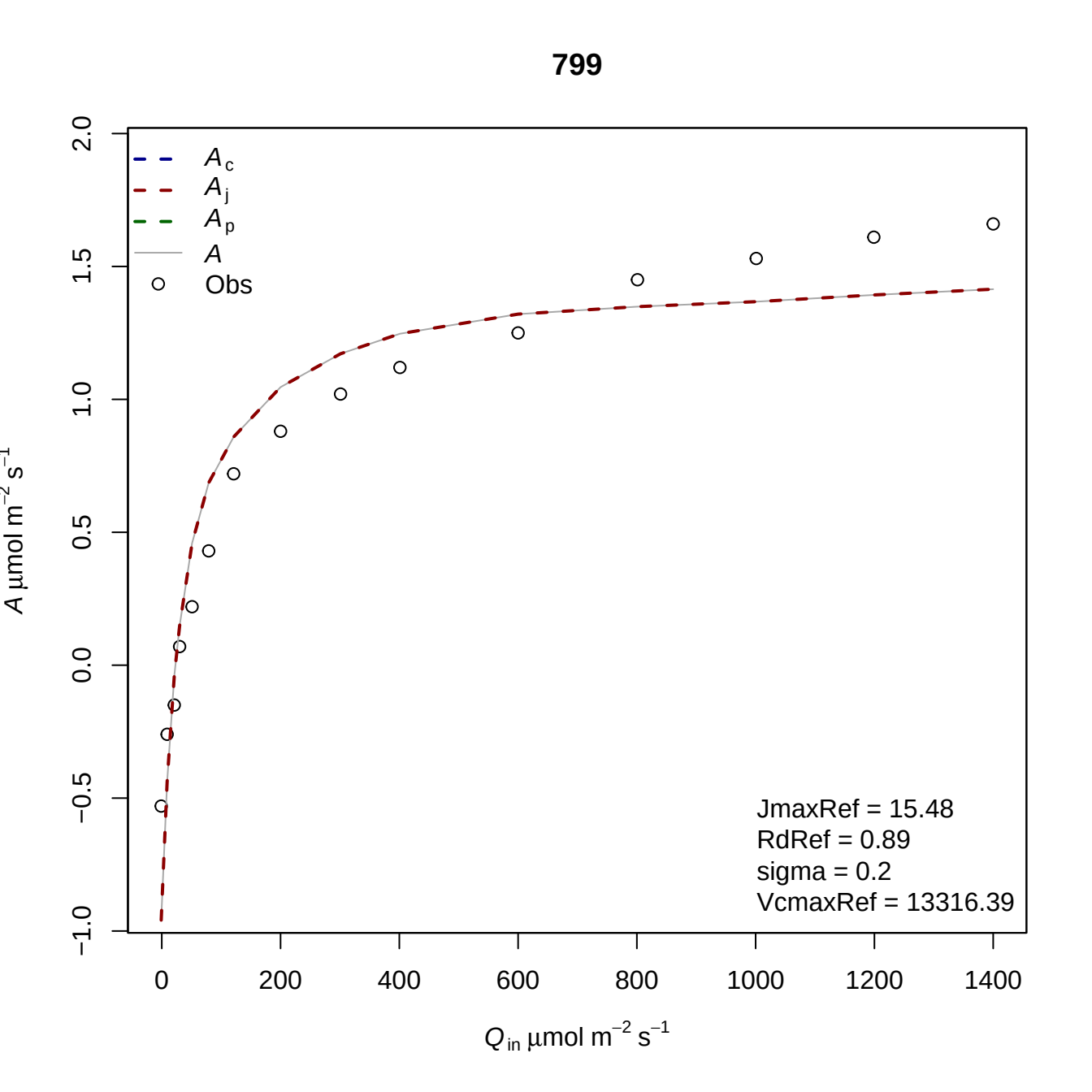

794
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.39
RdRef = 1.18
sigma = 0.1
VcmaxRef = 29.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

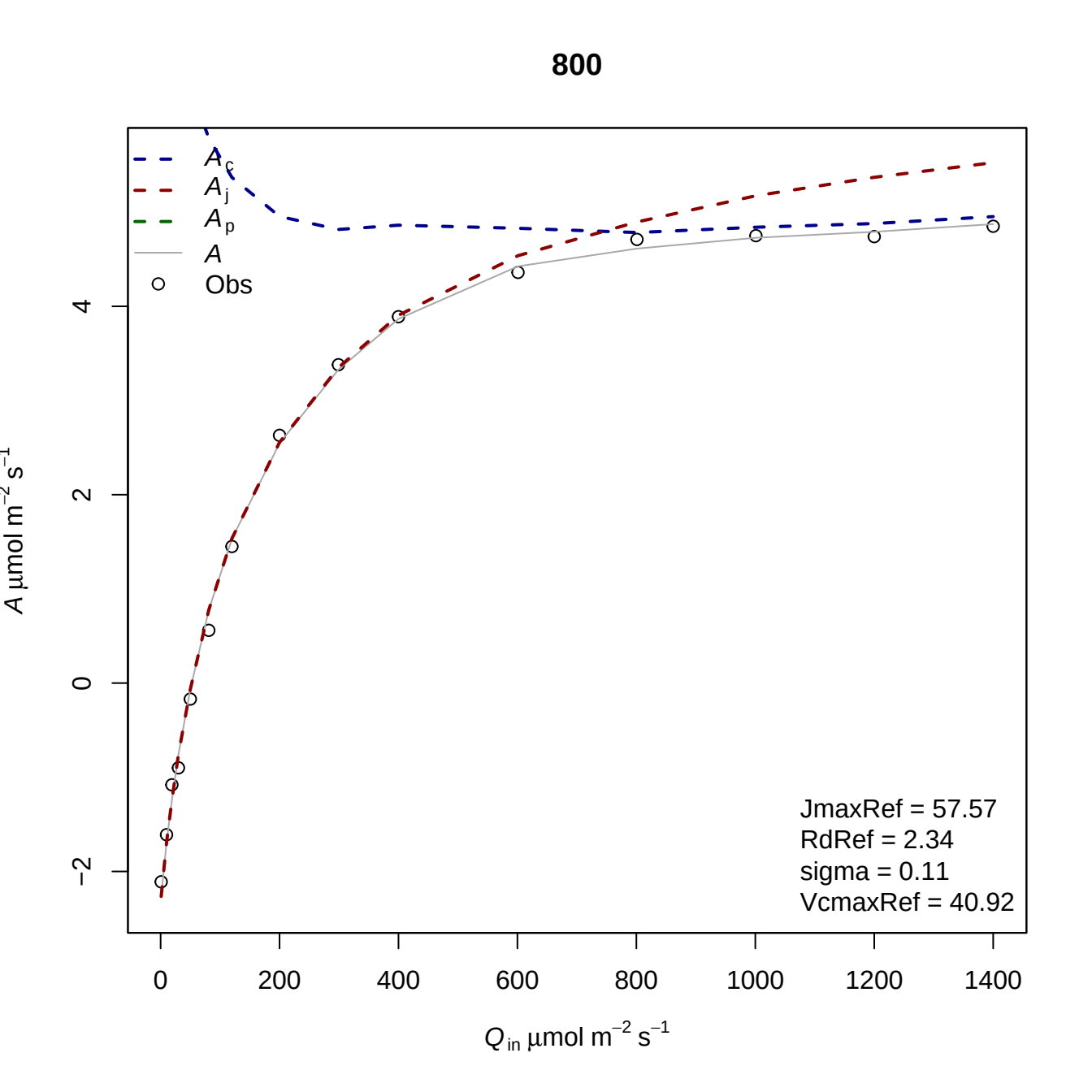

795
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.95
RdRef = 0.34
sigma = 0.19
VcmaxRef = 32.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

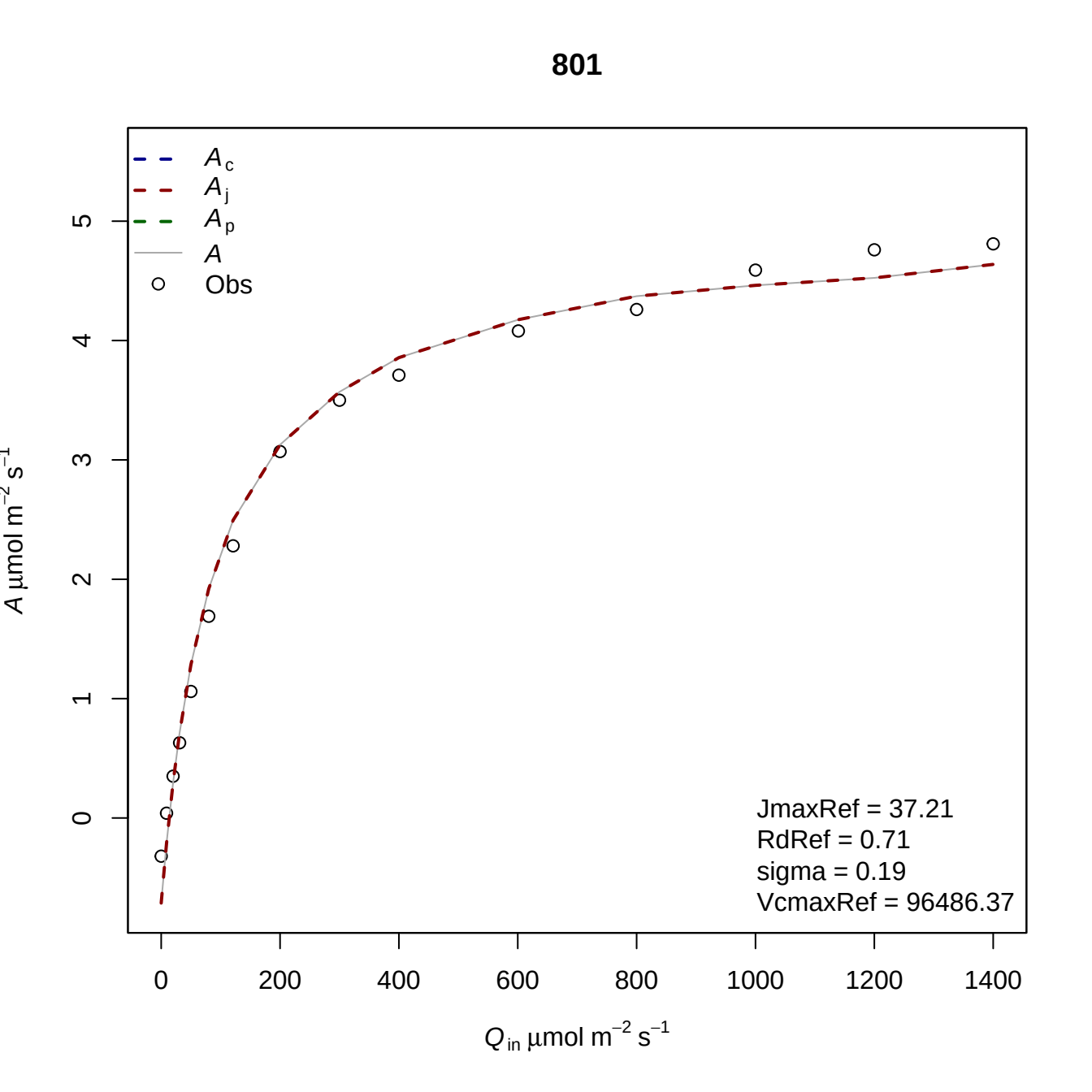

796
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.91
RdRef = 2.13
sigma = 0.21
VcmaxRef = 463693.06
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

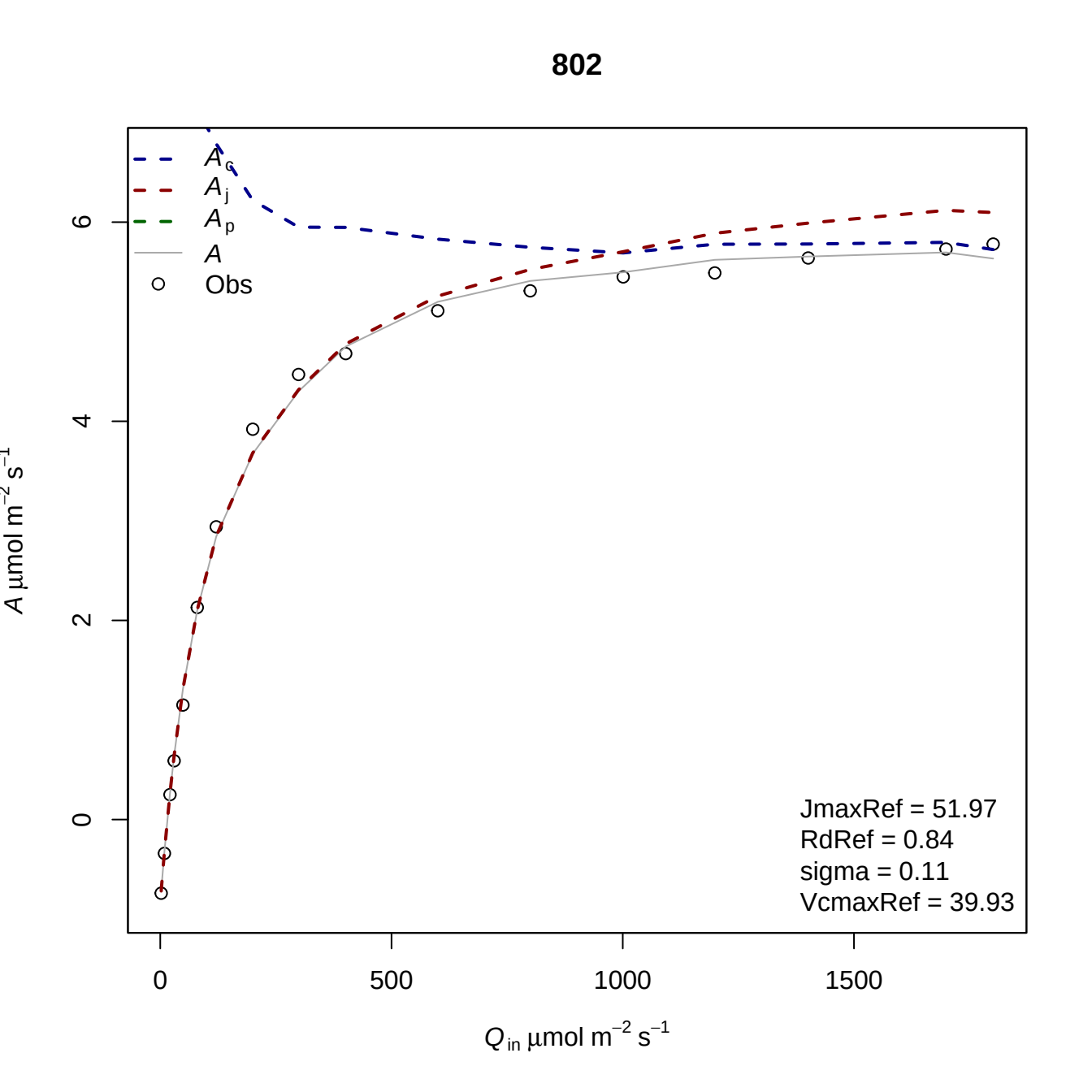

797
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.2
RdRef = 1.76
sigma = 0.14
VcmaxRef = 75.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

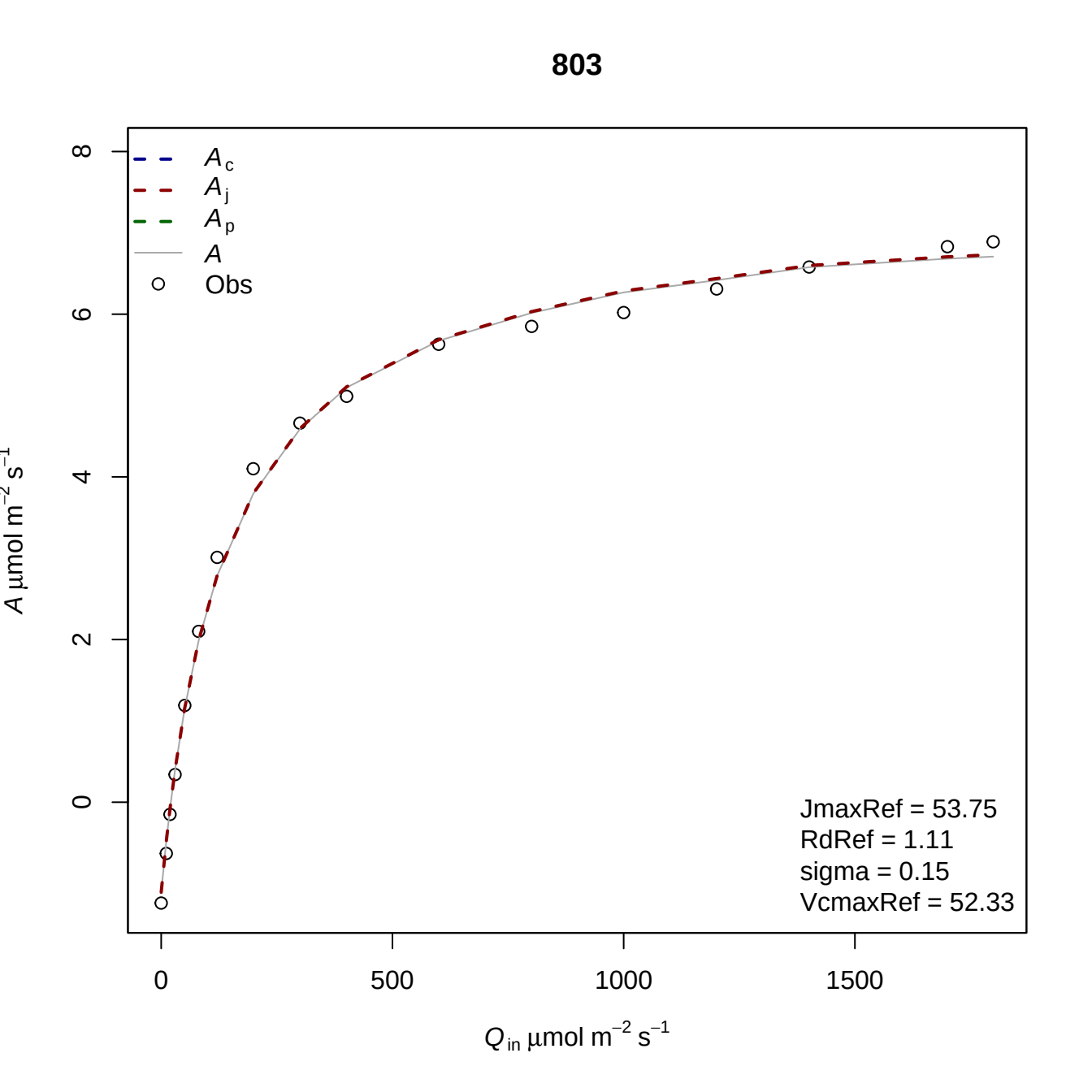

798
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 201.94
RdRef = 1.68
sigma = 0.43
VcmaxRef = 95.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

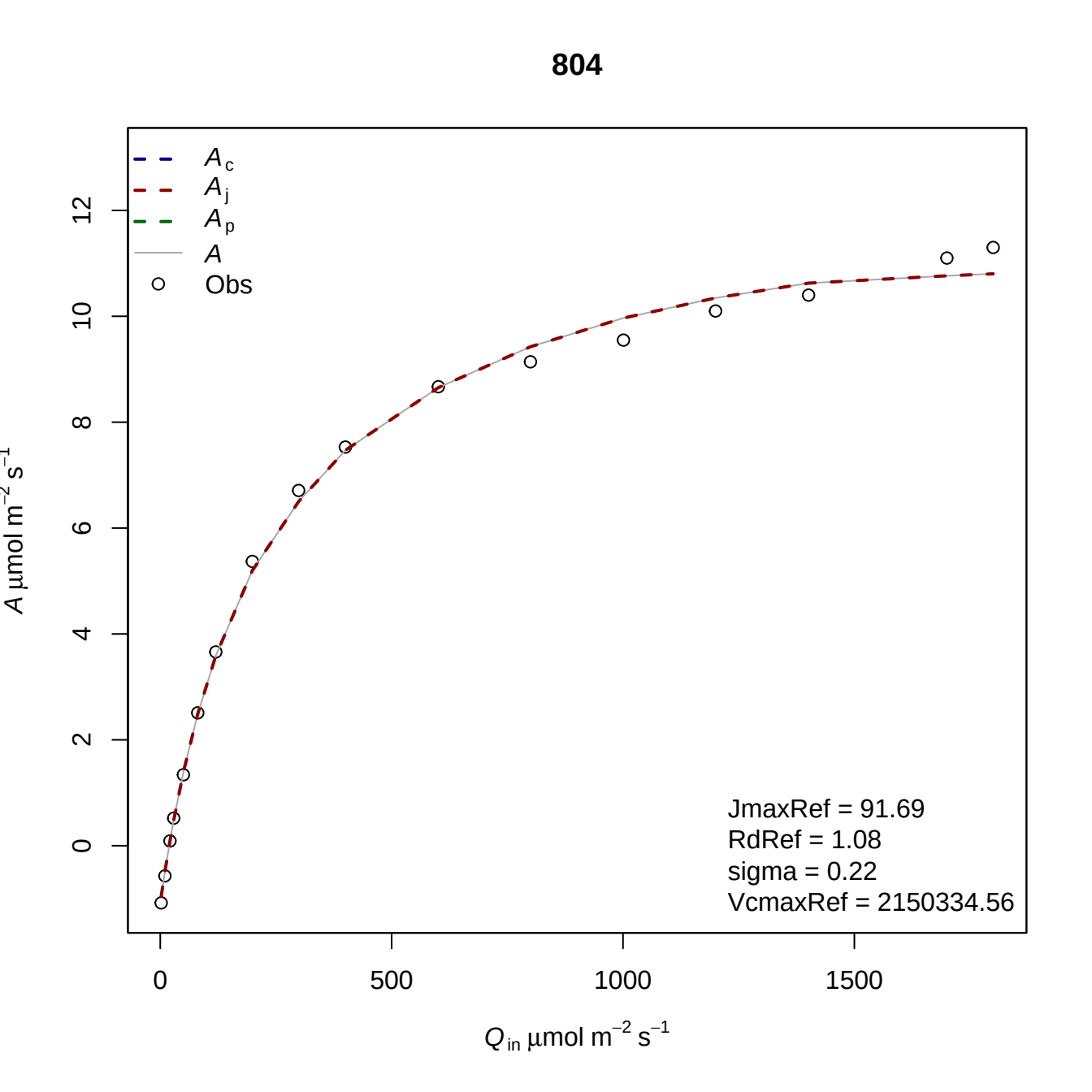

799
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 15.48
RdRef = 0.89
sigma = 0.2
VcmaxRef = 13316.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400

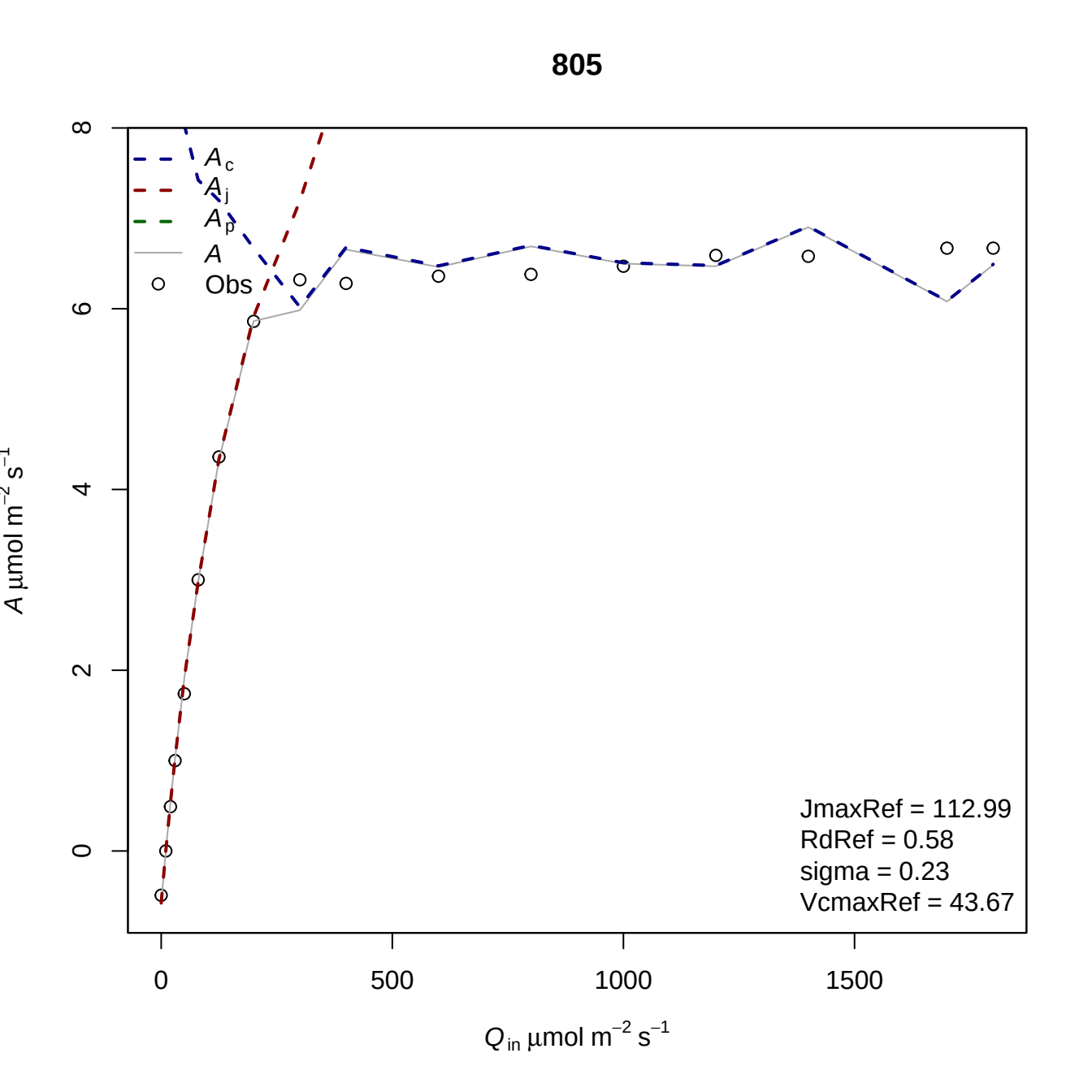

800
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.57
RdRef = 2.34
sigma = 0.11
VcmaxRef = 40.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000
1200
1400

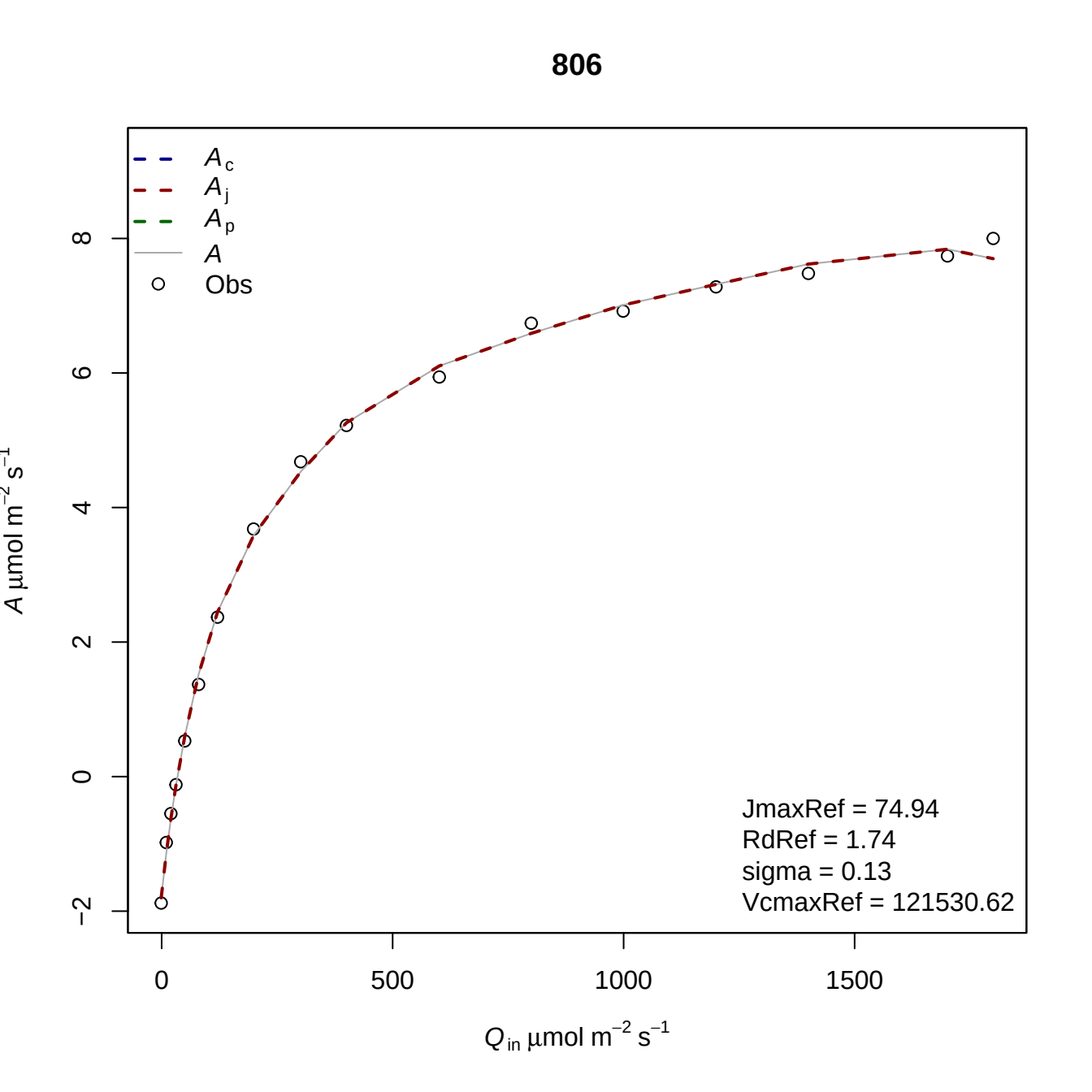

801
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.21
RdRef = 0.71
sigma = 0.19
VcmaxRef = 96486.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

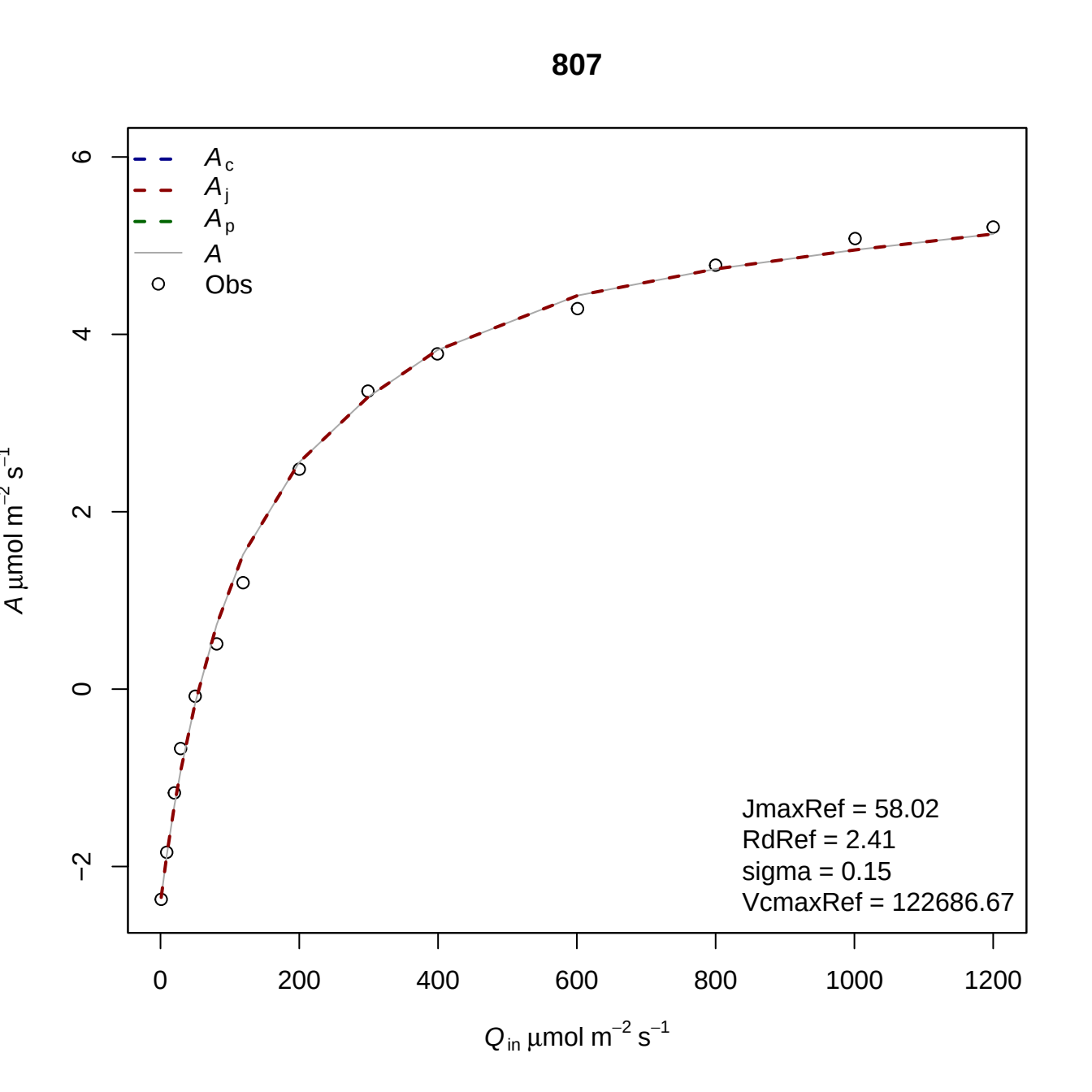

802
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.97
RdRef = 0.84
sigma = 0.11
VcmaxRef = 39.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

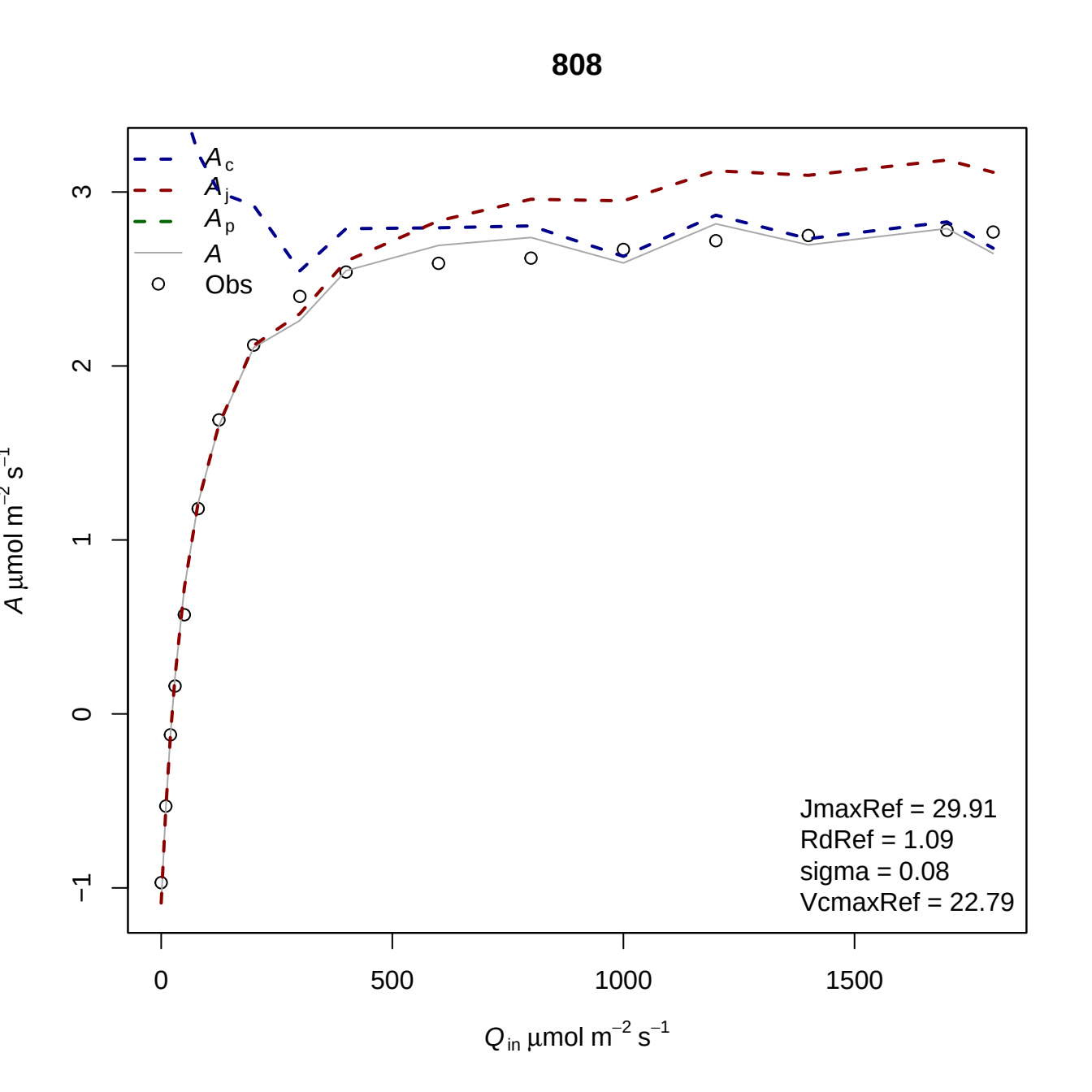

803
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 53.75
RdRef = 1.11
sigma = 0.15
VcmaxRef = 52.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

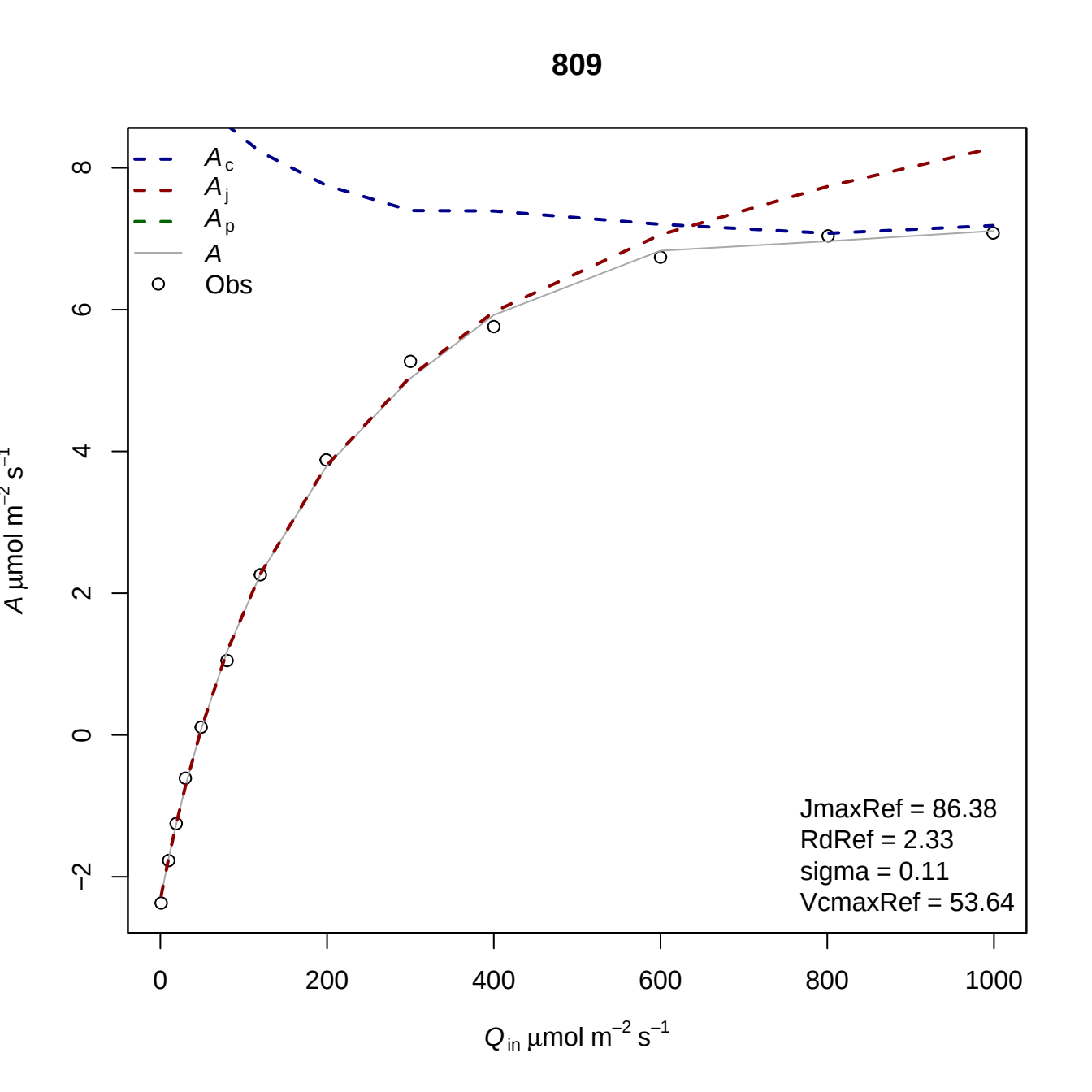

804
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.69
RdRef = 1.08
sigma = 0.22
VcmaxRef = 2150334.56
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

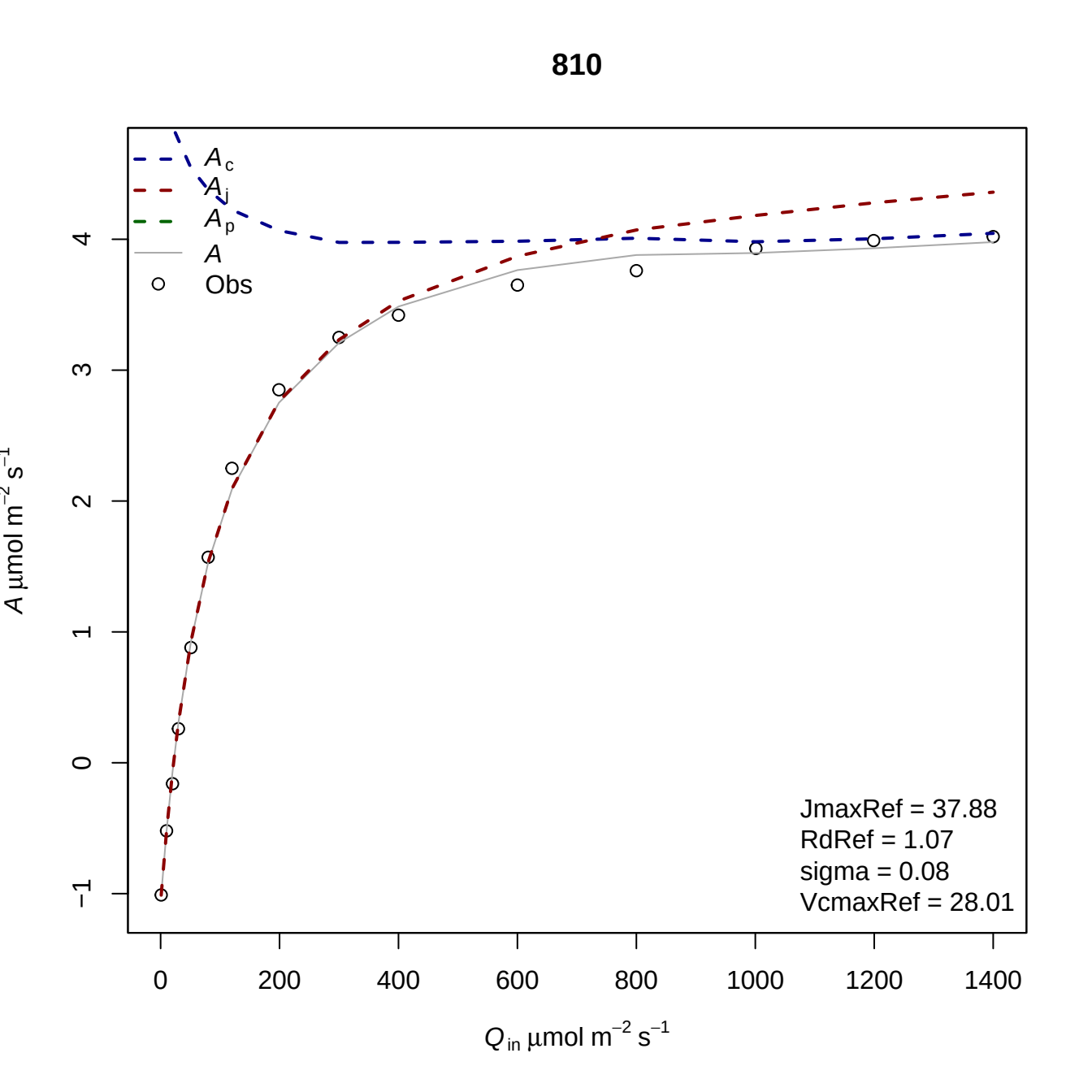

805
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.99
RdRef = 0.58
sigma = 0.23
VcmaxRef = 43.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

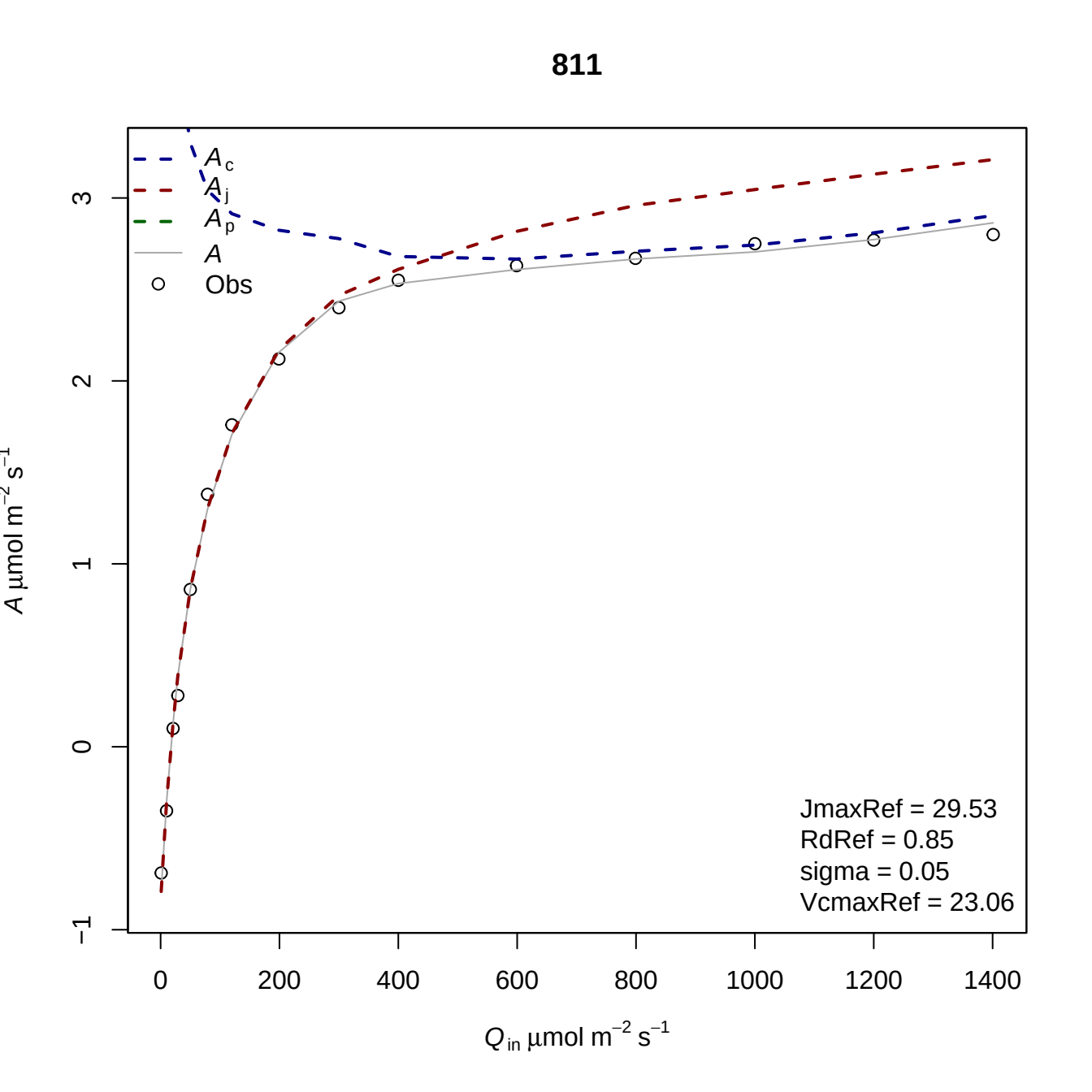

806
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.94
RdRef = 1.74
sigma = 0.13
VcmaxRef = 121530.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

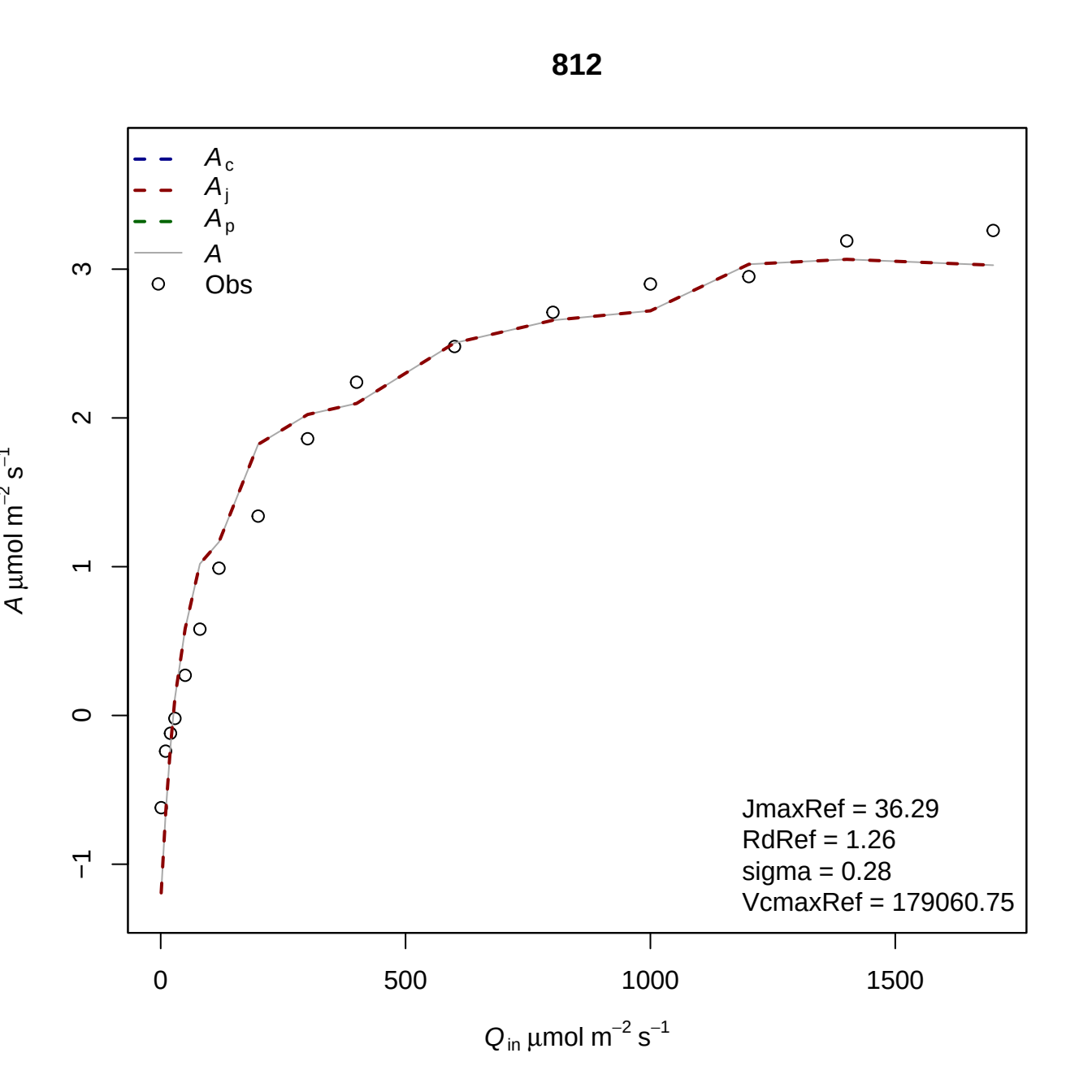

807
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.02
RdRef = 2.41
sigma = 0.15
VcmaxRef = 122686.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800
1000
1200

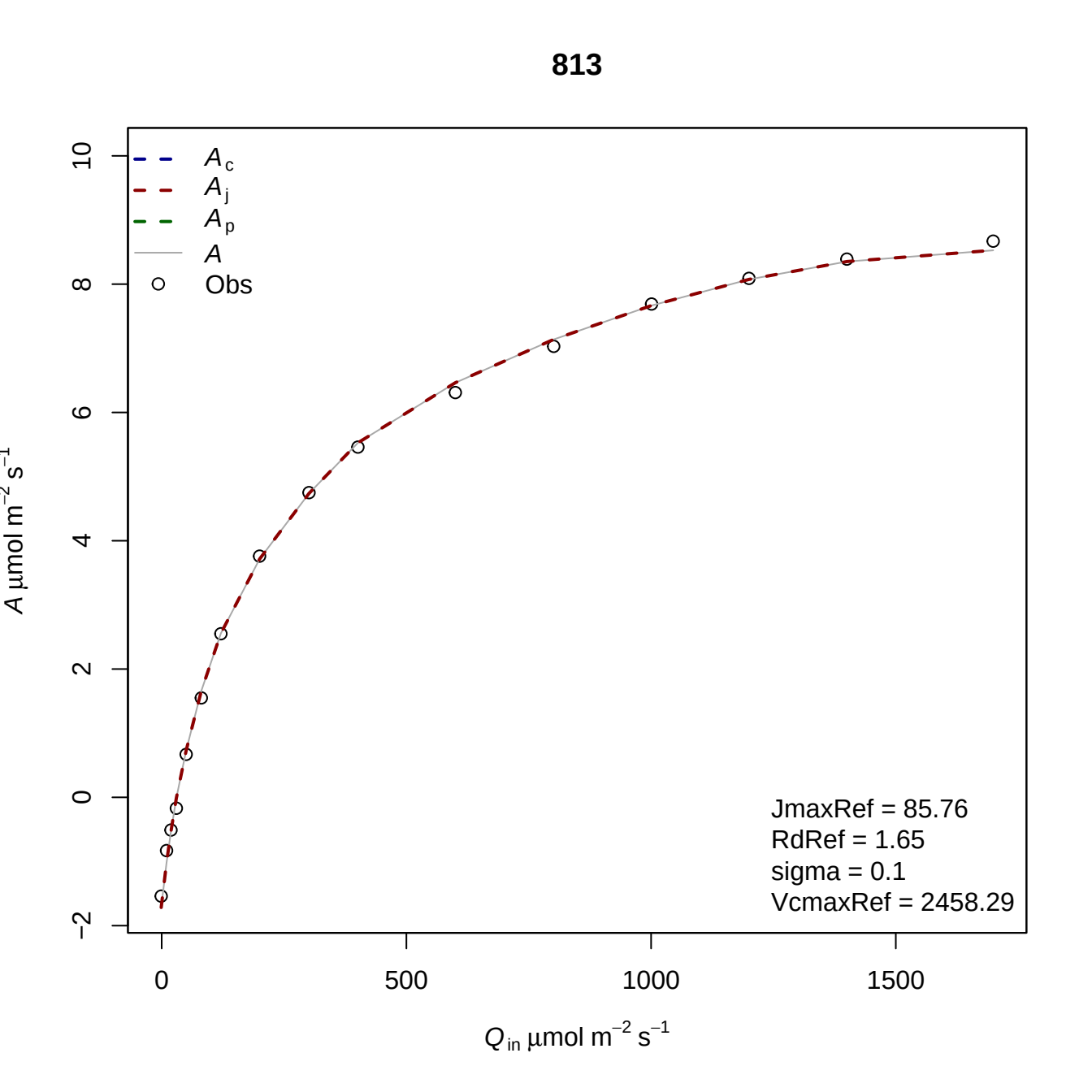

808
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.91
RdRef = 1.09
sigma = 0.08
VcmaxRef = 22.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

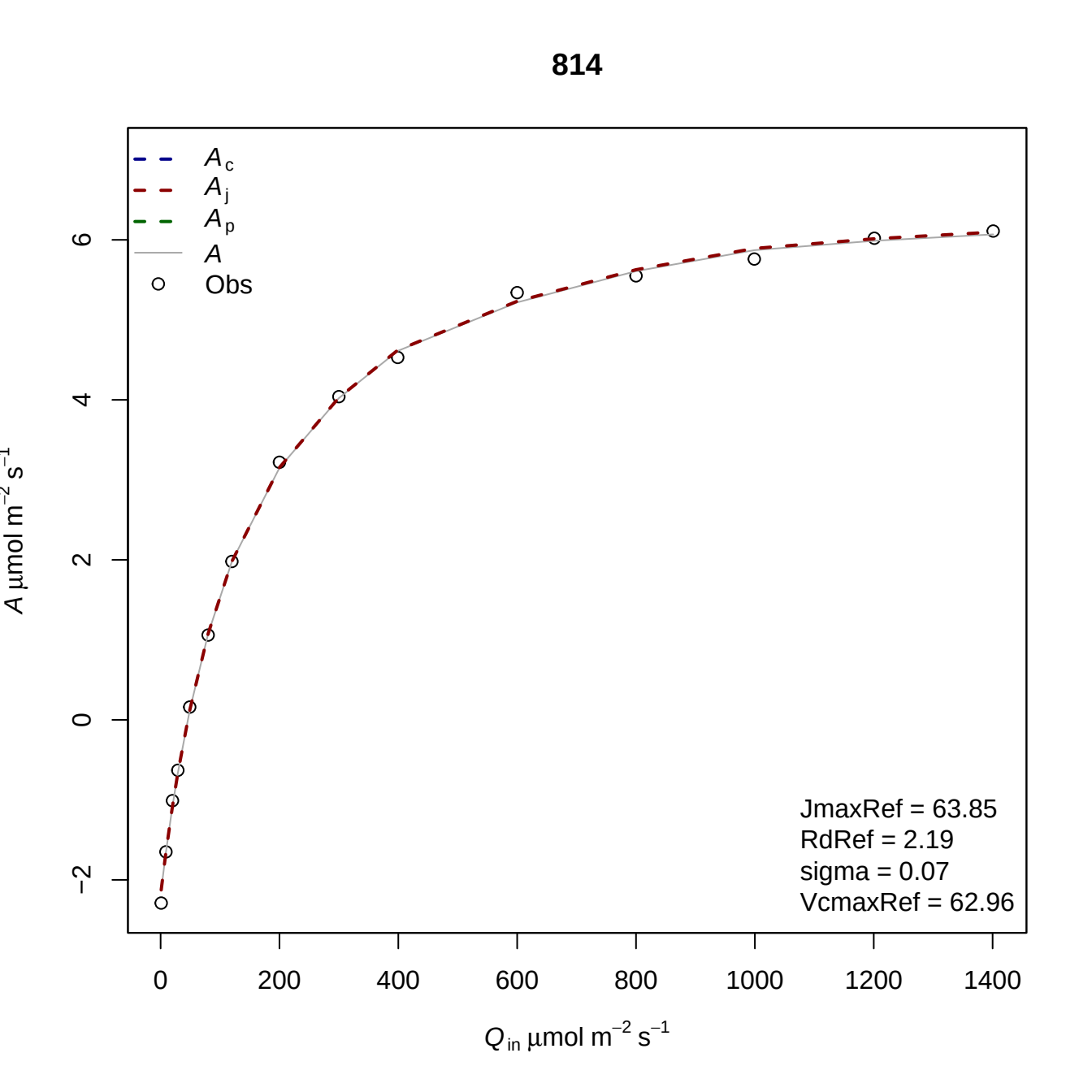

809
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.38
RdRef = 2.33
sigma = 0.11
VcmaxRef = 53.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
200
400
600
800
1000

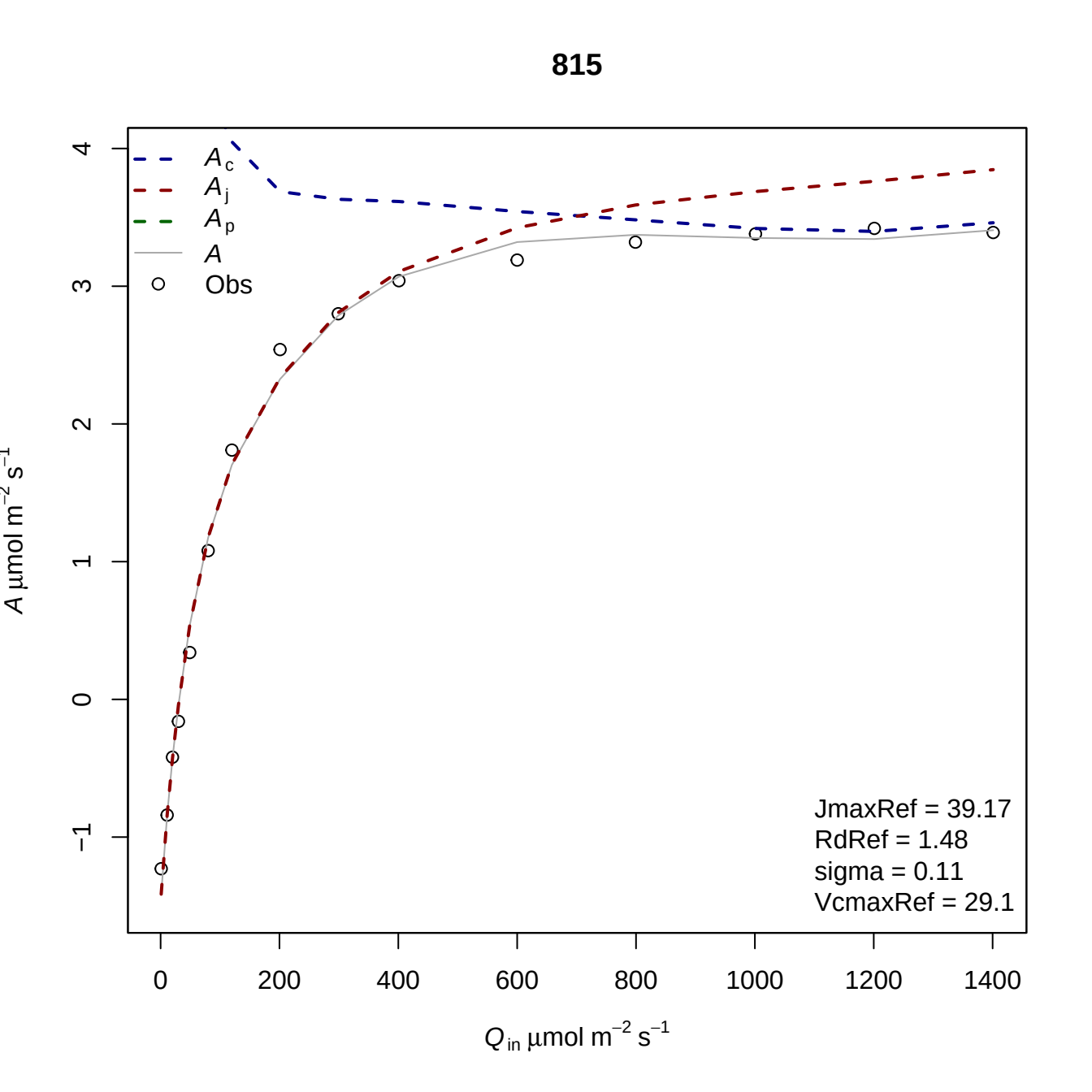

810
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.88
RdRef = 1.07
sigma = 0.08
VcmaxRef = 28.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

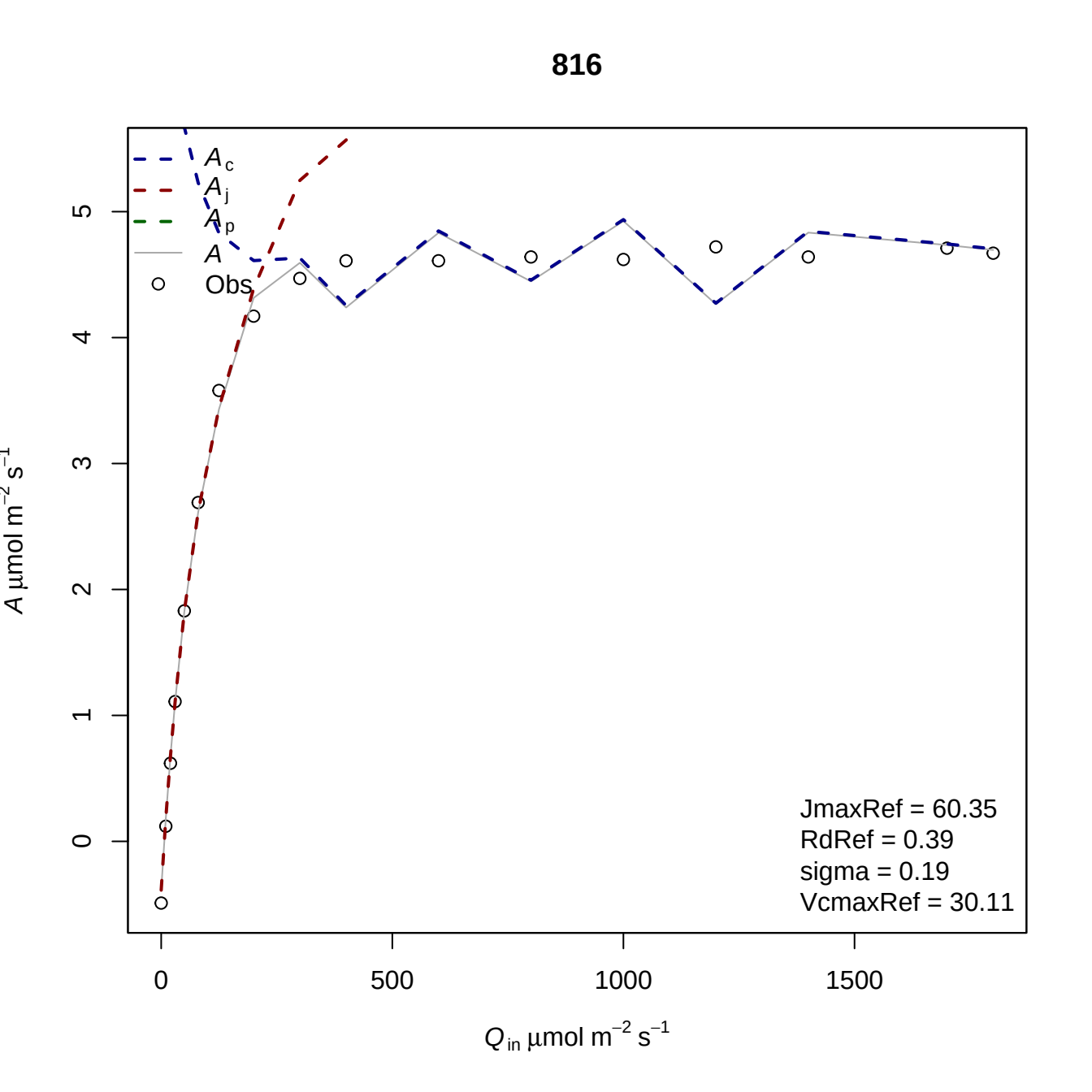

811
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.53
RdRef = 0.85
sigma = 0.05
VcmaxRef = 23.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

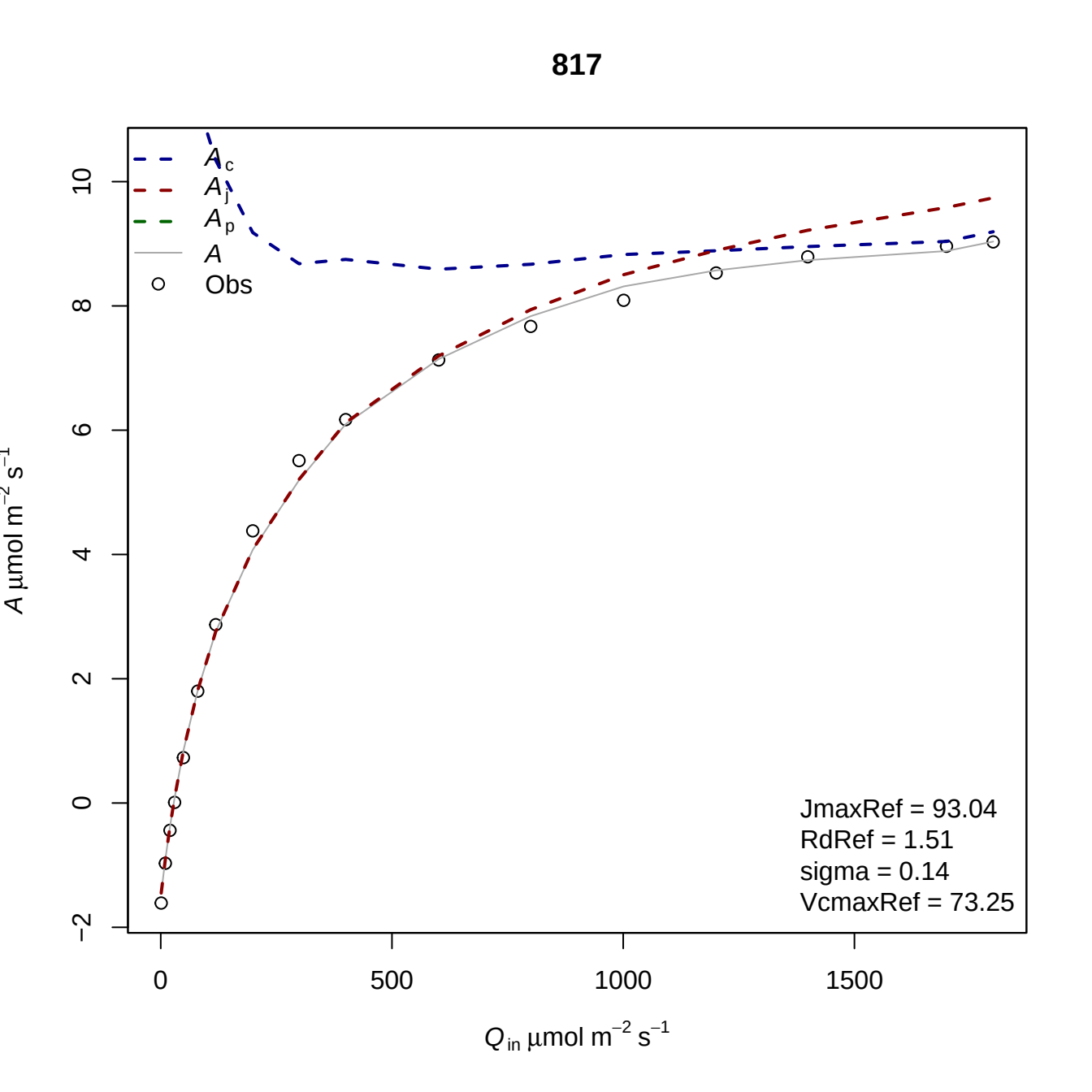

812
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.29
RdRef = 1.26
sigma = 0.28
VcmaxRef = 179060.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

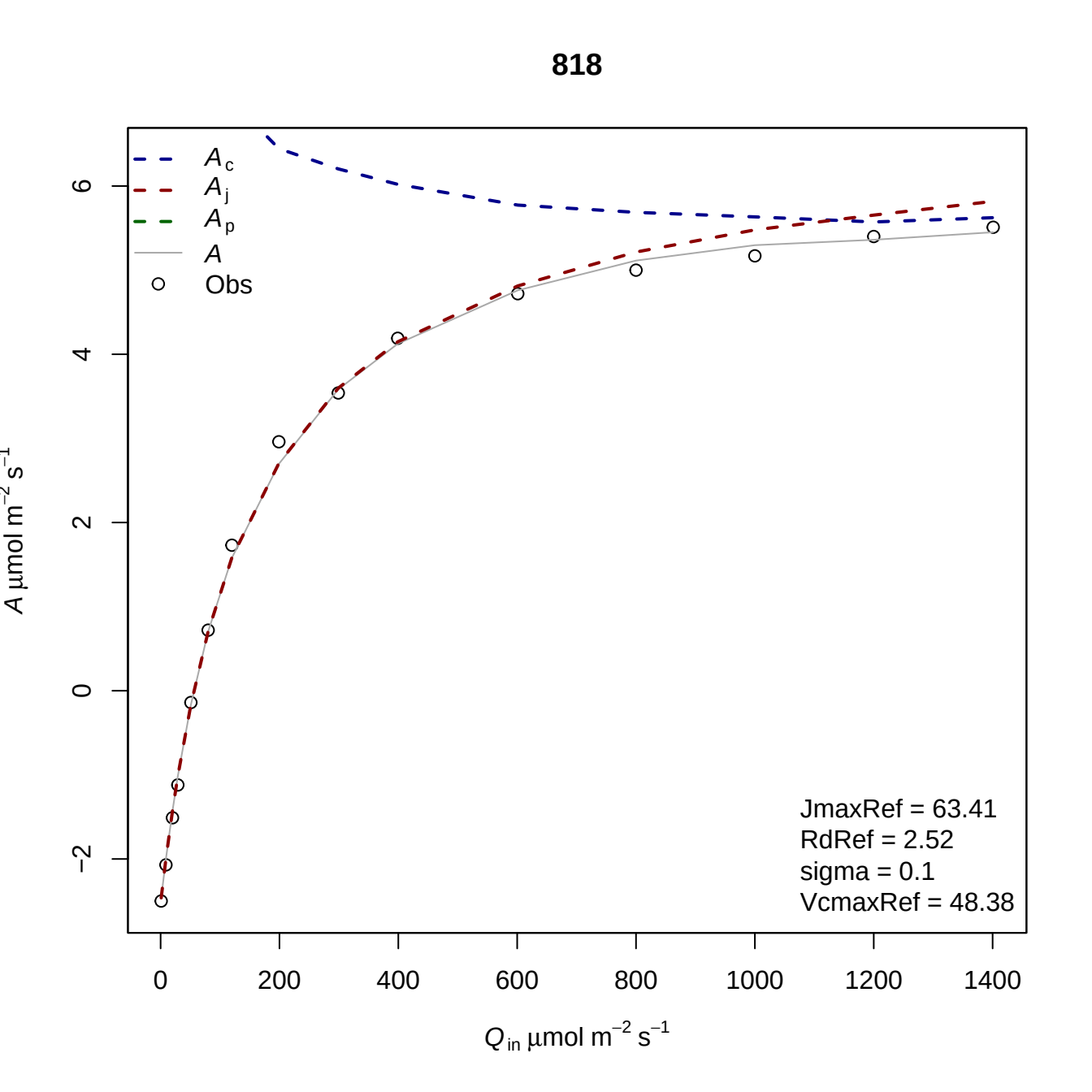

813
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.76
RdRef = 1.65
sigma = 0.1
VcmaxRef = 2458.29
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

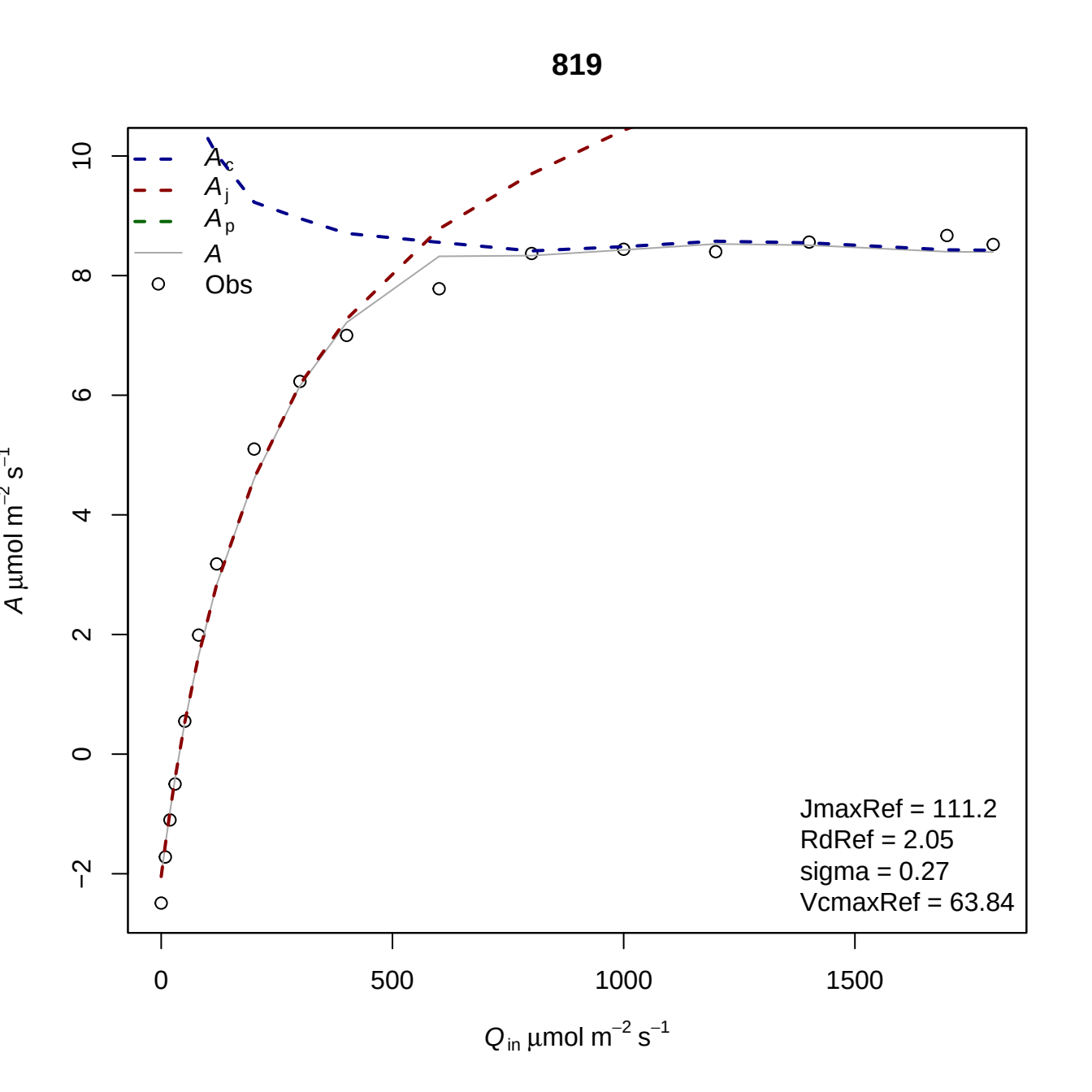

814
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.85
RdRef = 2.19
sigma = 0.07
VcmaxRef = 62.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

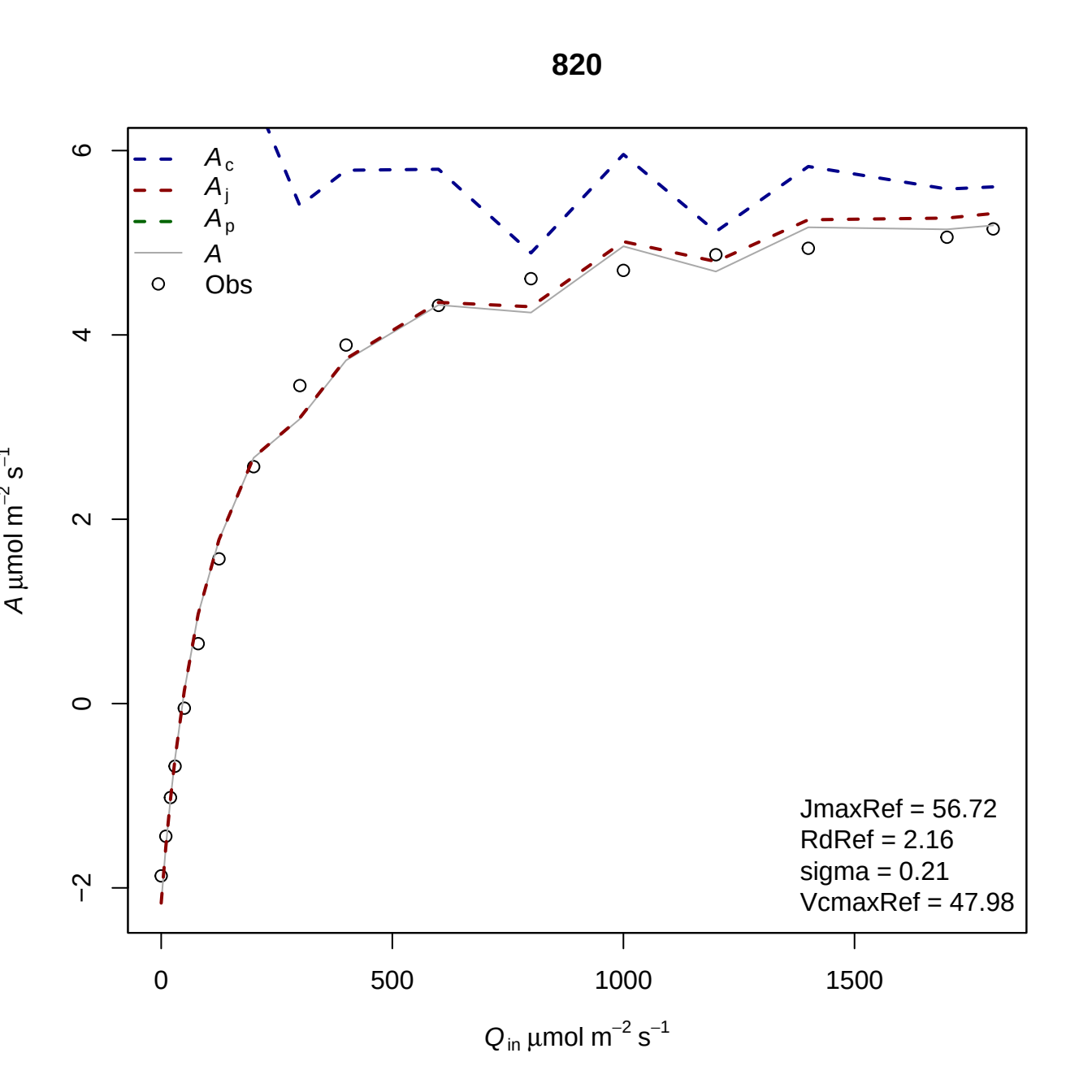

815
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 39.17
RdRef = 1.48
sigma = 0.11
VcmaxRef = 29.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

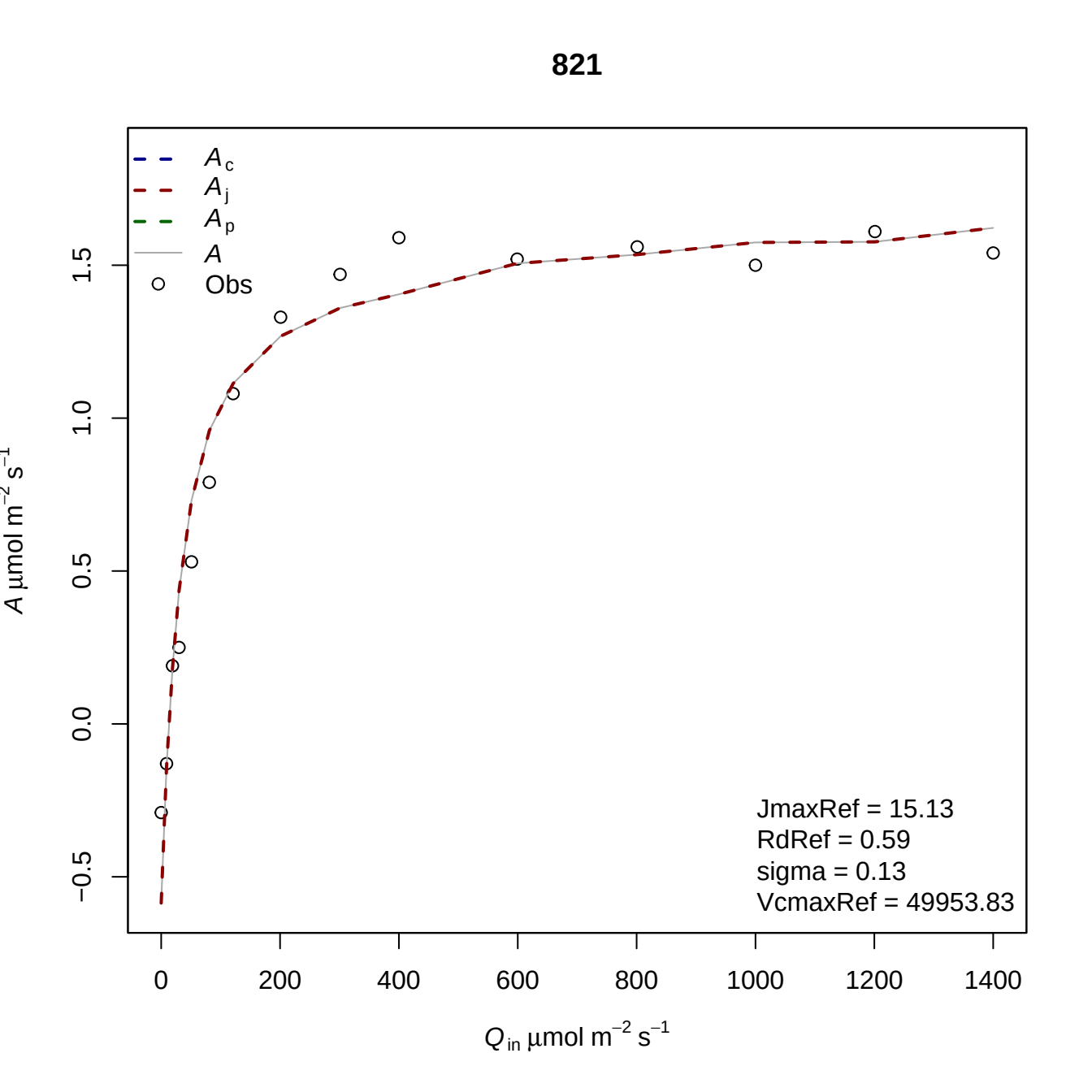

816
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.35
RdRef = 0.39
sigma = 0.19
VcmaxRef = 30.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

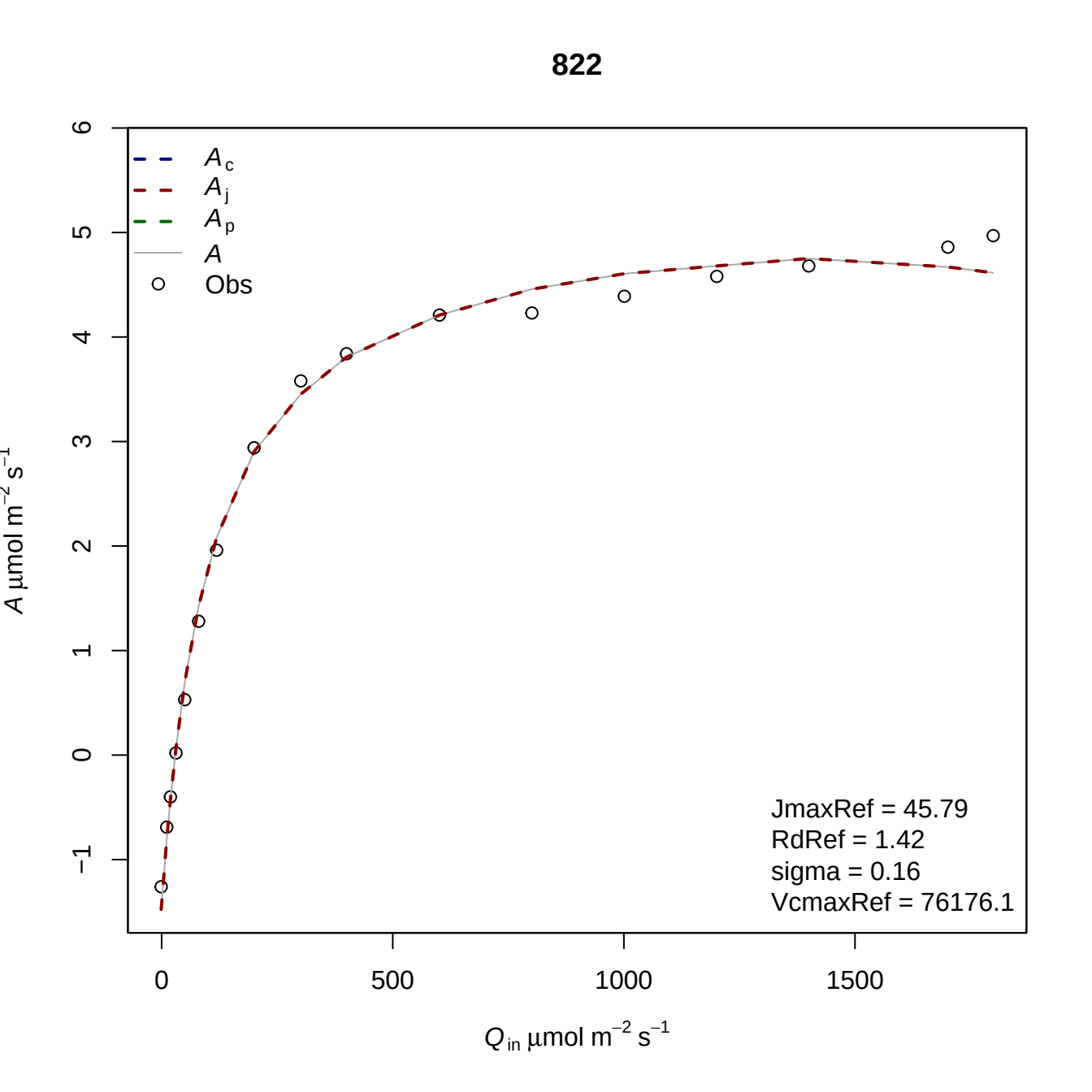

817
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.04
RdRef = 1.51
sigma = 0.14
VcmaxRef = 73.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

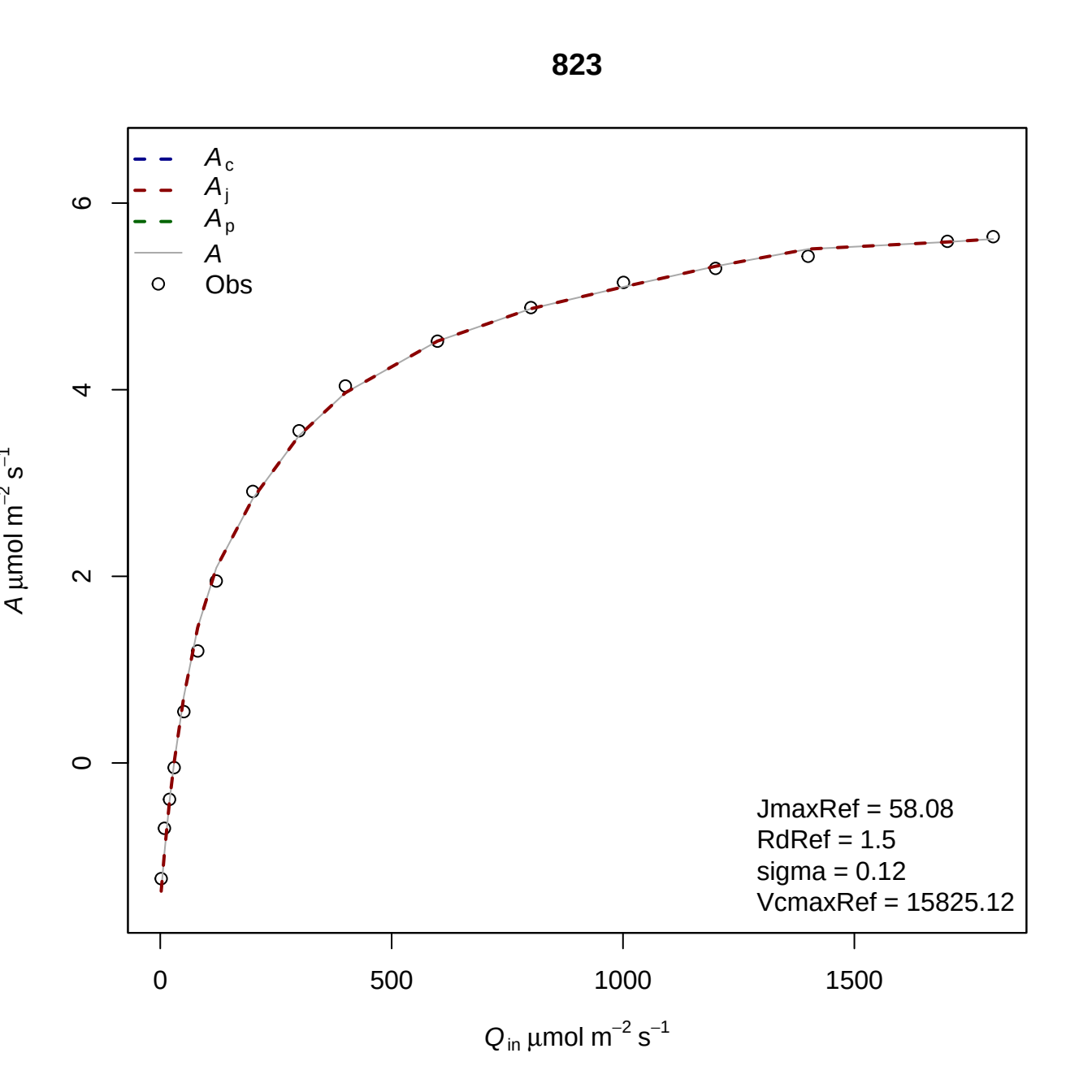

818
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.41
RdRef = 2.52
sigma = 0.1
VcmaxRef = 48.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

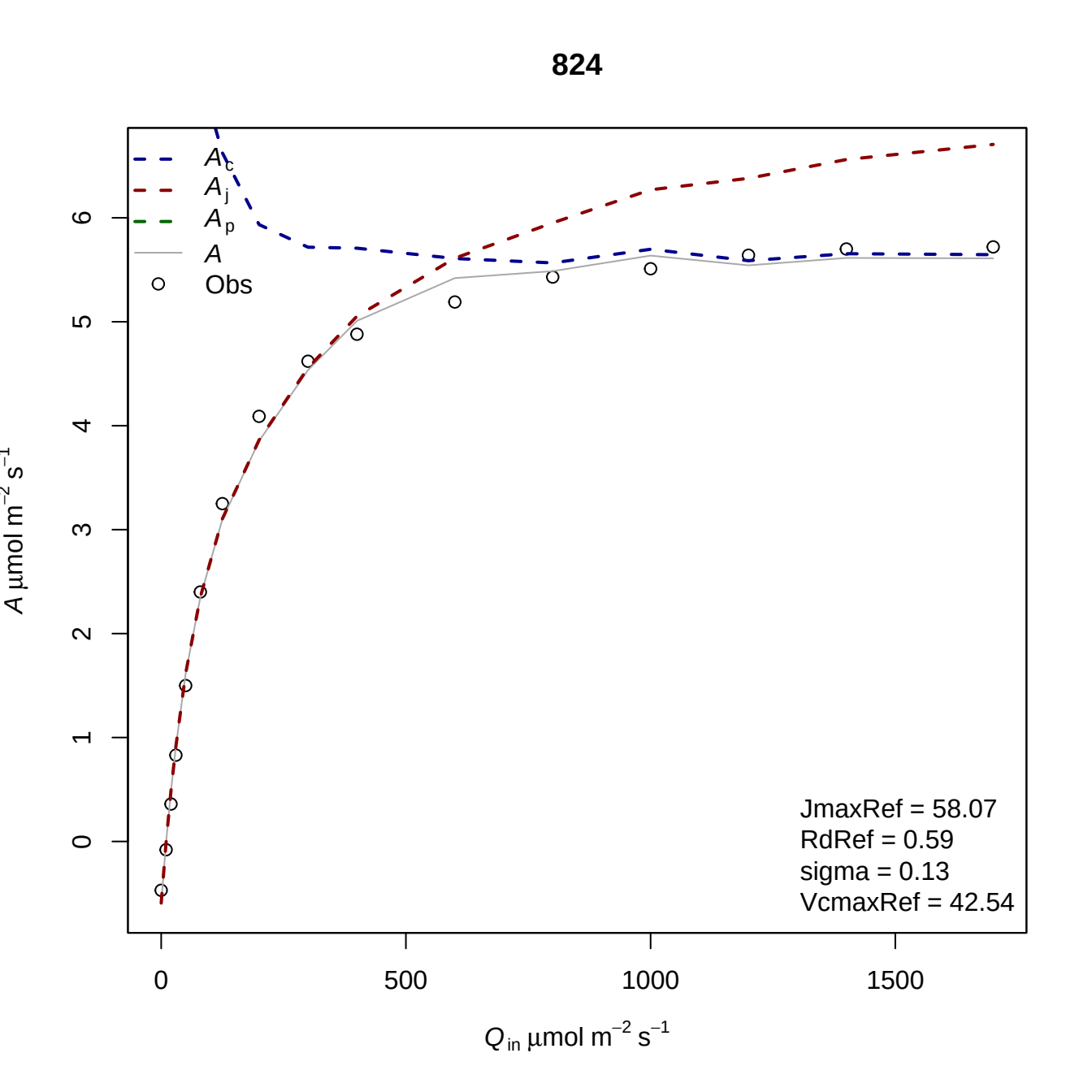

819
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.2
RdRef = 2.05
sigma = 0.27
VcmaxRef = 63.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

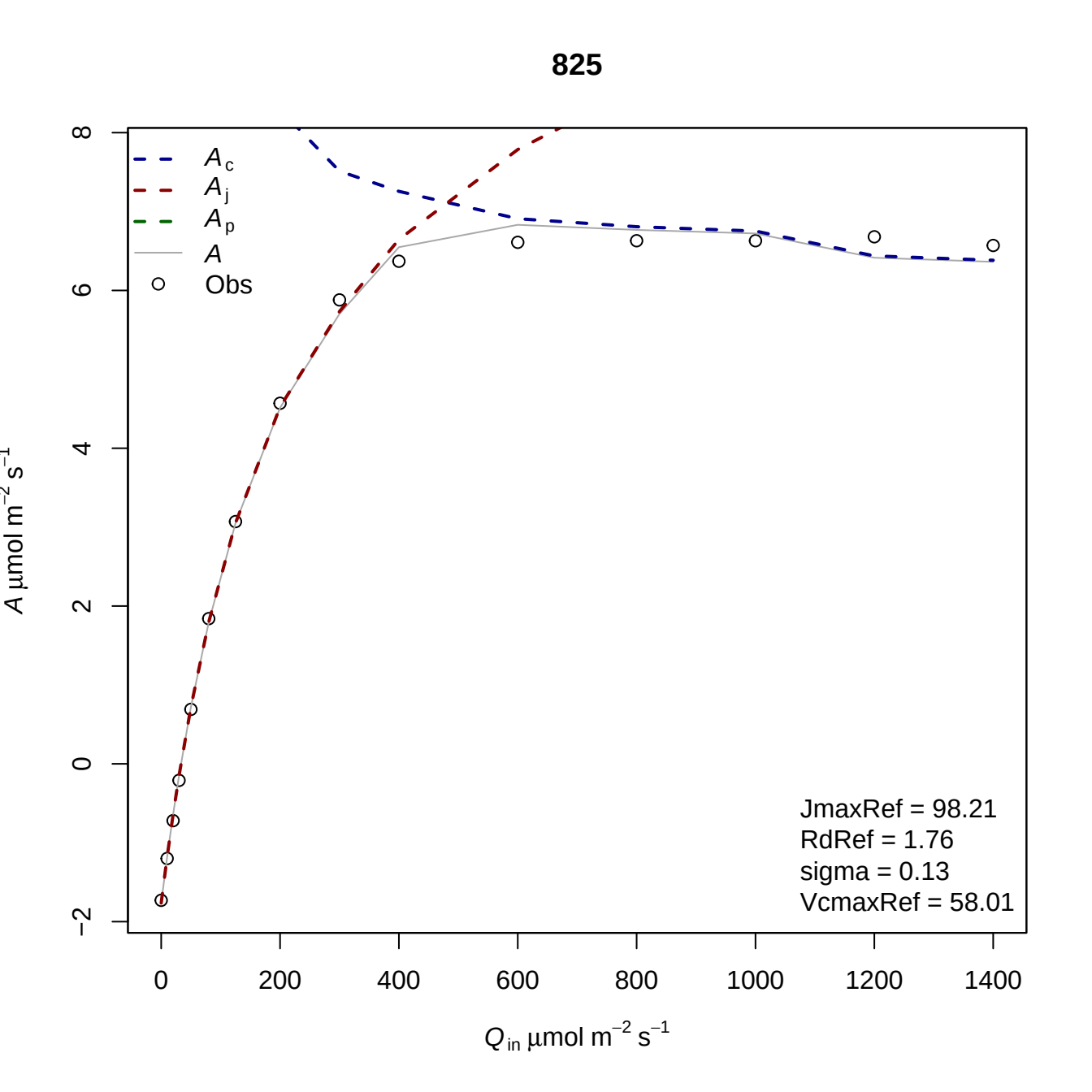

820
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.72
RdRef = 2.16
sigma = 0.21
VcmaxRef = 47.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

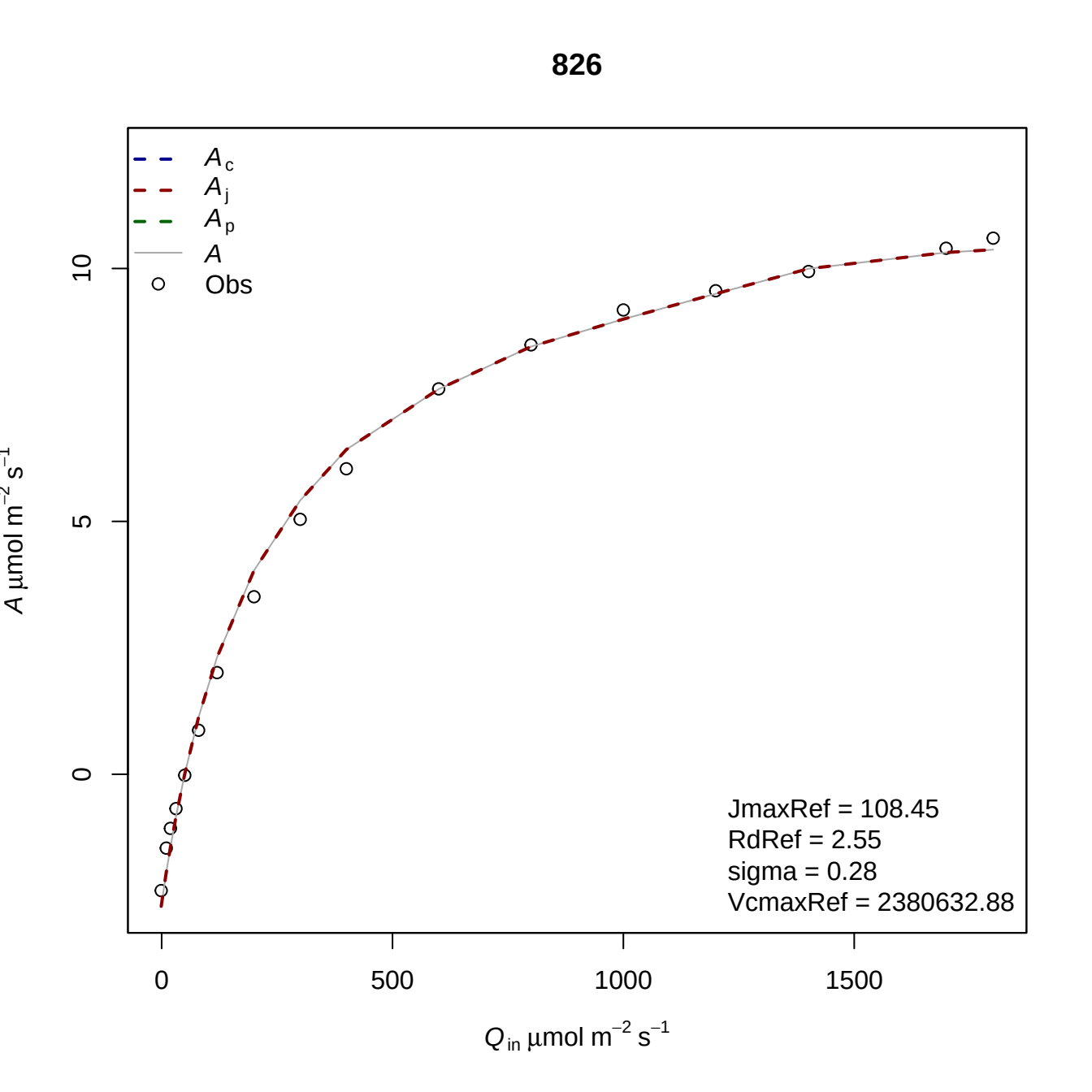

821
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 15.13
RdRef = 0.59
sigma = 0.13
VcmaxRef = 49953.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

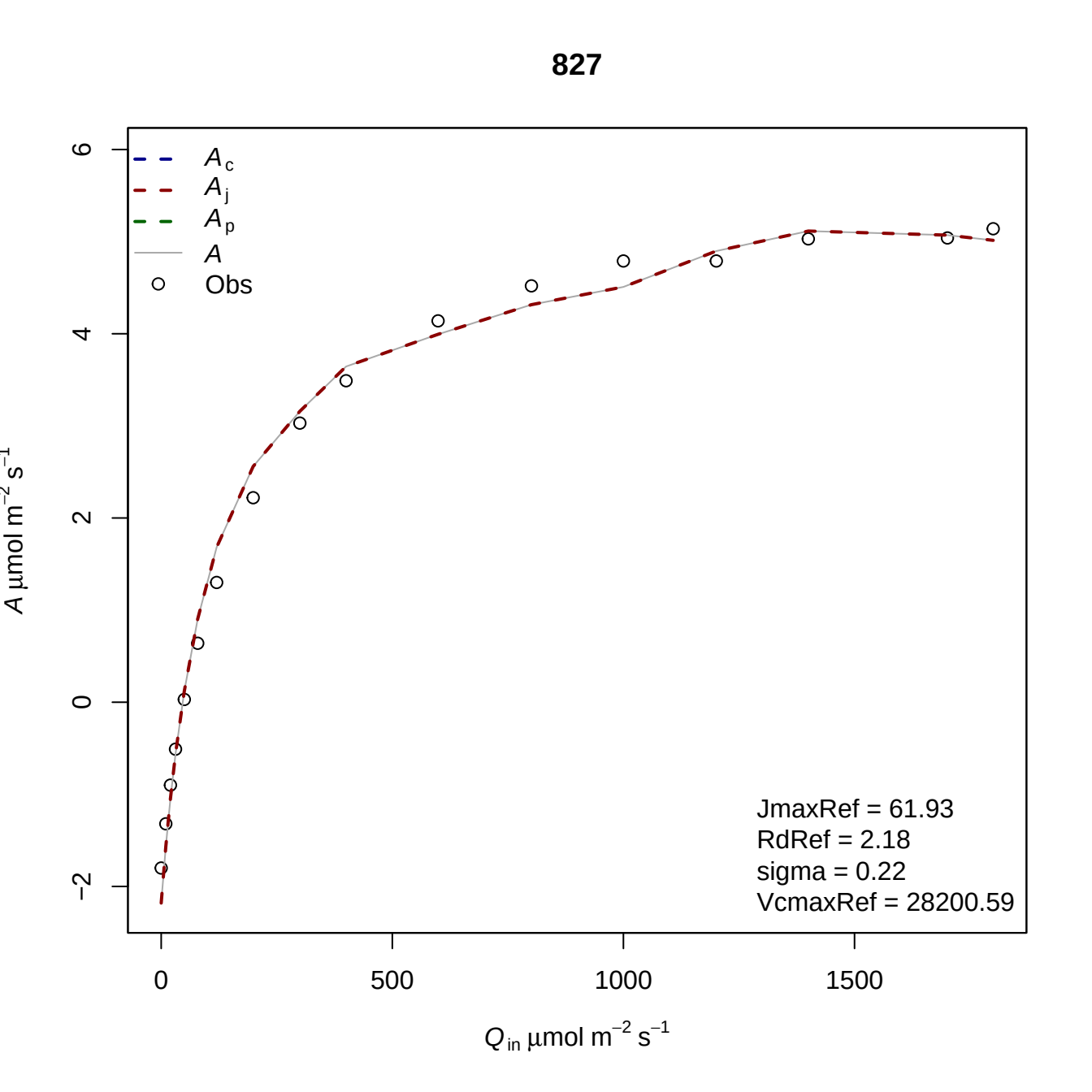

822
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.79
RdRef = 1.42
sigma = 0.16
VcmaxRef = 76176.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

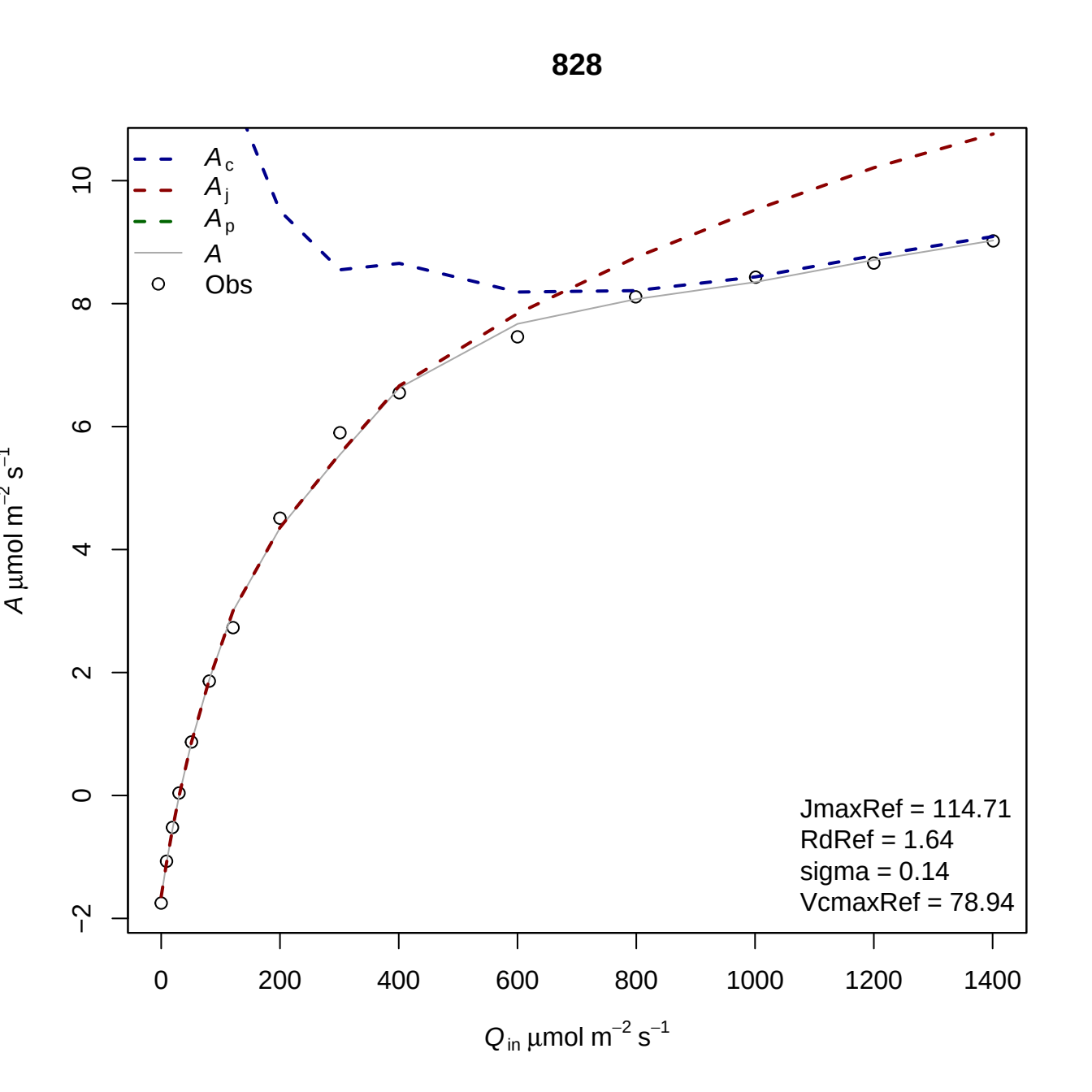

823
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.08
RdRef = 1.5
sigma = 0.12
VcmaxRef = 15825.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

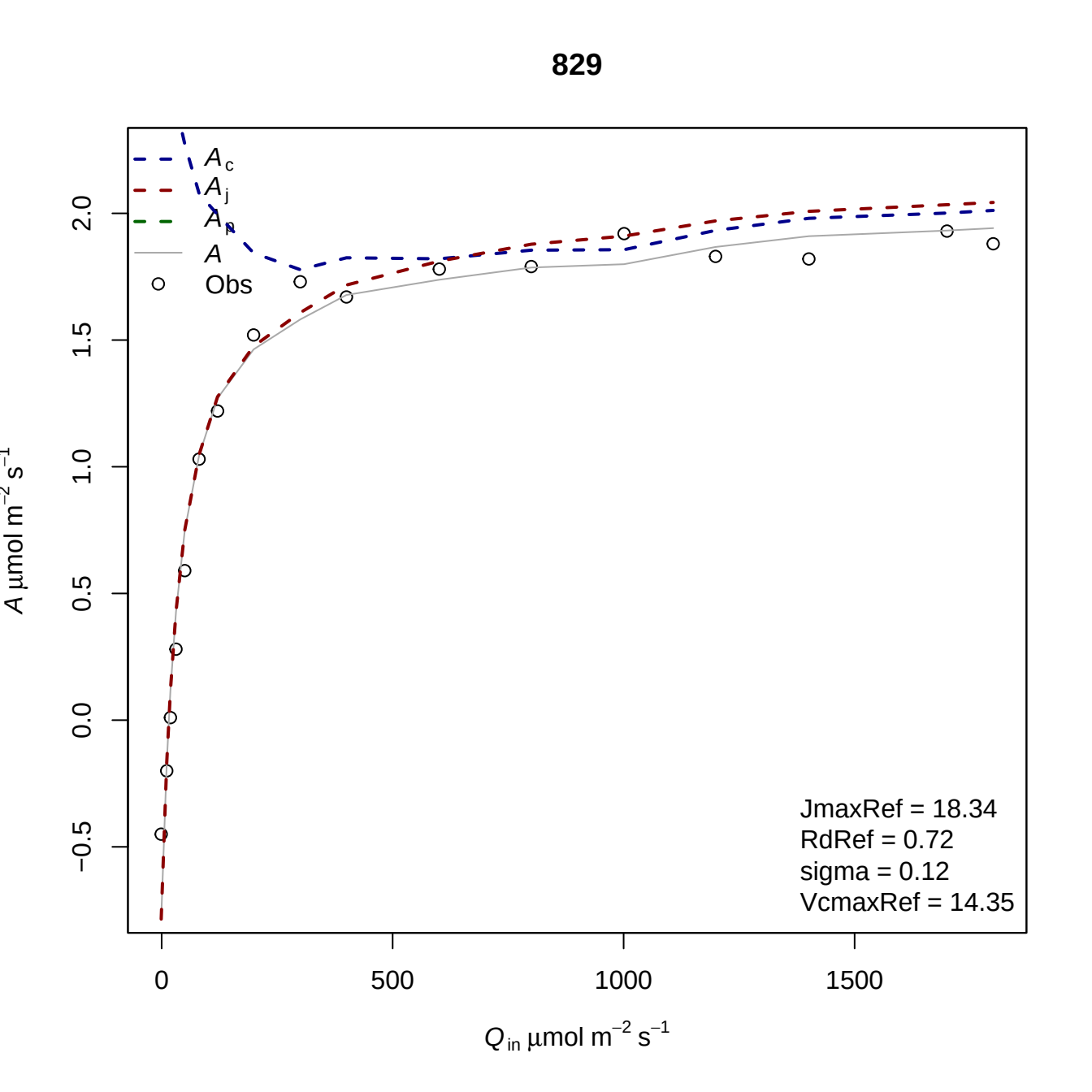

824
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.07
RdRef = 0.59
sigma = 0.13
VcmaxRef = 42.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

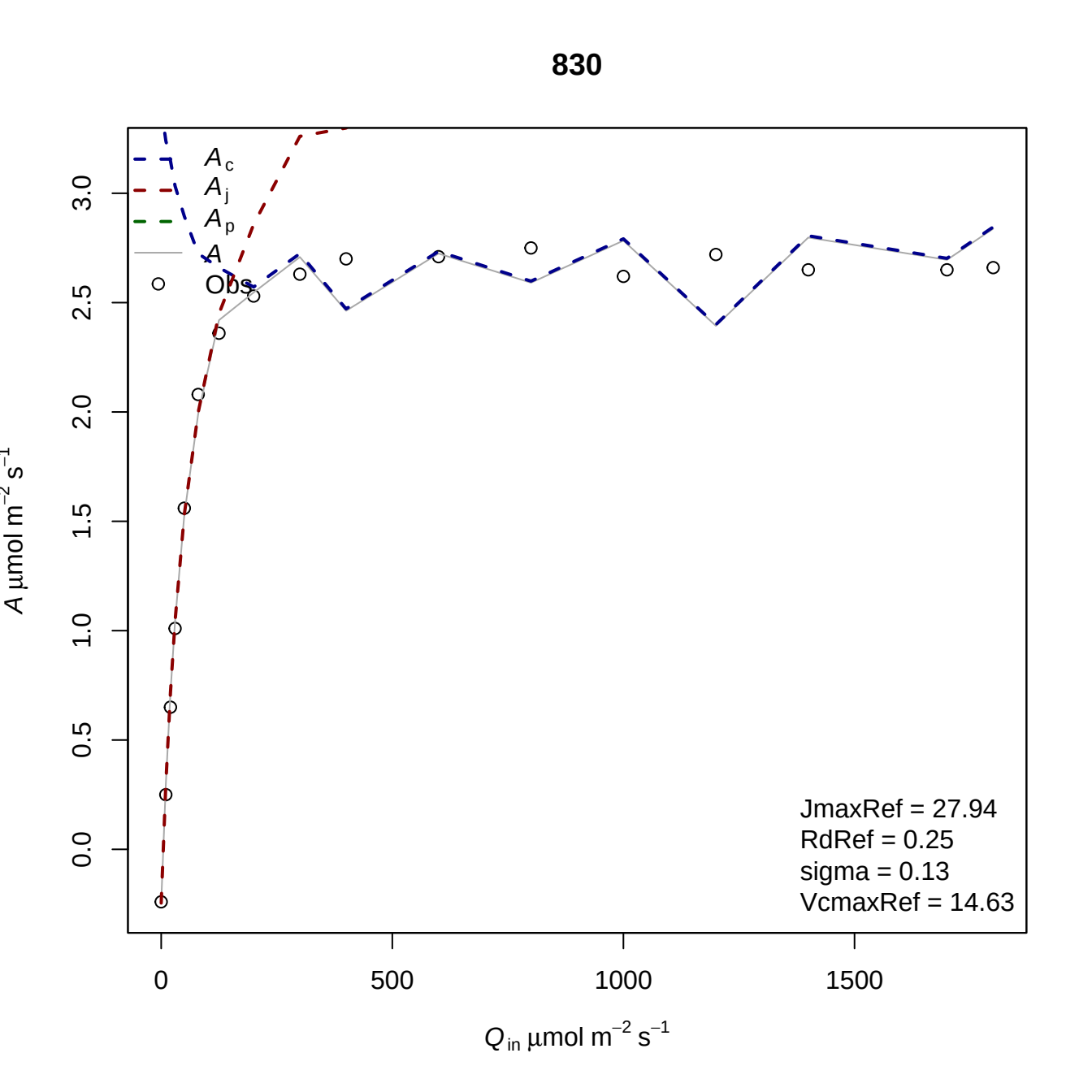

825
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.21
RdRef = 1.76
sigma = 0.13
VcmaxRef = 58.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

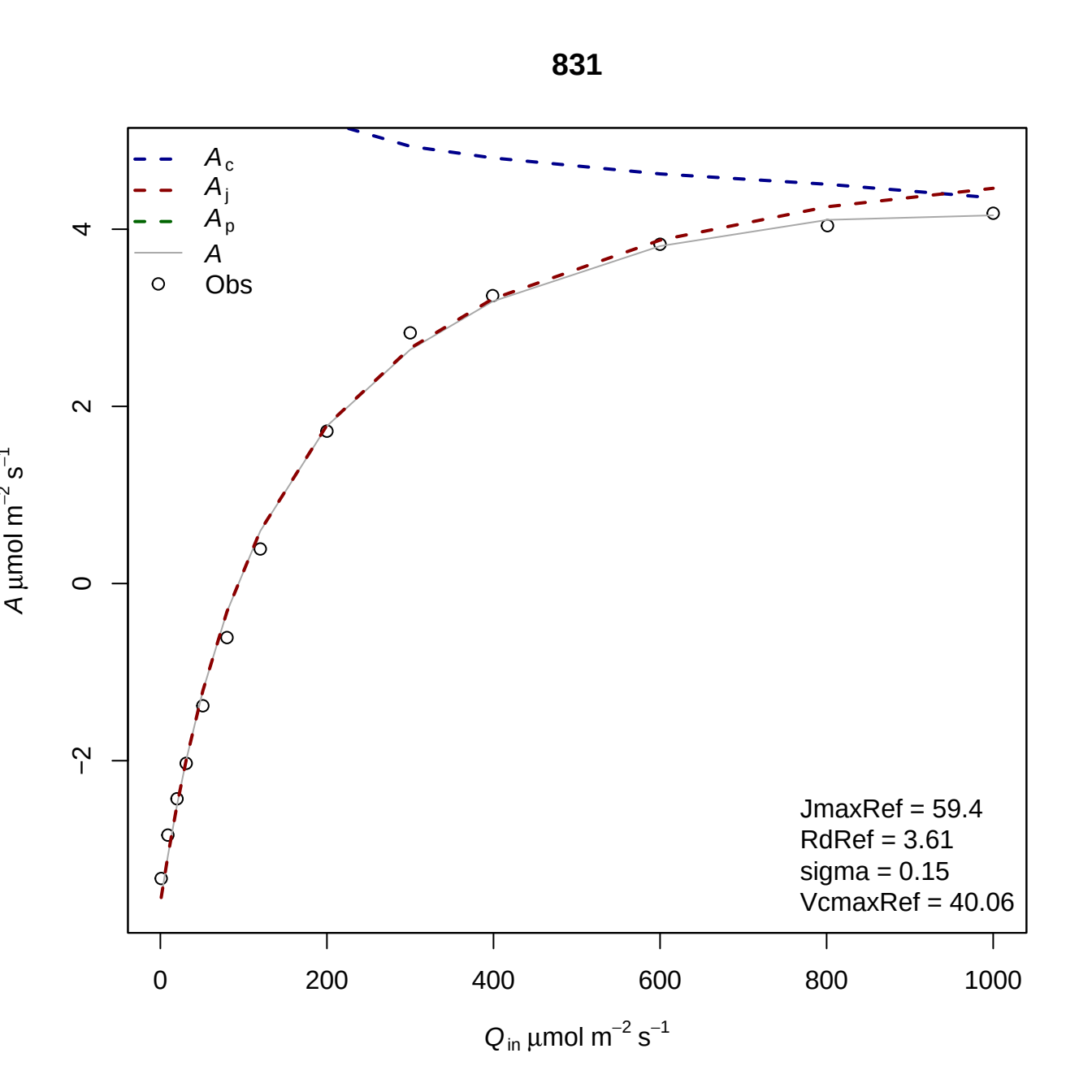

826
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.45
RdRef = 2.55
sigma = 0.28
VcmaxRef = 2380632.88
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

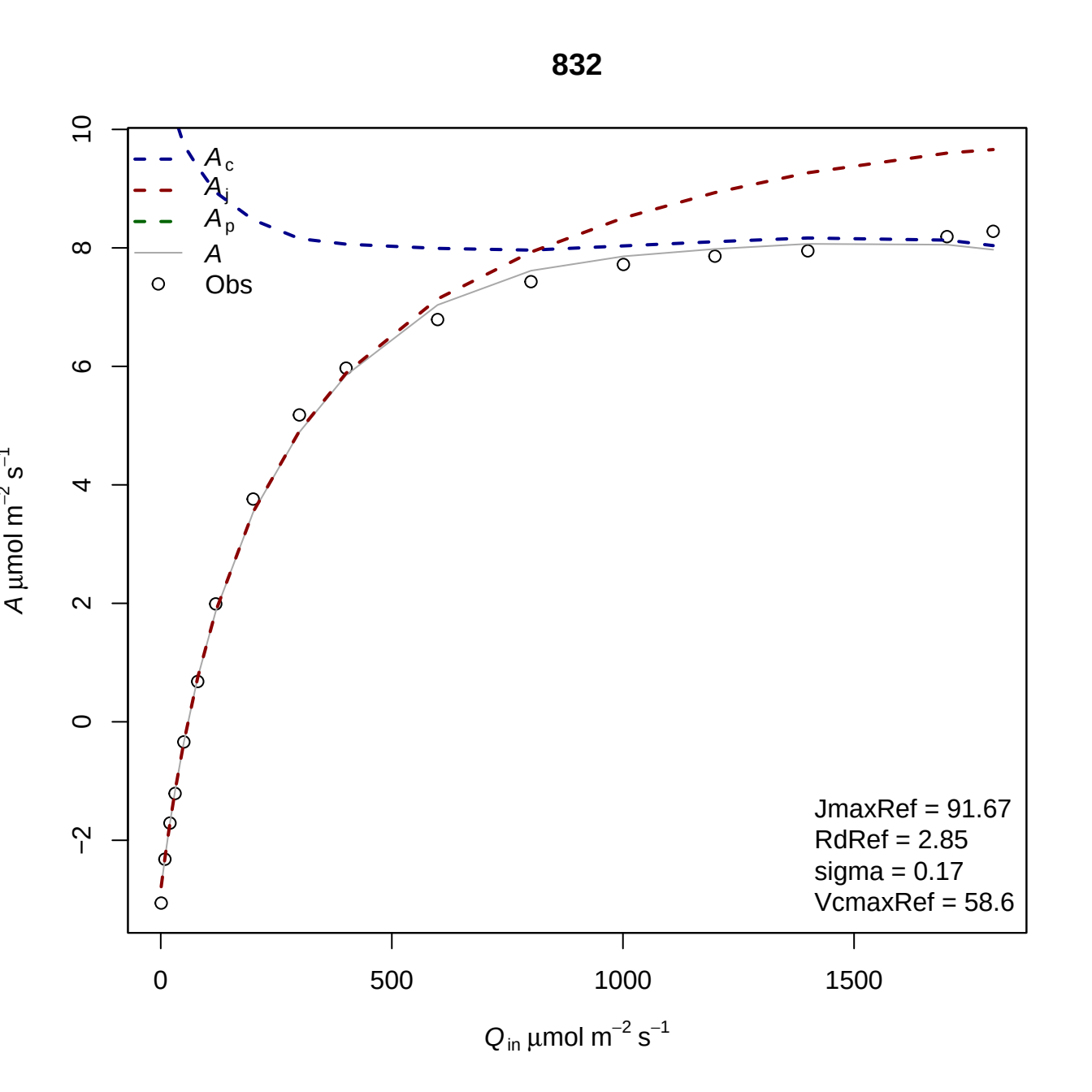

827
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.93
RdRef = 2.18
sigma = 0.22
VcmaxRef = 28200.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

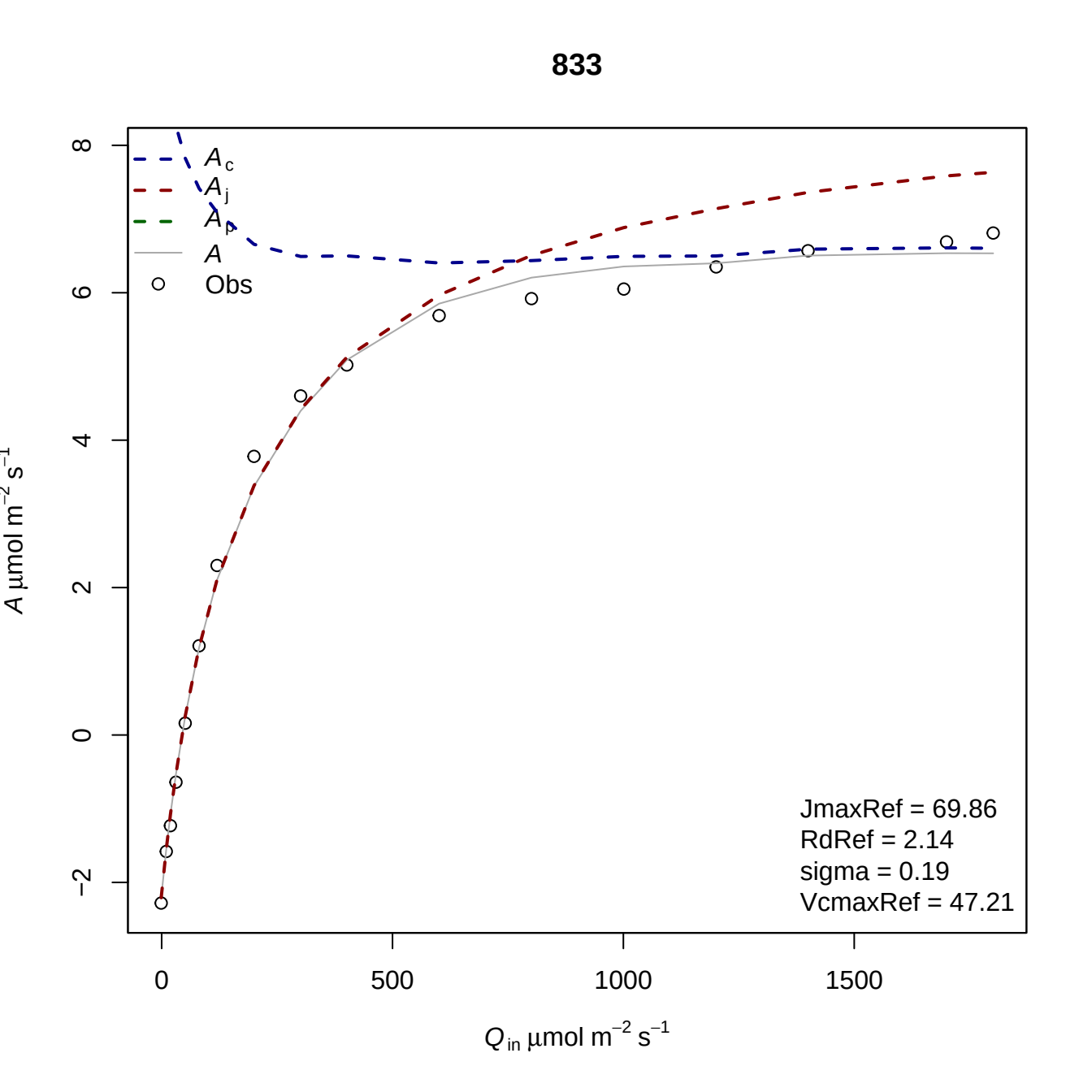

828
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.71
RdRef = 1.64
sigma = 0.14
VcmaxRef = 78.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

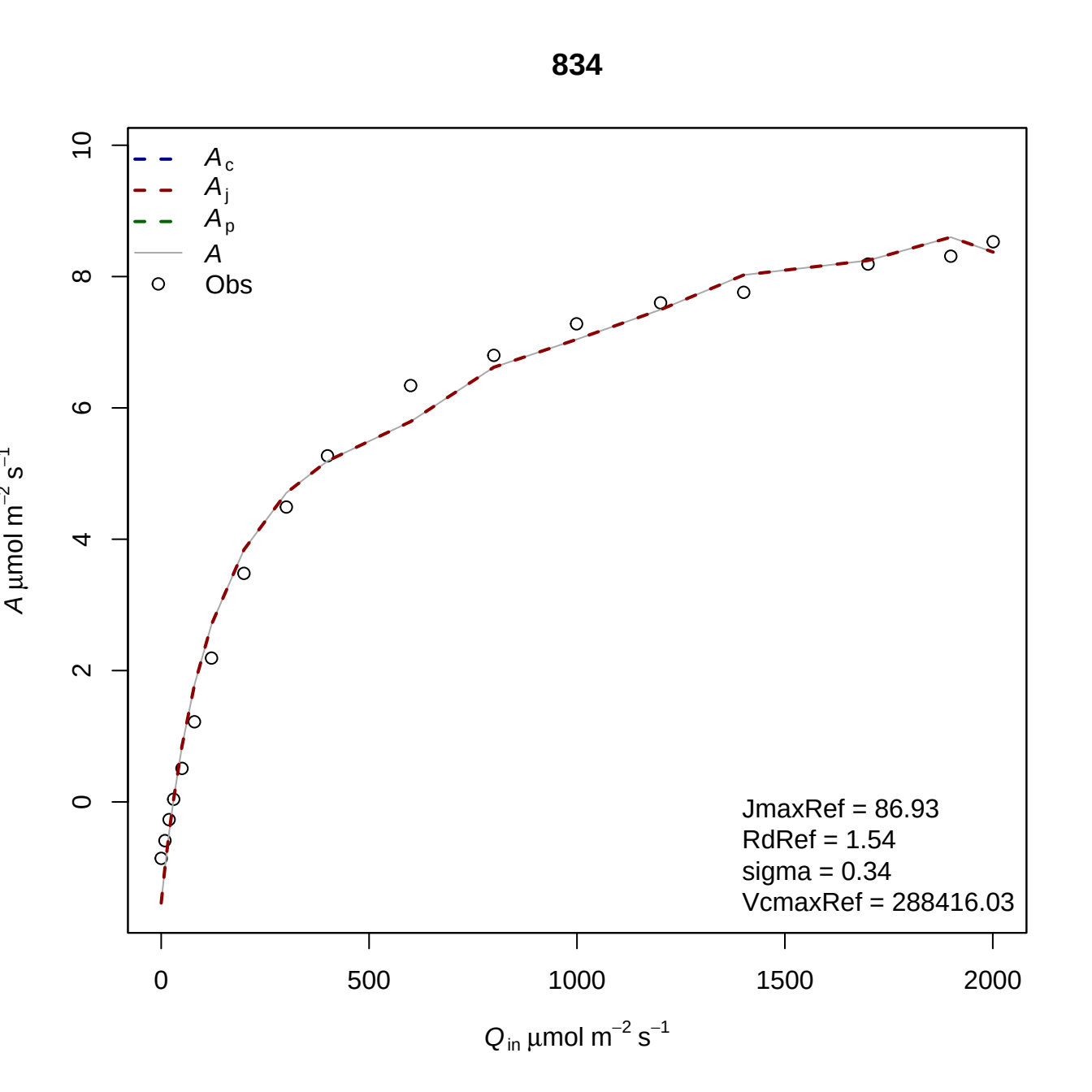

829
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 18.34
RdRef = 0.72
sigma = 0.12
VcmaxRef = 14.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

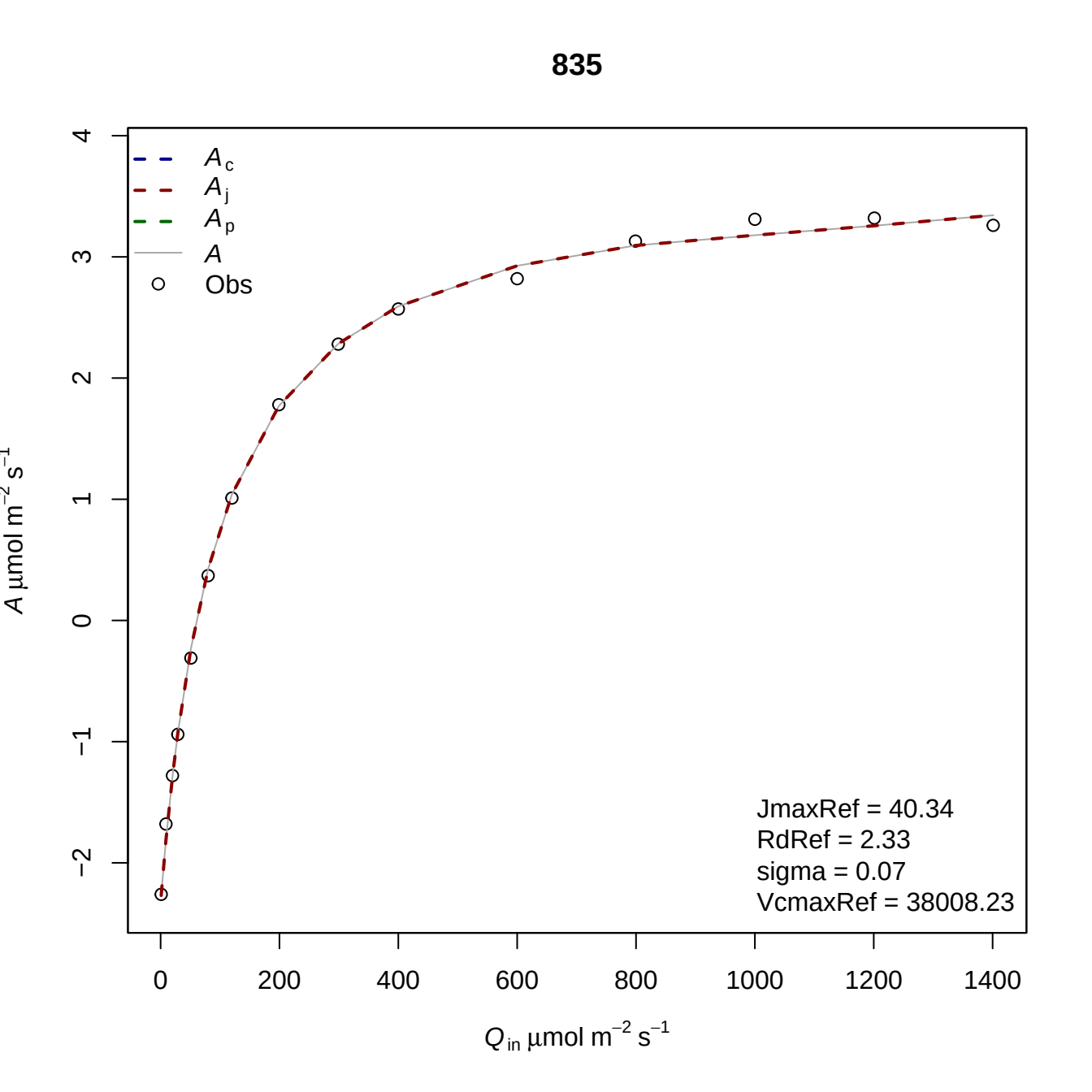

830
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.94
RdRef = 0.25
sigma = 0.13
VcmaxRef = 14.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

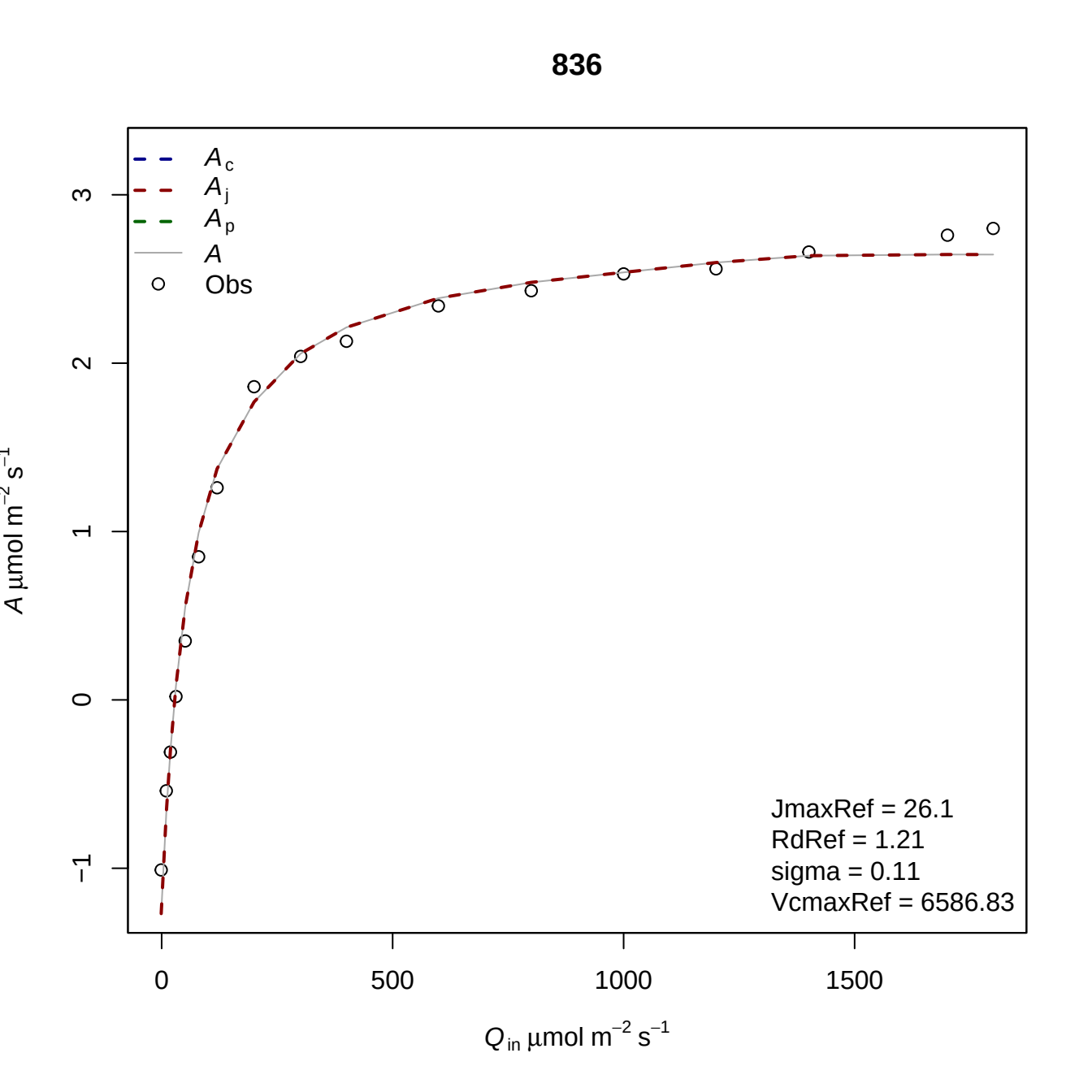

831
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.4
RdRef = 3.61
sigma = 0.15
VcmaxRef = 40.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
200
400
600
800
1000

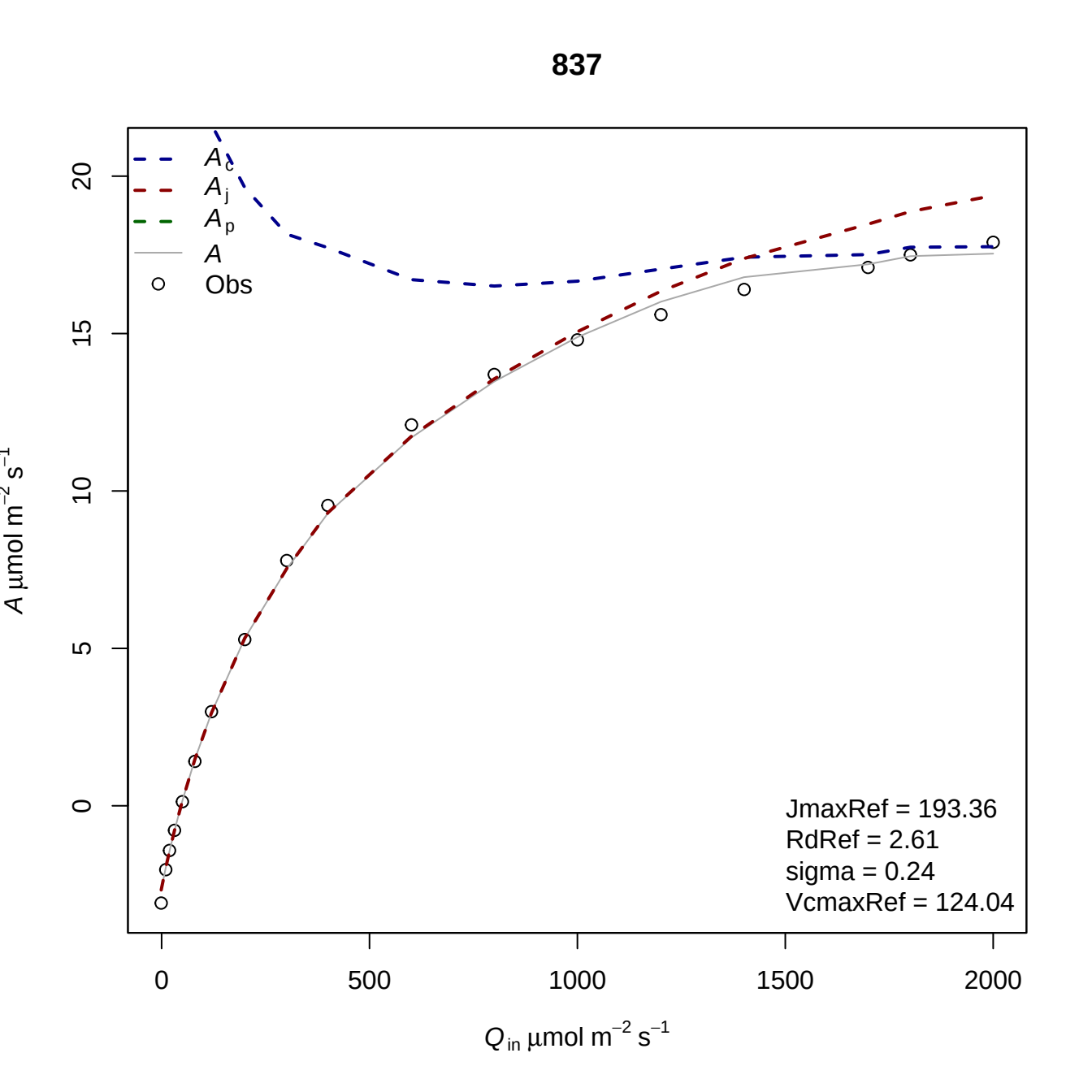

832
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.67
RdRef = 2.85
sigma = 0.17
VcmaxRef = 58.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

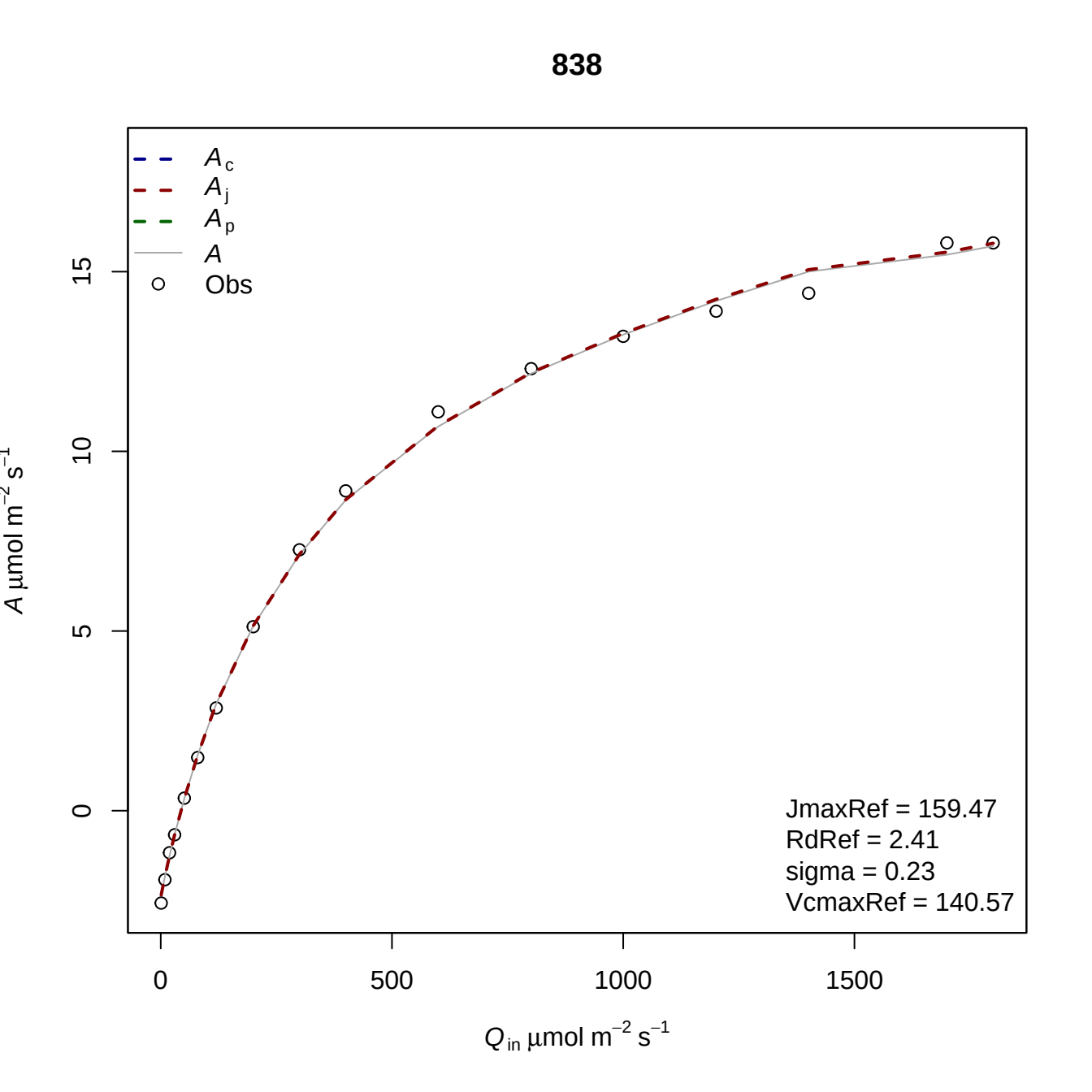

833
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.86
RdRef = 2.14
sigma = 0.19
VcmaxRef = 47.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

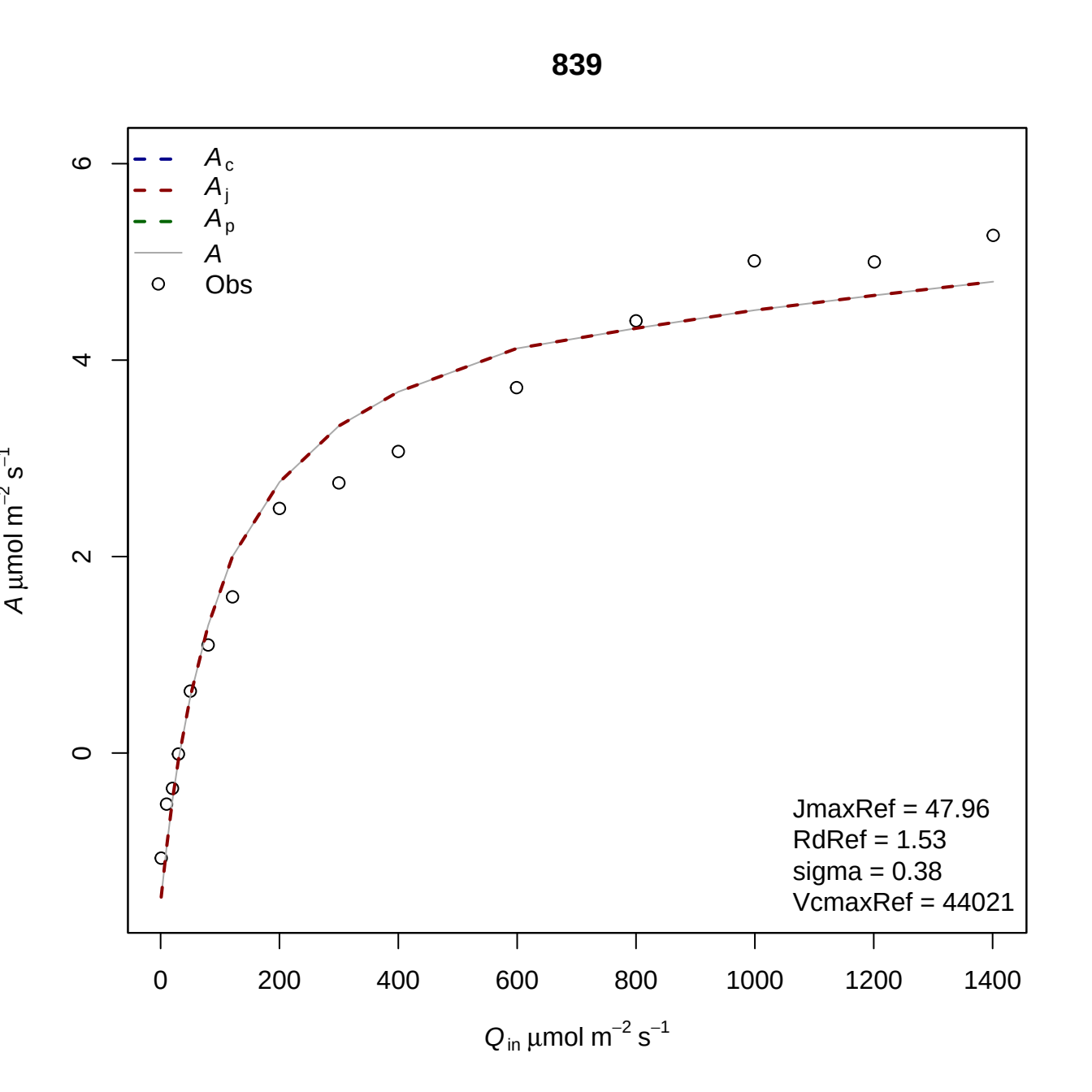

834
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.93
RdRef = 1.54
sigma = 0.34
VcmaxRef = 288416.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

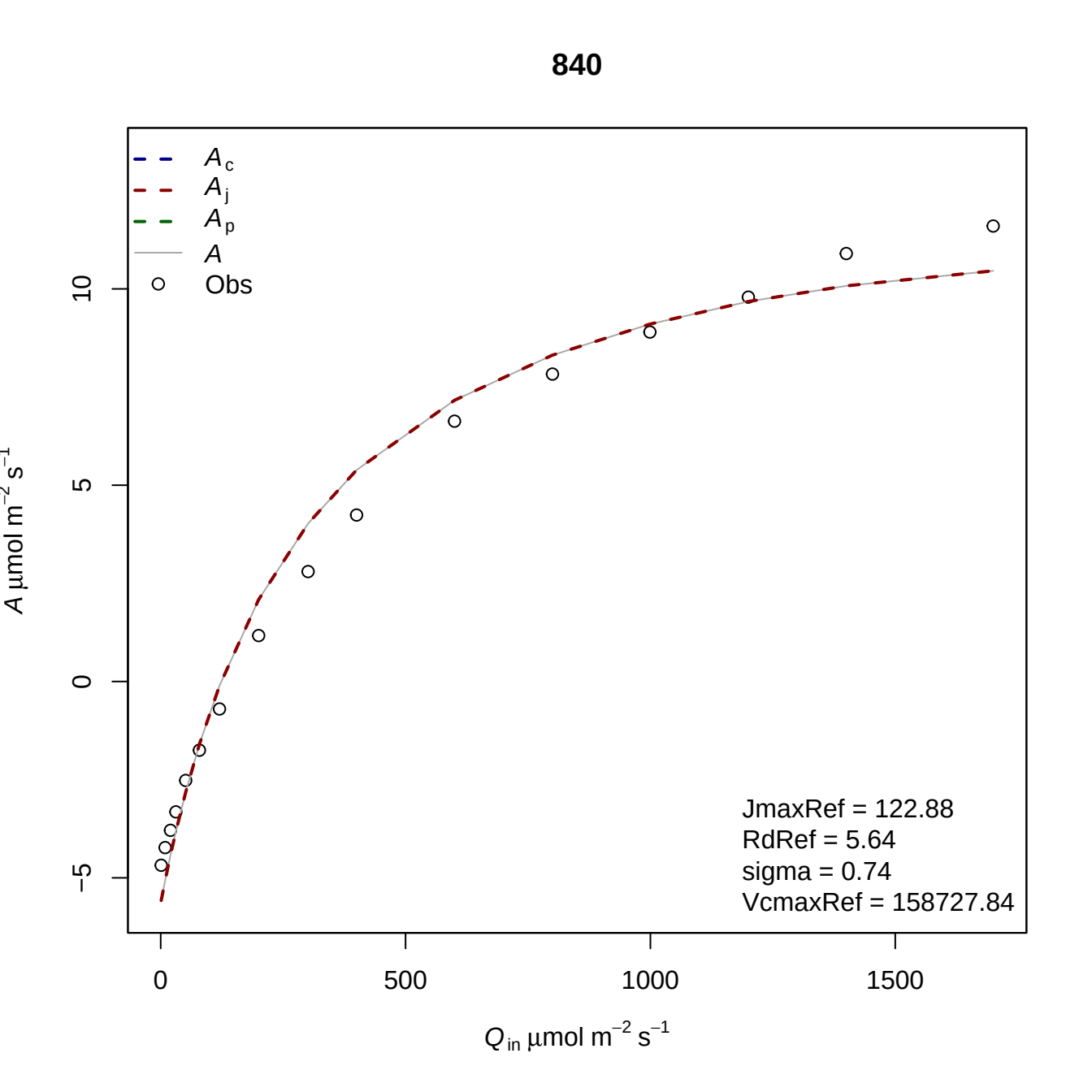

835
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.34
RdRef = 2.33
sigma = 0.07
VcmaxRef = 38008.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

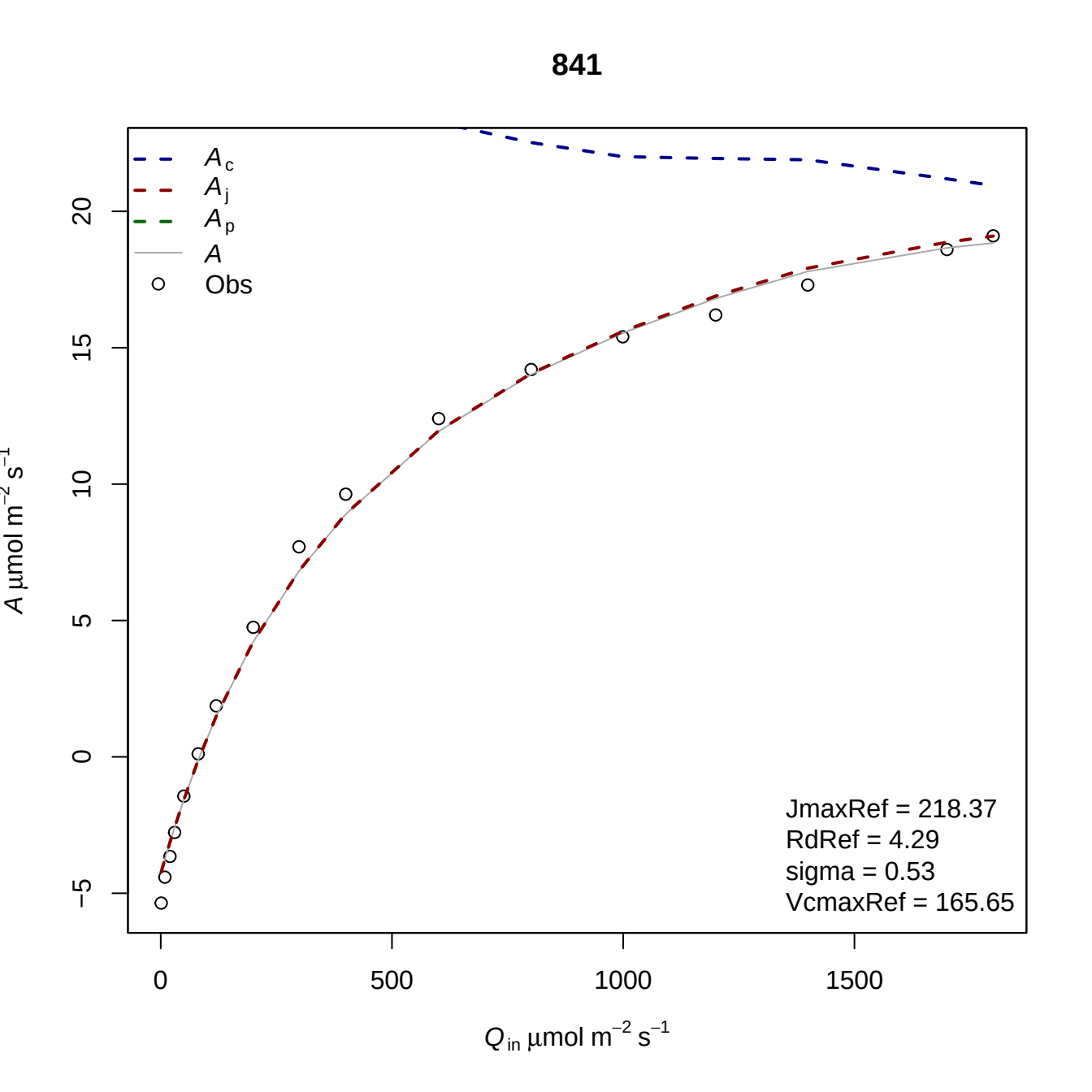

836
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.1
RdRef = 1.21
sigma = 0.11
VcmaxRef = 6586.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

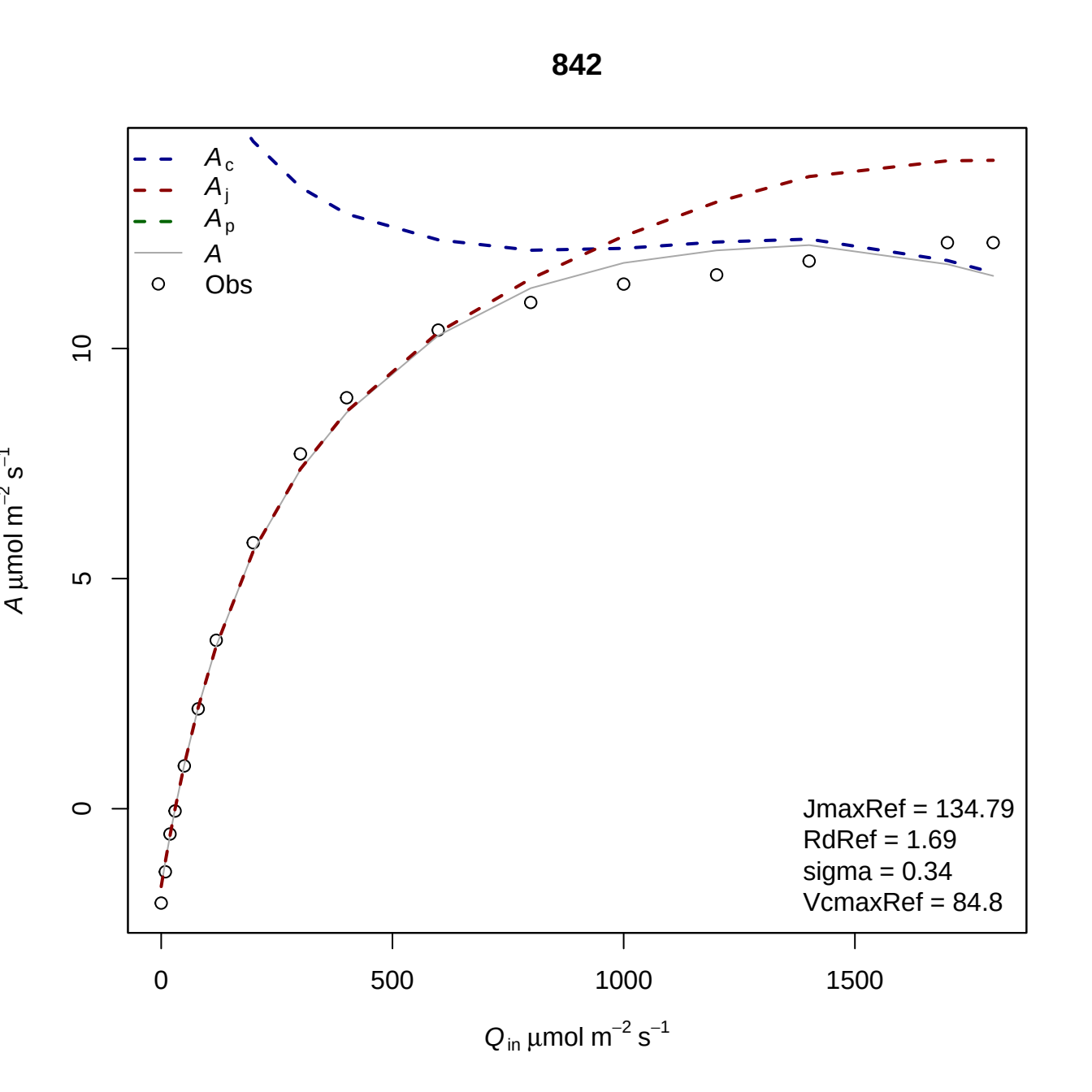

837
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 193.36
RdRef = 2.61
sigma = 0.24
VcmaxRef = 124.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

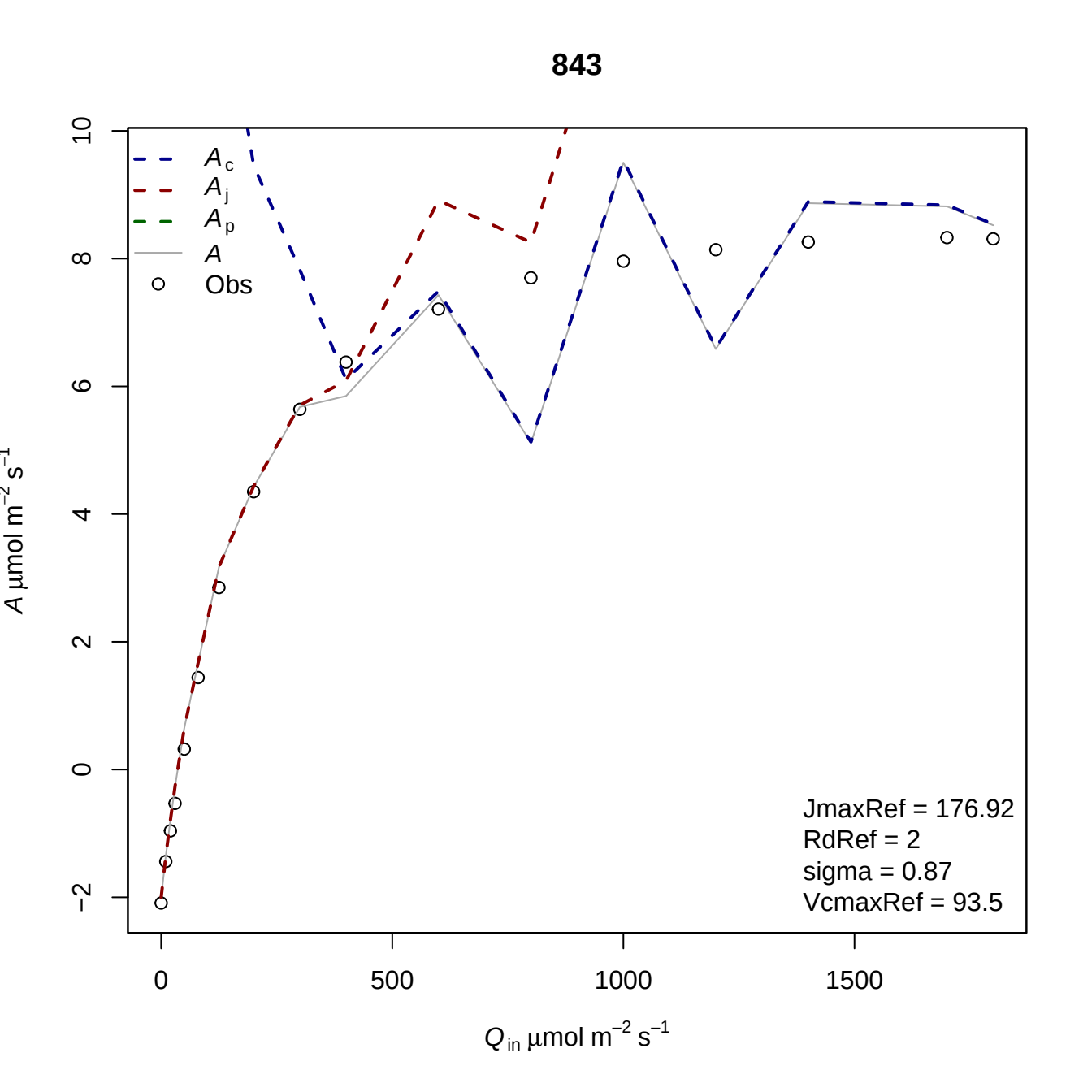

838
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 159.47
RdRef = 2.41
sigma = 0.23
VcmaxRef = 140.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

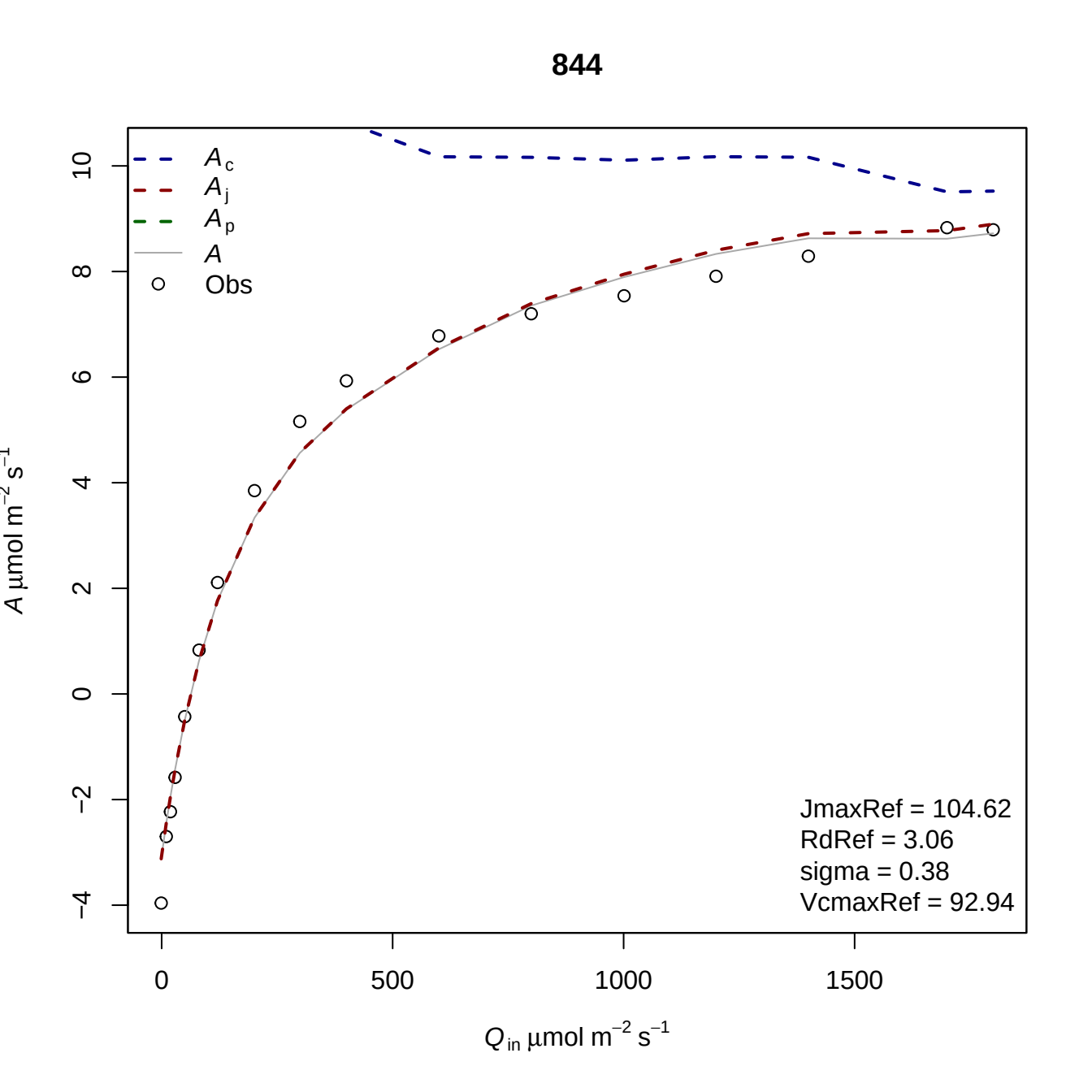

839
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.96
RdRef = 1.53
sigma = 0.38
VcmaxRef = 44021
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
200
400
600
800
1000
1200
1400

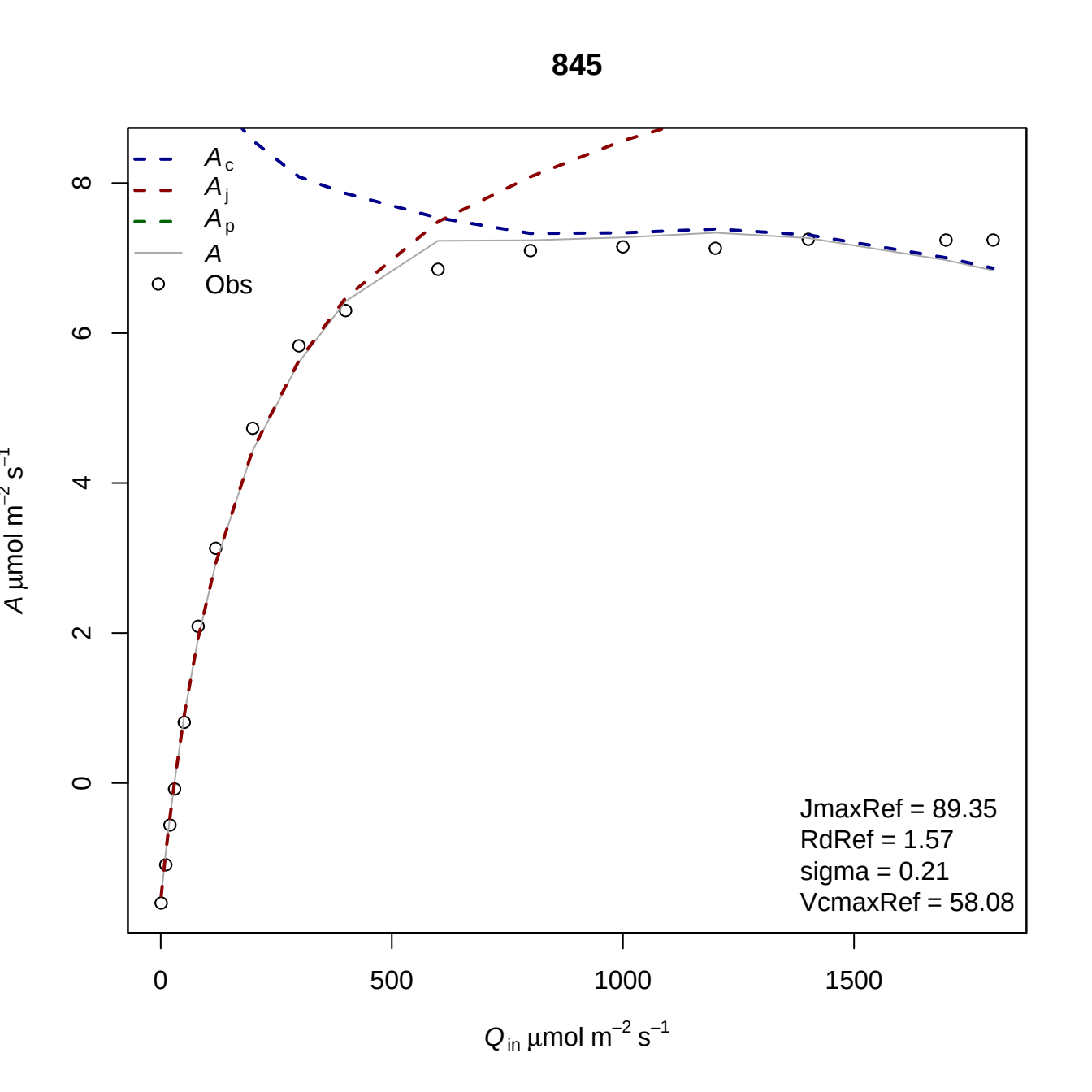

840
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.88
RdRef = 5.64
sigma = 0.74
VcmaxRef = 158727.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500

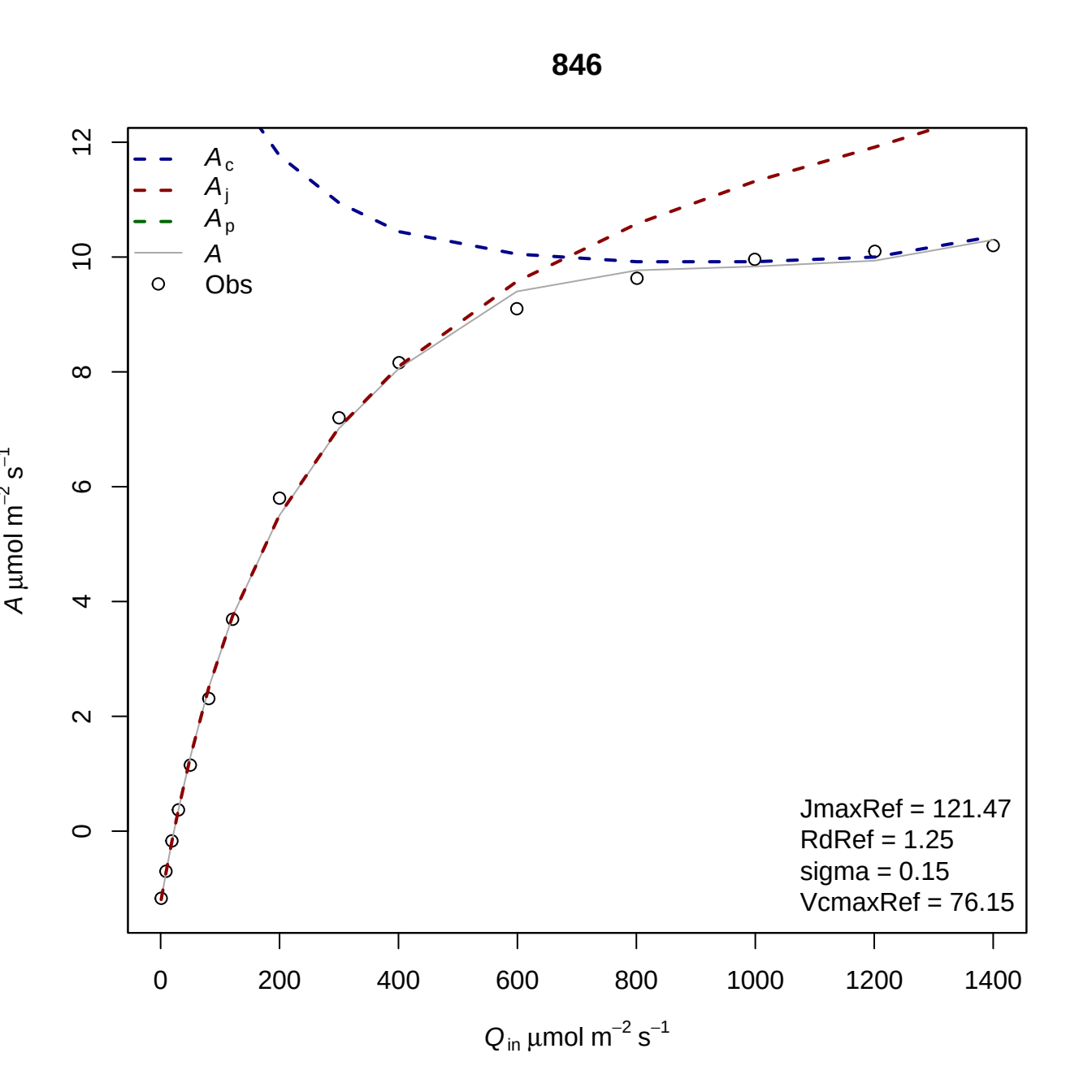

841
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 218.37
RdRef = 4.29
sigma = 0.53
VcmaxRef = 165.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500

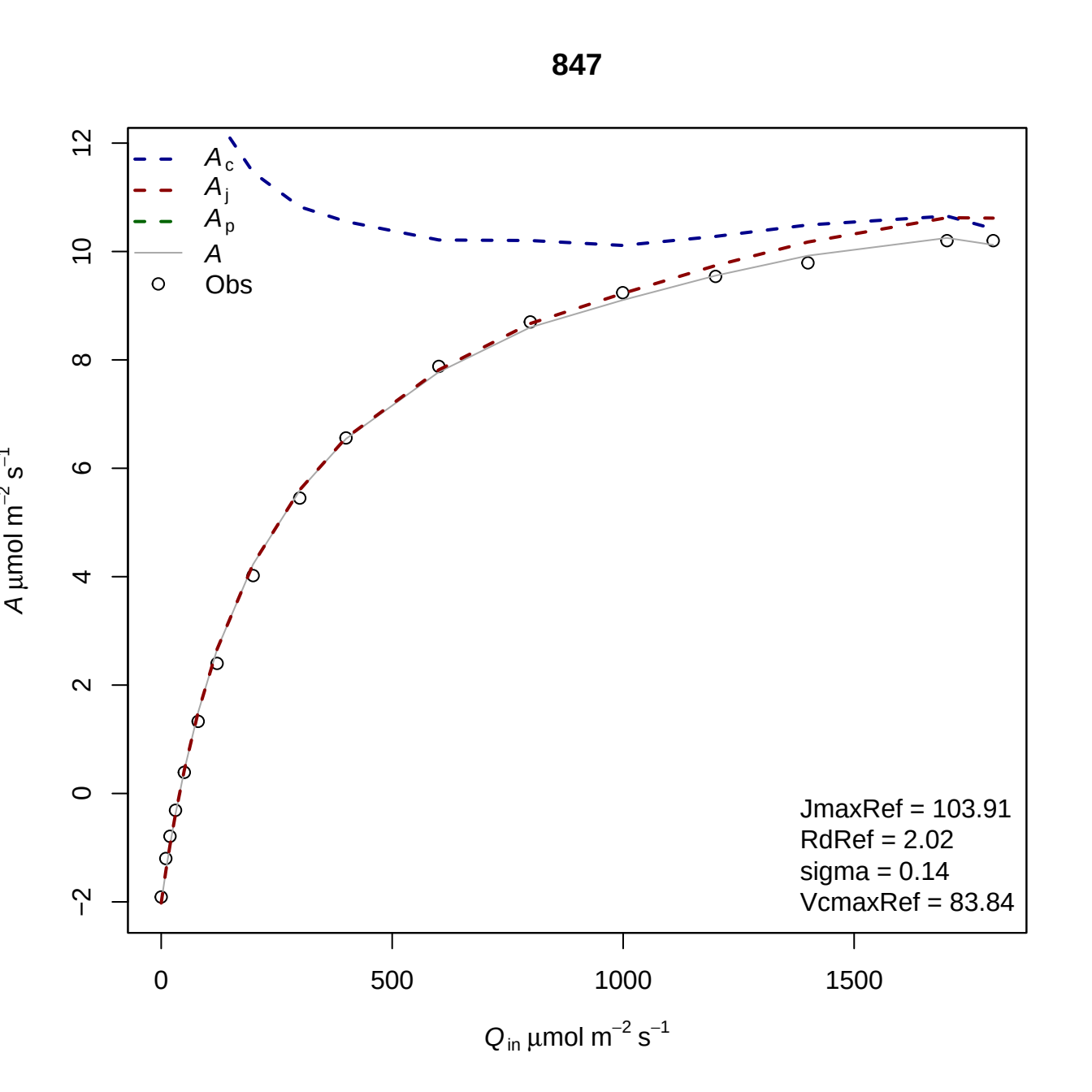

842
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.79
RdRef = 1.69
sigma = 0.34
VcmaxRef = 84.8
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

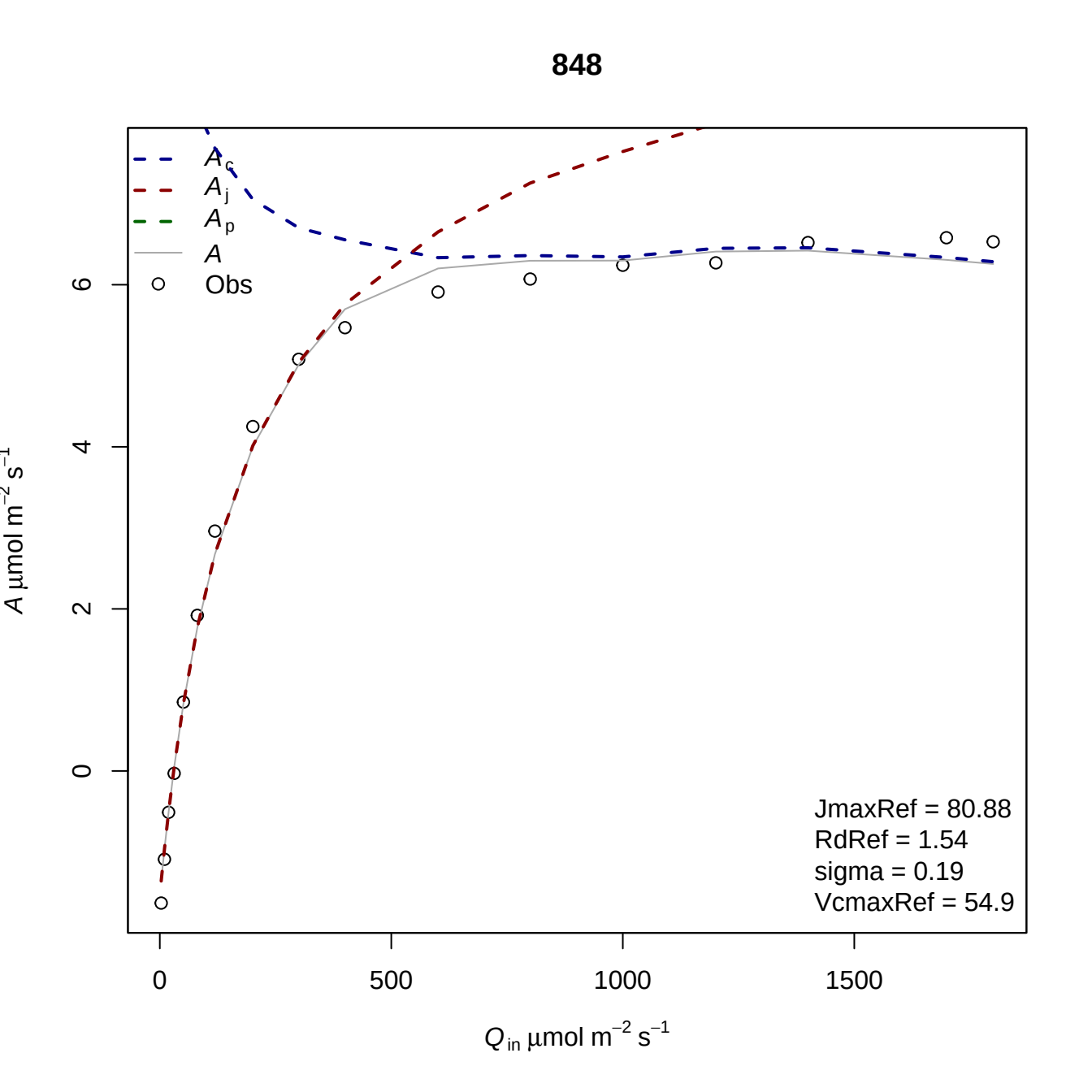

843
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 176.92
RdRef = 2
sigma = 0.87
VcmaxRef = 93.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

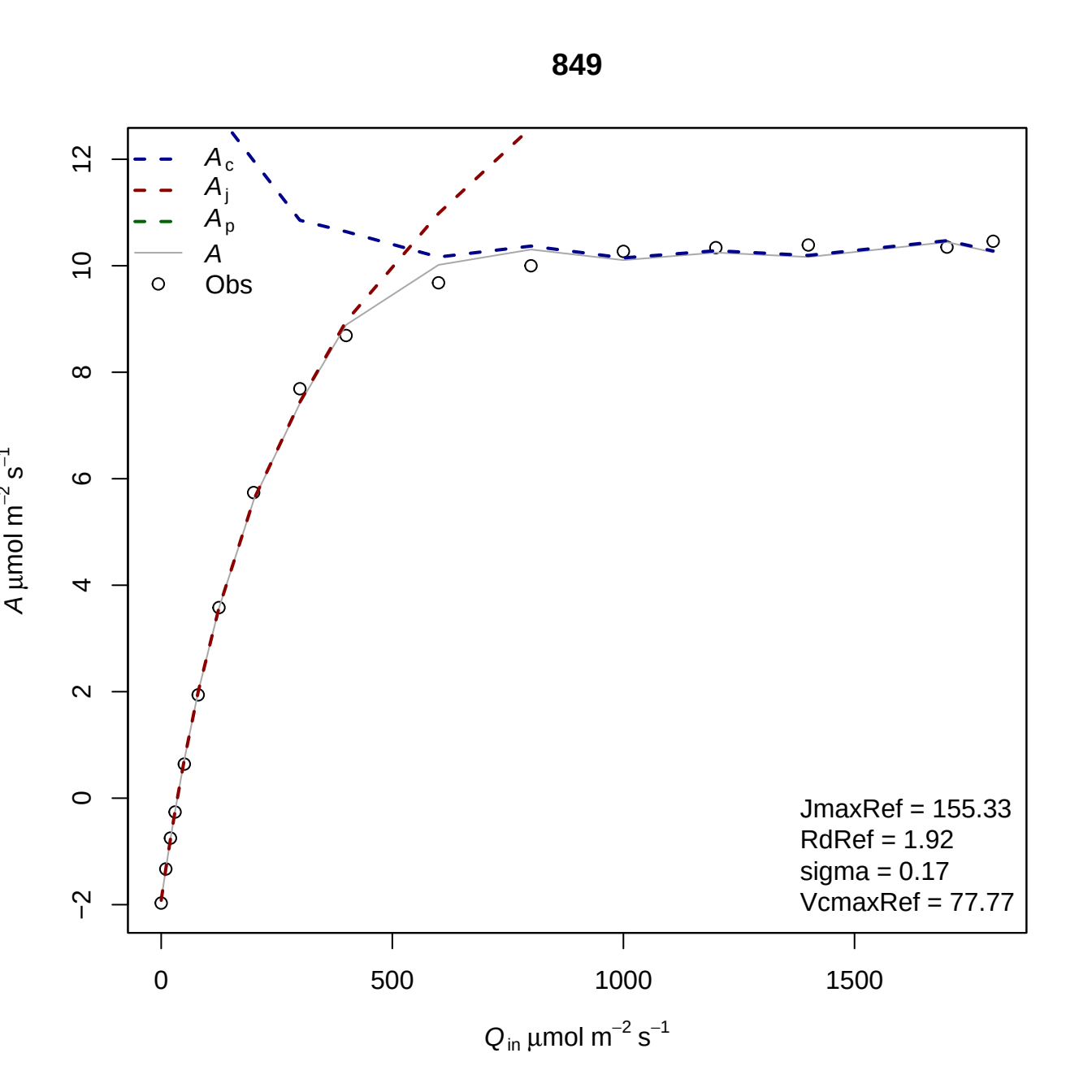

844
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.62
RdRef = 3.06
sigma = 0.38
VcmaxRef = 92.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500

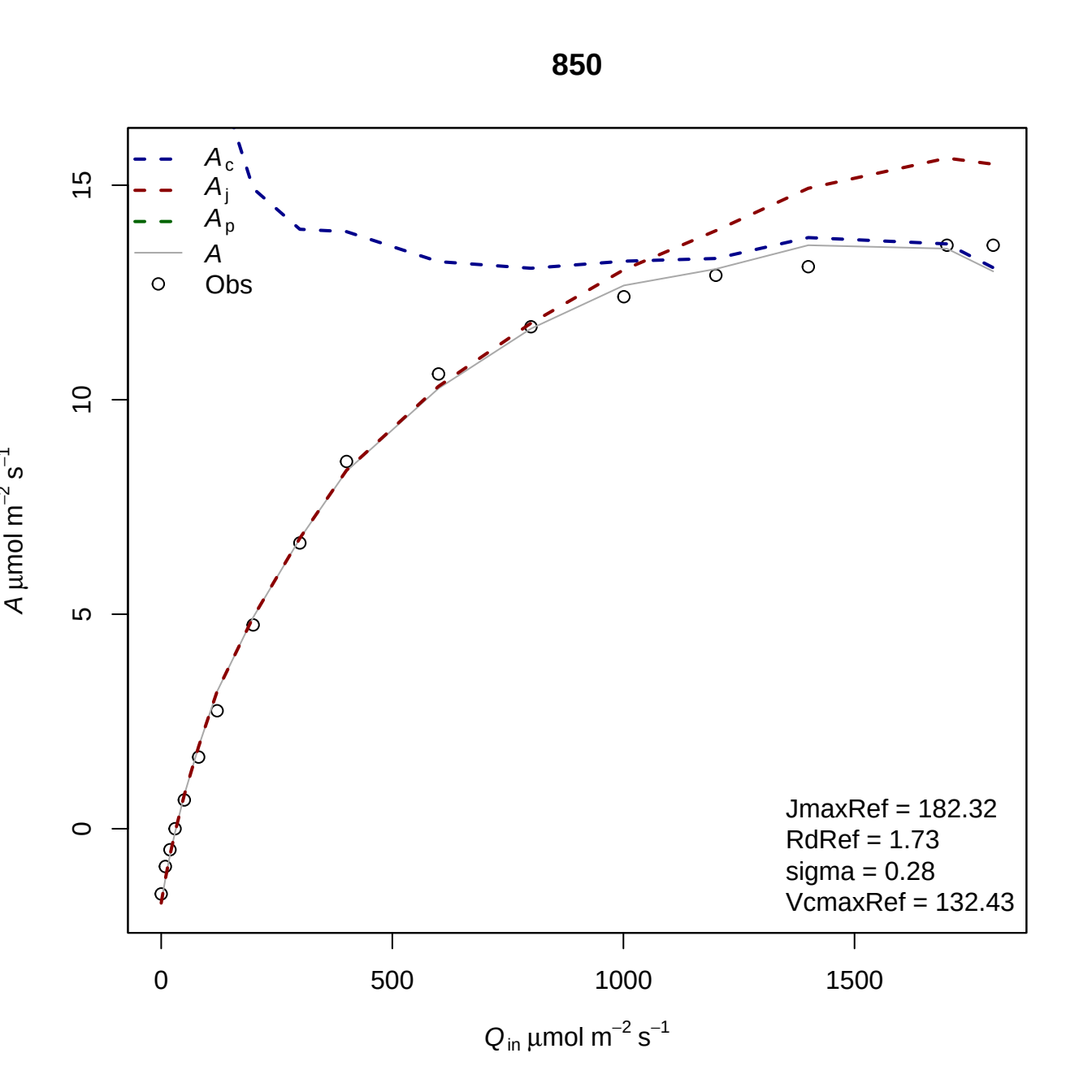

845
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.35
RdRef = 1.57
sigma = 0.21
VcmaxRef = 58.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

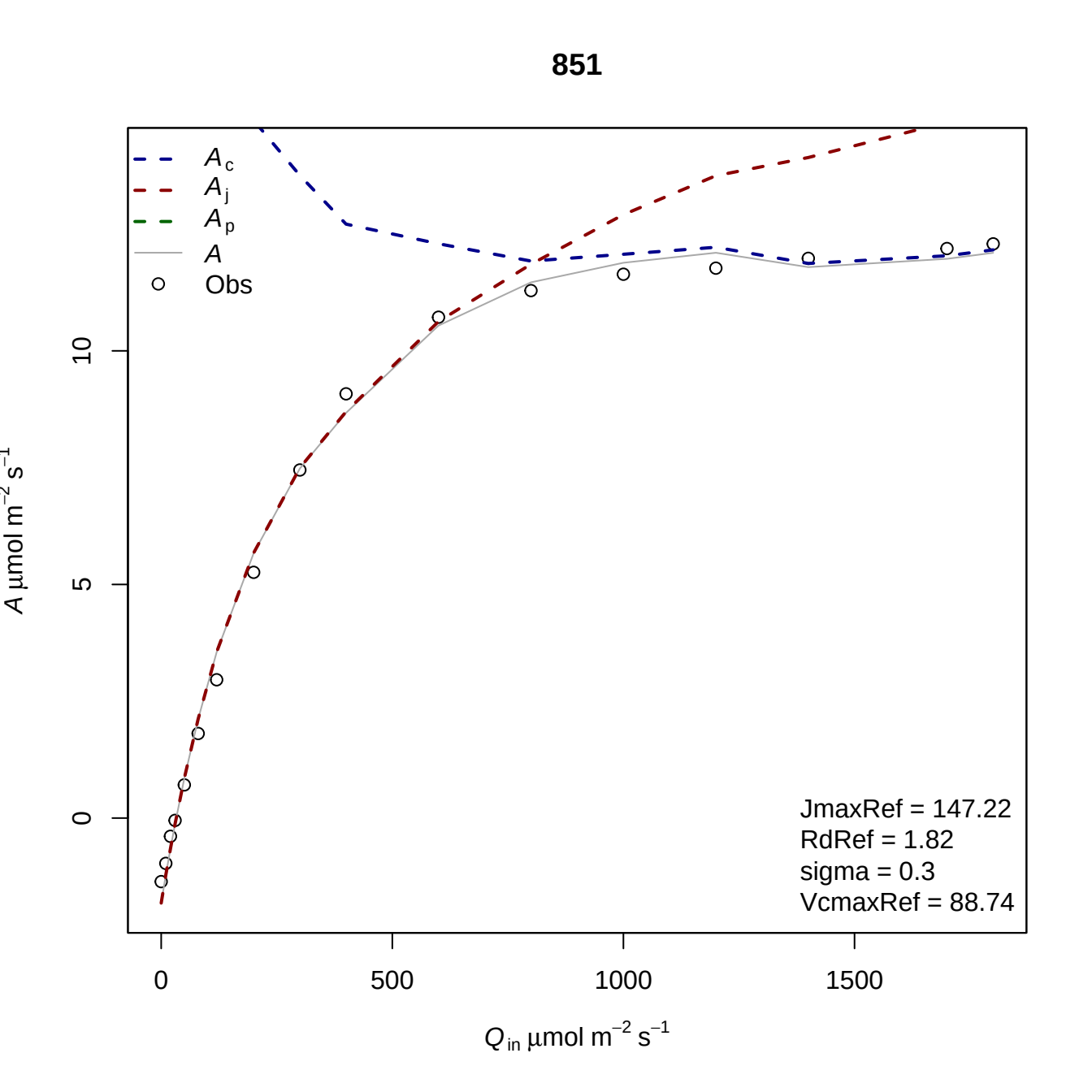

846
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.47
RdRef = 1.25
sigma = 0.15
VcmaxRef = 76.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

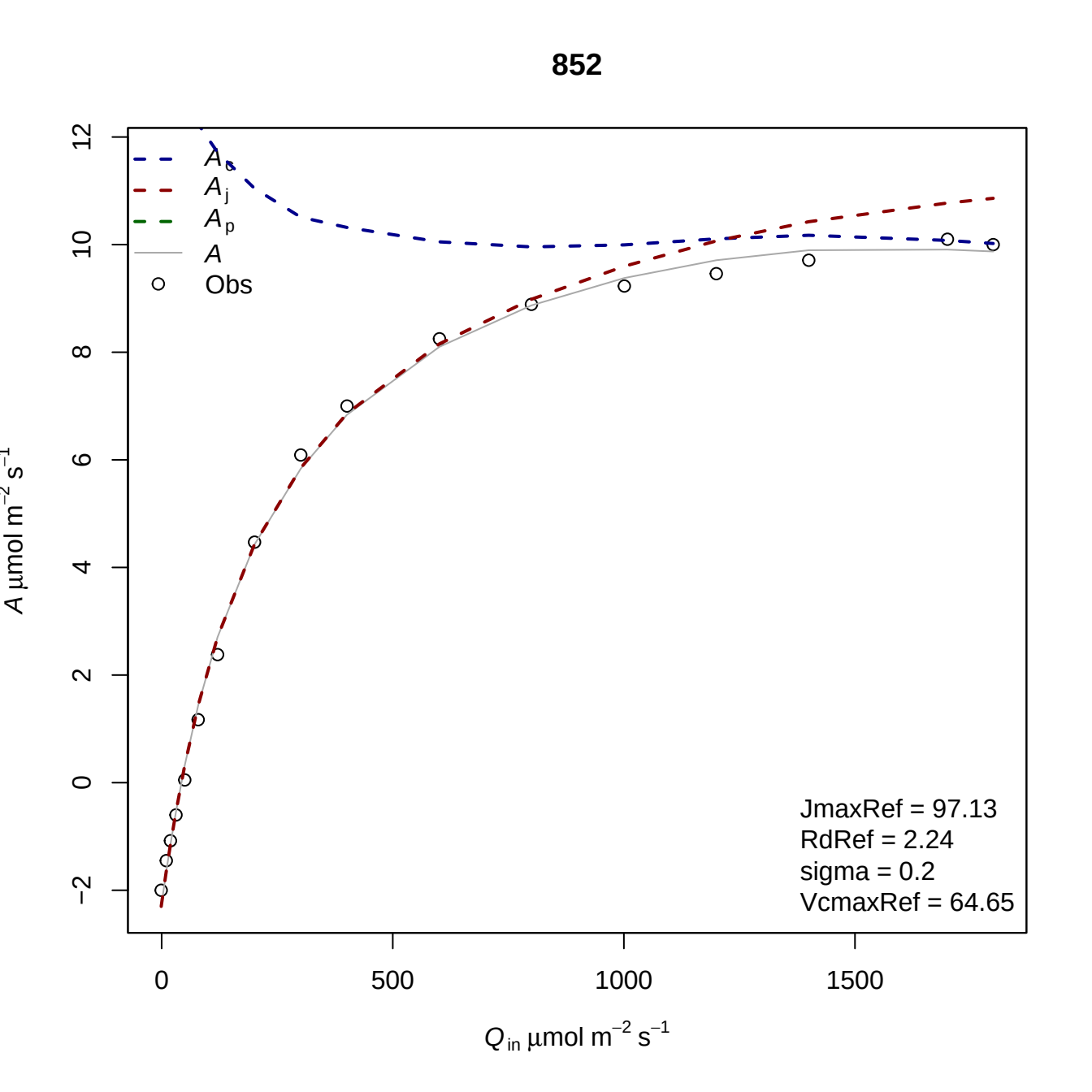

847
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.91
RdRef = 2.02
sigma = 0.14
VcmaxRef = 83.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

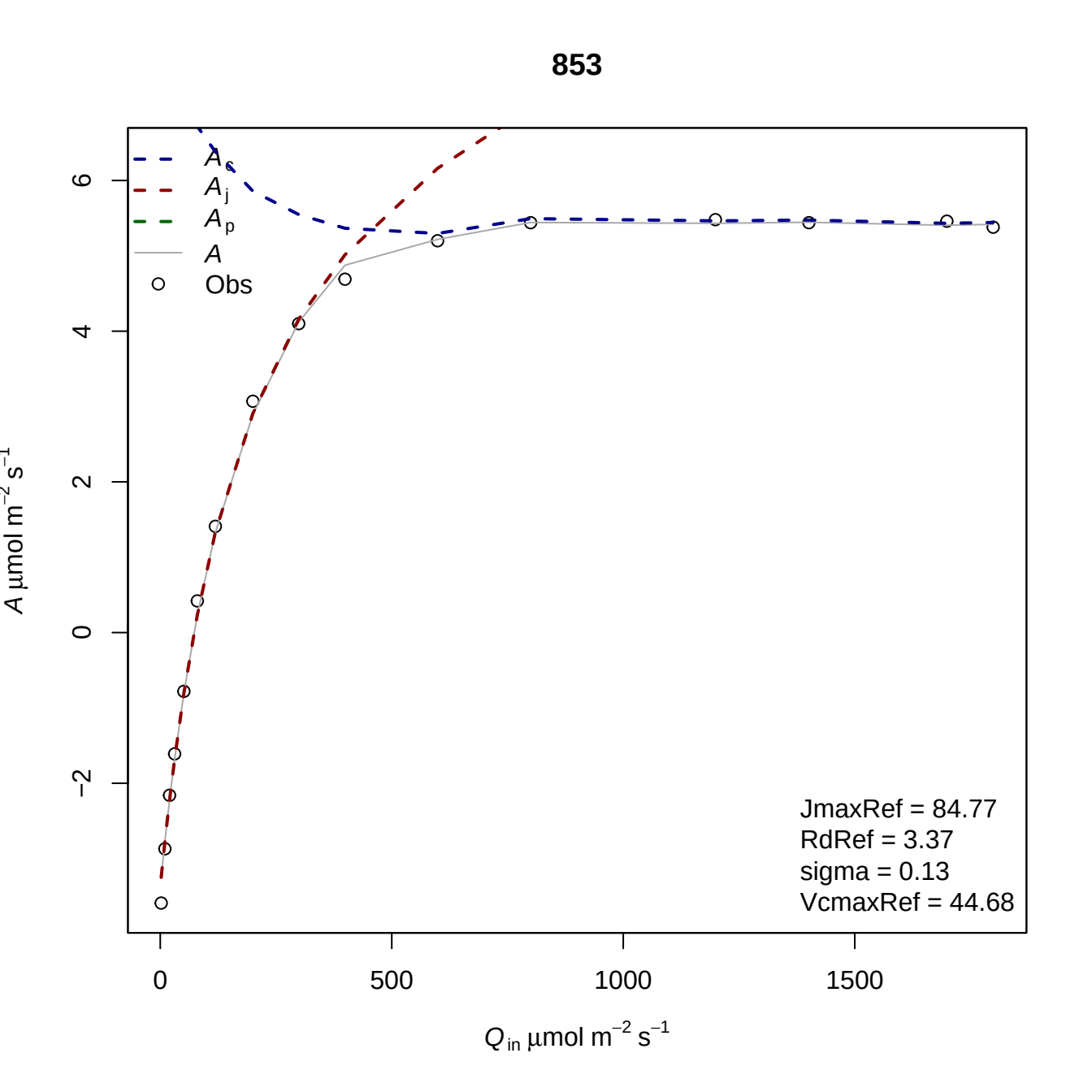

848
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.88
RdRef = 1.54
sigma = 0.19
VcmaxRef = 54.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

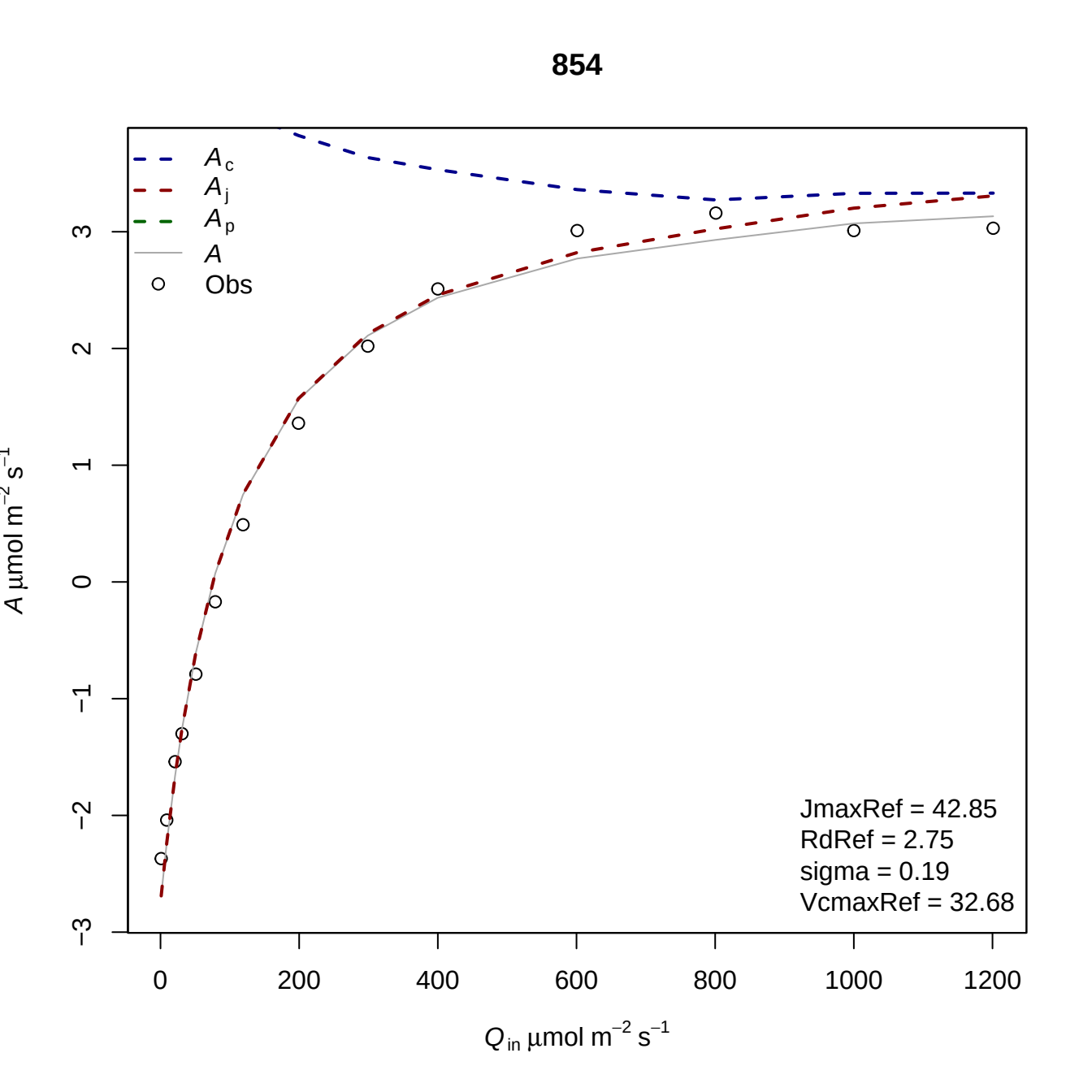

849
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 155.33
RdRef = 1.92
sigma = 0.17
VcmaxRef = 77.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

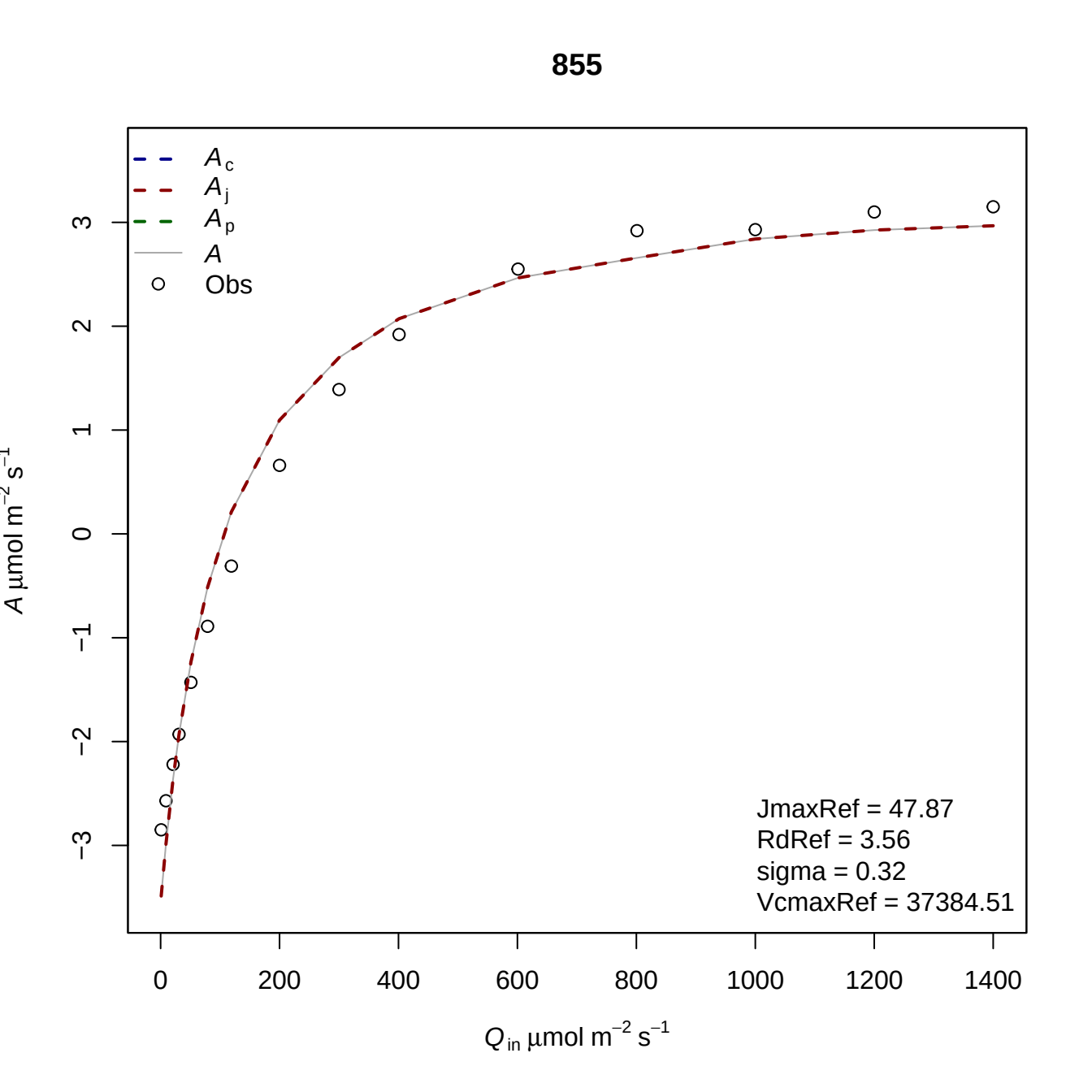

850
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 182.32
RdRef = 1.73
sigma = 0.28
VcmaxRef = 132.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

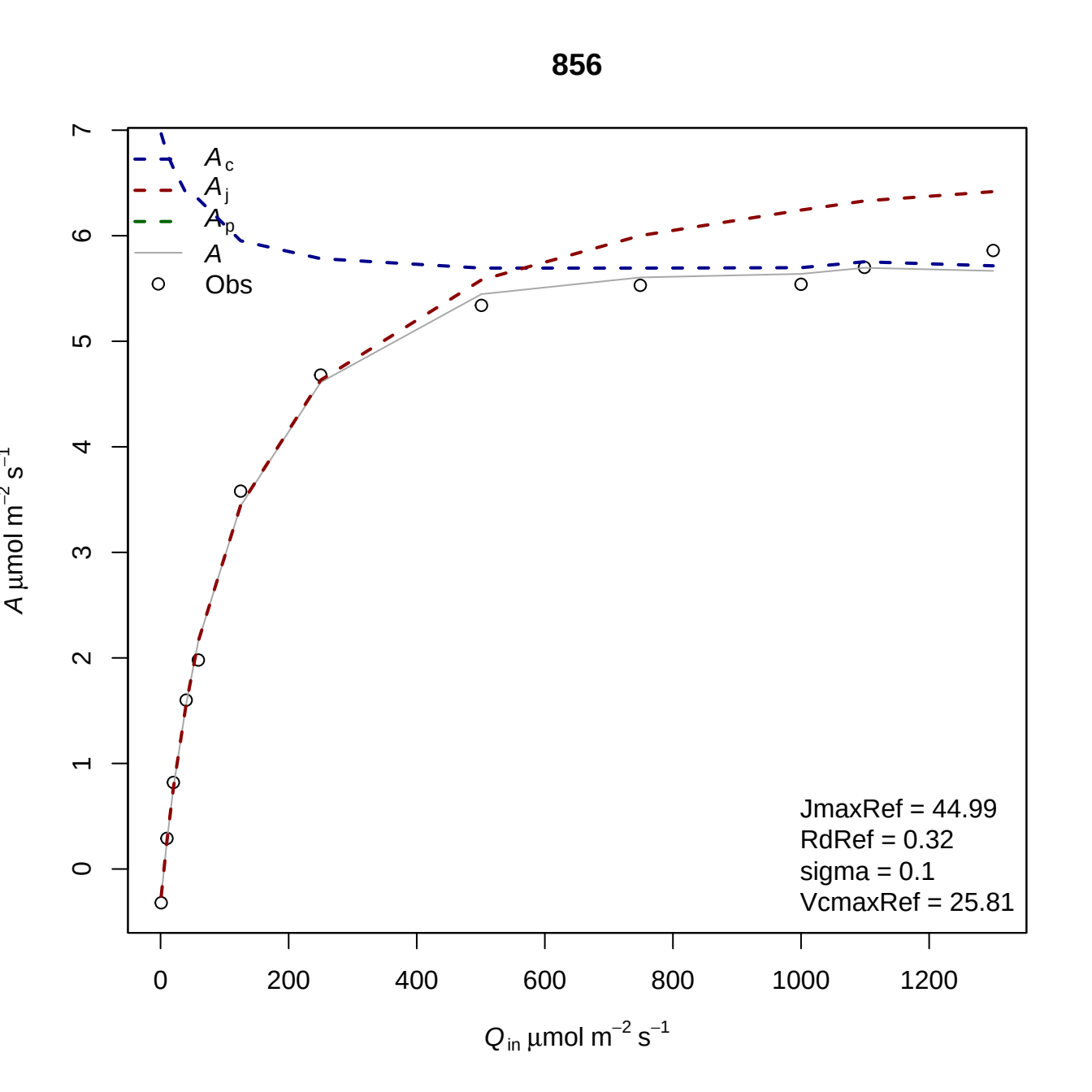

851
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.22
RdRef = 1.82
sigma = 0.3
VcmaxRef = 88.74
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

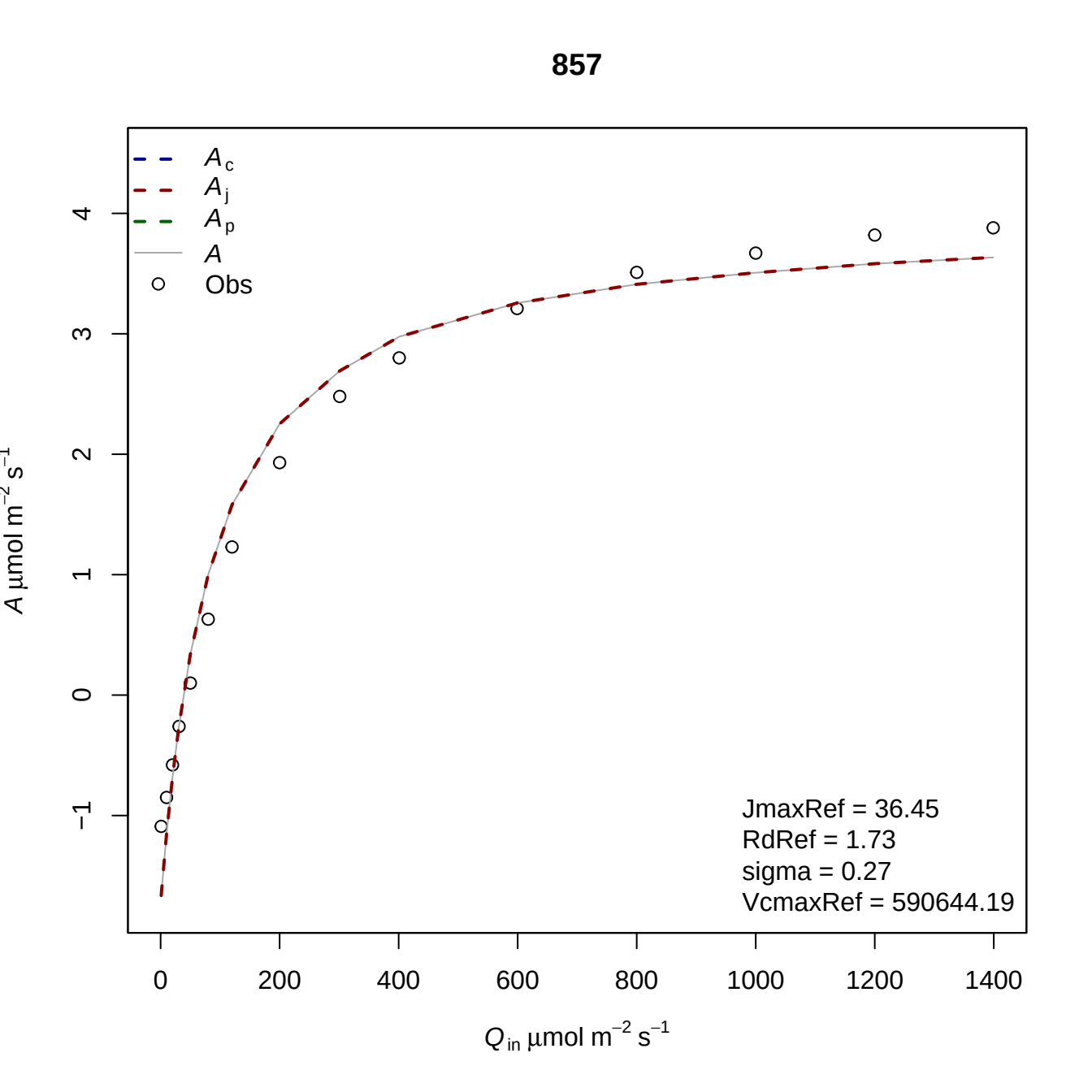

852
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.13
RdRef = 2.24
sigma = 0.2
VcmaxRef = 64.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

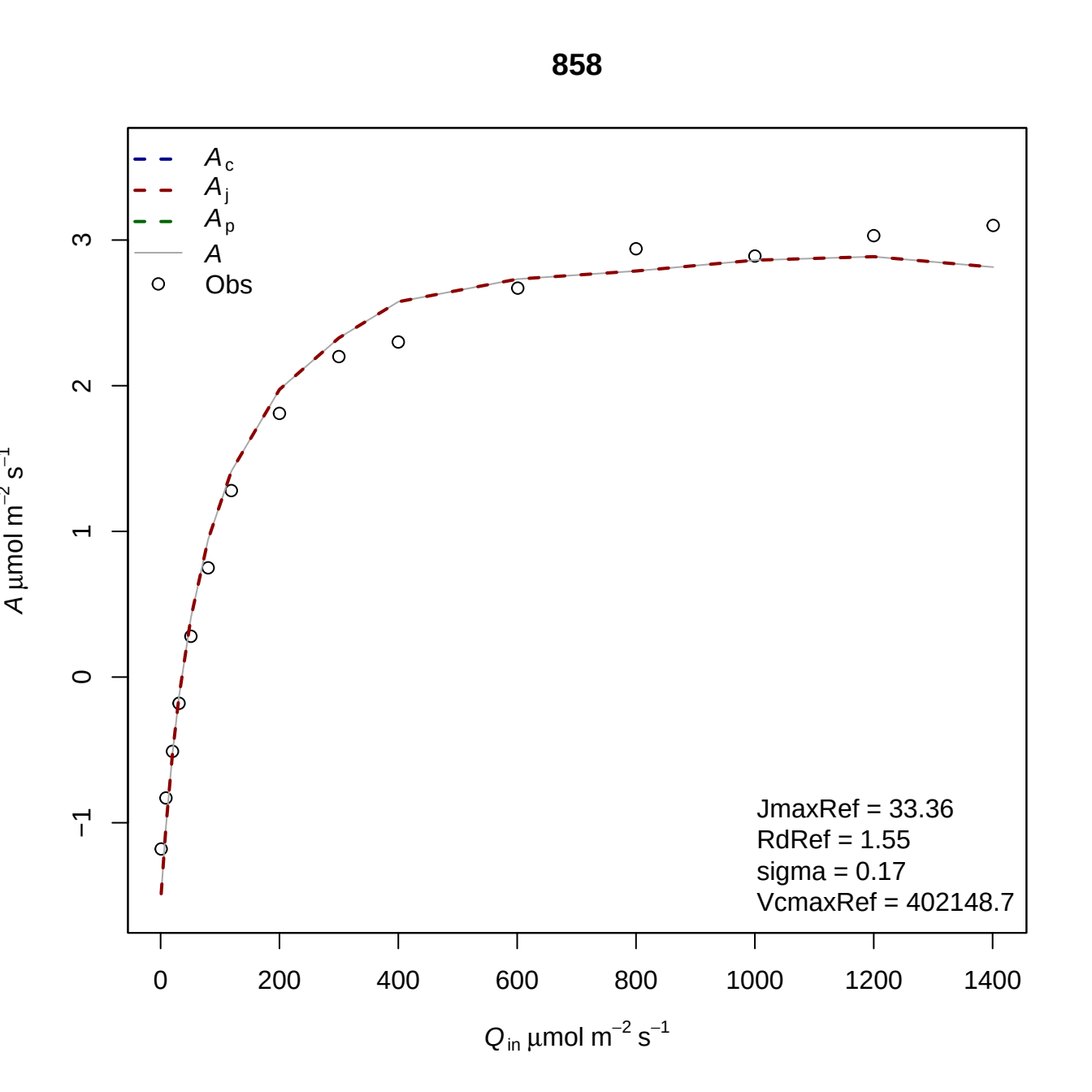

853
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.77
RdRef = 3.37
sigma = 0.13
VcmaxRef = 44.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

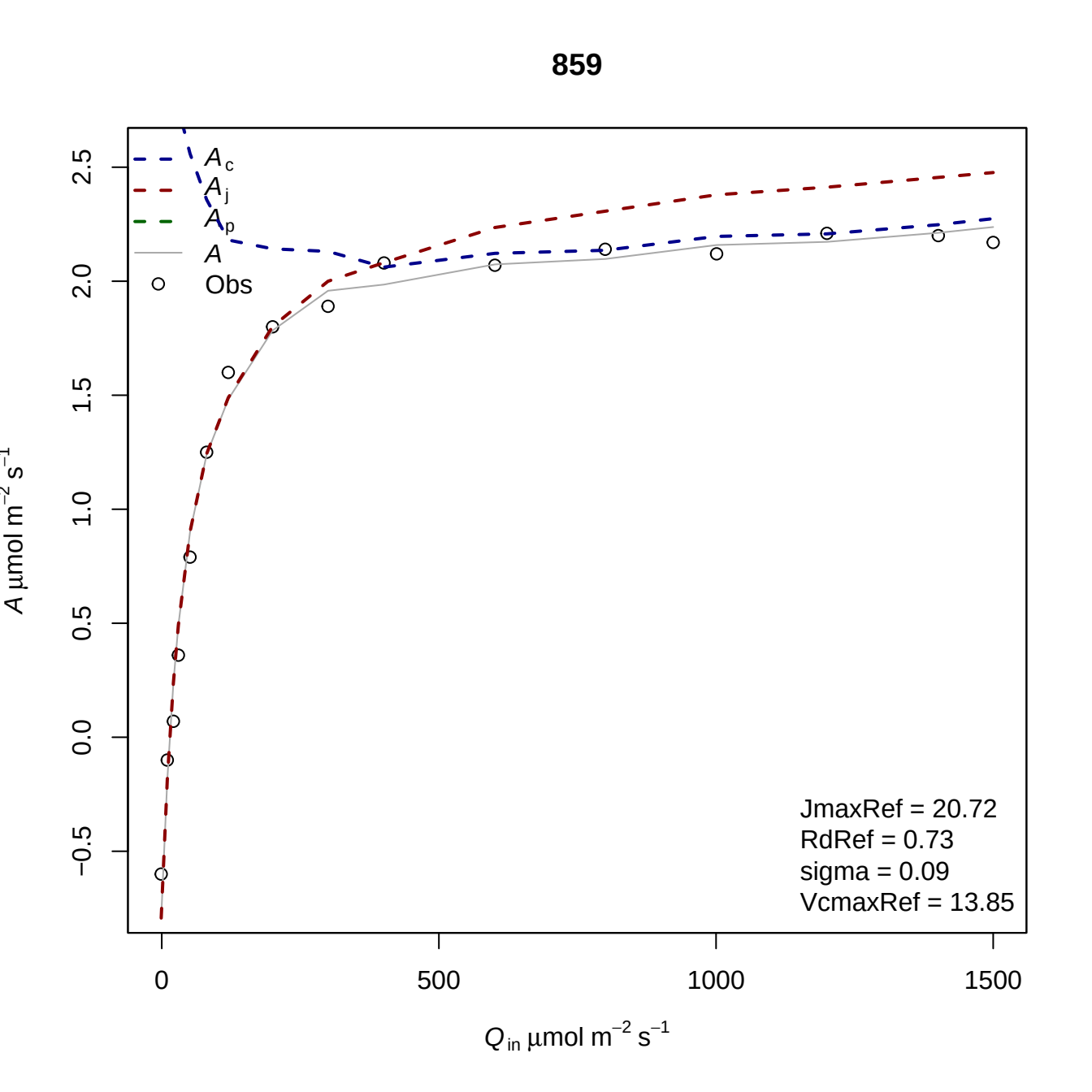

854
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.85
RdRef = 2.75
sigma = 0.19
VcmaxRef = 32.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

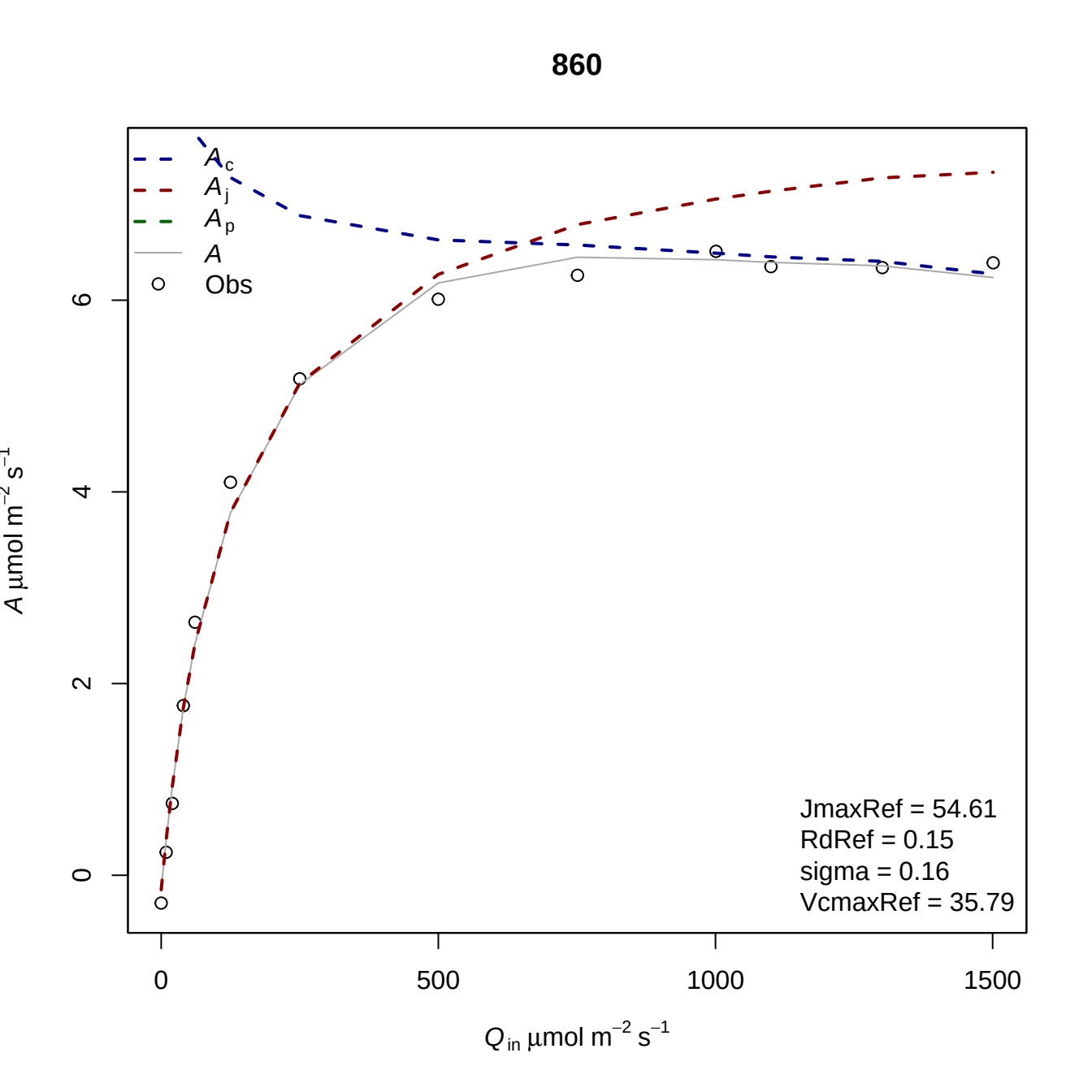

855
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.87
RdRef = 3.56
sigma = 0.32
VcmaxRef = 37384.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

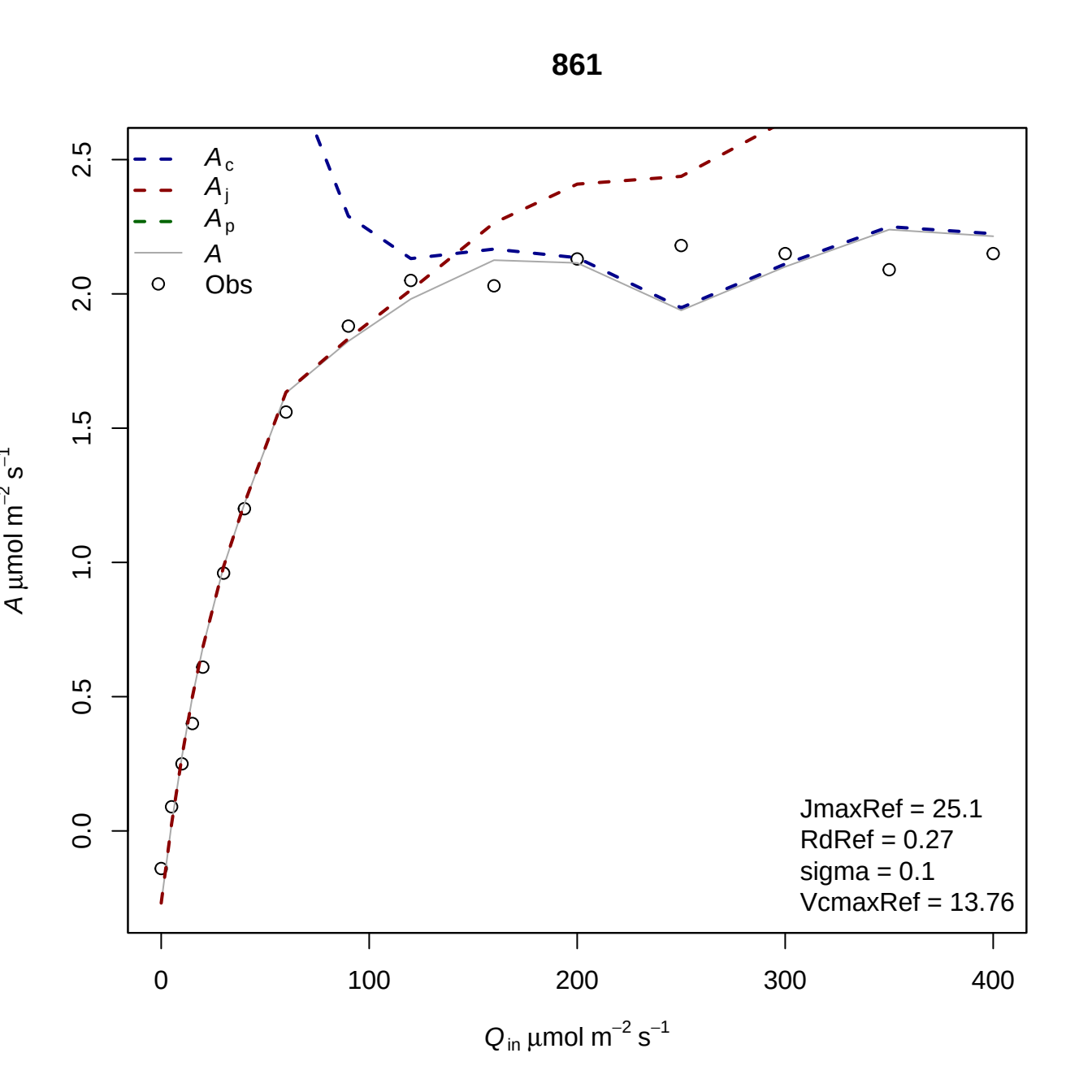

856
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.99
RdRef = 0.32
sigma = 0.1
VcmaxRef = 25.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

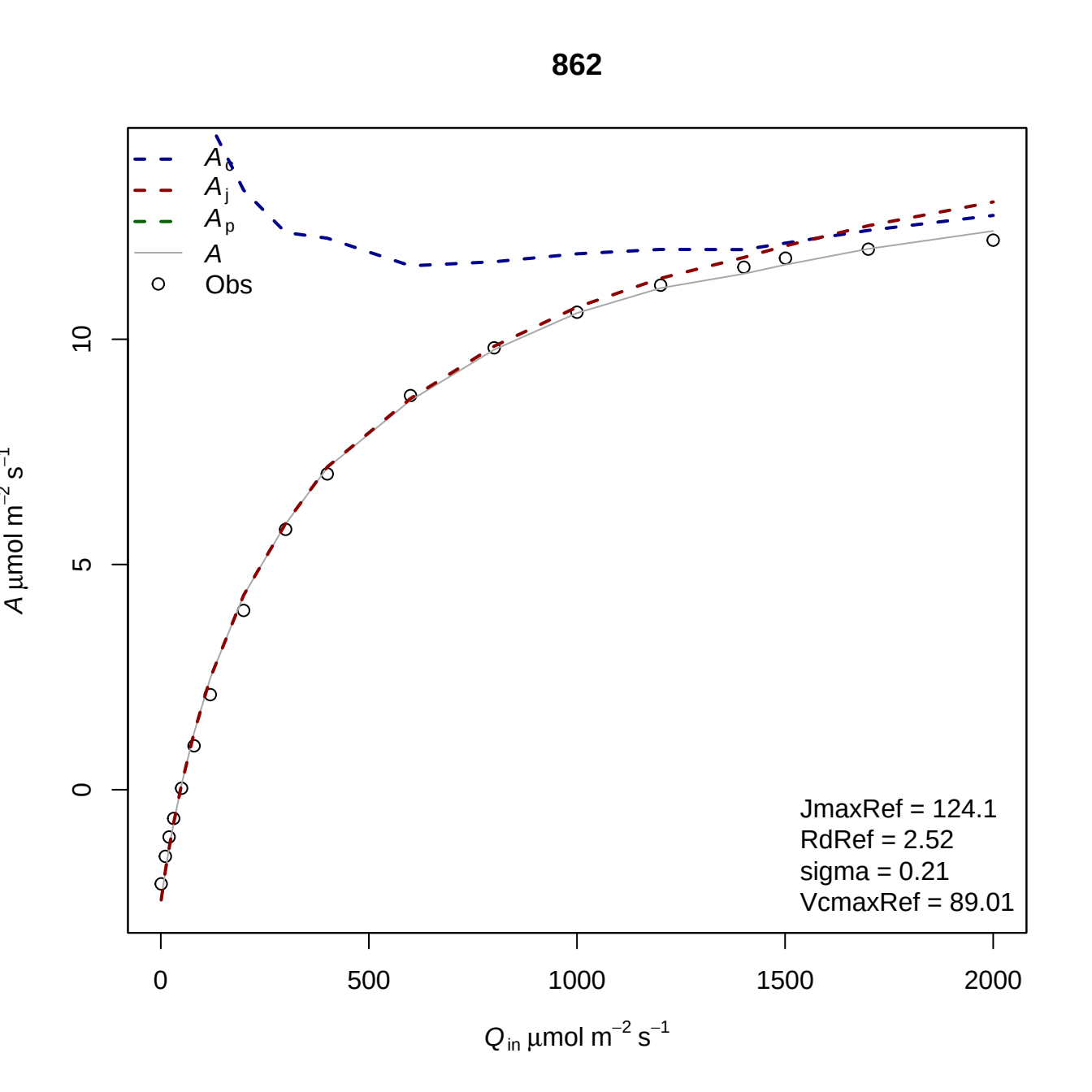

857
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.45
RdRef = 1.73
sigma = 0.27
VcmaxRef = 590644.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

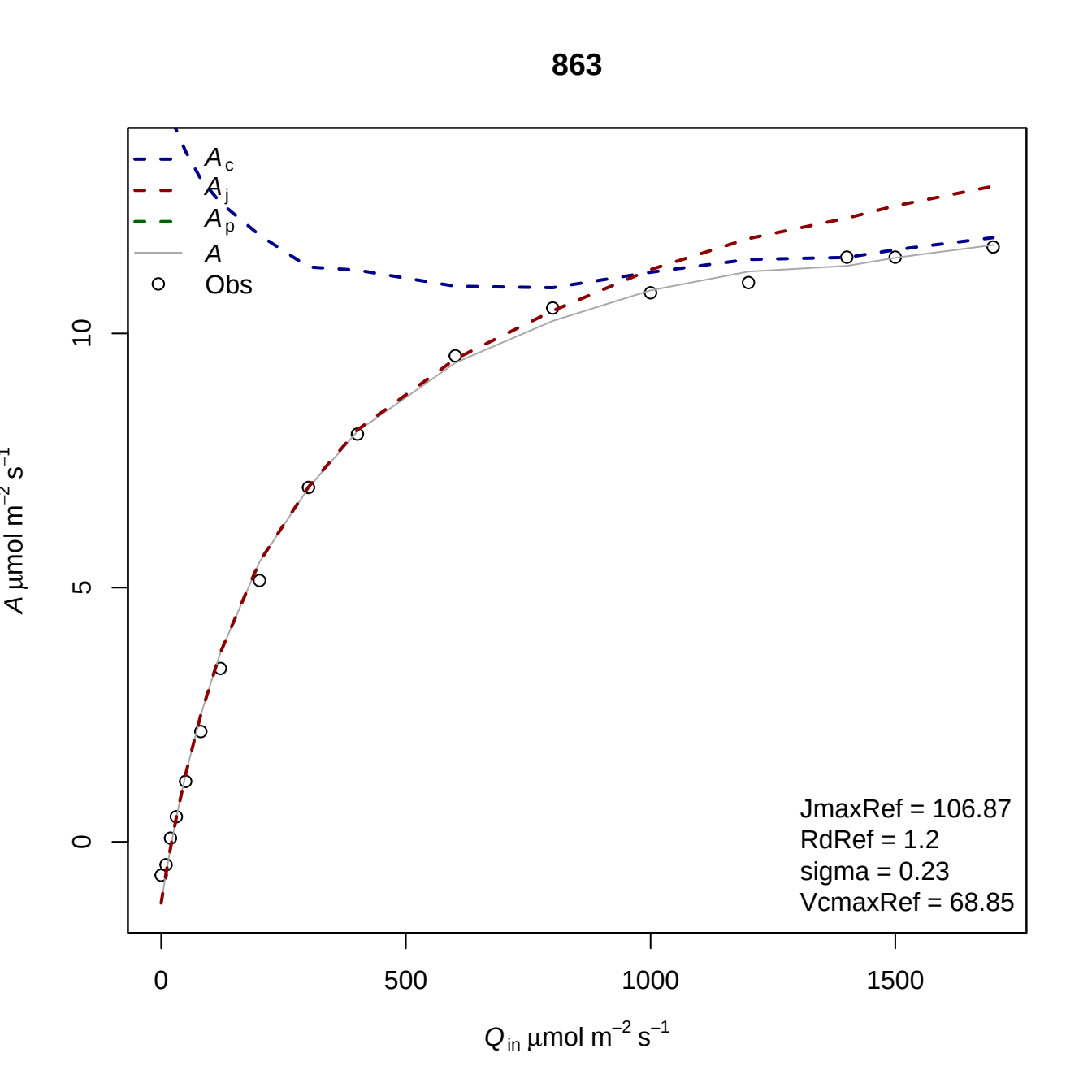

858
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 33.36
RdRef = 1.55
sigma = 0.17
VcmaxRef = 402148.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

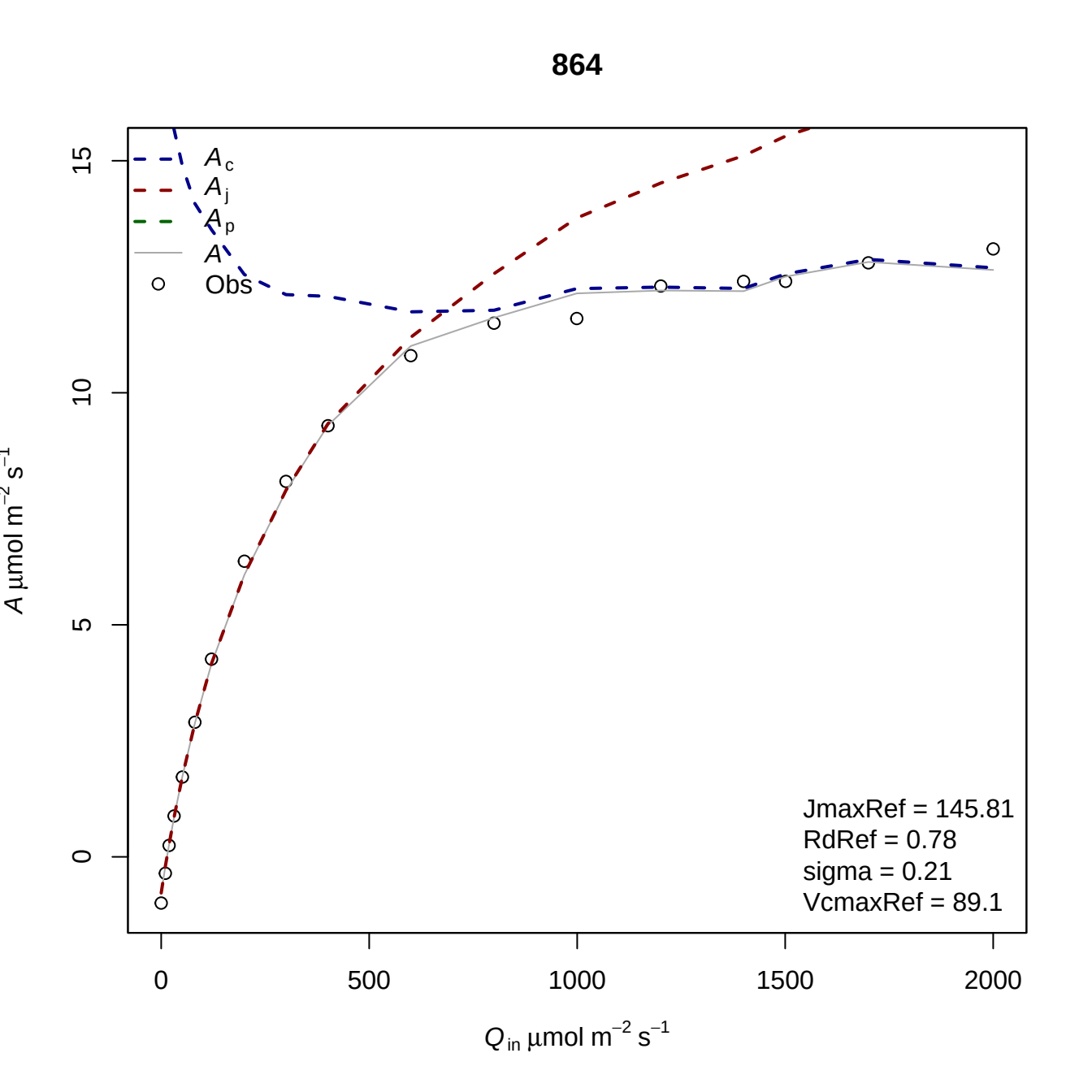

859
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 20.72
RdRef = 0.73
sigma = 0.09
VcmaxRef = 13.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

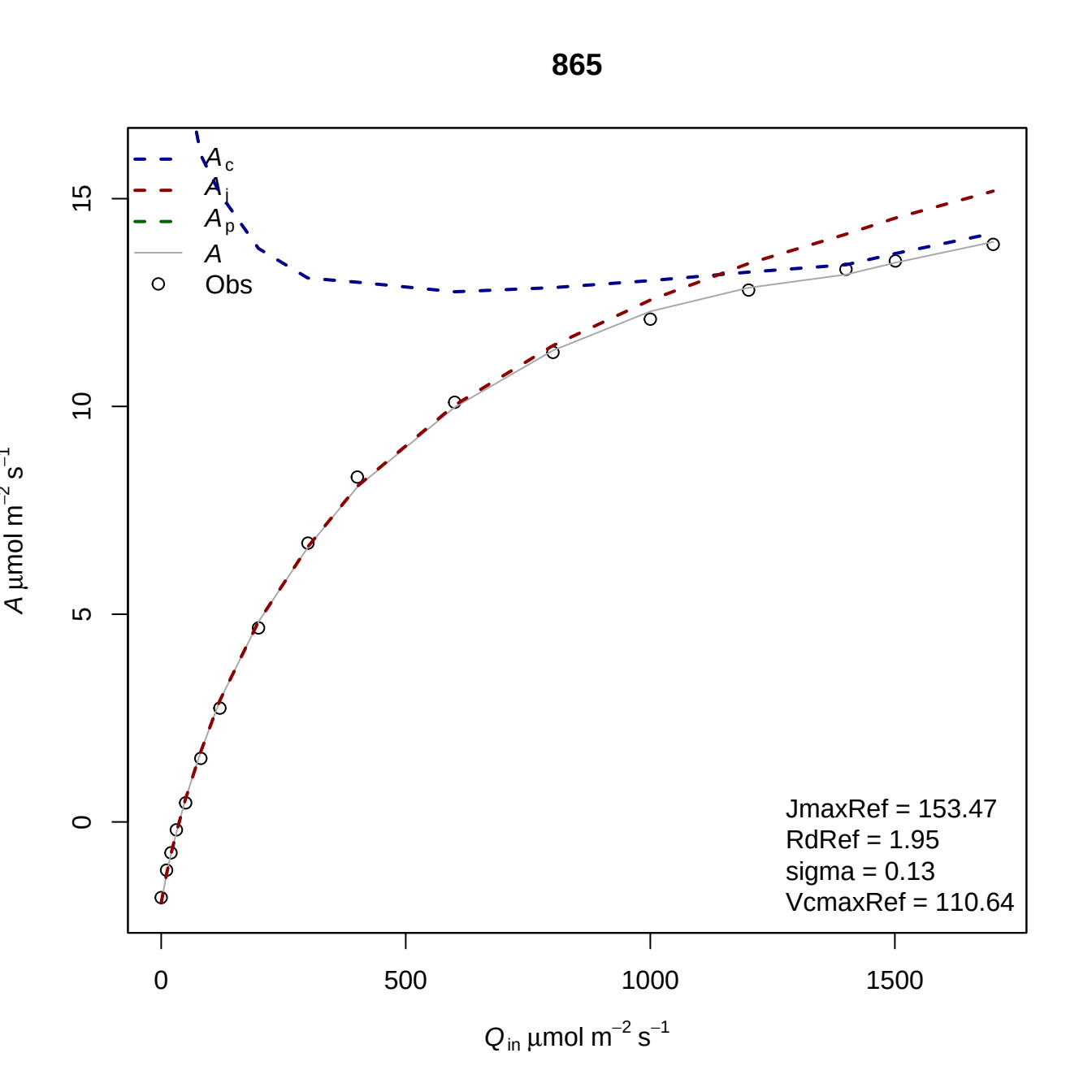

860
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.61
RdRef = 0.15
sigma = 0.16
VcmaxRef = 35.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

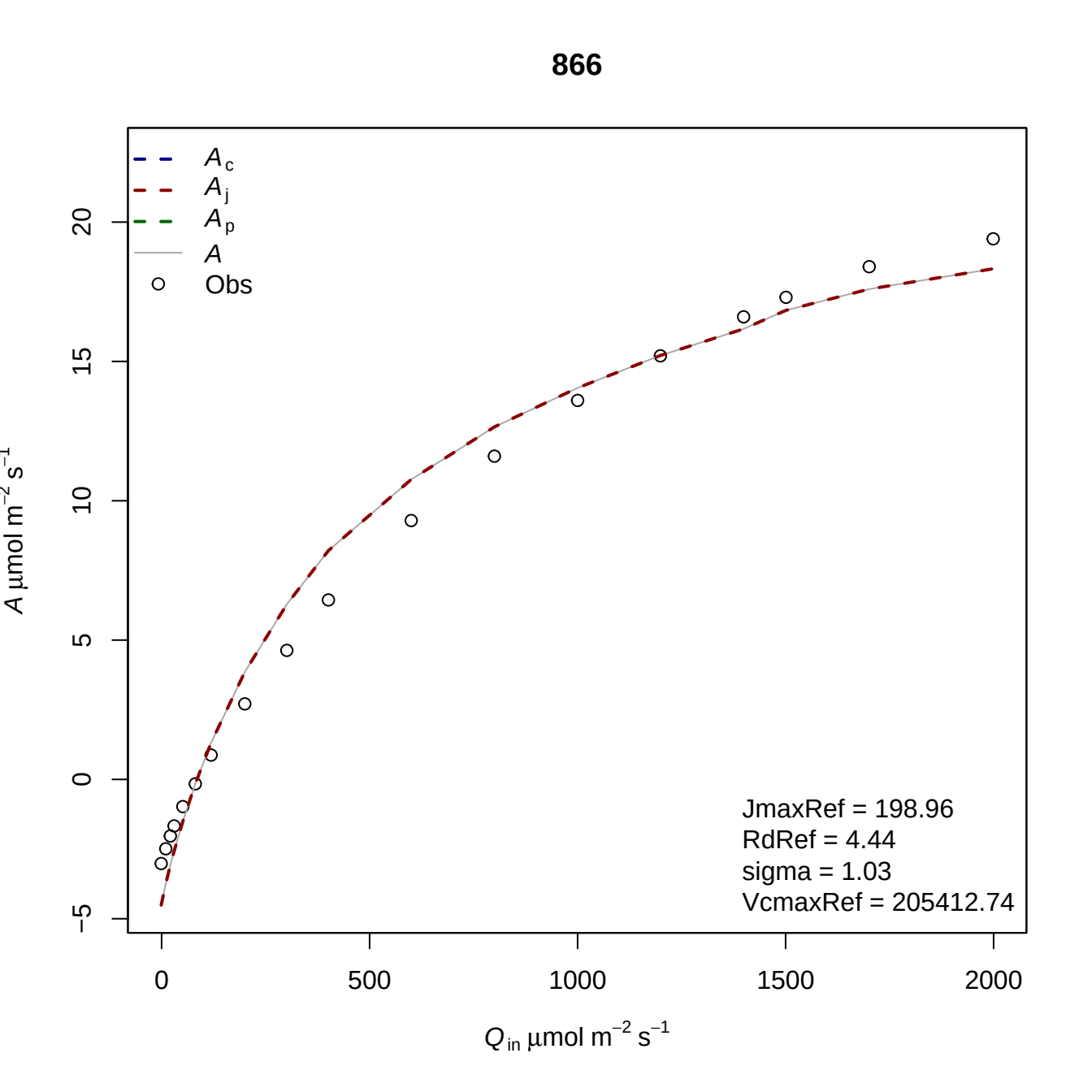

861
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.1
RdRef = 0.27
sigma = 0.1
VcmaxRef = 13.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

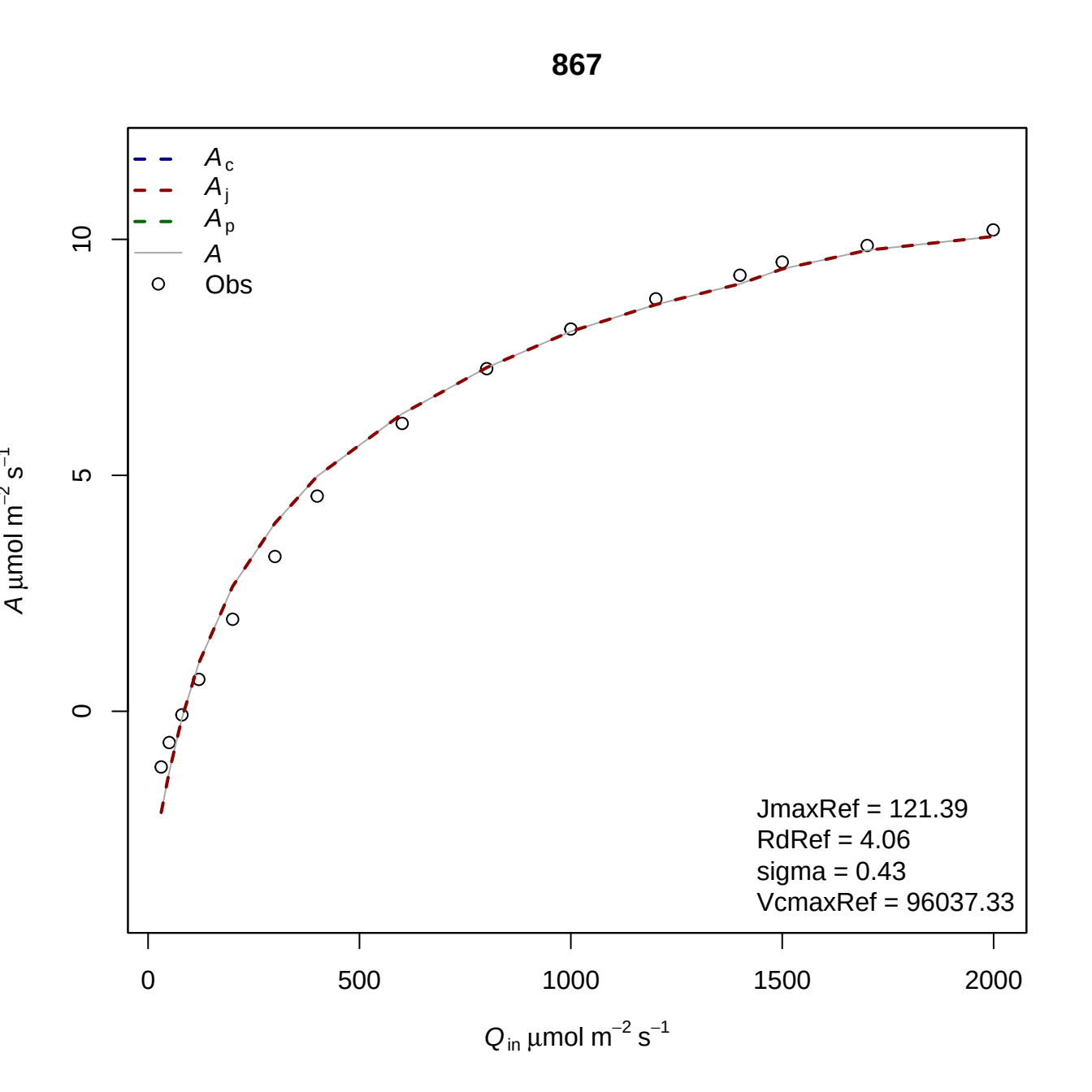

862
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.1
RdRef = 2.52
sigma = 0.21
VcmaxRef = 89.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

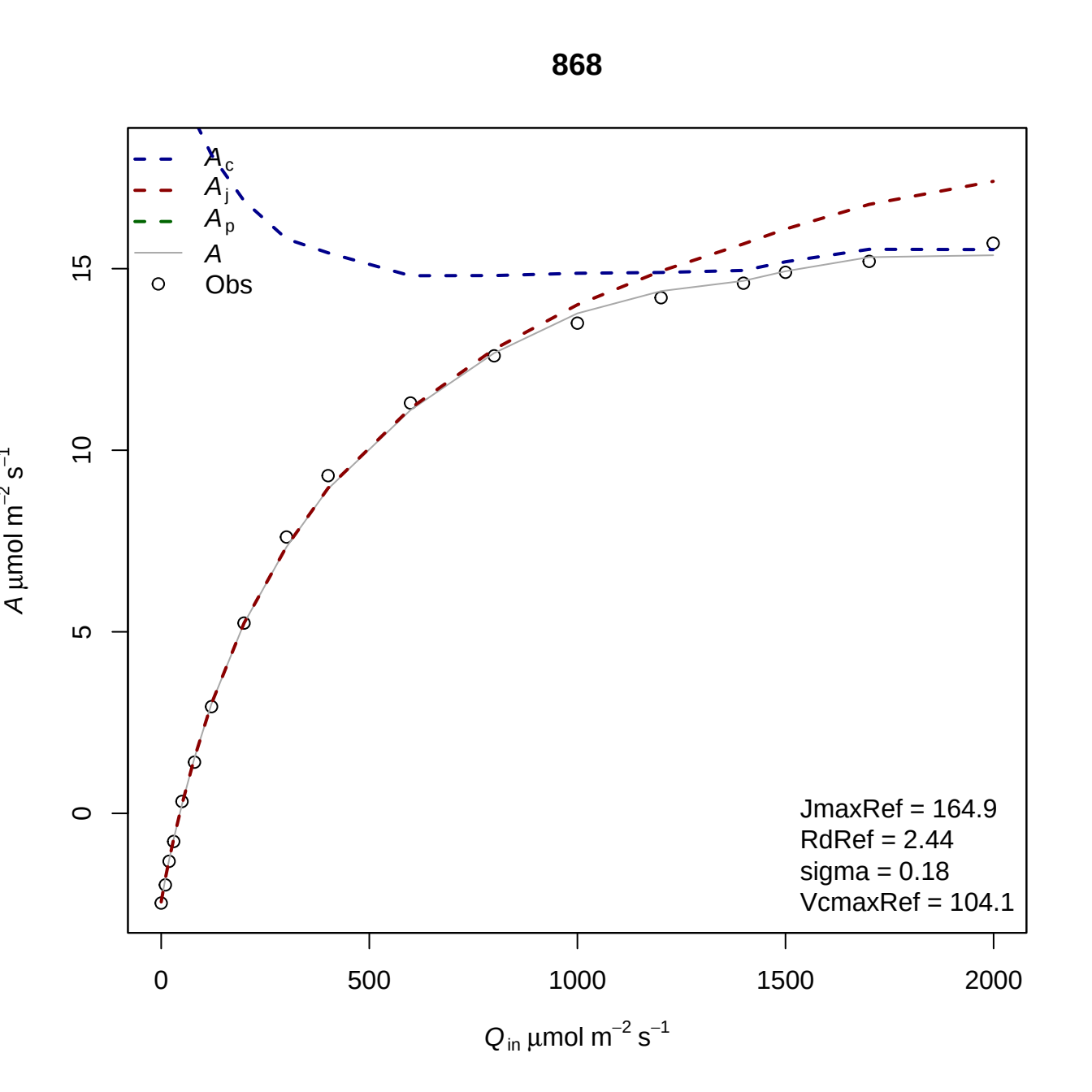

863
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 106.87
RdRef = 1.2
sigma = 0.23
VcmaxRef = 68.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

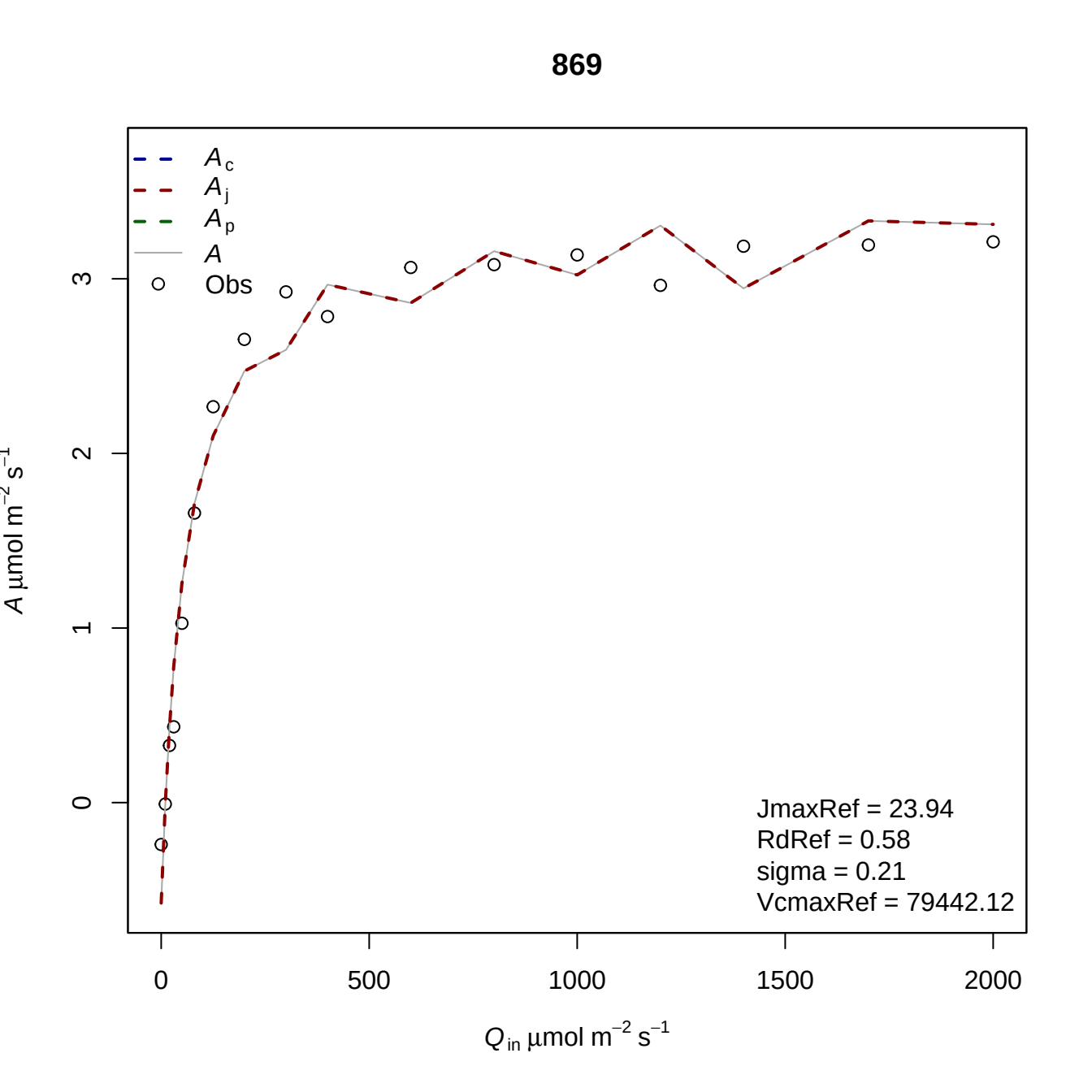

864
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.81
RdRef = 0.78
sigma = 0.21
VcmaxRef = 89.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

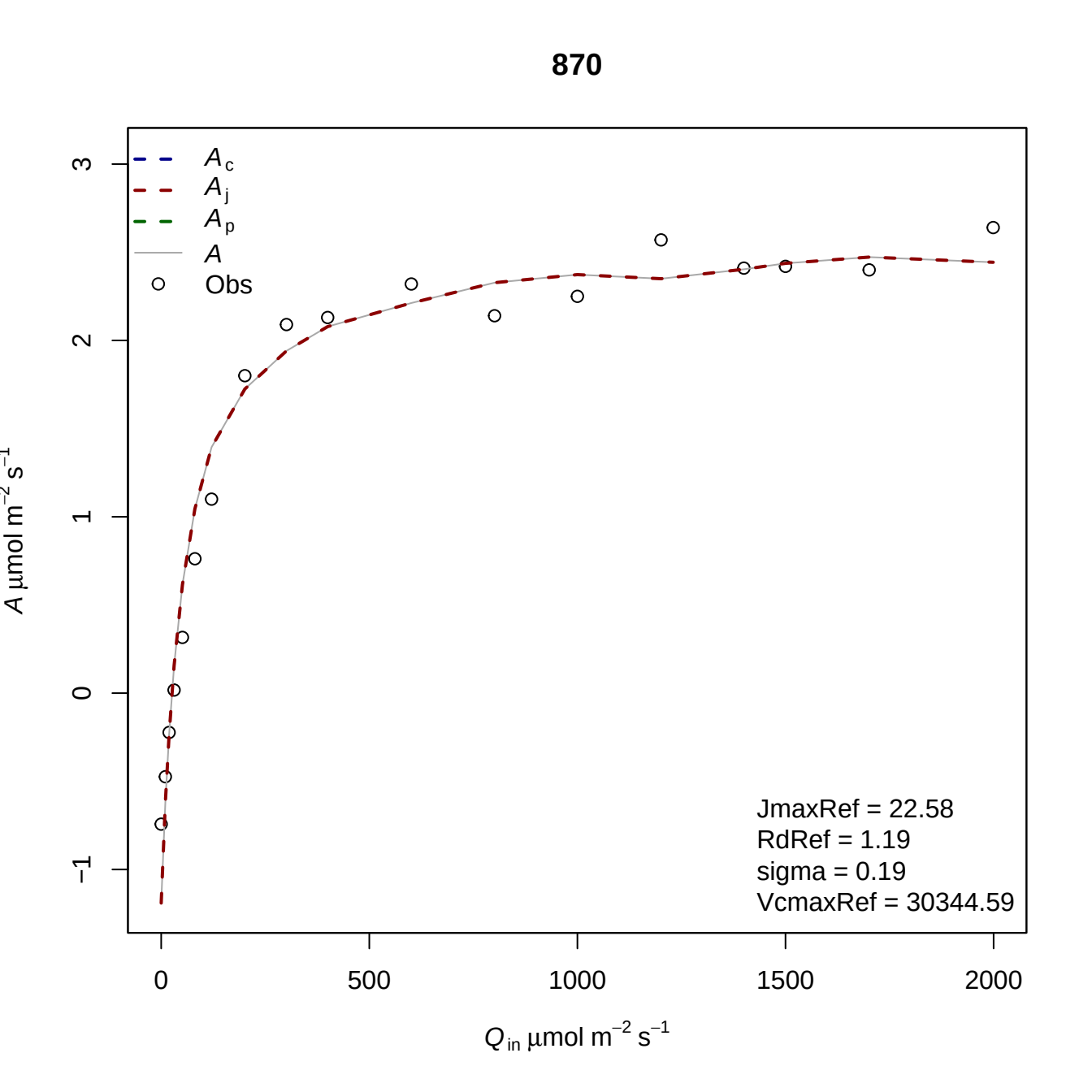

865
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.47
RdRef = 1.95
sigma = 0.13
VcmaxRef = 110.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

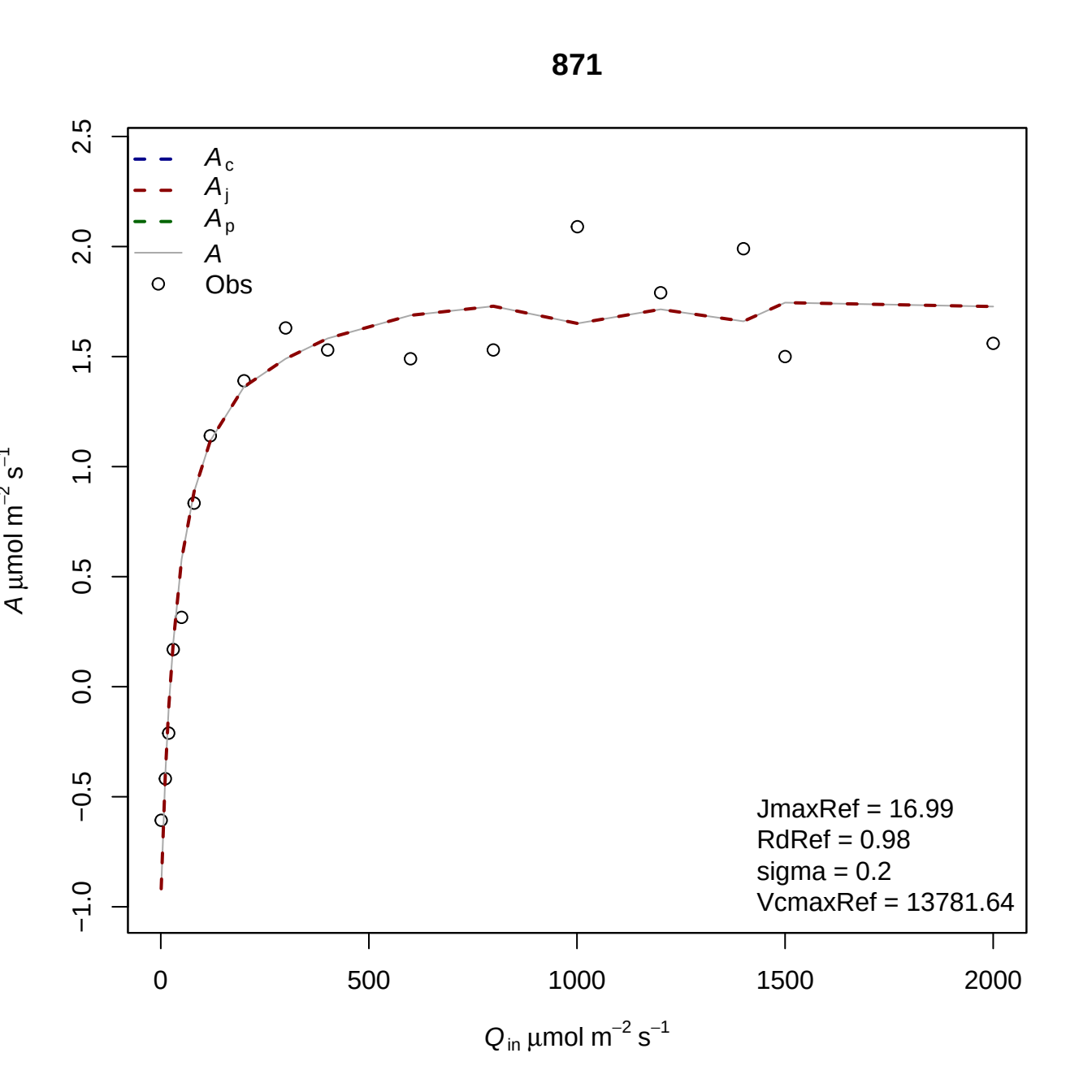

866
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 198.96
RdRef = 4.44
sigma = 1.03
VcmaxRef = 205412.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

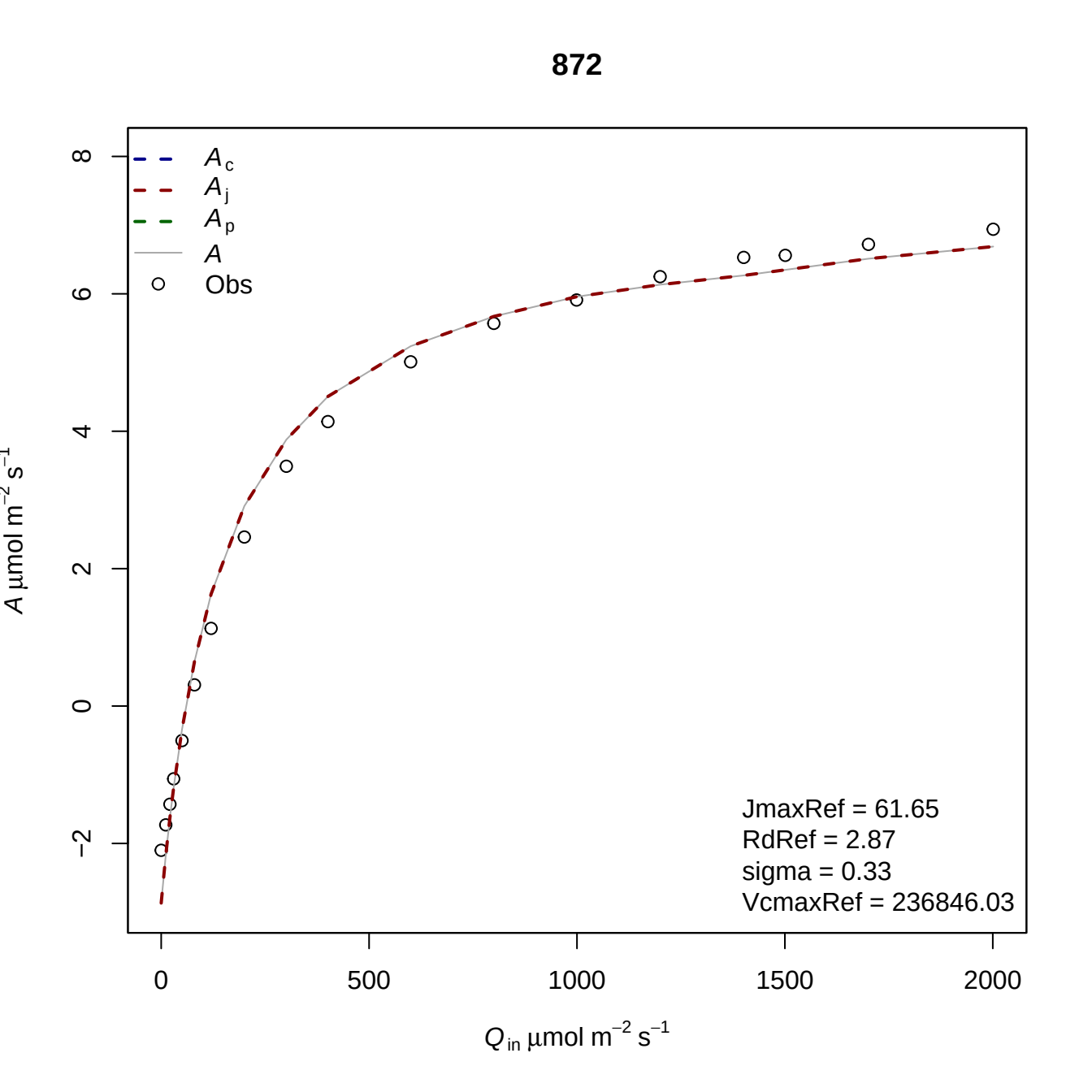

867
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.39
RdRef = 4.06
sigma = 0.43
VcmaxRef = 96037.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

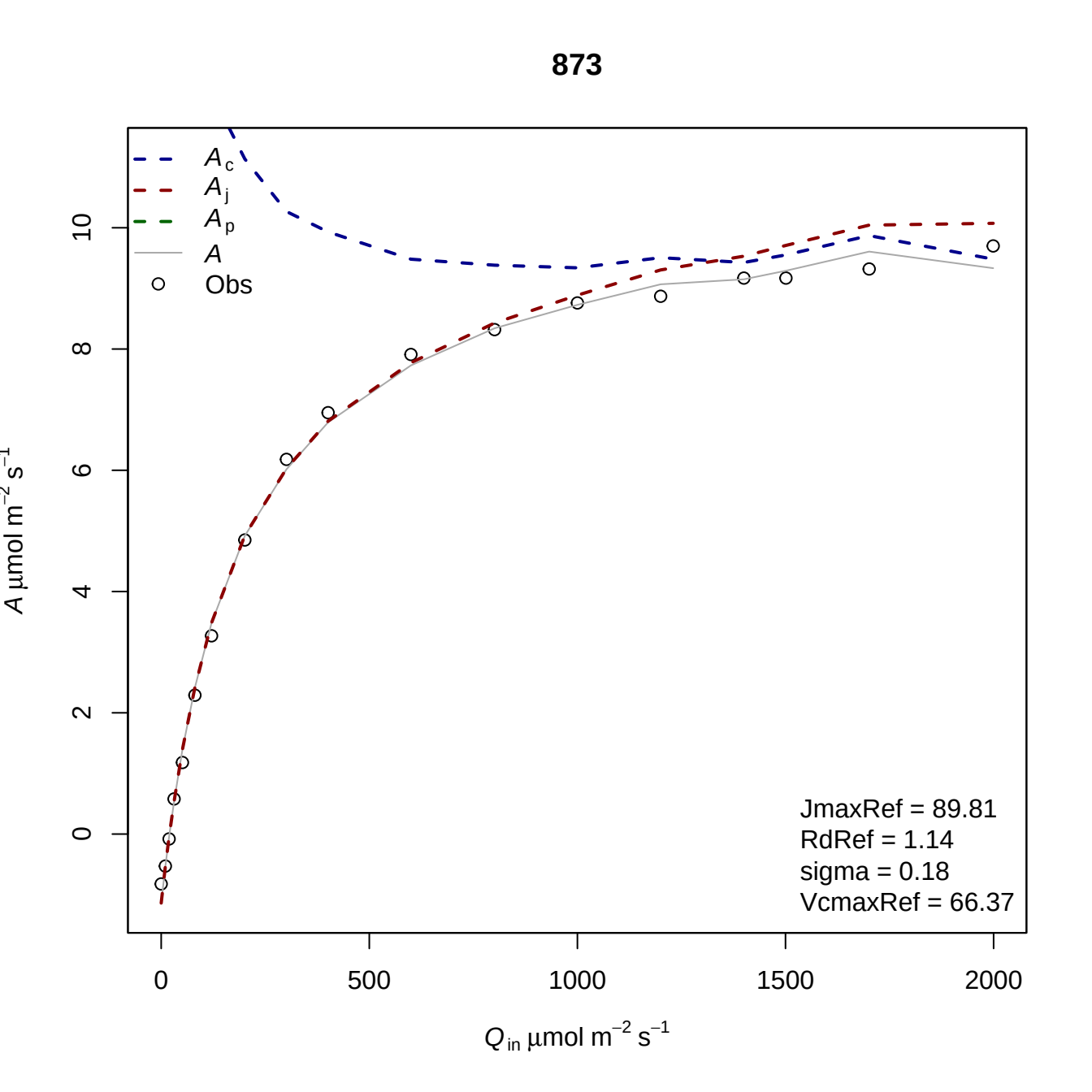

868
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.9
RdRef = 2.44
sigma = 0.18
VcmaxRef = 104.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

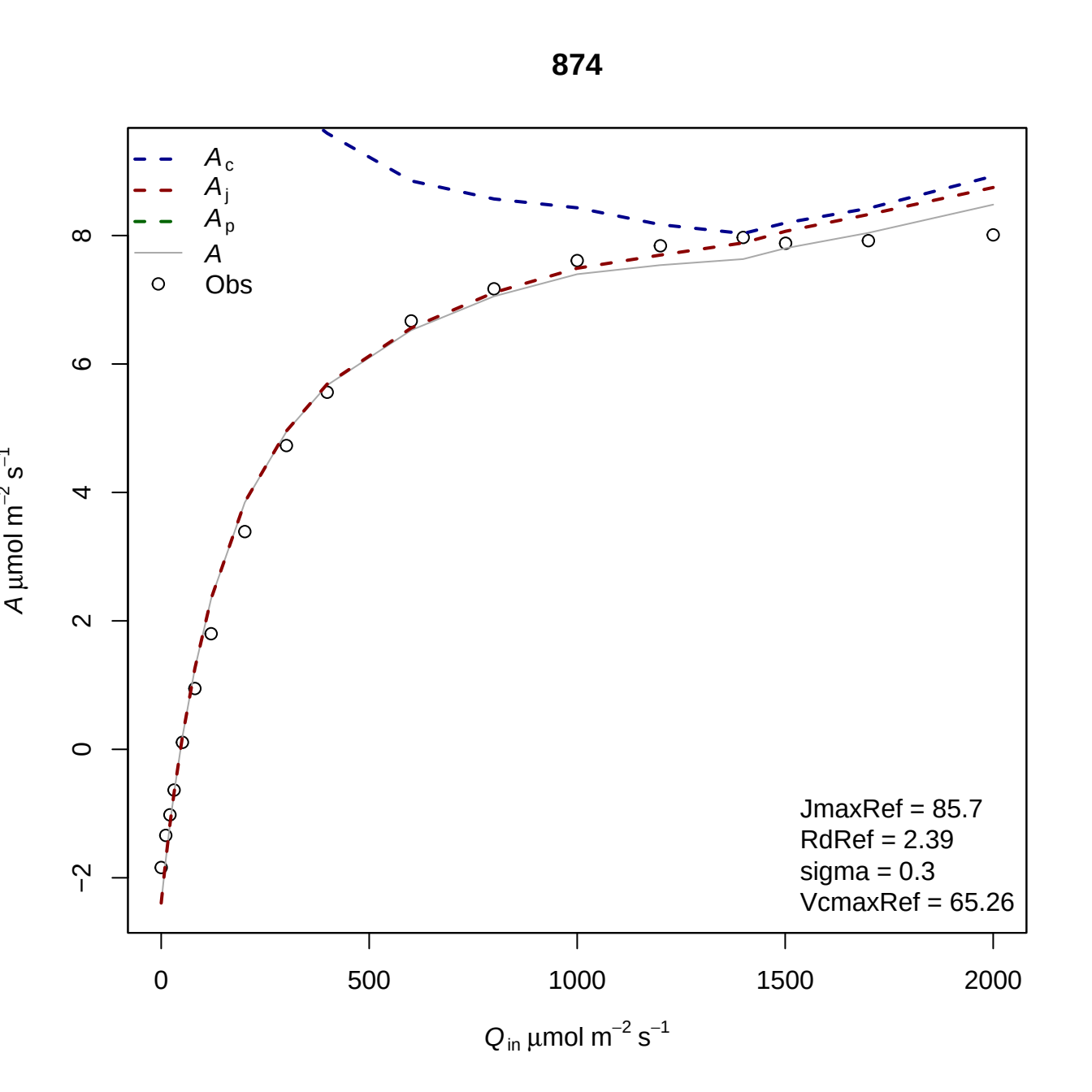

869
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 23.94
RdRef = 0.58
sigma = 0.21
VcmaxRef = 79442.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500
2000

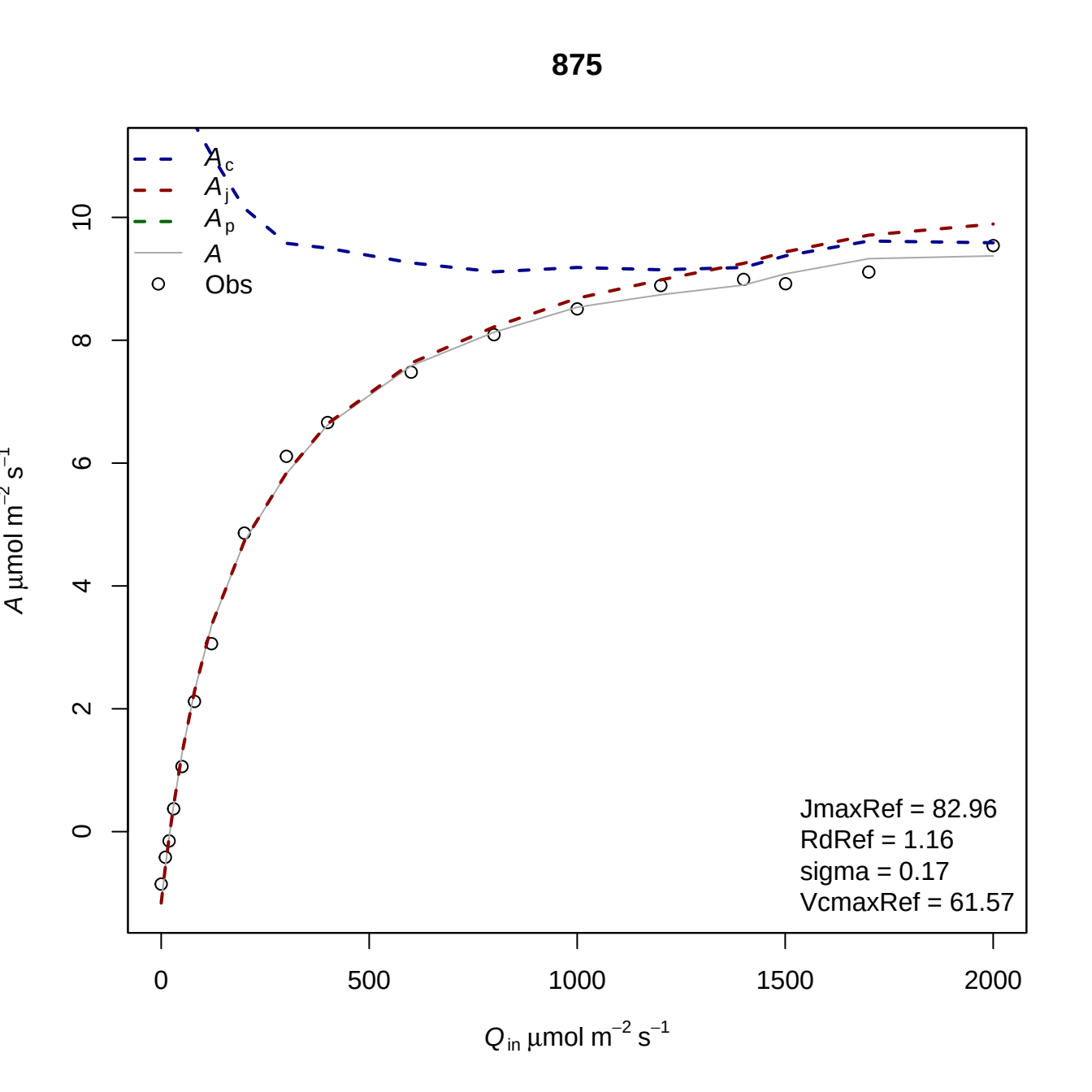

870
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.58
RdRef = 1.19
sigma = 0.19
VcmaxRef = 30344.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
0
500
1000
1500
2000

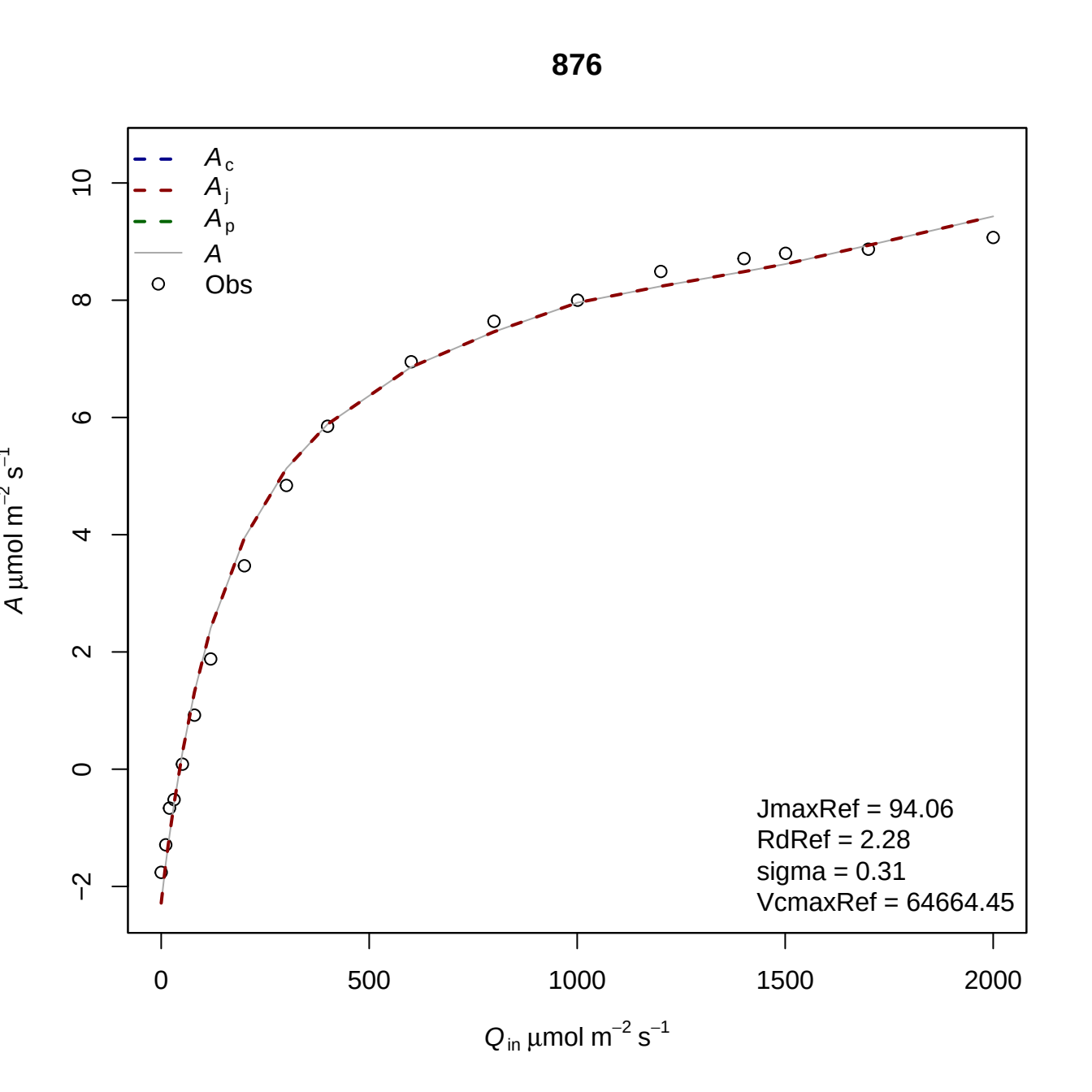

871
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 16.99
RdRef = 0.98
sigma = 0.2
VcmaxRef = 13781.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
2000

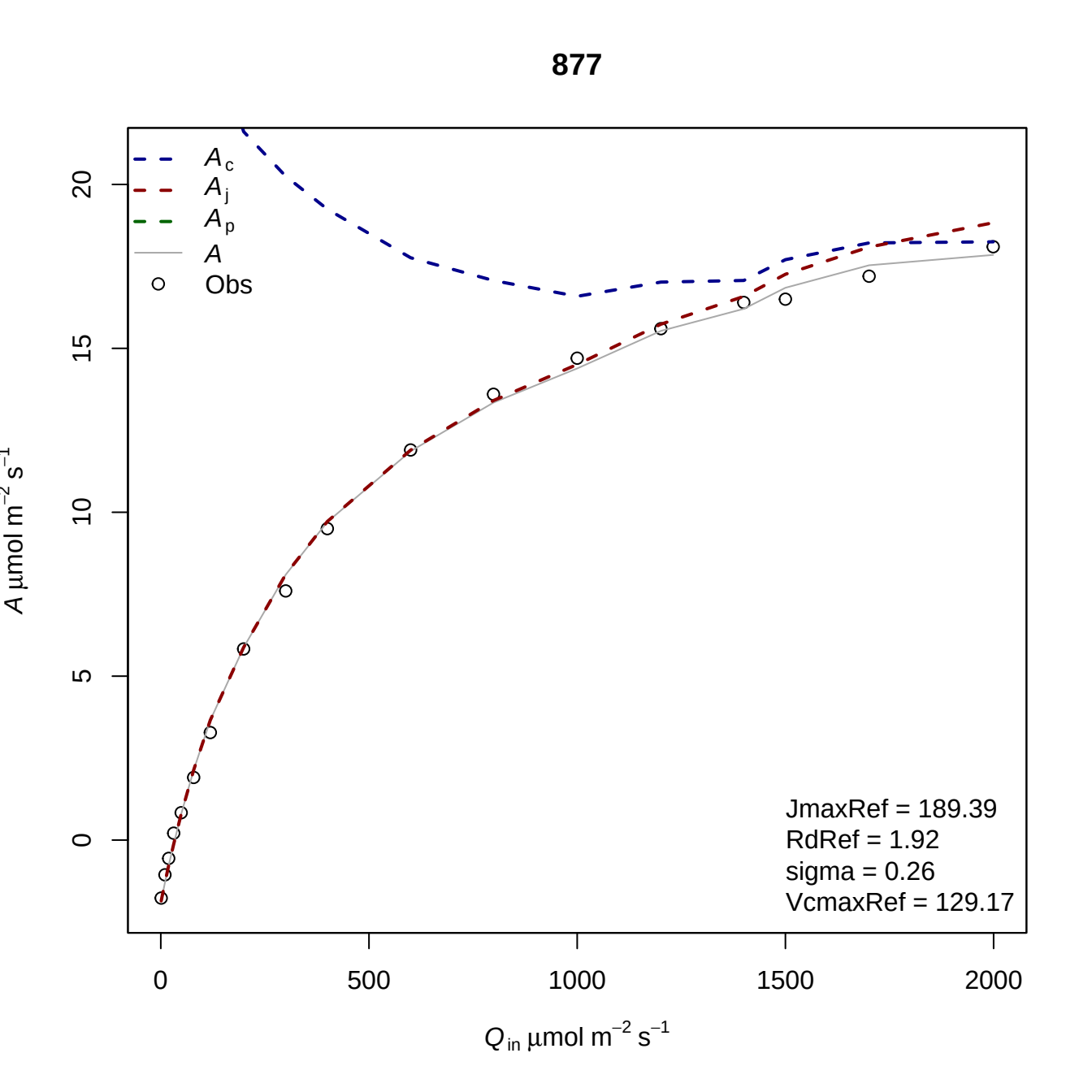

872
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.65
RdRef = 2.87
sigma = 0.33
VcmaxRef = 236846.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

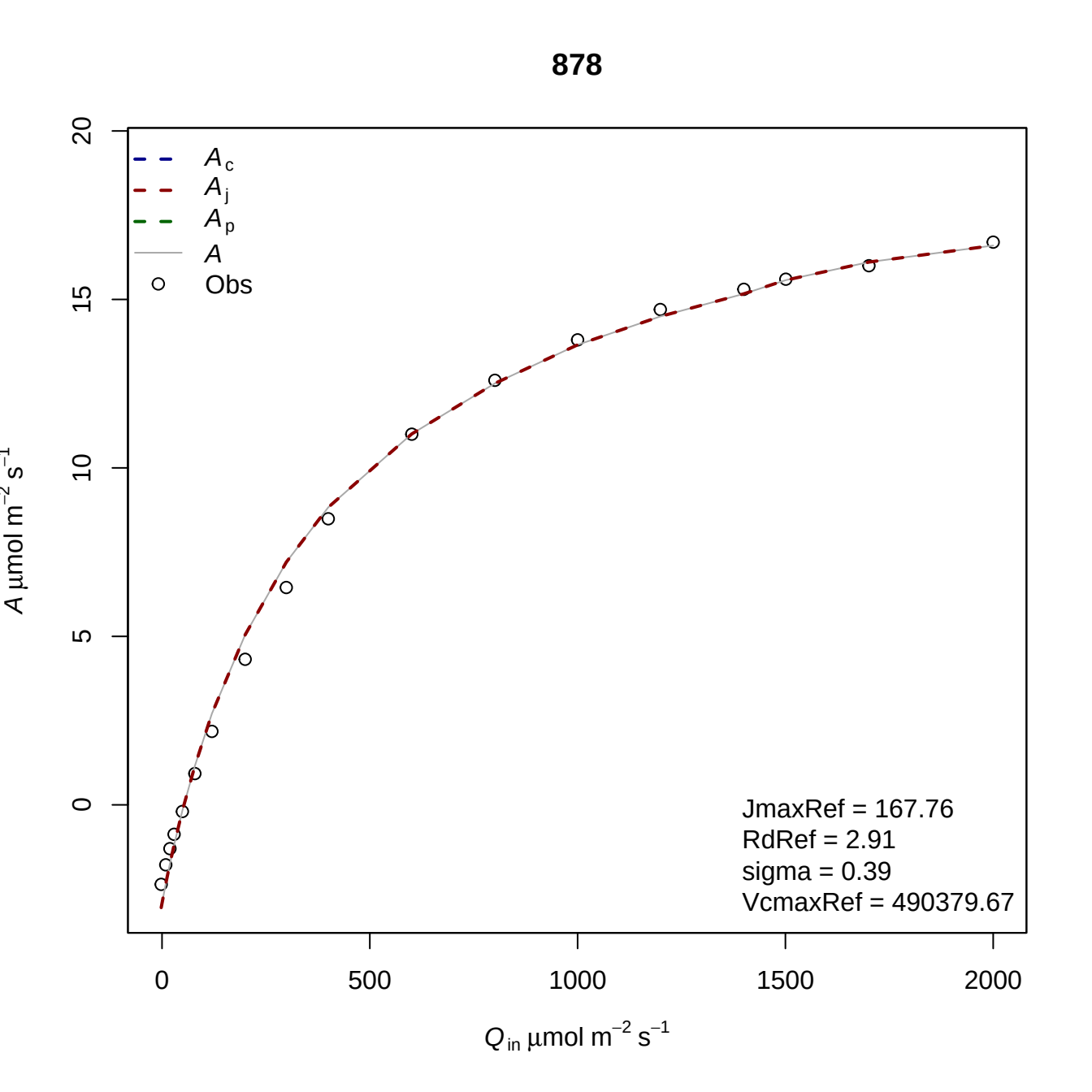

873
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.81
RdRef = 1.14
sigma = 0.18
VcmaxRef = 66.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

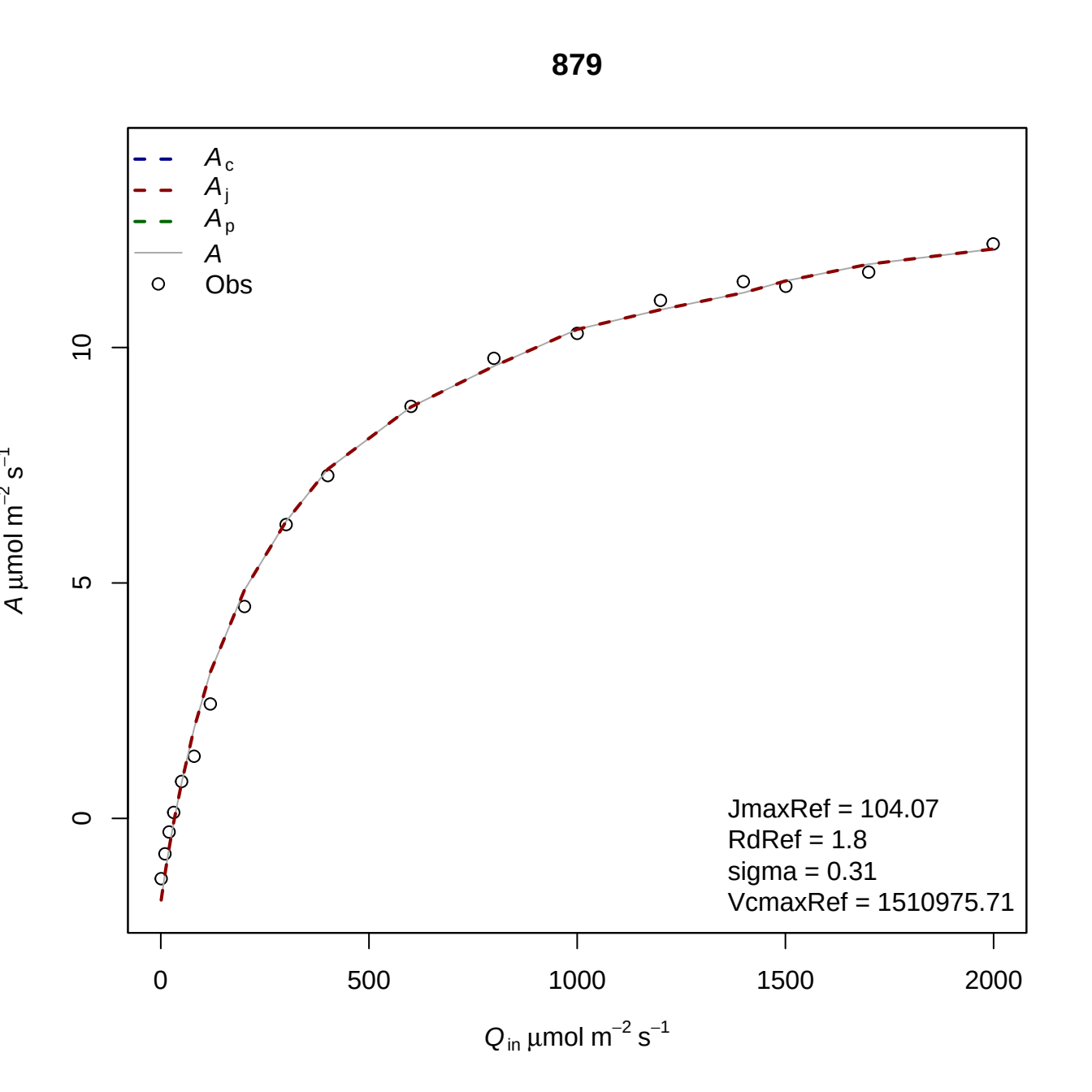

874
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.7
RdRef = 2.39
sigma = 0.3
VcmaxRef = 65.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

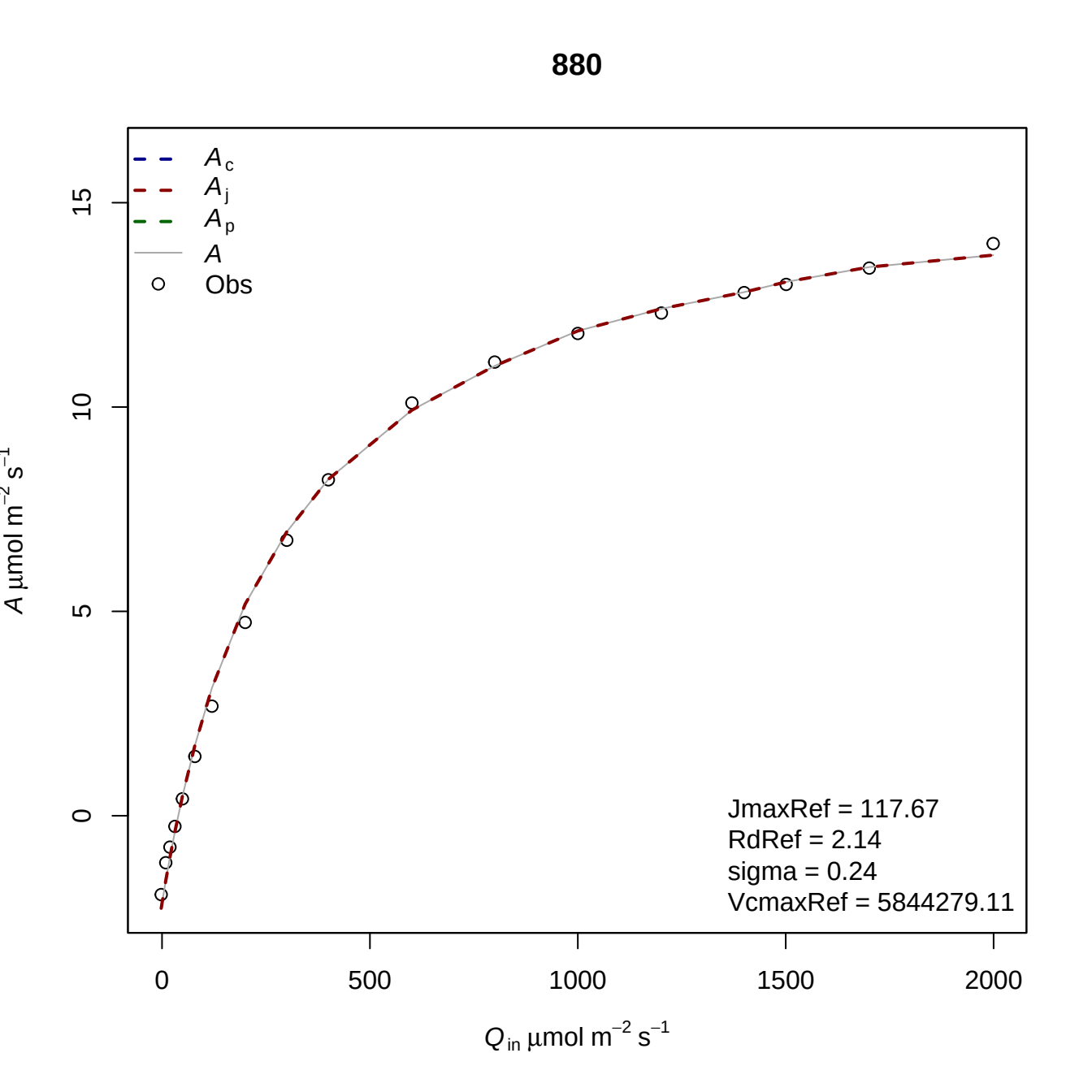

875
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.96
RdRef = 1.16
sigma = 0.17
VcmaxRef = 61.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

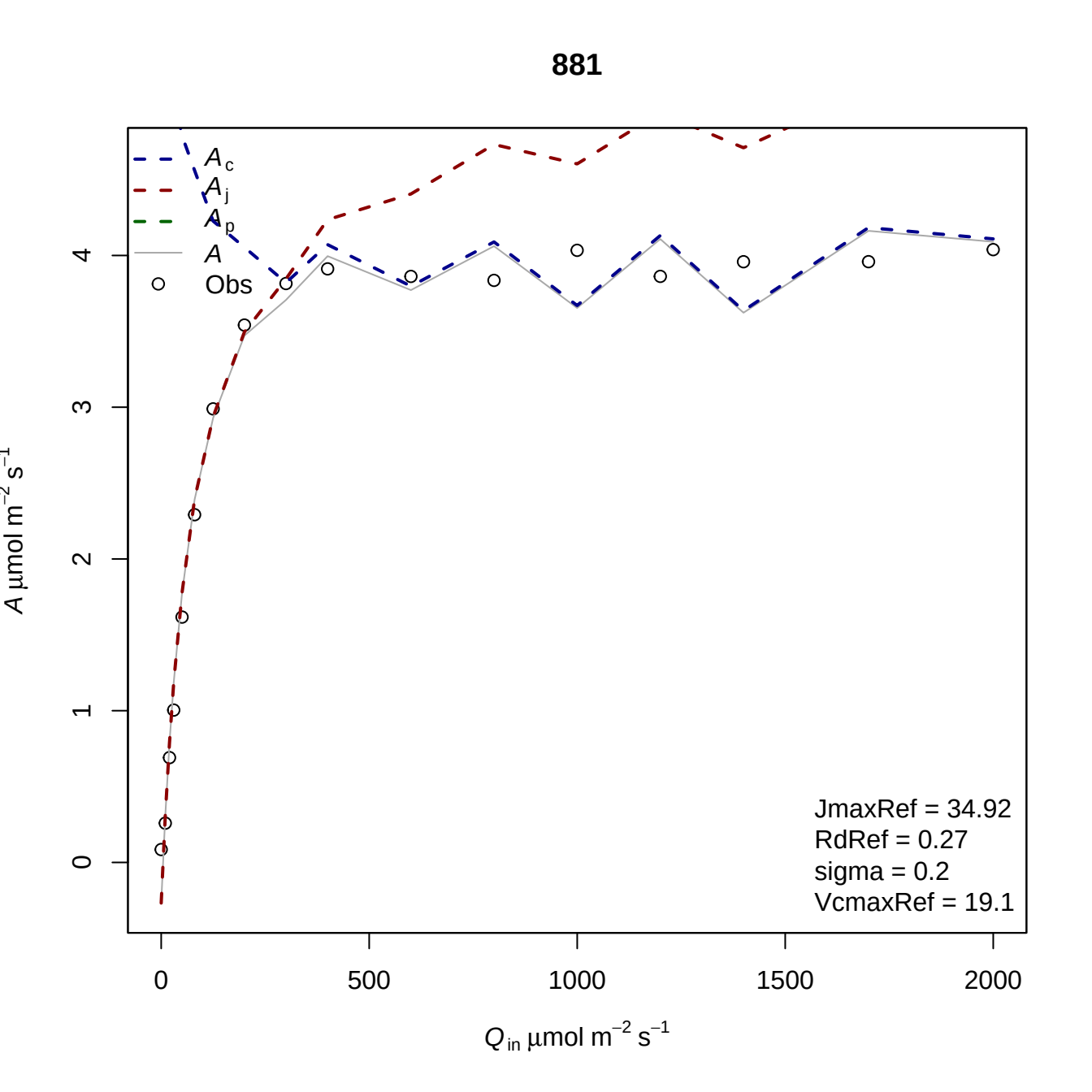

876
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.06
RdRef = 2.28
sigma = 0.31
VcmaxRef = 64664.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

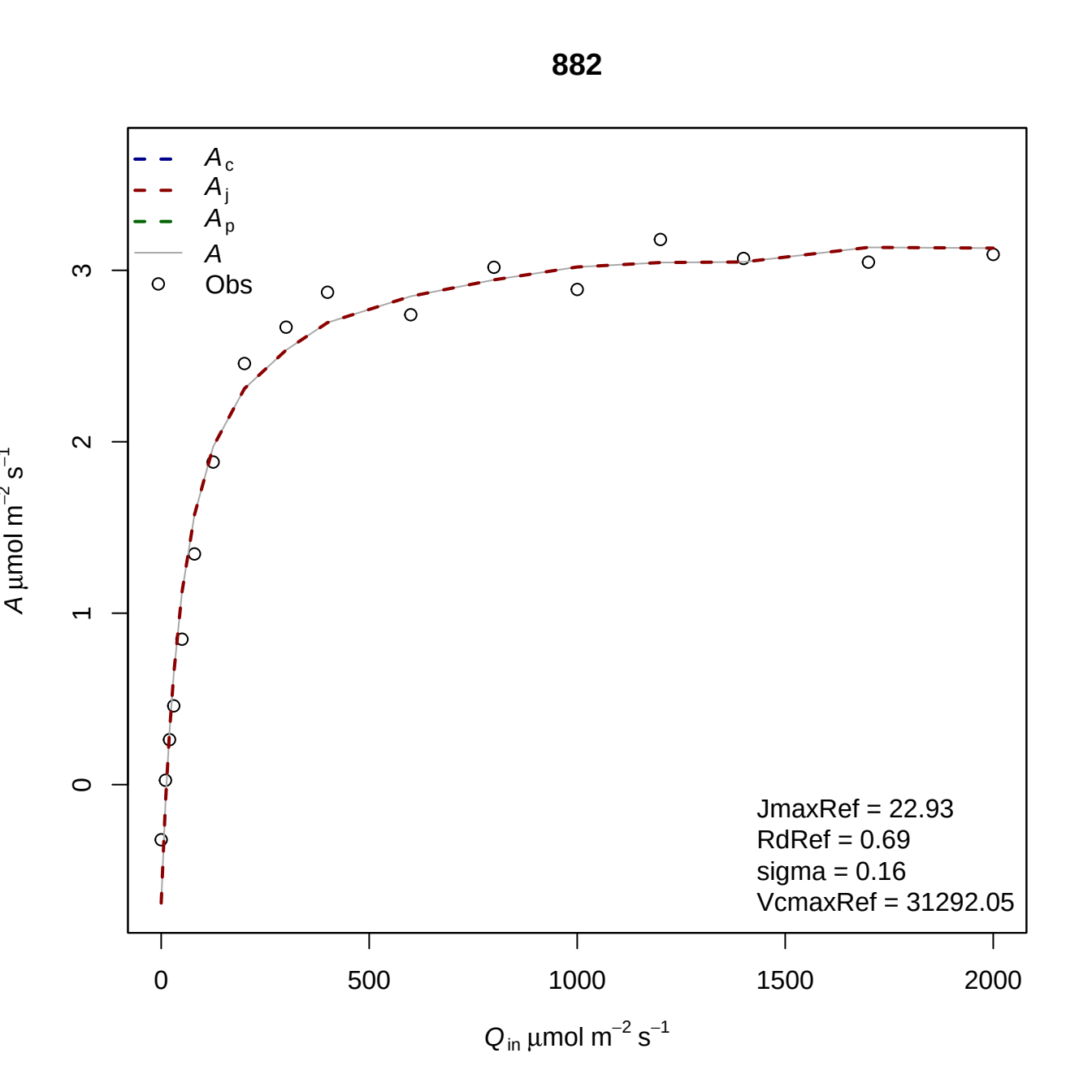

877
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 189.39
RdRef = 1.92
sigma = 0.26
VcmaxRef = 129.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

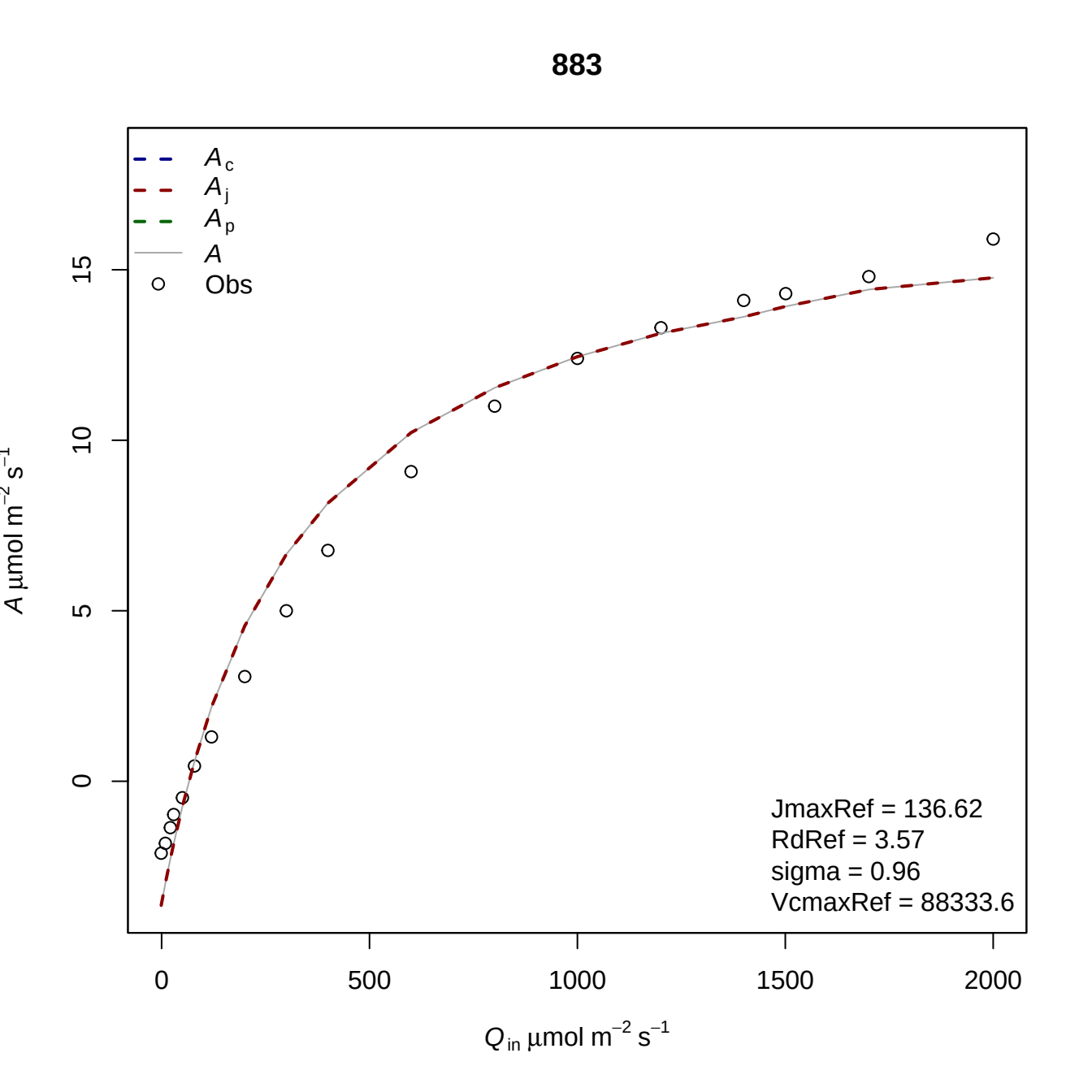

878
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 167.76
RdRef = 2.91
sigma = 0.39
VcmaxRef = 490379.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

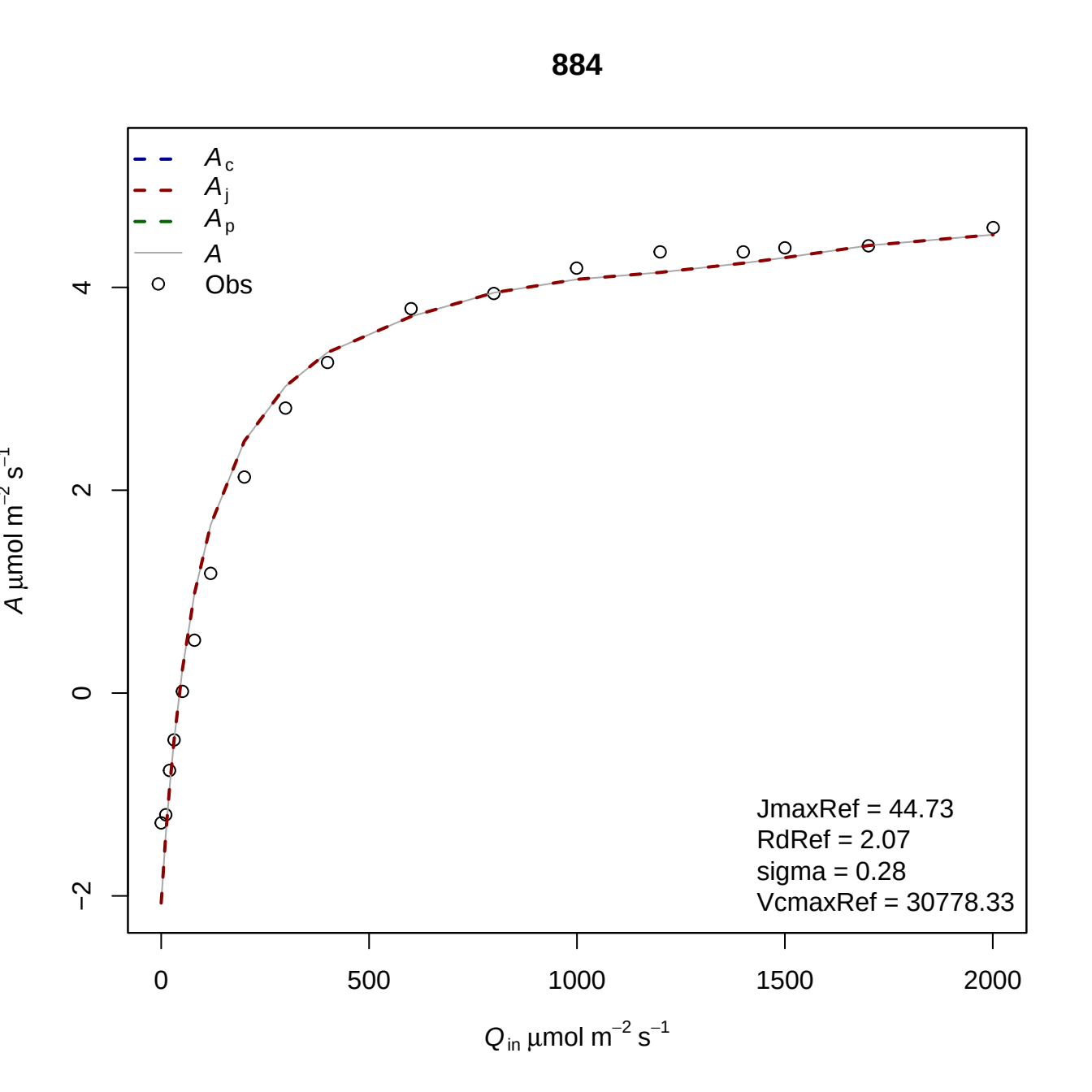

879
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.07
RdRef = 1.8
sigma = 0.31
VcmaxRef = 1510975.71
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

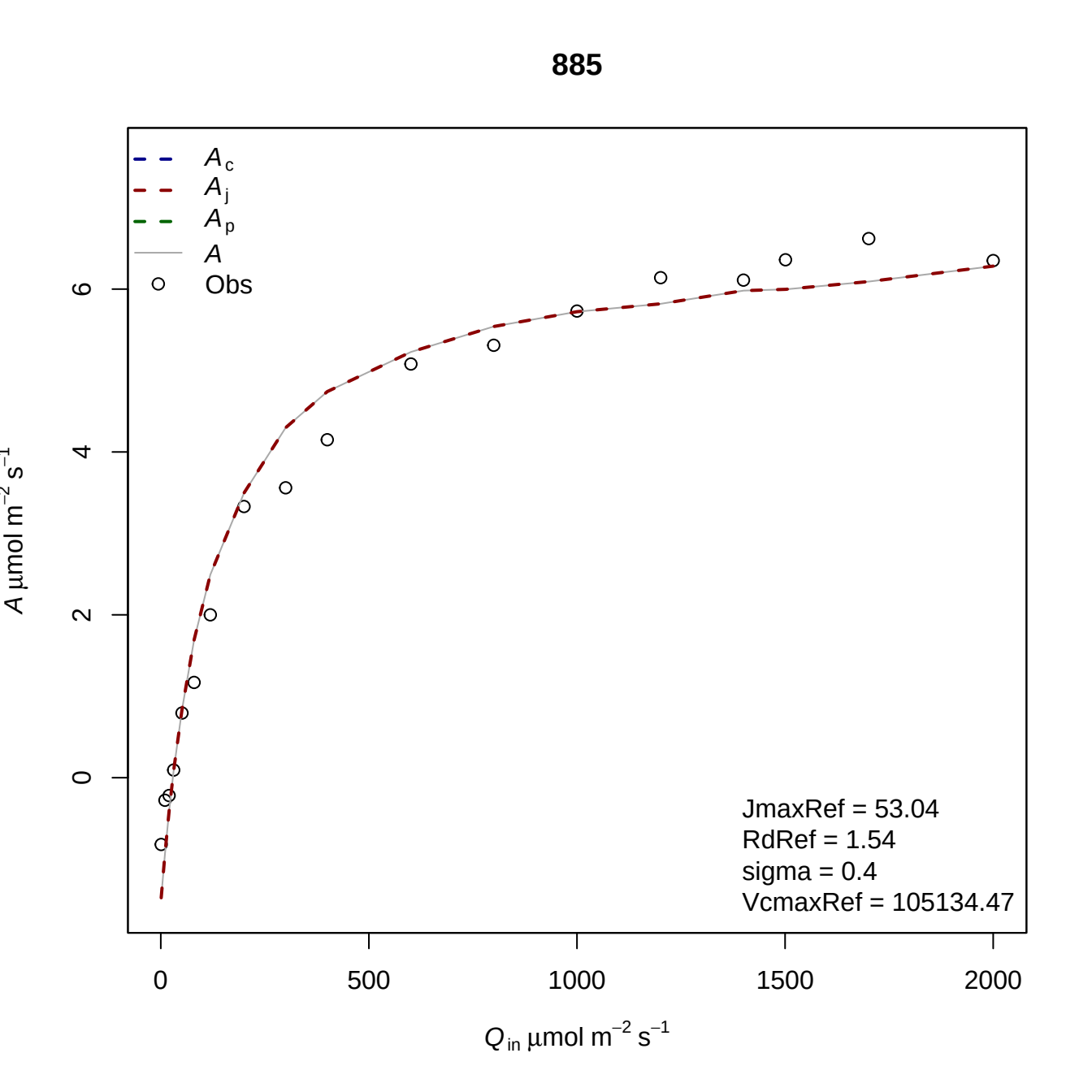

880
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.67
RdRef = 2.14
sigma = 0.24
VcmaxRef = 5844279.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

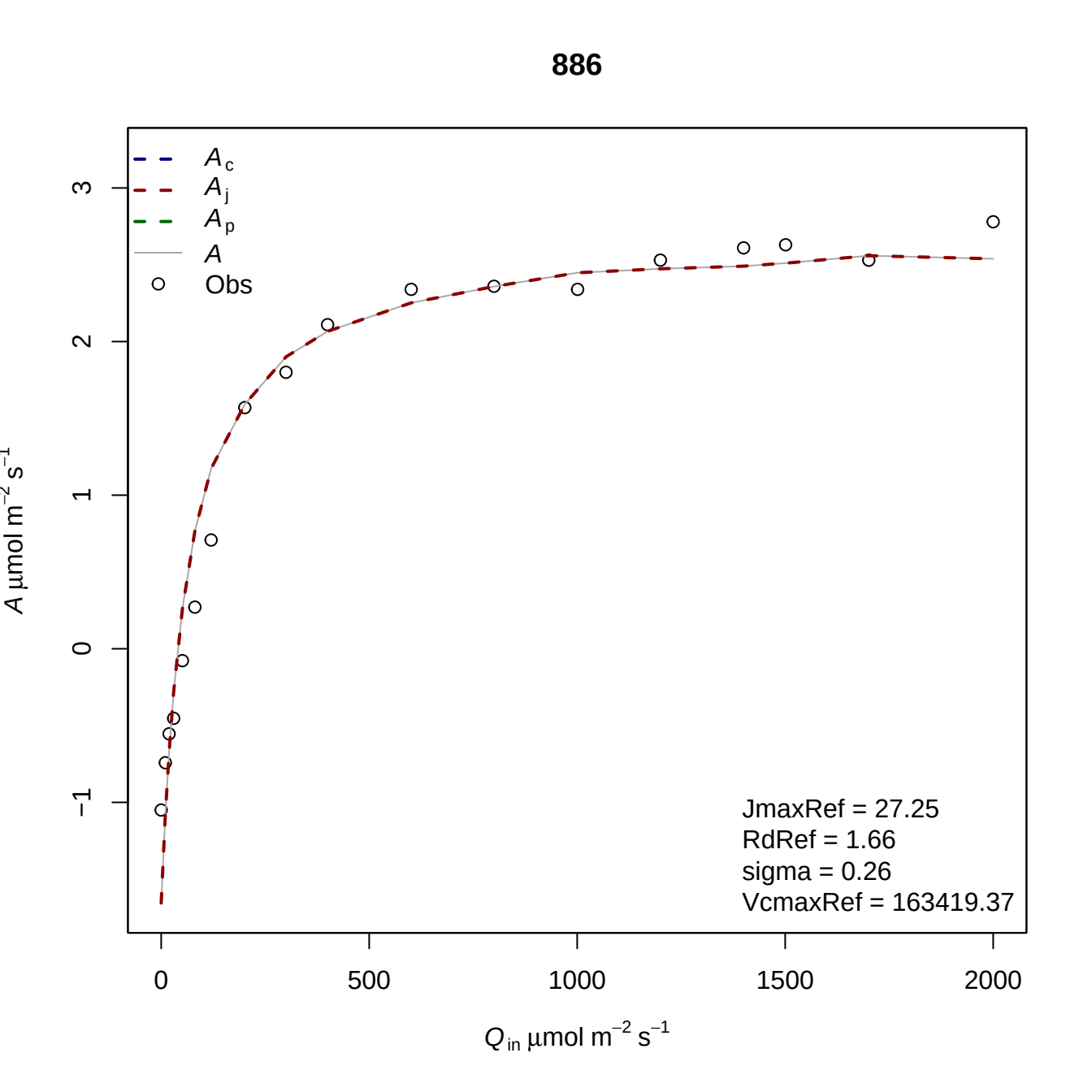

881
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 34.92
RdRef = 0.27
sigma = 0.2
VcmaxRef = 19.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500
2000

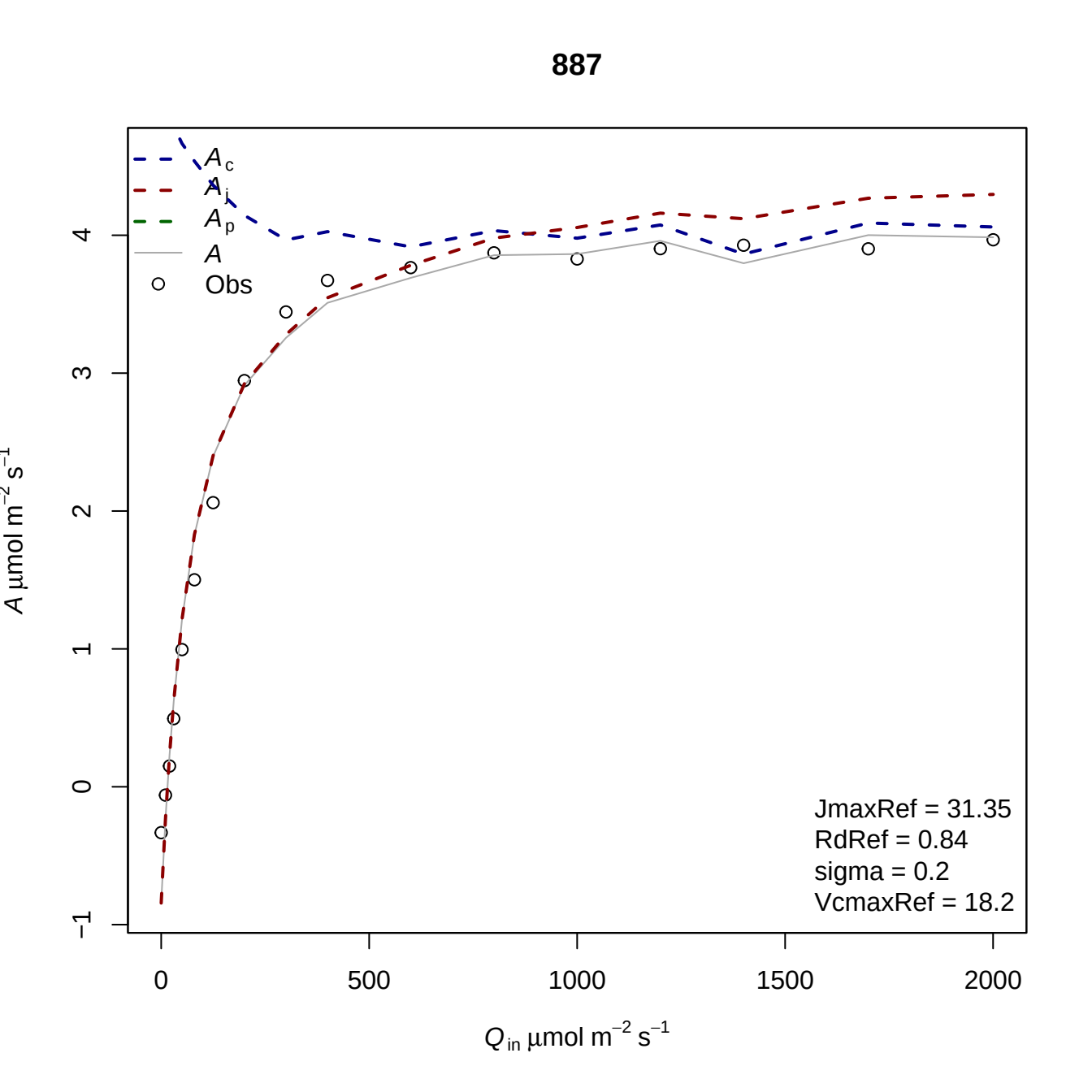

882
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.93
RdRef = 0.69
sigma = 0.16
VcmaxRef = 31292.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500
2000

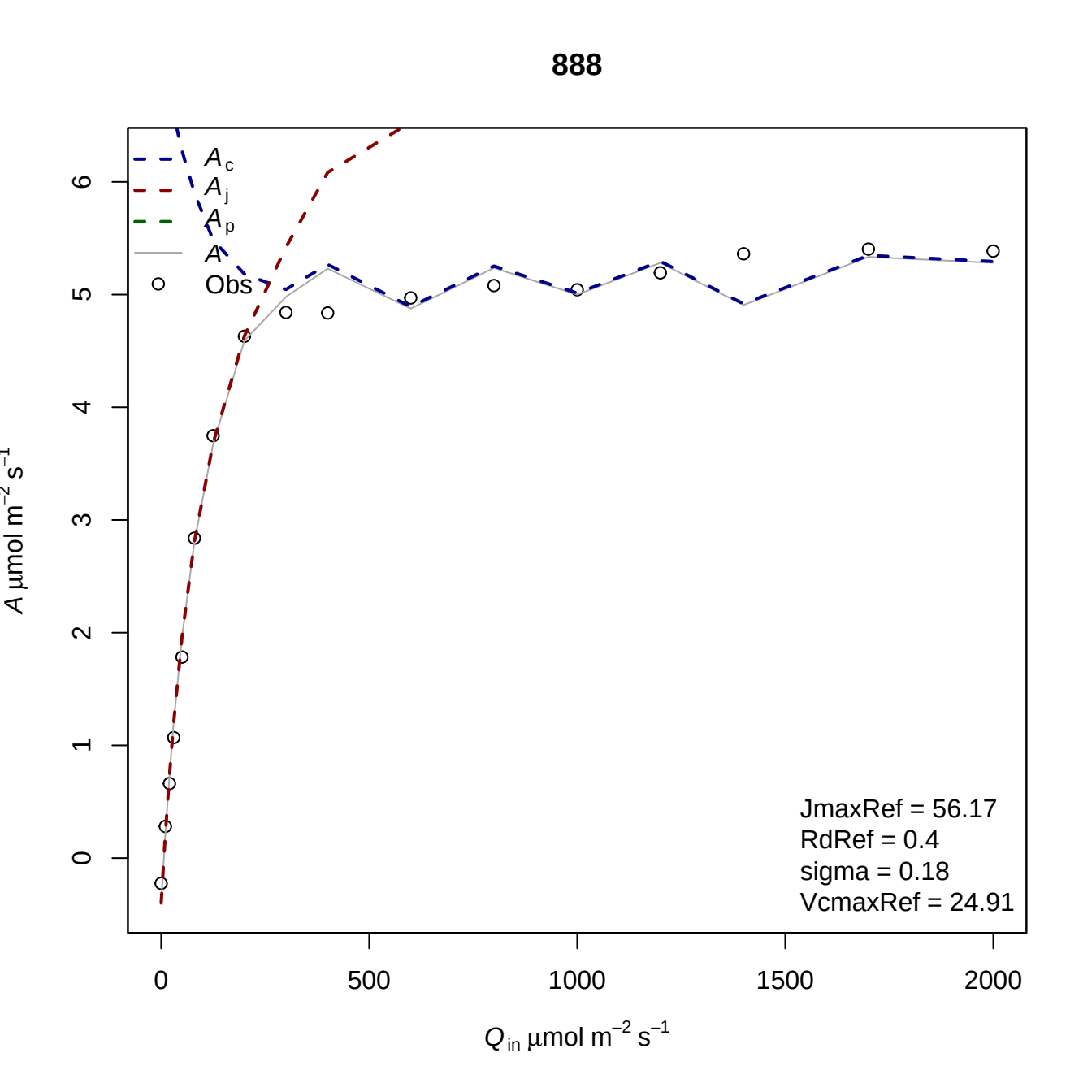

883
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.62
RdRef = 3.57
sigma = 0.96
VcmaxRef = 88333.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

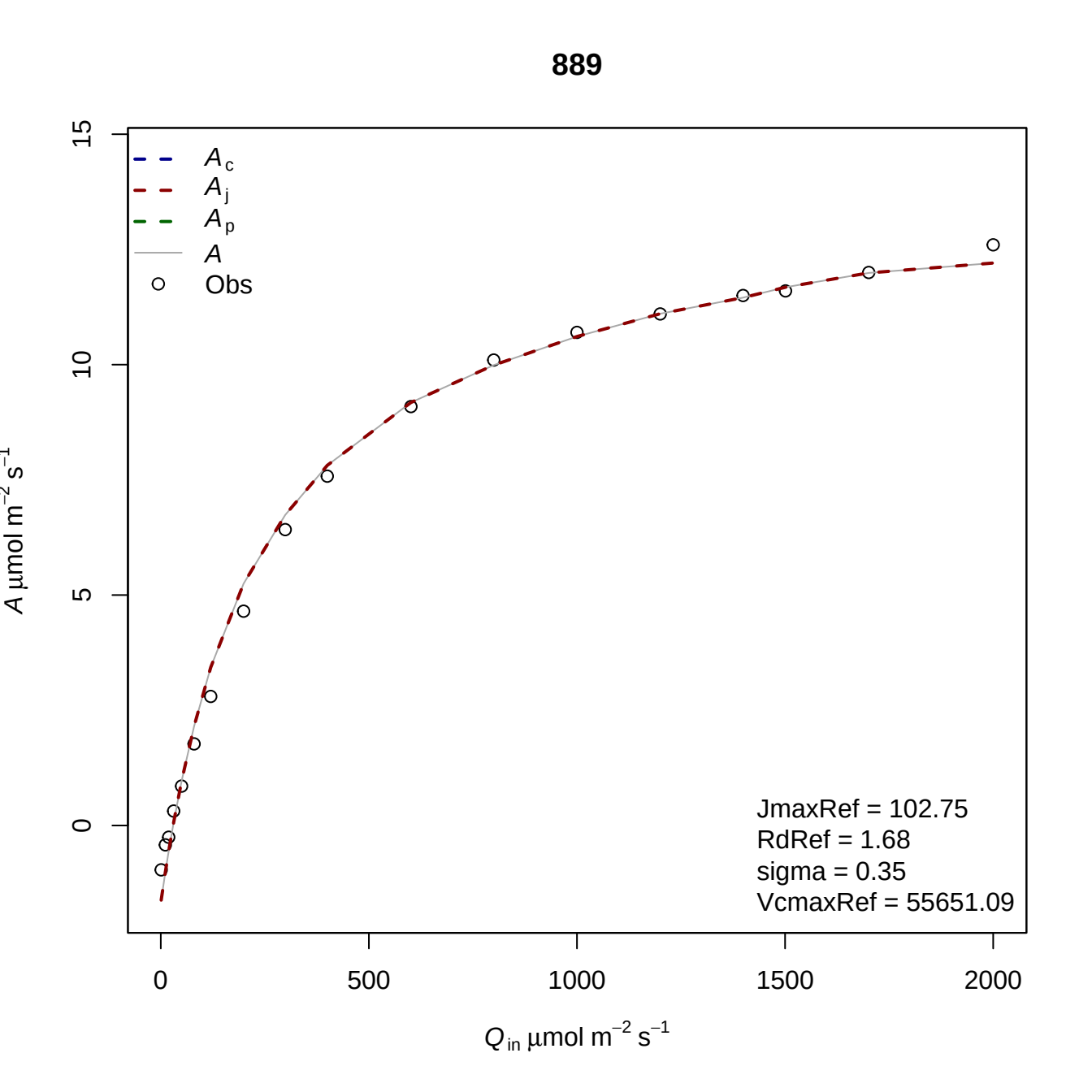

884
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.73
RdRef = 2.07
sigma = 0.28
VcmaxRef = 30778.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

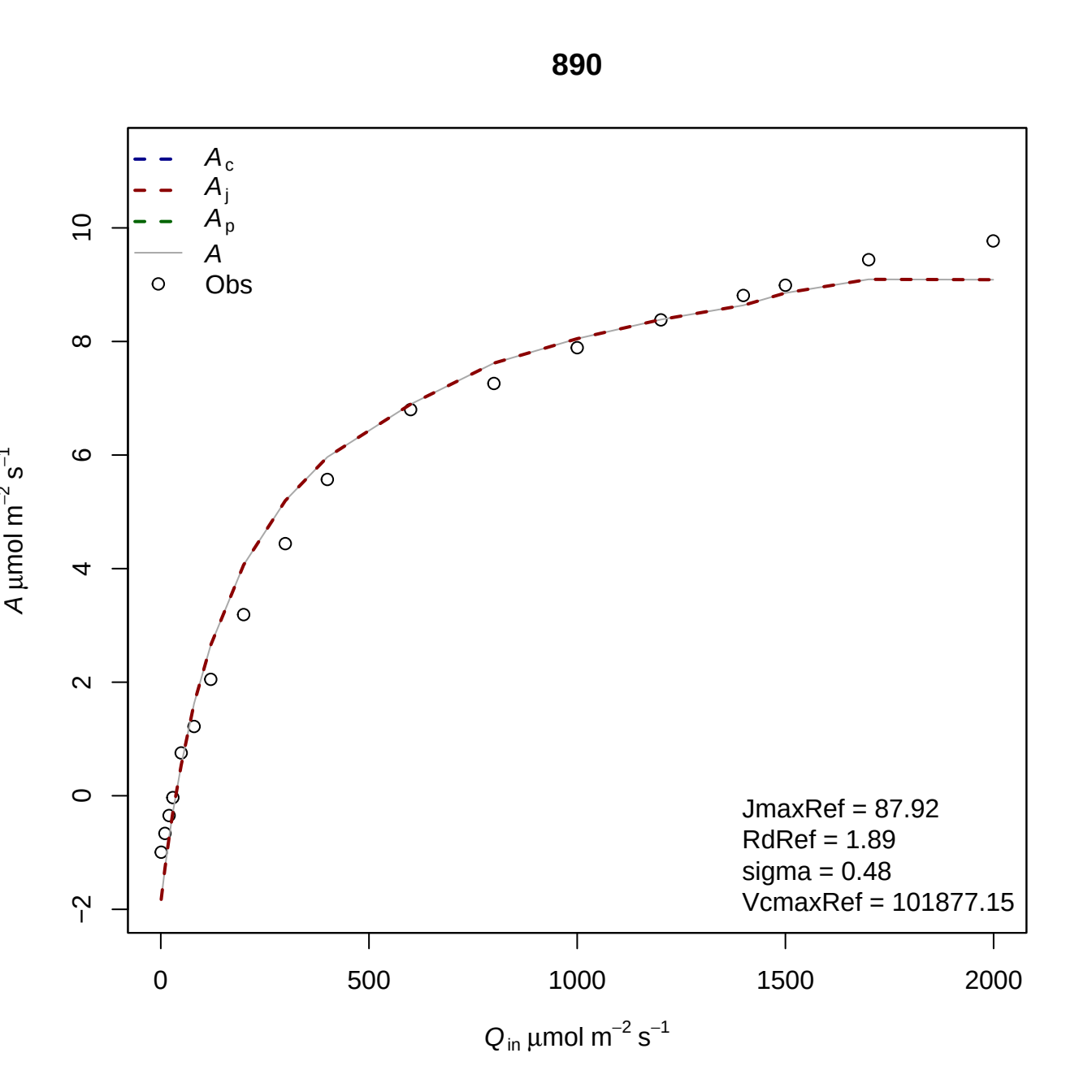

885
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 53.04
RdRef = 1.54
sigma = 0.4
VcmaxRef = 105134.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

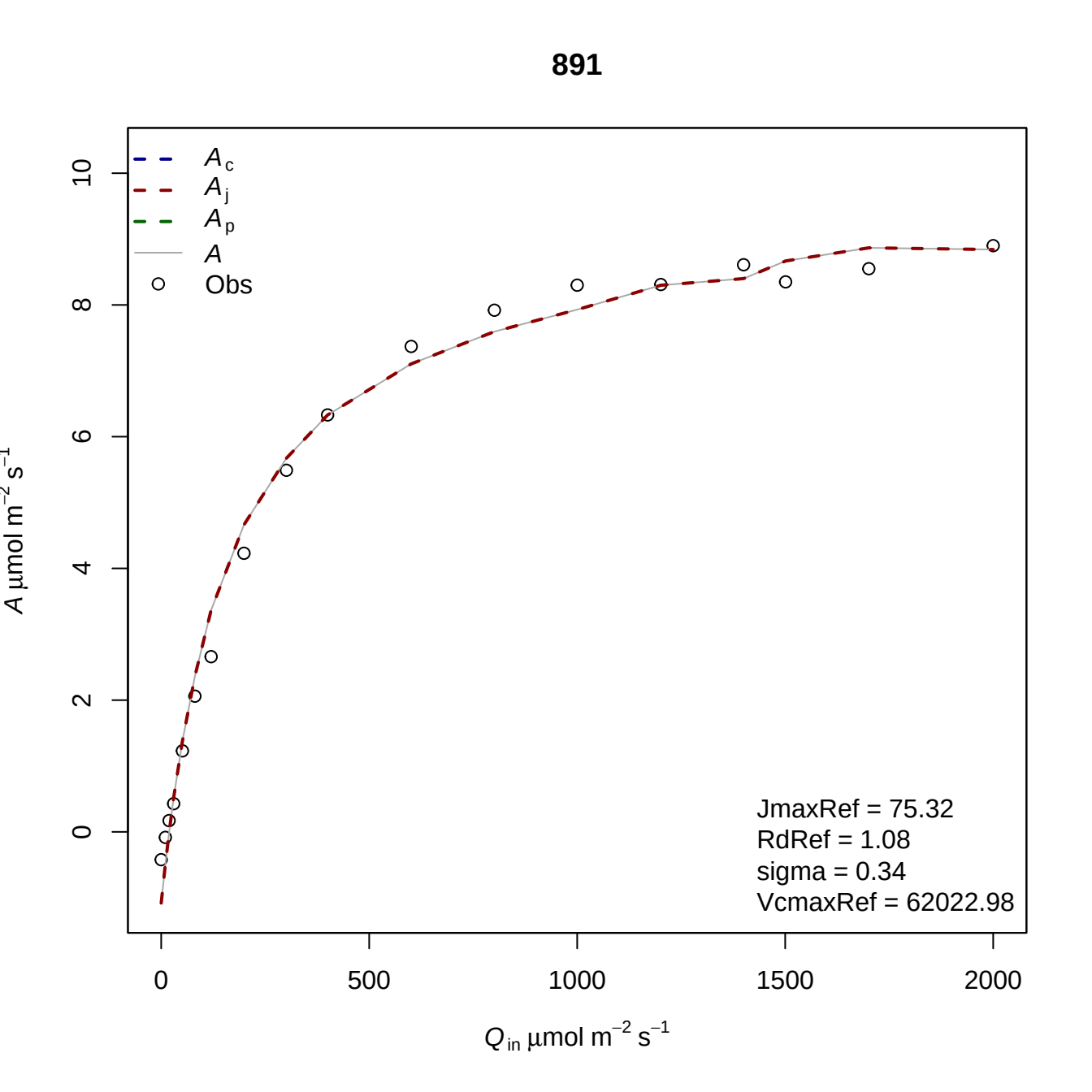

886
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.25
RdRef = 1.66
sigma = 0.26
VcmaxRef = 163419.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

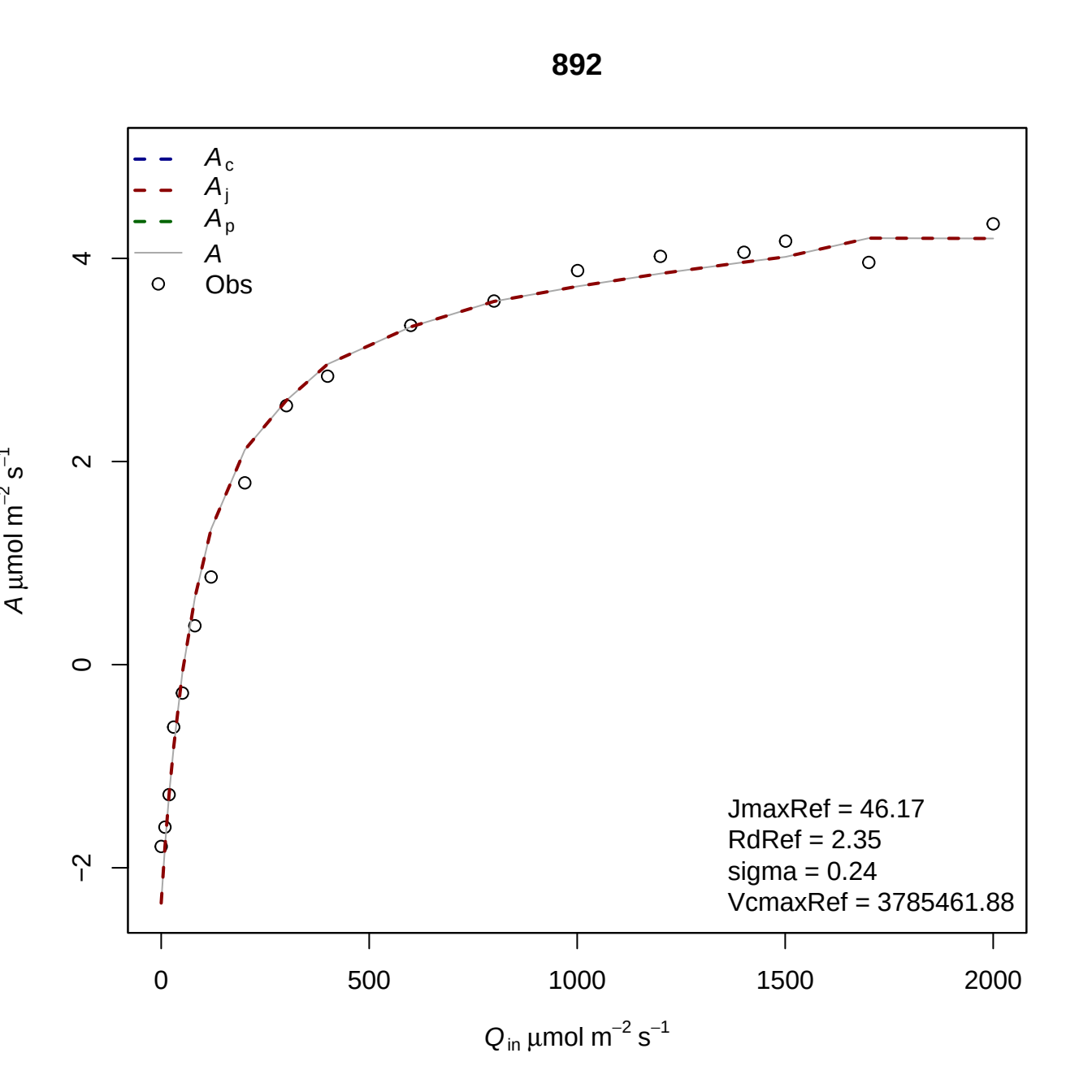

887
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.35
RdRef = 0.84
sigma = 0.2
VcmaxRef = 18.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

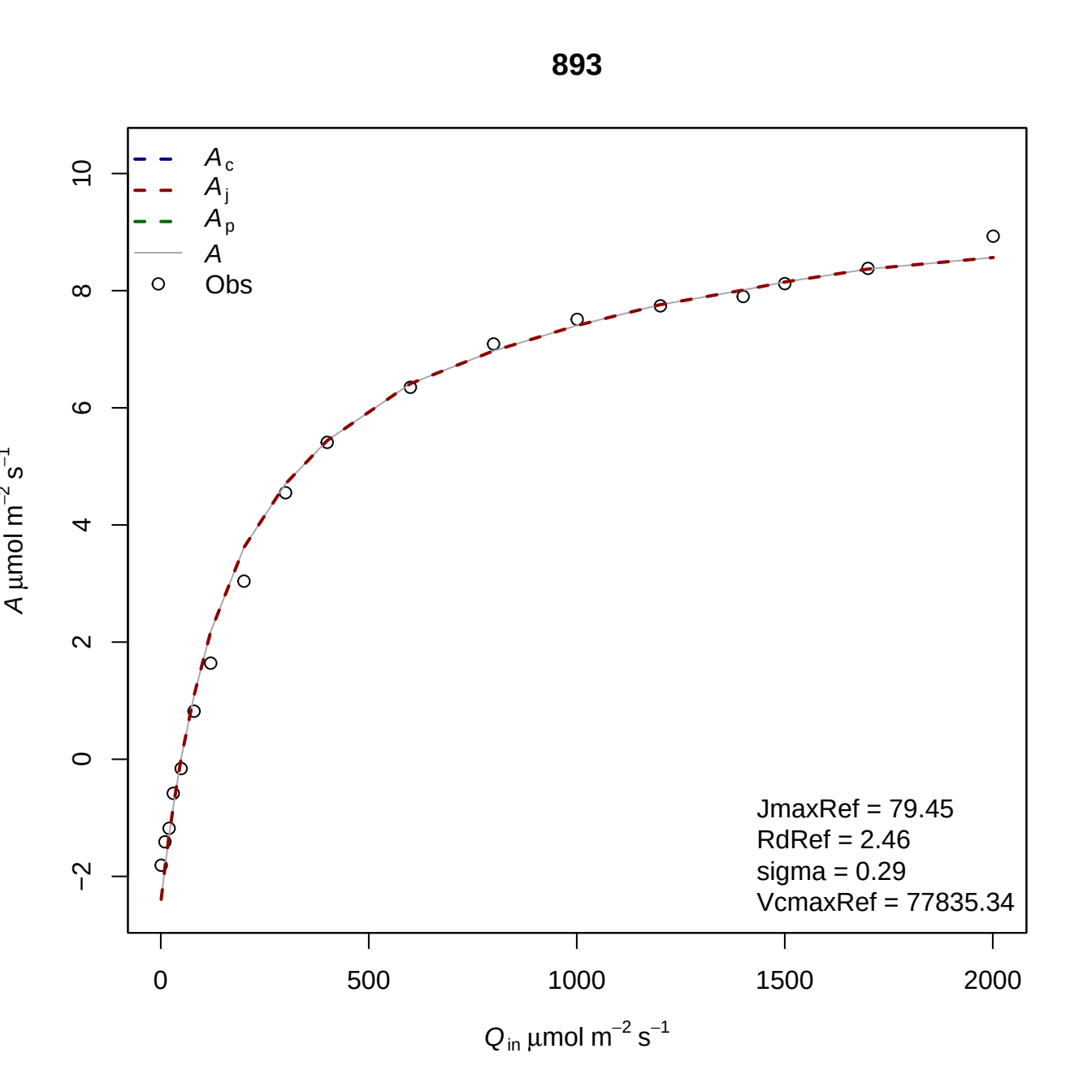

888
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.17
RdRef = 0.4
sigma = 0.18
VcmaxRef = 24.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500
2000

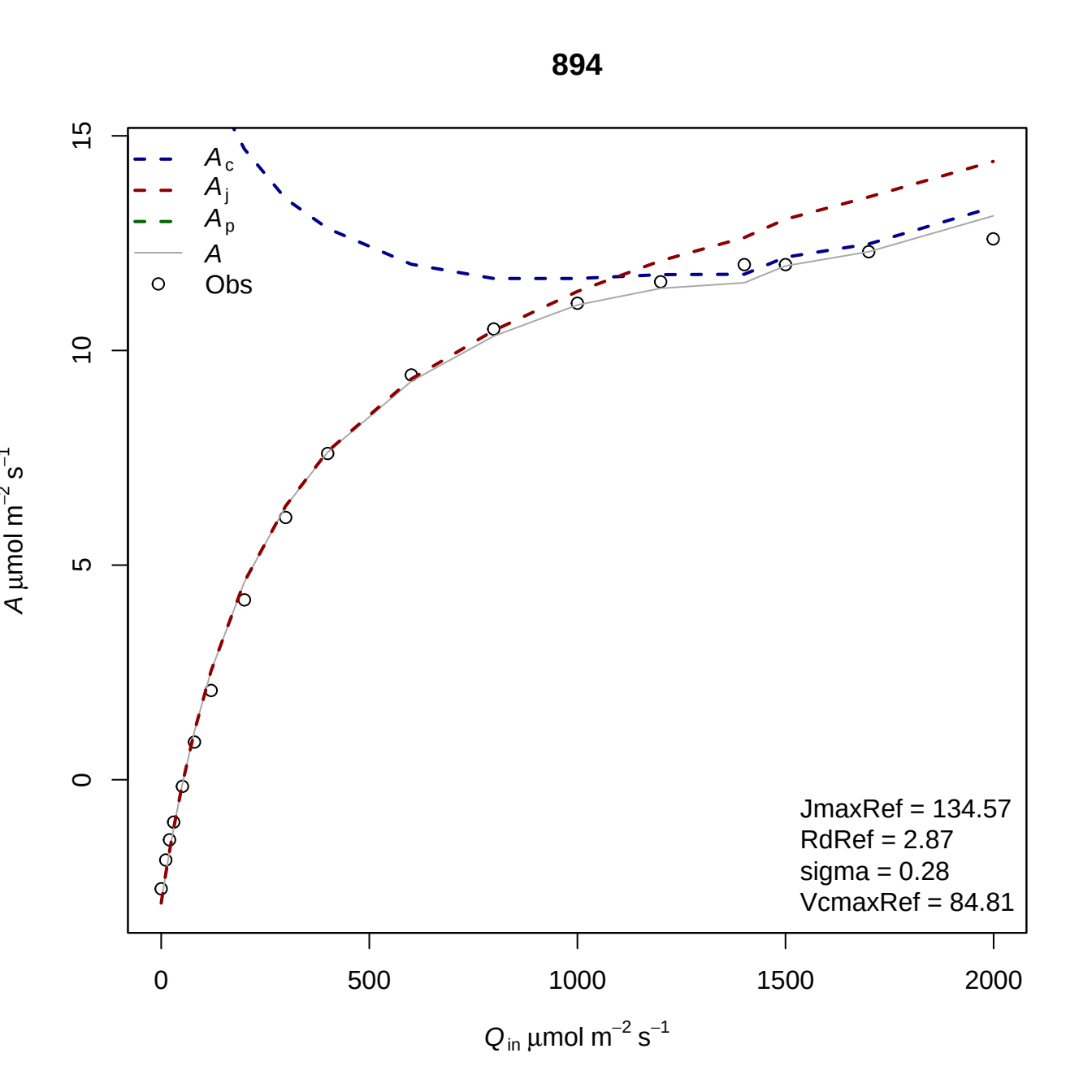

889
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.75
RdRef = 1.68
sigma = 0.35
VcmaxRef = 55651.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

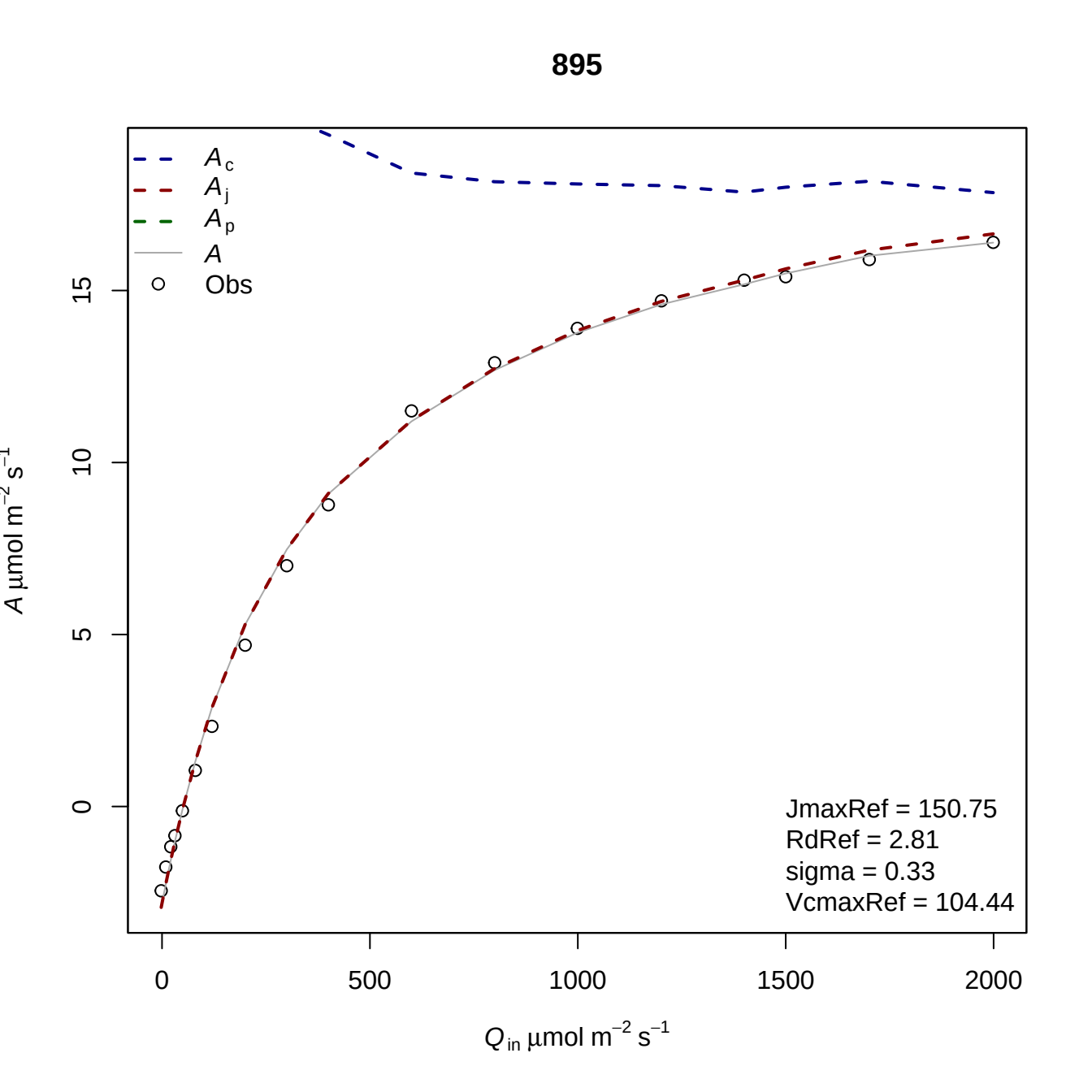

890
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.92
RdRef = 1.89
sigma = 0.48
VcmaxRef = 101877.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

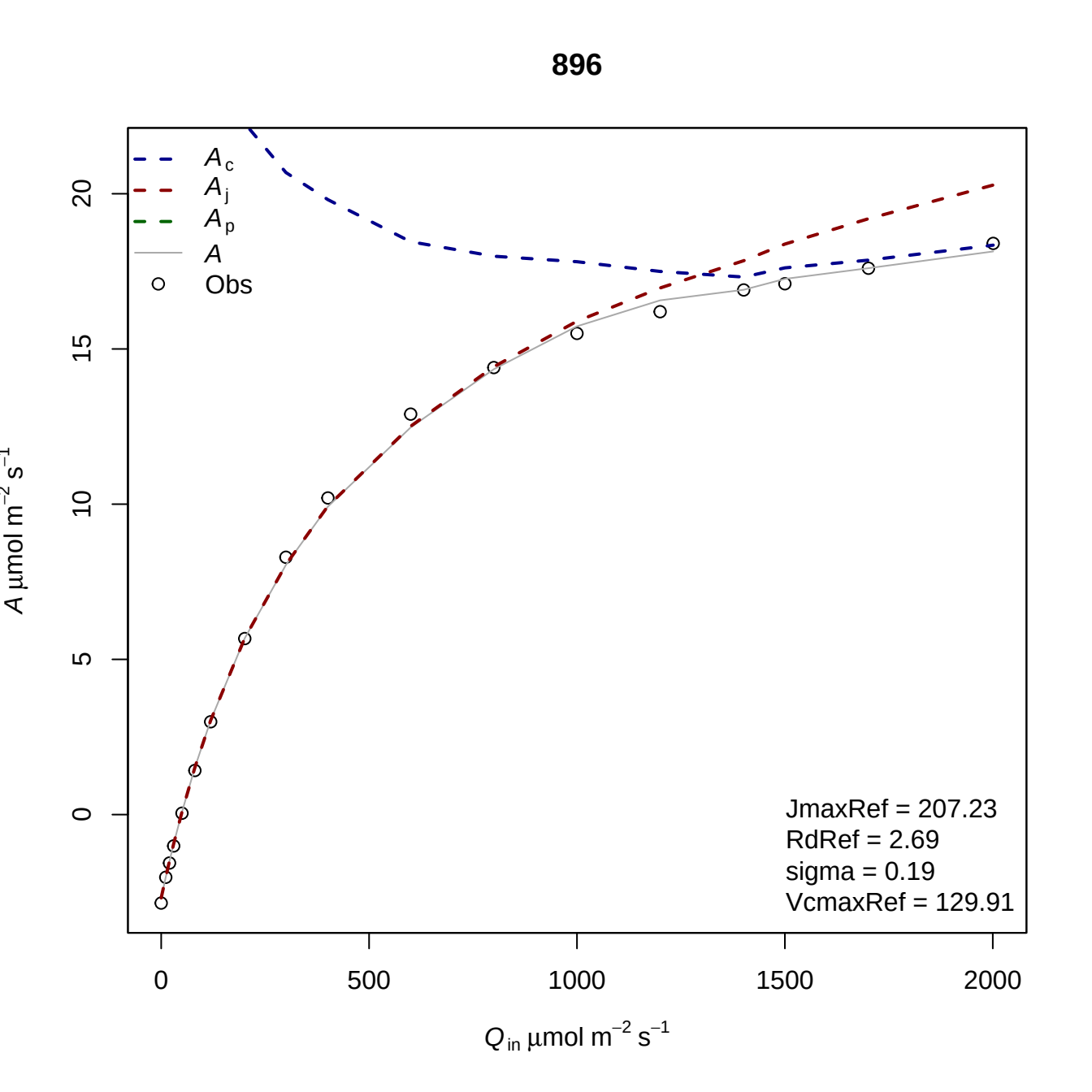

891
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.32
RdRef = 1.08
sigma = 0.34
VcmaxRef = 62022.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

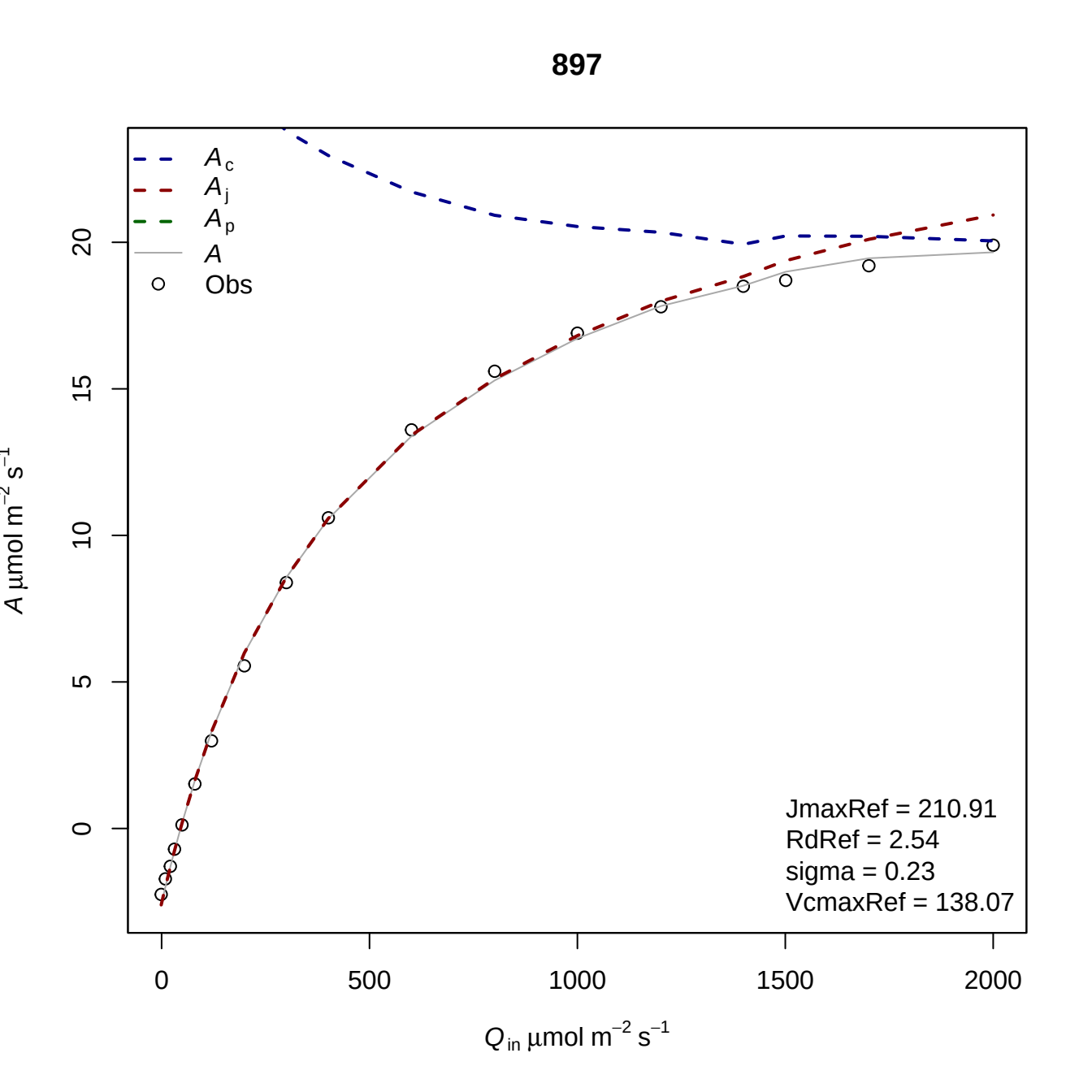

892
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.17
RdRef = 2.35
sigma = 0.24
VcmaxRef = 3785461.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

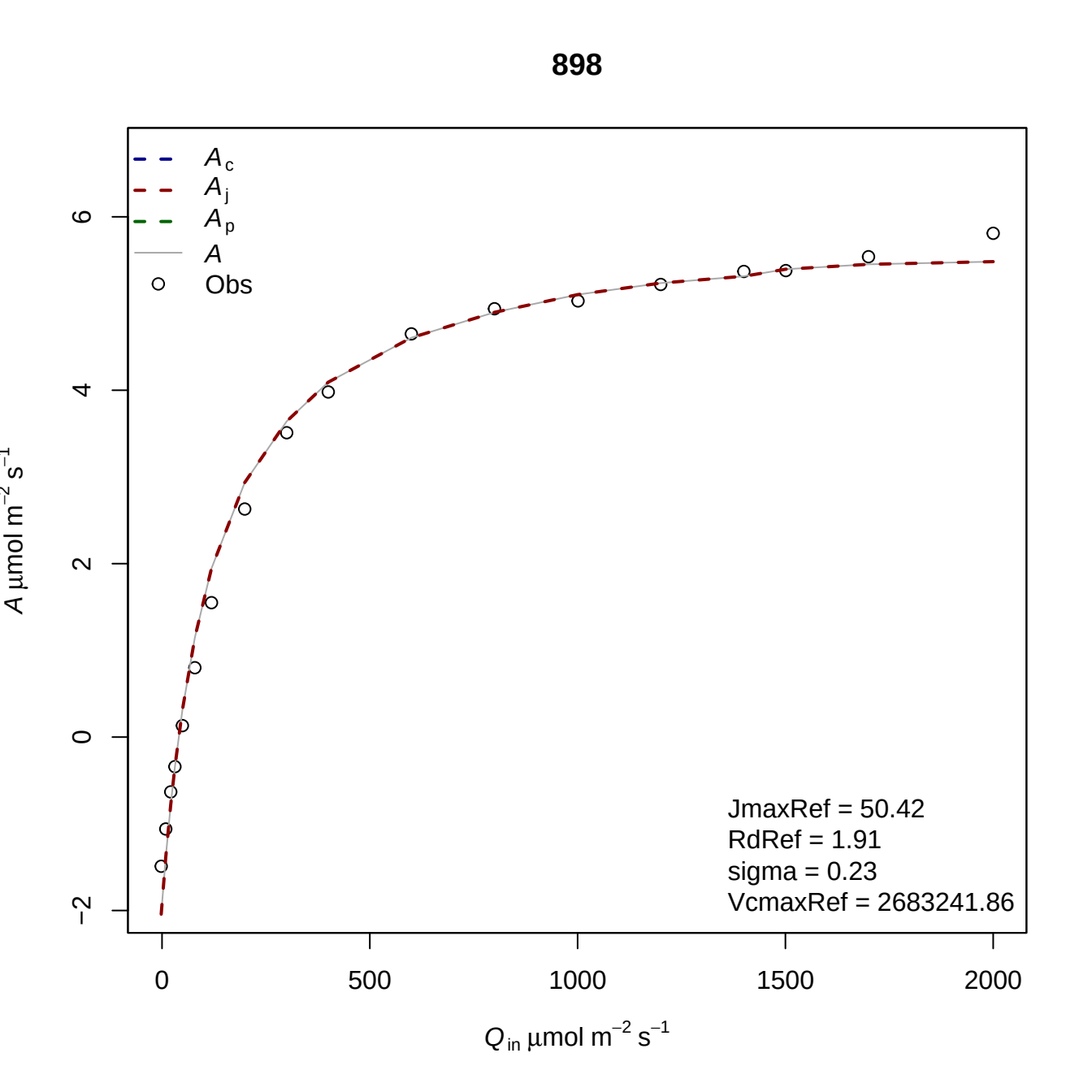

893
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.45
RdRef = 2.46
sigma = 0.29
VcmaxRef = 77835.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

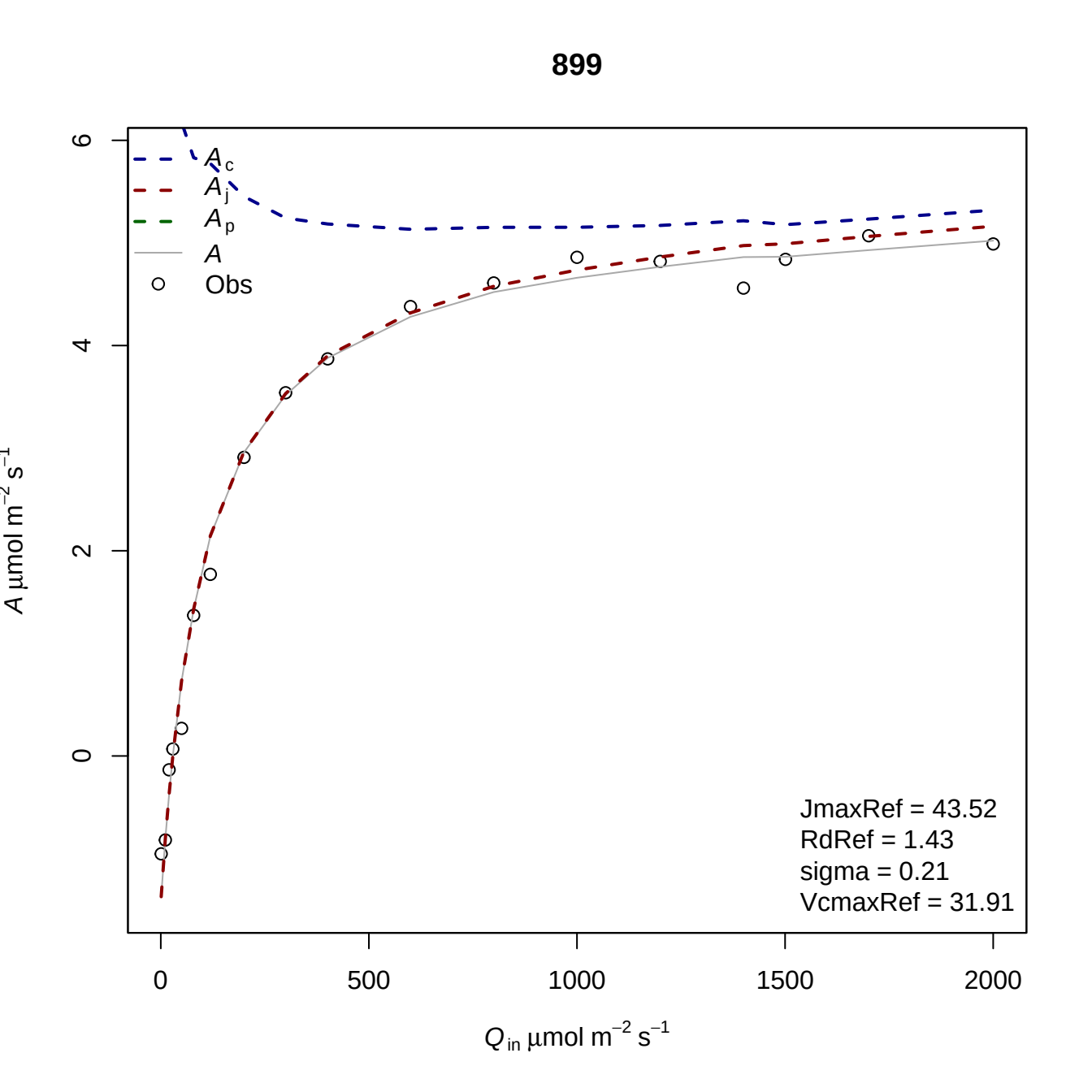

894
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.57
RdRef = 2.87
sigma = 0.28
VcmaxRef = 84.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

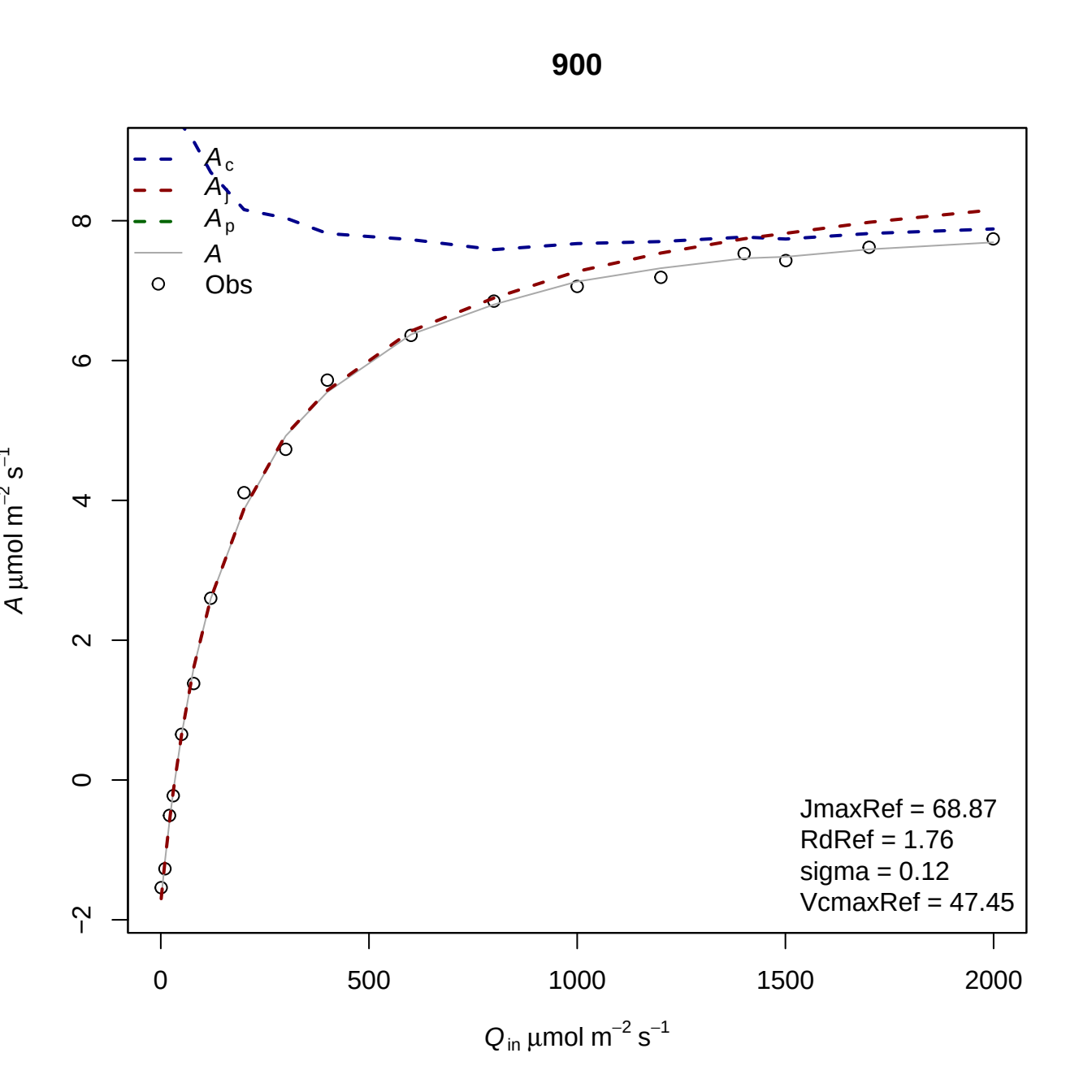

895
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 150.75
RdRef = 2.81
sigma = 0.33
VcmaxRef = 104.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

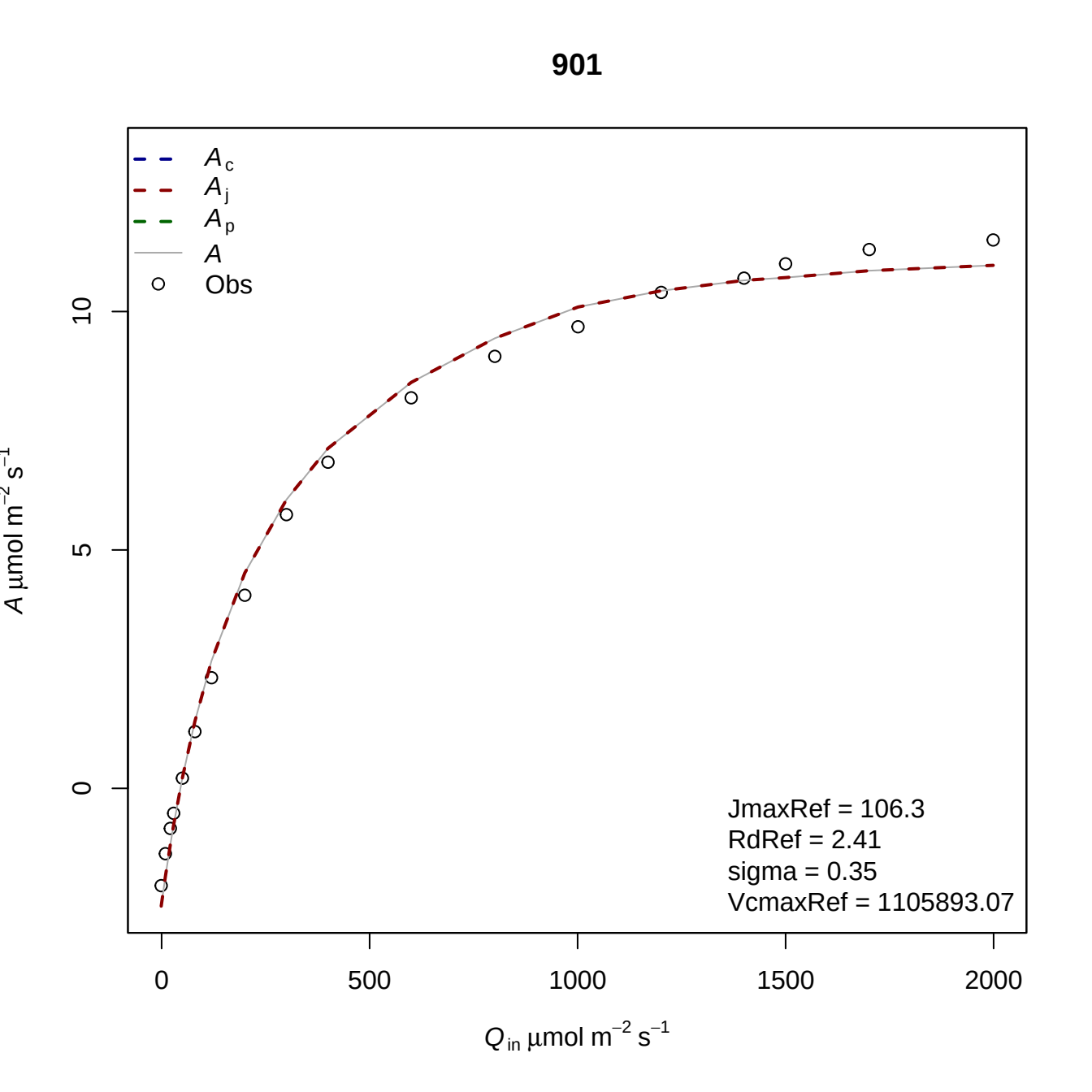

896
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 207.23
RdRef = 2.69
sigma = 0.19
VcmaxRef = 129.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

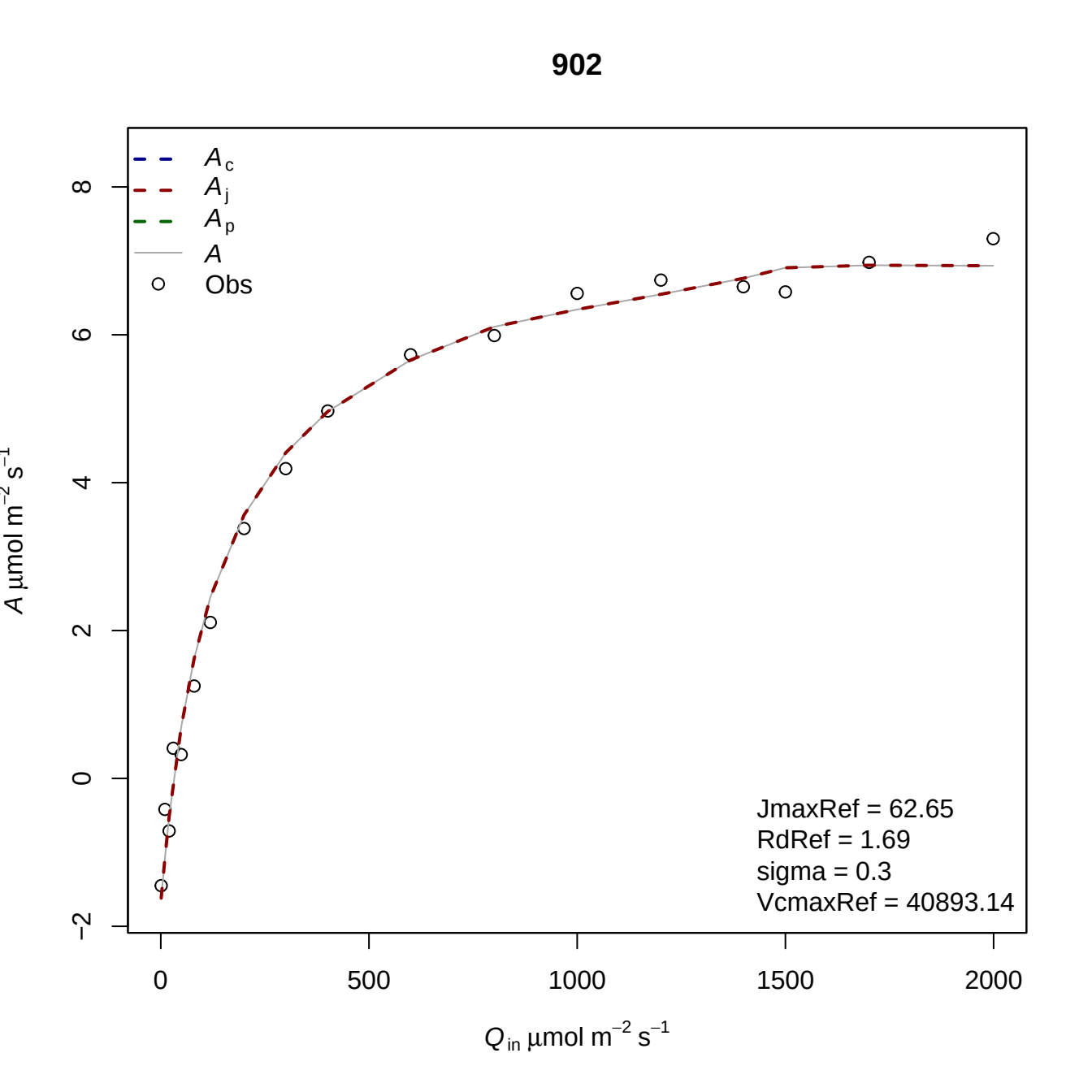

897
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 210.91
RdRef = 2.54
sigma = 0.23
VcmaxRef = 138.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

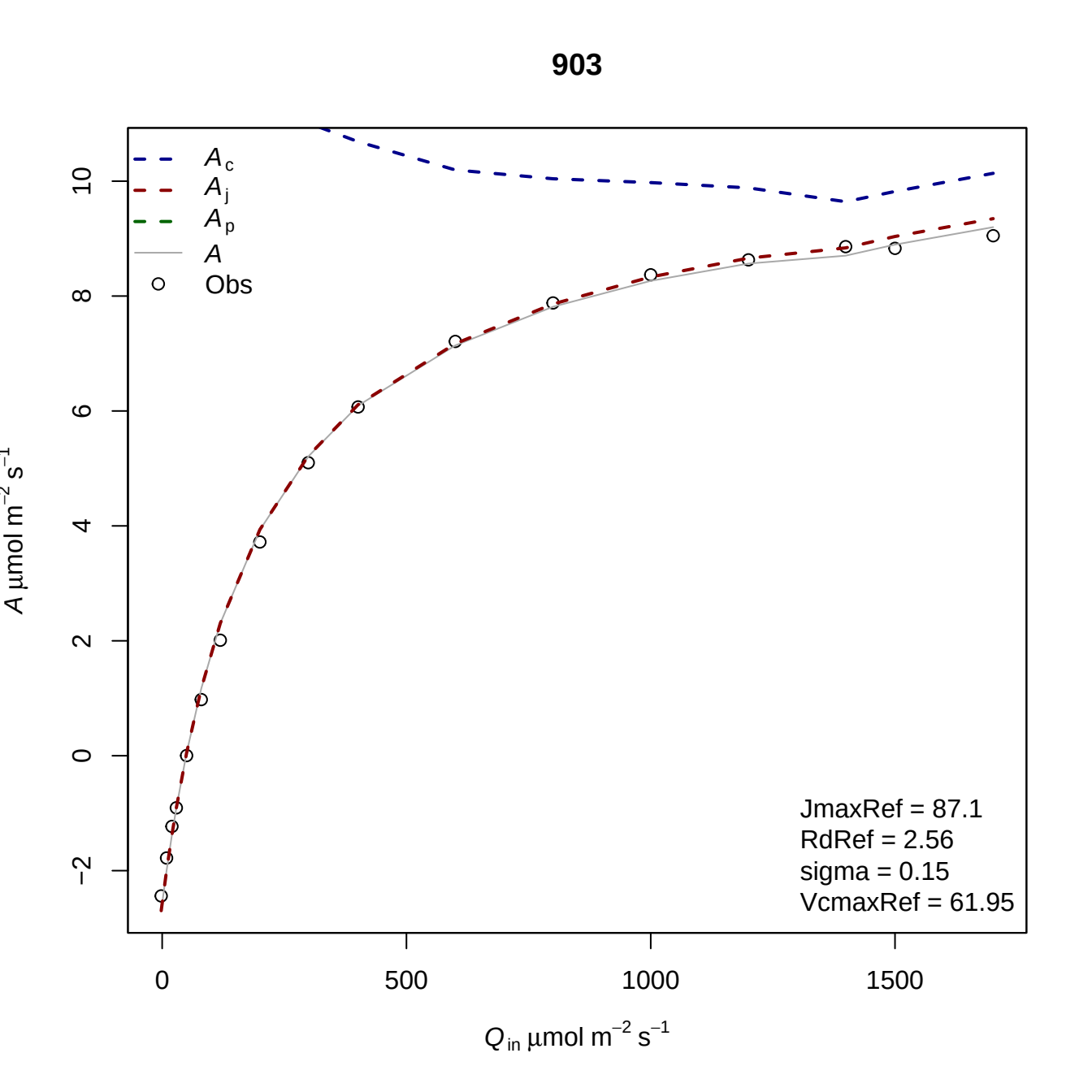

898
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.42
RdRef = 1.91
sigma = 0.23
VcmaxRef = 2683241.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

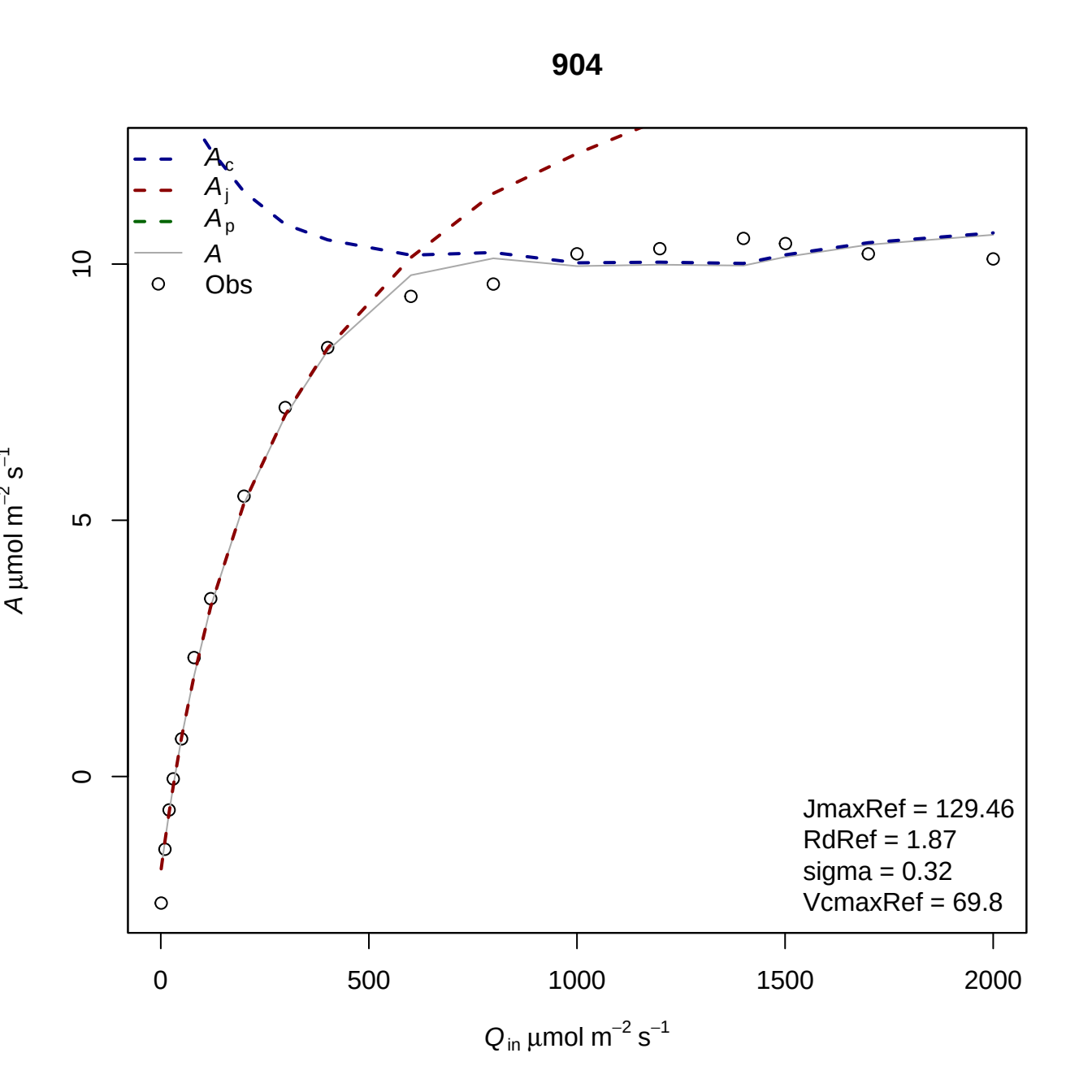

899
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.52
RdRef = 1.43
sigma = 0.21
VcmaxRef = 31.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

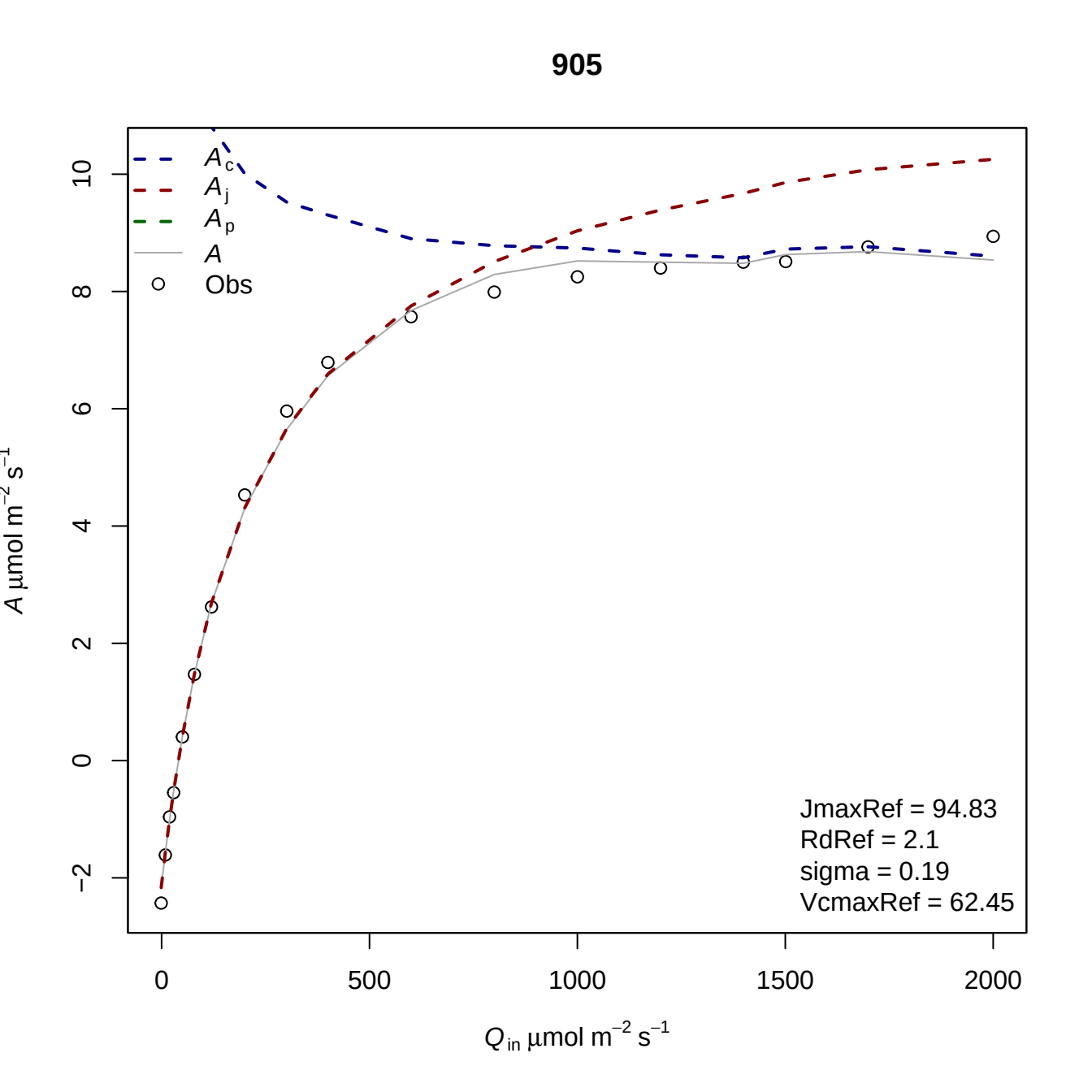

900
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.87
RdRef = 1.76
sigma = 0.12
VcmaxRef = 47.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

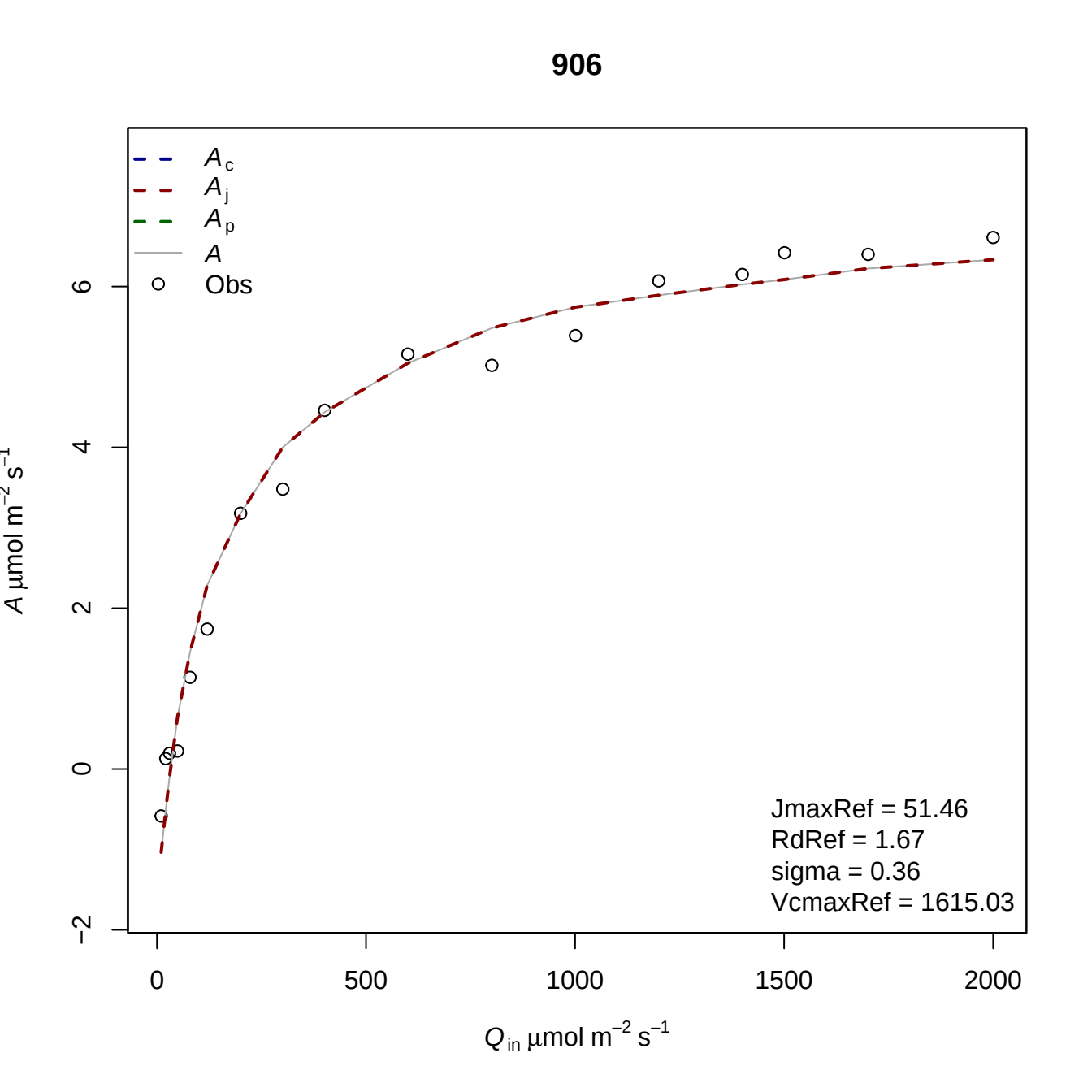

901
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 106.3
RdRef = 2.41
sigma = 0.35
VcmaxRef = 1105893.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

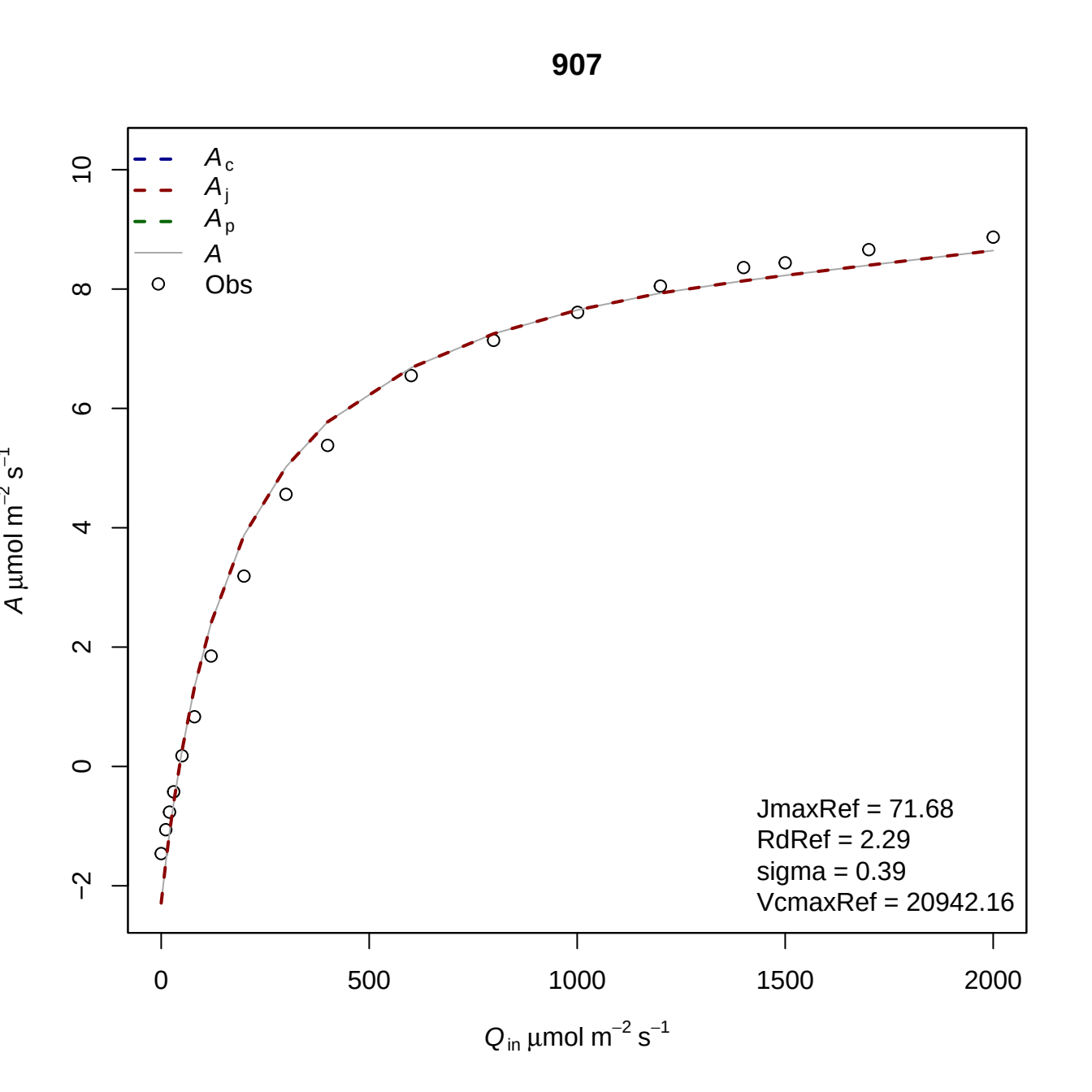

902
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.65
RdRef = 1.69
sigma = 0.3
VcmaxRef = 40893.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

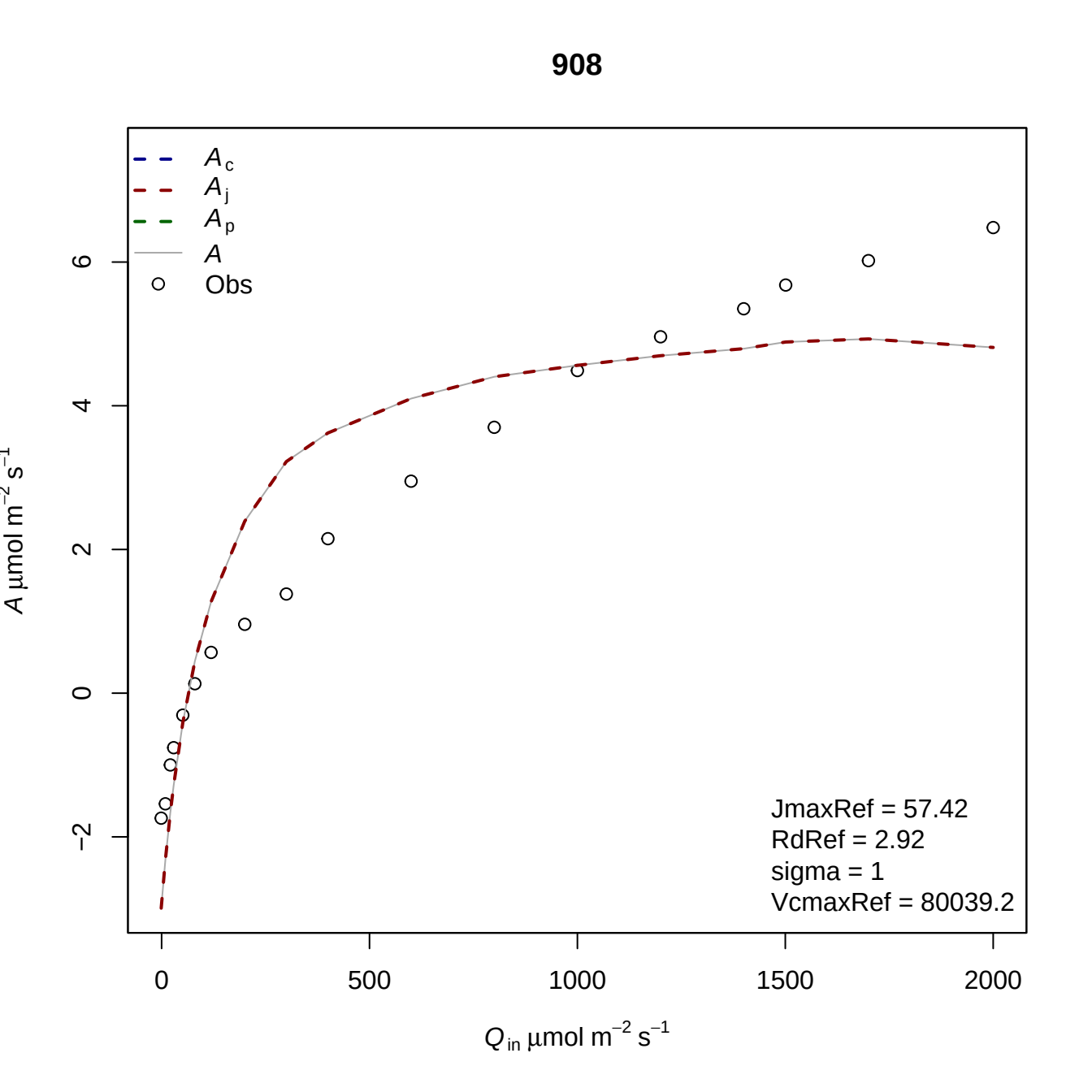

903
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.1
RdRef = 2.56
sigma = 0.15
VcmaxRef = 61.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

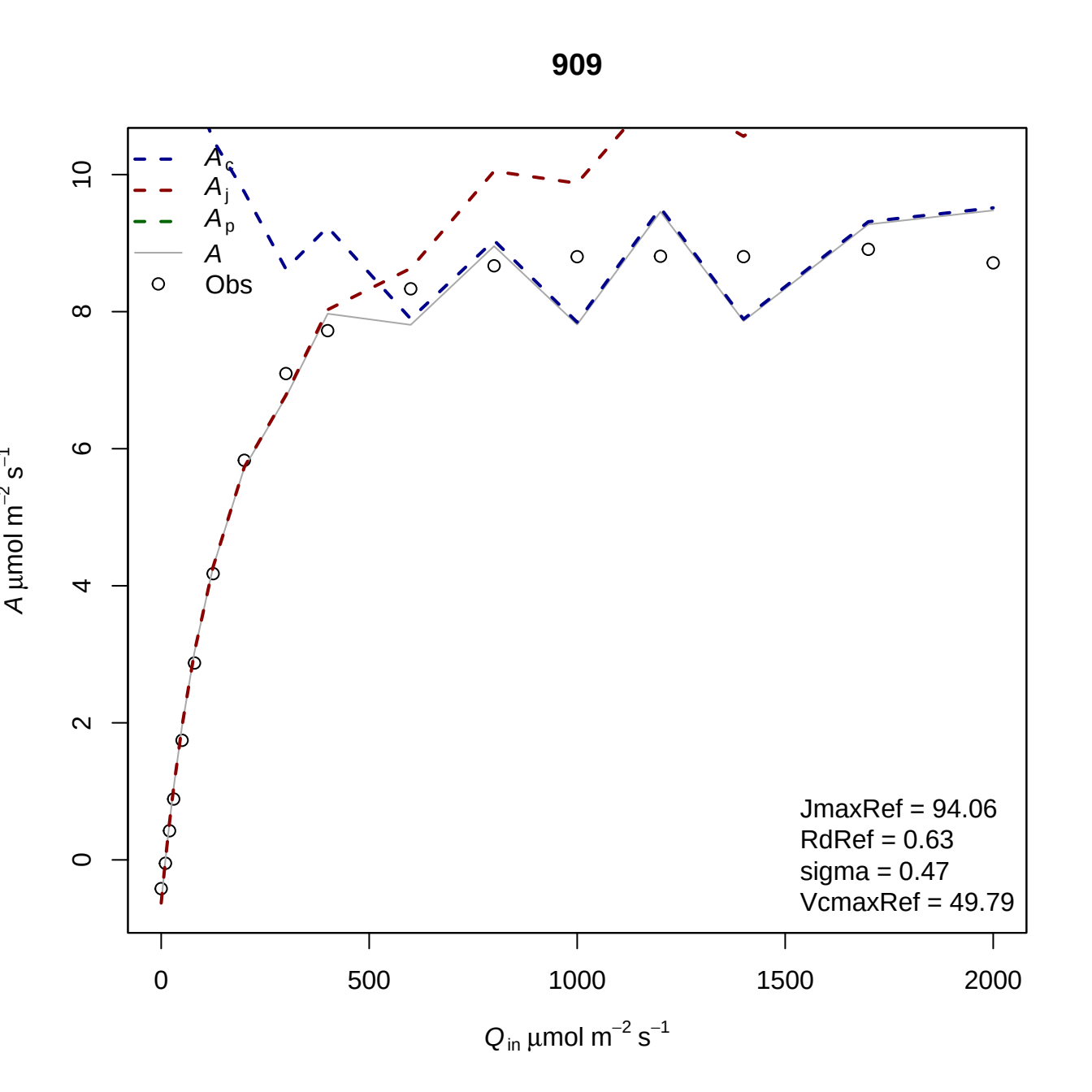

904
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.46
RdRef = 1.87
sigma = 0.32
VcmaxRef = 69.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

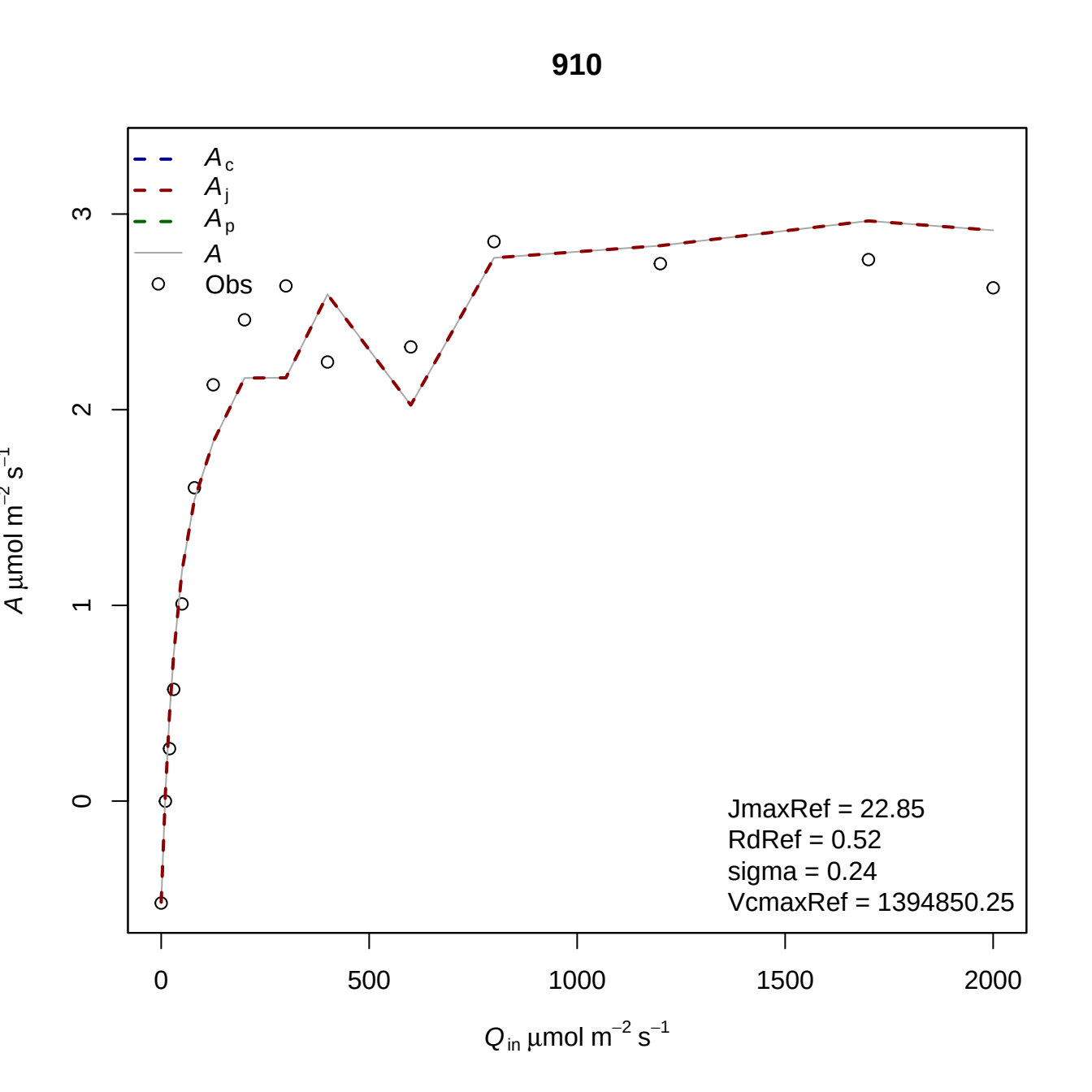

905
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.83
RdRef = 2.1
sigma = 0.19
VcmaxRef = 62.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

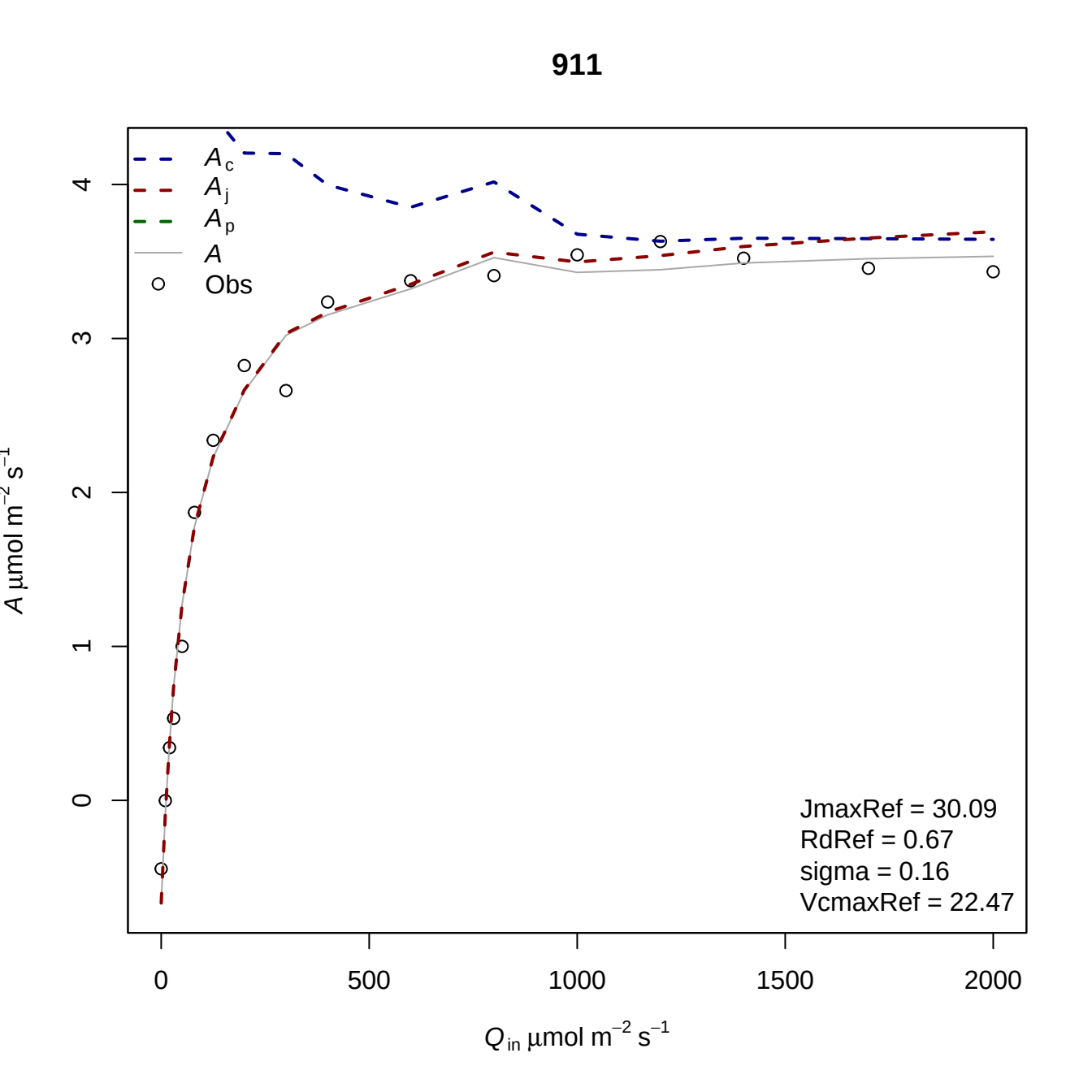

906
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.46
RdRef = 1.67
sigma = 0.36
VcmaxRef = 1615.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

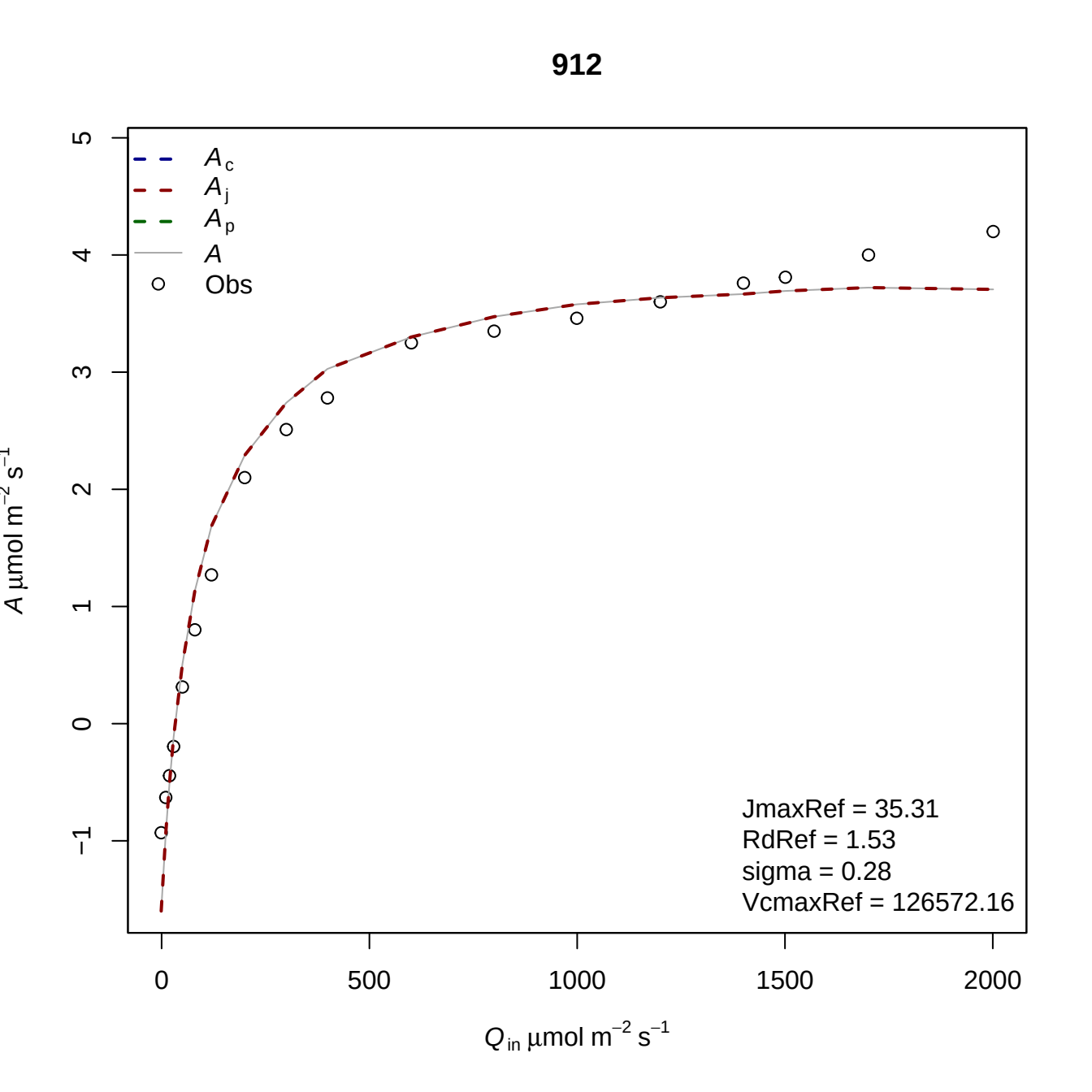

907
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.68
RdRef = 2.29
sigma = 0.39
VcmaxRef = 20942.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

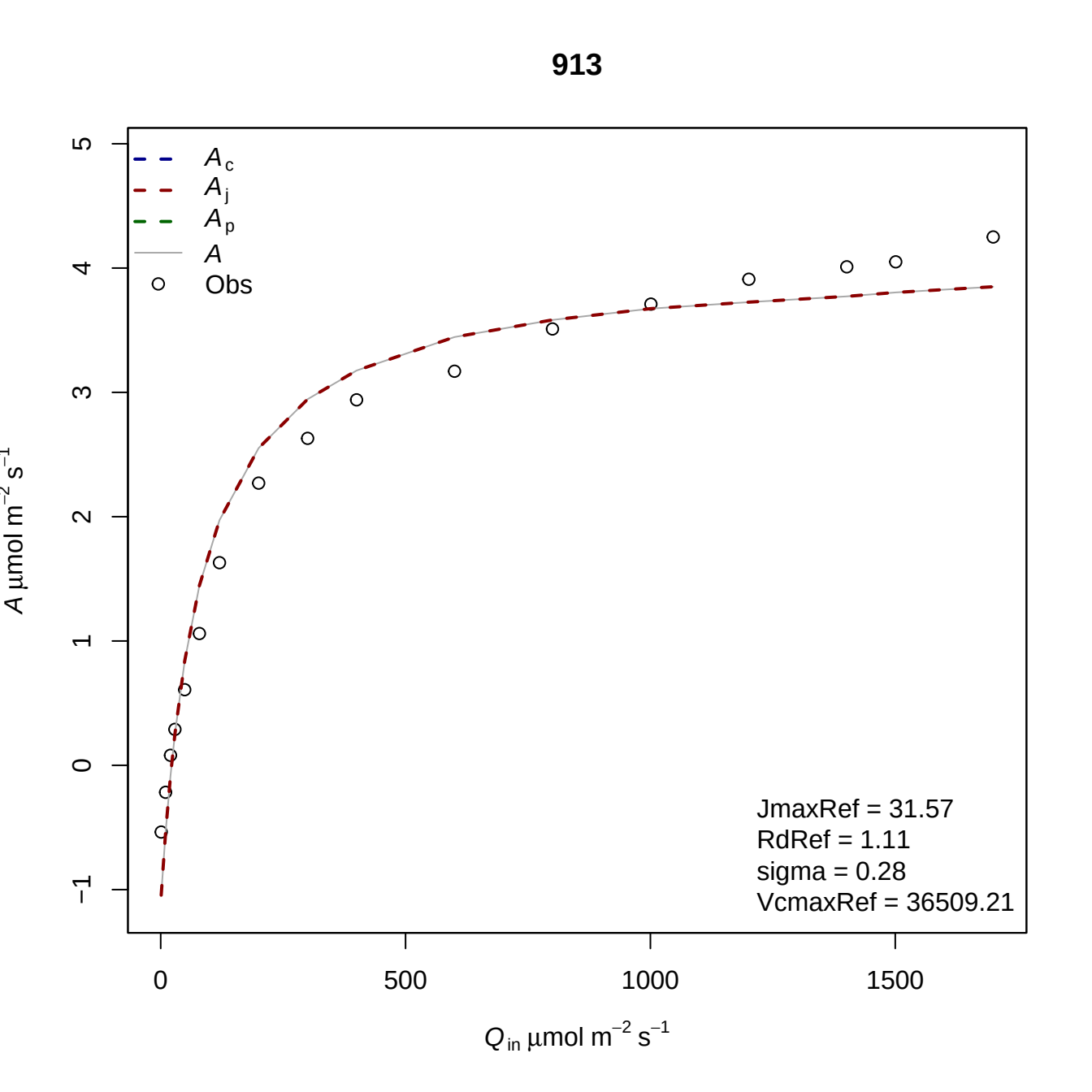

908
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.42
RdRef = 2.92
sigma = 1
VcmaxRef = 80039.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

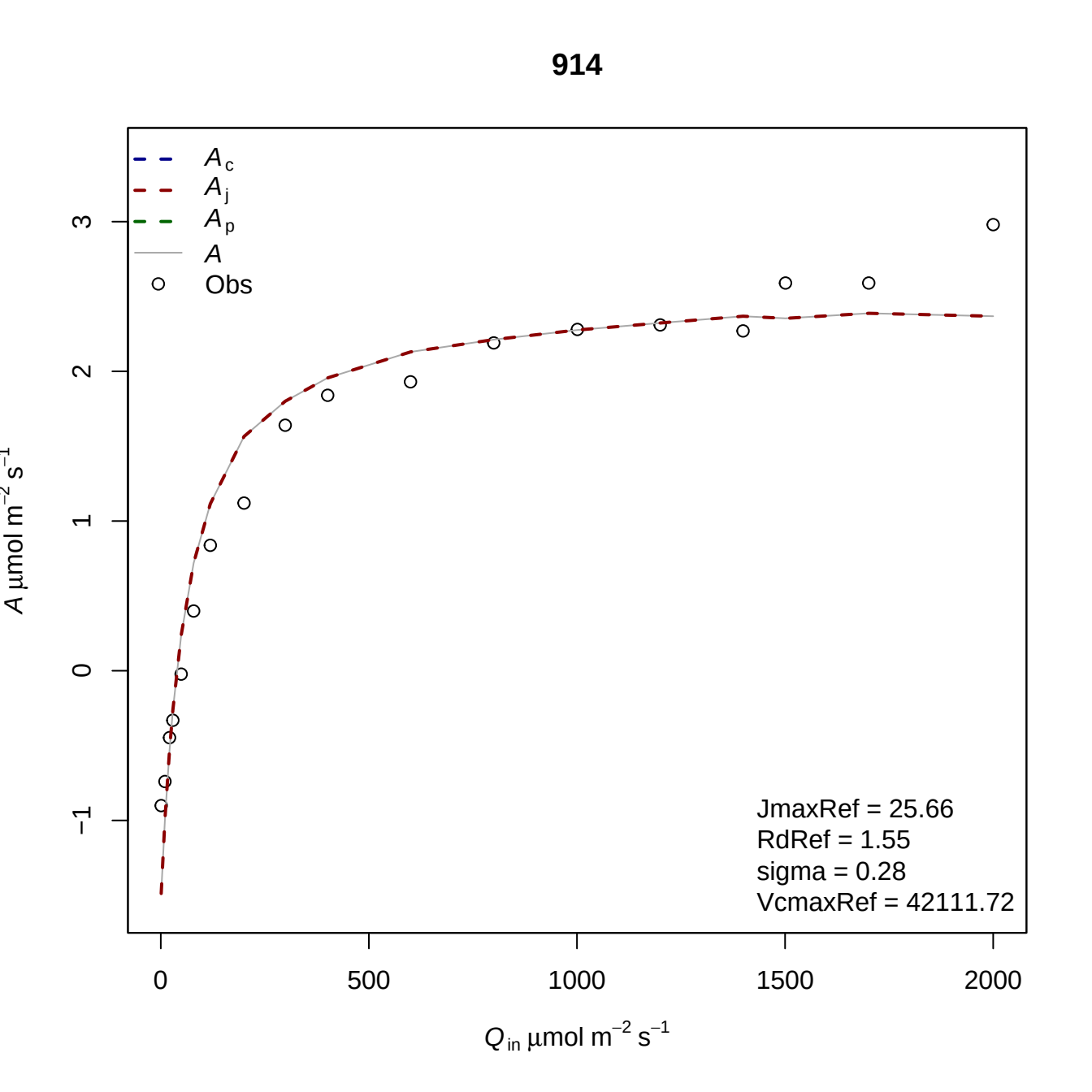

909
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.06
RdRef = 0.63
sigma = 0.47
VcmaxRef = 49.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

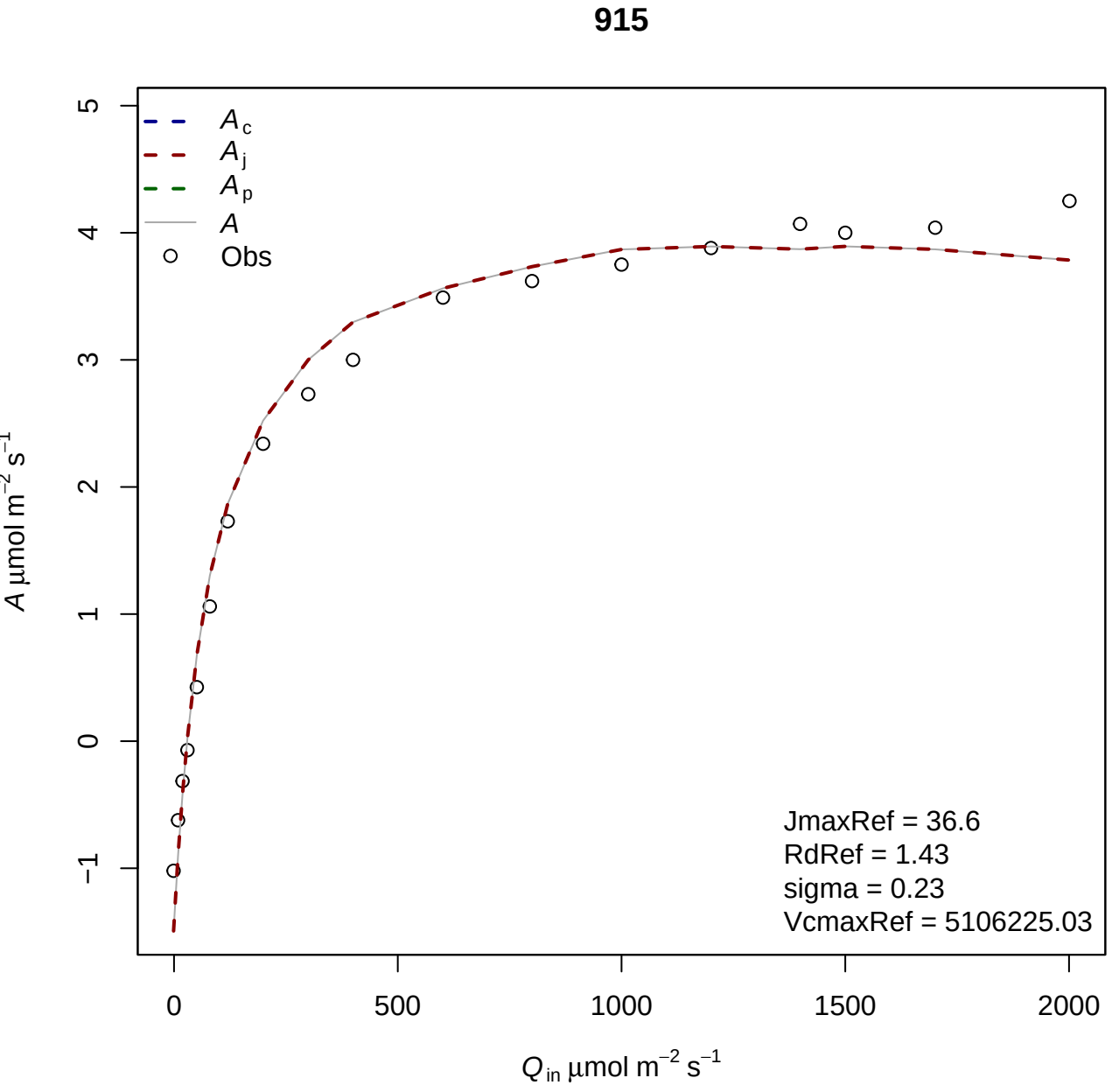

910
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.85
RdRef = 0.52
sigma = 0.24
VcmaxRef = 1394850.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500
2000

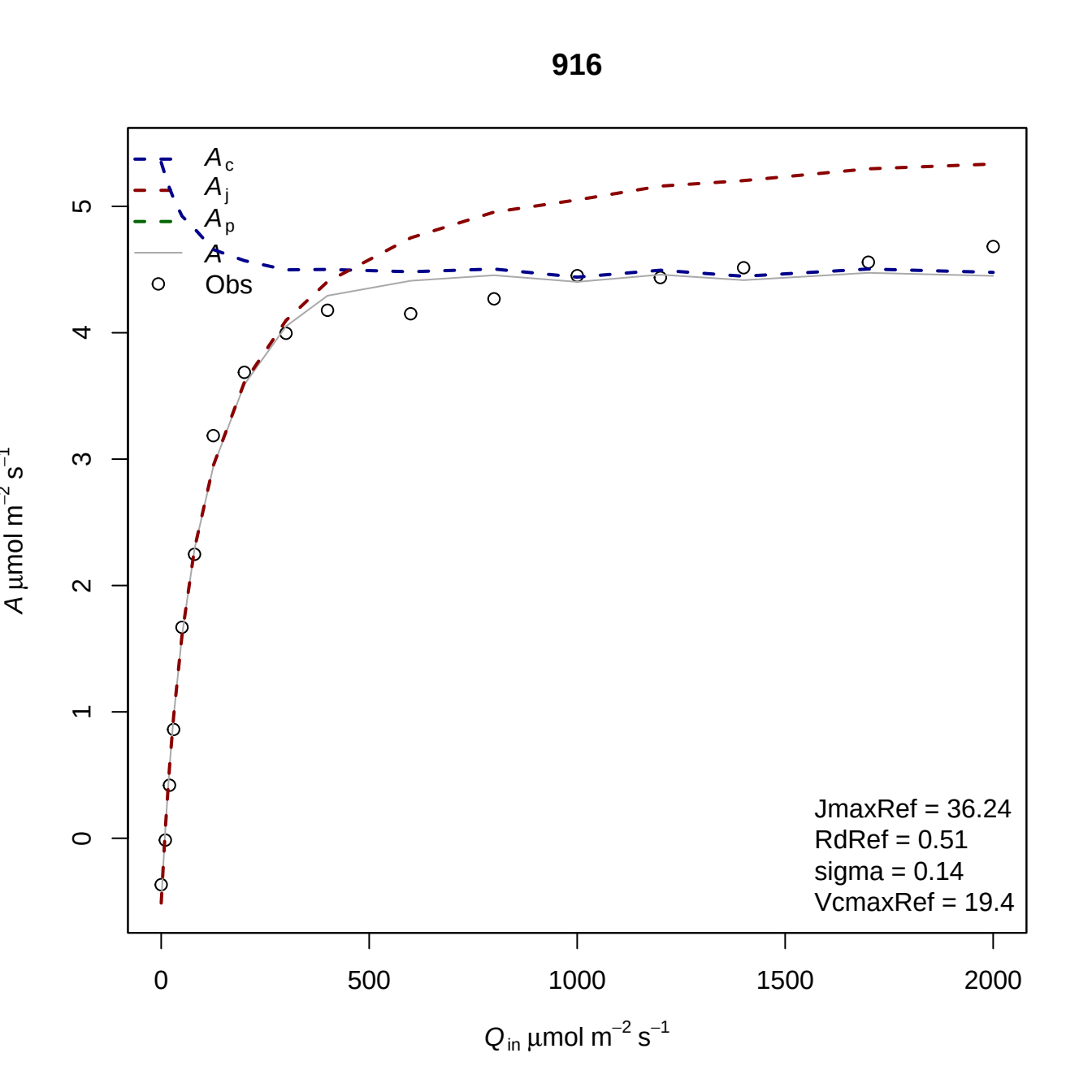

911
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.09
RdRef = 0.67
sigma = 0.16
VcmaxRef = 22.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500
2000

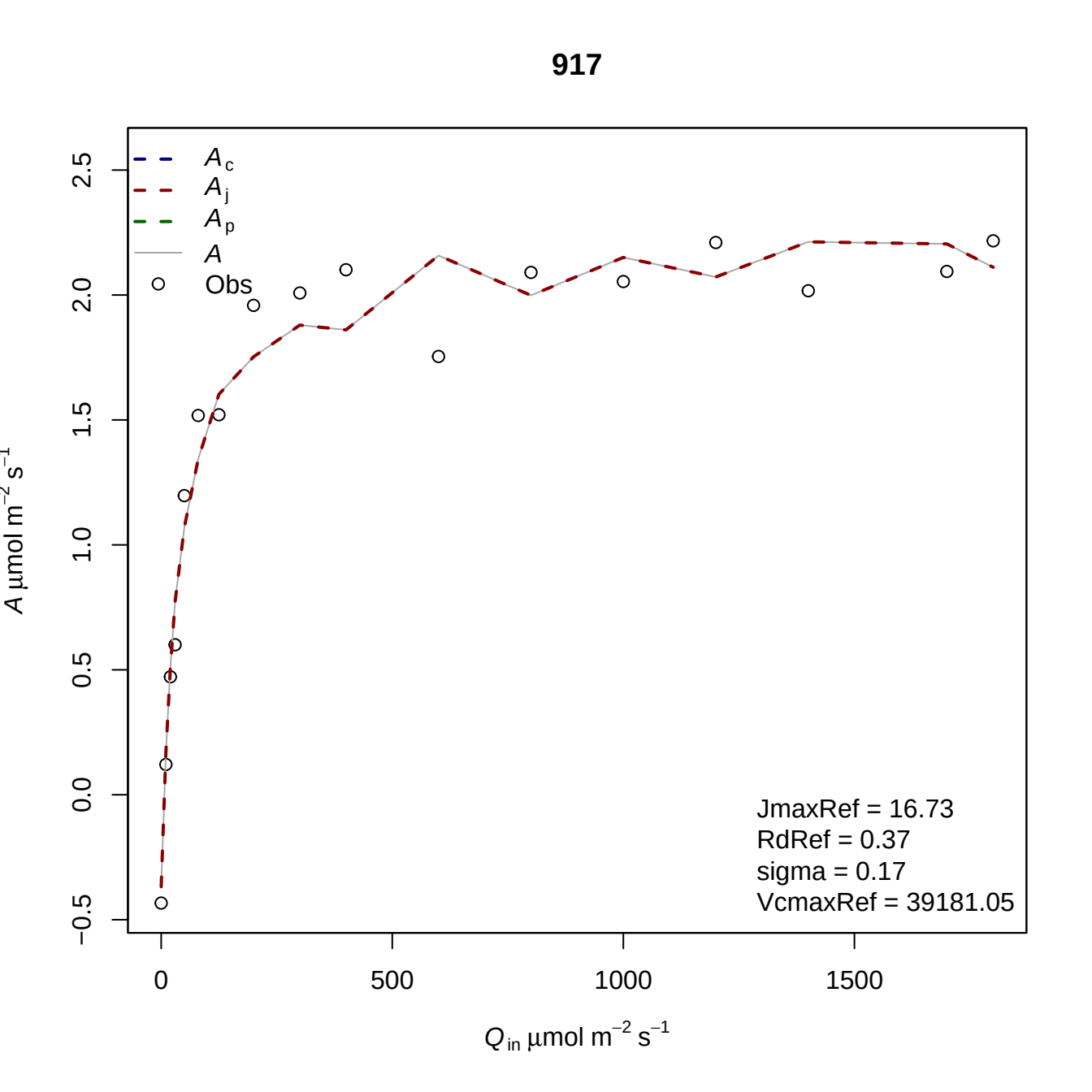

912
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 35.31
RdRef = 1.53
sigma = 0.28
VcmaxRef = 126572.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
5
4
3
2
1
0
−1
0
500
1000
1500
2000

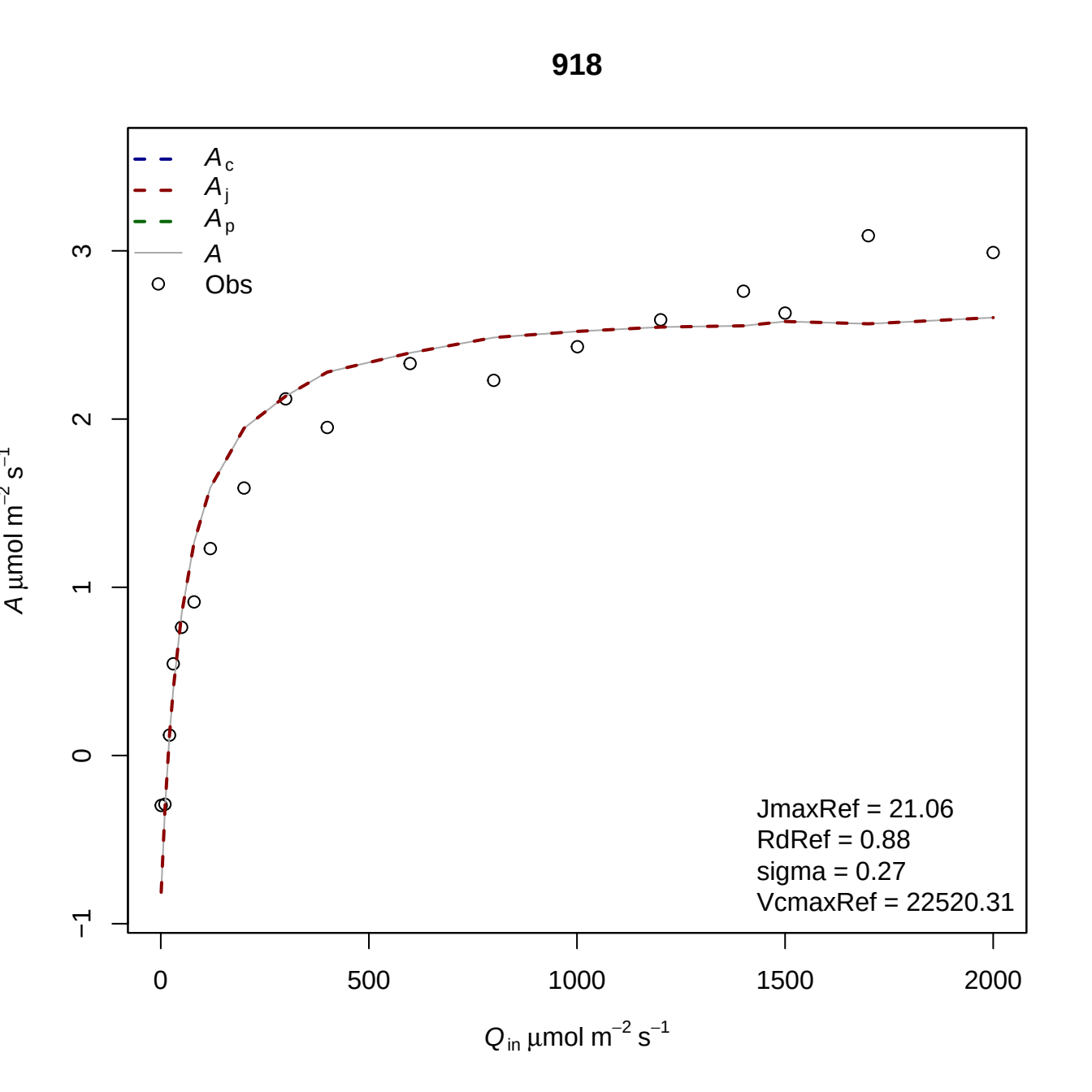

913
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.57
RdRef = 1.11
sigma = 0.28
VcmaxRef = 36509.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

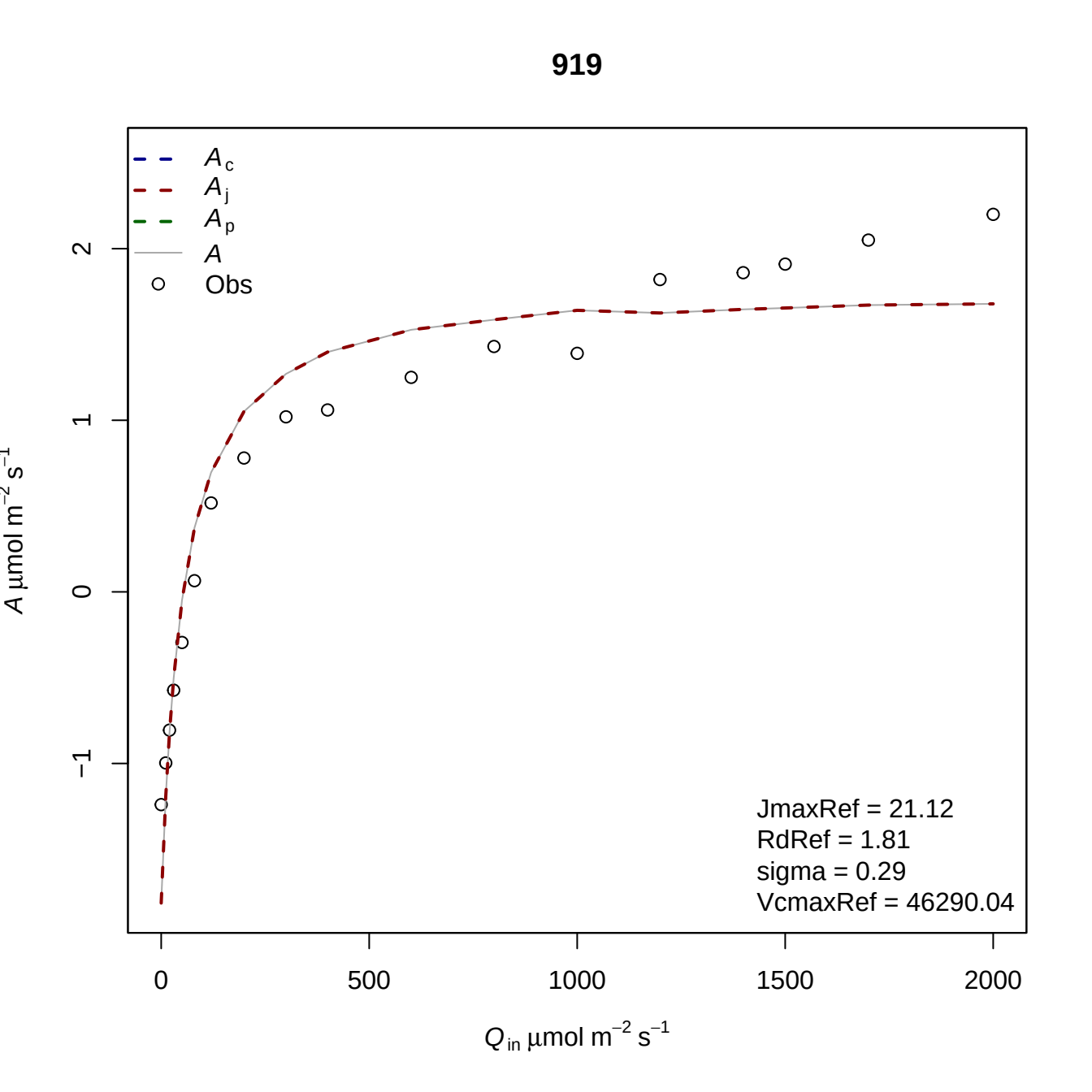

914
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.66
RdRef = 1.55
sigma = 0.28
VcmaxRef = 42111.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

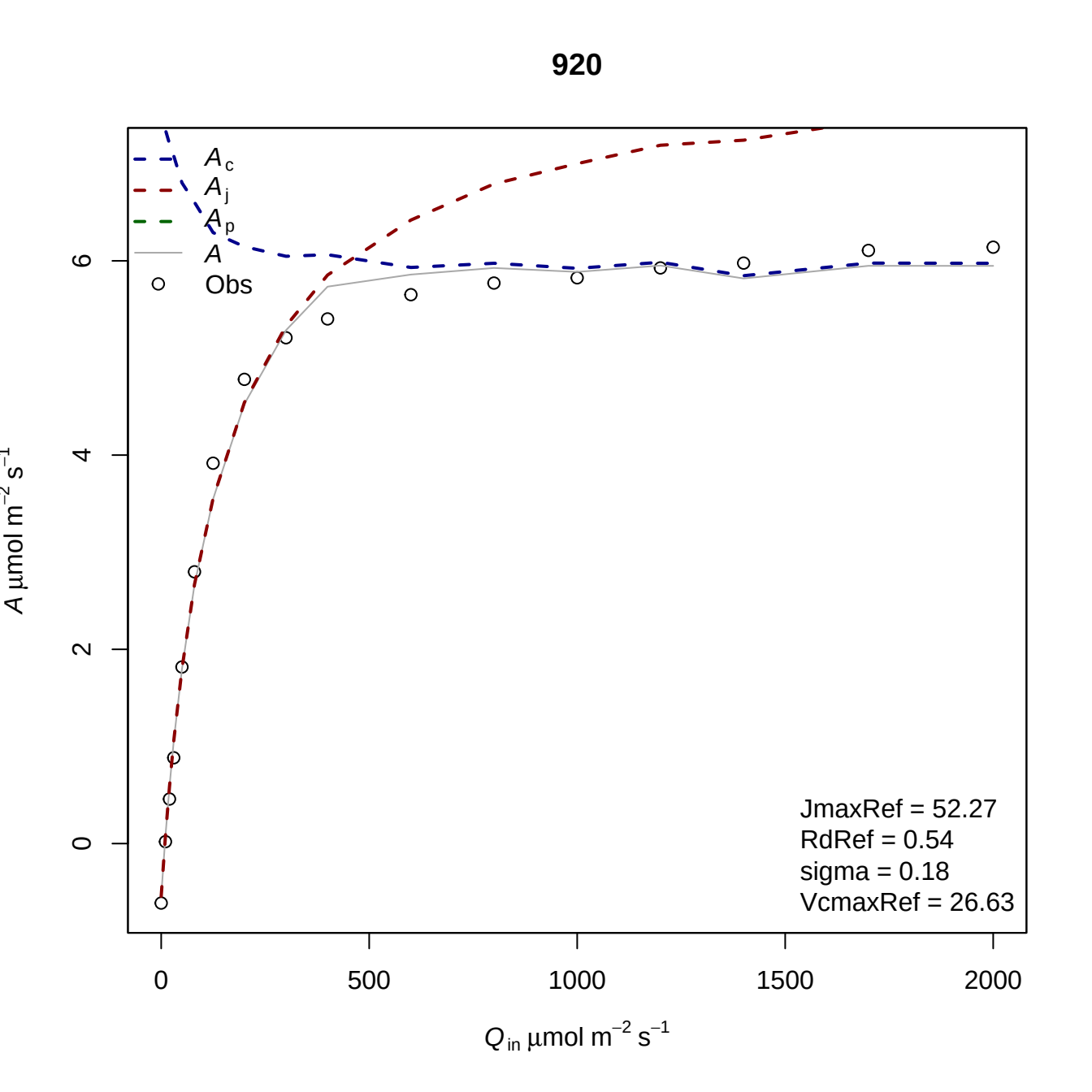

915
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.6
RdRef = 1.43
sigma = 0.23
VcmaxRef = 5106225.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
5
4
3
2
1
0
−1
0
500
1000
1500
2000

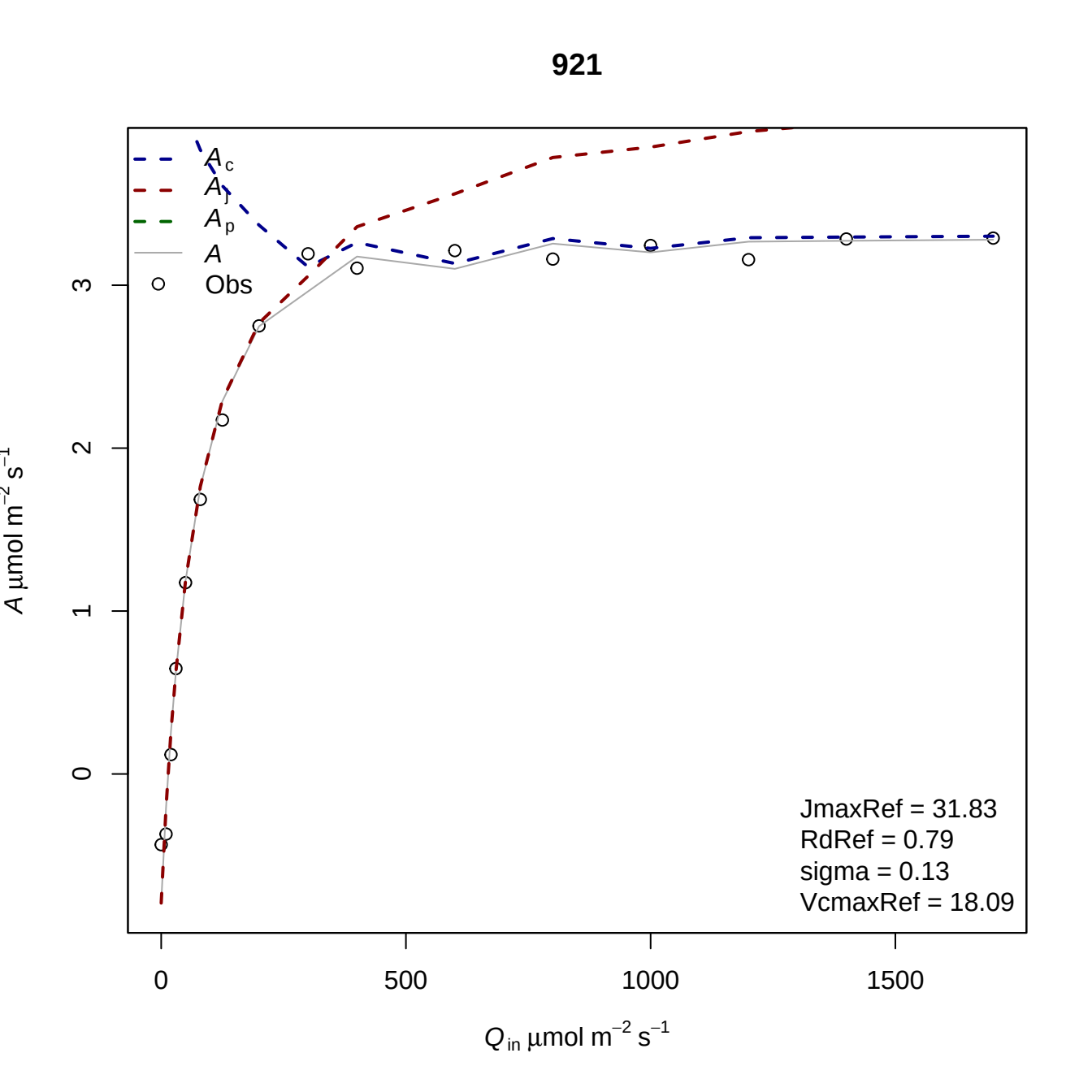

916
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.24
RdRef = 0.51
sigma = 0.14
VcmaxRef = 19.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500
2000

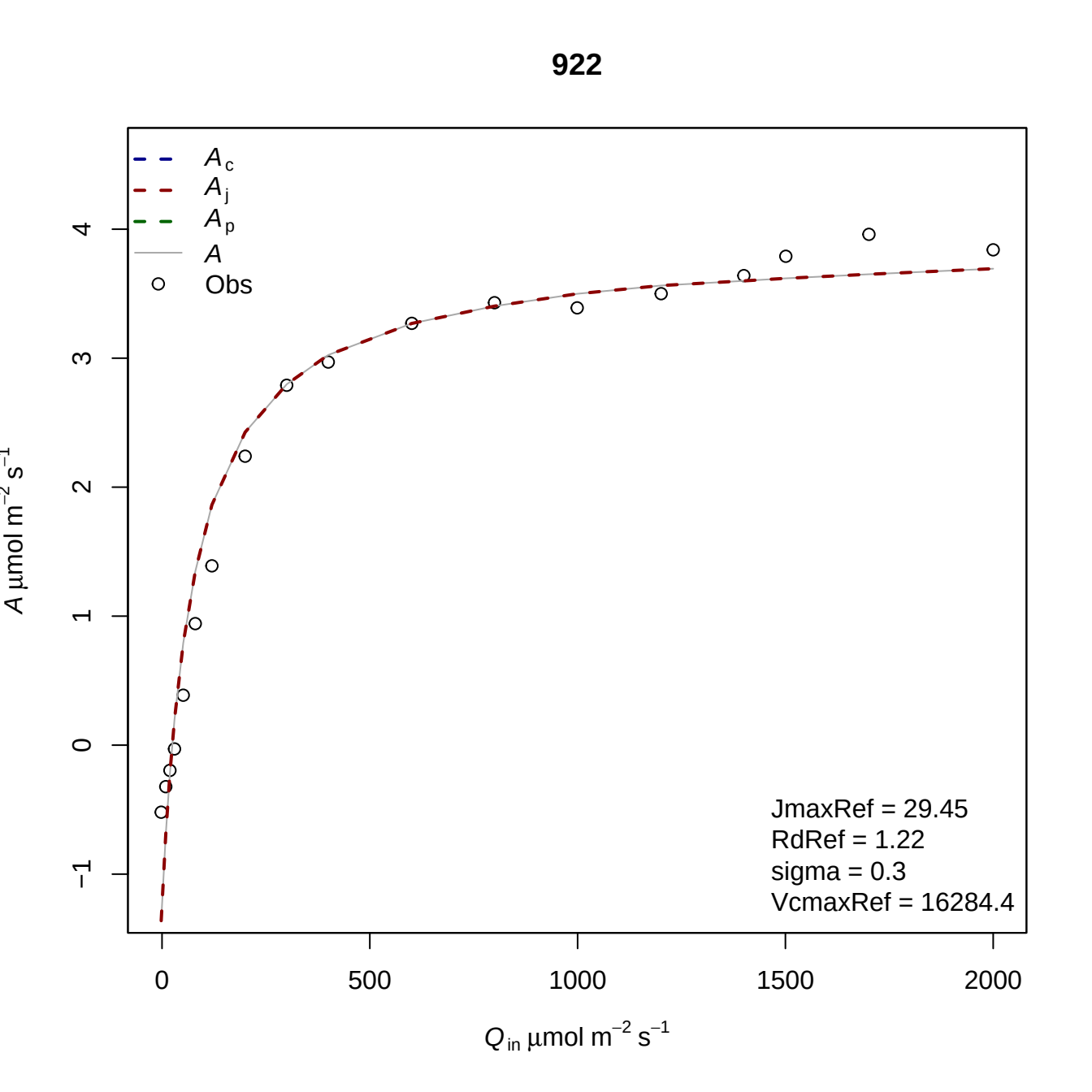

917
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 16.73
RdRef = 0.37
sigma = 0.17
VcmaxRef = 39181.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

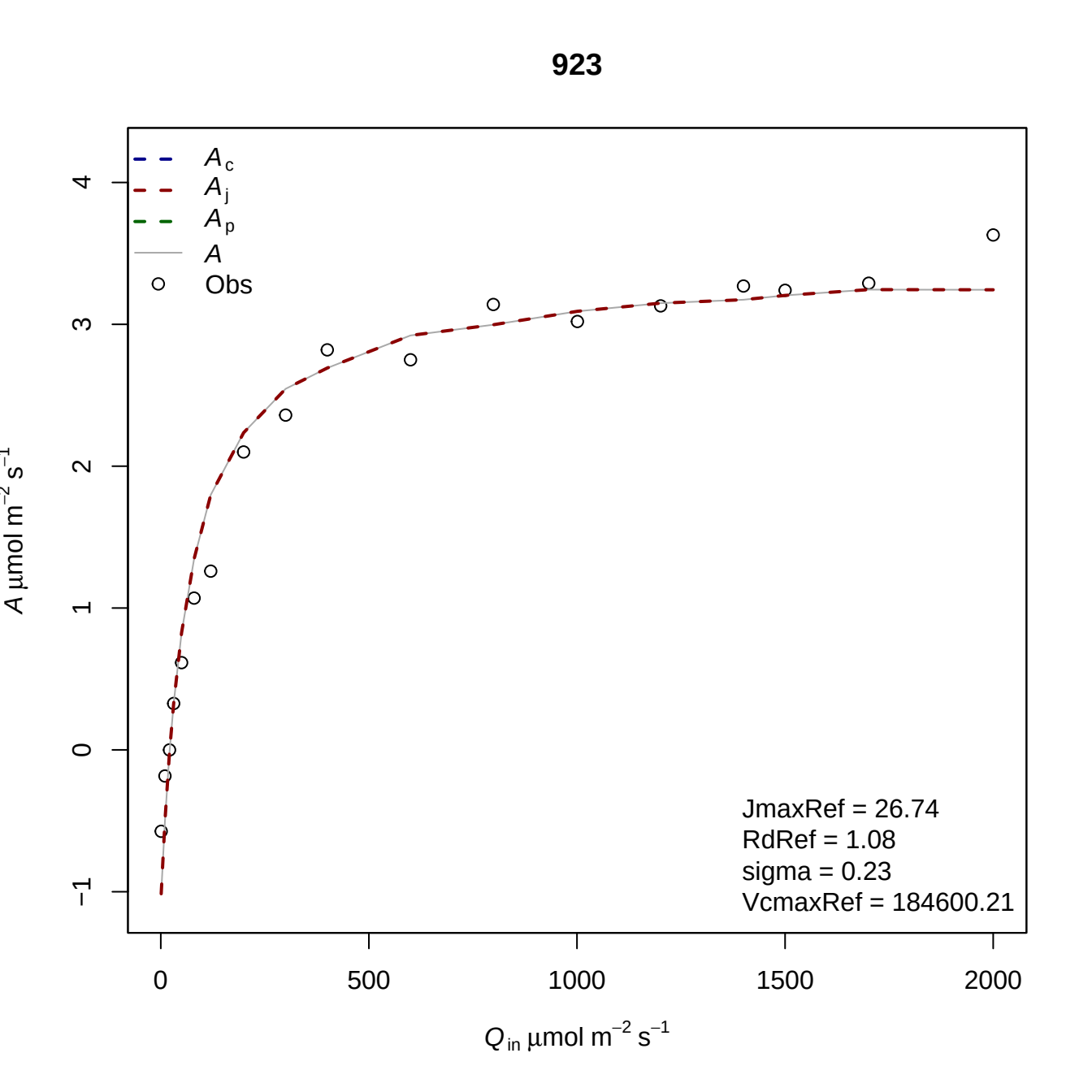

918
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.06
RdRef = 0.88
sigma = 0.27
VcmaxRef = 22520.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

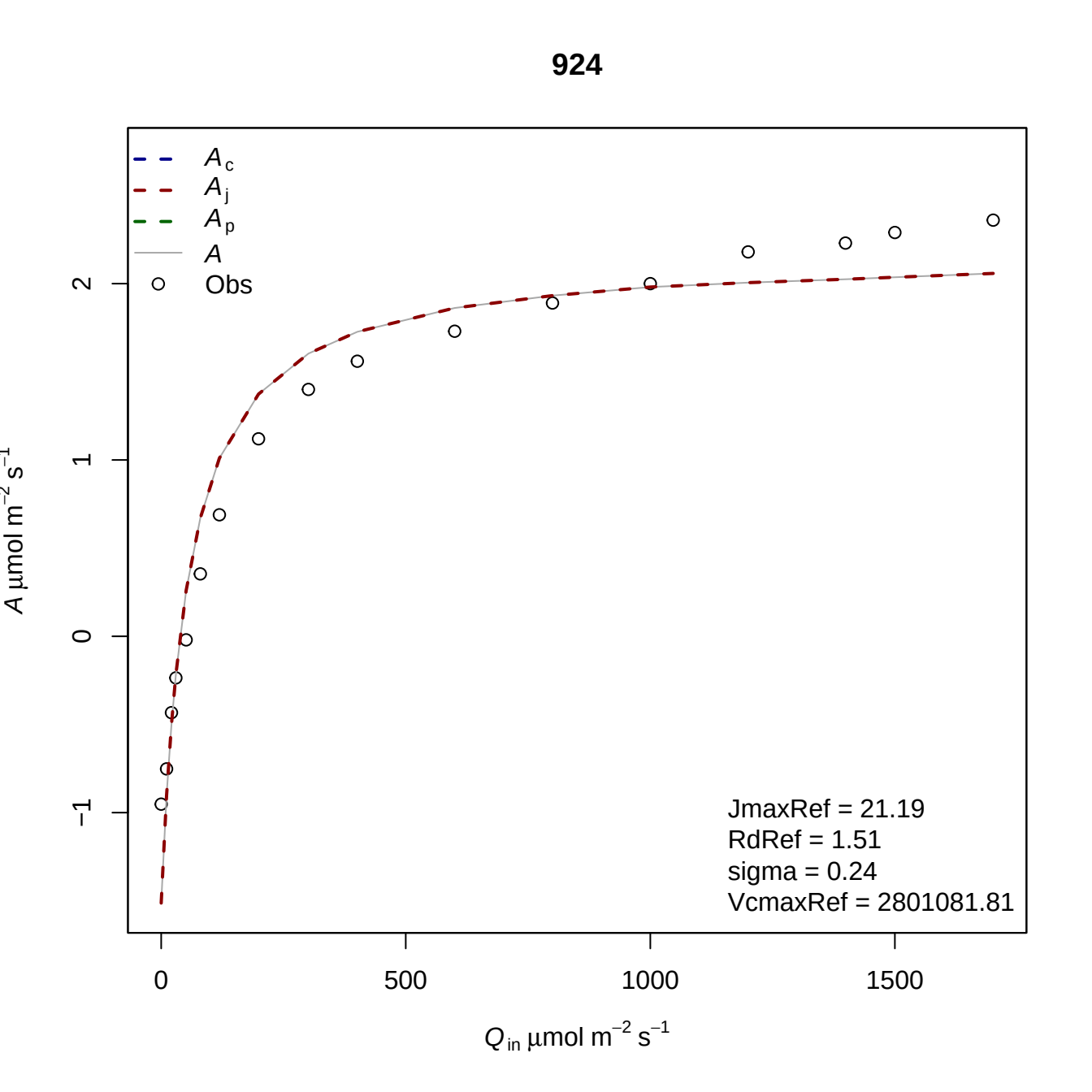

919
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.12
RdRef = 1.81
sigma = 0.29
VcmaxRef = 46290.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
500
1000
1500
2000

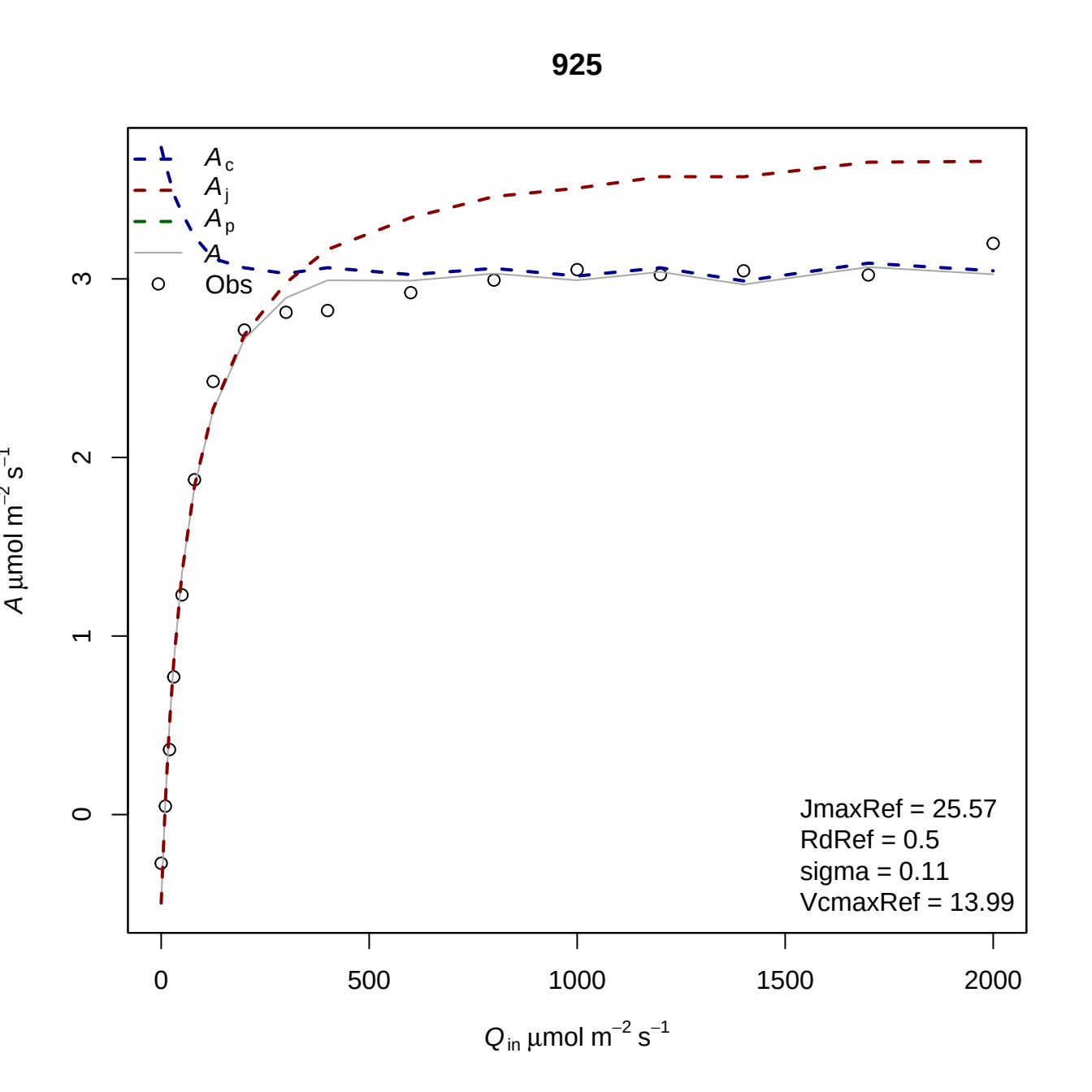

920
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 52.27
RdRef = 0.54
sigma = 0.18
VcmaxRef = 26.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

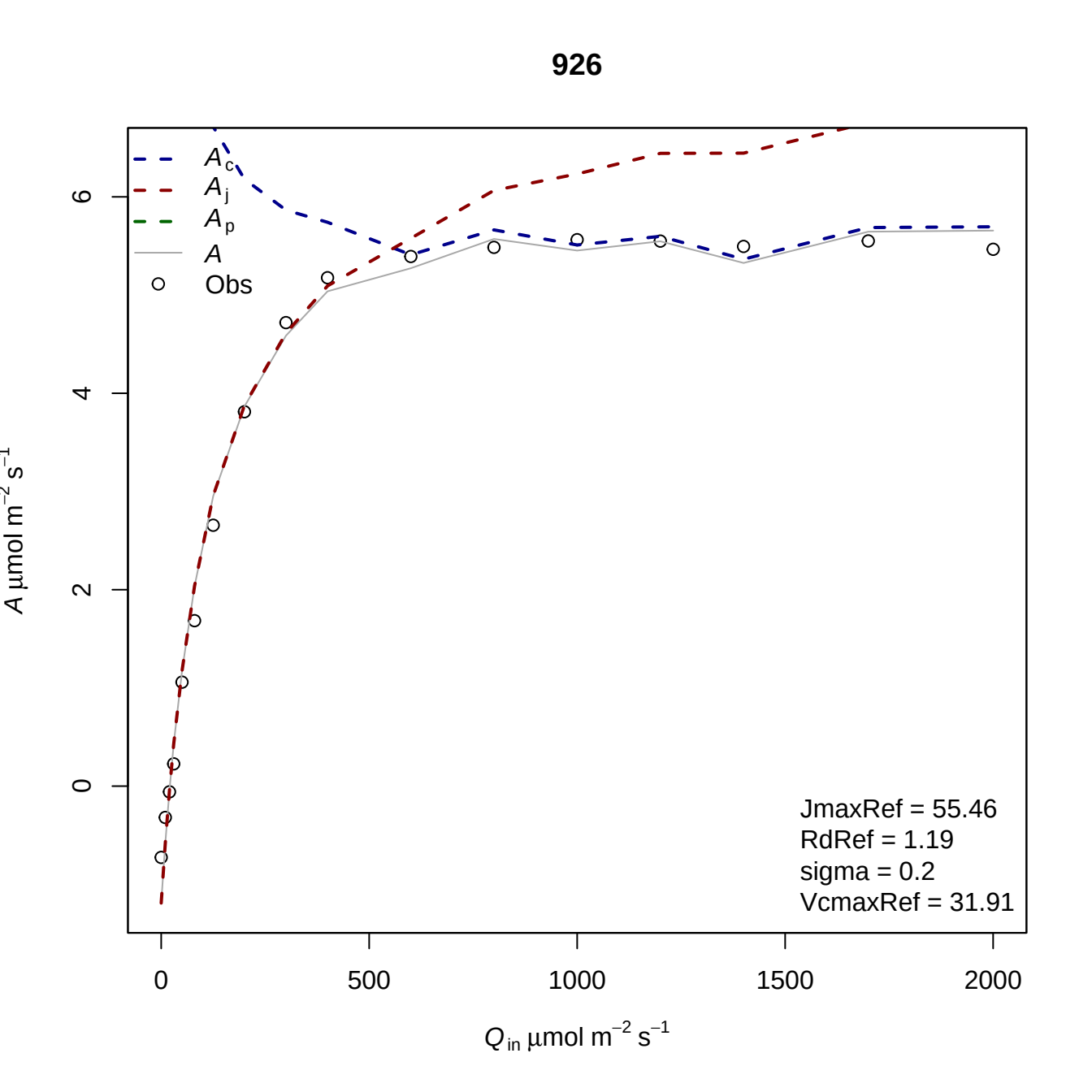

921
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 31.83
RdRef = 0.79
sigma = 0.13
VcmaxRef = 18.09
0
1
2
3
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

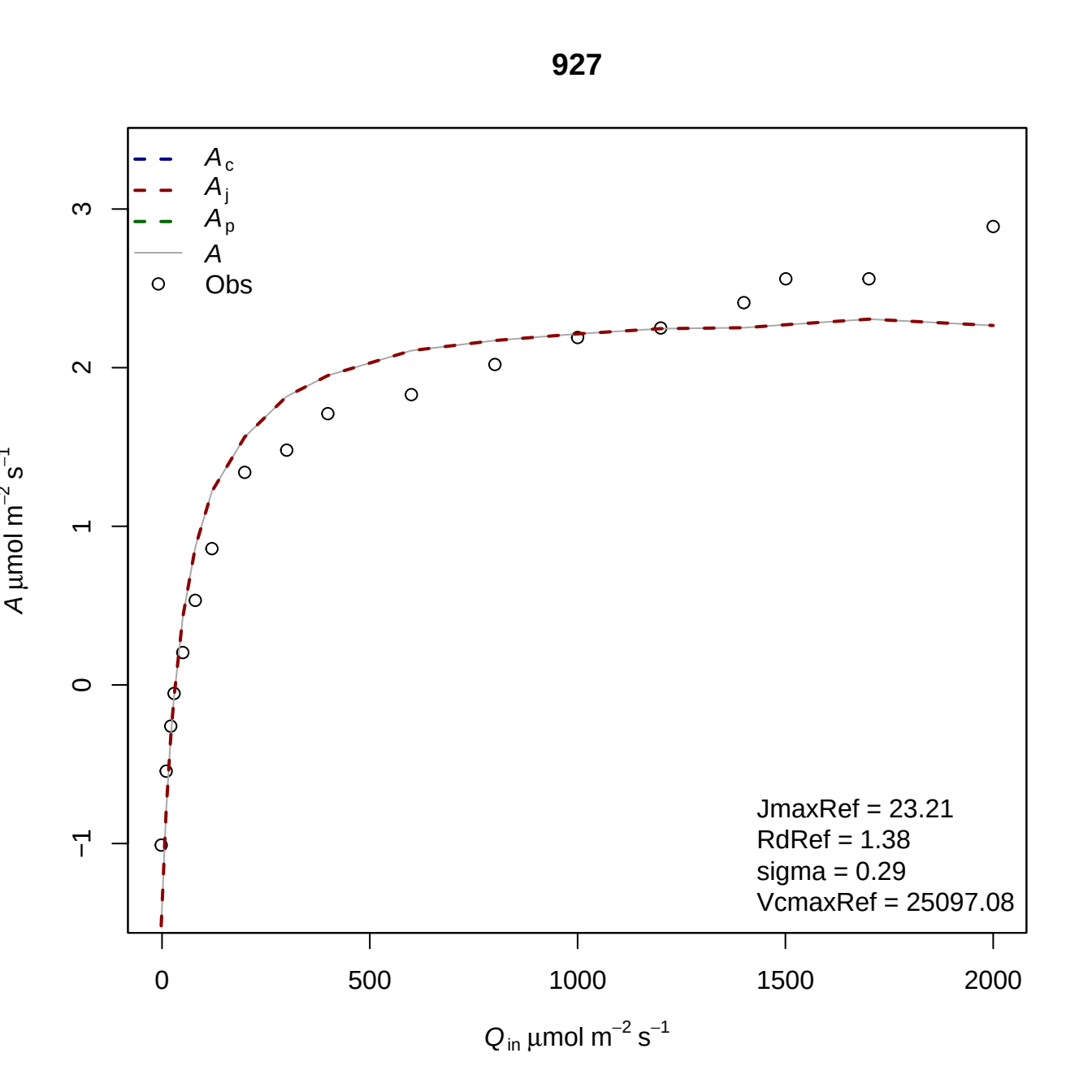

922
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.45
RdRef = 1.22
sigma = 0.3
VcmaxRef = 16284.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

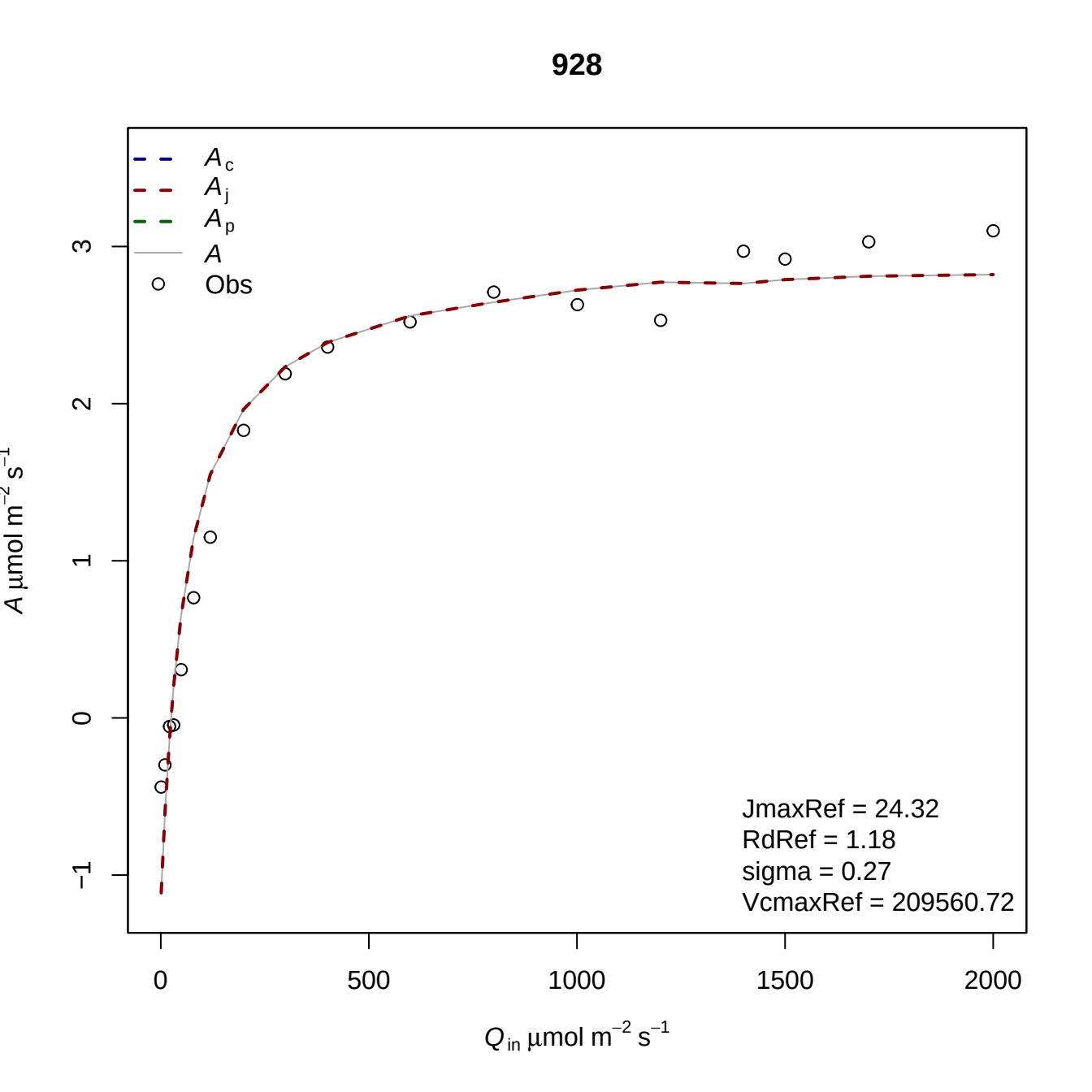

923
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.74
RdRef = 1.08
sigma = 0.23
VcmaxRef = 184600.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

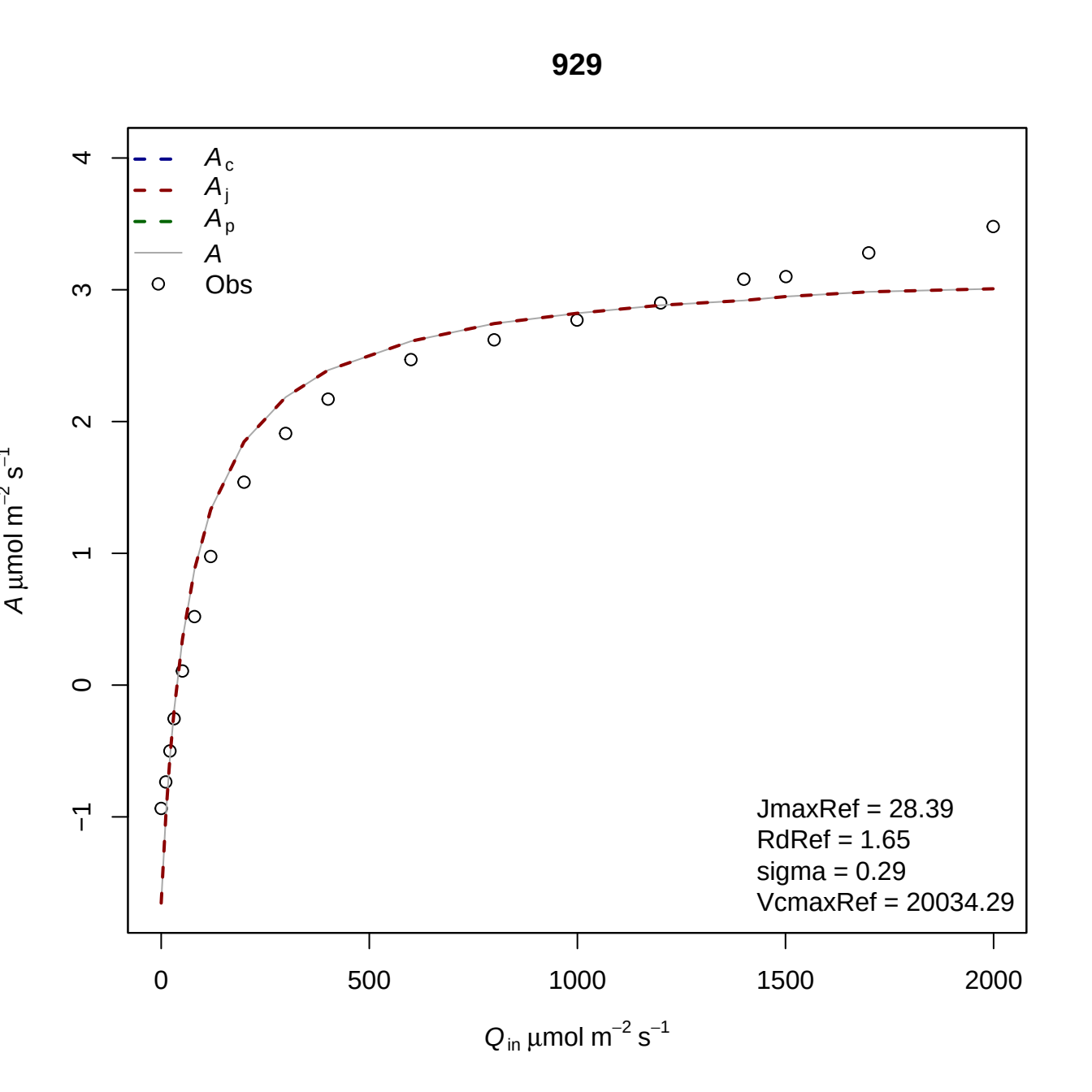

924
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.19
RdRef = 1.51
sigma = 0.24
VcmaxRef = 2801081.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
500
1000
1500

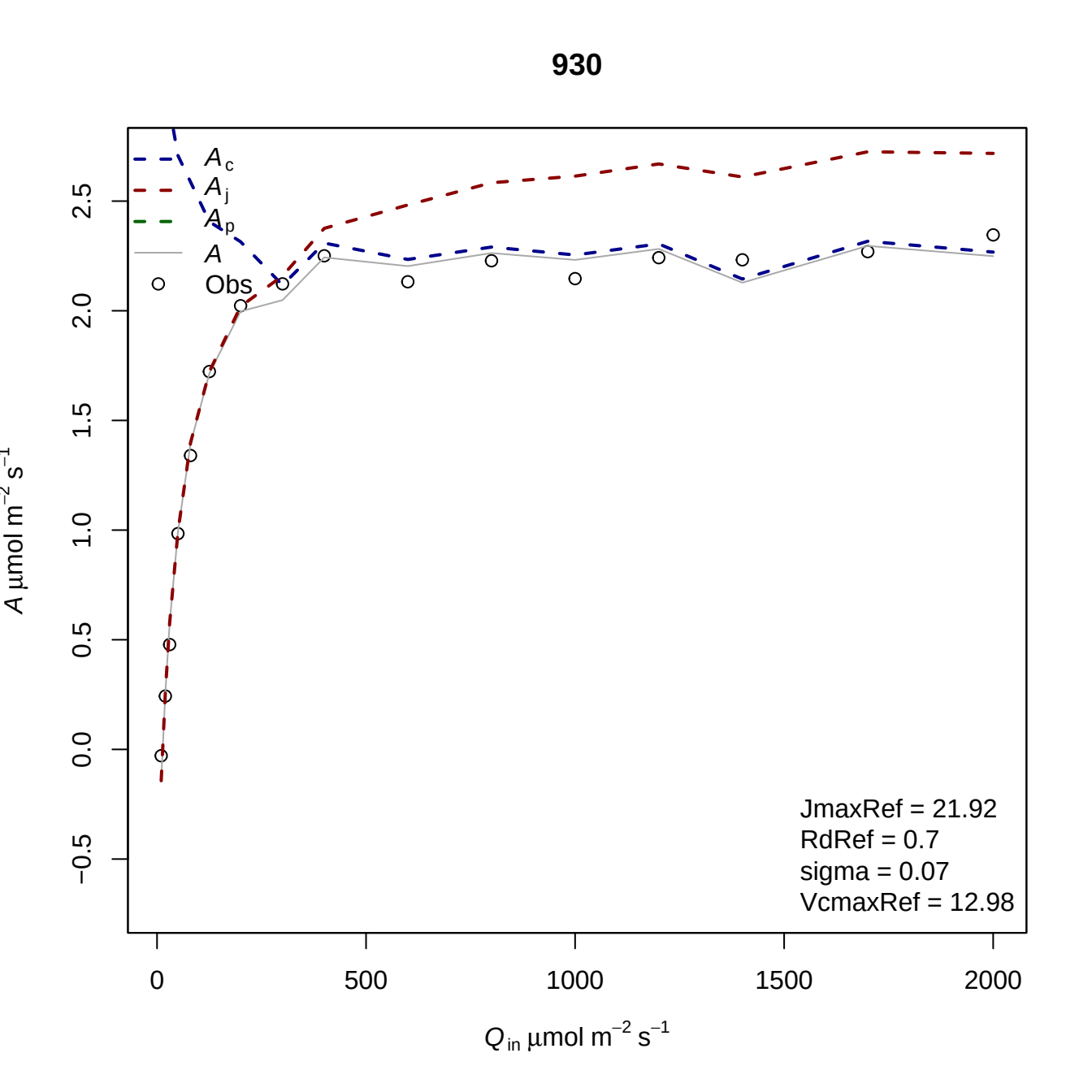

925
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.57
RdRef = 0.5
sigma = 0.11
VcmaxRef = 13.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500
2000

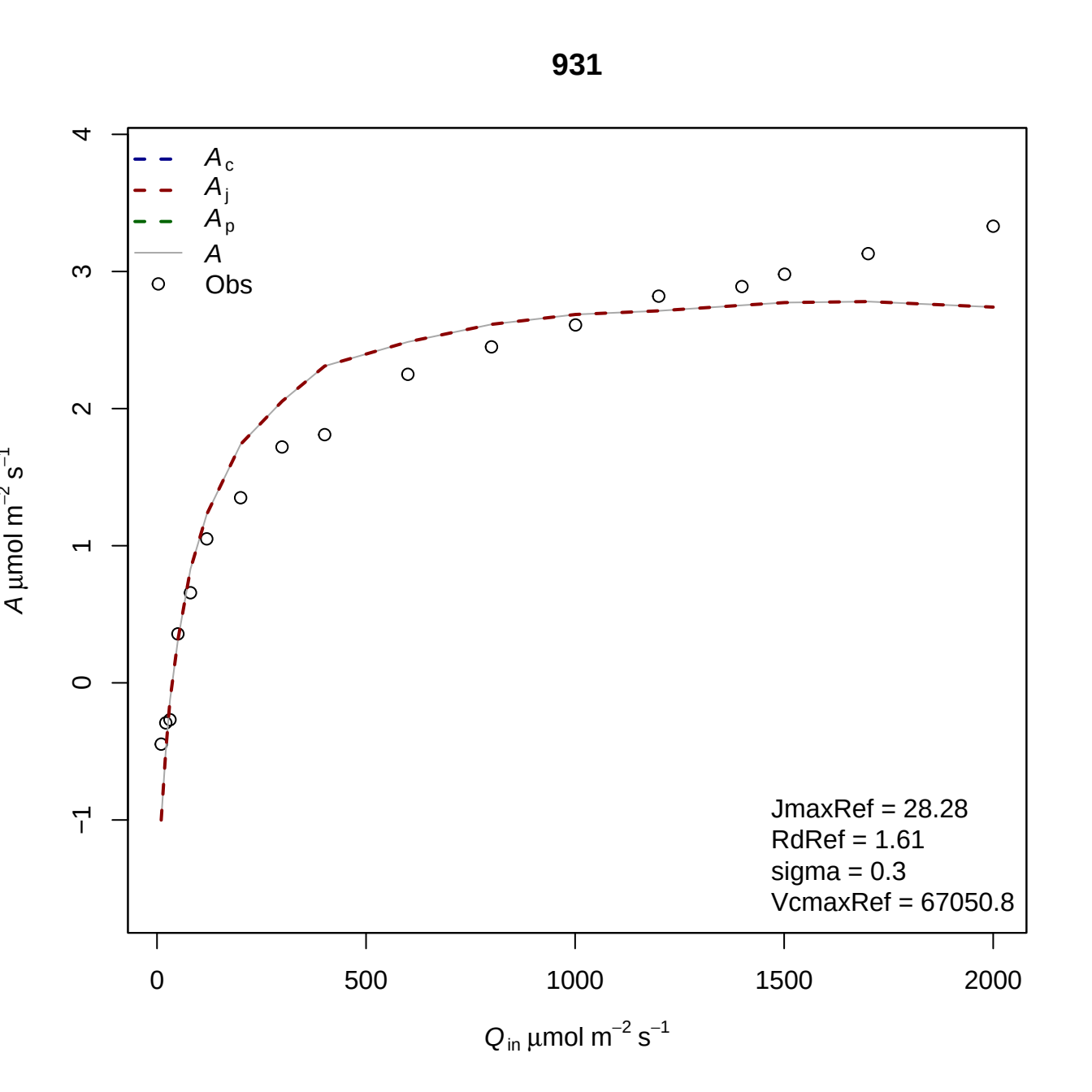

926
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.46
RdRef = 1.19
sigma = 0.2
VcmaxRef = 31.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

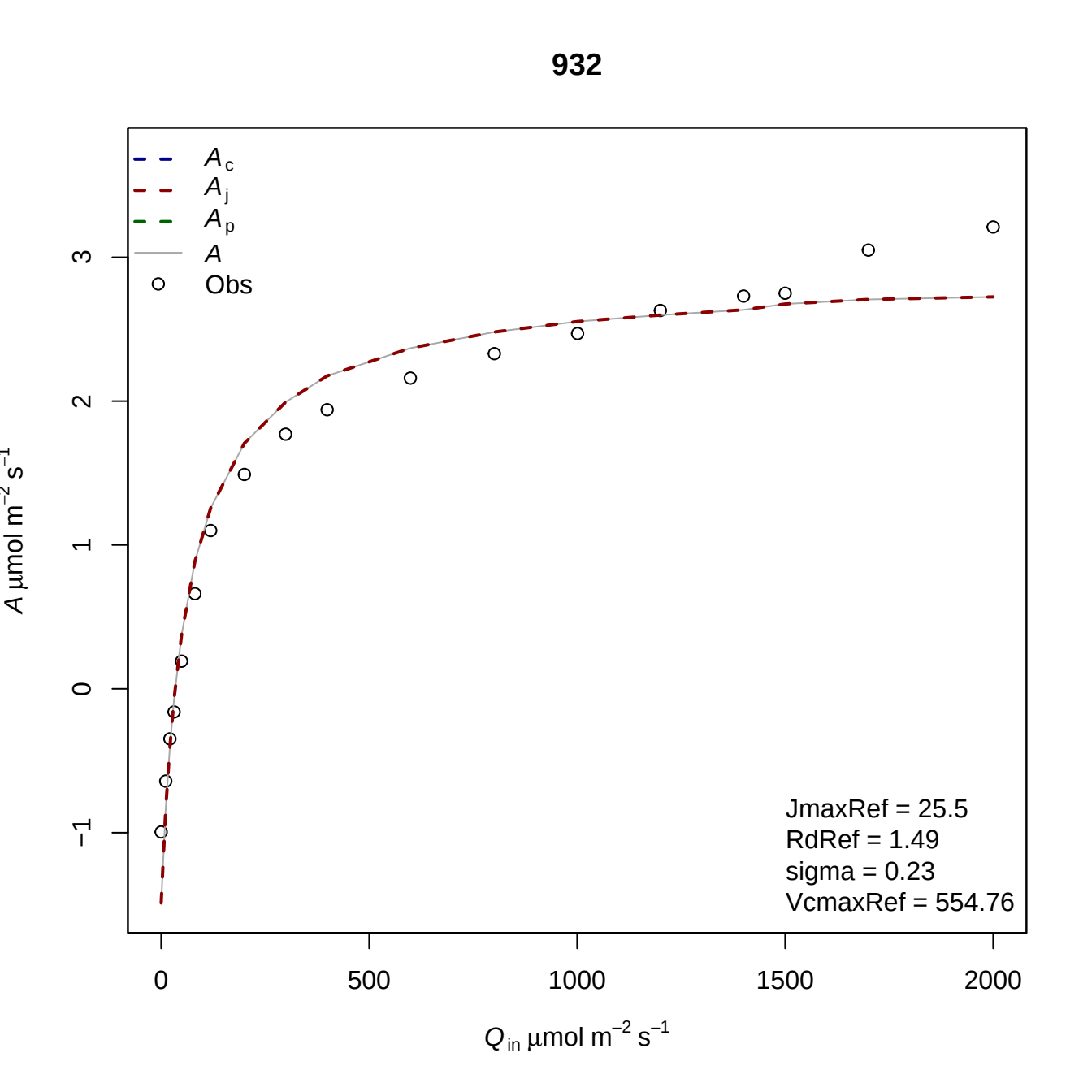

927
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 23.21
RdRef = 1.38
sigma = 0.29
VcmaxRef = 25097.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

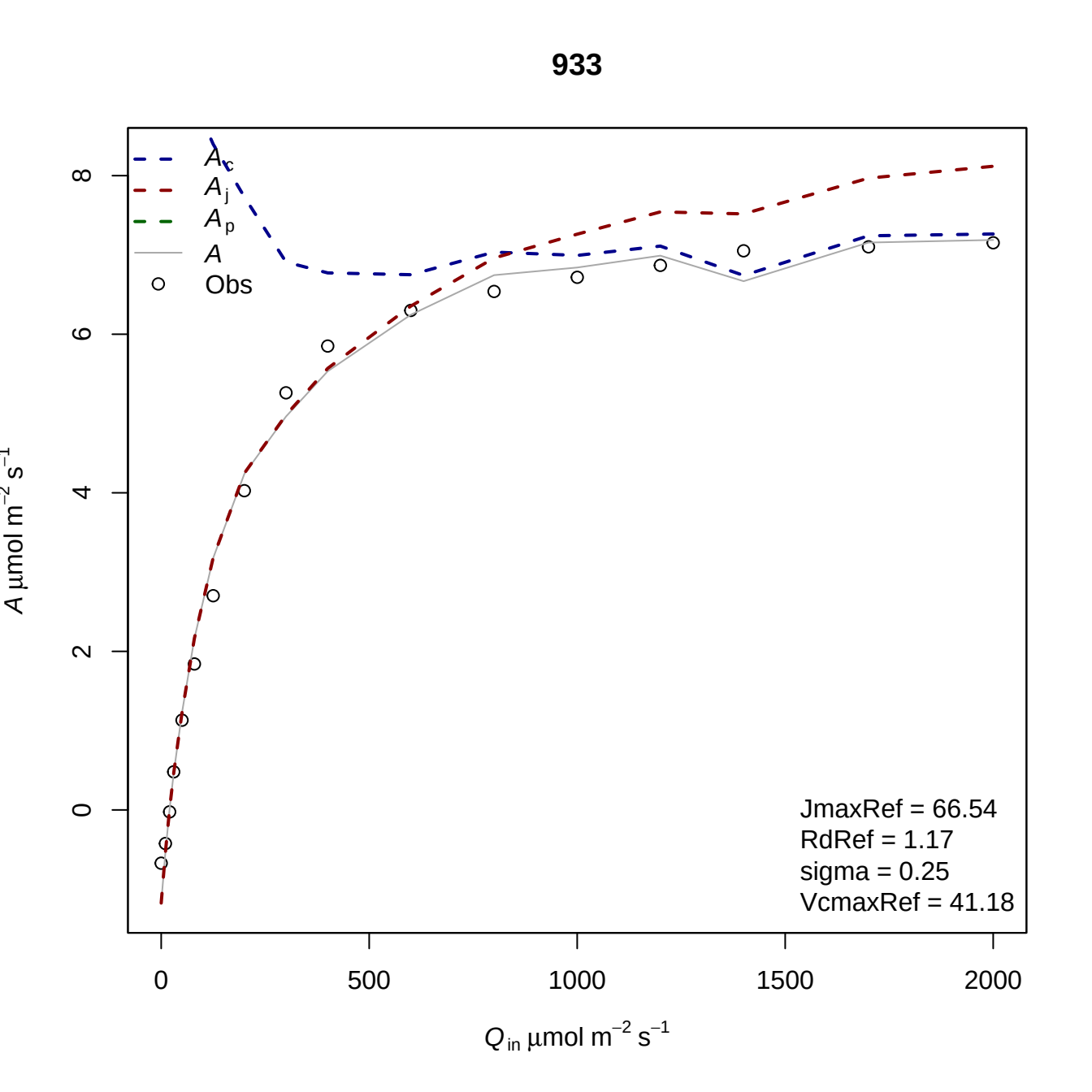

928
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 24.32
RdRef = 1.18
sigma = 0.27
VcmaxRef = 209560.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

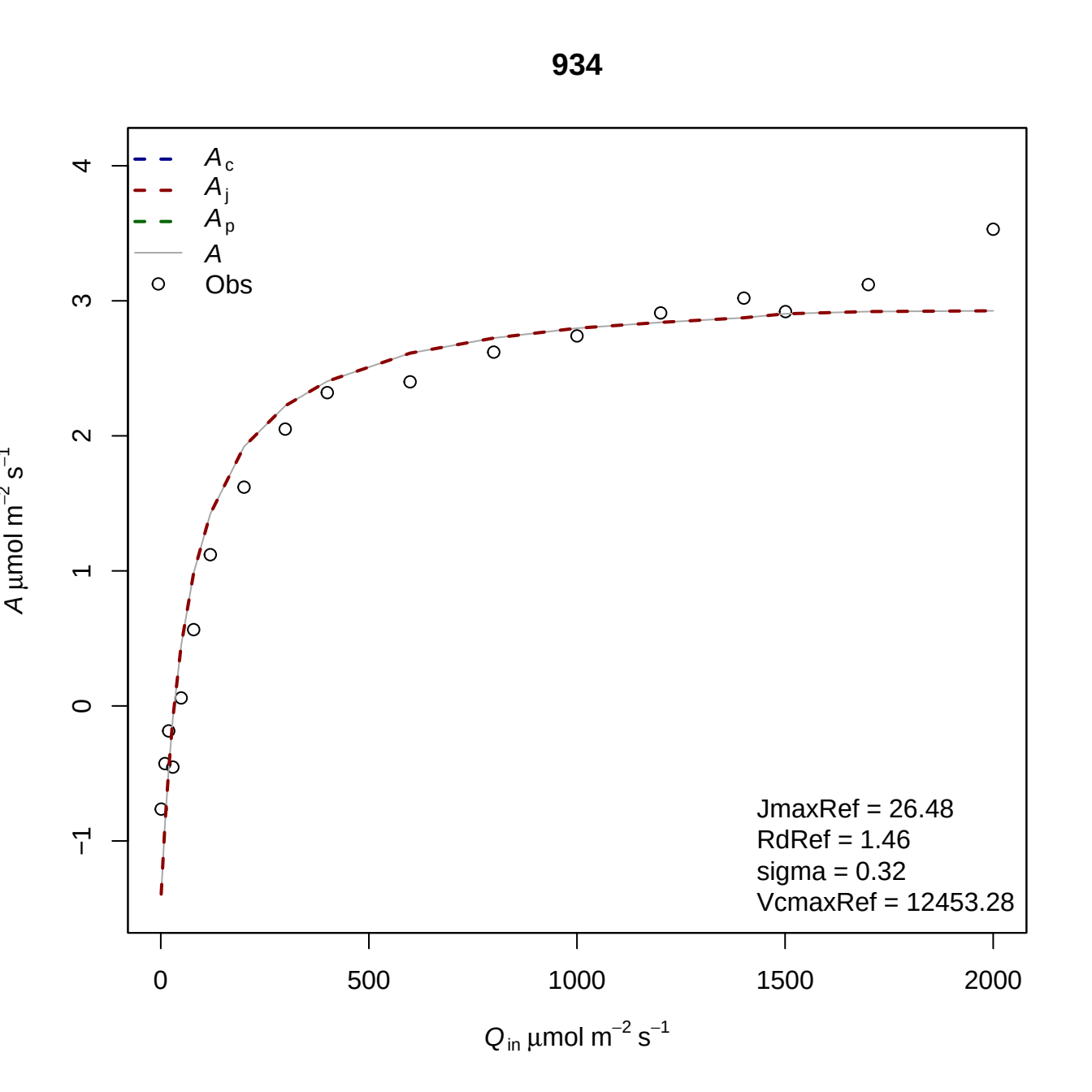

929
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 28.39
RdRef = 1.65
sigma = 0.29
VcmaxRef = 20034.29
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

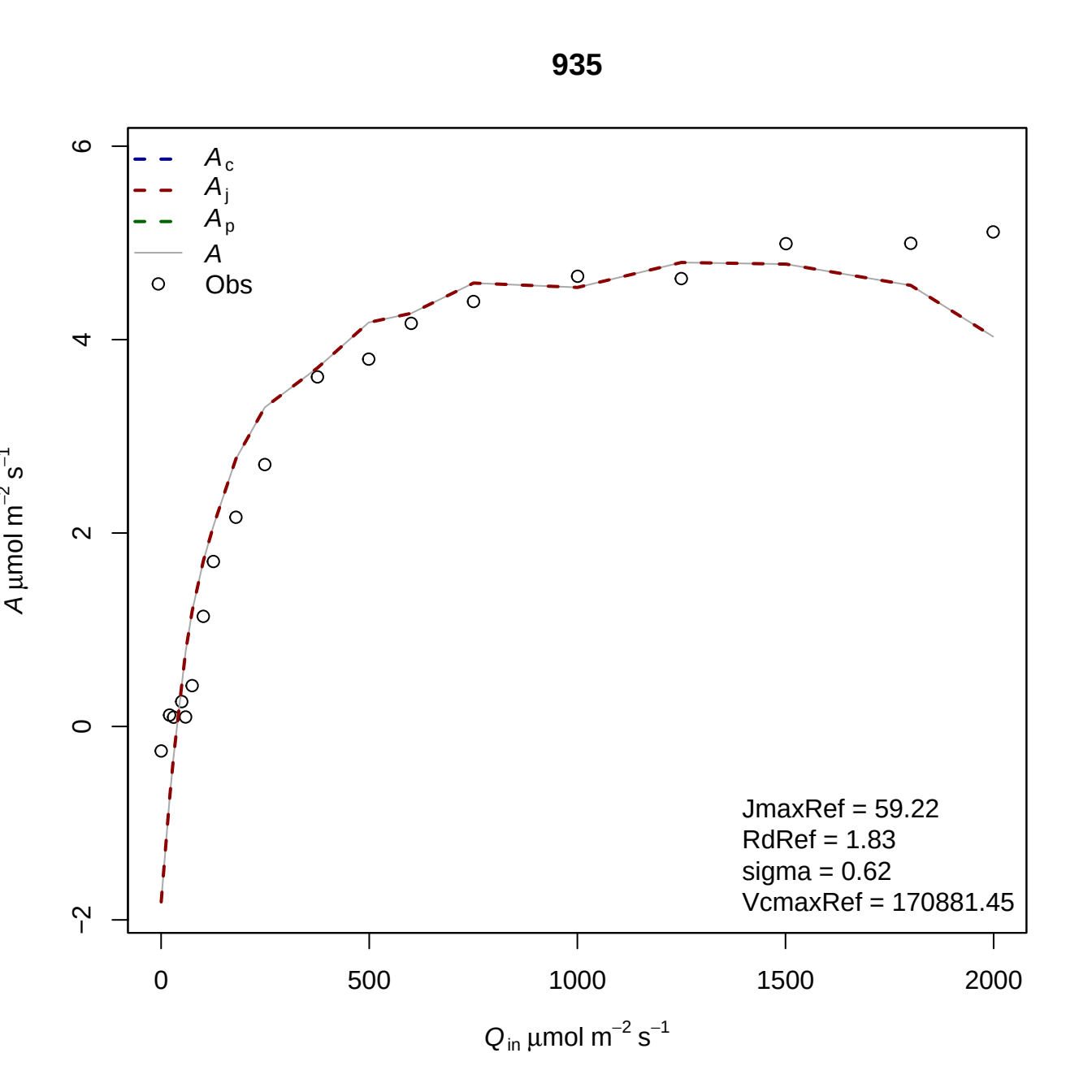

930
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.92
RdRef = 0.7
sigma = 0.07
VcmaxRef = 12.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

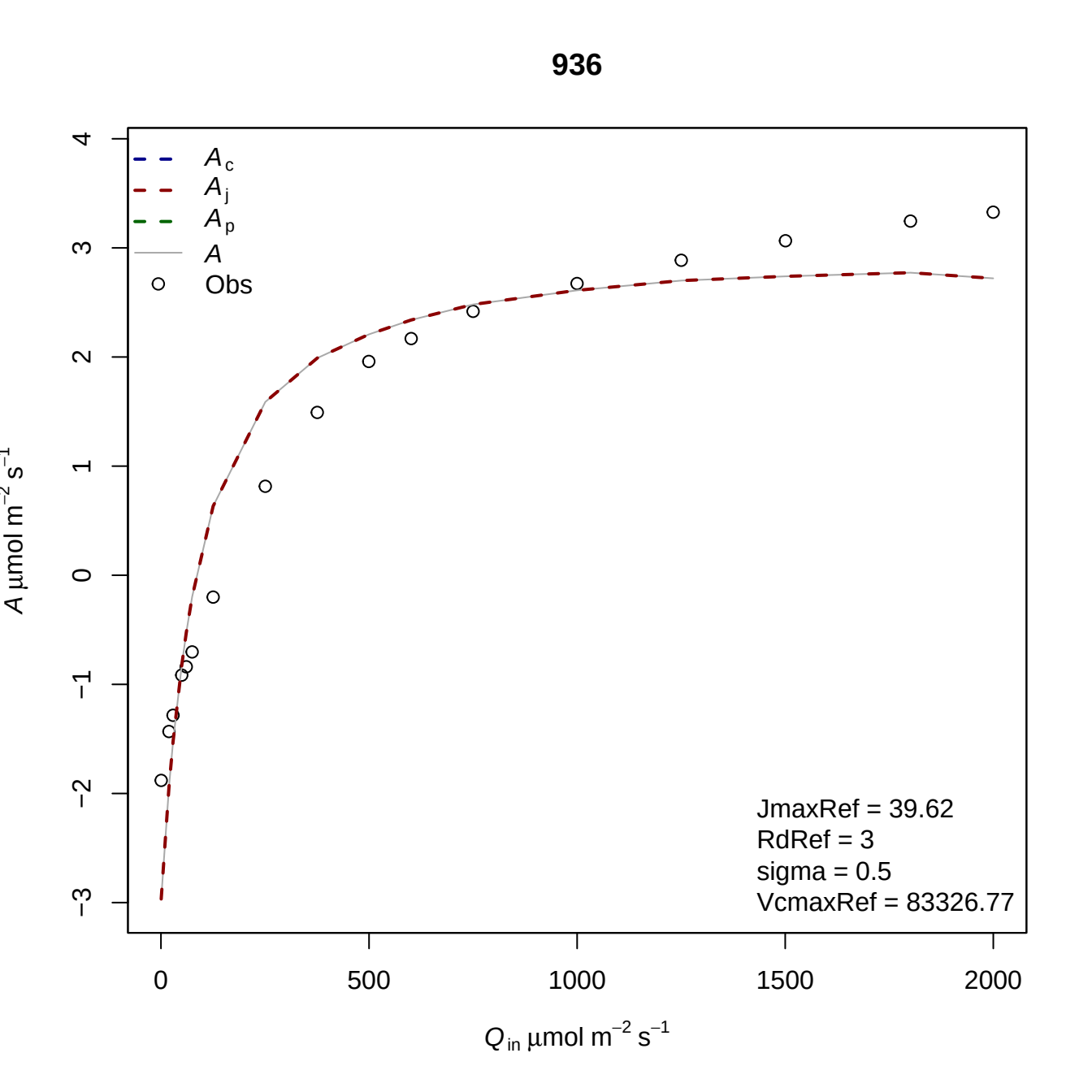

931
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 28.28
RdRef = 1.61
sigma = 0.3
VcmaxRef = 67050.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
0
500
1000
1500
2000

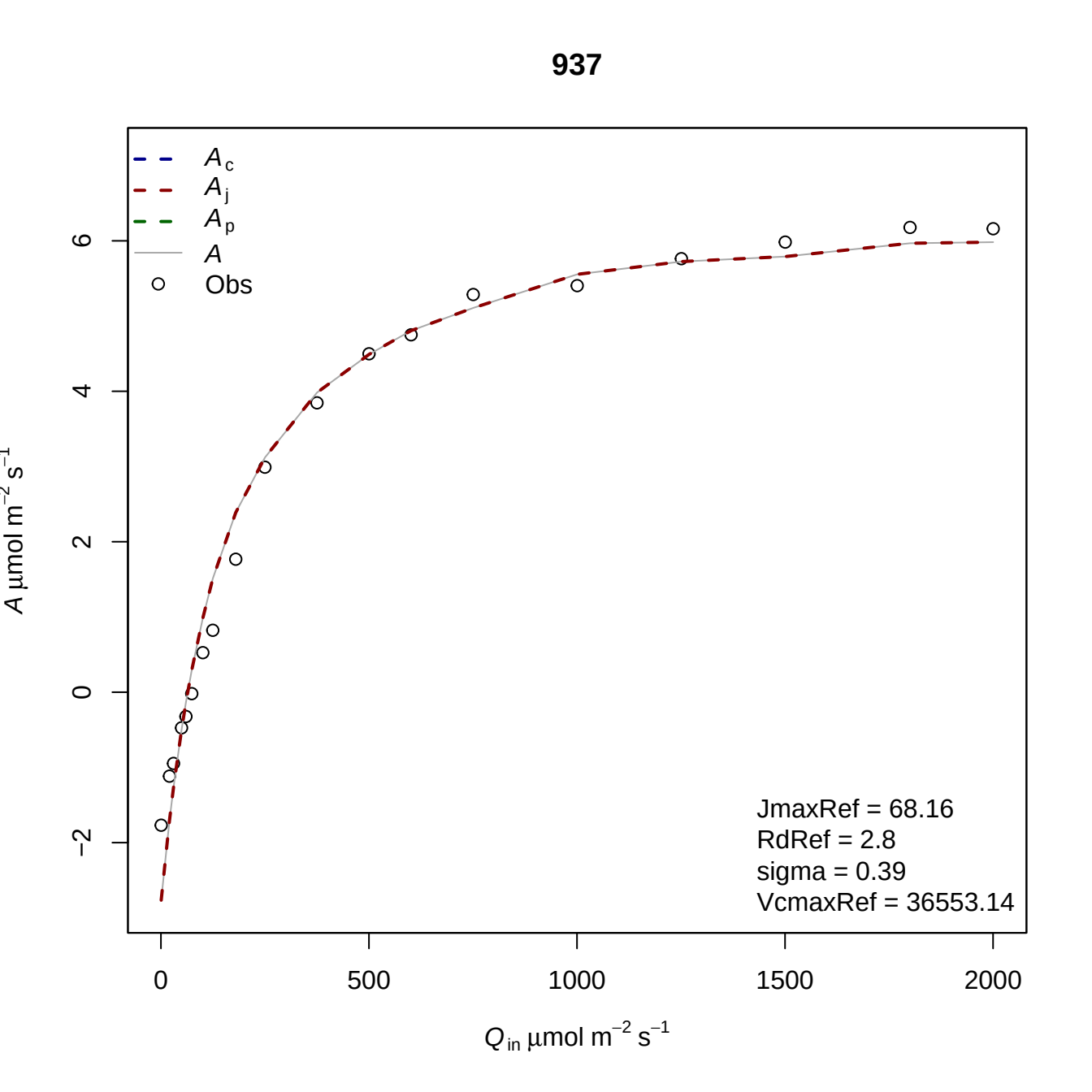

932
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.5
RdRef = 1.49
sigma = 0.23
VcmaxRef = 554.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

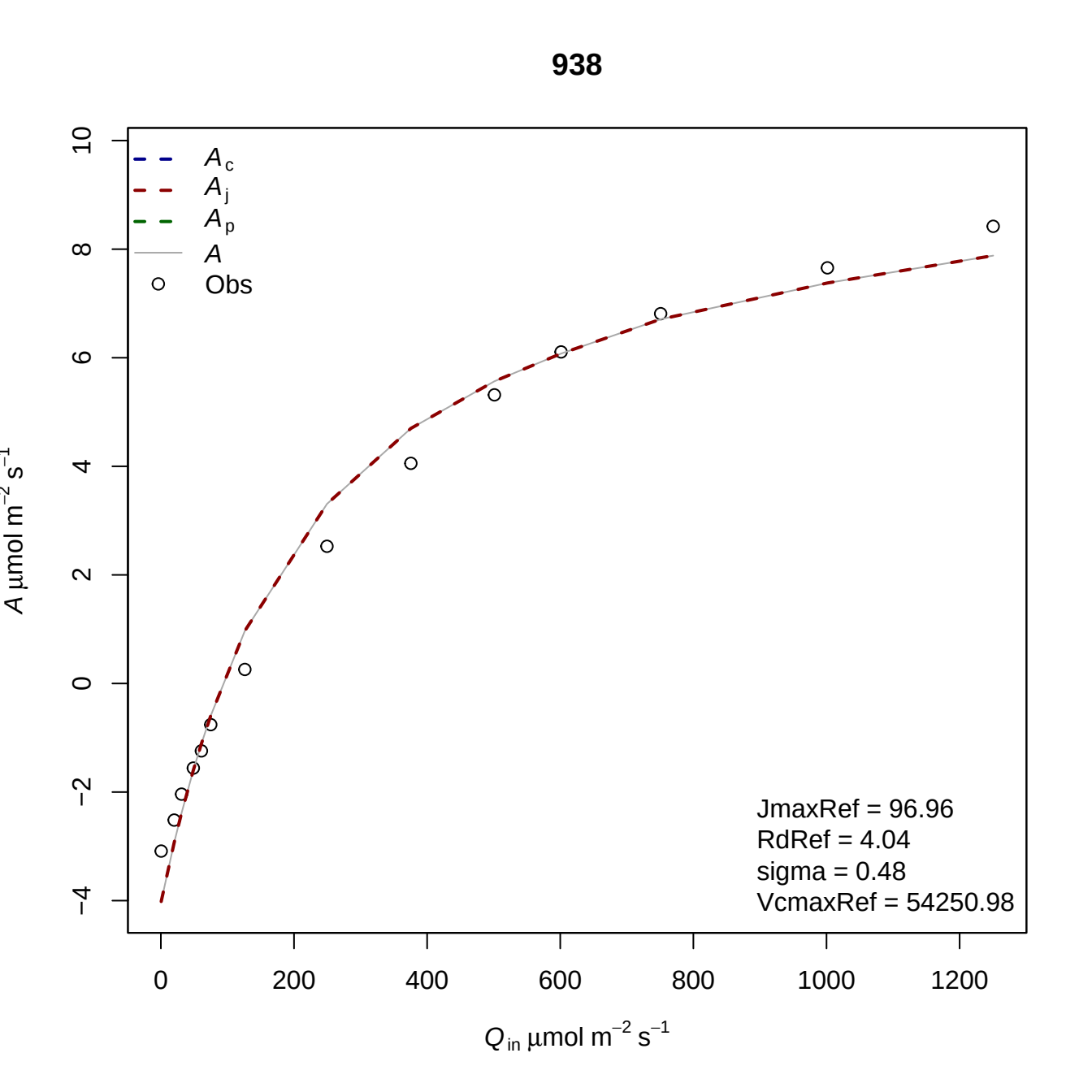

933
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.54
RdRef = 1.17
sigma = 0.25
VcmaxRef = 41.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

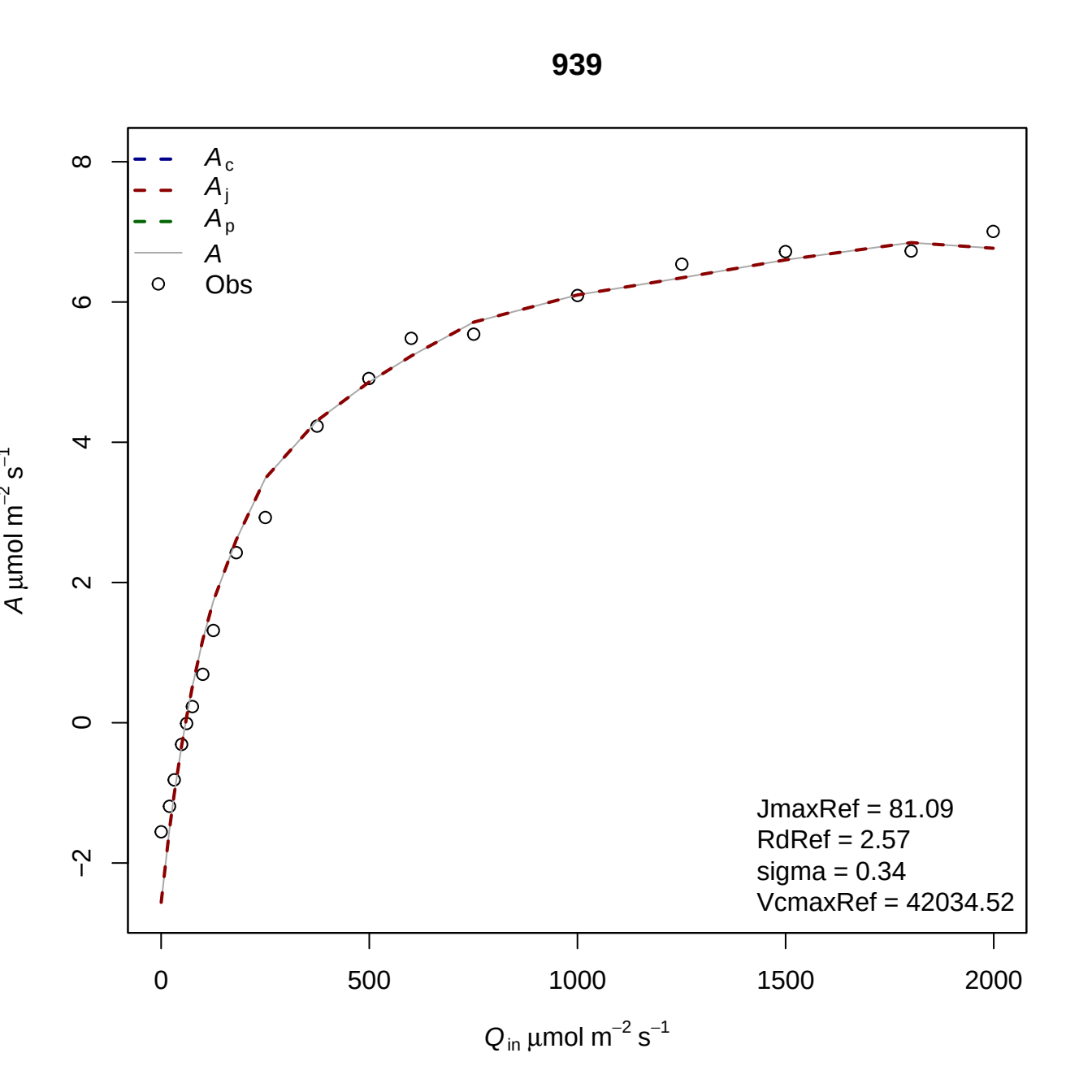

934
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.48
RdRef = 1.46
sigma = 0.32
VcmaxRef = 12453.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

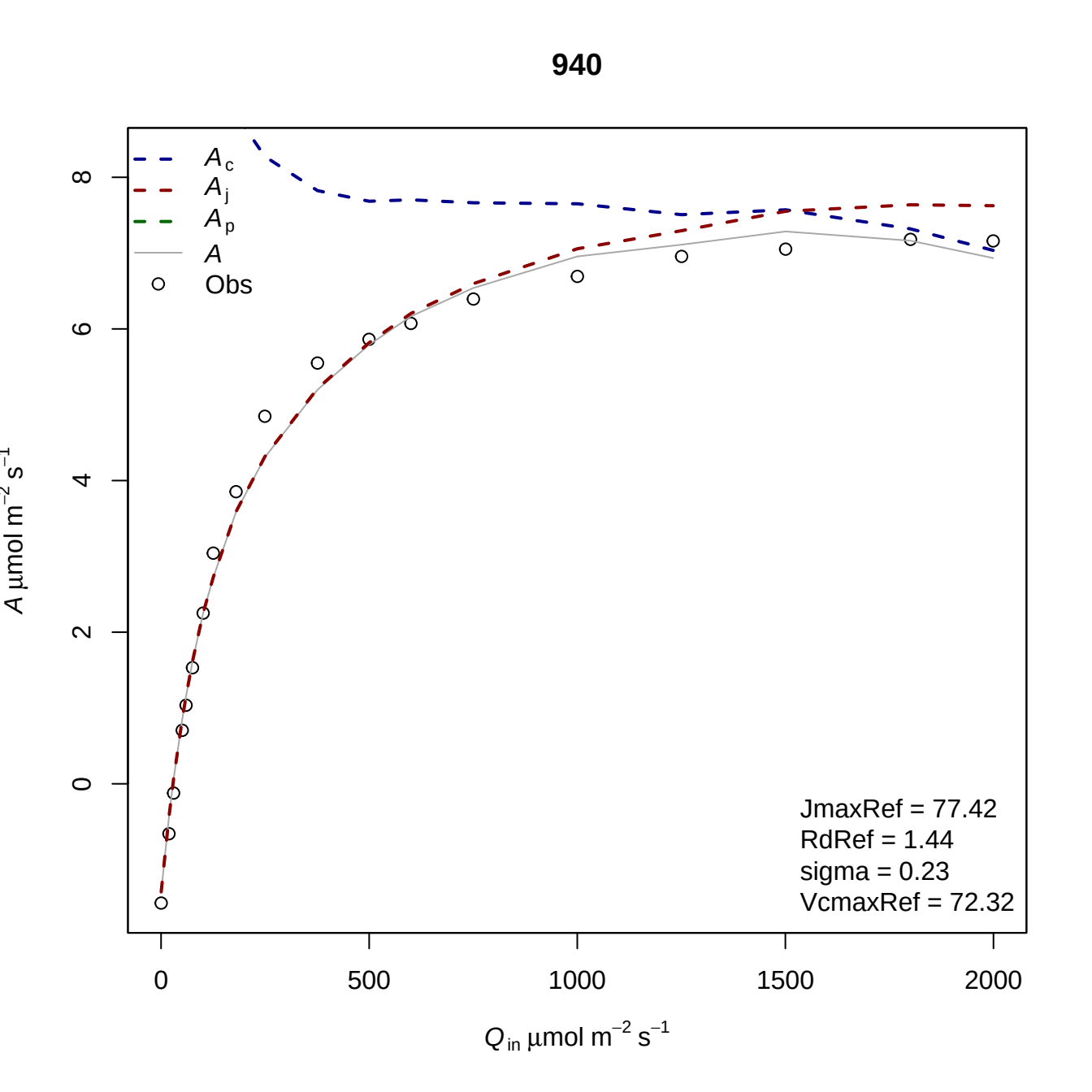

935
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.22
RdRef = 1.83
sigma = 0.62
VcmaxRef = 170881.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500
2000

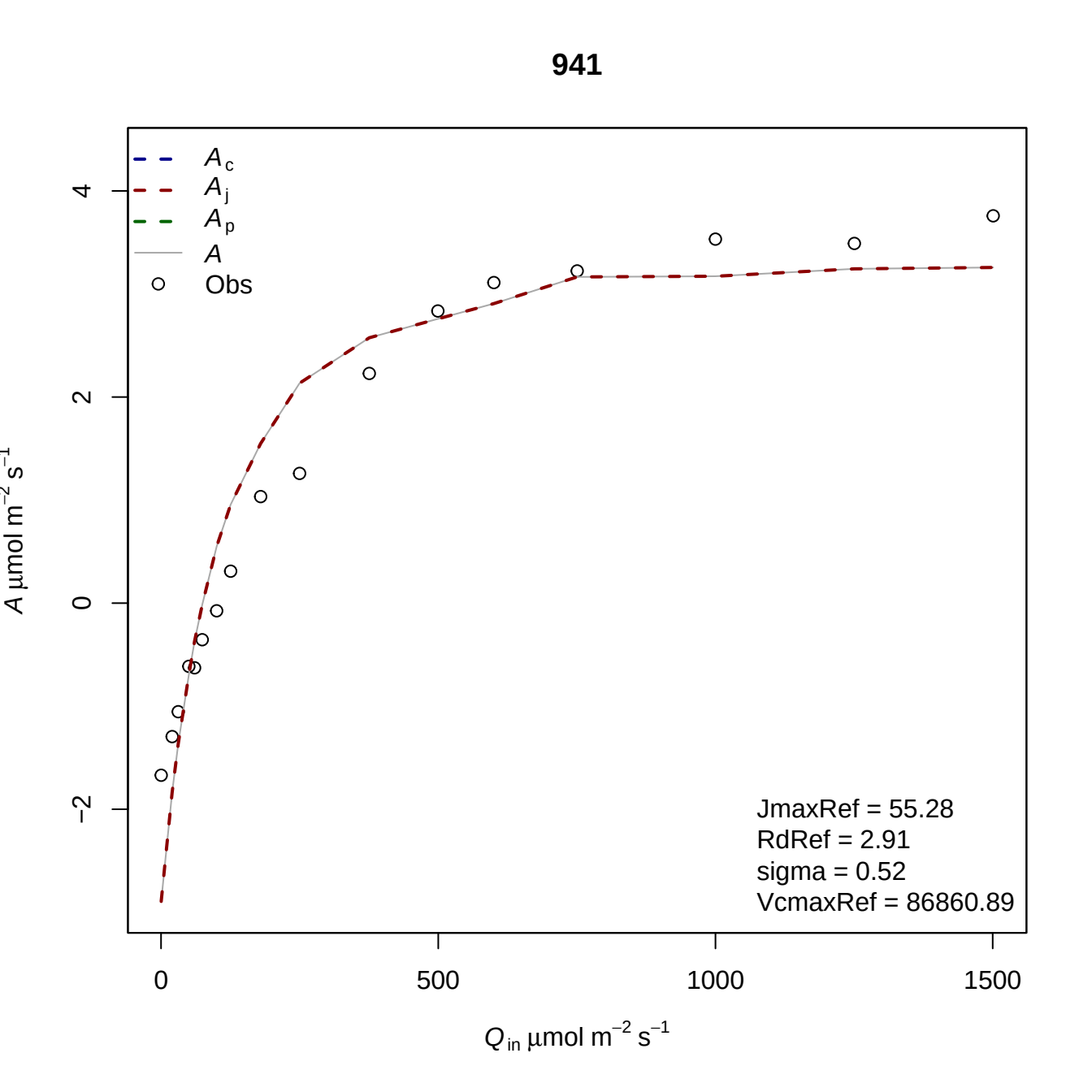

936
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 39.62
RdRef = 3
sigma = 0.5
VcmaxRef = 83326.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

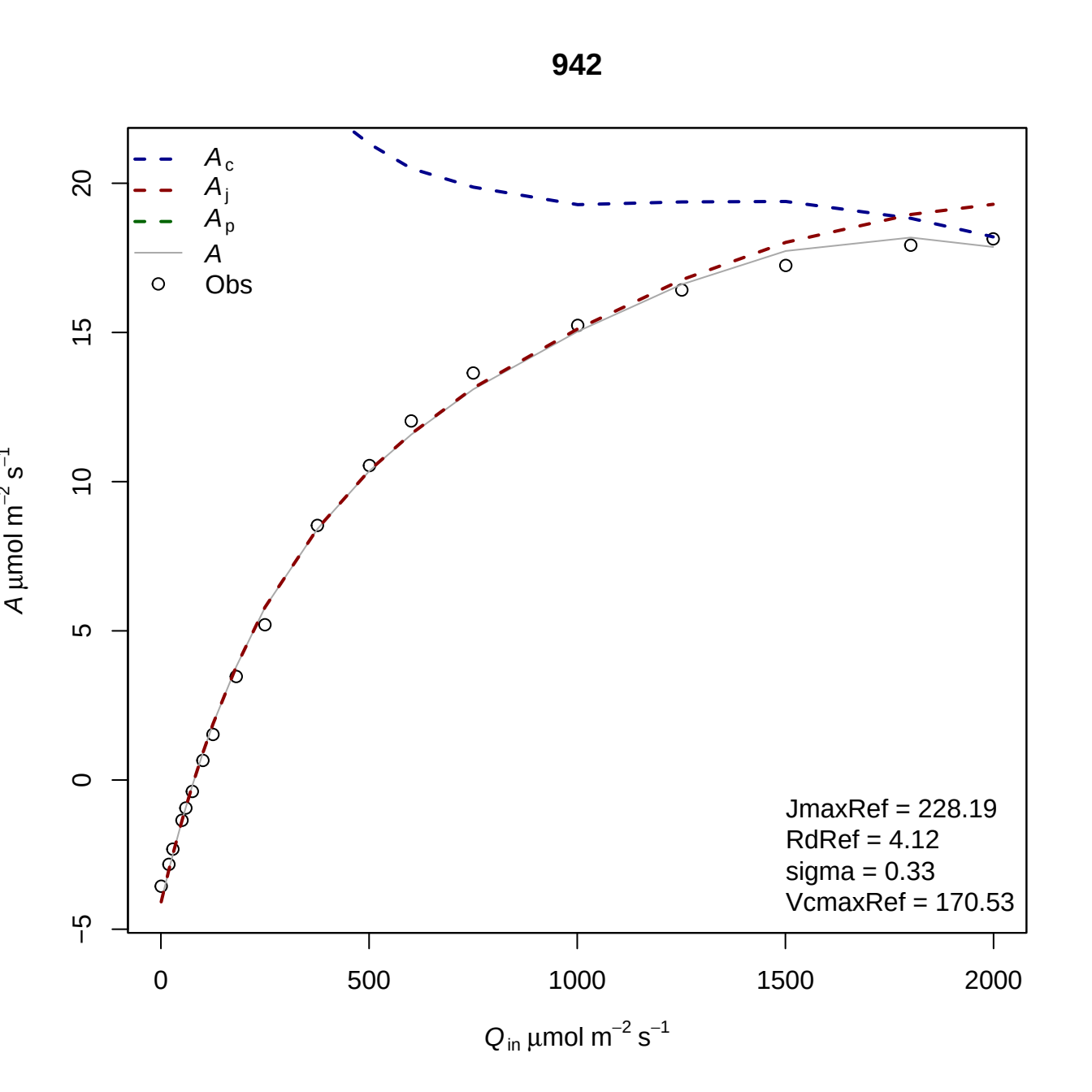

937
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.16
RdRef = 2.8
sigma = 0.39
VcmaxRef = 36553.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

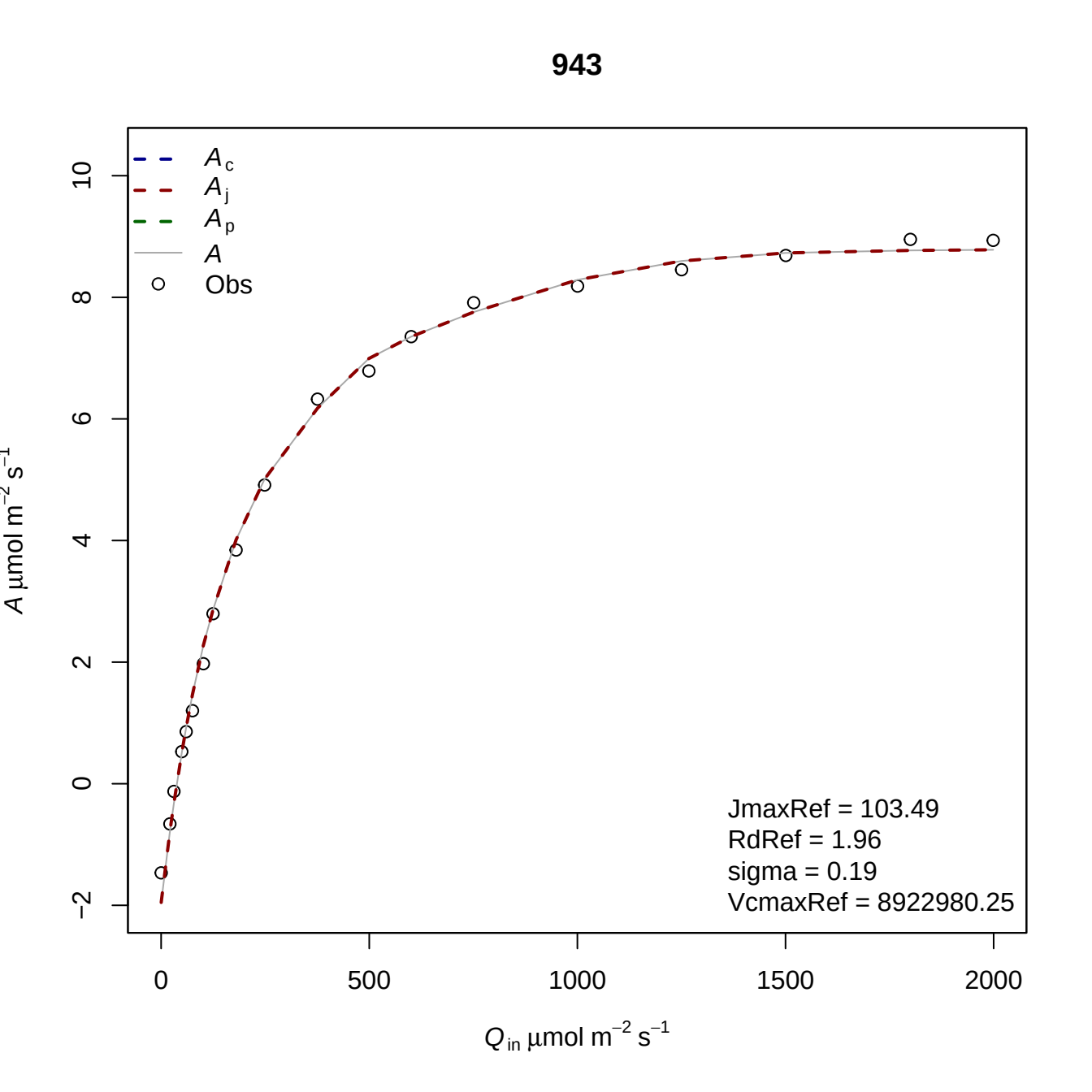

938
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.96
RdRef = 4.04
sigma = 0.48
VcmaxRef = 54250.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

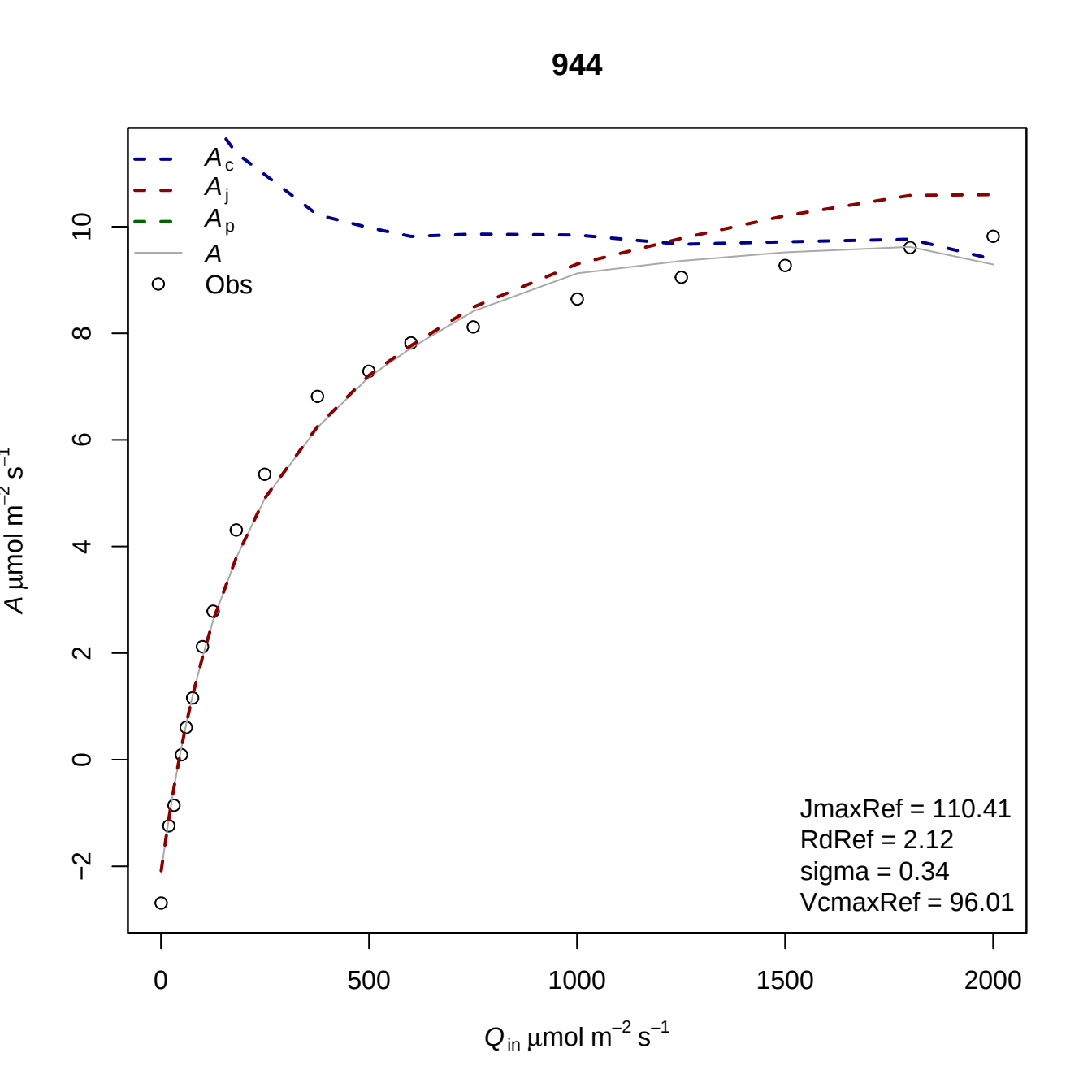

939
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.09
RdRef = 2.57
sigma = 0.34
VcmaxRef = 42034.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

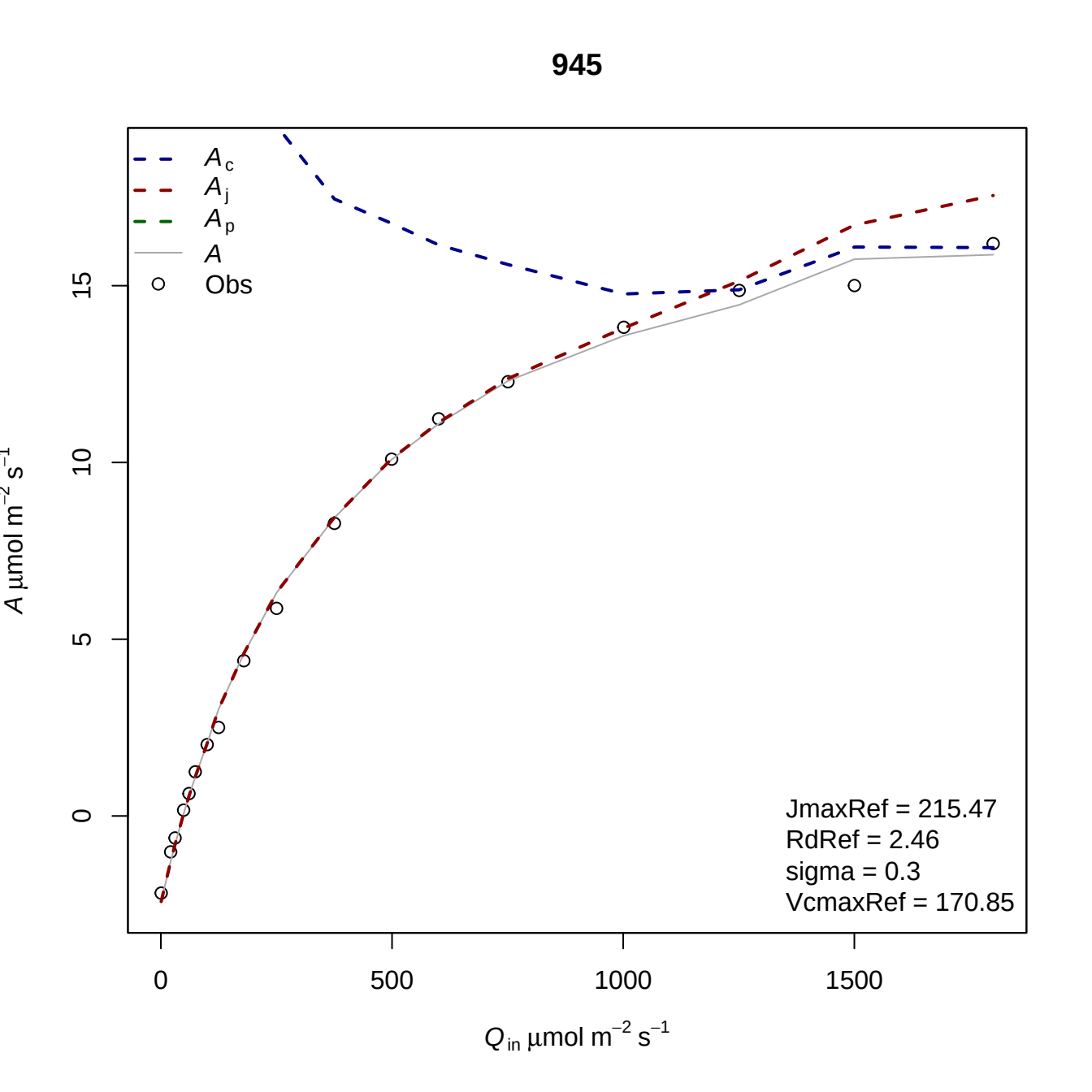

940
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.42
RdRef = 1.44
sigma = 0.23
VcmaxRef = 72.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

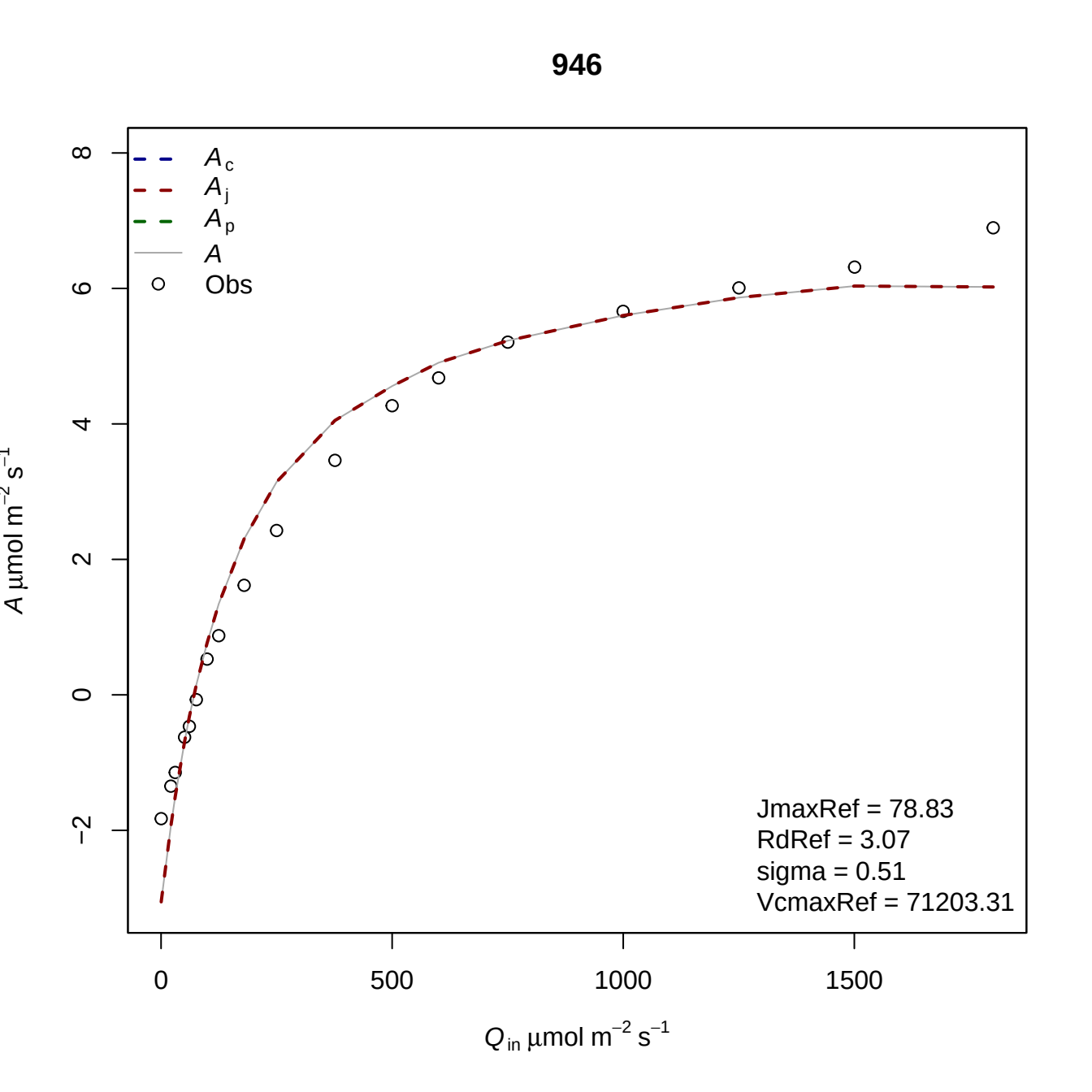

941
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.28
RdRef = 2.91
sigma = 0.52
VcmaxRef = 86860.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
0
500
1000
1500

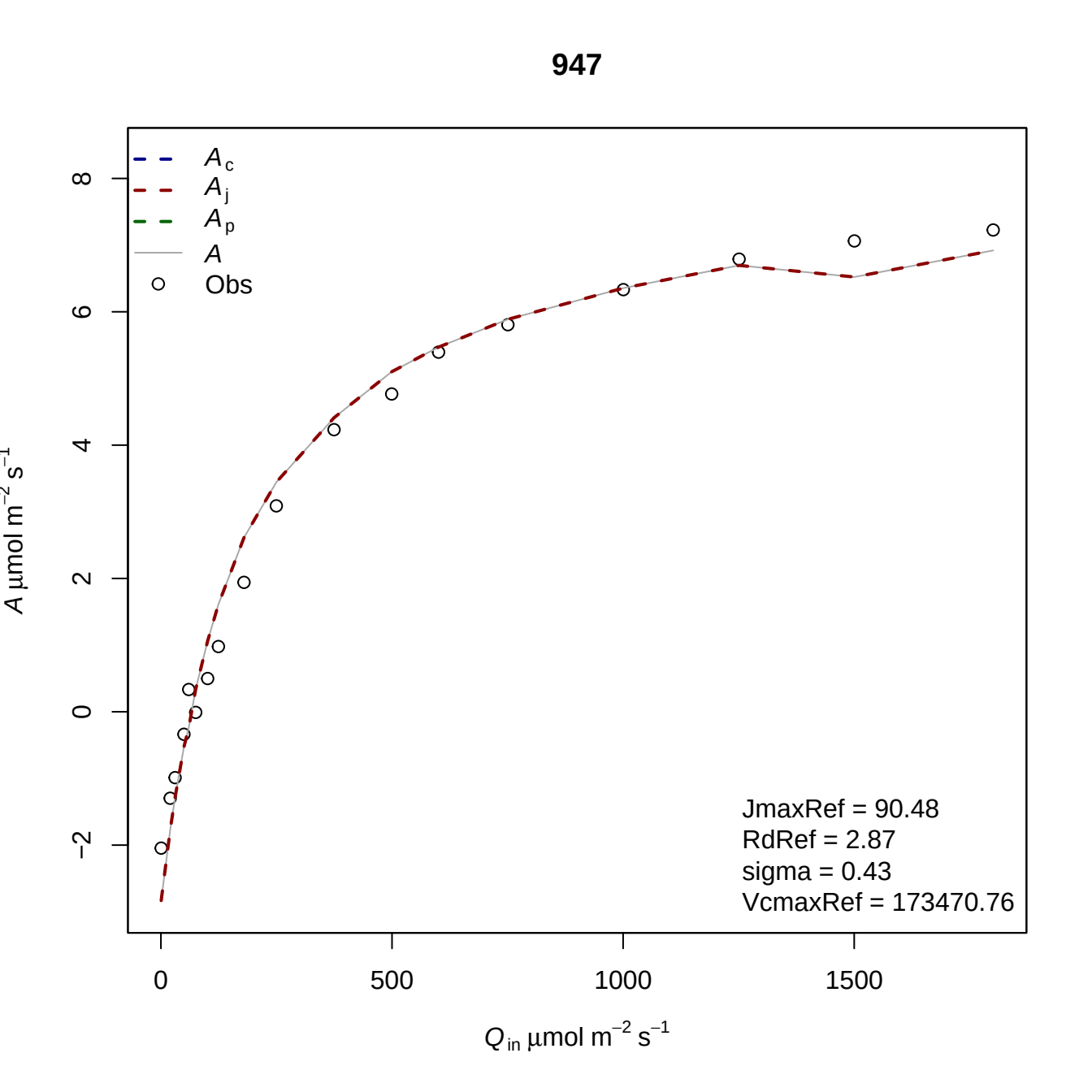

942
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 228.19
RdRef = 4.12
sigma = 0.33
VcmaxRef = 170.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

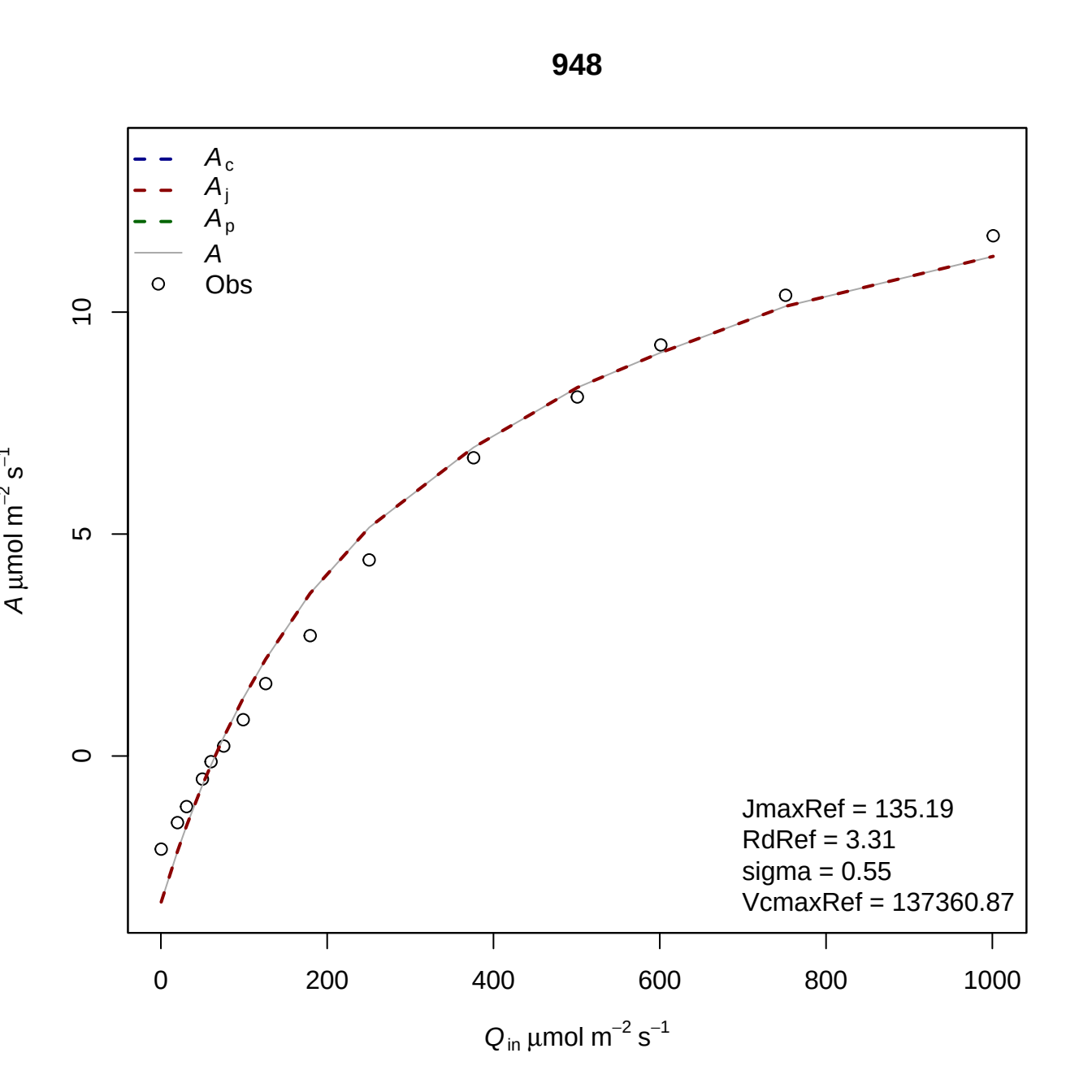

943
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.49
RdRef = 1.96
sigma = 0.19
VcmaxRef = 8922980.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

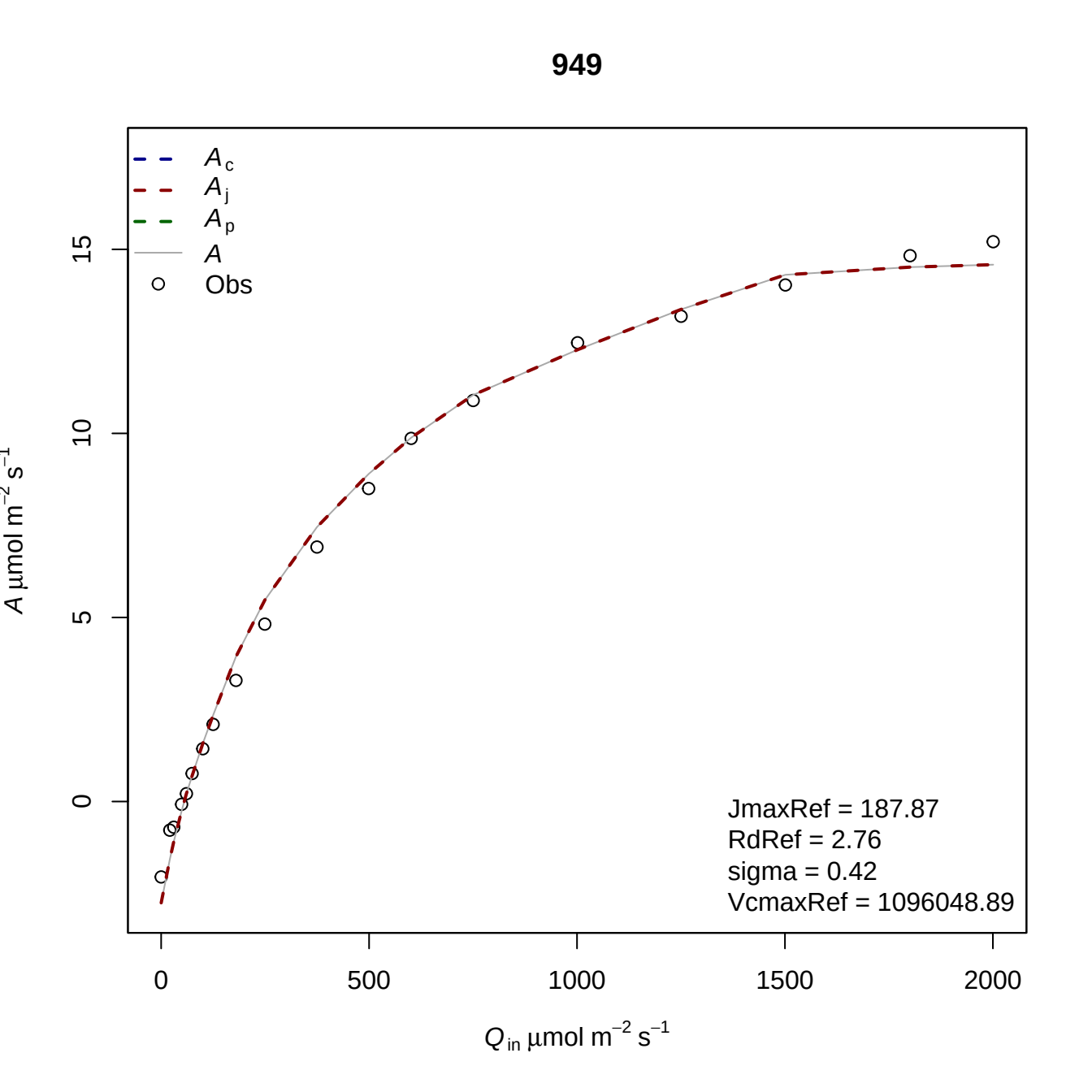

944
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.41
RdRef = 2.12
sigma = 0.34
VcmaxRef = 96.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

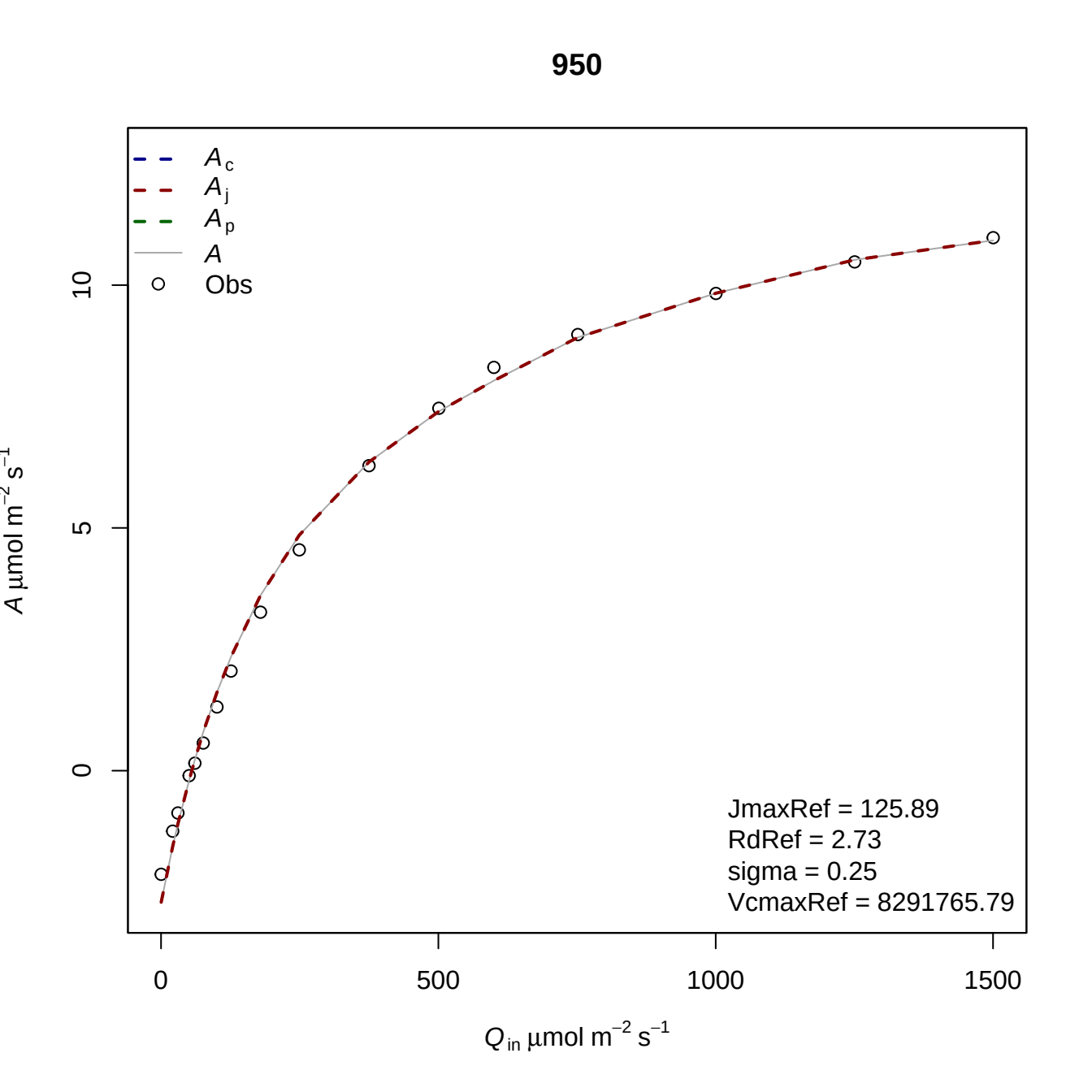

945
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 215.47
RdRef = 2.46
sigma = 0.3
VcmaxRef = 170.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

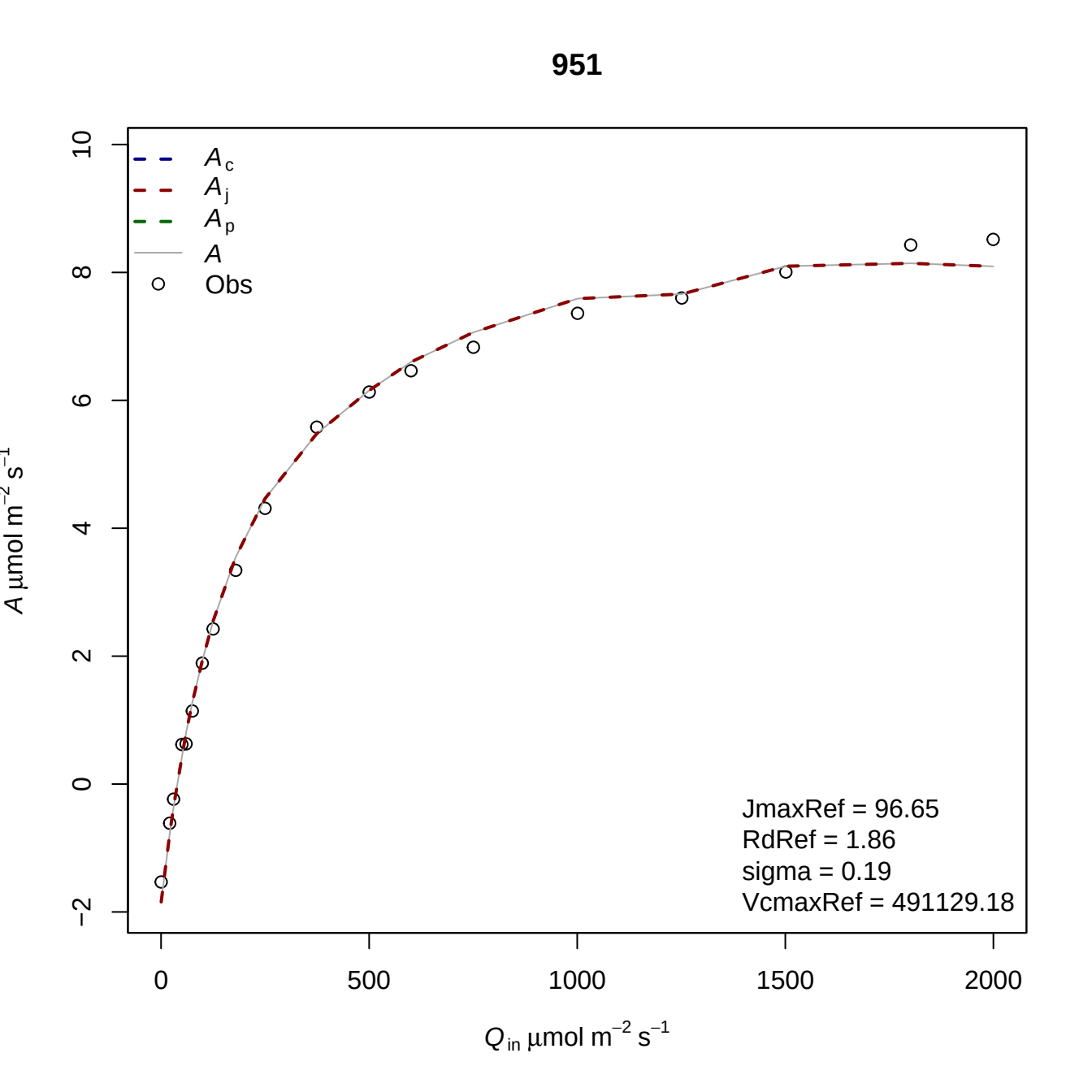

946
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.83
RdRef = 3.07
sigma = 0.51
VcmaxRef = 71203.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

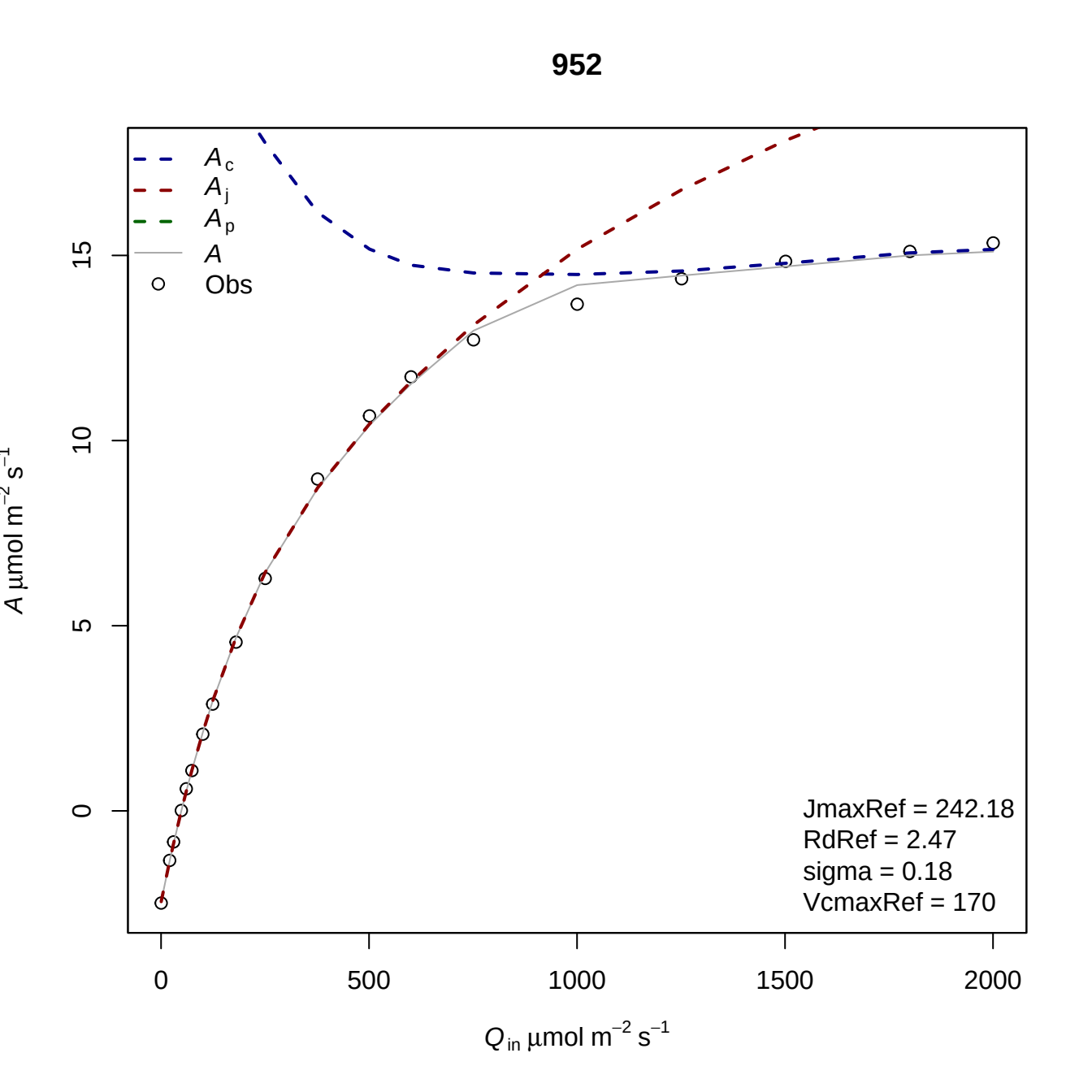

947
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.48
RdRef = 2.87
sigma = 0.43
VcmaxRef = 173470.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

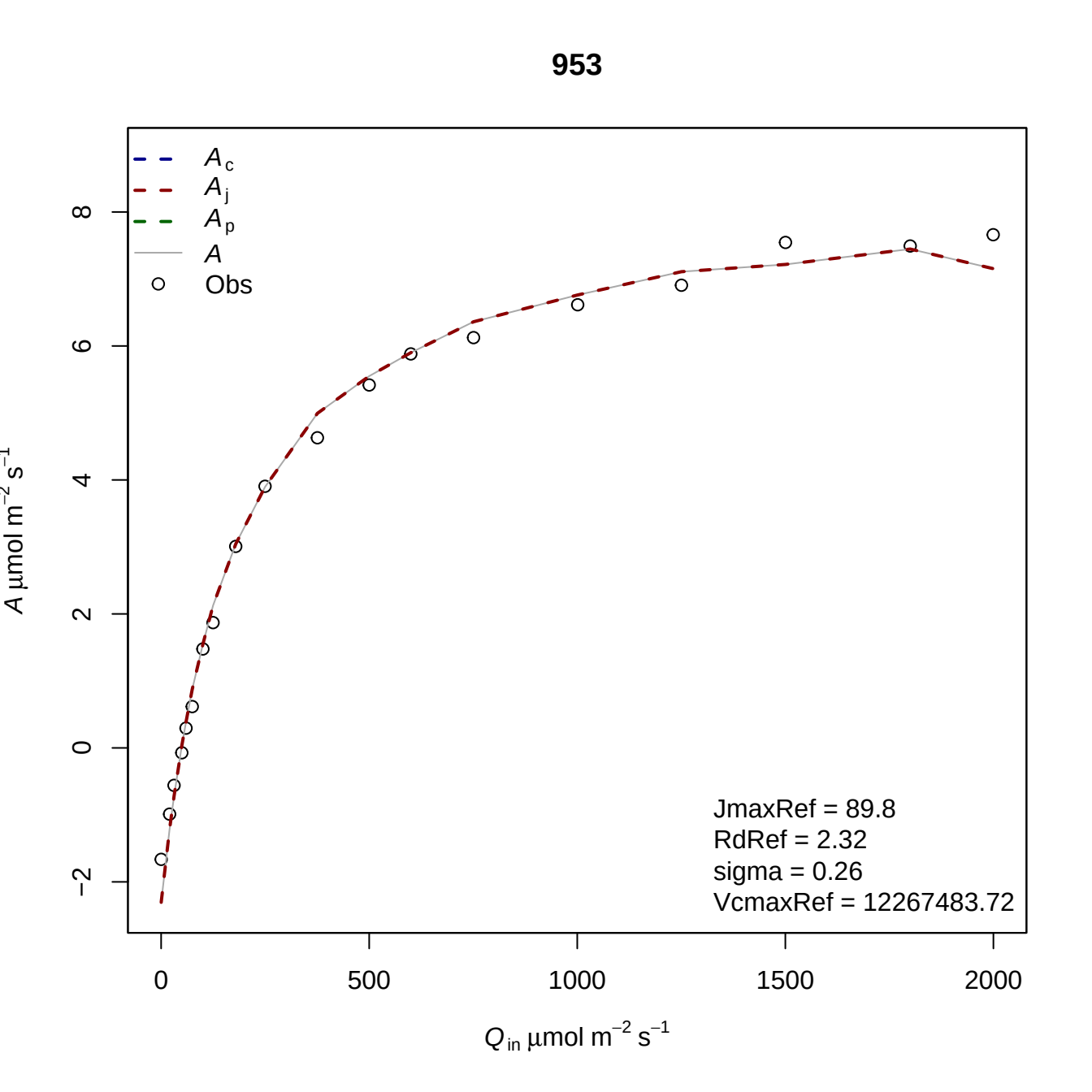

948
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.19
RdRef = 3.31
sigma = 0.55
VcmaxRef = 137360.87
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

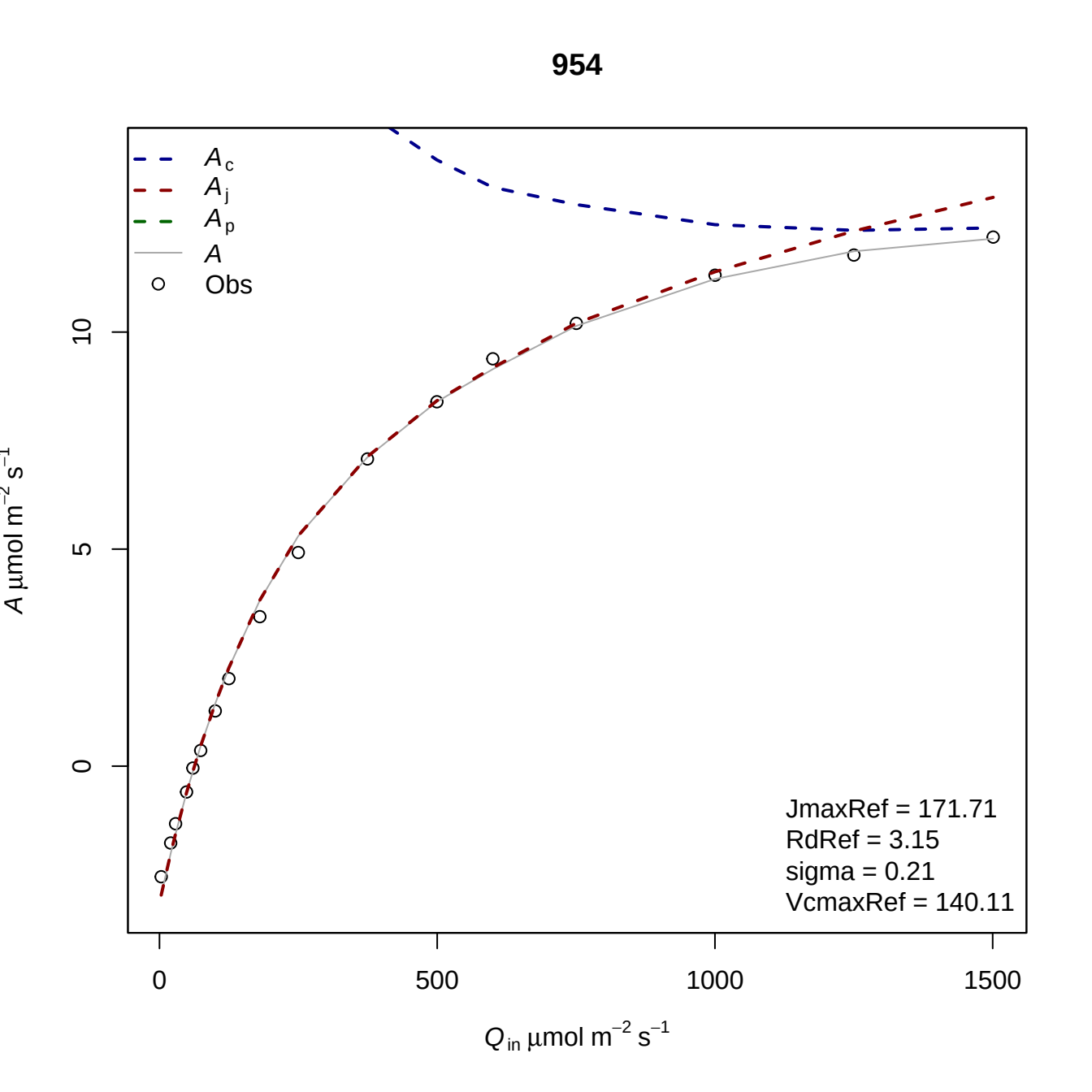

949
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 187.87
RdRef = 2.76
sigma = 0.42
VcmaxRef = 1096048.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

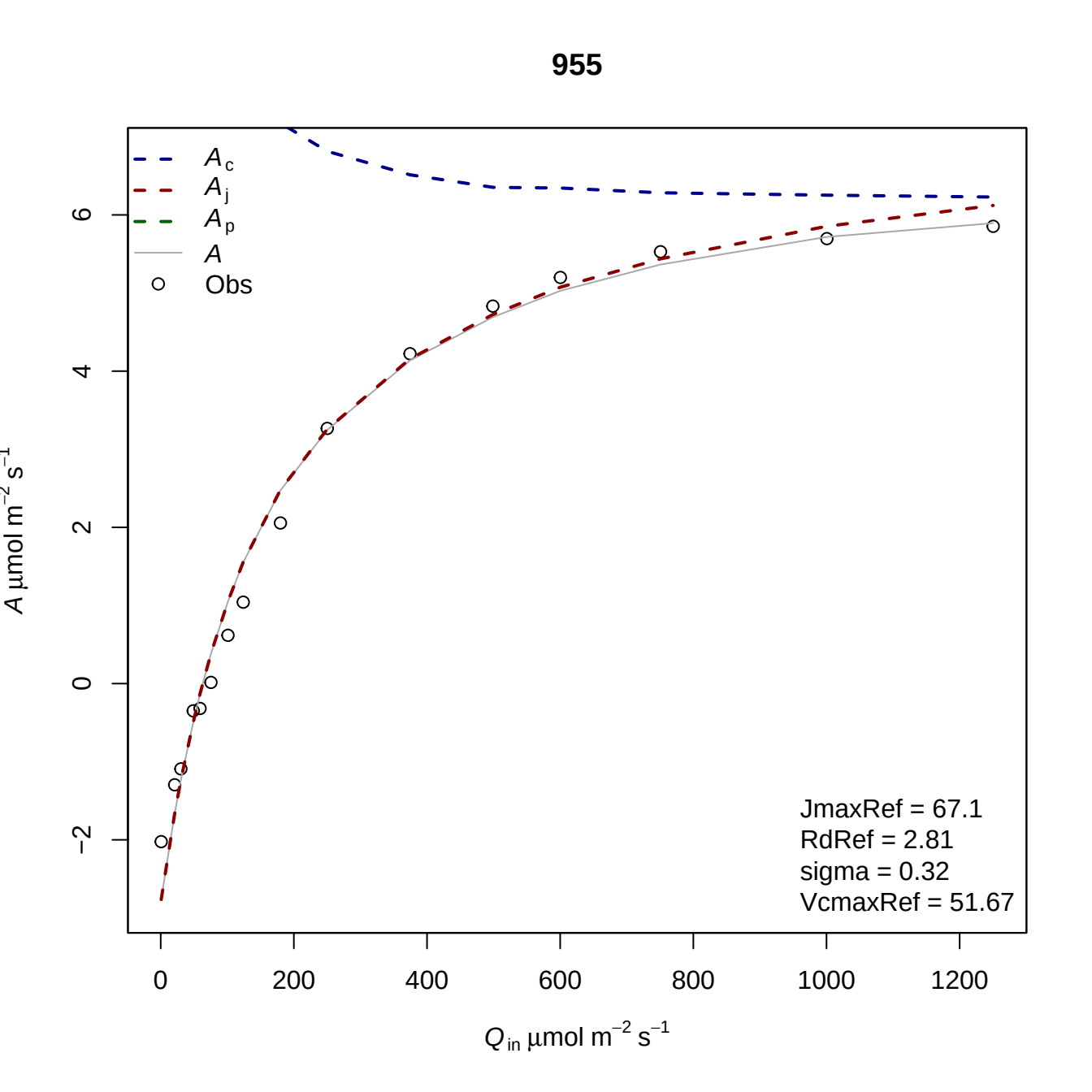

950
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.89
RdRef = 2.73
sigma = 0.25
VcmaxRef = 8291765.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

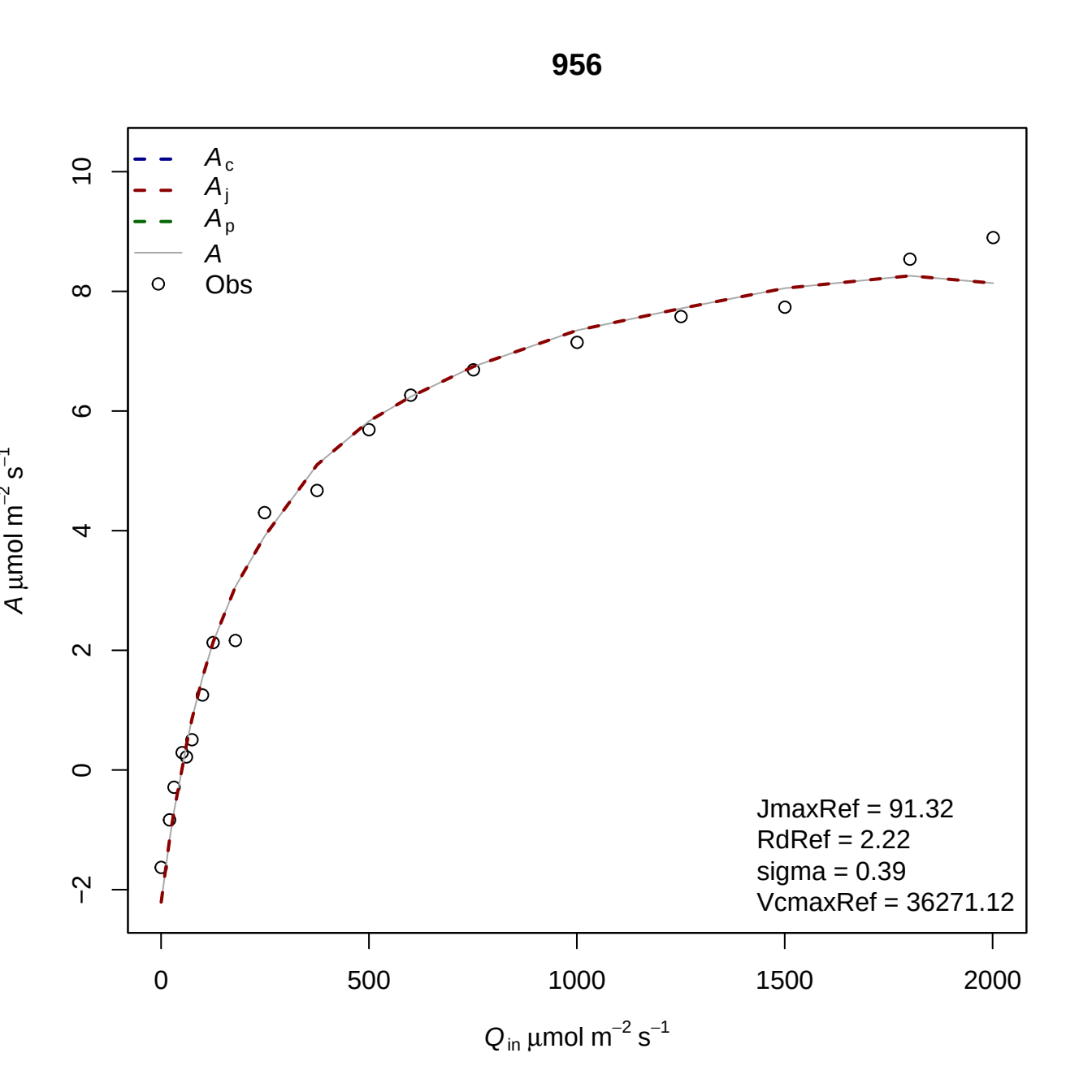

951
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.65
RdRef = 1.86
sigma = 0.19
VcmaxRef = 491129.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

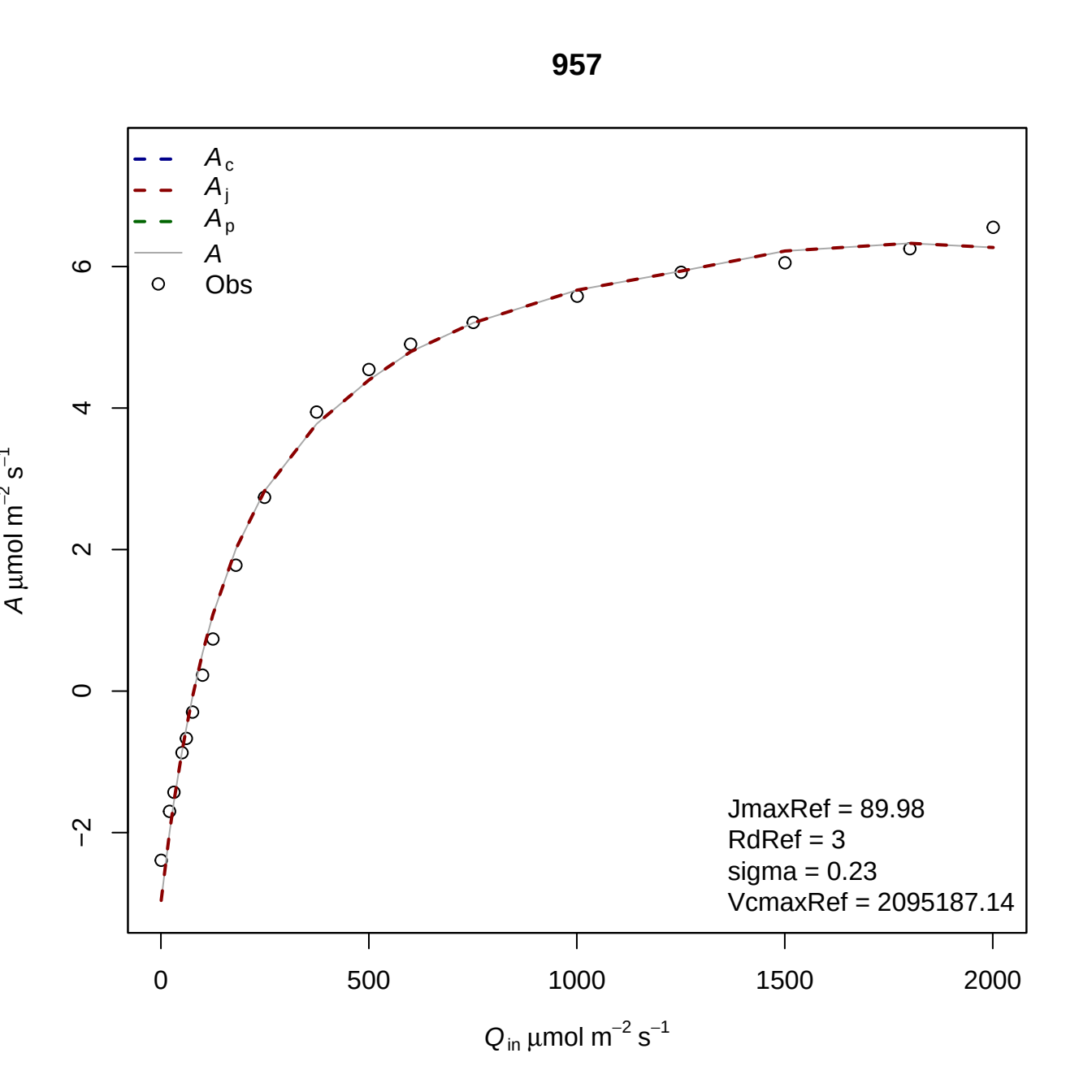

952
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 242.18
RdRef = 2.47
sigma = 0.18
VcmaxRef = 170
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

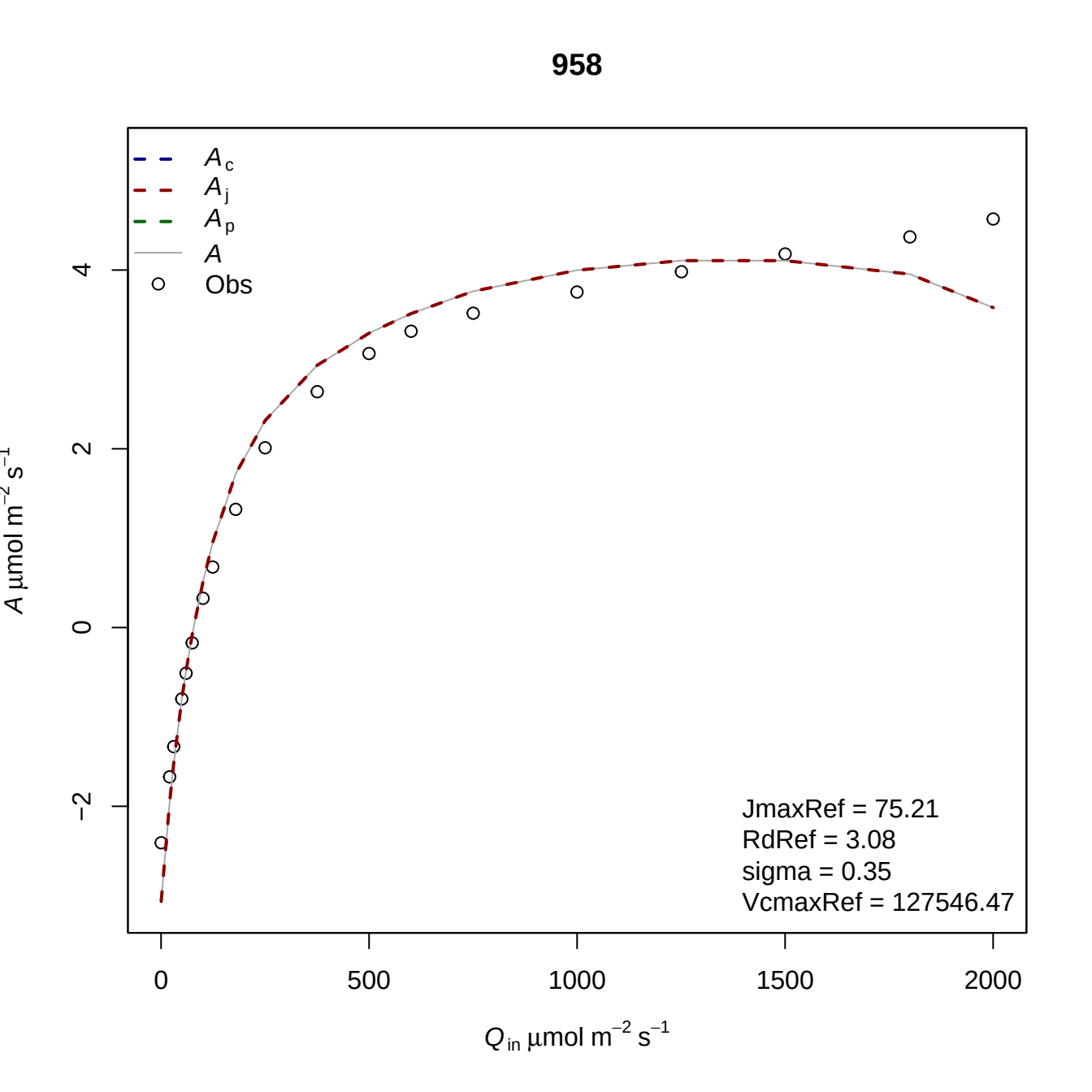

953
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.8
RdRef = 2.32
sigma = 0.26
VcmaxRef = 12267483.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

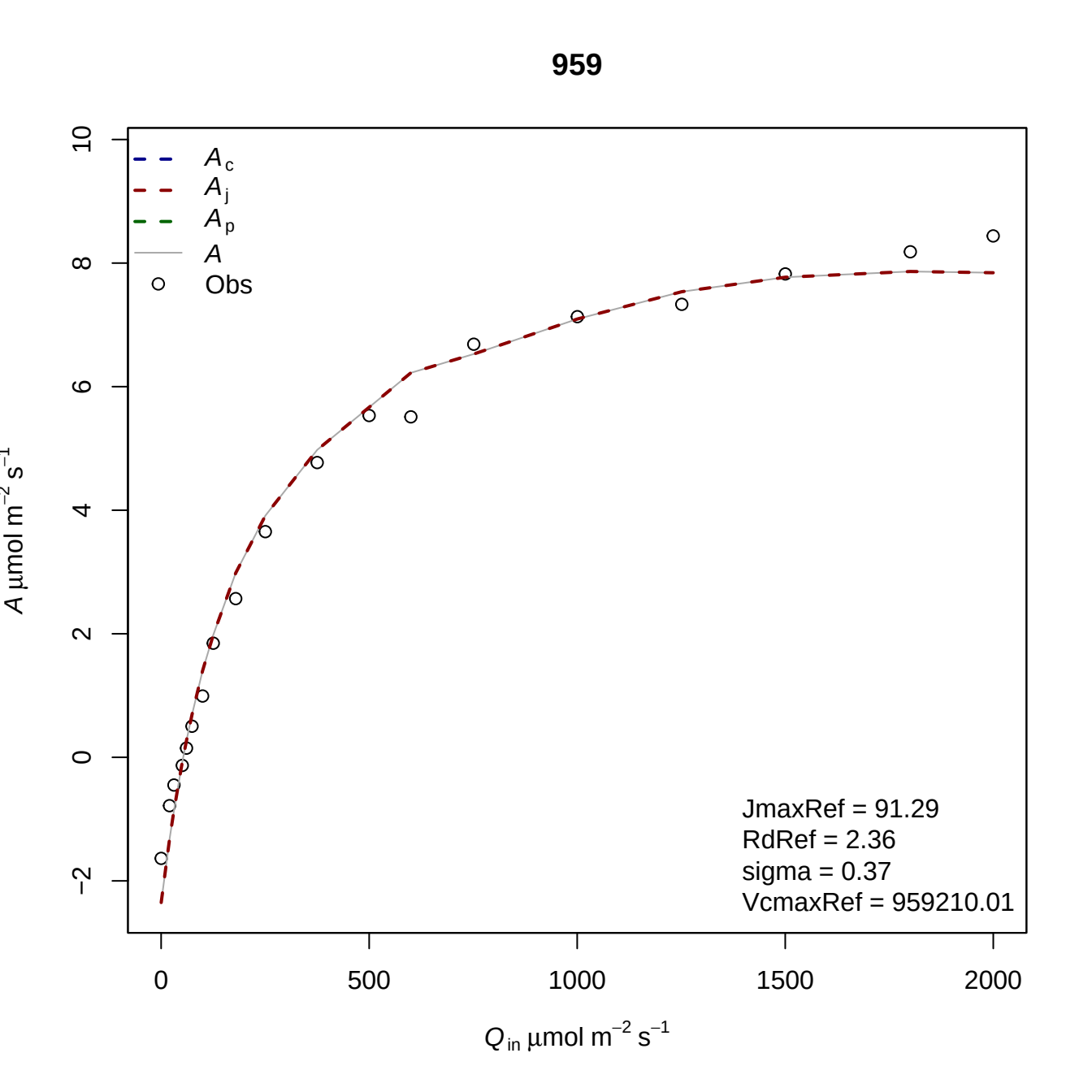

954
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 171.71
RdRef = 3.15
sigma = 0.21
VcmaxRef = 140.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

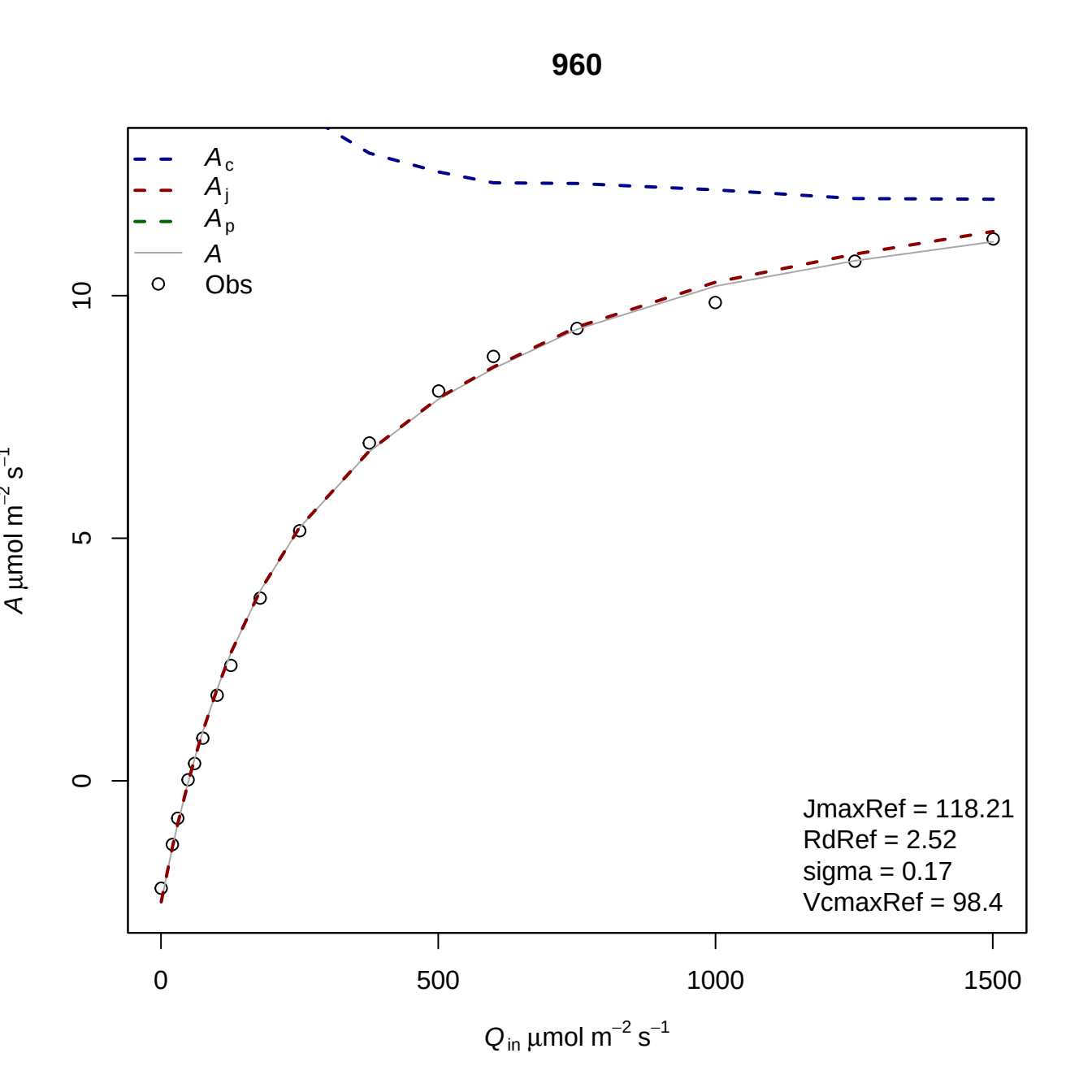

955
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.1
RdRef = 2.81
sigma = 0.32
VcmaxRef = 51.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200

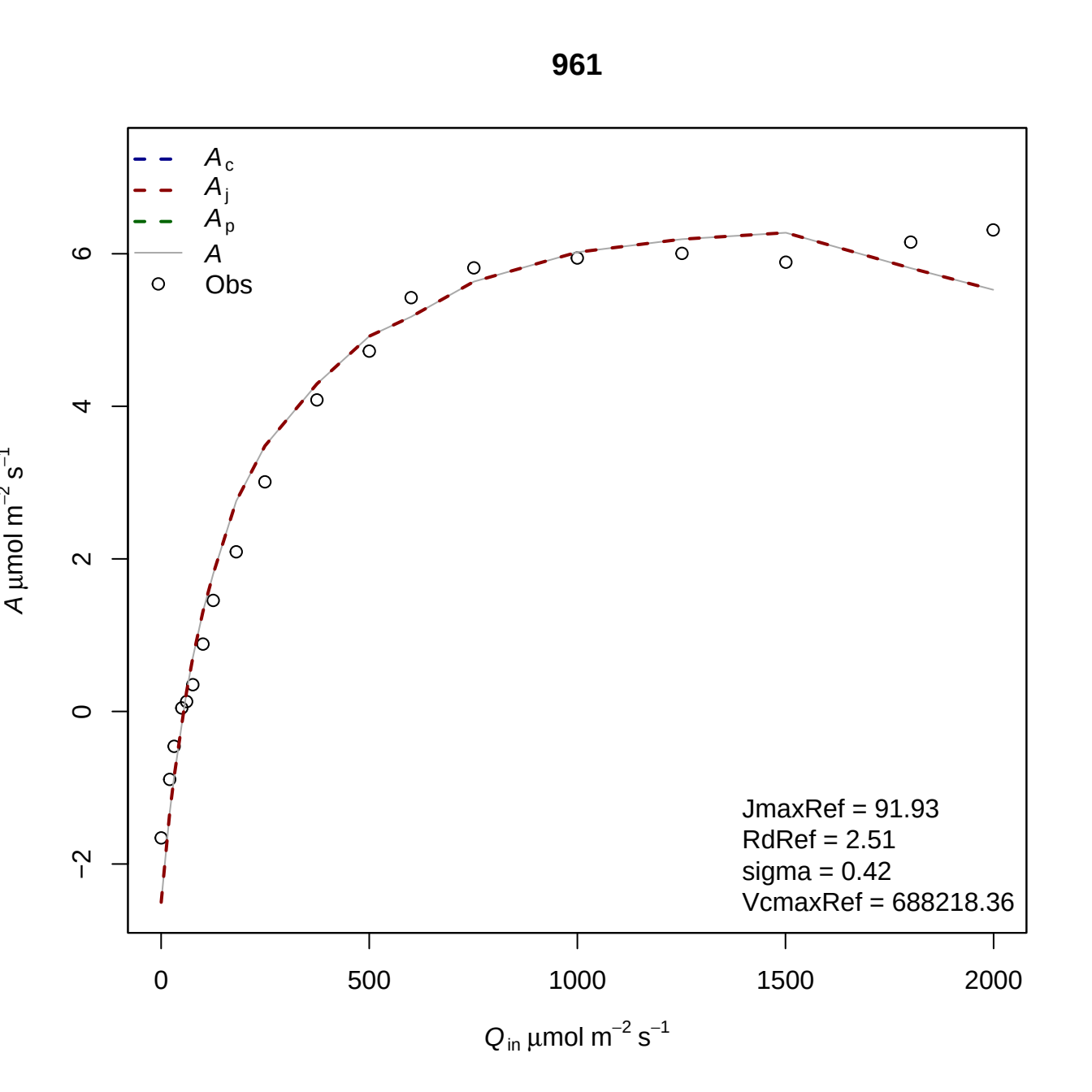

956
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.32
RdRef = 2.22
sigma = 0.39
VcmaxRef = 36271.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

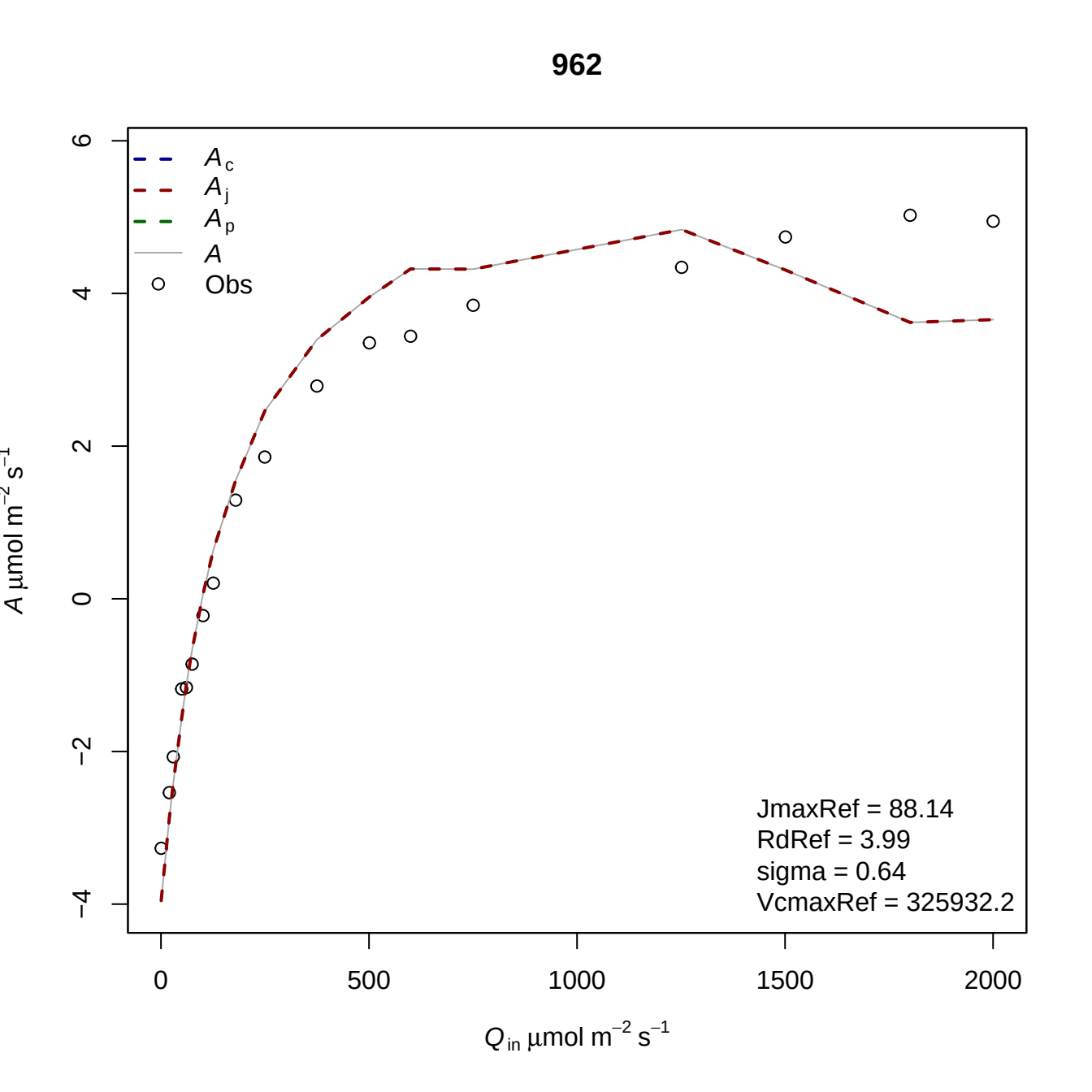

957
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.98
RdRef = 3
sigma = 0.23
VcmaxRef = 2095187.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

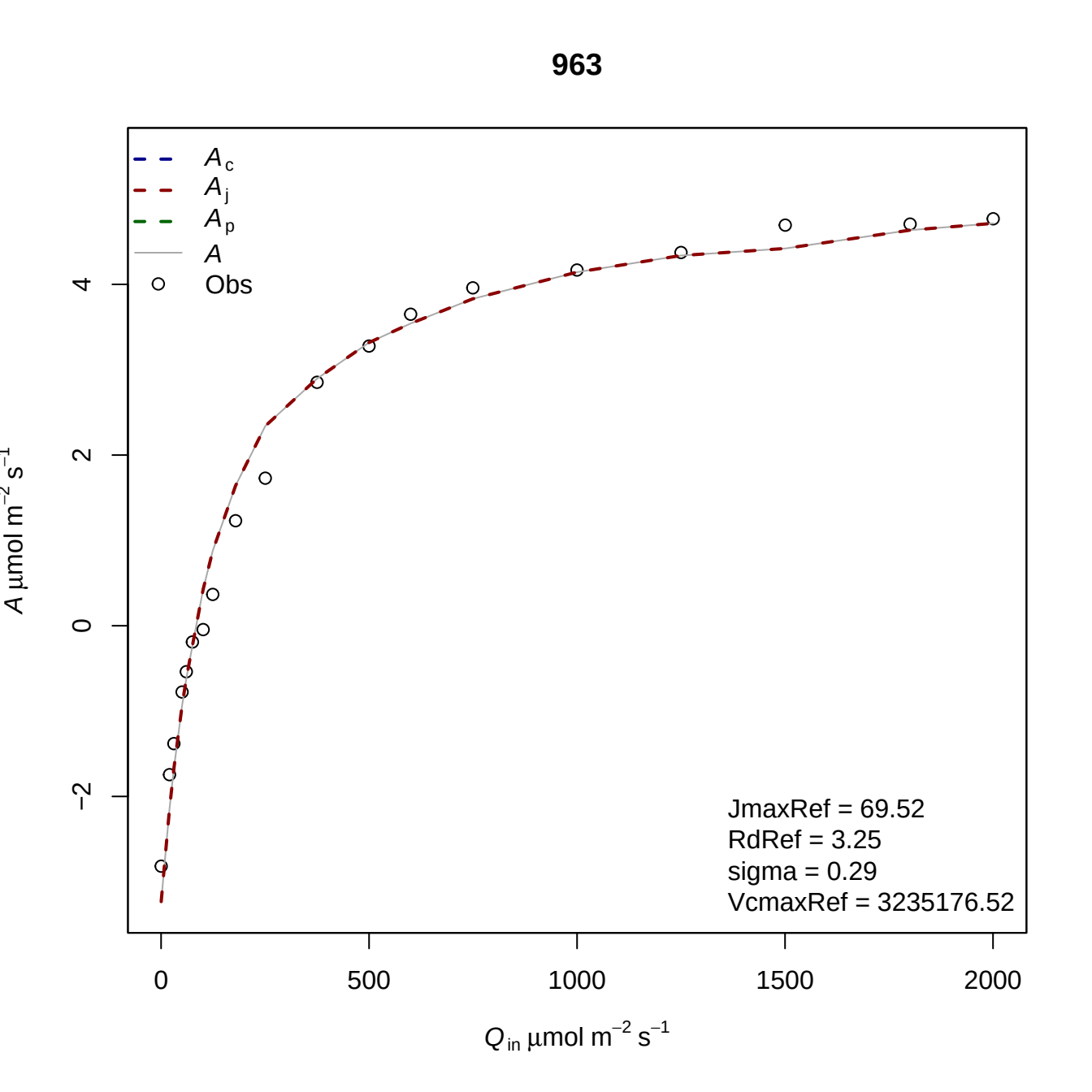

958
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.21
RdRef = 3.08
sigma = 0.35
VcmaxRef = 127546.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

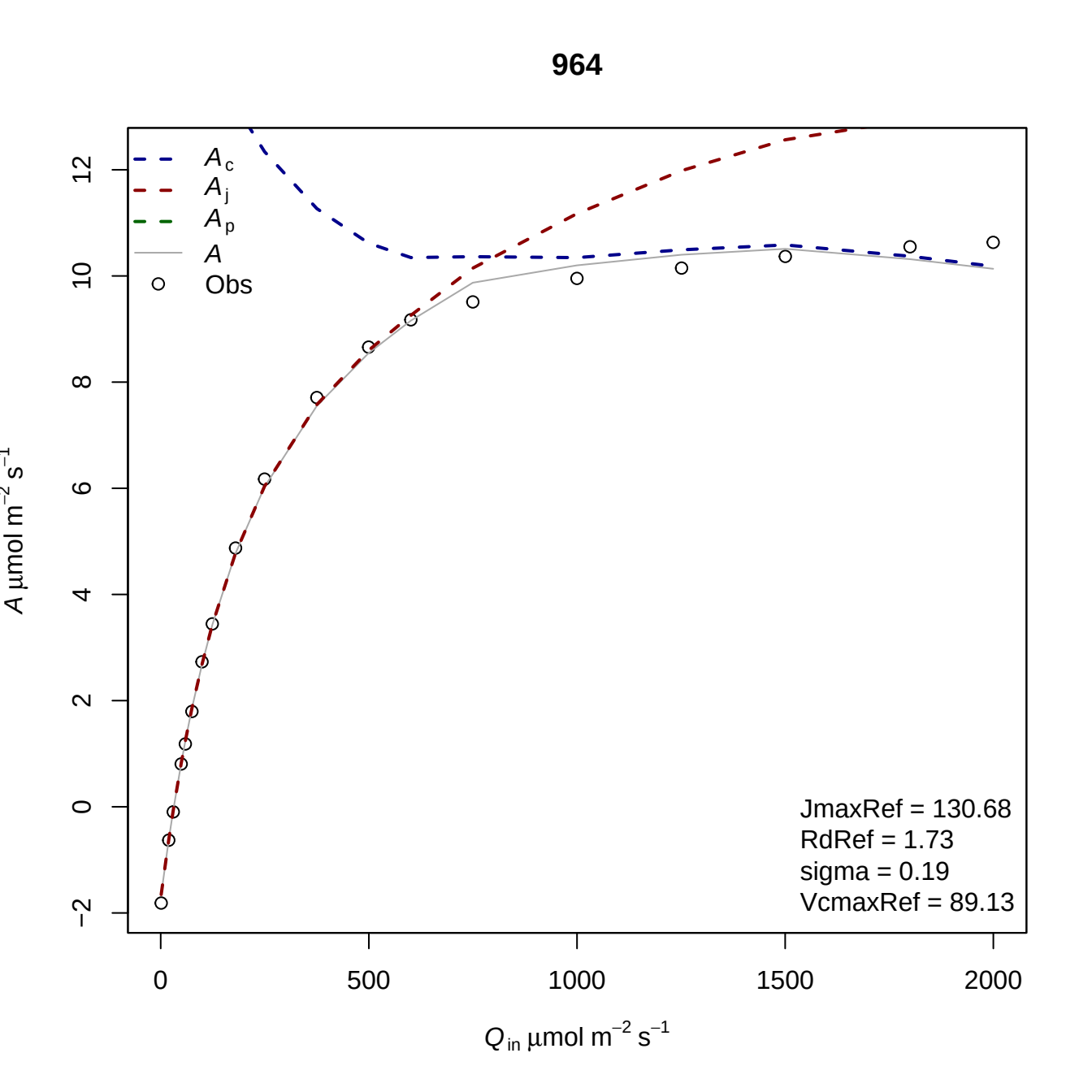

959
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.29
RdRef = 2.36
sigma = 0.37
VcmaxRef = 959210.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

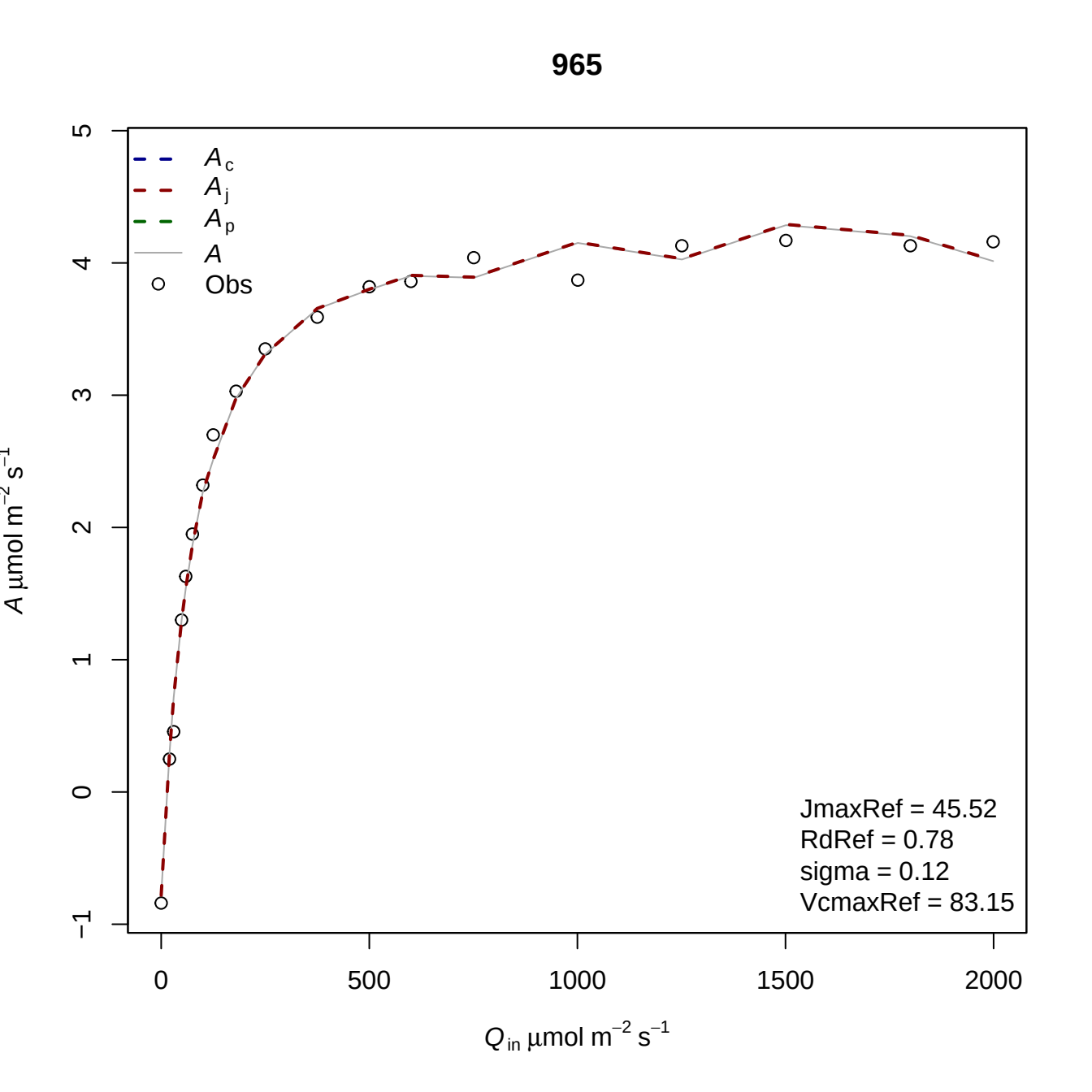

960
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.21
RdRef = 2.52
sigma = 0.17
VcmaxRef = 98.4
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

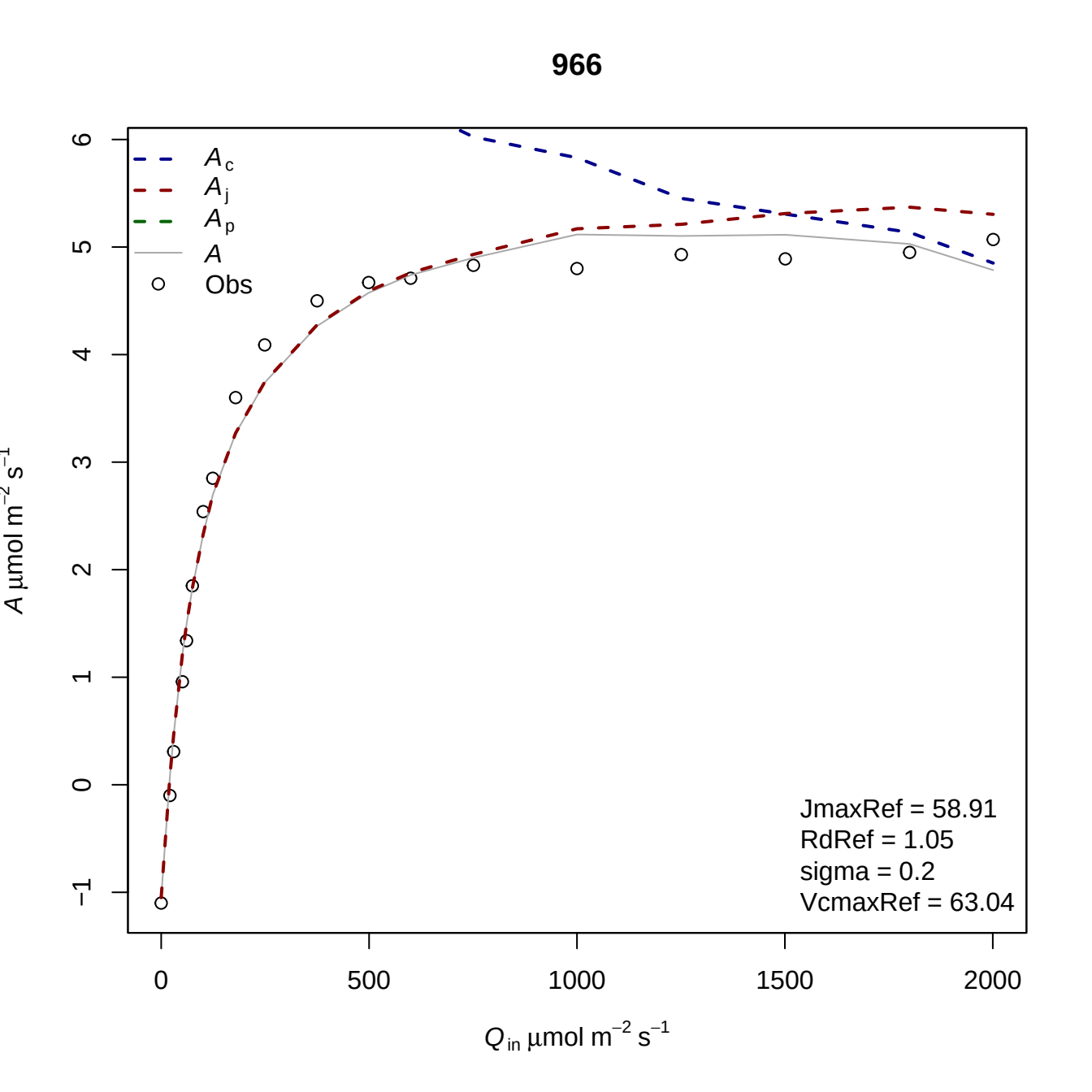

961
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.93
RdRef = 2.51
sigma = 0.42
VcmaxRef = 688218.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

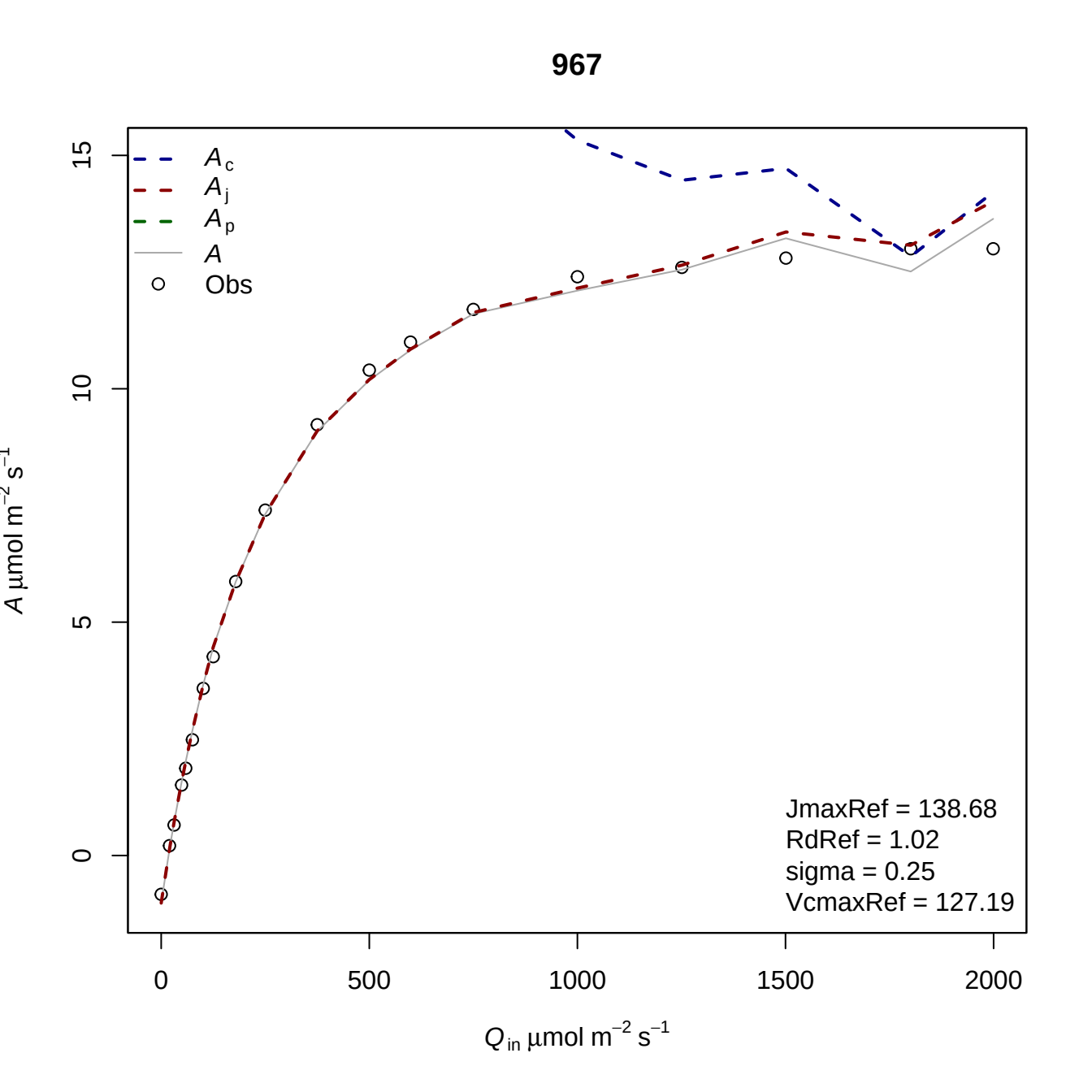

962
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.14
RdRef = 3.99
sigma = 0.64
VcmaxRef = 325932.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500
2000

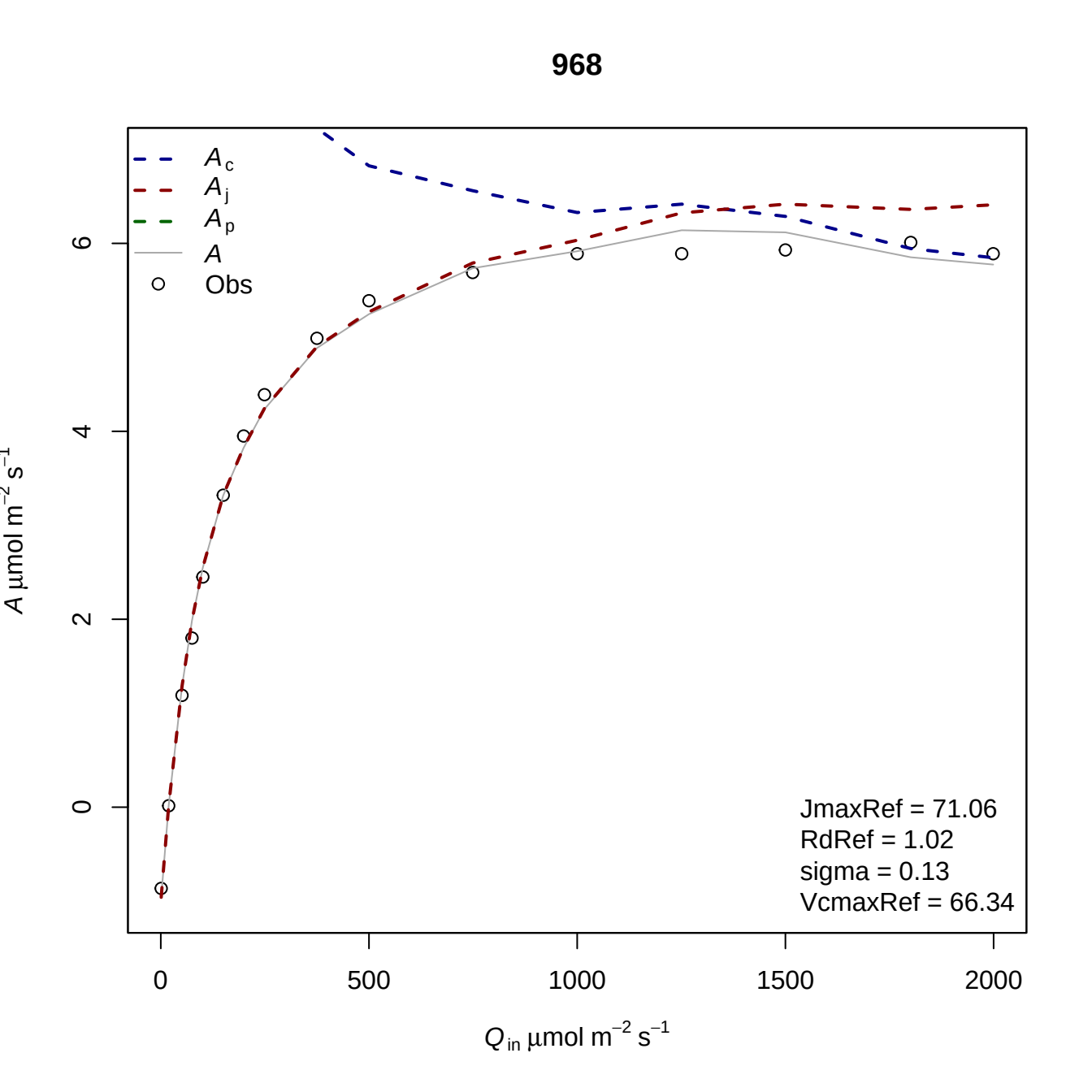

963
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.52
RdRef = 3.25
sigma = 0.29
VcmaxRef = 3235176.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500
2000

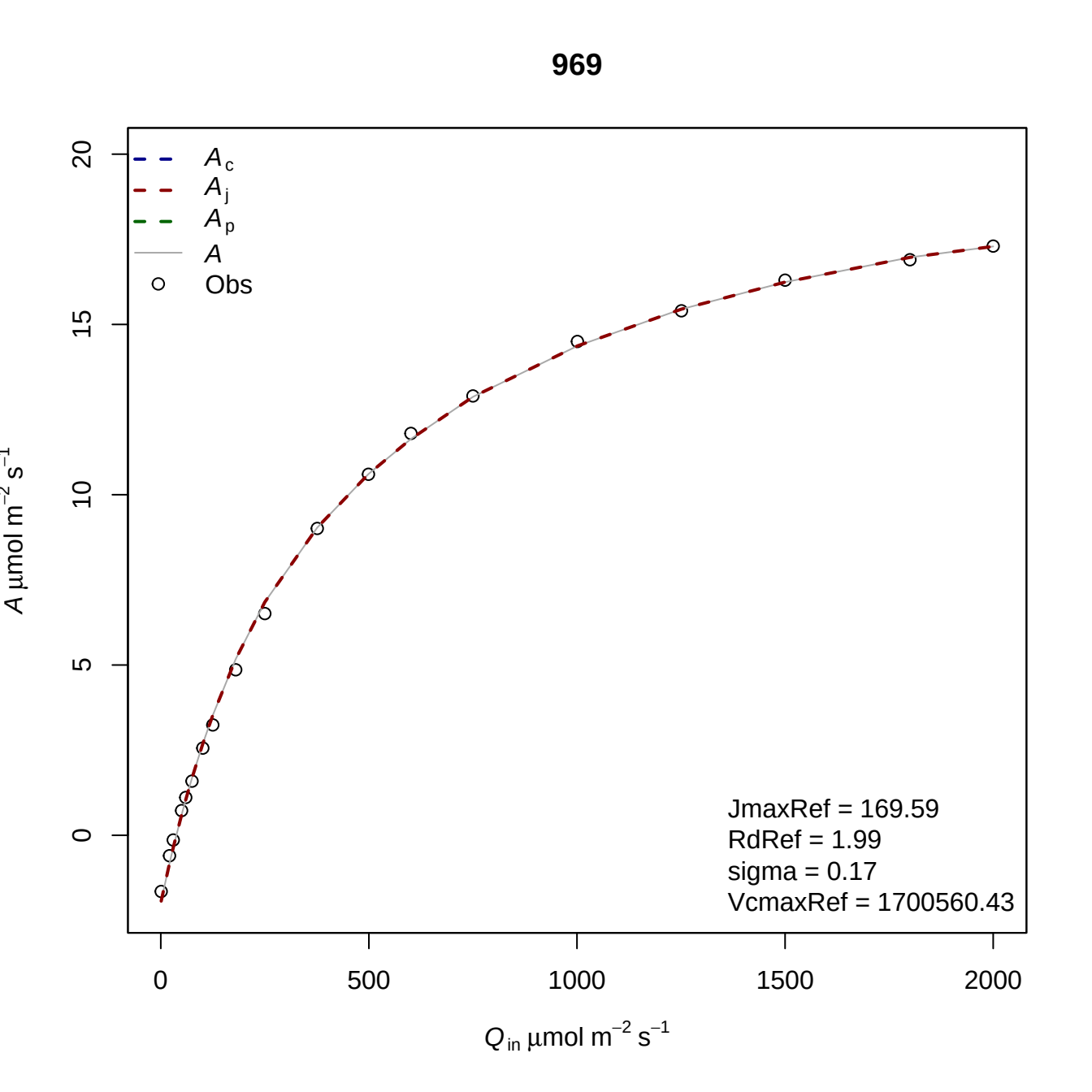

964
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.68
RdRef = 1.73
sigma = 0.19
VcmaxRef = 89.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

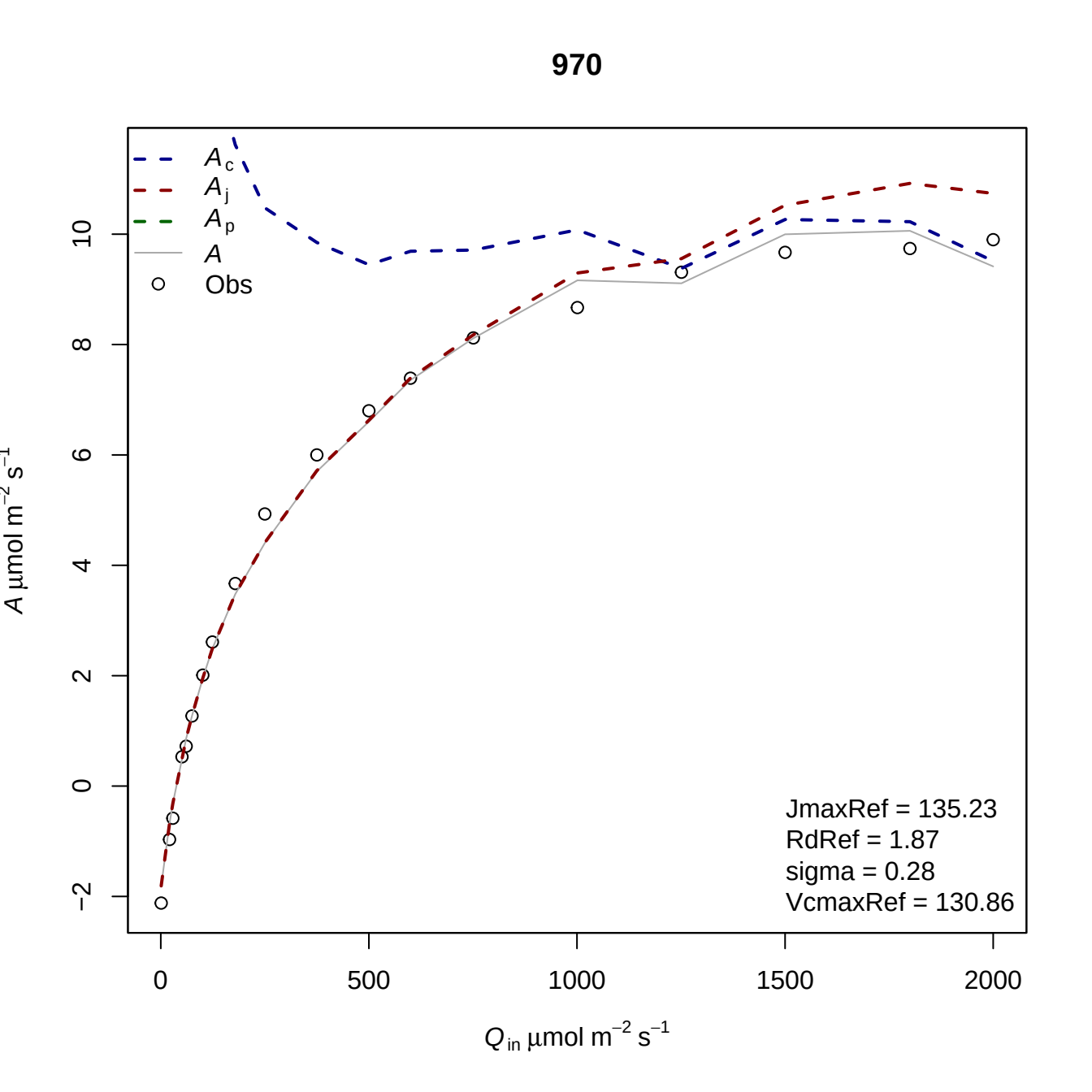

965
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.52
RdRef = 0.78
sigma = 0.12
VcmaxRef = 83.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500
2000

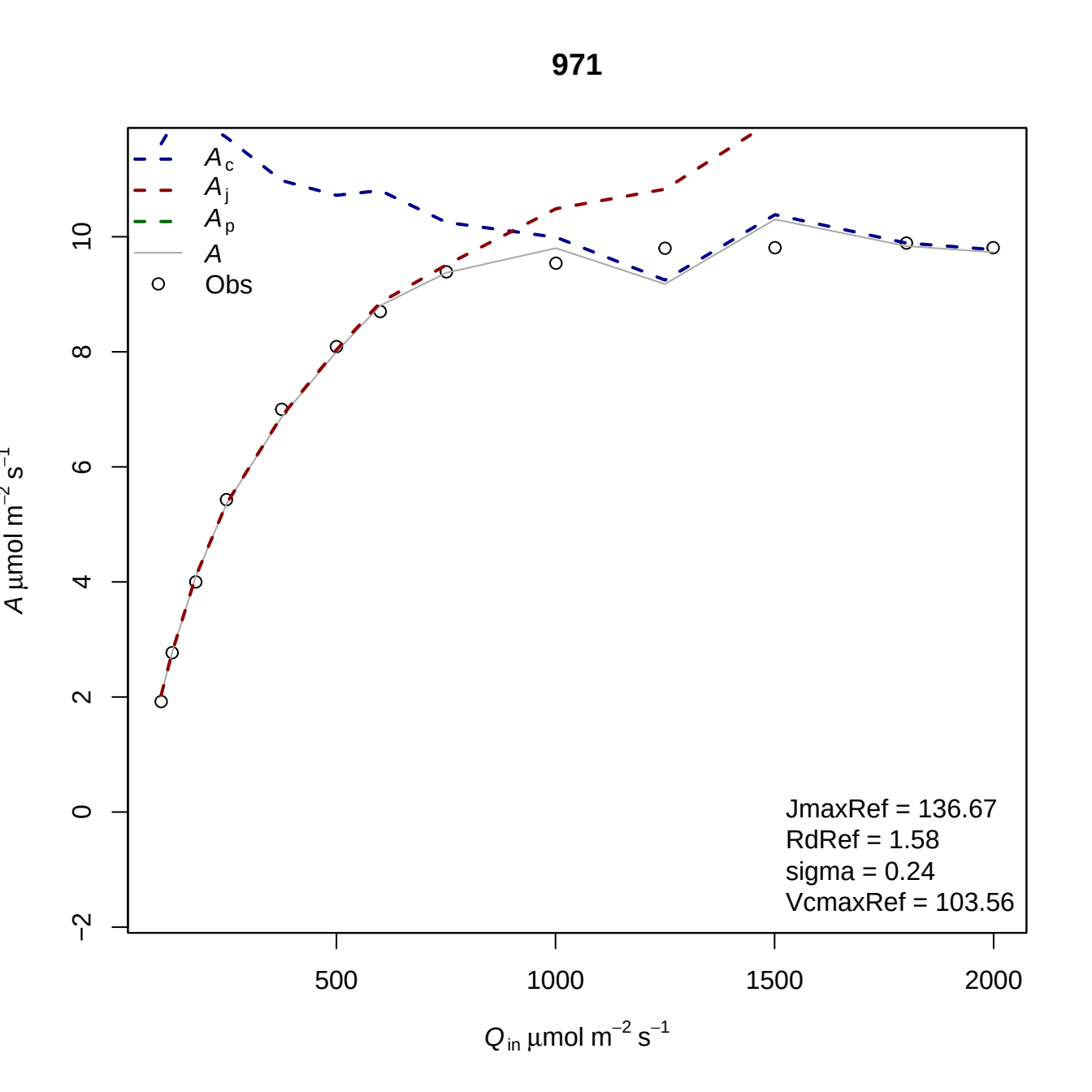

966
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.91
RdRef = 1.05
sigma = 0.2
VcmaxRef = 63.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

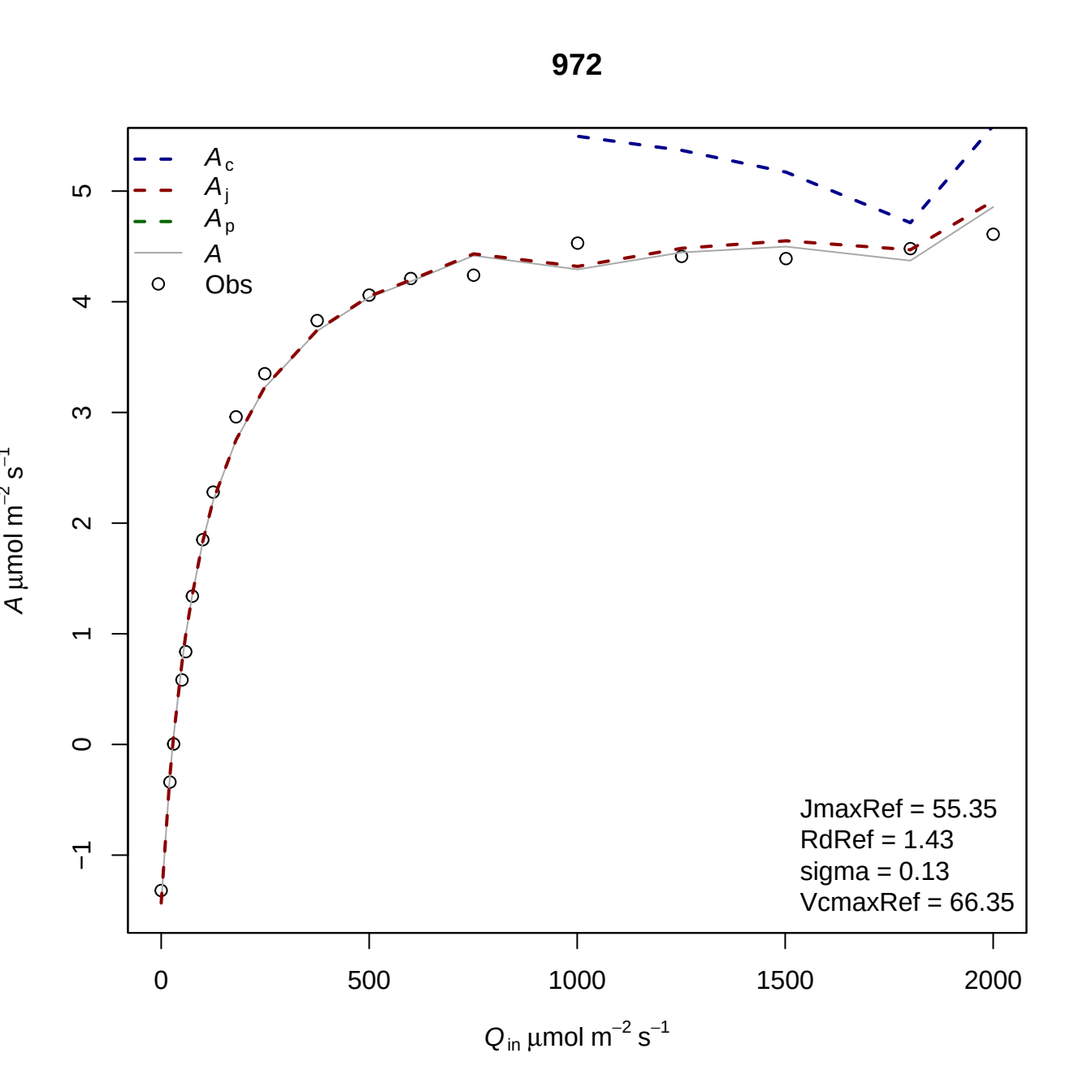

967
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 138.68
RdRef = 1.02
sigma = 0.25
VcmaxRef = 127.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

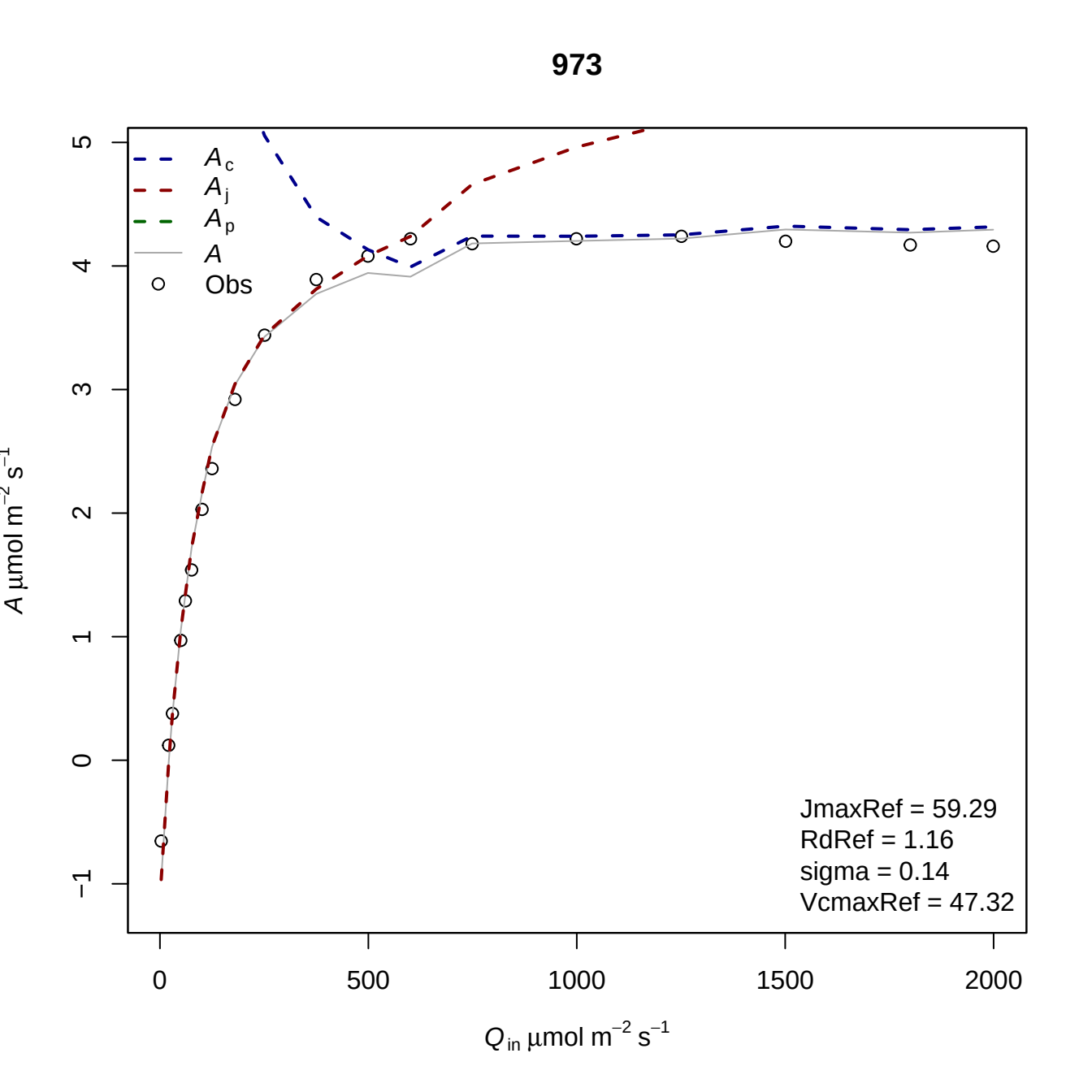

968
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.06
RdRef = 1.02
sigma = 0.13
VcmaxRef = 66.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

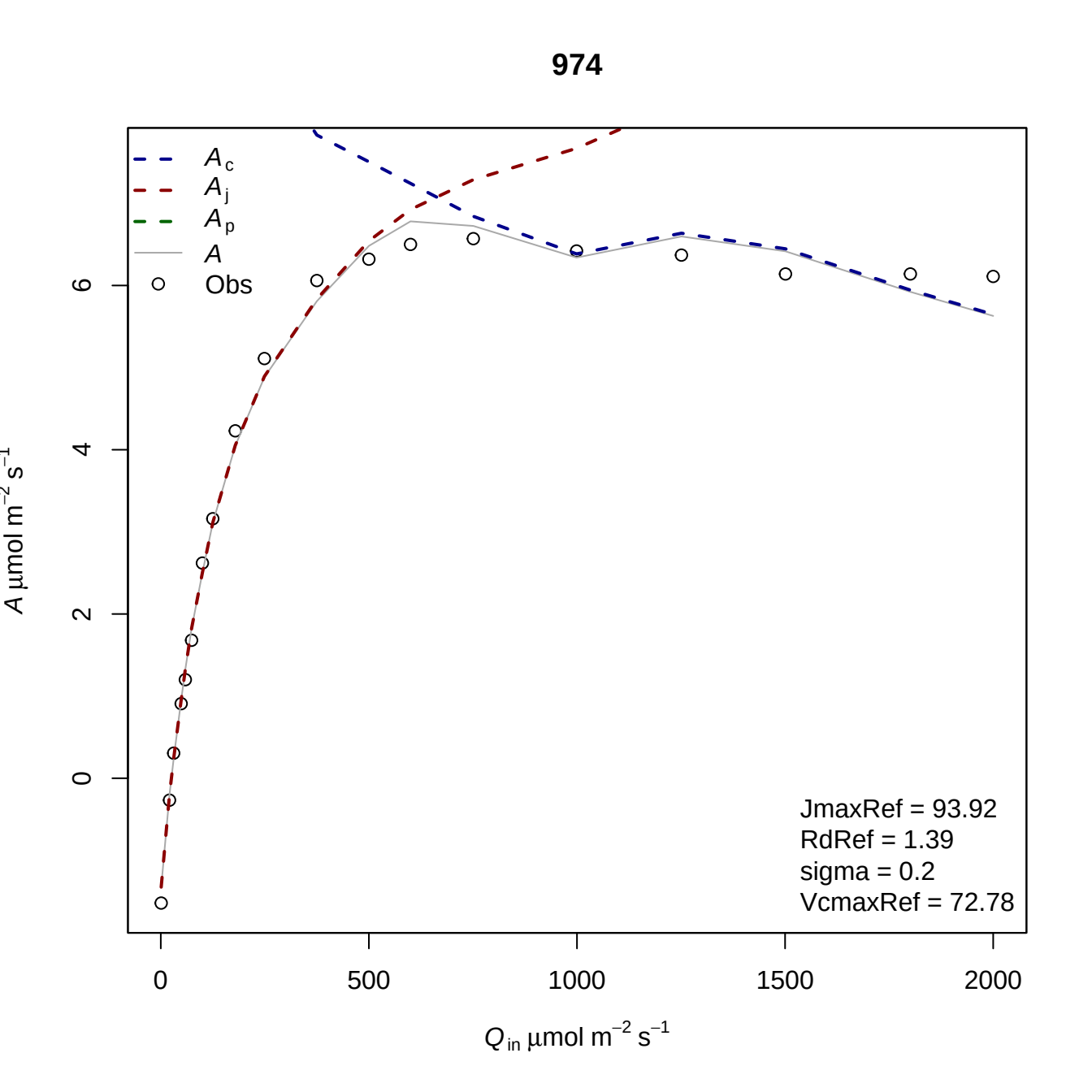

969
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 169.59
RdRef = 1.99
sigma = 0.17
VcmaxRef = 1700560.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

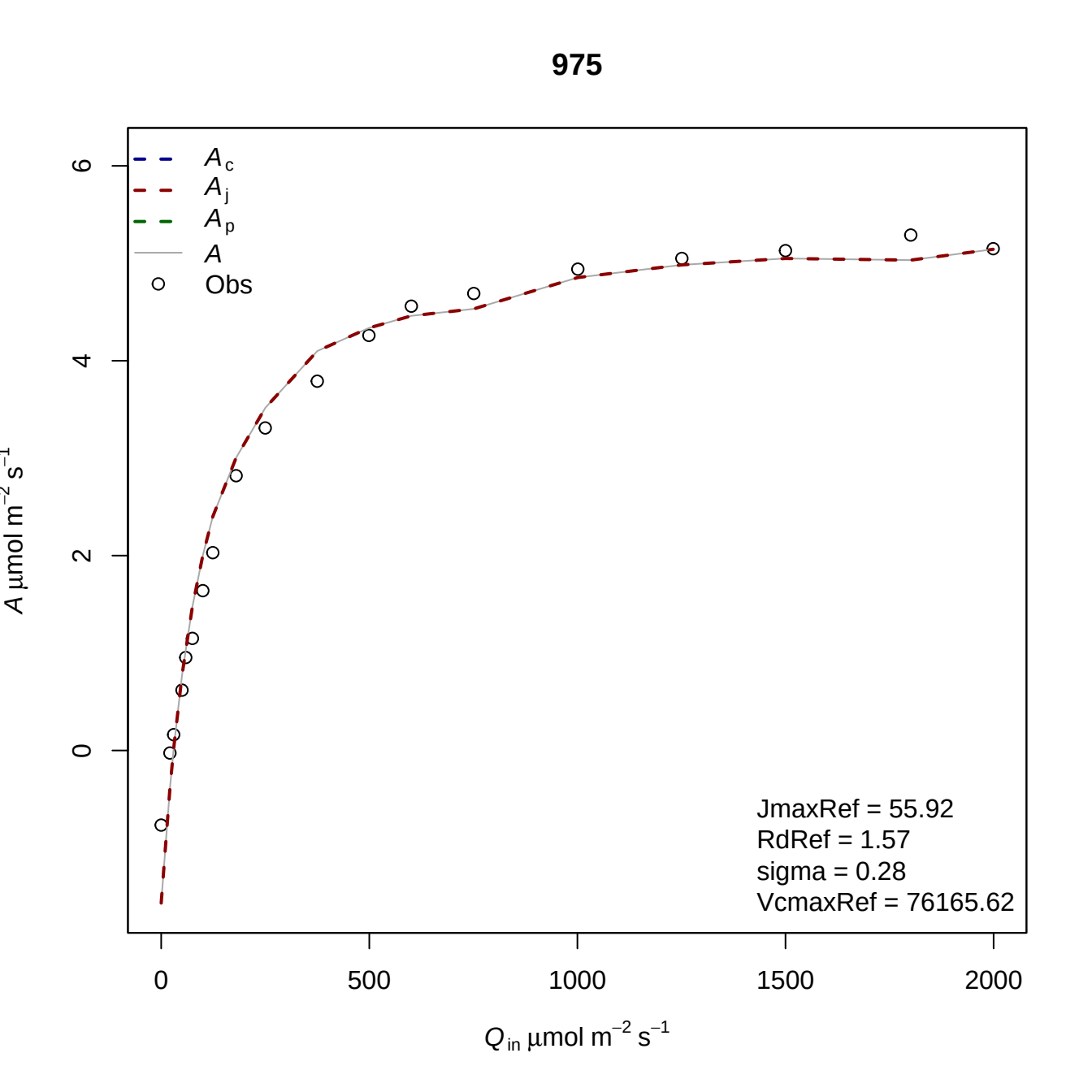

970
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.23
RdRef = 1.87
sigma = 0.28
VcmaxRef = 130.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

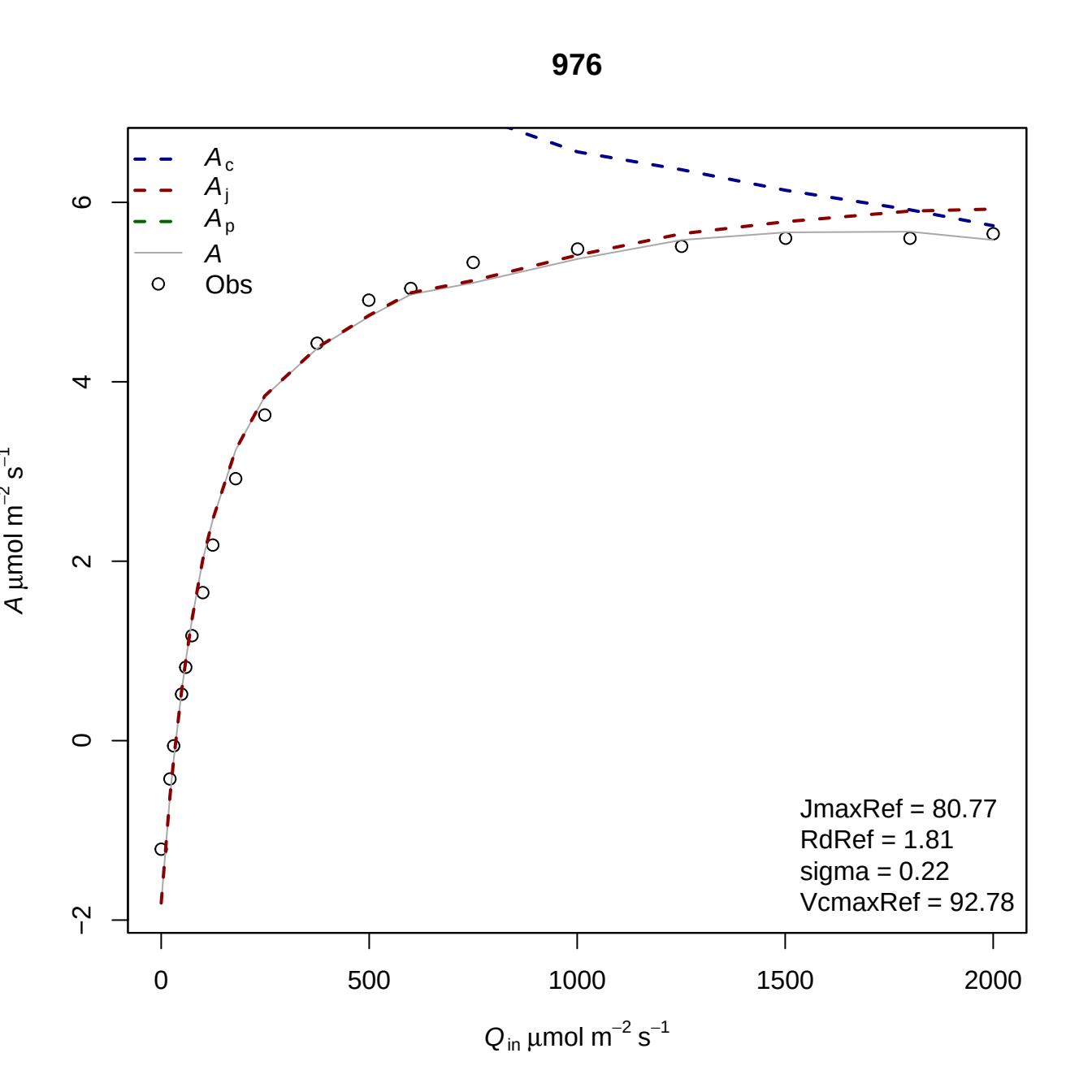

971
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.67
RdRef = 1.58
sigma = 0.24
VcmaxRef = 103.56
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
500
1000
1500
2000

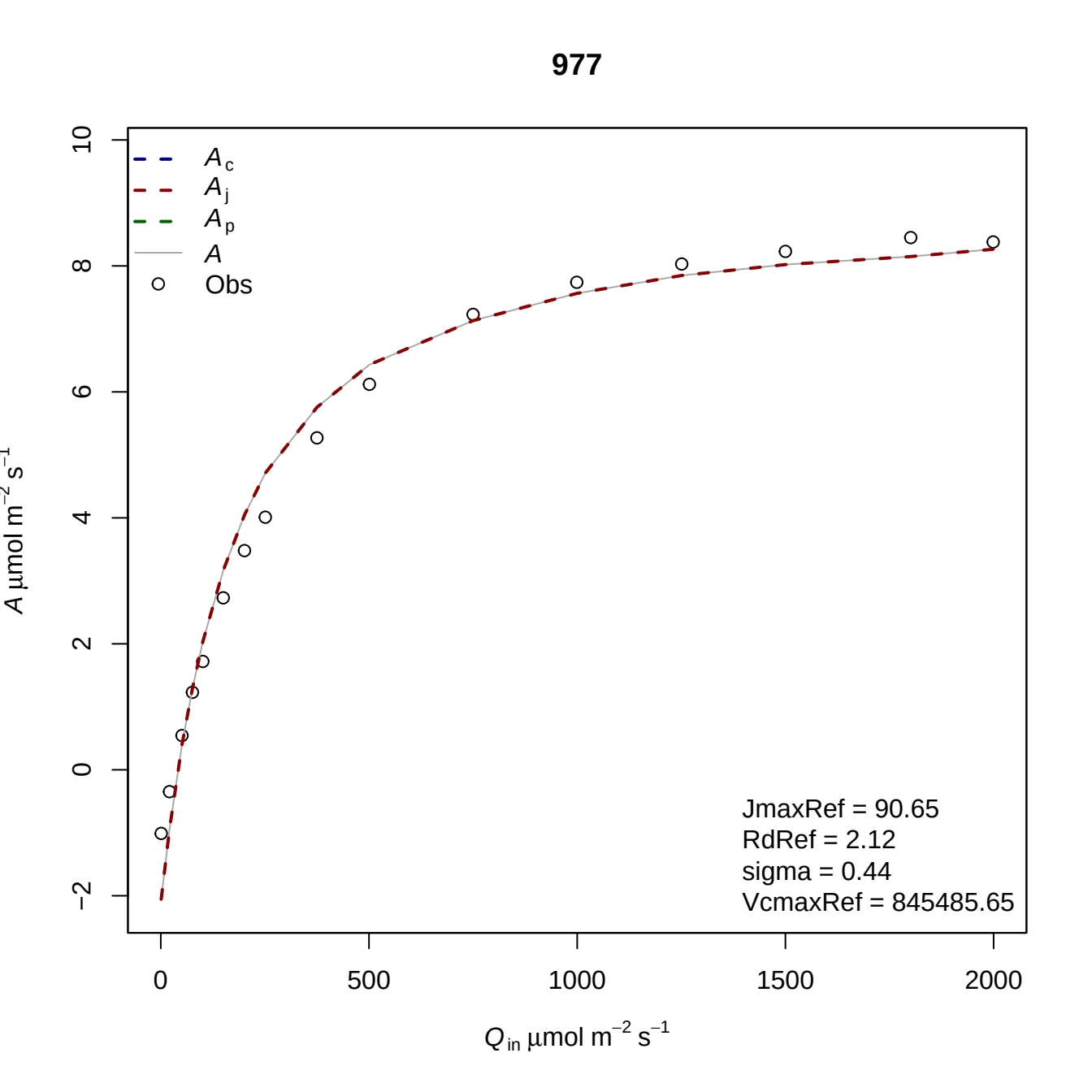

972
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.35
RdRef = 1.43
sigma = 0.13
VcmaxRef = 66.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500
2000

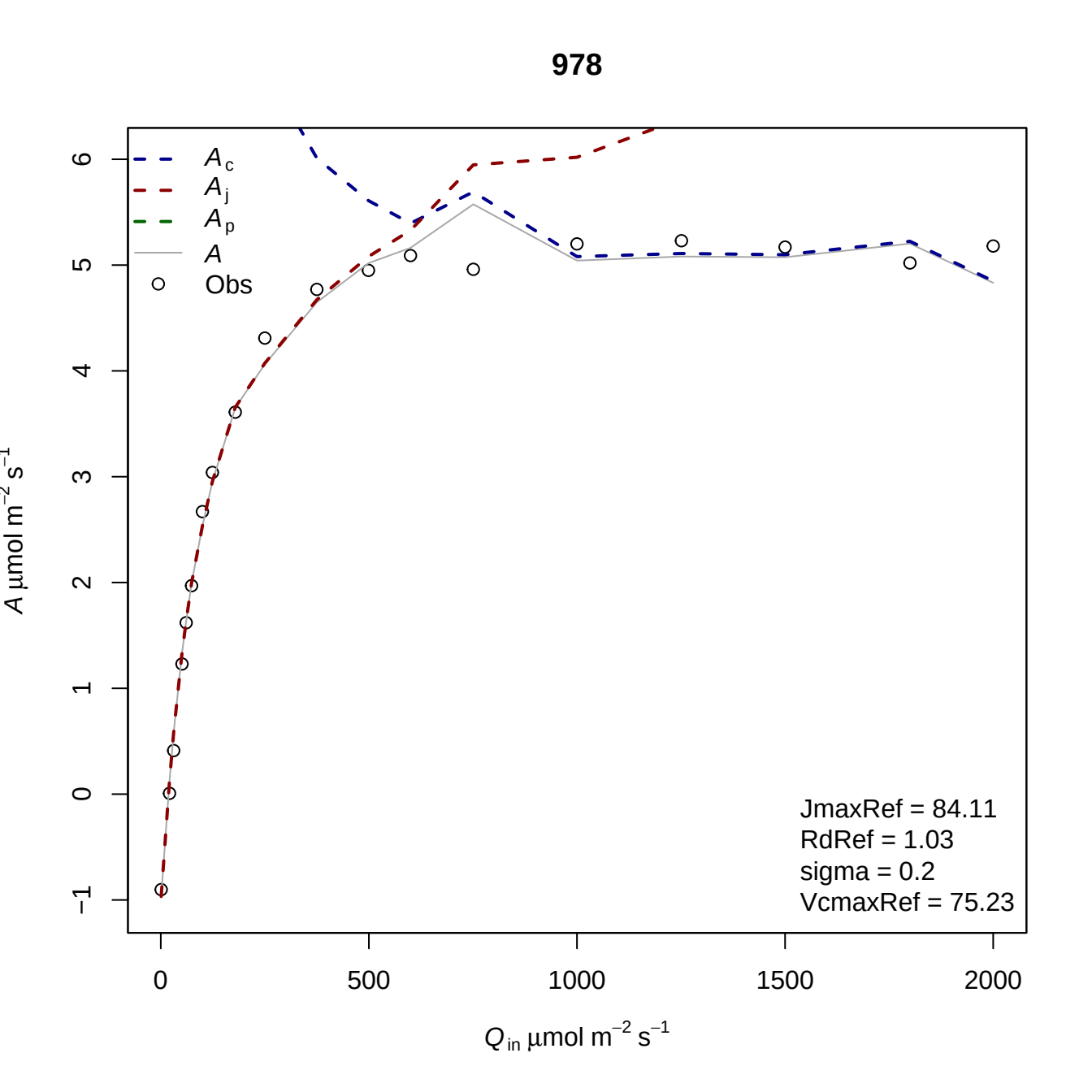

973
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.29
RdRef = 1.16
sigma = 0.14
VcmaxRef = 47.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500
2000

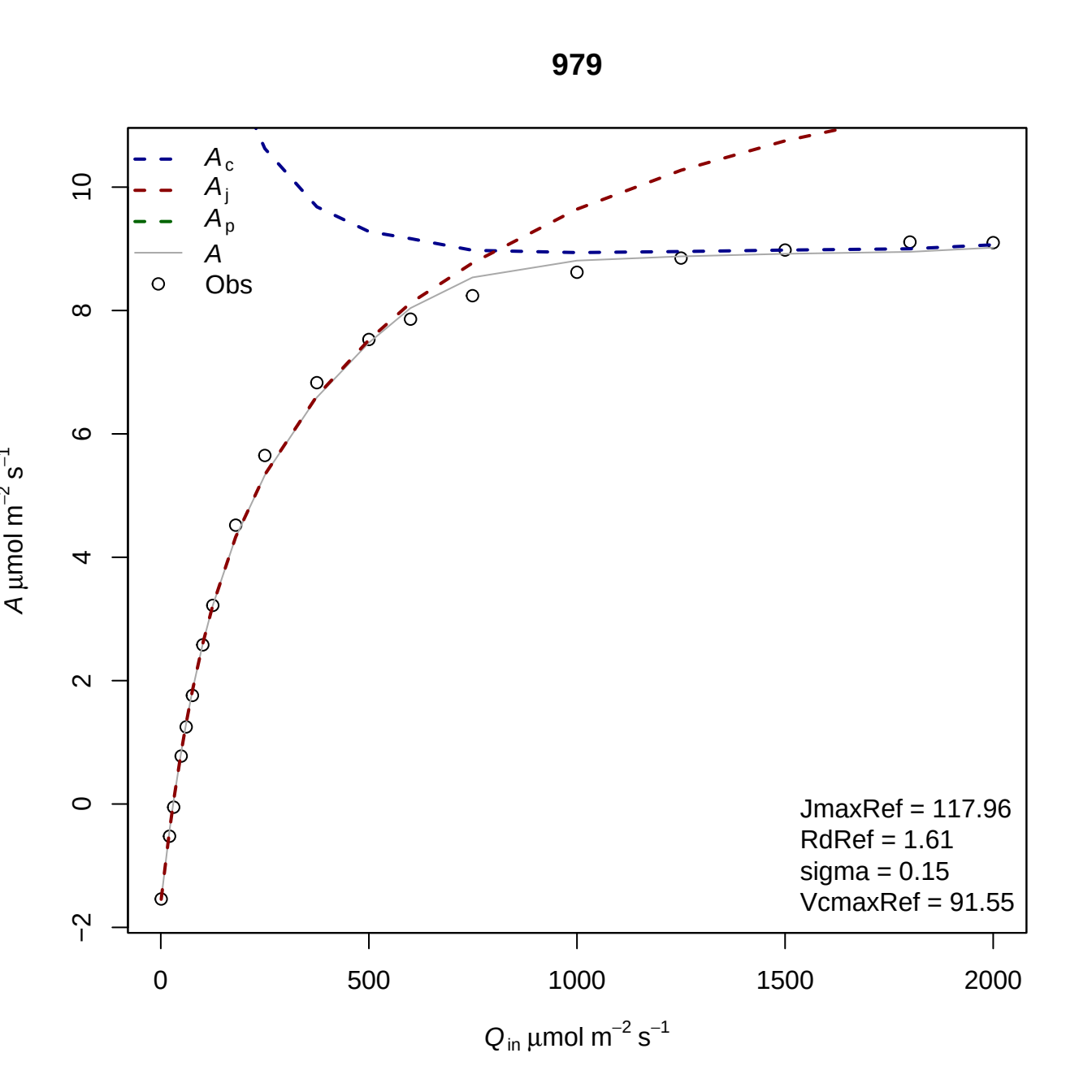

974
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.92
RdRef = 1.39
sigma = 0.2
VcmaxRef = 72.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

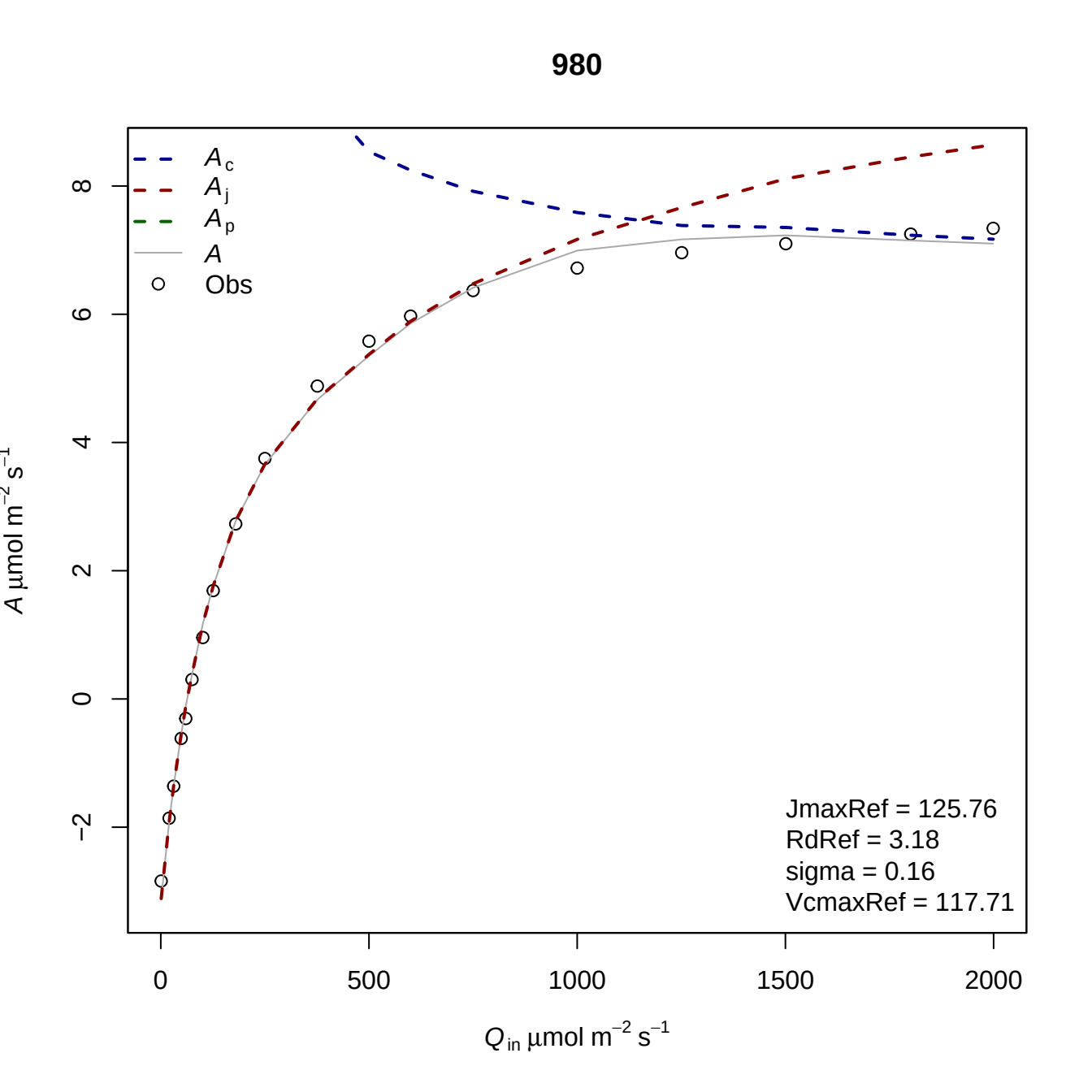

975
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.92
RdRef = 1.57
sigma = 0.28
VcmaxRef = 76165.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

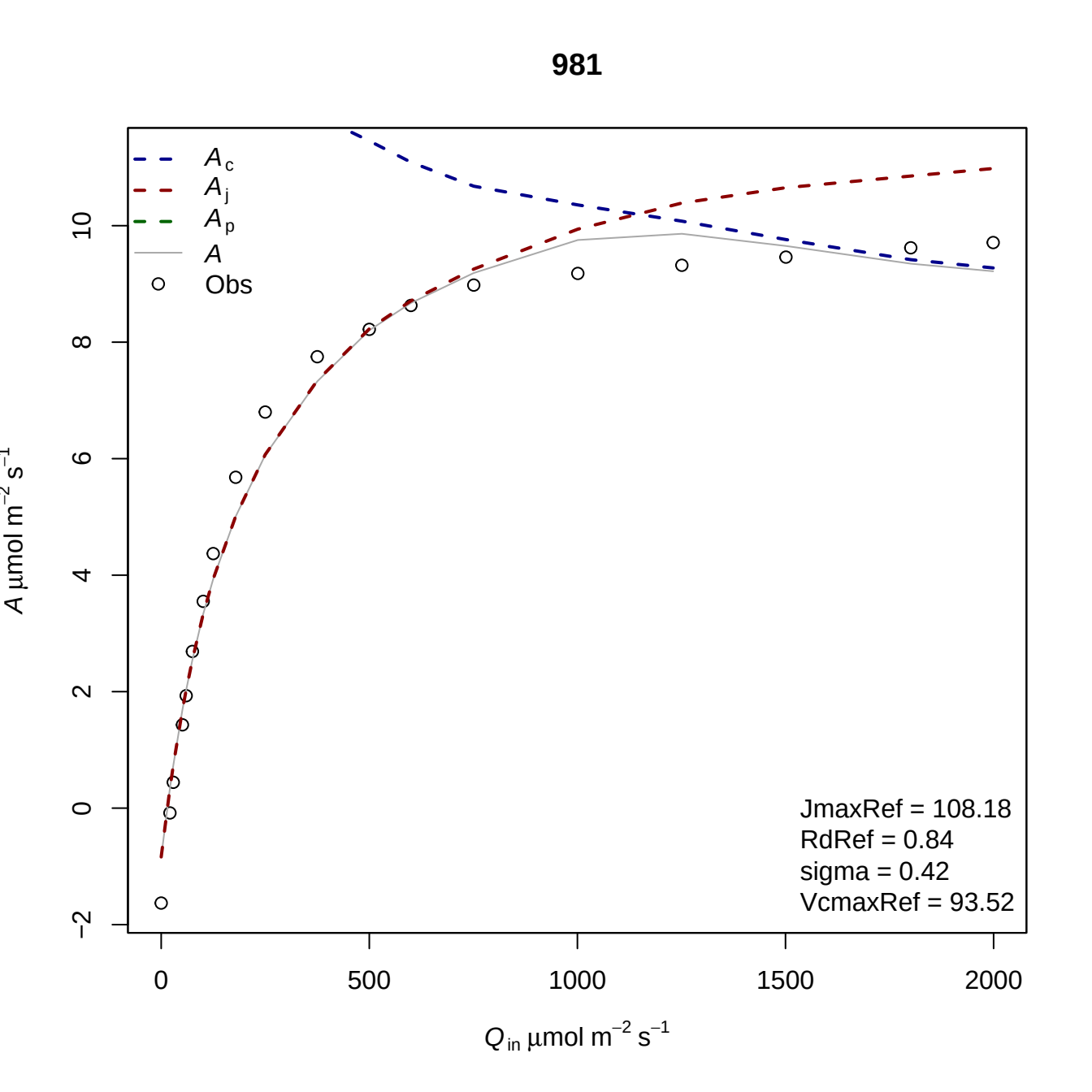

976
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.77
RdRef = 1.81
sigma = 0.22
VcmaxRef = 92.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

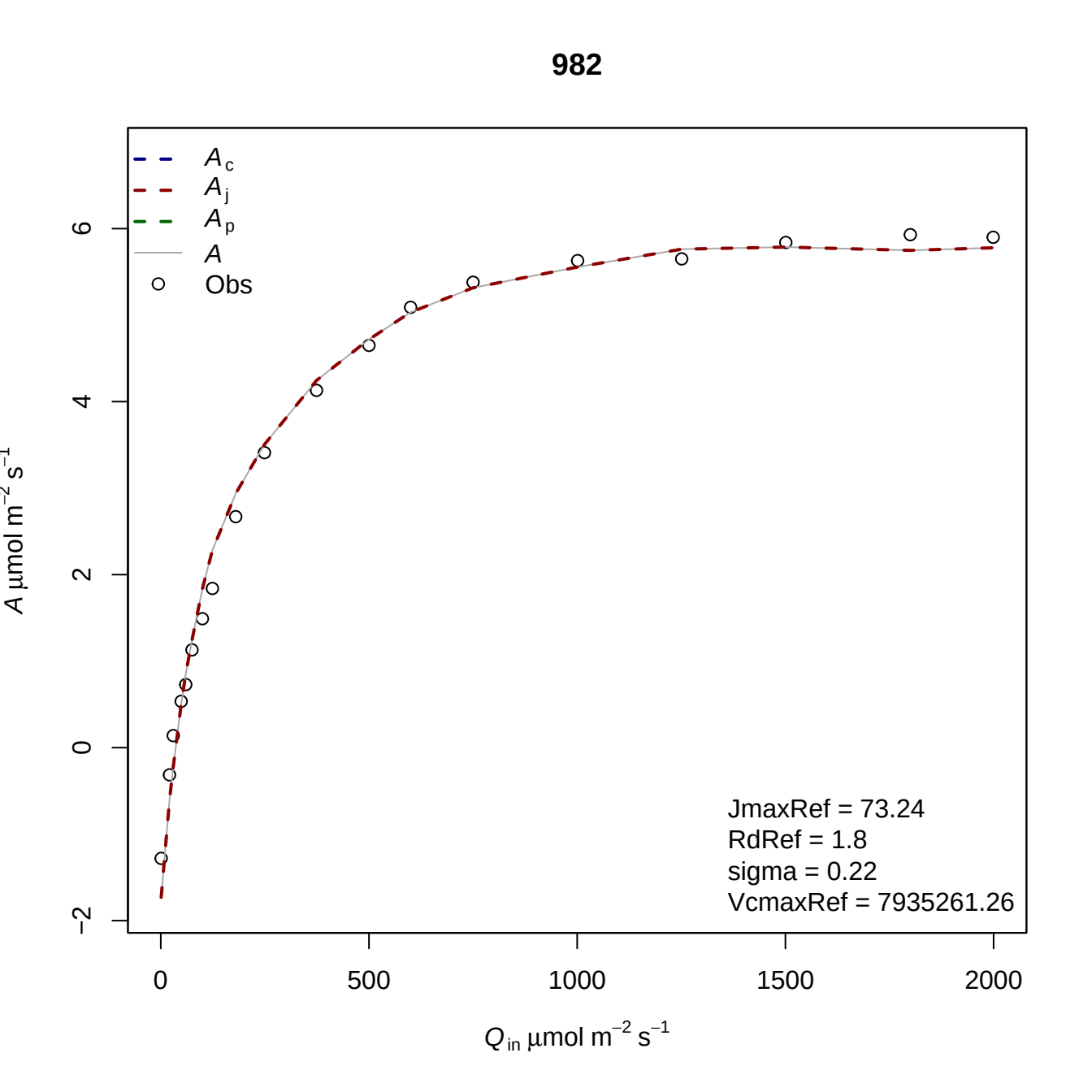

977
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.65
RdRef = 2.12
sigma = 0.44
VcmaxRef = 845485.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

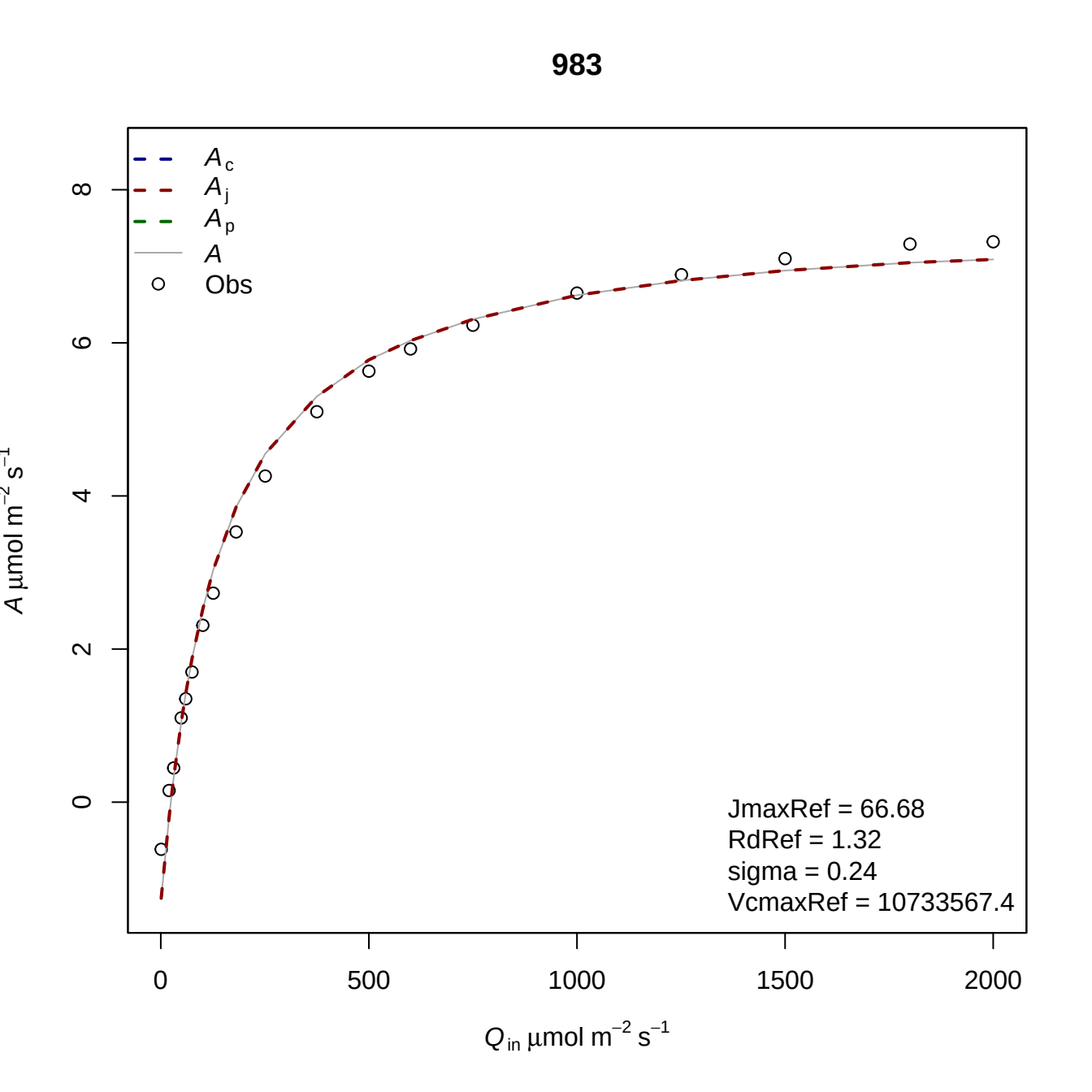

978
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.11
RdRef = 1.03
sigma = 0.2
VcmaxRef = 75.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

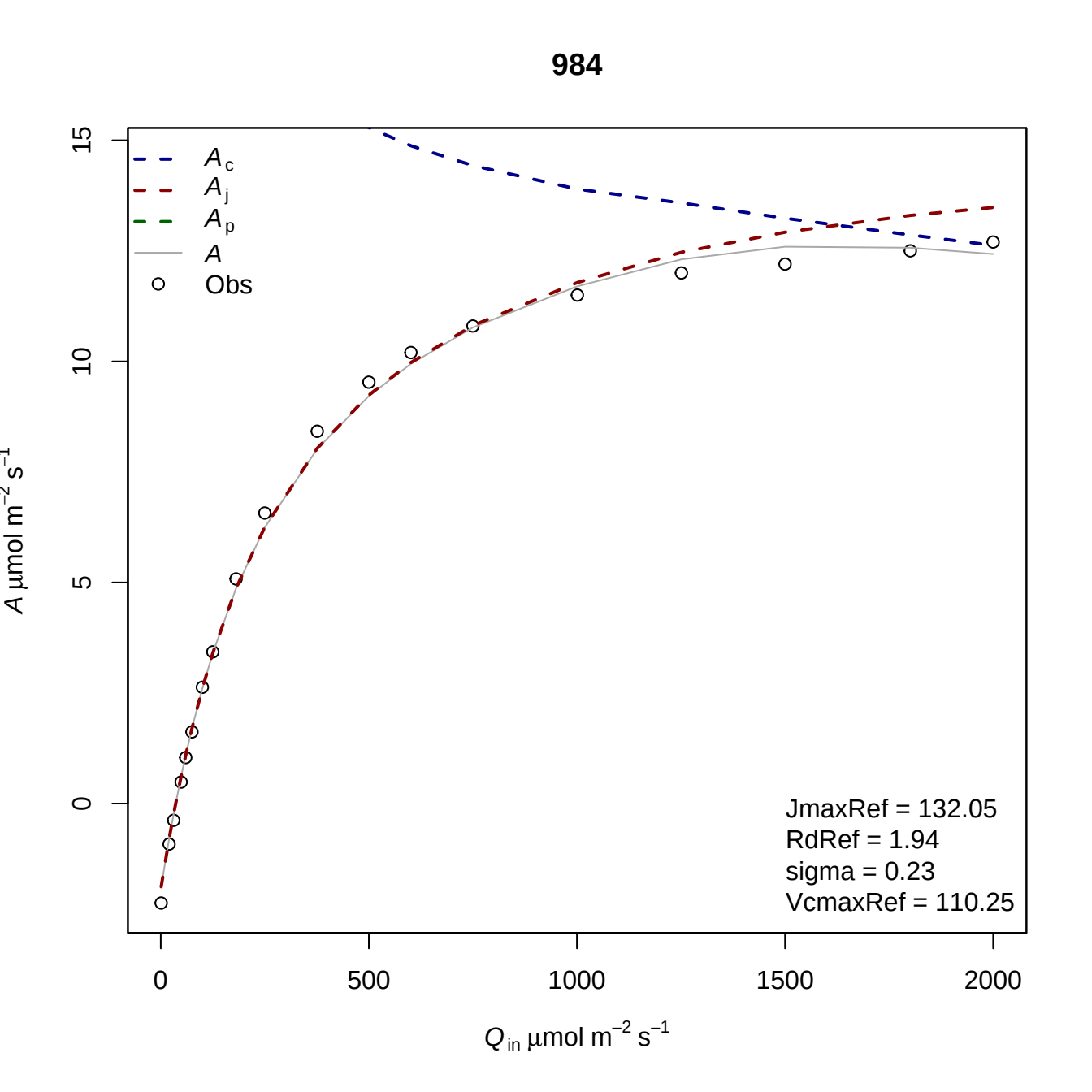

979
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.96
RdRef = 1.61
sigma = 0.15
VcmaxRef = 91.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

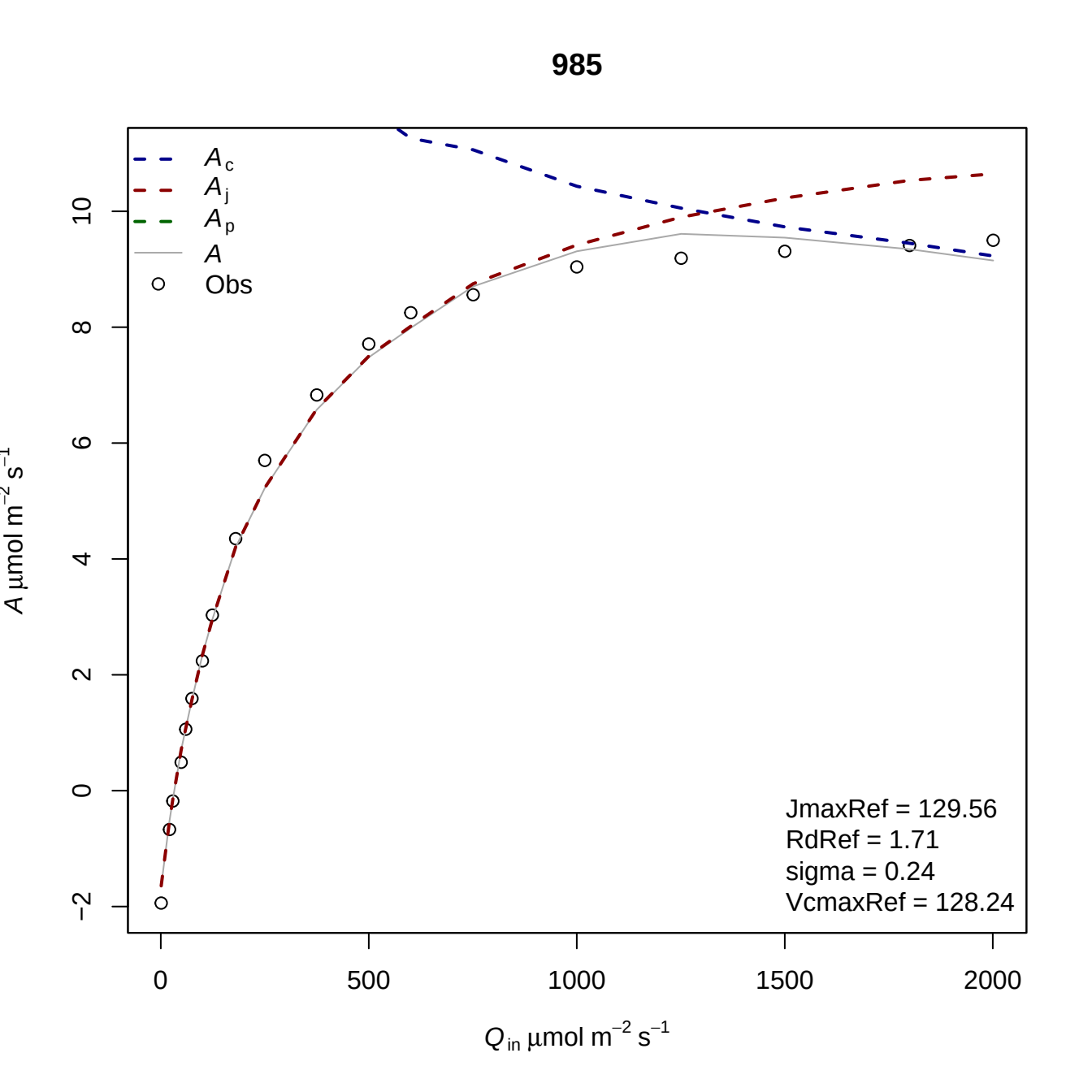

980
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.76
RdRef = 3.18
sigma = 0.16
VcmaxRef = 117.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

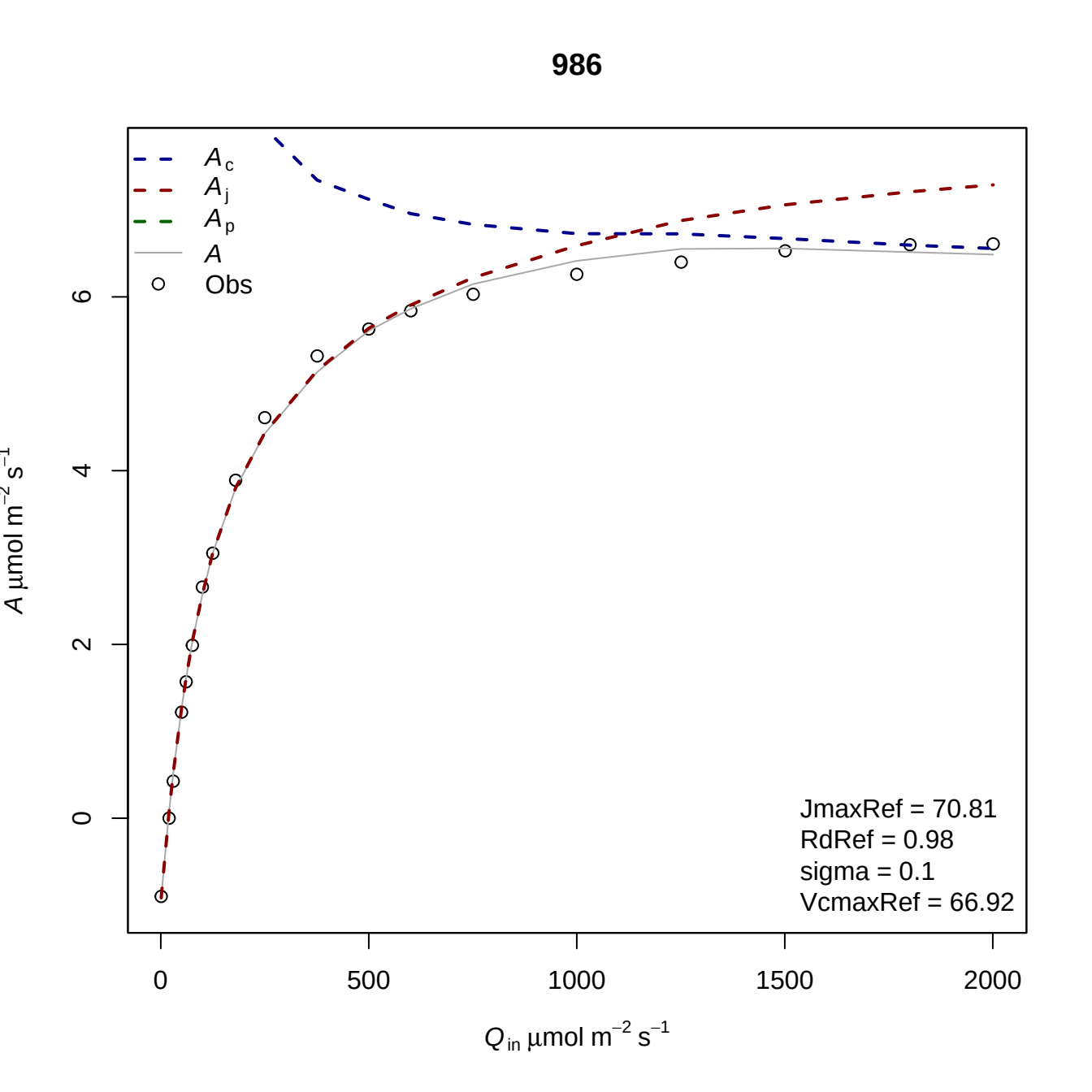

981
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.18
RdRef = 0.84
sigma = 0.42
VcmaxRef = 93.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

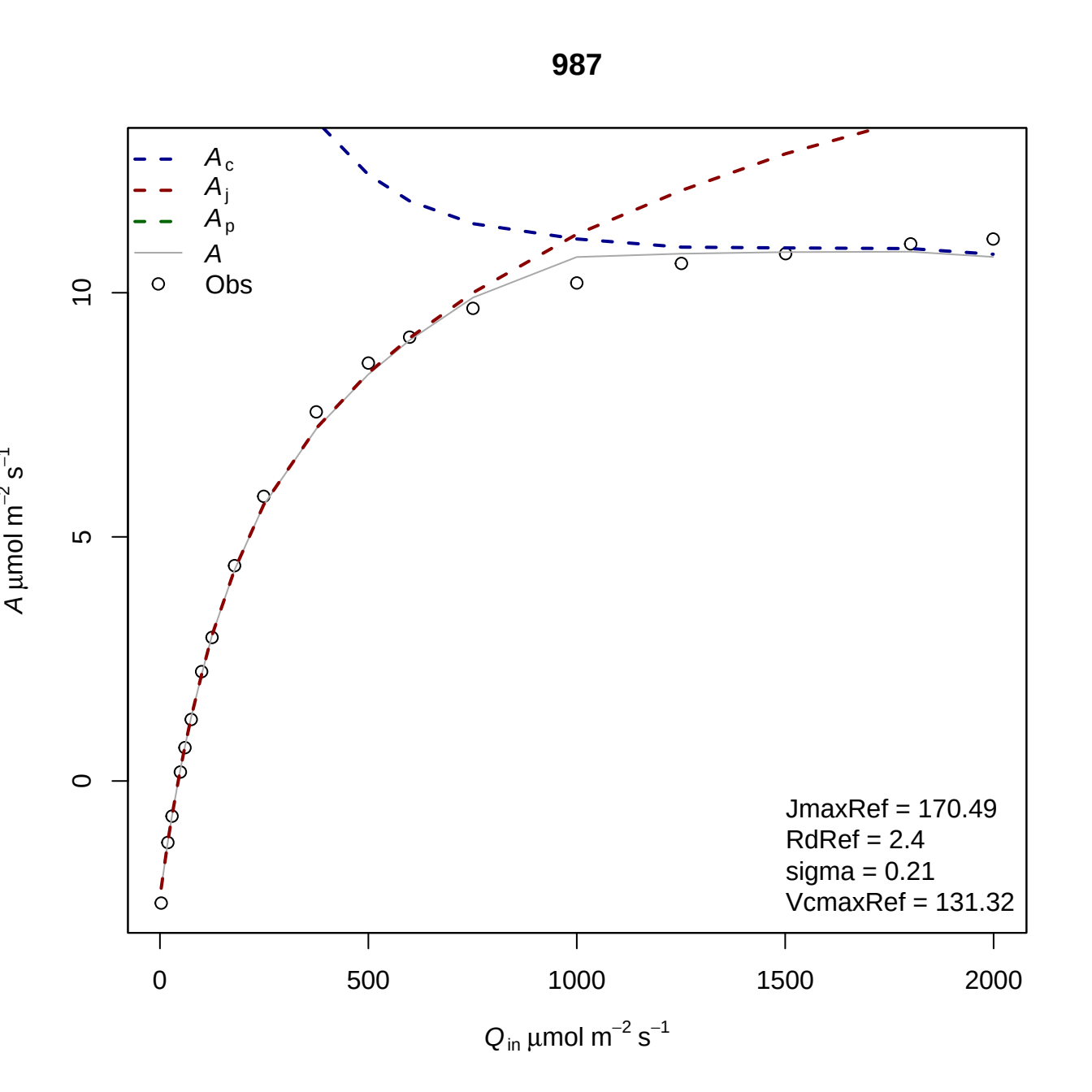

982
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.24
RdRef = 1.8
sigma = 0.22
VcmaxRef = 7935261.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

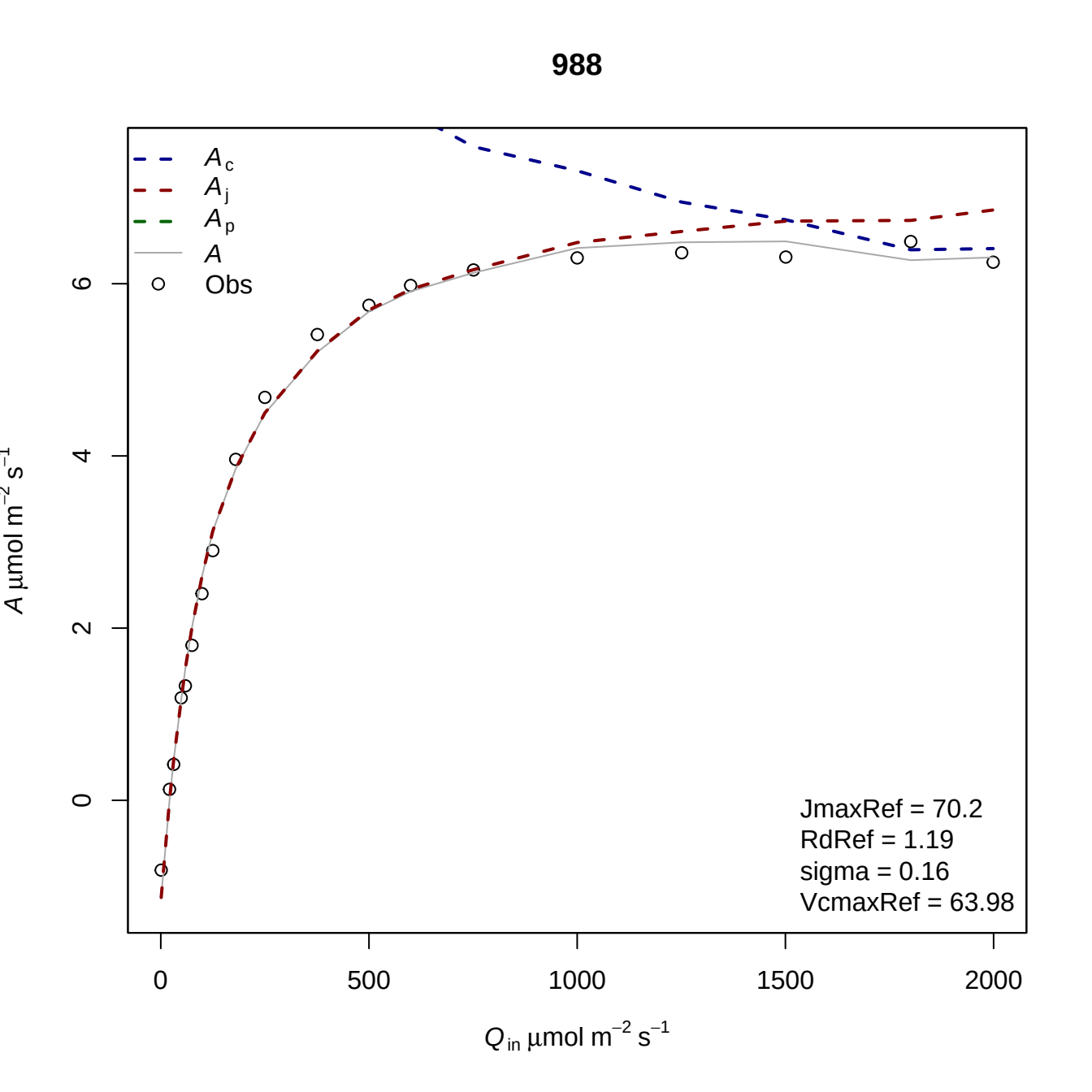

983
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.68
RdRef = 1.32
sigma = 0.24
VcmaxRef = 10733567.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

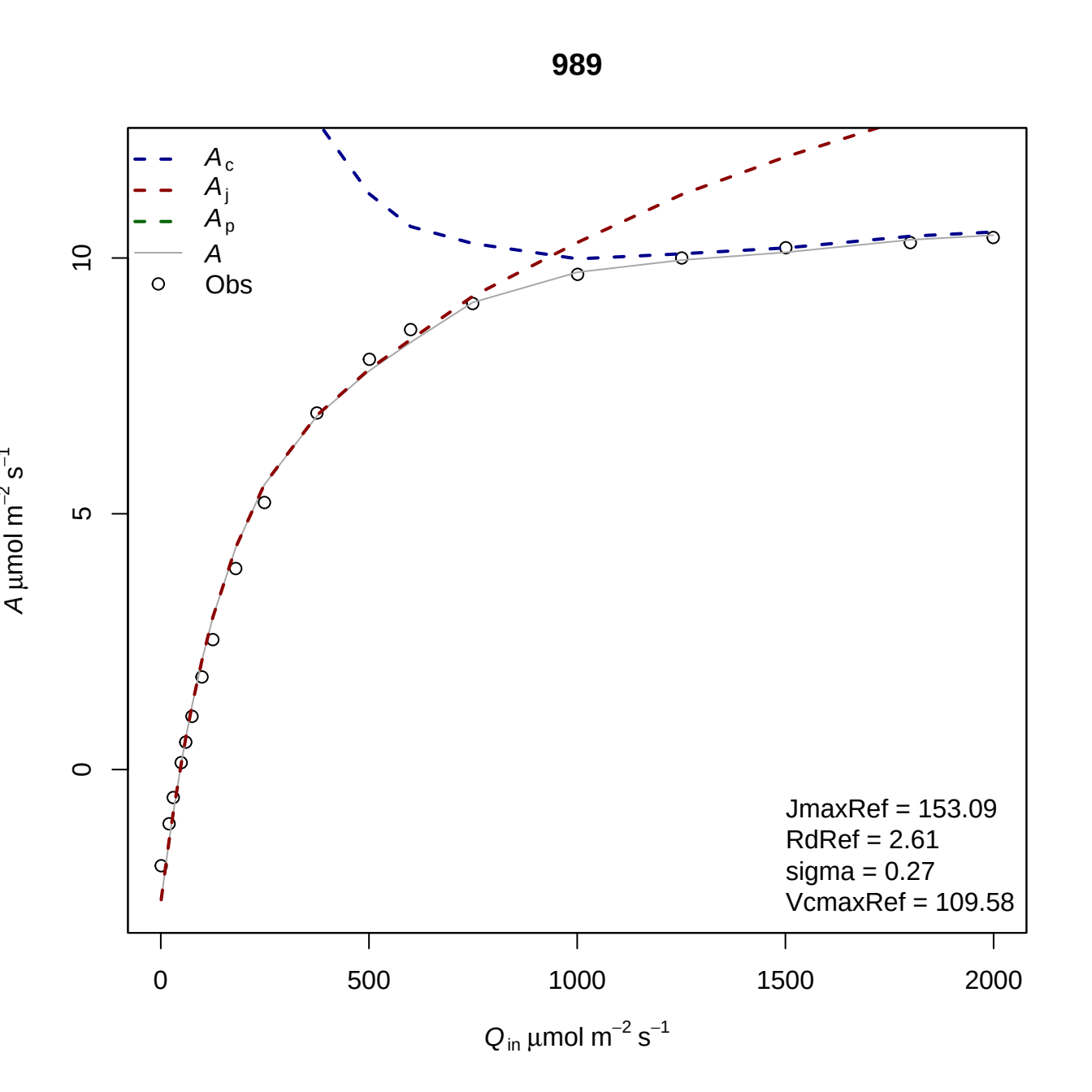

984
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.05
RdRef = 1.94
sigma = 0.23
VcmaxRef = 110.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

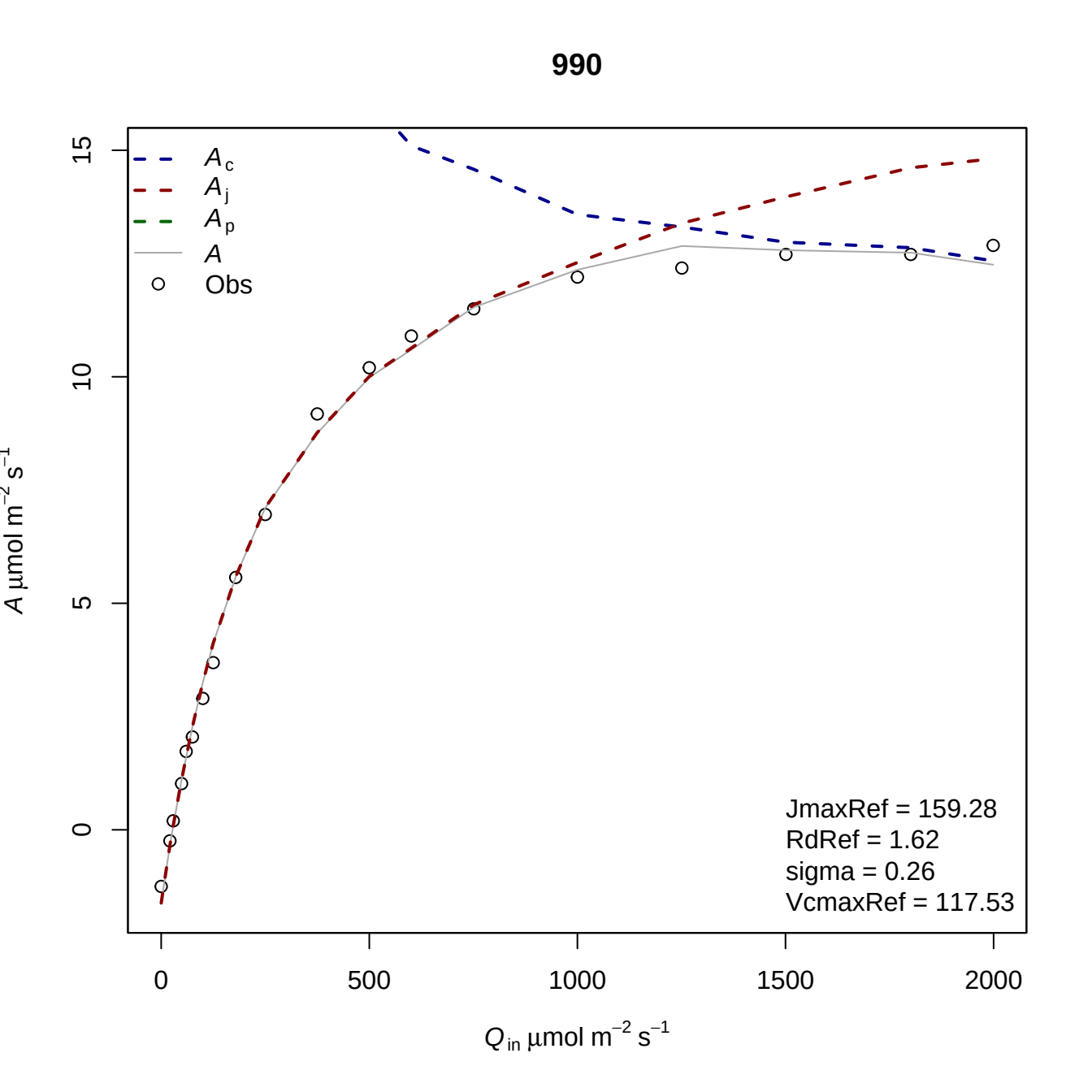

985
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.56
RdRef = 1.71
sigma = 0.24
VcmaxRef = 128.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

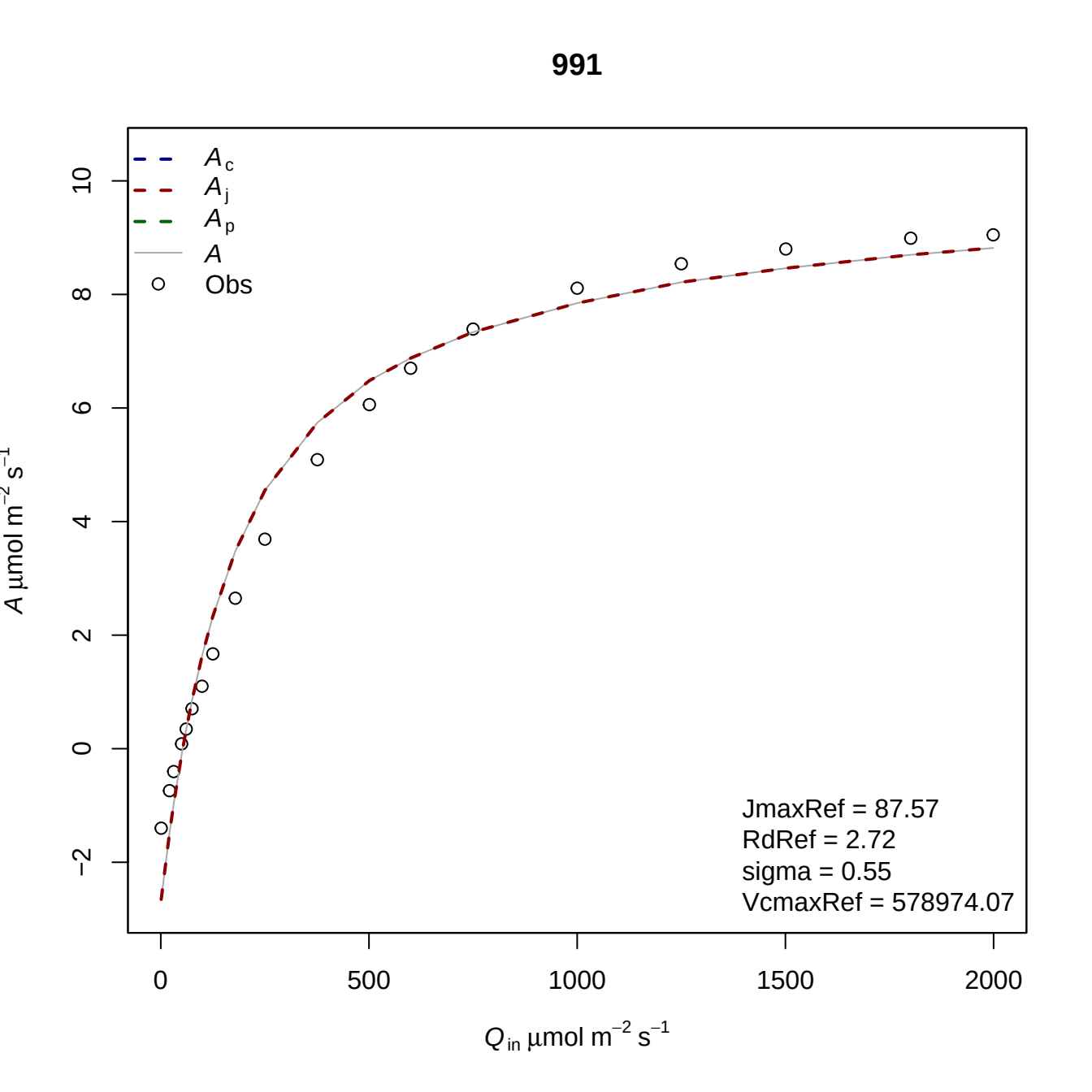

986
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.81
RdRef = 0.98
sigma = 0.1
VcmaxRef = 66.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

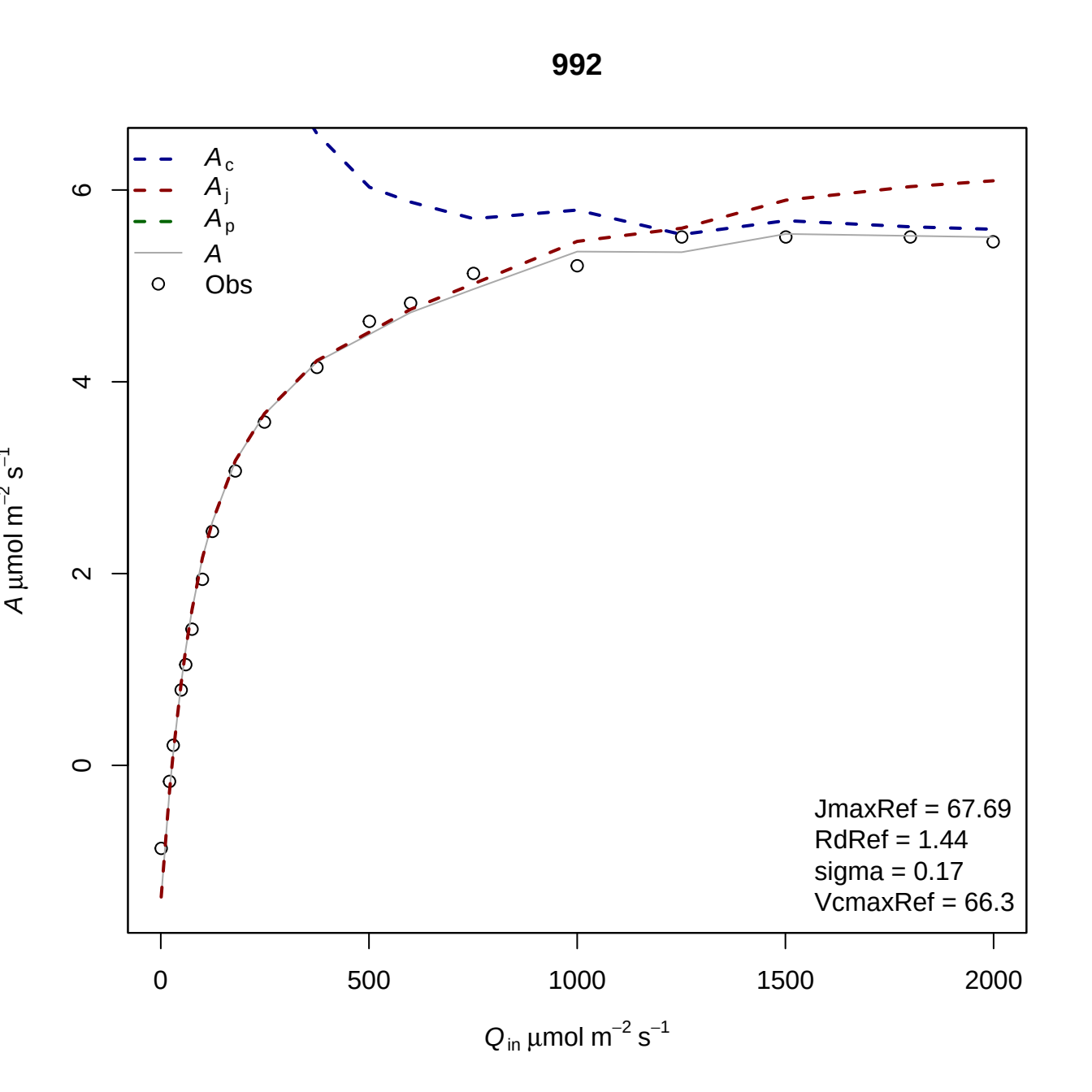

987
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 170.49
RdRef = 2.4
sigma = 0.21
VcmaxRef = 131.32
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

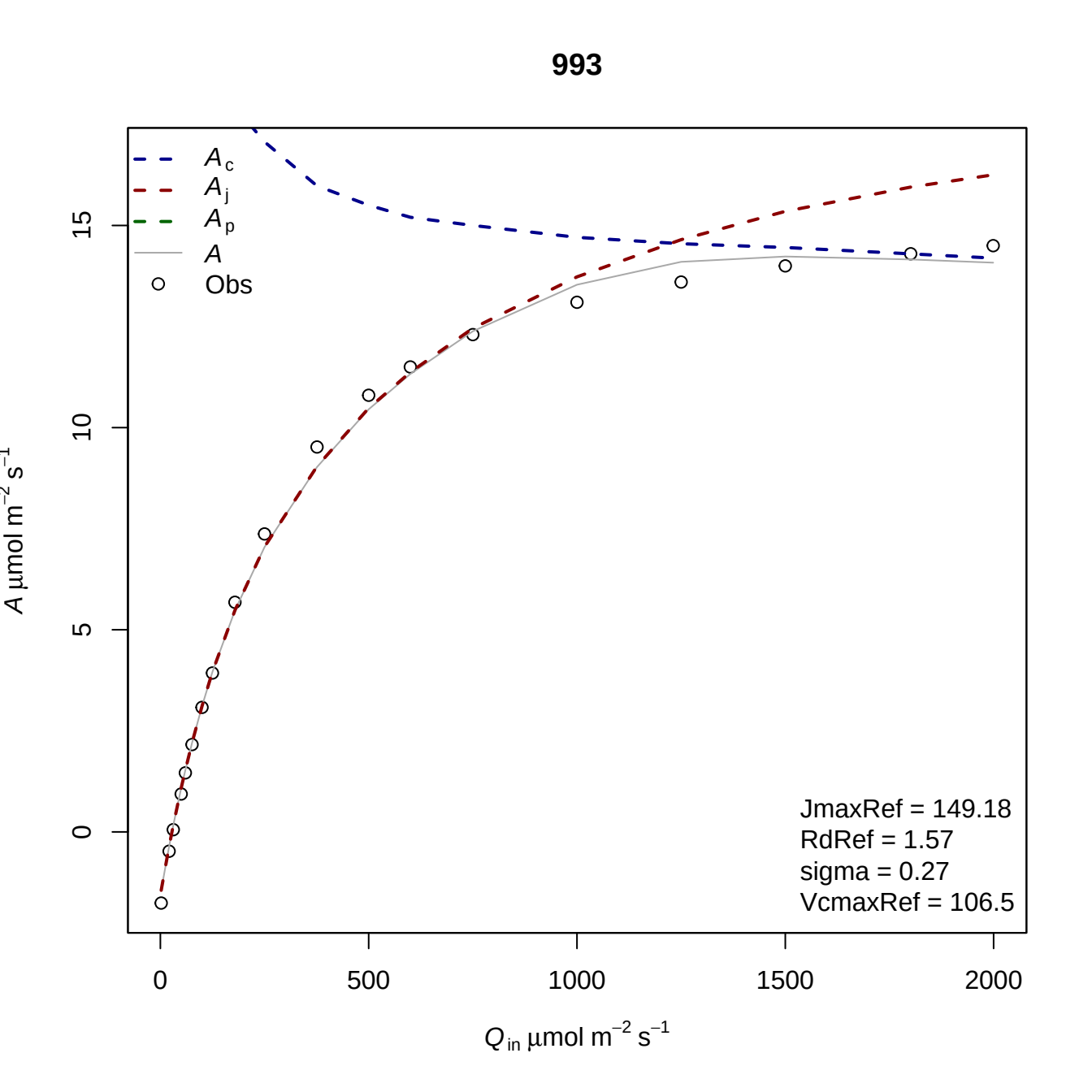

988
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.2
RdRef = 1.19
sigma = 0.16
VcmaxRef = 63.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

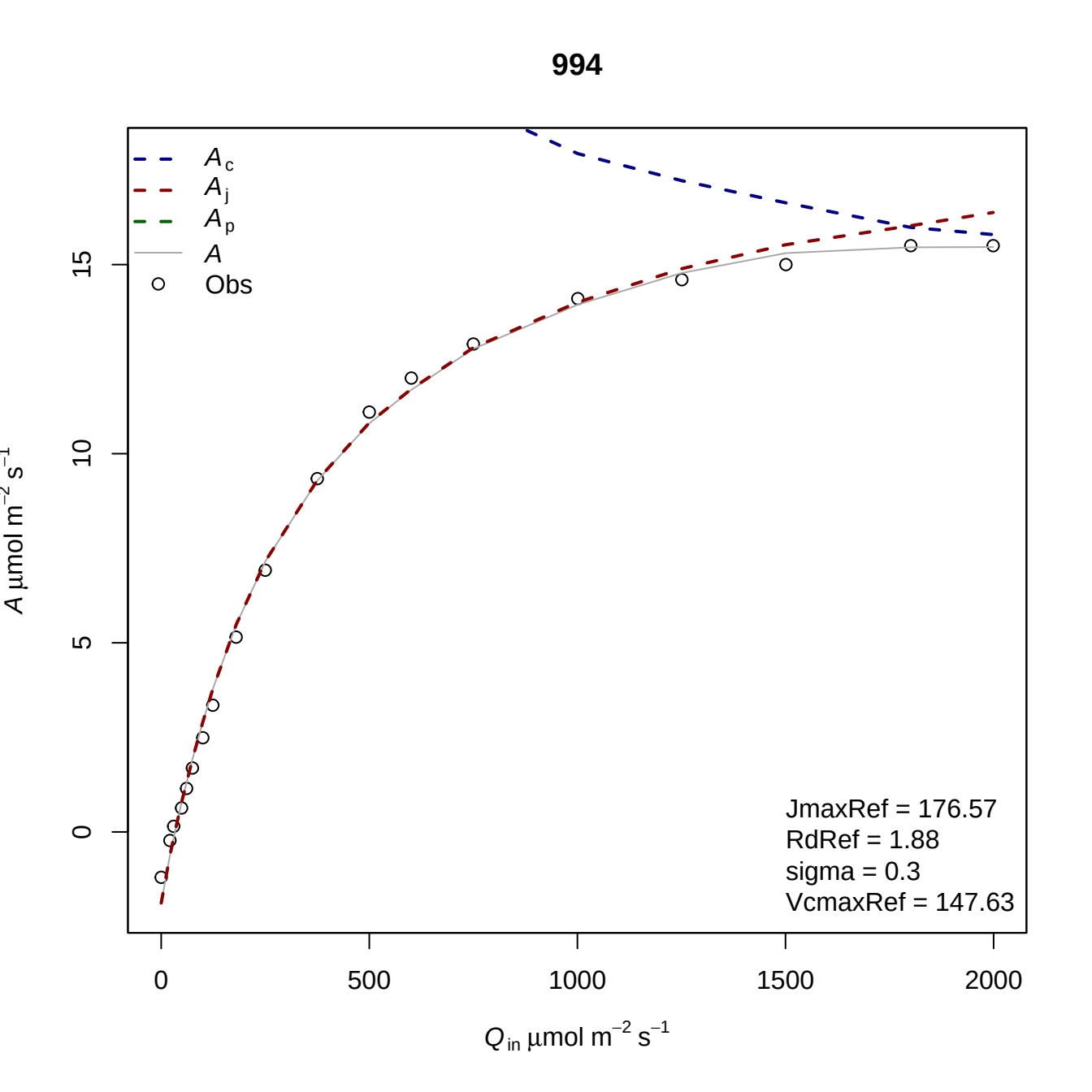

989
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.09
RdRef = 2.61
sigma = 0.27
VcmaxRef = 109.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

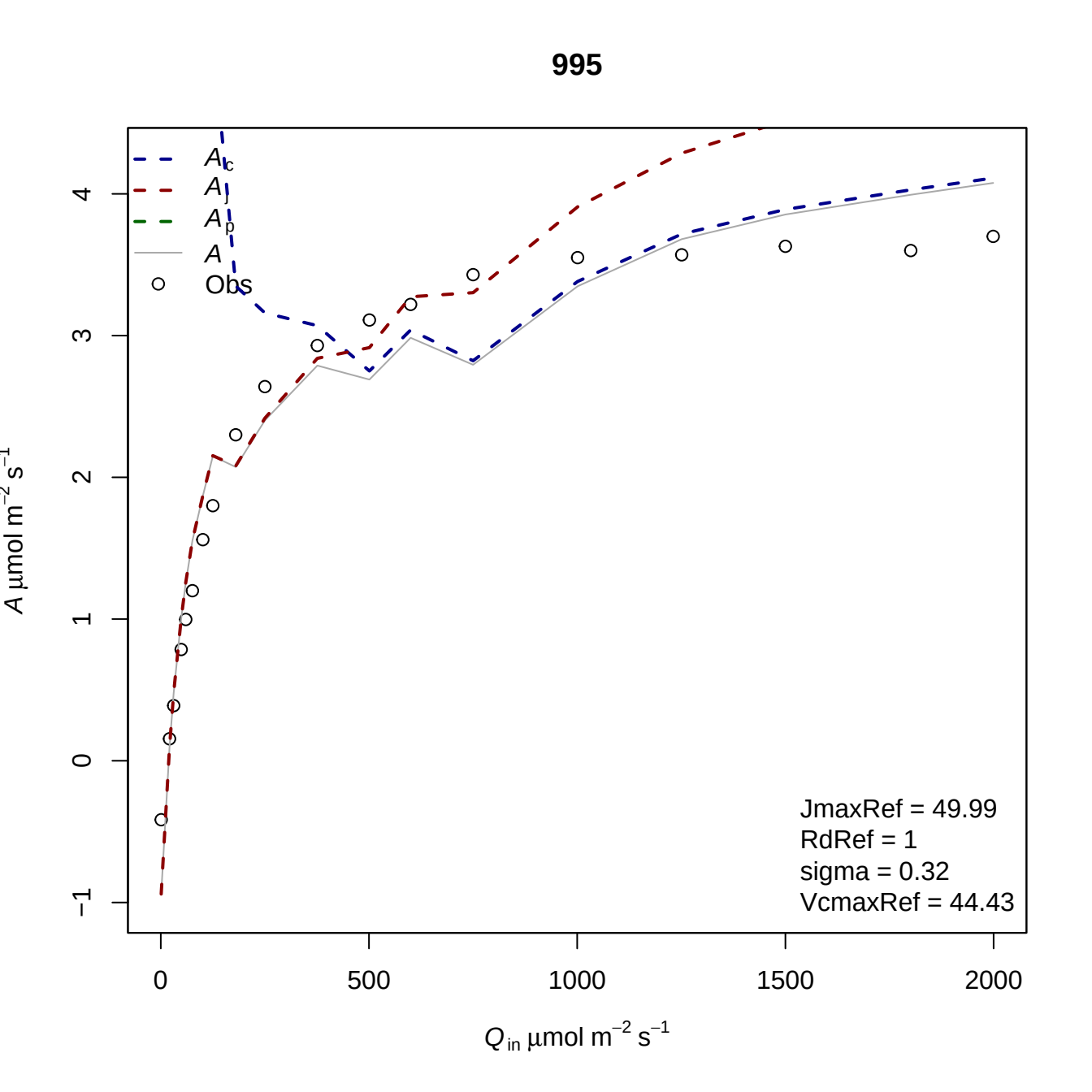

990
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 159.28
RdRef = 1.62
sigma = 0.26
VcmaxRef = 117.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

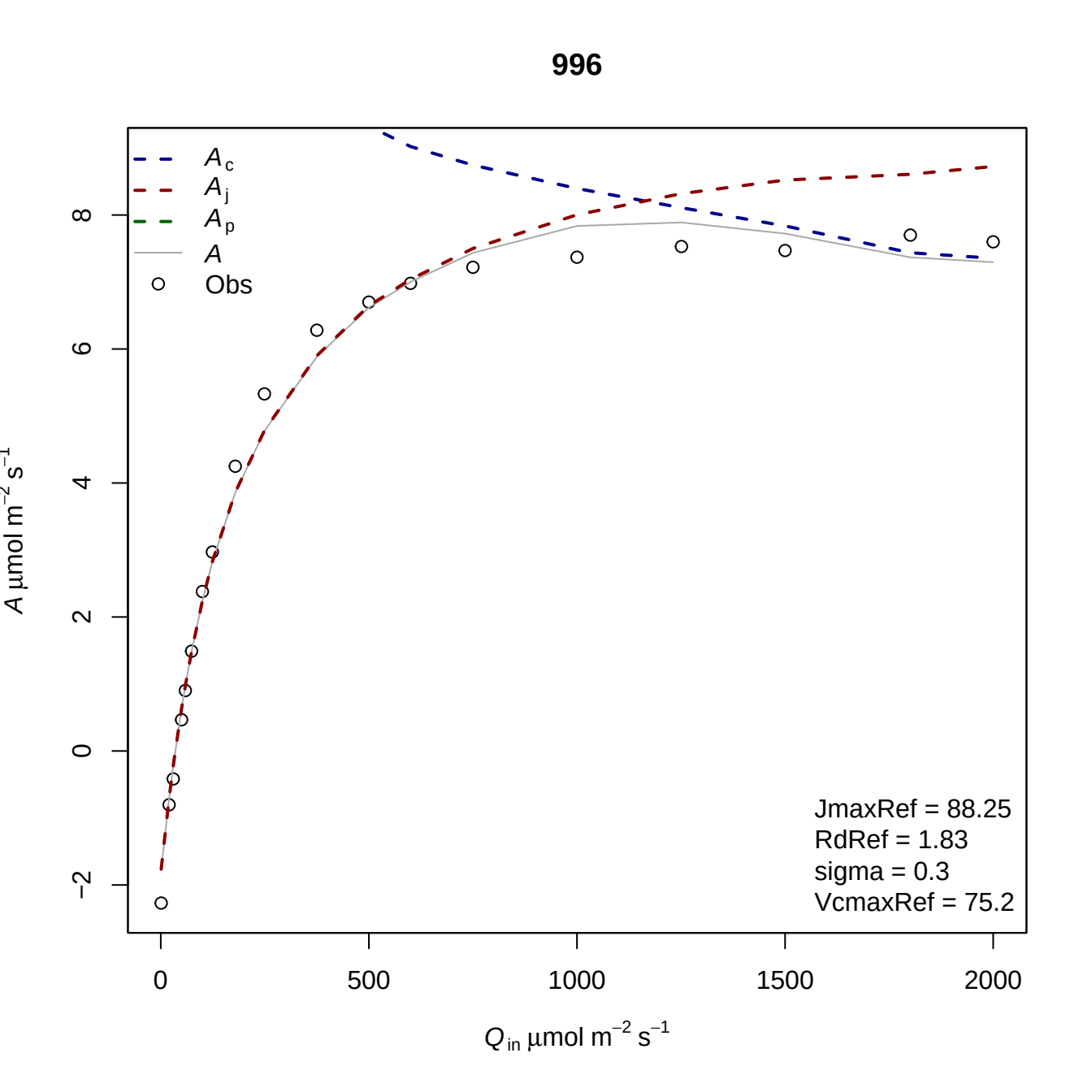

991
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.57
RdRef = 2.72
sigma = 0.55
VcmaxRef = 578974.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

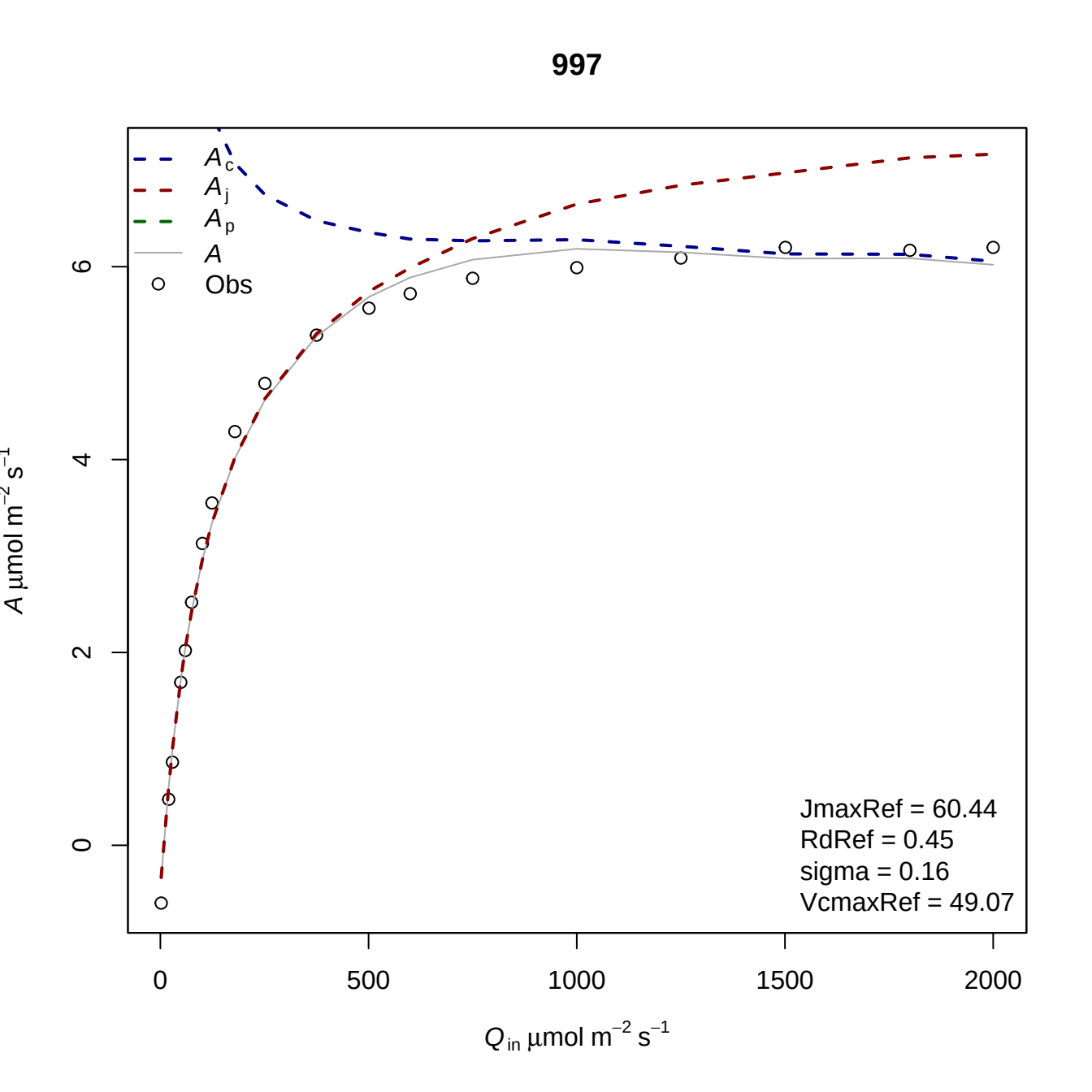

992
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.69
RdRef = 1.44
sigma = 0.17
VcmaxRef = 66.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

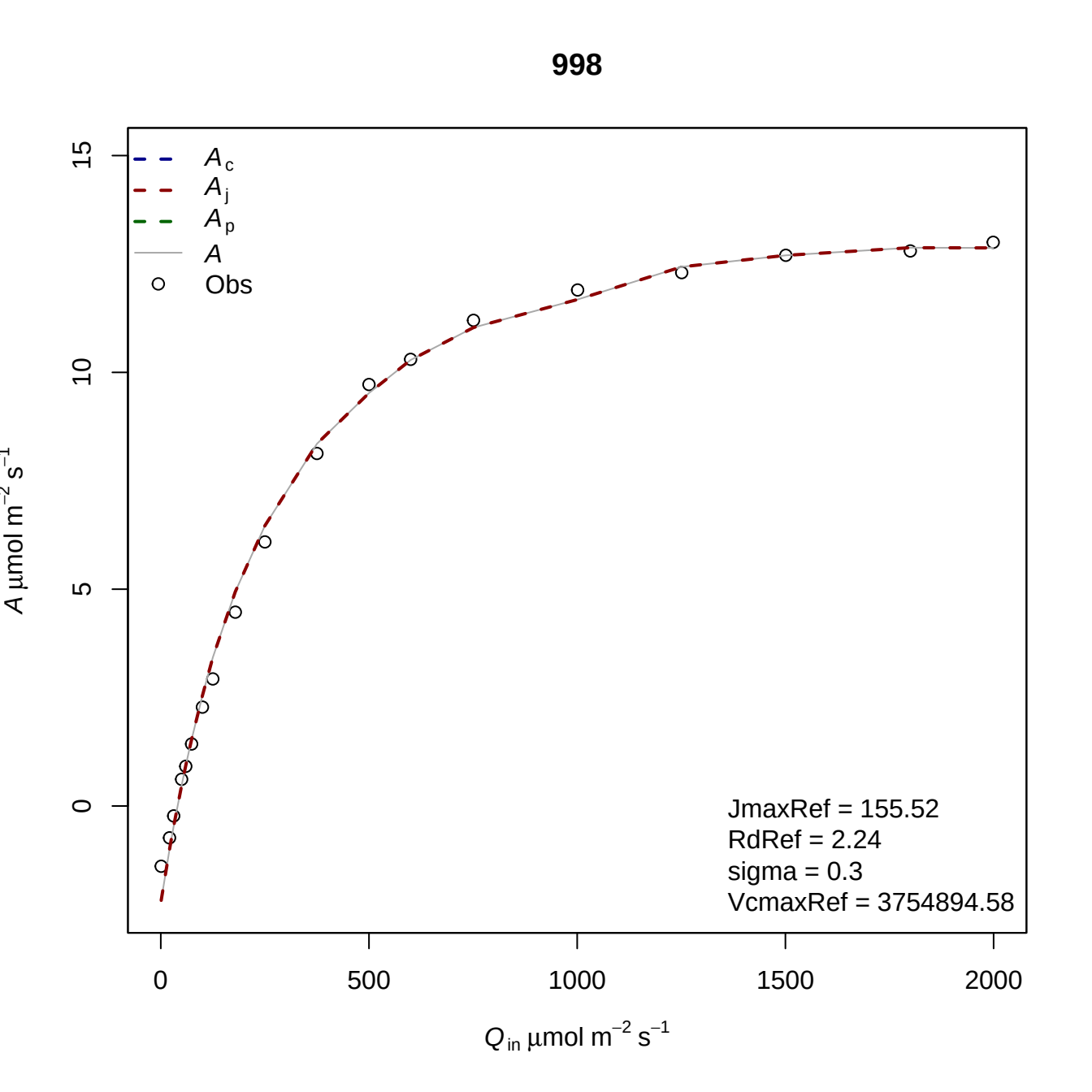

993
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 149.18
RdRef = 1.57
sigma = 0.27
VcmaxRef = 106.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

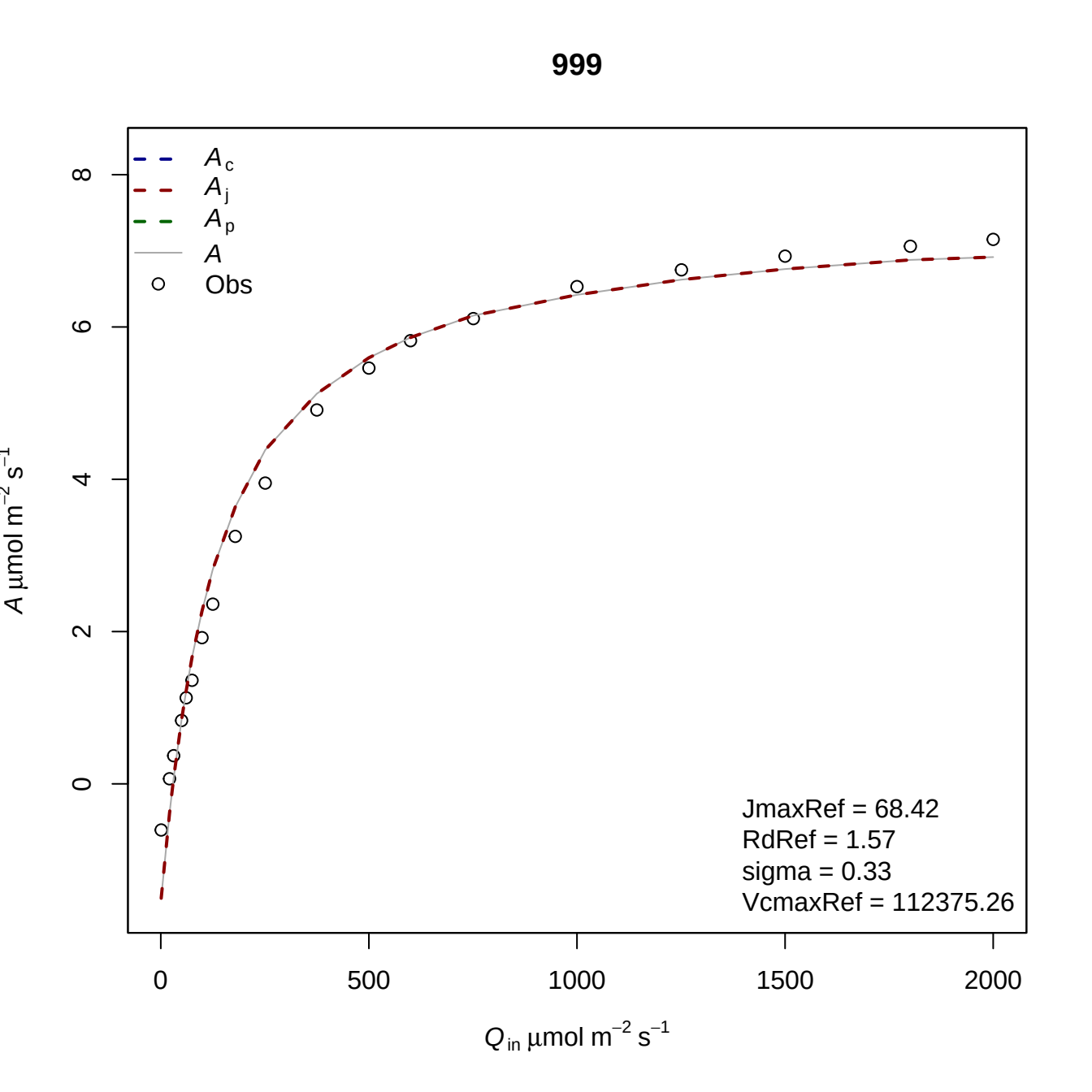

994
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 176.57
RdRef = 1.88
sigma = 0.3
VcmaxRef = 147.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

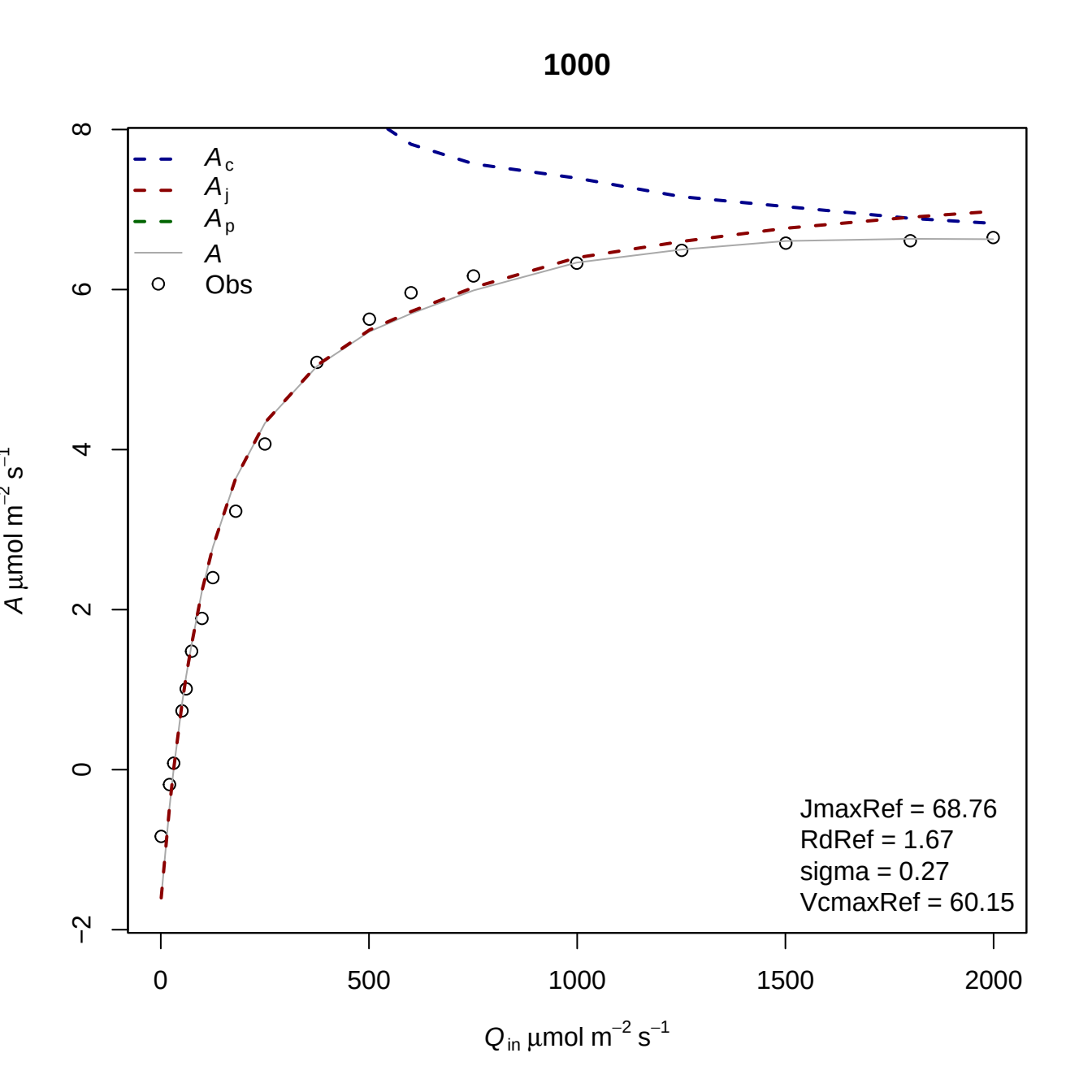

995
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.99
RdRef = 1
sigma = 0.32
VcmaxRef = 44.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500
2000

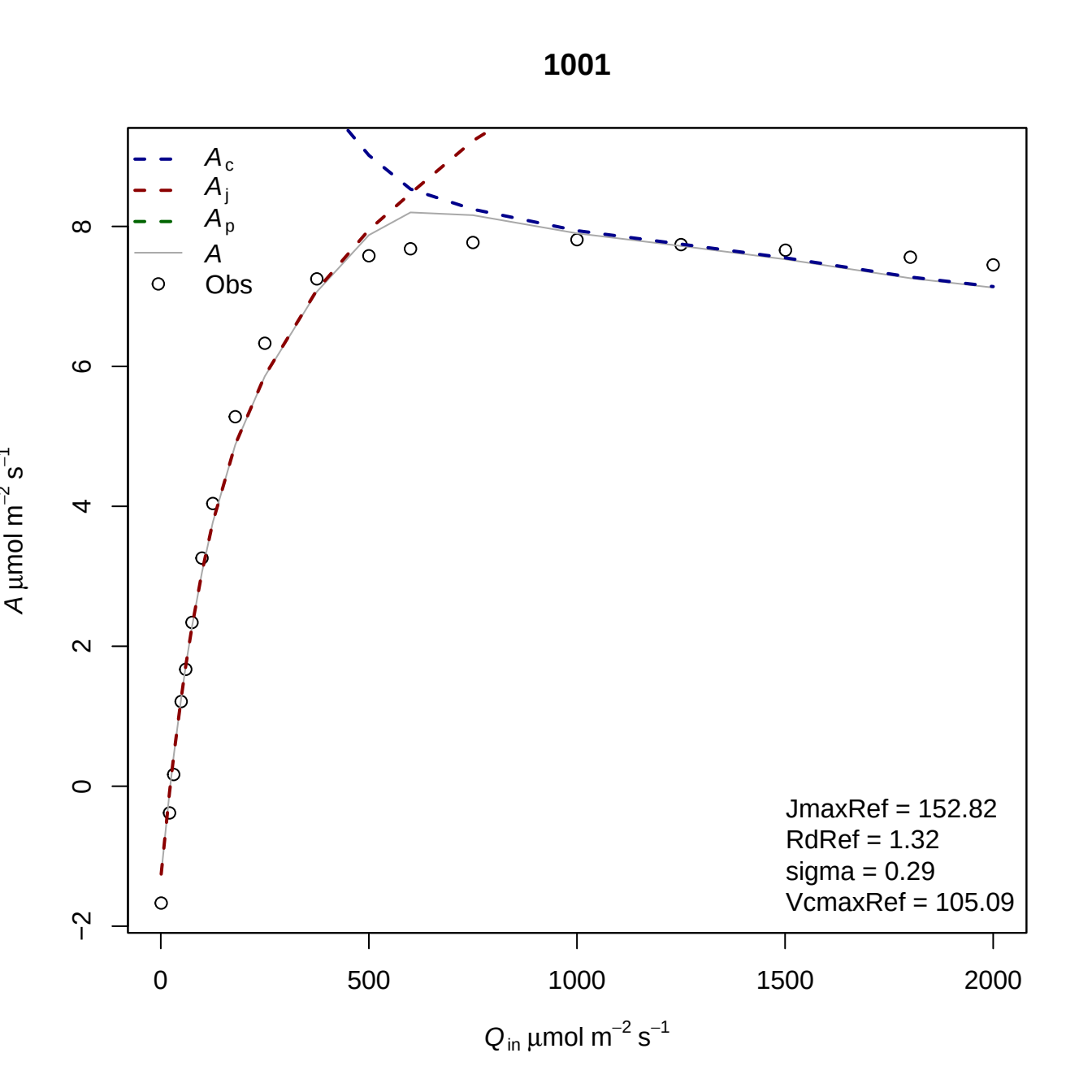

996
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.25
RdRef = 1.83
sigma = 0.3
VcmaxRef = 75.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

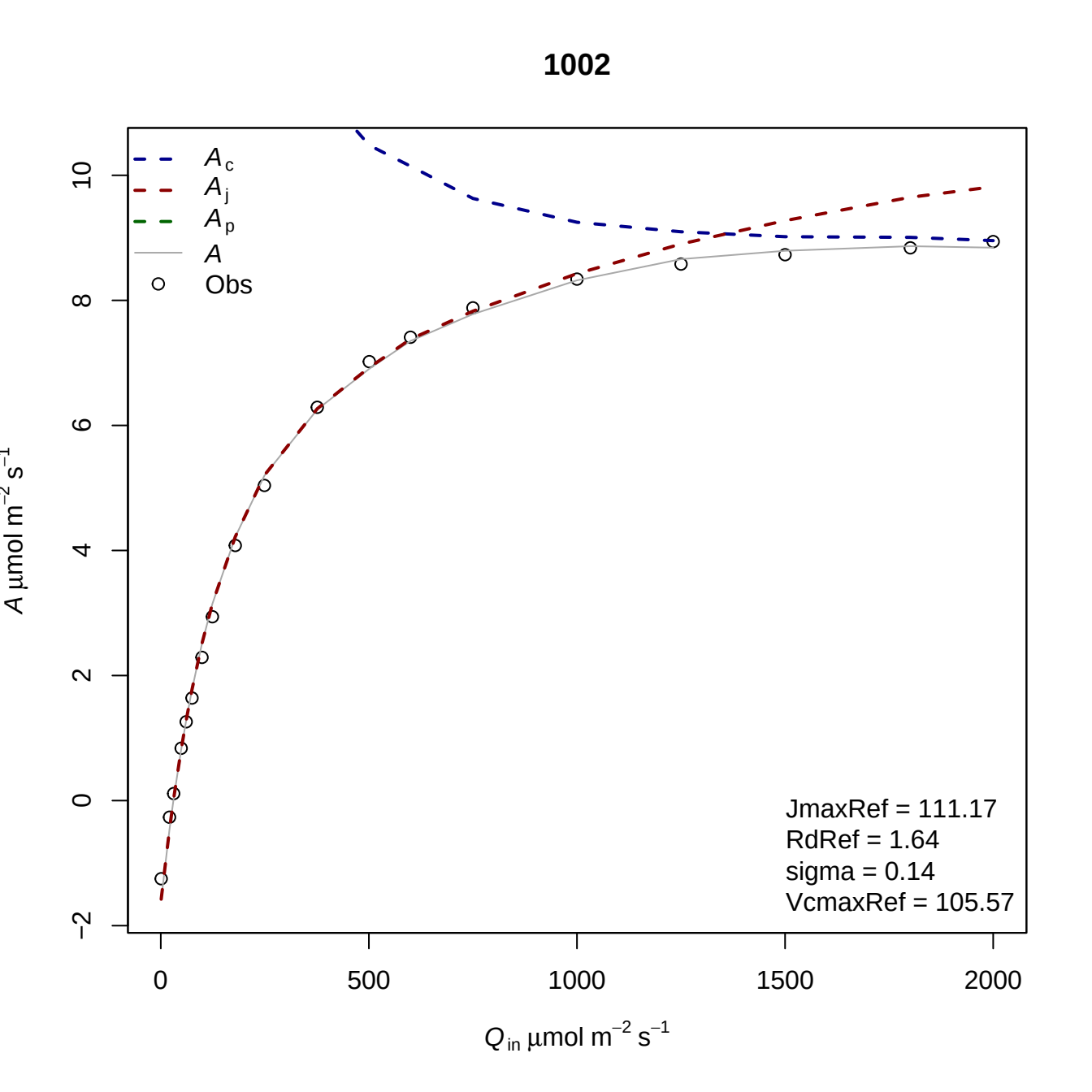

997
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.44
RdRef = 0.45
sigma = 0.16
VcmaxRef = 49.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

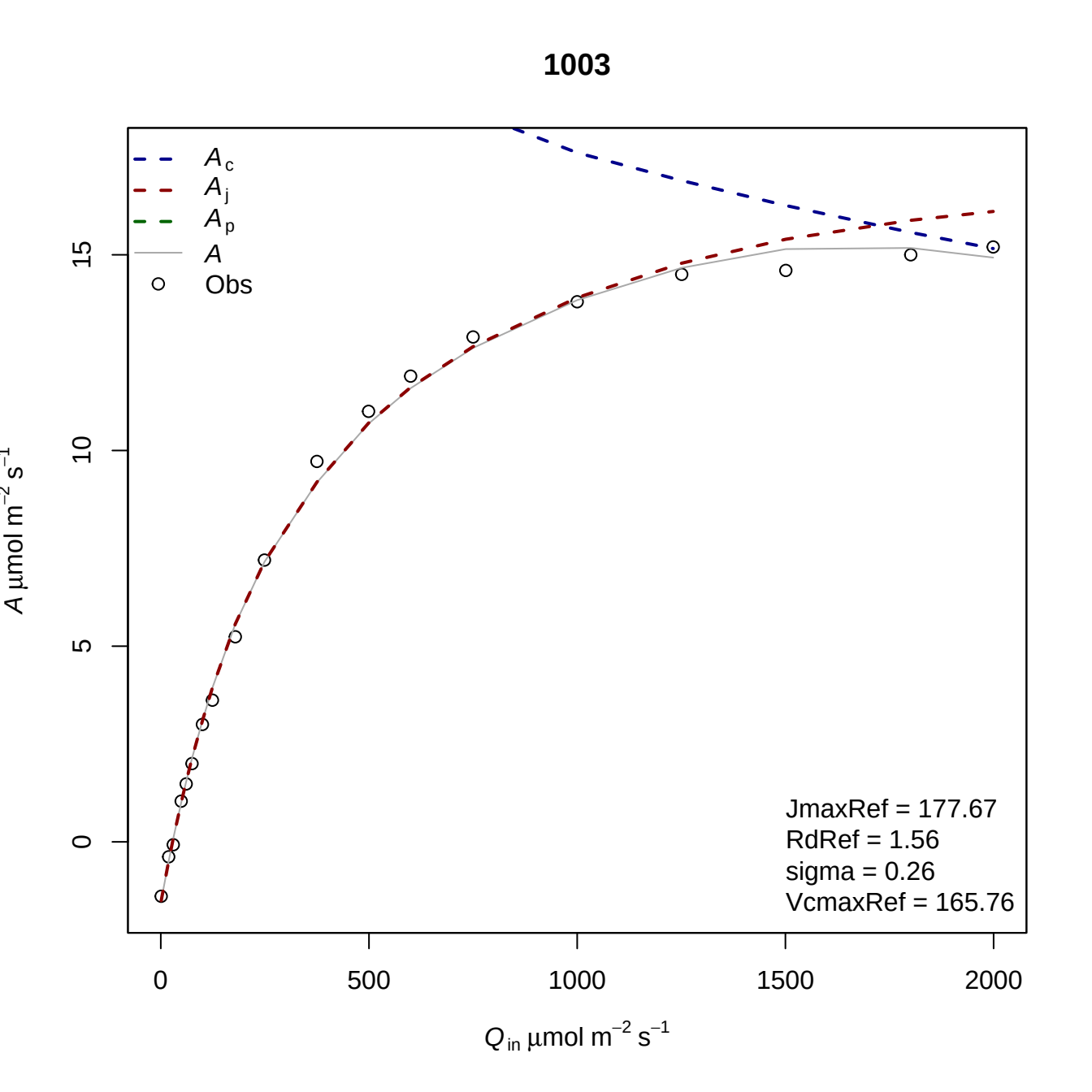

998
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 155.52
RdRef = 2.24
sigma = 0.3
VcmaxRef = 3754894.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

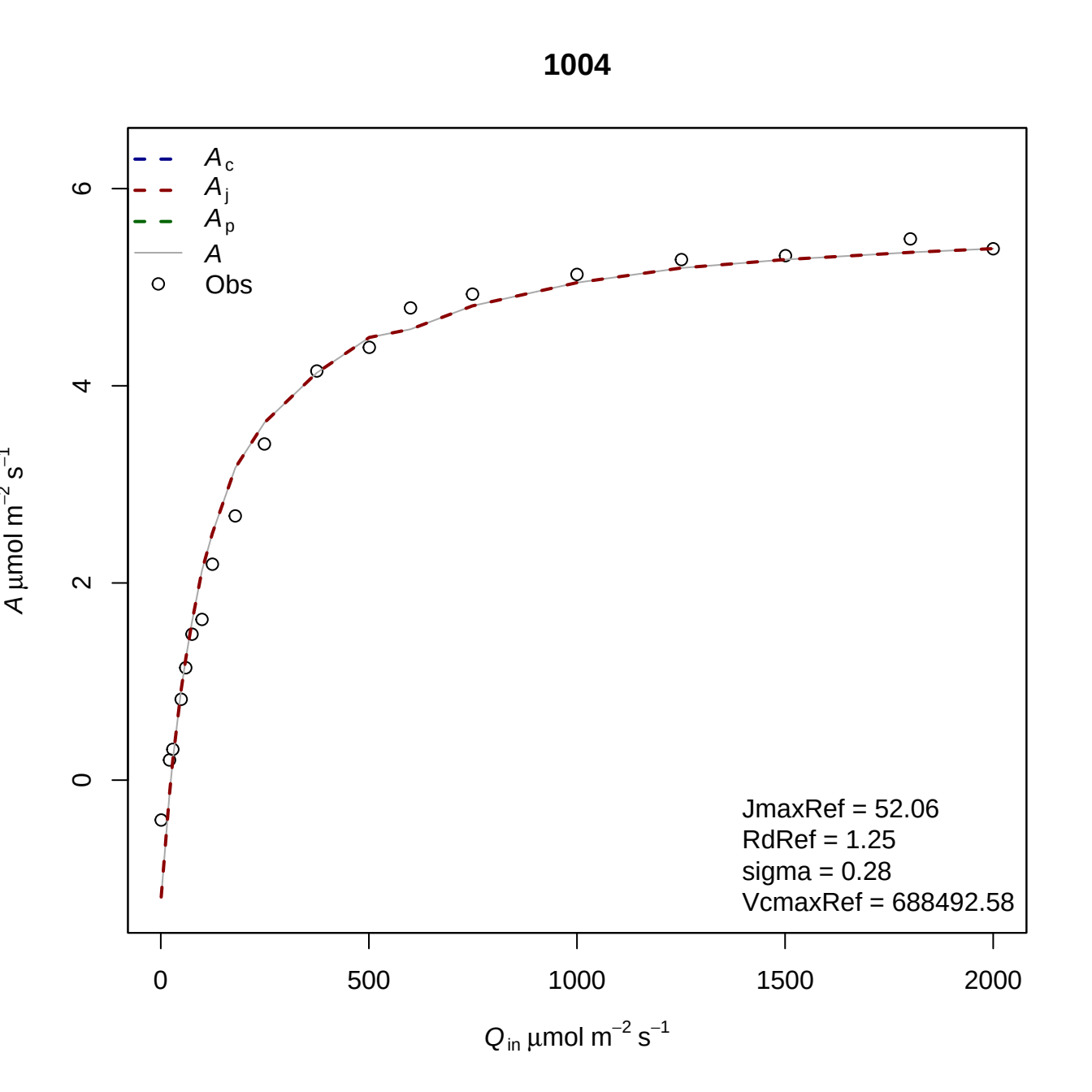

999
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.42
RdRef = 1.57
sigma = 0.33
VcmaxRef = 112375.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

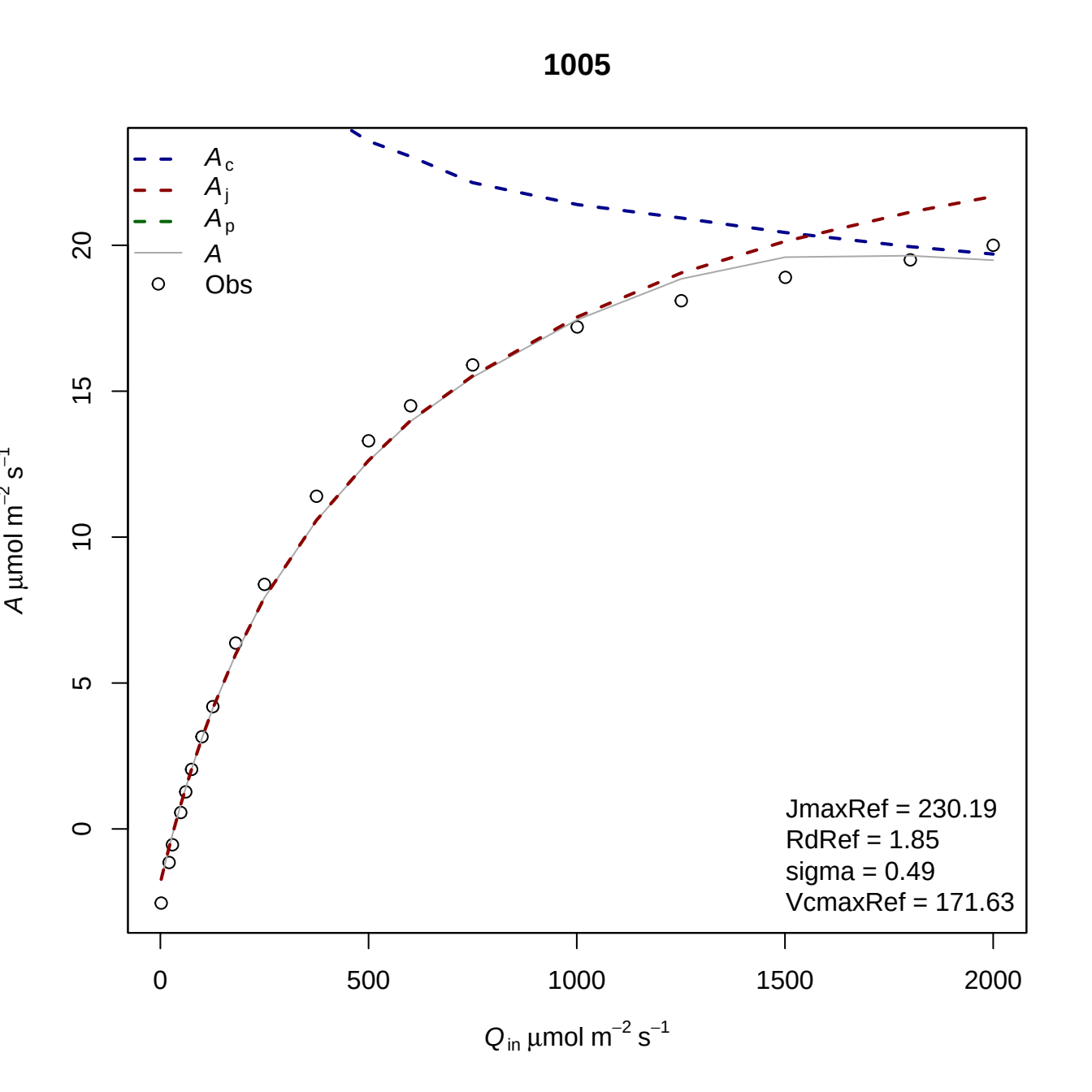

1000
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.76
RdRef = 1.67
sigma = 0.27
VcmaxRef = 60.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

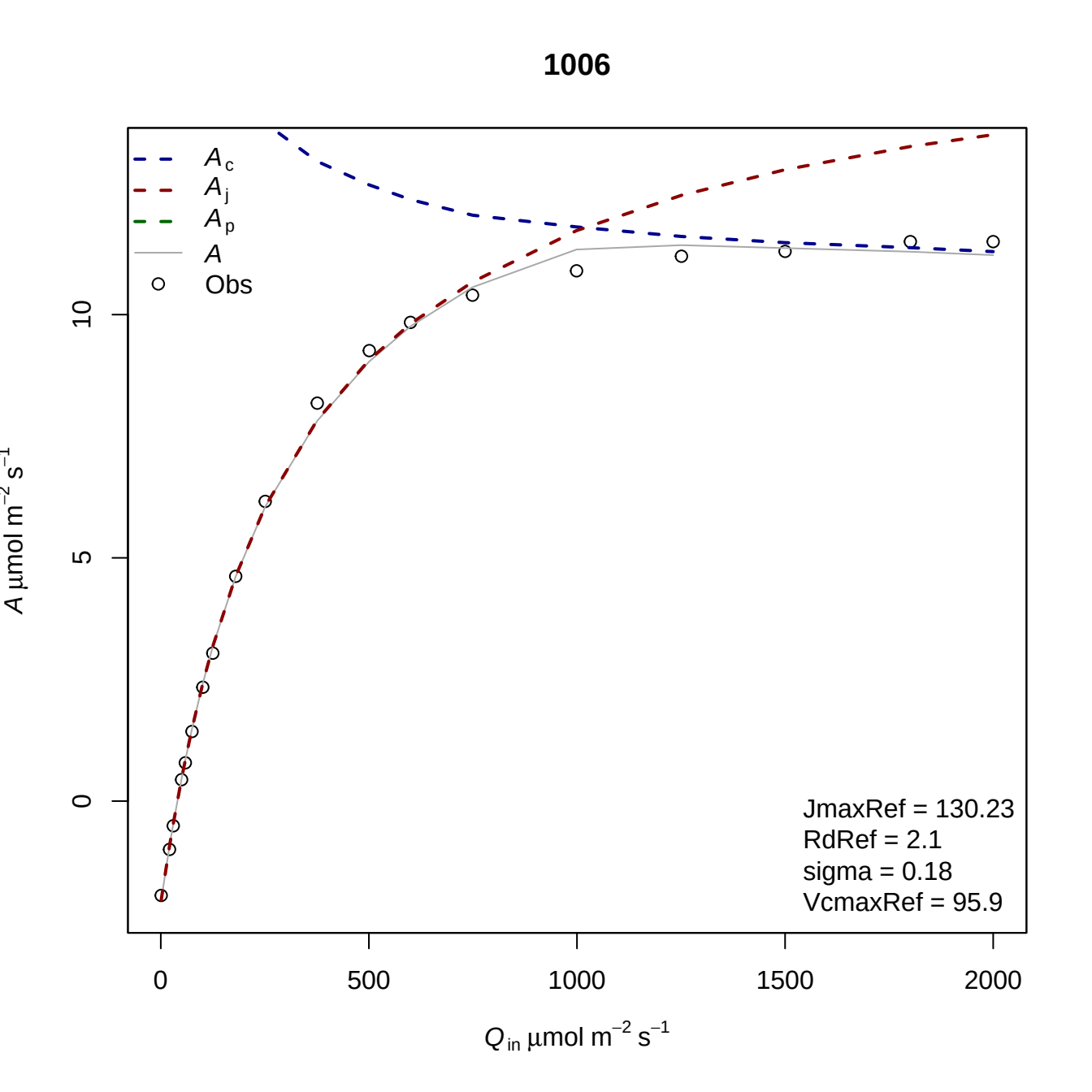

1001
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.82
RdRef = 1.32
sigma = 0.29
VcmaxRef = 105.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

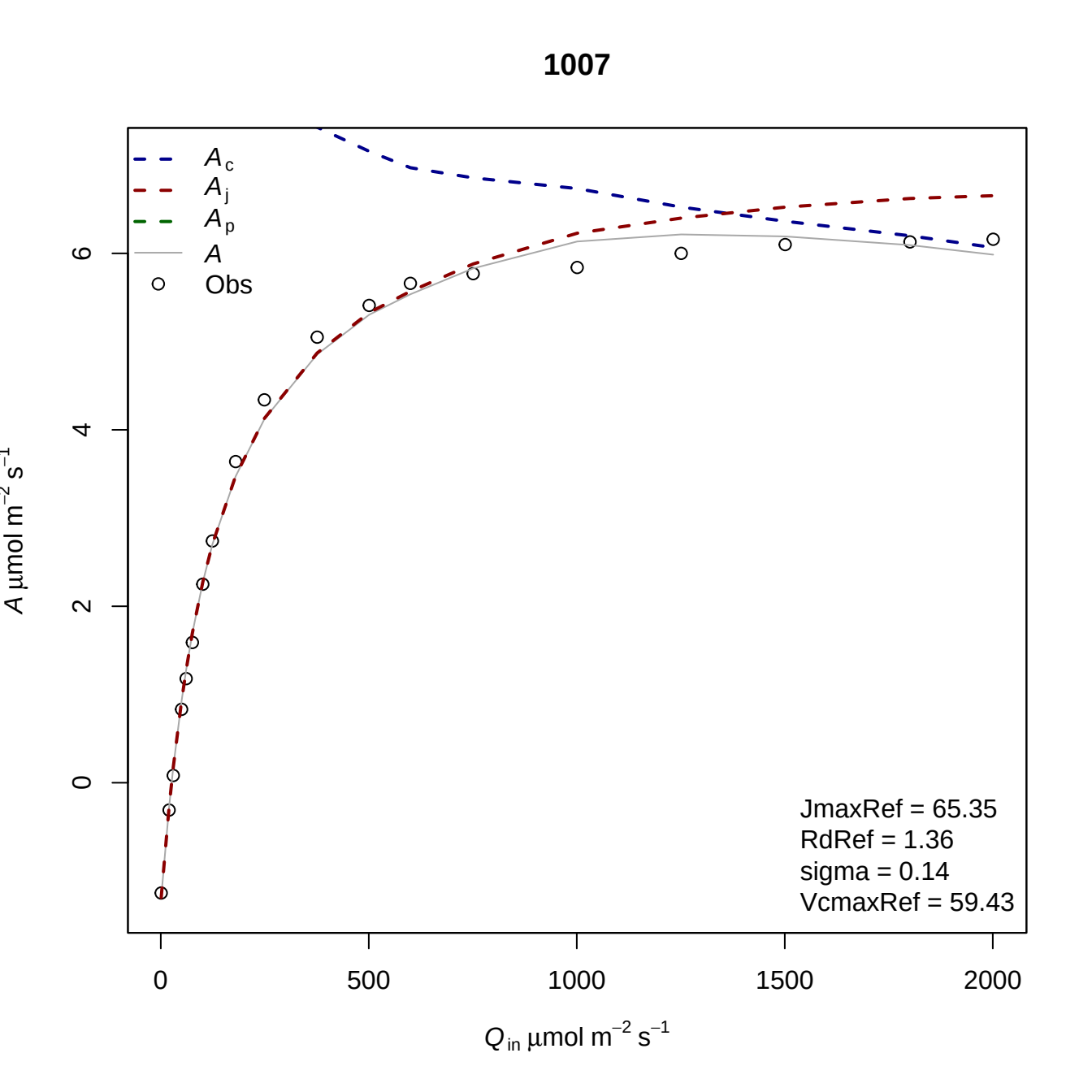

1002
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 111.17
RdRef = 1.64
sigma = 0.14
VcmaxRef = 105.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

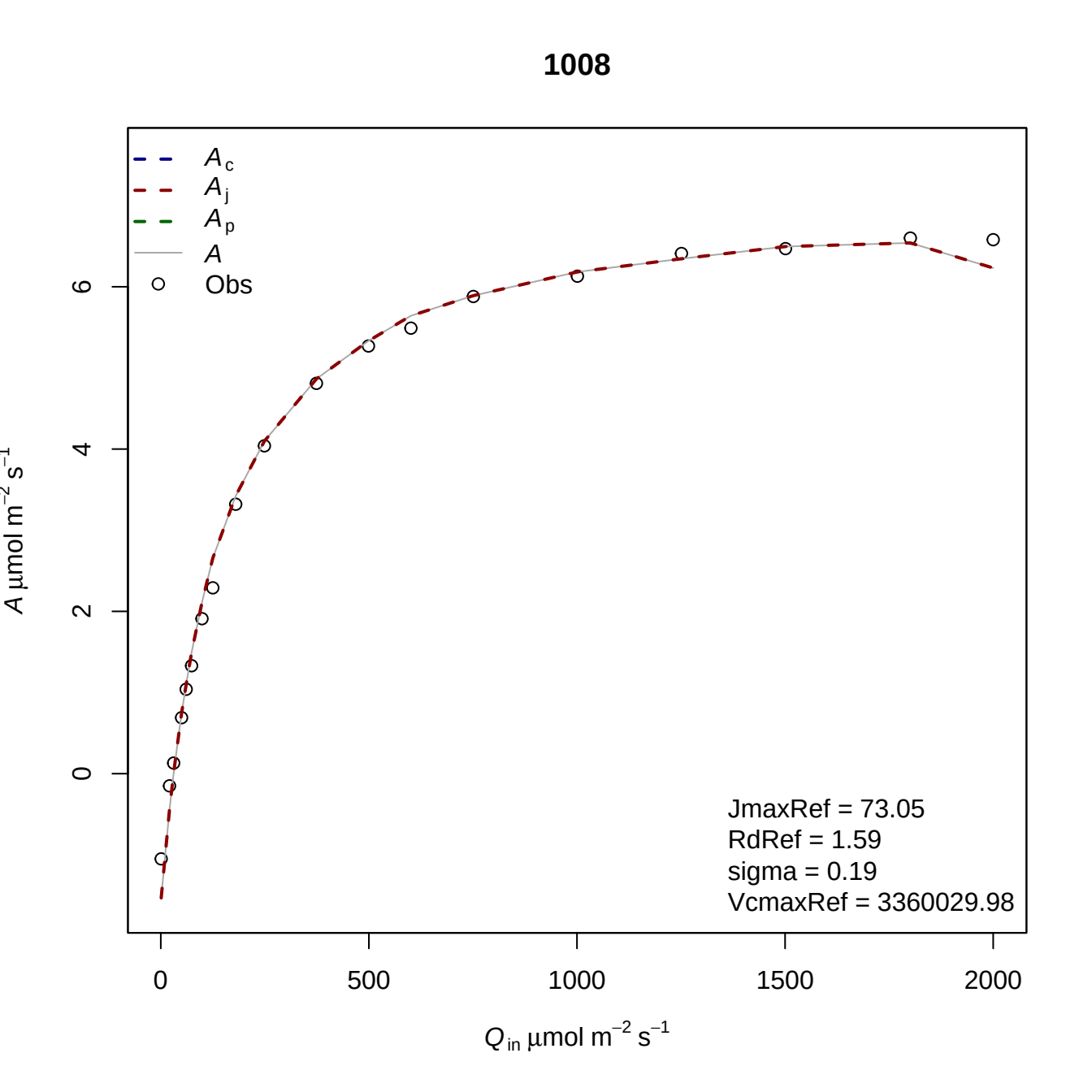

1003
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 177.67
RdRef = 1.56
sigma = 0.26
VcmaxRef = 165.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

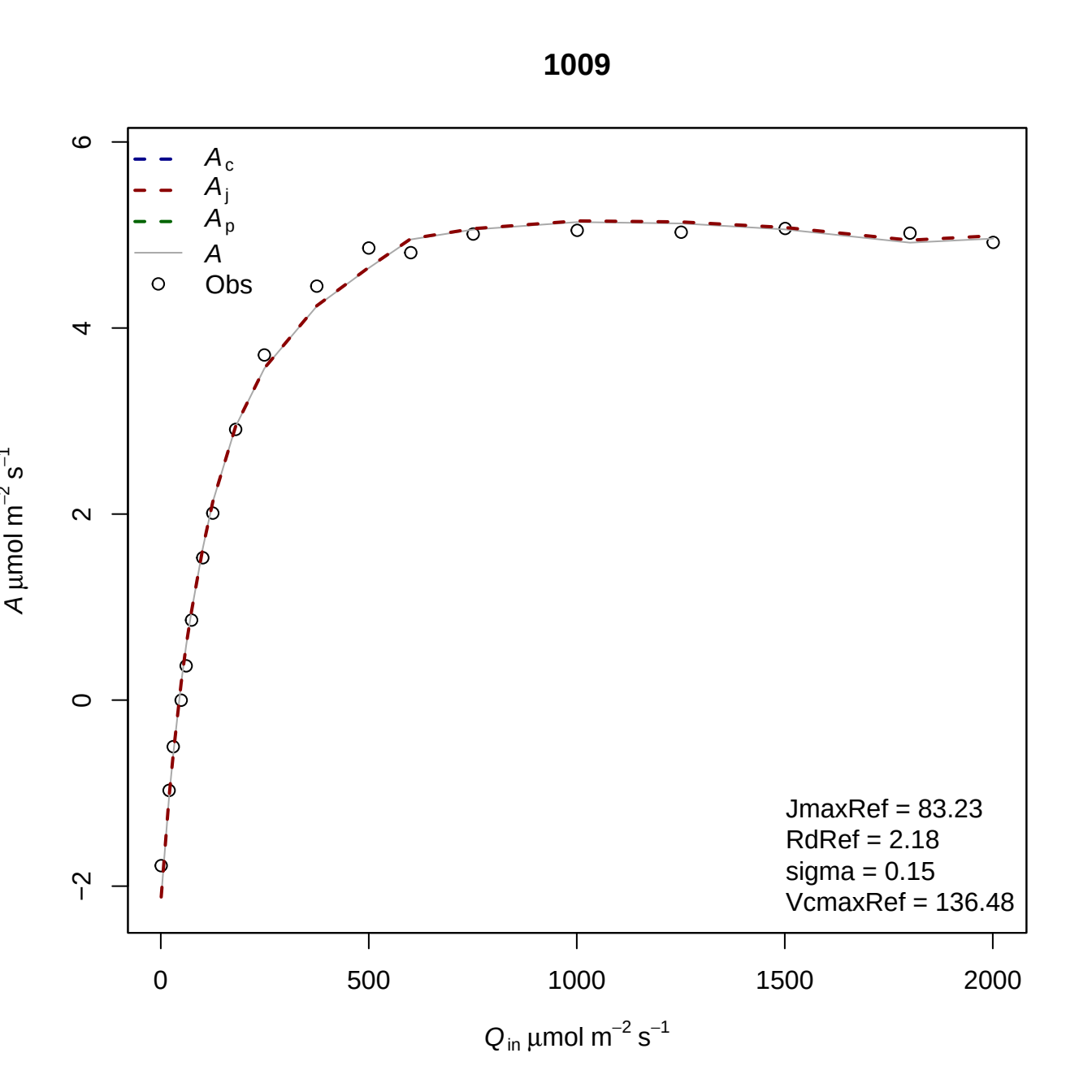

1004
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 52.06
RdRef = 1.25
sigma = 0.28
VcmaxRef = 688492.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

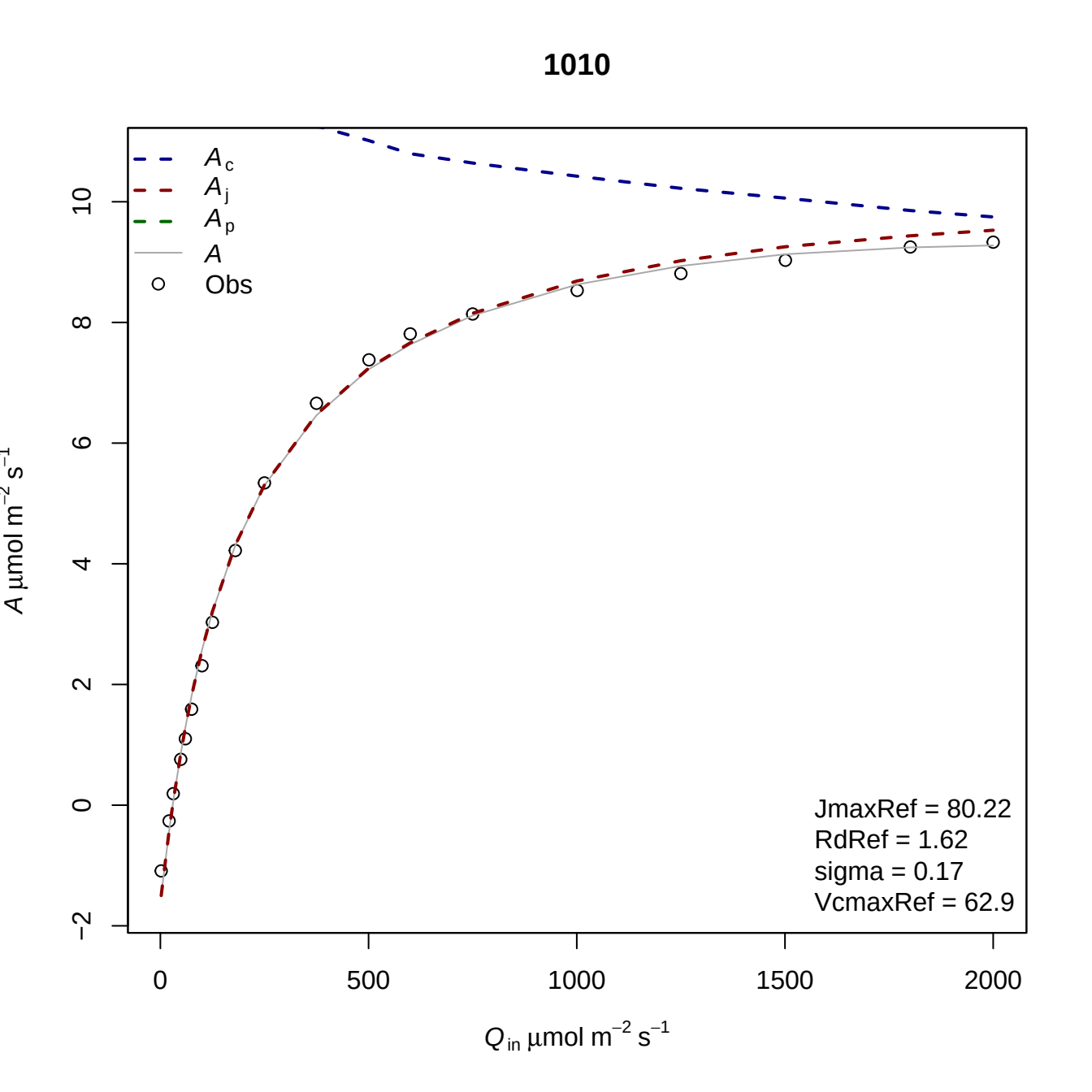

1005
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 230.19
RdRef = 1.85
sigma = 0.49
VcmaxRef = 171.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

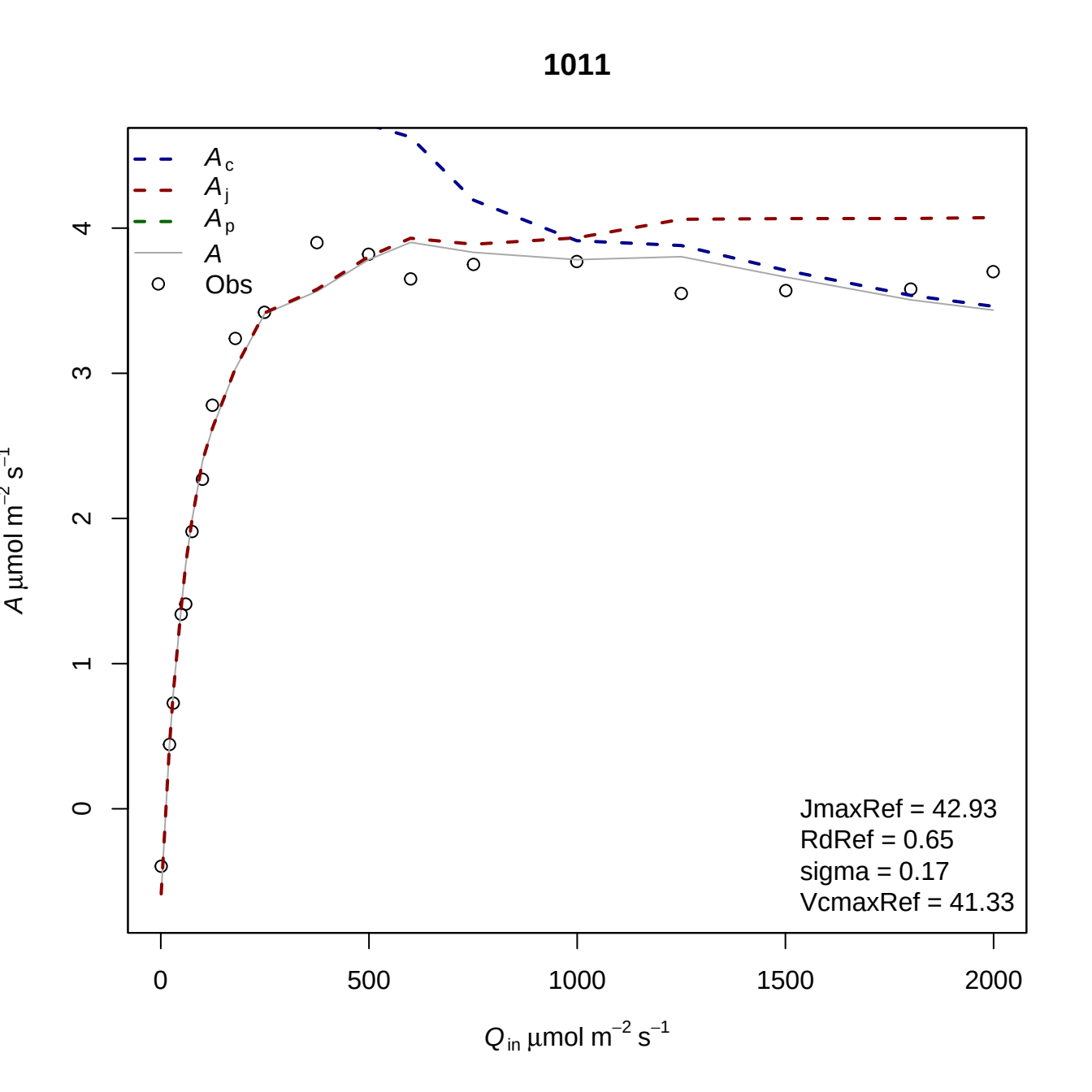

1006
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.23
RdRef = 2.1
sigma = 0.18
VcmaxRef = 95.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

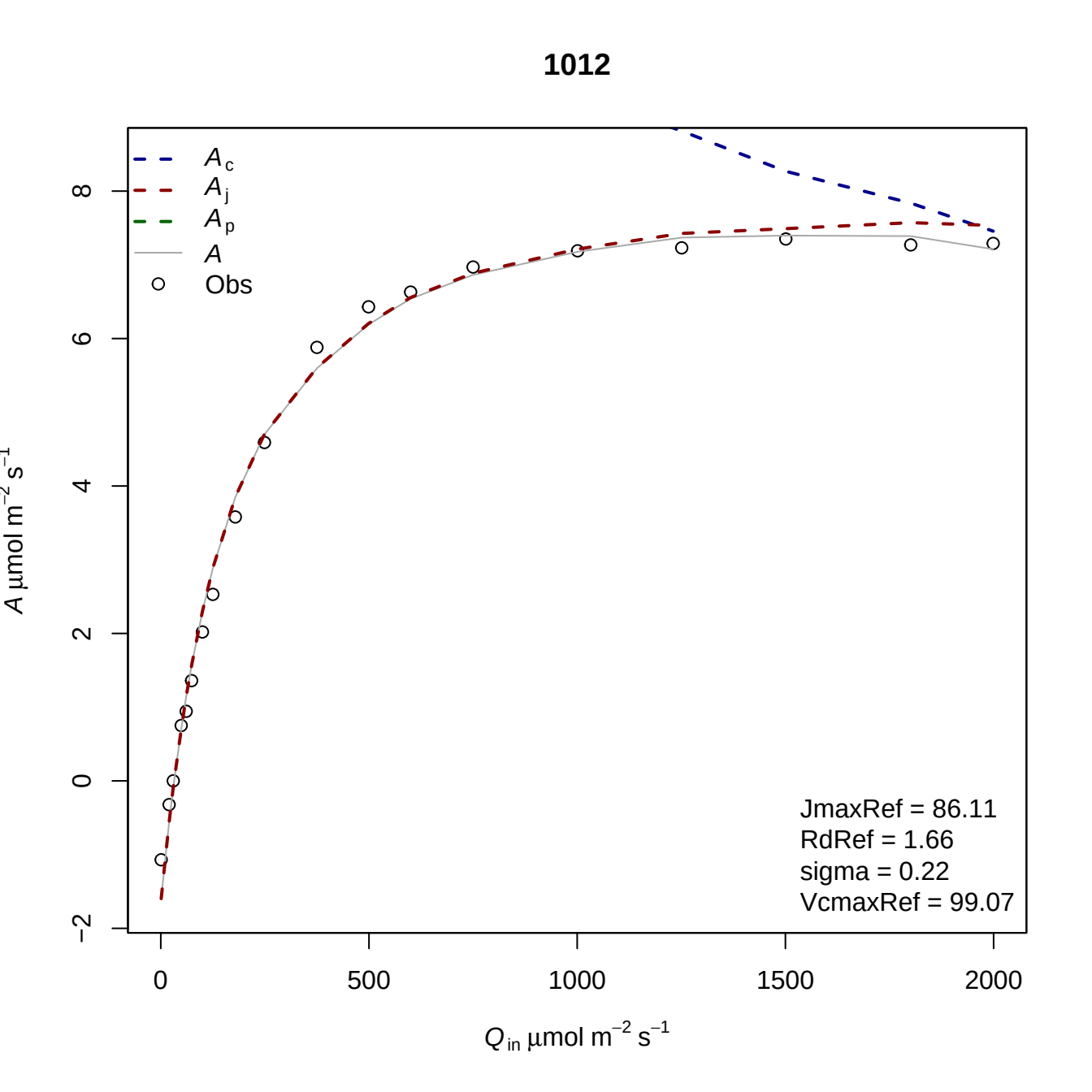

1007
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.35
RdRef = 1.36
sigma = 0.14
VcmaxRef = 59.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

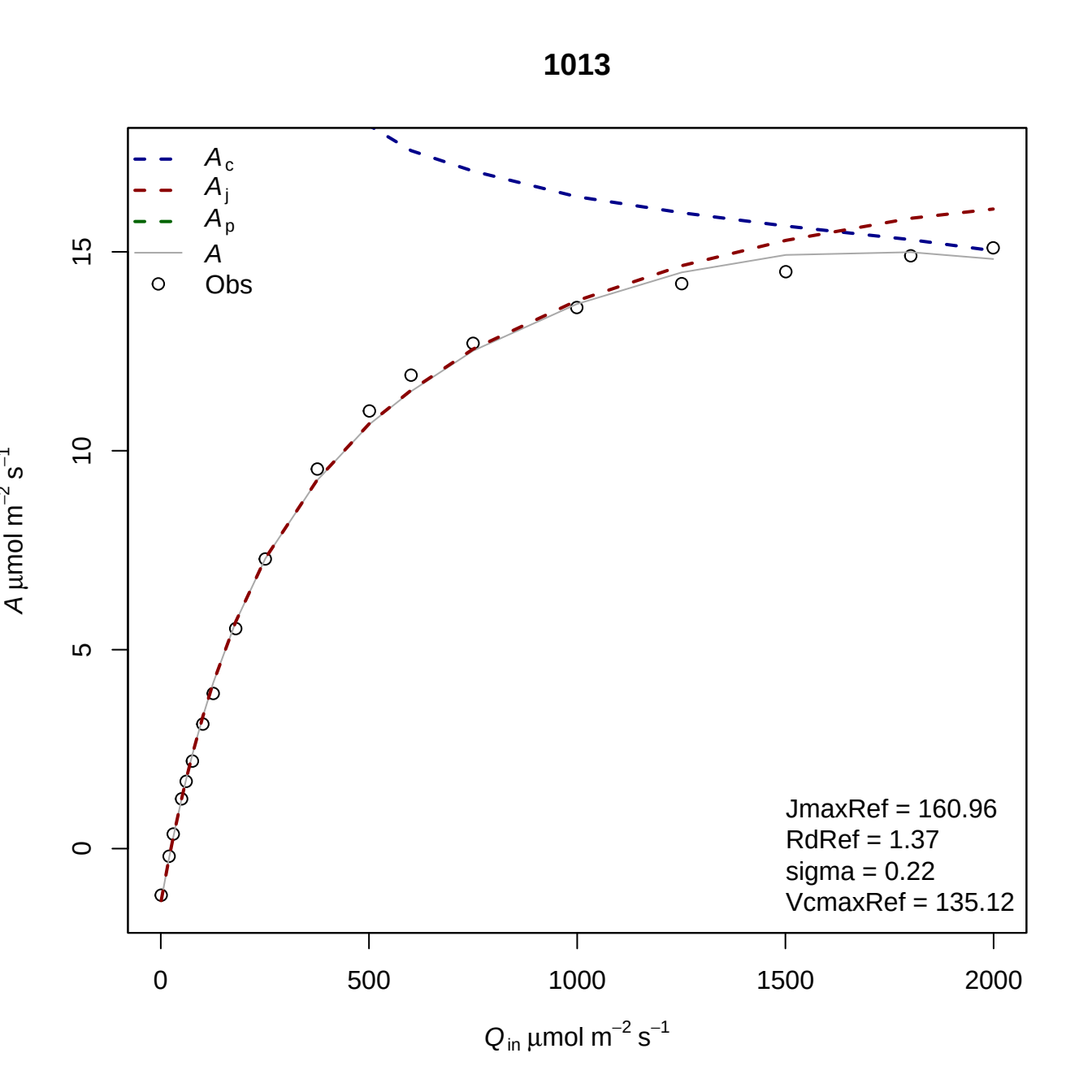

1008
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.05
RdRef = 1.59
sigma = 0.19
VcmaxRef = 3360029.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

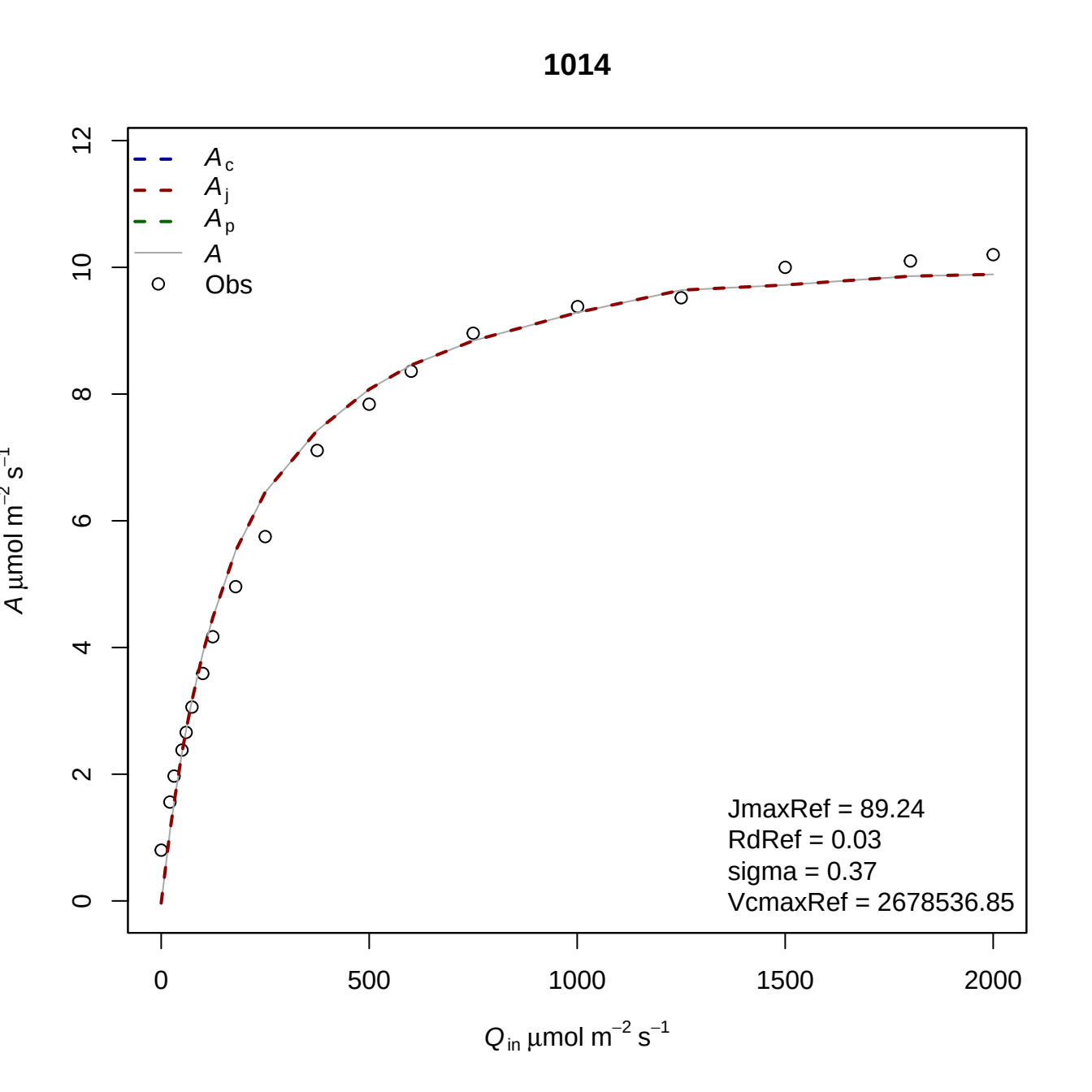

1009
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.23
RdRef = 2.18
sigma = 0.15
VcmaxRef = 136.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500
2000

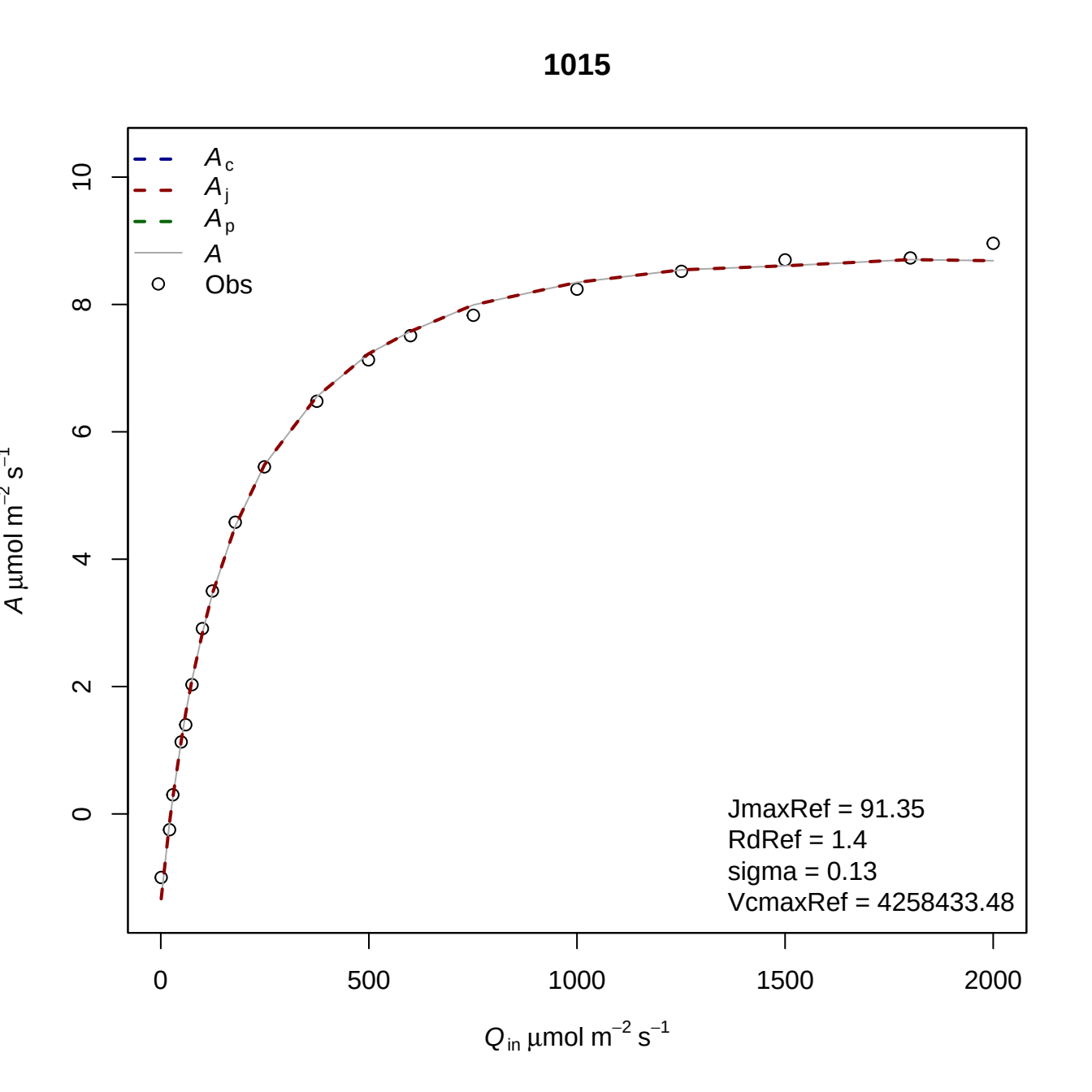

1010
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.22
RdRef = 1.62
sigma = 0.17
VcmaxRef = 62.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

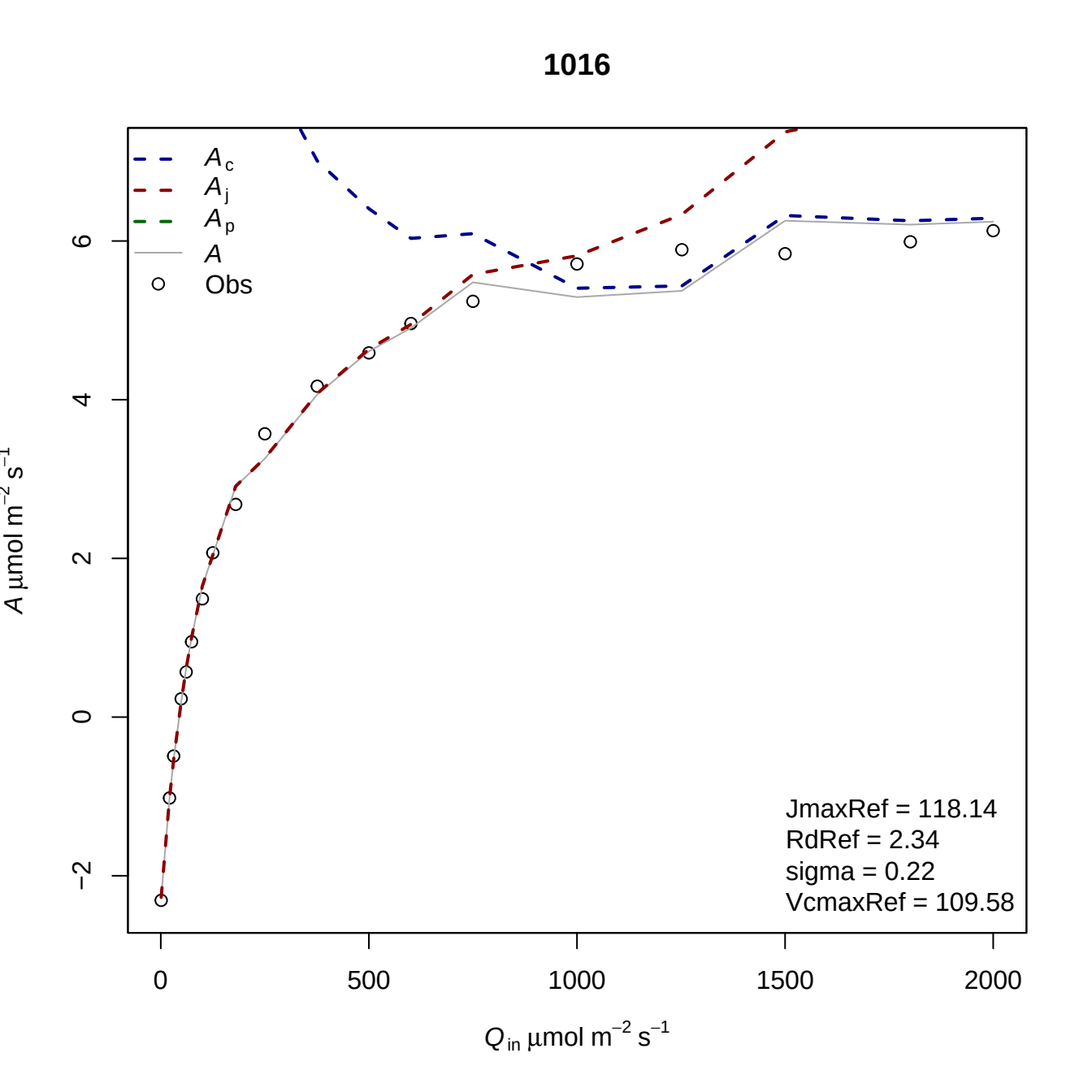

1011
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.93
RdRef = 0.65
sigma = 0.17
VcmaxRef = 41.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500
2000

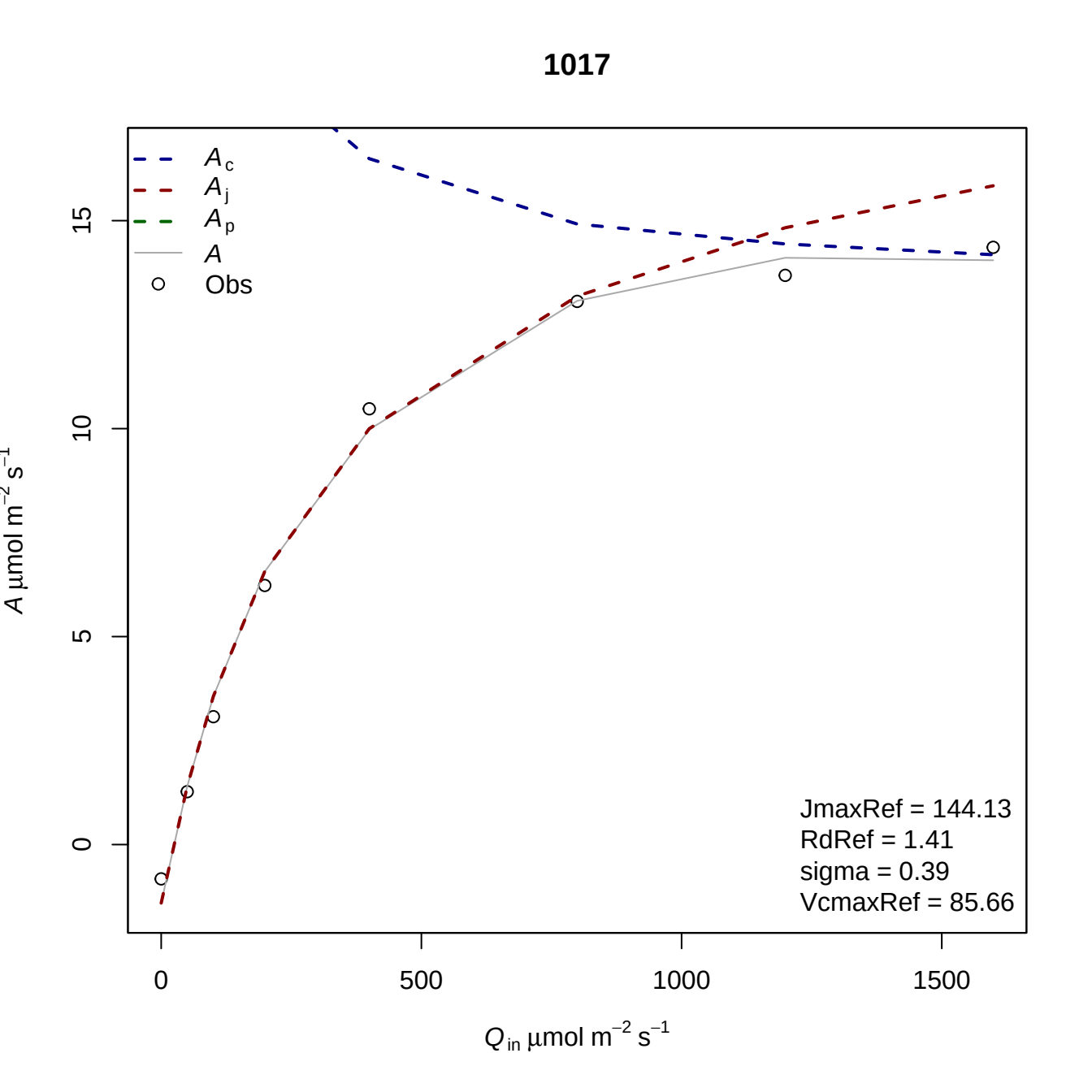

1012
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.11
RdRef = 1.66
sigma = 0.22
VcmaxRef = 99.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

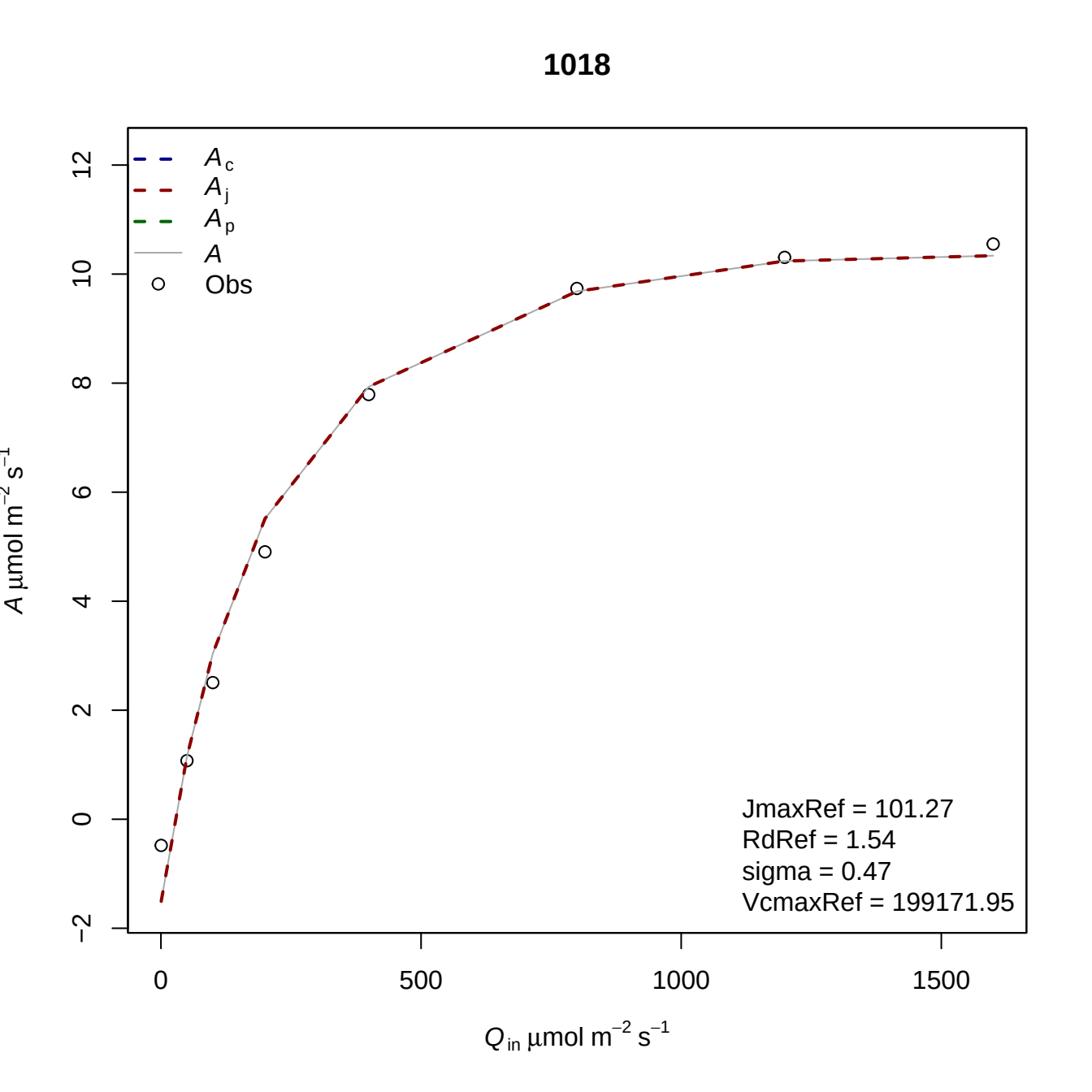

1013
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.96
RdRef = 1.37
sigma = 0.22
VcmaxRef = 135.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

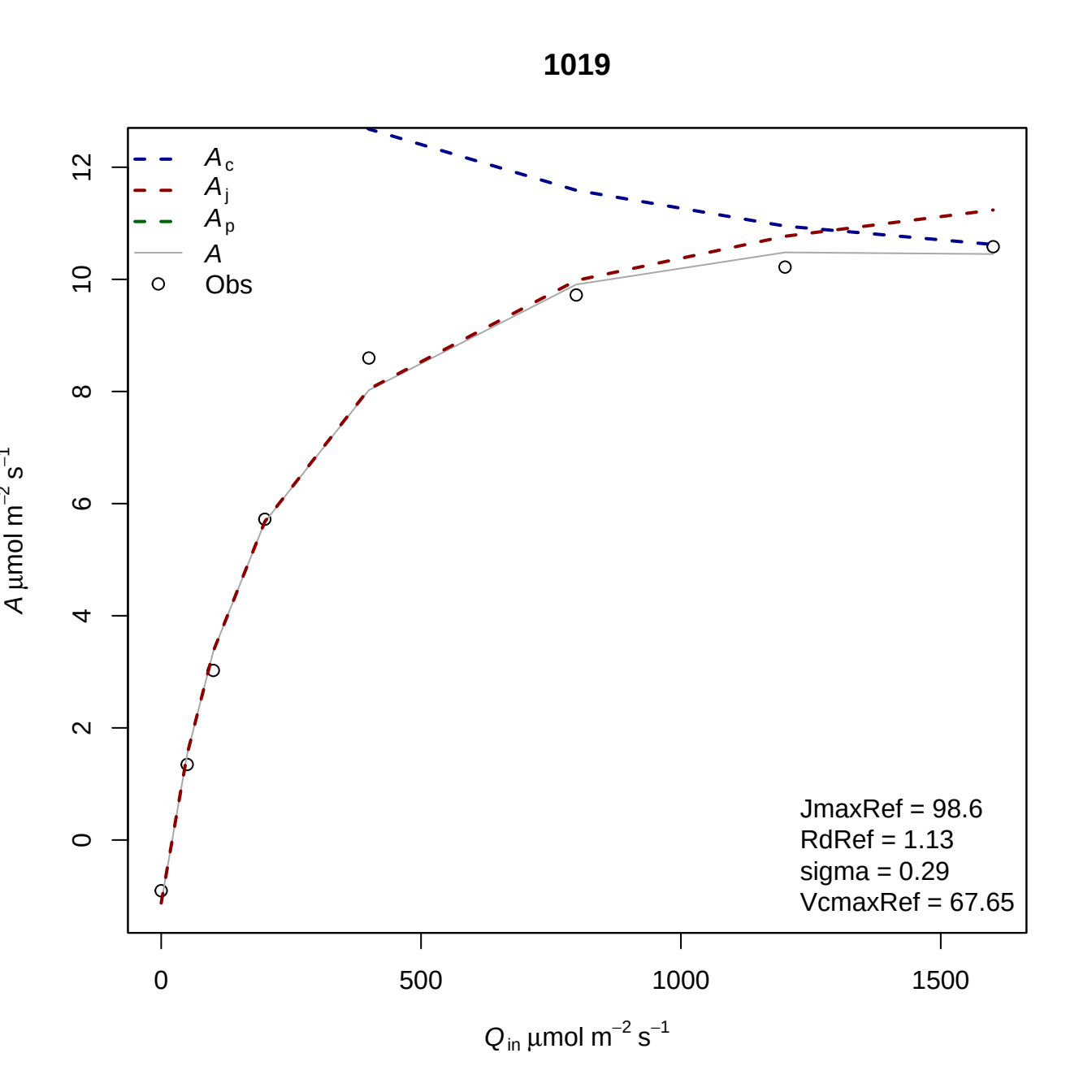

1014
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.24
RdRef = 0.03
sigma = 0.37
VcmaxRef = 2678536.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

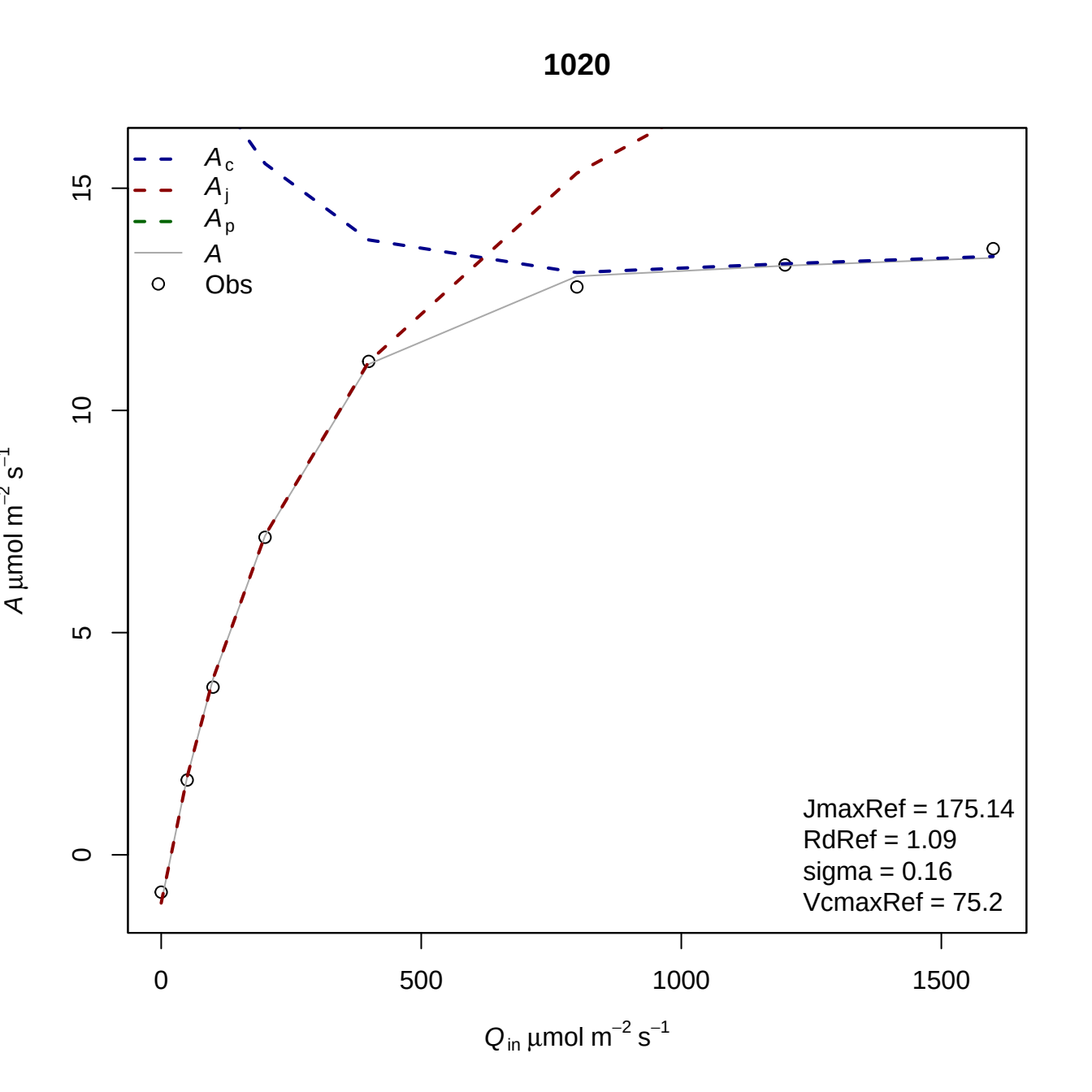

1015
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 91.35
RdRef = 1.4
sigma = 0.13
VcmaxRef = 4258433.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

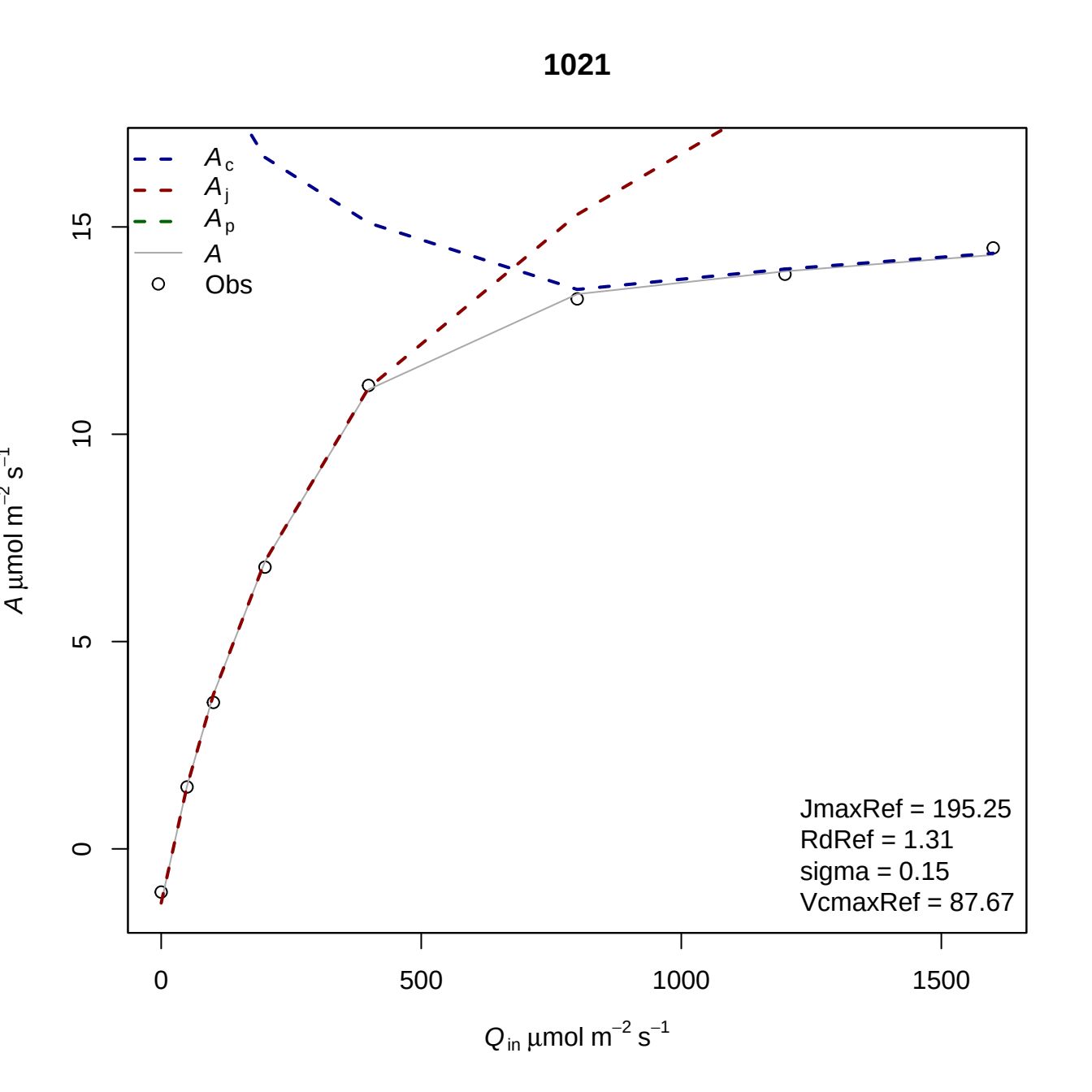

1016
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.14
RdRef = 2.34
sigma = 0.22
VcmaxRef = 109.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

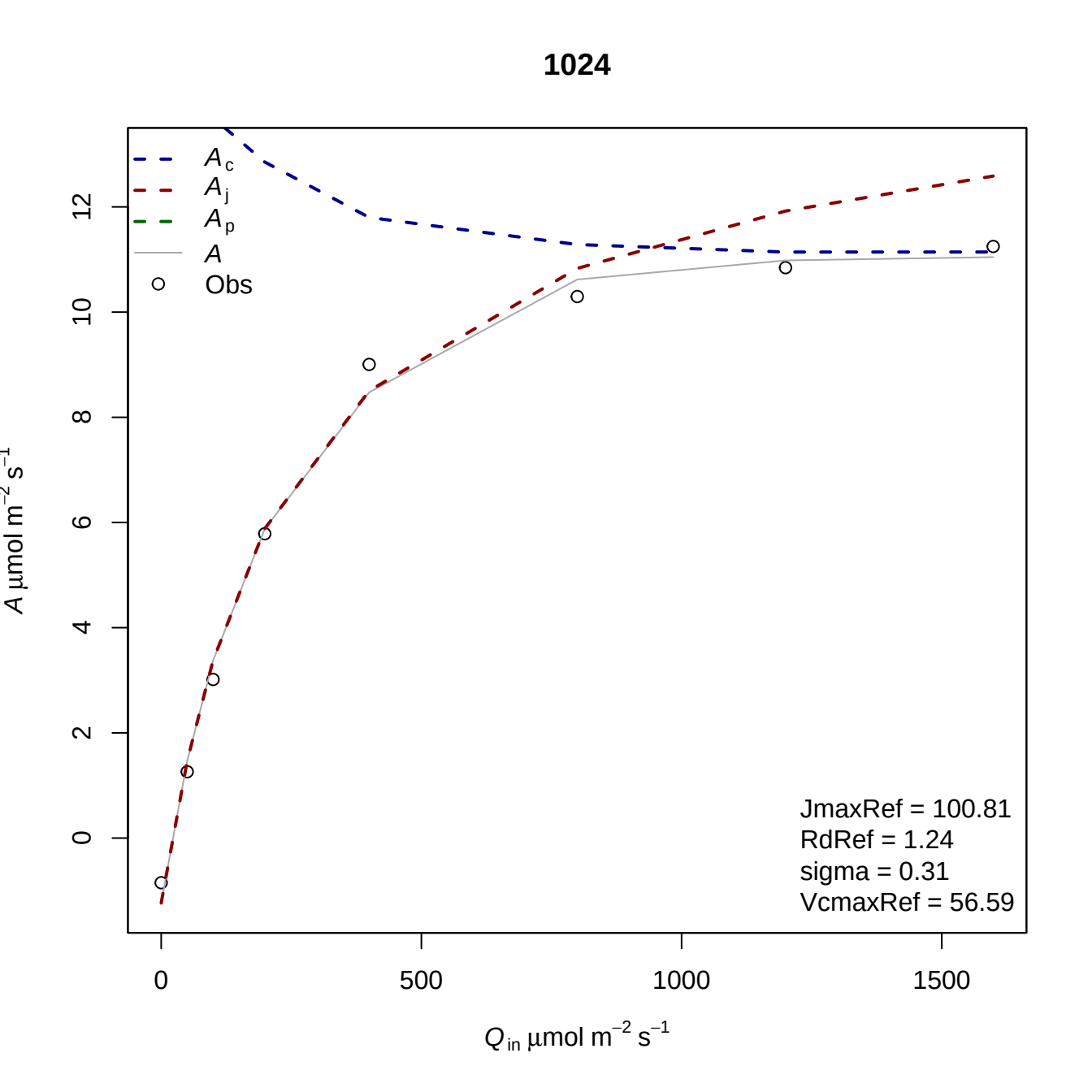

1017
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 144.13
RdRef = 1.41
sigma = 0.39
VcmaxRef = 85.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

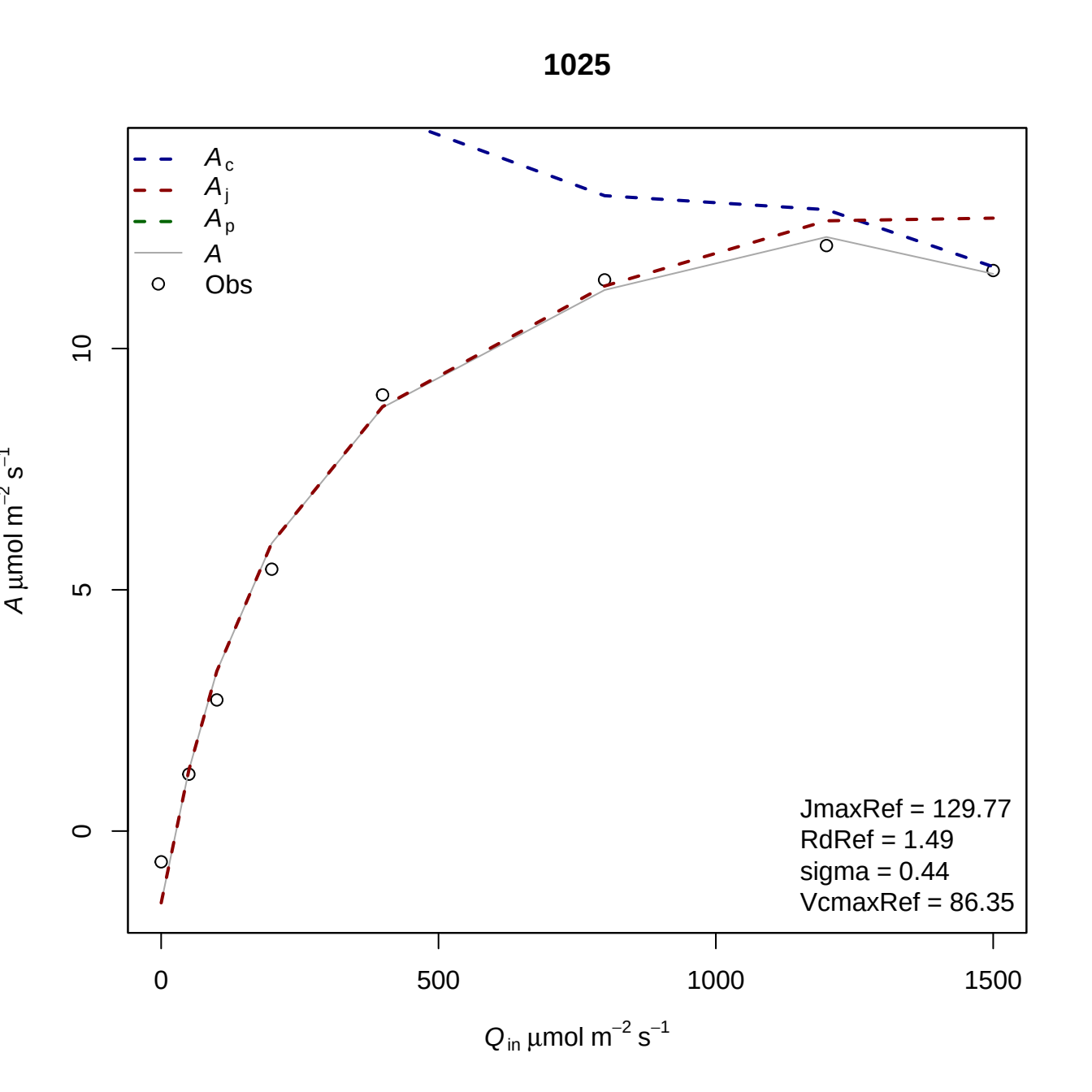

1018
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 101.27
RdRef = 1.54
sigma = 0.47
VcmaxRef = 199171.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500

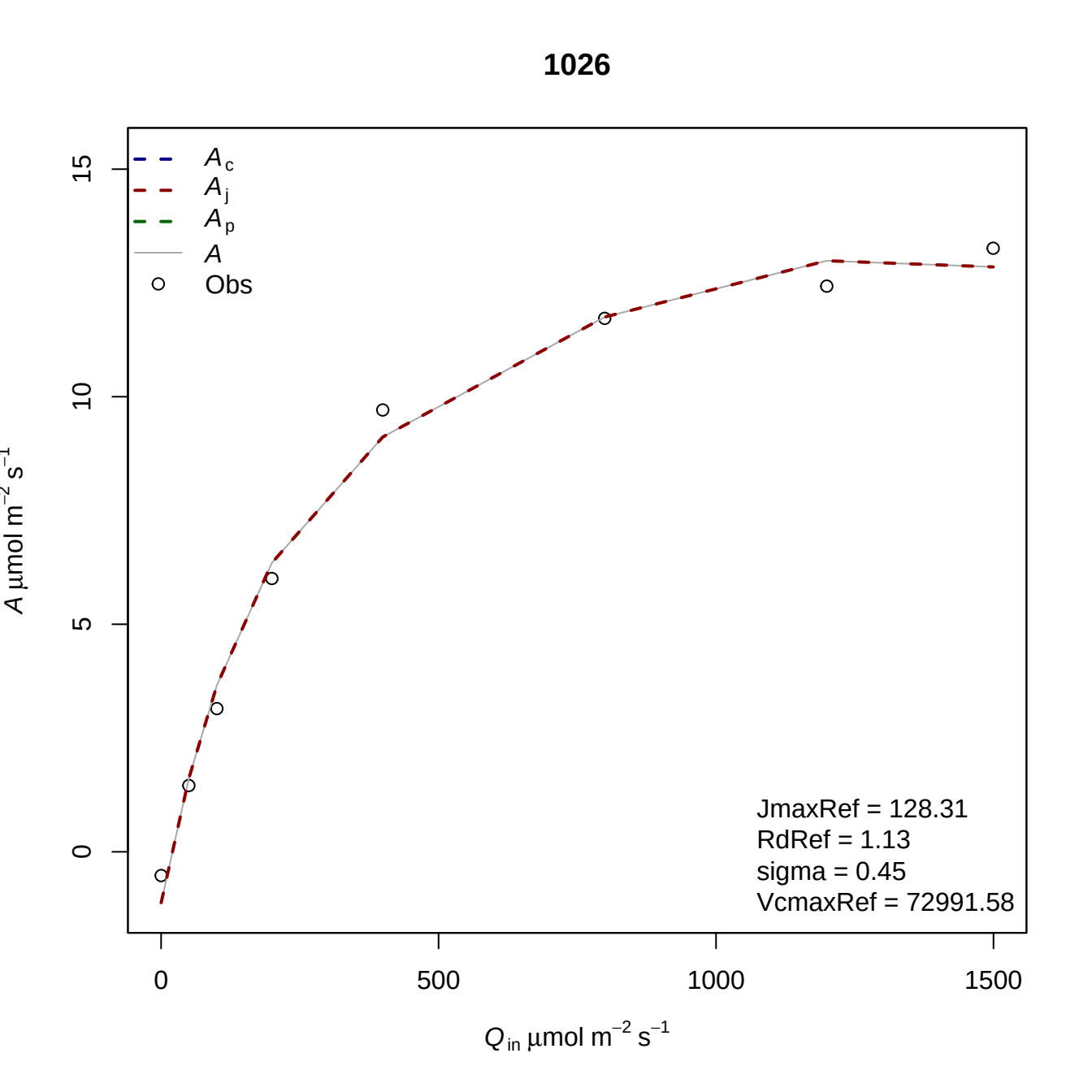

1019
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.6
RdRef = 1.13
sigma = 0.29
VcmaxRef = 67.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

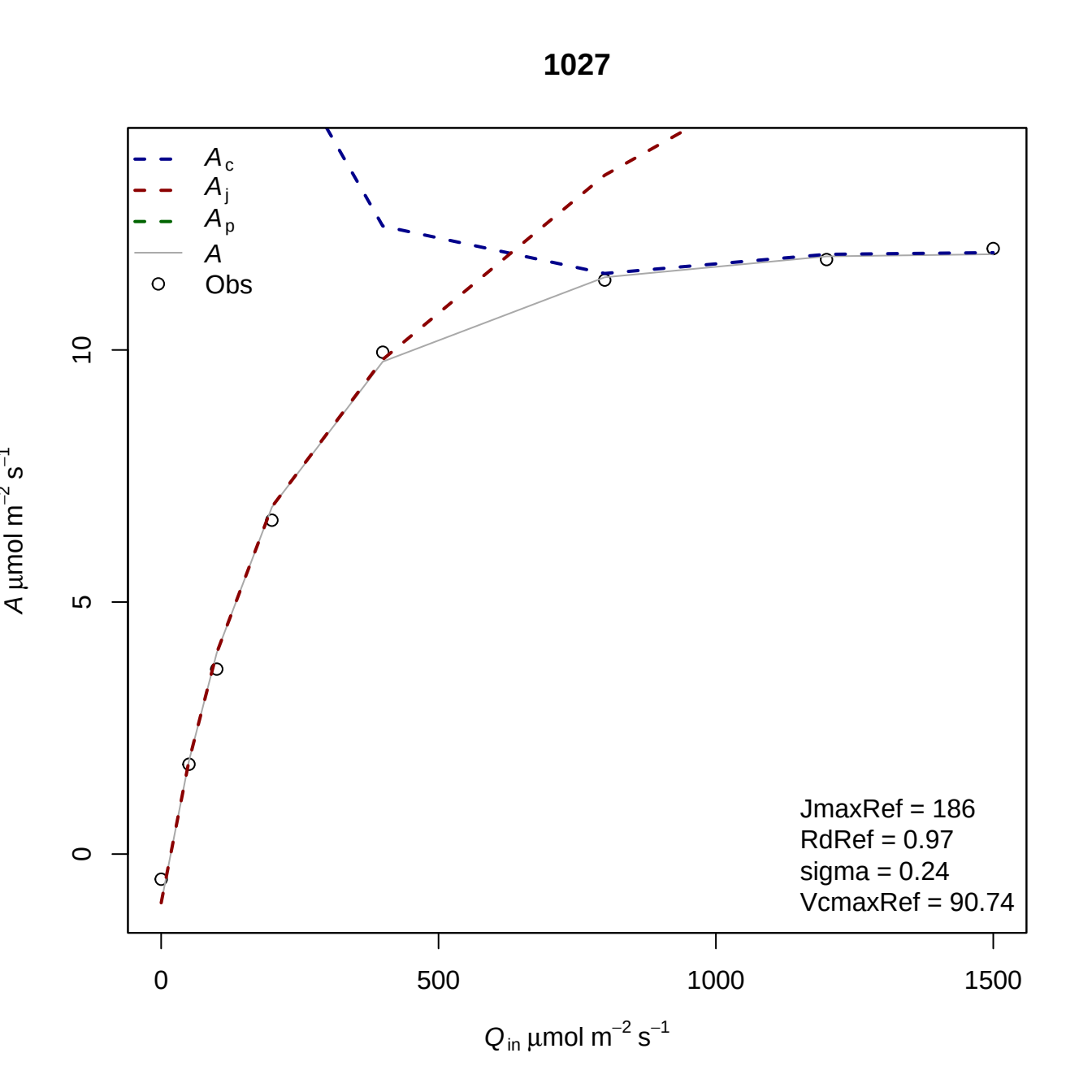

1020
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 175.14
RdRef = 1.09
sigma = 0.16
VcmaxRef = 75.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

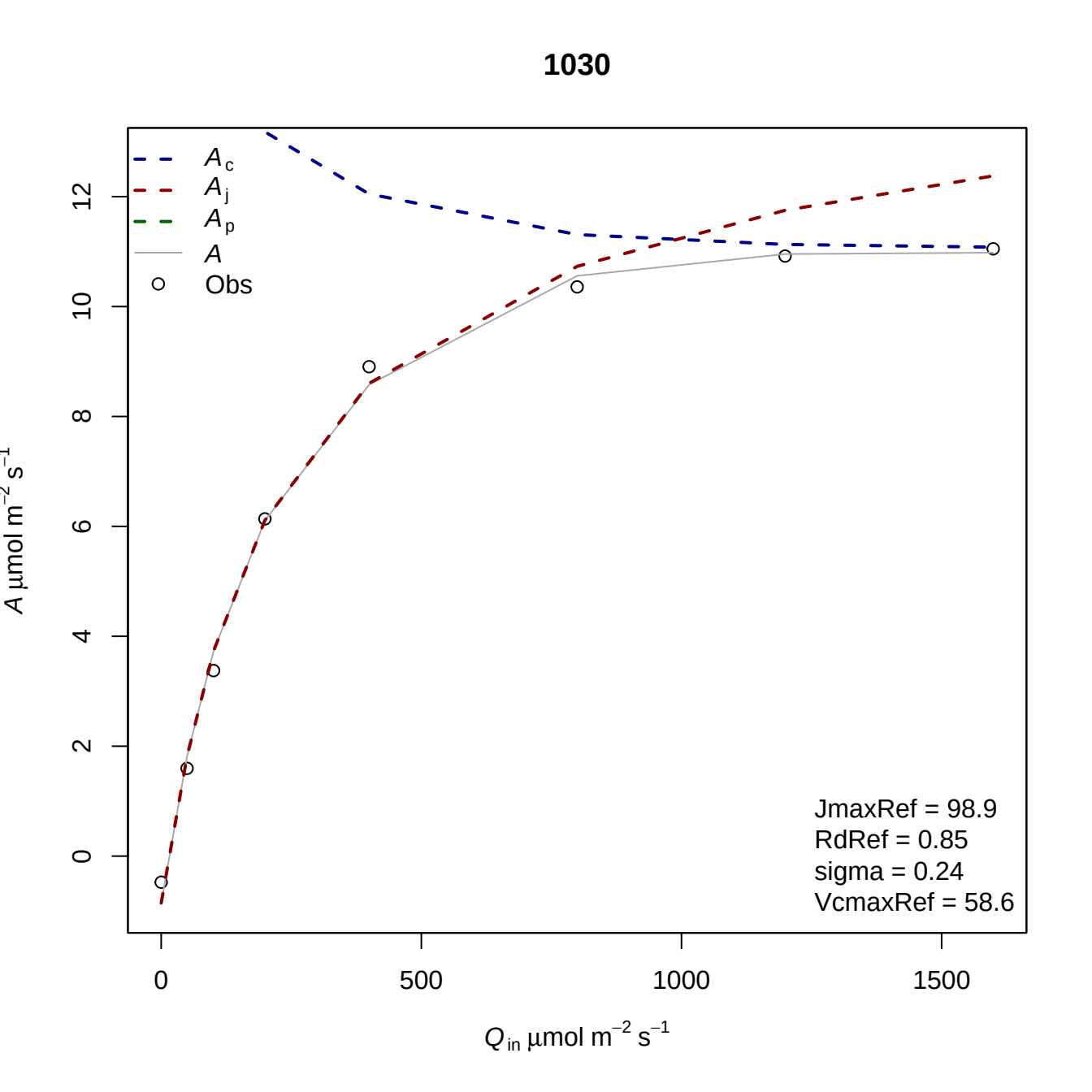

1021
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 195.25
RdRef = 1.31
sigma = 0.15
VcmaxRef = 87.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

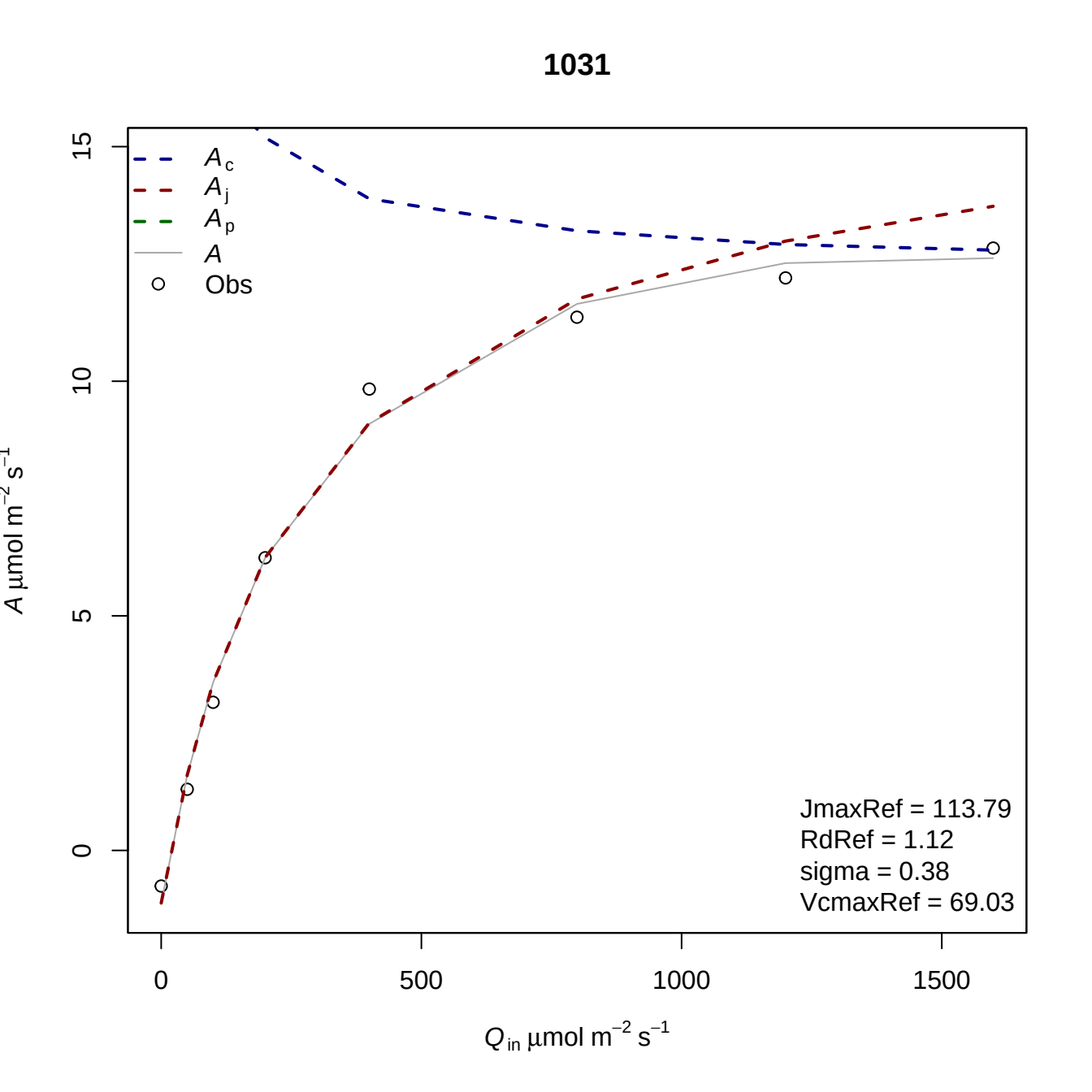

1024
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.81
RdRef = 1.24
sigma = 0.31
VcmaxRef = 56.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

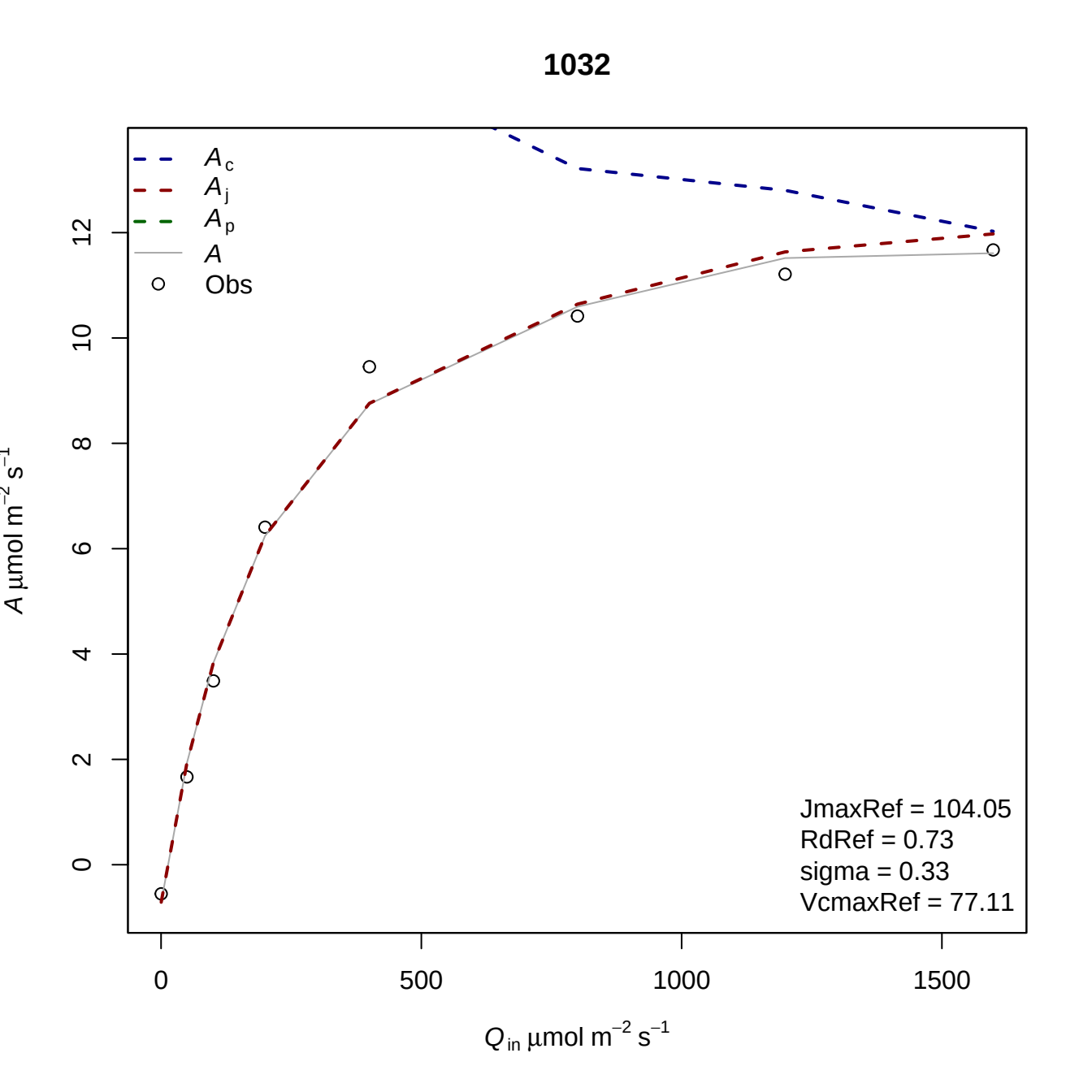

1025
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.77
RdRef = 1.49
sigma = 0.44
VcmaxRef = 86.35
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

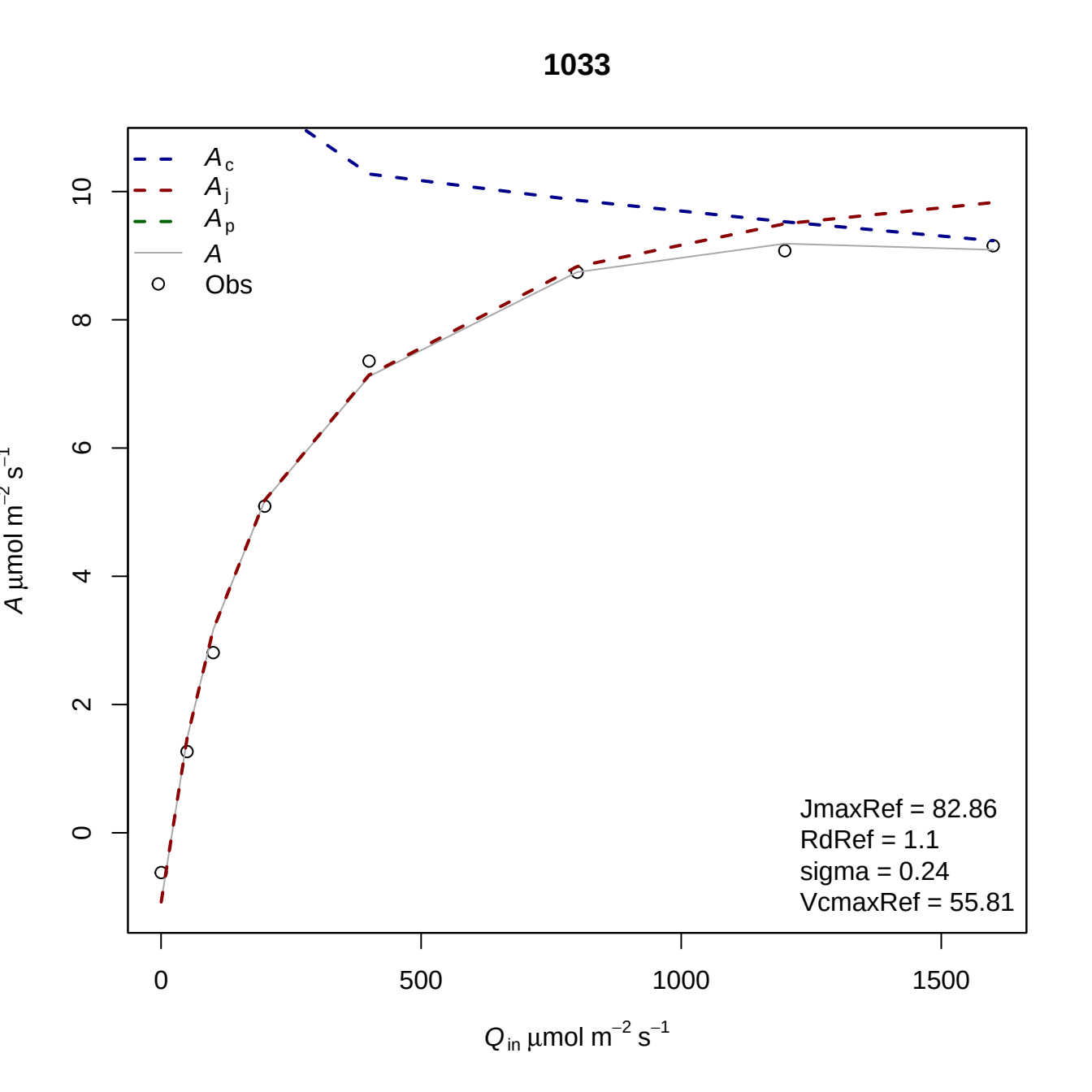

1026
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.31
RdRef = 1.13
sigma = 0.45
VcmaxRef = 72991.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

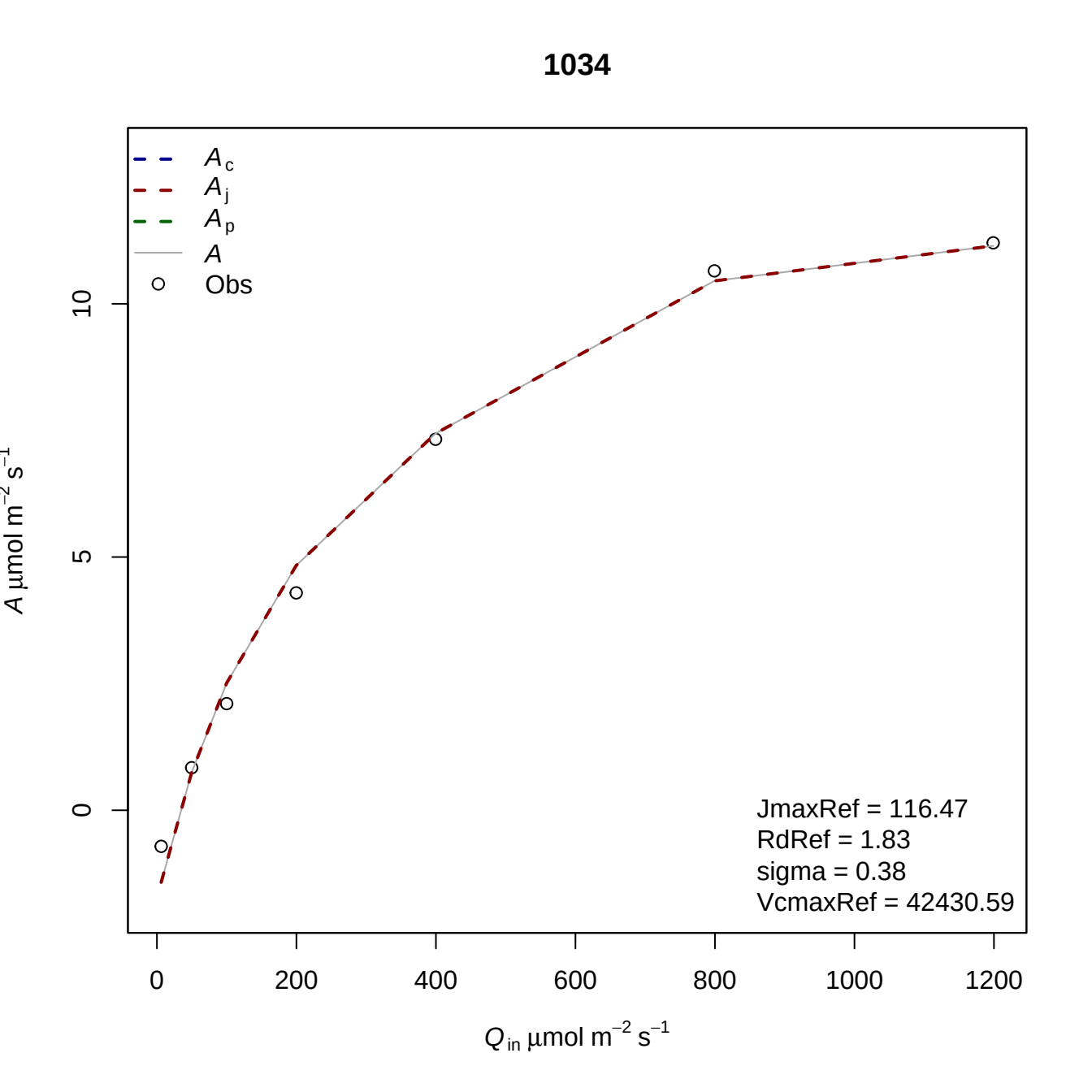

1027
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 186
RdRef = 0.97
sigma = 0.24
VcmaxRef = 90.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

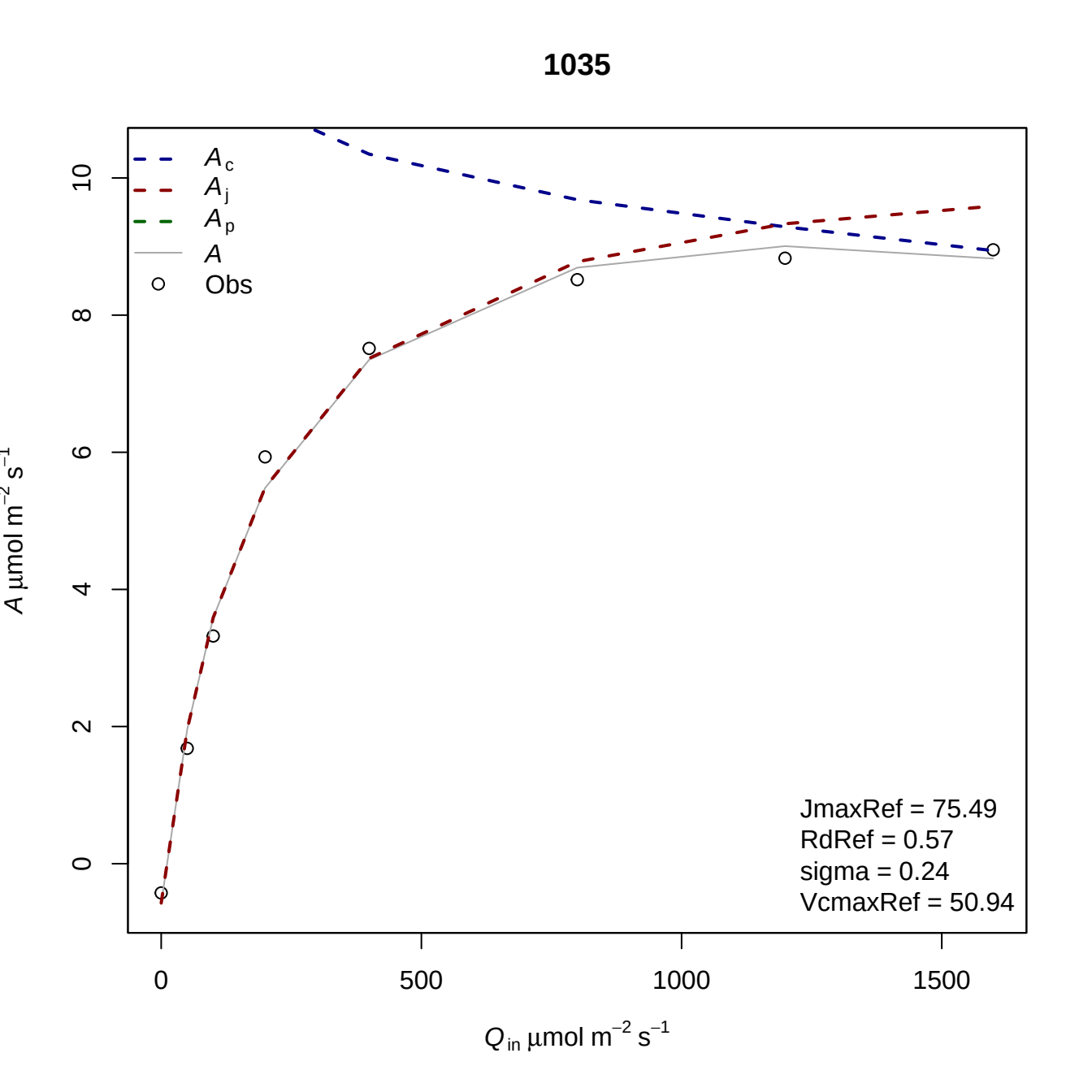

1030
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.9
RdRef = 0.85
sigma = 0.24
VcmaxRef = 58.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

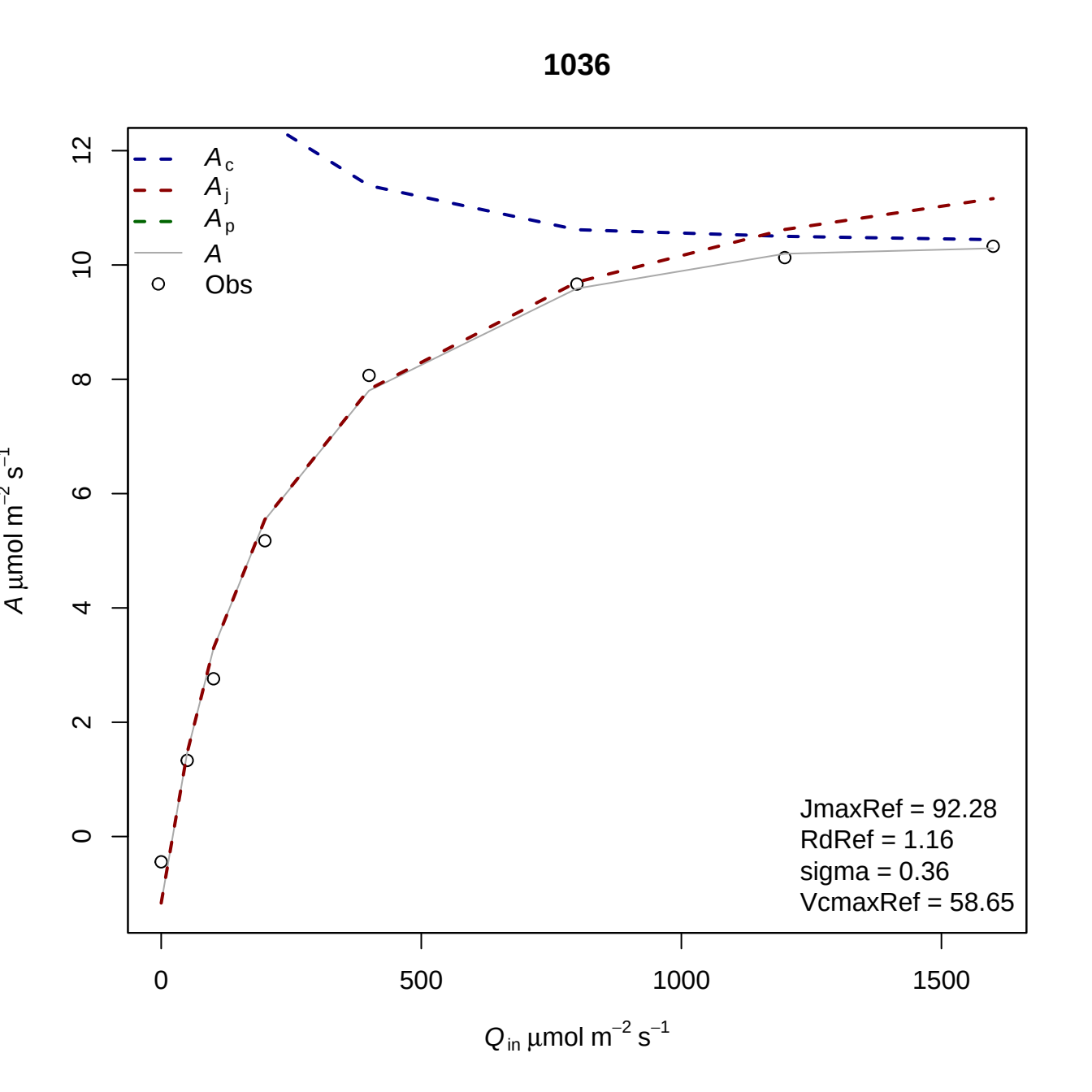

1031
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 113.79
RdRef = 1.12
sigma = 0.38
VcmaxRef = 69.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

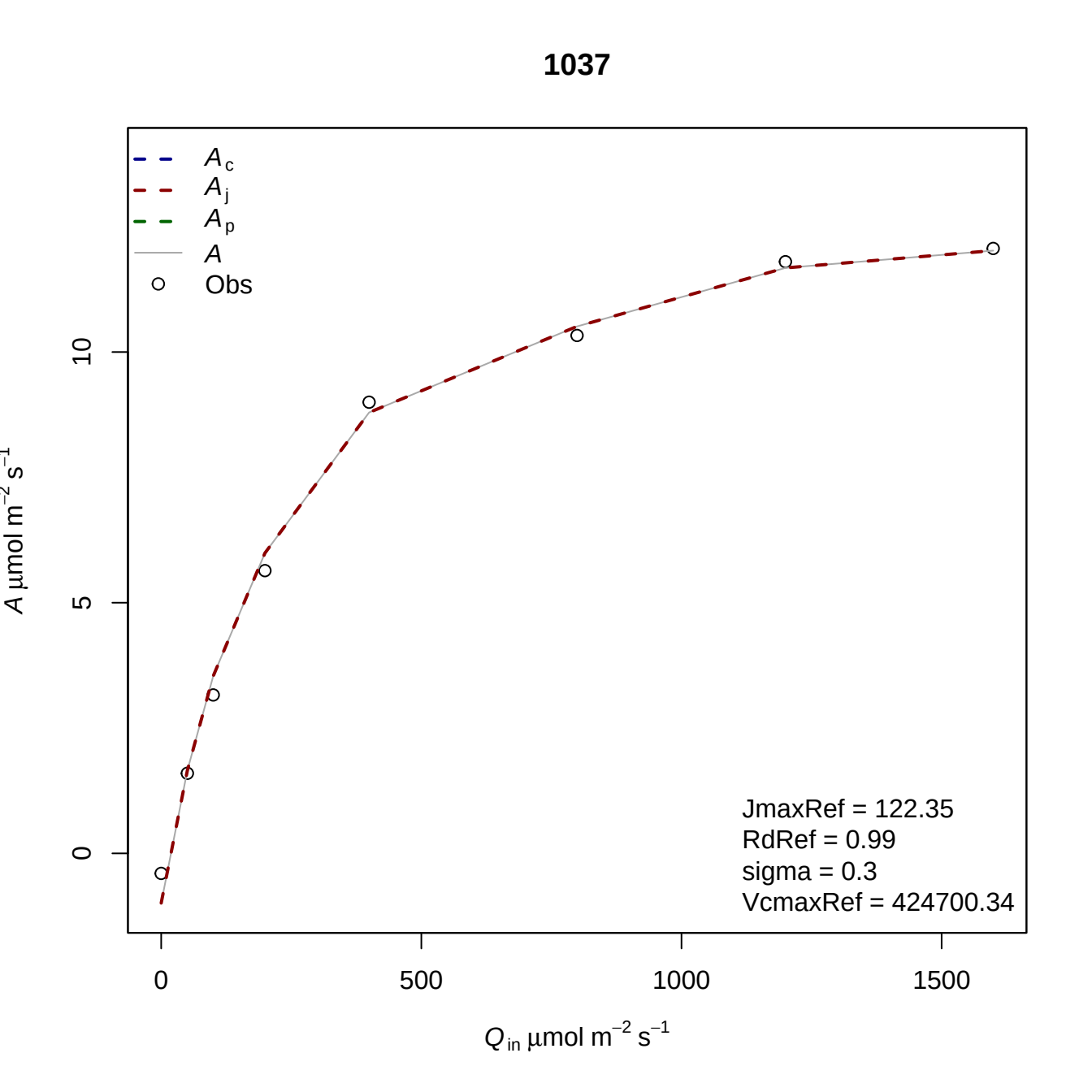

1032
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.05
RdRef = 0.73
sigma = 0.33
VcmaxRef = 77.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

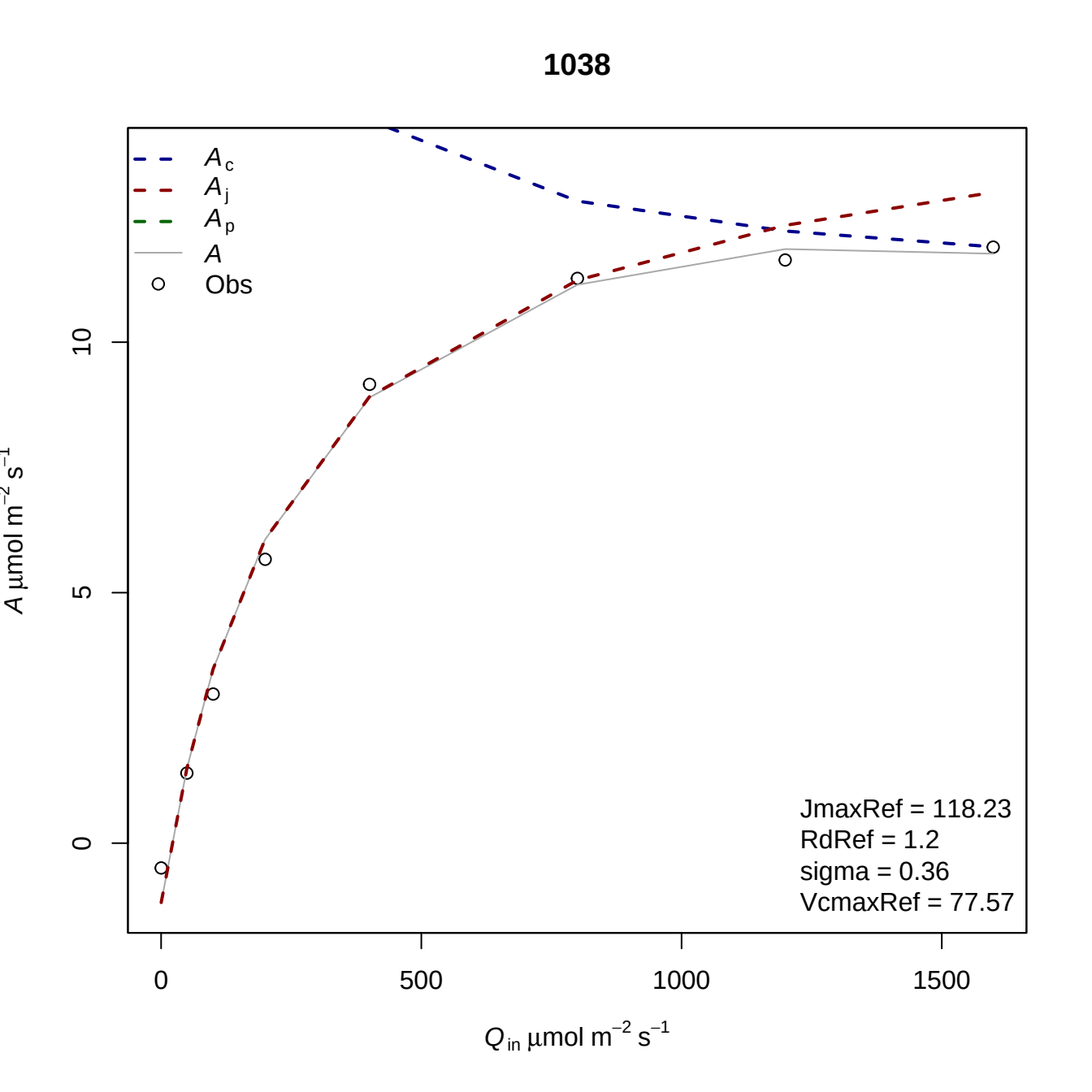

1033
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.86
RdRef = 1.1
sigma = 0.24
VcmaxRef = 55.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

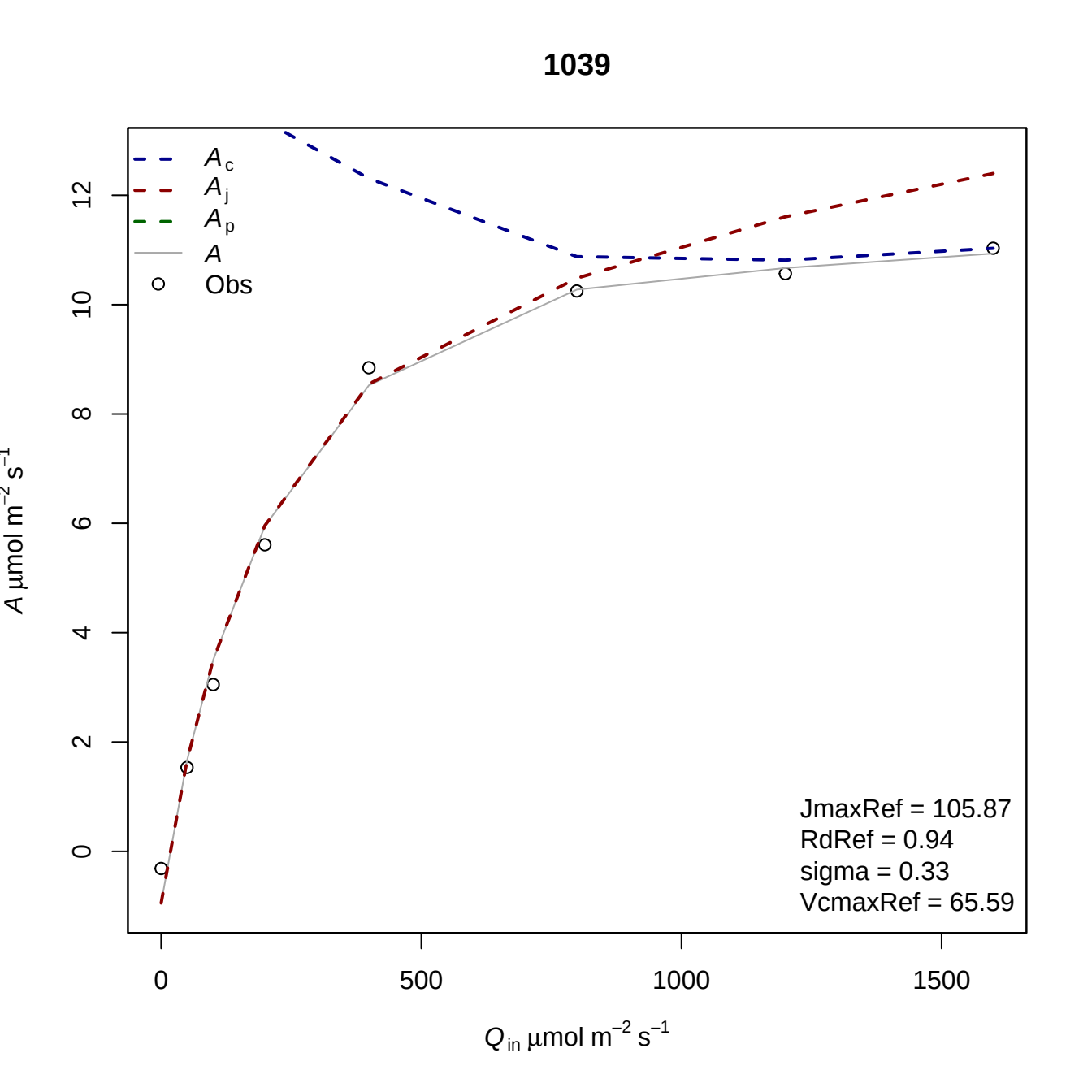

1034
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.47
RdRef = 1.83
sigma = 0.38
VcmaxRef = 42430.59
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000
1200
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

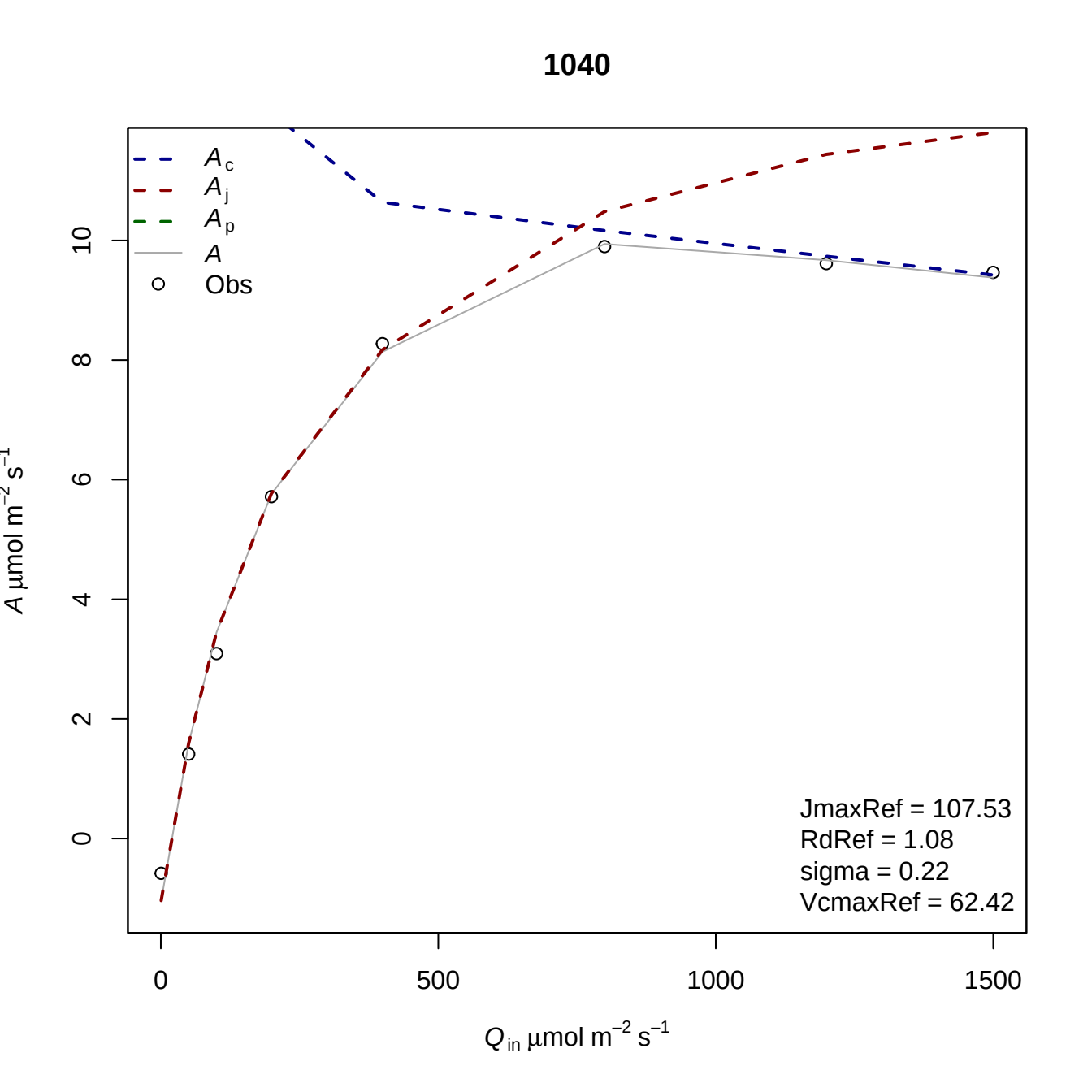

1035
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.49
RdRef = 0.57
sigma = 0.24
VcmaxRef = 50.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

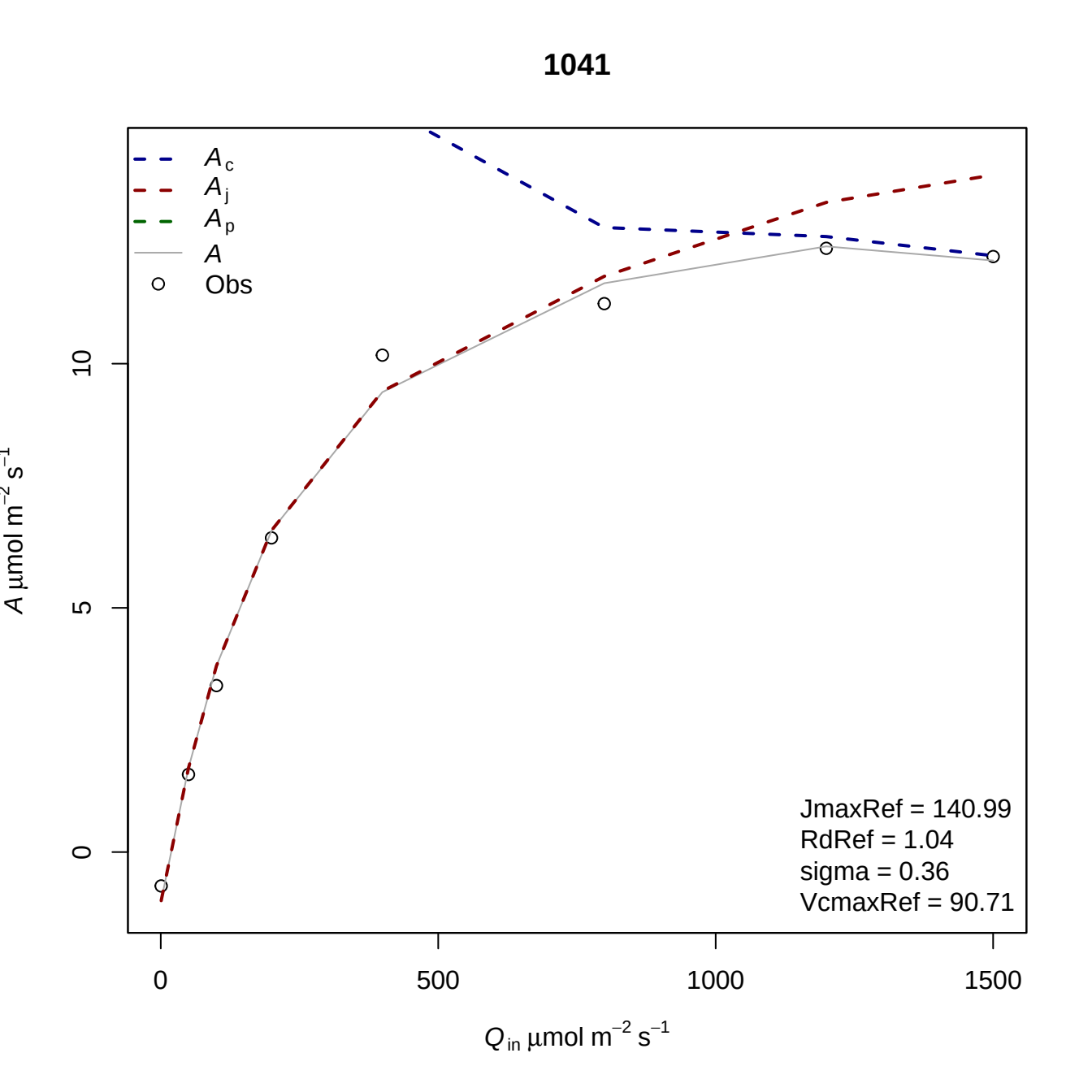

1036
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.28
RdRef = 1.16
sigma = 0.36
VcmaxRef = 58.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

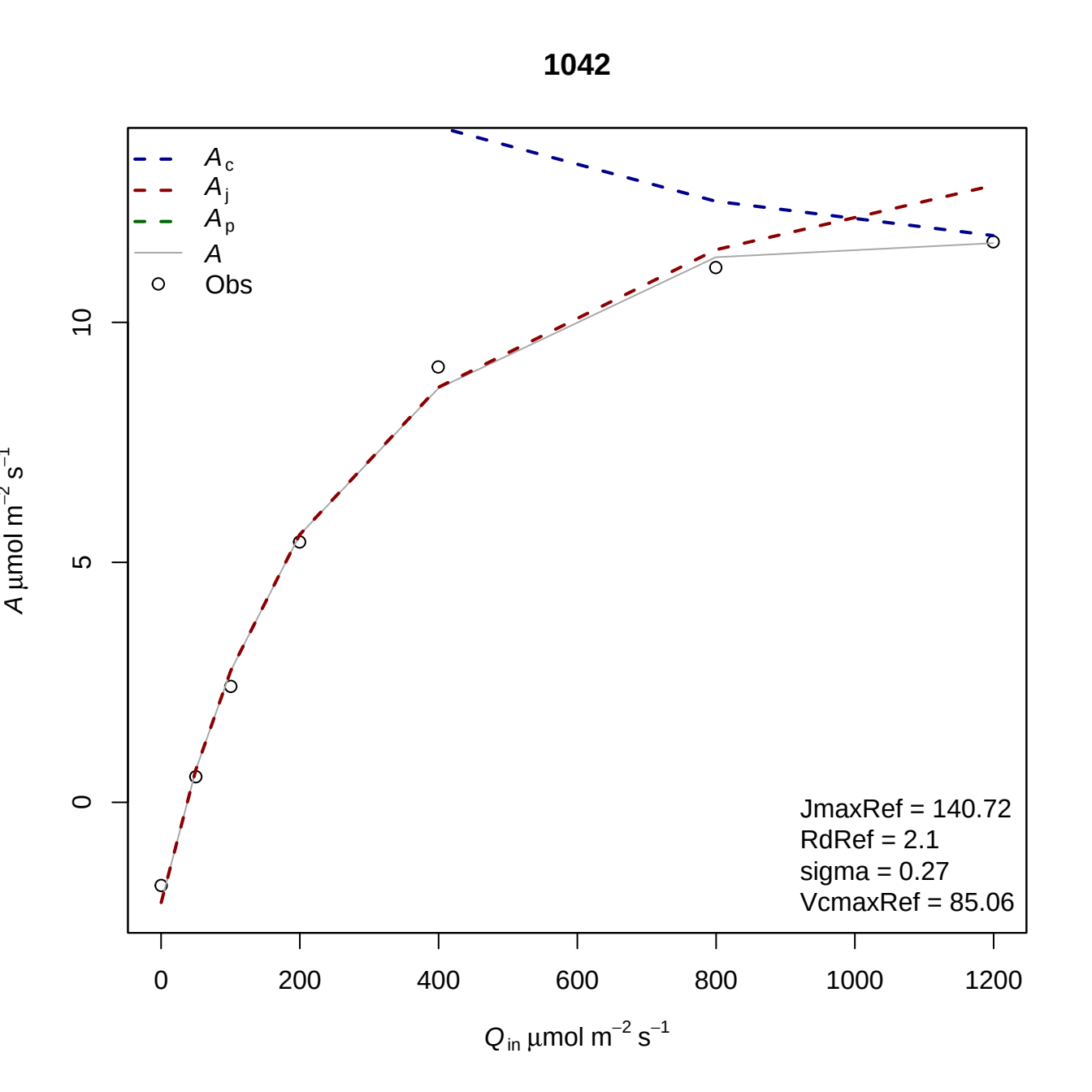

1037
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.35
RdRef = 0.99
sigma = 0.3
VcmaxRef = 424700.34
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

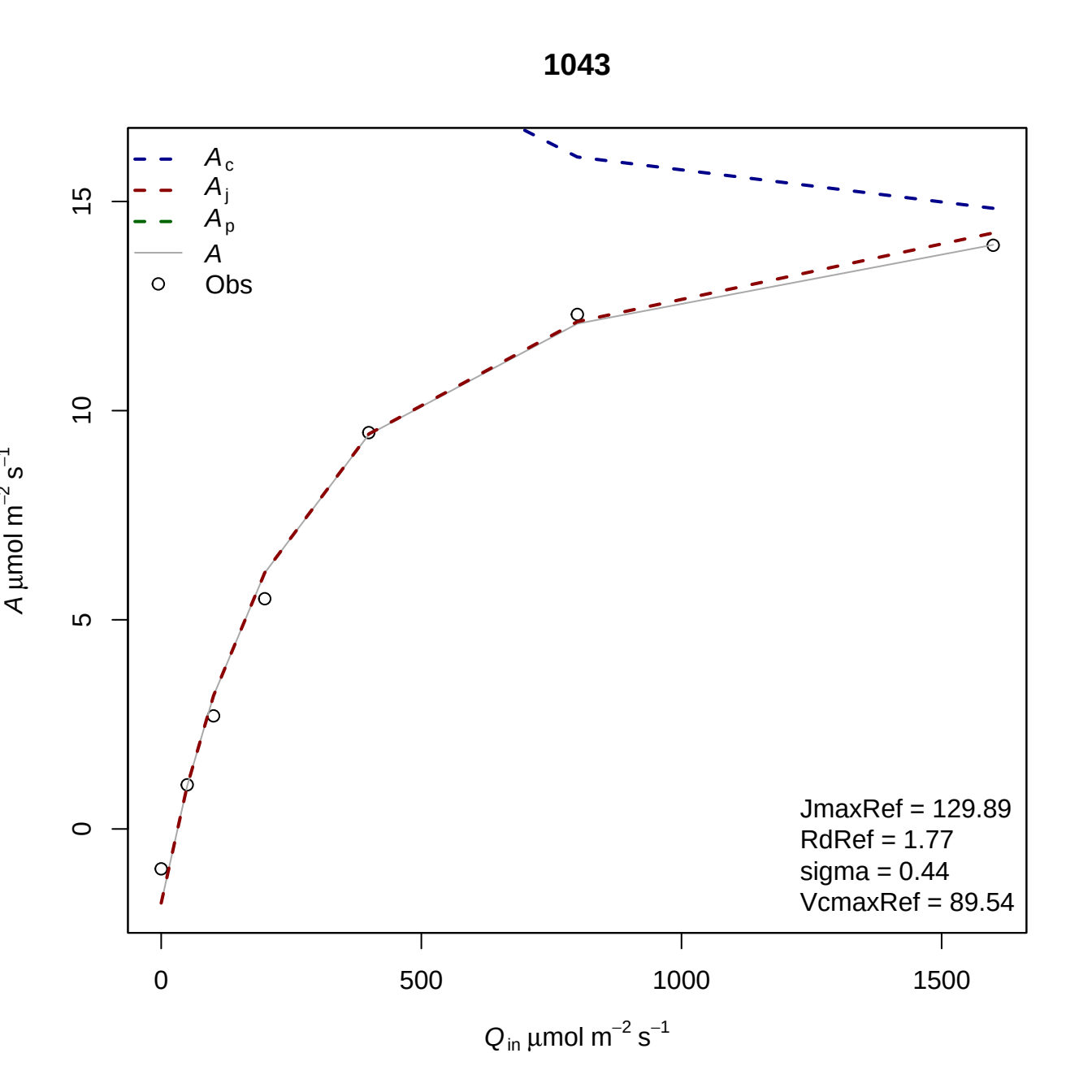

1038
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.23
RdRef = 1.2
sigma = 0.36
VcmaxRef = 77.57
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

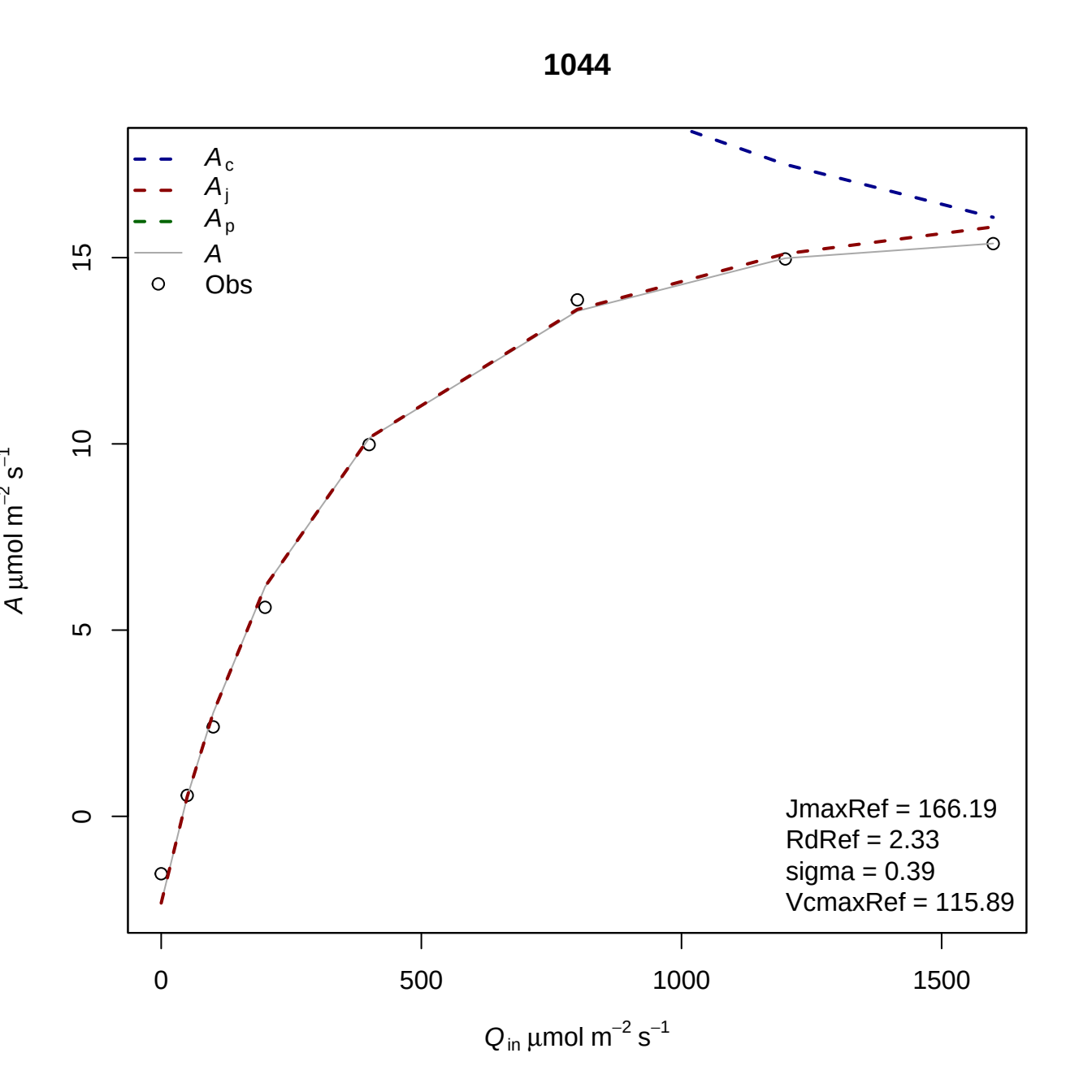

1039
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.87
RdRef = 0.94
sigma = 0.33
VcmaxRef = 65.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

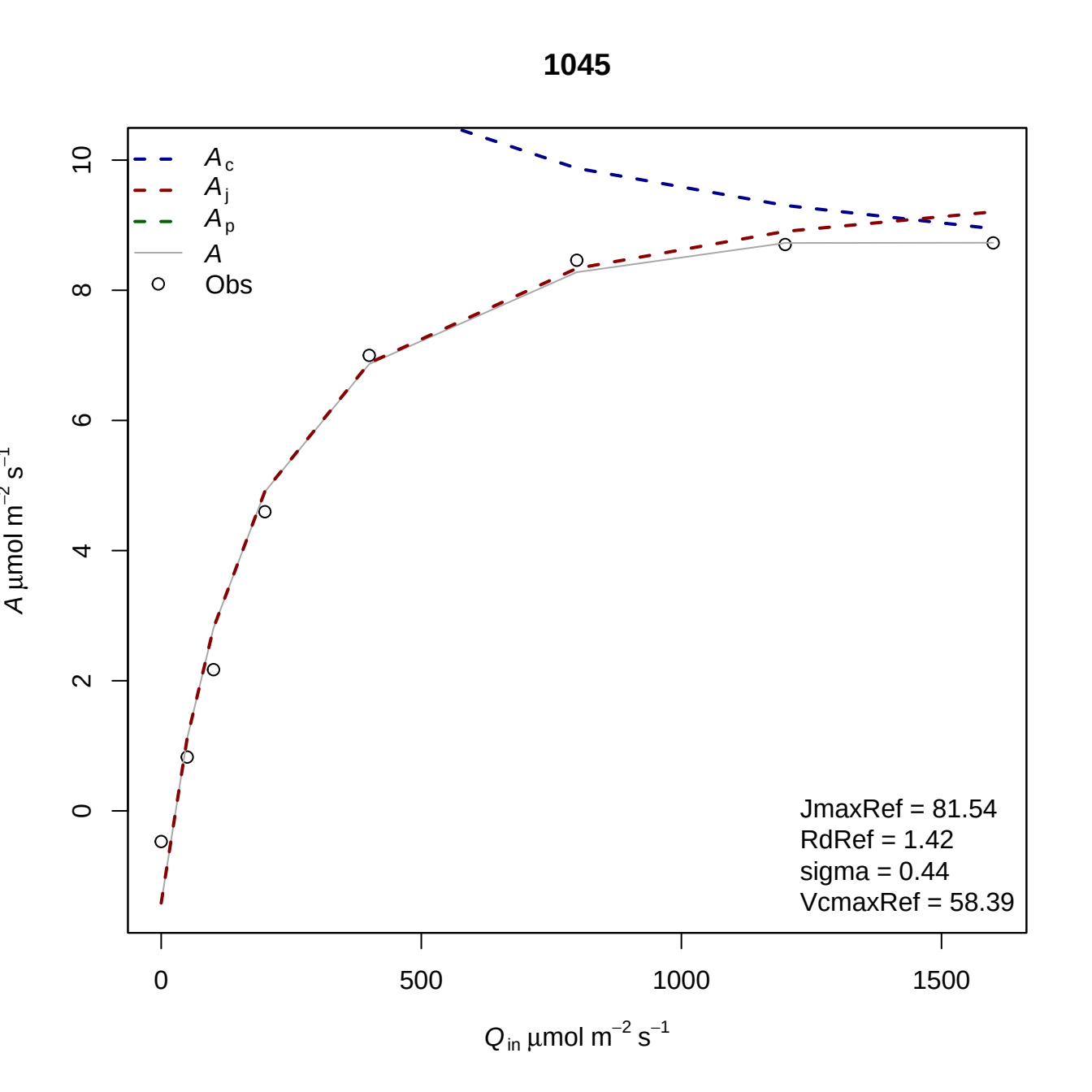

1040
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.53
RdRef = 1.08
sigma = 0.22
VcmaxRef = 62.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

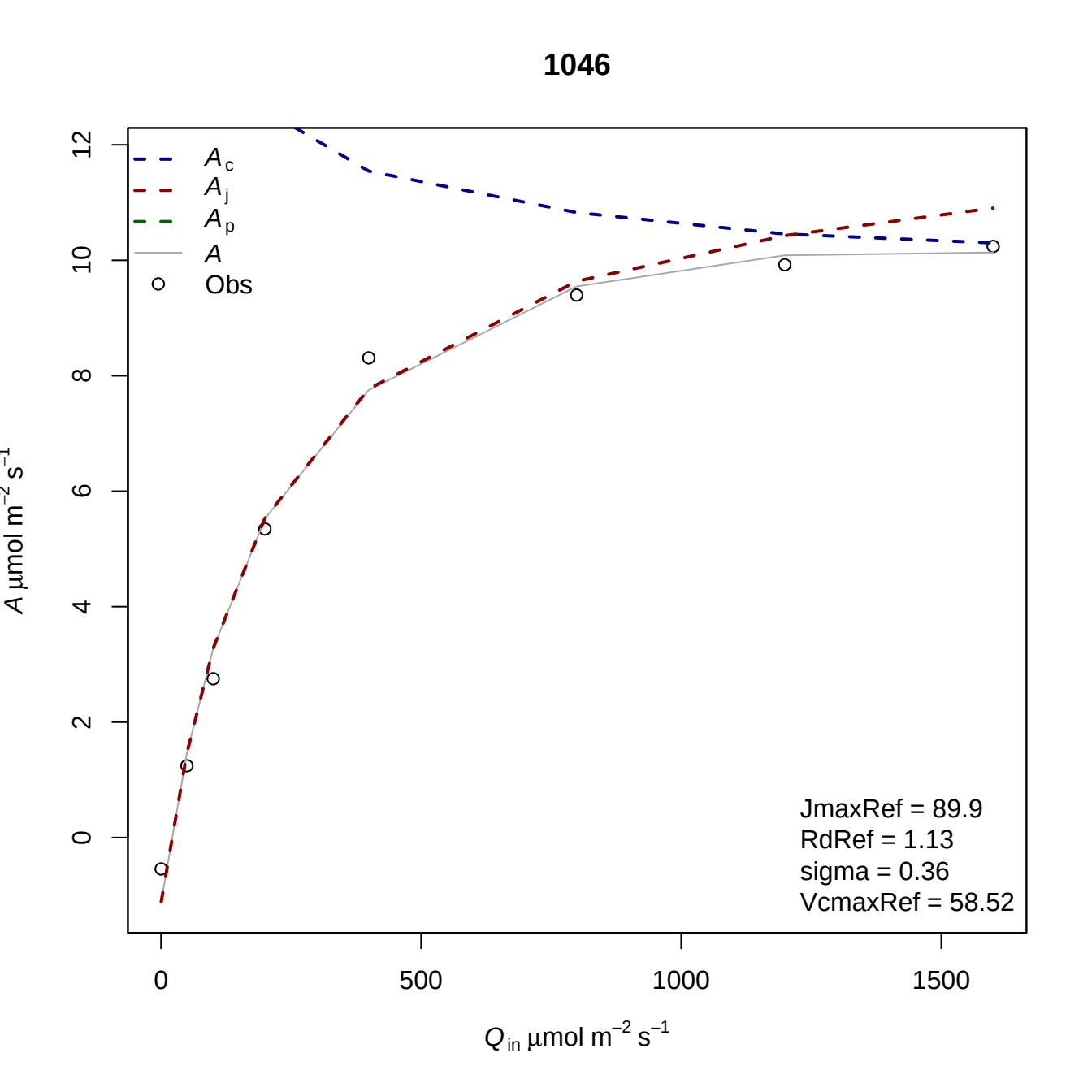

1041
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.99
RdRef = 1.04
sigma = 0.36
VcmaxRef = 90.71
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

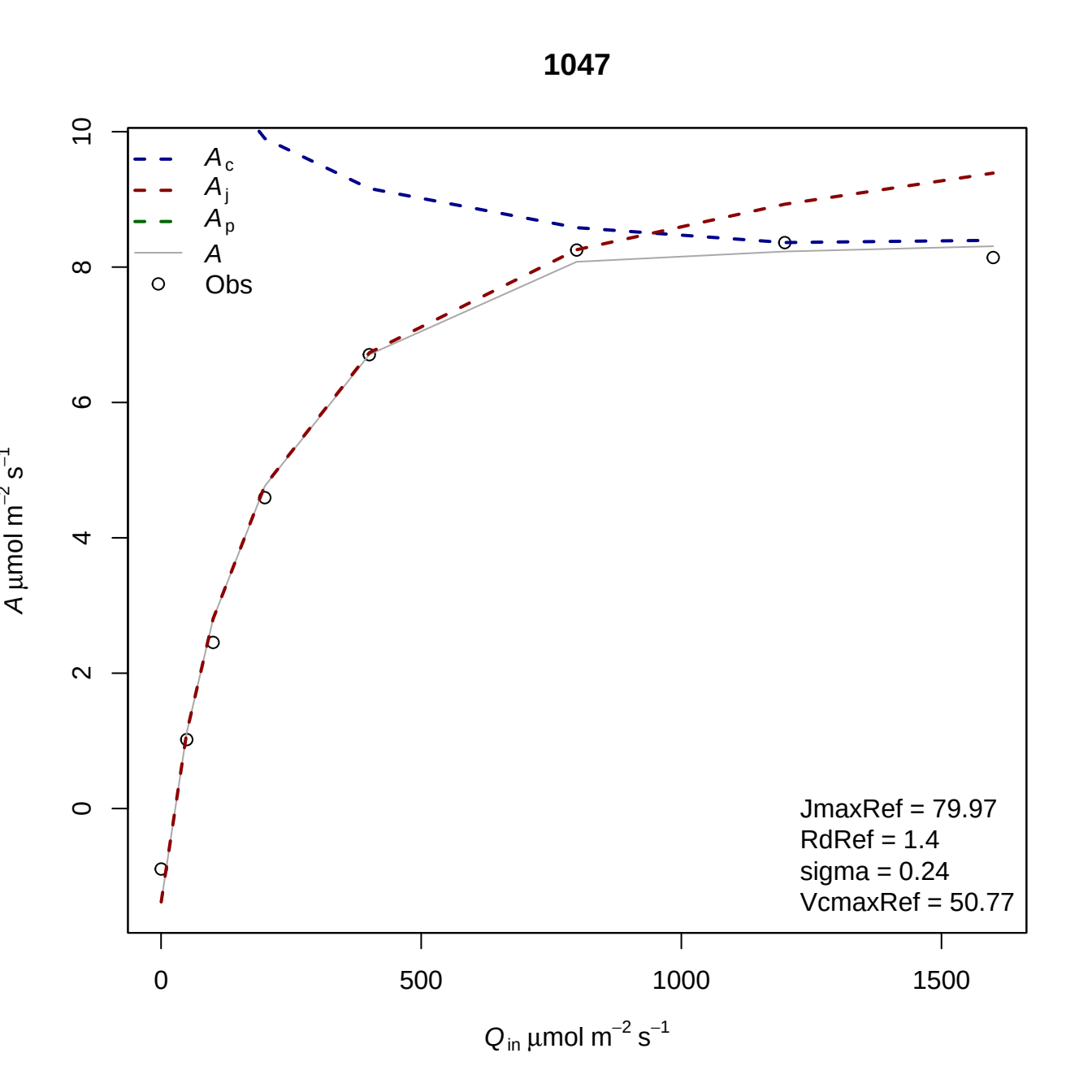

1042
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.72
RdRef = 2.1
sigma = 0.27
VcmaxRef = 85.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000
1200

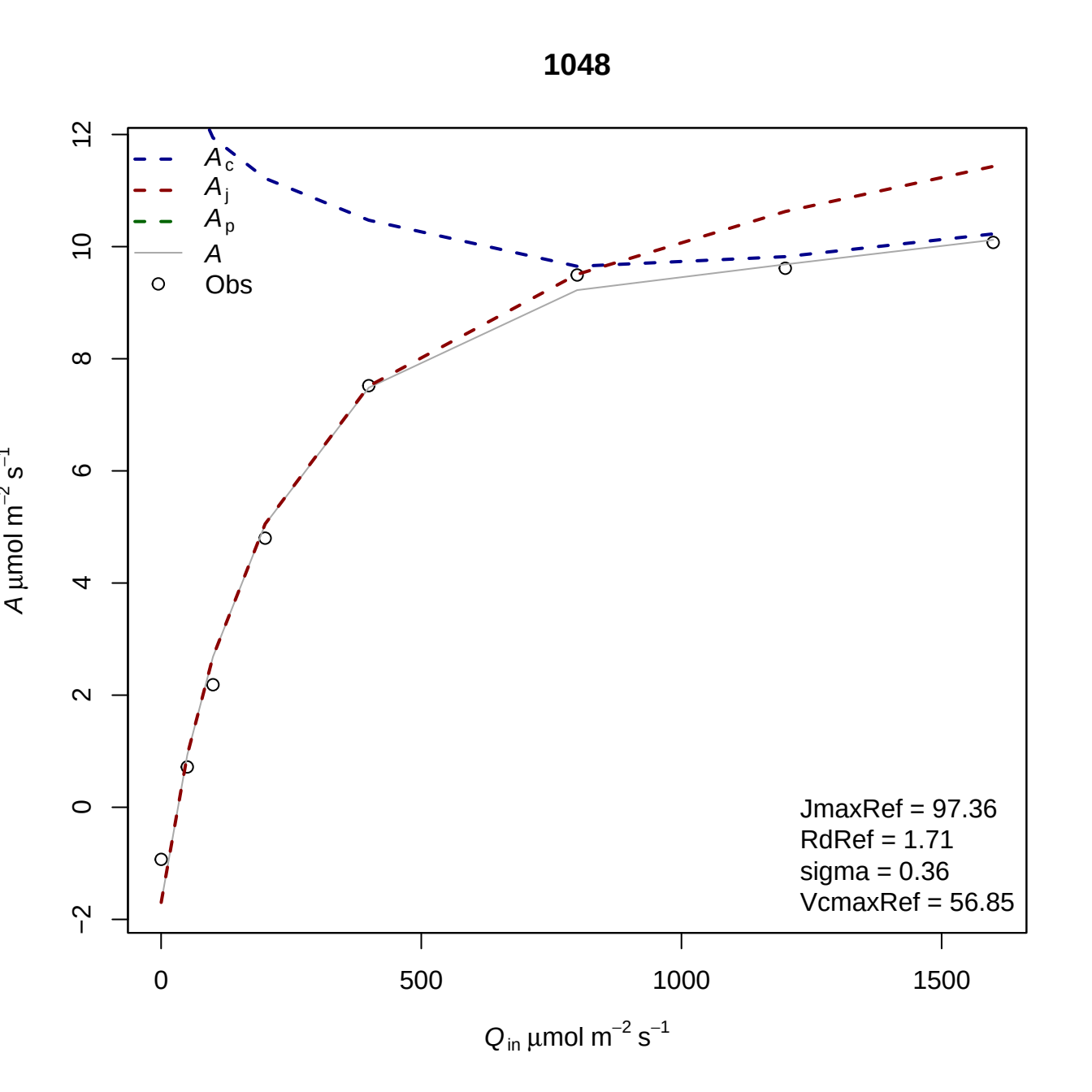

1043
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.89
RdRef = 1.77
sigma = 0.44
VcmaxRef = 89.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

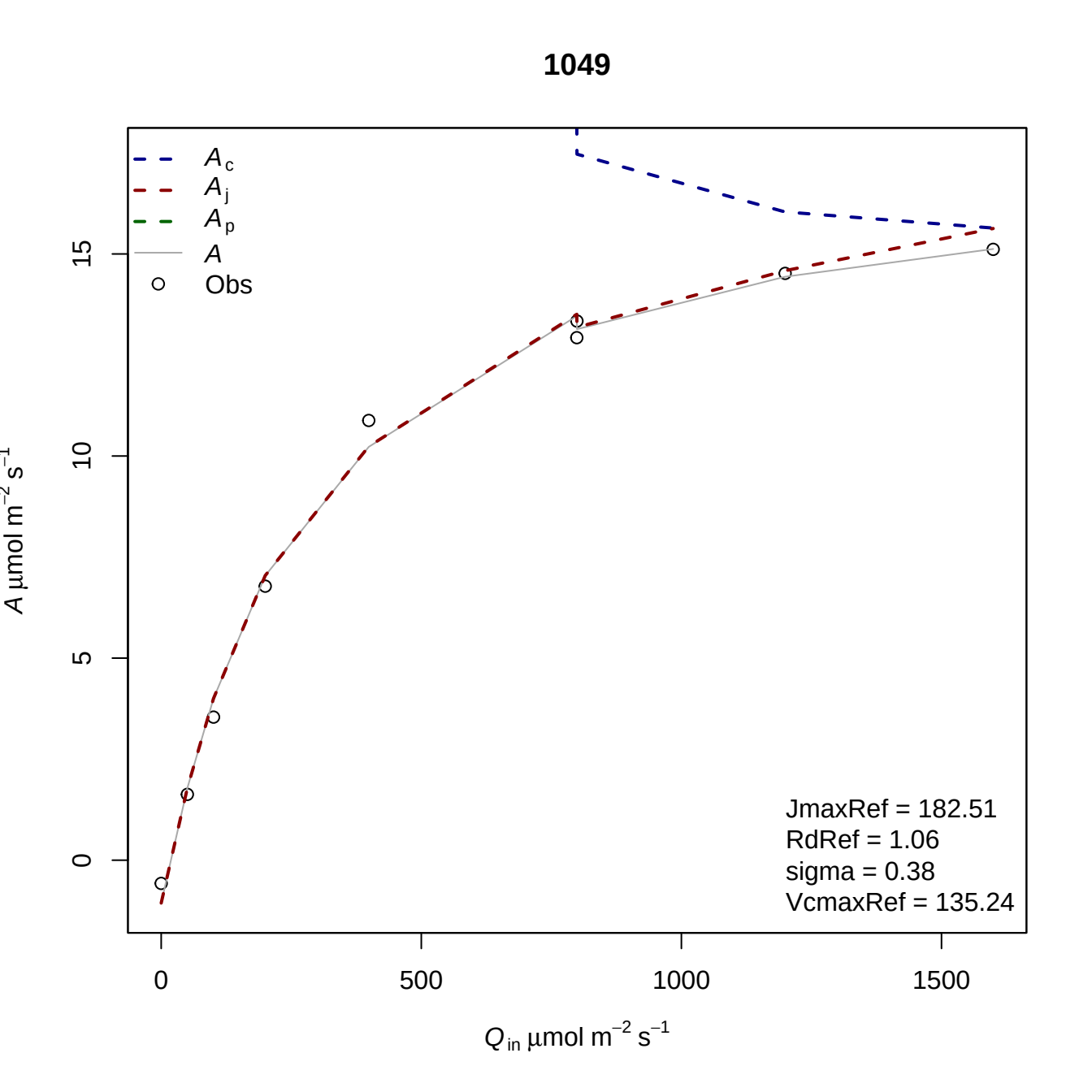

1044
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 166.19
RdRef = 2.33
sigma = 0.39
VcmaxRef = 115.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

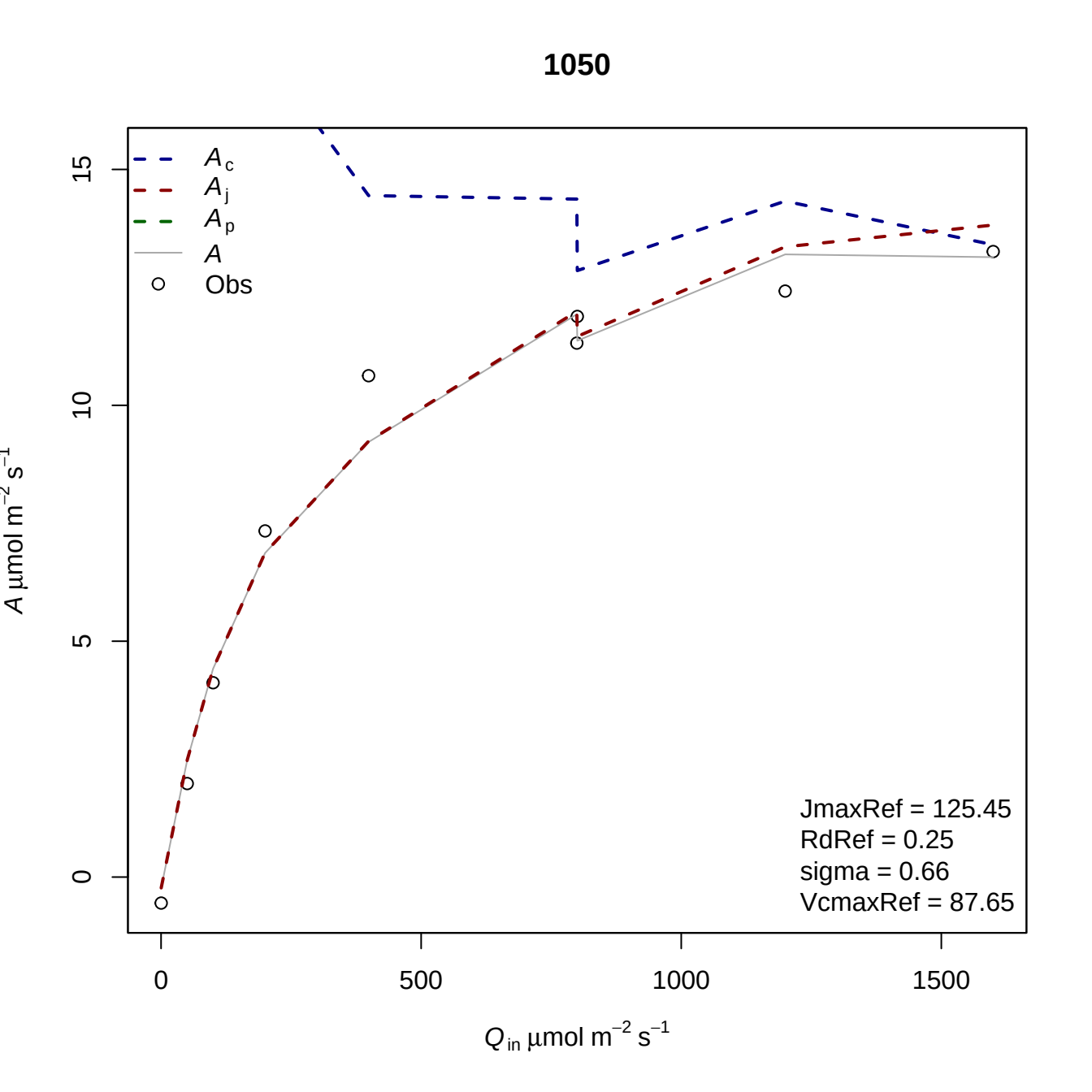

1045
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.54
RdRef = 1.42
sigma = 0.44
VcmaxRef = 58.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

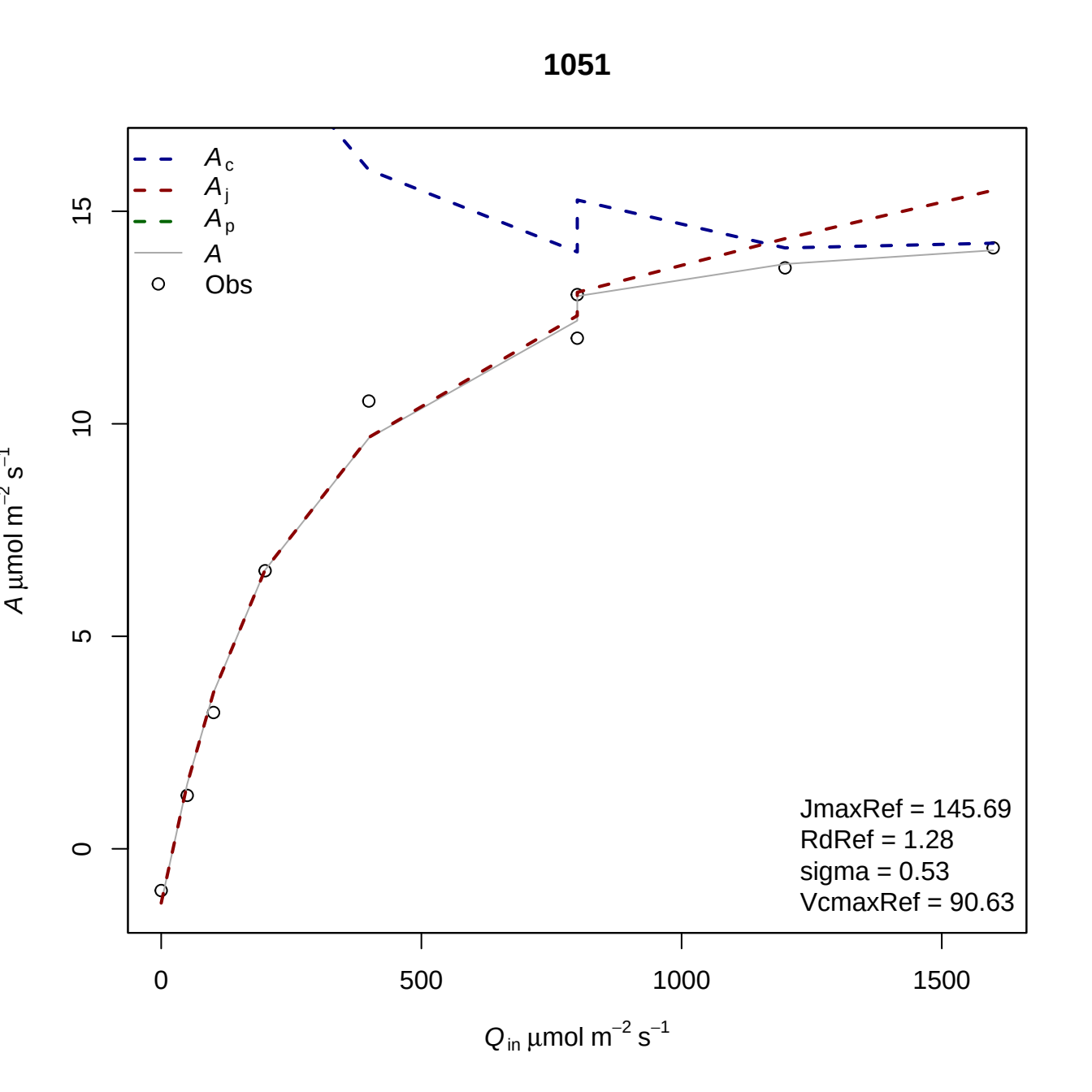

1046
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.9
RdRef = 1.13
sigma = 0.36
VcmaxRef = 58.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

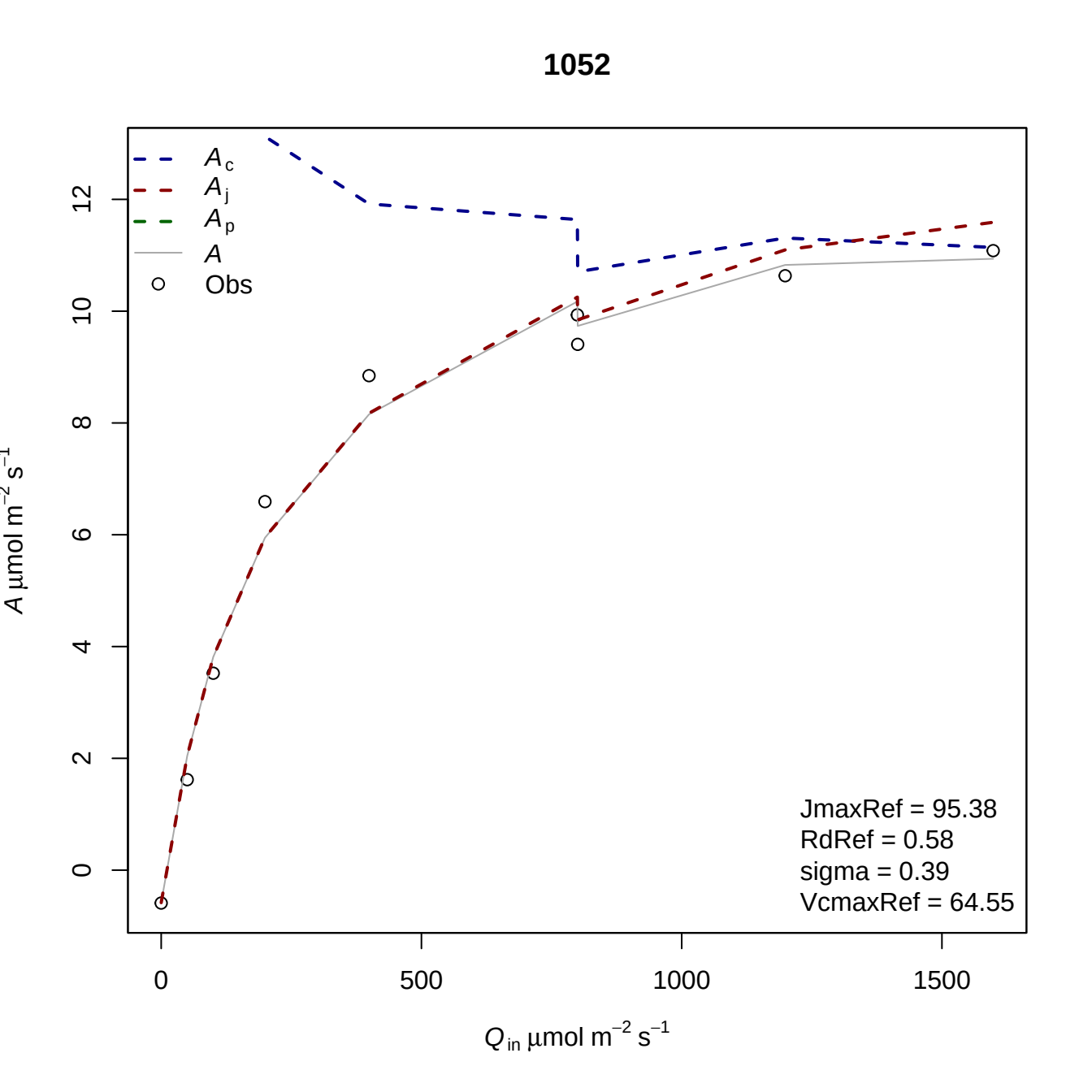

1047
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.97
RdRef = 1.4
sigma = 0.24
VcmaxRef = 50.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

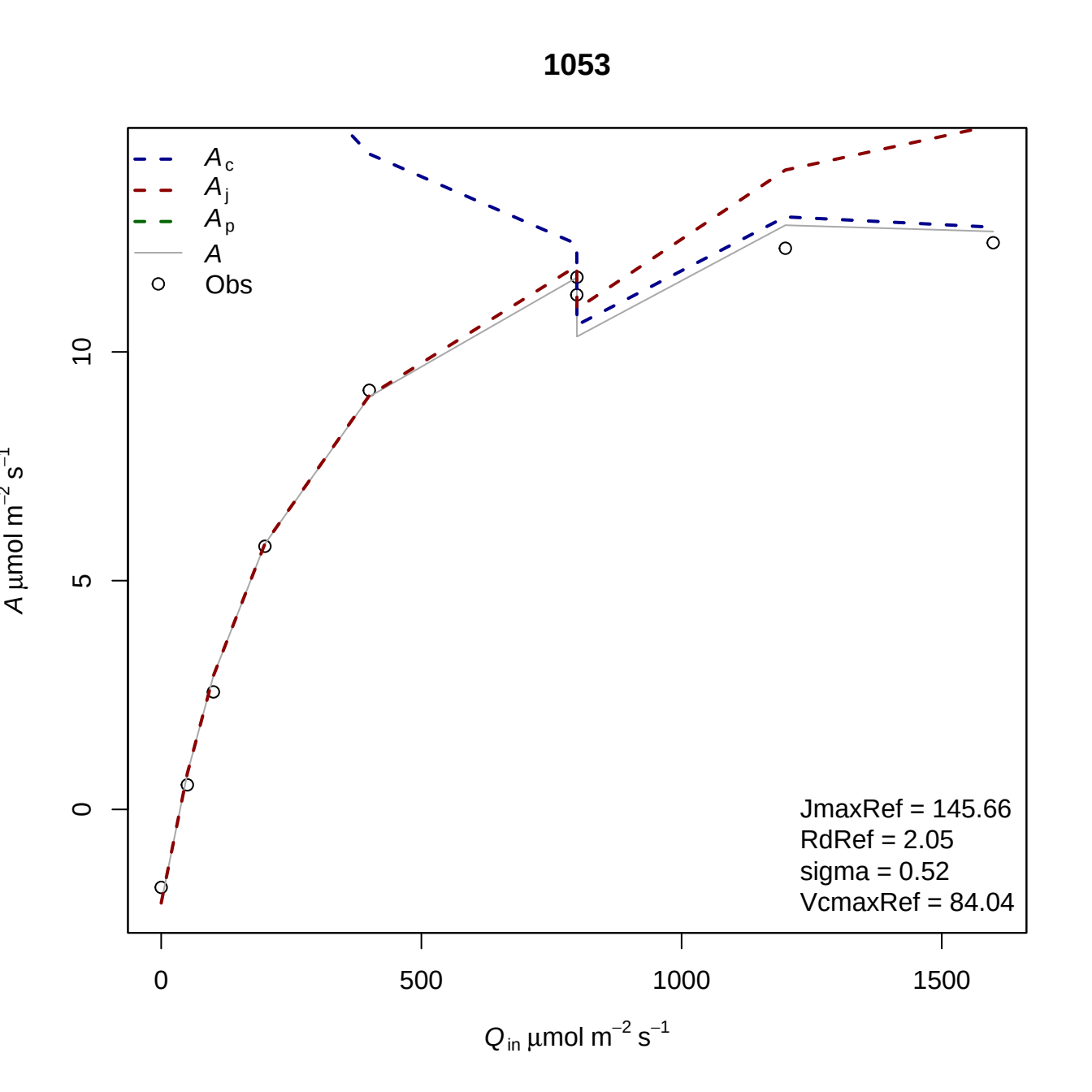

1048
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.36
RdRef = 1.71
sigma = 0.36
VcmaxRef = 56.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

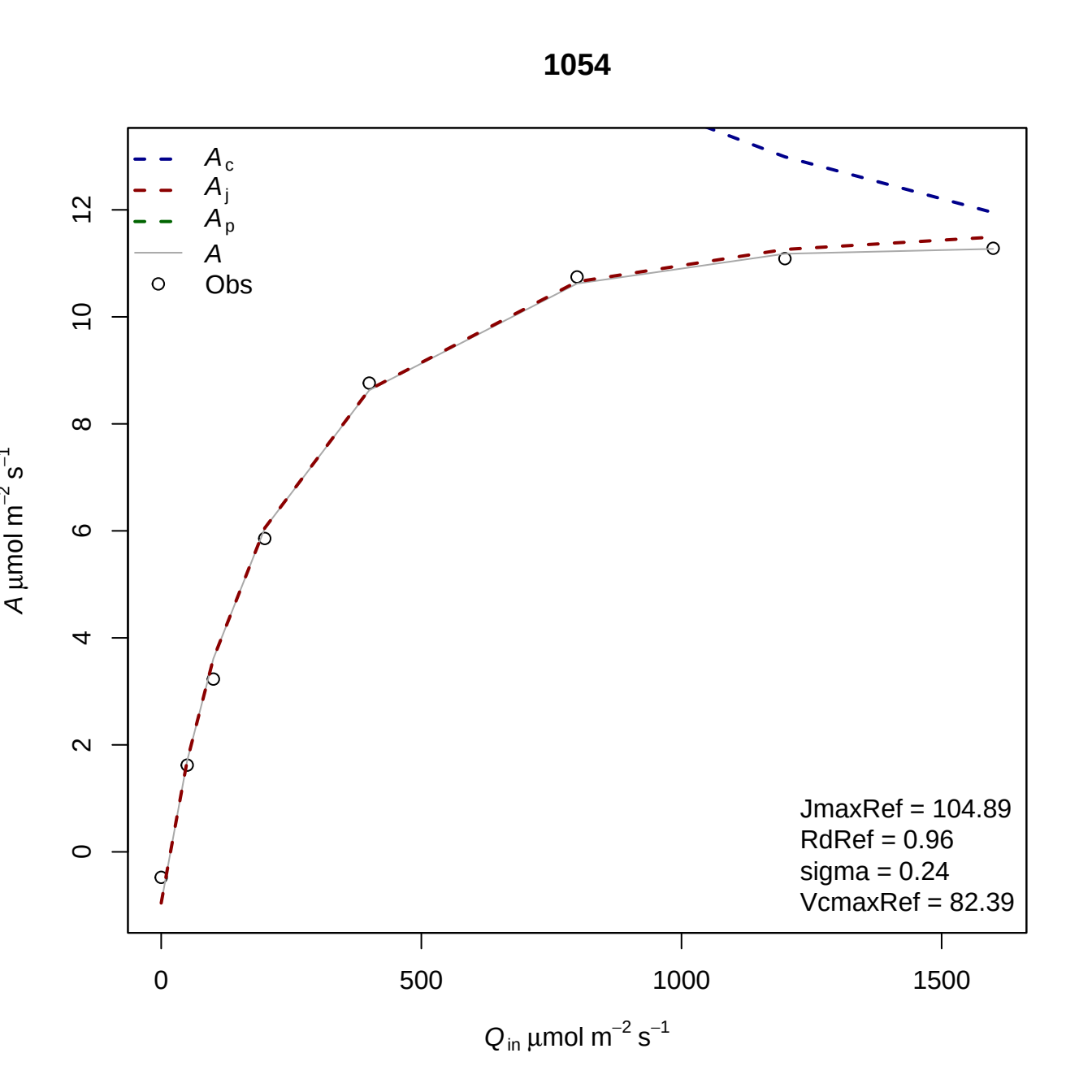

1049
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 182.51
RdRef = 1.06
sigma = 0.38
VcmaxRef = 135.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

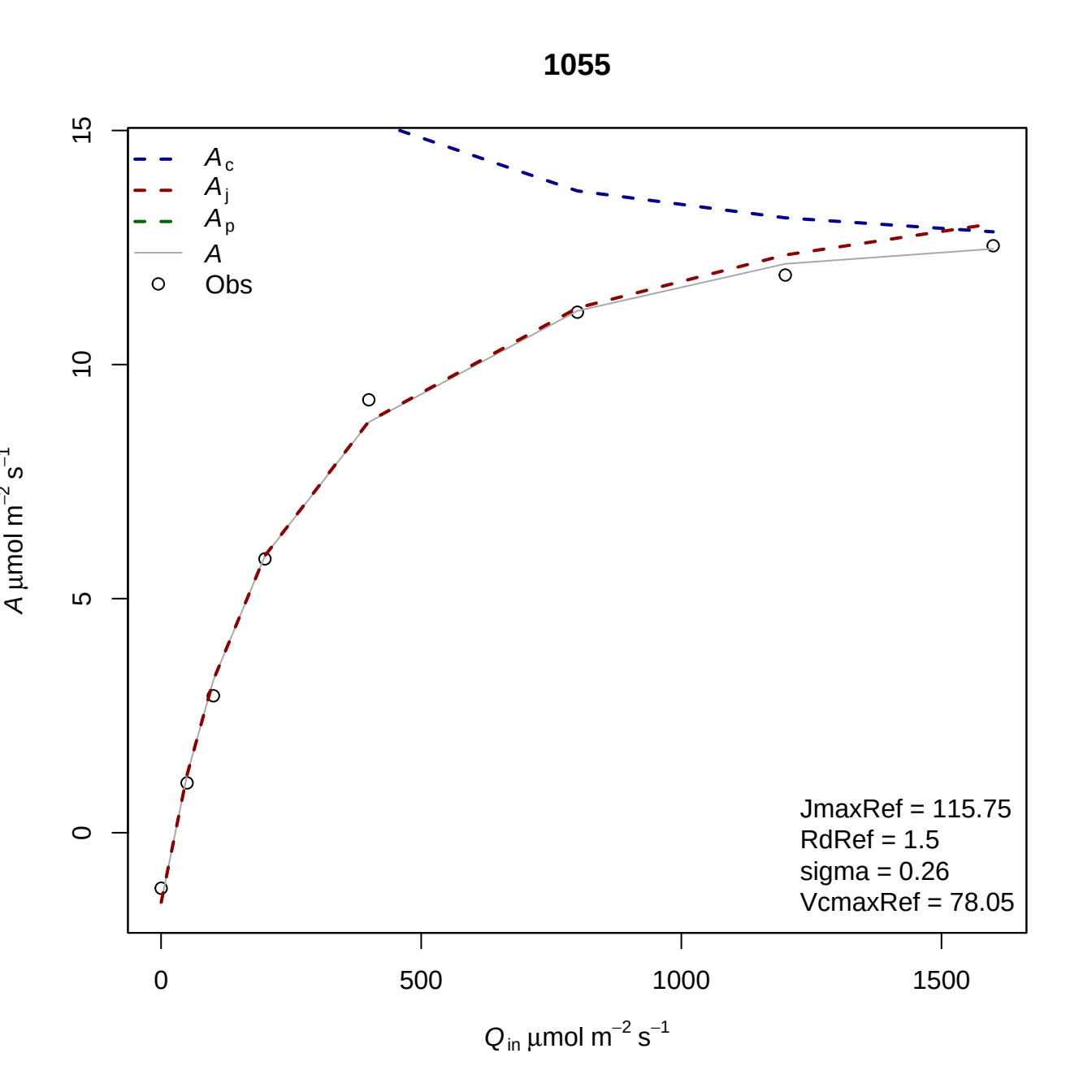

1050
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.45
RdRef = 0.25
sigma = 0.66
VcmaxRef = 87.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

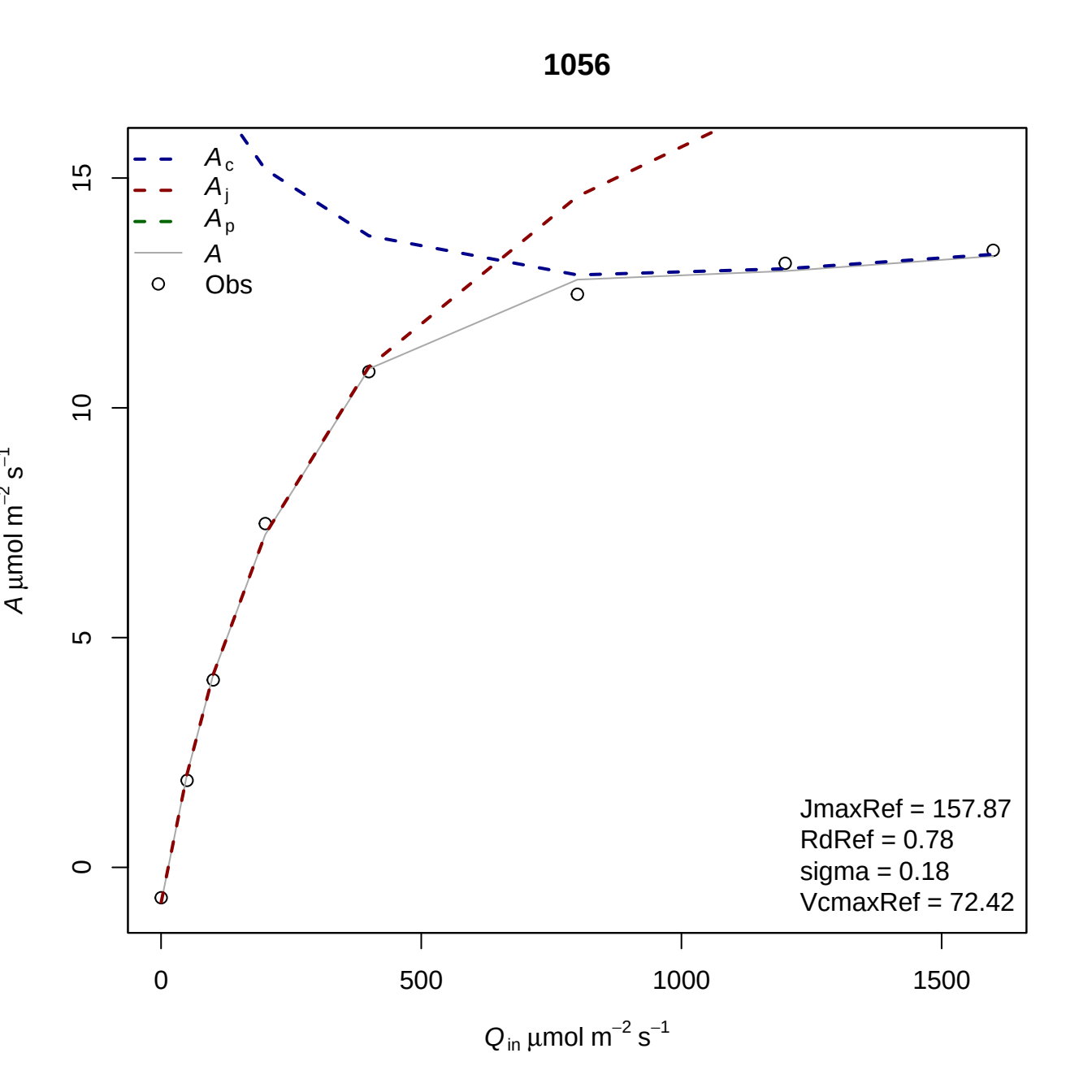

1051
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.69
RdRef = 1.28
sigma = 0.53
VcmaxRef = 90.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

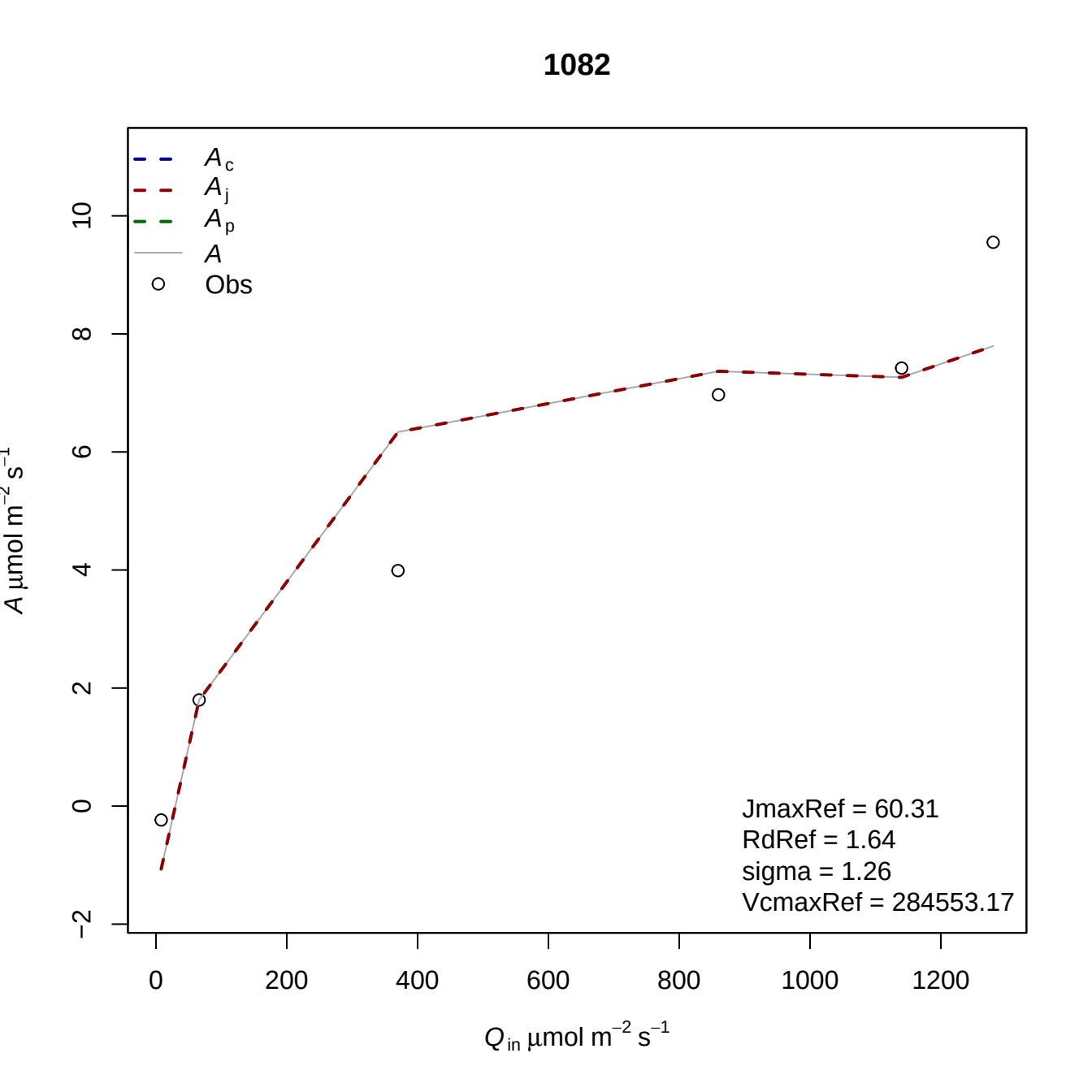

1052
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.38
RdRef = 0.58
sigma = 0.39
VcmaxRef = 64.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

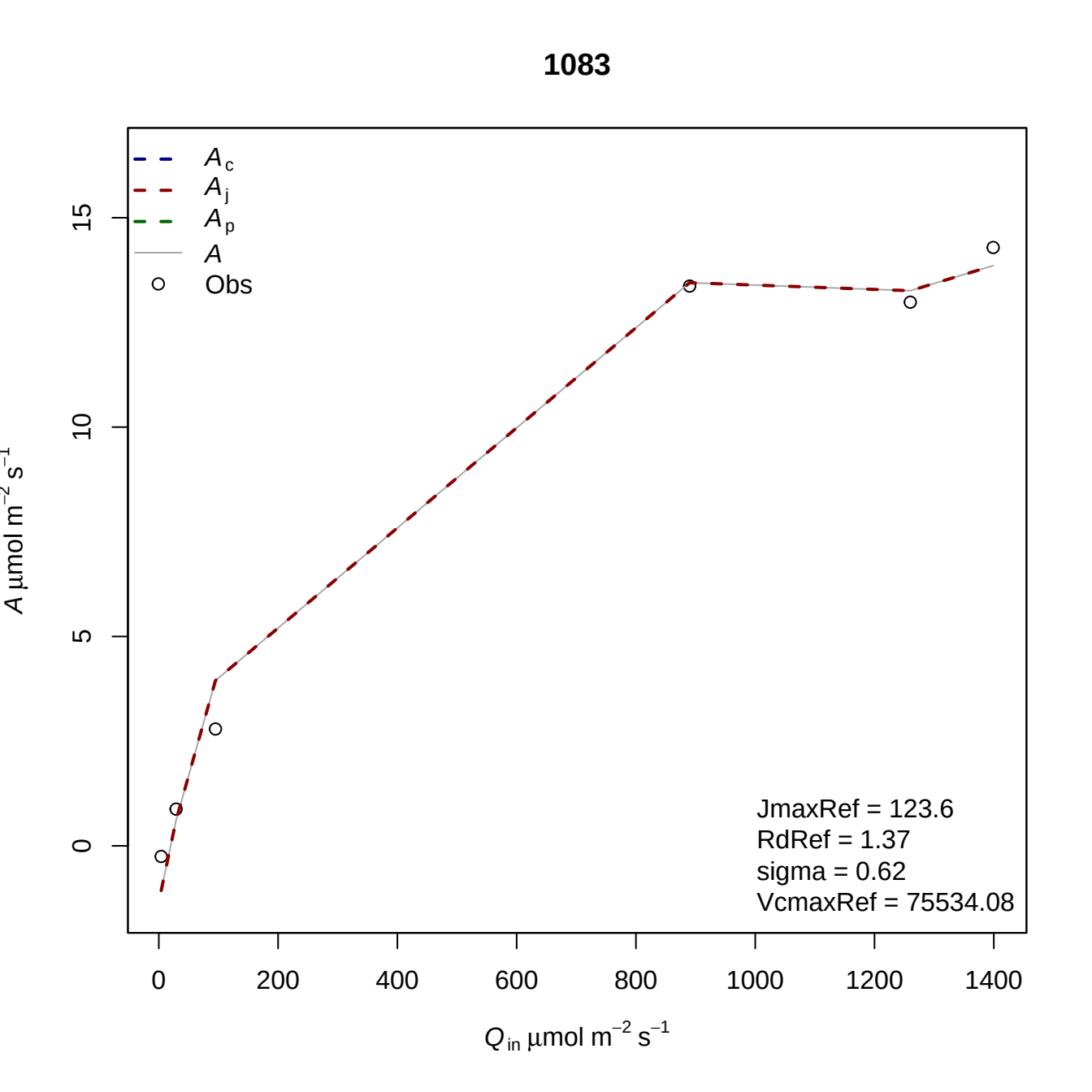

1053
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.66
RdRef = 2.05
sigma = 0.52
VcmaxRef = 84.04
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

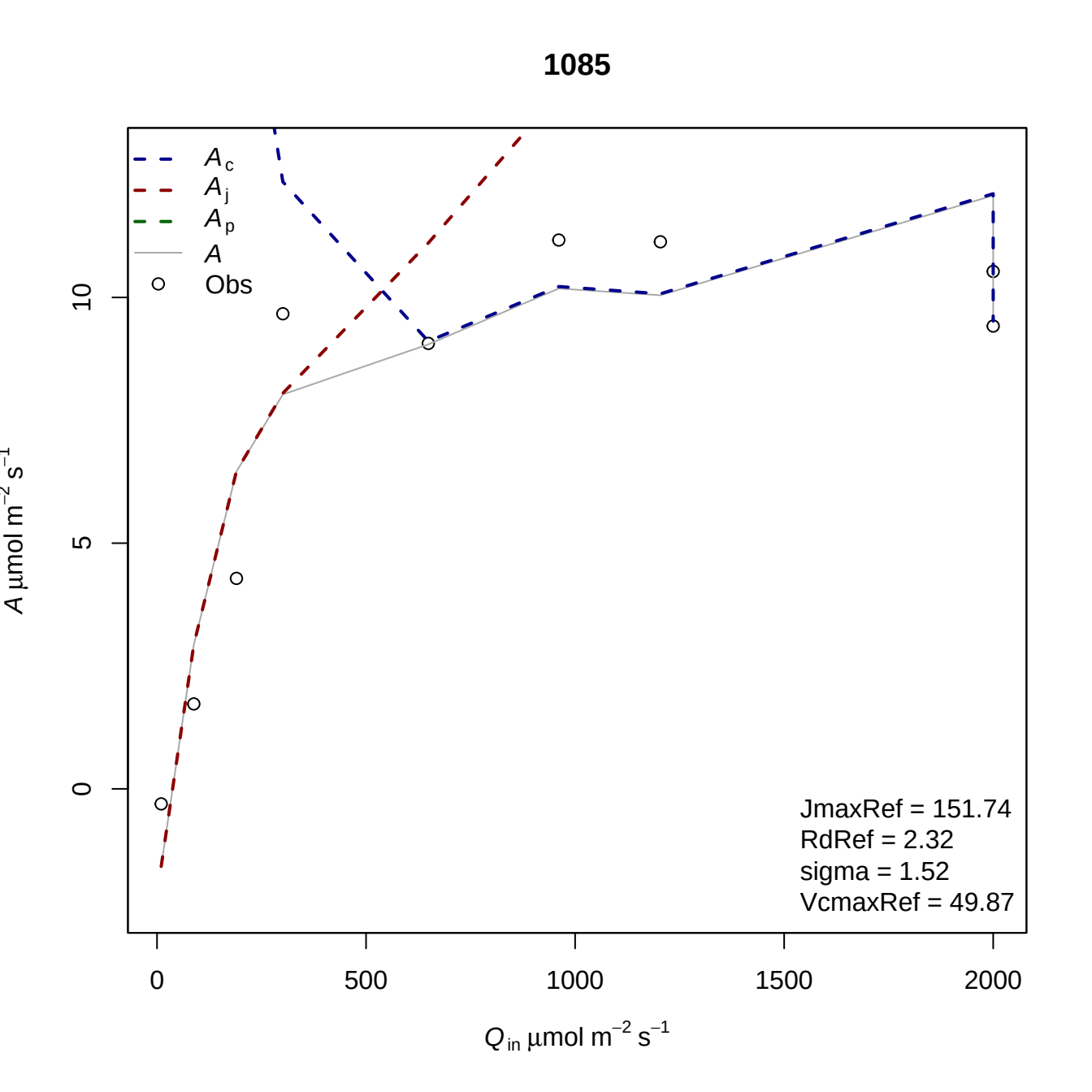

1054
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.89
RdRef = 0.96
sigma = 0.24
VcmaxRef = 82.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

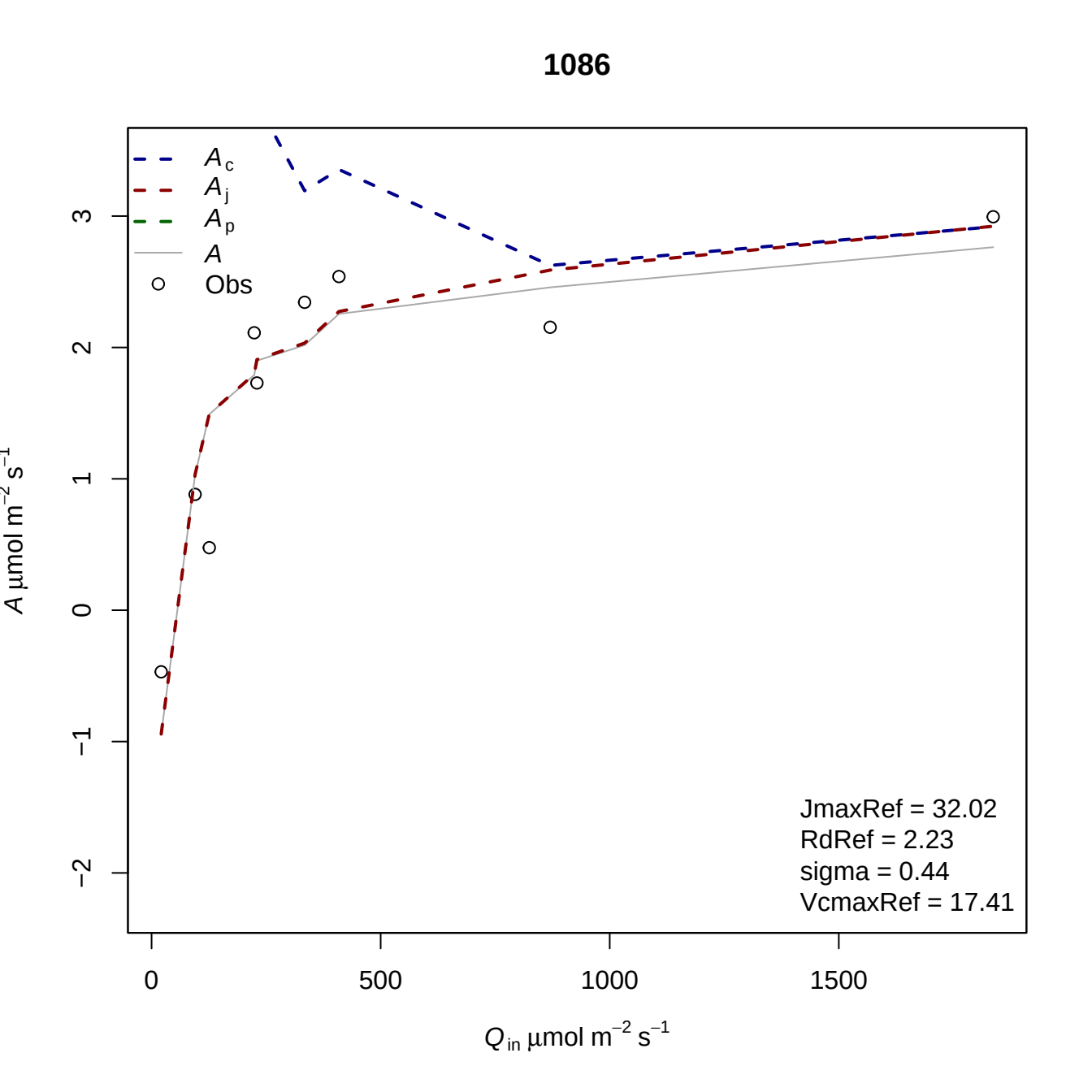

1055
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 115.75
RdRef = 1.5
sigma = 0.26
VcmaxRef = 78.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

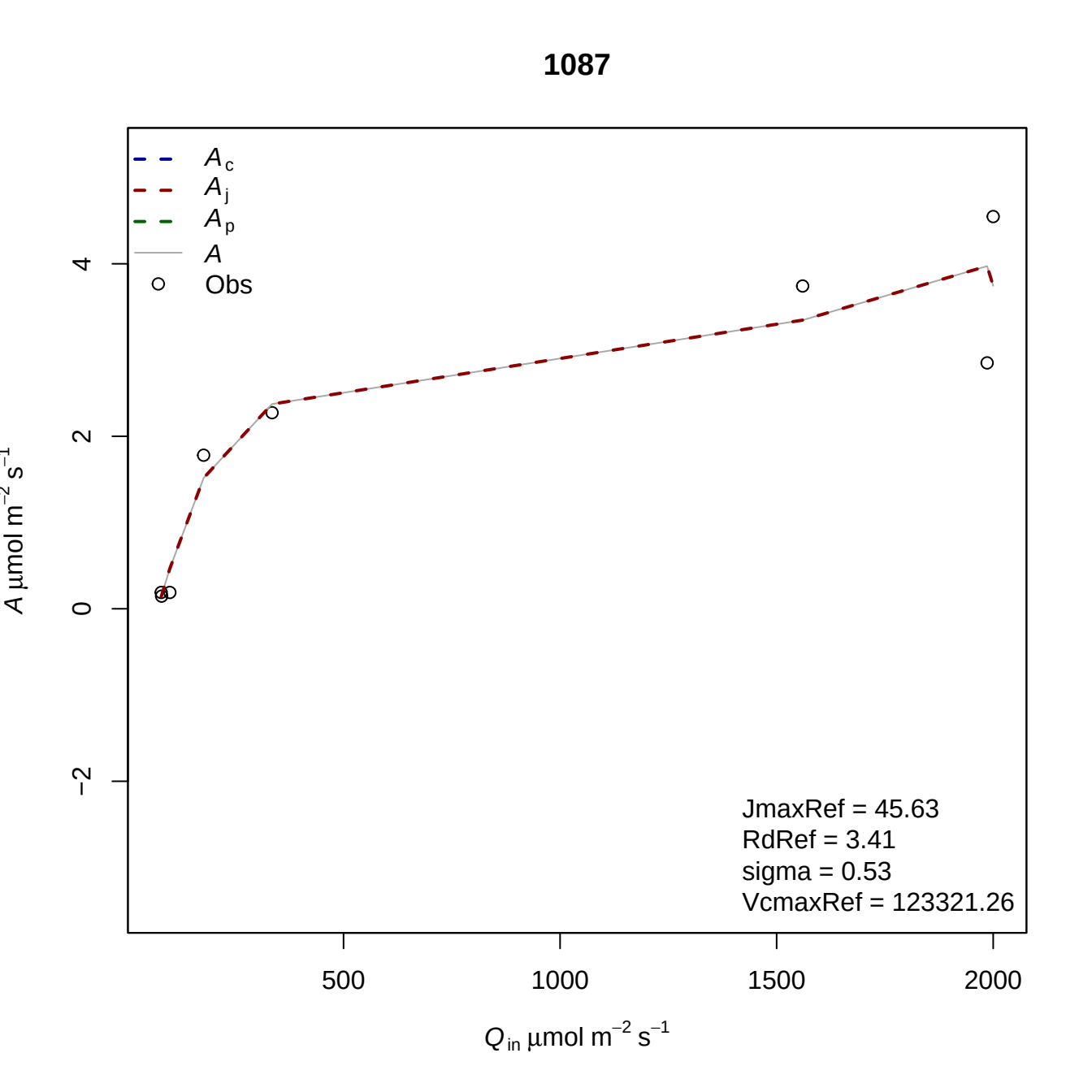

1056
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 157.87
RdRef = 0.78
sigma = 0.18
VcmaxRef = 72.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

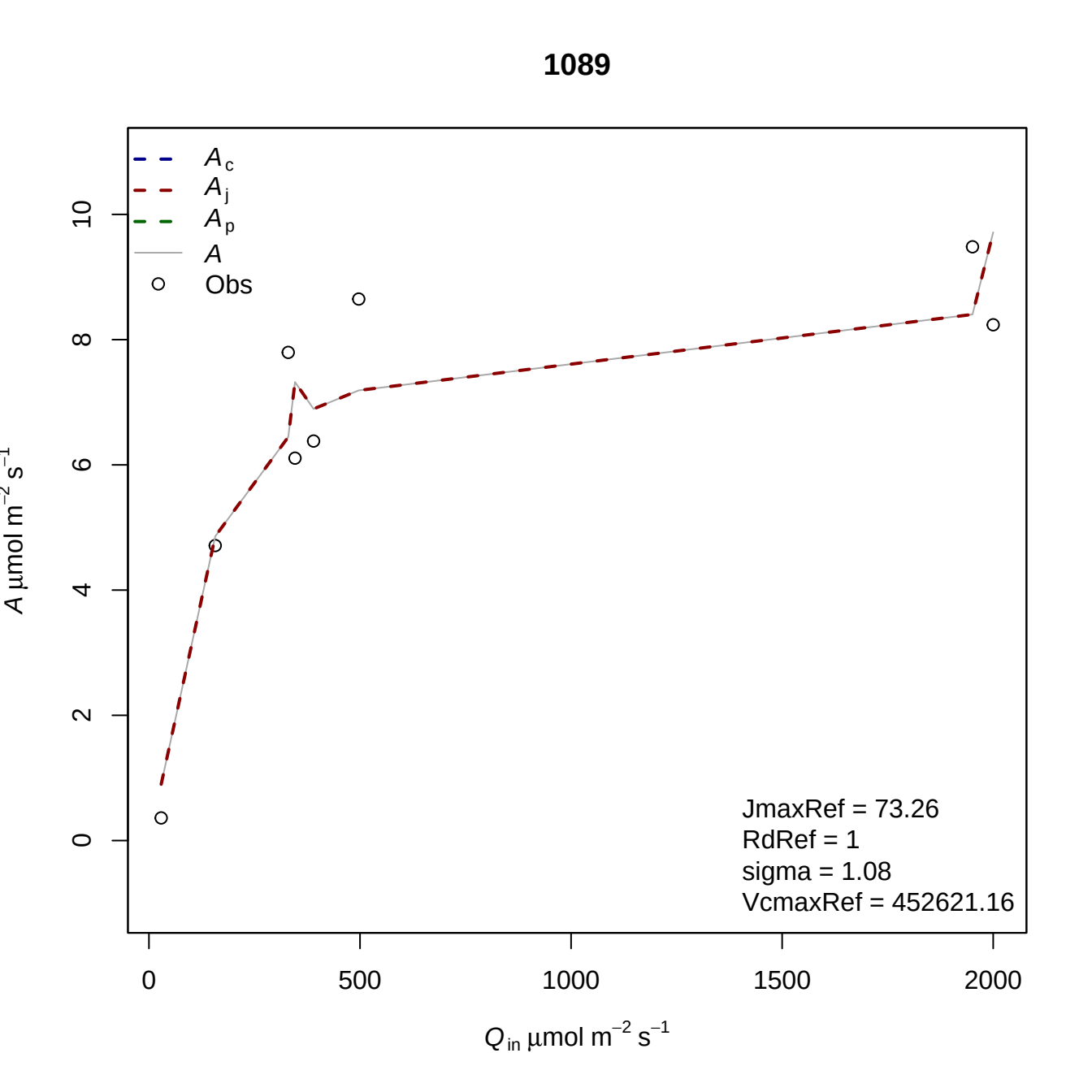

1082
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.31
RdRef = 1.64
sigma = 1.26
VcmaxRef = 284553.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

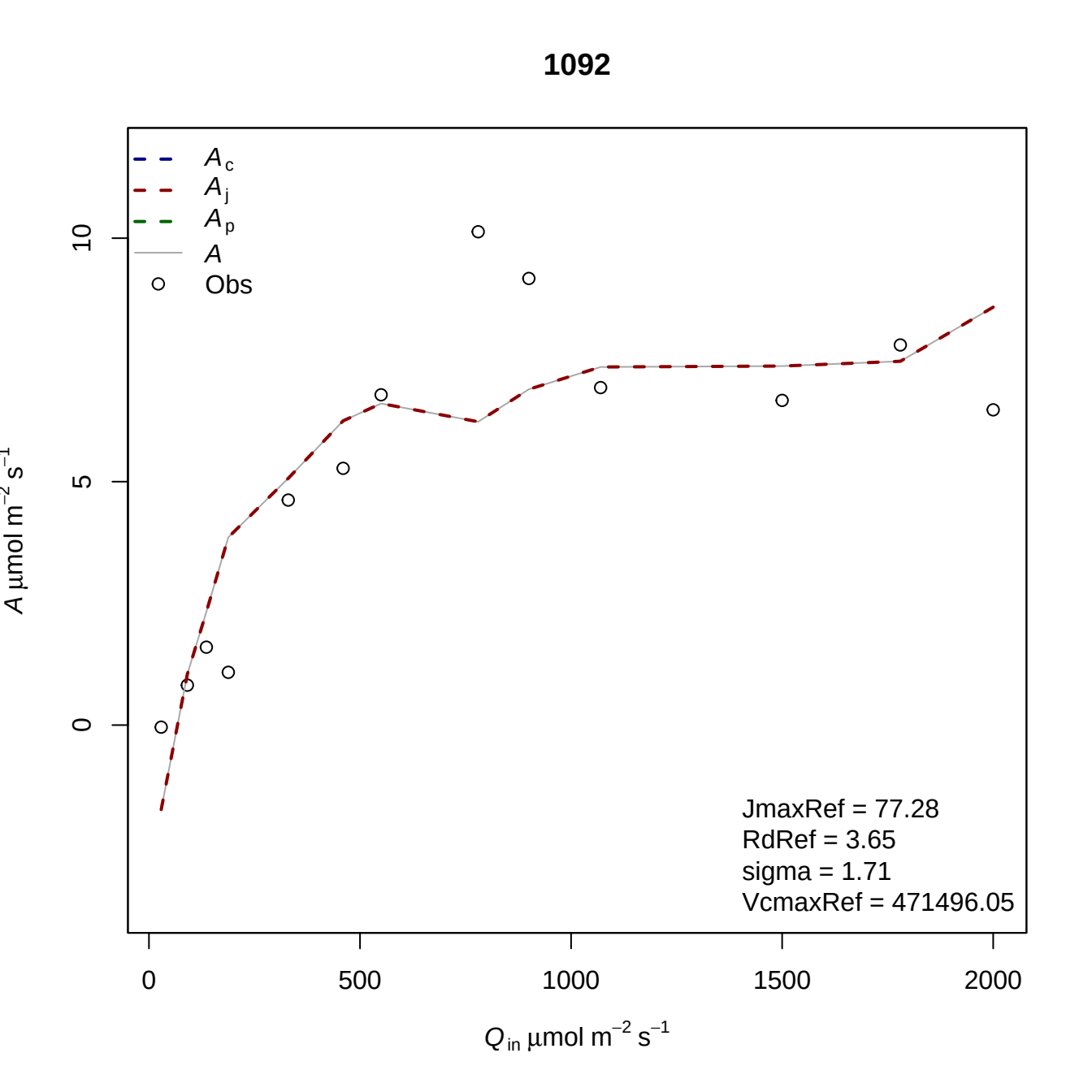

1083
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.6
RdRef = 1.37
sigma = 0.62
VcmaxRef = 75534.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000
1200
1400

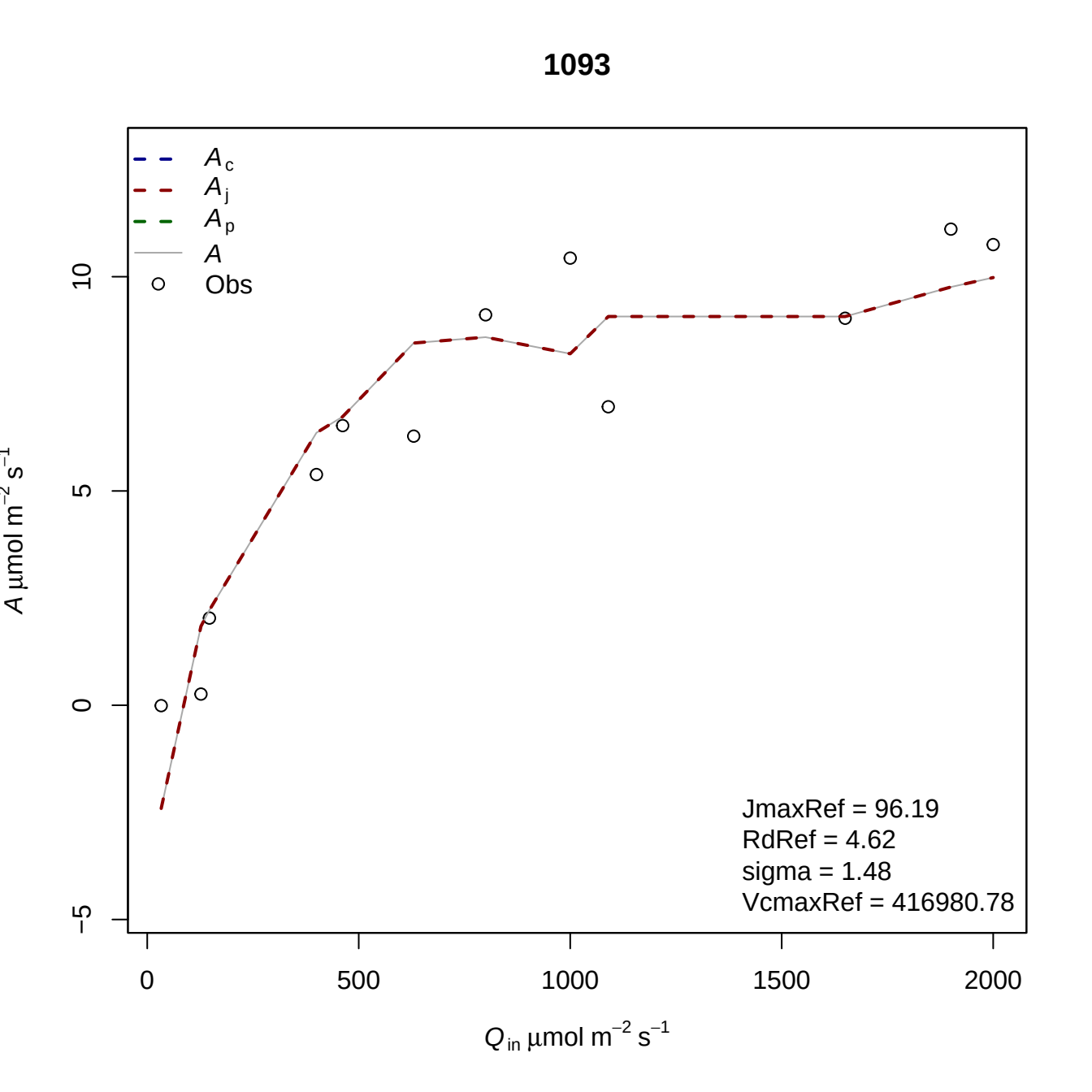

1085
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 151.74
RdRef = 2.32
sigma = 1.52
VcmaxRef = 49.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

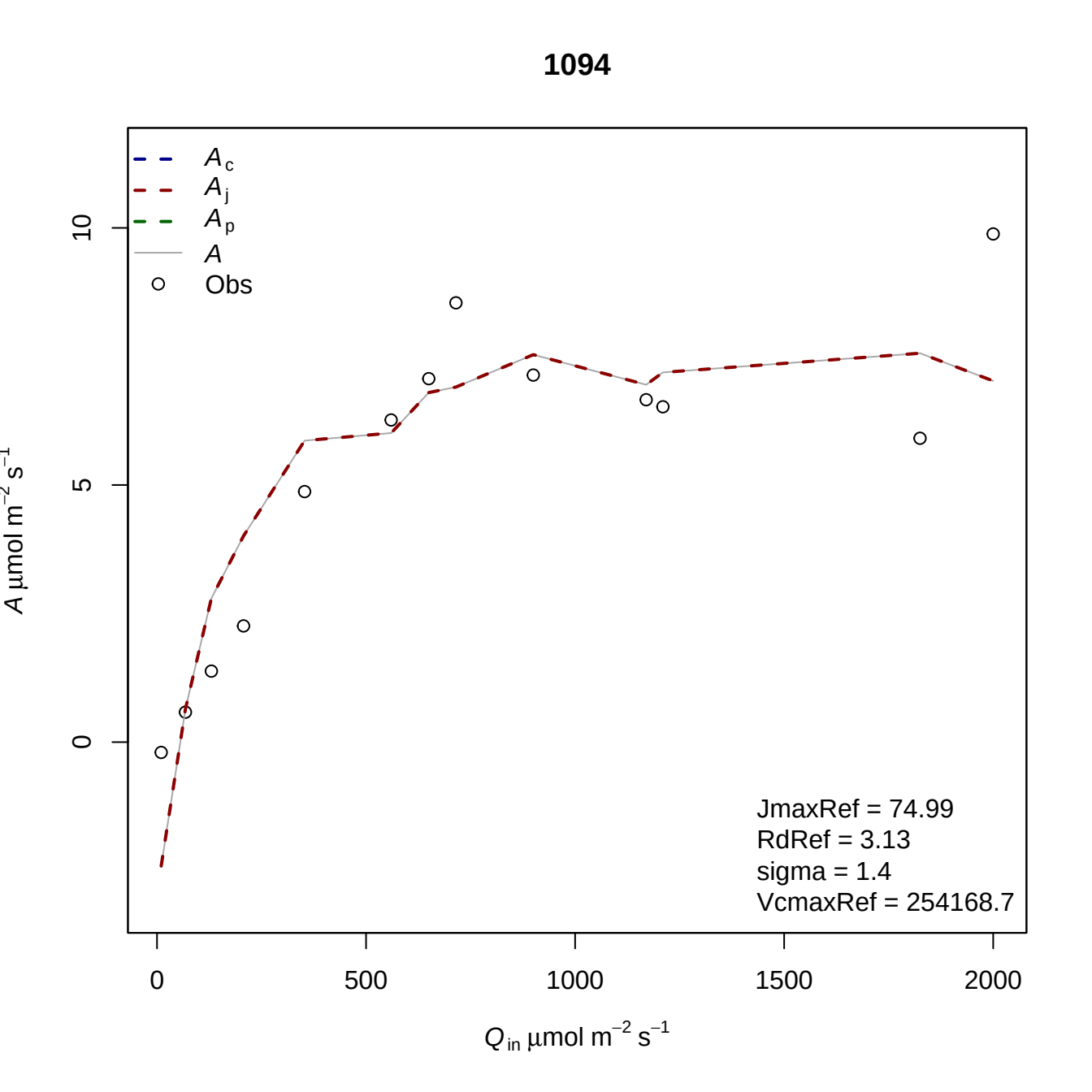

1086
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 32.02
RdRef = 2.23
sigma = 0.44
VcmaxRef = 17.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
−2
0
500
1000
1500

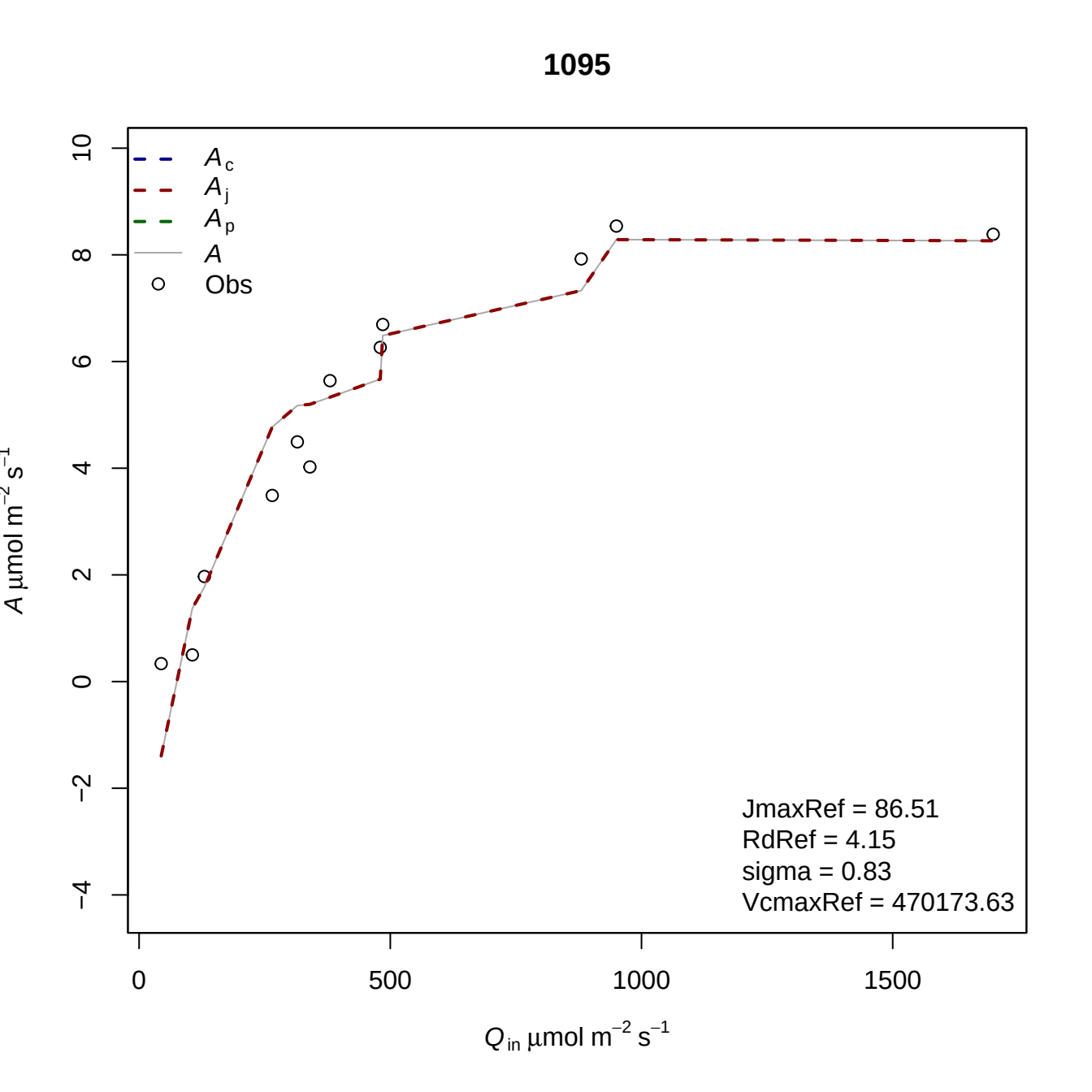

1087
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.63
RdRef = 3.41
sigma = 0.53
VcmaxRef = 123321.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
500
1000
1500
2000

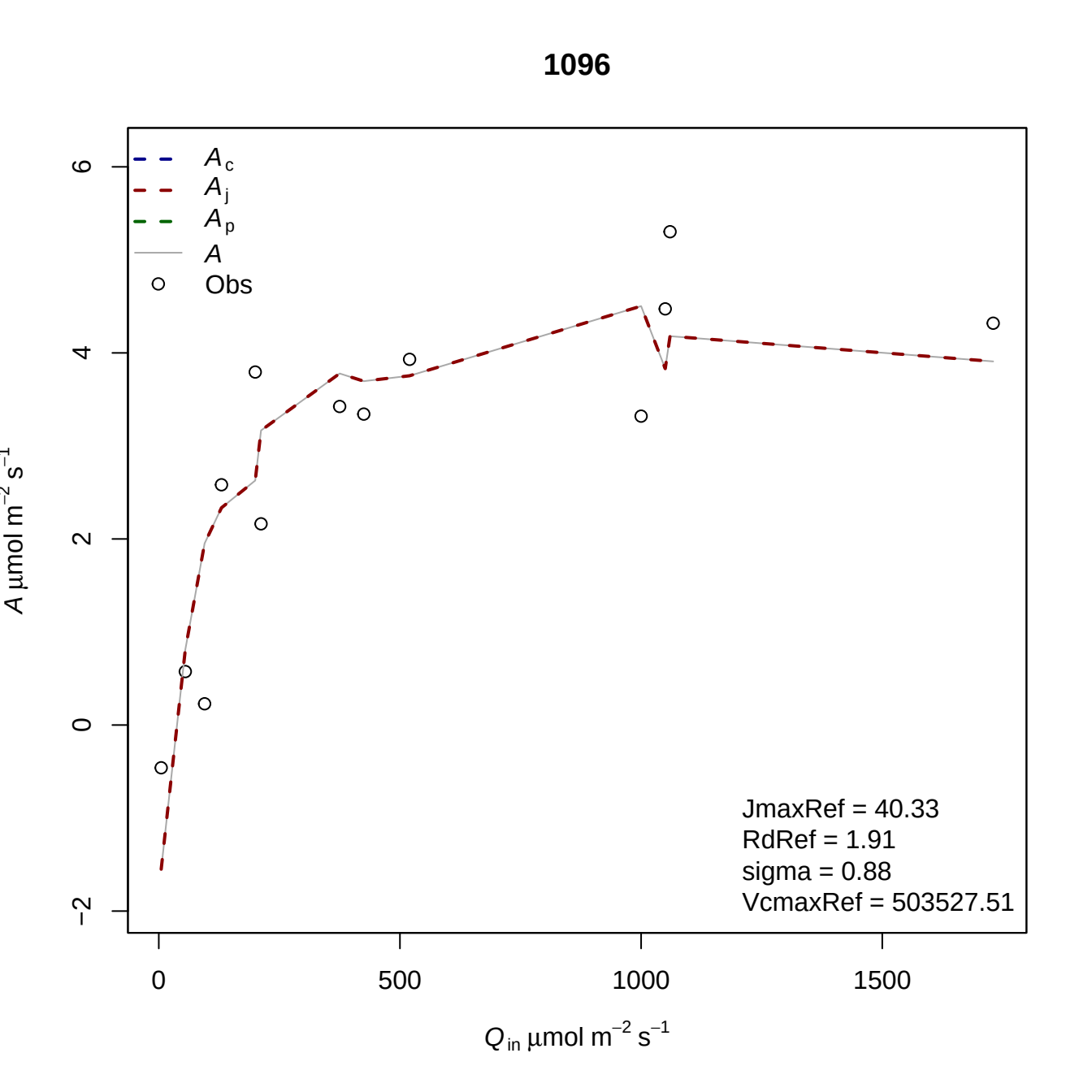

1089
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 73.26
RdRef = 1
sigma = 1.08
VcmaxRef = 452621.16
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

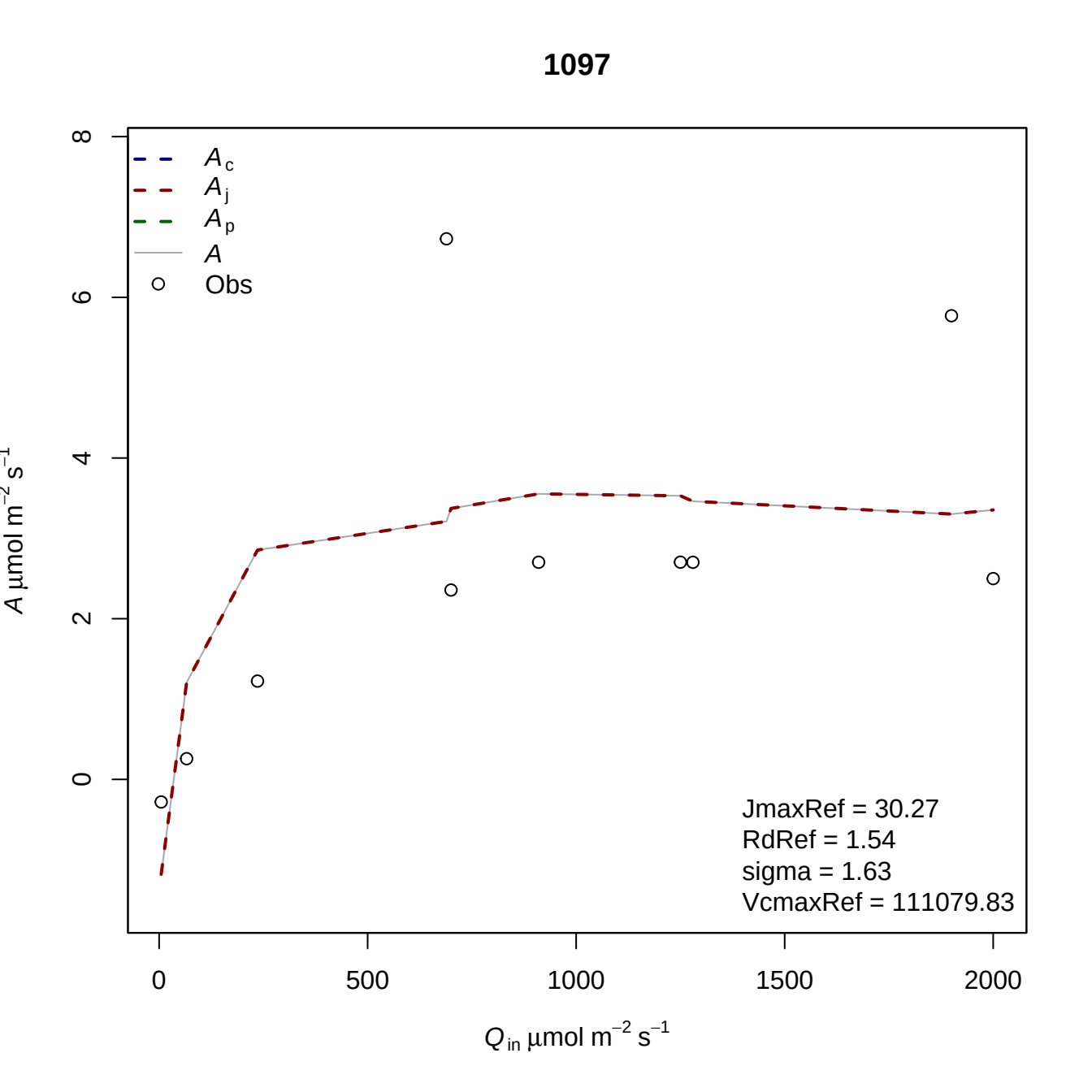

1092
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.28
RdRef = 3.65
sigma = 1.71
VcmaxRef = 471496.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

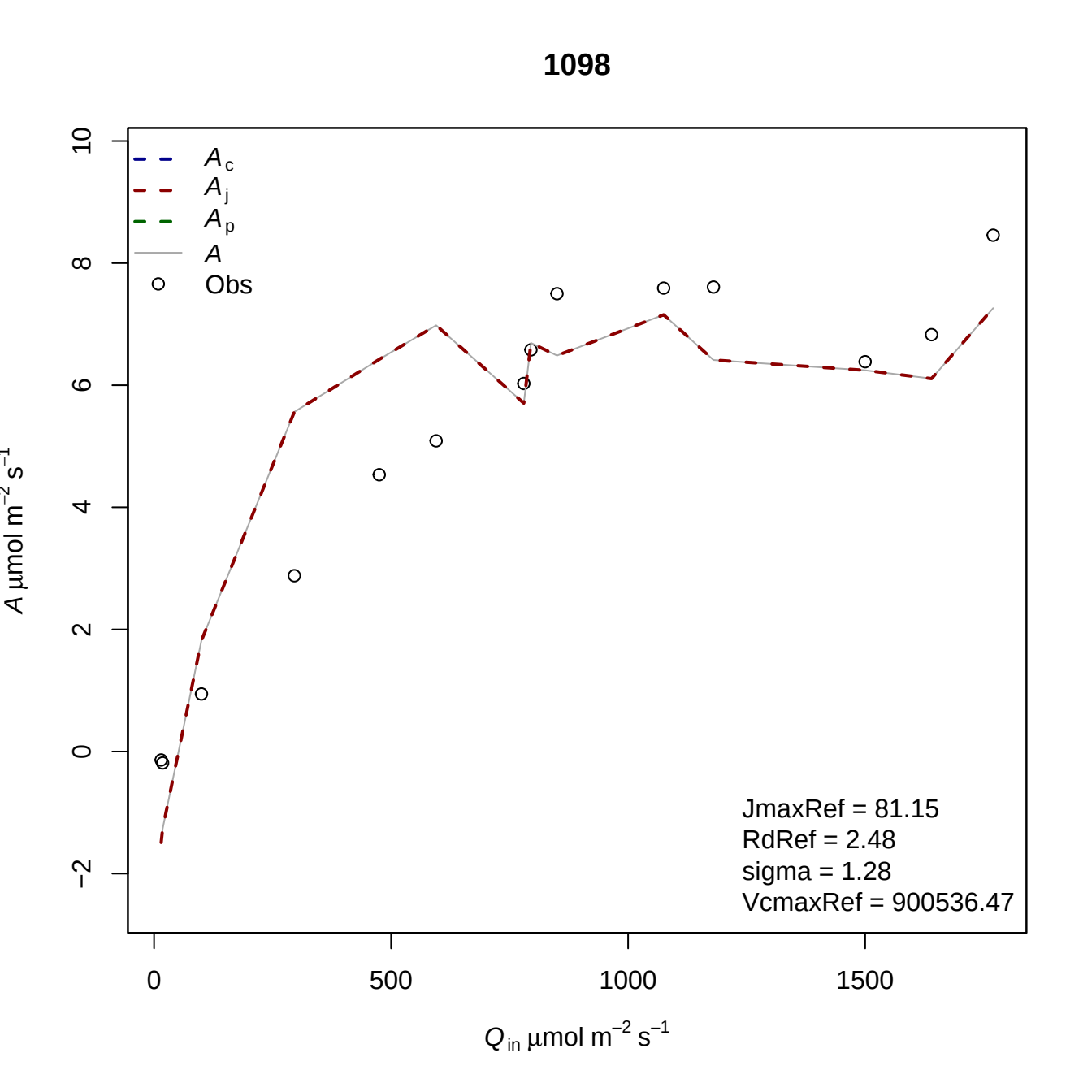

1093
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.19
RdRef = 4.62
sigma = 1.48
VcmaxRef = 416980.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

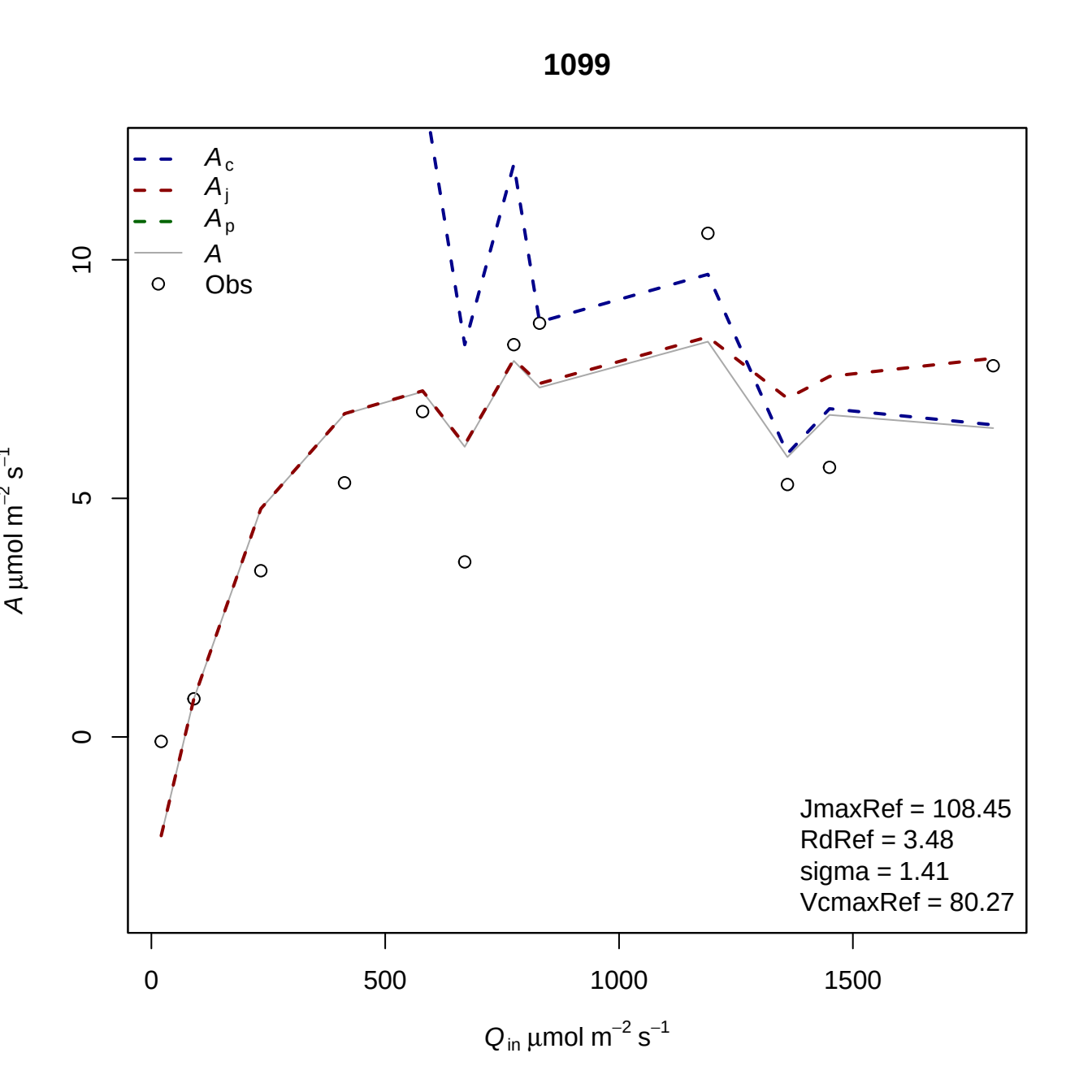

1094
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.99
RdRef = 3.13
sigma = 1.4
VcmaxRef = 254168.7
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

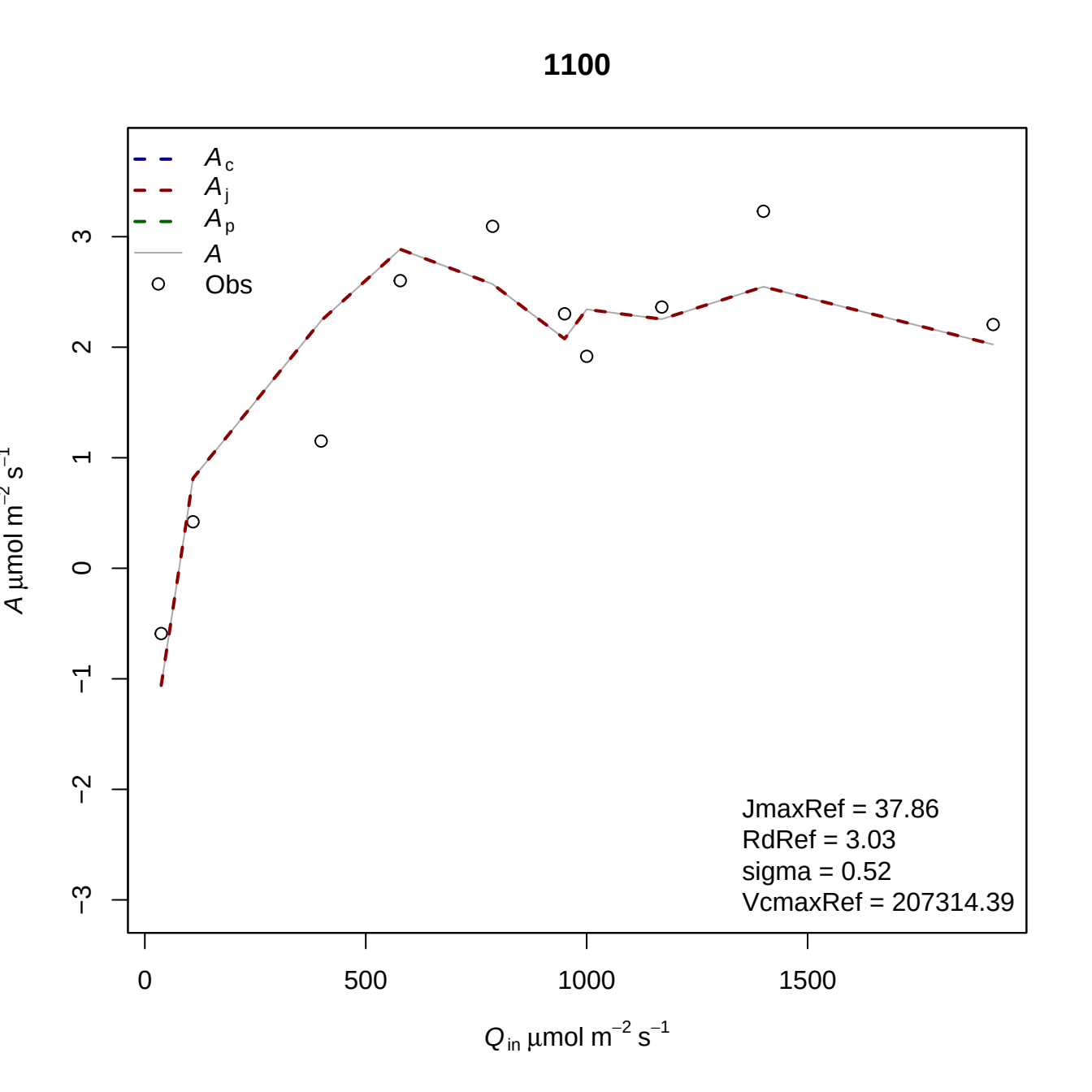

1095
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.51
RdRef = 4.15
sigma = 0.83
VcmaxRef = 470173.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500

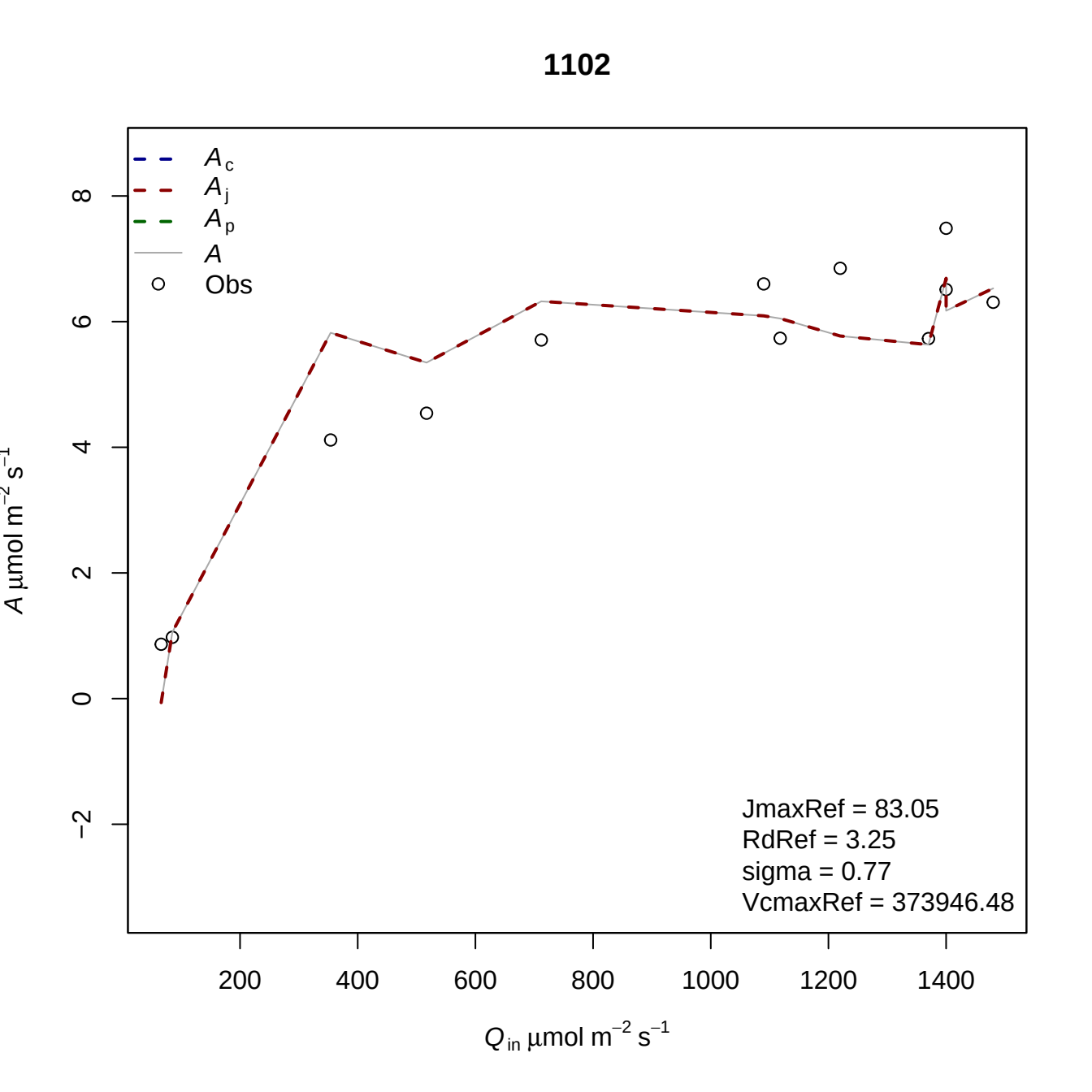

1096
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.33
RdRef = 1.91
sigma = 0.88
VcmaxRef = 503527.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500

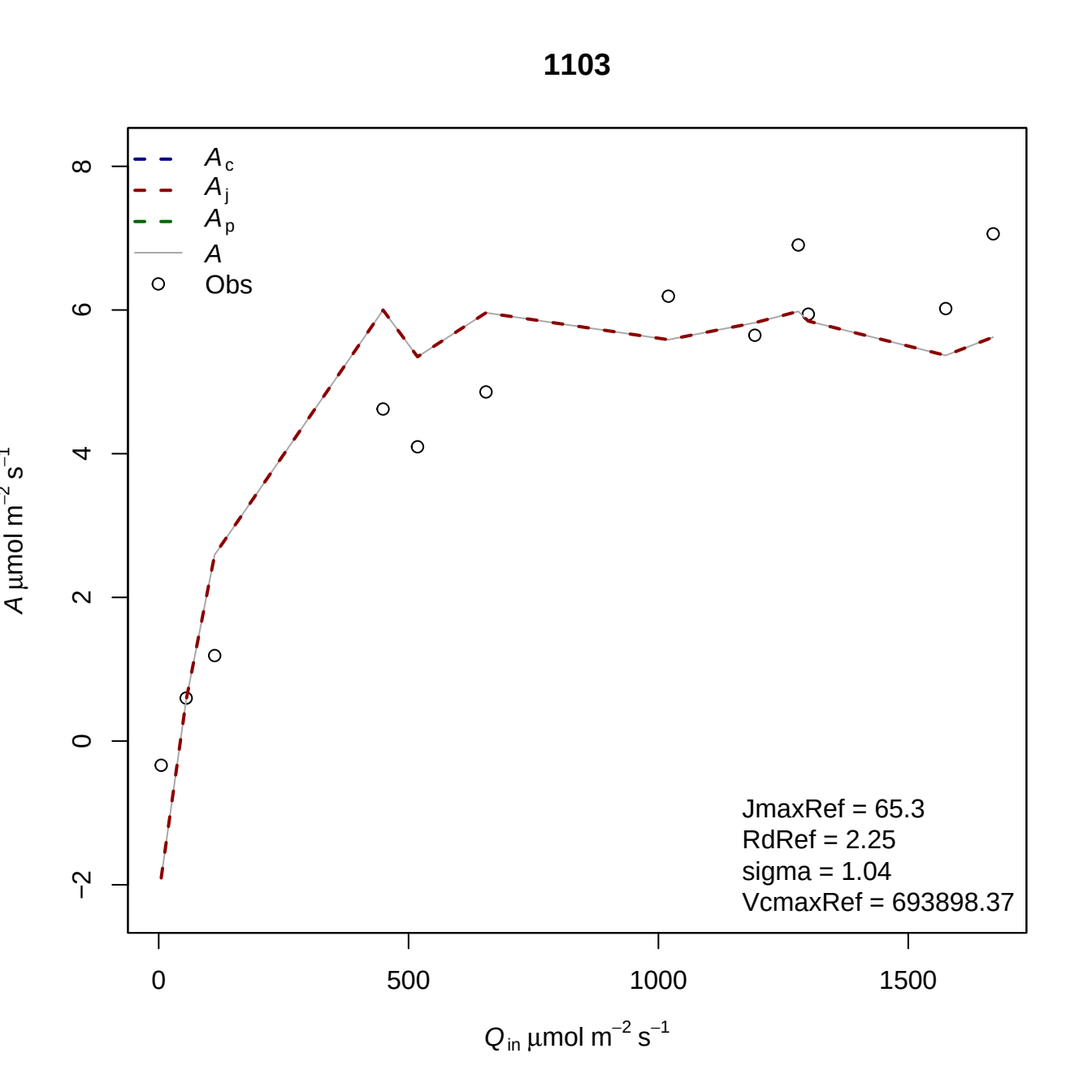

1097
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.27
RdRef = 1.54
sigma = 1.63
VcmaxRef = 111079.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

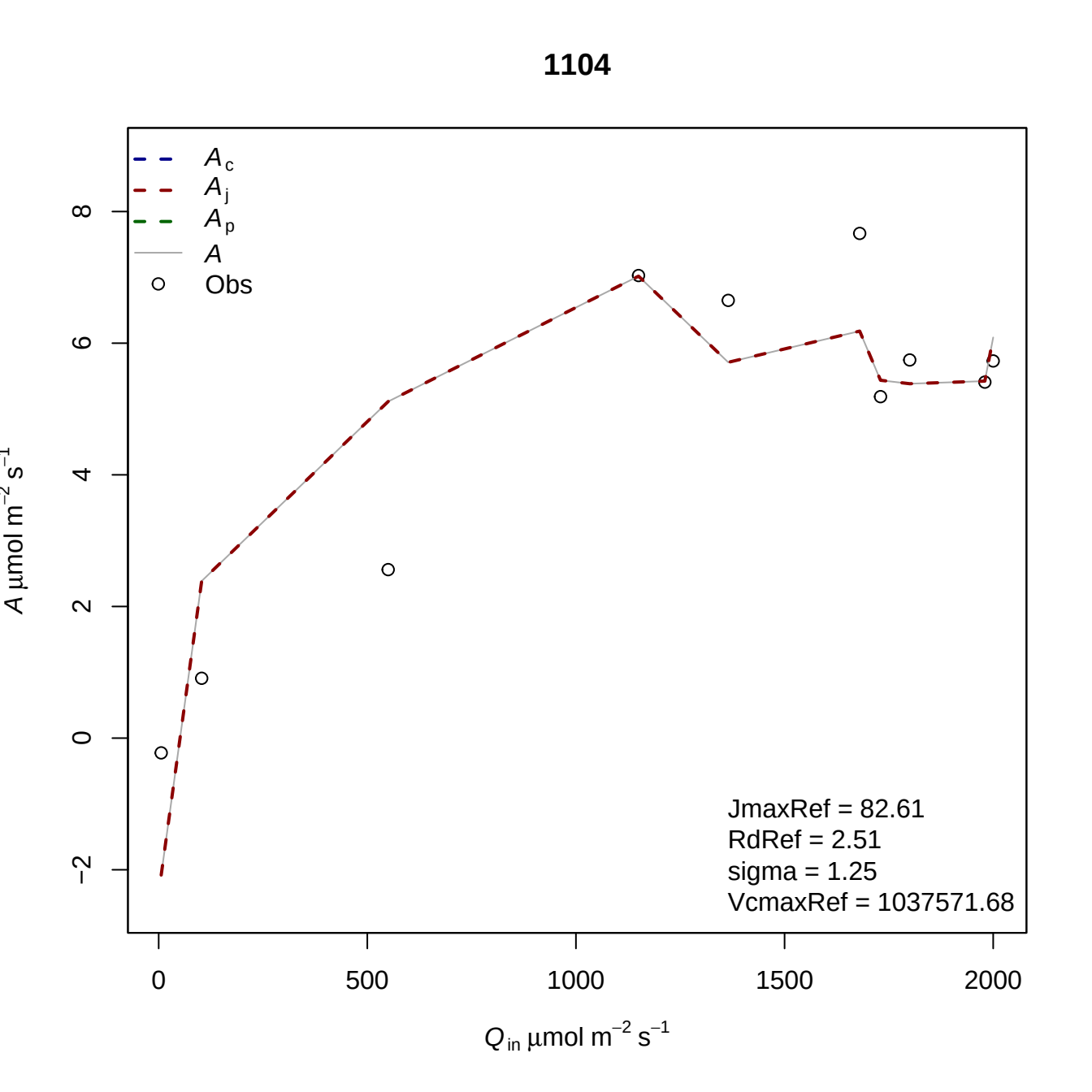

1098
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.15
RdRef = 2.48
sigma = 1.28
VcmaxRef = 900536.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

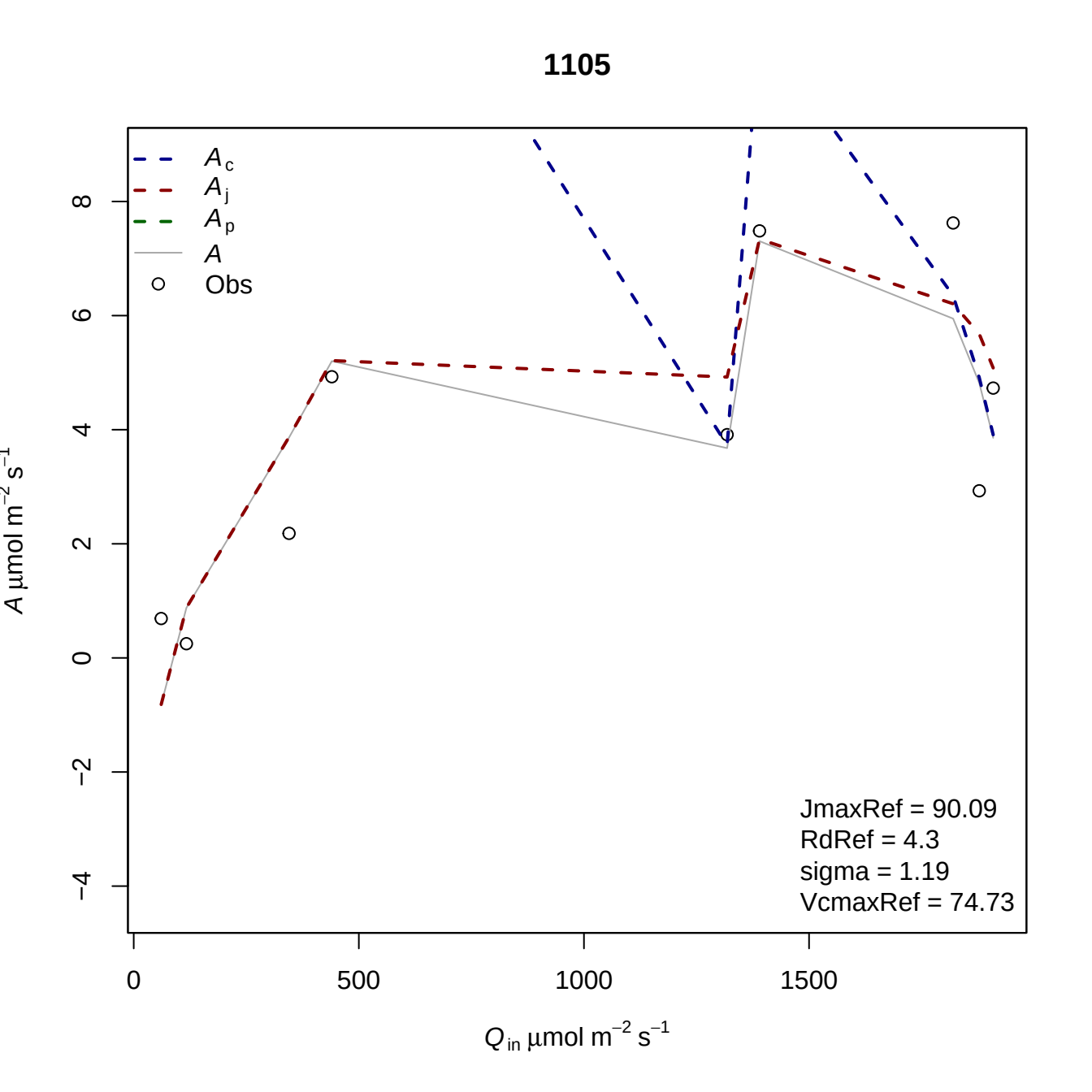

1099
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.45
RdRef = 3.48
sigma = 1.41
VcmaxRef = 80.27
0
5
10
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

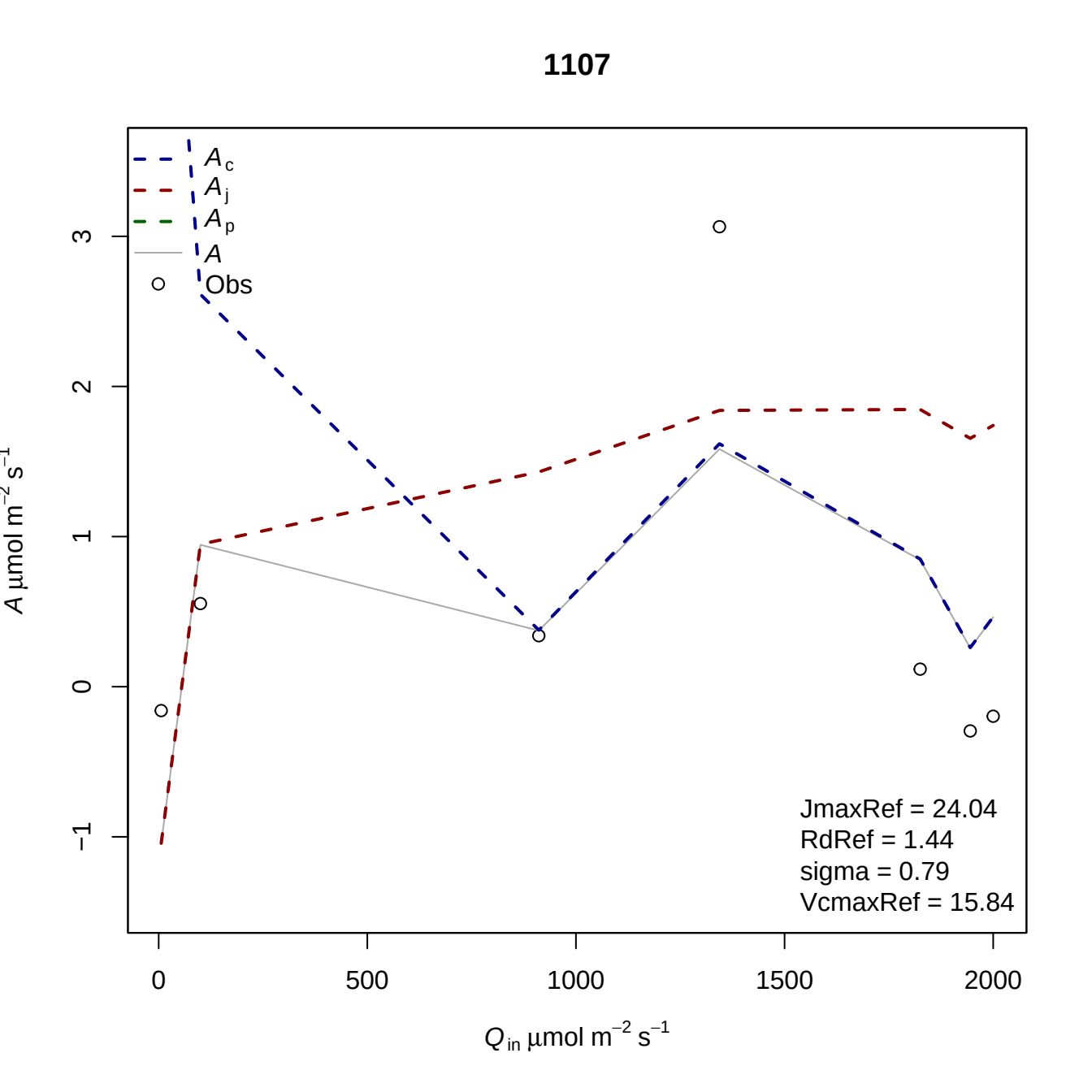

1100
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.86
RdRef = 3.03
sigma = 0.52
VcmaxRef = 207314.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
−2
−3
0
500
1000
1500

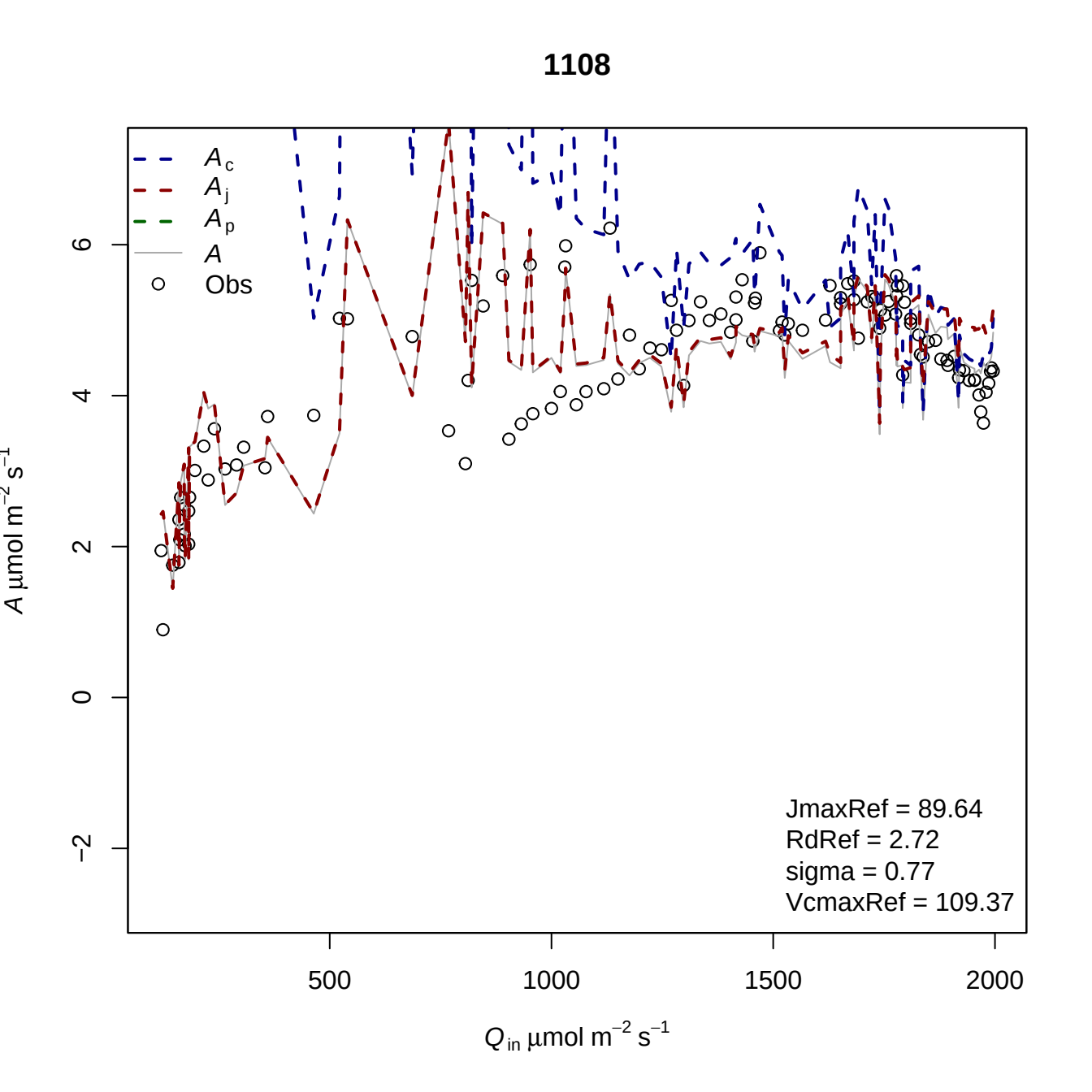

1102
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.05
RdRef = 3.25
sigma = 0.77
VcmaxRef = 373946.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

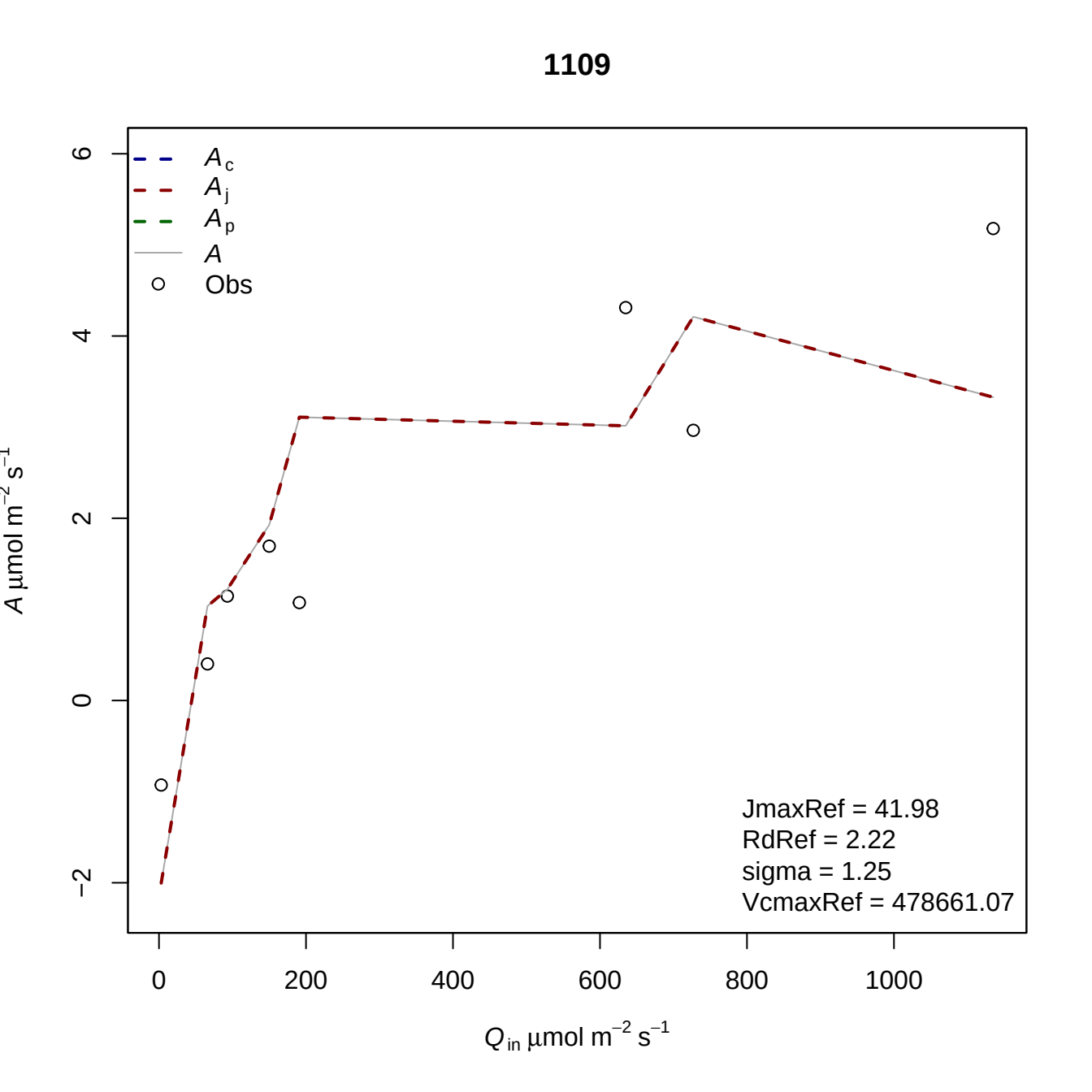

1103
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.3
RdRef = 2.25
sigma = 1.04
VcmaxRef = 693898.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

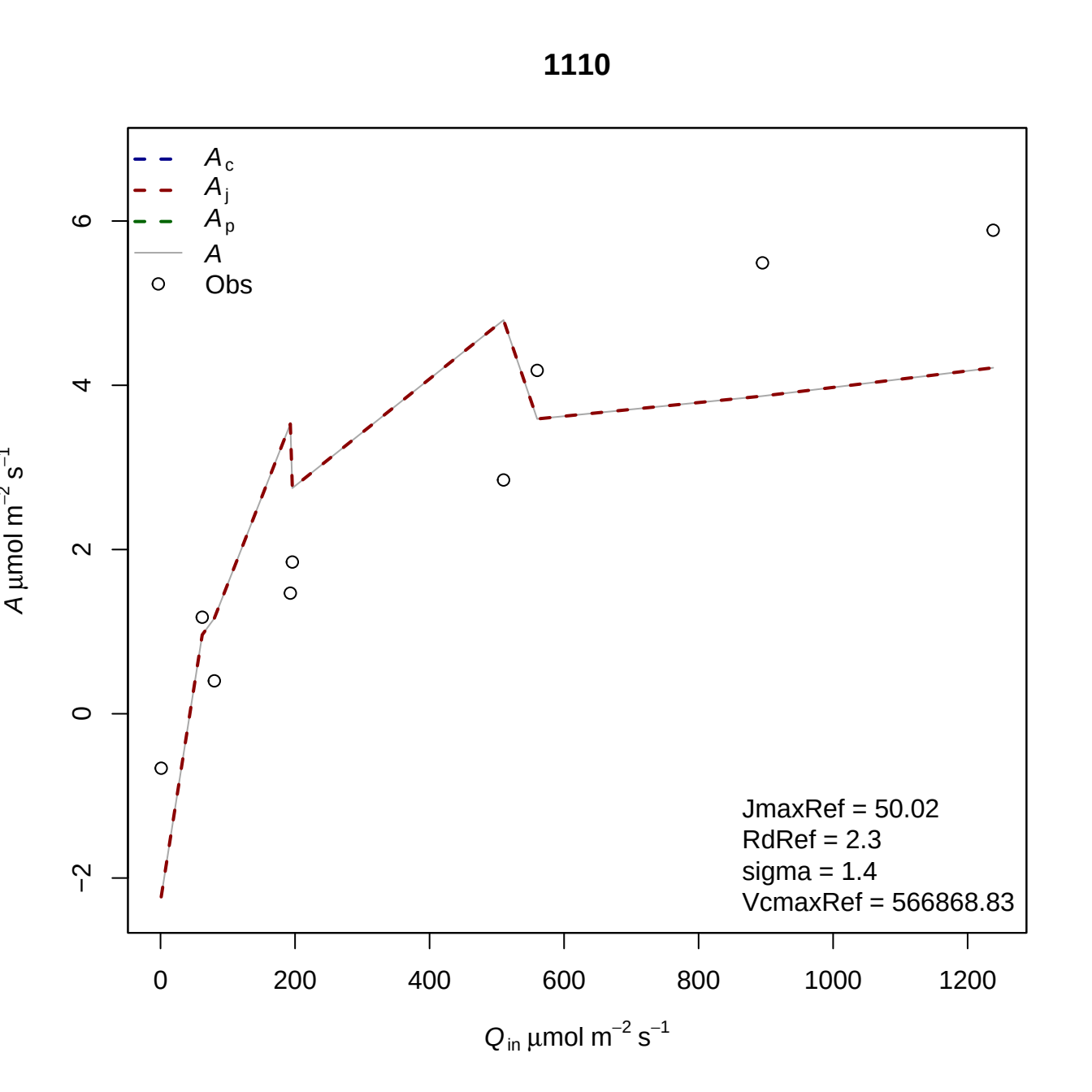

1104
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.61
RdRef = 2.51
sigma = 1.25
VcmaxRef = 1037571.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

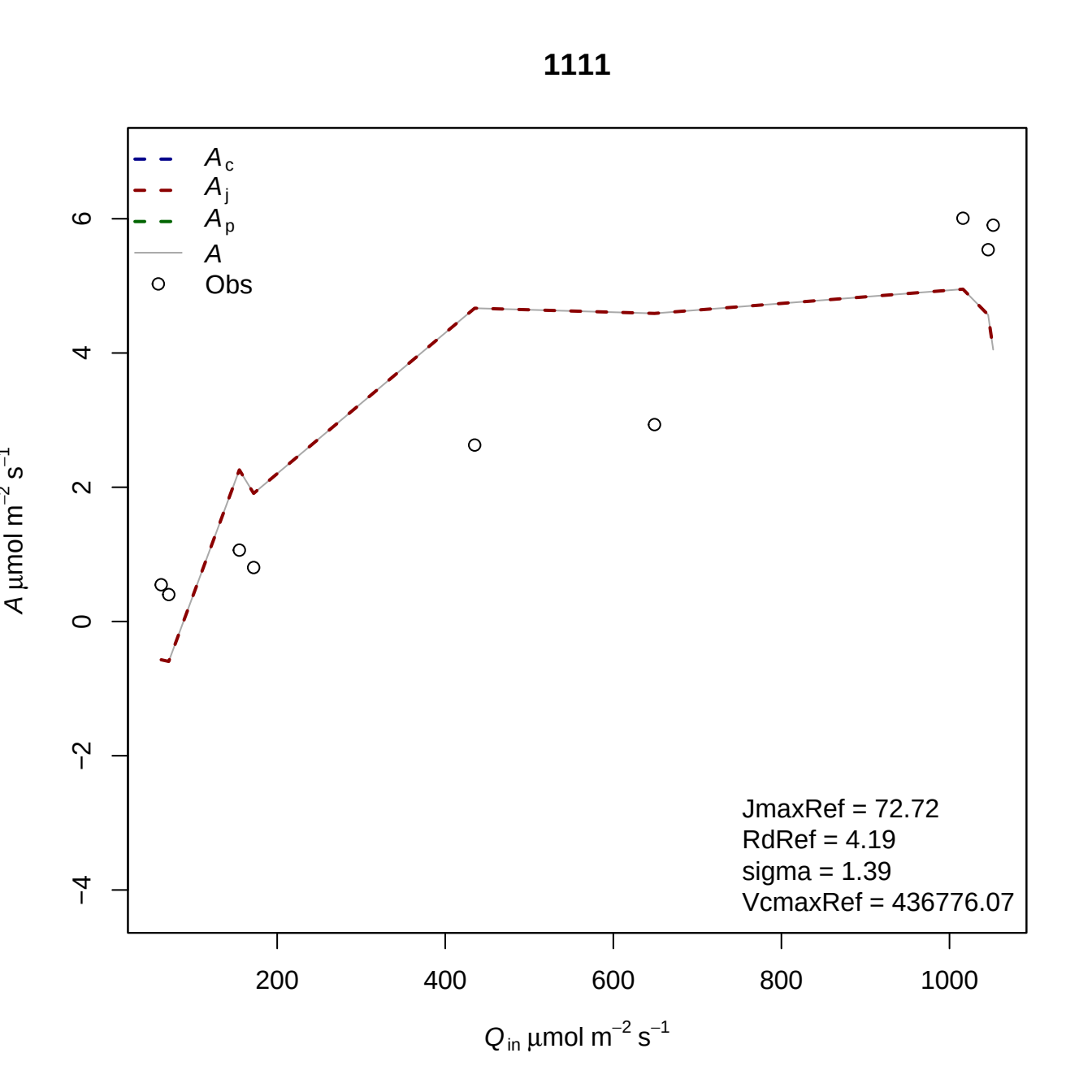

1105
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.09
RdRef = 4.3
sigma = 1.19
VcmaxRef = 74.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

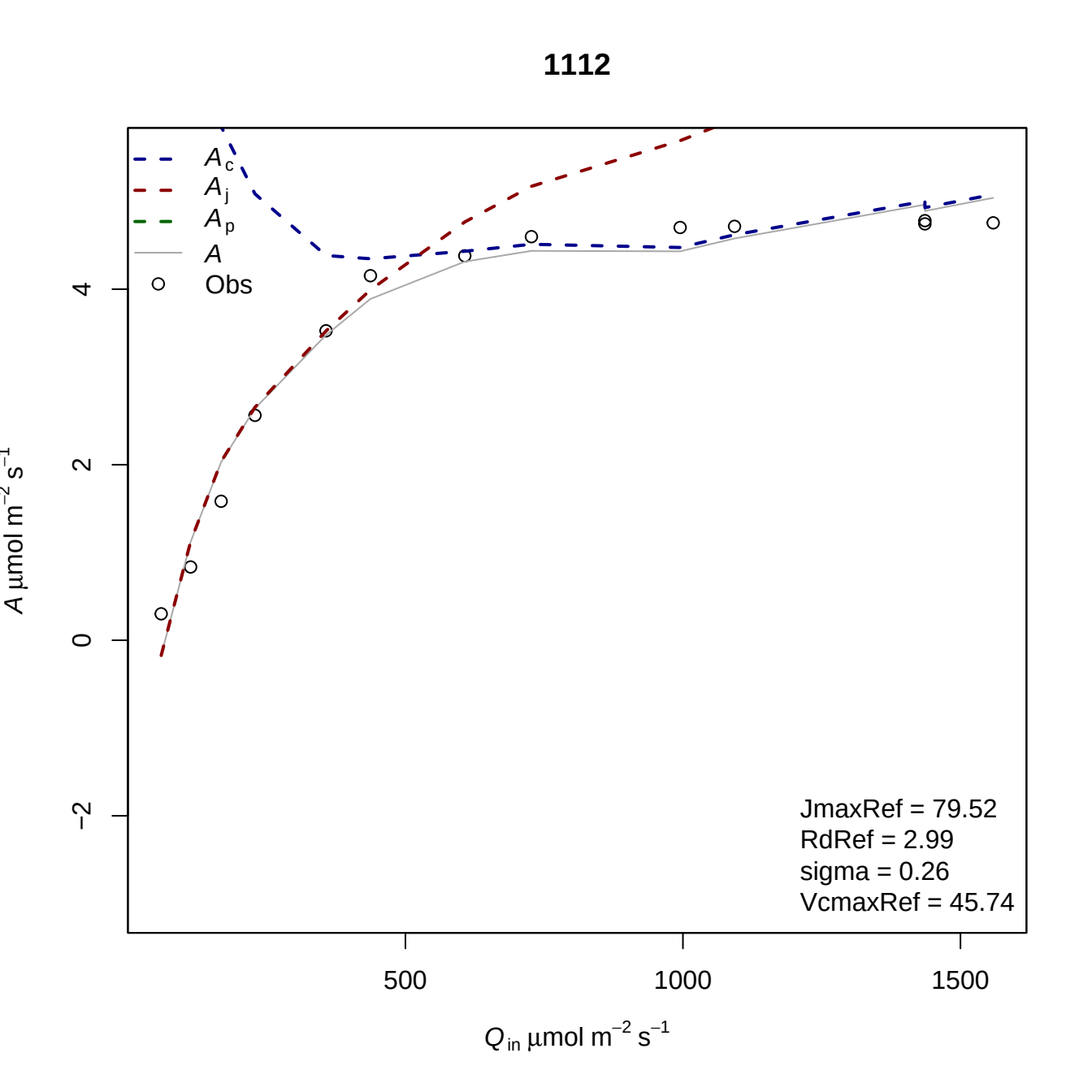

1107
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 24.04
RdRef = 1.44
sigma = 0.79
VcmaxRef = 15.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500
2000

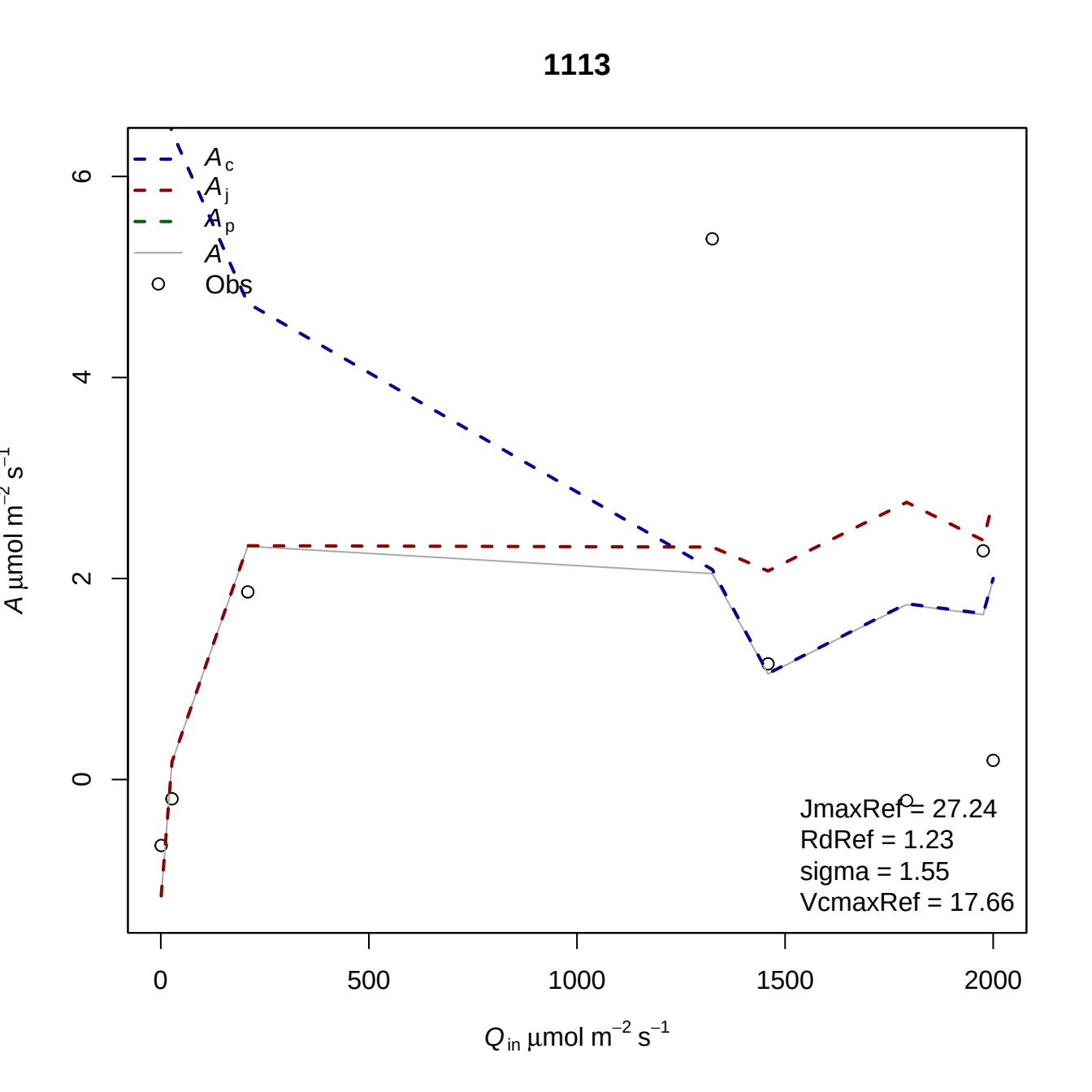

1108
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.64
RdRef = 2.72
sigma = 0.77
VcmaxRef = 109.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
500
1000
1500
2000

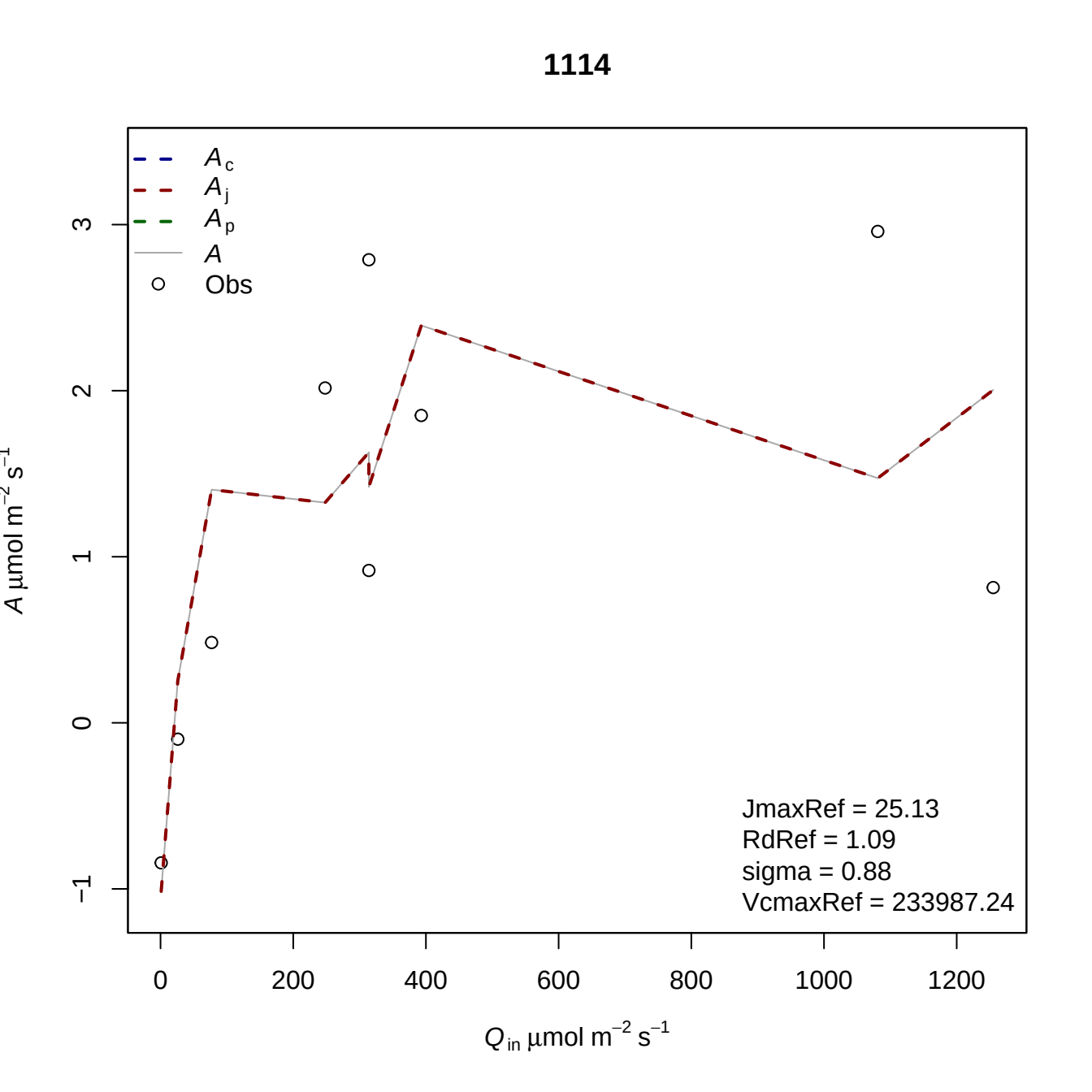

1109
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.98
RdRef = 2.22
sigma = 1.25
VcmaxRef = 478661.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800
1000

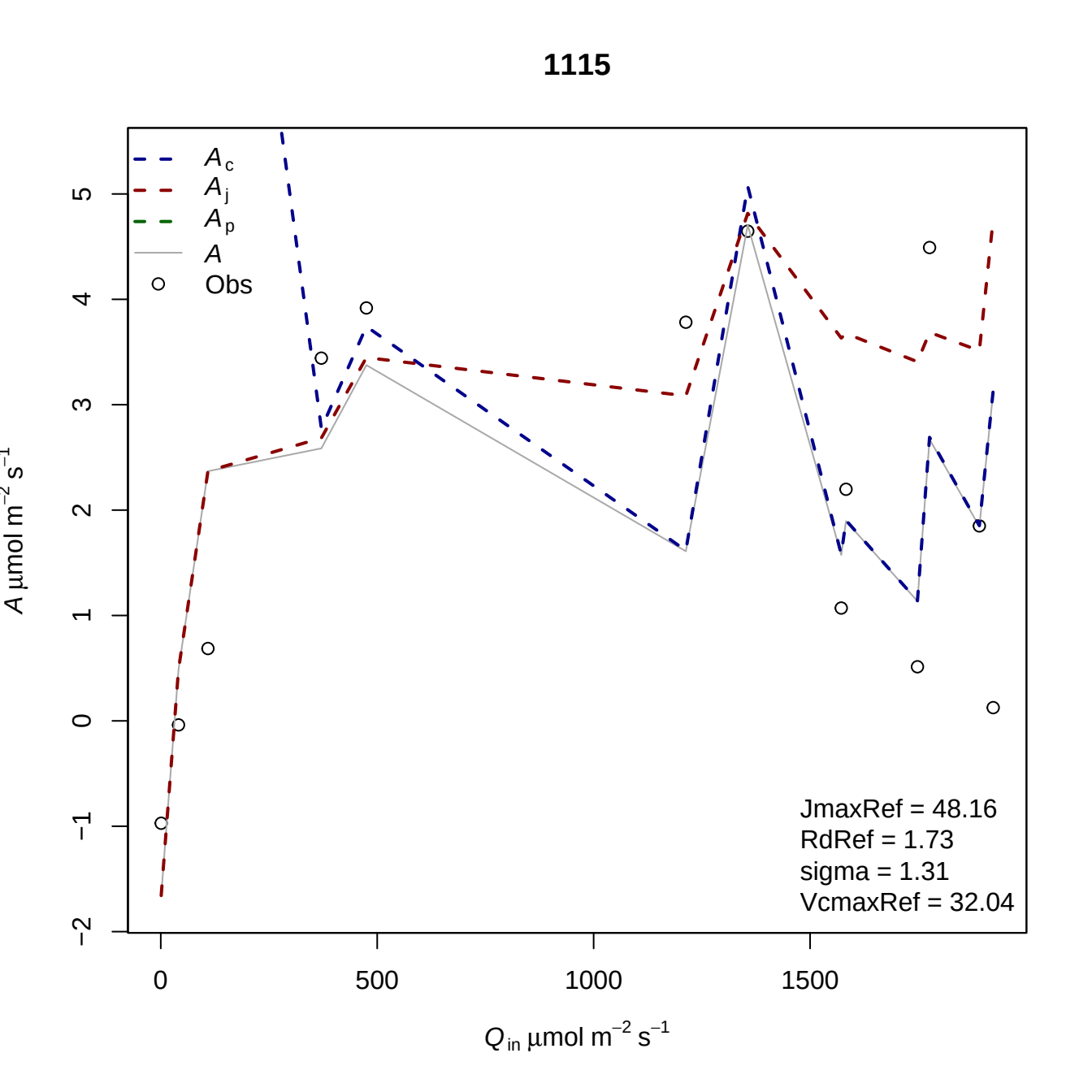

1110
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.02
RdRef = 2.3
sigma = 1.4
VcmaxRef = 566868.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
200
400
600
800
1000
1200

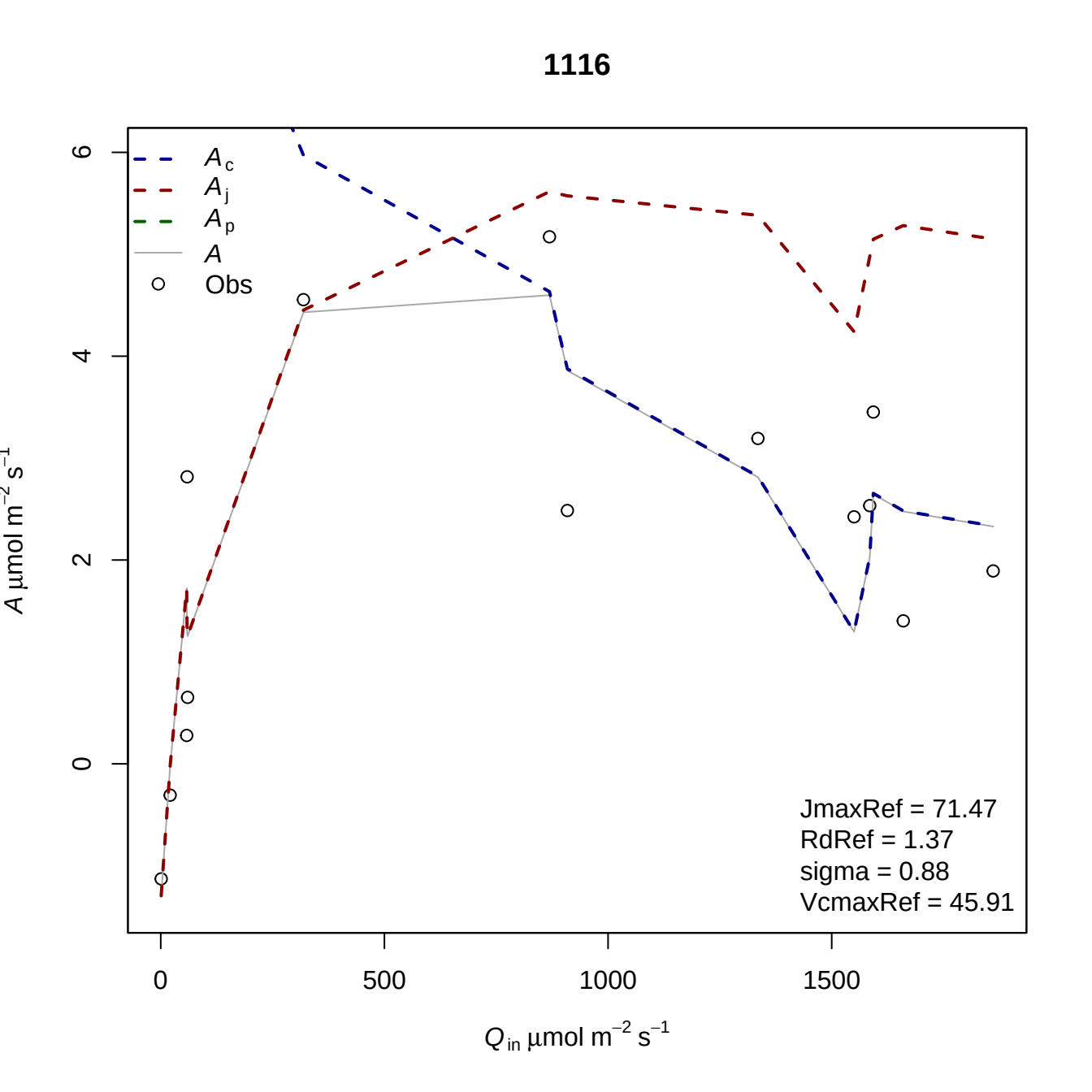

1111
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.72
RdRef = 4.19
sigma = 1.39
VcmaxRef = 436776.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
200
400
600
800
1000

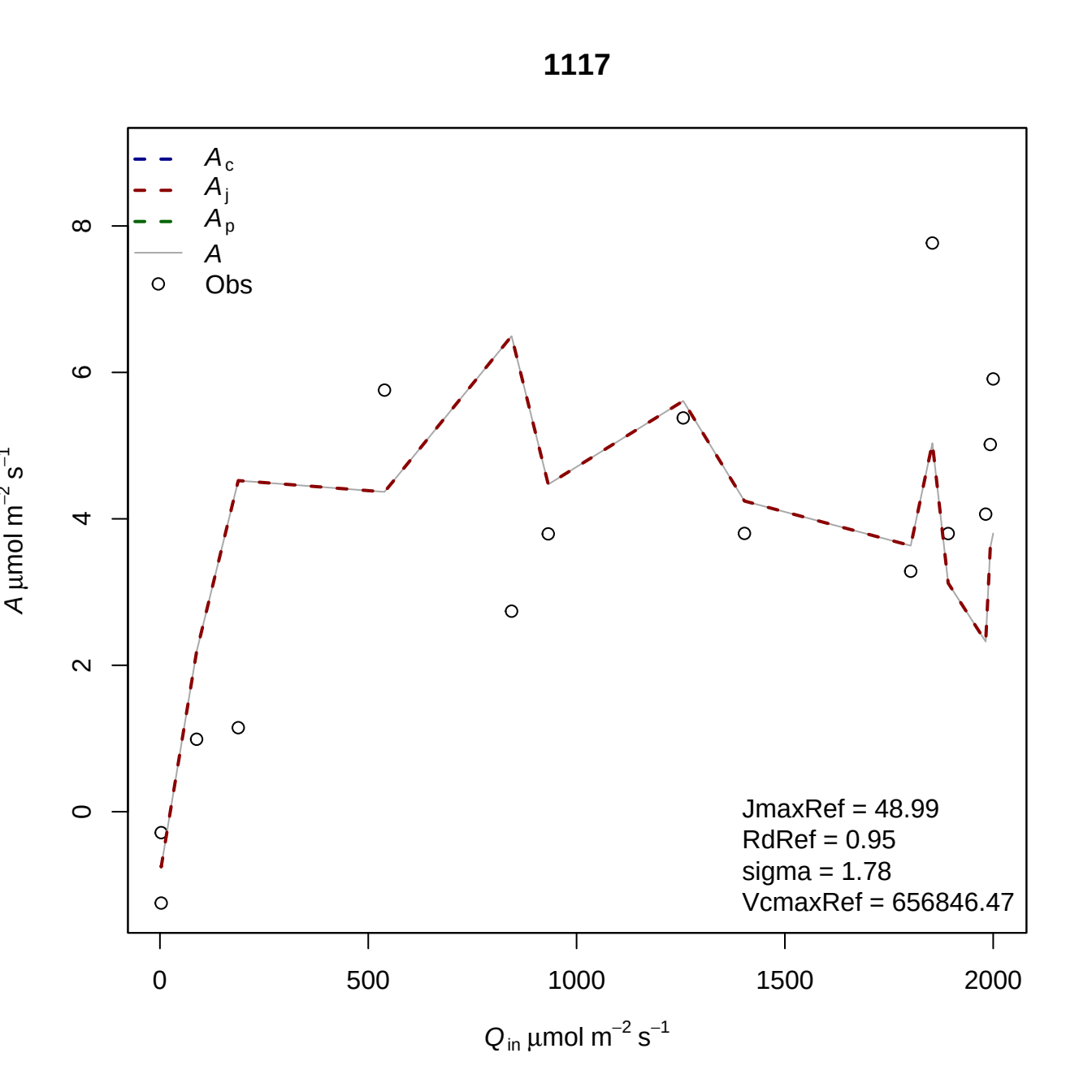

1112
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.52
RdRef = 2.99
sigma = 0.26
VcmaxRef = 45.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
500
1000
1500

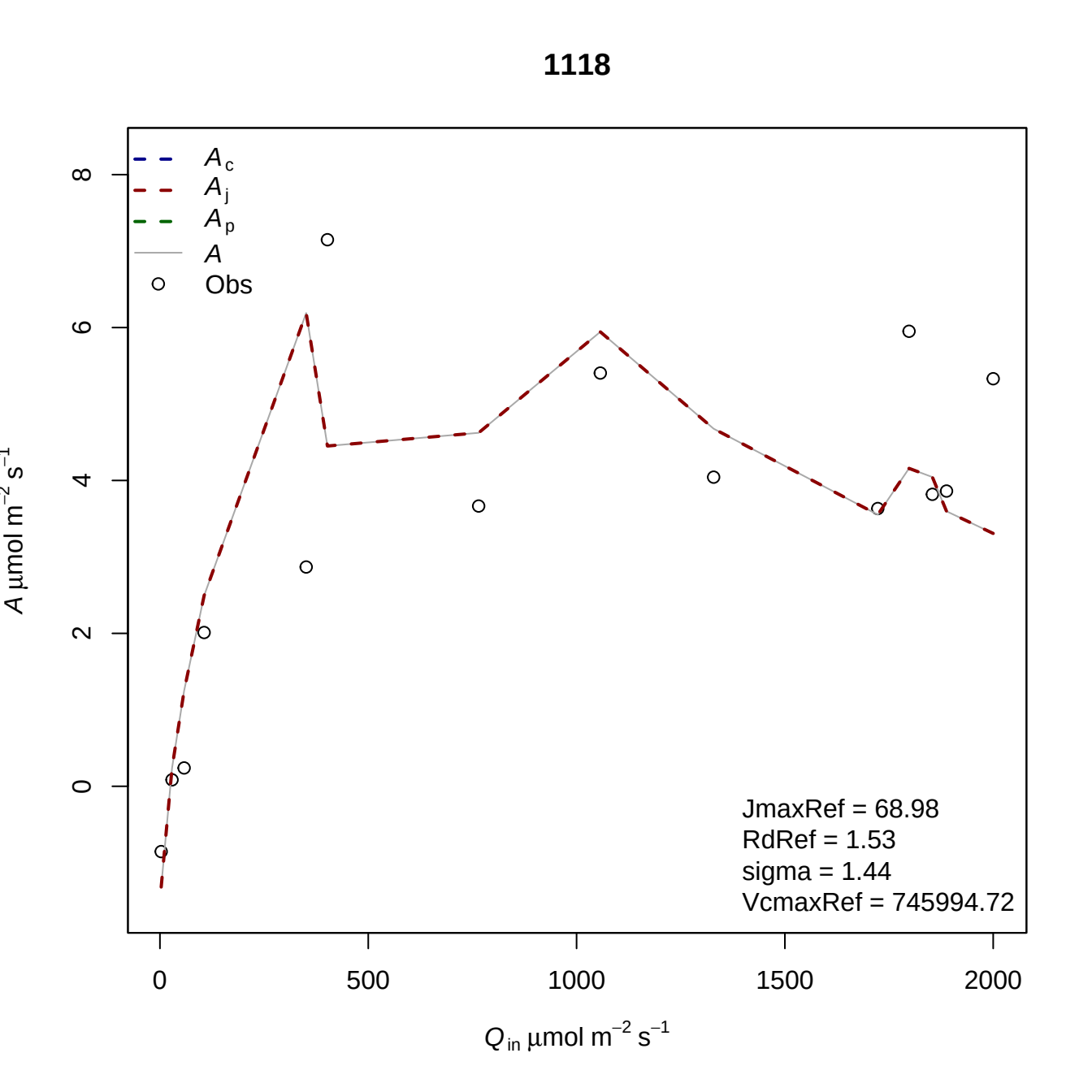

1113
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.24
RdRef = 1.23
sigma = 1.55
VcmaxRef = 17.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

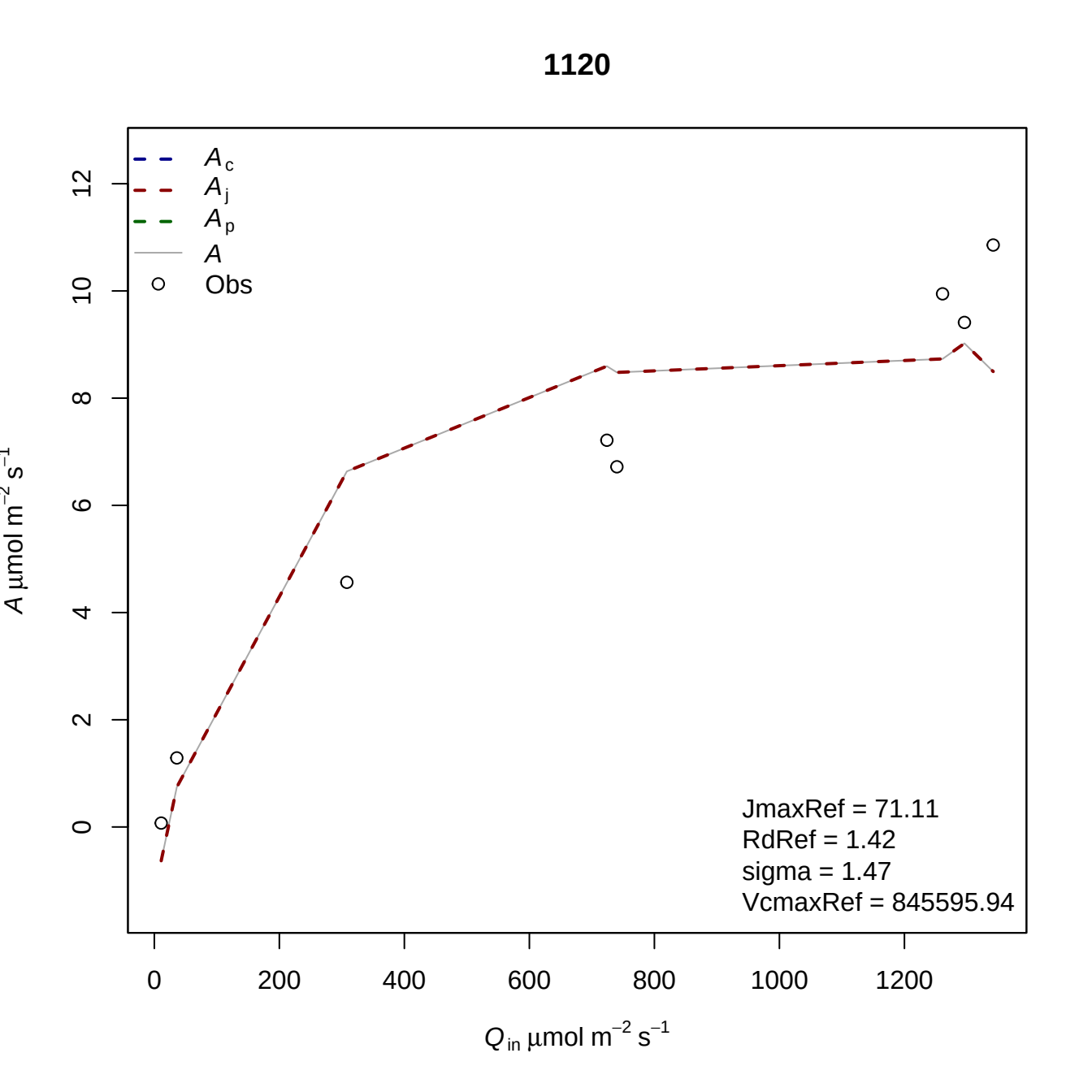

1114
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 25.13
RdRef = 1.09
sigma = 0.88
VcmaxRef = 233987.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000
1200

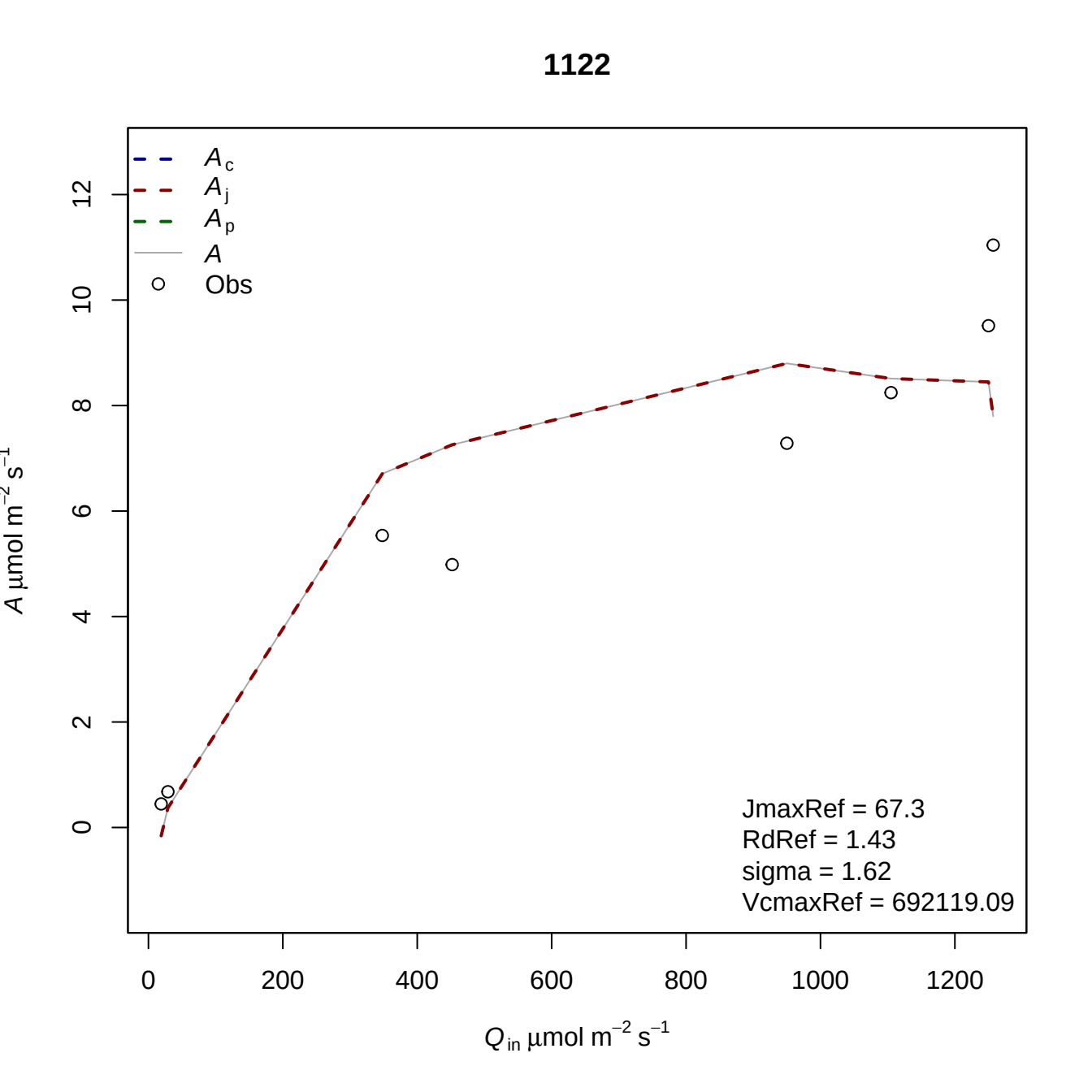

1115
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.16
RdRef = 1.73
sigma = 1.31
VcmaxRef = 32.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
−1
0
1
2
3
4
5
0
500
1000
1500

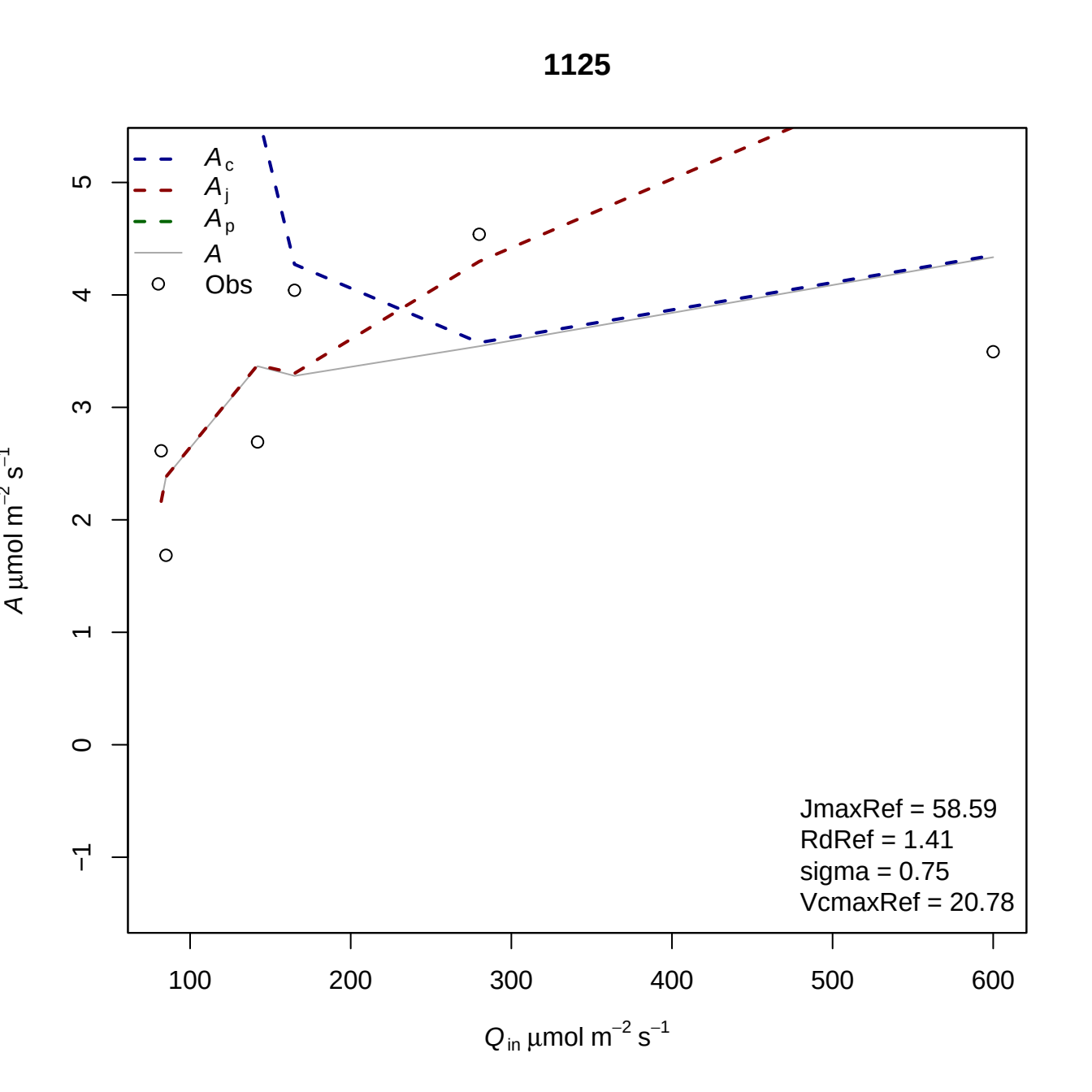

1116
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.47
RdRef = 1.37
sigma = 0.88
VcmaxRef = 45.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

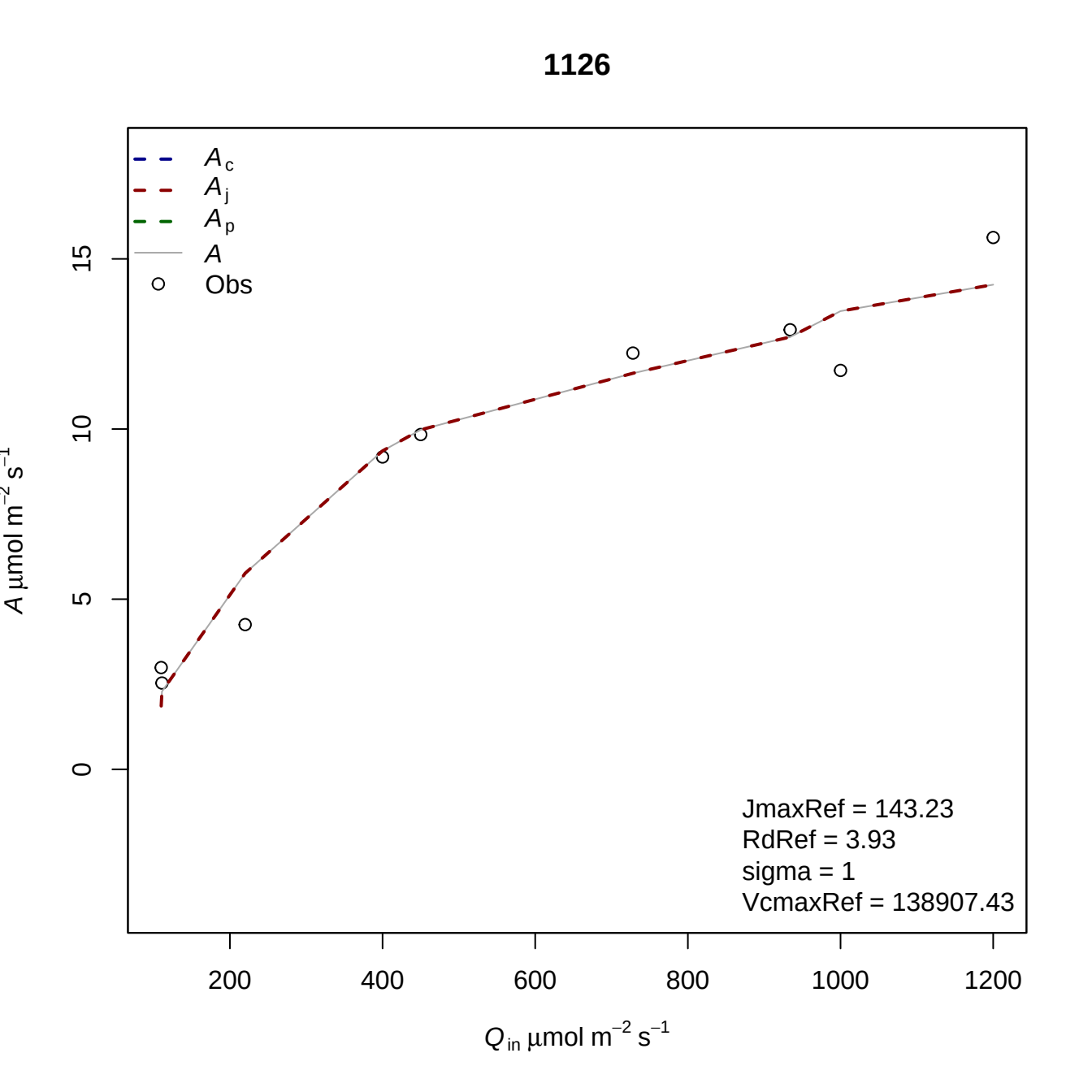

1117
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.99
RdRef = 0.95
sigma = 1.78
VcmaxRef = 656846.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

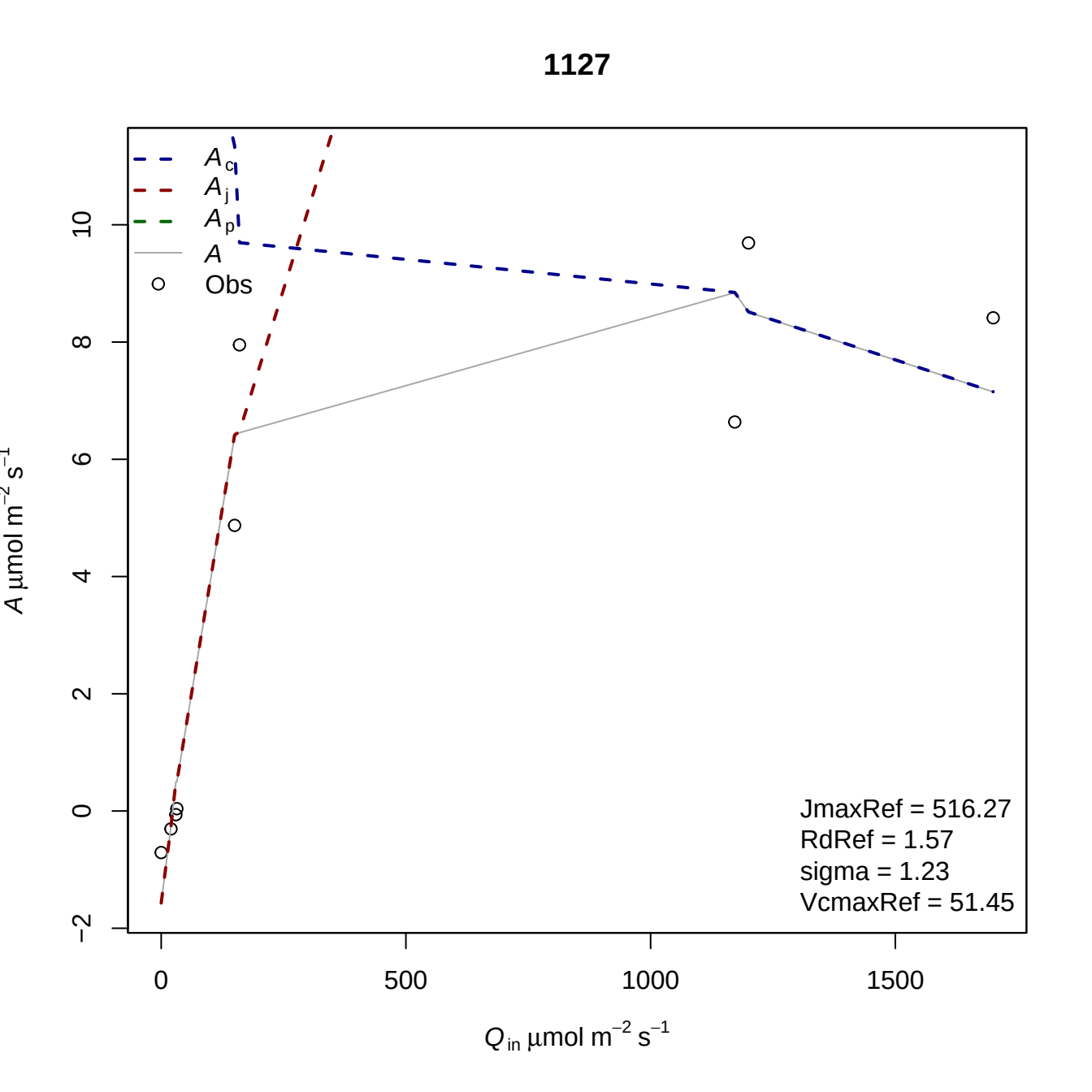

1118
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.98
RdRef = 1.53
sigma = 1.44
VcmaxRef = 745994.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

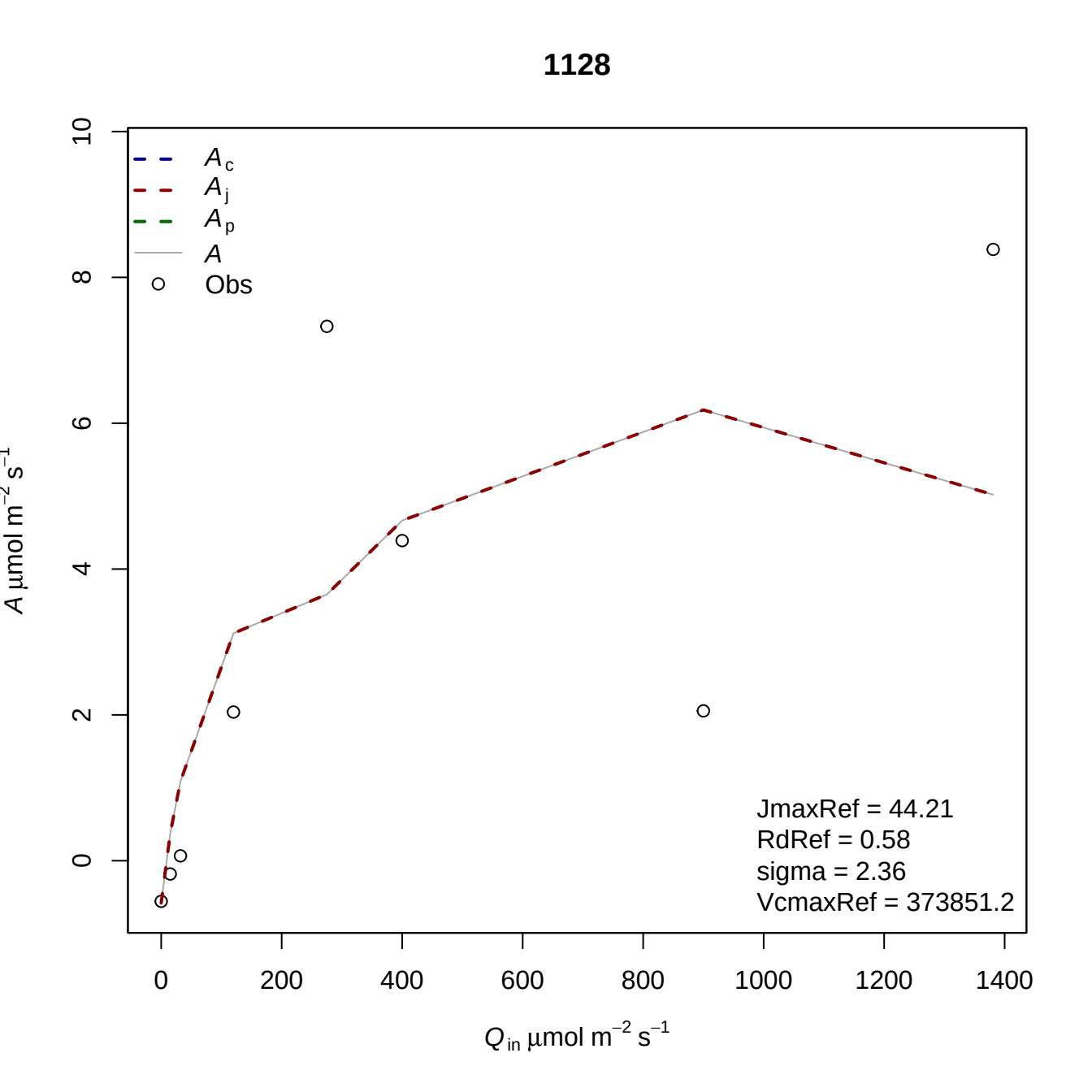

1120
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.11
RdRef = 1.42
sigma = 1.47
VcmaxRef = 845595.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

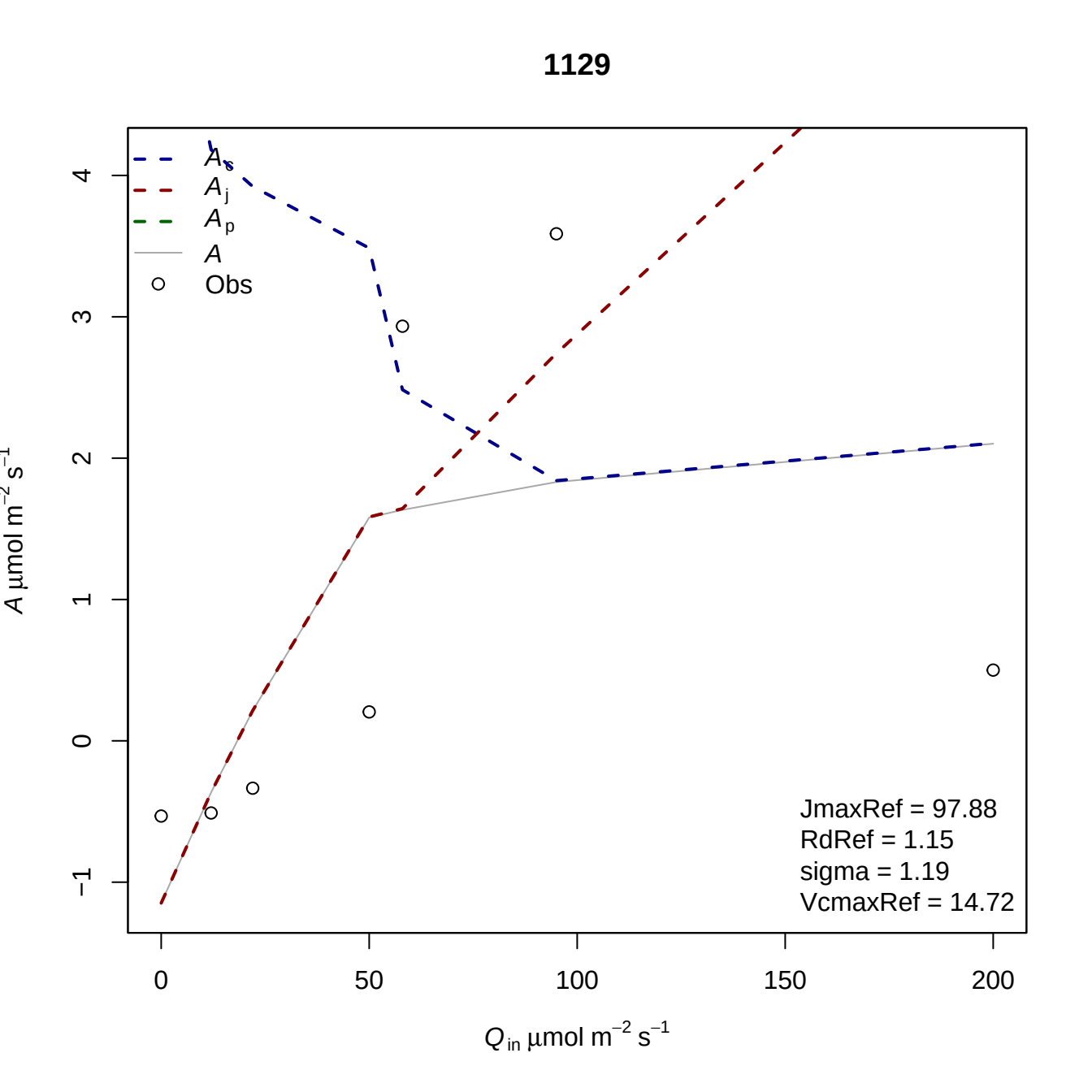

1122
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.3
RdRef = 1.43
sigma = 1.62
VcmaxRef = 692119.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

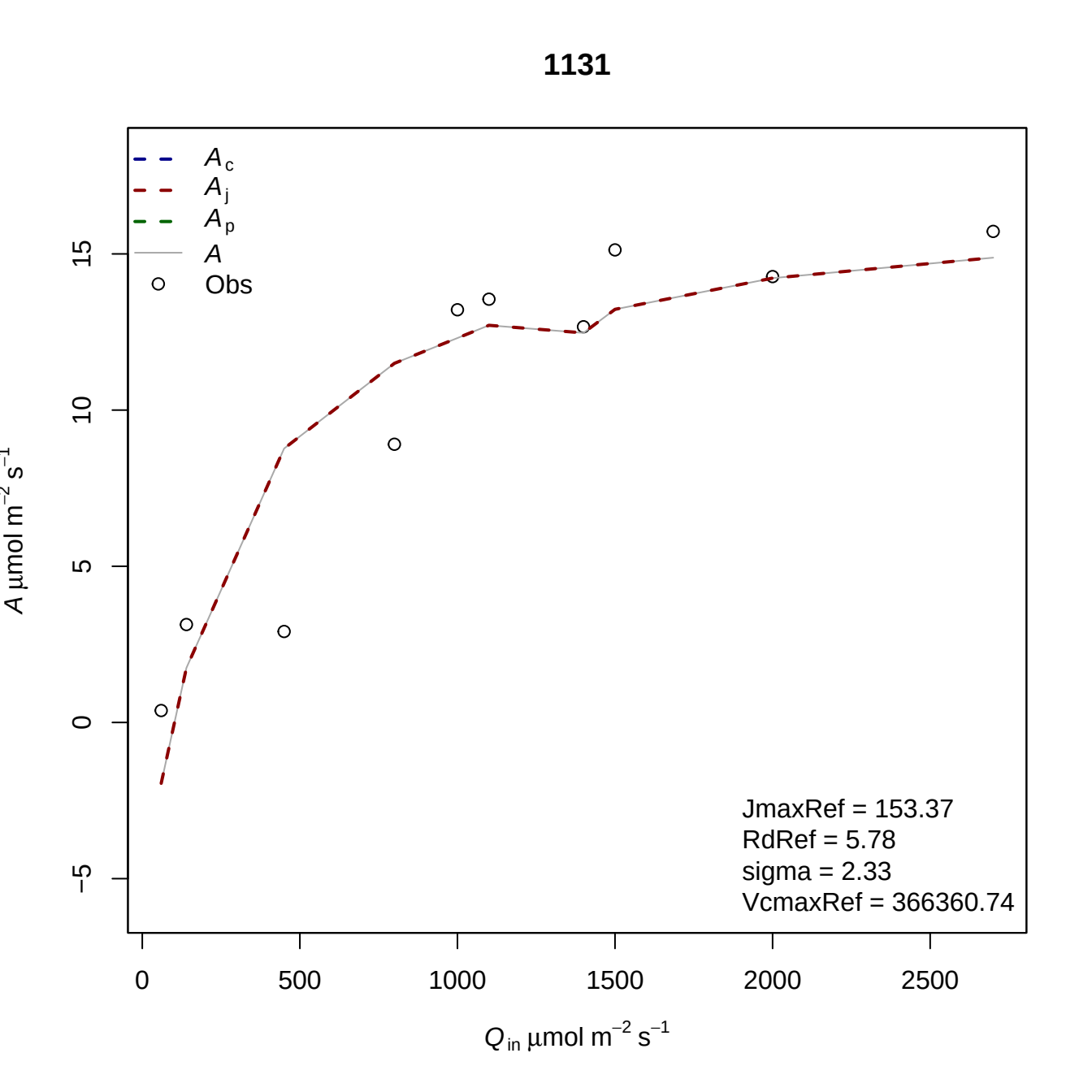

1125
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.59
RdRef = 1.41
sigma = 0.75
VcmaxRef = 20.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
100
200
300
400
500
600

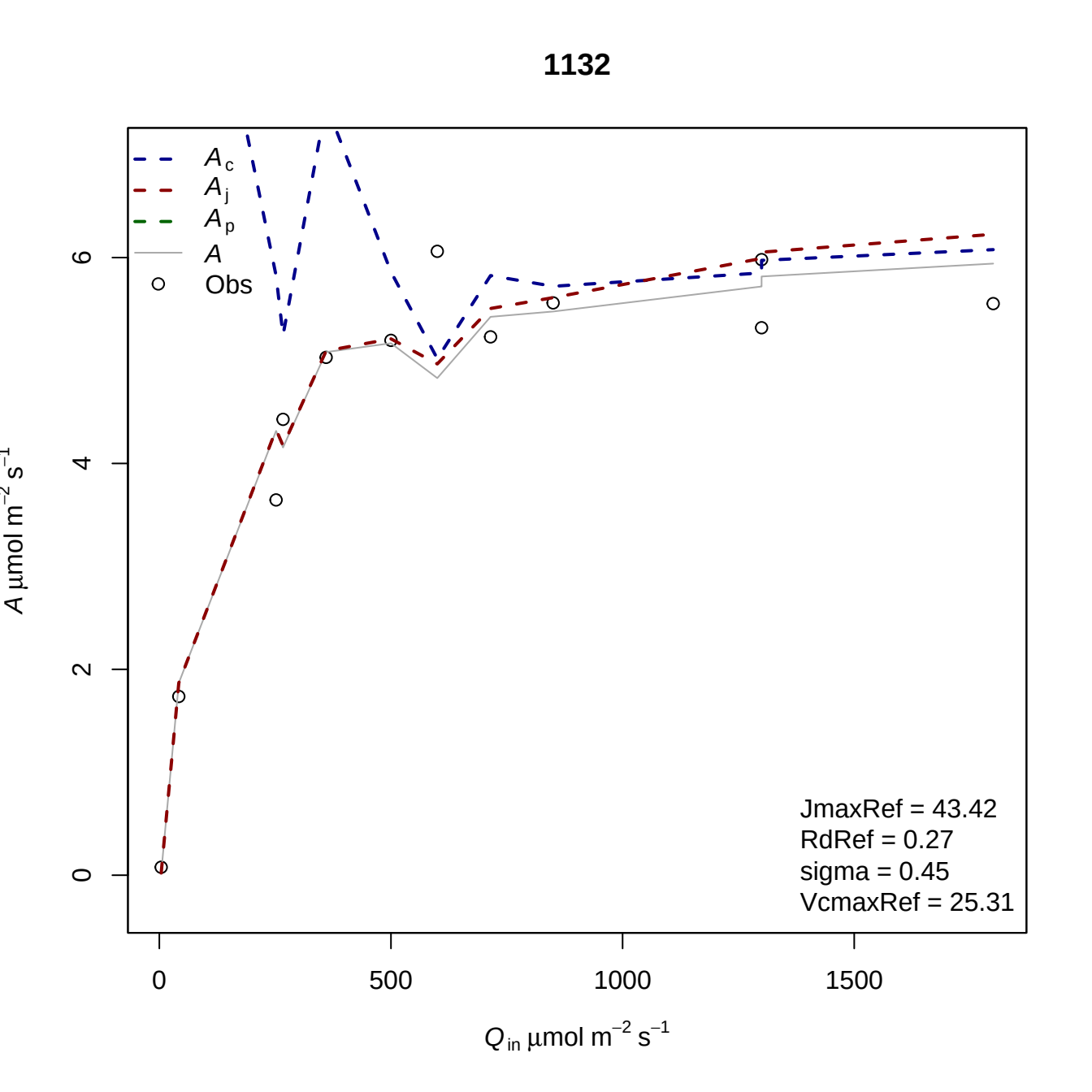

1126
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.23
RdRef = 3.93
sigma = 1
VcmaxRef = 138907.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
200
400
600
800
1000
1200

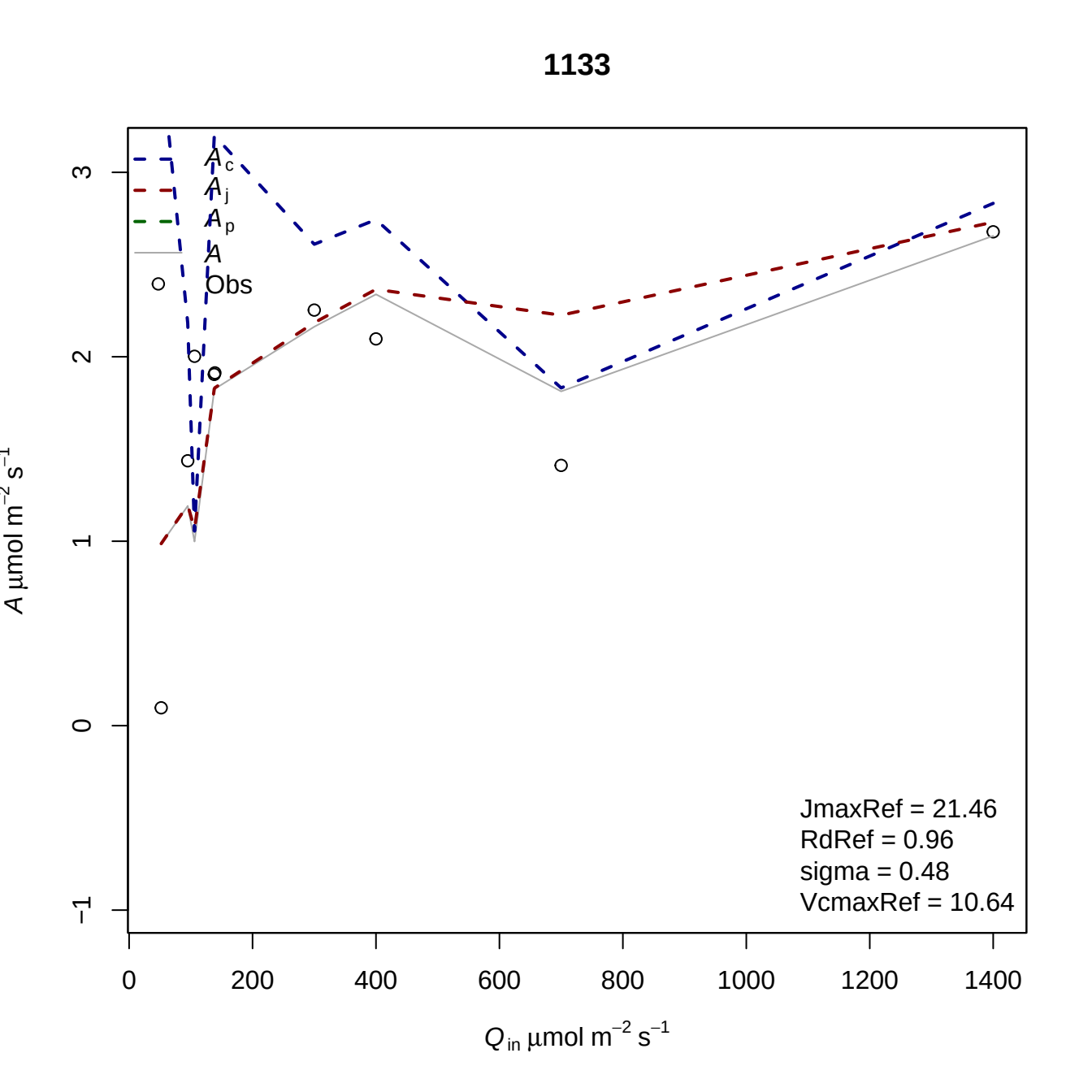

1127
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 516.27
RdRef = 1.57
sigma = 1.23
VcmaxRef = 51.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

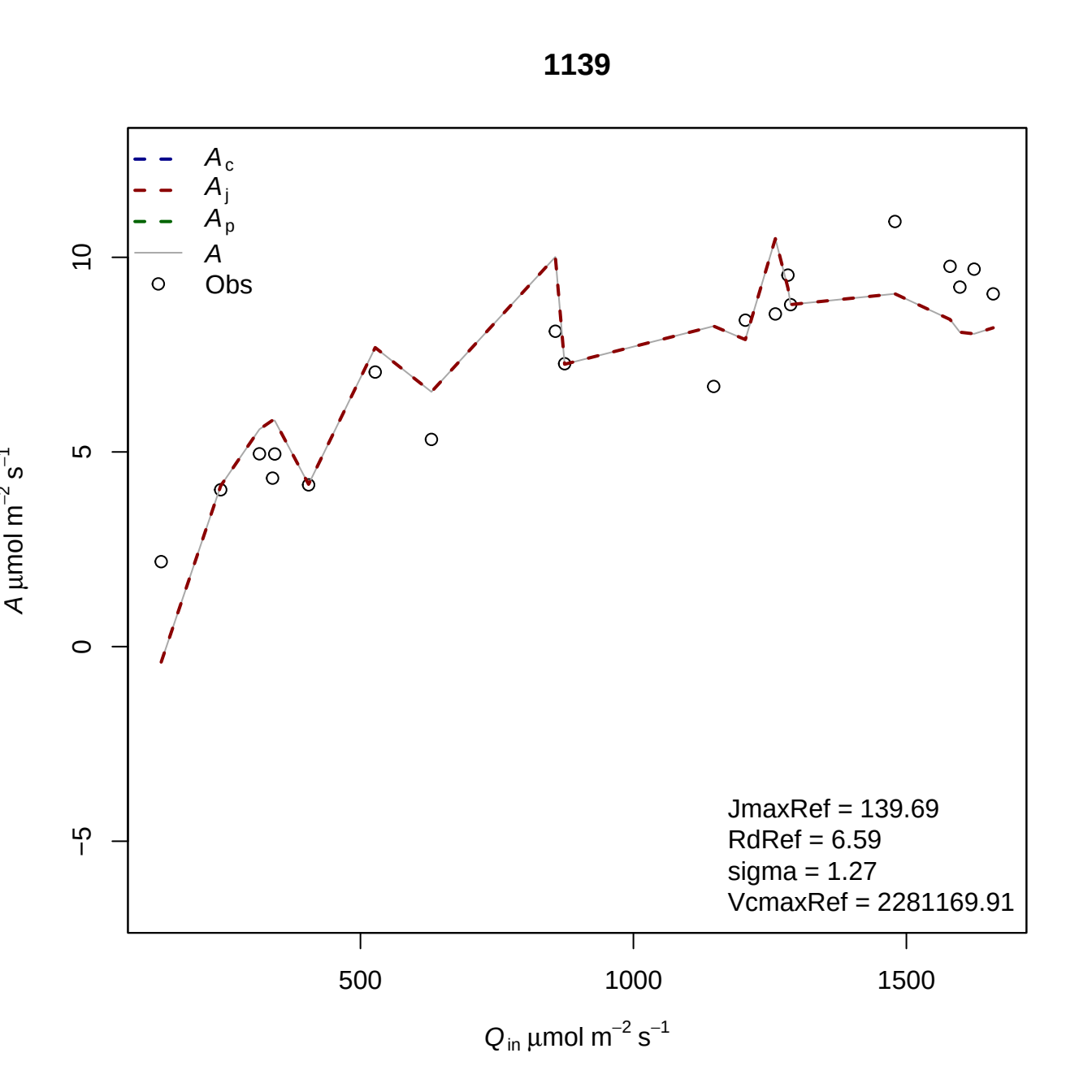

1128
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.21
RdRef = 0.58
sigma = 2.36
VcmaxRef = 373851.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

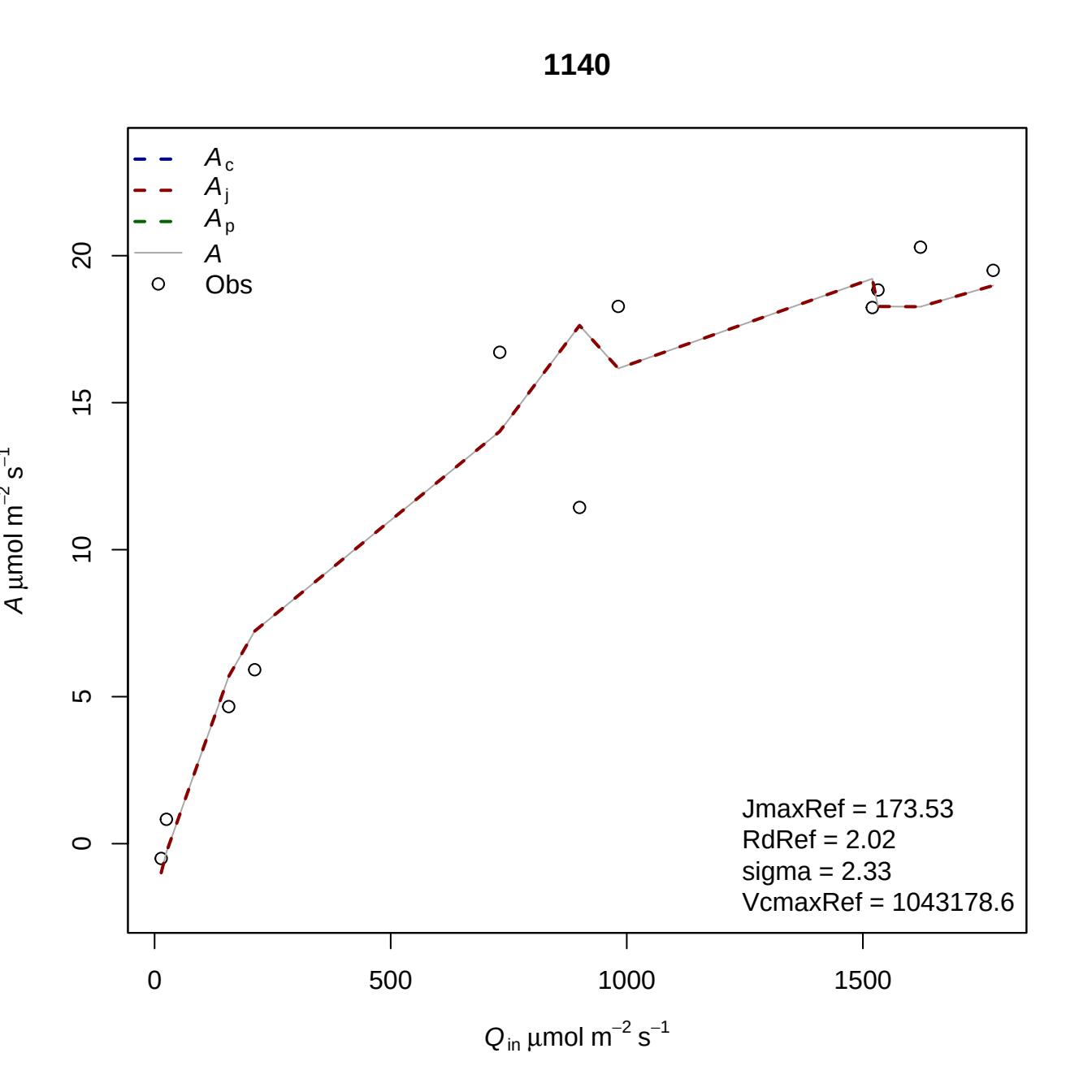

1129
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.88
RdRef = 1.15
sigma = 1.19
VcmaxRef = 14.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
50
100
150
200

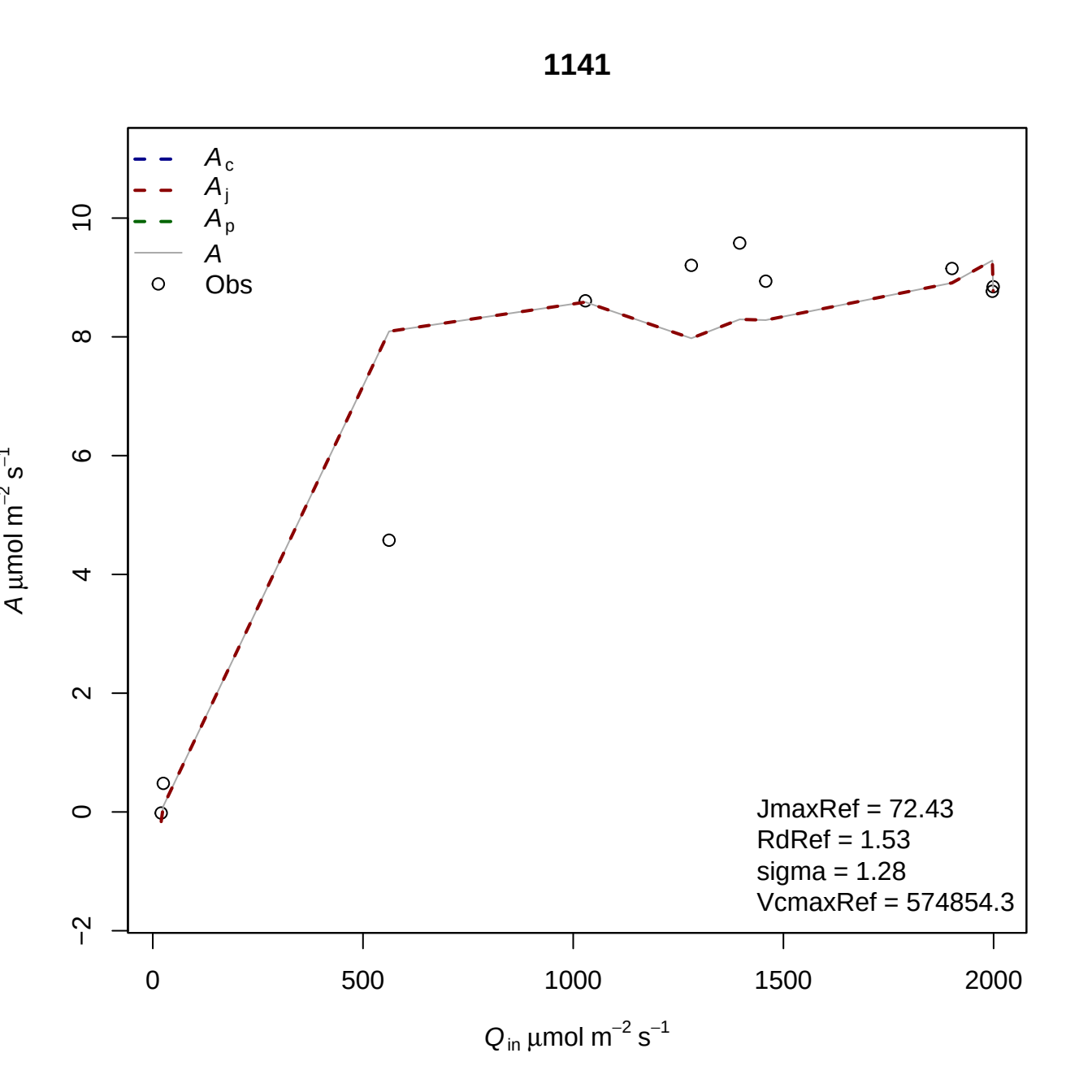

1131
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.37
RdRef = 5.78
sigma = 2.33
VcmaxRef = 366360.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000
2500

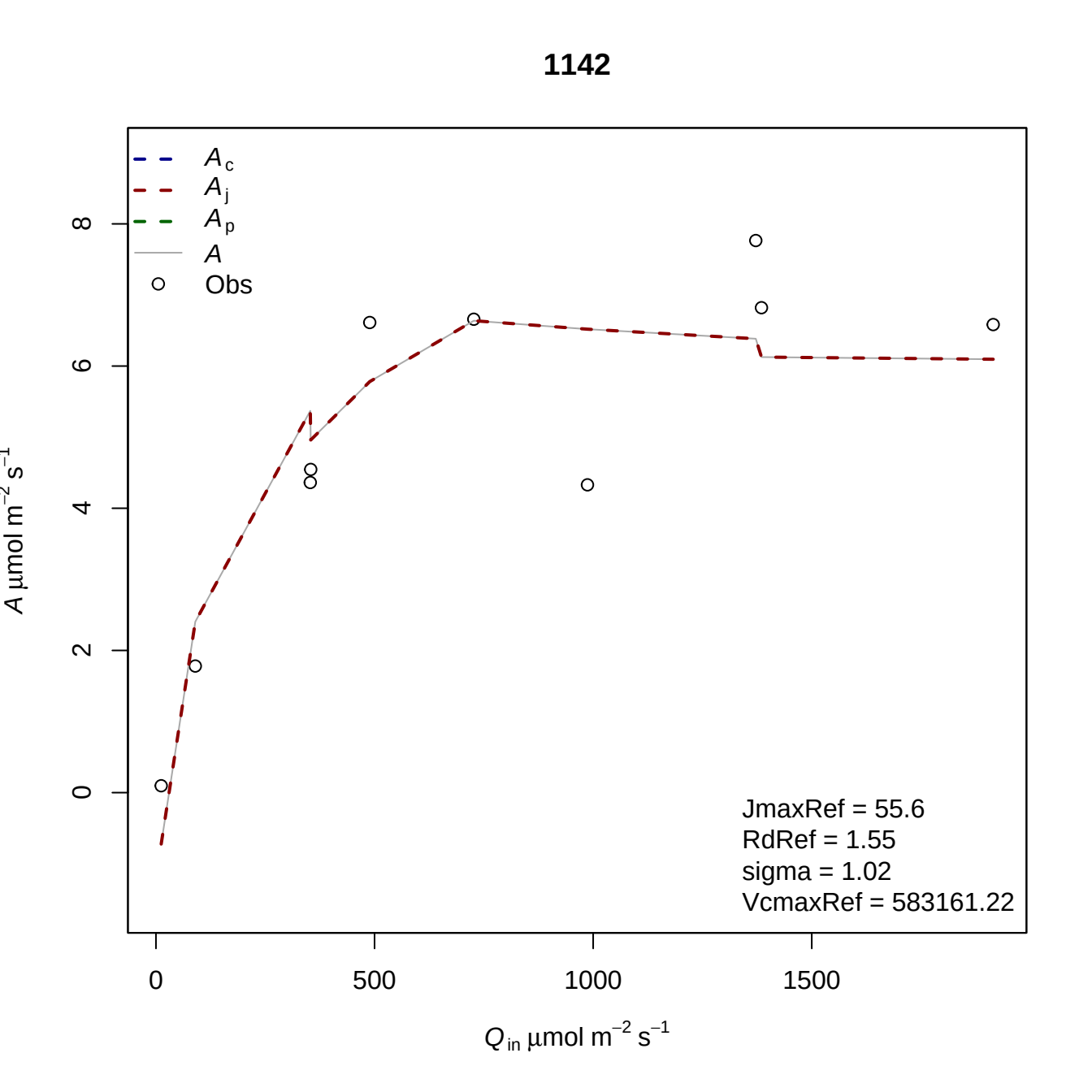

1132
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 43.42
RdRef = 0.27
sigma = 0.45
VcmaxRef = 25.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

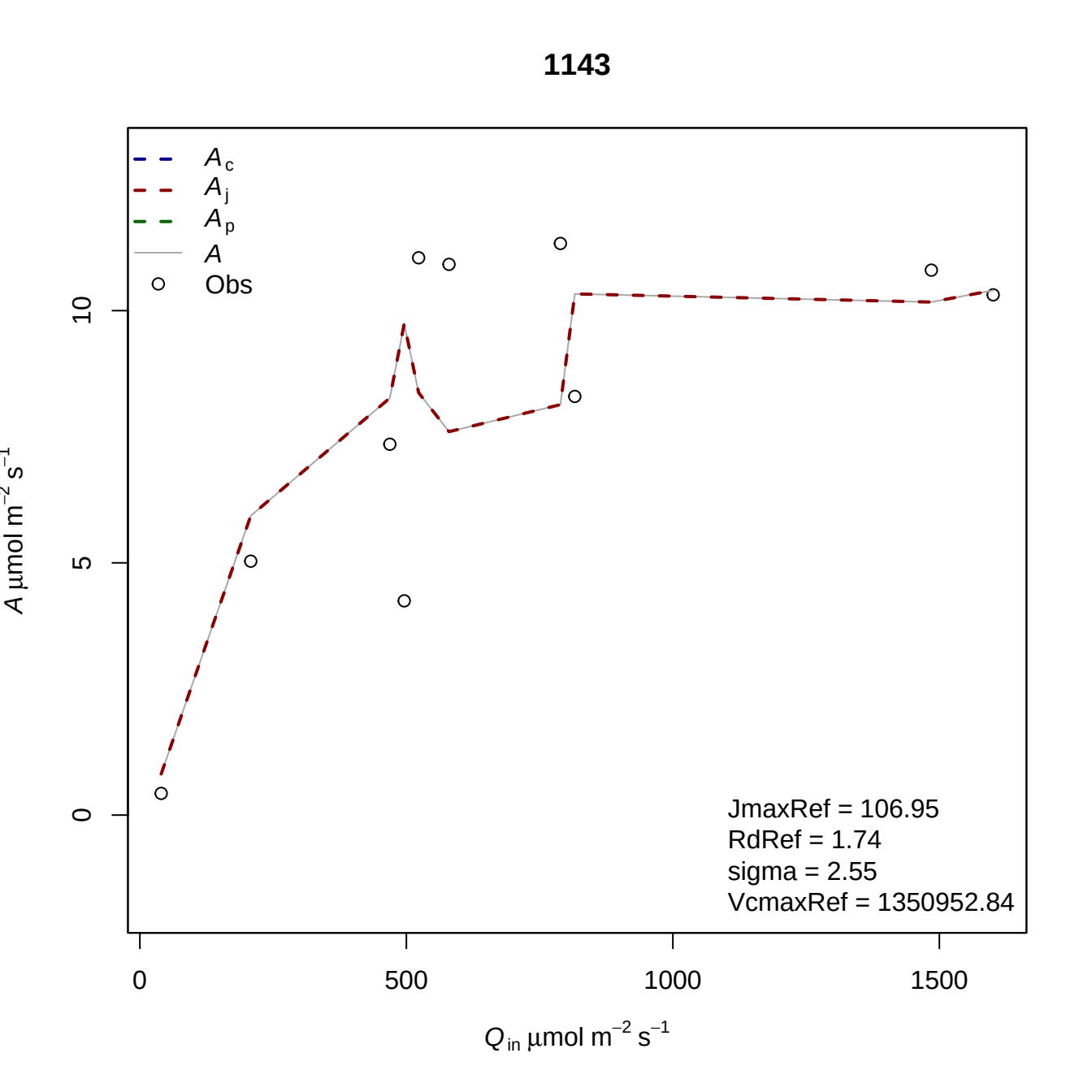

1133
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.46
RdRef = 0.96
sigma = 0.48
VcmaxRef = 10.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

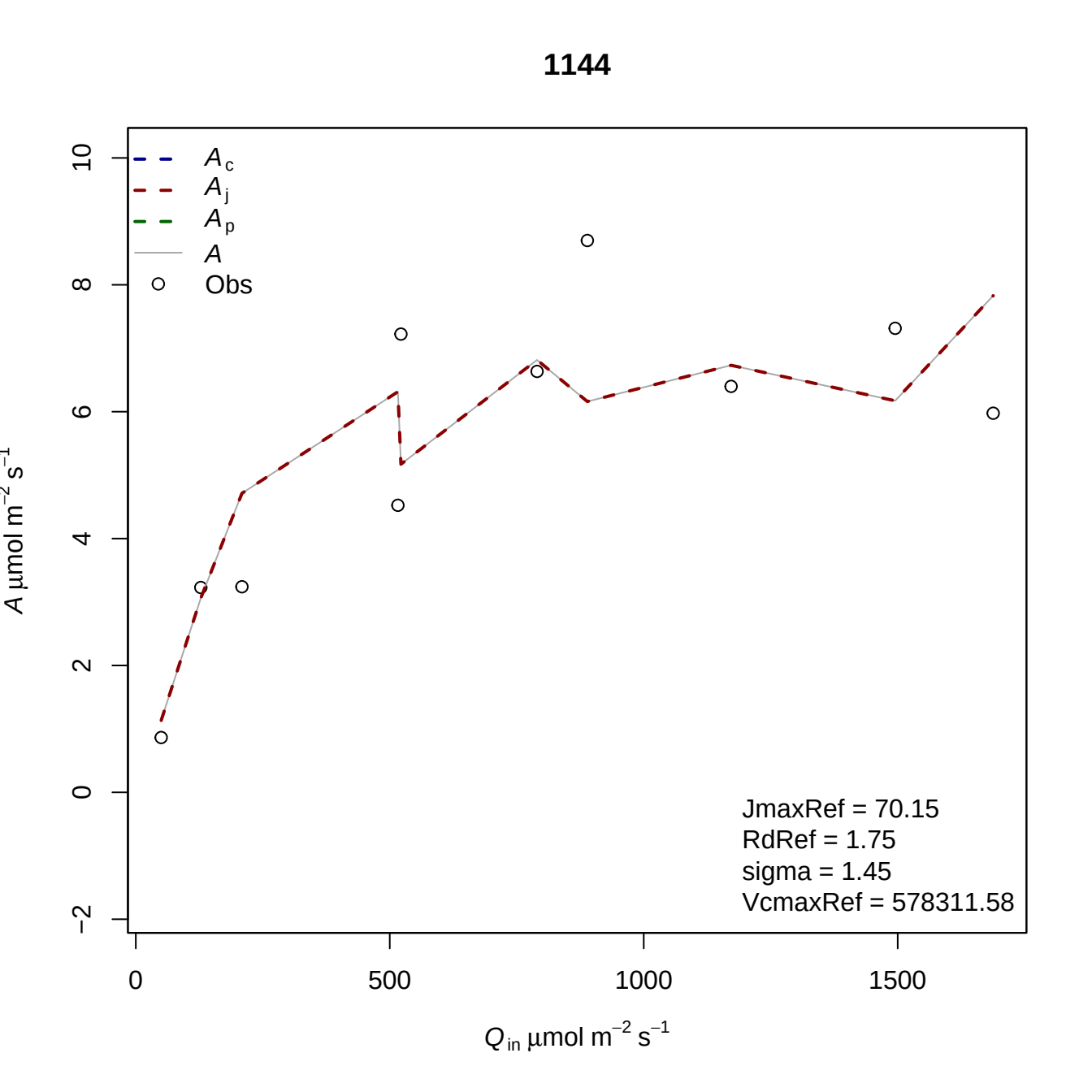

1139
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.69
RdRef = 6.59
sigma = 1.27
VcmaxRef = 2281169.91
10
5
0
−5
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

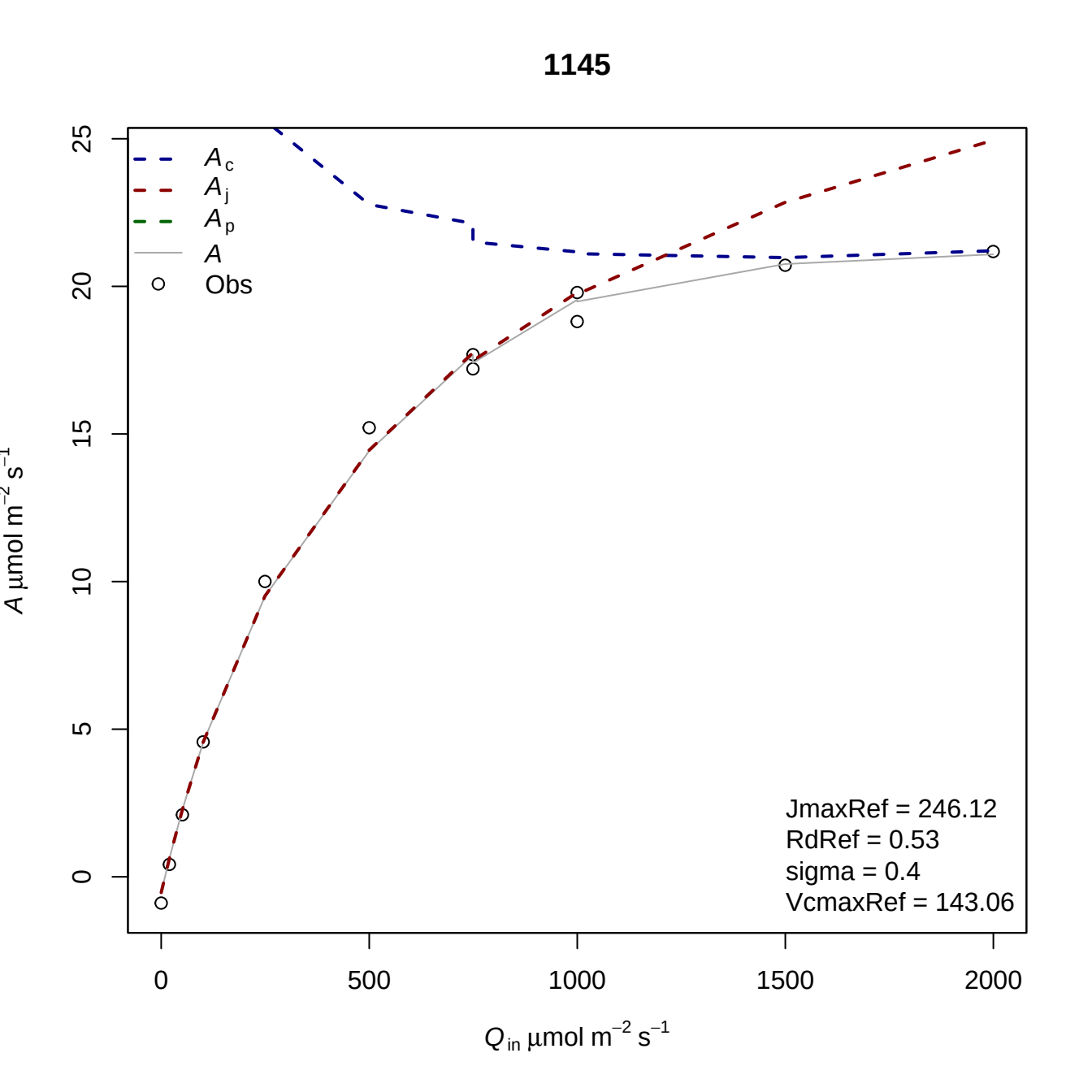

1140
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 173.53
RdRef = 2.02
sigma = 2.33
VcmaxRef = 1043178.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

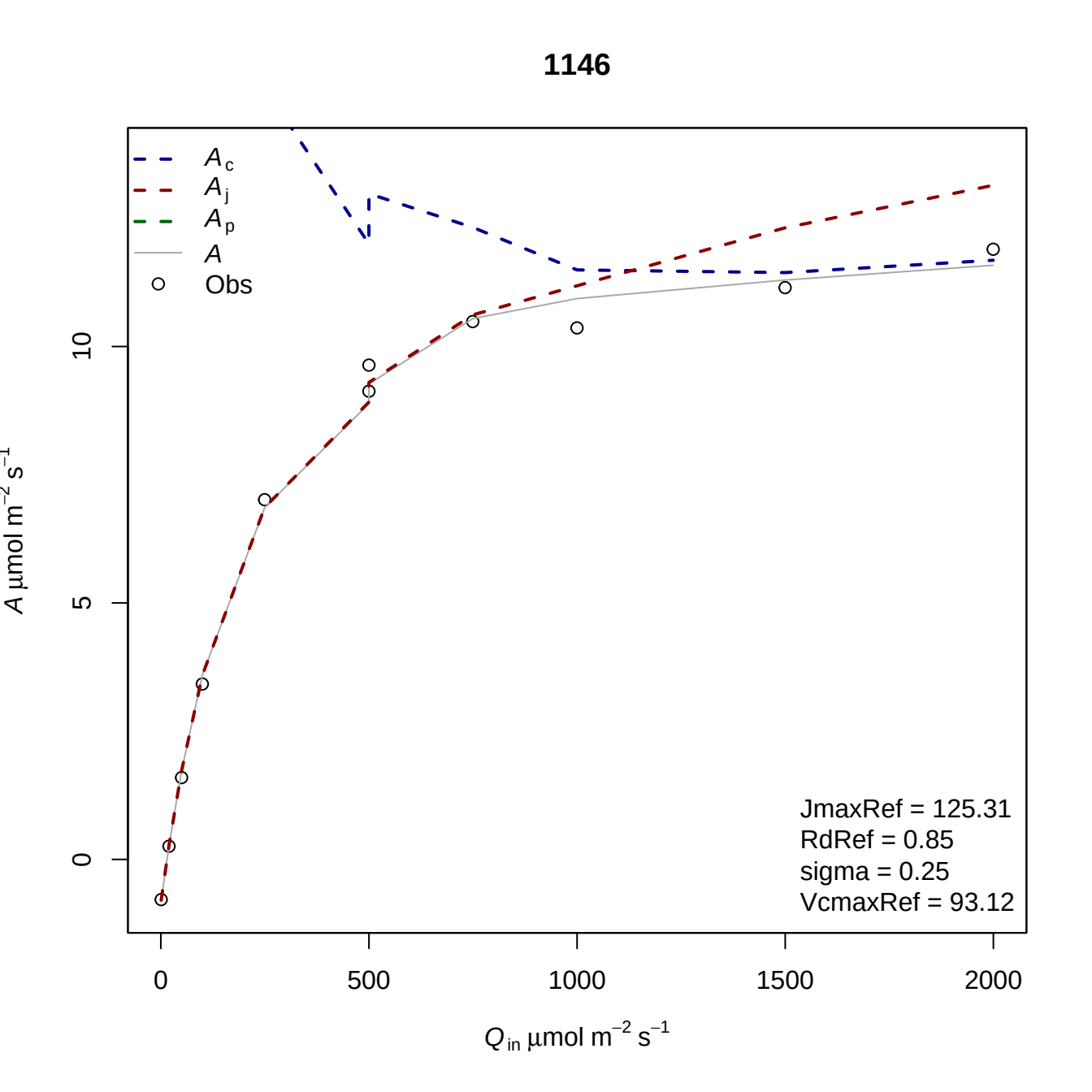

1141
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.43
RdRef = 1.53
sigma = 1.28
VcmaxRef = 574854.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

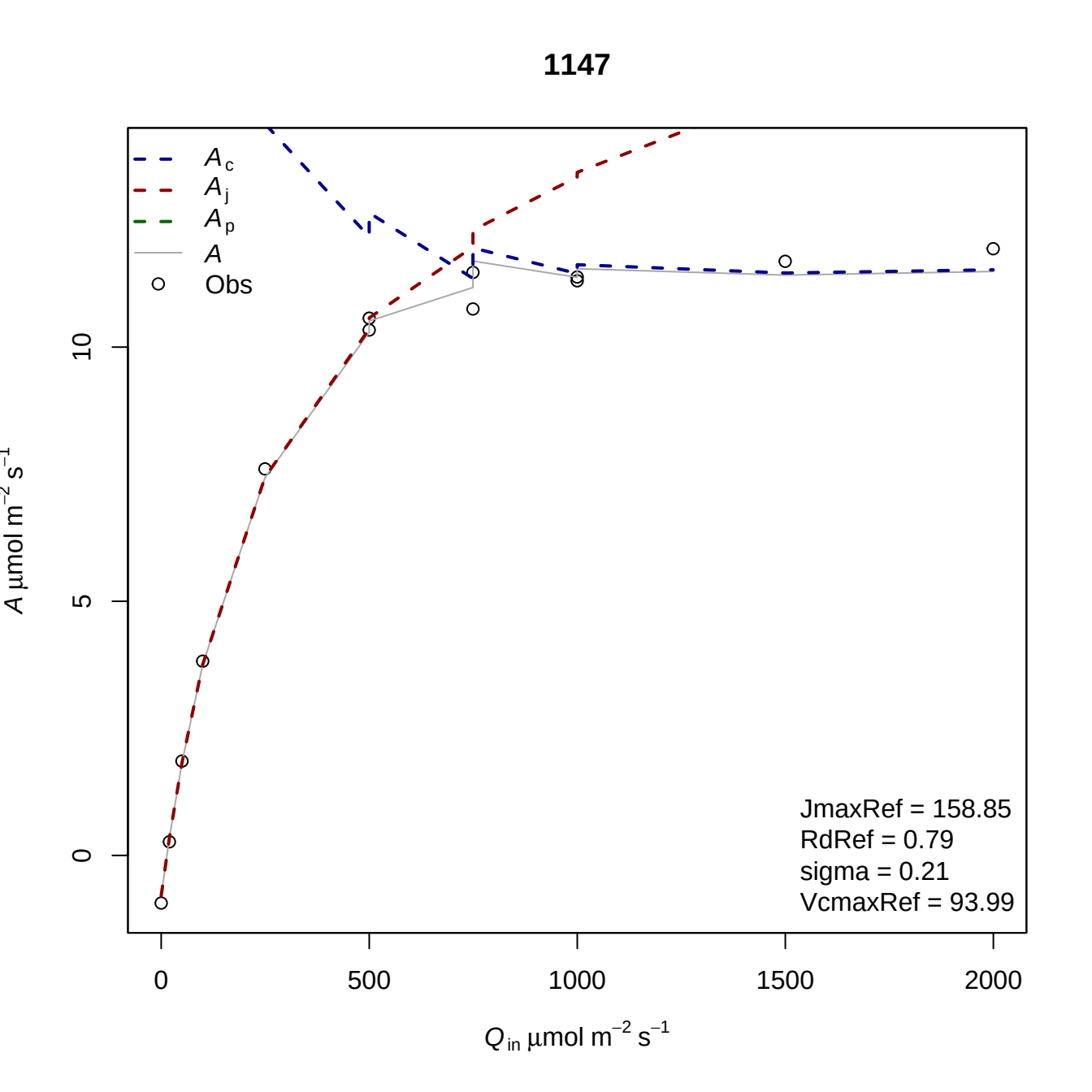

1142
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.6
RdRef = 1.55
sigma = 1.02
VcmaxRef = 583161.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

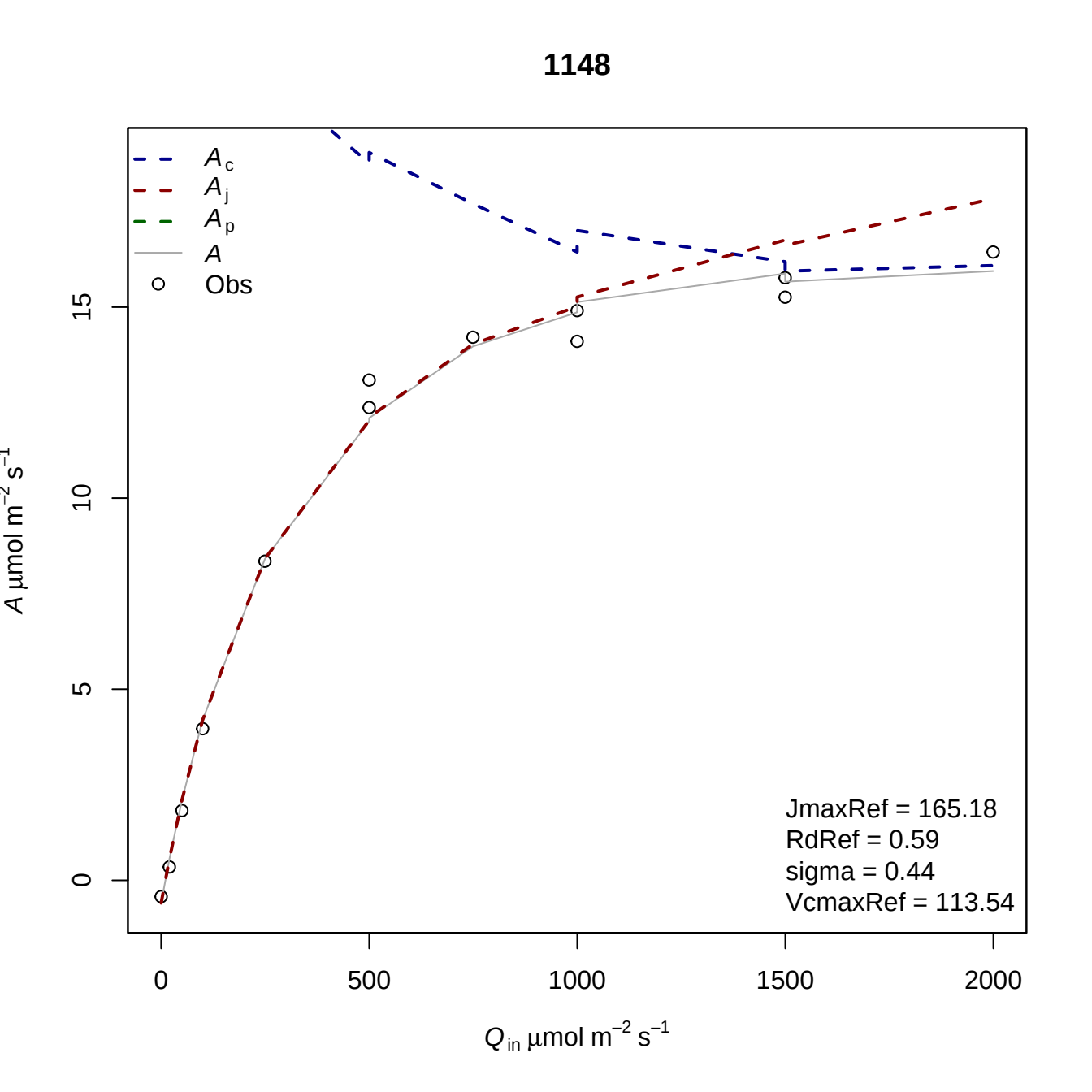

1143
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 106.95
RdRef = 1.74
sigma = 2.55
VcmaxRef = 1350952.84
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

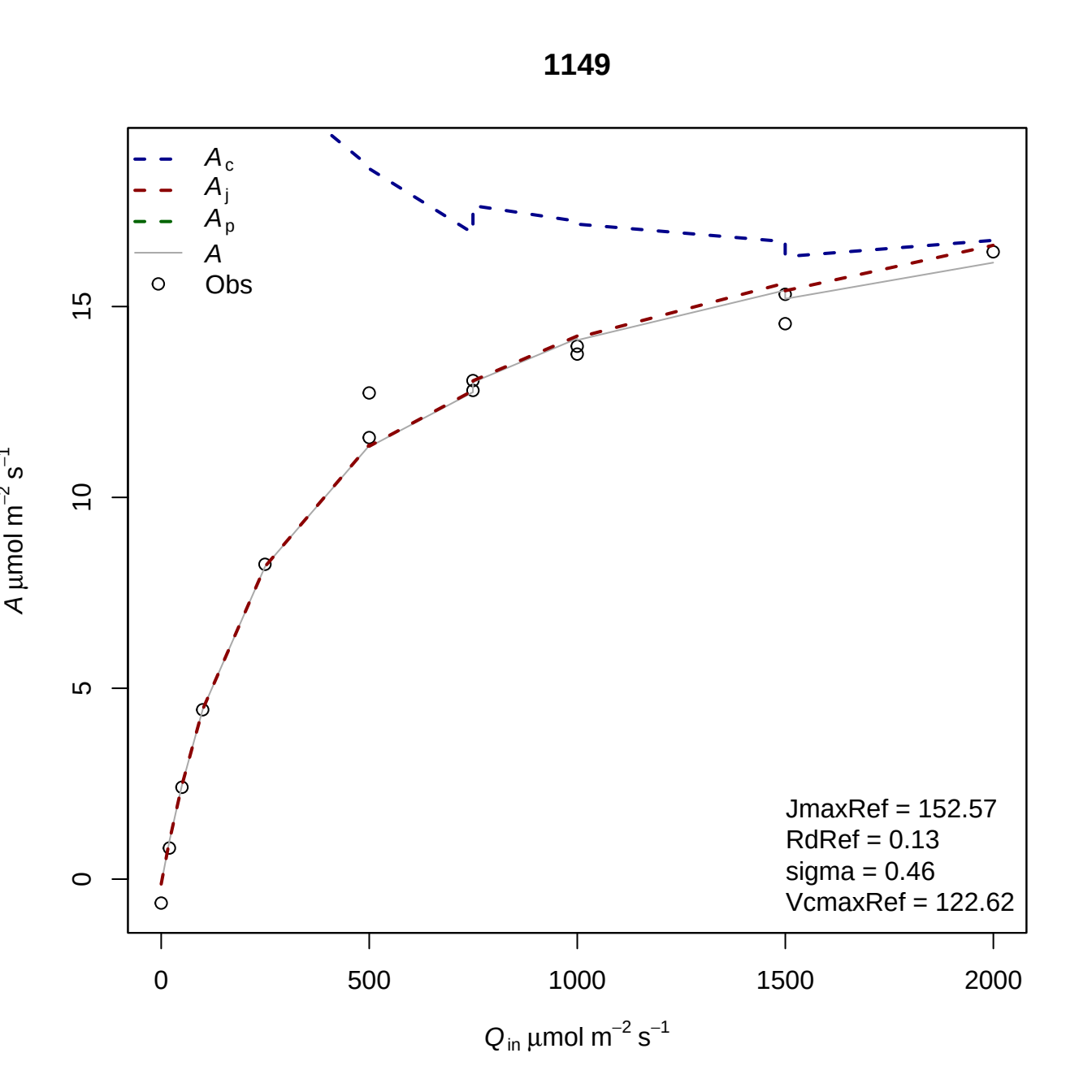

1144
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.15
RdRef = 1.75
sigma = 1.45
VcmaxRef = 578311.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

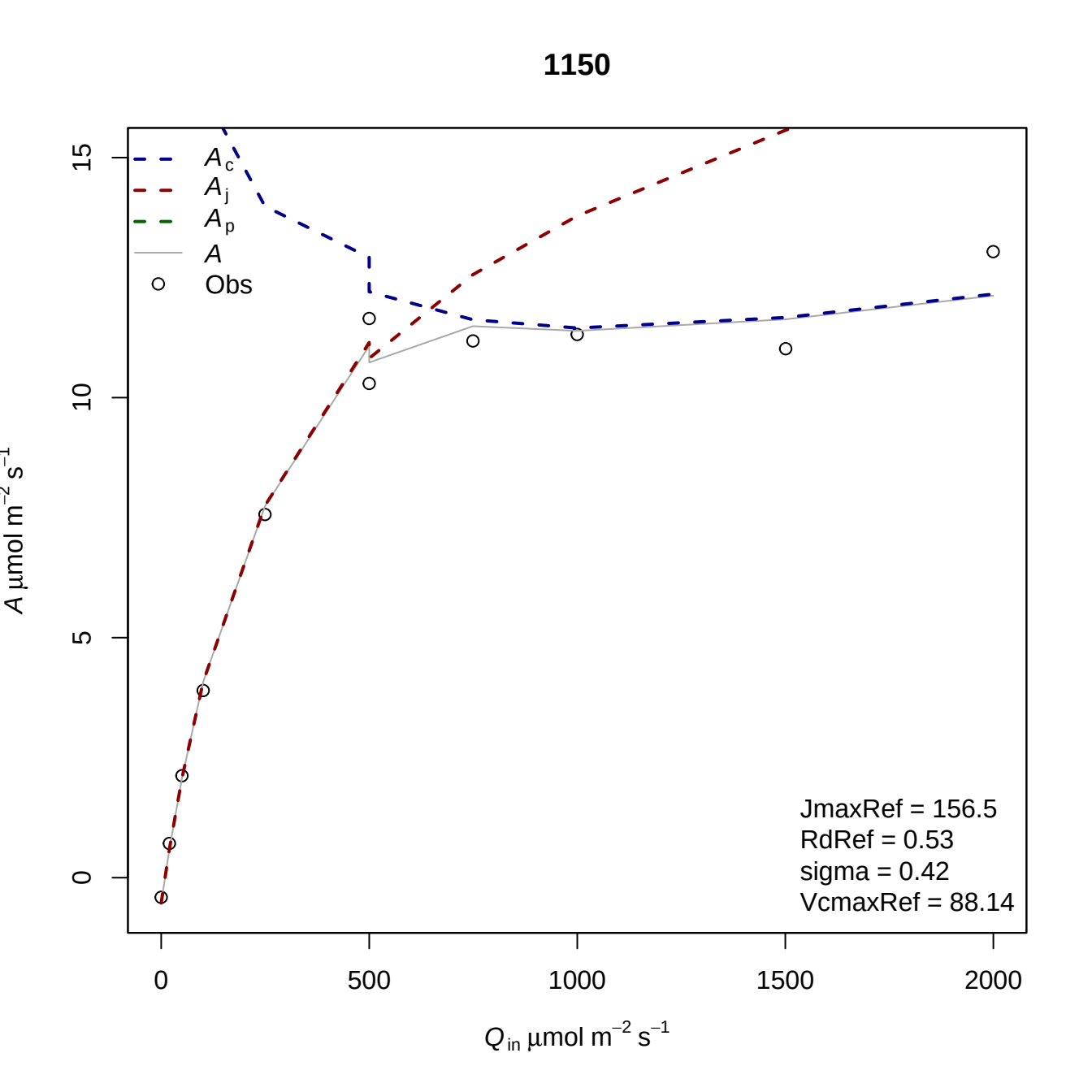

1145
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 246.12
RdRef = 0.53
sigma = 0.4
VcmaxRef = 143.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

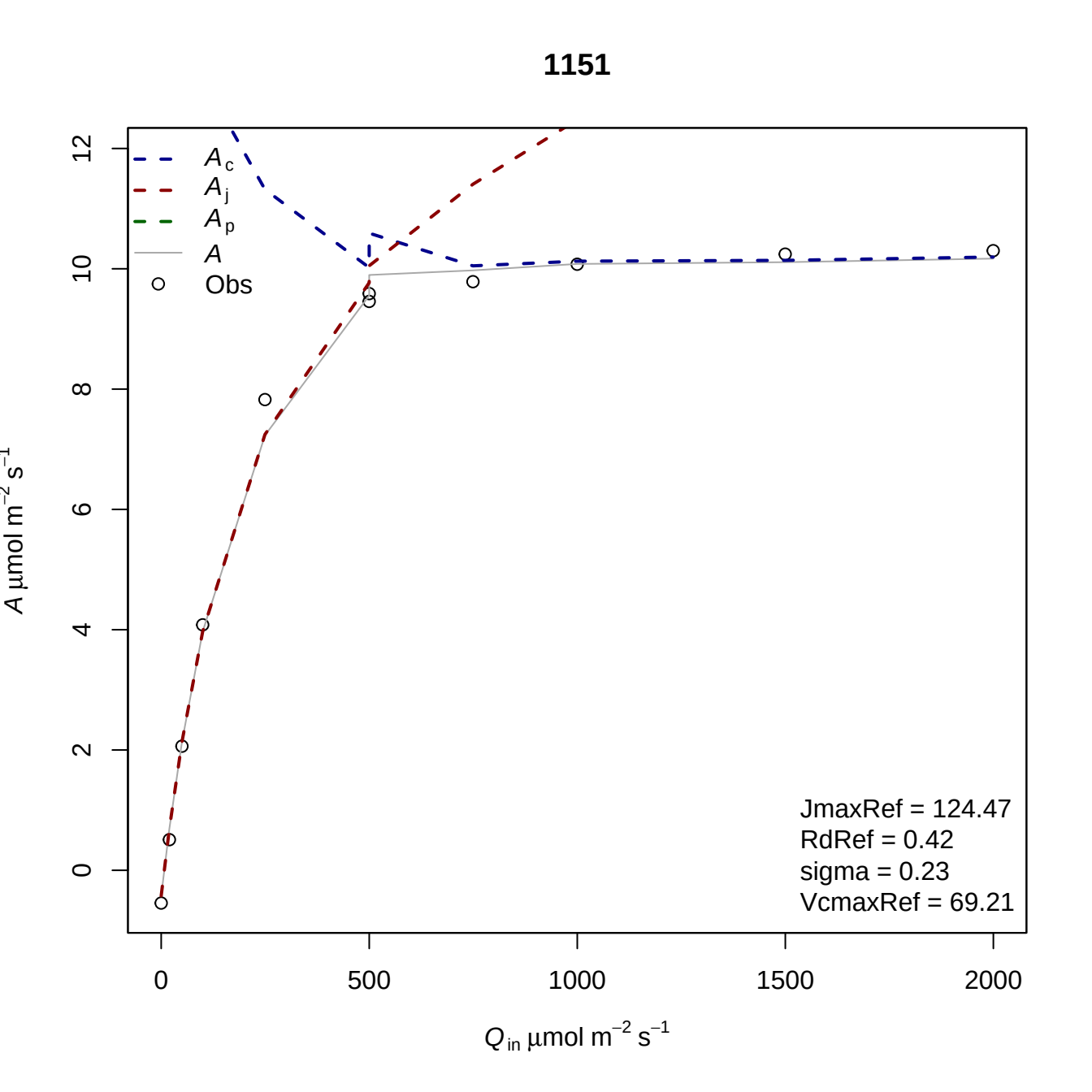

1146
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.31
RdRef = 0.85
sigma = 0.25
VcmaxRef = 93.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

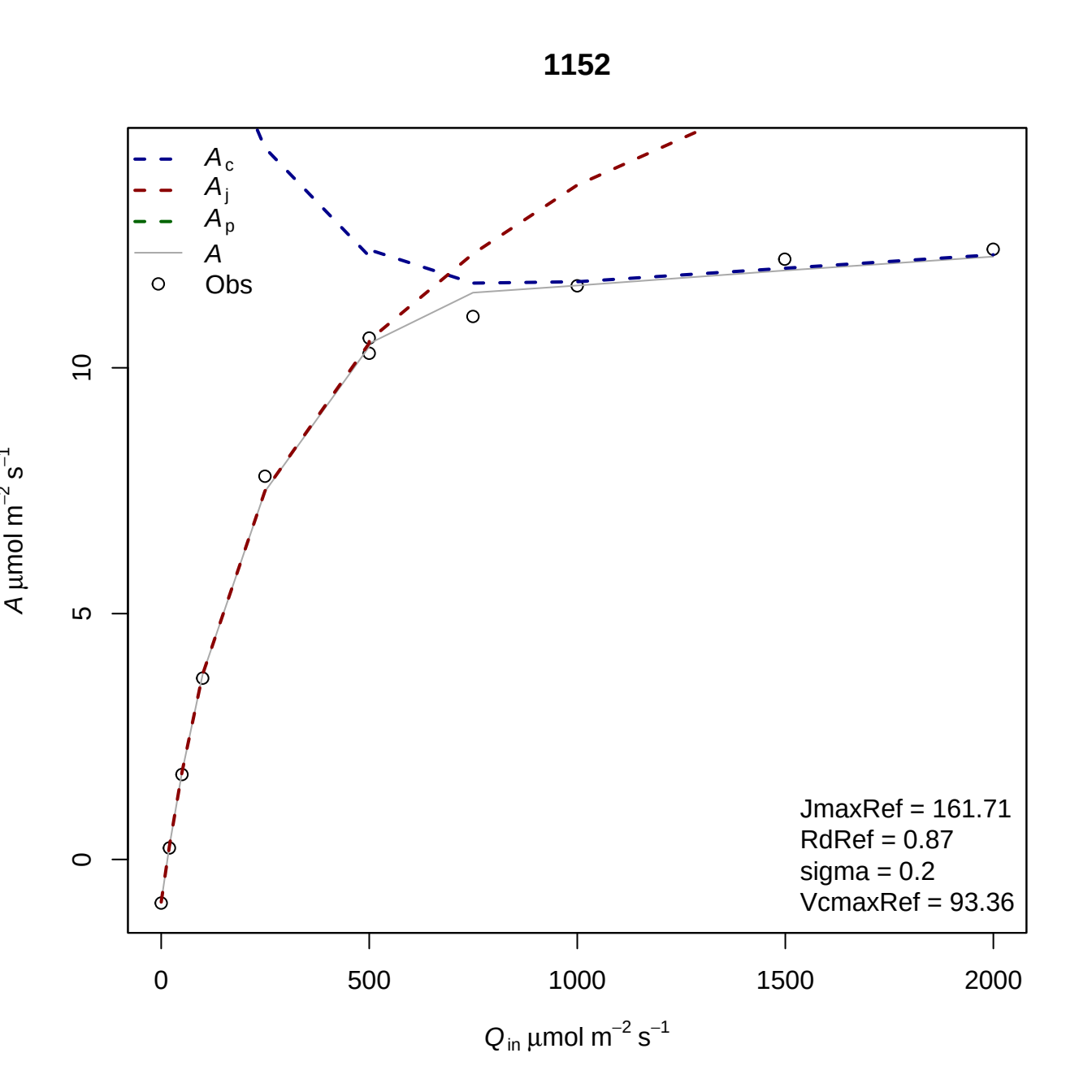

1147
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.85
RdRef = 0.79
sigma = 0.21
VcmaxRef = 93.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

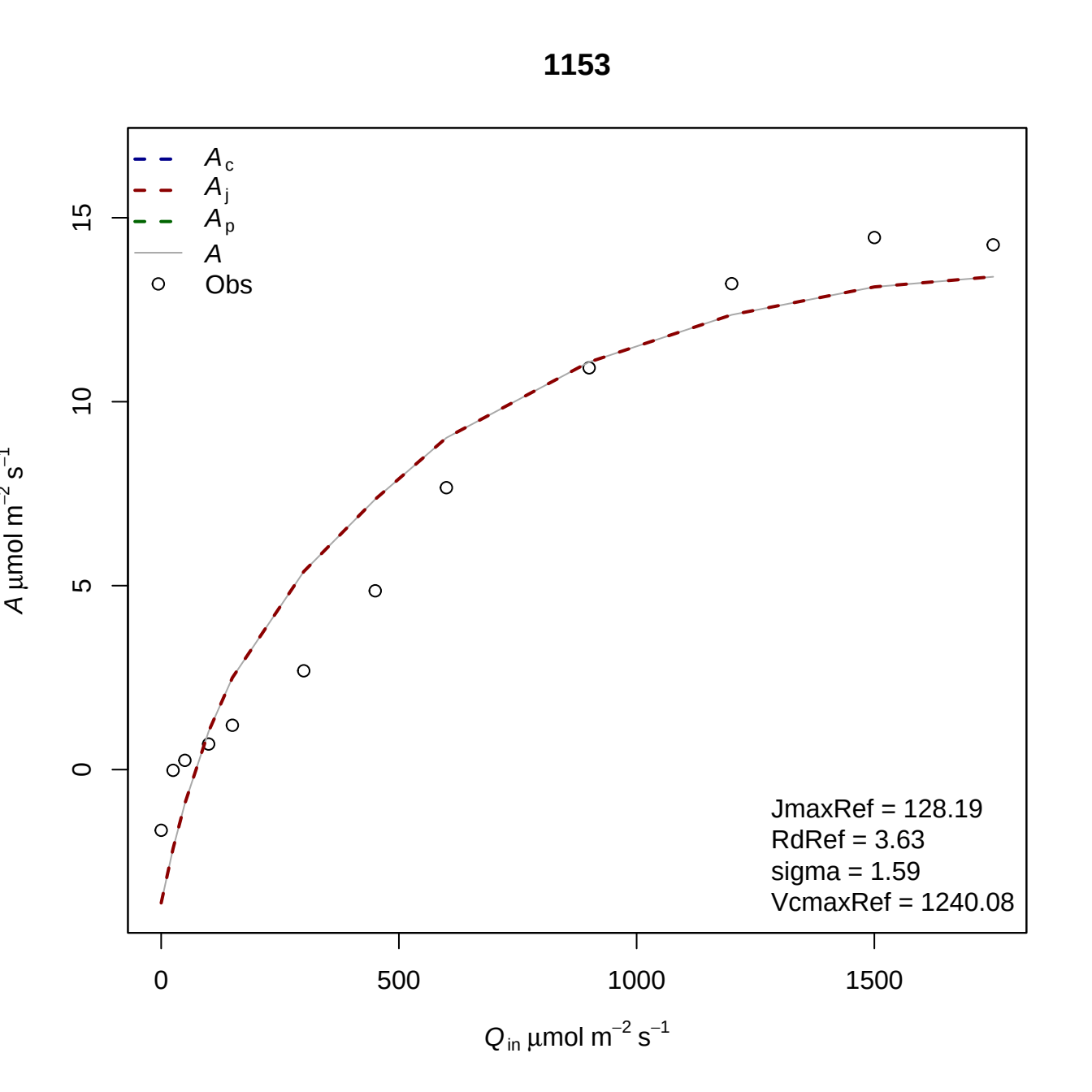

1148
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 165.18
RdRef = 0.59
sigma = 0.44
VcmaxRef = 113.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

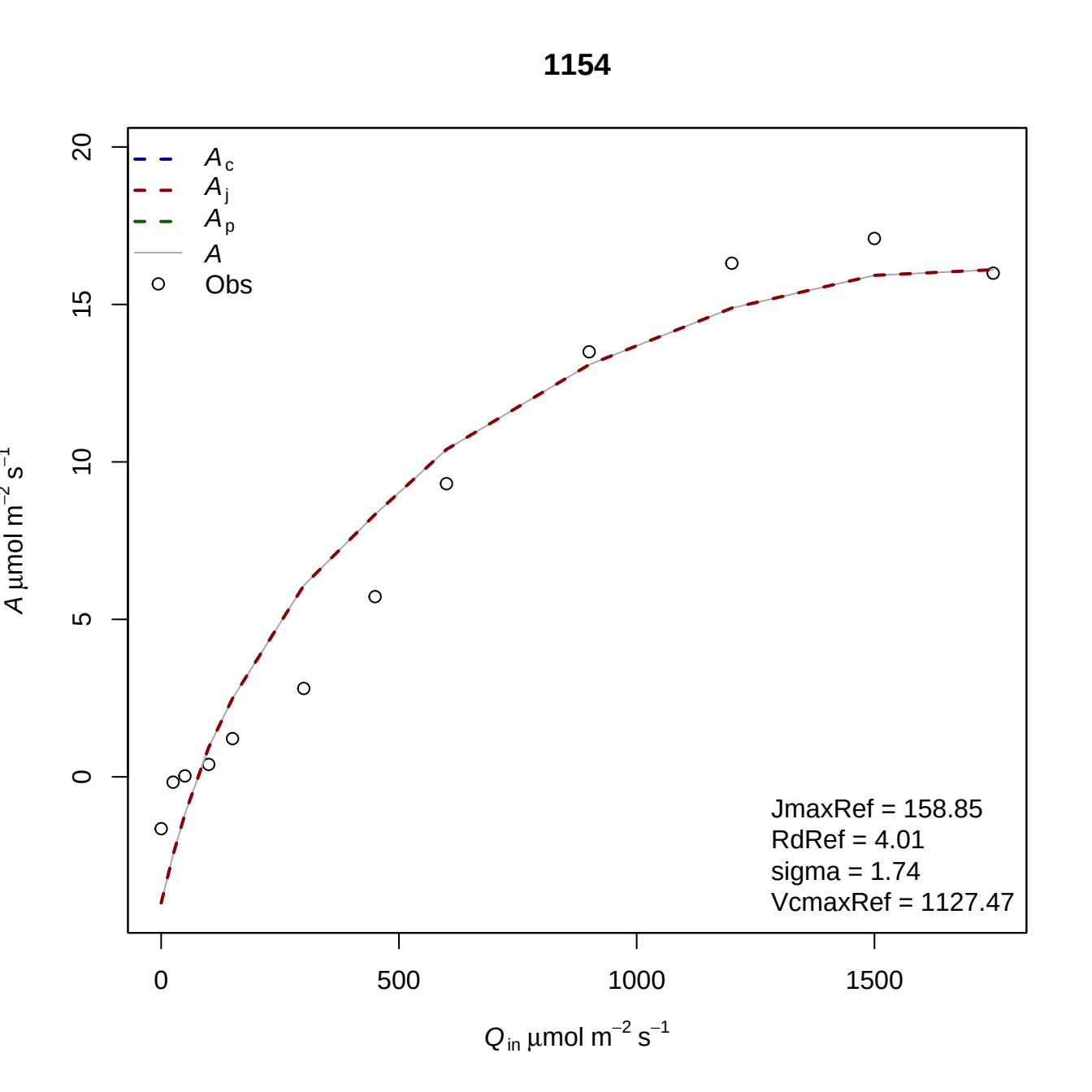

1149
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.57
RdRef = 0.13
sigma = 0.46
VcmaxRef = 122.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

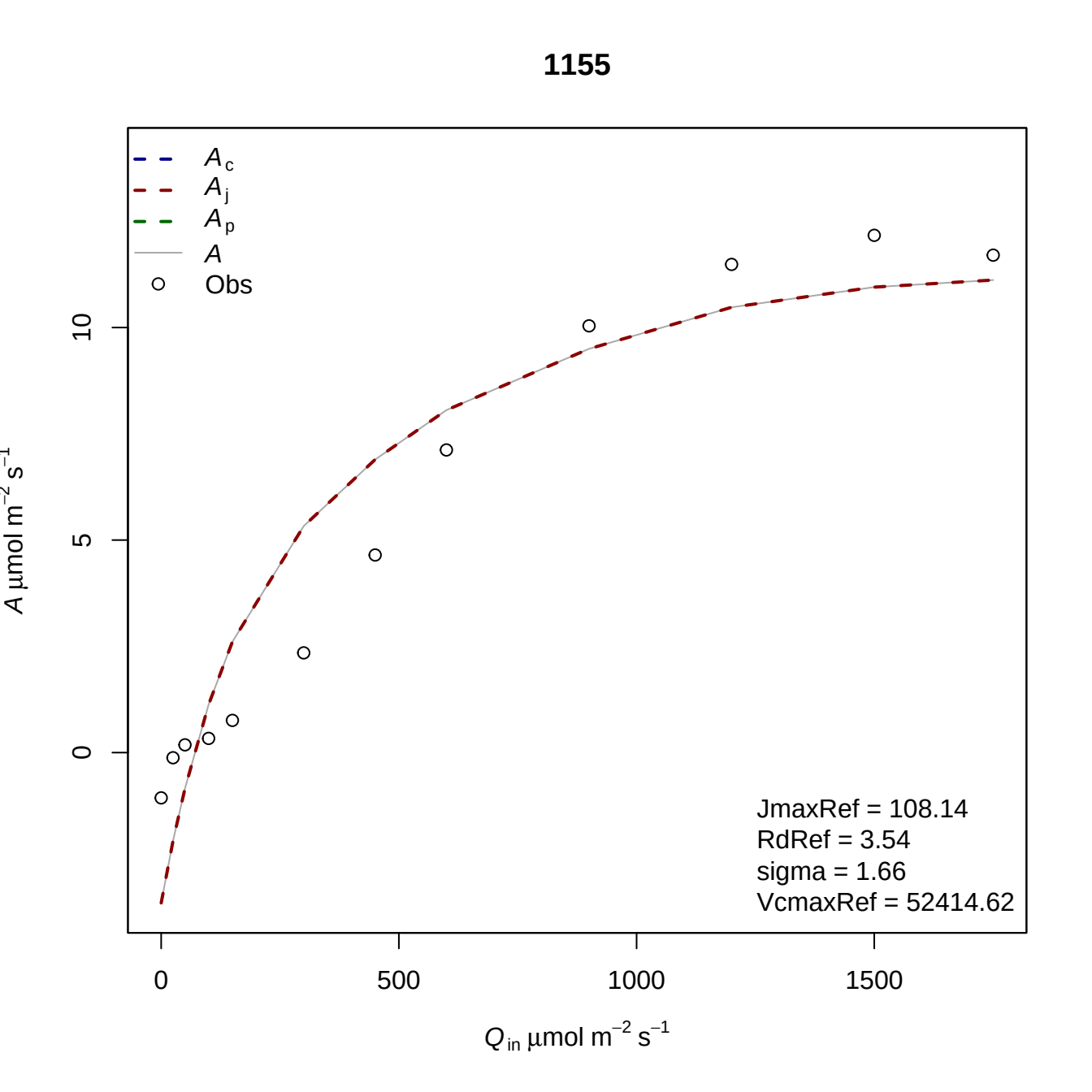

1150
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 156.5
RdRef = 0.53
sigma = 0.42
VcmaxRef = 88.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

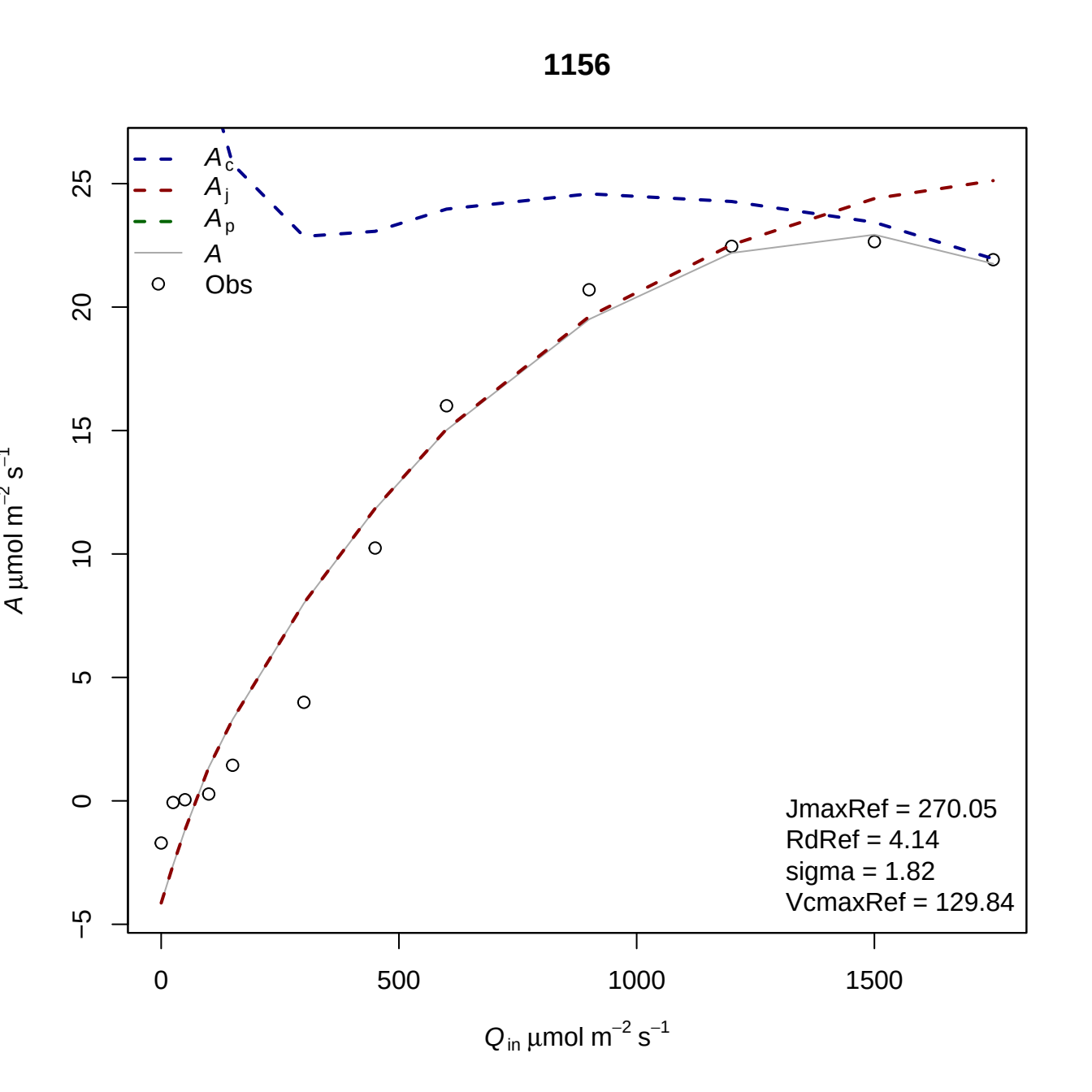

1151
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.47
RdRef = 0.42
sigma = 0.23
VcmaxRef = 69.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

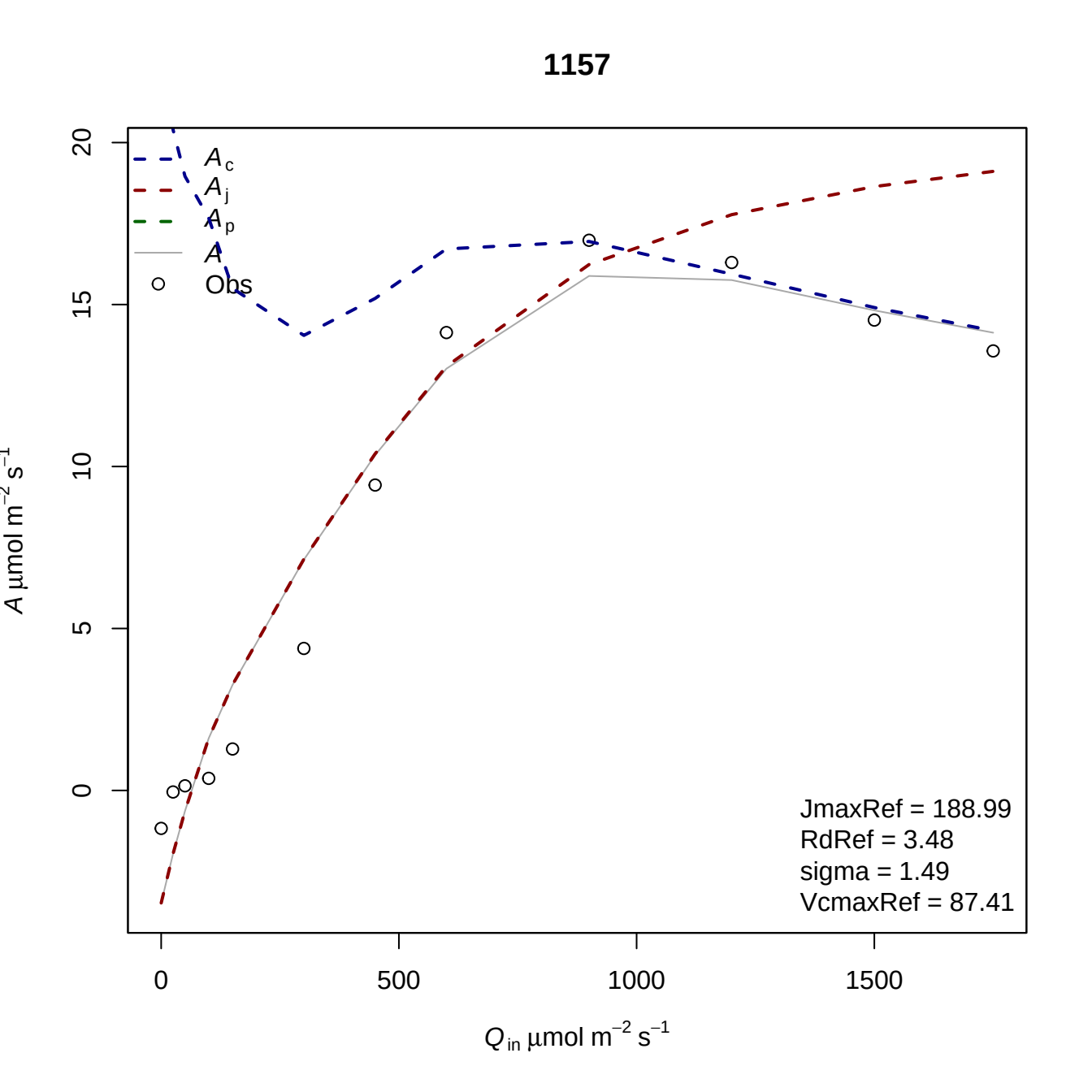

1152
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.71
RdRef = 0.87
sigma = 0.2
VcmaxRef = 93.36
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

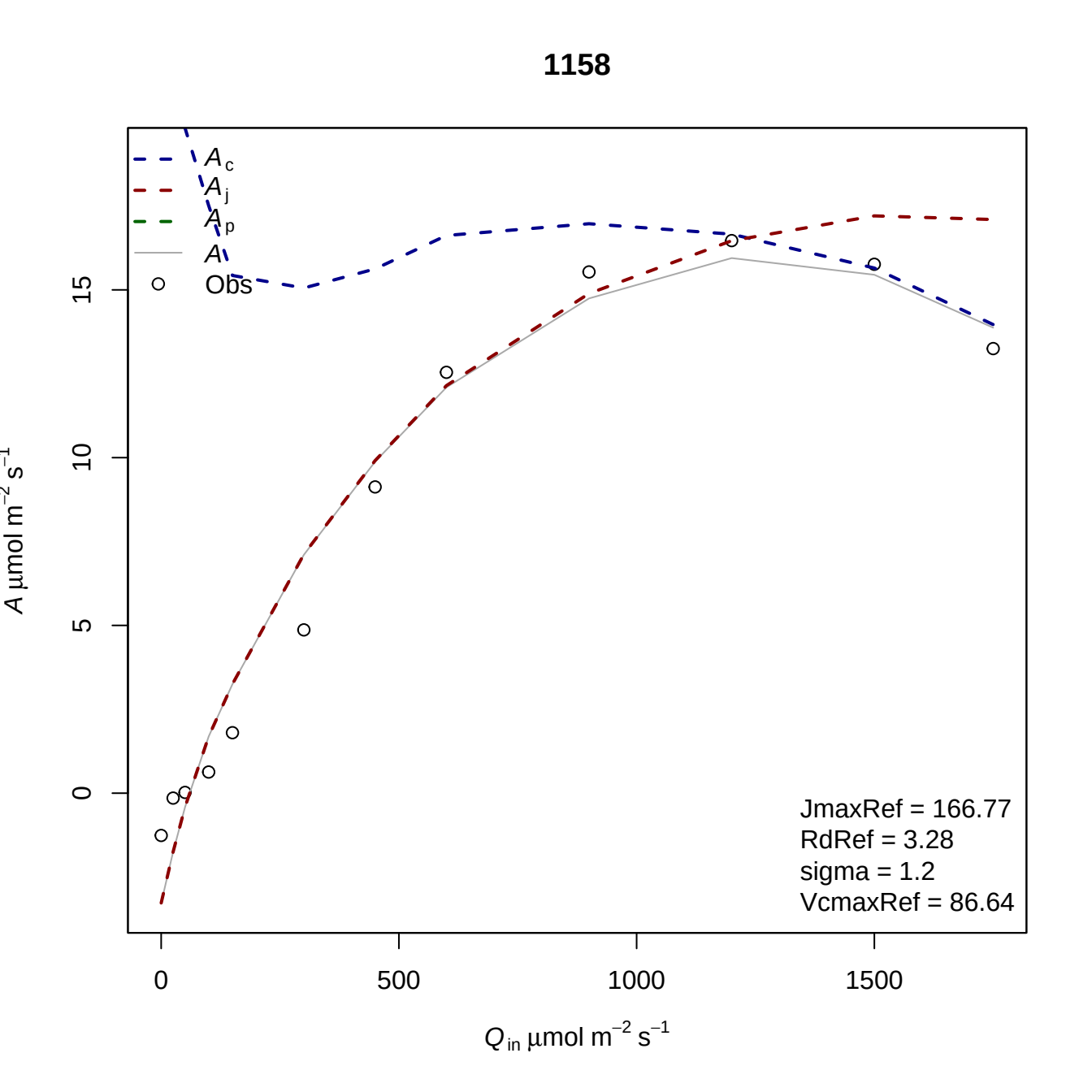

1153
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.19
RdRef = 3.63
sigma = 1.59
VcmaxRef = 1240.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

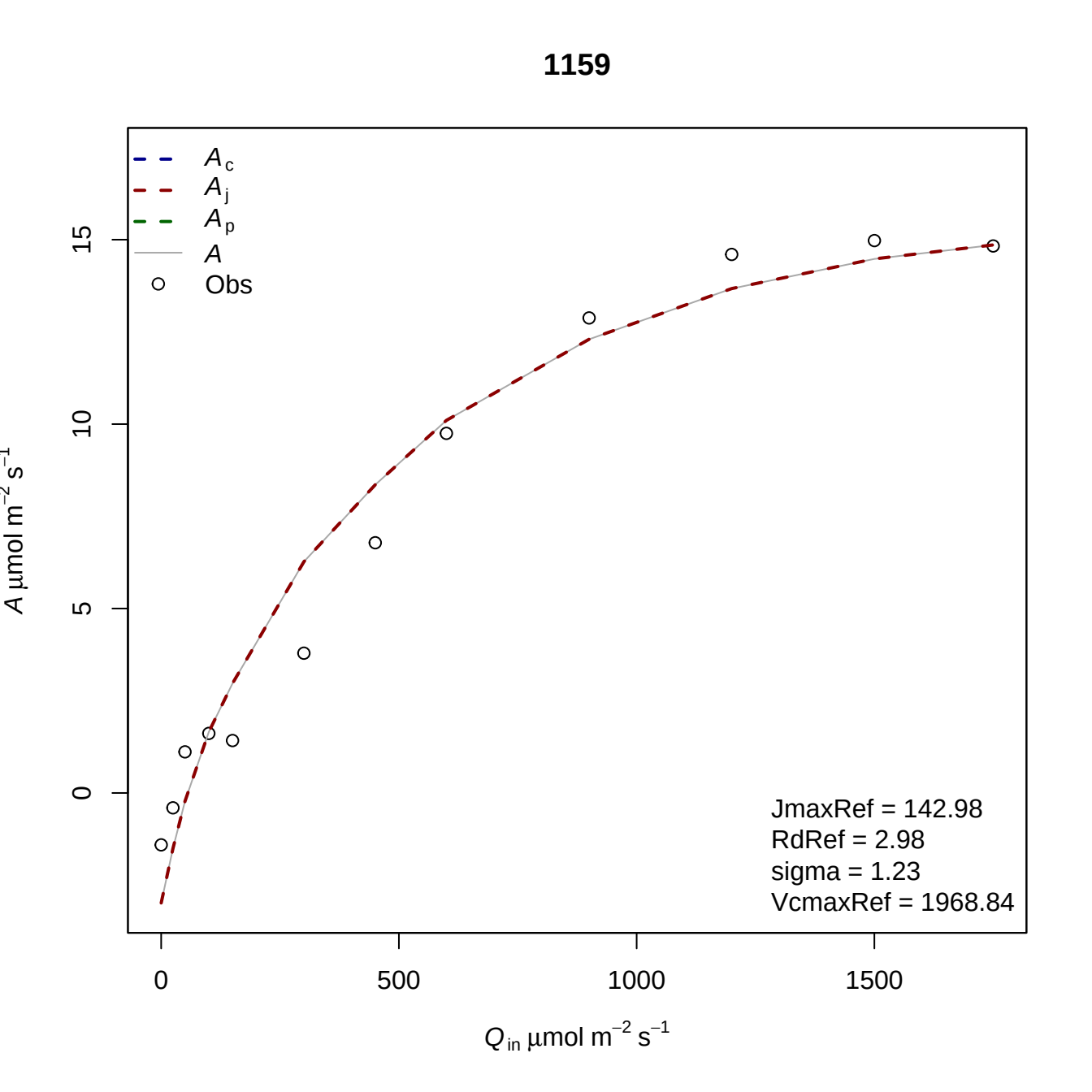

1154
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.85
RdRef = 4.01
sigma = 1.74
VcmaxRef = 1127.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

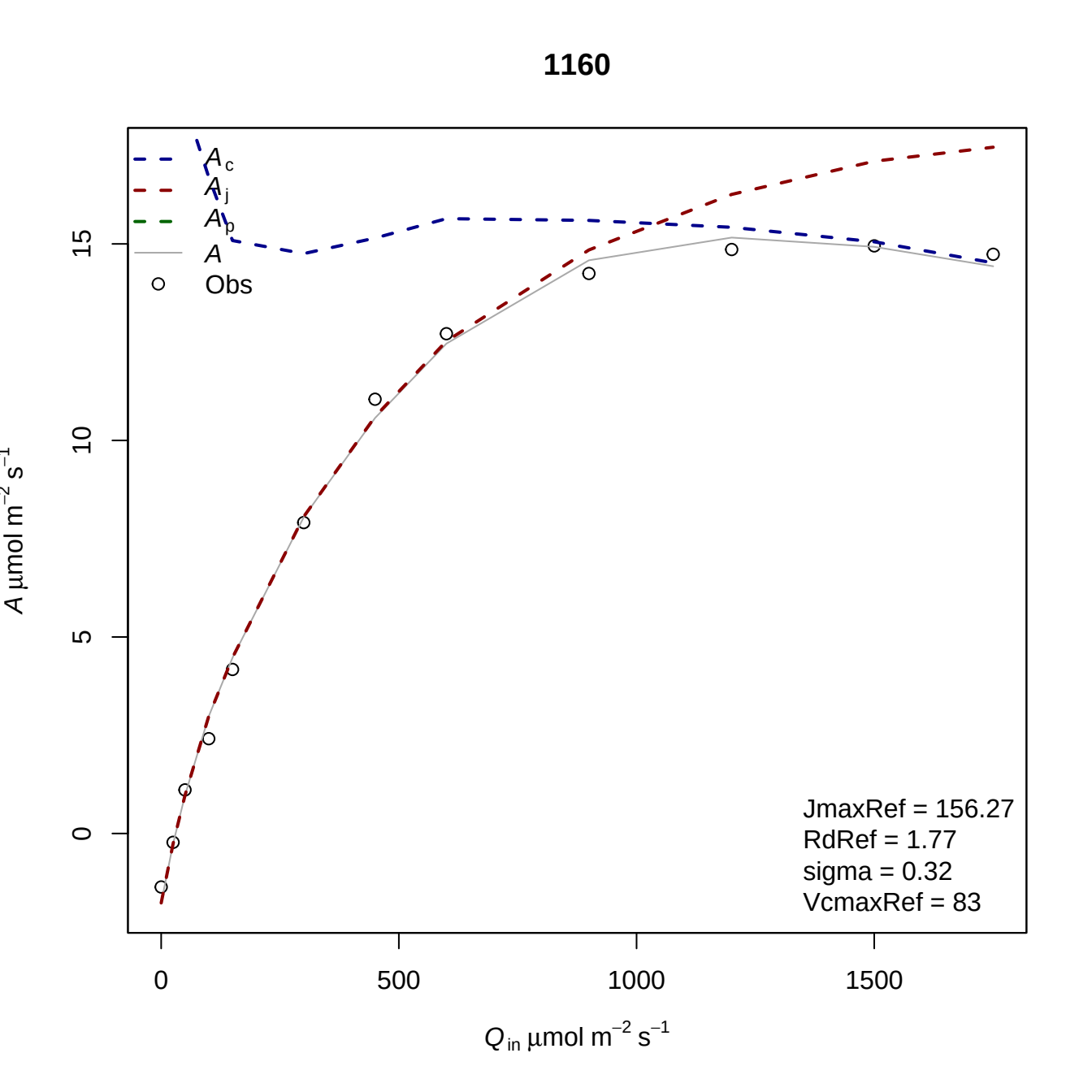

1155
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.14
RdRef = 3.54
sigma = 1.66
VcmaxRef = 52414.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

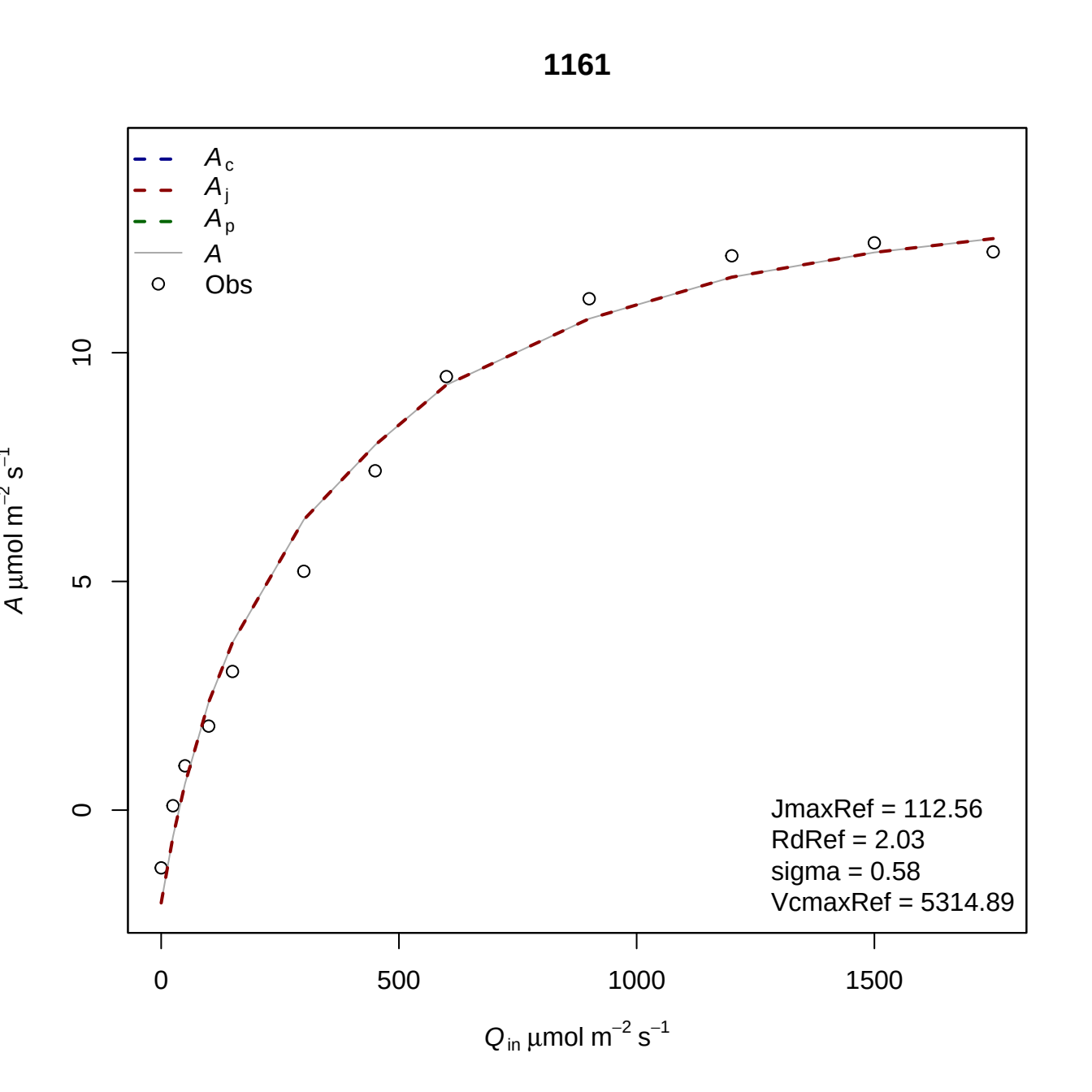

1156
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 270.05
RdRef = 4.14
sigma = 1.82
VcmaxRef = 129.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
25
0
500
1000
1500

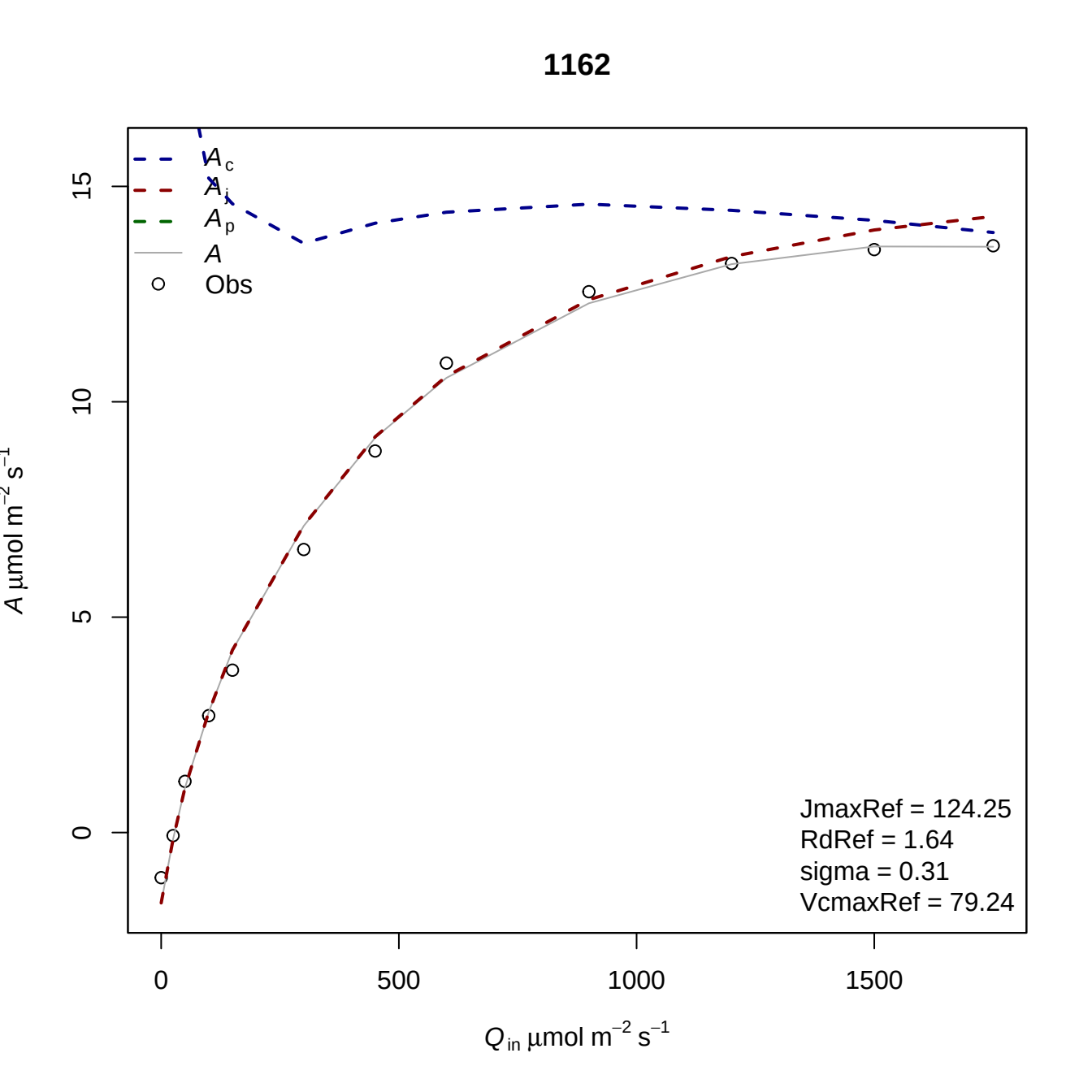

1157
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 188.99
RdRef = 3.48
sigma = 1.49
VcmaxRef = 87.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

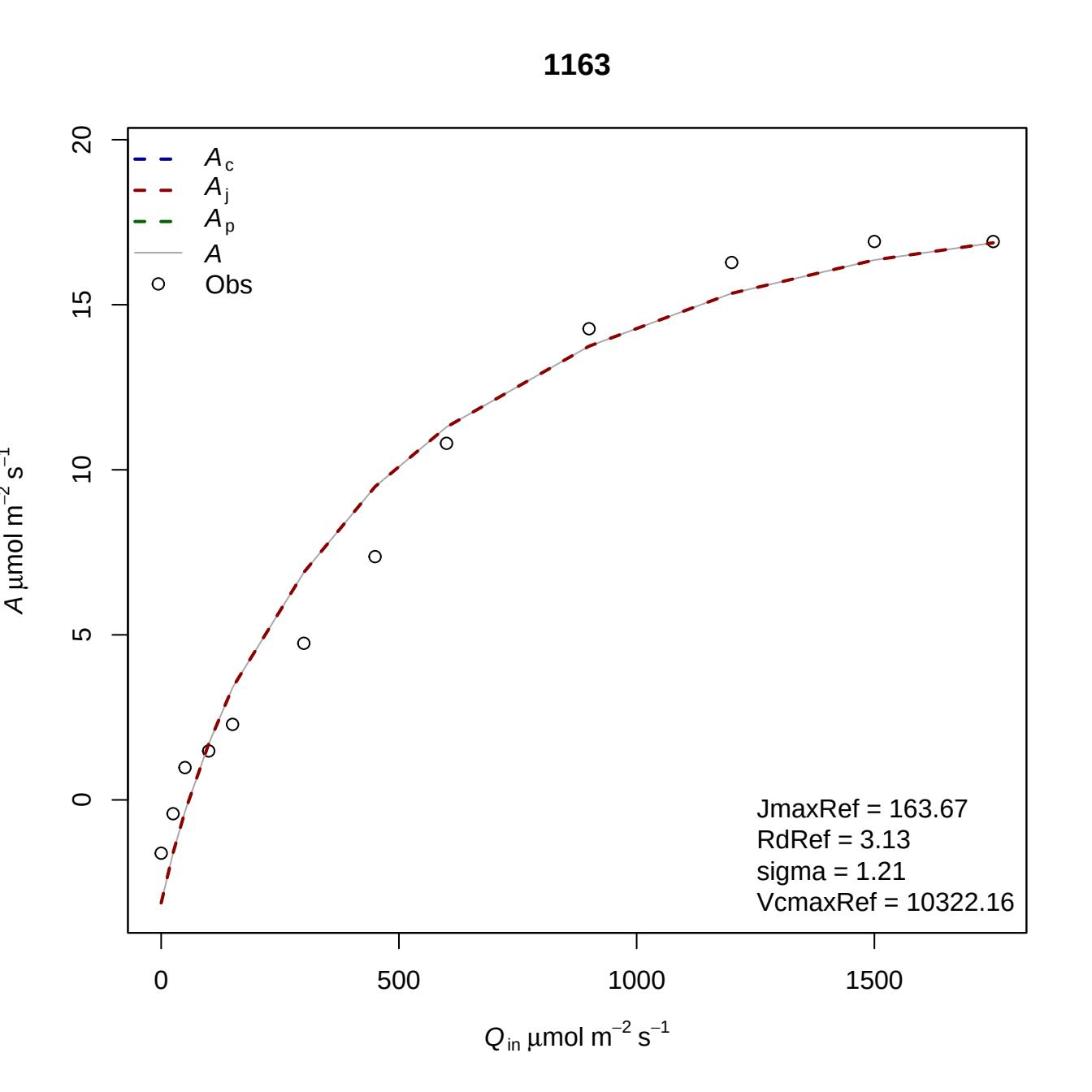

1158
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 166.77
RdRef = 3.28
sigma = 1.2
VcmaxRef = 86.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

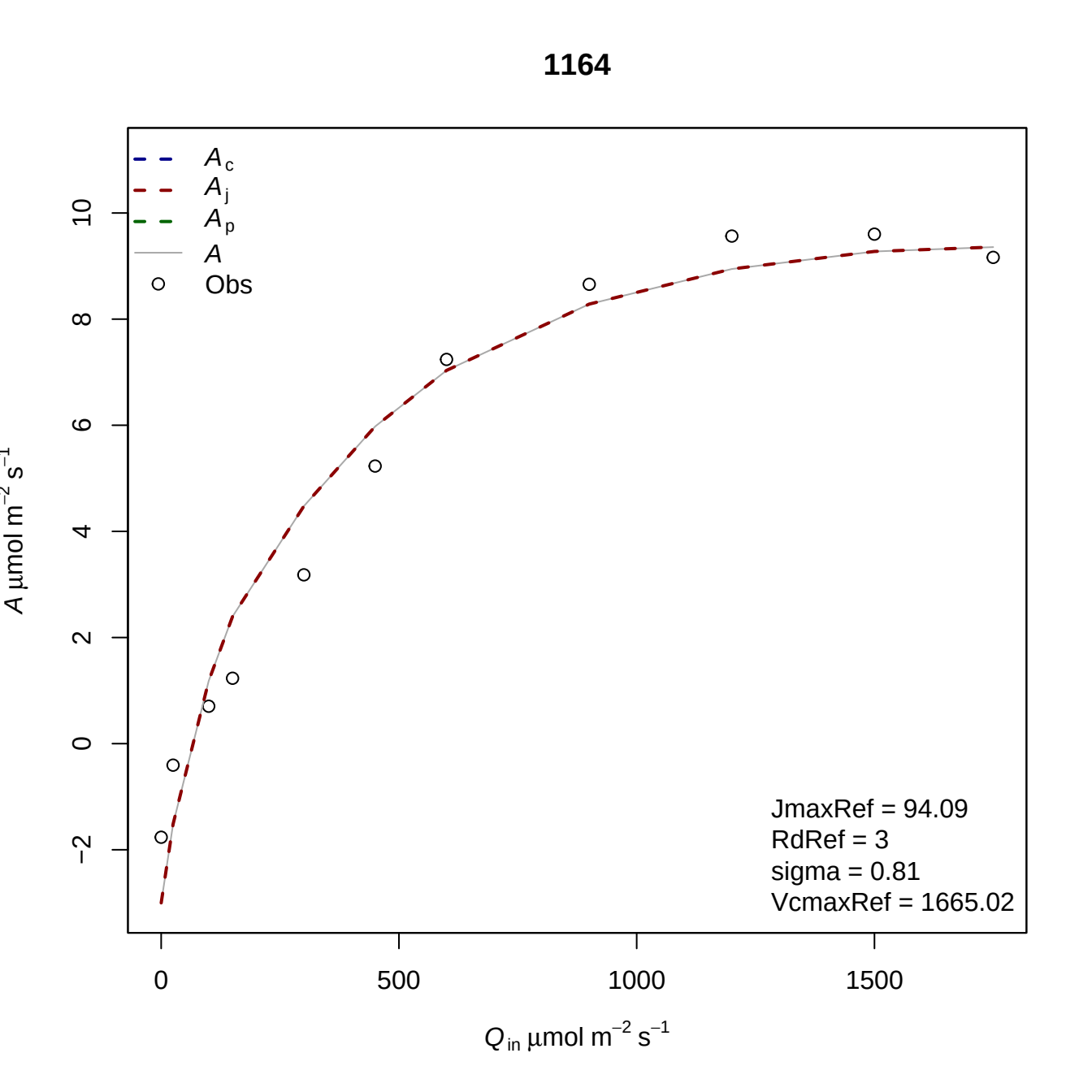

1159
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.98
RdRef = 2.98
sigma = 1.23
VcmaxRef = 1968.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

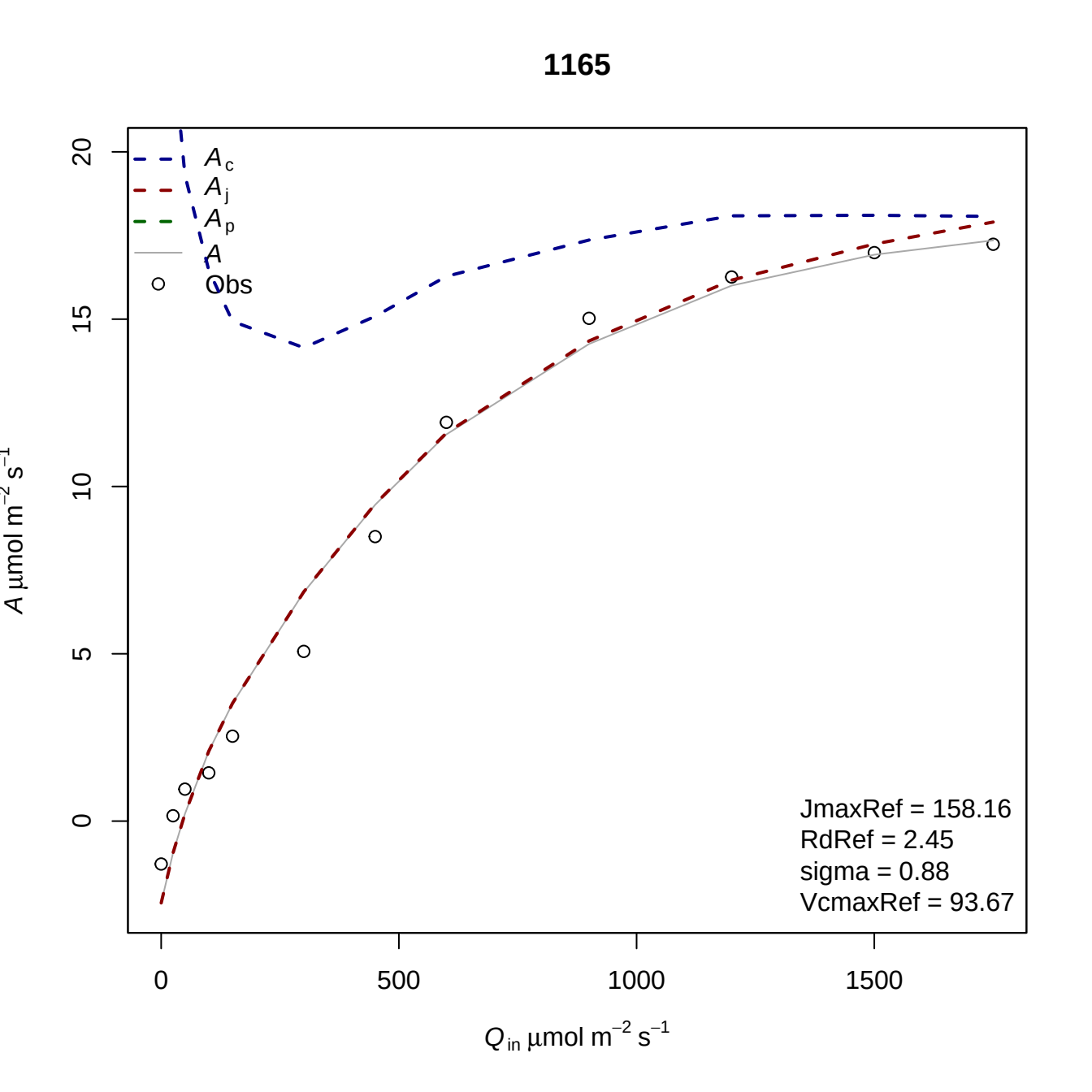

1160
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 156.27
RdRef = 1.77
sigma = 0.32
VcmaxRef = 83
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

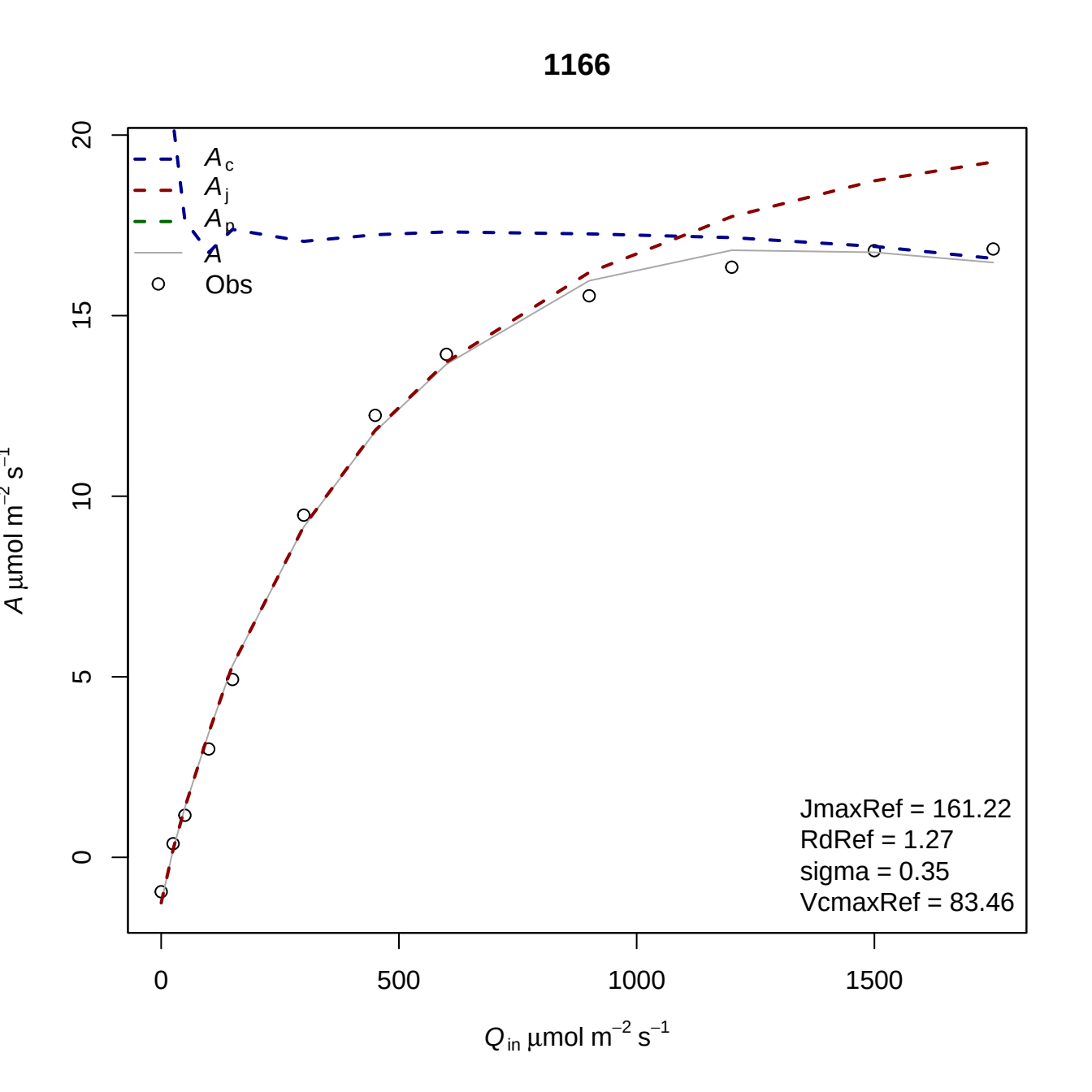

1161
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.56
RdRef = 2.03
sigma = 0.58
VcmaxRef = 5314.89
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

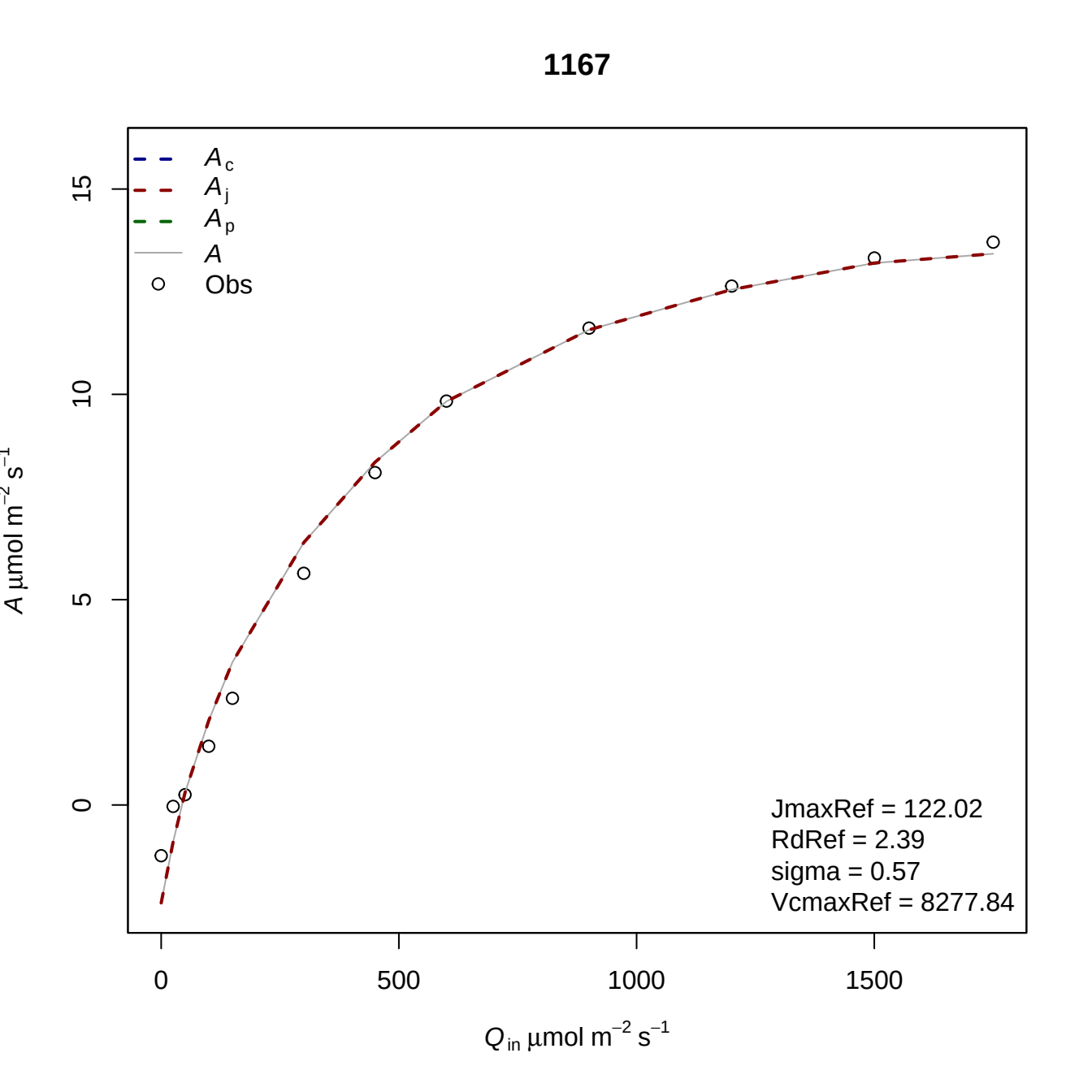

1162
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.25
RdRef = 1.64
sigma = 0.31
VcmaxRef = 79.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

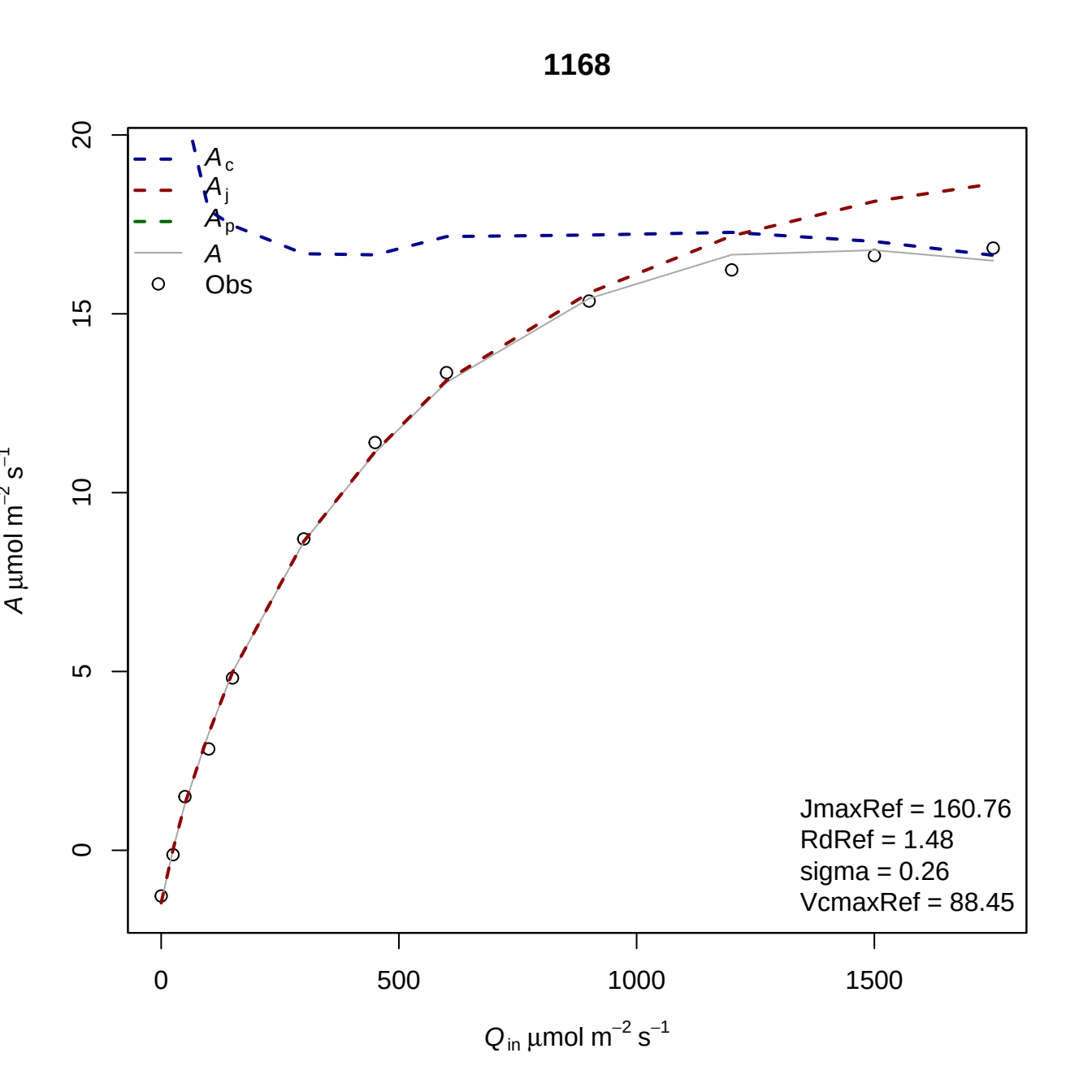

1163
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 163.67
RdRef = 3.13
sigma = 1.21
VcmaxRef = 10322.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

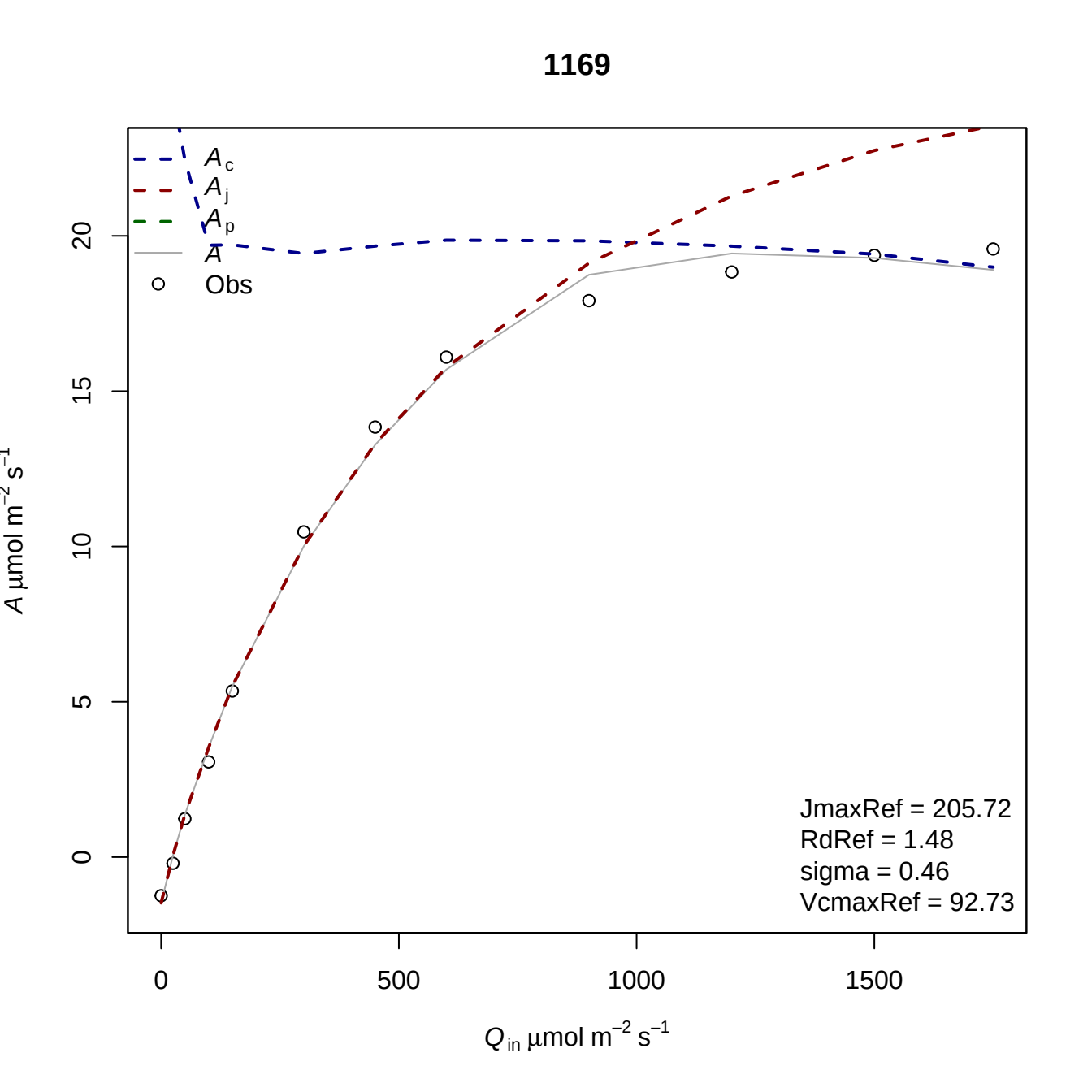

1164
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.09
RdRef = 3
sigma = 0.81
VcmaxRef = 1665.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

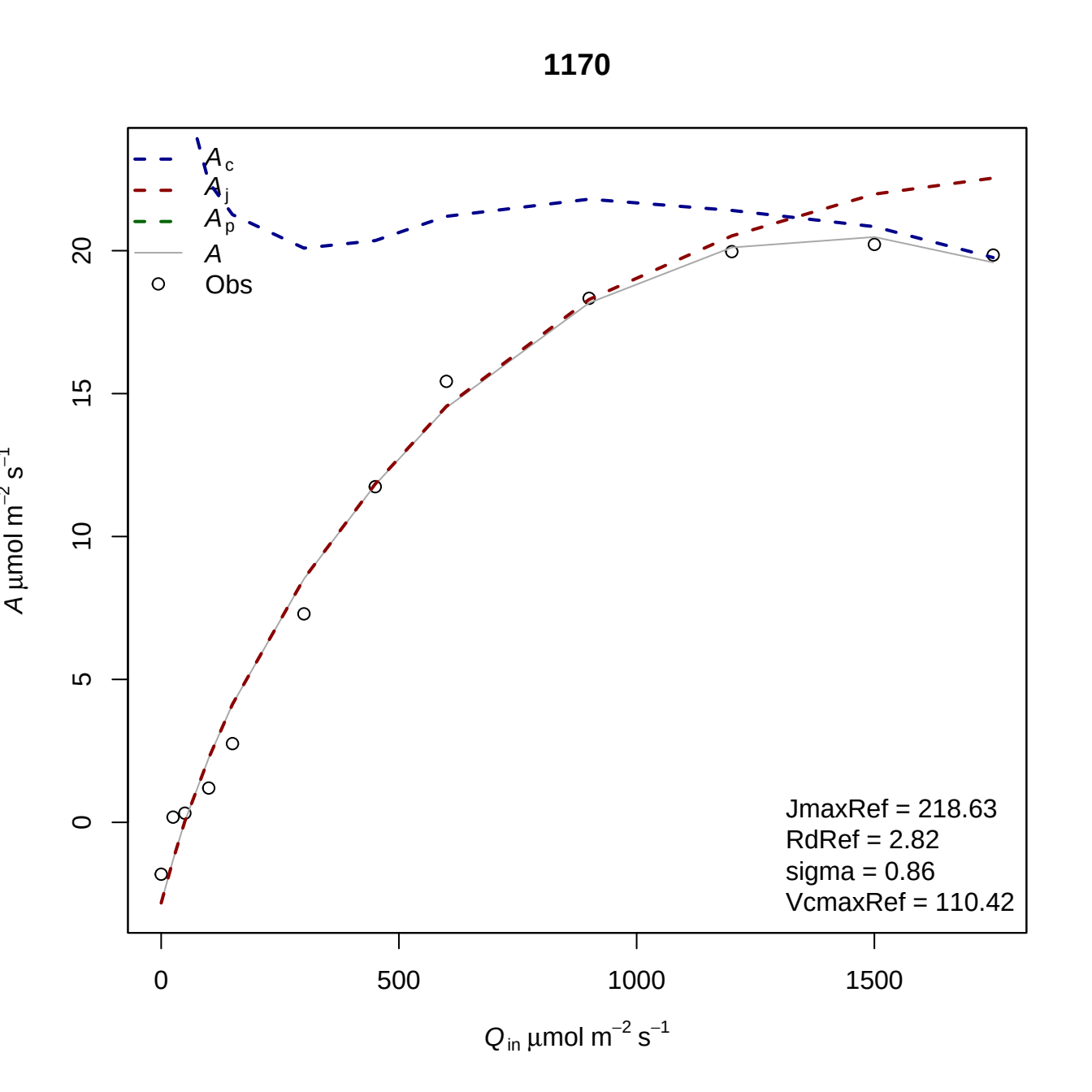

1165
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 158.16
RdRef = 2.45
sigma = 0.88
VcmaxRef = 93.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

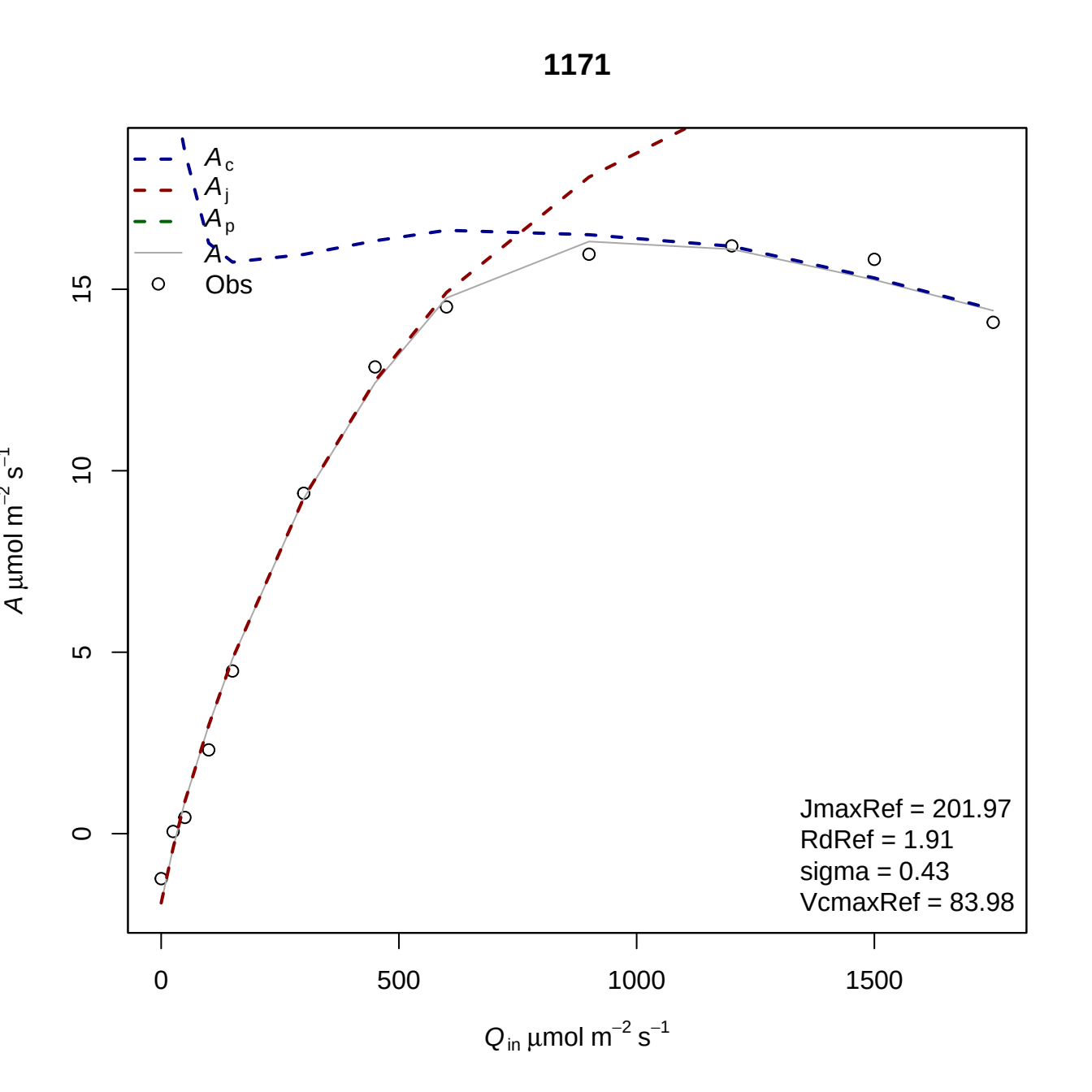

1166
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.22
RdRef = 1.27
sigma = 0.35
VcmaxRef = 83.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

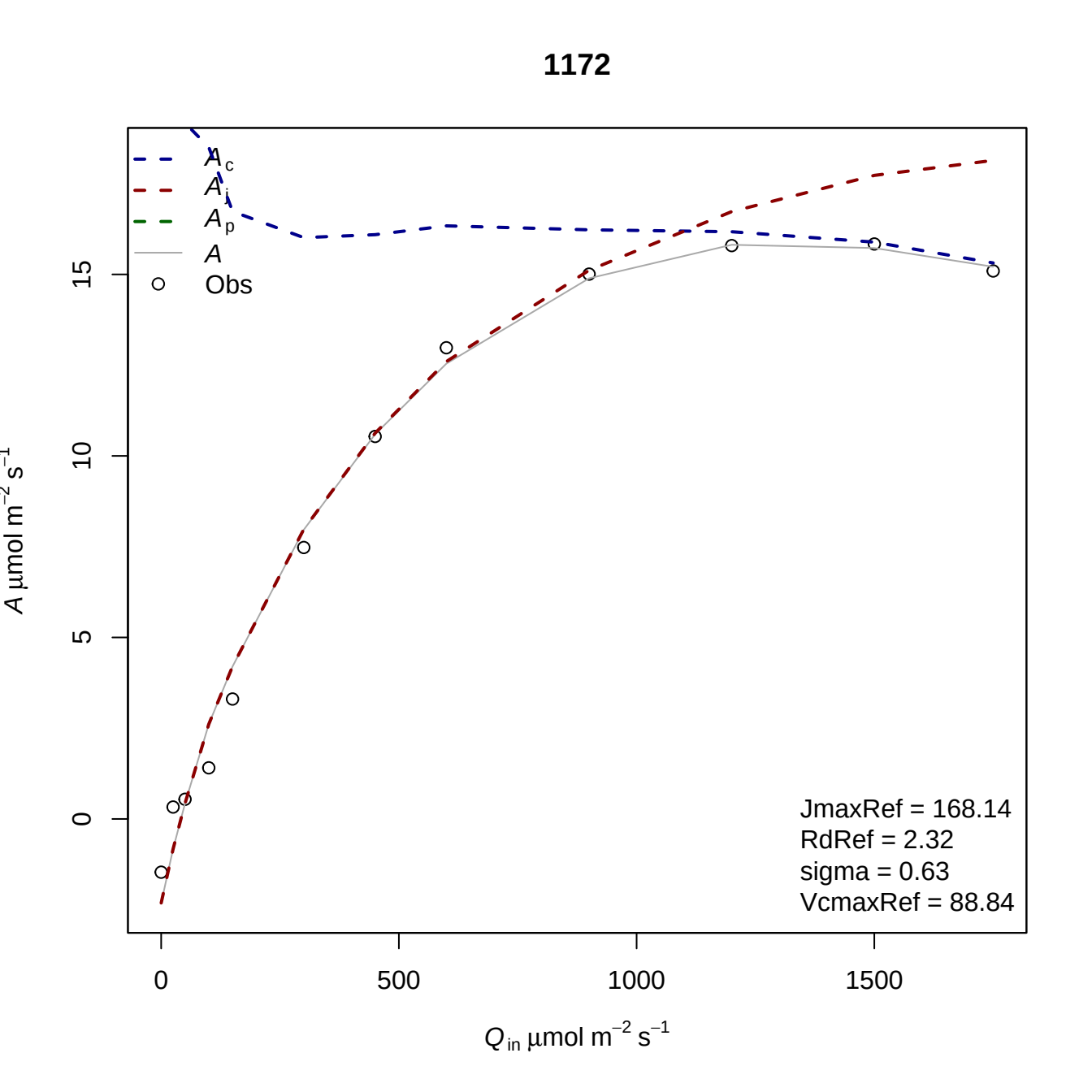

1167
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.02
RdRef = 2.39
sigma = 0.57
VcmaxRef = 8277.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

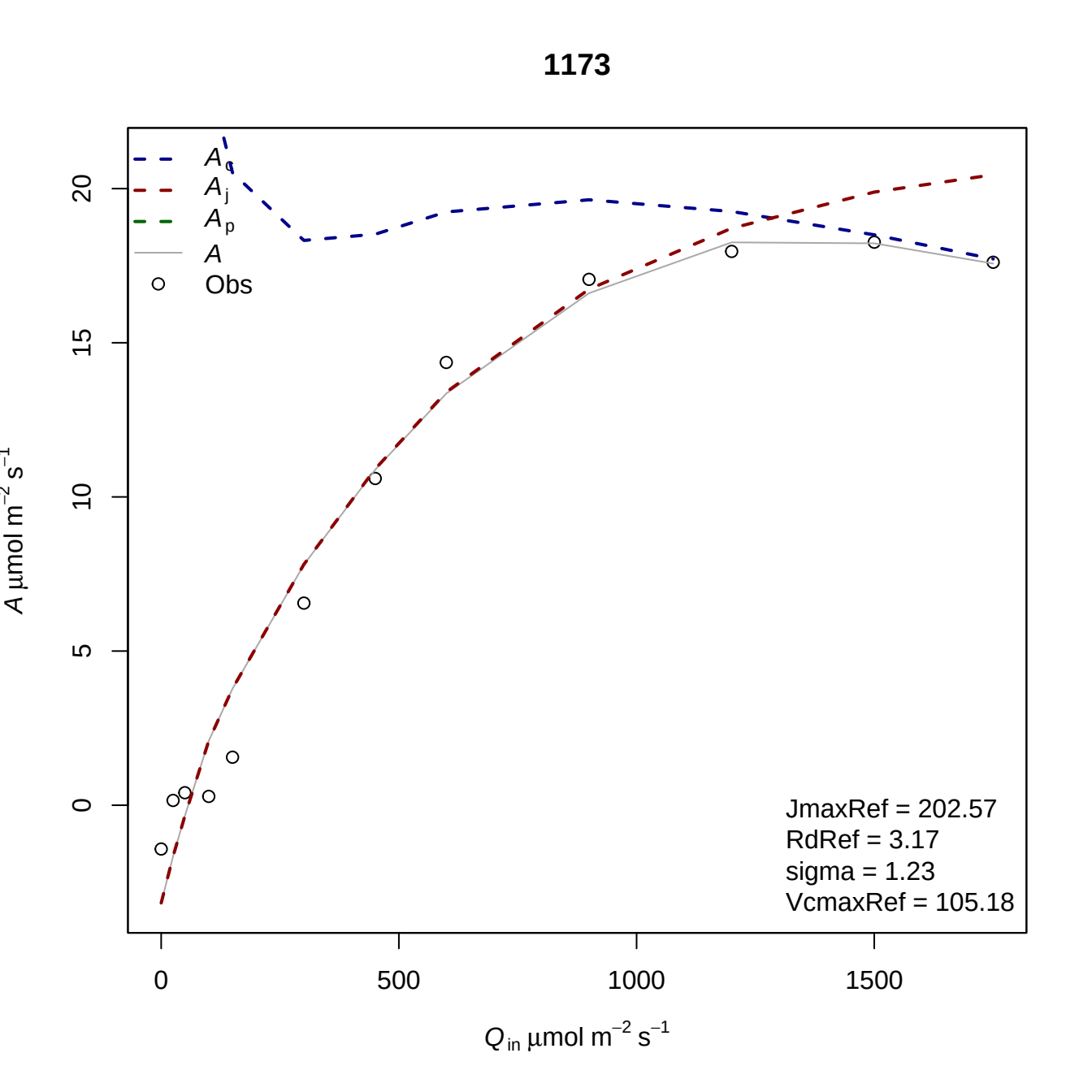

1168
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.76
RdRef = 1.48
sigma = 0.26
VcmaxRef = 88.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

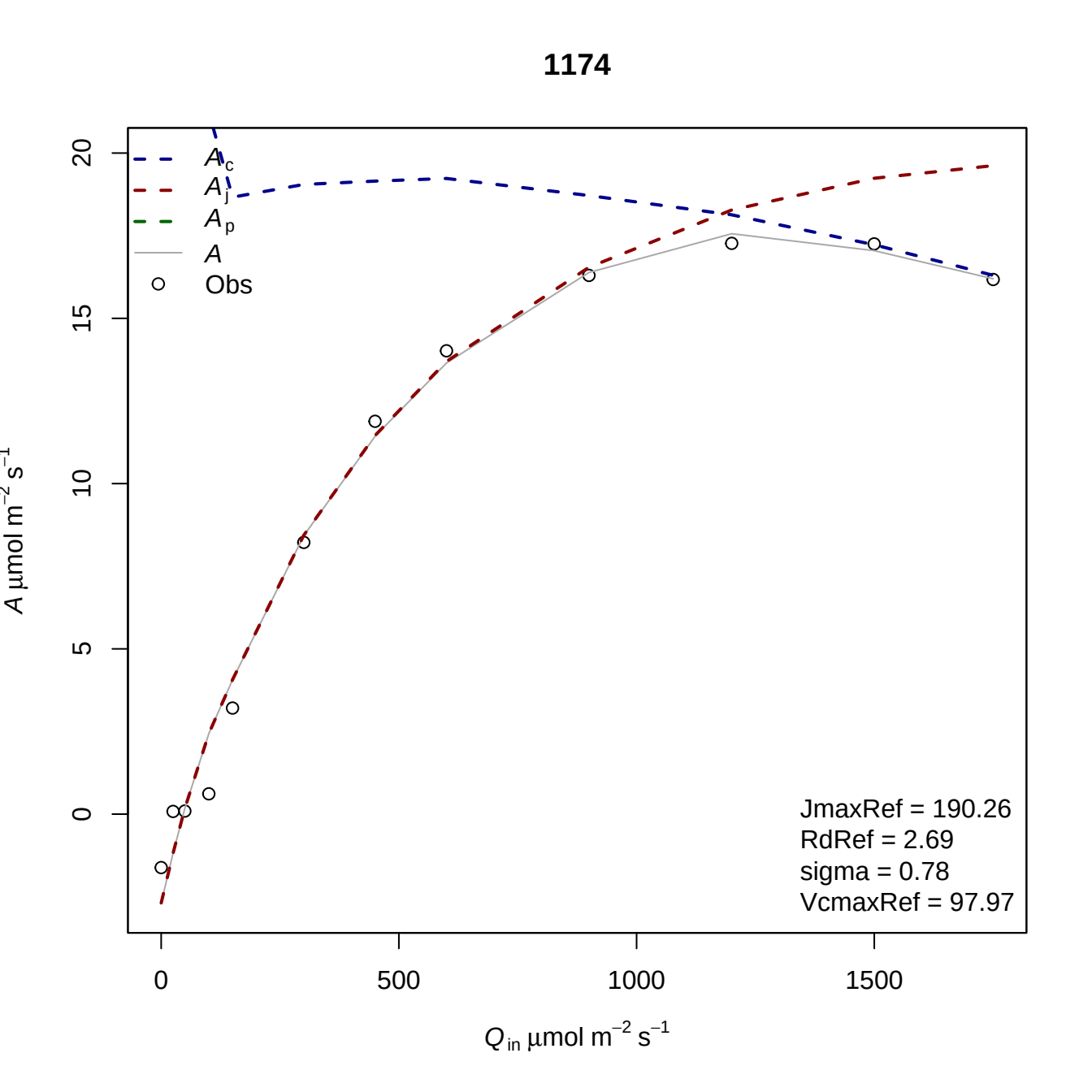

1169
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 205.72
RdRef = 1.48
sigma = 0.46
VcmaxRef = 92.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

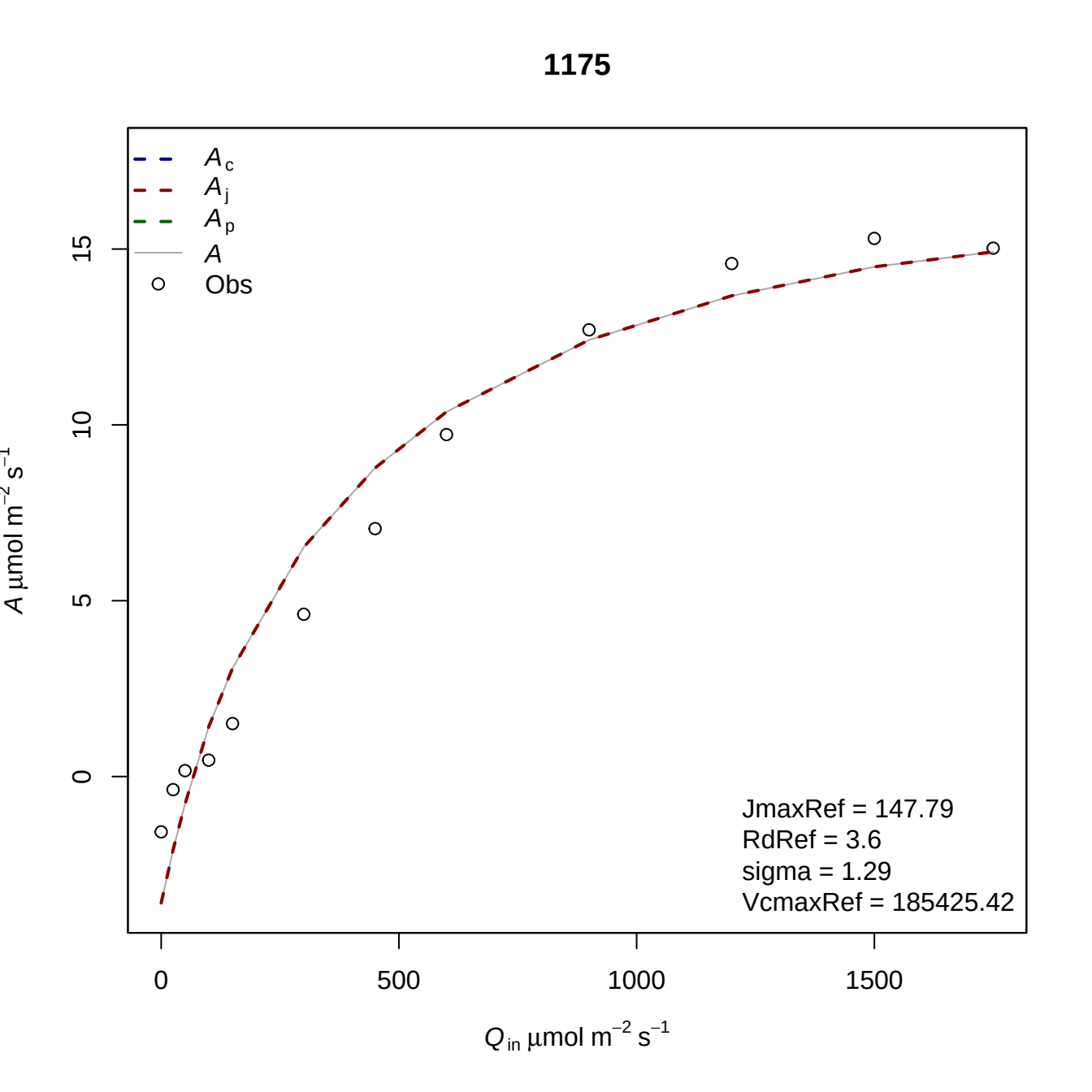

1170
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 218.63
RdRef = 2.82
sigma = 0.86
VcmaxRef = 110.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

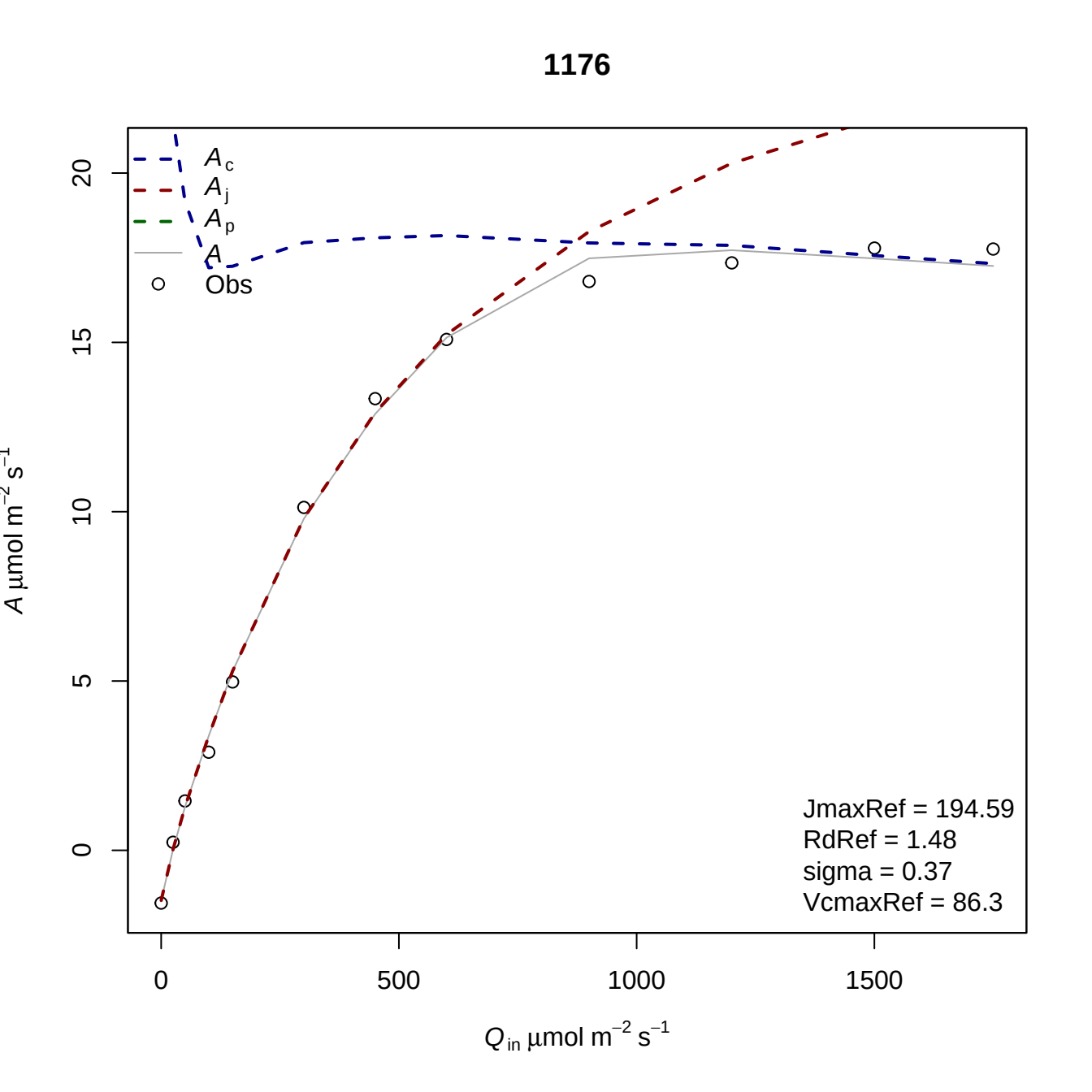

1171
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 201.97
RdRef = 1.91
sigma = 0.43
VcmaxRef = 83.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

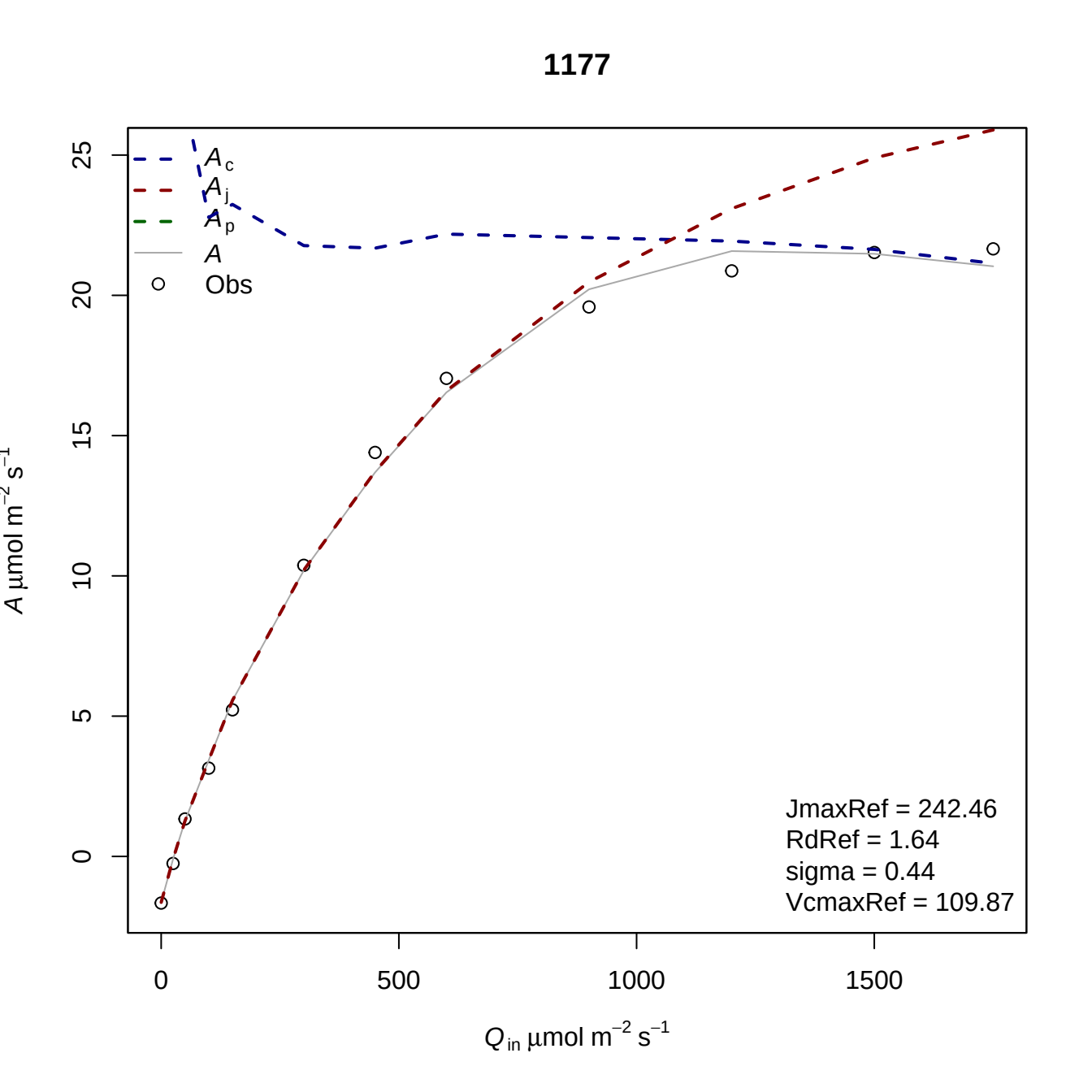

1172
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 168.14
RdRef = 2.32
sigma = 0.63
VcmaxRef = 88.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

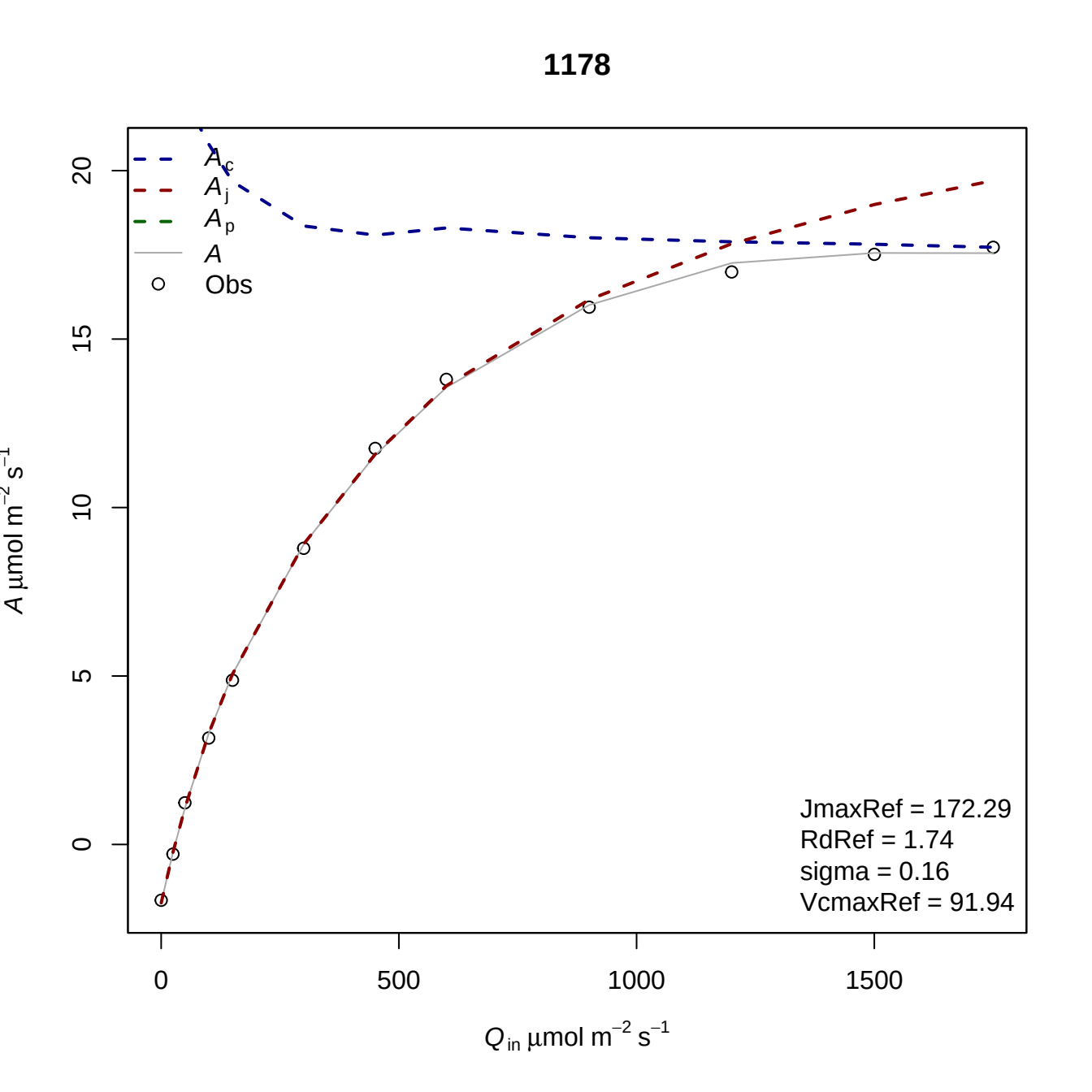

1173
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 202.57
RdRef = 3.17
sigma = 1.23
VcmaxRef = 105.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

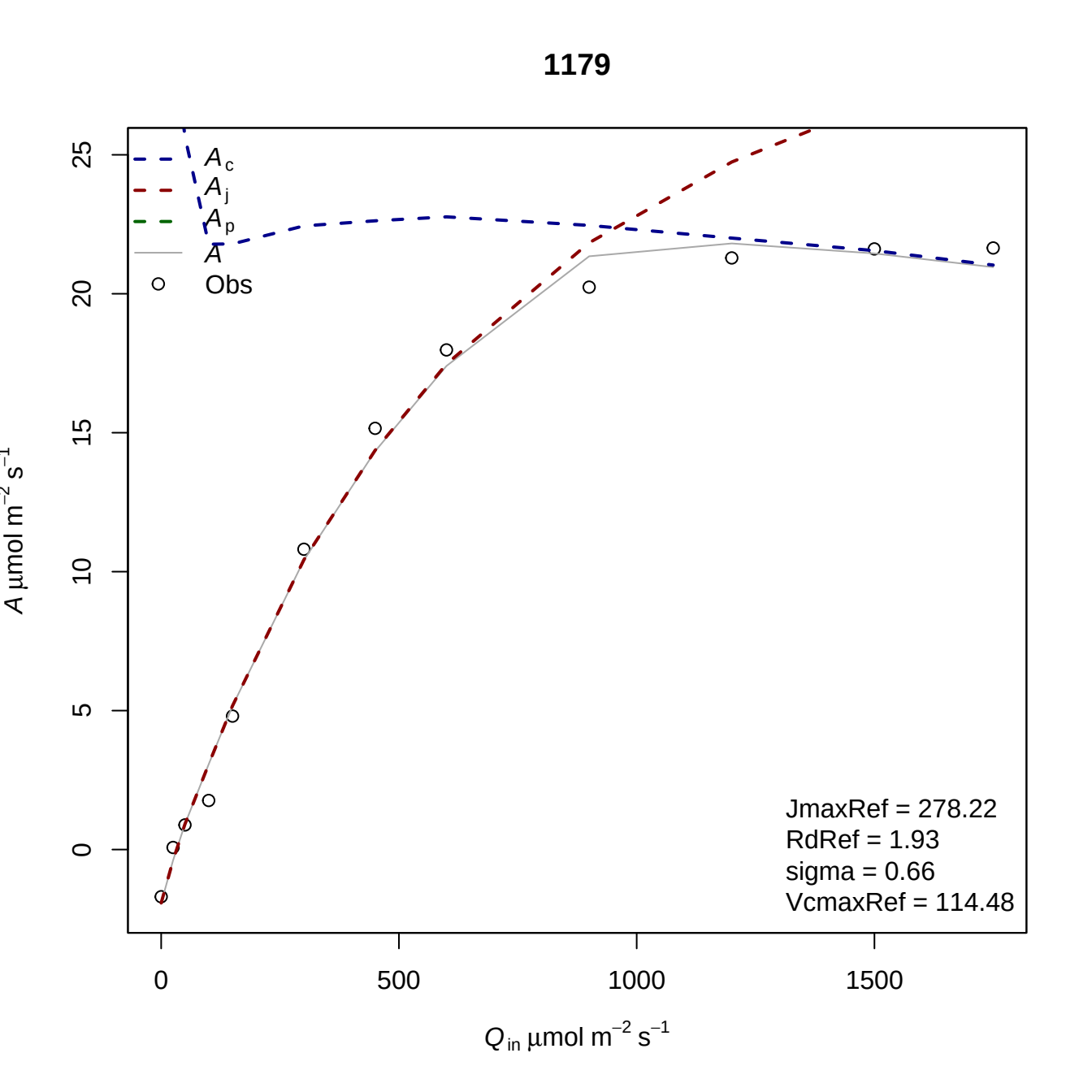

1174
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 190.26
RdRef = 2.69
sigma = 0.78
VcmaxRef = 97.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

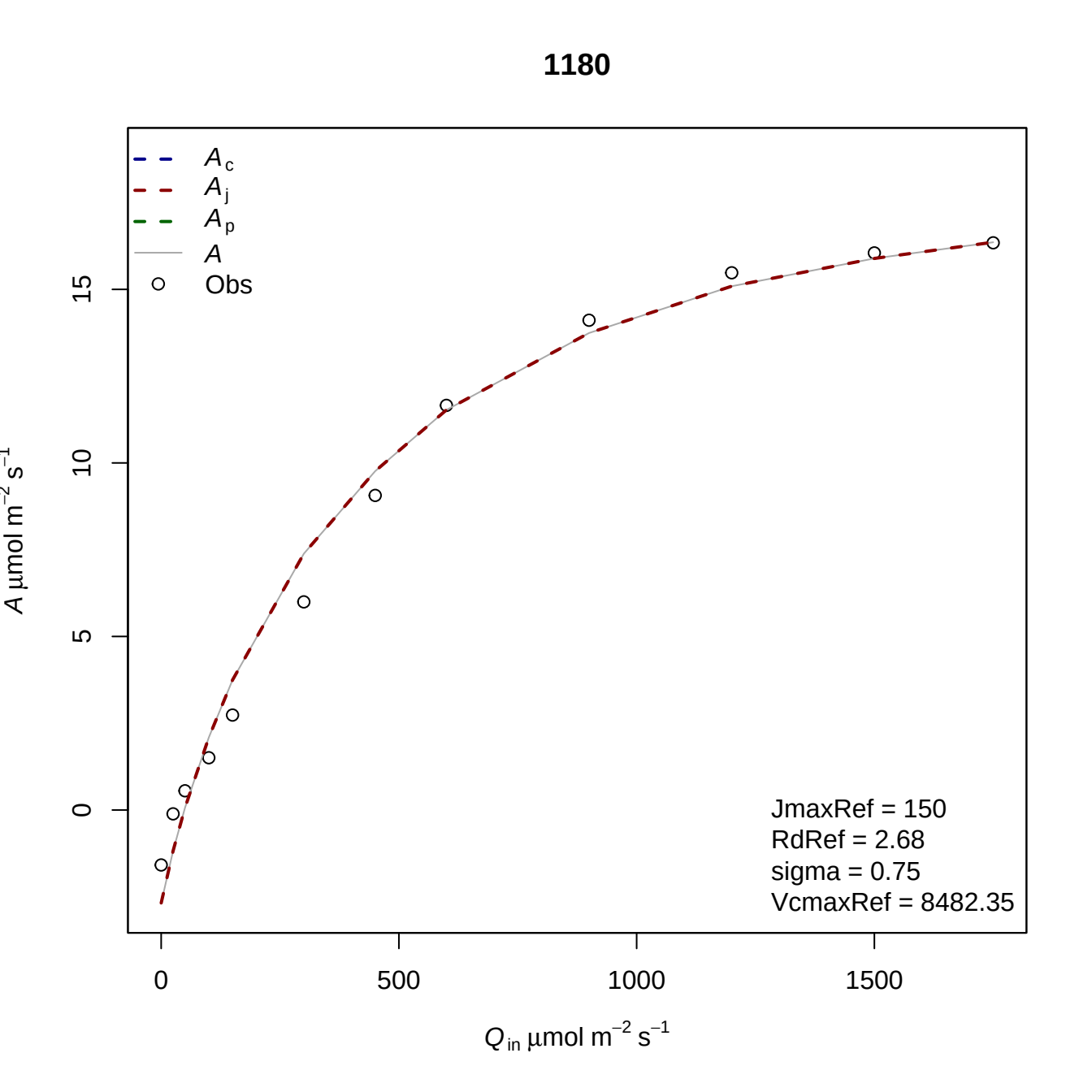

1175
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.79
RdRef = 3.6
sigma = 1.29
VcmaxRef = 185425.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

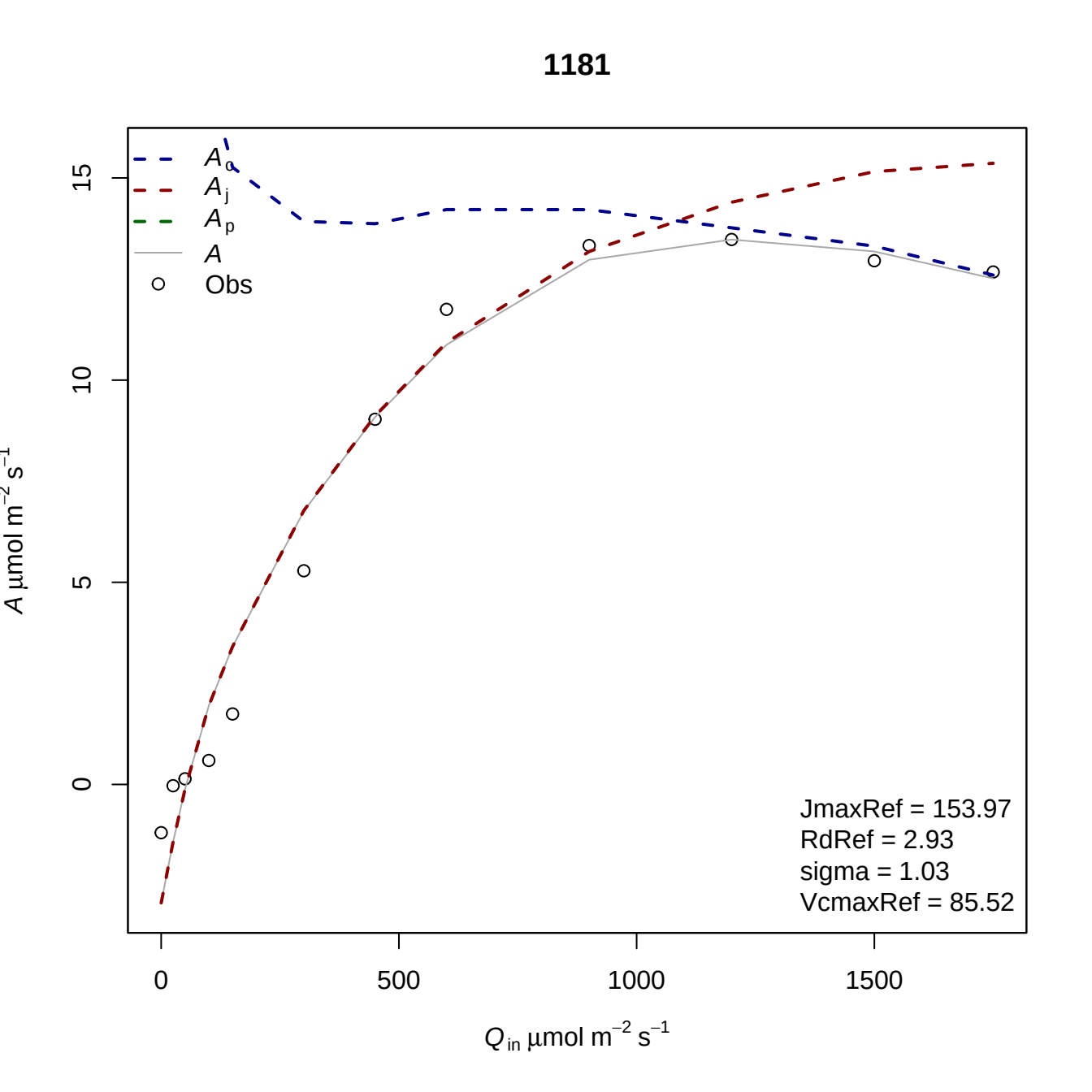

1176
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 194.59
RdRef = 1.48
sigma = 0.37
VcmaxRef = 86.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

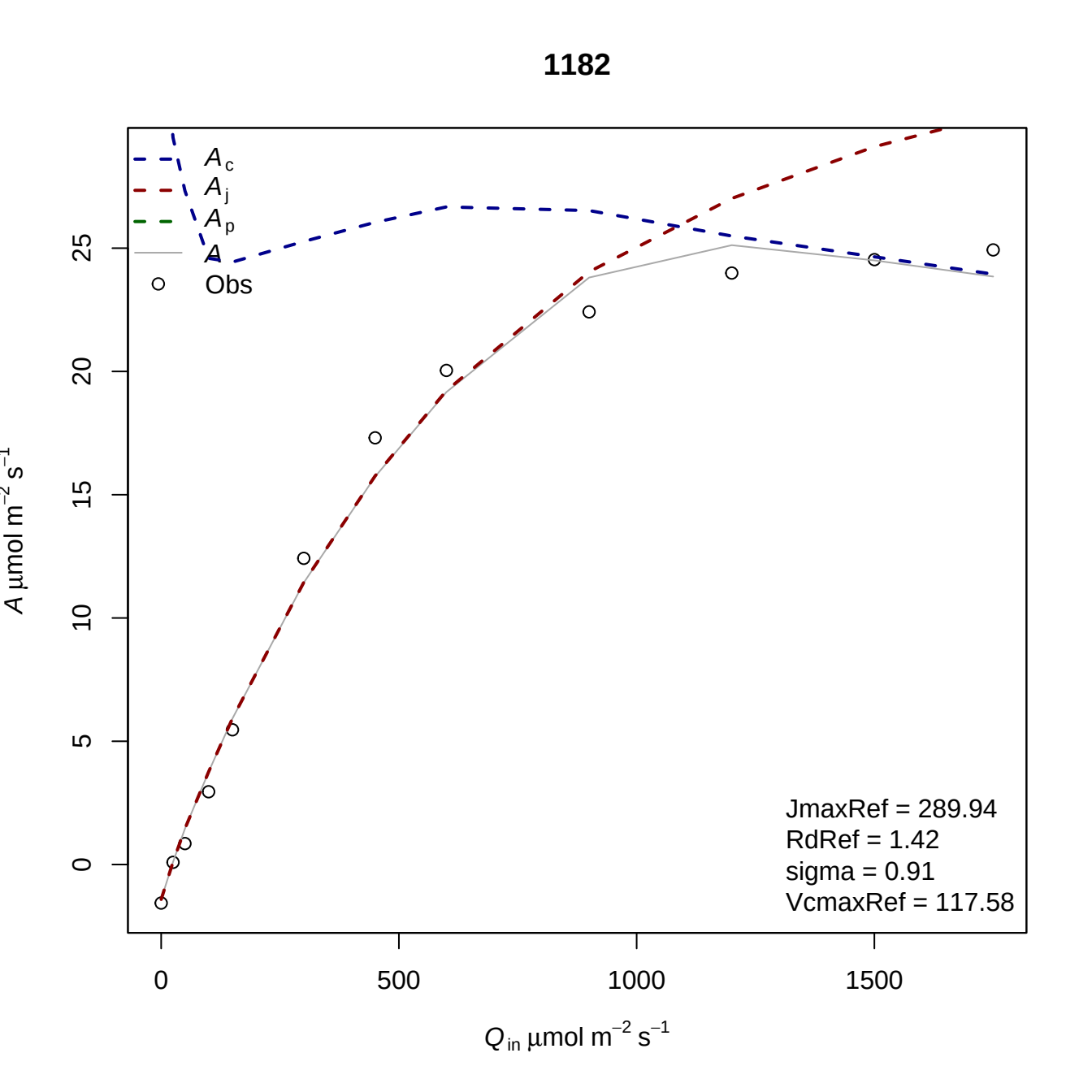

1177
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 242.46
RdRef = 1.64
sigma = 0.44
VcmaxRef = 109.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

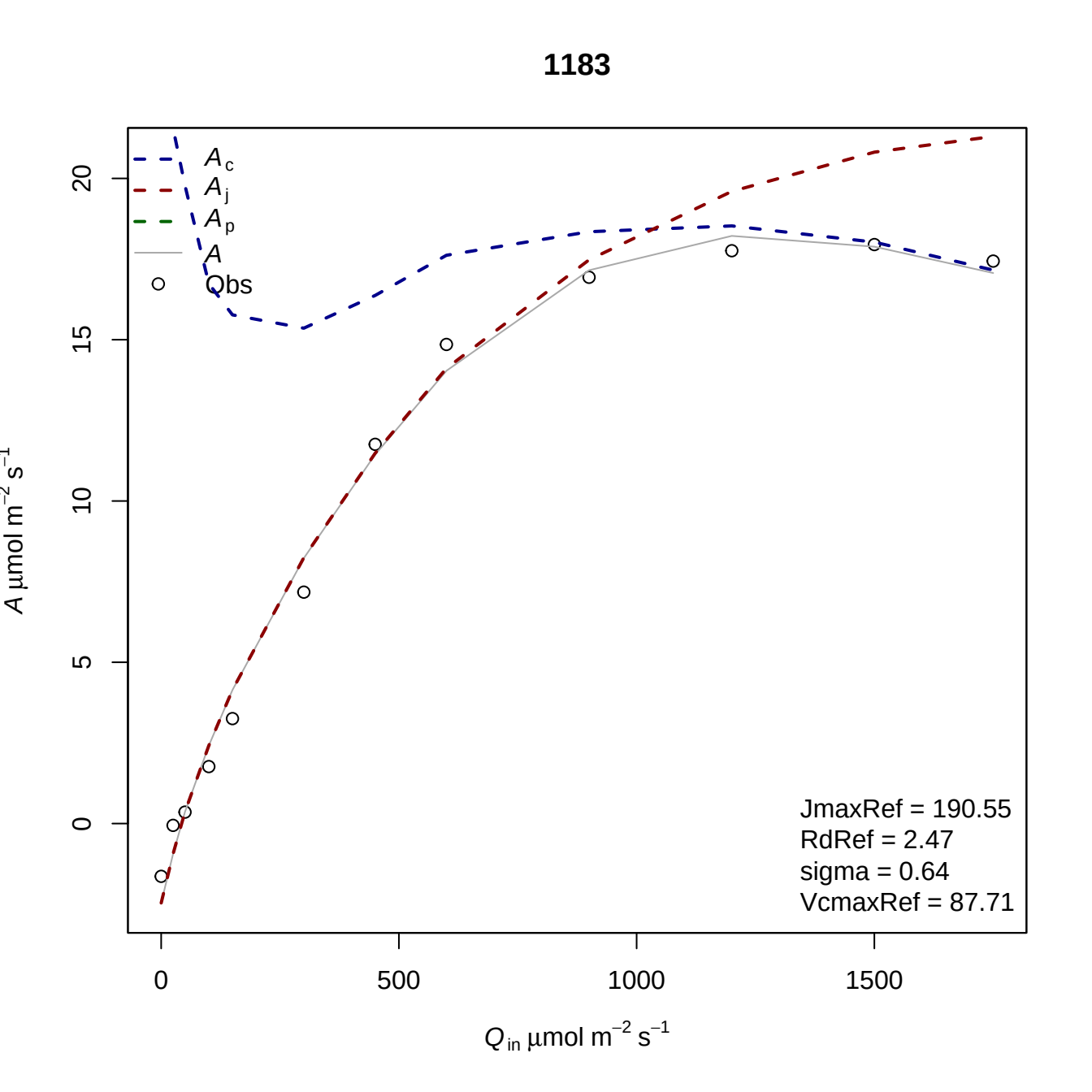

1178
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 172.29
RdRef = 1.74
sigma = 0.16
VcmaxRef = 91.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

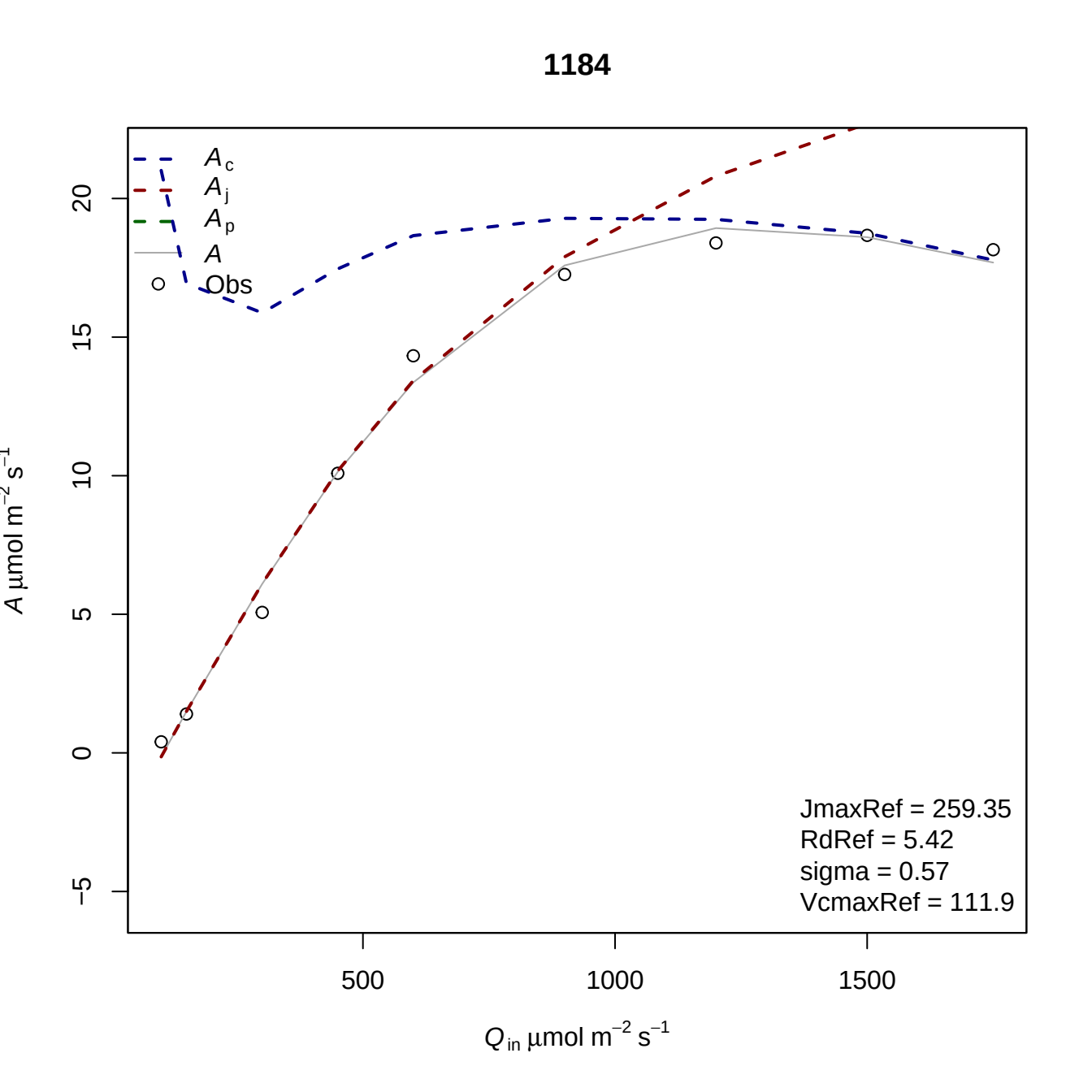

1179
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 278.22
RdRef = 1.93
sigma = 0.66
VcmaxRef = 114.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

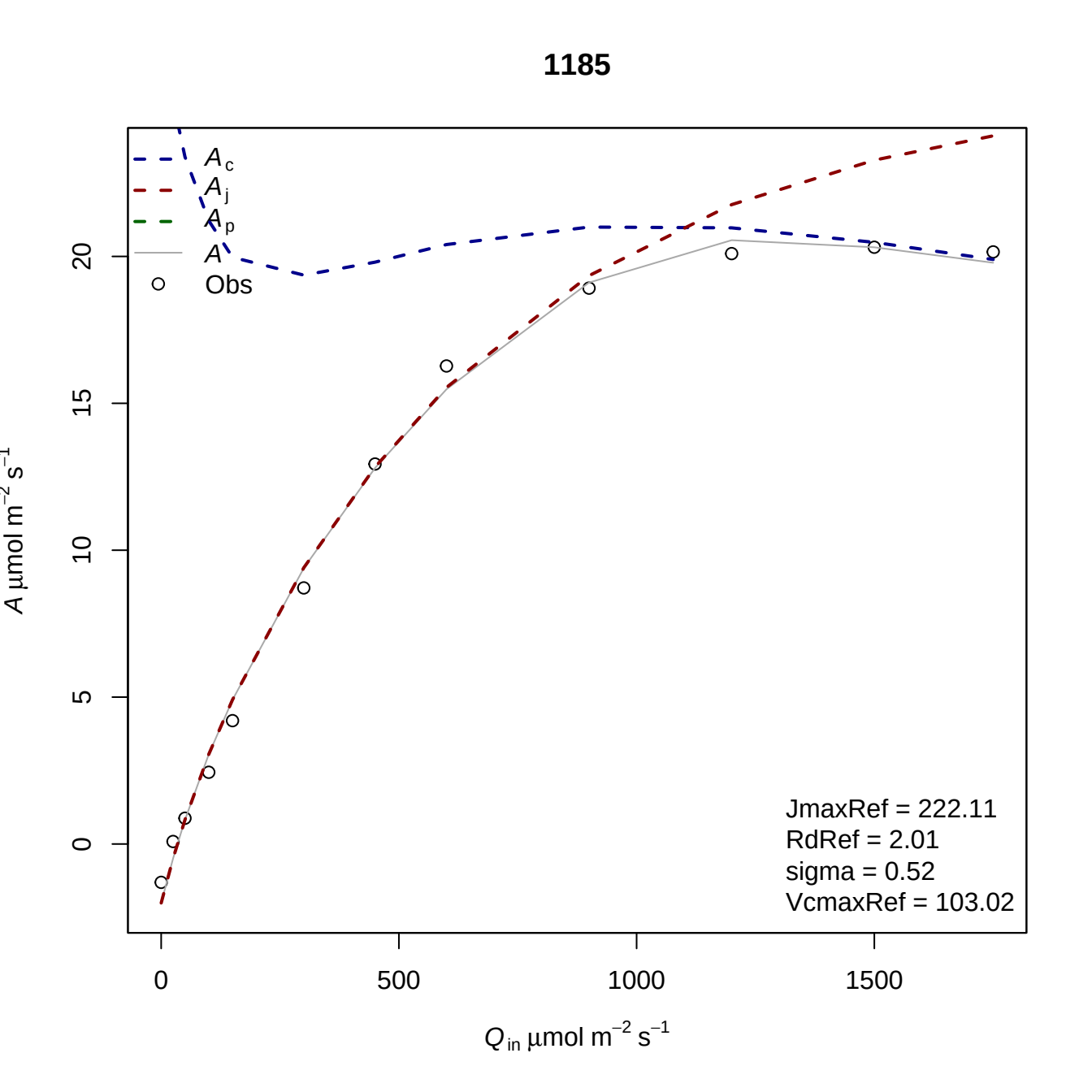

1180
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 150
RdRef = 2.68
sigma = 0.75
VcmaxRef = 8482.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

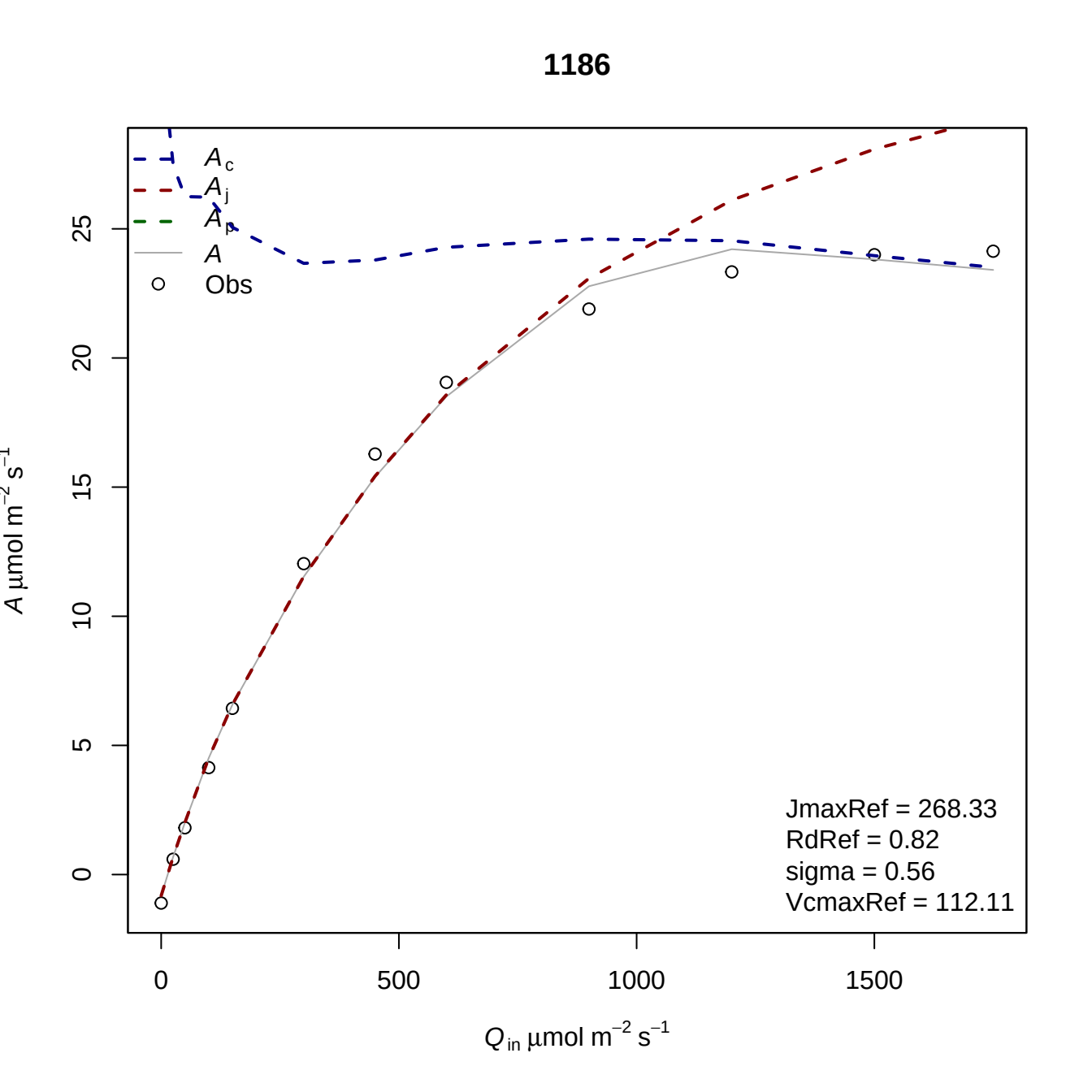

1181
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.97
RdRef = 2.93
sigma = 1.03
VcmaxRef = 85.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

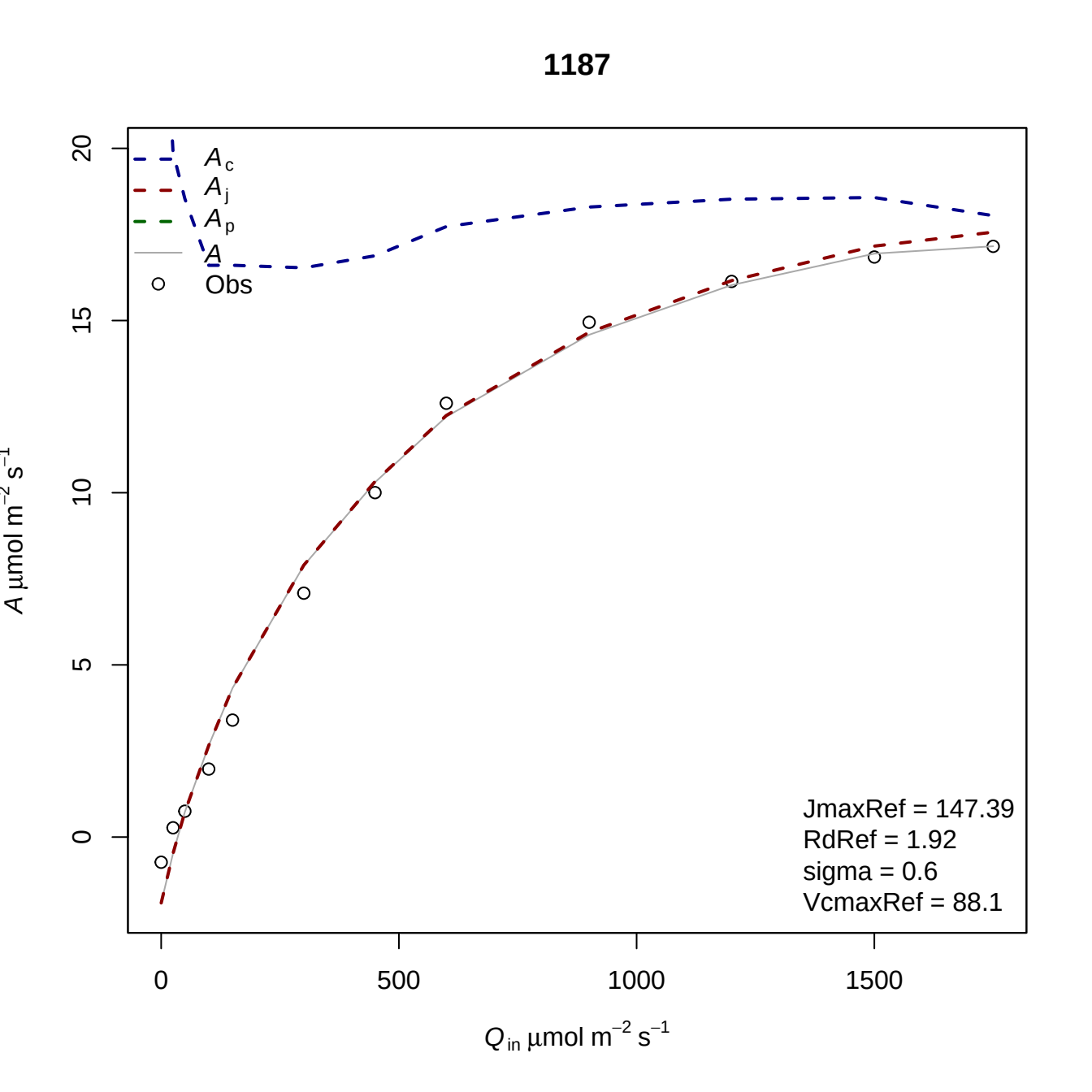

1182
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 289.94
RdRef = 1.42
sigma = 0.91
VcmaxRef = 117.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

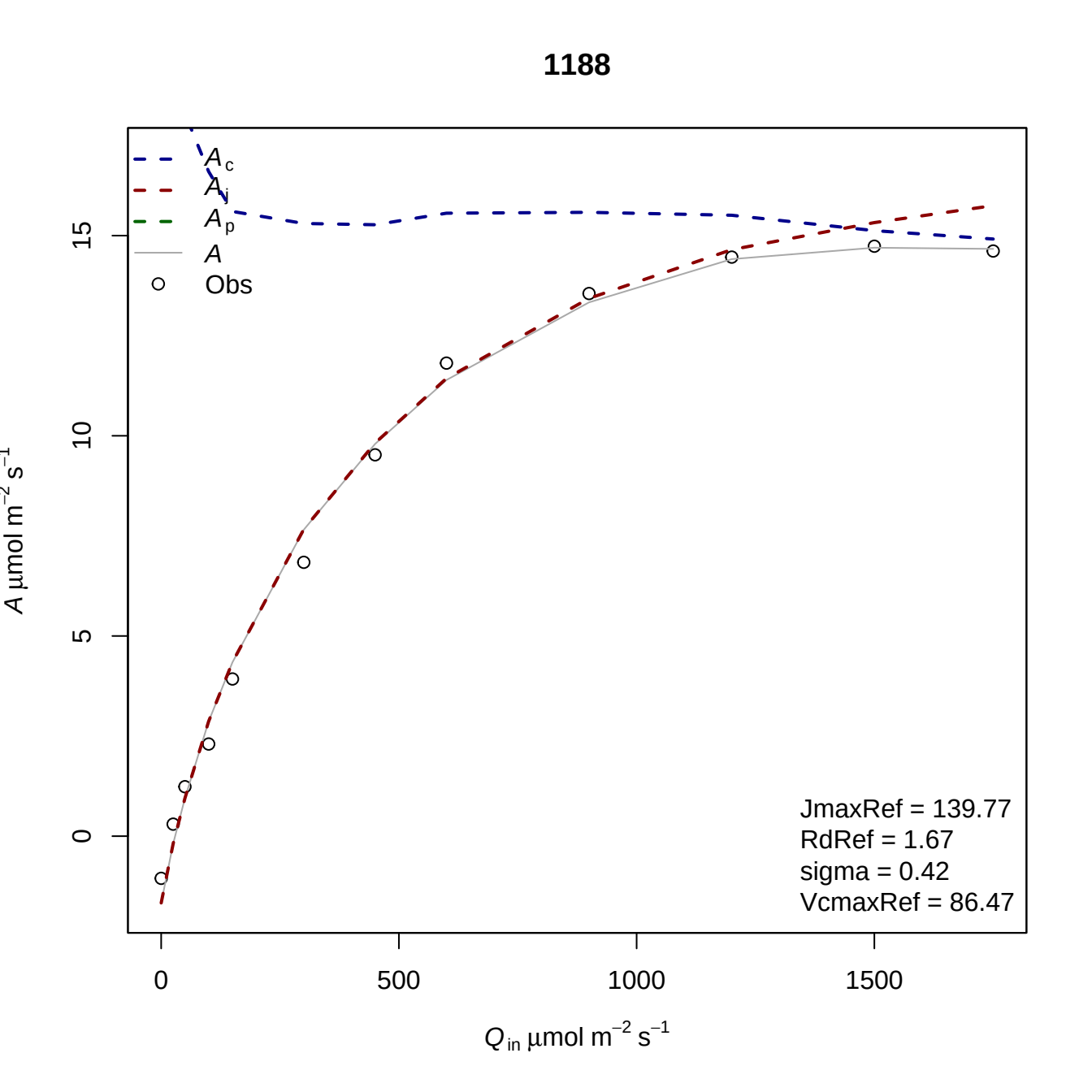

1183
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 190.55
RdRef = 2.47
sigma = 0.64
VcmaxRef = 87.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

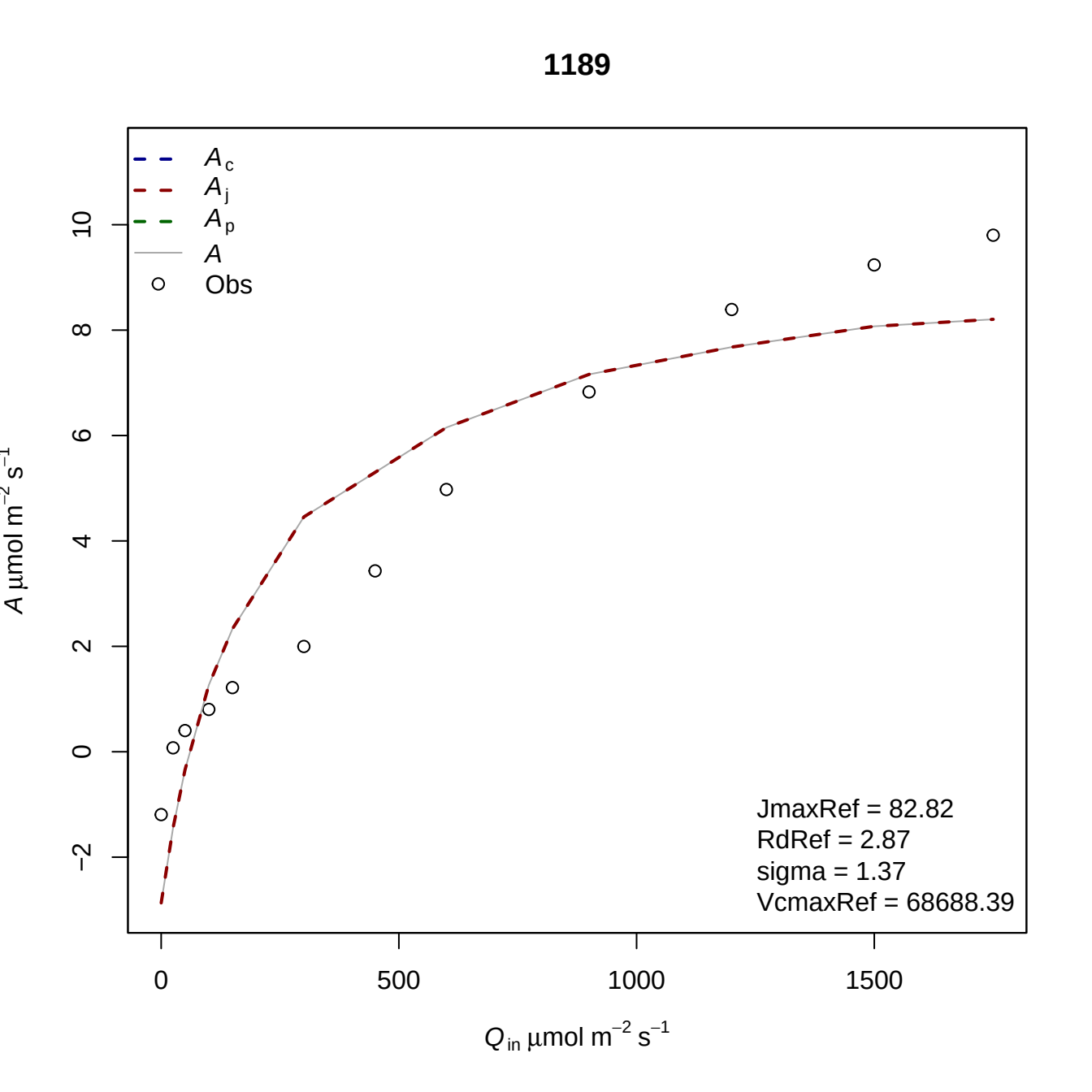

1184
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 259.35
RdRef = 5.42
sigma = 0.57
VcmaxRef = 111.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
500
1000
1500

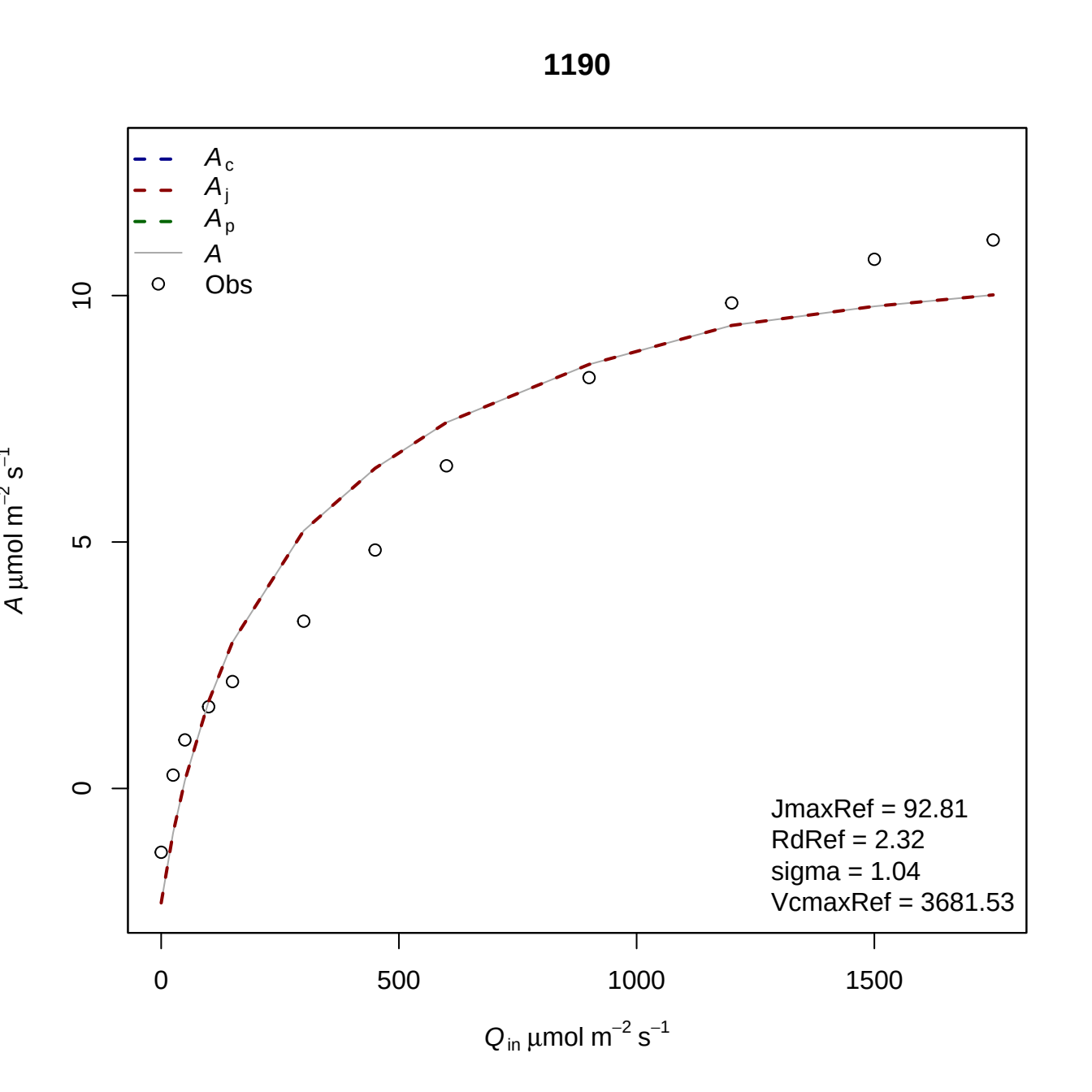

1185
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 222.11
RdRef = 2.01
sigma = 0.52
VcmaxRef = 103.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

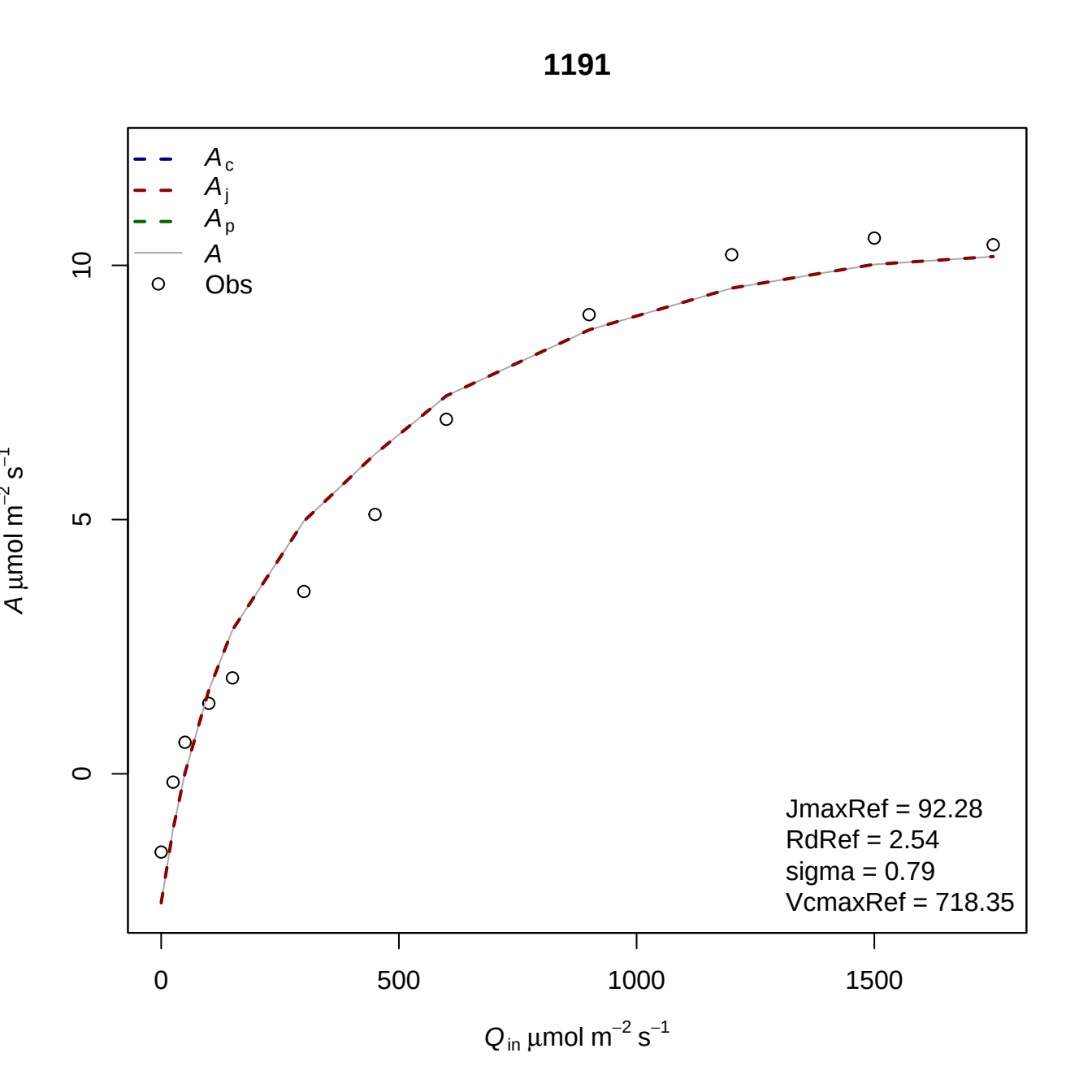

1186
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 268.33
RdRef = 0.82
sigma = 0.56
VcmaxRef = 112.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

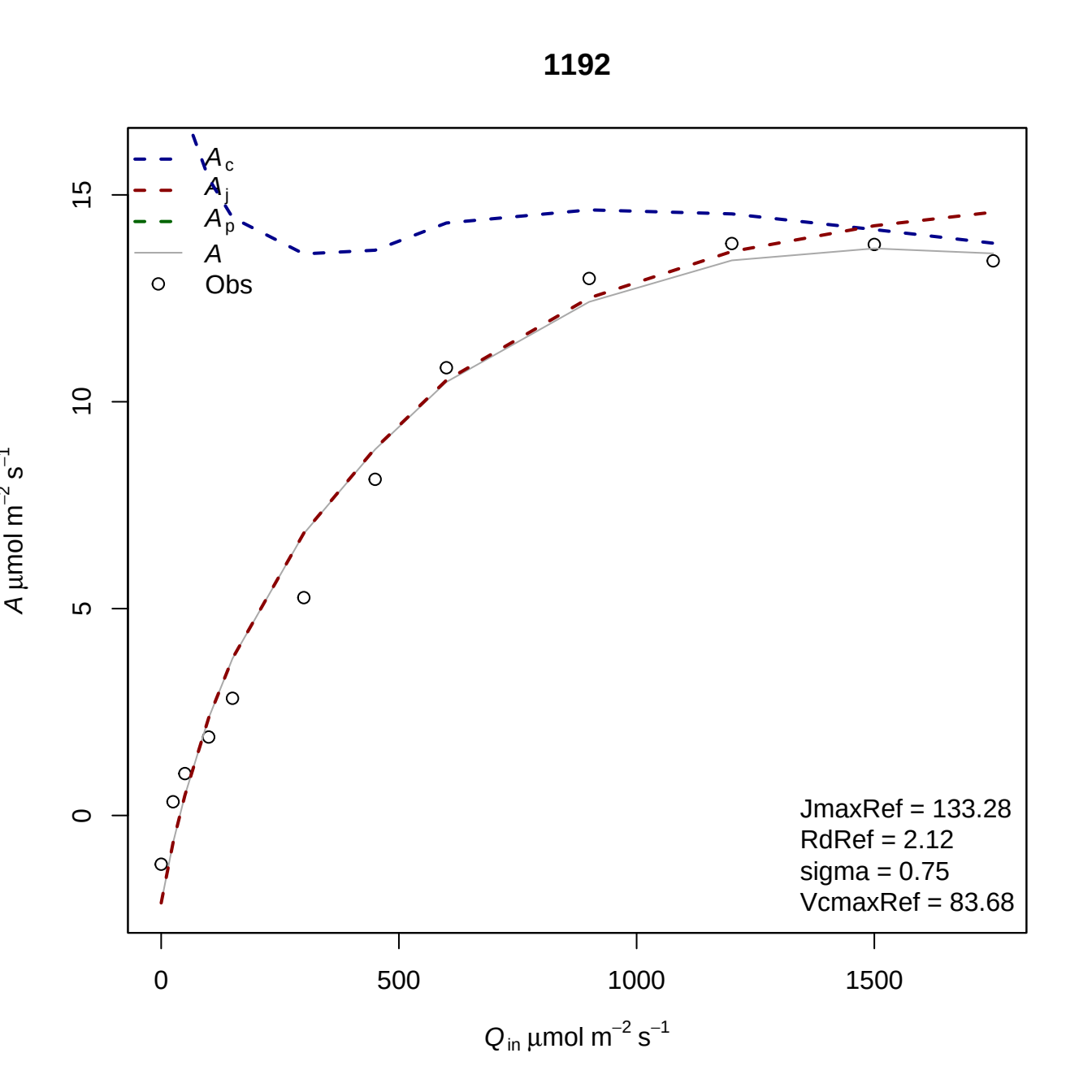

1187
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 147.39
RdRef = 1.92
sigma = 0.6
VcmaxRef = 88.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

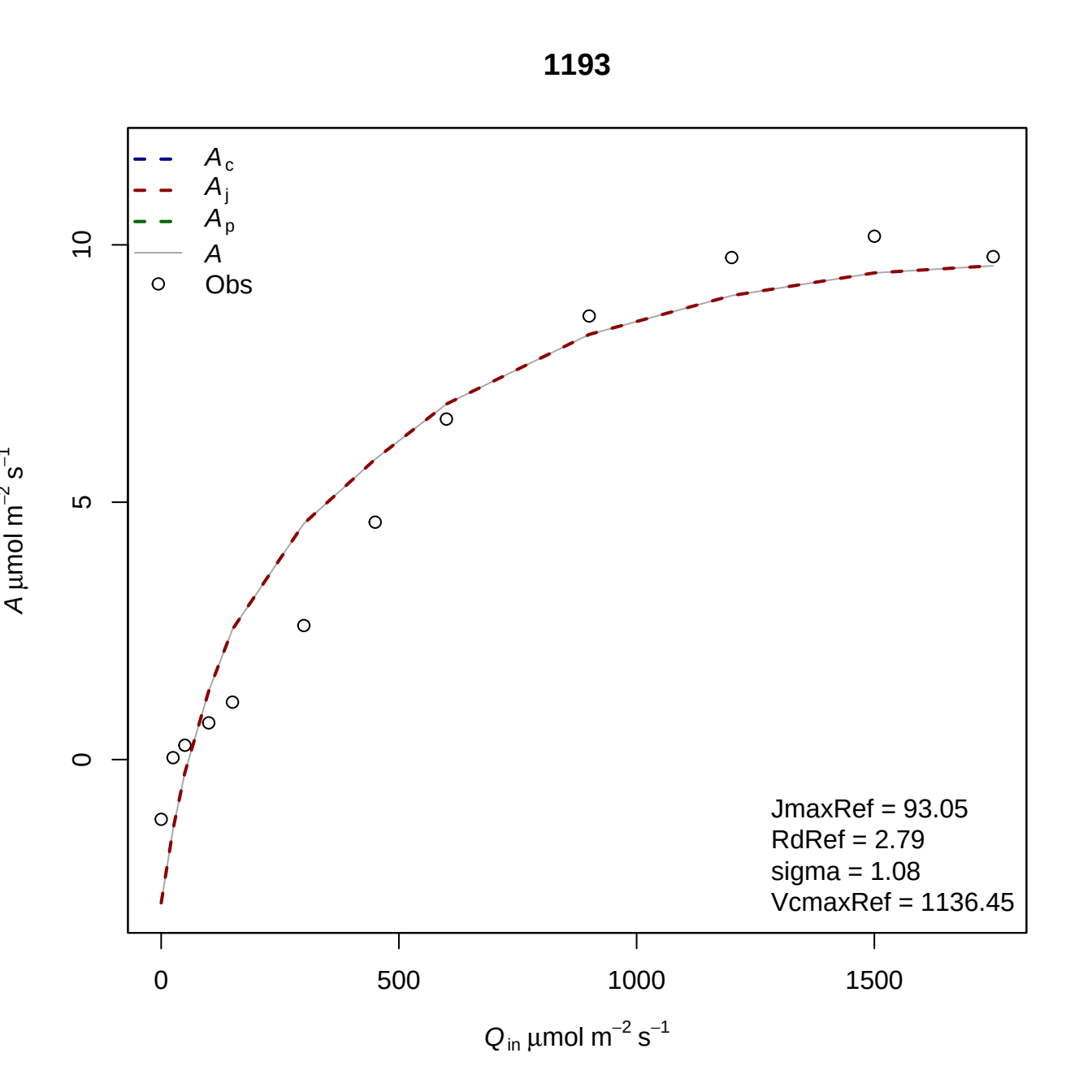

1188
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.77
RdRef = 1.67
sigma = 0.42
VcmaxRef = 86.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

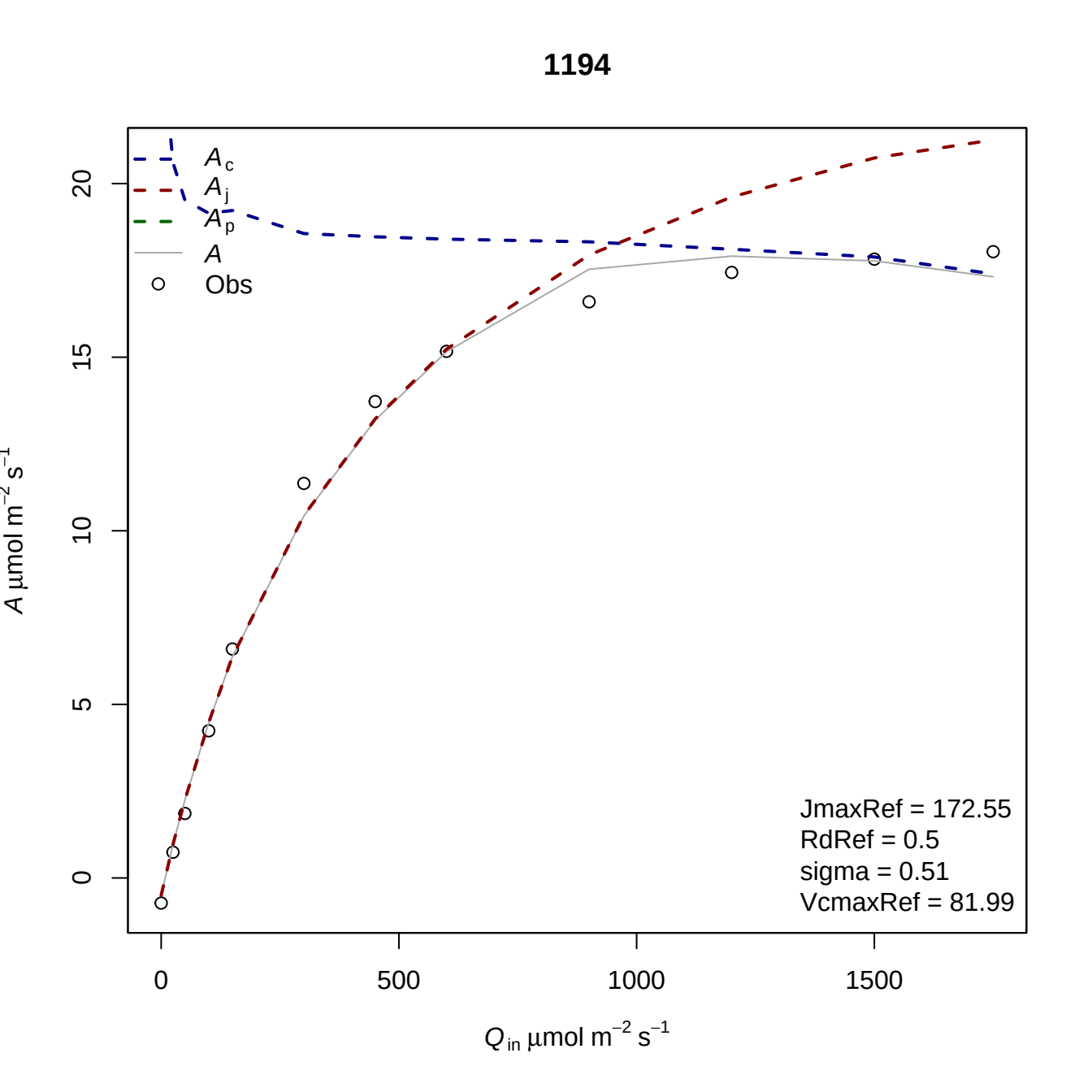

1189
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.82
RdRef = 2.87
sigma = 1.37
VcmaxRef = 68688.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

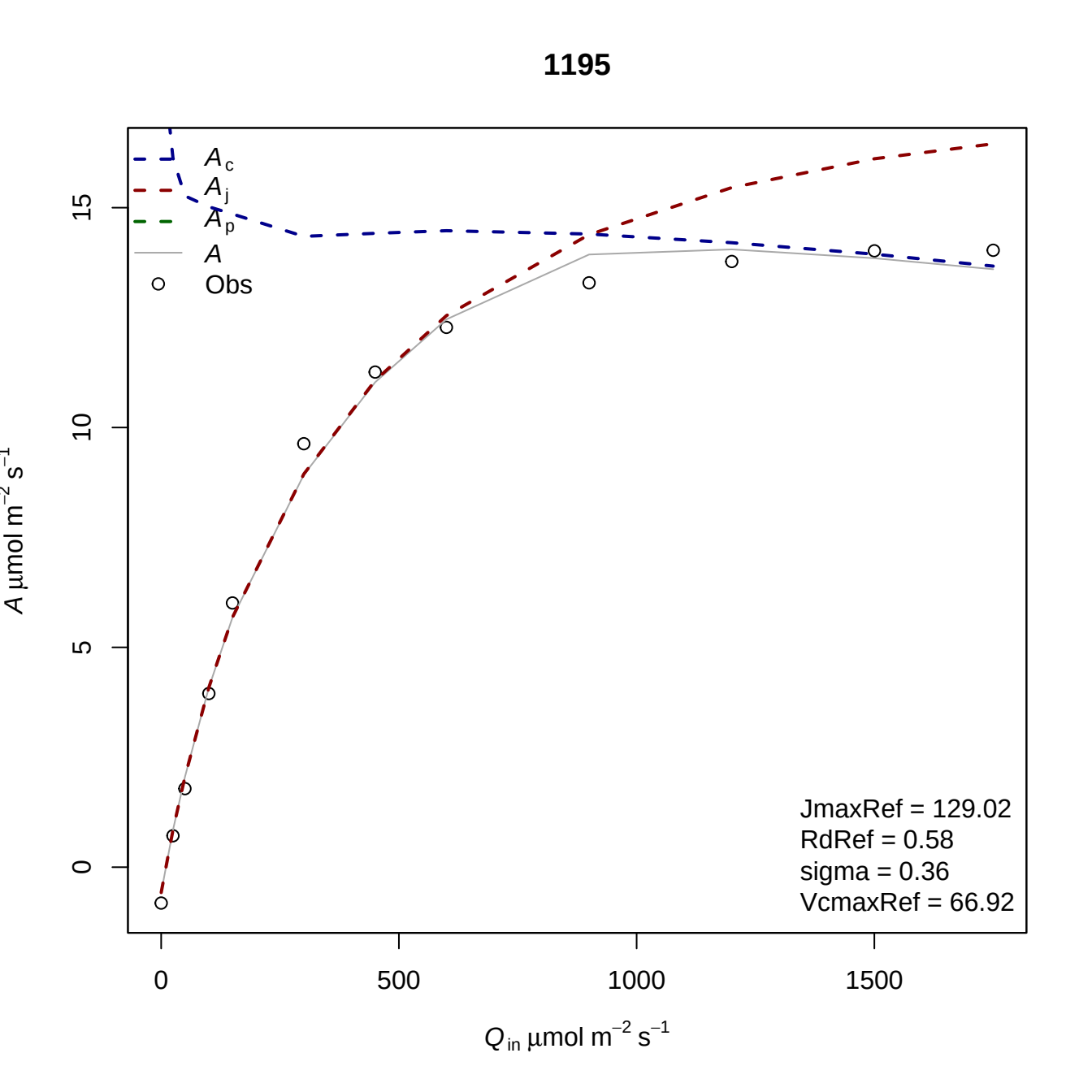

1190
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.81
RdRef = 2.32
sigma = 1.04
VcmaxRef = 3681.53
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

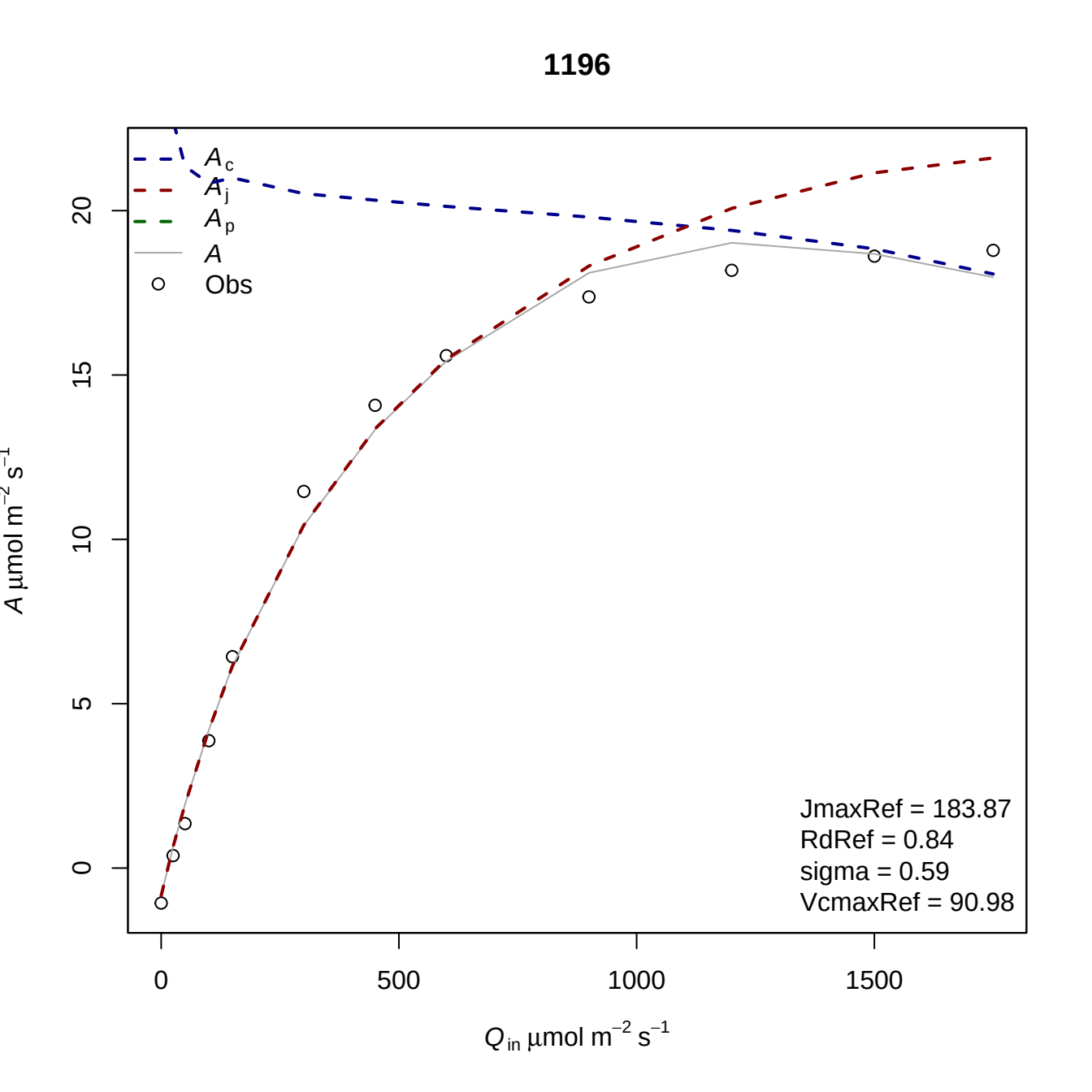

1191
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.28
RdRef = 2.54
sigma = 0.79
VcmaxRef = 718.35
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

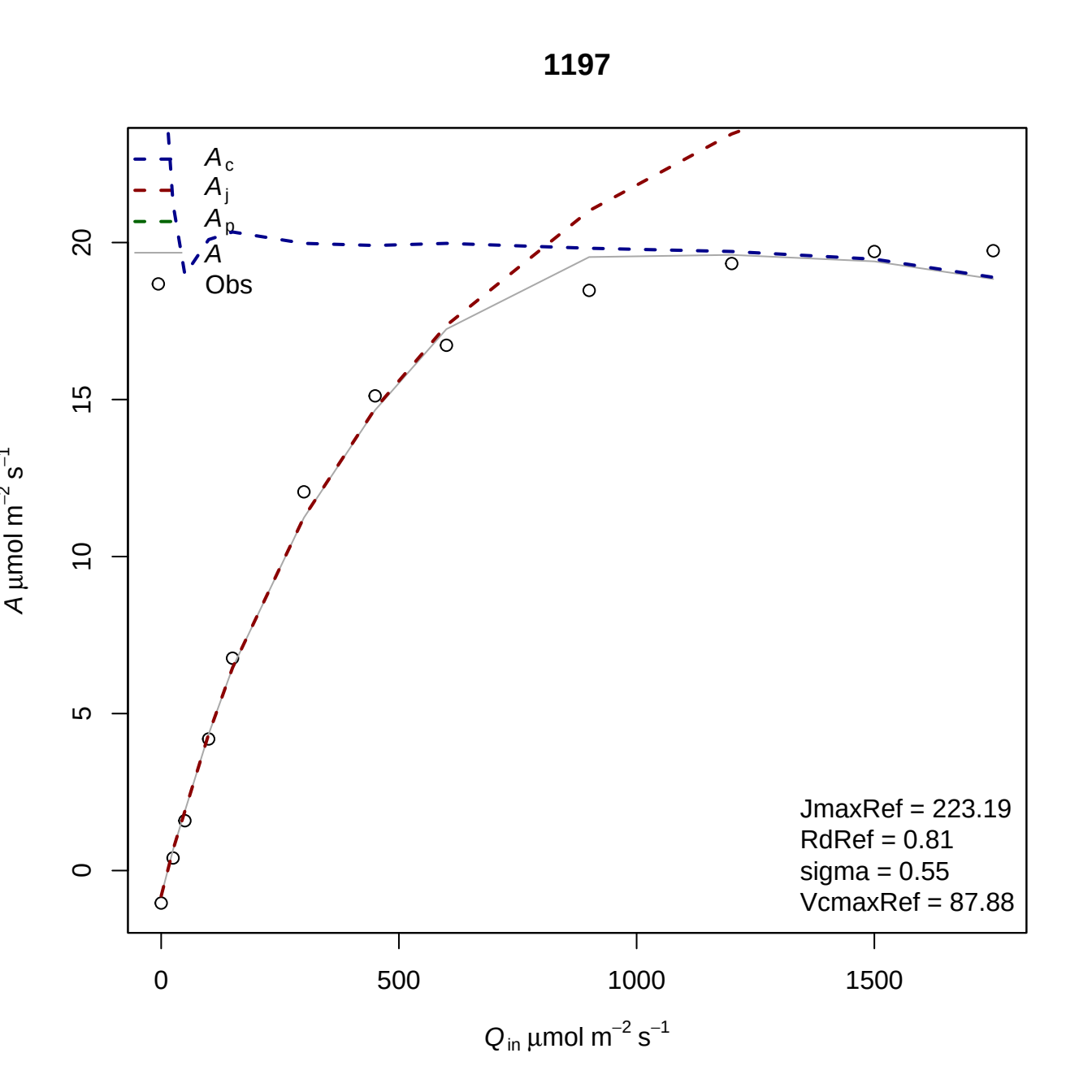

1192
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.28
RdRef = 2.12
sigma = 0.75
VcmaxRef = 83.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

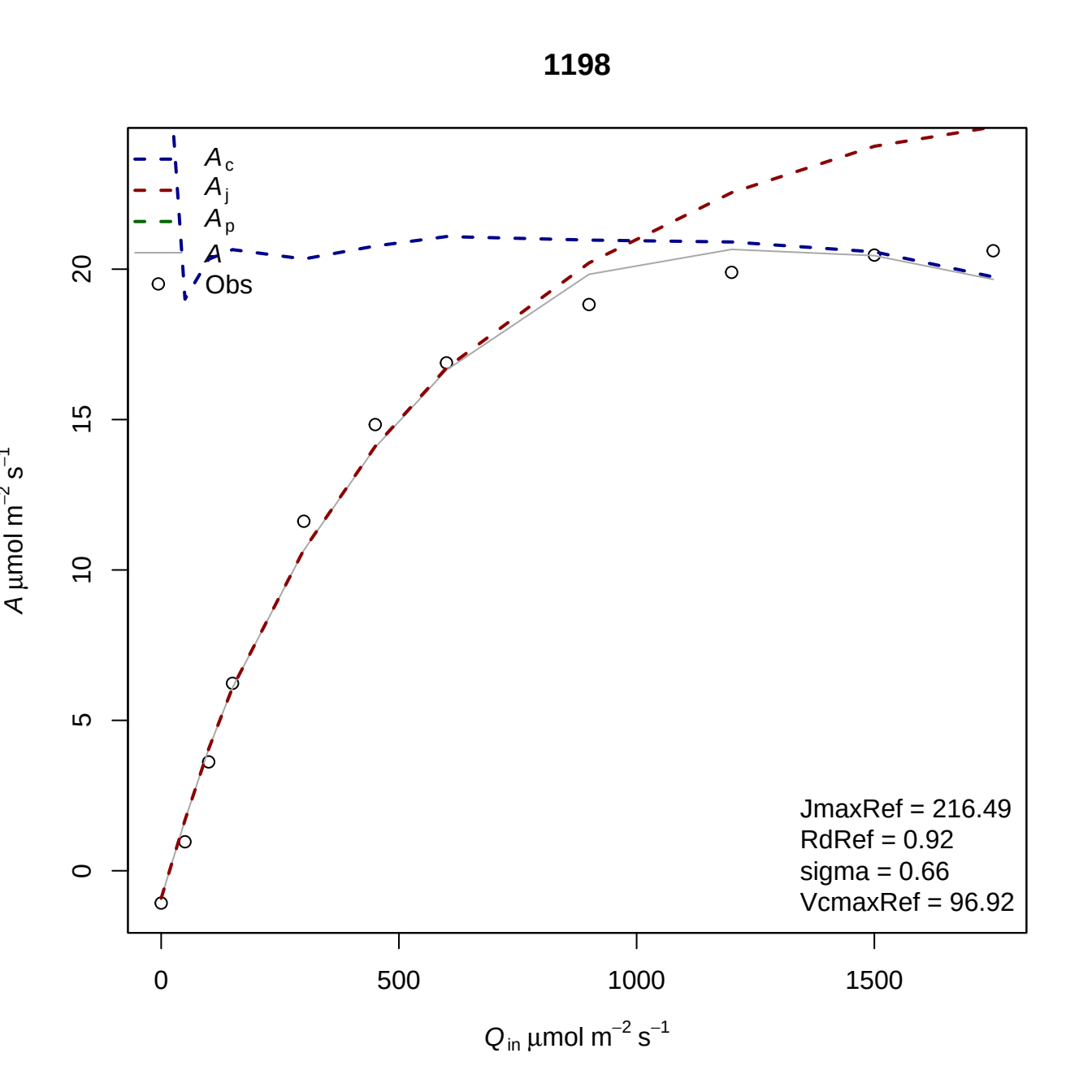

1193
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.05
RdRef = 2.79
sigma = 1.08
VcmaxRef = 1136.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

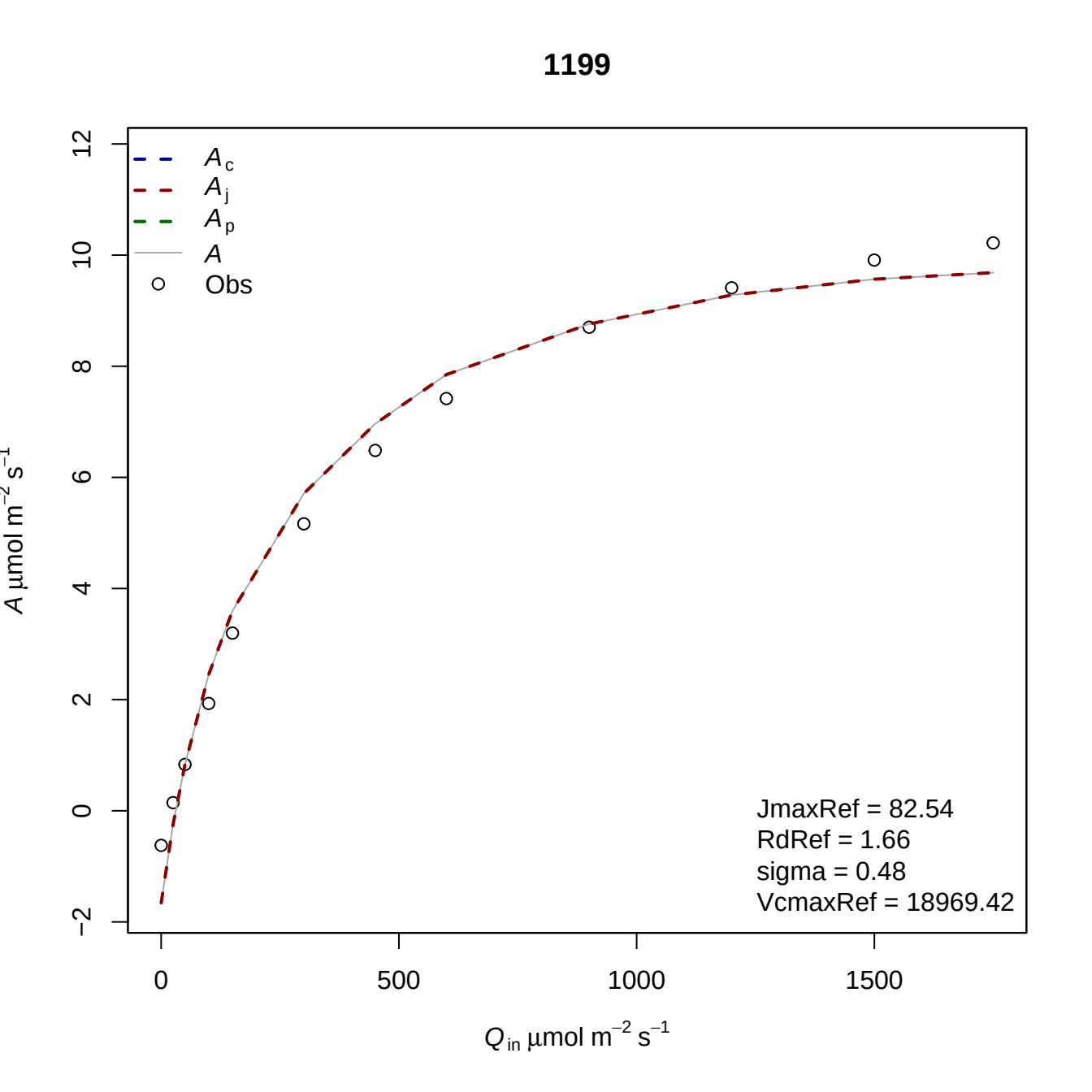

1194
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 172.55
RdRef = 0.5
sigma = 0.51
VcmaxRef = 81.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

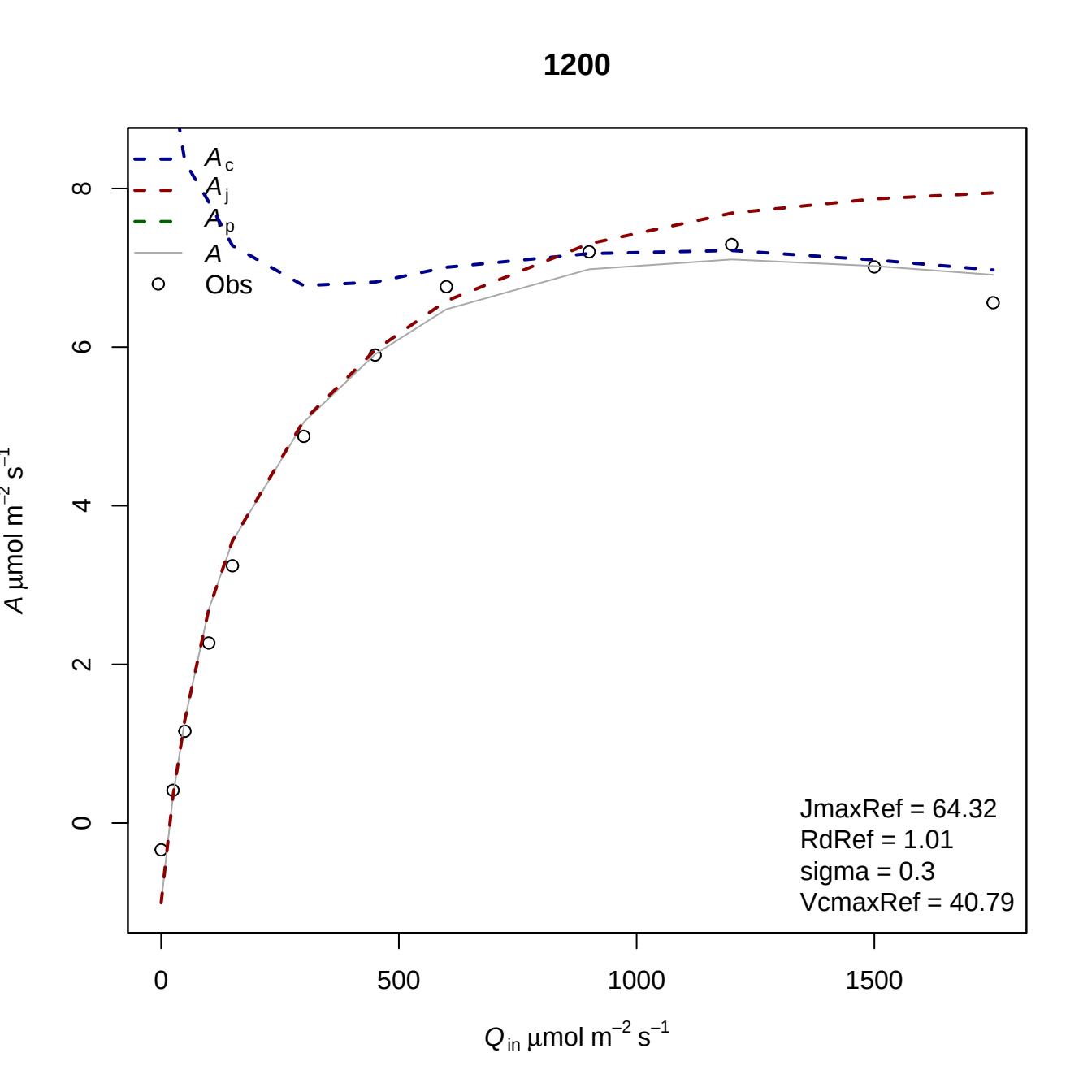

1195
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.02
RdRef = 0.58
sigma = 0.36
VcmaxRef = 66.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

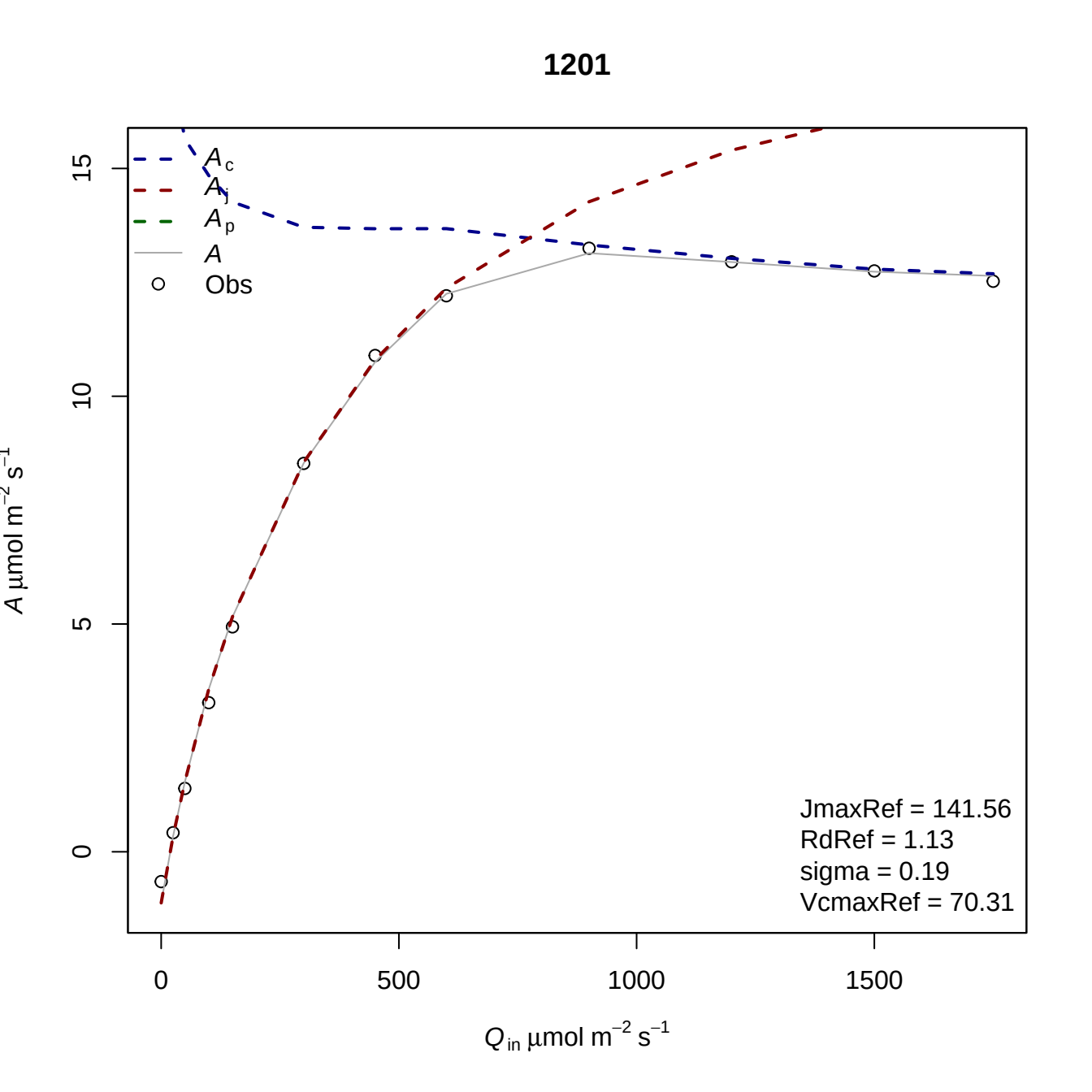

1196
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 183.87
RdRef = 0.84
sigma = 0.59
VcmaxRef = 90.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

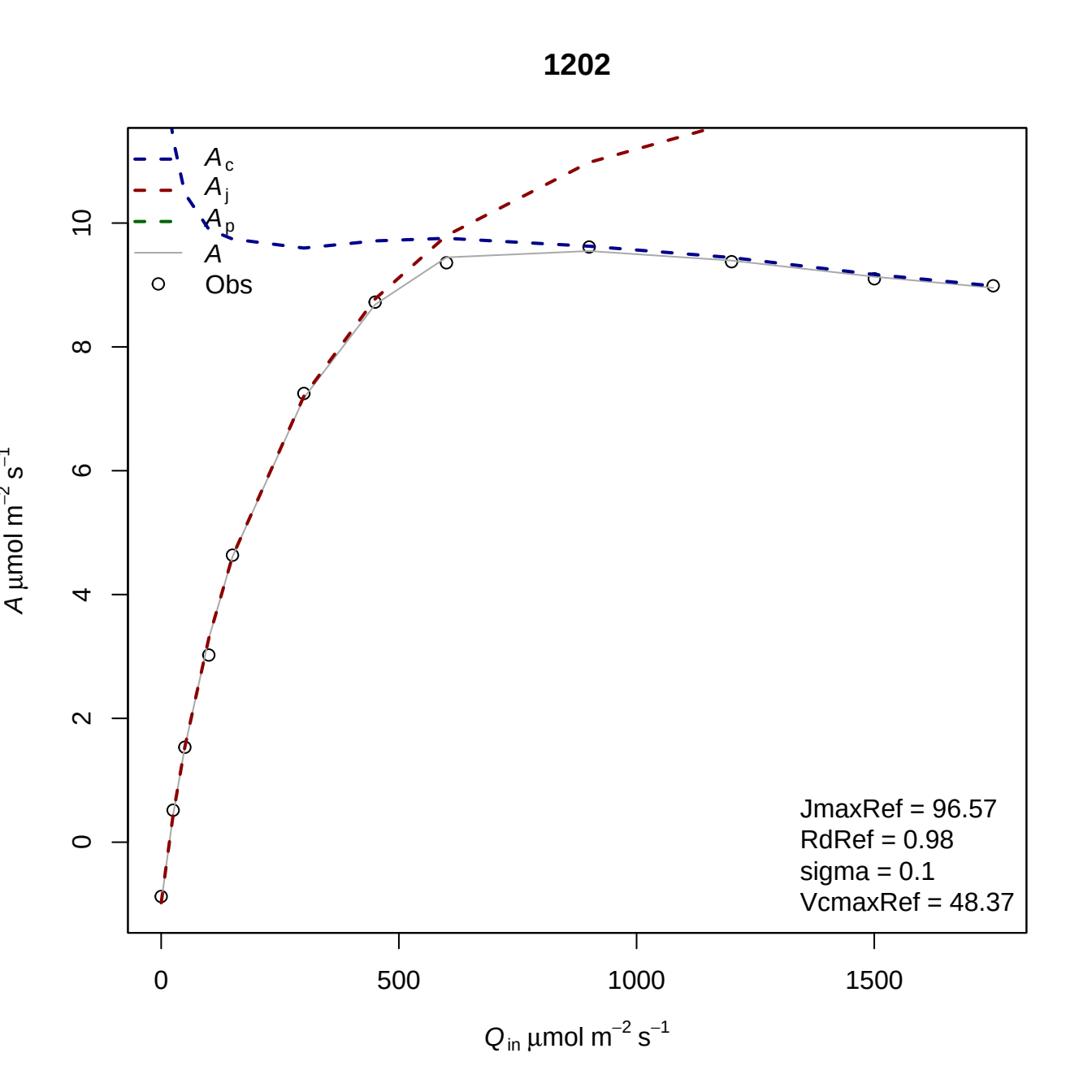

1197
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 223.19
RdRef = 0.81
sigma = 0.55
VcmaxRef = 87.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

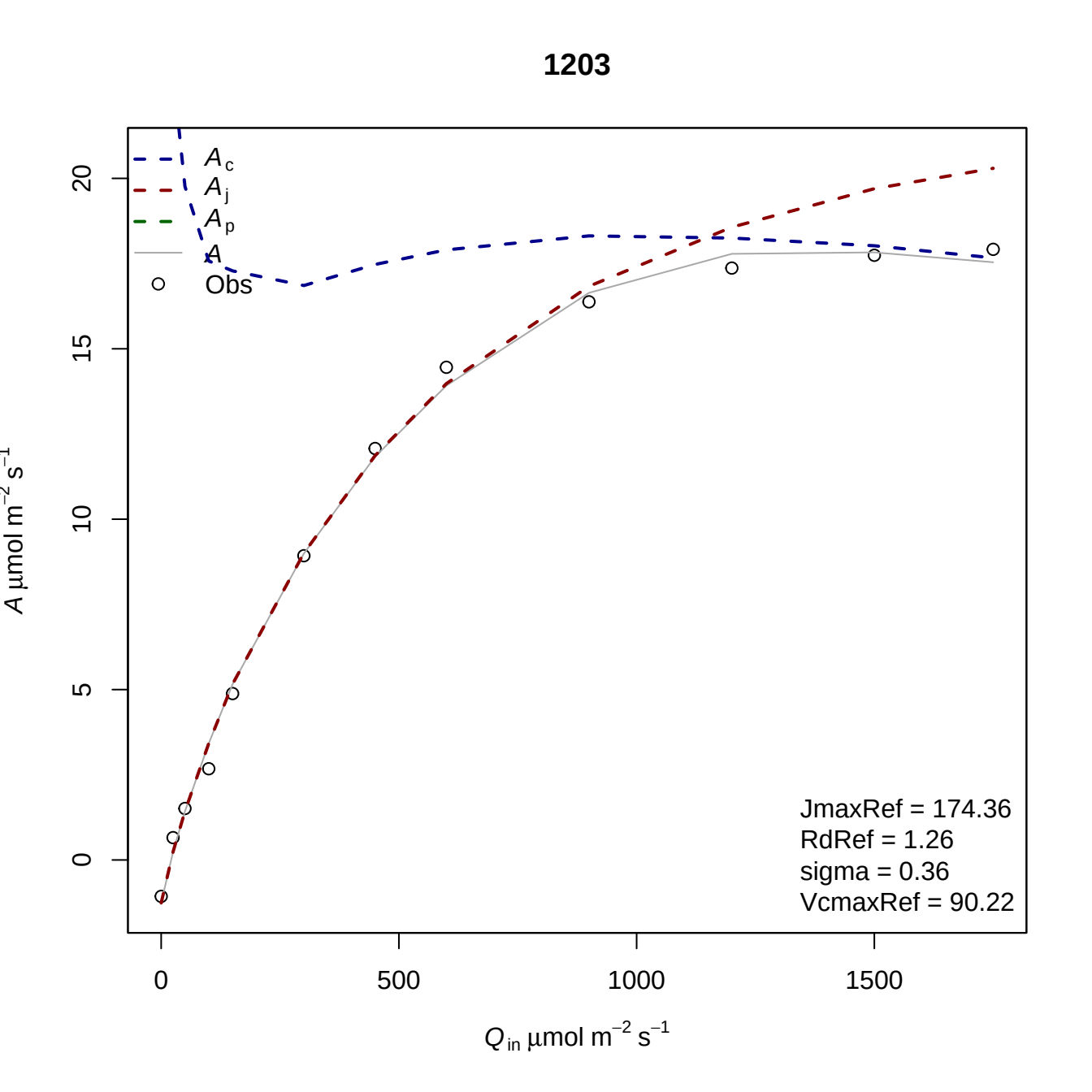

1198
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 216.49
RdRef = 0.92
sigma = 0.66
VcmaxRef = 96.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

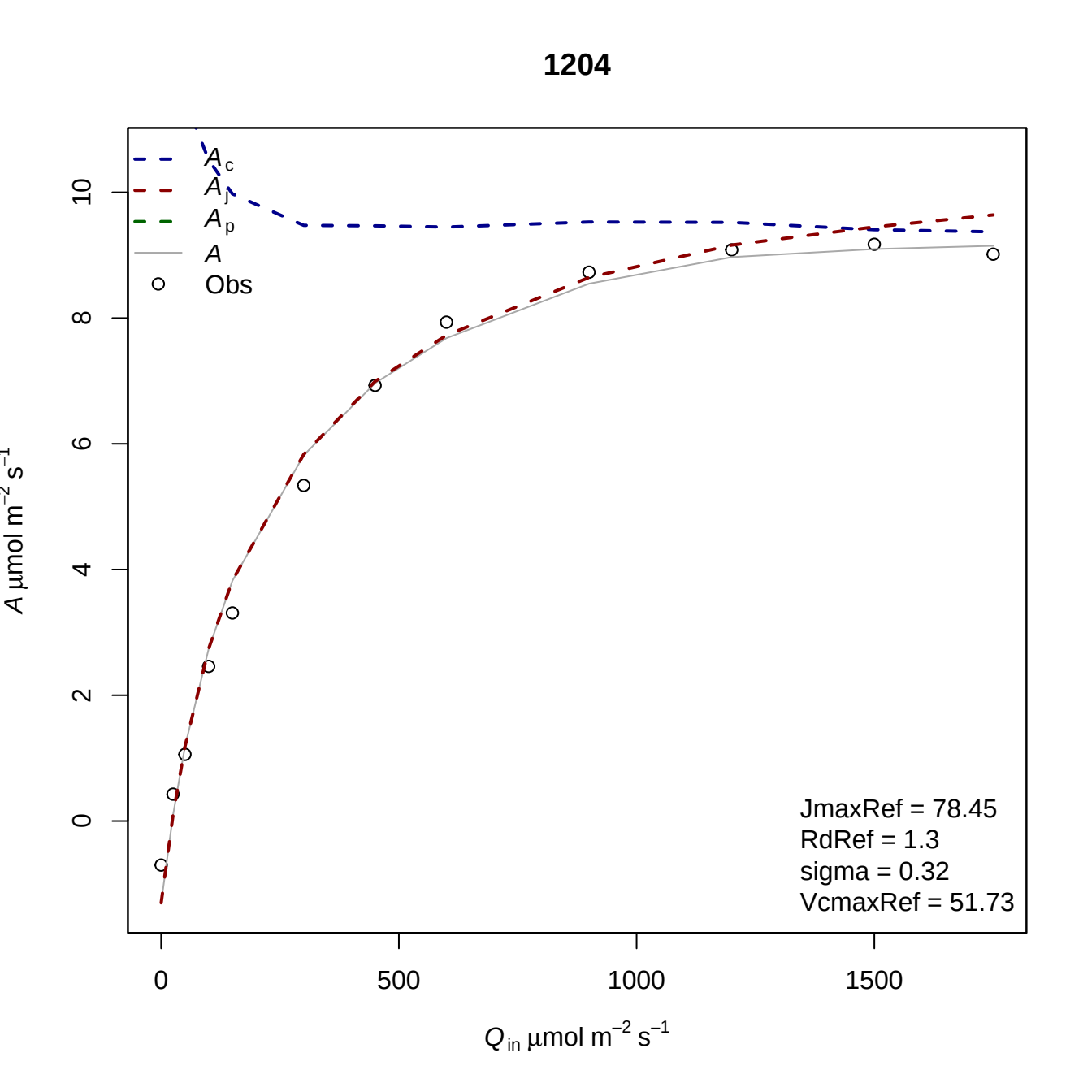

1199
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.54
RdRef = 1.66
sigma = 0.48
VcmaxRef = 18969.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500

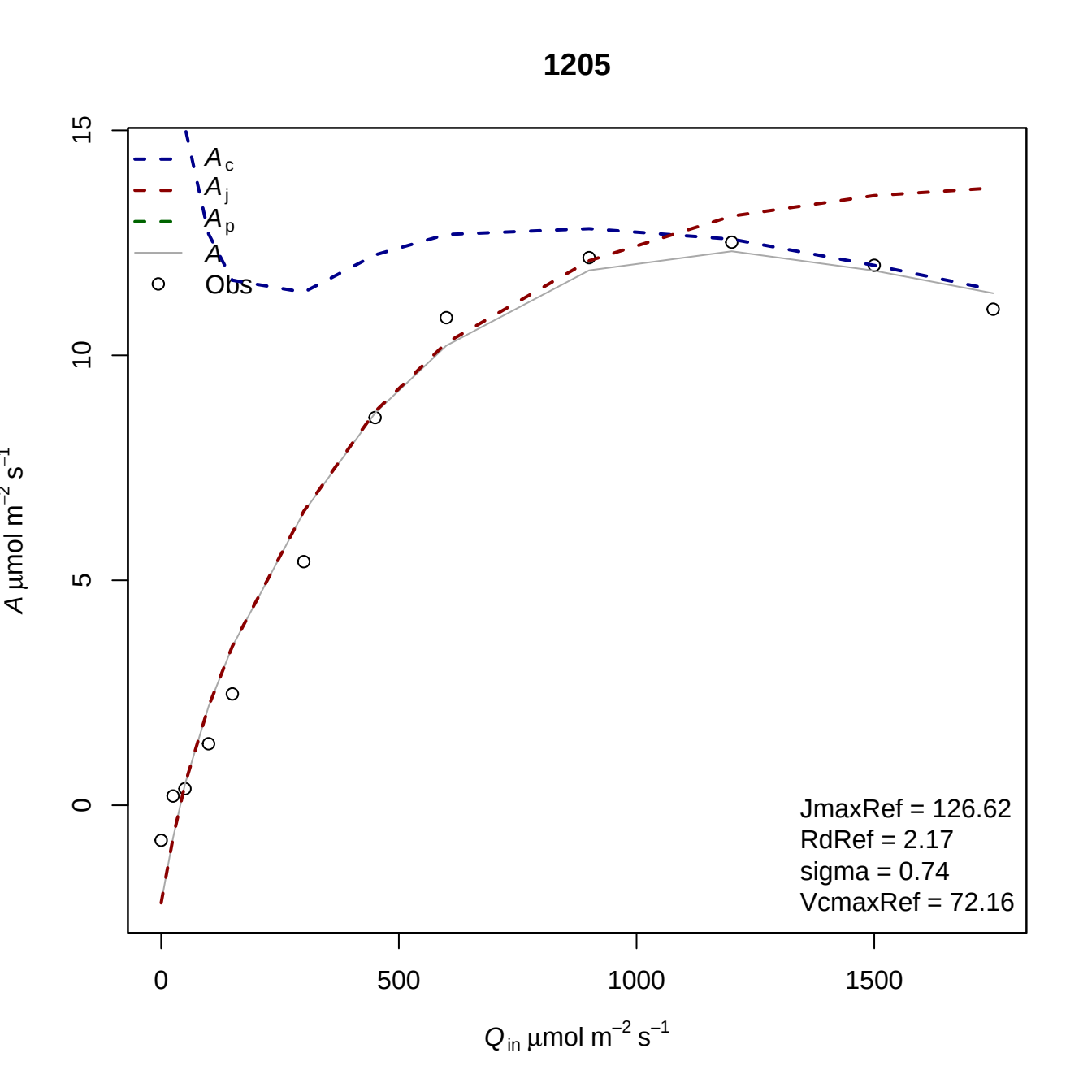

1200
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.32
RdRef = 1.01
sigma = 0.3
VcmaxRef = 40.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

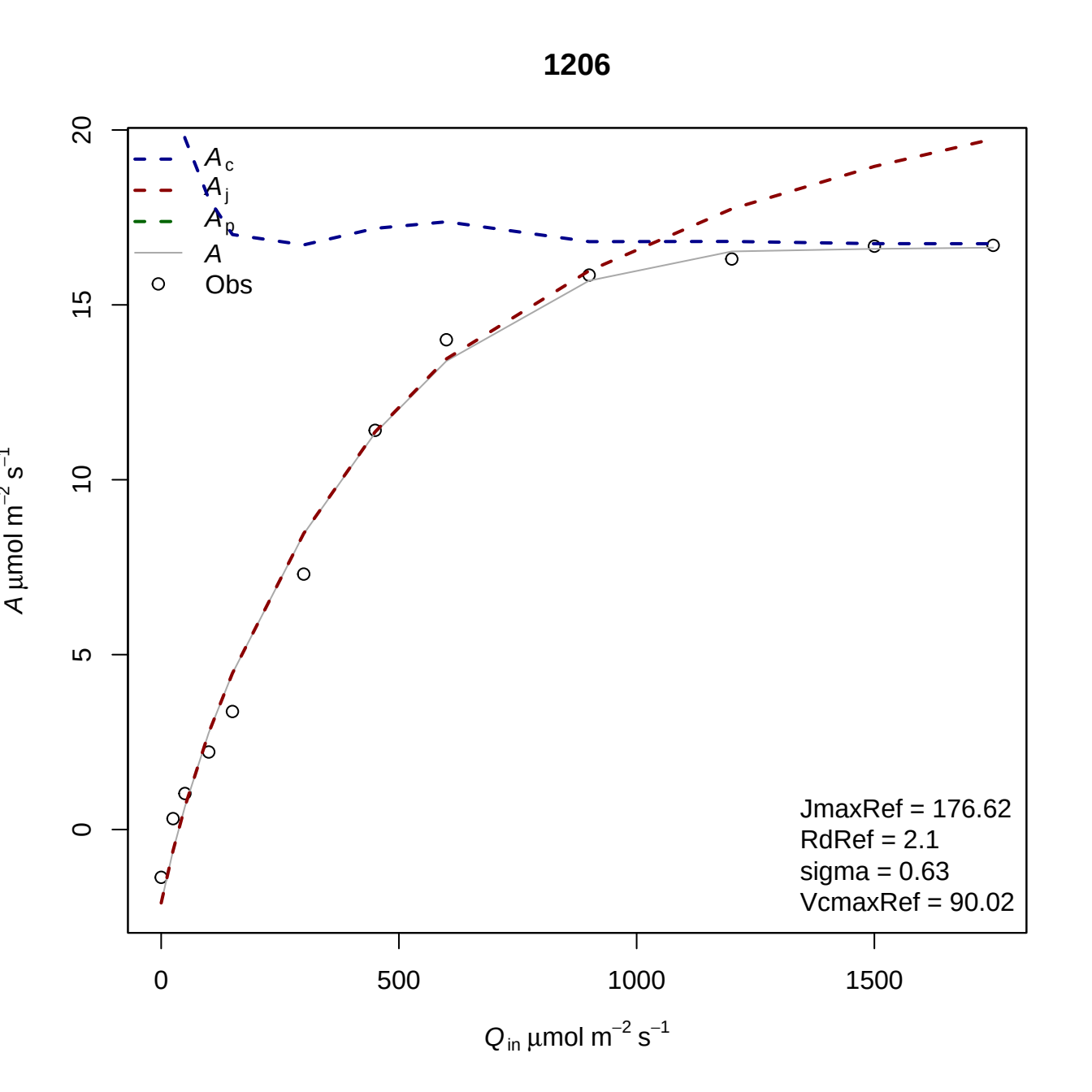

1201
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 141.56
RdRef = 1.13
sigma = 0.19
VcmaxRef = 70.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

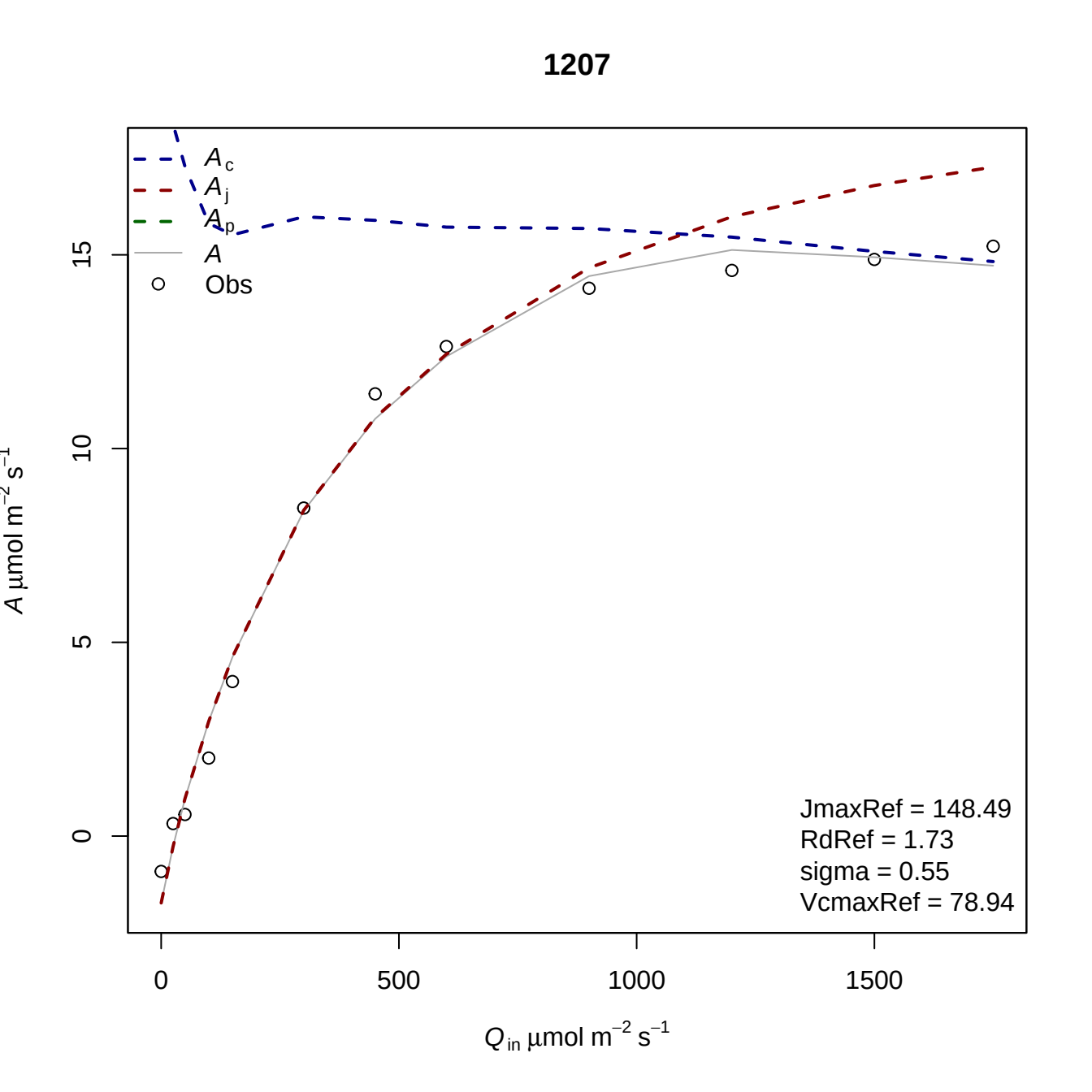

1202
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.57
RdRef = 0.98
sigma = 0.1
VcmaxRef = 48.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

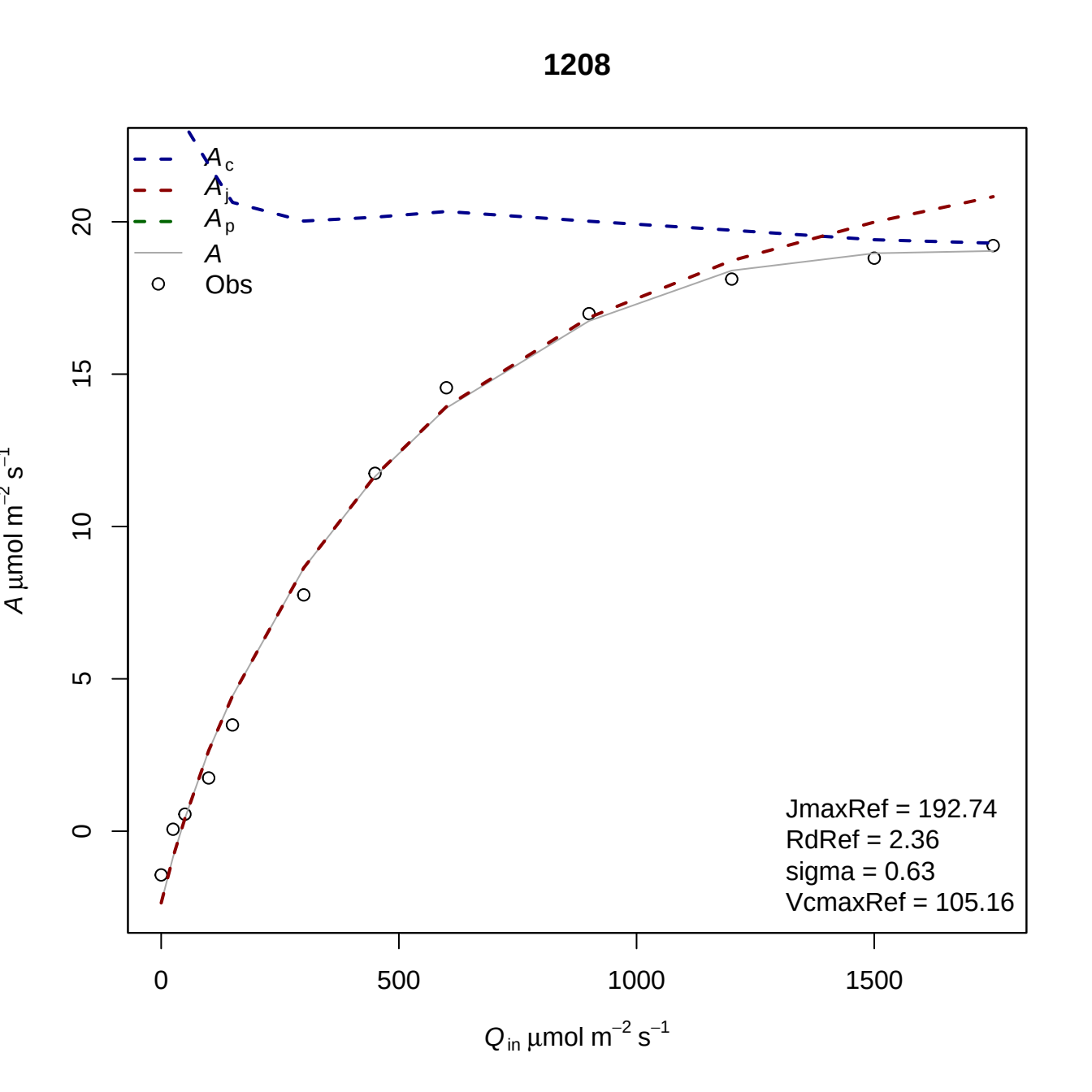

1203
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 174.36
RdRef = 1.26
sigma = 0.36
VcmaxRef = 90.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

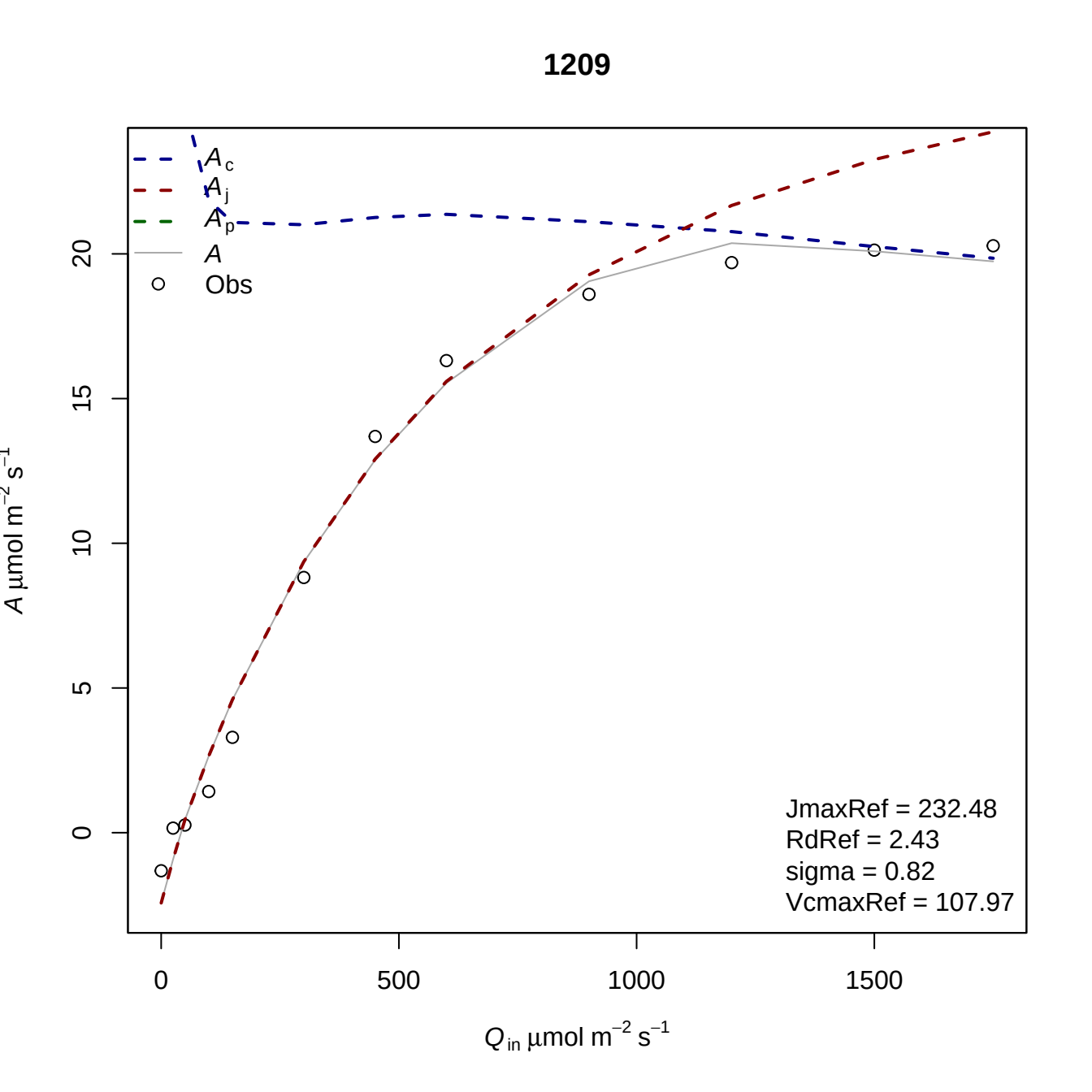

1204
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.45
RdRef = 1.3
sigma = 0.32
VcmaxRef = 51.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

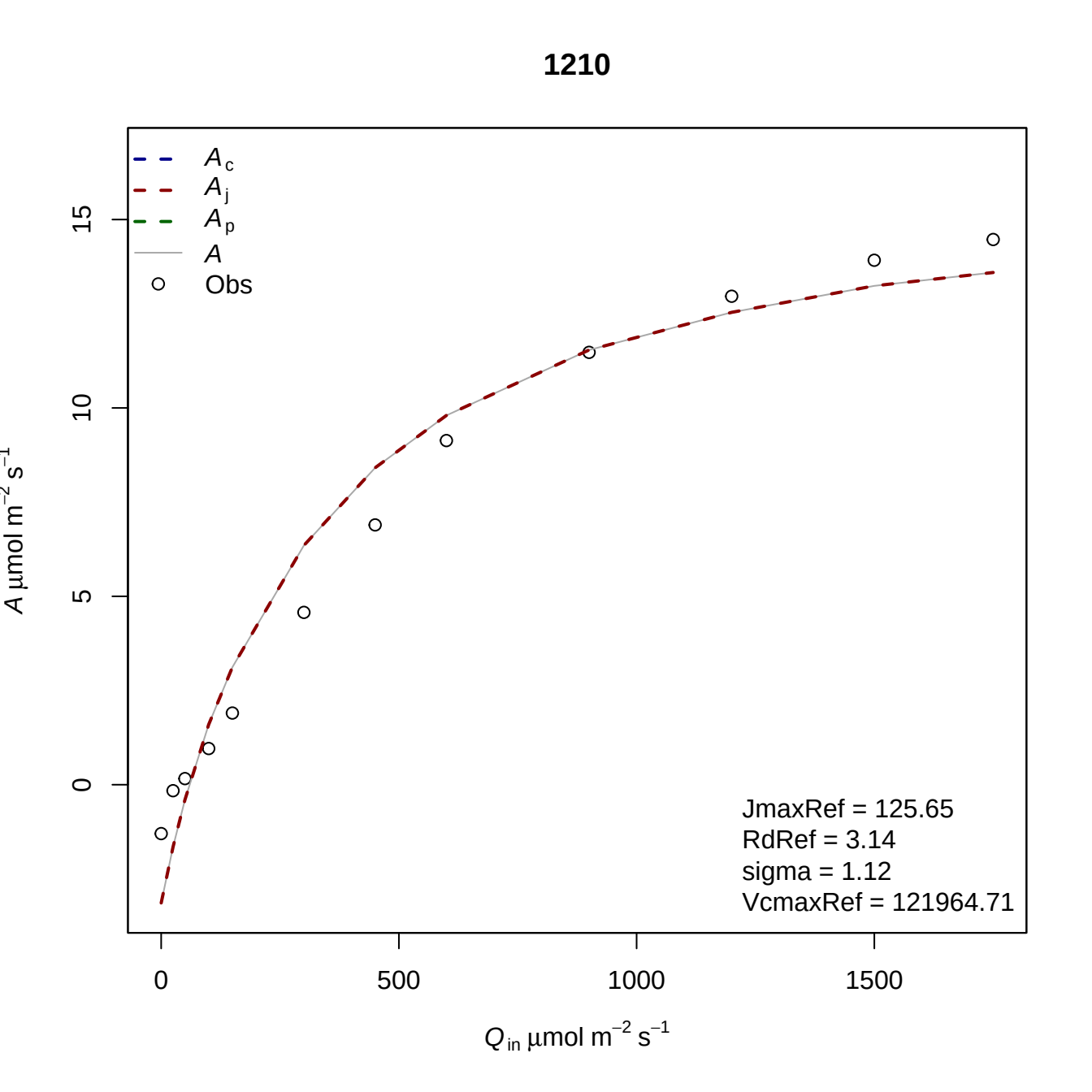

1205
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 126.62
RdRef = 2.17
sigma = 0.74
VcmaxRef = 72.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

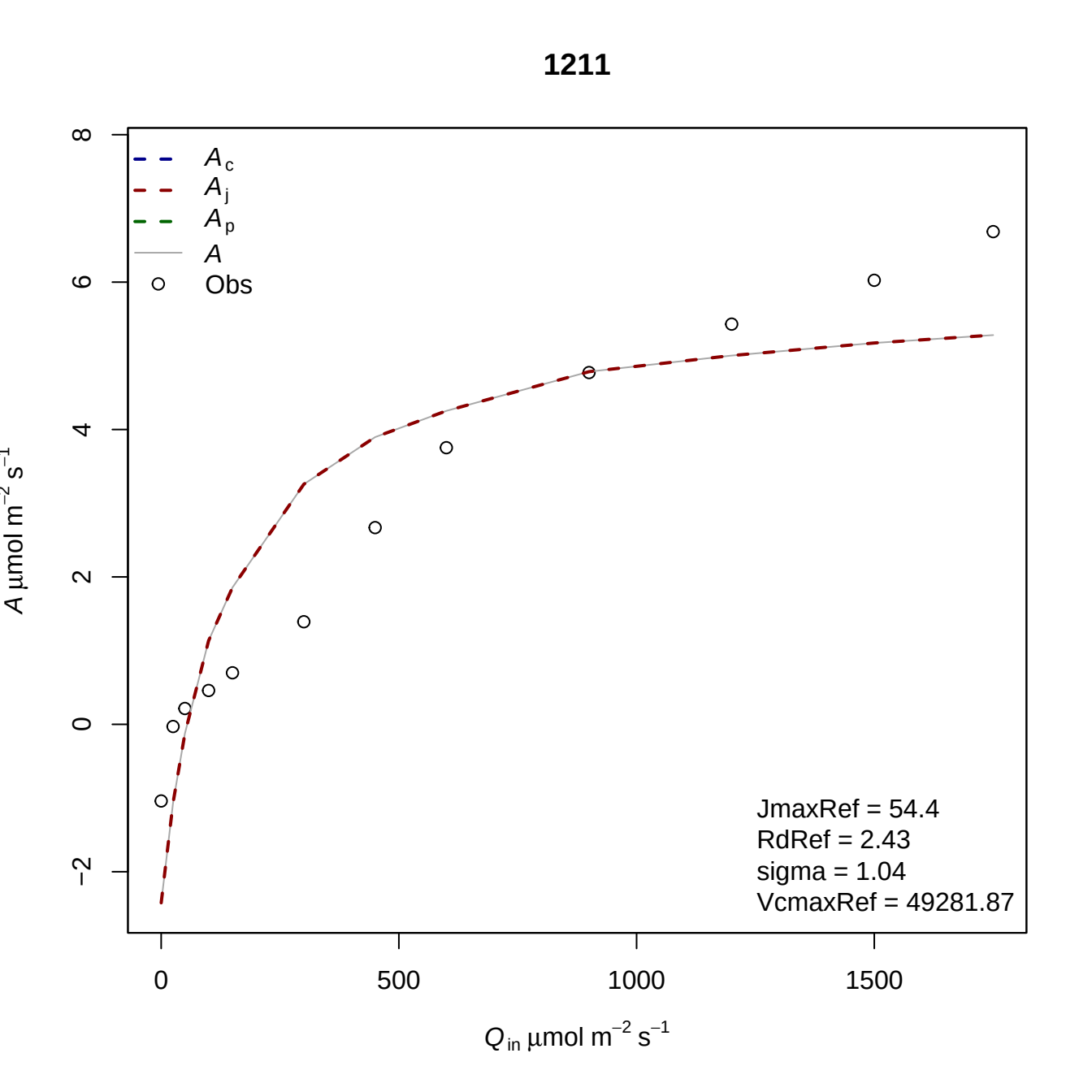

1206
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 176.62
RdRef = 2.1
sigma = 0.63
VcmaxRef = 90.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

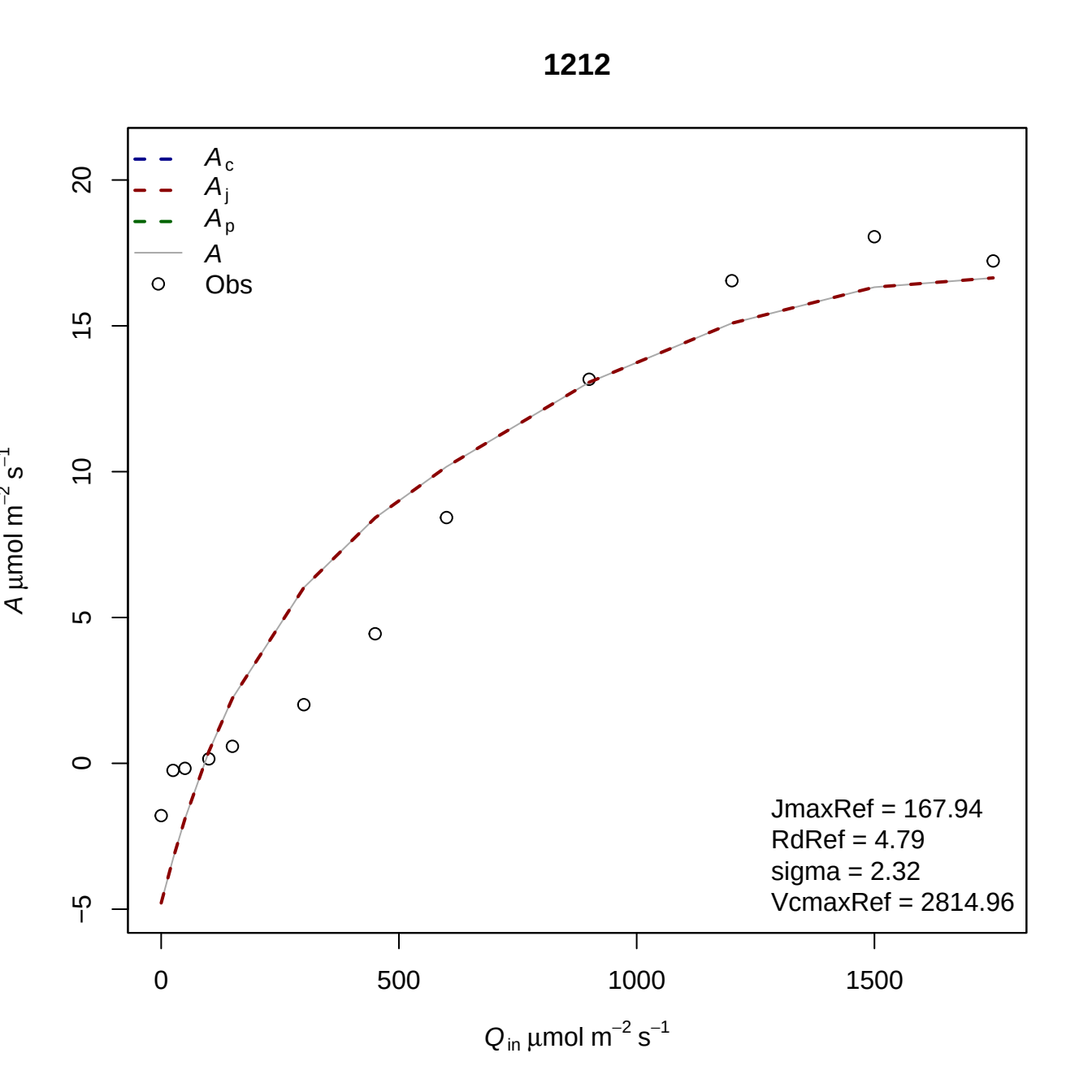

1207
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 148.49
RdRef = 1.73
sigma = 0.55
VcmaxRef = 78.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

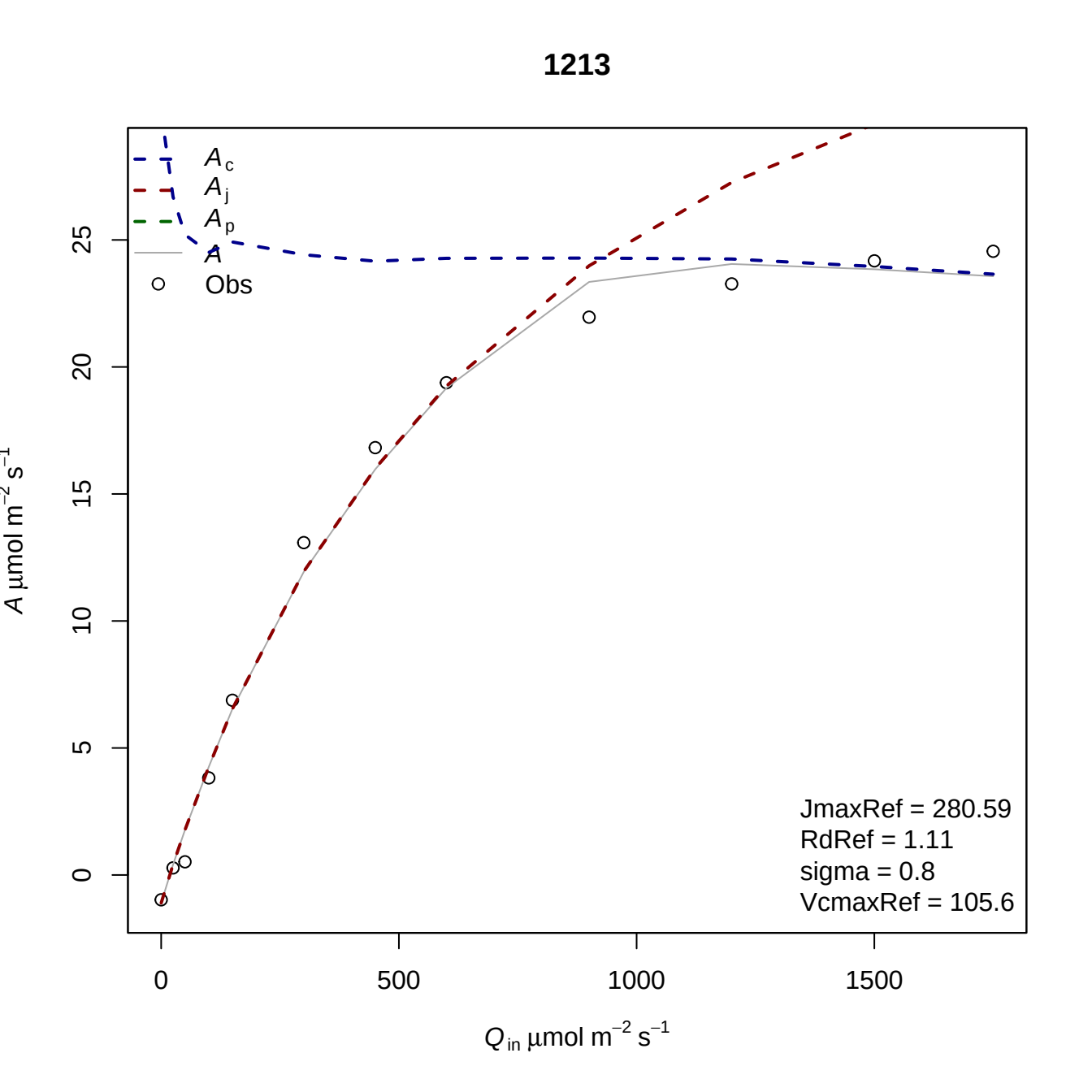

1208
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 192.74
RdRef = 2.36
sigma = 0.63
VcmaxRef = 105.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

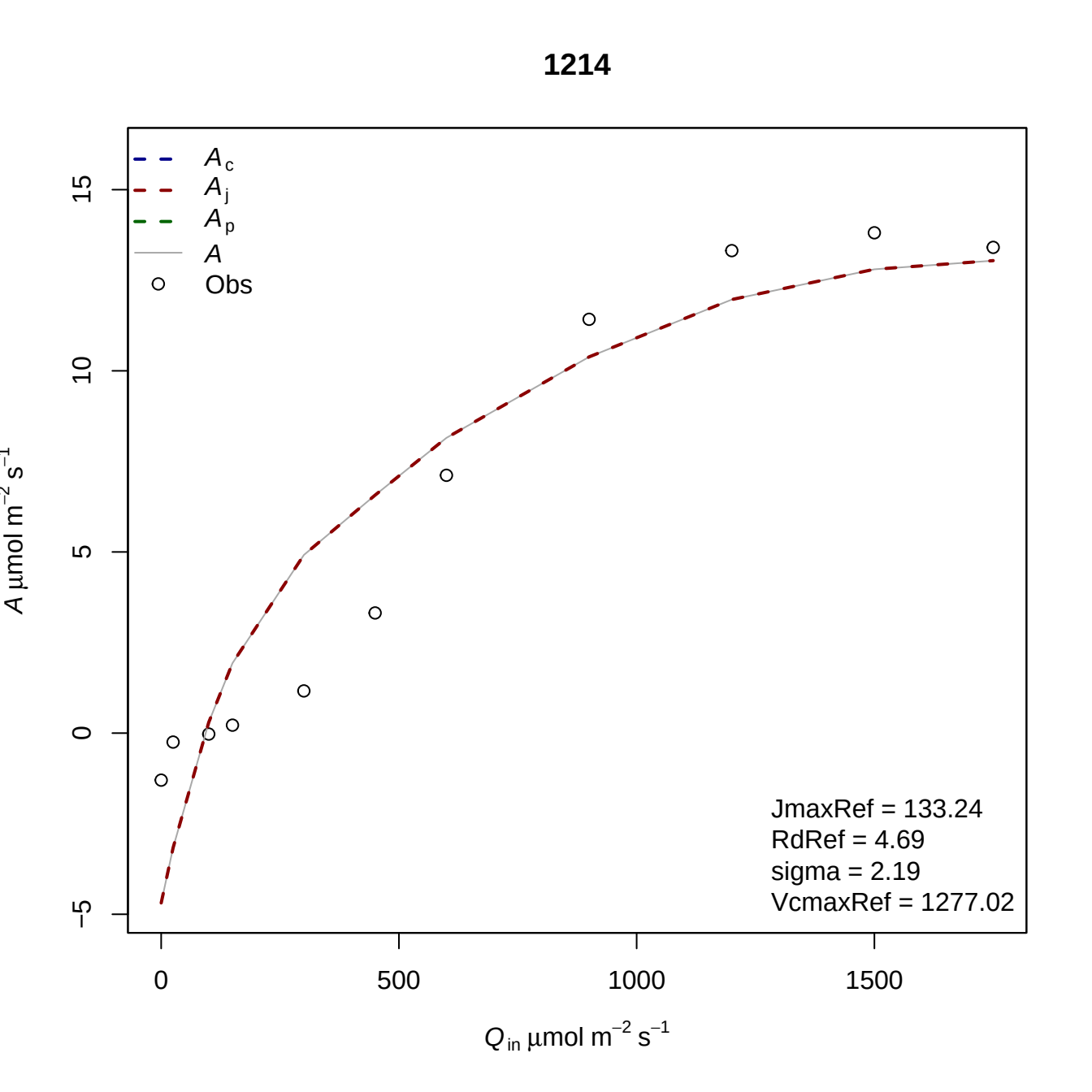

1209
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 232.48
RdRef = 2.43
sigma = 0.82
VcmaxRef = 107.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

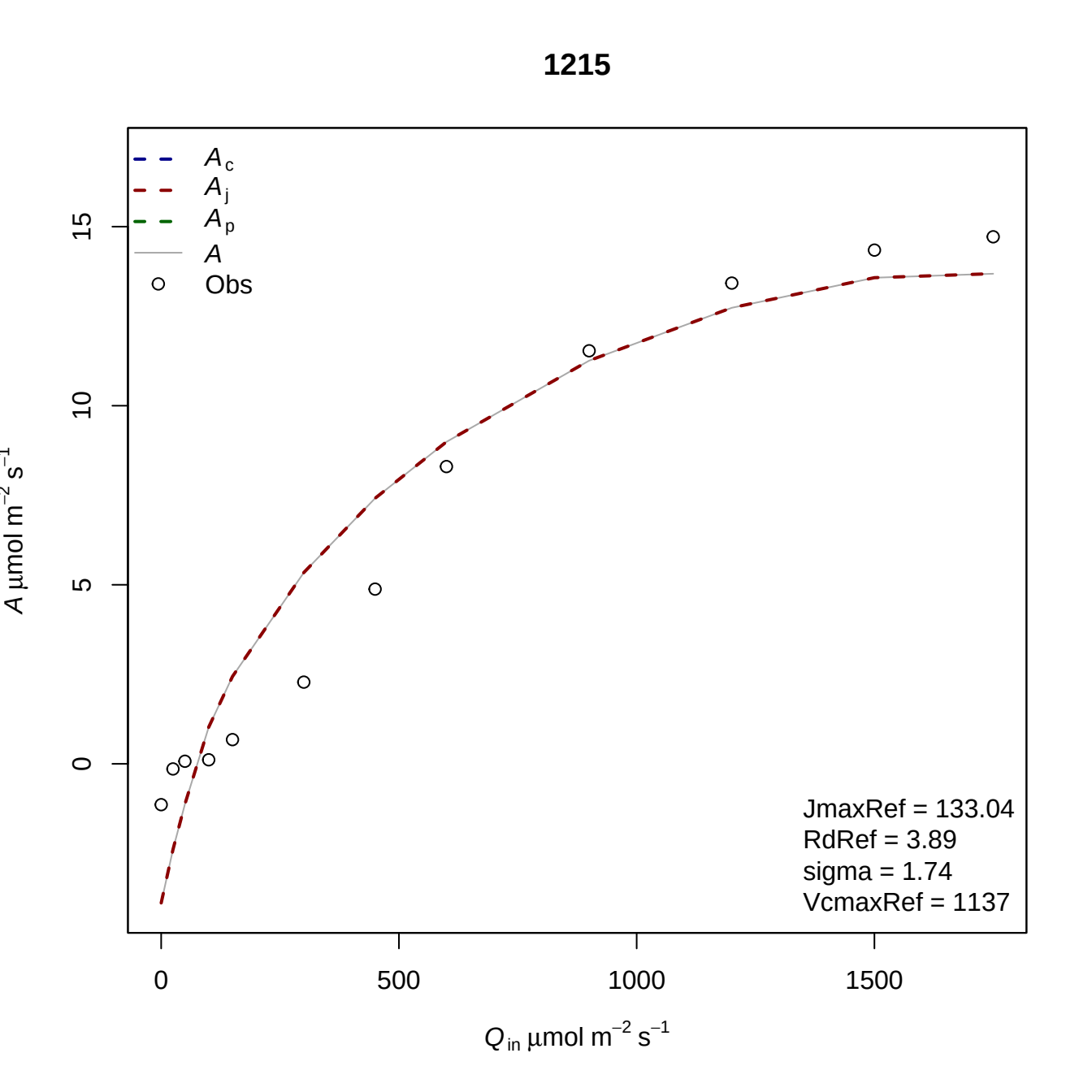

1210
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.65
RdRef = 3.14
sigma = 1.12
VcmaxRef = 121964.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

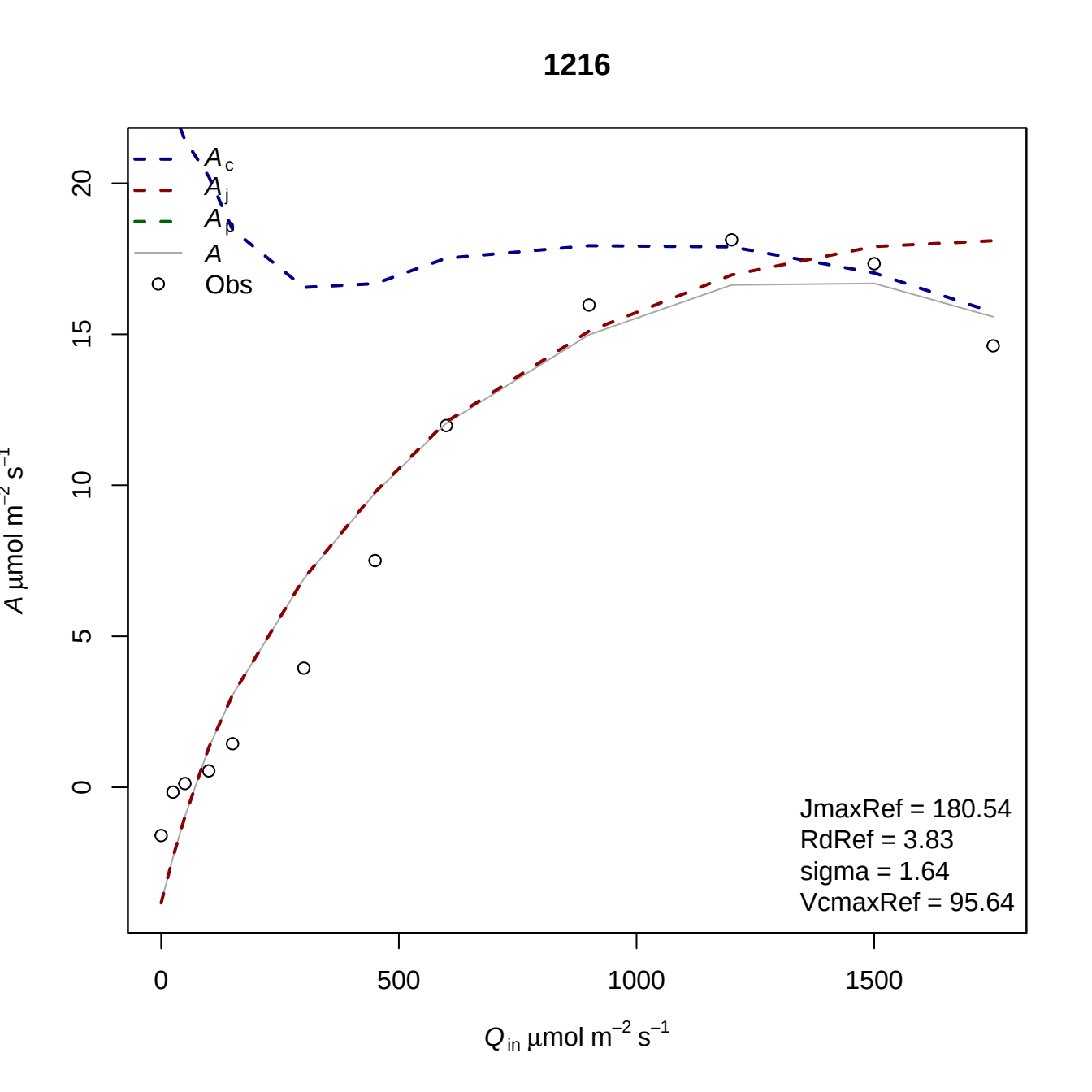

1211
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.4
RdRef = 2.43
sigma = 1.04
VcmaxRef = 49281.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

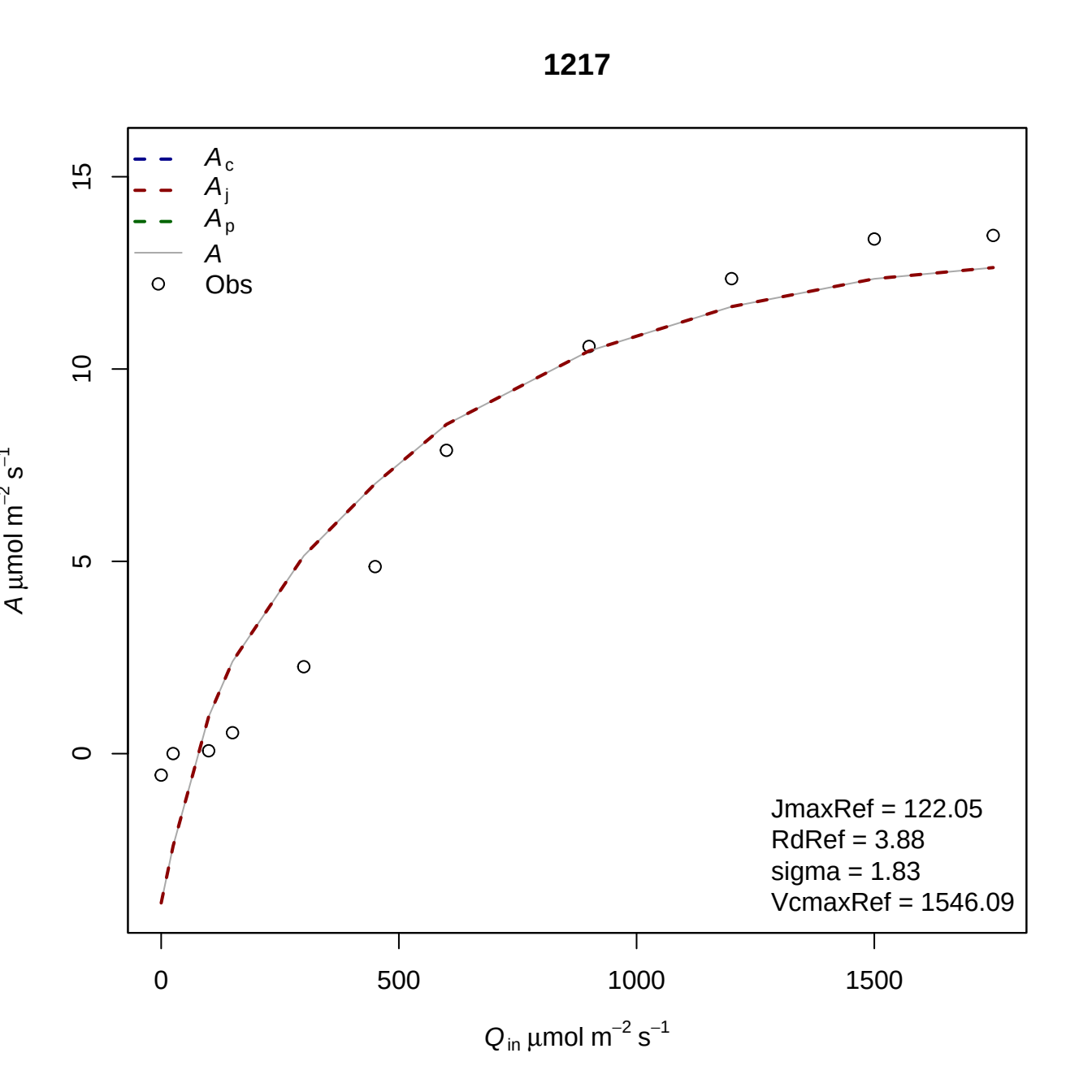

1212
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 167.94
RdRef = 4.79
sigma = 2.32
VcmaxRef = 2814.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
20
15
10
5
0
−5
0
500
1000
1500

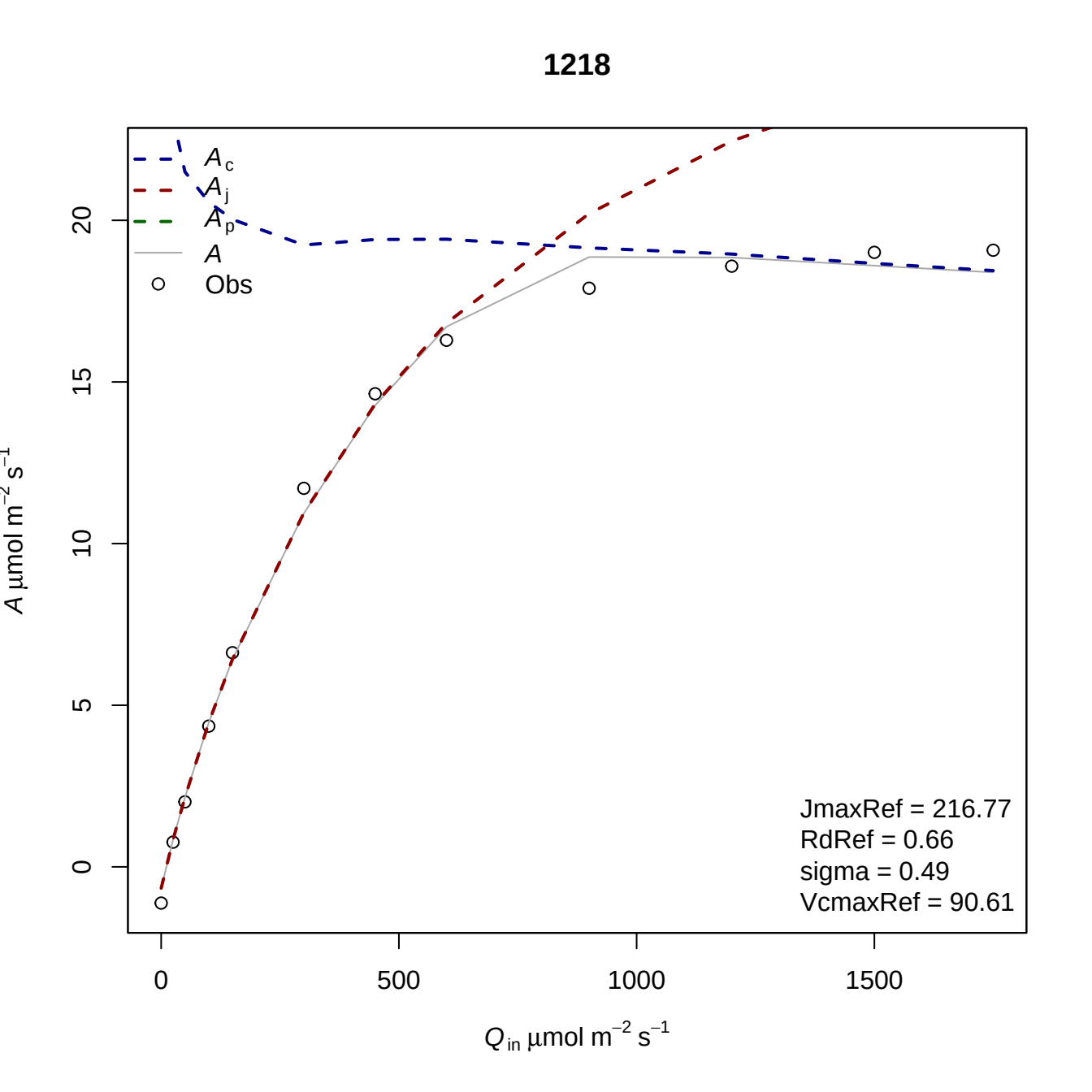

1213
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 280.59
RdRef = 1.11
sigma = 0.8
VcmaxRef = 105.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

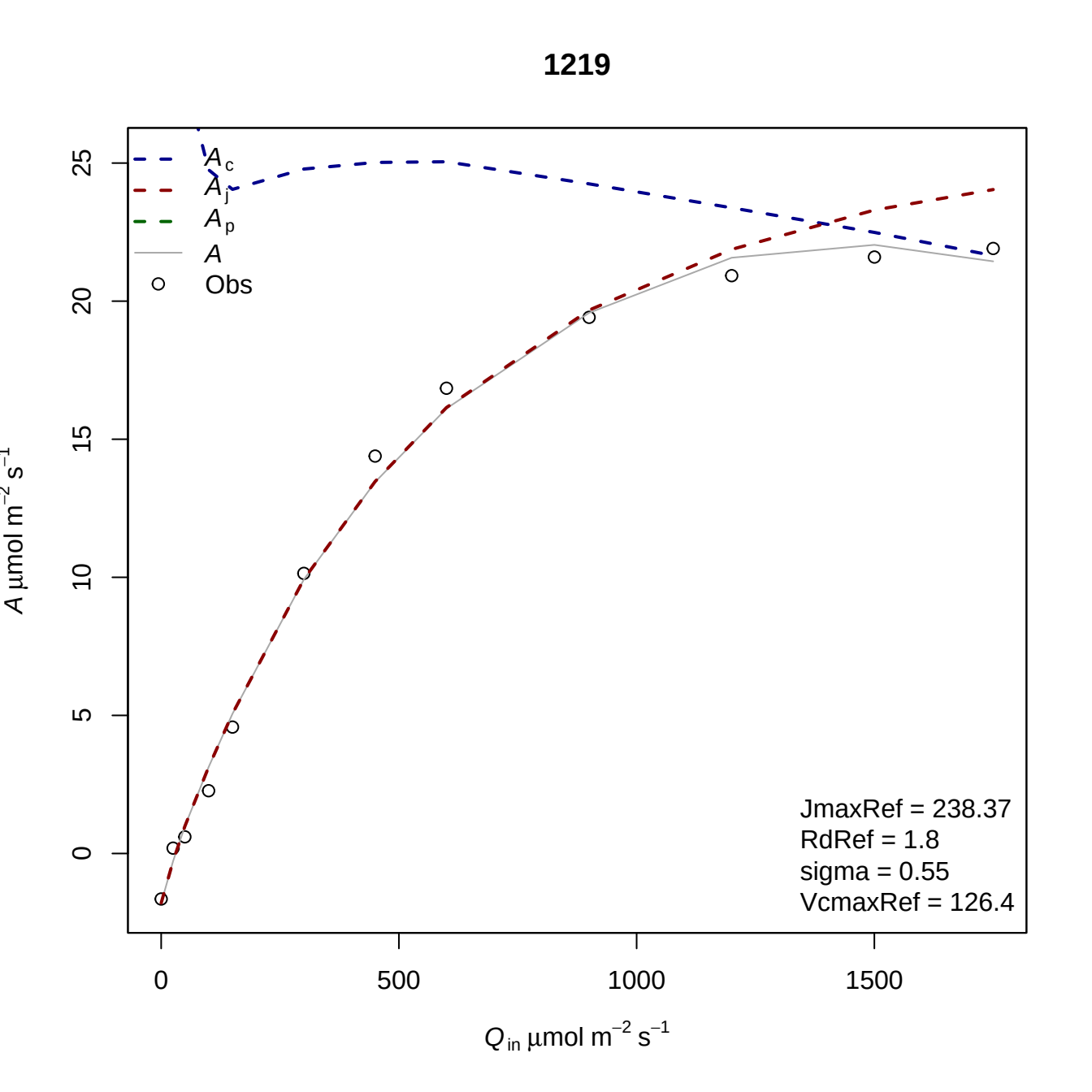

1214
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.24
RdRef = 4.69
sigma = 2.19
VcmaxRef = 1277.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
15
10
5
0
−5
0
500
1000
1500

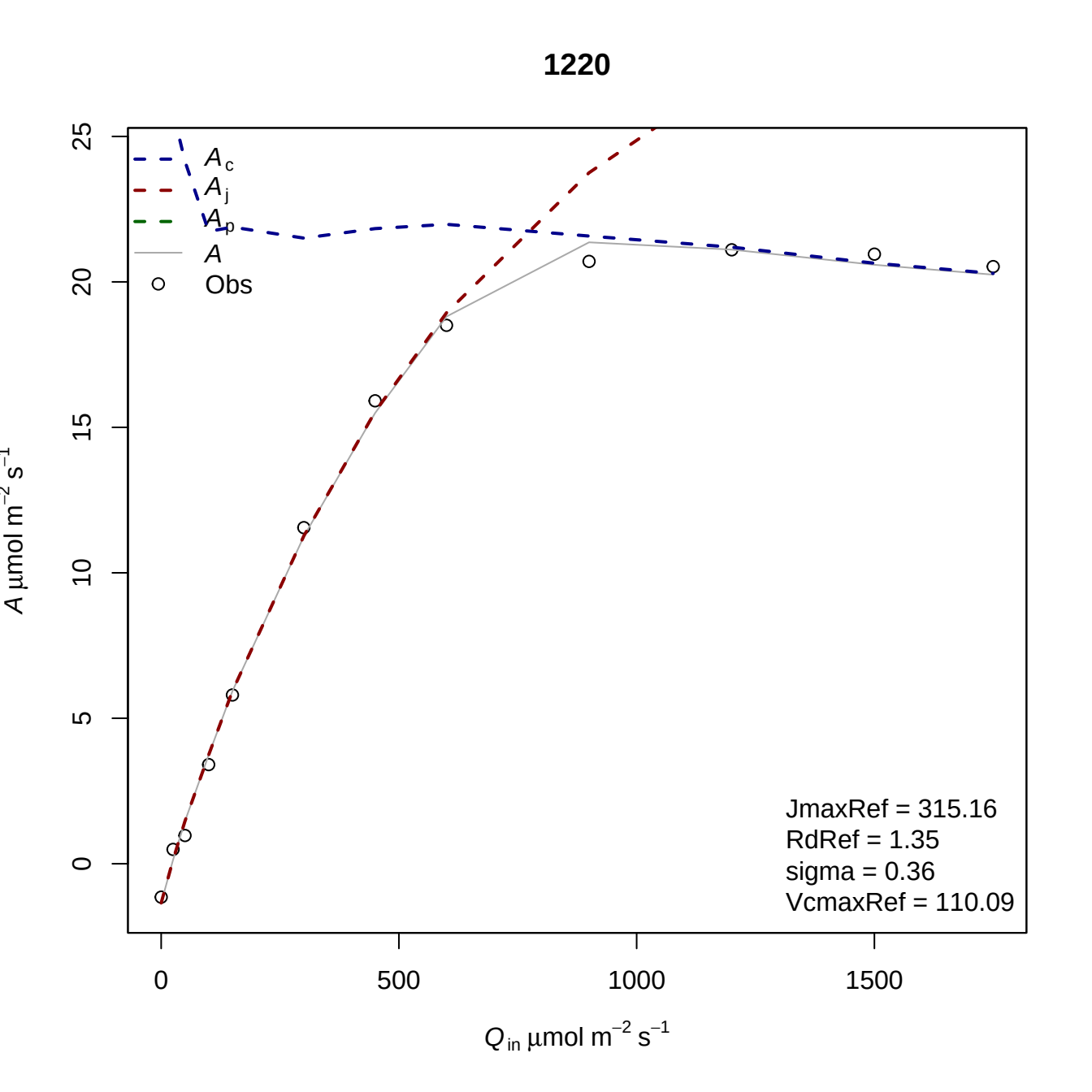

1215
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.04
RdRef = 3.89
sigma = 1.74
VcmaxRef = 1137
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

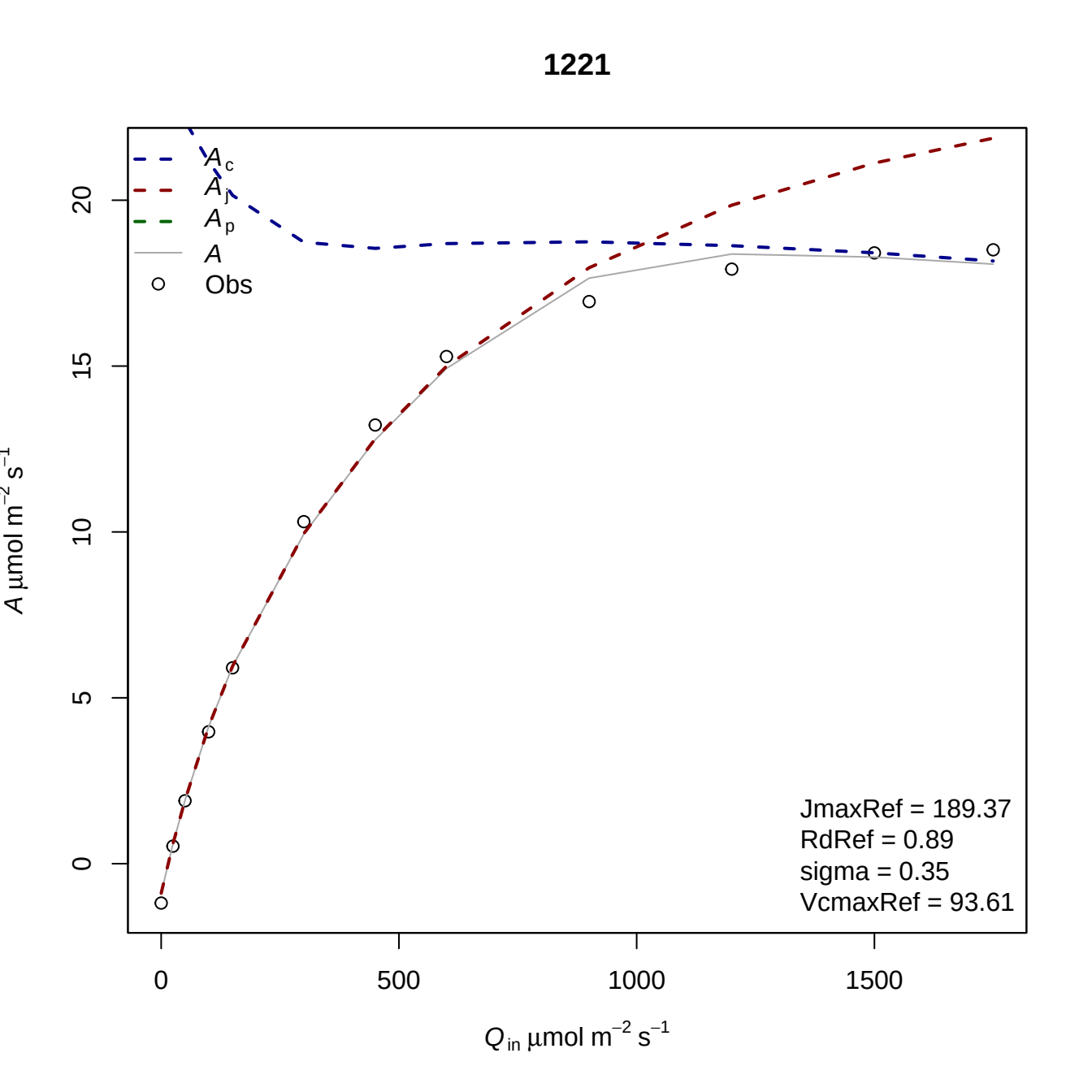

1216
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 180.54
RdRef = 3.83
sigma = 1.64
VcmaxRef = 95.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

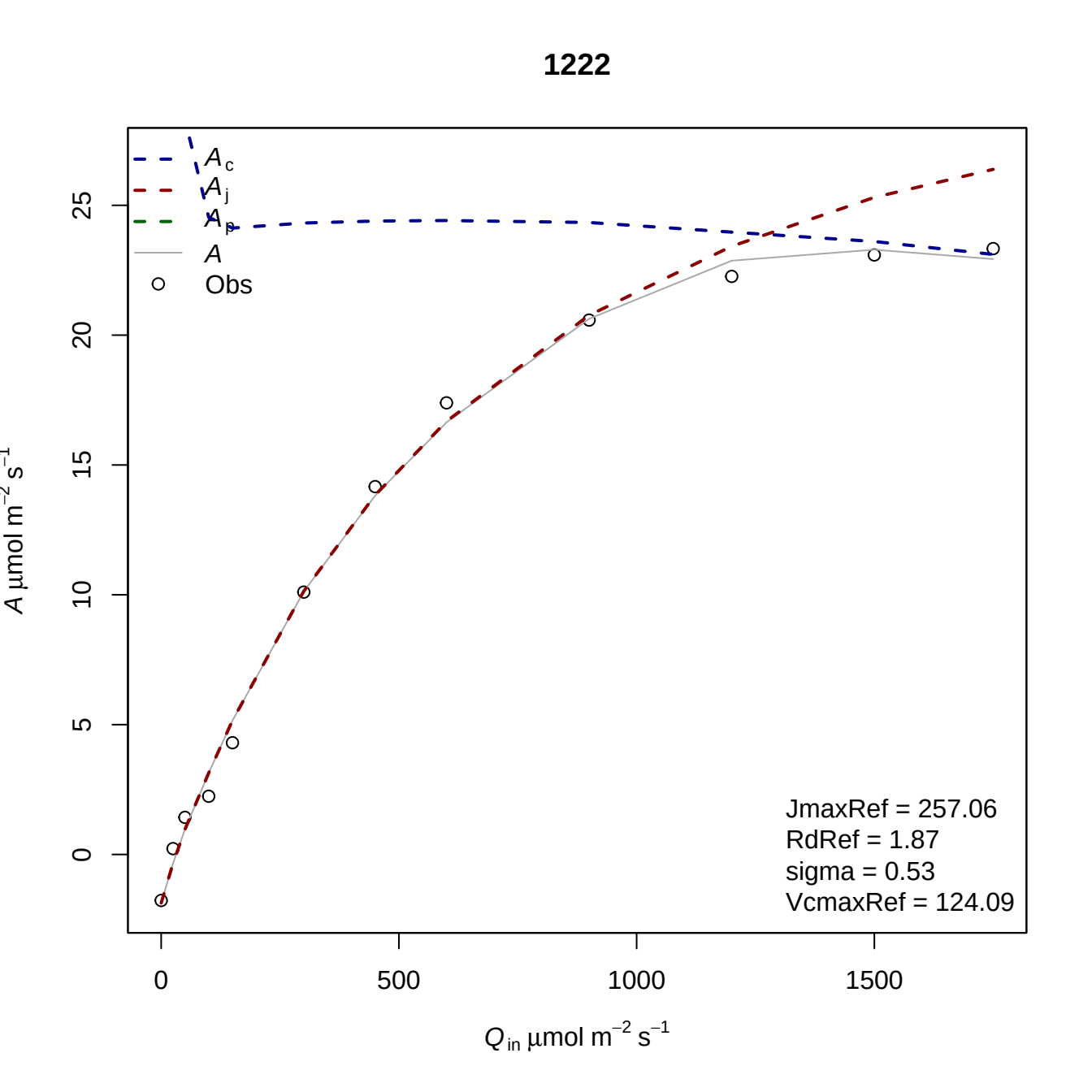

1217
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.05
RdRef = 3.88
sigma = 1.83
VcmaxRef = 1546.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

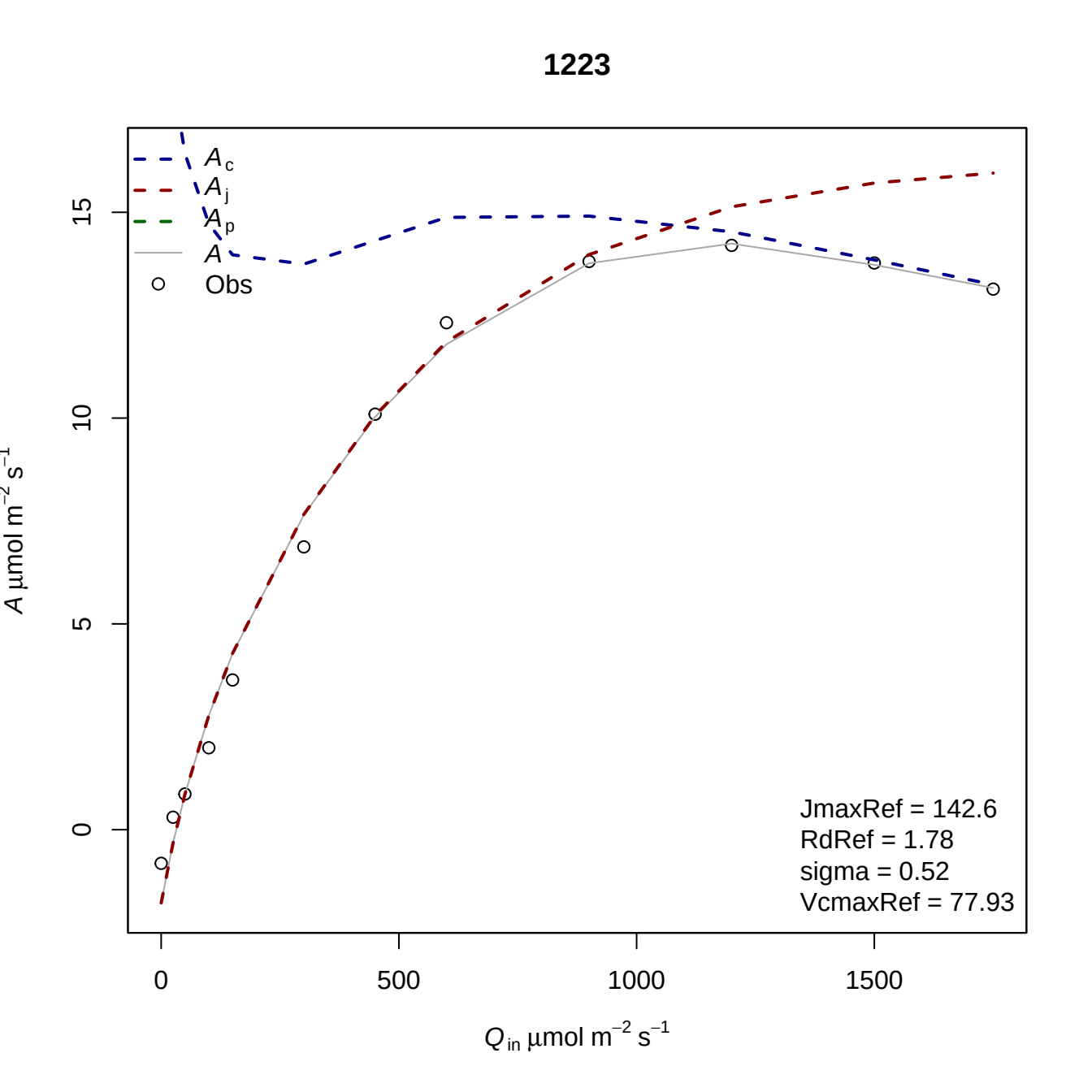

1218
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 216.77
RdRef = 0.66
sigma = 0.49
VcmaxRef = 90.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

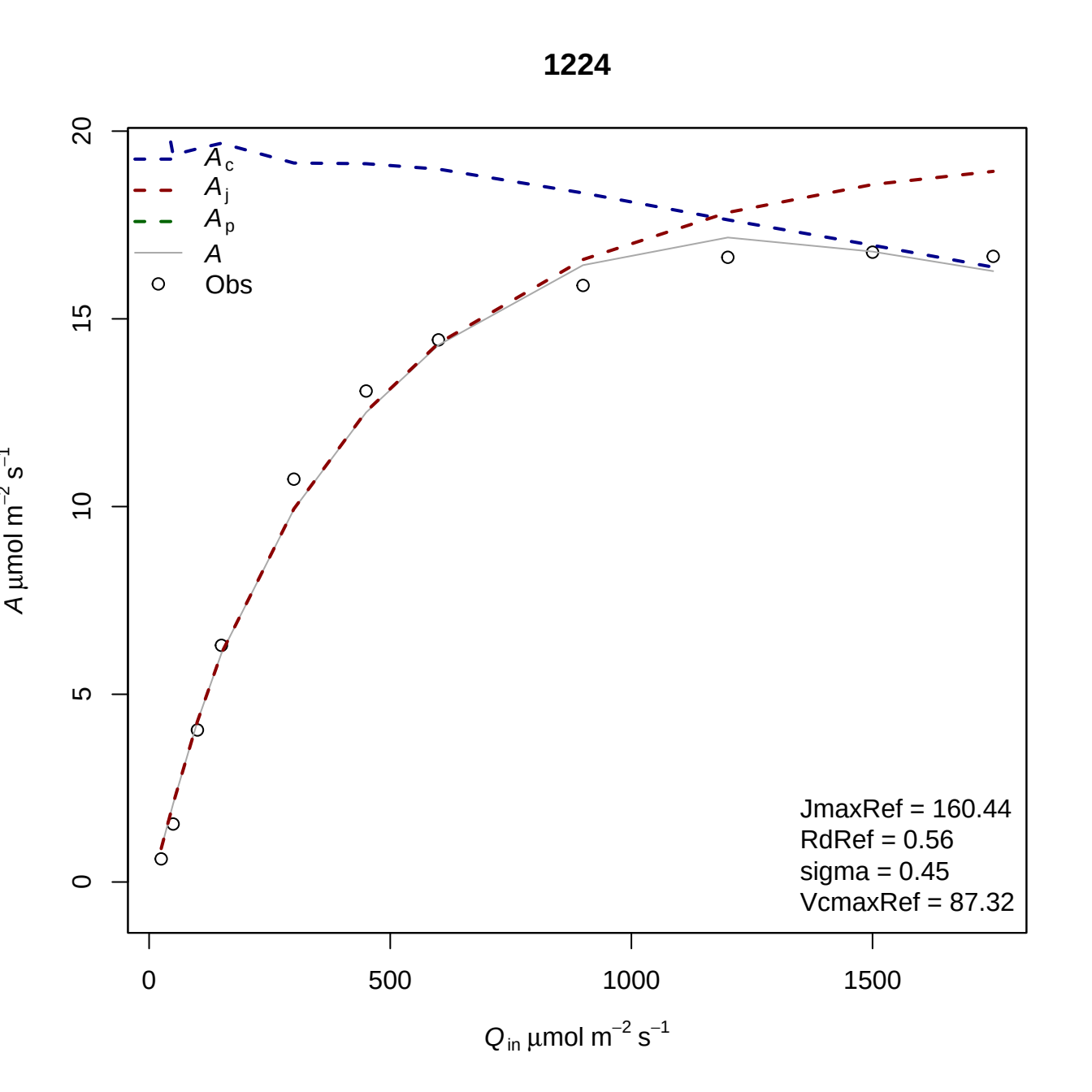

1219
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 238.37
RdRef = 1.8
sigma = 0.55
VcmaxRef = 126.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

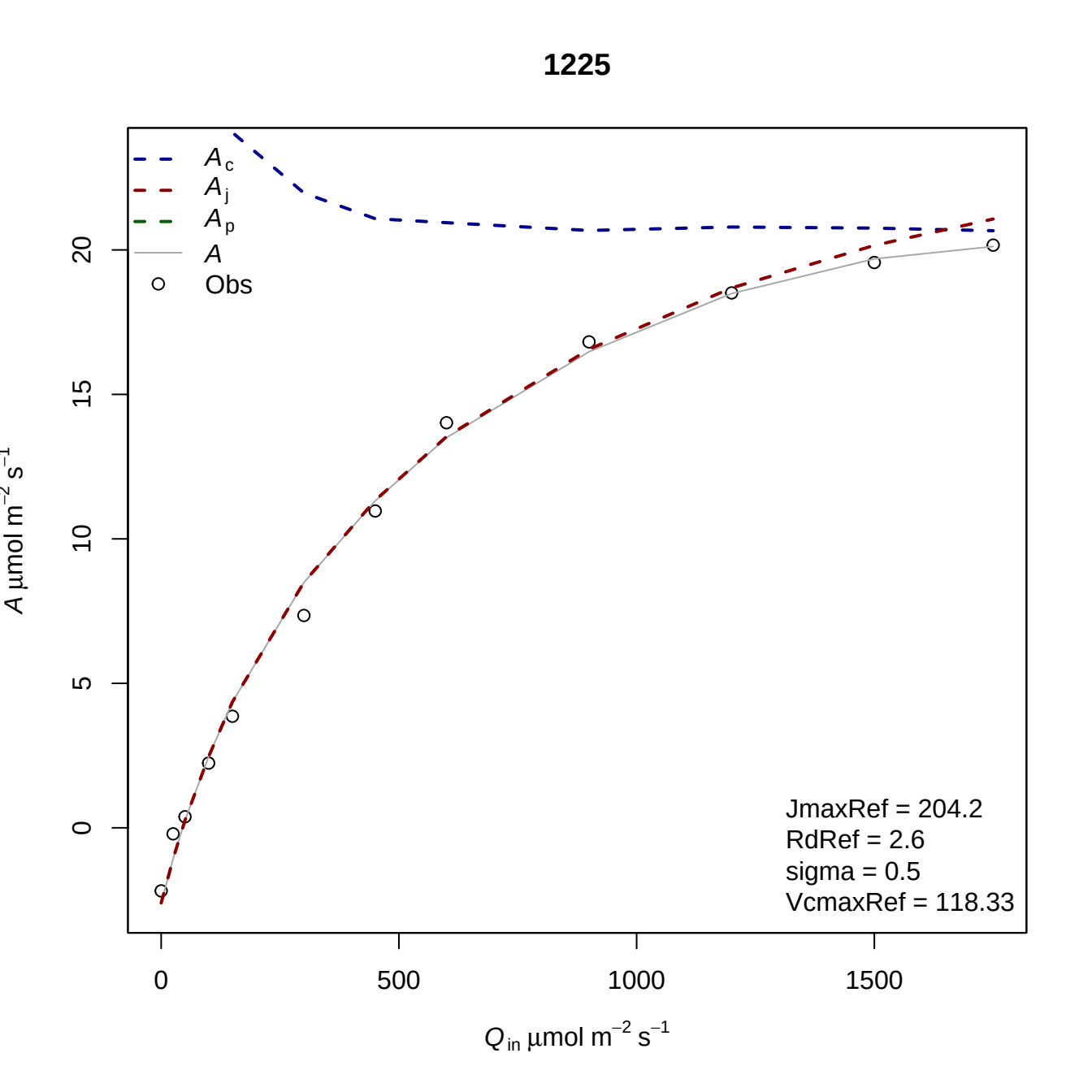

1220
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 315.16
RdRef = 1.35
sigma = 0.36
VcmaxRef = 110.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

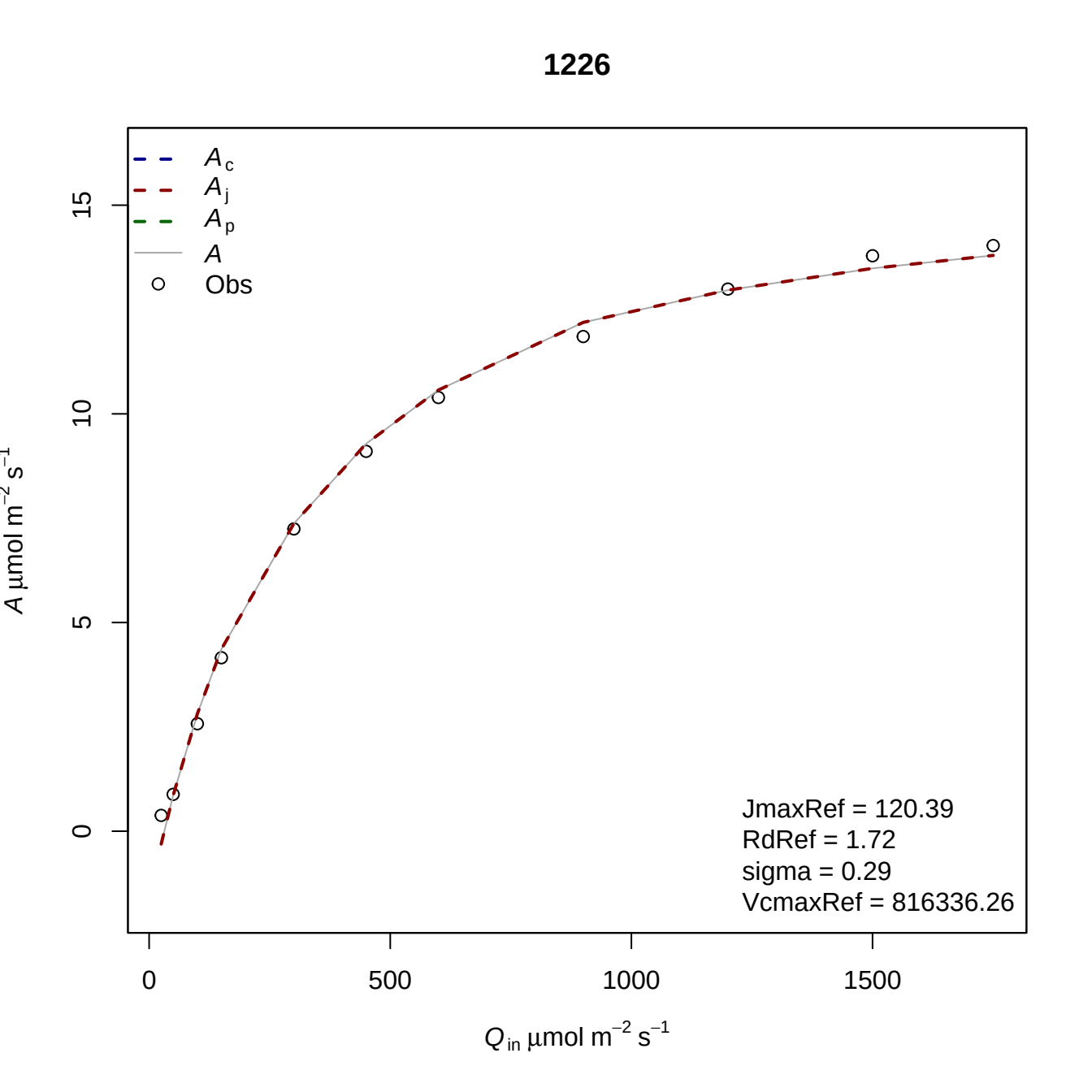

1221
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 189.37
RdRef = 0.89
sigma = 0.35
VcmaxRef = 93.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

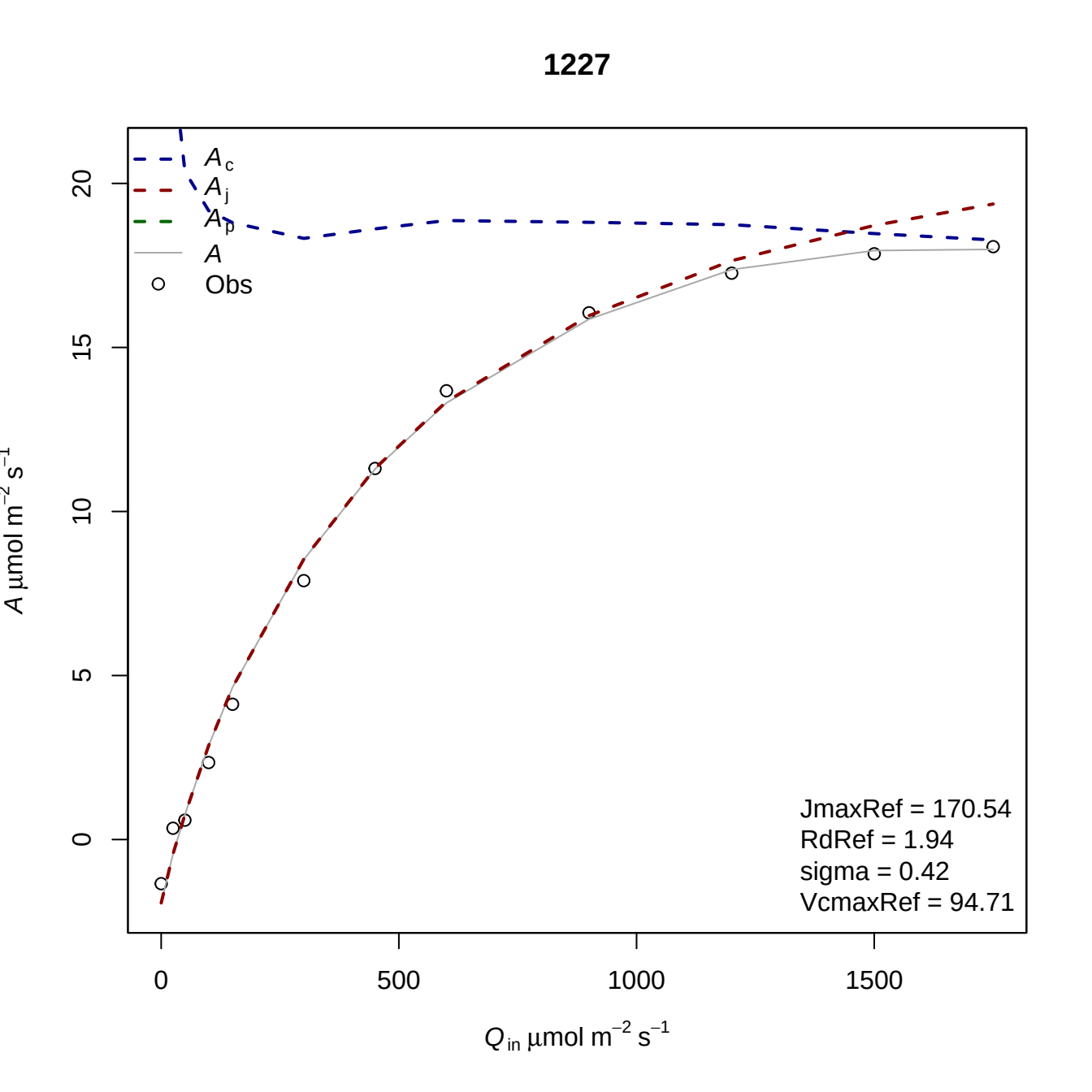

1222
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 257.06
RdRef = 1.87
sigma = 0.53
VcmaxRef = 124.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

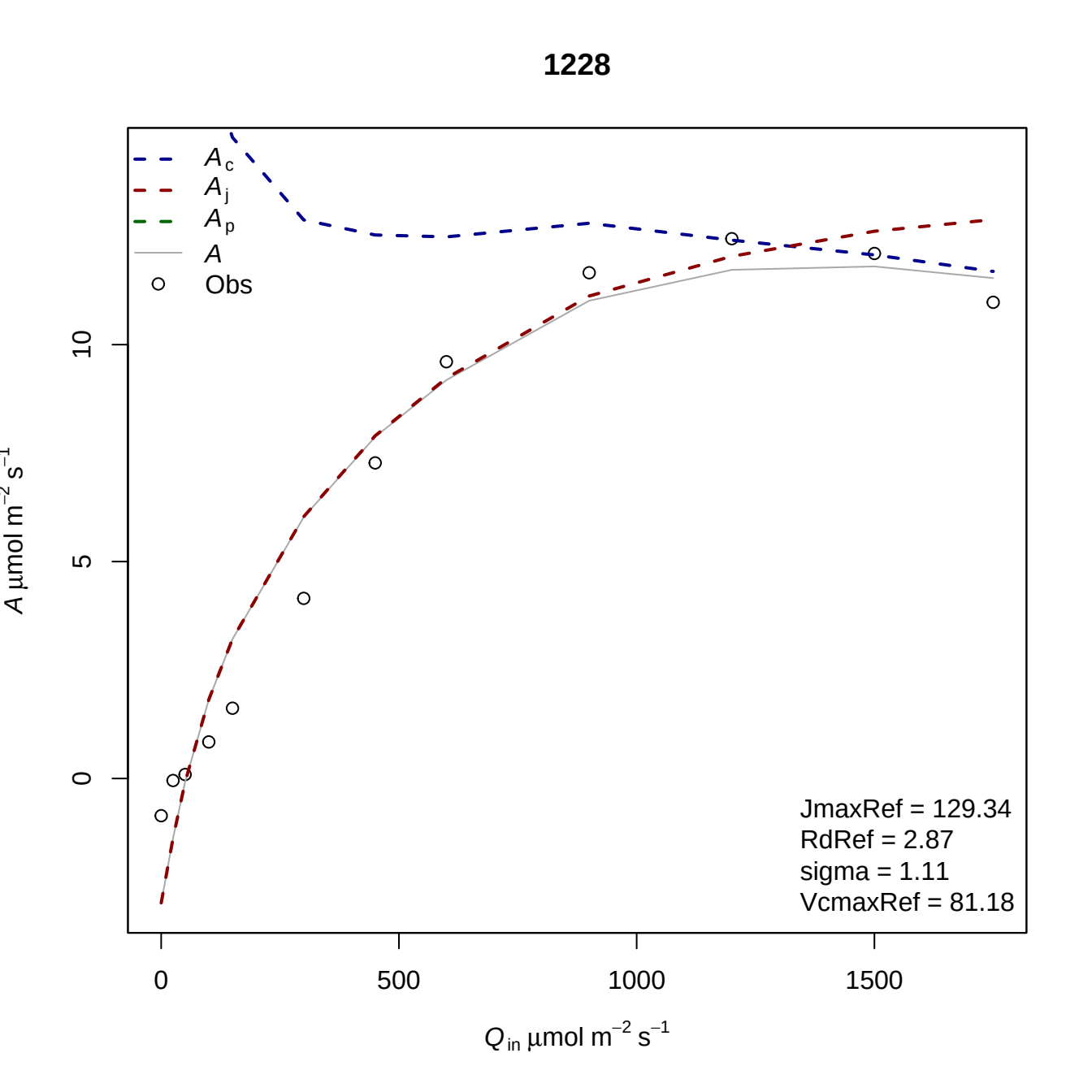

1223
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.6
RdRef = 1.78
sigma = 0.52
VcmaxRef = 77.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

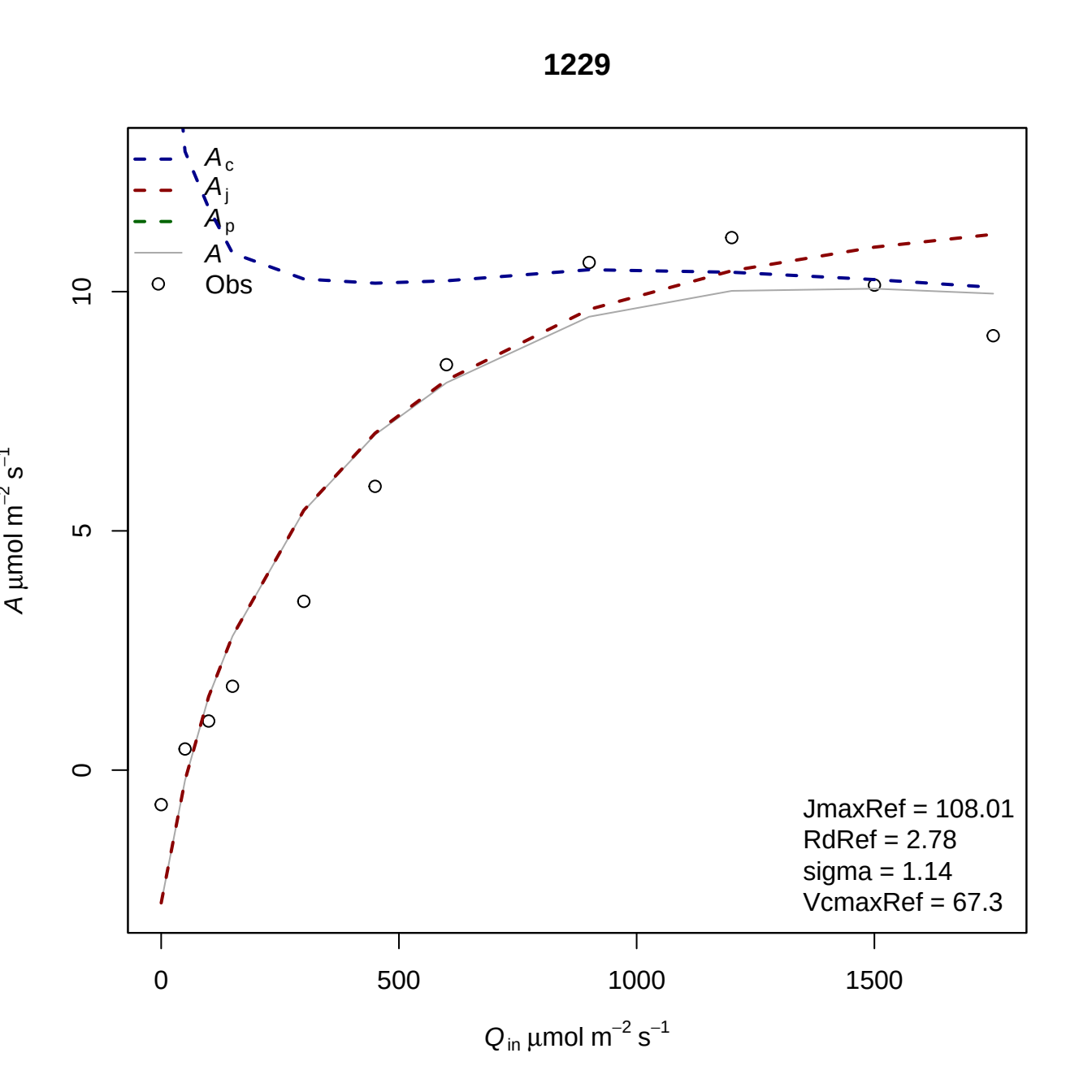

1224
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.44
RdRef = 0.56
sigma = 0.45
VcmaxRef = 87.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

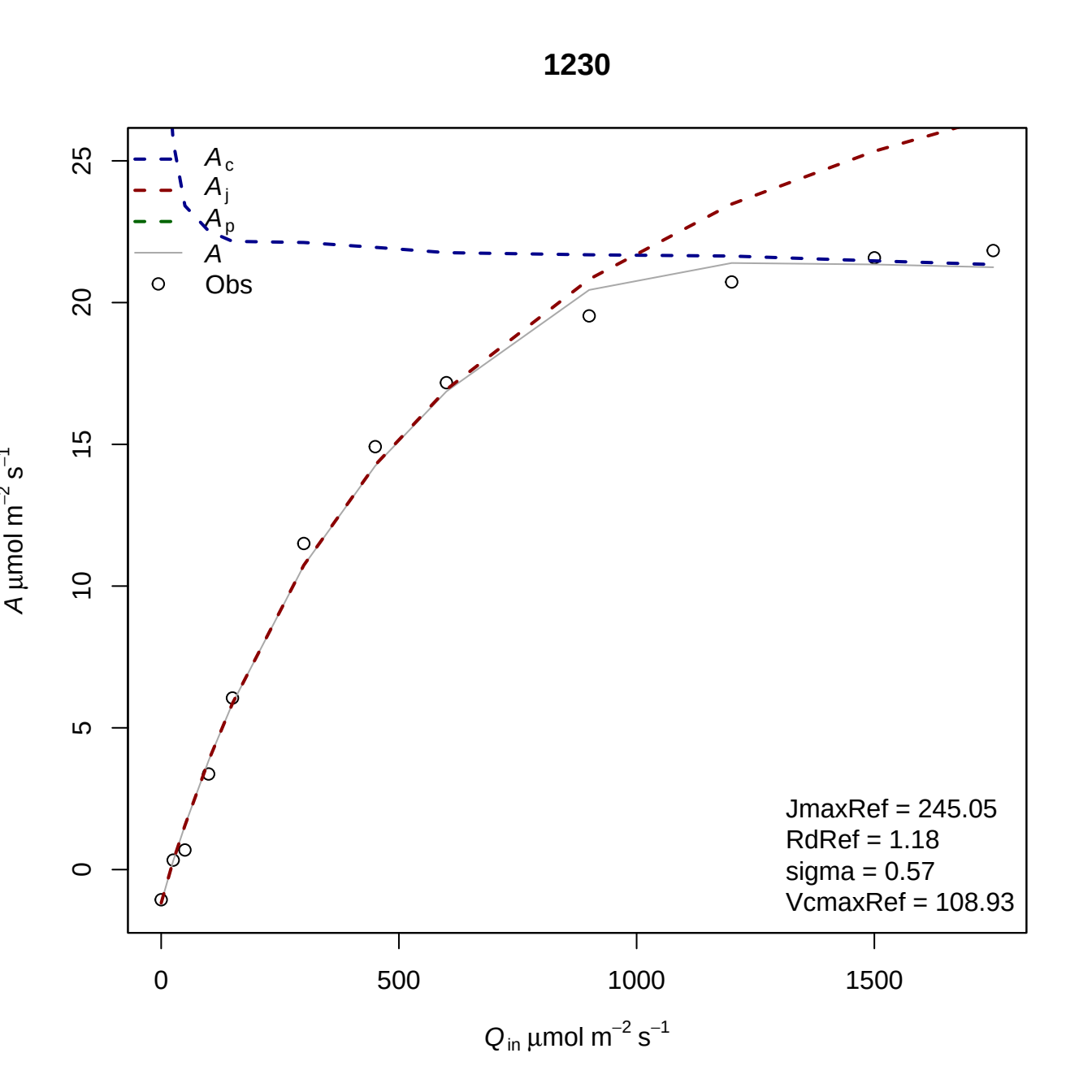

1225
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 204.2
RdRef = 2.6
sigma = 0.5
VcmaxRef = 118.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

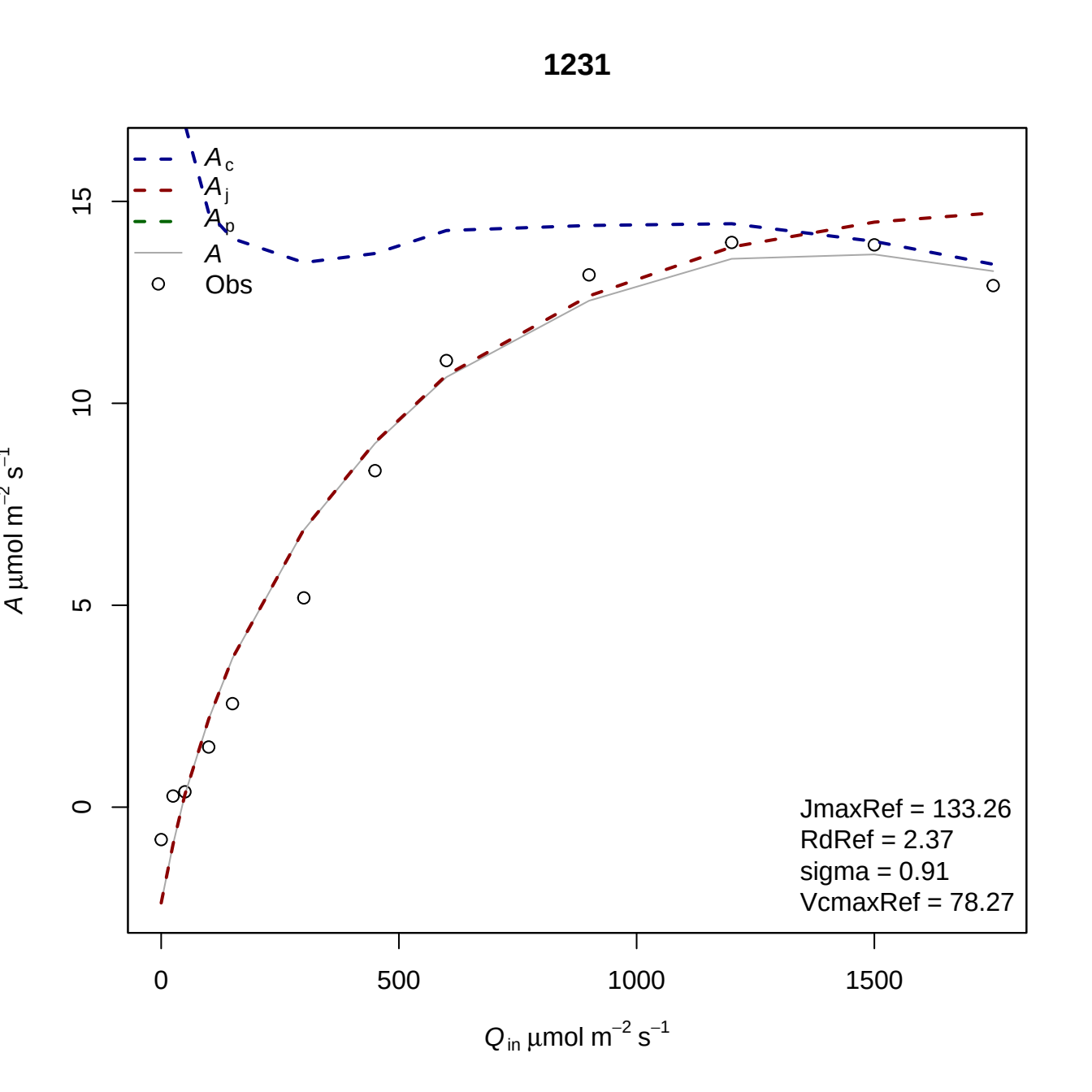

1226
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.39
RdRef = 1.72
sigma = 0.29
VcmaxRef = 816336.26
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

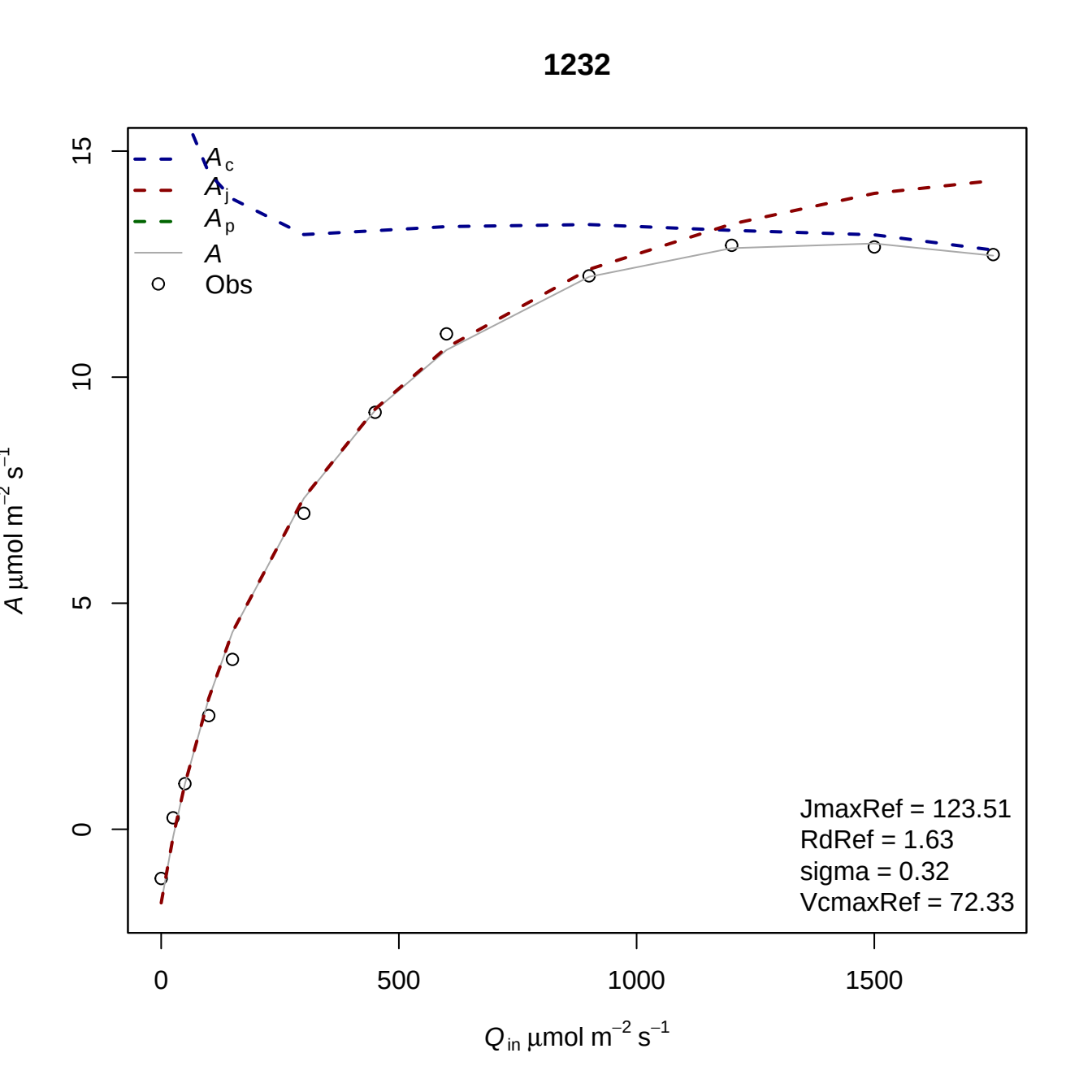

1227
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 170.54
RdRef = 1.94
sigma = 0.42
VcmaxRef = 94.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

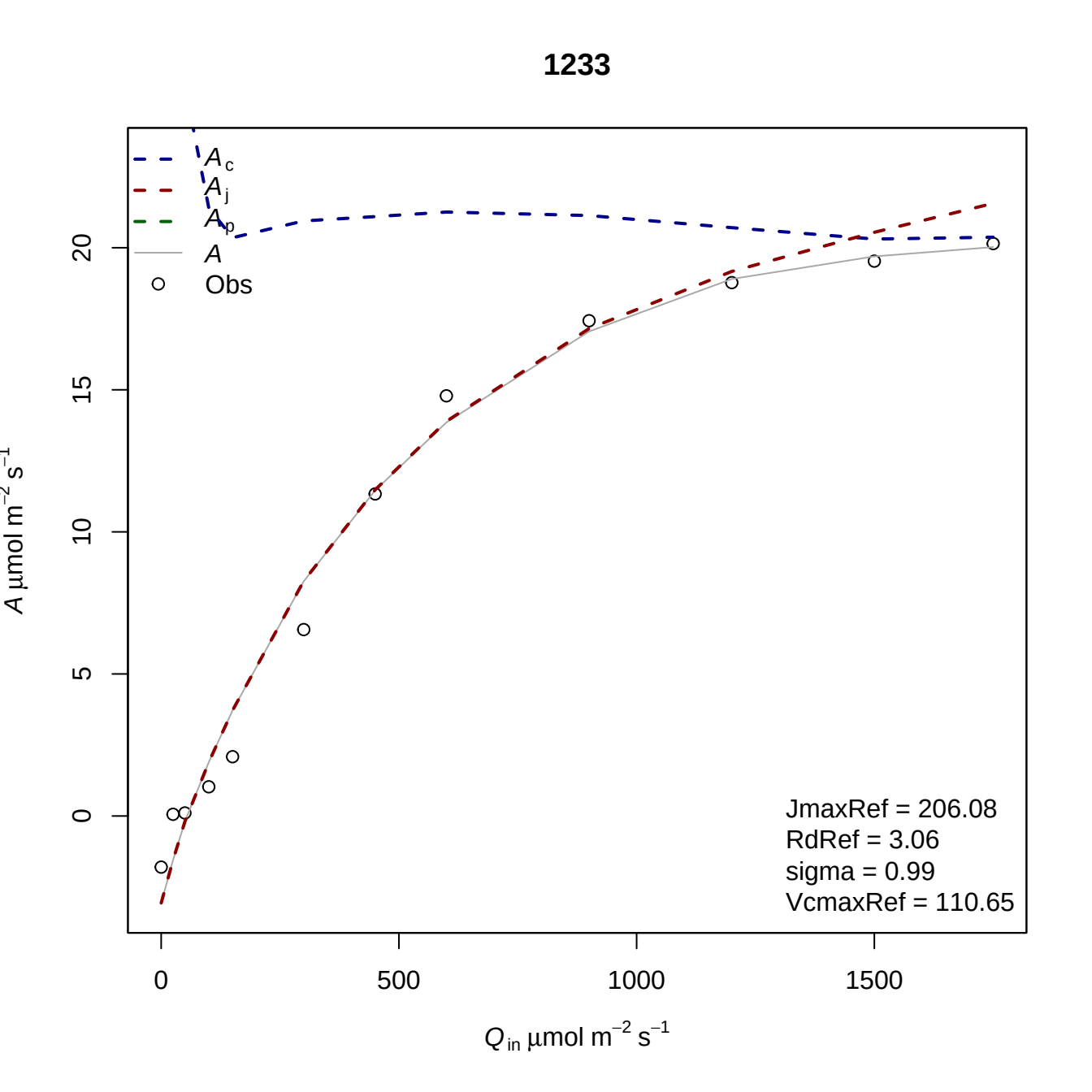

1228
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.34
RdRef = 2.87
sigma = 1.11
VcmaxRef = 81.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

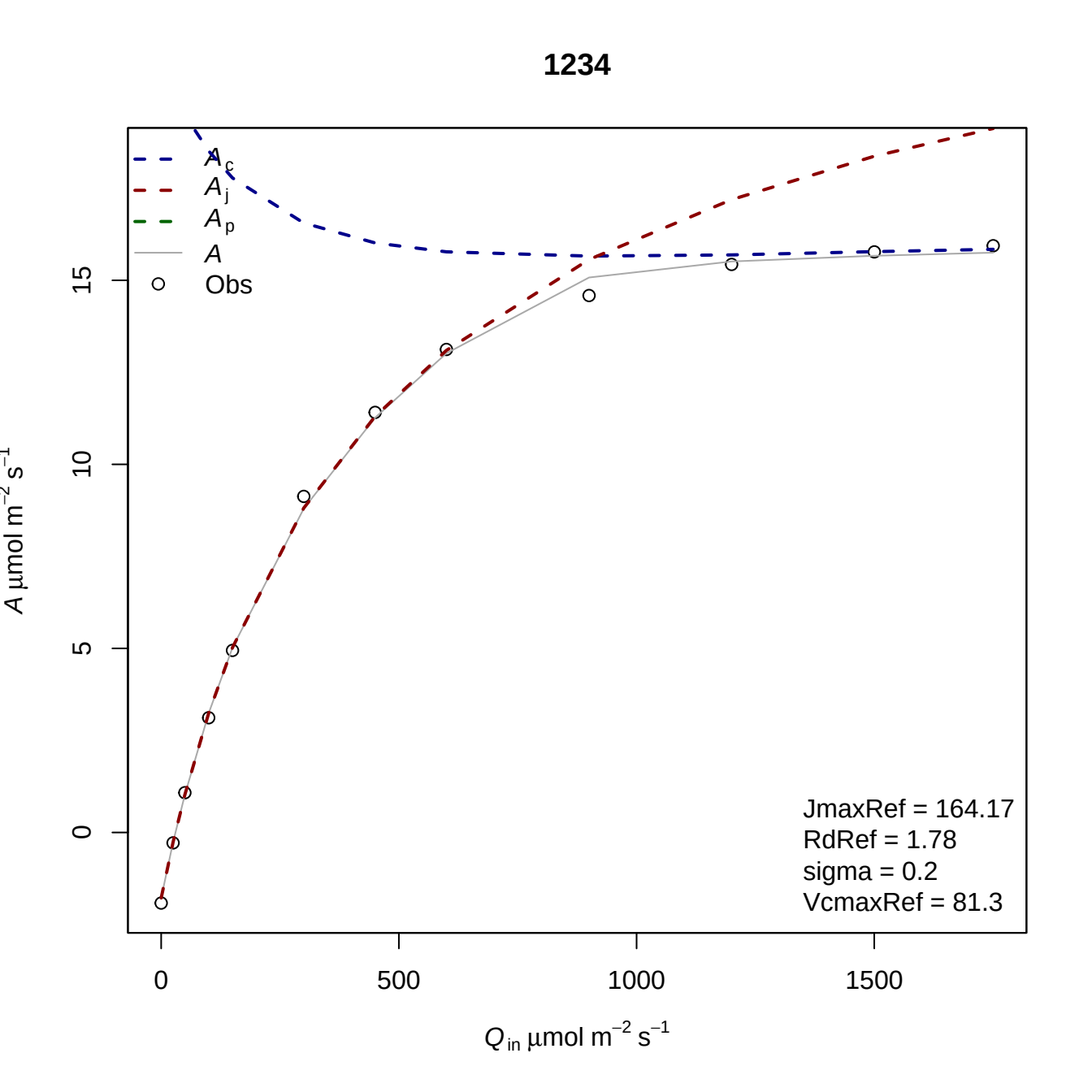

1229
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.01
RdRef = 2.78
sigma = 1.14
VcmaxRef = 67.3
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

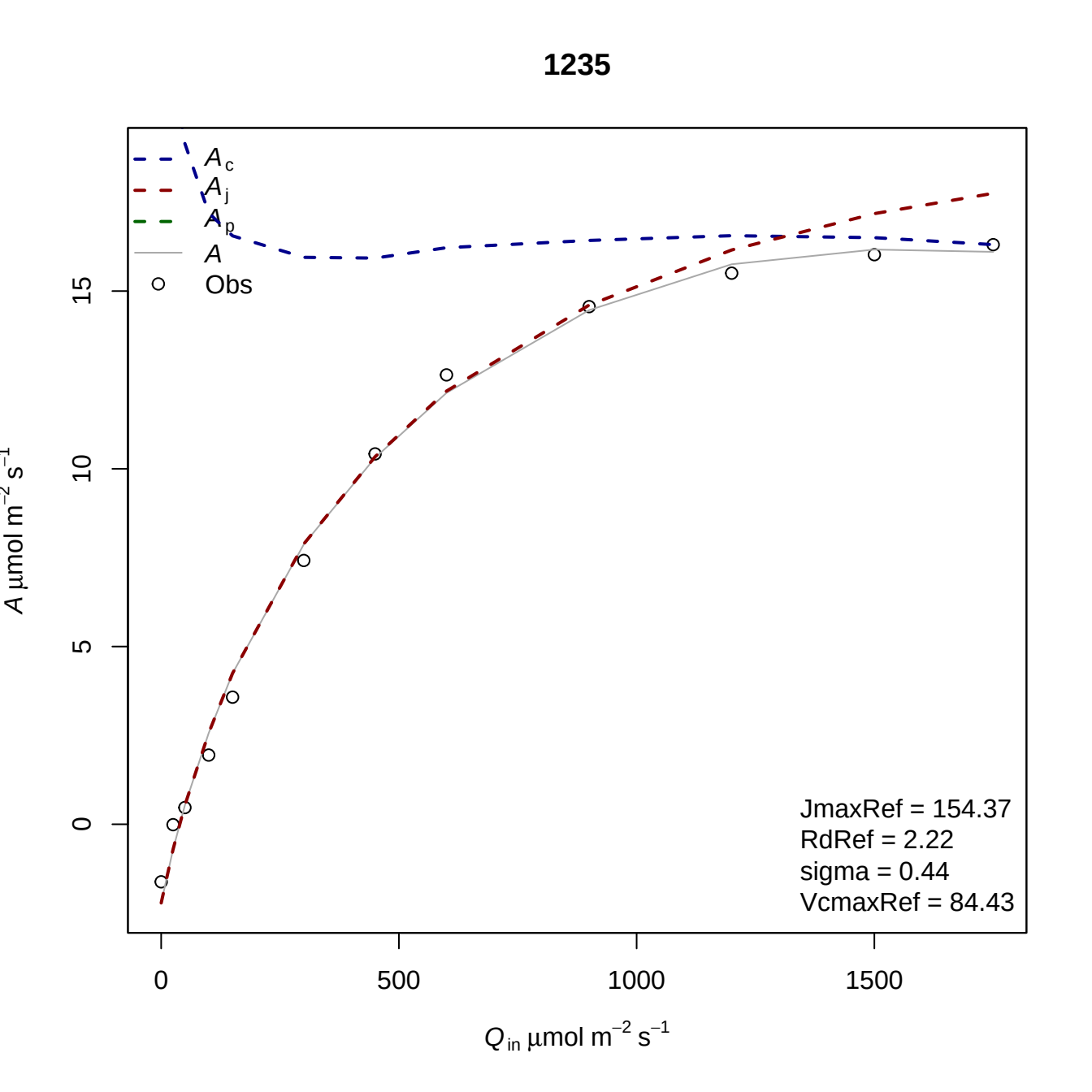

1230
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 245.05
RdRef = 1.18
sigma = 0.57
VcmaxRef = 108.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

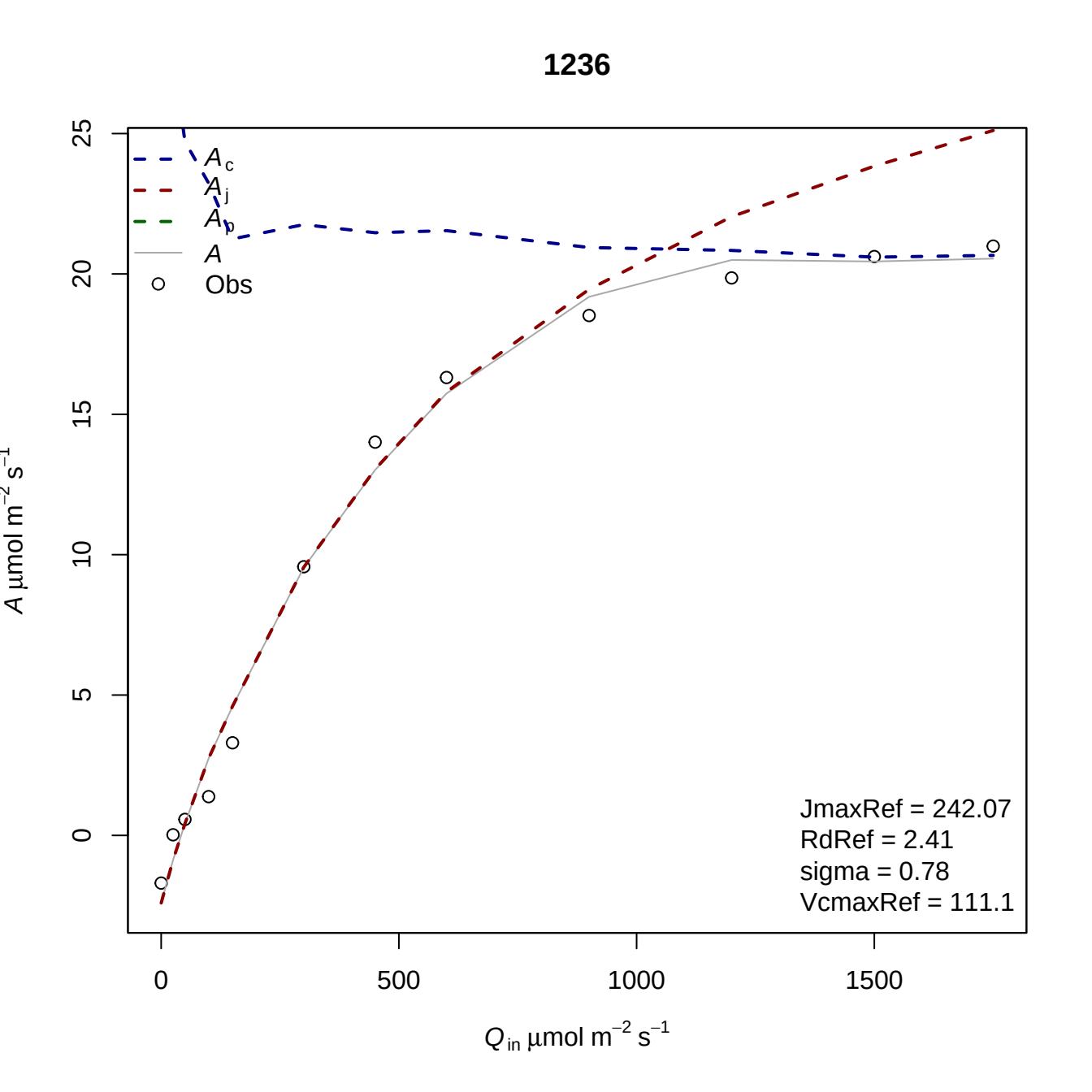

1231
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.26
RdRef = 2.37
sigma = 0.91
VcmaxRef = 78.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

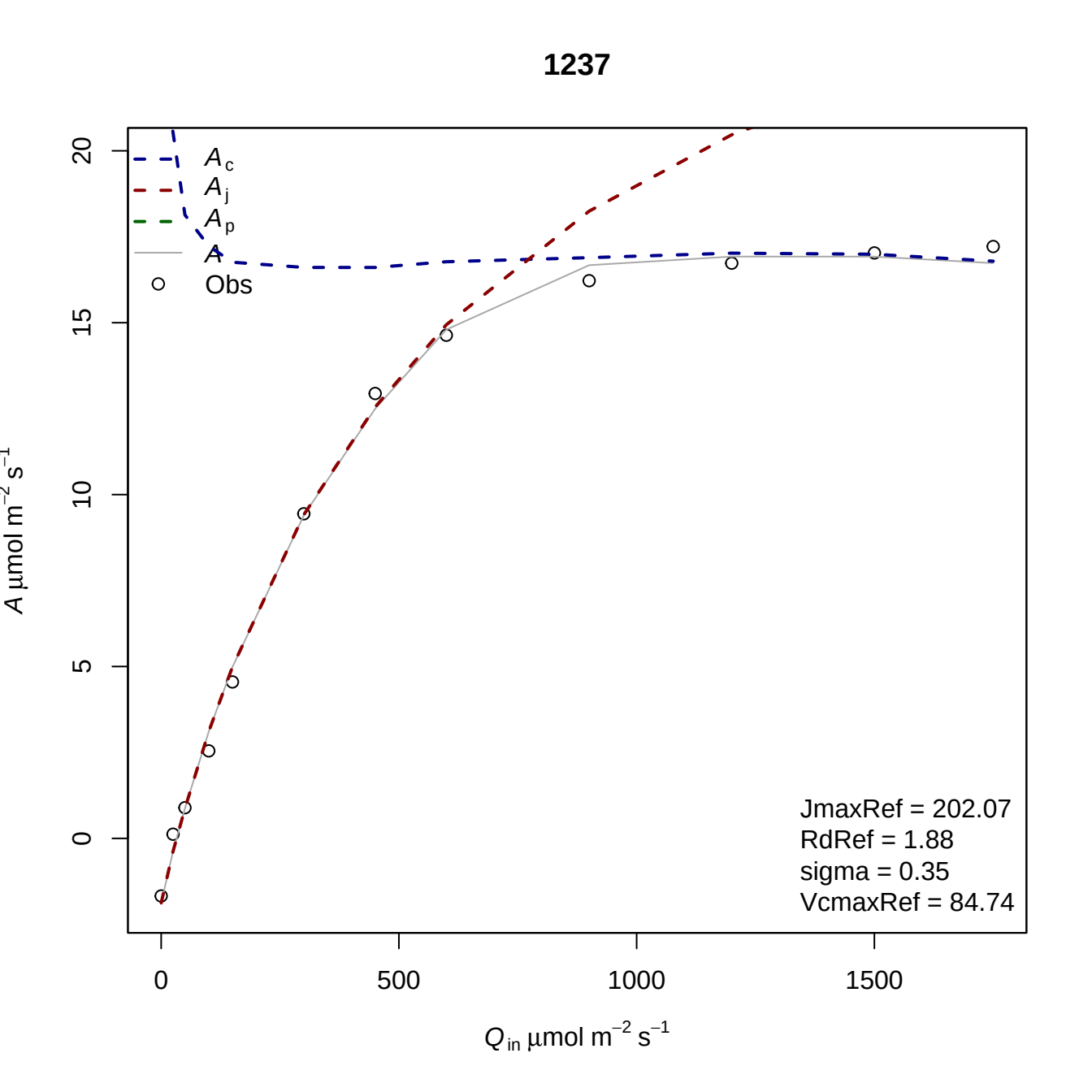

1232
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.51
RdRef = 1.63
sigma = 0.32
VcmaxRef = 72.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

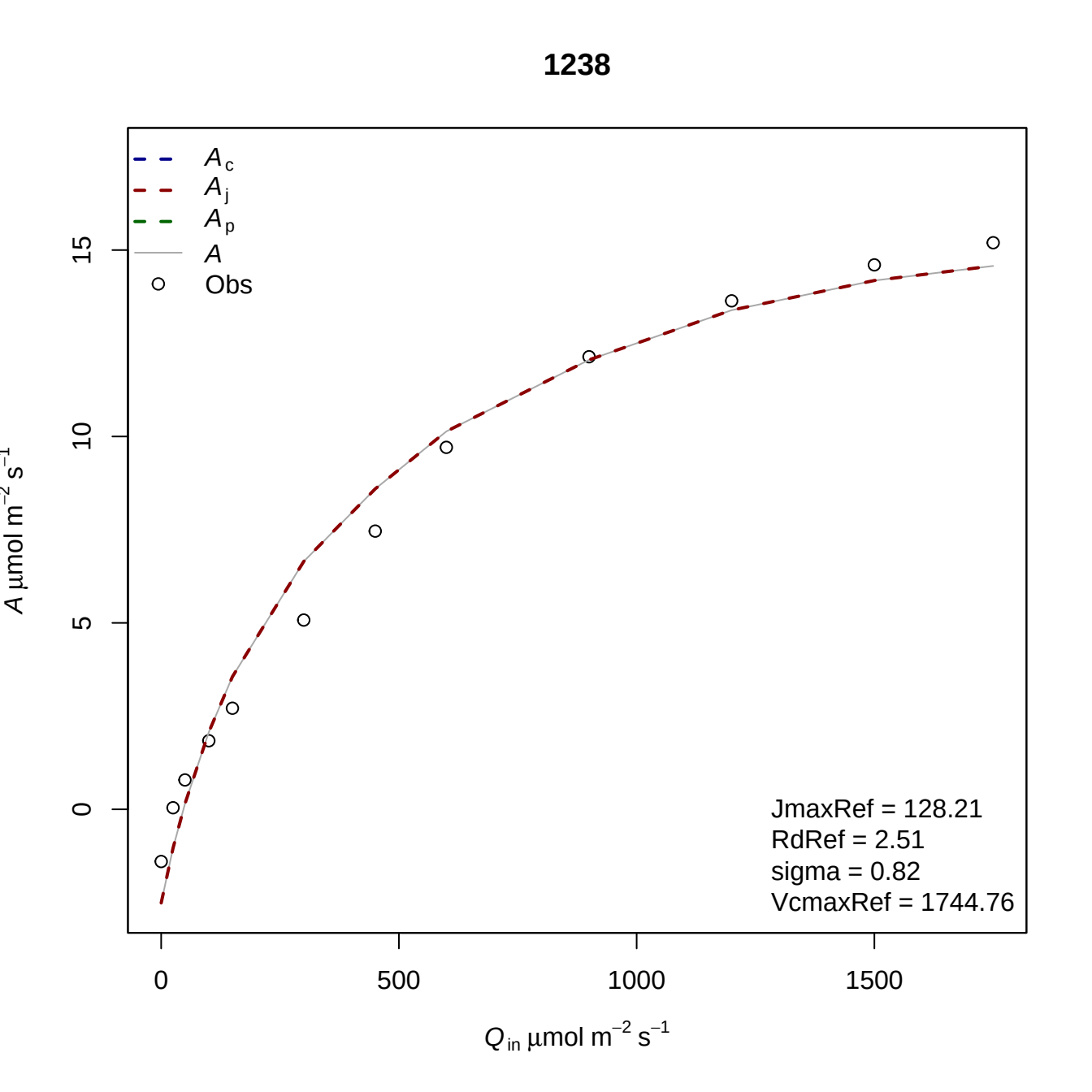

1233
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 206.08
RdRef = 3.06
sigma = 0.99
VcmaxRef = 110.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

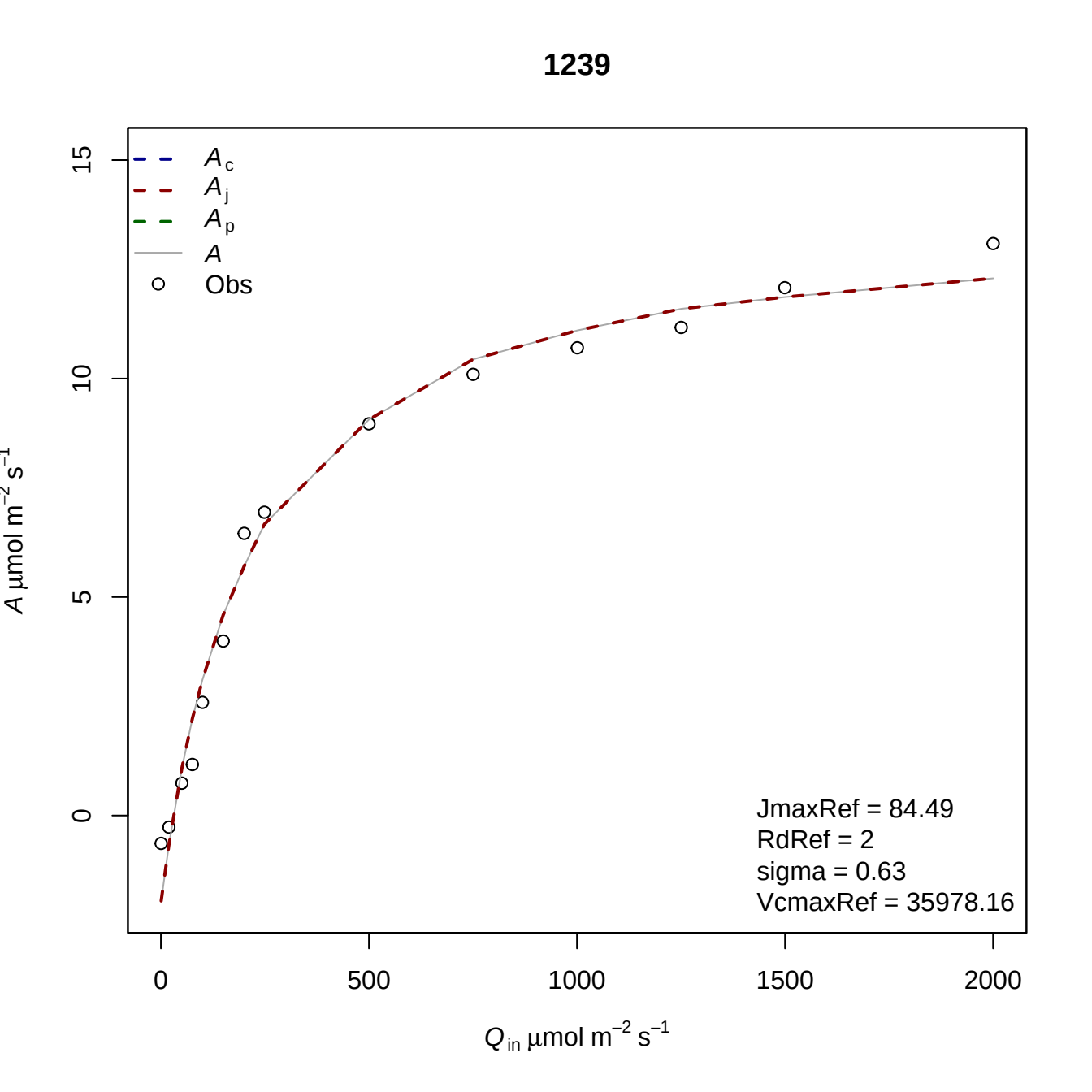

1234
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.17
RdRef = 1.78
sigma = 0.2
VcmaxRef = 81.3
0
5
10
15
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

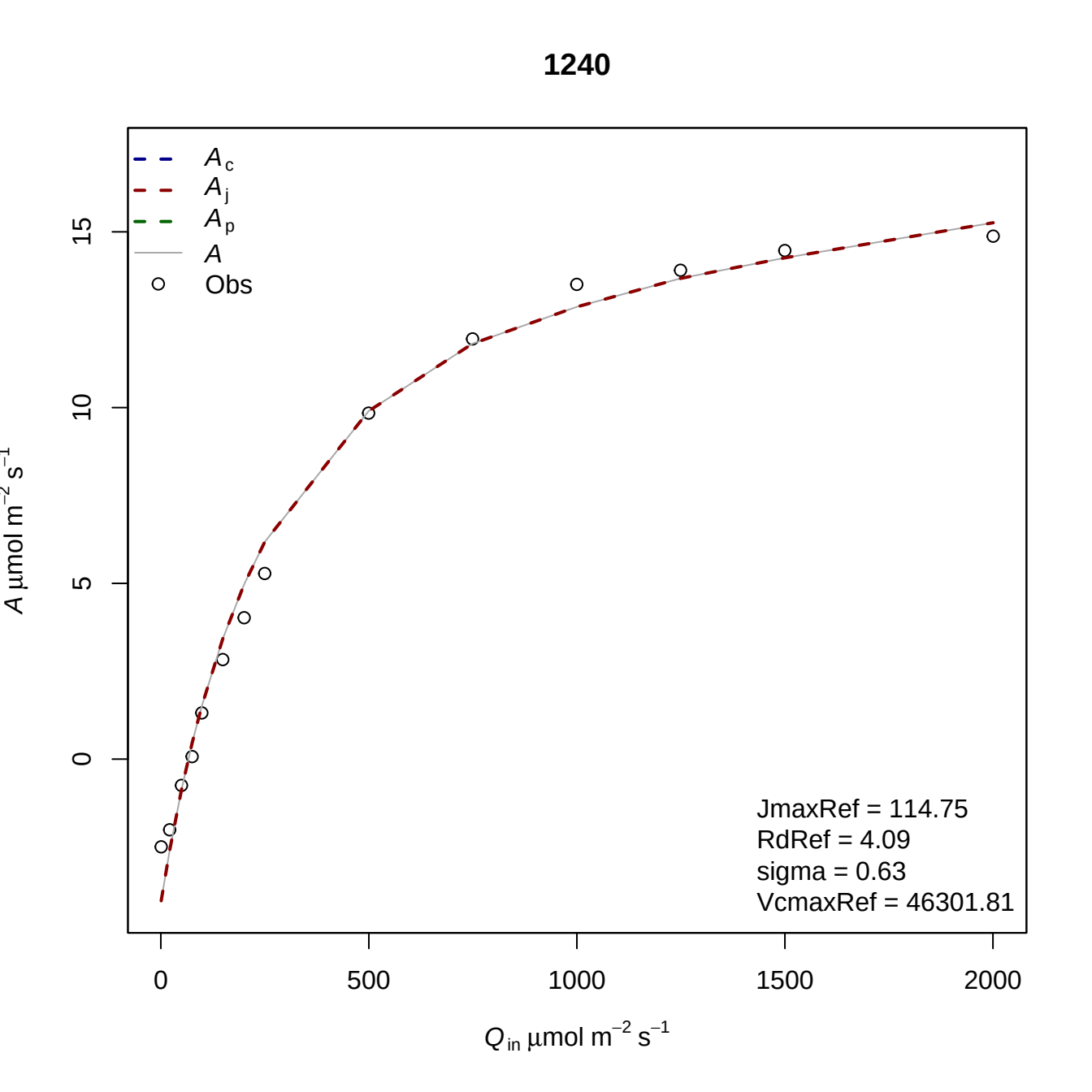

1235
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 154.37
RdRef = 2.22
sigma = 0.44
VcmaxRef = 84.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

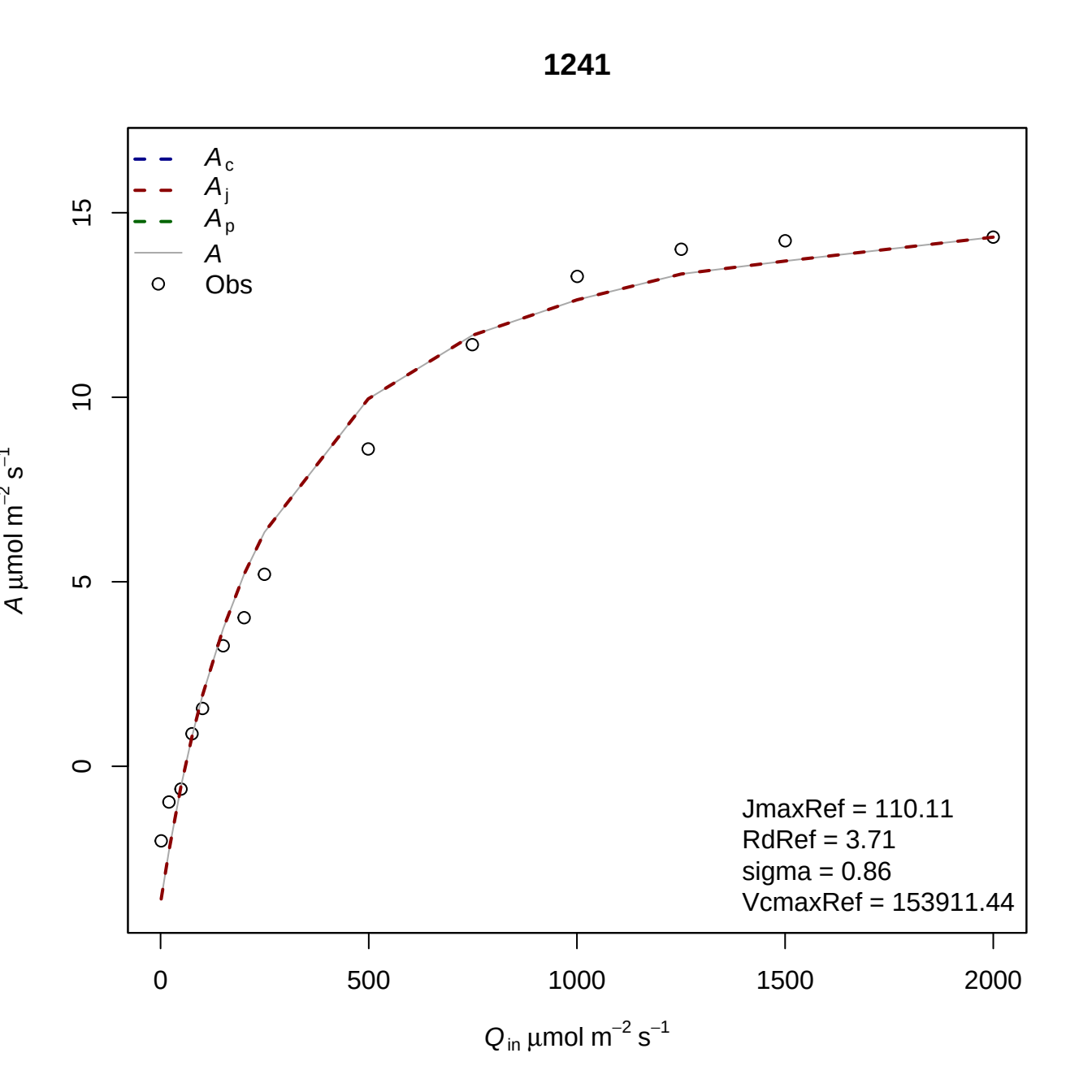

1236
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 242.07
RdRef = 2.41
sigma = 0.78
VcmaxRef = 111.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

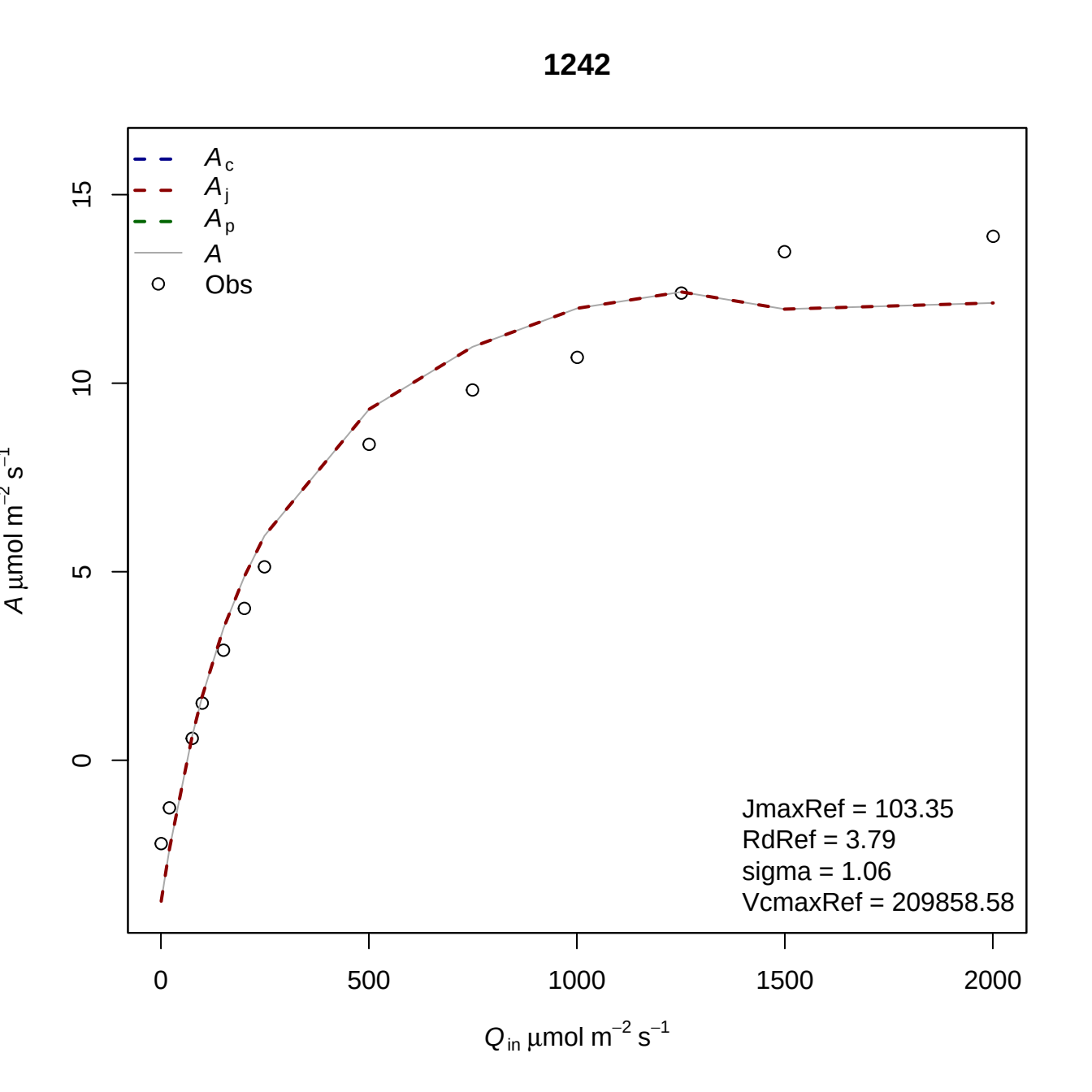

1237
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 202.07
RdRef = 1.88
sigma = 0.35
VcmaxRef = 84.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

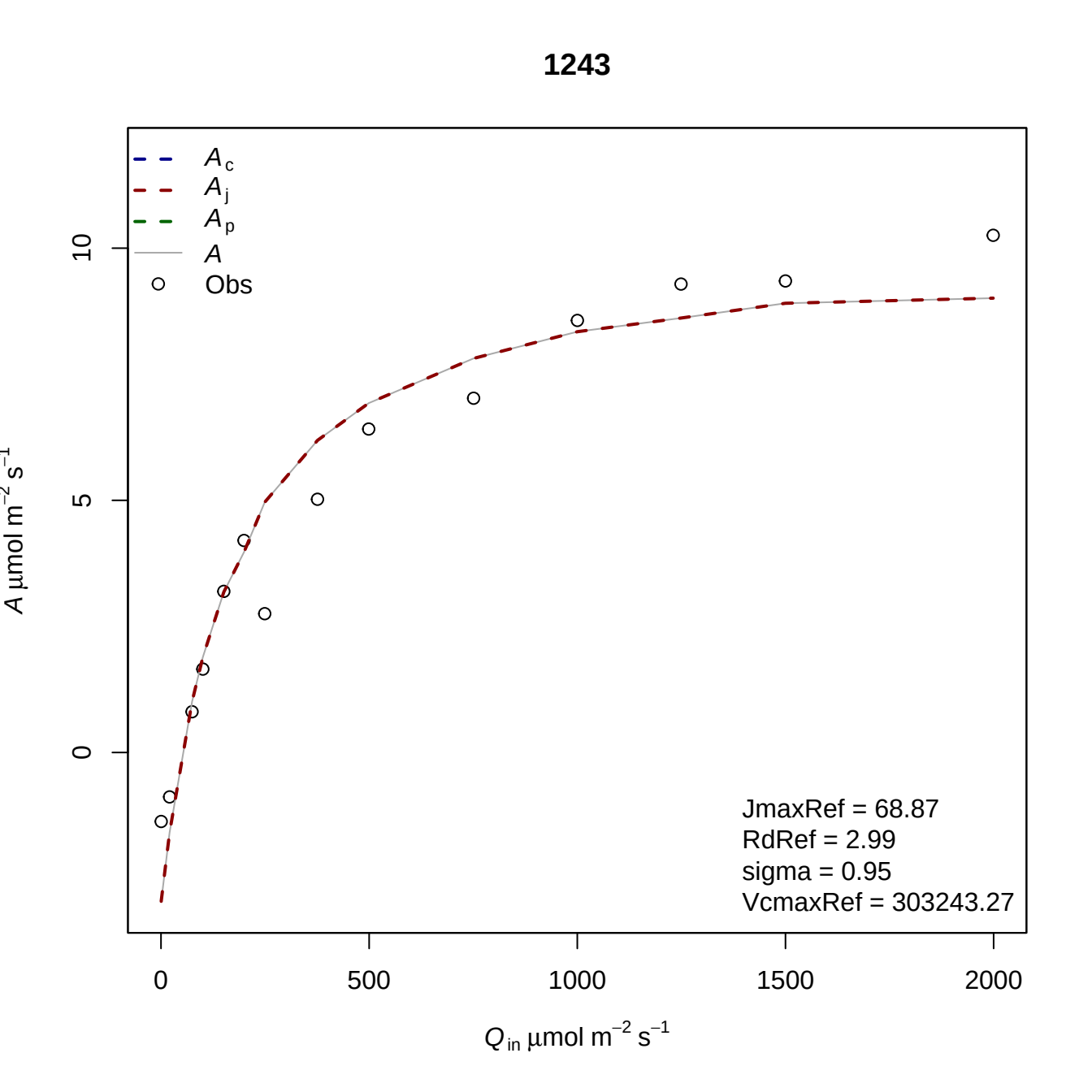

1238
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 128.21
RdRef = 2.51
sigma = 0.82
VcmaxRef = 1744.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

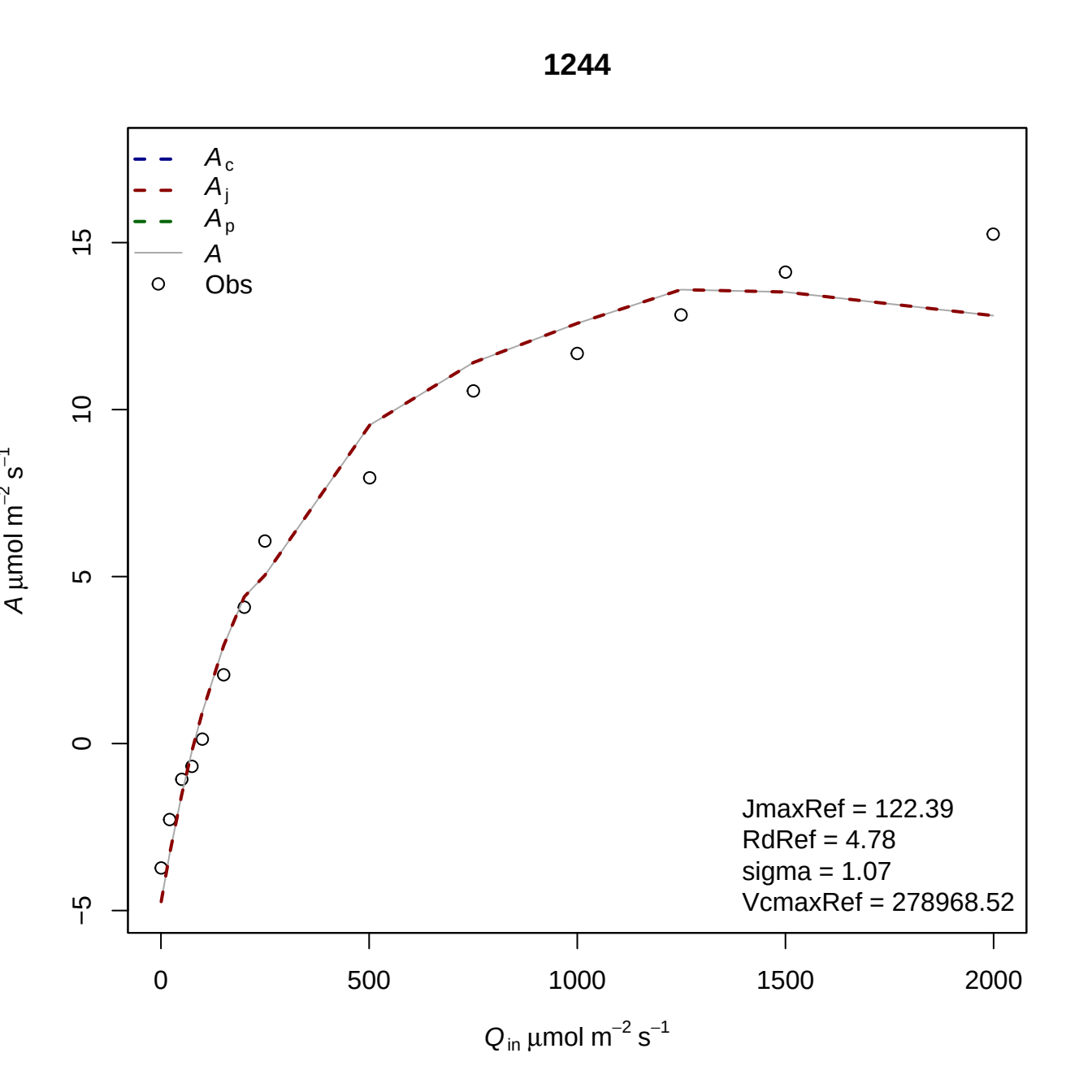

1239
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.49
RdRef = 2
sigma = 0.63
VcmaxRef = 35978.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

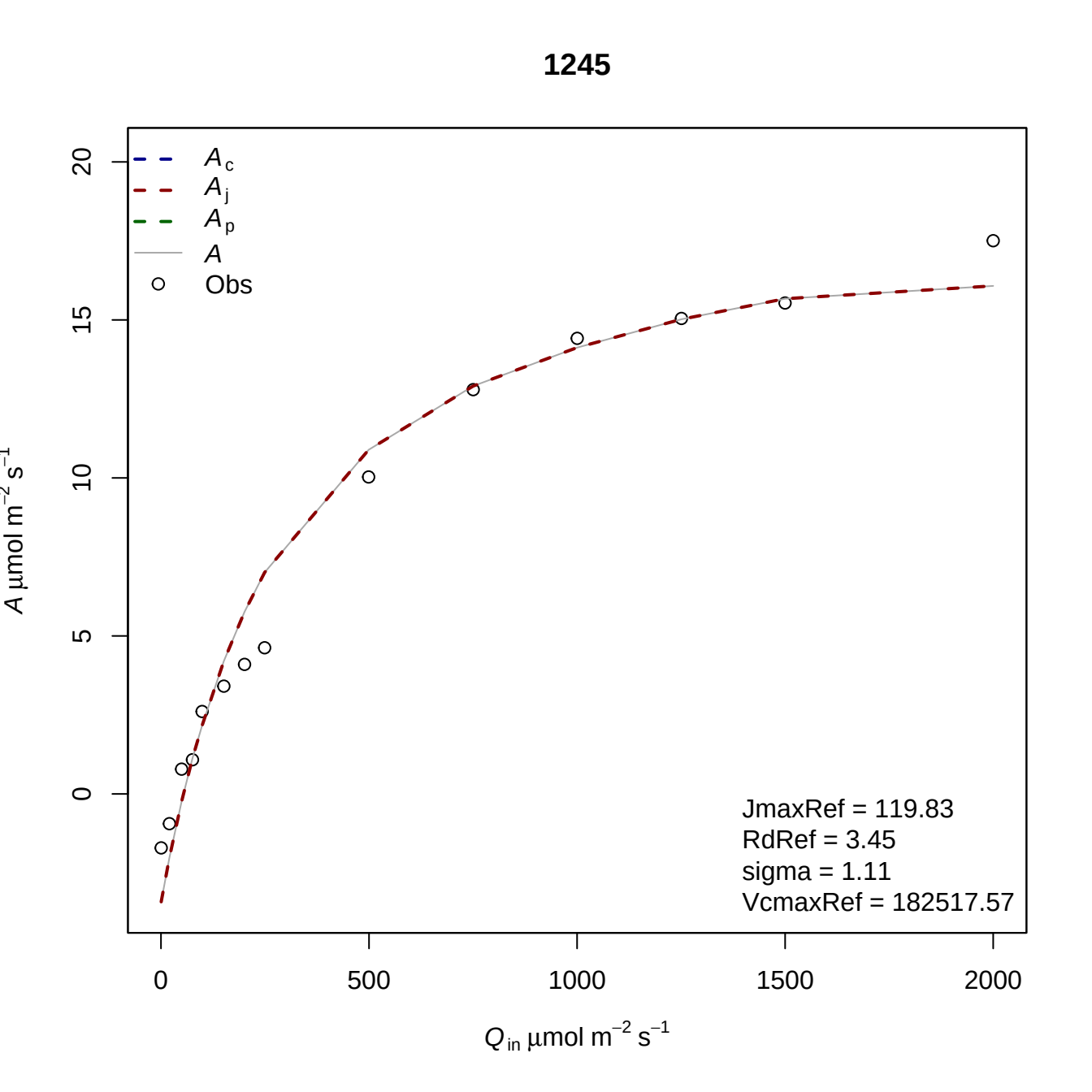

1240
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.75
RdRef = 4.09
sigma = 0.63
VcmaxRef = 46301.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

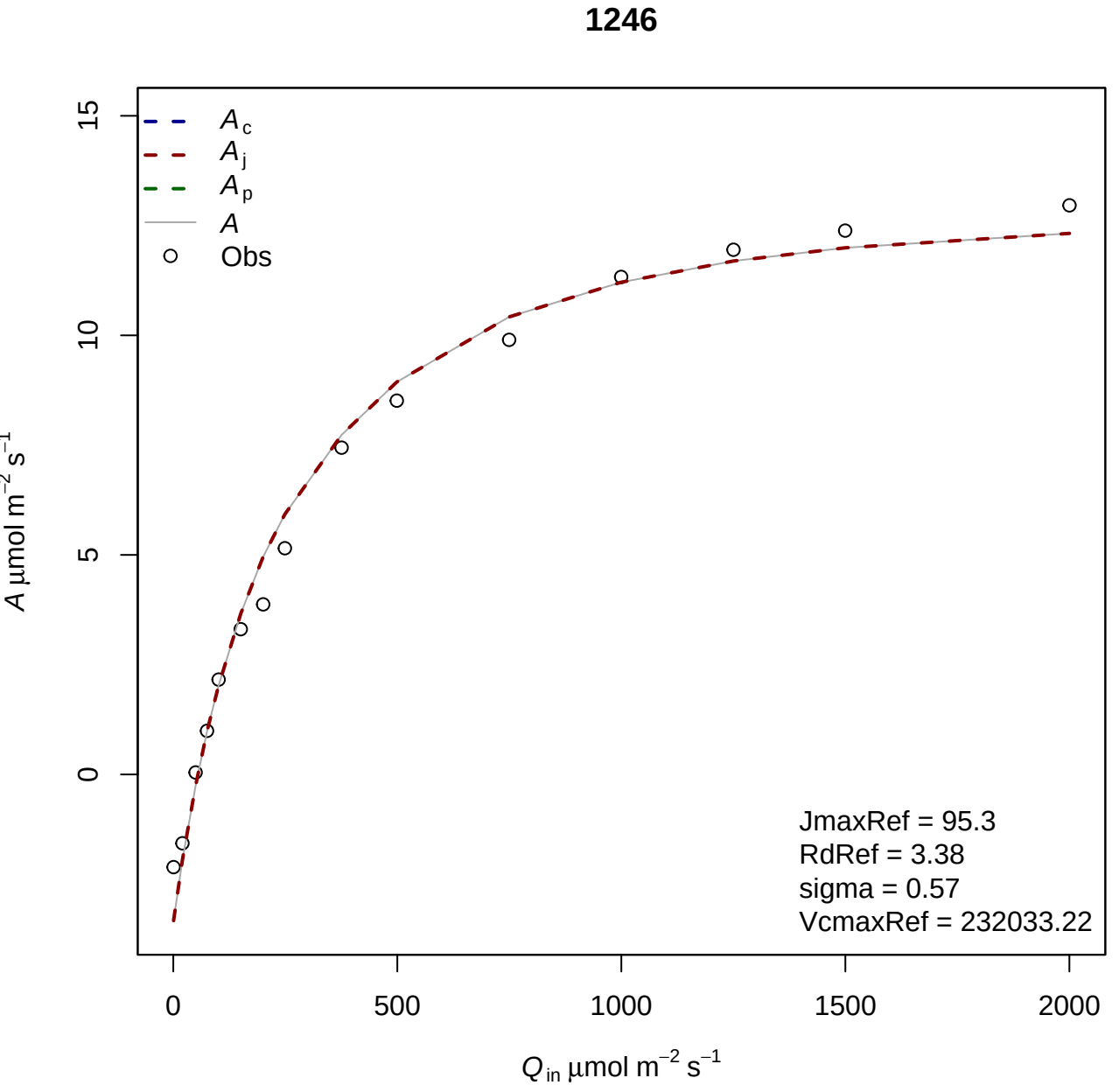

1241
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.11
RdRef = 3.71
sigma = 0.86
VcmaxRef = 153911.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

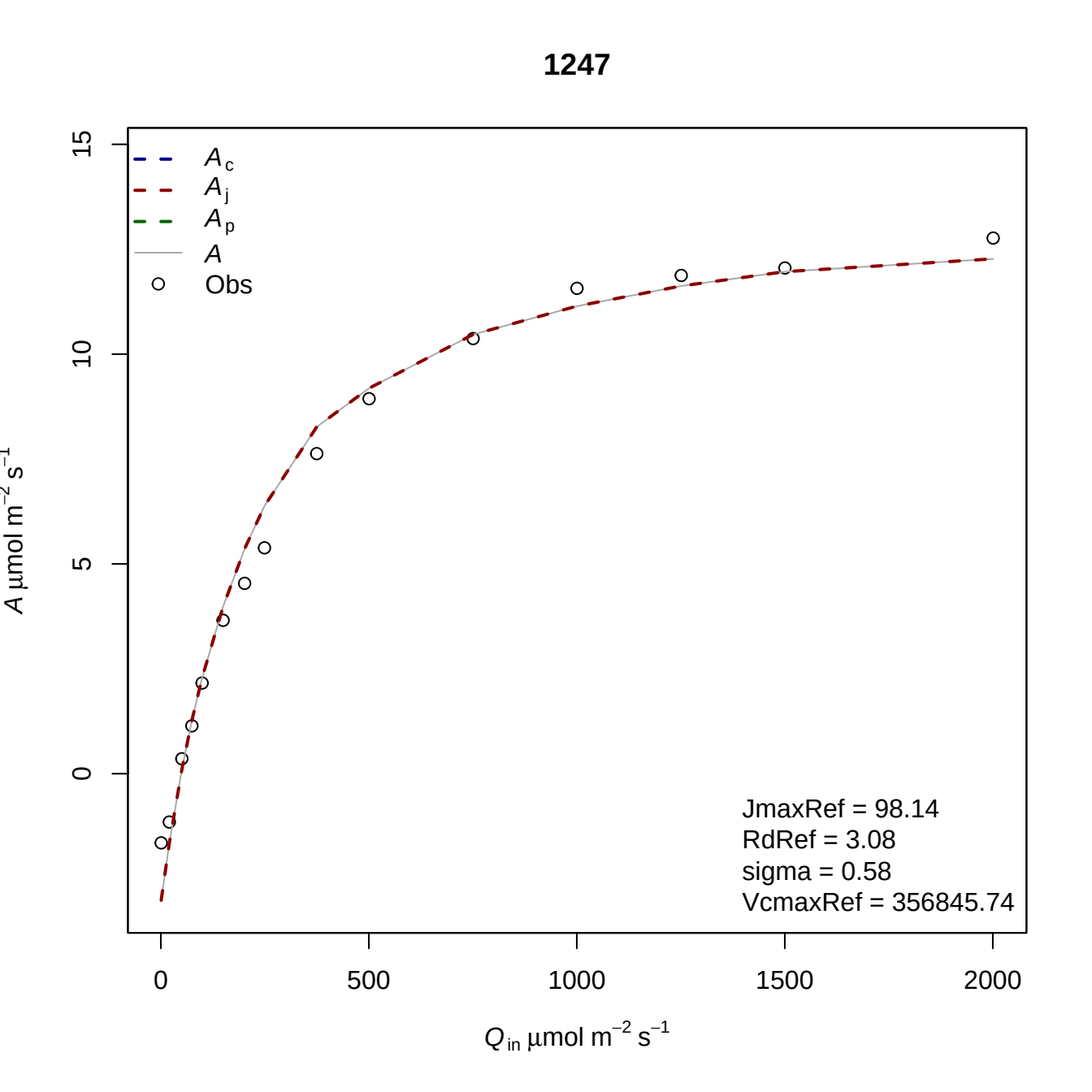

1242
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.35
RdRef = 3.79
sigma = 1.06
VcmaxRef = 209858.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

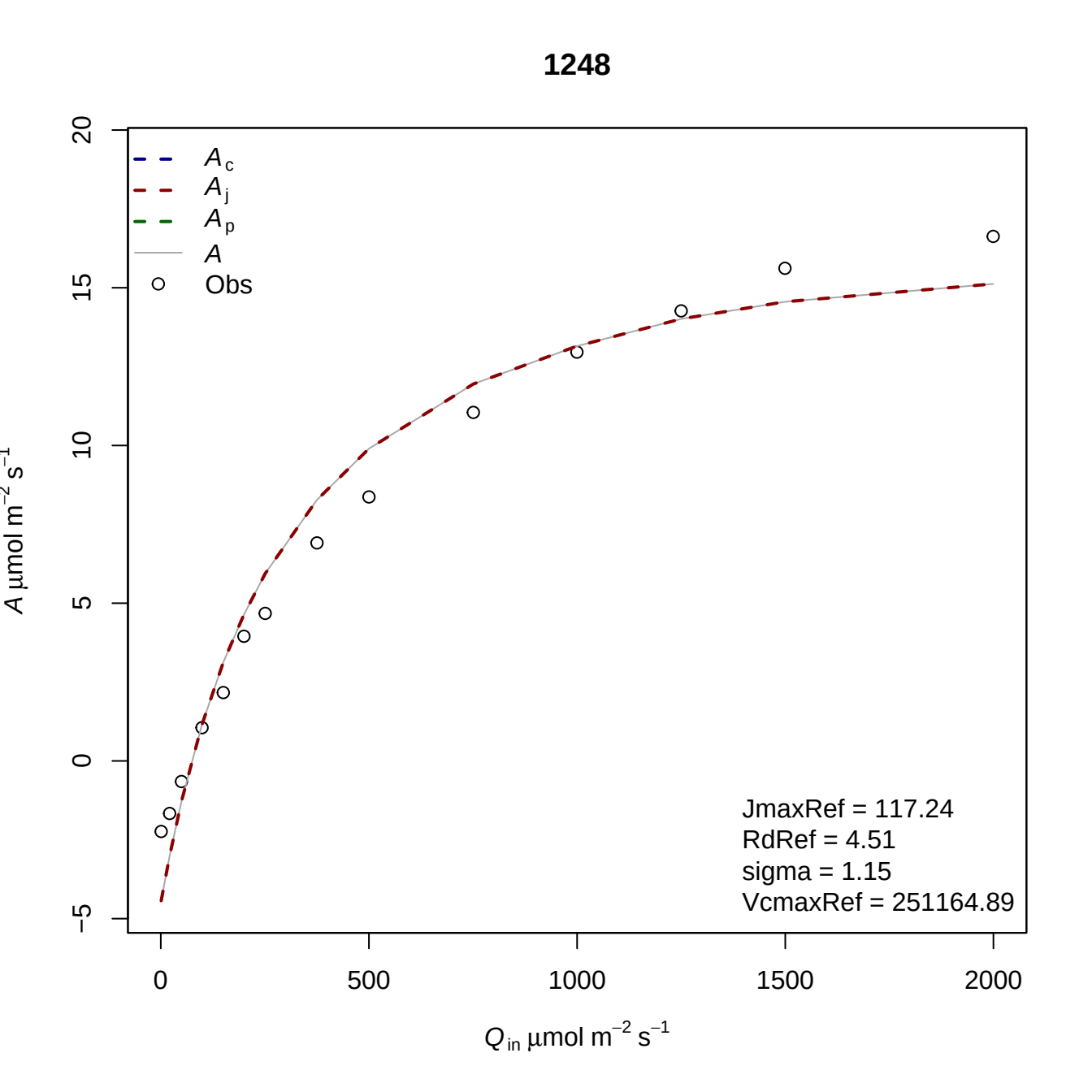

1243
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.87
RdRef = 2.99
sigma = 0.95
VcmaxRef = 303243.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

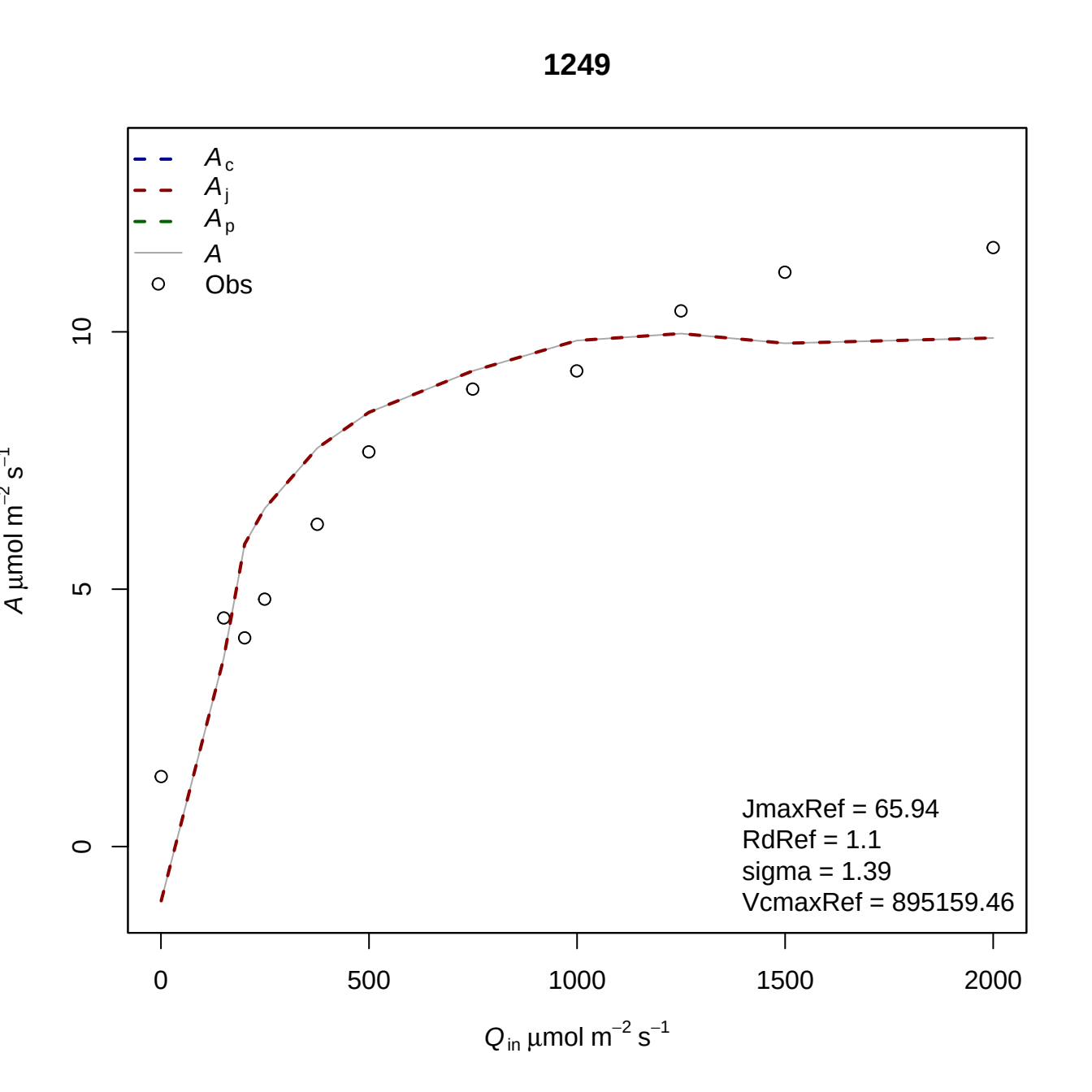

1244
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.39
RdRef = 4.78
sigma = 1.07
VcmaxRef = 278968.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

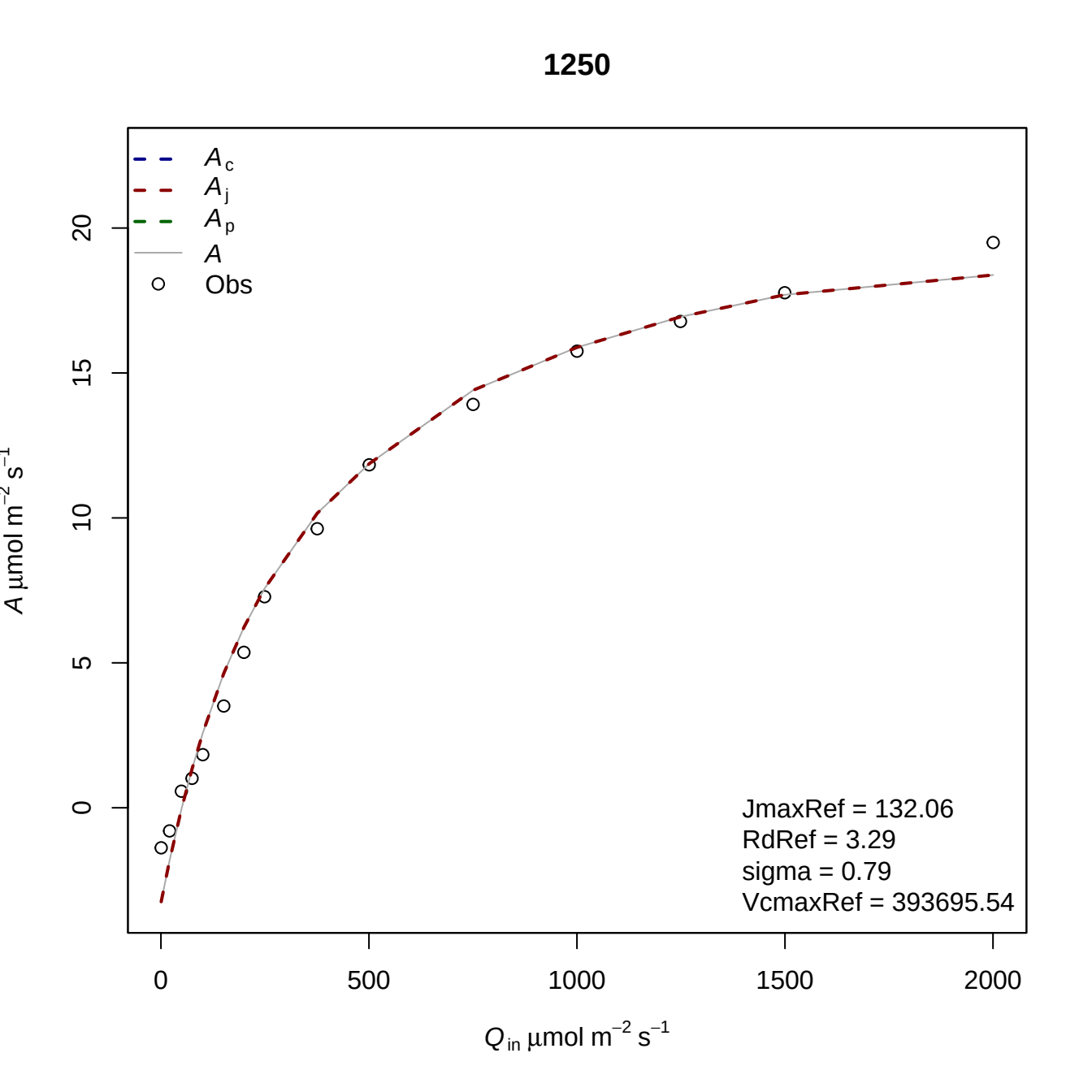

1245
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.83
RdRef = 3.45
sigma = 1.11
VcmaxRef = 182517.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

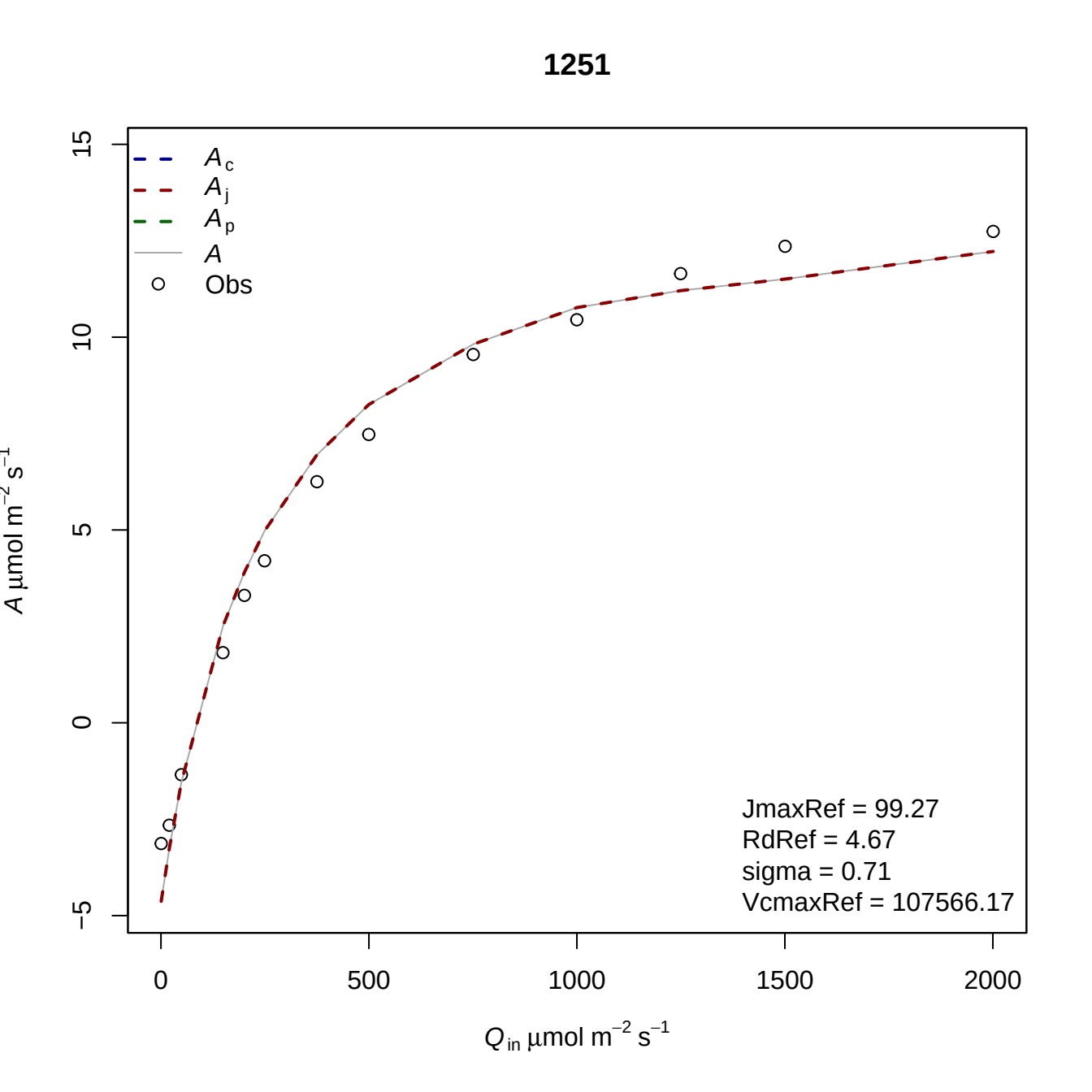

1246
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.3
RdRef = 3.38
sigma = 0.57
VcmaxRef = 232033.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

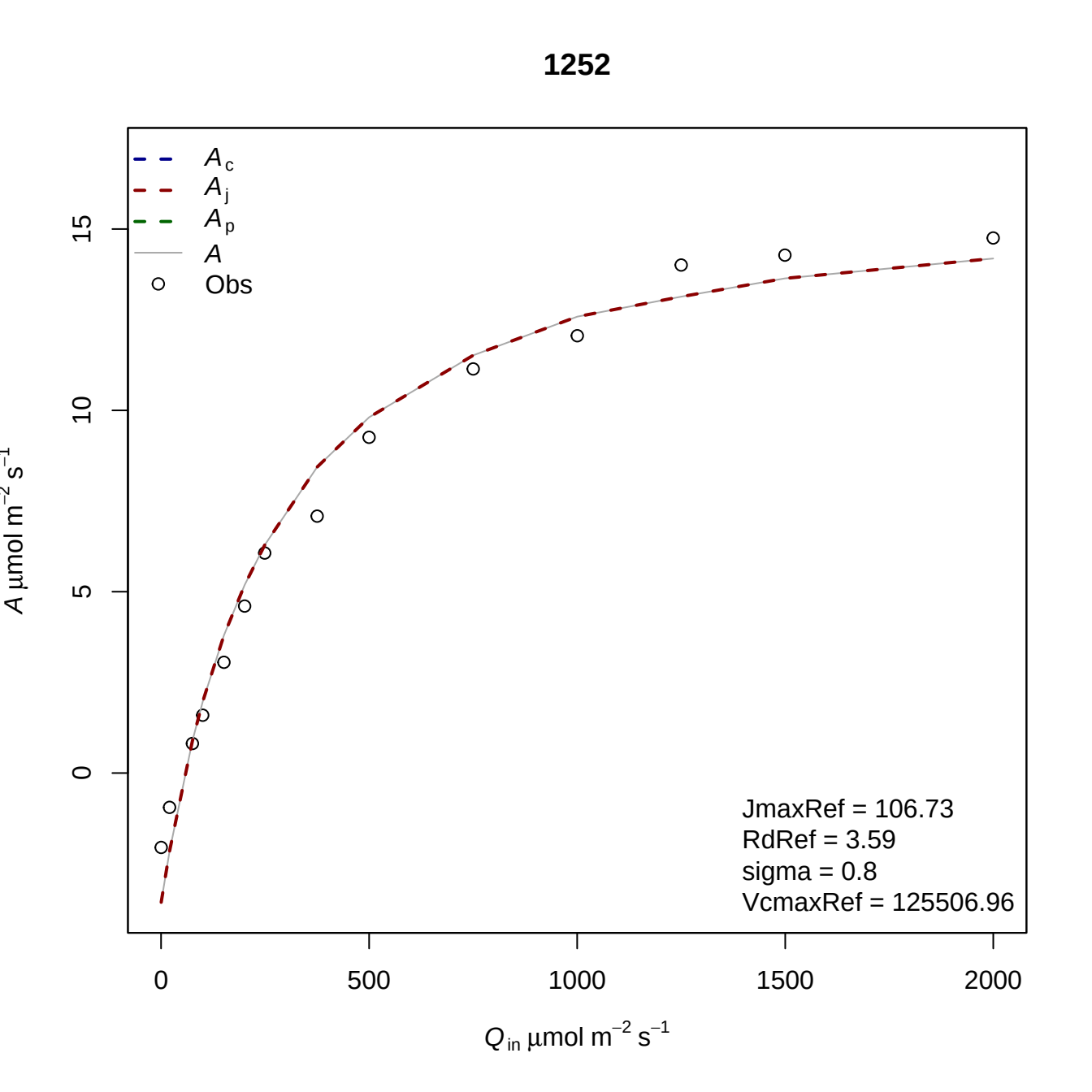

1247
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.14
RdRef = 3.08
sigma = 0.58
VcmaxRef = 356845.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

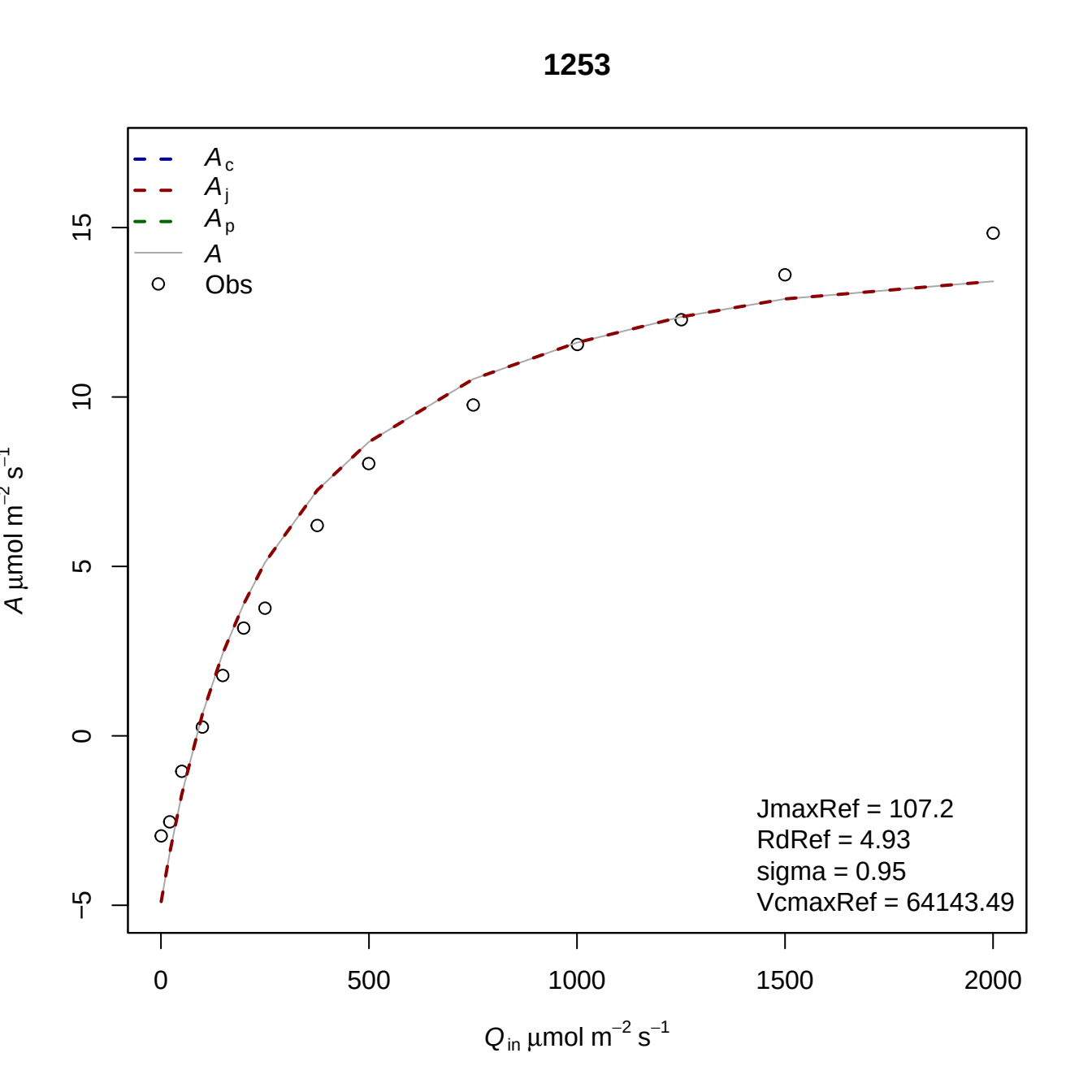

1248
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.24
RdRef = 4.51
sigma = 1.15
VcmaxRef = 251164.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
20
15
10
5
0
−5
0
500
1000
1500
2000

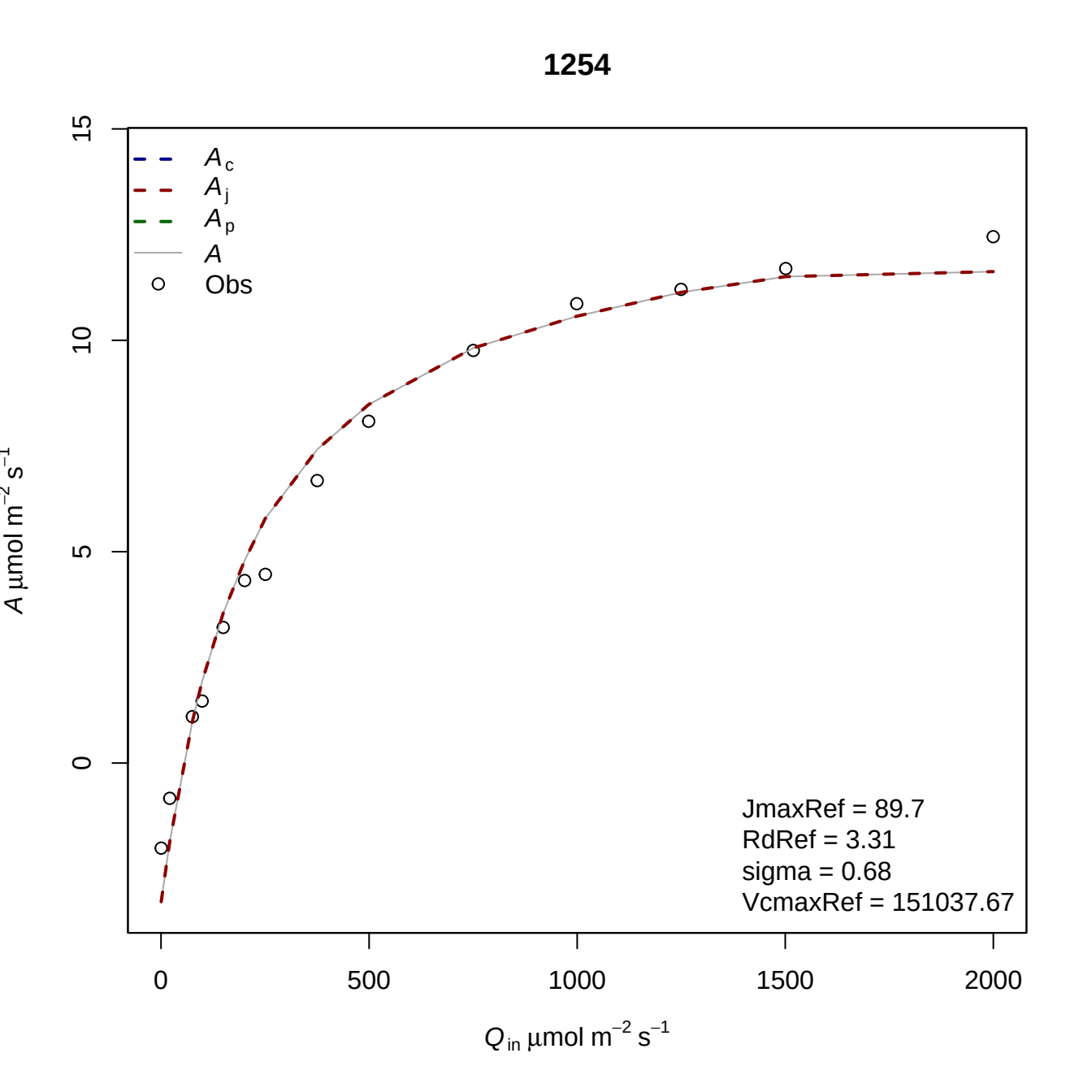

1249
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.94
RdRef = 1.1
sigma = 1.39
VcmaxRef = 895159.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

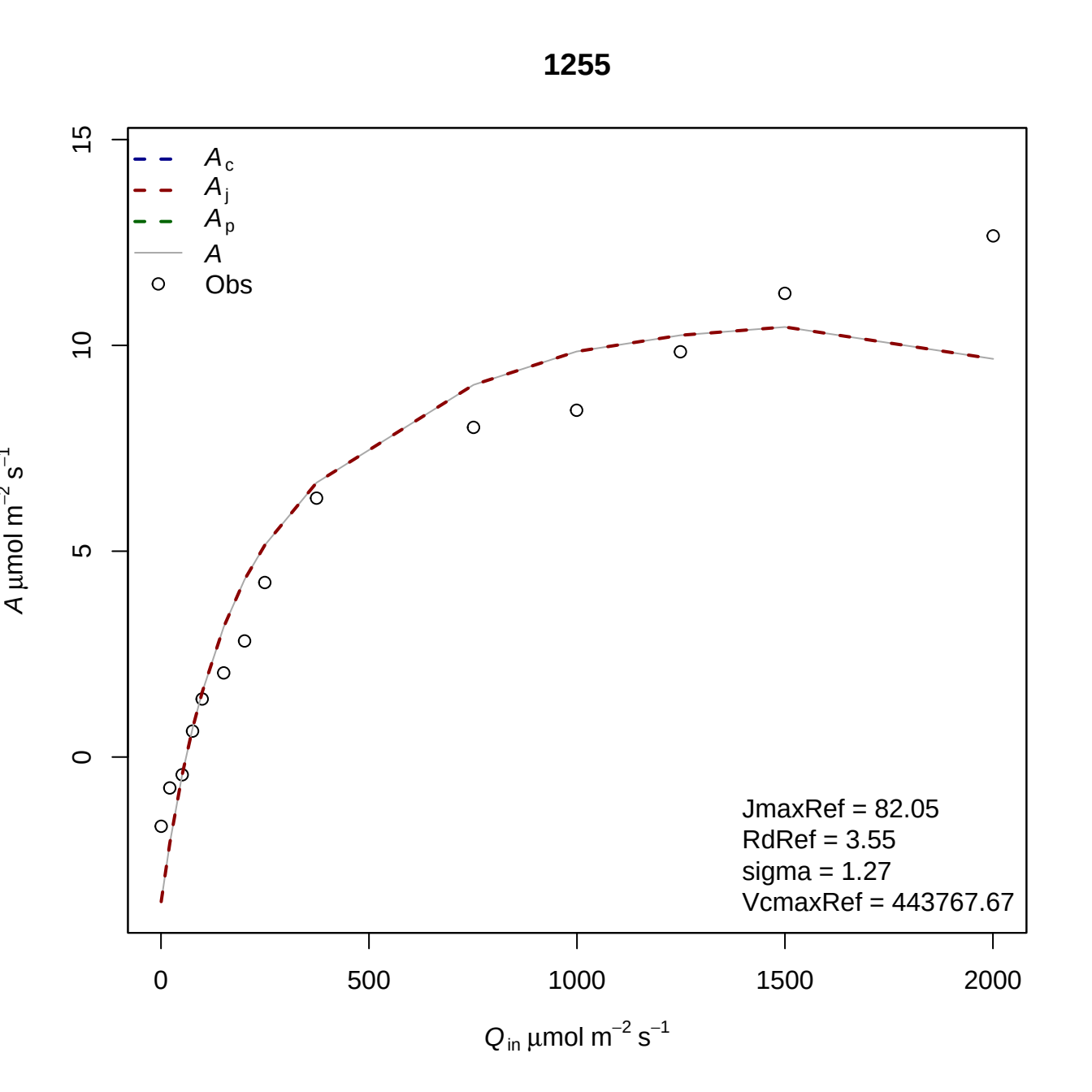

1250
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.06
RdRef = 3.29
sigma = 0.79
VcmaxRef = 393695.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

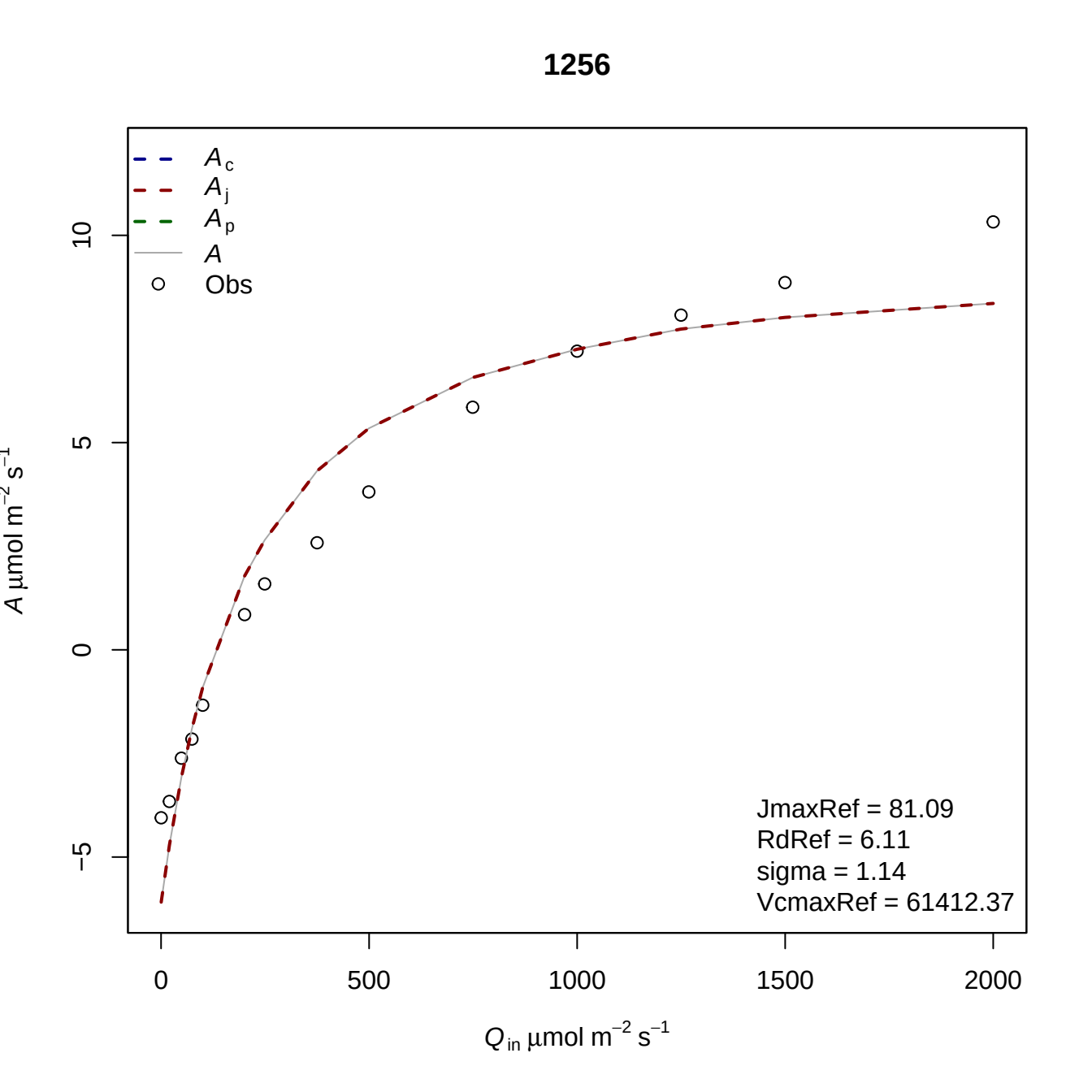

1251
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.27
RdRef = 4.67
sigma = 0.71
VcmaxRef = 107566.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
15
10
5
0
−5
0
500
1000
1500
2000

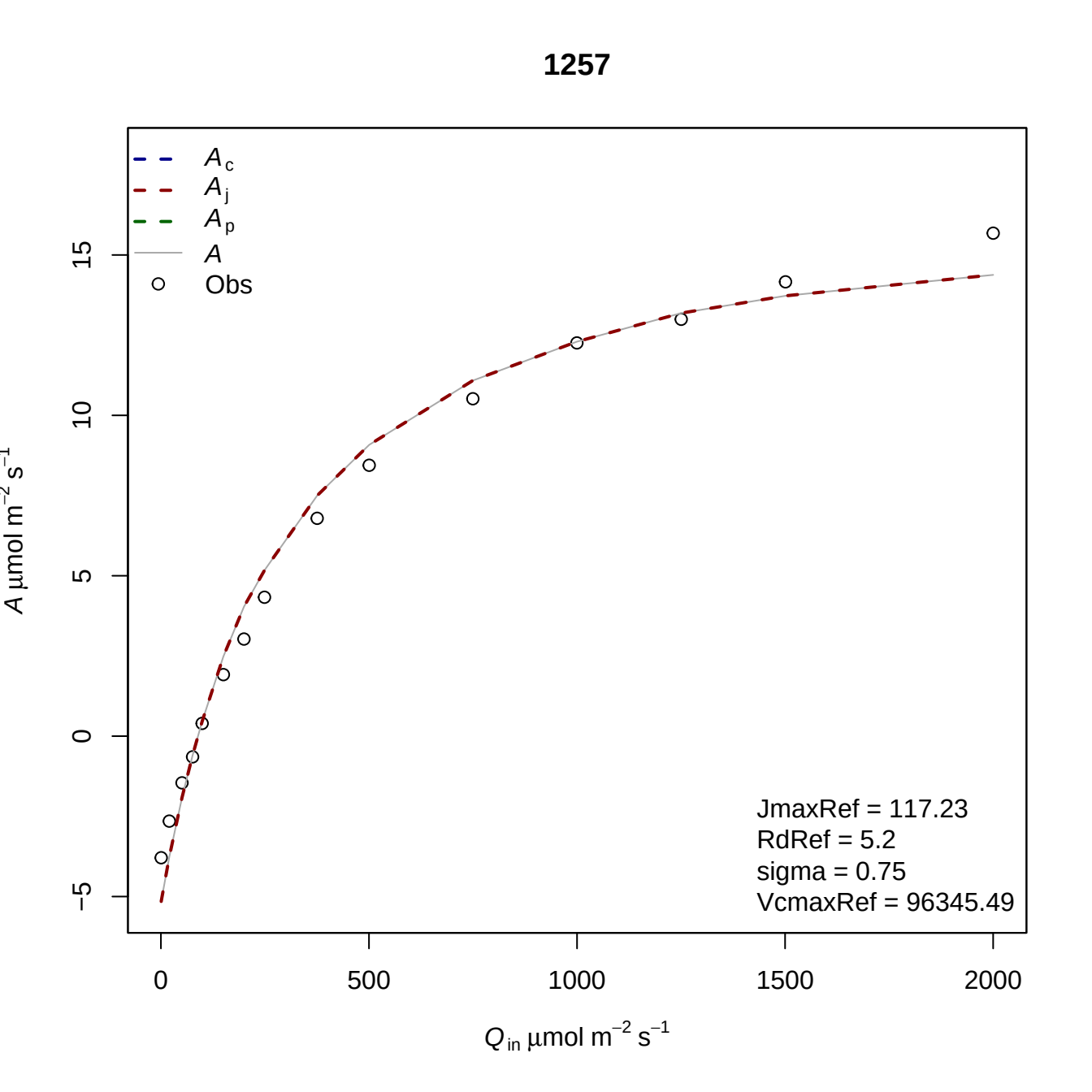

1252
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 106.73
RdRef = 3.59
sigma = 0.8
VcmaxRef = 125506.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

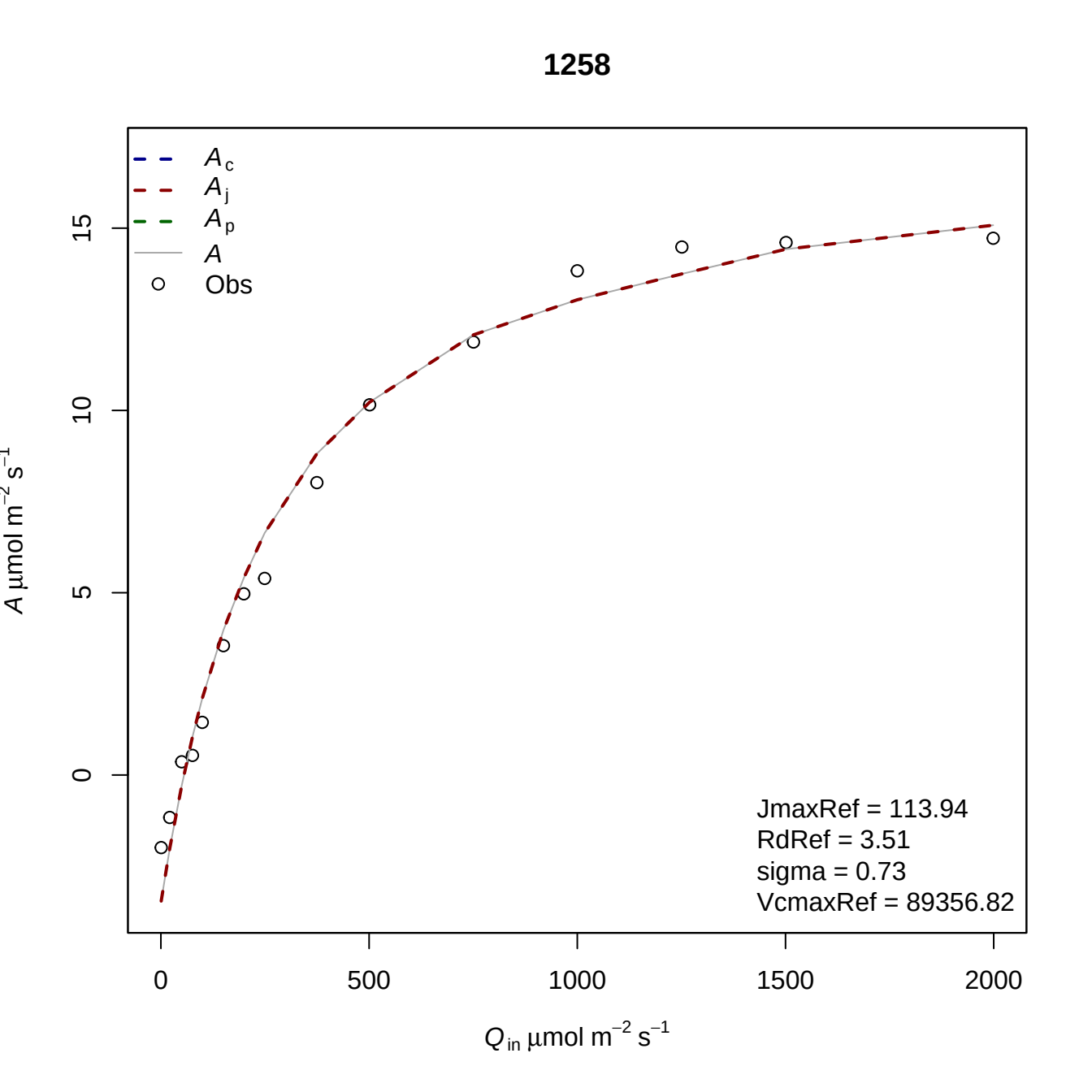

1253
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.2
RdRef = 4.93
sigma = 0.95
VcmaxRef = 64143.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
15
10
5
0
−5
0
500
1000
1500
2000

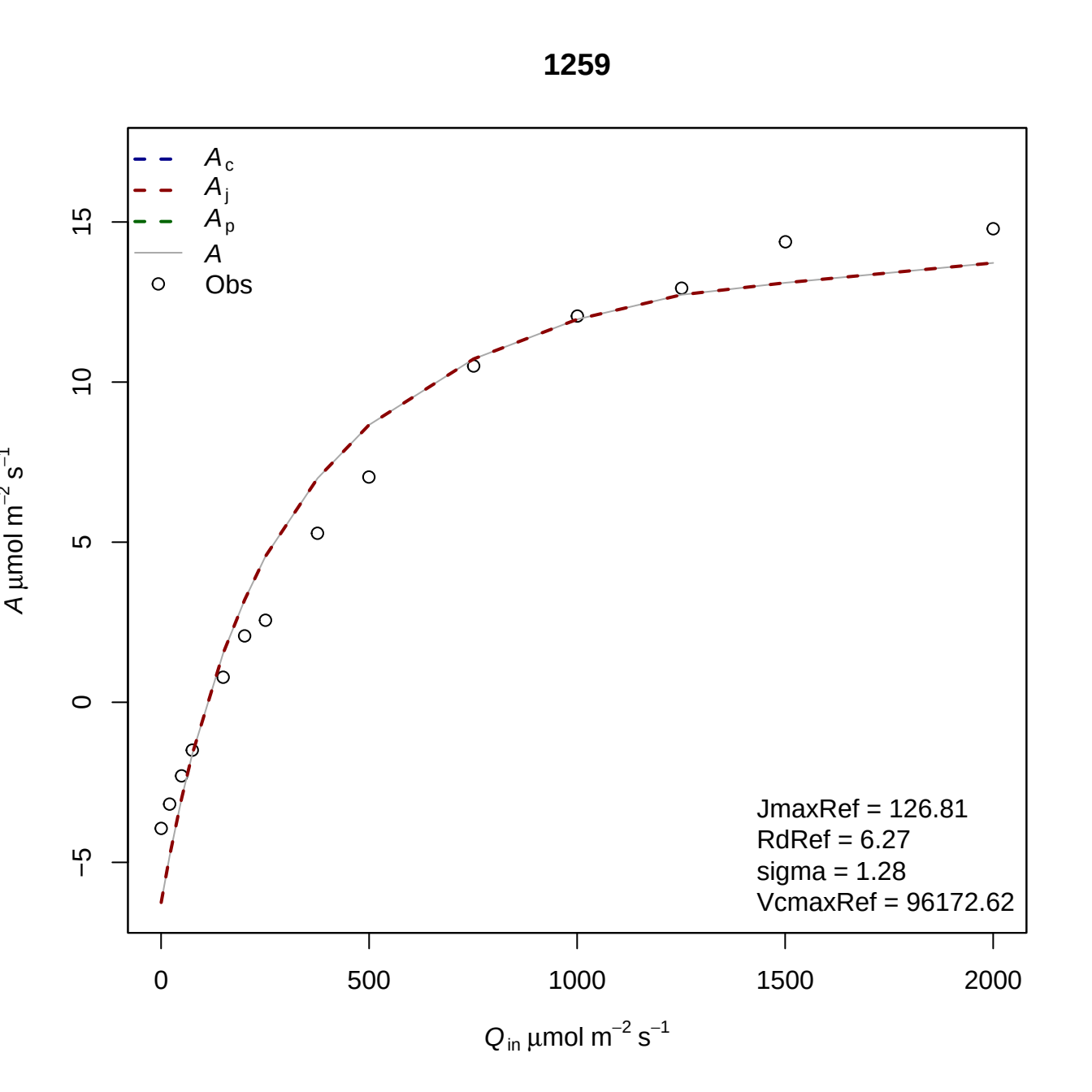

1254
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.7
RdRef = 3.31
sigma = 0.68
VcmaxRef = 151037.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

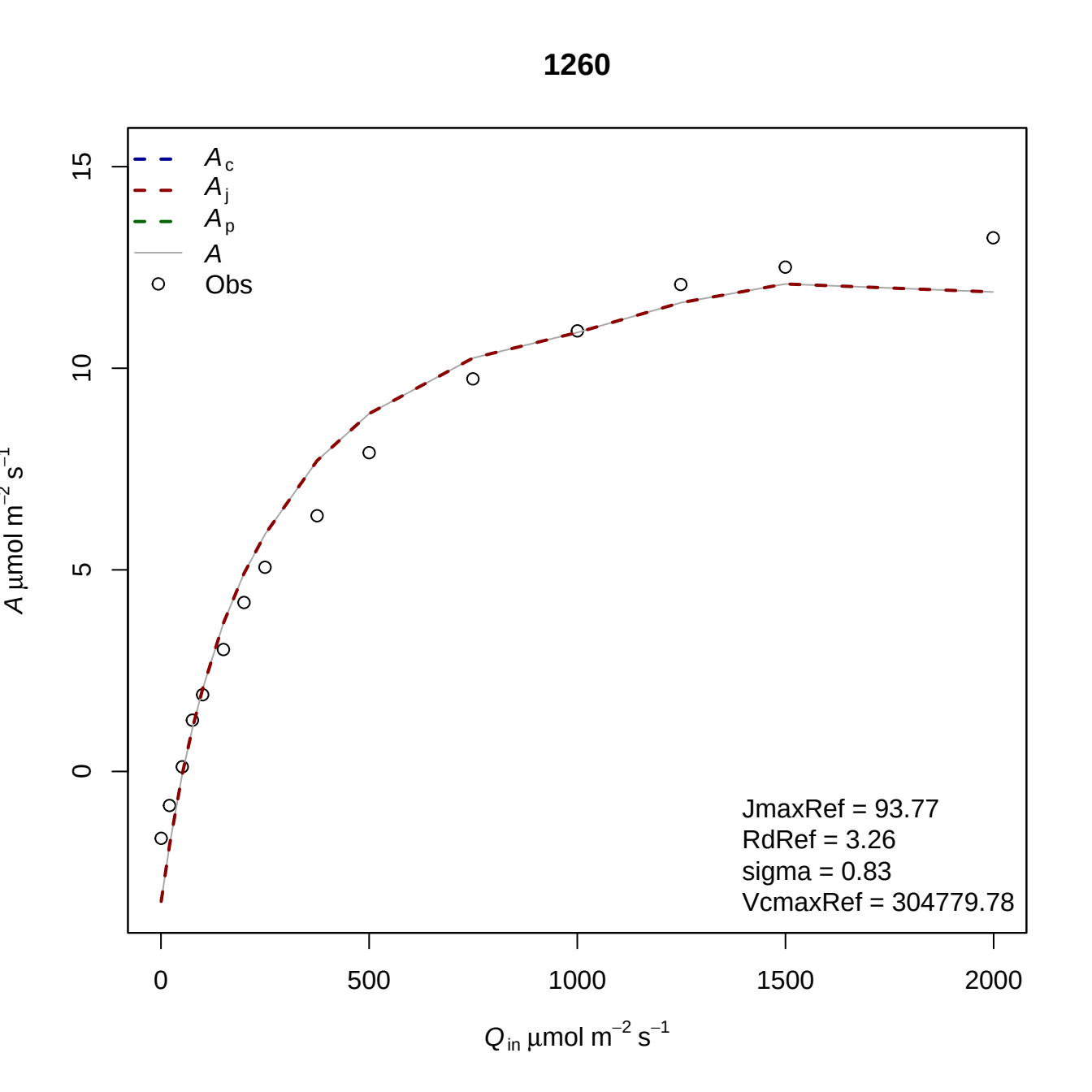

1255
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.05
RdRef = 3.55
sigma = 1.27
VcmaxRef = 443767.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

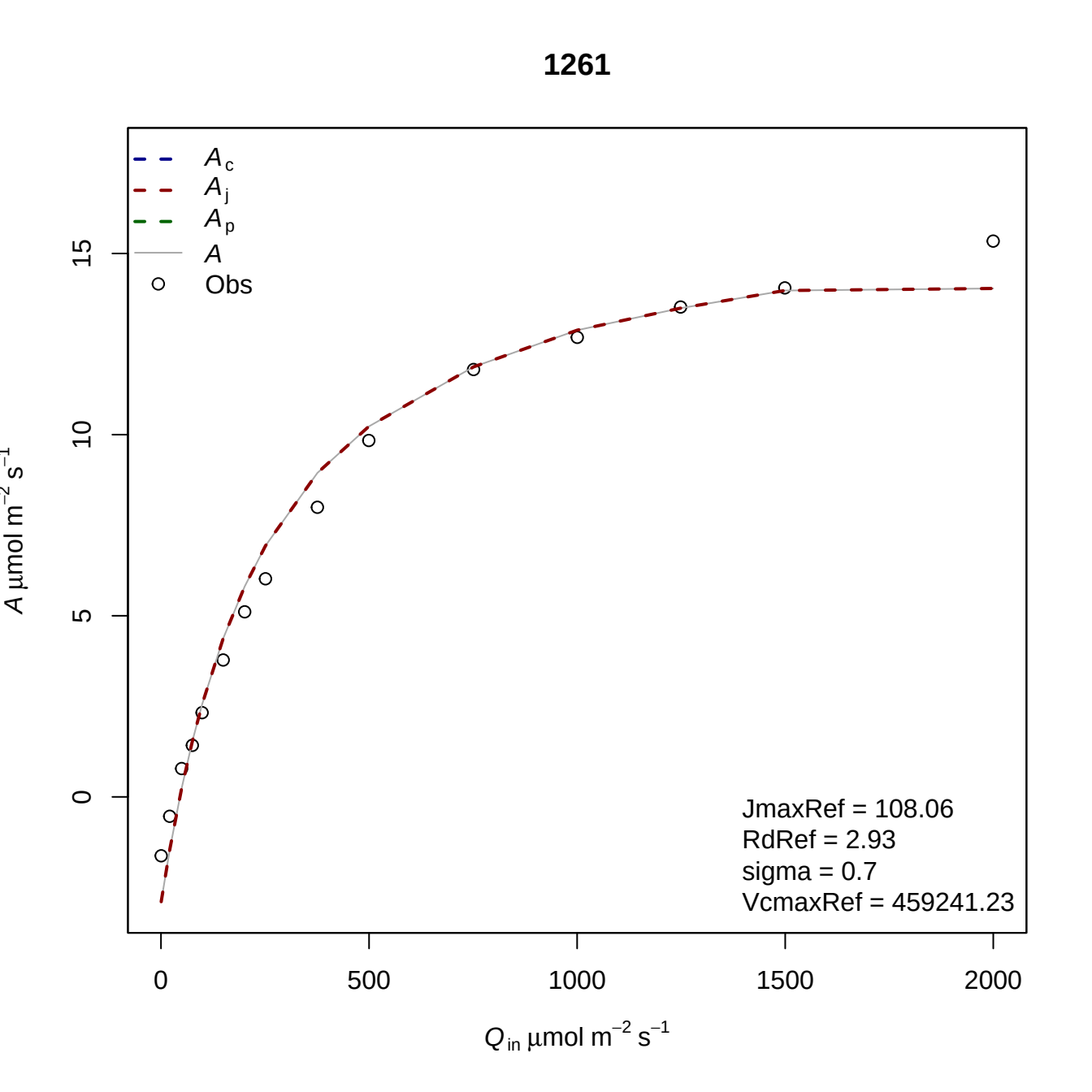

1256
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.09
RdRef = 6.11
sigma = 1.14
VcmaxRef = 61412.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
5
0
−5
0
500
1000
1500
2000

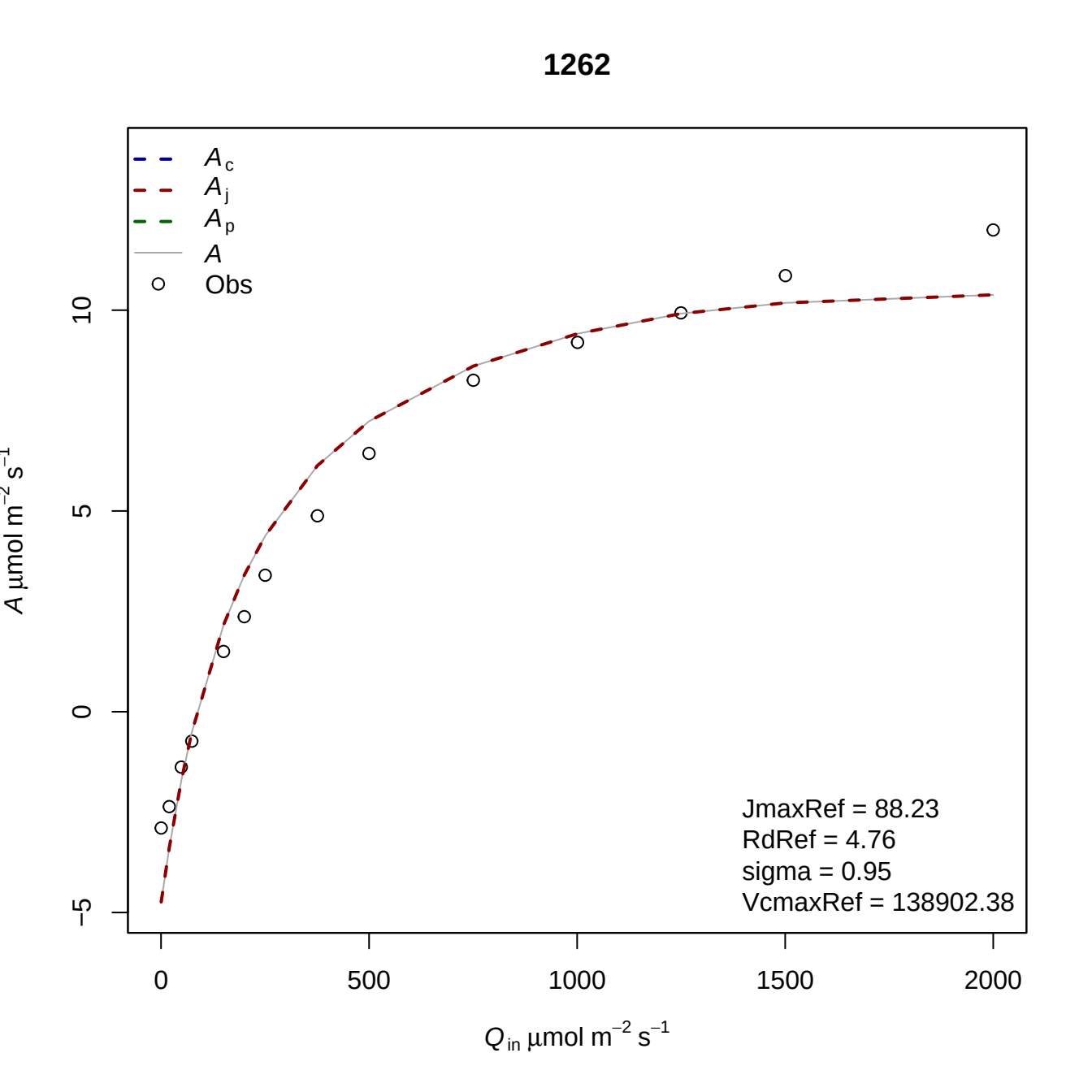

1257
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.23
RdRef = 5.2
sigma = 0.75
VcmaxRef = 96345.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

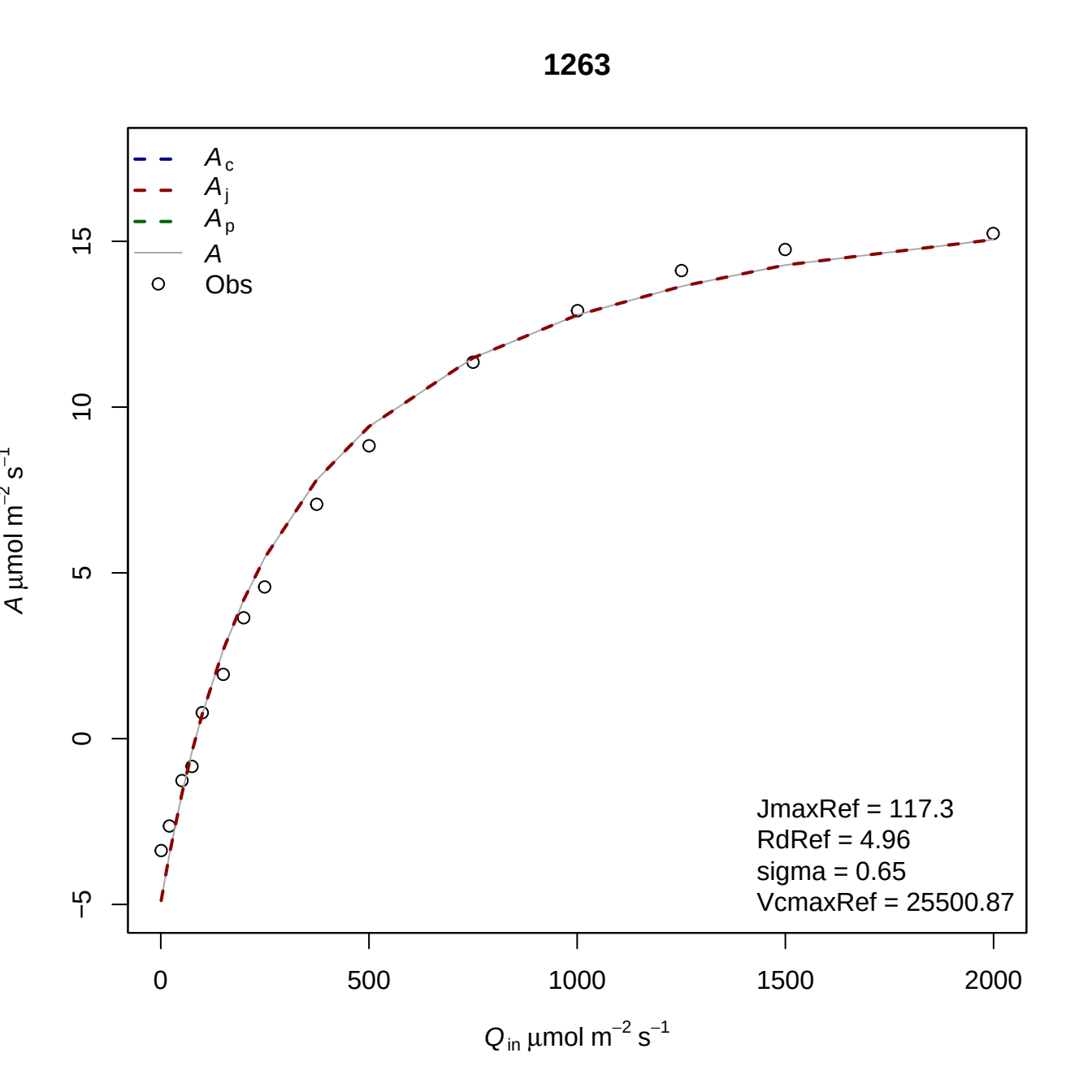

1258
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 113.94
RdRef = 3.51
sigma = 0.73
VcmaxRef = 89356.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

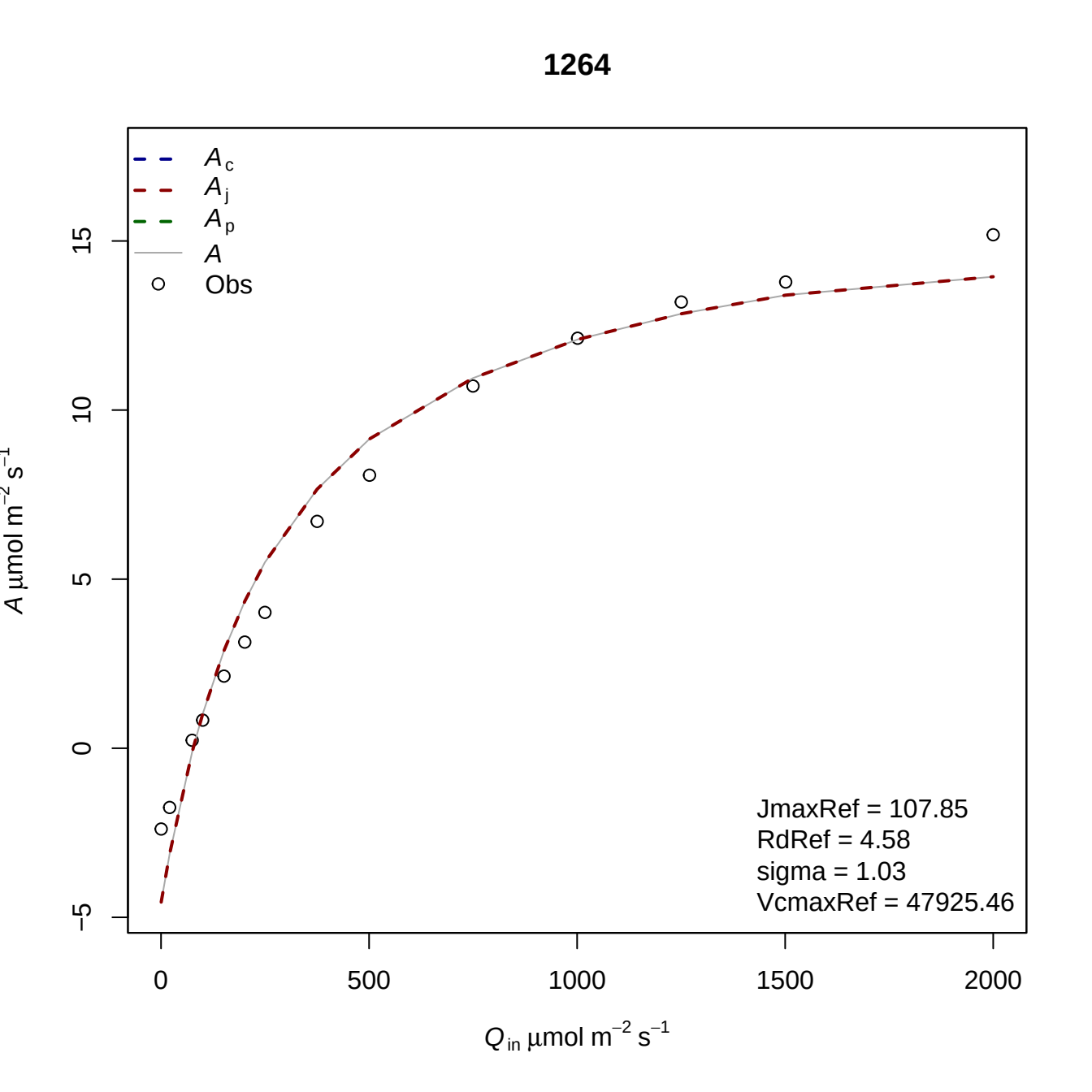

1259
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 126.81
RdRef = 6.27
sigma = 1.28
VcmaxRef = 96172.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
15
10
5
0
−5
0
500
1000
1500
2000

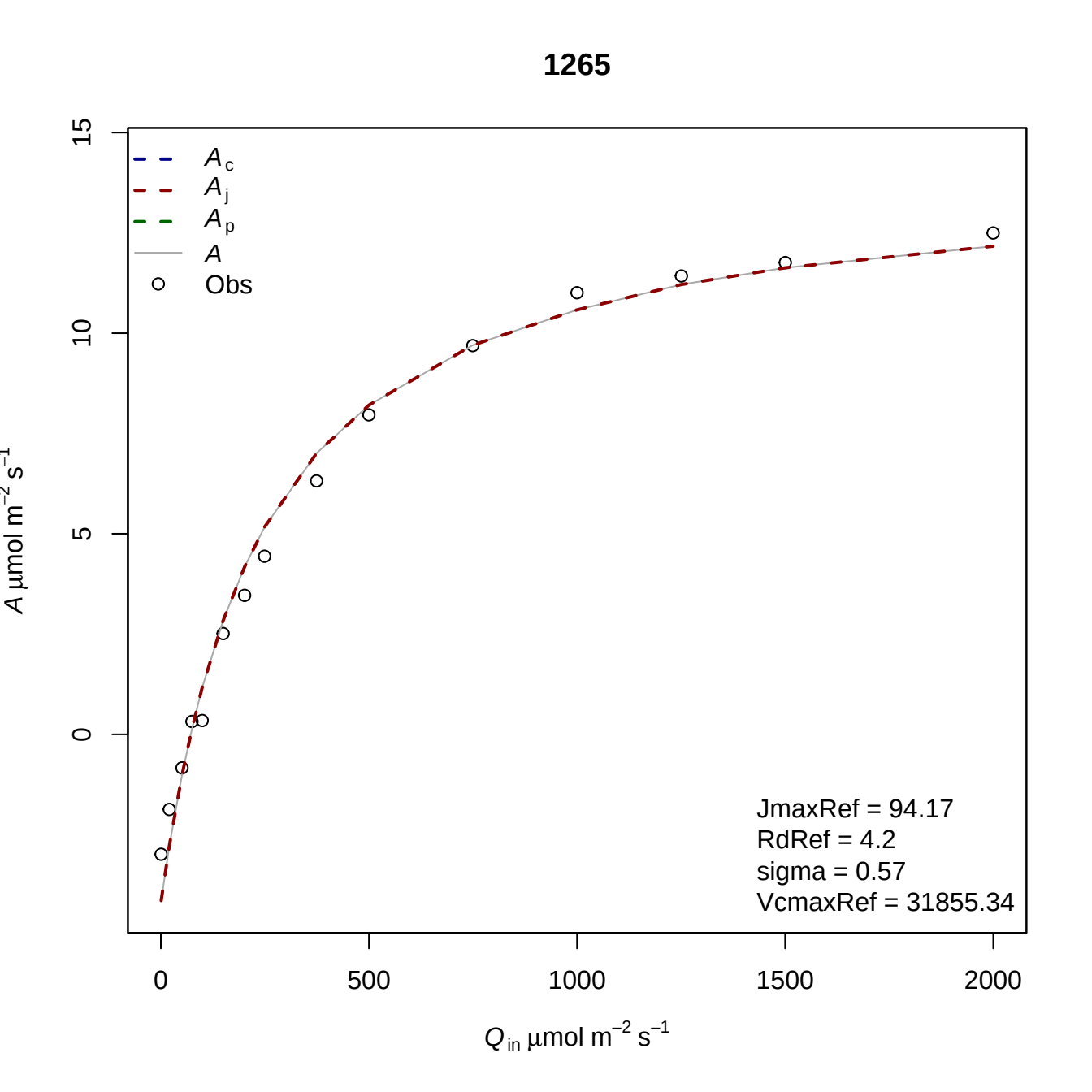

1260
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.77
RdRef = 3.26
sigma = 0.83
VcmaxRef = 304779.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

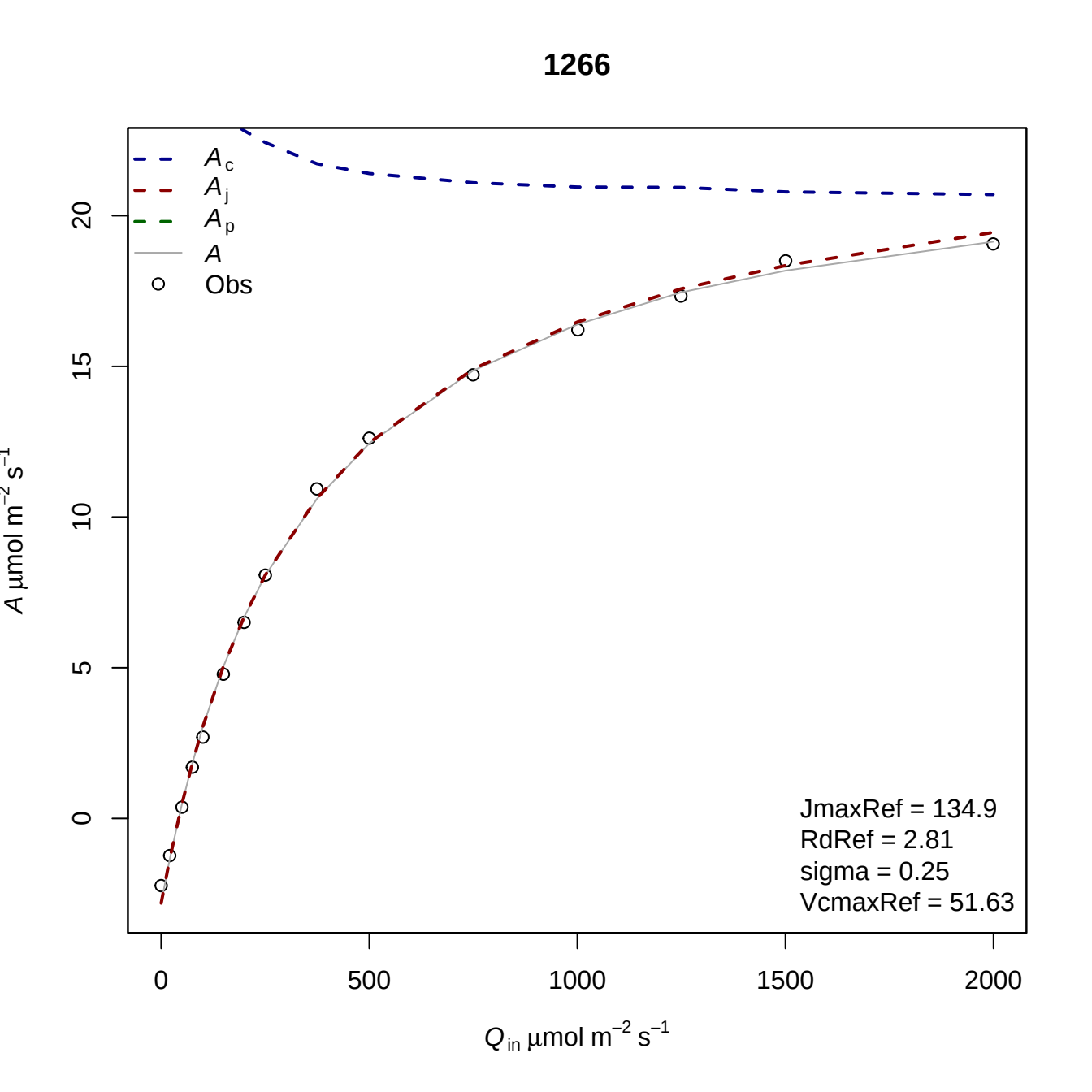

1261
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.06
RdRef = 2.93
sigma = 0.7
VcmaxRef = 459241.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

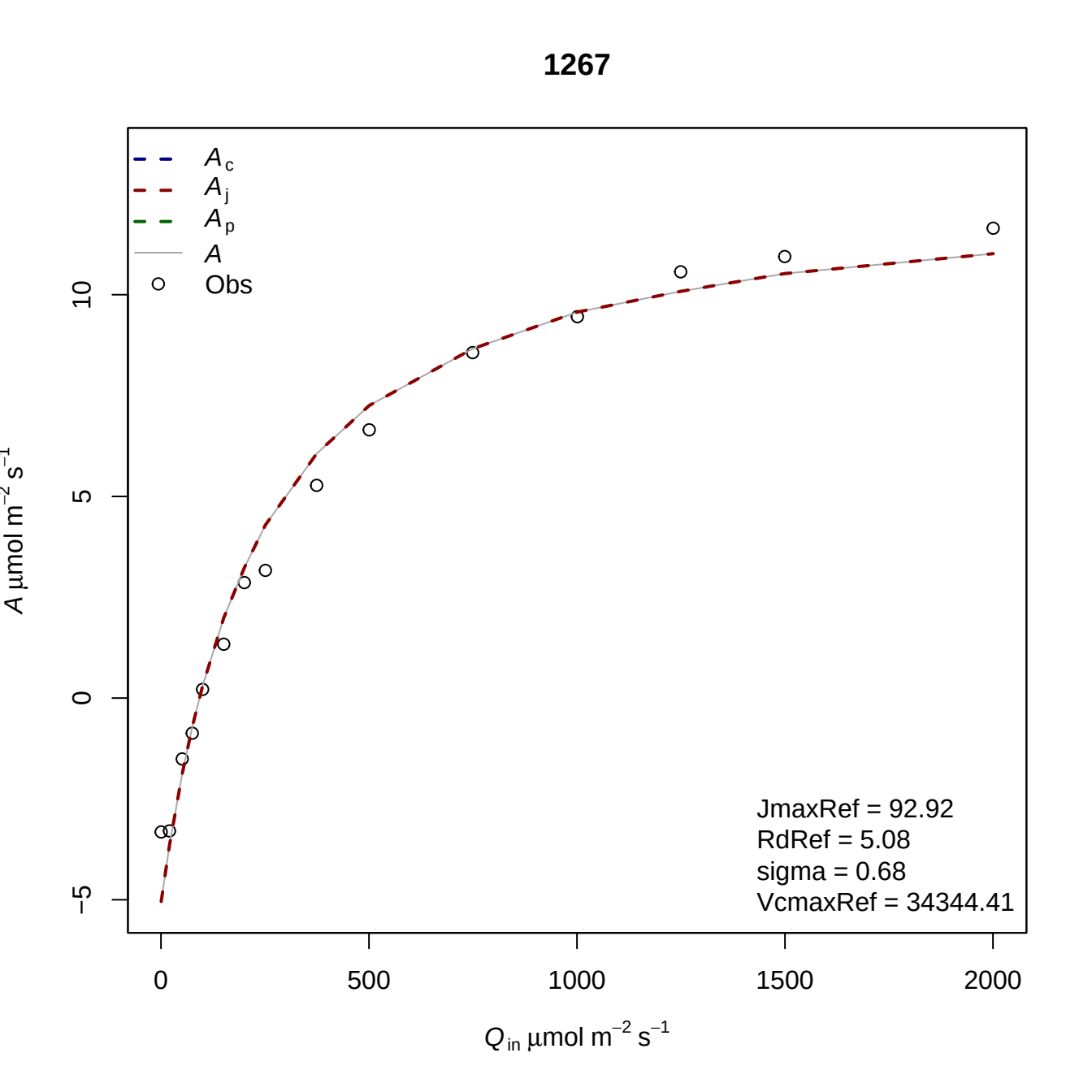

1262
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.23
RdRef = 4.76
sigma = 0.95
VcmaxRef = 138902.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

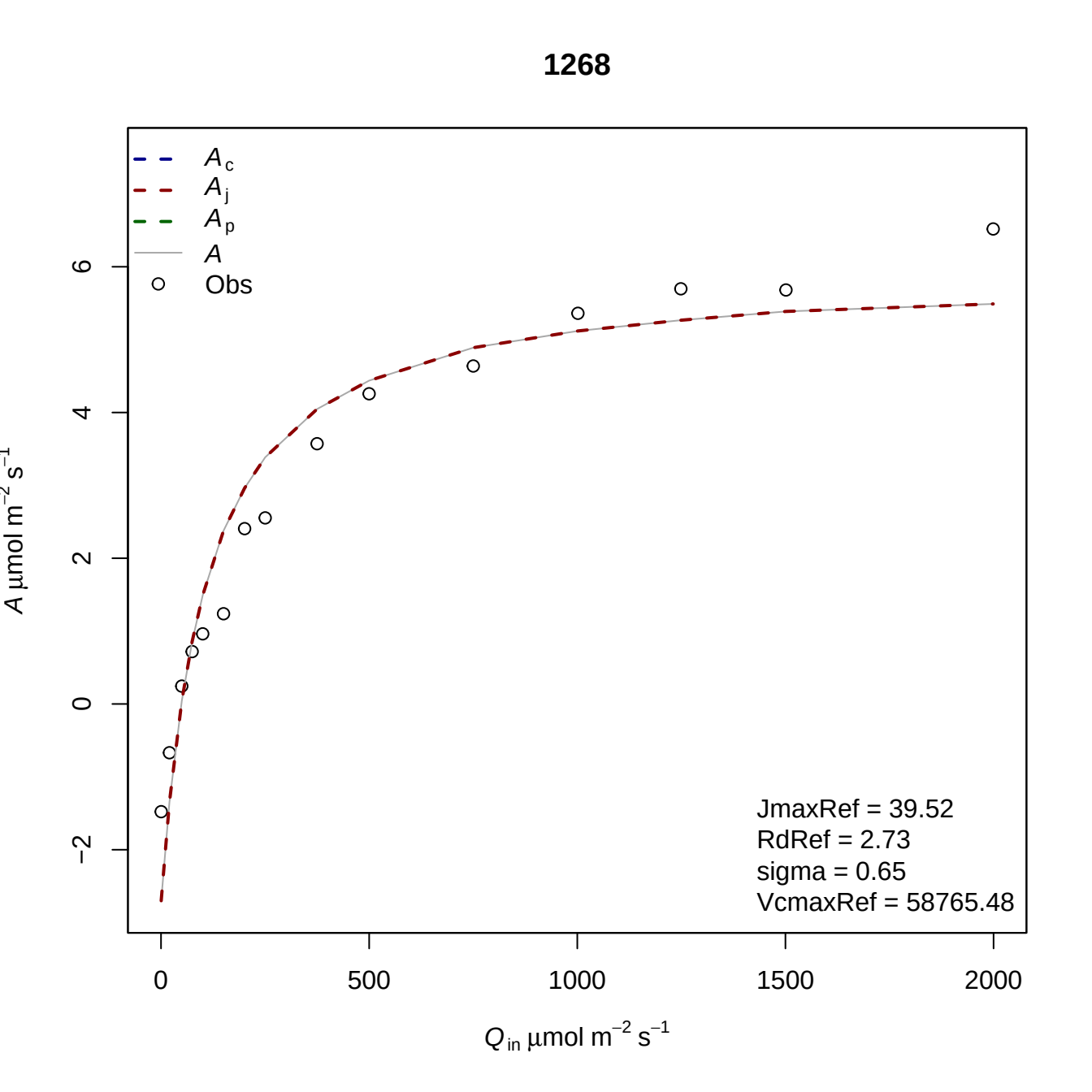

1263
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.3
RdRef = 4.96
sigma = 0.65
VcmaxRef = 25500.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

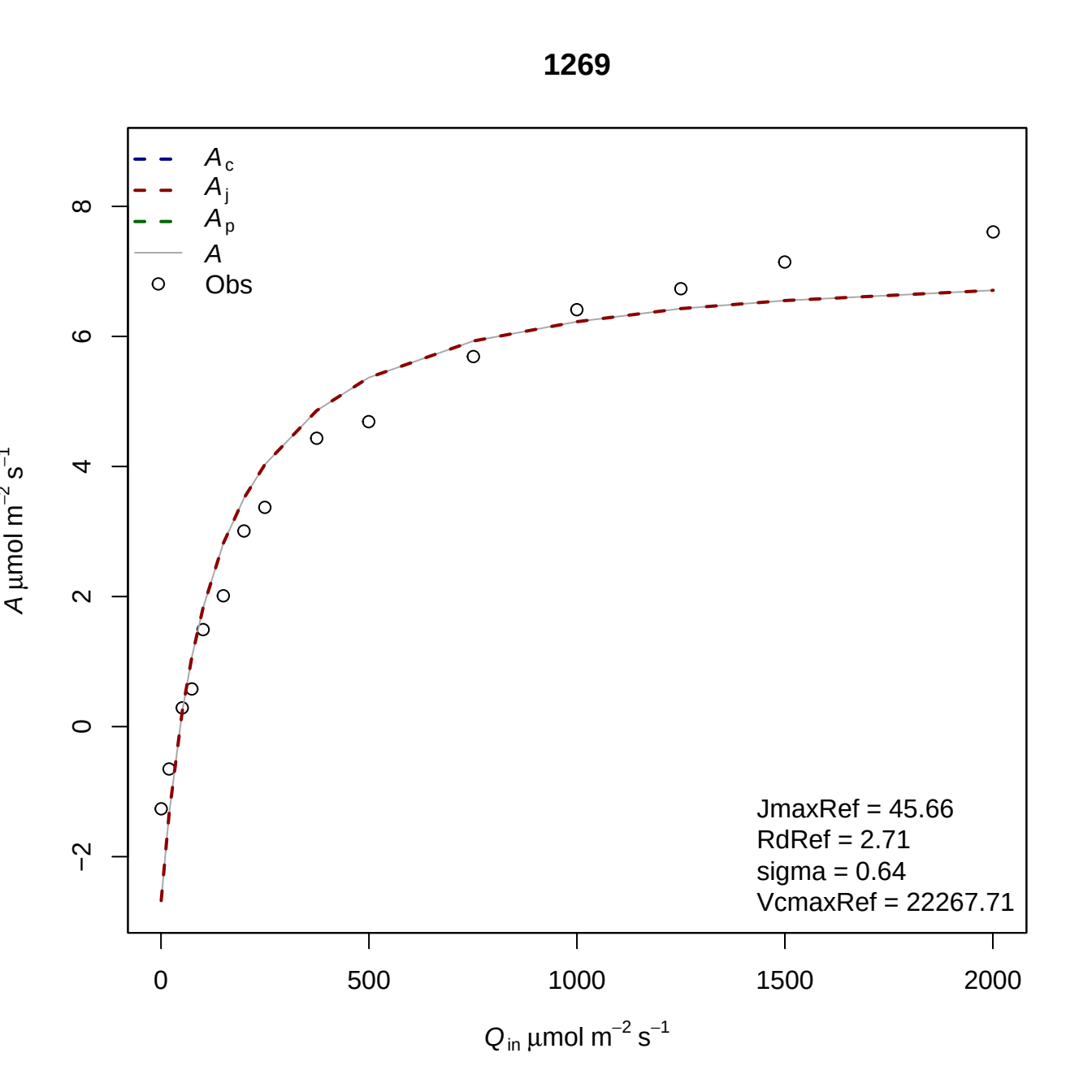

1264
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.85
RdRef = 4.58
sigma = 1.03
VcmaxRef = 47925.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

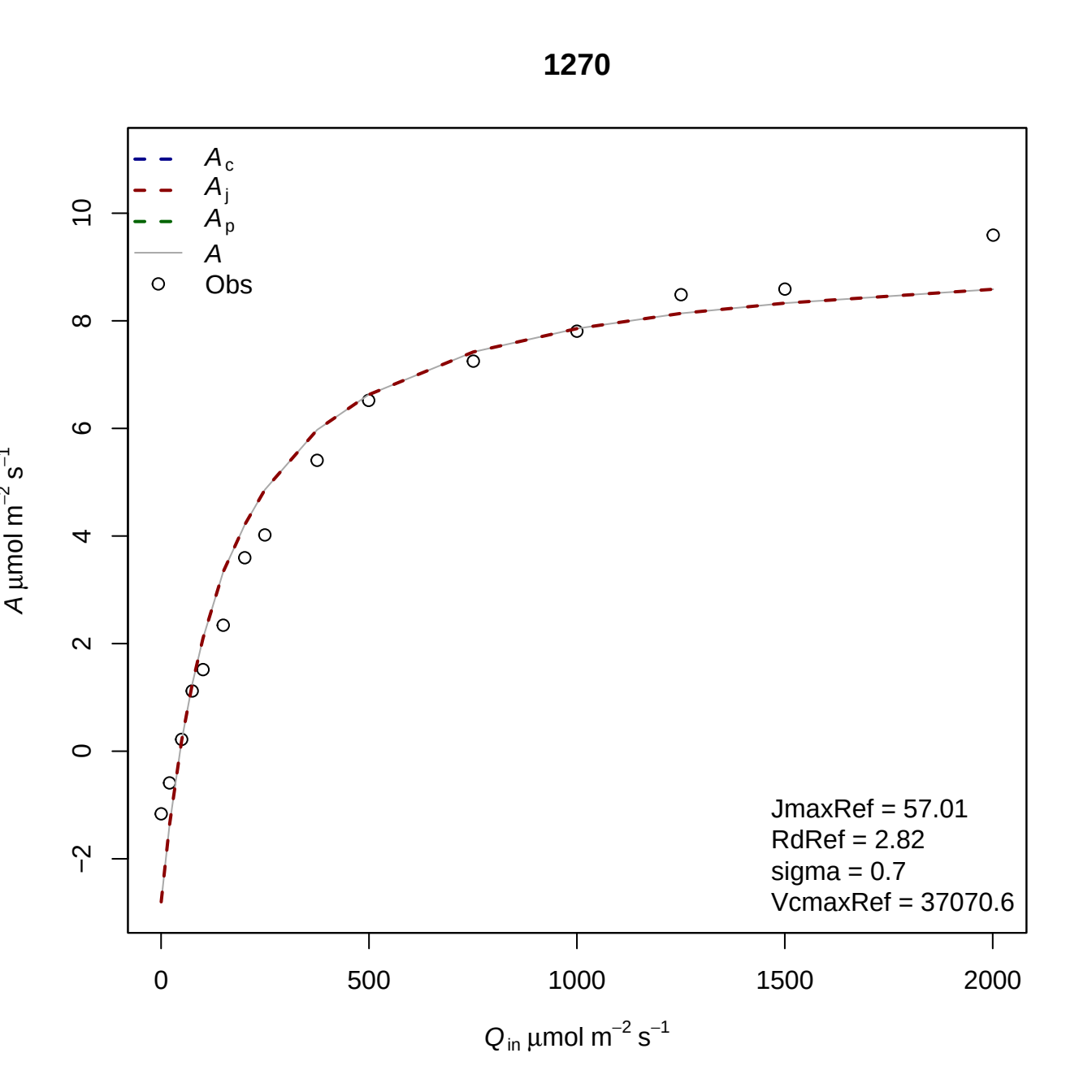

1265
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.17
RdRef = 4.2
sigma = 0.57
VcmaxRef = 31855.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

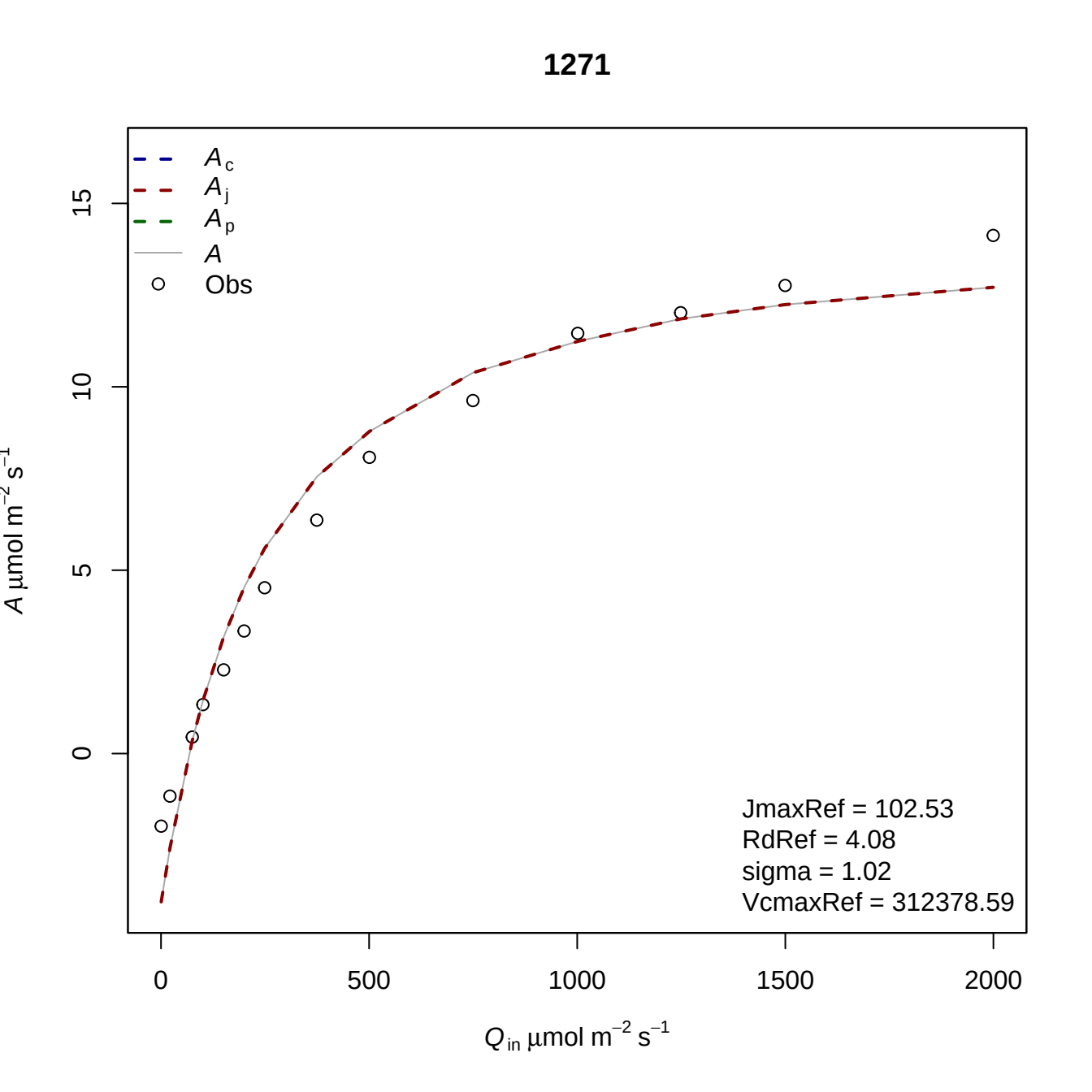

1266
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.9
RdRef = 2.81
sigma = 0.25
VcmaxRef = 51.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

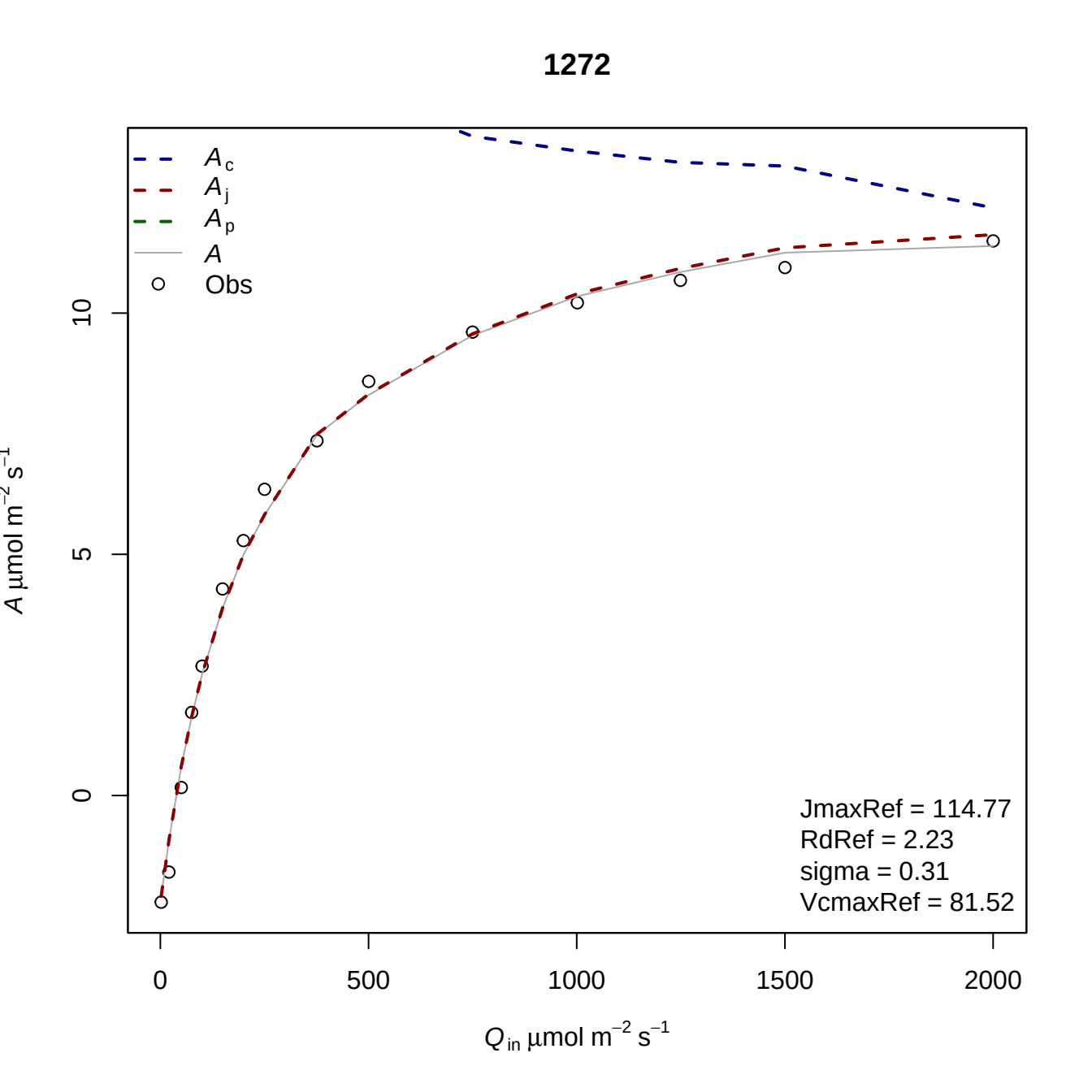

1267
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.92
RdRef = 5.08
sigma = 0.68
VcmaxRef = 34344.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

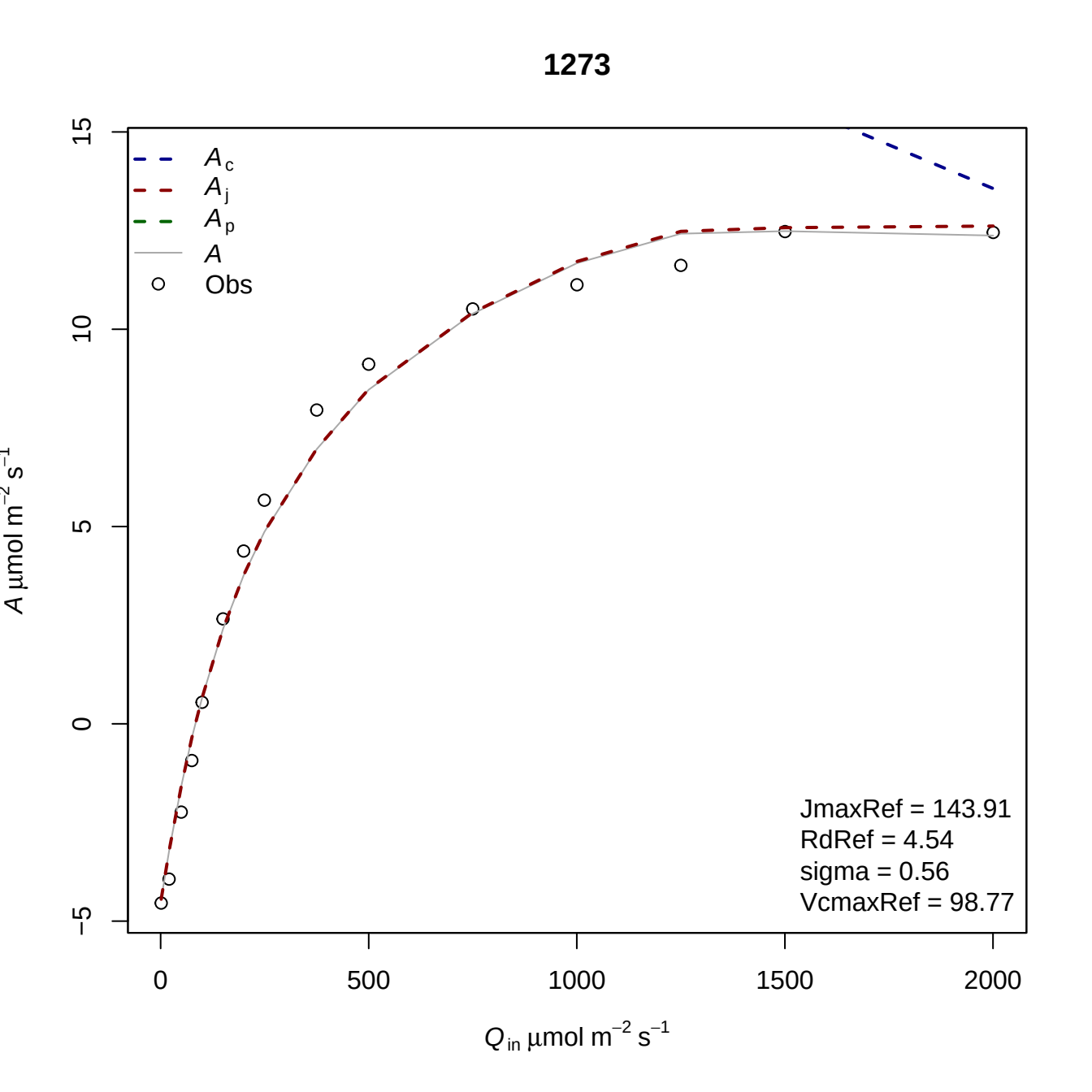

1268
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 39.52
RdRef = 2.73
sigma = 0.65
VcmaxRef = 58765.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

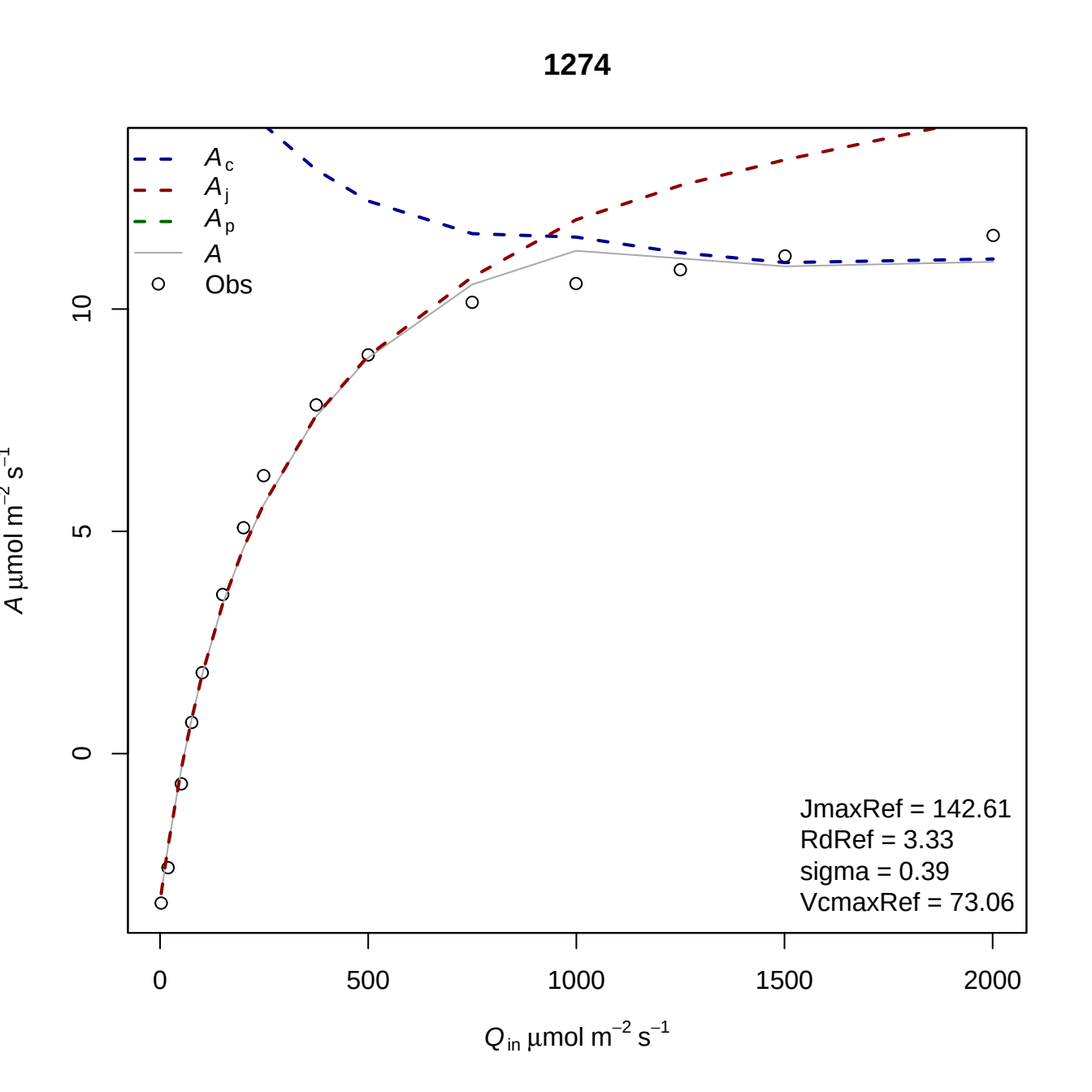

1269
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.66
RdRef = 2.71
sigma = 0.64
VcmaxRef = 22267.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

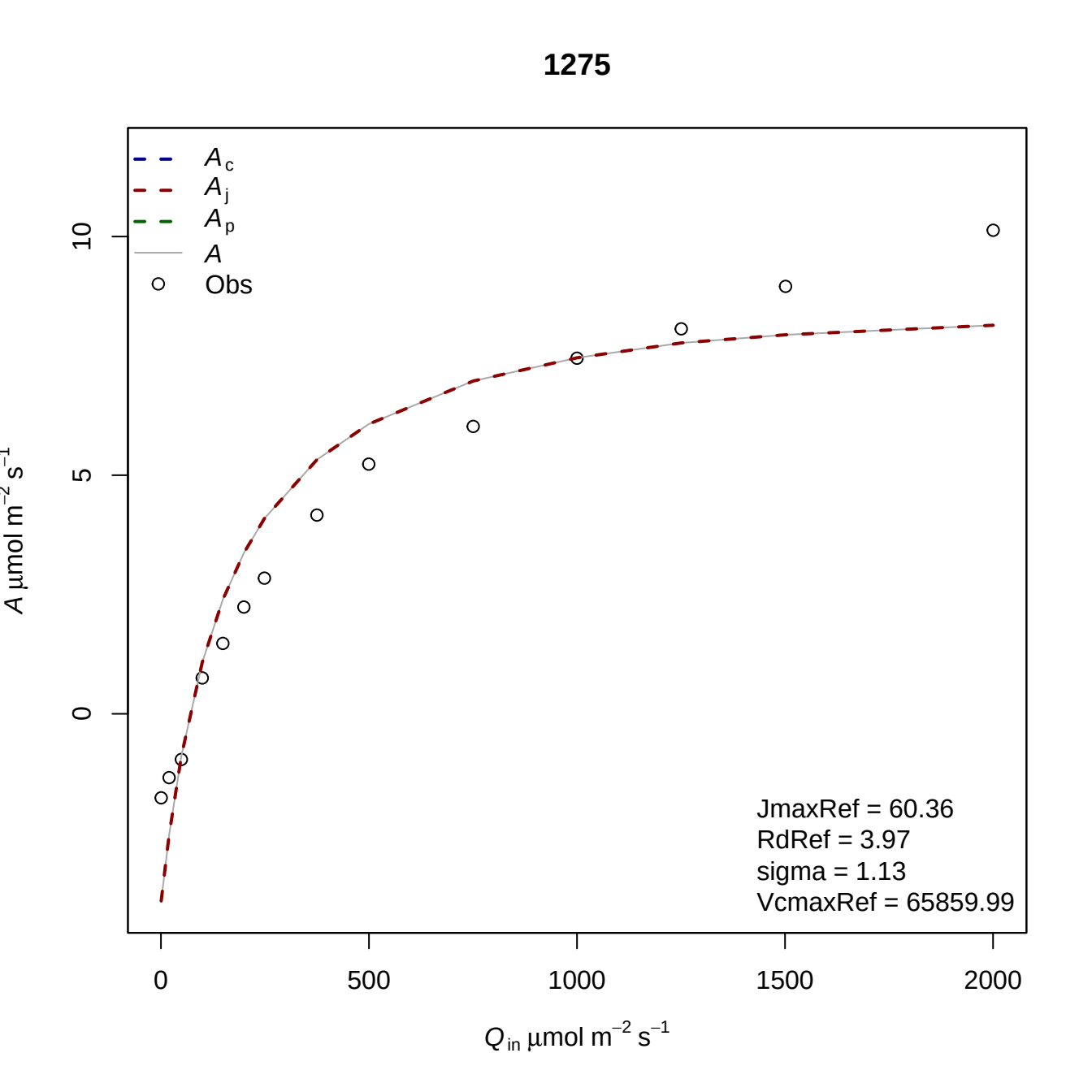

1270
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 57.01
RdRef = 2.82
sigma = 0.7
VcmaxRef = 37070.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

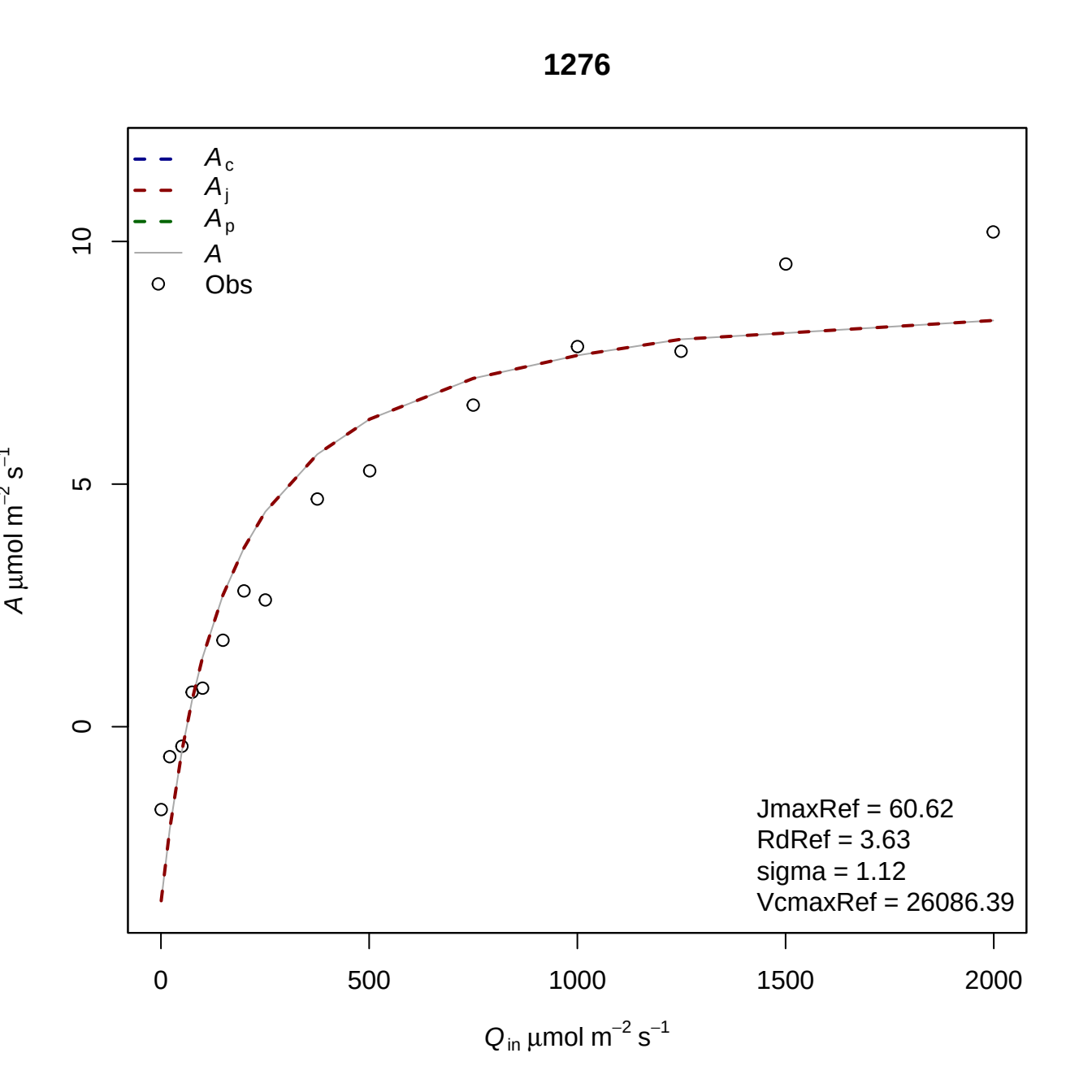

1271
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.53
RdRef = 4.08
sigma = 1.02
VcmaxRef = 312378.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

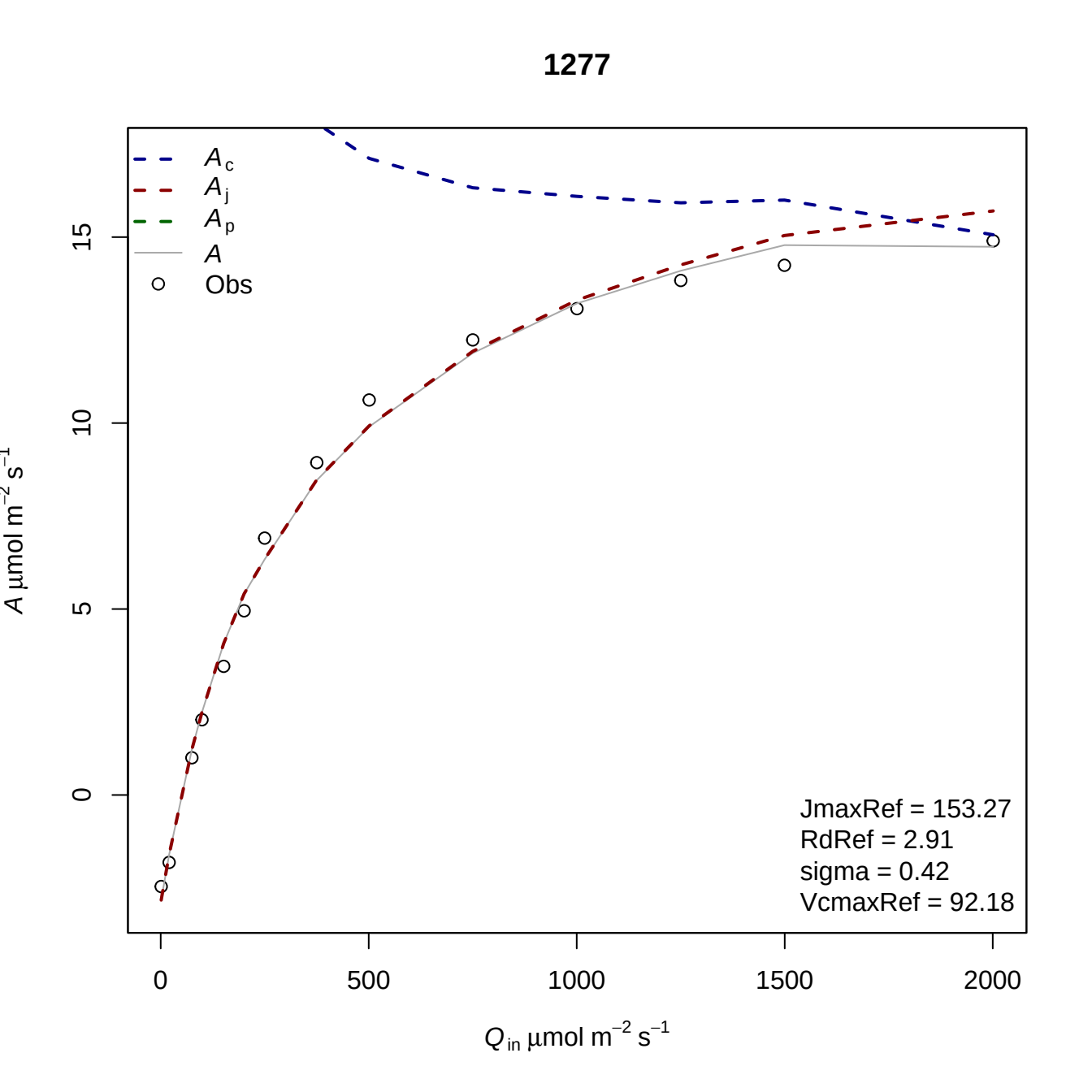

1272
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.77
RdRef = 2.23
sigma = 0.31
VcmaxRef = 81.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

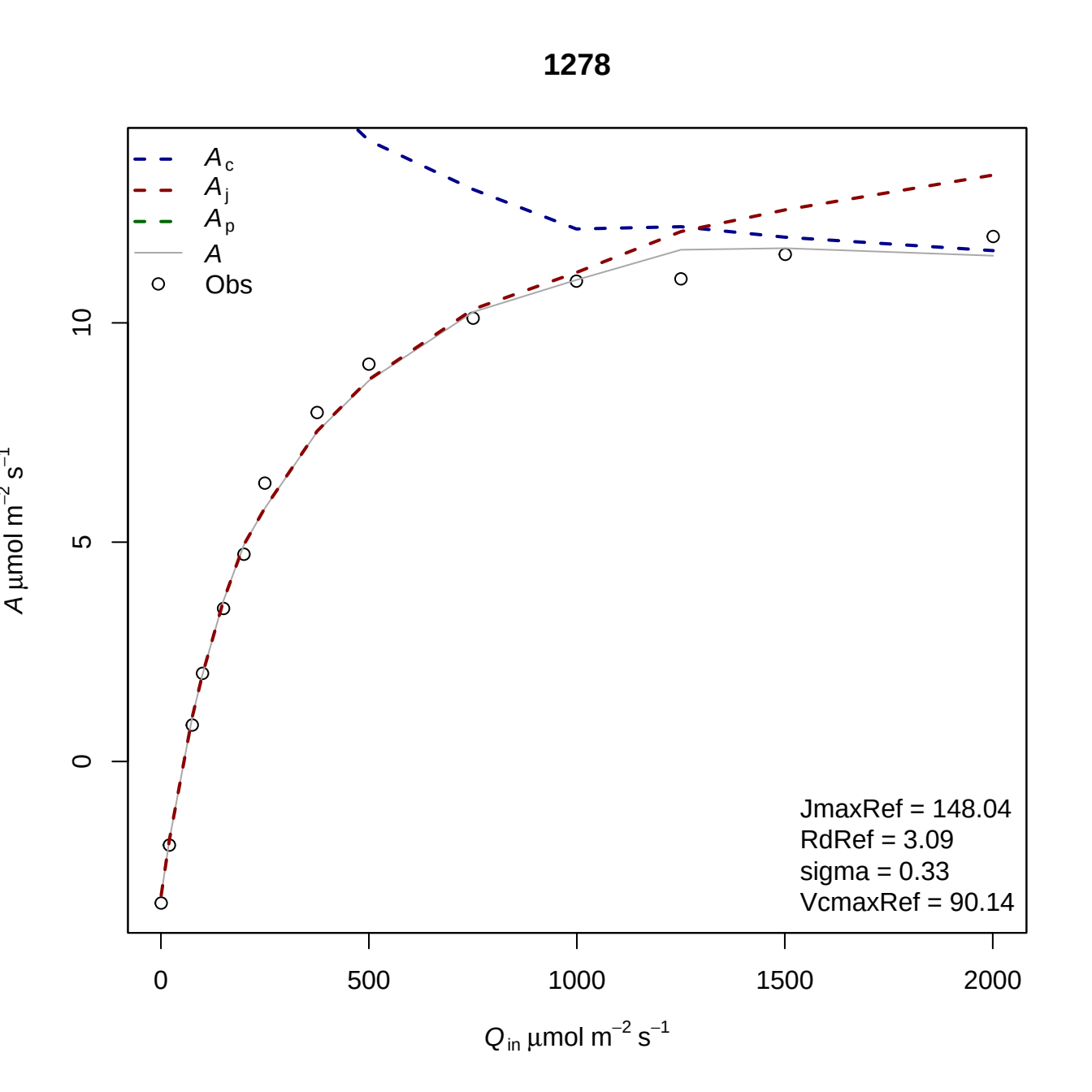

1273
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.91
RdRef = 4.54
sigma = 0.56
VcmaxRef = 98.77
15
10
5
0
−5
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

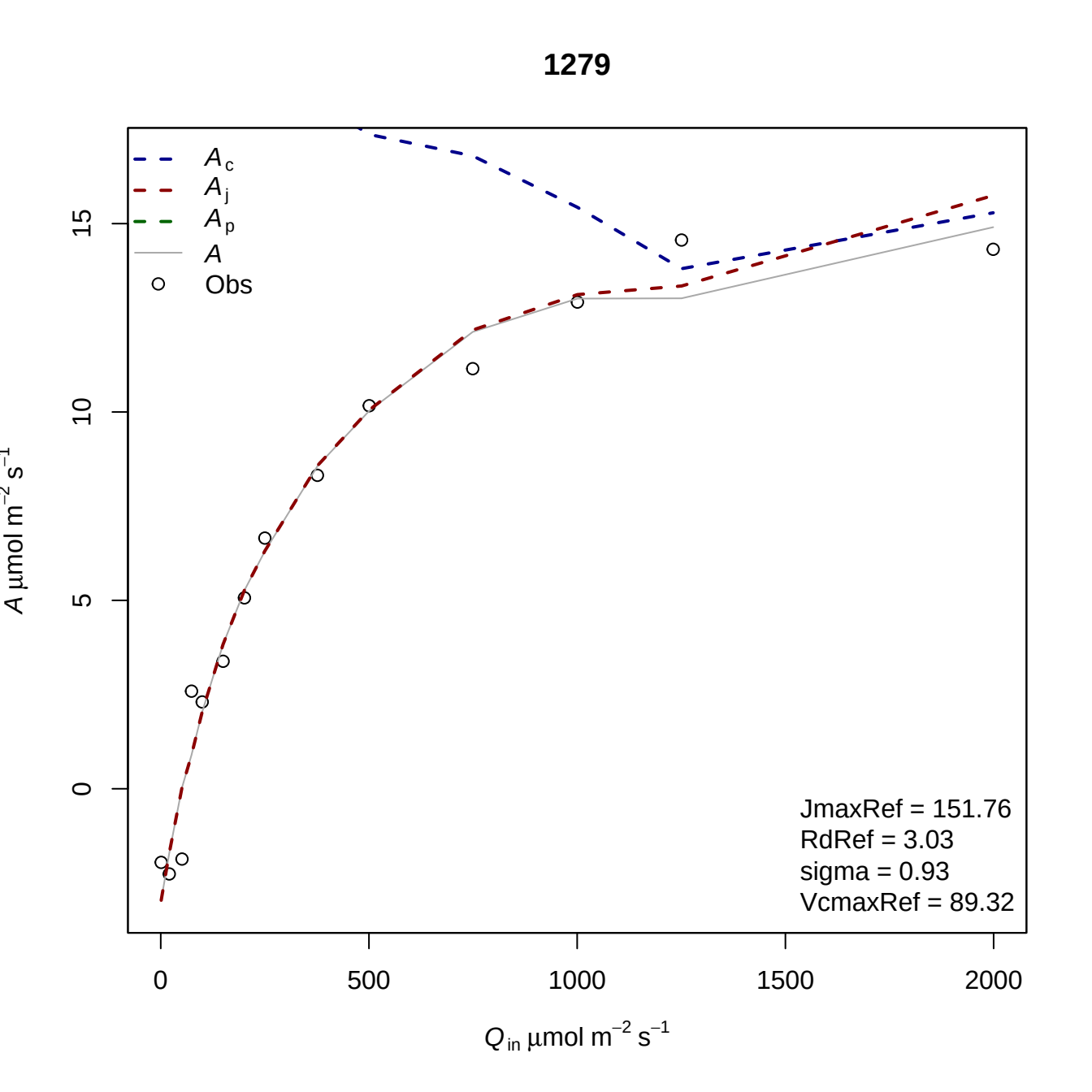

1274
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.61
RdRef = 3.33
sigma = 0.39
VcmaxRef = 73.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

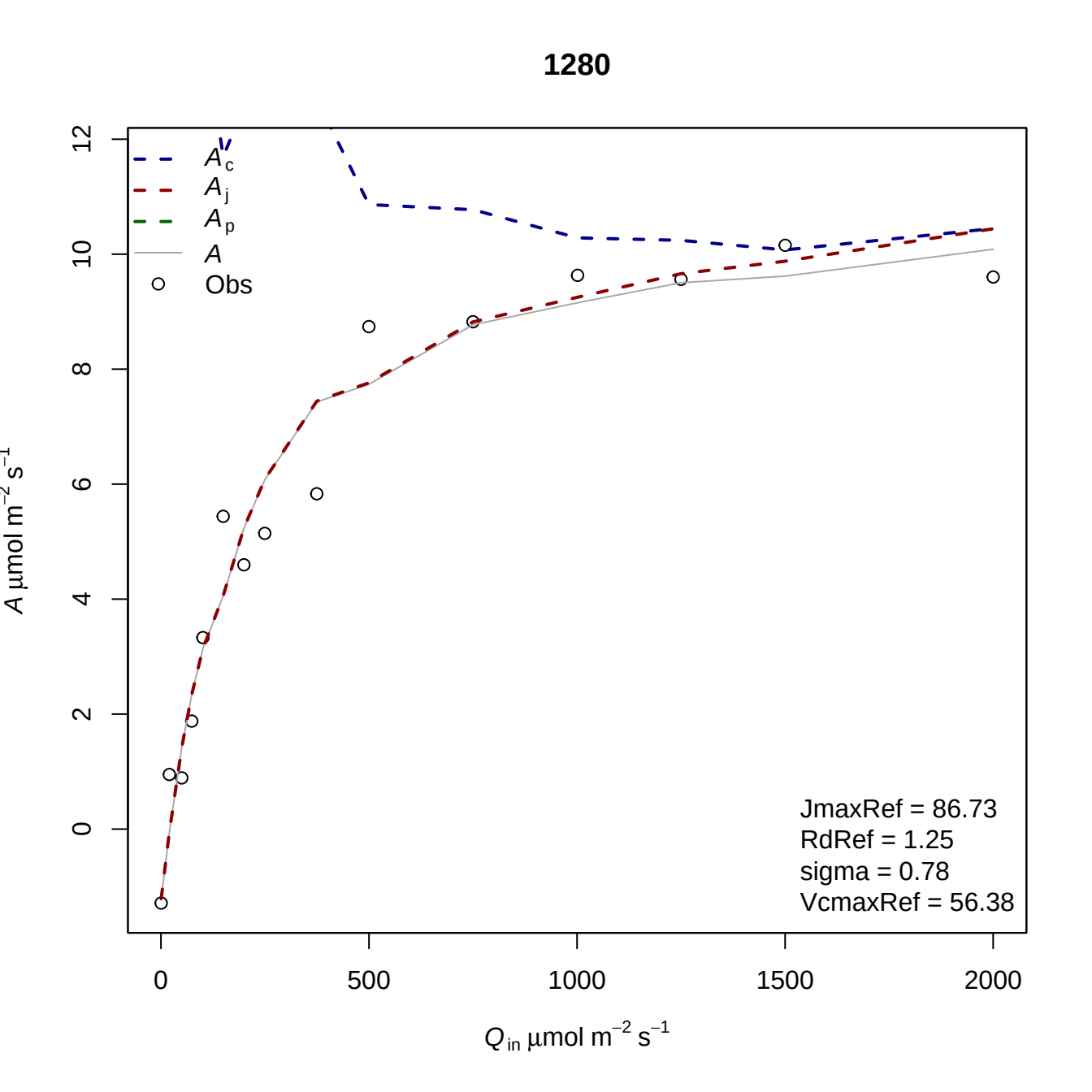

1275
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.36
RdRef = 3.97
sigma = 1.13
VcmaxRef = 65859.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

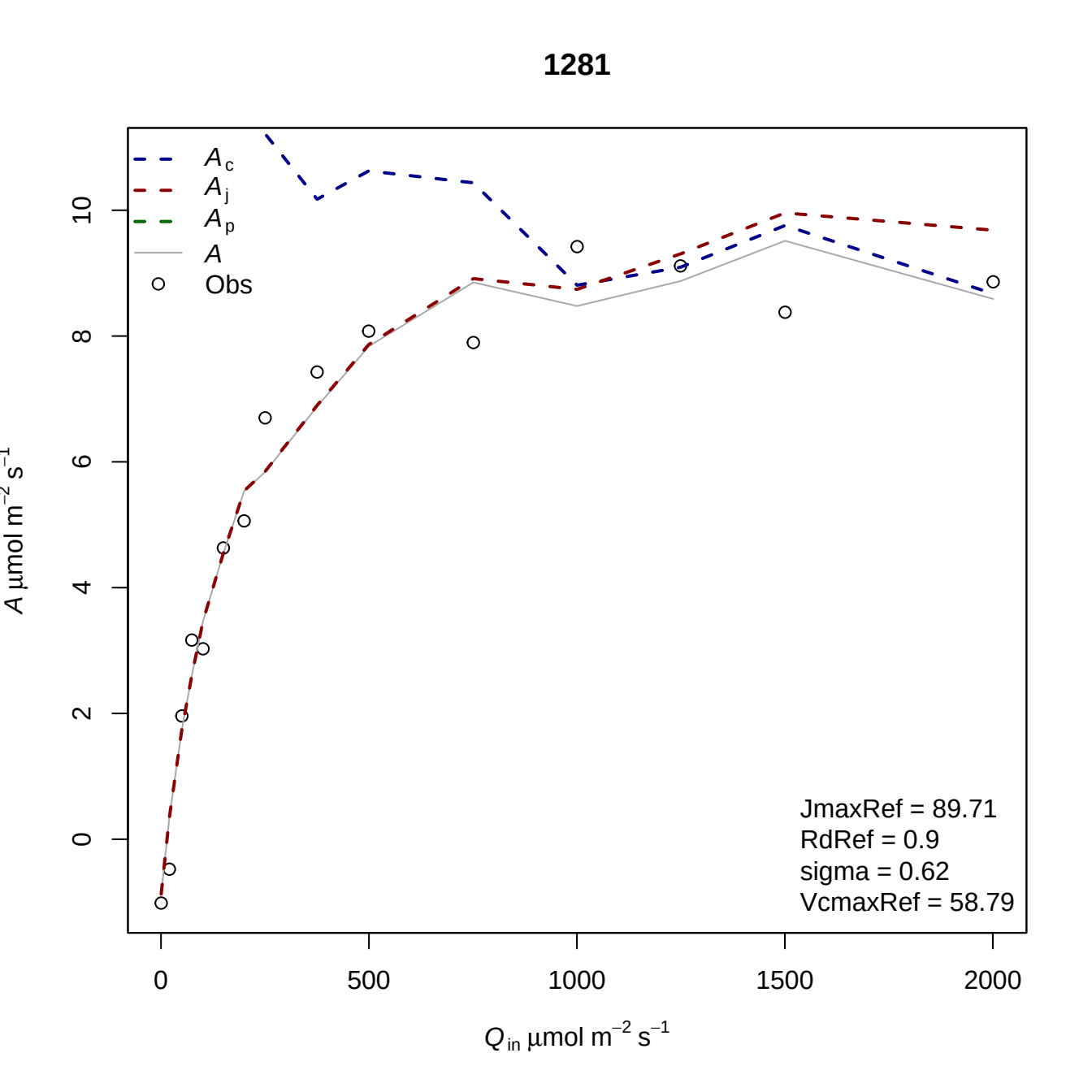

1276
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.62
RdRef = 3.63
sigma = 1.12
VcmaxRef = 26086.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

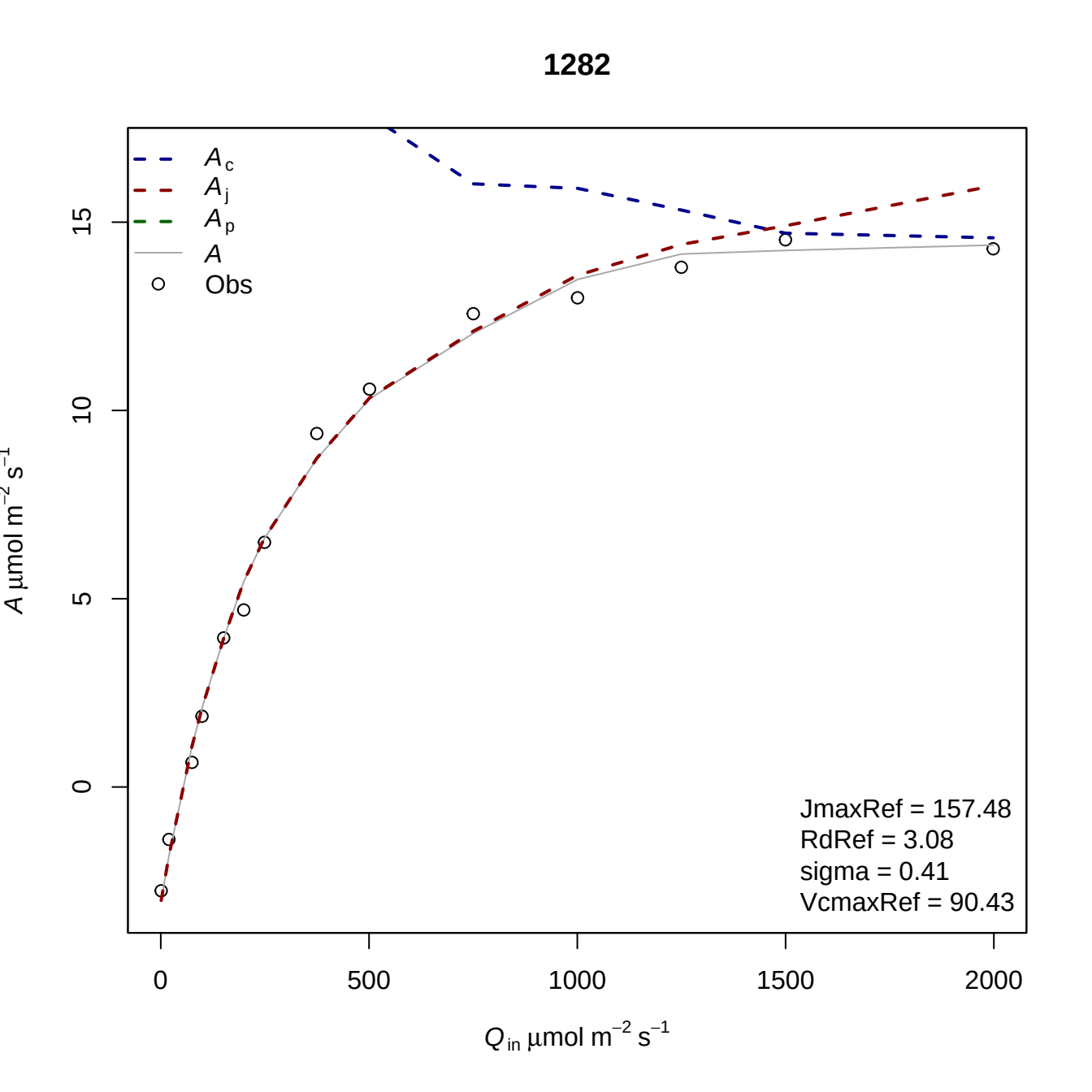

1277
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 153.27
RdRef = 2.91
sigma = 0.42
VcmaxRef = 92.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

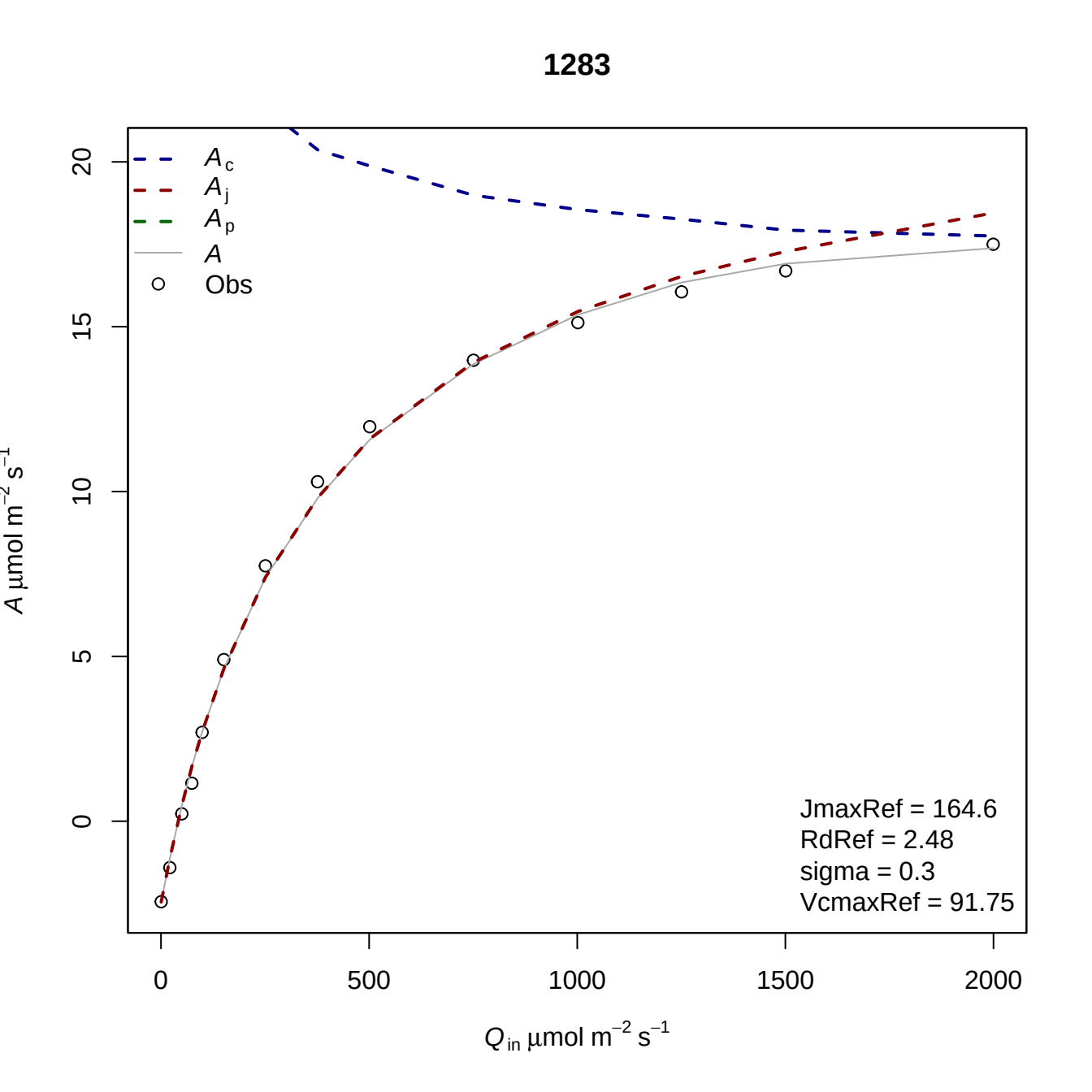

1278
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 148.04
RdRef = 3.09
sigma = 0.33
VcmaxRef = 90.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

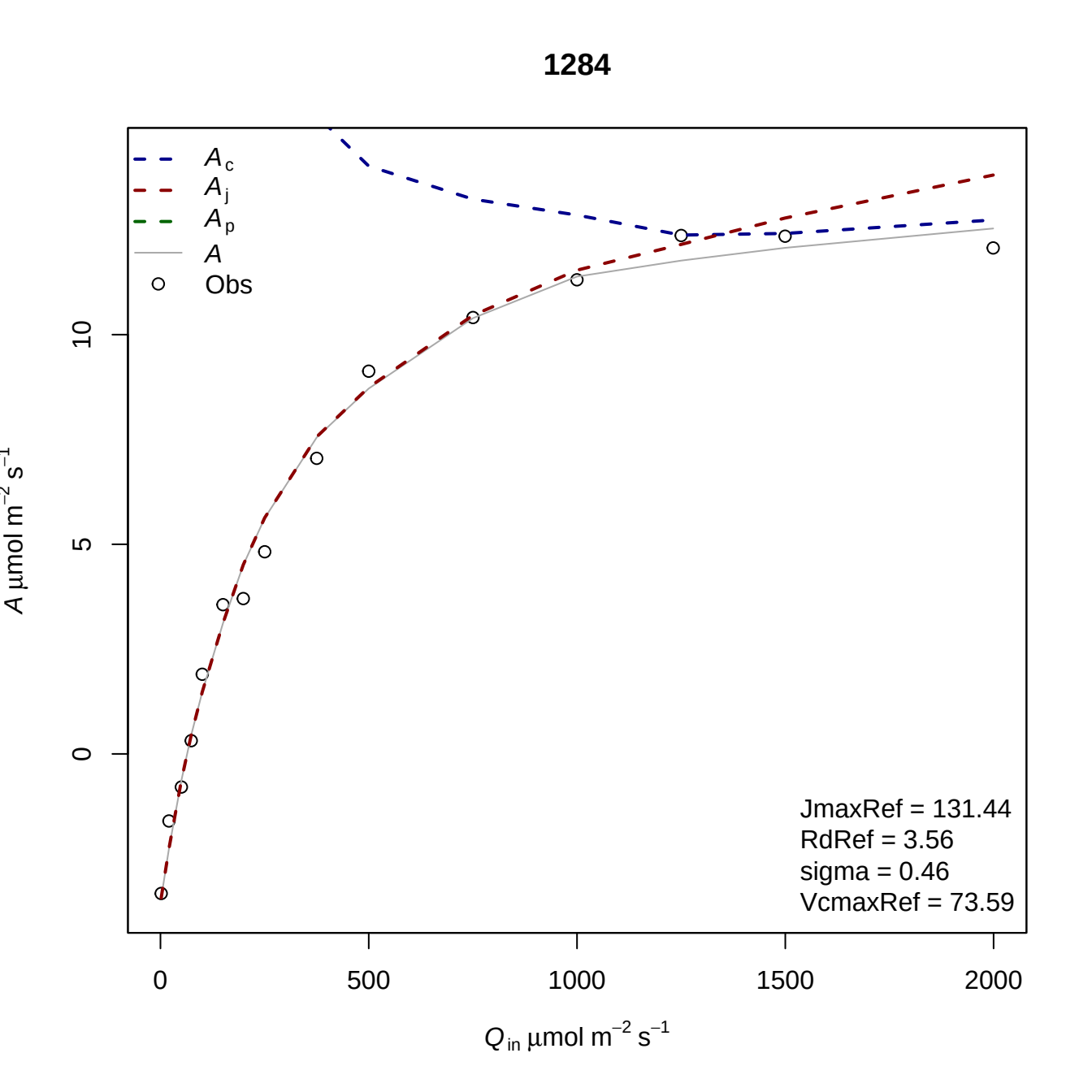

1279
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 151.76
RdRef = 3.03
sigma = 0.93
VcmaxRef = 89.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

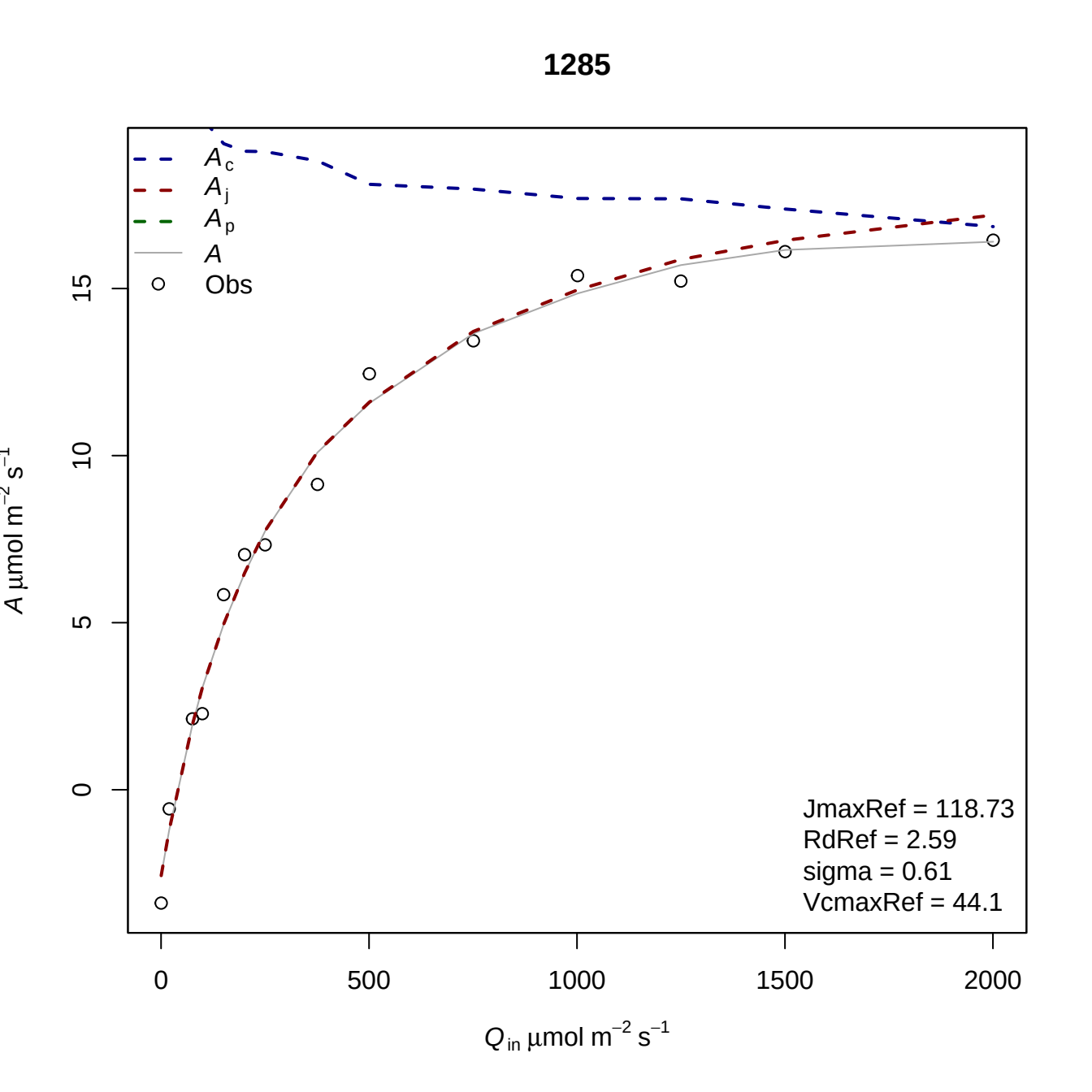

1280
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.73
RdRef = 1.25
sigma = 0.78
VcmaxRef = 56.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

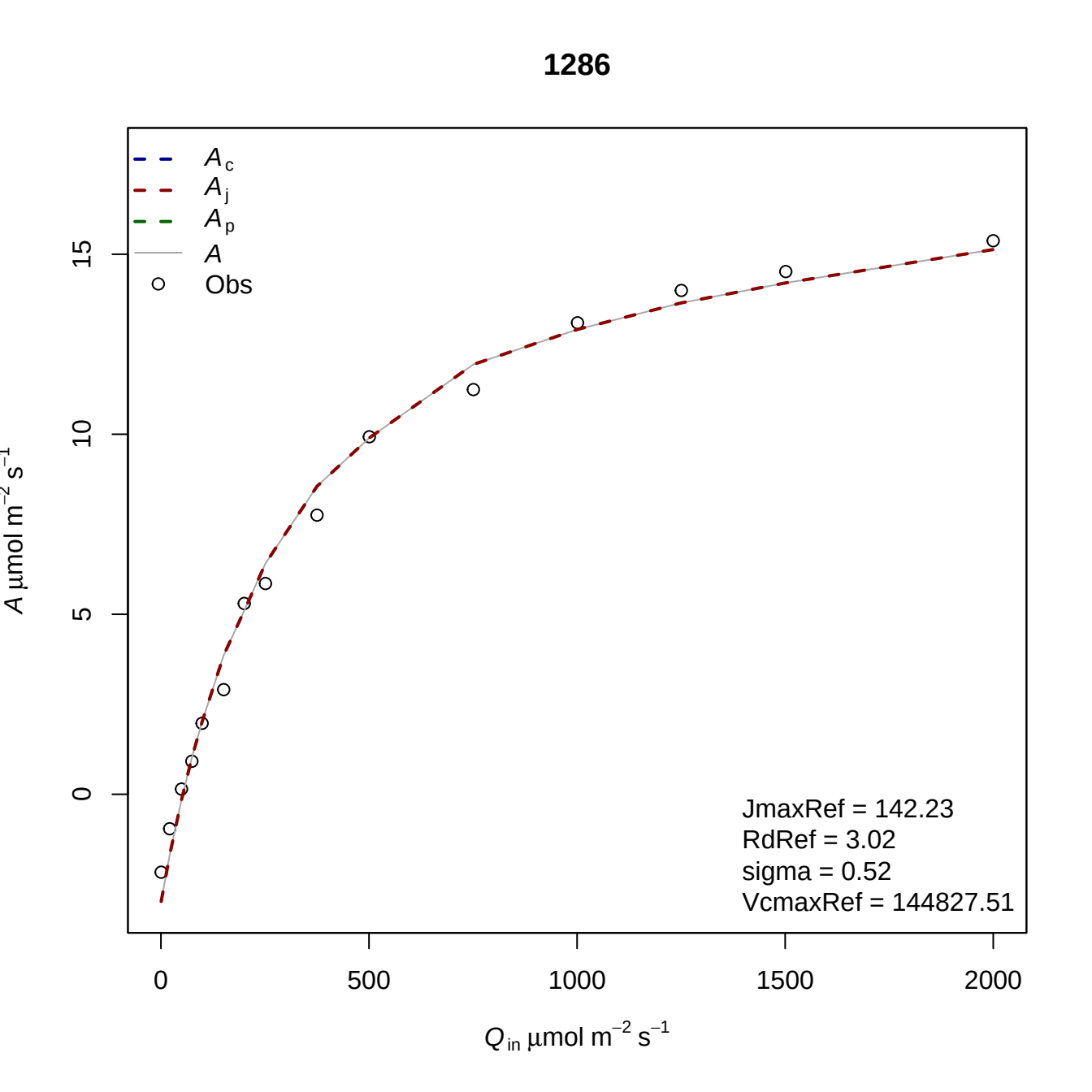

1281
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.71
RdRef = 0.9
sigma = 0.62
VcmaxRef = 58.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

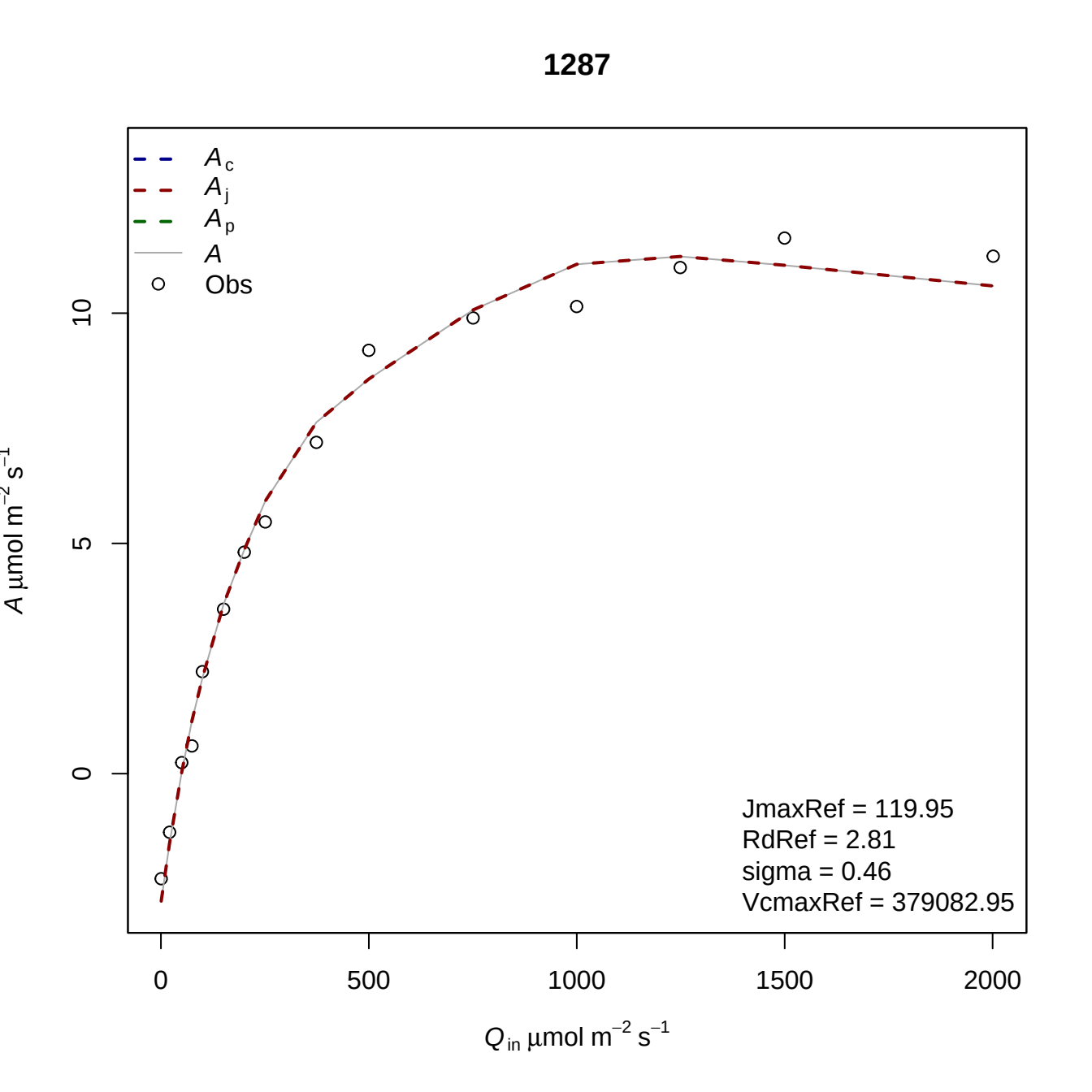

1282
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 157.48
RdRef = 3.08
sigma = 0.41
VcmaxRef = 90.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

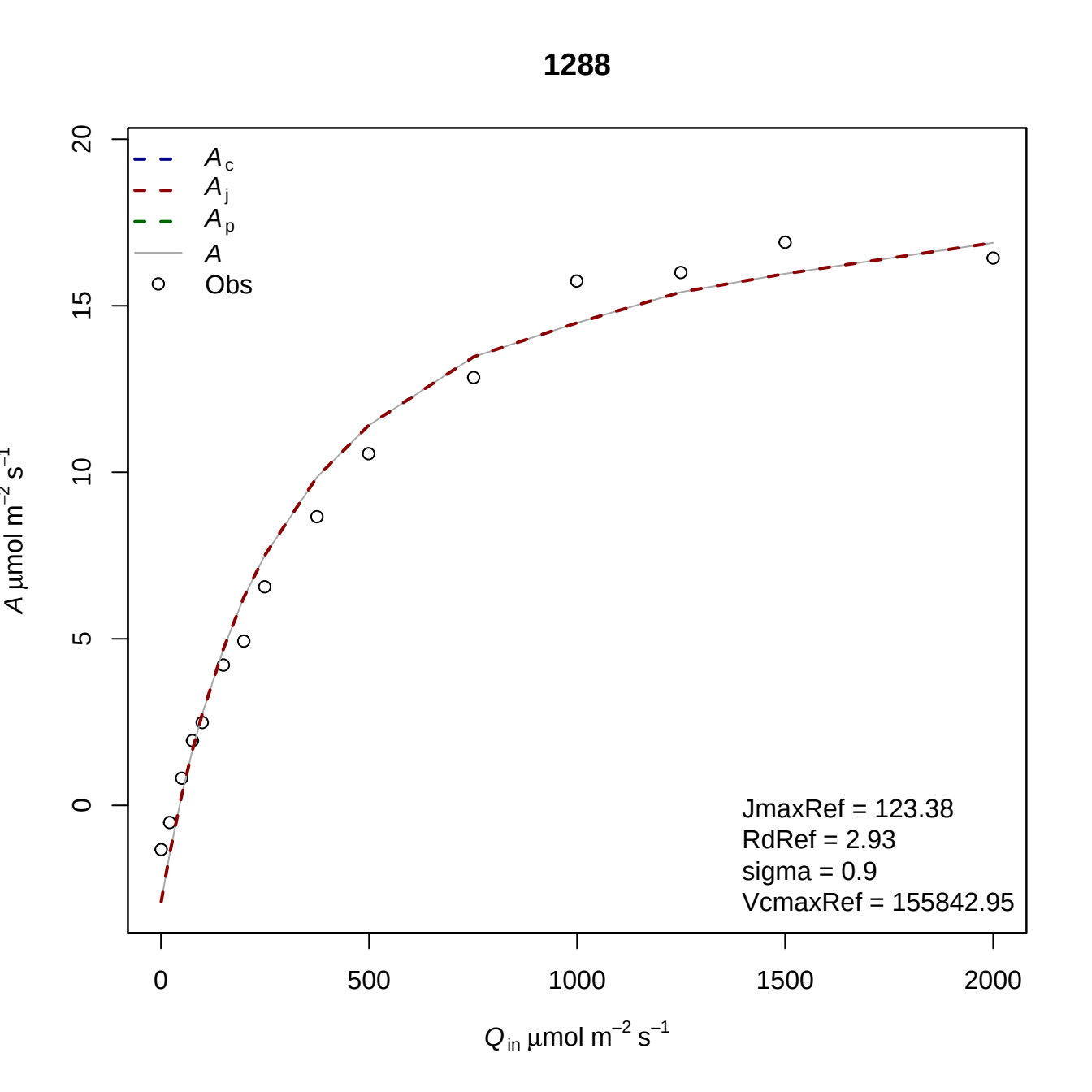

1283
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.6
RdRef = 2.48
sigma = 0.3
VcmaxRef = 91.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

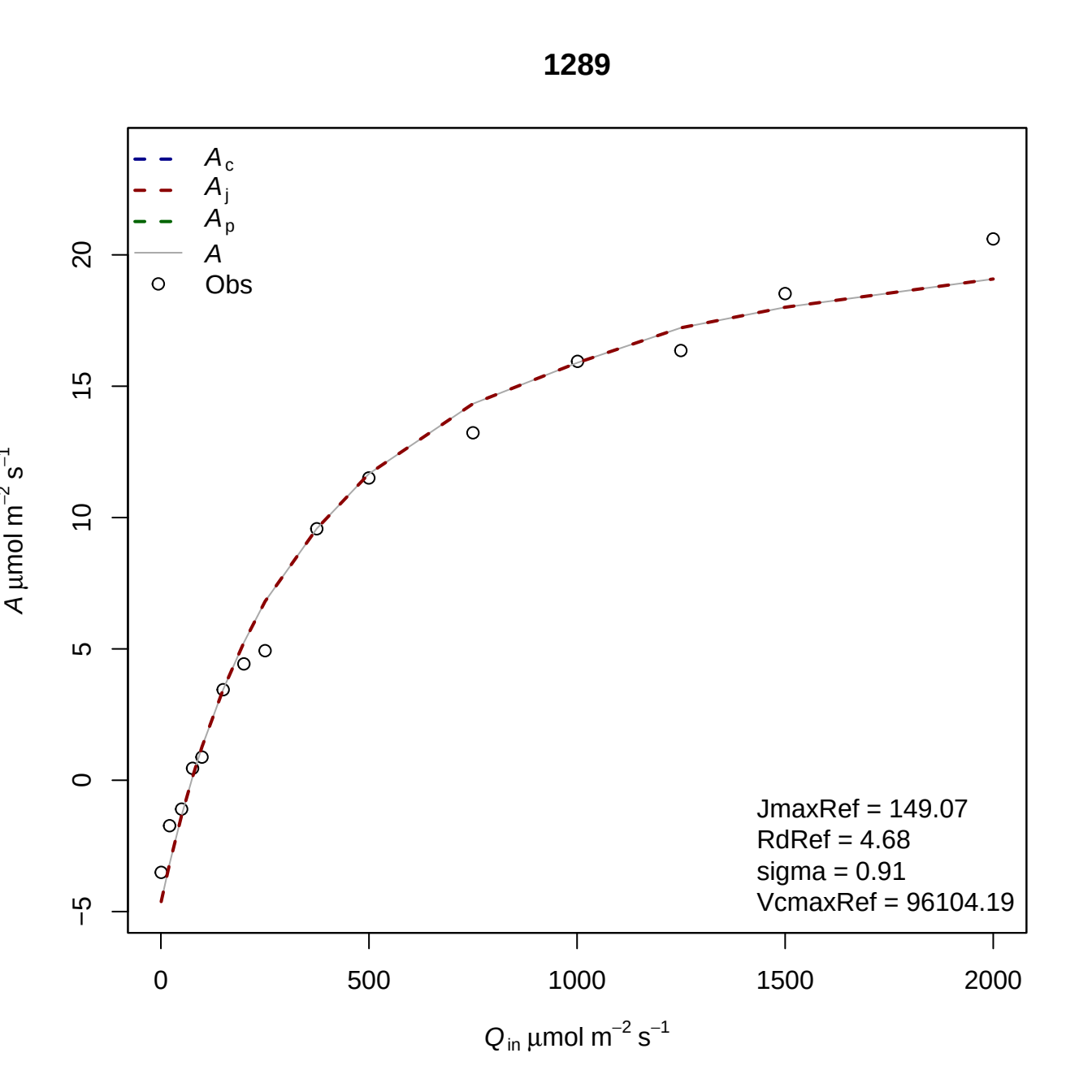

1284
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.44
RdRef = 3.56
sigma = 0.46
VcmaxRef = 73.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

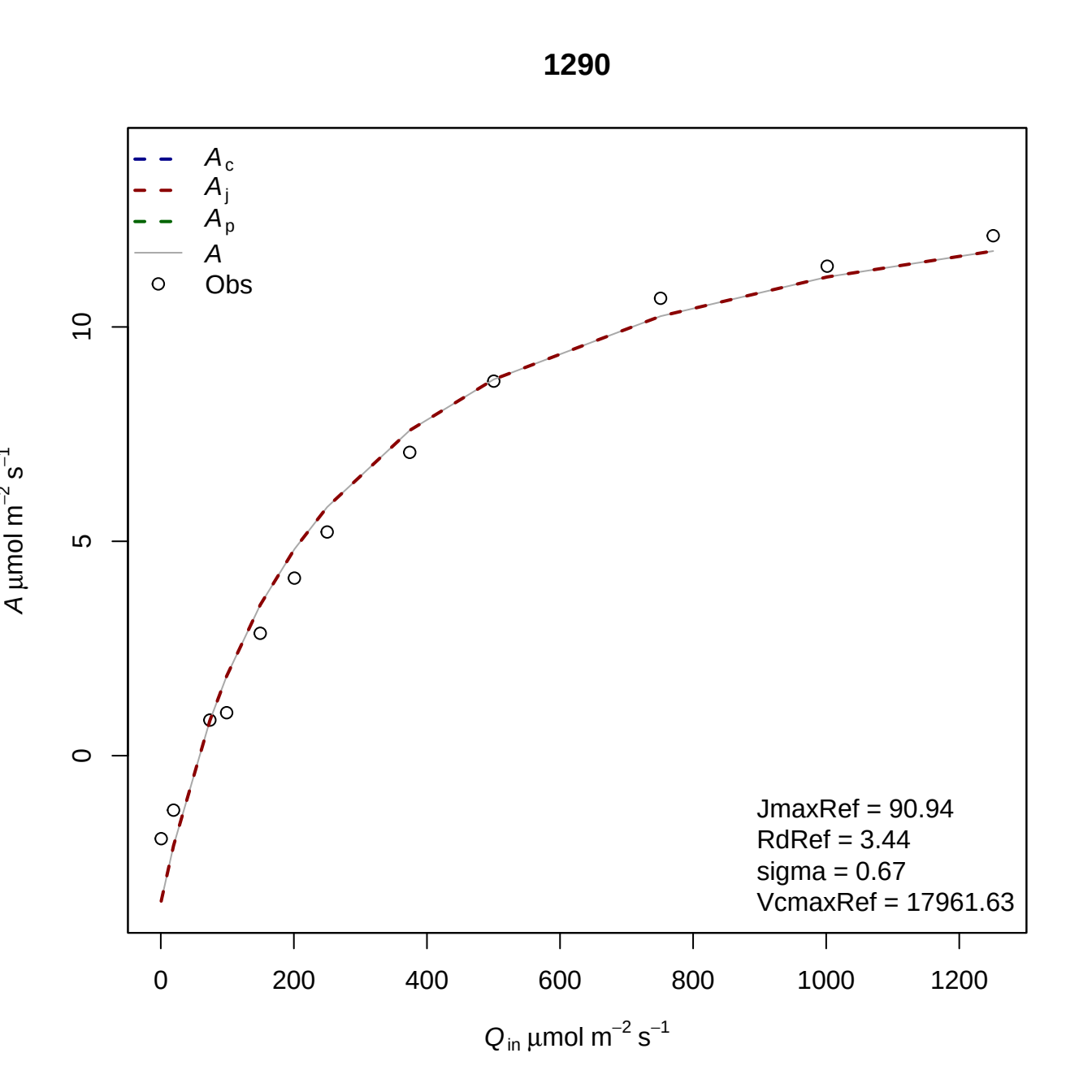

1285
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.73
RdRef = 2.59
sigma = 0.61
VcmaxRef = 44.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

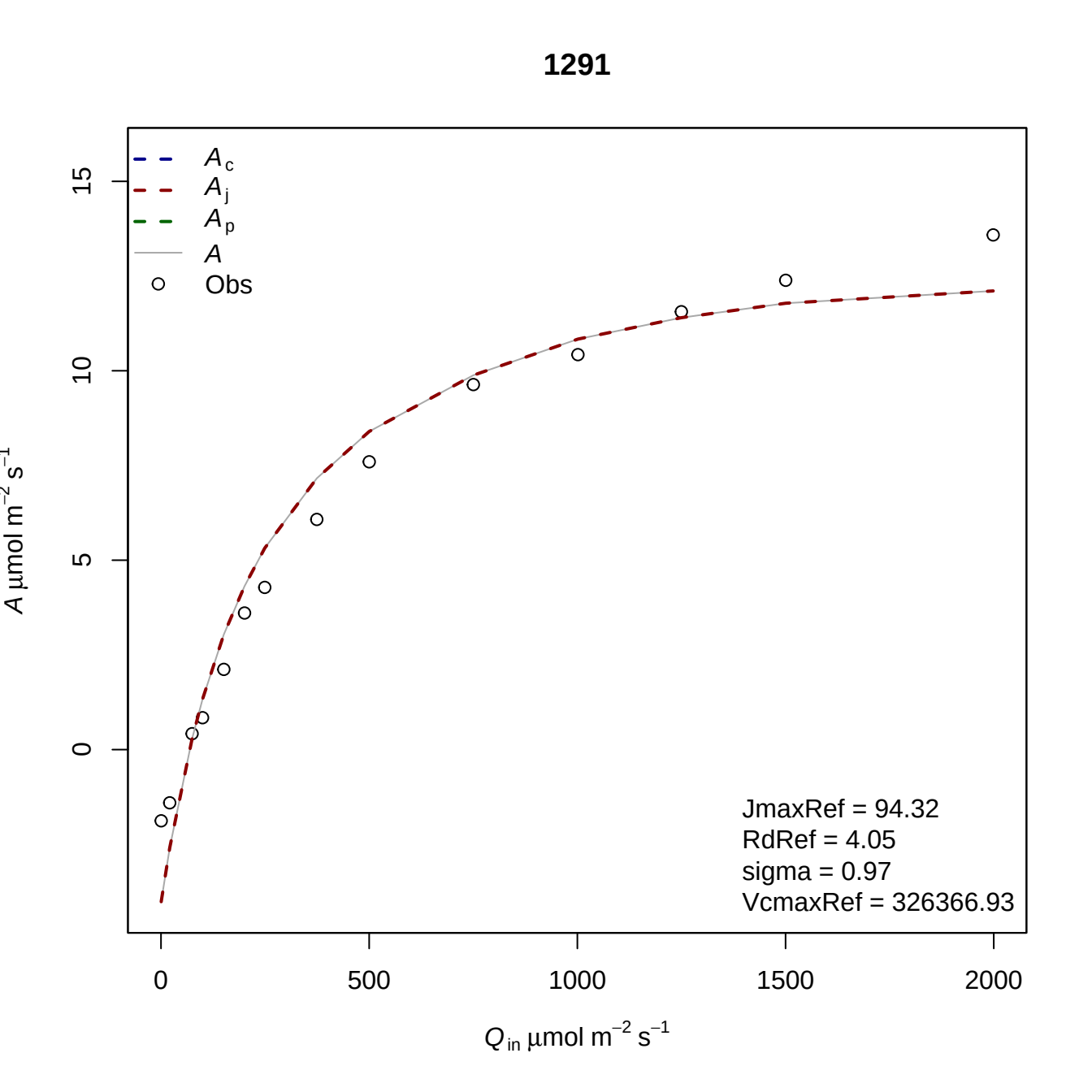

1286
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.23
RdRef = 3.02
sigma = 0.52
VcmaxRef = 144827.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

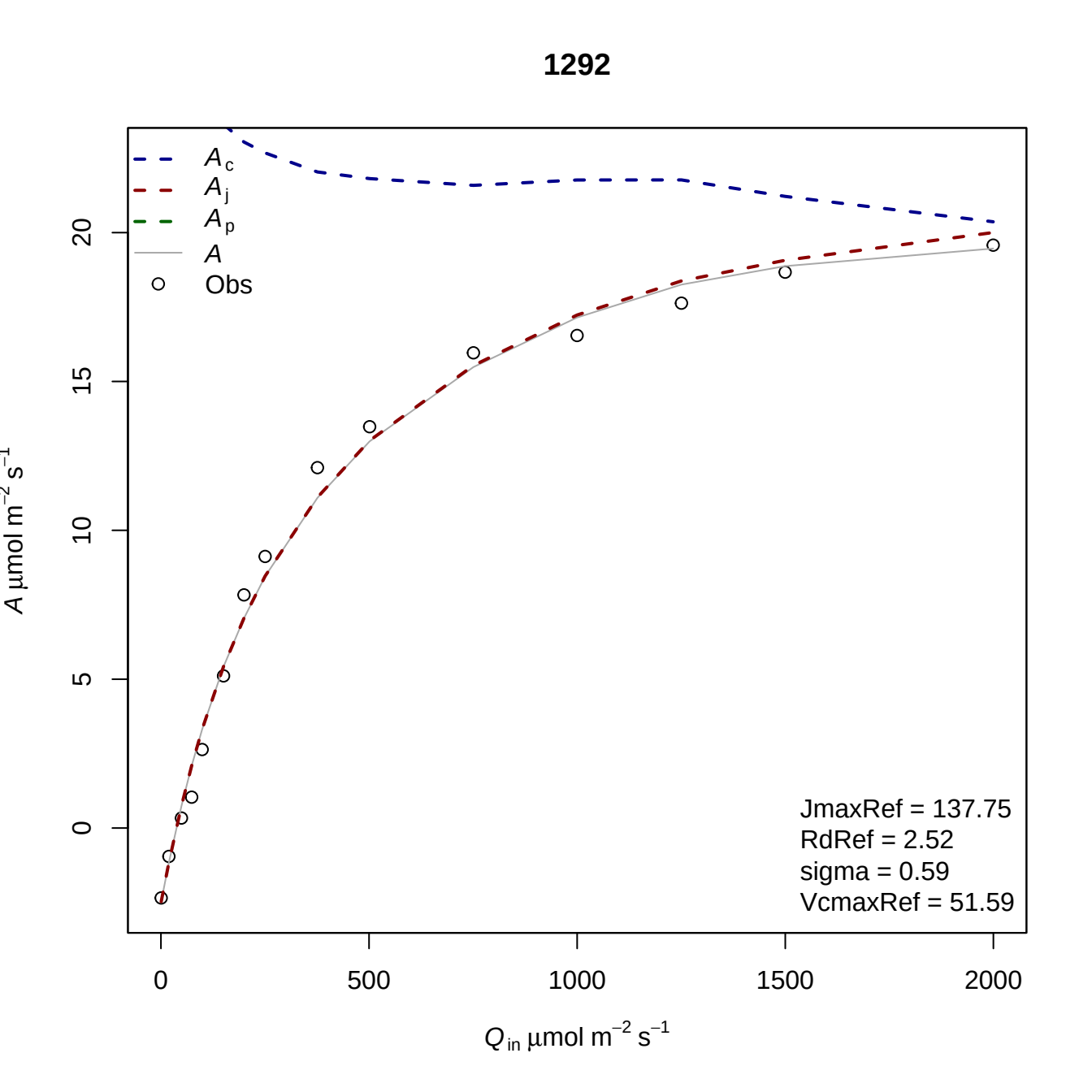

1287
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.95
RdRef = 2.81
sigma = 0.46
VcmaxRef = 379082.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

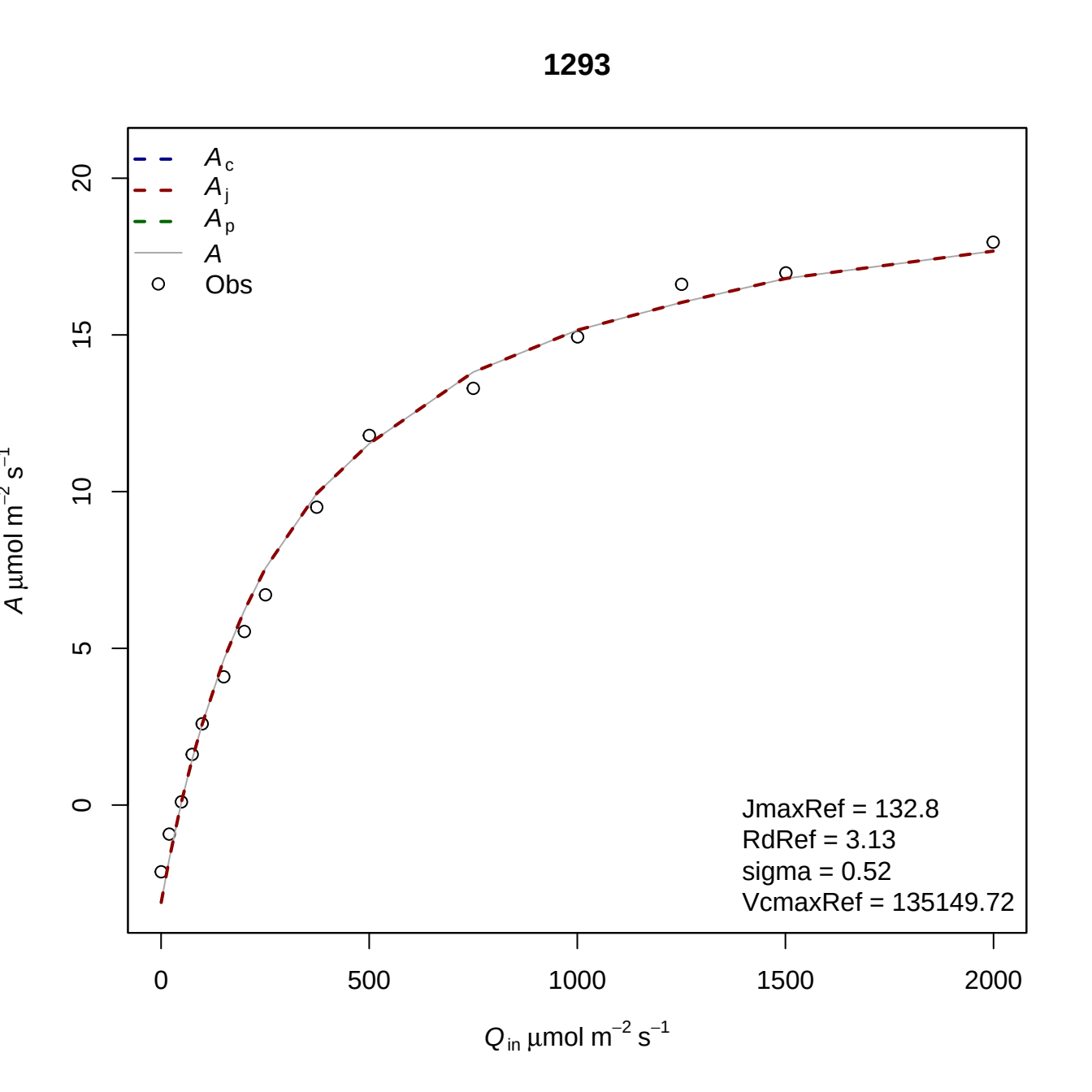

1288
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.38
RdRef = 2.93
sigma = 0.9
VcmaxRef = 155842.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

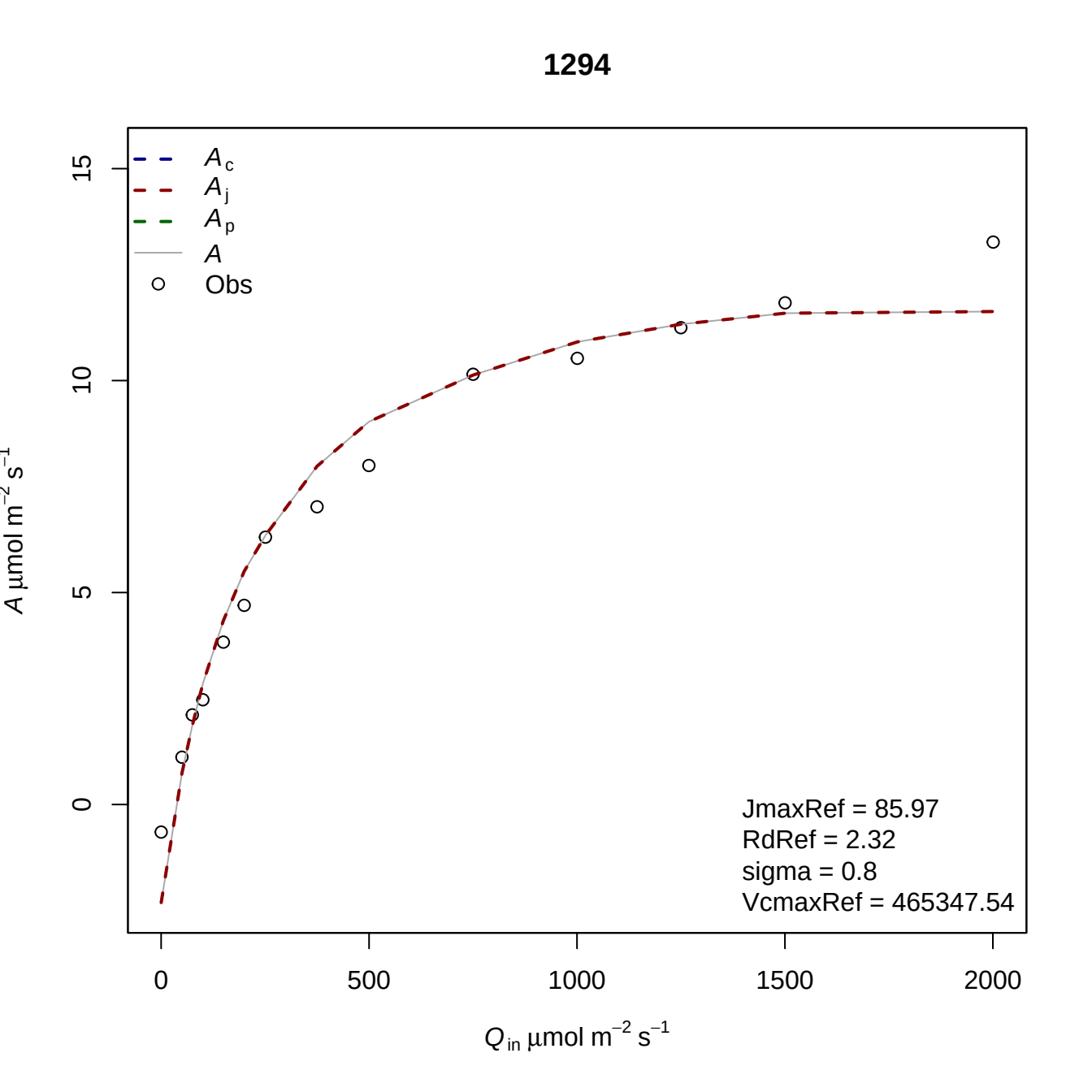

1289
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 149.07
RdRef = 4.68
sigma = 0.91
VcmaxRef = 96104.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

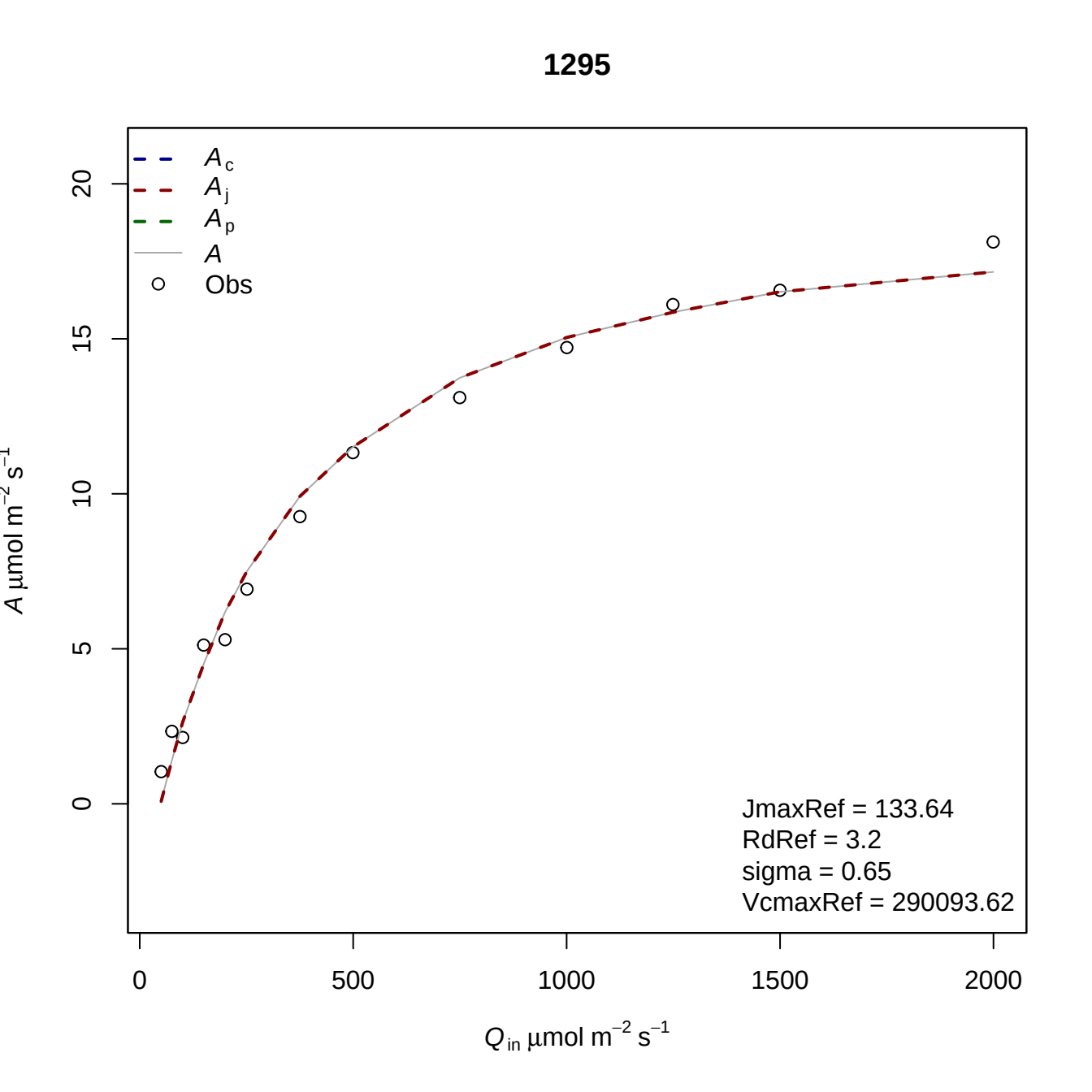

1290
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.94
RdRef = 3.44
sigma = 0.67
VcmaxRef = 17961.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000
1200

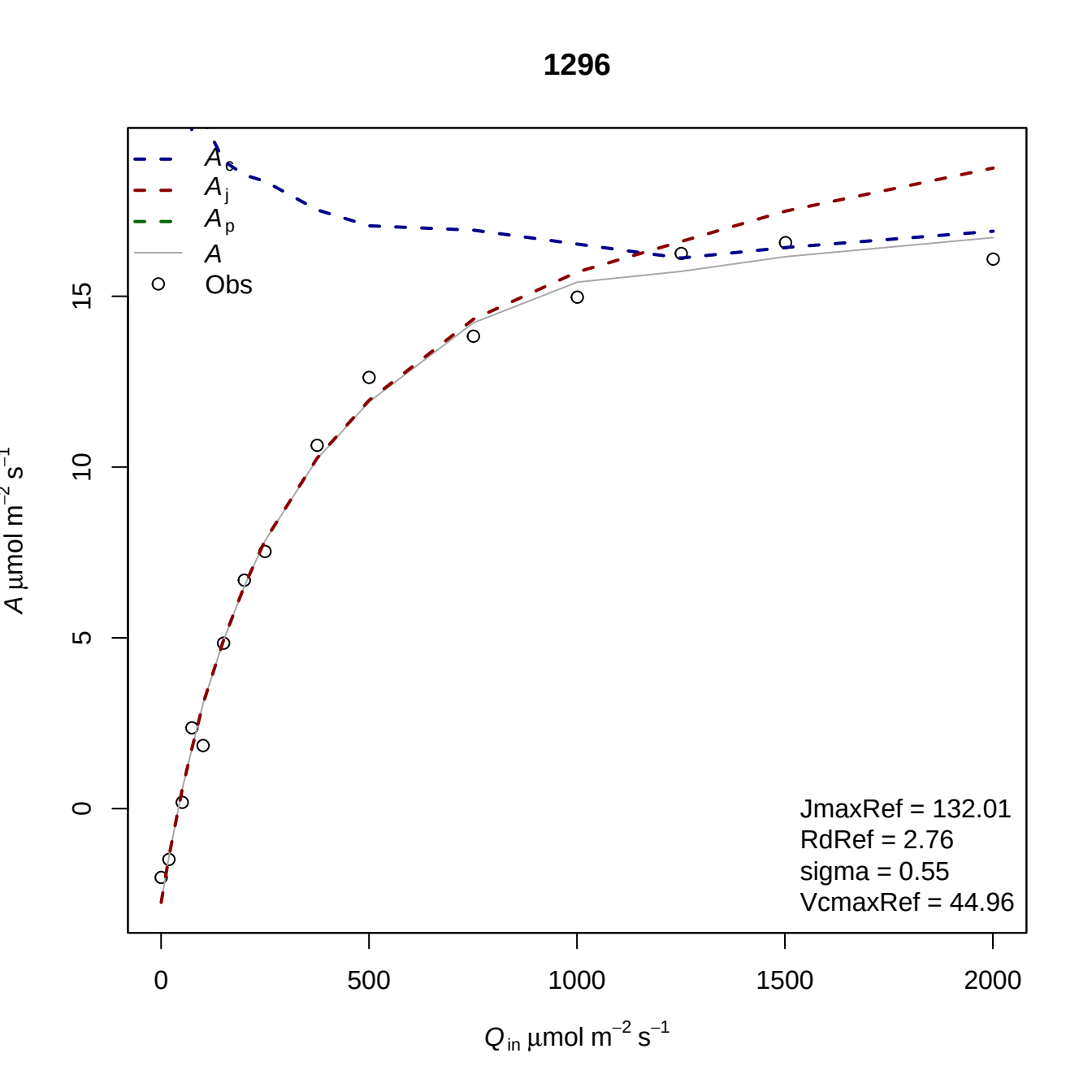

1291
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.32
RdRef = 4.05
sigma = 0.97
VcmaxRef = 326366.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

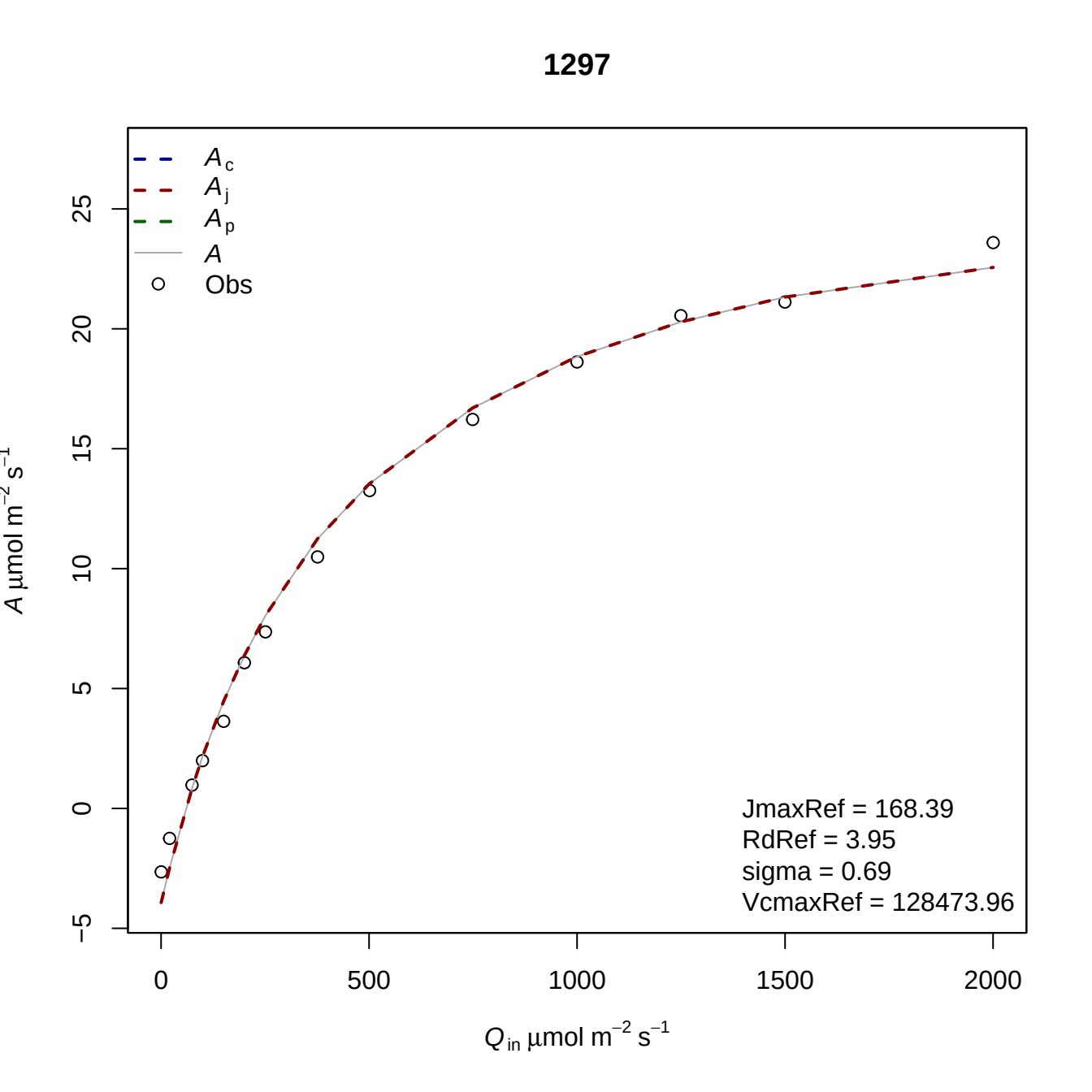

1292
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.75
RdRef = 2.52
sigma = 0.59
VcmaxRef = 51.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

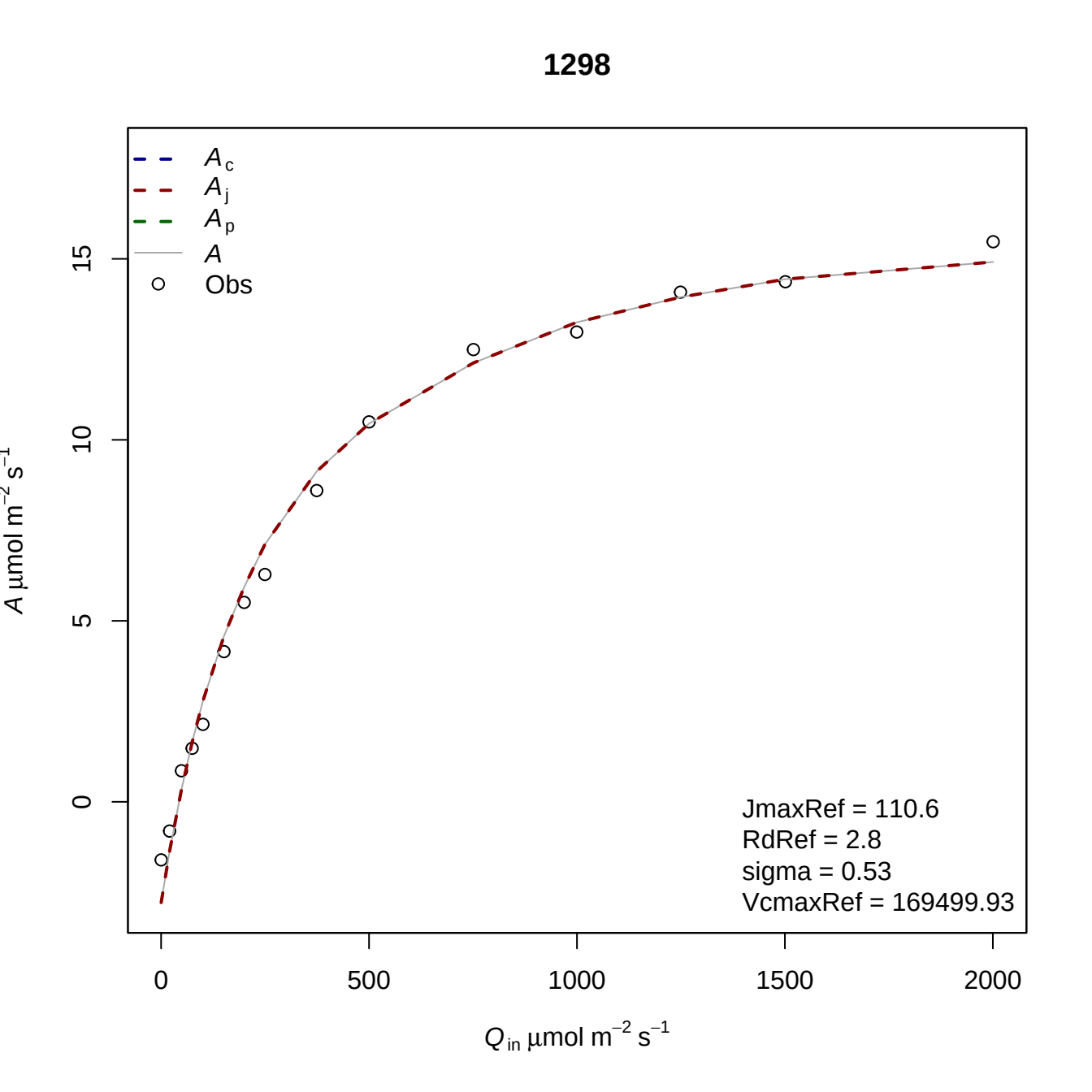

1293
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.8
RdRef = 3.13
sigma = 0.52
VcmaxRef = 135149.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

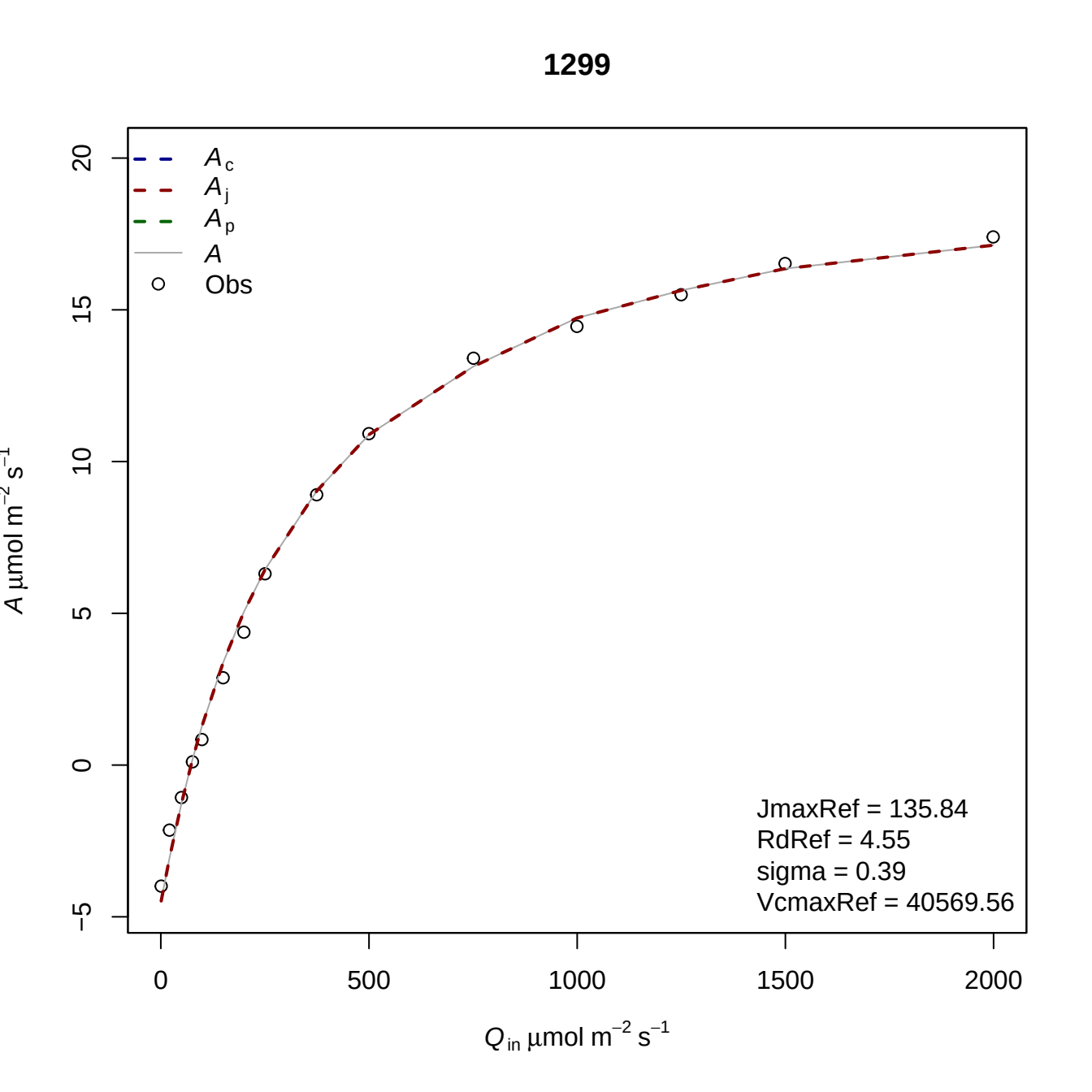

1294
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.97
RdRef = 2.32
sigma = 0.8
VcmaxRef = 465347.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

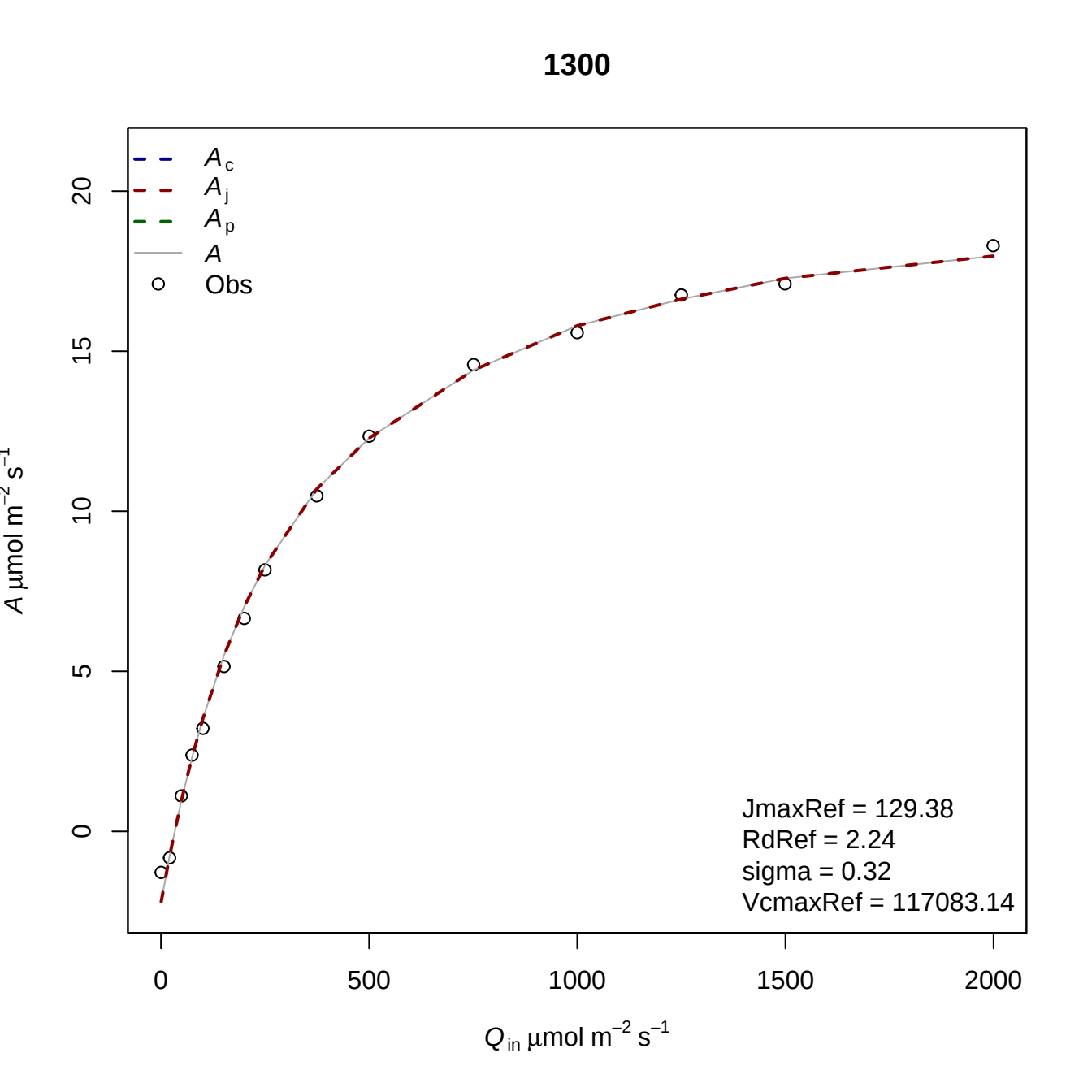

1295
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.64
RdRef = 3.2
sigma = 0.65
VcmaxRef = 290093.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

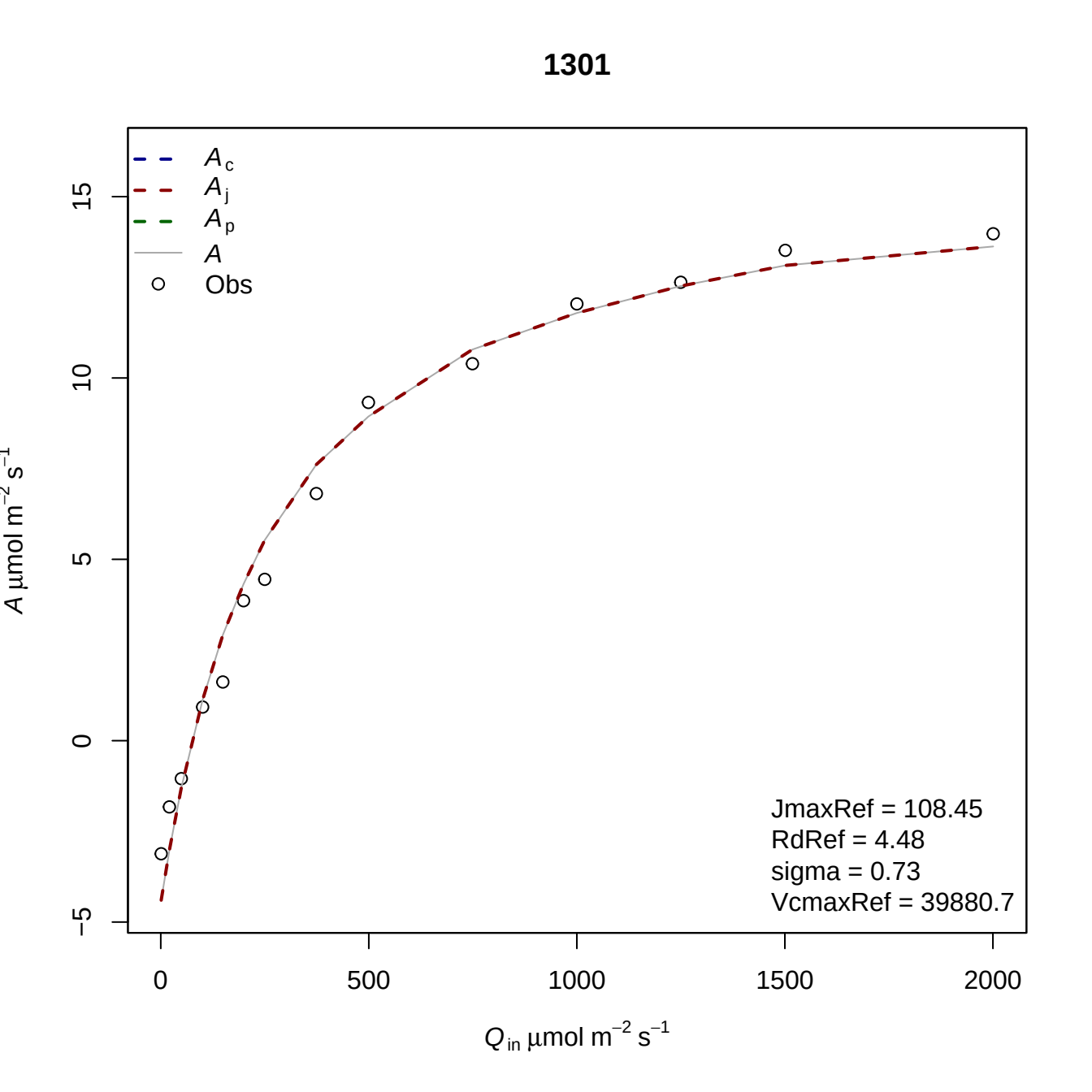

1296
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 132.01
RdRef = 2.76
sigma = 0.55
VcmaxRef = 44.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

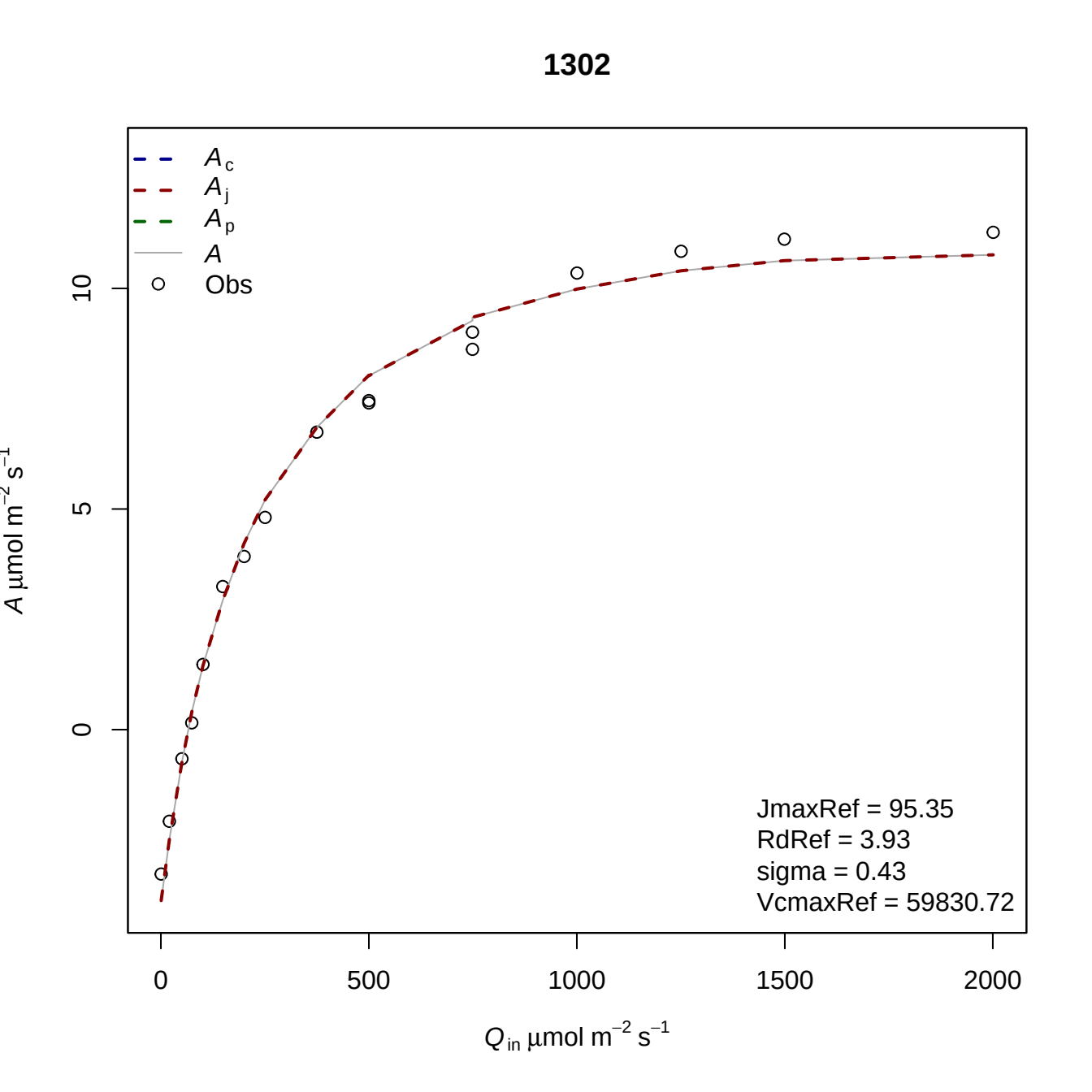

1297
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 168.39
RdRef = 3.95
sigma = 0.69
VcmaxRef = 128473.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
25
0
500
1000
1500
2000

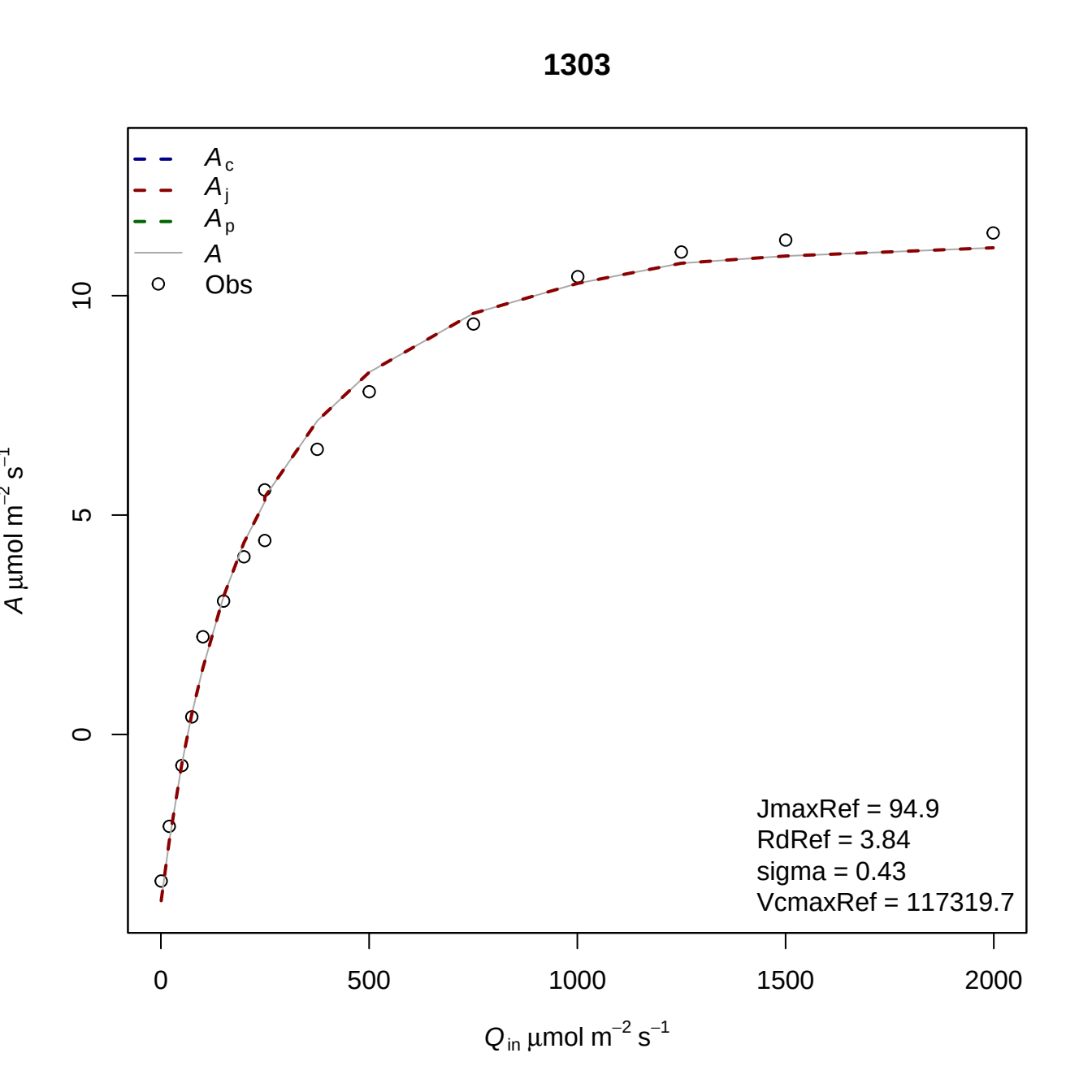

1298
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.6
RdRef = 2.8
sigma = 0.53
VcmaxRef = 169499.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

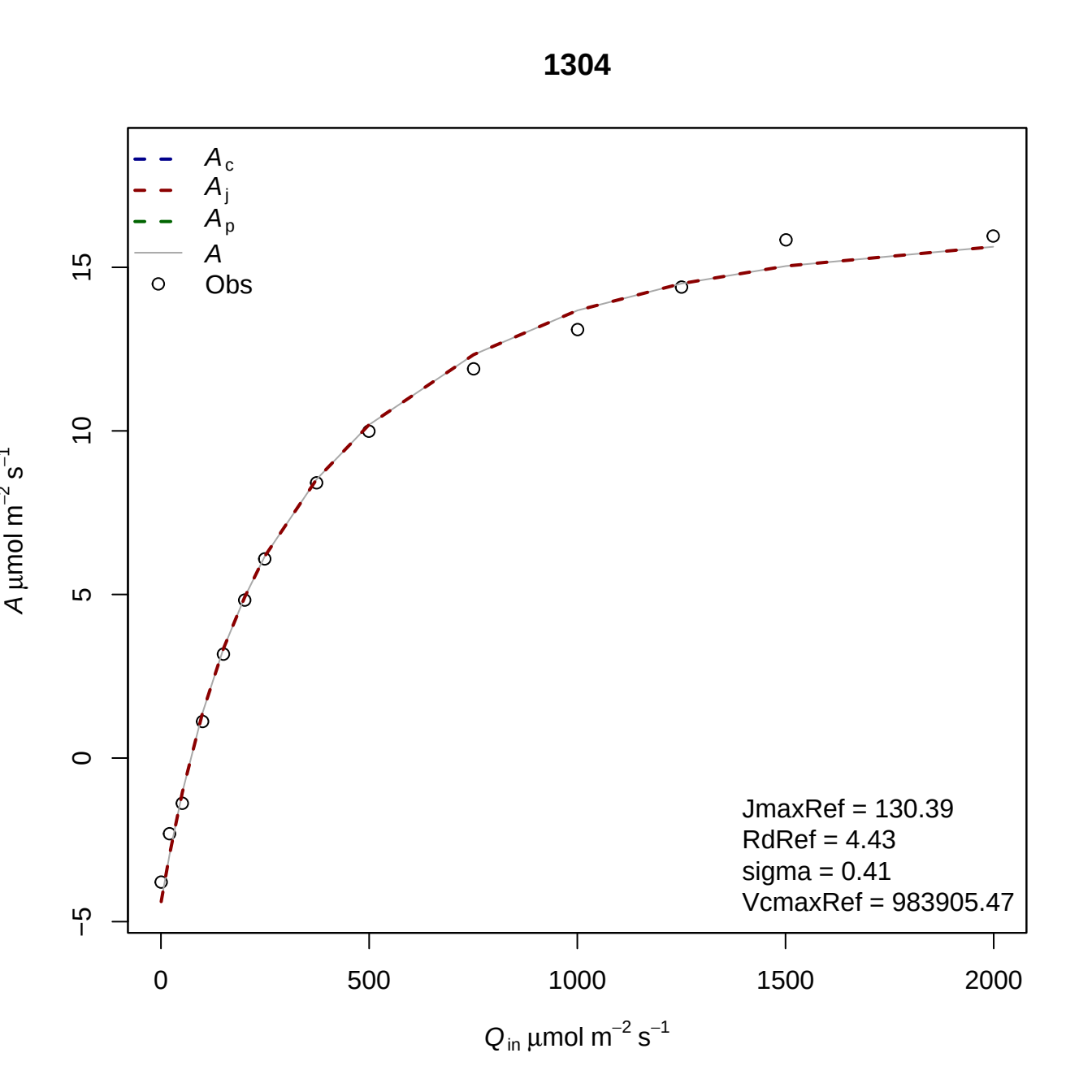

1299
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 135.84
RdRef = 4.55
sigma = 0.39
VcmaxRef = 40569.56
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
20
15
10
5
0
−5
0
500
1000
1500
2000

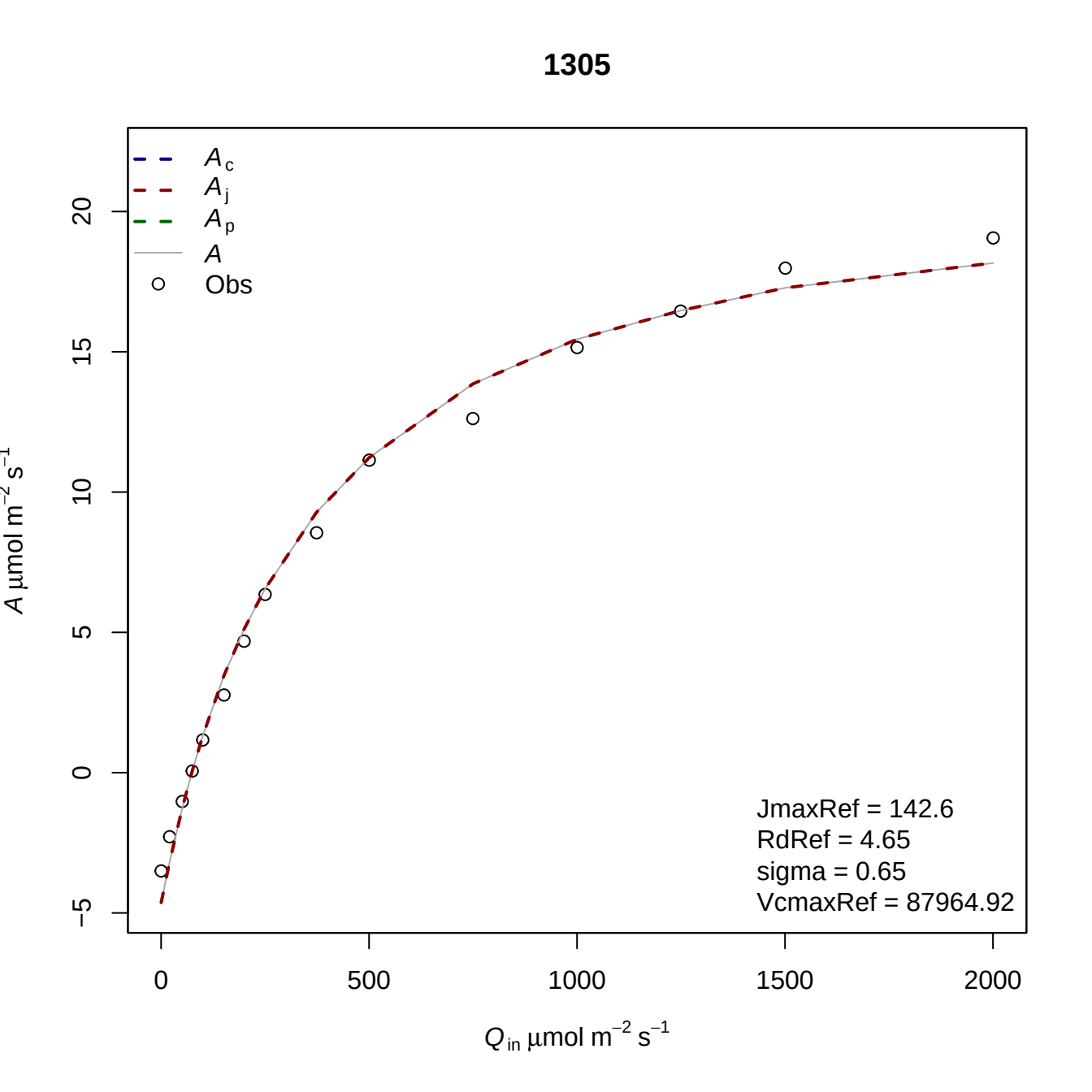

1300
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.38
RdRef = 2.24
sigma = 0.32
VcmaxRef = 117083.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

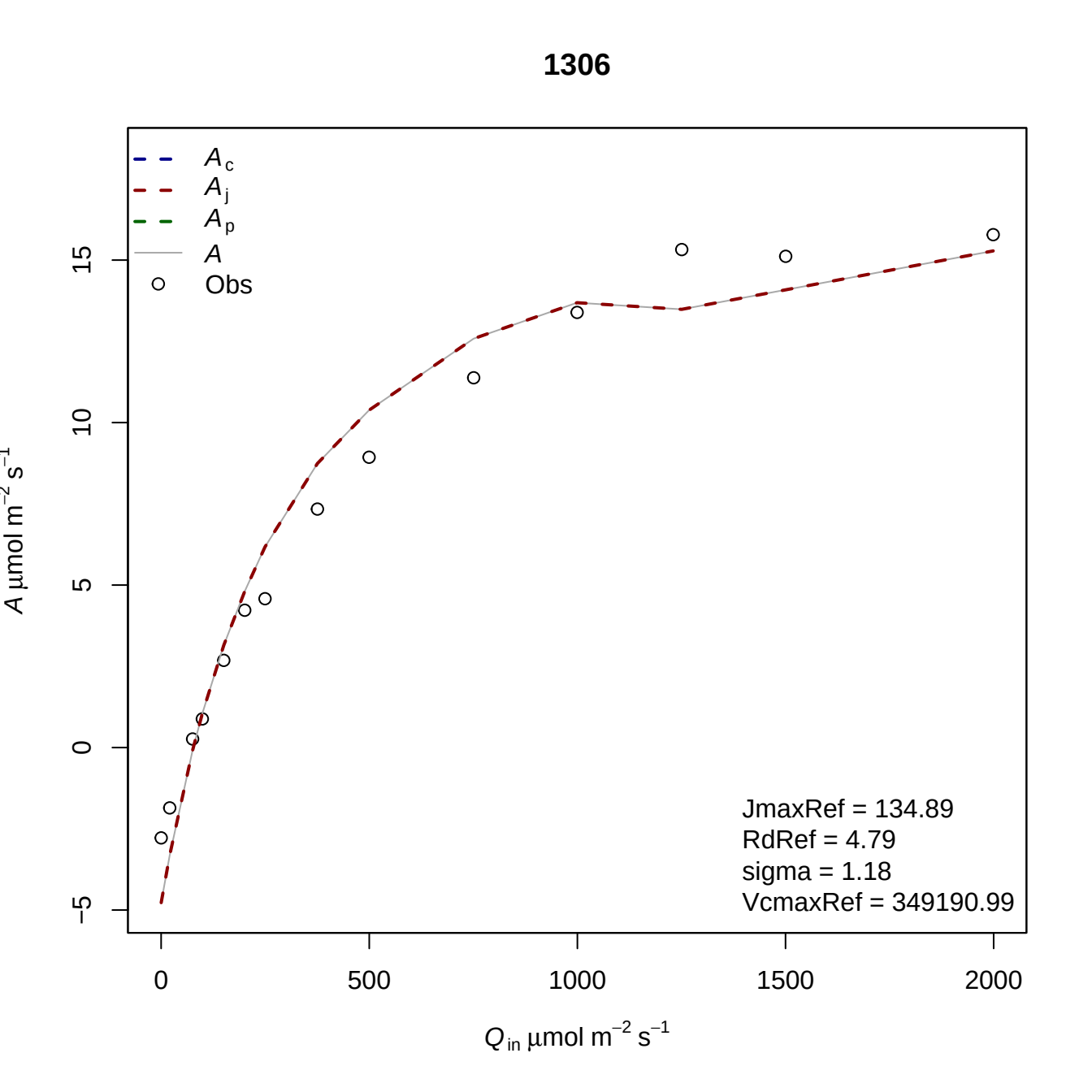

1301
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.45
RdRef = 4.48
sigma = 0.73
VcmaxRef = 39880.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

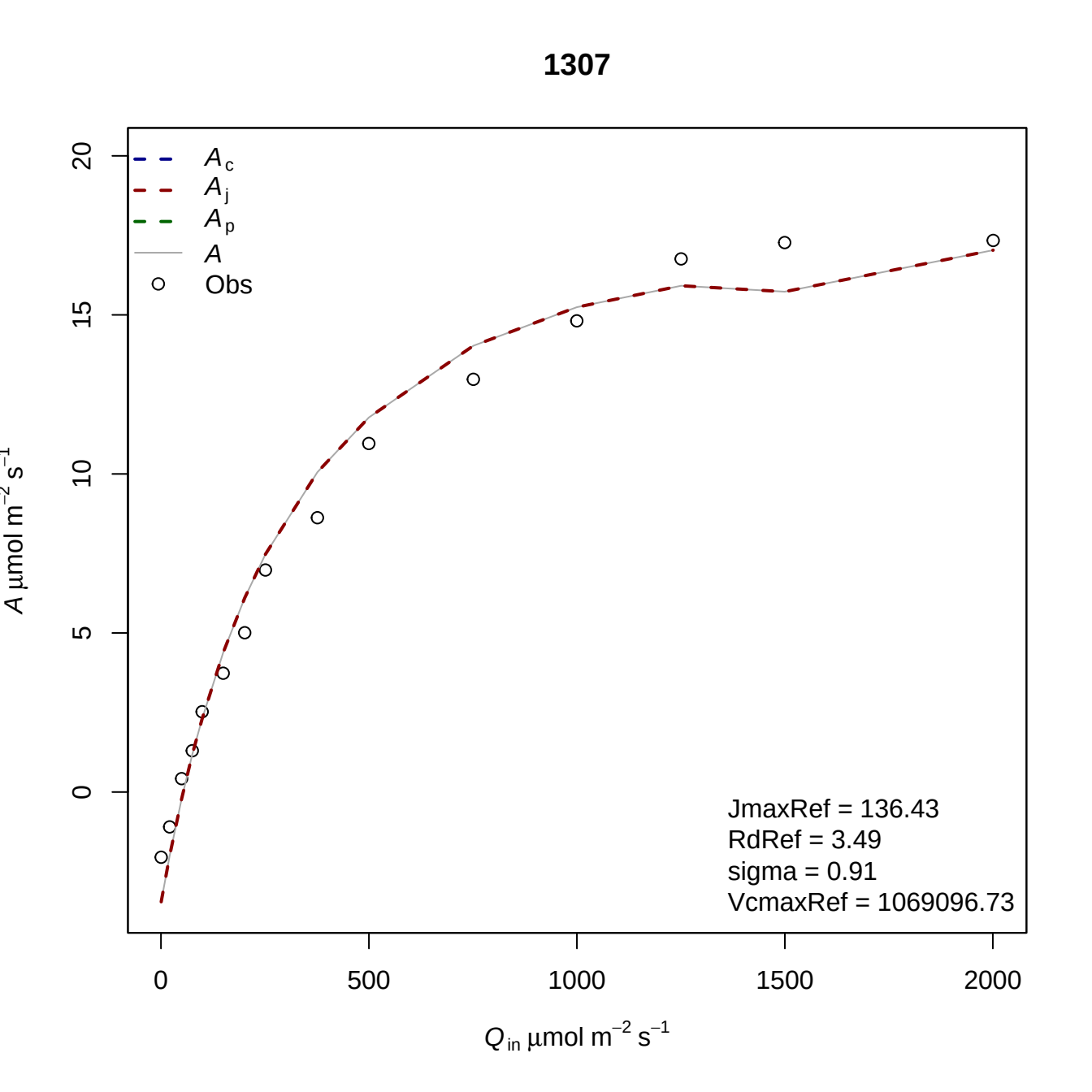

1302
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.35
RdRef = 3.93
sigma = 0.43
VcmaxRef = 59830.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

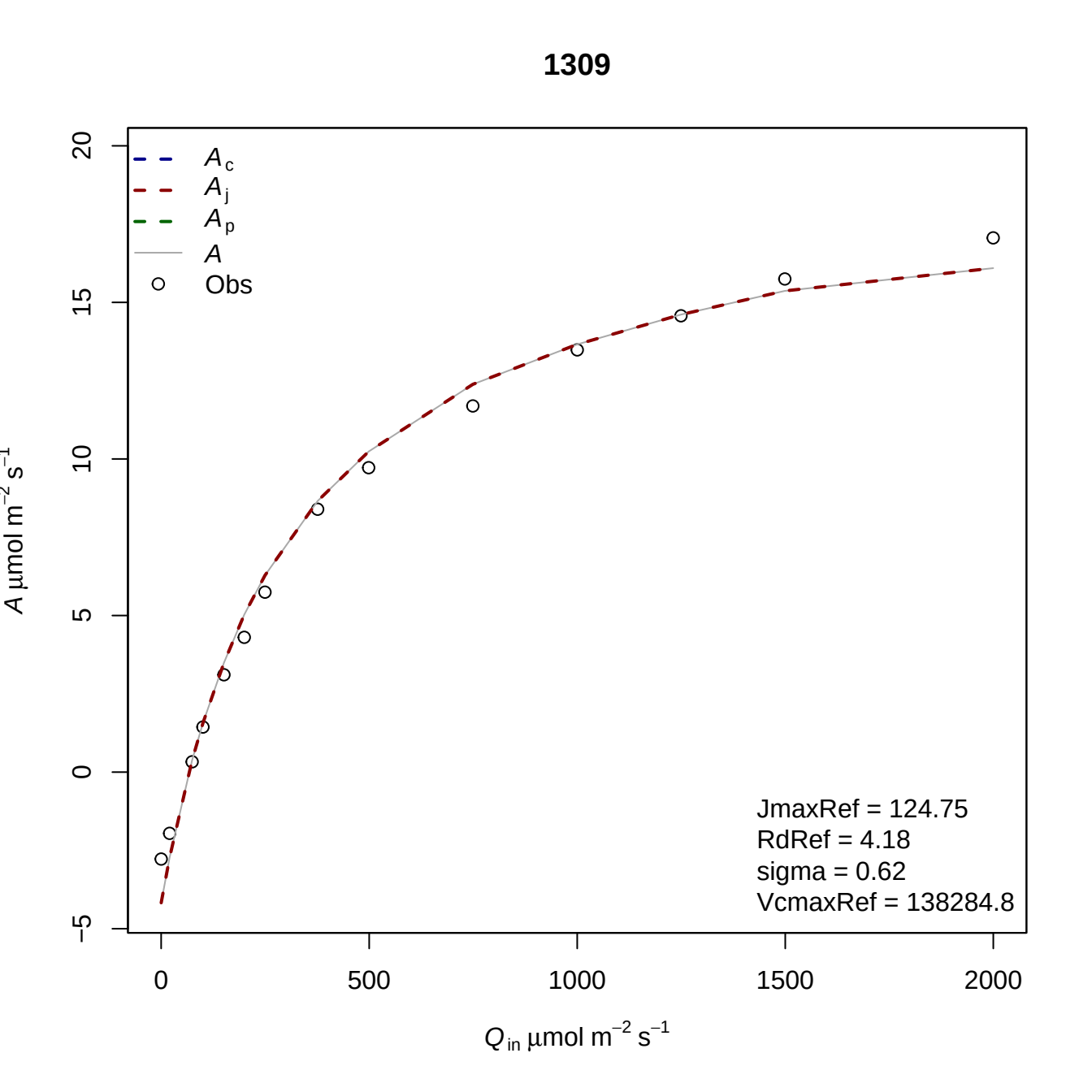

1303
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.9
RdRef = 3.84
sigma = 0.43
VcmaxRef = 117319.7
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

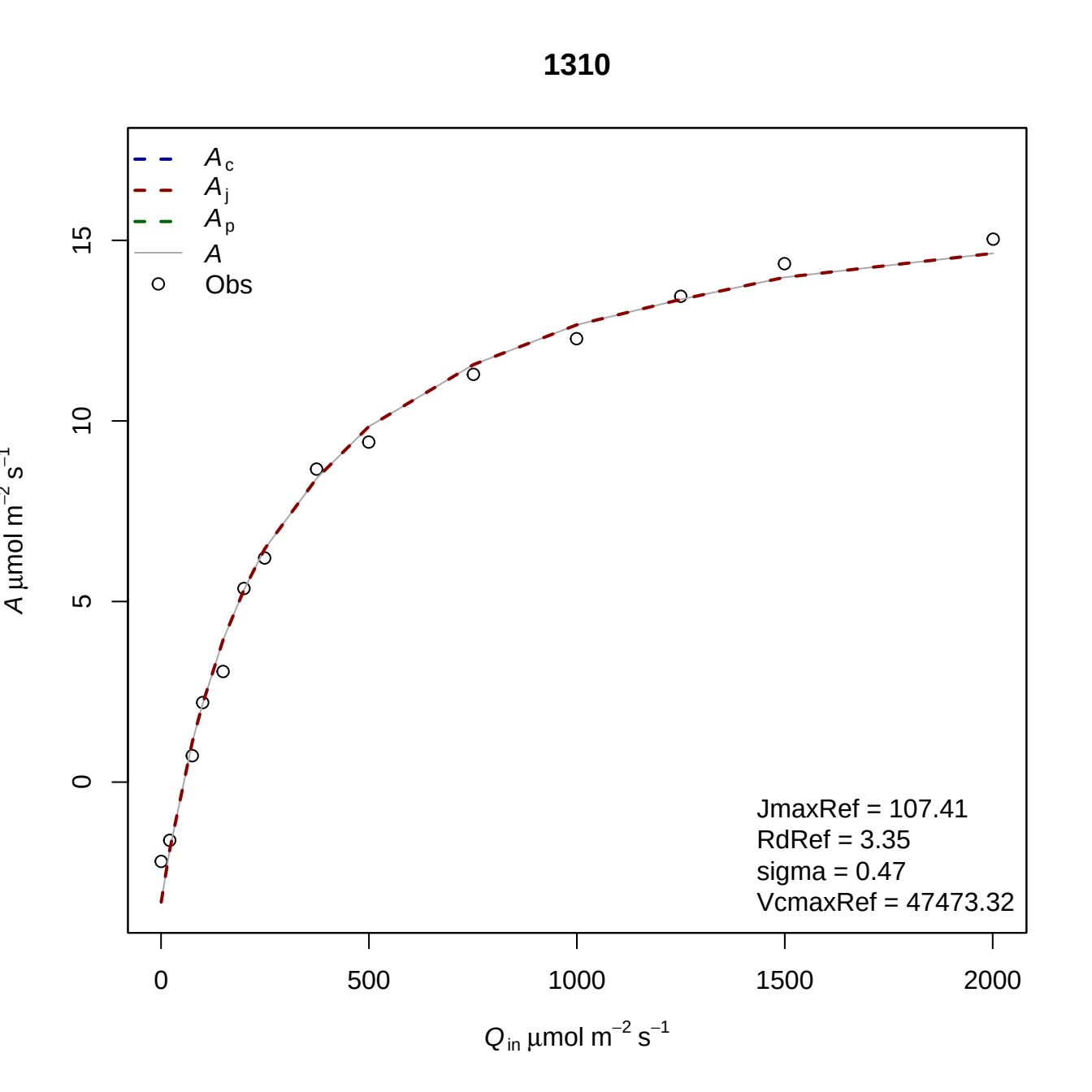

1304
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.39
RdRef = 4.43
sigma = 0.41
VcmaxRef = 983905.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

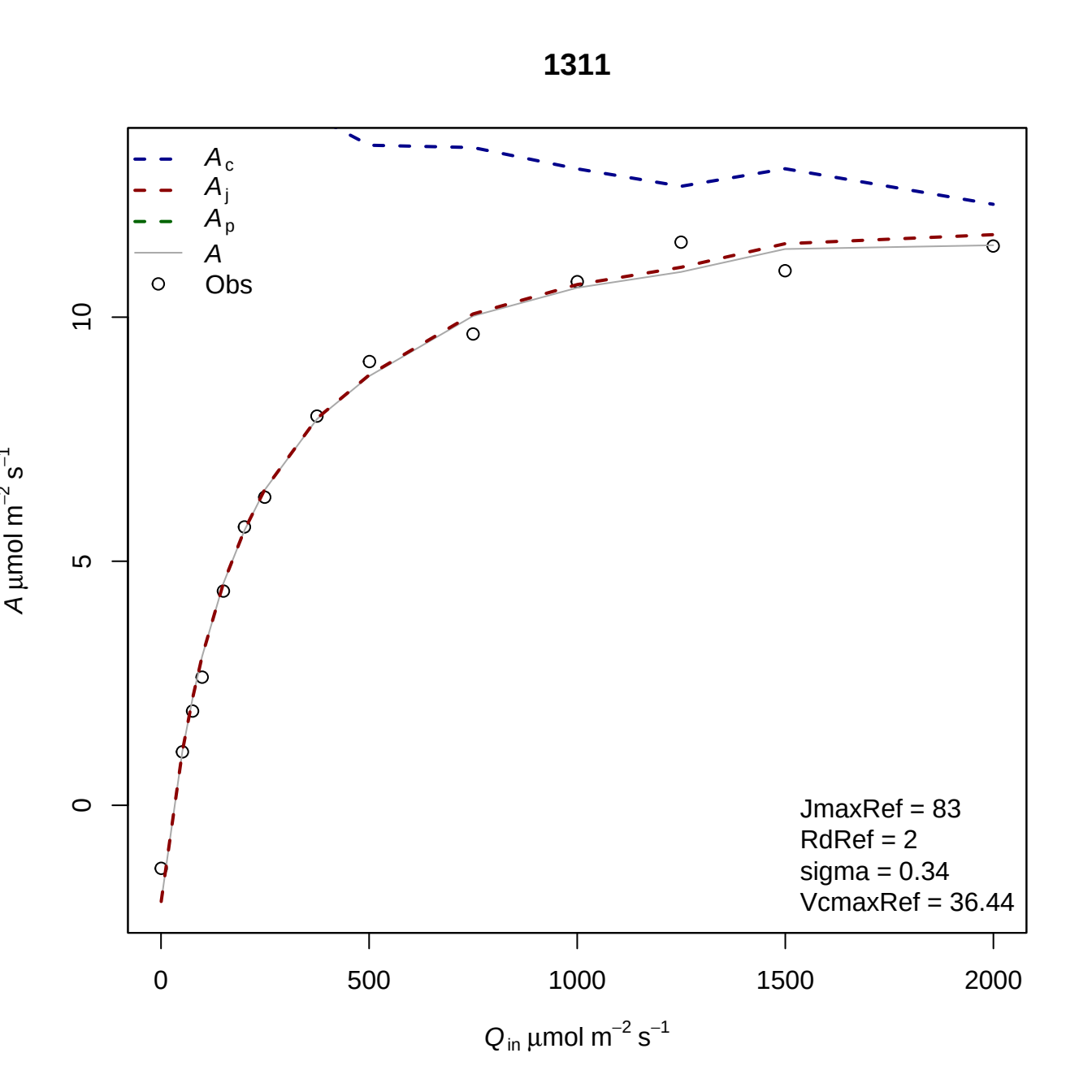

1305
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 142.6
RdRef = 4.65
sigma = 0.65
VcmaxRef = 87964.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
20
0
500
1000
1500
2000

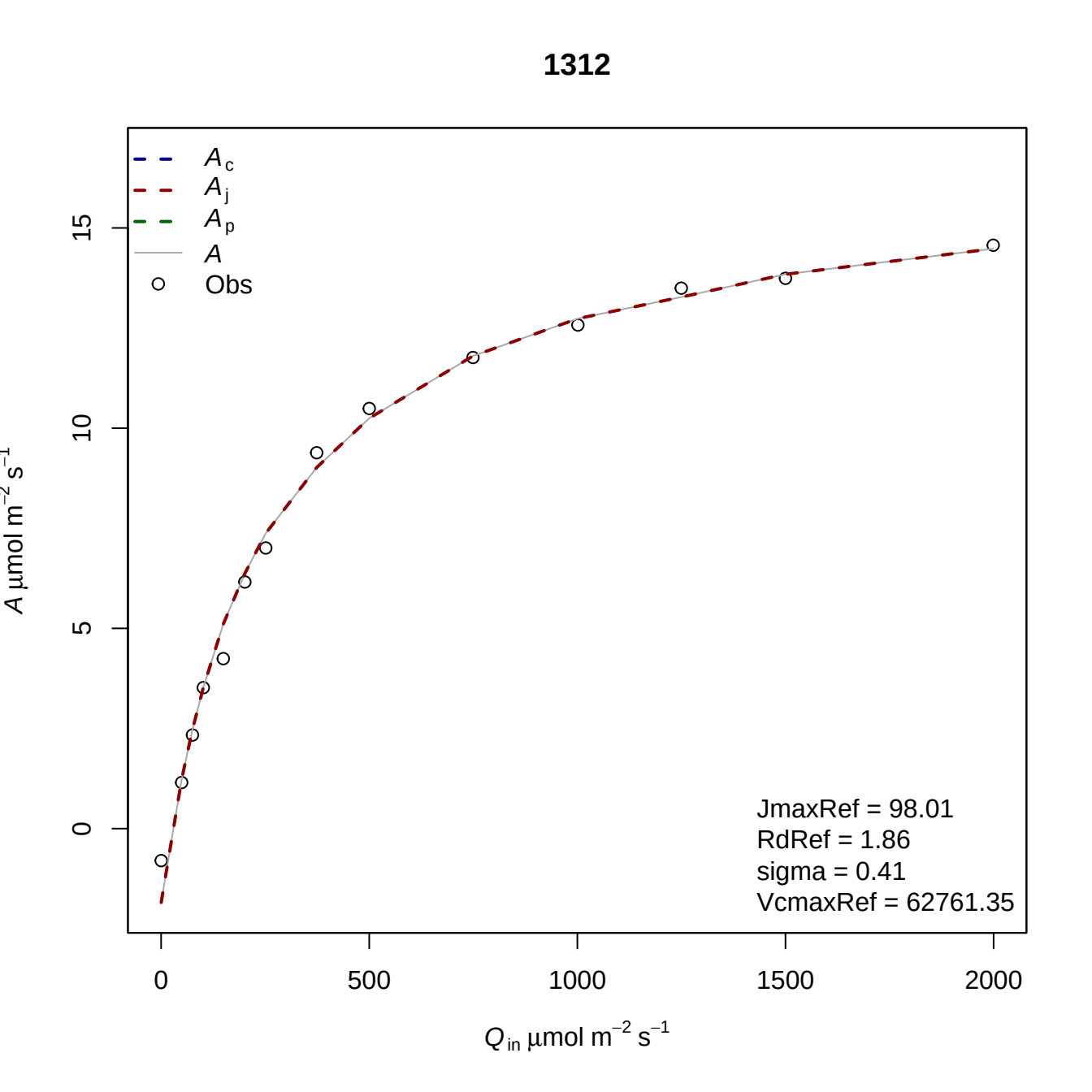

1306
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.89
RdRef = 4.79
sigma = 1.18
VcmaxRef = 349190.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

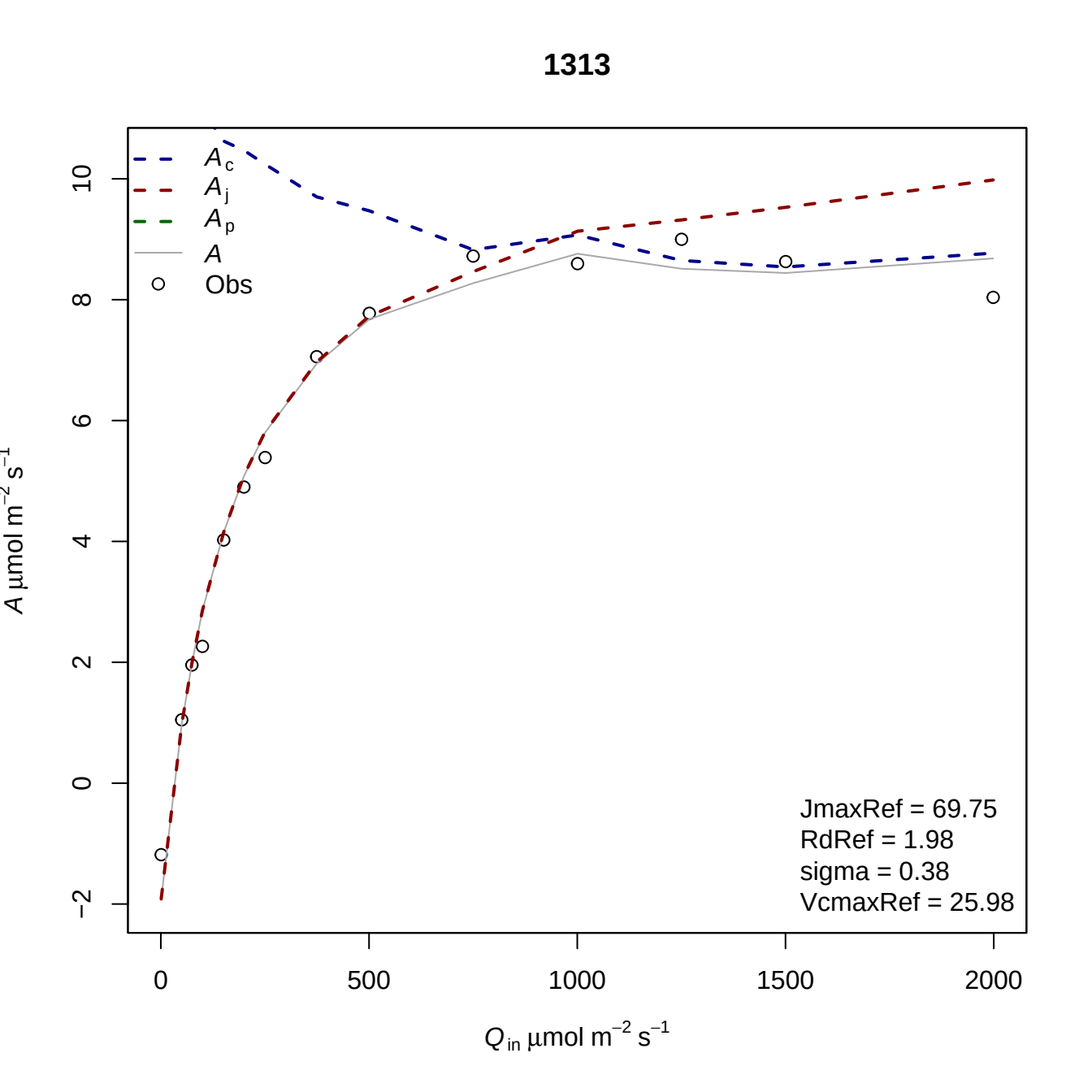

1307
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.43
RdRef = 3.49
sigma = 0.91
VcmaxRef = 1069096.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

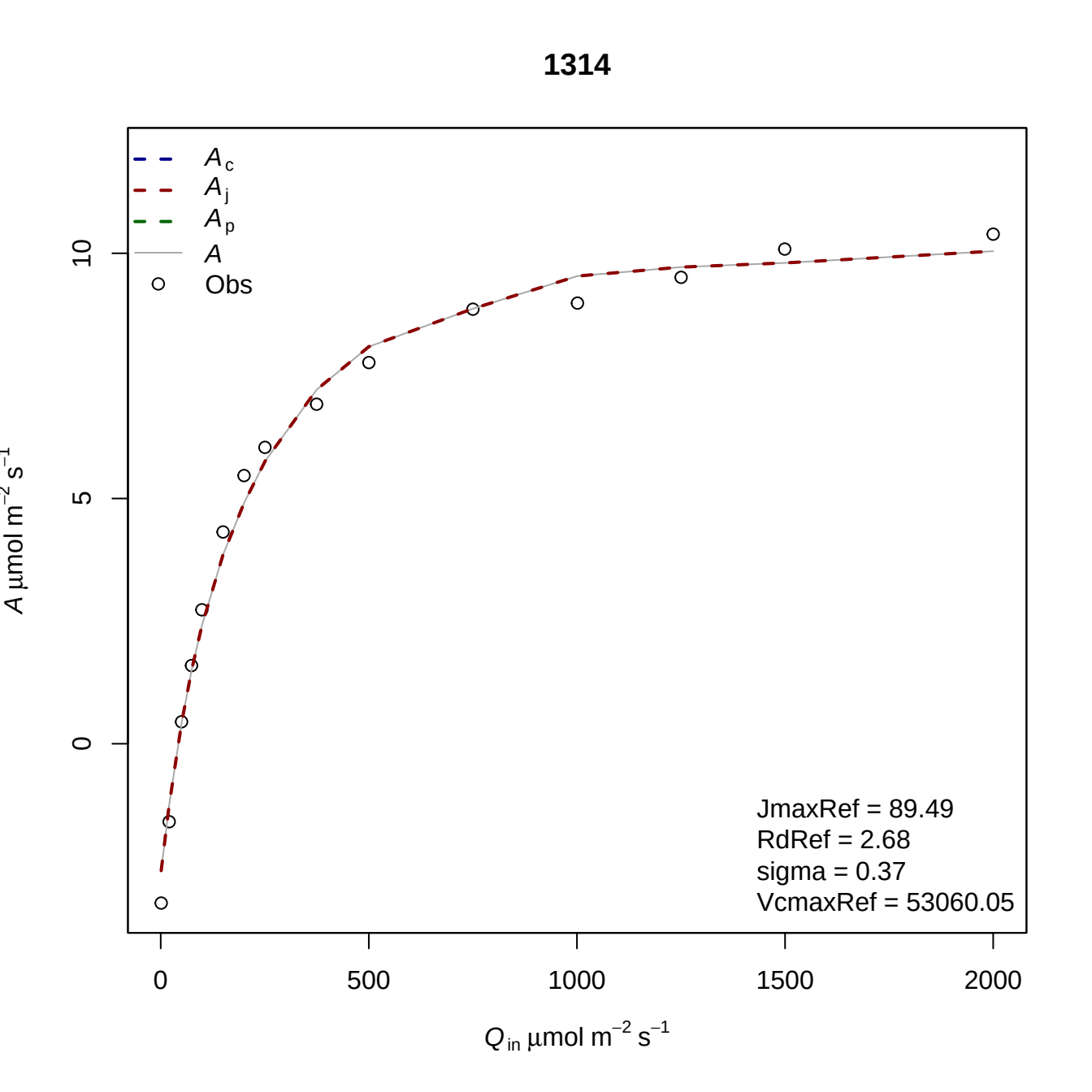

1309
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.75
RdRef = 4.18
sigma = 0.62
VcmaxRef = 138284.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
20
15
10
5
0
−5
0
500
1000
1500
2000

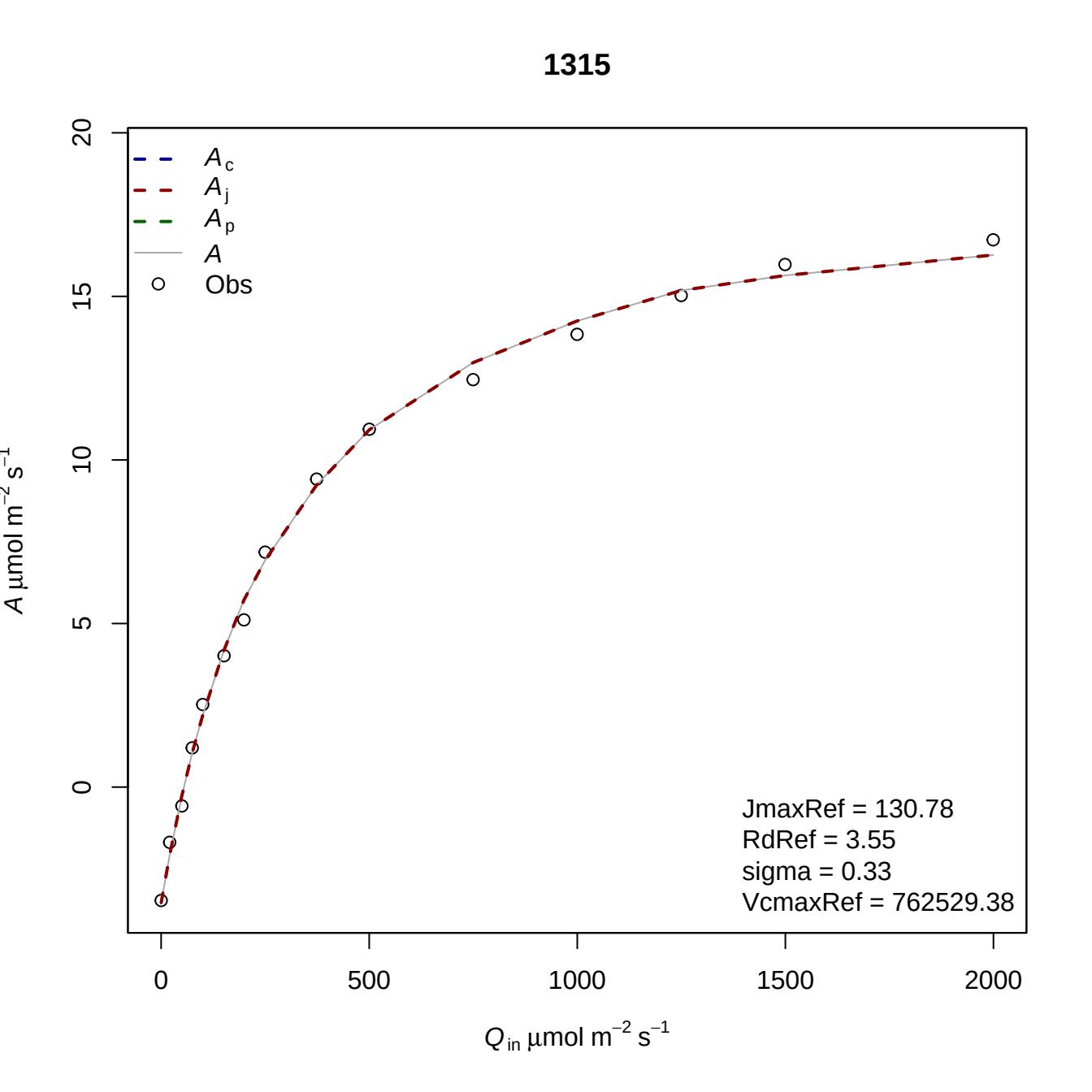

1310
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.41
RdRef = 3.35
sigma = 0.47
VcmaxRef = 47473.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

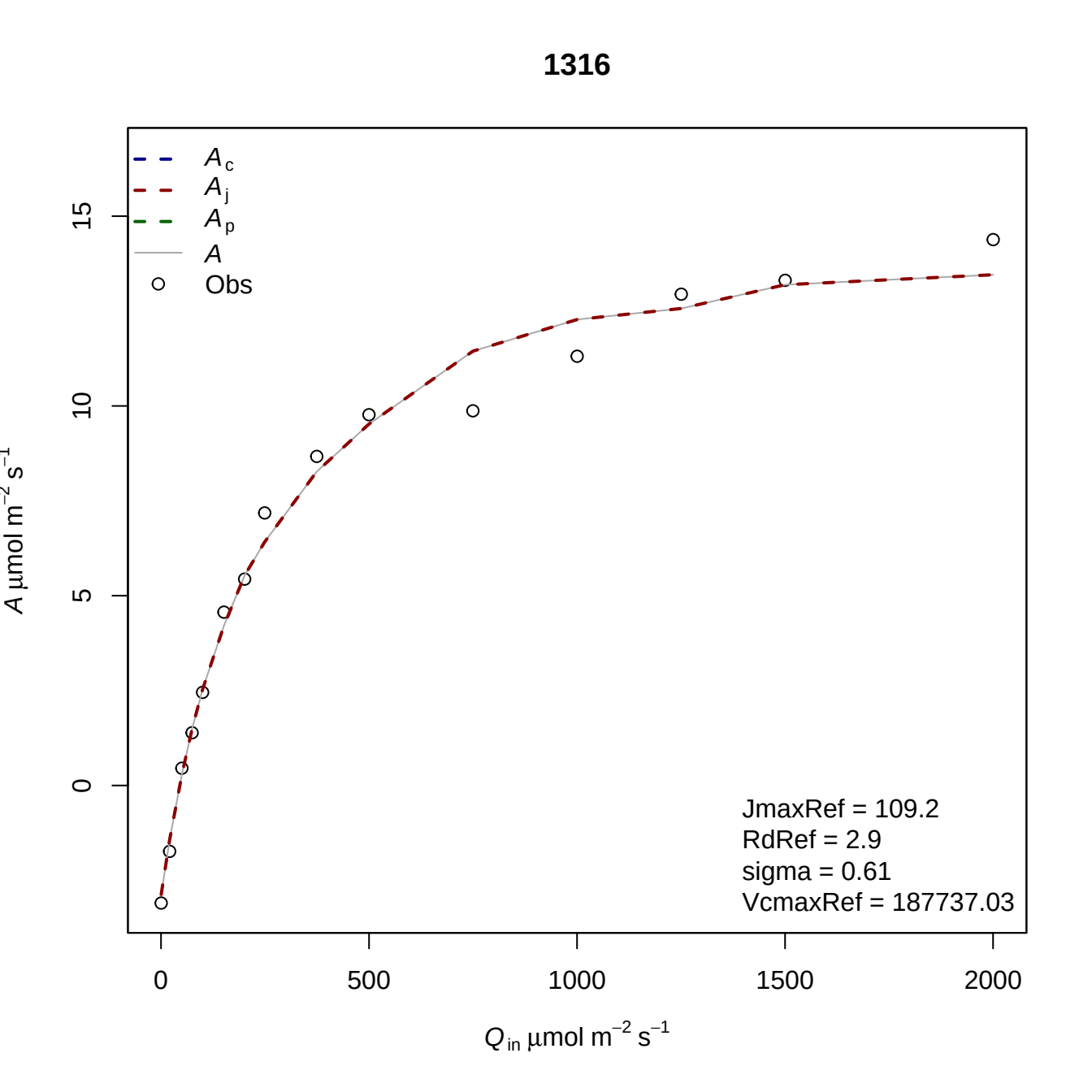

1311
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83
RdRef = 2
sigma = 0.34
VcmaxRef = 36.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

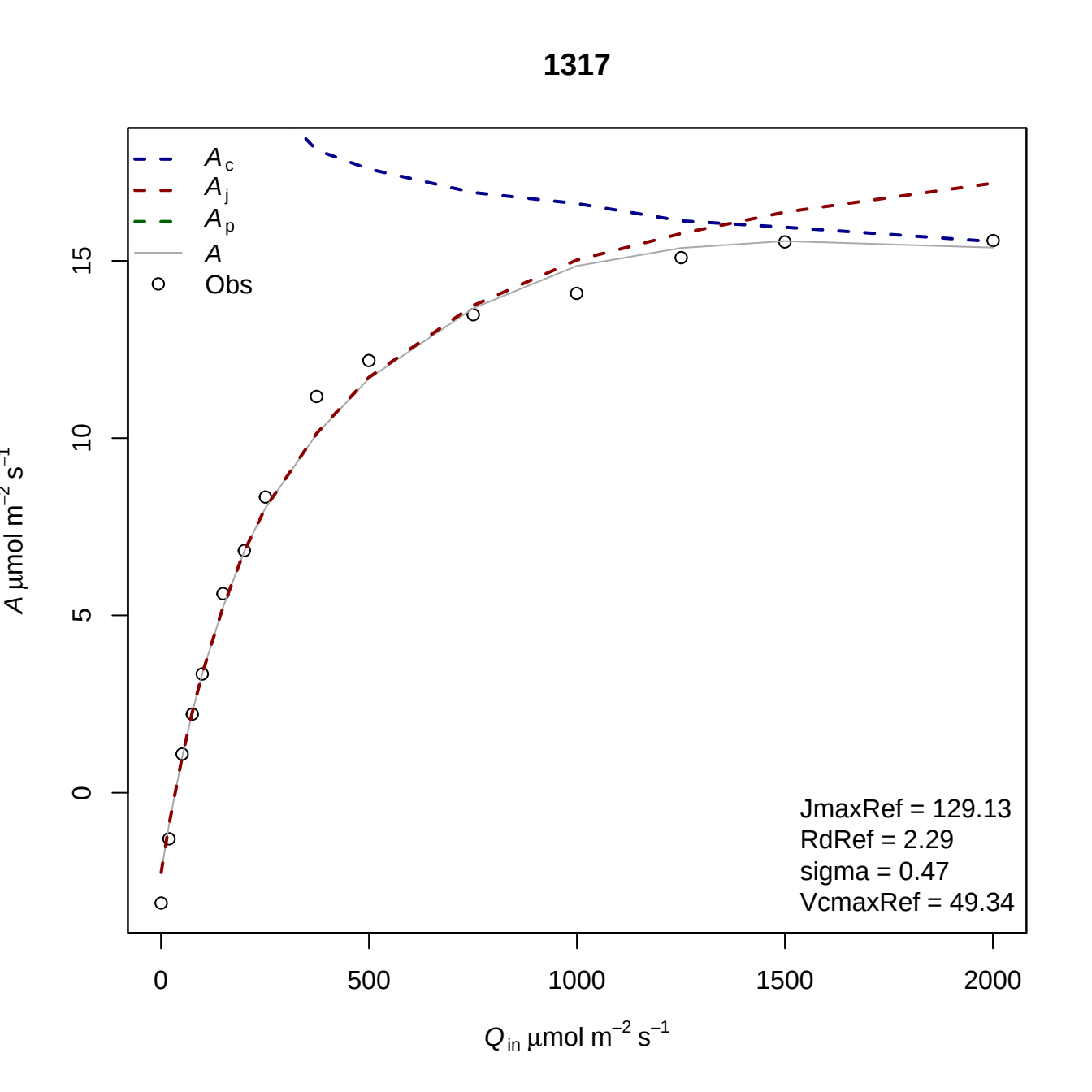

1312
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.01
RdRef = 1.86
sigma = 0.41
VcmaxRef = 62761.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

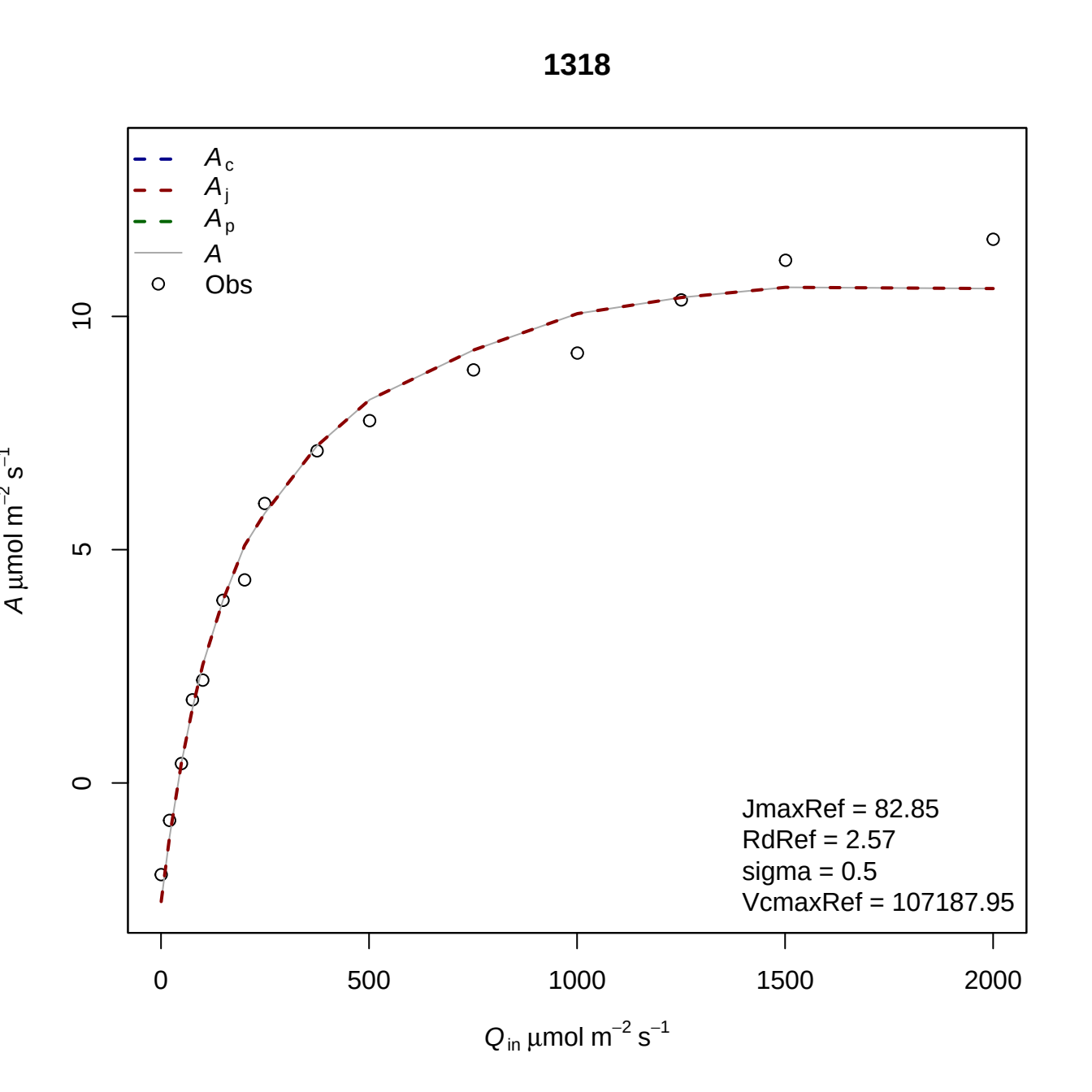

1313
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.75
RdRef = 1.98
sigma = 0.38
VcmaxRef = 25.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

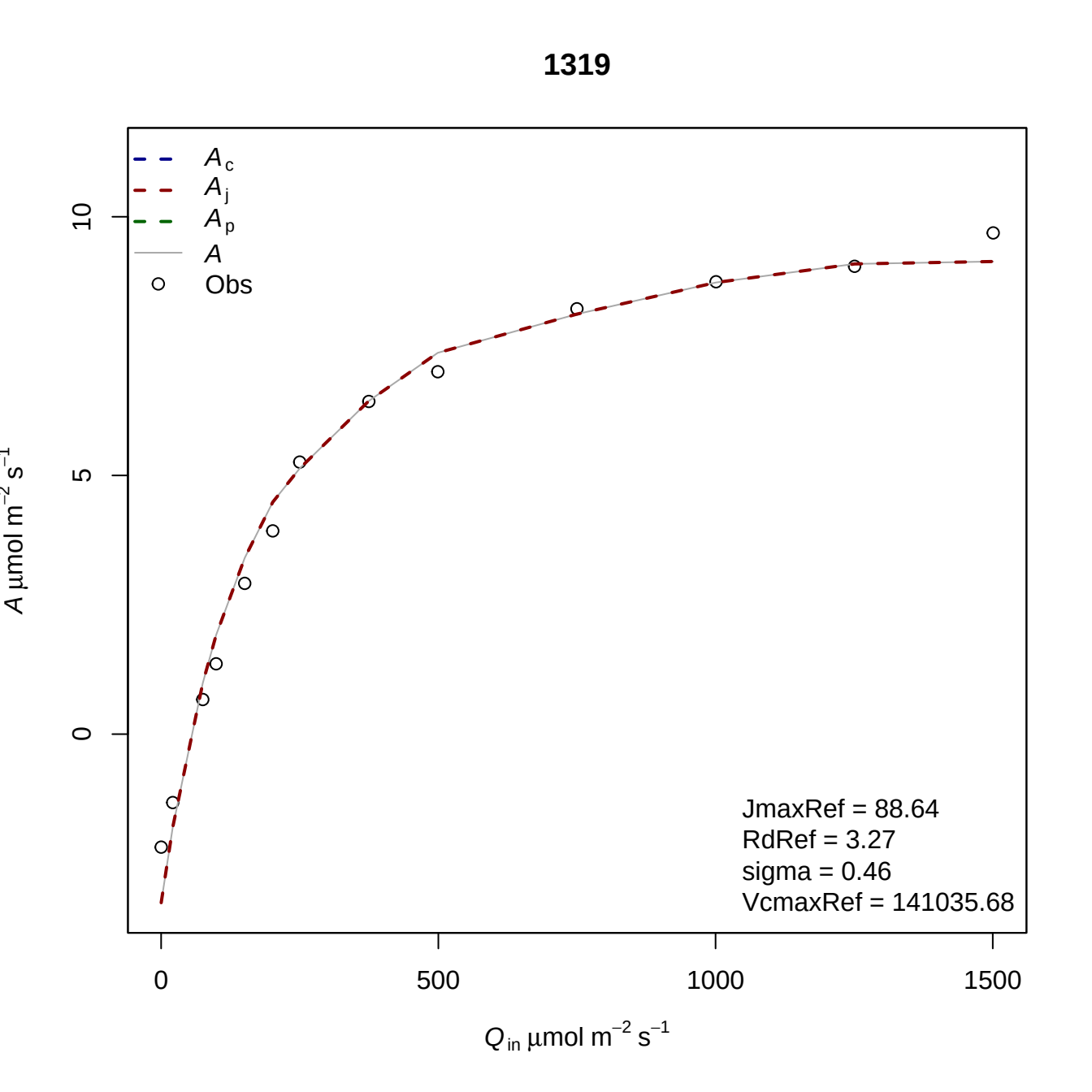

1314
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.49
RdRef = 2.68
sigma = 0.37
VcmaxRef = 53060.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

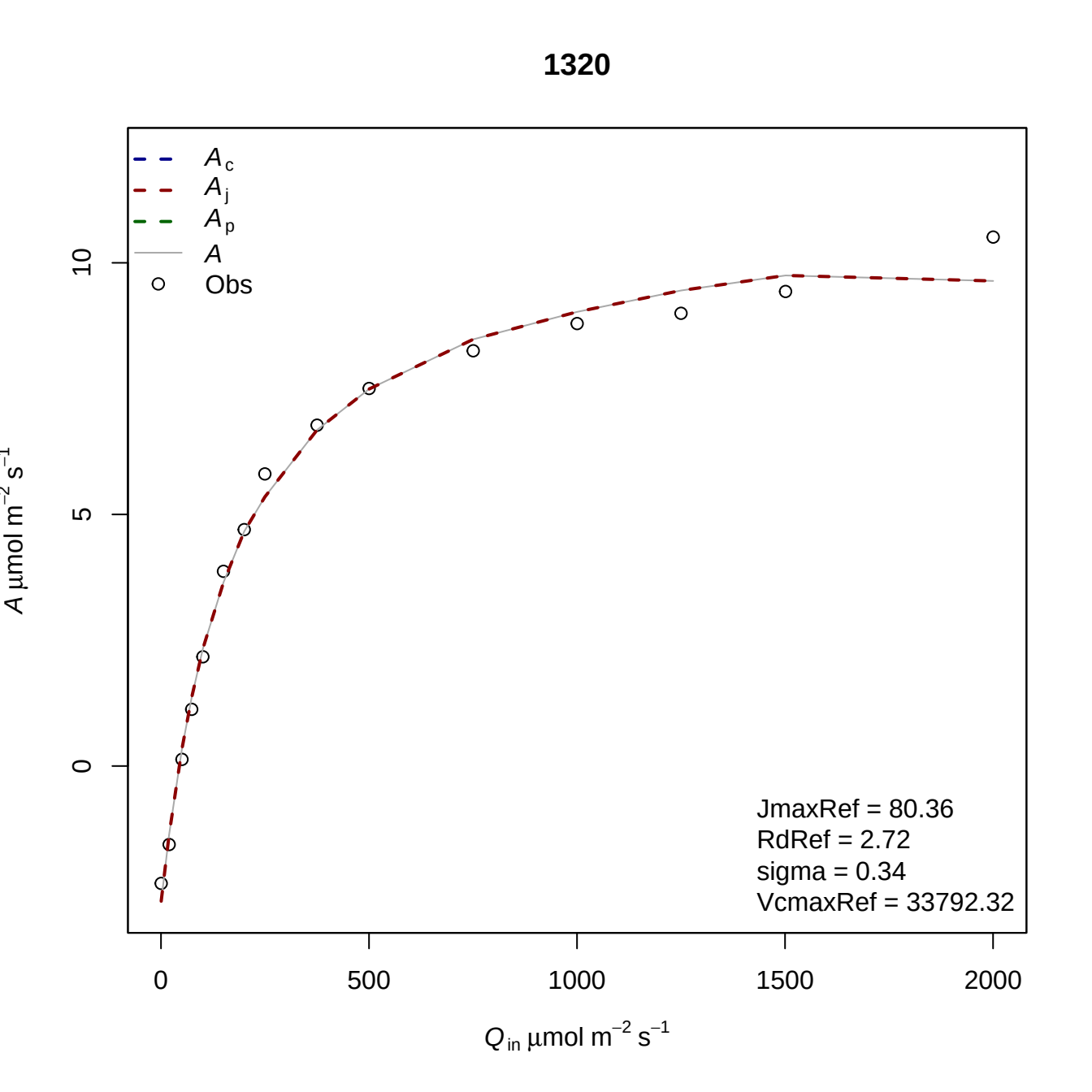

1315
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 130.78
RdRef = 3.55
sigma = 0.33
VcmaxRef = 762529.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

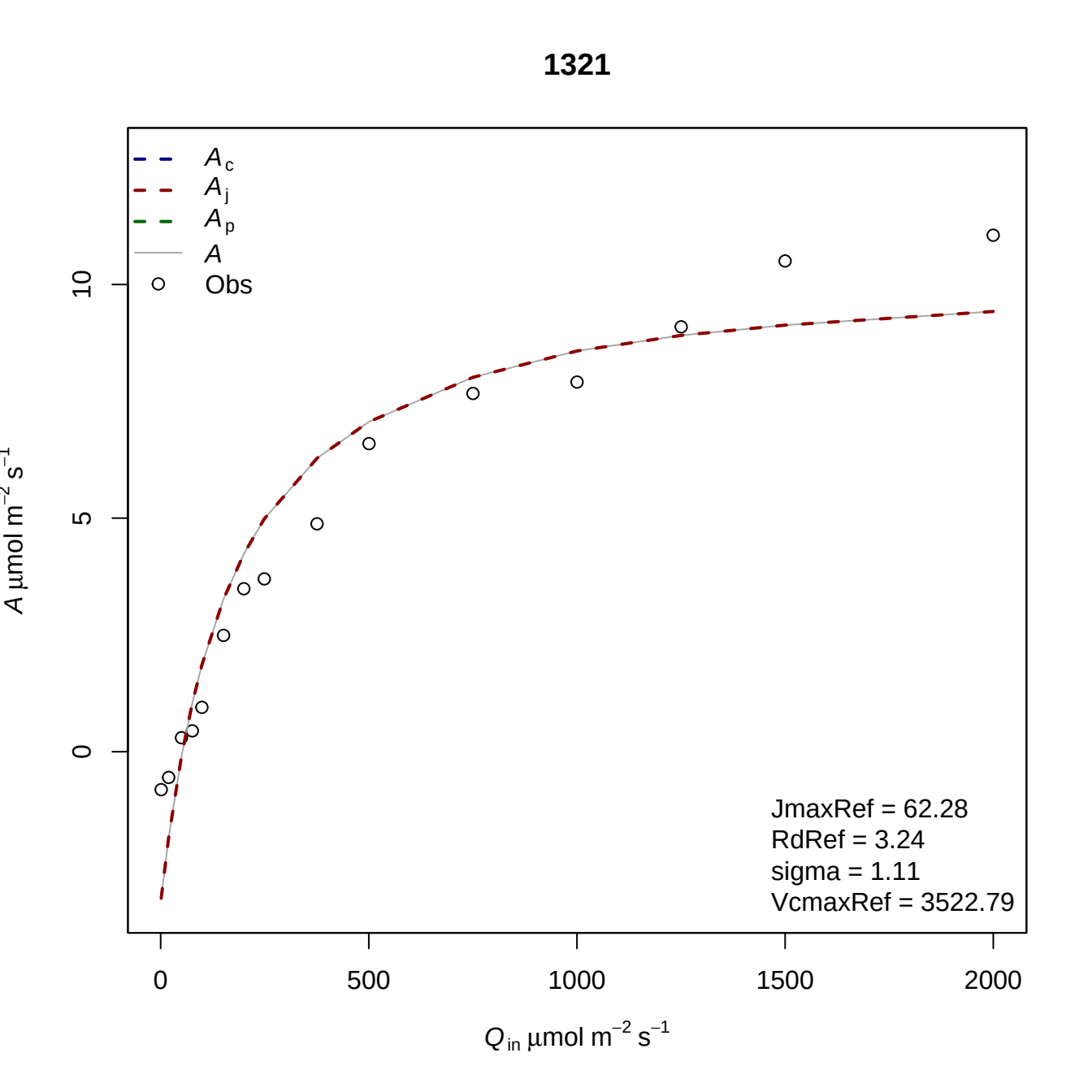

1316
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 109.2
RdRef = 2.9
sigma = 0.61
VcmaxRef = 187737.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

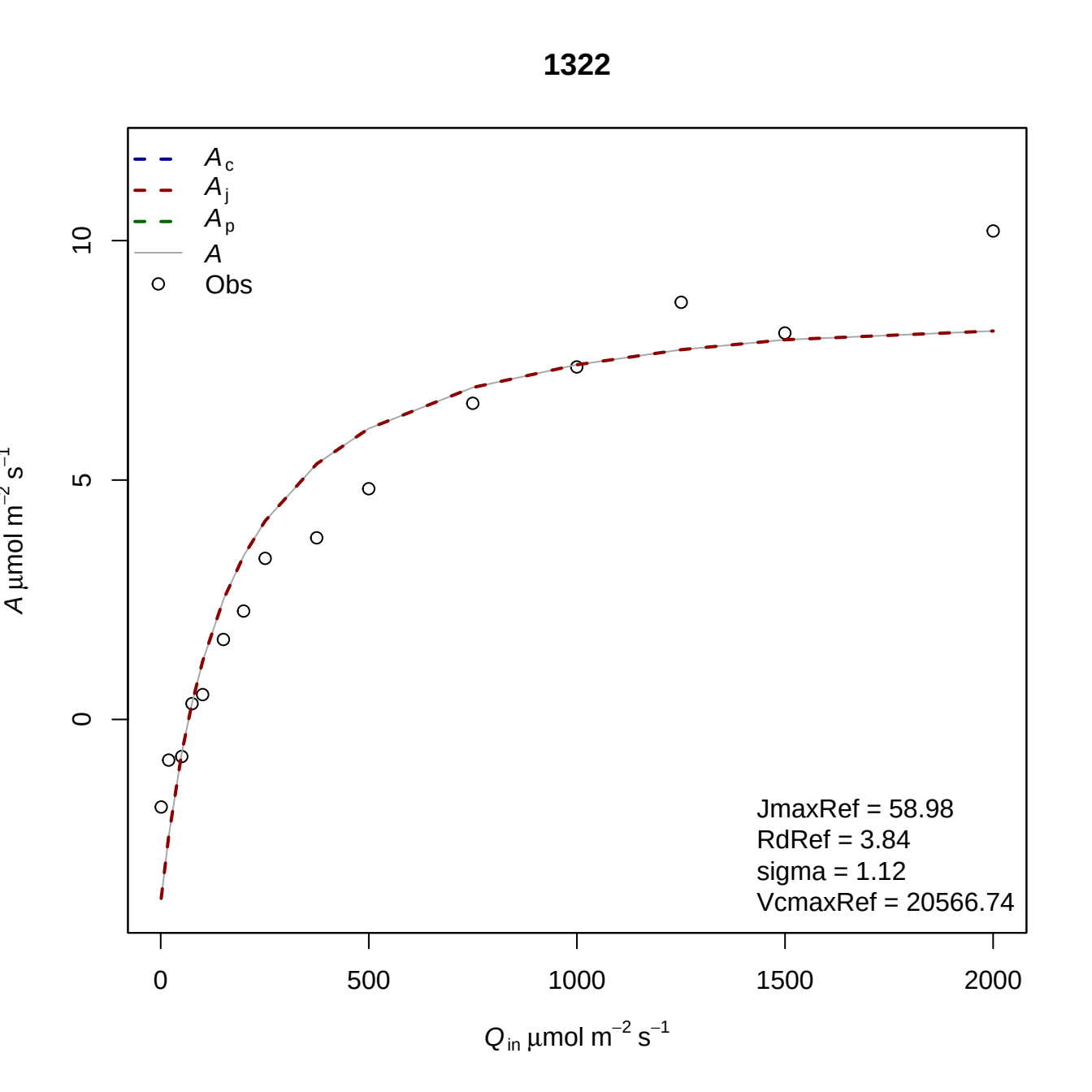

1317
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 129.13
RdRef = 2.29
sigma = 0.47
VcmaxRef = 49.34
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

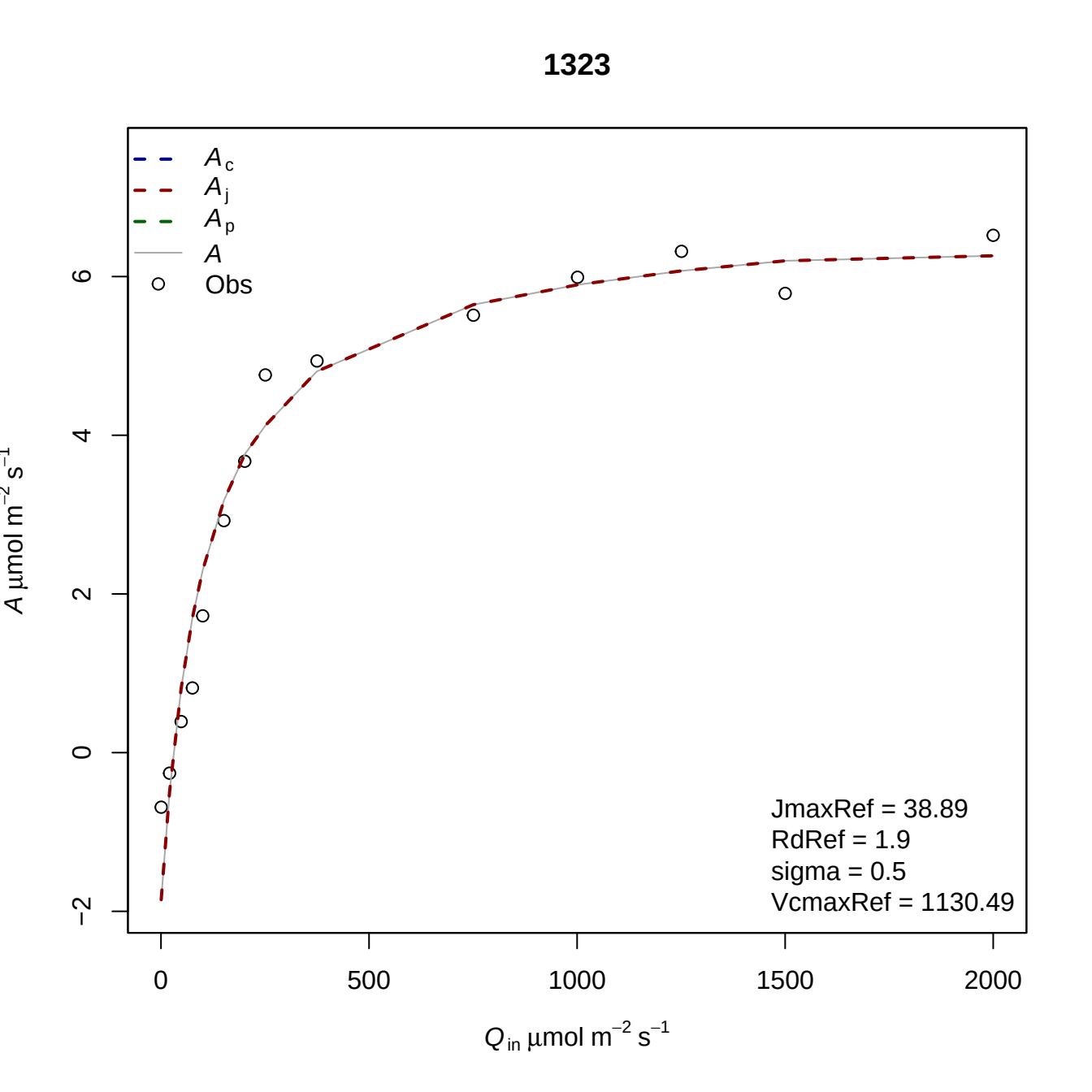

1318
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.85
RdRef = 2.57
sigma = 0.5
VcmaxRef = 107187.95
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

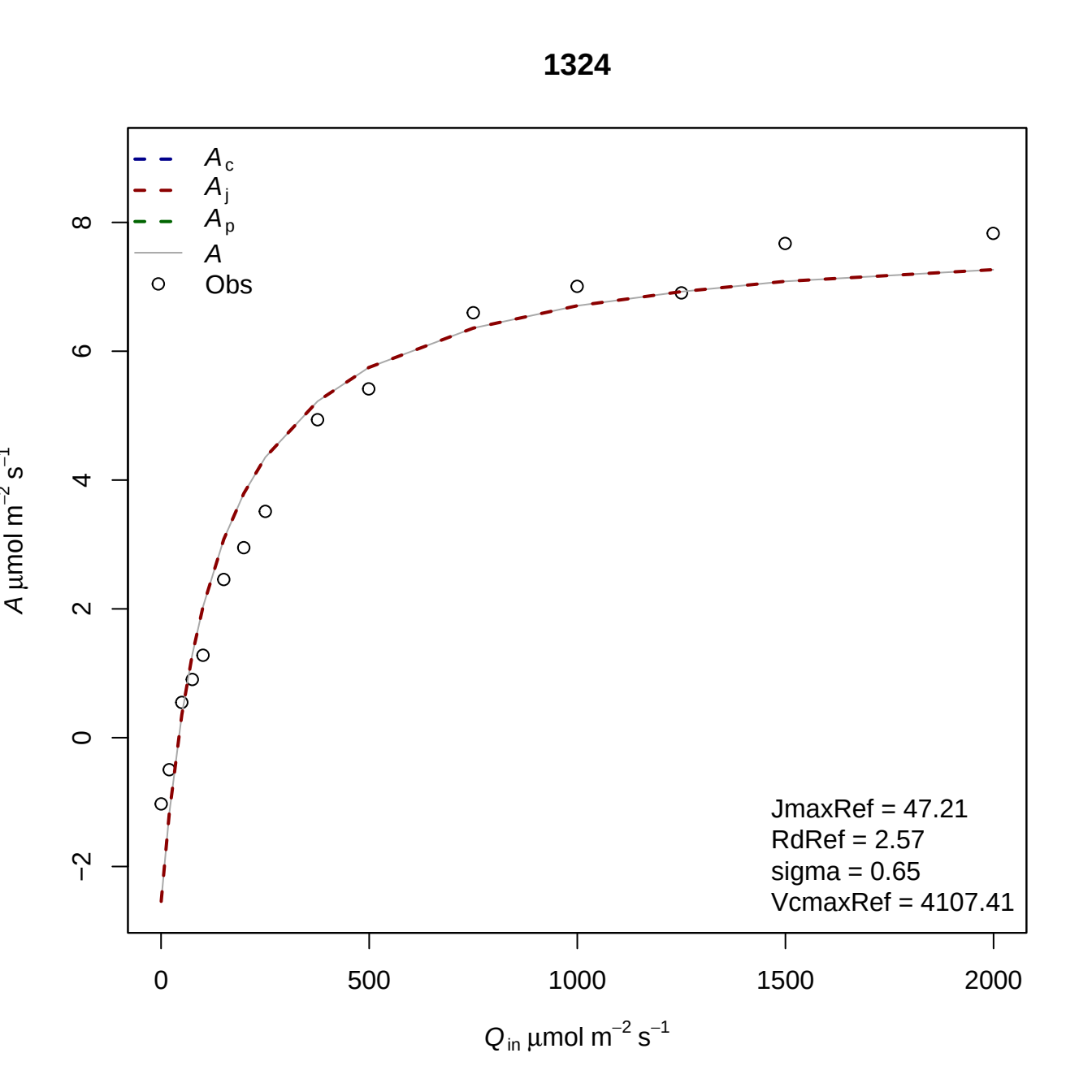

1319
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.64
RdRef = 3.27
sigma = 0.46
VcmaxRef = 141035.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

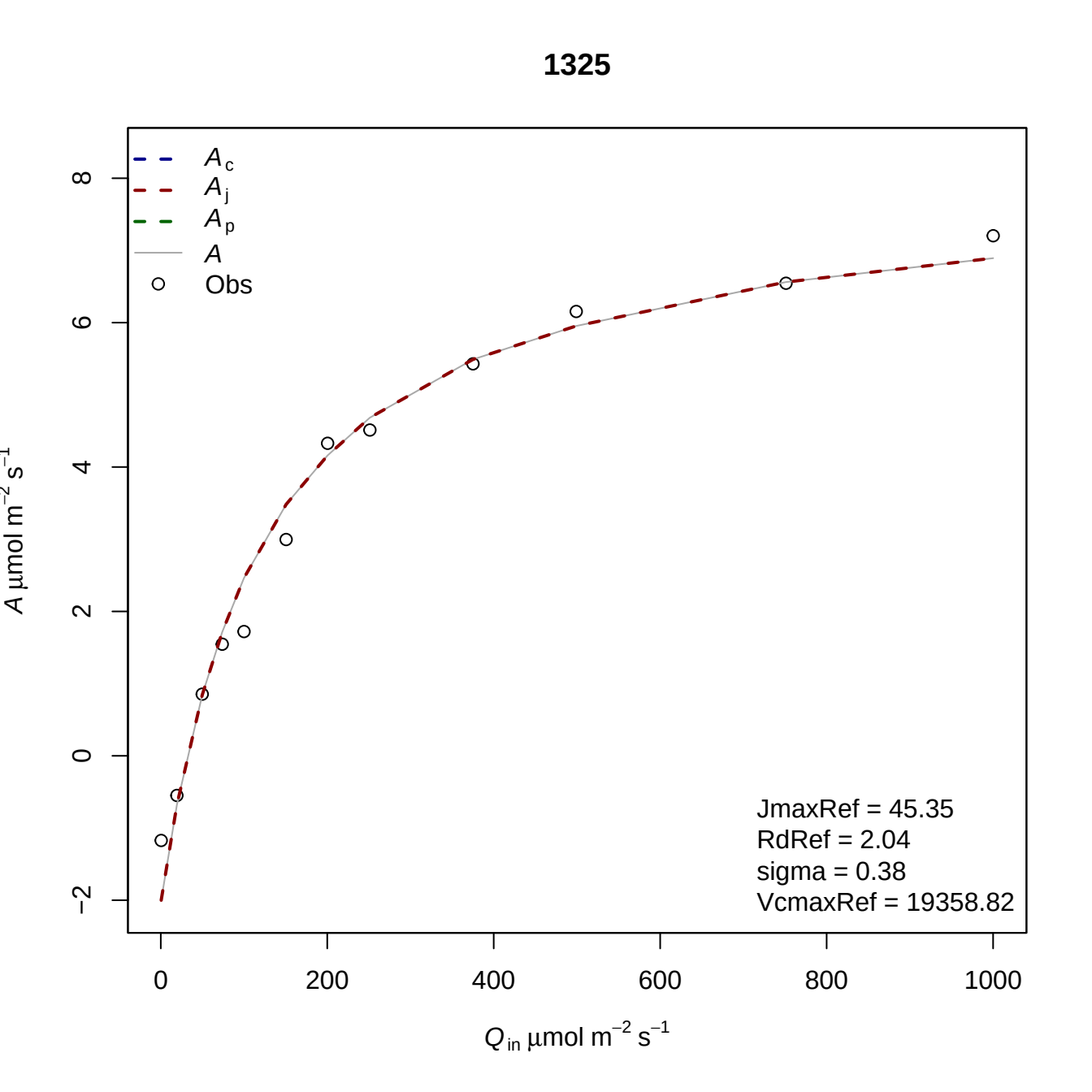

1320
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.36
RdRef = 2.72
sigma = 0.34
VcmaxRef = 33792.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

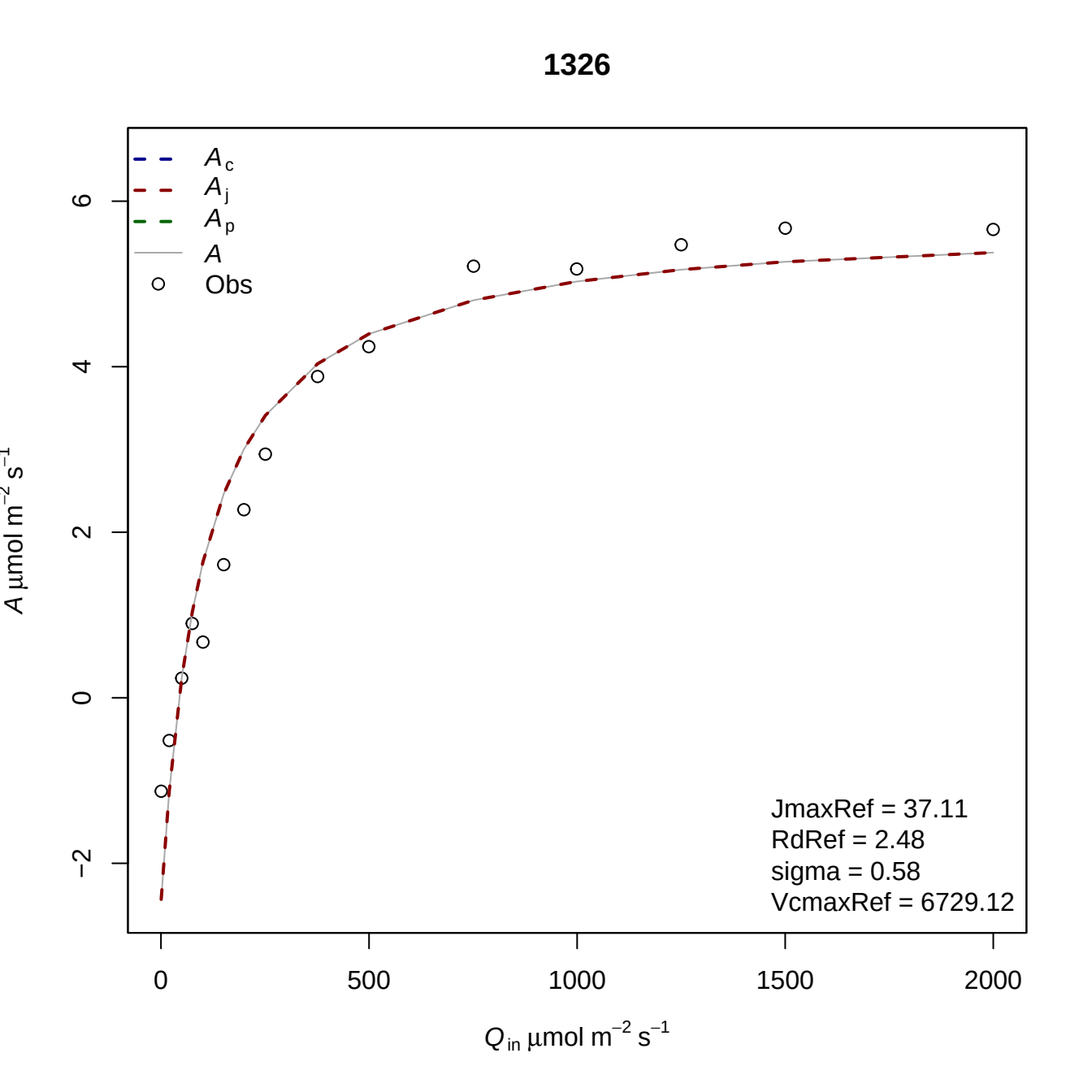

1321
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.28
RdRef = 3.24
sigma = 1.11
VcmaxRef = 3522.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

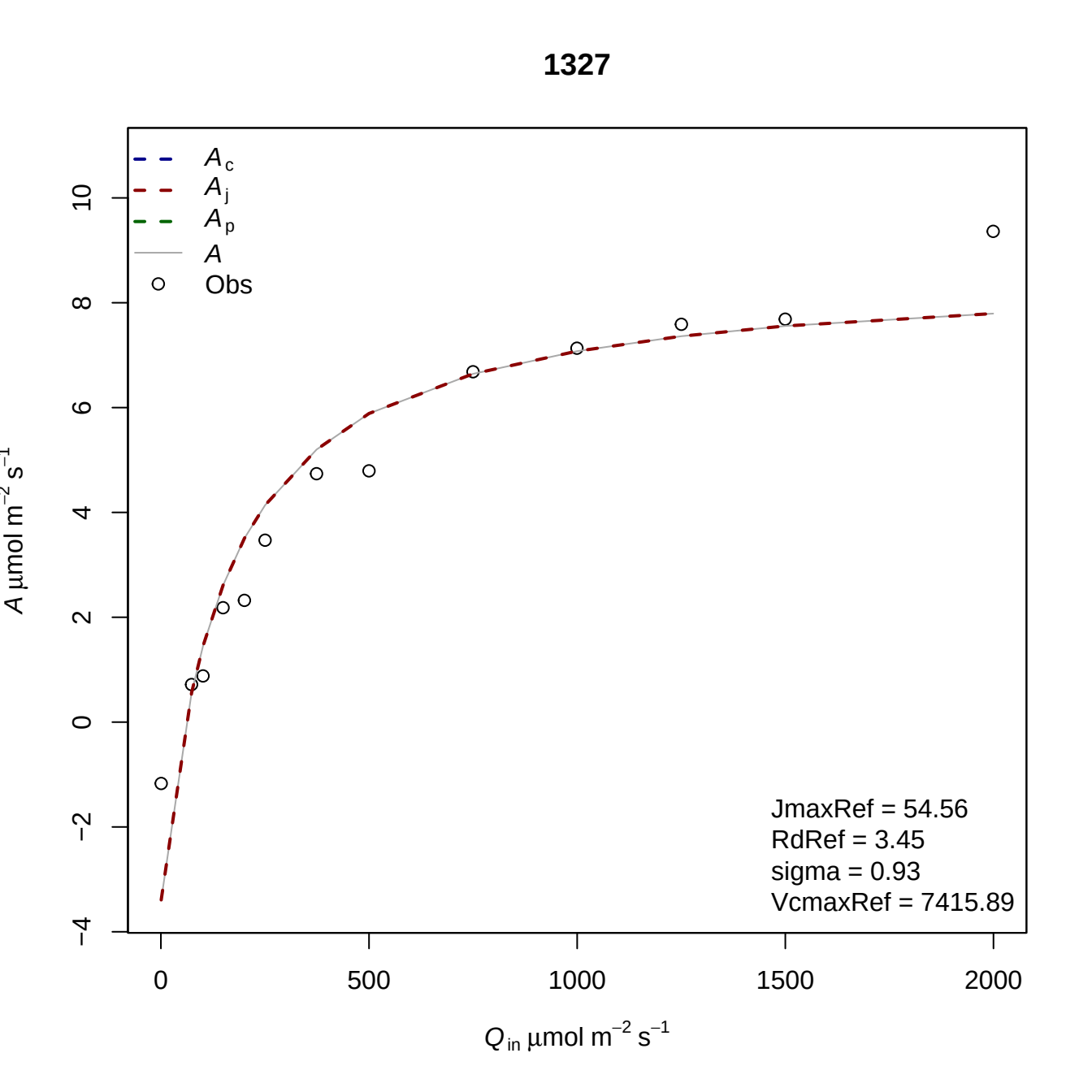

1322
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.98
RdRef = 3.84
sigma = 1.12
VcmaxRef = 20566.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

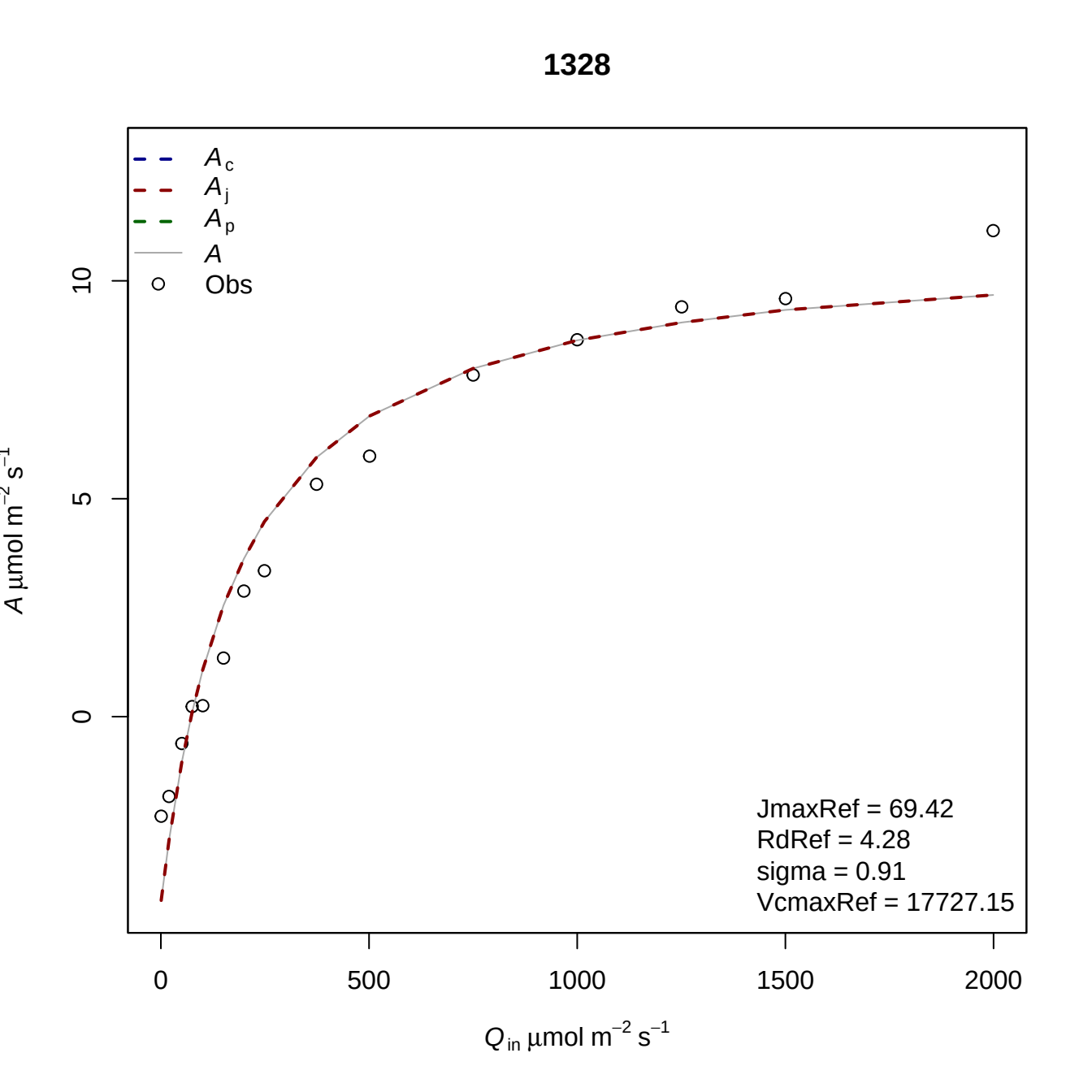

1323
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 38.89
RdRef = 1.9
sigma = 0.5
VcmaxRef = 1130.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

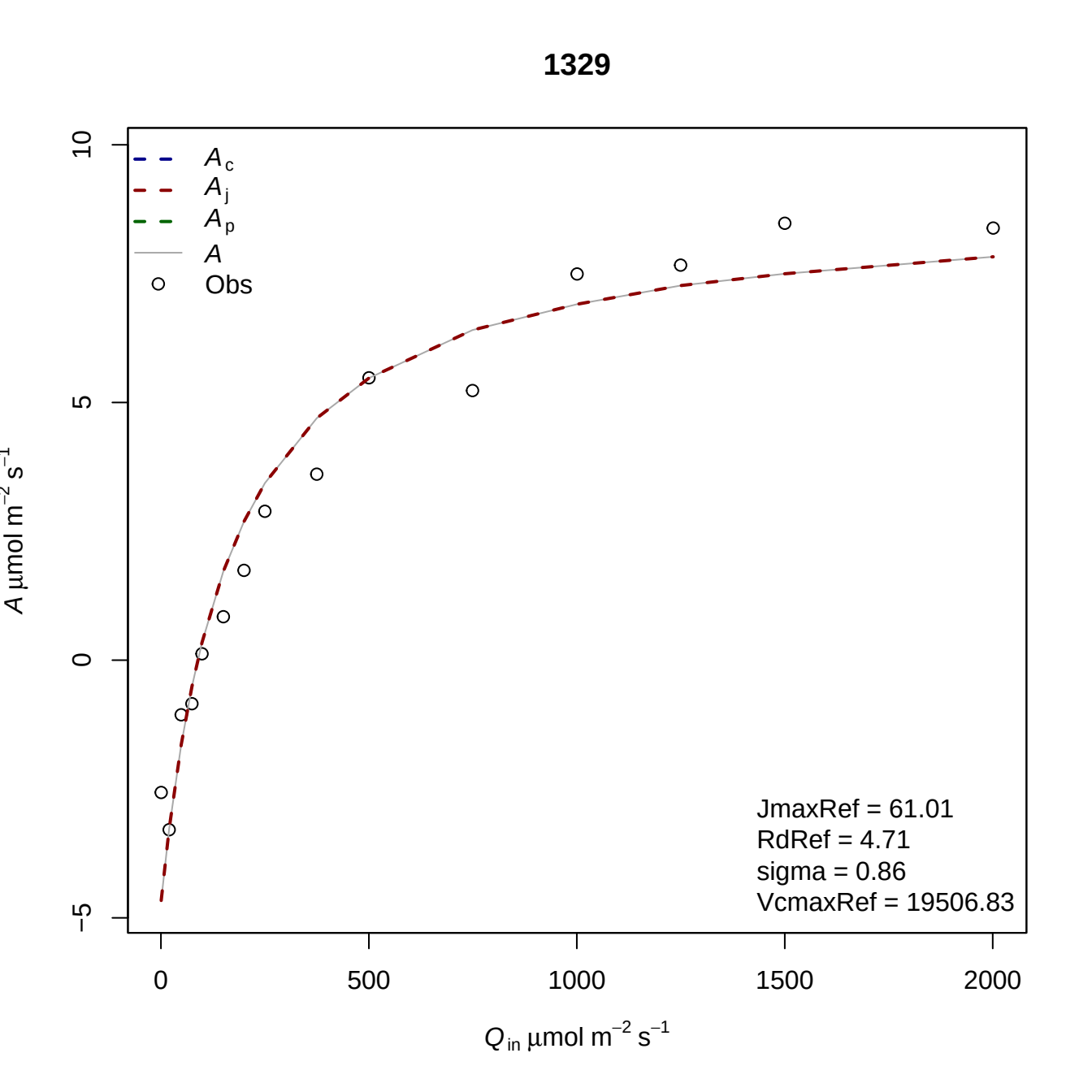

1324
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.21
RdRef = 2.57
sigma = 0.65
VcmaxRef = 4107.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

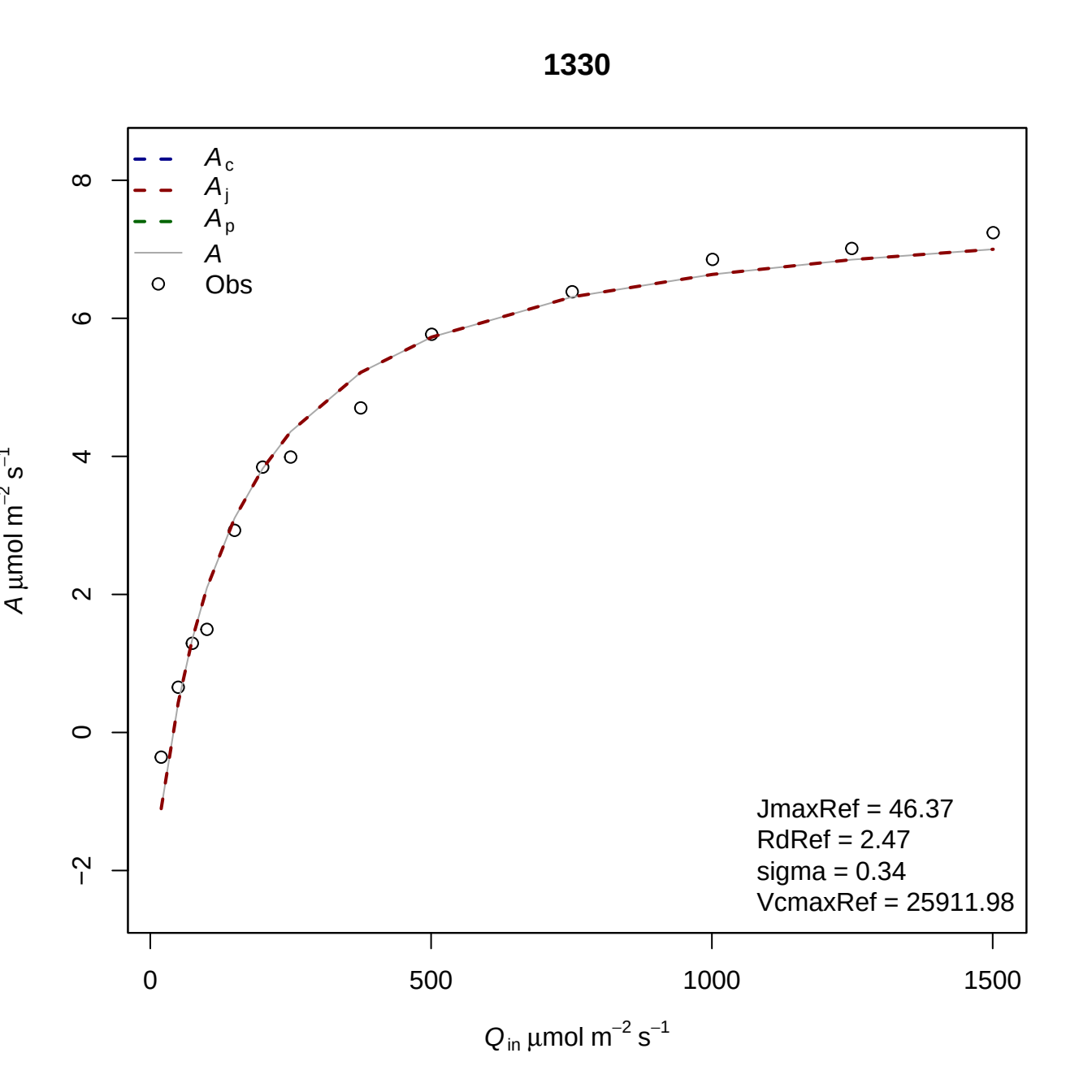

1325
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.35
RdRef = 2.04
sigma = 0.38
VcmaxRef = 19358.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
200
400
600
800
1000

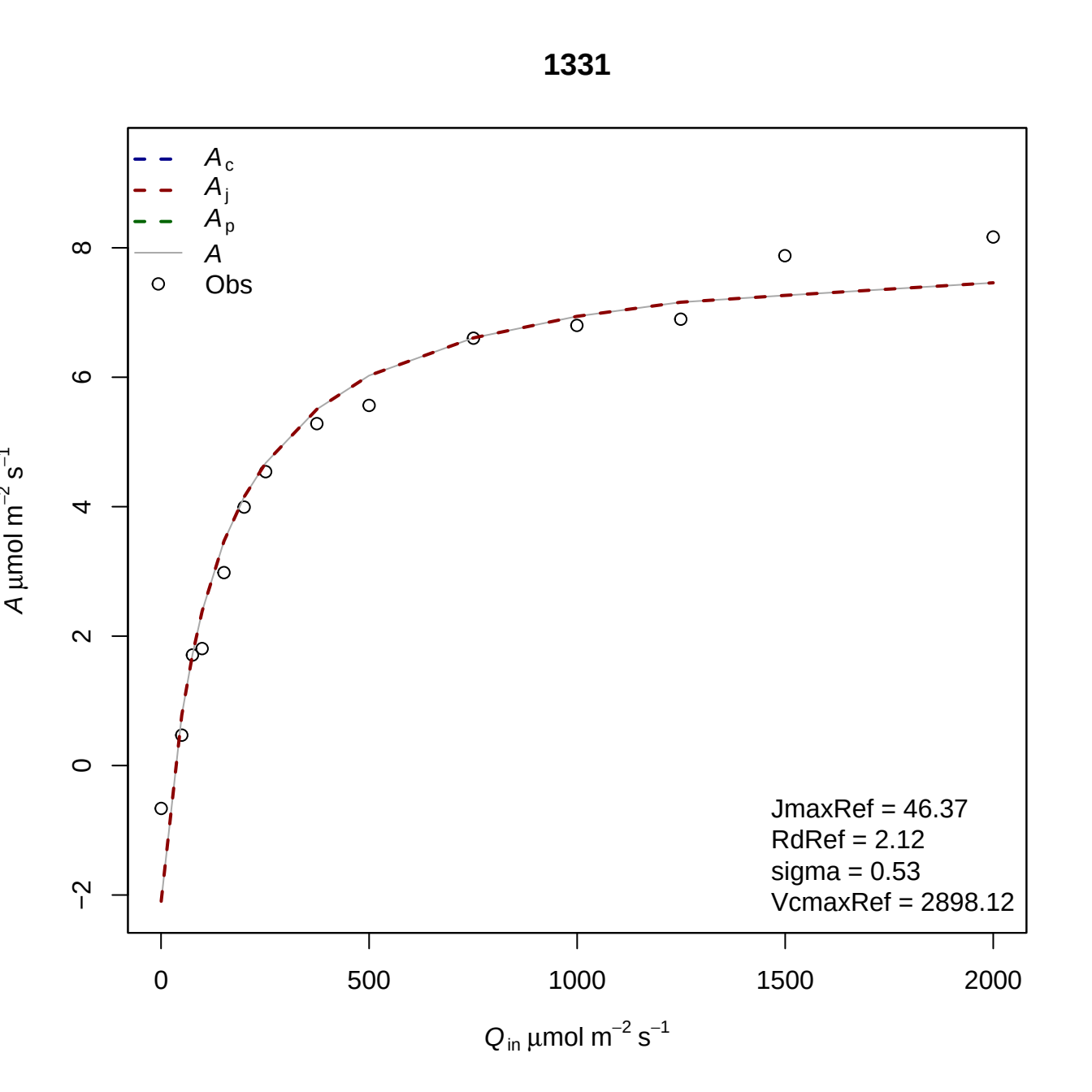

1326
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.11
RdRef = 2.48
sigma = 0.58
VcmaxRef = 6729.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

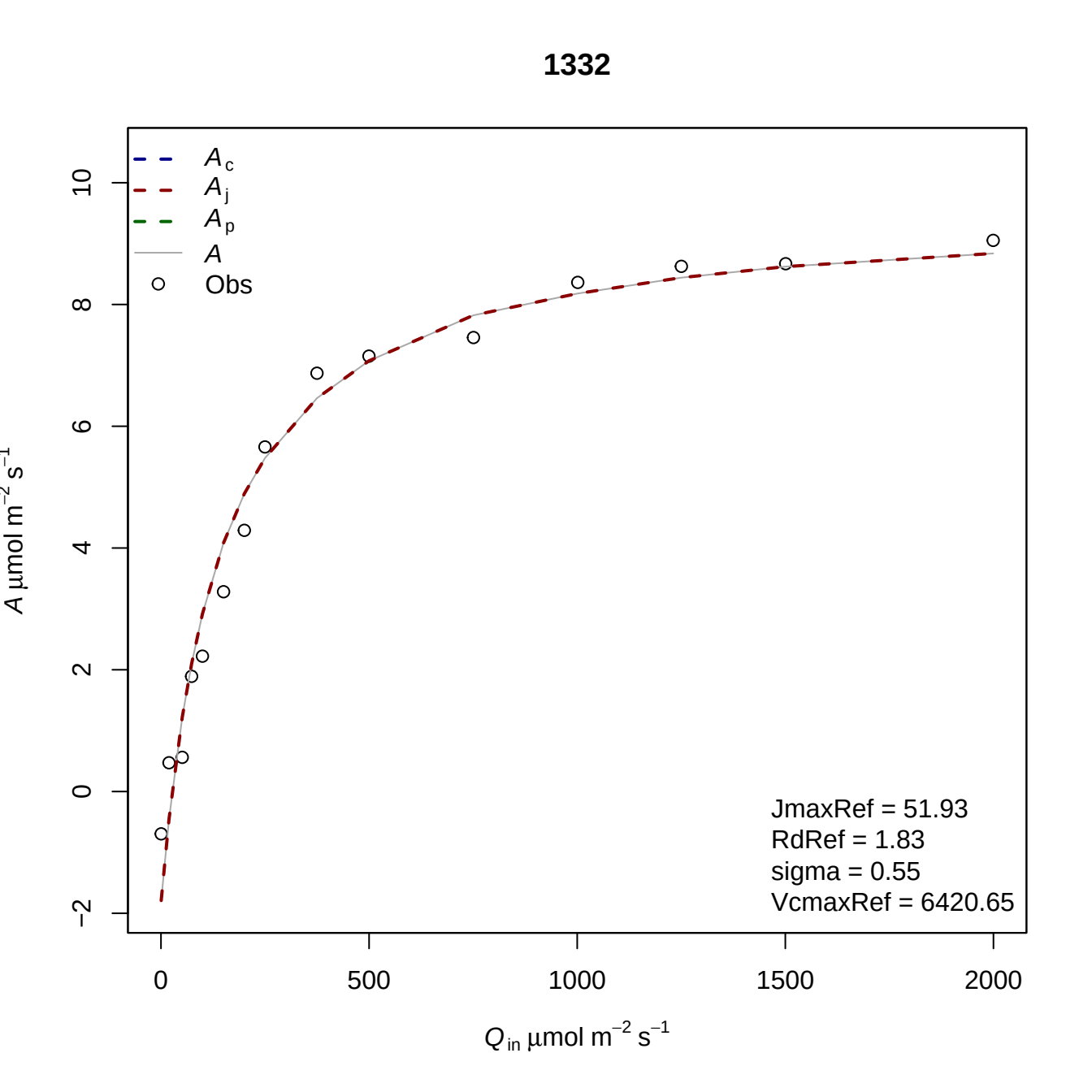

1327
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.56
RdRef = 3.45
sigma = 0.93
VcmaxRef = 7415.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

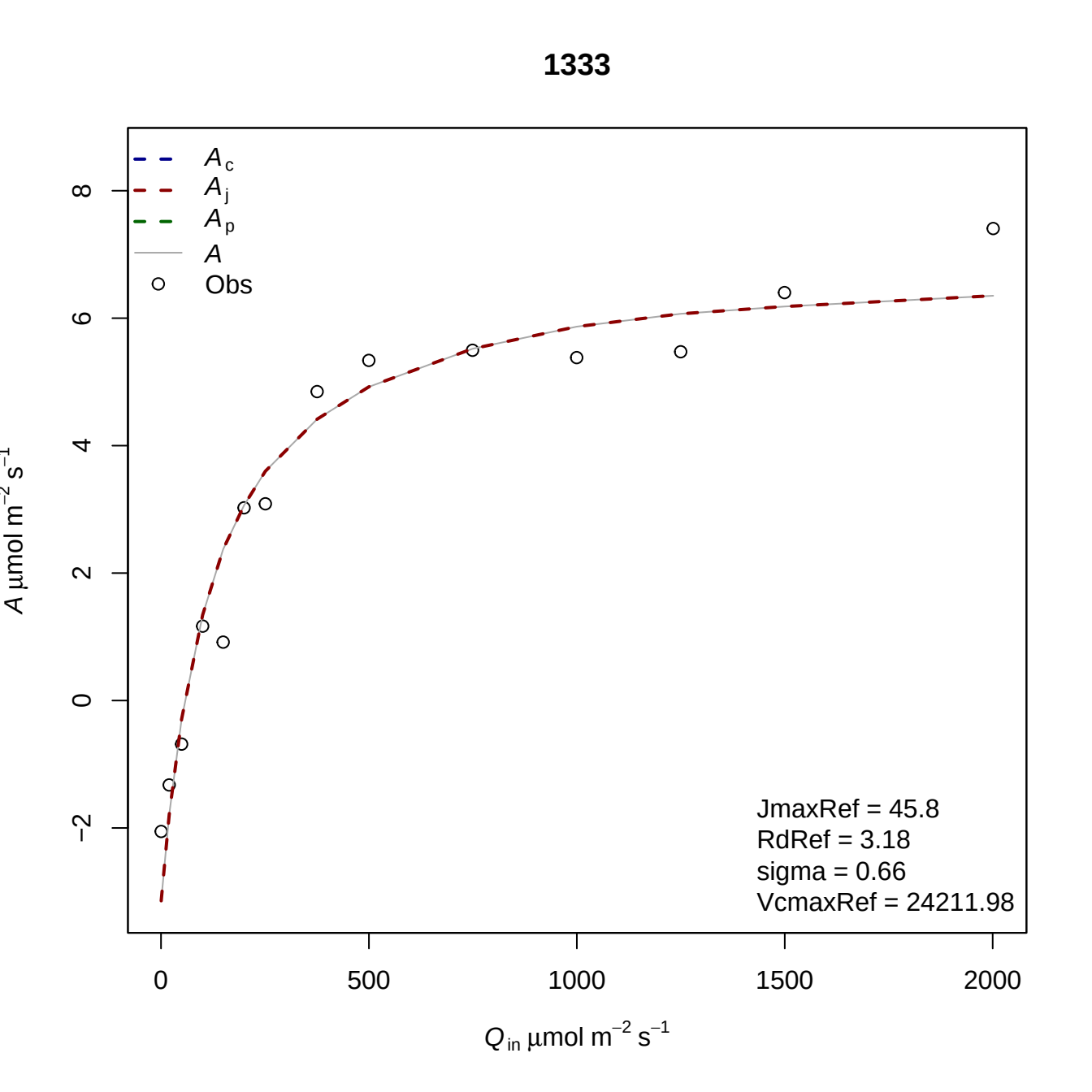

1328
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.42
RdRef = 4.28
sigma = 0.91
VcmaxRef = 17727.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

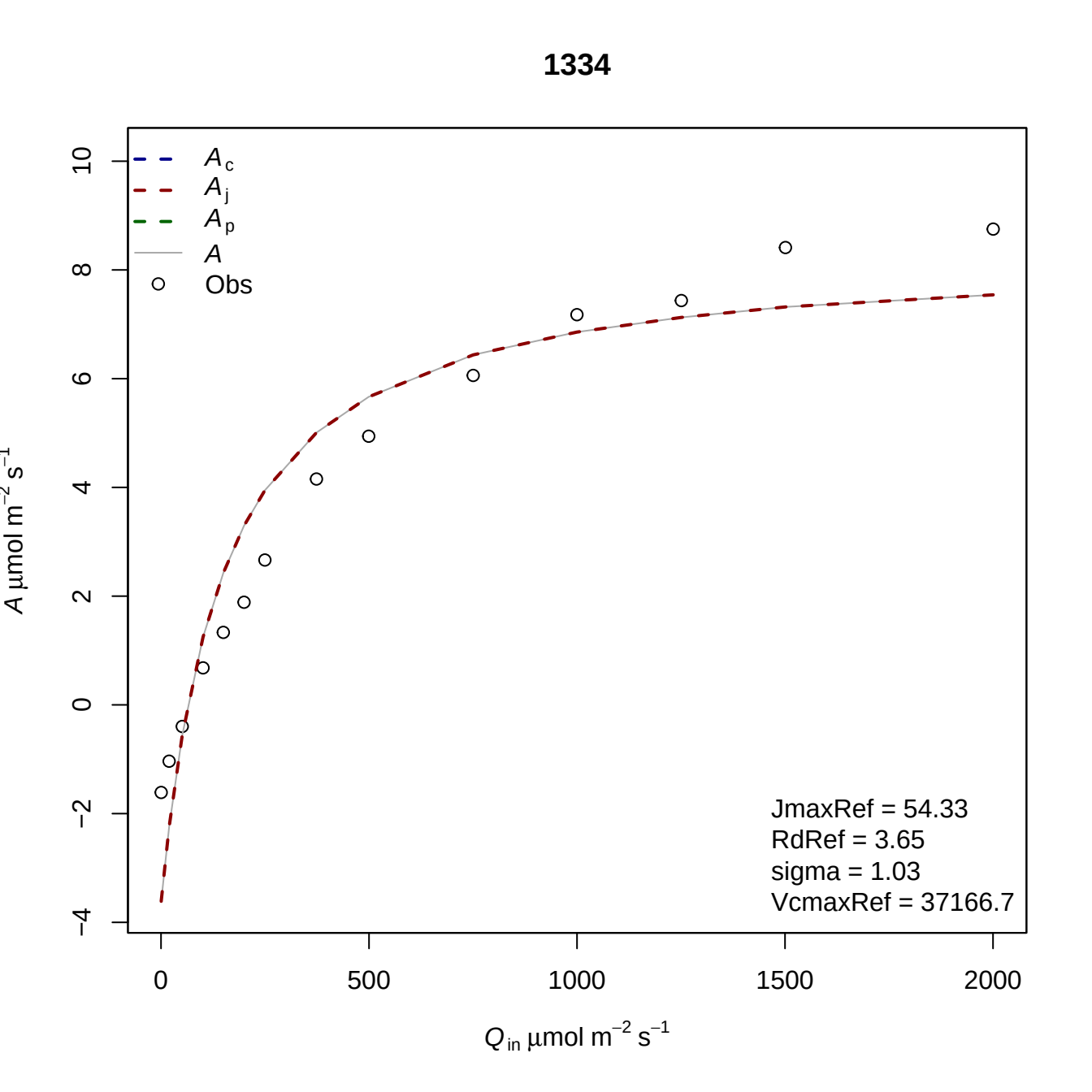

1329
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.01
RdRef = 4.71
sigma = 0.86
VcmaxRef = 19506.83
10
5
0
−5
0
500
1000
1500
2000
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

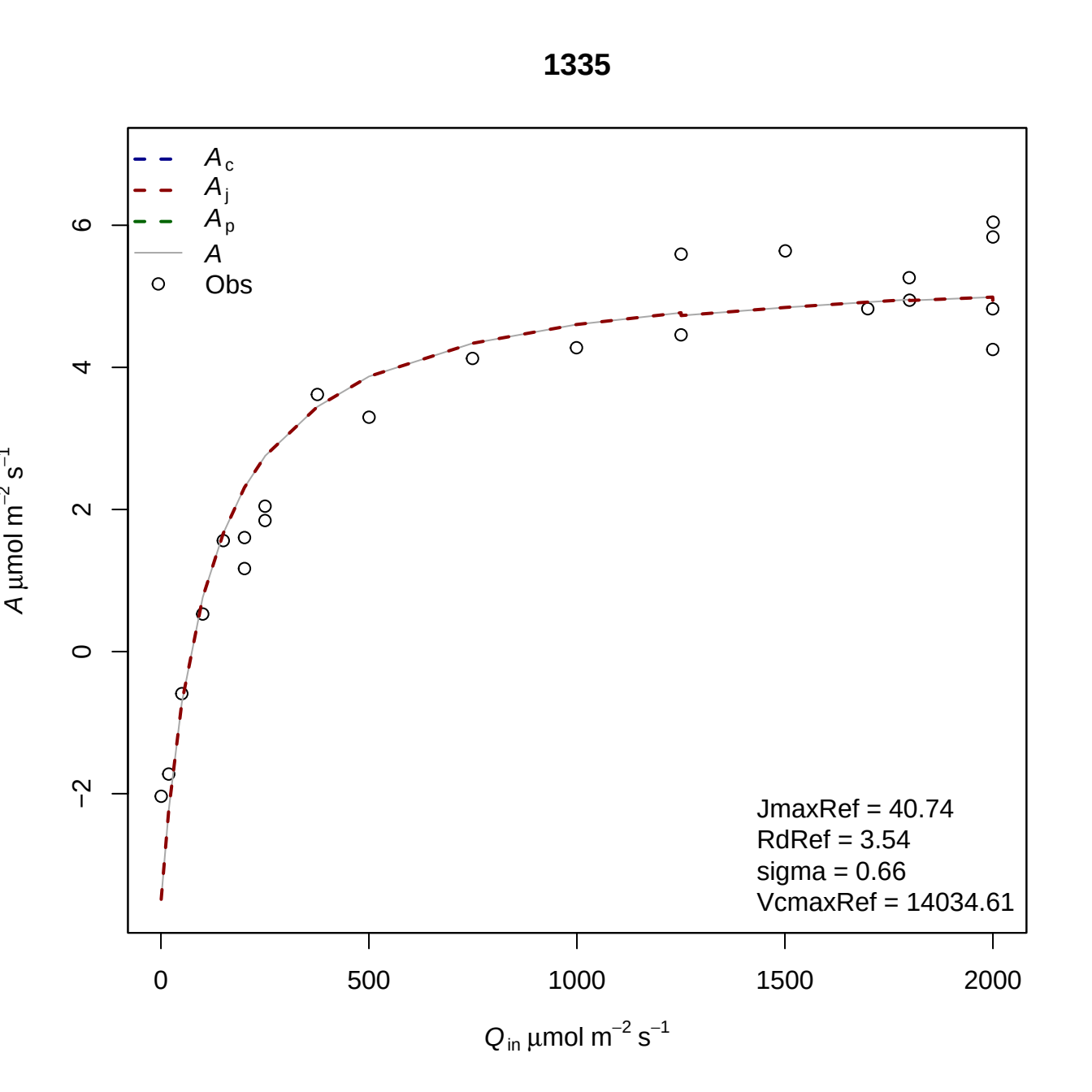

1330
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.37
RdRef = 2.47
sigma = 0.34
VcmaxRef = 25911.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

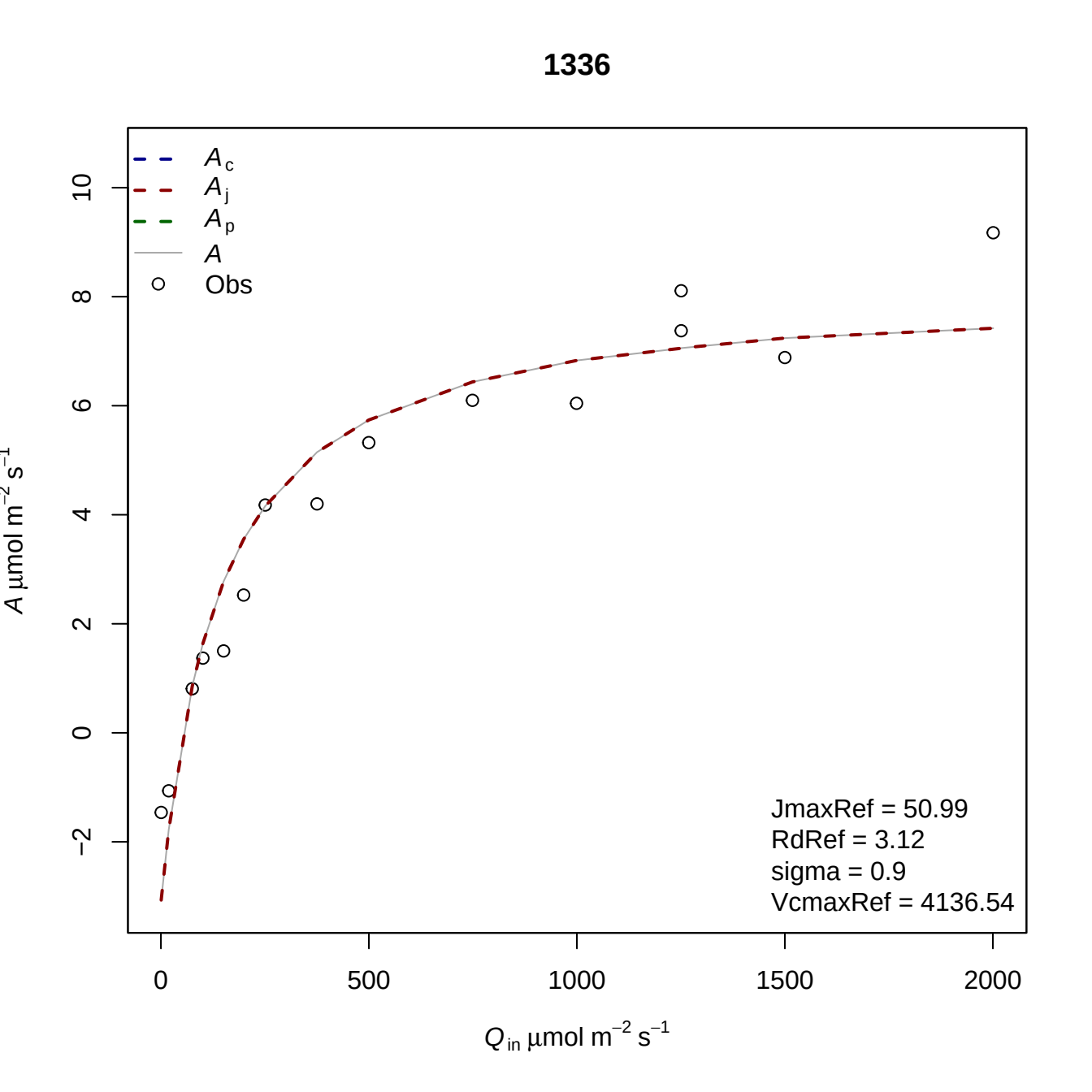

1331
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.37
RdRef = 2.12
sigma = 0.53
VcmaxRef = 2898.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

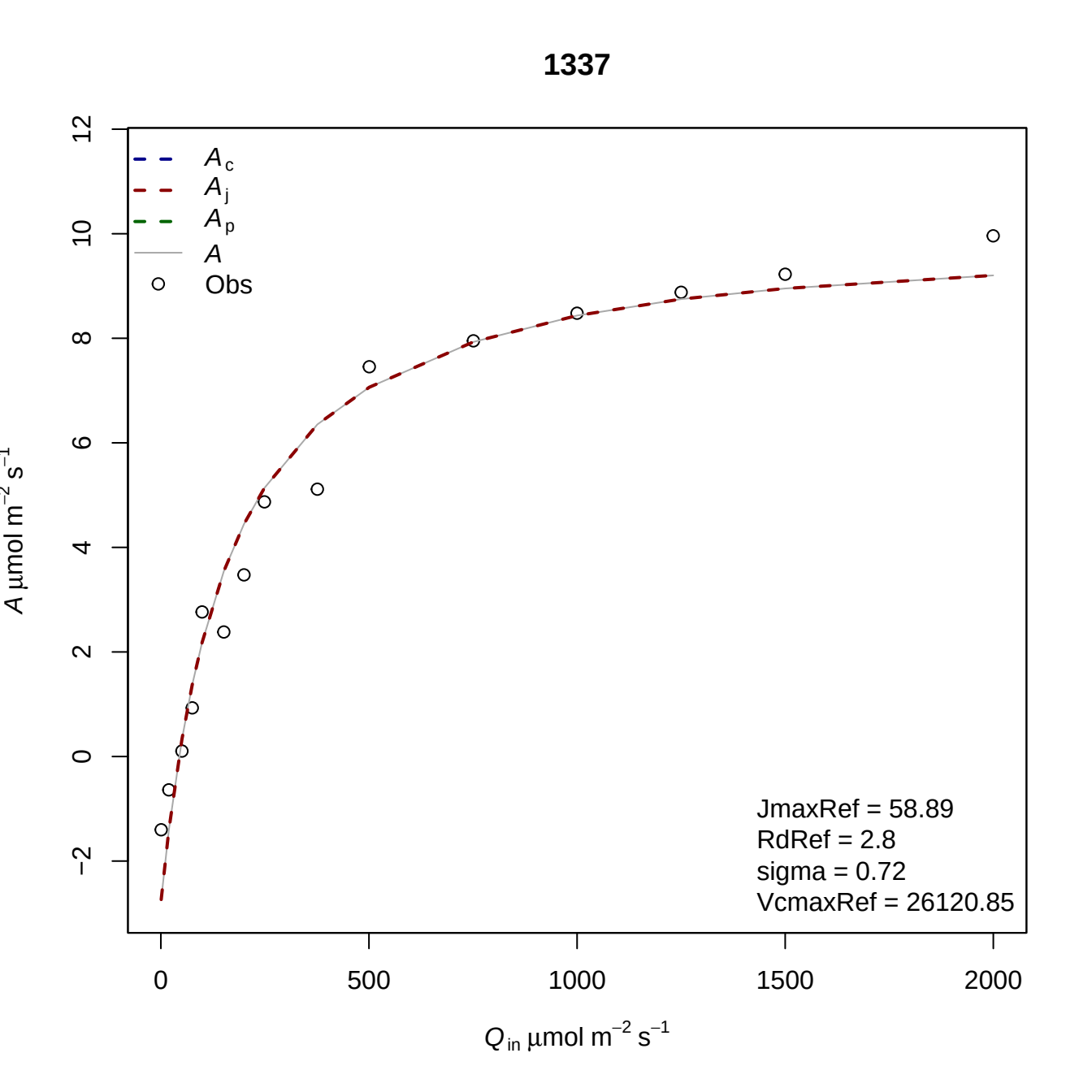

1332
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.93
RdRef = 1.83
sigma = 0.55
VcmaxRef = 6420.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

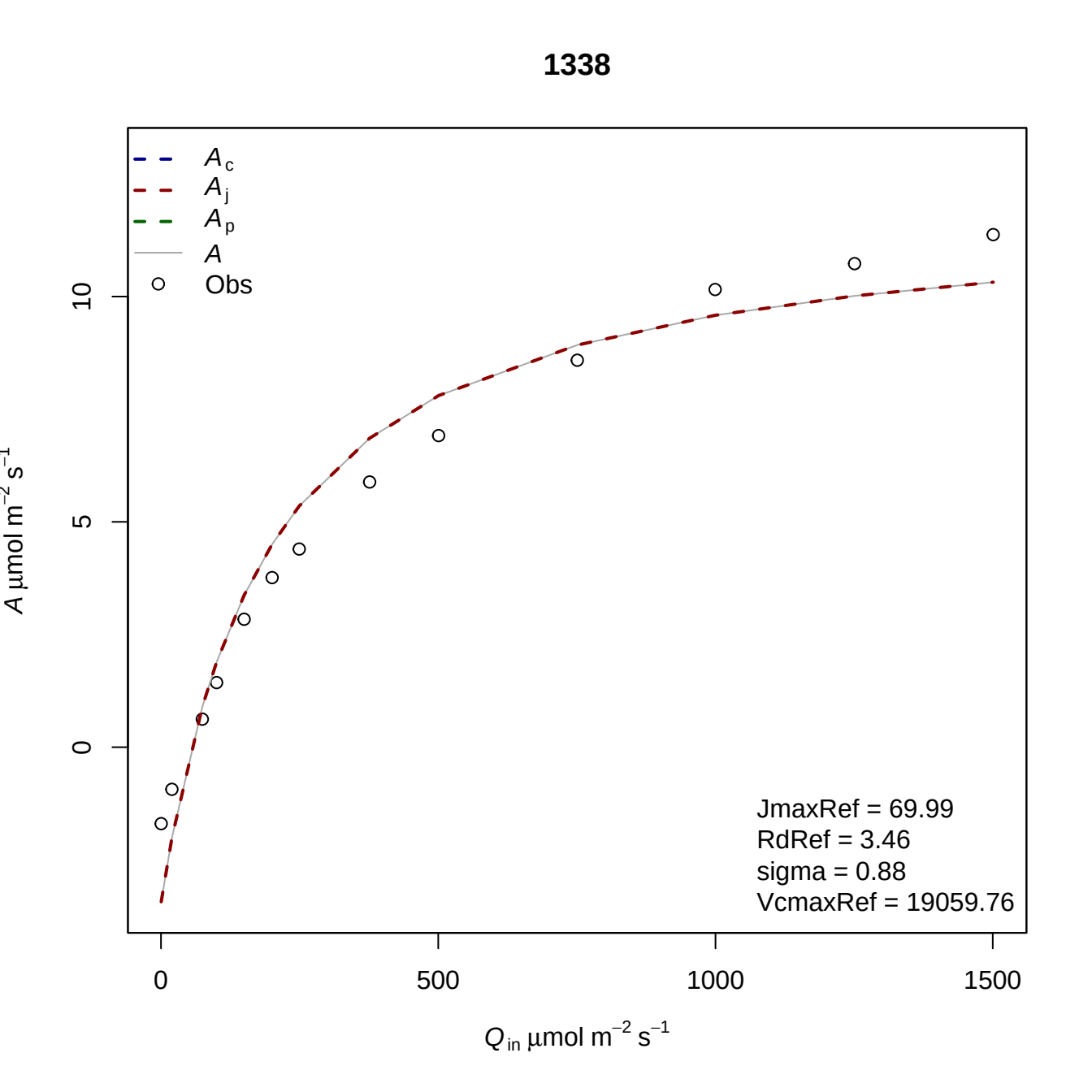

1333
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.8
RdRef = 3.18
sigma = 0.66
VcmaxRef = 24211.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

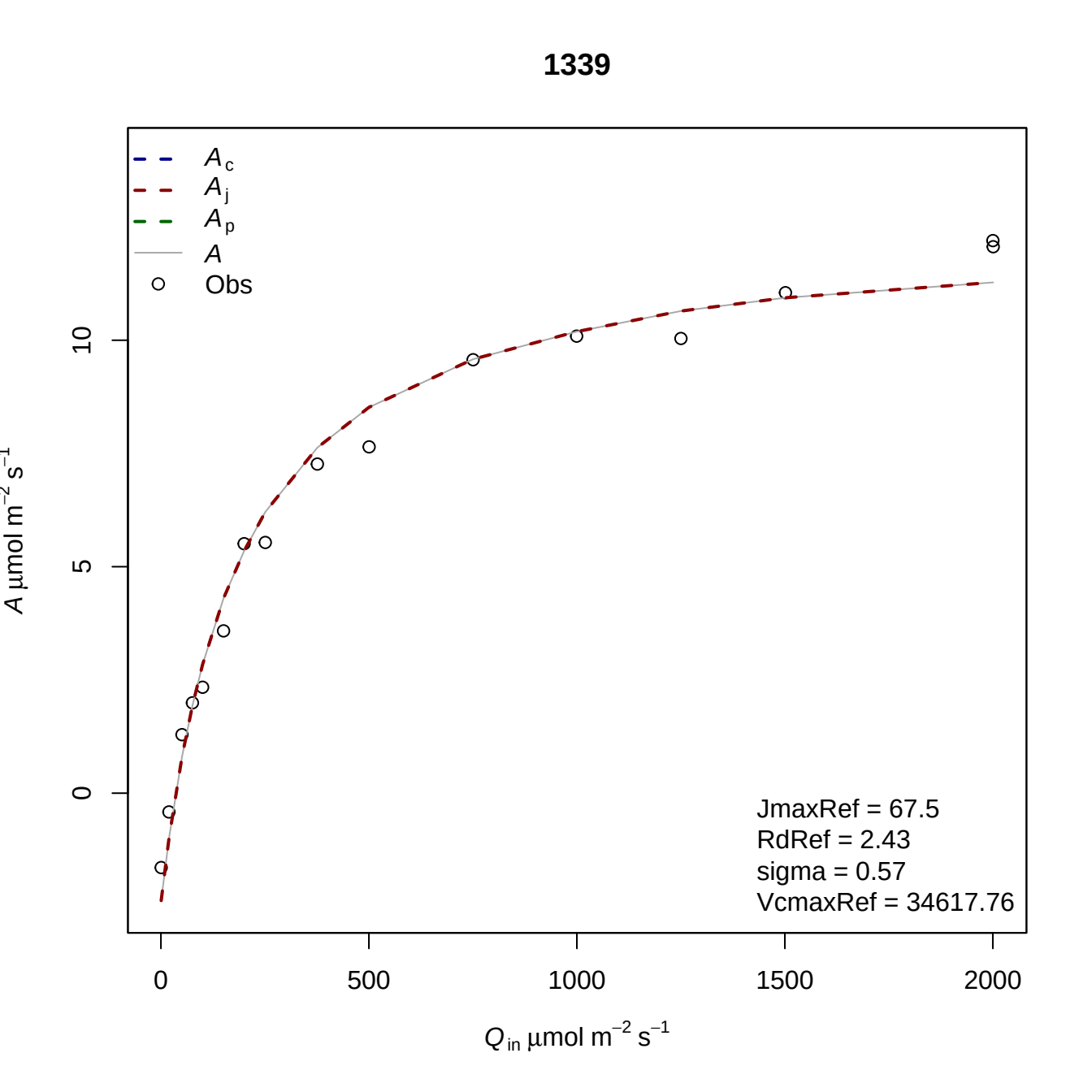

1334
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 54.33
RdRef = 3.65
sigma = 1.03
VcmaxRef = 37166.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

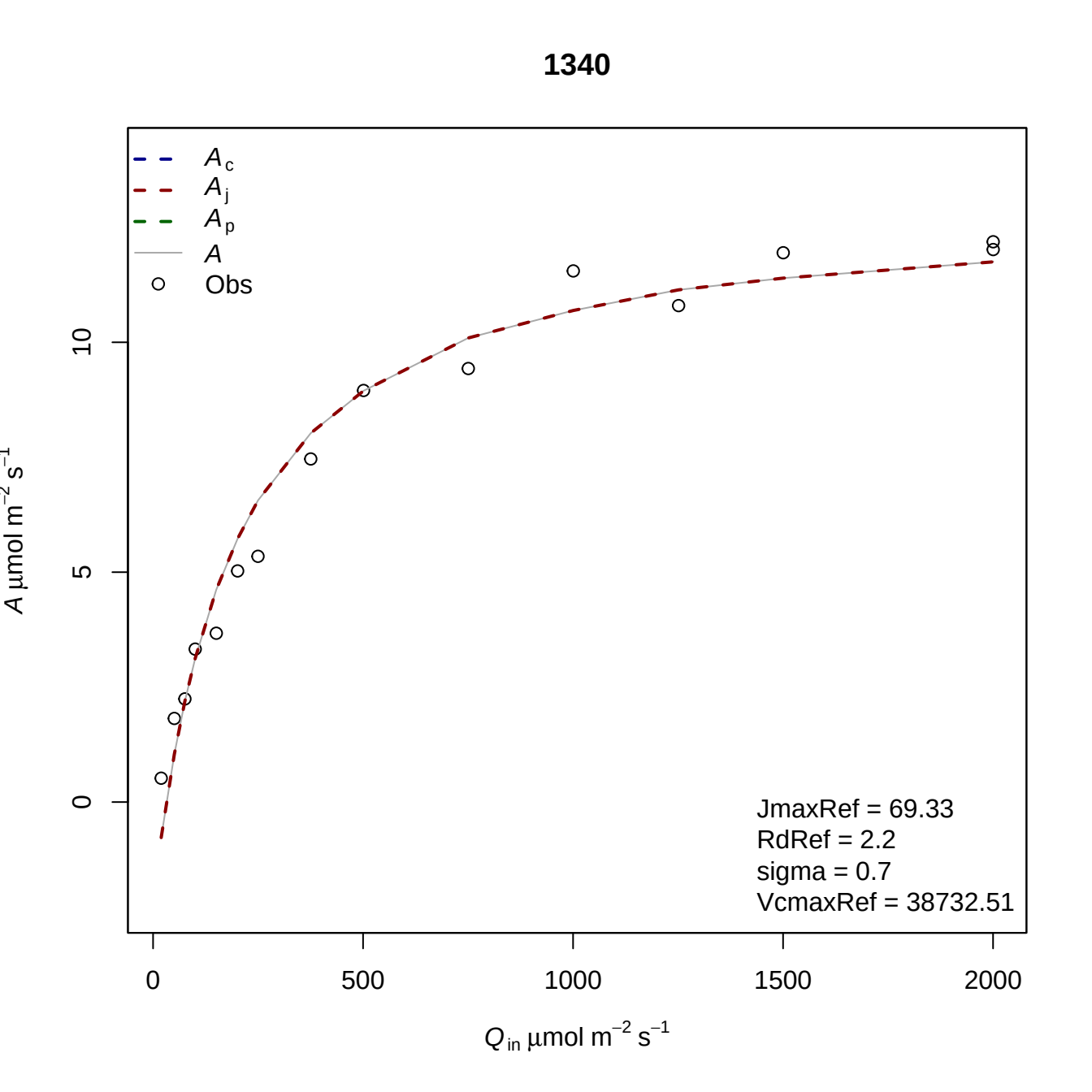

1335
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.74
RdRef = 3.54
sigma = 0.66
VcmaxRef = 14034.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500
2000

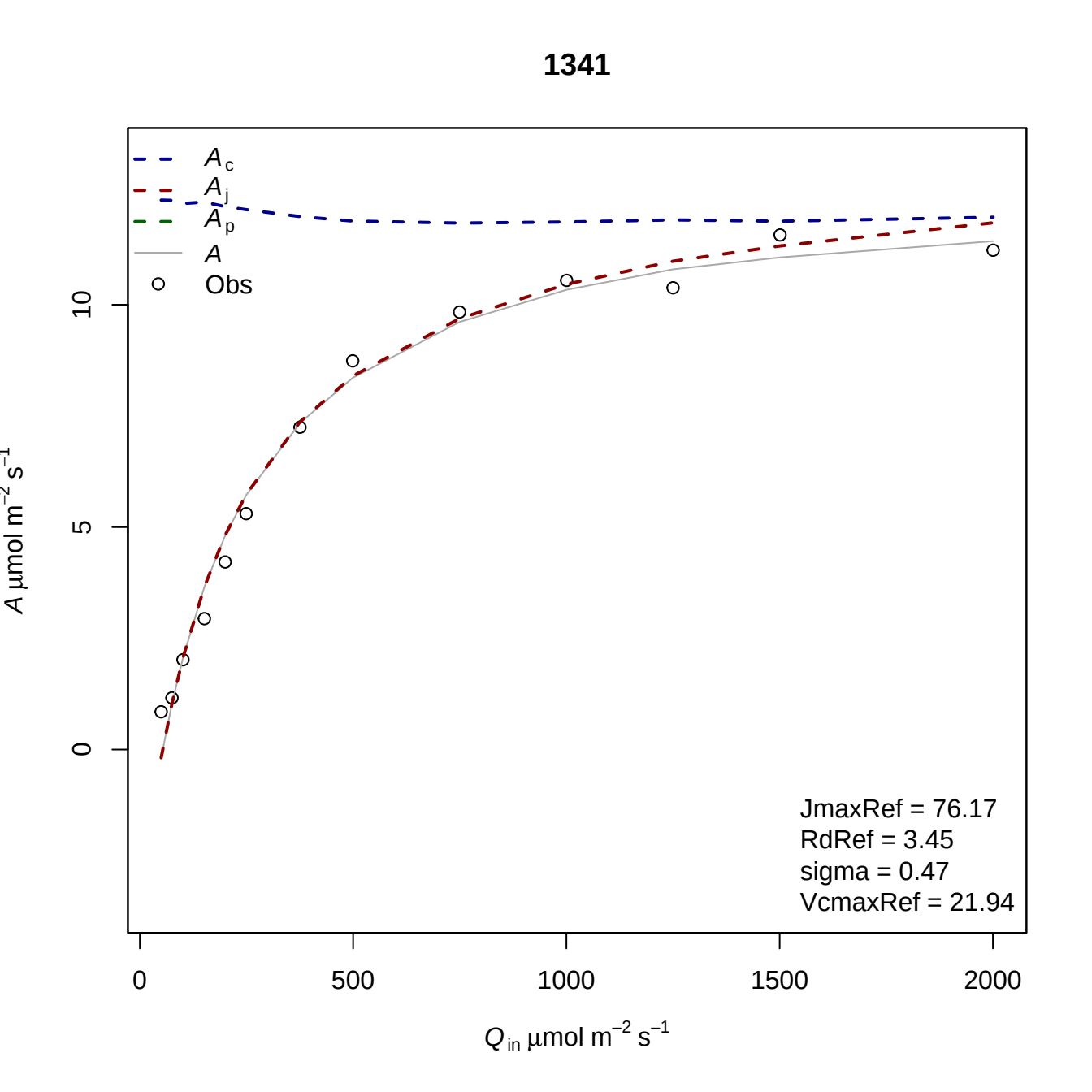

1336
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.99
RdRef = 3.12
sigma = 0.9
VcmaxRef = 4136.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

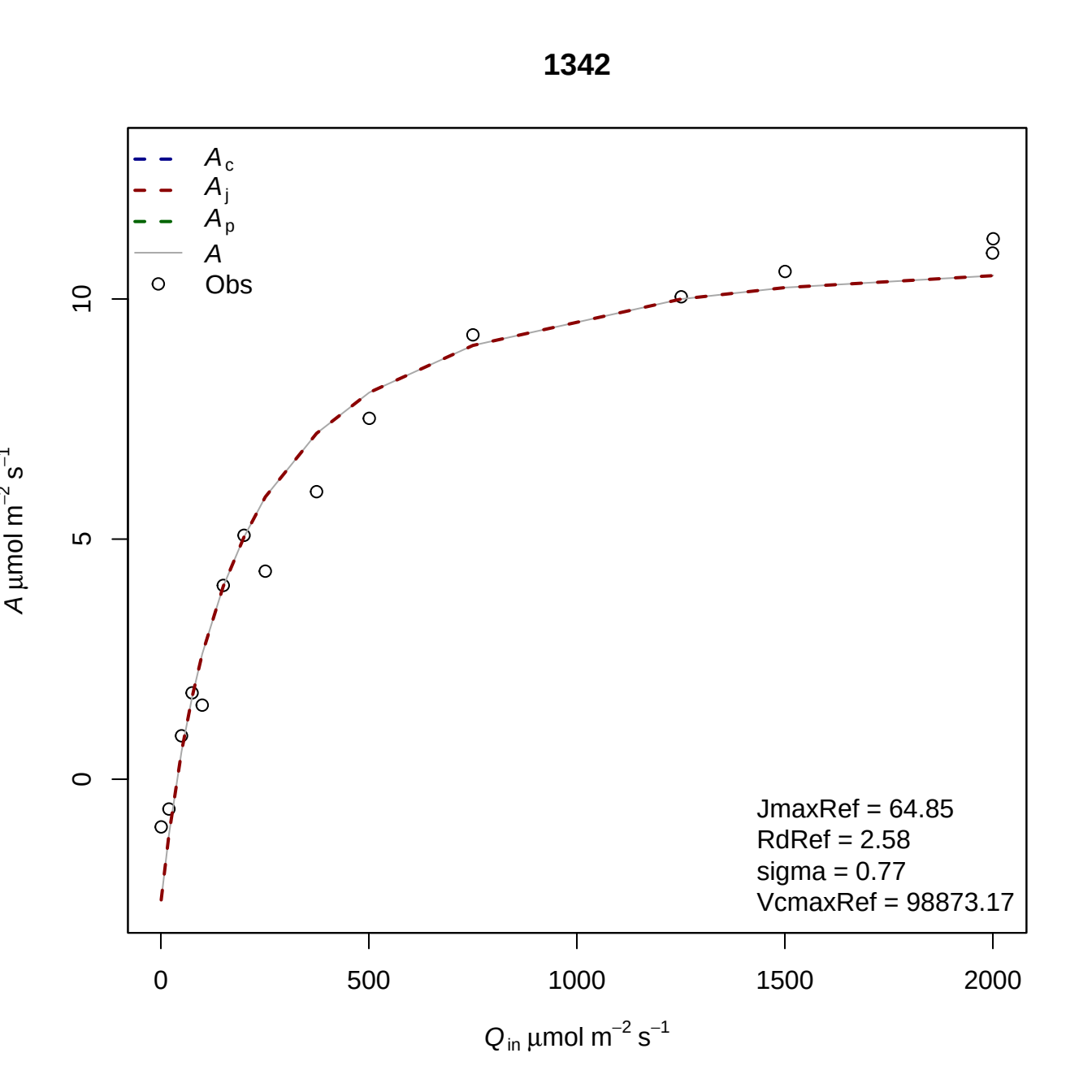

1337
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.89
RdRef = 2.8
sigma = 0.72
VcmaxRef = 26120.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

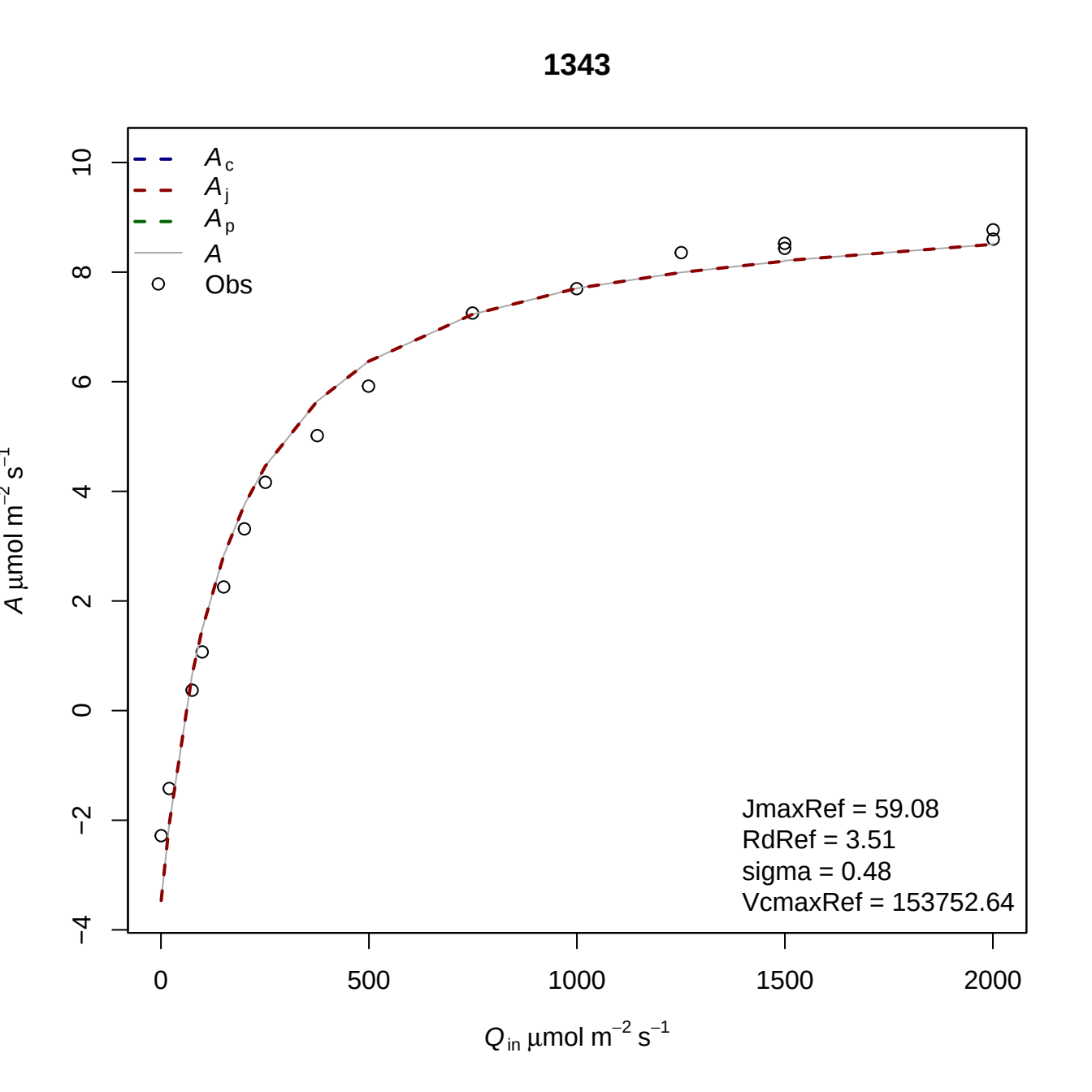

1338
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.99
RdRef = 3.46
sigma = 0.88
VcmaxRef = 19059.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

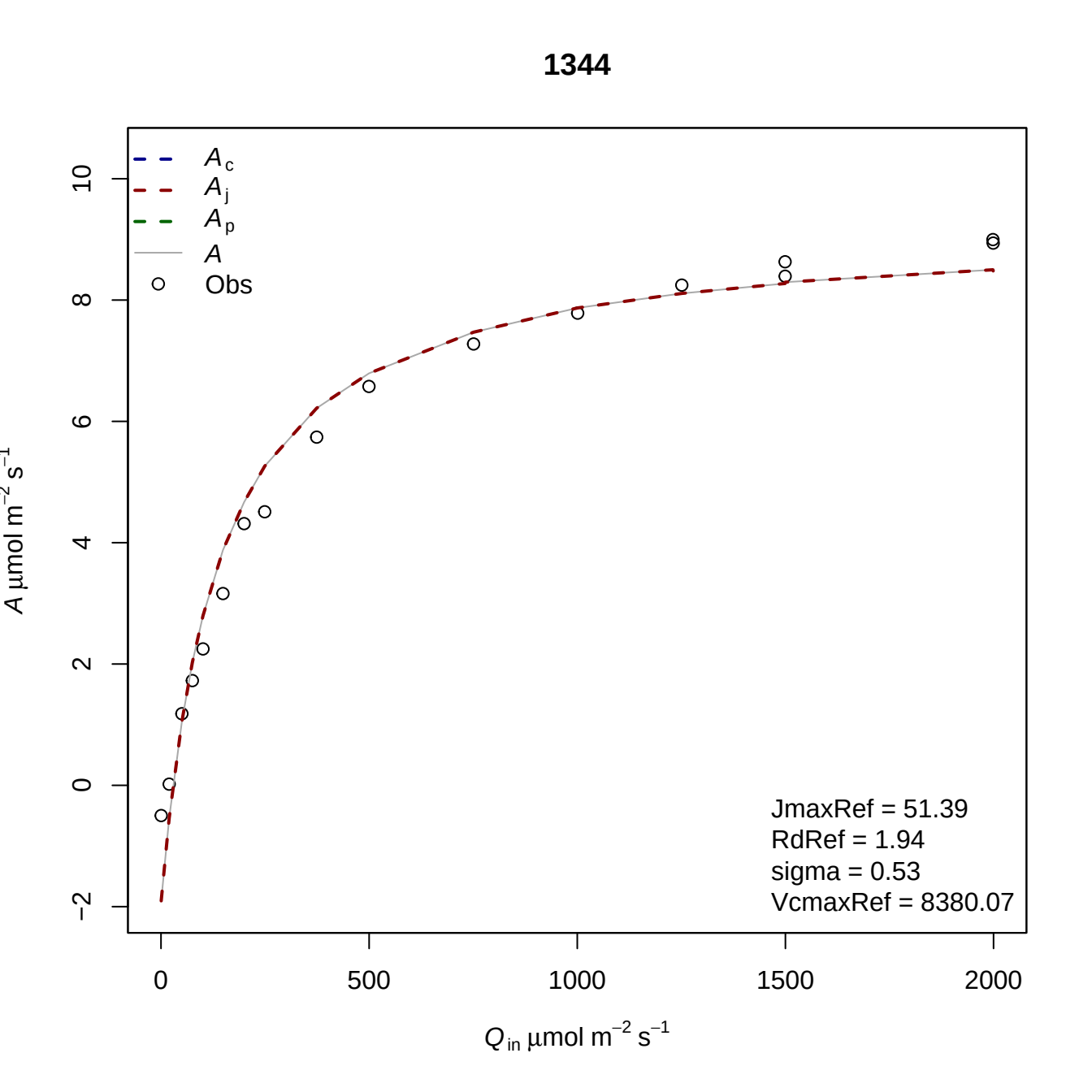

1339
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.5
RdRef = 2.43
sigma = 0.57
VcmaxRef = 34617.76
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

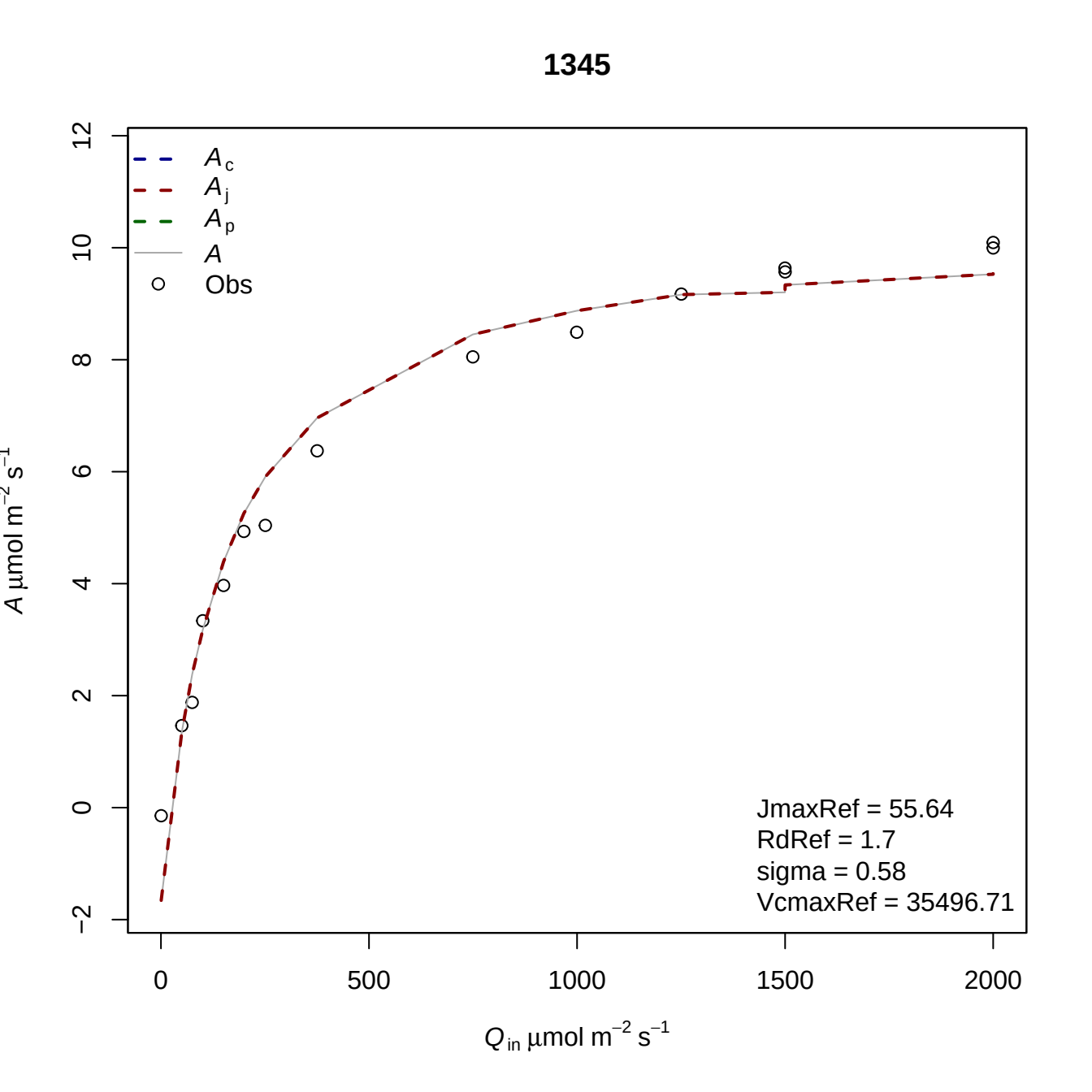

1340
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.33
RdRef = 2.2
sigma = 0.7
VcmaxRef = 38732.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

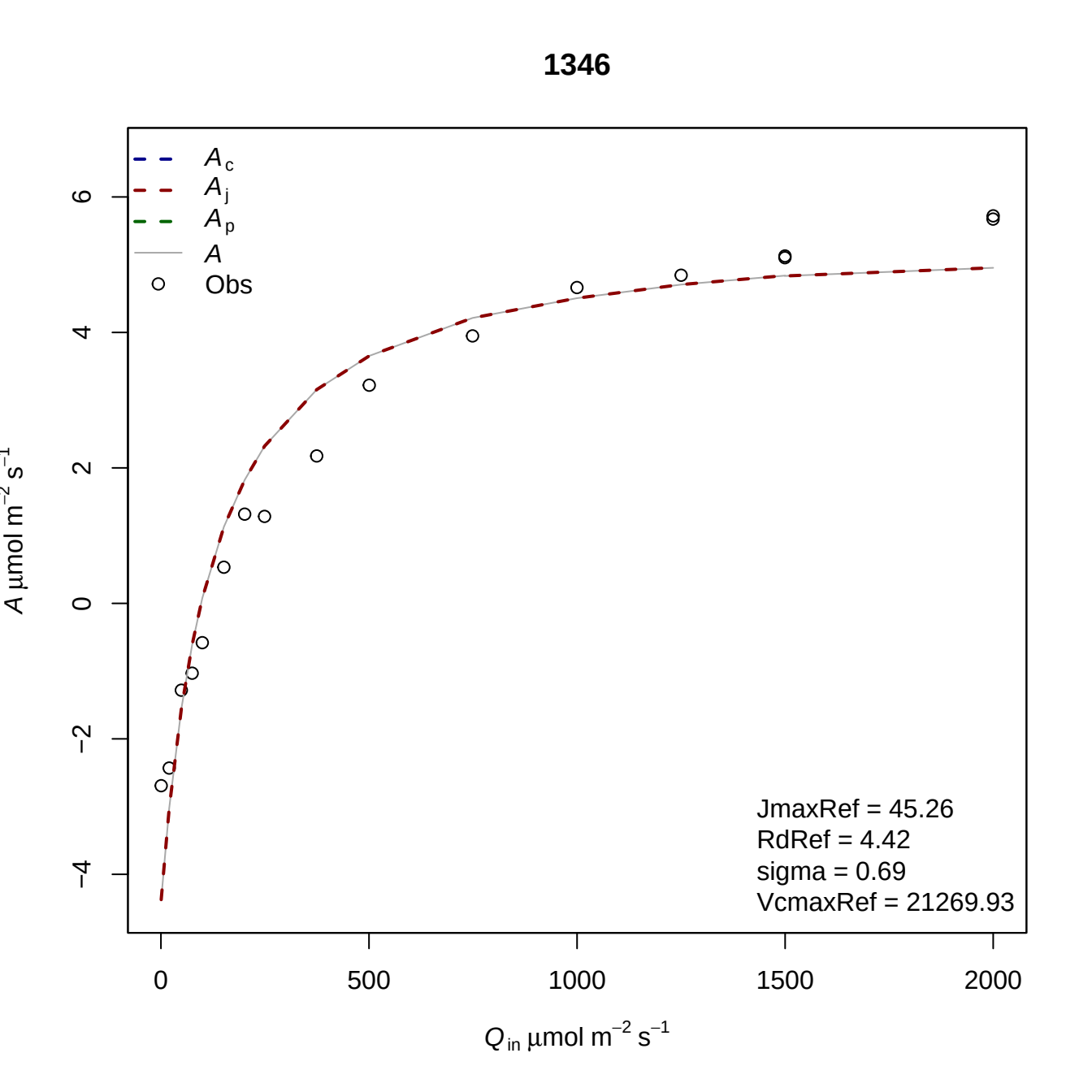

1341
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.17
RdRef = 3.45
sigma = 0.47
VcmaxRef = 21.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

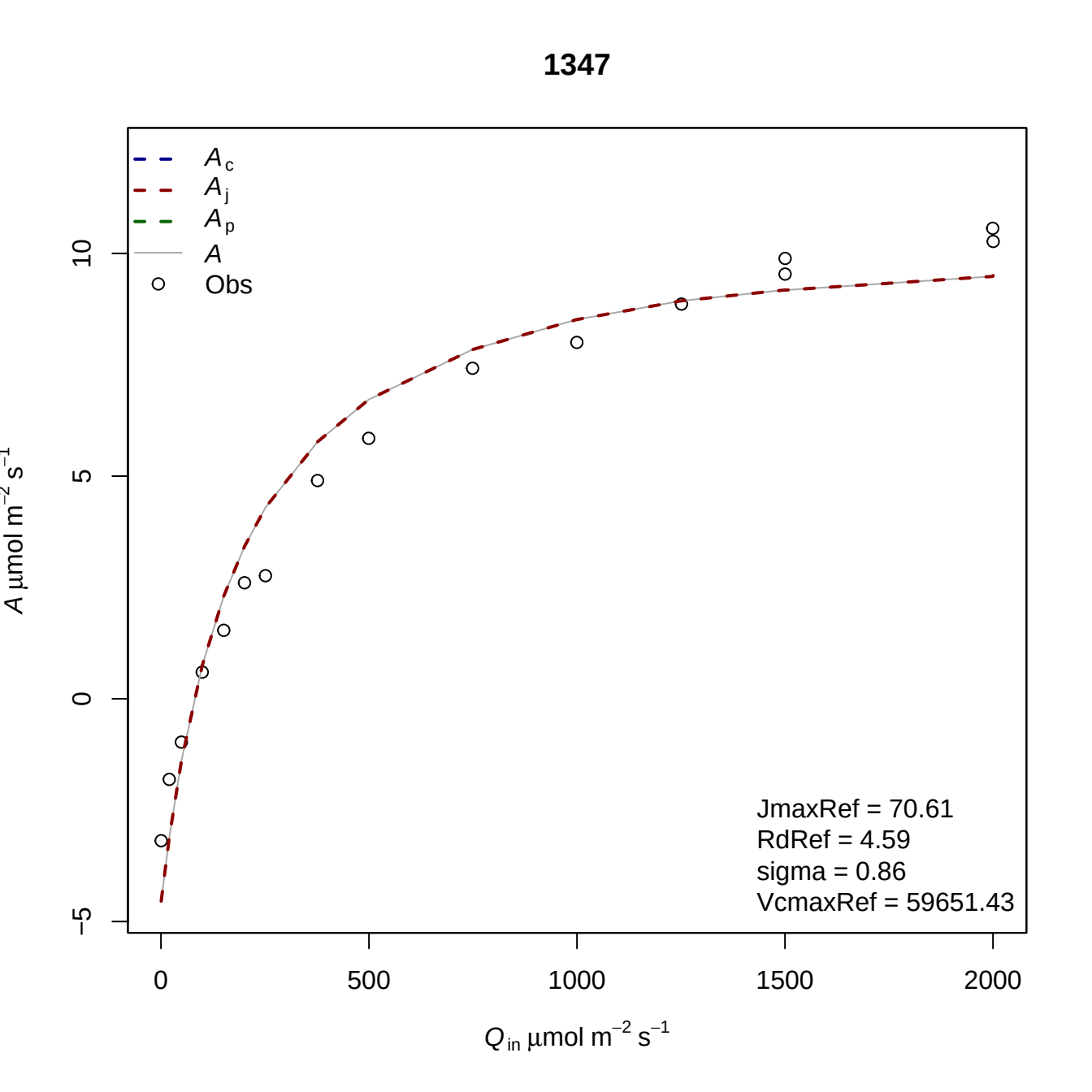

1342
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.85
RdRef = 2.58
sigma = 0.77
VcmaxRef = 98873.17
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

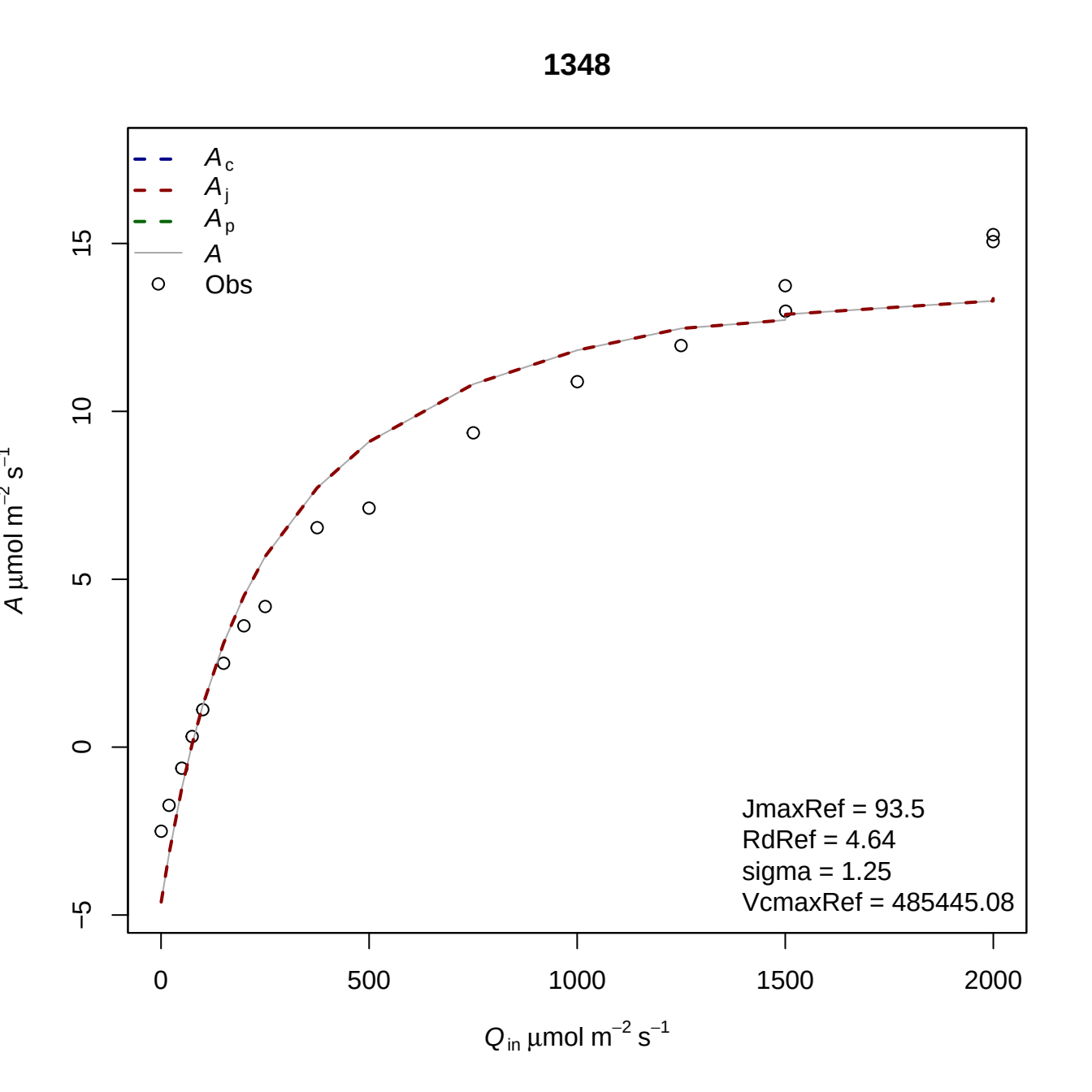

1343
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.08
RdRef = 3.51
sigma = 0.48
VcmaxRef = 153752.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

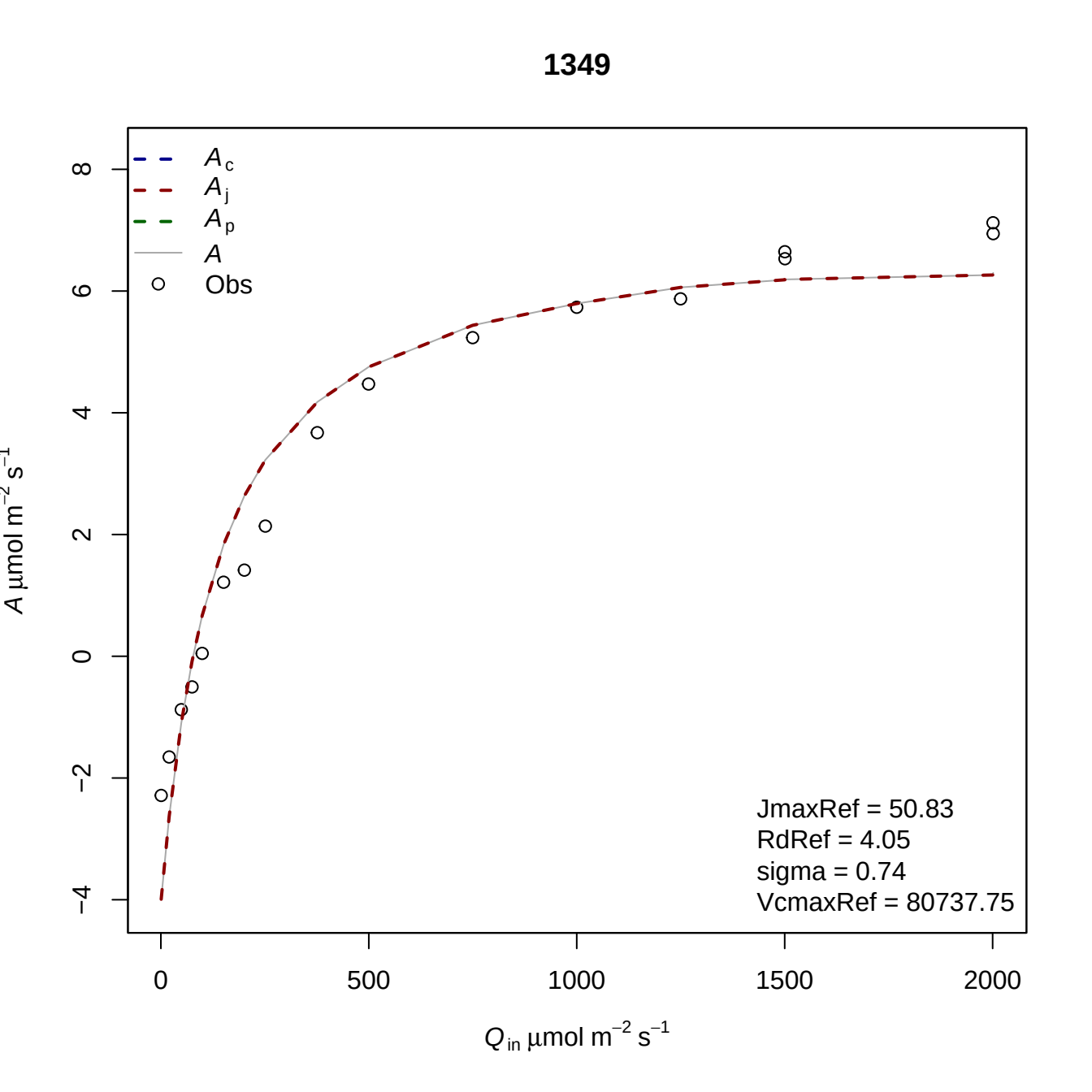

1344
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 51.39
RdRef = 1.94
sigma = 0.53
VcmaxRef = 8380.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500
2000

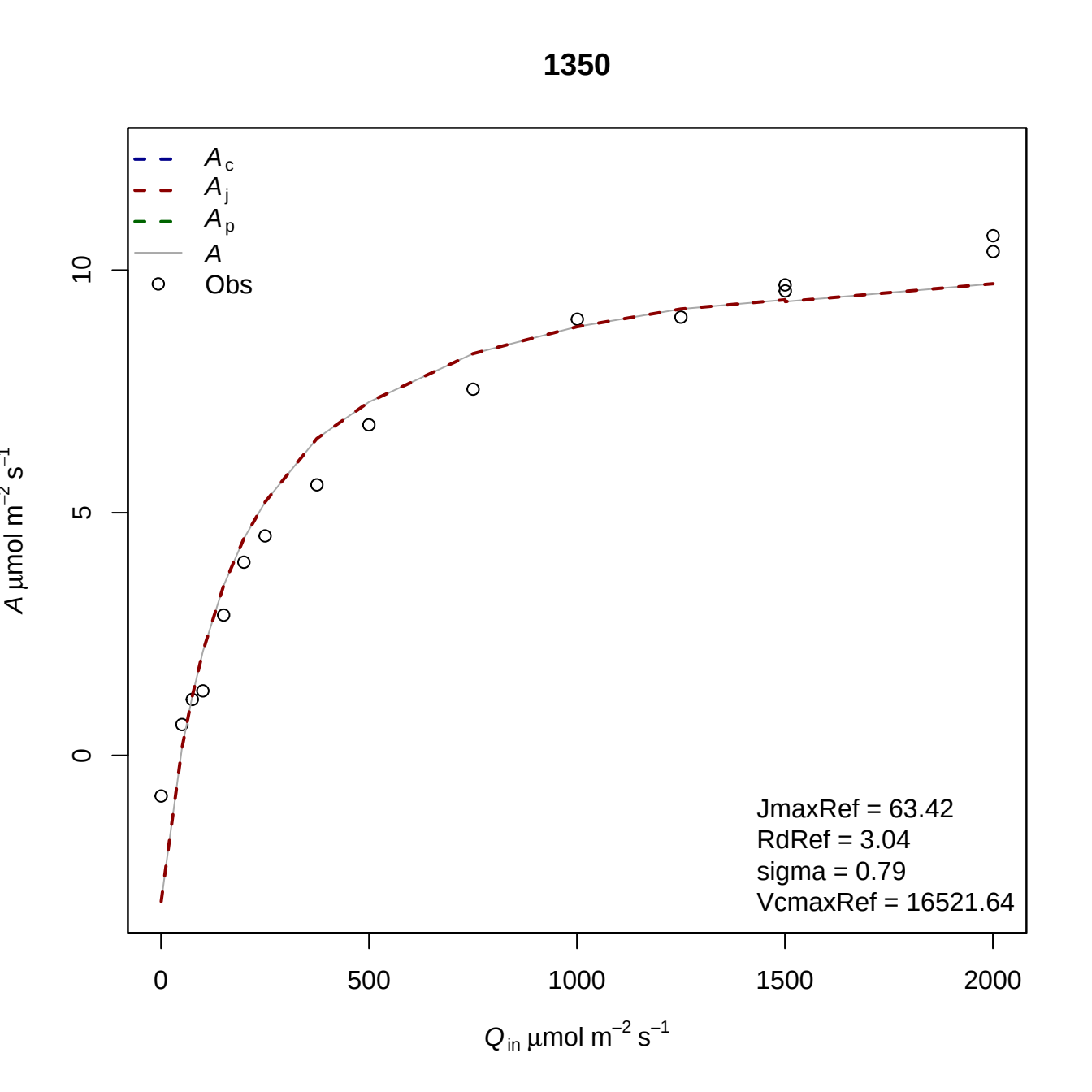

1345
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.64
RdRef = 1.7
sigma = 0.58
VcmaxRef = 35496.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

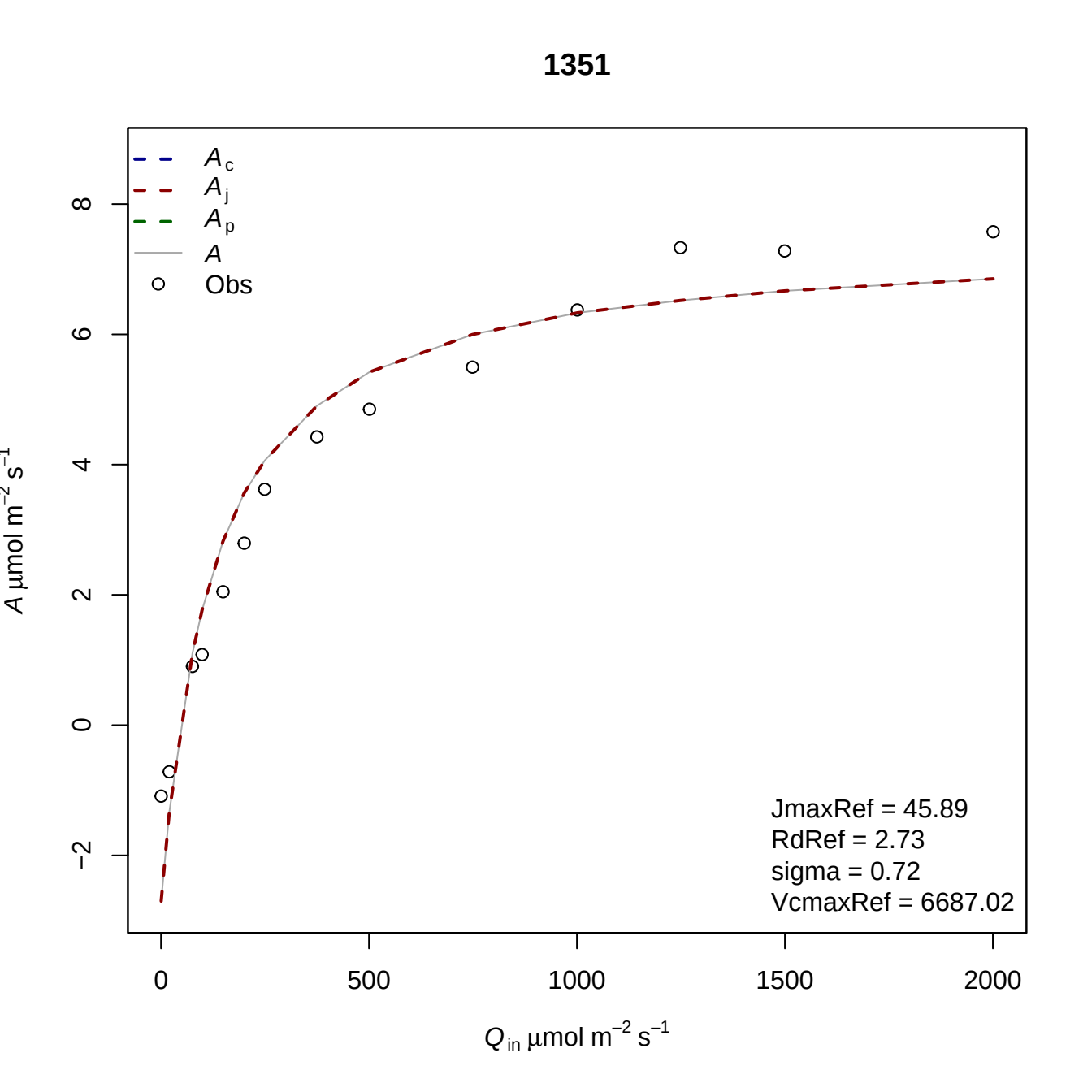

1346
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.26
RdRef = 4.42
sigma = 0.69
VcmaxRef = 21269.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500
2000

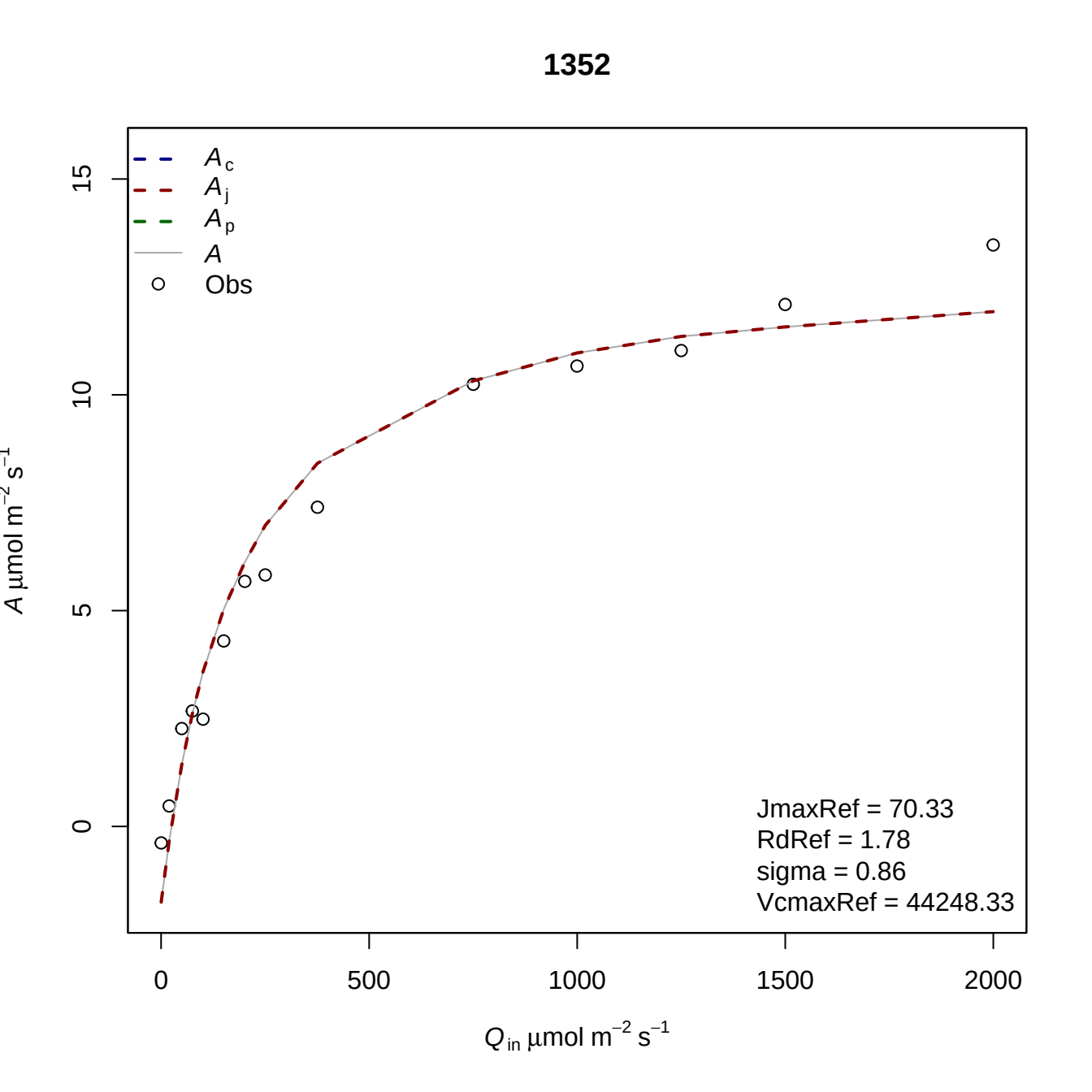

1347
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.61
RdRef = 4.59
sigma = 0.86
VcmaxRef = 59651.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

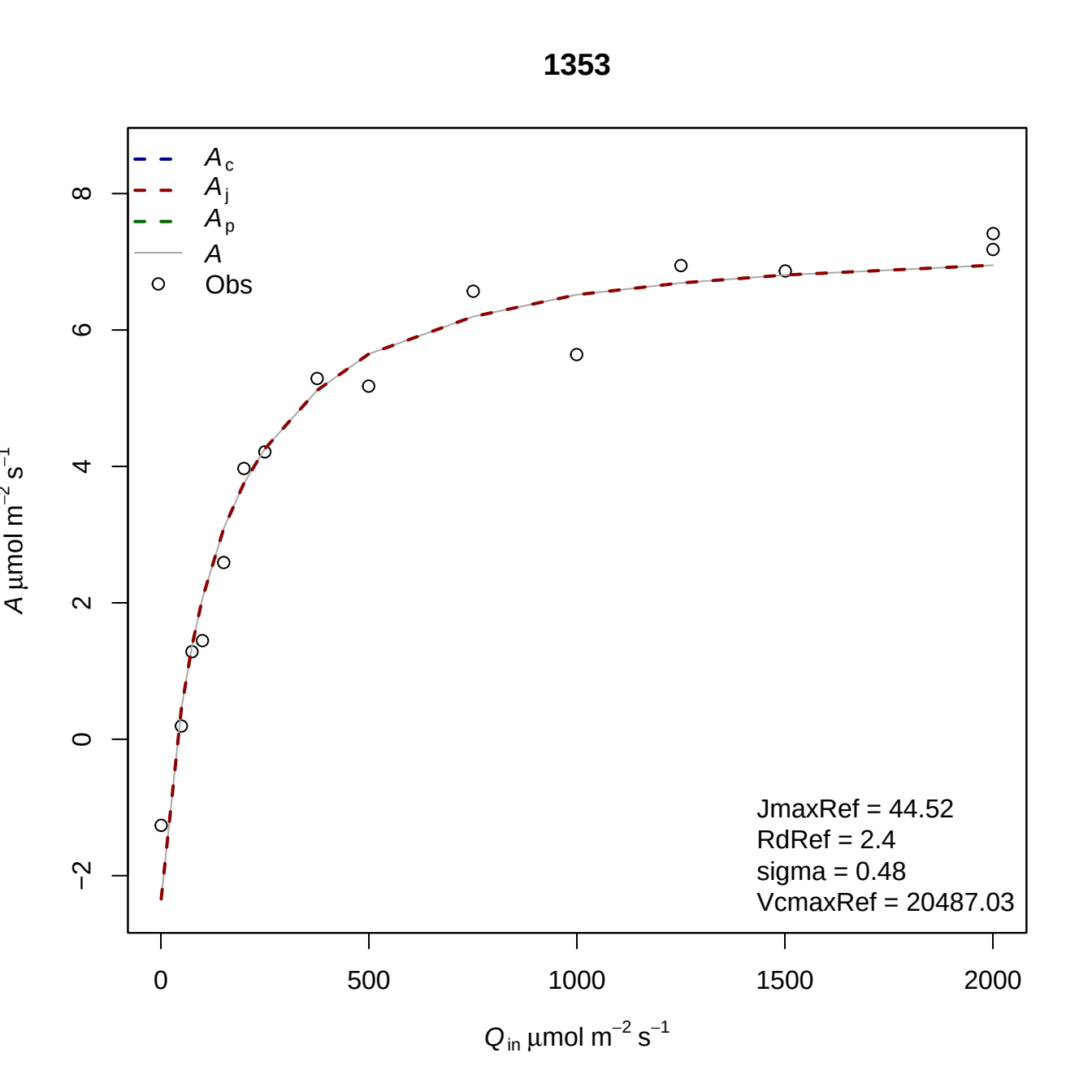

1348
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.5
RdRef = 4.64
sigma = 1.25
VcmaxRef = 485445.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
15
0
500
1000
1500
2000

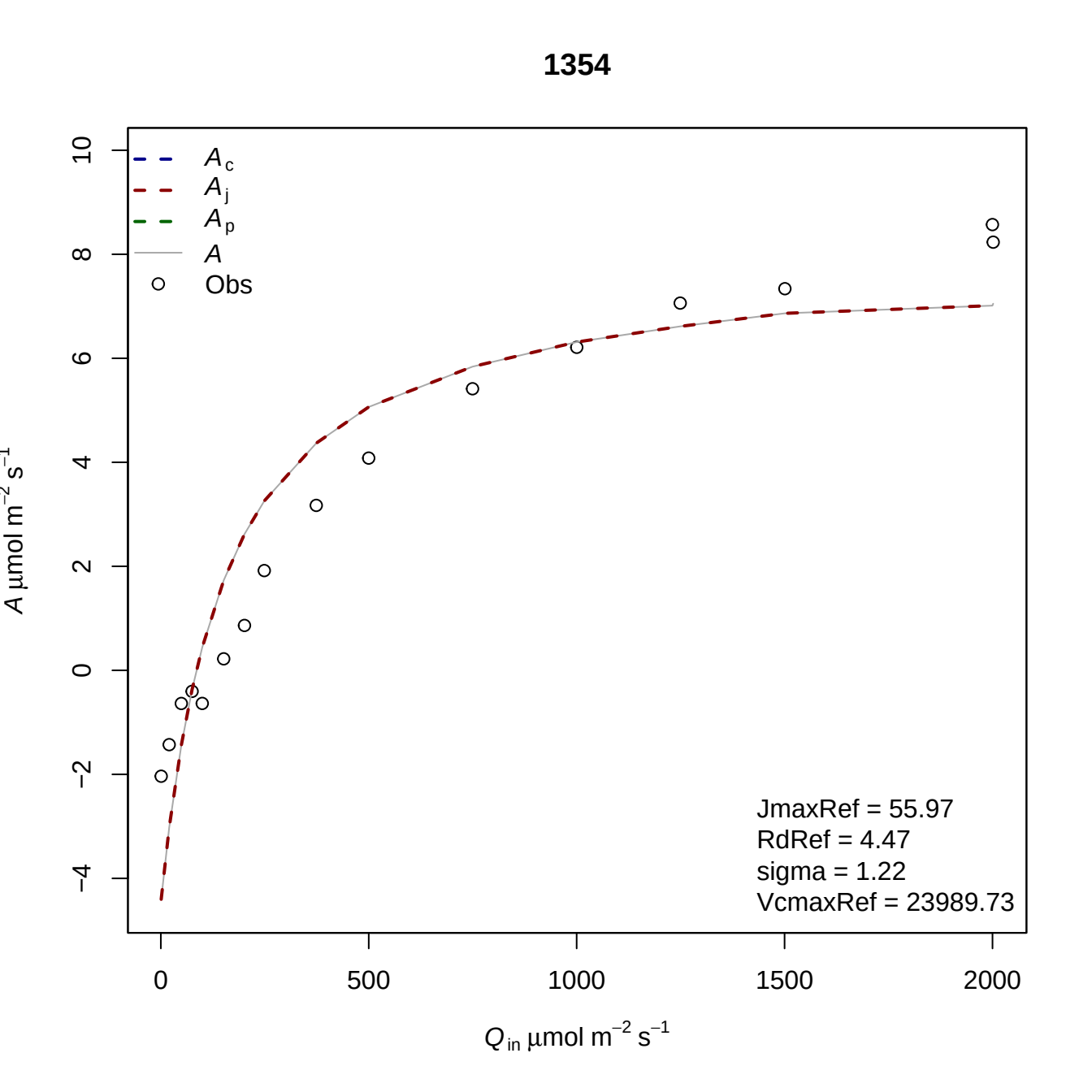

1349
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.83
RdRef = 4.05
sigma = 0.74
VcmaxRef = 80737.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

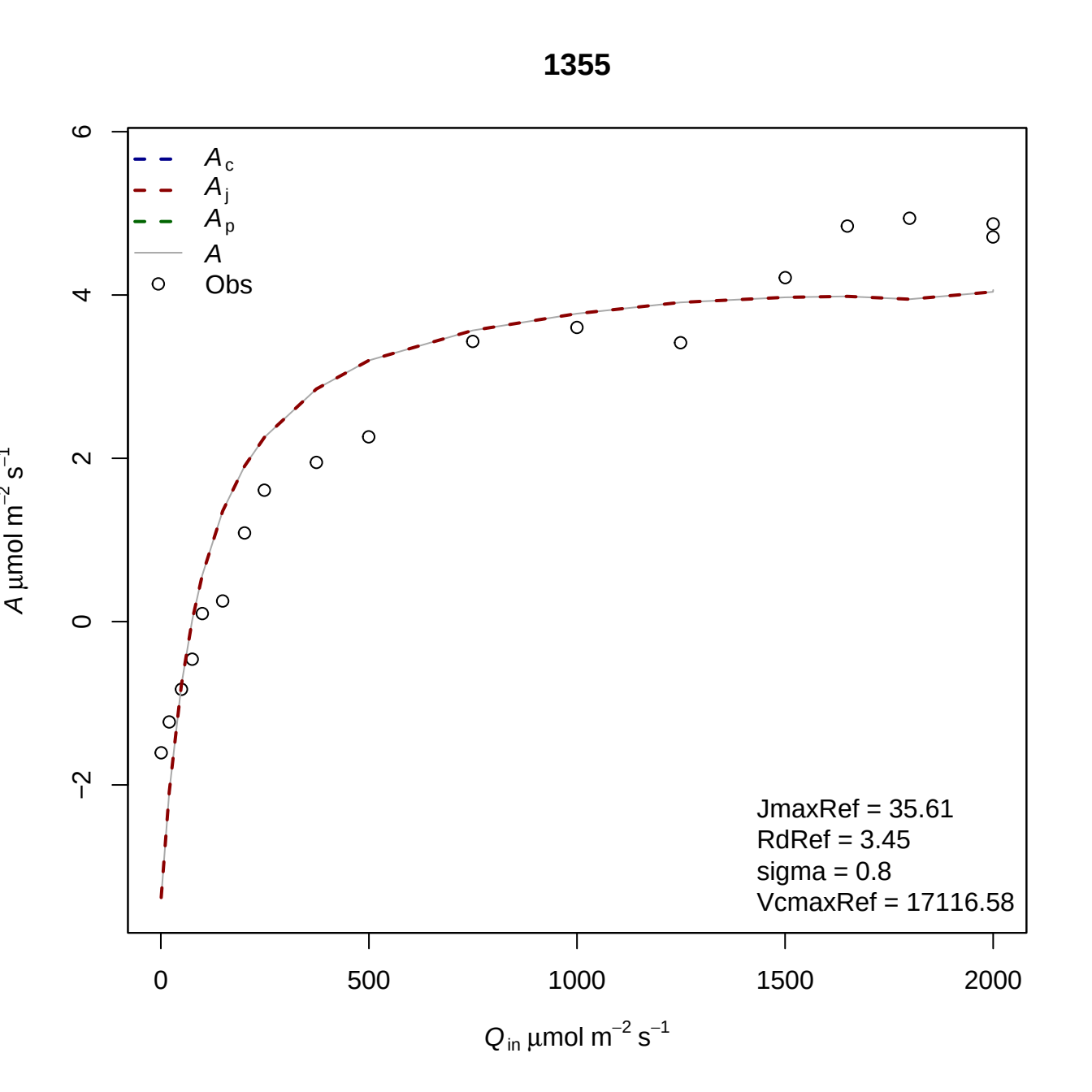

1350
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 63.42
RdRef = 3.04
sigma = 0.79
VcmaxRef = 16521.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

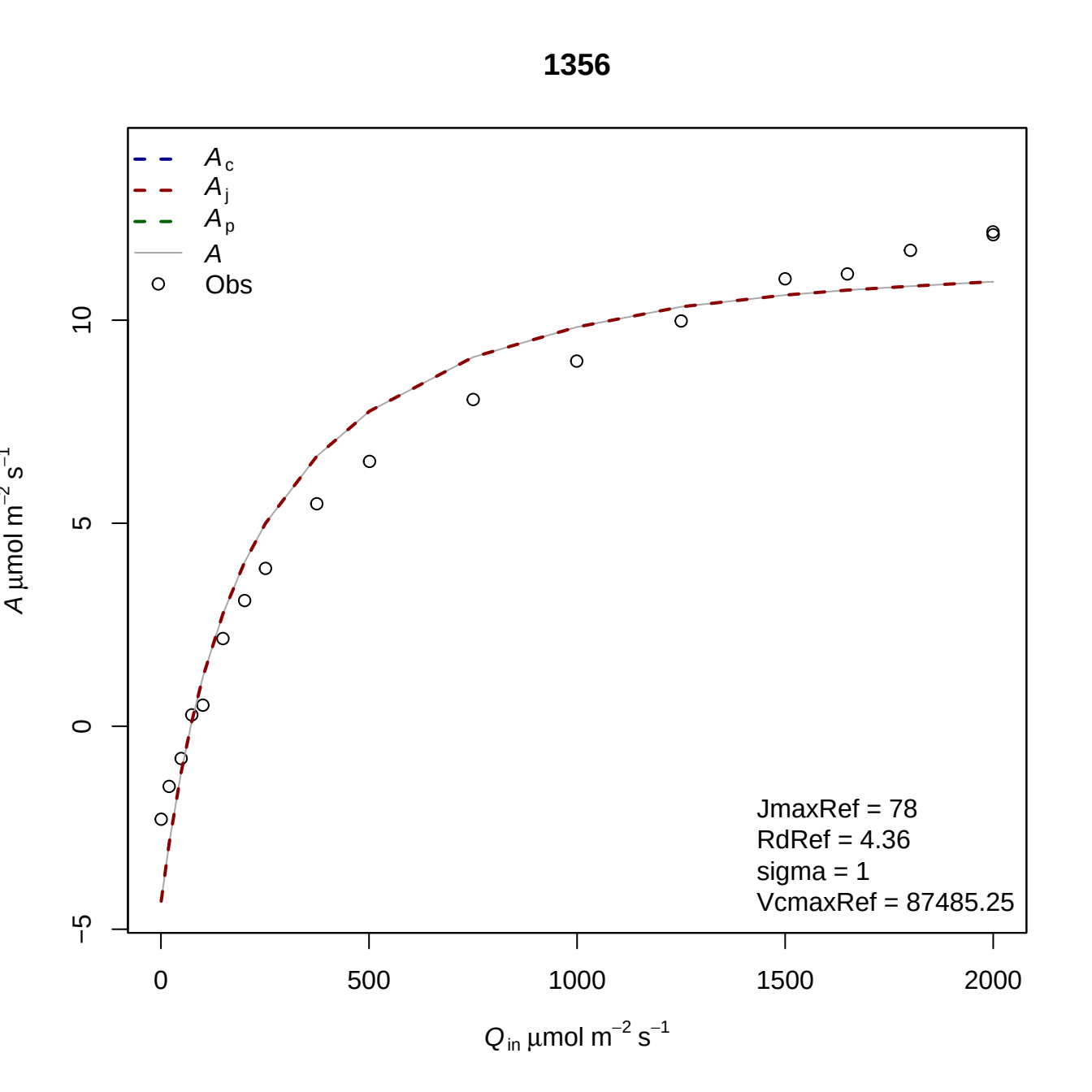

1351
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.89
RdRef = 2.73
sigma = 0.72
VcmaxRef = 6687.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

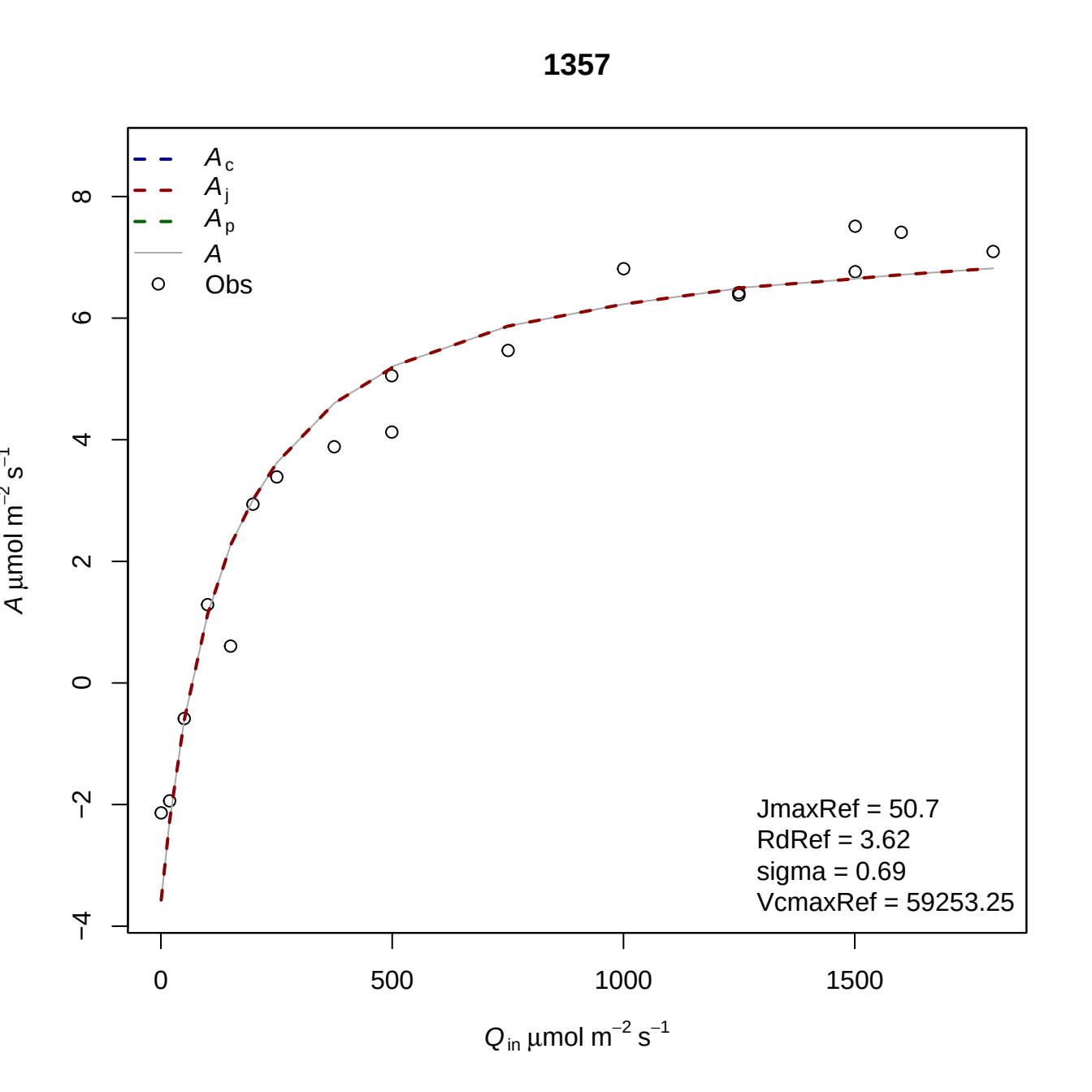

1352
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 70.33
RdRef = 1.78
sigma = 0.86
VcmaxRef = 44248.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

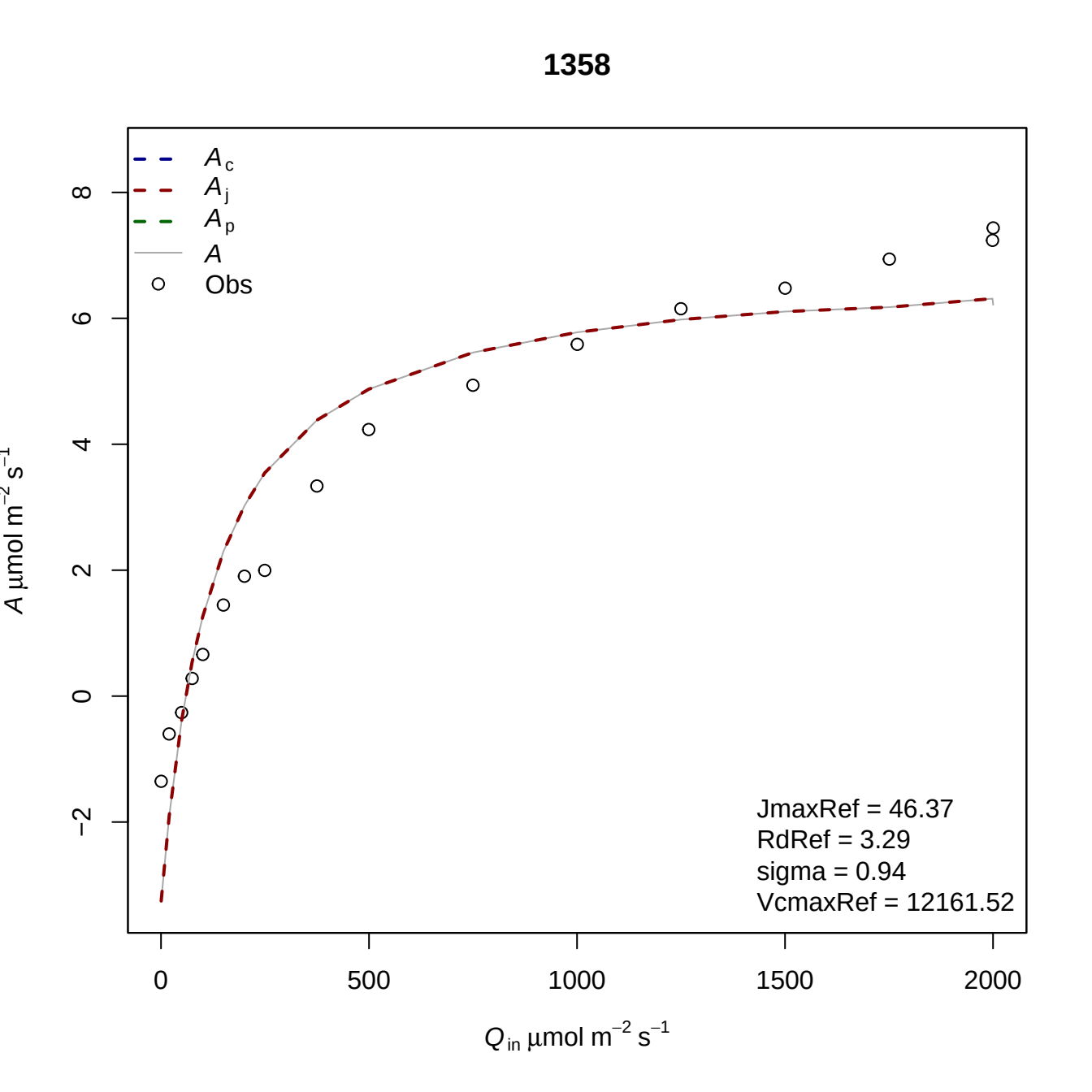

1353
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.52
RdRef = 2.4
sigma = 0.48
VcmaxRef = 20487.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

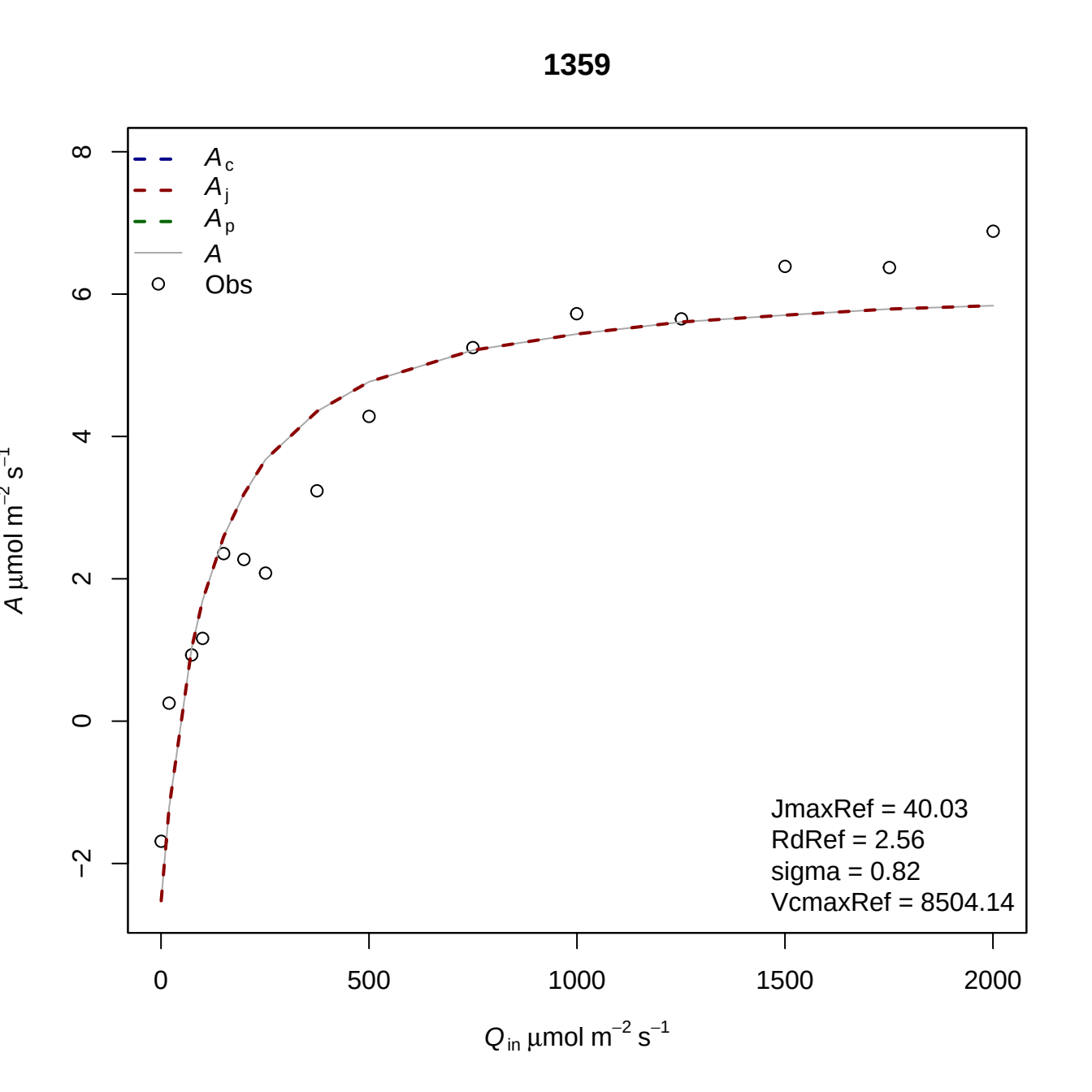

1354
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.97
RdRef = 4.47
sigma = 1.22
VcmaxRef = 23989.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

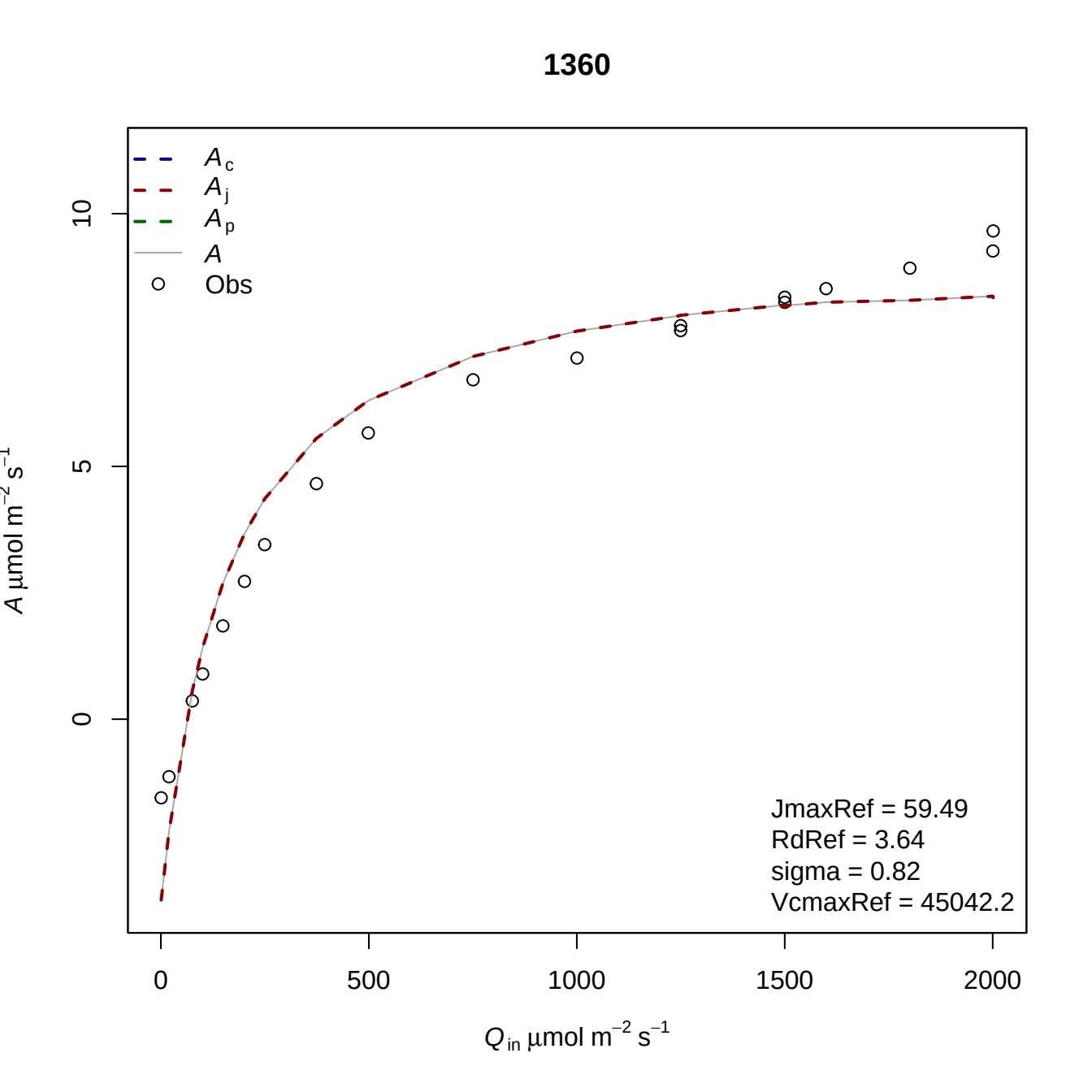

1355
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 35.61
RdRef = 3.45
sigma = 0.8
VcmaxRef = 17116.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
500
1000
1500
2000

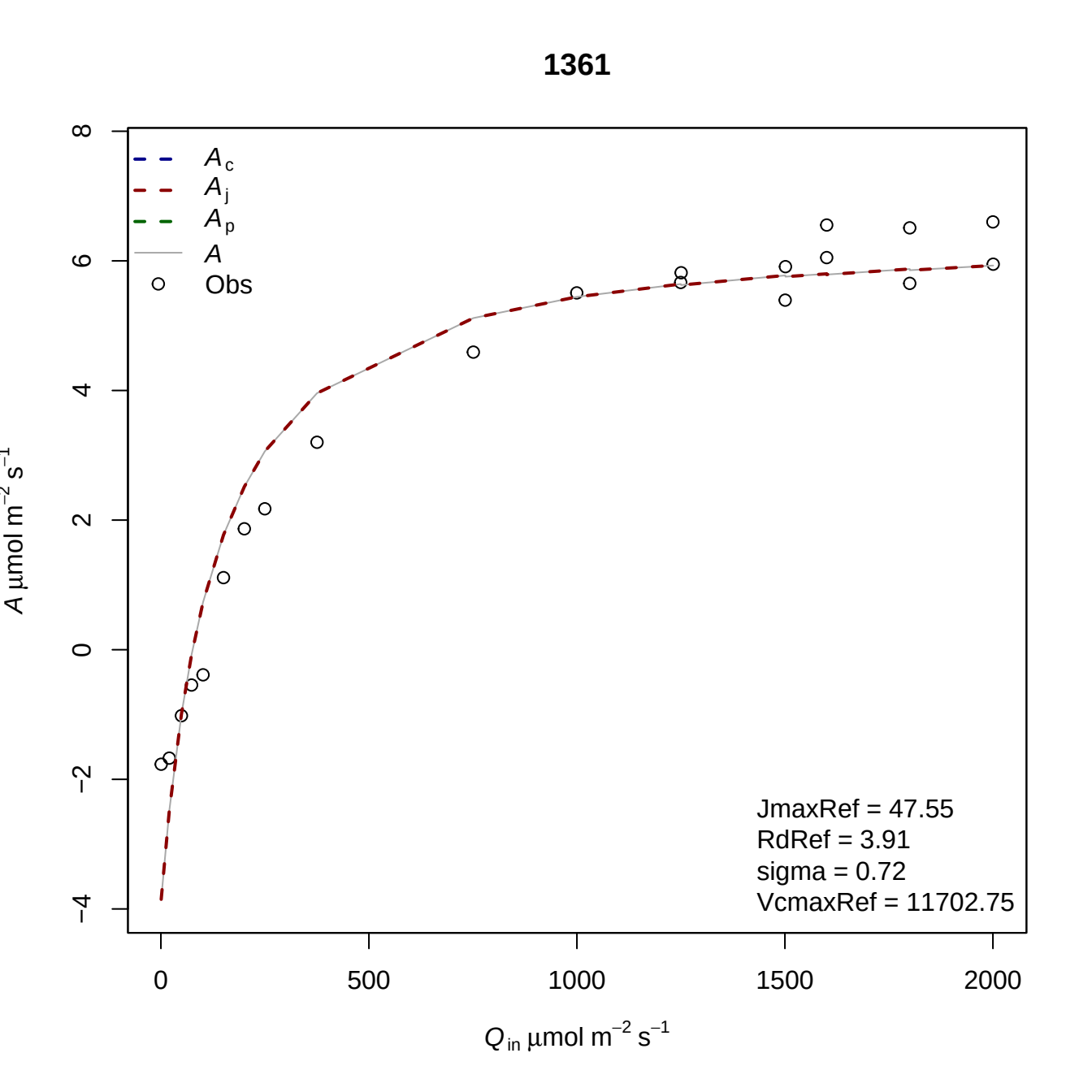

1356
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78
RdRef = 4.36
sigma = 1
VcmaxRef = 87485.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500
2000

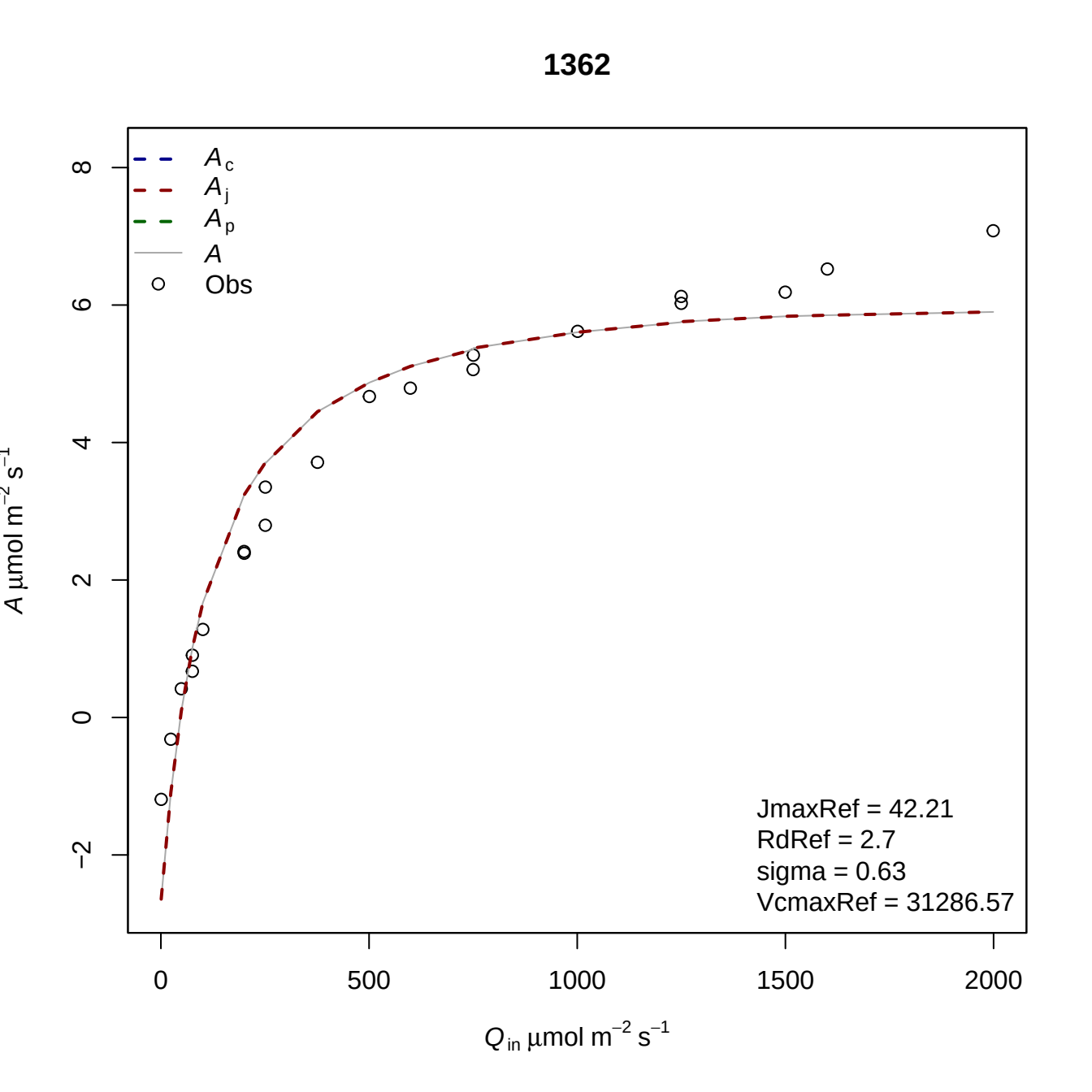

1357
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 50.7
RdRef = 3.62
sigma = 0.69
VcmaxRef = 59253.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

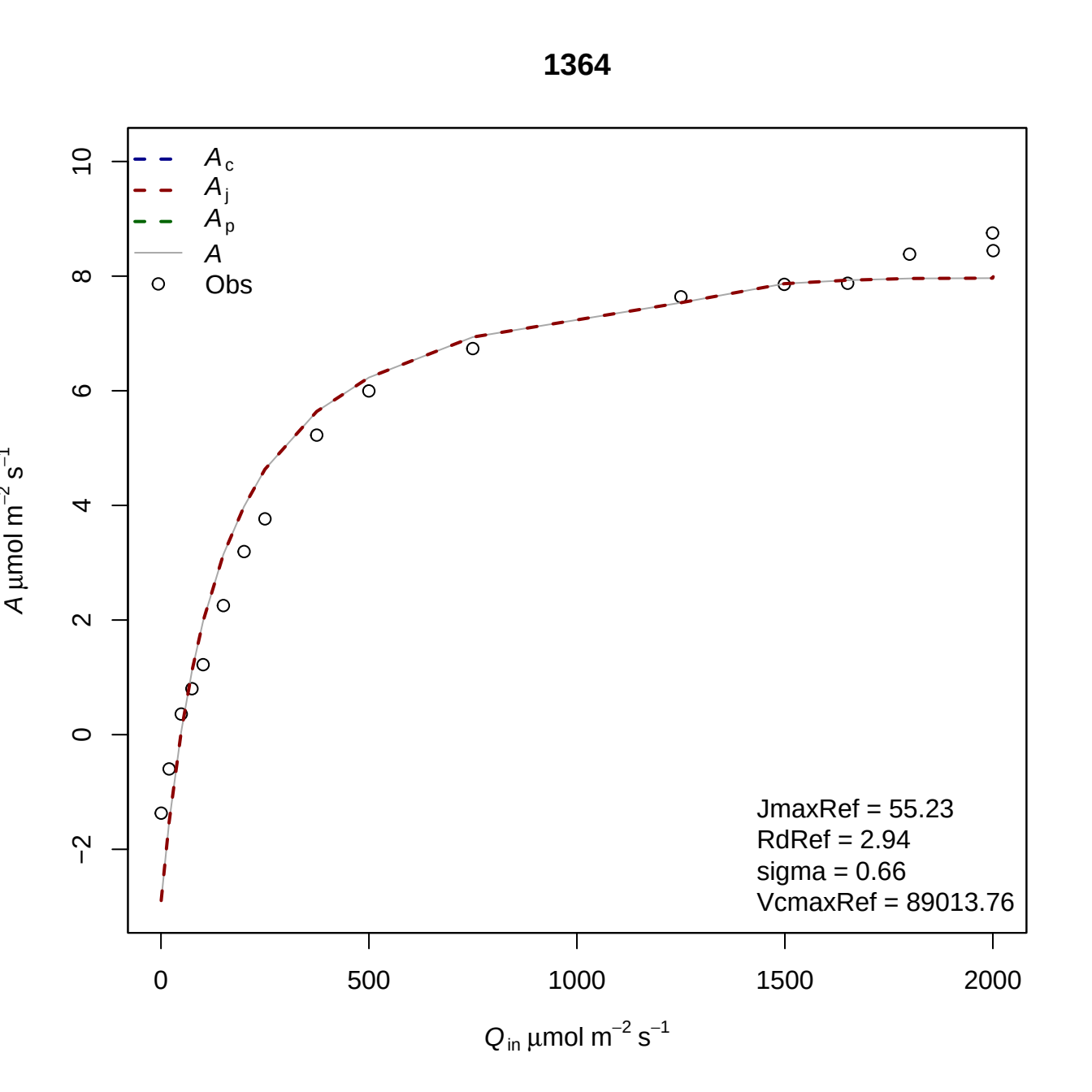

1358
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.37
RdRef = 3.29
sigma = 0.94
VcmaxRef = 12161.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500
2000

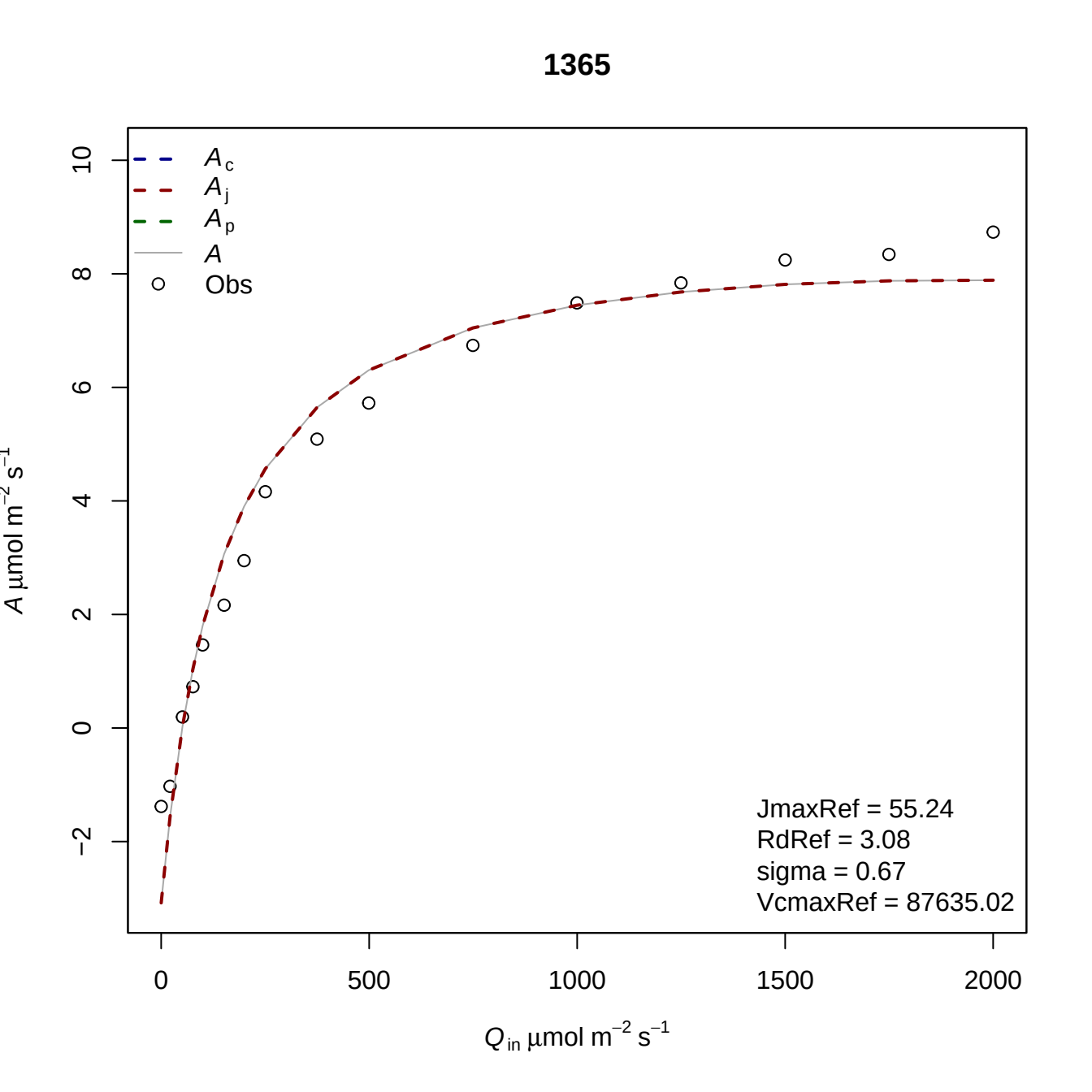

1359
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.03
RdRef = 2.56
sigma = 0.82
VcmaxRef = 8504.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

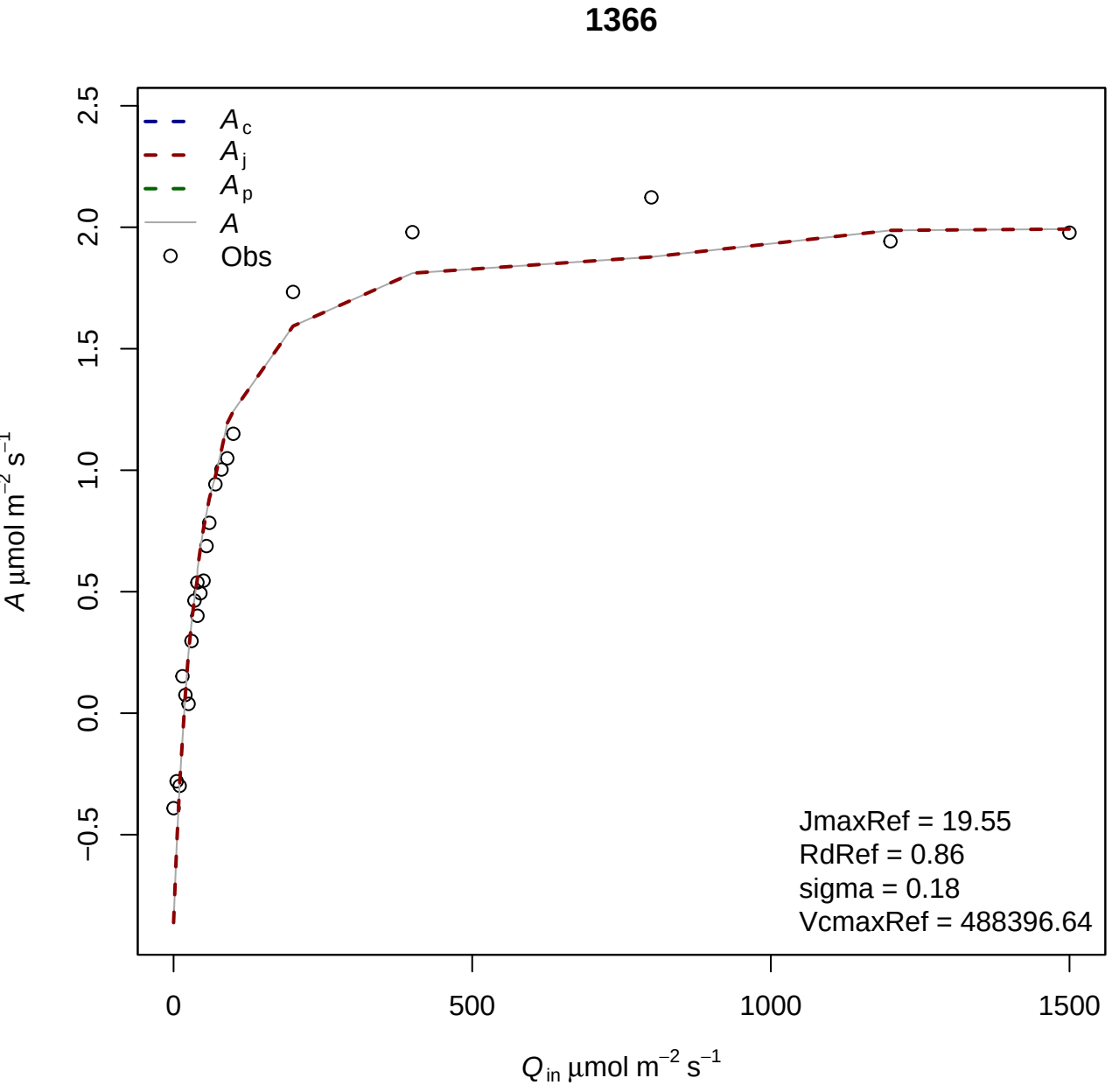

1360
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 59.49
RdRef = 3.64
sigma = 0.82
VcmaxRef = 45042.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

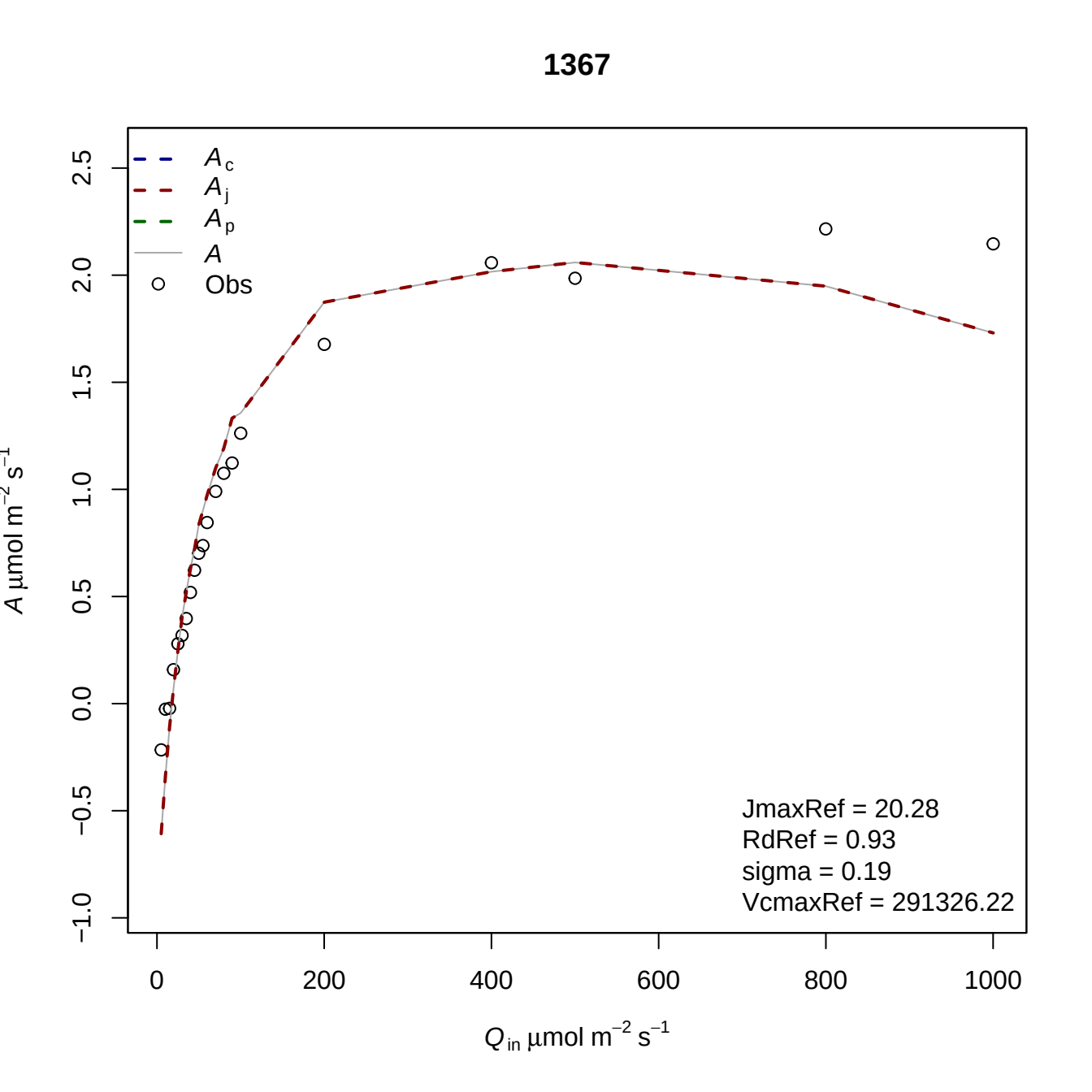

1361
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.55
RdRef = 3.91
sigma = 0.72
VcmaxRef = 11702.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

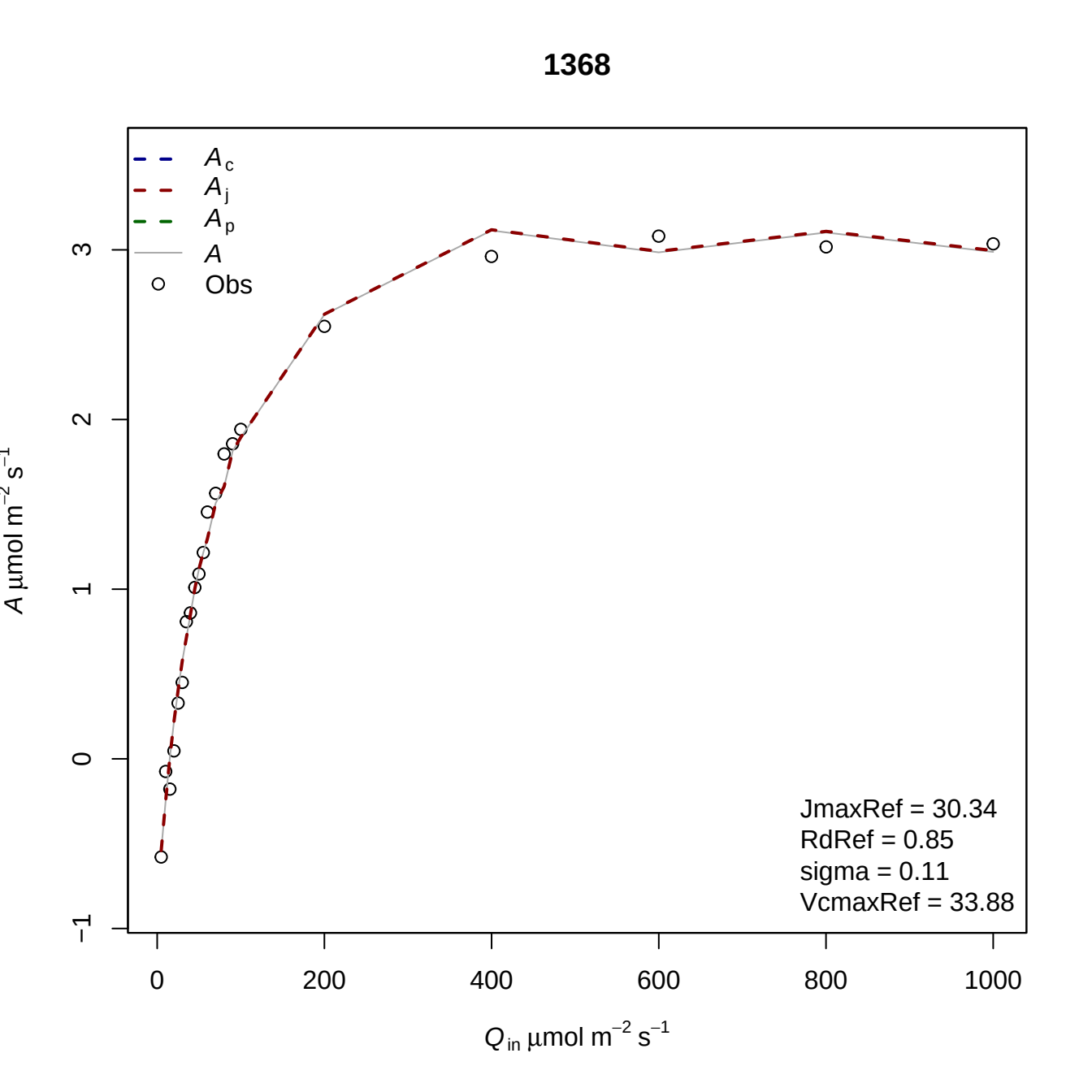

1362
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.21
RdRef = 2.7
sigma = 0.63
VcmaxRef = 31286.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500
2000

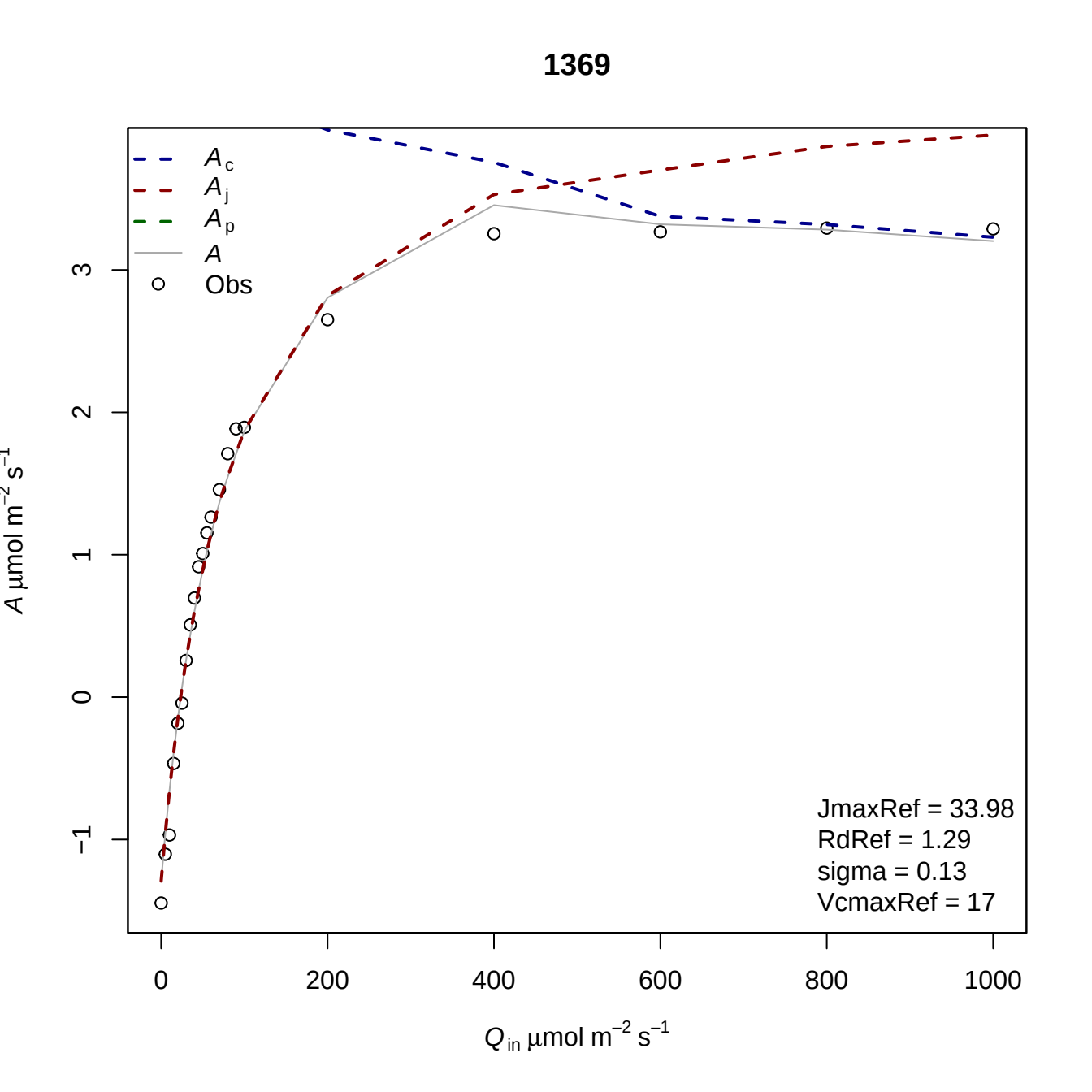

1364
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.23
RdRef = 2.94
sigma = 0.66
VcmaxRef = 89013.76
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

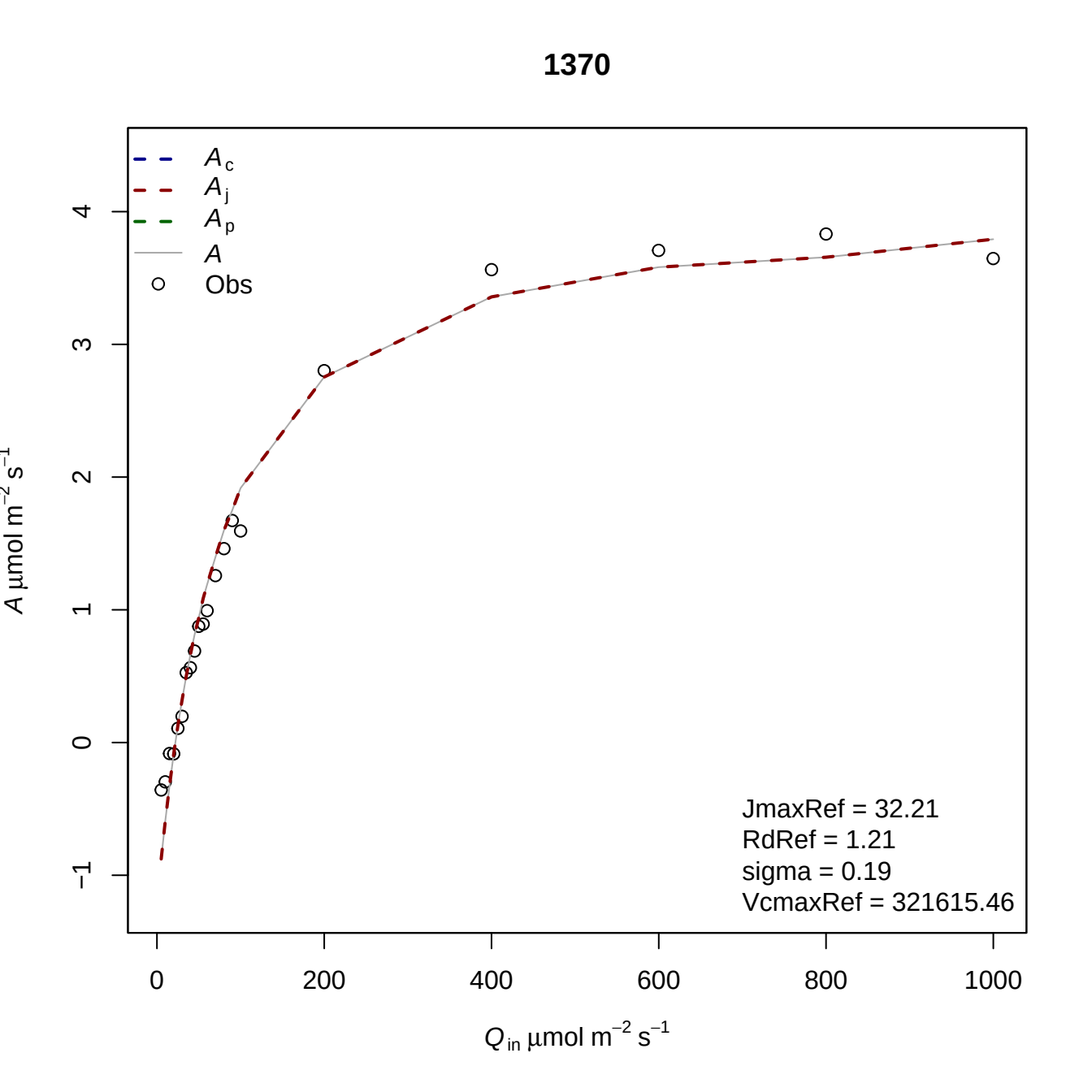

1365
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.24
RdRef = 3.08
sigma = 0.67
VcmaxRef = 87635.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500
2000

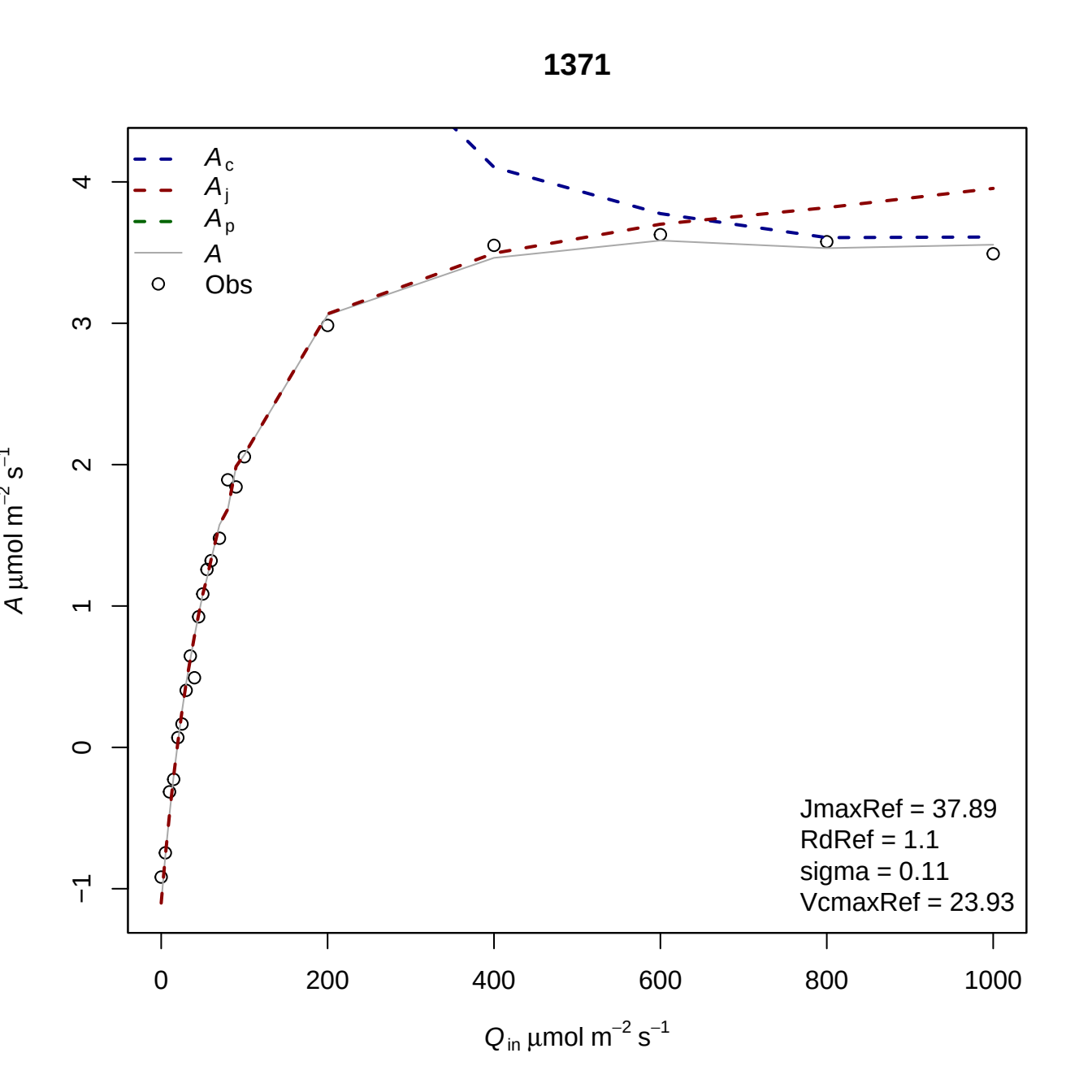

1366
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 19.55
RdRef = 0.86
sigma = 0.18
VcmaxRef = 488396.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

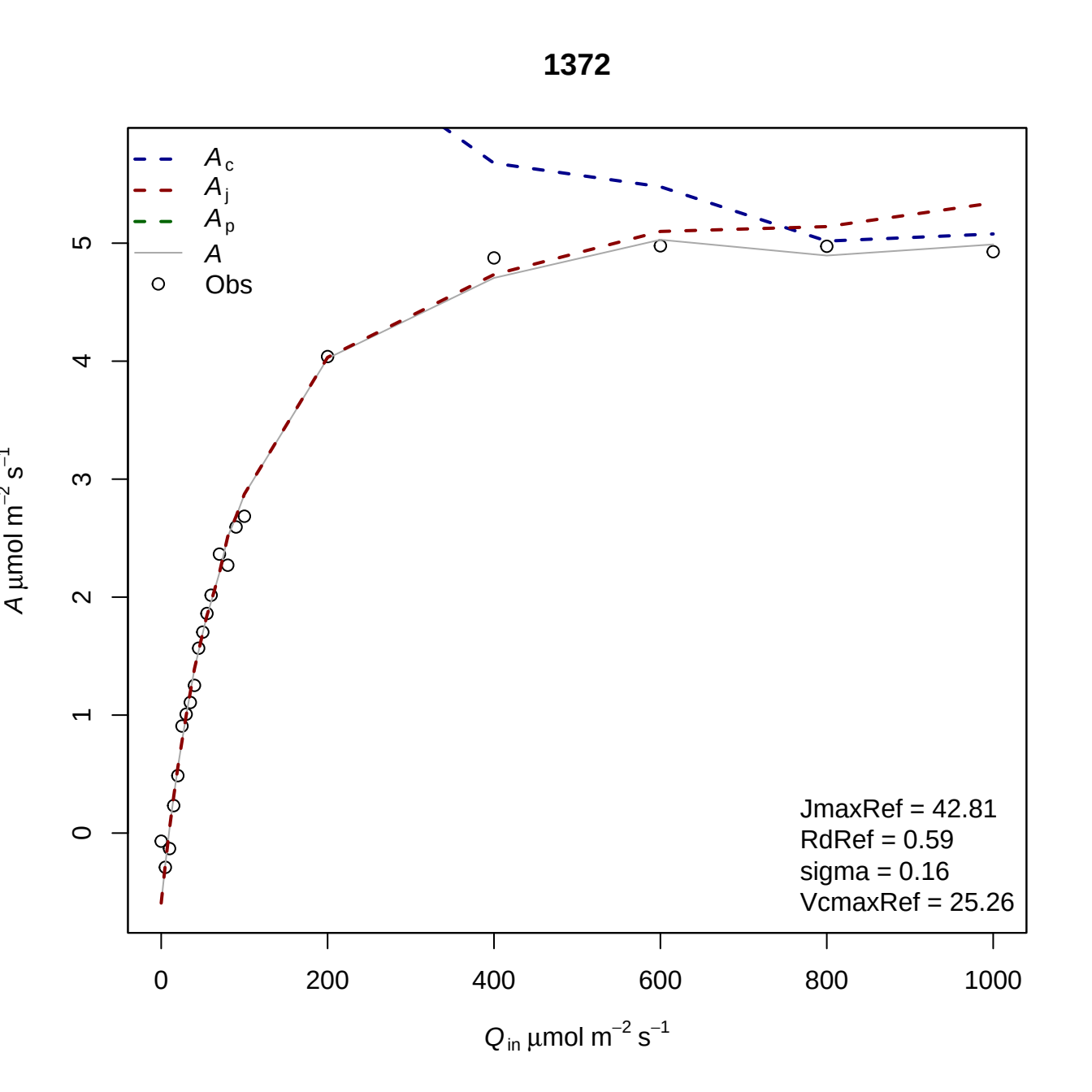

1367
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 20.28
RdRef = 0.93
sigma = 0.19
VcmaxRef = 291326.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000

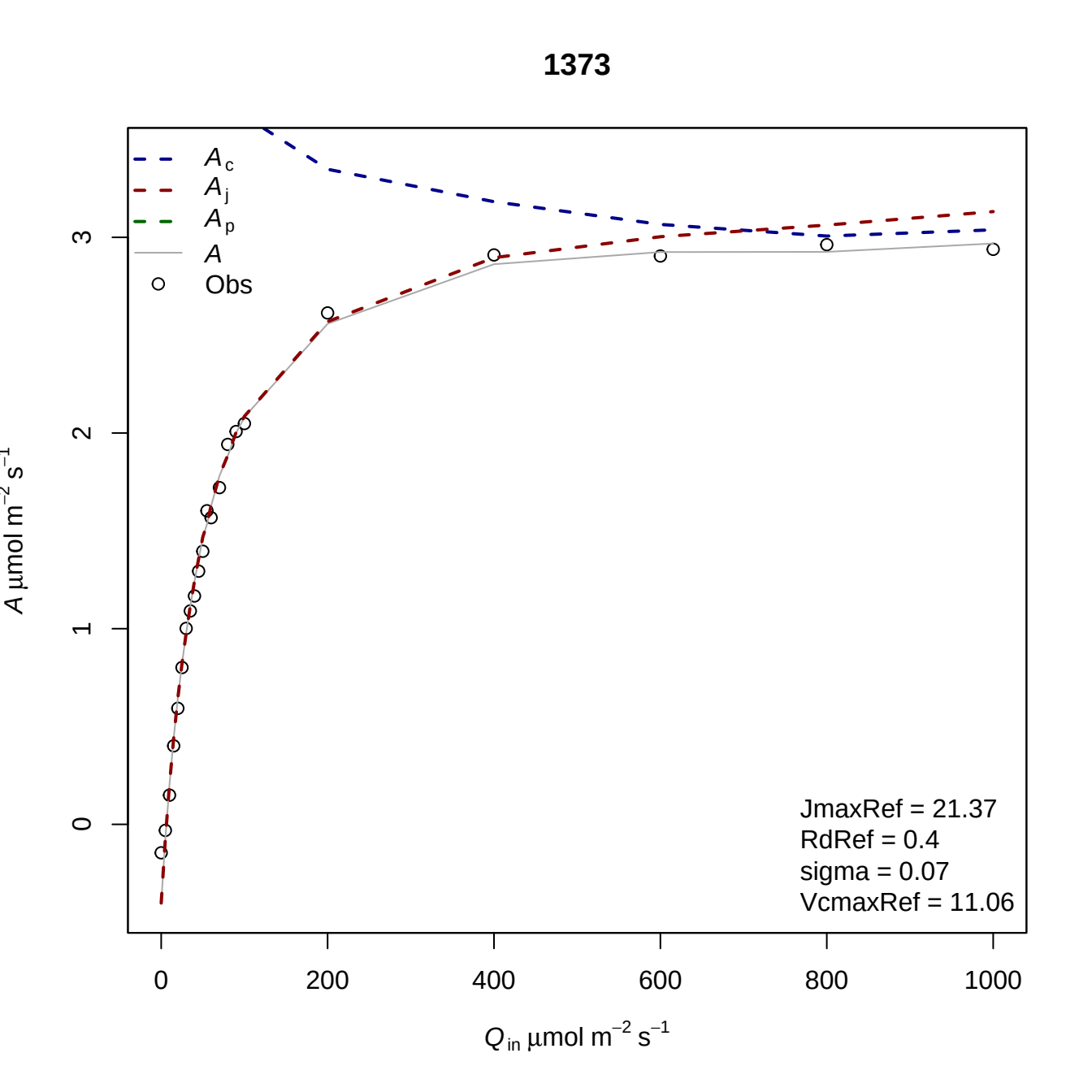

1368
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.34
RdRef = 0.85
sigma = 0.11
VcmaxRef = 33.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000

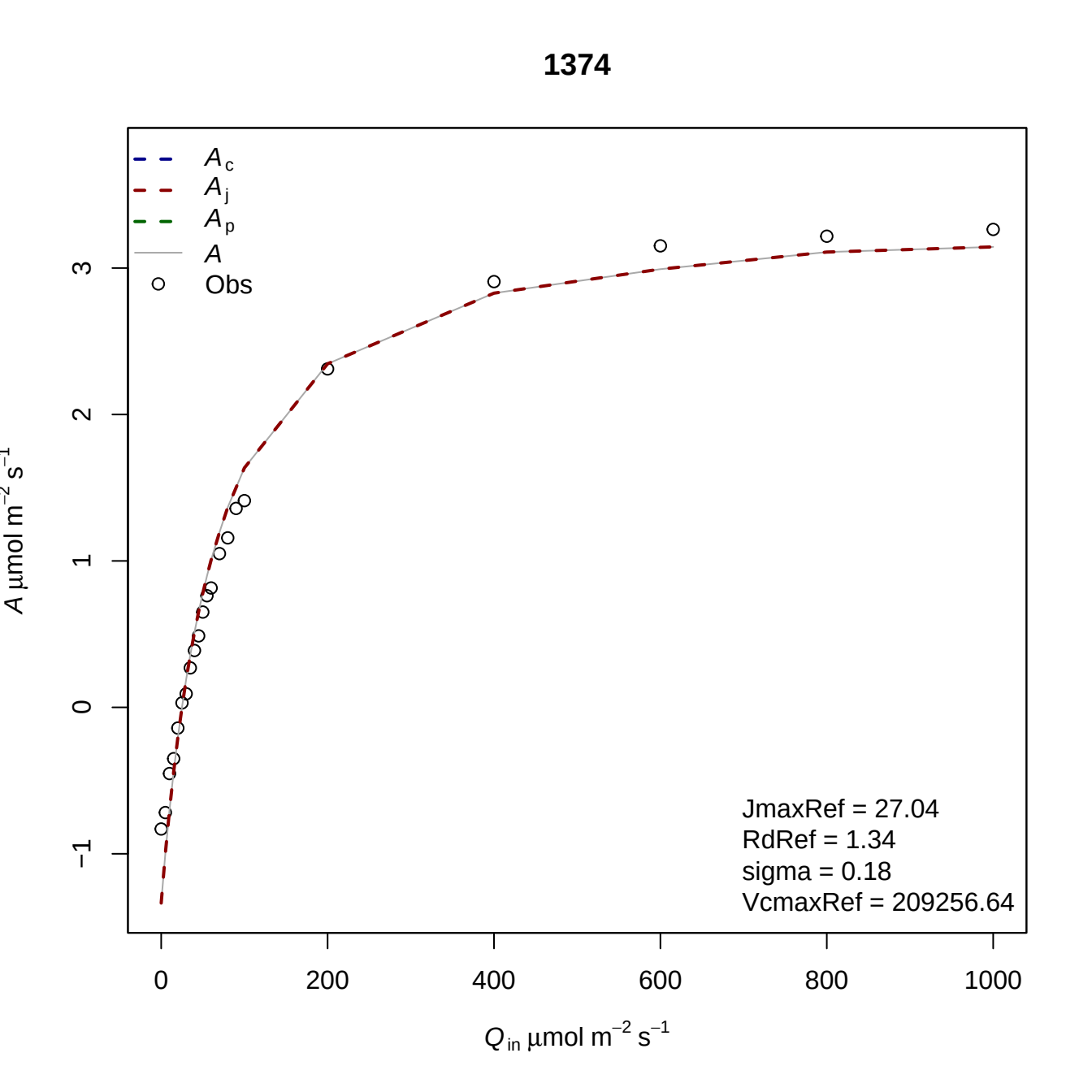

1369
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 33.98
RdRef = 1.29
sigma = 0.13
VcmaxRef = 17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000

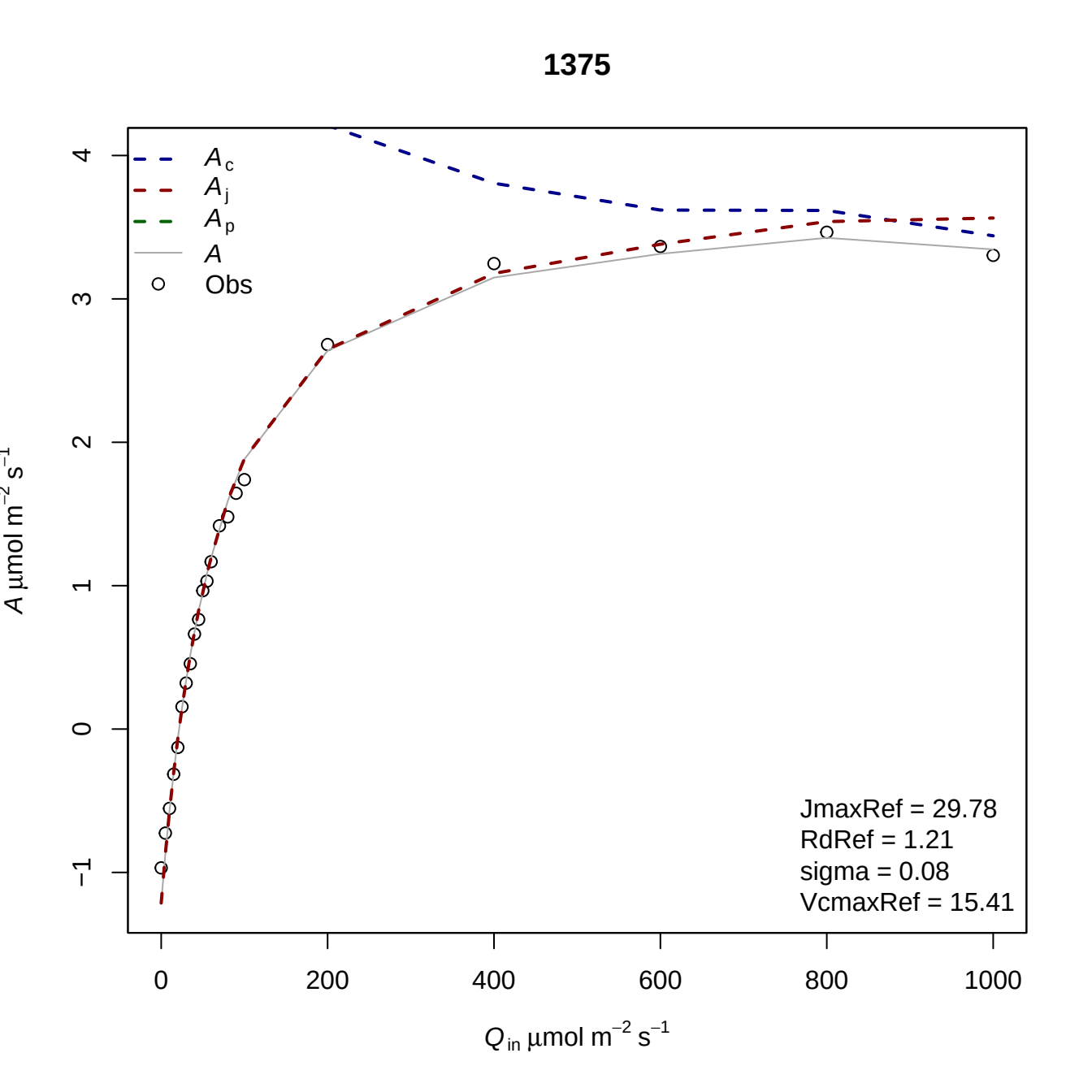

1370
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 32.21
RdRef = 1.21
sigma = 0.19
VcmaxRef = 321615.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000

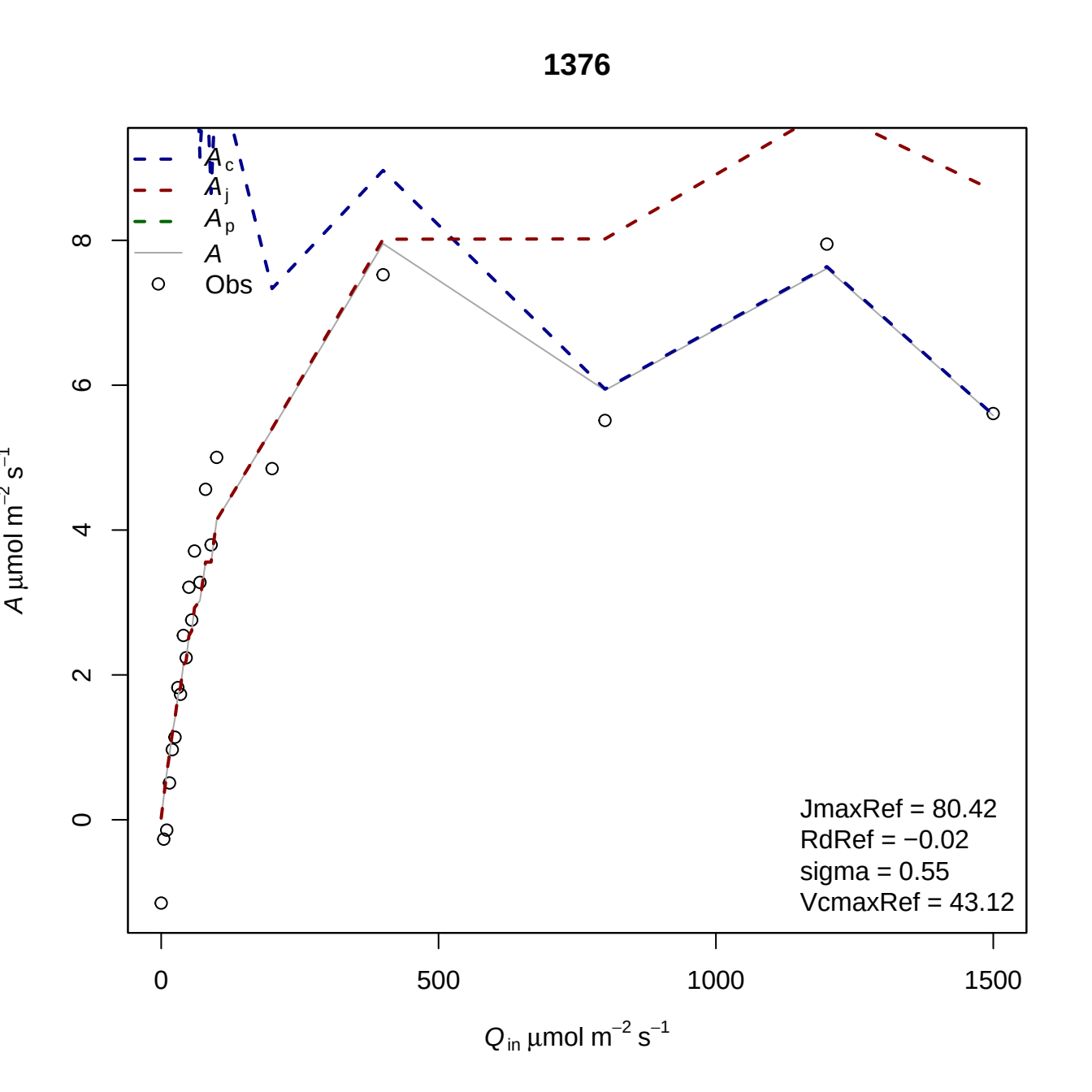

1371
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.89
RdRef = 1.1
sigma = 0.11
VcmaxRef = 23.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000

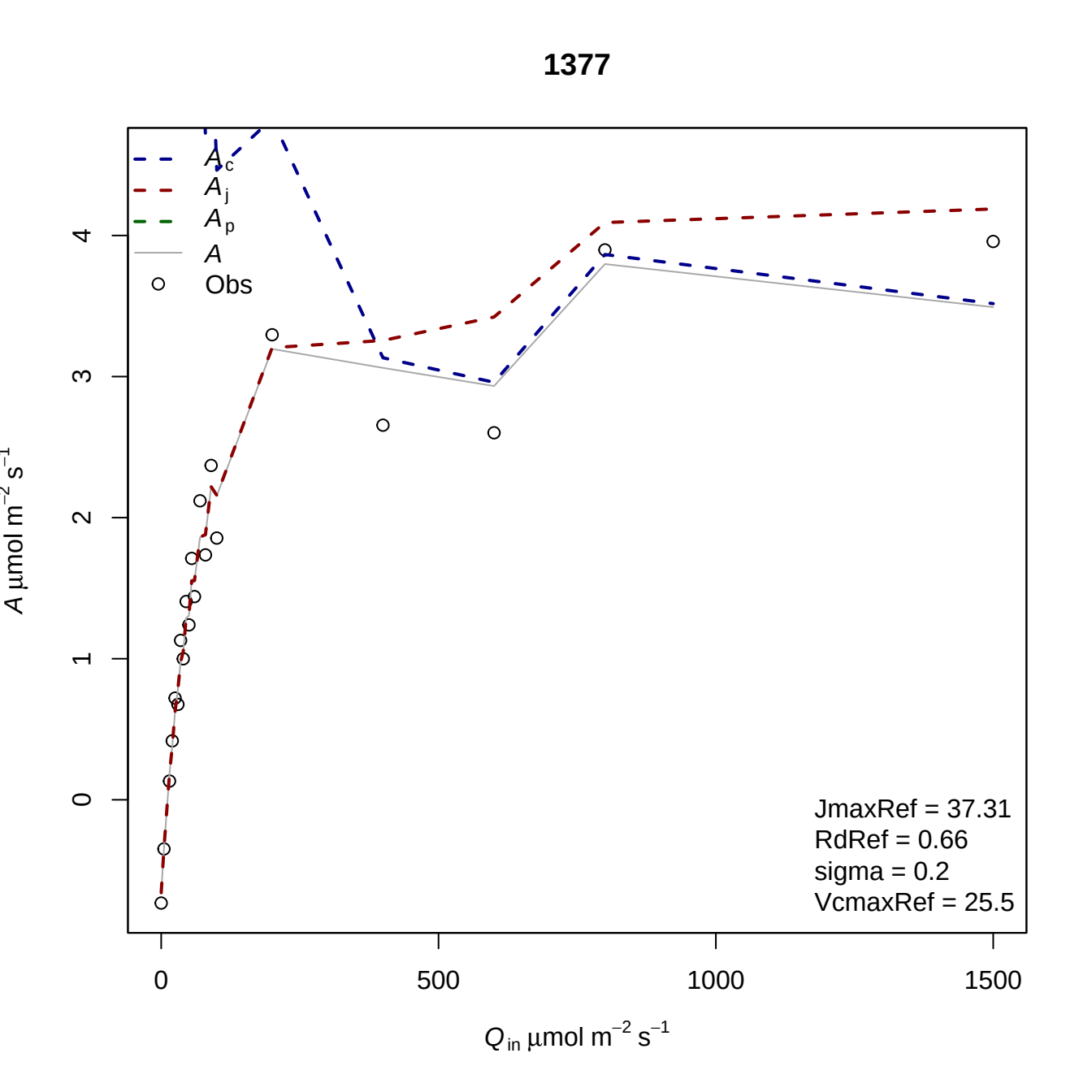

1372
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.81
RdRef = 0.59
sigma = 0.16
VcmaxRef = 25.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
200
400
600
800
1000

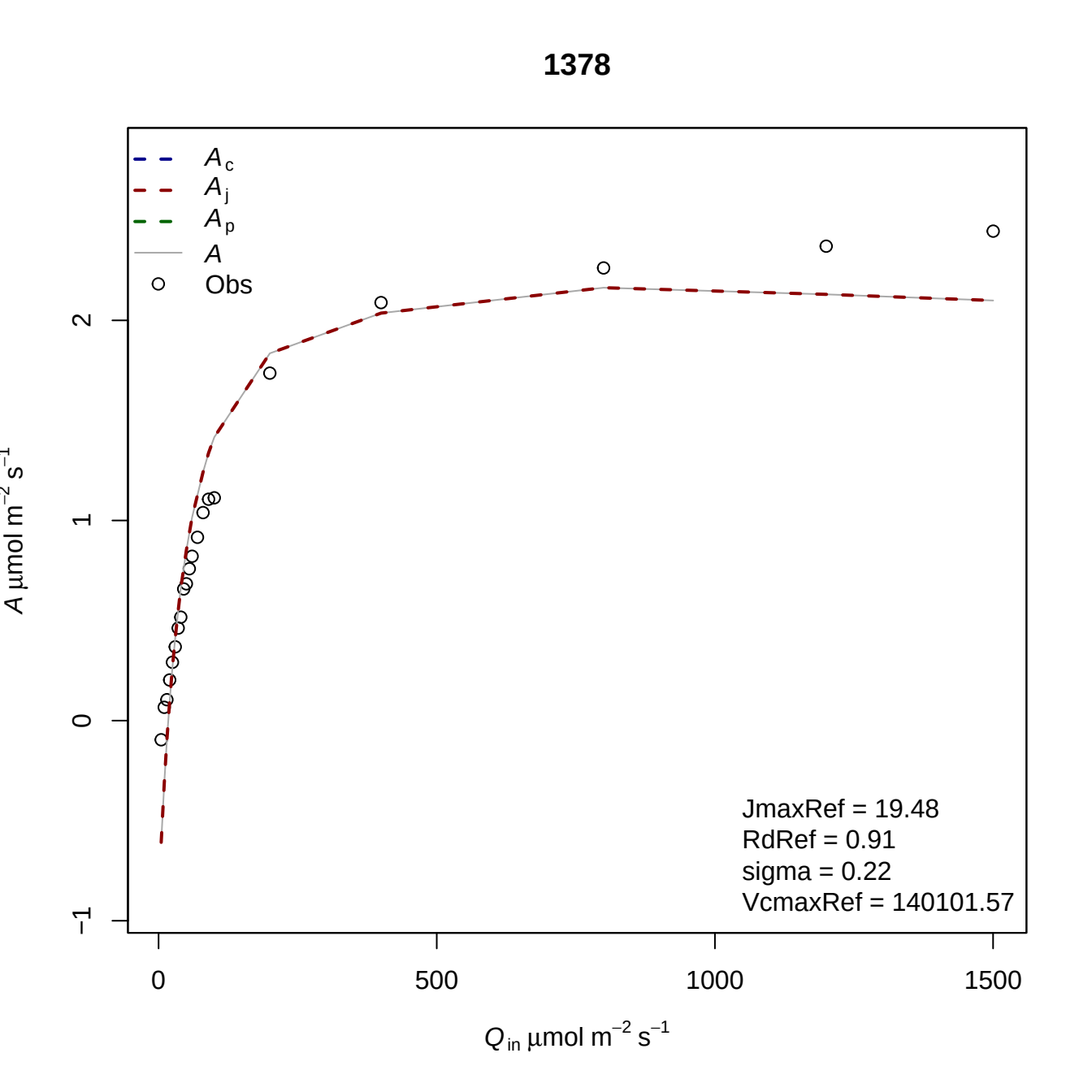

1373
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.37
RdRef = 0.4
sigma = 0.07
VcmaxRef = 11.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
200
400
600
800
1000

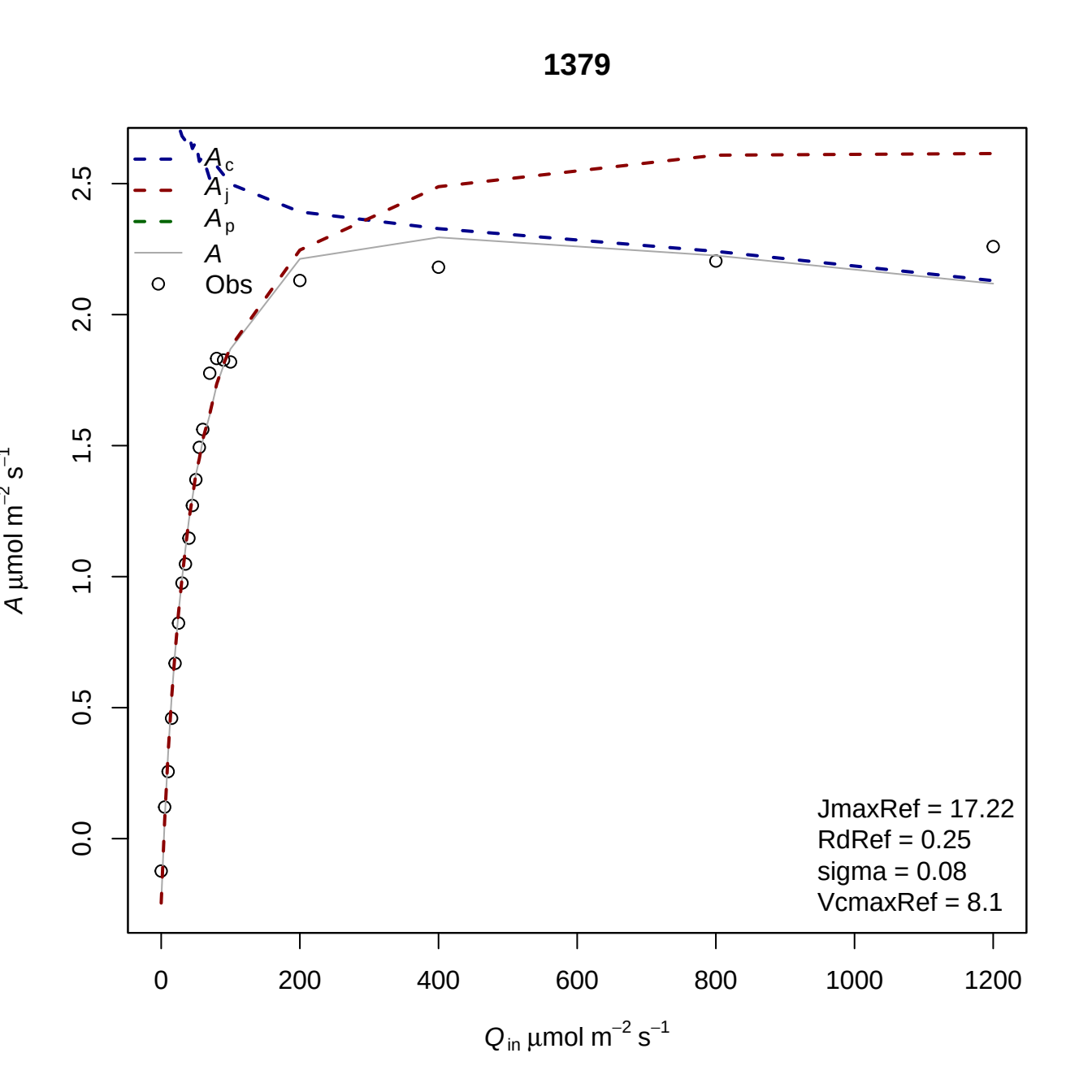

1374
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.04
RdRef = 1.34
sigma = 0.18
VcmaxRef = 209256.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800
1000

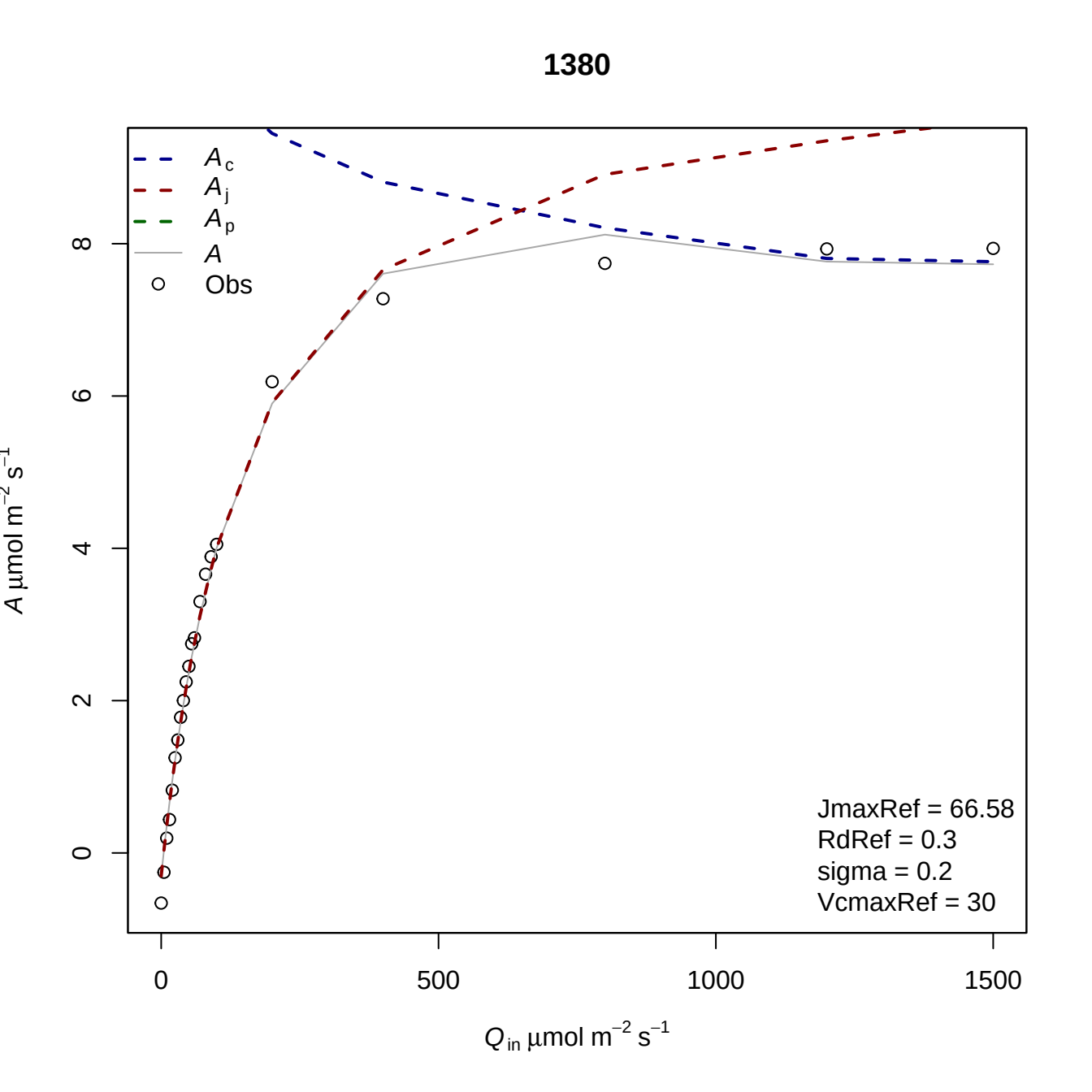

1375
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.78
RdRef = 1.21
sigma = 0.08
VcmaxRef = 15.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
0
200
400
600
800
1000

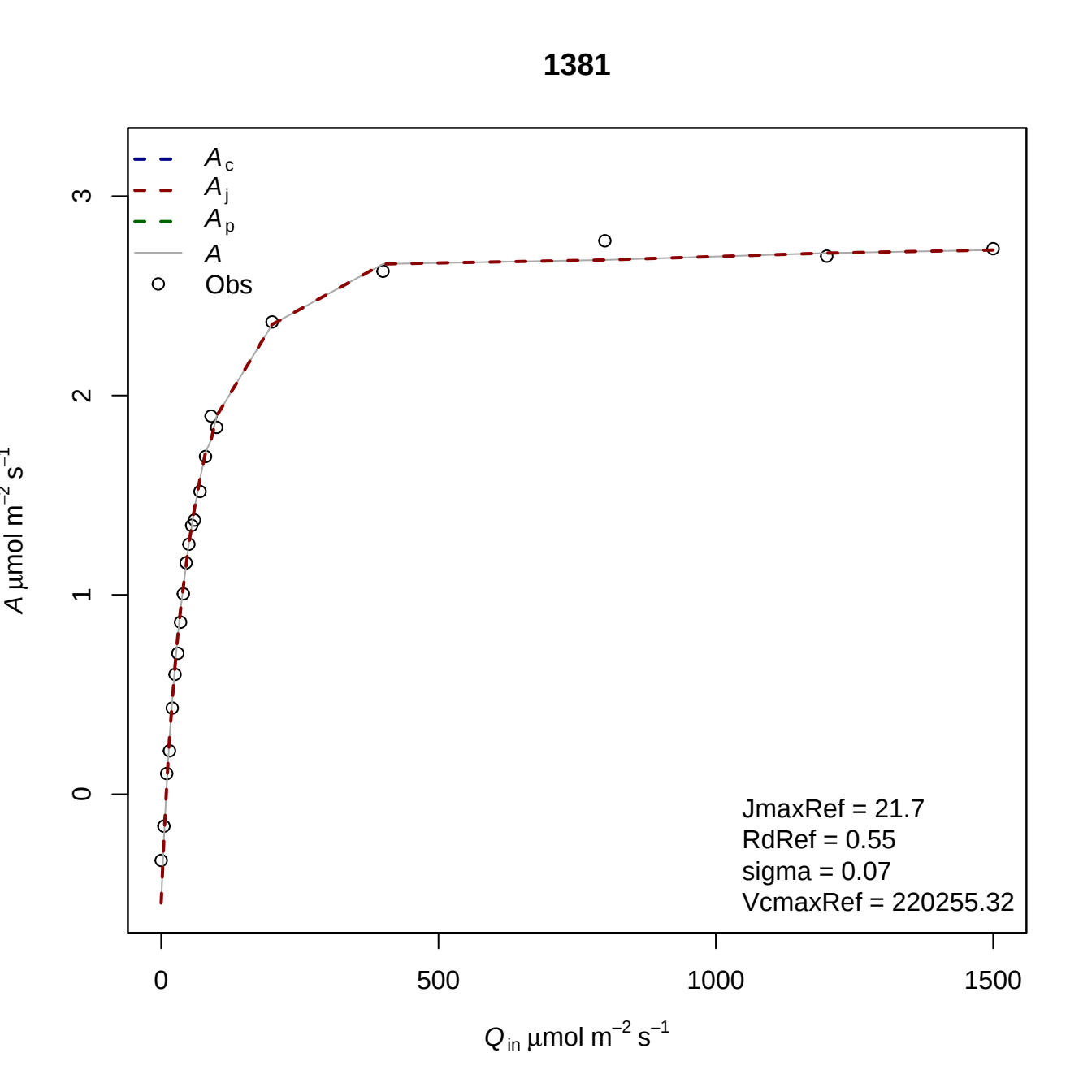

1376
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.42
RdRef = −0.02
sigma = 0.55
VcmaxRef = 43.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

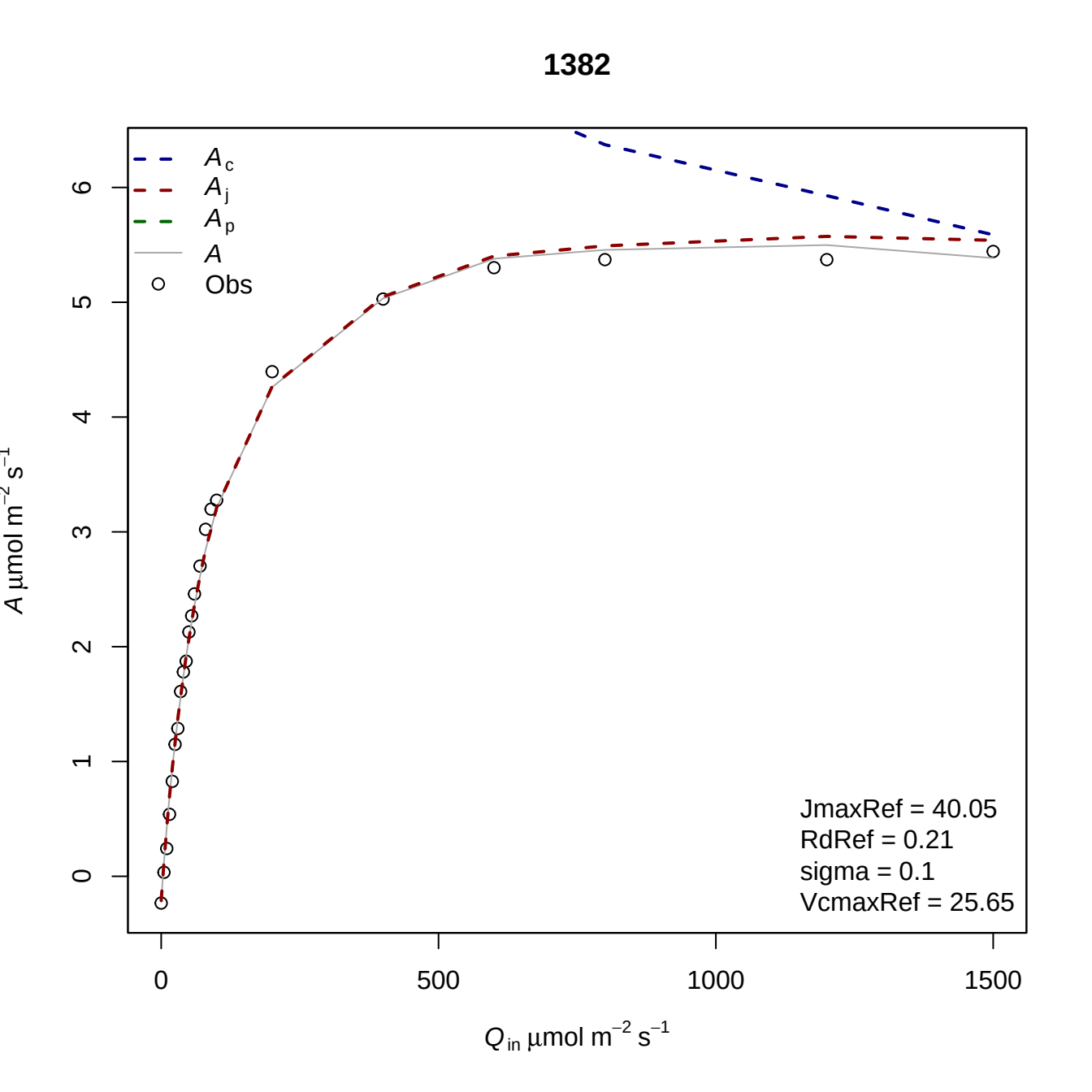

1377
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.31
RdRef = 0.66
sigma = 0.2
VcmaxRef = 25.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

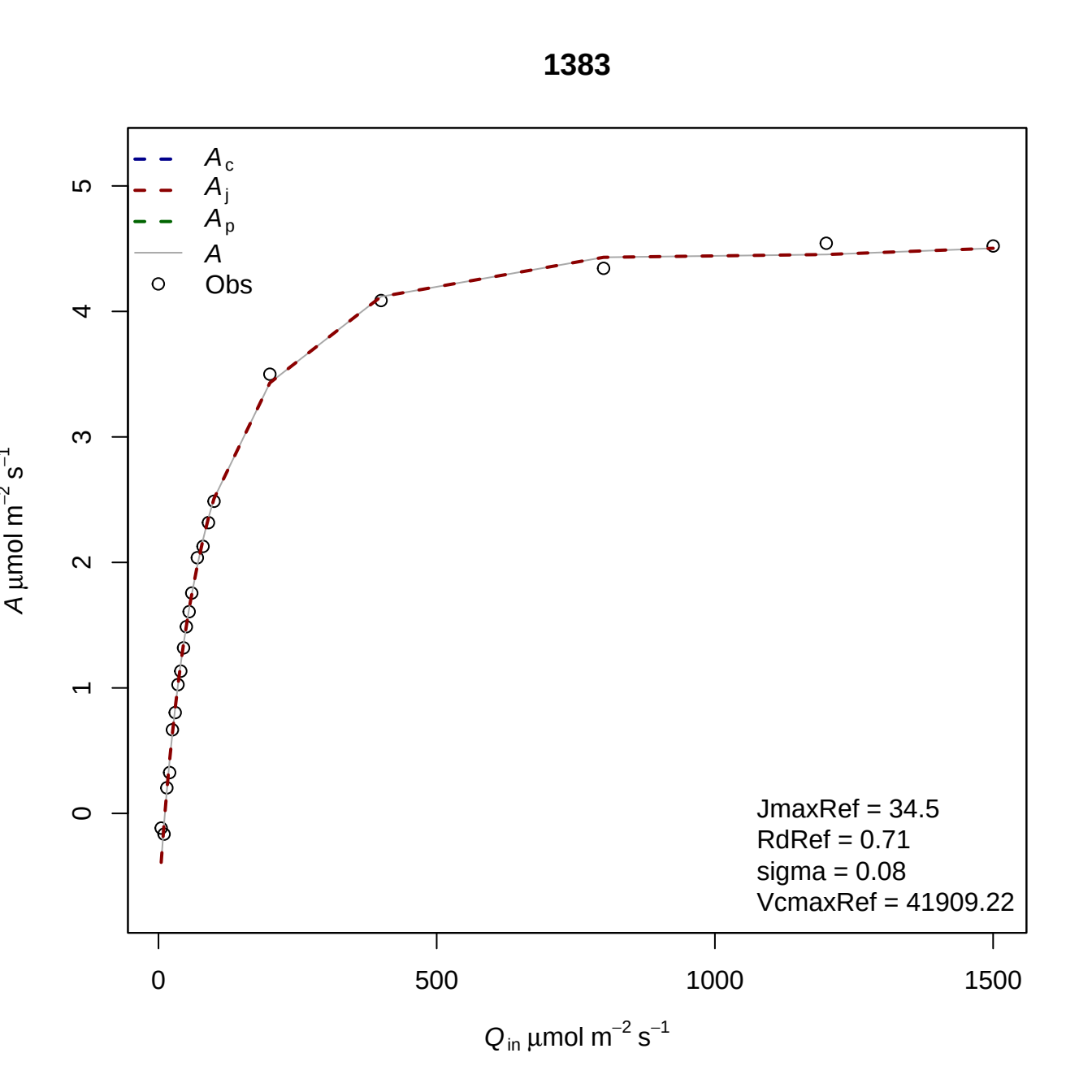

1378
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 19.48
RdRef = 0.91
sigma = 0.22
VcmaxRef = 140101.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
500
1000
1500

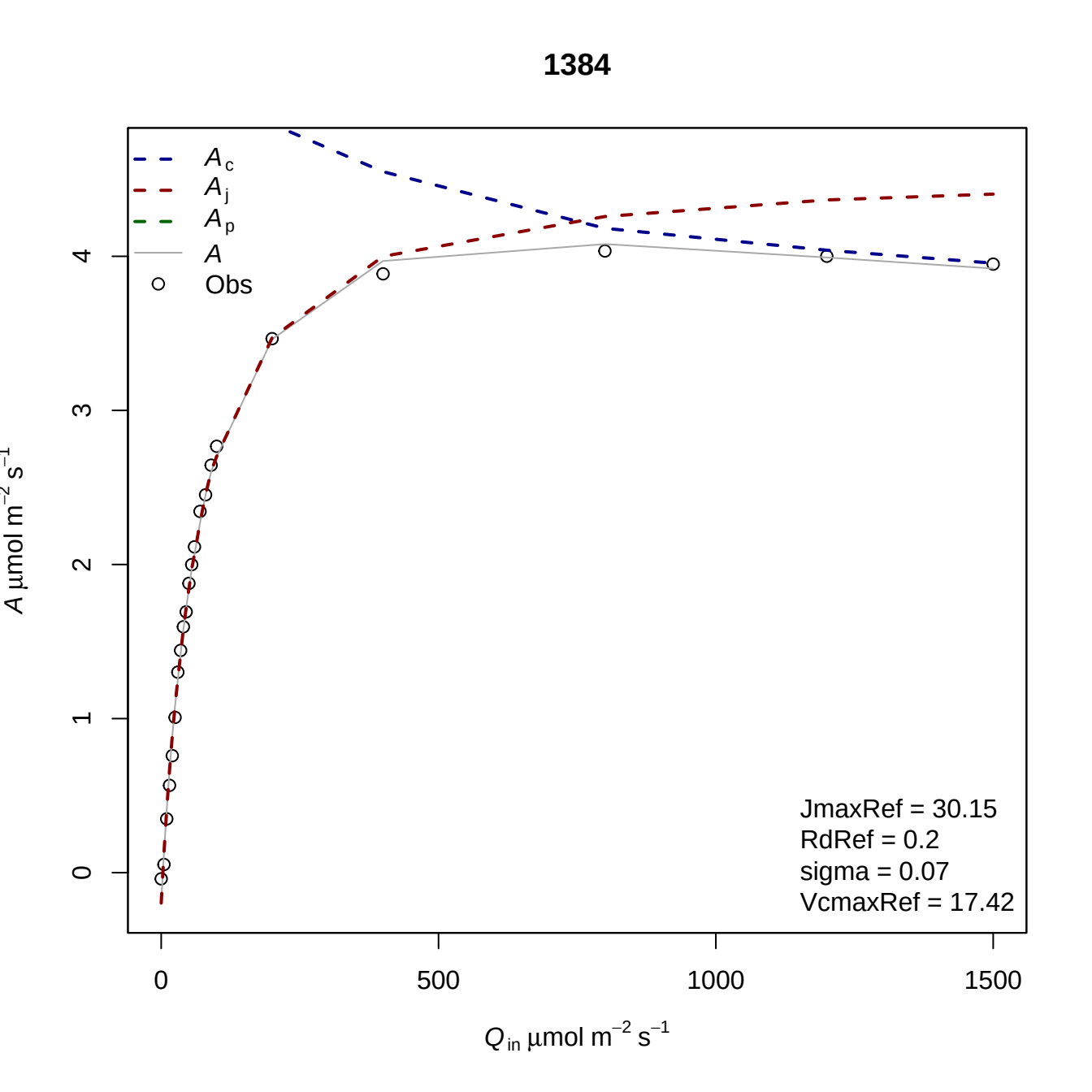

1379
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 17.22
RdRef = 0.25
sigma = 0.08
VcmaxRef = 8.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

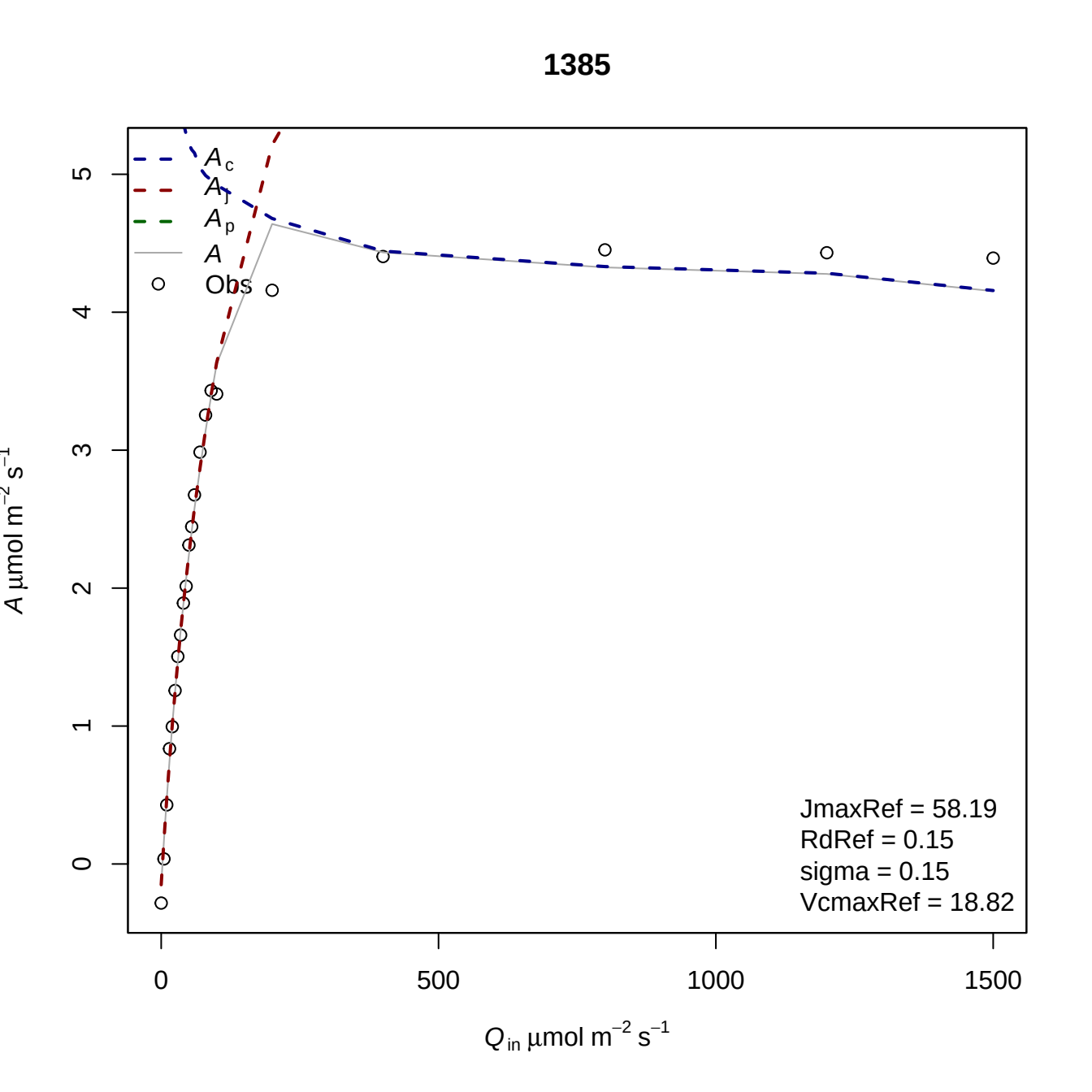

1380
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.58
RdRef = 0.3
sigma = 0.2
VcmaxRef = 30
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

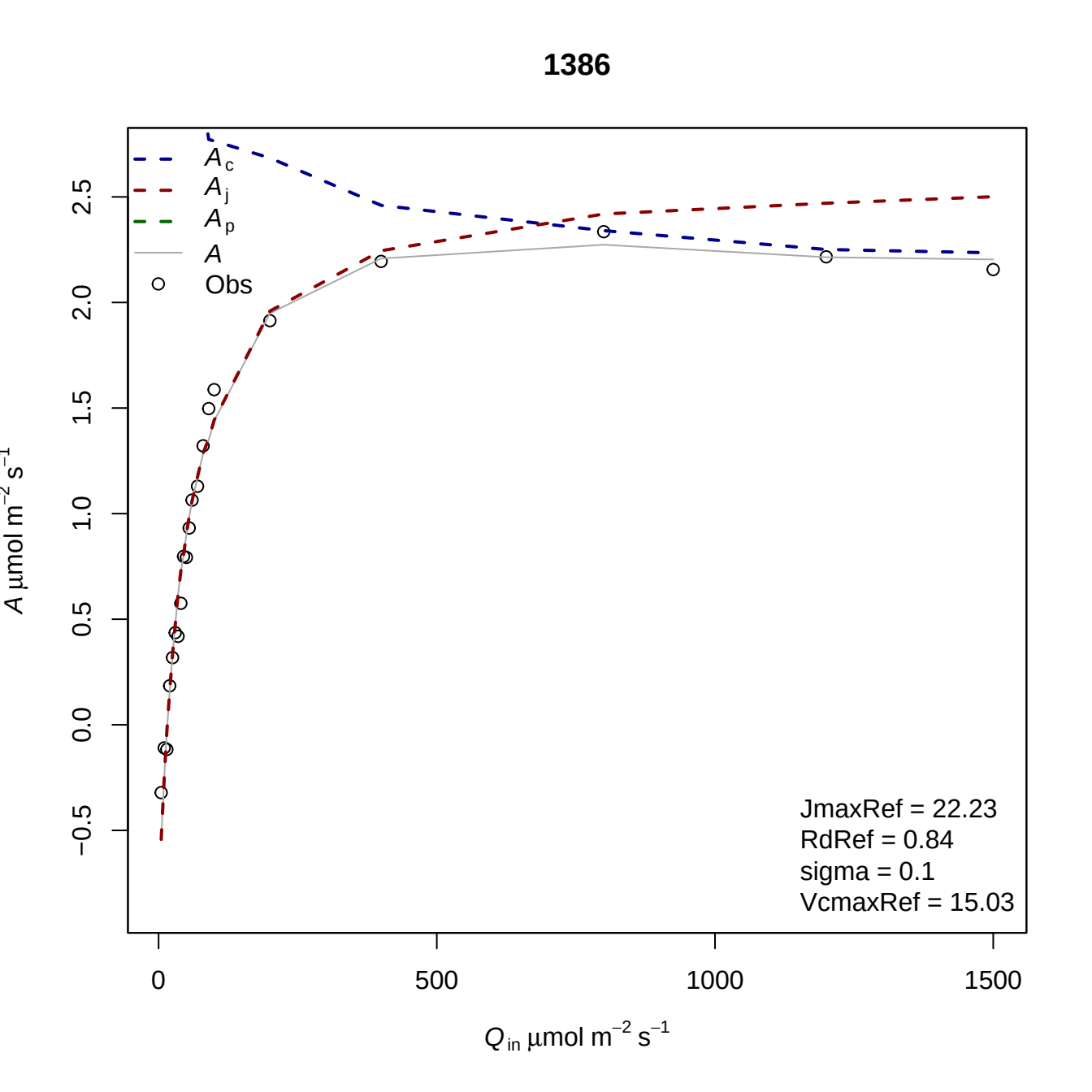

1381
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 21.7
RdRef = 0.55
sigma = 0.07
VcmaxRef = 220255.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500

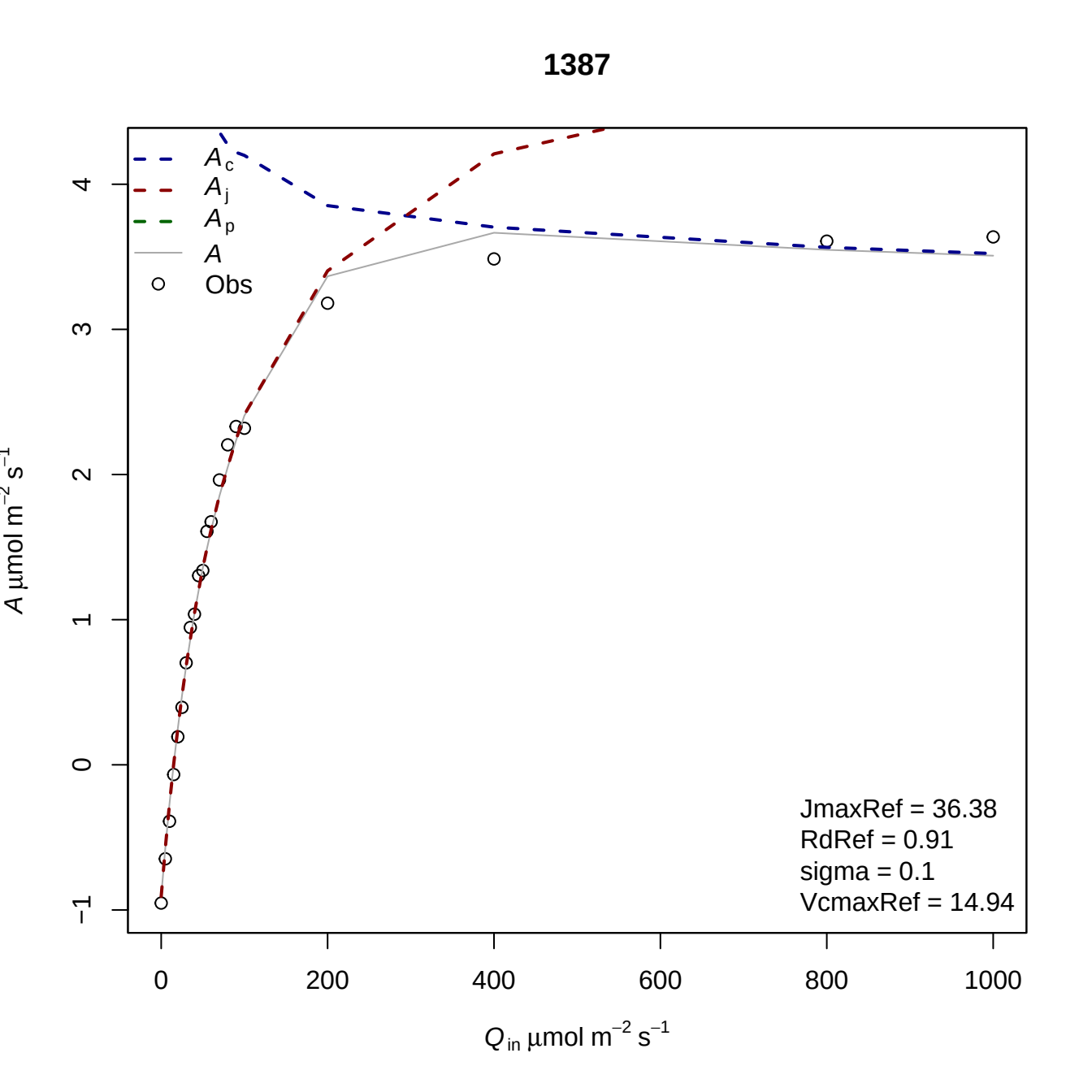

1382
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.05
RdRef = 0.21
sigma = 0.1
VcmaxRef = 25.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

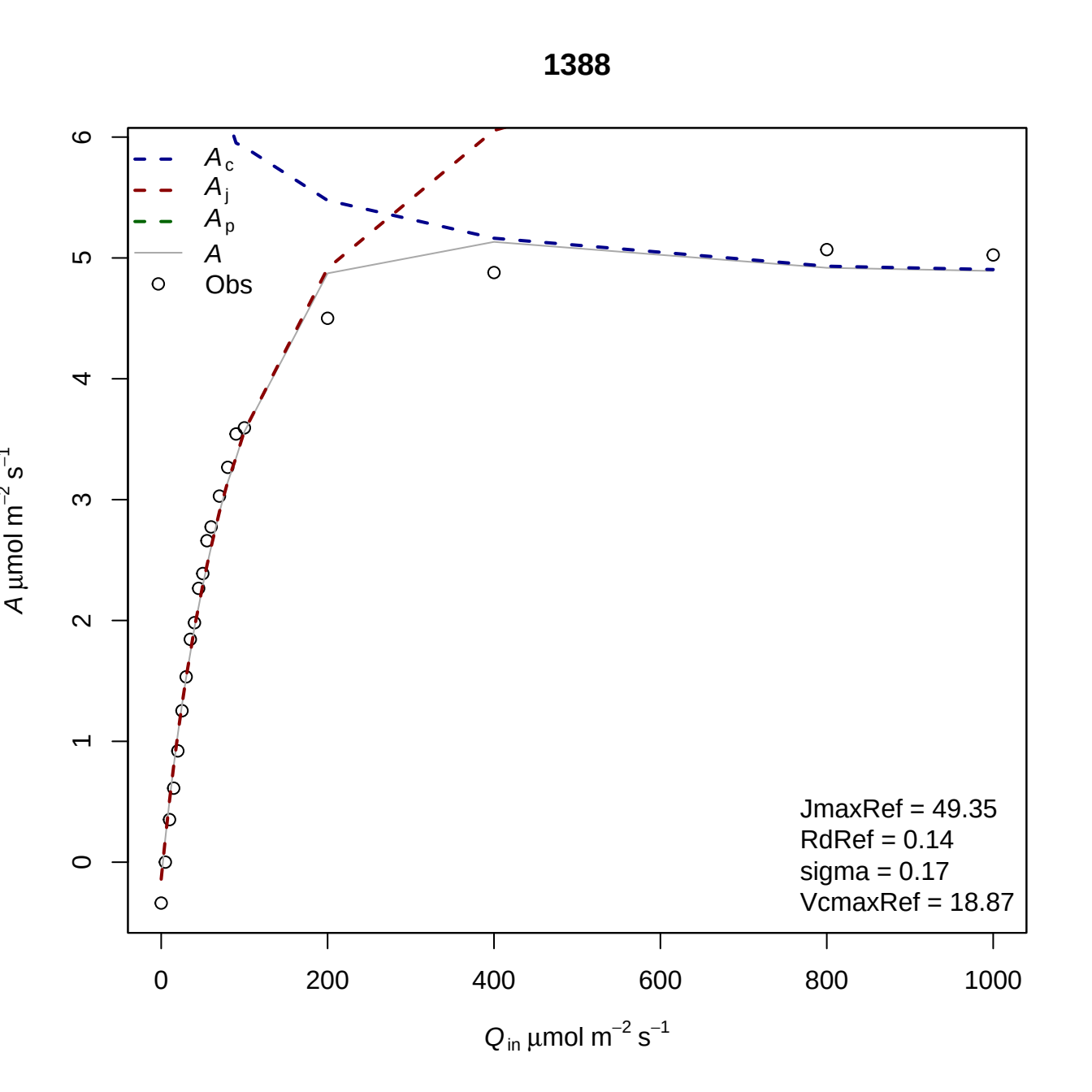

1383
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 34.5
RdRef = 0.71
sigma = 0.08
VcmaxRef = 41909.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

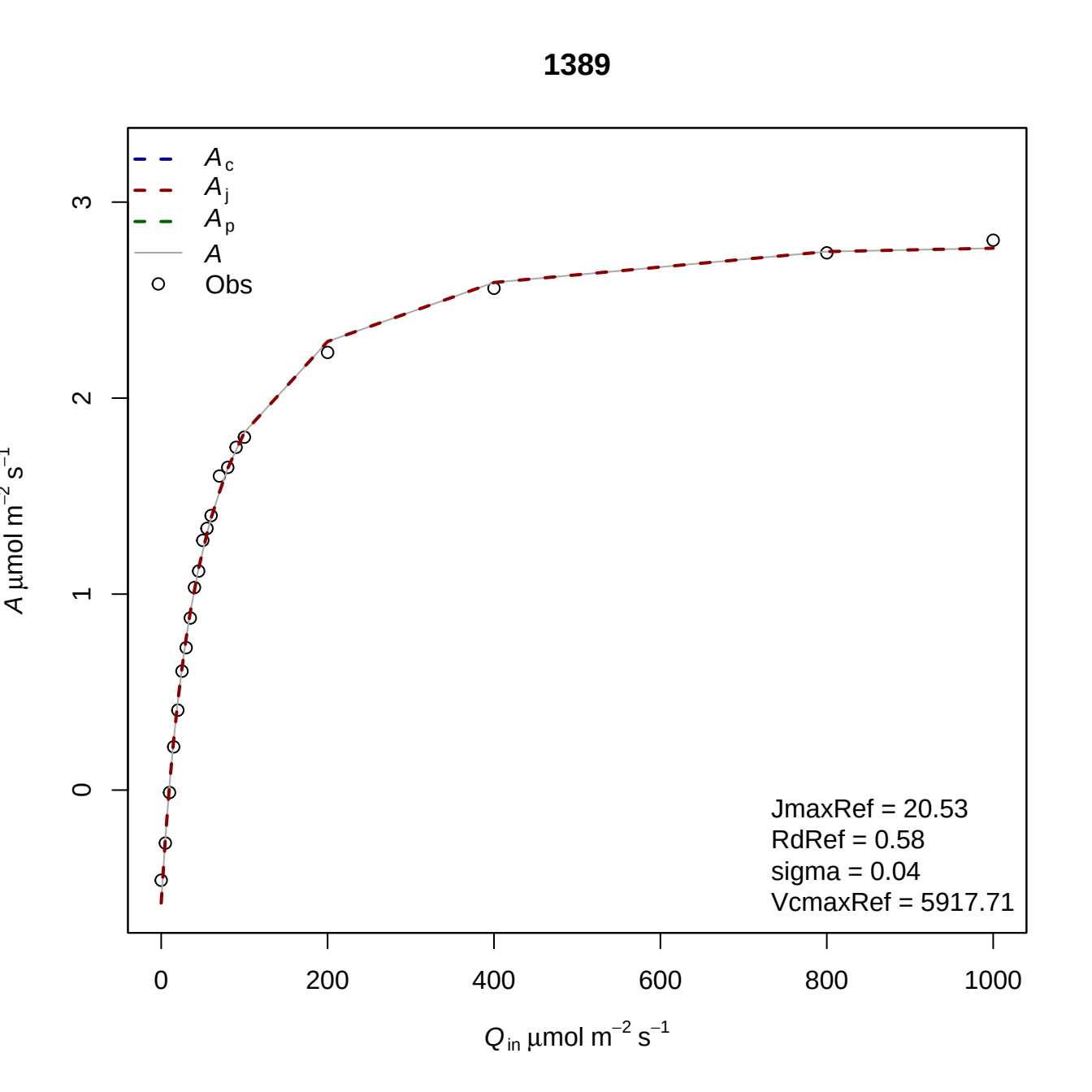

1384
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.15
RdRef = 0.2
sigma = 0.07
VcmaxRef = 17.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

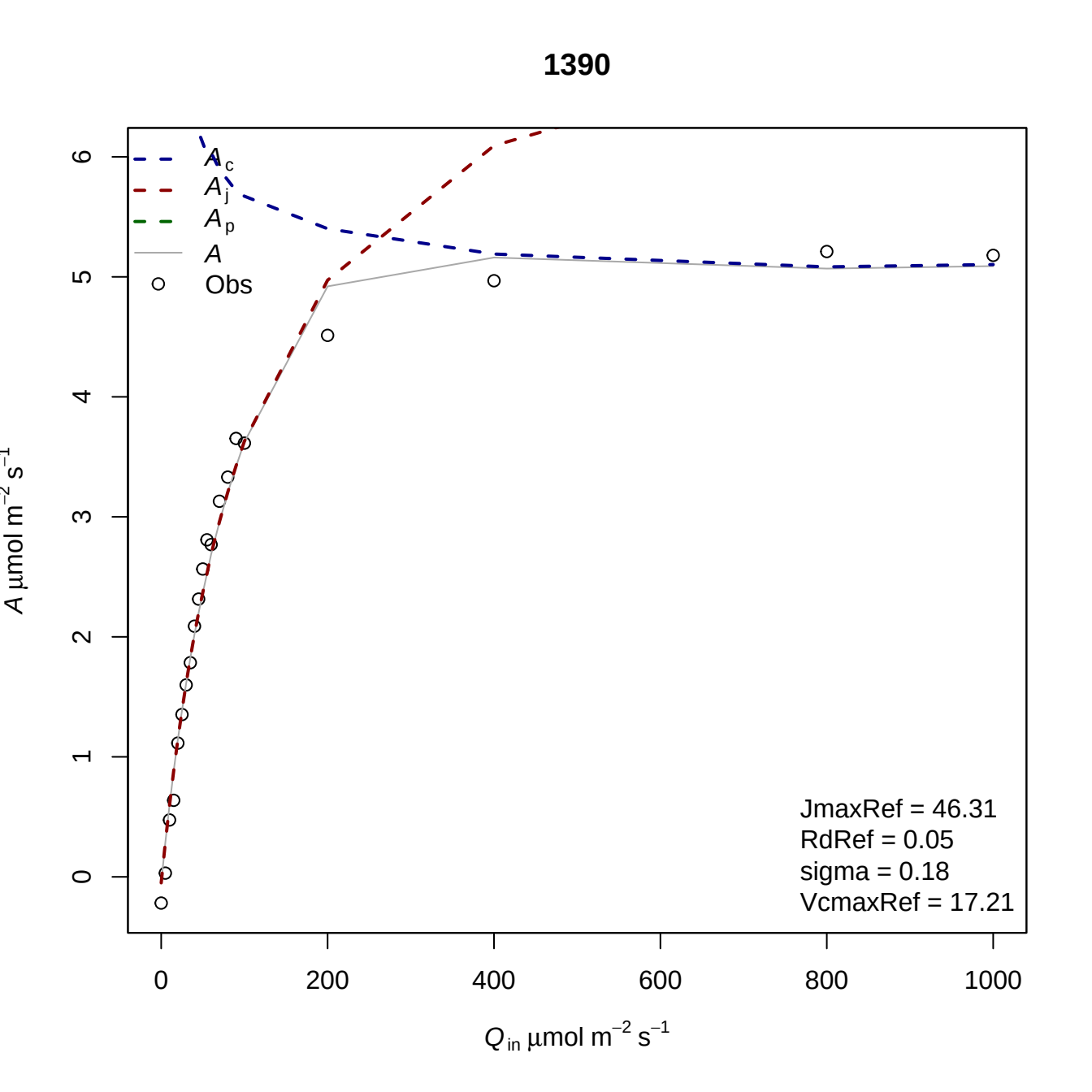

1385
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.19
RdRef = 0.15
sigma = 0.15
VcmaxRef = 18.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

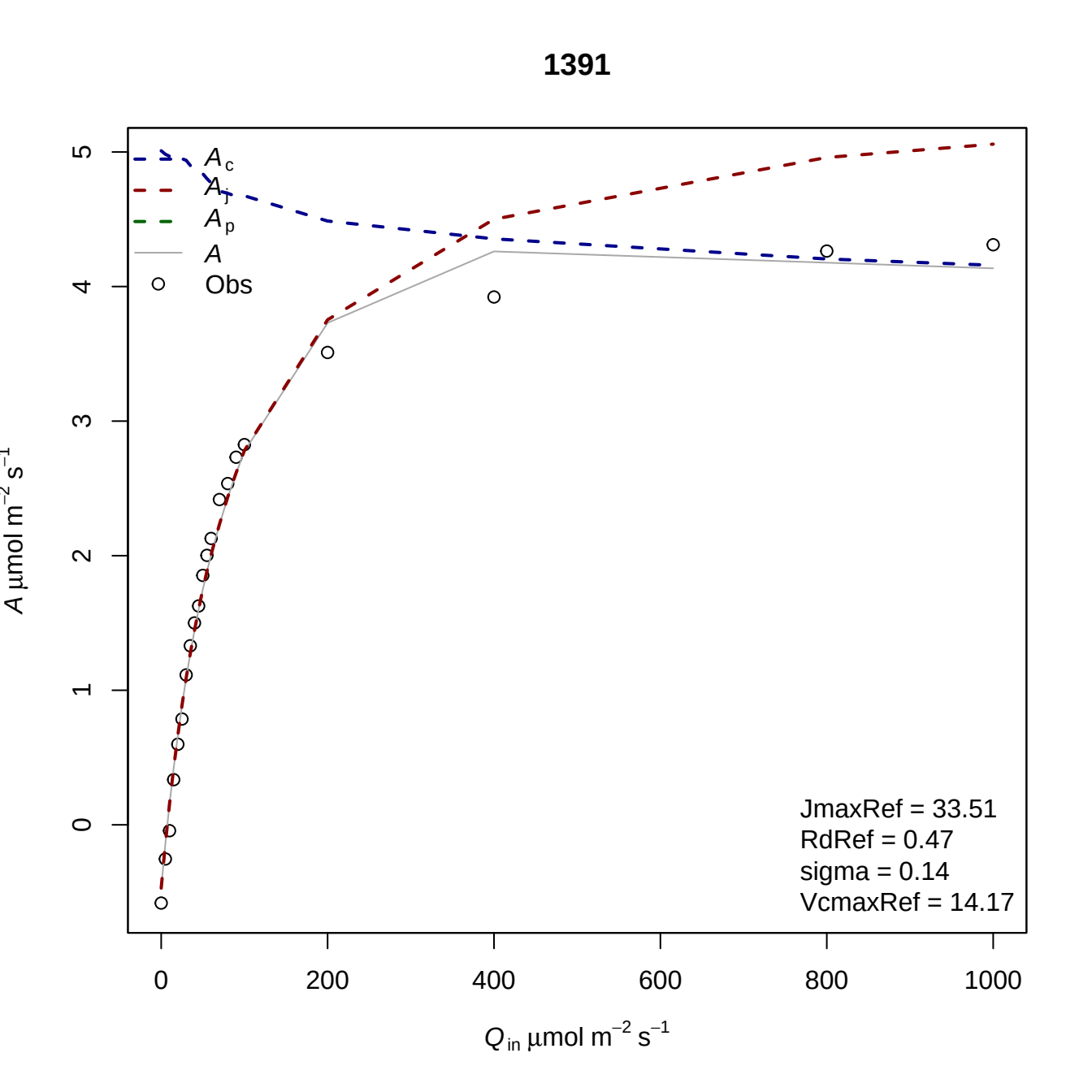

1386
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 22.23
RdRef = 0.84
sigma = 0.1
VcmaxRef = 15.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

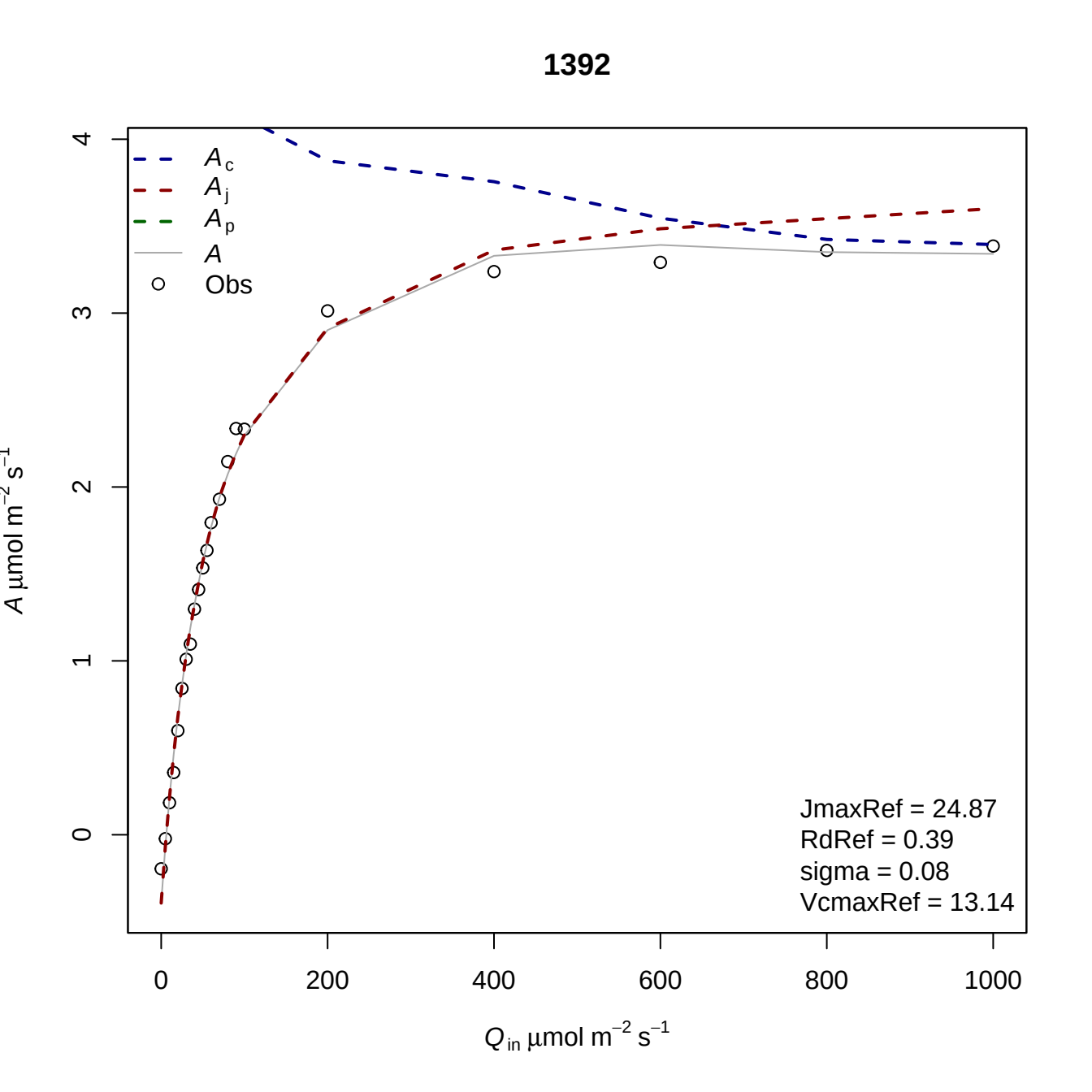

1387
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.38
RdRef = 0.91
sigma = 0.1
VcmaxRef = 14.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
200
400
600
800
1000

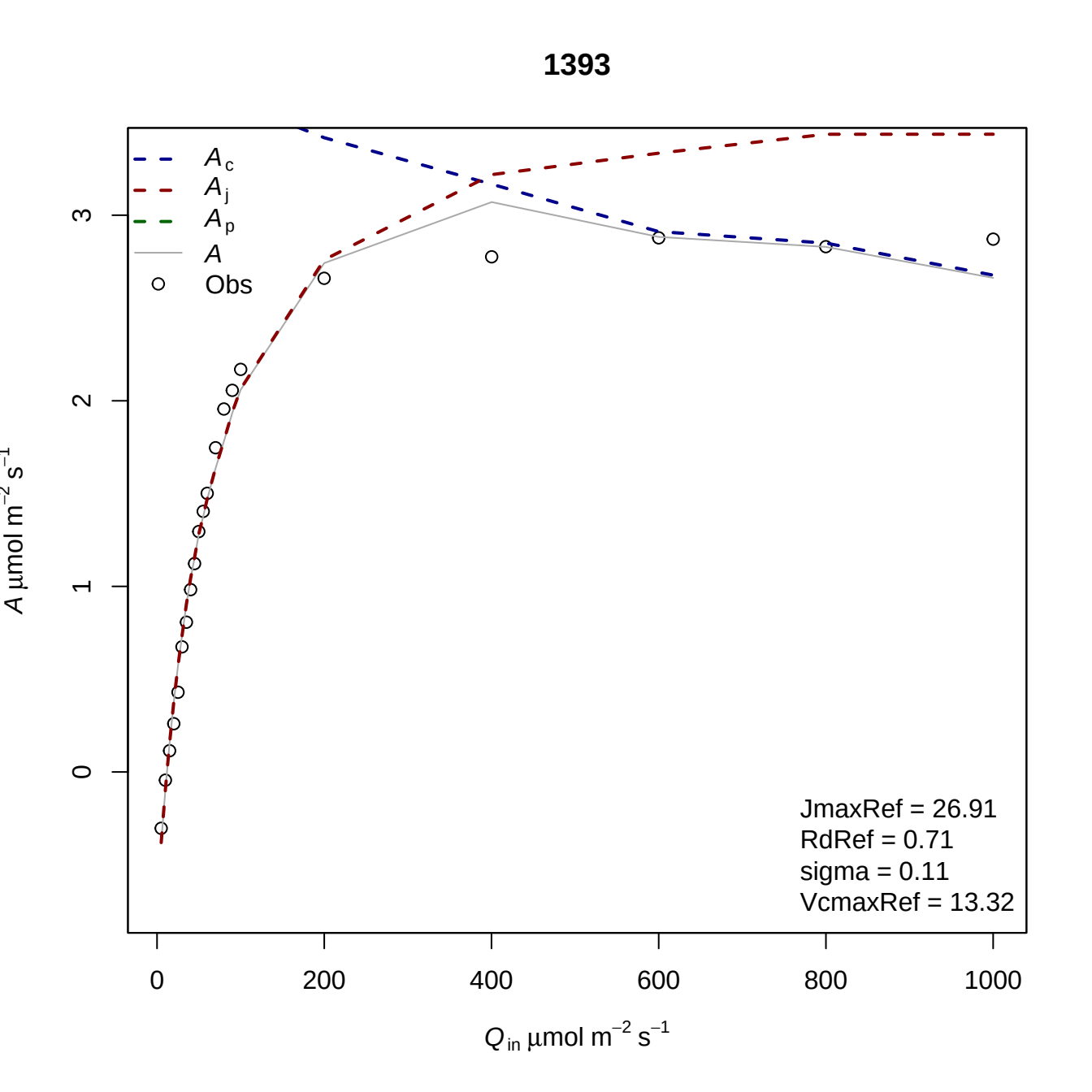

1388
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 49.35
RdRef = 0.14
sigma = 0.17
VcmaxRef = 18.87
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
200
400
600
800
1000

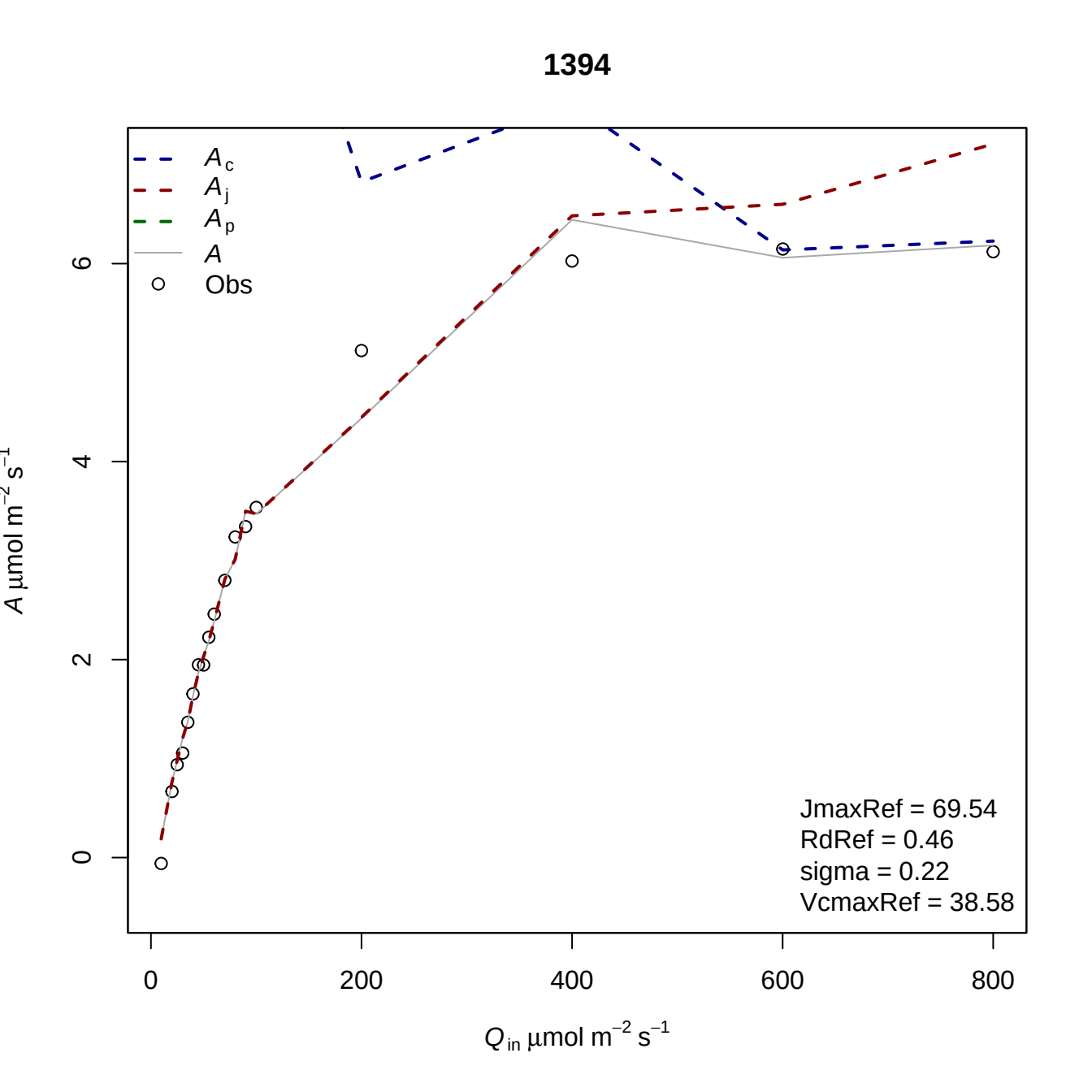

1389
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 20.53
RdRef = 0.58
sigma = 0.04
VcmaxRef = 5917.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
200
400
600
800
1000

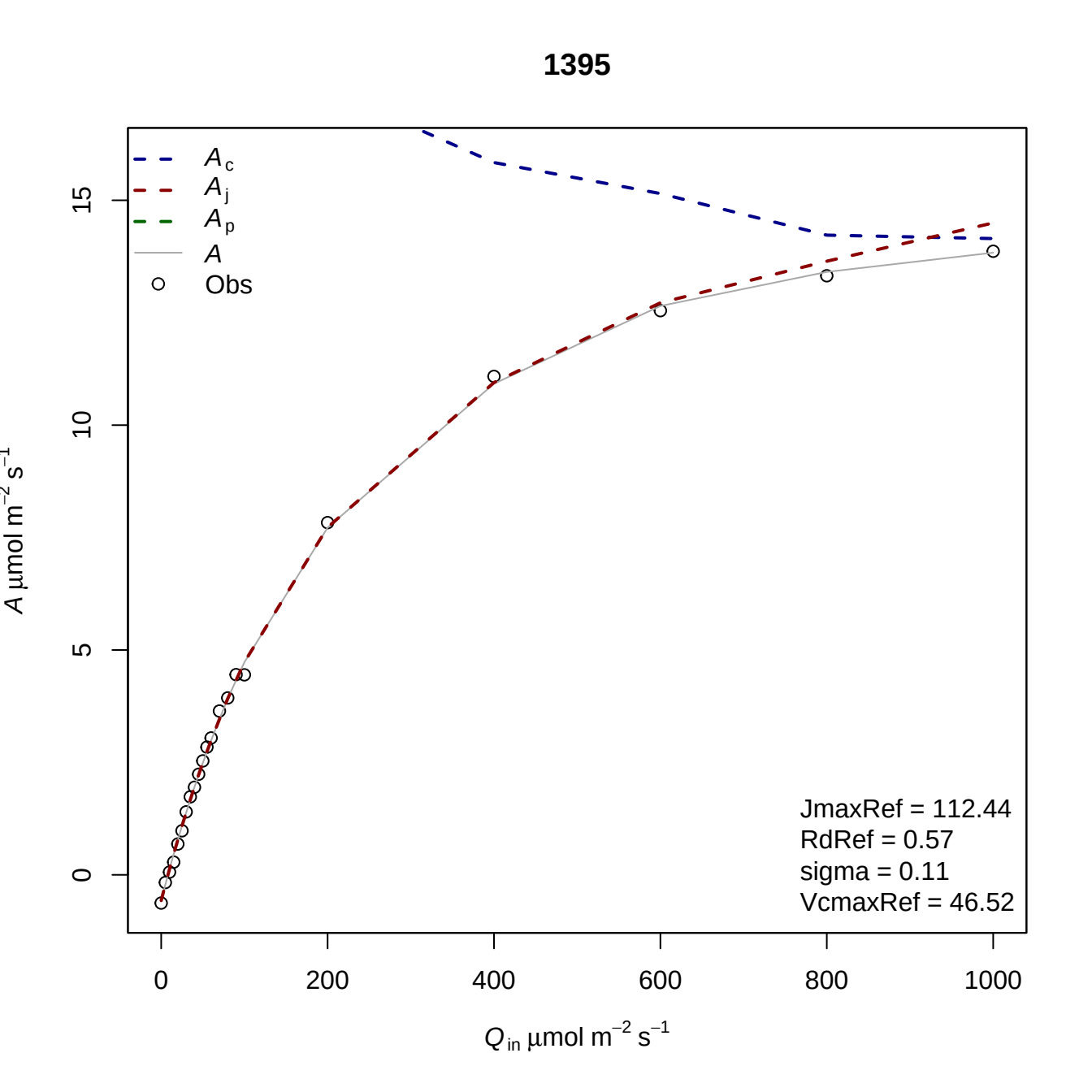

1390
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.31
RdRef = 0.05
sigma = 0.18
VcmaxRef = 17.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
200
400
600
800
1000

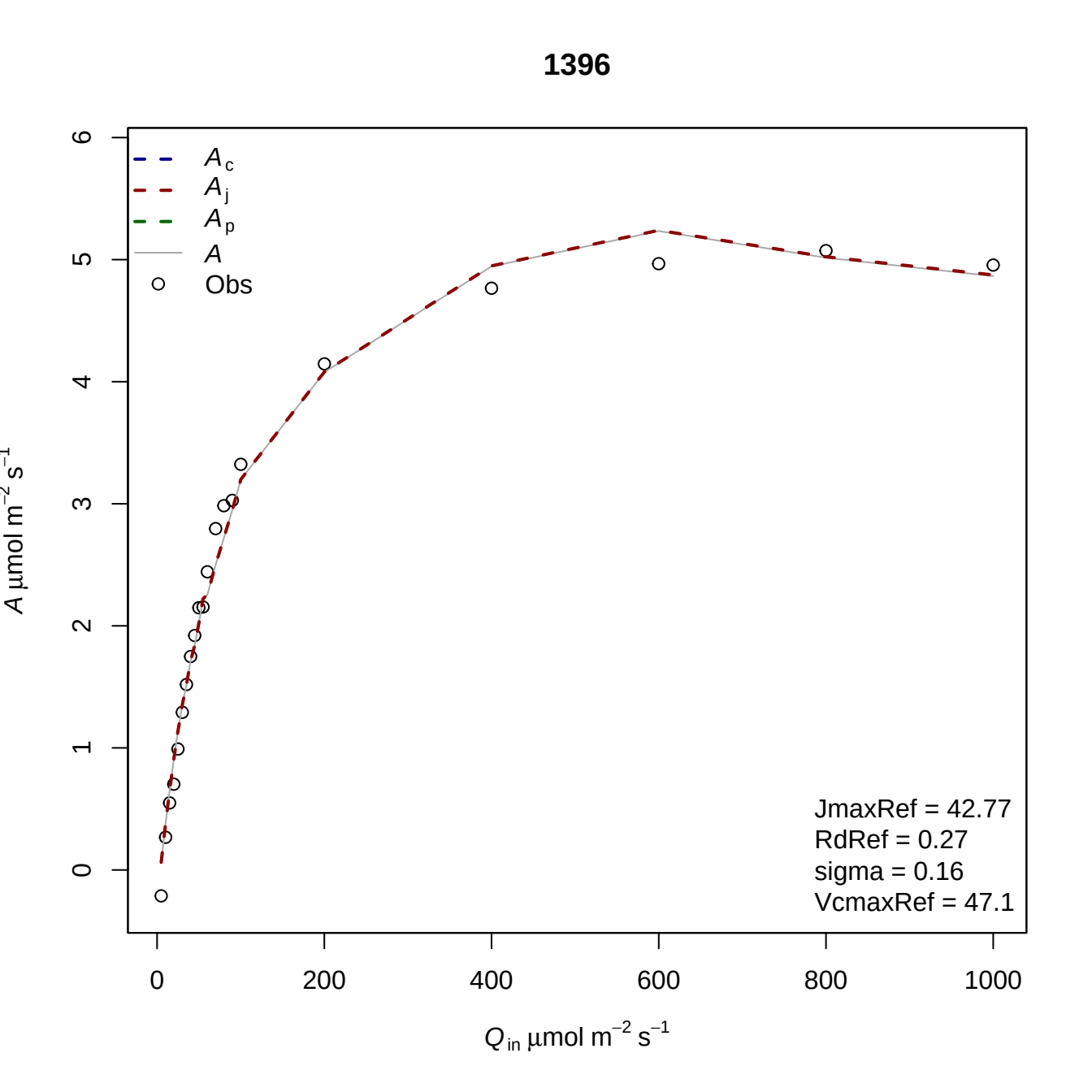

1391
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 33.51
RdRef = 0.47
sigma = 0.14
VcmaxRef = 14.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
200
400
600
800
1000

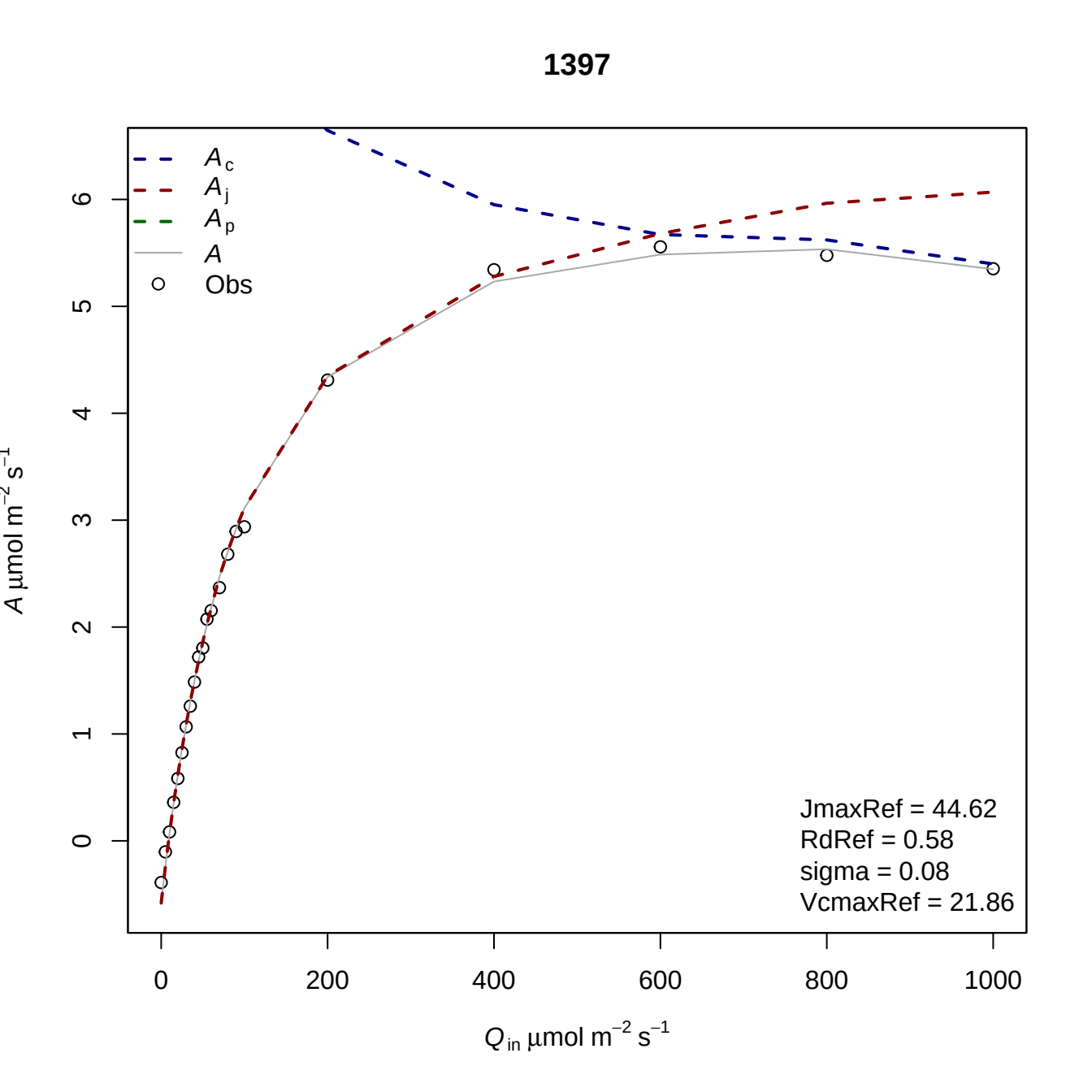

1392
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 24.87
RdRef = 0.39
sigma = 0.08
VcmaxRef = 13.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
200
400
600
800
1000

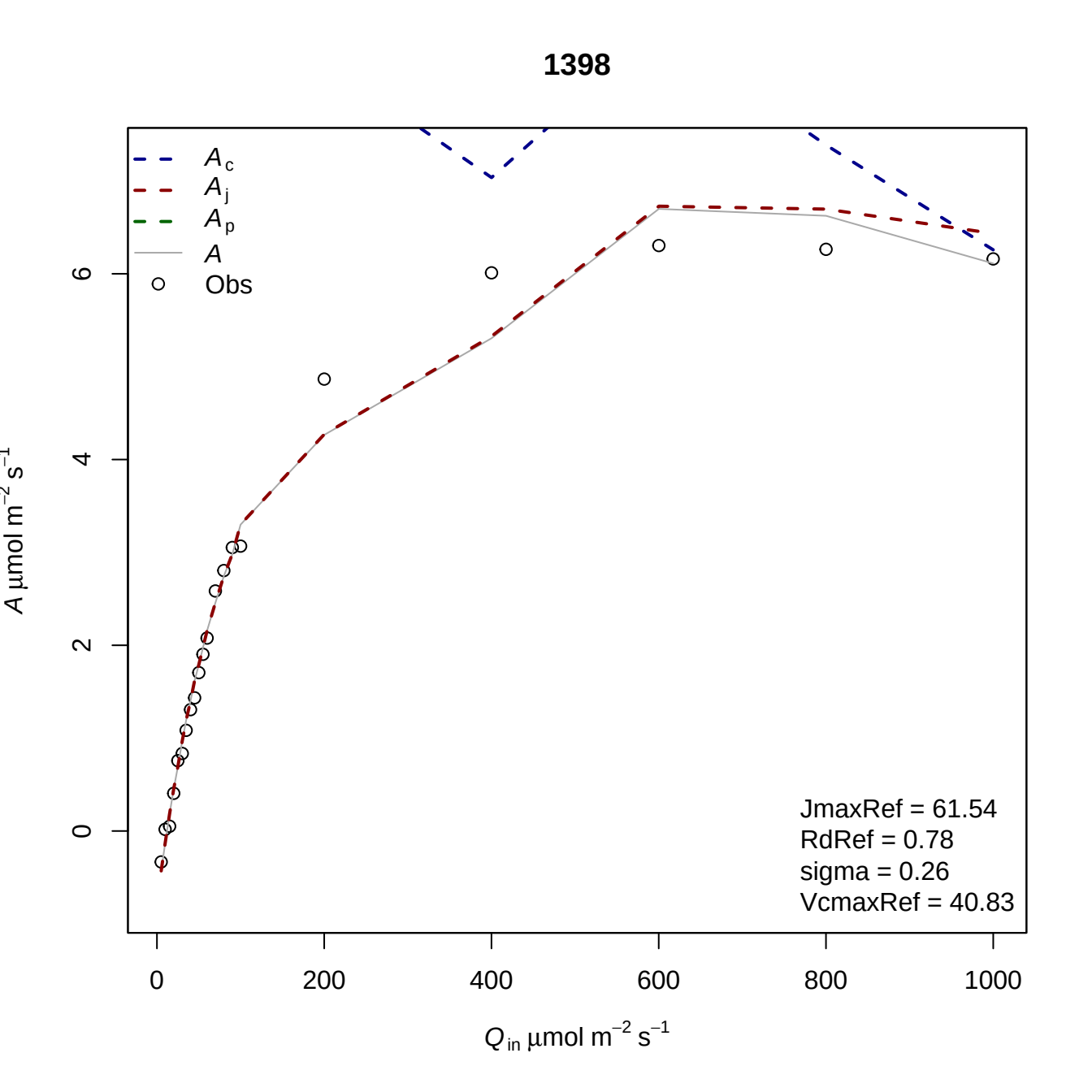

1393
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.91
RdRef = 0.71
sigma = 0.11
VcmaxRef = 13.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
200
400
600
800
1000

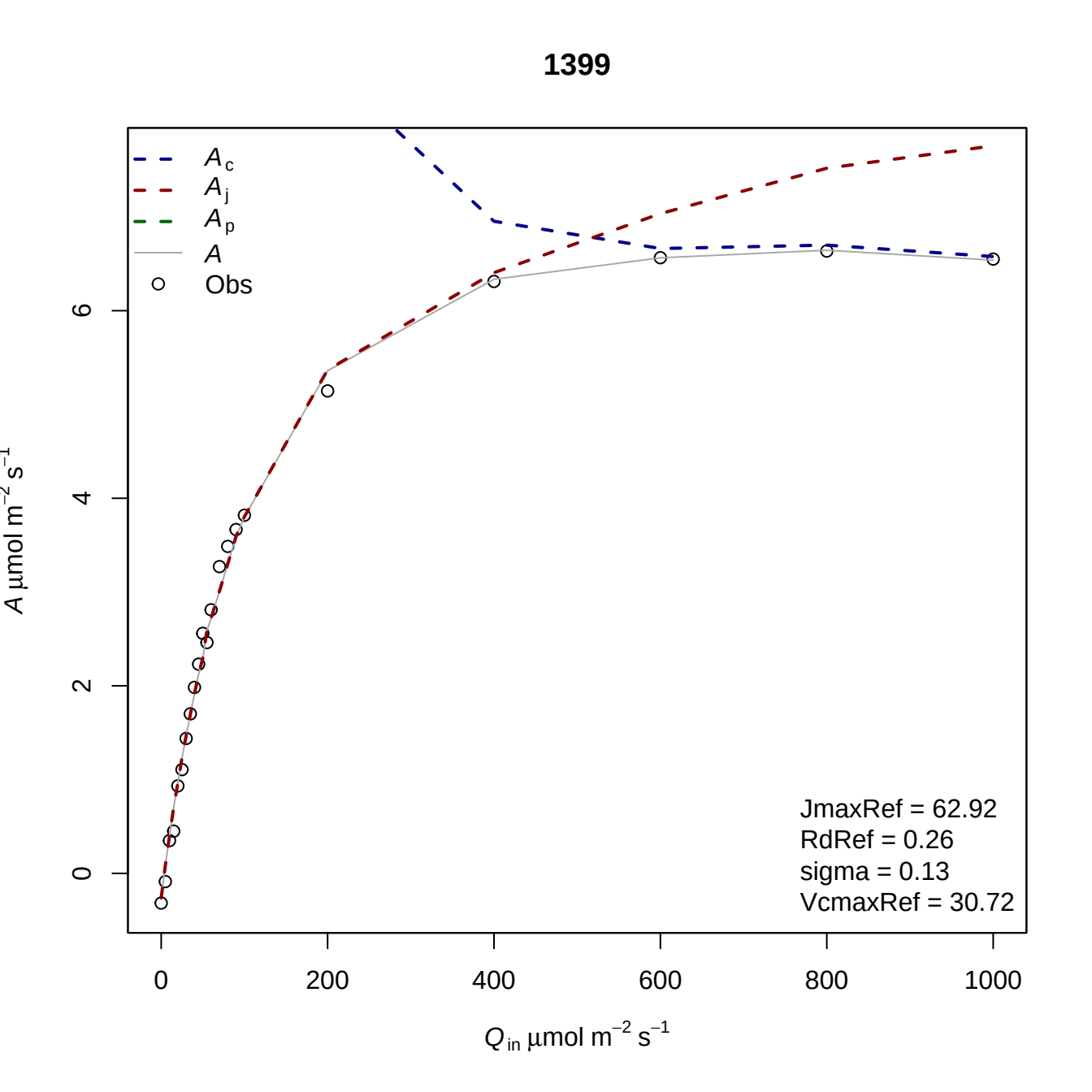

1394
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.54
RdRef = 0.46
sigma = 0.22
VcmaxRef = 38.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
200
400
600
800

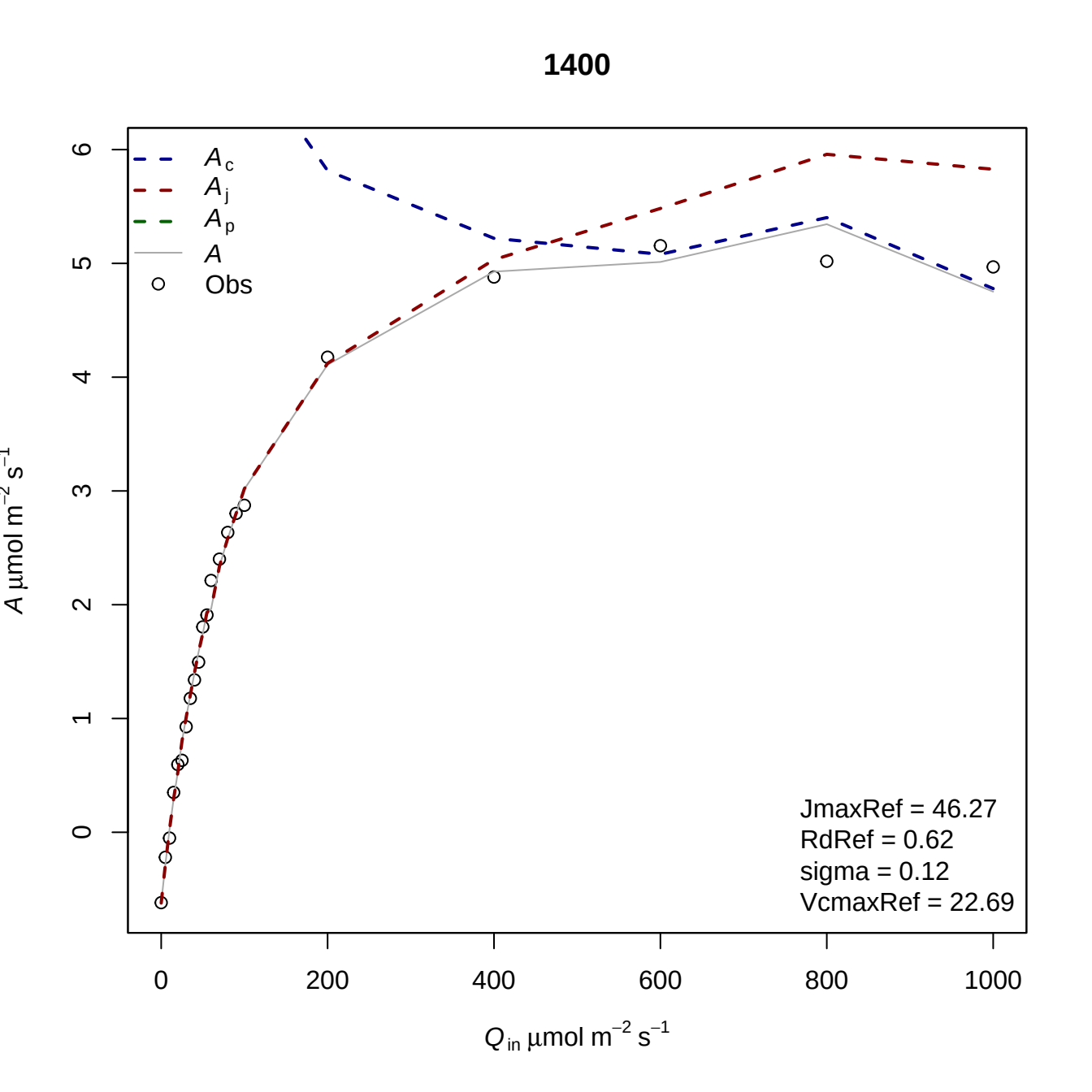

1395
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.44
RdRef = 0.57
sigma = 0.11
VcmaxRef = 46.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
200
400
600
800
1000

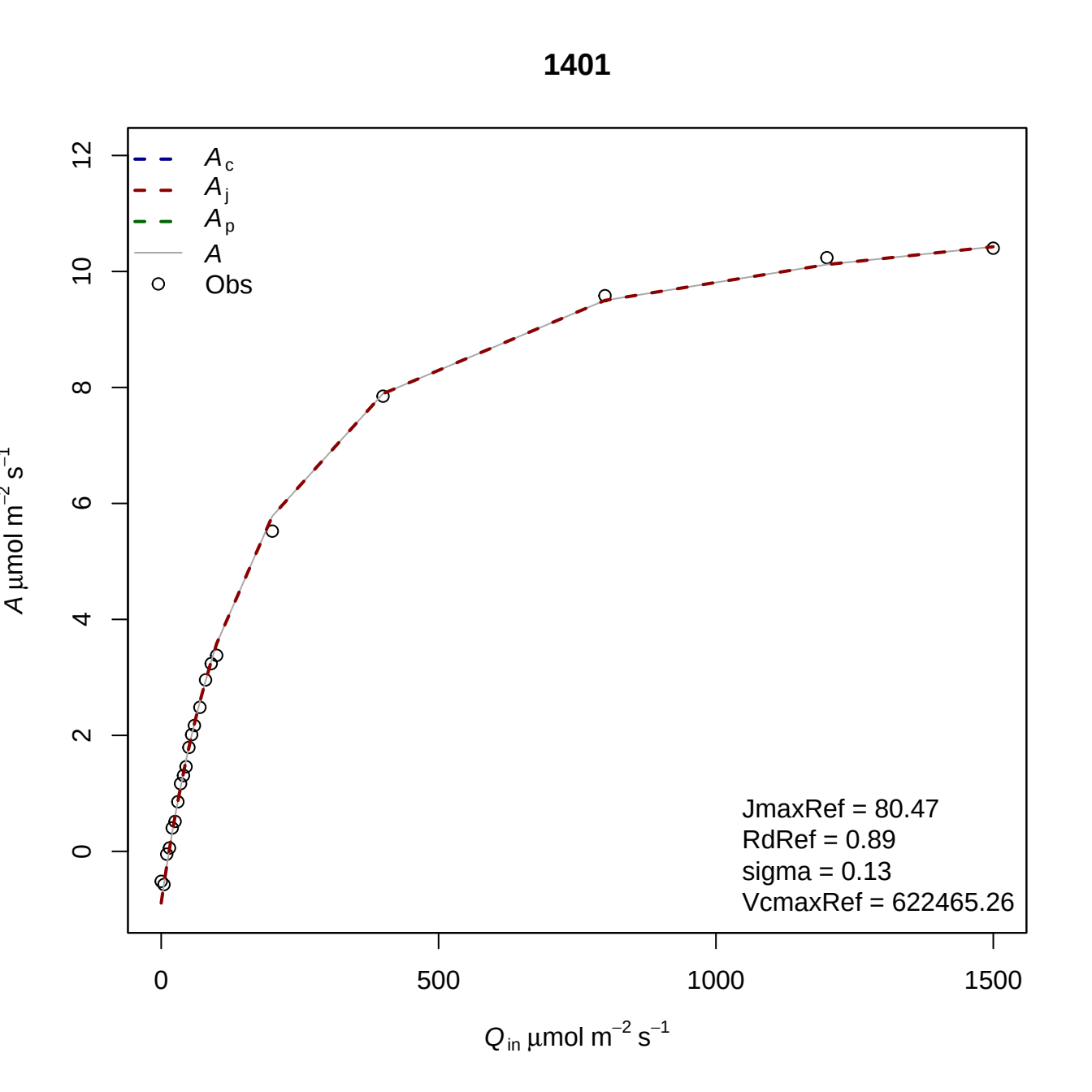

1396
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.77
RdRef = 0.27
sigma = 0.16
VcmaxRef = 47.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
200
400
600
800
1000

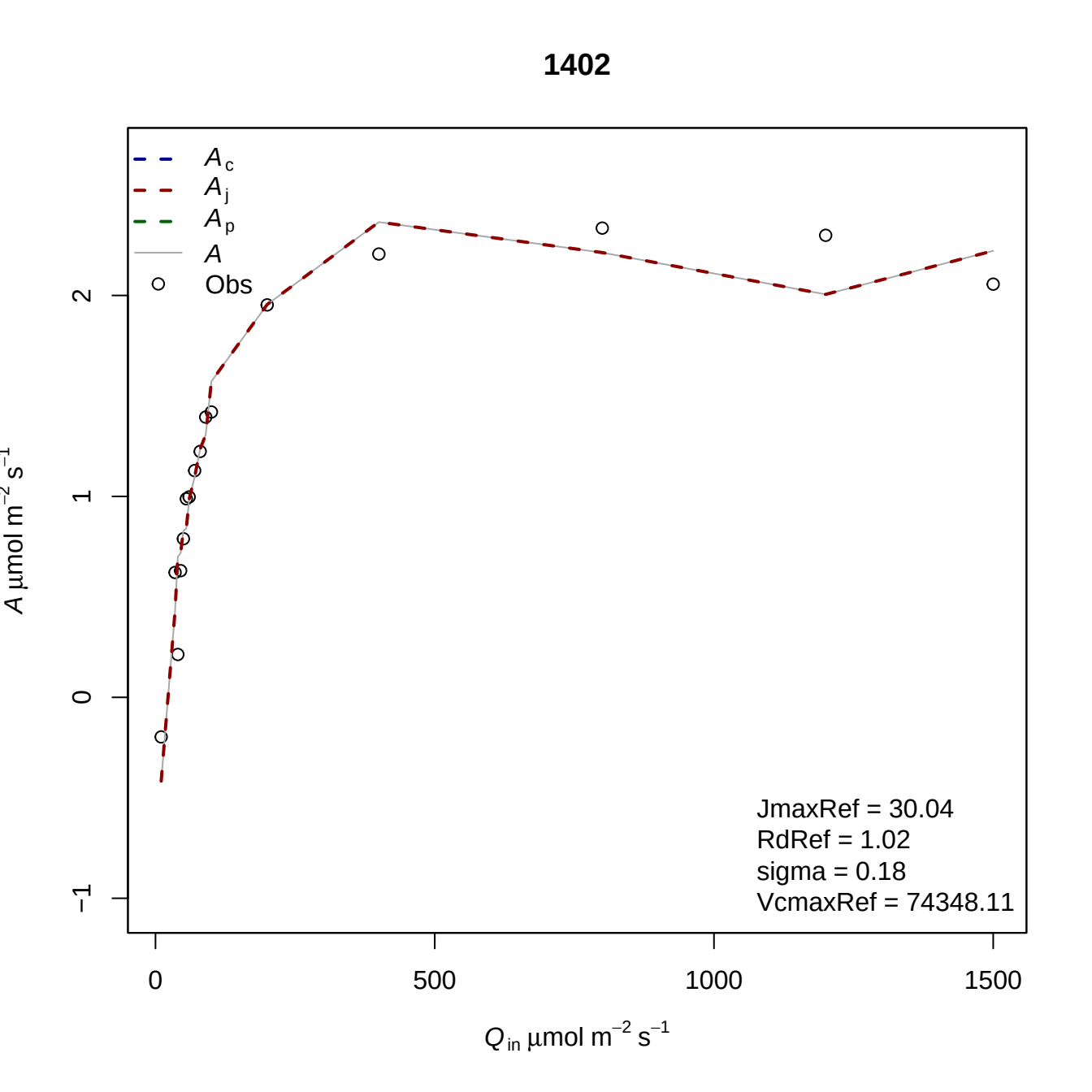

1397
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.62
RdRef = 0.58
sigma = 0.08
VcmaxRef = 21.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
200
400
600
800
1000

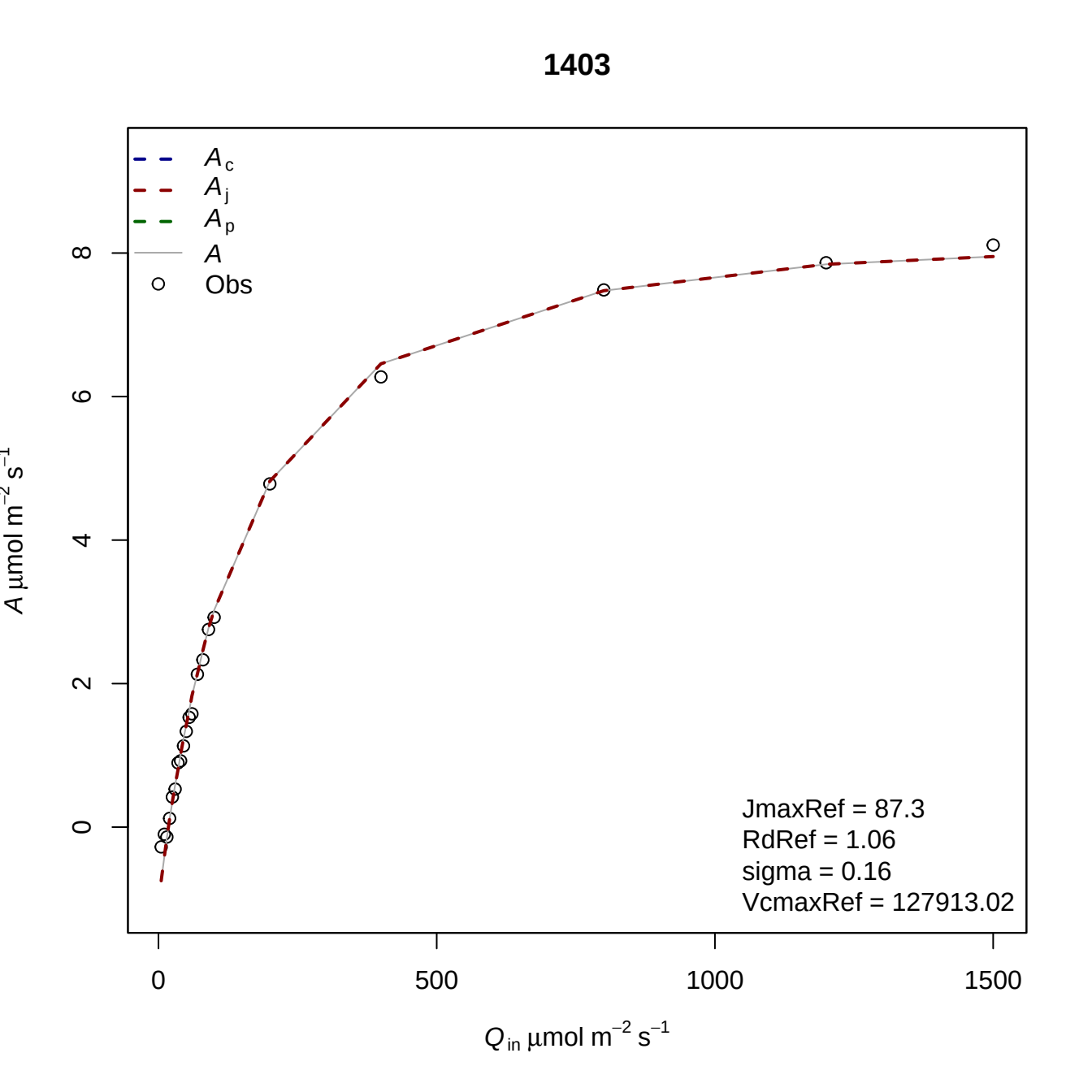

1398
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.54
RdRef = 0.78
sigma = 0.26
VcmaxRef = 40.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
200
400
600
800
1000

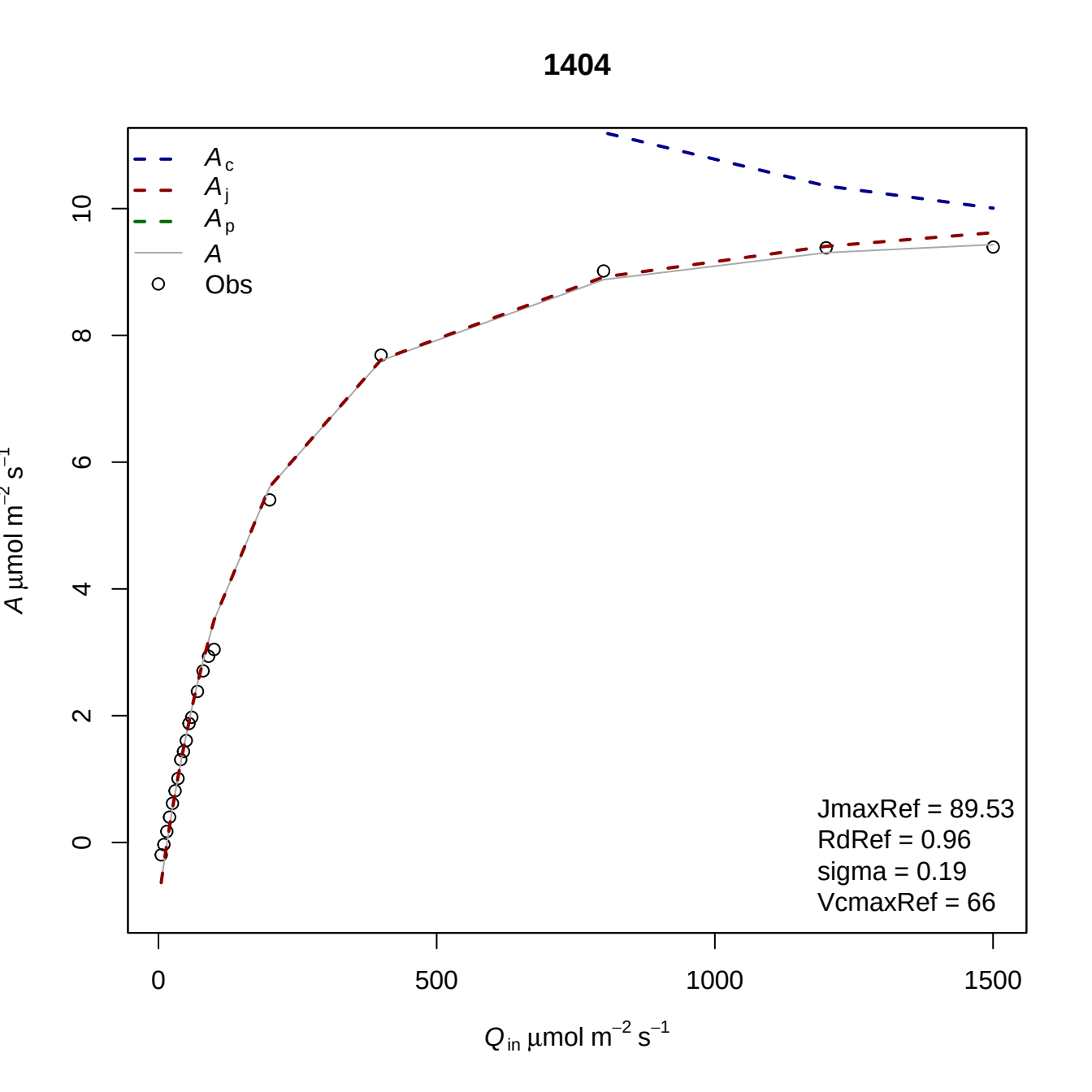

1399
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.92
RdRef = 0.26
sigma = 0.13
VcmaxRef = 30.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
200
400
600
800
1000

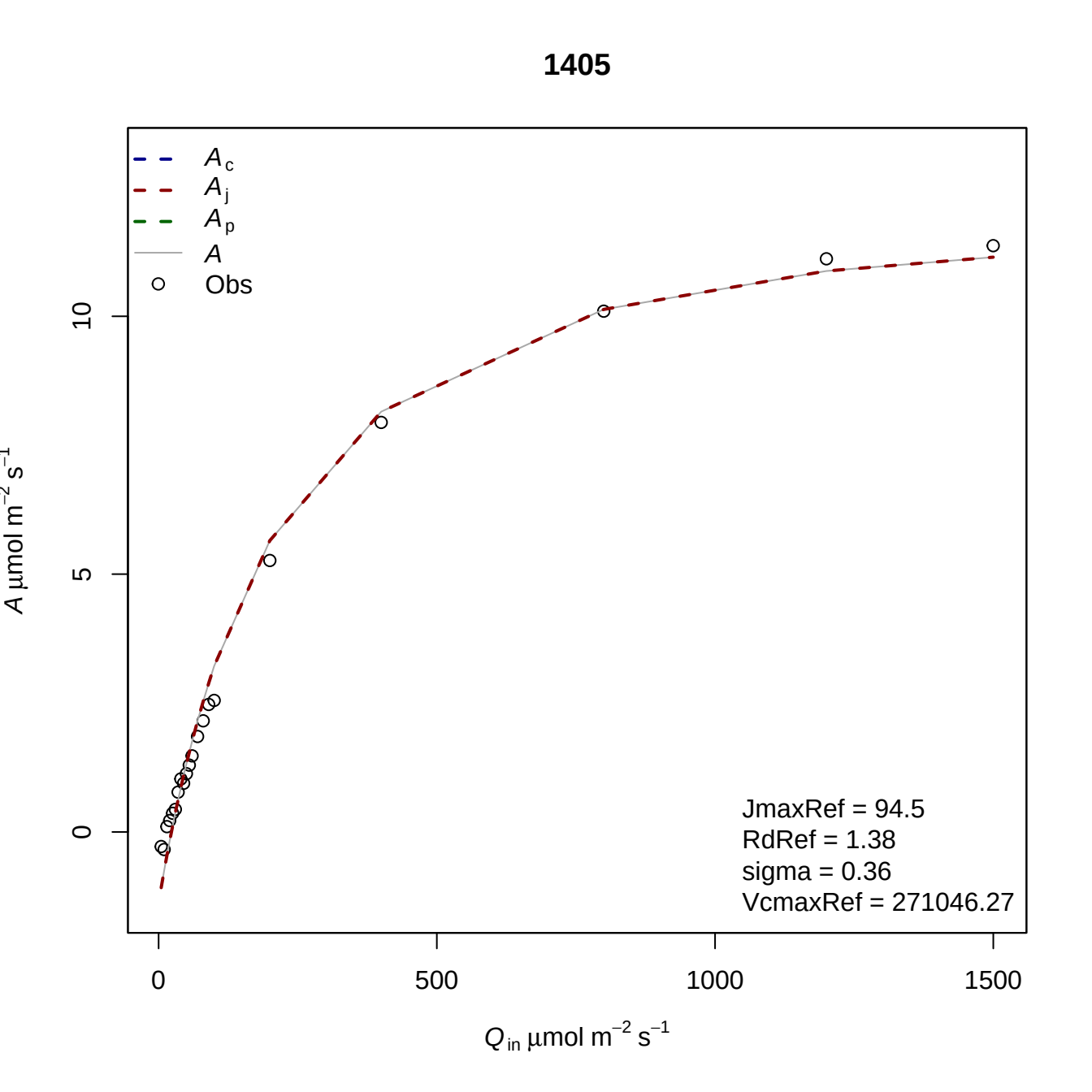

1400
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 46.27
RdRef = 0.62
sigma = 0.12
VcmaxRef = 22.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
200
400
600
800
1000

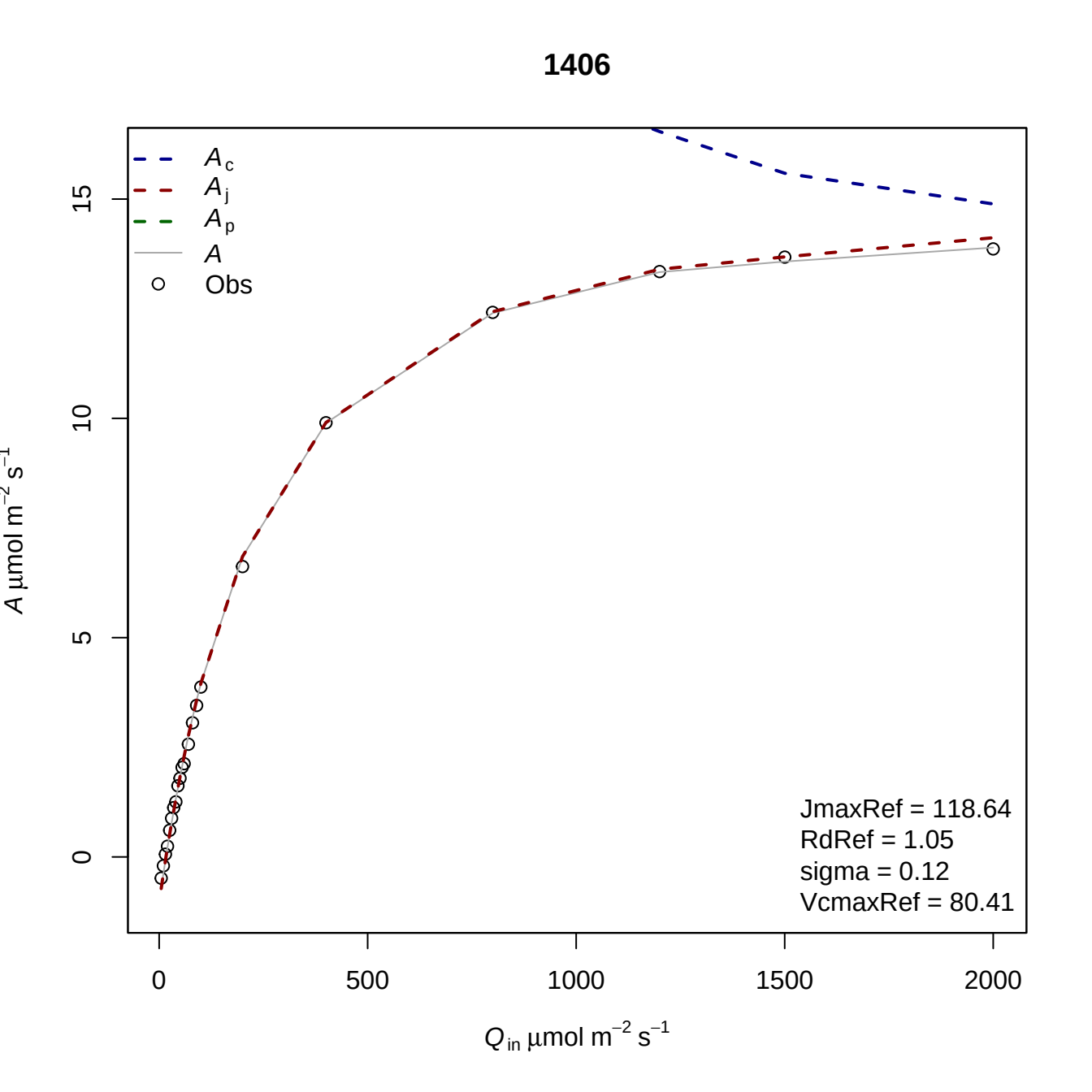

1401
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.47
RdRef = 0.89
sigma = 0.13
VcmaxRef = 622465.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

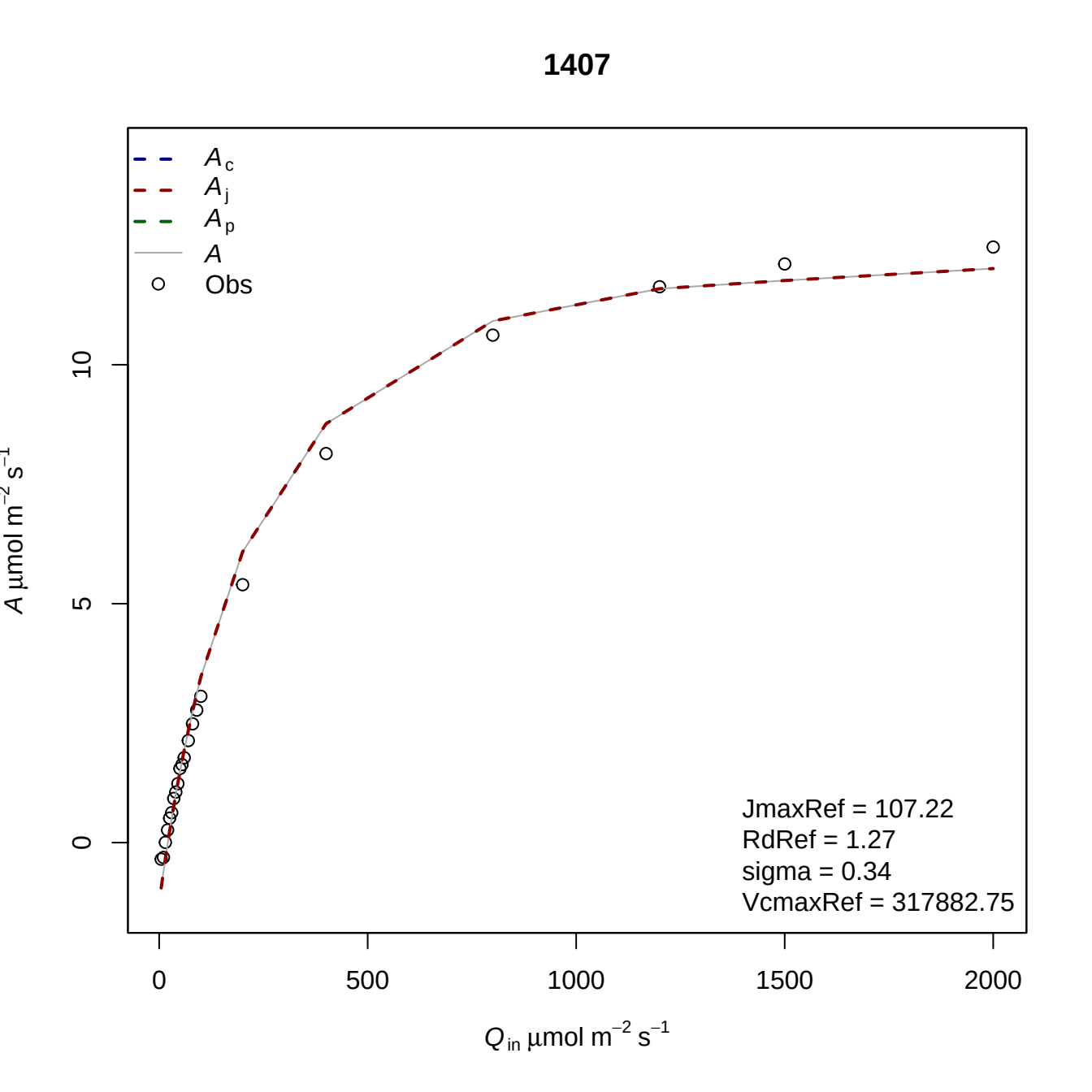

1402
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 30.04
RdRef = 1.02
sigma = 0.18
VcmaxRef = 74348.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
0
500
1000
1500

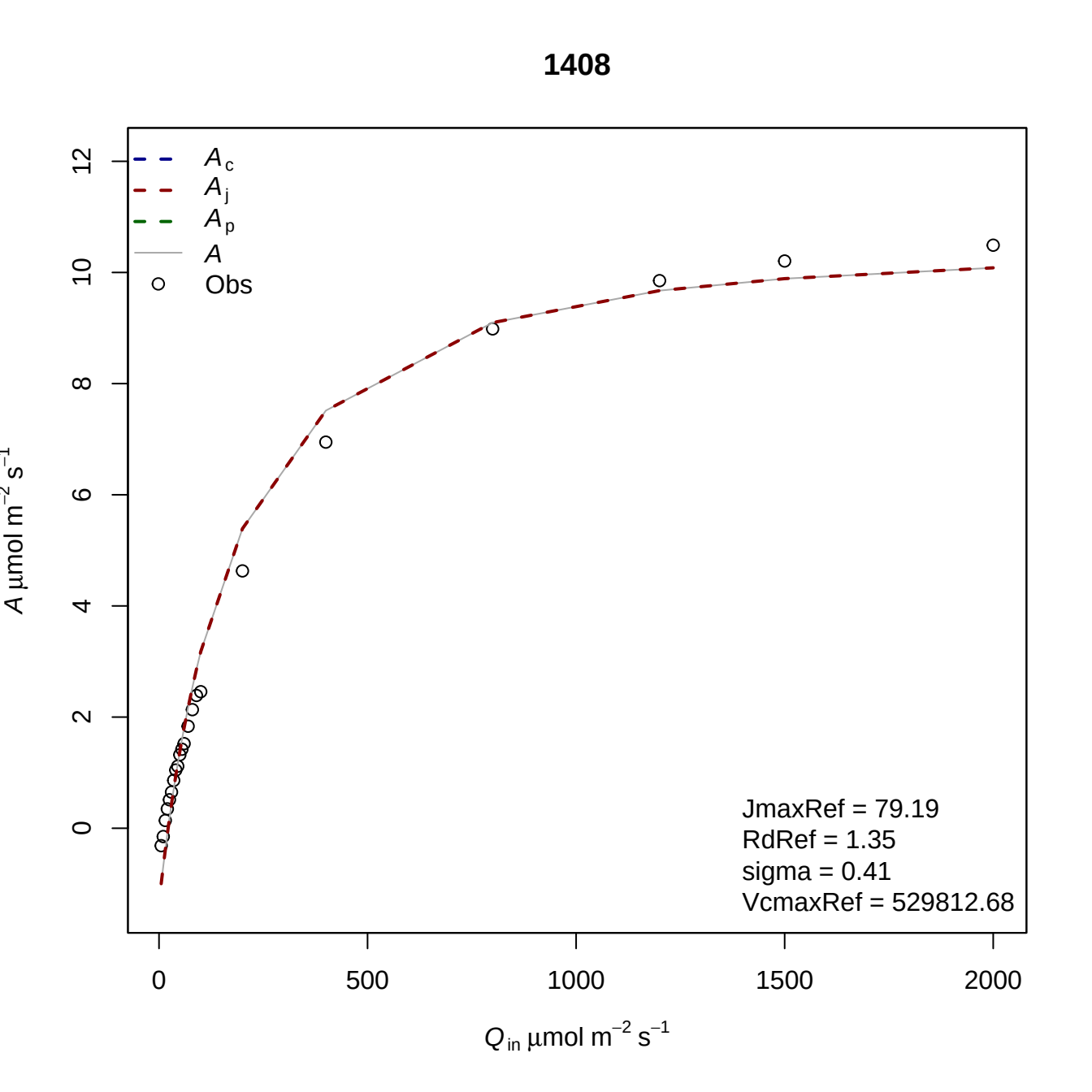

1403
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.3
RdRef = 1.06
sigma = 0.16
VcmaxRef = 127913.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

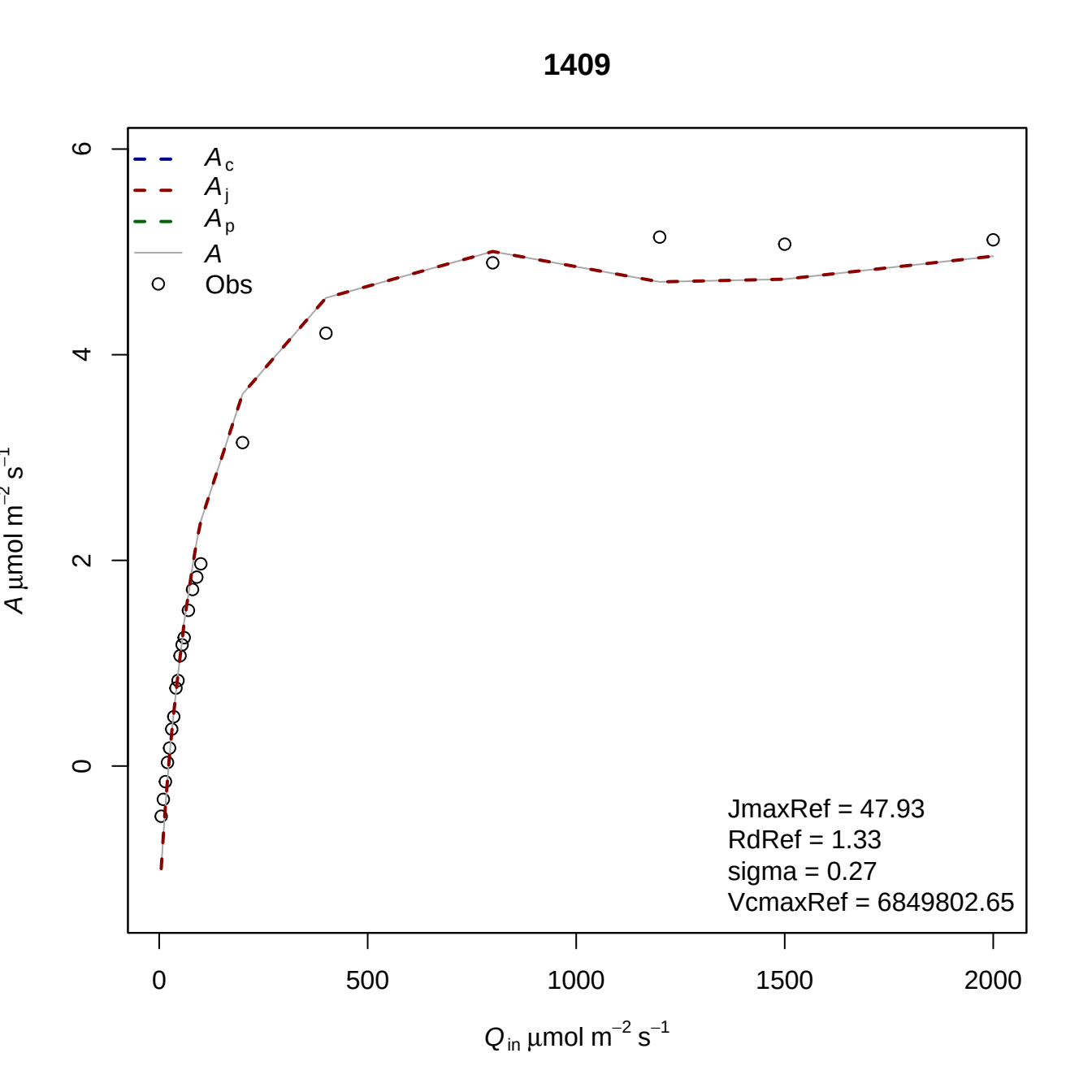

1404
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.53
RdRef = 0.96
sigma = 0.19
VcmaxRef = 66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

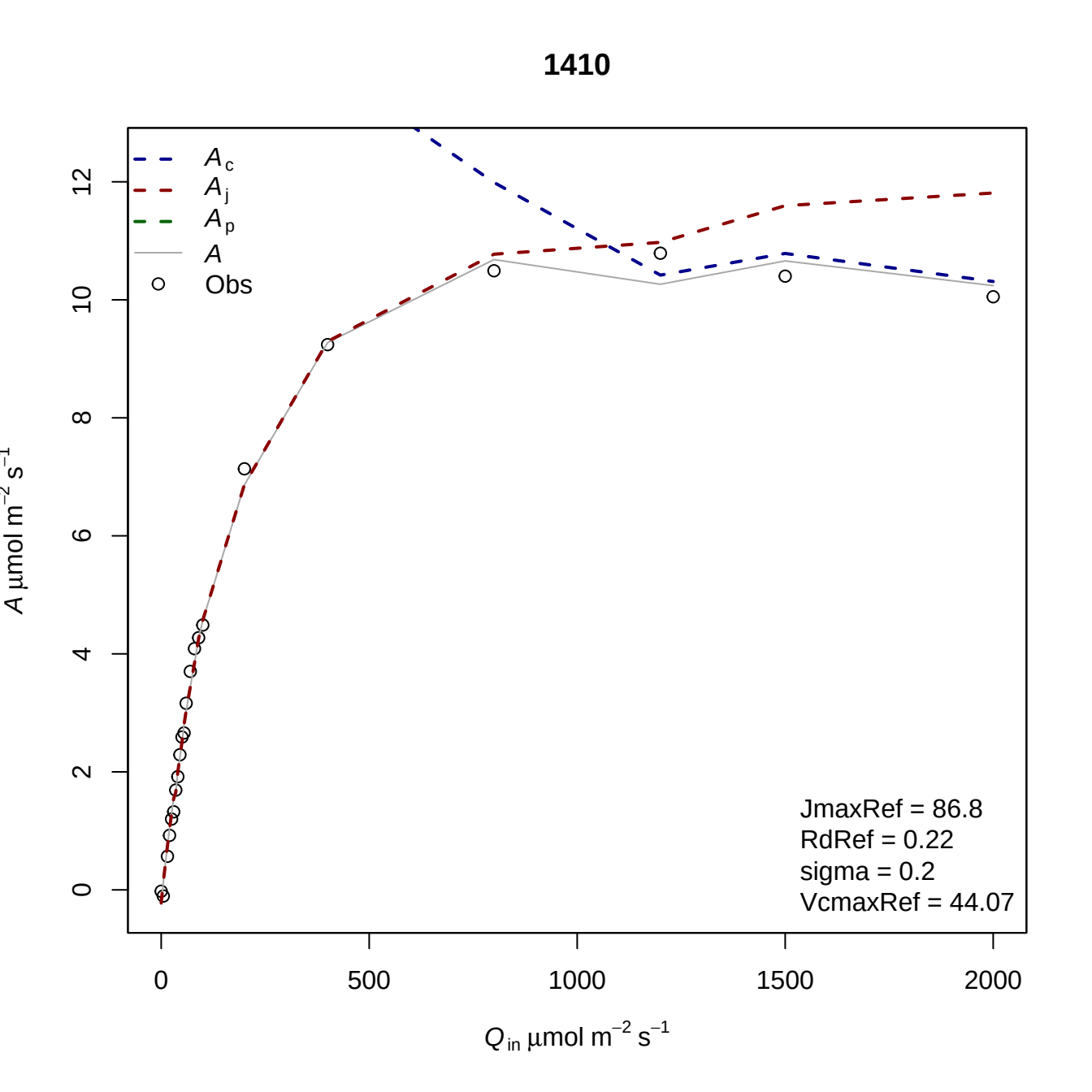

1405
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.5
RdRef = 1.38
sigma = 0.36
VcmaxRef = 271046.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

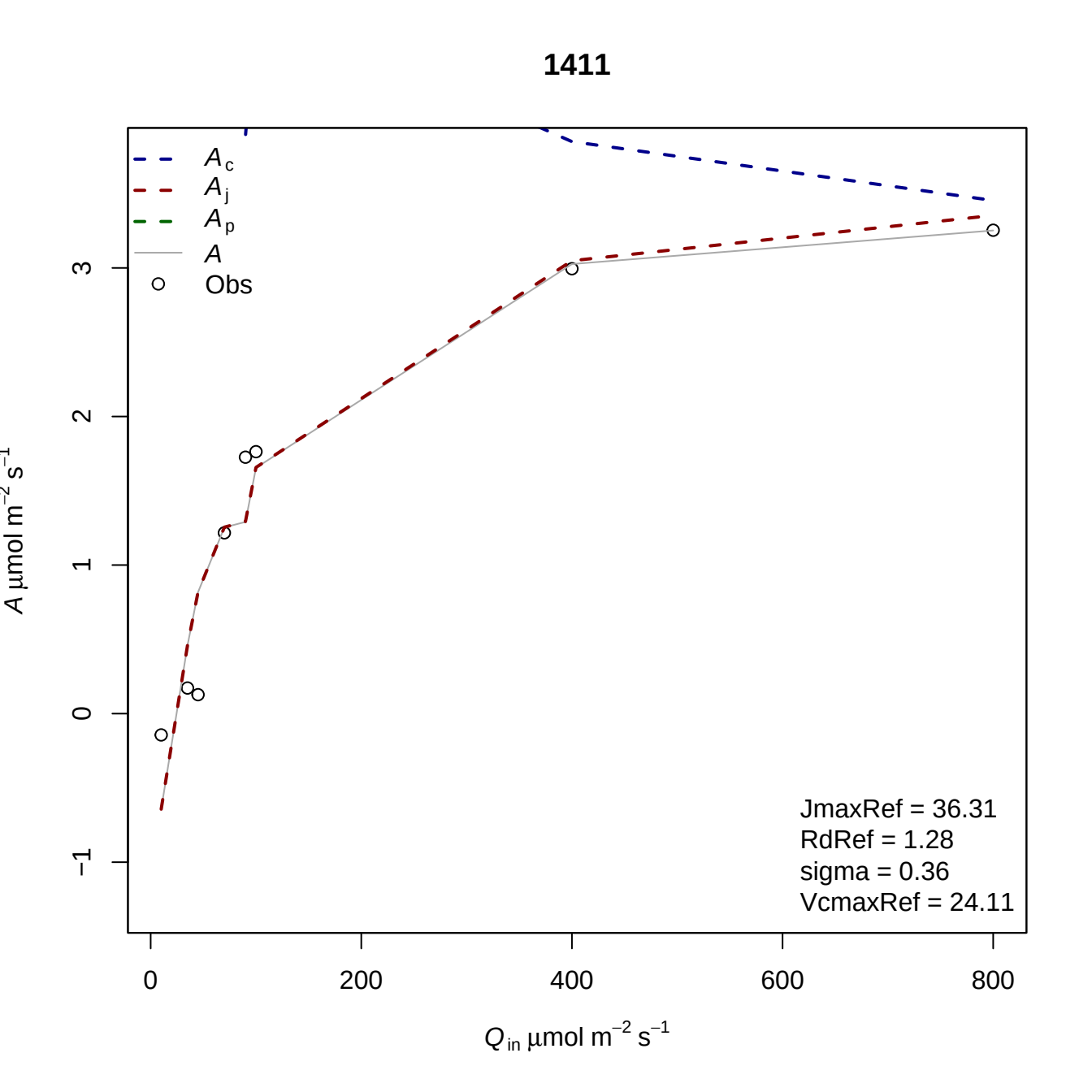

1406
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.64
RdRef = 1.05
sigma = 0.12
VcmaxRef = 80.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

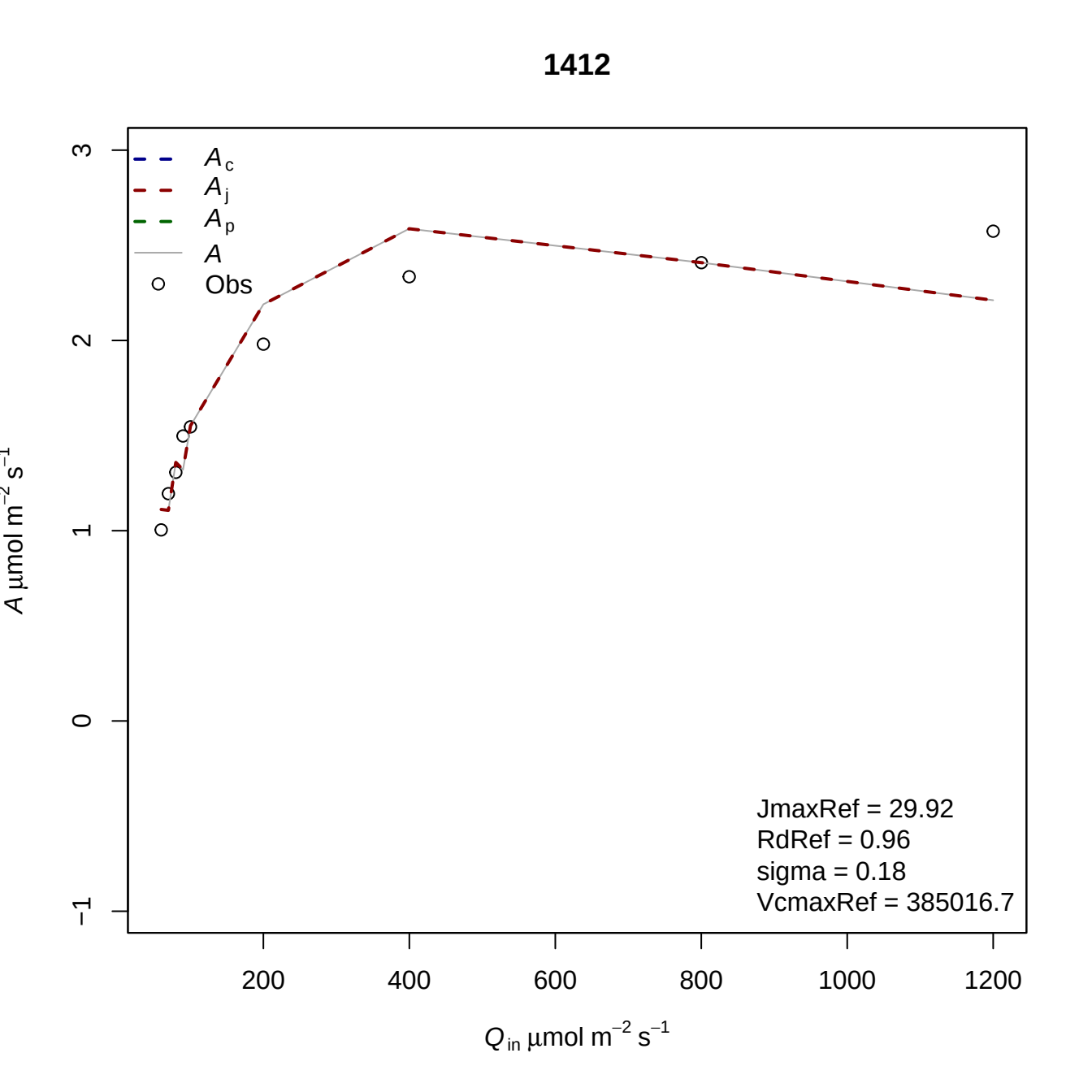

1407
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.22
RdRef = 1.27
sigma = 0.34
VcmaxRef = 317882.75
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

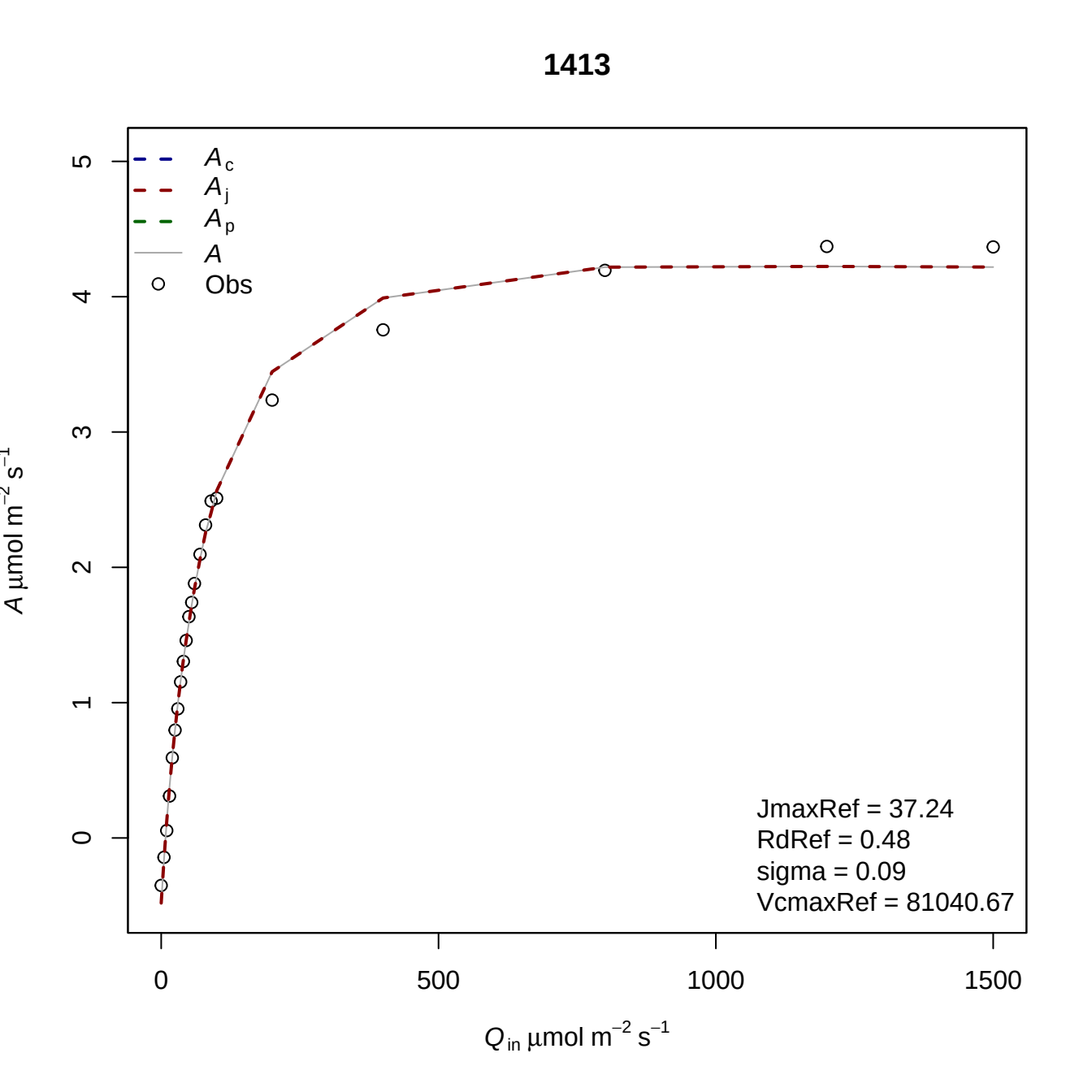

1408
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.19
RdRef = 1.35
sigma = 0.41
VcmaxRef = 529812.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

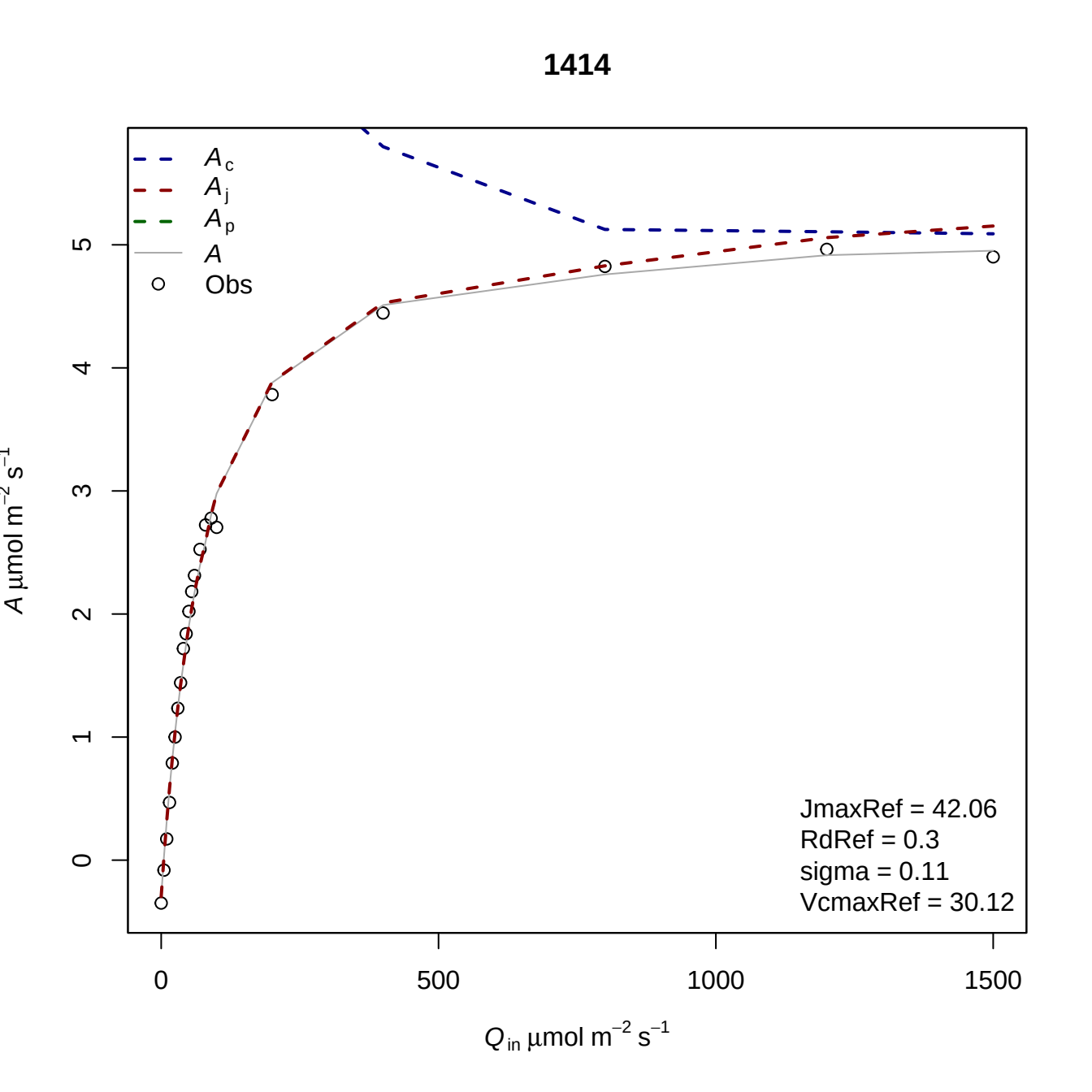

1409
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.93
RdRef = 1.33
sigma = 0.27
VcmaxRef = 6849802.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

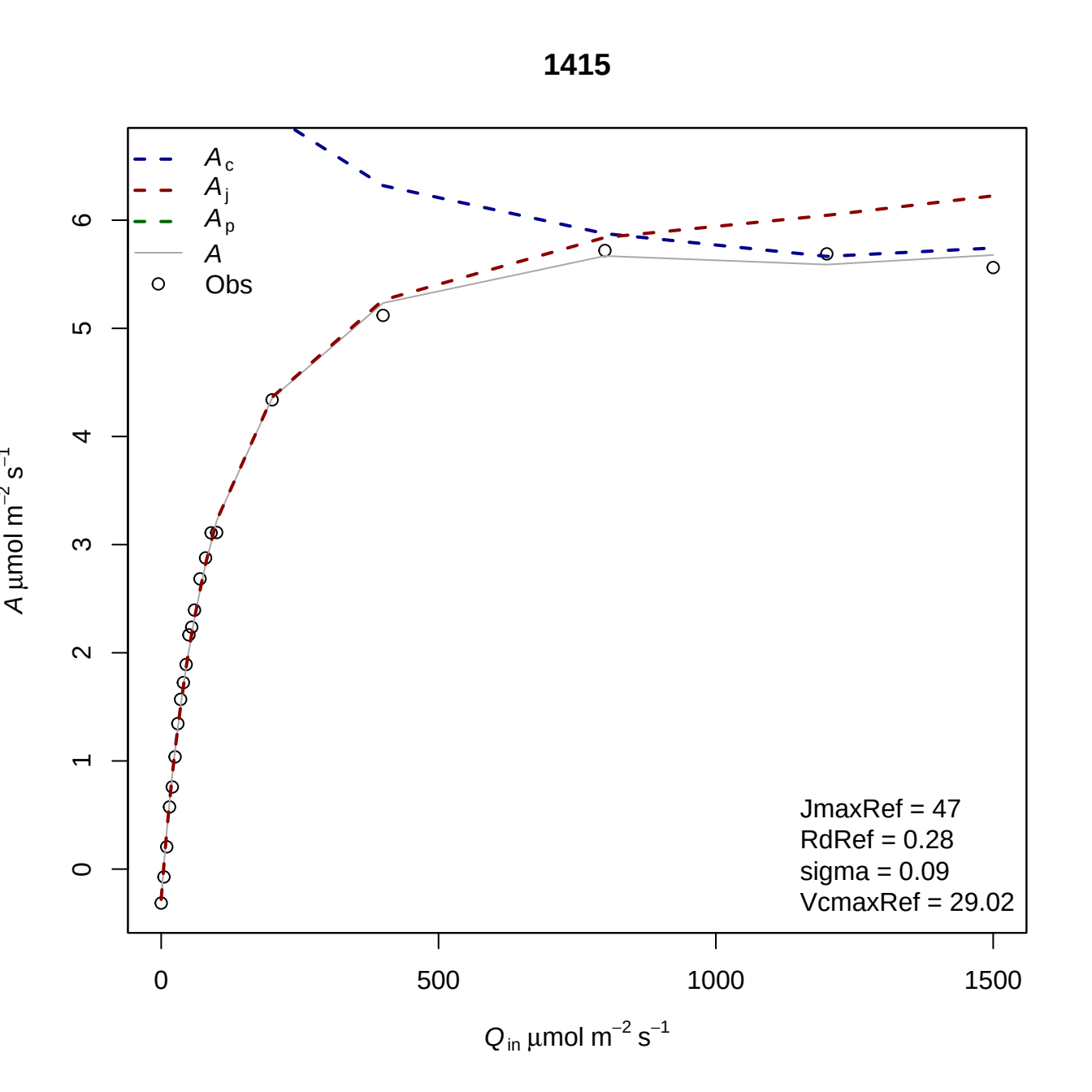

1410
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.8
RdRef = 0.22
sigma = 0.2
VcmaxRef = 44.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

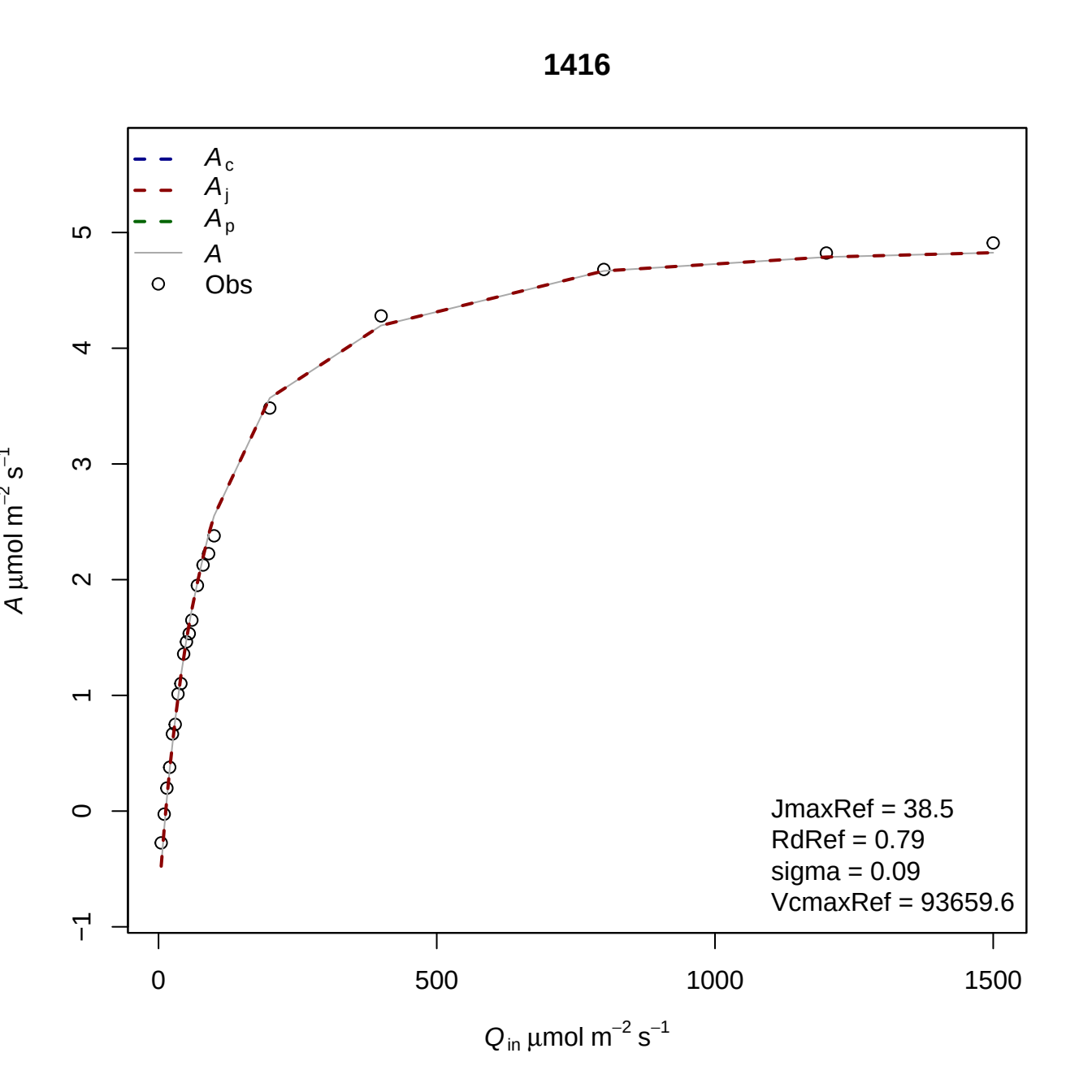

1411
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.31
RdRef = 1.28
sigma = 0.36
VcmaxRef = 24.11
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
200
400
600
800

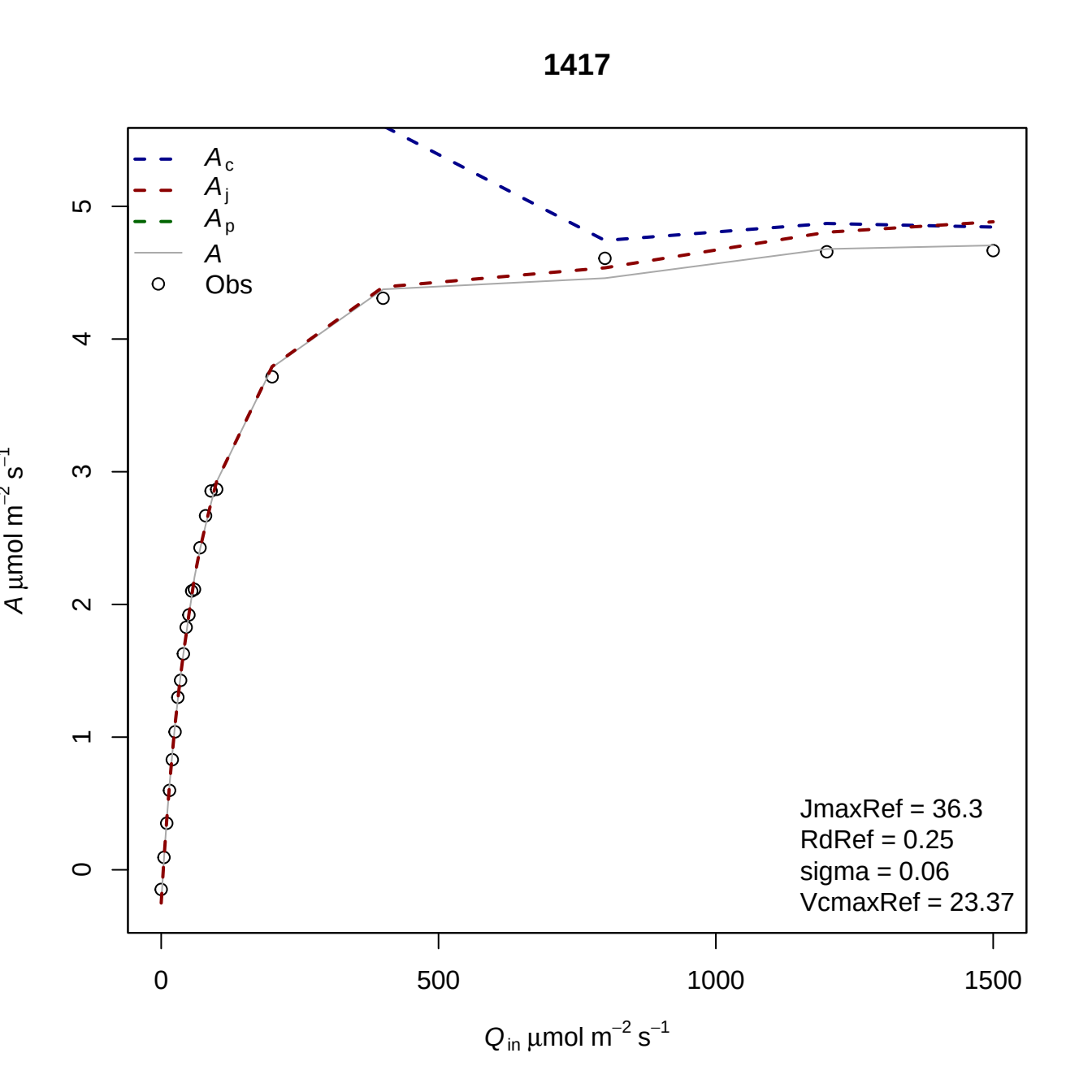

1412
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.92
RdRef = 0.96
sigma = 0.18
VcmaxRef = 385016.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
200
400
600
800
1000
1200

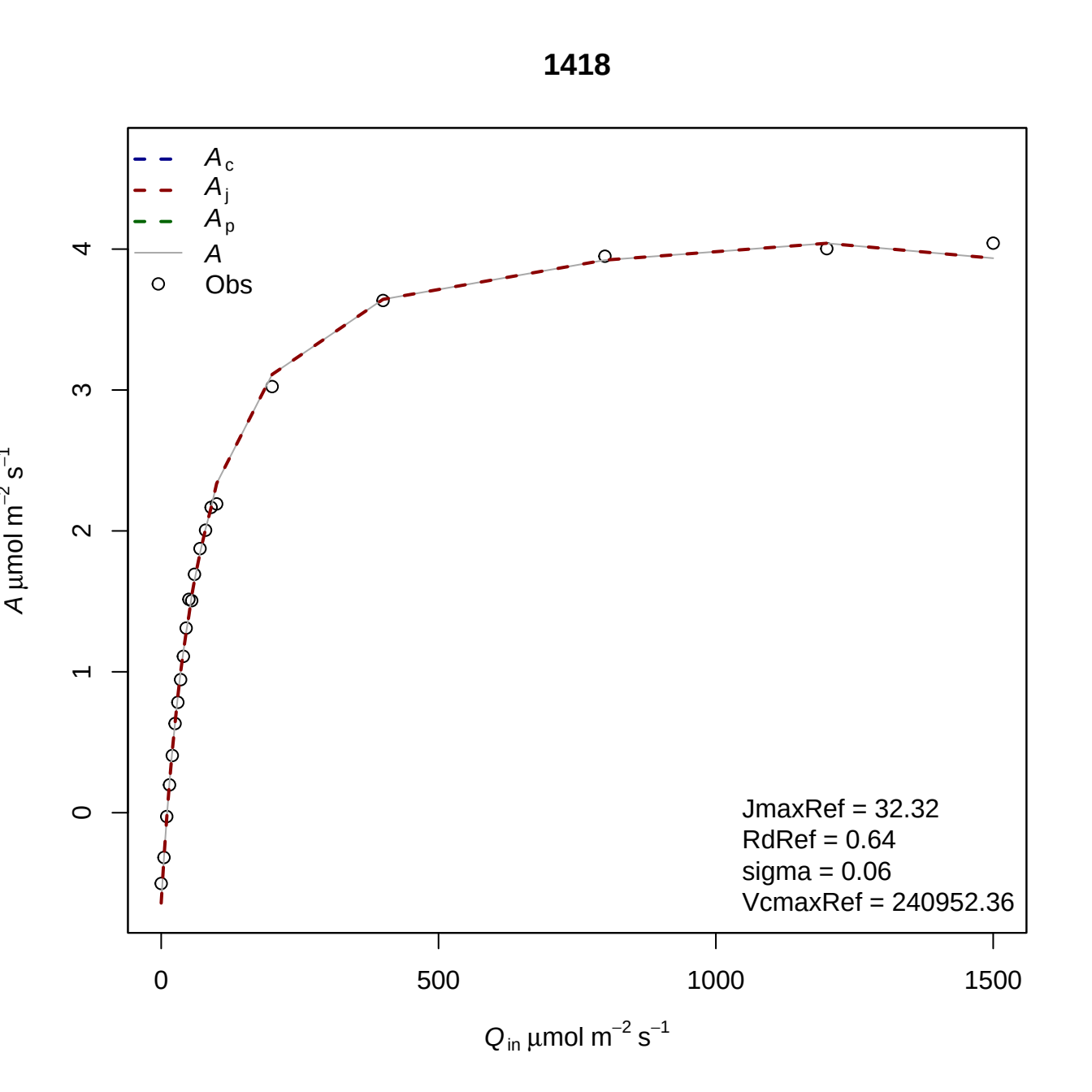

1413
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.24
RdRef = 0.48
sigma = 0.09
VcmaxRef = 81040.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

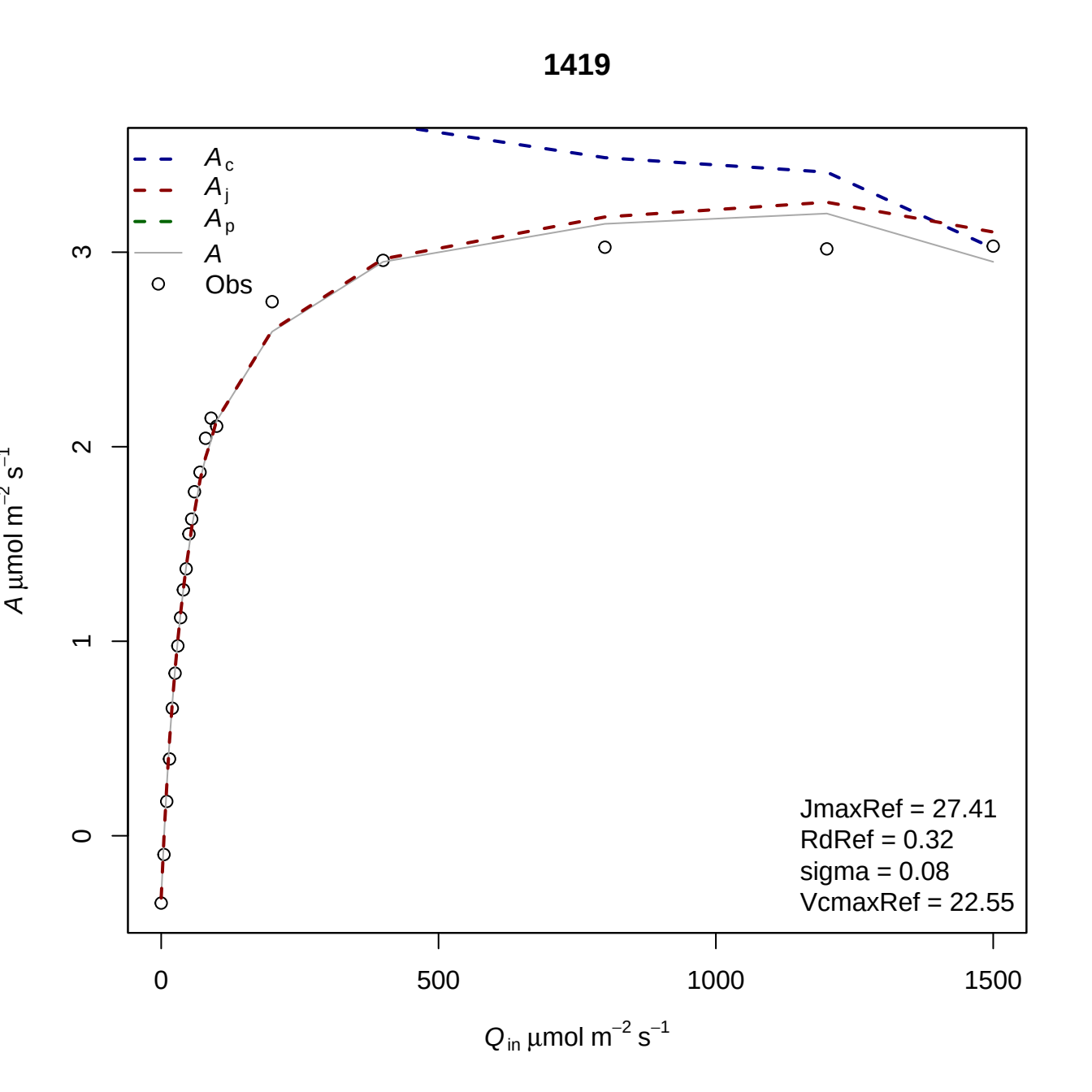

1414
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.06
RdRef = 0.3
sigma = 0.11
VcmaxRef = 30.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

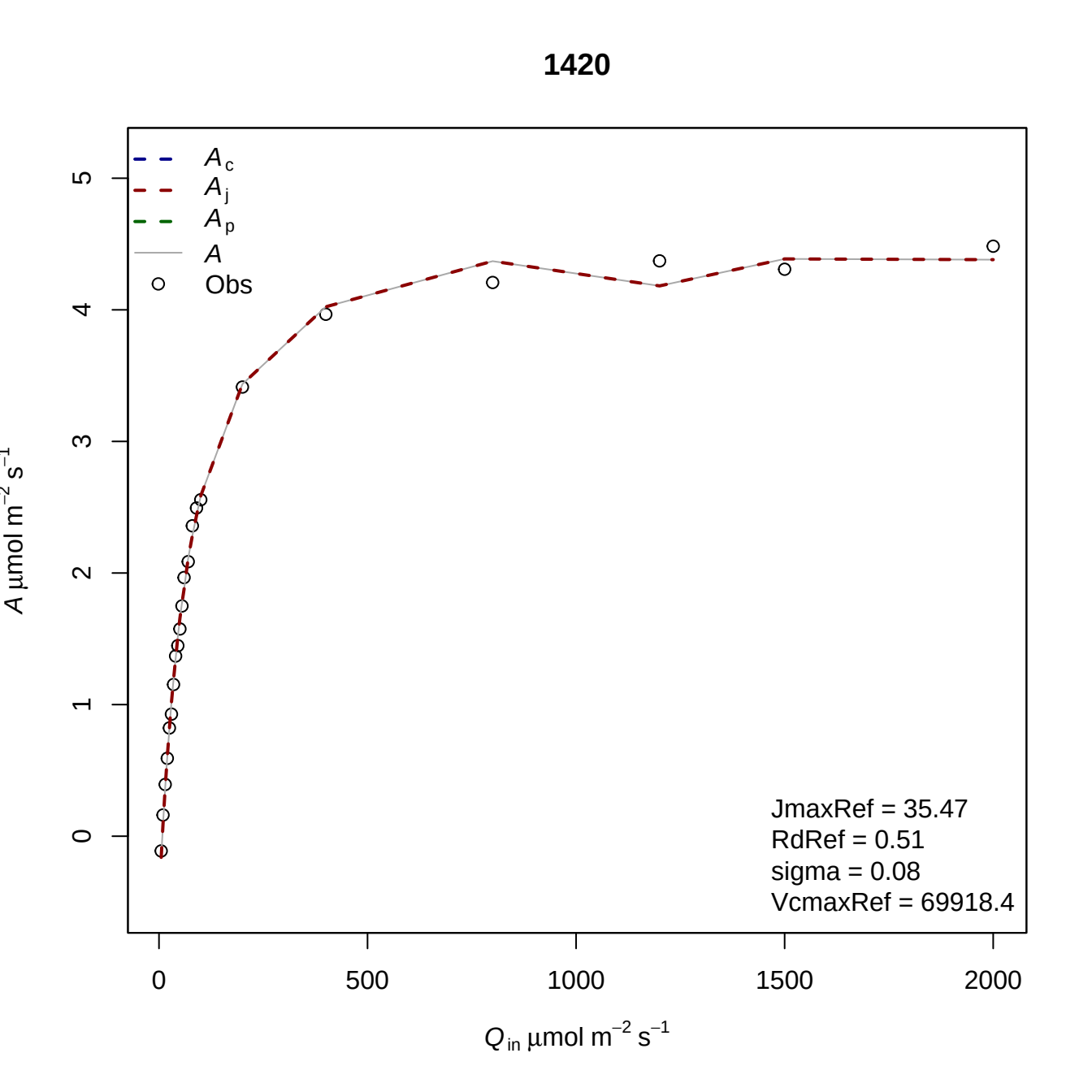

1415
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47
RdRef = 0.28
sigma = 0.09
VcmaxRef = 29.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

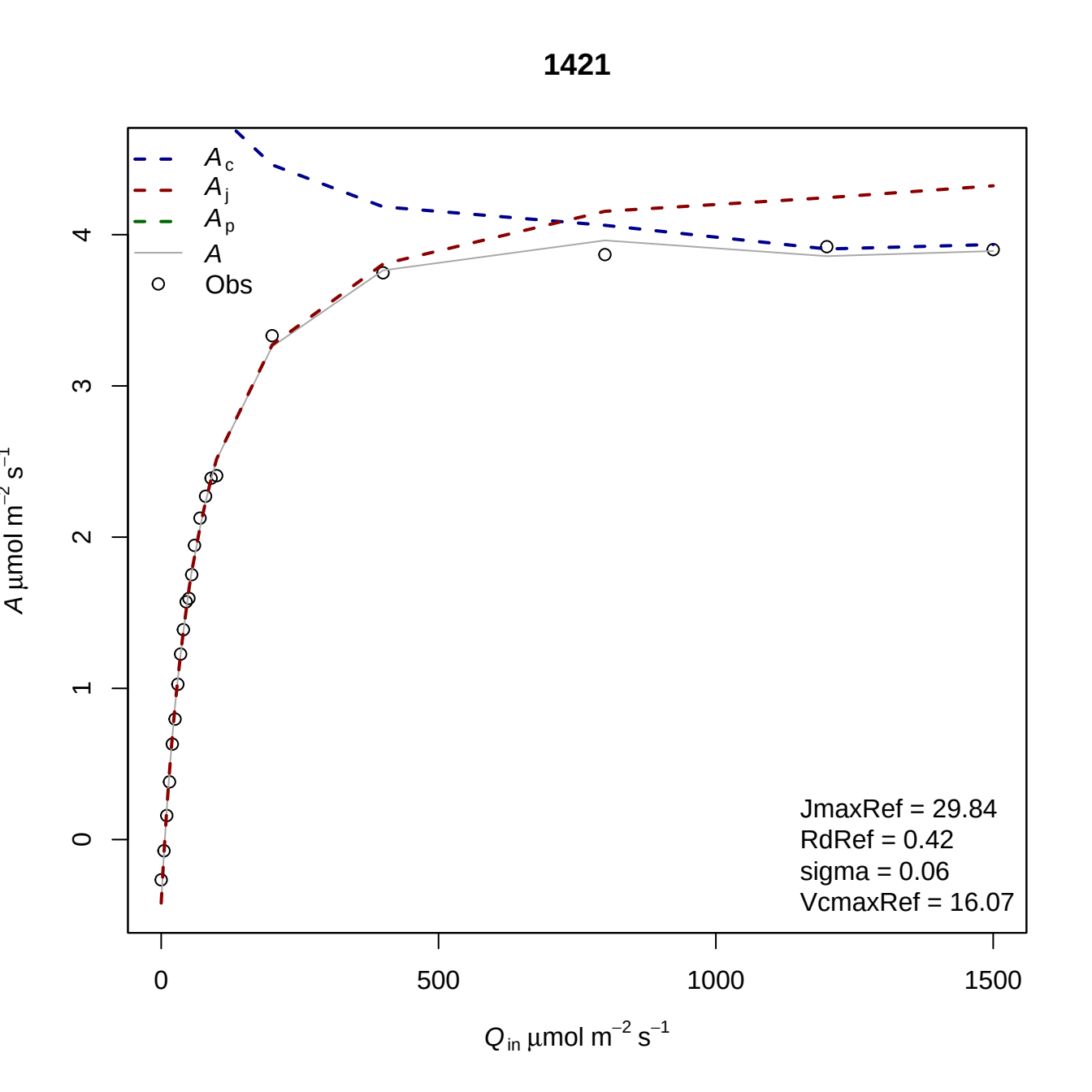

1416
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 38.5
RdRef = 0.79
sigma = 0.09
VcmaxRef = 93659.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

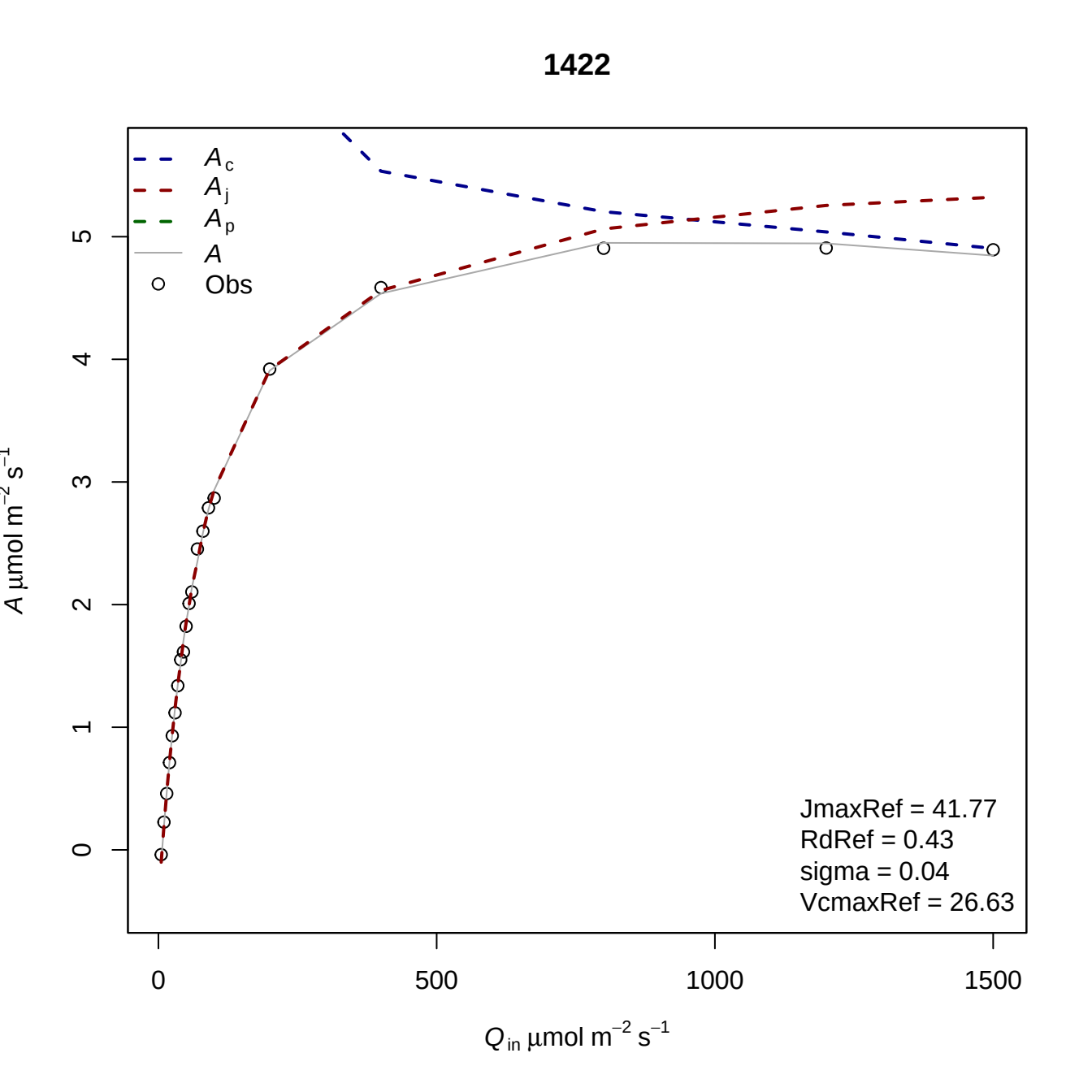

1417
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.3
RdRef = 0.25
sigma = 0.06
VcmaxRef = 23.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

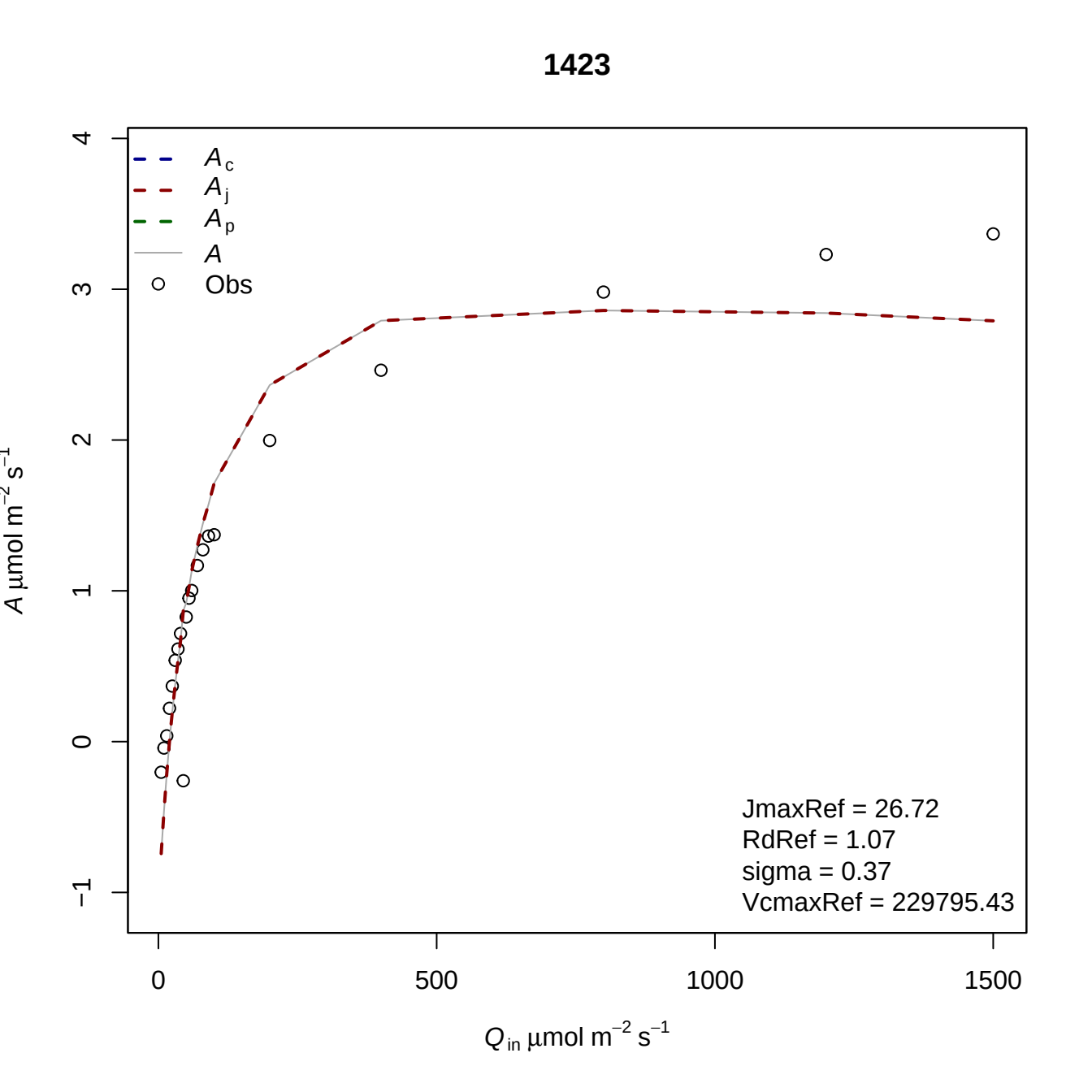

1418
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 32.32
RdRef = 0.64
sigma = 0.06
VcmaxRef = 240952.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

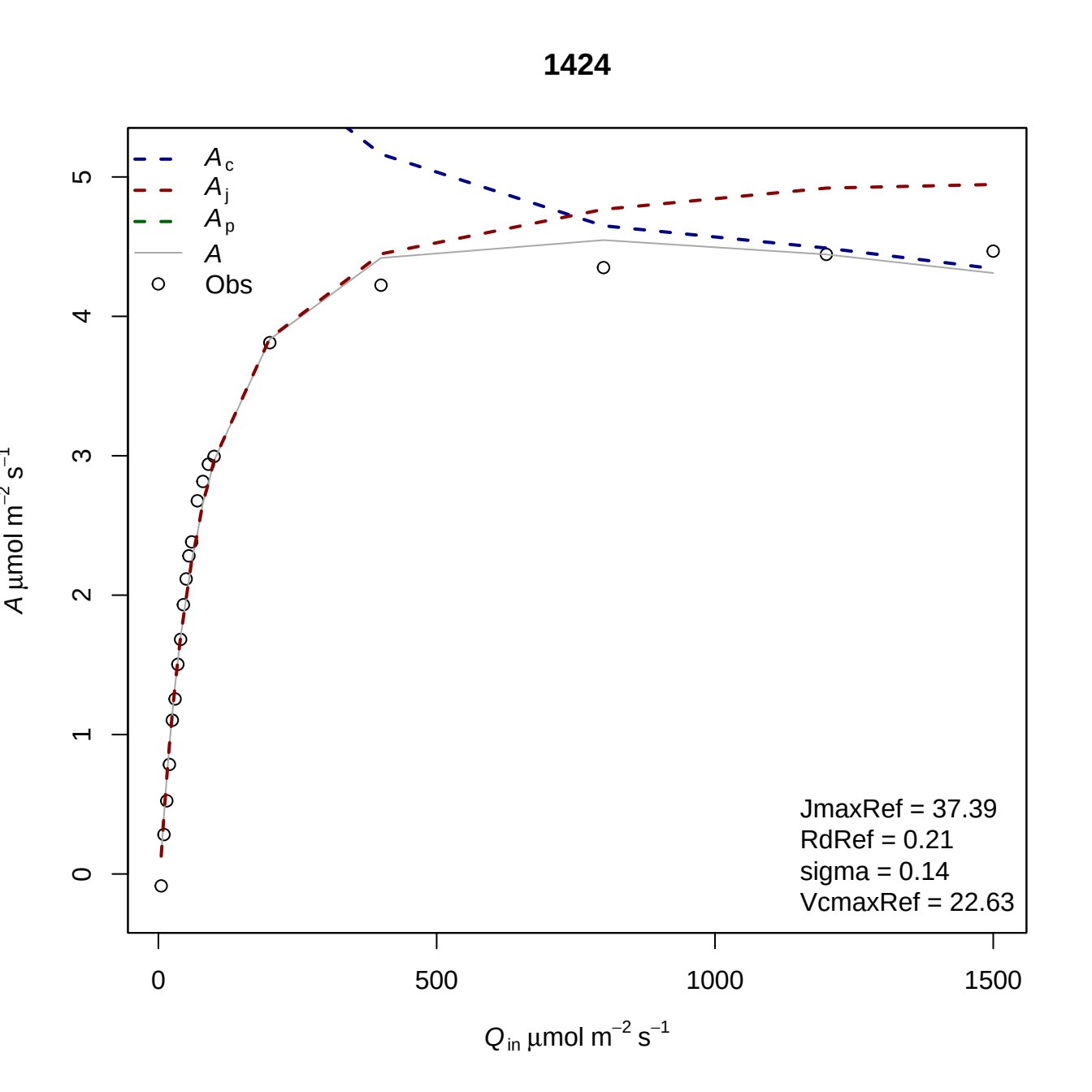

1419
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 27.41
RdRef = 0.32
sigma = 0.08
VcmaxRef = 22.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
0
500
1000
1500

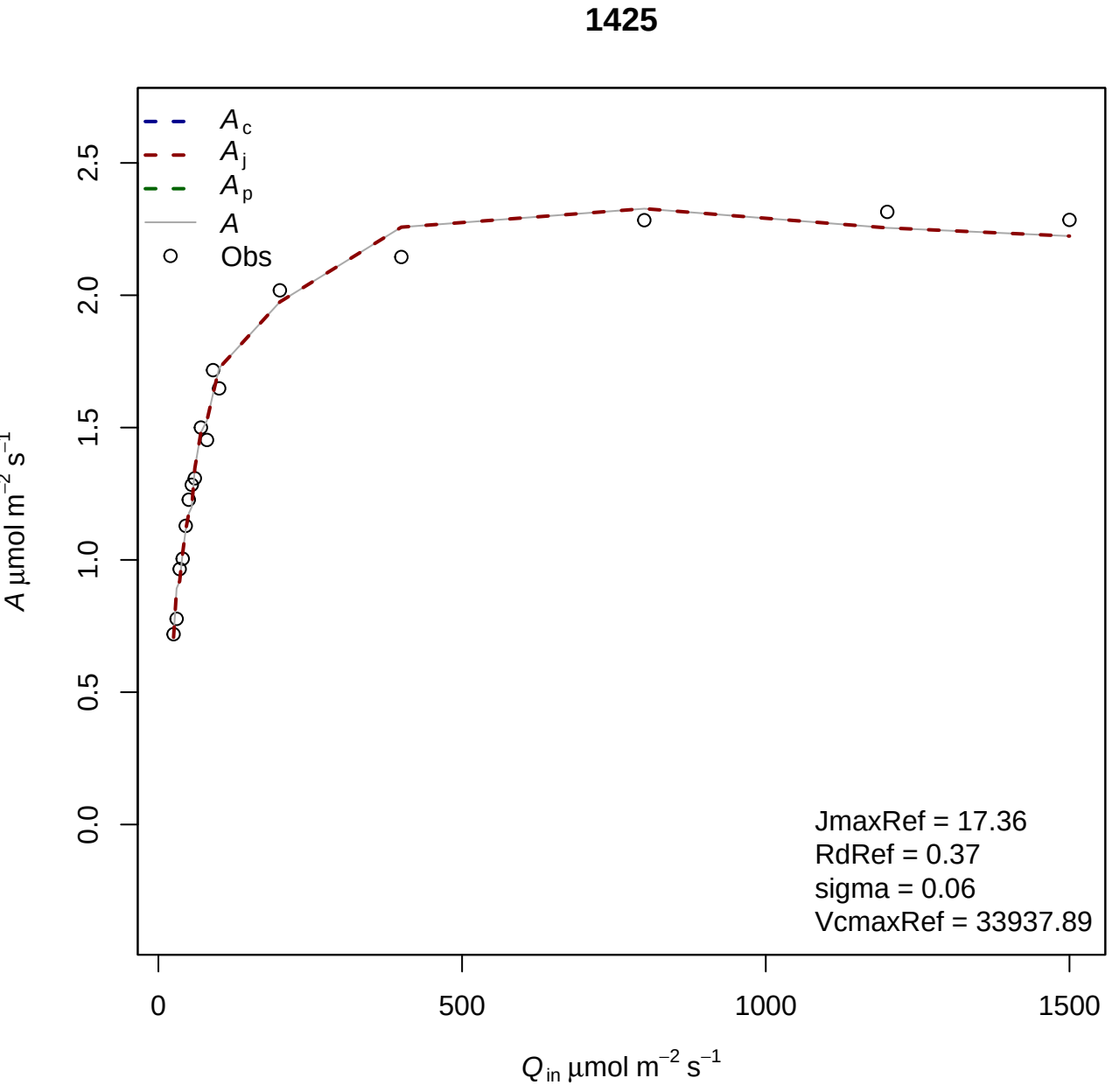

1420
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 35.47
RdRef = 0.51
sigma = 0.08
VcmaxRef = 69918.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500
2000

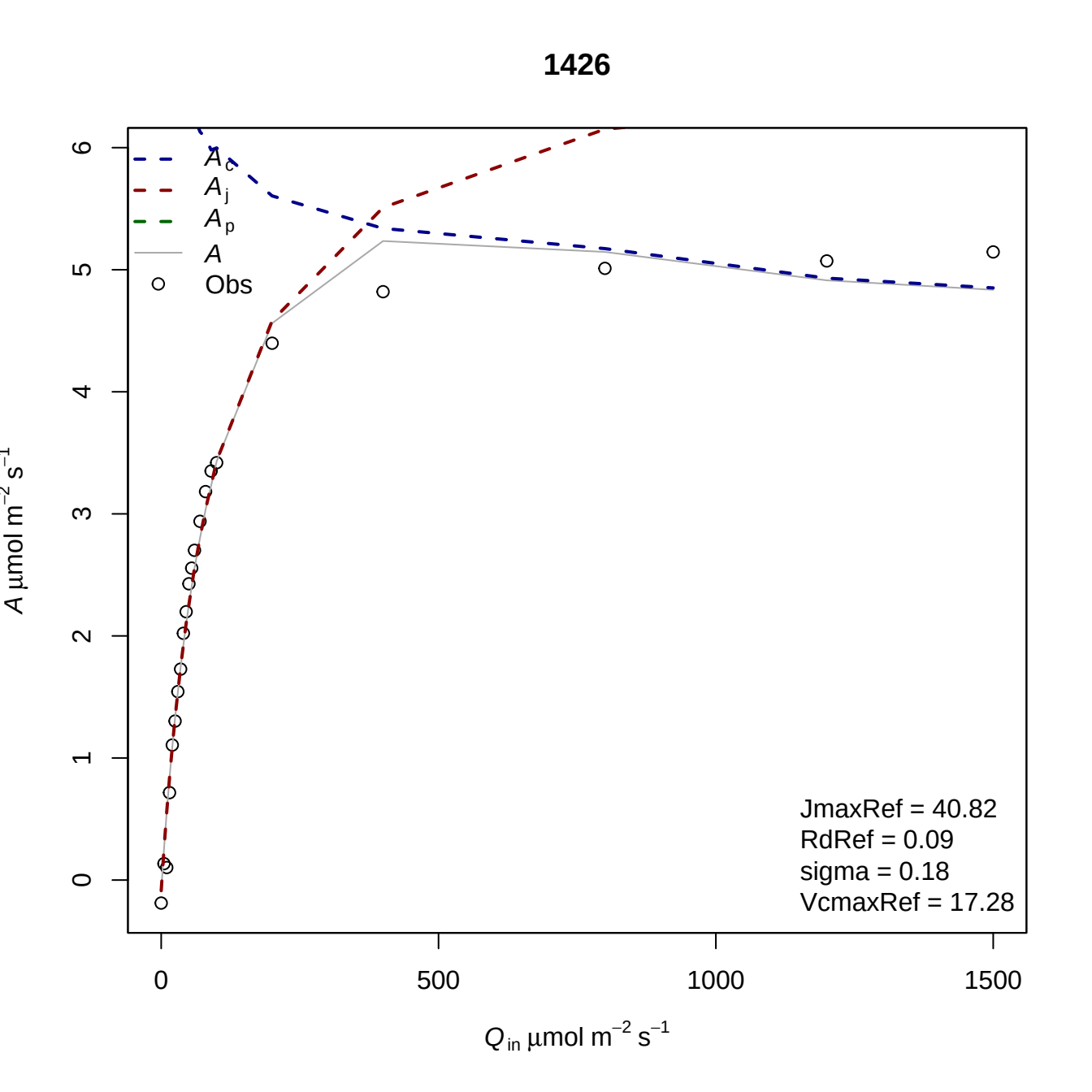

1421
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 29.84
RdRef = 0.42
sigma = 0.06
VcmaxRef = 16.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

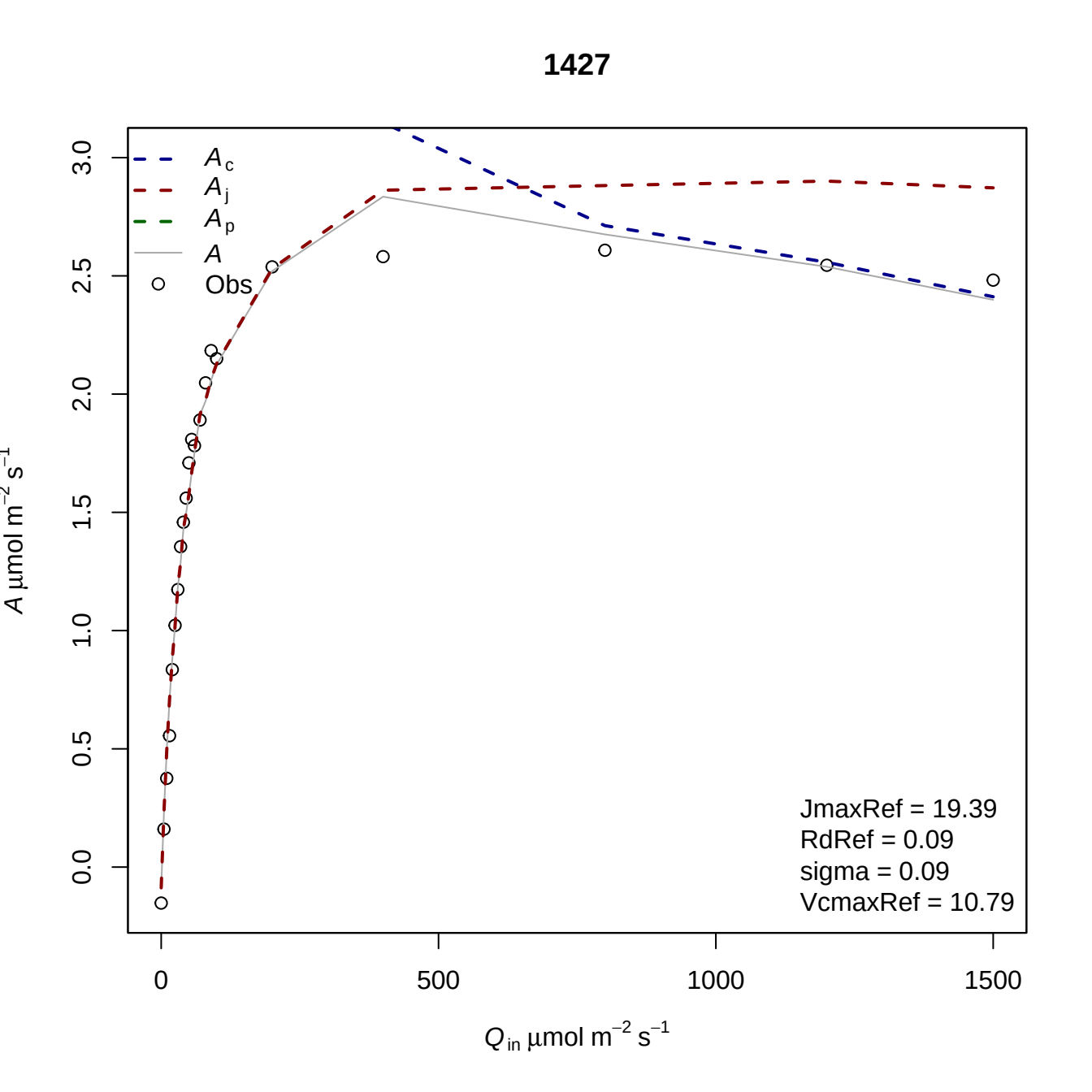

1422
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 41.77
RdRef = 0.43
sigma = 0.04
VcmaxRef = 26.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

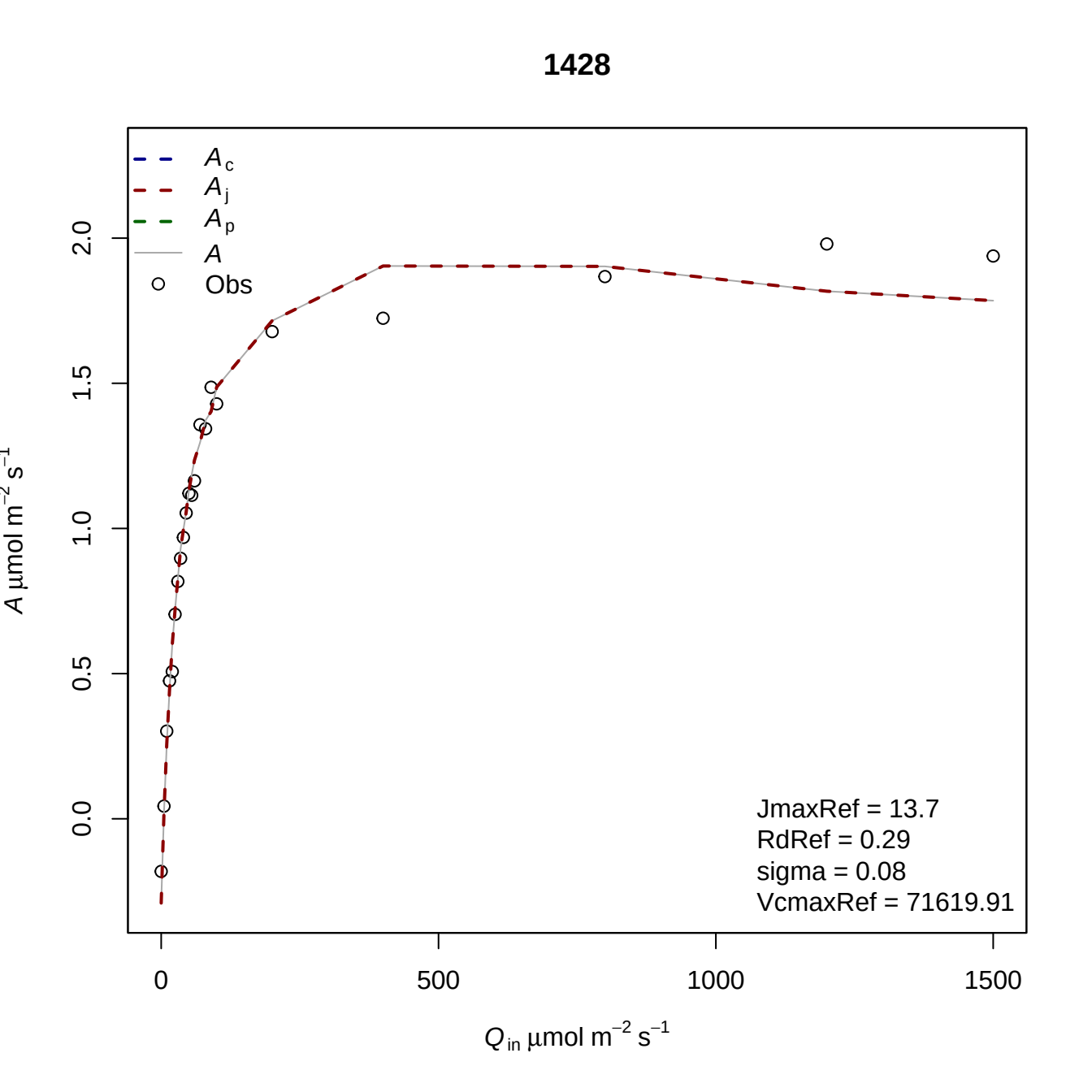

1423
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 26.72
RdRef = 1.07
sigma = 0.37
VcmaxRef = 229795.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

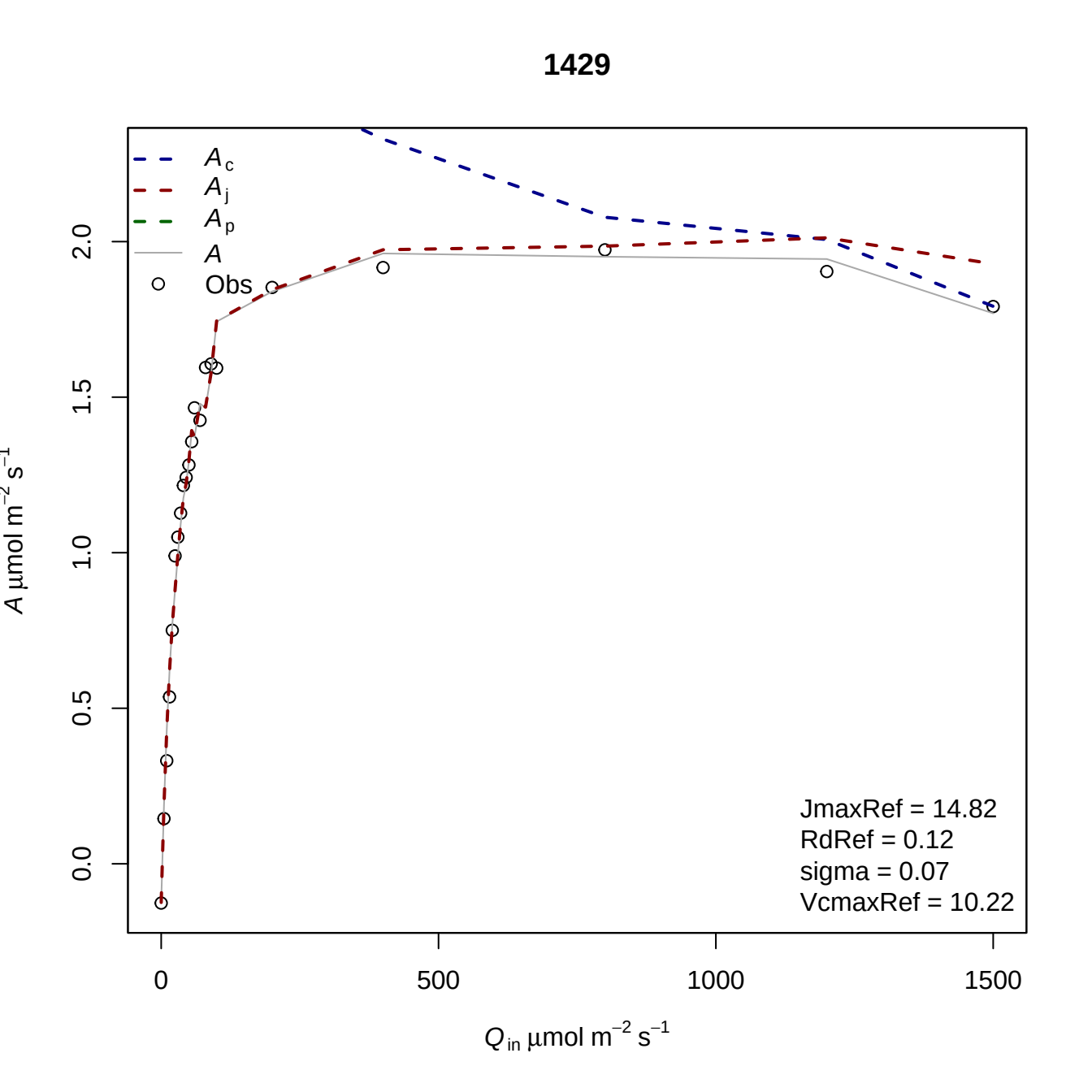

1424
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 37.39
RdRef = 0.21
sigma = 0.14
VcmaxRef = 22.63
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

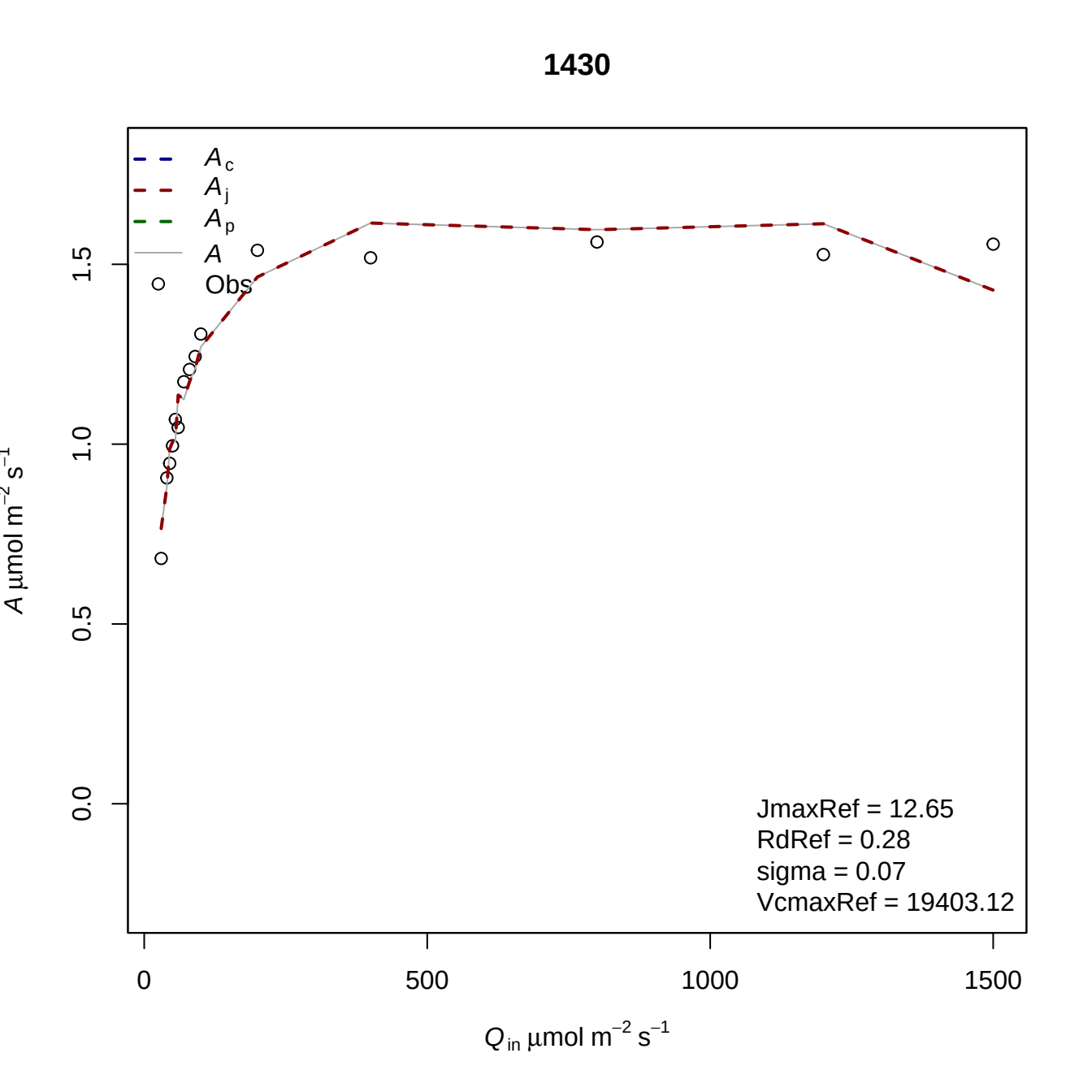

1425
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 17.36
RdRef = 0.37
sigma = 0.06
VcmaxRef = 33937.89
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

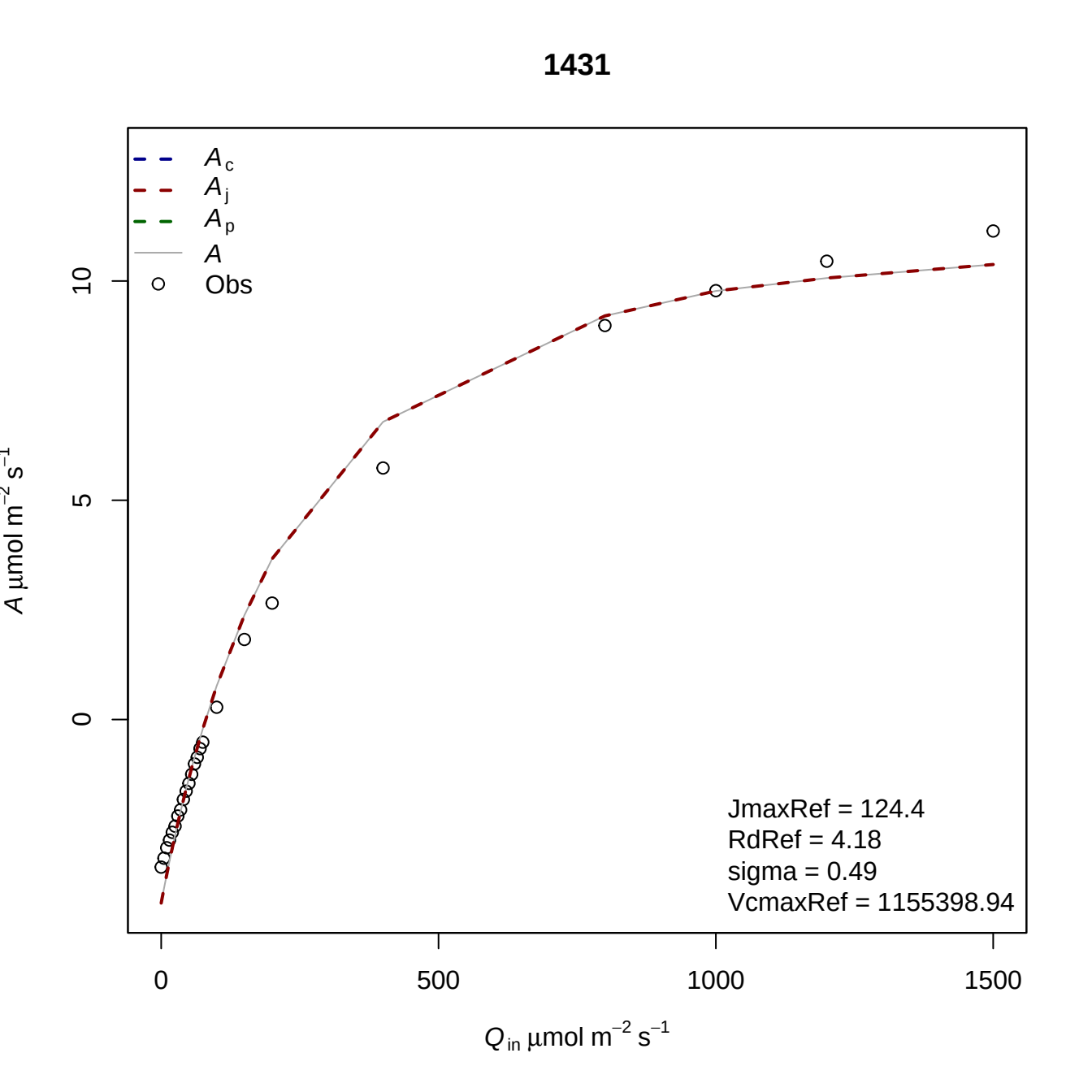

1426
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.82
RdRef = 0.09
sigma = 0.18
VcmaxRef = 17.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

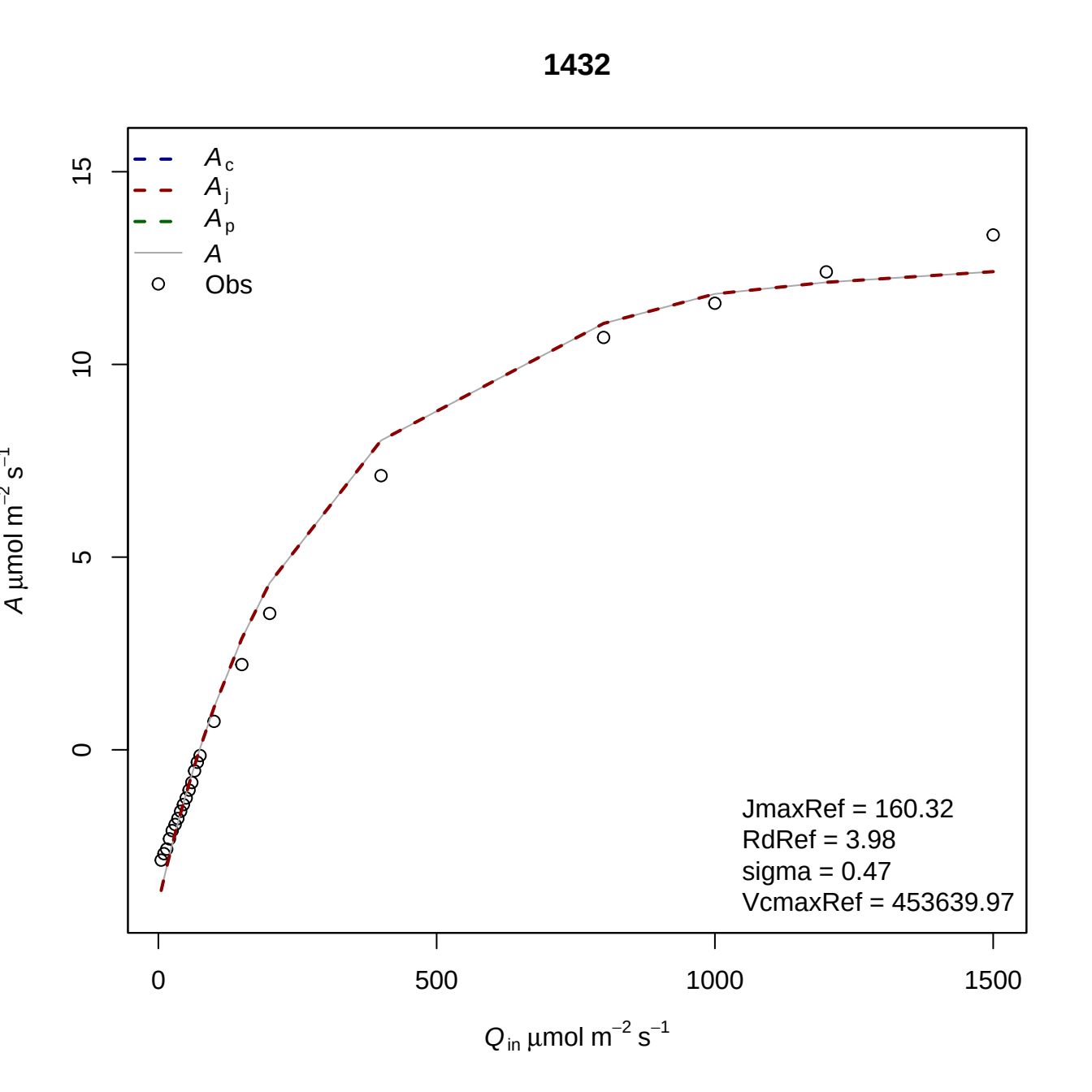

1427
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 19.39
RdRef = 0.09
sigma = 0.09
VcmaxRef = 10.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

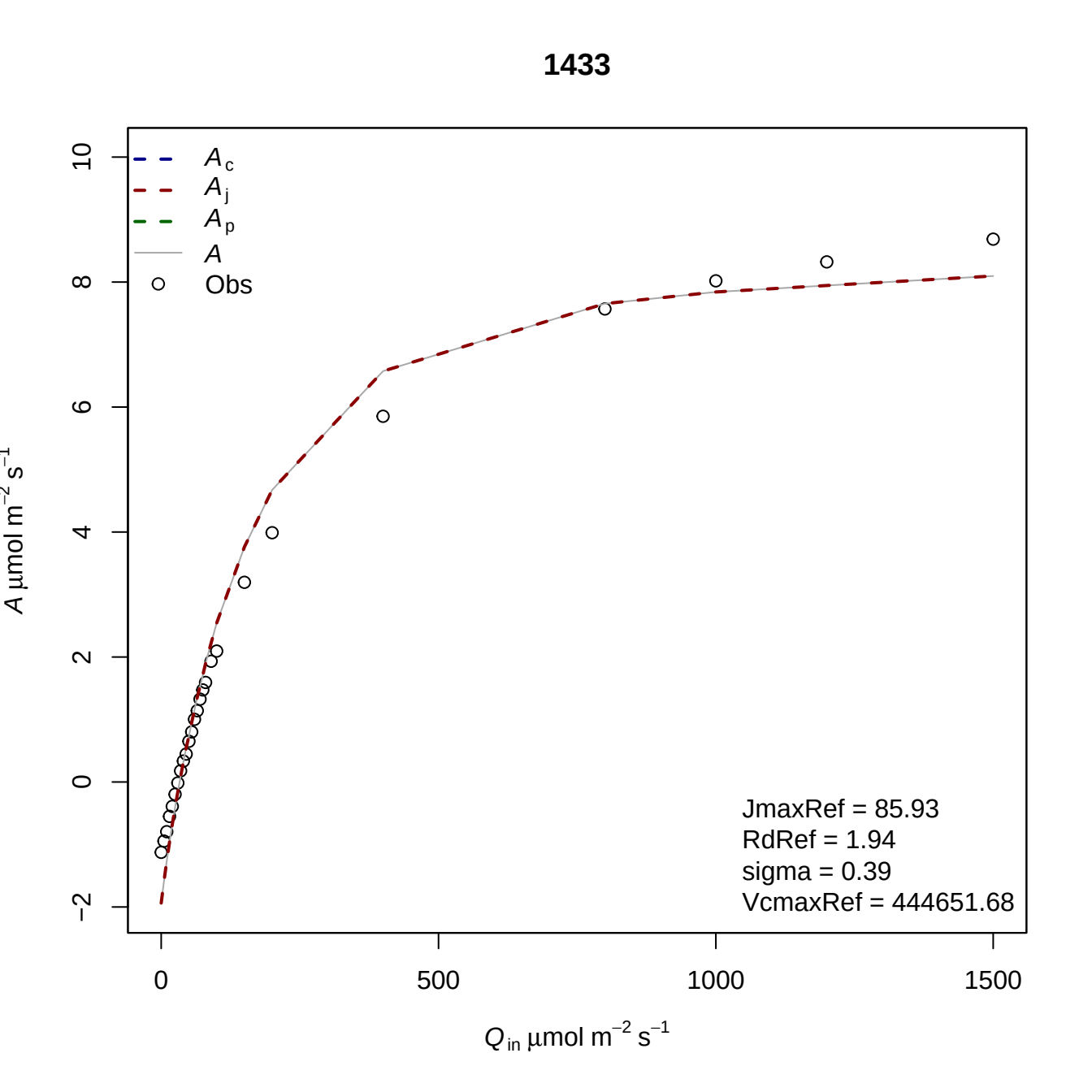

1428
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 13.7
RdRef = 0.29
sigma = 0.08
VcmaxRef = 71619.91
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

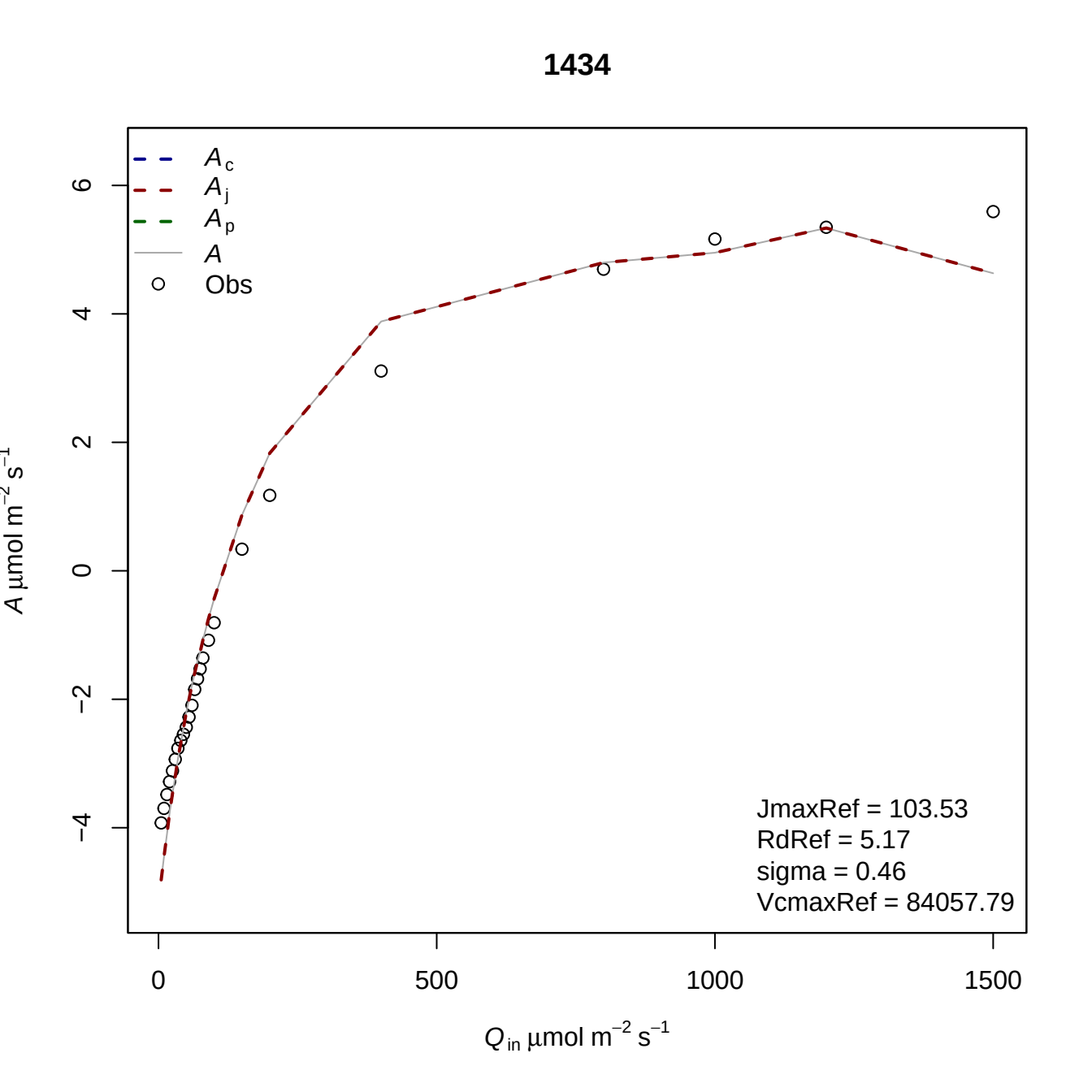

1429
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 14.82
RdRef = 0.12
sigma = 0.07
VcmaxRef = 10.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

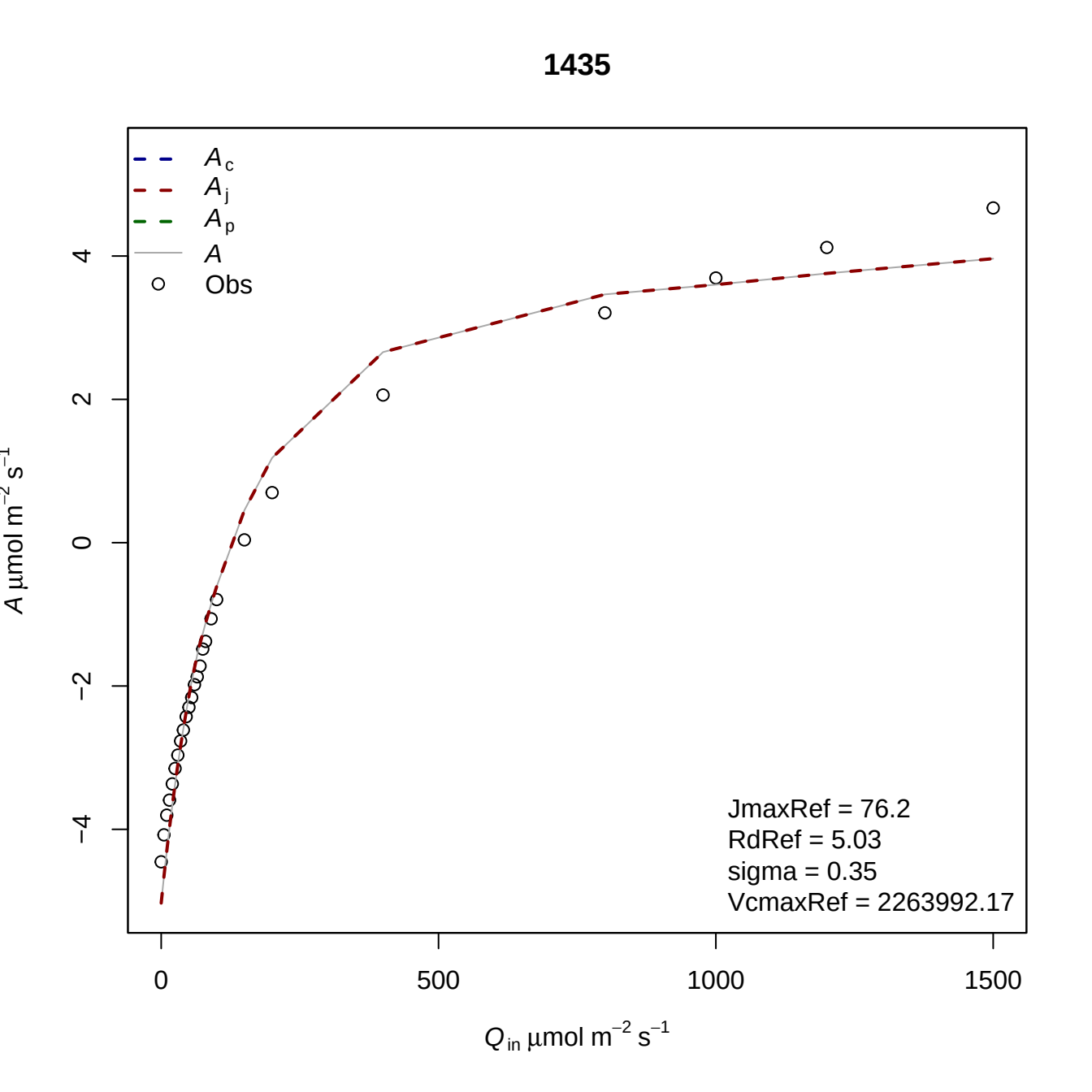

1430
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 12.65
RdRef = 0.28
sigma = 0.07
VcmaxRef = 19403.12
0.0
0.5
1.0
1.5
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

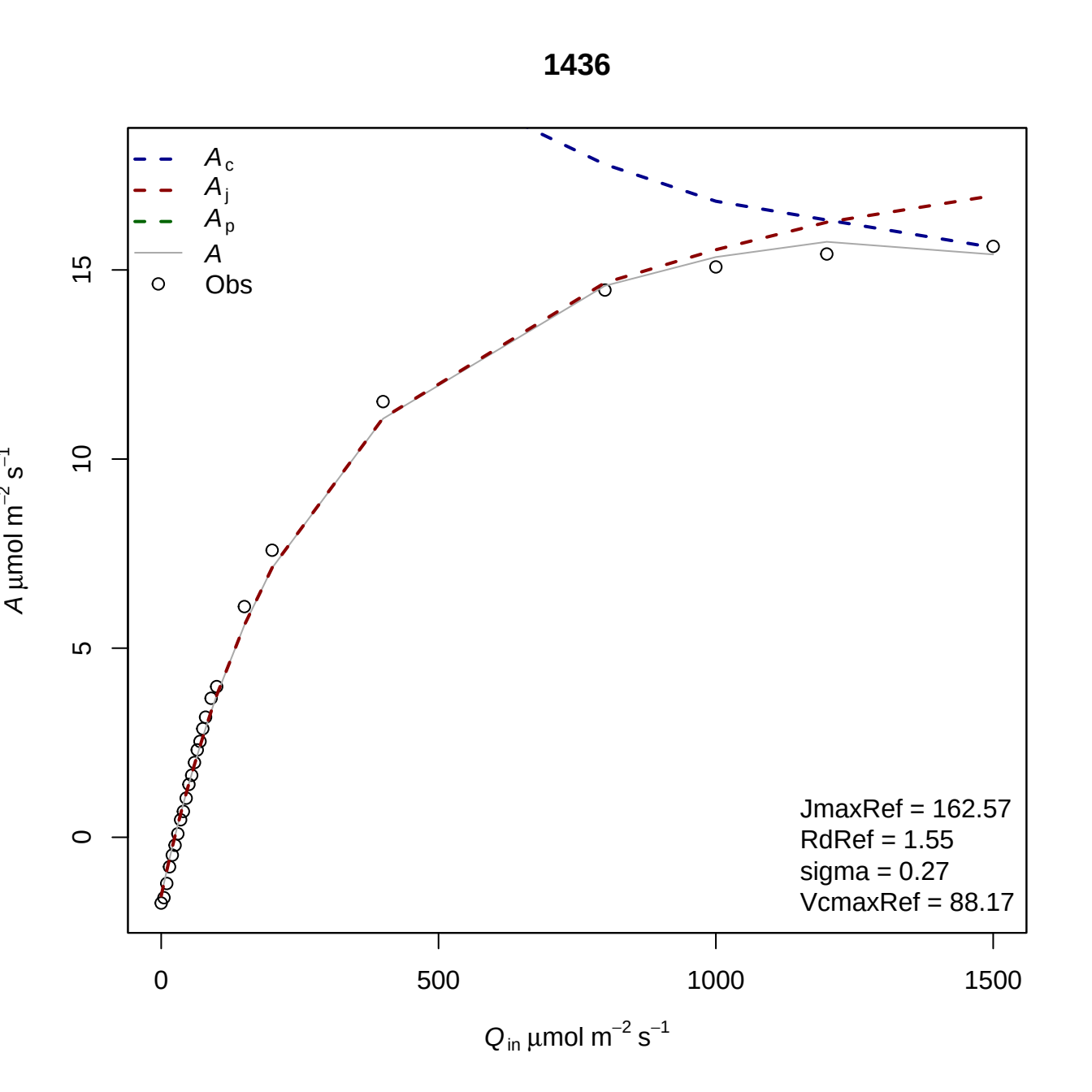

1431
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.4
RdRef = 4.18
sigma = 0.49
VcmaxRef = 1155398.94
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

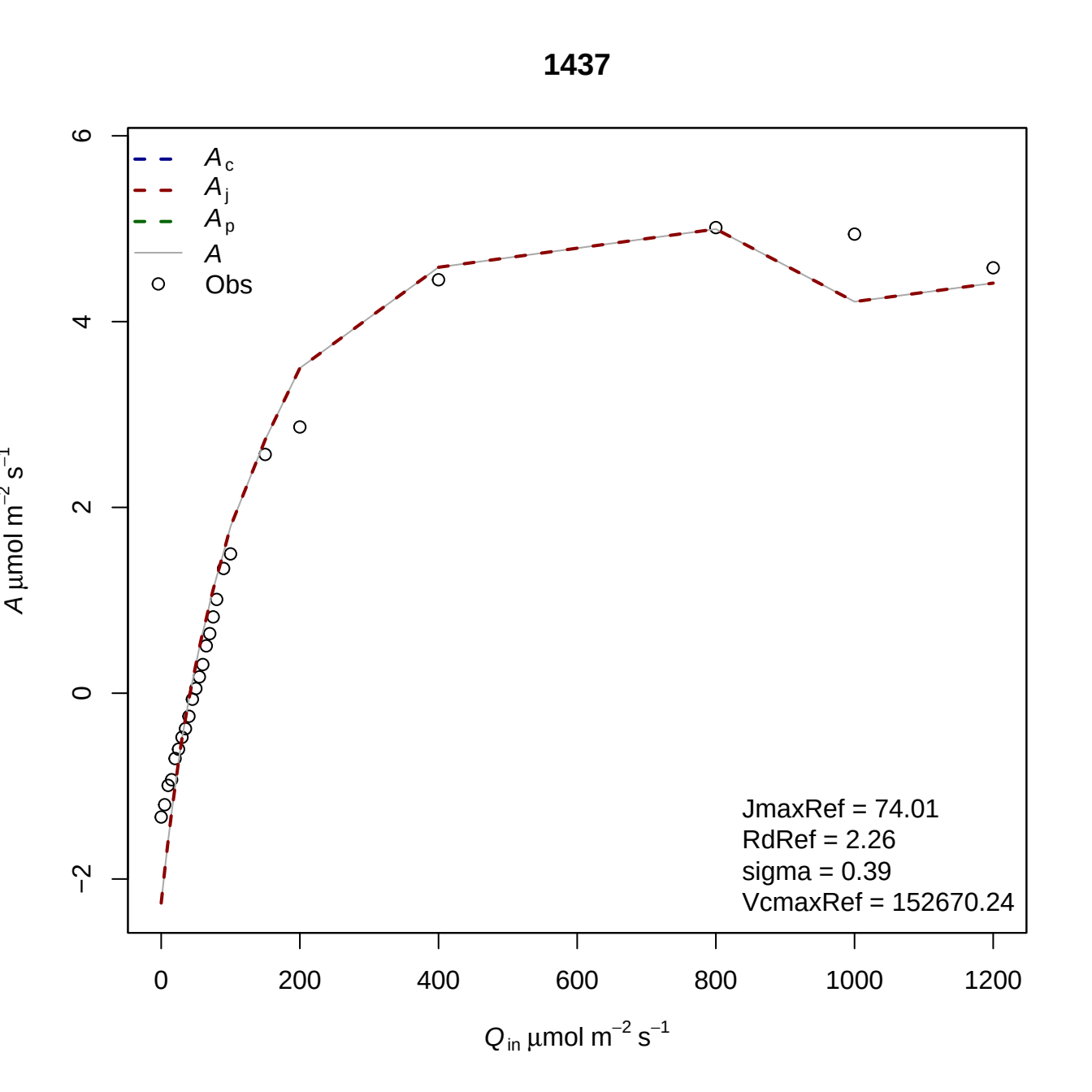

1432
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.32
RdRef = 3.98
sigma = 0.47
VcmaxRef = 453639.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

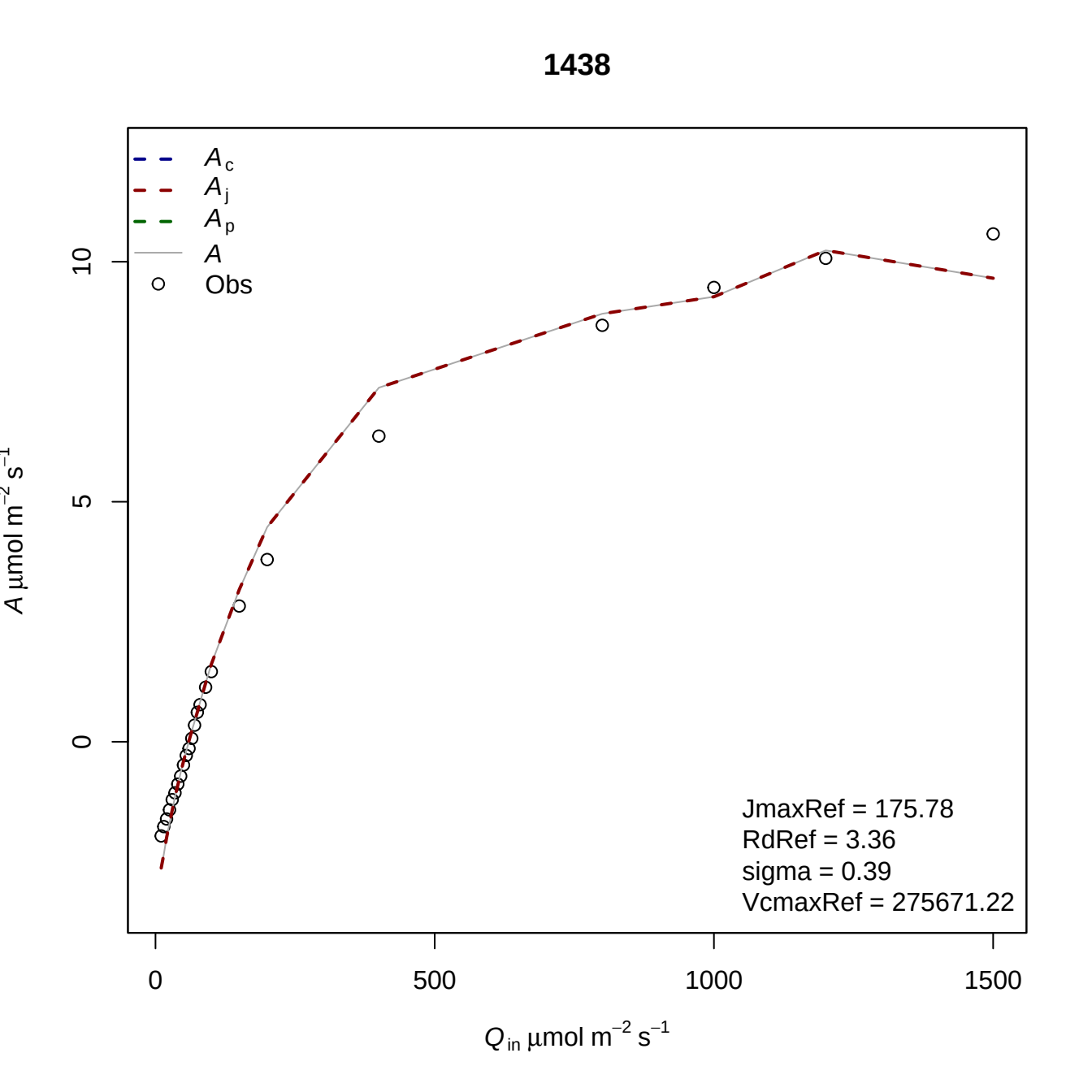

1433
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.93
RdRef = 1.94
sigma = 0.39
VcmaxRef = 444651.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

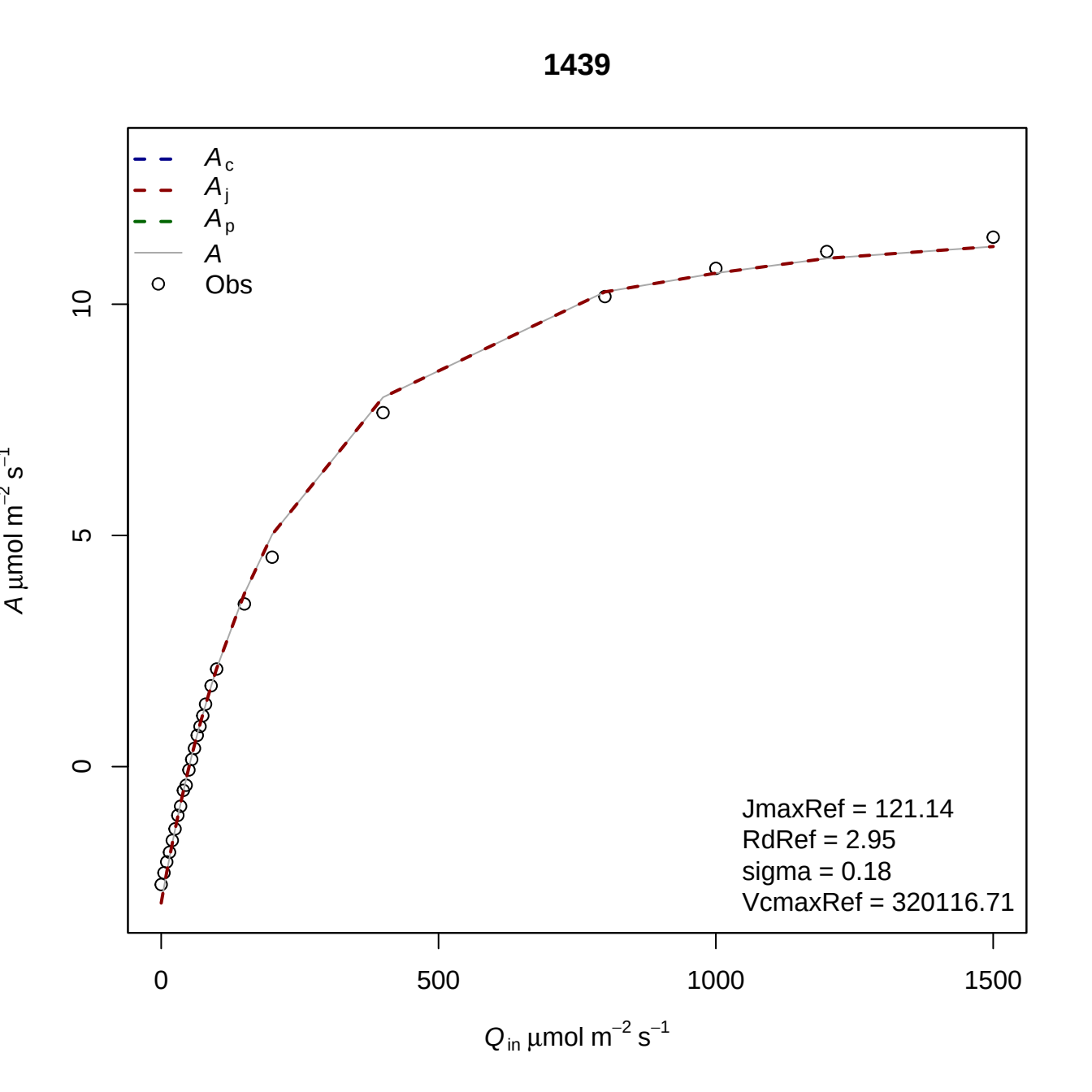

1434
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.53
RdRef = 5.17
sigma = 0.46
VcmaxRef = 84057.79
$A$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

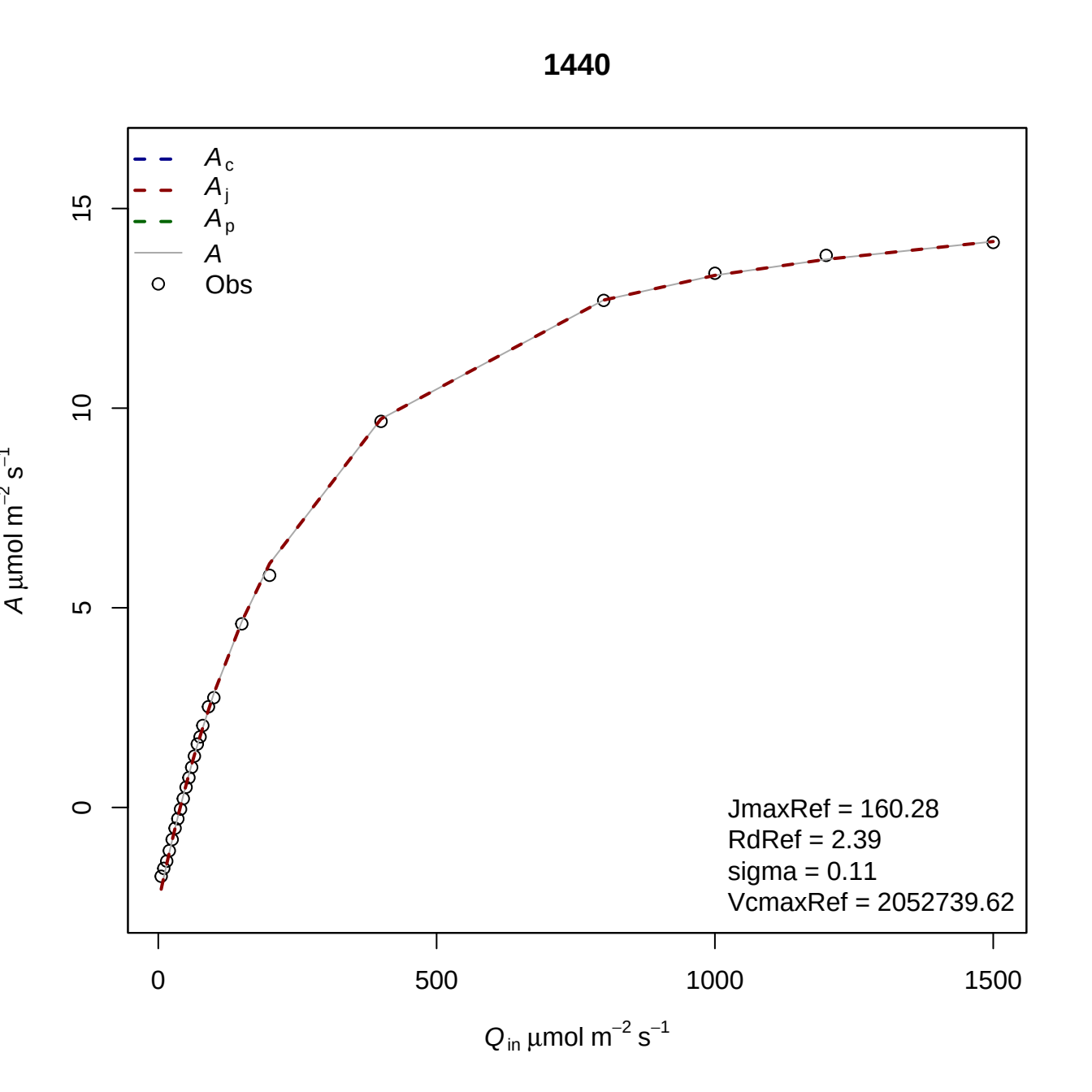

1435
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.2
RdRef = 5.03
sigma = 0.35
VcmaxRef = 2263992.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
−4
0
500
1000
1500

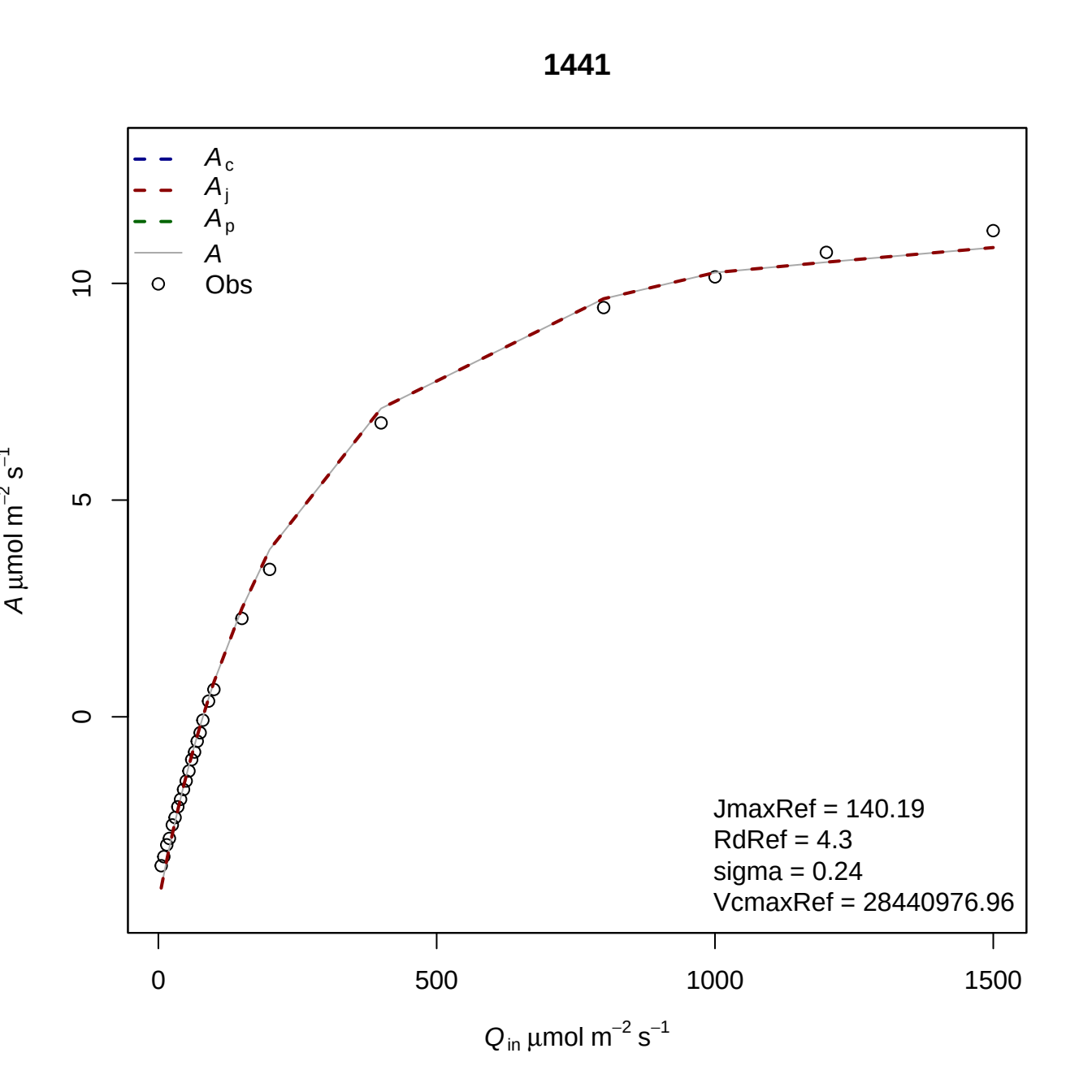

1436
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 162.57
RdRef = 1.55
sigma = 0.27
VcmaxRef = 88.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

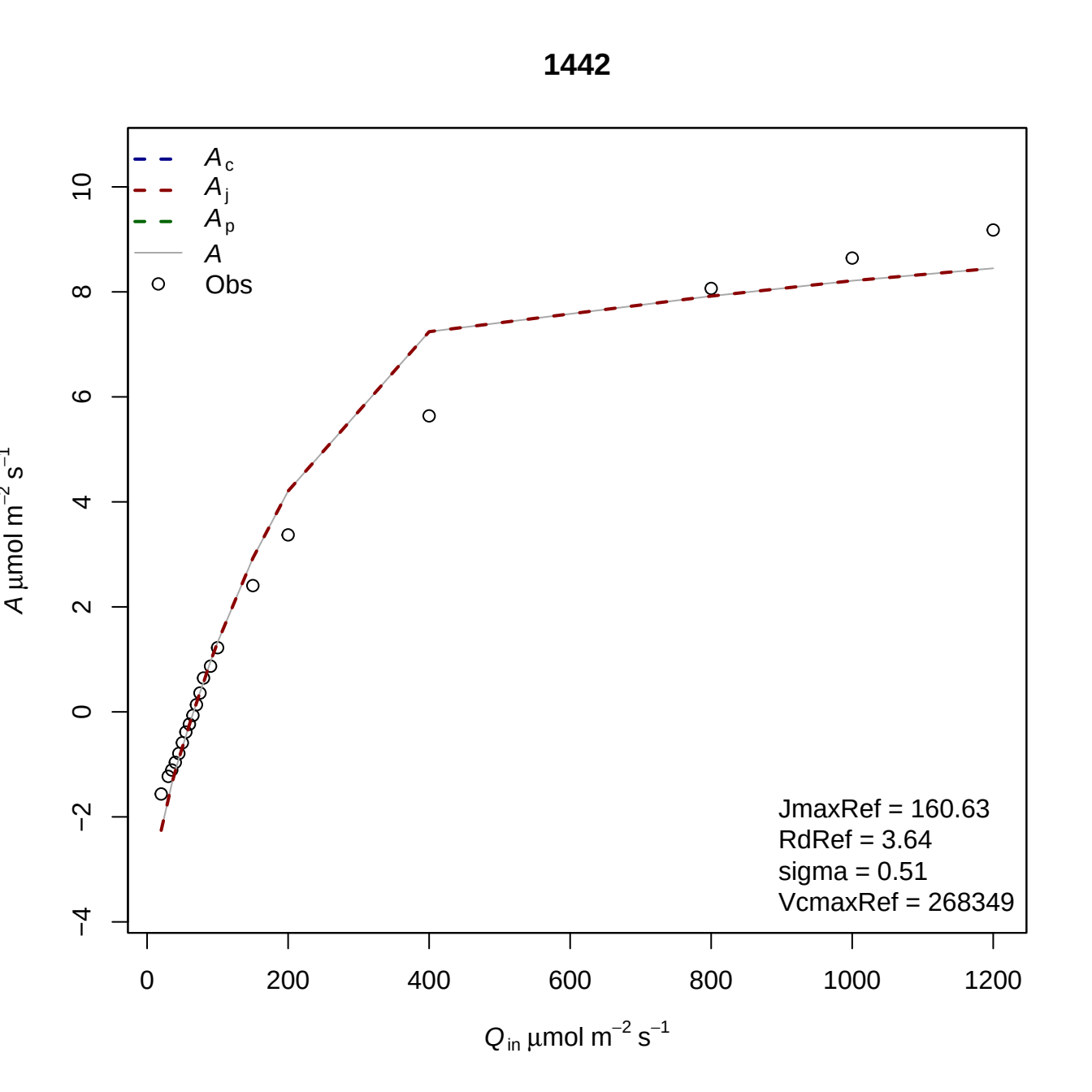

1437
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.01
RdRef = 2.26
sigma = 0.39
VcmaxRef = 152670.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800
1000
1200

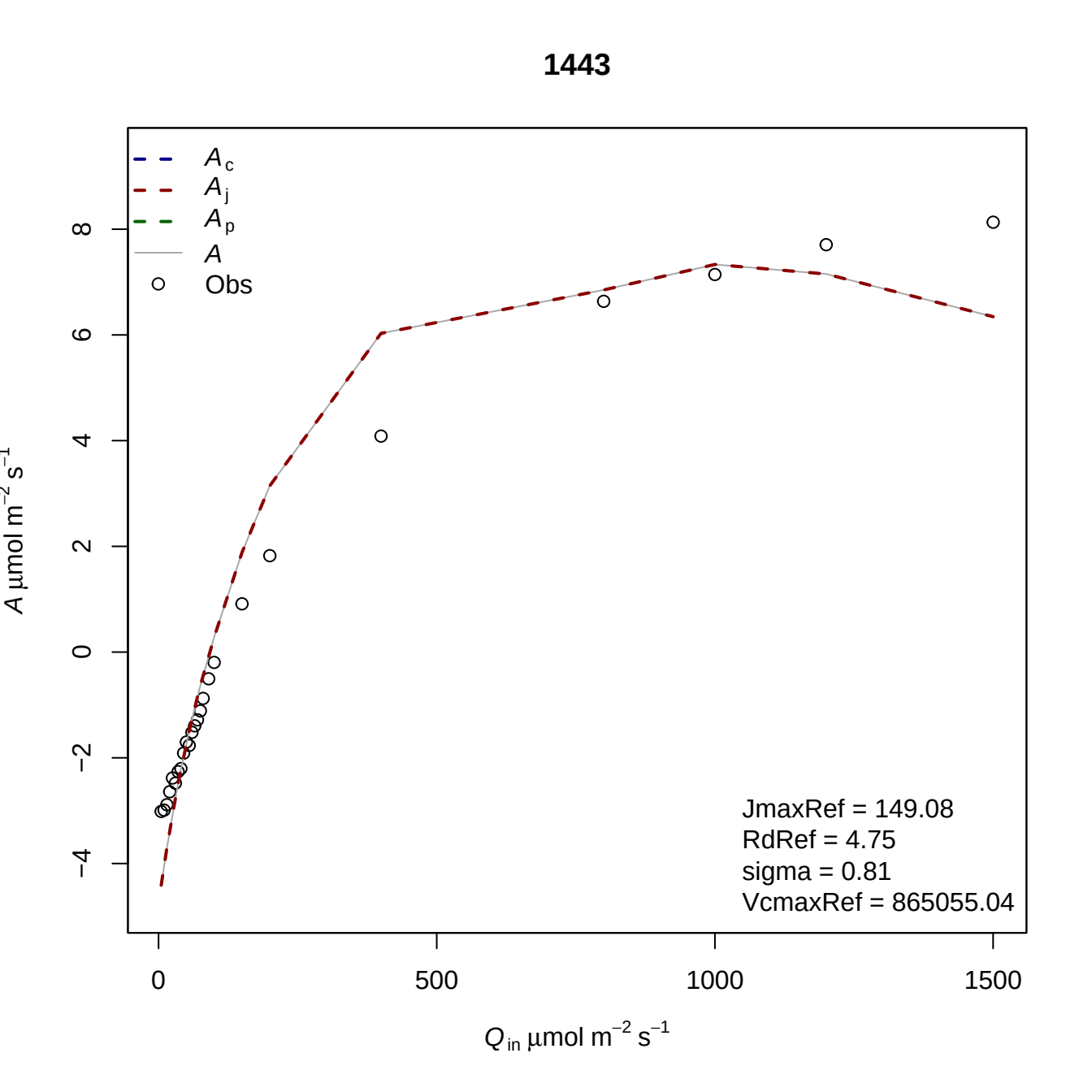

1438
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 175.78
RdRef = 3.36
sigma = 0.39
VcmaxRef = 275671.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

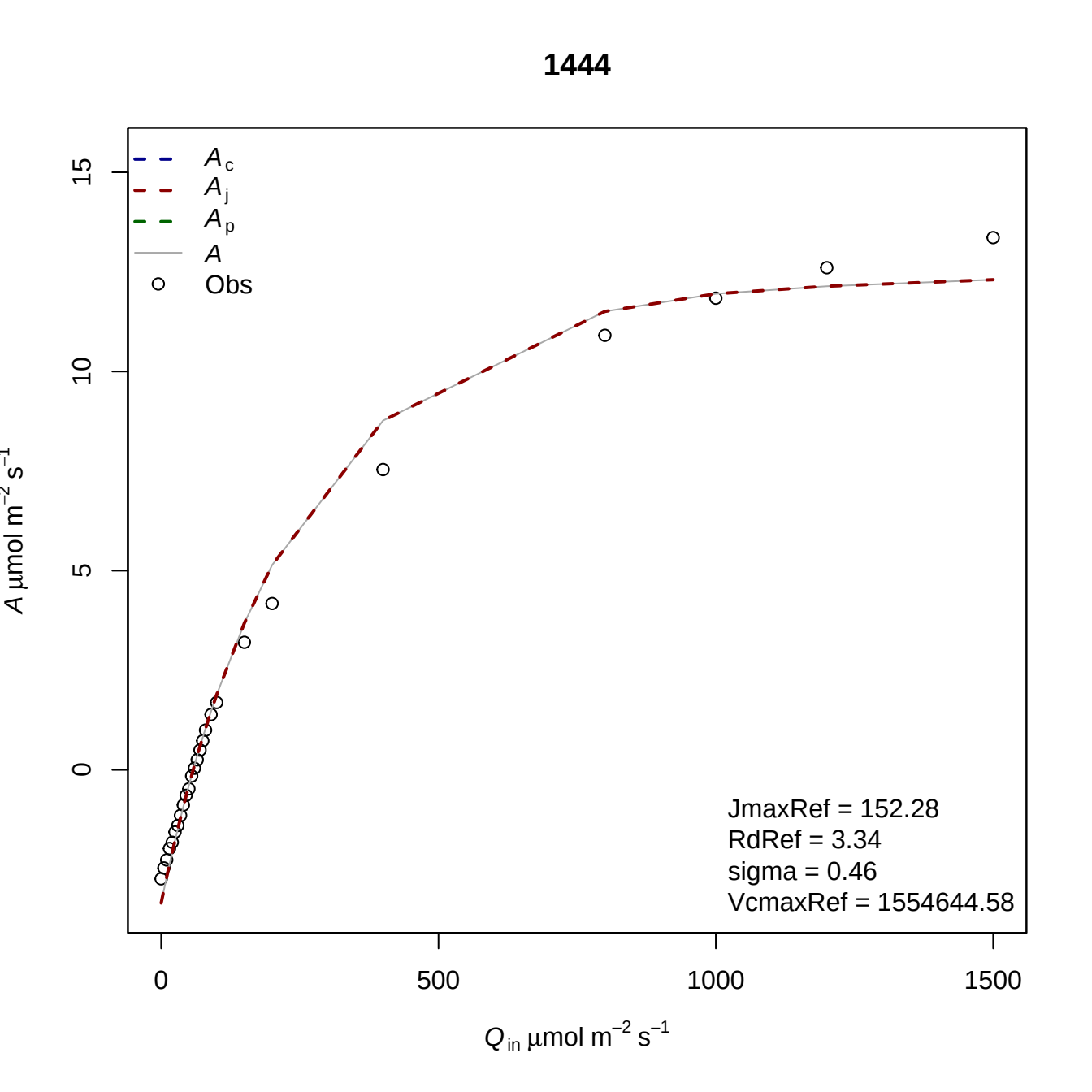

1439
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.14
RdRef = 2.95
sigma = 0.18
VcmaxRef = 320116.71
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

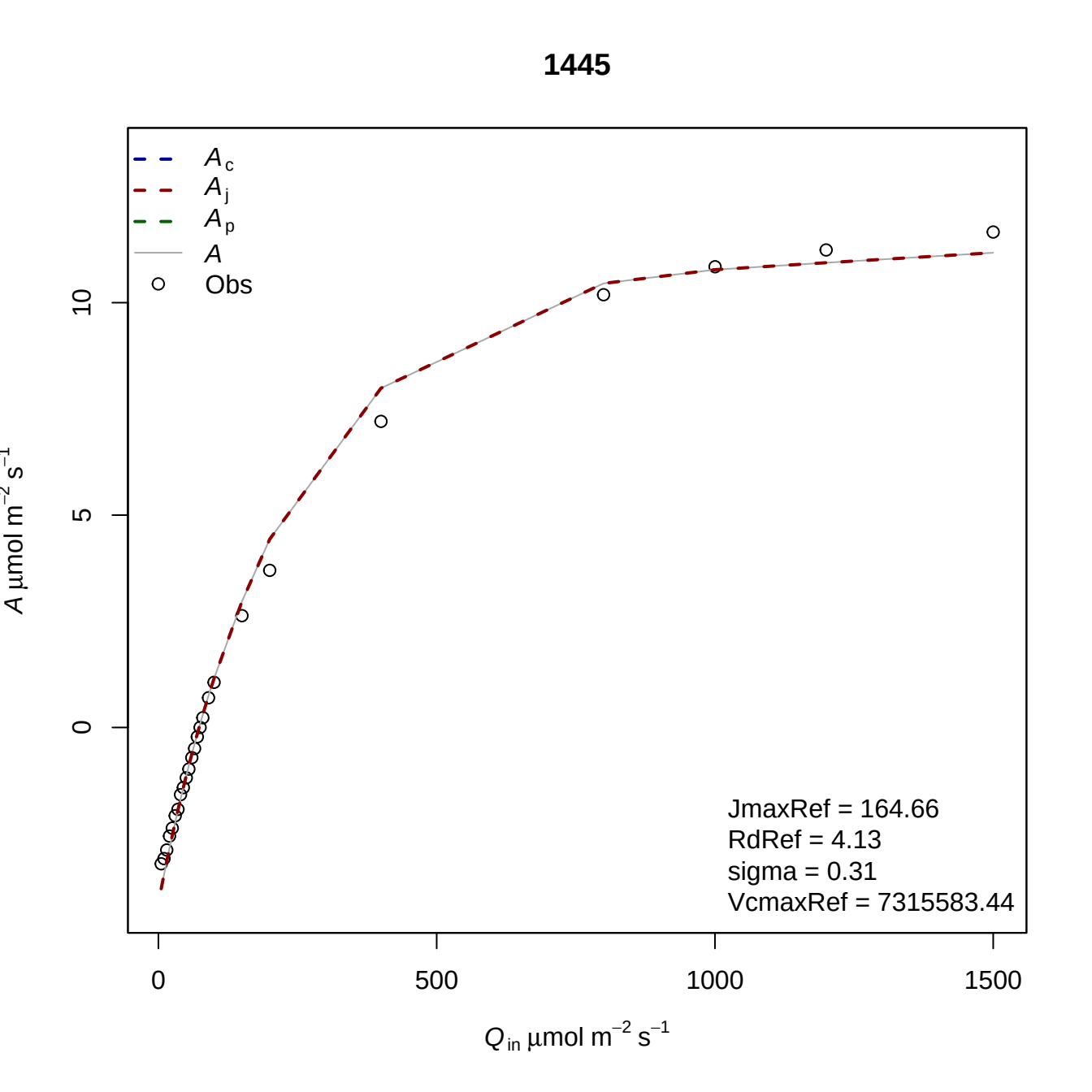

1440
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.28
RdRef = 2.39
sigma = 0.11
VcmaxRef = 2052739.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

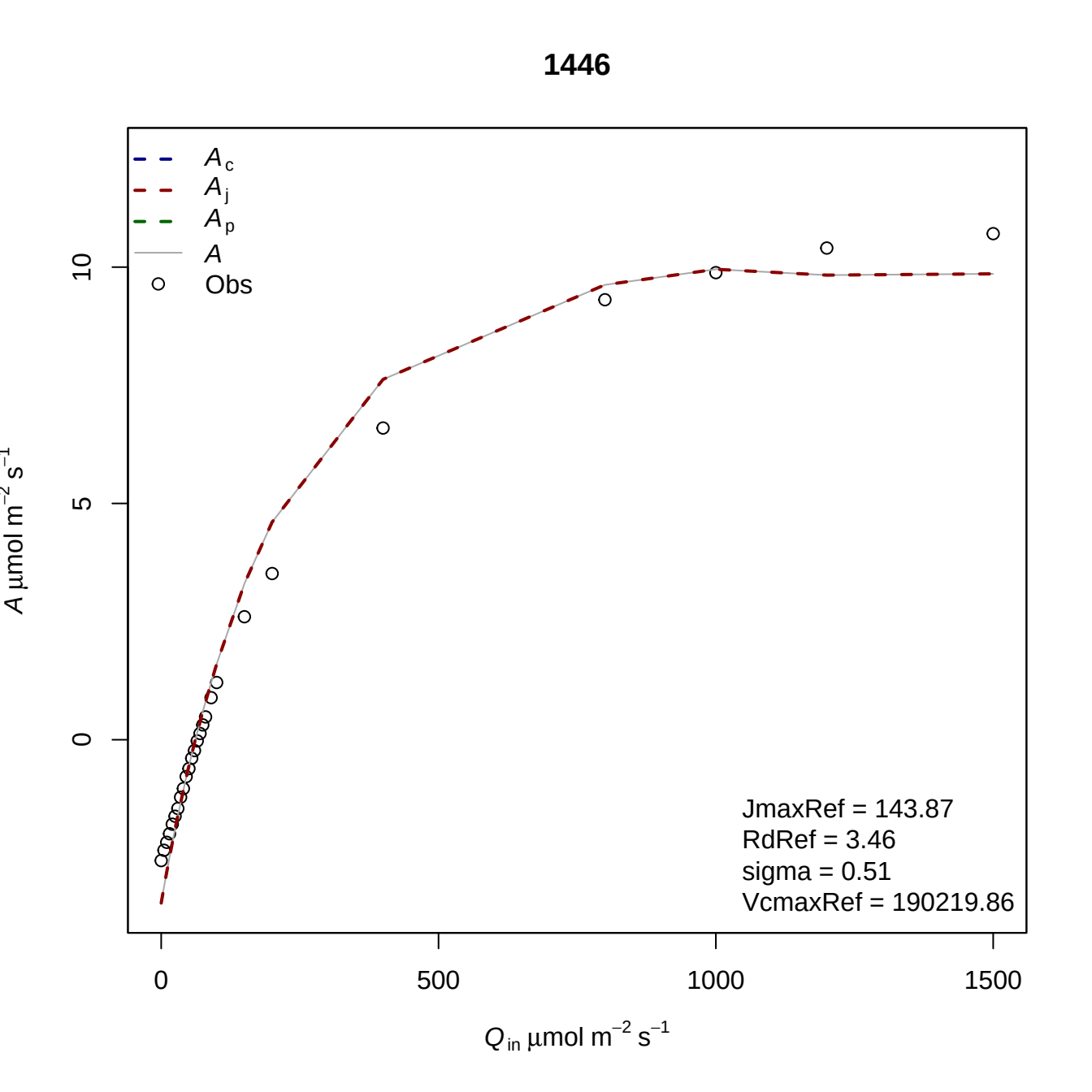

1441
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 140.19
RdRef = 4.3
sigma = 0.24
VcmaxRef = 28440976.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

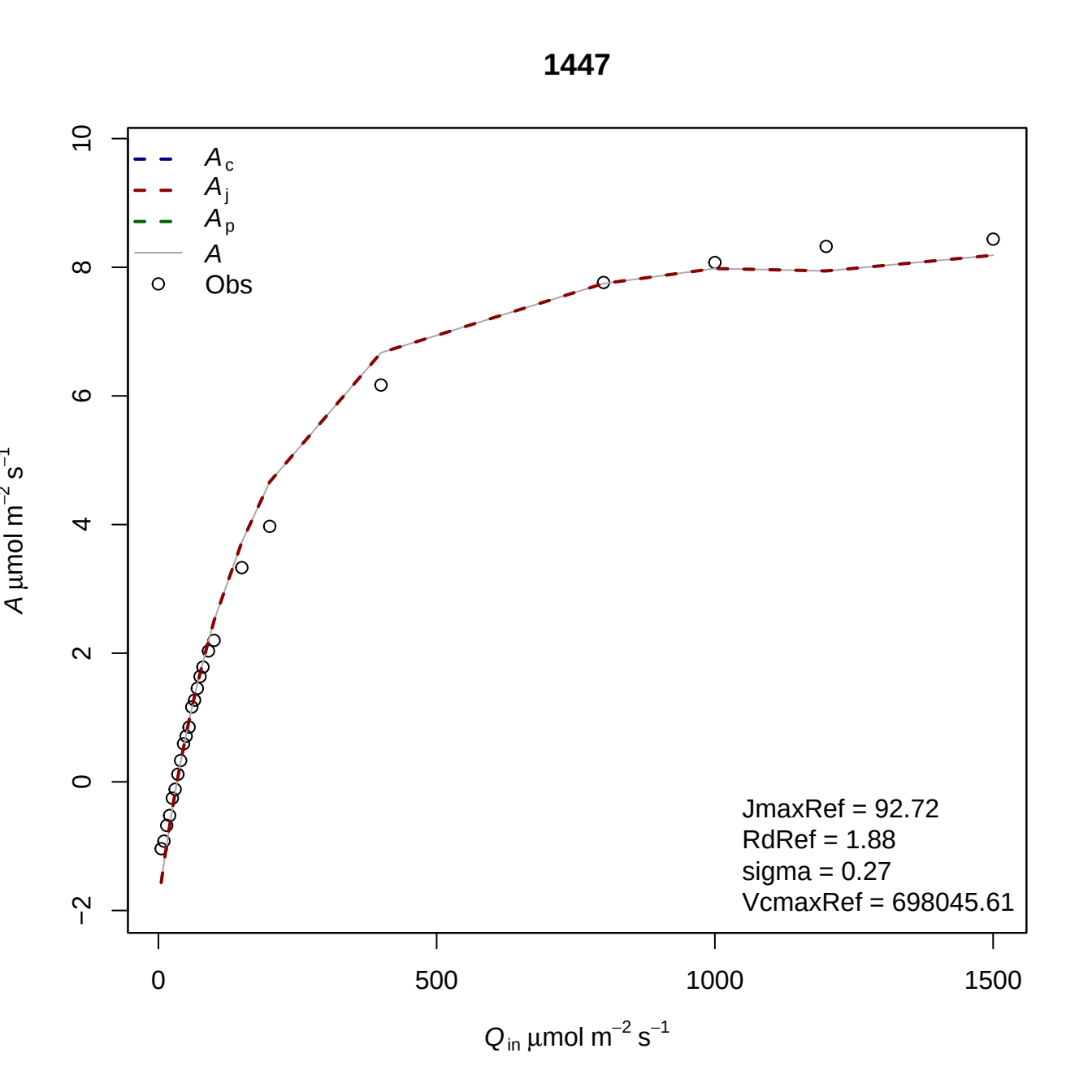

1442
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 160.63
RdRef = 3.64
sigma = 0.51
VcmaxRef = 268349
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

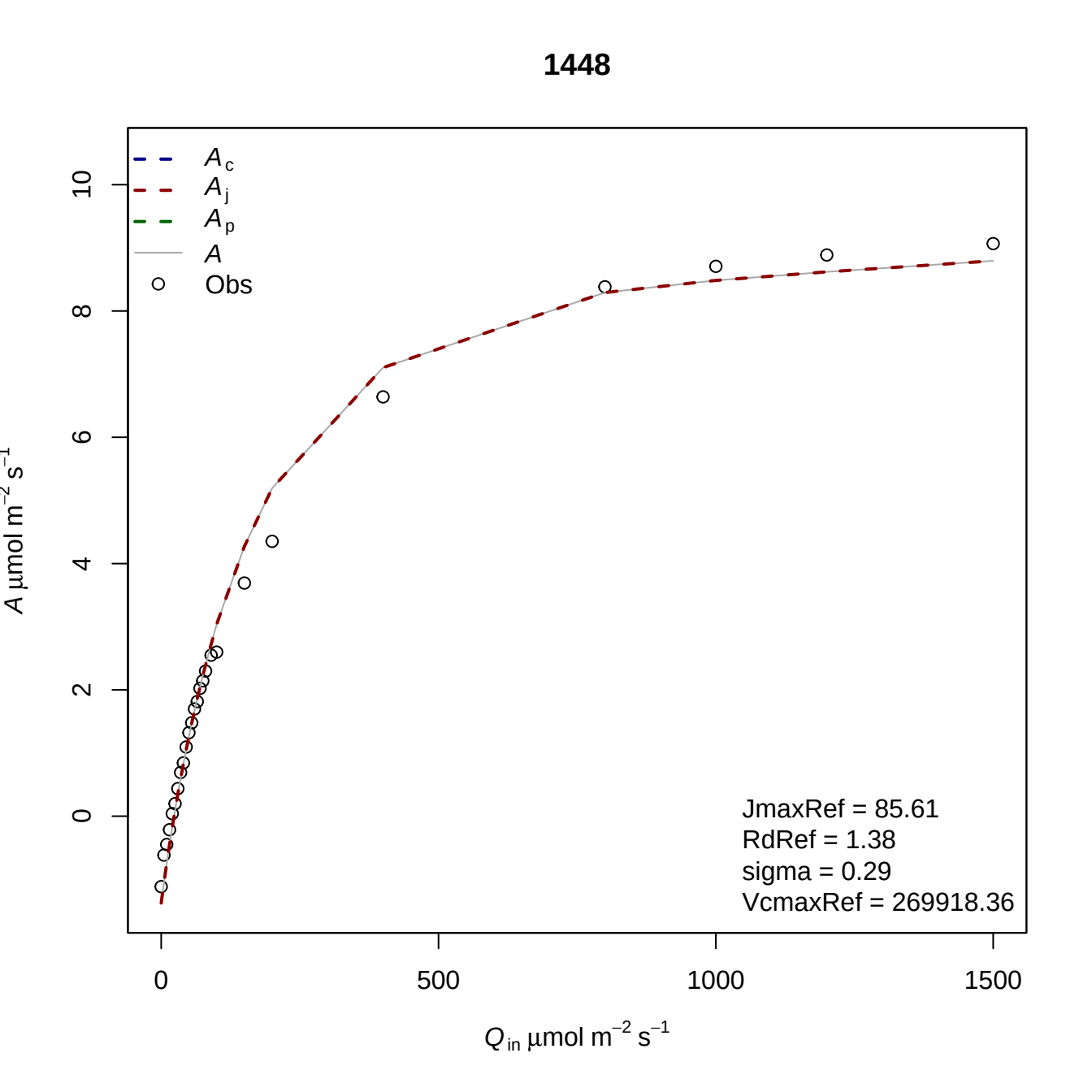

1443
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 149.08
RdRef = 4.75
sigma = 0.81
VcmaxRef = 865055.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

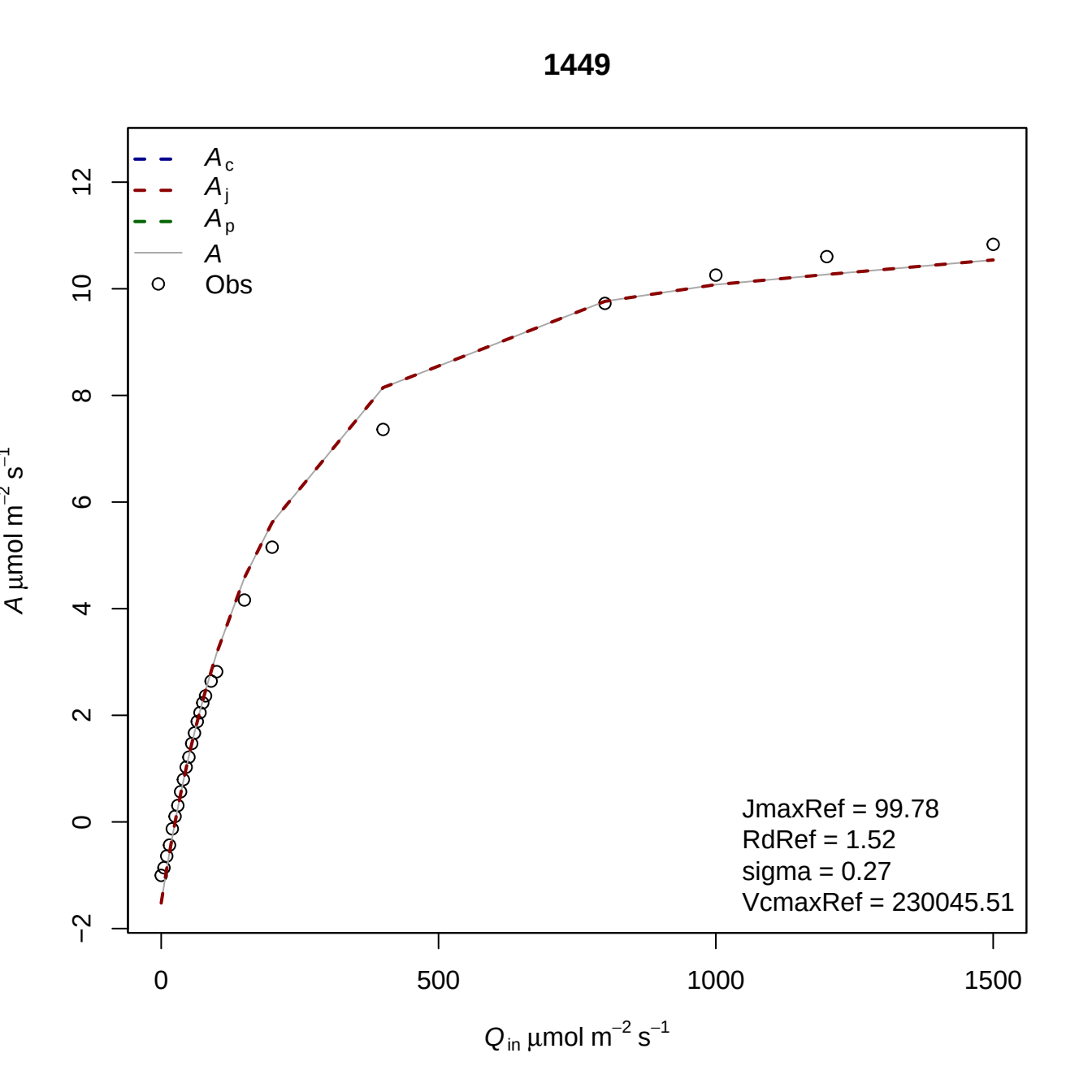

1444
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 152.28
RdRef = 3.34
sigma = 0.46
VcmaxRef = 1554644.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

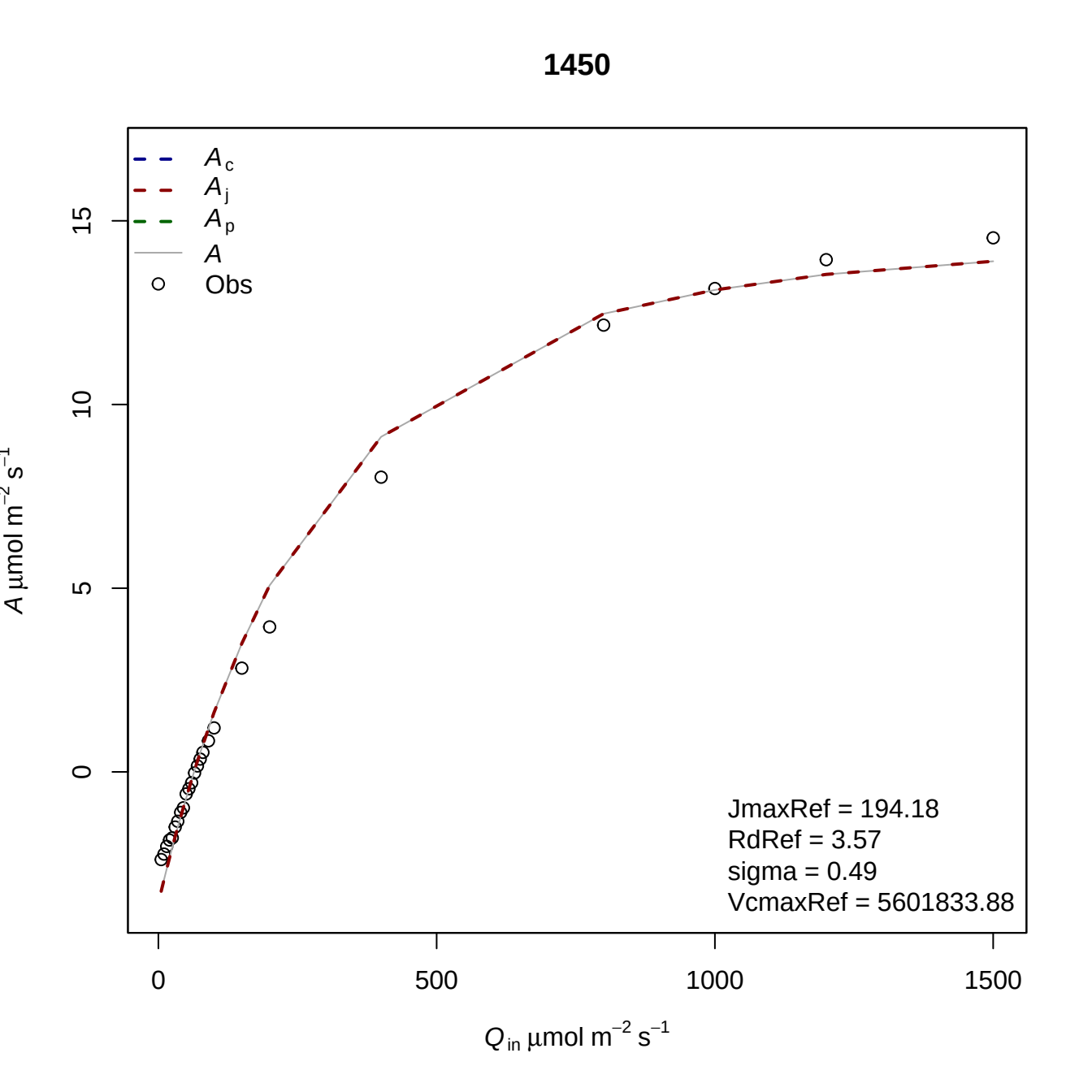

1445
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.66
RdRef = 4.13
sigma = 0.31
VcmaxRef = 7315583.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

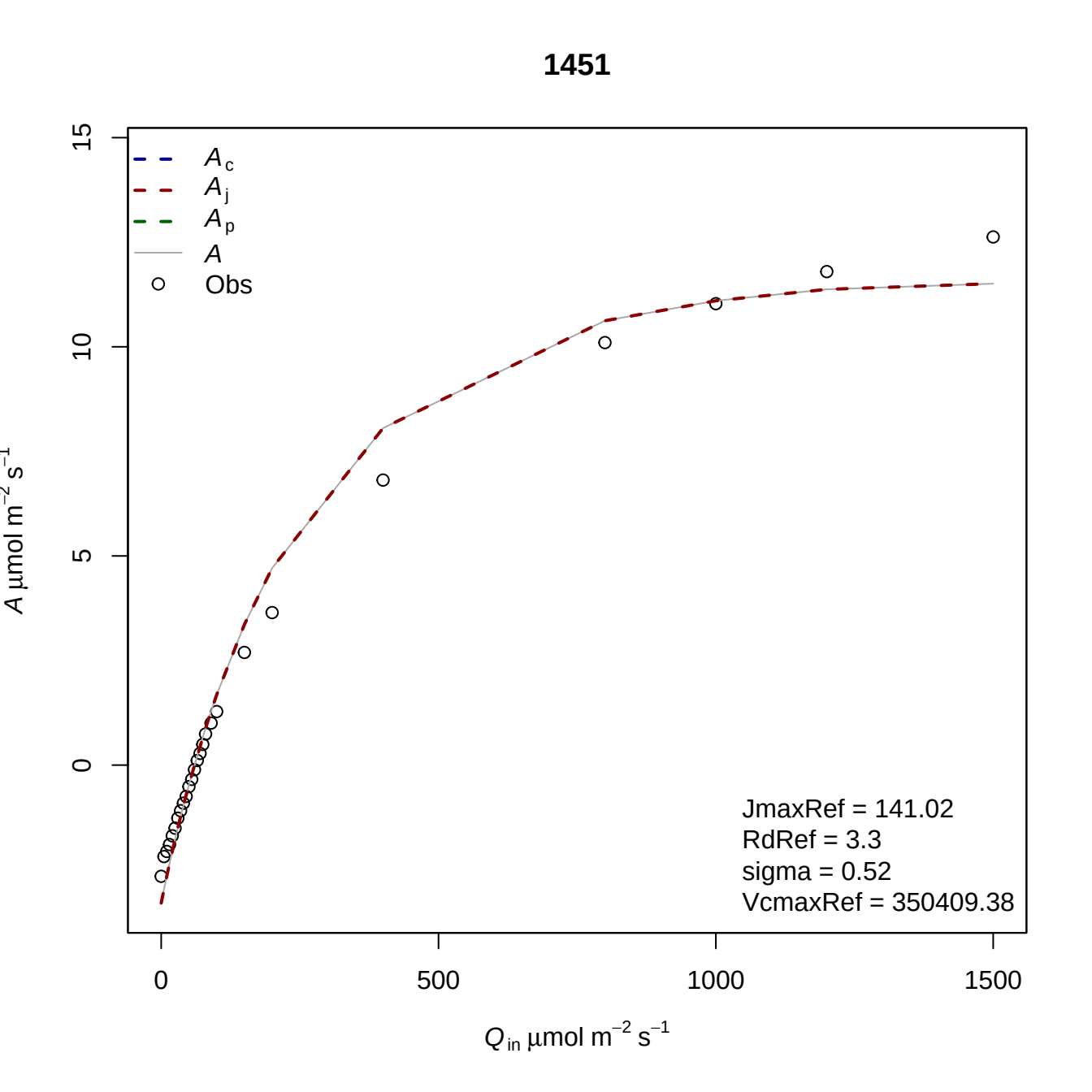

1446
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.87
RdRef = 3.46
sigma = 0.51
VcmaxRef = 190219.86
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

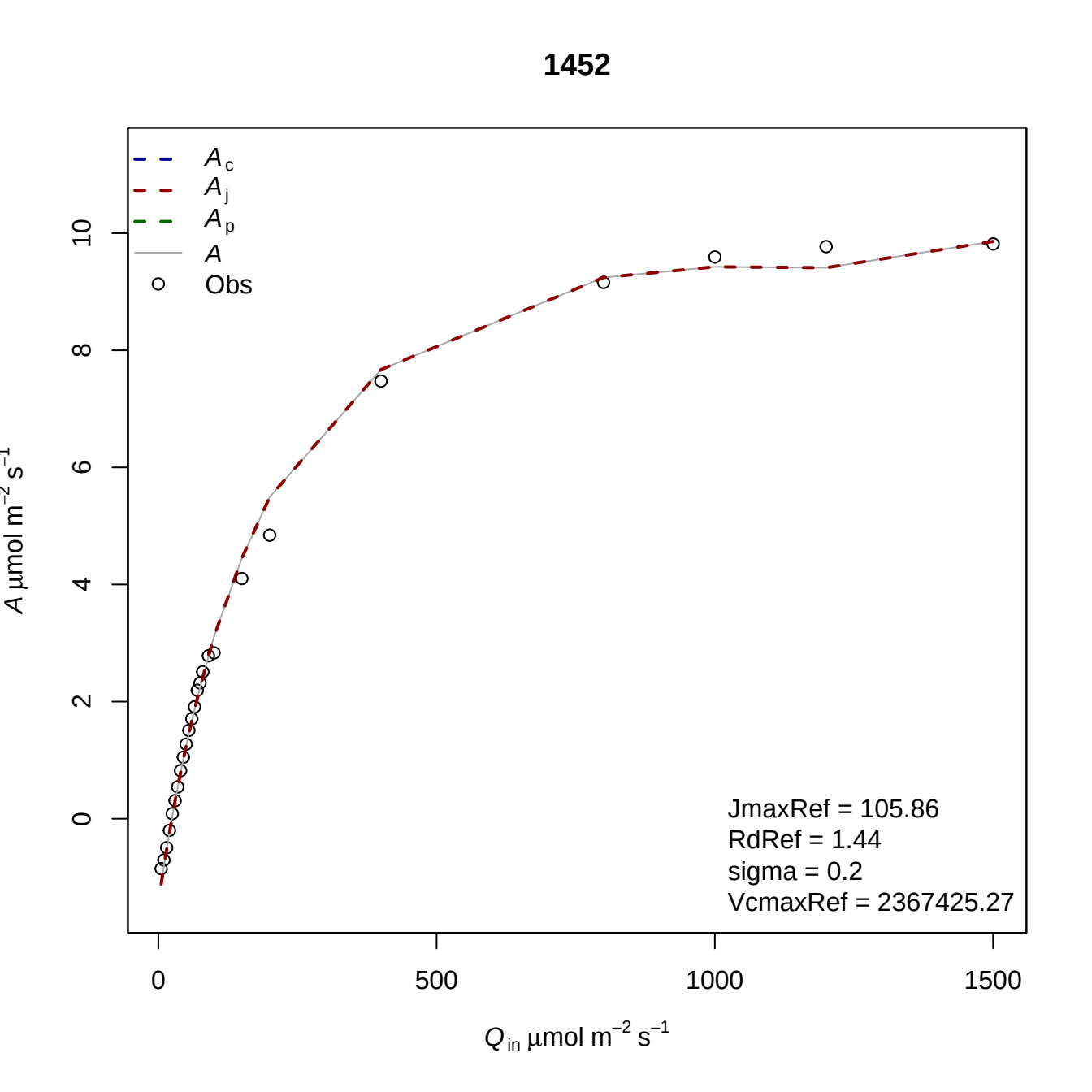

1447
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.72
RdRef = 1.88
sigma = 0.27
VcmaxRef = 698045.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

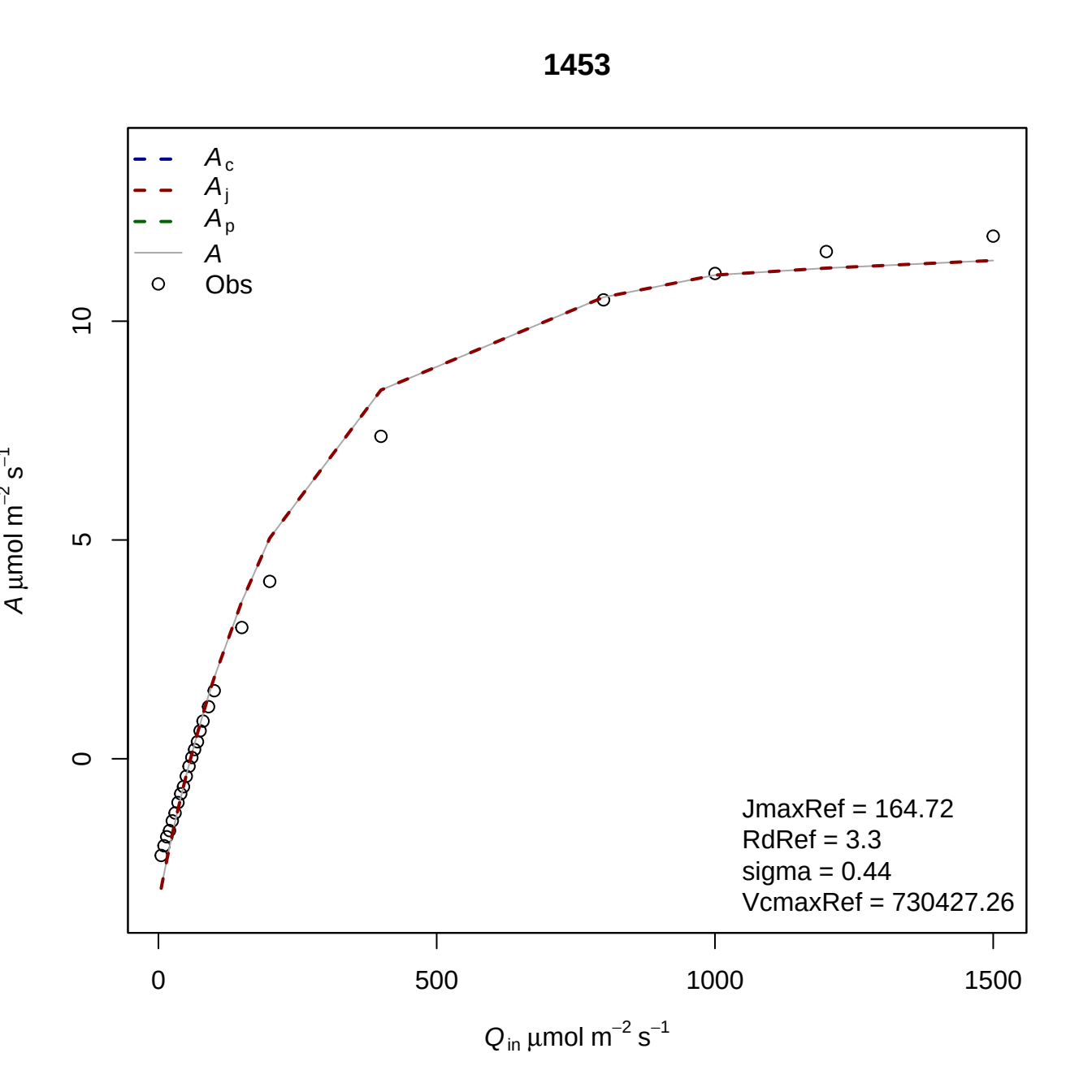

1448
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.61
RdRef = 1.38
sigma = 0.29
VcmaxRef = 269918.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

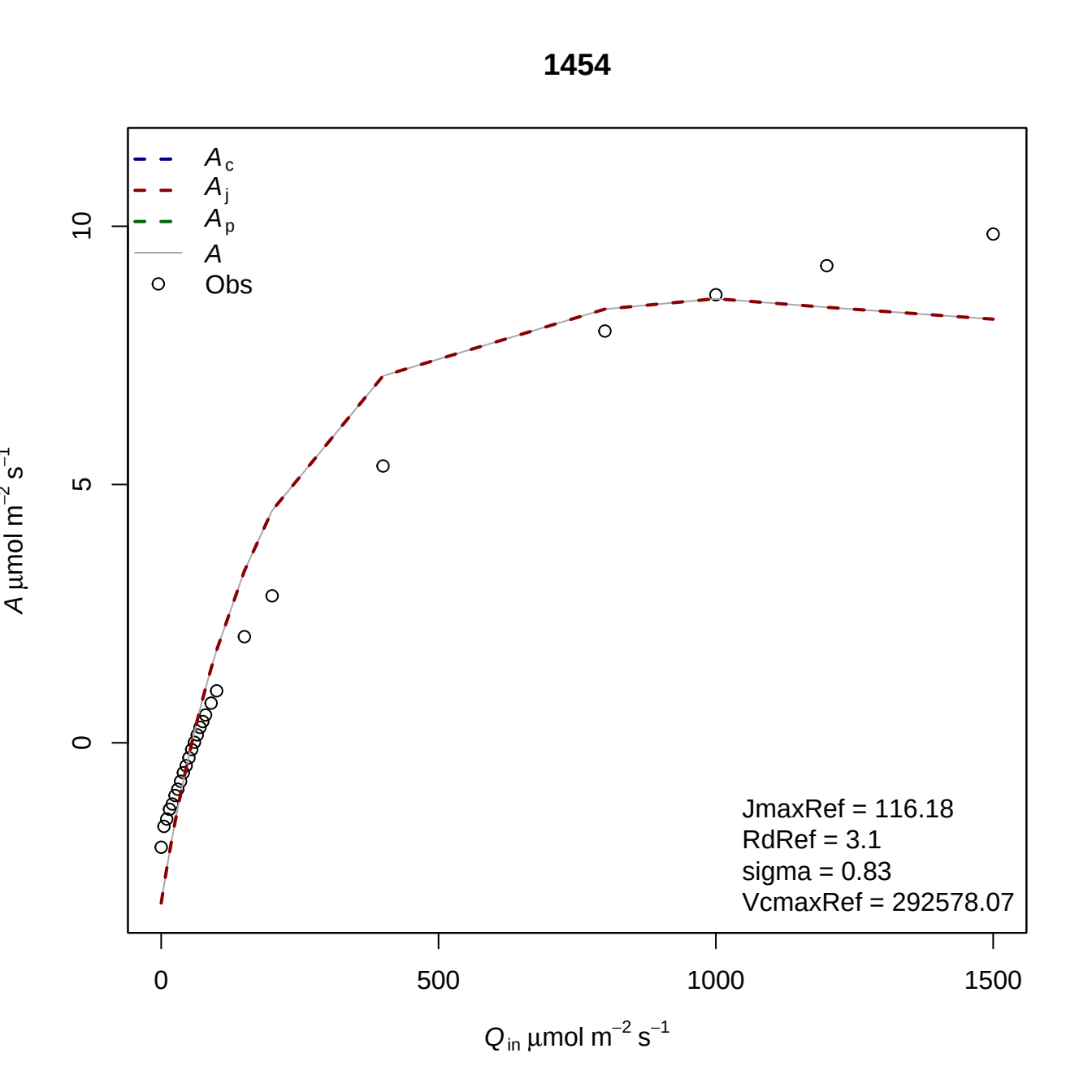

1449
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.78
RdRef = 1.52
sigma = 0.27
VcmaxRef = 230045.51
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
12
0
500
1000
1500

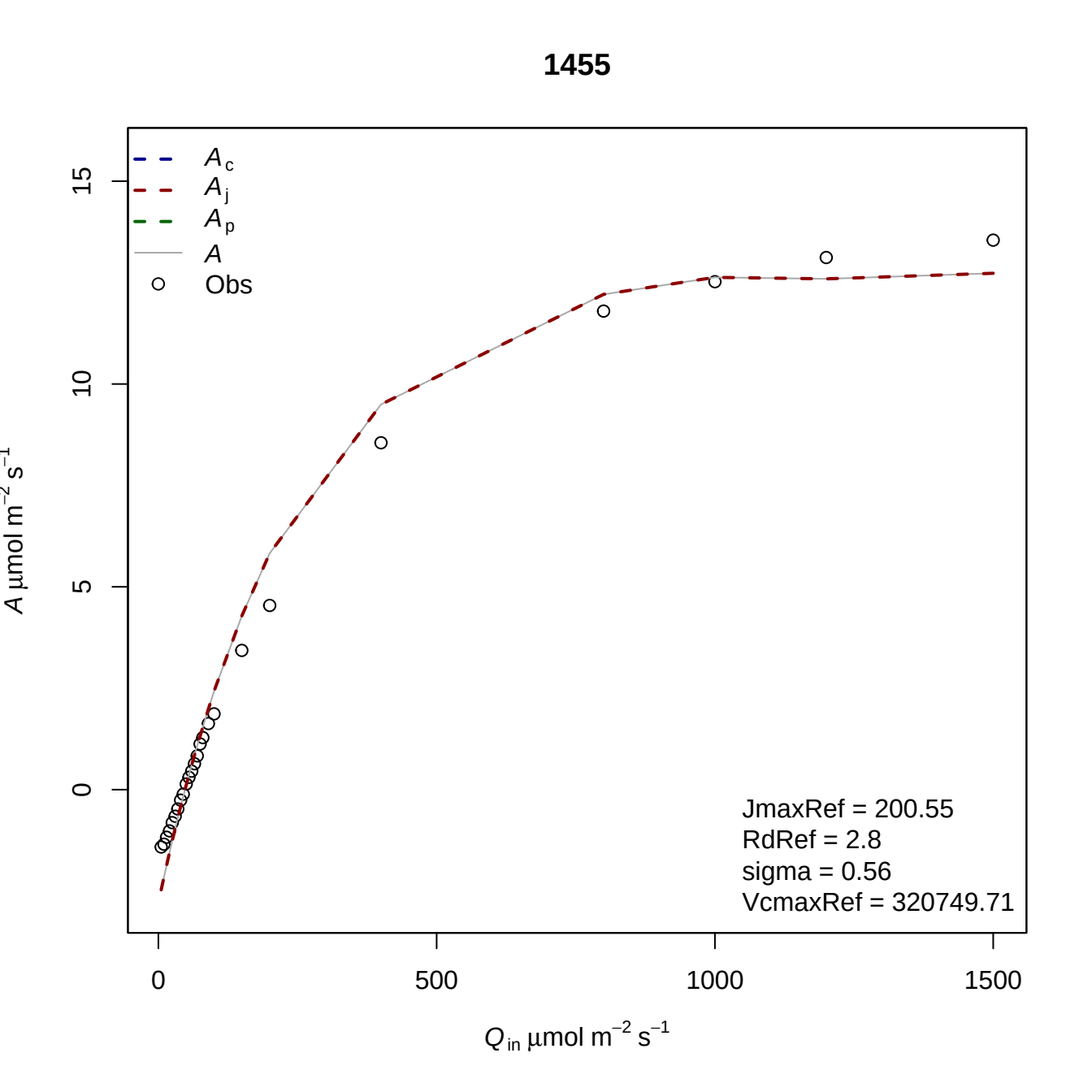

1450
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 194.18
RdRef = 3.57
sigma = 0.49
VcmaxRef = 5601833.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

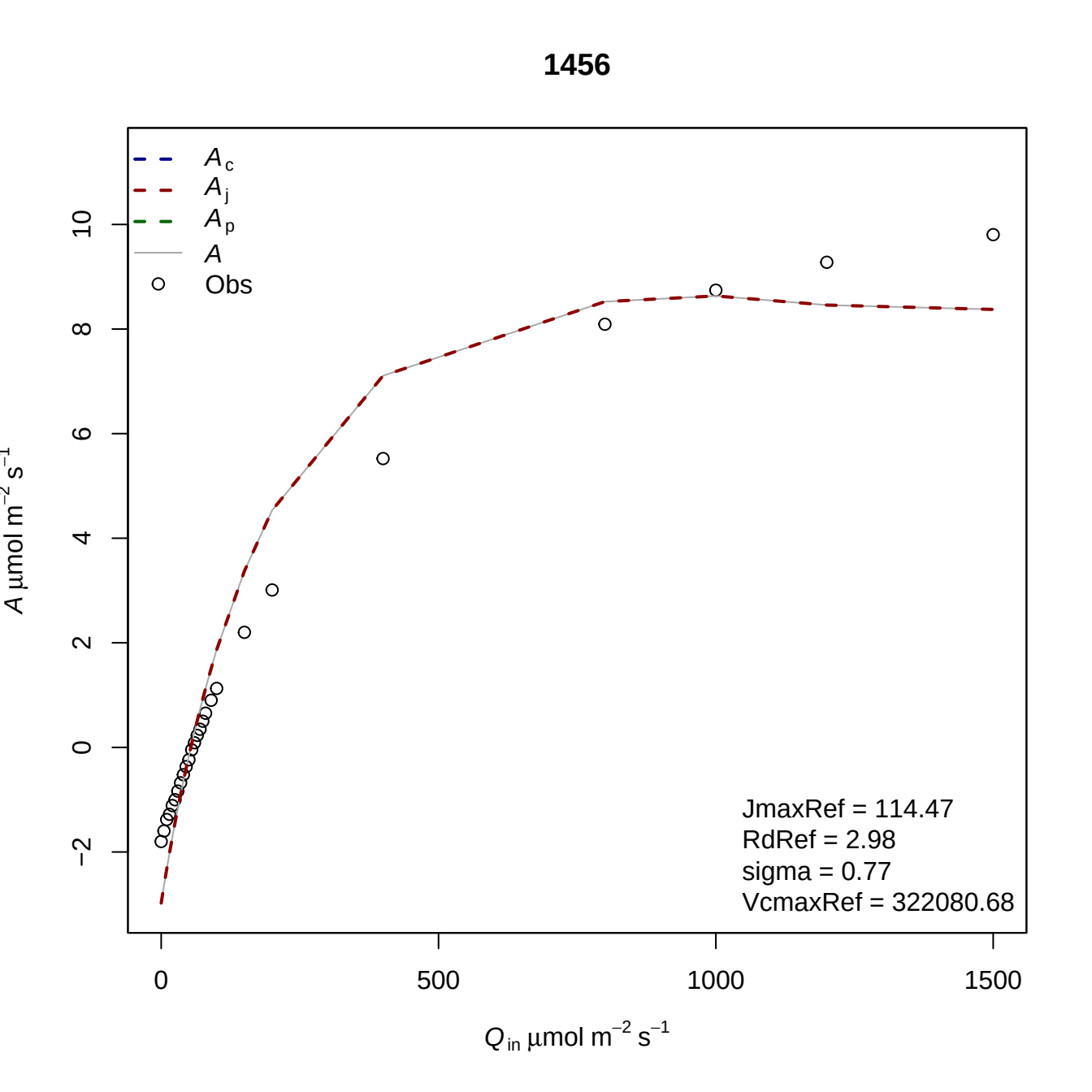

1451
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 141.02
RdRef = 3.3
sigma = 0.52
VcmaxRef = 350409.38
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

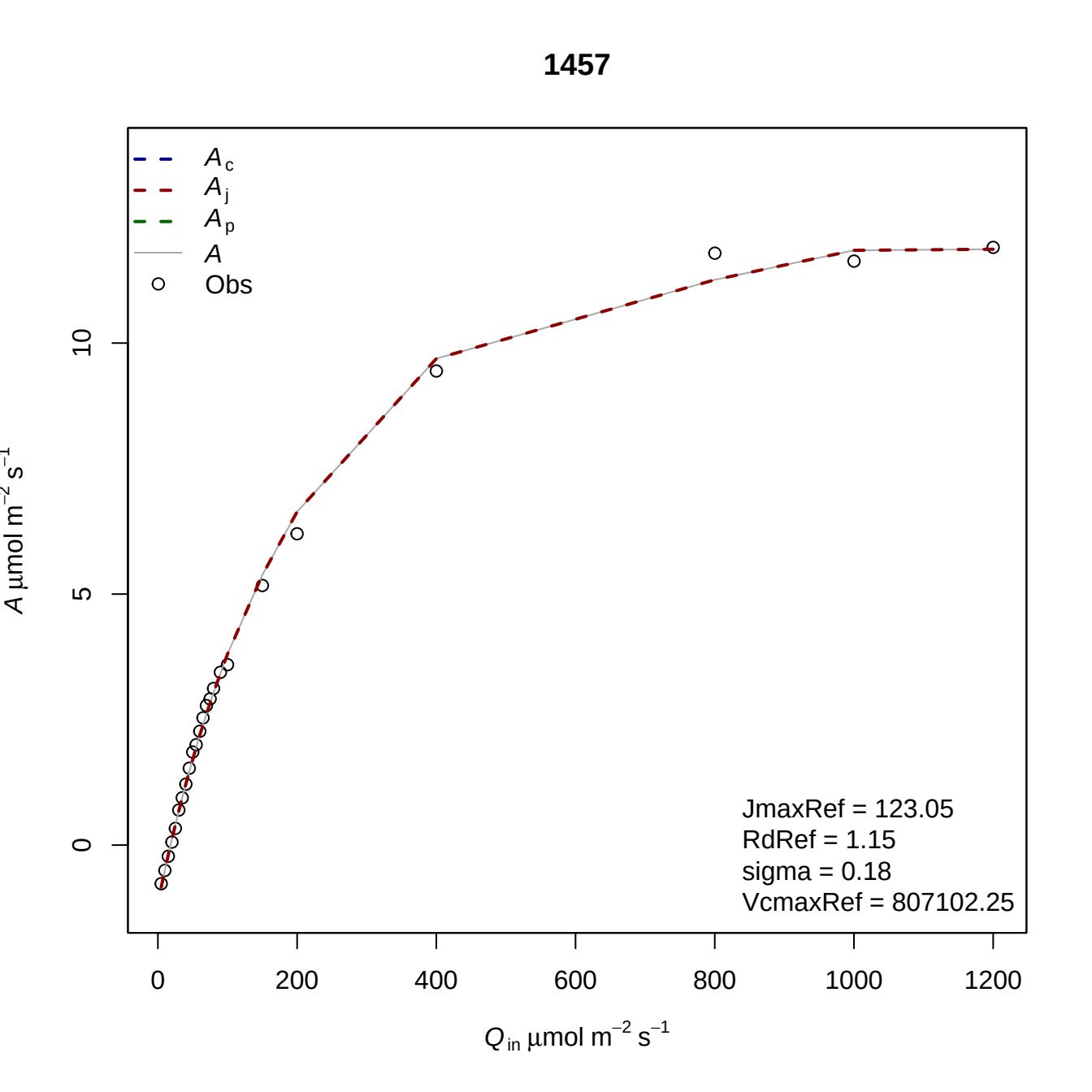

1452
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.86
RdRef = 1.44
sigma = 0.2
VcmaxRef = 2367425.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

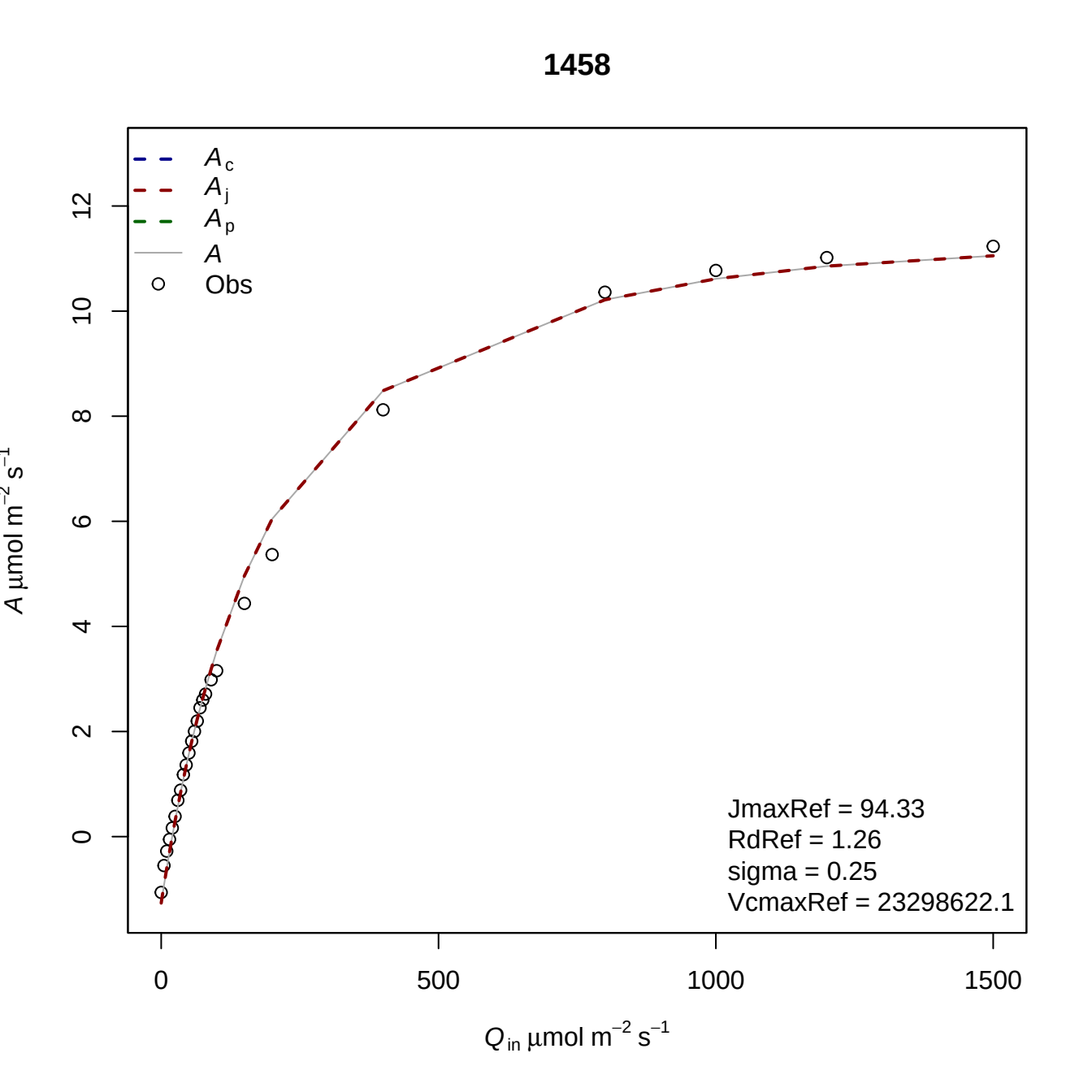

1453
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 164.72
RdRef = 3.3
sigma = 0.44
VcmaxRef = 730427.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

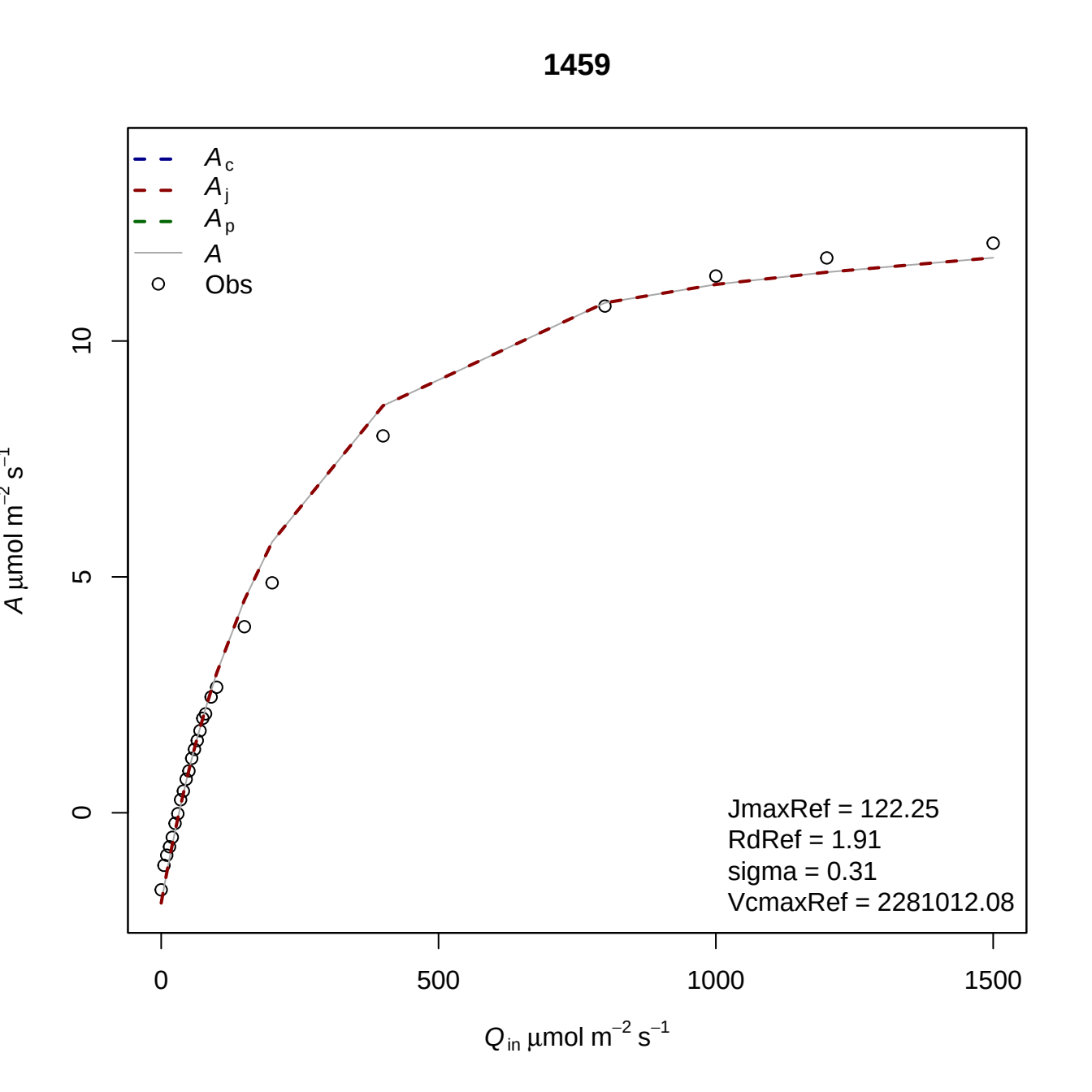

1454
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.18
RdRef = 3.1
sigma = 0.83
VcmaxRef = 292578.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

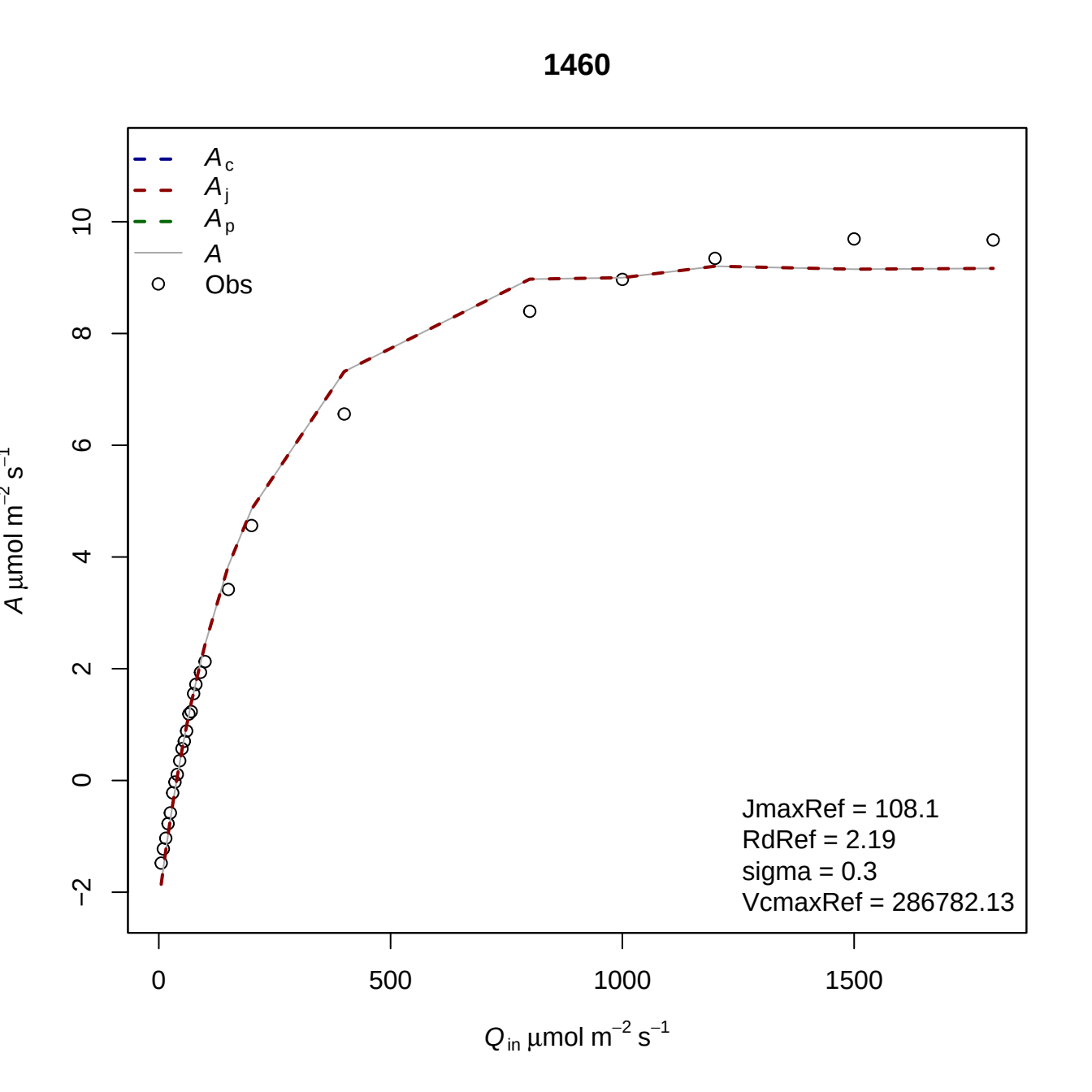

1455
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 200.55
RdRef = 2.8
sigma = 0.56
VcmaxRef = 320749.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

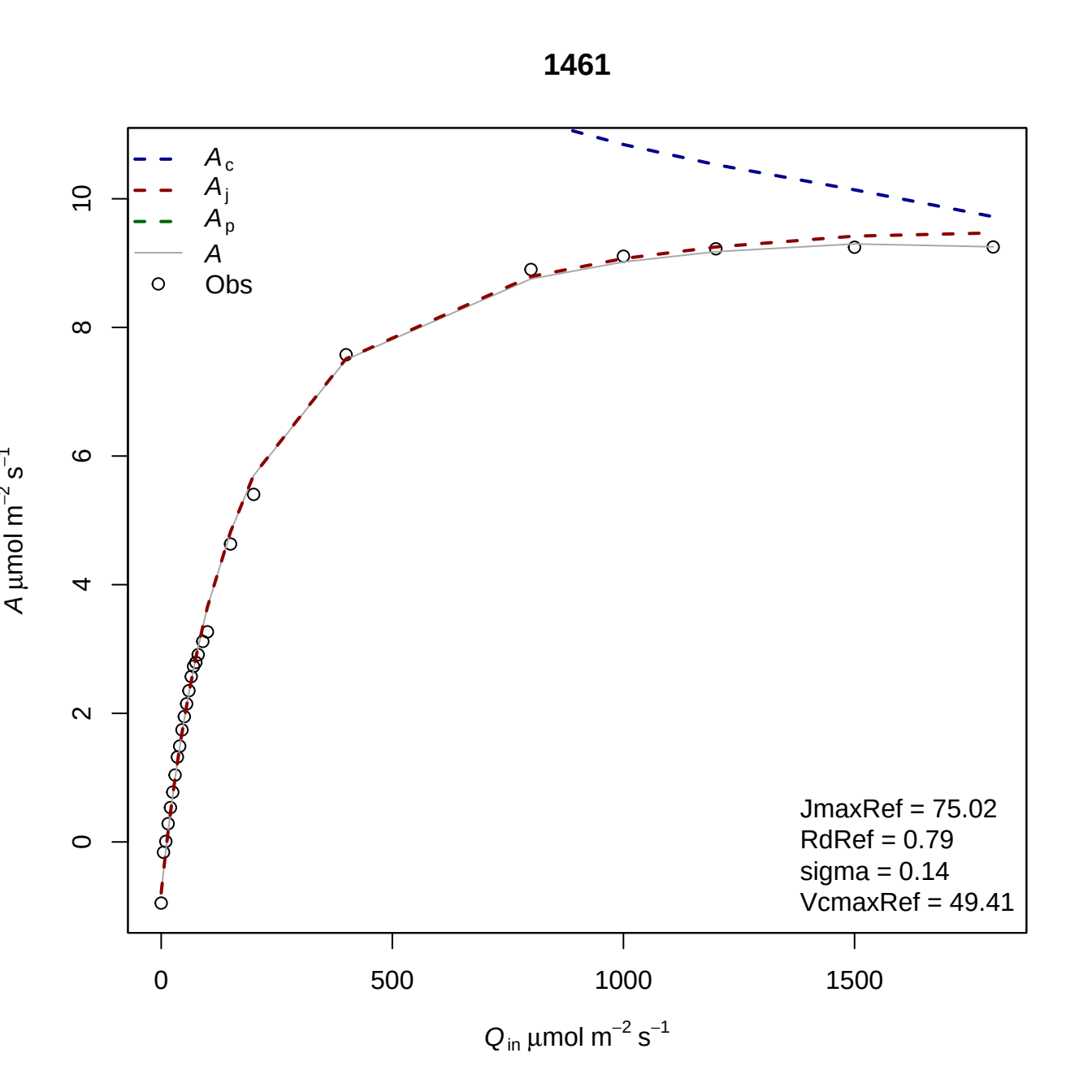

1456
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.47
RdRef = 2.98
sigma = 0.77
VcmaxRef = 322080.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

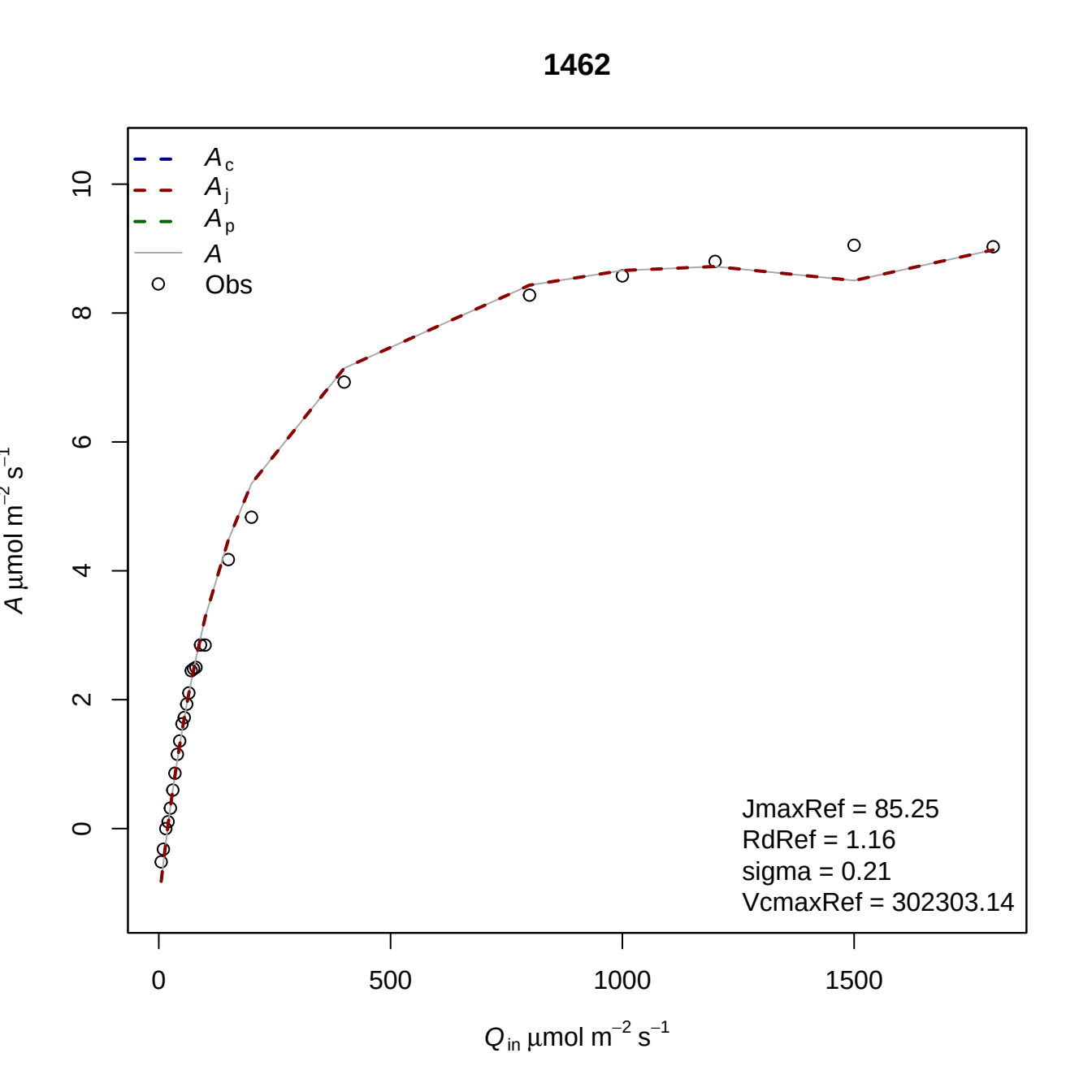

1457
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.05
RdRef = 1.15
sigma = 0.18
VcmaxRef = 807102.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
200
400
600
800
1000
1200

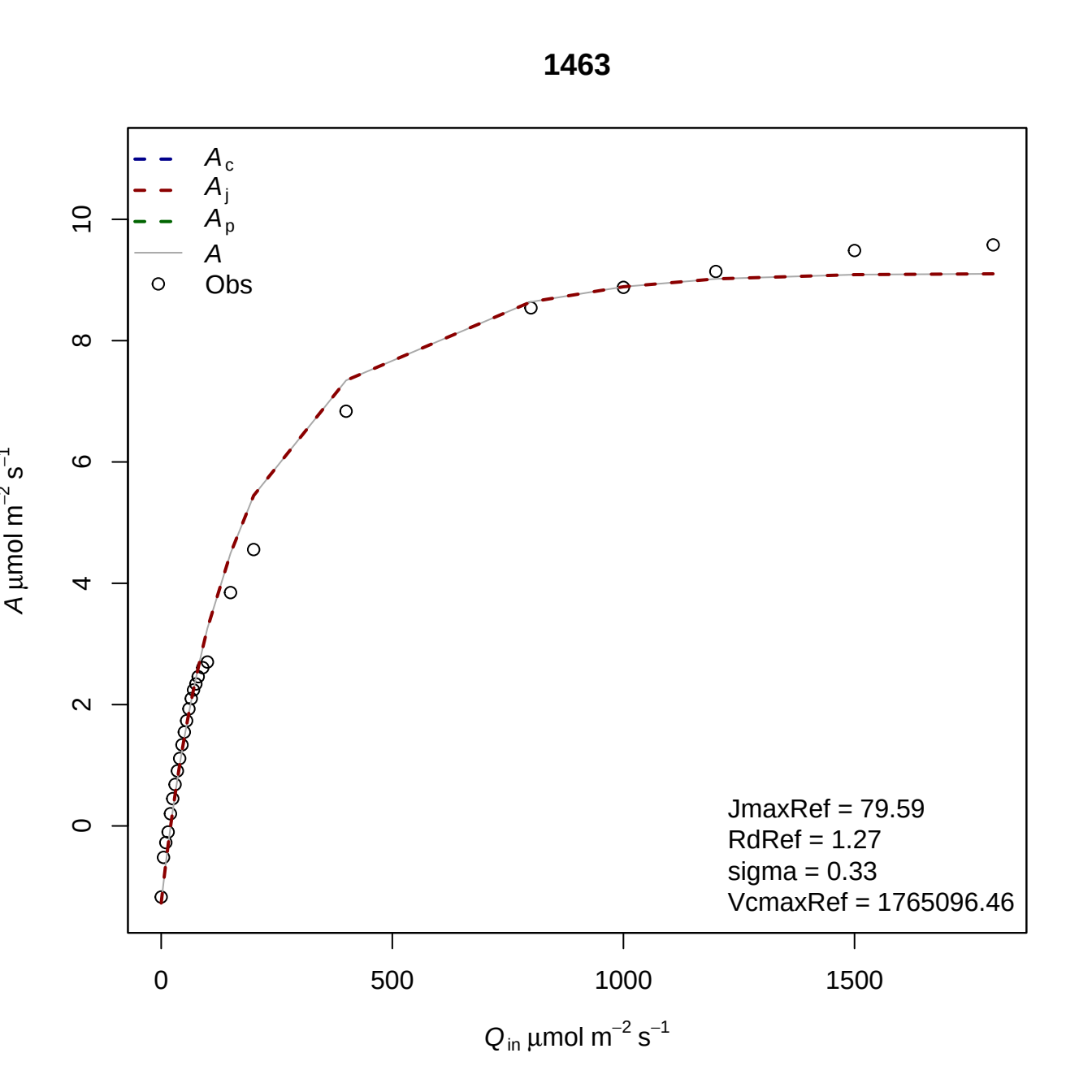

1458
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.33
RdRef = 1.26
sigma = 0.25
VcmaxRef = 23298622.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

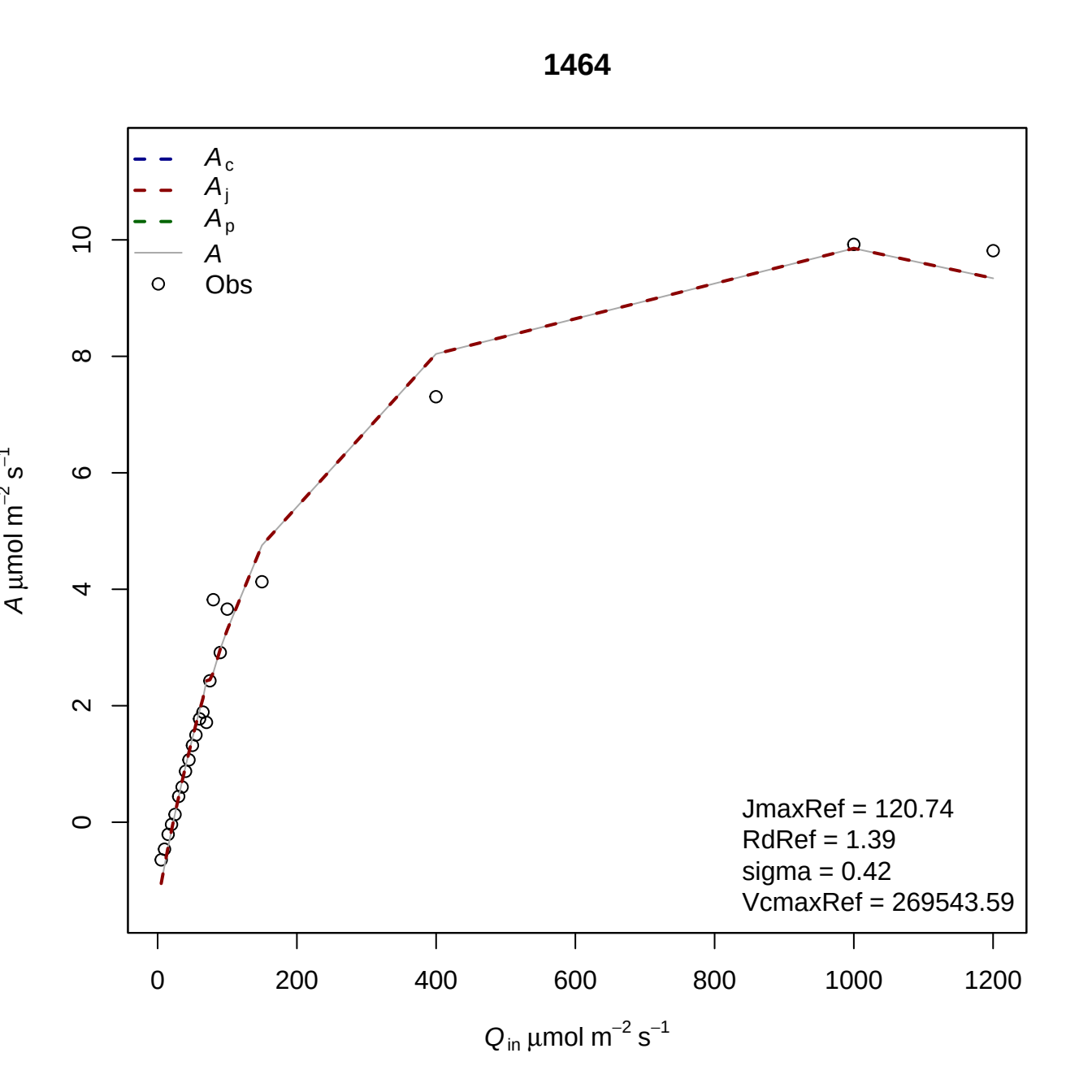

1459
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.25
RdRef = 1.91
sigma = 0.31
VcmaxRef = 2281012.08
0
5
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

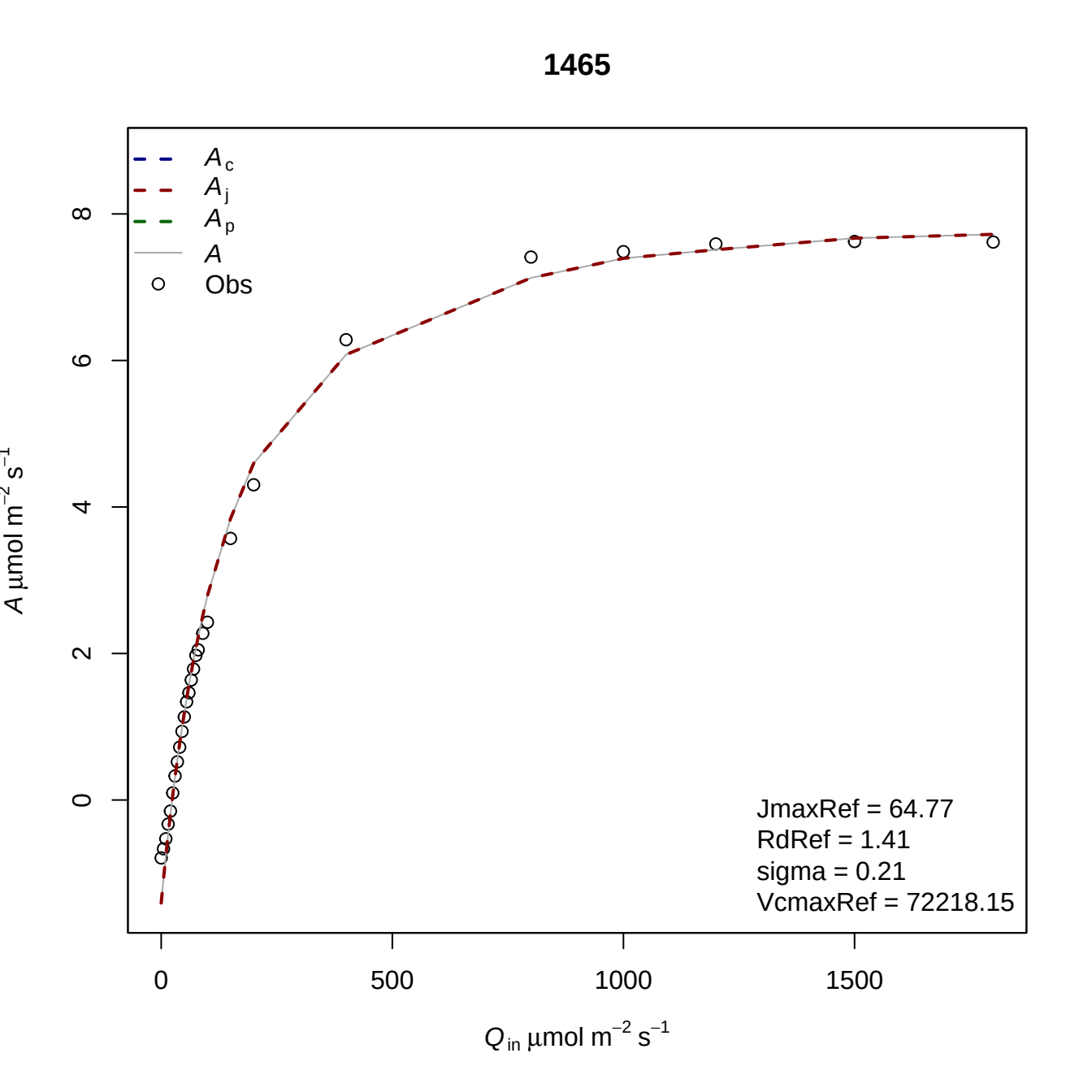

1460
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 108.1
RdRef = 2.19
sigma = 0.3
VcmaxRef = 286782.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

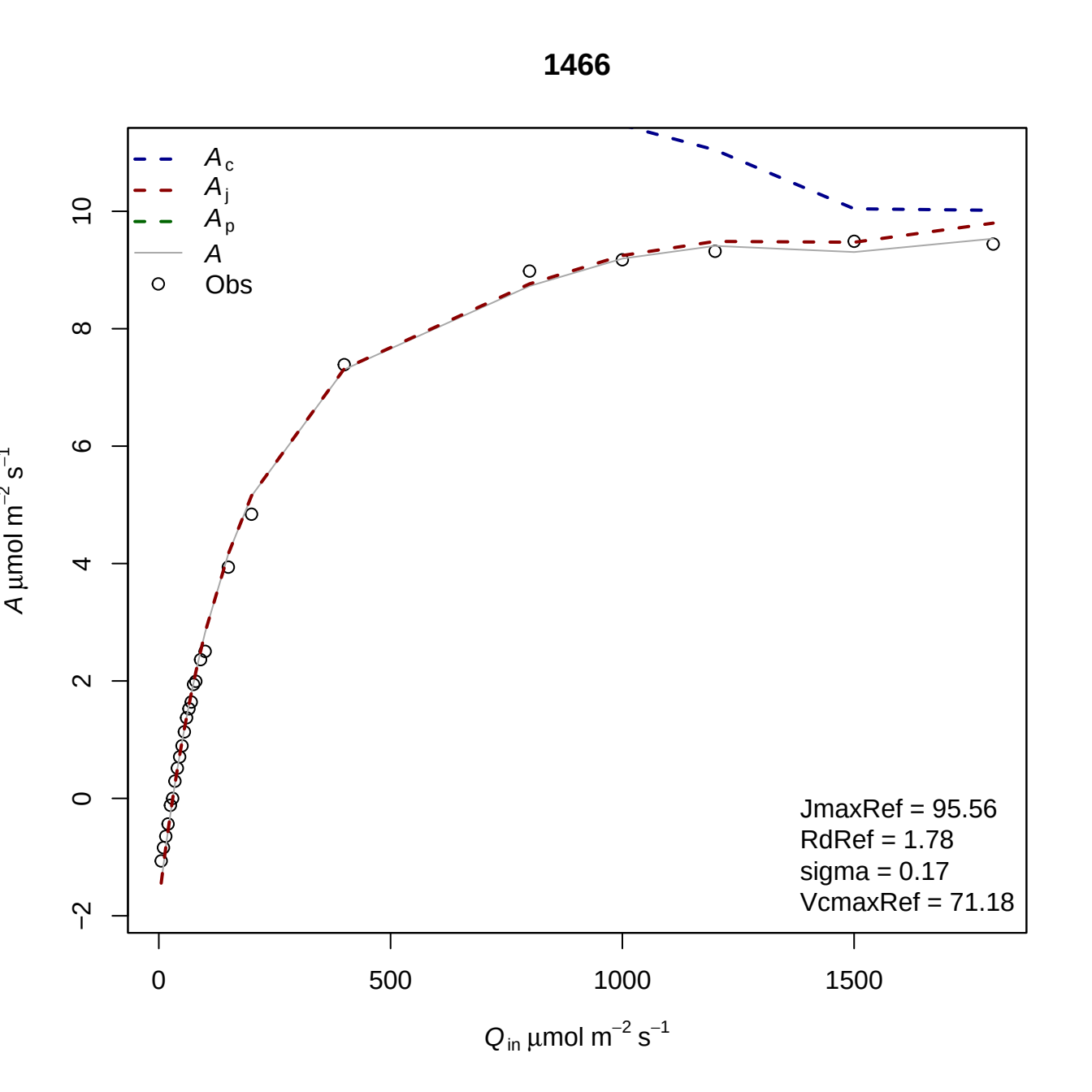

1461
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.02
RdRef = 0.79
sigma = 0.14
VcmaxRef = 49.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

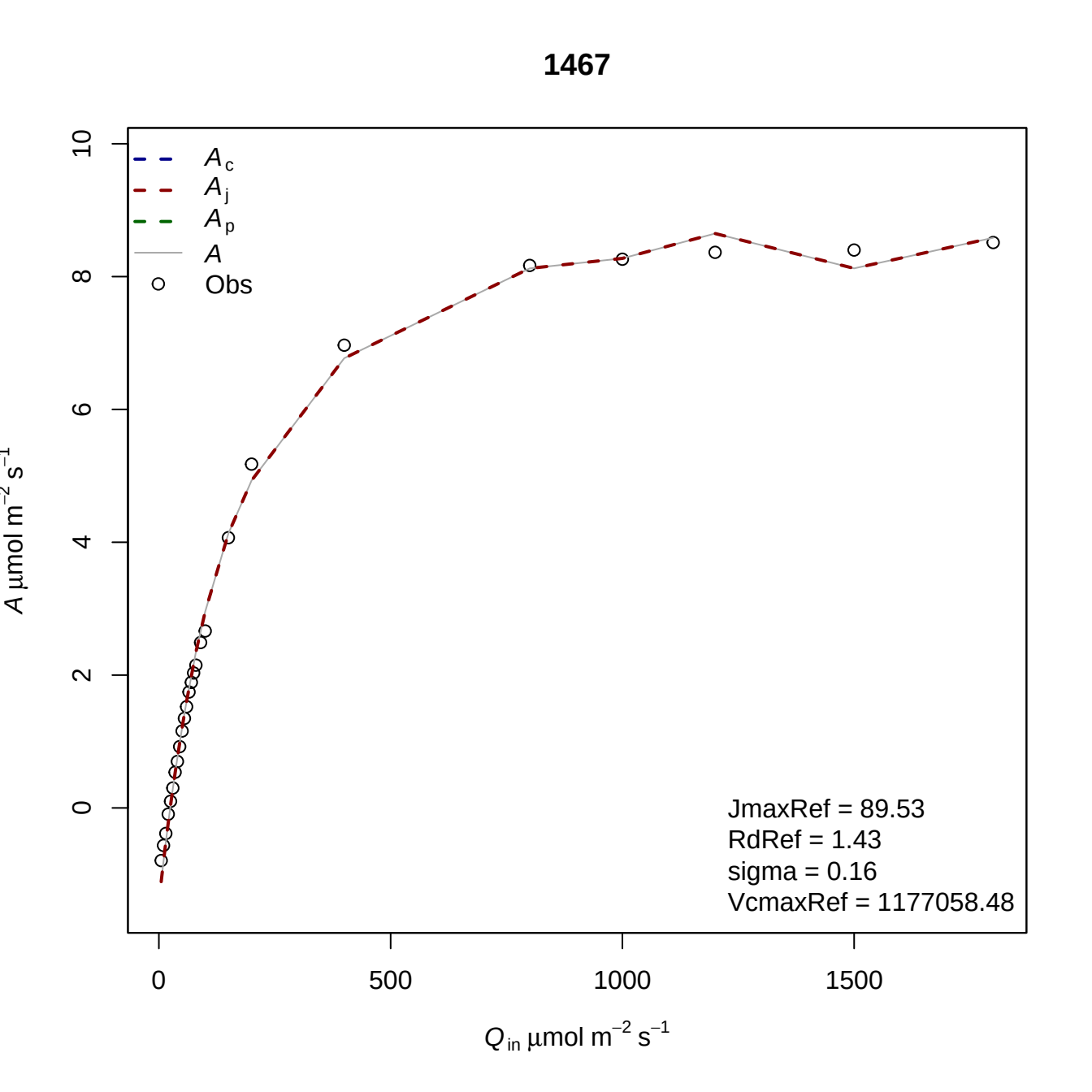

1462
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.25
RdRef = 1.16
sigma = 0.21
VcmaxRef = 302303.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

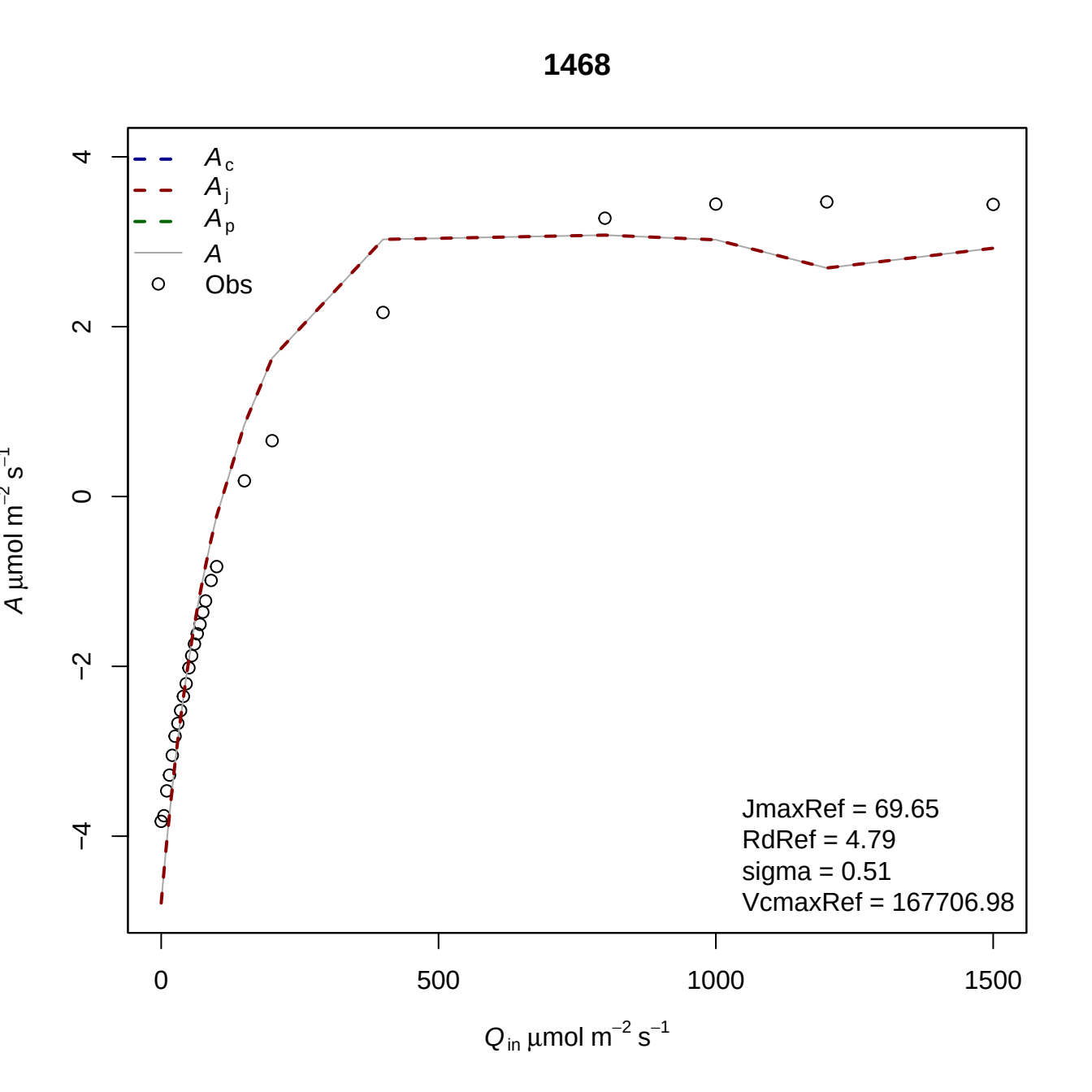

1463
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.59
RdRef = 1.27
sigma = 0.33
VcmaxRef = 1765096.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

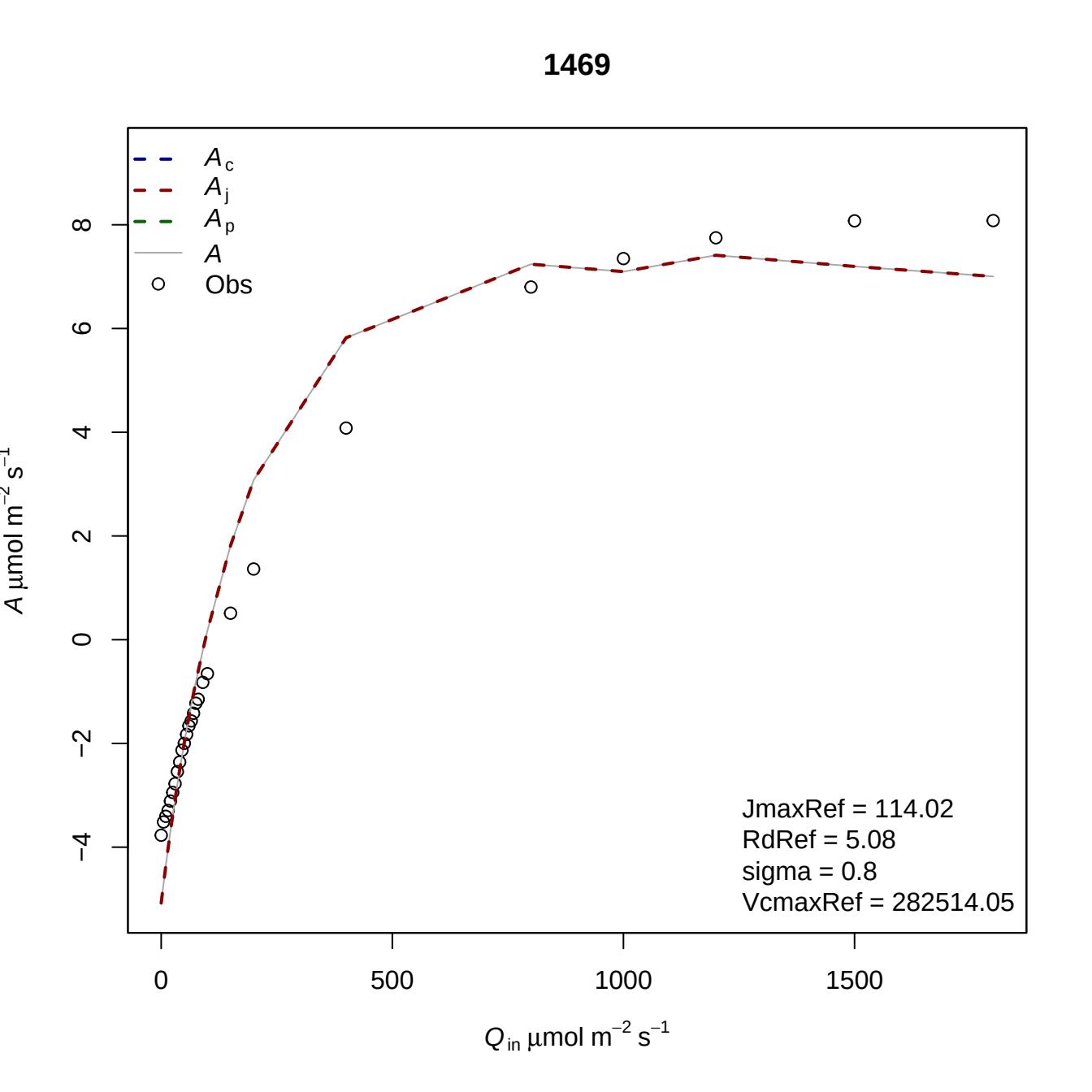

1464
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.74
RdRef = 1.39
sigma = 0.42
VcmaxRef = 269543.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
200
400
600
800
1000
1200

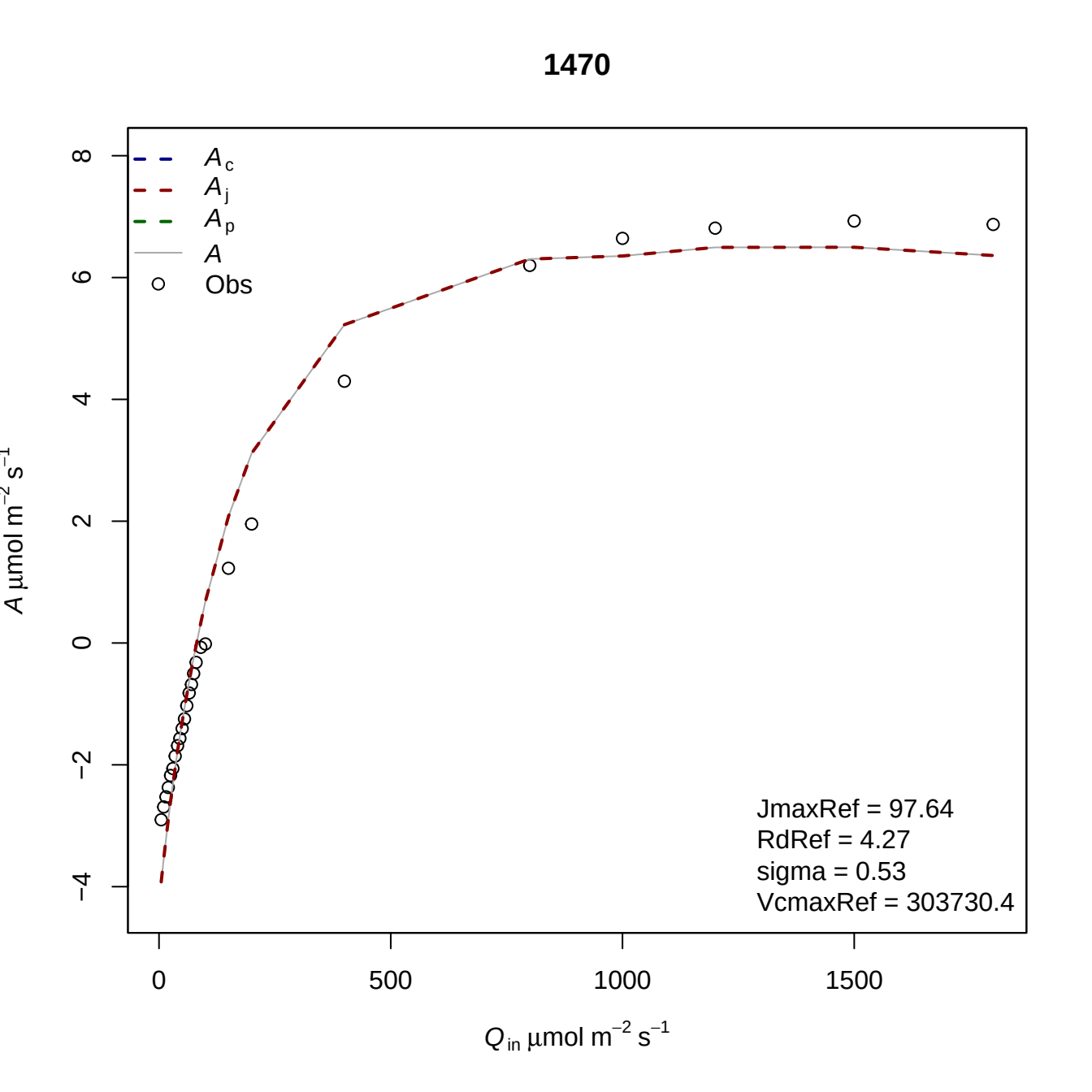

1465
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.77
RdRef = 1.41
sigma = 0.21
VcmaxRef = 72218.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

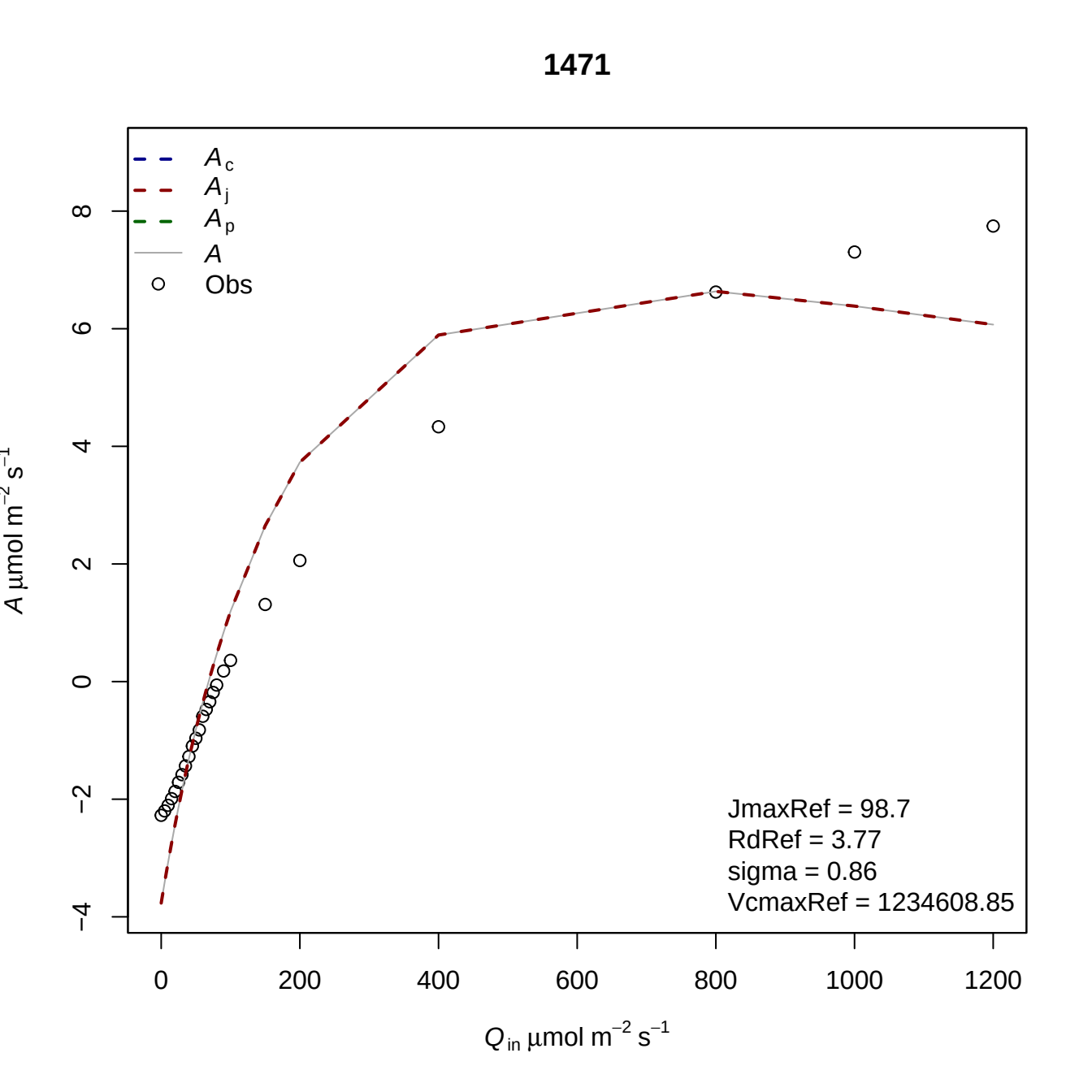

1466
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.56
RdRef = 1.78
sigma = 0.17
VcmaxRef = 71.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
10
0
500
1000
1500

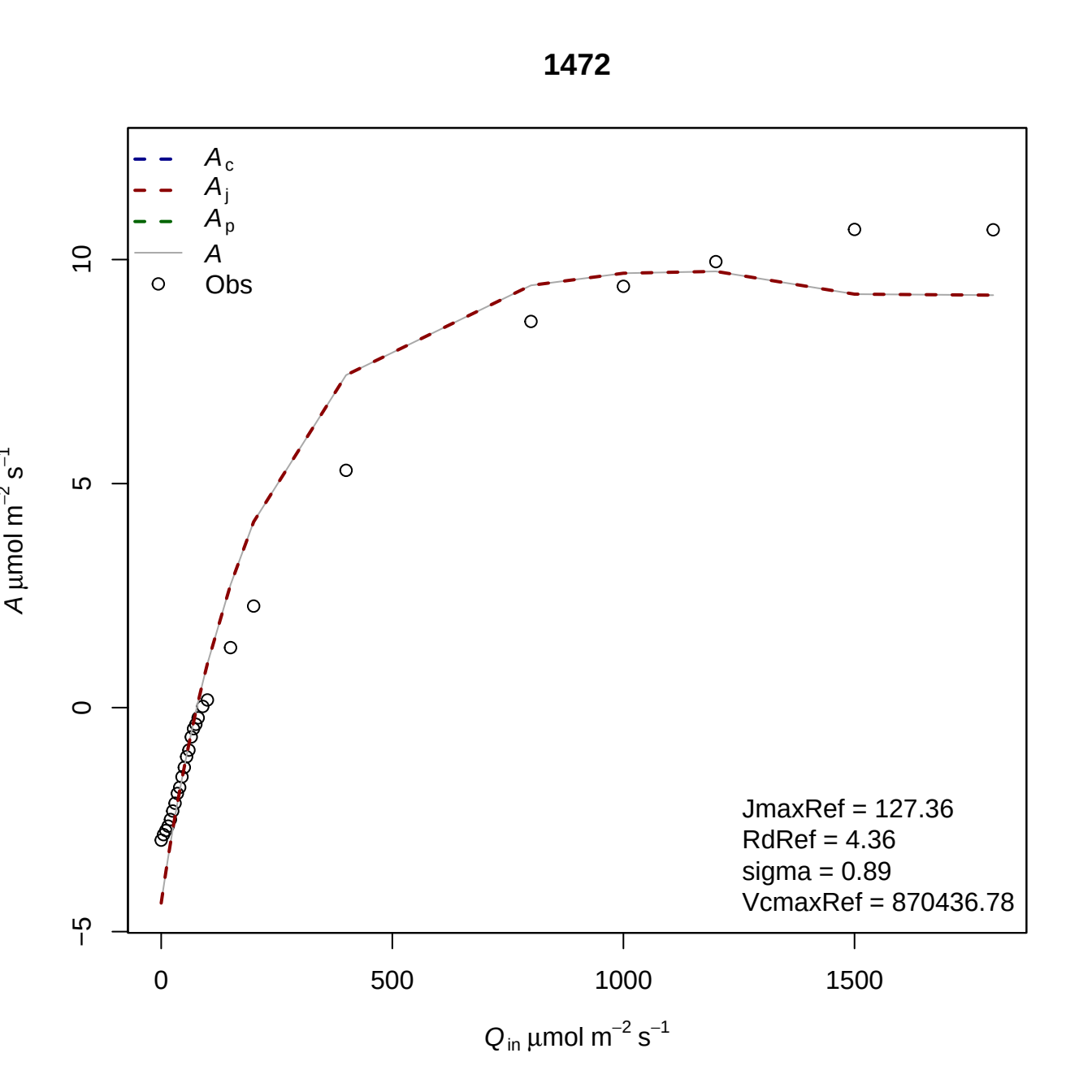

1467
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.53
RdRef = 1.43
sigma = 0.16
VcmaxRef = 1177058.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

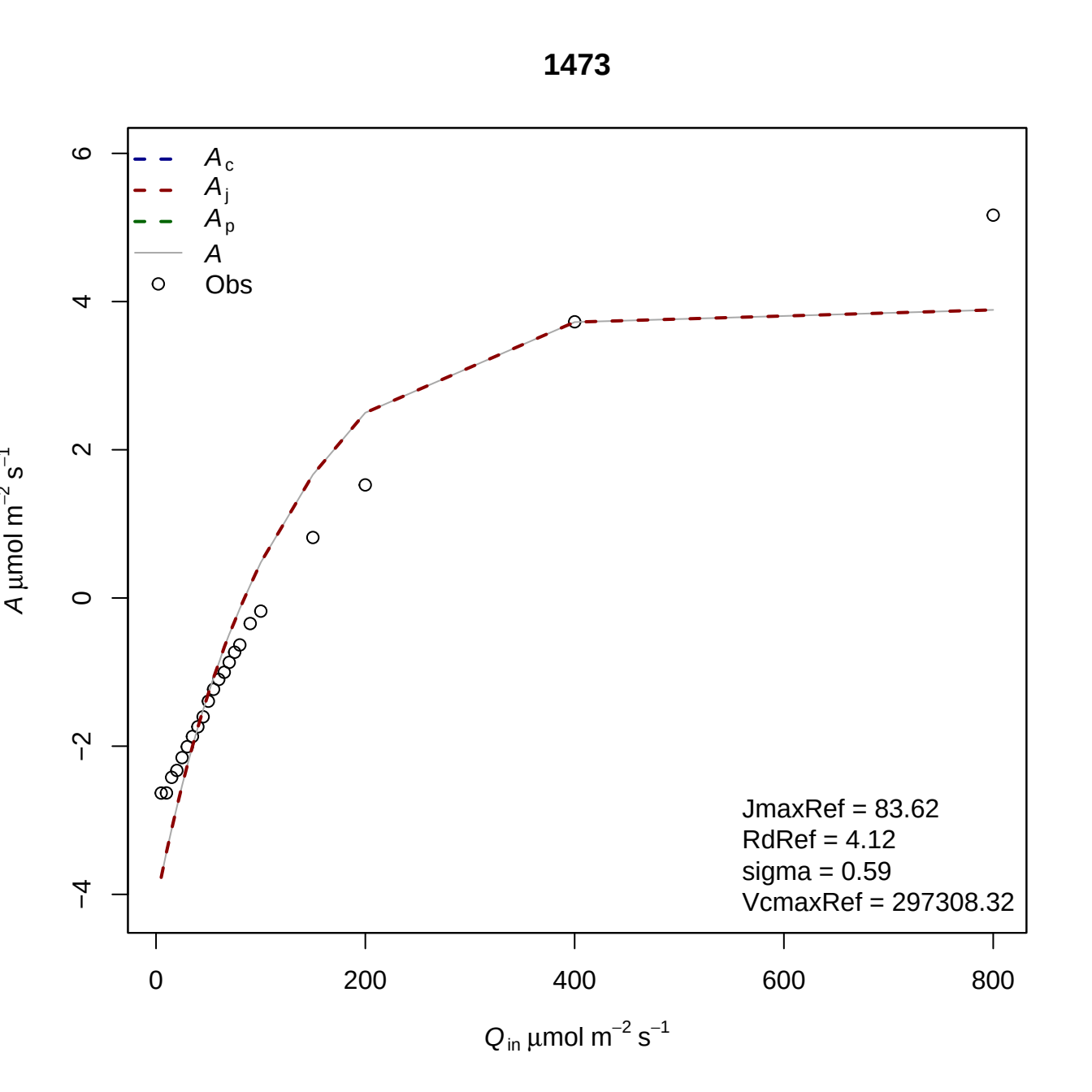

1468
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 69.65
RdRef = 4.79
sigma = 0.51
VcmaxRef = 167706.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
−4
0
500
1000
1500

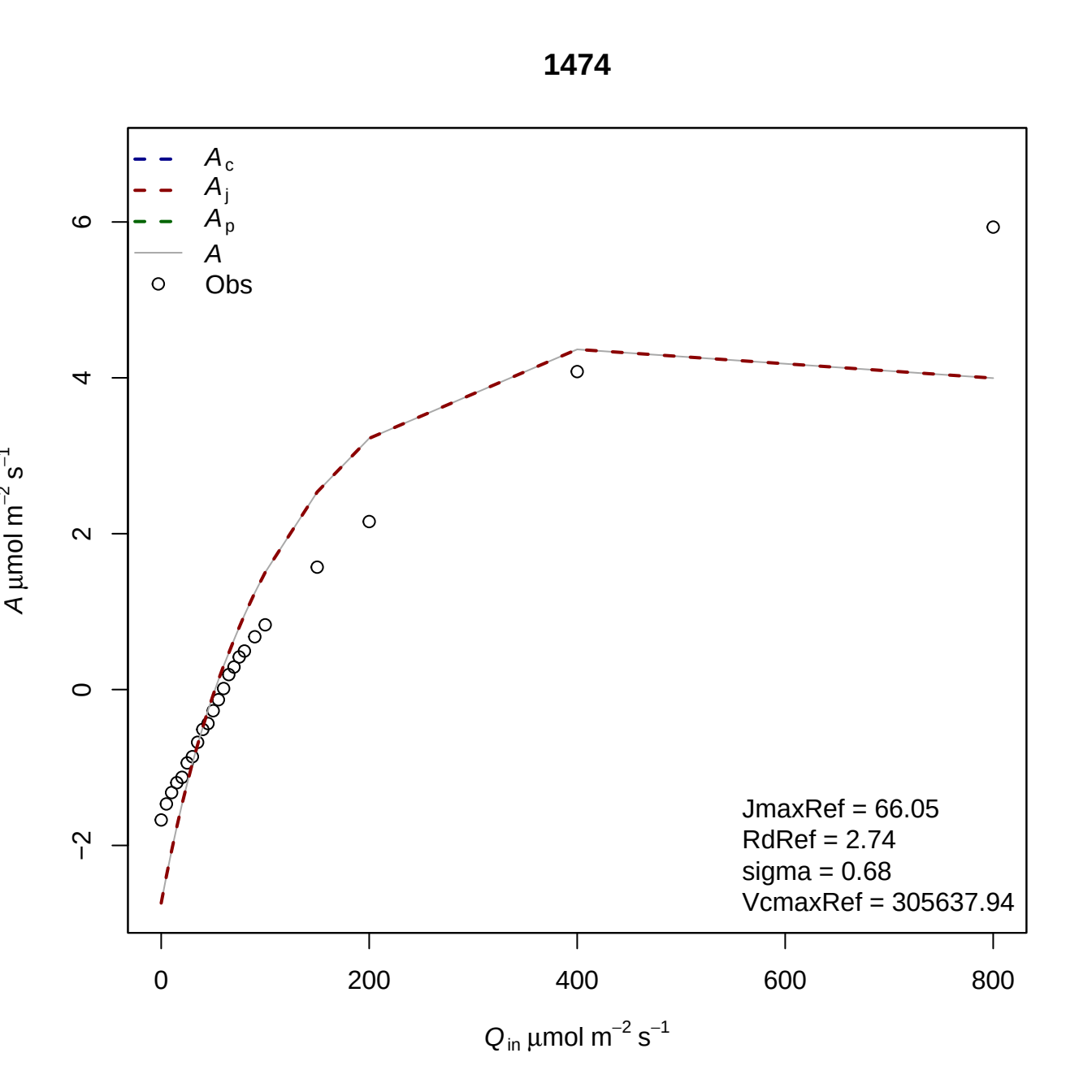

1469
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.02
RdRef = 5.08
sigma = 0.8
VcmaxRef = 282514.05
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

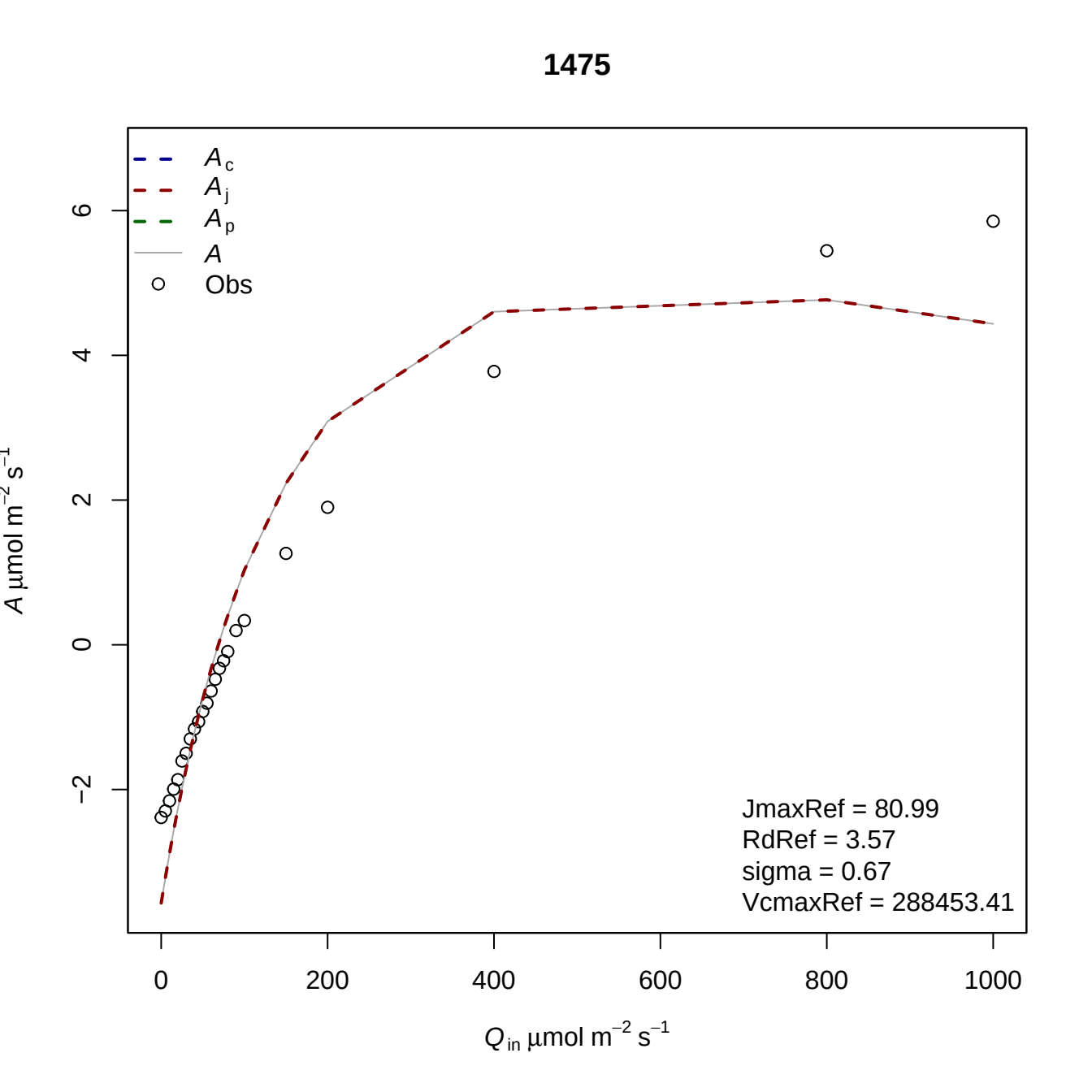

1470
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.64
RdRef = 4.27
sigma = 0.53
VcmaxRef = 303730.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

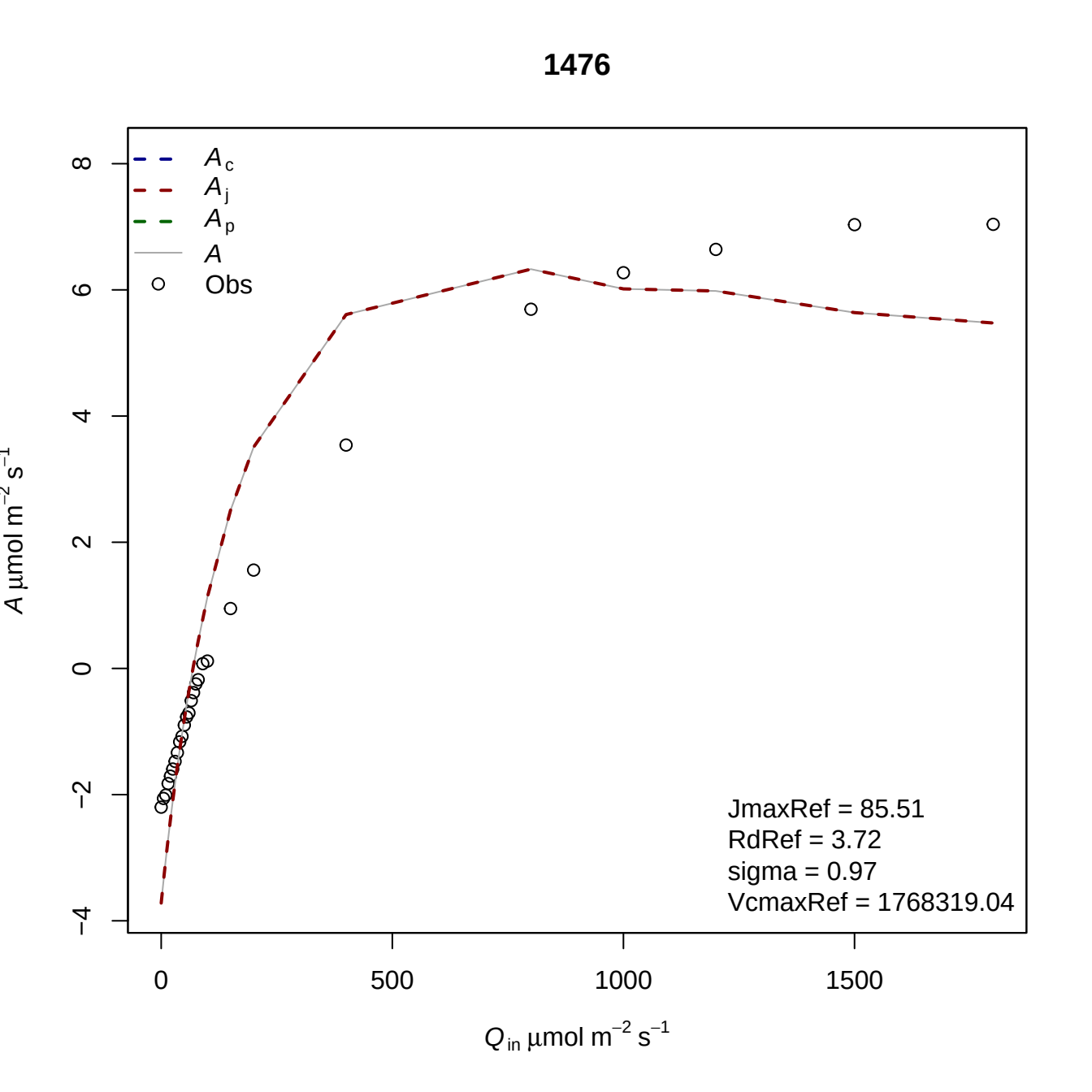

1471
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.7
RdRef = 3.77
sigma = 0.86
VcmaxRef = 1234608.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

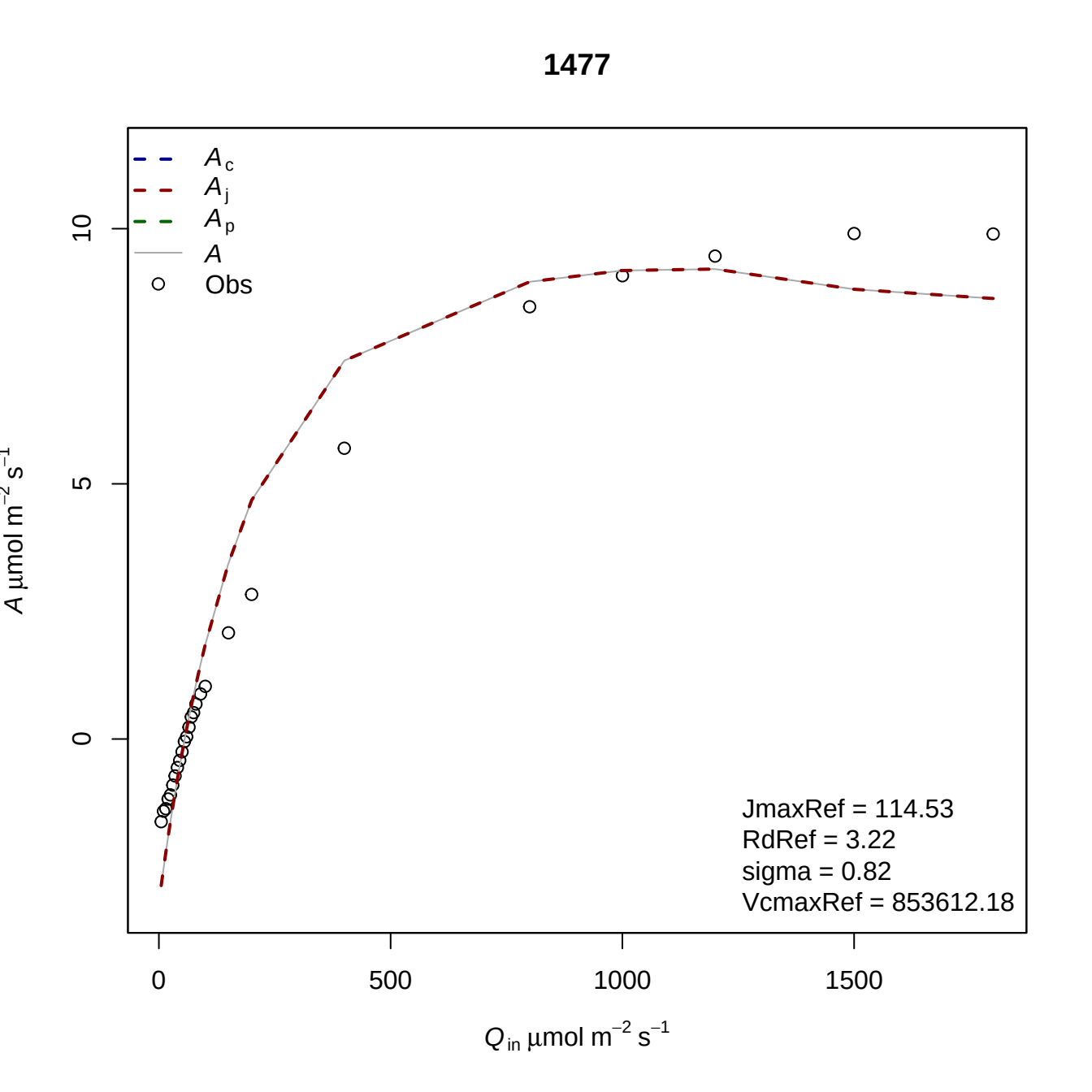

1472
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 127.36
RdRef = 4.36
sigma = 0.89
VcmaxRef = 870436.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−5
0
5
10
0
500
1000
1500

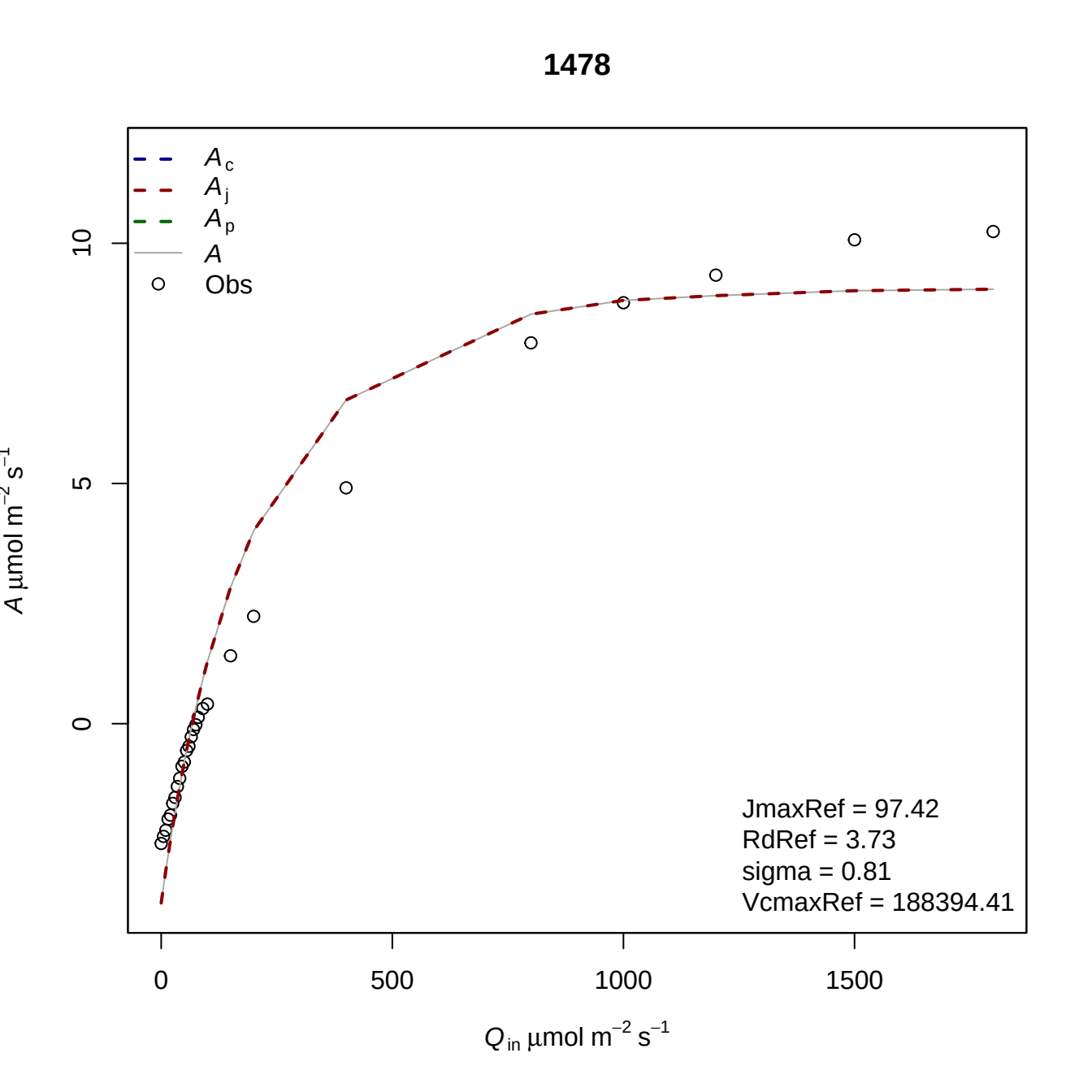

1473
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.62
RdRef = 4.12
sigma = 0.59
VcmaxRef = 297308.32
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
200
400
600
800

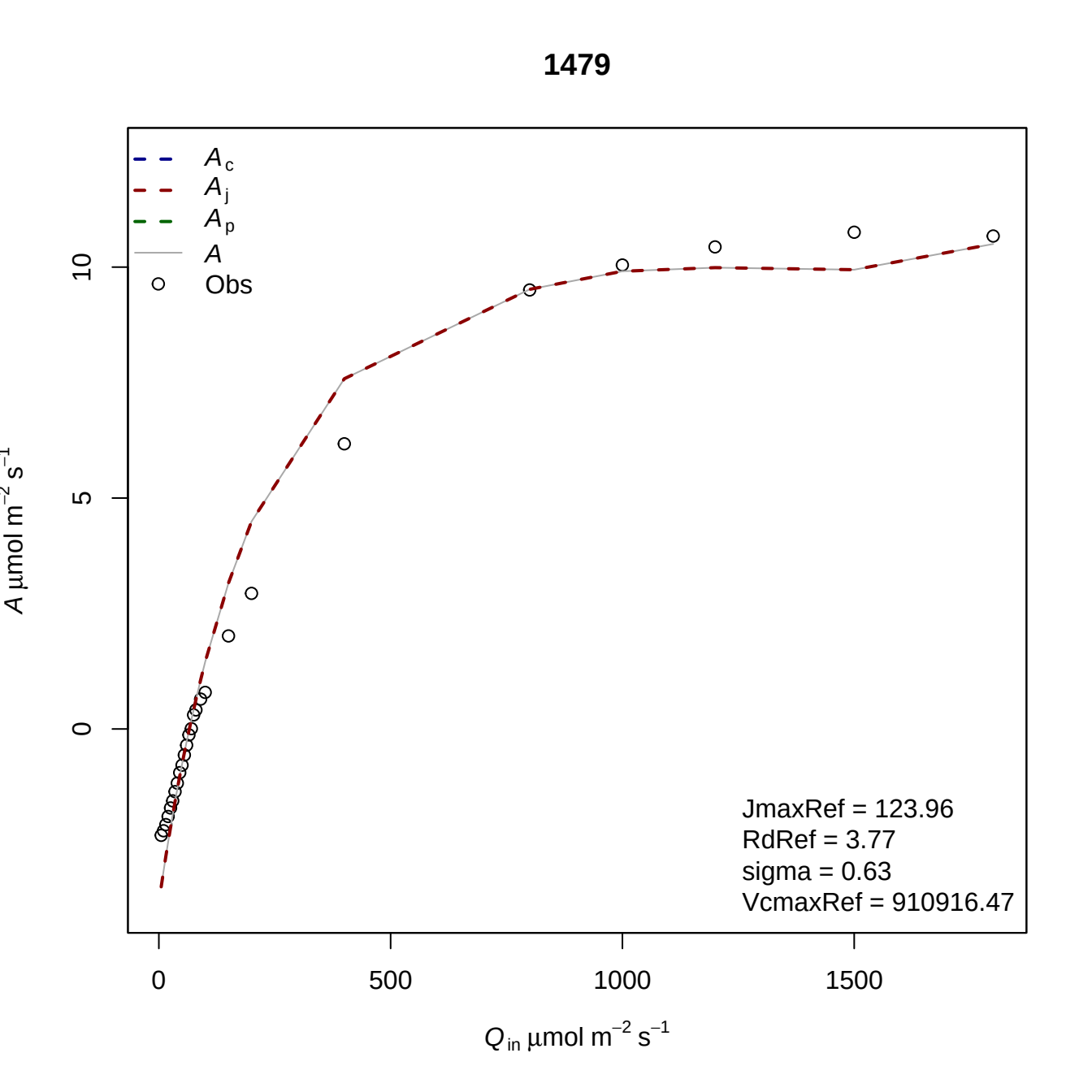

1474
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 66.05
RdRef = 2.74
sigma = 0.68
VcmaxRef = 305637.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800

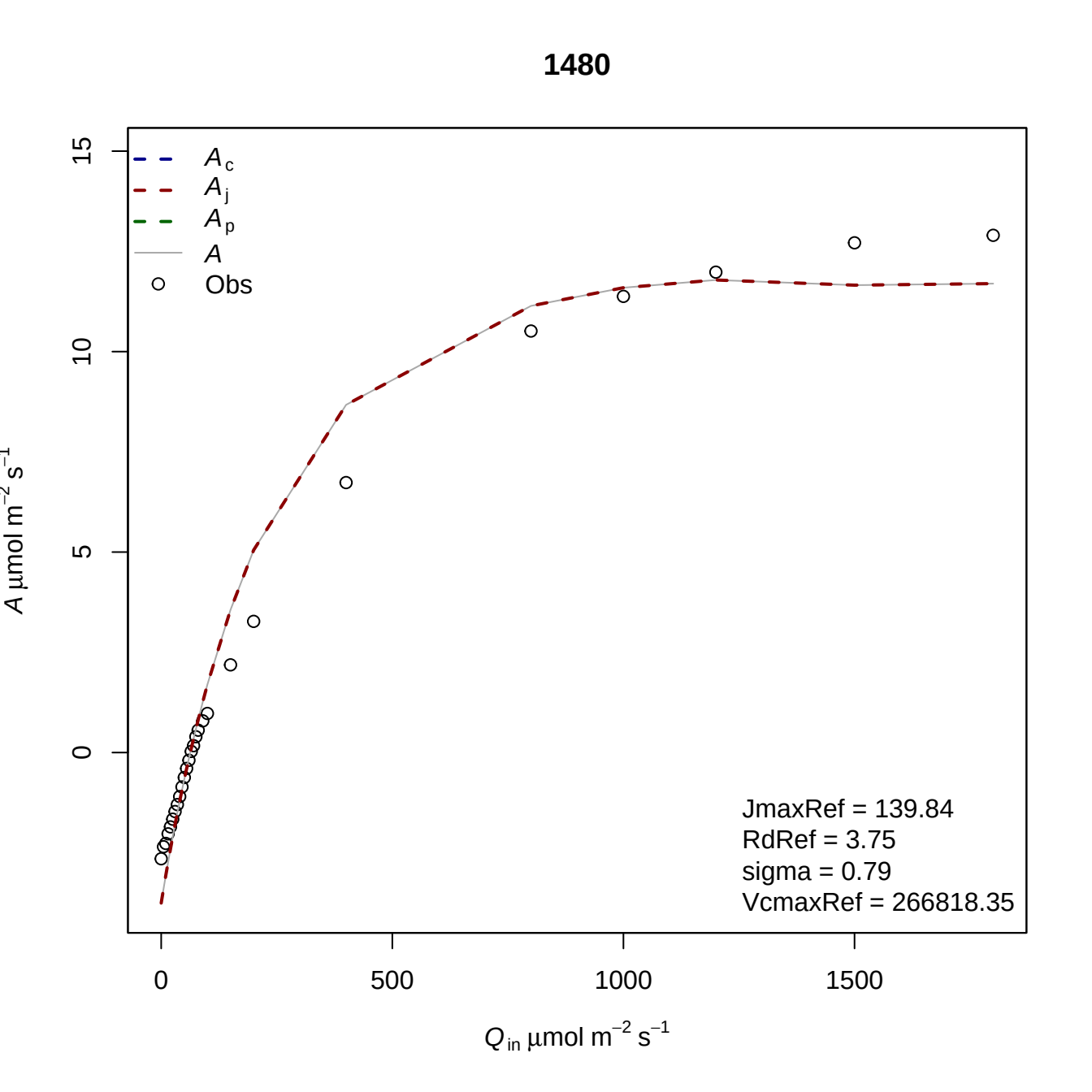

1475
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.99
RdRef = 3.57
sigma = 0.67
VcmaxRef = 288453.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
0
200
400
600
800
1000

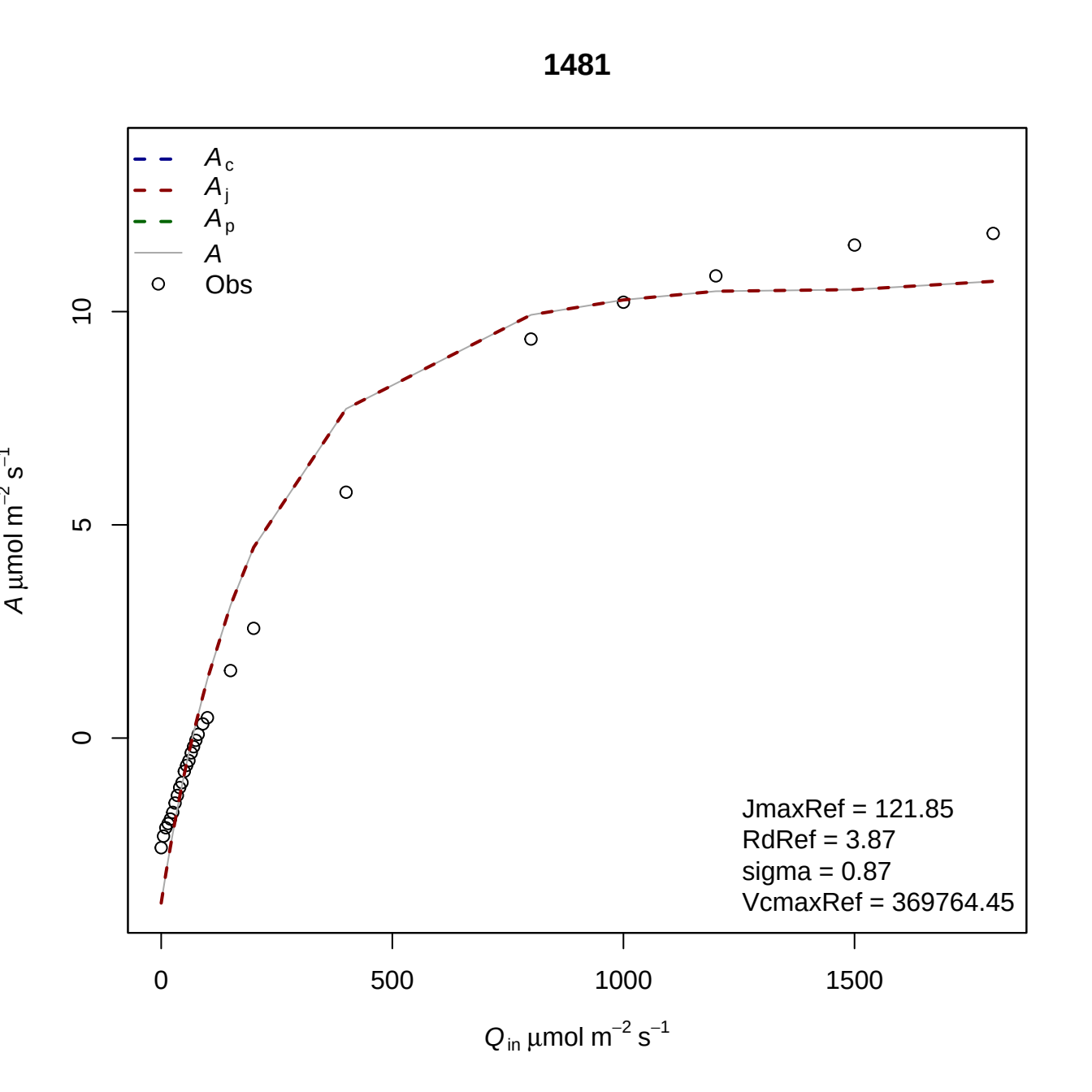

1476
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.51
RdRef = 3.72
sigma = 0.97
VcmaxRef = 1768319.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

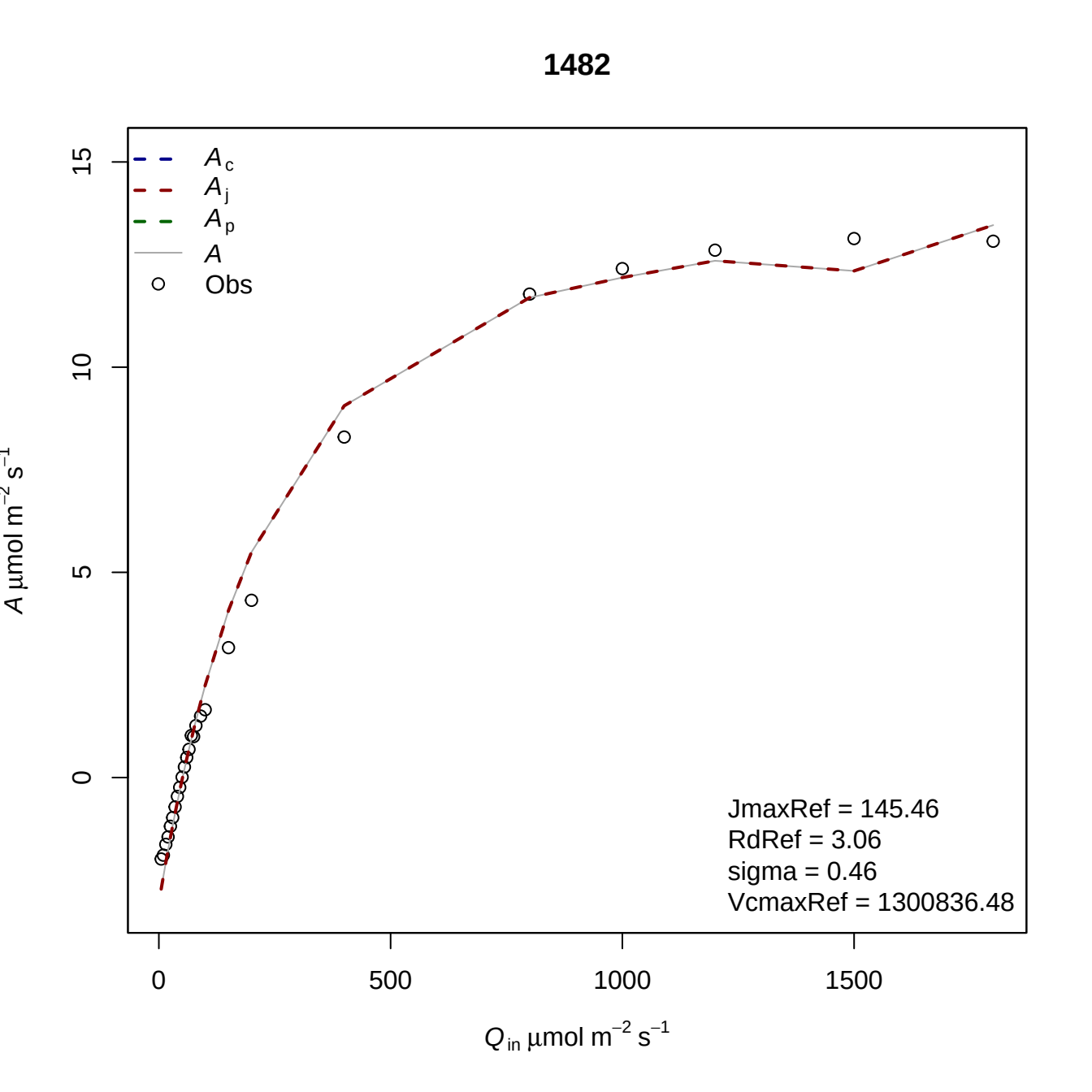

1477
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.53
RdRef = 3.22
sigma = 0.82
VcmaxRef = 853612.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

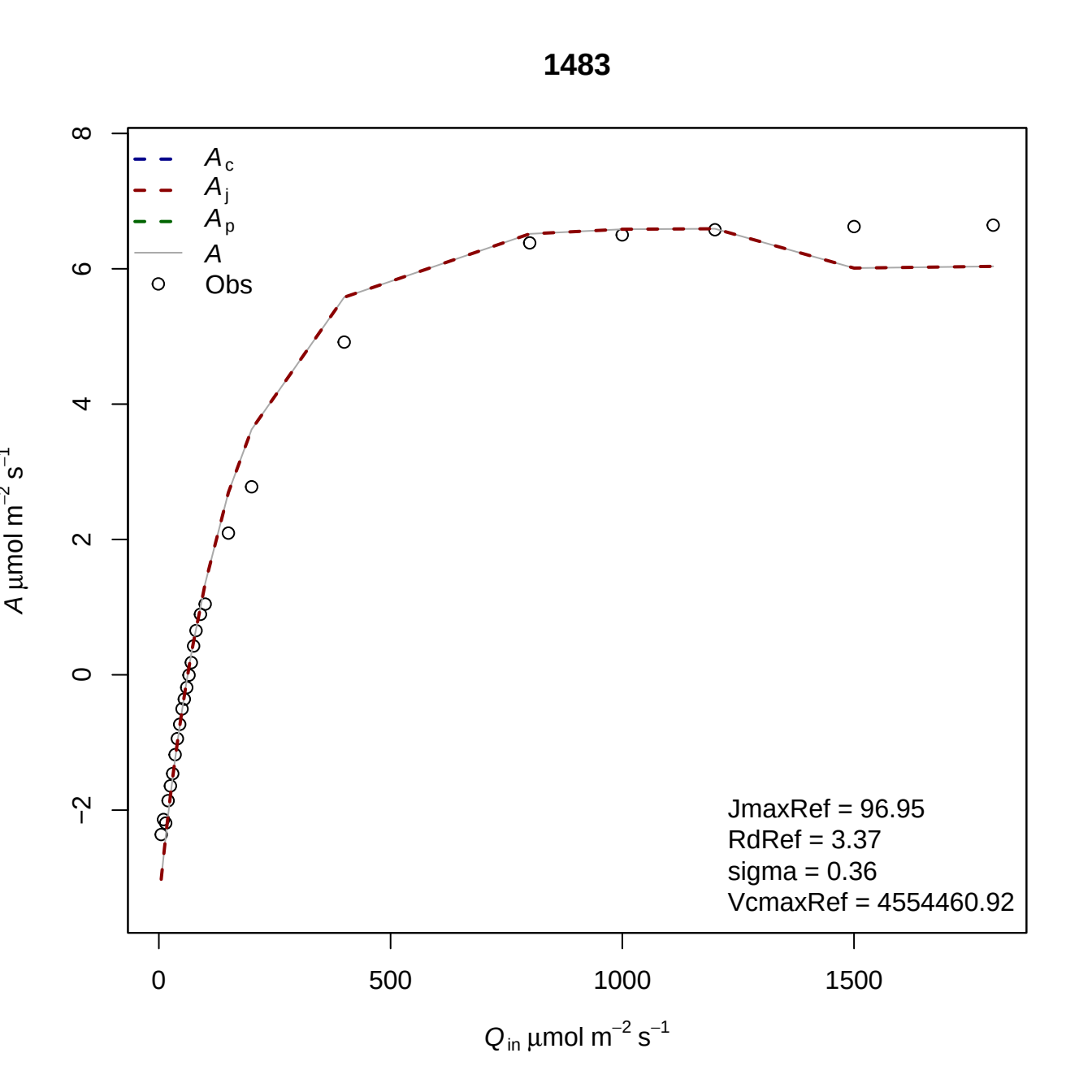

1478
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.42
RdRef = 3.73
sigma = 0.81
VcmaxRef = 188394.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

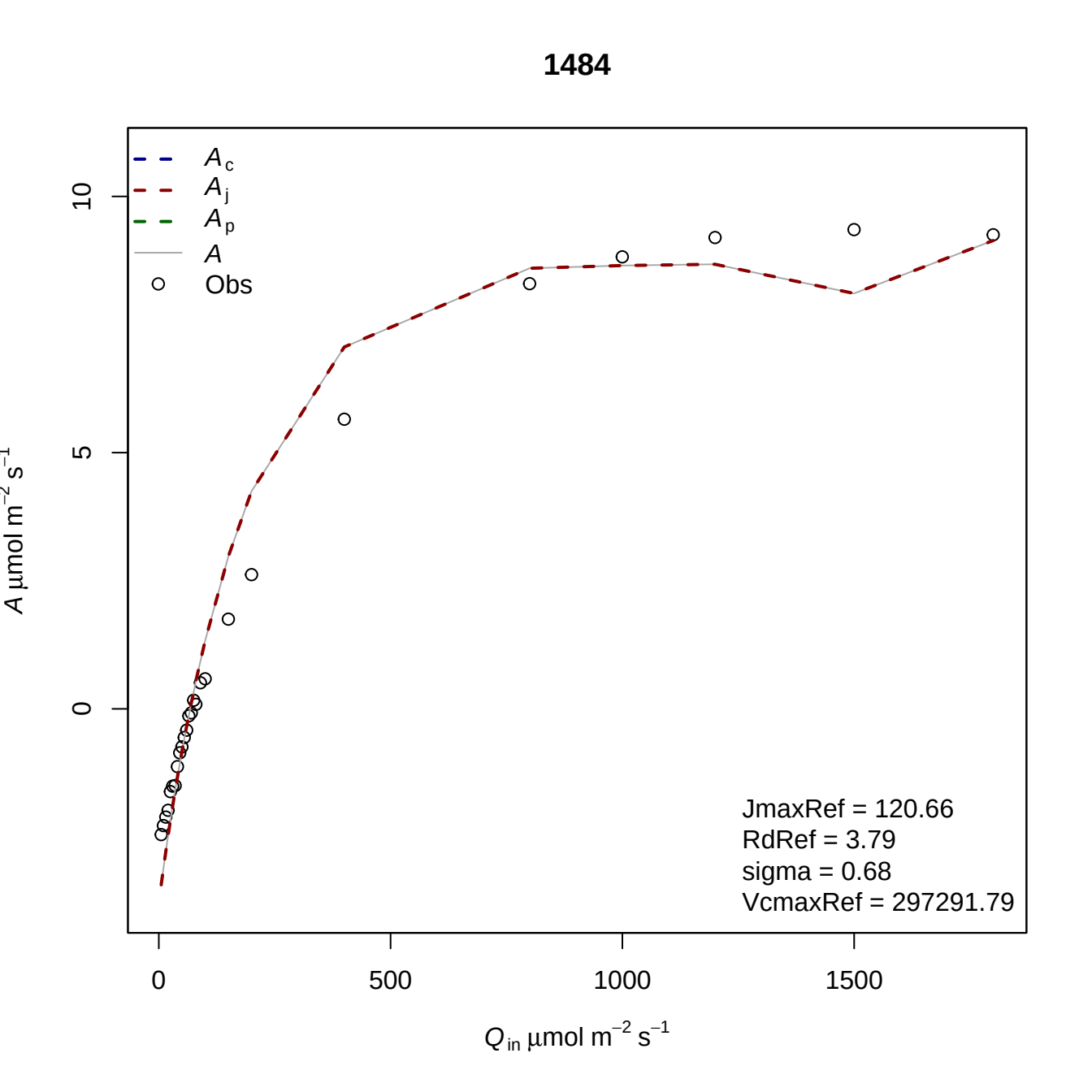

1479
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 123.96
RdRef = 3.77
sigma = 0.63
VcmaxRef = 910916.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

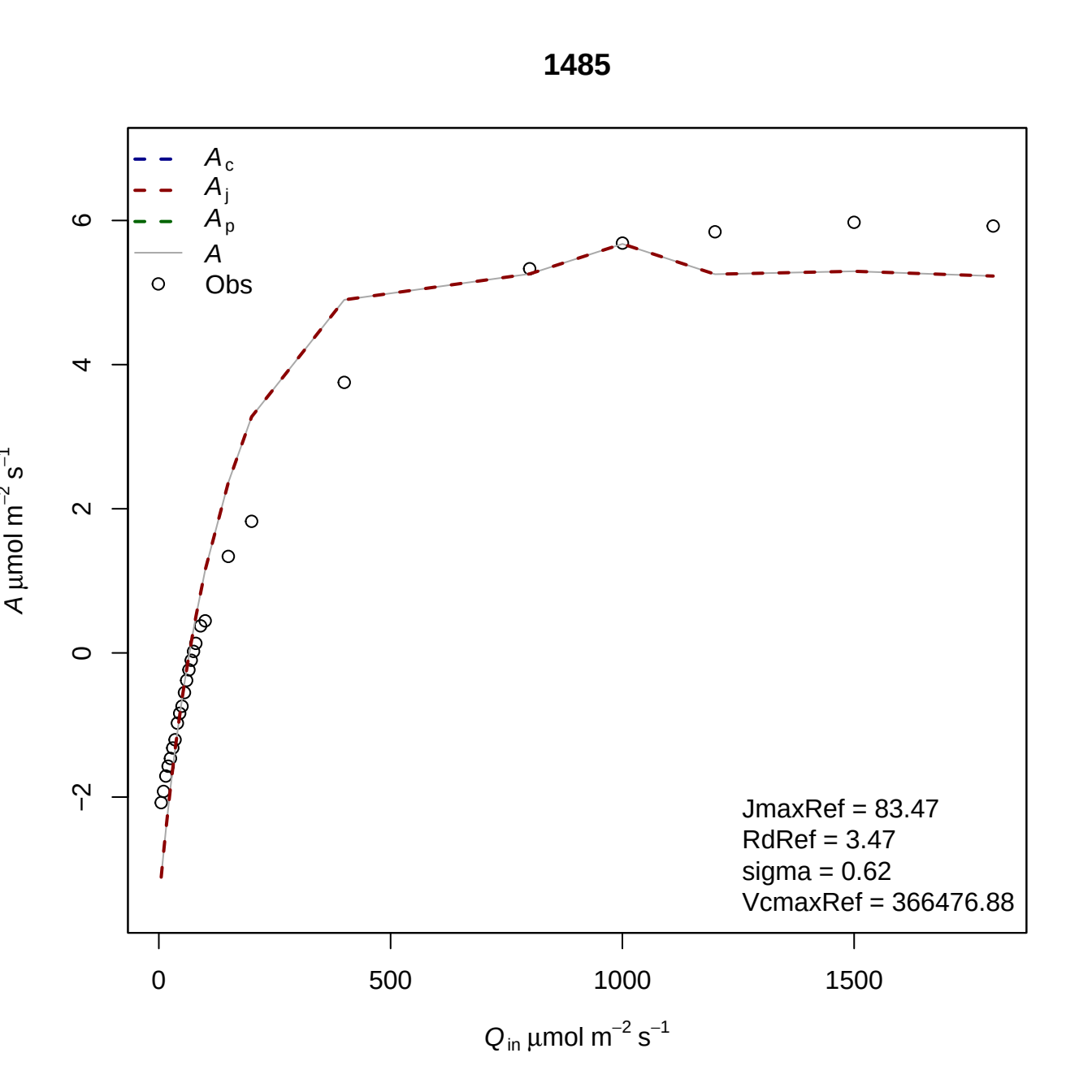

1480
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 139.84
RdRef = 3.75
sigma = 0.79
VcmaxRef = 266818.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

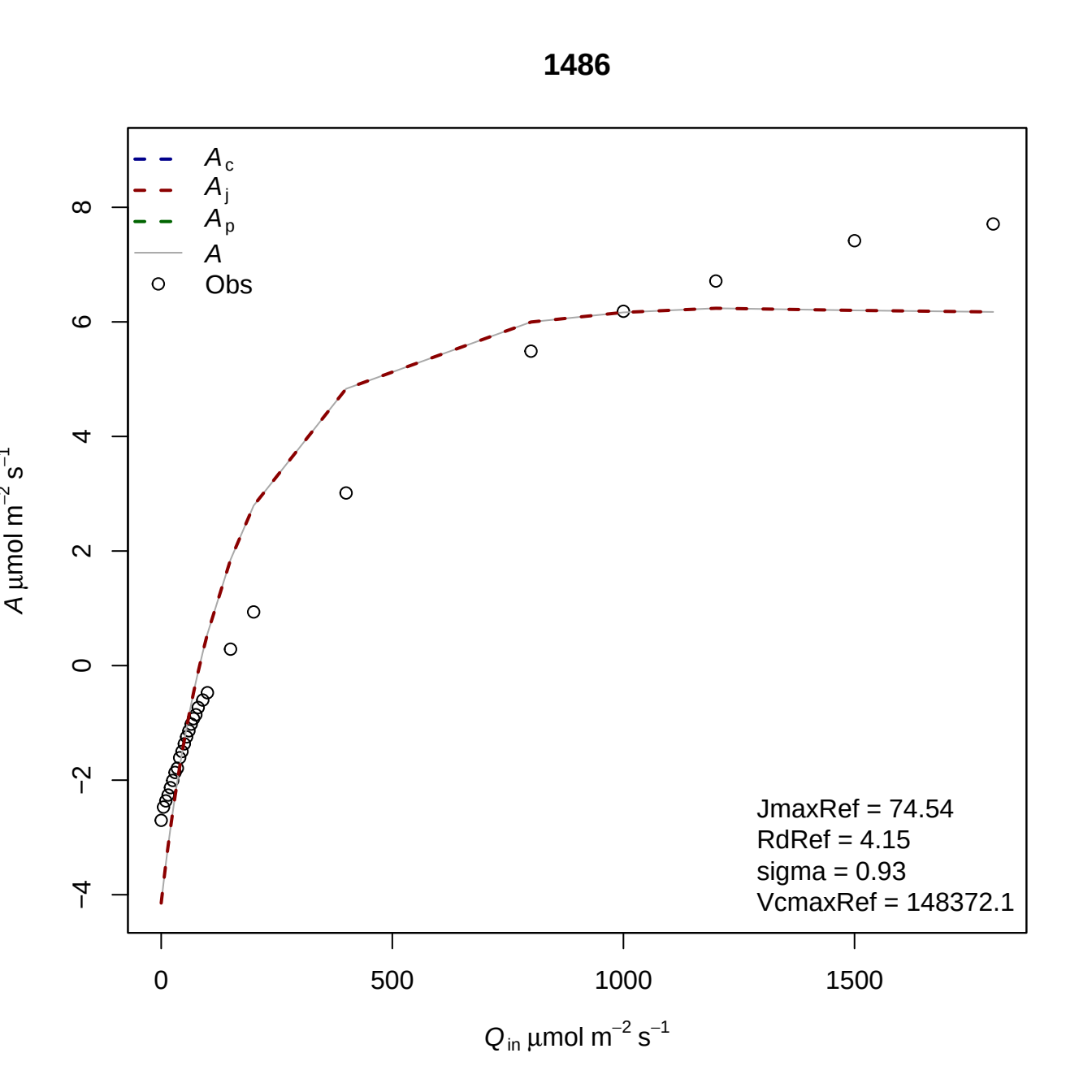

1481
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.85
RdRef = 3.87
sigma = 0.87
VcmaxRef = 369764.45
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

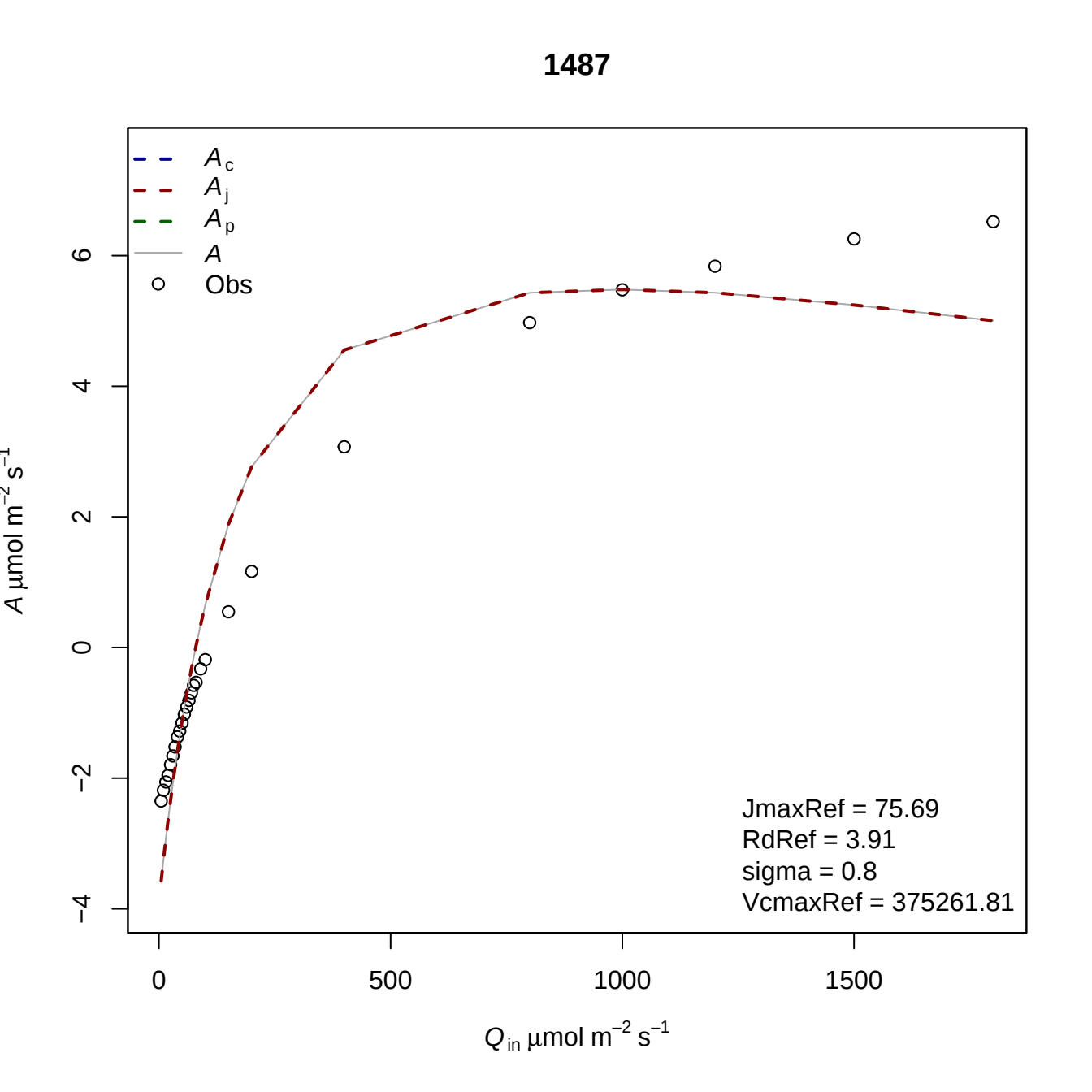

1482
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 145.46
RdRef = 3.06
sigma = 0.46
VcmaxRef = 1300836.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

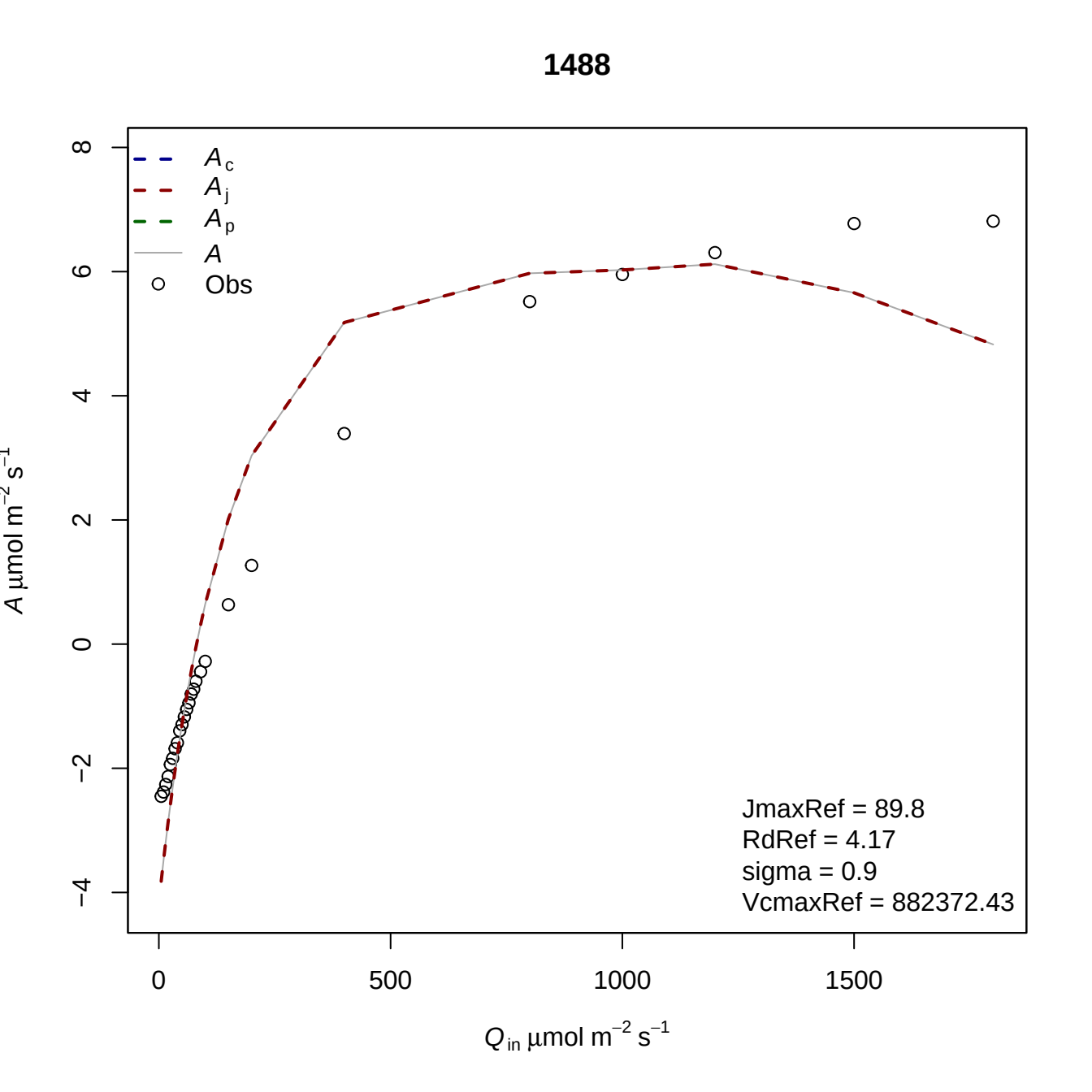

1483
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.95
RdRef = 3.37
sigma = 0.36
VcmaxRef = 4554460.92
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

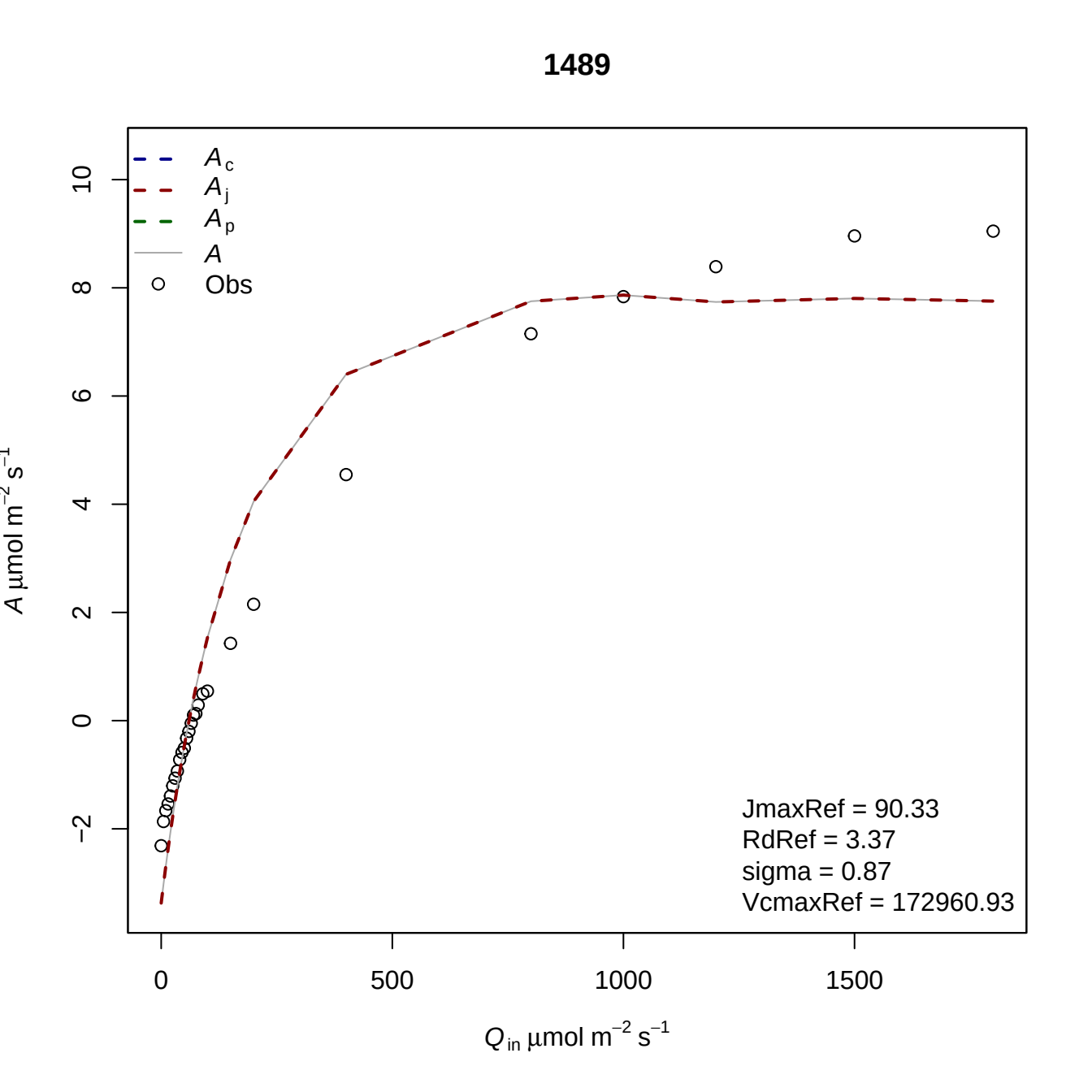

1484
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 120.66
RdRef = 3.79
sigma = 0.68
VcmaxRef = 297291.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

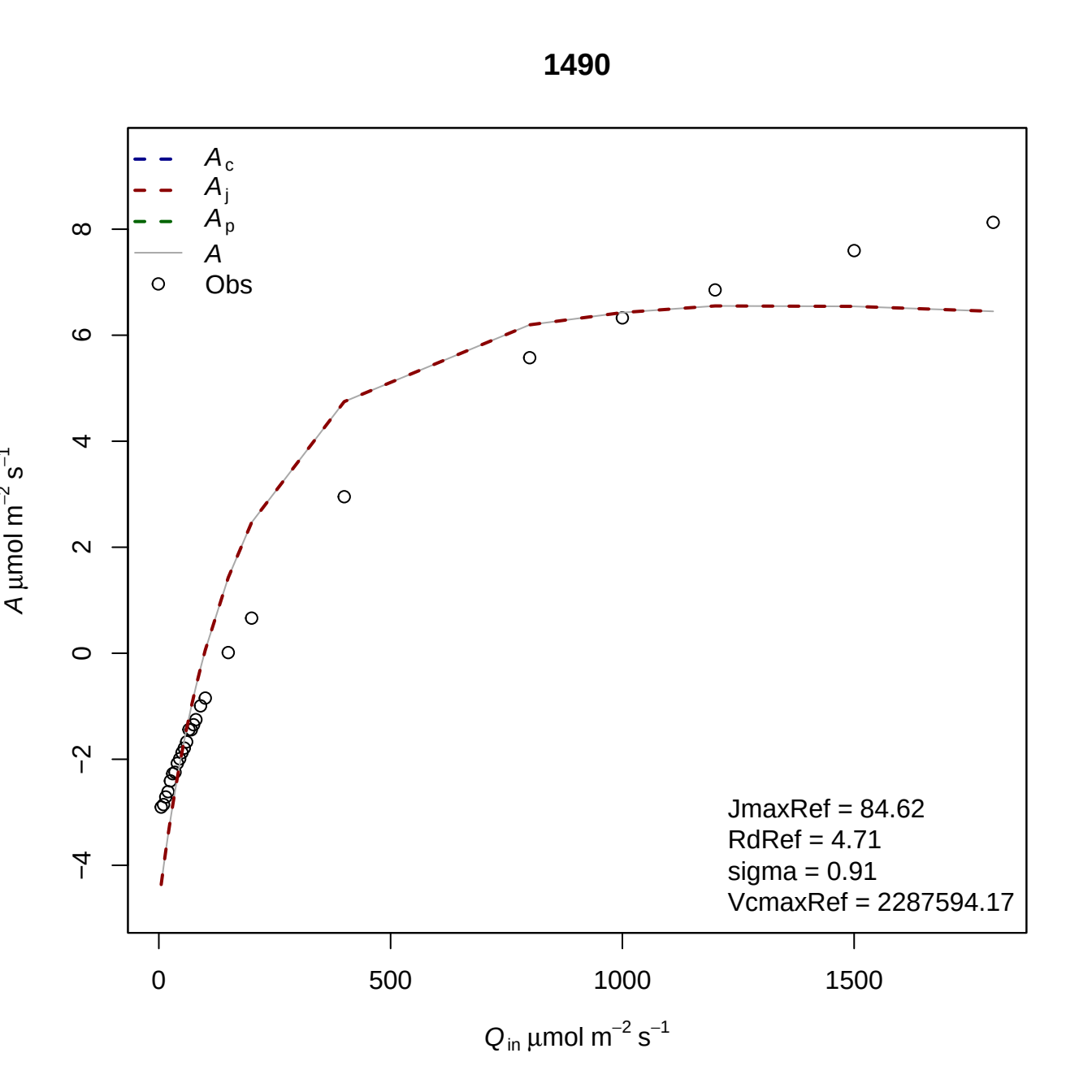

1485
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.47
RdRef = 3.47
sigma = 0.62
VcmaxRef = 366476.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
0
500
1000
1500

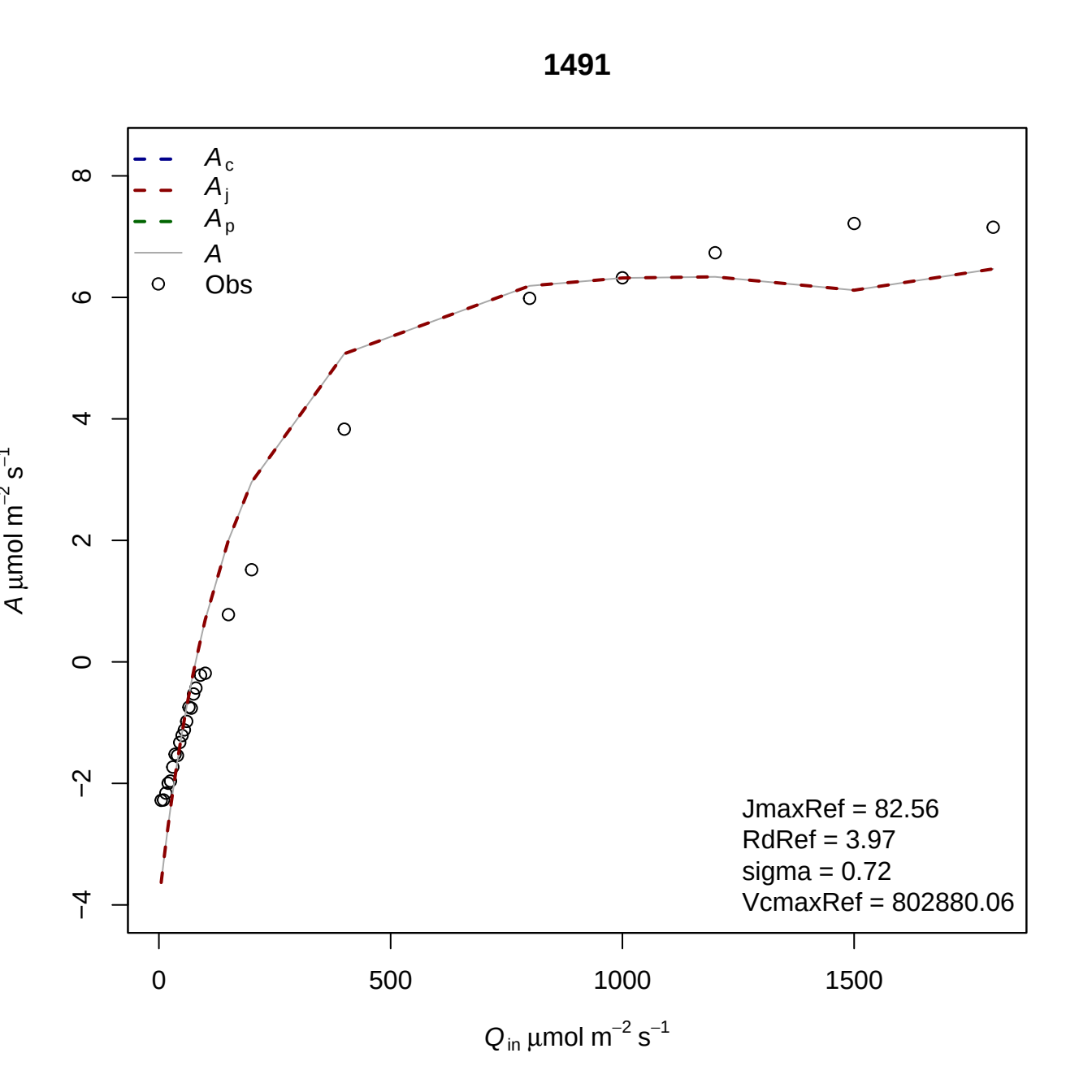

1486
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.54
RdRef = 4.15
sigma = 0.93
VcmaxRef = 148372.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

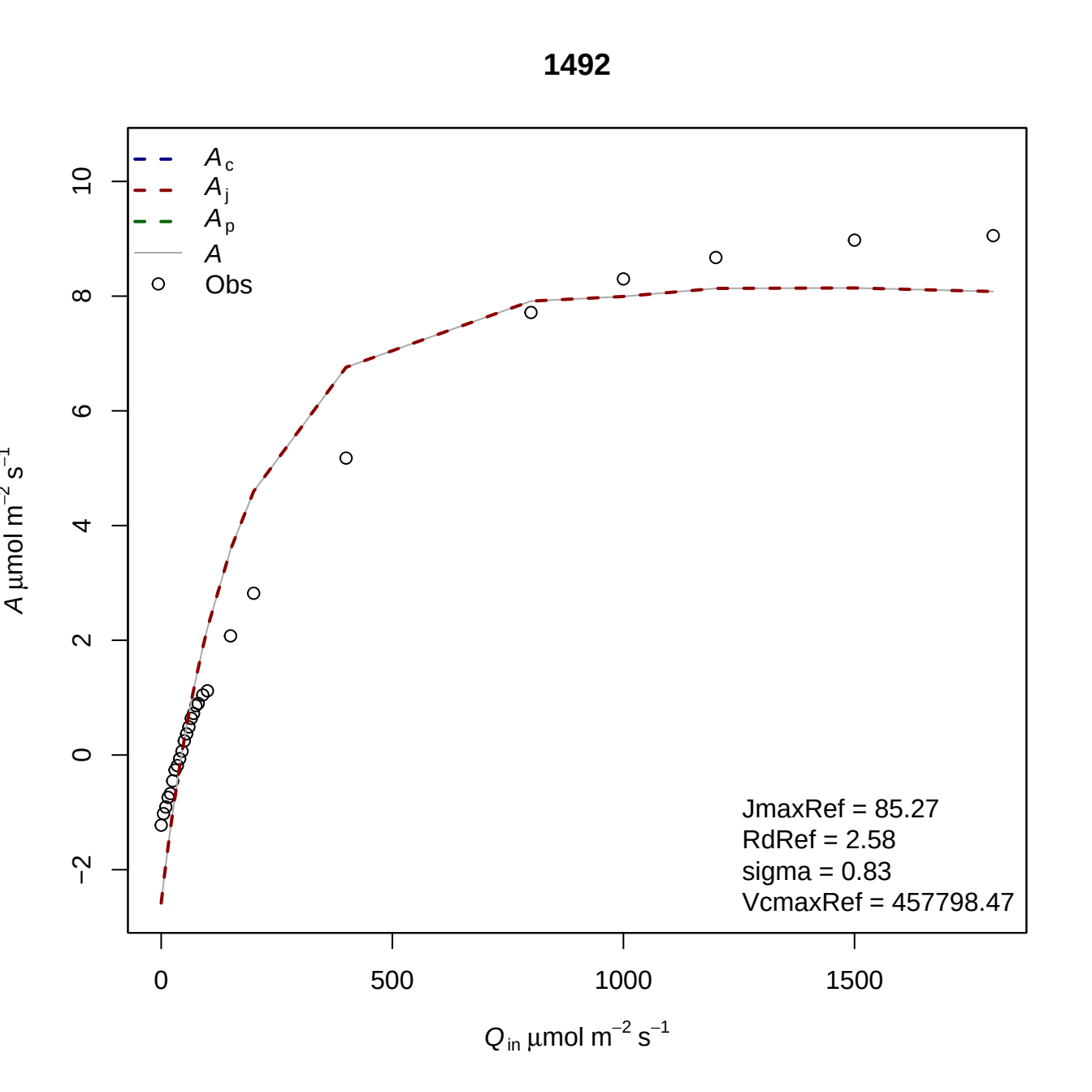

1487
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.69
RdRef = 3.91
sigma = 0.8
VcmaxRef = 375261.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500

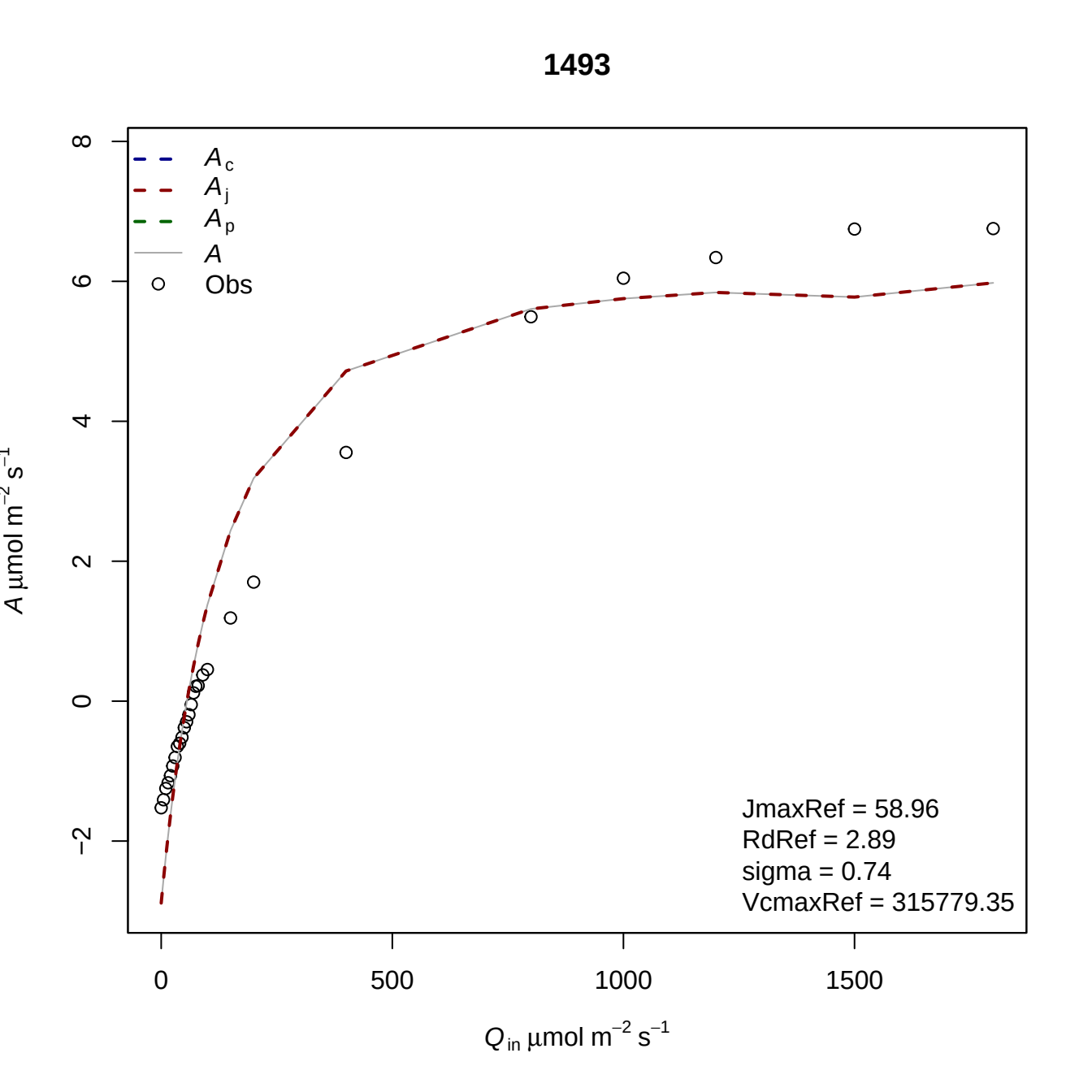

1488
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.8
RdRef = 4.17
sigma = 0.9
VcmaxRef = 882372.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

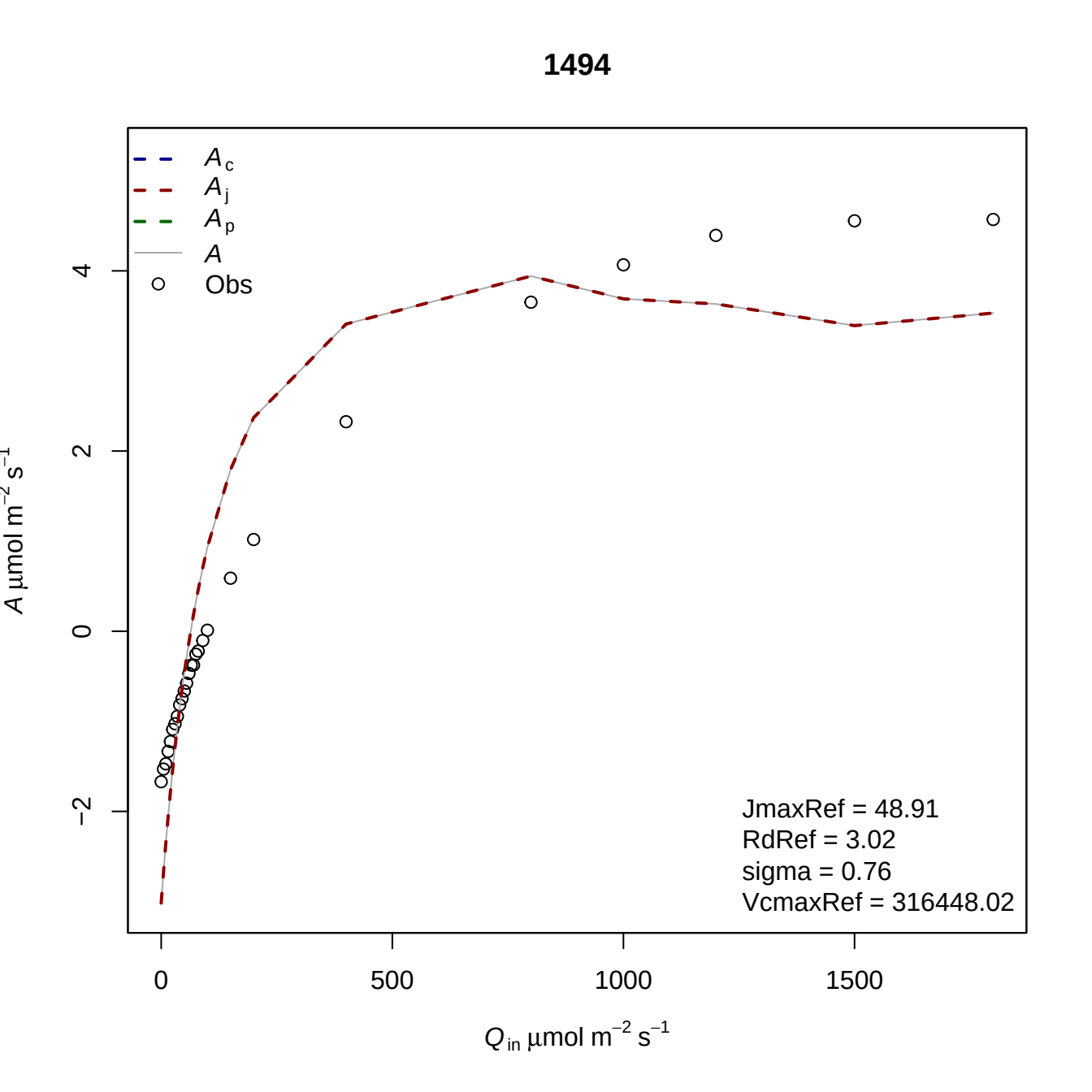

1489
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.33
RdRef = 3.37
sigma = 0.87
VcmaxRef = 172960.93
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

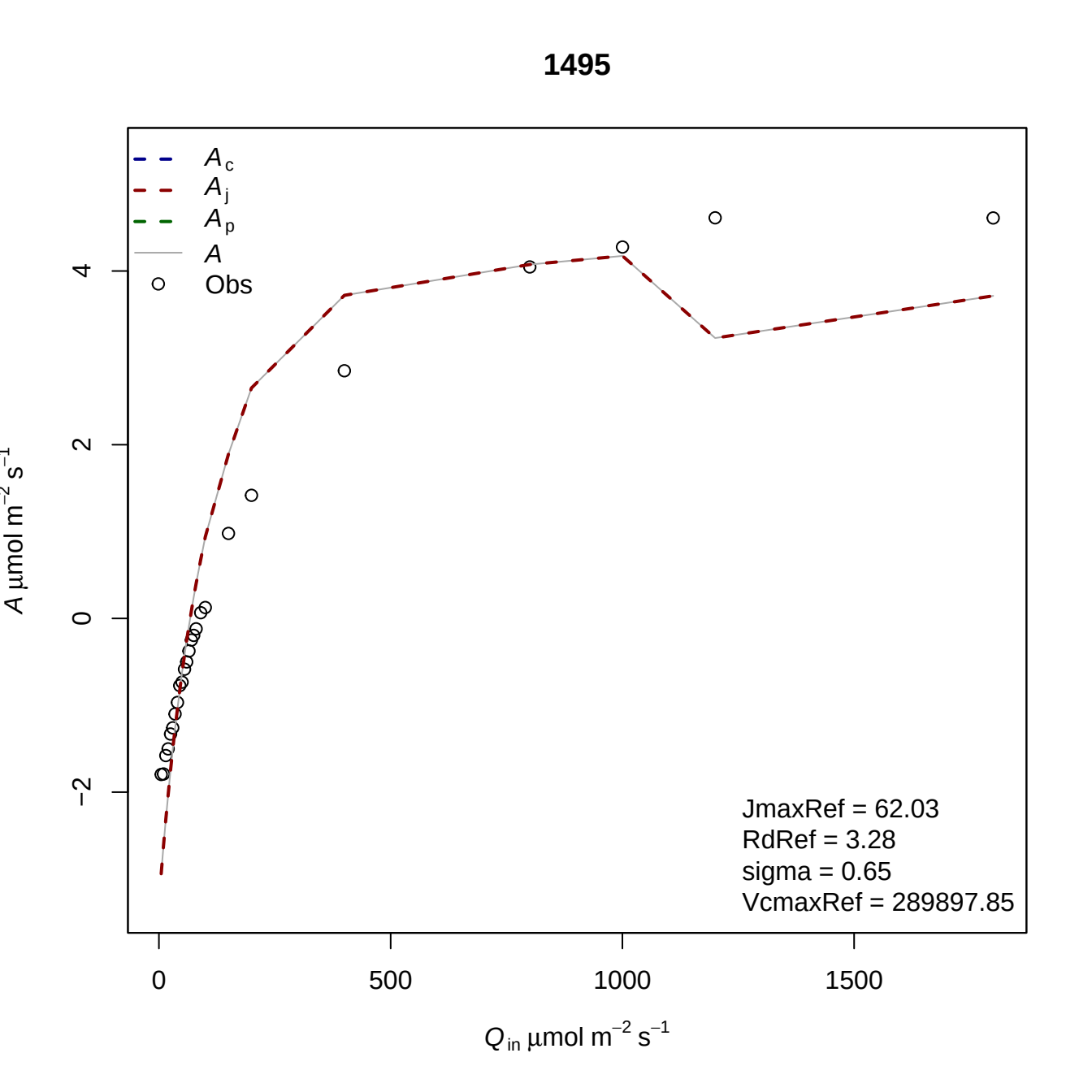

1490
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.62
RdRef = 4.71
sigma = 0.91
VcmaxRef = 2287594.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

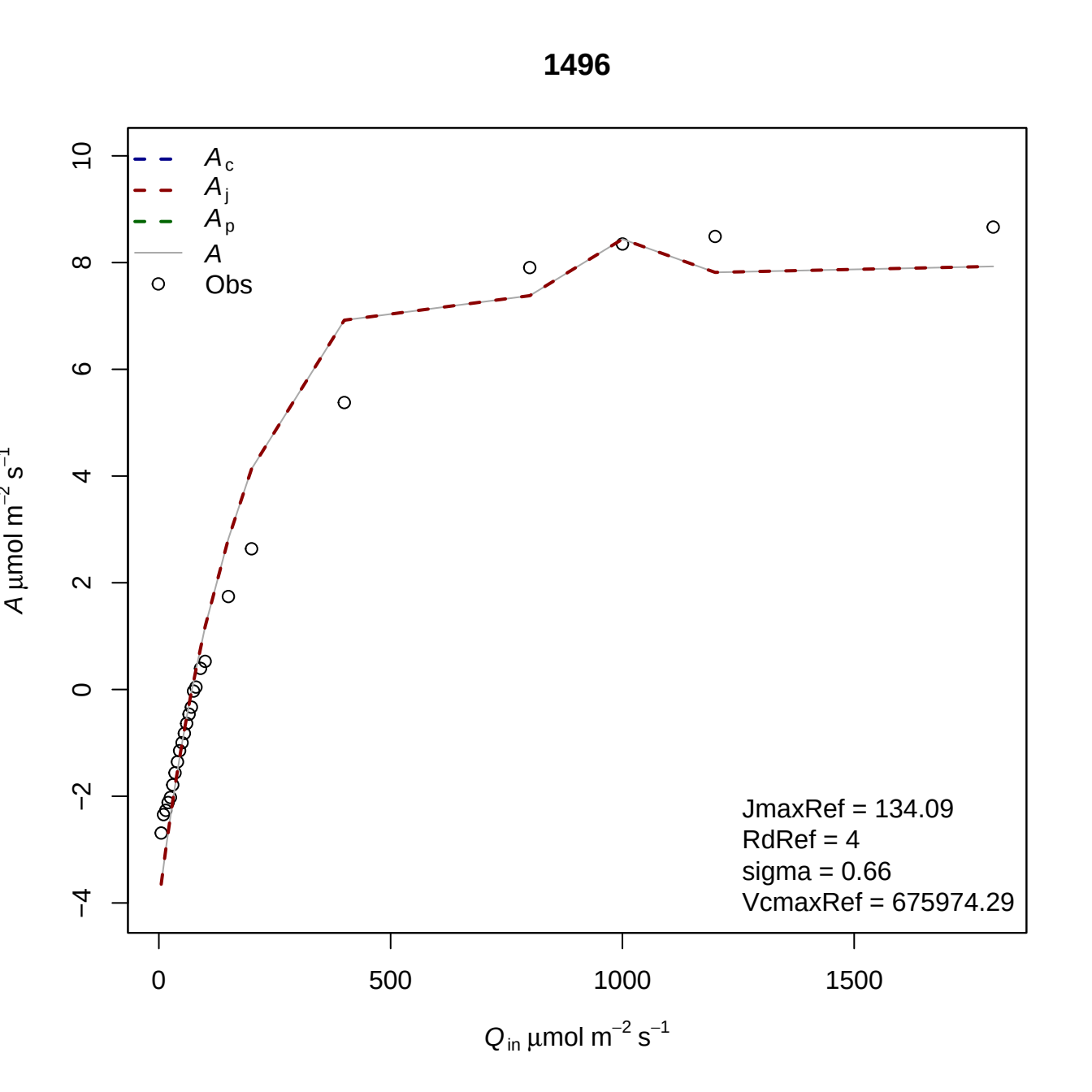

1491
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.56
RdRef = 3.97
sigma = 0.72
VcmaxRef = 802880.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

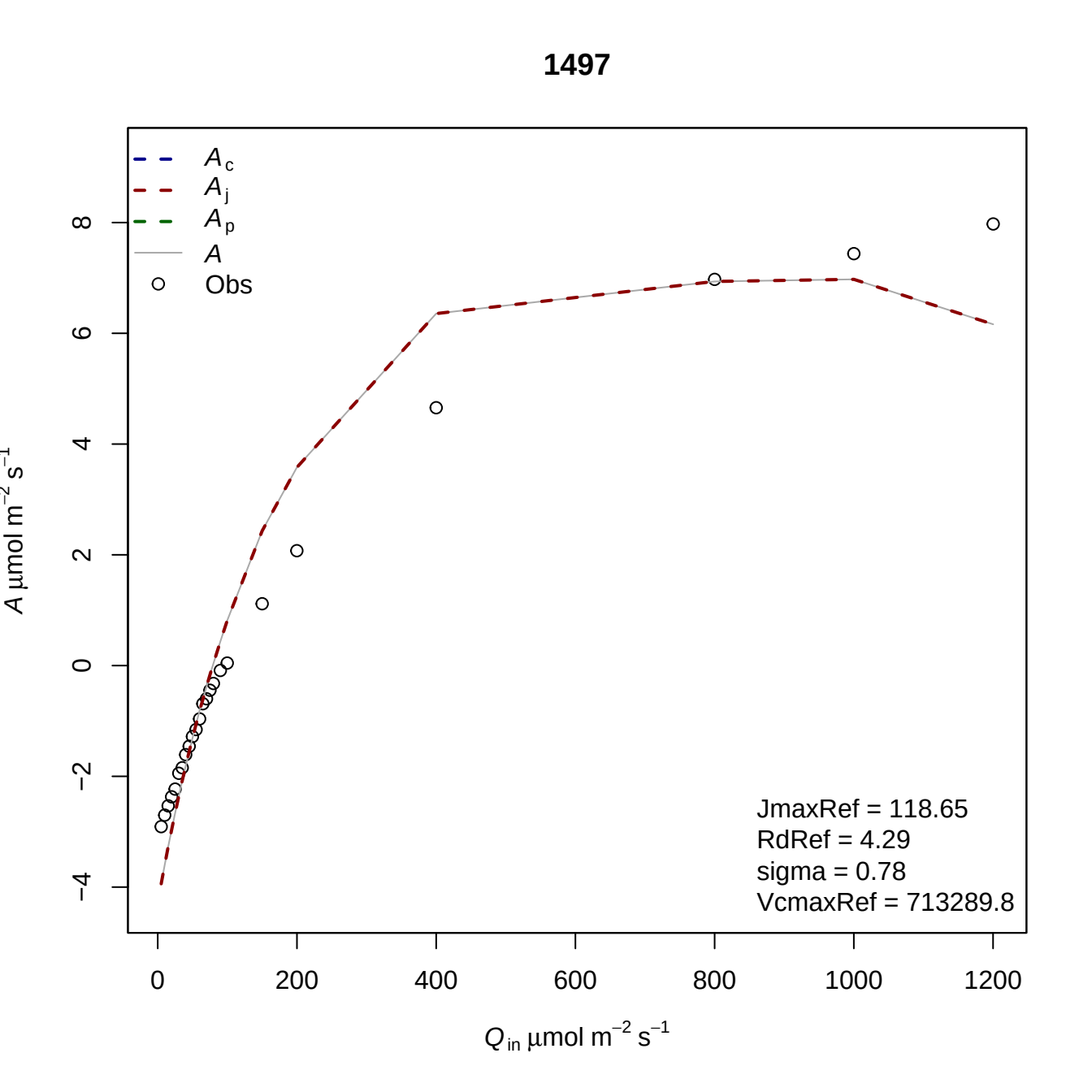

1492
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.27
RdRef = 2.58
sigma = 0.83
VcmaxRef = 457798.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
0
500
1000
1500

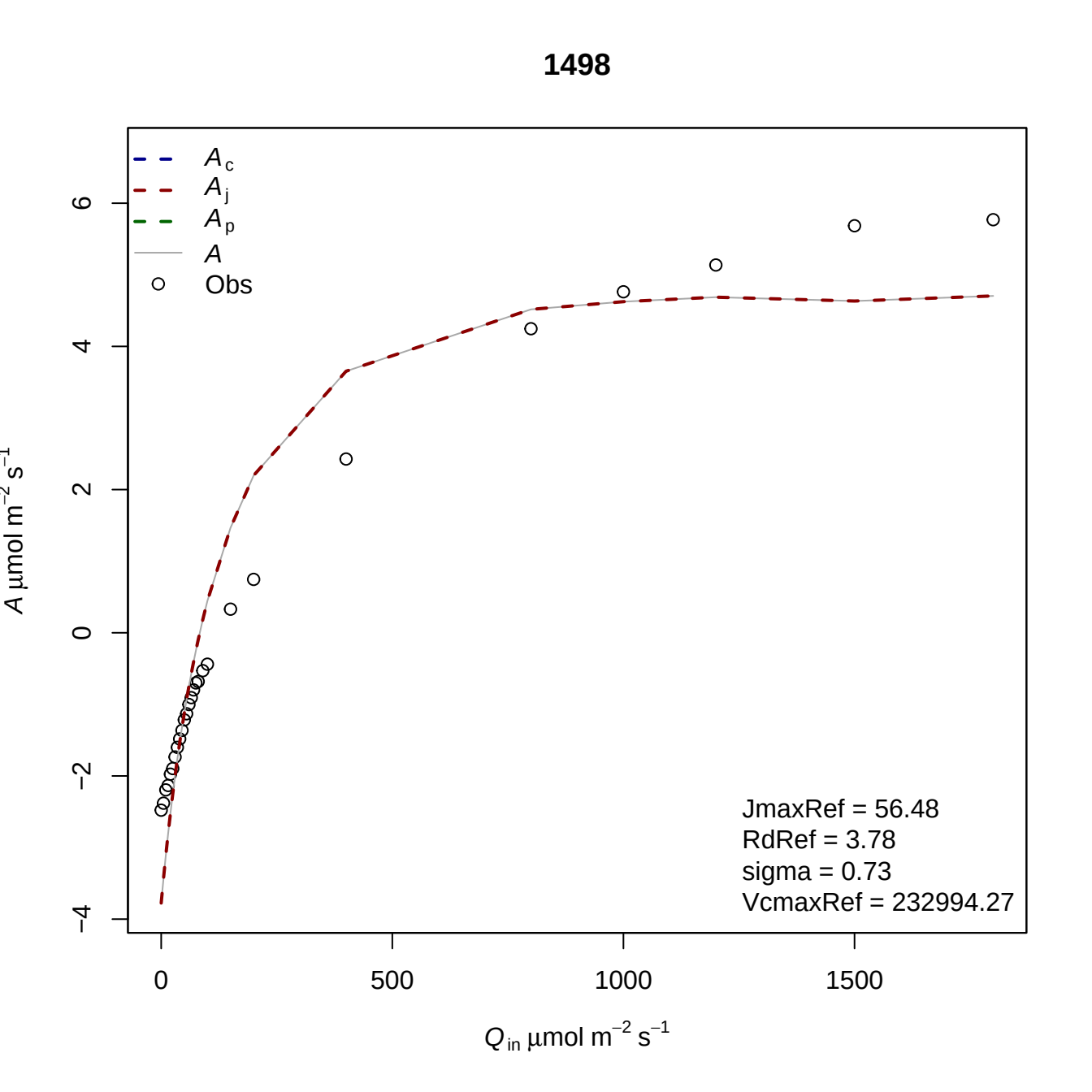

1493
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.96
RdRef = 2.89
sigma = 0.74
VcmaxRef = 315779.35
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
0
500
1000
1500

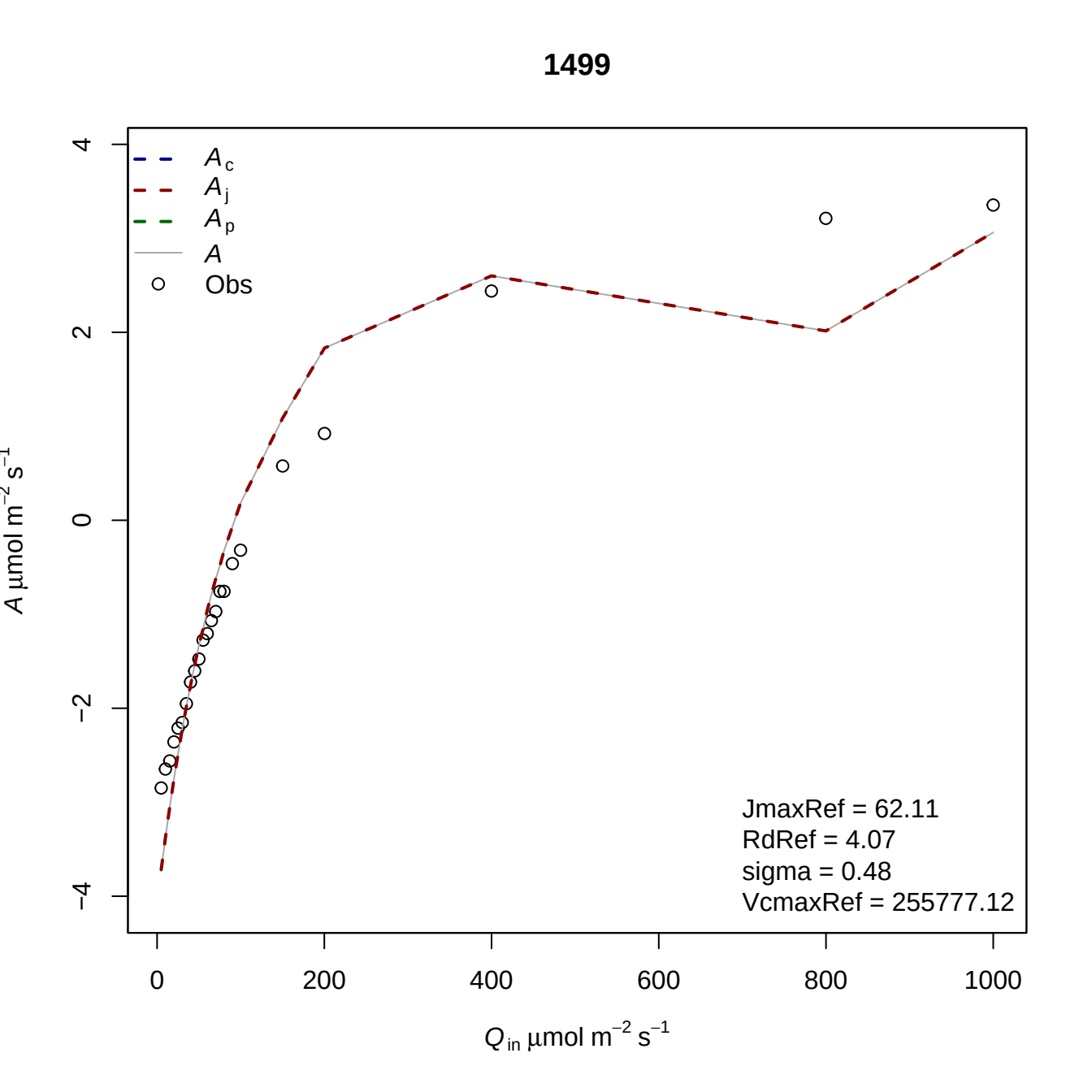

1494
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 48.91
RdRef = 3.02
sigma = 0.76
VcmaxRef = 316448.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500

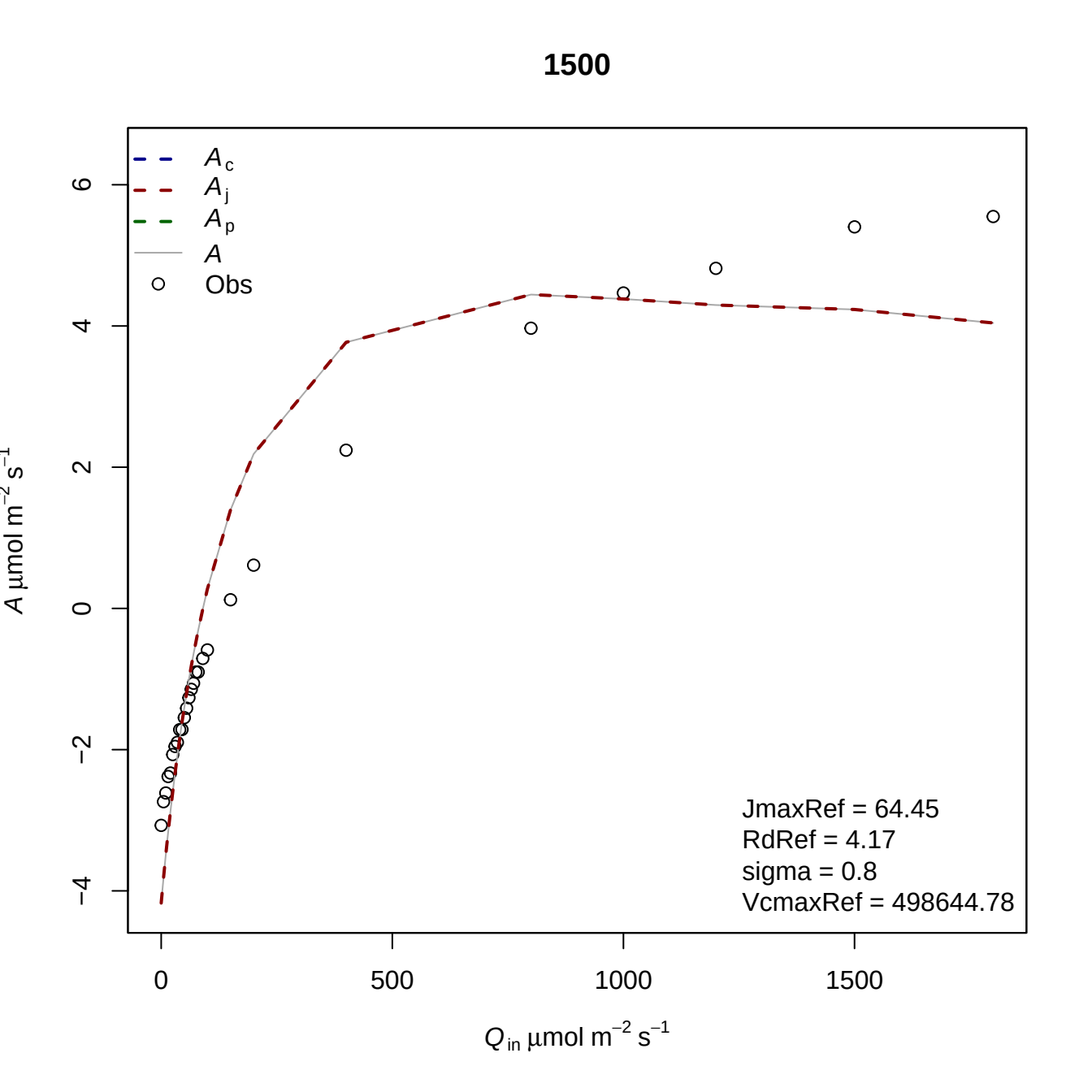

1495
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.03
RdRef = 3.28
sigma = 0.65
VcmaxRef = 289897.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
0
500
1000
1500

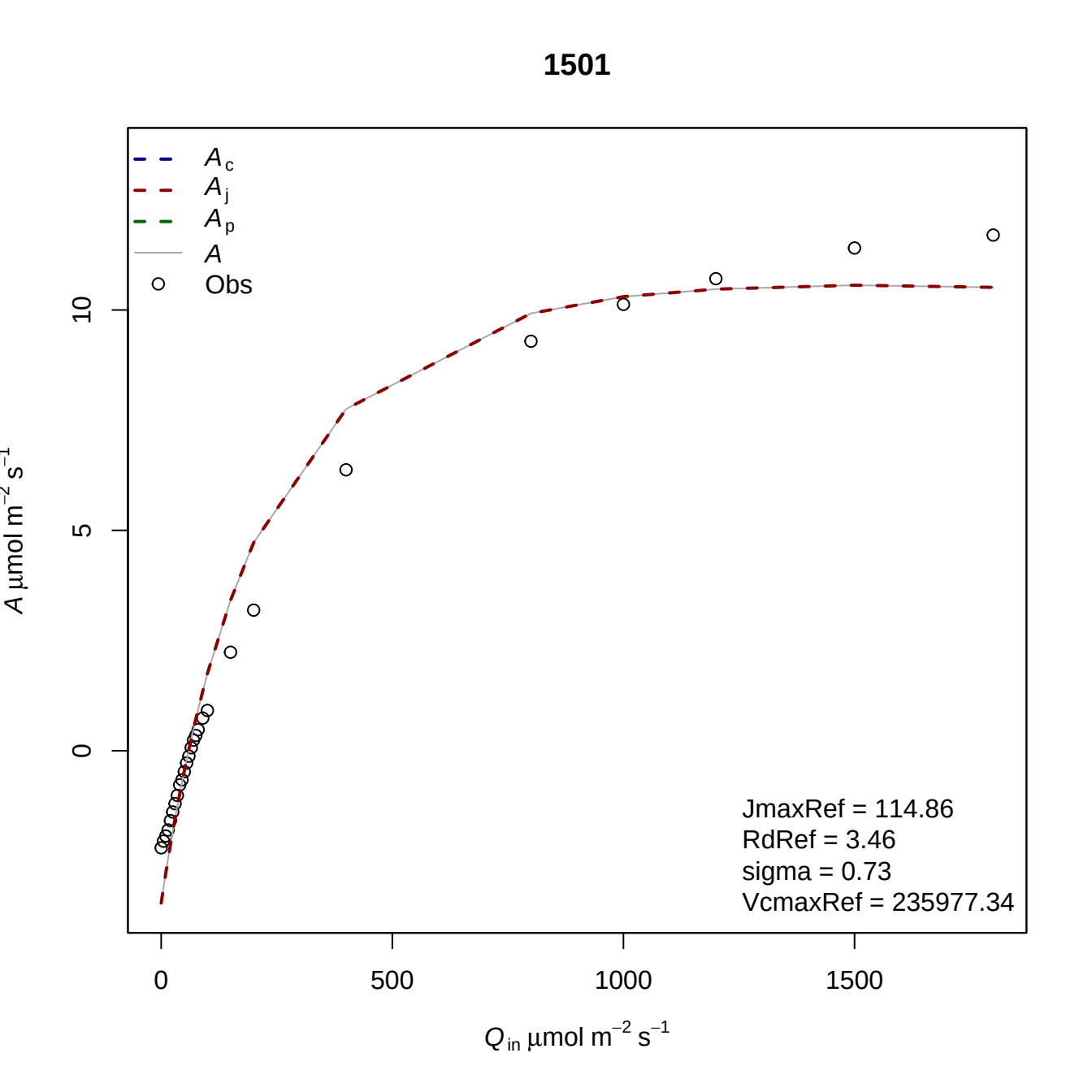

1496
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 134.09
RdRef = 4
sigma = 0.66
VcmaxRef = 675974.29
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
10
8
6
4
2
0
−2
−4
0
500
1000
1500

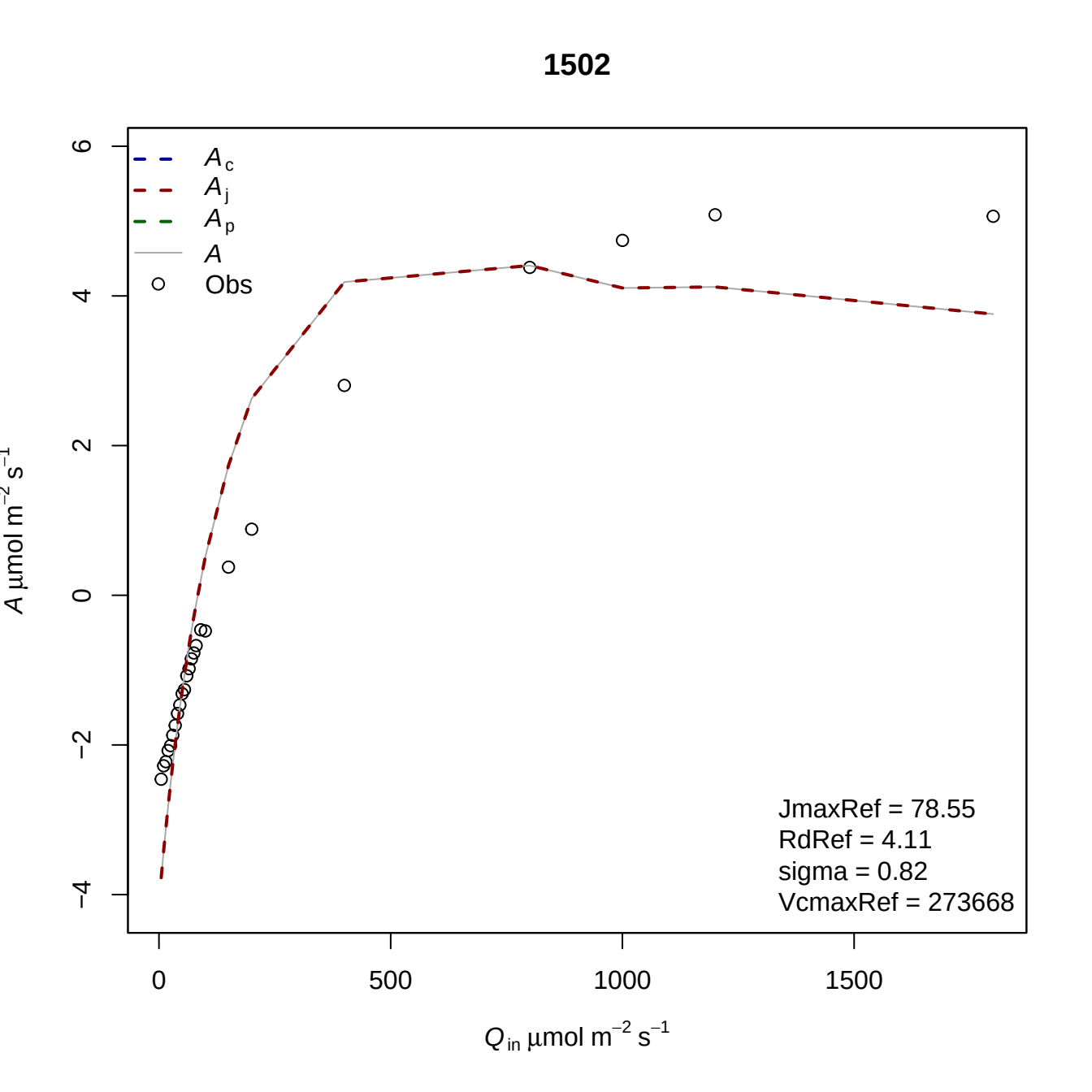

1497
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 118.65
RdRef = 4.29
sigma = 0.78
VcmaxRef = 713289.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

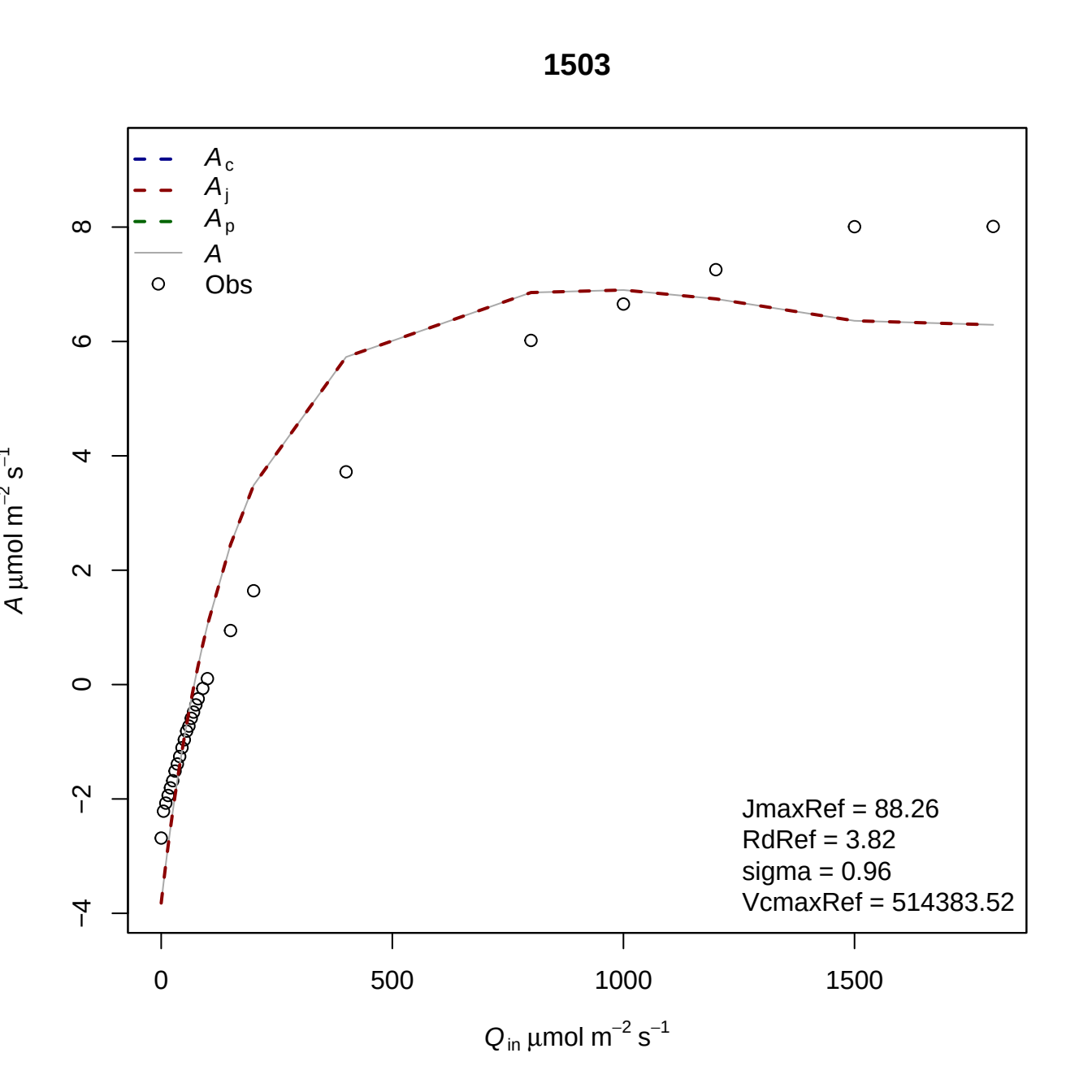

1498
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.48
RdRef = 3.78
sigma = 0.73
VcmaxRef = 232994.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500

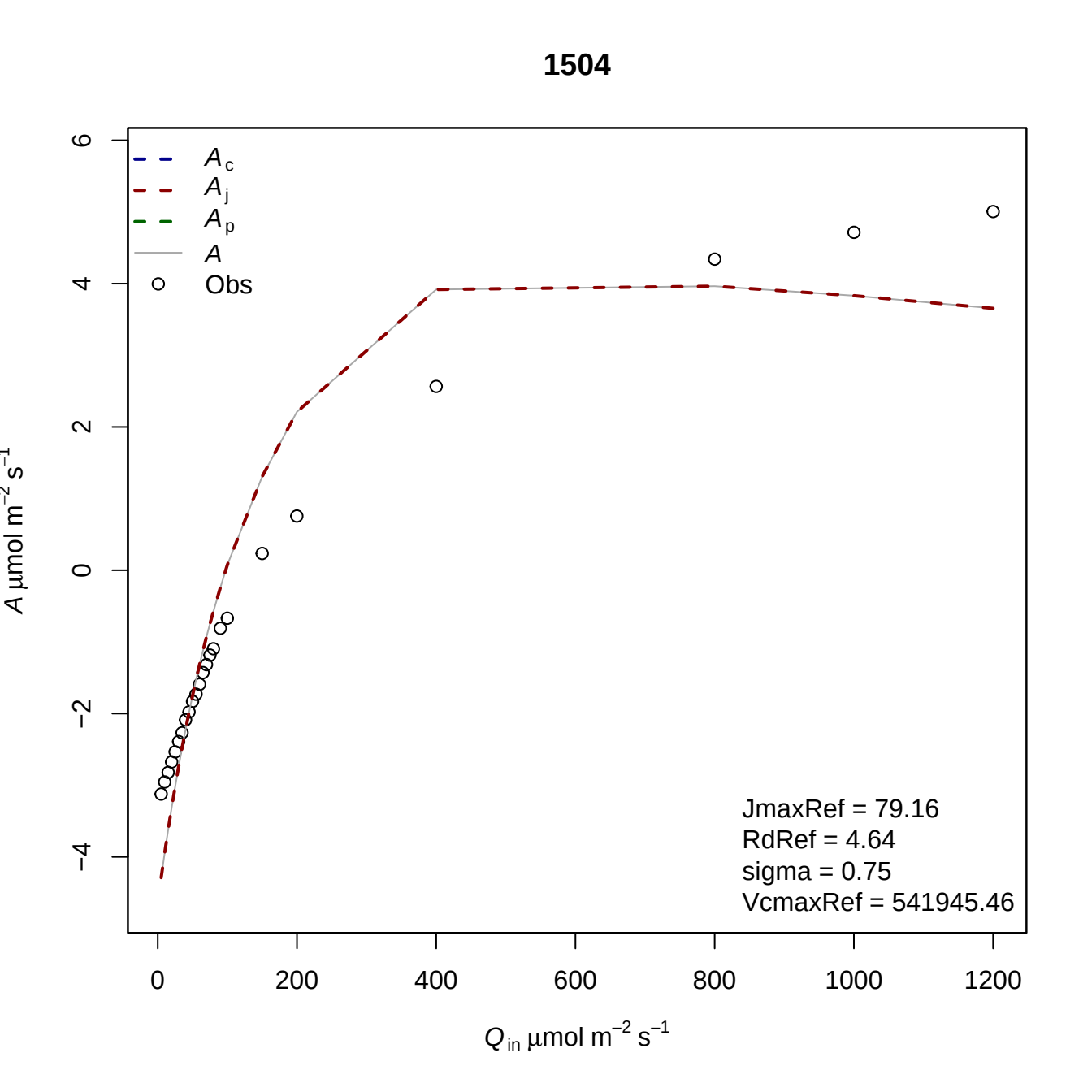

1499
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.11
RdRef = 4.07
sigma = 0.48
VcmaxRef = 255777.12
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
2
0
−2
−4
0
200
400
600
800
1000

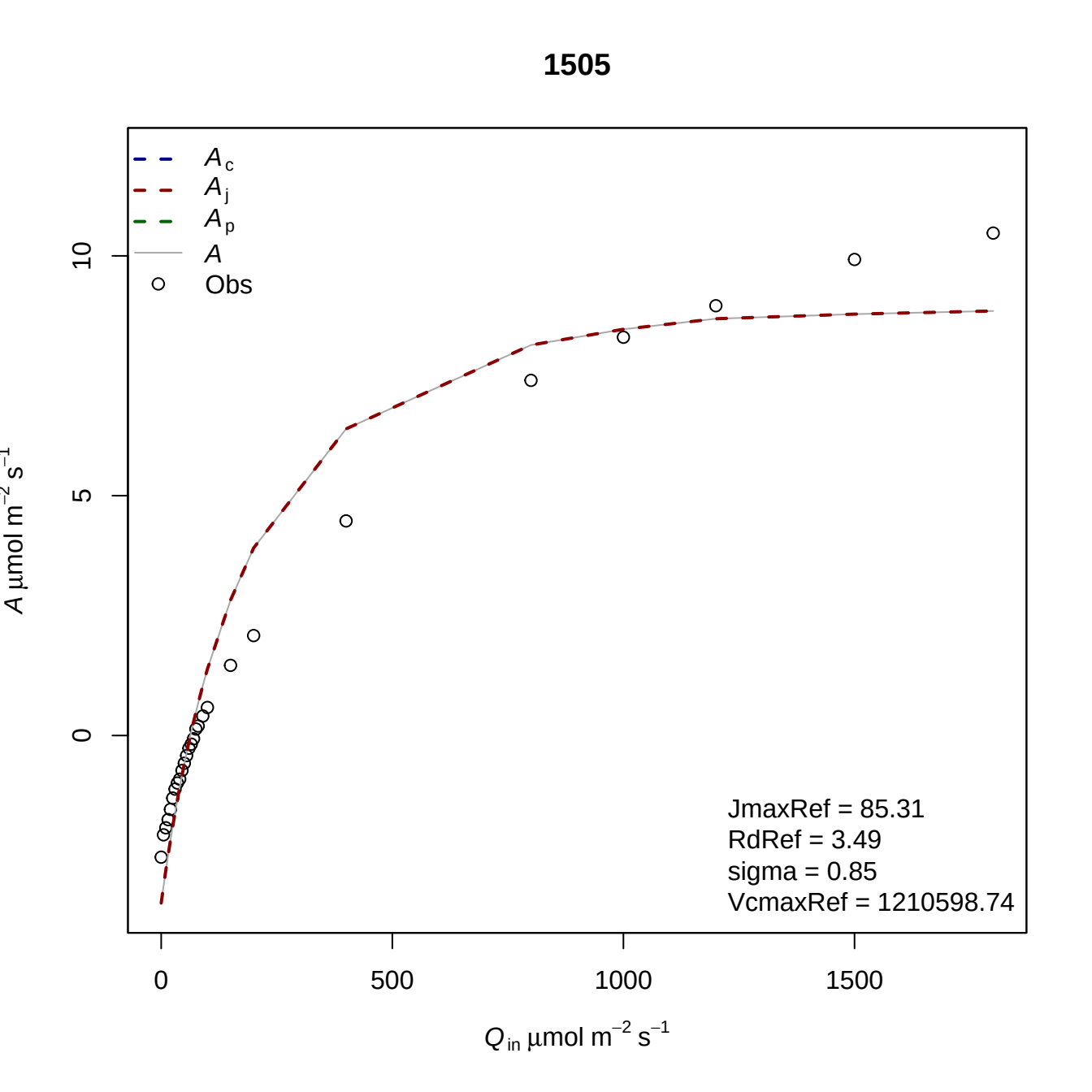

1500
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 64.45
RdRef = 4.17
sigma = 0.8
VcmaxRef = 498644.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500

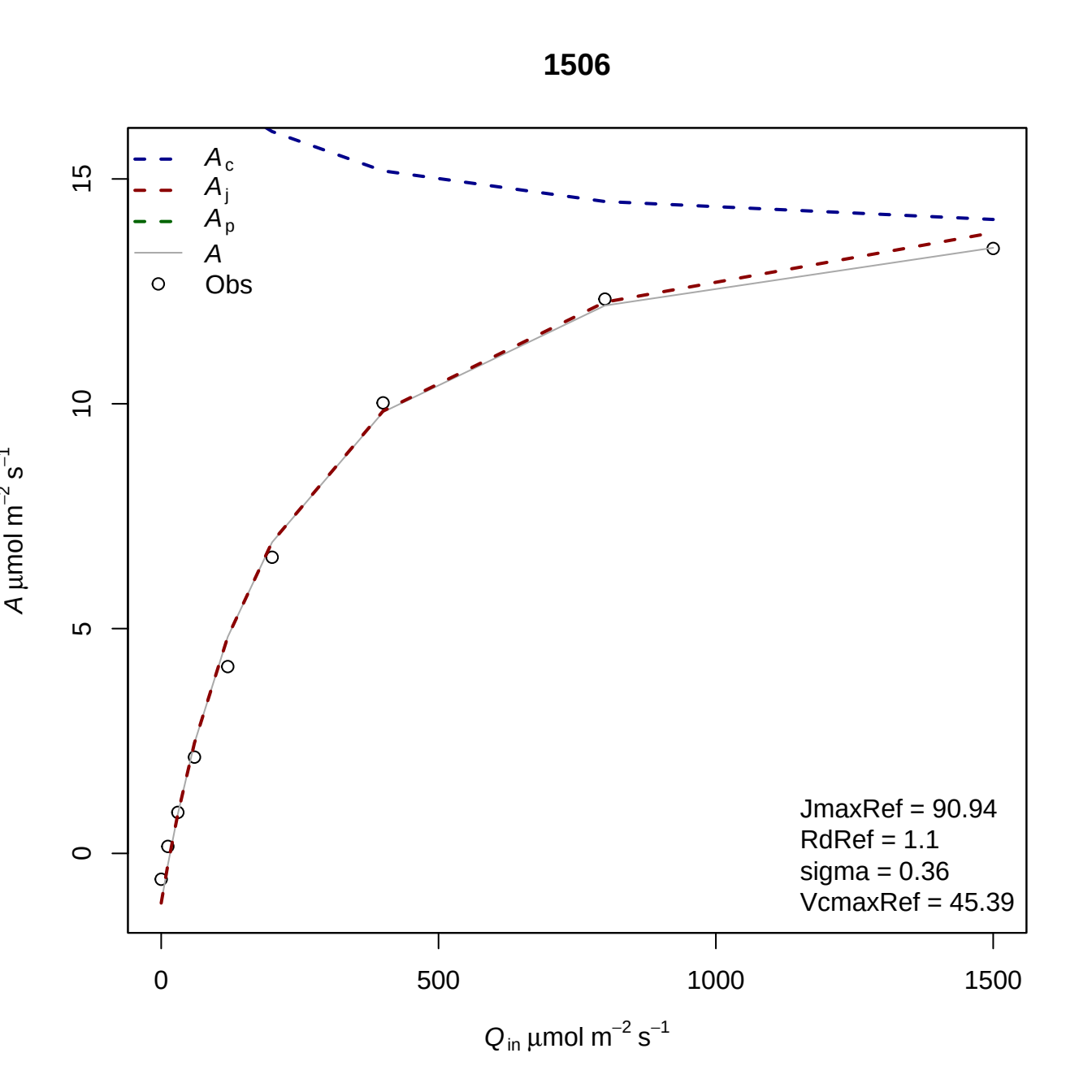

1501
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 114.86
RdRef = 3.46
sigma = 0.73
VcmaxRef = 235977.34
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

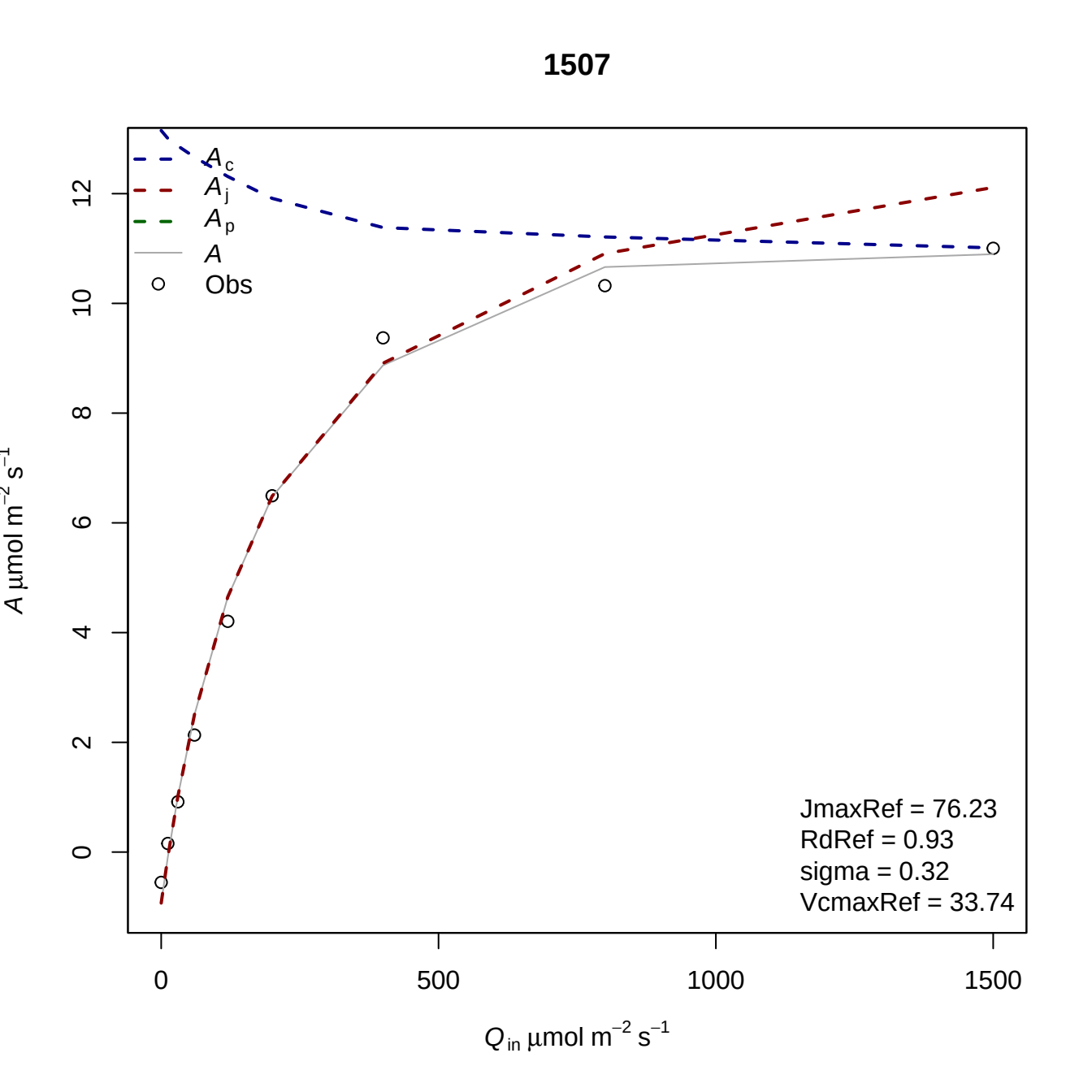

1502
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.55
RdRef = 4.11
sigma = 0.82
VcmaxRef = 273668
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
500
1000
1500

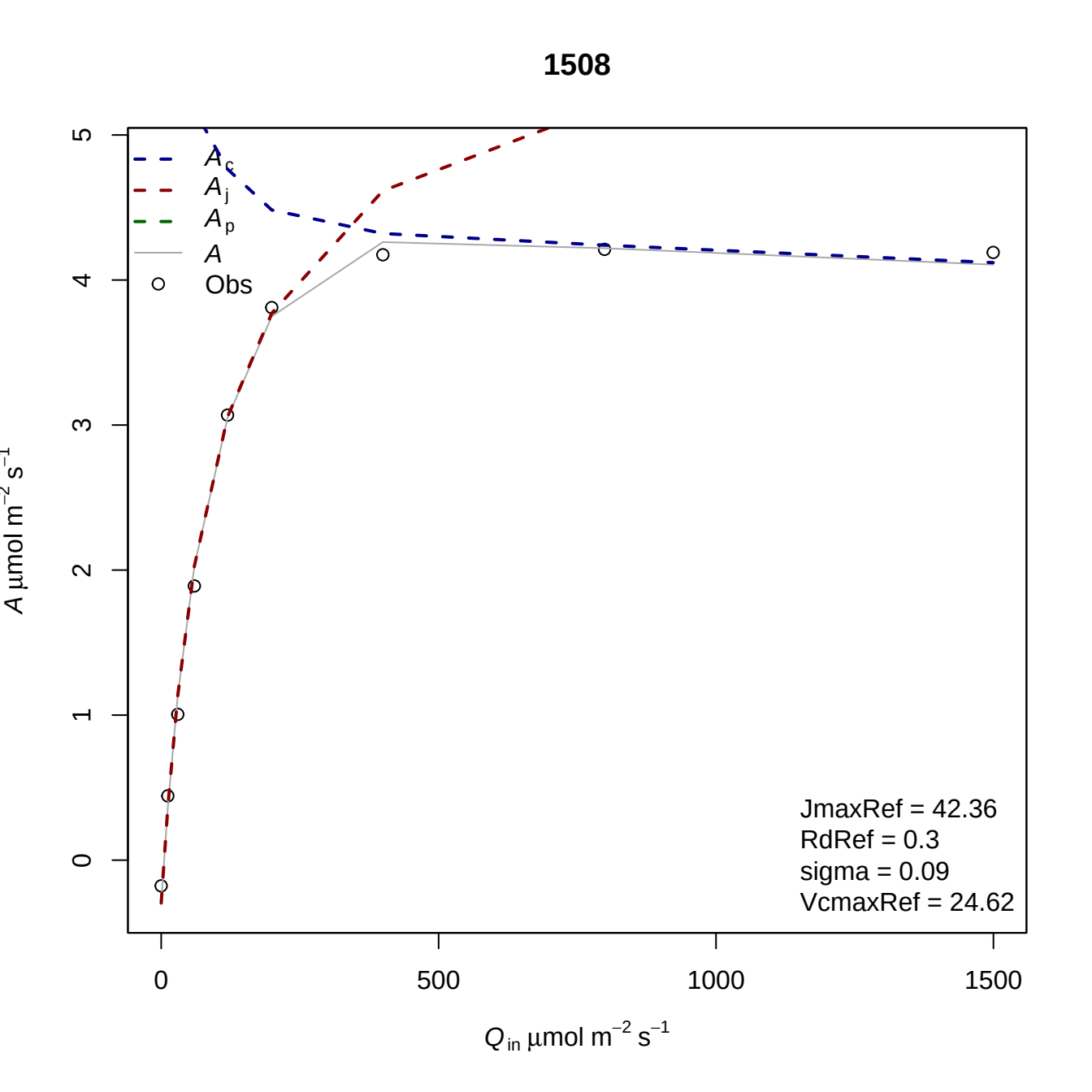

1503
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.26
RdRef = 3.82
sigma = 0.96
VcmaxRef = 514383.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
8
6
4
2
0
−2
−4
0
500
1000
1500

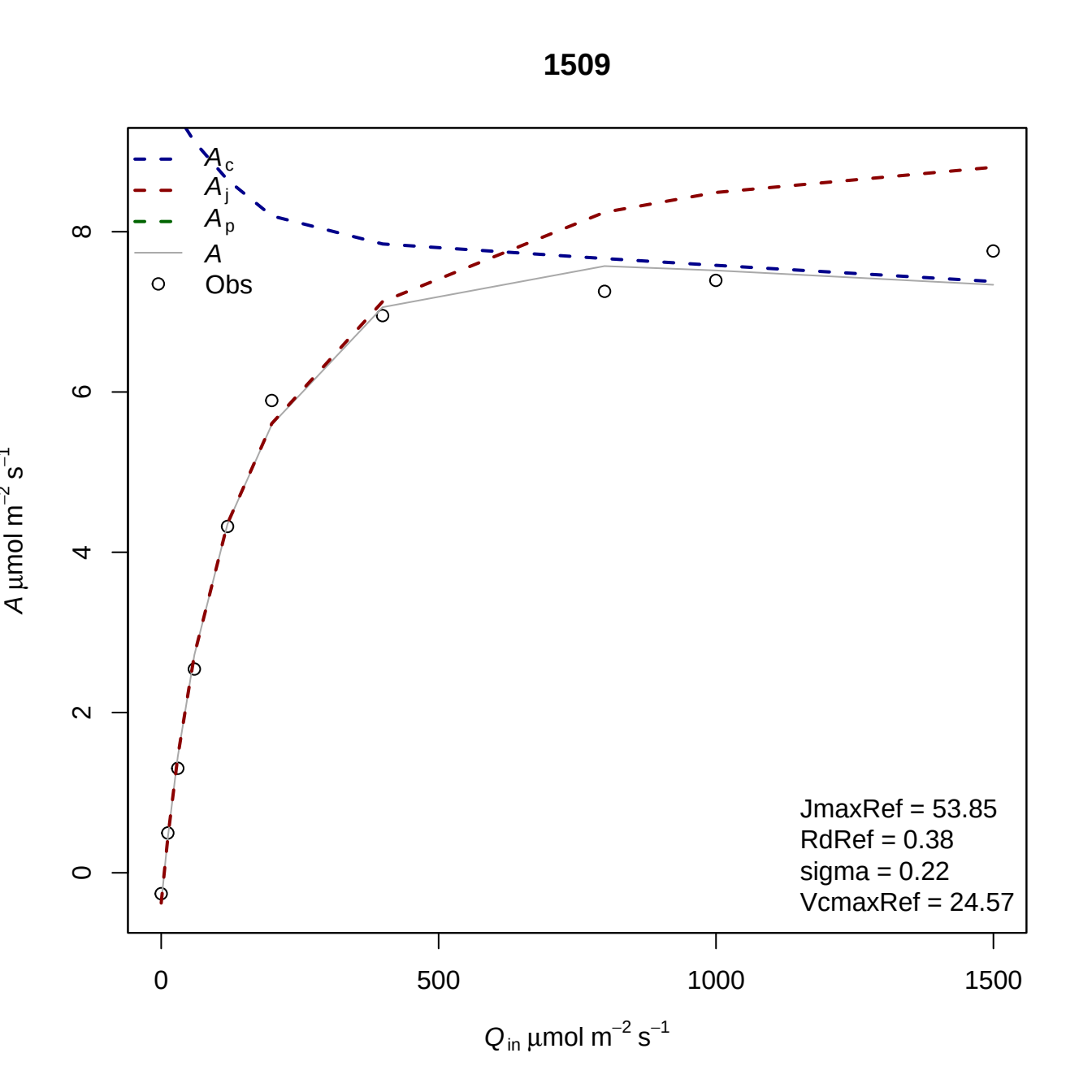

1504
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 79.16
RdRef = 4.64
sigma = 0.75
VcmaxRef = 541945.46
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

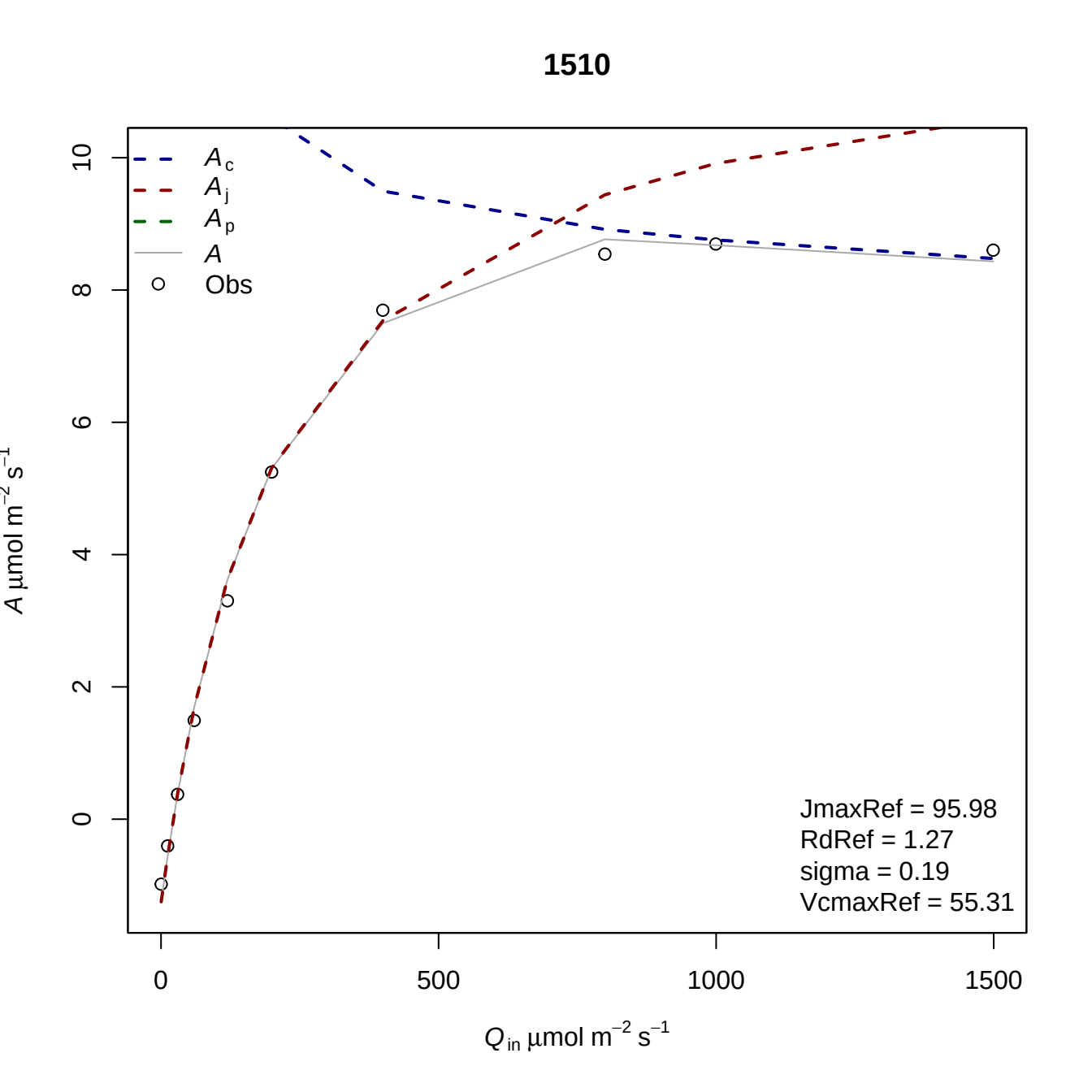

1505
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.31
RdRef = 3.49
sigma = 0.85
VcmaxRef = 1210598.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

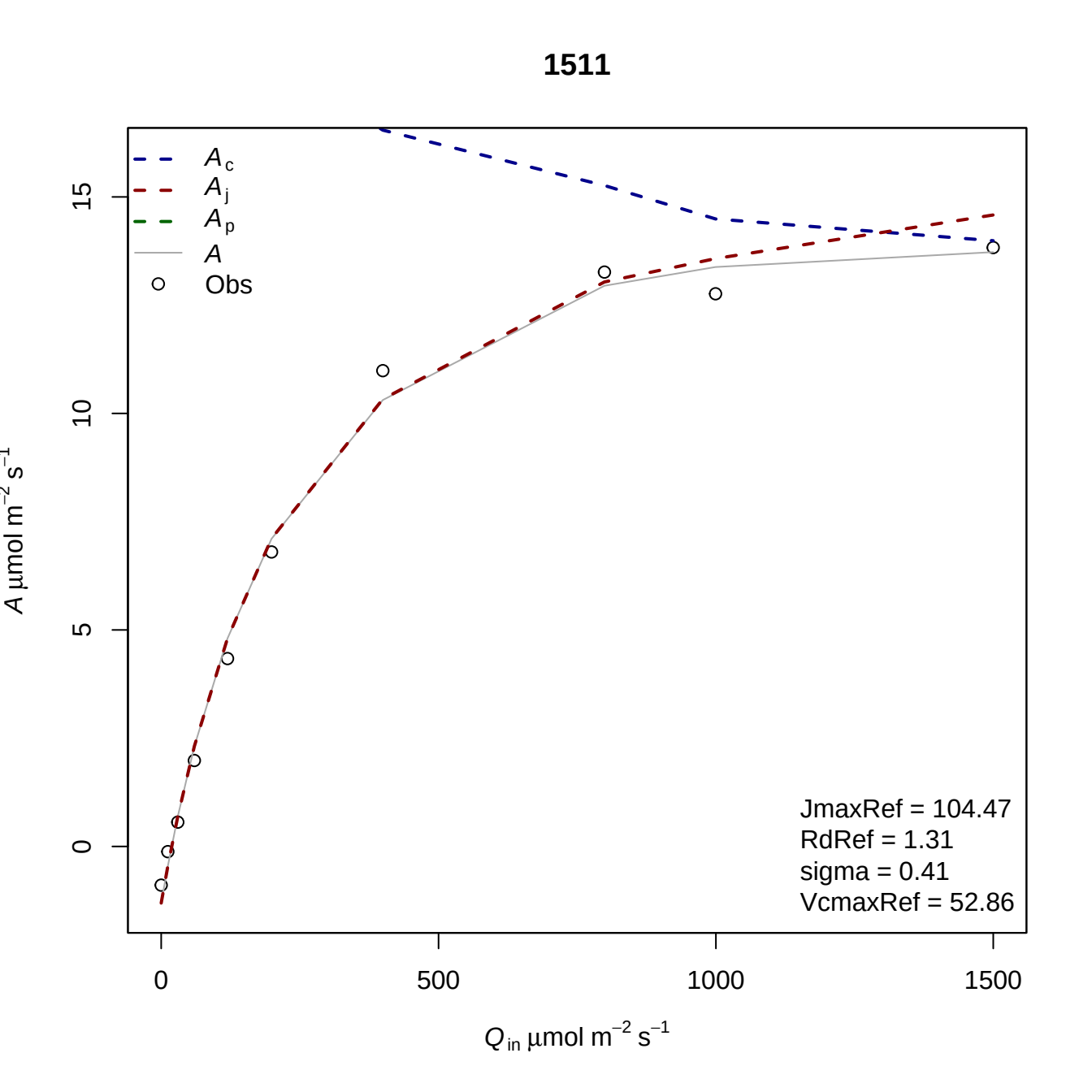

1506
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.94
RdRef = 1.1
sigma = 0.36
VcmaxRef = 45.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

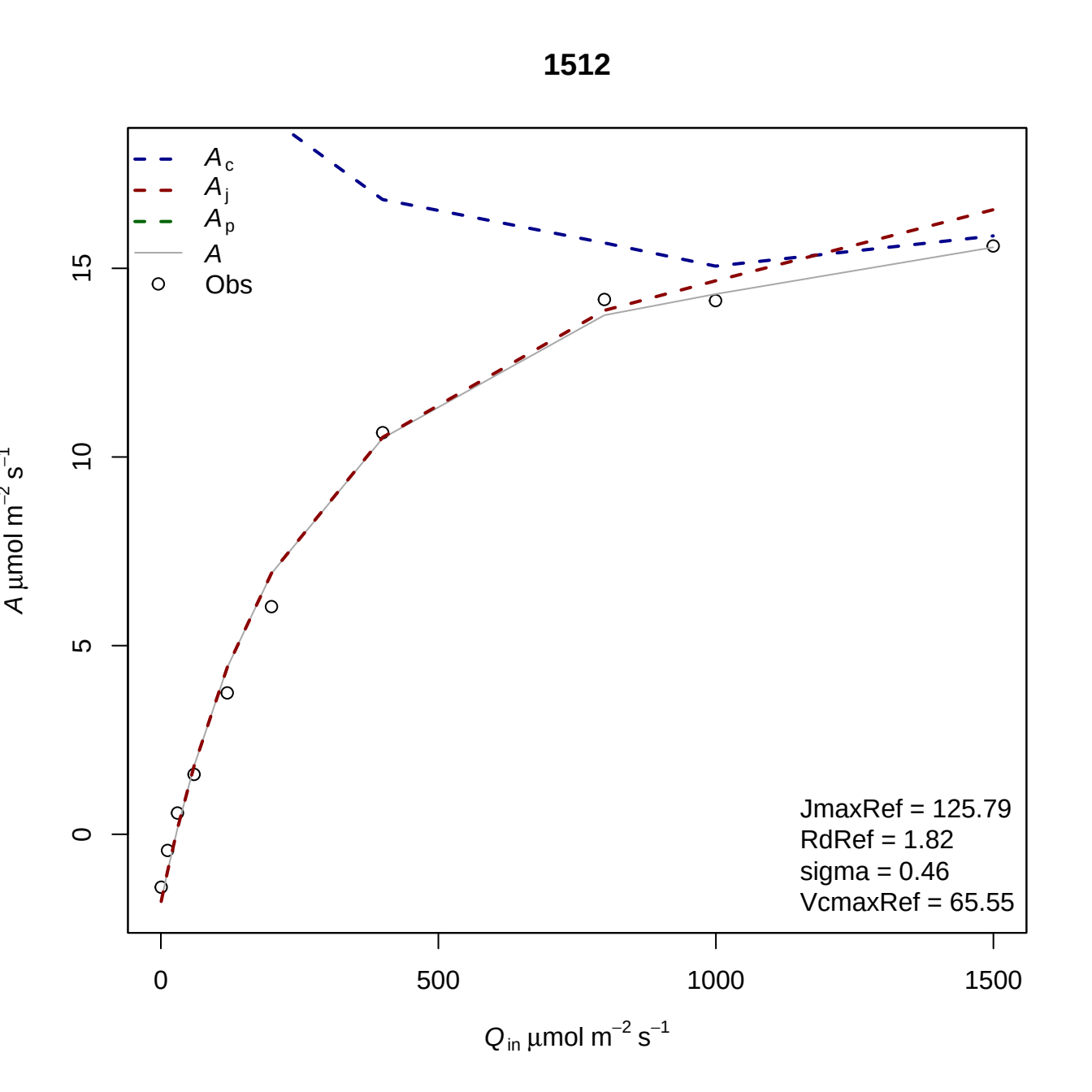

1507
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.23
RdRef = 0.93
sigma = 0.32
VcmaxRef = 33.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

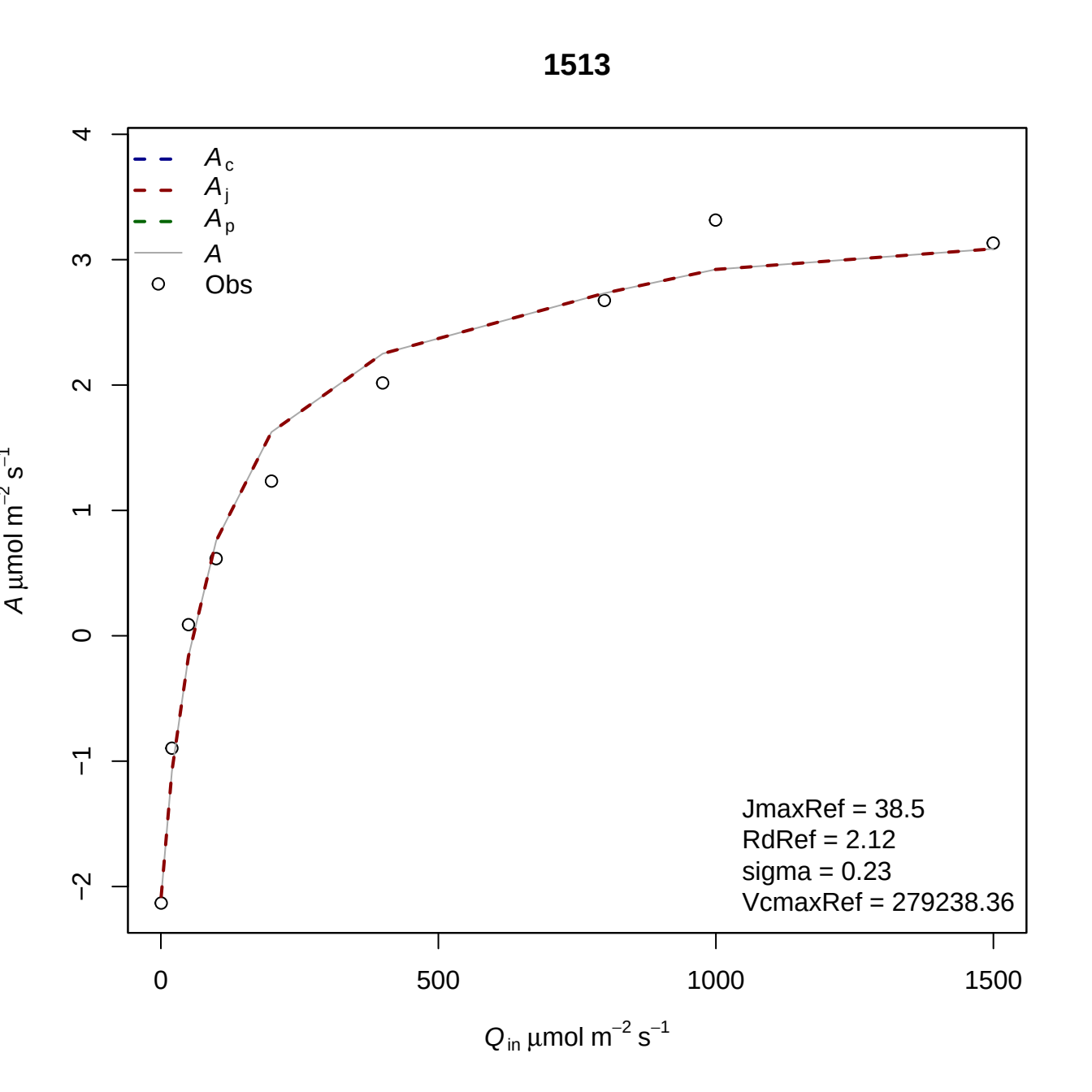

1508
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.36
RdRef = 0.3
sigma = 0.09
VcmaxRef = 24.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

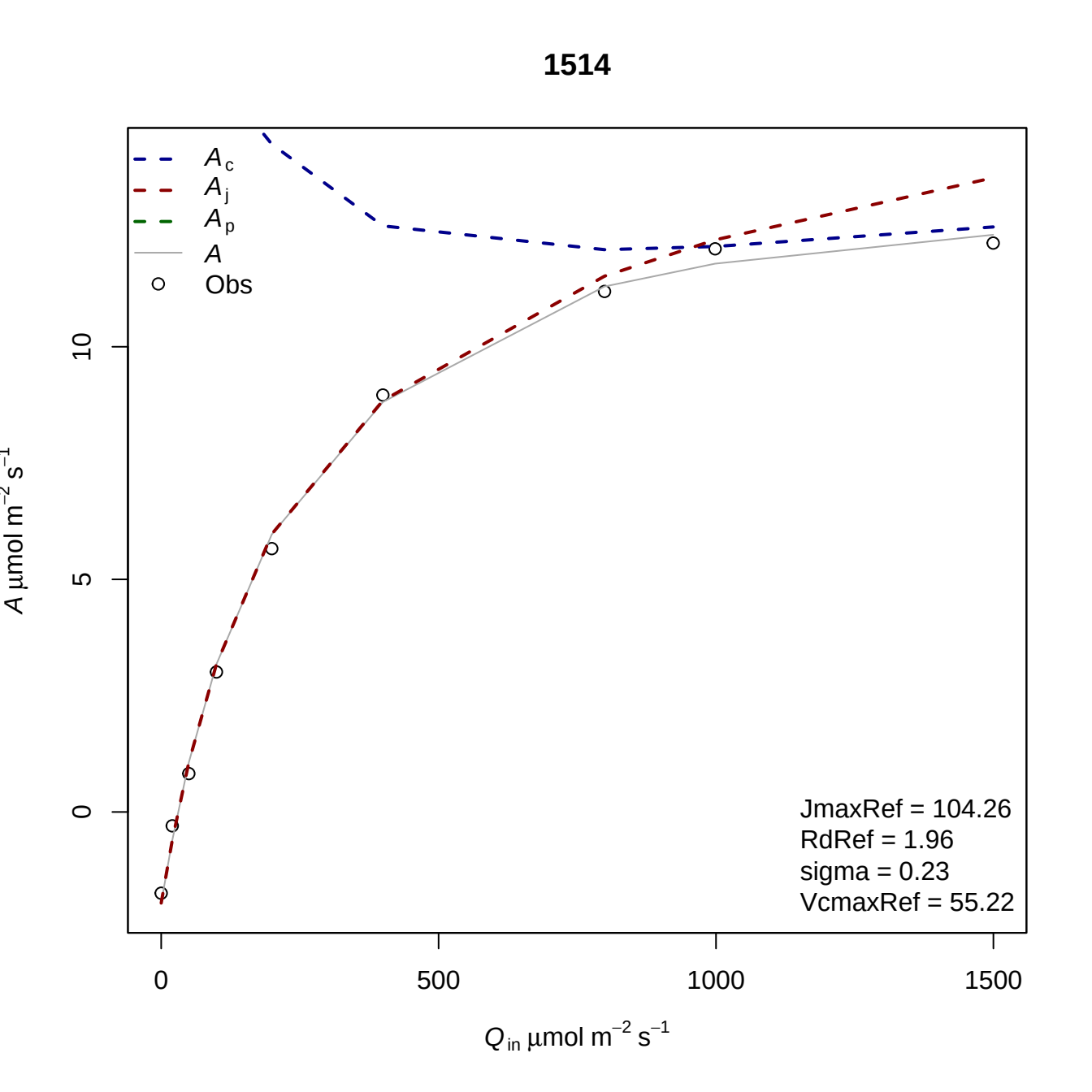

1509
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 53.85
RdRef = 0.38
sigma = 0.22
VcmaxRef = 24.57
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

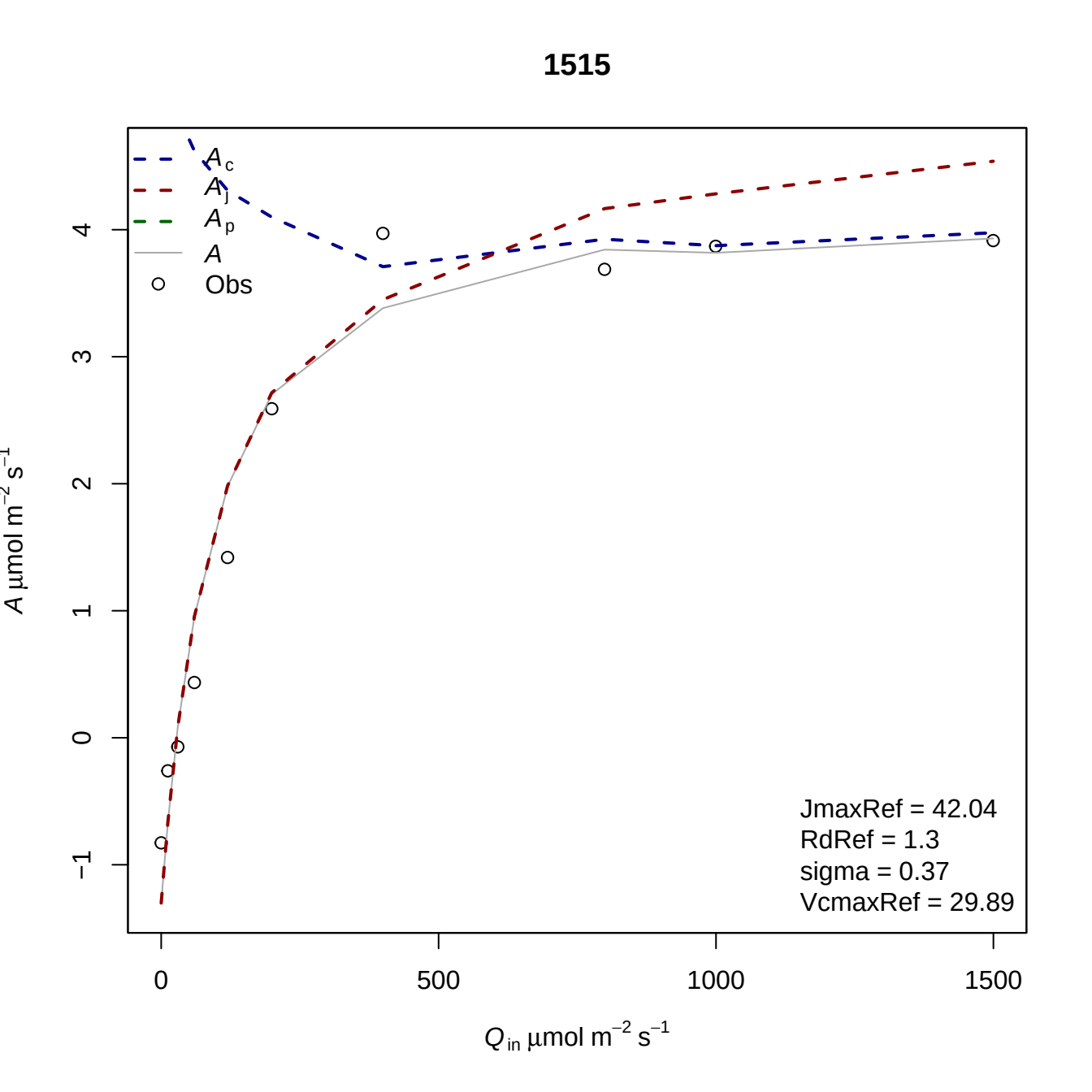

1510
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 95.98
RdRef = 1.27
sigma = 0.19
VcmaxRef = 55.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

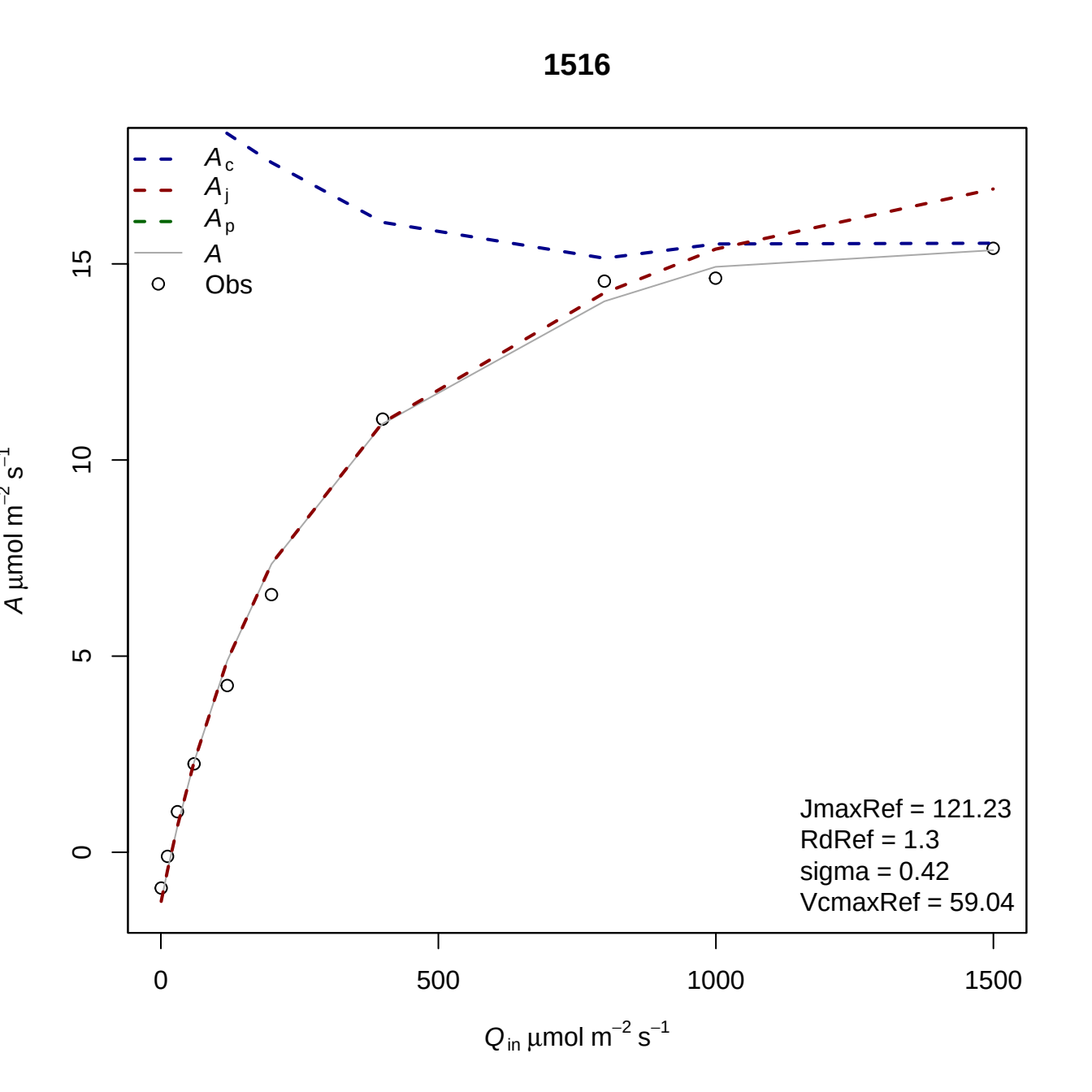

1511
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.47
RdRef = 1.31
sigma = 0.41
VcmaxRef = 52.86
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

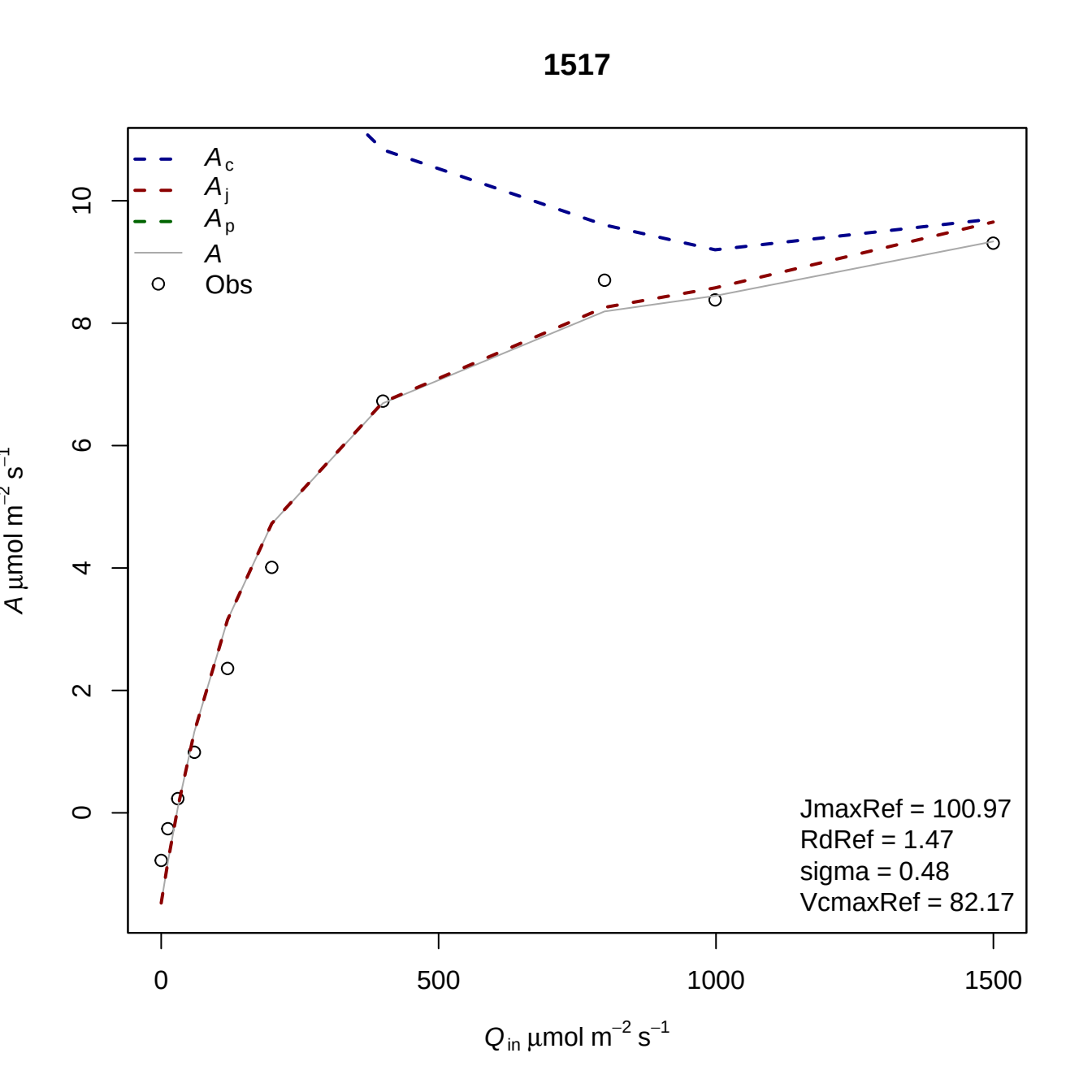

1512
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 125.79
RdRef = 1.82
sigma = 0.46
VcmaxRef = 65.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

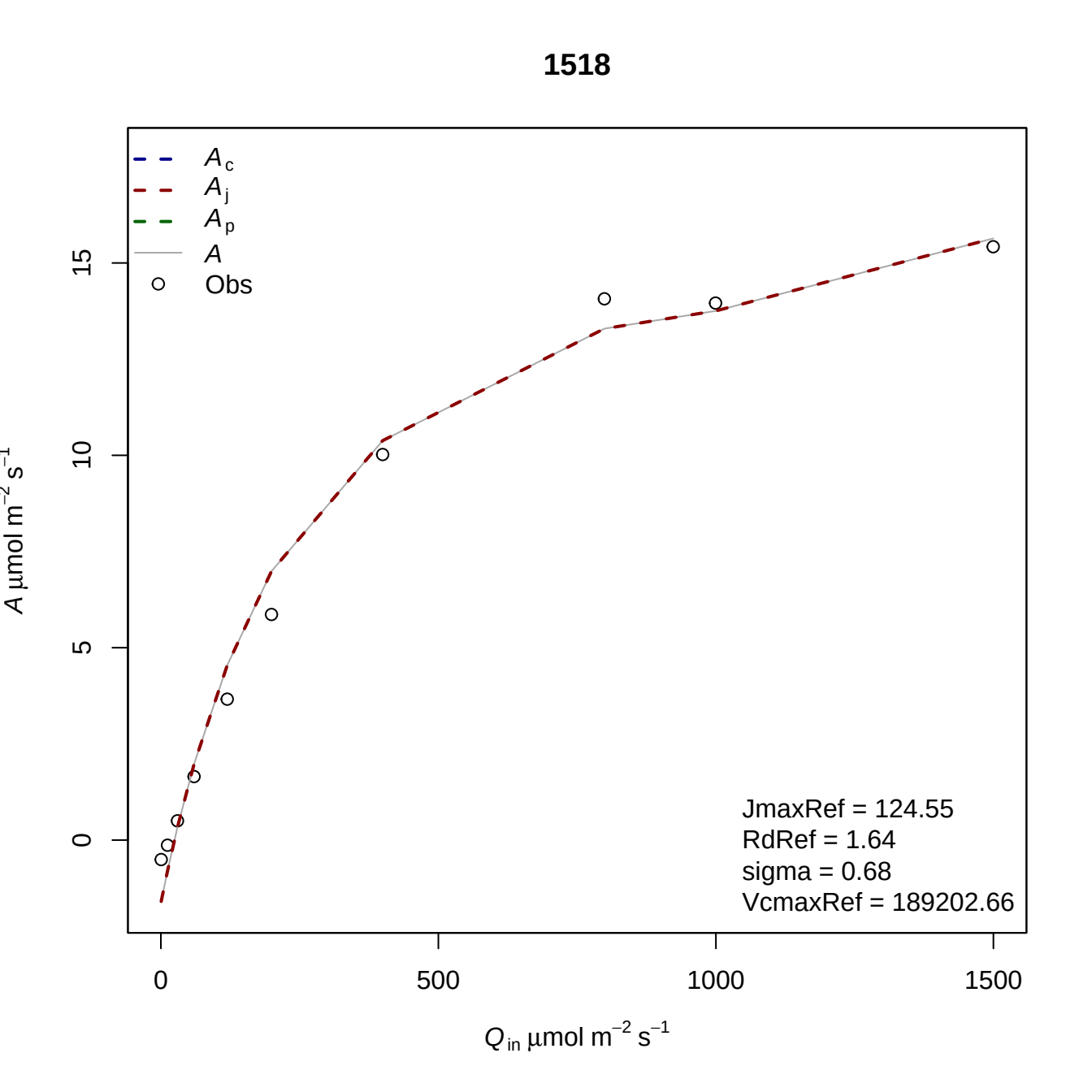

1513
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 38.5
RdRef = 2.12
sigma = 0.23
VcmaxRef = 279238.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
−2
0
500
1000
1500

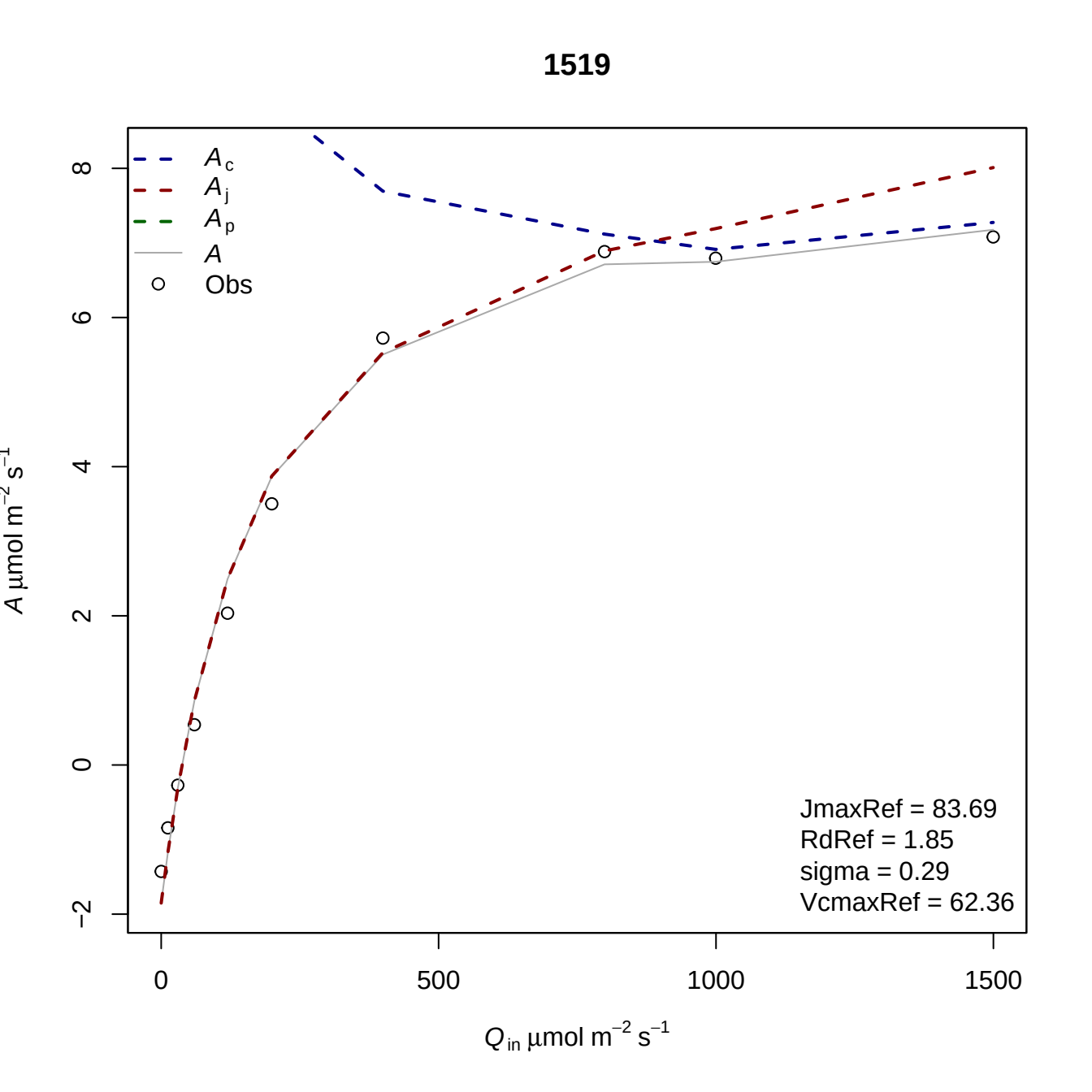

1514
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.26
RdRef = 1.96
sigma = 0.23
VcmaxRef = 55.22
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

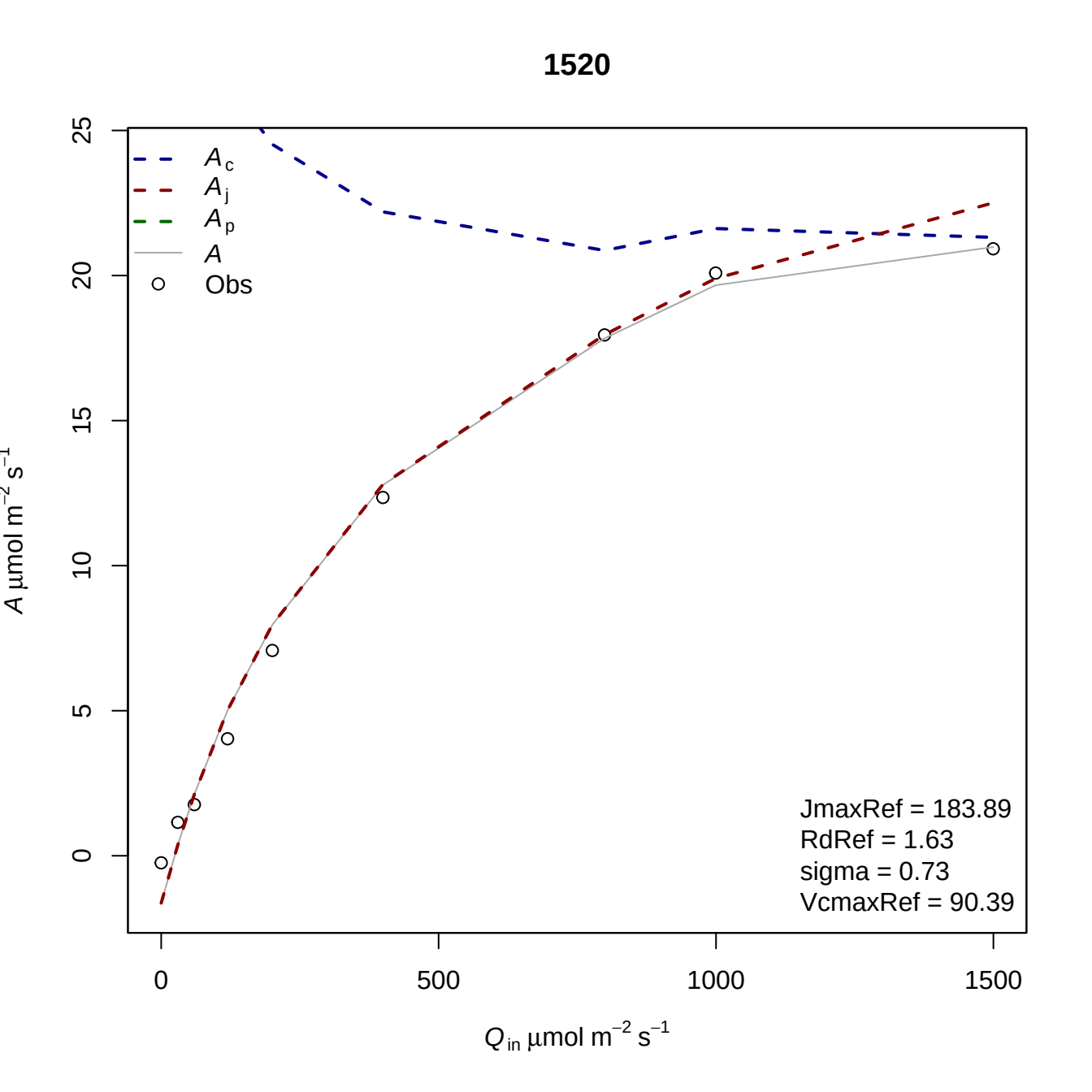

1515
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.04
RdRef = 1.3
sigma = 0.37
VcmaxRef = 29.89
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
0
500
1000
1500

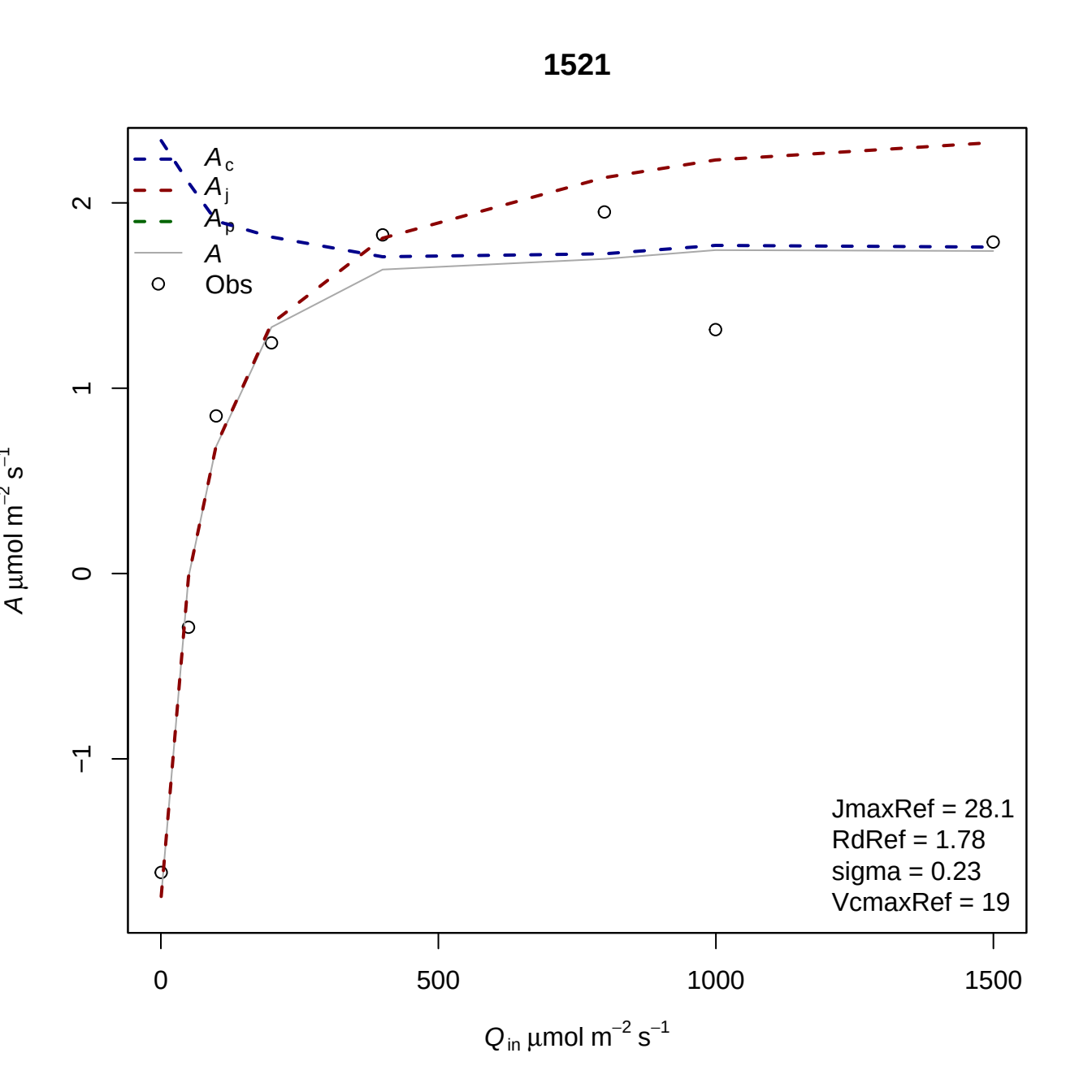

1516
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 121.23
RdRef = 1.3
sigma = 0.42
VcmaxRef = 59.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

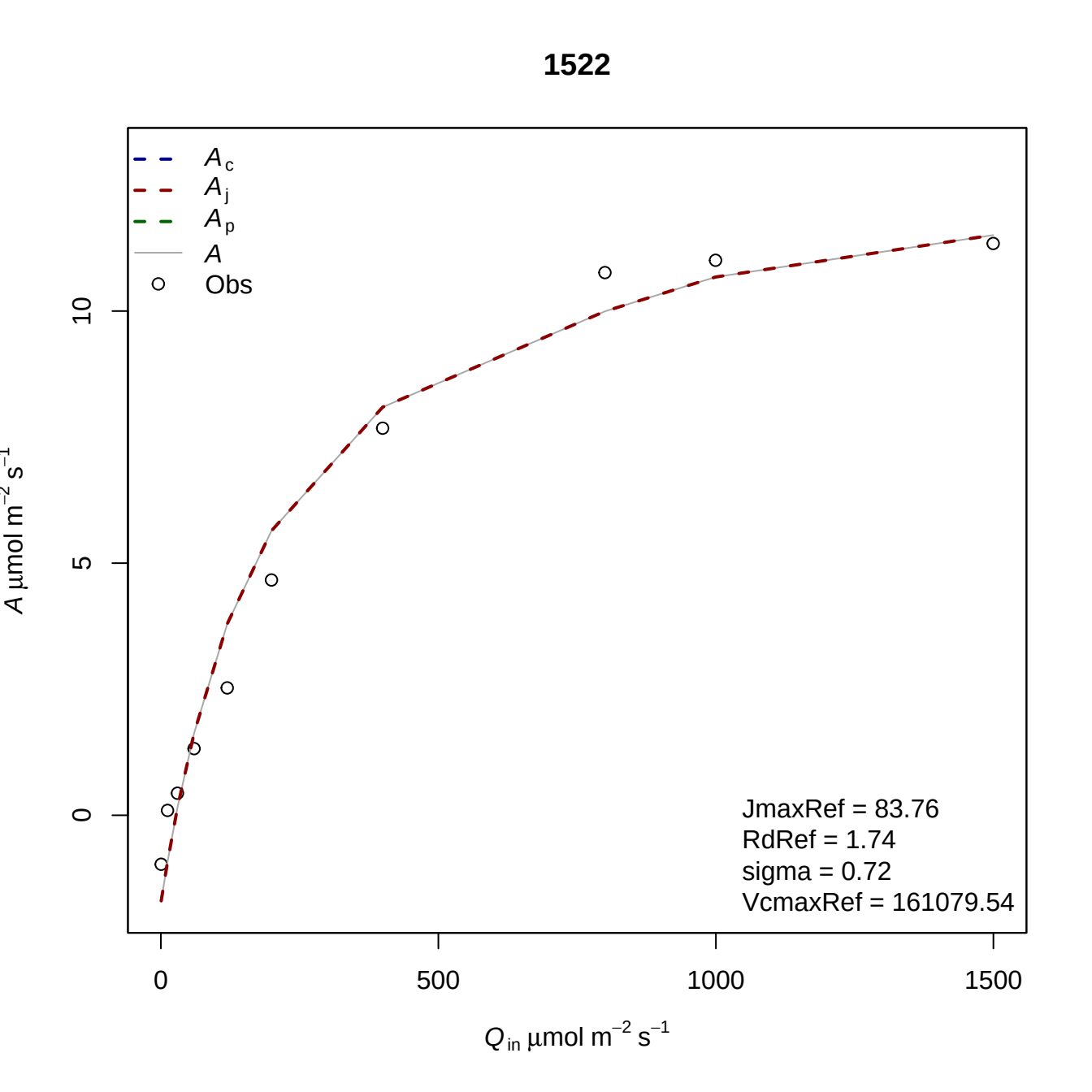

1517
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.97
RdRef = 1.47
sigma = 0.48
VcmaxRef = 82.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

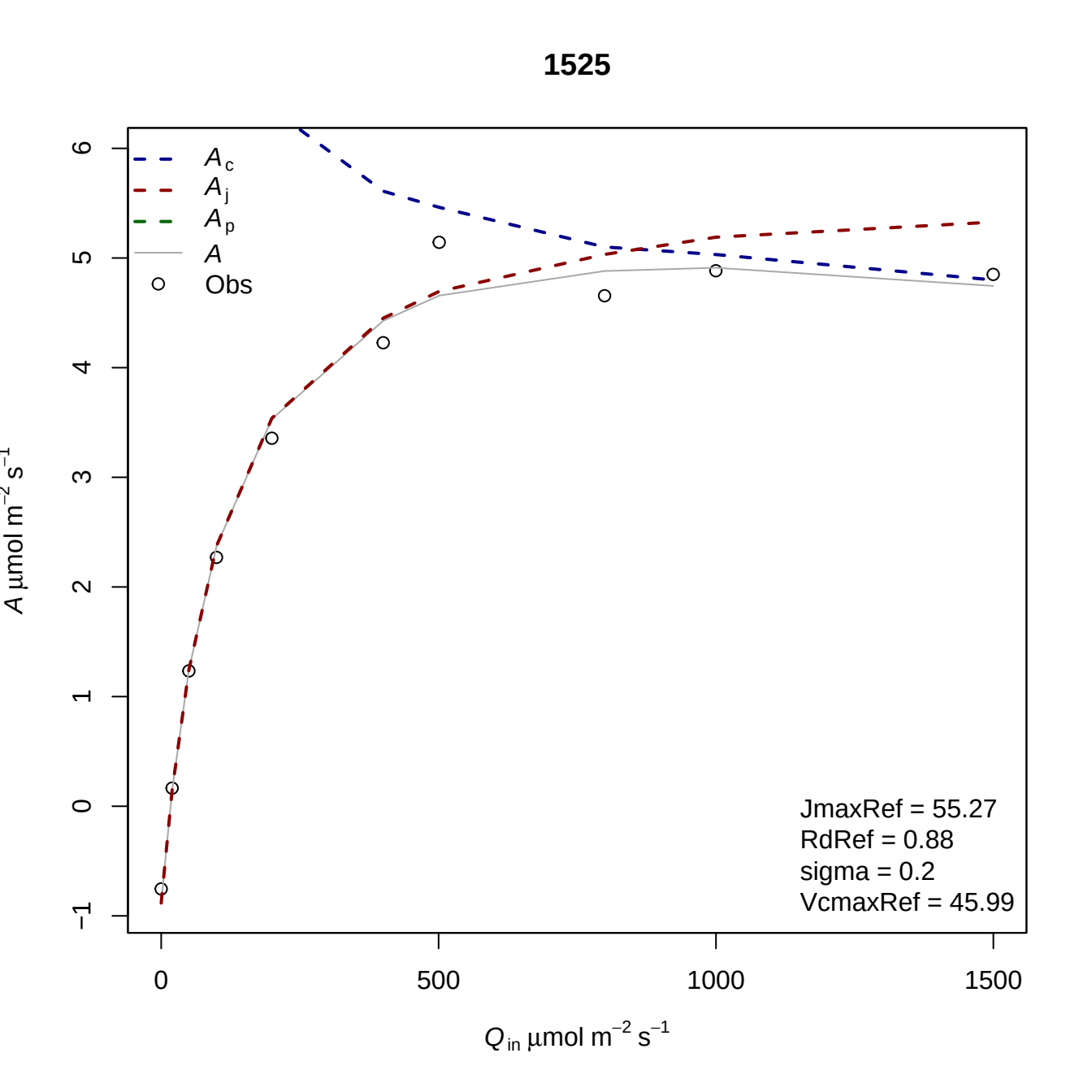

1518
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 124.55
RdRef = 1.64
sigma = 0.68
VcmaxRef = 189202.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

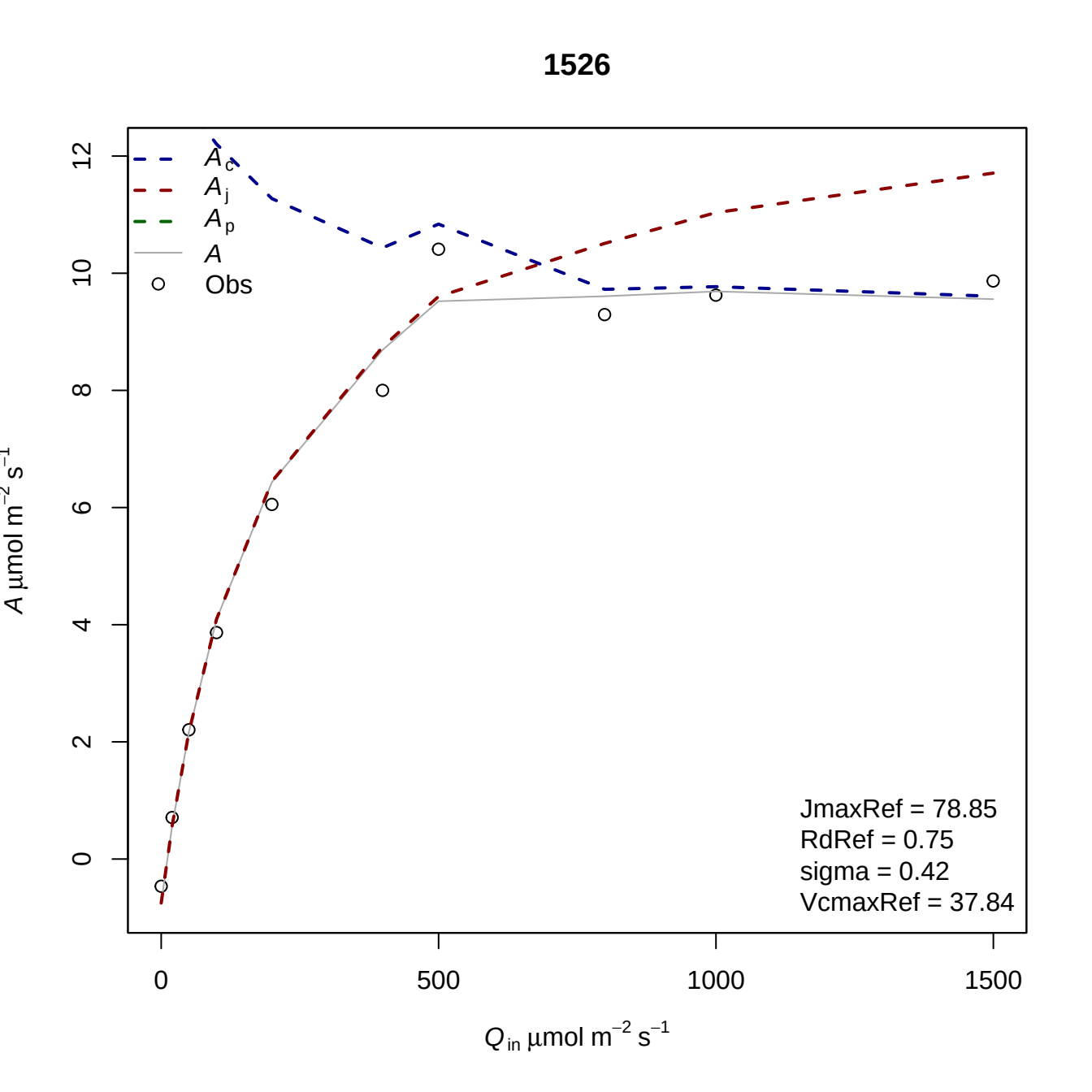

1519
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.69
RdRef = 1.85
sigma = 0.29
VcmaxRef = 62.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−2
0
2
4
6
8
0
500
1000
1500

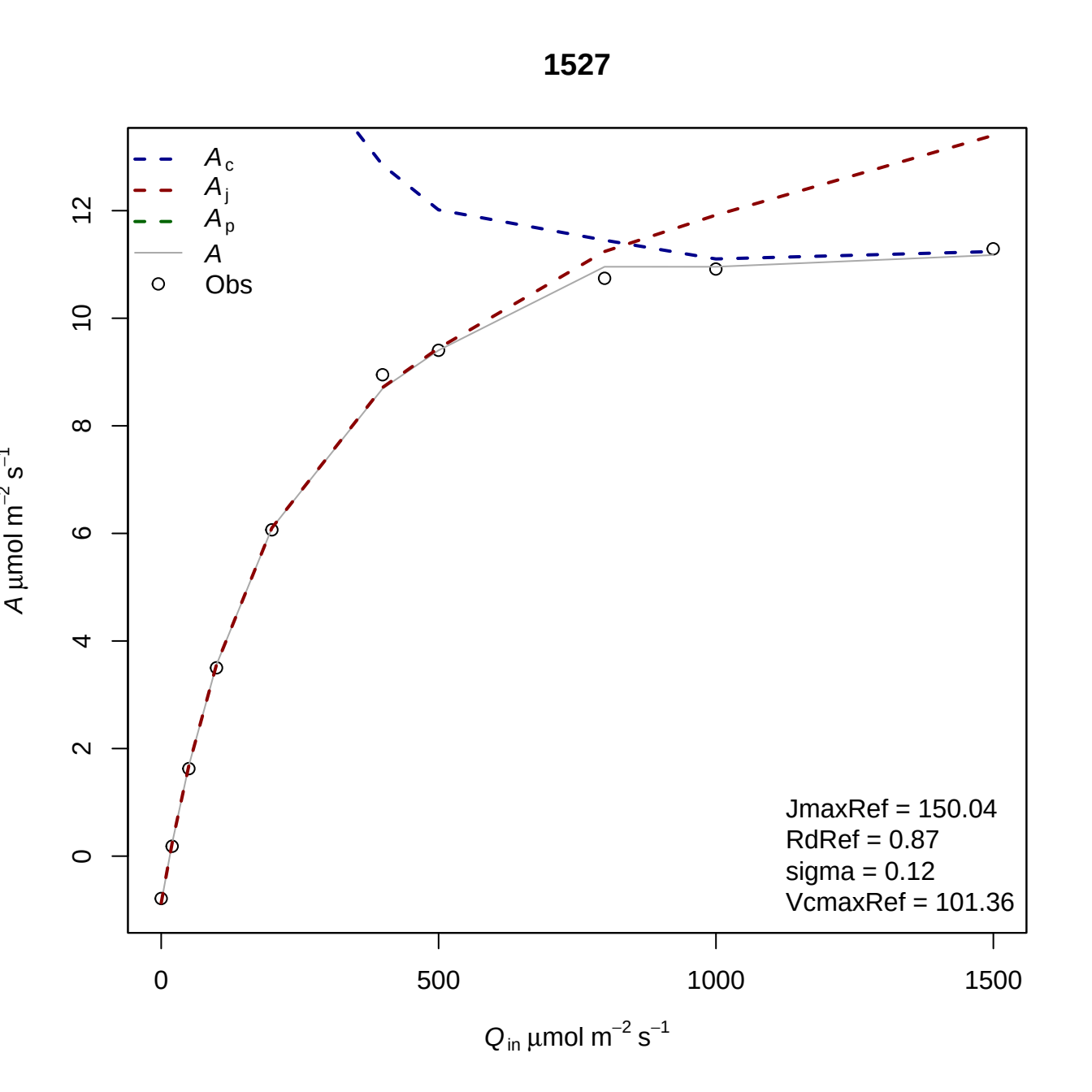

1520
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 183.89
RdRef = 1.63
sigma = 0.73
VcmaxRef = 90.39
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

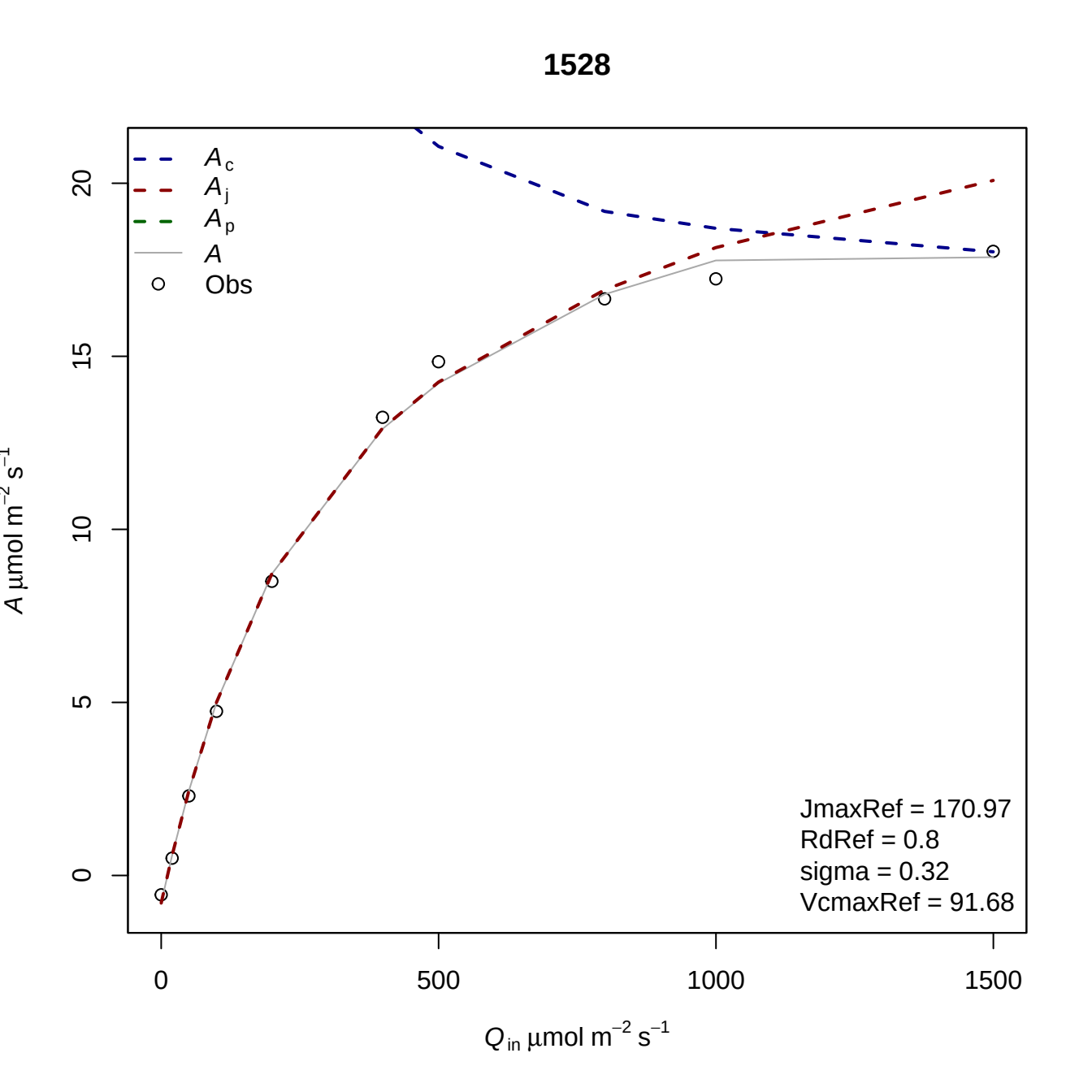

1521
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 28.1
RdRef = 1.78
sigma = 0.23
VcmaxRef = 19
2
1
0
−1
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

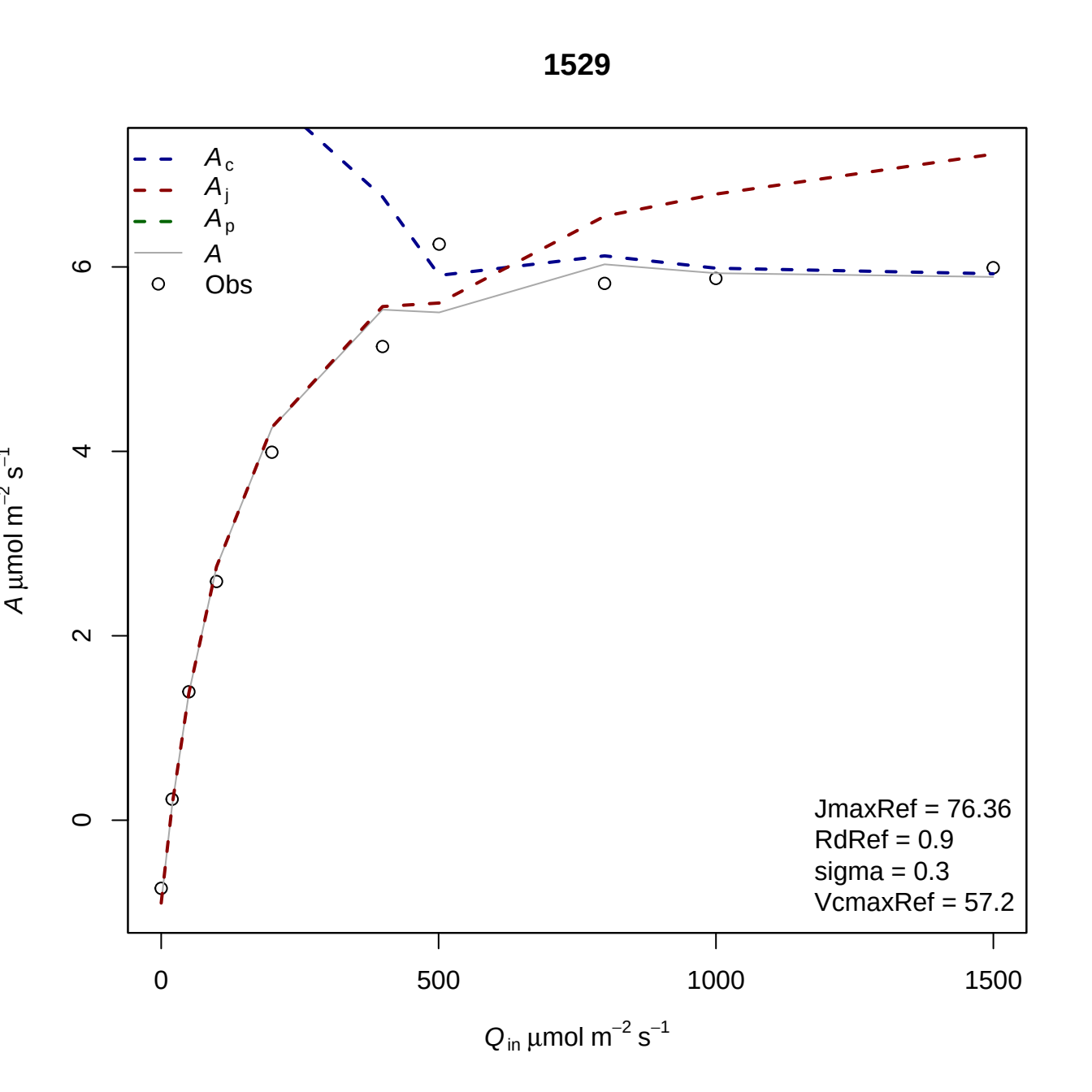

1522
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 83.76
RdRef = 1.74
sigma = 0.72
VcmaxRef = 161079.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500

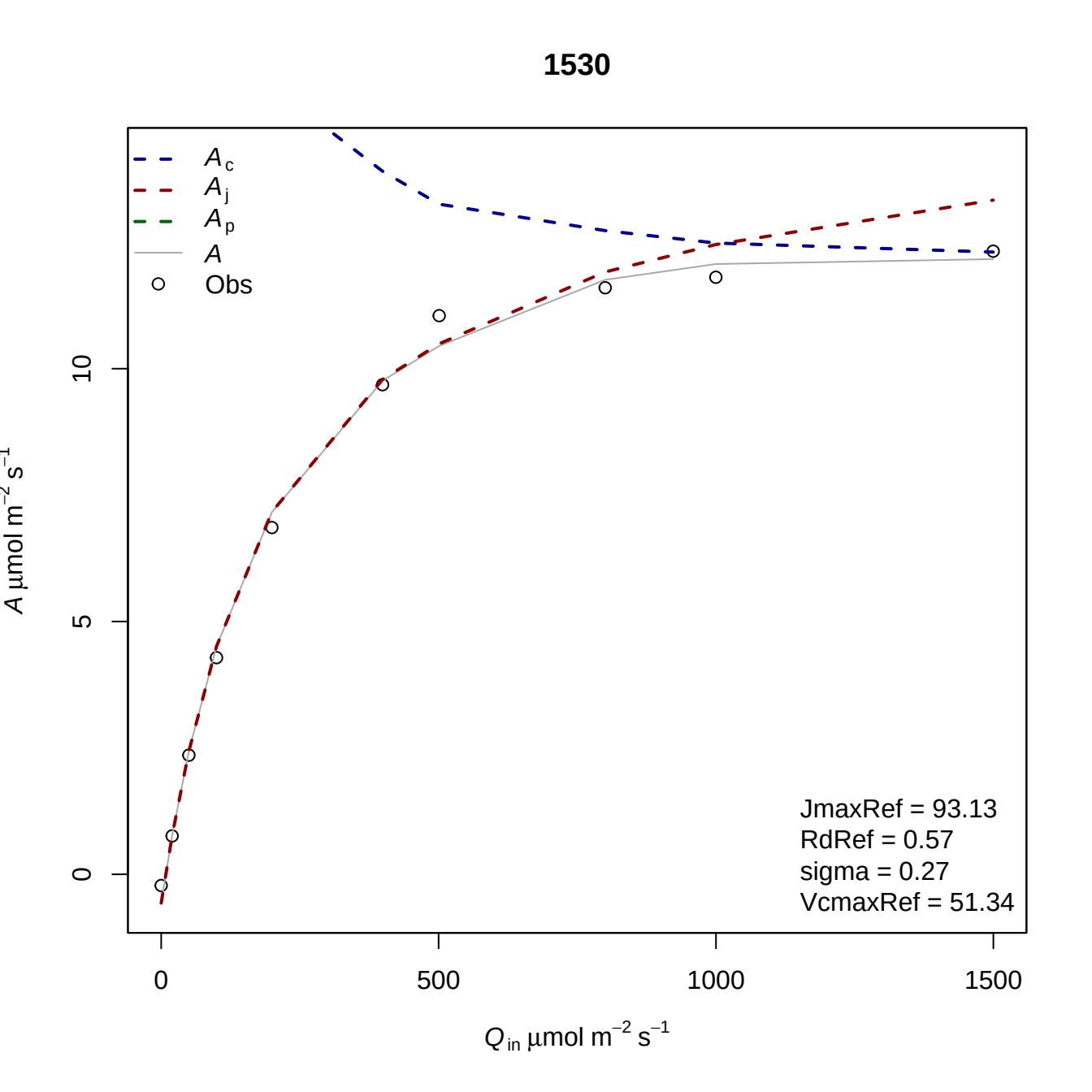

1525
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.27
RdRef = 0.88
sigma = 0.2
VcmaxRef = 45.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
6
0
500
1000
1500

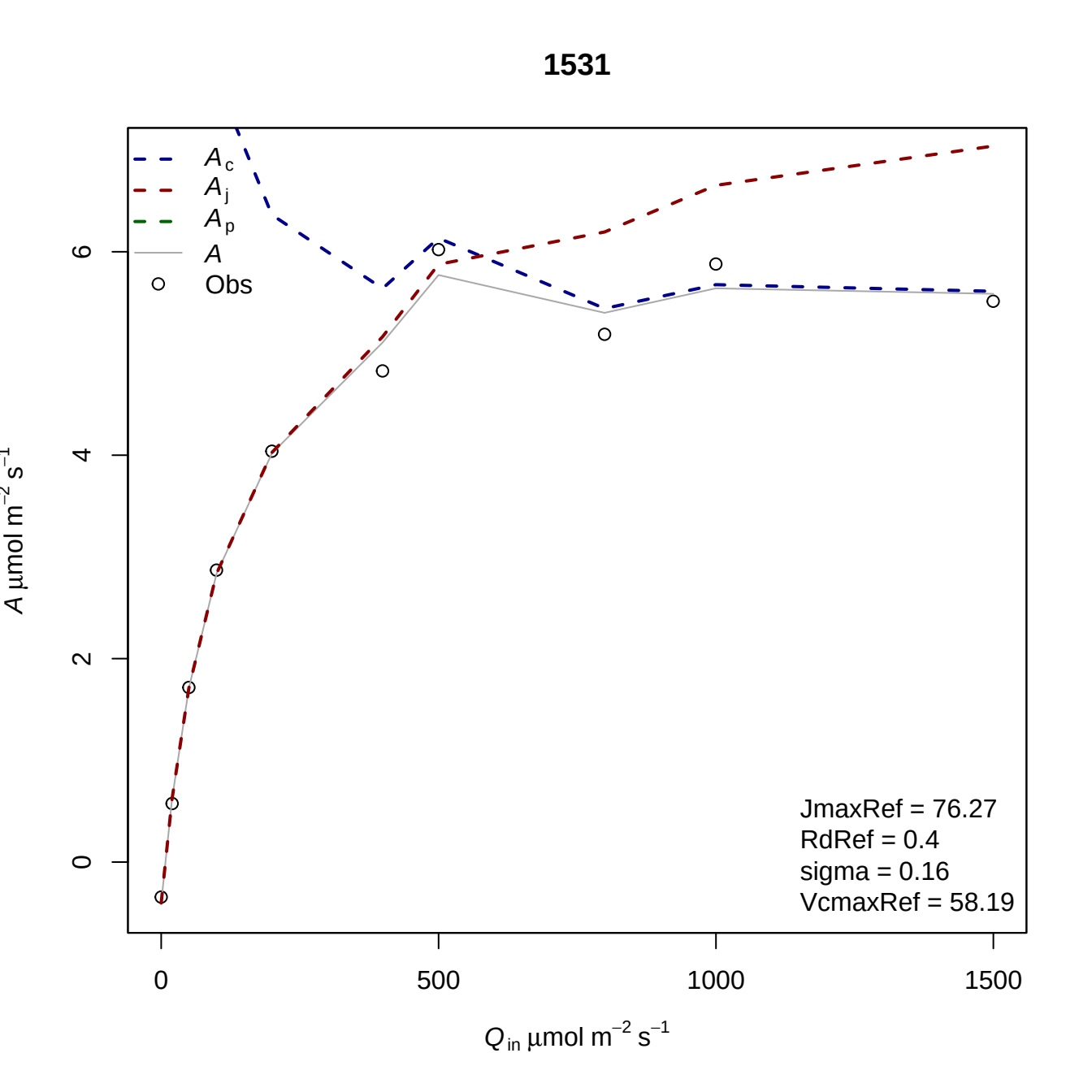

1526
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.85
RdRef = 0.75
sigma = 0.42
VcmaxRef = 37.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

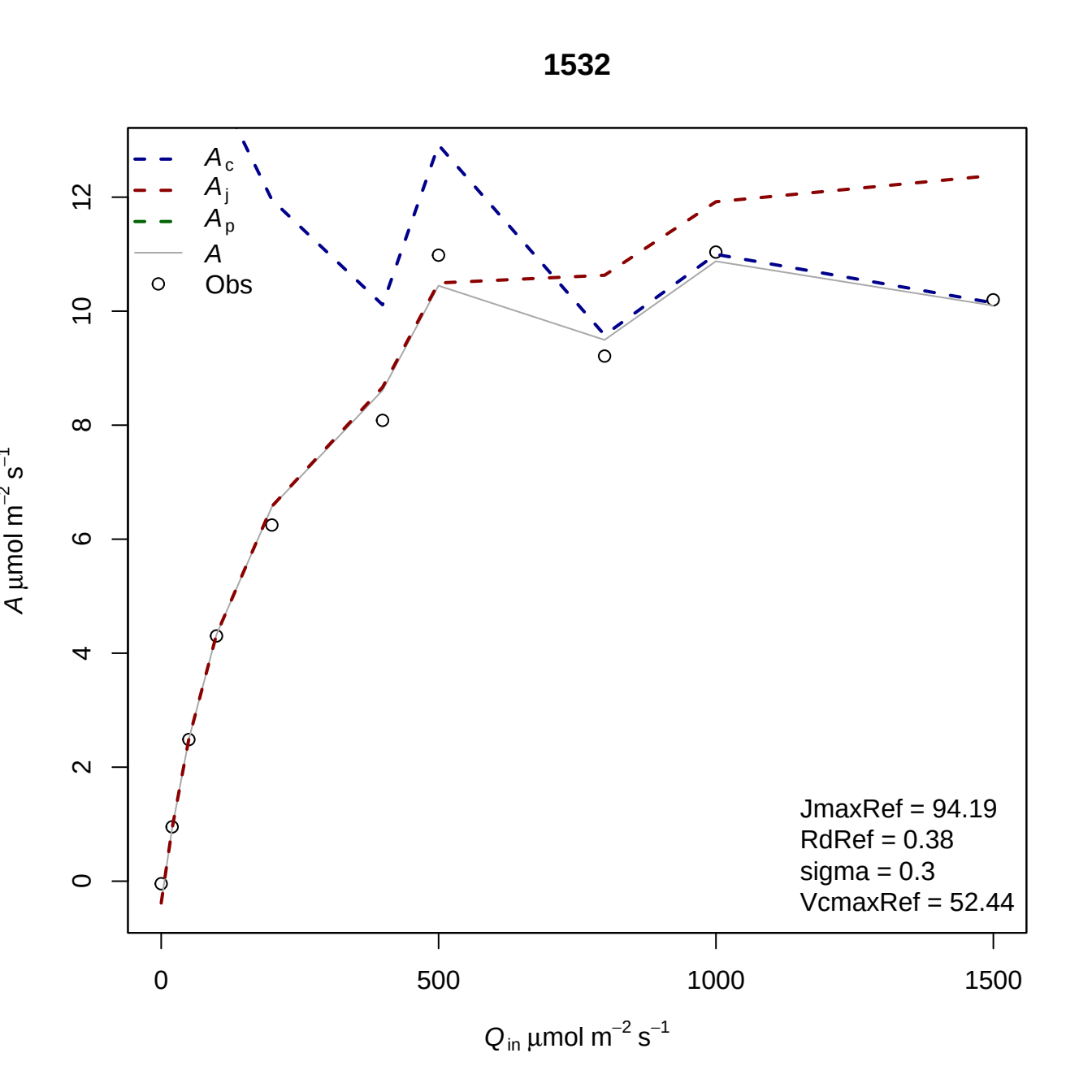

1527
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 150.04
RdRef = 0.87
sigma = 0.12
VcmaxRef = 101.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

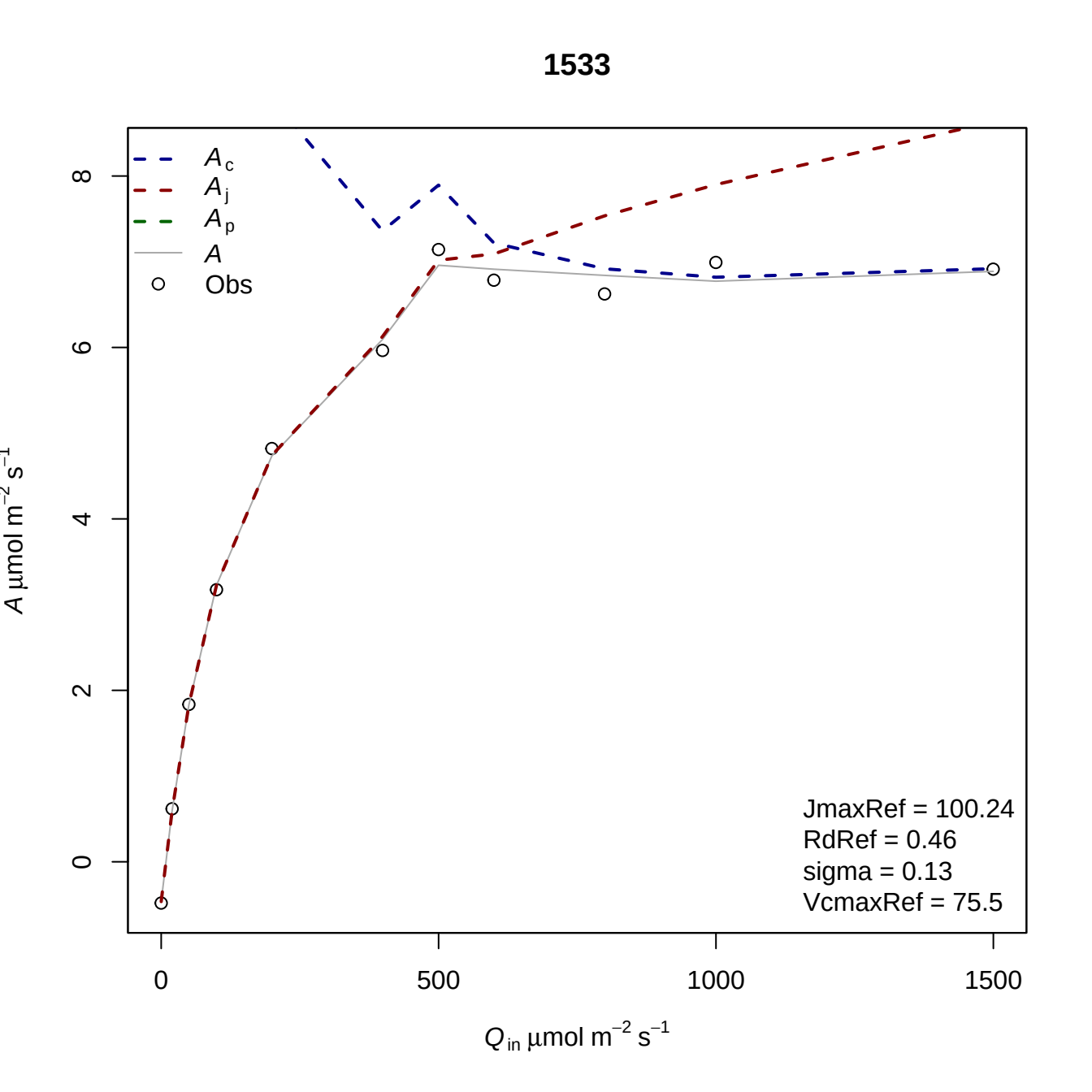

1528
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 170.97
RdRef = 0.8
sigma = 0.32
VcmaxRef = 91.68
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500

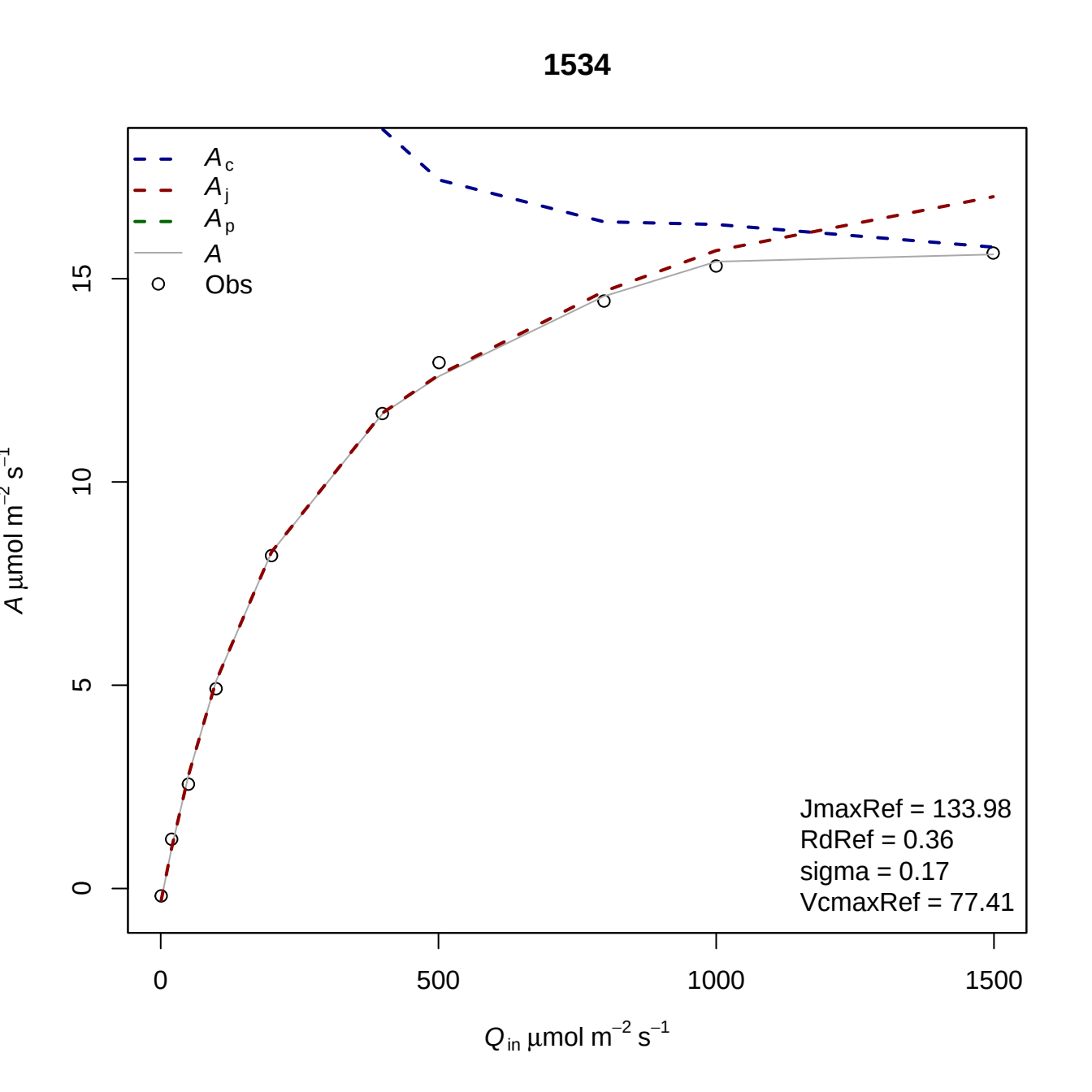

1529
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.36
RdRef = 0.9
sigma = 0.3
VcmaxRef = 57.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

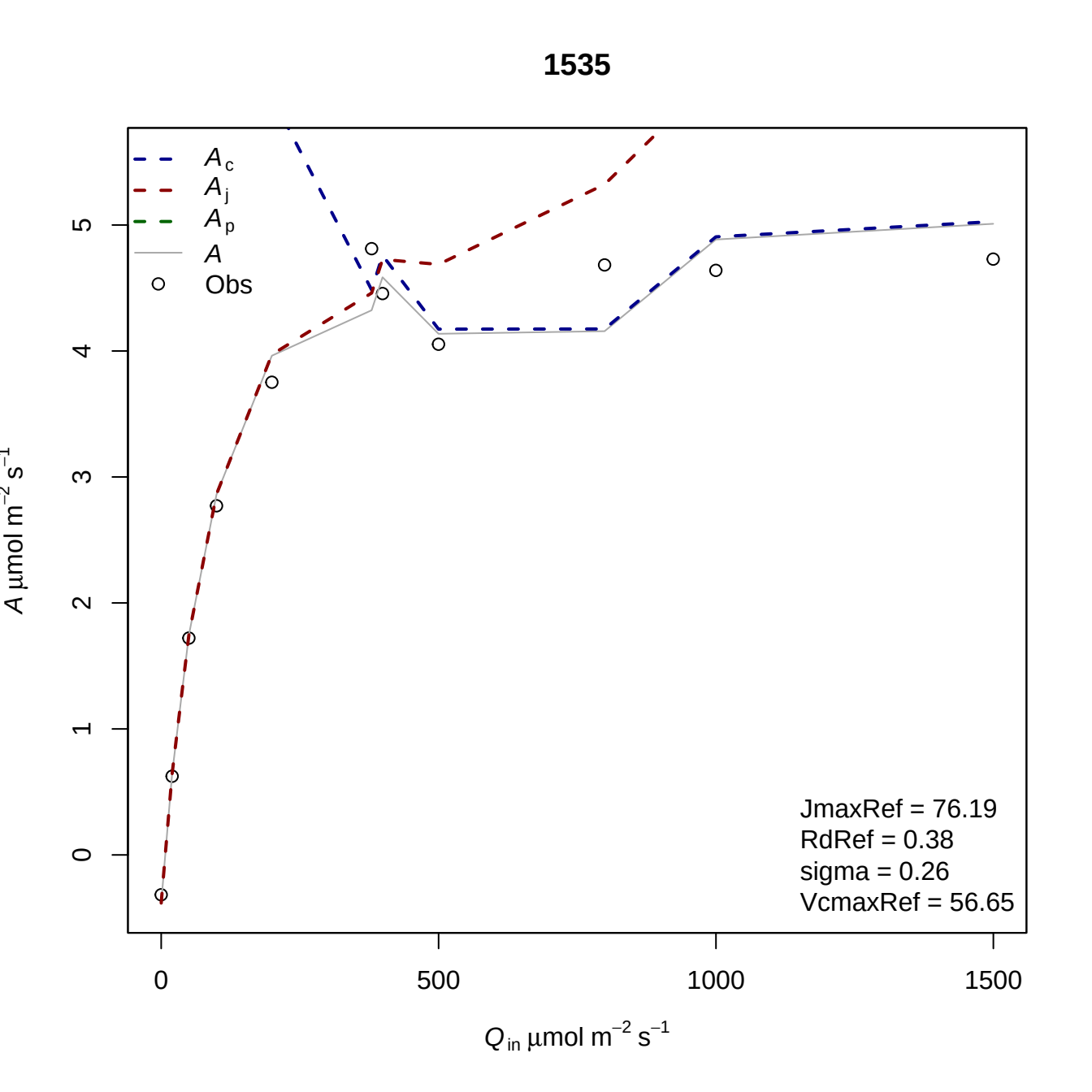

1530
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 93.13
RdRef = 0.57
sigma = 0.27
VcmaxRef = 51.34
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

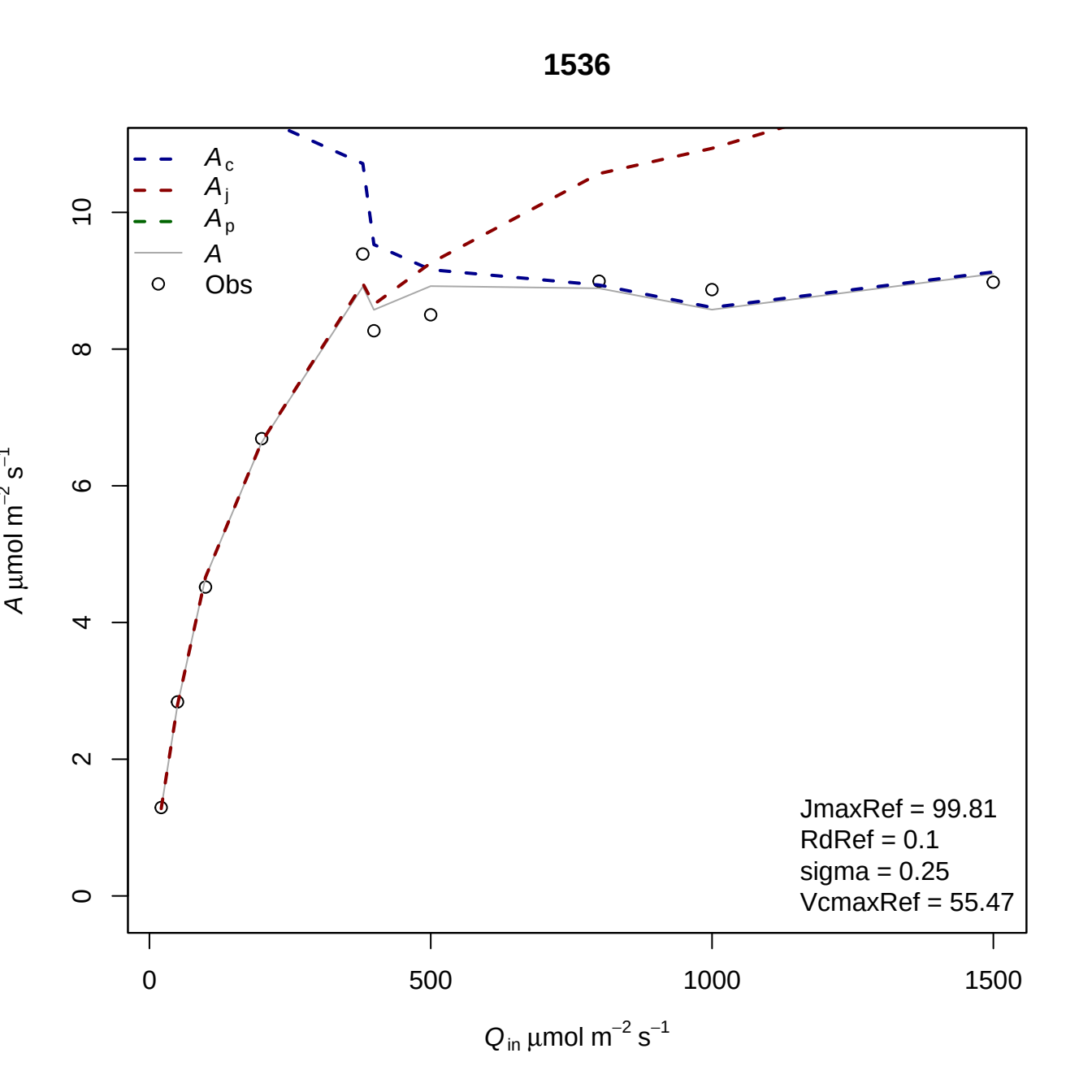

1531
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.27
RdRef = 0.4
sigma = 0.16
VcmaxRef = 58.19
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

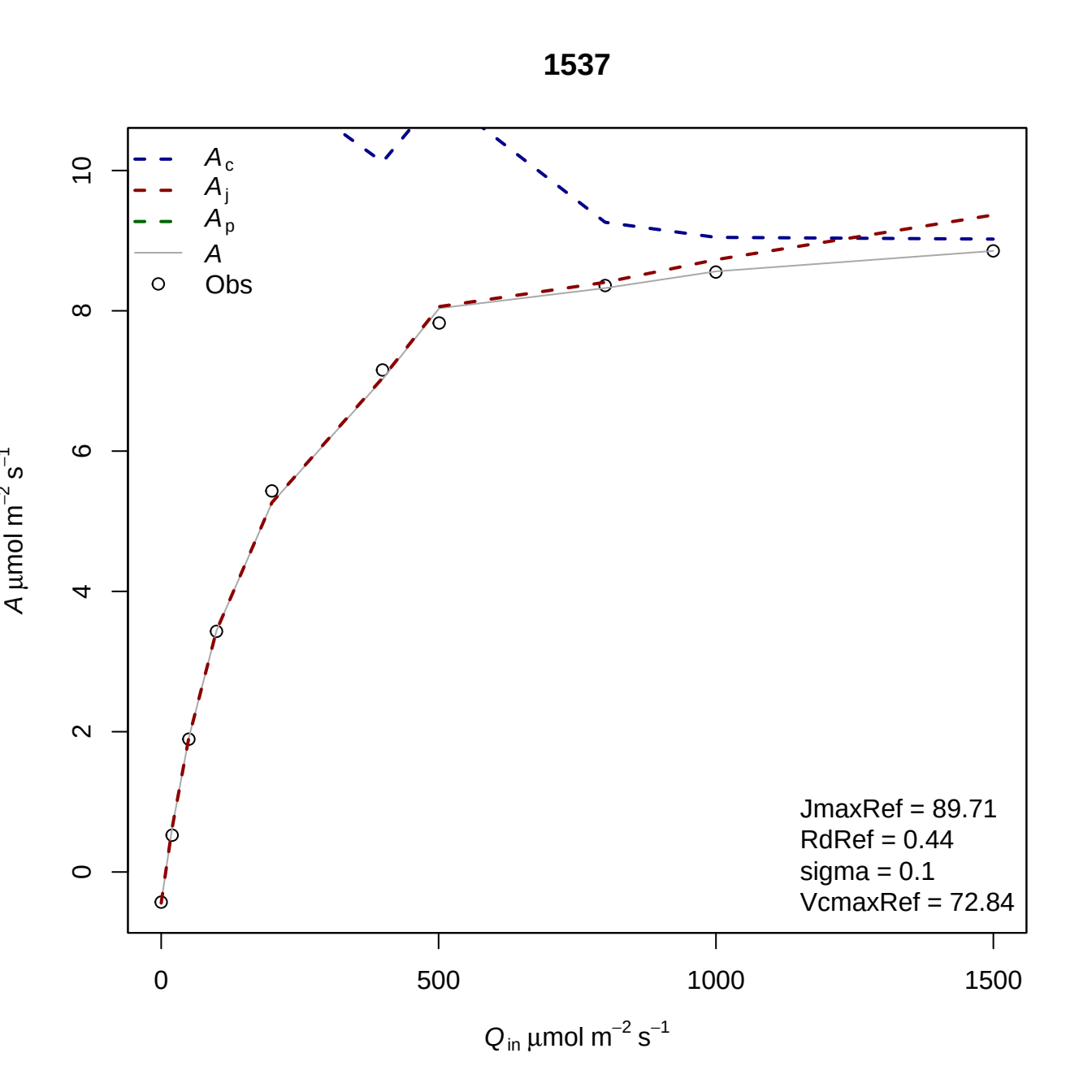

1532
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.19
RdRef = 0.38
sigma = 0.3
VcmaxRef = 52.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

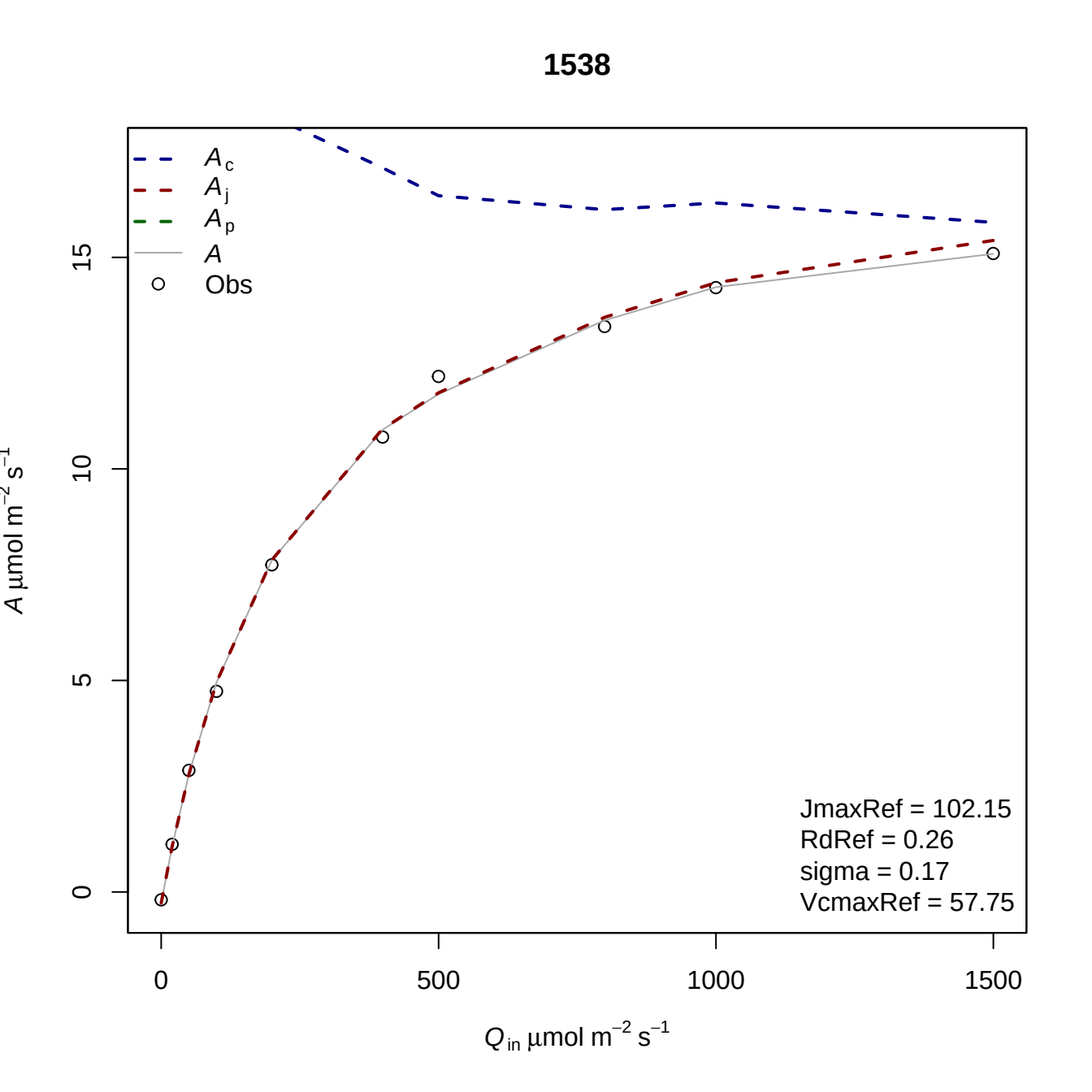

1533
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.24
RdRef = 0.46
sigma = 0.13
VcmaxRef = 75.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

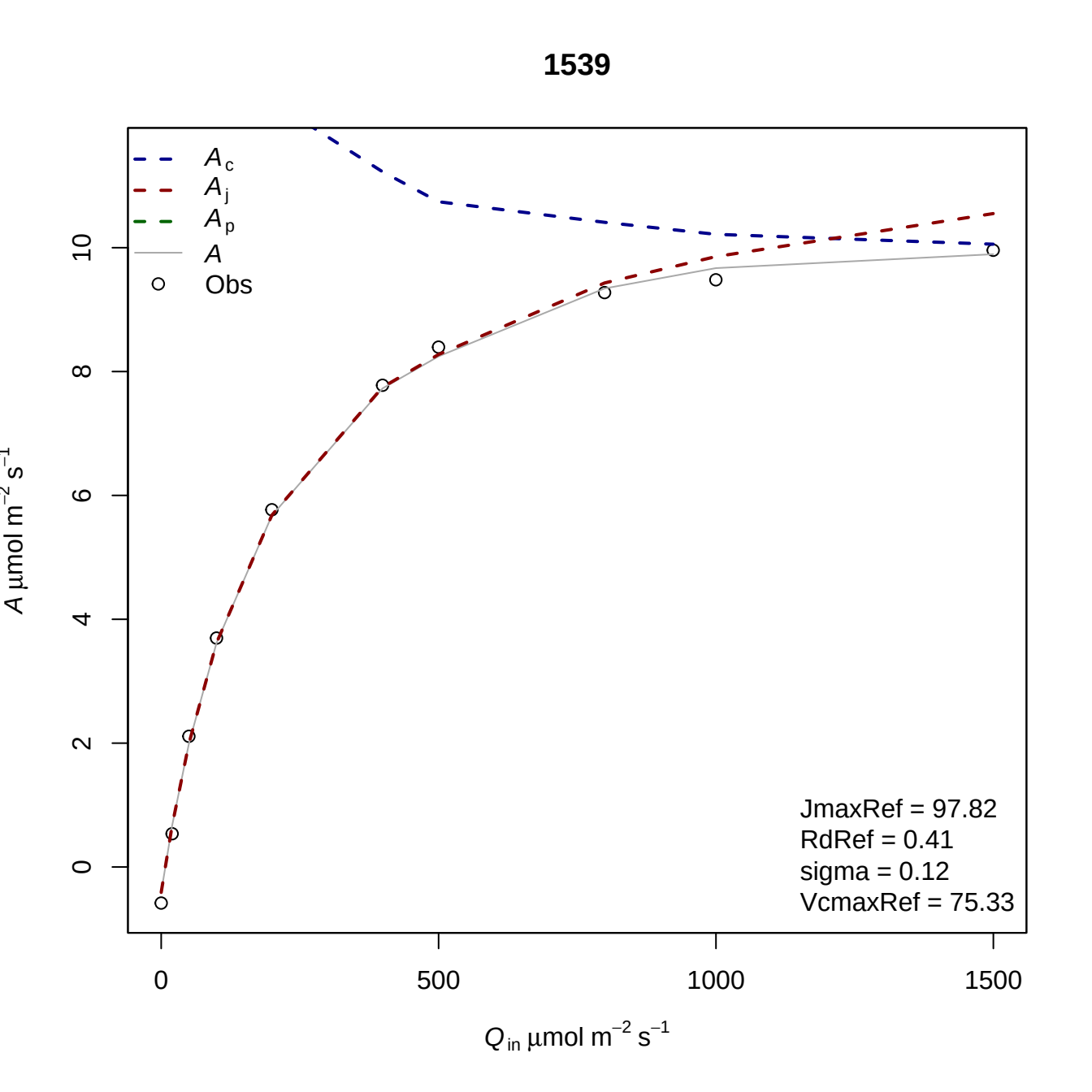

1534
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 133.98
RdRef = 0.36
sigma = 0.17
VcmaxRef = 77.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

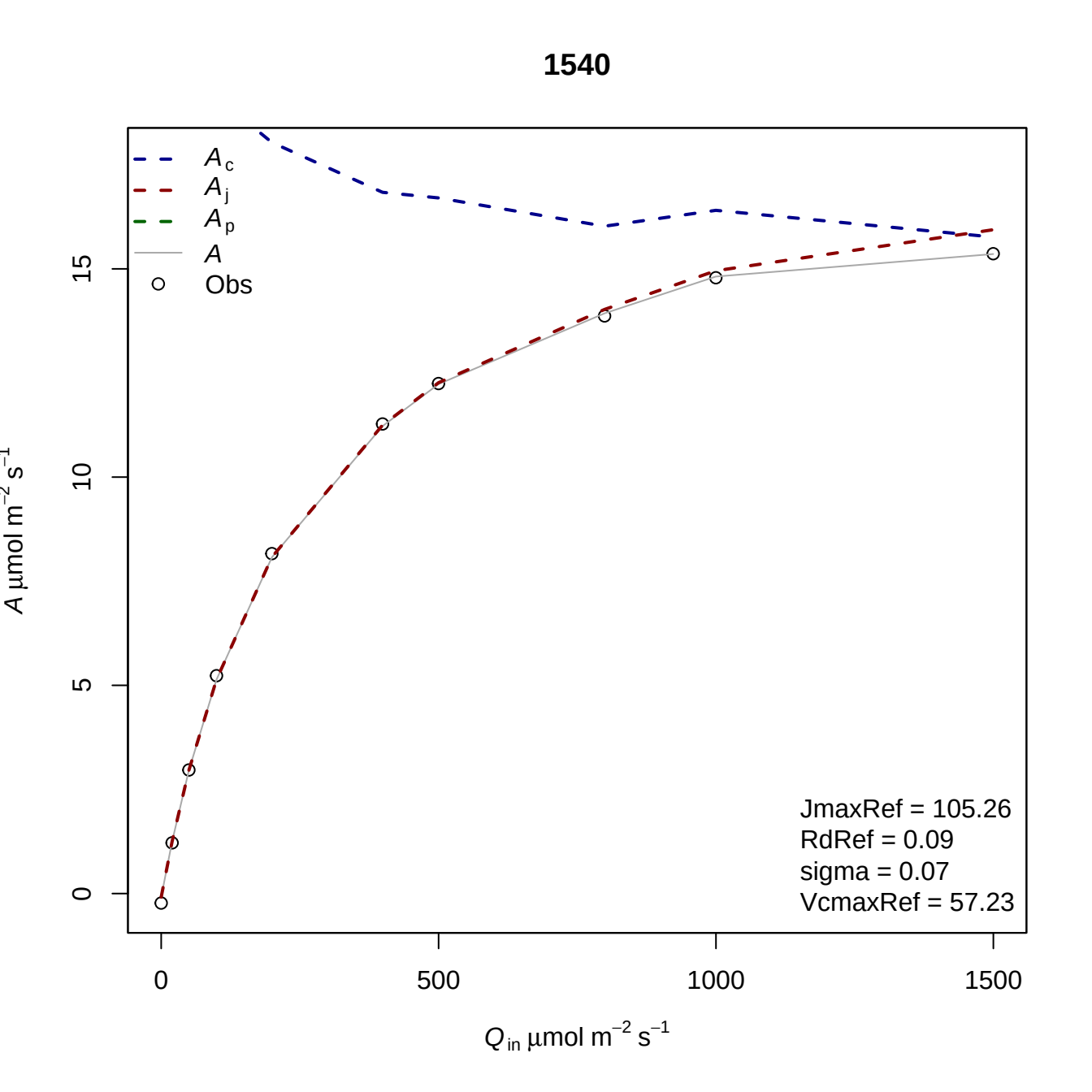

1535
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.19
RdRef = 0.38
sigma = 0.26
VcmaxRef = 56.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

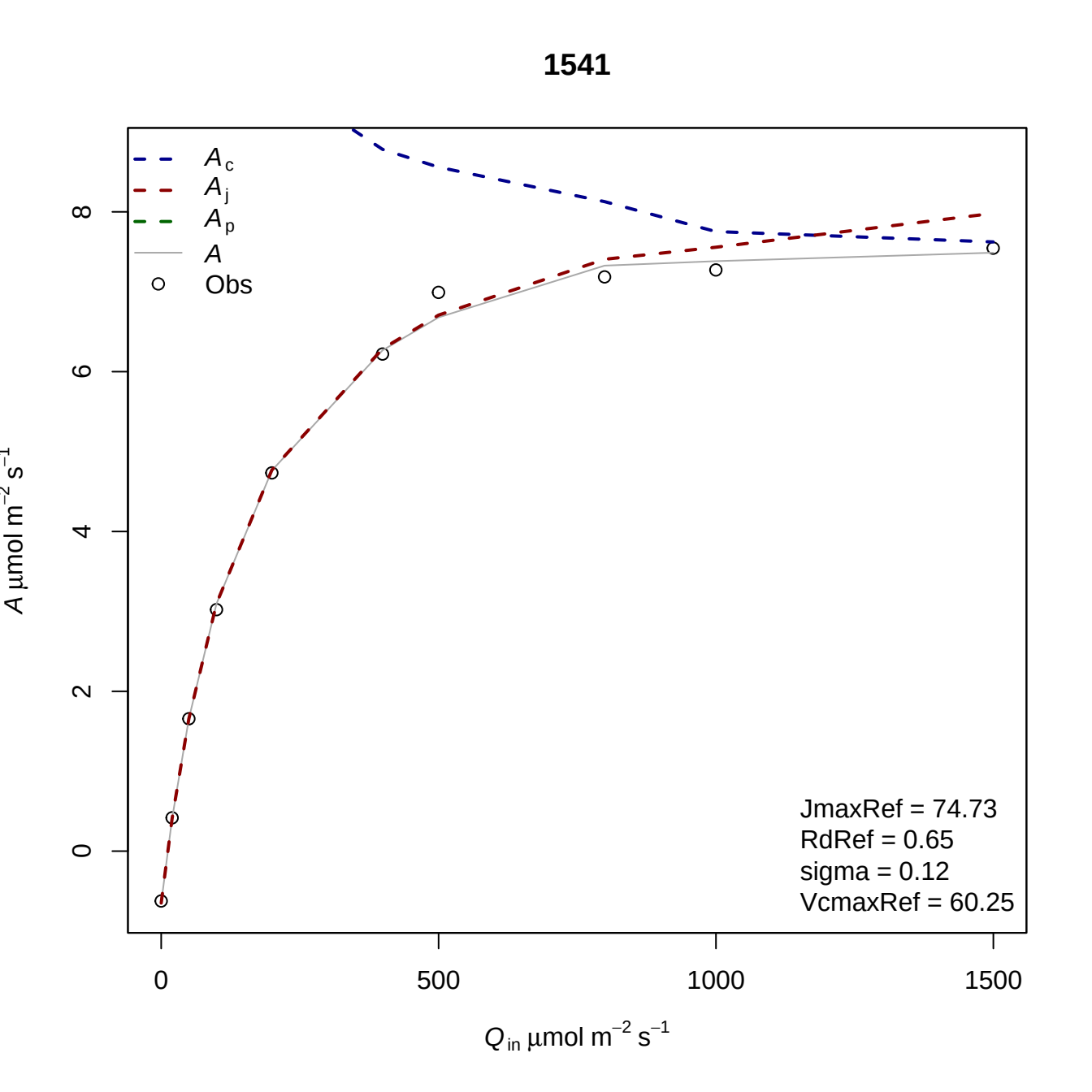

1536
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.81
RdRef = 0.1
sigma = 0.25
VcmaxRef = 55.47
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

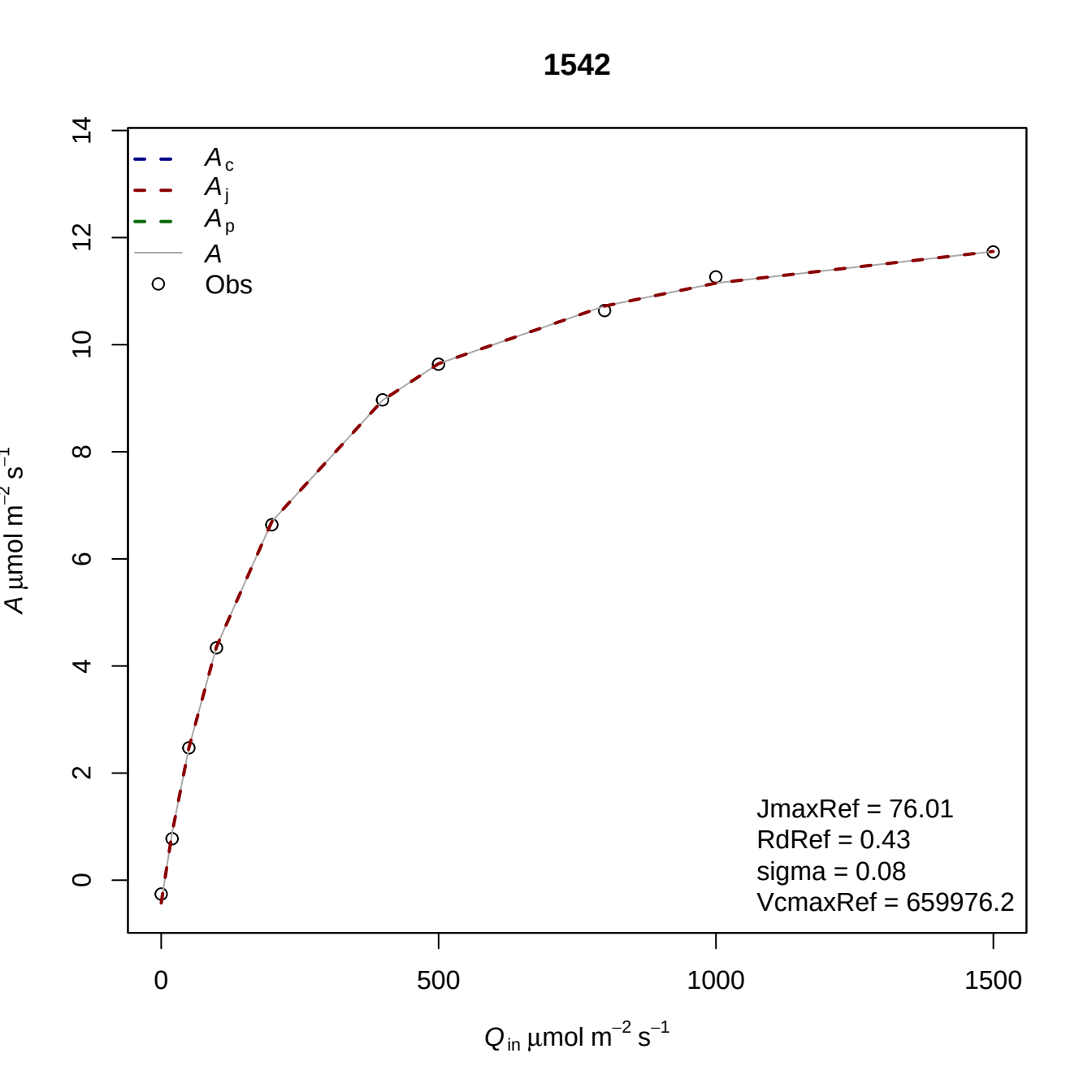

1537
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.71
RdRef = 0.44
sigma = 0.1
VcmaxRef = 72.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

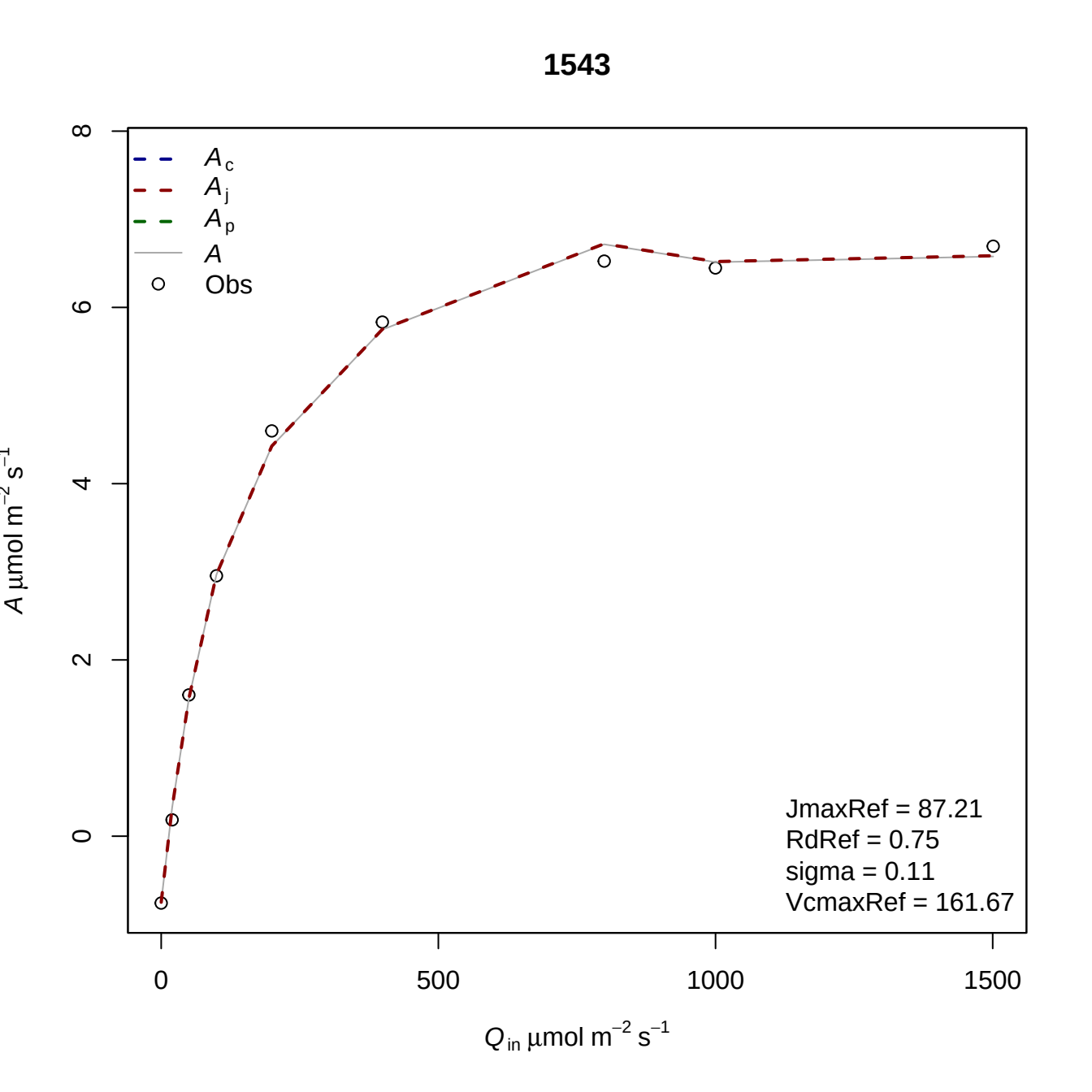

1538
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.15
RdRef = 0.26
sigma = 0.17
VcmaxRef = 57.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

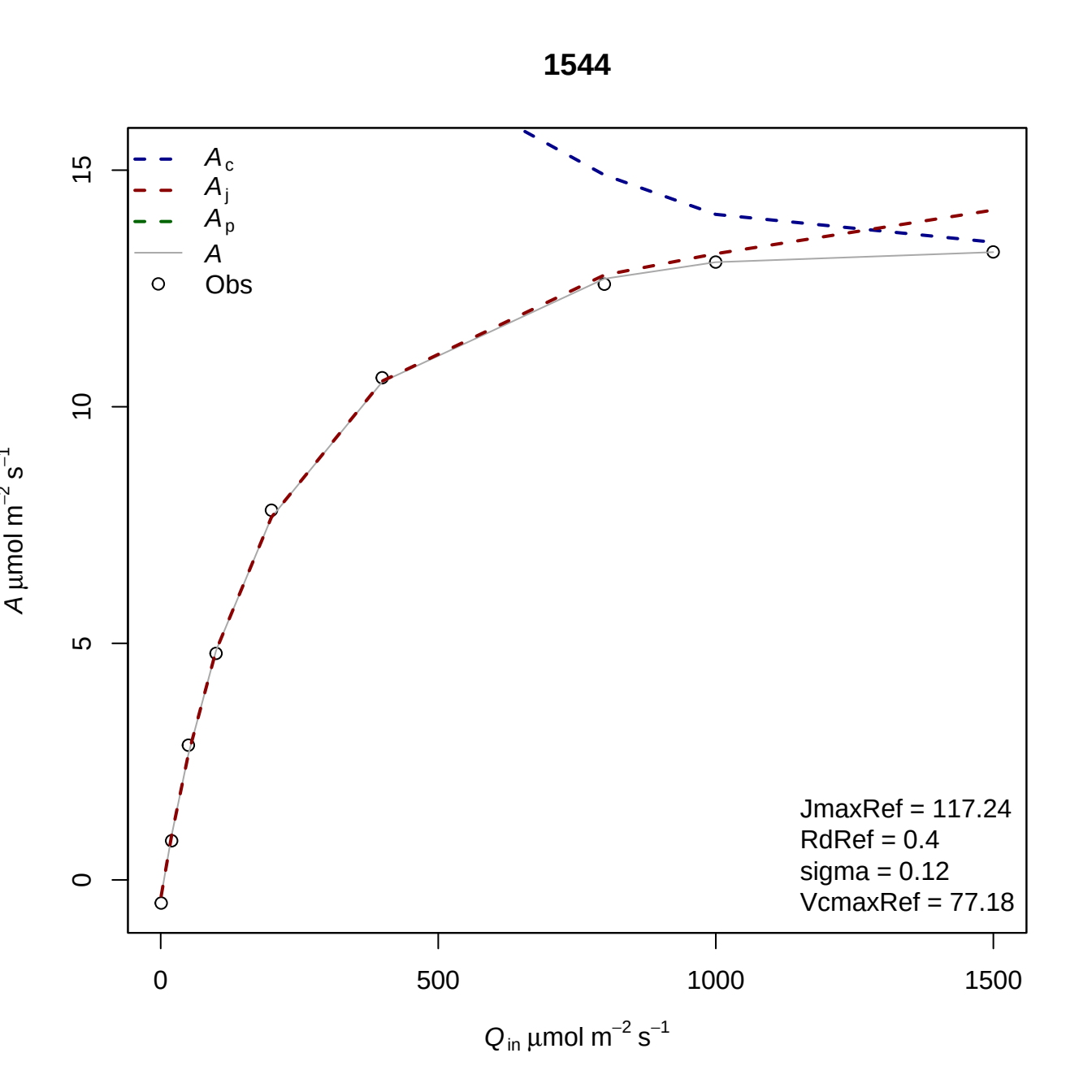

1539
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 97.82
RdRef = 0.41
sigma = 0.12
VcmaxRef = 75.33
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

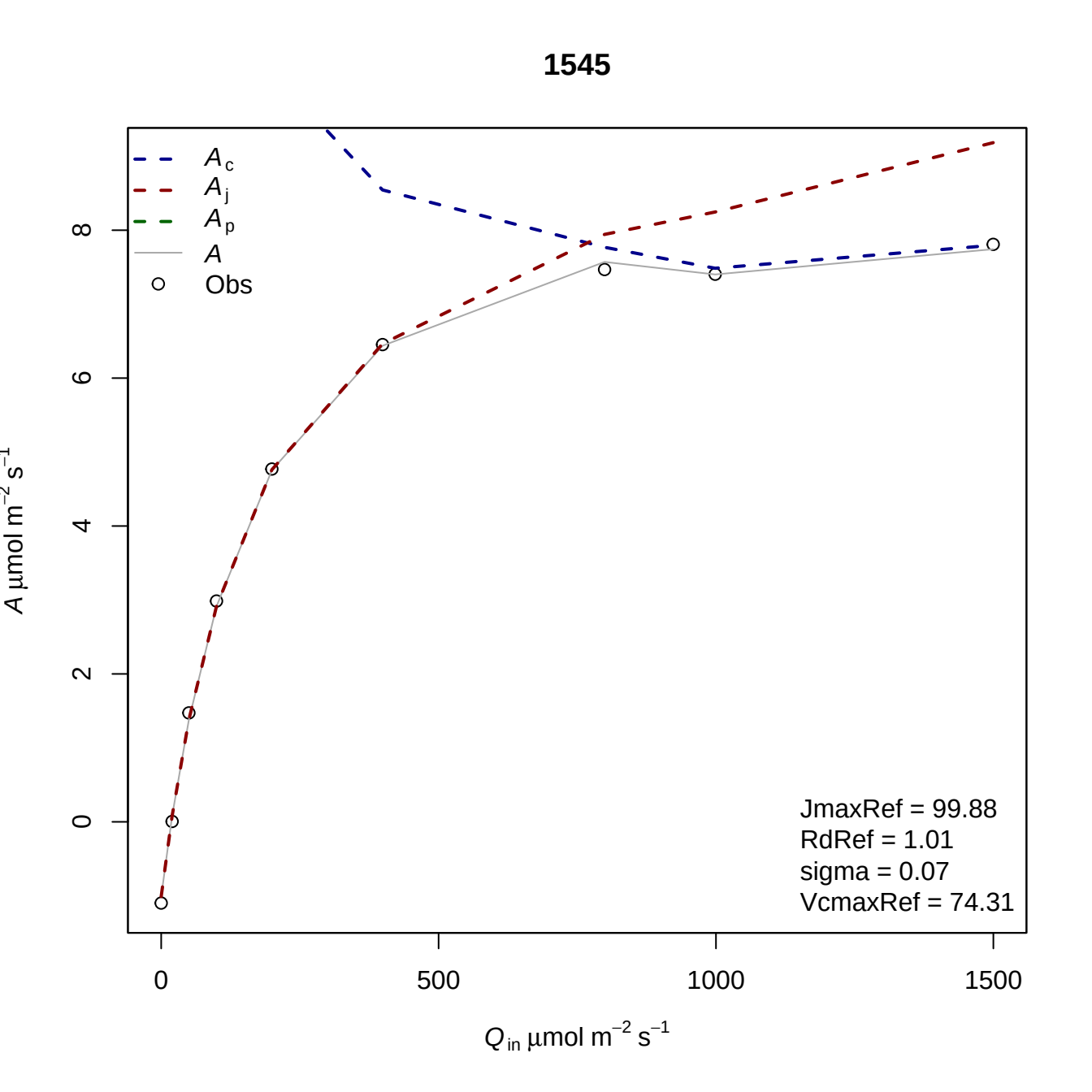

1540
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.26
RdRef = 0.09
sigma = 0.07
VcmaxRef = 57.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

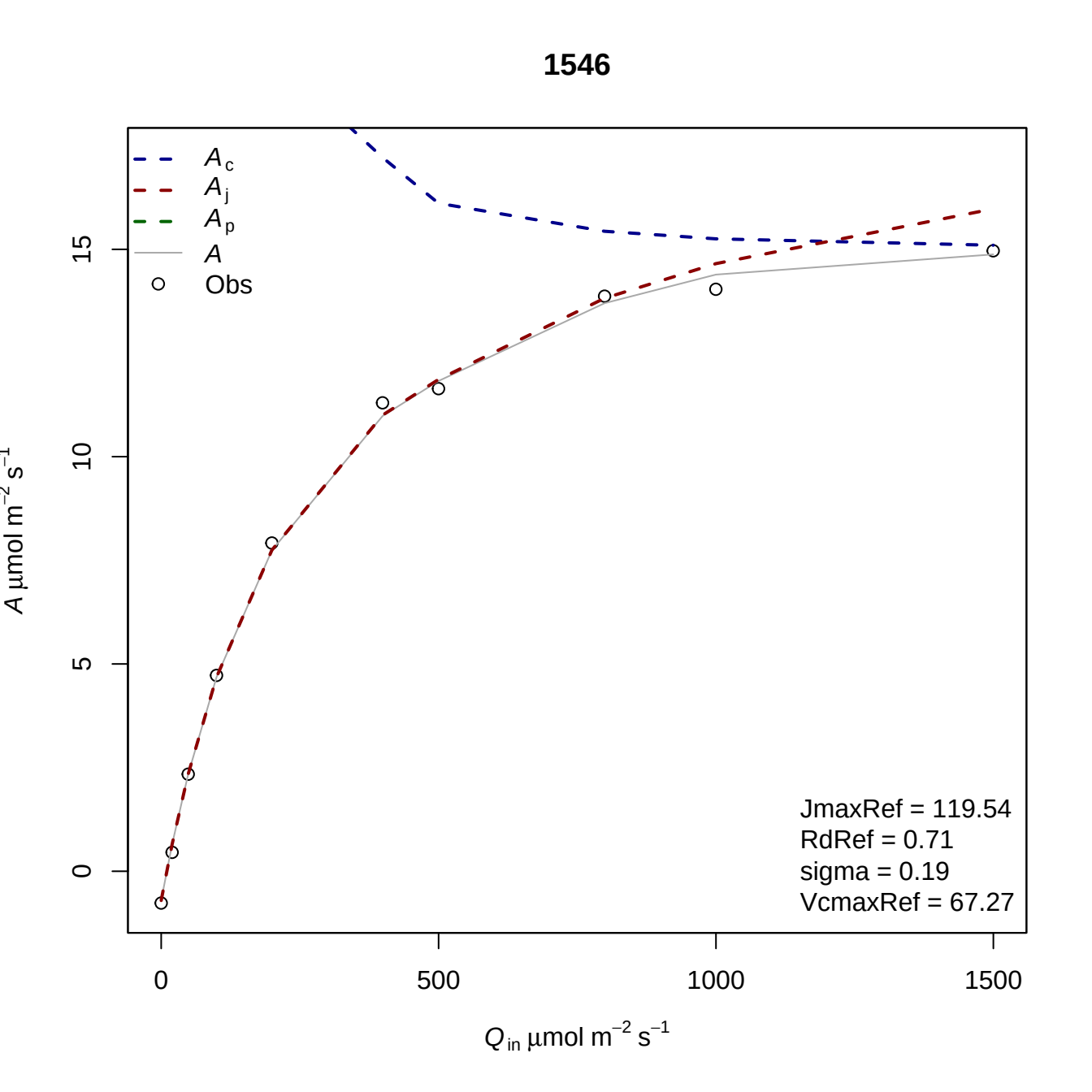

1541
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.73
RdRef = 0.65
sigma = 0.12
VcmaxRef = 60.25
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

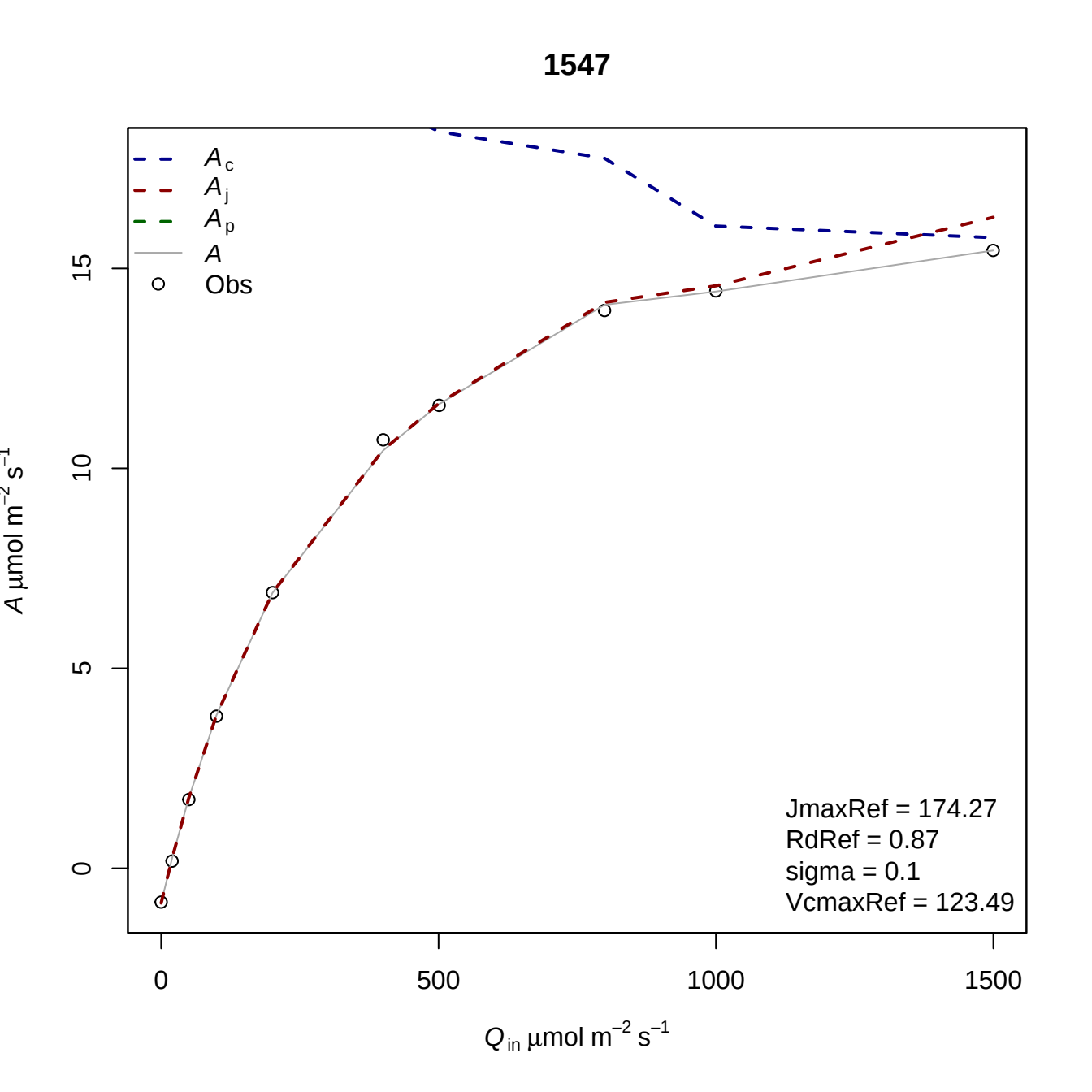

1542
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.01
RdRef = 0.43
sigma = 0.08
VcmaxRef = 659976.2
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

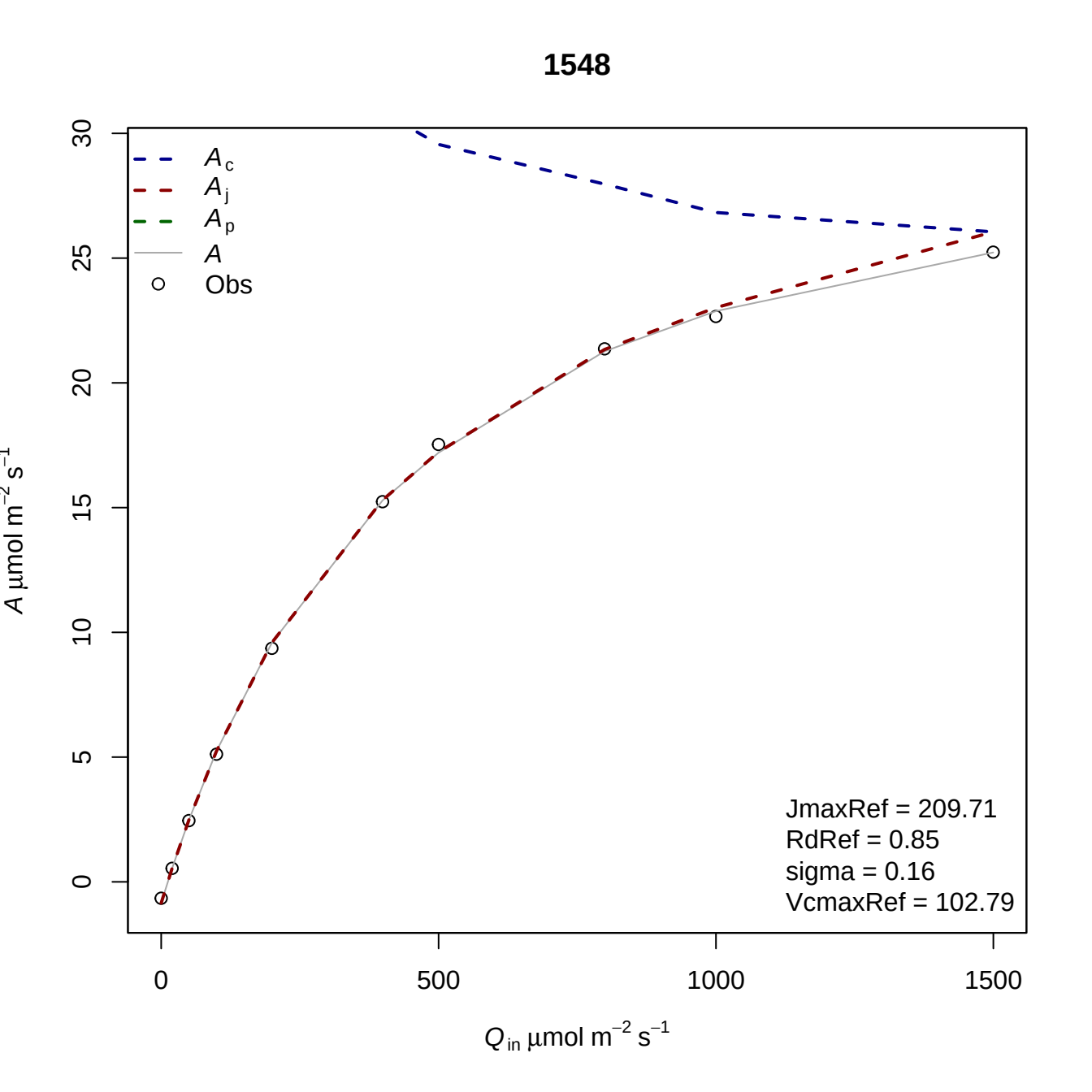

1543
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.21
RdRef = 0.75
sigma = 0.11
VcmaxRef = 161.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

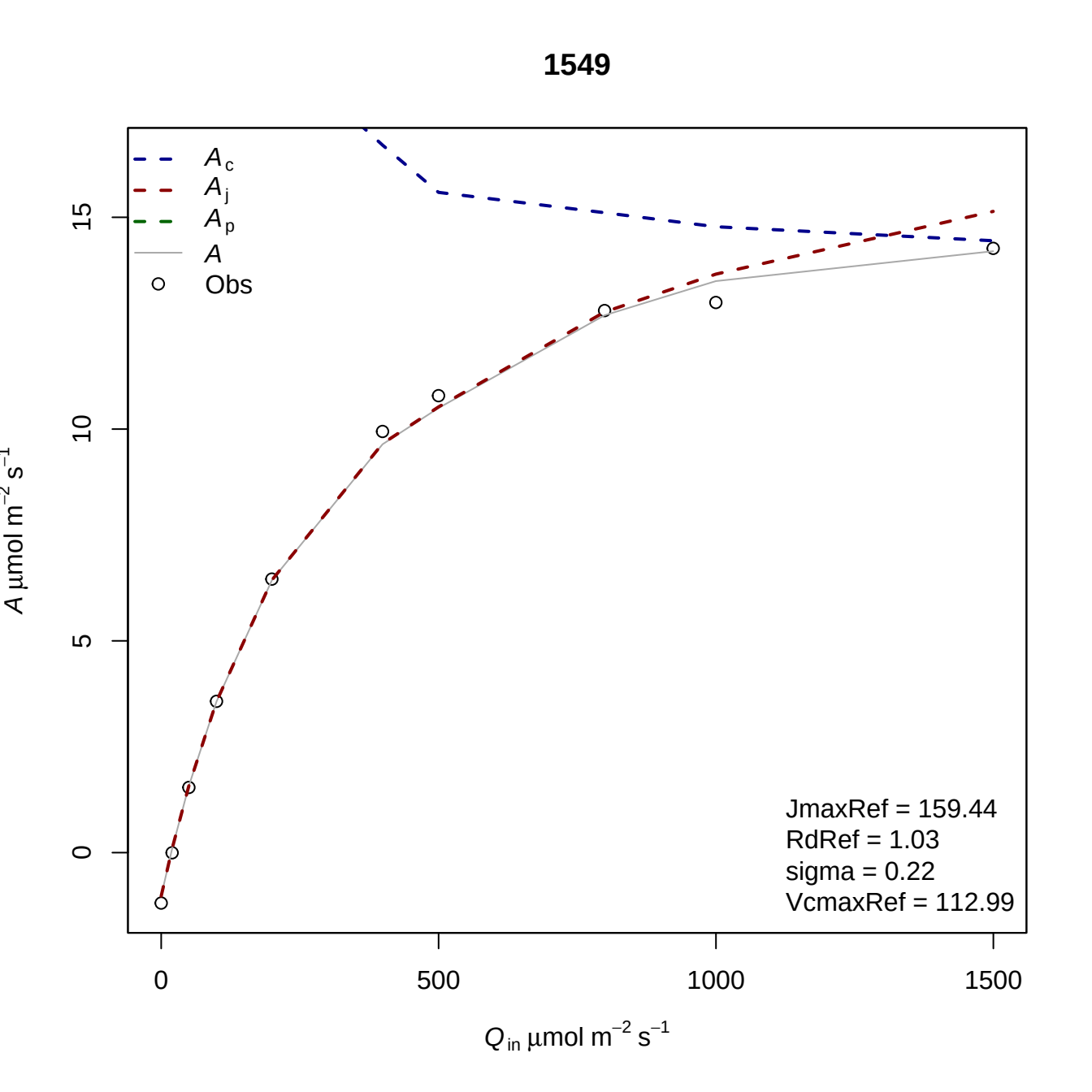

1544
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 117.24
RdRef = 0.4
sigma = 0.12
VcmaxRef = 77.18
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

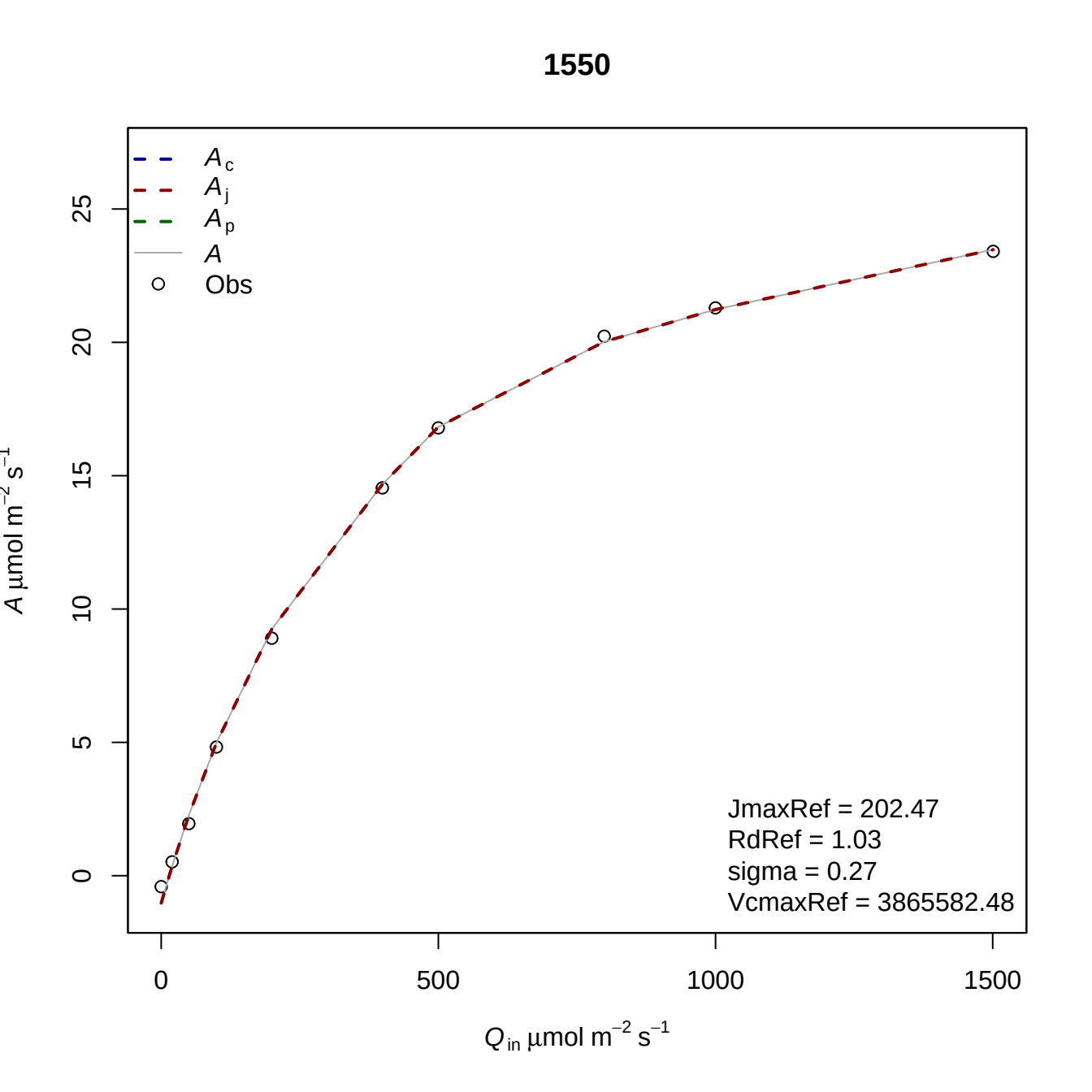

1545
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.88
RdRef = 1.01
sigma = 0.07
VcmaxRef = 74.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

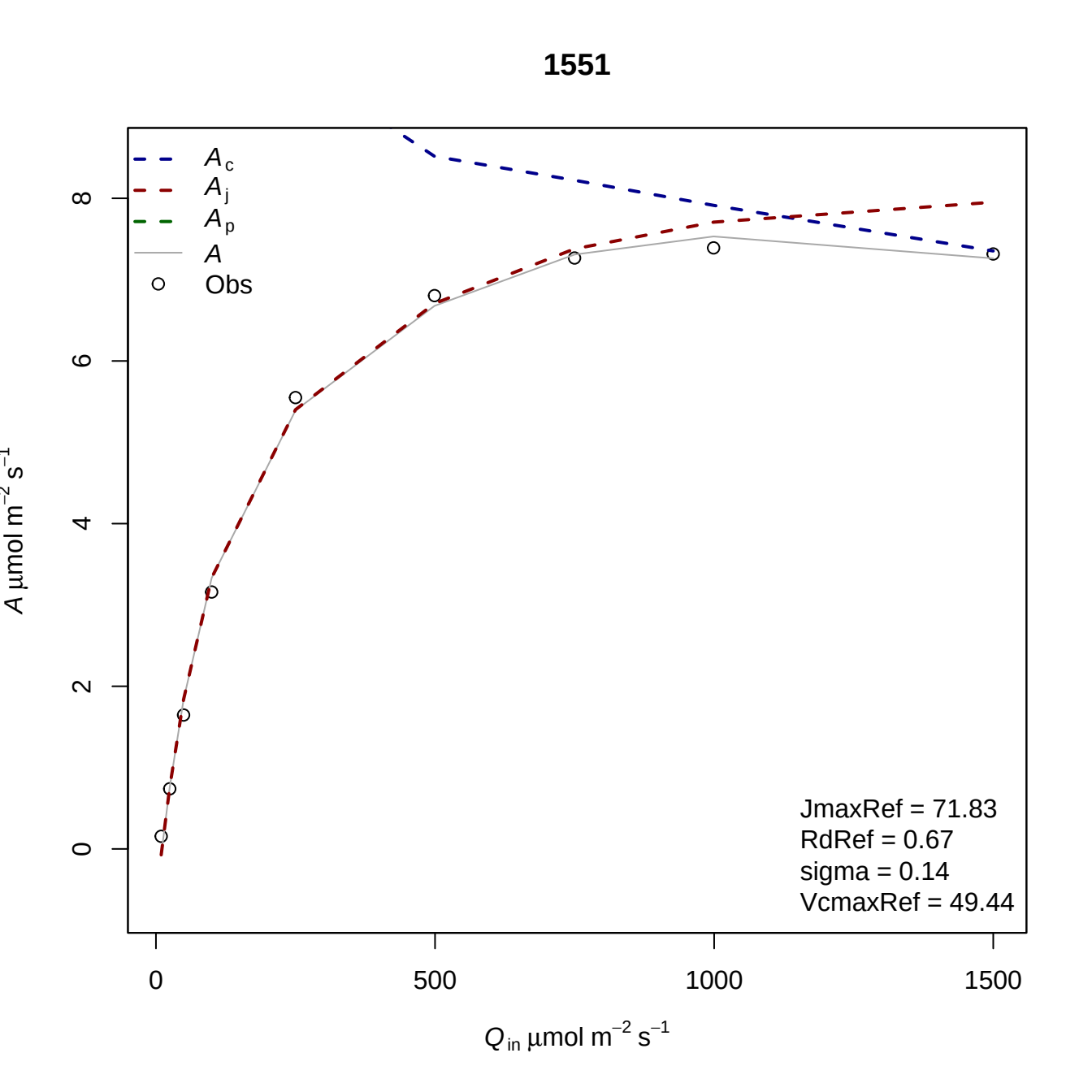

1546
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 119.54
RdRef = 0.71
sigma = 0.19
VcmaxRef = 67.27
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

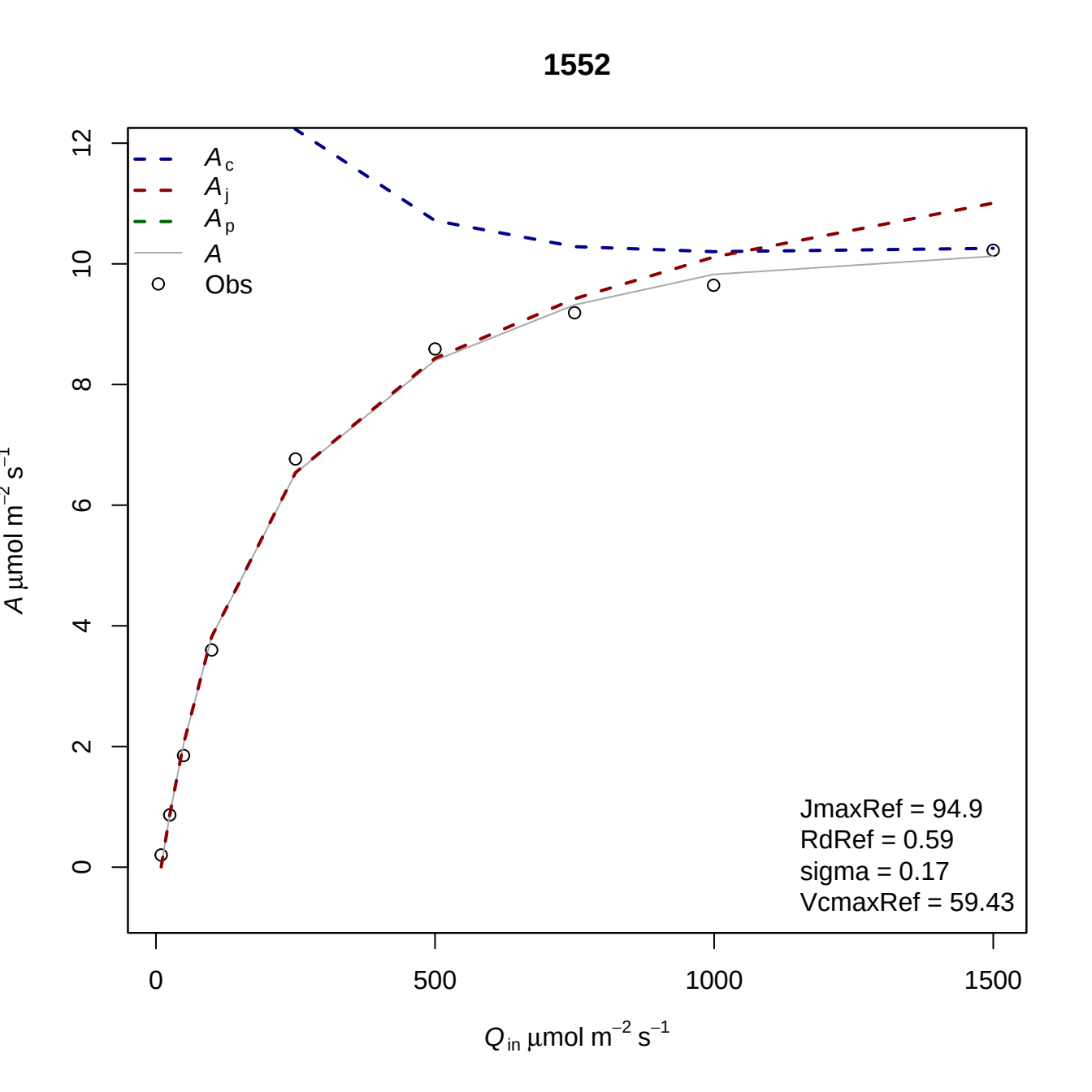

1547
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 174.27
RdRef = 0.87
sigma = 0.1
VcmaxRef = 123.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

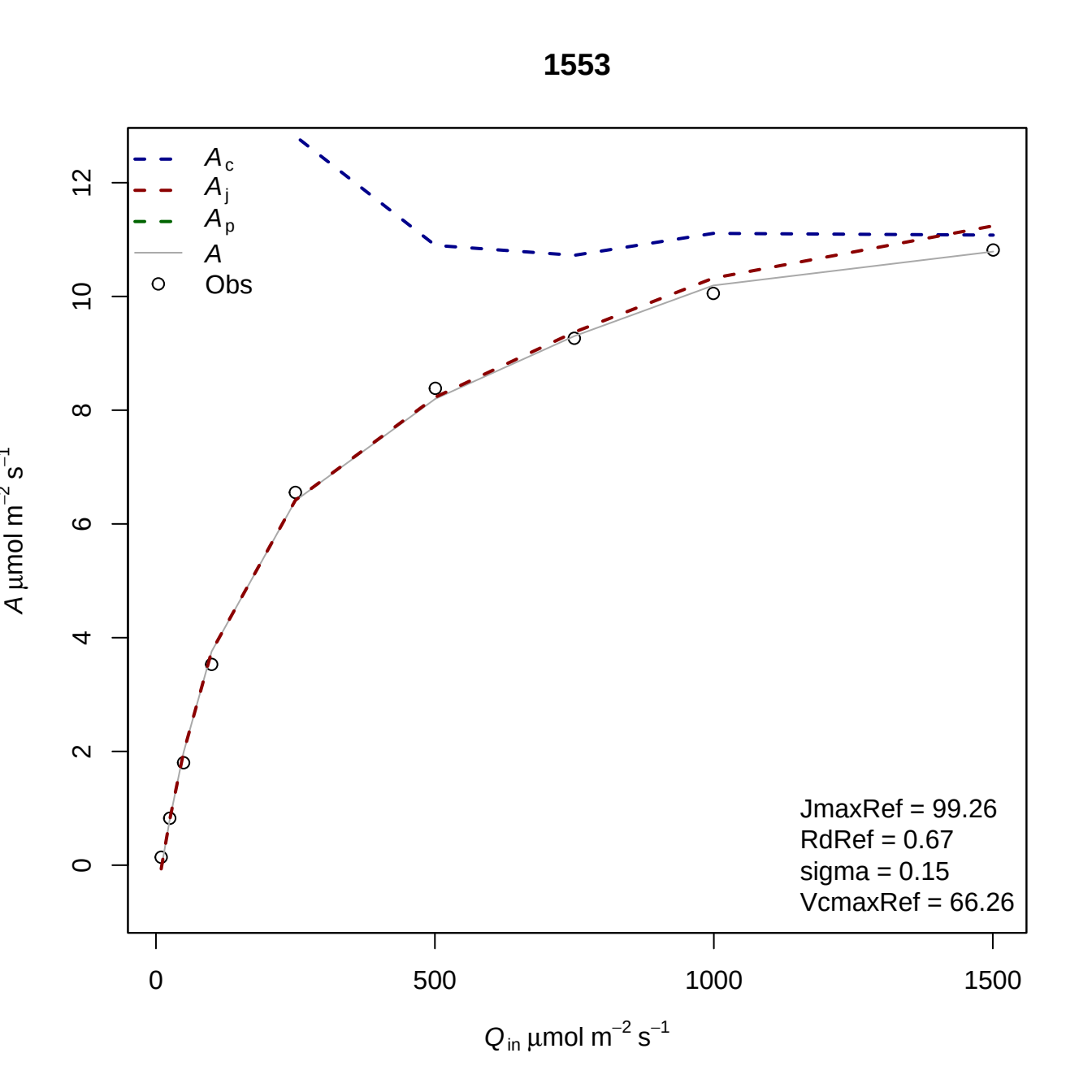

1548
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 209.71
RdRef = 0.85
sigma = 0.16
VcmaxRef = 102.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500

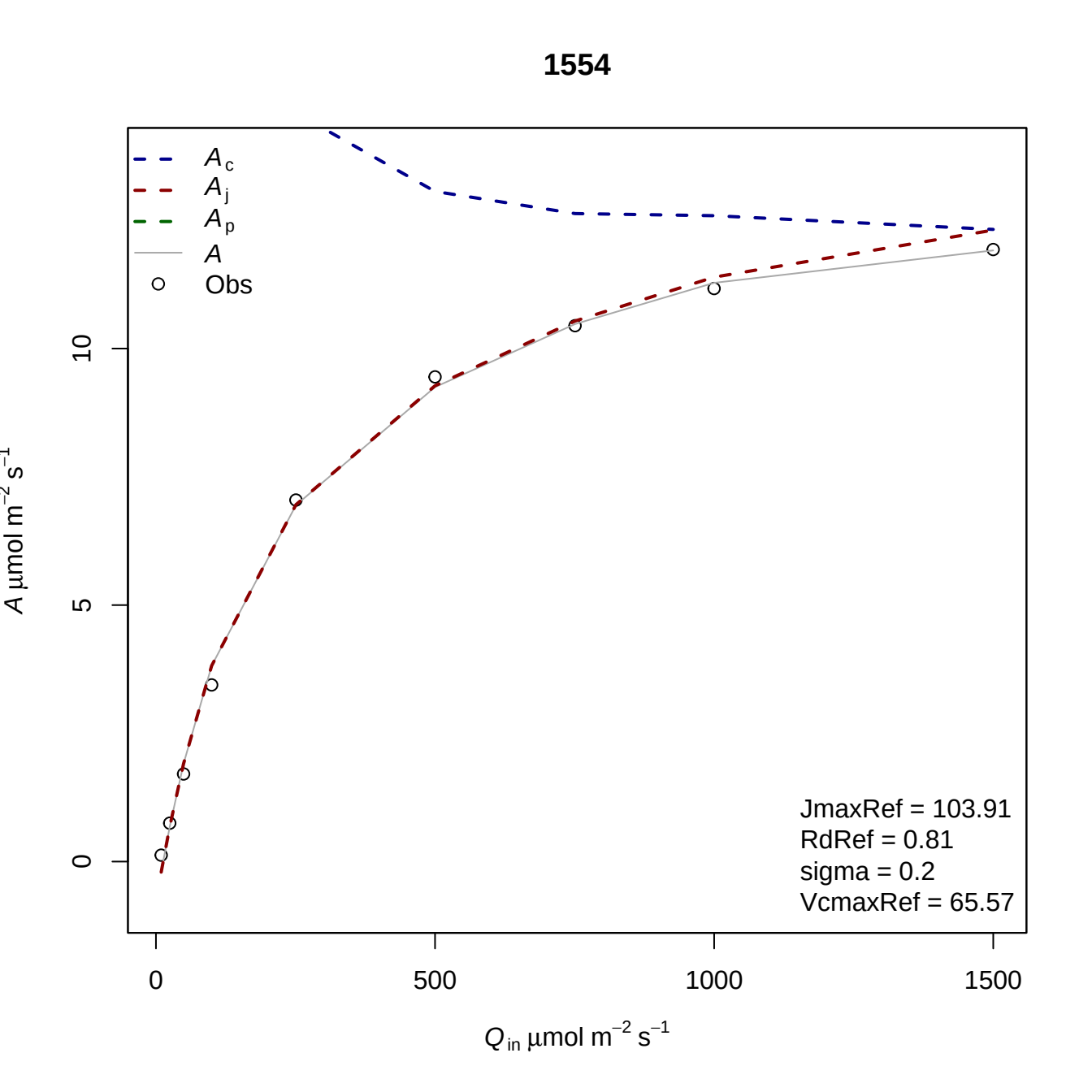

1549
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 159.44
RdRef = 1.03
sigma = 0.22
VcmaxRef = 112.99
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

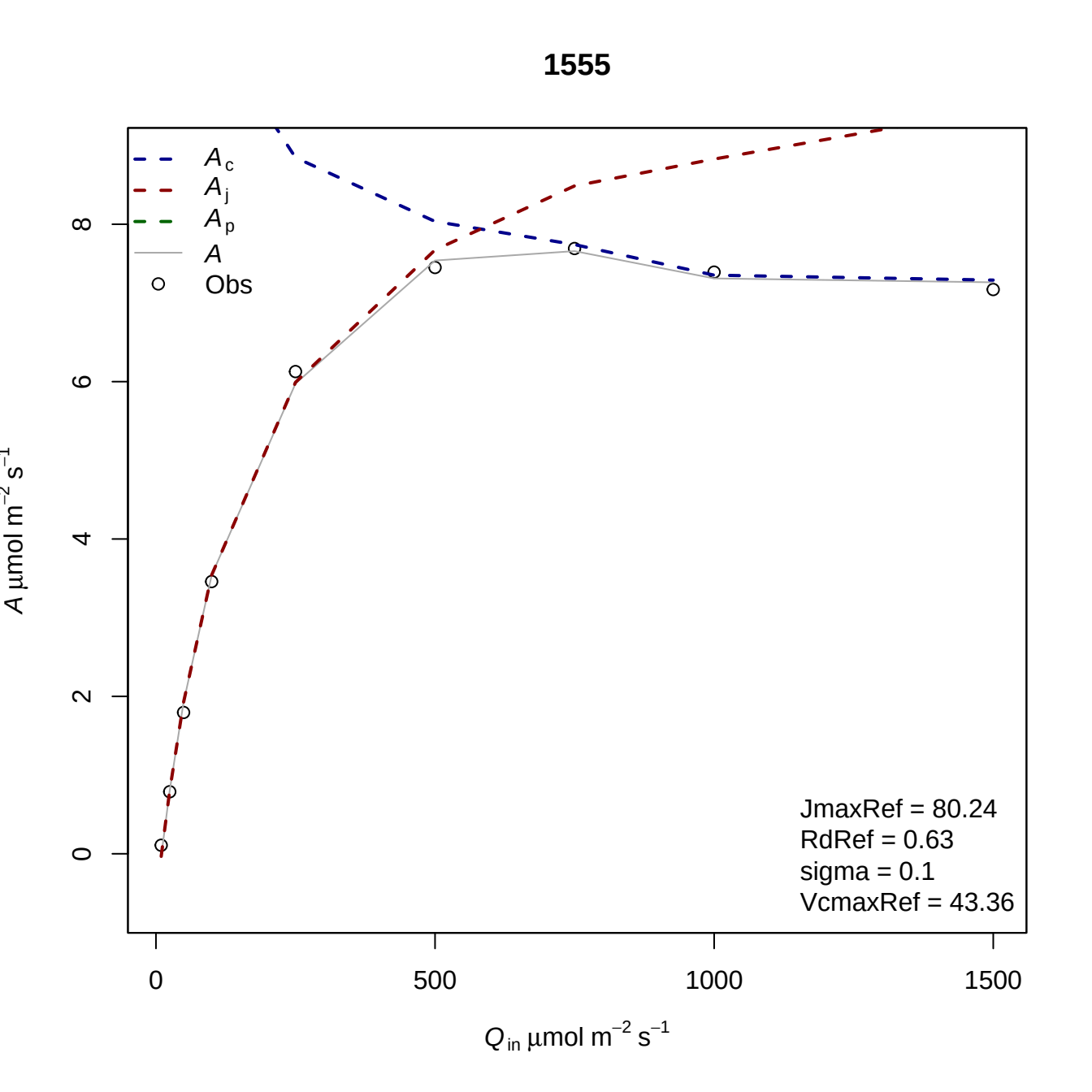

1550
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 202.47
RdRef = 1.03
sigma = 0.27
VcmaxRef = 3865582.48
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500

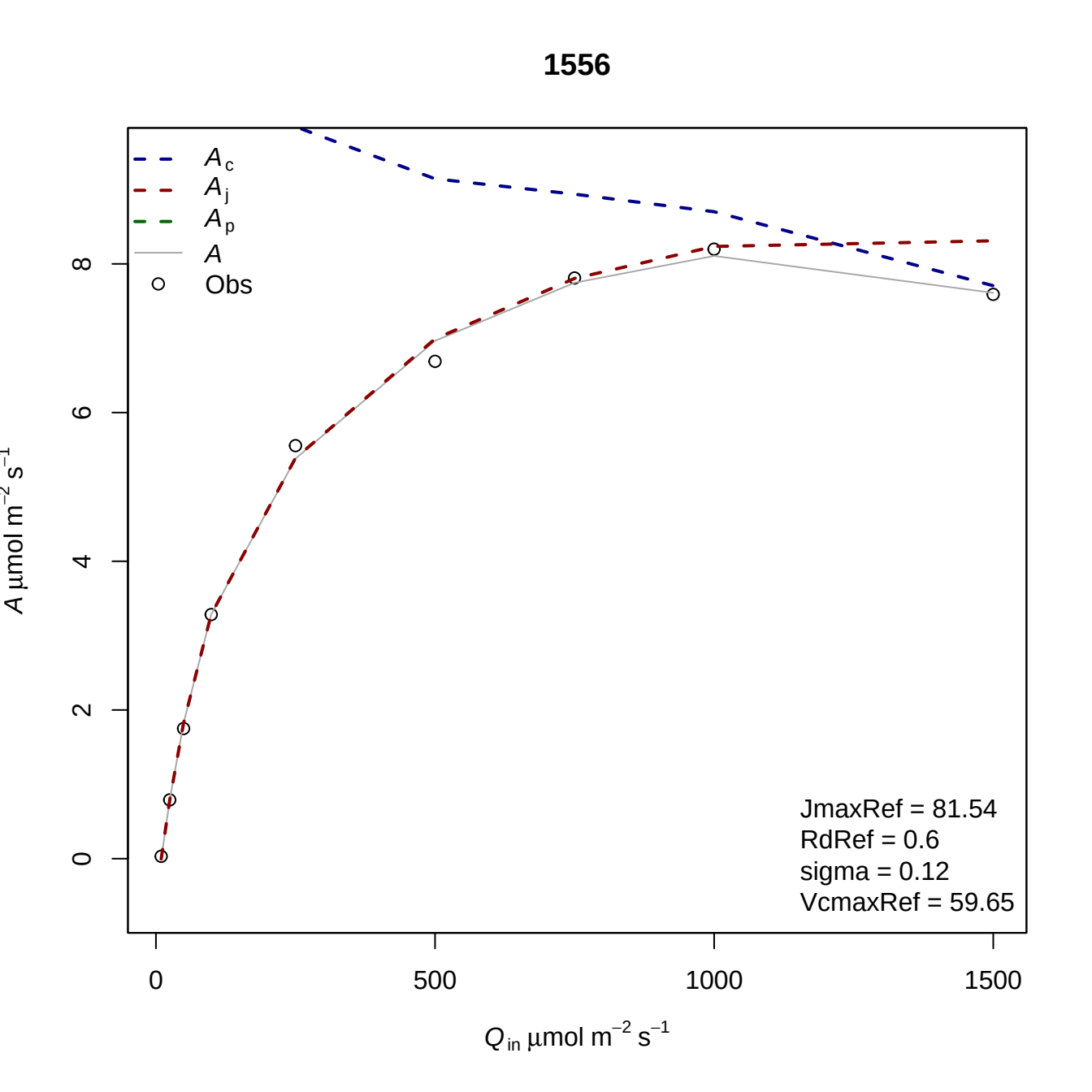

1551
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 71.83
RdRef = 0.67
sigma = 0.14
VcmaxRef = 49.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

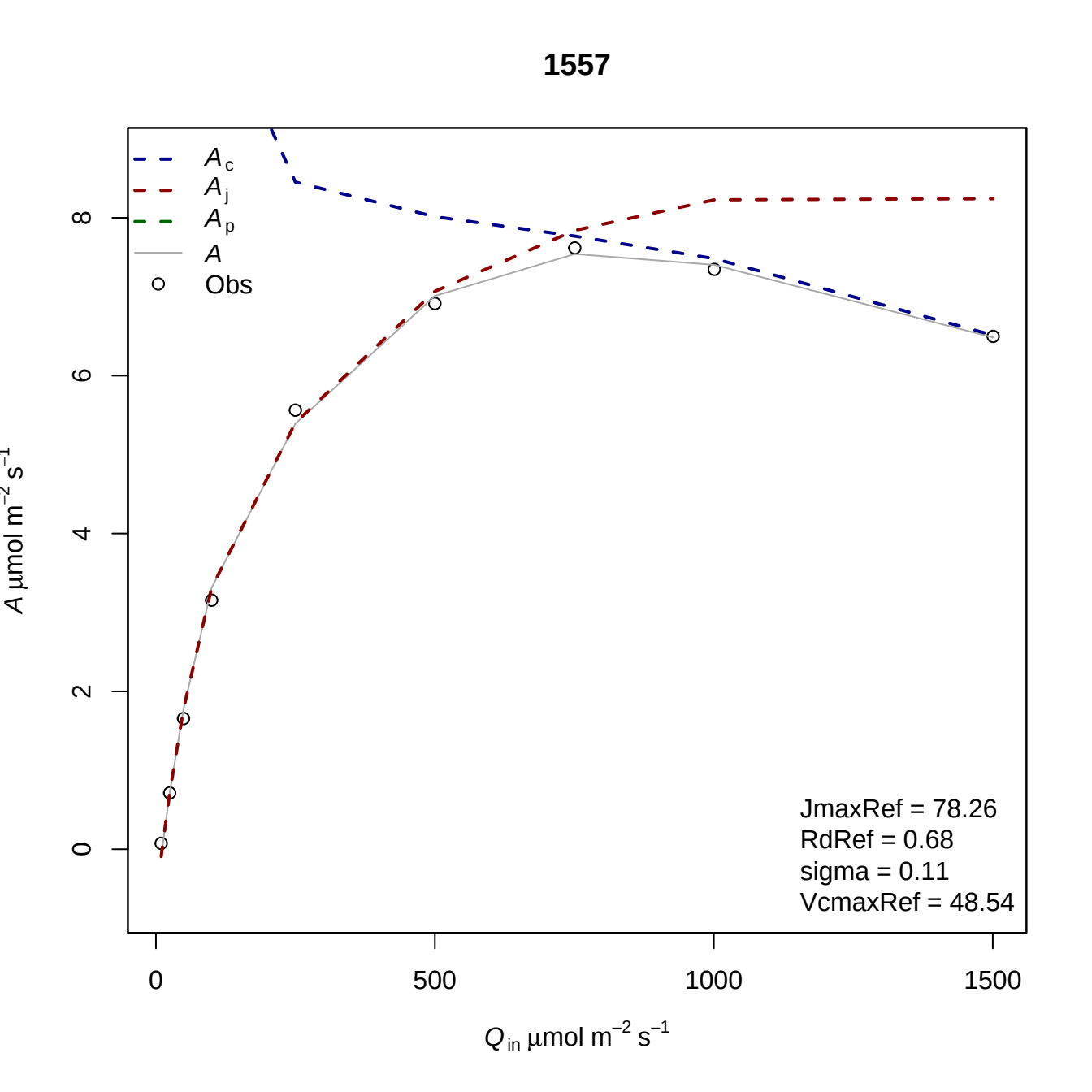

1552
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.9
RdRef = 0.59
sigma = 0.17
VcmaxRef = 59.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

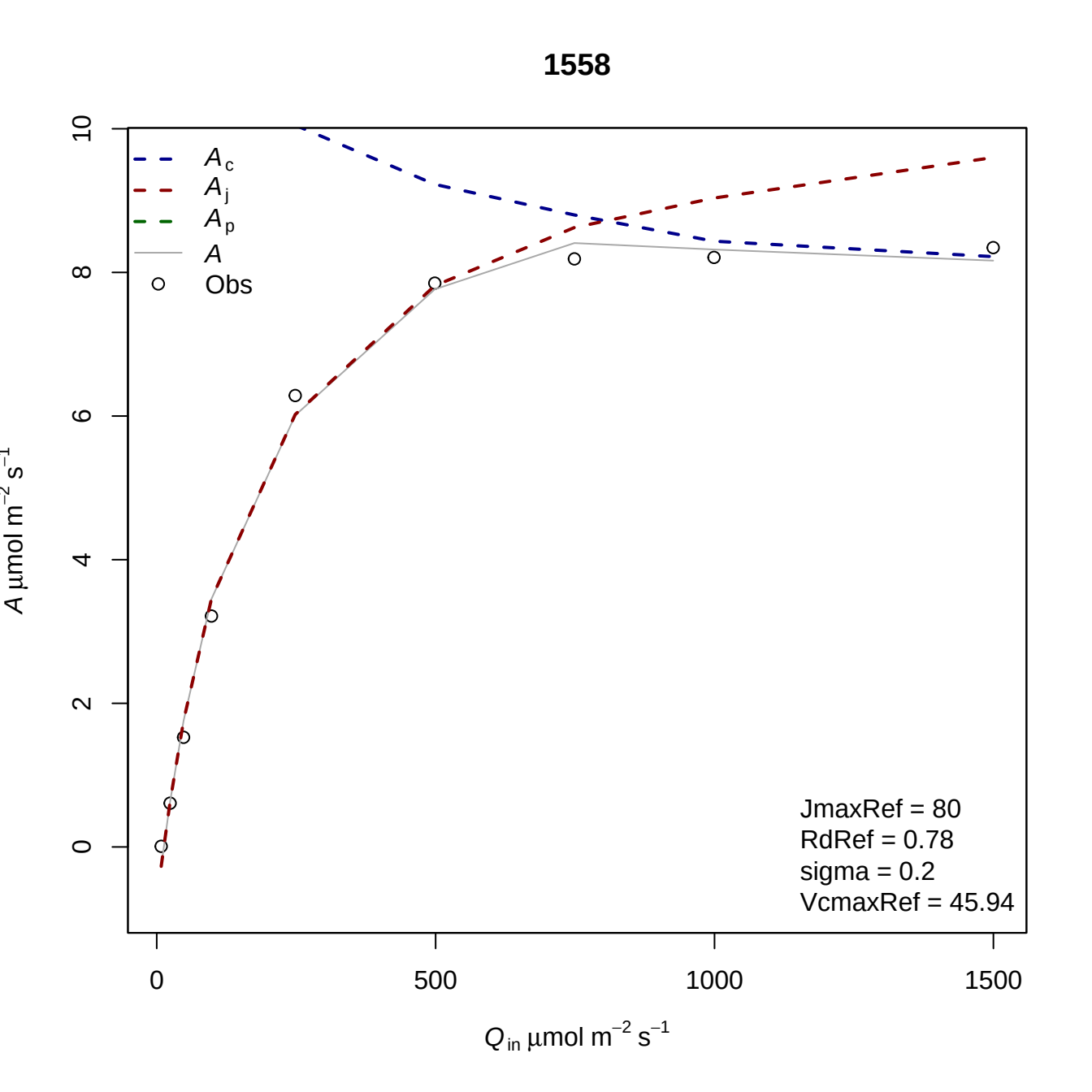

1553
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 99.26
RdRef = 0.67
sigma = 0.15
VcmaxRef = 66.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

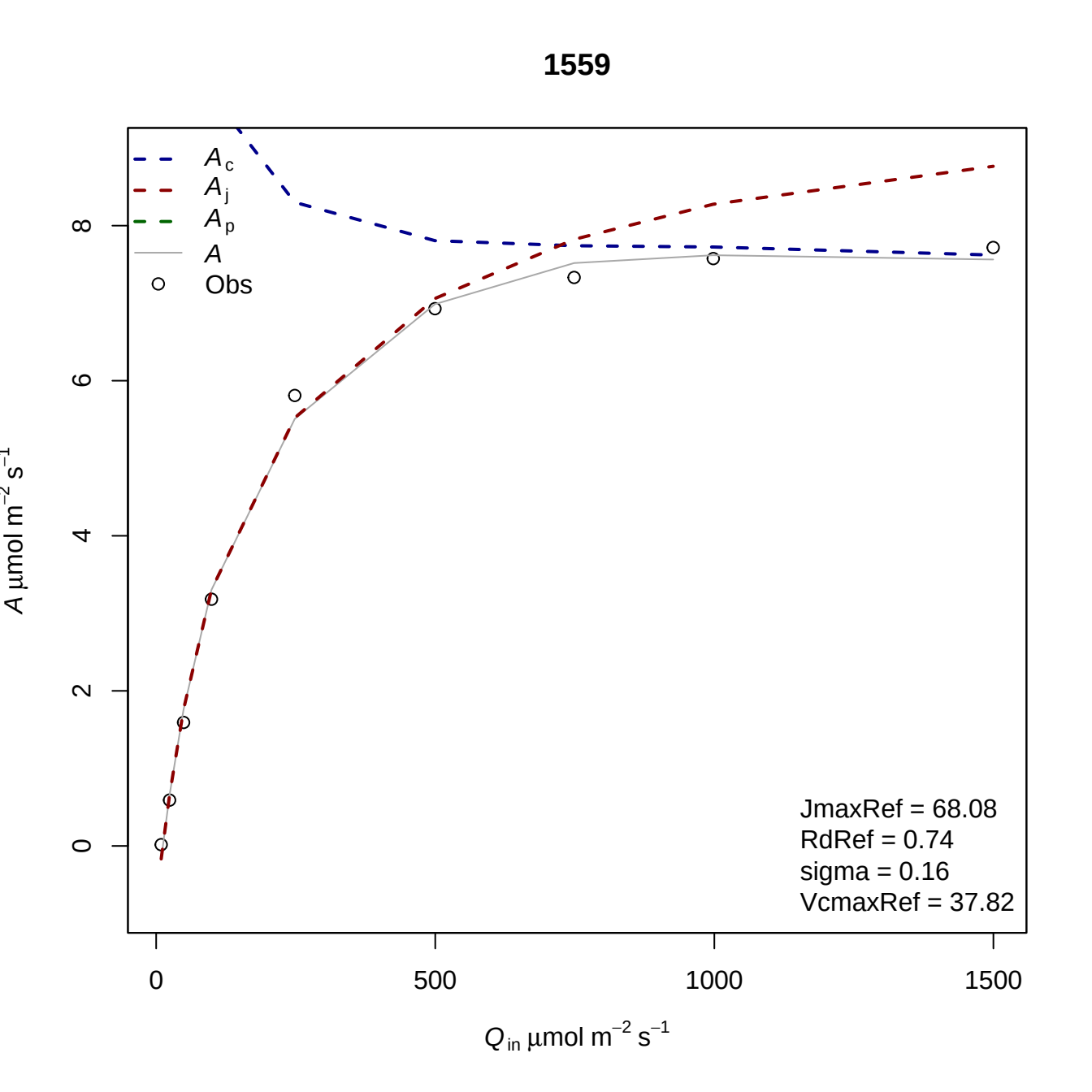

1554
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 103.91
RdRef = 0.81
sigma = 0.2
VcmaxRef = 65.57
$A$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

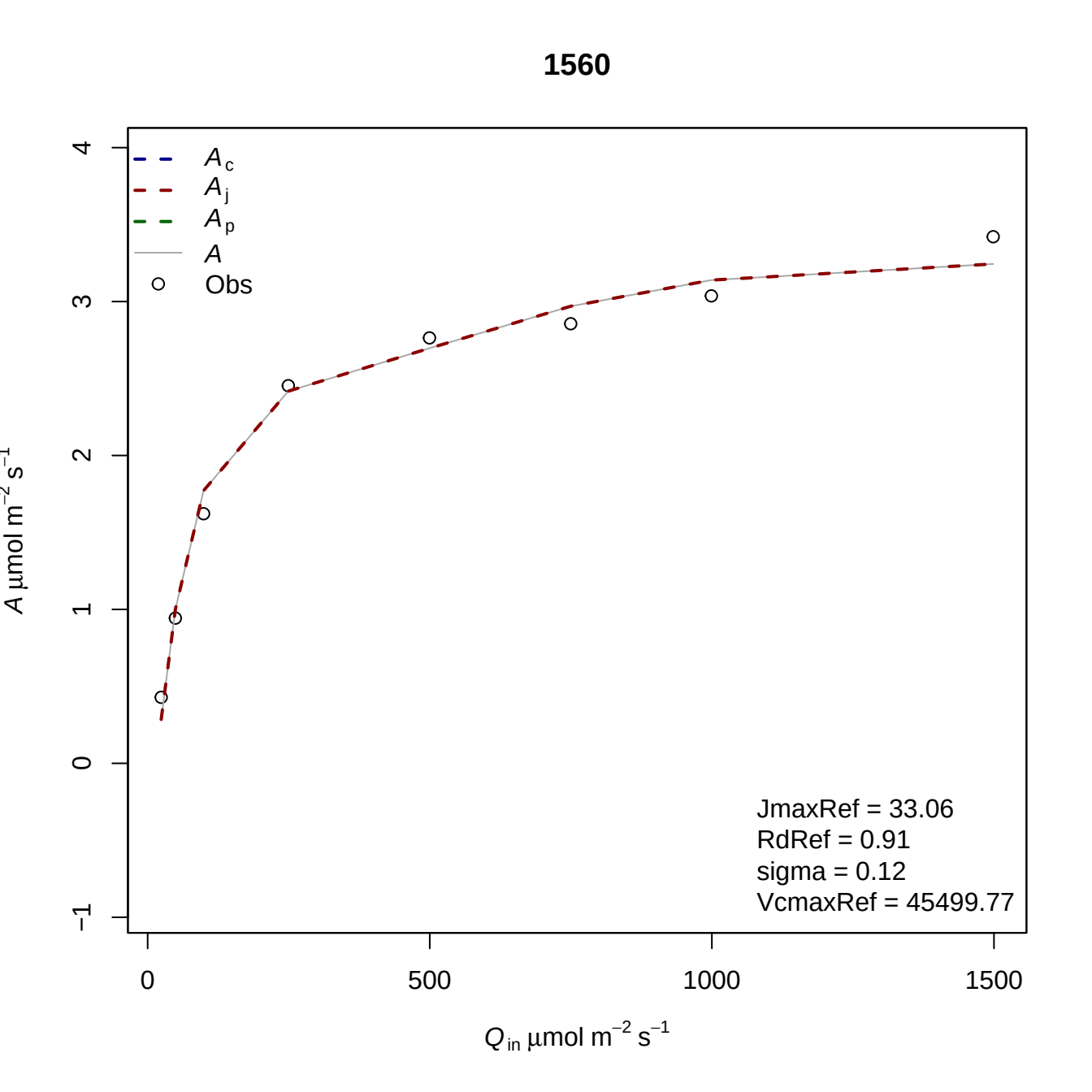

1555
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.24
RdRef = 0.63
sigma = 0.1
VcmaxRef = 43.36
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

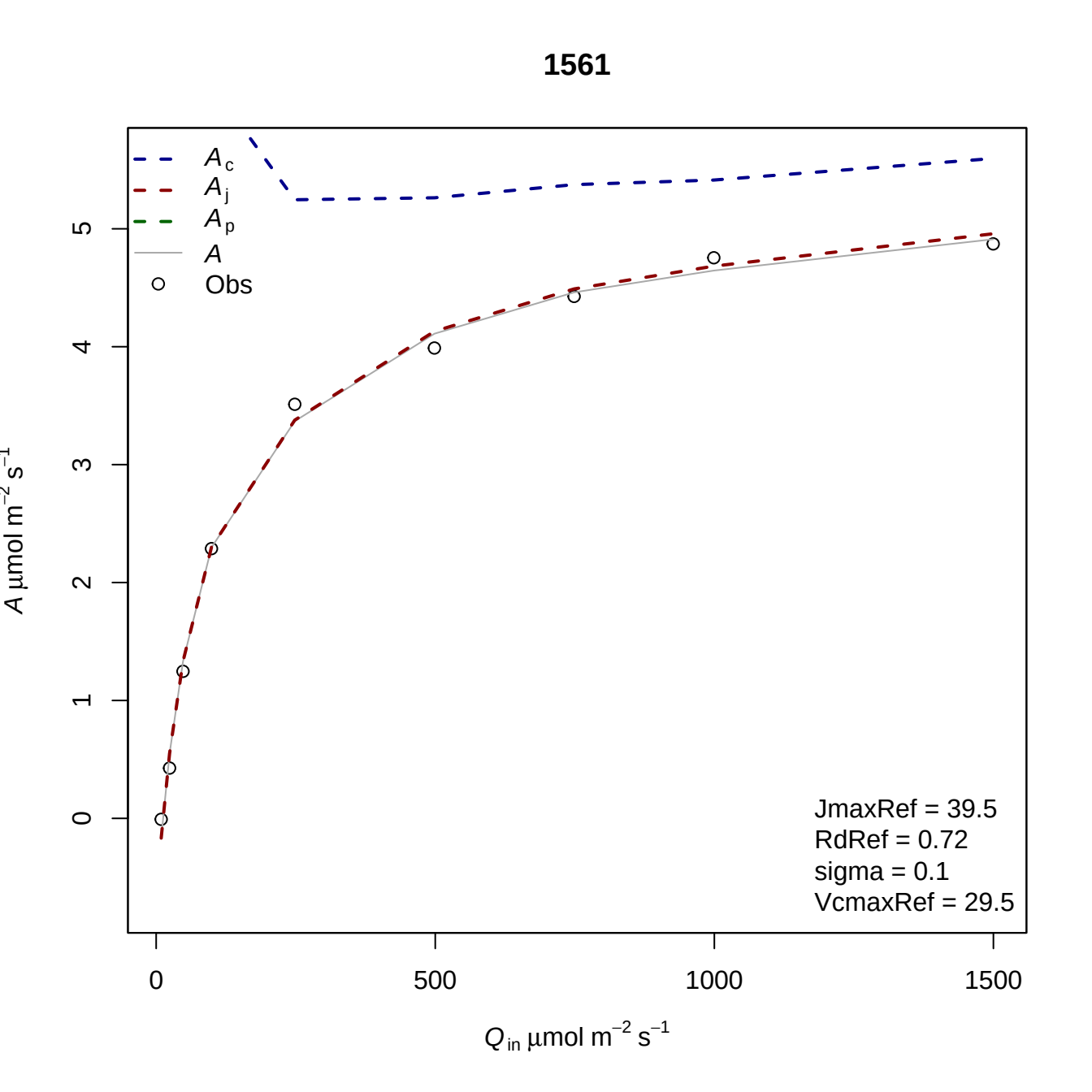

1556
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 81.54
RdRef = 0.6
sigma = 0.12
VcmaxRef = 59.65
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

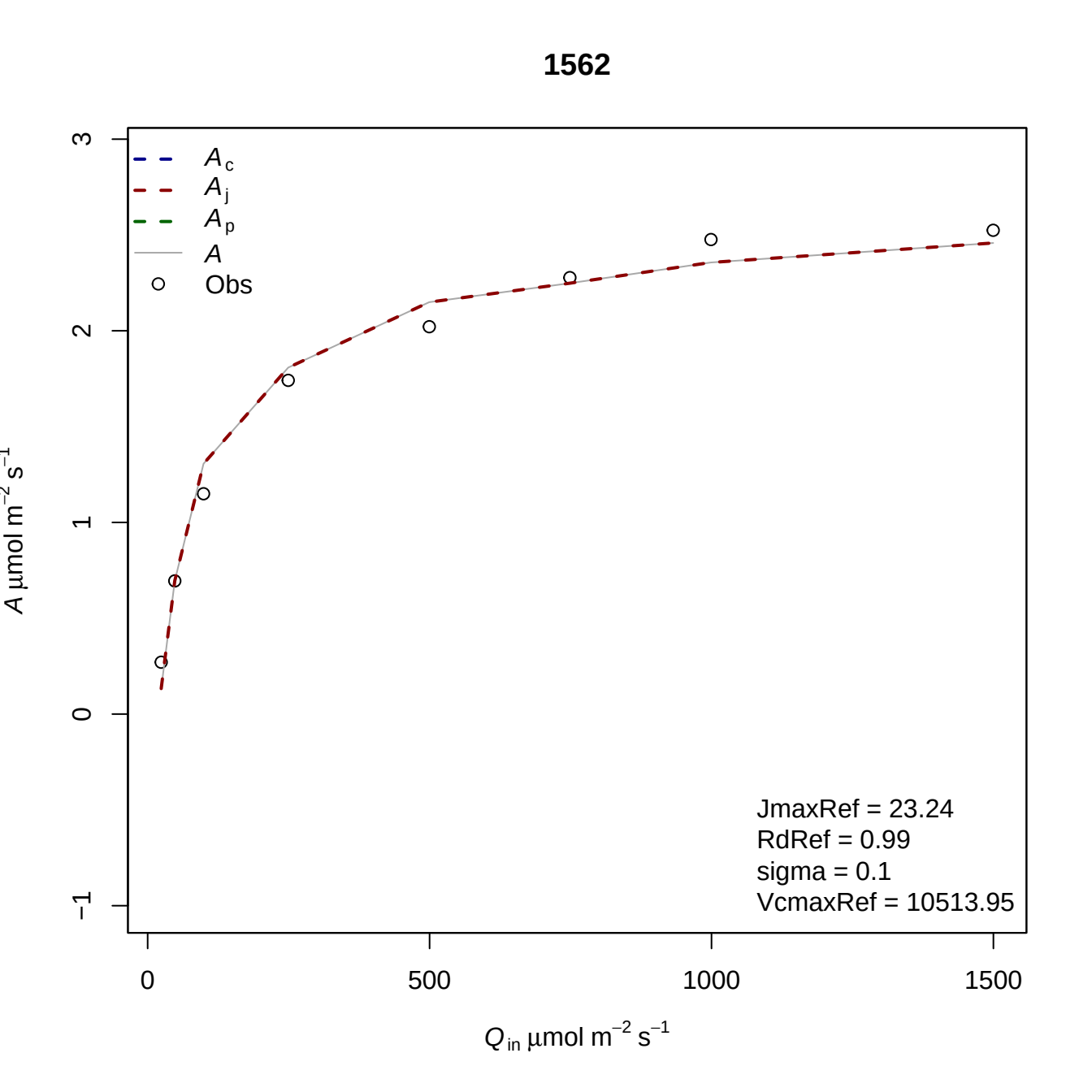

1557
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 78.26
RdRef = 0.68
sigma = 0.11
VcmaxRef = 48.54
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

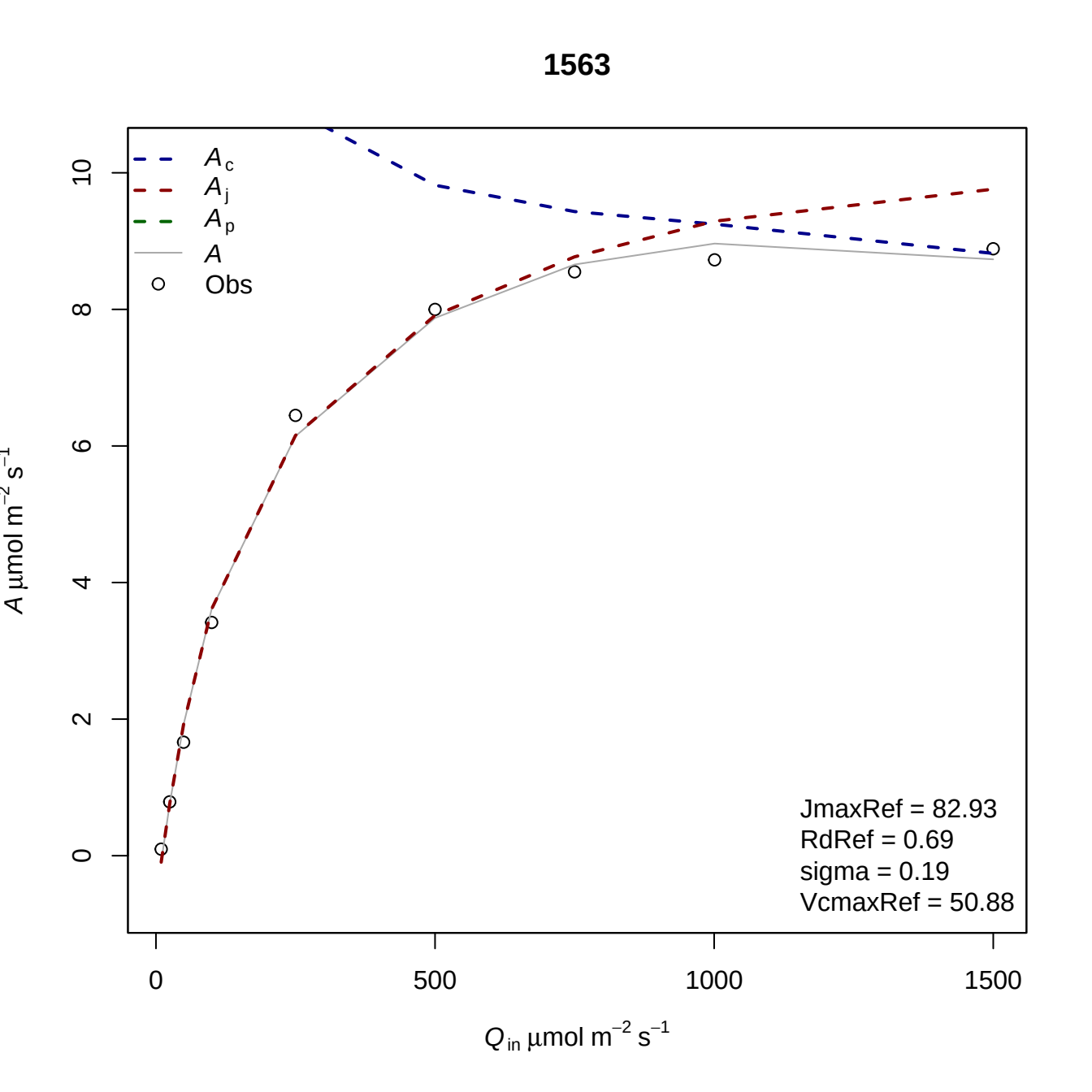

1558
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80
RdRef = 0.78
sigma = 0.2
VcmaxRef = 45.94
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

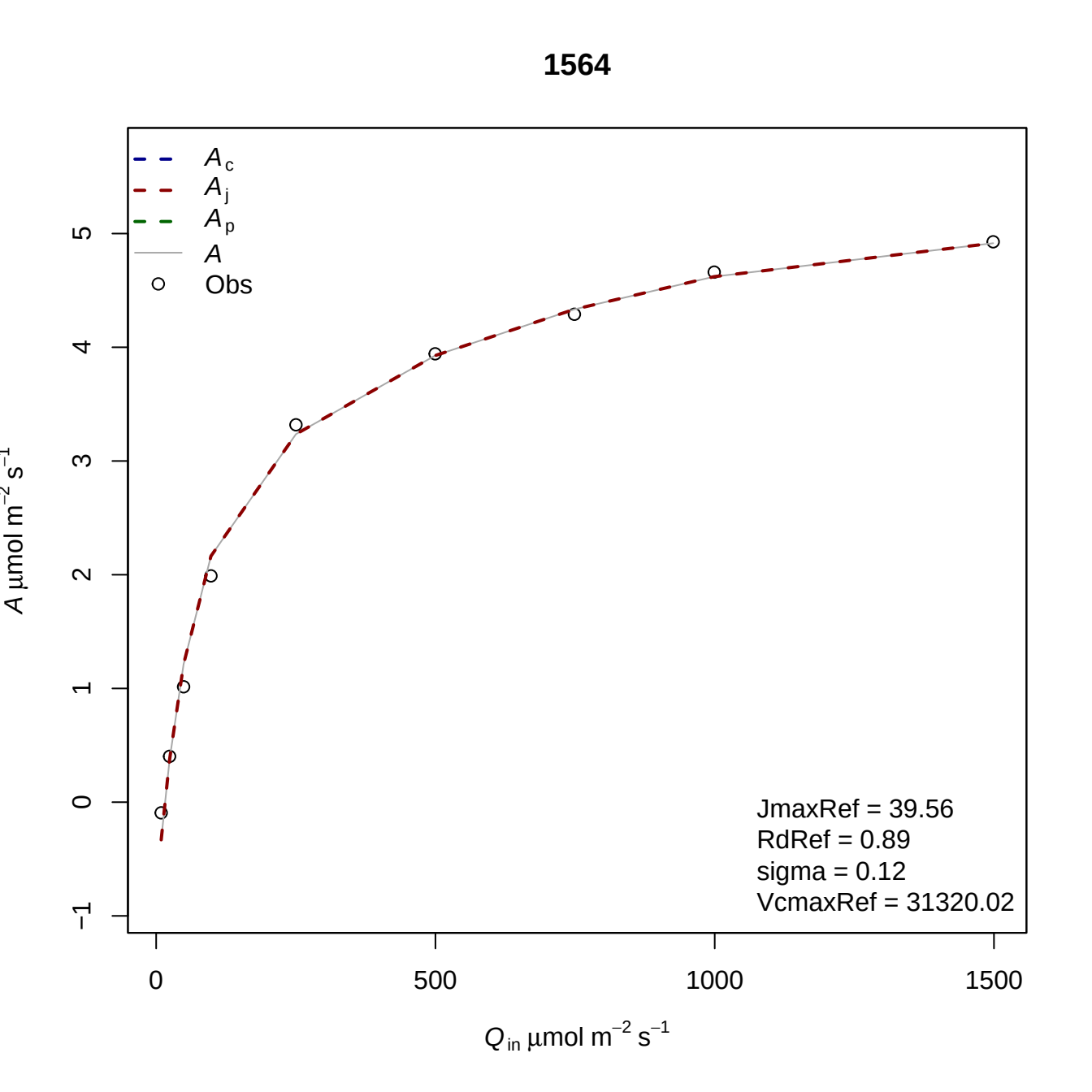

1559
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 68.08
RdRef = 0.74
sigma = 0.16
VcmaxRef = 37.82
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

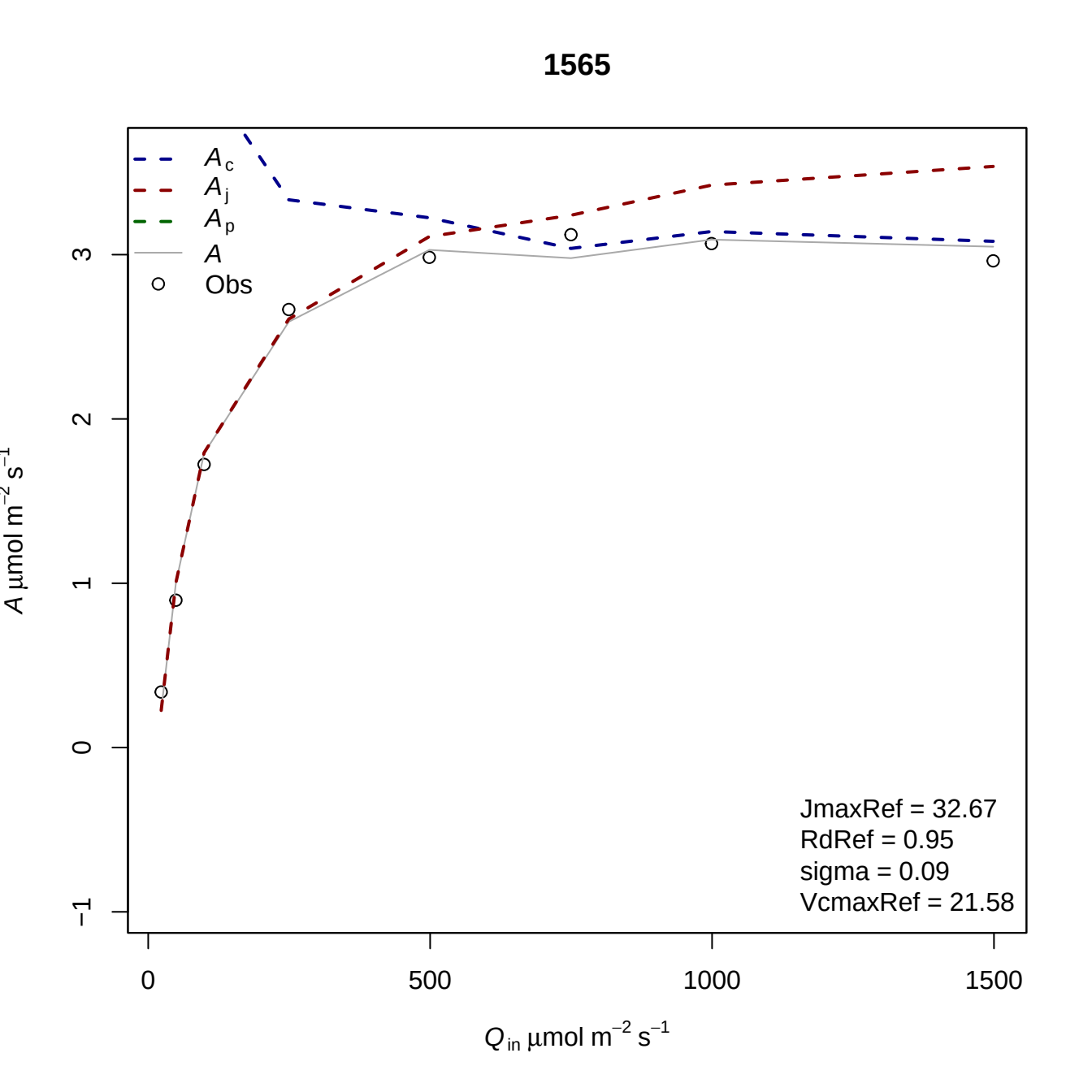

1560
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 33.06
RdRef = 0.91
sigma = 0.12
VcmaxRef = 45499.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
4
3
2
1
0
−1
0
500
1000
1500

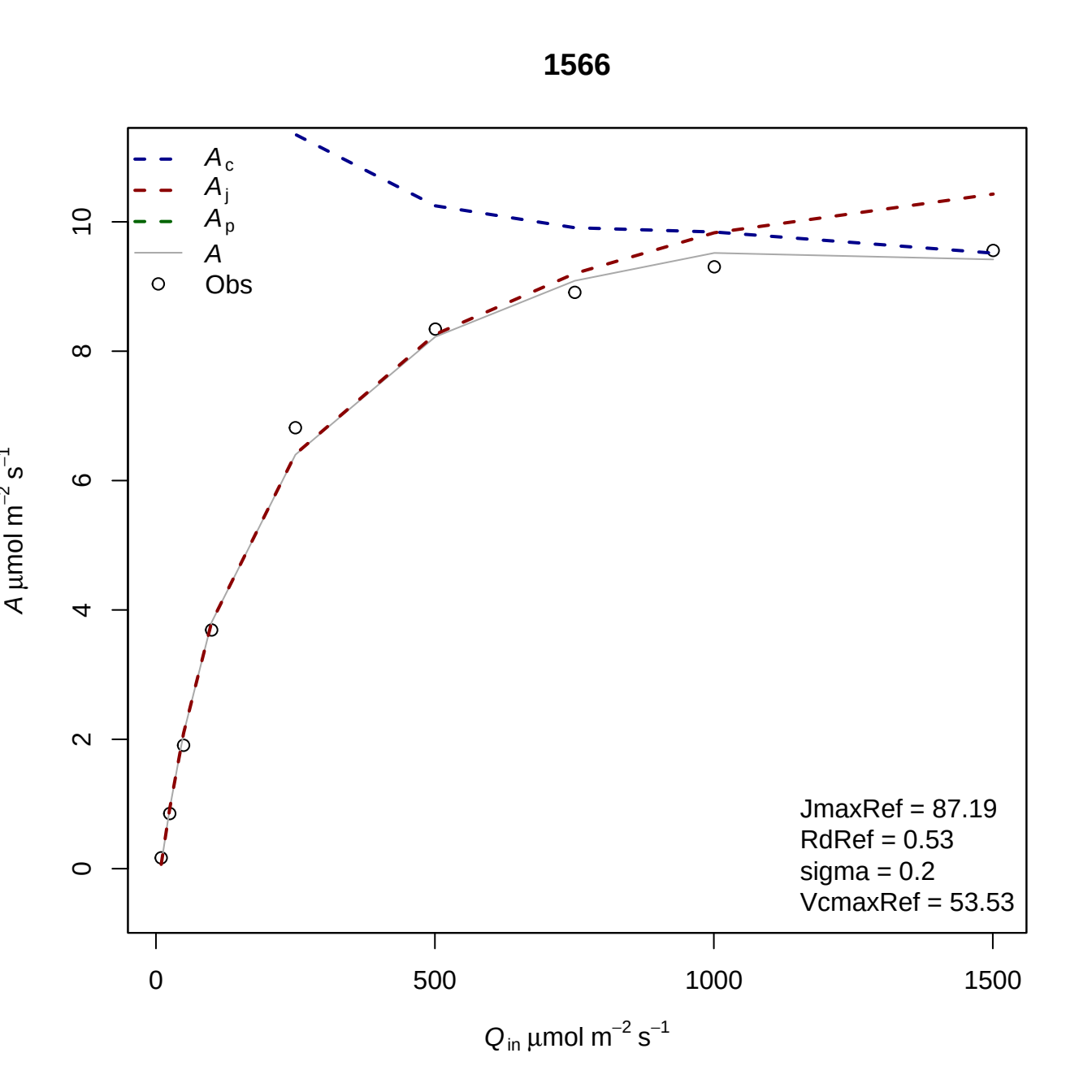

1561
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 39.5
RdRef = 0.72
sigma = 0.1
VcmaxRef = 29.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
0
500
1000
1500

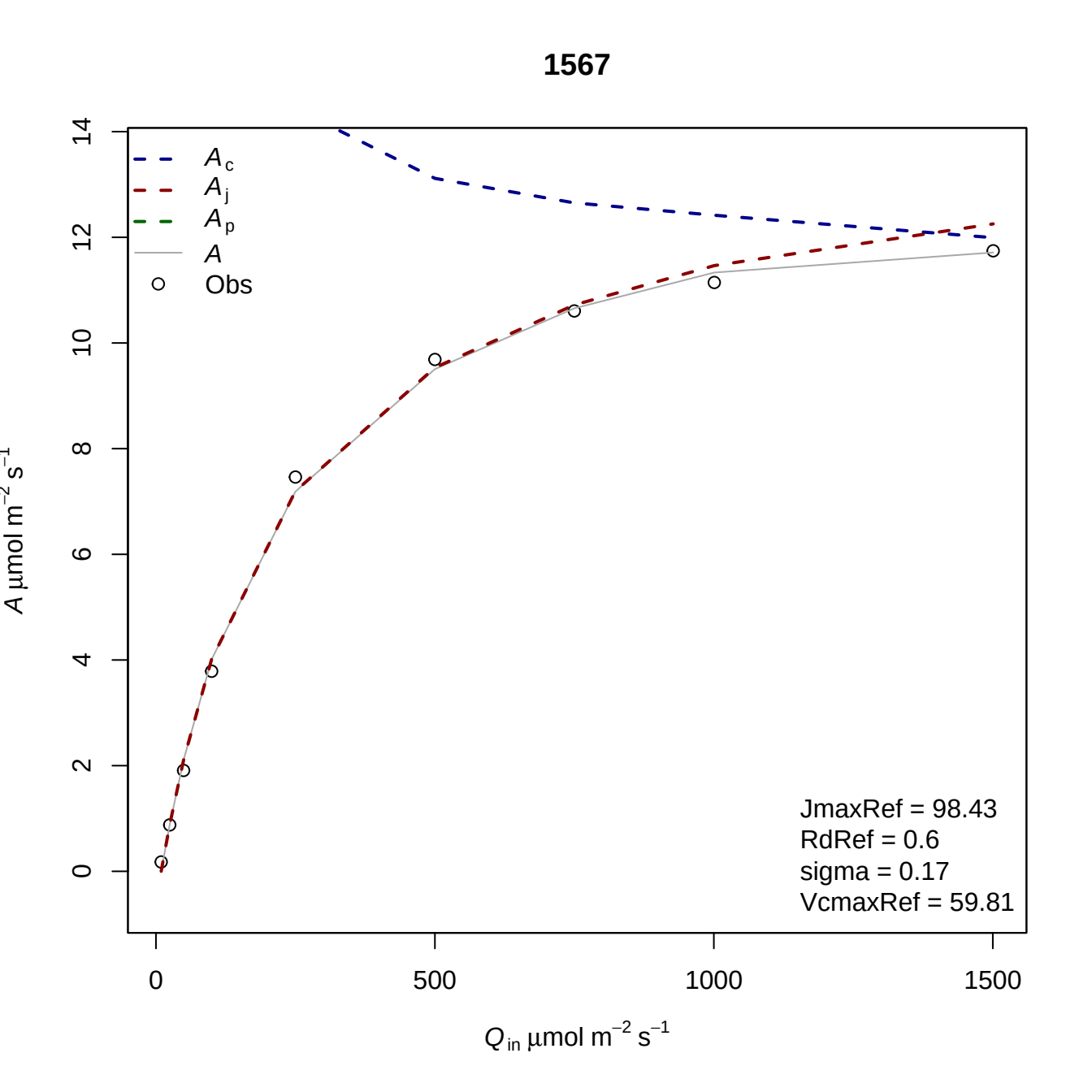

1562
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 23.24
RdRef = 0.99
sigma = 0.1
VcmaxRef = 10513.95
3
2
1
0
−1
0
500
1000
1500
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

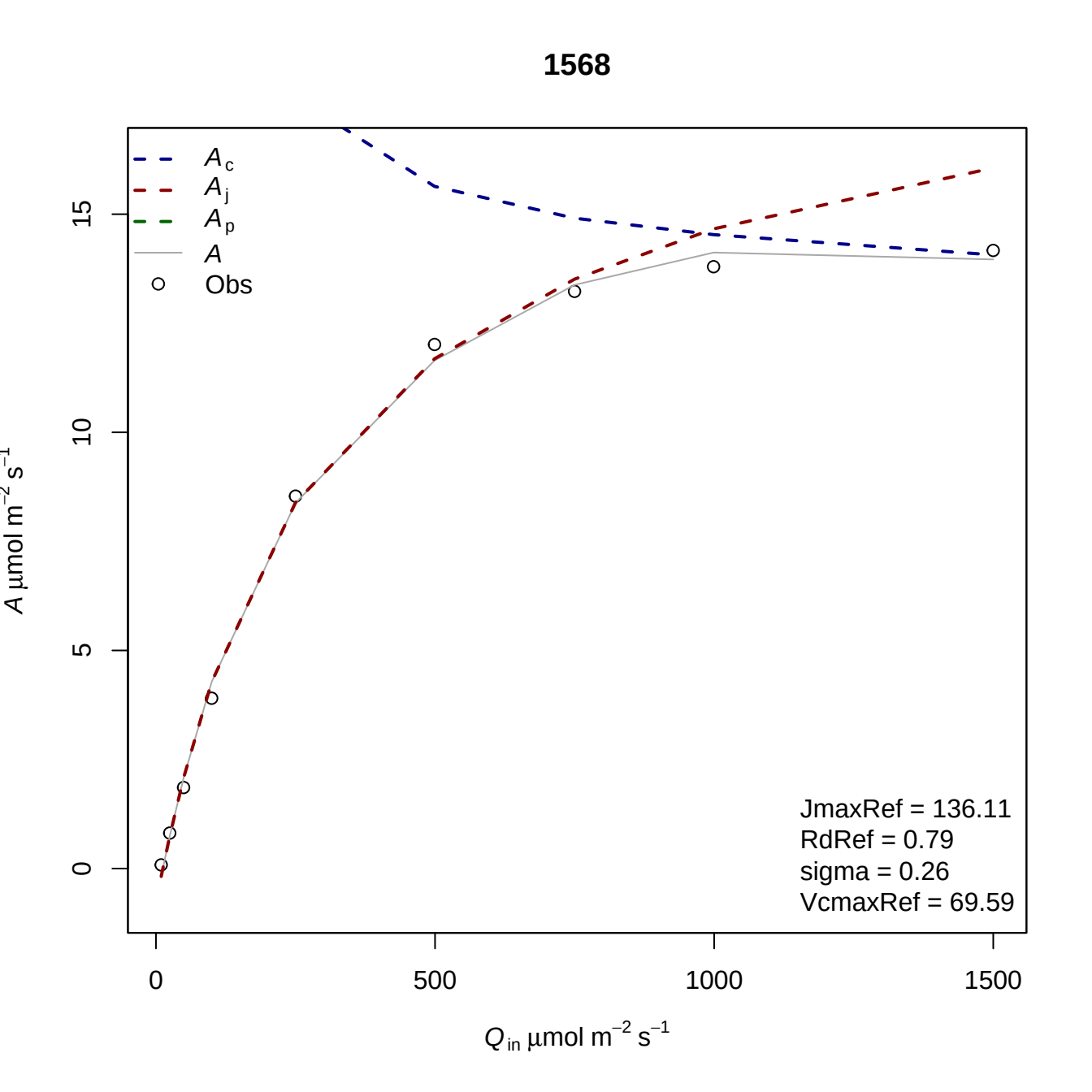

1563
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.93
RdRef = 0.69
sigma = 0.19
VcmaxRef = 50.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

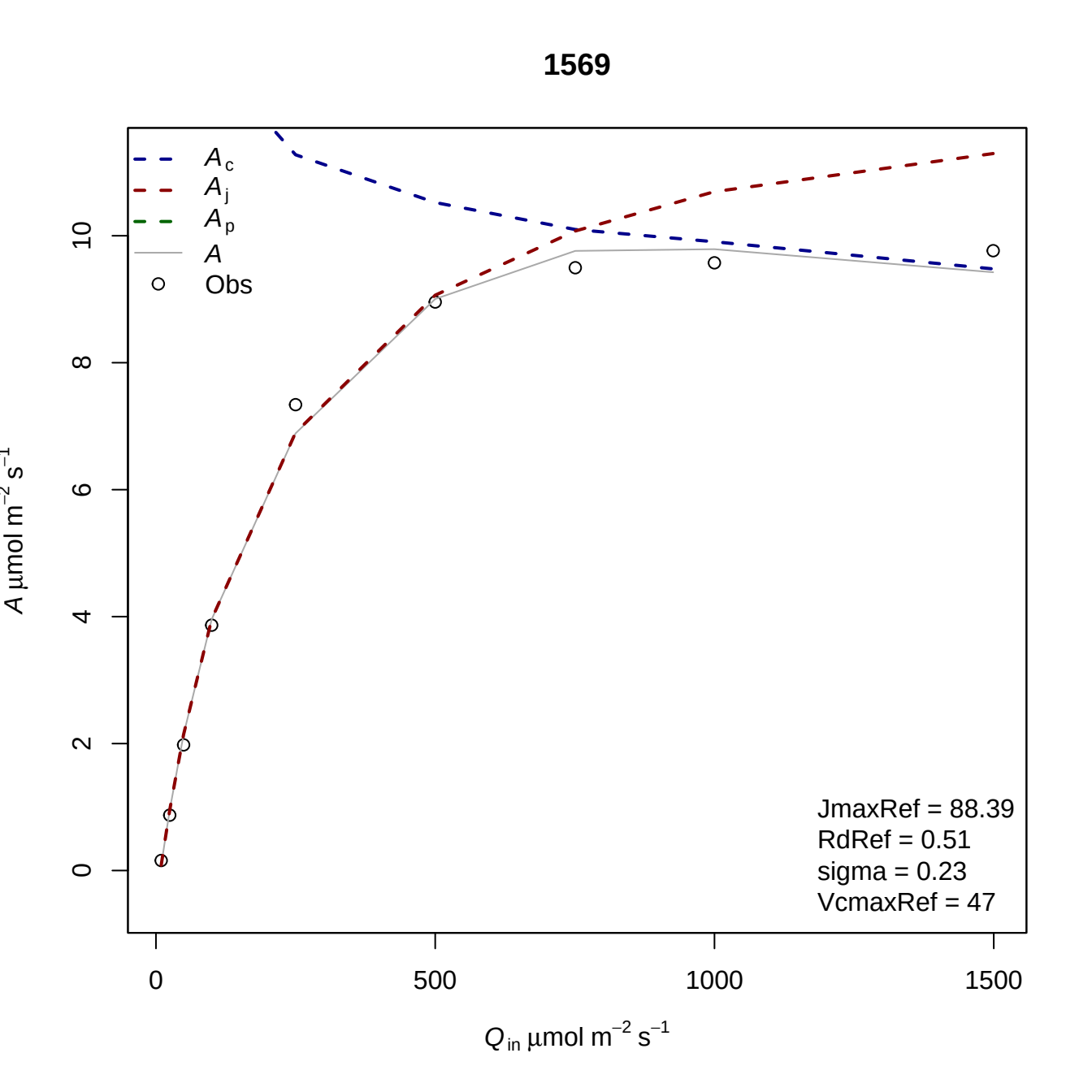

1564
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 39.56
RdRef = 0.89
sigma = 0.12
VcmaxRef = 31320.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

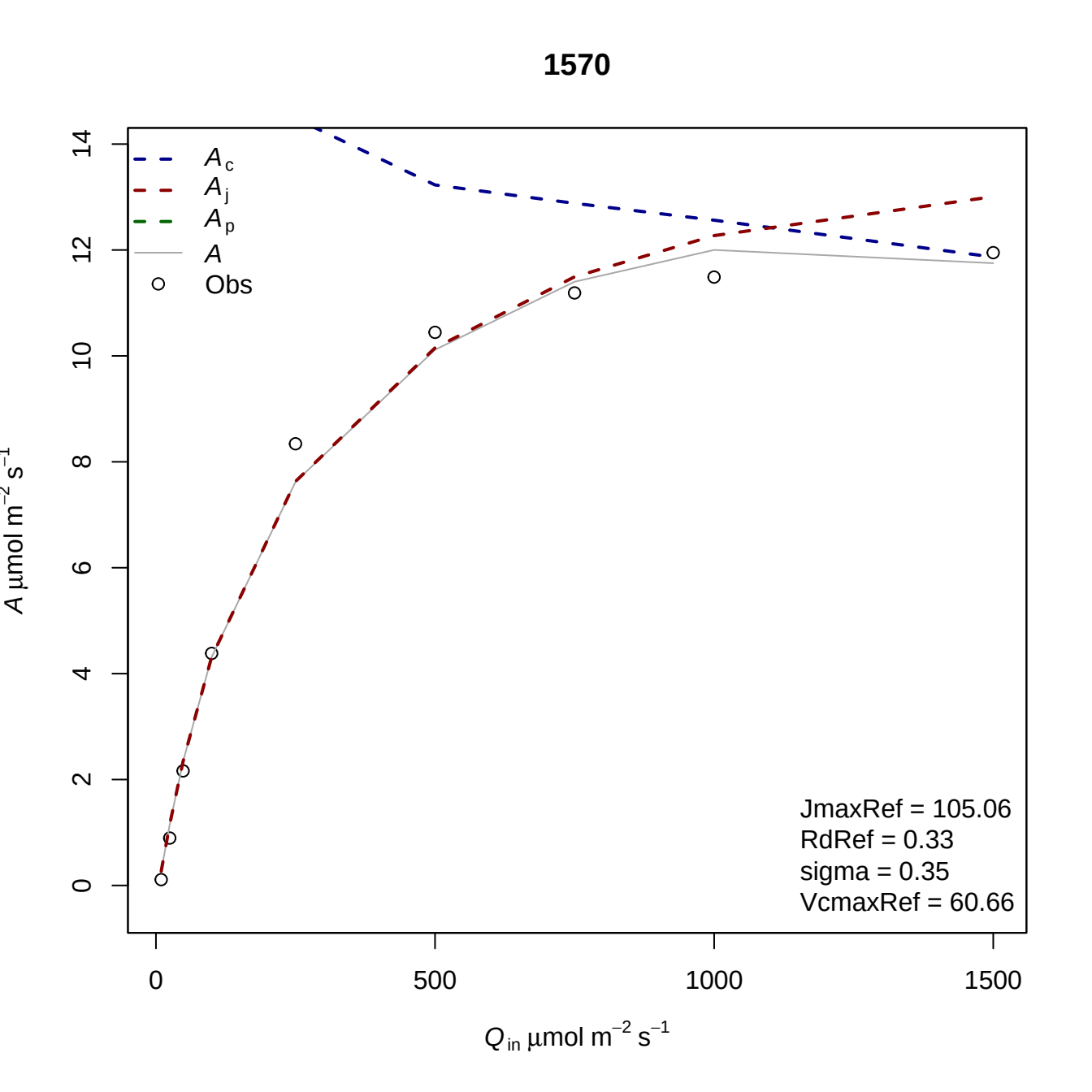

1565
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 32.67
RdRef = 0.95
sigma = 0.09
VcmaxRef = 21.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
0
500
1000
1500

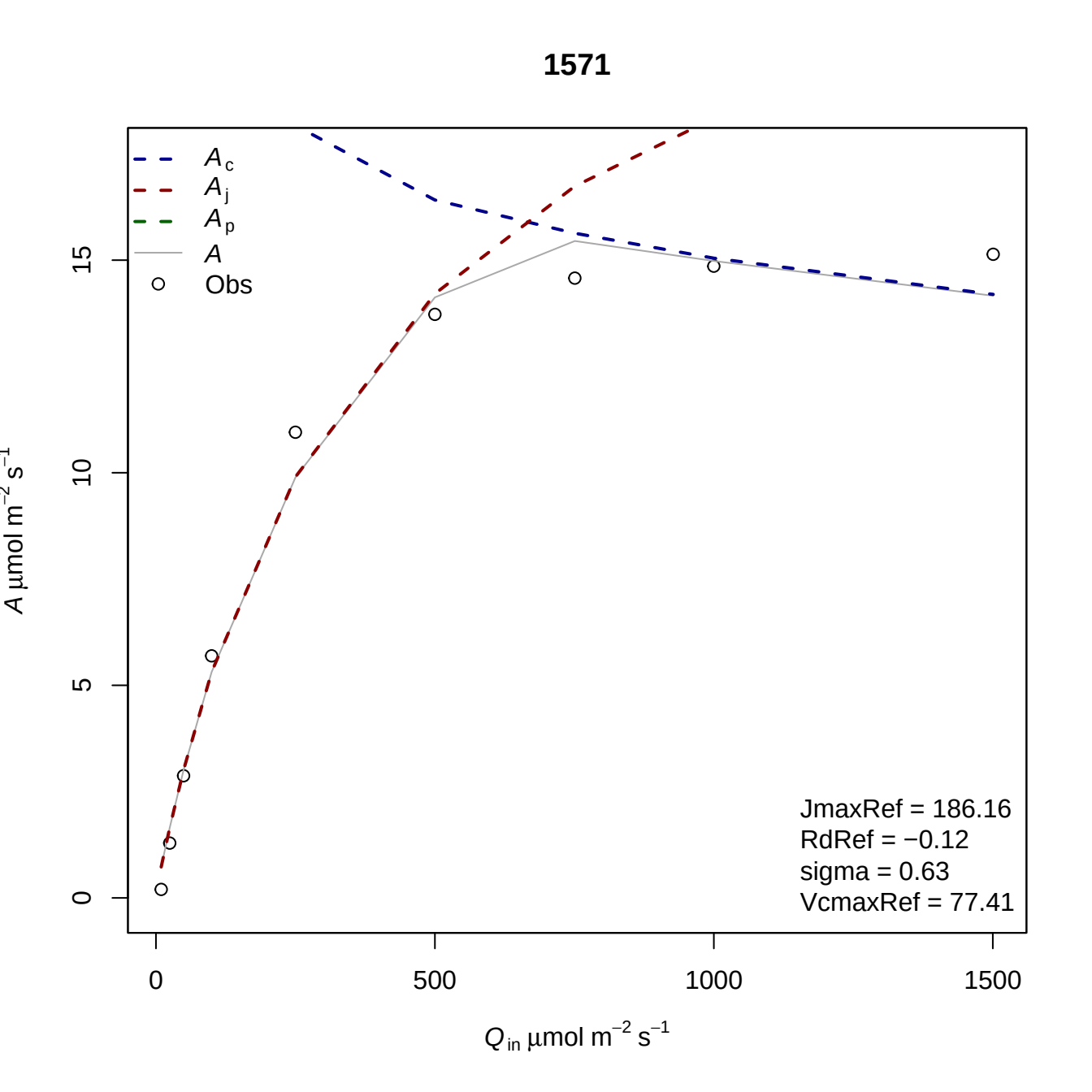

1566
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 87.19
RdRef = 0.53
sigma = 0.2
VcmaxRef = 53.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

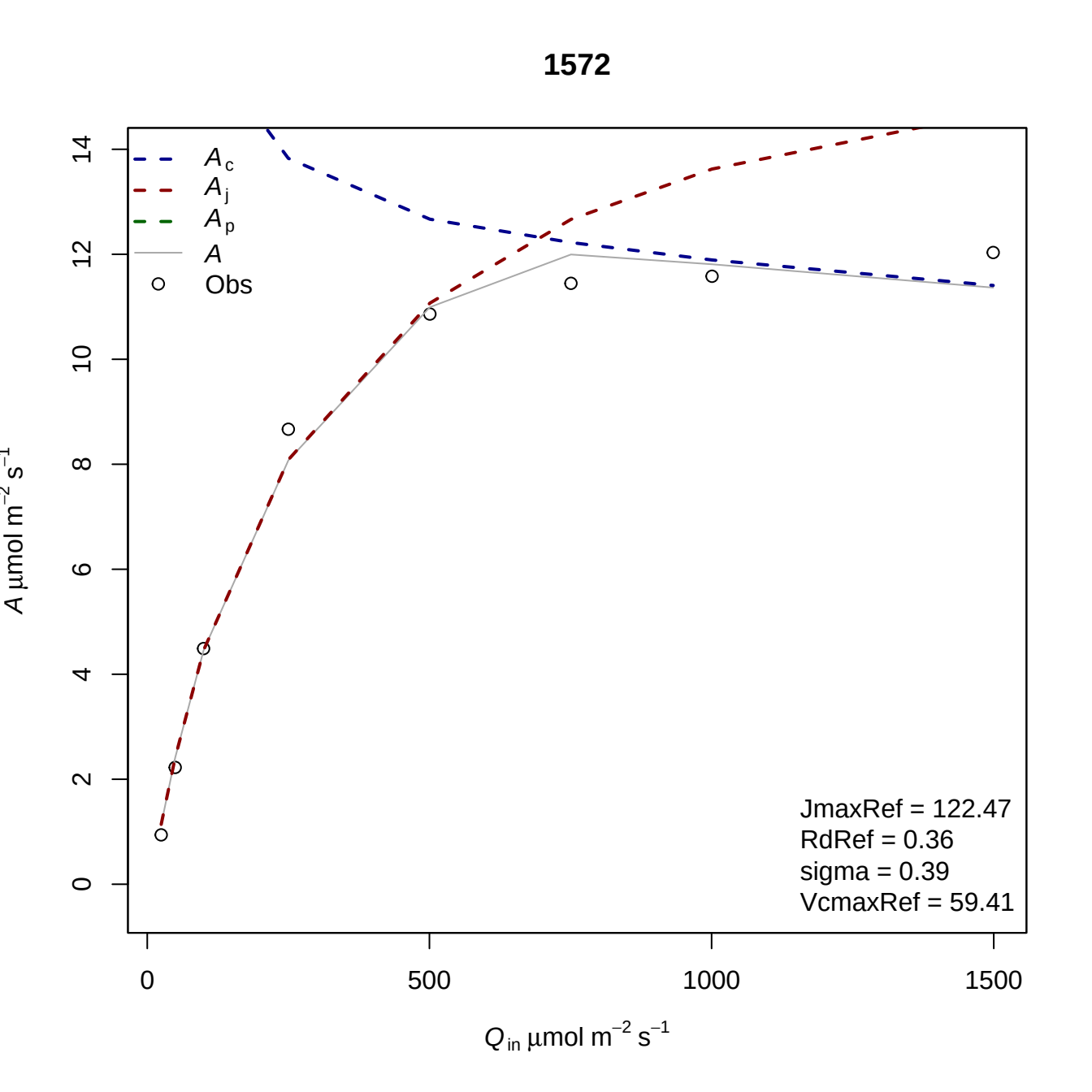

1567
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 98.43
RdRef = 0.6
sigma = 0.17
VcmaxRef = 59.81
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

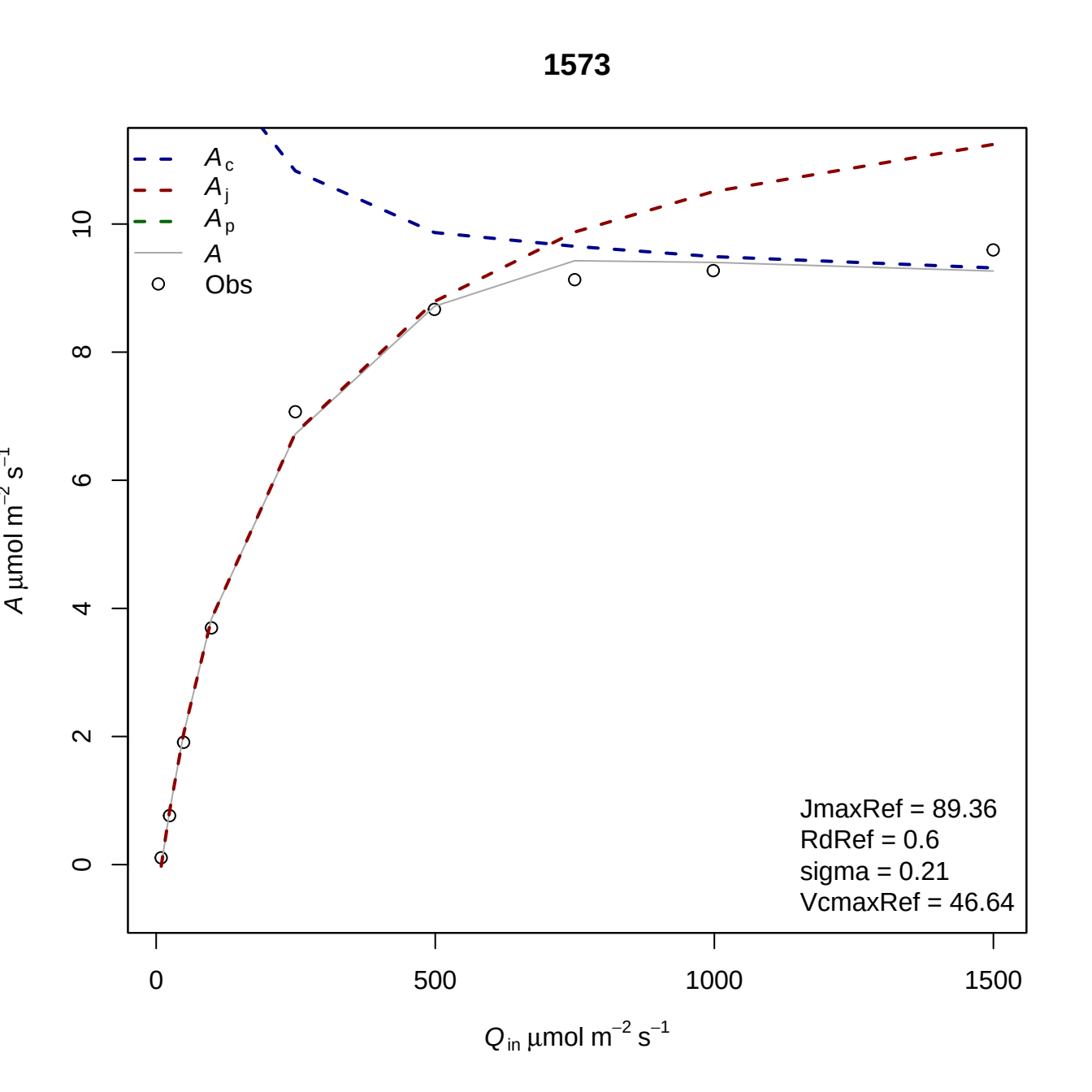

1568
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 136.11
RdRef = 0.79
sigma = 0.26
VcmaxRef = 69.59
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

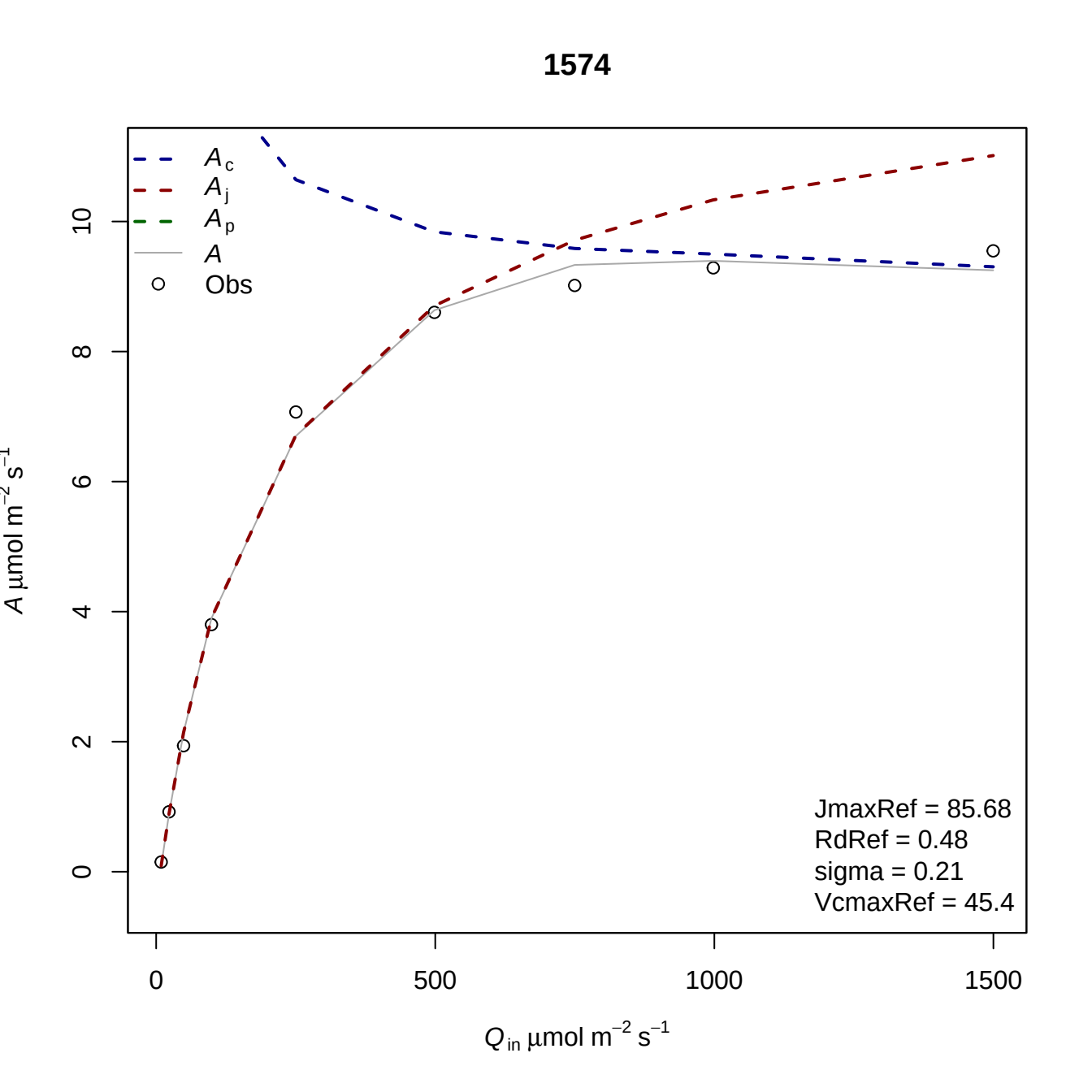

1569
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 88.39
RdRef = 0.51
sigma = 0.23
VcmaxRef = 47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

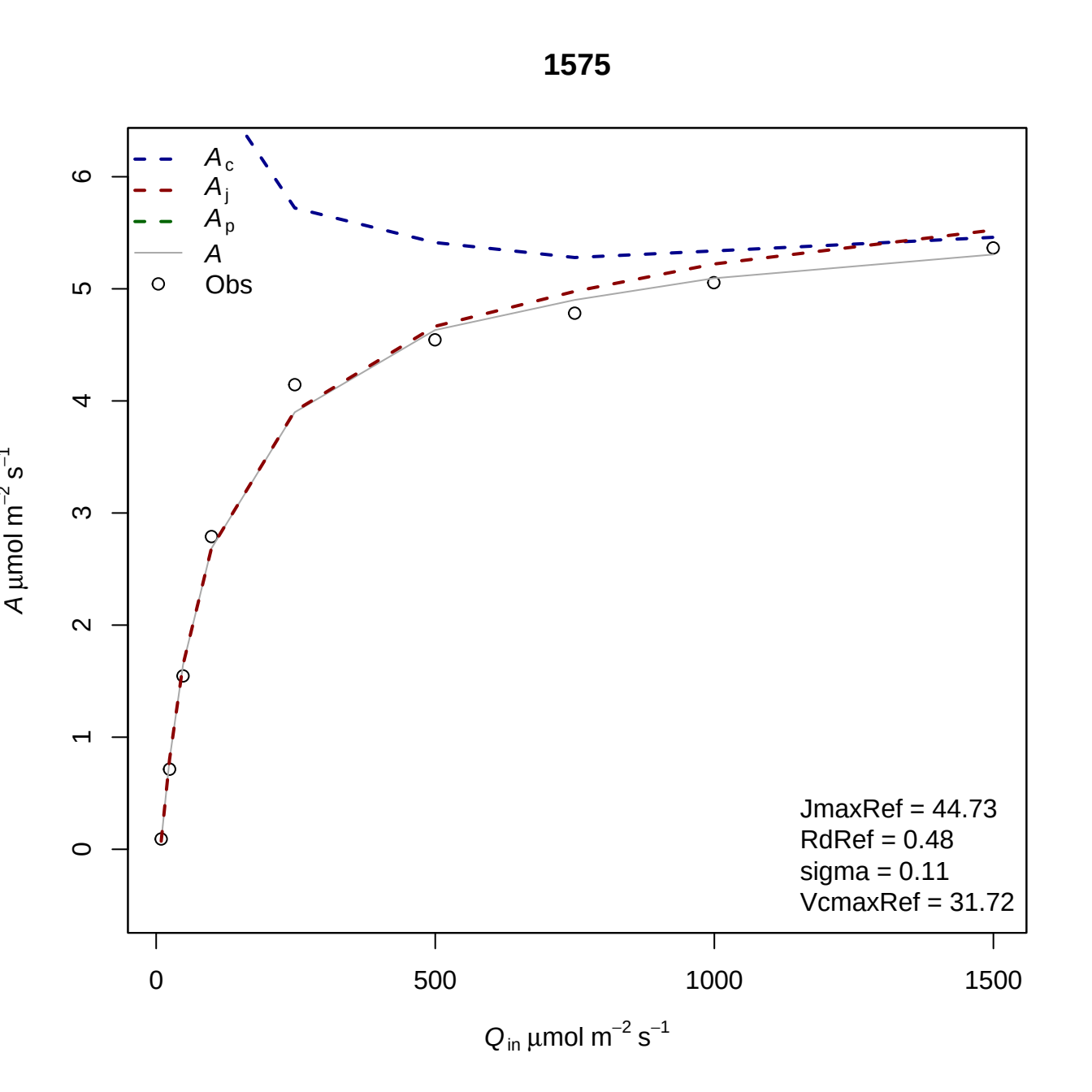

1570
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 105.06
RdRef = 0.33
sigma = 0.35
VcmaxRef = 60.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

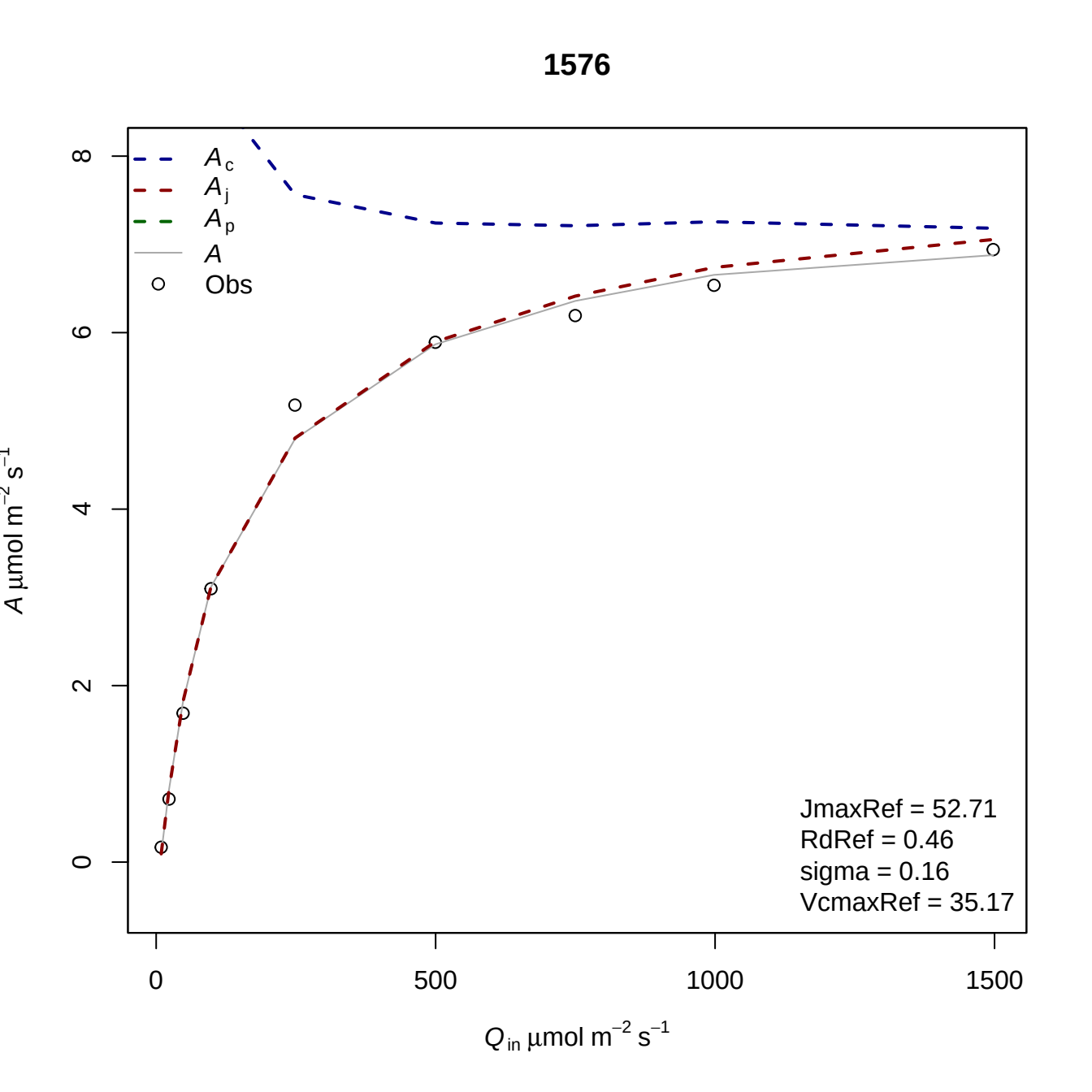

1571
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 186.16
RdRef = −0.12
sigma = 0.63
VcmaxRef = 77.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

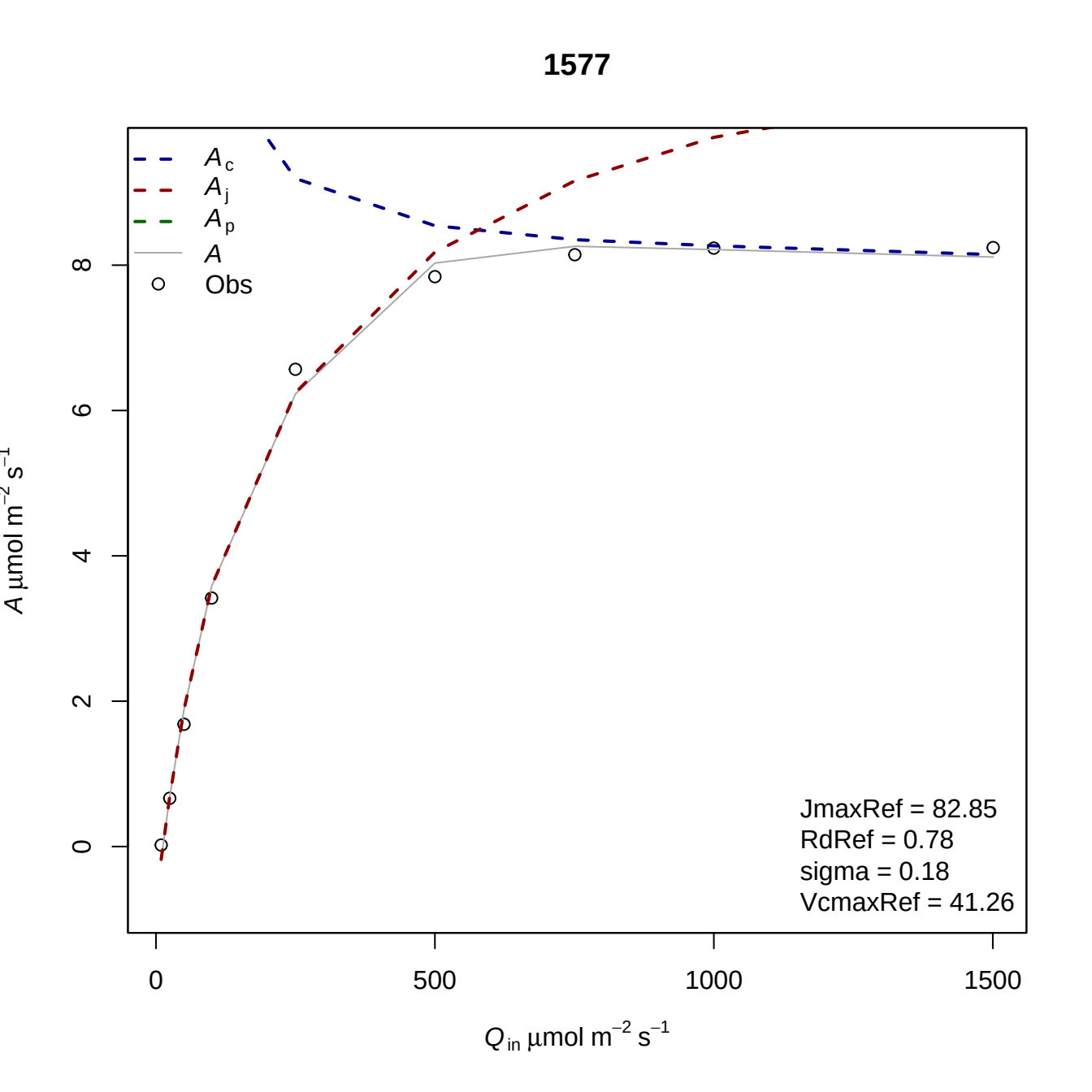

1572
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 122.47
RdRef = 0.36
sigma = 0.39
VcmaxRef = 59.41
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

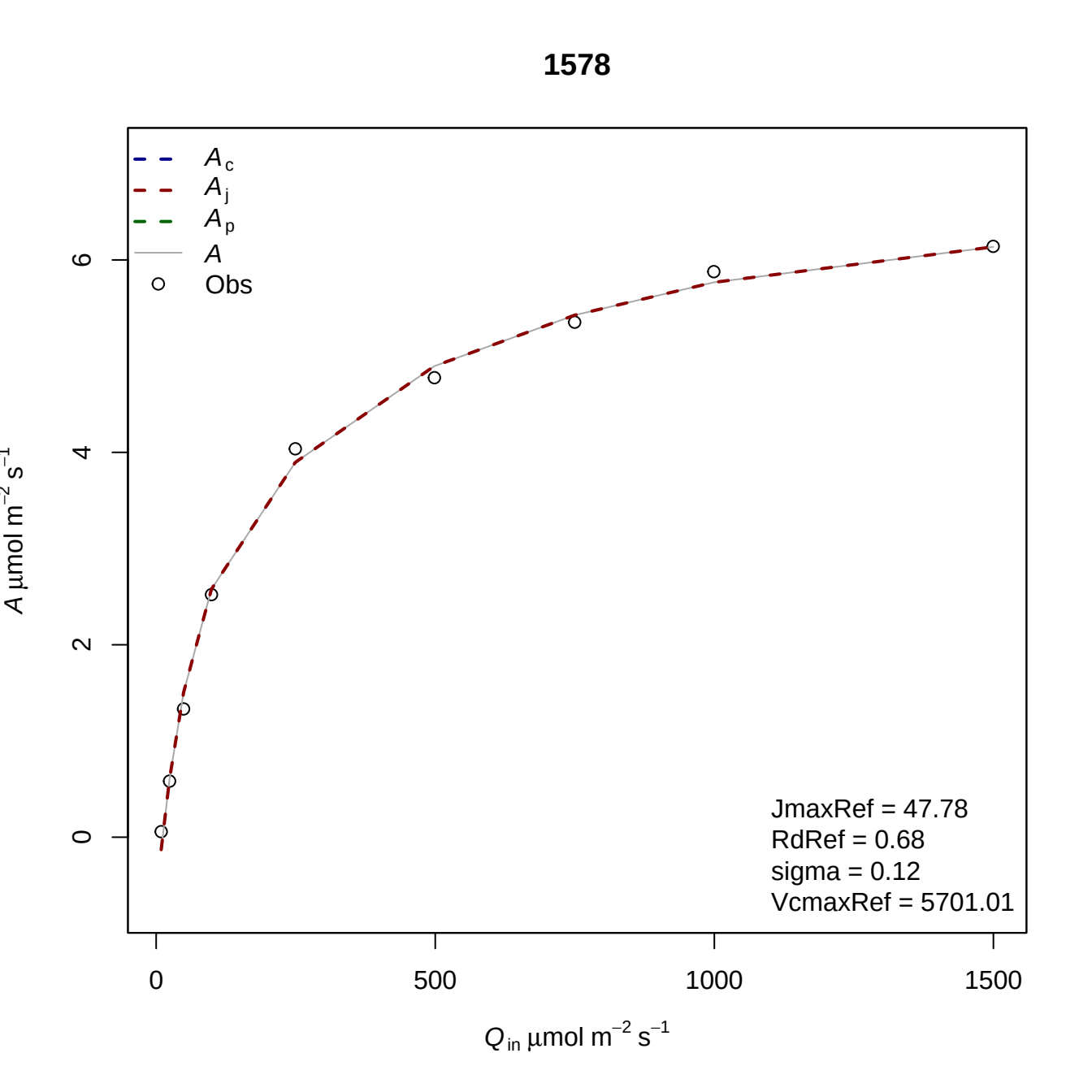

1573
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.36
RdRef = 0.6
sigma = 0.21
VcmaxRef = 46.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

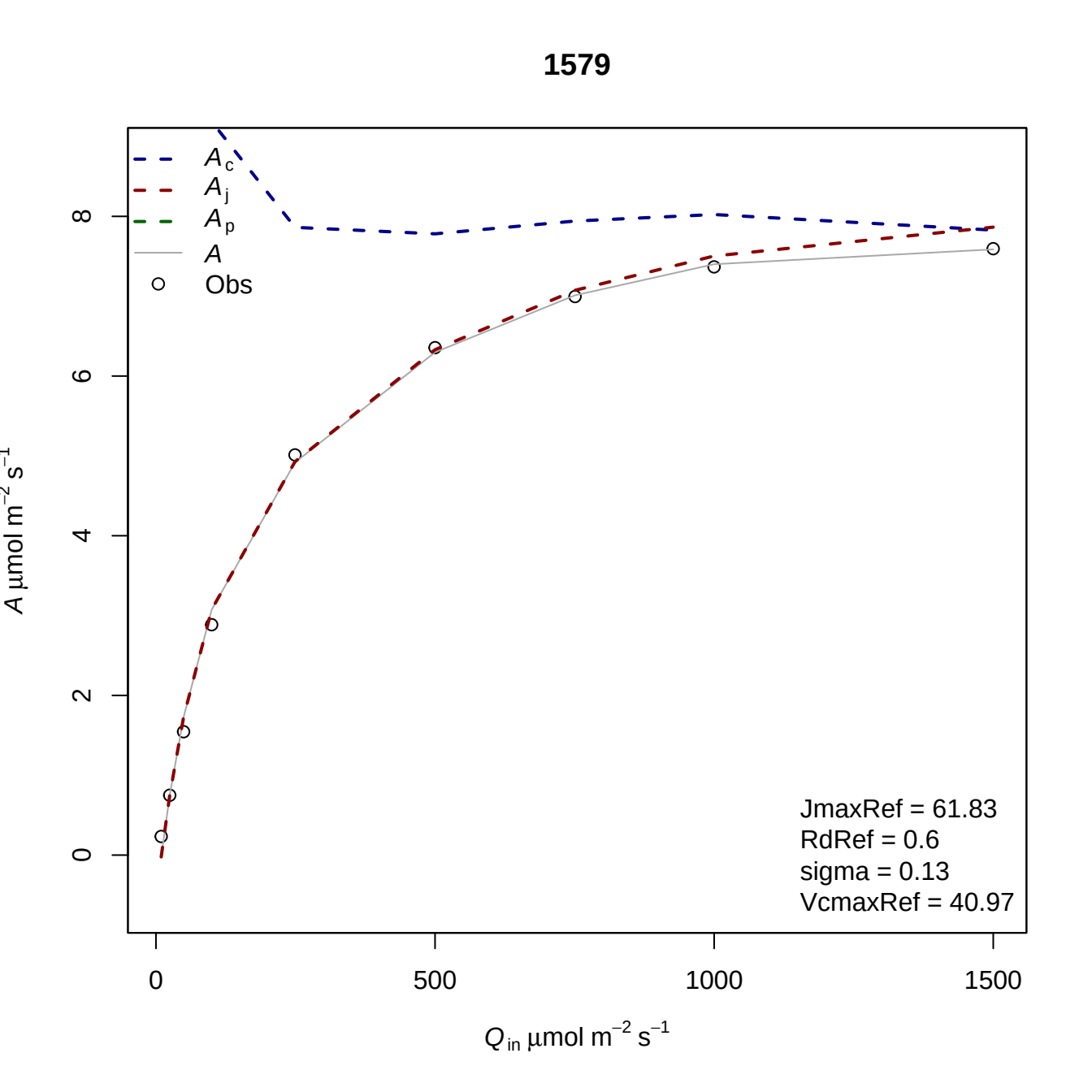

1574
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 85.68
RdRef = 0.48
sigma = 0.21
VcmaxRef = 45.4
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

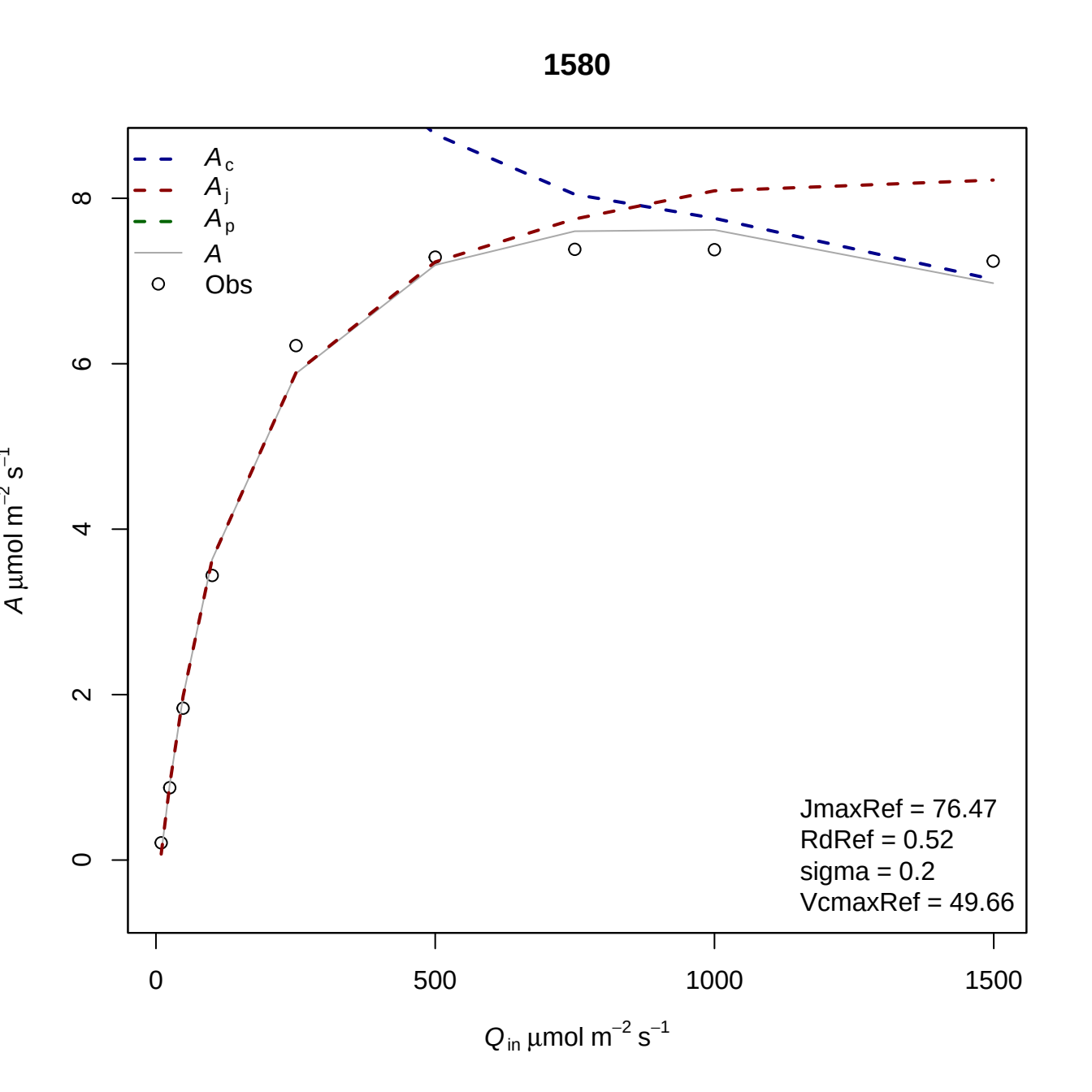

1575
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 44.73
RdRef = 0.48
sigma = 0.11
VcmaxRef = 31.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

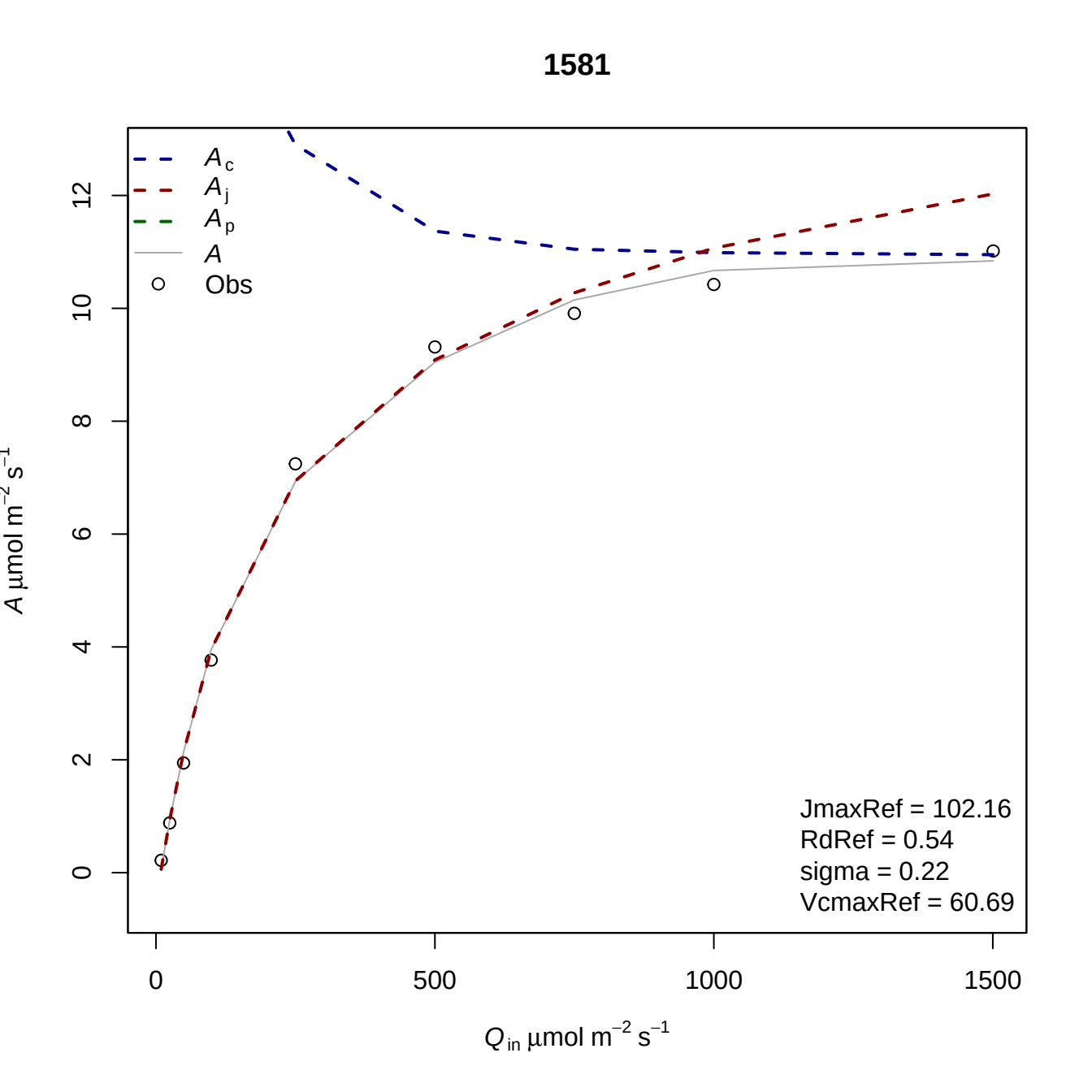

1576
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 52.71
RdRef = 0.46
sigma = 0.16
VcmaxRef = 35.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

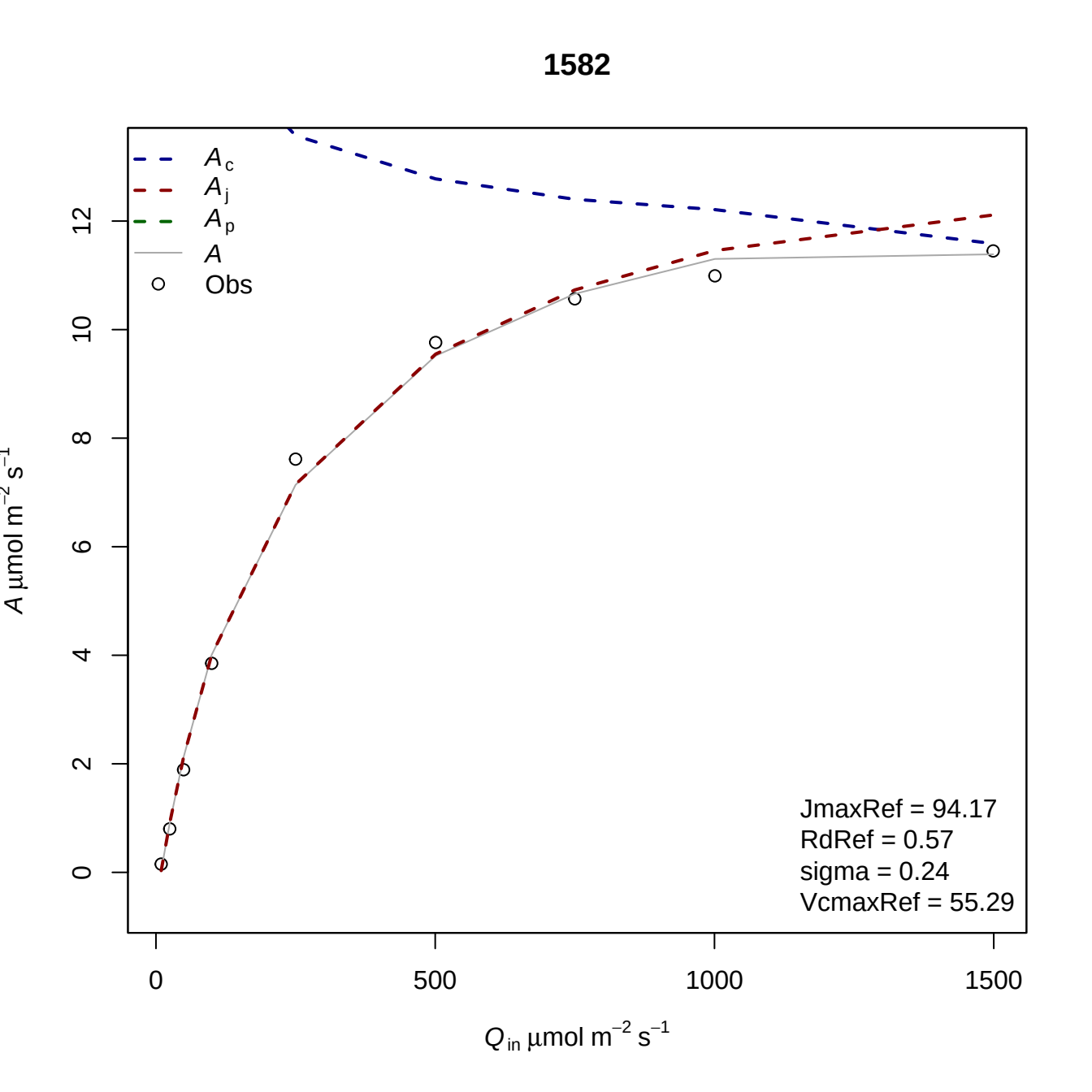

1577
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.85
RdRef = 0.78
sigma = 0.18
VcmaxRef = 41.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

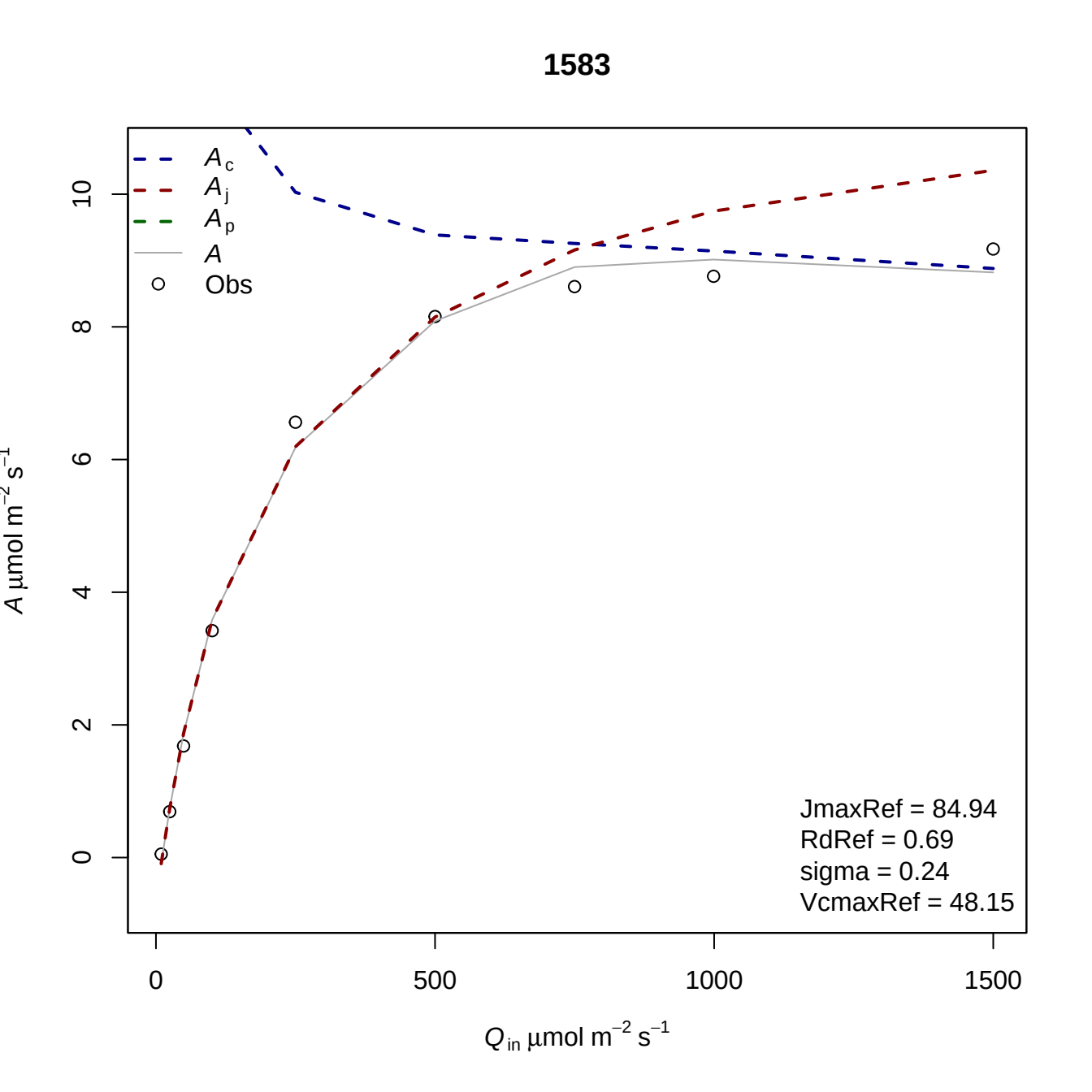

1578
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.78
RdRef = 0.68
sigma = 0.12
VcmaxRef = 5701.01
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

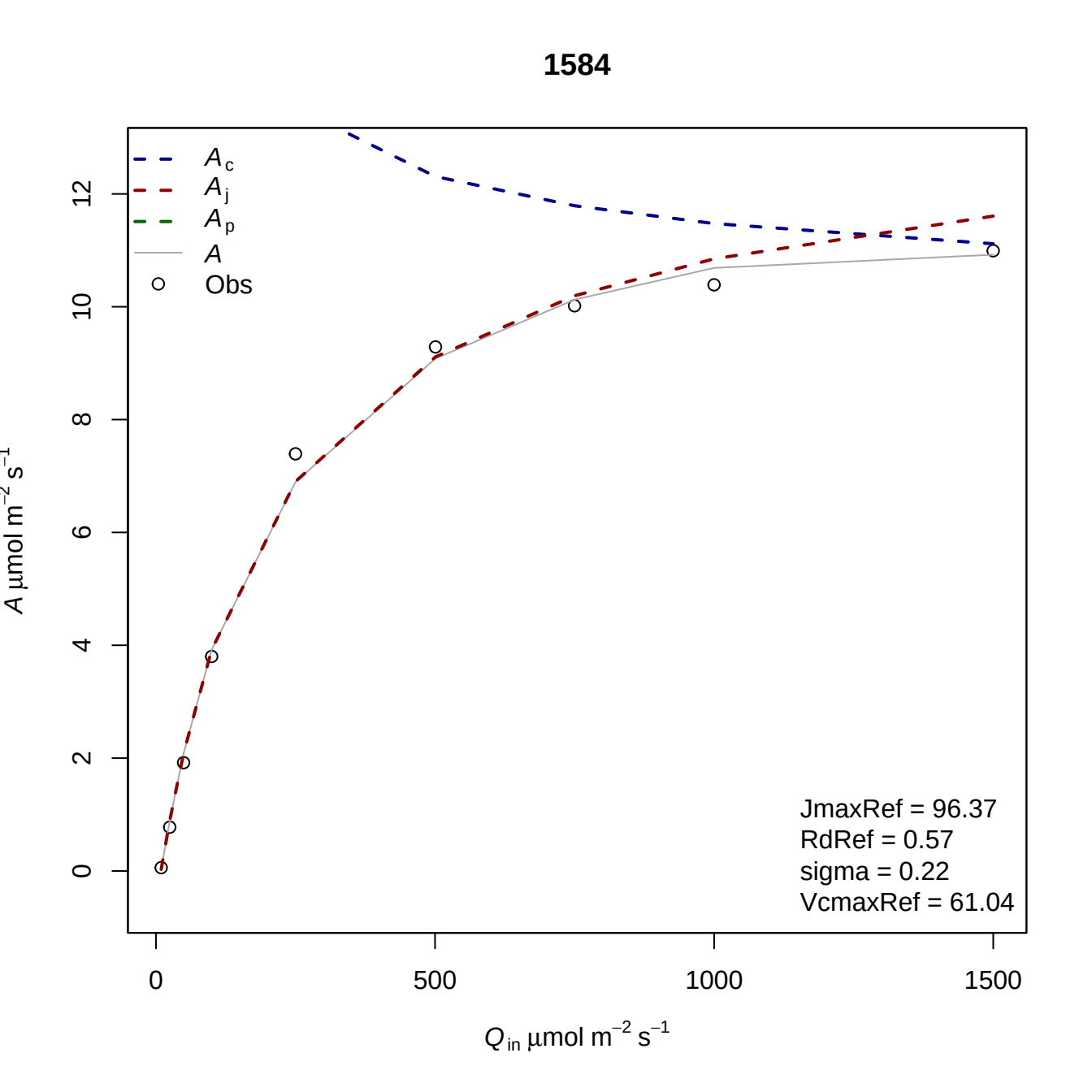

1579
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 61.83
RdRef = 0.6
sigma = 0.13
VcmaxRef = 40.97
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

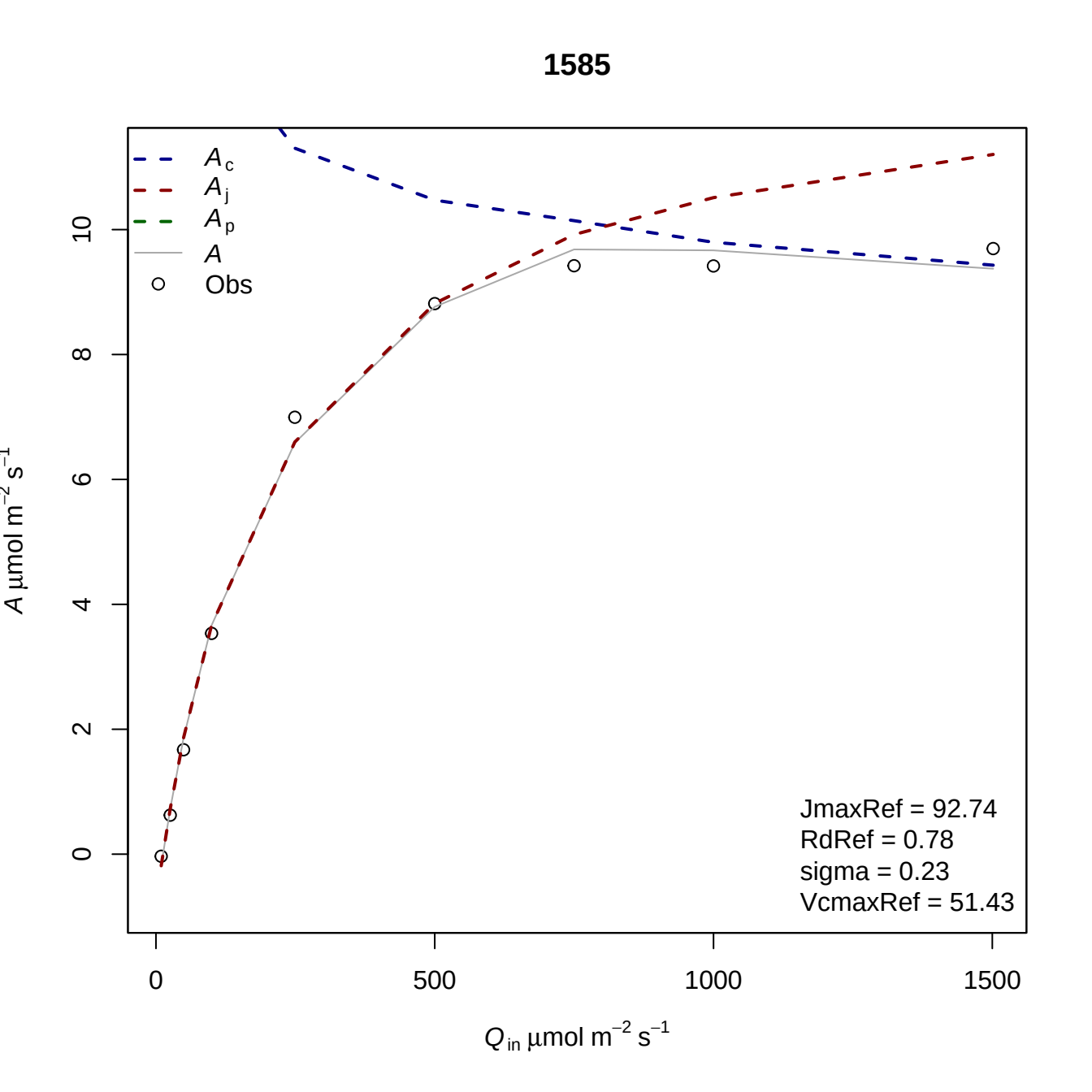

1580
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 76.47
RdRef = 0.52
sigma = 0.2
VcmaxRef = 49.66
0
2
4
6
8
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

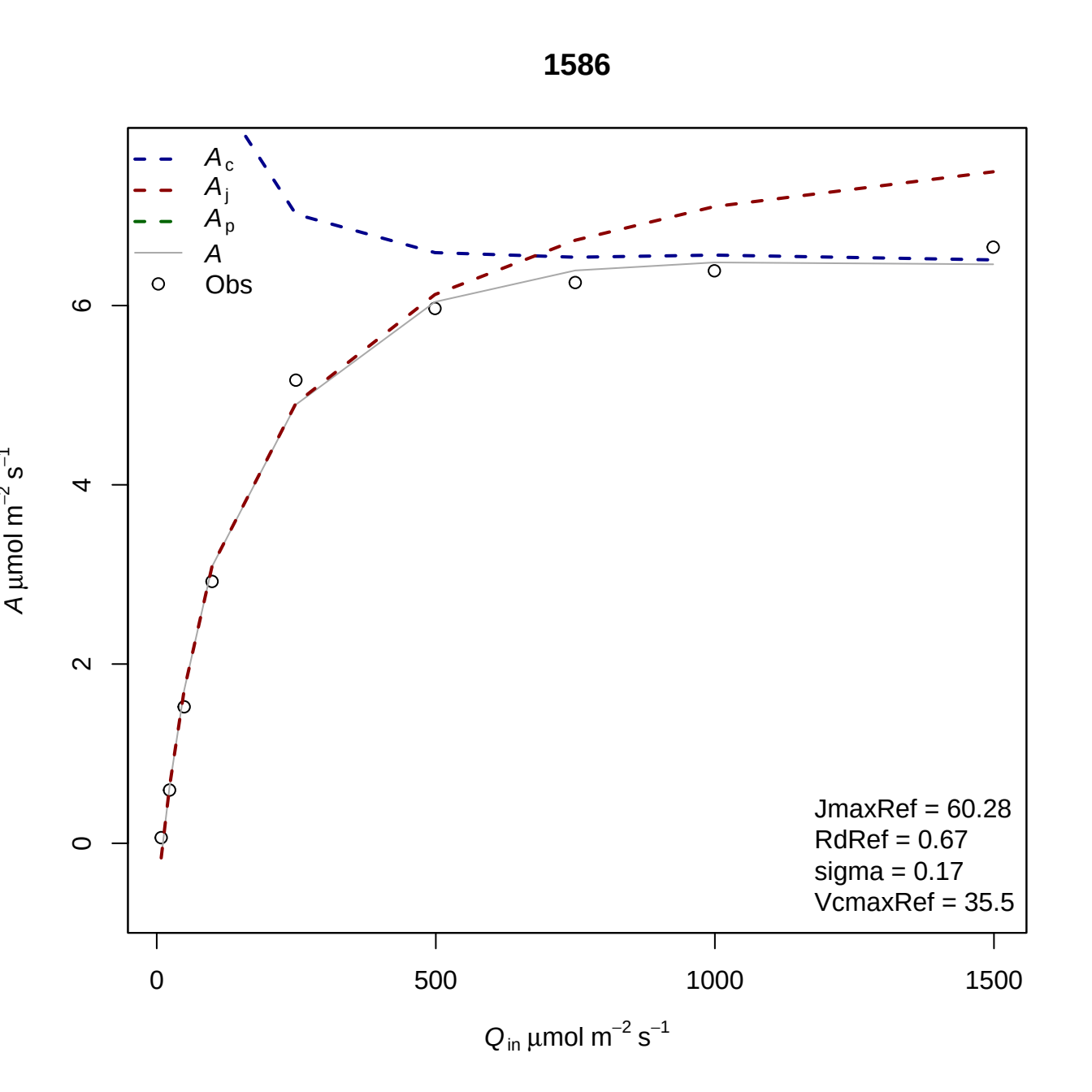

1581
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 102.16
RdRef = 0.54
sigma = 0.22
VcmaxRef = 60.69
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

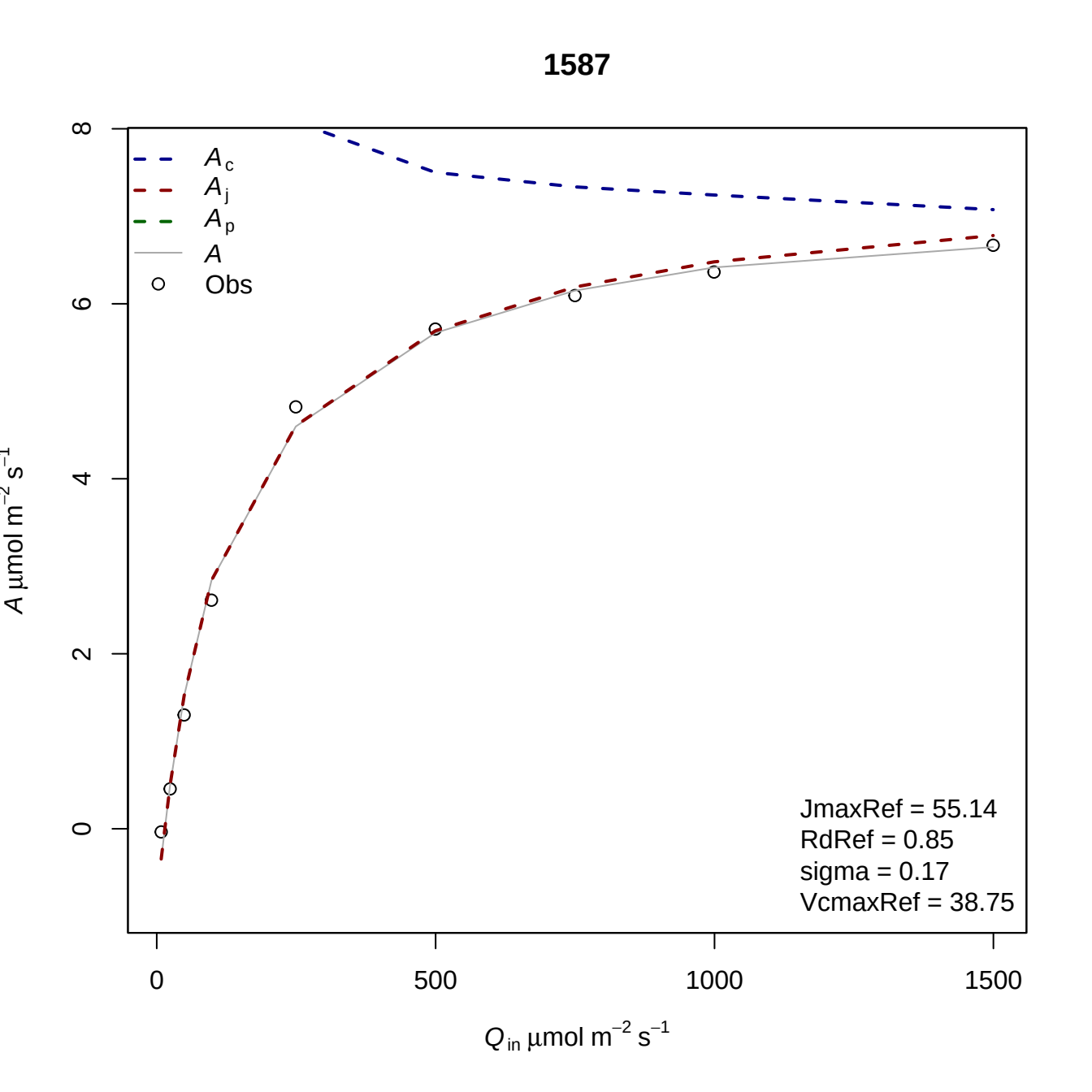

1582
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 94.17
RdRef = 0.57
sigma = 0.24
VcmaxRef = 55.29
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

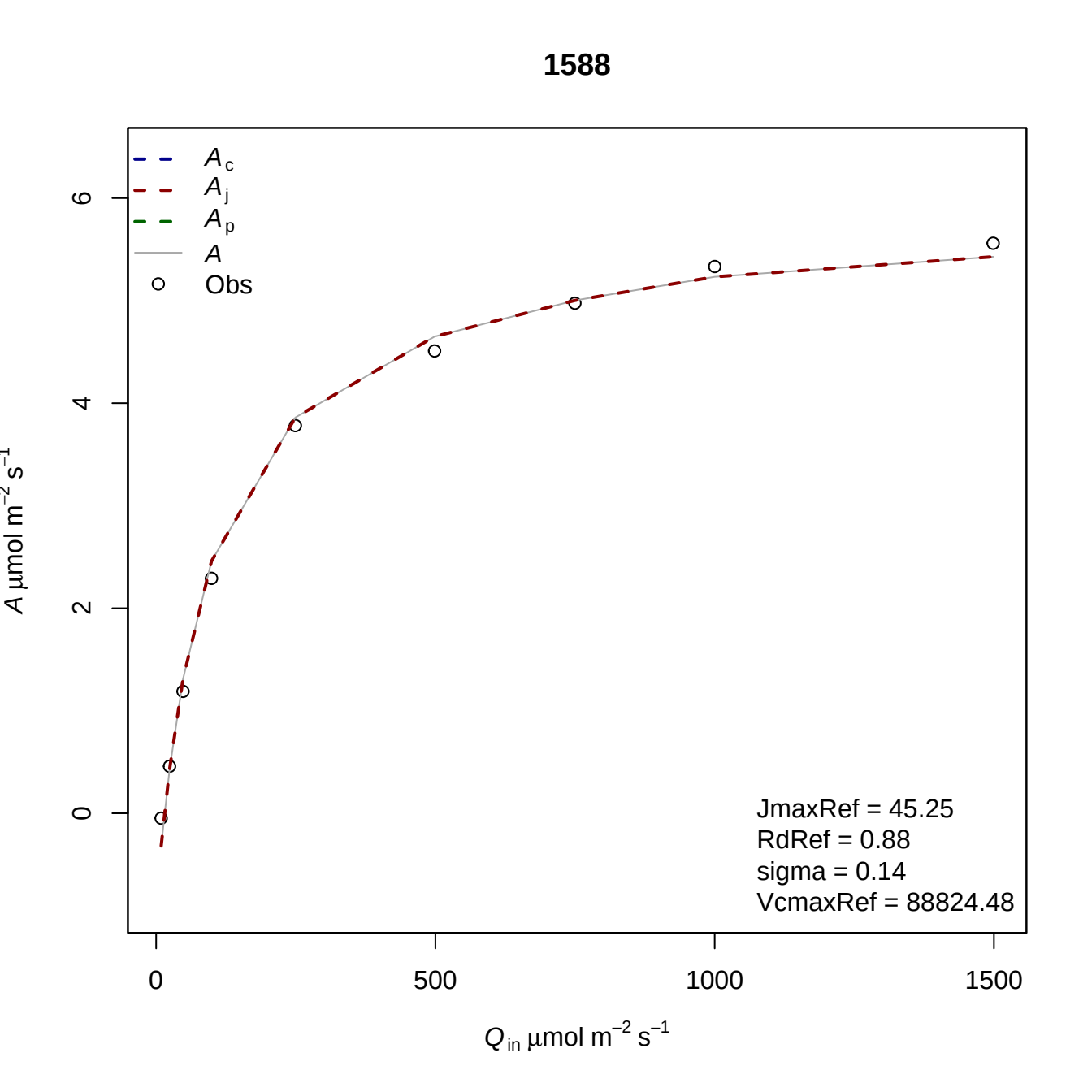

1583
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 84.94
RdRef = 0.69
sigma = 0.24
VcmaxRef = 48.15
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

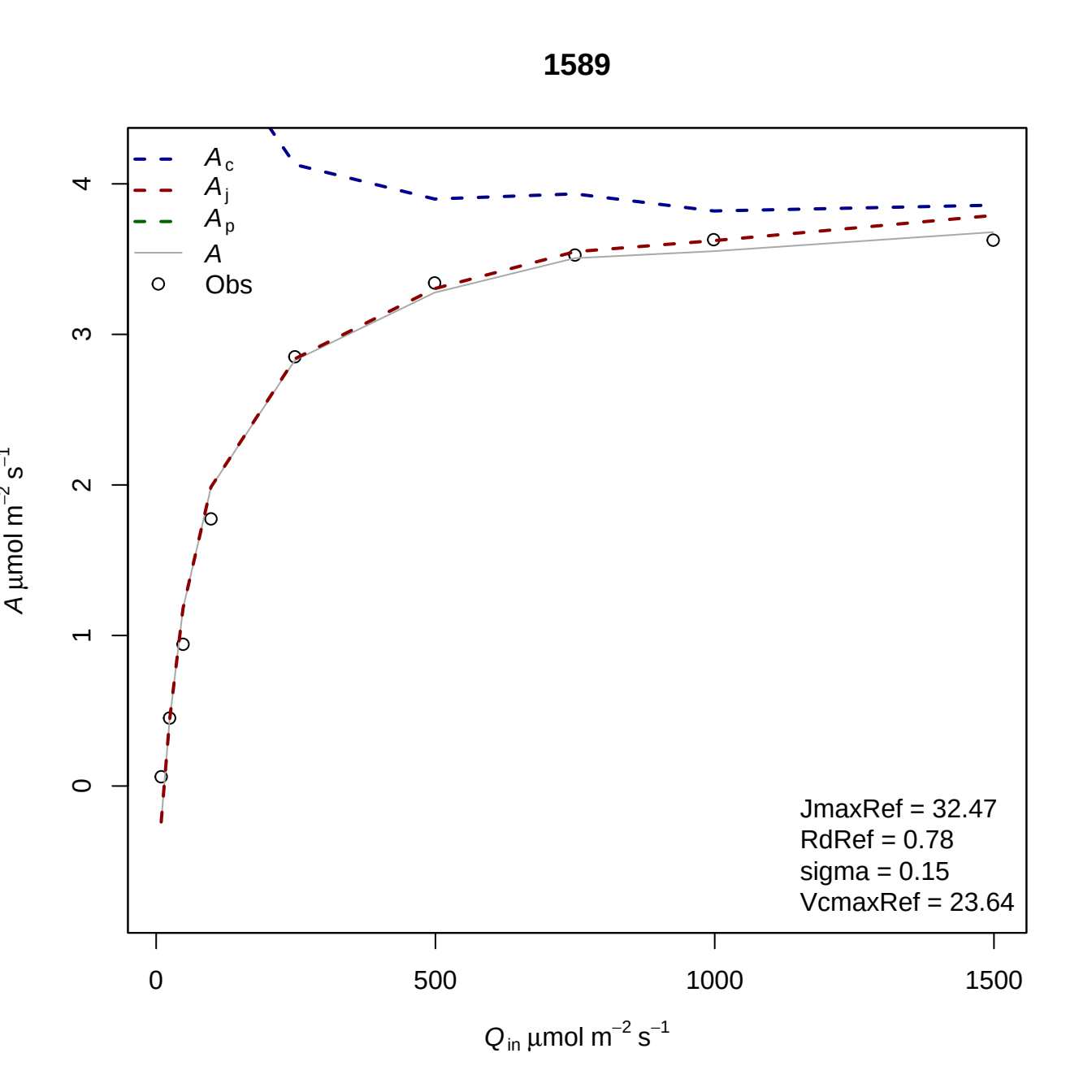

1584
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 96.37
RdRef = 0.57
sigma = 0.22
VcmaxRef = 61.04
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

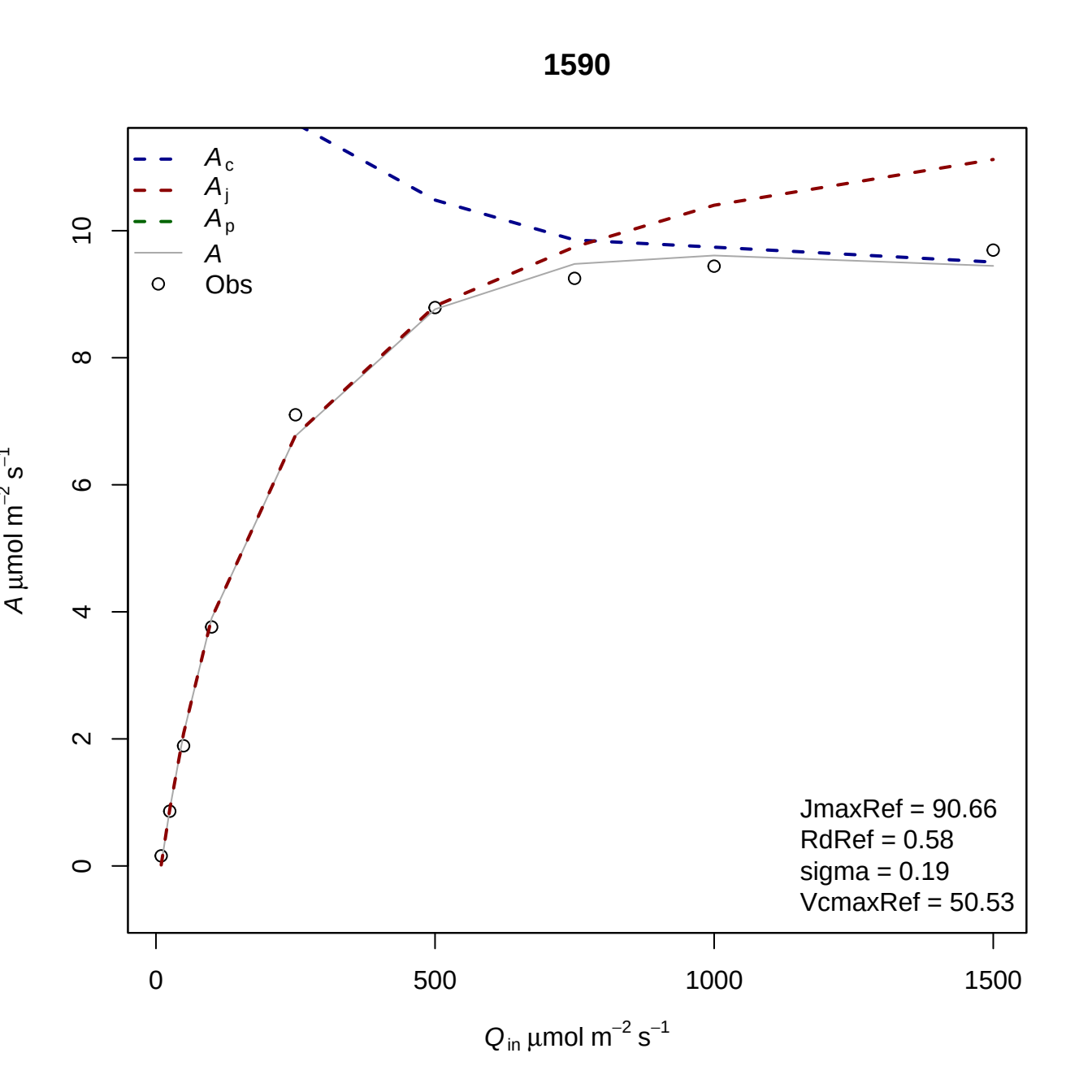

1585
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 92.74
RdRef = 0.78
sigma = 0.23
VcmaxRef = 51.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

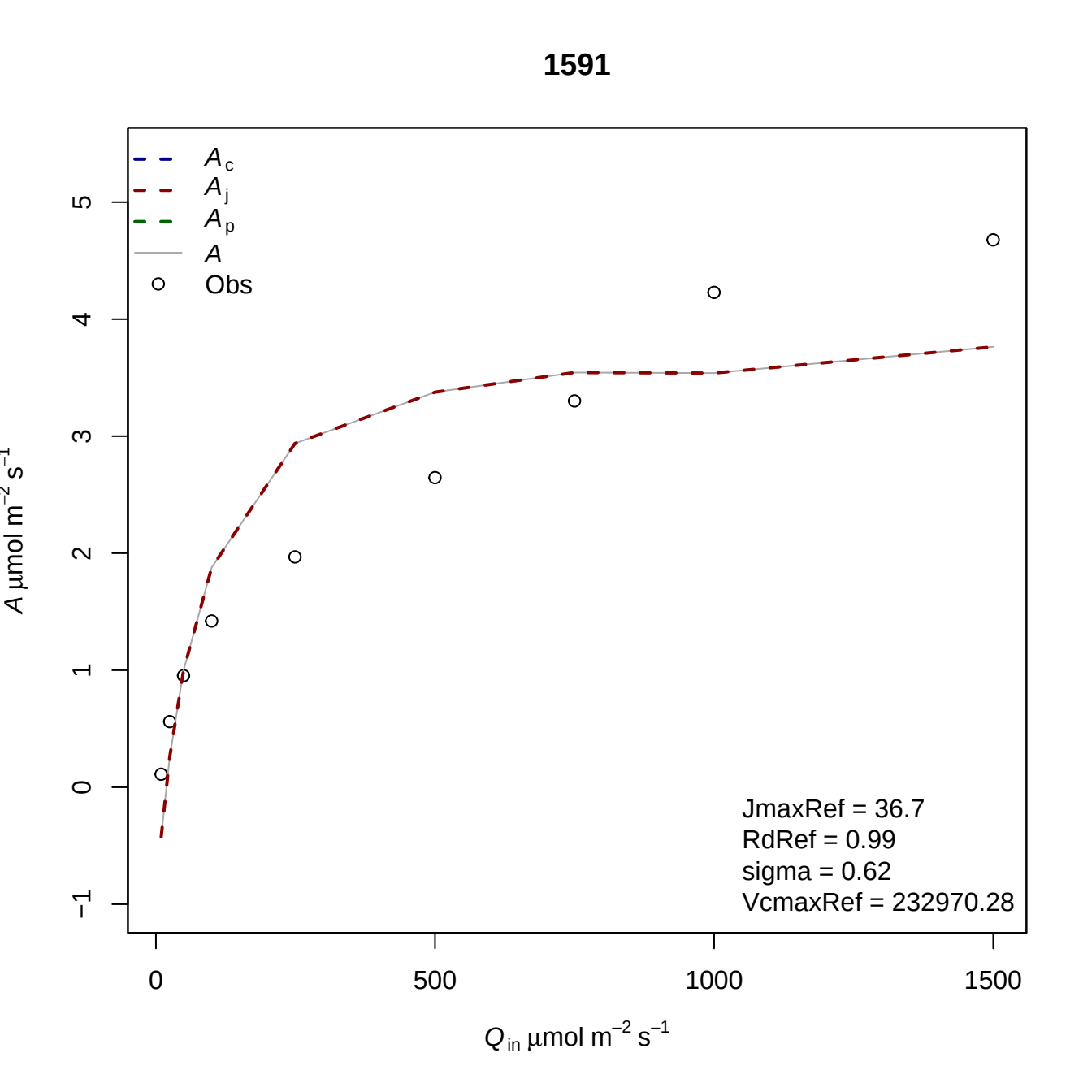

1586
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 60.28
RdRef = 0.67
sigma = 0.17
VcmaxRef = 35.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

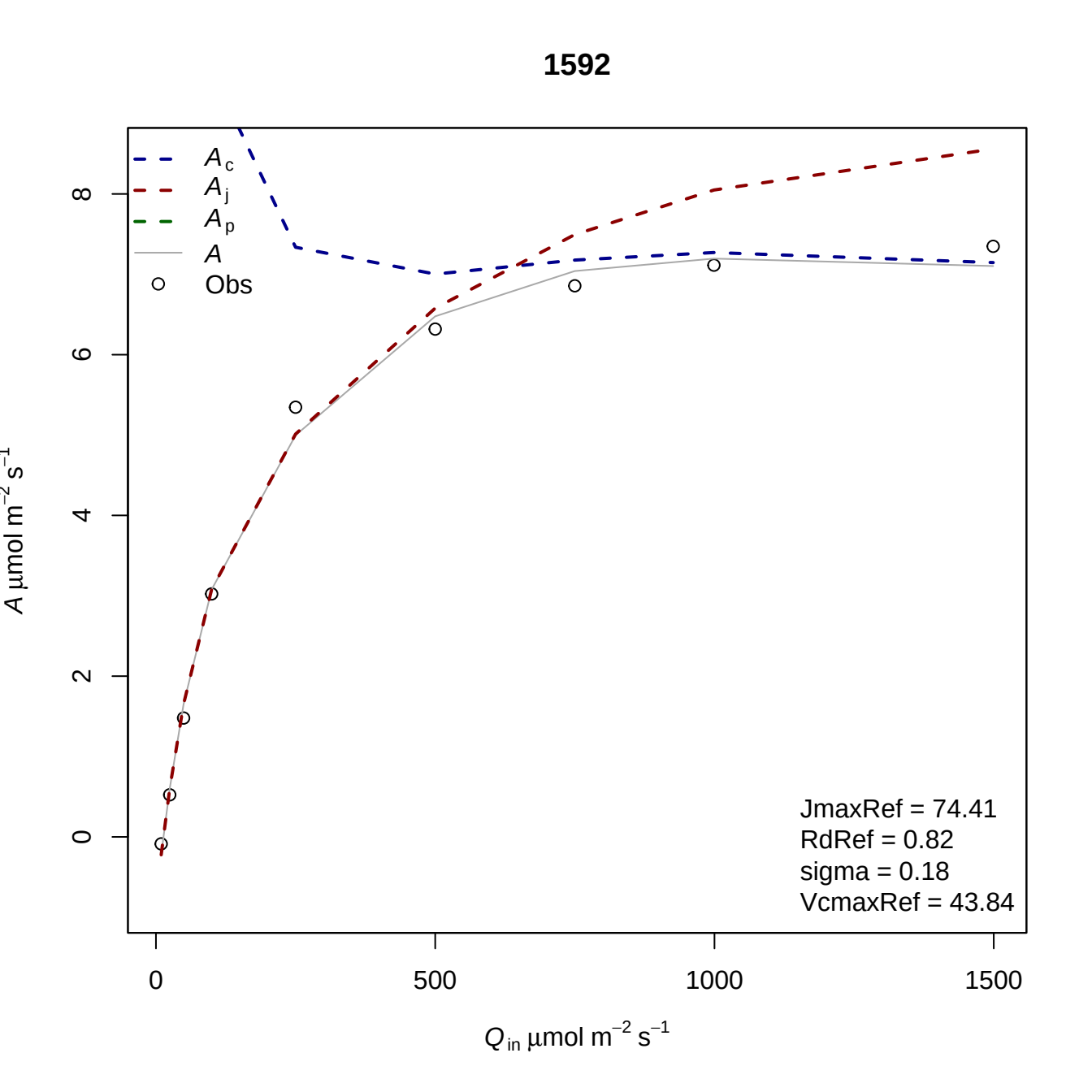

1587
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 55.14
RdRef = 0.85
sigma = 0.17
VcmaxRef = 38.75
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

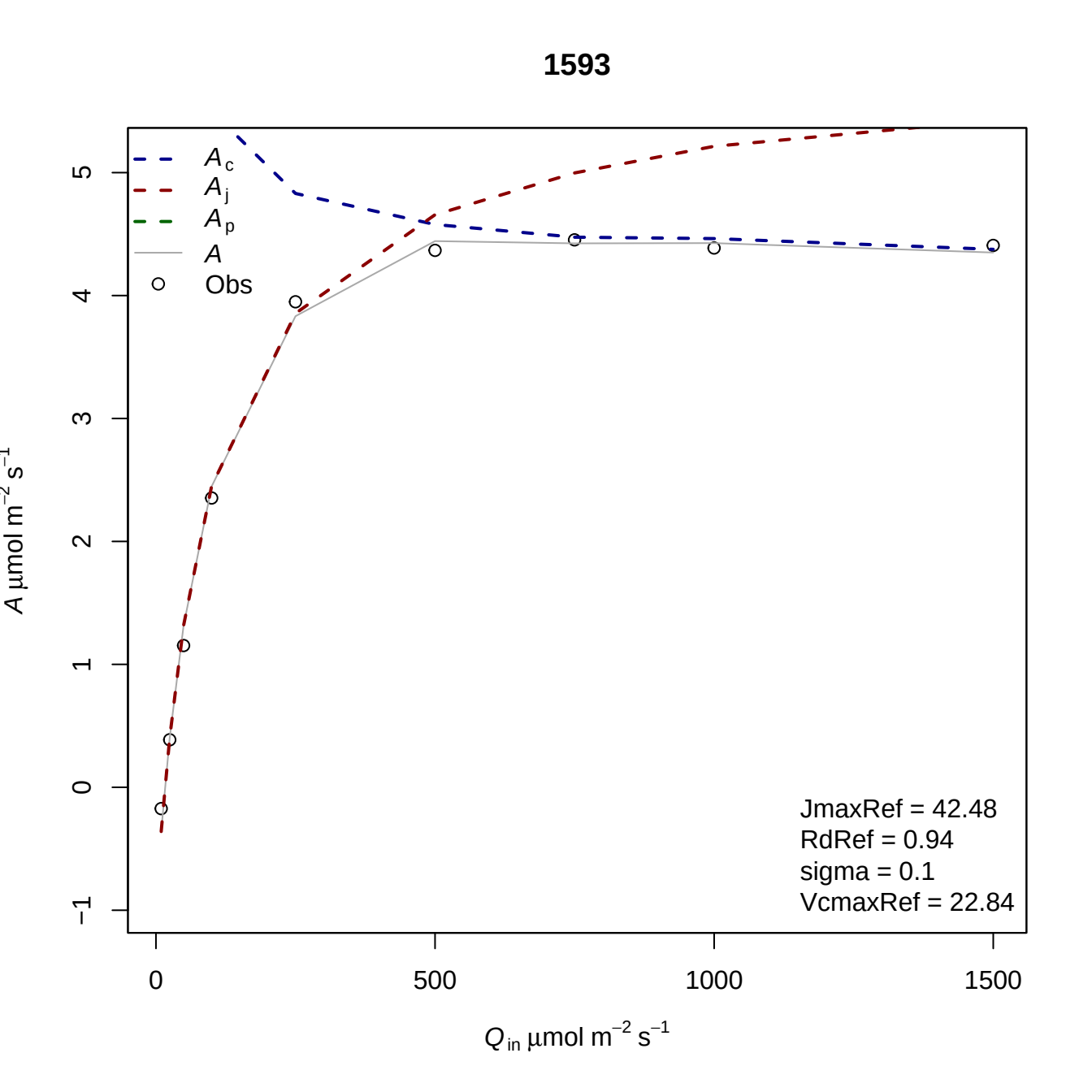

1588
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 45.25
RdRef = 0.88
sigma = 0.14
VcmaxRef = 88824.48
$A$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

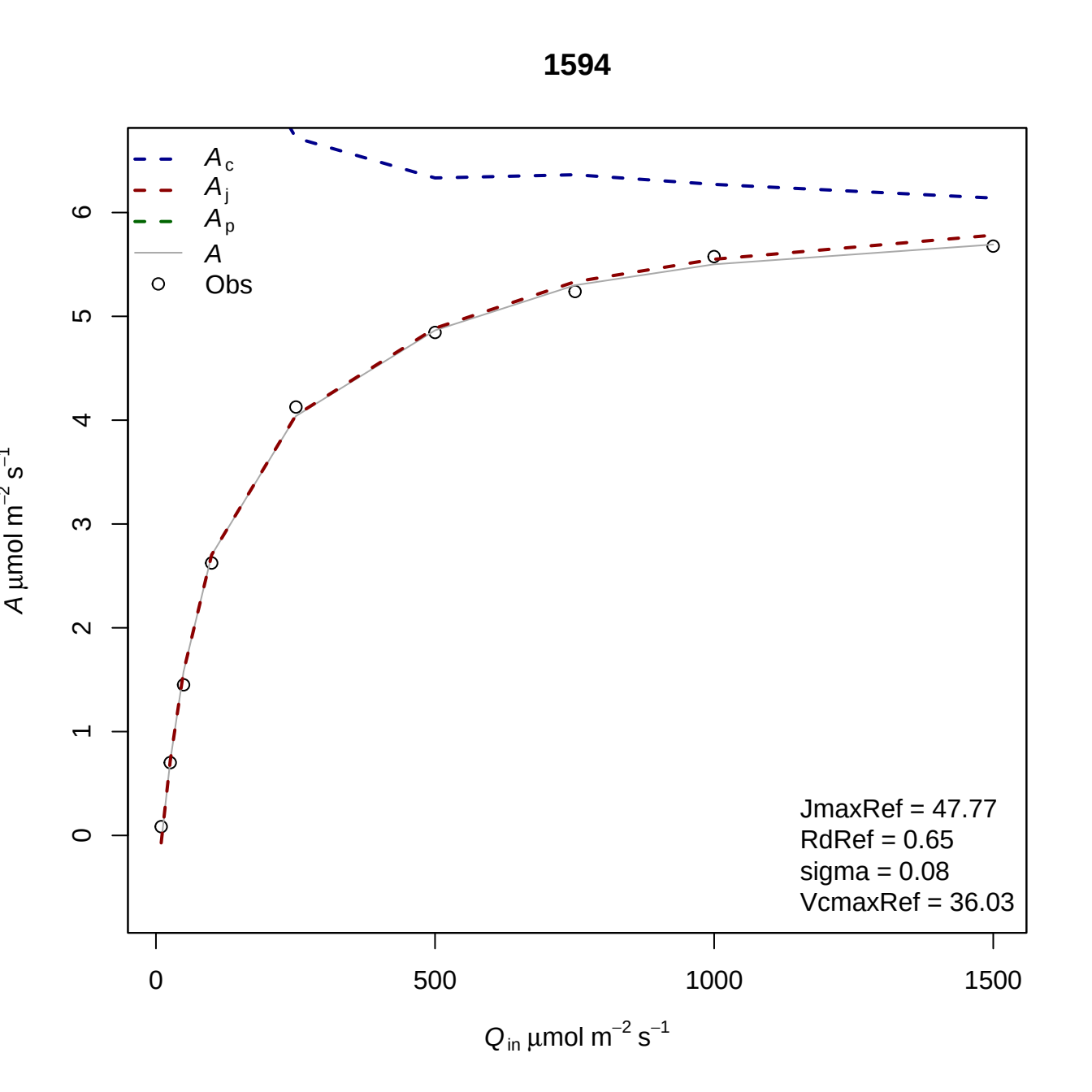

1589
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 32.47
RdRef = 0.78
sigma = 0.15
VcmaxRef = 23.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
0
500
1000
1500

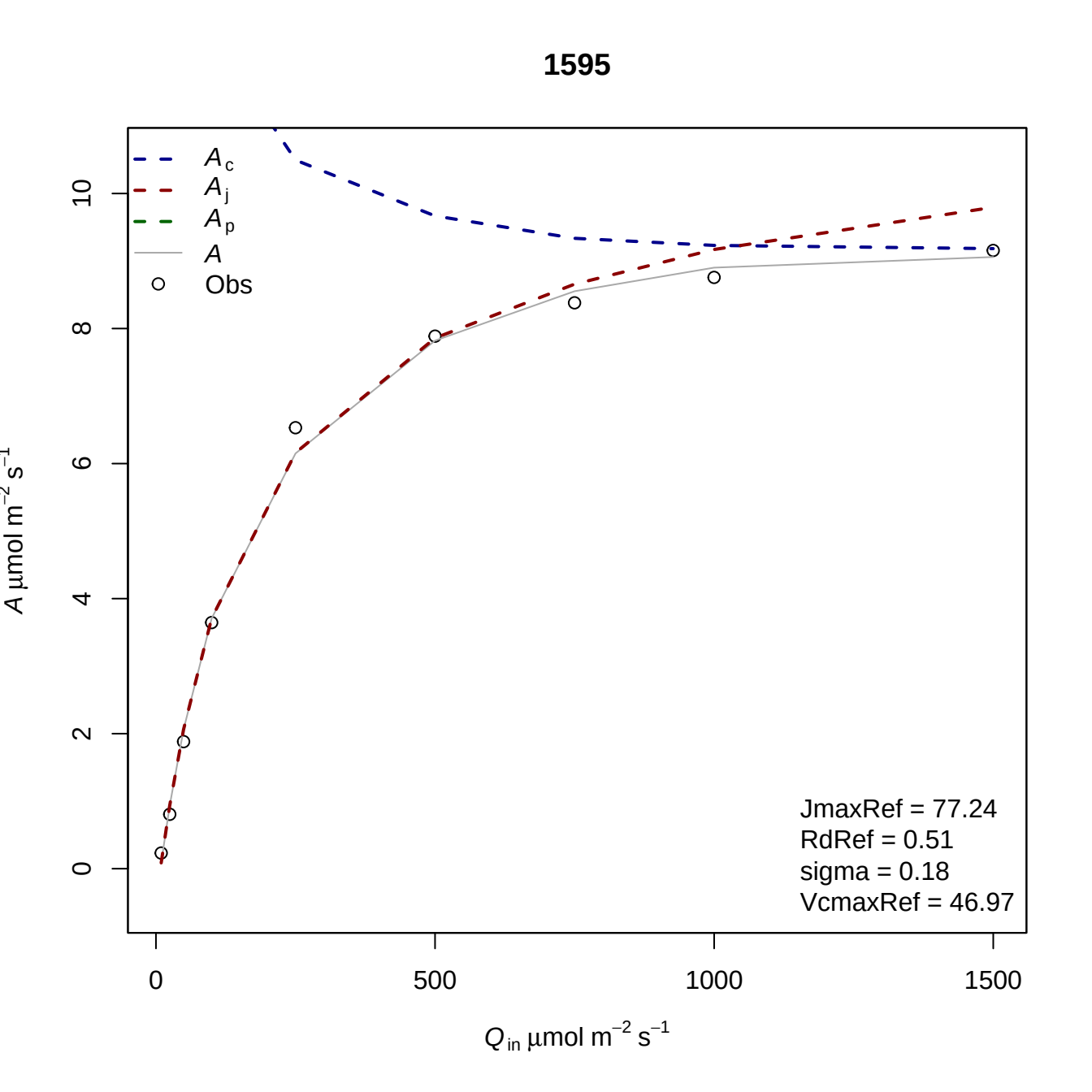

1590
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 90.66
RdRef = 0.58
sigma = 0.19
VcmaxRef = 50.53
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

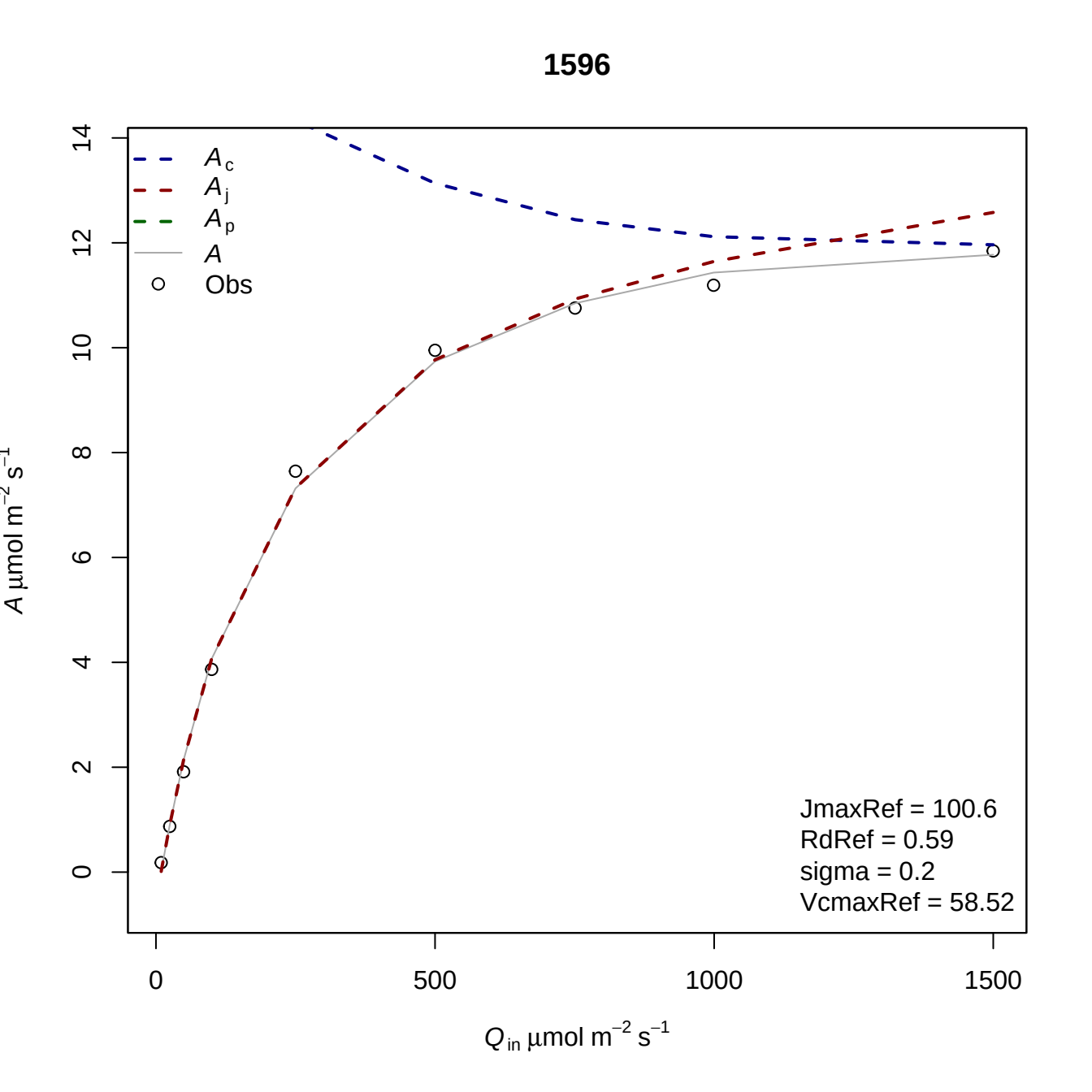

1591
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 36.7
RdRef = 0.99
sigma = 0.62
VcmaxRef = 232970.28
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

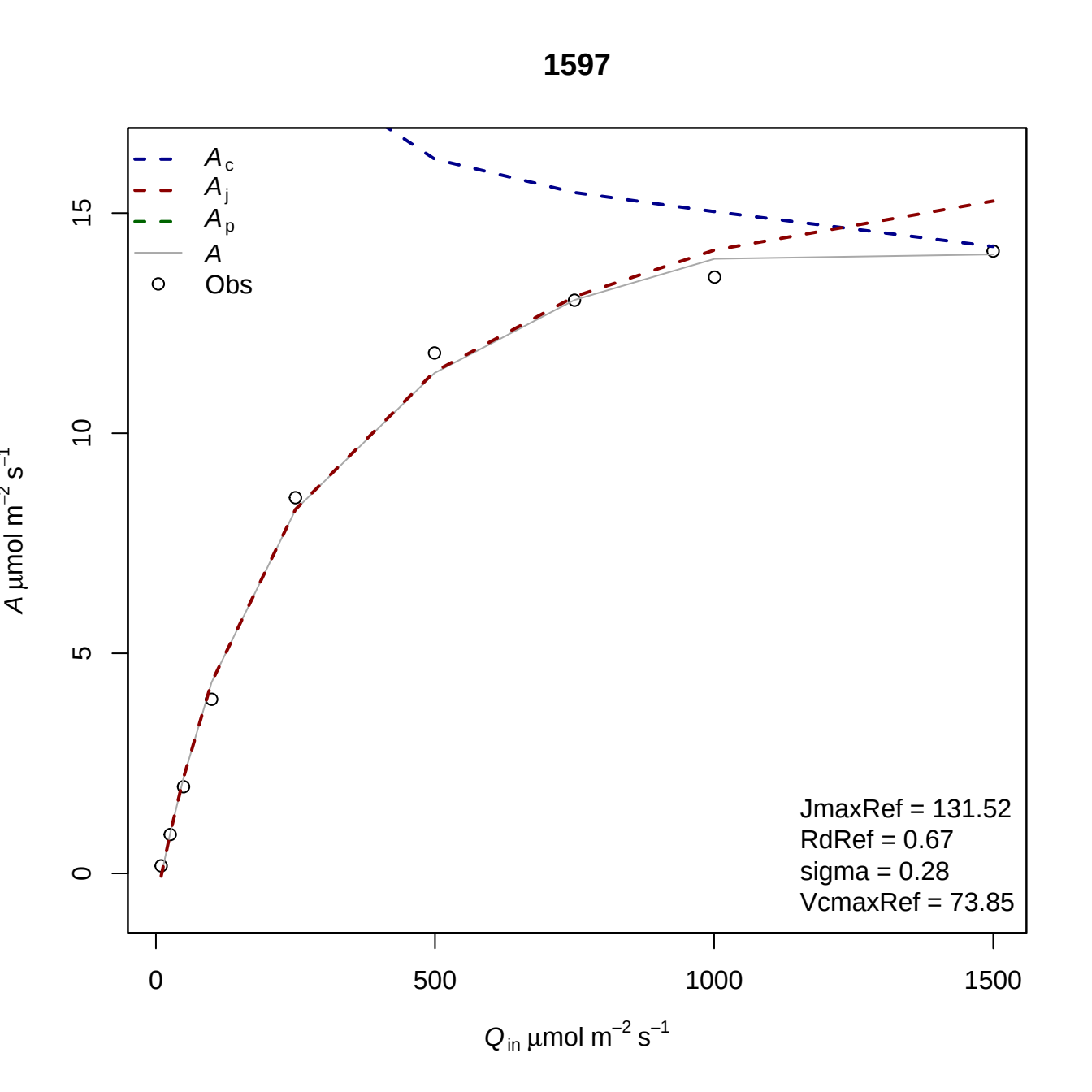

1592
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.41
RdRef = 0.82
sigma = 0.18
VcmaxRef = 43.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

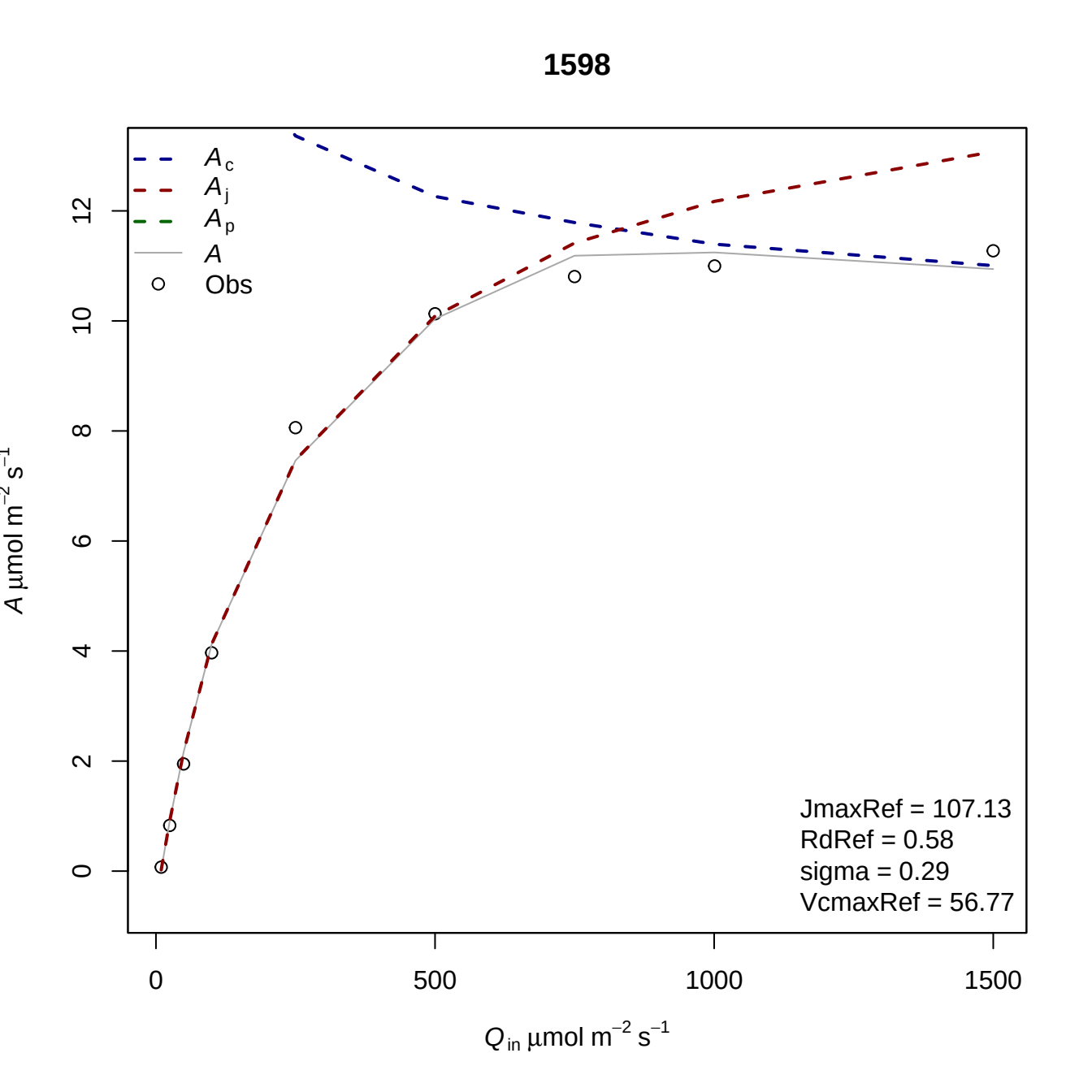

1593
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 42.48
RdRef = 0.94
sigma = 0.1
VcmaxRef = 22.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
−1
0
1
2
3
4
5
0
500
1000
1500

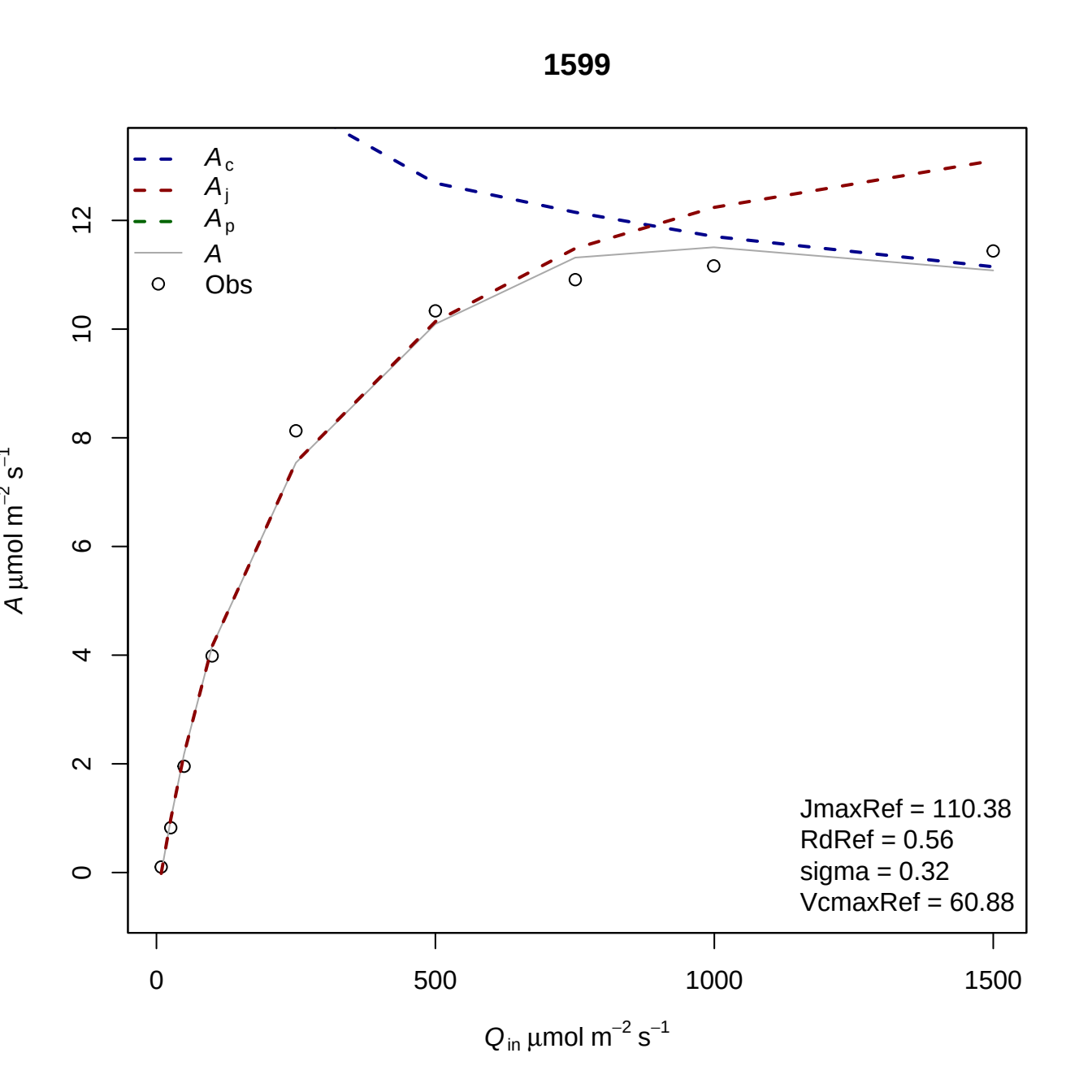

1594
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 47.77
RdRef = 0.65
sigma = 0.08
VcmaxRef = 36.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
1
2
3
4
5
6
0
500
1000
1500

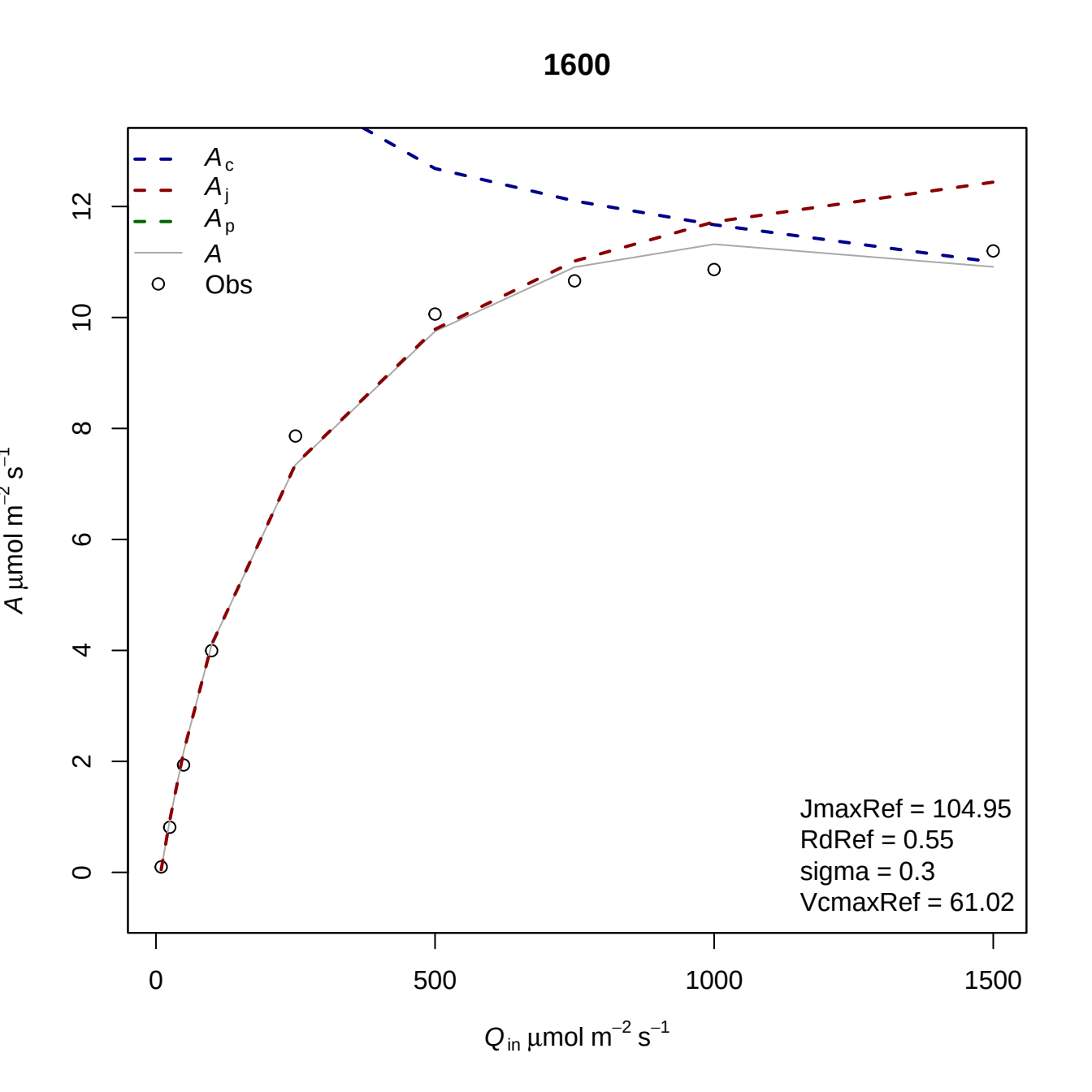

1595
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 77.24
RdRef = 0.51
sigma = 0.18
VcmaxRef = 46.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

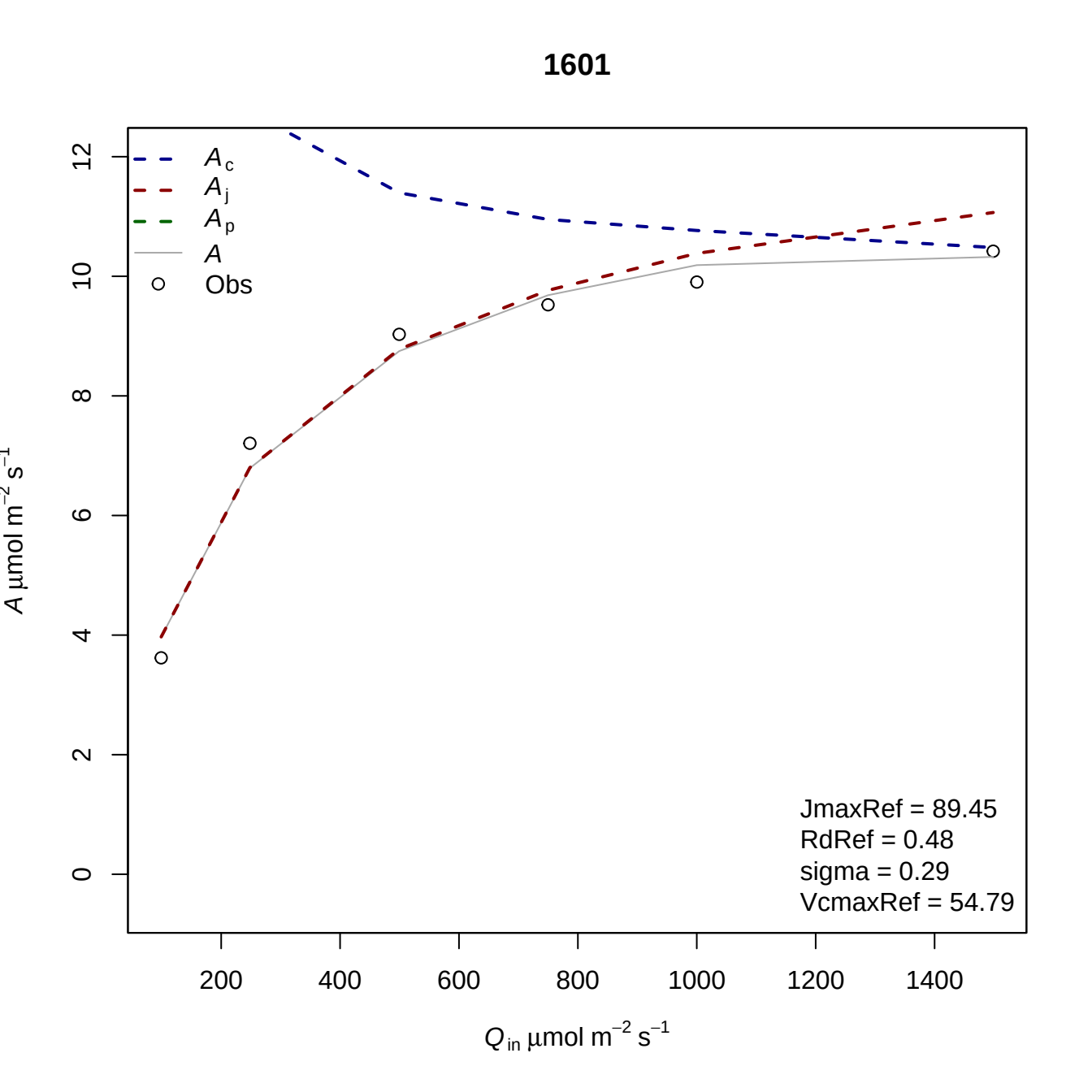

1596
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 100.6
RdRef = 0.59
sigma = 0.2
VcmaxRef = 58.52
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
14
0
500
1000
1500

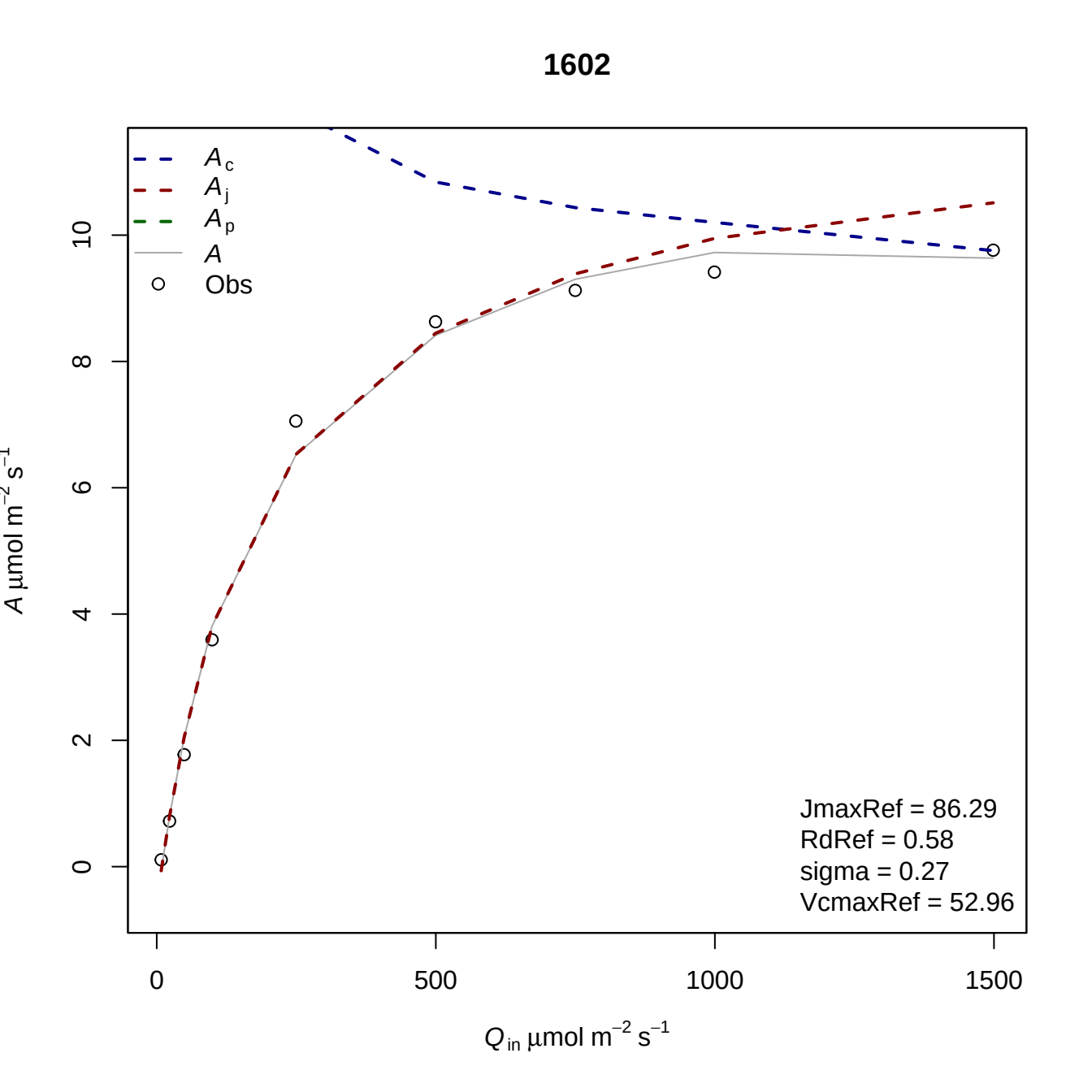

1597
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 131.52
RdRef = 0.67
sigma = 0.28
VcmaxRef = 73.85
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500

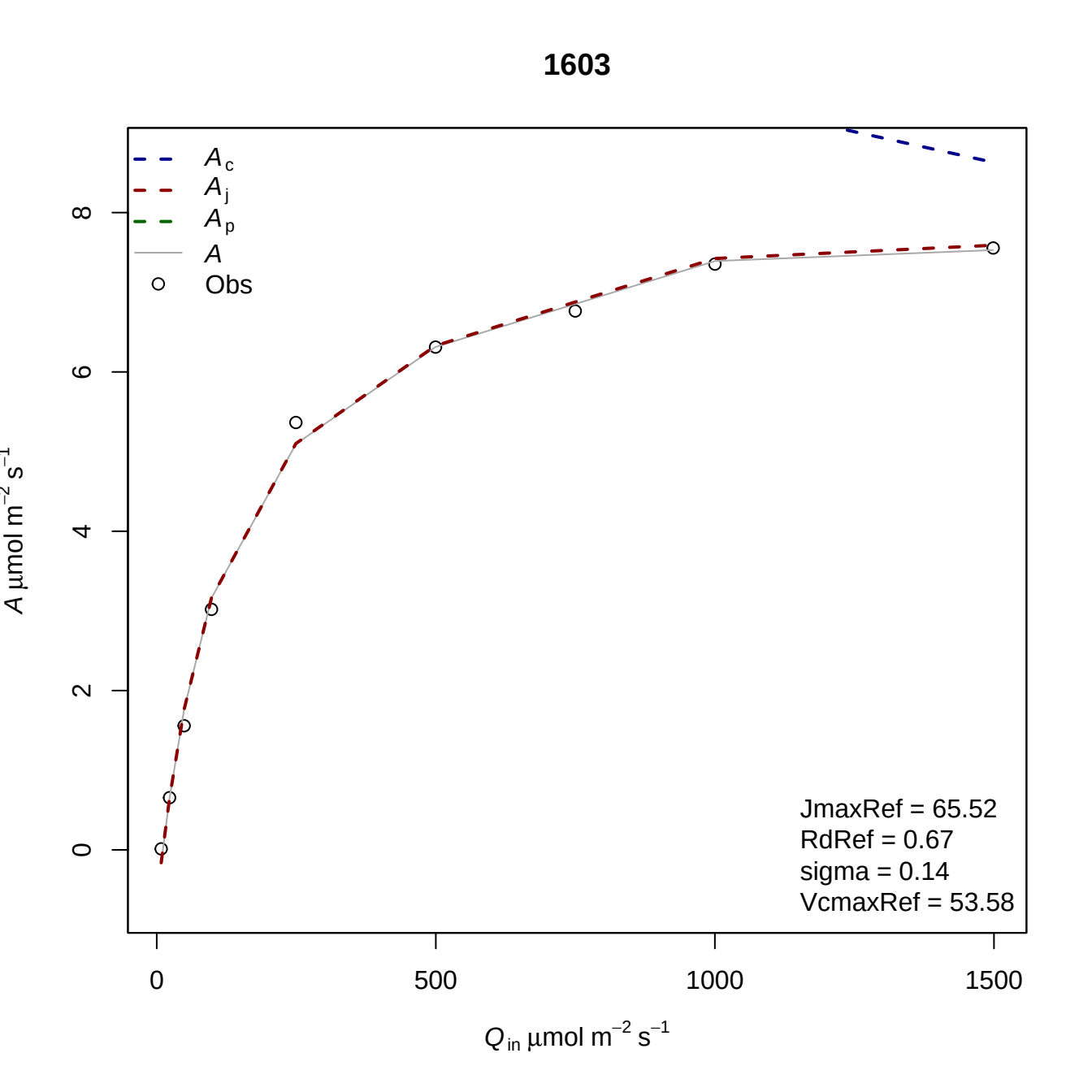

1598
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 107.13
RdRef = 0.58
sigma = 0.29
VcmaxRef = 56.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

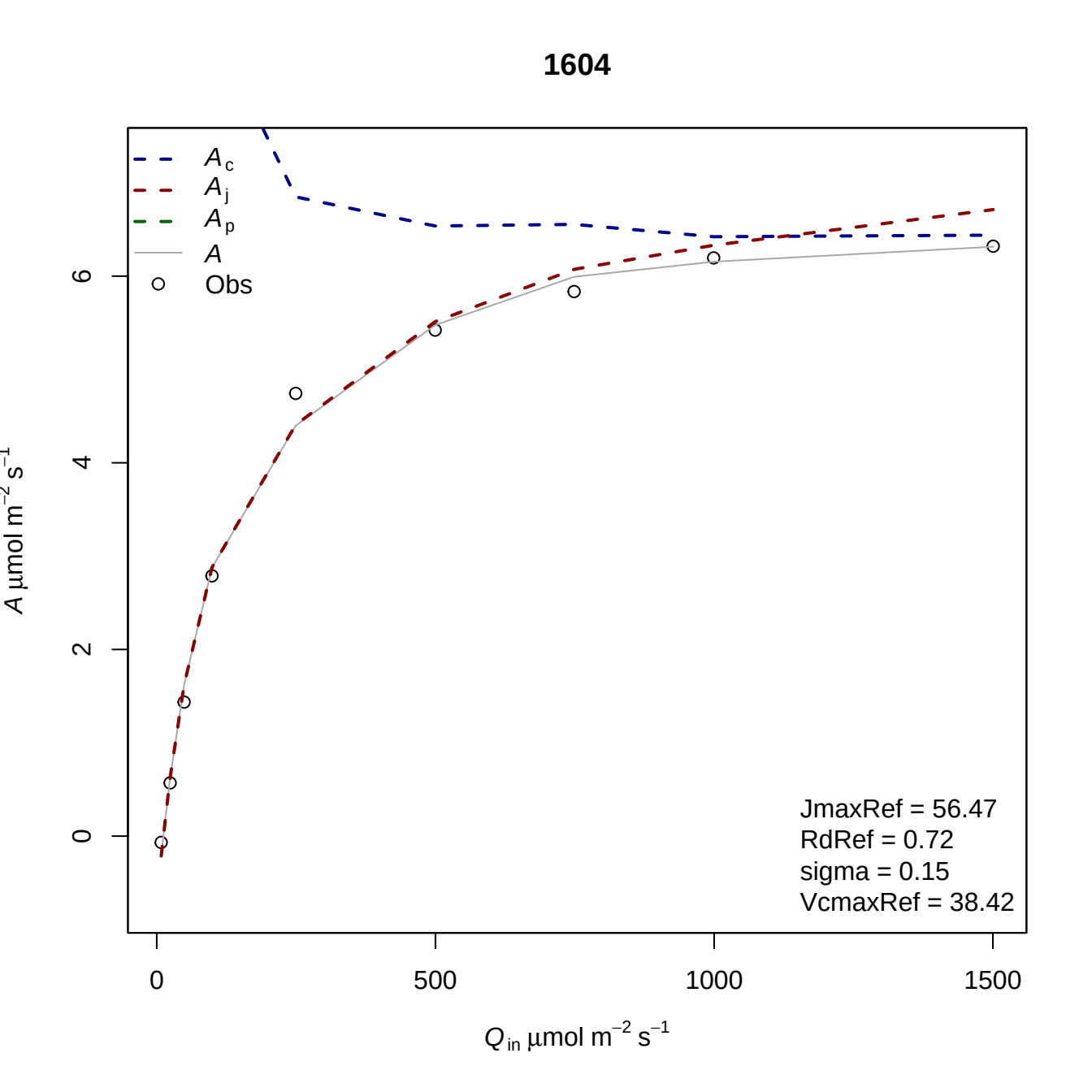

1599
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 110.38
RdRef = 0.56
sigma = 0.32
VcmaxRef = 60.88
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

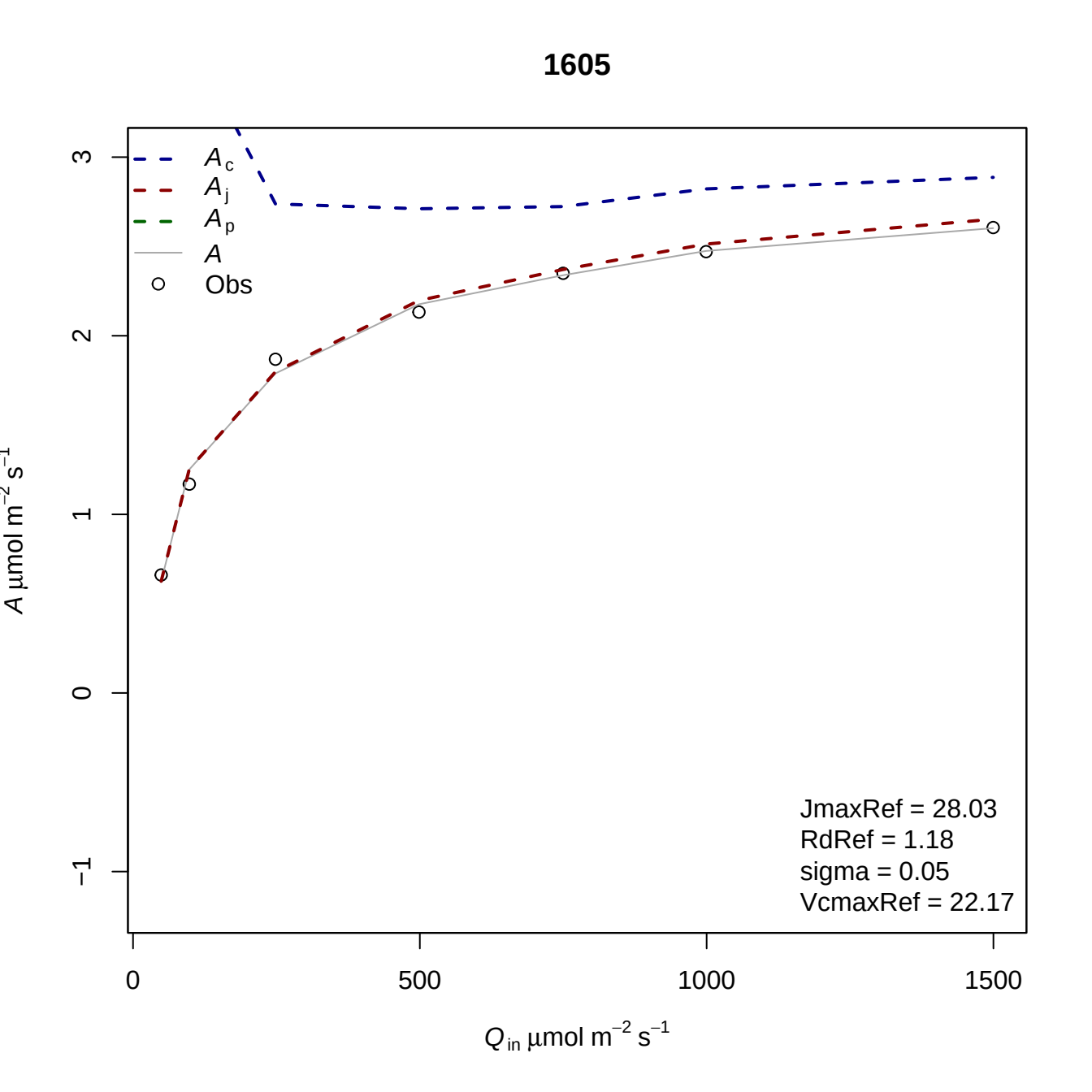

1600
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 104.95
RdRef = 0.55
sigma = 0.3
VcmaxRef = 61.02
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500

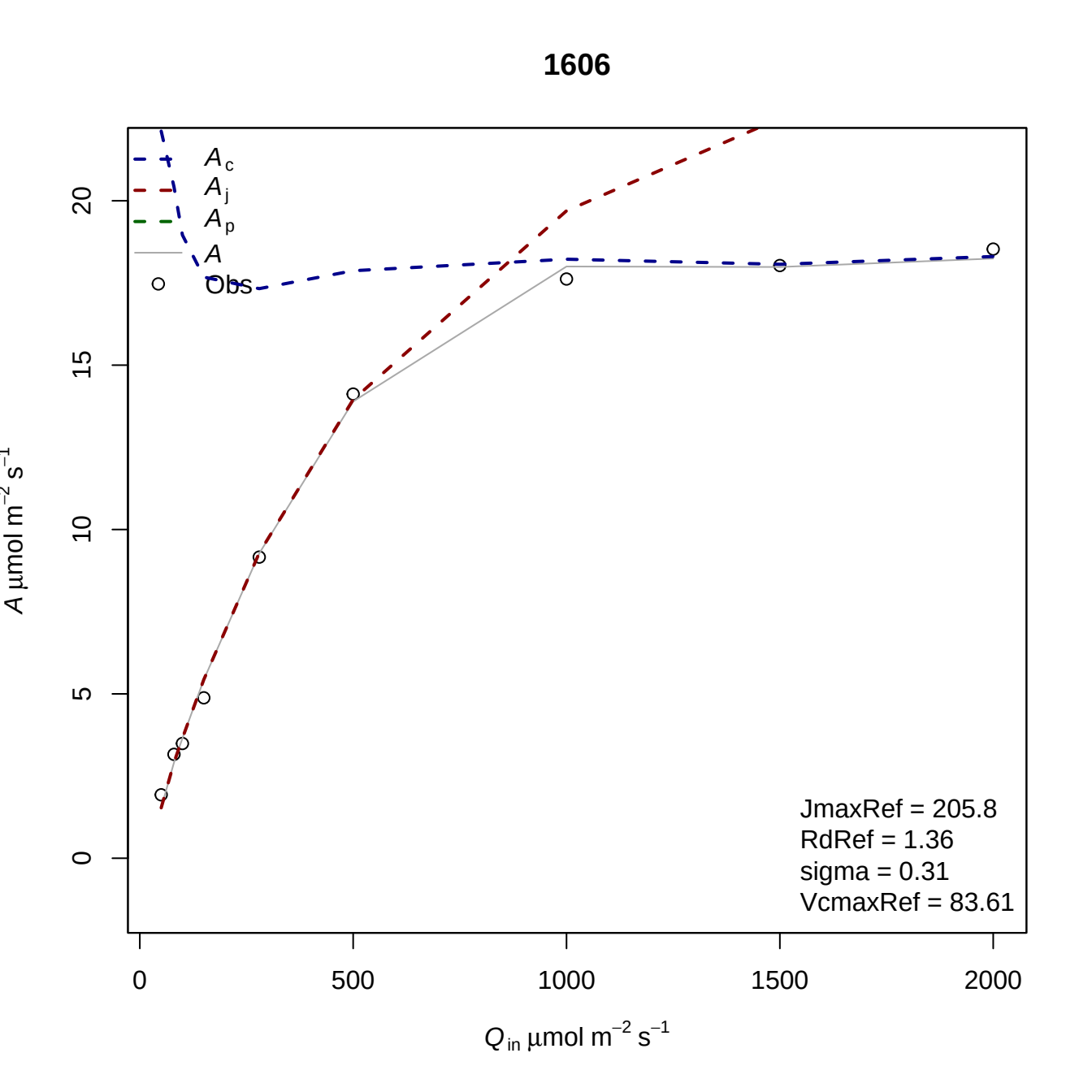

1601
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.45
RdRef = 0.48
sigma = 0.29
VcmaxRef = 54.79
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

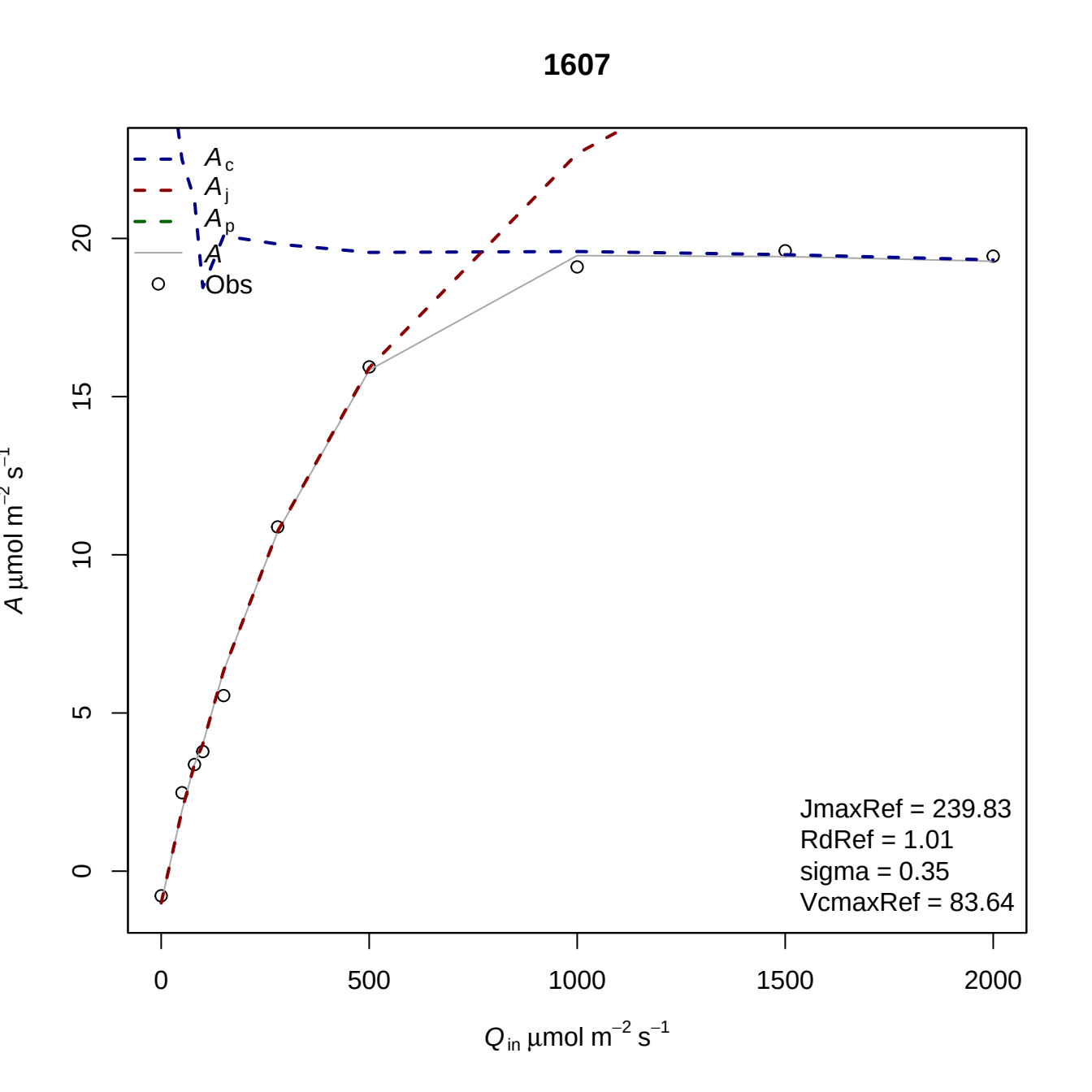

1602
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 86.29
RdRef = 0.58
sigma = 0.27
VcmaxRef = 52.96
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500

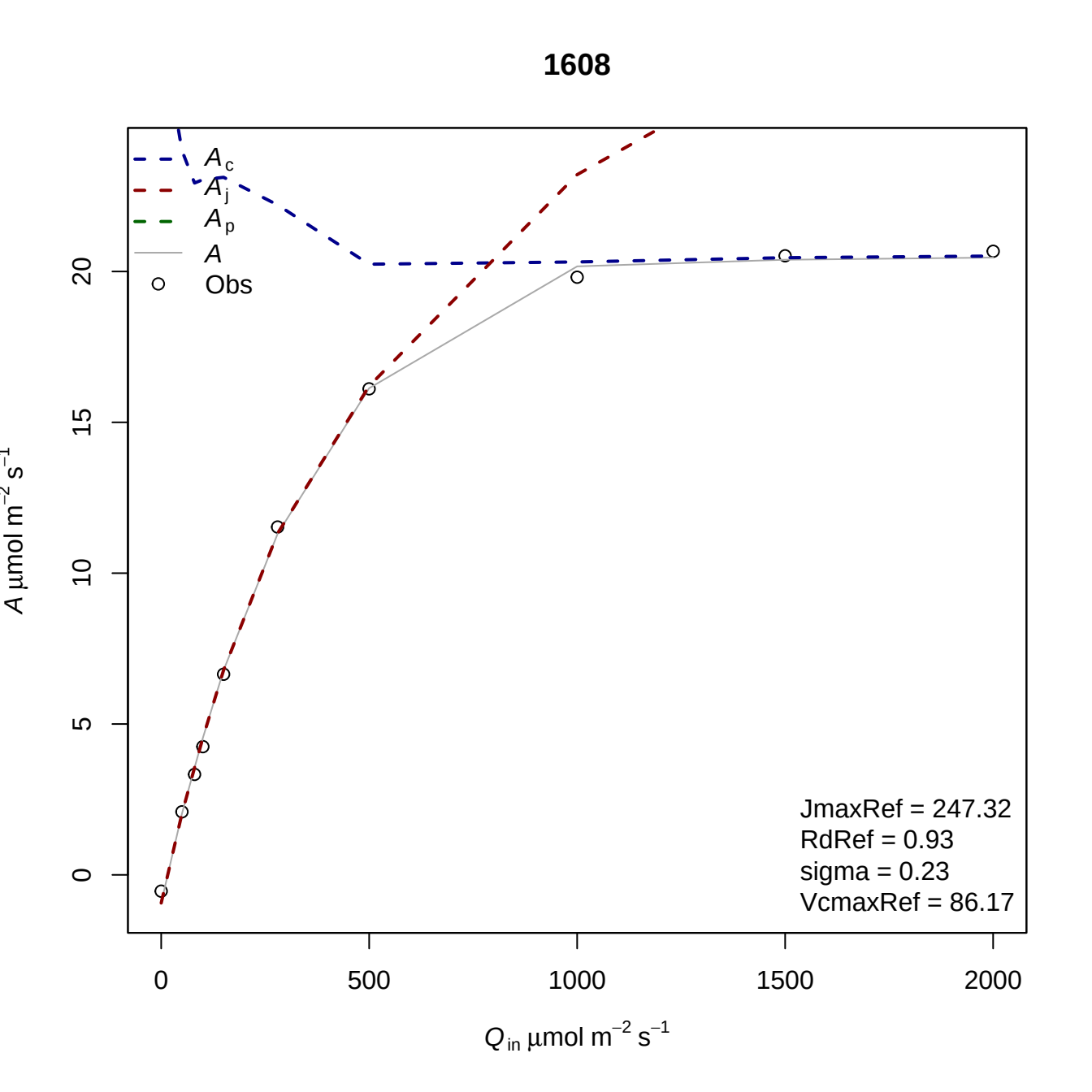

1603
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 65.52
RdRef = 0.67
sigma = 0.14
VcmaxRef = 53.58
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500

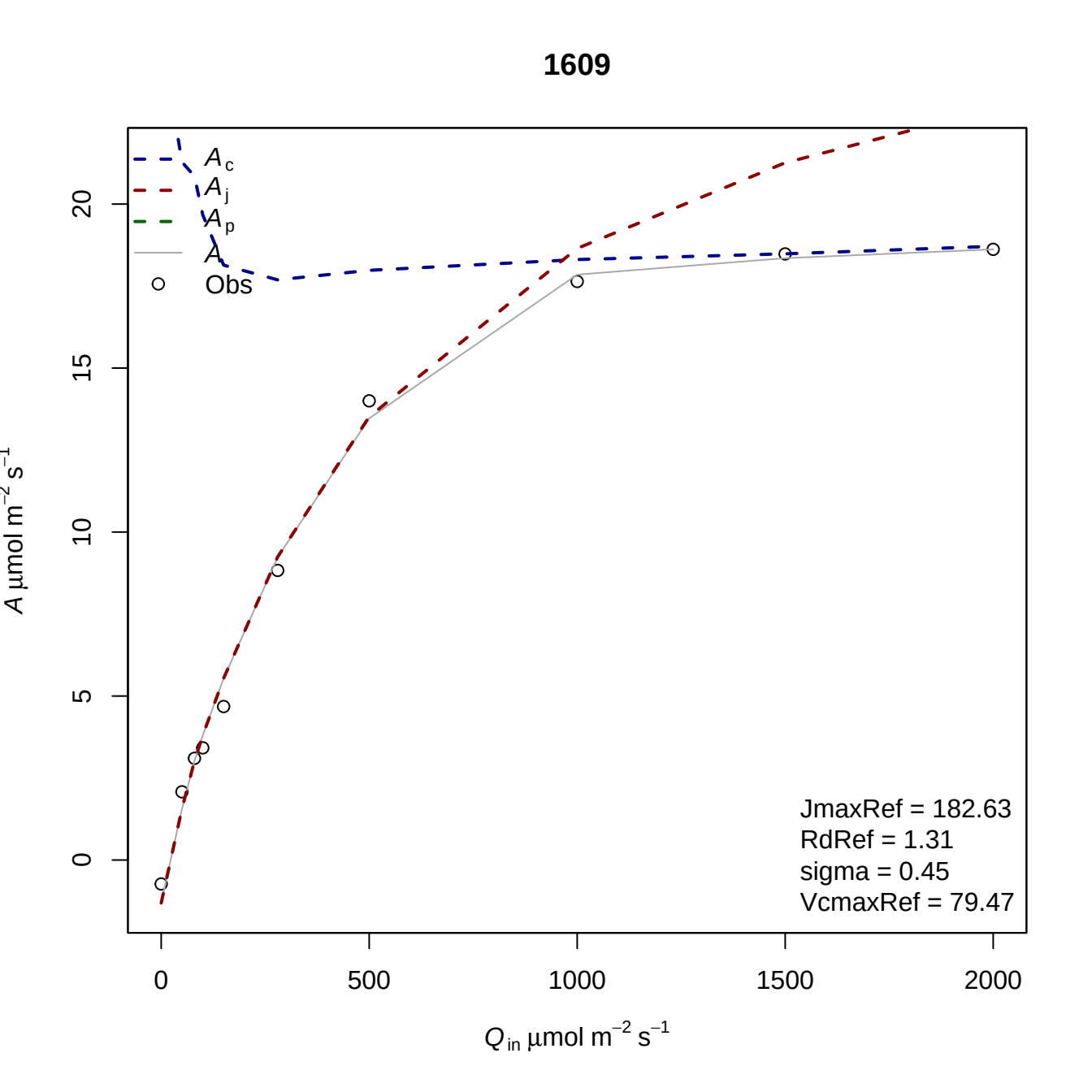

1604
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 56.47
RdRef = 0.72
sigma = 0.15
VcmaxRef = 38.42
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500

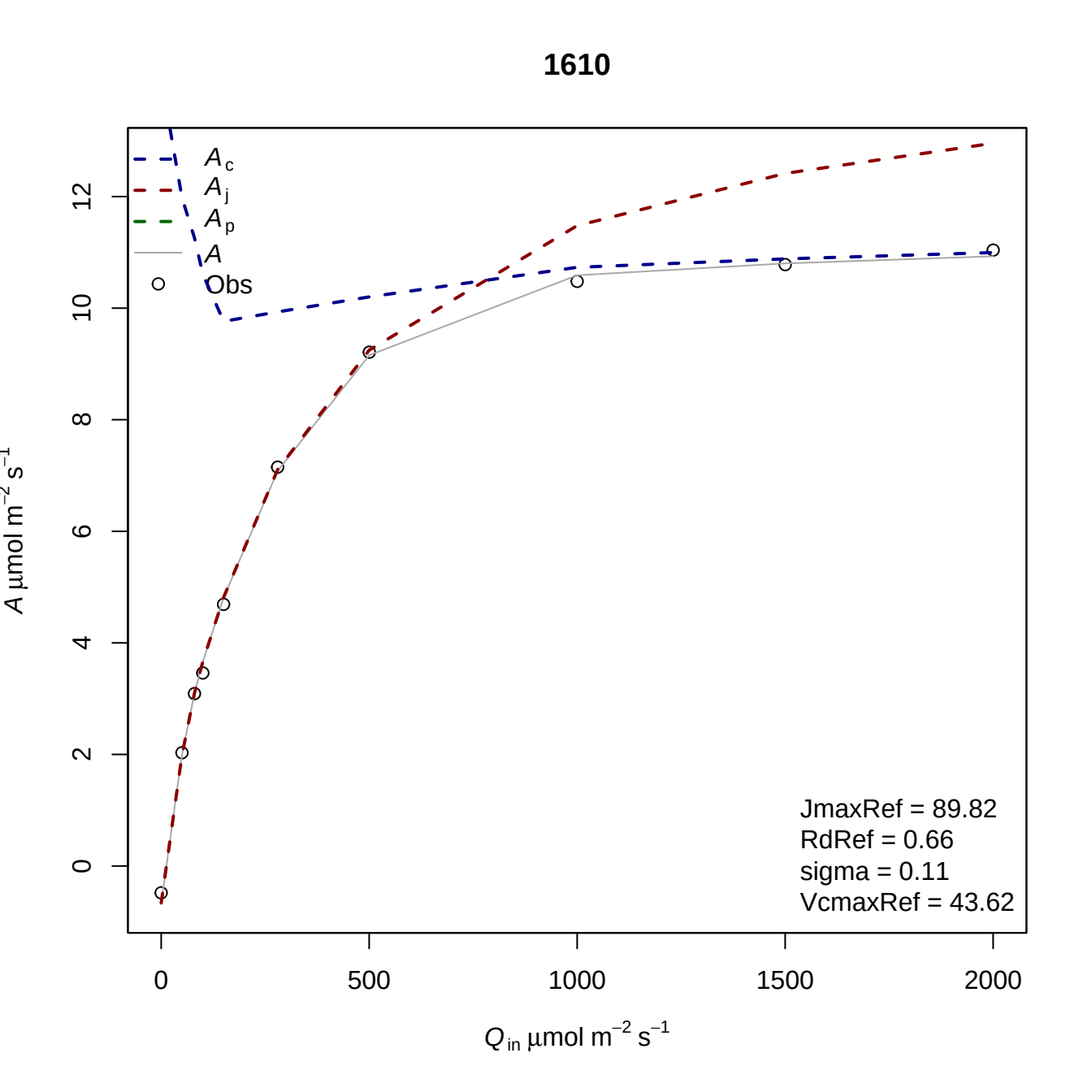

1605
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 28.03
RdRef = 1.18
sigma = 0.05
VcmaxRef = 22.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
3
2
1
0
−1
0
500
1000
1500

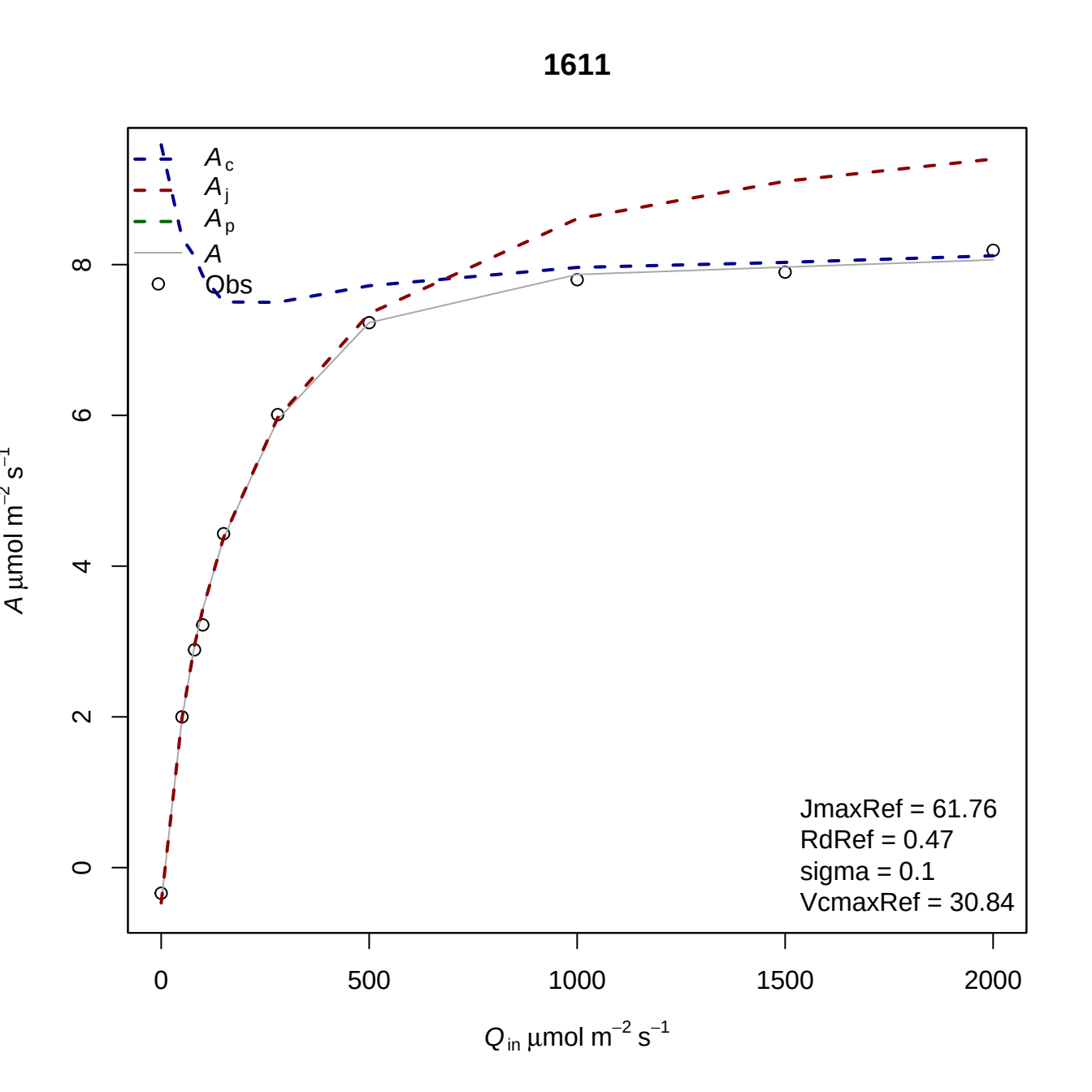

1606
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 205.8
RdRef = 1.36
sigma = 0.31
VcmaxRef = 83.61
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

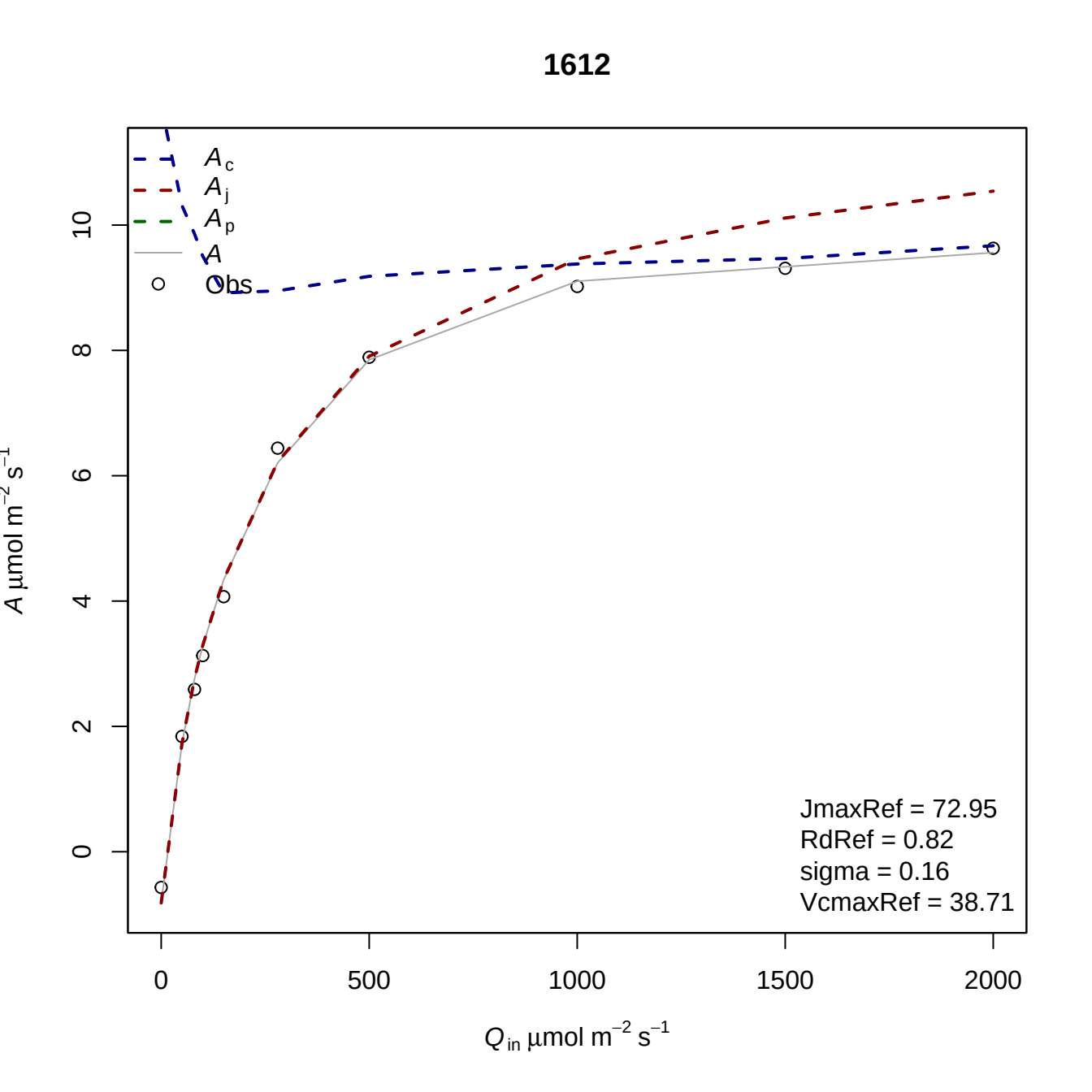

1607
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 239.83
RdRef = 1.01
sigma = 0.35
VcmaxRef = 83.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

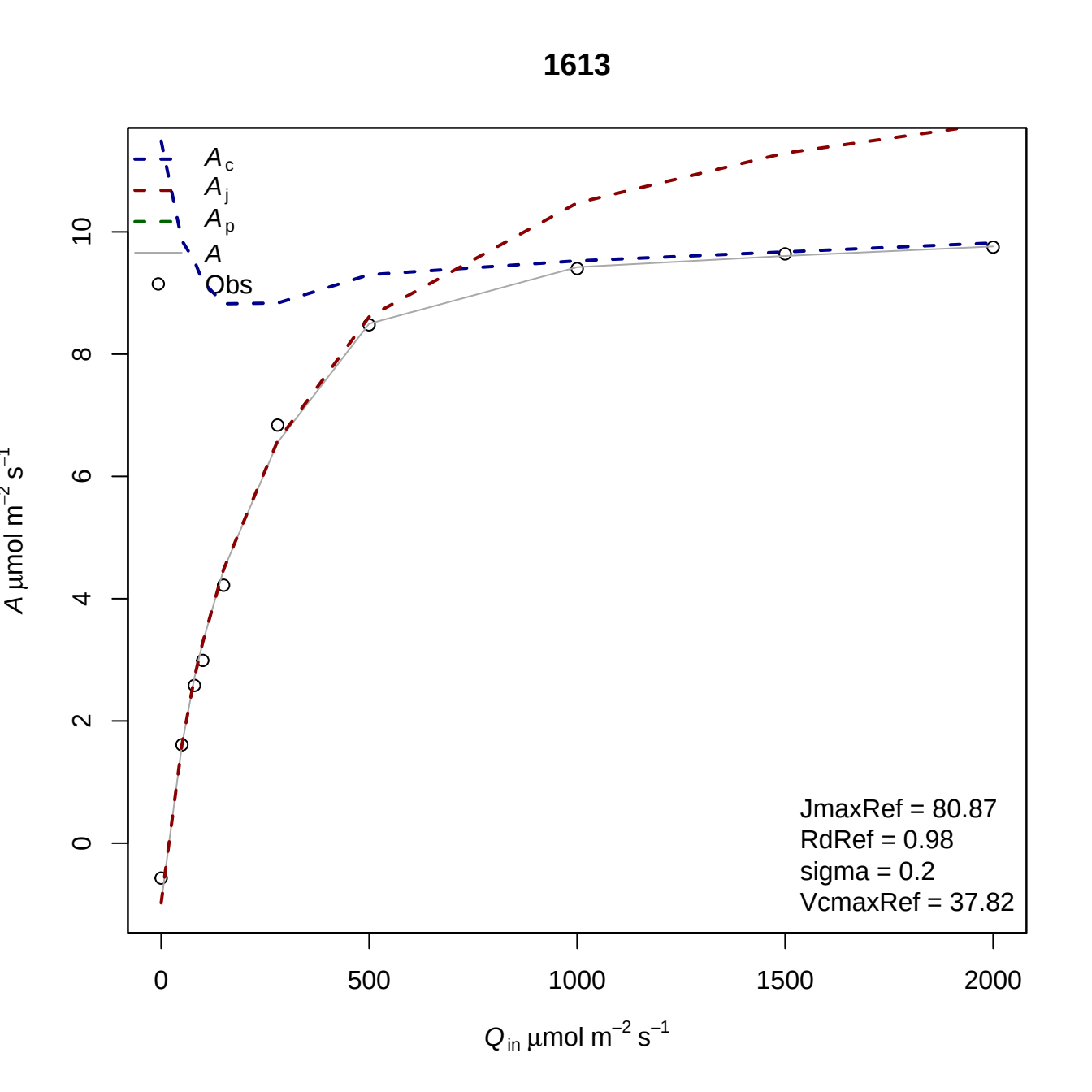

1608
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 247.32
RdRef = 0.93
sigma = 0.23
VcmaxRef = 86.17
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

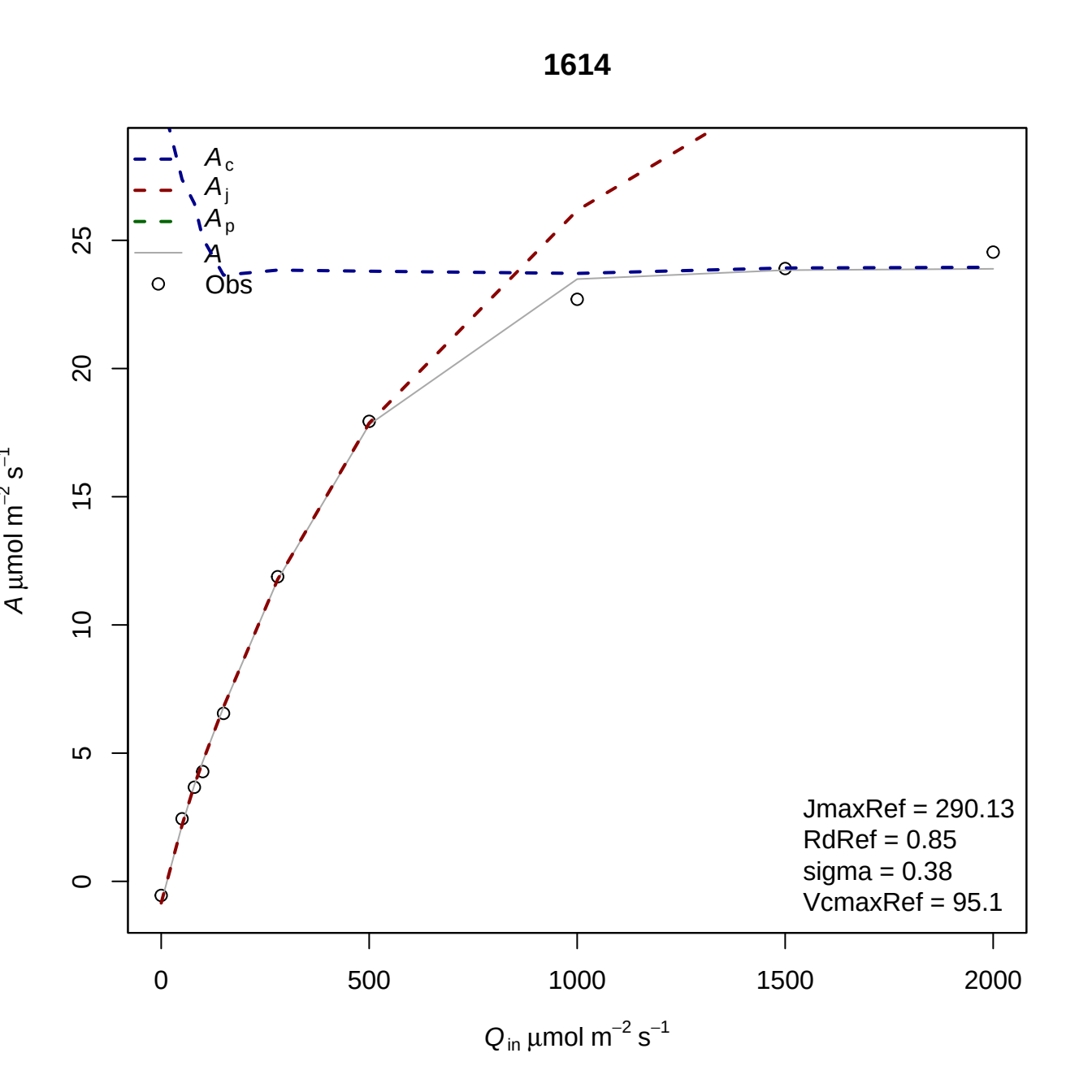

1609
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 182.63
RdRef = 1.31
sigma = 0.45
VcmaxRef = 79.47
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

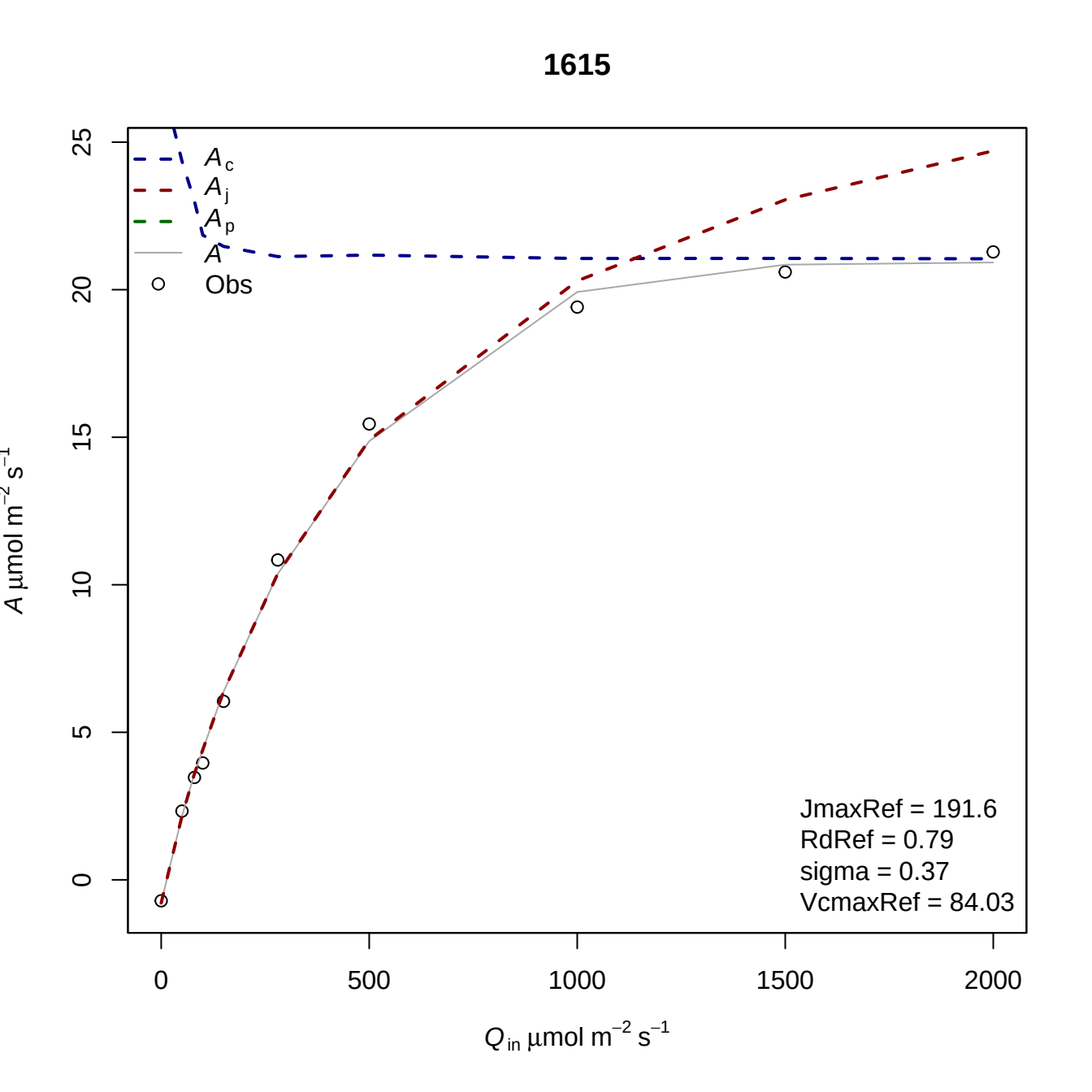

1610
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 89.82
RdRef = 0.66
sigma = 0.11
VcmaxRef = 43.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

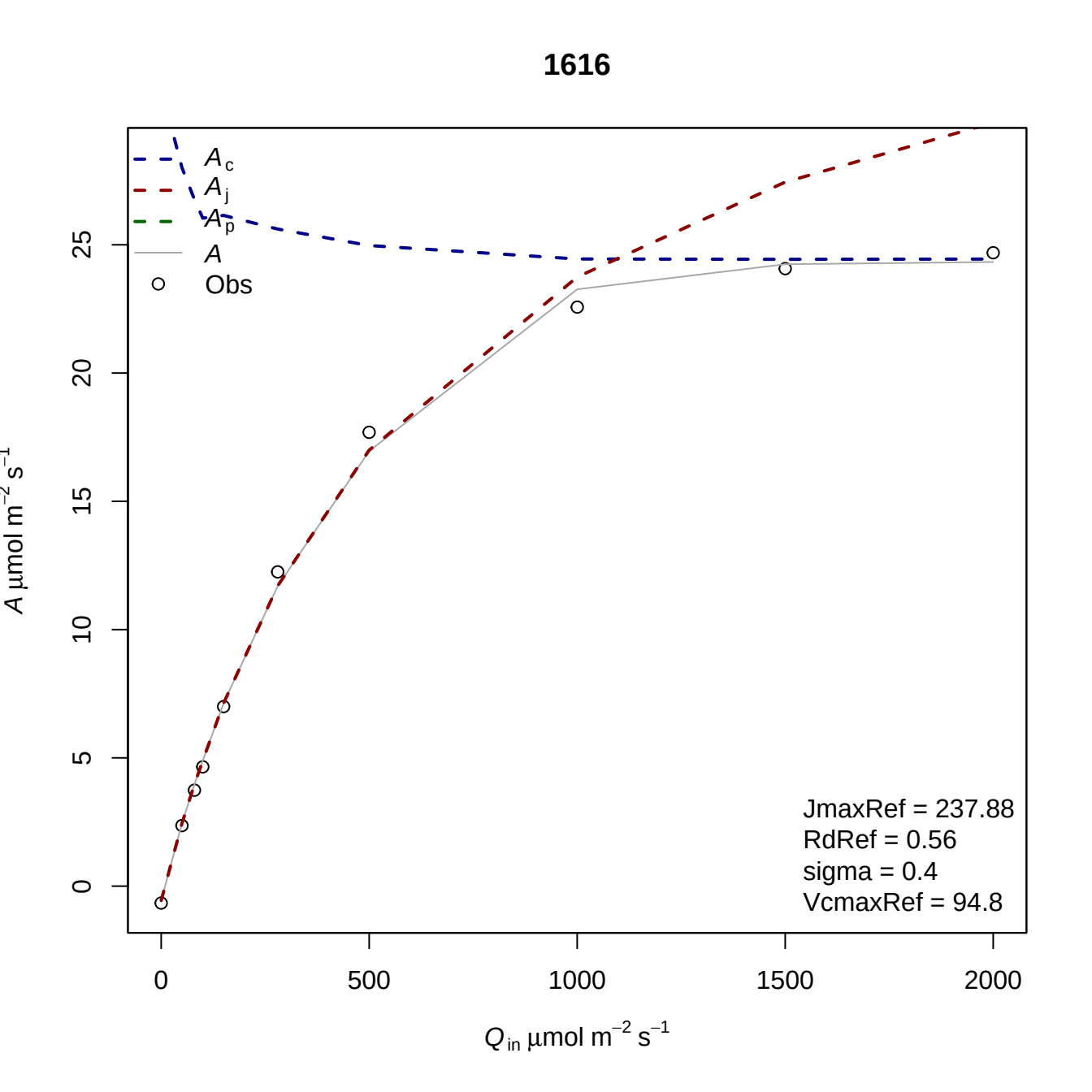

1611
$A_c$
$A_j$
$A_p$
$A$
Qbs
JmaxRef = 61.76
RdRef = 0.47
sigma = 0.1
VcmaxRef = 30.84
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

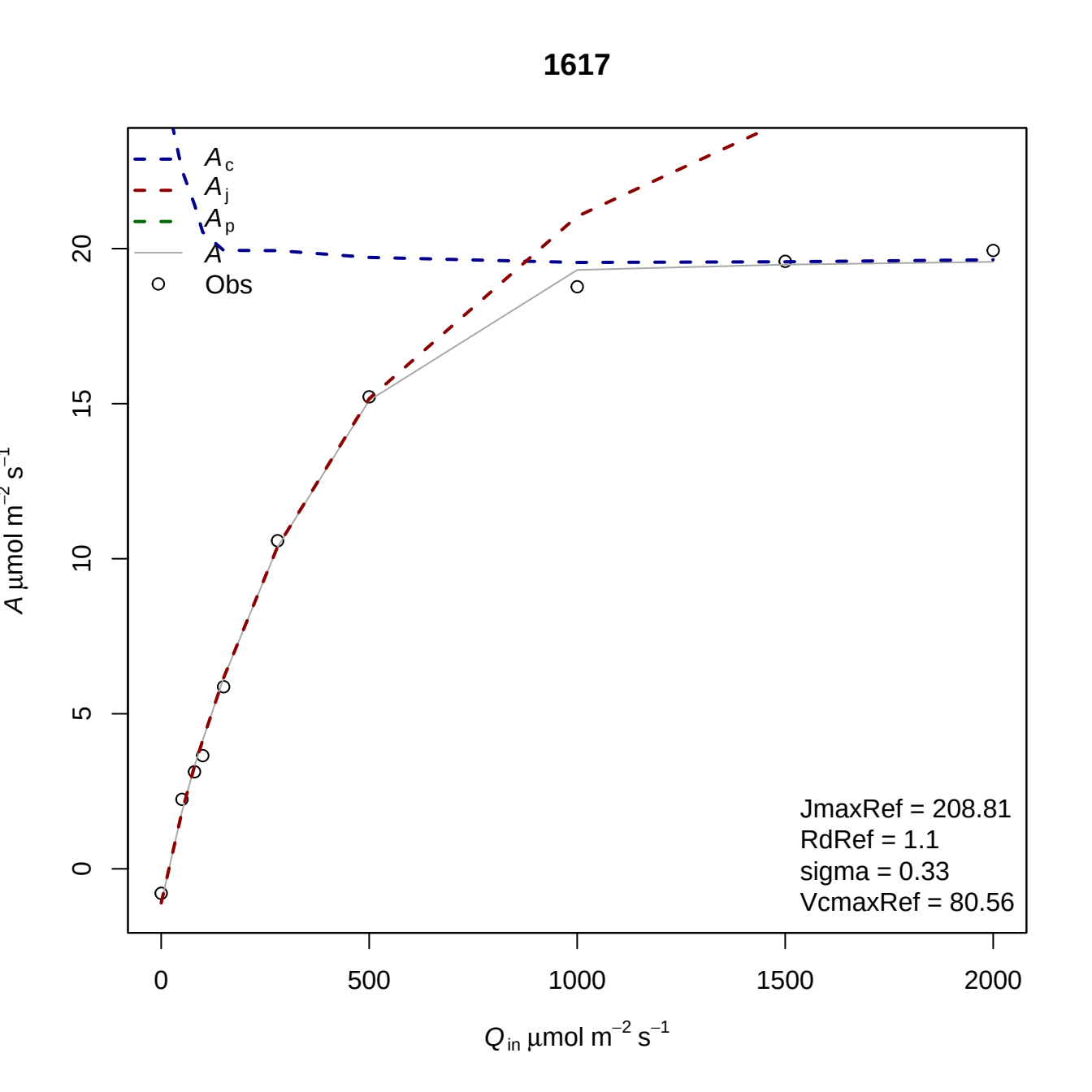

1612
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 72.95
RdRef = 0.82
sigma = 0.16
VcmaxRef = 38.71
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

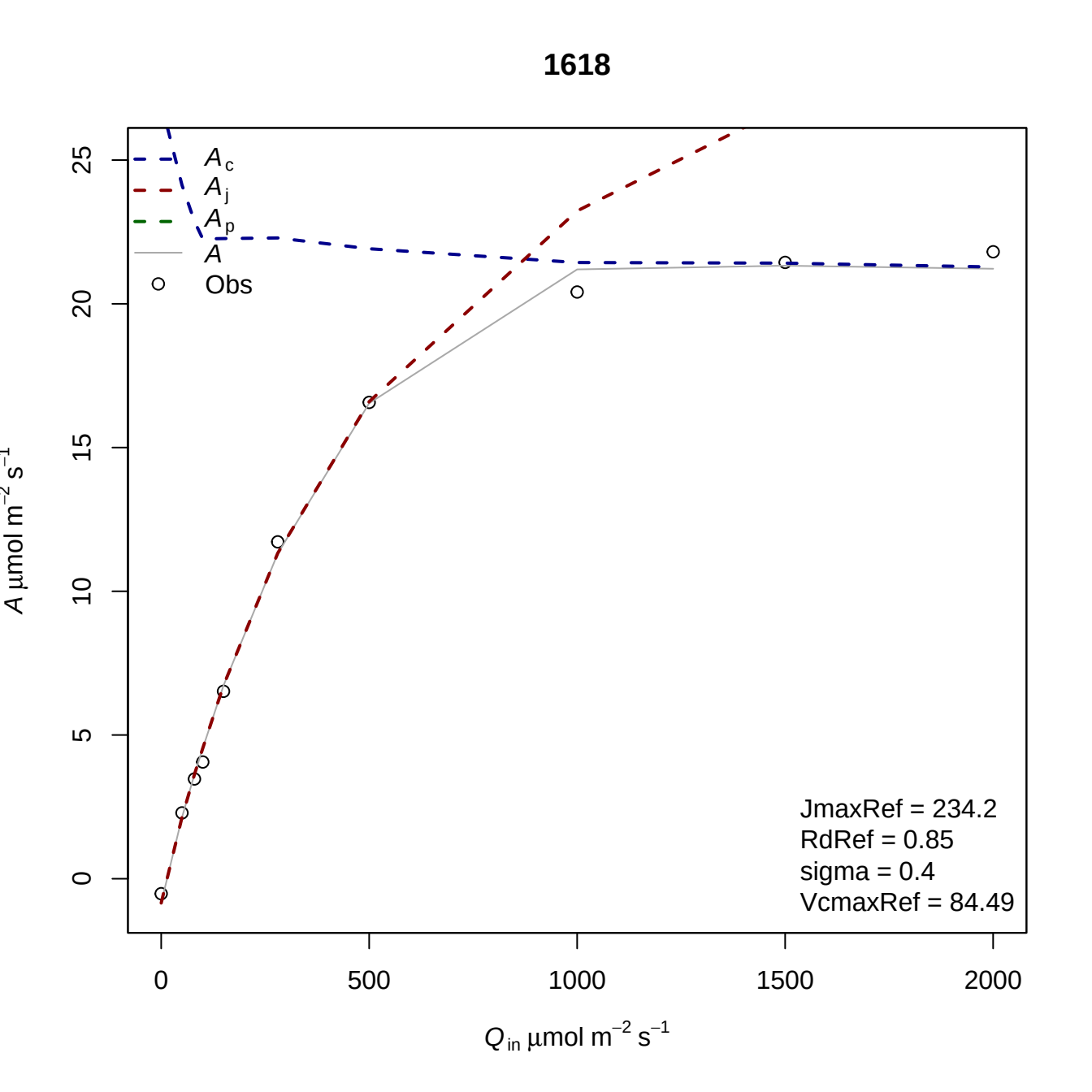

1613
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 80.87
RdRef = 0.98
sigma = 0.2
VcmaxRef = 37.82
0
2
4
6
8
10
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$

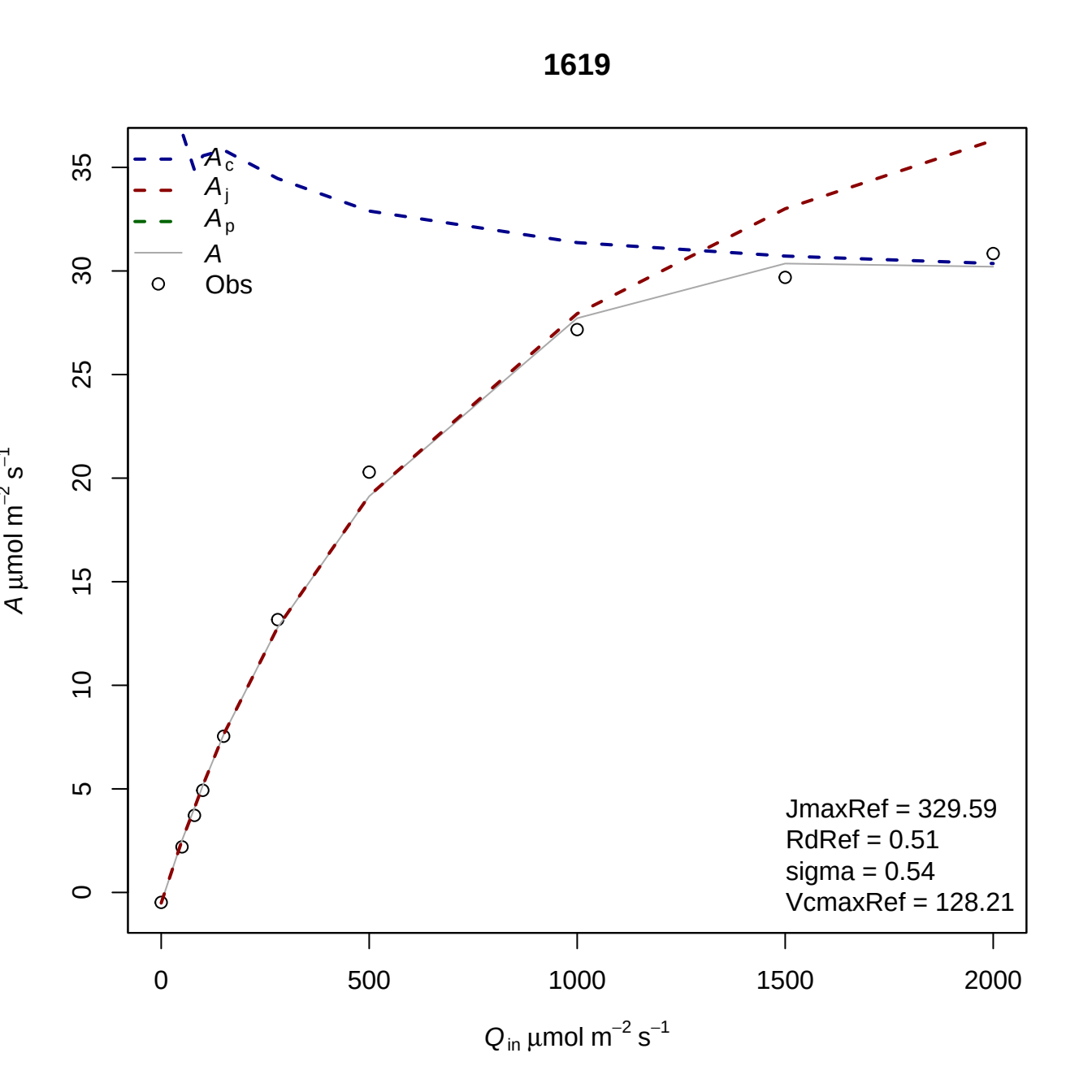

1614
$A_c$
$A_j$
$A_p$
A
Obs
JmaxRef = 290.13
RdRef = 0.85
sigma = 0.38
VcmaxRef = 95.1
A µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

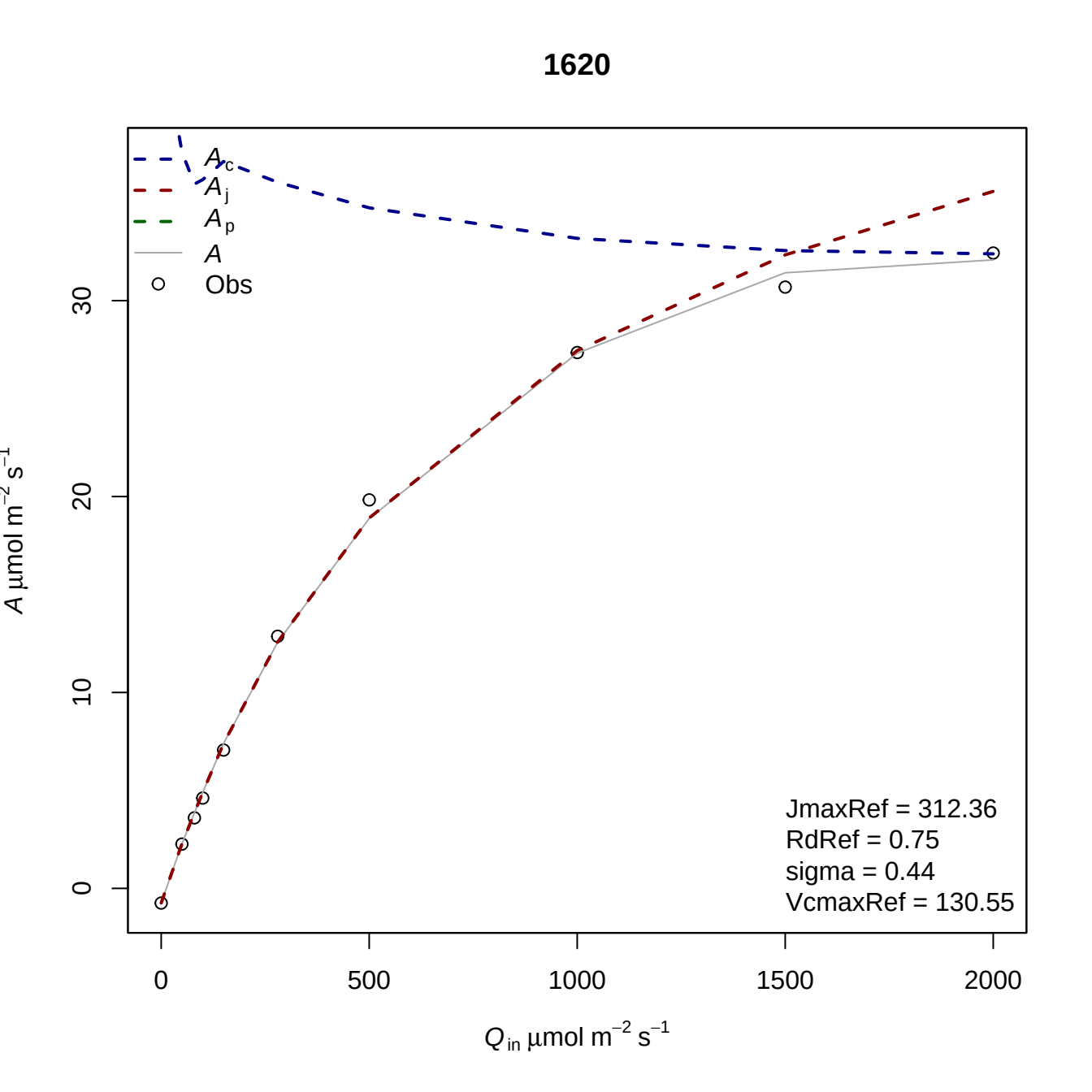

1615
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 191.6
RdRef = 0.79
sigma = 0.37
VcmaxRef = 84.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

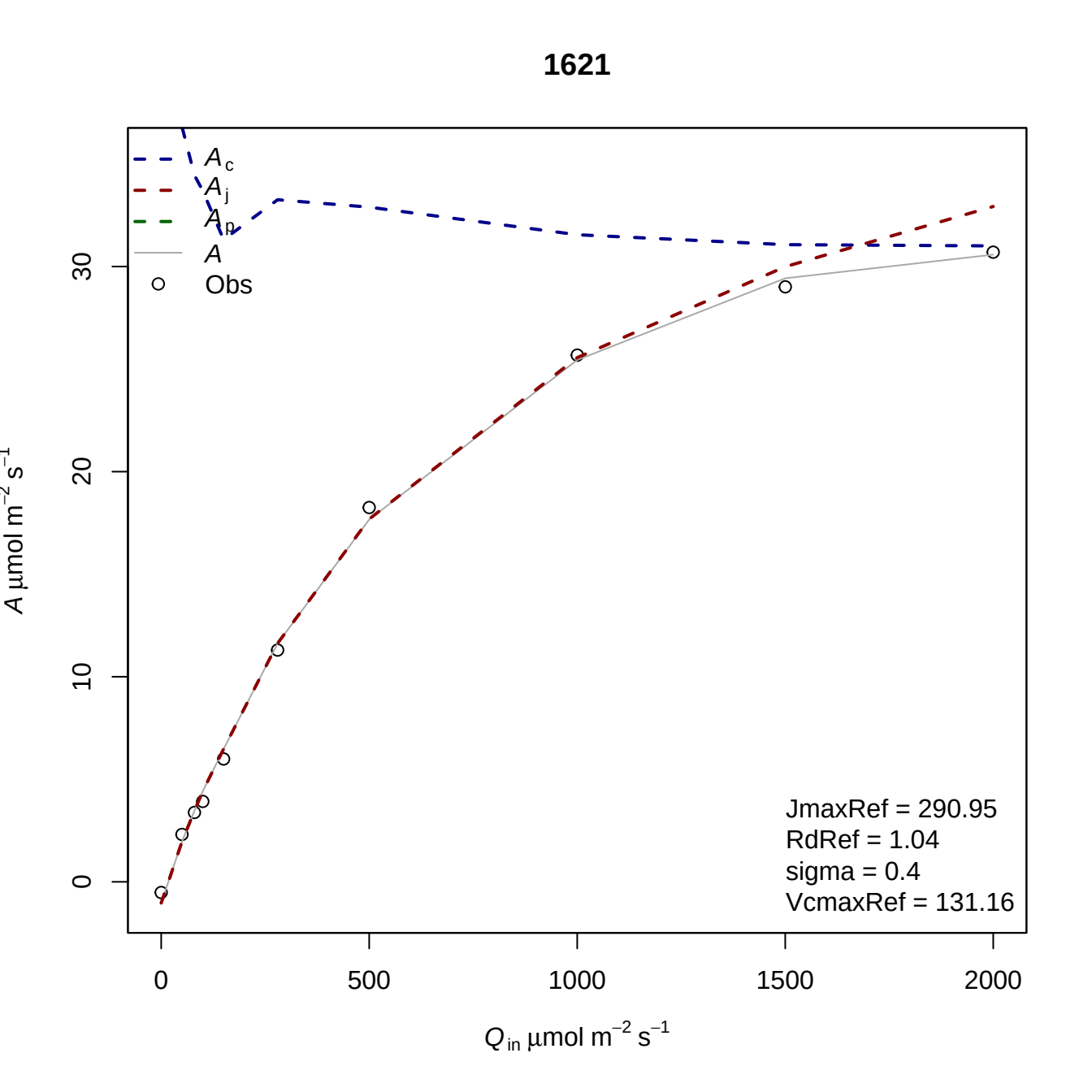

1616
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 237.88
RdRef = 0.56
sigma = 0.4
VcmaxRef = 94.8
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

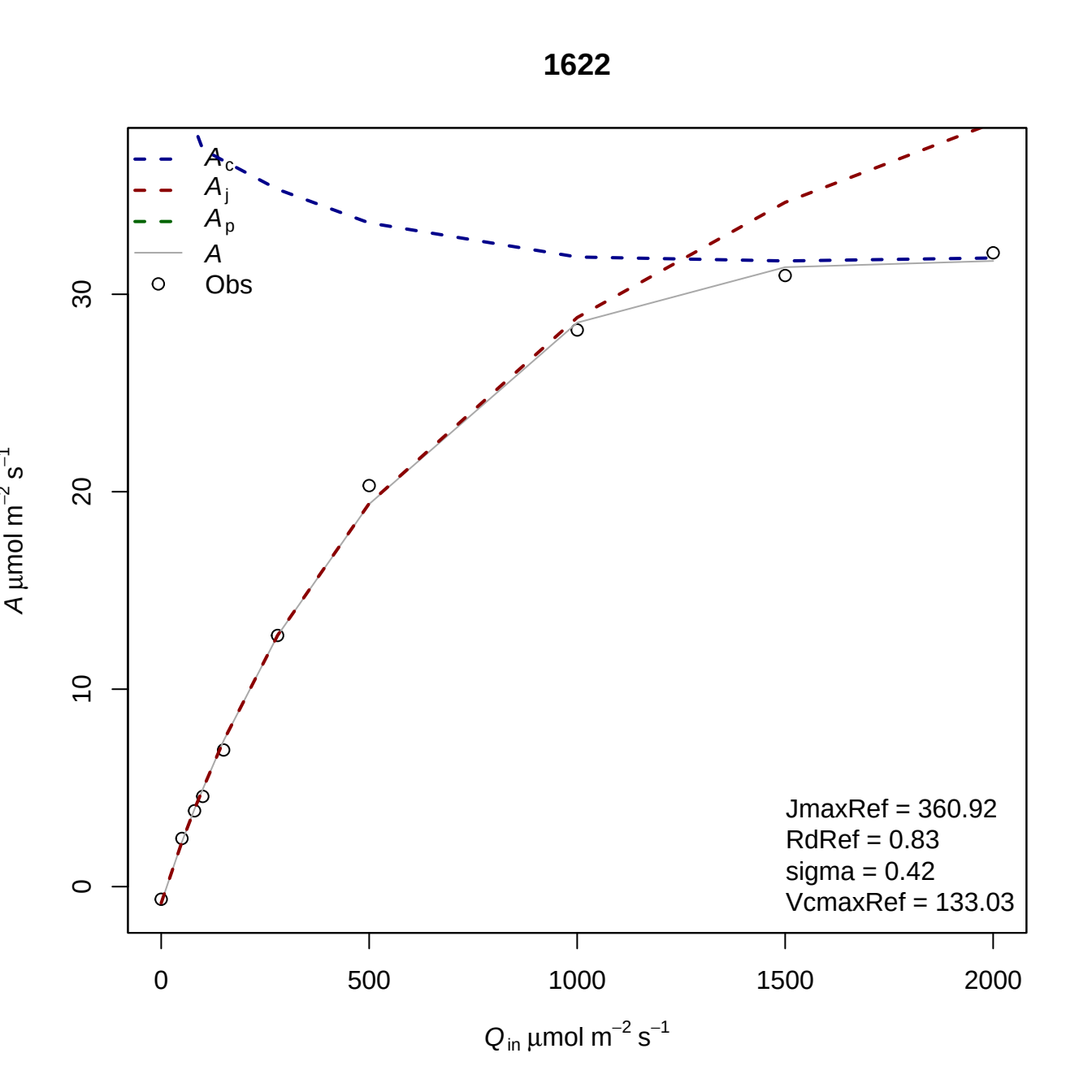

1617
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 208.81
RdRef = 1.1
sigma = 0.33
VcmaxRef = 80.56
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

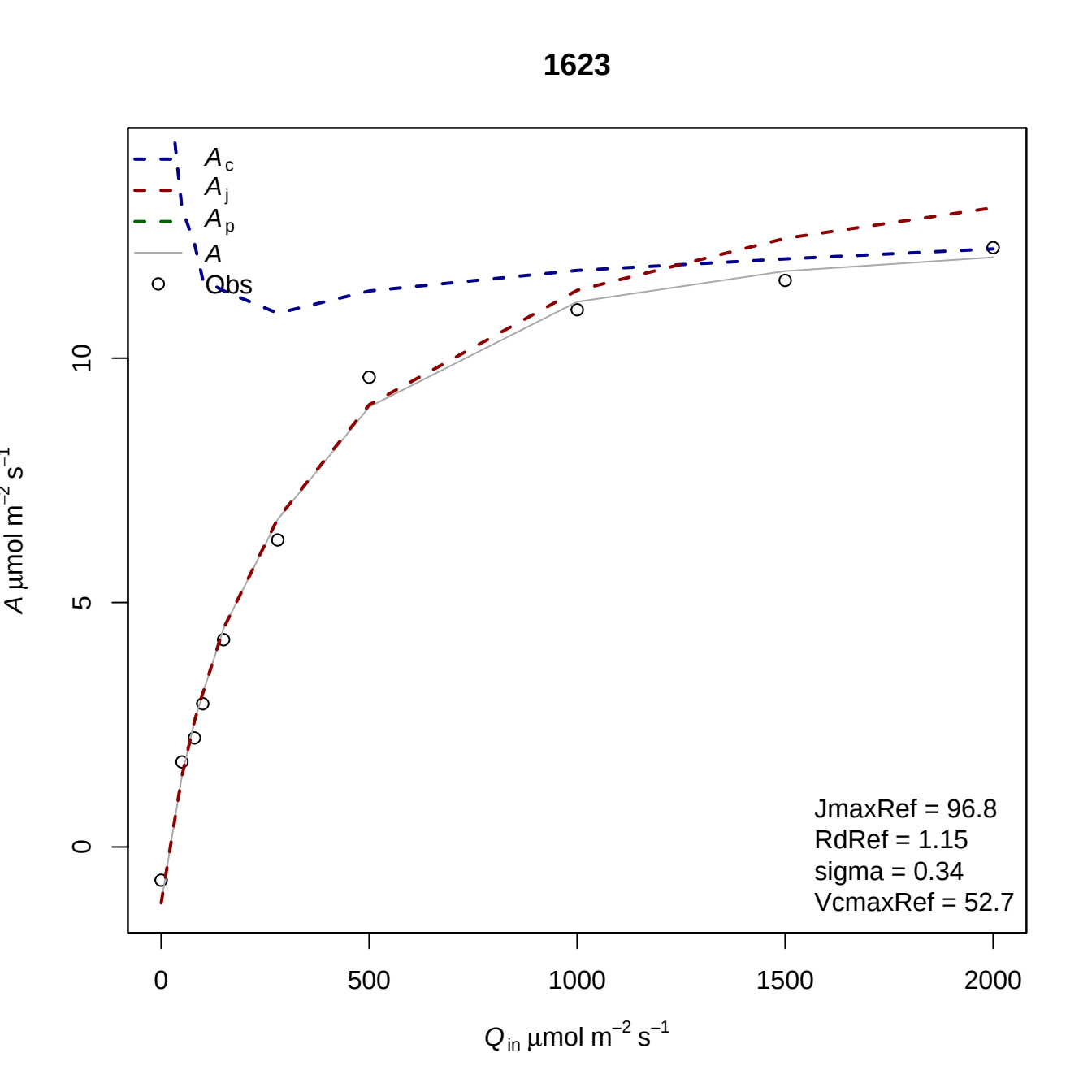

1618
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 234.2
RdRef = 0.85
sigma = 0.4
VcmaxRef = 84.49
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

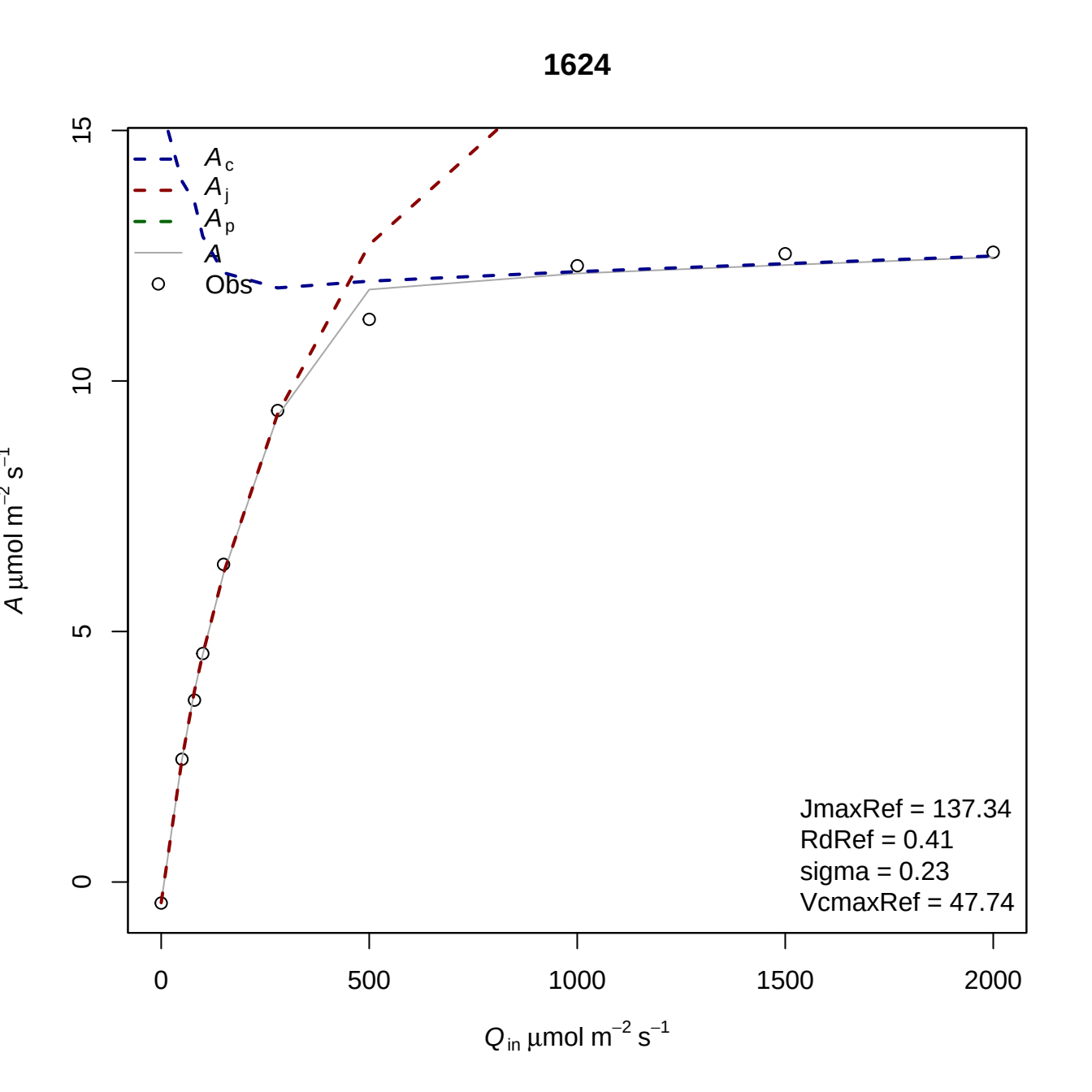

1619
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 329.59
RdRef = 0.51
sigma = 0.54
VcmaxRef = 128.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

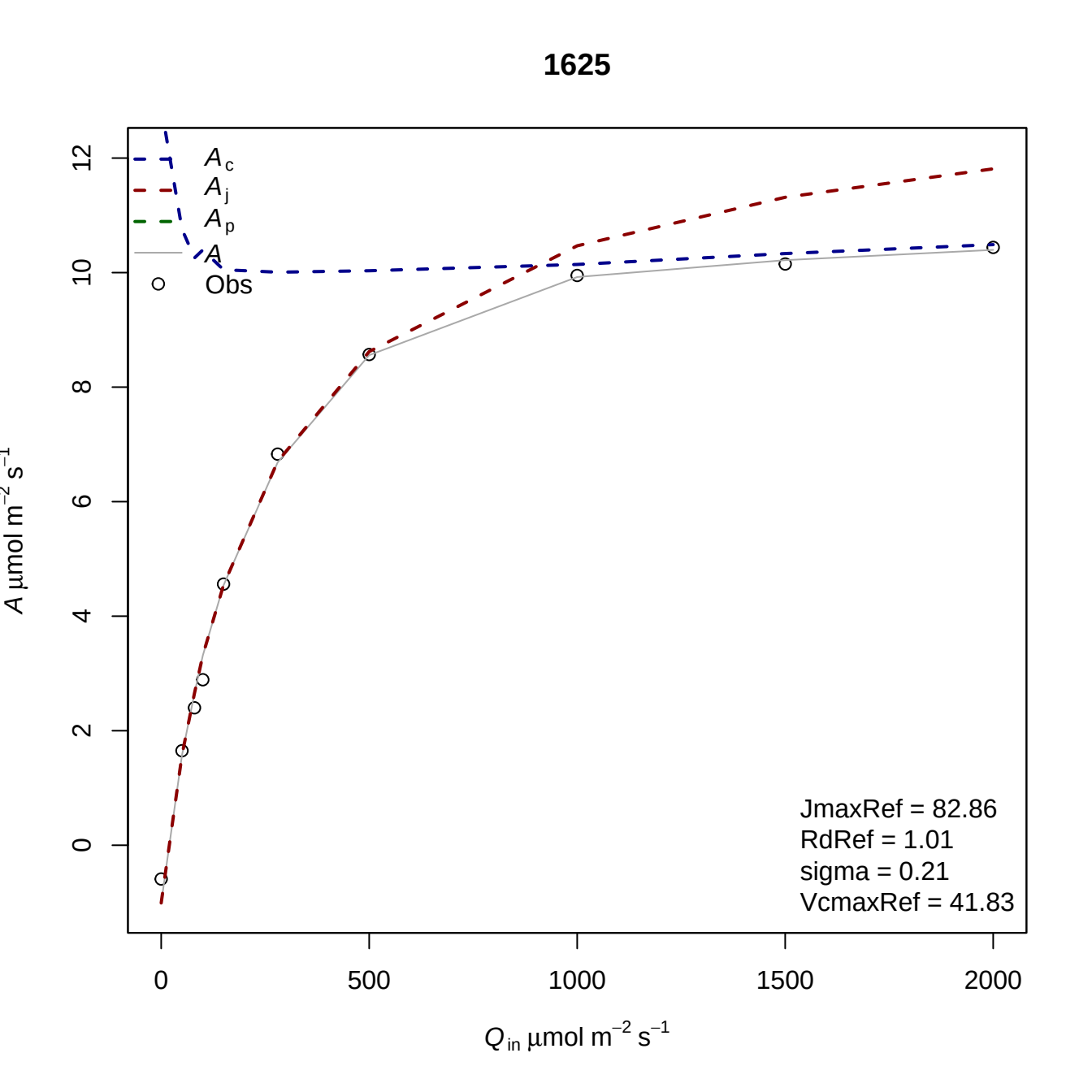

1620
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 312.36
RdRef = 0.75
sigma = 0.44
VcmaxRef = 130.55
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

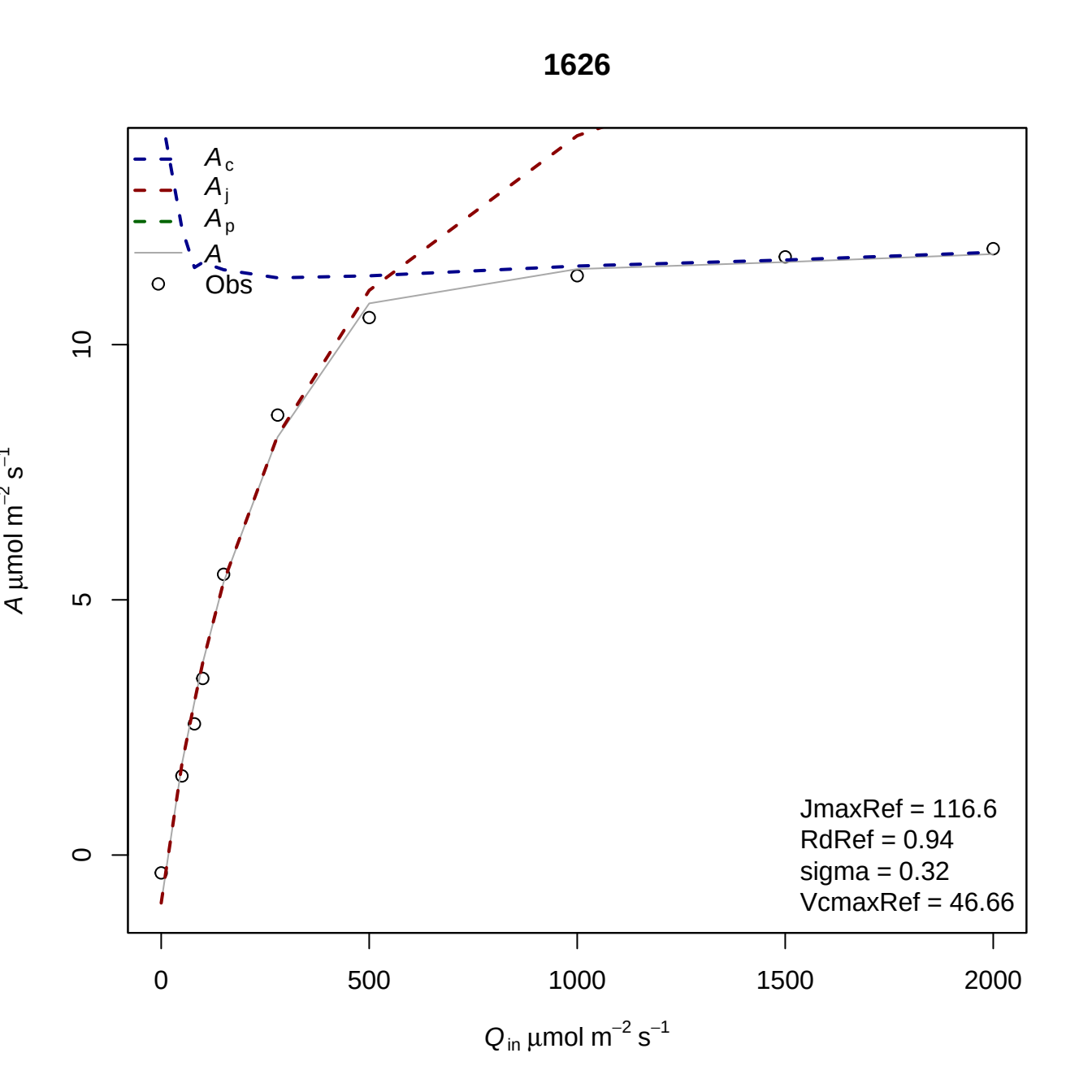

1621
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 290.95
RdRef = 1.04
sigma = 0.4
VcmaxRef = 131.16
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

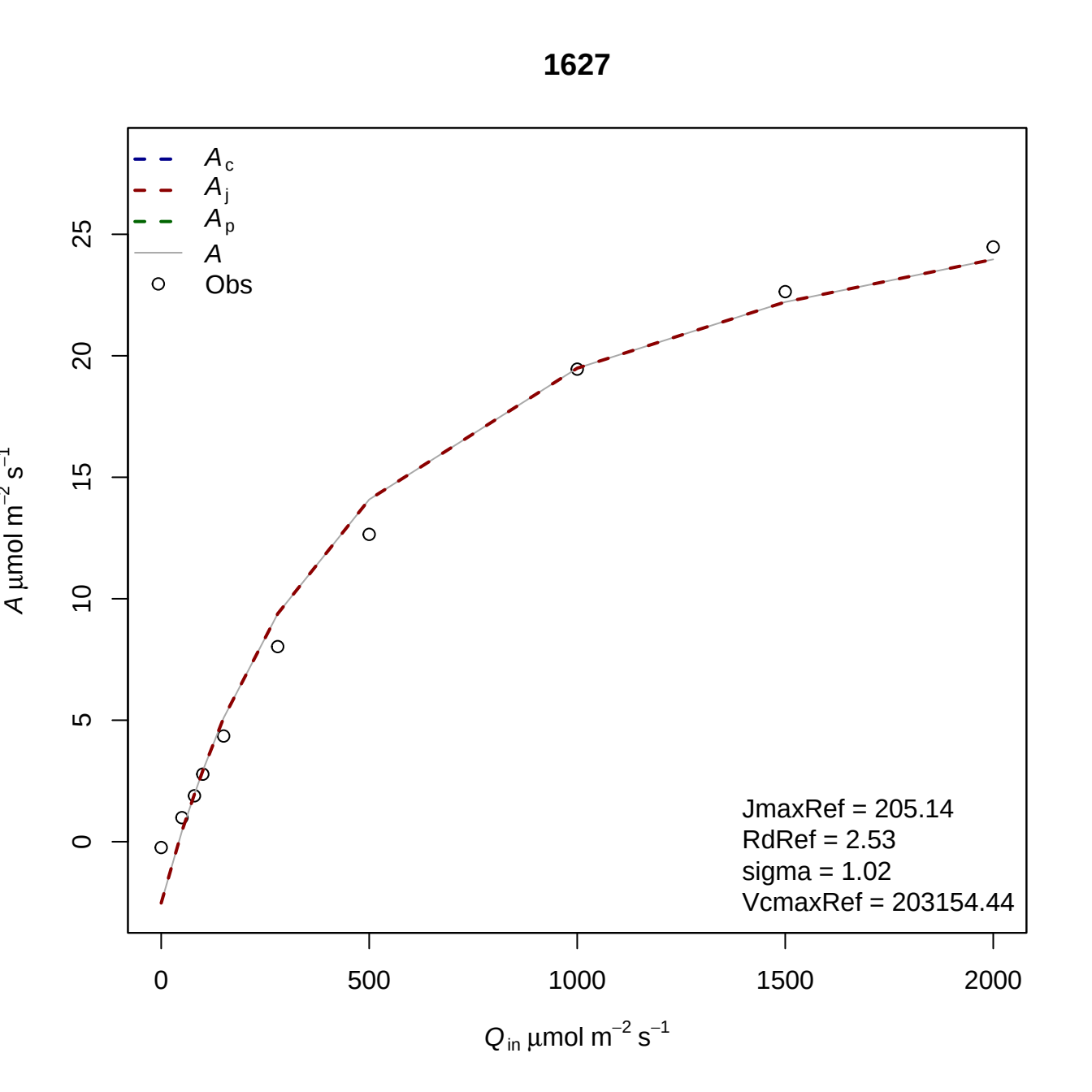

1622
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 360.92
RdRef = 0.83
sigma = 0.42
VcmaxRef = 133.03
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

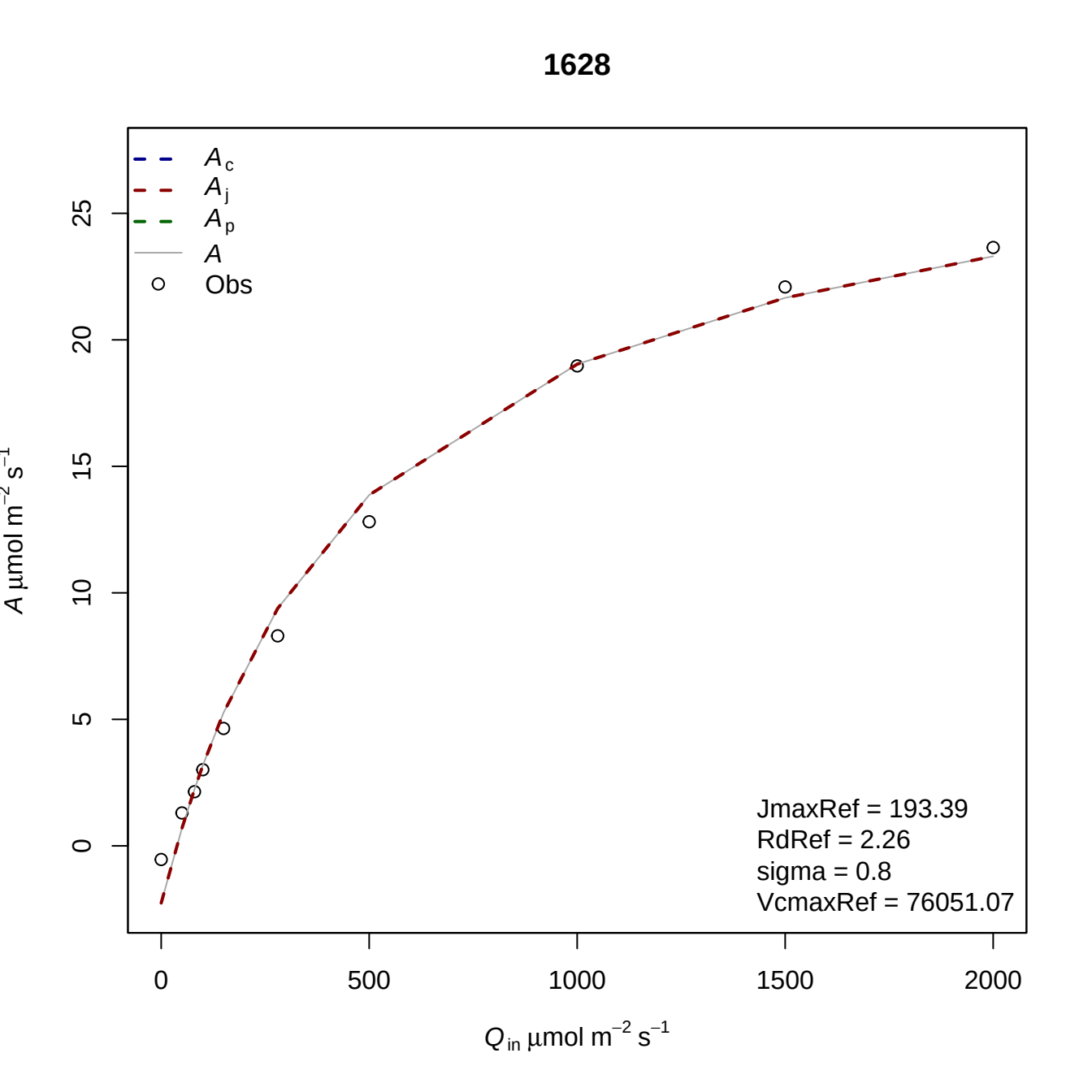

1623
$A_c$
$A_j$
$A_p$
$A$
Qbs
JmaxRef = 96.8
RdRef = 1.15
sigma = 0.34
VcmaxRef = 52.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

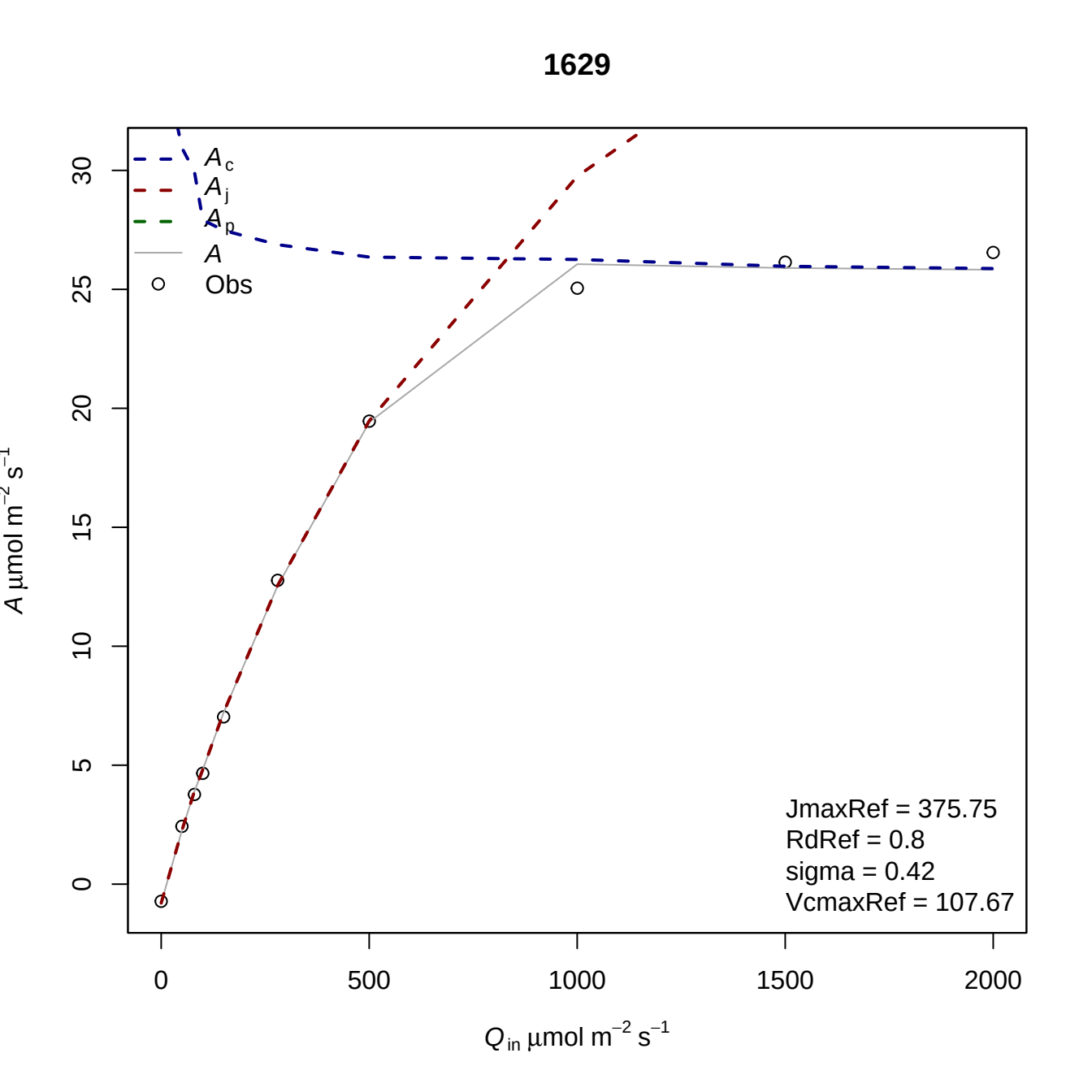

1624
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 137.34
RdRef = 0.41
sigma = 0.23
VcmaxRef = 47.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

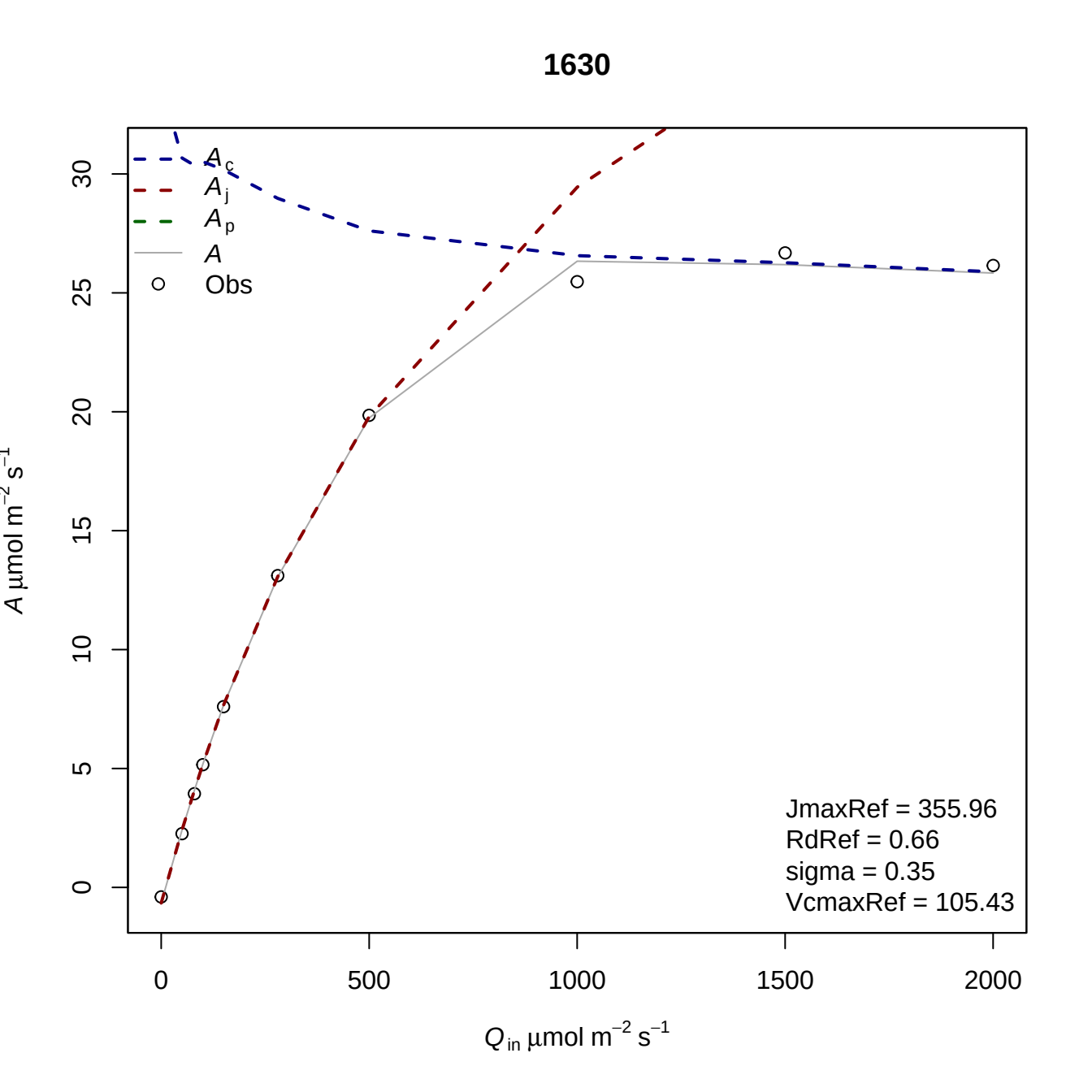

1625
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 82.86
RdRef = 1.01
sigma = 0.21
VcmaxRef = 41.83
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

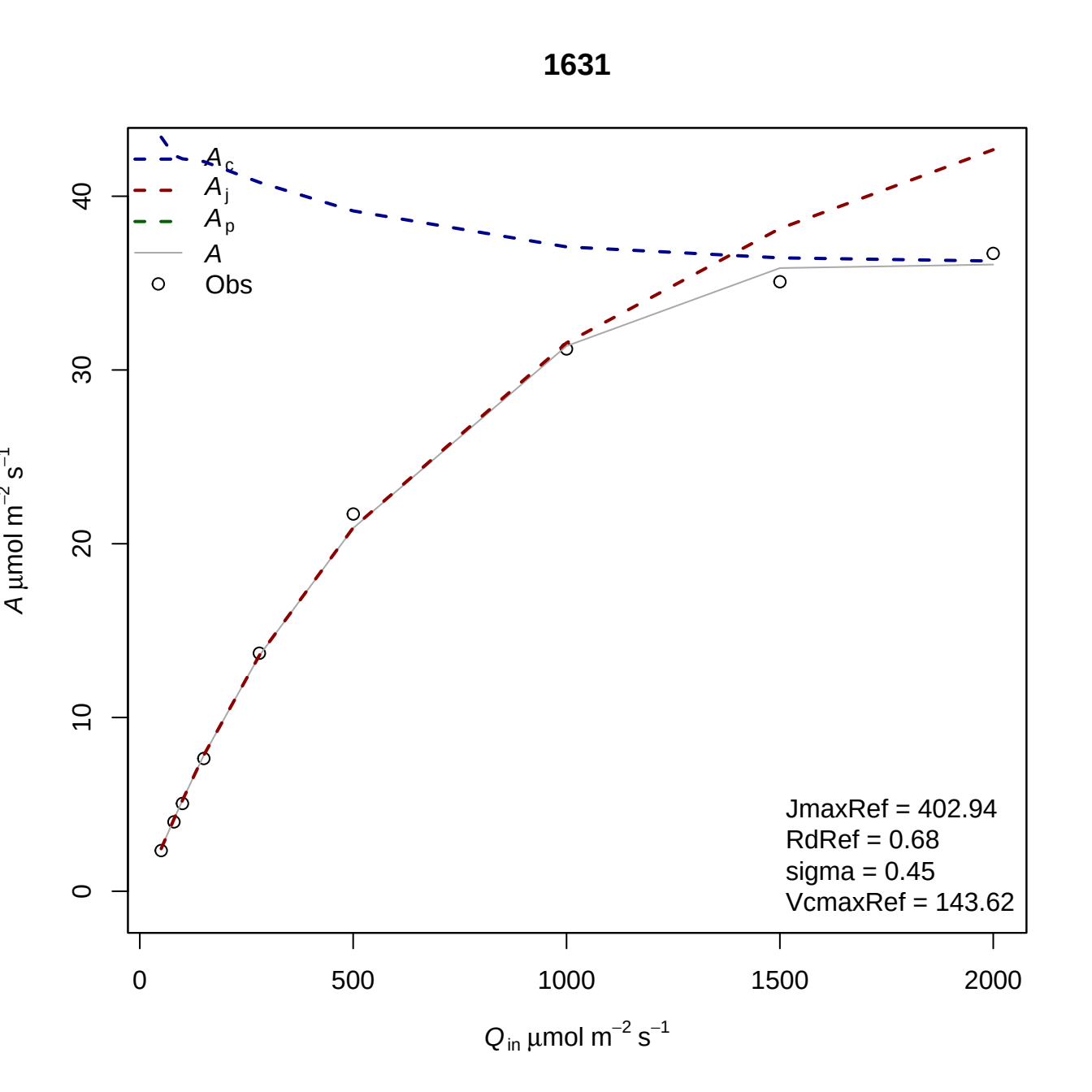

1626
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 116.6
RdRef = 0.94
sigma = 0.32
VcmaxRef = 46.66
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
0
500
1000
1500
2000

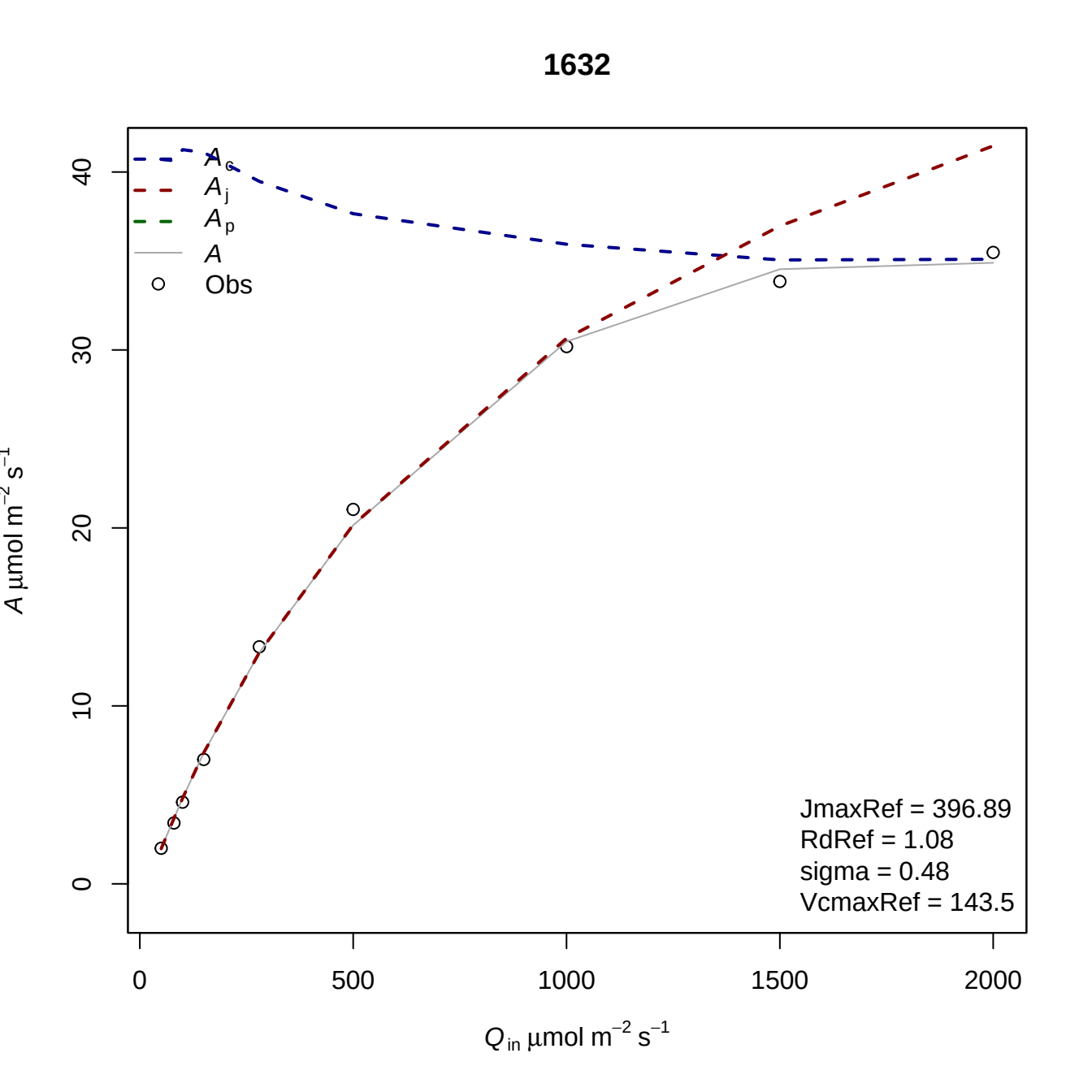

1627
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 205.14
RdRef = 2.53
sigma = 1.02
VcmaxRef = 203154.44
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

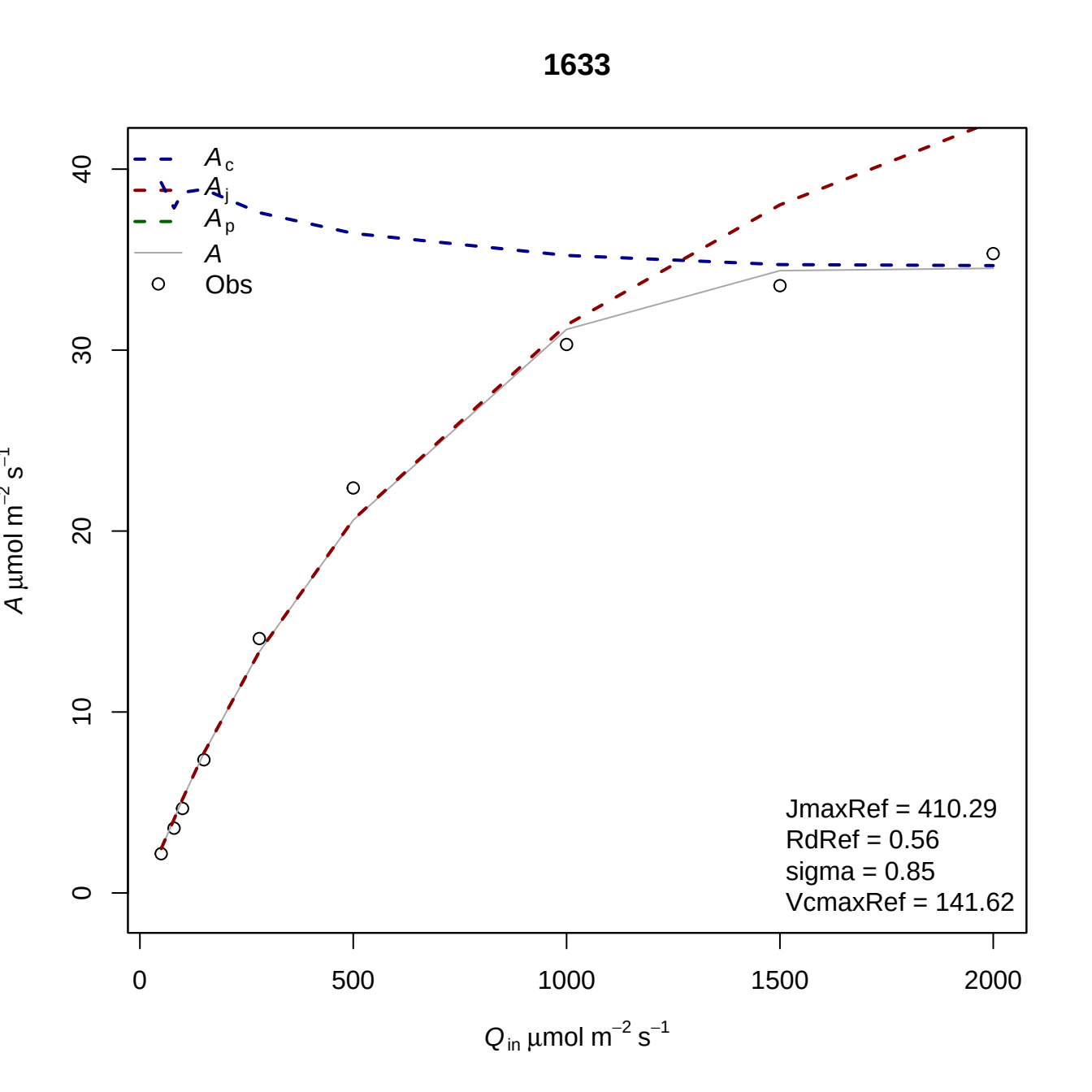

1628
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 193.39
RdRef = 2.26
sigma = 0.8
VcmaxRef = 76051.07
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

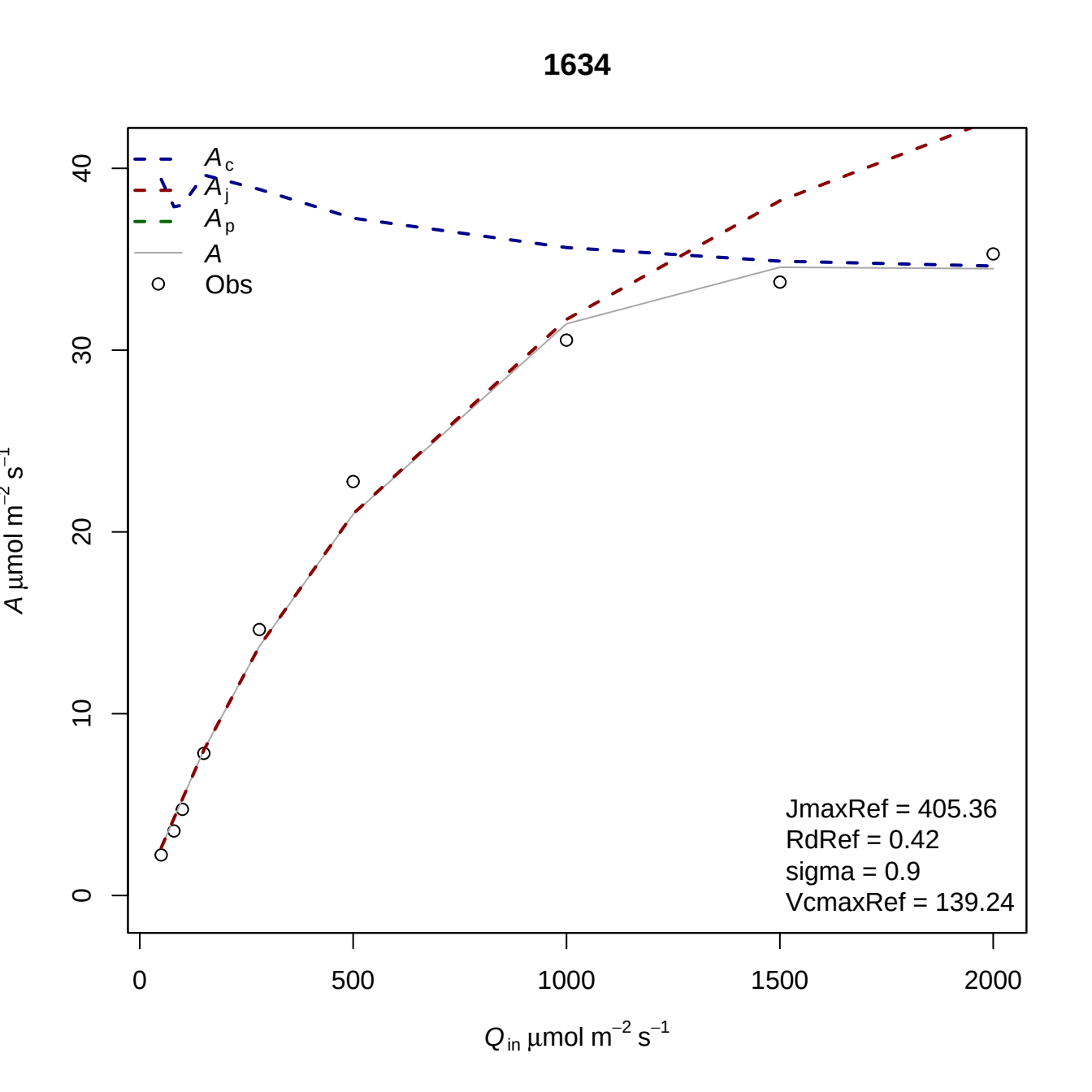

1629
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 375.75
RdRef = 0.8
sigma = 0.42
VcmaxRef = 107.67
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

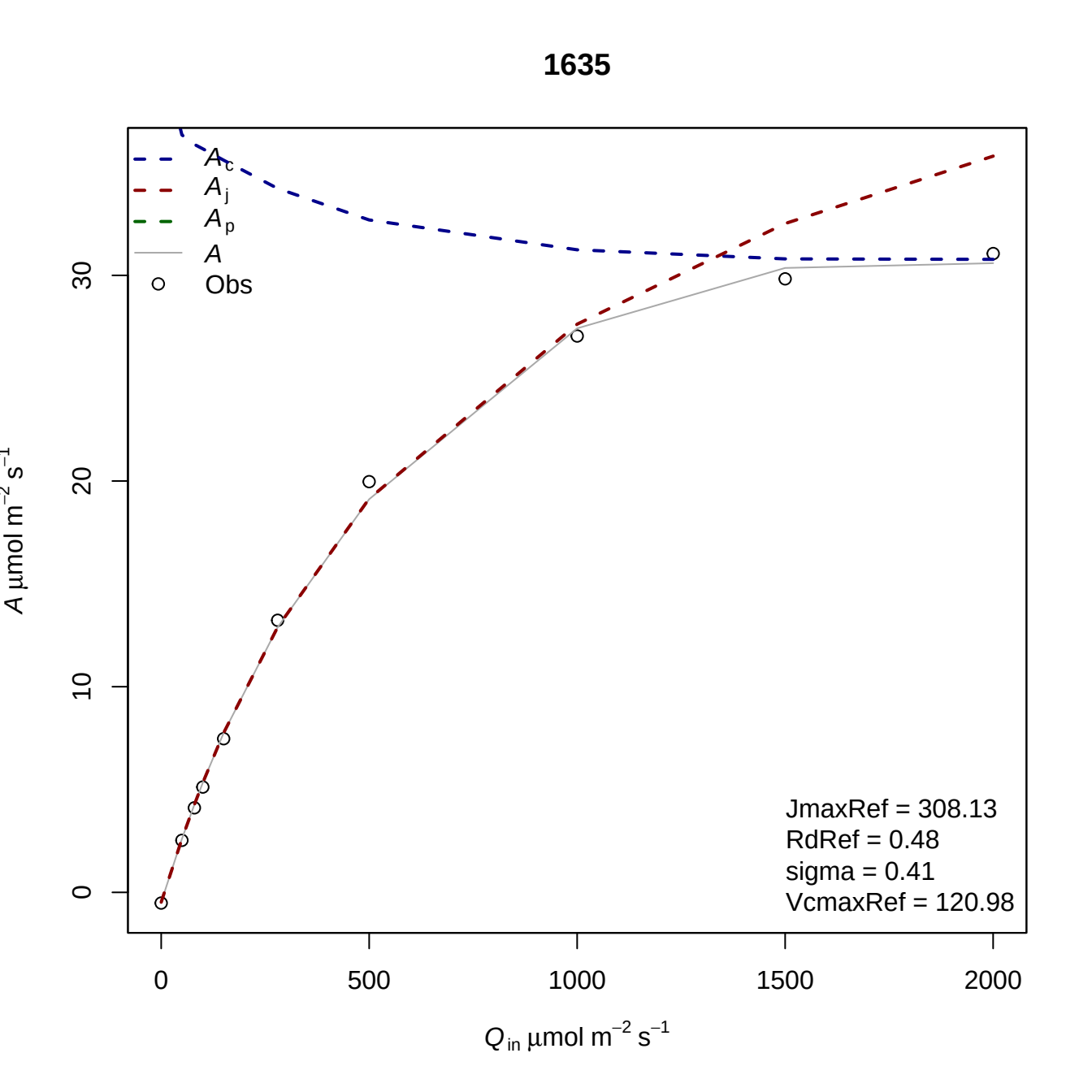

1630
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 355.96
RdRef = 0.66
sigma = 0.35
VcmaxRef = 105.43
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

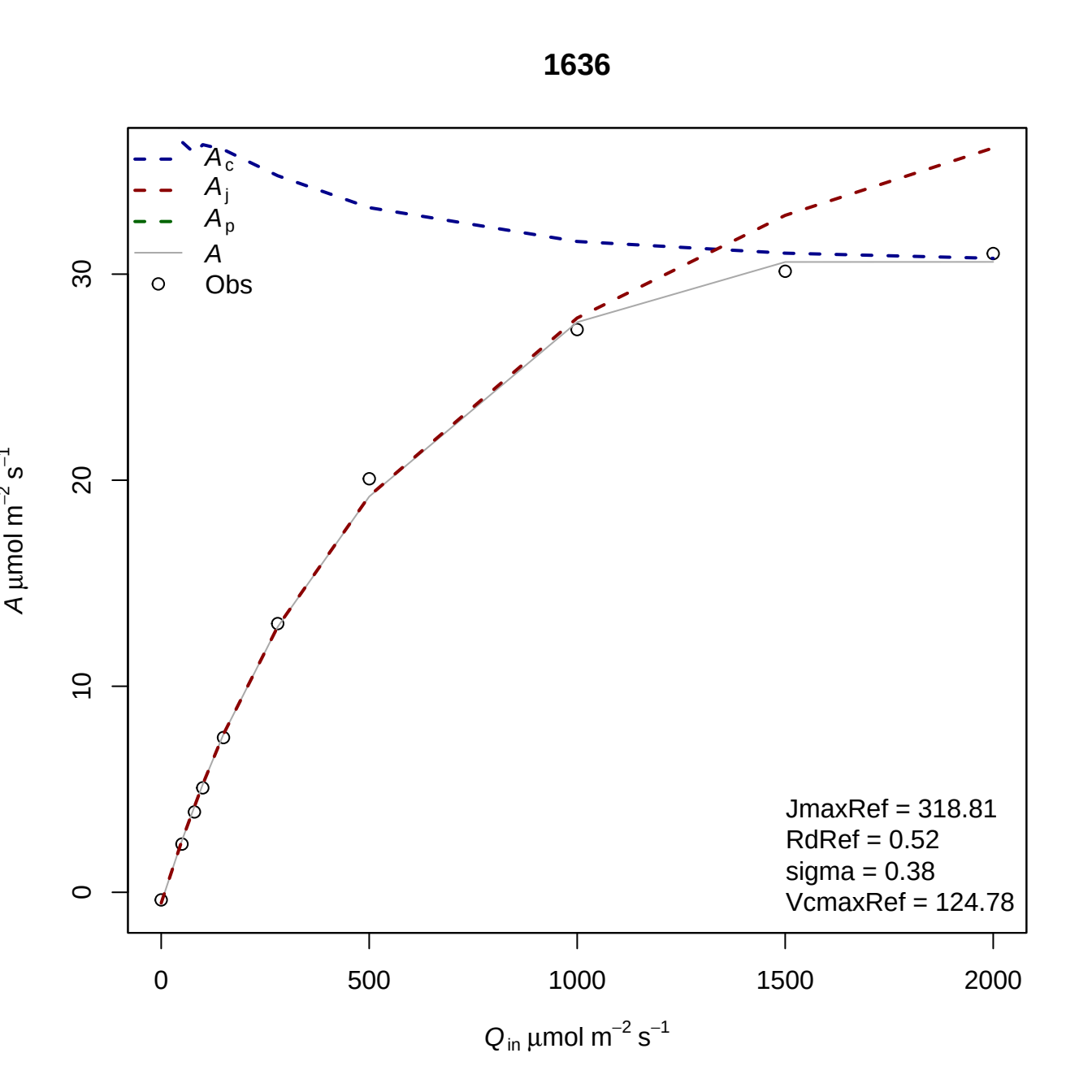

1631
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 402.94
RdRef = 0.68
sigma = 0.45
VcmaxRef = 143.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000

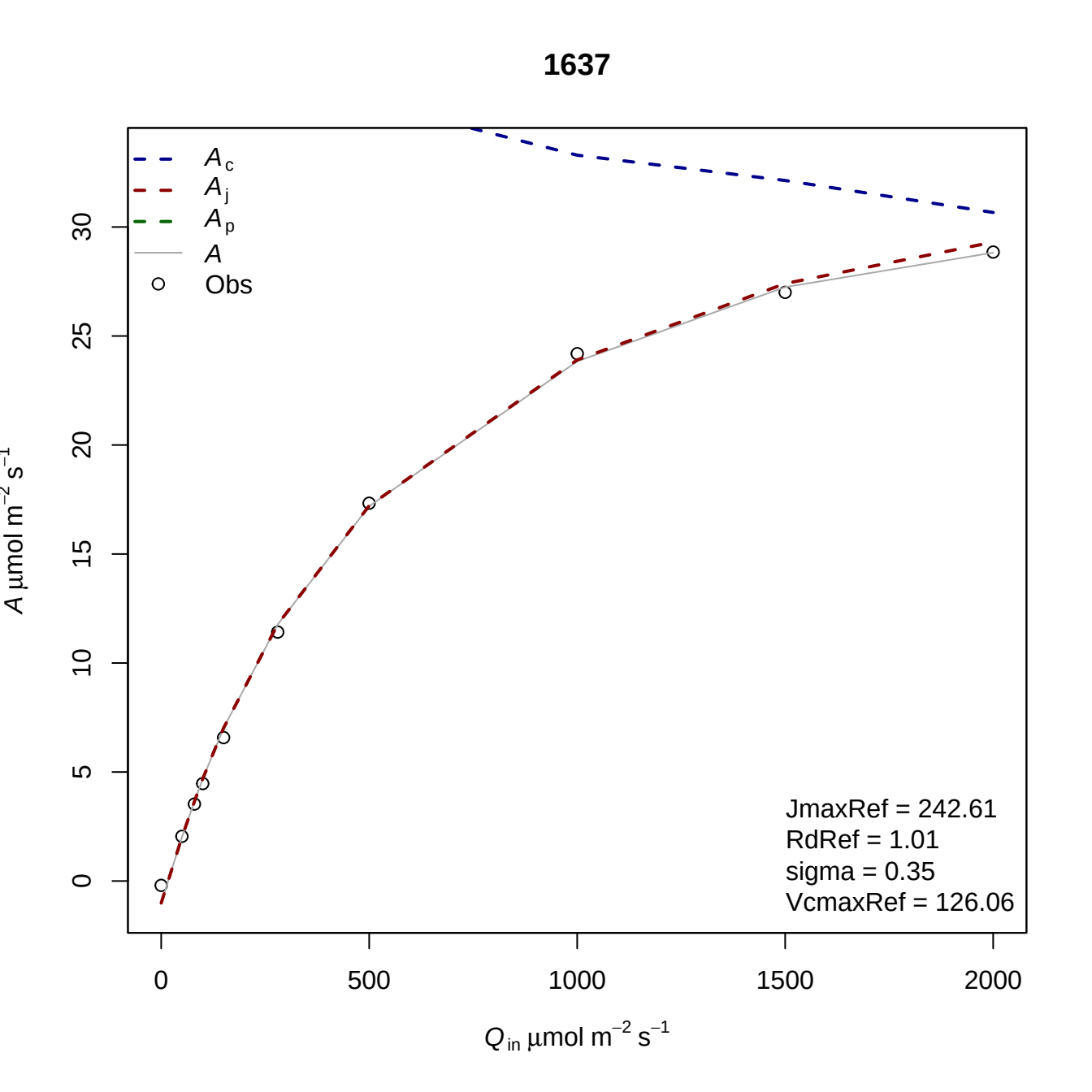

1632
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 396.89
RdRef = 1.08
sigma = 0.48
VcmaxRef = 143.5
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000

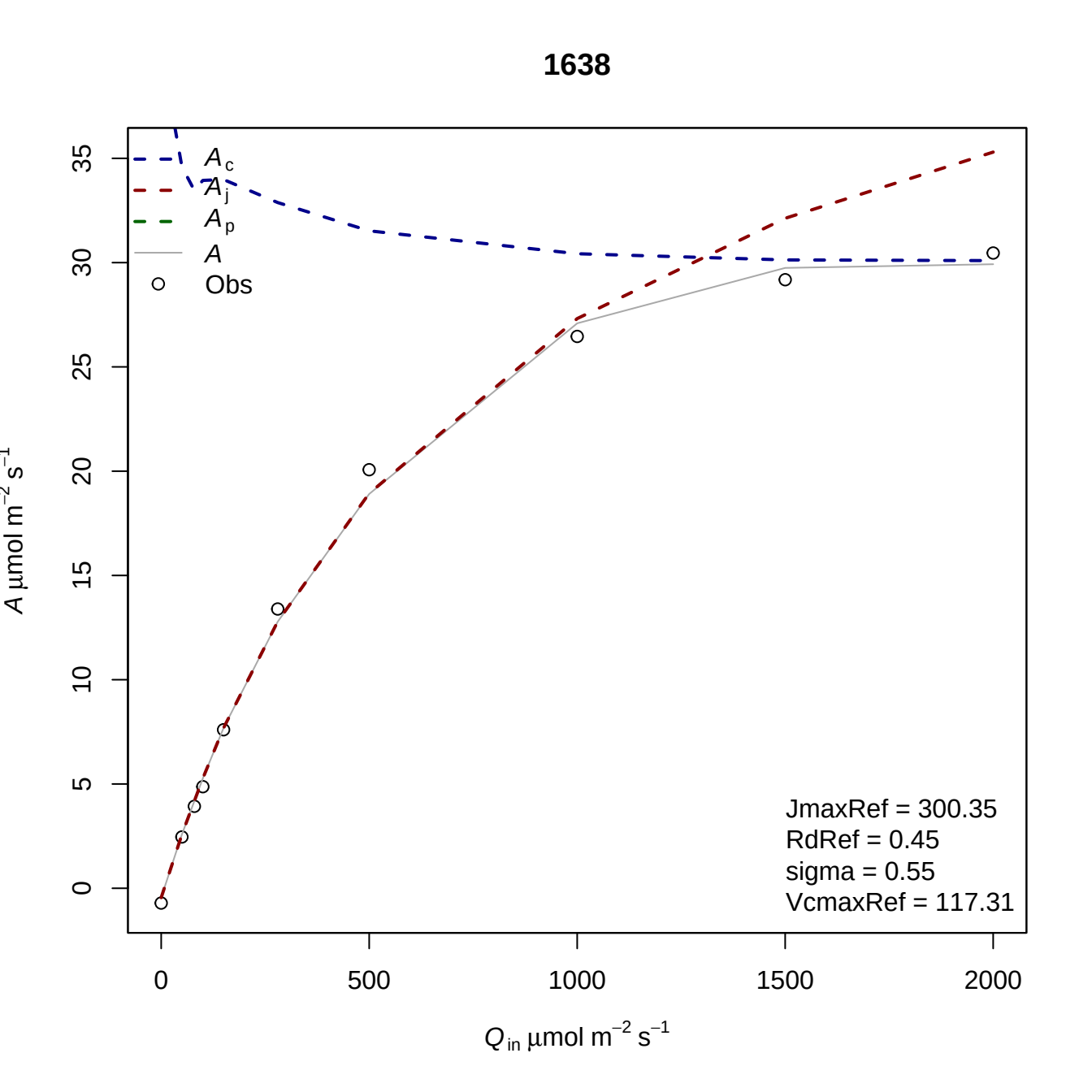

1633
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 410.29
RdRef = 0.56
sigma = 0.85
VcmaxRef = 141.62
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000

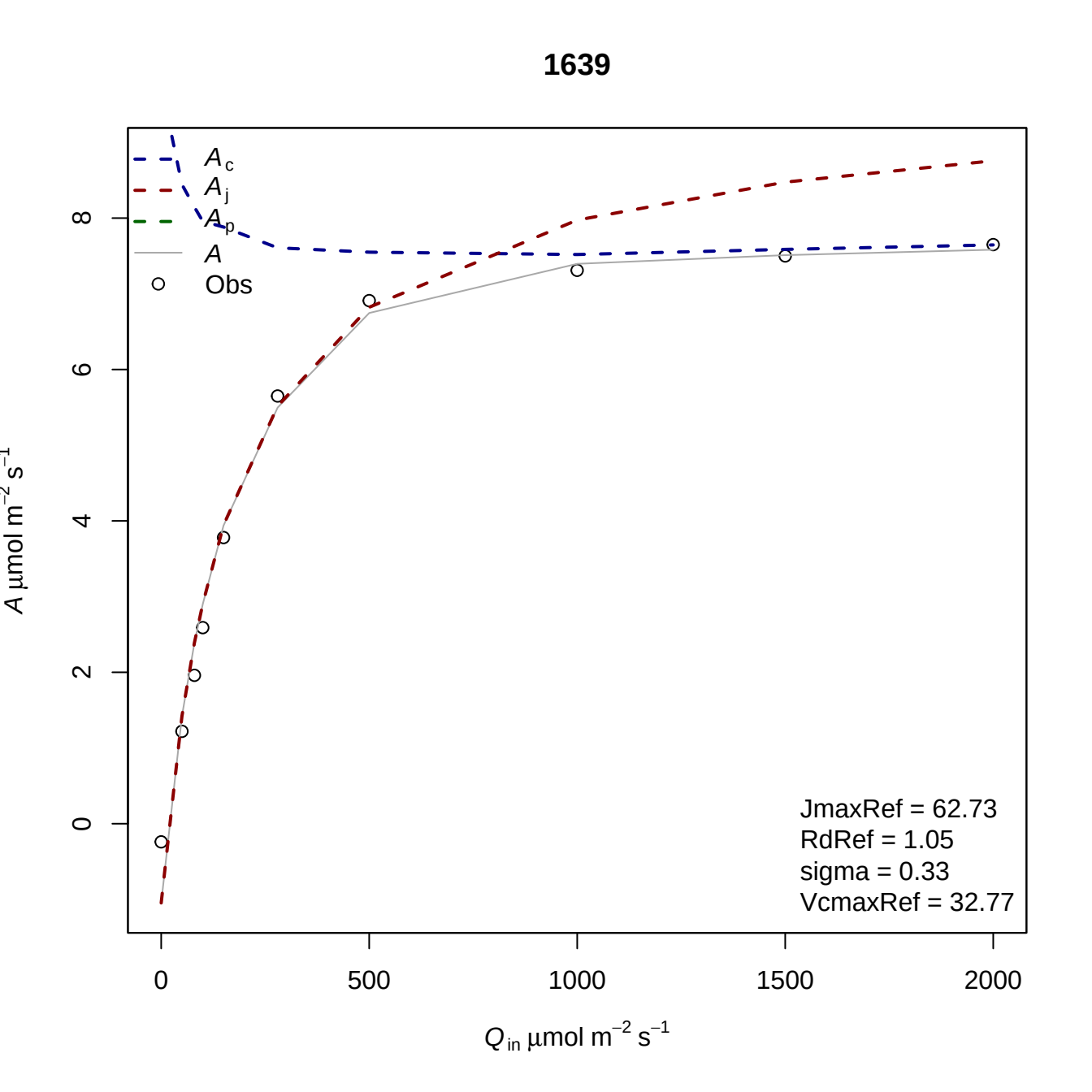

1634
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 405.36
RdRef = 0.42
sigma = 0.9
VcmaxRef = 139.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000

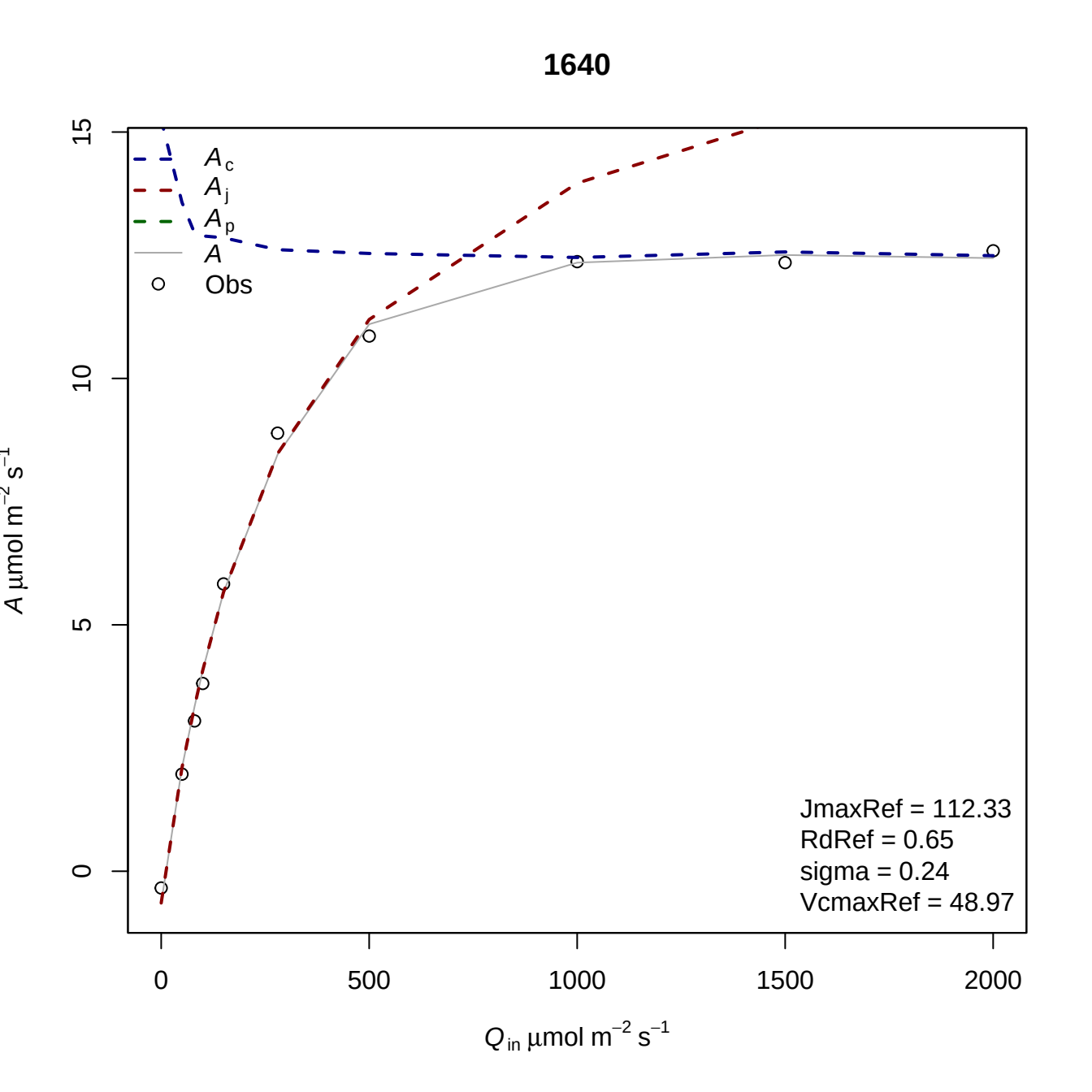

1635
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 308.13
RdRef = 0.48
sigma = 0.41
VcmaxRef = 120.98
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

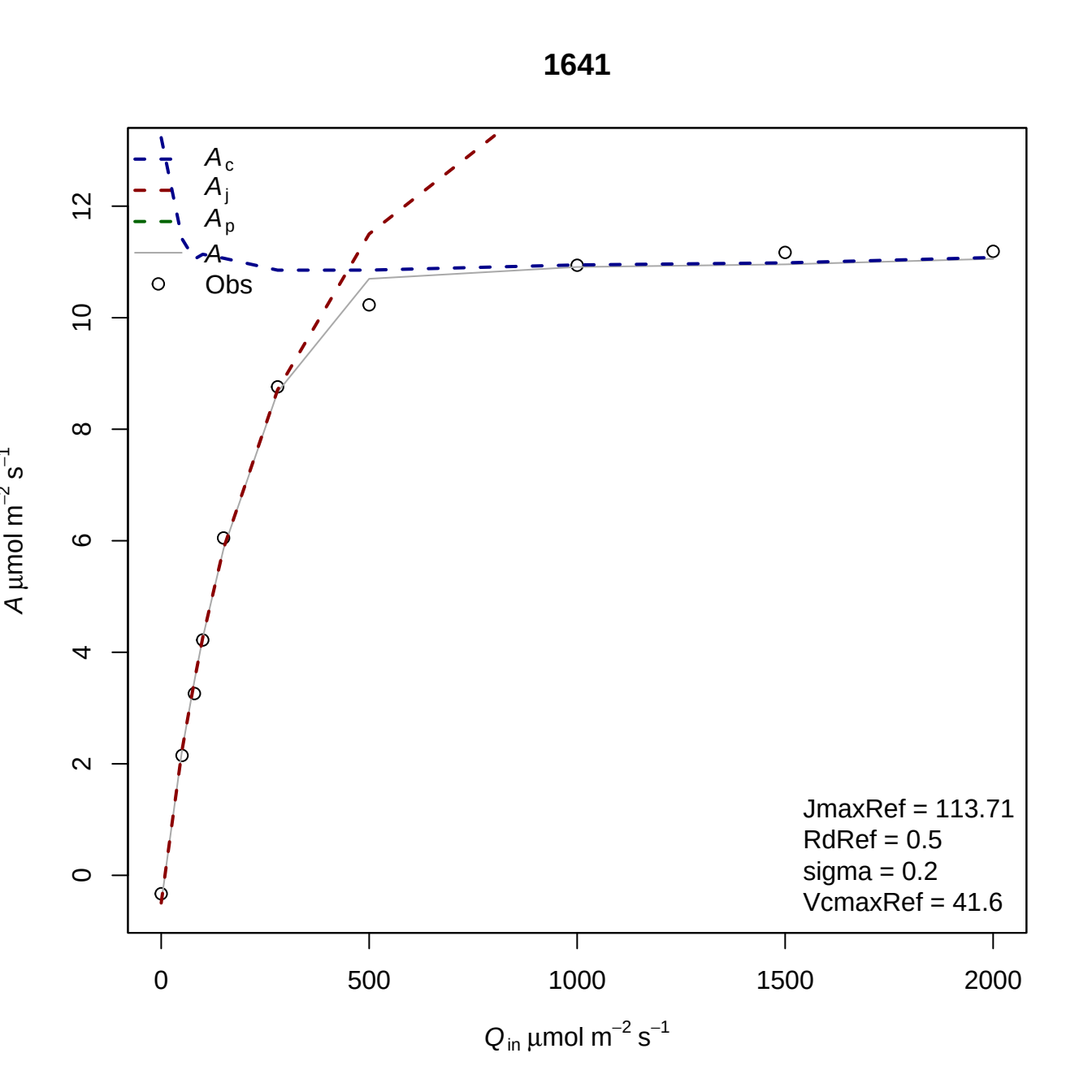

1636
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 318.81
RdRef = 0.52
sigma = 0.38
VcmaxRef = 124.78
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
0
500
1000
1500
2000

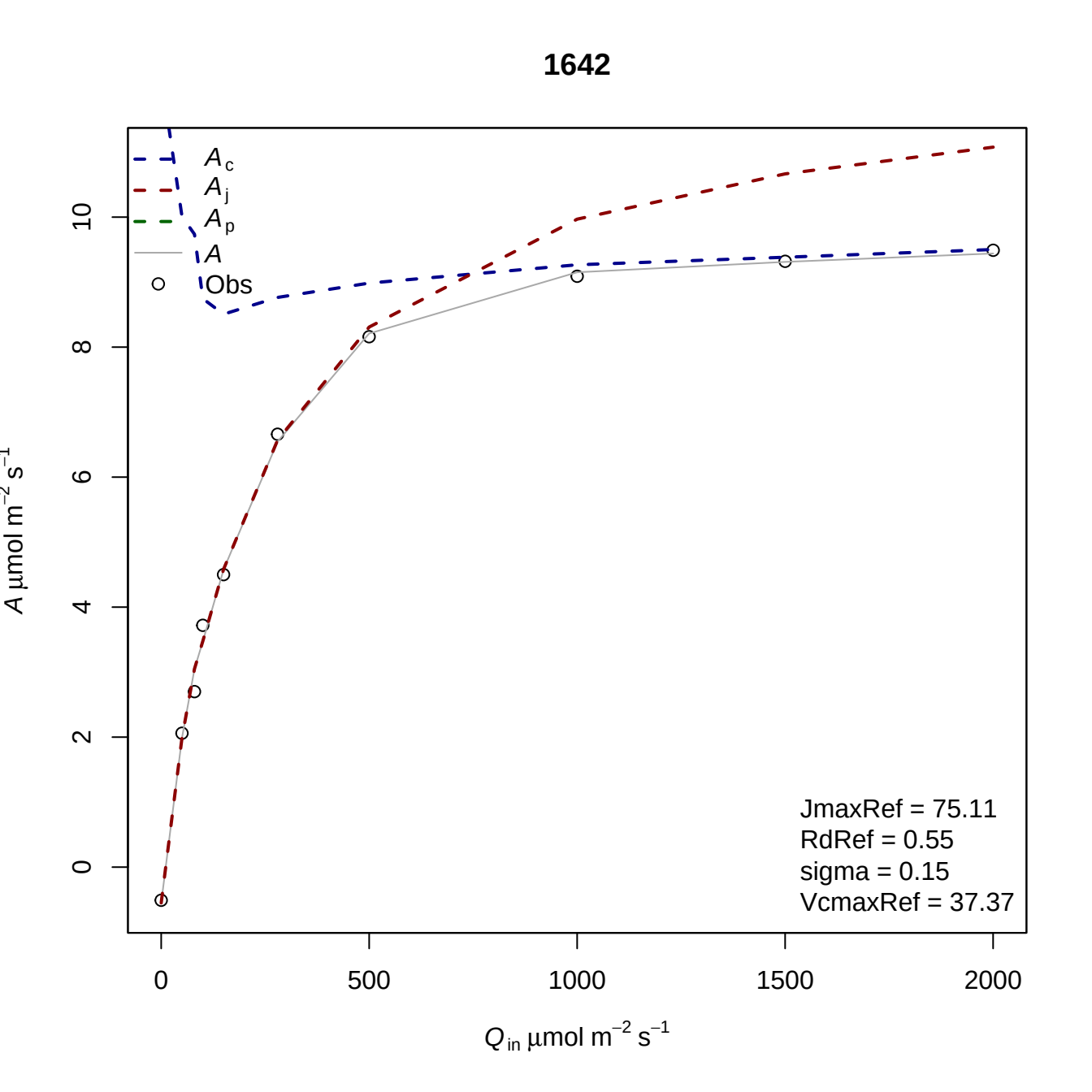

1637
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 242.61
RdRef = 1.01
sigma = 0.35
VcmaxRef = 126.06
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

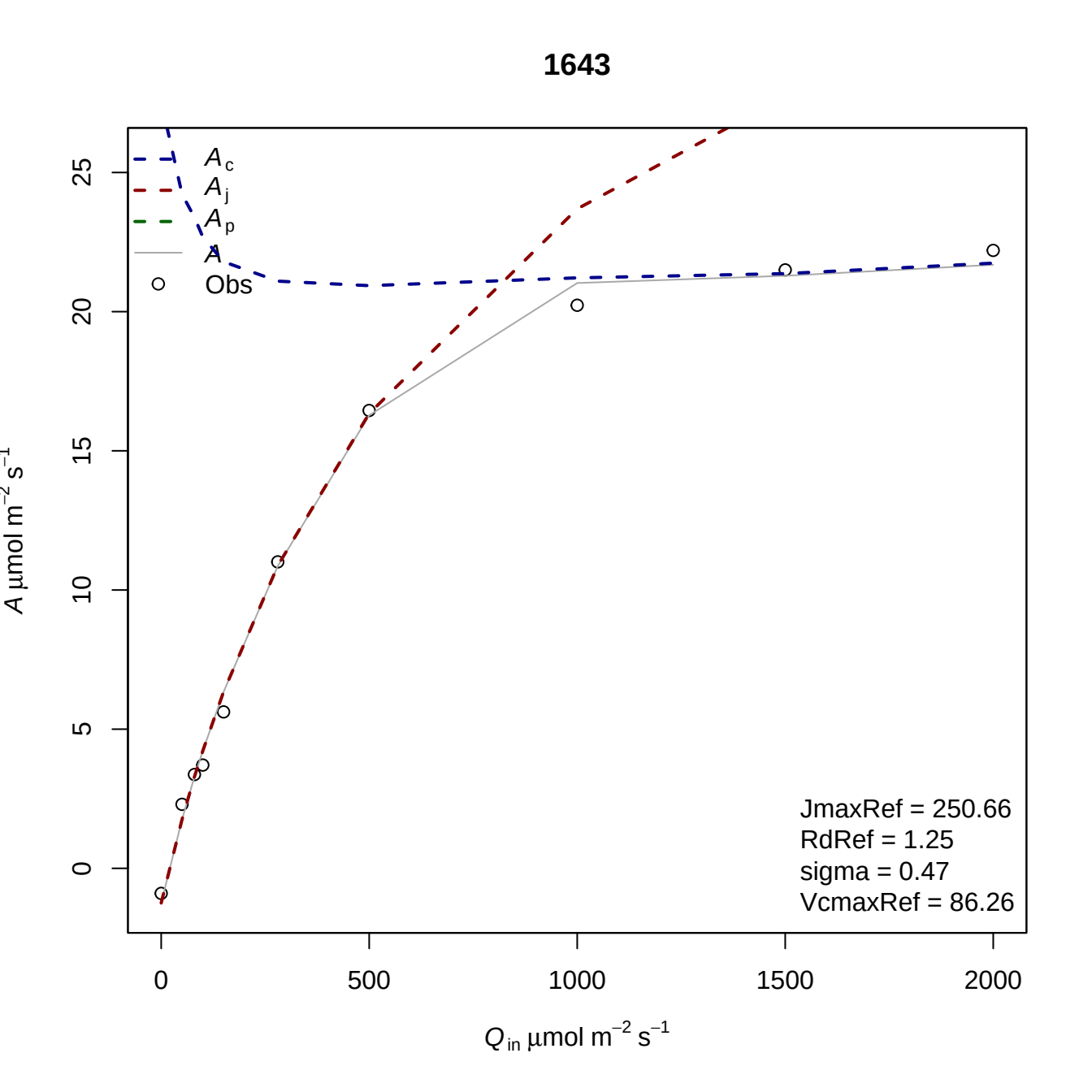

1638
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 300.35
RdRef = 0.45
sigma = 0.55
VcmaxRef = 117.31
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

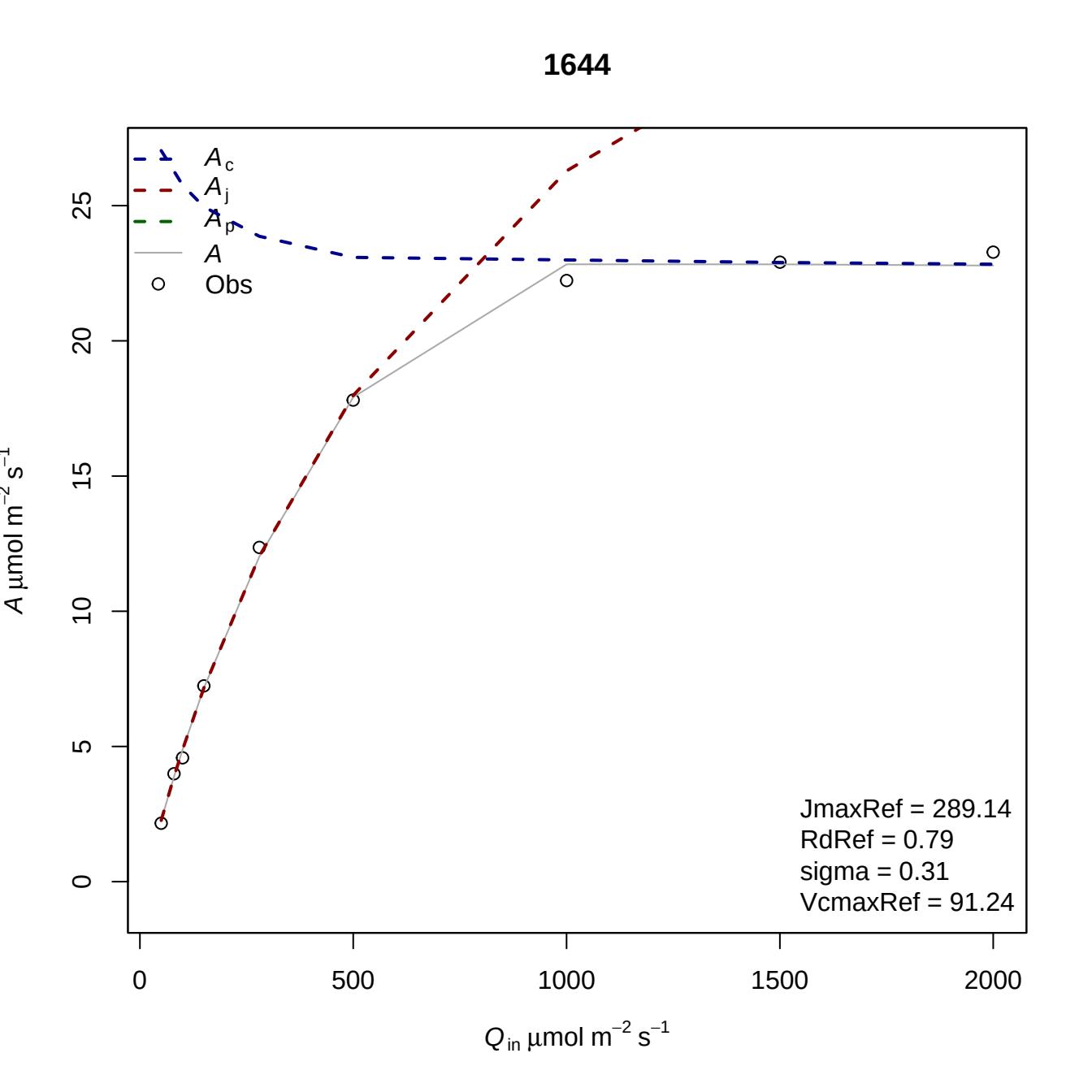

1639
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 62.73
RdRef = 1.05
sigma = 0.33
VcmaxRef = 32.77
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

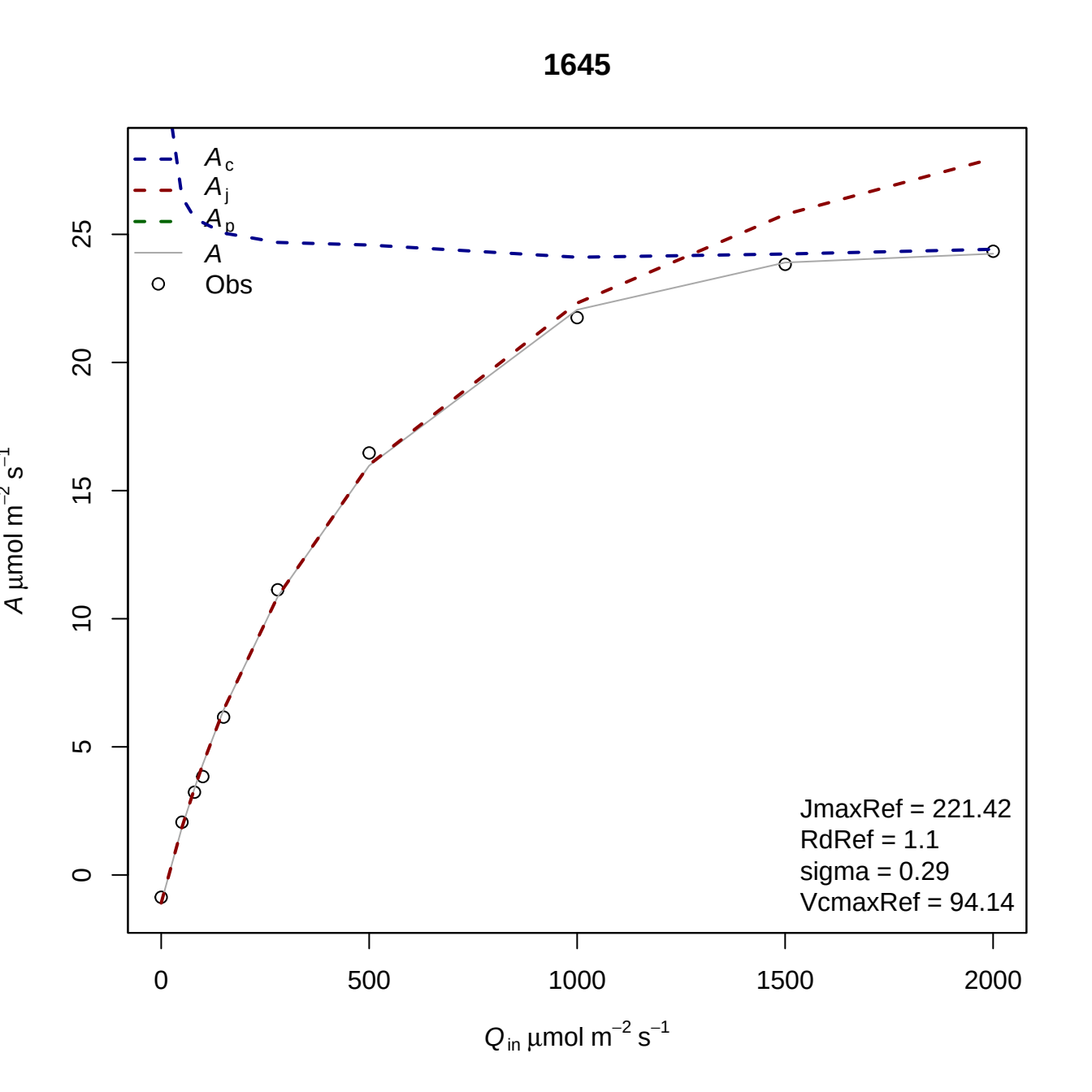

1640
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 112.33
RdRef = 0.65
sigma = 0.24
VcmaxRef = 48.97
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

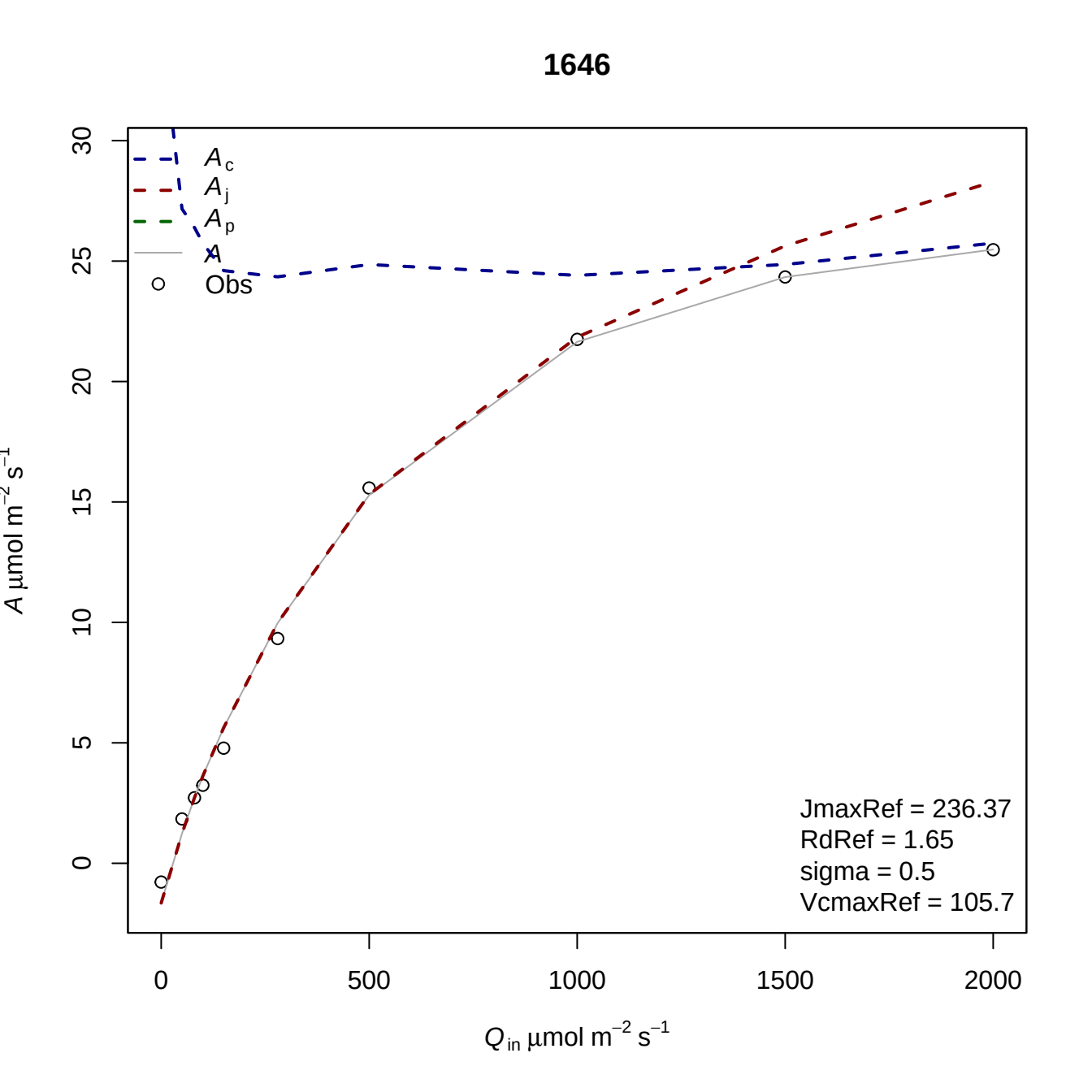

1641
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 113.71
RdRef = 0.5
sigma = 0.2
VcmaxRef = 41.6
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
12
0
500
1000
1500
2000

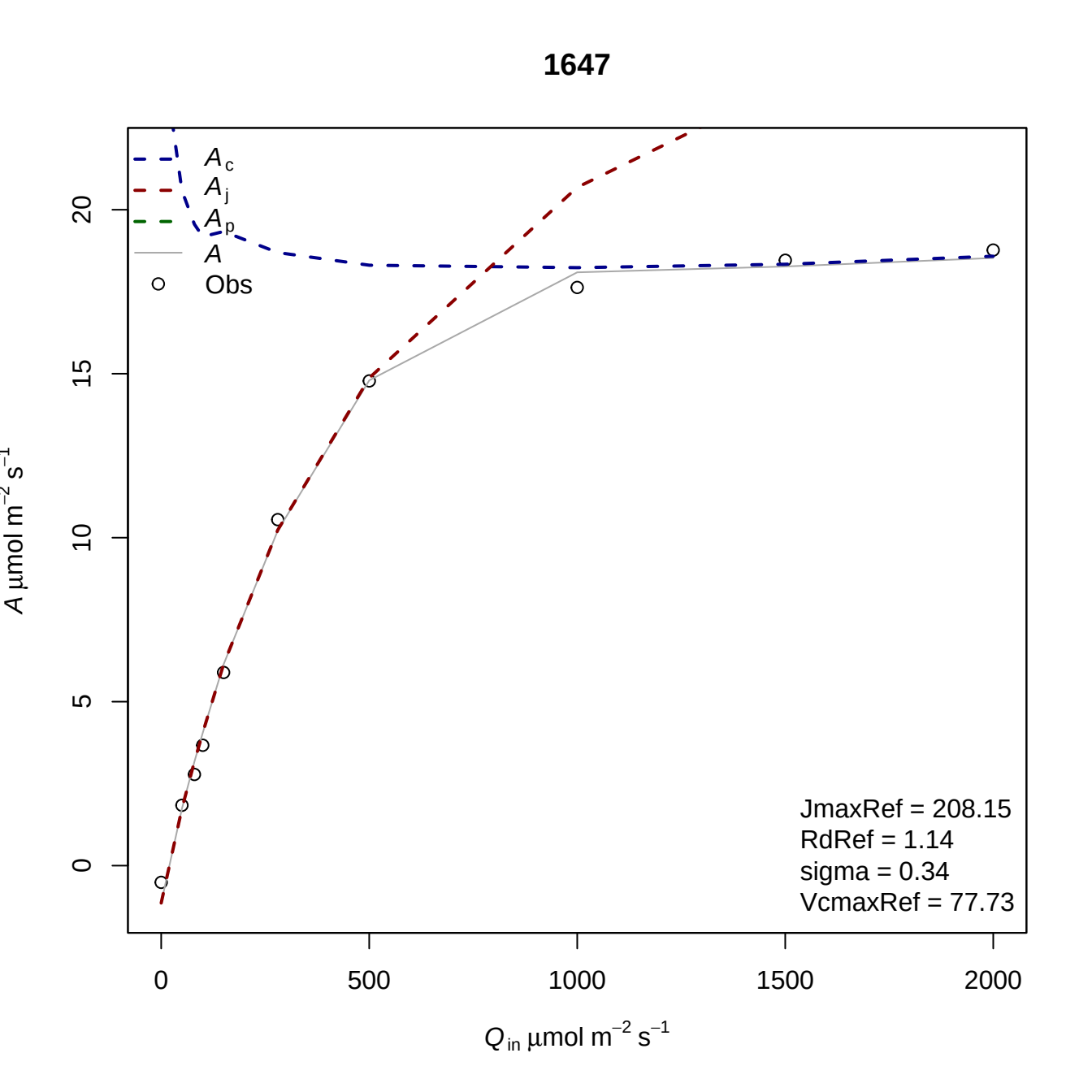

1642
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 75.11
RdRef = 0.55
sigma = 0.15
VcmaxRef = 37.37
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

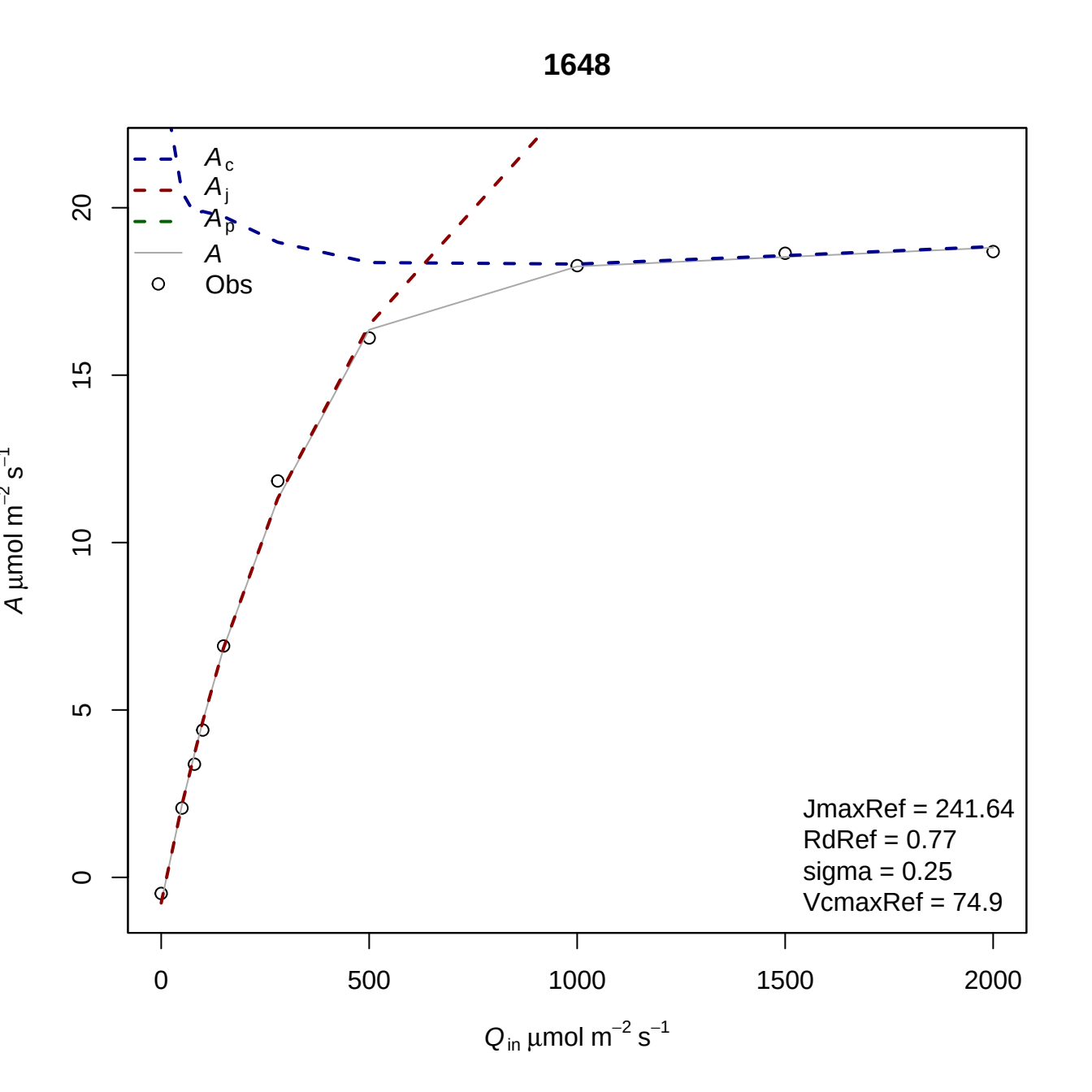

1643
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 250.66
RdRef = 1.25
sigma = 0.47
VcmaxRef = 86.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

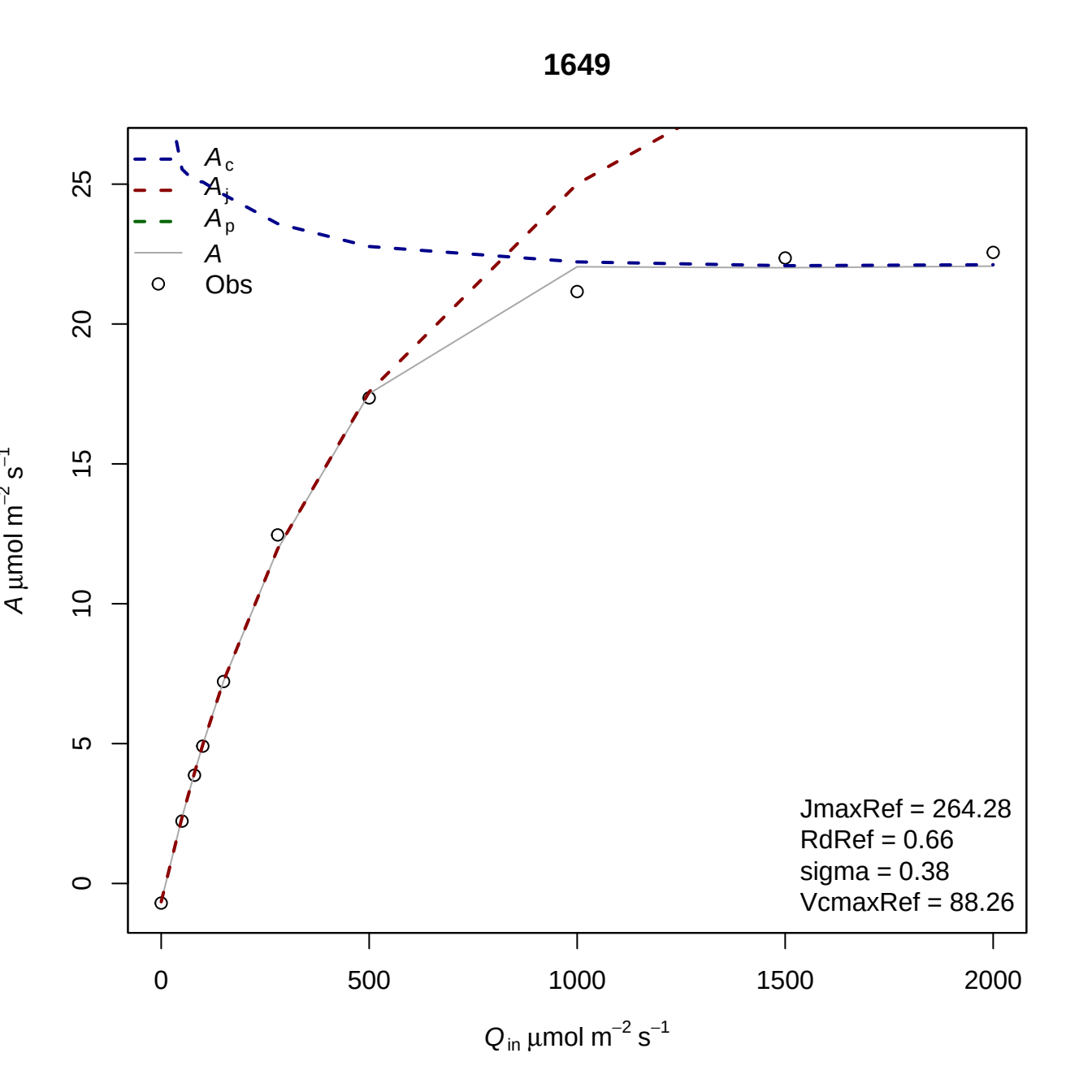

1644
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 289.14
RdRef = 0.79
sigma = 0.31
VcmaxRef = 91.24
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

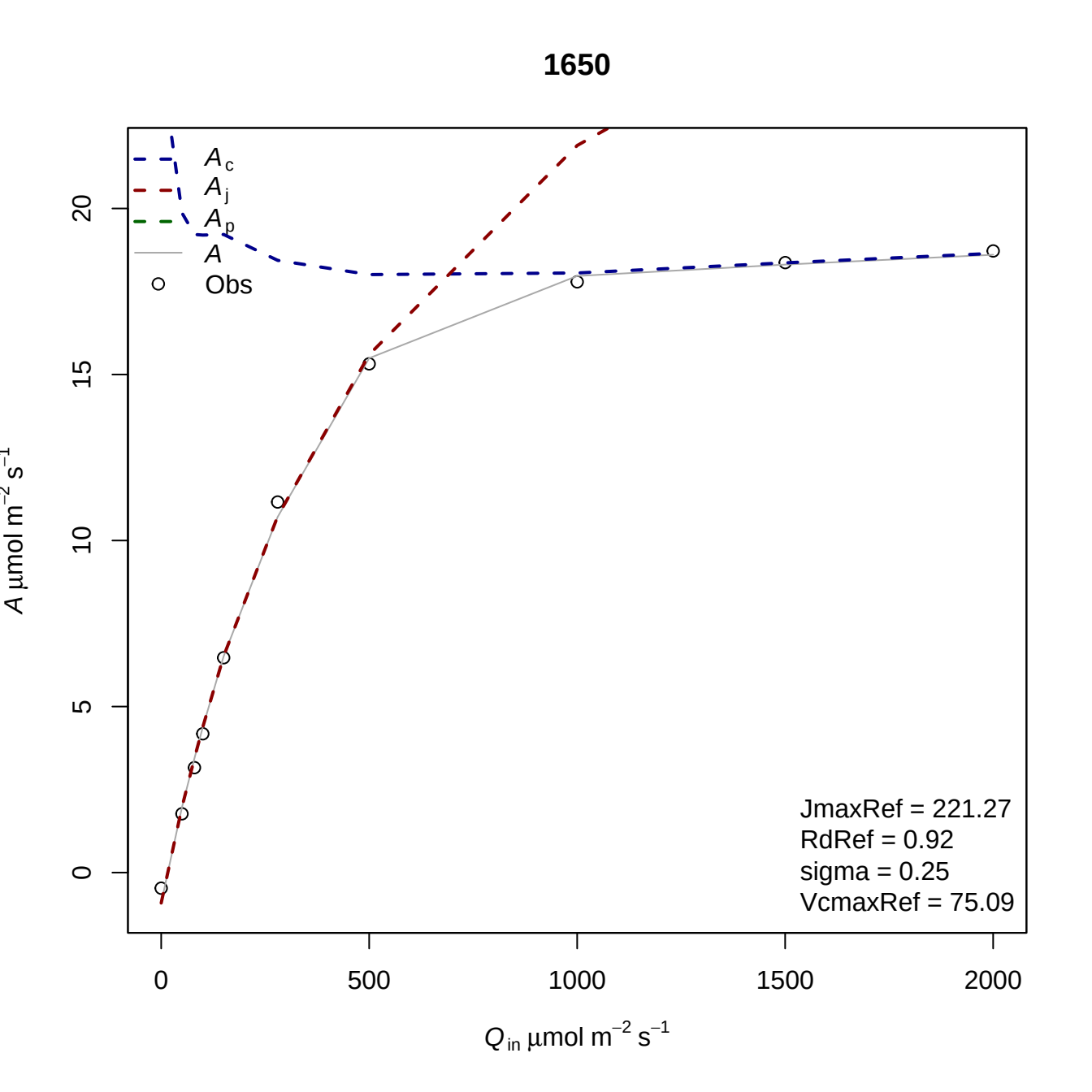

1645
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 221.42
RdRef = 1.1
sigma = 0.29
VcmaxRef = 94.14
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

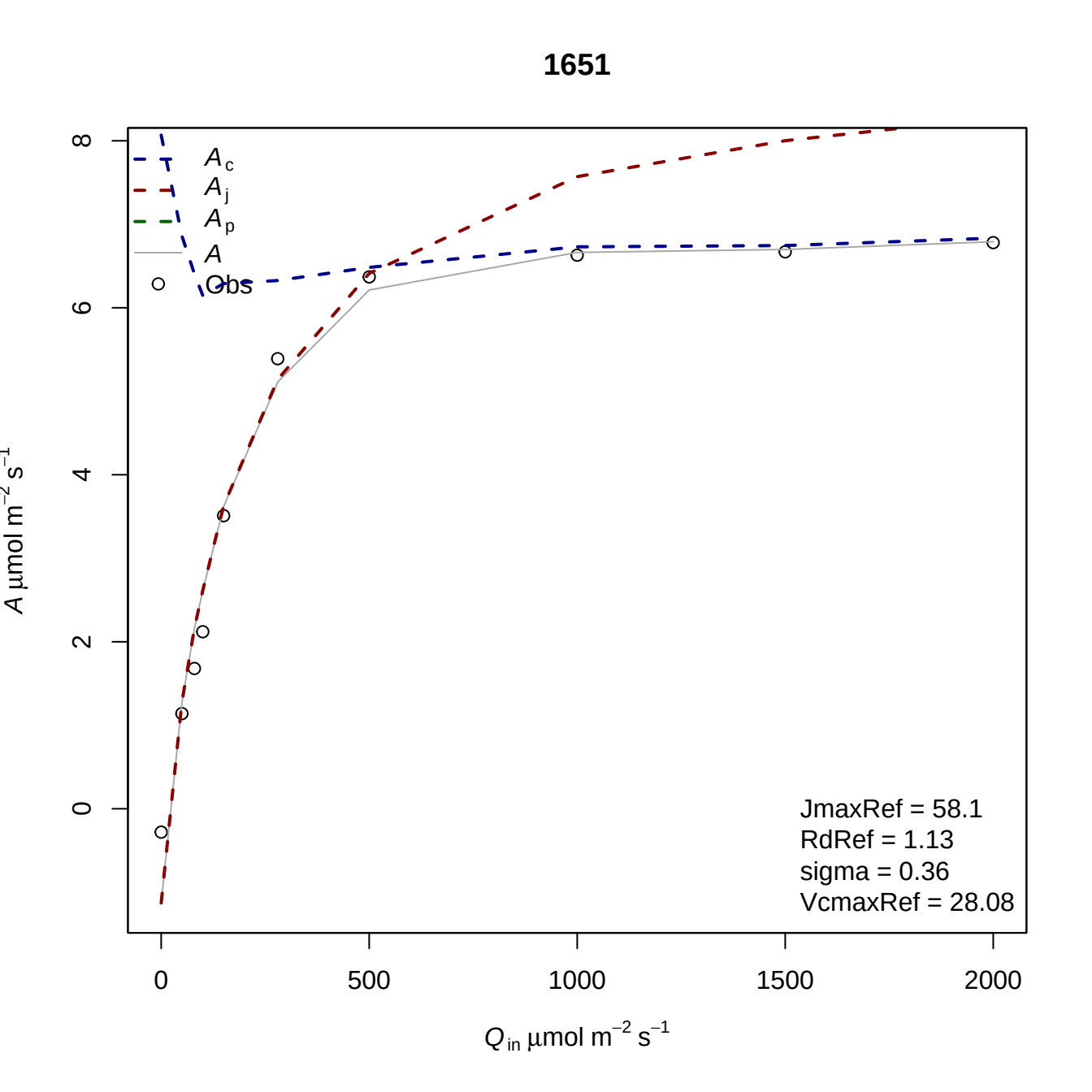

1646
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 236.37
RdRef = 1.65
sigma = 0.5
VcmaxRef = 105.7
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
30
0
500
1000
1500
2000

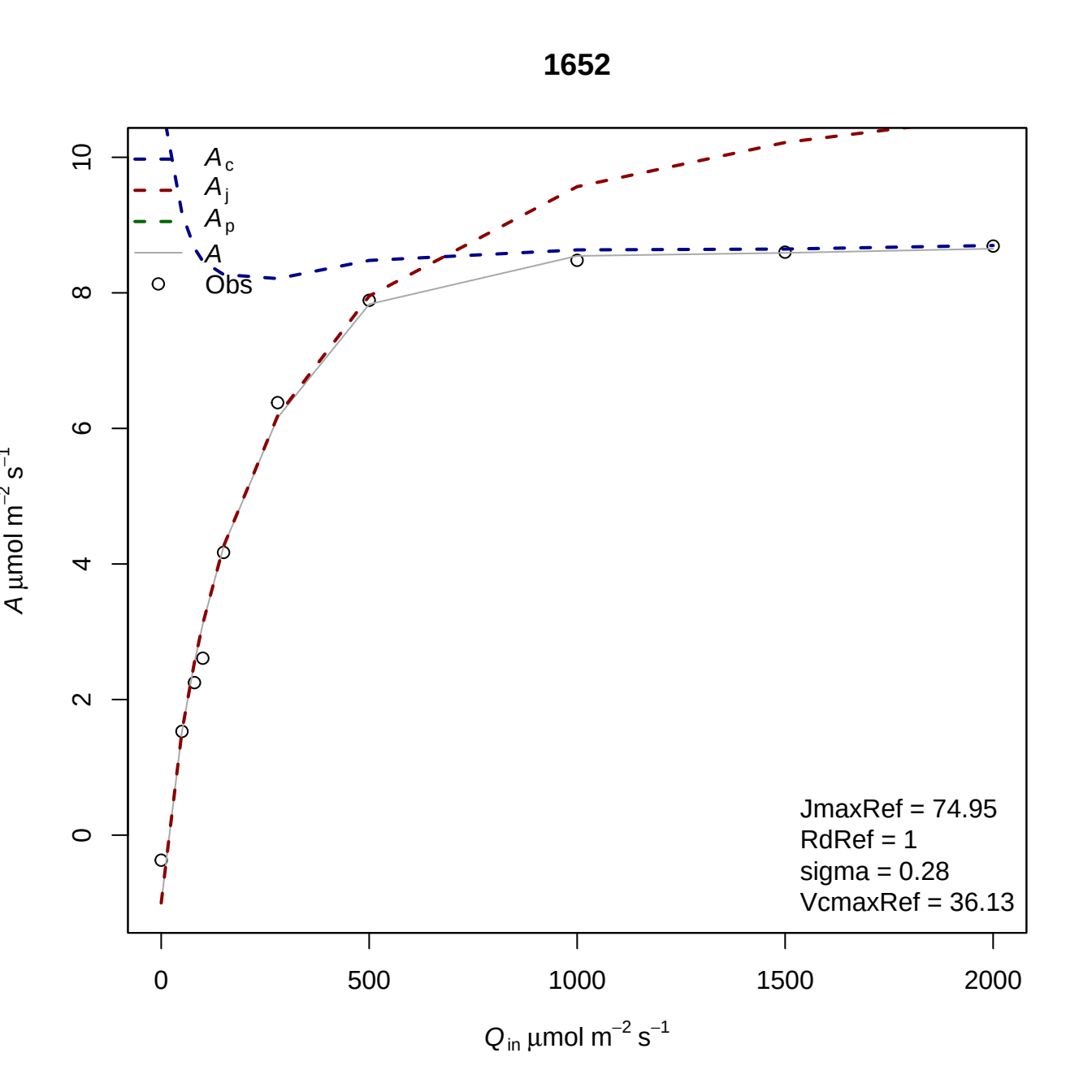

1647
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 208.15
RdRef = 1.14
sigma = 0.34
VcmaxRef = 77.73
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

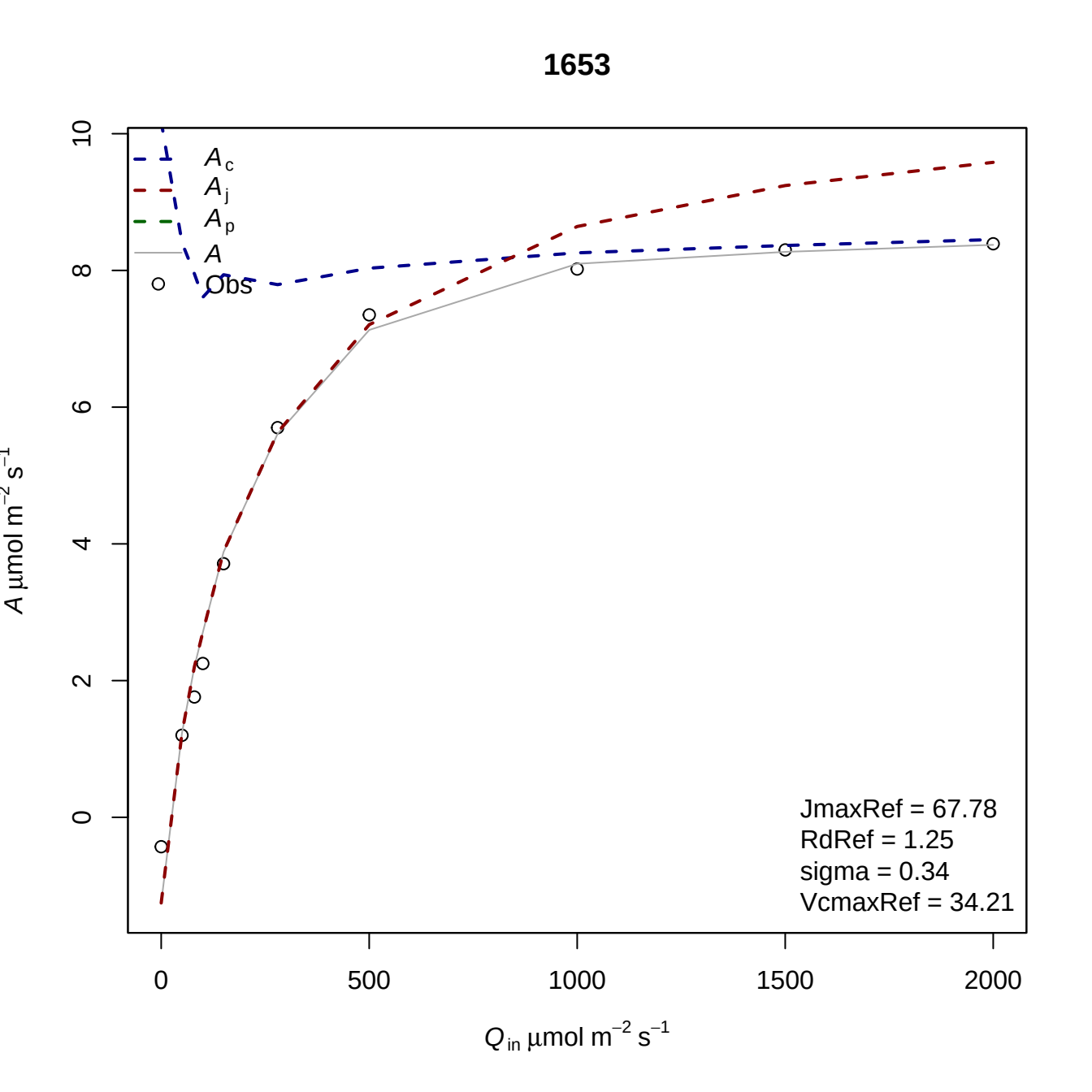

1648
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 241.64
RdRef = 0.77
sigma = 0.25
VcmaxRef = 74.9
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

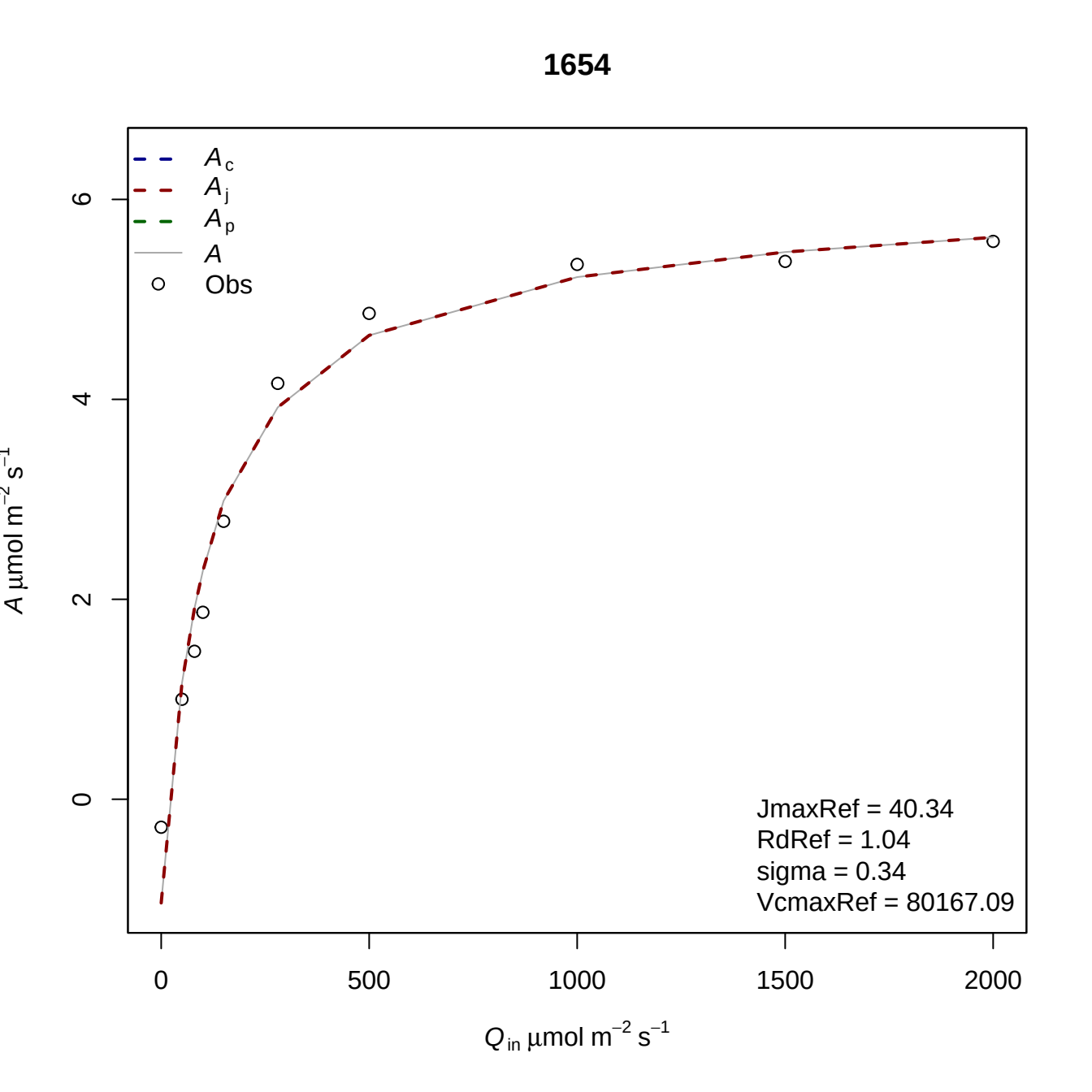

1649
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 264.28
RdRef = 0.66
sigma = 0.38
VcmaxRef = 88.26
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
25
0
500
1000
1500
2000

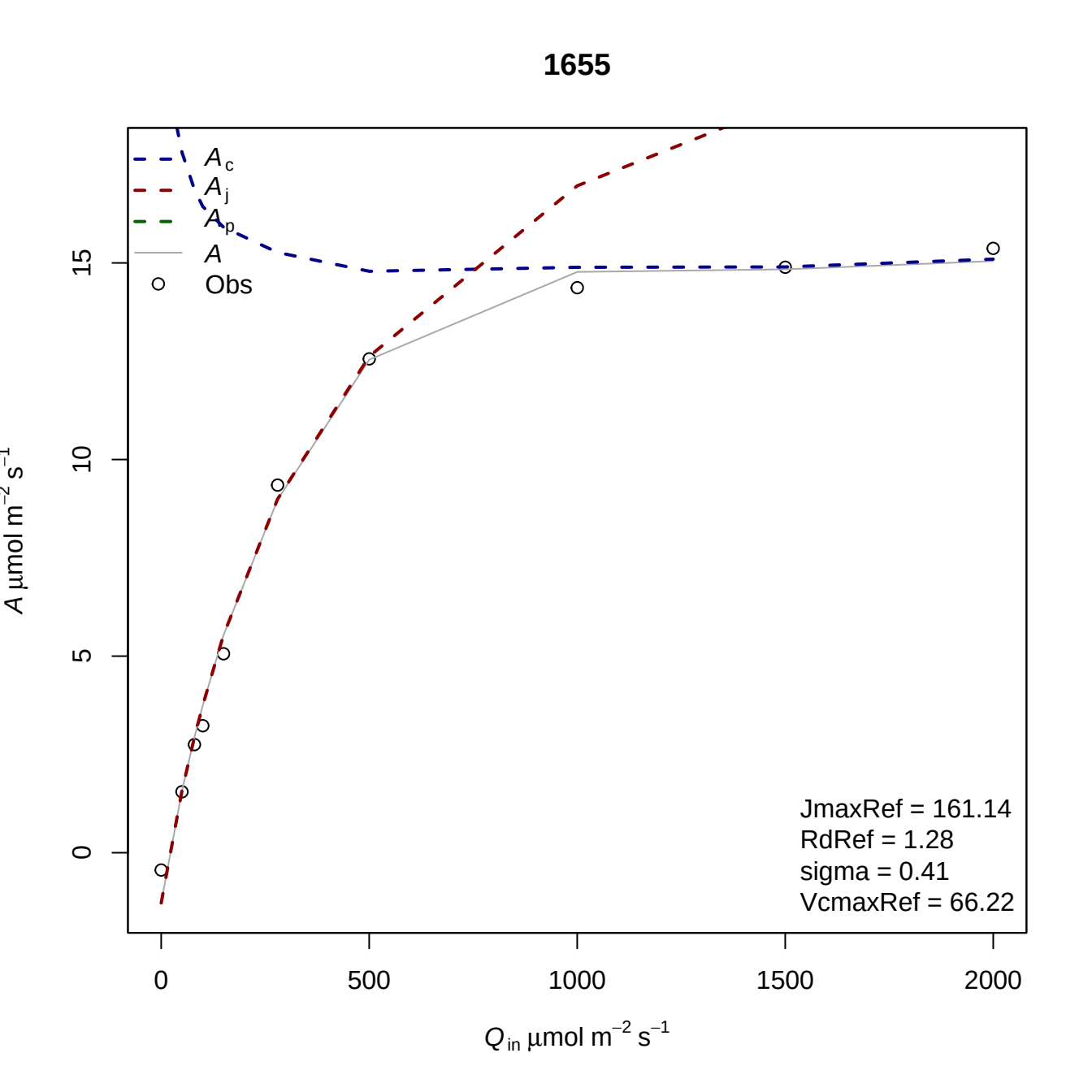

1650
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 221.27
RdRef = 0.92
sigma = 0.25
VcmaxRef = 75.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000

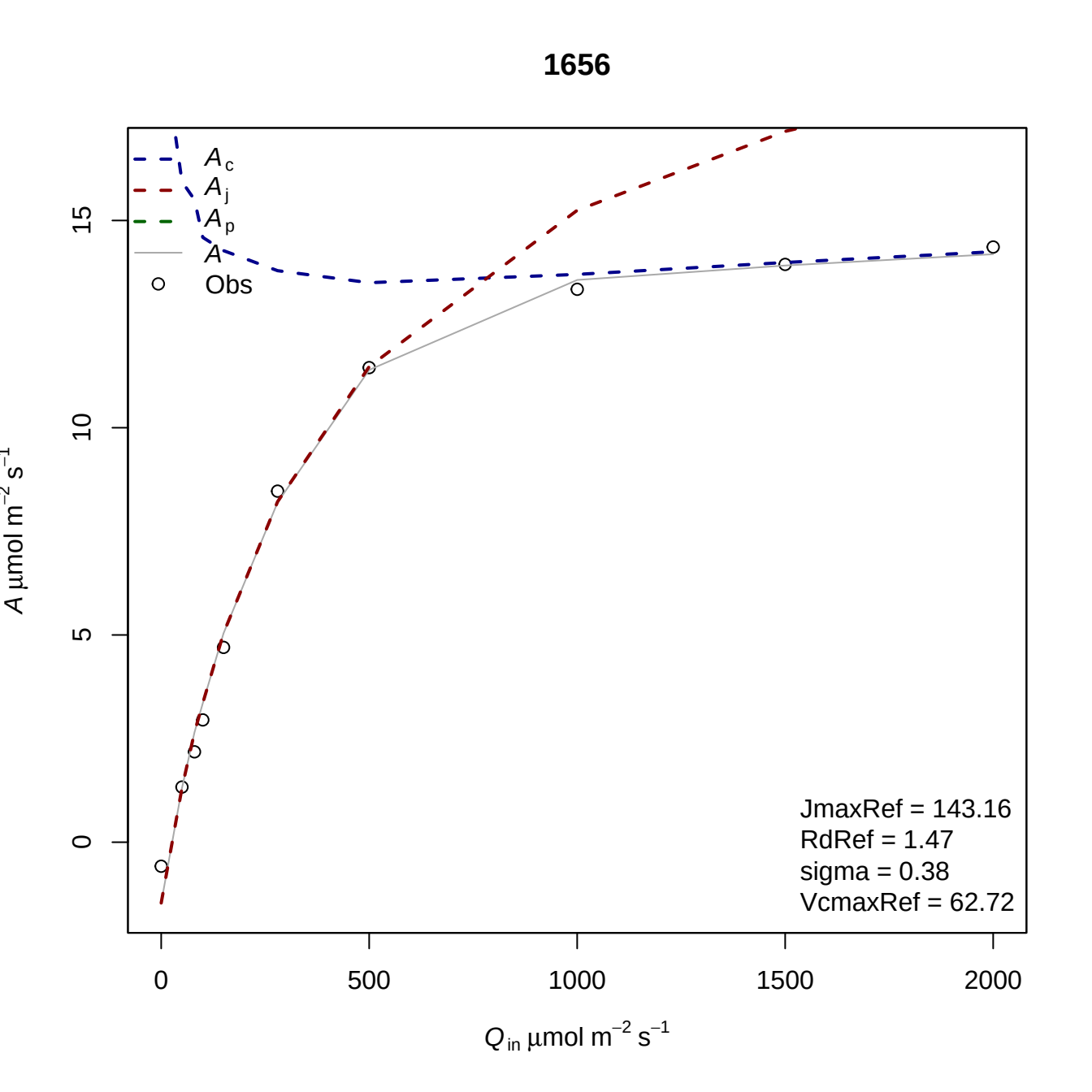

1651
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 58.1
RdRef = 1.13
sigma = 0.36
VcmaxRef = 28.08
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
0
500
1000
1500
2000

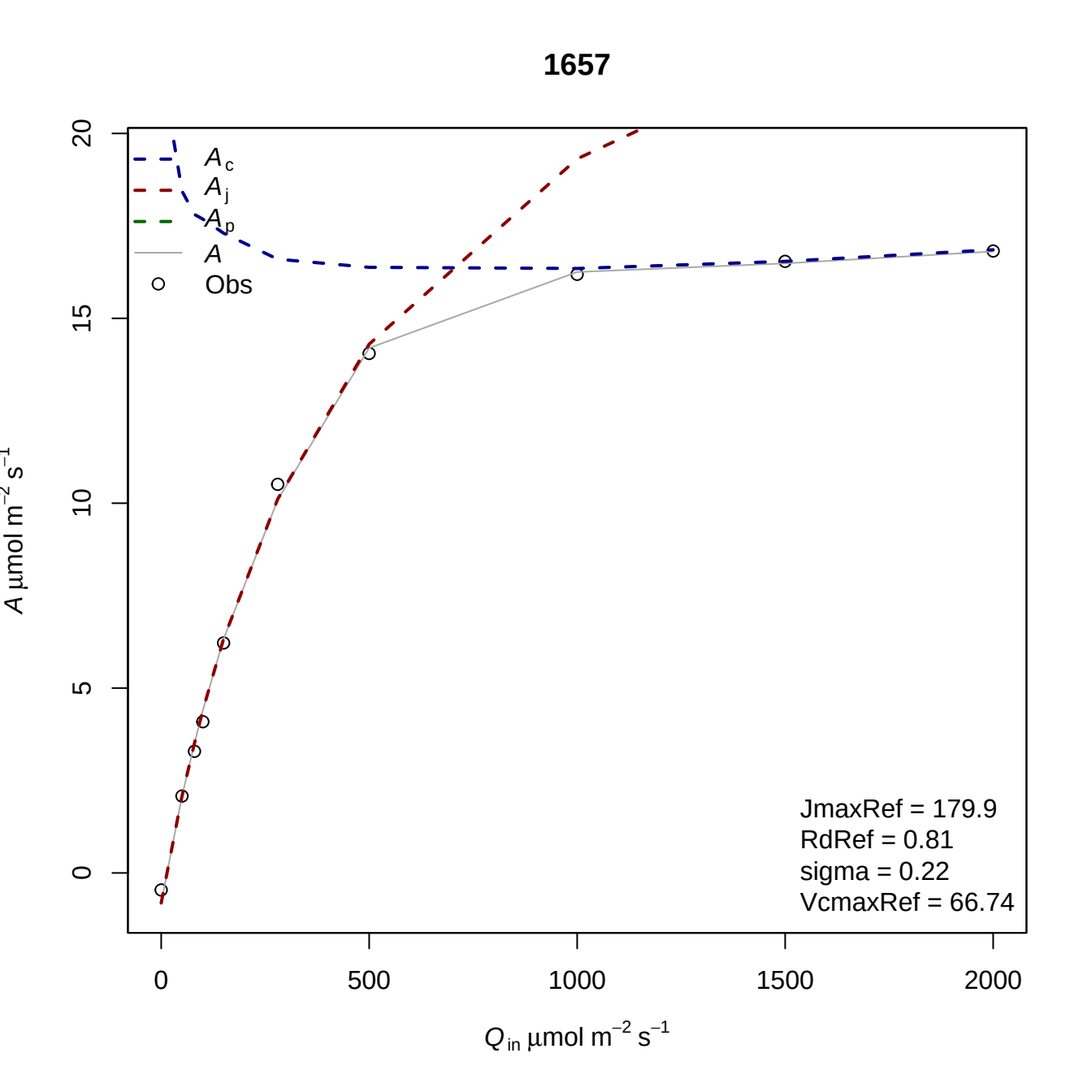

1652
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 74.95
RdRef = 1
sigma = 0.28
VcmaxRef = 36.13
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

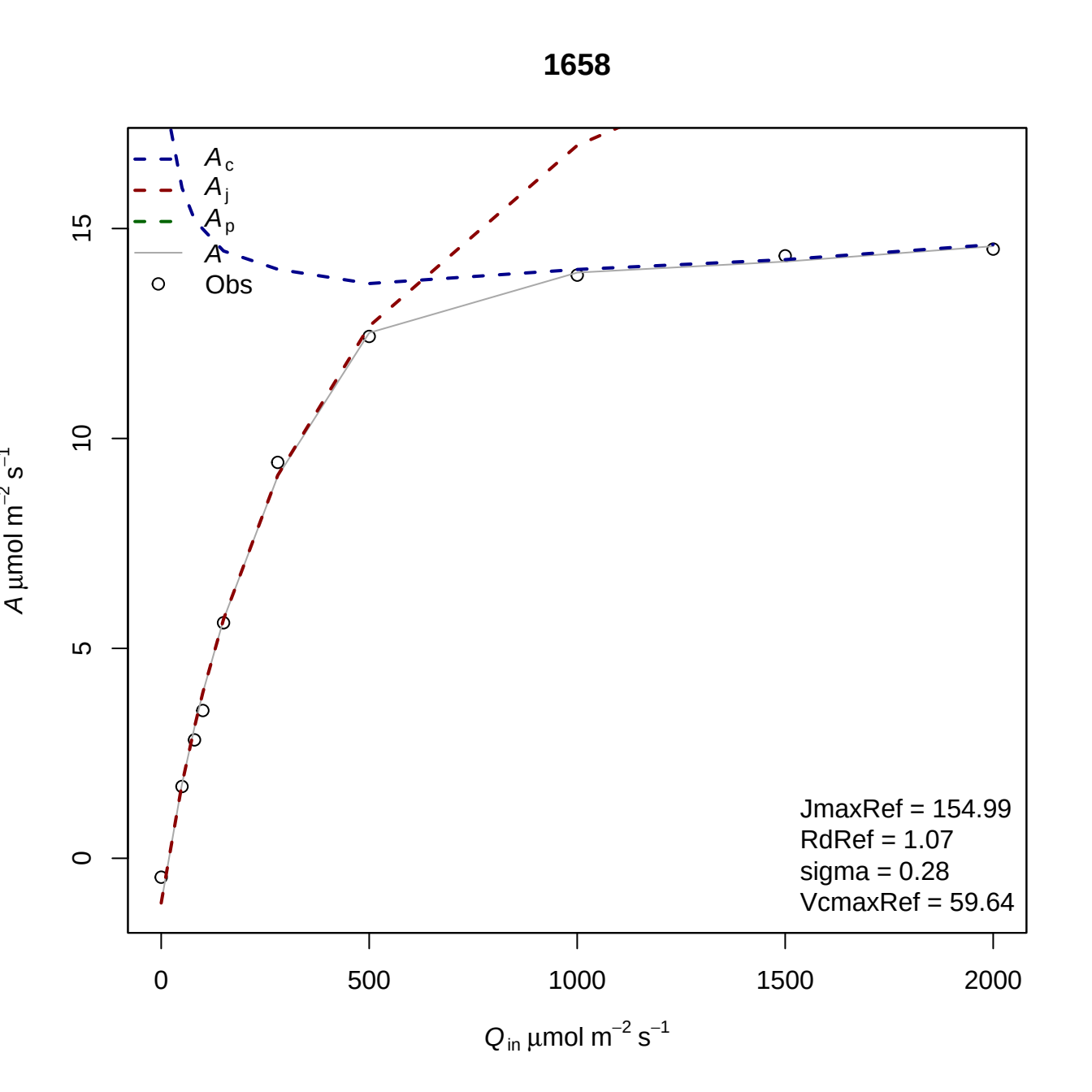

1653
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 67.78
RdRef = 1.25
sigma = 0.34
VcmaxRef = 34.21
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
8
10
0
500
1000
1500
2000

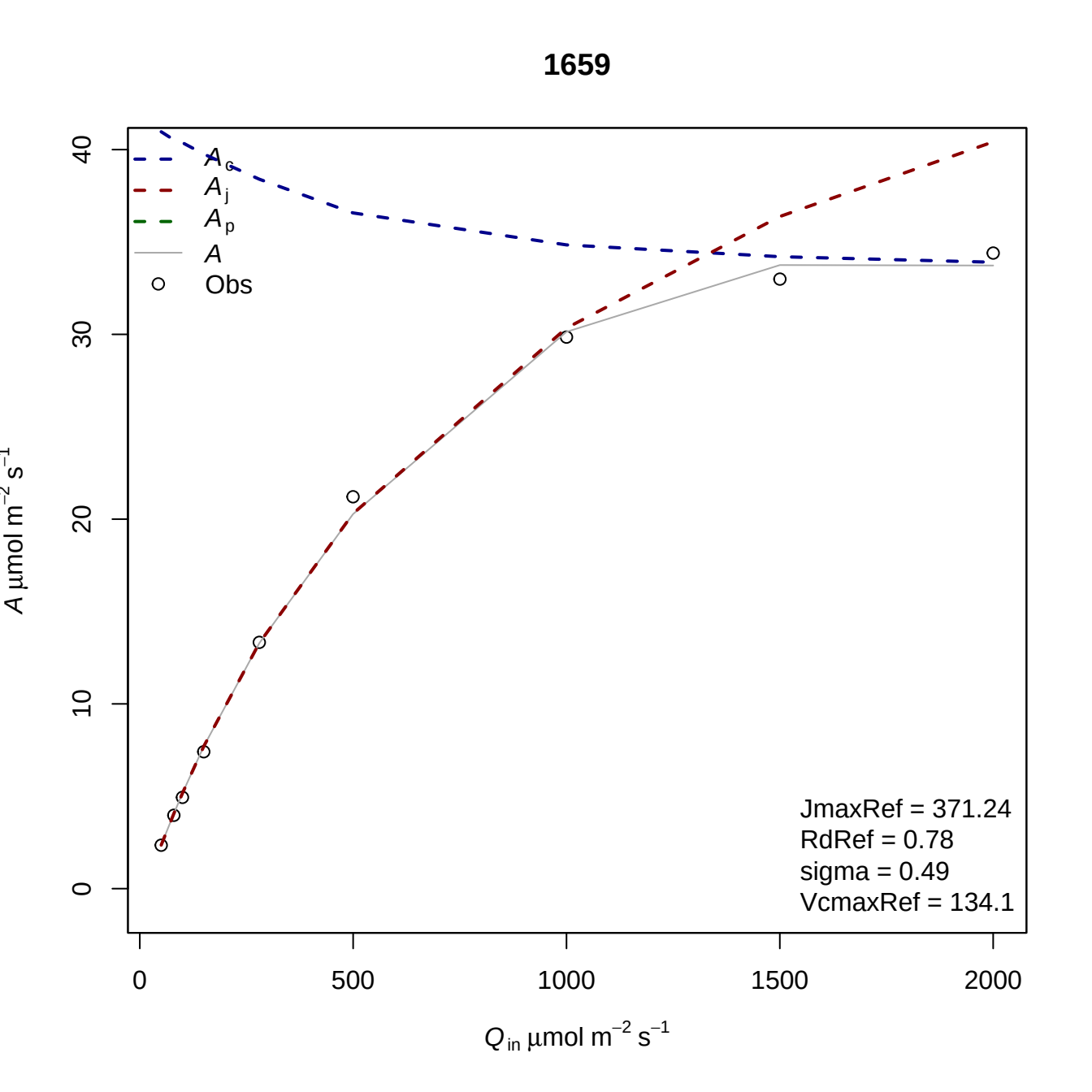

1654
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 40.34
RdRef = 1.04
sigma = 0.34
VcmaxRef = 80167.09
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
2
4
6
0
500
1000
1500
2000

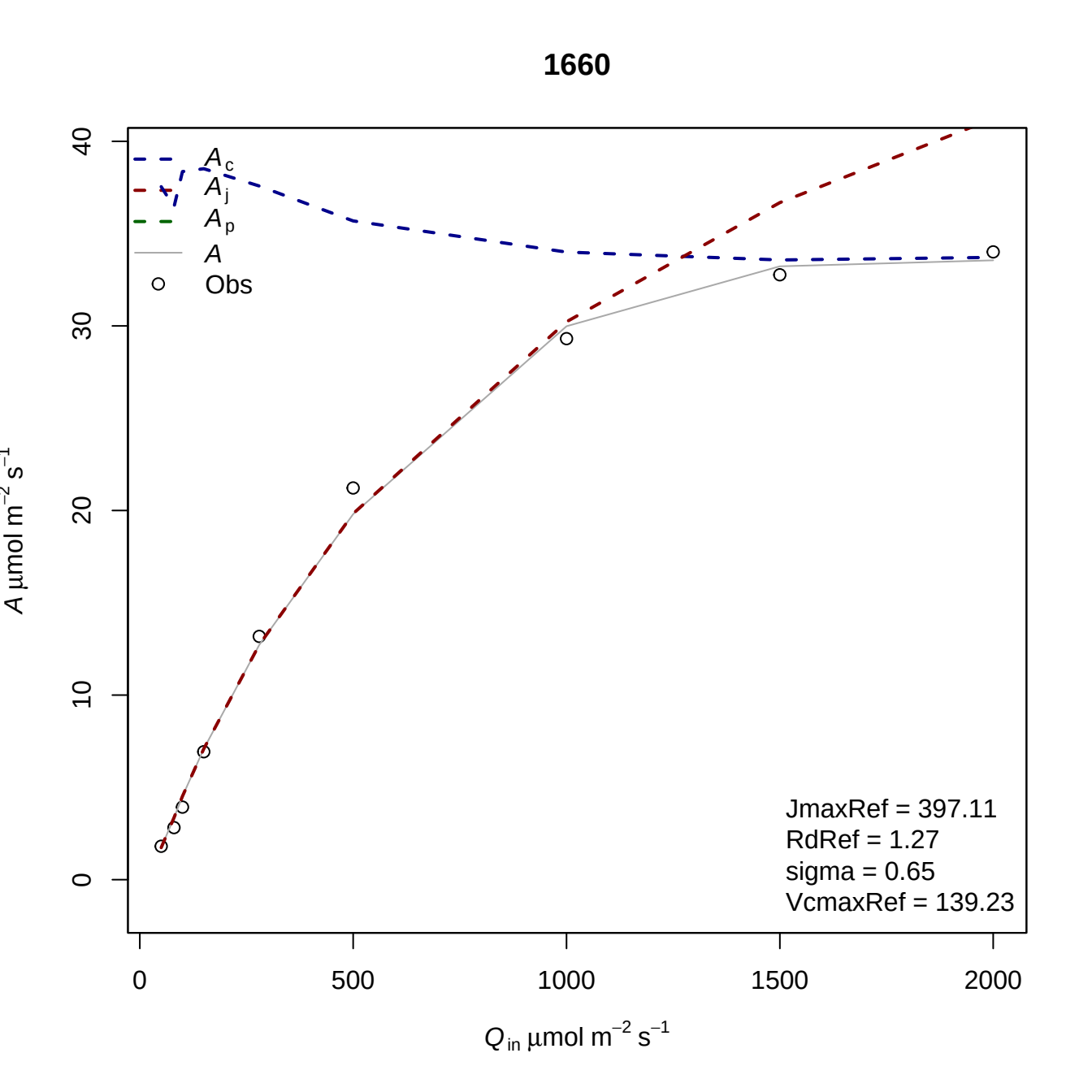

1655
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 161.14
RdRef = 1.28
sigma = 0.41
VcmaxRef = 66.22
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

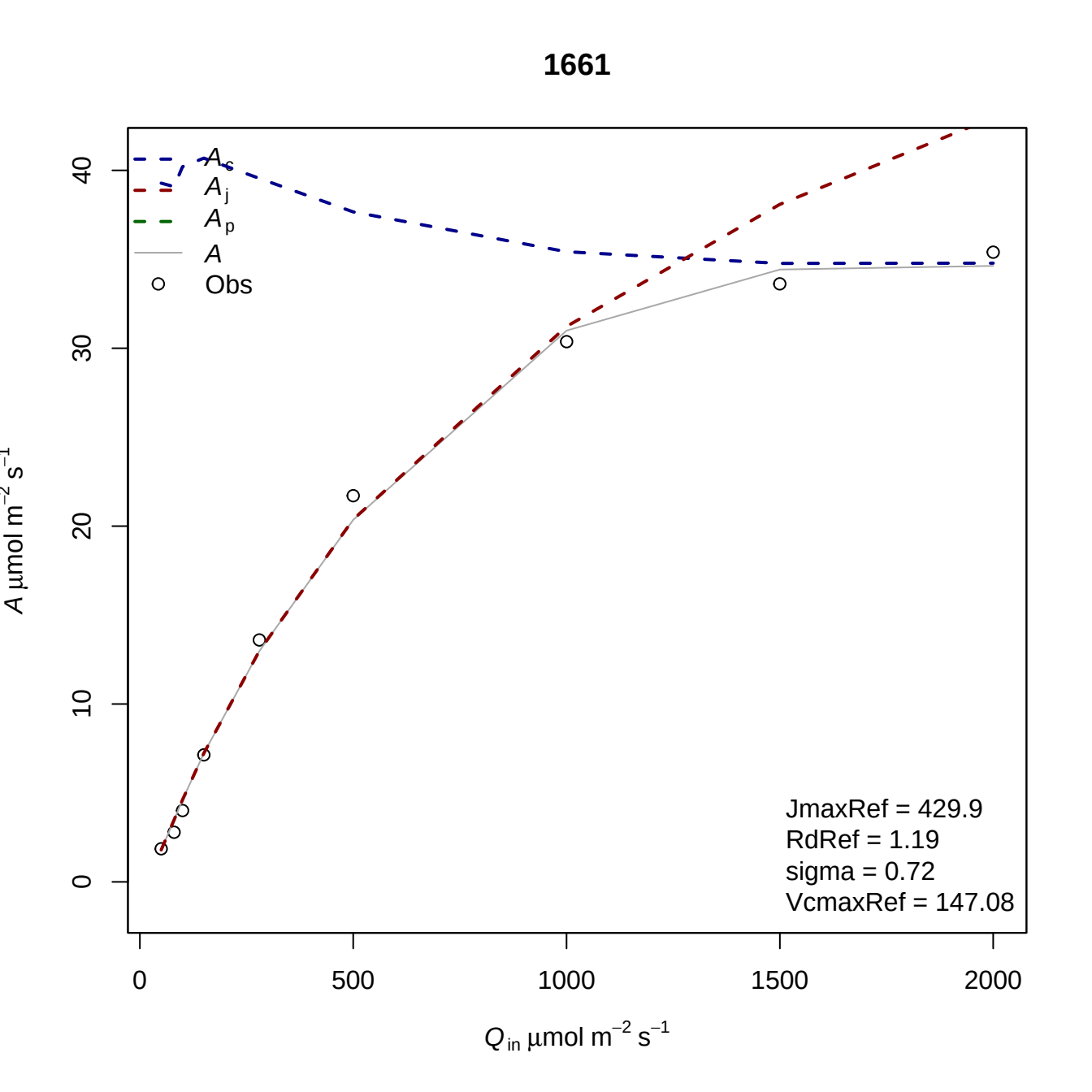

1656
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 143.16
RdRef = 1.47
sigma = 0.38
VcmaxRef = 62.72
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000

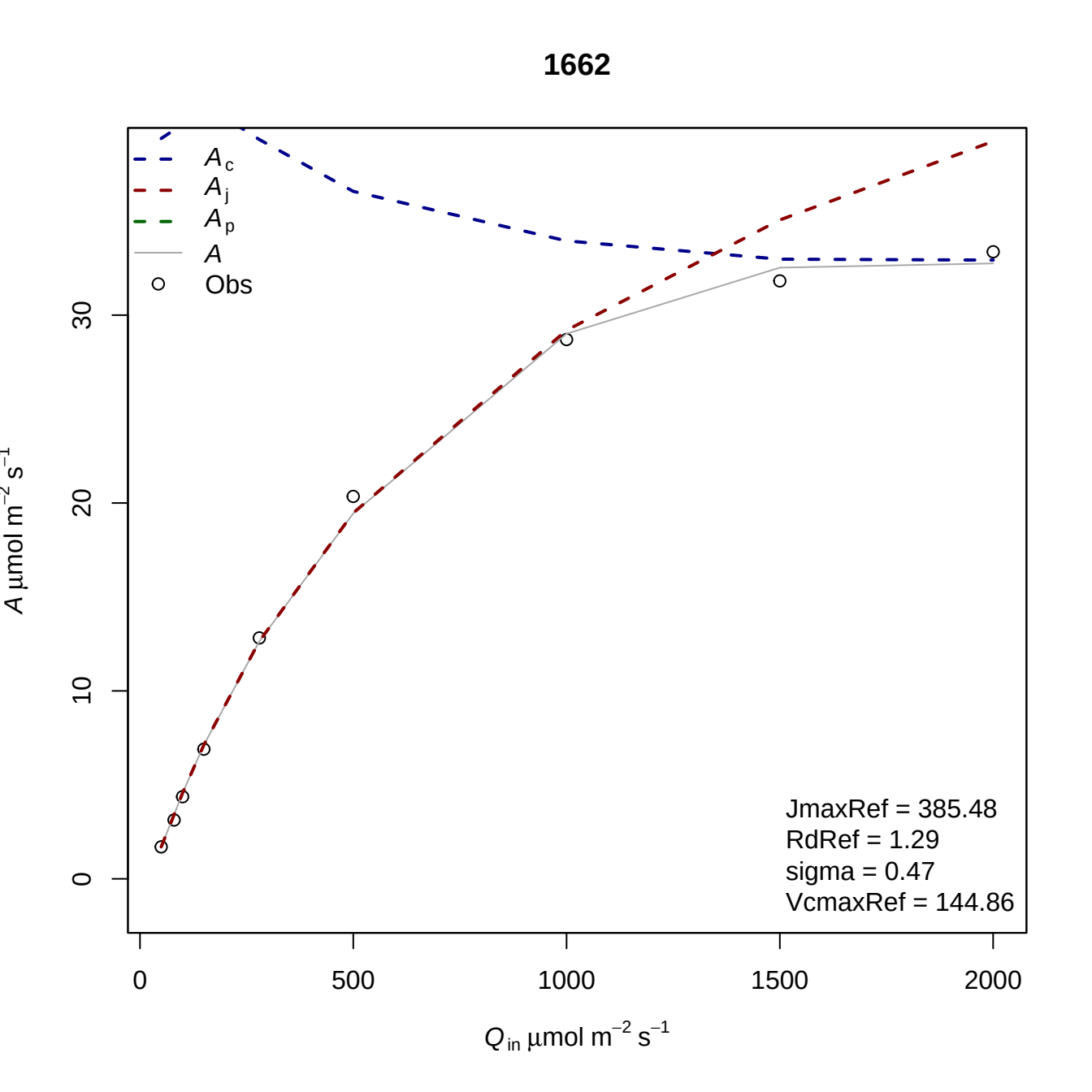

1657
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 179.9
RdRef = 0.81
sigma = 0.22
VcmaxRef = 66.74
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
20
0
500
1000
1500
2000


1658
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 154.99
RdRef = 1.07
sigma = 0.28
VcmaxRef = 59.64
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
5
10
15
0
500
1000
1500
2000


1659
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 371.24
RdRef = 0.78
sigma = 0.49
VcmaxRef = 134.1
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000


1660
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 397.11
RdRef = 1.27
sigma = 0.65
VcmaxRef = 139.23
$A$ µmol m$^{-2}$ s$^{-1}$
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$
0
10
20
30
40
0
500
1000
1500
2000


1661
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 429.9
RdRef = 1.19
sigma = 0.72
VcmaxRef = 147.08
0
10
20
30
40
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$


1662
$A_c$
$A_j$
$A_p$
$A$
Obs
JmaxRef = 385.48
RdRef = 1.29
sigma = 0.47
VcmaxRef = 144.86
0
10
20
30
$A$ µmol m$^{-2}$ s$^{-1}$
0
500
1000
1500
2000
$Q_{in}$ µmol m$^{-2}$ s$^{-1}$